SCALAR FIELDS FROM THREE-DIMENSIONAL LATTICE YANG–MILLS THEORY

Abstract. We construct continuum correlations of the centered scalar plaquette-energy observable for the three-dimensional SU(2) Wilson measure. Its formal classical counterpart is the Yang–Mills action density $\mathrm{tr}(F^2)$. On each fixed torus, connected correlations converge along the full cutoff sequence at every positive physical inverse coupling. Infinite-volume limits are subsequential. Limits through periodic Gibbs states admit canonical clustering components whose Schwinger functionals satisfy the Osterwalder–Schrader axioms on Schwartz tests flat at the coincidence diagonals. A point-insertion bound reconstructs a Hermitian scalar Wightman field in each component. At sufficiently small reference coupling, a uniform short-distance two-point law determines the physical inverse coupling and proves nontriviality. The proof combines a local complex-source expansion, reflection and chessboard estimates, comparison of consecutive cutoffs, and analytic continuation of the separated Euclidean kernels.

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1. Introduction

We study a scalar observable whose formal classical counterpart is $\operatorname{tr}(F^2)$, the Yang–Mills action density. Its quantum correlations are defined through centered Wilson plaquette energies.

Let $L \geq 13$ be odd and $a_K = L^{-K}$ for $K \in \mathbb{Z}$. On $a_K\mathbb{Z}^3$, a configuration assigns $U(b) \in \mathrm{SU}(2)$ to every oriented bond, with $U(\bar{b}) = U(b)^{-1}$. For a positive plaquette p, let $x(p)$ be its barycentre and put

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{tr}\partial U(p).$$

Here $\partial U(p)$ is the ordered product around p, and a_p is the gauge-invariant scalar plaquette energy. The Wilson action is normalized as in [Wil74, §III]. On the infinite lattice $a_K\mathbb{Z}^3$, a periodic Gibbs state is a probability law invariant under a finite-index subgroup of lattice translations whose conditional law on each finite set of bonds has density proportional to the Wilson factor for the plaquettes meeting that set. The boundary bonds are held fixed. Such states exist at every K and $\beta > 0$. For a compactly supported test f, put

$$\mathcal{O}_K(f) = a_K^{-1}\sum_p f(x(p))a_p, \qquad \mathcal{O}_K^{\nu}(f) = \mathcal{O}_K(f) - \nu(\mathcal{O}_K(f)).$$

The sum runs over positive plaquettes in all three orientations. We call $\mathcal{O}_K^{\nu}$ the smeared, centered scalar plaquette-energy density. The plaquette definition fixes the observable and its normalization; (74) gives the exact lattice relation. The notation $\operatorname{tr}(F^2)$ does not denote the pointwise square of a separately constructed continuum field strength.

For $n \geq 1$, let $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ consist of complex Schwartz functions whose derivatives of every order vanish whenever two arguments coincide. Let $\mathcal{S}_{+}(\mathbb{R}^{3n})$ be its subspace of functions also flat outside $\{x_{13} > 0, \dots, x_{n3} > 0\}$, and set $\mathcal{S}_{\neq}(\mathbb{R}^0) = \mathcal{S}_{+}(\mathbb{R}^0) = \mathbb{C}$. We use the seminorms

$$|\varphi|_k = \sup_{x,\,|\alpha|\leq k} (1+|x|^2)^{k/2}|\partial^{\alpha}\varphi(x)|.$$

For a Euclidean isometry h and a vector v, put

$$(h\cdot\varphi)(x_1,\dots,x_n) = \varphi(h^{-1}x_1,\dots,h^{-1}x_n),$$
$$(\tau_v\varphi)(x_1,\dots,x_n) = \varphi(x_1-v,\dots,x_n-v).$$

Write $\theta(x_1,x_2,x_3) = (x_1,x_2,-x_3)$ and $(\Theta\varphi)(x_1,\dots,x_n) = \varphi(\theta x_1,\dots,\theta x_n)$. These actions are the identity in degree zero. The following theorem states the Euclidean axioms of Osterwalder and Schrader [OS73, §§2–3], with an explicit growth bound.

Theorem 0 (Osterwalder–Schrader theorem for the plaquette-energy density)**.** There exist an odd integer $L \geq 13$, a physical inverse coupling $b > 0$, integers $K_j \to \infty$, and periodic Gibbs states ρ_j, ν_j at spacing $a_{K_j} = L^{-K_j}$ and inverse coupling $\beta_j = L^{K_j}b$ with the following properties. For every tuple of smooth compactly supported functions $f_1,\dots,f_n$ with pairwise disjoint supports, the limit

$$\mathsf{S}_n(f_1\otimes\cdots\otimes f_n) = \lim_{j\to\infty}\rho_j\left(\prod_{i=1}^{n}\mathcal{O}_{K_j}^{\nu_j}(f_i)\right)$$

exists. These moments extend uniquely to continuous linear functionals on $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$, with $\mathsf{S}_0(z) = z$ for $z \in \mathbb{C}$. They have the following properties.

(E0′) *Regularity.* There is a finite $C > 0$ such that, for every $n \geq 1$ and $\varphi \in \mathcal{S}_{\neq}(\mathbb{R}^{3n})$,

$$|\mathsf{S}_n(\varphi)| \leq 2C^n(n!)^8|\varphi|_{8n}.$$

Moreover, $\mathsf{S}_n(\bar{\varphi}) = \overline{\mathsf{S}_n(\varphi)}$.

(E1) *Euclidean invariance.* For every Euclidean isometry h, $\mathsf{S}_n(h\cdot\varphi) = \mathsf{S}_n(\varphi)$.

(E2) *Reflection positivity.* For every $N \geq 0$ and $\varphi_n \in \mathcal{S}_+(\mathbb{R}^{3n})$, $0 \leq n \leq N$,

$$\sum_{n,n'=0}^{N} \mathsf{S}_{n+n'}(\Theta\bar{\varphi}_n \otimes \varphi_{n'}) \geq 0.$$

(E3) *Symmetry.* The functional S_n is unchanged by permutations of its n arguments.

(E4) *Clustering.* For $n, n' \geq 0$, $\varphi \in \mathcal{S}_+(\mathbb{R}^{3n})$, $\psi \in \mathcal{S}_+(\mathbb{R}^{3n'})$ and $0 \neq v \in \mathbb{R}^2 \times \{0\}$,

$$\lim_{s\to\infty} \mathsf{S}_{n+n'}(\Theta\bar{\varphi} \otimes \tau_{sv}\psi) = \mathsf{S}_n(\Theta\bar{\varphi})\mathsf{S}_{n'}(\psi).$$

On their reflection Hilbert space these functionals reconstruct a Hermitian scalar Wightman field on $\mathbb{R}^{1+2}$ with positive energy, a unique invariant vacuum, a dense common Poincaré-invariant polynomial domain, covariance, local commutativity and tempered vacuum distributions. Its Euclidean continuations recover the separated moments and the functionals S_n on the stated spaces. The Hilbert space has dimension at least two. ■□

Proof. Fix a positive source radius and choose the reference coupling $g > 0$ below the ceiling in theorem C. Choose b in the nonempty interval $I(g)$ of theorem B and take a volume-first continuum limit at this label. Select one of its canonical clustering components. Its integrating states ρ_j are obtained by invariant conditioning of the original states ν_j, which remain in the centering. The component construction and the Euclidean properties are propositions 93.5 and 111.2 and theorem AP. The point-insertion bounds supplied there give the reconstruction by theorem AO; nontriviality is corollary 127.11. □

The Euclidean assertions use the flat Schwartz spaces specified above and leave coincident-point contact terms unspecified. The following theorems give the lattice limits, parameter ranges and short-distance normalization.

For $m \in \mathbb{Z}$, $K + m \geq 1$, and $\ell = 2L^m$, the torus lattice is

$$\Lambda_{K,\ell} = L^{-K}\mathbb{Z}^3/\ell\mathbb{Z}^3 \subset T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3.$$

On this finite lattice, put $A(c, U) = \sum_p c(x(p))a_p(U)$ and use normalized product Haar measure dU. A complex source w changes the partition function to

$$Z_{K,\ell,\beta}[w] = \int \exp[-A(\beta - L^K w, U)]\, dU. \tag{1}$$

Theorem A (The continuum limit on a fixed torus)**.** There is an odd integer $L \geq 13$ for which the following holds. Let $m \in \mathbb{Z}$, $\ell = 2L^m$, $b > 0$ and $n \geq 2$. For real Lipschitz functions $f_1, \ldots, f_n$ on T_ℓ whose supports have empty common intersection, the connected correlation

$$S_n^{(\ell,b)}(f_1, \ldots, f_n) = \lim_{K\to\infty} \partial_{z_1} \cdots \partial_{z_n} \log \frac{Z_{K,\ell,L^K b}[\sum_i z_i f_i]}{Z_{K,\ell,L^K b}[0]}\bigg|_{z=0}$$

exists along the full sequence of sufficiently fine lattices. Convergence is locally uniform in $b > 0$, with the summable cutoff rate of (2211) after a fixed dilation. The limit is smooth in b, and

$$\partial_b^\nu S_n^{(\ell,b)}(f_1, \ldots, f_n) = (-1)^\nu S_{n+\nu}^{(\ell,b)}(\underbrace{1, \ldots, 1}_{\nu}, f_1, \ldots, f_n).$$

For every $s \in \mathbb{Z}$, with $f^{(s)}(x) = f(L^{-s}x)$,

$$S_n^{(\ell,b)}(f_1, \ldots, f_n) = L^{sn} S_n^{(L^s\ell, L^s b)}(f_1^{(s)}, \ldots, f_n^{(s)}).$$

If $K_j \to \infty$ and $L^{-K_j}\beta_j \to b > 0$, the same limit is obtained along (K_j, β_j). For pairwise disjoint test supports these correlations are the restrictions of unique distributions on the space of distinct points of T_ℓ^n. Different positive labels b give different limiting families on the same torus. ■□

The proof is given by Theorem AS and Corollary 119.2; the distributional assertion follows from Section 93 and finite dilation. Corollary 127.7 proves the distinction of positive labels. The powers of the side in the cutoff estimate are retained throughout that proof.

The lattice conventions and block average are specified in sections 3 to 5. All distances on T_ℓ in the source norm are physical distances. For $M = L^{m'}$, define

$$\|w\|_{\flat;T_\ell} = \max\left\{\sup_{T_\ell}|w|, \frac{40M}{\sqrt{3}}\operatorname{Lip}_\infty(w;T_\ell)\right\}, \qquad \operatorname{Lip}_\infty(w;T_\ell) = \sup_{x\neq y}\frac{|w(x)-w(y)|}{|x-y|_\infty}. \tag{2}$$

Holomorphy on a ball of sources means holomorphy on every finite-dimensional affine slice of that ball.

At a fixed admissible threshold, we use the coefficient-determined real coupling recursion of Theorem D, with its canonical auxiliary data. Write it as

$$x_0(\beta) = \beta, \qquad x_j(\beta) = L^{-1}x_{j-1}(\beta) - b_j, \qquad |b_j| \le B_\sharp.$$

The increments in this recursion are computed at zero source. Set $\bar{B} = LB_\sharp/(L-1)$ and, for $g > 0$, set $X_g = L(g^{-2} + B_\sharp)$. The first-passage index at g is the least index $K \ge 0$ for which $x_K(\beta) \le X_g$. We call (K, β) a first-passage datum when this index is positive.

A separated tuple consists of real $C_c^1(\mathbb{R}^3)$ functions with pairwise disjoint supports. Its limiting moment is denoted by $\sigma(f_1, \dots, f_n)$, and $\sigma(\varnothing) = 1$.

Theorem B (Infinite-volume limits and reconstruction)**.** The canonical construction constants can be fixed, with L as above, so that for every $\varrho > 0$ there is $\gamma_\varrho > 0$ with the following properties. Fix $0 < g \le \gamma_\varrho$, using this one assignment of the coupling recursion throughout.

Let (K_j, β_j) be first-passage data at g, with $K_j \to \infty$, and let ν_j be any periodic Gibbs states at these data. There is a common subsequence on which

$$\sigma(f_1, \dots, f_n) = \lim_j \nu_j\left(\prod_{i=1}^n \mathcal{O}_{K_j}^{\nu_j}(f_i)\right)$$

exists for every separated tuple, and on which the positive labels $a_{K_j}\beta_j$ converge. The limit has zero one-point function, is invariant under all Euclidean isometries, and is reflection positive at every plane. Its reflection Hilbert space is separable and its joint energy–momentum spectrum lies in $\{(E, q) : E \ge |q|\}$.

Every such limit has a canonical component measure $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi\in\mathsf{X}})$ as in Theorem AL: X is compact metrizable, λ has full support, $\xi \mapsto \sigma_\xi(X)$ is continuous for each tuple, and $\sigma(X) = \int \sigma_\xi(X)\, d\lambda(\xi)$. The components are limits of the same lattice sequence after invariant conditioning, with the original laws ν_j retained for centering. Each component is Euclidean invariant and reflection positive, has $\sigma_\xi(f) = c_\xi \int f$ with a common bound on $|c_\xi|$, and satisfies

$$|\sigma_\xi(X \cdot \tau_v Y) - \sigma_\xi(X)\sigma_\xi(Y)| \le C_g(X, Y)|v|^{-1} \qquad (|v| \ge r(X, Y)),$$

uniformly in ξ. The measure is unique in the precise sense of Proposition 101.1.

The component moments extend uniquely to Schwinger functionals on $\mathcal{S}_{\neq}$ satisfying

$$|\mathsf{S}_n^{\sigma_\xi}(\varphi)| \le 2(C_E^g)^n (n!)^8 |\varphi|_{8n},$$

with the common constant in (2100) and the Schwartz seminorm (1871). They obey Euclidean invariance, symmetry, reflection positivity and Schwartz clustering in the exact test-space formulation of Proposition 111.2. On each component's reflection Hilbert space they reconstruct one Hermitian scalar Wightman field on $\mathbb{R}^{1+2}$, with positive energy, a unique invariant vacuum, a dense common Poincaré-invariant polynomial domain, covariance, local commutativity and tempered vacuum distributions. Its Euclidean continuations recover all the separated moments and the flat-Schwartz Schwinger functionals.

The original limit also has this reconstruction, except that vacuum uniqueness and clustering are not asserted. Its zero-energy space is $L^2(\mathsf{X},\lambda)$, with the vacuum vector corresponding to the constant 1. All growth and insertion constants are independent of the sequence and of its components.

The interval $I(g)=(g^{-2}+\bar B, X_g-\bar B]$ is nonempty. For every $b\in I(g)$ there are such limits with $a_{K_j}\beta_j=b$ at every index, obtained by taking the volume limit of cubic torus measures first. They are also limits along a suitable sequence of growing tori with the same (K_j,β_j). ■□

The infinite-volume conclusion is subsequential. The fixed-torus estimate retains a power of the volume and does not give uniqueness of these infinite-volume limits. Full rotation invariance for an arbitrary growing-torus limit has the additional criterion in Section 112.

The observable throughout is the scalar plaquette-energy density defined above. The reconstructed field is a scalar field. The construction makes no identification with a tensor-valued field strength and asserts no mass gap.

Theorem AP proves the component and reconstruction assertions. Existence and the comparison of limiting orders are proved in Section 92.

For $0<\vartheta<1$, let $\mathfrak{T}_E(d;\vartheta)$ consist of pairs of nonzero nonnegative Lipschitz functions (f,h) on $E=\mathbb{R}^3$ or T_ℓ such that

$$\operatorname{supp} f\subset \overline{B}(x,d),\qquad \operatorname{supp} h\subset \overline{B}(y,d),\qquad d\le \operatorname{dist}(\operatorname{supp} f,\operatorname{supp} h)\le 6d,$$
$$\int f\ge \vartheta\|f\|_\infty d^3,\qquad \int h\ge \vartheta\|h\|_\infty d^3,\qquad \operatorname{Lip} f\le \|f\|_\infty/(\vartheta d),\qquad \operatorname{Lip} h\le \|h\|_\infty/(\vartheta d).$$

On a torus also require $20d\le\ell$. The two supports then lie in one ball of radius $5d$, and the kernel integral

$$\mathcal{K}_{\mathrm{free}}(f,h)=\iint \frac{f(x)h(y)}{|x-y|^6}\,dx\,dy$$

uses their common Euclidean lift in the torus case. For a moment functional σ, write $\kappa_2^\sigma(f,h)=\sigma(f,h)-\sigma(f)\sigma(h)$.

Theorem C (The short-distance two-point law)**.** There is a further ceiling $0<\gamma_W\le\gamma_\varrho$, independent of ϑ, with the following property. Fix $0<g\le\gamma_W$ and $0<\vartheta<1$. There are $d_0=d_0(g,\vartheta)>0$ and a nondecreasing function $\varepsilon_{g,\vartheta}:(0,d_0]\to[0,1/3]$ with $\varepsilon_{g,\vartheta}(d)\to 0$ as $d\downarrow 0$ such that every limit σ of Theorem B and every one of its canonical components obeys

$$\kappa_2^\sigma(f,h)=\frac{9}{16\pi^2}\,b^{-2}\mathcal{K}_{\mathrm{free}}(f,h)(1+e_{\sigma,f,h}),\qquad |e_{\sigma,f,h}|\le\varepsilon_{g,\vartheta}(d),$$

for every $0<d\le d_0$ and every C^1 pair $(f,h)\in\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$. Here b is the limiting label of the defining lattice sequence; the components have the same label as their parent. The error bound is uniform in the limit, component, test pair and its position.

The label is uniquely determined by the connected two-point function. Along every defining sequence at g the labels converge to this same value. The connected two-point function is positive on each indicated pair, and each reflection Hilbert space has dimension at least two. ■□

Theorem AW proves the law and label recovery; Corollary 127.11 gives their component versions and nontriviality. The finite-cutoff estimate and its torus and growing-torus limits are proved in Section 127.

1.1. **The source estimates behind the construction.** The finite-lattice construction begins in section 2. Regular critical backgrounds, their Gaussian fluctuation measures and a positive history decomposition give the exact renormalisation theorem, theorem D; its proof is completed in section 71. The complex-source construction begins in section 72 and extends these measures to a complex plaquette coefficient. The quadratic part of a source changes the Gaussian damping and must be retained separately; the remaining terms are expanded about the conditional mean.

The resulting local source energies have uniform bounds, although their complete sum need not be bounded uniformly in the cutoff.

Separated source derivatives avoid this last difficulty. A local energy contributes to a mixed derivative only when its localization domain meets all the source supports. At sufficiently fine scales, this forces a large domain and its exponential decay makes the scale sum converge. Reflection positivity and chessboard estimates turn these local bounds into bounds on cumulants, uniform in the cutoff and volume. A countable determining family then gives simultaneous subsequential limits on all separated tuples; flatness at the coincidence diagonals permits their Schwartz extensions.

Full convergence on a fixed torus requires a separate comparison. We discard histories whose first large-field region occurs too early and compare the remaining flows after their early interactions have decayed. Classical backgrounds on consecutive meshes are compared in a common gauge. The resulting real-source estimate is continued to complex sources and differentiated after subtracting the local contact terms. This gives a summable cutoff difference. Dilation places any positive physical inverse coupling in the interval where the comparison applies.

For infinite-volume Gibbs states, reflection at coordinate and diagonal planes restricts the translation spectrum. The slab insertion bound yields analytic point fields and controls their growth near a narrow slab. Angular continuation then proves full rotation invariance. These estimates hold for every limit in a Gibbs frame, including limits obtained by invariant conditioning. The canonical component measure therefore has rotation-invariant components with a uniform clustering bound.

Finally the point-insertion estimate gives polynomial energy bounds for smeared fields. Their ordered products have tempered boundary values. Complex rotation gives Lorentz covariance, and continuation through spacelike configurations gives locality. The clustering estimate makes the vacuum of each component unique.

For the short-distance law we transport the covariance through the small-field history. The linearized curvature and its Gaussian covariance produce the squared dipole kernel. Four fluctuation cumulants control the departure from this Gaussian calculation; locality and the rarity of early large-field regions control the remaining history. A choice of intermediate scales makes all these errors small before the two-state comparison transfers the estimate to arbitrary admissible volumes and periodic Gibbs states. Shrinking one fixed test pair then recovers the physical label.

2. Finite renormalisation and critical backgrounds

For the finite renormalisation construction, write $A(U) = A(1, U)$, so the bare measure is $e^{-\beta A(U)}\, dU$. Blocking by an odd integer L gives a tower $\mathbb{T}^{(0)}, \dots, \mathbb{T}^{(K)}$; on $\mathbb{T}^{(j)}$ each side has $2L^{m+K-j}$ sites. Write $N_j = \#\mathbb{T}^{(j)}$. Thus the final side length is $\ell = 2L^m$. We use the nonabelian block average $\mathcal{M}$ of section 5: the symmetrized contour mean on its small-curvature domain, with the global real extension specified in (37)–(38).

The theorem concerns a finite number of steps on each such torus. Its local expansion uses regular critical backgrounds and retains the fluctuation coordinates outside the integration regions.

Theorem D (Finite renormalisation)**.** There is a choice of an odd blocking factor $L \geq 13$, integers $r \geq 2$, $\mathfrak{p}_0 \geq 6r + 1$, a power $M = L^{m'}$, $m' \in \mathbb{Z}_{\geq 1}$, and positive construction constants, including

$$A_0,\ A_1,\ B_\sharp,\ \kappa,\ E_0,\ B,\ B_{\mathrm{fr}},\ C_{\mathrm{maj}},\ c_\star,\ c_{\mathrm{low}},\ \gamma, \qquad 0 < \gamma < 1,$$

fixed independently of the lattice and the number of steps, for which the following holds. Let $K \geq 1$, $m \geq 1$, and $x_0, \dots, x_K$ satisfy

$$x_j \geq \gamma^{-2} \quad (0 \leq j \leq K), \qquad x_j = L^{-1} x_{j-1} - b_j, \quad |b_j| \leq B_\sharp \quad (1 \leq j \leq K).$$

Put $\beta = x_0$, $g_j = x_j^{-1/2}$, $p_0(g) = A_0(\log g^{-2})^{\mathfrak{p}_0}$, and

$$R(x) = \min\{L^a : a \in \mathbb{Z}_{\geq 0},\ L^a \geq (\log x)^r\}.$$

Assume $MR(x_K) \le L^m$. For every $0 \le k \le K$, the effective measure $\rho_k\, dV_k = (\mathcal{M}^k)_*(e^{-\beta A}\, dU)$ has the following properties.

(i) It is the sum of finite positive history measures,

$$\rho_k\, dV_k = \sum_{h \in \mathcal{H}_k} \tau_h, \qquad \sum_h \mathsf{t}(h) = Z_{K,\ell,\beta} := \int e^{-\beta A}\, dU \in (0,1],$$

where $\mathsf{t}(h) = \tau_h(\mathcal{V}_k)$. The histories and their regions are defined by the local tests in section 44.

(ii) On its completed free and stored coordinates, each history measure is the marginal of the nonnegative integrable density

$$\mathsf{Stay}_h^{\mathrm{T}}\, \chi_k(\Omega_k) \exp(A_{k,h} + \Phi_{k,h})\, d\mathscr{M}_h.$$

The exponent $A_{k,h}$ is

$$-A(\beta_k(\cdot), \mathsf{U}) + \mathbf{E}_k(\mathsf{U}) + \mathbf{E}_k^{\mathrm{bd}}(\mathsf{U}, \mathfrak{q}) - \mathcal{E}_{k,\mathcal{S}(h)},$$

with the finite sums in (1472)–(1474); $\Phi_{k,h}$ is the sum of the retained quadratic and pivot exponents. On the support of the tests these two exponents are defined outside a Borel $\mathscr{M}_h$-null set and are independent of the admitted adapted representative U. The density is zero off that support.

(iii) The pointed regular terms are holomorphic on their fixed analytic cores, depend only on the background pair, are gauge invariant and conjugation real, and satisfy

$$|E^{(j)}(X, P, z)| \le E_0 e^{-(\kappa+1)d_j(X)}.$$

They depend only on $x_0, \dots, x_{j-1}$. The created boundary terms have norm at most $Be^{-\kappa d_j(X)}$ on their layered output domains. At fixed stored data, the sum of the oscillations of the retained exponents at (j, X) is at most $B_{\mathrm{fr}} \mathfrak{b}_{j-1}^2 e^{-\kappa d_j(X)}$, where $\mathfrak{b}_i = A_1(\log x_i)^{\mathfrak{p}_0}$. Earlier functions and their domains are unchanged on extending a history or the coupling prefix.

(iv) For $1 \le k \le K$, each history obeys

$$e^{\mathcal{E}_{k,\mathcal{S}(h)}} \mathsf{t}(h) \le \exp\left[C_{\mathrm{maj}} \left(N_k + \sum_{j=1}^{k} \#_j Z_j(h) \right) - c_\star \sum_{j=1}^{k} p_0(g_j)^2 n_j^{\mathrm{cr}}(h) \right].$$

Here $Z_j = \Lambda_j^c$, and n_j^{cr} counts the cells created at step j. The vacuum-normalized partition function satisfies

$$e^{-C_{\mathrm{low}}(x_k) N_k} \le e^{-E_{(k)}} Z_{K,\ell,\beta} \le e^{(C_{\mathrm{maj}}+2)N_k}, \qquad C_{\mathrm{low}}(x) = c_{\mathrm{low}} + \tfrac{9}{2} \log x.$$

Here $E_{(k)}$ is the accumulated vacuum constant defined in (1407).

There is in addition a unique coefficient-determined real coupling sequence from every $\beta \ge \gamma^{-2}$. It is defined successively by

$$x_0 = \beta, \qquad b_{j+1} = \mathsf{cf}(\widehat{E}^{(j+1)}[x_0, \dots, x_j]), \qquad x_{j+1} = L^{-1} x_j - b_{j+1}.$$

Each coefficient is defined before the next coupling, is independent of the lattice, and satisfies $|b_{j+1}| \le B_\sharp$. Here $\widehat{E}^{(j+1)}$ is the regular family of theorem Y, and cf is its second-moment coefficient (940). The sequence reaches $1 < x_J < \gamma^{-2}$ in finitely many steps. For every $K < J$, the preceding conclusions apply on every torus with $MR(x_K) \le L^m$. ■□

The principal issue is preservation of the density under a change of critical background. The small-field integral gives local regular terms, while the terms meeting a large-field boundary remain attached to their original scale. The kept quadratic forms retain coercivity on the stored fluctuations. A positive change of variables identifies the resulting density and proves its integrability at each step. The failed local tests then provide a gain for each created large-field cell; that gain absorbs the number of histories. The proof of theorem D is completed in section 71. The weighted bounds

(1484) and (1485) retain quantitative gains for large-field components and their earlier deposits; their parameter ranges and thresholds are stated with those bounds.

2.1. **Critical backgrounds.** Fix an odd $L \geq 13$ and a finite periodic cubic lattice tower $\mathbb{T}^{(0)} \supset \cdots \supset \mathbb{T}^{(K)}$. Successive spacings have ratio L. At each level the three sides have the same number of sites, twice a power of L. In level-k units, $1 \leq k \leq K$,

$$\eta = L^{-k}, \qquad \xi_j = L^{j-k} \quad (0 \leq j \leq k).$$

A configuration assigns $U(b) \in \mathrm{SU}(2)$ to oriented bonds, with $U(\bar{b}) = U(b)^{-1}$. Write $\partial U(p)$ for the ordered plaquette product and

$$A_{\Omega_0}(U) = \sum_{p \subset \Omega_0} \left(1 - \tfrac{1}{2} \operatorname{tr} \partial U(p)\right).$$

Here inner plaquettes have all corners in Ω_0. Use operator norms, and let $\mathfrak{g}$ be the traceless Hermitian 2×2 matrices, $\mathfrak{g}^c = \mathfrak{g} \oplus i\mathfrak{g}$.

A decreasing sequence $\Omega_0 \supset \cdots \supset \Omega_k$, with $\Omega_{-1} = \Omega_0$, $\Omega_{k+1} = \emptyset$, is *separated* with parameters $\mathfrak{M}, R_1$, powers of L, if each Ω_j is a union of cubes of side $\mathfrak{M}\xi_j$ in the fixed centred partition, is empty when no such cubes exist, and

$$\operatorname{dist}_\infty(\Omega_j^c, \Omega_{j+1}) > R_1 \mathfrak{M} \xi_j \qquad (0 \leq j < k).$$

Complements are closed; distance to the empty set is infinite. Let $\Gamma_j^\flat$ be the set of level-j bonds with both endpoints in Ω_0, at least one in Ω_j, and neither in Ω_{j+1}. The reduced determining set is $\mathbf{B}^\flat = \bigcup_j \Gamma_j^\flat$.

For the iterated nonabelian mean $\mathcal{M}^j$ defined in sections 5 and 7, put

$$M_{\mathbf{B}^\flat}(U) = ((\mathcal{M}^j U)|_{\Gamma_j^\flat})_{j=0}^k.$$

A datum $V = (V_j)$ assigns group elements to these rows, with inversion under reversal. A level-j plaquette is *tested* if it has a bond in $\Gamma_j^\flat$ and every bond outside $\Gamma_j^\flat$ has both endpoints in $\Omega_{j-1} \setminus \Omega_j$. Its holonomy uses V_j on retained bonds and $\mathcal{M}(V_{j-1})$ elsewhere. Criticality over V means membership in $M_{\mathbf{B}^\flat}^{-1}(V)$ and vanishing derivative of A_{Ω_0} along every differentiable curve therein. Residual gauges equal I at all retained-bond endpoints. The axial condition is $\mathrm{Ax}^\circ(\mathbf{B}^\flat, I)$ of definition 8.2, with its fixed coordinate combs.

For a bond or plaquette $\mathfrak{e}$, let $\mathsf{L}(\mathfrak{e}) = \xi_j$ for the largest j such that Ω_j contains an endpoint or corner. A bond is *full* if all incident plaquettes are inner. The class $\mathfrak{U}_k'(r)$ consists of real configurations on inner bare bonds satisfying

$$|\partial U(p) - I| < r\eta^2 \mathsf{L}(p)^{-2}, \qquad |(D_U^\mathsf{T} \partial U)(b)| < r\eta^2 \mathsf{L}(b)^{-3}$$

at inner plaquettes and full bonds. Here D_U^T is the covariant plaquette divergence (68), including scalar matrix parts. The action current is

$$J(U) = D_U^\mathsf{T} F_U, \qquad F_U(p) = \frac{\partial U(p) - \partial U(p)^{-1}}{2i\eta^2}.$$

Relative to U_0, put

$$F(A) = \eta^{-1} A_{\Omega_0}(e^{i\eta A} U_0), \qquad \|A\|_{X,\mathbf{B}} = \max\{|A|_{(-1)}, |\nabla^{U_0} A|_{(-2)}\}.$$

Here $|f|_{(-s)} = \sup \mathsf{L}^s |f|$. Use only inner differences, weighted by the larger local scale of their two bonds, with the zero-extension convention of proposition 29.1. Use the mean-Landau projection R_m of (479), the dimensionless relative straightened mean $\mathcal{Q}$ of section 33, and the inverse datum twist $B_\natural$ of (580). Transposes use bare weight η^3 and row weight ξ_c^{-1}, where $\xi_c = \xi_j$ at level j.

Theorem E (Canonical critical background)**.** There are constants

$$a_{\mathrm{orb}} > 0, \qquad B_{\mathrm{rad}} \geq \max\{4, L^2\}, \qquad C_\partial \geq 4$$

and powers $\mathfrak{M}_{\mathrm{G}}, \rho_{\mathrm{G}}$ of L, depending only on L, with the following property. For every power $\rho \geq \rho_{\mathrm{G}}$ of L, there are positive numbers $\varepsilon_{\mathrm{G}}(\rho), r_{\mathrm{G}}(\rho), K_{\mathrm{G}}(\rho)$, depending only on L, ρ. Suppose

$$\mathfrak{M} \geq \mathfrak{M}_{\mathrm{G}}, \qquad R_1 \geq L, \qquad R_1 \mathfrak{M} \geq L^2 \rho,$$

and $\mathbb{T}^{(k)}$ has at least 6ρ sites in each direction. Let the region sequence be separated with these parameters. Let V_0 be real reduced data whose tested holonomies are defined and satisfy

$$|\partial V_0 - I| < \varepsilon_1, \qquad 0 < \varepsilon_1 \leq \varepsilon_{\mathrm{G}}(\rho).$$

For every real row field B with $\sup|B| < r_{\mathrm{G}}(\rho)$, the critical configurations over $e^{iB}V_0$ in $\mathfrak{U}'_k(a_{\mathrm{orb}})$ form one nonempty residual gauge orbit. This orbit is contained in

$$\mathfrak{U}'_k(B_{\mathrm{rad}}(\varepsilon_1 + C_\partial \sup|B|)).$$

Here the perturbed datum is defined on positive rows by $(e^{iB}V_0)(c) = e^{iB(c)}V_0(c)$, and on reversed rows by inversion. Denote the unique axial representative by $U^\circ(B)$, and set $U_0 = U^\circ(0)$.

There is exactly one holomorphic map Φ_{V_0} from the complex row ball $\mathcal{B} = \{B : \sup|B| < r_{\mathrm{G}}(\rho)\}$ to $\mathrm{SL}(2,\mathbb{C})$-valued configurations on the inner bare bonds which equals $U^\circ(B)$ at real B. It has the form

$$\Phi_{V_0}(B) = (e^{i\eta A(B)}U_0)^{h(B)}, \qquad A(0) = 0, \quad h(0) = I,$$

where A, h are holomorphic and are respectively real and unitary at real B. There is a holomorphic multiplier $\lambda(B)$, real at real B, such that

$$\mathsf{R}_{\mathsf{m}} D^{\mathsf{T}} A = 0, \qquad \mathcal{Q}(A) = B_\natural, \qquad F'(A) = \mathcal{Q}'(A)^{\mathsf{T}}\lambda(B),$$
$$\|A(B)\|_{X,\mathbf{B}} + |D^{\mathsf{T}} D A(B)|_{(-3)} + \sup_x |h(B)(x) - I| \leq K_{\mathrm{G}}(\rho) \sup|B|,$$
$$\sup|\lambda(B)| \leq K_{\mathrm{G}}(\rho)(\varepsilon_1 + \sup|B|).$$

The covariant difference operators use the reference U_0, and $D^{\mathsf{T}}D$ is the inner-plaquette curl followed by its transpose. All actual complex averages are defined, and

$$M_{\mathbf{B}^\flat}(\Phi_{V_0}(B)) = e^{iB}V_0.$$

At every inner plaquette and full bond, respectively,

$$|\partial\Phi_{V_0}(B)(p) - I| < (B_{\mathrm{rad}}\varepsilon_1 + K_{\mathrm{G}}(\rho)\sup|B|)\eta^2 \mathsf{L}(p)^{-2},$$
$$|J(\Phi_{V_0}(B))(b)| < (B_{\mathrm{rad}}\varepsilon_1 + K_{\mathrm{G}}(\rho)\sup|B|)\mathsf{L}(b)^{-3}.$$

■□

The uniqueness assertion concerns the specified regular class. It does not require the critical point to minimize the action over the entire fibre. All radii and bounds in the theorem are independent of the number of layers, the lattice spacing and the volume, once L, ρ are fixed. The proof is given in section 38.

2.2. **Fluctuations, response and Gaussian coefficients.** The fluctuation theorem applies to the natural regular pair class of definition 27.1, which controls curvature, a complex potential and its first covariant differences, the current, and real gauges on cubes contained in the union of two successive layers. For an additional separated region and any subset of its coarse sites, theorem L constructs both the Landau and axial fluctuation covariances on the corresponding nonpivot coordinates. Under its stated geometric thresholds and smallness bounds, either covariance $\mathcal{C}$ satisfies

$$|\mathcal{C}(b, b')| \leq B_0 e^{-\delta_{\mathrm{cv}}|b_- - b'_-|_1}.$$

It is the sum of an absolutely convergent expansion over tree walks. Each term is holomorphic in the pair restricted to its carrier, and the sum of the terms with m edges is bounded by $2^{-m-1}B_0 e^{-\delta_{\mathrm{cv}}|b_- - b'_-|_1}$. The constants are independent of the number of layers, the volume and the chosen subset. These are covariances of the constrained fluctuation form specified in that theorem.

The same local expansions allow parameters to turn off couplings across prescribed walls. The nonlinear response of theorem M is jointly holomorphic in those parameters, the local regular pair and the small coarse datum. It has weighted Lipschitz bounds, hence exponential decay of the response to a change in a distant datum row. When wall parameters vanish, the solution splits into the corresponding components; each component depends only on the pair and datum on its stated local region. The parameter domains and their associated datum radii are specified in that theorem. At parameters equal to one, the response is identified with the critical background, in its mean-Landau coordinates, by proposition 41.2 and corollary 41.3. The converse uses the reduced determining set and the common small ball fixed by the criticality radius $\varepsilon_{\mathrm{crit}}$ and the response radius.

There is a separate uniform statement for the Gaussian normalizers near the trivial configuration. For the one-step normalizer $\Phi_l^\natural$, let $\pi_{\mu\nu}^{(l)}(u)$ be its infinite-volume second variation on individual unit-lattice bonds, with the colour factor removed. Theorem O proves that this kernel is real and exponentially decreasing, and that

$$b_l^0 = -\tfrac{1}{2}\sum_{u\in\mathbb{Z}^3} u_\mu^2 \pi_{\nu\nu}^{(l)}(u), \qquad \mu \neq \nu,$$

converges absolutely and is independent of the distinct directions. It depends only on l, L, and

$$\sup_{k\geq 0}\left|\sum_{l=1}^{k} L^{-(k-l)} b_l^0\right| \leq c_2(L) < \infty.$$

The nonlinear Gaussian normalizers are germs at the trivial configuration; their neighbourhoods may depend on the height and the finite volume. The bound displayed above is uniform in the height.

2.3. **The proof.** The block averages and gauge conditions follow the multiscale framework of Bałaban [Bał85a; Bał85c]; his background propagators provide the corresponding analytic setting [Bał85b]. We begin with the local averaging maps and gauge representatives, keeping the actual averaging constraint throughout the construction.

The linear estimates are proved in local gauges and assembled over separated regions. Coercivity controls the gauge and average constraints, while the gap between successive regions makes the localization error small. This gives a constrained inverse with exponential decay and uniform control of one covariant difference. The nonlinear equation is then solved with its initial current retained. A correction imposes the gauge condition and the prescribed coarse averages.

To construct the real background, we descend the given data to the fine lattice and then solve upward through the levels. After each step a regularity improvement restores a fixed radius. Comparing two solutions in a common gauge proves uniqueness of the regular orbit. The real background supplies the local regular gauges needed by the complex response, and the axial condition identifies its continuation with Φ_{V_0}.

The covariance expansion adds the fluctuation coordinate to the same constrained system. Local terms and current multipliers can then be organized into convergent tree walks. Their carrier dependence is what makes the decoupled response local. Finally, Gaussian gluing makes the accumulated coefficients telescope. A Ward identity reduces their quadratic part to a potential response; the remainder is controlled by a trace with three propagators. The remaining massive coefficient is computed in a constant abelian field and bounded uniformly as the mass decreases.

3. The finite-lattice model and its sources

3.1. **Configurations and the Wilson measure.** Let $L \geq 3$ be odd, let K, m be positive integers, and put

$$a_j = L^{-j} \quad (j \geq 0), \qquad \ell = 2L^m, \qquad T_\ell = \mathbb{R}^3/\ell\mathbb{Z}^3, \qquad \Lambda_{K,\ell} = a_K\mathbb{Z}^3/\ell\mathbb{Z}^3.$$

The group is $G = \mathrm{SU}(2)$, and tr denotes the ordinary matrix trace, so that $\operatorname{tr}\mathbf{1} = 2$. Distances used in Lipschitz norms on T_ℓ are the quotient Euclidean distances in physical units. Write $a = a_K$ and $N = \ell/a = 2L^{m+K}$ when the parameters are fixed.

A bond is an ordered pair $b = \langle x, x + \epsilon a e_\mu\rangle$, where $x \in \Lambda_{K,\ell}$, $\epsilon \in \{-1, 1\}$ and $1 \leq \mu \leq 3$. It is positive when $\epsilon = 1$, and $\bar{b}$ denotes its reverse. A configuration assigns $U_b \in G$ to each bond, subject to $U_{\bar{b}} = U_b^{-1}$. Thus the configuration space $\mathcal{V}_0$ is the finite product of copies of G indexed by the positive bonds. We equip it with the product Haar probability measure dU.

For $\mu < \nu$, let $p = p_{\mu\nu}(x)$ be the elementary plaquette based at x, with barycentre $x(p) = x + \frac{a}{2}(e_\mu + e_\nu)$. Its boundary product is

$$\begin{aligned} U_p = U(\langle x, x + ae_\mu\rangle)U(\langle x + ae_\mu, x + ae_\mu + ae_\nu\rangle) \\ \cdot U(\langle x + ae_\nu, x + ae_\mu + ae_\nu\rangle)^{-1}U(\langle x, x + ae_\nu\rangle)^{-1}. \end{aligned} \tag{3}$$

Each plaquette is counted once in sums over p. There are $\mathsf{n} = 3N^3$ plaquettes and the same number of positive bonds. We use Wilson's plaquette action [Wil74, §III], with the following normalization:

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{tr} U_p, \qquad A_K(U) = \sum_p a_p(U). \tag{4}$$

For $V \in G$, $V + V^{-1} = (\operatorname{tr} V)\mathbf{1}$ and $V^{-1} = V^*$, so $\operatorname{tr} V \in [-2, 2]$ and $0 \leq A_K \leq 2\mathsf{n}$. For $\beta \in \mathbb{R}$, define

$$\rho_0(U) = e^{-\beta A_K(U)}, \qquad Z_{K,\ell,\beta} = \int \rho_0\, dU, \qquad \mu_{K,\ell,\beta}(dU) = Z_{K,\ell,\beta}^{-1}\rho_0(U)\, dU. \tag{5}$$

The continuous density on the compact configuration space satisfies $e^{-2|\beta|\mathsf{n}} \leq \rho_0, Z_{K,\ell,\beta} \leq e^{2|\beta|\mathsf{n}}$; for $\beta \geq 0$ the bounds improve to $e^{-2\beta\mathsf{n}} \leq \rho_0, Z_{K,\ell,\beta} \leq 1$. Thus $\mu_{K,\ell,\beta}$ is a Borel probability measure. We write $\langle F\rangle_{K,\ell,\beta} = \int F\, d\mu_{K,\ell,\beta}$.

A gauge transformation $u : \Lambda_{K,\ell} \to G$ acts by

$$U^u(\langle x, y\rangle) = u(x)U(\langle x, y\rangle)u(y)^{-1}.$$

Cancellation along a plaquette gives $U_p^u = u(x)U_p u(x)^{-1}$, so a_p, A_K and ρ_0 are invariant. Left and right Haar invariance on each bond then gives gauge invariance of dU and $\mu_{K,\ell,\beta}$.

3.2. **The observable and its source.** For a complex weight c on plaquettes, or a function on T_ℓ evaluated at their barycentres, set

$$A(c, U) = \sum_p c(p)a_p(U), \qquad A^\xi(c, U) = \xi^{-1}A(c, U). \tag{6}$$

Here ξ is the spacing of the same lattice in the chosen units. The scalar observable used below is the weighted action in physical units,

$$\mathcal{O}(f) = A^{a_K}(f, U) = a_K^{-1}\sum_p f(x(p))a_p(U). \tag{7}$$

For real continuous f it is a real continuous function on $\mathcal{V}_0$, and

$$|\mathcal{O}(f)| \leq 2a_K^{-1}\mathsf{n}\|f\|_\infty. \tag{8}$$

From now on take $\beta > 0$. Its physical form is

$$\mathsf{b} = a_K\beta, \qquad \beta A_K(U) = \mathsf{b}A^{a_K}(1, U). \tag{9}$$

For real Lipschitz functions $f_1, \dots, f_n$ on T_ℓ, $n \ge 1$, and $z \in \mathbb{C}^n$, put $w = \sum_i z_i f_i$. The source changes the physical inverse coupling to $\mathsf{b} - w$:

$$\rho_0^w(U) = \exp[-A^{a_K}(\mathsf{b} - w, U)] = \rho_0(U) \exp\Big(\sum_i z_i \mathcal{O}(f_i)\Big). \tag{10}$$

Write $Z(z) = Z_{K,\ell,\beta}[w] = \int \rho_0^w \, dU$. The positive sign of the source term in (10) fixes the cumulant convention. In particular a constant source $w = \zeta$ replaces the bare inverse coupling by $\beta - L^K \zeta$.

3.3. **Generating functions and cumulants.** We recall the finite-measure argument that justifies differentiation of Z. Let $X_1, \dots, X_n$ be bounded complex measurable functions on a probability space (Ω, ν), and put

$$\Gamma(z) = \int e^{\sum_i z_i X_i} \, d\nu, \qquad c_i = \sup |X_i|.$$

The logarithm near the origin is the branch with value zero there. The joint cumulant is

$$\langle X_1; \dots; X_n \rangle_\nu = \partial_{z_1} \cdots \partial_{z_n} \log \Gamma(z)\big|_{z=0}. \tag{11}$$

Lemma 3.1. The function Γ is entire and satisfies

$$|\Gamma(z)| \le e^{\sum_i |z_i| c_i}, \qquad |\Gamma(z) - 1| \le e^{\sum_i |z_i| c_i} - 1.$$

On $D = \{z : \sum_i |z_i| c_i < \log 2\}$ the logarithmic series defines a holomorphic logarithm of Γ with value zero at the origin. In particular (11) is well defined.

Let $\mathcal{P}$ range over the partitions of $\{1, \dots, n\}$ into nonempty blocks, and write $m_B = \int \prod_{i \in B} X_i \, d\nu$ and $\kappa_B = \langle X_i; i \in B \rangle_\nu$. Then

$$\langle X_1; \dots; X_n \rangle_\nu = \sum_{\mathcal{P}} (-1)^{|\mathcal{P}|-1} (|\mathcal{P}| - 1)! \prod_{B \in \mathcal{P}} m_B, \tag{12}$$

$$\int \prod_{i=1}^n X_i \, d\nu = \sum_{\mathcal{P}} \prod_{B \in \mathcal{P}} \kappa_B. \tag{13}$$

Thus cumulants are symmetric and complex multilinear, are real for real variables, and are unchanged by adding constants when $n \ge 2$. For centered variables, (13) reduces to the sum over partitions with no singleton blocks.

If $\nu = T_* \nu'$ for a measurable map $T : \Omega' \to \Omega$ and a probability measure ν' on Ω', then

$$\langle X_1; \dots; X_n \rangle_\nu = \langle X_1 \circ T; \dots; X_n \circ T \rangle_{\nu'}.$$

For fixed continuous X_i on a compact metrizable space, the cumulants are also continuous under weak convergence of Borel probability measures. ∎

Proof. Normal convergence of the exponential series proves analyticity and the bounds; on D, the logarithmic series also converges normally. The coefficient of $z_1 \cdots z_n$ in $(\Gamma - 1)^j$ is $j! \sum_{|\mathcal{P}|=j} \prod_{B \in \mathcal{P}} m_B$. Substitution in the logarithmic and exponential series gives (12) and (13). These imply symmetry, multilinearity, reality and weak continuity, since each m_B converges under the stated hypotheses. Adding constants adds only a linear term to $\log \Gamma$; centering removes singleton cumulants. Pushforwards preserve Γ. □

Apply the lemma with $\nu = \mu_{K,\ell,\beta}$ and $X_i = \mathcal{O}(f_i)$. Since $Z(z) = Z(0)\Gamma(z)$, Z is entire and

$$\langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K,\ell,\beta} = \partial_{z_1} \cdots \partial_{z_n} \log Z(z)\big|_{z=0}. \tag{14}$$

Here the branch is chosen so that $\log Z(0) = \log Z_{K,\ell,\beta}$. By (8), one sufficient domain for this logarithm is

$$2 a_K^{-1} \mathsf{n} \sum_i |z_i| \|f_i\|_\infty < \log 2.$$

Its size depends on both the cutoff and the volume.

3.4. **Source bounds and units of length.** Fix $M > 0$ independently of K, ℓ and β, and set

$$\|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}. \tag{15}$$

The Lipschitz constant is taken in physical units. For $\varrho > 0$ define

$$\mathbb{D}_\varrho(f_1, \dots, f_n) = \left\{ z \in \mathbb{C}^n : \sum_i |z_i| \|f_i\|_\sharp < \varrho \right\}.$$

This is the source domain used in the following density estimates.

In level-k units, $0 \le k \le K$, a physical length r is $L^{K-k}r$, and the bare lattice has spacing $\eta_k = L^{-k}$. Thus $A^{\eta_k}(c, U) = L^k A(c, U)$.

Lemma 3.2. Let $\beta > 0$, $\mathsf{b} = a_K\beta$, and let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ. For $z \in \mathbb{D}_\varrho(f_1, \dots, f_n)$ and $w = \sum_i z_i f_i$,

$$\|w\|_\infty < \varrho, \qquad \operatorname{Lip} w < \frac{\varrho}{40M}. \tag{16}$$

For every configuration U,

$$\rho_0(U)^{1+\varrho/\mathsf{b}} \le |\rho_0^w(U)| \le \rho_0(U)^{1-\varrho/\mathsf{b}}. \tag{17}$$

In level-k units the same density is

$$\rho_0^w(U) = \exp[-A(\beta - L^K w, U)] = \exp[-A^{\eta_k}(L^{-k}\beta - L^{K-k}w, U)]. \tag{18}$$

The Lipschitz constant of $L^{K-k}w$, measured in level-k units, is $\operatorname{Lip} w$. Its variation between any two points of a closed cube of side M in those units is less than $\sqrt{3}\varrho/40$. ∎

Proof. The triangle inequality gives (16), and linearity in the weight gives (18). Since $a_p \ge 0$,

$$|\rho_0^w| = \rho_0 e^{A^{a_K}(\operatorname{Re} w, U)}, \qquad |A^{a_K}(\operatorname{Re} w, U)| \le \varrho A^{a_K}(1, U),$$

which proves (17). Rescaling source and distance by L^{K-k} preserves the Lipschitz quotient; a cube of side M has diameter at most $\sqrt{3}M$. □

For $\xi' = \upsilon\xi$, one has $A^{\xi'}(c, U) = \upsilon^{-1}A^\xi(c, U)$ and $A^{\xi'}(\upsilon c, U) = A^\xi(c, U)$. The physical weight corresponding to a level-k weight c is $a_{K-k}c$; its weight in the unscaled plaquette sum is $L^k c$.

3.5. **Dilation of the finite lattice.**

Lemma 3.3. Let $\beta > 0$, let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ, and put $w = \sum_i z_i f_i$ for $z \in \mathbb{C}^n$. Let $b \in \mathbb{Z}$ with $0 \le b < K$. For $h : T_\ell \to \mathbb{C}$ write $h^{(b)}(x) = h(L^{-b}x)$ on $T_{L^b\ell}$. Then

$$Z_{K,\ell,\beta}[w] = Z_{K-b,L^b\ell,\beta}[L^b w^{(b)}], \tag{19}$$

and

$$\langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_n) \rangle_{K,\ell,\beta} = L^{bn} \langle \mathcal{O}(f_1^{(b)}); \dots; \mathcal{O}(f_n^{(b)}) \rangle_{K-b,L^b\ell,\beta}. \tag{20}$$

On the right, the observables use spacing a_{K-b} and the dilated torus. With the same M on both tori, $\|f_i^{(b)}\|_\sharp \le \|f_i\|_\sharp$, so that

$$z \in \mathbb{D}_\varrho(f_1, \dots, f_n) \quad \Longrightarrow \quad L^b z \in \mathbb{D}_{L^b\varrho}(f_1^{(b)}, \dots, f_n^{(b)}).$$

The physical inverse coupling is multiplied by L^b. ∎

Proof. Dilation preserves orientations, plaquette products and product Haar measure. At corresponding barycentres, $w^{(b)}(L^b x(p)) = w(x(p))$ and $a_{K-b}^{-1}L^b = a_K^{-1}$, so the sourced coefficients agree, proving (19). Differentiating $Z(z) = \widetilde{Z}(L^b z)$ in the local logarithms gives (20). The remaining assertions follow from $\|f_i^{(b)}\|_\infty = \|f_i\|_\infty$, $\operatorname{Lip} f_i^{(b)} = L^{-b} \operatorname{Lip} f_i$ and $a_{K-b}\beta = L^b\mathsf{b}$. □

4. Block averaging and image measures

The block map replaces fine bond variables by coarse ones while preserving gauge covariance. On configurations with small plaquette curvature, contour transports lie close together and their logarithmic mean depends analytically on the fine field. Two further rules extend the map to all real configurations. We prove its measurability before using it to transform measures.

4.1. **Blocks and contour comparison.** For integers $j \geq -m$, put

$$a_j = L^{-j}, \qquad \Lambda_{j,\ell} = a_j \mathbb{Z}^3 / \ell \mathbb{Z}^3.$$

Thus the level-k lattice is $\mathbb{T}^{(k)} = \Lambda_{K-k,\ell}$, $0 \leq k \leq K$. Its configuration space $\mathcal{V}_k$ consists of the G-valued bond assignments satisfying $U(\bar{b}) = U(b)^{-1}$. The norm $|x|_\infty$ on T_ℓ is the least sup norm of a lift of x. The cube centered at $y \in \Lambda_{j,\ell}$ and its block of order $n \geq 0$ are

$$\Delta_j(y) = \{x : |x - y|_\infty \leq a_j/2\}, \qquad B^n(y) = \Delta_j(y) \cap \Lambda_{j+n,\ell}.$$

Write $B(y) = B^1(y)$ and $h = (L-1)/2$. The specified lattice of centers determines the scale suppressed in this notation. In particular,

$$B^n(y) = \left\{ y + a_{j+n} \sum_{\mu=1}^{3} r_\mu e_\mu : r_\mu \in \mathbb{Z}, \quad |r_\mu| \leq \frac{L^n - 1}{2} \right\}.$$

These L^{3n} points are distinct. The blocks centered on $\Lambda_{j,\ell}$ partition $\Lambda_{j+n,\ell}$, and

$$B^{n+n'}(y) = \bigcup_{x \in B^n(y)} B^{n'}(x), \qquad \Delta_j(y) = \bigcup_{x \in B^n(y)} \Delta_{j+n}(x).$$

For $j > -m$, every fine bond either has both endpoints in one block of order n, or joins two such blocks whose centers are the endpoints of a unique coarse bond of the same orientation and direction. In the second case we say that the fine bond lies over the coarse bond.

These assertions follow from the unique decomposition $N = L^n q + r$, $|r| \leq (L^n - 1)/2$, of an integer N. Applied in each coordinate, it gives the partition into blocks. Two admissible coordinate vectors cannot represent the same site: their differences have absolute value at most $L^n - 1$, whereas the period in fine lattice units is $2L^{m+j+n} \geq 2L^n$. Applying the decomposition twice gives the nesting identity. The intervals of length a_{j+n} centered at the consecutive fine sites fill the interval of length a_j, proving the cube identity. Finally, increasing one fine coordinate by one stays in the same block except at $r_\mu = (L^n - 1)/2$; there it enters the block centered at $y + a_j e_\mu$. Reversal handles negative bonds. Since the coarse lattice has at least six sites along each axis when $j > -m$, the coarse bond is determined by its endpoints.

If X is a union of cubes $\Delta_j(y)$, a fine site $x \in B^n(y)$ belongs to X exactly when its center y does. Indeed, $|x - y|_\infty \leq (a_j - a_{j+n})/2 < a_j/2$, whereas distinct cube centers are at sup distance at least a_j. Thus x lies in the interior of $\Delta_j(y)$ and in no other coarse cube. This observation permits the same block construction inside a region.

Let $j > -m$ and $y \in \Lambda_{j,\ell}$. For $x = y + a_{j+1} \sum_\mu r_\mu e_\mu \in B(y)$, write $S = \{\mu : r_\mu \neq 0\}$. For each ordering σ of S, let $\Gamma_\sigma(y, x)$ traverse the $|r_{\sigma(1)}|$ bonds in direction $\operatorname{sign}(r_{\sigma(1)}) e_{\sigma(1)}$, then those in direction $\operatorname{sign}(r_{\sigma(2)}) e_{\sigma(2)}$, and so on. Following [Bał87, p. 252], the contour family is

$$\mathcal{G}(y, x) = \{\Gamma_\sigma(y, x) : \sigma \text{ orders } S\}.$$

It consists of $|S|! \leq 6$ paths; $\mathcal{G}(y, y)$ consists of the empty path. For a path $\Gamma = (b_1, \ldots, b_r)$ put $U(\Gamma) = U(b_1) \cdots U(b_r)$, with value $\mathbf{1}$ for the empty path. If c is a coarse bond and $x \in B(c_-)$, set $x' = x + (c_+ - c_-)$ and write $[x, x']$ for the straight fine path between them. We abbreviate $U[c] = U([c_-, c_+])$.

To compare these products, take the spacing of a lattice $\Lambda_{r,\ell}$, $r > -m$, as the unit of length. A step sequence $s = (s_1, \dots, s_q)$ has entries in $\{\pm e_1, \pm e_2, \pm e_3\}$. Its path from z is denoted by $\Gamma(z; s)$. Define

$$\Sigma(z; s) = \left\{ z + \sum_{i \in I} s_i : I \subset \{1, \dots, q\} \right\}. \tag{21}$$

All rearrangements of s have the same endpoint and the same set $\Sigma(z; s)$, which contains every site visited by their paths. We use the operator norm for matrices.

Lemma 4.1 (Rearrangement of steps)**.** Let U be a G-valued configuration, let s have length q, and let s' be a rearrangement of s. If $\omega \geq 0$ and $|U_p - \mathbf{1}| \leq \omega$ for every plaquette whose four corners lie in $\Sigma(z; s)$, then

$$\left| U(\Gamma(z; s'))U(\Gamma(z; s))^{-1} - \mathbf{1} \right| \leq (1 + \omega)^{q(q-1)/2} - 1. \tag{22}$$

■

Proof. First exchange two adjacent steps. The initial path before them is unchanged, and the common final path cancels from the quotient of the two transports. If P is the initial transport, this quotient is PHP^{-1}, where H traverses the four-edge path formed by the two orders of the exchanged steps. Equal or opposite steps give $H = \mathbf{1}$. Otherwise H is a conjugate of U_p or U_p^{-1} for the plaquette with those four corners. All corners belong to $\Sigma(z; s)$. Unitary invariance of the operator norm and $|V^{-1} - \mathbf{1}| = |V - \mathbf{1}|$ for $V \in G$ therefore give

$$U(\Gamma(z; s'')) = (\mathbf{1} + E)U(\Gamma(z; s)), \qquad |E| \leq \omega. \tag{23}$$

Any permutation of q entries is a product of at most $q(q-1)/2$ adjacent exchanges: sorting a permutation removes one inversion at each exchange. Each intermediate sequence has the same set $\Sigma(z; s)$, so the plaquette hypothesis continues to apply. After T exchanges the quotient is $(\mathbf{1} + E_T) \cdots (\mathbf{1} + E_1)$. Expanding this product and using submultiplicativity gives

$$|(\mathbf{1} + E_T) \cdots (\mathbf{1} + E_1) - \mathbf{1}| \leq \prod_{t=1}^{T} (1 + |E_t|) - 1 \leq (1 + \omega)^{q(q-1)/2} - 1.$$

□

Lemma 4.2 (Contours in a double block)**.** Let $j > -m$ and express lengths in units of a_{j+1}. If $x = y + \sum_\mu r_\mu e_\mu \in B(y)$, all paths in $\mathcal{G}(y, x)$ are rearrangements of the same $q = \sum_\mu |r_\mu| \leq 3(L-1)/2$ steps. Their common subset-sum set is

$$\left\{ y + \sum_{\mu \in S} t_\mu \operatorname{sign}(r_\mu) e_\mu : 0 \leq t_\mu \leq |r_\mu|, \quad t_\mu \in \mathbb{Z} \right\} \subset B(y).$$

The $|S|!$ paths are distinct and shortest.

For a positive coarse bond $c = \langle c_-, c_- + Le_\nu \rangle$, translation by Le_ν identifies $\mathcal{G}(c_-, x)$ with $\mathcal{G}(c_+, x')$, where $x' = x + Le_\nu$. If Γ' is the translate of Γ, the paths $\Gamma \cup [x, x']$ and $[c_-, c_+] \cup \Gamma'$ are rearrangements of at most $(5L-3)/2$ steps. Their common subset-sum set is contained in $B(c_-) \cup B(c_+)$; in particular, the straight paths have L bonds and lie in this double block. The straight path associated with the reversed bond is the reverse of $[x, x']$. ■

Proof. Each ordering uses $|r_\mu|$ copies of the signed step in direction μ. Choosing a subset of these copies gives exactly the displayed site set. Distinct orderings first differ by steps in different coordinate directions and hence yield distinct paths. The period of the fine lattice is $N' = 2L^{m+j+1} \geq 2L^2$. A path from y to x has a signed displacement congruent to r_μ modulo N' in direction μ. If it used fewer than $|r_\mu|$ steps in that direction, the difference from r_μ would be a nonzero multiple of N' of absolute value less than $2|r_\mu| < N'$. This is impossible, so the prescribed paths are shortest.

The translated endpoints have the same displacement vector, which proves the correspondence of contour families. The two concatenations place the L steps in direction e_ν after or before the same q contour steps. Relative to c_-, every coordinate of their common subset-sum set lies in $[-h,h]$ except the ν coordinate, which lies in $[-h,L+h]$. These are exactly the integer coordinate ranges of $B(c_-)\cup B(c_+)$. They describe distinct sites because $N'\ge 2L^2>2L$. The length is $q+L\le (5L-3)/2$, and reversing the straight path reverses its bonds in reverse order. □

4.2. **The average on a double block.** Continue to use the operator norm on $\mathbb{C}^{2\times 2}$. Put $G^c=\mathrm{SL}(2,\mathbb{C})$, and $\mathfrak{g}=\{A : A^*=A,\ \operatorname{tr}A=0\}$; its complexification $\mathfrak{g}^c$ is the space of traceless complex matrices. The series

$$\log X=\sum_{r\ge 1}\frac{(-1)^{r-1}}{r}(X-\mathbf{1})^r \qquad (|X-\mathbf{1}|<1)$$

will be used near the identity. On $G\setminus\{-\mathbf{1}\}$ we also use the intrinsic logarithm, the unique traceless skew-Hermitian logarithm whose eigenvalues lie in $i(-\pi,\pi)$. These logarithms agree on their common domain. The matrix estimates and logarithm properties needed here are proved in section 5.

For a finite family $\mathbf{V}=(V_1,\dots,V_n)$ of invertible matrices, put

$$d(\mathbf{V})=\max_{i,j}|V_iV_j^{-1}-\mathbf{1}|. \tag{24}$$

A mean is an invertible matrix V satisfying

$$\max_i|V_iV^{-1}-\mathbf{1}|<1, \qquad \sum_i\log(V_iV^{-1})=0. \tag{25}$$

For a family in G or G^c, a mean in that group is required to belong to the same group. We use the logarithmic mean in Bałaban's presentation [Bał87, (0.10), p. 253], where the equivalent definition is attributed to Federbush.

Proposition 4.3 (The mean of a small-diameter family)**.** Let $n\ge 1$, $\mathbf{V}\in\mathrm{GL}(2,\mathbb{C})^n$ and $d=d(\mathbf{V})\le 1/40$. On the ball $|Y|\le 1/8$, the map

$$\Phi_{\mathbf{V}}(Y)=Y+\frac{1}{n}\sum_i\log(V_iV_1^{-1}e^{-Y}) \tag{26}$$

is defined, maps the ball into itself, and has Lipschitz constant at most $2/5$. Its unique fixed point Y_* satisfies $|Y_*|\le 2d$. The matrix

$$\mathrm{av}(\mathbf{V})=e^{Y_*}V_1 \tag{27}$$

satisfies

$$\max_i|V_i\,\mathrm{av}(\mathbf{V})^{-1}-\mathbf{1}|\le 4d, \qquad \sum_i\log(V_i\,\mathrm{av}(\mathbf{V})^{-1})=0. \tag{28}$$

It is the only invertible solution of the mean equation for which $\min_i|V_iV^{-1}-\mathbf{1}|\le 1/10$ and $\max_i|V_iV^{-1}-\mathbf{1}|<1$. It is independent of the ordering of the family and equals V on the constant family $(V,\dots,V)$.

If $V_i\in G^c$ for every i, then $Y_*\in\mathfrak{g}^c$ and $\mathrm{av}(\mathbf{V})\in G^c$. If $V_i\in G$ for every i, then $Y_*\in i\mathfrak{g}$ and $\mathrm{av}(\mathbf{V})\in G$; in this case it is the only mean in G satisfying (25), without the additional $1/10$ condition. We denote this mean also by $\mathrm{Av}(\mathbf{V})$.

The maps $\mathbf{V}\mapsto Y_*$ and $\mathbf{V}\mapsto\mathrm{av}(\mathbf{V})$ are holomorphic in the matrix entries on the open set

$$\mathcal{O}_n=\{\mathbf{V}\in\mathrm{GL}(2,\mathbb{C})^n : d(\mathbf{V})<1/40\}.$$

Their restrictions to $G^n\cap\mathcal{O}_n$ are real-analytic. For $\mathbf{V}\in(G^c)^n\cap\mathcal{O}_n$ and $W,W'\in G$,

$$\mathrm{av}(W\mathbf{V}W')=W\,\mathrm{av}(\mathbf{V})W'. \tag{29}$$

■

The proof in section 5 constructs the mean by a contraction with constant $2/5$, uniform in the number of matrices. The uniqueness statement makes the average independent of the matrix chosen as the reference point in that construction.

Fix $0 \le k < K$ and a coarse bond c of $\mathbb{T}^{(k+1)}$. Let P_c be the bonds of $\mathbb{T}^{(k)}$ whose endpoints lie in $B(c_-) \cup B(c_+)$, and let P_c^+ be its positive bonds. Write $U[c] = U([c_-, c_+])$. For $0 < \omega \le 1$, set

$$\mathcal{S}_c(\omega) = \{U \in \mathcal{V}_k : |\partial U(p) - \mathbf{1}| < \omega \text{ for every plaquette with all corners in } B(c_-) \cup B(c_+)\}.$$

Here $\partial U(p) = U_p$. We fix

$$\omega_M = \frac{1}{200L^2}, \qquad \mathcal{S}_c = \mathcal{S}_c(\omega_M). \tag{30}$$

Whenever the contour family has a unique mean in G, define

$$U(y,x) = \mathrm{Av}(\{U(\Gamma) : \Gamma \in \mathcal{G}(y,x)\}), \qquad U(x,y) = U(y,x)^{-1}. \tag{31}$$

The notation Av denotes that unique group mean; on families of diameter at most $1/40$ it equals av. This is the averaged contour variable of [Bał87, (0.11), p. 253].

Lemma 4.4. Let $0 \le k < K$ and $0 < \omega \le \omega_M$. For $U \in \mathcal{S}_c(\omega)$, $y \in \{c_-, c_+\}$ and $x \in B(y)$, the contour family has diameter at most $1/160$. Its unique mean satisfies

$$|U(\Gamma)U(y,x)^{-1} - \mathbf{1}| \le 1/40 \quad (\Gamma \in \mathcal{G}(y,x)). \tag{32}$$

If c is positive and $x' = x + (c_+ - c_-)$, then

$$W_x = U(c_-, x)U([x, x'])U(c_+, x')^{-1}U[c]^{-1} \tag{33}$$

belongs to G and satisfies $|W_x - \mathbf{1}| < 1/14$. The sets $\mathcal{S}_c(\omega)$ are open, and the contour means are real-analytic there. ∎

Proof. Two contour paths from y to x have rearranged step sequences of common length $r \le 3(L-1)/2$. All subset sums of these steps stay in $B(y)$. Thus lemma 4.1 gives

$$|U(\Gamma')U(\Gamma)^{-1} - \mathbf{1}| \le (1+\omega)^{9L^2/8} - 1 \le e^{9/1600} - 1 < 1/160. \tag{34}$$

The mean assertion and (32) follow from proposition 4.3.

Choose $\Gamma \in \mathcal{G}(c_-, x)$ and let Γ' be its translate in $\mathcal{G}(c_+, x')$. The paths $\Gamma \cup [x, x']$ and $[c_-, c_+] \cup \Gamma'$ have rearranged step sequences of length at most $(5L-3)/2$. Their subset sums stay in the double block. Applying lemma 4.1 again gives

$$P_x = U(\Gamma)U([x, x'])U(\Gamma')^{-1}U[c]^{-1}, \qquad |P_x - \mathbf{1}| \le e^{1/64} - 1 < 1/60. \tag{35}$$

Write $U(c_-, x) = (\mathbf{1} + E_1)U(\Gamma)$ and $U(c_+, x')^{-1} = U(\Gamma')^{-1}(\mathbf{1} + E_2)$. Equation (32) and unitarity give $|E_1|, |E_2| \le 1/40$. Consequently

$$W_x = (\mathbf{1} + E_1)P_x(\mathbf{1} + U[c]E_2U[c]^{-1}),$$

and expansion of this product gives

$$|W_x - \mathbf{1}| \le \left(\frac{41}{40}\right)^2 \frac{61}{60} - 1 = \frac{6541}{96000} < \frac{1}{14}. \tag{36}$$

Each plaquette transport is continuous, so its finitely many strict curvature inequalities define an open set. Path transports extend holomorphically to invertible complex bond variables. Their contour families depend holomorphically on these variables, and the condition $d < 1/40$ is open. Composition with the mean of proposition 4.3 therefore gives a holomorphic extension of each contour mean near every point of $\mathcal{S}_c(\omega)$. On the real configurations the values lie in G, proving the claimed real-analyticity. □

4.3. **The global real map and its analytic branch.** For a positive coarse bond c, define $\mathcal{M}(U)(c)$ by the following three cases, and set $\mathcal{M}(U)(\bar{c}) = \mathcal{M}(U)(c)^{-1}$. On $\mathcal{S}_c$, put

$$\mathcal{M}(U)(c) = \exp\left(L^{-3} \sum_{x \in B(c_-)} \log W_x\right) U[c]. \tag{37}$$

The logarithms and means in this formula exist by lemma 4.4. It is the symmetrized average of [Bał87, (0.12), p. 254], in dimension three.

For $x \in B(c_-)$, $\Gamma \in \mathcal{G}(c_-, x)$ and $\Gamma' \in \mathcal{G}(c_+, x')$, let

$$Y_{x,\Gamma,\Gamma'} = U(\Gamma)U([x, x'])U(\Gamma')^{-1}U[c]^{-1},$$

and let $\mathcal{N}_c$ be the set where at least one of these matrices equals $-\mathbf{1}$. On $\mathcal{V}_k \setminus (\mathcal{S}_c \cup \mathcal{N}_c)$, put

$$\begin{aligned} \mathcal{M}(U)(c) &= \exp\left(L^{-3} \sum_{x \in B(c_-)} H_x(U)\right) U[c], \\ H_x(U) &= \frac{\displaystyle\sum_{\Gamma \in \mathcal{G}(c_-,x)} \sum_{\Gamma' \in \mathcal{G}(c_+,x')} \log Y_{x,\Gamma,\Gamma'}}{|\mathcal{G}(c_-, x)|\,|\mathcal{G}(c_+, x')|}. \end{aligned} \tag{38}$$

Here the logarithm is the intrinsic logarithm on $G \setminus \{-\mathbf{1}\}$; this formula follows [Bał87, (0.4), p. 253]. On the remaining set $\mathcal{N}_c \setminus \mathcal{S}_c$, put $\mathcal{M}(U)(c) = U[c]$. Each exponent is traceless and skew-Hermitian, so all three formulas take values in G.

For complex configurations, we use (37) on its analytic domain. More precisely, let $\mathcal{O}_c$ be the configurations in $(G^c)^{\mathsf{P}_c^+}$ whose contour families all have diameter less than $1/40$. Define their contour means using av, and let

$$\mathcal{O}_c' = \{\mathbf{U} \in \mathcal{O}_c : |W_x(\mathbf{U}) - \mathbf{1}| < 1 \text{ for all } x \in B(c_-)\}.$$

On this open set, define

$$\mathcal{M}(\mathbf{U})(c) = \exp\left(L^{-3} \sum_{x \in B(c_-)} \log W_x(\mathbf{U})\right) \mathbf{U}[c]. \tag{39}$$

Define the value on a reversed coarse bond by inversion. The map is holomorphic: transports, inversion, the contour means and the logarithmic series are holomorphic in the positive-bond coordinates. The logarithms of the G^c-valued matrices have trace zero, so (39) lies in G^c. The restriction of every $U \in \mathcal{S}_c$ lies in $\mathcal{O}_c'$, where (39) agrees with (37). For $U_0 \in \mathcal{S}_c$, assign traceless complex matrices $A(b)$ on P_c^+ and define $(e^{iA}U_0)(b) = e^{iA(b)}U_0(b)$ there, with the values on reversed bonds defined by inversion. Then $A \mapsto \mathcal{M}(e^{iA}U_0)(c)$ is holomorphic near zero.

Proposition 4.5. For $0 \le k < K$, the map $\mathcal{M} : \mathcal{V}_k \to \mathcal{V}_{k+1}$ is Borel measurable. At a positive bond c it is continuous on each of the three sets defining it, and real-analytic on $\mathcal{S}_c$. Its value and the applicable case depend only on $U|_{\mathsf{P}_c}$. For every G-valued gauge transformation u,

$$\mathcal{M}(U^u) = \mathcal{M}(U)^{u|_{k+1}}, \tag{40}$$

and the applicable case is unchanged. For complex configurations, $\mathcal{O}_c'$ is invariant under such gauge transformations, and the same covariance holds on that domain. ∎

Proof. The set $\mathcal{S}_c$ is open. Each $Y_{x,\Gamma,\Gamma'}$ is continuous and there are finitely many of them, so $\mathcal{N}_c$ is closed. The three defining sets are therefore Borel. On the first, analyticity follows from the complex extension. On the second, continuity follows from continuity of the intrinsic logarithm away from $-\mathbf{1}$; on the third the formula is the path product $U[c]$. The preimage of an open subset of G is thus a finite union of Borel sets. The finite product of these bond maps is Borel as a map to $\mathcal{V}_{k+1}$.

All paths and plaquettes used at c have their bonds in P_c, proving locality. For a path from y to x, cancellation of the internal gauge factors gives

$$U^u(\Gamma) = u(y)U(\Gamma)u(x)^{-1}.$$

Equation (29) therefore gives $U^u(y,x) = u(y)U(y,x)u(x)^{-1}$ wherever the contour mean is used. The quantities W_x and $Y_{x,\Gamma,\Gamma'}$ transform by conjugation by $u(c_-)$, and each plaquette transforms by conjugation at its base point. Unitary conjugation preserves the operator norm, the equation $Y = -\mathbf{1}$, and both logarithms. It follows that $\mathcal{S}_c$, $\mathcal{N}_c$ and all three cases are gauge invariant. Conjugating the exponent and using $U^u[c] = u(c_-)U[c]u(c_+)^{-1}$ proves (40). The value at the reversed bond follows by inversion. The same calculation for complex configurations preserves the contour diameters and $|W_x - \mathbf{1}|$, proving invariance of $\mathcal{O}'_c$ and covariance there. □

4.4. **Partial configurations and iteration.** If U is given on a reversal-invariant set $\mathcal{B}$ of fine bonds, its average is defined at precisely the coarse bonds c with $\mathsf{P}_c \subset \mathcal{B}$, using any extension of U to the whole lattice. Locality makes this independent of the extension. Define $\mathcal{M}^n$ by composing the maps one level at a time, with $\mathcal{M}^0 = \mathrm{id}$.

Let $0 \le j \le j+n \le K$, and let c be a bond of $\mathbb{T}^{(j+n)}$. Denote by $\mathsf{D}_n(c)$ the bonds of $\mathbb{T}^{(j)}$ with both endpoints in $B^n(c_-) \cup B^n(c_+)$. Then $(\mathcal{M}^n U)(c)$ is defined exactly when $\mathsf{D}_n(c) \subset \mathcal{B}$, and depends only on that restriction. To prove this by induction, the required geometric identity is

$$\mathsf{D}_{n+1}(c) = \bigcup_{c' \in \mathsf{P}_c} \mathsf{D}_n(c'). \tag{41}$$

The nesting of blocks gives inclusion from right to left. Conversely, each fine bond in the left side either joins two order-n blocks with adjacent centers in the double block of c, or lies within one such block. In the first case it belongs to $\mathsf{D}_n(c')$ for the bond between those centers. In the second, choose a neighbor of that center within the double block; the bond to this neighbor provides the required c'. Every site of a double block has such a neighbor. This proves the identity and the induction, starting with $\mathsf{D}_0(c) = \{c, \bar{c}\}$.

Iteration of (40) gives, for a bare gauge transformation,

$$\mathcal{M}^n(U^u) = (\mathcal{M}^n U)^{u|_n}.$$

The identity also holds for partial configurations on their domains, by extending a configuration and using locality. For a region made of closed coarse cubes, all fine sites in the block of a coarse site of the region also lie in the region. Hence these partial averages are defined on every inner coarse bond when the original field is given on the inner fine bonds. Here an inner bond has both endpoints in the region.

Complex iteration uses (39) at every participating bond. For any finite family of iterates, the common domain is open and the iterates are holomorphic there. Indeed, the domain of a single step is open in $(G^c)^{\mathsf{P}_c^+}$. Inductively, intersect the domains of the previous-level values on P_c and take the inverse image of $\mathcal{O}'_c$ under their joint holomorphic map. On this open set the next iterate is its composition with (39). Applying the same strict inequalities and formulas in the ambient invertible positive-bond coordinates gives an open set on which the iterates are holomorphic. The actual domain is its intersection with the product of copies of G^c. The complex and real iterates agree whenever every participating real configuration lies in the corresponding $\mathcal{S}_c$.

4.5. **Linearization.** Fix $0 \le k < K$ and a positive bond c of $\mathbb{T}^{(k+1)}$. Regard a bond field A as an antisymmetric function on bonds, $A(\bar{b}) = -A(b)$, with values in $\mathfrak{g}^c$, and write $A(\Gamma)$ for its sum along a path. Put

$$\bar{A}(y,x) = \frac{1}{|\mathcal{G}(y,x)|} \sum_{\Gamma \in \mathcal{G}(y,x)} A(\Gamma).$$

The derivative as a map on bond variables is

$$(\mathbf{Q}^{\mathrm{lin}}A)(c) = L^{-3} \sum_{x \in B(c_-)} [\bar{A}(c_-, x) + A([x, x']) - \bar{A}(c_+, x')]. \tag{42}$$

When fine bonds have length one and coarse bonds have length L, the normalized operator on one-form components is

$$Q = L^{-1}\mathbf{Q}^{\mathrm{lin}}. \tag{43}$$

It fixes constant components: the two contour sums cancel by translation, and $A([x, x']) = LA_\nu$ at a bond in direction ν.

Lemma 4.6. Suppose $t \mapsto \mathbf{U}_t$ is a curve of G^c-valued configurations on P_c, differentiable at zero, with $\mathbf{U}_0 = \mathbf{1}$ and $\dot{\mathbf{U}}_0(b) = iA(b)$. For small $|t|$ the complex average is defined, and

$$\mathcal{M}(\mathbf{1})(c) = \mathbf{1}, \qquad \frac{d}{dt}\mathcal{M}(\mathbf{U}_t)(c)\Big|_{t=0} = i(\mathbf{Q}^{\mathrm{lin}}A)(c). \tag{44}$$

For a curve on $\mathsf{D}_n(c)$ the same assertion holds for the nth iterate, with the composition $\mathbf{Q}^{\mathrm{lin}}_{j\to j+n}$ on the right. For real curves these averages agree, near zero, with the global real averages. ■

Proof. The identity configuration lies in $\mathcal{O}'_c$, and all its contour means and relative loop variables equal $\mathbf{1}$. Openness gives the first assertion. A path product has derivative $iA(\Gamma)$ at zero. Differentiating the equation defining the contour mean, whose value is $\mathbf{1}$ at zero, gives

$$0 = \sum_{\Gamma \in \mathcal{G}(y,x)} [iA(\Gamma) - \dot{\mathbf{U}}_0(y, x)], \qquad \dot{\mathbf{U}}_0(y, x) = i\bar{A}(y, x).$$

Here the differential of log at $\mathbf{1}$ is the identity. The derivative of W_x, and of $\log W_x$, is therefore

$$i[\bar{A}(c_-, x) + A([x, x']) - \bar{A}(c_+, x') - A([c_-, c_+])].$$

After averaging over the L^3 sites, the exponent has derivative $i[\mathbf{Q}^{\mathrm{lin}}A(c) - A([c_-, c_+])]$. The final factor $\mathbf{U}_t[c]$ contributes the opposite reference-path term, proving (44). For iterates apply this calculation successively on the finitely many determining bonds and use the chain rule. Each intermediate configuration equals the identity at zero. For real curves, continuity puts every intermediate plaquette inside its fixed small-curvature bound for sufficiently small $|t|$, so the real and complex branches agree throughout the finite iteration. □

There is one useful coefficient in (42). The corridor $B(c)$ consists of the fine bonds directed from $B(c_-)$ to $B(c_+)$. It has L^2 members, all directed as c. In fine-lattice units its initial sites have ν-coordinate $(L-1)/2$ relative to c_- and transverse coordinates between $-(L-1)/2$ and $(L-1)/2$. Contour paths stay within a block, so only the straight paths $[x, x']$ contribute to a corridor bond. Exactly the L starting sites on its longitudinal line contribute, each once. Thus, writing $\mathbf{Q}^{\mathrm{lin}}A(c) = \sum_b \mathsf{w}_c(b)A(b)$ over positive fine bonds, its unique real coefficients vanish outside P_c and satisfy

$$\mathsf{w}_c(b) = L^{-2} \quad (b \in B(c)), \qquad \sum_{b \in B(c)} \mathsf{w}_c(b) = 1. \tag{45}$$

4.6. **Image measures and block fields.** For $0 \le k < K$ and a finite complex Borel measure ν on $\mathcal{V}_k$, define $(\mathbf{T}\nu)(E) = \nu(\mathcal{M}^{-1}E)$ for Borel $E \subset \mathcal{V}_{k+1}$.

Lemma 4.7 (Image measures)**.** The measure $\mathbf{T}\nu$ is finite and complex, depends linearly on ν, preserves its total mass, and satisfies

$$\int F\, d(\mathbf{T}\nu) = \int F(\mathcal{M}U)\, d\nu \tag{46}$$

for every bounded Borel F on $\mathcal{V}_{k+1}$, with $|\mathbf{T}\nu| \le \mathbf{T}|\nu|$. For $0 \le j \le l \le K$ and ν on $\mathcal{V}_j$, $\mathbf{T}^{l-j}\nu$ is the image under $\mathcal{M}^{l-j}$, so these iterates preserve mass and compose according to their levels. In particular, for ν on $\mathcal{V}_0$,

$$\int F\, d(\mathbf{T}^k\nu) = \int F(\mathcal{M}^k U)\, d\nu, \qquad 0 \le k \le K. \tag{47}$$

■

Proof. The map $\mathcal{M}$ is Borel. The integral formula follows from indicators by bounded approximation; taking total variations over Borel partitions gives $|\mathbf{T}\nu| \le \mathbf{T}|\nu|$. Linearity, mass preservation and iteration follow from the corresponding properties of inverse images. □

Positivity is preserved. If ν is gauge invariant, extend a coarse gauge transformation v by the identity to a fine gauge u. Then $\mathcal{M}(U^u) = (\mathcal{M}U)^v$ and (46) give gauge invariance of $\mathbf{T}\nu$. For a probability measure ν on $\mathcal{V}_0$, $\mathbf{T}^k\nu$ is the law of $\mathcal{M}^k U$ when U has law ν; in particular the Wilson measure gives a gauge-invariant probability measure at each coarse level. A representation by a Haar density requires a further argument.

5. The nonabelian mean

Writing $T_i = V_i V_1^{-1}$, the linear terms in $Y + \log(T_i e^{-Y})$ cancel when $T_i = \mathbf{1}$. The following estimates quantify the error when T_i is close to $\mathbf{1}$. They also give the uniqueness of the mean on G. At the end of the section, lemma 5.3 establishes the Borel and covariance properties of arbitrary unique group means needed for gauge normalization in section 6.

Lemma 5.1 (Exponential and logarithm estimates)**.** For complex 2×2 matrices, the following hold. If $|Z|, |Z'| \le r$, then

$$|e^Z| \le e^{|Z|}, \qquad |e^Z - \mathbf{1}| \le e^{|Z|} - 1,$$
$$|e^Z - e^{Z'}| \le e^r |Z - Z'|, \qquad |e^Z - e^{Z'} - (Z - Z')| \le (e^r - 1)|Z - Z'|.$$

Also $|We^Z - \mathbf{1}| \le (1 + |W - \mathbf{1}|)e^{|Z|} - 1$. For $|E| < 1$,

$$|\log(\mathbf{1} + E)| \le -\ln(1 - |E|) \le \frac{|E|}{1 - |E|}, \qquad |(\mathbf{1} + E)^{-1} - \mathbf{1}| \le \frac{|E|}{1 - |E|},$$

and $e^{\log(\mathbf{1}+E)} = \mathbf{1} + E$. For $|Z| < \ln 2$, one has $|e^Z - \mathbf{1}| < 1$ and $\log(e^Z) = Z$.

If $|A - \mathbf{1}|, |B - \mathbf{1}| \le \varepsilon < 1$, then

$$\log A - \log B = \int_0^1 (\mathbf{1} + t(A - \mathbf{1}))^{-1}(A - B)(\mathbf{1} + t(B - \mathbf{1}))^{-1}\, dt \tag{48}$$

and

$$|\log A - \log B - (A - B)| \le \frac{\varepsilon}{1 - \varepsilon}|A - B|,$$
$$|\log A - \log B| \le \frac{1}{1 - \varepsilon}|A - B|. \tag{49}$$

For unitary A, B with $|A - \mathbf{1}|, |B - \mathbf{1}| < 1$, the sharper uniform estimate is

$$|\log A - \log B| \le 2|A - B|.$$

Finally, $\det e^Z = e^{\operatorname{tr} Z}$. If $X \in G^c$ and $|X - \mathbf{1}| < 1$, then $\log X \in \mathfrak{g}^c$; if $X \in G$ as well, then $\log X \in i\mathfrak{g}$. ■

Proof. The operator norm is submultiplicative and unchanged by unitary multiplication. All power series used below converge absolutely in this norm. Multiplying the exponential series gives $e^{Z}e^{-Z} = \mathbf{1}$; conjugation commutes termwise with both the exponential and the logarithm whenever the latter series converges before and after conjugation. The identity

$$Z^q - Z'^q = \sum_{j=0}^{q-1} Z^j (Z - Z') Z'^{q-1-j}$$

bounds the qth difference by $qr^{q-1}|Z - Z'|$. Summing the exponential series, and then omitting its linear term, gives the exponential estimates. The estimate with W follows by writing $We^{Z} - \mathbf{1} = (W - \mathbf{1})e^{Z} + (e^{Z} - \mathbf{1})$. The logarithmic and inverse bounds follow from their power series and the Neumann series.

For completeness, the inverse identities can be checked along $0 \le t \le 1$. Put $g(t) = \log(\mathbf{1}+tE)$ and $R(t) = (\mathbf{1} + tE)^{-1}$. Termwise differentiation is justified by $|E| < 1$, and gives $g' = ER$, $R' = -ER^2$. These matrices commute, being limits of polynomials in E. It follows that $(e^{g(t)}R(t))' = 0$, and its value at 0 is $\mathbf{1}$. Thus $e^{\log(\mathbf{1}+E)} = \mathbf{1} + E$. If $|Z| < \ln 2$, the bound $|e^{tZ} - \mathbf{1}| \le e^{|Z|} - 1 < 1$ permits the same differentiation of the logarithmic series; it yields $(\log(e^{tZ}))' = Z$ and hence $\log(e^{Z}) = Z$.

Set $E = A - \mathbf{1}$, $F = B - \mathbf{1}$, $R_t = (\mathbf{1} + tE)^{-1}$, and $S_t = (\mathbf{1} + tF)^{-1}$. Integrating the Neumann series gives $\log A = \int_0^1 ER_t\,dt$, and similarly for B. Since E commutes with R_t,

$$ER_t - FS_t = R_t(E(\mathbf{1} + tF) - (\mathbf{1} + tE)F)S_t = R_t(A - B)S_t,$$

which proves (48). Moreover,

$$|R_t|, |S_t| \le (1 - t\varepsilon)^{-1}, \qquad |R_t - \mathbf{1}|, |S_t - \mathbf{1}| \le t\varepsilon(1 - t\varepsilon)^{-1}.$$

The decomposition $R_t(A-B)S_t - (A-B) = (R_t - \mathbf{1})(A-B)S_t + (A-B)(S_t - \mathbf{1})$ therefore bounds its norm by $[(1 - t\varepsilon)^{-2} - 1]|A - B|$. Integration, using $\int_0^1 (1 - t\varepsilon)^{-2}\,dt = (1 - \varepsilon)^{-1}$, gives both estimates in (49).

When A is unitary and $|A - \mathbf{1}| < 1$, each eigenvalue $e^{i\theta}$ has $\cos\theta > 1/2$. Unitary diagonalization gives

$$|(\mathbf{1} + t(A - \mathbf{1}))^{-1}| \le (1 - t/2)^{-1}.$$

The same bound applies to B, so (48) gives $|\log A - \log B| \le |A - B| \int_0^1 (1 - t/2)^{-2}\,dt = 2|A - B|$.

Triangularizing Z proves $\det e^{Z} = e^{\operatorname{tr} Z}$. If $|X - \mathbf{1}| < 1$, its two eigenvalues lie in the disc $|\lambda - 1| < 1$, where the scalar logarithm has imaginary part in $(-\pi/2, \pi/2)$. The trace of the matrix logarithm is the sum of these scalar logarithms, as follows by taking the trace termwise in its series. For $\det X = 1$, this sum belongs to $2\pi i\mathbb{Z}$ and has imaginary part in $(-\pi, \pi)$, so it is zero. If X is also unitary, unitary diagonalization shows that its logarithm is skew-Hermitian. Thus it belongs to $i\mathfrak{g}$. □

The exponential and the local logarithm are holomorphic in the matrix entries by local uniform convergence of their power series; products and inversion on $\mathrm{GL}(2, \mathbb{C})$ are holomorphic as well. Local uniform limits preserve holomorphy by Cauchy's formula in each coordinate. These facts apply in the charts $A \mapsto e^{A}X_0$ of G^c. A matrix-valued holomorphic map with values in G^c is therefore holomorphic into G^c; its restriction to G is real-analytic when it takes values in G.

For the definition of the averaging map away from the small-field set, we also use the intrinsic logarithm on $G \setminus \{-\mathbf{1}\}$.

Lemma 5.2 (The intrinsic logarithm)**.** Every U in this set has a unique representation $U = e^{iH}$ with $H \in \mathfrak{g}$ and $|H| < \pi$. Put $\log U = iH$. Writing $s(\theta) = \sin\theta/\theta$, with $s(0) = 1$, this gives

$$H = \frac{U - U^{-1}}{2i\, s(\arccos(\operatorname{tr} U/2))}.$$

In particular, the intrinsic logarithm is continuous, agrees with the logarithmic series when $|U-\mathbf{1}|<1$, and satisfies

$$\log(WUW^{-1}) = W(\log U)W^{-1}, \qquad \log(U^{-1}) = -\log U \qquad (W \in G).$$

■

Proof. A trace-zero Hermitian H satisfies $H^2 = |H|^2\mathbf{1}$, so $e^{iH} = \cos|H|\,\mathbf{1} + i\,s(|H|)H$. The characteristic equation of $U \in G$ gives $U + U^{-1} = (\operatorname{tr} U)\mathbf{1}$. Thus $U = t\mathbf{1} + iJ$, where $t = \operatorname{tr} U/2 \in \mathbb{R}$, $J \in \mathfrak{g}$, and $t^2 + |J|^2 = 1$. For $U \neq -\mathbf{1}$, set $\theta = \arccos t \in [0,\pi)$. Then $H = J/s(\theta)$ satisfies $|H| = \theta$ and $e^{iH} = U$. Conversely the exponential formula determines θ from the trace and then determines H, proving uniqueness and the displayed formula. Continuity follows from this formula, while conjugation and inversion follow from uniqueness. For $|U-\mathbf{1}|<1$, unitary diagonalization shows that the eigenphases lie in $(-\pi/3,\pi/3)$; the logarithmic series has these eigenphases multiplied by i, and therefore gives the same logarithm. □

The numerical bounds follow by substituting the indicated rational arguments into

$$e^t \le 1 + t + \frac{t^2}{2} + \frac{t^3}{6(1-t/4)} \qquad (0 \le t < 4),$$

and using $-\ln(1-x) \le x + x^2/2 + x^3/(3(1-x))$ for $0 \le x < 1$.

Proof of proposition 4.3. Write $T_i = V_i V_1^{-1}$, so that $|T_i - \mathbf{1}| \le d$ and $|T_i| \le 1 + d$. For $|Y| \le 1/8$ put $M_i(Y) = T_i e^{-Y}$. Then

$$|M_i(Y) - \mathbf{1}| \le (1+d)e^{|Y|} - 1 \le \frac{41}{40}e^{1/8} - 1 < \frac{1}{6}.$$

In particular, all logarithms in (26) are defined. For $\Delta = Y - Y'$, the exponential estimate gives

$$e^{-Y} - e^{-Y'} = -\Delta + \sigma, \qquad |\sigma| \le (e^{1/8} - 1)|\Delta|.$$

By (49) with $\varepsilon = 1/6$,

$$\log M_i(Y) - \log M_i(Y') = M_i(Y) - M_i(Y') + \rho_i, \qquad |\rho_i| \le \frac{1}{5}(1+d)e^{1/8}|\Delta|.$$

Consequently,

$$\Delta + \log M_i(Y) - \log M_i(Y') = -(T_i - \mathbf{1})\Delta + T_i\sigma + \rho_i.$$

Averaging yields the contraction estimate

$$\begin{aligned} |\Phi_{\mathbf{V}}(Y) - \Phi_{\mathbf{V}}(Y')| &\le \left[d + (1+d)(e^{1/8}-1) + \frac{1}{5}(1+d)e^{1/8}\right]|Y - Y'| \\ &\le \frac{2}{5}|Y - Y'|. \end{aligned} \tag{50}$$

Here $e^{1/8} < 1.1332$ makes the bracket less than $2/5$. At the origin, $|\Phi_{\mathbf{V}}(0)| \le d/(1-d) \le 1/39$; hence

$$|\Phi_{\mathbf{V}}(Y)| \le \frac{1}{39} + \frac{2}{5}\frac{1}{8} = \frac{1}{39} + \frac{1}{20} < \frac{1}{8}.$$

The ball is complete, so the contraction principle gives its unique fixed point. The same estimates give

$$|Y_*| \le \frac{5}{3}\frac{d}{1-d} \le 2d.$$

Since e^{Y_*} is invertible with inverse e^{-Y_*}, $V_i \operatorname{av}(\mathbf{V})^{-1} = M_i(Y_*)$. The fixed-point equation is the logarithmic equation in (28). Its bound follows from

$$|M_i(Y_*) - \mathbf{1}| \le (1+d)e^{2d} - 1 \le e^{3d} - 1 \le 3de^{3d} \le 4d,$$

where $e^{3/40} < 1.078$.

Suppose an invertible V' satisfies the two mean conditions and $|V_1V'^{-1} - \mathbf{1}| \le 1/10$. For $X = V'V_1^{-1}$, the inverse estimate gives $|X - \mathbf{1}| \le 1/9$. Thus

$$Y = \log X, \qquad e^Y = X, \qquad |Y| \le \ln(9/8) < 1/8.$$

The mean equation says $\Phi_{\mathbf{V}}(Y) = Y$. It follows that $Y = Y_*$ and $V' = \mathrm{av}(\mathbf{V})$. If the $1/10$ condition holds at V_j instead, apply this argument first with V_j numbered first. The mean constructed with that numbering lies within $4d \le 1/10$ of every member, so the argument with V_1 numbered first identifies the two constructions. This also proves independence of the numbering. For a constant family, $T_i = \mathbf{1}$ and $\Phi_{\mathbf{V}}(Y) = Y + \log(e^{-Y}) = 0$ on the ball; hence $Y_* = 0$ and its mean is V_1.

If every $V_i \in G^c$, then the ball's intersection with $\mathfrak{g}^c$ is preserved by $\Phi_{\mathbf{V}}$: for trace-zero Y, each $T_i e^{-Y}$ has determinant one and its logarithm has trace zero. If every $V_i \in G$, the same argument preserves the intersection with $i\mathfrak{g}$, since e^{-Y} and $T_i e^{-Y}$ are unitary there and their logarithms are skew-Hermitian. Starting the contraction at zero proves the assertions about Y_* and the mean in both cases.

Now suppose the family and an arbitrary admissible mean V' lie in G. Set $U_i = V_i V'^{-1}$. Then $|U_i - U_1| = |(T_i - \mathbf{1})U_1| \le d$, and the unitary logarithm estimate gives $|\log U_i - \log U_1| \le 2d$. Since $\sum_i \log U_i = 0$,

$$|\log U_1| = \left| \frac{1}{n} \sum_i (\log U_1 - \log U_i) \right| \le 2d \le \frac{1}{20}.$$

It follows that $|U_1 - \mathbf{1}| \le e^{1/20} - 1 < 1/10$. The preceding local uniqueness argument therefore applies to every admissible mean in G.

The diameter is continuous on $\mathrm{GL}(2,\mathbb{C})^n$, so $\mathcal{O}_n$ is open. On $\mathcal{O}_n \times \{|Y| < 1/5\}$,

$$|V_i V_1^{-1} e^{-Y} - \mathbf{1}| < \frac{41}{40} e^{1/5} - 1 < 1.$$

Thus $\Phi_{\mathbf{V}}(Y)$ is jointly holomorphic there. The iterates $Y_0 = 0$, $Y_{q+1} = \Phi_{\mathbf{V}}(Y_q)$ are holomorphic on $\mathcal{O}_n$ and remain in $|Y| \le 1/8$. Their convergence is uniform on that whole open set, because

$$|Y_q(\mathbf{V}) - Y_*(\mathbf{V})| \le \frac{1}{8} \left(\frac{2}{5} \right)^q.$$

Hence Y_* and $e^{Y_*} V_1$ are holomorphic, with the stated real-analytic restrictions.

Finally, unitary left and right multiplication leave the diameter unchanged. They replace each relative matrix by

$$(W V_i W')(W V W')^{-1} = W (V_i V^{-1}) W^{-1}.$$

Unitary conjugation preserves its distance from $\mathbf{1}$ and commutes with the logarithmic series. Thus $W\,\mathrm{av}(\mathbf{V})W'$ is a mean of $W\mathbf{V}W'$ within $4d \le 1/10$ of every member. The local uniqueness already proved gives (29). □

Lemma 5.3 (Unique group means)**.** Let $n \ge 1$. The set $\mathsf{E}_n \subset G^n$ of families having exactly one group mean is Borel, and $\mathrm{Av} : \mathsf{E}_n \to G$ is Borel. Consequently the domain of each averaged contour variable $U(y,x)$ is Borel, and that variable is Borel on its domain. These contour assertions hold for configurations on any reversal-invariant bond set containing all bonds of $B(y)$.

The set E_n is invariant under left and right multiplication of the family by fixed elements $W, W' \in G$, and

$$\mathrm{Av}(W\mathbf{V}W') = W\,\mathrm{Av}(\mathbf{V})W'.$$

■

Proof. For $q \ge 2$, let

$$K_q = \left\{ (\mathbf{V}, V) \in G^n \times G : \max_i |V_i V^{-1} - \mathbf{1}| \le 1 - q^{-1}, \quad \sum_i \log(V_i V^{-1}) = 0 \right\}.$$

These are increasing compact sets: the logarithms are continuous on the indicated closed subdomain of their open domain. Every admissible mean belongs to one of these sets. Also set

$$K'_q = \{(\mathbf{V}, V, V') : (\mathbf{V}, V), (\mathbf{V}, V') \in K_q,\ |V - V'| \geq q^{-1}\}.$$

These sets are compact as well. Projection onto G^n maps each of them to a compact set. Thus the families with at least one mean form $\bigcup_q \operatorname{pr} K_q$, and those with at least two form $\bigcup_q \operatorname{pr} K'_q$. The latter identity follows by taking q large enough for both distinct means and their positive separation. Their difference is E_n, hence is Borel. For closed $F \subset G$,

$$\mathrm{Av}^{-1}(F) = \mathsf{E}_n \cap \bigcup_q \operatorname{pr}(K_q \cap (G^n \times F))$$

is Borel. Finally, the contour transports form a continuous map from the configuration space to G^n, depending only on the bonds of $B(y)$. Taking the inverse image of E_n and composing with Av proves the assertions about the contour variables.

For covariance, the map $V \mapsto WVW'$ is a bijection between the admissible group means of $\mathbf{V}$ and those of $W\mathbf{V}W'$. Indeed, the relative matrices are conjugated by W, which preserves their norms and commutes with the logarithmic series in (25). The inverse bijection uses W^{-1} and $(W')^{-1}$. It preserves the number of means and gives the asserted formula when this number is one. □

6. Gauge normalization and lifts

The contour mean permits a separate integration over each noncentral gauge variable in a block. This gives a normalized gauge factor. An explicit lift then realizes arbitrary coarse bond values as fine configurations whose block averages recover those values exactly. Both constructions are used in Bałaban's renormalization expansion [Bał88, (1.3), (1.5)–(1.7)].

6.1. **Axial gauge and its normalized factor.**

Definition 6.1 (Block axial gauge)**.** Let $0 \leq k < K$ and let P'' be a set of sites of $\mathbb{T}^{(k+1)}$. Write $B(P'') = \bigcup_{y \in P''} B(y)$. A configuration V on a reversal-invariant bond set containing all bonds of these blocks is in the block axial gauge on P'' if all its contour means are defined and

$$V(y, x) = \mathbf{1} \qquad (y \in P'',\ x \in B(y) \setminus \{y\}).$$

■

We use the averaged-contour gauge factor of Bałaban [Bał87, (0.14)–(0.16), pp. 254–255]. Its normalization integrates independently over the gauge value at every noncentral site of these blocks.

For $\mathsf{x} > 0$ and $0 < \epsilon \leq 1$, put

$$\mathsf{z}(\mathsf{x}, \epsilon) = \int_G \mathbf{1}_{\{|v - \mathbf{1}| < \epsilon\}} e^{-\mathsf{x}(1 - \frac{1}{2}\operatorname{tr} v)}\, dv \tag{51}$$

and define

$$\mathfrak{G}(V; P''; \mathsf{x}, \epsilon) = \prod_{y \in P''} \prod_{x \in B(y) \setminus \{y\}} \frac{\mathbf{1}_{\{|V(y,x) - \mathbf{1}| < \epsilon\}}}{\mathsf{z}(\mathsf{x}, \epsilon)} e^{-\mathsf{x}(1 - \frac{1}{2}\operatorname{tr} V(y,x))}. \tag{52}$$

Set this factor equal to zero when any required mean is undefined. Here $\mathsf{z} > 0$: the integrand is positive on a nonempty open set, and such a set has positive Haar measure since finitely many of its translates cover G. The factor is bounded and Borel by lemma 5.3; each of its factors lies between zero and z^{-1}.

For parameters $x_k > 0$ and $\varepsilon_k \in (0, 1]$, write

$$\mathfrak{G}_k(V; P'') = \mathfrak{G}(V; P''; x_k, \varepsilon_k). \tag{53}$$

On the domain where the means exist, the associated gauge-fixing function is

$$G_k(P''; V) = \sum_{y \in P''} \sum_{x \in B(y) \setminus \{y\}} \left(1 - \tfrac{1}{2}\operatorname{tr} V(y, x)\right). \tag{54}$$

Every summand is $\frac{1}{2}|V(y,x)-\mathbf{1}|^2$, so this function is nonnegative and vanishes precisely in axial gauge.

We use invariance of normalized Haar measure under inversion. Indeed, its image κ under inversion is right invariant: for bounded Borel φ and fixed v, left invariance gives

$$\int \varphi(uv)\,\kappa(du) = \int \varphi((v^{-1}u)^{-1})\,du = \int \varphi(u)\,\kappa(du).$$

Integrating $\varphi(uv)$ against $\kappa(du)\,dv$ in the two orders, right invariance of κ and left invariance of dv give $\int \varphi\,d\kappa = \int \varphi\,dv$. Thus $\kappa = dv$.

Proposition 6.2. Suppose all the means $V(y,x)$ in (52) exist. Let u equal $\mathbf{1}$ outside $B(P'')\setminus P''$. Then

$$\int \mathfrak{G}(V^u;P'';\mathsf{x},\epsilon) \prod_{x\in B(P'')\setminus P''} du(x) = 1. \tag{55}$$

■

Proof. By lemma 5.3, left and right multiplication by elements of G preserve the unique-mean domain. Consequently all transformed means exist and $V^u(y,x) = V(y,x)u(x)^{-1}$, since $u(y)=\mathbf{1}$. The blocks are disjoint, and their centers are their only sites of $\mathbb{T}^{(k+1)}$. Hence each integration variable occurs in precisely one factor. Haar invariance under inversion and translation makes the integral of this factor equal to $\mathsf{z}/\mathsf{z}=1$. Taking the product proves (55). □

6.2. **Lifting coarse configurations.** An explicit lift preserves a coarse value on every crossing bond and assigns the identity within each block. Its averages recover the same assignment at every intermediate scale.

Let $0\le k\le K$, and let V be a G- or G^c-valued configuration on a reversal-invariant set $\mathcal{C}$ of bonds of $\mathbb{T}^{(k)}$. Define its bare lift by

$$(Q_k^{s*}V)(b) = \begin{cases} \mathbf{1}, & b_-,b_+\in B^k(y) \text{ for some } y\in\mathbb{T}^{(k)},\\ V(c), & b \text{ lies over } c\in\mathcal{C}. \end{cases} \tag{56}$$

It is defined on precisely these bonds. Reversal leaves the first case unchanged and replaces c by $\bar c$ in the second, so it is a configuration. At $k=0$ this is the identity map.

Proposition 6.3. For $0\le j\le k$ the partial iterate $Q^{s*}_{k\to j}V := \mathcal{M}^j(Q_k^{s*}V)$ is defined on exactly the bonds of $\mathbb{T}^{(j)}$ lying inside an order-$(k-j)$ block or over a bond of $\mathcal{C}$. There,

$$(Q^{s*}_{k\to j}V)(b) = \begin{cases} \mathbf{1}, & b_-,b_+\in B^{k-j}(y),\\ V(c), & b \text{ lies over } c\in\mathcal{C}. \end{cases} \tag{57}$$

In particular $\mathcal{M}^k(Q_k^{s*}V)=V$ on $\mathcal{C}$. Every participating complex average lies in its analytic domain. For real V, the small-curvature case applies at every step and the real and complex averages agree. ■

Proof. Let W_j denote the right side of (57) on the stated domain. We have $W_0 = Q_k^{s*}V$ and $W_k = V$. To prove the assertion, fix $j<k$ and a positive bond c' of $\mathbb{T}^{(j+1)}$. Its endpoints lie in unique order-$(k-j-1)$ blocks centered at $y_-,y_+\in\mathbb{T}^{(k)}$. Block nesting gives $B(c'_\pm)\subset B^{k-j}(y_\pm)$.

If $y_-=y_+$, then $W_j=\mathbf{1}$ on the entire double block of c', and its average is $\mathbf{1}=W_{j+1}(c')$. Otherwise c' lies over $c=\langle y_-,y_+\rangle$. Every bond of its double block is internal to one of $B(c'_-),B(c'_+)$ or is a corridor bond or its reverse. Internal bonds lie within order-$(k-j)$ blocks; corridor bonds lie over c, and their reverses over $\bar c$. Since the corridor is nonempty, all the required fine bonds belong to the domain of W_j if and only if $c\in\mathcal{C}$. This is exactly the domain condition for $W_{j+1}(c')$.

When this condition holds, internal values are $\mathbf{1}$ and all corridor values are $V(c)$. A plaquette within either block has holonomy $\mathbf{1}$; a plaquette crossing between them has two crossing edges whose

factors cancel, so again has holonomy **1**. Each contour stays in one block, hence has transport **1**. Each straight path $[x, x']$ crosses the corridor once, hence has transport $V(c)$. Thus all contour families are constant with mean **1**, and every relative loop W_x in the averaging formula is **1**. The complex formula is defined and gives $V(c) = W_{j+1}(c')$. For real values the plaquette calculation places the field in $\mathcal{S}_{c'}$. Negative bonds follow by inversion. This proves the induction. □

7. Regions and multiscale constraints

Constraints at different scales are assigned to the successive layers of a decreasing sequence of regions. To evaluate a constraint on a coarse bond, both of its fine endpoint blocks must stay in the initial region. We give this domain criterion, then describe the removal of inward bonds and a choice of nested regions that supplies the required separation. The determining data and region towers follow Bałaban's construction [Bał88, (2.10)–(2.13), pp. 256–257].

7.1. **Regions and determining data.** Write $\pi[a_r] = \{\Delta_r(y) : y \in \Lambda_{r,\ell}\}$, $r \geq -m$. A region for this partition is a union of its closed cubes, including the empty union. Its complement X^c means the union of the other cubes. As a point set this is the closure of $T_\ell \setminus X$ and is independent of refinement. Indeed, excluded cube interiors are disjoint from X and dense in their cubes; the union of the excluded cubes is closed and contains the point-set complement. A lattice site at the partition scale or a finer scale lies in exactly one of X, X^c, by the block membership established above. Unions, intersections and differences of regions mean the corresponding operations on their cube families at a specified common refinement. Two cubes touch when their intersection is nonempty. We use $\operatorname{dist}_\infty(\cdot, \emptyset) = +\infty$.

For two cubes in the same standard partition, of side s centered at y, y',

$$\operatorname{dist}_\infty(\Delta(y), \Delta(y')) = \max\{0, |y - y'|_\infty - s\}. \tag{58}$$

For the lower bound use the triangle inequality. For equality, choose a least lift v of $y' - y$ and move from each center toward the other by $t_\mu = \operatorname{sign}(v_\mu) \min\{s/2, |v_\mu|/2\}$ in coordinate μ. The resulting points belong to their cubes and their difference has coordinates of modulus $\max\{0, |v_\mu| - s\}$. Distances between nonempty regions of this partition are consequently attained integer multiples of s. The diameter of a cube in the sup metric is at most s.

Definition 7.1. Let $0 \leq k \leq K$ and let $\Omega_0 \supset \Omega_1 \supset \cdots \supset \Omega_k$ be regions, with Ω_j a union of cubes of side at least a_{K-j}. Empty members are allowed. Set $\Omega_{k+1} = \emptyset$ and, for $0 \leq j \leq k$, let

$$\Lambda_j(\mathbf{B}) = (\Omega_j \setminus \Omega_{j+1})^{(j)}, \qquad \Gamma_j = \{\text{inner bonds of } \Omega_0 \text{ at level } j \text{ with an endpoint in } \Lambda_j(\mathbf{B})\}.$$

Here $X^{(j)} = X \cap \mathbb{T}^{(j)}$, and the region difference is taken at cube side a_{K-j}. Thus its sites at level j are precisely those in Ω_j and outside Ω_{j+1}. The determining set is $\mathbf{B} = \bigcup_{j=0}^k \Gamma_j$; its support is Ω_1 (empty when $k = 0$). A datum is a family $V = (V_j)_{j=0}^k$ of G-valued configurations on Γ_j. For a configuration U on the inner bare bonds of Ω_0, put

$$M_{\mathbf{B}}(U) = \big((\mathcal{M}^j U)|_{\Gamma_j}\big)_{j=0}^k$$

whenever all indicated values exist. A node is an endpoint of one of the bonds in $\mathbf{B}$. ■

Proposition 7.2 (Domain of the determining map)**.** For real configurations, the map $M_{\mathbf{B}}$ is defined on every configuration on the inner bare bonds of Ω_0 if and only if

$$B^j(c_-), B^j(c_+) \subset \Omega_0 \qquad (0 \leq j \leq k,\ c \in \Gamma_j). \tag{59}$$

This is a condition on the sequence of regions. Each of the following conditions suffices:

(i) $\Omega_0 = T_\ell$;
(ii) Ω_0 is a region for $\pi[a_{K-k}]$;
(iii) in level-0 units, $\operatorname{dist}_\infty(\Omega_{j-1}^c, \Omega_j) > L^j$ for $1 \leq j \leq k$.

Complex configurations use the complex averaging domains at every participating bond, in addition to the geometric condition.

■

Proof. The exact determining bond set $\mathsf{D}_j(c)$ consists of the bare bonds with both endpoints in $B^j(c_-) \cup B^j(c_+)$. Every site of this double block is an endpoint of such a bond: for $j = 0$ use c, and for $j \geq 1$ use a neighbor within its block. Thus every determining bond is inner to Ω_0 precisely when both blocks lie there. Conditions (i) and (ii) give this by the previously proved block membership property. Under (iii), choose an endpoint $y \in \Lambda_j(\mathbf{B})$ of c and denote the other by y'. For $j = 0$ both endpoints already lie in Ω_0. For $j \geq 1$, the level-j cube centered at y belongs to Ω_j and touches the cube centered at y'. Take a common point z. Every point of the latter cube is within distance L^j of $z \in \Omega_j$. The strict gap excludes this entire cube from Ω_{j-1}^c, so it lies in $\Omega_{j-1} \subset \Omega_0$. Both required blocks are therefore contained in Ω_0. □

We also use the coordinate formula for $M_{\mathbf{B}}$ when the configuration is given on a larger bond domain, evaluating each average on that domain. In particular, a full configuration $V \in \mathcal{V}_k$ induces the data

$$M_{\mathbf{B}}(Q_k^{s*}V) = \big((\mathcal{M}^j(Q_k^{s*}V))|_{\Gamma_j}\big)_{j=0}^{k} = \big((Q_{k\to j}^{s*}V)|_{\Gamma_j}\big)_{j=0}^{k}, \qquad \mathbf{B} = \bigcup_{j=0}^{k} \Gamma_j. \tag{60}$$

All these values are defined by proposition 6.3. When (59) holds, the same datum is obtained from the restriction of the lift to the inner bare bonds of Ω_0, by locality of the averages.

The constraint set consists of the configurations on the inner bare bonds of Ω_0 satisfying $M_{\mathbf{B}}(U) = V$:

$$C(\mathbf{B}, V) = \{U : M_{\mathbf{B}}(U) = V\}.$$

Its residual gauge group is

$$\mathcal{G}^{\mathrm{res}}(\mathbf{B}) = \{u : \Omega_0^{(0)} \to G : \ u = \mathbf{1} \text{ at every node}\}.$$

Set $C(\mathbf{B}, V) = \emptyset$ if the real map is not defined on this domain. The restricted action is $A_{\Omega_0}(U) = \sum_{p \subset \Omega_0} a_p(U)$, with the sum over inner bare plaquettes. The residual group preserves the constraint set: iterated covariance gives

$$(\mathcal{M}^j(U^u))(c) = u(c_-)(\mathcal{M}^j U)(c)u(c_+)^{-1} = (\mathcal{M}^j U)(c) \qquad (c \in \Gamma_j).$$

Extend u by the identity outside Ω_0 when applying that covariance formula. The action is invariant under every gauge transformation, since every plaquette transforms by conjugation.

Lemma 7.3. In level-k units put $\xi = \eta_k = L^{-k}$ and $A_{\Omega_0}^{\eta_k} = L^k A_{\Omega_0}$. For $k < K$, $A_{\Omega_0}^{\eta_{k+1}} = L A_{\Omega_0}^{\eta_k}$. If $r \geq 0$, U is real and $|\partial U(p) - \mathbf{1}| \leq r\xi^2$ for every inner bare plaquette of Ω_0, then

$$A_{\Omega_0}^{\eta_k}(U) \leq \tfrac{1}{2} r^2 \xi^3 \#\{p \subset \Omega_0\} \leq \tfrac{3}{2} r^2 \operatorname{vol}(\Omega_0),$$

where the volume is in level-k units. ■

Proof. The rescaling identity follows from $\eta_{k+1}^{-1} = L\eta_k^{-1}$. For the bound, $a_p(U) = \frac{1}{2}|\partial U(p) - \mathbf{1}|^2 \leq \frac{1}{2} r^2 \xi^4$. Sum over inner plaquettes and multiply by ξ^{-1}. Each plaquette has a base site in Ω_0 and one of three direction pairs, so there are at most three per site. The region contains one bare cube of volume ξ^3 per bare site, proving the last inequality. □

7.2. **Removing inward bonds.** When $k \geq 1$, removing inward bonds leaves the noncentral sites of blocks centered in $\Omega_k^{(k)}$ outside the node set, as proved in lemma 7.5(iv). The residual group can then vary the gauge values at these sites while preserving the retained constraints. The sequence of regions is kept fixed.

Definition 7.4 (Bond sets)**.** Let $\mathbf{B} = \bigcup_{j=0}^{k} \Gamma_j$ be the determining set of $\Omega_0 \supset \cdots \supset \Omega_k$, and put $\Omega_{k+1} = \emptyset$. If $\mathbf{C} \subset \mathbf{B}$ is closed under reversal, its members are $\mathbf{C} \cap \Gamma_j$. Data, the averaging map, nodes, the constraint set and the residual gauge group are defined by restricting the corresponding definitions to these members. Thus

$$M_{\mathbf{C}}(U) = \big((\mathcal{M}^j U)|_{\mathbf{C}\cap\Gamma_j}\big)_{j=0}^{k}, \qquad C(\mathbf{C}, V) = \{U : M_{\mathbf{C}}(U) = V\},$$

where U is a configuration on the inner bare bonds of Ω_0. The map is defined when every displayed average is defined. The site sets remain those of the given sequence.

For $j < k$, a bond of Γ_j is *inward* if it has an endpoint in Ω_{j+1}. Write $\mathfrak{F}_{\mathrm{in}}$ for the set of all inward bonds. For a reversal-invariant subset $\mathfrak{F} \subset \mathfrak{F}_{\mathrm{in}}$, the associated *bond set* and the *reduced determining set* are

$$\mathbf{B}_{\mathfrak{F}} = \mathbf{B} \setminus \mathfrak{F}, \qquad \mathbf{B}^{\flat} = \mathbf{B} \setminus \mathfrak{F}_{\mathrm{in}}, \qquad \Gamma_j^{\flat} = \Gamma_j \setminus \mathfrak{F}_{\mathrm{in}}.$$

The full set is $\mathbf{B}_{\emptyset} = \mathbf{B}$. ■

The residual group preserves $C(\mathbf{C}, V)$ by the same covariance calculation: each retained bond has both endpoints among the nodes fixed by this group. The restricted action remains gauge invariant.

Lemma 7.5 (Members and nodes)**.** Let $\mathbf{B}$ be as above.

(i) An ordered pair of sites is a bond of at most one level $0 \le j \le K$. Hence the members Γ_j are pairwise disjoint. Each member and $\mathfrak{F}_{\mathrm{in}}$ are closed under reversal.

(ii) The set $\Gamma_j^{\flat}$ consists of the bonds of $\mathbb{T}^{(j)}$ with both endpoints in Ω_0, at least one endpoint in Ω_j, and neither endpoint in Ω_{j+1}. Every set $\mathbf{B}_{\mathfrak{F}}$ contains $\mathbf{B}^{\flat}$, and therefore contains every level-j bond with both endpoints in $\Lambda_j(\mathbf{B})$.

(iii) Suppose $k < K$, and append a region $\Omega'_{k+1} \subset \Omega_k$ whose partition cubes have side at least the spacing of $\mathbb{T}^{(k+1)}$. Write Γ'_j and $\Gamma'^{\flat}_j$ for the new full and reduced members. For $j < k$ these agree with the old members. The top member $\Gamma'_{k+1} = \Gamma'^{\flat}_{k+1}$ consists of the level-$(k+1)$ bonds with both endpoints in Ω_0 and an endpoint in Ω'_{k+1}. Moreover,

$$\Gamma_k^{\flat} \setminus \Gamma'^{\flat}_k = \{b \text{ of level } k : b_-, b_+ \in \Omega_0,\ \{b_-, b_+\} \cap \Omega'_{k+1} \neq \emptyset\}.$$

For the full sets, $\Gamma_k \setminus \Gamma'_k$ instead consists of those $b \in \Gamma_k$ whose endpoints each lie in Ω'_{k+1} or outside Ω_k. If

$$\operatorname{dist}_\infty(\Omega_k^c, \Omega'_{k+1}) > L^k$$

in level-0 units, this is exactly the set of level-k bonds with both endpoints in Ω'_{k+1}.

(iv) Every node of $\mathbf{B}^{\flat}$ belongs to $\bigcup_{j=0}^{k} \Lambda_j(\mathbf{B})$. In particular, if $k \ge 1$ and $Y \subset \Omega_k^{(k)}$, then $B(Y) \setminus Y$ contains no node of $\mathbf{B}^{\flat}$. Here $B(Y)$ is the union of the level-$(k-1)$ blocks centered at the sites of Y.

■

Proof. Use level-0 units. If the same ordered pair is a bond of levels q, q', then $\epsilon L^q e_\mu - \epsilon' L^{q'} e_{\mu'}$ is a period of the torus. Every coordinate has modulus at most $2L^K < 2L^{m+K}$, so this period vanishes. Thus $q = q'$. The endpoint conditions for the full members and for inward bonds are invariant under reversal, proving (i).

Removing the inward bonds from Γ_j leaves exactly the bonds described in (ii), including at $j = k$ because $\Omega_{k+1} = \emptyset$. Since $\mathfrak{F} \subset \mathfrak{F}_{\mathrm{in}}$, every $\mathbf{B}_{\mathfrak{F}}$ contains $\mathbf{B}^{\flat}$. Both endpoints of a bond in $\Lambda_j(\mathbf{B})$ lie in Ω_j and outside Ω_{j+1}, giving the last claim.

The members with $j < k$ depend only on $\Omega_0, \Omega_j, \Omega_{j+1}$, which have not changed. At level k, the old reduced member equals the old full member, and its new reduced member deletes exactly the bonds meeting Ω'_{k+1}. The new full member deletes a bond precisely when neither endpoint belongs to $\Omega_k \setminus \Omega'_{k+1}$. This gives the two descriptions in (iii). Under the distance condition, a deleted full-set

bond has an endpoint in Ω'_{k+1}, and its other endpoint, at distance L^k, cannot lie outside Ω_k. Both therefore lie in Ω'_{k+1}. The converse follows from the same descriptions.

For (iv), let x be an endpoint of a bond of $\Gamma_i^{\flat}$. Then $x \in \Omega_0$ and $x \notin \Omega_{i+1}$. There is a unique $i' \le i$ with $x \in \Omega_{i'}$ and $x \notin \Omega_{i'+1}$. The inclusion $\mathbb{T}^{(i)} \subset \mathbb{T}^{(i')}$ gives $x \in \Lambda_{i'}(\mathbf{B})$. If $y \in Y$ and $x \in B(y) \setminus \{y\}$, block containment gives $x \in \Omega_k$. It cannot belong to a site set of index less than k, and it is not a site of $\mathbb{T}^{(k)}$, since $B(y) \cap \mathbb{T}^{(k)} = \{y\}$. It is therefore not a node. □

7.3. **Separated towers.** Fix powers $\mathfrak{M}, R_1, M_1$ of L, with nonnegative integer exponents, $R_1 \ge L$ and $M_1 \ge R_1\mathfrak{M}$. For $s \in \{L^r : r \in \mathbb{N}_0\}$ and $0 \le q \le K$, an s-cube at level q has physical side sa_{K-q}; such cubes exist exactly when $s \le L^{m+K-q}$. Let Ω be a union of cubes of side at least $M_1 a_{K-i}$, with $0 \le i \le k \le K$. A tower to an integer depth $n \ge 0$, $i + n \le k$, is a decreasing sequence $\Omega = \Omega_i \supset \cdots \supset \Omega_{i+n}$ built by either of the following rules, for $i \le q < i + n$. In these rules lengths are in level-i units.

(i) Under $\mathfrak{r}_1$, take all $\mathfrak{M}$-cubes at level $q+1$ contained in Ω_q whose distance from Ω_q^c is strictly greater than $R_1\mathfrak{M}L^{q-i}$.

(ii) Under $\mathfrak{r}_2$, take all M_1-cubes at level $q+1$ contained in Ω_q whose distance from Ω_q^c is at least $M_1 L^{q+1-i}$.

The union is empty when the required cubes do not exist. Each rule selects the largest union of its specified cubes satisfying its own condition.

More generally, a decreasing sequence satisfies the separation condition of $\mathfrak{r}_1$ if each member is a union of $\mathfrak{M}$-cubes at its level, is empty when no such cubes exist, and

$$\operatorname{dist}_\infty(\Omega_q^c, \Omega_{q+1}) > R_1\mathfrak{M}L^{q-i}. \tag{61}$$

Both tower rules satisfy this condition. Indeed, the gap of a finite union is the minimum of the gaps of its cubes, and $M_1 L^{q+1-i} \ge R_1\mathfrak{M}L^{q+1-i} > R_1\mathfrak{M}L^{q-i}$. Every M_1-cube is a union of $\mathfrak{M}$-cubes. The constant whole-torus sequence satisfies the condition exactly when $\mathfrak{M} \le L^{m+K-i-n}$; then its complementary regions are empty and all gaps are infinite. The constant whole-torus sequence will also be allowed when this size condition fails; in that case it is not called separated.

For $i = 0$, separation gives

$$\operatorname{dist}_\infty(\Omega_{j-1}^c, \Omega_j) > R_1\mathfrak{M}L^{j-1} \ge L^j,$$

so (59) holds. The region $\Omega_q \setminus \Omega_{q+1}$ is the seam of level q. Write $\mathbf{B}_j(\Omega)$ for the determining set of the $\mathfrak{r}_2$-tower from level zero to depth j; at $j = 0$ it is the determining set of Ω alone.

For a region X in a fixed partition, let $X^{\sim -1}$ be the union of its cubes that do not touch X^c, and set $X^{\sim -2} = (X^{\sim -1})^{\sim -1}$.

Proposition 7.6. Let $1 \le j \le k \le K$, suppose $M_1 \le L^{m+K-j}$, and let Ω be a union of M_1-cubes at level j. In level-zero units set $s_q = M_1 L^q$. Its $\mathfrak{r}_2$-tower from level zero satisfies

$$\Omega_1 = \bigcup\{\Delta \text{ of side } s_1 : \Delta \subset \Omega,\ \operatorname{dist}_\infty(\Delta, \Omega^c) \ge s_1\},$$

and, for $2 \le q \le j$,

$$\Omega_q = \bigcup\{\Delta \text{ of side } s_q : \Delta \subset \Omega,\ \operatorname{dist}_\infty(\Delta, \Omega^c) \ge 2s_q\}.$$

With layers measured in M_1-cubes at level j, $\Omega^{\sim -2} \subset \Omega_j$, with equality for $j \ge 2$. ■

Proof. For a cube Δ of side s_q, put $D(\Delta) = \operatorname{dist}_\infty(\Delta, \Omega^c)$. Both Ω and Ω^c are unions of cubes at this scale. Thus $D(\Delta)$ belongs to $s_q\mathbb{Z} \cup \{+\infty\}$, and every point of Δ lies within $D(\Delta) + s_q$ of Ω^c when the distance is finite. Set $t_1 = s_1$ and $t_q = 2s_q$ for $q \ge 2$, and let $\Omega_q^\star$ be the union of the cubes with $\Delta \subset \Omega$ and $D(\Delta) \ge t_q$. The first step is the definition of $\mathfrak{r}_2$. Suppose $q \ge 2$ and $\Omega_{q-1} = \Omega_{q-1}^\star$.

Every point at distance at most s_{q-1} from Ω^c lies in Ω_{q-1}^c. For strict inequality each containing cube fails the threshold. In the equality case take a shortest segment toward Ω^c; its remaining points

satisfy the strict inequality, and Ω^c_{q-1} is closed. Conversely every point of Ω^c_{q-1} is within $2s_{q-1}$ of Ω^c: a cube there either lies in Ω^c or has $D < t_{q-1} \le 2s_{q-1}$, hence $D \le s_{q-1}$ by quantization. Its diameter supplies the remaining s_{q-1}.

If $D(\Delta) \ge 2s_q$, its subcubes meet the previous threshold, so $\Delta \subset \Omega_{q-1}$. The preceding distance bound gives

$$\operatorname{dist}_\infty(\Delta, \Omega^c_{q-1}) \ge 2s_q - 2s_{q-1} \ge s_q,$$

so Δ is selected by the rule. Conversely, if selected Δ had $D(\Delta) < 2s_q$, quantization would give $\delta = D(\Delta) \le s_q$. Choose points $x \in \Delta$ and $z \in \Omega^c$ realizing δ. If $\delta \le s_{q-1}$, then $x \in \Omega^c_{q-1}$. Otherwise the point on a shortest segment from z to x at distance s_{q-1} from z lies in Ω^c_{q-1} and is at distance $\delta - s_{q-1} < s_q$ from x. Both cases contradict the required gap s_q. This proves $\Omega_q = \Omega^\star_q$ by induction.

At the final scale, a cube remains after one layer is removed exactly when its center is at least $2s_j$ from every center of a complementary cube. It remains after two removals exactly when those distances are at least $3s_j$. For the latter criterion, a complementary center within $2s_j$ gives an intermediate center within s_j of both centers: move each coordinate toward the complementary center by the smaller of s_j and that coordinate's displacement in a least lift. The intermediate cube is removed at the first step and touches the original cube. Conversely, if all complementary centers are at least $3s_j$ away, every cube touching the original has center at least $2s_j$ from them and remains at the first step. Formula (58) identifies this two-layer criterion with $D(\Delta) \ge 2s_j$. The claimed inclusion and equality now follow from the formulas for Ω_j. □

8. Axial gauge representatives

8.1. **Combs and tower gauge conditions.** A gauge transformation changes a transport only at its endpoints. We shall use this fact successively inside the blocks of a tower. Here the paths are fixed coordinate-order paths. The block gauge of definition 6.1 instead uses averaged contour variables.

We use Bałaban's coordinate-order combs [Bał84, (1.7), p. 18] and the relative axial gauge conditions imposed at successive scales [Bał85a, (64)–(69), p. 29] [Bał85c, (1.15), (1.19), pp. 78–79], with the centered blocks and bond sets defined above.

Definition 8.1. On $\mathbb{T}^{(q)}$, write ξ for the spacing in the chosen units and $N_q = 2L^{m+K-q}$ for the number of sites in each direction. If

$$x - y = \xi \sum_{\mu=1}^{3} n_\mu e_\mu, \qquad |n_\mu| < N_q/2,$$

the integers n_μ are uniquely determined. The comb from y to x adjusts the coordinates in the order $3, 2, 1$:

$$\Gamma_{y,x} = [y, (y_1, y_2, x_3)] \cup [(y_1, y_2, x_3), (y_1, x_2, x_3)] \cup [(y_1, x_2, x_3), x]. \tag{62}$$

Each segment is the straight lattice path with the prescribed signed displacement. An empty segment contributes no bonds. Coordinates in this formula may be taken in a lift of any containing box whose sides are less than half the torus side.

Let W_3 be the group of signed coordinate permutations. For $r \in W_3$, the comb in the frame r is

$$\Gamma^r_{y,x} = r\Gamma_{r^{-1}y, r^{-1}x}.$$

Every $r \in W_3$ preserves the lattices, bonds, and standard cube partitions. Unless a frame is specified, we use the coordinate frame in (62).

Let $\mathbf{B}$ be any bond set of the decreasing sequence $\Omega_0 \supset \cdots \supset \Omega_k$, and let U_0, U be G-valued configurations on the inner bare bonds of Ω_0. Here $\mathbf{B}$ may be full, reduced, or obtained by deleting any reversal-invariant subset of the inward bonds; $\Lambda_j(\mathbf{B})$ always denotes the site set of the underlying sequence. We write $U \in \operatorname{Ax}(\mathbf{B}, U_0)$ if

$$(\mathcal{M}^n U)(\Gamma_{z,x}) = (\mathcal{M}^n U_0)(\Gamma_{z,x}) \tag{63}$$

whenever

$$1 \le j \le k, \qquad x_j \in \Lambda_j(\mathbf{B}), \qquad 0 \le n < j,$$
$$z \in \mathbb{T}^{(n+1)} \cap B^{j-n-1}(x_j), \qquad x \in B(z).$$

These are the tower axial gauge conditions relative to U_0. For $U_0 = \mathbf{1}$ we call them the absolute tower axial gauge conditions. Replacing every comb by its image in the frame r defines $\mathrm{Ax}^r(\mathbf{B}, U_0)$. ■

The strict half-period restriction makes the comb unambiguous: two possible displacements differ in each coordinate by an integer multiple of N_q, while their difference has modulus less than N_q. If $z \in \mathbb{T}^{(n+1)}$ and $x \in B(z)$, the offsets have modulus at most $(L-1)/2 < N_n/2$. Thus $\Gamma_{z,x}$ is one of the contour paths already defined in the block, and it stays in $B(z)$.

The transports in (63) are defined on these blocks, as verified in the proof of proposition 8.3 below. At $x = z$ the path is empty and the condition is an identity.

8.2. **A representative of each residual orbit.** Fix a bond set $\mathbf{B}$ of a decreasing sequence $\Omega_0 \supset \cdots \supset \Omega_k$, and put $\Omega_{k+1} = \emptyset$. Its site sets are $\Lambda_j = \Lambda_j(\mathbf{B})$. The tower gauge fixes transports within the blocks associated with these site sets. We first impose its conditions only at endpoints that the residual gauge group is free to change.

Definition 8.2. For $x_j \in \Lambda_j$, call $B^j(x_j)$ its *pillar*, with top x_j. If $x \ne x_j$ belongs to this pillar, define its level and parent by

$$x \in \mathbb{T}^{(n(x))} \setminus \mathbb{T}^{(n(x)+1)}, \qquad x \in B(z(x)), \quad z(x) \in \mathbb{T}^{(n(x)+1)}.$$

Here $0 \le n(x) < j$. The ascent from x is obtained by successively taking parents until reaching x_j. Its *free part* stops at the first node of $\mathbf{B}$ or at the top.

For configurations U_0, U on the inner bare bonds of Ω_0, write $U \in \mathrm{Ax}^\circ(\mathbf{B}, U_0)$ when the conditions (63) hold for every tower triple (n, z, x) with x not a node. Thus the full tower gauge $\mathrm{Ax}(\mathbf{B}, U_0)$ imposes these conditions also when x is a node. The choice $U_0 = \mathbf{1}$ gives the absolute gauges. ■

The definitions of level, parent and ascent are unambiguous; their geometric properties are proved below. The following condition suffices for a representative in every residual orbit:

$$\begin{gathered}\text{every site of } \Lambda_j,\ 1 \le j \le k, \text{ is a node of } \mathbf{B},\\ \text{and every site of } \Lambda_0 \text{ incident to an inner bare bond of } \Omega_0 \text{ is a node.}\end{gathered} \tag{T}$$

The second condition permits isolated bare sites, on which a gauge transformation has no effect on the configuration.

Proposition 8.3 (Tower gauge in a residual orbit)**.** Suppose (T) holds, and let U_0 be any configuration on the inner bare bonds of Ω_0. Every orbit of $\mathcal{G}^{\mathrm{res}}(\mathbf{B})$ on these configurations meets $\mathrm{Ax}^\circ(\mathbf{B}, U_0)$ in exactly one configuration.

For a given configuration U, its representative is U^u. The gauge u is uniquely specified by $u = \mathbf{1}$ at the nodes and at Λ_0, and by the recursion

$$u(x) = (\mathcal{M}^n U_0)(\Gamma_{z,x})^{-1} u(z) (\mathcal{M}^n U)(\Gamma_{z,x}), \qquad (n, z) = (n(x), z(x)), \tag{64}$$

at all remaining sites. This recursion is performed in each pillar in decreasing order of level. If a residual gauge w also satisfies $U^w \in \mathrm{Ax}^\circ(\mathbf{B}, U_0)$, then $w = u$ except possibly at isolated sites of Λ_0 that are not nodes. ■

Proof. We first verify the geometry used in the recursion. The regions Ω_j and Ω_{j+1} are unions of partition cubes at level j. The bare sites in the cube centered at $x_j \in \mathbb{T}^{(j)}$ form $B^j(x_j)$. Consequently, the bare sites of the layer $\Omega_j \setminus \Omega_{j+1}$ are the disjoint union of $B^j(x_j)$ over $x_j \in \Lambda_j$. As these layers partition the bare sites of Ω_0, so do all the pillars.

A non-top site of $B^j(x_j)$ lies at a unique level $0 \le n < j$. Nesting of blocks gives

$$B^j(x_j) \cap \mathbb{T}^{(n)} = B^{j-n}(x_j) = \bigcup_{z \in B^{j-n-1}(x_j)} B(z).$$

The last union is disjoint, so the parent exists uniquely and belongs to the same pillar. It is either the top or a site of strictly greater level. Hence every ascent reaches the top after at most j parent steps.

The nontrivial tower triples (n, z, x), with $x \neq z$, are exactly $(n(x), z(x), x)$ for non-top bare sites x. Indeed $B(z) \cap \mathbb{T}^{(n+1)} = \{z\}$ identifies the level of x, and disjointness of the pillars identifies its top. Conversely the displayed decomposition of a pillar gives the tower triple of every non-top site. A triple with $x = z$ imposes only the identity transport along an empty path.

For a triple belonging to $B^j(x_j)$, the comb lies in $B(z) \subset B^{j-n}(x_j)$. Every bond c of this comb has $B^n(c_\pm) \subset B^j(x_j) \subset \Omega_0$ at the level of bare sites. Thus $\mathcal{M}^n U$ and $\mathcal{M}^n U_0$ are defined on the comb by the partial-domain criterion proved with (41).

Extend any gauge v by $\mathbf{1}$ outside Ω_0. Gauge covariance and cancellation of the factors at internal vertices give

$$(\mathcal{M}^n(U^v))(\Gamma_{z,x}) = v(z)(\mathcal{M}^n U)(\Gamma_{z,x})v(x)^{-1}.$$

It follows that $U^v \in \mathrm{Ax}^\circ(\mathbf{B}, U_0)$ if and only if v satisfies (64), with v in place of u, at every site that is neither a top nor a node. Prescribing v at the tops and nodes determines it at every other site, since the parent has greater level.

Under (T), prescribe $u = \mathbf{1}$ at the nodes and at Λ_0. Every top is now assigned a value. Use (64) at each remaining site after its parent has been assigned. The pillars are disjoint, so every site receives one value. The resulting gauge is residual and its transform lies in the required gauge.

If w is another residual gauge with this property, it agrees with u at all positive-level tops and at all nodes. The same recursion gives equality throughout their pillars. The only possible exceptions are non-node sites of Λ_0. By (T), none is an endpoint of an inner bare bond. Thus w and u agree at the endpoints of every bond of the configuration, and $U^w = U^u$. □

The recursion also records the dependence on U and U_0. For a site x that is neither a node nor a top, let $y_0 = x, \dots, y_f$ be its free ascent and put

$$\tau_s(W) = (\mathcal{M}^{n(y_s)}W)(\Gamma_{y_{s+1},y_s}), \qquad 0 \le s < f, \qquad \mathcal{T}_W(x) = \tau_{f-1}(W)\cdots\tau_0(W).$$

For any gauge v satisfying the recursion with v in place of u, successive substitution gives

$$v(x) = \mathcal{T}_{U_0}(x)^{-1} v(y_f) \mathcal{T}_U(x). \tag{65}$$

The ordering is from the endpoint of the free ascent down to x. For the normalized gauge of proposition 8.3, $u(y_f) = \mathbf{1}$, and hence $u(x) = \mathcal{T}_{U_0}(x)^{-1}\mathcal{T}_U(x)$. If $x \in B^j(x_j)$, this expression uses only averages of orders $n < j$ on the indicated combs, whose sites lie in $B^{j-n(y_s)}(x_j)$. At the prescribed nodes and at Λ_0, u is constant. This is the locality of the gauge construction.

Lemma 8.4. Condition (T) holds for every bond set of a sequence with $\Omega_0 = T_\ell$, and for every bond set of a decreasing sequence satisfying (61). For the full determining set, the level-zero clause of (T) always holds. ■

Proof. Use level-zero units. For $x_j \in \Lambda_j$, choose its block centered at $z \in \Lambda_{K-j-1,\ell}$. This auxiliary coarser lattice exists also at $j = K$: it is $\Lambda_{-1,\ell}$, and $m \ge 1$. Write

$$x_j = z + L^j \sum_{\mu=1}^{3} n_\mu e_\mu, \qquad |n_\mu| \le h = (L-1)/2.$$

Put $\epsilon = -1$ if $n_1 > 0$ and $\epsilon = 1$ otherwise. Then $x' = x_j + \epsilon L^j e_1$ is a different site of the same next-level block, because $h \ge 1$. Since $x_j \notin \Omega_{j+1}$ and this region is a union of next-level cubes,

$x' \notin \Omega_{j+1}$; at $j = k$ this follows from $\Omega_{k+1} = \emptyset$. Whenever $x' \in \Omega_0$, the bond joining x_j to x' lies in $\Gamma_j^\flat$ and therefore in every bond set of the sequence. It makes x_j a node.

For $\Omega_0 = T_\ell$ the required inclusion is automatic. For a separated sequence and $j \geq 1$,

$$|x' - x_j|_\infty = L^j < \operatorname{dist}_\infty(\Omega_{j-1}^c, \Omega_j),$$

so $x' \in \Omega_{j-1} \subset \Omega_0$. For $j = 0$, suppose x_0 has an inner-bond neighbor x''. If $x'' \notin \Omega_1$, that bond itself belongs to $\Gamma_0^\flat$. If $x'' \in \Omega_1$, then

$$|x' - x''|_\infty \leq 2 < L \leq R_1 \mathfrak{M} < \operatorname{dist}_\infty(\Omega_0^c, \Omega_1),$$

so again $x' \in \Omega_0$. This proves both clauses of (T) for every bond set in the stated cases. For the full determining set, any inner bare bond with an endpoint in Λ_0 already belongs to Γ_0, which proves the final assertion without separation. □

8.3. **Conditions at fixed nodes.** For a separated sequence and the full determining set, an inward bond fixes a site inside the next block, and the block center is also a node. The transport between them is therefore invariant under the residual gauge group. The following proposition identifies exactly the tower conditions that remain after gauge fixing outside the nodes.

Under (T), let U^u be the representative in $\mathrm{Ax}^\circ(\mathbf{B}, U_0)$. Its orbit meets the full tower gauge $\mathrm{Ax}(\mathbf{B}, U_0)$ if and only if U^u satisfies the omitted conditions at nontrivial triples whose endpoint x is a node. The intersection then consists only of U^u. This follows because the full gauge is a subset of the gauge outside the nodes.

Proposition 8.5. Suppose that the decreasing sequence satisfies the separation condition (61), in level-zero units, and let $\mathbf{B}$ be any of its bond sets. The nontrivial conditions in (63) omitted from $\mathrm{Ax}^\circ(\mathbf{B}, U_0)$ are exactly those at triples

$$(n, z, x) = (j - 1, x_j, b_+), \qquad 1 \leq j \leq k,$$

where $b \in \mathbf{B}$ is an inward bond of level $j - 1$, oriented with $b_- \in \Omega_{j-1} \setminus \Omega_j$ and $b_+ \in \Omega_j$, and $b_+ \in B(x_j)$. For every such bond,

$$B(x_j) \subset \Omega_j, \qquad x_j \in \Lambda_j(\mathbf{B}), \qquad b_+ \neq x_j.$$

Both x_j and b_+ are nodes of $\mathbf{B}$. No bond of Γ_{x_j, b_+} belongs to the full determining set, and the transport

$$(\mathcal{M}^{j-1} U)(\Gamma_{x_j, b_+})$$

is constant on the residual-gauge orbit of U.

In particular, a residual orbit meets the absolute tower axial gauge if and only if all these transports are $\mathbf{1}$. The intersection then consists of its unique representative in $\mathrm{Ax}^\circ(\mathbf{B}, \mathbf{1})$. ■

Proof. Write $h = (L - 1)/2$. Since Ω_j is a union of cubes of side at least L^j, the block containing b_+ lies in Ω_j. Thus $b_- \notin B(x_j)$. Every level-$(j - 1)$ neighbor of x_j lies in this block, so $b_+ \neq x_j$. If $j < k$, then

$$|x_j - b_-|_\infty \leq (h + 1) L^{j-1} < L^j \leq R_1 \mathfrak{M} L^j < \operatorname{dist}_\infty(\Omega_j^c, \Omega_{j+1}).$$

As $b_- \in \Omega_j^c$, this excludes $x_j \in \Omega_{j+1}$. The same exclusion is automatic at $j = k$, because $\Omega_{k+1} = \emptyset$. Consequently $x_j \in \Lambda_j(\mathbf{B})$ and is a node by lemma 8.4. The endpoint b_+ is a node because b is retained.

The comb stays inside $B(x_j) \subset \Omega_j$, so its bonds have no endpoint in $\Lambda_{j-1}(\mathbf{B})$. They therefore do not belong to the full member Γ_{j-1}. A bond of this lattice is a bond of no other level, so none belongs to the full determining set. Gauge covariance of the averages gives

$$(\mathcal{M}^{j-1} U^w)(\Gamma_{x_j, b_+}) = w(x_j)(\mathcal{M}^{j-1} U)(\Gamma_{x_j, b_+}) w(b_+)^{-1}.$$

For a residual transformation both endpoint factors are $\mathbf{1}$, proving the invariance.

It remains to identify all the omitted conditions. An inward bond as above gives a triple belonging to the pillar with top x_j, and its terminal point is a node. Conversely, suppose that a triple (n, z, x) belongs to the pillar with top x_j and that $x \neq z$ is a node. The pillar decomposition gives $n = n(x) < j$ and $x \in \Omega_j \setminus \Omega_{j+1}$. Some retained bond c of level i has endpoint x. Since $x \in \mathbb{T}^{(i)}$ but $x \notin \mathbb{T}^{(n(x)+1)}$, we have $i \leq n(x) < j$. The other endpoint of c must lie in $\Omega_i \setminus \Omega_{i+1}$, whereas $x \in \Omega_j \subset \Omega_{i+1}$. Thus c is inward. Applying the preceding geometry to c shows that x lies in a pillar with top $x_{i+1} \in \Lambda_{i+1}$, and is a noncentral site of $B(x_{i+1})$. Distinct pillars are disjoint; hence $i + 1 = j$, $x_{i+1} = x_j$, $n = j - 1$, and $z = x_j$.

By proposition 8.3, the orbit has a unique representative satisfying the conditions outside the nodes. It satisfies the full tower conditions precisely when it satisfies the remaining conditions just classified. Their orbit invariance proves the final assertion. □

For $\mathbf{B}^\flat$, all nodes belong to $\bigcup_j \Lambda_j$ by lemma 7.5, whereas a nontrivial tower triple has an endpoint that is not a pillar top. Thus no nontrivial condition is omitted:

$$\mathrm{Ax}^\circ(\mathbf{B}^\flat, U_0) = \mathrm{Ax}(\mathbf{B}^\flat, U_0).$$

Every ascent for the reduced set has its entire length free. When (T) holds, each residual orbit therefore has exactly one representative in the full tower gauge.

For the constant sequence $\Omega_0 = \cdots = \Omega_k = T_\ell$, the layers below k are empty, its bond set consists of all level-k bonds, and its nodes are precisely $\mathbb{T}^{(k)}$. A nontrivial tower triple has endpoint in $\mathbb{T}^{(n)} \setminus \mathbb{T}^{(n+1)}$ for $n < k$, so again no condition is omitted. Condition (T) holds by lemma 8.4.

Finally, if $U_0 \in \mathrm{Ax}^\circ(\mathbf{B}, \mathbf{1})$, the defining transports of U_0 at all the retained triples are $\mathbf{1}$. Hence $\mathrm{Ax}^\circ(\mathbf{B}, U_0) = \mathrm{Ax}^\circ(\mathbf{B}, \mathbf{1})$, and the representatives in these two gauges agree.

9. The current and regular critical configurations

The current is the first variation of the Wilson action with respect to a bond potential. We establish this identity with the spacing explicit, and then use the current to define regular configurations in a multiscale constraint set. The covariant derivatives and current follow Bałaban [Bał85b, pp. 390–392, (3.3)–(3.12)]; we use the ordinary two-by-two trace and display its factor one-half explicitly.

9.1. Forms and the current.

Let Λ be one of the lattices $\Lambda_{j,\ell}$ with $j > -m$, and let ξ be its spacing in the units in use. It has more than two sites along every coordinate axis, so each ordered nearest-neighbor pair has a unique direction and orientation.

Definition 9.1. A one-form B assigns an element of $\mathfrak{g}^c$ to each positive bond and is extended by $B(\bar{b}) = -B(b)$. Write $B_\mu(x) = B(\langle x, x + \xi e_\mu\rangle)$. A two-form assigns a value $F_{\mu\nu}(x) \in \mathfrak{g}^c$ to each plaquette $p_{\mu\nu}(x)$, $\mu < \nu$, and is extended by $F_{\nu\mu} = -F_{\mu\nu}$. A site function has values in $\mathfrak{g}^c$. These objects are real when their values lie in $\mathfrak{g}$.

For $X, Y \in \mathfrak{g}^c$, put $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$. On site functions, one-forms and two-forms, respectively, use the pairings

$$(\lambda, \lambda')_\xi = \xi^3 \sum_x \langle \lambda(x), \lambda'(x)\rangle, \quad (B, C)_\xi = \xi^3 \sum_b \langle B(b), C(b)\rangle, \quad (F, H)_\xi = \xi^3 \sum_p \langle F(p), H(p)\rangle.$$

The bond sum is over positive bonds, and the plaquette sum counts each plaquette once. These pairings are complex bilinear. Their restrictions to real functions and forms are real inner products.

For a real or complex configuration U, set

$$(\nabla^U_\mu f)(x) = \xi^{-1}[\mathrm{Ad}(U(\langle x, x + \xi e_\mu\rangle))f(x + \xi e_\mu) - f(x)].$$

The derivative acts componentwise on forms. Its exterior derivative on one-forms is

$$(D_U B)_{\mu\nu} = \nabla^U_\mu B_\nu - \nabla^U_\nu B_\mu.$$

The transpose D_U^T is defined by $(D_U B, F)_\xi = (B, D_U^\mathsf{T} F)_\xi$. ■

The pairings are nondegenerate: if $X \neq 0$ then $X^* \in \mathfrak{g}^c$ and $\langle X, X^* \rangle = \frac{1}{2}\operatorname{tr}(XX^*) > 0$. Thus the transpose exists and is unique on these finite dimensional spaces. For real U it is the real adjoint, also denoted D_U^* on two-forms.

For $p = p_{\mu\nu}(x)$, $\mu < \nu$, denote its positive bonds by

$$\begin{aligned} b_1 &= \langle x, x + \xi e_\mu \rangle, & b_2 &= \langle x + \xi e_\mu, x + \xi e_\mu + \xi e_\nu \rangle, \\ b_3 &= \langle x + \xi e_\nu, x + \xi e_\mu + \xi e_\nu \rangle, & b_4 &= \langle x, x + \xi e_\nu \rangle. \end{aligned} \tag{66}$$

Thus $\partial p = (b_1, b_2, \bar{b}_3, \bar{b}_4)$ and

$$(D_U B)(p) = \xi^{-1}\big[B(b_1) + \operatorname{Ad}(U(b_1))B(b_2) - \operatorname{Ad}(U(b_4))B(b_3) - B(b_4)\big]. \tag{67}$$

The transpose is

$$(D_U^{\mathsf{T}} F)_\mu(x) = \xi^{-1} \sum_{\nu \neq \mu} \Big[F_{\mu\nu}(x) - \operatorname{Ad}(U(\langle x - \xi e_\nu, x \rangle)^{-1}) F_{\mu\nu}(x - \xi e_\nu)\Big]. \tag{68}$$

Indeed, write $(D_U B, F)_\xi$ as the sum over ordered pairs $\mu \neq \nu$ of $\xi^3 \langle \nabla_\nu^U B_\mu, F_{\nu\mu} \rangle$. In its translated term use $\langle \operatorname{Ad}(V)X, Y \rangle = \langle X, \operatorname{Ad}(V^{-1})Y \rangle$ and replace x by $x - \xi e_\nu$. The coefficient of $B_\mu(x)$ is the right side of (68), since $F_{\nu\mu} = -F_{\mu\nu}$.

For a complex configuration U, define its field strength and current by

$$F_U(p) = \frac{\partial U(p) - \partial U(p)^{-1}}{2i\xi^2}, \qquad J(U) = D_U^{\mathsf{T}} F_U. \tag{69}$$

For $X \in \mathrm{SL}(2, \mathbb{C})$, Cayley–Hamilton gives

$$X + X^{-1} = (\operatorname{tr} X)\mathbf{1}. \tag{70}$$

Hence F_U is traceless, and for real U it is Hermitian. The formulas for D_U and its transpose preserve real forms, so $J(U)$ is real in this case.

Each positive bond belongs to four plaquettes: for each transverse direction ν, the two plaquettes based at x and $x - \xi e_\nu$. Formula (68) shows that $J(U)$ at this bond uses only the variables on those four plaquettes. If U is given on a reversal-invariant partial bond set, $F_U(p)$ is defined when all four bonds of p are given, and $J(U)(b)$ is defined when all four plaquettes containing b are defined. These domain conditions will be retained for regional configurations.

9.2. **Variation of the action.** For a complex configuration, use the same polynomial action $a_p(U) = 1 - \frac{1}{2}\operatorname{tr} \partial U(p)$ and $A^\xi(c, U) = \xi^{-1} \sum_p c(p) a_p(U)$. An inverse in G^c is polynomial in the matrix entries by (70). The perturbation $e^{it\xi B}U$ is defined at positive bonds by

$$(e^{it\xi B}U)(b) = e^{it\xi B(b)}U(b),$$

and at reverse bonds by inversion. It is real for real t, B and U.

Proposition 9.2 (The current as a first variation)**.** Let U be a complex configuration on Λ, let B be a $\mathfrak{g}^c$-valued one-form, and let c be a complex plaquette weight. Then

$$\begin{aligned} \frac{d}{dt}\Big|_{t=0} A^\xi(c, e^{it\xi B}U) &= (D_U B, cF_U)_\xi = (B, D_U^{\mathsf{T}}(cF_U))_\xi, \\ \frac{d}{dt}\Big|_{t=0} A^\xi(e^{it\xi B}U) &= (B, J(U))_\xi. \end{aligned} \tag{71}$$

At the identity configuration,

$$\frac{d^2}{dt^2}\Big|_{t=0} A^\xi(c, e^{it\xi B}) = (cD_{\mathbf{1}} B, D_{\mathbf{1}} B)_\xi.$$

Let Ω_0 be a region whose cubes have side at least the lattice spacing, and let U' be defined on its inner bonds. If B vanishes outside the bonds where $J(U')$ is defined, then

$$\left.\frac{d}{dt}\right|_{t=0} A^{\xi}_{\Omega_0}(e^{it\xi B}U') = (B, J(U'))_{\xi}, \tag{72}$$

where the pairing is summed over that current domain.

For a complex gauge transformation u,

$$F_{U^u}(p) = \mathrm{Ad}(u(x_{\mathrm{b}}(p)))F_U(p), \qquad J(U^u)(b) = \mathrm{Ad}(u(b_-))J(U)(b)$$

at positive bonds. The weighted action is gauge invariant. The same covariance holds on every partial domain where these quantities are defined. ∎

Proof. The SU(2) matrix identity underlying the calculation extends to G^c. For $X \in G^c$, put $\theta = \frac{1}{2}\operatorname{tr} X$ and $H = (X - X^{-1})/(2i)$. Equation (70) gives $X = \theta\mathbf{1} + iH$ and $H^2 = (1-\theta^2)\mathbf{1}$. Applying this to the plaquette variable gives

$$\begin{aligned} \partial U(p) &= (1 - a_p(U))\mathbf{1} + i\xi^2 F_U(p), \\ \xi^4\langle F_U(p), F_U(p)\rangle &= a_p(U)(2 - a_p(U)). \end{aligned} \tag{73}$$

Multiplication by $c(p)/(2\xi)$ and summation yield

$$A^{\xi}(c, U) = \tfrac{1}{2}(cF_U, F_U)_{\xi} + \frac{1}{2\xi}\sum_p c(p)a_p(U)^2. \tag{74}$$

For the first variation, use the four positive bonds in (66), and write $U_i = U(b_i)$, $B_i = B(b_i)$ and $P = \partial U(p)$. The perturbed plaquette variable is

$$e^{it\xi B_1}U_1 e^{it\xi B_2}U_2 U_3^{-1} e^{-it\xi B_3}U_4^{-1} e^{-it\xi B_4}.$$

Its derivative at zero equals

$$i\xi\big[B_1 P + (\mathrm{Ad}(U_1)B_2)P - P\,\mathrm{Ad}(U_4)B_3 - PB_4\big].$$

By cyclicity of the trace and (67), the trace of this expression is $i\xi^2 \operatorname{tr}(P(D_U B)(p))$. The scalar part of P in (73) has zero pairing with the traceless matrix $(D_U B)(p)$. The trace is therefore $-\xi^4 \operatorname{tr}(F_U(p)(D_U B)(p))$, and

$$\left.\frac{d}{dt}\right|_{t=0} a_p(e^{it\xi B}U) = \xi^4\langle (D_U B)(p), F_U(p)\rangle.$$

Multiplying by $\xi^{-1}c(p)$ and summing proves (71), first by the definition of the transpose and then by (69).

For the regional identity, extend U' by $\mathbf{1}$ outside the inner bonds and take c to be the indicator of the inner plaquettes. Then $A^{\xi}_{\Omega_0}(e^{it\xi B}U') = A^{\xi}(c, e^{it\xi B}U)$. By (67), $D_U B$ vanishes at a plaquette unless one of its bonds carries a nonzero value of B. Every such bond is in the current domain of U', so all four plaquettes containing it are inner. Thus $c = 1$ wherever $D_U B$ can be nonzero, and

$$(D_U B, cF_U)_{\xi} = (D_U B, F_U)_{\xi} = (B, J(U))_{\xi}.$$

On the support of B, the four plaquettes defining the current use only U', so $J(U) = J(U')$ there. This proves (72).

At $U = \mathbf{1}$, expansion of the same four factors gives

$$\partial(e^{it\xi B})(p) = \mathbf{1} + it\xi^2(D_{\mathbf{1}}B)(p) + O(t^2), \qquad F_{e^{it\xi B}}(p) = t(D_{\mathbf{1}}B)(p) + O(t^2).$$

Also $a_p(e^{it\xi B}) = O(t^2)$, because its value and first derivative vanish at zero. Equation (74) therefore gives

$$A^{\xi}(c, e^{it\xi B}) = \tfrac{1}{2}t^2(cD_{\mathbf{1}}B, D_{\mathbf{1}}B)_{\xi} + O(t^3),$$

which proves the second-variation formula. These finite products of matrix exponentials are entire in t.

Finally, a plaquette variable is conjugated at its base point by a gauge transformation. This proves the action invariance and the transformation of F_U. In (68), set $y = x - \xi e_\nu$ and use

$$U^u(\langle y, x\rangle)^{-1} u(y) = u(x) U(\langle y, x\rangle)^{-1}.$$

Each summand defining $J(U^u)_\mu(x)$ is conjugated by $u(x)$, which proves the current covariance. The calculation uses only the four incident plaquettes, so it also applies on the stated partial domains. □

For a general variation on the inner bonds of a region, the weighted identity (71), with the indicator of its inner plaquettes, remains the regional variation formula. The unweighted pairing with $J(U')$ in (72) uses the stated support condition.

If the spacing in a second system of units is $\xi' = \upsilon\xi$, the same bond perturbation is written with $B' = \upsilon^{-1} B$. Directly from the formulas,

$$F'_U = \upsilon^{-2} F_U, \qquad J'(U) = \upsilon^{-3} J(U), \qquad (B', J'(U))_{\xi'} = \upsilon^{-1} (B, J(U))_\xi.$$

This agrees with $A^{\xi'}(c, U) = \upsilon^{-1} A^\xi(c, U)$: the pairing contributes the cell weight ξ'^3, while its two arguments contribute υ^{-1} and υ^{-3}.

9.3. **Regularity at the local block scale.** Let $\mathbf{B}$ be a bond set of $\Omega_0 \supset \cdots \supset \Omega_k$.

Definition 9.3. For a bare plaquette or bond $\mathfrak{e}$ meeting Ω_0 at a corner or endpoint, respectively, let j be the largest index for which such a corner or endpoint lies in Ω_j. Its local block side, in level-k units, is

$$\mathsf{L}_{\mathbf{B}}(\mathfrak{e}) = L^{j-k}.$$

Put $\eta_k = L^{-k}$. A real configuration on the inner bare bonds of Ω_0 is *regular at radius* $r > 0$ if

$$\begin{aligned} |\partial U(p) - \mathbf{1}| &< r\eta_k^2 \mathsf{L}_{\mathbf{B}}(p)^{-2} \quad \text{for every inner bare plaquette } p, \\ |J(U)(b)| &< r \mathsf{L}_{\mathbf{B}}(b)^{-3} \quad \text{wherever the current is defined.} \end{aligned} \tag{75}$$

The current is computed with bare spacing η_k. Denote this set by $\mathfrak{U}_k(\mathbf{B}, r)$.

■

For the constant whole-torus sequence all local block sides equal one. Thus $\mathfrak{U}_k(\mathbf{B}, r)$ consists of the configurations with

$$|\partial U(p) - \mathbf{1}| < r\eta_k^2, \qquad |J(U)(b)| < r$$

at every bare plaquette and positive bond. The following comparison relates this condition to a bound on the divergence of the plaquette variables.

Lemma 9.4. Let $1 \le k \le K$, put $\eta = \eta_k$, and let $\mathbf{B}$ be the constant-sequence determining set. Define $\mathfrak{U}'_k(\mathbf{B}, r)$ to be the set of real configurations satisfying

$$|\partial U(p) - \mathbf{1}| < r\eta^2, \qquad |(D_U^{\mathsf{T}} \partial U)(b)| < r\eta^2$$

at every bare plaquette and positive bond. Here D_U^{T} is the formula (68), extended to matrix-valued two-forms; the plaquette variables are extended by $X_{\nu\mu} = -X_{\mu\nu}$ after setting $X_{\mu\nu}(x) = \partial U(p_{\mu\nu}(x))$ for $\mu < \nu$. Then, for every $r > 0$,

$$\mathfrak{U}'_k(\mathbf{B}, r) \subset \mathfrak{U}_k(\mathbf{B}, r) \subset \mathfrak{U}'_k(\mathbf{B}, r + r^2). \tag{76}$$

■

Proof. Write $\partial U(p) = c(p)\mathbf{1} + i\eta^2 F_U(p)$, where $c(p) = \frac{1}{2}\operatorname{tr}\partial U(p)$ is real. Since conjugation fixes scalar matrices, the extension of (68) gives

$$(D_U^{\mathsf{T}}\partial U)(b) = a(b)\mathbf{1} + i\eta^2 J(U)(b).$$

Here $a(b)$ is the sum of two terms $\pm\eta^{-1}(c(p) - c(p'))$, one for each transverse direction, with p, p' the two incident plaquettes in that coordinate plane. For real a and $H \in \mathfrak{g}$, $|a\mathbf{1} + iH|^2 = a^2 + |H|^2$; indeed $H^2 = |H|^2\mathbf{1}$ and $(a\mathbf{1} - iH)(a\mathbf{1} + iH) = (a^2 + |H|^2)\mathbf{1}$. Thus

$$|(D_U^{\mathsf{T}}\partial U)(b)|^2 = a(b)^2 + \eta^4 |J(U)(b)|^2,$$

which proves the first inclusion.

If $U \in \mathfrak{U}_k(\mathbf{B}, r)$, the SU(2) identity $1 - c(p) = \frac{1}{2}|\partial U(p) - \mathbf{1}|^2$ gives $0 \le 1 - c(p) < \frac{1}{2}r^2\eta^4$. Consequently $|c(p) - c(p')| < \frac{1}{2}r^2\eta^4$ and $|a(b)| < r^2\eta^3$. Since $\eta \le 1$,

$$|(D_U^{\mathsf{T}}\partial U)(b)| \le |a(b)| + \eta^2|J(U)(b)| < (r^2\eta + r)\eta^2 \le (r + r^2)\eta^2.$$

The plaquette bound with radius $r + r^2$ follows from the bound with radius r, proving the second inclusion. □

9.4. **Critical orbits.** Return to a general bond set $\mathbf{B}$ of $\Omega_0 \supset \cdots \supset \Omega_k$ and its constraint set $C(\mathbf{B}, V)$.

Definition 9.5. A configuration $U \in C(\mathbf{B}, V)$ is *critical* if

$$\left.\frac{d}{dt}\right|_{t=0} A_{\Omega_0}(U_t) = 0$$

for every differentiable curve $t \mapsto U_t$ in $C(\mathbf{B}, V)$ with $U_0 = U$. Here differentiability means entrywise differentiability on an open real interval containing zero. Fix a radius $\varepsilon_{\mathrm{v}} > 0$, and set

$$\mathcal{C}(\mathbf{B}, V; r) = \{U \in C(\mathbf{B}, V) : U \text{ is critical and } U \in \mathfrak{U}_k(\mathbf{B}, r)\}, \qquad \mathcal{C}(\mathbf{B}, V) = \mathcal{C}(\mathbf{B}, V; \varepsilon_{\mathrm{v}}).$$

If $\mathcal{C}(\mathbf{B}, V)$ is nonempty and consists of one orbit of $\mathcal{G}^{\mathrm{res}}(\mathbf{B})$, call it the *minimal orbit*. This term specifies an orbit of regular critical points and does not assert global action minimization. Write $U_{\mathbf{B}}(V)$ for any element of this orbit; an invariant function of $U_{\mathbf{B}}(V)$ has its common value on the orbit. ■

Criticality is unchanged if the action is multiplied by the positive constant L^k, and hence can be tested with $A_{\Omega_0}^{\eta_k}$. The definition uses curves in the constraint set and does not require it to have been identified as a smooth manifold.

Proposition 9.6. The set $\mathcal{C}(\mathbf{B}, V; r)$ is invariant under $\mathcal{G}^{\mathrm{res}}(\mathbf{B})$. Suppose the minimal orbit exists and condition (T) holds. For every real reference configuration U_0 on the inner bare bonds of Ω_0, there is exactly one element of this orbit in $\mathrm{Ax}^\circ(\mathbf{B}, U_0)$. Denote it by $U_{\mathbf{B}}^\circ(V; U_0)$, and set $U_{\mathbf{B}}^\circ(V) = U_{\mathbf{B}}^\circ(V; \mathbf{1})$.

The absolute tower representative $U_{\mathbf{B}}^{\mathrm{ax}}(V)$ exists precisely when $U_{\mathbf{B}}^\circ(V)$ satisfies the omitted node conditions, and then the two representatives agree. For a reduced bond set under (T), or for the constant whole-torus sequence, they are both defined and equal. ■

Proof. Plaquette and current covariance, with invariance of the operator norm under unitary conjugation, preserve regularity. A residual gauge maps $C(\mathbf{B}, V)$ and its differentiable curves bijectively onto themselves and preserves the action; it therefore preserves criticality. If the minimal orbit exists, proposition 8.3 gives its unique outside-node representative. The full tower-gauge criterion and its reduced and constant cases are the corresponding conclusions of section 8. □

If $U_0 \in \mathrm{Ax}^\circ(\mathbf{B}, \mathbf{1})$, then $\mathrm{Ax}^\circ(\mathbf{B}, U_0) = \mathrm{Ax}^\circ(\mathbf{B}, \mathbf{1})$, so $U_{\mathbf{B}}^\circ(V; U_0) = U_{\mathbf{B}}^\circ(V)$. For a full or partially reduced separated set, the omitted conditions are exactly the retained seam conditions described in proposition 8.5.

9.5. **Backgrounds induced by block fields.** The use of constrained critical orbits as background fields follows Bałaban [Bał87, (0.21), p. 256]; localization by a tower determining set follows [Bał88, (2.12)–(2.16), pp. 256–257]. For $V \in \mathcal{V}_k$, take the constant sequence $\Omega_0 = \cdots = \Omega_k = T_\ell$. Its determining map is $\mathcal{M}^k$ and its bond set consists of all level-k bonds. Whenever the minimal orbit at V exists, write

$$U_k(V) = U_{\mathbf{B}}(V).$$

This notation specifies the orbit through any of its elements; its invariant quantities, including the action, are functions of V. The unique absolute tower representative is $U^{\mathrm{ax}}_{\mathbf{B}}(V) = U^{\circ}_{\mathbf{B}}(V)$. More generally, a sequence beginning at T_ℓ and block fields $V_j \in \mathcal{V}_j$ supply the datum $V = ((V_j)|_{\mathbf{B}\cap\Gamma_j})_{j=0}^{k}$ on a specified bond set. The same notation $U_k(V)$ is used when that datum has a minimal orbit.

Let Ω be a union of M_1-cubes at level k and let $\mathbf{B}_k(\Omega)$ be the full determining set of its $\mathfrak{r}_2$-tower. The localized background is

$$U_{k,\Omega}(V) = U_{\mathbf{B}_k(\Omega)}\big(M_{\mathbf{B}_k(\Omega)}(Q_k^{s*}V)\big), \tag{77}$$

provided the indicated minimal orbit exists. This is a configuration on the inner bare bonds of Ω, specified up to the residual group. Its unique outside-node tower representative is obtained by replacing $U_{\mathbf{B}_k(\Omega)}$ in (77) by $U^{\circ}_{\mathbf{B}_k(\Omega)}$.

The datum in (77) is defined for every block field V. Indeed the tower gaps, in level-zero units, satisfy

$$\operatorname{dist}_\infty(\Omega_{j-1}^c, \Omega_j) \geq M_1 L^j > L^j,$$

so proposition 7.2 applies. By locality of averaging, the restriction of $Q_k^{s*}V$ to the inner bare bonds gives the datum (60). At a member bond b of level j, its value is either $\mathbf{1}$ or $V(c)$ for the level-k bond above b. In the latter case $b_\pm \in B^{k-j}(c_\pm)$; since $b_\pm \in \Omega$ and Ω is a union of level-k partition cubes, $c_\pm \in \Omega$. Thus the induced datum, and the conditional background orbit and its selected representative, depend only on the inner level-k bonds of V in Ω.

The reduced background $U^{\flat}_{k,\Omega}(V)$ is defined by replacing $\mathbf{B}_k(\Omega)$ with $\mathbf{B}_k(\Omega)^{\flat}$ in (77), whenever the corresponding minimal orbit exists. The domain and locality arguments are unchanged; its absolute and outside-node tower representatives agree when the minimal orbit exists.

10. Gauge averages and the Landau condition

To specify a Landau representative relative to a background, we normalize its gauge transformation by transported block averages. Their derivative gives the admissible infinitesimal gauge transformations and hence the projection in the Landau condition. The averages follow Bałaban [Bał85a, (78)–(81), p. 30].

10.1. **Transported gauge averages.** Let $\mathbf{B}$ be a bond set of $\Omega_0 \supset \cdots \supset \Omega_k$, let ξ be the bare spacing, and let U be a real configuration on the inner bare bonds of Ω_0. The averages $\mathcal{M}^n U$ needed below are defined on the inner bonds of $B^{j-n}(y)$ for $y \in \Lambda_j(\mathbf{B})$ and $n < j$. Indeed, for a bond c with endpoints in $B^{j-n}(y)$, nesting gives $B^n(c_\pm) \subset B^j(y) \subset \Omega_0$ at the level of bare sites. The partial-domain criterion proved with (41) therefore applies.

Definition 10.1. For $0 \leq n < K$ and $z \in \mathbb{T}^{(n+1)}$, let W be a configuration on the bonds of $B(z)$ and let v be a G^c-valued function on its sites. Define

$$(\bar{R}_W v)(z) = v(z) \exp\left\{ L^{-3} \sum_{x \in B(z)} \log\Big[v(z)^{-1} \operatorname{Ad}(W(\Gamma_{z,x}))v(x) \Big] \right\}. \tag{78}$$

Here and below the logarithm is its series at $\mathbf{1}$. This average is defined when every argument X of a logarithm satisfies $|X - \mathbf{1}| < 1$. Its value lies in G^c, and lies in G when W and v are real.

Put $\bar{R}_U^{(0)} v = v$ and

$$\bar{R}_U^{(n+1)} v = \bar{R}_{\mathcal{M}^n U}(\bar{R}_U^{(n)} v).$$

Every stage of an iterated average must be defined. A gauge transformation v satisfies the *averaged normalization relative to* $(\mathbf{B}, U)$ if

$$(\bar{R}_U^{(j)} v)(y) = \mathbf{1}, \qquad y \in \Lambda_j(\mathbf{B}), \quad 0 \le j \le k. \tag{79}$$

In particular, $v = \mathbf{1}$ on $\Lambda_0(\mathbf{B})$.

For a site function λ, set

$$(q_W \lambda)(z) = L^{-3} \sum_{x \in B(z)} \mathrm{Ad}(W(\Gamma_{z,x}))\lambda(x). \tag{80}$$

Put $q_U^{(0)} = \mathrm{id}$ and $q_U^{(n+1)} = q_{\mathcal{M}^n U} q_U^{(n)}$. The admissible infinitesimal gauge transformations form the space

$$\mathcal{N}(\mathbf{B}, U) = \left\{ \lambda : \ (q_U^{(j)}\lambda)(y) = 0 \text{ for } y \in \Lambda_j(\mathbf{B}),\ 0 \le j \le k \right\}, \tag{81}$$

where λ is a site function on the bare sites of Ω_0. ■

Proposition 10.2. With the background averages fixed, $(\bar{R}_U^{(j)} v)(y)$ depends on v only through its values on the pillar $B^j(y)$. Moreover, for every site function λ, the averages of $e^{it\lambda}$ and their logarithms are defined for sufficiently small $|t|$ wherever all required background averages exist, and

$$\frac{d}{dt}\Big|_{t=0} \frac{1}{i} \log \bar{R}_U^{(n)}(e^{it\lambda}) = q_U^{(n)} \lambda.$$

This is the linearization used in the Landau condition [Bał85c, p. 80]. ■

Proof. Each logarithm in (78) is traceless, so its exponential has determinant one. For real W and v, the logarithms lie in $i\mathfrak{g}$; their average exponentiates to an element of G. This proves the group-valued assertions. Locality follows inductively from

$$\bigcup_{x \in B(z)} B^n(x) = B^{n+1}(z).$$

For the derivative, suppose $w_t = \bar{R}_U^{(n)}(e^{it\lambda})$ satisfies $w_0 = \mathbf{1}$ and $\dot{w}_0 = i\mu$, where $\mu = q_U^{(n)}\lambda$; this holds at $n = 0$. Set $W = \mathcal{M}^n U$. Every argument of a logarithm in $\bar{R}_W w_t$ equals $\mathbf{1}$ at $t = 0$ and is therefore in the logarithm domain for small $|t|$. Its derivative is

$$i\left[\mathrm{Ad}(W(\Gamma_{z,x}))\mu(x) - \mu(z)\right].$$

The differential of log at $\mathbf{1}$ is the identity. Since $B(z)$ has L^3 sites, the exponent in (78) has derivative $i[(q_W\mu)(z) - \mu(z)]$. The prefactor $w_t(z)$ contributes $i\mu(z)$, and these base-point terms cancel:

$$\frac{d}{dt}\Big|_{t=0} (\bar{R}_W w_t)(z) = i\mu(z) + i[(q_W\mu)(z) - \mu(z)] = i(q_U^{(n+1)}\lambda)(z).$$

This proves the induction. Taking the logarithm at $\mathbf{1}$ gives the asserted derivative. □

10.2. **Constraint mismatch.** For a one-form A on the inner bare bonds of Ω_0 and $c \in \mathbf{B} \cap \Gamma_j$, define

$$Q_j(U, \xi A)(c) = \frac{1}{i} \log\left[\mathcal{M}^j(e^{i\xi A}U)(c)\mathcal{M}^j(U)(c)^{-1}\right]. \tag{82}$$

Both averages must be defined, and the logarithm argument must lie within distance one of $\mathbf{1}$. For real A use the global real block average; for nonreal A use the complex branch for the first average. When the directional derivative exists, put

$$Q_j'(U)A = \frac{d}{dt}\Big|_{t=0} Q_j(U, t\xi A).$$

It follows that

$$\mathcal{M}^j(e^{it\xi A}U)(c) = \exp\{it(Q_j'(U)A)(c) + o(t)\}\mathcal{M}^j(U)(c).$$

Indeed, $Q_j(U,0)=0$, and exponentiating the defining logarithm gives this equality from $Q_j(U,t\xi A)=tQ_j'(U)A+o(t)$.

The derivative includes the coarse bond length [Bał85c, (1.37), p. 82], [Bał85b, (3.110), p. 417]. For $\Omega_0=T_\ell$, $U=\mathbf{1}$, and a constant one-form $A_\mu(x)=a_\mu$, it is

$$(Q_j'(\mathbf{1})A)(c)=L^j\xi a_\mu \quad \text{if } c \text{ has direction } \mu.$$

To see this, apply lemma 4.6 to $e^{it\xi A}$. One application of $\mathbf{Q}^{\mathrm{lin}}$ multiplies a constant component by L, so j iterations give $L^j\xi A$. The real and complex block averages agree near the identity for real A by the same lemma.

10.3. **The Landau condition.** The Landau condition requires the divergence to be orthogonal to the image of the admissible infinitesimal gauge transformations under the Dirichlet Laplacian. The site operators follow Bałaban [Bał85b, pp. 390–393, (3.3), (3.8)]. Let Ω_0 be a region whose cubes have side at least the bare spacing ξ. A bond of Ω_0 has at least one endpoint there; an inner bond has both endpoints there. Initially let U be a real or complex configuration on all bonds of Ω_0. Site functions on Ω_0 and one-forms on its bonds are extended by zero outside their respective domains.

Definition 10.3. The covariant gradient sends site functions to one-forms on the bonds of Ω_0 and is given on positive bonds by

$$(D_U\lambda)(b)=\xi^{-1}\big[\mathrm{Ad}(U(b))\lambda(b_+)-\lambda(b_-)\big]. \tag{83}$$

Its transpose for the bilinear pairings is denoted D_U^*: $(D_U^*A,\lambda)_\xi=(A,D_U\lambda)_\xi$. At a site $x\in\Omega_0$ it is

$$(D_U^*A)(x)=\xi^{-1}\sum_{\mu=1}^{3}\Big[\mathrm{Ad}(U(\langle x-\xi e_\mu,x\rangle)^{-1})A_\mu(x-\xi e_\mu)-A_\mu(x)\Big]. \tag{84}$$

The covariant Laplacian on site functions is $\Delta_U=D_U^*D_U$. Here D_U acts on site functions, whereas the exterior derivative in (67) acts on one-forms; the domains also distinguish their transposes. For real U, D_U^* is the adjoint on the real spaces. ■

Every bond entering (84) has an endpoint in Ω_0. The formula follows from (83), trace invariance and a shift of the summation variable in the translated term. The zero extension of λ leaves only sites of Ω_0 in the resulting sum; nondegeneracy of the pairings gives uniqueness of the transpose. Extend a gauge u on Ω_0 by the identity outside, and let $\mathrm{Ad}(u)$ act at a site on site functions and at the initial point of a positive bond on one-forms. Then

$$D_{U^u}\,\mathrm{Ad}(u)=\mathrm{Ad}(u)D_U,\qquad D_{U^u}^*\,\mathrm{Ad}(u)=\mathrm{Ad}(u)D_U^*,\qquad \Delta_{U^u}\,\mathrm{Ad}(u)=\mathrm{Ad}(u)\Delta_U.$$

Cancellation of endpoint factors proves the first identity; trace invariance gives the second, and composition the third. The same algebra applies to complex gauges.

Let $\mathbf{B}$ be a bond set of a decreasing sequence $\Omega_0\supset\cdots\supset\Omega_k$ satisfying the separation condition at scale $\mathfrak{r}_1$, or the determining set of the constant sequence $\Omega_0=\cdots=\Omega_k=T_\ell$. Let U_0 be a real configuration on the inner bare bonds of Ω_0. No regularity assumption on U_0 is needed for the following definitions.

Extend U_0 by the identity to bonds with exactly one endpoint in Ω_0, site functions by zero outside Ω_0, and inner one-forms by zero to the other bonds. The intermediate gradient in $\Delta_{U_0}=D_{U_0}^*D_{U_0}$ is still computed on all incident bonds. At a site of Ω_0,

$$\begin{aligned}(\Delta_{U_0}\lambda)(x)=\xi^{-2}\sum_{\mu=1}^{3}\big[&2\lambda(x)-\mathrm{Ad}(U_0(\langle x-\xi e_\mu,x\rangle)^{-1})\lambda(x-\xi e_\mu)\\&-\mathrm{Ad}(U_0(\langle x,x+\xi e_\mu\rangle))\lambda(x+\xi e_\mu)\big].\end{aligned}$$

Each background value on a non-inner bond multiplies a vanishing exterior value of λ, so this expression does not depend on the chosen background extension. The same holds for $D^*_{U_0}A$, since A vanishes on non-inner bonds, and, at $x \in \Omega_0$, for

$$(\nabla^{U_0}_\mu A_\nu)(x) = \xi^{-1}\big[\mathrm{Ad}(U_0(\langle x, x+\xi e_\mu\rangle))A_\nu(x+\xi e_\mu) - A_\nu(x)\big] :$$

if $x+\xi e_\mu$ is outside Ω_0, the one-form component there vanishes. We use these Dirichlet conventions throughout the Landau construction.

Definition 10.4. Following Bałaban [Bał85c, p. 80, (1.27), p. 82, (1.38)], let $R(U_0) = R(\mathbf{B}, U_0)$ be the orthogonal projection in the real space of site functions, with inner product $(\cdot,\cdot)_\xi$, onto

$$\Delta_{U_0}(\mathcal{N}(\mathbf{B}, U_0) \cap \{\text{real site functions}\}).$$

Extend $R(U_0)$ complex-linearly to complex site functions. A one-form A satisfies the Landau condition relative to $(\mathbf{B}, U_0)$ if

$$R(U_0)D^*_{U_0}A = 0. \tag{85}$$

For real A, this is equivalent to

$$(D^*_{U_0}A, \Delta_{U_0}\lambda)_\xi = 0 \quad \text{for every real } \lambda \in \mathcal{N}(\mathbf{B}, U_0).$$

■

For real U_0, the averages and site operators preserve real functions. Their displayed image is a subspace of a finite-dimensional real inner-product space, so its orthogonal projection exists and has the orthogonal complement as kernel, giving the equivalence.

Let U be a real configuration on the inner bare bonds of Ω_0. A *Landau pair of U relative to* $(\mathbf{B}, U_0)$ is a real one-form A on these bonds and a G-valued gauge transformation u on the bare sites of Ω_0 such that

$$\begin{aligned} U = (e^{i\xi A}U_0)^u, \qquad & R(U_0)D^*_{U_0}A = 0, \\ (\bar{R}^{(j)}_{U_0}u)(y) = \mathbf{1}, \qquad & y \in \Lambda_j(\mathbf{B}), \quad 0 \le j \le k. \end{aligned} \tag{86}$$

The last condition includes the definedness of every averaged normalization. The first two relations are equivalent to

$$e^{i\xi A}U_0 = U^{u^{-1}}, \qquad R(U_0)D^*_{U_0}A = 0. \tag{87}$$

We call $e^{i\xi A}$ a *Landau representative of U relative to* $(\mathbf{B}, U_0)$. As in section 9, the product $e^{i\xi A}U_0$ is defined on positive bonds and extended to reversed bonds by inversion of the whole product.

Set $\Omega_{k+1} = \emptyset$. The small-field conditions at radius $\alpha > 0$ are the following inequalities: for every $0 \le j \le k$ and every positive inner bare bond $b = \langle x, x+\xi e_\nu\rangle$ with $\{b_-, b_+\} \cap (\Omega_j \setminus \Omega_{j+1}) \neq \emptyset$,

$$L^j\xi\,|A(b)| < \alpha, \qquad (L^j\xi)^2 \max_{1\le\mu\le 3} |(\nabla^{U_0}_\mu A_\nu)(x)| < \alpha. \tag{88}$$

The direction ν is fixed by b; the maximum is only over μ. Both inequalities are imposed for every layer containing an endpoint of b.

The constraint mismatch of the Landau representative is the family $Q_j(U_0, \xi A)$ on $\mathbf{B} \cap \Gamma_j$, $0 \le j \le k$, whenever these maps are defined. Fix a radius $\alpha > 0$. If there is exactly one Landau pair whose first member satisfies (88) at that radius, denote this member by $A^{\mathrm{La}}(U; \mathbf{B}, U_0)$.

For a real one-form A on the inner bare bonds of Ω_0, the Wilson Hessian at U_0 is the quadratic form

$$\begin{aligned} (A, \Delta^{\mathrm{W}}_{U_0}A)_\xi &= \frac{d^2}{dt^2}\bigg|_{t=0} A^\xi_{\Omega_0}(e^{it\xi A}U_0) \\ &= \xi^{-1}\,\frac{d^2}{dt^2}\bigg|_{t=0} A_{\Omega_0}(e^{it\xi A}U_0). \end{aligned} \tag{89}$$

At the identity background, the weighted second-variation formula of proposition 9.2, with the indicator of the inner plaquettes as weight, gives

$$(A, \Delta_{\mathbf{1}}^{\mathrm{W}} A)_\xi = \xi^3 \sum_{p \text{ inner to } \Omega_0} \langle (D_{\mathbf{1}} A)(p), (D_{\mathbf{1}} A)(p) \rangle .$$

Here $D_{\mathbf{1}}$ is the exterior derivative on one-forms.

11. Continuation from real background data

Complex tower configurations enter the domains for analytic fluctuations, following Bałaban [Bał87, pp. 261–263, (1.10)–(1.16)]. They are obtained by holomorphic continuation of the selected representatives of real regular critical orbits. The two existence hypotheses below define this continuation; the identity principle then proves its uniqueness and continues the block constraint on the specified domains. For the reduced determining sets and radii of theorem E, these existence hypotheses are established in section 38.

11.1. **Perturbations of a real datum.** Let $\mathbf{B}$ be a bond set of a decreasing sequence $\Omega_0 \supset \cdots \supset \Omega_k$ satisfying the separation condition of the rule $\mathfrak{r}_1$, or the determining set of the constant sequence $\Omega_0 = \cdots = \Omega_k = T_\ell$. A complex datum on $\mathbf{B}$ is a family $\mathbf{V} = (\mathbf{V}_j)_{j \le k}$ of G^c-valued functions on its members, satisfying $\mathbf{V}_j(\bar{b}) = \mathbf{V}_j(b)^{-1}$. It is real when its values lie in G.

Fix a real datum V_0. A perturbation $B = (B_j)_{j \le k}$ assigns an element of $\mathfrak{g}^c$ to each positive bond of each member. Define $e^{iB} V_0$ on positive bonds by

$$(e^{iB} V_0)_j(b) = e^{iB_j(b)} V_{0,j}(b), \qquad b \in \mathbf{B} \cap \Gamma_j,$$

and on reversed bonds by inversion of the whole product. Real perturbations have values in $\mathfrak{g}$ and give real data.

For $r > 0$, put

$$\mathcal{B}_r = \left\{ B : \sup_{j,b} |B_j(b)| < r \right\}.$$

Since $\mathfrak{g}^c = \mathfrak{g} \oplus i\mathfrak{g}$ over $\mathbb{R}$, a real basis identifies the finite-dimensional perturbation space with $\mathbb{C}^N$ and its real slice with $\mathbb{R}^N$. The set $\mathcal{B}_r$ is open and convex, hence connected. Holomorphy means holomorphy of the matrix entries on positive inner bonds; inversion gives the values on their reverses.

Definition 11.1. The continuation hypotheses at (V_0, r_{B}), where $r_{\mathrm{B}} > 0$, are the following:

(i) For every real $B \in \mathcal{B}_{r_{\mathrm{B}}}$, the set $\mathcal{C}(\mathbf{B}, e^{iB} V_0)$ is nonempty and consists of one orbit of $\mathcal{G}^{\mathrm{res}}(\mathbf{B})$.

(ii) The map $B \mapsto U^\circ_{\mathbf{B}}(e^{iB} V_0)$ on those real perturbations is the restriction of a holomorphic map from $\mathcal{B}_{r_{\mathrm{B}}}$ to the G^c-valued configurations on the inner bare bonds of Ω_0.

Under these hypotheses, write $\mathcal{B}(V_0) = \mathcal{B}_{r_{\mathrm{B}}}$ and denote the holomorphic extension by Φ_{V_0}. Its value is the *representative continued from* V_0:

$$U_{\mathbf{B}}(V_0; B) = \Phi_{V_0}(B) = U^\circ_{\mathbf{B}}(V_0; B), \qquad B \in \mathcal{B}(V_0). \tag{90}$$

■

The real representatives used in (ii) are defined by proposition 9.6: the stated geometry implies (T). They are the outside-node representatives. At real data where the absolute tower representative exists, it agrees with this choice; this holds in particular for a reduced bond set and for the constant whole-torus sequence.

Proposition 11.2. Under the continuation hypotheses, Φ_{V_0} is unique. If those hypotheses hold at two radii, the two extensions agree on the smaller parameter neighborhood.

Let $\mathcal{D} \subset \mathcal{B}(V_0)$ be the set on which all the complex block averages $M_{\mathbf{B}}(\Phi_{V_0}(B))$ are defined. Then $\mathcal{D}$ is open and this map is holomorphic there. Let $\mathcal{D}' \subset \mathcal{D}$ be connected and open. Suppose there is

a nonempty relatively open set $E \subset \mathcal{D}' \cap \mathbb{R}^N$ such that the real and complex averages of $U^\circ_{\mathbf{B}}(e^{iB}V_0)$ agree for every $B \in E$. Then, with the averages on the left taken in the complex branch,

$$M_{\mathbf{B}}(U_{\mathbf{B}}(V_0; B)) = e^{iB}V_0, \qquad B \in \mathcal{D}'.$$

■

Proof. A holomorphic map on a connected open subset of $\mathbb{C}^N$ that vanishes on a nonempty relatively open real slice vanishes identically. At a point of that slice its Taylor coefficients are the real derivatives divided by their factorials, hence vanish. The set where the map vanishes locally is open and closed: at a limit point all derivatives vanish by continuity, so the local power series is again zero. Connectedness proves the claim, entrywise for a matrix-valued map.

Apply this principle to the difference of two proposed extensions on $\mathcal{B}(V_0)$, or on the smaller ball when two radii are given. This proves uniqueness and radius compatibility.

By the finite-iteration construction in section 4, the domain of $M_{\mathbf{B}}$ is $(G^c)^{\mathsf{P}} \cap \mathcal{W}$, where P is the set of positive inner bare bonds of Ω_0 and $\mathcal{W}$ is open in the ambient matrix coordinates; the averages extend holomorphically to $\mathcal{W}$. Thus $\mathcal{D} = \Phi_{V_0}^{-1}(\mathcal{W})$ is open and $M_{\mathbf{B}} \circ \Phi_{V_0}$ is holomorphic there.

For real $B \in E$, the continued representative belongs to $C(\mathbf{B}, e^{iB}V_0)$, so its real block averages equal $e^{iB}V_0$. By the assumed agreement of branches, its complex averages have the same value. The two holomorphic maps therefore agree on the nonempty relatively open real slice E of the connected domain $\mathcal{D}'$. The preceding principle proves their equality on $\mathcal{D}'$; inversion gives the values on reversed bonds. □

11.2. **Tower configurations.** Let $\mathbf{B}'$ be a bond set of a decreasing sequence with first region X, depth n, and the separation condition at scale $\mathfrak{r}_1$. Let U be a real configuration and A' a complex one-form on the inner bare bonds of X, and put $\mathbf{U} = e^{i\xi A'}U$. Thus $\mathbf{U}(b) = e^{i\xi A'(b)}U(b)$ on positive bonds, with the inverse value on reversed bonds.

We distinguish two averaged data. The datum $M'(\mathbf{U}) = M^{\mathrm{C}}_{\mathbf{B}'}(\mathbf{U})$ uses the complex branch at every step, including when $\mathbf{U}$ is real. The real datum $V_0 = M'(U) = M^{\mathrm{R}}_{\mathbf{B}'}(U)$ uses the global real averages. Assume that the following conditions hold:

(i) The continuation of definition 11.1 exists at the real datum V_0 with radius $r_{\mathrm{B}} > 0$.

(ii) Every average defining $M'(\mathbf{U})$ exists, and

$$|M'(\mathbf{U})_j(b)V_{0,j}(b)^{-1} - \mathbf{1}| < 1$$

for every bond b of every member of $\mathbf{B}'$, $0 \le j \le n$. At positive bonds define

$$B_j(b) = \frac{1}{i}\log[M'(\mathbf{U})_j(b)V_{0,j}(b)^{-1}].$$

(iii) The resulting family satisfies $B \in \mathcal{B}_{r_{\mathrm{B}}}$.

The tower configuration and its current are

$$U_n(M'(\mathbf{U})) := U_{\mathbf{B}'}(V_0; B), \qquad J_n(M'(\mathbf{U})) := J(U_n(M'(\mathbf{U}))). \tag{91}$$

The current is taken on its domain from (69), using the bare spacing $\eta_n = L^{-n}$ in level-n units. By proposition 11.2, uniqueness on the smaller of two admissible balls makes (91) independent of the radius used in these conditions.

These quantities are functions of the pair (A', U), through $M'(\mathbf{U})$ and the real base datum $M'(U)$. The complex configuration alone does not determine that pair: for a real one-form A'' and a real configuration W,

$$e^{i\xi A''}(e^{-i\xi A''}W) = W = e^{i\xi 0}W.$$

Proposition 11.3. Suppose that the conditions defining (91) hold with $A' = 0$, and write $V_{\mathrm{C}} = M^{\mathrm{C}}_{\mathbf{B}'}(U)$ and $V_0 = M^{\mathrm{R}}_{\mathbf{B}'}(U)$. Then V_{C} is real, the family B is real, and $e^{iB}V_0 = V_{\mathrm{C}}$. The regular critical set $\mathcal{C}(\mathbf{B}', V_{\mathrm{C}})$ is nonempty and consists of one residual orbit. Moreover,

$$U_n(M'(\mathbf{U})) = U^{\circ}_{\mathbf{B}'}(V_{\mathrm{C}}),$$

so the tower configuration is real and its current is the current of this representative. Finally,

$$U_n(M'(\mathbf{U})) = U^{\circ}_{\mathbf{B}'}(V_0) \quad \Longleftrightarrow \quad V_{\mathrm{C}} = V_0 \quad \Longleftrightarrow \quad B = 0.$$

These equalities hold if every configuration averaged in computing $M^{\mathrm{R}}_{\mathbf{B}'}(U)$ belongs to the set $\mathcal{S}_c$ for the bond c at which it is averaged. ■

Proof. For a real input W in the domain $\mathcal{O}'_c$, the contour means lie in G by proposition 4.3. The relative matrices $W_x(W)$ therefore lie in G and satisfy $|W_x(W) - \mathbf{1}| < 1$; their logarithms lie in $i\mathfrak{g}$ by lemma 5.1. Formula (39), and inversion at reversed bonds, preserve G. Iteration proves that V_{C} is real.

At each positive member bond, $g = V_{\mathrm{C},j}(b)V_{0,j}(b)^{-1}$ belongs to G and satisfies $|g - \mathbf{1}| < 1$. The logarithm therefore gives $B_j(b) \in \mathfrak{g}$ and $e^{iB_j(b)} = g$. The inverse convention at reversed bonds yields $e^{iB}V_0 = V_{\mathrm{C}}$. Since B is a real point of the continuation ball, its defining hypotheses give the asserted critical orbit and

$$U_{\mathbf{B}'}(V_0; B) = U^{\circ}_{\mathbf{B}'}(e^{iB}V_0) = U^{\circ}_{\mathbf{B}'}(V_{\mathrm{C}}).$$

The logarithmic definition gives $B = 0$ exactly when $V_{\mathrm{C}} = V_0$, in which case the representatives coincide. Conversely, the real/global averages of $U^{\circ}_{\mathbf{B}'}(V)$ equal V for every real datum for which this representative is defined. Equal representatives for V_{C} and V_0 therefore imply equal data. If every real averaging step lies in its set $\mathcal{S}_c$, (37) and (39) agree at that step. Induction through the levels gives $V_{\mathrm{C}} = V_0$. □

12. Summability in the multiscale distance

We next develop quantitative bounds for the action and averaging map, followed by real gauge-fixed inverse estimates. We first show that exponential decay in the multiscale distance can be summed over all determining bonds with a bound independent of the volume and the number of layers. Separation between successive regions compensates for the increasing density of sites at finer scales.

Assume $L \geq 13$ throughout this section. Fix $1 \leq n \leq K$ and a decreasing sequence $\Omega_0 \supset \cdots \supset \Omega_n$ satisfying the separation condition of (61) with initial level zero. We measure lengths in level-n units and set

$$\xi_q = L^{q-n} \quad (0 \leq q \leq n), \qquad \Omega_{n+1} = \emptyset.$$

Thus Ω_q is a union of $\mathfrak{M}$-cubes of side $\mathfrak{M}\xi_q$, and

$$\operatorname{dist}_\infty(\Omega_q^c, \Omega_{q+1}) > R_1\mathfrak{M}\xi_q \quad (0 \leq q < n).$$

Here $\mathfrak{M}, R_1$ are powers of L, with $\mathfrak{M} \geq 1$ and $R_1 \geq L$, as in the separation condition. Empty members are allowed. Denote the full determining set by $\mathbf{B}$ and its reduced set by $\mathbf{B}^\flat$, with members $\Gamma^\flat_q$.

Let

$$\mathcal{L}_q = \Omega_q \setminus \Omega_{q+1}, \qquad \Lambda_q = \Lambda_q(\mathbf{B}) = \mathbb{T}^{(q)} \cap \mathcal{L}_q, \qquad \Lambda_* = \bigcup_{q=0}^{n} \Lambda_q.$$

The region difference is cube-wise. Its point set is the closure of the point-set difference, and the sets Λ_q are disjoint; these facts are proved in lemma 12.2 below. For each positively oriented $c \in \Gamma^\flat_q$, choose

$$y_c = \begin{cases} c_-, & c_- \in \Lambda_q, \\ c_+, & c_- \notin \Lambda_q, \end{cases} \qquad y_{\bar{c}} = y_c.$$

By lemma 7.5, one endpoint of c belongs to Λ_q; hence $y_c \in \Lambda_q$.

For torus points, $|y-y'|_1$ and $|y-y'|_\infty$ denote the minima of the corresponding vector norms over lifts of $y-y'$, measured in these units. Both minima are attained by choosing each coordinate of the lift with least absolute value. The sum of least lifts is a lift of the sum, so the vector triangle inequalities give the two metric triangle inequalities. The same least lift also gives $|y-y'|_\infty \le |y-y'|_1$.

For $x \in T_\ell$ and $v \in \mathbb{R}^3$, write $\sigma(x,v) = \{x+sv : 0 \le s \le 1\}$. The length of a set $S \subset T_\ell$ is

$$\operatorname{length}(S) = \inf\left\{ \sum_{i=1}^{r} |v_i|_1 : r \ge 0,\ S \subset \bigcup_{i=1}^{r} \sigma(x_i, v_i) \right\}, \qquad \inf \emptyset = +\infty, \tag{92}$$

where $x_i \in T_\ell$, $v_i \in \mathbb{R}^3$, and the vectors are expressed in level-n units. In particular, the covering is finite and $|v_i|_1 = \sum_\mu |v_{i,\mu}|$. For a finite union Γ of closed segments, set

$$|\Gamma|_{\mathbf{B}^\flat} = \sum_{q=1}^{n} \xi_q^{-1} \operatorname{length}(\Gamma \cap \mathcal{L}_q) + \xi_0^{-1} \operatorname{length}(\Gamma \setminus \Omega_1), \tag{93}$$

with point-set intersections and difference. Define

$$d_{\mathbf{B}^\flat}(y,y') = \inf\{ |\Gamma|_{\mathbf{B}^\flat} : \Gamma \text{ is a connected finite union of closed segments containing } y, y' \}.$$

The competitors may lie anywhere in T_ℓ. The closed layers may meet along faces, and (93) retains each of their contributions there.

Proposition 12.1 (Sums over determining bonds)**.** For $a > 0$, the series

$$\mathsf{S}(a) = 54L^3 \sum_{r \ge 1} (2r+1)^3 e^{-a(r-1)/2}$$

converges. For the separated sequence above and $y \in \Lambda_*$, the following estimates hold.

(a) For every $y' \in \Lambda_*$,

$$d_{\mathbf{B}^\flat}(y,y') \ge |y-y'|_\infty.$$

(b) If $aR_1\mathfrak{M} \ge 4(3\log L + \log 2)$, then

$$\sum_{c \in (\mathbf{B}^\flat)^+} e^{-a d_{\mathbf{B}^\flat}(y,y_c)} \le \mathsf{S}(a).$$

(c) If $aR_1\mathfrak{M} \ge 8(3\log L + \log 2)$ and $D \ge 0$, then

$$\sum_{\substack{c \in (\mathbf{B}^\flat)^+ \\ d_{\mathbf{B}^\flat}(y,y_c) \ge D}} e^{-a d_{\mathbf{B}^\flat}(y,y_c)} \le e^{-aD/2} \mathsf{S}(a/2).$$

The sums count positively oriented bonds, with no scale weight. The bound $\mathsf{S}(a)$ depends only on L and a; in particular, it is independent of the torus size, the number of layers, and the choice of separated sequence. ∎

The proof in section 12.2 combines two estimates for a connecting set: its displacement costs length at the coarsest scale it reaches, and each intermediate layer costs at least $R_1\mathfrak{M}$ in weighted length. The resulting exponential factors control both the spatial shell count and the sum over scales.

12.1. **Layers and weighted length.**

Lemma 12.2. For the sequence above, the point set of $\mathcal{L}_q$ is

$$\overline{\{x \in T_\ell : x \in \Omega_q,\ x \notin \Omega_{q+1}\}}.$$

It is independent of the standard partition used to take the cube-wise difference. Moreover,

$$\Omega_0 = \bigcup_{q=0}^{n} \mathcal{L}_q, \qquad \mathcal{L}_q \subset \Omega_q, \qquad \mathcal{L}_q \subset \Omega_{q+1}^c.$$

Every bare site of Ω_0 belongs to exactly one layer, and the sets Λ_q are pairwise disjoint. Every bond of $\Gamma_q^\flat$ has an endpoint in Λ_q. ■

Proof. Let X_1, X_2 be regions for one standard partition, and let D be their cube-wise difference. If $x \in X_1$ and $x \notin X_2$, a cube of X_1 containing x cannot be a cube of X_2, so $x \in D$. Since D is a finite union of closed cubes, it contains the closure of this point-set difference. Conversely, let Δ be a cube of D. Its interior points belong to no other cube of the partition and therefore do not belong to X_2. Every point of Δ is a limit of these interior points: writing $\Delta = \{y + w : |w|_\infty \le s/2\}$, use $y + (1 - j^{-1})w$ as $j \to \infty$. Thus D is contained in the same closure. This description does not refer to the partition. Applying it to $X_1 = \Omega_q$, $X_2 = \Omega_{q+1}$ proves the first assertion.

The cubes of $\mathcal{L}_q$ are cubes of Ω_q and of Ω_{q+1}^c, giving the two inclusions. If $x \in \Omega_0$, choose the largest q such that $x \in \Omega_q$. Then $x \notin \Omega_{q+1}$, so $x \in \mathcal{L}_q$. This proves the covering assertion.

For $0 \le i \le q$, a site of $\mathbb{T}^{(i)}$ belongs to $\mathcal{L}_q$ exactly when it belongs to Ω_q and not to Ω_{q+1}, by the site convention in definition 7.1. Taking $i = 0$ gives a unique layer for each bare site. If $q < q'$ and $y \in \Lambda_q \cap \Lambda_{q'}$, then $y \notin \Omega_{q+1}$ but $y \in \Omega_{q'} \subset \Omega_{q+1}$, a contradiction. Finally, lemma 7.5 characterizes a reduced member bond as having an endpoint in Ω_q and neither endpoint in Ω_{q+1}. That endpoint is a site of $\mathbb{T}^{(q)}$ in $\mathcal{L}_q$, hence in Λ_q. □

Lemma 12.3. For finite unions Γ, Γ' of closed segments,

$$|\Gamma \cup \Gamma'|_{\mathbf{B}^\flat} \le |\Gamma|_{\mathbf{B}^\flat} + |\Gamma'|_{\mathbf{B}^\flat}.$$

The distance is symmetric and satisfies

$$d_{\mathbf{B}^\flat}(y, y'') \le d_{\mathbf{B}^\flat}(y, y') + d_{\mathbf{B}^\flat}(y', y'') \qquad (y, y', y'' \in T_\ell).$$

■

Proof. The length in (92) is monotone under inclusion, since a cover of a larger set covers every subset. Combining covers of S and S' and taking infima gives

$$\operatorname{length}(S \cup S') \le \operatorname{length}(S) + \operatorname{length}(S').$$

Apply this inequality to the intersections with each layer and to the point-set differences from Ω_1, multiply by the positive weights in (93), and add. This proves weighted subadditivity.

The competitors defining $d_{\mathbf{B}^\flat}(y, y')$ are unchanged when y, y' are interchanged, proving symmetry. If Γ connects y to y' and Γ' connects y' to y'', their union is connected and contains y, y''. Indeed, two connected sets with a common point cannot be separated into two nonempty relatively open subsets of their union. Weighted subadditivity therefore gives

$$d_{\mathbf{B}^\flat}(y, y'') \le |\Gamma \cup \Gamma'|_{\mathbf{B}^\flat} \le |\Gamma|_{\mathbf{B}^\flat} + |\Gamma'|_{\mathbf{B}^\flat}.$$

Taking the two infima separately proves the triangle inequality. □

12.2. **Proof of the summation estimate.**

Proof of proposition 12.1. For $a > 0$, the ratio of consecutive terms of the series defining $\mathsf{S}(a)$ tends to $e^{-a/2} < 1$, so the series converges. Set $w = R_1\mathfrak{M}$.

Lower bounds on connecting sets. We first record a consequence of the definition of length. Suppose $g : T_\ell \to \mathbb{R}$ satisfies $|g(z) - g(z')| \le |z - z'|_1$, and a bounded interval I is contained in $g(S)$ for some $S \subset T_\ell$. Then

$$\operatorname{length}(S) \ge |I|.$$

Indeed, let $S \subset \bigcup_{j=1}^{r} \sigma(x_j, v_j)$ be a finite segment cover. The image of each segment under g is an interval I_j of length at most $|v_j|_1$, since

$$|g(x_j + sv_j) - g(x_j + s'v_j)| \le |s - s'|\,|v_j|_1.$$

The intervals I_j cover I. Integrating $\mathbf{1}_I \le \sum_j \mathbf{1}_{I_j}$ gives $|I| \le \sum_j |I_j| \le \sum_j |v_j|_1$. Taking the infimum over covers proves the assertion; if there is no finite cover, the asserted lower bound holds because the length is infinite.

Let $y \in \Lambda_q$ and $y' \in \Lambda_{q'}$, and let Γ be a connected finite union of segments containing y and y'. Let t be the largest index for which $\Gamma \cap \Omega_t \neq \emptyset$. This index exists since $y \in \Omega_q$, and $t \ge \max\{q, q'\}$. Every point of Γ either lies outside Ω_1 or belongs to one of the layers $\mathcal{L}_i$, $1 \le i \le t$: for a point in Ω_1, take the largest index of a region containing it. Thus

$$\Gamma = \bigcup_{i=1}^{t} (\Gamma \cap \mathcal{L}_i) \cup (\Gamma \setminus \Omega_1).$$

The sets on the right can share layer faces. Since $\xi_i^{-1} \ge \xi_t^{-1}$ for $i \le t$, finite subadditivity of length gives

$$|\Gamma|_{\mathbf{B}^\flat} \ge \xi_t^{-1} \operatorname{length}(\Gamma).$$

Apply the preceding observation to $g(z) = |z - y|_1$. Its image on the connected set Γ contains the interval from 0 to $|y - y'|_1$, and hence

$$|\Gamma|_{\mathbf{B}^\flat} \ge \xi_t^{-1} |y - y'|_\infty. \tag{94}$$

In particular, $\xi_t \le 1$ implies the first assertion of the proposition after taking the infimum over Γ.

For an index $q < i < t$, consider $f(z) = \operatorname{dist}_\infty(z, \Omega_{i+1})$. The region Ω_{i+1} is nonempty because Γ meets $\Omega_t \subset \Omega_{i+1}$, and f is 1-Lipschitz for the sup metric, hence also for the ℓ^1 metric. The site y lies outside $\Omega_{q+1} \supset \Omega_i$, so $y \in \Omega_i^c$. Separation gives

$$f(y) \ge \operatorname{dist}_\infty(\Omega_i^c, \Omega_{i+1}) > w\xi_i.$$

The function f vanishes at a point of $\Gamma \cap \Omega_t$. By connectedness, it takes every value in $(0, w\xi_i)$ on Γ. Consequently the set

$$S_i = \{z \in \Gamma : 0 < f(z) < w\xi_i\}$$

has length at least $w\xi_i$. A point of S_i lies outside Ω_{i+1} and outside Ω_i^c: the latter follows from the strict separation inequality. It therefore lies in $\Omega_i \setminus \Omega_{i+1}$ as a point-set difference, and in the layer $\mathcal{L}_i$. Monotonicity of length gives

$$\xi_i^{-1} \operatorname{length}(\Gamma \cap \mathcal{L}_i) \ge w.$$

There are $(t - q - 1)_+$ indices strictly between q and t, where $s_+ = \max\{s, 0\}$. Applying the same argument from y' gives the two bounds

$$|\Gamma|_{\mathbf{B}^\flat} \ge w(t - q - 1)_+, \qquad |\Gamma|_{\mathbf{B}^\flat} \ge w(t - q' - 1)_+. \tag{95}$$

Take one half of the lower bound (94) and one quarter of each lower bound in (95). We obtain

$$|\Gamma|_{\mathbf{B}^\flat} \ge \frac{1}{2}\xi_t^{-1}|y - y'|_\infty + \frac{w}{4}\big((t - q - 1)_+ + (t - q' - 1)_+\big).$$

Exponentiating, and bounding the term for this t by the sum over all possible t, gives a bound independent of Γ. Taking the supremum of $e^{-a|\Gamma|_{\mathbf{B}^\flat}}$ over the connecting sets yields

$$e^{-ad_{\mathbf{B}^\flat}(y,y')} \le \sum_{t=\max\{q,q'\}}^{n} e^{-\frac{a}{2}\xi_t^{-1}|y-y'|_\infty} e^{-\frac{aw}{4}((t-q-1)_+ + (t-q'-1)_+)}. \tag{96}$$

Summing sites and levels. Fix $t \ge q'$ and a lift $\widetilde{y}$ of y. Every site $y' \in \mathbb{T}^{(q')}$ satisfying $|y-y'|_\infty < r\xi_t$, $r \ge 1$, has a lift in the open cube of radius $r\xi_t$ centered at $\widetilde{y}$. The lifted lattice has spacing $\xi_{q'} = L^{q'-t}\xi_t$. In each coordinate the cube contains at most $2rL^{t-q'}+1$ lattice values. Therefore

$$\#\{y' \in \mathbb{T}^{(q')} : |y-y'|_\infty < r\xi_t\} \le (2rL^{t-q'}+1)^3 \le (2r+1)^3 L^{3(t-q')}.$$

Partitioning $\Lambda_{q'}$ into the shells $r-1 \le \xi_t^{-1}|y-y'|_\infty < r$ gives

$$\sum_{y' \in \Lambda_{q'}} e^{-\frac{a}{2}\xi_t^{-1}|y-y'|_\infty} \le L^{3(t-q')} \sum_{r \ge 1} (2r+1)^3 e^{-a(r-1)/2}. \tag{97}$$

Under the hypothesis of the second assertion, $L^3 e^{-aw/4} \le \frac{1}{2}$. Hence

$$L^{3(t-q')} e^{-\frac{aw}{4}(t-q'-1)_+} \le L^3\, 2^{-(t-q'-1)_+},$$
$$e^{-\frac{aw}{4}(t-q-1)_+} \le 2^{-(t-q-1)_+}.$$

The layer site sets are disjoint. Sum (96) over $y' \in \Lambda_*$, use (97), and interchange the finite sums over q' and t. The result is

$$\begin{aligned}\sum_{y' \in \Lambda_*} e^{-ad_{\mathbf{B}^\flat}(y,y')} &\le L^3 \sum_{t=q}^{n} 2^{-(t-q-1)_+} \sum_{q'=0}^{t} 2^{-(t-q'-1)_+} \sum_{r \ge 1} (2r+1)^3 e^{-a(r-1)/2} \\ &\le 9L^3 \sum_{r \ge 1} (2r+1)^3 e^{-a(r-1)/2},\end{aligned}$$

since each of the two sums of powers of 2 is at most $\sum_{j \ge 0} 2^{-(j-1)_+} = 3$.

If $y' = y_c$ for a positive bond $c \in \Gamma^\flat_{q'}$, then $y' \in \Lambda_{q'}$. Disjointness of the layer site sets implies that all bonds anchored at a given site belong to this one level. A site of its lattice is incident to at most six positive bonds. It follows that

$$\sum_{c \in (\mathbf{B}^\flat)^+} e^{-ad_{\mathbf{B}^\flat}(y,y_c)} \le 6 \sum_{y' \in \Lambda_*} e^{-ad_{\mathbf{B}^\flat}(y,y')} \le \mathsf{S}(a).$$

The tail estimate. For $d \ge D$, one has $e^{-ad} \le e^{-aD/2} e^{-ad/2}$. The stronger lower bound on aw is precisely the hypothesis of the sum estimate with $a/2$ in place of a. Therefore

$$\begin{aligned}\sum_{\substack{c \in (\mathbf{B}^\flat)^+ \\ d_{\mathbf{B}^\flat}(y,y_c) \ge D}} e^{-ad_{\mathbf{B}^\flat}(y,y_c)} &\le e^{-aD/2} \sum_{c \in (\mathbf{B}^\flat)^+} e^{-\frac{a}{2} d_{\mathbf{B}^\flat}(y,y_c)} \\ &\le e^{-aD/2} \mathsf{S}(a/2).\end{aligned}$$

□

13. Local estimates for the action

The current formula gives the derivative of the action along multiplicative variations. We now estimate this derivative in exponential coordinates, separating its covariant second-order part, the correction due to the reference curvature, and a quadratic remainder. The estimates are local and uniform in the lattice spacing.

Throughout this section assume $L \geq 13$, fix $0 \leq k \leq K$, and use level-k scaling, with $\eta = L^{-k}$. Let U_0 be a real configuration on the bare lattice. Write $D = D_{U_0}$ and $\nabla = \nabla^{U_0}$; the transpose D^{T} below acts on two-forms. For a complex bare one-form A, set

$$F(A) = A^{\eta}(e^{i\eta A}U_0) = \eta^{-1}\sum_p a_p(e^{i\eta A}U_0).$$

At positive bonds the perturbed configuration has value $e^{i\eta A(b)}U_0(b)$; reverse bonds are defined by inversion. The gradient $F'(A)$ is characterized by

$$\frac{d}{dt}\Big|_{t=0} F(A + t\delta A) = (F'(A), \delta A)_\eta$$

for all complex one-forms δA. The pairing is the complex bilinear pairing of definition 9.1. Its nondegeneracy gives a unique gradient, as verified below.

For a plaquette p, let $\mathsf{b}(p)$ be its four positive bonds. For a positive bond b, let $\mathsf{pl}(b)$ be the four plaquettes containing it and put $\mathsf{st}(b) = \bigcup_{p\in\mathsf{pl}(b)} \mathsf{b}(p)$. For a set S of positive bonds, write

$$s_S(A) = \max_{b\in S} |A(b)|, \qquad g_S(A) = \max |\nabla_\lambda A_\kappa(x)|, \qquad \mathsf{n}_S(A) = \max\{s_S(A), g_S(A)\},$$

where the second maximum is over covariant differences between two bonds of S, and is zero if there are none. Thus the two bonds in such a difference are $\langle x, x+\eta e_\kappa\rangle$ and $\langle x+\eta e_\lambda, x+\eta e_\lambda+\eta e_\kappa\rangle$. Use subscripts p and b for $S = \mathsf{b}(p)$ and $S = \mathsf{st}(b)$, respectively, and set

$$\|A\|_X = \max\{\sup|A|, \sup|\nabla A|\}. \tag{98}$$

Proposition 13.1 (Local first-variation estimates)**.** For every real reference configuration U_0, the maps F and F' are entire in the complex form A, and $F'(0) = J(U_0)$. Put

$$\mathcal{W}(A) = F'(A) - J(U_0) - D^{\mathsf{T}}DA, \qquad \Delta' = \mathcal{W}'(0), \qquad \mathcal{V}(A) = \mathcal{W}(A) - \Delta' A.$$

The values $F'(A)(b)$, $(\Delta' A)(b)$ and $\mathcal{V}(A)(b)$ depend on A only through its restriction to $\mathsf{st}(b)$.

There is an absolute constant $C_4 \geq 64$ such that, with $\rho_1 = 1/128$, the following hold whenever

$$0 \leq \alpha_0 \leq 1, \qquad |\partial U_0(p) - \mathbf{1}| \leq \alpha_0\eta^2 \quad \text{at every bare plaquette } p.$$

The constant is independent of the volume, the cutoffs, the lattice spacing and U_0.

(i) The first variation decomposes as

$$F'(A) = J(U_0) + D^{\mathsf{T}}DA + \Delta' A + \mathcal{V}(A).$$

The operator Δ' is linear and symmetric for $(\cdot,\cdot)_\eta$, and

$$|(\Delta' A)(b)| \leq C_4\alpha_0 s_b(A)$$

for every complex A. Its value at b vanishes if $\partial U_0(p) = \mathbf{1}$ for all $p \in \mathsf{pl}(b)$. The operator $\Delta^{\mathrm{W}}_{U_0} = D^{\mathsf{T}}D + \Delta'$ represents the Wilson Hessian:

$$(A, \Delta^{\mathrm{W}}_{U_0} A)_\eta = \frac{d^2}{dt^2}\Big|_{t=0} F(tA).$$

The map $\mathcal{V}$ is entire, $\mathcal{V}(0) = 0$ and $\mathcal{V}'(0) = 0$. If A is complex and $s_b(A) \leq \rho_1$, then

$$|\mathcal{V}(A)(b)| \leq C_4 s_b(A)\mathsf{n}_b(A).$$

In particular, for every complex A,

$$\sup|\Delta' A| \leq C_4\alpha_0 \sup|A|.$$

For complex A with $\sup|A| \leq \rho_1$,

$$\sup|\mathcal{V}(A)| \leq C_4 \sup|A|\,\|A\|_X.$$

(ii) Let A be real and put $E = e^{i\eta A}U_0$. With $\operatorname{ad}_X Y = XY - YX$ and

$$\mathsf{f}(z) = \frac{e^z - 1}{z} = \sum_{q\ge 0} \frac{z^q}{(q+1)!},$$

where the power series defines the value at zero and on linear maps, one has

$$F'(A)(b) = \mathsf{f}(-i\eta \operatorname{ad}_{A(b)}) J(E)(b).$$

If $\eta \sup|A| \le 1/8$, then

$$|J(E)(b)| \le (1 + 4\eta|A(b)|)\,|F'(A)(b)|,$$
$$|F'(A)(b)| \le (1 + 2\eta|A(b)|)\,|J(E)(b)|.$$

(iii) Let A be real and put $E = e^{i\eta A}U_0$. At any plaquette p with $s_p(A) \le \rho_1$,

$$|\partial E(p) - \mathbf{1}| \le |\partial U_0(p) - \mathbf{1}| + (2g_p(A) + C_4 s_p(A)^2)\eta^2.$$

Consequently, if $\sup|A| \le \rho_1$, the same bound holds with $g_p(A)$ and $s_p(A)$ replaced by $\sup|\nabla A|$ and $\sup|A|$.

(iv) If A_1, A_2 are complex and $\|A_1\|_X, \|A_2\|_X \le r \le \rho_1/4$, then

$$\sup|\mathcal{V}(A_1) - \mathcal{V}(A_2)| \le 4C_4 r\,\|A_1 - A_2\|_X. \qquad \blacksquare$$

The nonlinear error of a single plaquette is quadratic in A. To estimate its exterior transpose uniformly in η, we compare adjacent plaquettes after transport to the same base point. Their difference contains a factor η, which cancels the inverse spacing in D^{T}. The calculation is given in section 13.2; section 13.3 then extracts the linear curvature correction and quadratic remainder.

13.1. **The coordinate gradient.**

Entire dependence and locality in proposition 13.1. The action is a finite polynomial in the entries of $e^{\pm i\eta A(b)}$, with coefficients determined by U_0. It is therefore entire. Choose a real basis t_1, t_2, t_3 of $\mathfrak{g}$ with $\langle t_a, t_{a'}\rangle = \delta_{aa'}$, and write $A(b) = \sum_a A^a(b)t_a$. This is also a complex basis of $\mathfrak{g}^c$. Since the pairing on one-forms has the factor η^3, the form

$$F'(A)(b) = \eta^{-3} \sum_a \frac{\partial F}{\partial A^a(b)}(A) t_a$$

satisfies the defining derivative identity and is entire. If a form pairs to zero with every variation, variations supported at one bond and equal to one basis vector show that each of its coordinates is zero. This proves uniqueness. Equation (71) at U_0 gives $F'(0) = J(U_0)$.

Let $\delta_b Y$ be supported at the positive bond b, with value $Y \in \mathfrak{g}^c$. Only the four plaquettes containing b contribute to its variation, so

$$\eta^3 \langle F'(A)(b), Y\rangle = \eta^{-1} \sum_{p\in \mathsf{pl}(b)} \frac{d}{dt}\Big|_{t=0} a_p(e^{i\eta(A + t\delta_b Y)}U_0). \tag{99}$$

Every summand uses A only on $\mathsf{b}(p)$. Testing against the basis vectors proves the asserted locality of F'. The formula for D^{T} in (68) uses only those same four plaquettes; each value of DA uses their four bonds. Thus $\mathcal{W}(A)(b)$ depends only on $A|_{\mathsf{st}(b)}$. Its derivative at zero and the remainder have the same dependence. Finally, $\mathcal{W}$ is entire and $\mathcal{W}(0) = 0$; hence $\mathcal{V} = \mathcal{W} - \mathcal{W}'(0)A$ is entire and $\mathcal{V}(0) = \mathcal{V}'(0) = 0$. $\square$

13.2. **The plaquette calculation.** We use the four positive bonds $b_1, \dots, b_4$ of (66), with $\xi = \eta$. Transport the values of A to the base point of $p = p_{\mu\nu}(x)$ by setting

$$\alpha_1 = A(b_1), \qquad \alpha_2 = \mathrm{Ad}(U_0(b_1))A(b_2), \qquad \alpha_3 = \mathrm{Ad}(U_0(b_4))A(b_3), \qquad \alpha_4 = A(b_4).$$

Since U_0 is unitary, $|\alpha_j| = |A(b_j)|$. Moving the reference transports past the exponentials gives

$$\partial(e^{i\eta A}U_0)(p) = e^{i\eta\alpha_1}e^{i\eta\alpha_2}\,\partial U_0(p)\,e^{-i\eta\alpha_3}e^{-i\eta\alpha_4}. \tag{100}$$

The covariant differences satisfy

$$\begin{aligned} \alpha_2 - \alpha_4 &= \eta(\nabla_\mu A_\nu)(x), \qquad & \alpha_3 - \alpha_1 &= \eta(\nabla_\nu A_\mu)(x), \\ \eta(DA)(p) &= \alpha_1 + \alpha_2 - \alpha_3 - \alpha_4. \end{aligned} \tag{101}$$

Consequently

$$|(DA)(p)| \le 2g_p(A), \qquad \eta|(DA)(p)| \le 4s_p(A). \tag{102}$$

These identities follow directly from the definition of ∇ and (67). They also show that $(D\,\delta_b Y)(p) = 0$ when $p \notin \mathsf{pl}(b)$, where $\delta_b Y$ has value Y at b and zero at the other positive bonds.

For complex matrices $X_1, \dots, X_4$ sufficiently close to zero, put

$$\Lambda(X_1, \dots, X_4) = \log(e^{-X_3}e^{-X_4}e^{X_1}e^{X_2}) - (X_1 + X_2 - X_3 - X_4),$$
$$\mathsf{N}(\beta) = \frac{\Lambda(i\eta\beta_1, \dots, i\eta\beta_4)}{i\eta^2}.$$

The order of the four factors comes from cyclically moving the last two factors of (100) to the front inside the trace. We write

$$N_p = \mathsf{N}(\alpha_1, \dots, \alpha_4), \qquad \Theta_p = (DA)(p) + N_p.$$

Lemma 13.2. If $s_p(A) \le s \le 1/64$, then $N_p, \Theta_p \in \mathfrak{g}^c$, and $i\eta^2\Theta_p$ is the series logarithm of $e^{-i\eta\alpha_3}e^{-i\eta\alpha_4}e^{i\eta\alpha_1}e^{i\eta\alpha_2}$. Moreover,

$$|N_p| \le 64s^2, \qquad \eta|\Theta_p| \le 5s, \qquad |\Theta_p| \le 2g_p(A) + 64s^2. \tag{103}$$

For $|\beta_j| \le s$, the function N satisfies

$$\left|\frac{\partial \mathsf{N}}{\partial \beta_j}(\beta)Z\right| \le 64s|Z|. \tag{104}$$

Let $\lambda \in \{\mu, \nu\}$, $p' = p_{\mu\nu}(x - \eta e_\lambda)$, $u = U_0(\langle x - \eta e_\lambda, x\rangle)^{-1}$, and $S = \mathsf{b}(p) \cup \mathsf{b}(p')$. Then

$$|\,\mathrm{Ad}(u)\alpha_j(p') - \alpha_j(p)| \le 3\eta g_S(A).$$

If $s_S(A) \le s \le 1/64$, then

$$|\,\mathrm{Ad}(u)N_{p'} - N_p| \le 768\eta s\, g_S(A). \tag{105}$$

■

Proof. First consider matrices $Z_1, \dots, Z_n$, $n \le 4$, and put $\sigma = \sum_j |Z_j| \le 1$. Multiplication of the exponential series bounds all terms of degree at least two by $e^\sigma - 1 - \sigma \le \sigma^2$. Hence

$$\left|e^{Z_1}\cdots e^{Z_n} - \mathbf{1} - \sum_j Z_j\right| \le \sigma^2, \qquad \left|e^{Z_1}\cdots e^{Z_n} - e^{\sum_j Z_j}\right| \le 2\sigma^2.$$

For $\sigma \le 1/4$, both products in the second difference are within $e^{1/4} - 1 < 1/2$ of $\mathbf{1}$. The logarithmic estimate in lemma 5.1, and $\log e^{\sum_j Z_j} = \sum_j Z_j$, therefore give

$$\left|\log(e^{Z_1}\cdots e^{Z_n}) - \sum_j Z_j\right| \le 4\sigma^2.$$

Applied in the order $-X_3, -X_4, X_1, X_2$, this proves that Λ is holomorphic on the ball $|X| := \sum_j |X_j| < 1/4$, with $|\Lambda(X)| \le 4|X|^2$. Conjugation by a unitary matrix commutes with this logarithm and preserves its domain. If the X_j are traceless, the product has determinant one; its series logarithm is traceless by lemma 5.1. Thus Λ is traceless as well.

We also need a derivative bound that vanishes at the origin. On $|X| < 1/4$, the preceding estimate gives $|\Lambda(X)| \le 1/4$. The one-variable Cauchy estimate along $X + t\,\delta X$ yields

$$|\Lambda'(X)\delta X| \le \frac{(1/4)|\delta X|}{1/4 - |X|} \le 2|\delta X|, \qquad |X| < 1/8,$$

where $|\delta X| = \sum_j |\delta X_j|$. The quadratic estimate gives $\Lambda'(0) = 0$. Apply the Schwarz lemma on the ball of radius $1/8$ to $X \mapsto \Lambda'(X)\delta X$, by restricting to a complex line through zero and then applying a linear functional of norm one. This gives

$$|\Lambda'(X)\delta X| \le 16|X|\,|\delta X|, \qquad |X| < 1/8. \tag{106}$$

If $|\beta_j| \le s \le 1/64$, then $\sum_j |i\eta\beta_j| \le 4\eta s \le 1/16$. The quadratic bound for Λ gives $|\mathsf{N}(\beta)| \le 64s^2$. For the derivative in β_j, the variation of $i\eta\beta$ has the single nonzero entry $i\eta Z$. Therefore (106) gives

$$\left|\frac{\partial \mathsf{N}}{\partial \beta_j}(\beta)Z\right| \le \eta^{-2}\,16(4\eta s)\eta|Z| = 64s|Z|.$$

At $\beta = \alpha$, tracelessness and (101) imply

$$\begin{aligned} i\eta^2\Theta_p &= i\eta(\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4) + \Lambda(i\eta\alpha) \\ &= \log(e^{-i\eta\alpha_3}e^{-i\eta\alpha_4}e^{i\eta\alpha_1}e^{i\eta\alpha_2}). \end{aligned}$$

The inverse identity for the series logarithm proves the assertion about its exponential. Together with (102), the bound on N_p gives $\eta|\Theta_p| \le 4s + 64\eta s^2 \le 5s$ and $|\Theta_p| \le 2g_p(A) + 64s^2$.

It remains to compare the two plaquettes. For $\kappa \in \{\mu, \nu\}$, the definition of the covariant difference gives

$$\mathrm{Ad}(u)A_\kappa(x - \eta e_\lambda) - A_\kappa(x) = -\eta\,\mathrm{Ad}(u)(\nabla_\lambda A_\kappa)(x - \eta e_\lambda).$$

The two bonds involved belong to S. This gives the bound $\eta g_S(A)$ for α_1 and α_4. For α_2, use (101) at both plaquettes:

$$\begin{aligned} \mathrm{Ad}(u)\alpha_2(p') - \alpha_2(p) = \mathrm{Ad}(u)\alpha_4(p') - \alpha_4(p) \\ + \eta\big[\mathrm{Ad}(u)(\nabla_\mu A_\nu)(x - \eta e_\lambda) - (\nabla_\mu A_\nu)(x)\big]. \end{aligned}$$

Each of the last two covariant differences is between bonds of one of the two plaquettes. Their norms are at most $g_S(A)$, so the bound is $3\eta g_S(A)$. The same computation with $\alpha_3 - \alpha_1$ gives the bound for α_3.

Unitary covariance of N gives

$$\mathrm{Ad}(u)N_{p'} = \mathsf{N}(\mathrm{Ad}(u)\alpha_1(p'), \dots, \mathrm{Ad}(u)\alpha_4(p')).$$

Every entry on the line segment joining this quadruple to $(\alpha_j(p))_j$ has norm at most s. Integrating (104) along that segment yields

$$|\,\mathrm{Ad}(u)N_{p'} - N_p| \le 64s \sum_{j=1}^{4} |\,\mathrm{Ad}(u)\alpha_j(p') - \alpha_j(p)| \le 768\eta s\, g_S(A).$$

□

To differentiate the plaquette action, write

$$c_p = \tfrac{1}{2}\,\mathrm{tr}\,\partial U_0(p), \qquad F_p = F_{U_0}(p).$$

The reference configuration is real. The SU(2) decomposition in (73) gives

$$\begin{aligned} \partial U_0(p) &= c_p \mathbf{1} + i\eta^2 F_p, \qquad -1 \le c_p \le 1, \\ 1 - c_p &= \tfrac{1}{2}|\partial U_0(p) - \mathbf{1}|^2, \qquad \eta^2 |F_p| \le |\partial U_0(p) - \mathbf{1}|. \end{aligned} \tag{107}$$

Indeed, the eigenvalues of a matrix in SU(2) are $e^{i\theta}, e^{-i\theta}$, so $c_p = \cos\theta$ and $|\partial U_0(p) - \mathbf{1}|^2 = 2(1 - c_p)$. Also $\eta^4 |F_p|^2 = 1 - c_p^2 \le 2(1 - c_p)$.

Introduce the entire scalar functions

$$\mathrm{co}(\zeta) = \sum_{q \ge 0} \frac{(-\zeta)^q}{(2q)!}, \qquad \mathrm{si}(\zeta) = \sum_{q \ge 0} \frac{(-\zeta)^q}{(2q+1)!},$$
$$\zeta_p = \eta^4 \langle \Theta_p, \Theta_p \rangle.$$

For a traceless two-by-two matrix Θ, Cayley–Hamilton gives $\Theta^2 = \langle \Theta, \Theta \rangle \mathbf{1}$. Hence

$$e^{i\eta^2 \Theta} = \mathrm{co}(\eta^4 \langle \Theta, \Theta \rangle) \mathbf{1} + i\eta^2 \mathrm{si}(\eta^4 \langle \Theta, \Theta \rangle) \Theta.$$

By cyclicity of the trace in (100), and by lemma 13.2,

$$\tfrac{1}{2} \operatorname{tr} \partial (e^{i\eta A} U_0)(p) = \tfrac{1}{2} \operatorname{tr} \big[\partial U_0(p) e^{i\eta^2 \Theta_p} \big].$$

Using (107), we obtain

$$a_p(e^{i\eta A} U_0) = 1 - c_p \mathrm{co}(\zeta_p) + \eta^4 \mathrm{si}(\zeta_p) \langle F_p, \Theta_p \rangle. \tag{108}$$

The logarithm calculation used here holds on the open set $4\eta s_p(A) < 1/4$, so the identity may be differentiated at each A with $s_p(A) \le 1/64$.

Set

$$\begin{aligned} M_p^{\mathrm{r}} &= -c_p \big(2\mathrm{co}'(\zeta_p) + 1\big) \Theta_p + \big(\mathrm{si}(\zeta_p) - 1\big) F_p \\ &\quad + 2\eta^4 \mathrm{si}'(\zeta_p) \langle F_p, \Theta_p \rangle \Theta_p, \qquad M_p = F_p + c_p \Theta_p + M_p^{\mathrm{r}}. \end{aligned}$$

All these matrices are traceless. Differentiation of (108), with $\delta\zeta_p = 2\eta^4 \langle \Theta_p, \delta\Theta_p \rangle$, gives

$$\begin{aligned} \delta a_p &= \eta^4 \left\langle -2c_p \mathrm{co}'(\zeta_p) \Theta_p + \mathrm{si}(\zeta_p) F_p, \delta\Theta_p \right\rangle \\ &\quad + 2\eta^8 \mathrm{si}'(\zeta_p) \langle F_p, \Theta_p \rangle \langle \Theta_p, \delta\Theta_p \rangle \\ &= \eta^4 \langle M_p, \delta\Theta_p \rangle. \end{aligned}$$

For $b = b_j \in \mathsf{b}(p)$ and a variation $\delta_b Y$, let

$$\mathsf{L}_{p,b} Y = \frac{\partial \mathsf{N}}{\partial \beta_j}(\alpha)\, Y_j,$$
$$Y_1 = Y_4 = Y, \quad Y_2 = \mathrm{Ad}(U_0(b_1)) Y, \quad Y_3 = \mathrm{Ad}(U_0(b_4)) Y.$$

Then $\delta\Theta_p = (D\, \delta_b Y)(p) + \mathsf{L}_{p,b} Y$. The map $\mathsf{L}_{p,b}$ preserves $\mathfrak{g}^c$ and, by (104), $|\mathsf{L}_{p,b} Y| \le 64 s |Y|$ if $s_p(A) \le s \le 1/64$. Its transpose is taken in the complex bilinear form $\langle \cdot, \cdot \rangle$. To bound its norm, we use

$$|X| \le 2 \sup_{\substack{Y \in \mathfrak{g}^c \\ |Y| \le 1}} |\langle X, Y \rangle|.$$

For $X \ne 0$, choose $Y = X^*/|X|$; then $\langle X, Y \rangle = \operatorname{tr}(XX^*)/(2|X|) \ge |X|/2$. Since $|\langle Z, W \rangle| \le |Z|\,|W|$, it follows that

$$|\mathsf{L}_{p,b}^{\mathsf{T}} Z| \le 2 \sup_{|Y| \le 1} |\langle Z, \mathsf{L}_{p,b} Y \rangle| \le 128 s |Z|.$$

We have proved the local differential formula

$$\begin{aligned}\frac{d}{dt}\,a_p(e^{i\eta(A+t\delta_b Y)}U_0)\Big|_{t=0} &= \eta^4\langle M_p, (D\,\delta_b Y)(p)\rangle \\ &\quad + \eta^4\langle \mathsf{L}^{\mathsf{T}}_{p,b} M_p, Y\rangle, \\ |\mathsf{L}^{\mathsf{T}}_{p,b} Z| &\le 128 s|Z|.\end{aligned} \tag{109}$$

Finally, $|\zeta_p| \le \eta^4|\Theta_p|^2 \le (5\eta s)^2 \le 1$. The scalar power series show, for $|\zeta| \le 1$, that

$$|2\mathrm{co}'(\zeta) + 1| \le |\zeta|, \qquad |\mathrm{si}(\zeta) - 1| \le |\zeta|, \qquad |\mathrm{si}'(\zeta)| \le 1.$$

For example, the absolute sums bounding the respective coefficients are $\sum_{q\ge2} 2q/(2q)!$, $\sum_{q\ge1} 1/(2q+1)!$, and $\sum_{q\ge1} q/(2q+1)!$, each less than one. As $|c_p| \le 1$, these inequalities give

$$|M^{\mathrm{r}}_p| \le \eta^4(|\Theta_p|^3 + 3|F_p|\,|\Theta_p|^2). \tag{110}$$

13.3. **The linear correction and nonlinear remainder.** We prove the expansion and local estimates in proposition 13.1. The entire-gradient and locality argument above defines $\mathcal{W}(A) = F'(A) - J(U_0) - D^{\mathsf{T}}DA$ and its linear and higher-order terms $\Delta' A$ and $\mathcal{V}(A)$. In the estimates below C is an absolute constant whose value may change.

Assume the reference curvature bound. By (107),

$$0 \le 1 - c_p \le \tfrac{1}{2}\alpha_0^2\eta^4, \qquad |F_p| \le \alpha_0. \tag{111}$$

For $s_b(A) \le 1/64$, insert (109) into the local gradient identity (99) and transpose D. The factors $\eta^{-1}\eta^4 = \eta^3$ agree with the pairing of forms, and nondegeneracy gives

$$\begin{aligned}F'(A)(b) &= (D^{\mathsf{T}}M)(b) + \mathcal{T}(A)(b), \\ \mathcal{T}(A)(b) &= \sum_{p\in\mathsf{pl}(b)} \mathsf{L}^{\mathsf{T}}_{p,b} M_p.\end{aligned} \tag{112}$$

Here M need only be specified on $\mathsf{pl}(b)$: extend it by zero before taking the transpose. The value at b is independent of this extension by (68). Also,

$$|\mathcal{T}(A)(b)| \le 4 \cdot 128 s_b(A) \max_{p\in\mathsf{pl}(b)} |M_p|.$$

Fix $b = \langle x, x + \eta e_\mu\rangle$ and abbreviate $s = s_b(A) \le 1/64$, $g = g_b(A)$, and $\mathsf{n} = \mathsf{n}_b(A)$. On the incident plaquettes,

$$M_p = F_p + (DA)(p) + (c_p - 1)(DA)(p) + c_p N_p + M^{\mathrm{r}}_p.$$

Since $J(U_0) = D^{\mathsf{T}}F_{U_0}$, subtracting the constant and exterior-Laplacian terms in (112) gives

$$\begin{aligned}\mathcal{W}(A)(b) &= (D^{\mathsf{T}}((c-1)DA))(b) + (D^{\mathsf{T}}(cN))(b) \\ &\quad + (D^{\mathsf{T}}M^{\mathrm{r}})(b) + \mathcal{T}(A)(b).\end{aligned} \tag{113}$$

Formula (68) expresses $(D^{\mathsf{T}}X)(b)$ as two terms of the form $\pm\eta^{-1}[\mathrm{Ad}(u_\nu)X(p') - X(p)]$, where p' is the translate of p by $-\eta e_\nu$ and $u_\nu = U_0(\langle x - \eta e_\nu, x\rangle)^{-1}$. Both plaquettes belong to $\mathsf{pl}(b)$. In particular,

$$|(D^{\mathsf{T}}X)(b)| \le 4\eta^{-1} \max_{p\in\mathsf{pl}(b)} |X(p)|.$$

For the first term of (113), (102) and (111) yield

$$|(D^{\mathsf{T}}((c-1)DA))(b)| \le 4\eta^{-1} \cdot \tfrac{1}{2}\alpha_0^2\eta^4 \cdot 4\eta^{-1}s \le 8\alpha_0 s.$$

For the second term, the bound on N_p alone would leave an inverse power of η. Compare the two values before applying the transpose:

$$c_{p'}\,\mathrm{Ad}(u_\nu)N_{p'} - c_pN_p = c_{p'}[\mathrm{Ad}(u_\nu)N_{p'} - N_p] + (c_{p'} - c_p)N_p.$$

The first bracket is at most $768\eta sg$ by (105). Both $1-c_p$ and $1-c_{p'}$ belong to $[0,\alpha_0^2\eta^4/2]$, so $|c_{p'}-c_p|\le\alpha_0^2\eta^4/2$. Since $|c_{p'}|\le 1$ and $|N_p|\le 64s^2$, the two differences in the transpose satisfy

$$\bigl|(D^{\mathsf T}(cN))(b)\bigr|\le 2\eta^{-1}(768\eta sg+32\alpha_0^2\eta^4 s^2)\le Cs\mathsf{n}.$$

For the third term, use (110), $|F_p|\le\alpha_0$, and $\eta|\Theta_p|\le 5s$:

$$\begin{aligned}|(D^{\mathsf T}M^{\mathrm r})(b)|&\le 4\eta^3\max_{p\in\mathsf{pl}(b)}(|\Theta_p|^3+3\alpha_0|\Theta_p|^2)\\&\le 4(125s^3+75\alpha_0 s^2)\le Cs(\alpha_0+\mathsf{n}).\end{aligned}$$

For the last term, $|\Theta_p|\le 2g+64s^2\le 3\mathsf{n}$. Furthermore,

$$|M_p^{\mathrm r}|\le(\eta|\Theta_p|)^2(|\Theta_p|+3\alpha_0)\le|\Theta_p|+3\alpha_0 s,$$

because $\eta\le 1$ and $(\eta|\Theta_p|)^2\le 25s^2\le s\le 1$. It follows that $|M_p|\le\alpha_0+C\mathsf{n}$, and the bound following (112) gives $|\mathcal{T}(A)(b)|\le Cs(\alpha_0+\mathsf{n})$. Thus the four terms establish

$$|\mathcal{W}(A)(b)|\le Cs_b(A)(\alpha_0+\mathsf{n}_b(A)),\qquad s_b(A)\le 1/64.\tag{114}$$

We now separate the linear term. For fixed A and b, write the entire expansion $\mathcal{W}(tA)(b)=\sum_{q\ge1}t^q w_q$. If $s_b(A)=0$, locality gives $\mathcal{W}(tA)(b)=\mathcal{W}(0)(b)=0$, and all coefficients vanish. Otherwise, put $T=1/(64s_b(A))$. For $|t|\le T$, (114) gives

$$|\mathcal{W}(tA)(b)|\le C|t|s_b(A)(\alpha_0+|t|\mathsf{n}_b(A)).$$

For any $0<R\le T$, the Cauchy estimate for the linear coefficient therefore gives

$$|w_1|\le Cs_b(A)(\alpha_0+R\mathsf{n}_b(A)).$$

Letting R tend to zero removes the quadratic part and yields $|w_1|\le C'\alpha_0 s_b(A)$. For the coefficients of order at least two, use the largest available circle, $R=T$:

$$|w_q|\le CT^{1-q}s_b(A)(\alpha_0+T\mathsf{n}_b(A)),\qquad q\ge2.$$

If $s_b(A)\le 1/128$, then $T\ge 2$ and $\sum_{q\ge2}T^{1-q}\le 2T^{-1}$. Consequently

$$\begin{aligned}|\mathcal{V}(A)(b)|&\le\sum_{q\ge2}|w_q|\\&\le 2Cs_b(A)(\alpha_0T^{-1}+\mathsf{n}_b(A))\le C''s_b(A)\mathsf{n}_b(A).\end{aligned}$$

The last inequality uses $\alpha_0T^{-1}=64\alpha_0 s_b(A)\le 64\mathsf{n}_b(A)$. Choose

$$\rho_1=1/128,\qquad C_4=\max\{64,C',C''\}.$$

This proves the stated local bounds for $\Delta' A=w_1$ and $\mathcal{V}(A)$. The global bounds follow from $s_b(A)\le\sup|A|$ and $\mathsf{n}_b(A)\le\|A\|_X$. No smallness of $g_b(A)$ was used.

The derivative of F' at zero is $D^{\mathsf T}D+\Delta'$, since $\mathcal{V}'(0)=0$. For complex A,B,

$$(B,(D^{\mathsf T}D+\Delta')A)_\eta=\frac{\partial^2}{\partial t\,\partial u}F(tA+uB)\Big|_{t=u=0}.$$

Mixed derivatives commute, so this form is symmetric. The form $(B,D^{\mathsf T}DA)_\eta=(DB,DA)_\eta$ is symmetric as well; hence Δ' is symmetric. Taking $B=A$ identifies $D^{\mathsf T}D+\Delta'$ with the Hessian.

Finally, suppose $\partial U_0(p)=\mathbf{1}$ for all $p\in\mathsf{pl}(b)$. Then $c_p=1$ and $F_p=0$ there. In (113) the first term vanishes. At tA, the bounds for N_p, $M_p^{\mathrm r}$ and $\mathsf{L}_{p,b}^{\mathsf T}M_p$ show that each remaining term is $O(|t|^2)$ as $t\to0$: N_p is quadratic, $|M_p^{\mathrm r}|\le\eta^4|\Theta_p|^3$, and $|\mathsf{L}_{p,b}^{\mathsf T}M_p|\le 128s|M_p|$ with $s=|t|s_b(A)$ and $M_p=\Theta_p+M_p^{\mathrm r}$. The constants here may depend on the fixed A and η. Thus the coefficient w_1 vanishes, which is the locally flat assertion.

13.4. **The current and the real plaquette bound.**

Completion of proposition 13.1. We prove (ii) by differentiating exponential coordinates. For complex matrices X, Y,

$$\frac{d}{dt}\Big|_{t=0} e^{X+tY} = [\mathsf{f}(\mathrm{ad}_X)Y]e^X.$$

Indeed, integrating the derivative of $s \mapsto e^{s(X+tY)}e^{(1-s)X}$ gives

$$e^{X+tY} - e^X = t\int_0^1 e^{s(X+tY)}Ye^{(1-s)X}\,ds.$$

After differentiation at $t=0$, the right side becomes $[\int_0^1 e^{sX}Ye^{-sX}\,ds]e^X$. Left and right multiplication by X commute, so termwise integration of the exponential series yields

$$\int_0^1 e^{sX}Ye^{-sX}\,ds = \sum_{q\ge 0}\frac{\mathrm{ad}_X^q\, Y}{(q+1)!} = \mathsf{f}(\mathrm{ad}_X)Y.$$

Now let A be real, $E = e^{i\eta A}U_0$, and let δA be complex. The formula at each positive bond shows that the curves

$$t \longmapsto e^{i\eta(A+t\delta A)}U_0, \qquad t \longmapsto e^{it\eta B}E, \qquad B(b) = \mathsf{f}(i\eta\,\mathrm{ad}_{A(b)})\delta A(b),$$

have the same derivative at zero. A commutator has trace zero, so B is a complex one-form. Applying the first variation formula (71) at E gives

$$(F'(A), \delta A)_\eta = \big(J(E), \mathsf{f}(i\eta\,\mathrm{ad}_A)\delta A\big)_\eta.$$

Cyclicity of the trace gives

$$\langle [A(b), X], Z\rangle = -\langle X, [A(b), Z]\rangle.$$

Thus the transpose of $\mathsf{f}(i\eta\,\mathrm{ad}_{A(b)})$ for the bilinear pairing is $\mathsf{f}(-i\eta\,\mathrm{ad}_{A(b)})$. The nondegeneracy of the pairing proves the identity in (ii).

For the norm bounds, use the operator norm on linear maps of the matrix space. If $\|z\| \le 1/4$, the power series and $(q+1)! \ge 2\cdot 3^{q-1}$ give

$$\|\mathsf{f}(z) - I\| \le \frac{1}{2}\|z\|\sum_{q\ge 1} 12^{-(q-1)} = \frac{6}{11}\|z\| \le \frac{3}{22}.$$

In particular $\|\mathsf{f}(z) - I\| \le \|z\|$. The Neumann series gives

$$\|\mathsf{f}(z)^{-1} - I\| \le \frac{(6/11)\|z\|}{1 - 3/22} = \frac{12}{19}\|z\| \le 2\|z\|.$$

If $\eta \sup|A| \le 1/8$, then $z = -i\eta\,\mathrm{ad}_{A(b)}$ satisfies $\|z\| \le 2\eta|A(b)| \le 1/4$. Applying these two bounds to the identity in (ii), and then to its inverse, gives the stated factors $1 + 2\eta|A(b)|$ and $1 + 4\eta|A(b)|$.

For (iii), fix a plaquette p with $s_p(A) \le \rho_1$ and use the transported variables of lemma 13.2. They are real. Put

$$X = e^{i\eta\alpha_1}e^{i\eta\alpha_2}, \qquad Y = e^{-i\eta\alpha_3}e^{-i\eta\alpha_4}.$$

These matrices are unitary, and (100) gives $\partial E(p) = X\partial U_0(p)Y$. Consequently,

$$\begin{aligned}|\partial E(p) - \mathbf{1}| &\le |\partial U_0(p) - \mathbf{1}| + |XY - \mathbf{1}| \\ &= |\partial U_0(p) - \mathbf{1}| + |YX - \mathbf{1}|.\end{aligned}$$

Here the last equality follows by unitary conjugation. Since $s_p(A) \le \rho_1 \le 1/64$, lemma 13.2 identifies the series logarithm of YX as $i\eta^2\Theta_p$. The real logarithm assertion in lemma 5.1 gives $\eta^2\Theta_p \in \mathfrak{g}$. For $H \in \mathfrak{g}$, its eigenvalues are $\pm|H|$, and hence

$$|e^{iH} - \mathbf{1}| = 2|\sin(|H|/2)| \le |H|.$$

The bound on Θ_p in lemma 13.2 now gives

$$|YX - \mathbf{1}| \le \eta^2 |\Theta_p| \le (2g_p(A) + 64 s_p(A)^2)\eta^2.$$

This proves the local estimate because $C_4 \ge 64$; the global form follows from $s_p(A) \le \sup|A|$ and $g_p(A) \le \sup|\nabla A|$.

Finally, let $\delta = A_1 - A_2$. If $\delta = 0$, (iv) is immediate. Otherwise put $A_t = tA_1 + (1-t)A_2$, $0 \le t \le 1$, and $R = r/\|\delta\|_X$. For $|z| = R$,

$$\|A_t + z\delta\|_X \le 2r \le \rho_1, \qquad \sup|\mathcal{V}(A_t + z\delta)| \le C_4 (2r)^2.$$

The Cauchy formula for the entire function $z \mapsto \mathcal{V}(A_t + z\delta)$ therefore gives

$$\sup|\mathcal{V}'(A_t)\delta| \le R^{-1} C_4 (2r)^2 = 4C_4 r \, \|\delta\|_X.$$

Integrating this estimate in $\mathcal{V}(A_1) - \mathcal{V}(A_2) = \int_0^1 \mathcal{V}'(A_t)\delta \, dt$ proves (iv). □

14. Quantitative bounds for averaging

We estimate the curvature of successive block averages and their distance from straight bare transports. The centered mean and corner comb constructions have the same leading curvature bound: one averaging step multiplies the fine curvature bound by L^2, with a quadratic error. After rescaling, these errors are summable over the averaging steps.

Assume $L \ge 13$ throughout this section. At one step, $0 \le n < K$, use level-n scaling, so that fine bonds have length one. For $y \in \mathbb{T}^{(n+1)}$, put

$$o_{\mathsf{m}}(y) = y, \qquad o_{\mathsf{c}}(y) = y - \mathsf{h}(e_1 + e_2 + e_3), \qquad \mathsf{h} = (L-1)/2.$$

The symbols m and c denote the centered mean and corner comb schemes. For $x \in B(y)$, the former uses the contour mean already constructed in section 4, while the latter uses the comb of (62):

$$W_{\mathsf{m}}(y,x) = \operatorname{av}\big((W(\Gamma))_{\Gamma \in \mathcal{G}(y,x)}\big), \qquad W_{\mathsf{c}}(y,x) = W(\Gamma_{o_{\mathsf{c}}(y),x}).$$

The centered variable is defined when the contour family has diameter less than $1/40$. The corner construction uses Bałaban's logarithmic contour average with the lowest corner as block origin [Bał85a, (42)–(43), pp. 23–24].

Let $c = \langle c_-, c_- + Le_\mu \rangle$ be a positive coarse bond. Write $\mathsf{D}_c = B(c_-) \cup B(c_+)$ and let P_c be all fine bonds whose endpoints belong to D_c. Write P_c^+ for its positive bonds. For $x \in B(c_-)$, put $x' = x + Le_\mu$. Denote straight transport from $o_\star(c_-)$ to $o_\star(c_+)$ by $W_\star[c]$, where $\star \in \{\mathsf{m}, \mathsf{c}\}$. For a G^c-valued configuration on P_c, set

$$W_x^\star = W_\star(c_-, x) W([x, x']) W_\star(c_+, x')^{-1} W_\star[c]^{-1},$$
$$\mathcal{M}_\star(W)(c) = \exp\left(L^{-3} \sum_{x \in B(c_-)} \log W_x^\star \right) W_\star[c]. \tag{115}$$

This formula is used when all its contour variables are defined and $|W_x^\star - \mathbf{1}| < 1$ for every x; all logarithms in it are series logarithms. Define the value at $\bar{c}$ by inversion. For the centered scheme this is (39). Write $\mathcal{M}_\star^0 = \mathrm{id}$ and $\mathcal{M}_\star^{j+1} = \mathcal{M}_\star \mathcal{M}_\star^j$ where defined.

Let $o_\star^{(j)}$ be the composition of the base maps at levels $j-1, \dots, 0$, with $o_\star^{(0)} = \mathrm{id}$. In bare units,

$$o_{\mathsf{m}}^{(j)}(y) = y, \qquad o_{\mathsf{c}}^{(j)}(y) = y - \tfrac{1}{2}(L^j - 1)(e_1 + e_2 + e_3),$$

so the latter is the lowest corner of $B^j(y)$. For a level-j bond c, write $U[\![c]\!]_\star$ for the transport of the bare configuration U along the straight path from $o_\star^{(j)}(c_-)$ to $o_\star^{(j)}(c_+)$. Finally, for real W, put

$$\varpi(W) = \max_p |\partial W(p) - \mathbf{1}|, \qquad \varpi_c(W) = \max_{p \text{ with corners in } \mathsf{D}_c} |\partial W(p) - \mathbf{1}|.$$

Proposition 14.1 (Iterated averages)**.** There are positive constants c_2^{SO}, ϖ_L' and κ_4, depending only on L, with the following property. Let $1 \le k \le K$, use level-k scaling with $\eta = L^{-k}$, and let $0 < \alpha_0 \le c_2^{\mathrm{SO}}$. Suppose the real bare configuration U satisfies

$$|\partial U(p) - \mathbf{1}| \le \alpha_0 \eta^2 \qquad \text{at every bare plaquette } p.$$

For each $\star \in \{\mathsf{m}, \mathsf{c}\}$ and $0 \le n \le k$, the iterate $\mathcal{M}_\star^n U$ is defined and real, and

$$\begin{aligned} \varpi(\mathcal{M}_\star^n U) &\le 2\alpha_0 (L^n \eta)^2 \le \varpi_L', \\ |\mathcal{M}_\star^n U(c) - U[\![c]\!]_\star| &\le \kappa_4 \alpha_0 (L^n \eta)^2 \qquad (c \text{ a bond of } \mathbb{T}^{(n)}). \end{aligned} \tag{116}$$

For the centered scheme, $\mathcal{M}_{\mathsf{m}}^n U = \mathcal{M}^n U$, and the small-curvature branch (37) applies at every participating bond of every step. Moreover, $\mathcal{M}^k$ is real-analytic on

$$\left\{ U \in \mathcal{V}_0 : |\partial U(p) - \mathbf{1}| < c_2^{\mathrm{SO}} \eta^2 \text{ for every bare plaquette } p \right\}.$$

Under the same global hypothesis, the estimates have the following local form. For a plaquette or bond $\mathfrak{e}$ of $\mathbb{T}^{(n)}$, put

$$\mathsf{Z}(\mathfrak{e}) = \bigcup_{y \text{ a vertex of } \mathfrak{e}} B^n(y), \qquad \alpha(\mathfrak{e}) = \max_{p \text{ bare, with corners in } \mathsf{Z}(\mathfrak{e})} \eta^{-2} |\partial U(p) - \mathbf{1}|,$$

with an empty maximum equal to zero. Then the curvature bound in (116) at a plaquette $\mathfrak{e}$ holds with $\alpha(\mathfrak{e})$ in place of α_0, and so does the transport bound at a bond $\mathfrak{e}$. The constants are chosen in sections 14.4 and 14.5. ∎

The one-step proof uses a local axial gauge and cancellation of the contour terms around a coarse plaquette, as in Bałaban's argument [Bał85a, (47)–(51), pp. 25–26]. We give the calculation for both schemes. The surviving linear term is an average of sums over $L \times L$ squares; it has total weight L^2. A uniform holomorphic neighborhood controls the quadratic error. The normalized curvature recursion then has a bounded product, while a separate recursion controls the straight transports.

14.1. **Domains and base sites.**

Lemma 14.2. The contours used by either scheme stay in their block, and the straight paths in (115) stay in the double block. The corner contours translate with the block, and $o_{\mathsf{c}}^{(j)}(y)$ is its lowest bare corner.

For either scheme, the domain of (115) is open in the positive-bond coordinates, and the contour variables and the average extend holomorphically to an open subset of the ambient matrix space. At real configurations in this domain their values lie in G.

Let $u : \mathsf{D}_c \to G$. The domain is invariant under the real gauge transformation u, and

$$\begin{aligned} (W^u)_\star(y, x) &= u(o_\star(y)) W_\star(y, x) u(x)^{-1}, \\ \mathcal{M}_\star(W^u)(c) &= u(o_\star(c_-)) \mathcal{M}_\star(W)(c) u(o_\star(c_+))^{-1}. \end{aligned} \tag{117}$$

For the centered scheme, if W is real and $\varpi_c(W) < \omega_M$, its average is the small-curvature branch (37); this restriction is real-analytic.

At a level-j bond c, the value and definedness of $\mathcal{M}_\star^j U(c)$ depend only on the bare bonds with endpoints in $B^j(c_-) \cup B^j(c_+)$. ∎

Proof. In level-n units, $B(y)$ has integer coordinates $y + \sum_\nu t_\nu e_\nu$, $|t_\nu| \le \mathsf{h}$. The corner comb adjusts coordinates monotonically from $y - \mathsf{h}\sum_\nu e_\nu$ to x, in the order $3, 2, 1$, and thus stays in $B(y)$. Its formula translates with the block. The centered contours have the same containment property by their definition. The double block has coordinate intervals $[-\mathsf{h}, \mathsf{h} + L]$ in direction μ and $[-\mathsf{h}, \mathsf{h}]$ in the other directions, relative to c_-. Each interval contains at most $2L$ sites, and $2L < L^{m+K-n}$,

so these charts are injective. Thus all the straight paths in (115) lie in it. Composing the corner translations in bare units gives

$$o_{\mathsf{c}}^{(j)}(y) = y - \mathsf{h}(1 + L + \cdots + L^{j-1}) \sum_{\nu} e_\nu = y - \tfrac{1}{2}(L^j - 1) \sum_{\nu} e_\nu,$$

the lowest site of $B^j(y)$.

On $\mathrm{GL}(2,\mathbb{C})^{\mathsf{P}_c^+}$, extend bond variables by inversion on reverse bonds. Each corner transport is then a product of matrix variables and their inverses. Its domain imposes only the finitely many open conditions $|W_x^{\mathsf{c}} - \mathbf{1}| < 1$, so the average is holomorphic there. For the centered scheme, these properties were proved for (39), using the holomorphic mean in proposition 4.3. At G^c-valued points the logarithms are traceless, so their exponential and the average lie in G^c. For real configurations the contour variables and loops lie in G, their series logarithms belong to $i\mathfrak{g}$ by lemma 5.1, and the average lies in G.

The first identity in (117) follows by cancellation along a path for the corner scheme and by (29) for the centered scheme. Writing $u_- = u(o_\star(c_-))$, it gives

$$(W^u)_x^\star = u_- W_x^\star u_-^{-1}.$$

Unitary conjugation preserves the norm condition defining each logarithm and commutes with its series. The centered contour diameters are preserved as well. Substitution in (115) proves the second identity and one direction of domain preservation; applying it to u^{-1} gives the reverse direction. Centered branch agreement and real-analyticity follow from lemma 4.4 and proposition 4.5.

Finally, the one-step formula uses only the double block. If locality holds after j steps, every value needed for the next step lies on a level-j bond c' whose endpoints belong to $B(c_-) \cup B(c_+)$. Block nesting gives

$$B^j(c'_-) \cup B^j(c'_+) \subset B^{j+1}(c_-) \cup B^{j+1}(c_+).$$

This proves the assertion, including definedness, by induction from $j = 0$. Reverse bonds follow by inversion. □

14.2. **Path estimates and one averaging step.** We use fine-lattice units in this subsection. A nonparallel exchange of two adjacent steps changes a path transport by one plaquette variable, up to unitary conjugation. Counting these exchanges gives bounds proportional to the curvature. The two uses below are a gauge with small bond variables on a box and a comparison of the average with straight transport between its block origins.

A bond function a takes values in $\mathfrak{g}^c$ on positive bonds and is extended by $a(\bar{b}) = -a(b)$. It is real when its values lie in $\mathfrak{g}$. We write $a(\Gamma) = \sum_{b\in\Gamma} a(b)$ and $e^{ia}(b) = e^{ia(b)}$, with no spacing factor in the exponent.

For $V \in G$, put $\theta(V) = \arccos(\frac{1}{2}\operatorname{tr} V) \in [0,\pi]$. Its eigenvalues are $e^{\pm i\theta(V)}$, so

$$|V - \mathbf{1}| = 2\sin\frac{\theta(V)}{2}, \qquad |V - \mathbf{1}| \le \theta(V) \le \frac{\pi}{2}|V - \mathbf{1}|.$$

By lemma 5.2, the intrinsic logarithm satisfies $|i^{-1}\log V| = \theta(V)$ when $V \neq -\mathbf{1}$. If $|V - \mathbf{1}| < 1$, this is the series logarithm and $\theta(V) < \pi/3$.

Lemma 14.3 (Path exchanges)**.** Let W be a G-valued configuration on a reversal-symmetric set of bonds of $\mathbb{T}^{(n)}$, $0 \le n \le K$. Suppose that $\Gamma_0, \dots, \Gamma_N$ arise successively by exchanging adjacent steps, with every intermediate path in the domain of W. Suppose also that every plaquette associated with a nonparallel exchange lies in that domain. List these plaquettes, with repetitions, as $p_1, \dots, p_{N'}$. Then

$$|W(\Gamma_N) - W(\Gamma_0)| \le \sum_{j=1}^{N'} |\partial W(p_j) - \mathbf{1}|. \tag{118}$$

■

Proof. The adjacent-step calculation in the proof of lemma 4.1 is local to the exchanged steps. Parallel exchanges leave the transport unchanged; for a nonparallel exchange the quotient of the two transports is a unitary conjugate of $\partial W(p_j)$ or its inverse. Thus its distance from $\mathbf{1}$ is $|\partial W(p_j) - \mathbf{1}|$. Summing the differences of consecutive transports proves (118). □

Lemma 14.4 (Comb gauge on a box)**.** Let $0 \le n \le K$. In level-n units, let Y be the lattice sites in a coordinate box whose sides are less than half the torus side. Suppose that Y has at most $s+1$ sites in each direction, where $s \ge 0$. Let W be G-valued on all bonds with endpoints in Y, and suppose, for some $\varpi \ge 0$, that $|\partial W(p) - \mathbf{1}| \le \varpi$ whenever all four corners of p lie in Y. For $y \in \mathsf{Y}$ define $h(x) = W(\Gamma_{y,x})$, using the comb (62). Then $h(y) = \mathbf{1}$, and for $b = \langle x, x + e_\nu \rangle$ with endpoints in Y,

$$|W^h(b) - \mathbf{1}| \le \varpi \sum_{\nu' < \nu} |x_{\nu'} - y_{\nu'}| \le \min\{2s, |x - y|_1\}\varpi. \tag{119}$$

If $2s\varpi < 2$, then $W^h = e^{ia^0}$ for a real bond function a^0 with $\sup |a^0| \le \pi s \varpi$. If Γ_1, Γ_2 are paths in Y with the same initial and final points and with ℓ_1, ℓ_2 bonds, respectively, then

$$|W(\Gamma_1) - W(\Gamma_2)| \le 2s\varpi(\ell_1 + \ell_2).$$

■

Proof. Use chart coordinates in the containing box. No bond with endpoints in Y wraps in these coordinates: its two lifts differ by the indicated unit step, since a nonzero period would make their separation at least half the torus side. In particular, the combs stay in Y. The comb from y to itself is empty, and

$$W^h(b) = W(\Gamma_{y,x} \cup b) W(\Gamma_{y,x+e_\nu})^{-1}.$$

The comb adjusts coordinates in the order $3, 2, 1$. Move the final step e_ν backwards through its $\sum_{\nu' < \nu} |x_{\nu'} - y_{\nu'}|$ predecessors in directions smaller than ν. It then prolongs the steps in direction ν, or cancels the last step $-e_\nu$. The resulting transport is that of $\Gamma_{y,x+e_\nu}$. Every intermediate site and every exchanged plaquette has coordinates between those of y and those of x or $x + e_\nu$; hence they remain in Y. Equation (118) proves the first bound. At most two coordinates occur in the sum, each of size at most s. Their total is also at most $|x - y|_1$: the chart differences are the least coordinate lifts because they are smaller than half a period.

If $2s\varpi < 2$, every positive bond value of W^h avoids $-\mathbf{1}$. Define $a^0(b) = i^{-1} \log W^h(b)$ by the intrinsic logarithm, and extend by oddness to reversed bonds. Then

$$|a^0(b)| = \theta(W^h(b)) \le \frac{\pi}{2} |W^h(b) - \mathbf{1}| \le \pi s \varpi.$$

For the last assertion, the common endpoints give the same unitary factors in the gauge transforms of both path transports. Thus

$$|W(\Gamma_1) - W(\Gamma_2)| = |W^h(\Gamma_1) - W^h(\Gamma_2)|.$$

Every factor in either transformed path is within $2s\varpi$ of $\mathbf{1}$, by (119), also for reversed bonds. Telescoping each unitary product gives $|W^h(\Gamma_i) - \mathbf{1}| \le 2s\varpi\ell_i$. The triangle inequality proves the claim. □

Lemma 14.5 (Spread about the contour mean)**.** Let $0 \le n < K$, $y \in \mathbb{T}^{(n+1)}$, and let V be G-valued on the bonds inside $B(y)$. Put

$$\varpi_y = \max_{p \subset B(y)} |\partial V(p) - \mathbf{1}|, \qquad \mathsf{i} = \frac{3}{4}(L-1)^2, \tag{120}$$

where $p \subset B(y)$ means that all four corners lie in $B(y)$. For $x \in B(y)$, the family $(V(\Gamma))_{\Gamma \in \mathcal{G}(y,x)}$ has diameter at most $\mathsf{i}\varpi_y$. If this family has exactly one mean $V(y,x)$, then for every $\Gamma \in \mathcal{G}(y,x)$,

$$\begin{aligned} |V(\Gamma)V(y,x)^{-1} - \mathbf{1}| &\le \theta(V(\Gamma)V(y,x)^{-1}) \\ &\le \frac{2\pi}{3\sqrt{3}}\,\mathsf{i}\varpi_y = \frac{\pi}{2\sqrt{3}}(L-1)^2\varpi_y. \end{aligned} \tag{121}$$

■

Proof. Write $x - y = \sum_\nu n_\nu e_\nu$, with $|n_\nu| \le (L-1)/2$. A contour takes the steps in each direction together; two contours differ only in the order of these groups. Changing one order to another exchanges each pair of steps in differently ordered directions once. The number of nonparallel exchanges is therefore at most

$$\sum_{\nu<\nu'} |n_\nu|\,|n_{\nu'}| \le 3\left(\frac{L-1}{2}\right)^2 = \mathsf{i}.$$

All intermediate coordinates lie between those of y and x, so the exchanges stay inside $B(y)$. Equation (118) gives the asserted diameter.

We need the following estimate for the inverse exponential: if $Y, Y' \in \mathfrak{g}$ and $|Y|, |Y'| \le \vartheta < \pi$, with $\vartheta > 0$, then

$$|Y - Y'| \le \frac{\vartheta}{\sin\vartheta}|e^{iY} - e^{iY'}|. \tag{122}$$

Put $a = |Y|$, $a' = |Y'|$, and $\langle Y, Y'\rangle = qaa'$, $-1 \le q \le 1$; if $aa' = 0$, take any q in this interval. The exponential formula in the proof of lemma 5.2 gives

$$\begin{aligned} f(q) &:= |e^{iY} - e^{iY'}|^2 = 2 - 2\cos a\cos a' - 2q\sin a\sin a', \\ g(q) &:= |Y - Y'|^2 = a^2 + a'^2 - 2qaa'. \end{aligned}$$

Both functions are affine in q, and

$$\begin{aligned} f(1) &= 4\sin^2\tfrac{a-a'}{2}, & g(1) &= (a-a')^2, \\ f(-1) &= 4\sin^2\tfrac{a+a'}{2}, & g(-1) &= (a+a')^2. \end{aligned}$$

The function $\sin u/u$ decreases on $(0,\pi)$: the numerator $u\cos u - \sin u$ of its derivative vanishes at zero and has derivative $-u\sin u < 0$. Since $|a-a'|/2 \le (a+a')/2 \le \vartheta$, we have $f(\pm1) \ge (\sin\vartheta/\vartheta)^2 g(\pm1)$. Affineness gives the same inequality for all $q \in [-1,1]$, proving (122). The same monotonicity gives $\vartheta/\sin\vartheta < 2\pi/(3\sqrt{3})$ when $0 < \vartheta < \pi/3$.

Let $M = V(y,x)$ and index the contour family by σ. The defining equation of its mean says that

$$Z_\sigma = i^{-1}\log(V(\Gamma_\sigma)M^{-1}) \in \mathfrak{g}, \qquad |Z_\sigma| < \pi/3, \qquad \sum_\sigma Z_\sigma = 0.$$

Take $\vartheta = \max_\sigma |Z_\sigma|$; the case $\vartheta = 0$ is immediate. Unitary invariance and (122) imply

$$|Z_\sigma - Z_{\sigma'}| \le \frac{2\pi}{3\sqrt{3}}|V(\Gamma_\sigma) - V(\Gamma_{\sigma'})| \le \frac{2\pi}{3\sqrt{3}}\,\mathsf{i}\varpi_y.$$

If the family has N members, then $Z_\sigma = N^{-1}\sum_{\sigma'}(Z_\sigma - Z_{\sigma'})$. This proves the angle bound in (121); the norm bound follows from $|V - \mathbf{1}| \le \theta(V)$. □

The mean contributes two endpoint errors when a coarse bond is compared with straight transport. The corner scheme uses actual path transports at these endpoints. The following constants cover both choices:

$$\kappa_1^{\mathrm{SO}} = \frac{\pi}{2\sqrt{3}}(L-1)^2, \qquad \kappa_2^{\mathrm{SO}} = 2L(L-1) + 2\kappa_1^{\mathrm{SO}}, \qquad \kappa_3 = \frac{\pi}{2}\kappa_2^{\mathrm{SO}}.$$

Lemma 14.6 (One-step transport estimate)**.** Let $0 \le n < K$, let c be a positive bond of $\mathbb{T}^{(n+1)}$, and let W be G-valued on P_c. Suppose

$$\varpi := \varpi_c(W) < \min\{\omega_M, (\kappa_2^{\mathrm{SO}})^{-1}\}.$$

For either scheme $\star$, its contour variables and $\mathcal{M}_\star(W)(c)$ are defined and lie in G. Moreover, for $y = c_-$ or c_+, $x \in B(y)$, and $\Gamma \in \mathcal{G}(y,x)$,

$$\theta(W(\Gamma)W_{\mathsf{m}}(y,x)^{-1}) \le \kappa_1^{\mathrm{SO}}\varpi.$$

For $x \in B(c_-)$ we have $|W_x^\star - \mathbf{1}| \le \kappa_2^{\mathrm{SO}}\varpi$, and

$$\mathcal{M}_\star(W)(c) = e^{iY}W_\star[c], \qquad Y \in \mathfrak{g}, \qquad |Y| \le \kappa_3\varpi.$$

■

Proof. The centered contour variables are the unique means by lemma 14.2, since $\varpi < \omega_M$. Their angle bound follows from lemma 14.5 on each endpoint block. The corner contours are transports along combs contained in those blocks.

Let μ be the direction of c and put $x' = x + Le_\mu$. For the centered scheme choose any $\Gamma \in \mathcal{G}(c_-, x)$; for the corner scheme take $\Gamma = \Gamma_{o_{\mathsf{c}}(c_-),x}$. In either case $\Gamma' = \Gamma + Le_\mu$ is the corresponding contour from $o_\star(c_+)$ to x'. Compare

$$\Gamma \cup [x, x'] \quad \text{and} \quad [o_\star(c_-), o_\star(c_+)] \cup \Gamma'.$$

Move the L final steps e_μ in the first path to its beginning. Each passes at most $2(L-1)$ steps in directions different from μ. Thus at most $2L(L-1)$ exchanges are nonparallel. All intermediate sites and exchanged plaquette corners are of the form $z + je_\mu$, where z lies on Γ and $0 \le j \le L$. They lie in D_c, so (118) gives

$$\bigl|W(\Gamma)W([x,x']) - W_\star[c]W(\Gamma')\bigr| \le 2L(L-1)\varpi. \tag{123}$$

For the corner scheme the two contour transports here are exactly its contour variables. Multiplication by their unitary inverses gives $|W_x^{\mathsf{c}} - \mathbf{1}| \le 2L(L-1)\varpi$. For the centered scheme, insert these two path transports in

$$|W_x^{\mathsf{m}} - \mathbf{1}| = |W_{\mathsf{m}}(c_-, x)W([x,x']) - W_{\mathsf{m}}[c]W_{\mathsf{m}}(c_+, x')|.$$

Each replacement costs at most $\kappa_1^{\mathrm{SO}}\varpi$ by the angle bound. Together with (123), this proves the bound with κ_2^{SO}.

In particular, $|W_x^\star - \mathbf{1}| < 1$ for every $x \in B(c_-)$. All logarithms defining the average therefore exist, including those for the corner scheme. They belong to $i\mathfrak{g}$, and

$$Y = L^{-3} \sum_{x \in B(c_-)} i^{-1} \log W_x^\star$$

satisfies

$$|Y| \le L^{-3} \sum_{x \in B(c_-)} \frac{\pi}{2} |W_x^\star - \mathbf{1}| \le \kappa_3\varpi.$$

Formula (115) gives the asserted factorization and its G-valuedness. □

14.3. **The logarithm near the trivial configuration.** The comb gauge in lemma 14.4 turns small plaquette variables on a box into small bond variables. In that gauge, holomorphy gives a quadratic remainder after linearizing the average. We first identify this linearization; its contour terms will cancel around a coarse plaquette.

For a bond function a and $x \in B(y)$, write

$$\bar{a}_{\mathsf{m}}(y,x) = \frac{1}{|\mathcal{G}(y,x)|} \sum_{\Gamma \in \mathcal{G}(y,x)} a(\Gamma), \qquad \bar{a}_{\mathsf{c}}(y,x) = a(\Gamma_{o_{\mathsf{c}}(y),x}),$$

where $a(\Gamma)$ is the oriented sum of a along the path.

Lemma 14.7 (Logarithm of a small average)**.** There is $r_* \in (0, 1/16]$, depending only on L, with the following property. Let $0 \le n < K$, let c be a positive bond of $\mathbb{T}^{(n+1)}$, and let $\star \in \{\mathsf{m}, \mathsf{c}\}$. On the space of complex bond functions a on P_c with $\sup|a| < r_*$, the series logarithm

$$\Phi_\star(a)(c) = i^{-1} \log \mathcal{M}_\star(e^{ia})(c)$$

is defined and holomorphic, vanishes at $a = 0$, and has norm at most 1. In the chart $x \mapsto x - c_-$, with fine bonds of length one, this map depends only on L, the scheme, and the direction of c. Its derivative at zero is

$$(\mathbf{Q}^{\mathrm{lin}}_\star a)(c) = L^{-3} \sum_{x \in B(c_-)} [\bar{a}_\star(c_-, x) + a([x, x']) - \bar{a}_\star(c_+, x')]. \tag{124}$$

For the centered scheme this is (42). For either scheme, $y \in \{c_-, c_+\}$, and $x \in B(y)$, the contour variable has derivative

$$\frac{d}{dt}(e^{ita})_\star(y, x)\Big|_{t=0} = i\bar{a}_\star(y, x).$$

■

Proof. In fine-lattice units the chart $x \mapsto x - c_-$ identifies D_c with the double block of $\langle 0, Le_\mu \rangle$ in $\mathbb{Z}^3$, where μ is the direction of c. Blocks, contour families, combs and straight paths become their fixed counterparts in this chart. Thus there are six families of finitely many functions to consider: two schemes and three directions. None depends on the level or on the location of c.

By lemma 14.2, the average is holomorphic in the matrix entries on an open set containing the trivial configuration. The map $a \mapsto e^{ia}$ is holomorphic and takes values in G^c. At $a = 0$, all contour variables and the average equal $\mathbf{1}$. The series logarithm of the average is therefore defined and holomorphic in a neighborhood of zero, where it vanishes at zero. Continuity gives a radius on which it has norm at most 1. A common radius for the six families, chosen at most $1/16$, gives the assertion about r_* and its dependence on L.

For the centered scheme, (44) and its proof give both derivatives. For the corner scheme, differentiation of the path product gives $\frac{d}{dt}(e^{ita})_\mathsf{c}(y, x)\Big|_{t=0} = i\bar{a}_\mathsf{c}(y, x)$. Let $a_\mathsf{c}[c]$ be the sum of a along the straight path from $o_\mathsf{c}(c_-)$ to $o_\mathsf{c}(c_+)$. The derivative at zero of the loop variable in (115) is

$$i[\bar{a}_\mathsf{c}(c_-, x) + a([x, x']) - \bar{a}_\mathsf{c}(c_+, x') - a_\mathsf{c}[c]].$$

The differential of log at $\mathbf{1}$ is the identity. Averaging this expression therefore gives $i(\mathbf{Q}^{\mathrm{lin}}_\mathsf{c} a)(c) - ia_\mathsf{c}[c]$ as the derivative of the exponent. Its last term cancels the derivative $ia_\mathsf{c}[c]$ of the final path factor in (115). Hence $\frac{d}{dt}\mathcal{M}_\star(e^{ita})(c)\Big|_{t=0} = i(\mathbf{Q}^{\mathrm{lin}}_\star a)(c)$ for both schemes. Applying the logarithm proves $\Phi'_\star(0)a = \mathbf{Q}^{\mathrm{lin}}_\star a$. □

When this lemma is applied after the real gauge transformation $W^h = e^{ia}$, the coarse endpoint frames are $h(o_\star(c_-))$ and $h(o_\star(c_+))$, as in (117). Consequently the plaquette of the coarse average changes by unitary conjugation. This permits the curvature estimate below to be proved using the small potential a on the four blocks surrounding the coarse plaquette.

14.4. **The one-step curvature bound.** Let $0 \le n < K$ and use level-n scaling. For a coarse plaquette $p' = p_{\mu\nu}(y_1)$ and $x \in B(y_1)$, let $S_x(p')$ consist of the fine plaquettes

$$p_{\mu\nu}(x + ie_\mu + je_\nu), \qquad 0 \le i, j < L.$$

Define the nonnegative kernel

$$\mathsf{W}(p', p) = L^{-3}\#\{x \in B(y_1) : p \in S_x(p')\}, \qquad (\mathsf{W}f)(p') = \sum_p \mathsf{W}(p', p) f(p). \tag{125}$$

Lemma 14.8. Every $S_x(p')$ has L^2 plaquettes, whose corners lie in the union of the four blocks of the corners of p'. Moreover,

$$(\mathsf{W}f)(p') = L^{-3} \sum_{x \in B(y_1)} \sum_{p \in S_x(p')} f(p), \qquad \sum_p \mathsf{W}(p',p) = L^2, \qquad \sum_{p'} \mathsf{W}(p',p) = L^{-1}.$$

Consequently, for $f \geq 0$,

$$\sup \mathsf{W}f \leq L^2 \sup f, \qquad \sum_{p'} (\mathsf{W}f)(p') = L^{-1} \sum_p f(p), \qquad \sum_{p'} (\mathsf{W}f)(p')^2 \leq L \sum_p f(p)^2.$$

■

Proof. The L^2 base points $x + ie_\mu + je_\nu$, $0 \leq i,j < L$, are distinct, since the fine torus has more than L sites in each direction. The corners of their plaquettes have $0 \leq i,j \leq L$. Relative to y_1, their coordinates therefore belong to $[-\mathsf{h}, \mathsf{h}+L]$ in the two plaquette directions and to $[-\mathsf{h}, \mathsf{h}]$ in the third. These are the coordinate intervals of the union of the four blocks. The formula for $\mathsf{W}f$ follows by exchanging the count over x with the sum over p. Taking $f = 1$ proves the row sum.

Fix $p = p_{\mu\nu}(z)$. Only coarse plaquettes with the same two directions contribute to its column. The condition $p \in S_x(p_{\mu\nu}(y_1))$ holds precisely when

$$x = z - ie_\mu - je_\nu \in B(y_1), \qquad 0 \leq i,j < L.$$

These are L^2 distinct sites, each in exactly one coarse block. Summing the count in (125) over y_1 therefore gives $L^{-3}L^2 = L^{-1}$.

The first two consequences follow by summation, since $\mathsf{W} \geq 0$. For the last, Cauchy–Schwarz and the row sum give

$$(\mathsf{W}f)(p')^2 \leq L^2 \sum_p \mathsf{W}(p',p) f(p)^2.$$

Summing over p' and using the column sum proves the factor $L^2 L^{-1} = L$. □

With r_* from lemma 14.7 and the constants of lemma 14.6, fix

$$\varpi_L = \min\left\{ \tfrac{1}{2}\omega_M, \tfrac{1}{2}(\kappa_2^{\mathrm{SO}})^{-1}, \frac{r_*}{4\pi L} \right\},$$
$$\varpi'_L = \min\left\{ \varpi_L, \frac{r_*}{8\pi L} \right\}, \qquad C_0^{\mathrm{SO}} = (2\pi L)^2 (35L^2 + 40 r_*^{-2}).$$

These are positive constants depending only on L, with $C_0^{\mathrm{SO}} \geq 1$.

Lemma 14.9 (One-step curvature estimate)**.** Let $p' = p_{\mu\nu}(y_1)$ be a plaquette of $\mathbb{T}^{(n+1)}$, with corners y_1, y_2, y_3, y_4 in cyclic order, and put $\mathsf{Y} = \bigcup_{i=1}^4 B(y_i)$. Let W be a real configuration on the fine bonds whose endpoints belong to Y, and suppose

$$\varpi = \max_{p \text{ with corners in } \mathsf{Y}} |\partial W(p) - \mathbf{1}| \leq \varpi'_L.$$

For either scheme the average is defined at the four bonds of p', and

$$|\partial \mathcal{M}_\star(W)(p') - \mathbf{1}| \leq \sum_p \mathsf{W}(p',p) |\partial W(p) - \mathbf{1}| + C_0^{\mathrm{SO}} \varpi^2. \tag{126}$$

Only plaquettes with corners in Y contribute to the sum. In particular,

$$|\partial \mathcal{M}_\star(W)(p') - \mathbf{1}| \leq L^2 \varpi + C_0^{\mathrm{SO}} \varpi^2. \tag{127}$$

■

Proof. The union Y is a coordinate box with $2L$ sites in directions μ,ν and L in the third direction. Its sides are less than half the fine torus side, since $2L < L^{m+K-n}$. Apply lemma 14.4 with $s = 2L-1$. Its logarithm condition holds because

$$2(2L-1)\varpi \le 4L\varpi_L \le r_*/\pi < 2.$$

We obtain a real gauge transformation u on Y such that $W^u = e^{ia}$, with a real and

$$\sup|a| \le \pi(2L-1)\varpi \le 2\pi L\varpi \le r_*/4.$$

The double block of each boundary bond of p' lies in Y. Its curvature bound is at most $\varpi_L < \min\{\omega_M, (\kappa_2^{\mathrm{SO}})^{-1}\}$, so lemma 14.6 defines the four coarse variables. By (117), gauge transformation conjugates their plaquette variable by $u(o_\star(y_1))$. It likewise conjugates each fine plaquette variable. Unitary conjugation preserves all norms in (126), so we may replace W by $W^u = e^{ia}$.

Write $s_a = \sup|a|$, and let

$$c^{(1)} = \langle y_1, y_2\rangle, \quad c^{(2)} = \langle y_2, y_3\rangle, \quad c^{(3)} = \langle y_4, y_3\rangle, \quad c^{(4)} = \langle y_1, y_4\rangle.$$

Put $\phi(c) = \Phi_\star(a)(c)$ at these four positive bonds. The average is real, and its series logarithm exists by lemma 14.7, so $\phi(c) \in \mathfrak{g}$. That lemma bounds $\Phi_\star$ by one on the complex ball of radius r_* and gives $\Phi_\star(0) = 0$. The one-variable Schwarz bound on each complex line, followed by the supremum over norm-one linear functionals, gives $|\Phi_\star(a)(c)| \le s_a/r_*$. The Cauchy coefficient formula on the same lines bounds the terms of degree q by $(s_a/r_*)^q$. Since the derivative is (124), summing from $q = 2$ gives

$$|\phi(c)| \le s_a/r_*, \qquad |\phi(c) - (\mathbf{Q}_\star^{\mathrm{lin}} a)(c)| \le \frac{s_a^2}{r_*(r_* - s_a)} \le 2s_a^2 r_*^{-2}.$$

We use the elementary exponential-product estimate

$$|e^{X_1}\cdots e^{X_q} - e^{X_1+\cdots+X_q}| \le 2\sigma^2, \qquad \sigma = \sum_j |X_j| \le 1, \quad q \le 4.$$

Indeed, multiplying the exponential series bounds the remainder after its constant and linear terms by $e^\sigma - 1 - \sigma \le \sigma^2$; the exponential of the sum has the same bound. For the coarse plaquette the four exponents are $i\phi(c^{(1)})$, $i\phi(c^{(2)})$, $-i\phi(c^{(3)})$ and $-i\phi(c^{(4)})$, whose norms sum to at most $4s_a/r_* \le 1$. Their sum is skew-Hermitian, and $|e^{iH} - \mathbf{1}| \le |H|$ for $H \in \mathfrak{g}$. It follows that

$$\begin{aligned}|\partial\mathcal{M}_\star(e^{ia})(p') - \mathbf{1}| &\le |\phi(c^{(1)}) + \phi(c^{(2)}) - \phi(c^{(3)}) - \phi(c^{(4)})| \\ &\quad + 32\max_j |\phi(c^{(j)})|^2.\end{aligned}$$

We next compute the same signed combination of the linear terms. For $x \in B(y_1)$ put

$$x^{(1)} = x, \quad x^{(2)} = x + Le_\mu, \quad x^{(3)} = x + Le_\mu + Le_\nu, \quad x^{(4)} = x + Le_\nu.$$

These translations are bijections from $B(y_1)$ onto the four corner blocks. Thus (124) gives

$$\begin{aligned}&(\mathbf{Q}_\star^{\mathrm{lin}} a)(c^{(1)}) + (\mathbf{Q}_\star^{\mathrm{lin}} a)(c^{(2)}) - (\mathbf{Q}_\star^{\mathrm{lin}} a)(c^{(3)}) - (\mathbf{Q}_\star^{\mathrm{lin}} a)(c^{(4)}) \\ &= L^{-3}\sum_{x\in B(y_1)} \Big\{ \bar a_\star(y_1, x) + a([x, x^{(2)}]) - \bar a_\star(y_2, x^{(2)}) \\ &\qquad + \bar a_\star(y_2, x^{(2)}) + a([x^{(2)}, x^{(3)}]) - \bar a_\star(y_3, x^{(3)}) \\ &\qquad - \bar a_\star(y_4, x^{(4)}) - a([x^{(4)}, x^{(3)}]) + \bar a_\star(y_3, x^{(3)}) \\ &\qquad - \bar a_\star(y_1, x) - a([x, x^{(4)}]) + \bar a_\star(y_4, x^{(4)}) \Big\}.\end{aligned}$$

At each corner the same contour sum occurs once with each sign, for either choice of scheme. The remaining expression is

$$L^{-3}\sum_{x\in B(y_1)} a(\partial S_x), \qquad a(\partial S_x)=a([x,x^{(2)}])+a([x^{(2)},x^{(3)}])-a([x^{(4)},x^{(3)}])-a([x,x^{(4)}]).$$

Every interior bond of S_x occurs once with each sign in the sum of its plaquette boundaries. Hence

$$a(\partial S_x)=\sum_{p\in S_x} a(\partial p), \qquad a(\partial p)=a(b_1)+a(b_2)-a(b_3)-a(b_4),$$

with the four positive bonds ordered as in (66).

At a fine plaquette, the exponential-product estimate gives

$$|e^{ia(\partial p)}-\partial e^{ia}(p)|\le 32 s_a^2,$$

since the four norms sum to at most $4s_a\le 1$. For $H\in\mathfrak{g}$ with $|H|\le\pi$, the spectral formula and $\sin u\ge u-u^3/6$ give

$$|H|\le |e^{iH}-\mathbf{1}|+\tfrac{1}{24}|H|^3.$$

Applying this with $H=a(\partial p)$, whose norm is at most $4s_a\le 1/16$, yields

$$\begin{aligned}|a(\partial p)| &\le |\partial W(p)-\mathbf{1}|+32s_a^2+\tfrac{64}{24}s_a^3\\ &\le |\partial W(p)-\mathbf{1}|+35 s_a^2.\end{aligned}$$

All plaquettes of S_x lie in Y by lemma 14.8, so this estimate applies to them. Combining the four Taylor errors, the coarse product error, and the square identity gives

$$\begin{aligned}|\partial\mathcal{M}_\star(W)(p')-\mathbf{1}| &\le L^{-3}\sum_{x\in B(y_1)}\sum_{p\in S_x}|a(\partial p)|+40 s_a^2 r_*^{-2}\\ &\le \sum_p \mathsf{W}(p',p)|\partial W(p)-\mathbf{1}|+(35L^2+40r_*^{-2})s_a^2.\end{aligned}$$

Since $s_a\le 2\pi L\varpi$, the last term is at most $C_0^{\mathrm{SO}}\varpi^2$, proving (126). Every nonzero kernel entry uses a plaquette in Y, where the fine curvature is at most ϖ. The row sum L^2 therefore gives (127). □

14.5. **Iteration and local estimates.** With the constants of lemmas 14.6 and 14.9, set

$$c_2^{\mathrm{SO}}=\min\{\tfrac12\varpi_L', L^2/(4C_0^{\mathrm{SO}})\}, \qquad \kappa_4=4\kappa_3 L^{-2}.$$

The quadratic curvature error contributes a factor proportional to $L^{-2(k-n)}$ at step n. Its sum is bounded independently of the number of steps. The transport error obeys a separate recursion, since each coarse straight path contains L fine bonds.

Proof of proposition 14.1. Write $W_n=\mathcal{M}_\star^n U$ whenever this iterate is defined, and set

$$\varpi_n=\varpi(W_n)=\alpha_0(L^n\eta)^2 p_n.$$

The hypothesis gives $p_0\le 1$. Suppose that $n<k$ and $p_n\le 2$. Then

$$\varpi_n\le 2\alpha_0(L^n\eta)^2\le 2c_2^{\mathrm{SO}}\le\varpi_L'.$$

In particular, $\varpi_c(W_n)\le\varpi_n\le\varpi_L<\min\{\omega_M,(\kappa_2^{\mathrm{SO}})^{-1}\}$ at every positive bond of level $n+1$. Thus lemma 14.6 defines the next real iterate. Apply lemma 14.9 to the four blocks at each level-$(n+1)$ plaquette. It gives

$$\varpi_{n+1}\le L^2\varpi_n(1+C_0^{\mathrm{SO}}L^{-2}\varpi_n),$$

and consequently

$$p_{n+1}\le p_n\bigl(1+2C_0^{\mathrm{SO}}L^{-2}\alpha_0 L^{-2(k-n)}\bigr).$$

If $p_j \le 2$ for $0 \le j \le n$, then $\sum_{\ell \ge 1} L^{-2\ell} \le 2L^{-2}$ yields

$$\begin{aligned} p_{n+1} &\le \prod_{j=0}^{n} (1 + 2C_0^{\mathrm{SO}} L^{-2} \alpha_0 L^{-2(k-j)}) \\ &\le \exp(4C_0^{\mathrm{SO}} L^{-4} \alpha_0) \le e^{1/L^2} < 2. \end{aligned}$$

Here $\alpha_0 \le L^2/(4C_0^{\mathrm{SO}})$ bounds the exponent by $1/L^2$. Induction proves existence of all iterates and the curvature bound in (116).

For the transport bound, let

$$\mathsf{d}_n = \max_c |W_n(c) - U[\![c]\!]_\star|.$$

We have $\mathsf{d}_0 = 0$. Reversing a bond replaces both matrices by their inverses; for unitary X, Y, $|X^{-1} - Y^{-1}| = |X - Y|$. It suffices, therefore, to estimate positive bonds. For a positive level-$(n+1)$ bond c, lemma 14.6 gives

$$W_{n+1}(c) = e^{iY} (W_n)_\star[c], \qquad |e^{iY} - \mathbf{1}| \le |Y| \le \kappa_3 \varpi_n.$$

The straight transport $(W_n)_\star[c]$ has L factors $W_n(c'')$, each within d_n of $U[\![c'']\!]_\star$. The corresponding bare paths join to the straight path defining $U[\![c]\!]_\star$: the origins satisfy $o_\star^{(n+1)} = o_\star^{(n)} \circ o_\star$, and $o_\star^{(n)}$ is the identity in the centered scheme and a fixed translation in the corner scheme. Telescoping the two products of unitary matrices gives

$$|W_{n+1}(c) - U[\![c]\!]_\star| \le 2\kappa_3 \alpha_0 (L^n \eta)^2 + L \mathsf{d}_n.$$

Hence, for $n \ge 1$,

$$\begin{aligned} \mathsf{d}_n &\le 2\kappa_3 \alpha_0 \eta^2 \sum_{j<n} L^{n-1-j} L^{2j} \\ &\le 2\kappa_3 \alpha_0 \eta^2 \frac{L^{2n-1}}{L-1} \\ &\le 4\kappa_3 L^{-2} \alpha_0 (L^n \eta)^2 = \kappa_4 \alpha_0 (L^n \eta)^2. \end{aligned}$$

The same bound holds at $n = 0$ because its left side vanishes.

For $\star = \mathsf{m}$, the curvature bound gives $\varpi(W_n) \le \varpi_L' < \omega_M$ at every step. The branch agreement in lemma 14.2, applied successively, therefore gives $\mathcal{M}_{\mathsf{m}}^n U = \mathcal{M}^n U$ and puts each bond in the branch (37). Every U in the open set stated in proposition 14.1 satisfies the hypotheses with $\alpha_0 = c_2^{\mathrm{SO}}$. All intermediate configurations there remain in the real-analytic domains of lemma 14.2. Thus $\mathcal{M}^k$ is real-analytic on that set as a composition of real-analytic maps. The set is open because it is defined by finitely many strict inequalities between continuous functions.

We now prove the local refinements. All iterates have already been defined by the global hypothesis. The local coefficients in the statement satisfy $0 \le \alpha(\mathfrak{e}) \le \alpha_0$. Set

$$\pi_0 = 1, \qquad \pi_{n+1} = \pi_n \big(1 + 2C_0^{\mathrm{SO}} L^{-2} \alpha_0 L^{-2(k-n)}\big).$$

The preceding product estimate gives $\pi_n < 2$ for $0 \le n \le k$. We claim that

$$|\partial W_n(p') - \mathbf{1}| \le \alpha(p') (L^n \eta)^2 \pi_n$$

at every level-n plaquette p'. At $n = 0$ this is the definition of $\alpha(p')$. Suppose it holds at level n, and let p' be a level-$(n+1)$ plaquette with corners y_i. If a level-n plaquette p has all its corners in $\bigcup_i B(y_i)$, block nesting gives

$$\mathsf{Z}(p) \subset \bigcup_i B^{n+1}(y_i) = \mathsf{Z}(p'), \qquad \alpha(p) \le \alpha(p').$$

The local curvature bound ϖ used in lemma 14.9 therefore satisfies

$$\varpi \le \alpha(p') (L^n \eta)^2 \pi_n \le 2\alpha_0 L^{-2(k-n)} \le \varpi_L'.$$

By (127),

$$\begin{aligned}|\partial W_{n+1}(p') - \mathbf{1}| &\le L^2\varpi(1 + C_0^{\mathrm{SO}}L^{-2}\varpi)\\ &\le \alpha(p')(L^{n+1}\eta)^2\pi_n(1 + 2C_0^{\mathrm{SO}}L^{-2}\alpha_0 L^{-2(k-n)})\\ &= \alpha(p')(L^{n+1}\eta)^2\pi_{n+1}.\end{aligned}$$

This proves the local curvature claim, including the case $\alpha(p') = 0$.

For bonds, put

$$\delta_0 = 0, \qquad \delta_{n+1} = L\delta_n + 2\kappa_3 L^{2n}.$$

The geometric sum used above gives $\delta_n \le 4\kappa_3 L^{-2}L^{2n}$. We claim that

$$|W_n(c) - U[\![c]\!]_\star| \le \delta_n\alpha(c)\eta^2$$

at every positive level-n bond. Both sides vanish for $n = 0$. Let c be a positive level-$(n+1)$ bond. Every level-n plaquette p with corners in D_c satisfies $\mathsf{Z}(p) \subset \mathsf{Z}(c)$, so the local curvature bound just proved gives

$$\varpi_c(W_n) \le 2\alpha(c)(L^n\eta)^2.$$

The L bonds c'' of the straight base path have their endpoints in D_c by lemma 14.2. Hence $\mathsf{Z}(c'') \subset \mathsf{Z}(c)$ and $\alpha(c'') \le \alpha(c)$. The induction hypothesis bounds each fine-bond transport error by $\delta_n\alpha(c)\eta^2$. Using the local estimate $|Y| \le \kappa_3\varpi_c(W_n)$ in lemma 14.6 and telescoping the same products as before, we obtain

$$\begin{aligned}|W_{n+1}(c) - U[\![c]\!]_\star| &\le 2\kappa_3\alpha(c)(L^n\eta)^2 + L\delta_n\alpha(c)\eta^2\\ &= \delta_{n+1}\alpha(c)\eta^2.\end{aligned}$$

Reversal preserves the left side and $\alpha(\bar c) = \alpha(c)$. The bounds $\pi_n < 2$ and $\delta_n\eta^2 \le \kappa_4(L^n\eta)^2$ give both local refinements. □

14.6. **Corner averages in a small gauge.** The local inverse construction uses corner averages. We compare their linear parts with the straight averages at the identity, keeping the entry bound needed for the weighted transpose. The endpoint correction and linearization follow Bałaban's averaging construction [Bał85a, pp. 34–39, (110), (121)–(127)].

We use the one-step blocks and corner transports defined above. For a complex Lie-algebra-valued function a on positive fine bonds, put $\widetilde W(b) = e^{ia(b)}W(b)$ and define its value on reversed bonds by inversion. Set

$$v_{\mathsf{c}}(W,a)(y) = \exp\left(L^{-3}\sum_{x\in B(y)}\log(\widetilde W_{\mathsf{c}}(y,x)W_{\mathsf{c}}(y,x)^{-1})\right),$$

$$\mathcal{Q}^1_{\mathsf{c}}(W,a)(c) = i^{-1}\log[v_{\mathsf{c}}(W,a)(c_-)^{-1}\mathcal{M}_{\mathsf{c}}(e^{ia}W)(c)v_{\mathsf{c}}(W,a)(c_+)\mathcal{M}_{\mathsf{c}}(W)(c)^{-1}].$$

All logarithms are series logarithms. The endpoint factors remove the infinitesimal comb averages from the bond average. Write $\mathbf{Q}_{\mathsf{c}}(W) = d_a\mathcal{Q}^1_{\mathsf{c}}(W,0)$. The following argument gives a uniform domain for this derivative at real configurations whose plaquette variables are sufficiently close to the identity.

Fix level-n units, $0 \le n \le K$, so $\eta = L^{-n}$, and put $\xi_i = L^i\eta$. For $1 \le i \le K$, let Y be one block $B^i(y)$ or the double block $\mathsf{Y}_c = B^i(c_-) \cup B^i(c_+)$ of a positive level-i bond. Assume that each coordinate interval of Y has at most half as many sites as the corresponding torus direction. Extend U by coordinatewise folding, as specified in the proof, and denote the extension by $\widetilde U$.

Let $W_q = \mathcal{M}^q_{\mathsf{c}}\widetilde U$. For $x \in B^i(y)$, let x_q be the unique level-q site with $x \in B^q(x_q)$, so $x_i = y$ and $x_0 = x$. Define

$$T^{\mathsf{c}}_U(y,x) = (W_{i-1})_{\mathsf{c}}(x_i,x_{i-1})\cdots(W_0)_{\mathsf{c}}(x_1,x_0).$$

On a double block, define the dimensionless bond kernel by

$$Q_i^{\mathsf{c}}[U] = \eta \mathbf{Q}_{\mathsf{c}}(W_{i-1}) \cdots \mathbf{Q}_{\mathsf{c}}(W_0),$$

restricted to the output c and to input bonds in Y_c. Equivalently, it is the derivative at zero of the iteration $a_0 = \eta A$, $a_{q+1} = \mathcal{Q}^1_{\mathsf{c}}(W_q, a_q)$. Only a neighborhood of zero is needed to take this derivative. All kernels use counting measure on positive bonds; row and column norms mean the supremum of the sums of entry norms. At level zero, set $T_U^{\mathsf{c}}(y,y) = \mathbf{1}$ and $Q_0^{\mathsf{c}}[U](c,b) = \eta\delta_{cb}$.

Proposition 14.10. There are positive constants $c_{\mathsf{c}}, \kappa_7^{\mathsf{c}}$ and κ_δ, depending only on L, with the following property. Let $1 \le i \le K$, $0 < \alpha \le c_{\mathsf{c}}$ and $\varepsilon > 0$. Suppose that $U = e^{i\eta A^\circ}$ on the positive bonds in Y, with A° real, and

$$\xi_i |A^\circ| \le \varepsilon, \qquad |\partial U(p) - \mathbf{1}| \le \alpha L^{-2i} \quad \text{for every plaquette in } \mathsf{Y}.$$

The local averages just defined exist and depend only on the data in Y. On a single block,

$$|T_U^{\mathsf{c}}(y,x) - \mathbf{1}| \le \kappa_7^{\mathsf{c}}\alpha + 3\varepsilon \qquad (x \in B^i(y)).$$

On a double block, $Q_i^{\mathsf{c}}[U](c,b) = 0$ unless both endpoints of b belong to Y_c, and

$$\sum_b |Q_i^{\mathsf{c}}[U](c,b) - Q_i^{\mathsf{c}}[\mathbf{1}](c,b)| \le \kappa_\delta(\alpha+\varepsilon)\xi_i,$$
$$|Q_i^{\mathsf{c}}[U](c,b) - Q_i^{\mathsf{c}}[\mathbf{1}](c,b)| \le \kappa_\delta(\alpha+\varepsilon)\eta^3\xi_i^{-2}.$$

The kernels also satisfy

$$\sum_b |Q_i^{\mathsf{c}}[U](c,b)| \le 2\xi_i, \qquad |Q_i^{\mathsf{c}}[U](c,b)| \le 2\eta^3\xi_i^{-2}.$$

At level zero both defects vanish. ∎

Proof. In bare lattice coordinates, write each coordinate interval as $\{s, \dots, s+N_0-1\} \subset \mathbb{Z}/N\mathbb{Z}$, where $2N_0 \le N$. The folding map in that coordinate is

$$s+t \longmapsto \begin{cases} s+t, & 0 \le t < N_0, \\ s+2N_0-1-t, & N_0 \le t < 2N_0, \\ s, & 2N_0 \le t < N. \end{cases}$$

Apply it in each coordinate and denote the resulting map by F. A collapsed bond receives the value $\mathbf{1}$; every other bond receives the value of U on its image, in the image orientation. Denote this extension by $\widetilde{U} = F^*U$. Its restriction to Y is U.

Each coordinate increment of F is 0, 1 or -1. Thus a folded plaquette collapses, giving holonomy $\mathbf{1}$, or maps onto a plaquette in Y with a possible change of orientation and base point. Its holonomy is then a conjugate of that plaquette holonomy or its inverse. The curvature bound therefore holds everywhere for $\widetilde{U}$. In level-i units the bare spacing is $\eta_i = L^{-i}$; proposition 14.1, with $\alpha \le c_2^{\mathrm{SO}}$, gives

$$\varpi_q := \varpi(W_q) \le 2\alpha L^{-2(i-q)}, \qquad \sum_{q<i} \varpi_q \le 4\alpha L^{-2}.$$

Nested blocks and the locality of one averaging step imply that all factors contributing to a value on Y use only its original data. Indeed, a contributing level-q bond has its order-q endpoint blocks inside Y; the same holds for every bond used in the next finer step. This also proves the support assertion for the derivative kernel.

We first estimate one derivative. Put

$$(\mathbf{S}_{\mathsf{c}}(W)a)(c) = L^{-3} \sum_{x \in B(c_-)} \sum_{b \in [x,x']} \mathrm{Ad}(W_{\mathsf{c}}(c_-,x)W([x,b_-]))a(b), \qquad x' = x + Le_\mu,$$

where μ is the direction of c. For real W, the transports are unitary. There are L^4 summands at a fixed output c, and L pairs (c, x) for a fixed input bond. Hence

$$\|\mathbf{S}_{\mathsf{c}}(W)\|_{\mathrm{row}} \le L, \qquad \|\mathbf{S}_{\mathsf{c}}(W)\|_{\mathrm{col}} \le L^{-2}.$$

At $W = \mathbf{1}$, differentiation of the two endpoint factors cancels the endpoint contour means in (124). Thus $\mathbf{Q}_{\mathsf{c}}(\mathbf{1}) = \mathbf{S}_{\mathsf{c}}(\mathbf{1}) = \mathbf{Q}^s$, where $\mathbf{Q}^s a = L^{-3} \sum_{x \in B(c_-)} a([x, x'])$.

For $W = e^{ia^0}$, use the finite double-block charts of lemma 14.7, now for the straightening expressions in both variables (a^0, a). At $(0, 0)$ every logarithm has argument $\mathbf{1}$. Holomorphy and continuity therefore give a common radius $r_{\mathsf{c}} \in (0, r_*]$, depending only on L, such that on $\sup|a^0|, \sup|a| < r_{\mathsf{c}}$ the logarithms have norm at most one and the adjoint transports in $\mathbf{S}_{\mathsf{c}}$ have norm at most two. Cauchy's formula in $a(b)$ bounds the kernel of $\mathbf{Q}_{\mathsf{c}}$ by r_{c}^{-1} there. The kernel of $\mathbf{S}_{\mathsf{c}}$ is bounded by $2L^{-2}$. Their difference is holomorphic in a^0 and vanishes at zero; the Schwarz bound on each complex line gives

$$|(\mathbf{Q}_{\mathsf{c}}(e^{ia^0}) - \mathbf{S}_{\mathsf{c}}(e^{ia^0}))(c, b)| \le (r_{\mathsf{c}}^{-1} + 2) r_{\mathsf{c}}^{-1} \sup|a^0|.$$

For real W with curvature at most ϖ, the comb gauge of lemma 14.4 on D_c gives $W^h = e^{ia^0}$ and $\sup|a^0| \le 2\pi L\varpi$, provided $\varpi \le r_{\mathsf{c}}/(4\pi L)$. The gauge transforms of both kernels are obtained by left composition with $\mathrm{Ad}(h(o_{\mathsf{c}}(c_-)))$ and right composition with $\mathrm{Ad}(h(b_-))^{-1}$. This follows directly from (117) in the definitions; the maps are isometries, so their norms are unchanged. The derivative is consequently defined at W and

$$\|\mathbf{Q}_{\mathsf{c}}(W) - \mathbf{S}_{\mathsf{c}}(W)\|_{\mathrm{row,col}} \le \kappa_9^{\mathsf{c}} \varpi, \qquad \kappa_9^{\mathsf{c}} = \kappa_9^{\mathsf{c}}(L).$$

Here a row has at most $6L^3$ entries; a fine bond belongs to at most six coarse double blocks. The same estimates hold after restriction to local boxes.

Choose c_{c} small enough that

$$c_{\mathsf{c}} \le c_2^{\mathrm{SO}}, \qquad 2c_{\mathsf{c}} \le r_{\mathsf{c}}/(4\pi L), \qquad 8\kappa_9^{\mathsf{c}} c_{\mathsf{c}} \le \log 2.$$

All one-step derivatives at W_q are now defined. For real a close to zero, the arguments of the series logarithms in the straightening formula are G-valued; their logarithms belong to $i\mathfrak{g}$ by lemma 5.1. Thus $\mathcal{Q}_{\mathsf{c}}^1(W_q, a)$ is $\mathfrak{g}$-valued. Its trace is holomorphic and vanishes on the real bond space; the power series in a real basis therefore vanishes also for complex a near zero. Each step consequently takes values in $\mathfrak{g}^c$ and vanishes at zero. Their finite composition is defined near zero, and the chain rule gives the stated product formula for Q_i^{c}. Products are ordered from $q = i - 1$ to $q = 0$. Expanding the product of $\mathbf{Q}_{\mathsf{c}}(W_q) = \mathbf{S}_{\mathsf{c}}(W_q) + E_q$ and using the preceding row and column bounds gives

$$\left\| \prod_{q<i} \mathbf{Q}_{\mathsf{c}}(W_q) - \prod_{q<i} \mathbf{S}_{\mathsf{c}}(W_q) \right\|_{\mathrm{row}} \le 8\kappa_9^{\mathsf{c}} L^{-3} \alpha L^i,$$

$$\left\| \prod_{q<i} \mathbf{Q}_{\mathsf{c}}(W_q) - \prod_{q<i} \mathbf{S}_{\mathsf{c}}(W_q) \right\|_{\mathrm{col}} \le 8\kappa_9^{\mathsf{c}} \alpha L^{-2i}.$$

For example, the row norm of the difference is bounded by $L^i[\exp(\kappa_9^{\mathsf{c}} L^{-1} \sum_q \varpi_q) - 1]$; the column estimate uses L^2 in place of L^{-1}. The chosen threshold permits $e^t - 1 \le 2t$. The products without subtraction are bounded by $2L^i$ and $2L^{-2i}$, respectively.

It remains to compare the leading transports with the identity. An entry of $\prod_{q<i} \mathbf{S}_{\mathsf{c}}(W_q)$ is a sum of $L^{-3i}\,\mathrm{Ad}(\tau)$ over nested choices of a starting site in a block and a bond on its straight path. There are L^{4i} such choices from one output. There are at most L^i choices ending at a fixed input bond: at each upward step its starting site is $b_- - se_\mu$, $0 \le s < L$, in the units of that step, and its parent block and parent bond are then determined.

Each factor of τ follows a corner comb and an initial straight segment at level q, of total length at most $4L$. Replace every coarse bond of that path by the bare straight path between its order-q corners. By (116), the change in its transport is bounded by $4L\kappa_4\alpha(L^q\eta_i)^2$. The resulting bare paths concatenate, stay in Y_c by block nesting, and have total length at most $\sum_{q<i} 4L^{q+1} \le 5L^i$. The sum of the transport errors is at most $8\kappa_4 L^{-1}\alpha$. The same construction for $T^{\mathsf{c}}_U(y,x)$ uses only the comb portions and stays in $B^i(y)$. For either construction, compare the resulting path with a bare path Γ in the same box, with the same endpoints and at most $16L^i$ bonds. The path estimate of lemma 14.4, with $s = 2L^i - 1$ and curvature αL^{-2i}, bounds the change by 84α. Thus there is $\kappa^{\mathsf{c}}_7 = \kappa^{\mathsf{c}}_7(L)$ such that

$$|\tau - U(\Gamma)| \le \kappa^{\mathsf{c}}_7\alpha, \qquad |T^{\mathsf{c}}_U(y,x) - U(\Gamma)| \le \kappa^{\mathsf{c}}_7\alpha.$$

For the scalar transport choose the bare comb from the lowest corner of $B^i(y)$ to x, of length at most $3L^i$. For the bond transport choose the comb from the lowest corner of Y_c to b_-, of length at most $4L^i$. Telescoping products and $|e^{iX} - \mathbf{1}| \le |X|$ for real X give $|U(\Gamma) - \mathbf{1}| \le 3\varepsilon$ and 4ε, respectively. This proves the scalar claim. For the leading bond kernel, the corresponding adjoint transport differs from the identity by at most $2\kappa^{\mathsf{c}}_7\alpha + 8\varepsilon$. For the difference between the leading bond kernel and its value at the identity, the two counts give a row bound $(2\kappa^{\mathsf{c}}_7\alpha + 8\varepsilon)L^i$ and an entry bound $(2\kappa^{\mathsf{c}}_7\alpha + 8\varepsilon)L^{-2i}$.

Add the product errors and multiply by η in the definition of Q^{c}_i. Since $\eta L^i = \xi_i$ and $\eta L^{-2i} = \eta^3\xi_i^{-2}$, these estimates give the asserted defect bounds for a constant κ_δ depending only on L. The bounds without subtraction follow from the product bounds above. The level-zero identities give the last assertion. □

14.7. **Solving the averaging constraint at one bond.** Fix a positive bond $c = \langle c_-, c_- + Le_\nu\rangle$ of $\mathbb{T}^{(k+1)}$, where $0 \le k < K$, and measure lengths in level-k units. Put $h = (L-1)/2$. The middle bond

$$b_0(c) = \langle c_- + he_\nu, c_- + (h+1)e_\nu\rangle$$

is the pivot bond. For a reversal-symmetric bond set S, write $\mathcal{V}(S) = G^{S^+}$, with inverse values on reversed bonds. Write P°_c for the bonds of the double block other than $b_0(c)$ and its reverse. For a configuration w on P°_c and $x \in G$, let $w \sqcup x$ denote its completion with value x at $b_0(c)$ and x^{-1} at the reverse. Define the partial map $\pi_c(w,v)$ to be the unique $x \in G$ satisfying $\mathcal{M}(w \sqcup x)(c) = v$, whenever exactly one such x exists. The average here is the global real map, with all three branches of section 4.

The local analytic inverse alone does not exclude other solutions on the other branches. The next proposition supplies this uniqueness and the measurability needed when the eliminated variables are recovered. Set

$$\mathsf{c}_{\mathrm{piv}} = \frac{5}{2}L^3, \qquad \omega_{\mathrm{p}} = \frac{\omega_M}{16L^4} = \frac{1}{3200L^6}. \tag{128}$$

Proposition 14.11 (The pivot variable)**.** Let $L \ge 13$ be odd. Suppose $\omega, \omega' \ge 0$, $\omega + \omega' \le \omega_{\mathrm{p}}$, and G_1 is a real configuration on P_c with $|\partial G_1(p) - \mathbf{1}| \le \omega$ on every plaquette of the double block. For every real configuration w on P°_c such that

$$|w(b)G_1(b)^{-1} - \mathbf{1}| \le \omega' \qquad (b \in \mathsf{P}^\circ_c),$$

there is exactly one $x \in G$ satisfying

$$\mathcal{M}(w \sqcup x)(c) = \mathcal{M}(G_1)(c). \tag{129}$$

It obeys

$$|xG_1(b_0)^{-1} - \mathbf{1}| \le \frac{5}{2}L^3\omega', \tag{130}$$

and $\varpi_c(w \sqcup x) < \omega_M$, so that the small-curvature branch applies to the solution. In particular, these conclusions hold for every W whose average agrees with that of G_1 and whose nonpivot values satisfy the displayed bound, without any initial assumption on $W(b_0)$ or on the plaquettes of W.

On the full space $\mathcal{V}(\mathsf{P}_c^\circ) \times G$, the domain of π_c is Borel and π_c is a Borel map on its domain. This last assertion requires no smallness hypothesis. All constants in (128) are independent of the level and the torus. ∎

Proof. We first record the estimates used below. For $g \in G$, let $\theta(g) \in [0, \pi]$ be its angle. Then

$$|g - \mathbf{1}| \le \theta(g) \le \frac{\pi}{2}|g - \mathbf{1}|, \qquad \theta(gg') \le \theta(g) + \theta(g').$$

The angle is invariant under inversion and conjugation, and $\theta(e^{iA}) \le |A|$ for $A \in \mathfrak{g}$, by its eigenvalues. For the second inequality, only $\theta(g) + \theta(g') < \pi$ needs proof. Use

$$gg' - \mathbf{1} = \tfrac{1}{2}(g - \mathbf{1})(g' + \mathbf{1}) + \tfrac{1}{2}(g + \mathbf{1})(g' - \mathbf{1})$$

and $|g \mp \mathbf{1}| = 2\sin(\theta(g)/2), 2\cos(\theta(g)/2)$, respectively. Monotonicity of sine on $[0, \pi/2]$ gives the result. Unitary multiplication preserves the norm, and telescoping gives a bound by the sum of the bond errors for any path product. Also

$$|e^{iA} - e^{iB}| \le |A - B| \qquad (A, B \in \mathfrak{g}),$$

by integrating $\frac{d}{ds}(e^{isA}e^{i(1-s)B}) = e^{isA}i(A - B)e^{i(1-s)B}$. Together with (122), this gives the corresponding logarithm Lipschitz bound $\vartheta/\sin\vartheta$ when both angles are at most $\vartheta < \pi$.

We also need a sharper comparison of two means. Suppose $A, B \in G$ are means of the families (U_j) and (V_j) in G, and

$$|U_jA^{-1} - \mathbf{1}|,\ |V_jB^{-1} - \mathbf{1}| \le \rho \le 1/12, \qquad \delta = \max_j |U_jV_j^{-1} - \mathbf{1}|.$$

Put $X_j = U_jA^{-1}$, $Y_j = V_jB^{-1}$, $E_j = U_jV_j^{-1}$, $g = BA^{-1}$ and $a = |g - \mathbf{1}|$. Since $X_j = E_jY_jg$, $|X_j - Y_j| \le \delta + a$. The mean equations and (49) give

$$0 = \sum_j (E_j - \mathbf{1})Y_jg + n(g - \mathbf{1}) + \sum_j (Y_j - \mathbf{1})(g - \mathbf{1}) + \sum_j R_j, \qquad |R_j| \le \frac{\rho}{1 - \rho}(\delta + a).$$

Consequently $a \le \delta + \rho a + \rho(\delta + a)/(1 - \rho)$, and

$$|AB^{-1} - \mathbf{1}| \le \frac{\delta}{(1 - \rho)^2 - \rho} \le (1 + 4\rho)\delta. \tag{131}$$

The last inequality follows from $(1 + 4\rho)(1 - 3\rho) \ge 1$.

Let Γ_- and Γ_+ be the portions of $[c_-, c_+]$ before and after the pivot. They each have h bonds, all nonpivot. Use the straight transport u as unknown:

$$\begin{aligned} u_0 = G_1[c], \qquad v_0 = \mathcal{M}(G_1)(c), \\ x(u) = w(\Gamma_-)^{-1}u\,w(\Gamma_+)^{-1}, \qquad U_u = w \sqcup x(u). \end{aligned} \tag{132}$$

This parametrizes every completion once, and $U_u[c] = u$. Telescoping the two half-paths gives

$$|x(u) - G_1(b_0)| \le (L - 1)\omega' + |u - u_0|. \tag{133}$$

The L sites on the axis through c have identical contour transports along that axis, and their loops W_x and $Y_{x,\Gamma,\Gamma'}$ equal $\mathbf{1}$ by cancellation. Write B' for the other $L^3 - L$ sites and put $q = 1 - L^{-2}$. For $x \in B'$, the path $[x, x']$ avoids the pivot, as do the contours inside either block. Thus

$$T_x(w) = w(c_-, x)w([x, x'])w(c_+, x')^{-1}, \qquad T_{x,\Gamma,\Gamma'}(w) = w(\Gamma)w([x, x'])w(\Gamma')^{-1}$$

depend only on w, and the corresponding loops of U_u are $T_x(w)u^{-1}$ and $T_{x,\Gamma,\Gamma'}(w)u^{-1}$.

Here all the means defining $w(y, x)$ exist. Indeed,

$$16L^4\omega + 8L^3\omega' \le \omega_M, \tag{134}$$

so $\omega + 4\omega' \le \omega_M/L^3$. The internal plaquettes of w are bounded by $\omega + 4\omega'$. By lemma 14.5 and proposition 4.3, their contour families have unique means, and their distances from those means are at most

$$\kappa_1^{\mathrm{SO}}(\omega + 4\omega') \le \rho_L := \frac{1}{200L^3}.$$

The same holds for G_1. Applying (131) to the two families and telescoping their paths yields, with $C_L = 1 + 4\rho_L$,

$$|w(y,x) - G_1(y,x)| \le C_L|x-y|_1\omega', \qquad |T_x(w) - T_x(G_1)| \le (2C_L|x-c_-|_1 + L)\omega' \le 4L\omega'.$$

For the individual path products the angle error is at most $2\pi L\omega'$, since their paths have at most $4L-3$ bonds. Moreover

$$\sum_{x\in B'} |x-c_-|_1 = \tfrac{1}{4}(L^2-1)(3L^2-1).$$

Since $C_L(3L^2-1) < 3L^2$, the averaged error satisfies

$$d_w := L^{-3}\sum_{x\in B'} |T_x(w) - T_x(G_1)| \le \frac{5}{2}\frac{L^2-1}{L}\omega'. \tag{135}$$

To exclude distant solutions, write $\mathcal{M}(U)(c) = e^{i\zeta(U)}U[c]$. On the first branch, ζ is L^{-3} times the sum over B' of the intrinsic logarithms divided by i; on the second branch use the mean over the individual contour pairs instead; on the third put $\zeta = 0$. The omitted axial terms vanish. For G_1, lemmas 14.2 and 14.6 give the first branch and

$$\theta(W_x(G_1)) \le \kappa_3\omega, \quad \theta(Y_{x,\Gamma,\Gamma'}(G_1)) \le (2\kappa_1^{\mathrm{SO}} + \kappa_3)\omega, \quad |\zeta(G_1)| \le q\kappa_3\omega.$$

The second bound follows by inserting the two endpoint contour errors around $W_x(G_1)$. The constants defined above satisfy $\kappa_1^{\mathrm{SO}} < 0.907L^2$, $\kappa_3 < 5.992L^2$, and $2\kappa_1^{\mathrm{SO}} + 2\kappa_3 < 14L^2$. For every $u \in G$, each of the first two branches therefore gives

$$|\zeta(U_u)| \le q\big[\theta(uu_0^{-1}) + 2\pi L\omega' + (2\kappa_1^{\mathrm{SO}} + \kappa_3)\omega\big],$$

and the third satisfies this inequality because its exponent is zero. Apply angle subadditivity to $uu_0^{-1} = e^{-i\zeta(U_u)}[\mathcal{M}(U_u)(c)v_0^{-1}]e^{i\zeta(G_1)}$ to obtain

$$L^{-2}\theta(uu_0^{-1}) \le \theta(\mathcal{M}(U_u)(c)v_0^{-1}) + q(2\pi L\omega' + 14L^2\omega). \tag{136}$$

Every solution thus lies in the closed ball

$$\mathcal{U} = \{u \in G : |u - u_0| \le t\}, \qquad t = (L^2-1)(14L^2\omega + 2\pi L\omega'). \tag{137}$$

For $u \in \mathcal{U}$, a plaquette has at most one pivot edge. Using (133) on that edge gives

$$\varpi_c(U_u) \le \omega + (L+2)\omega' + t < \omega_M.$$

Indeed, the coefficients of ω and ω' are respectively less than $16L^4$ and $8L^3$; if both vanish the conclusion is immediate. Thus only the first branch occurs throughout $\mathcal{U}$. The loop angles there obey

$$\theta(W_x(U_u)) \le 2\pi L\omega' + \kappa_3\omega + \tfrac{\pi}{2}t \le 2\omega_M =: \vartheta_L, \qquad \theta(W_x(G_1)) \le \vartheta_L.$$

For the middle inequality, the two coefficients are at most $22L^4$ and $9.87L^3$, bounded by twice those in (134).

Set $\varkappa = \vartheta_L/\sin\vartheta_L$ and $\Psi(u) = e^{-i\zeta(U_u)}v_0$. The logarithm bound now gives

$$|\Psi(u) - \Psi(u')| \le q\varkappa|u-u'|, \qquad |\Psi(u_0) - u_0| \le \varkappa d_w. \tag{138}$$

Here $\vartheta_L = 1/(100L^2)$. The inequality $\sin s \ge s - s^3/6$ for $0 \le s \le 1$ gives $s/\sin s \le 1 + s^2/5$, hence

$$\varkappa \le 1 + \frac{1}{50000L^4}, \quad 1 - q\varkappa \ge L^{-2}\left(1 - \frac{1}{50000L^2}\right) > 0, \quad \frac{\varkappa}{1-q\varkappa} \le L^2 + 10^{-4}.$$

For the last bound, put $\epsilon_1 = 1/(50000L^2)$ and use $(1+\epsilon_1)/(1-\epsilon_1) \le 1+3\epsilon_1$. Equations (137) and (135) also give

$$(1-q\varkappa)t \ge \left(1-\frac{1}{50000L^2}\right)2\pi\frac{L^2-1}{L}\omega' \ge \left(1+\frac{1}{50000L^4}\right)\frac{5}{2}\frac{L^2-1}{L}\omega' \ge \varkappa d_w.$$

Thus Ψ maps the complete space $\mathcal{U}$ into itself and is a contraction. Its unique fixed point u_* satisfies

$$|u_* - u_0| \le \frac{\varkappa d_w}{1-q\varkappa} \le (L^2+10^{-4})\frac{5}{2}\frac{L^2-1}{L}\omega'. \tag{139}$$

On $\mathcal{U}$, the fixed point equation is precisely $\mathcal{M}(U_u)(c) = v_0$, and (136) puts every solution on every branch in this ball. This proves existence and global uniqueness. Finally, (133) and (139) give

$$|x(u_*) - G_1(b_0)| \le \left[L - 1 + \frac{5}{2}(L^2+10^{-4})\frac{L^2-1}{L}\right]\omega' \le \frac{5}{2}L^3\omega'.$$

For measurability, no reference configuration is needed. The space $\mathcal{V}(\mathsf{P}_c)$ is compact. With $\mathcal{N}_c$ as in section 4, exhaust its three branches by

$$\begin{aligned} K_n^{\mathrm{a}} &= \{U : \varpi_c(U) \le (1-1/n)\omega_M\},\\ K_n^{\mathrm{b}} &= \{U : \varpi_c(U) \ge \omega_M,\ |Y_{x,\Gamma,\Gamma'}(U) + \mathbf{1}| \ge 1/n \text{ for every } x,\Gamma,\Gamma'\},\\ K^{\mathrm{c}} &= \{U : \varpi_c(U) \ge \omega_M\} \cap \mathcal{N}_c. \end{aligned} \tag{140}$$

These sets are compact, and the average is continuous on each separately by proposition 4.5. Put $F = \mathcal{V}(\mathsf{P}_c^\circ) \times G$, and let $\Sigma_n \subset F \times G$ consist of the solutions (w,v,x) with $w \sqcup x \in K_n^{\mathrm{a}} \cup K_n^{\mathrm{b}} \cup K^{\mathrm{c}}$. Each Σ_n is a finite union of compact graphs, and these sets increase to the full solution set Σ. The pairs admitting at least one solution form $F_{\ge 1} = \bigcup_n \mathrm{pr}_F \Sigma_n$, a Borel set. The pairs admitting two distinct solutions form the Borel set

$$F_{\ge 2} = \bigcup_n \mathrm{pr}_F\{(w,v,x,x') : (w,v,x),(w,v,x') \in \Sigma_n,\ |x-x'| \ge 1/n\},$$

since each set being projected is compact. The use of one index n loses no pair: the Σ_n increase, and any positive distance eventually exceeds $1/n$. Therefore $\mathrm{dom}\,\pi_c = F_{\ge 1} \setminus F_{\ge 2}$ is Borel. For any closed $E \subset G$,

$$\pi_c^{-1}(E) = \mathrm{dom}\,\pi_c \cap \bigcup_n \mathrm{pr}_F(\Sigma_n \cap (F \times E)) \tag{141}$$

is Borel, again by compactness. Since closed sets generate the Borel sets, this proves the last assertion. □

15. Energy under repeated block averaging

Throughout this section, $L \ge 13$ is odd. The small-field decomposition tests the curvature of block averages. A failed test must therefore impose a cost on the original Wilson action. The estimate below gives this implication without assuming that the original configuration is regular. Its proof uses the one-step curvature estimate only while the energy remains below a fixed threshold.

For a plaquette p' of $\mathbb{T}^{(t)}$ and $0 \le l \le t$, let

$$\mathsf{Y}_{t,l}(p') = \bigcup_{y \text{ a corner of } p'} B^{t-l}(y).$$

Let $\mathcal{N}_l(p')$ be the set of plaquettes of $\mathbb{T}^{(l)}$ whose four corners lie in this set, and put $\mathcal{N}_l(S) = \bigcup_{p' \in S}\mathcal{N}_l(p')$ for a set S of plaquettes of $\mathbb{T}^{(t)}$. In this notation the level t is determined by the argument. In particular, membership in $\mathcal{N}_l(S)$ requires all four corners to lie in the cone of one plaquette of S.

With the constants C_0^{SO} and ϖ_L' of lemma 14.9, fix

$$C_{\mathrm{en}} = \sqrt{12}\, C_0^{\mathrm{SO}} L^{-1/2},$$
$$\mathsf{e}_L = \min\left\{ \frac{L}{2}(\varpi_L')^2, \frac{(\log 2)^2(\sqrt{L}-1)^2}{8C_{\mathrm{en}}^2} \right\}, \qquad e_* = \frac{1}{2}\mathsf{e}_L. \tag{142}$$

These positive constants depend only on L.

Proposition 15.1 (Energy of block averages)**.** Let $1 \le t \le K$, let S be a set of plaquettes of $\mathbb{T}^{(t)}$, and let $U \in \mathcal{V}_0$. Then

$$L^t \sum_{p \in \mathcal{N}_0(S)} a_p(U) \ge \min\left\{ \frac{1}{4} \sum_{p' \in S} |\partial \mathcal{M}^t U(p') - \mathbf{1}|^2, e_* \right\}. \tag{143}$$

If S consists of inner plaquettes of a region X for the level-t cube partition, the sum on the left is at most $L^t A_X(U)$. Consequently, let R be a set of bare sites and let $\sigma_{\mathsf{R}}(U)$ be the sum of $a_p(U)$ over bare plaquettes whose corners lie in R. If $1 \le l \le K$ and $L^l \sigma_{\mathsf{R}}(U) < e_*$, then every plaquette p' of $\mathbb{T}^{(l)}$ with $\mathsf{Y}_{l,0}(p') \subset \mathsf{R}$ satisfies

$$|\partial \mathcal{M}^l U(p') - \mathbf{1}| \le 2\big(L^l \sigma_{\mathsf{R}}(U)\big)^{1/2}. \tag{144}$$

■

Proof. We first record the geometry needed to iterate the one-step estimate. For $l < t$,

$$\mathcal{N}_t(S) = S, \qquad \mathcal{N}_l(S) = \bigcup_{p'' \in \mathcal{N}_{l+1}(S)} \mathcal{N}_l(p''). \tag{145}$$

Here the cones on the right are those of one step. To check the second identity, fix $p' = p_{\mu\nu}(y) \in S$ and put $r = L^{t-l}$. In units of a_{K-l}, its cone is a rectangular box with $2r$ sites in the μ and ν directions and r sites in the third direction. It does not wrap around the torus: $2r$ is less than half the number of sites in each direction. At consecutive levels the boxes are related by replacing each coarse site by its block of L^3 sites. Thus a fine cone of a coarse plaquette in the coarser box is contained in the finer box.

Conversely, use coordinates s_ρ starting from a corner of the finer box. If a fine plaquette has directions κ, κ', and the coarser box has R_ρ sites in direction ρ, choose its coarse base coordinates to be

$$q_\rho^* = \min\{\lfloor s_\rho/L \rfloor, R_\rho - 2\} \quad (\rho = \kappa, \kappa'), \qquad q_\rho^* = \lfloor s_\rho/L \rfloor \quad (\rho \ne \kappa, \kappa').$$

When $l+1 < t$, all $R_\rho \ge L$, so these are coordinates of a plaquette in the coarser box. The two fine coordinates of the plaquette in either of its directions belong to the union of the blocks based at q_ρ^* and $q_\rho^* + 1$. Its remaining coordinate belongs to the block based at q_ρ^*. This proves the reverse inclusion. When $l + 1 = t$, the coarser box consists of the four corners of p' and the identity is immediate. Taking unions over $p' \in S$ proves (145).

A fine plaquette belongs to the cones of at most twelve coarse plaquettes at any fixed higher level. Indeed, its base point belongs to a unique block of that level, whose center must be a corner of each such coarse plaquette. There are at most four plaquettes through a given corner in each of the three coordinate planes. Therefore, for every nonnegative function g on level-l plaquettes,

$$\sum_{p' \in S} \sum_{p \in \mathcal{N}_l(p')} g(p) \le 12 \sum_{p \in \mathcal{N}_l(S)} g(p). \tag{146}$$

If the corners of p' belong to a region X made of level-t cubes, their blocks of every finer level belong to X. This proves the asserted inclusion of the fine plaquettes in X.

We next prove a one-step estimate. For a real configuration W, write $f_W(p) = |\partial W(p) - \mathbf{1}|$ and use the counting ℓ^2 norm on sets of plaquettes. Let S now consist of plaquettes of $\mathbb{T}^{(n+1)}$, and suppose

$$\varpi = \max_{p \in \mathcal{N}_n(S)} f_W(p) \le \varpi'_L.$$

For $p' \in S$, put $\varpi_{p'} = \max_{p \in \mathcal{N}_n(p')} f_W(p)$. The four double blocks at the bonds of p' lie in its cone. Hence lemma 14.9 applies to the restriction of W there and agrees with the regular branch of the globally defined average:

$$f_{\mathcal{M}(W)}(p') \le (\mathsf{W} f_W)(p') + C_0^{\mathrm{SO}} \varpi_{p'}^2.$$

The kernel W is supported in the one-step cone. Its row and column sums in lemma 14.8 give $\|\mathsf{W} f_W\|_S \le \sqrt{L}\, \|f_W\|_{\mathcal{N}_n(S)}$, by applying the full operator estimate to f_W restricted to that set and extended by zero. Moreover, (146) gives

$$\sum_{p' \in S} \varpi_{p'}^4 \le \varpi^2 \sum_{p' \in S} \sum_{p \in \mathcal{N}_n(p')} f_W(p)^2 \le 12 \varpi^2 \|f_W\|^2_{\mathcal{N}_n(S)}.$$

The triangle inequality therefore yields

$$\|f_{\mathcal{M}(W)}\|_S \le \sqrt{L}(1 + C_{\mathrm{en}} \varpi) \|f_W\|_{\mathcal{N}_n(S)}. \tag{147}$$

Return to the original t, S, U, and put

$$E_l = \sum_{p \in \mathcal{N}_l(S)} |\partial \mathcal{M}^l U(p) - \mathbf{1}|^2, \qquad \varpi_l = \max_{p \in \mathcal{N}_l(S)} |\partial \mathcal{M}^l U(p) - \mathbf{1}|.$$

Empty sums and maxima are zero. We have $\varpi_l \le \sqrt{E_l}$, $E_0 = 2 \sum_{\mathcal{N}_0(S)} a_p(U)$, and $E_t = \sum_{p' \in S} |\partial \mathcal{M}^t U(p') - \mathbf{1}|^2$. By (145) and (147), as long as $\varpi_j \le \varpi'_L$ for $j < l$,

$$E_l^{1/2} \le L^{l/2} \exp\left(C_{\mathrm{en}} \sum_{j<l} \varpi_j \right) E_0^{1/2}. \tag{148}$$

Suppose $L^t E_0 \le \mathsf{e}_L$. We prove $E_l \le 2 L^l E_0$ by induction. It holds at $l = 0$. If it holds through $l < t$, then, for $0 \le j \le l$,

$$\varpi_j \le \sqrt{2\mathsf{e}_L}\, L^{-(t-j)/2} \le \varpi'_L, \qquad C_{\mathrm{en}} \sum_{j \le l} \varpi_j \le \frac{C_{\mathrm{en}} \sqrt{2\mathsf{e}_L}}{\sqrt{L} - 1} \le \frac{1}{2} \log 2.$$

The first and last inequalities use the two bounds in (142). Equation (148) now gives the claimed bound at $l + 1$. In particular $E_t \le 2 L^t E_0$. Thus, if $L^t E_0 \le \mathsf{e}_L$, then $\frac{1}{4} E_t \le \frac{1}{2} L^t E_0$; otherwise $\frac{1}{2} L^t E_0 > e_*$. These alternatives prove (143).

Finally, apply (143) to the single plaquette p' at level l. Its fine cone lies in R, so

$$\min\left\{ \tfrac{1}{4} |\partial \mathcal{M}^l U(p') - \mathbf{1}|^2, e_* \right\} \le L^l \sigma_{\mathsf{R}}(U) < e_*.$$

The minimum must be its first entry, proving (144). □

16. Regional gauge fixing and the minimizing lift

We construct a field with prescribed straight block averages that minimizes the curl energy. The ambiguity consists precisely of gradients whose scalar block averages vanish. A relative Landau condition selects one field from this affine space. The scalar projection, gauge-fixed inverse and minimizing-lift formula follow Bałaban's construction [Bał84, pp. 24–35, especially (1.103)]. We give the proof for separated regions and for both choices of boundary bonds.

Fix an odd integer $L \geq 13$, positive integers m, K, and an integer $0 \leq t \leq K$. Use the torus of section 3 and the nested lattices of section 4, with physical torus side $\ell = 2L^m$, and measure lengths in level-t units. Thus

$$\xi_n = L^{n-t}, \qquad \xi = \xi_0 = L^{-t}.$$

Let $\mathfrak{M} = L^p$ for an integer $p \geq 0$, and let R_1 be a power of L with $R_1 \geq L$. Suppose that the required partitions into $\mathfrak{M}\xi_n$-cubes exist, and fix regions $\Omega_0 \supset \cdots \supset \Omega_t$ which are unions of these cubes and satisfy

$$\Omega_0 \neq \emptyset, \qquad \operatorname{dist}_\infty(\Omega_n^c, \Omega_{n+1}) > R_1 \mathfrak{M} \xi_n \quad (0 \leq n < t).$$

Here the complement is the closed-cube complement used in section 12; set $\Omega_{t+1} = \emptyset$. Empty later regions, the whole torus, and disconnected regions are allowed. Write

$$\mathcal{L}_n = \overline{\Omega_n \setminus \Omega_{n+1}}, \qquad \Lambda_n = \mathbb{T}^{(n)} \cap \mathcal{L}_n, \qquad \Lambda_* = \bigcup_{n=0}^{t} \Lambda_n,$$

where the difference inside the closure is a point-set difference. The sets Λ_n are disjoint by lemma 12.2. Retain the centered cubes and blocks

$$\Delta^n(y) = \{x : |x - y|_\infty \leq \tfrac{1}{2}\xi_n\}, \qquad B^n(y) = \Delta^n(y) \cap \mathbb{T}^{(0)}, \qquad B(y) = \Delta^n(y) \cap \mathbb{T}^{(n-1)} \quad (n \geq 1).$$

The level in $B(y)$ will always be specified. Also write $B^n(Y) = \bigcup_{y \in Y} B^n(y)$.

Let $\mathsf{S} = \mathbb{T}^{(0)}$, let E be the positive bare bonds $(x, \mu) = \langle x, x + \xi e_\mu \rangle$, and let F be the bare plaquettes $p_{\mu\nu}(x)$ with $\mu < \nu$. The real spaces $\mathcal{H}^0, \mathcal{H}^1, \mathcal{H}^2$ of functions on these sets have inner products $\langle f, g \rangle = \xi^3 \sum fg$. Extend a bond function by $A(\bar{b}) = -A(b)$ and put $A(\Gamma) = \sum_{b \in \Gamma} A(b)$. The ambient differences are

$$\partial_\mu f(x) = \xi^{-1}(f(x + \xi e_\mu) - f(x)), \qquad (\partial\lambda)_\mu = \partial_\mu \lambda, \qquad (\partial A)_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

Adjoints refer to these inner products, and $\Delta = \partial^*\partial$ on $\mathcal{H}^0$.

Put $\mathsf{S}_0 = \Omega_0 \cap \mathsf{S}$. Choose one of the two sets of positive bonds

$$\mathsf{E}_0^{\mathrm{in}} = \{b : b_-, b_+ \in \mathsf{S}_0\}, \qquad \mathsf{E}_0^{\mathrm{st}} = \{b : \{b_-, b_+\} \cap \mathsf{S}_0 \neq \emptyset\}, \qquad \mathsf{E}_0 \in \{\mathsf{E}_0^{\mathrm{in}}, \mathsf{E}_0^{\mathrm{st}}\}.$$

We call these the inner and star frames. Let χ_0, χ_1 be multiplication by the indicators of $\mathsf{S}_0, \mathsf{E}_0$, respectively, and identify $\mathcal{H}^i(\Omega_0) = \chi_i \mathcal{H}^i$, $i = 0, 1$, with functions extended by zero to the whole lattice. In particular, every ambient derivative below is taken after this extension. Define

$$\mathrm{d} = \chi_1 \partial|_{\mathcal{H}^0(\Omega_0)}, \qquad \mathrm{d}^* = \chi_0 \partial^*|_{\mathcal{H}^1(\Omega_0)}, \qquad \Delta_\Omega = \chi_0 \Delta \chi_0|_{\mathcal{H}^0(\Omega_0)}.$$

We also use the compressed curl operator $(\partial^*\partial)(\Omega_0) = \chi_1 \partial^* \partial \chi_1|_{\mathcal{H}^1(\Omega_0)}$. Thus

$$\langle \lambda, \Delta_\Omega \lambda \rangle = \|\partial\lambda\|^2, \qquad \langle A, (\partial^*\partial)(\Omega_0) A \rangle = \|\partial A\|^2.$$

The second norm includes every plaquette having a bond in the frame. In the inner frame Δ_Ω need not equal $\mathrm{d}^*\mathrm{d}$.

For $1 \leq n \leq t$, take the coarse positive rows

$$\Lambda_n^{\mathrm{b}} = \{c \text{ positive in } \mathbb{T}^{(n)} : \{c_-, c_+\} \cap \Omega_n \neq \emptyset,\ \{c_-, c_+\} \cap \Omega_{n+1} = \emptyset\},$$

and set $\Lambda_0^{\mathrm{b}} = \{c \in \mathsf{E}_0 : \{c_-, c_+\} \cap \Omega_1 = \emptyset\}$. In particular, a row may point out of Ω_n; rows entering Ω_{n+1} are excluded. The level-zero rows include boundary bonds in the star frame. Let $\mathcal{K}^0, \mathcal{K}^1$ be the real function spaces on Λ_* and $\bigcup_{n=0}^t \Lambda_n^{\mathrm{b}}$, with inner products

$$\langle \omega, \omega' \rangle = \sum_{n=0}^{t} \xi_n^3 \sum_{y \in \Lambda_n} \omega(y)\omega'(y), \qquad \langle B, B' \rangle = \sum_{n=0}^{t} \xi_n^3 \sum_{c \in \Lambda_n^{\mathrm{b}}} B(c)B'(c).$$

The row sets are disjoint because their bond lengths ξ_n are distinct.

For ambient scalar and bond functions define the straight averages

$$\begin{aligned}(\mathsf{Q}'_n\lambda)(y) &= L^{-3n}\sum_{x\in B^n(y)}\lambda(x),\\ (\mathsf{Q}_nA)(c) &= L^{-4n}\sum_{x\in B^n(c_-)}A([x,x+\xi_ne_\mu]),\qquad c=\langle c_-,c_-+\xi_ne_\mu\rangle.\end{aligned}\tag{149}$$

Here the path contains L^n bare bonds. Restrict these values to $y\in\Lambda_n$ and $c\in\Lambda_n^{\mathrm{b}}$ to obtain

$$\mathsf{Q}':\mathcal{H}^0(\Omega_0)\longrightarrow\mathcal{K}^0,\qquad \mathsf{Q}:\mathcal{H}^1(\Omega_0)\longrightarrow\mathcal{K}^1.$$

These symbols always denote (149) in this section. At level zero both averages are the identity.

Fix $a>0$ and put

$$\begin{aligned}a_0=a,\qquad a_n&=\frac{a(1-L^{-2})}{1-L^{-2n}}\quad(n\ge1),\\ (\mathsf{a}'\omega)(y)=a_n\xi_n^{-2}\omega(y)\ (y\in\Lambda_n),&\qquad(\mathsf{a}B)(c)=a\xi_n^{-2}B(c)\ (c\in\Lambda_n^{\mathrm{b}}).\end{aligned}$$

Let $N=\ker\mathsf{Q}'$, let R be the orthogonal projection in $\mathcal{H}^0(\Omega_0)$ onto $\Delta_\Omega N$, and put $\mathsf{P}=1-\mathsf{R}$. Define

$$\Delta'_a=\Delta_\Omega+\mathsf{Q}'^*\mathsf{a}'\mathsf{Q}',\qquad \Delta_a=(\partial^*\partial)(\Omega_0)+\mathrm{d}\mathsf{R}\mathrm{d}^*+\mathsf{Q}^*\mathsf{a}\mathsf{Q}.$$

Proposition 16.1. For the separated sequence and either frame above, the following hold.

(a) For every $\lambda\in\mathcal{H}^0(\Omega_0)$,

$$\begin{aligned}\langle\lambda,\Delta'_a\lambda\rangle&\ge\sum_{n=0}^{t}\min\{a_n,4\}\xi_n^{-2}\sum_{x\in B^n(\Lambda_n)}\xi^3|\lambda(x)|^2,\\ \langle\lambda,\Delta_\Omega\lambda\rangle&\ge4\sum_{n=1}^{t}\xi_n^{-2}\sum_{x\in B^n(\Lambda_n)}\xi^3|\lambda(x)|^2\qquad(\lambda\in N).\end{aligned}\tag{150}$$

In a term with $n\ge1$, the number 4 may be replaced by $4L^{2n}\sin^2(\pi/(2L^n))\in[4,\pi^2)$. Consequently Δ'_a is positive definite, its inverse $\mathsf{G}'=(\Delta'_a)^{-1}$ exists and is positive definite, and Δ_Ω is injective on N.

(b) If $y\in\Lambda_n$ and $x\in B^n(y)$, then $(\mathsf{Q}'^*\omega)(x)=\omega(y)$. Moreover $\mathsf{Q}'\mathsf{Q}'^*=1$, Q' is onto, and both $\mathsf{Q}'\mathsf{G}'^2\mathsf{Q}'^*$ and $\mathsf{Q}'\mathsf{G}'\mathsf{Q}'^*$ are positive definite.

(c)
$$\{A\in\mathcal{H}^1(\Omega_0):\partial A=0,\ \mathsf{Q}A=0\}=\mathrm{d}N,$$
and d is injective on N.

(d) Δ_a is positive definite. Its inverse $\mathsf{G}=\Delta_a^{-1}$ exists and is positive definite.

(e) Q is onto, and $\mathsf{Q}\mathsf{G}\mathsf{Q}^*$ is positive definite on $\mathcal{K}^1$.

■

The scalar assertions follow from the gap of the Neumann difference operator on each block; their proof is in section 16.2. The proof of (c) in section 16.3 starts with an ambient closed one-form and its affine potential on the covering lattice. The zero-average equations make its block means agree across successive layers; subtracting those means cancels the affine increments. This identifies the joint kernel of curl and the coarse average, including the boundary restrictions in each frame. The kernel identity and scalar positivity give the vector positivity in (d). A triangular construction on block-face corridors proves surjectivity in (e), and hence positivity of $\mathsf{Q}\mathsf{G}\mathsf{Q}^*$.

Proposition 16.2. In the same setting, the projection, gauge correction and minimizing lift have the following properties.

(i) Every $\lambda \in N$ is supported in Ω_1. Its gradient and the gradient of its ambient Laplacian are supported on inner bonds of Ω_0; $\Delta\lambda$ is supported on the bare sites at graph distance at most one from Ω_1. Moreover

$$\mathrm{d}\lambda = \partial\lambda, \qquad \mathrm{d}^*\mathrm{d}\lambda = \Delta_\Omega\lambda = \Delta'_a\lambda = \Delta\lambda, \qquad \mathrm{d}\Delta_\Omega\lambda = \partial\Delta\lambda.$$

The map $\Delta_\Omega : N \longrightarrow \operatorname{ran}\mathsf{R}$ is bijective.

(ii) The projection is given by

$$\mathsf{P} = \mathsf{G}'\mathsf{Q}'^*(\mathsf{Q}'\mathsf{G}'^2\mathsf{Q}'^*)^{-1}\mathsf{Q}'\mathsf{G}', \qquad \mathsf{R} = 1 - \mathsf{P}.$$

Furthermore

$$\mathsf{R}\mathsf{G}'\mathsf{Q}'^* = 0, \qquad \mathsf{Q}'\mathsf{G}'\mathsf{R} = 0, \qquad \mathsf{G}'\mathsf{R}\mathcal{H}^0(\Omega_0) = N.$$

For fixed geometry and frame, R, P are independent of a.

(iii) For $A \in \mathcal{H}^1(\Omega_0)$, the unique $\lambda \in N$ such that $\mathsf{R}\mathrm{d}^*(A - \mathrm{d}\lambda) = 0$ is

$$\lambda = \mathsf{G}'\mathsf{R}\mathrm{d}^*A.$$

It is also the unique minimizer over N of the full ambient norm $\|\partial^*(A - \partial\lambda)\|^2$. The correction preserves $\mathsf{Q}A$ and ∂A.

(iv) For $\lambda \in N$, $\Delta_a\mathrm{d}\lambda = \mathrm{d}\Delta'_a\lambda$. The inverse identities are

$$\mathsf{G}\mathrm{d}\mathsf{R} = \mathrm{d}\mathsf{G}'\mathsf{R}, \qquad \mathsf{R}\mathrm{d}^*\mathsf{G} = \mathsf{R}\mathsf{G}'\mathrm{d}^*. \tag{151}$$

They imply

$$\mathsf{R}\mathrm{d}^*\mathsf{G}\mathrm{d} = \mathrm{d}^*\mathsf{G}\mathrm{d}\mathsf{R} = \mathsf{R}, \qquad \mathsf{R}\mathrm{d}^*\mathsf{G}\mathsf{Q}^* = 0, \qquad \mathsf{Q}\mathsf{G}\mathrm{d}\mathsf{R} = 0.$$

(v) Define

$$\mathsf{H} = \mathsf{G}\mathsf{Q}^*(\mathsf{Q}\mathsf{G}\mathsf{Q}^*)^{-1}.$$

Then $\mathsf{Q}\mathsf{H} = 1$ and $\mathsf{R}\mathrm{d}^*\mathsf{H} = 0$. For $B \in \mathcal{K}^1$, $\mathsf{H}B$ is the unique minimizer of $\|\partial A\|^2$ subject to $\mathsf{Q}A = B$ and $\mathsf{R}\mathrm{d}^*A = 0$. With only the constraint $\mathsf{Q}A = B$, the set of minimizers is $\mathsf{H}B + \mathrm{d}N$. For fixed geometry and frame, H is independent of a, and

$$\begin{aligned} (\mathsf{Q}\mathsf{G}\mathsf{Q}^*)^{-1} &= \mathsf{a} + \mathsf{H}^*(\partial^*\partial)(\Omega_0)\mathsf{H}, \\ \langle B, (\mathsf{Q}\mathsf{G}\mathsf{Q}^*)^{-1}B\rangle &= \langle B, \mathsf{a}B\rangle + \|\partial\mathsf{H}B\|^2. \end{aligned} \tag{152}$$

■

The support identities are proved in section 16.1, and the projection formula in section 16.2. The gauge correction and minimizing-lift assertions are proved in section 16.4.

16.1. **Blocks and the regional boundary.** For a level-n positive bond $c = (y, \mu)$, let

$$\mathrm{cor}(c) = \{(z, \mu) \in \mathsf{E} : z \in B^n(y),\ z + \xi e_\mu \in B^n(y + \xi_n e_\mu)\}.$$

These are the bare bonds crossing the common face of the two blocks. In particular, $\mathrm{cor}(c) = \{c\}$ when $n = 0$. Write S_0^∂ for the sites of S_0 having a bare nearest neighbour outside S_0. For $r \geq 0$, let Ω_1^{+r} be the set of bare sites reachable from a bare site of Ω_1 by at most r bonds. It is empty if Ω_1 is empty.

Lemma 16.3. The separated sequence and either frame have the following properties.

(i) The blocks $B^n(y)$, $y \in \Lambda_n$, $0 \leq n \leq t$, partition S_0. Write $n(x), y(x)$ for the unique indices such that $x \in B^{n(x)}(y(x))$. If two sites of S_0 are bare nearest neighbours, their indices $n(x)$ differ by at most one.

(ii) Let $1 \le n \le t$ and let $y \in \mathbb{T}^{(n)} \setminus \Omega_n$ be a level-n nearest neighbour of a site of Ω_n. Then

$$\Delta^n(y) \subset \Omega_{n-1}, \qquad B(y) \subset \Lambda_{n-1}, \qquad B^n(y) = \bigsqcup_{z \in B(y)} B^{n-1}(z).$$

Every endpoint of a row in Λ_n^{b} either belongs to Λ_n or is such an outer site. For $n = 0$, each endpoint belongs to Λ_0 or lies outside S_0; the latter case occurs only in the star frame. A positive level-n bond with both endpoints in Λ_n belongs to Λ_n^{b}.

(iii) All bare bonds on the paths defining $(\mathsf{Q}_n A)(c)$ have endpoints in $B^n(c_-) \cup B^n(c_+)$. For $c \in \Lambda_n^{\mathrm{b}}$ and $n \ge 1$, this union lies in $\Omega_{n-1} \setminus \Omega_{n+1}$, and all these bonds are inner to Ω_0. The corridor contains L^{2n} bonds. Each of them occurs in exactly L^n of the paths defining $(\mathsf{Q}_n A)(c)$ and in none of the paths for another positive level-n bond.

(iv)

$$\Omega_1^{+3} \subset \mathsf{S}_0, \qquad \Omega_1^{+2} \cap \mathsf{S}_0^{\partial} = \emptyset, \qquad \mathsf{S}_0^{\partial} \subset \Lambda_0.$$

Every bare bond with an endpoint in Ω_1^{+2} is inner to Ω_0.

■

Proof. The nonempty region Ω_0 contains a defining $\mathfrak{M}$-cube. Its center is a bare site, so $\mathsf{S}_0 \neq \emptyset$. The coordinate description of blocks in section 4 shows that a site of $\mathbb{T}^{(j)}$, $j \le n$, lies in a unique level-n cube, and that its membership in Ω_n is the membership of that cube's center. Indeed, L is odd, so no such site lies on a face of a level-n cube. The regions Ω_n and Ω_{n+1} are unions of level-n cubes. By the layer-site description in lemma 12.2, a bare site x of Ω_0 has a unique index n with $x \in \Omega_n \setminus \Omega_{n+1}$; the center of its level-n block then belongs to Λ_n. Conversely, $B^n(y) \subset \Omega_n \setminus \Omega_{n+1}$ when $y \in \Lambda_n$. Blocks at one level are disjoint, and the point-set differences for different levels are disjoint. This proves the partition. For $t = 0$ it is the partition into singleton blocks.

If $x, x' \in \mathsf{S}_0$ are bare neighbours and $n = n(x) > n(x')$, then $x \in \Omega_n$ and $x' \notin \Omega_n$. Separation gives

$$\operatorname{dist}_\infty(x', \Omega_n) \le \xi \le \xi_{n-1} < R_1 \mathfrak{M} \xi_{n-1} < \operatorname{dist}_\infty(\Omega_{n-1}^c, \Omega_n).$$

Thus $x' \in \Omega_{n-1} \setminus \Omega_n$, and $n(x') = n - 1$.

For (ii), choose a neighbour $y' \in \Omega_n \cap \mathbb{T}^{(n)}$ of the outer site y. Translation by $y' - y$ maps $\Delta^n(y)$ to $\Delta^n(y') \subset \Omega_n$. Every point of $\Delta^n(y)$ is consequently within sup distance ξ_n of Ω_n, whereas

$$\operatorname{dist}_\infty(\Omega_{n-1}^c, \Omega_n) > R_1 \mathfrak{M} \xi_{n-1} \ge \xi_n.$$

The cube therefore lies in Ω_{n-1}. Its level-$(n-1)$ sites lie outside Ω_n, since its center y does. Hence $B(y) \subset \Lambda_{n-1}$, and the decomposition of $B^n(y)$ follows from block nesting. The assertions about row endpoints now follow from the definition of Λ_n^{b}.

For (iii), put $h_n = (L^n - 1)/2$ and write $x = c_- + \xi \sum_\nu k_\nu e_\nu$ with $|k_\nu| \le h_n$. Along $[x, x + \xi_n e_\mu]$, only k_μ changes. The sites before crossing the common face lie in $B^n(c_-)$; after crossing it they lie in $B^n(c_+)$, since subtracting L^n from the μ-coordinate brings it back into $[-h_n, h_n]$. If an endpoint of c lies in Λ_n, its block is contained in $\Omega_n \setminus \Omega_{n+1}$. The block at an outer endpoint lies in Ω_{n-1} by (ii), and lies outside Ω_n by the block membership property. This proves the support assertion.

A bond (z, μ) belongs to $\operatorname{cor}(c)$ precisely when

$$z = c_- + \xi \left(h_n e_\mu + \sum_{\nu \neq \mu} k_\nu e_\nu \right), \qquad |k_\nu| \le h_n.$$

There are L^{2n} choices. Such a bond lies on the path from x to $x + \xi_n e_\mu$ precisely for $x = z - i\xi e_\mu$, $0 \le i < L^n$. These are the L^n sites in $B^n(c_-)$ on the line through z in direction μ. They are distinct, since the torus period in each direction is at least $2\xi_n$. A path of another direction cannot contain this bond. If a path of direction μ from another level-n block contains it, its initial site is one of these same L^n sites. Disjointness of the blocks forces the initial block, and hence the coarse bond, to be the original one.

Finally, the assertions in (iv) hold when Ω_1 is empty. Otherwise $t \geq 1$, and each site of Ω_1^{+3} is within sup distance 3ξ of Ω_1. Since

$$3\xi \leq L\xi \leq R_1 \mathfrak{M}\xi < \operatorname{dist}_\infty(\Omega_0^c, \Omega_1),$$

it lies in S_0. All neighbours of a site of Ω_1^{+2} therefore lie in S_0. Such a site cannot belong to S_0^∂, and every bond incident to it is inner. In particular, a boundary site lies in $\mathsf{S}_0 \setminus \Omega_1 = \Lambda_0$. □

For $\lambda \in N$, the identity $\mathsf{Q}_0' = 1$ implies that λ vanishes on $\Lambda_0 = \mathsf{S}_0 \setminus \Omega_1$. Consequently $\partial\lambda$ is supported on bonds incident to Ω_1, $\Delta\lambda$ is supported in Ω_1^{+1}, and $\partial\Delta\lambda$ is supported on bonds incident to Ω_1^{+1}. The collar in lemma 16.3 places all these sites and bonds inside their respective regional spaces. Thus

$$\begin{gathered} \mathrm{d}\lambda = \partial\lambda, \qquad \mathrm{d}^*\mathrm{d}\lambda = \Delta_\Omega\lambda = \Delta_a'\lambda = \Delta\lambda, \\ \mathrm{d}\Delta_\Omega\lambda = \partial\Delta\lambda \qquad (\lambda \in N). \end{gathered} \tag{153}$$

Here $\Delta_a'\lambda = \Delta_\Omega\lambda$ also uses $\mathsf{Q}'\lambda = 0$.

16.2. **The scalar inverse and the Landau projection.** The nesting of blocks expresses a scalar average at one level as the average of the preceding level's averages:

$$(\mathsf{Q}_n'\lambda)(y) = L^{-3} \sum_{z \in B(y)} (\mathsf{Q}_{n-1}'\lambda)(z) \quad (n \geq 1).$$

Indeed, $B^n(y)$ is the disjoint union of $B^{n-1}(z)$ over $z \in B(y)$, and the averaging factors multiply. Telescoping the differences along each straight path also gives

$$\begin{aligned} (\mathsf{Q}_n\partial\lambda)(c) &= L^{-4n}\xi^{-1} \sum_{x \in B^n(c_-)} (\lambda(x + \xi_n e_\mu) - \lambda(x)) \\ &= \xi_n^{-1}((\mathsf{Q}_n'\lambda)(c_+) - (\mathsf{Q}_n'\lambda)(c_-)). \end{aligned} \tag{154}$$

For the second equality, translation by $\xi_n e_\mu$ maps $B^n(c_-)$ onto $B^n(c_+)$, and $L^{-n}\xi^{-1} = \xi_n^{-1}$. Finally, the inner products and weights give

$$\langle \lambda, \mathsf{Q}'^* \mathsf{a}' \mathsf{Q}' \lambda \rangle = \sum_{n=0}^{t} a_n \xi_n \sum_{y \in \Lambda_n} |(\mathsf{Q}_n'\lambda)(y)|^2,$$

$$\langle A, \mathsf{Q}^* \mathsf{a} \mathsf{Q} A \rangle = \sum_{n=0}^{t} a \xi_n \sum_{c \in \Lambda_n^{\mathrm{b}}} |(\mathsf{Q}_n A)(c)|^2.$$

In each term the product of the coarse volume ξ_n^3 and the inverse-square weight is $a_n\xi_n$ or $a\xi_n$, respectively.

Proof of proposition 16.1(a),(b). Fix a level n and put $J = L^n$. The graph Laplacian of the path with vertices $0, \dots, J-1$ has eigenvectors

$$u_j^{(k)} = \cos\left(\frac{\pi k(j + \frac{1}{2})}{J}\right), \qquad 0 \leq k < J,$$

with eigenvalues $4\sin^2(\pi k/(2J))$. Indeed, the cosine identity gives $u_{j+1}^{(k)} + u_{j-1}^{(k)} = 2\cos(\pi k/J)u_j^{(k)}$, while $u_{-1}^{(k)} = u_0^{(k)}$ and $u_J^{(k)} = u_{J-1}^{(k)}$ give the same eigenvalue equation at the two endpoints. The eigenvalues are distinct and $u_0^{(k)} = \cos(\pi k/(2J)) > 0$, so these vectors form a basis. Their threefold tensor products form an eigenbasis for the nearest-neighbour box $\{0, \dots, J-1\}^3$. The zero eigenspace consists of constants; for $J > 1$ the first positive eigenvalue is

$$4\sin^2\left(\frac{\pi}{2J}\right) \geq 4J^{-2}, \qquad 4J^2\sin^2\left(\frac{\pi}{2J}\right) < \pi^2.$$

The inequalities follow from $\sin s \geq 2s/\pi$ on $[0, \pi/2]$ and $\sin s < s$ for $s > 0$.

The coordinate description of $B^n(y)$ identifies it with the translated box $\{-(J-1)/2, \dots, (J-1)/2\}^3$. Internal box edges map to distinct positive bare bonds. Write $b \subset B^n(y)$ for these bonds and $\bar{\lambda}_y = (\mathsf{Q}'_n \lambda)(y)$. Subtracting this mean makes the function orthogonal to constants and leaves its differences unchanged. The box gap therefore gives

$$\sum_{b \subset B^n(y)} \xi^3 |(\partial\lambda)(b)|^2 \geq 4\xi_n^{-2} \sum_{x \in B^n(y)} \xi^3 |\lambda(x) - \bar{\lambda}_y|^2.$$

For $n = 0$ both sides vanish, since the block is a singleton. For $n \geq 1$ we may retain the exact eigenvalue, replacing the coefficient 4 by $4L^{2n} \sin^2(\pi/(2L^n))$.

By lemma 16.3, the blocks partition S_0. Their internal bonds are distinct, so the sum of the left sides is at most $\|\partial\lambda\|^2 = \langle \lambda, \Delta_\Omega \lambda \rangle$. Since $|B^n(y)|\xi^3 = L^{3n}\xi^3 = \xi_n^3$,

$$\langle \lambda, \mathsf{Q}'^* \mathsf{a}' \mathsf{Q}' \lambda \rangle = \sum_{n=0}^{t} \sum_{y \in \Lambda_n} a_n \xi_n^{-2} \sum_{x \in B^n(y)} \xi^3 |\bar{\lambda}_y|^2.$$

On each block the mean and the mean-zero part are orthogonal:

$$\sum_{x \in B^n(y)} \xi^3 |\lambda(x)|^2 = \sum_{x \in B^n(y)} \xi^3 |\lambda(x) - \bar{\lambda}_y|^2 + \sum_{x \in B^n(y)} \xi^3 |\bar{\lambda}_y|^2.$$

Adding the two bounds proves the first inequality of (150). For $\lambda \in N$ all block means vanish, and the block gaps with $n \geq 1$ give the second inequality. The exact block gap gives its stated refinement and the corresponding refinement of the first inequality.

The operator Δ'_a is symmetric, and every coefficient $\min\{a_n, 4\}$ is positive. The block partition makes the first lower bound strictly positive for nonzero λ. Thus Δ'_a and its inverse G' are positive definite. If $\lambda \in N$ and $\Delta_\Omega \lambda = 0$, the second bound makes λ vanish on every block of positive level. It also vanishes on Λ_0 since $\mathsf{Q}'_0 = 1$. This proves injectivity on N.

For $\omega \in \mathcal{K}^0$, the definition of the coarse inner product gives

$$\begin{aligned} \langle \omega, \mathsf{Q}' \lambda \rangle &= \sum_{n=0}^{t} \xi_n^3 \sum_{y \in \Lambda_n} \omega(y) L^{-3n} \sum_{x \in B^n(y)} \lambda(x) \\ &= \xi^3 \sum_{x \in \mathsf{S}_0} \omega(y(x)) \lambda(x). \end{aligned}$$

Hence $(\mathsf{Q}'^* \omega)(x) = \omega(y(x))$. Averaging this block-constant function gives $\mathsf{Q}' \mathsf{Q}'^* \omega = \omega$, proving surjectivity. In particular, Q'^* is injective. For $\omega \neq 0$,

$$\langle \omega, \mathsf{Q}' \mathsf{G}' \mathsf{Q}'^* \omega \rangle = \langle \mathsf{Q}'^* \omega, \mathsf{G}' \mathsf{Q}'^* \omega \rangle > 0, \qquad \langle \omega, \mathsf{Q}' \mathsf{G}'^2 \mathsf{Q}'^* \omega \rangle = \|\mathsf{G}' \mathsf{Q}'^* \omega\|^2 > 0.$$

Both operators are symmetric, which proves (b). □

Proof of proposition 16.2(i),(ii). The support identities were proved in (153). The injectivity just established, together with the definition of R, shows that Δ_Ω maps N bijectively onto $\operatorname{ran} \mathsf{R}$.

The projection calculation applies to any positive definite symmetric operator D on $\mathcal{H}^0(\Omega_0)$ satisfying $D\lambda = \Delta_\Omega \lambda$ for $\lambda \in N$. Put $S = D^{-1} \mathsf{Q}'^*$. Since Q'^* is injective, $S^* S = \mathsf{Q}' D^{-2} \mathsf{Q}'^*$ is positive definite. The operator $S(S^*S)^{-1} S^*$ is symmetric, is the identity on $\operatorname{ran} S$, and vanishes on its orthogonal complement. Moreover

$$(\operatorname{ran} S)^\perp = \ker S^* = \{f : \mathsf{Q}' D^{-1} f = 0\} = DN = \Delta_\Omega N.$$

Thus

$$\mathsf{P} = D^{-1} \mathsf{Q}'^* (\mathsf{Q}' D^{-2} \mathsf{Q}'^*)^{-1} \mathsf{Q}' D^{-1}, \qquad \mathsf{R} = 1 - \mathsf{P}. \tag{155}$$

For $D = \Delta_a'$ the hypotheses hold by proposition 16.1(a) and (153), and this gives the formula in (ii) for every $a > 0$. The space $\Delta_\Omega N$ is independent of a, so its orthogonal projections R, P are also independent of a.

If $\Omega_0 \neq T_\ell$, one may also take $D = \Delta_\Omega$. Indeed, a nonempty complementary cube contains its bare center, so $\mathsf{S}_0 \neq \mathsf{S}$. If $\langle \lambda, \Delta_\Omega \lambda \rangle = \|\partial\lambda\|^2 = 0$, then λ is constant on the connected bare torus and vanishes outside S_0; hence $\lambda = 0$. This proves the required positive definiteness.

With $D = \Delta_a'$, the preceding calculation identifies $\operatorname{ran}(\mathsf{G}'\mathsf{Q}'^*)$ with $\operatorname{ran}\mathsf{P}$. Therefore $\mathsf{R}\mathsf{G}'\mathsf{Q}'^* = 0$; taking adjoints gives $\mathsf{Q}'\mathsf{G}'\mathsf{R} = 0$. Finally, $\operatorname{ran}\mathsf{R} = \Delta_\Omega N = \Delta_a' N$, so applying G' gives $\mathsf{G}'\mathsf{R}\mathcal{H}^0(\Omega_0) = N$. □

16.3. **The curl kernel and the coarse averages.**

Proof of proposition 16.1(c)–(e). Let $\lambda \in N$. By (153), $\mathrm{d}\lambda = \partial\lambda$, so $\partial\mathrm{d}\lambda = 0$. For $c \in \Lambda_n^{\mathrm{b}}$, telescoping gives

$$(\mathsf{Q}\mathrm{d}\lambda)(c) = \xi_n^{-1}\big((\mathsf{Q}_n'\lambda)(c_+) - (\mathsf{Q}_n'\lambda)(c_-)\big).$$

Both endpoint averages vanish. At an endpoint in Λ_n this is the condition $\mathsf{Q}'\lambda = 0$. At an outer endpoint y with $n \geq 1$,

$$(\mathsf{Q}_n'\lambda)(y) = L^{-3} \sum_{z \in B(y)} (\mathsf{Q}_{n-1}'\lambda)(z) = 0,$$

because $B(y) \subset \Lambda_{n-1}$ by lemma 16.3(ii). At a level-zero endpoint outside S_0, the average is the zero extension of λ. Thus $\mathsf{Q}\mathrm{d}\lambda = 0$. If $\mathrm{d}\lambda = \partial\lambda = 0$, then λ is constant on the connected bare torus. Its mean on any layer block is zero, and at least one such block exists because $\mathsf{S}_0 \neq \emptyset$. Hence $\lambda = 0$. This proves injectivity and one inclusion in the kernel formula.

An affine potential on the covering lattice. Suppose now that $A \in \mathcal{H}^1(\Omega_0)$ satisfies $\partial A = 0$ and $\mathsf{Q}A = 0$. We first write its zero extension as $A = \partial\mu + h$ on the full torus, with h constant in each component. The ambient summation-by-parts identity [Bał84, (1.21), p. 21] is

$$\|\partial B\|^2 + \|\partial^* B\|^2 = \sum_{\mu,\nu} \|\partial_\mu B_\nu\|^2 \qquad (B \in \mathcal{H}^1). \tag{156}$$

To check it with the present normalizations, translations preserve the ξ^3-weighted sums, and therefore

$$\partial_\mu^* f(x) = \xi^{-1}\big(f(x - \xi e_\mu) - f(x)\big).$$

All forward and backward difference operators commute. Consequently

$$\langle \partial_\mu B_\nu, \partial_\nu B_\mu \rangle = \langle \partial_\nu^* B_\nu, \partial_\mu^* B_\mu \rangle.$$

Expanding $\frac{1}{2}\sum_{\mu,\nu} \|\partial_\mu B_\nu - \partial_\nu B_\mu\|^2$ and summing this identity gives $\|\partial B\|^2 = \sum_{\mu,\nu} \|\partial_\mu B_\nu\|^2 - \|\sum_\nu \partial_\nu^* B_\nu\|^2$, which is (156).

The symmetric nonnegative operator $\Delta^{(1)} = \partial^*\partial + \partial\partial^*$ on ambient one-forms consequently has kernel exactly the forms with constant components. Let h be the orthogonal projection of A onto this kernel. The vector $A - h$ belongs to the range of $\Delta^{(1)}$, so $A - h = \Delta^{(1)} B$ for some B. Since $\partial(A - h) = 0$ and $\partial\partial = 0$,

$$\partial\partial^*\partial B = 0, \qquad \|\partial^*\partial B\|^2 = \langle \partial B, \partial\partial^*\partial B \rangle = 0.$$

Thus $A = \partial\mu + h$, where $\mu = \partial^* B$. Write $h_\nu = c_\nu$.

Let $\ell_t = \ell L^{K-t}$ be the side length in level-t units, and let $\mathsf{p} : \mathbb{R}^3 \to T_\ell$ be the corresponding quotient map, with period group $\mathsf{V} = \ell_t \mathbb{Z}^3$. On the covering lattice $\xi\mathbb{Z}^3$, define

$$\widetilde{\mu}(\widetilde{x}) = \mu(\mathsf{p}\widetilde{x}) + c \cdot \widetilde{x}.$$

Then

$$\partial\widetilde{\mu} = A \circ \mathsf{p}, \qquad \widetilde{\mu}(\widetilde{x} + v) = \widetilde{\mu}(\widetilde{x}) + c \cdot v \quad (v \in \mathsf{V}).$$

The potential and its block means have the same affine increments. Subtracting the means will give a periodic potential.

For $\widetilde{y} \in \xi_n \mathbb{Z}^3$, put

$$\widetilde{B}^n(\widetilde{y}) = \left\{ \widetilde{x} \in \xi \mathbb{Z}^3 : |\widetilde{x} - \widetilde{y}|_\infty \le \tfrac{1}{2}(\xi_n - \xi) \right\},$$
$$\widetilde{\varphi}_n(\widetilde{y}) = L^{-3n} \sum_{\widetilde{x} \in \widetilde{B}^n(\widetilde{y})} \widetilde{\mu}(\widetilde{x}).$$

The covering map sends this lifted block bijectively onto $B^n(\mathsf{p}\widetilde{y})$. At a fixed level these lifted blocks partition $\xi\mathbb{Z}^3$: each integer has a unique representation $L^n i + i'$ with $|i'| \le (L^n - 1)/2$. Two bare neighbours belong either to the same lifted block or to blocks with nearest-neighbour centers. Indeed, if two such integer representations differ by one, then $L^n|i - j| \le 1 + |i' - j'| \le L^n$; the other coordinates of the two sites agree.

If $\widetilde{c}$ is a lift of a positive level-n bond c of direction ν, telescoping on its parallel bare paths gives

$$(\mathsf{Q}_n A)(c) = \xi_n^{-1}(\widetilde{\varphi}_n(\widetilde{c}_+) - \widetilde{\varphi}_n(\widetilde{c}_-)). \tag{157}$$

In fact, each path sum is $\xi^{-1}(\widetilde{\mu}(\widetilde{x} + \xi_n e_\nu) - \widetilde{\mu}(\widetilde{x}))$. The factor L^{-4n} in (149), followed by the block sum, leaves $L^{-n}\xi^{-1} = \xi_n^{-1}$.

Matching the means across levels. For $\widetilde{x} \in \mathsf{p}^{-1}(\mathsf{S}_0)$, let $n = n(\mathsf{p}\widetilde{x})$ and let $\widetilde{y}(\widetilde{x}) \in \xi_n \mathbb{Z}^3$ be the center of its lifted level-n block. Its image is $y(\mathsf{p}\widetilde{x}) \in \Lambda_n$. Define

$$\widetilde{\varphi}(\widetilde{x}) = \widetilde{\varphi}_n(\widetilde{y}(\widetilde{x})).$$

We show that this function has equal values at any two bare neighbours $\widetilde{x}, \widetilde{x}'$ whose images lie in S_0. Write $n = n(\mathsf{p}\widetilde{x})$, $n' = n(\mathsf{p}\widetilde{x}')$, and $\widetilde{y} = \widetilde{y}(\widetilde{x})$, $\widetilde{y}' = \widetilde{y}(\widetilde{x}')$; exchange the sites if necessary so that $n \ge n'$.

If $n = n'$, there is nothing to prove when $\widetilde{y} = \widetilde{y}'$. Otherwise the centers are nearest neighbours at level n, and their images lie in Λ_n. The bond between them, with positive orientation, belongs to Λ_n^{b} by lemma 16.3(ii). The condition $\mathsf{Q}A = 0$ and (157) give equality of the means.

Suppose $n' < n$. By lemma 16.3(i), $n' = n - 1$. Let $\widetilde{y}''$ be the center of the lifted level-n block containing $\widetilde{x}'$. The image of $\widetilde{x}'$ lies outside Ω_n, so the image of $\widetilde{y}''$ also lies outside Ω_n. In particular $\widetilde{y}'' \ne \widetilde{y}$. These centers are nearest neighbours, since their blocks contain the neighbours $\widetilde{x}'$ and $\widetilde{x}$. The level-n bond between their images has one endpoint in Λ_n and the other outside Ω_n; it belongs to Λ_n^{b}. Therefore

$$\widetilde{\varphi}_n(\widetilde{y}'') = \widetilde{\varphi}_n(\widetilde{y}).$$

By lemma 16.3(ii), the outer block at $\mathsf{p}\widetilde{y}''$ is a union of level-$(n-1)$ blocks with centers in Λ_{n-1}. The corresponding lifted decomposition is

$$\widetilde{B}^n(\widetilde{y}'') = \bigsqcup_{\substack{\widetilde{z} = \widetilde{y}'' + \xi_{n-1} \sum_\nu i_\nu e_\nu \\ |i_\nu| \le (L-1)/2}} \widetilde{B}^{n-1}(\widetilde{z}).$$

To see the bounds in this decomposition directly, write each coordinate k of $\xi^{-1}(\widetilde{x} - \widetilde{y}'')$ as $L^{n-1}i + i'$ with $|i'| \le (L^{n-1} - 1)/2$. Then

$$|i| \le (L-1)/2 \implies |k| \le \tfrac{1}{2}L^{n-1}(L-1) + \tfrac{1}{2}(L^{n-1} - 1) = \tfrac{1}{2}(L^n - 1),$$
$$|i| \ge (L+1)/2 \implies |k| \ge \tfrac{1}{2}L^{n-1}(L+1) - \tfrac{1}{2}(L^{n-1} - 1) = \tfrac{1}{2}(L^n + 1).$$

Thus precisely these L^3 lifted centers occur. Their images are the distinct sites of $B(\mathsf{p}\widetilde{y}'') \subset \Lambda_{n-1}$. The rectangular grid of centers is connected by level-$(n-1)$ nearest-neighbour bonds. Each such bond projects, after choosing positive orientation, to a row in $\Lambda_{n-1}^{\mathrm{b}}$. Equation (157) and $\mathsf{Q}A = 0$ therefore show that $\widetilde{\varphi}_{n-1}$ is constant on these centers. Its mean over them is $\widetilde{\varphi}_n(\widetilde{y}'')$. The center

$\widetilde{y}'$ is among them, because $\widetilde{x}'$ belongs to its level-$(n-1)$ block and to the displayed level-n block. Hence

$$\widetilde{\varphi}(\widetilde{x}') = \widetilde{\varphi}_{n-1}(\widetilde{y}') = \widetilde{\varphi}_n(\widetilde{y}'') = \widetilde{\varphi}_n(\widetilde{y}) = \widetilde{\varphi}(\widetilde{x}).$$

This proves equality along every bare bond with both endpoints above S_0.

The regional potential and its periods. Set

$$\widetilde{\lambda}(\widetilde{x}) = \begin{cases} \widetilde{\mu}(\widetilde{x}) - \widetilde{\varphi}(\widetilde{x}), & \mathsf{p}\widetilde{x} \in \mathsf{S}_0, \\ 0, & \mathsf{p}\widetilde{x} \notin \mathsf{S}_0. \end{cases}$$

On a bond with both endpoints above S_0, the equality of neighbouring means gives $\partial\widetilde{\lambda} = A \circ \mathsf{p}$. On a bond with neither endpoint there, both sides vanish: the image bond lies outside even the star frame. For a boundary bond, its endpoint x inside S_0 belongs to $\mathsf{S}_0^{\partial} \subset \Lambda_0$. At any lift $\widetilde{x}$ of this endpoint, the level-zero block is a singleton, so $\widetilde{\varphi}(\widetilde{x}) = \widetilde{\mu}(\widetilde{x})$. Thus $\widetilde{\lambda}$ vanishes at both endpoints. The field A also vanishes on this boundary bond. In the inner frame this follows from zero extension. In the star frame the bond belongs to Λ_0^{b}, since neither endpoint lies in Ω_1; there $\mathsf{Q}_0 A = A$ and $\mathsf{Q}A = 0$. We have therefore proved $\partial\widetilde{\lambda} = A \circ \mathsf{p}$ on every bare bond.

For $v \in \mathsf{V}$, the ratio ℓ_t/ξ_n is an integer at every level. Translating a lifted level-n block by v translates its center by v, and the affine periodicity of $\widetilde{\mu}$ gives

$$\widetilde{\varphi}_n(\widetilde{y} + v) = \widetilde{\varphi}_n(\widetilde{y}) + c \cdot v.$$

The layer assignment is unchanged by a period, so $\widetilde{\varphi}(\widetilde{x}+v) = \widetilde{\varphi}(\widetilde{x}) + c\cdot v$ above S_0. These increments cancel those of $\widetilde{\mu}$. Outside the invariant set $\mathsf{p}^{-1}(\mathsf{S}_0)$, $\widetilde{\lambda}$ is zero. Hence it is periodic everywhere and descends to $\lambda \in \mathcal{H}^0(\Omega_0)$ with $\partial\lambda = A$ and $\mathrm{d}\lambda = A$. On a lifted layer block it is the difference between $\widetilde{\mu}$ and the mean of $\widetilde{\mu}$ on that block. Its mean is zero. The block projects bijectively, so $\mathsf{Q}'\lambda = 0$. This completes the reverse inclusion in (c).

Surjectivity of the coarse average. For $c \in \Lambda_n^{\mathrm{b}}$ let E_c be the indicator one-form of $\mathrm{cor}(c)$. It belongs to $\mathcal{H}^1(\Omega_0)$: for $n \geq 1$ the corridor bonds are inner by lemma 16.3(iii), and for $n = 0$ the corridor is $\{c\}$. The path count in that lemma gives, for $c' \in \Lambda_n^{\mathrm{b}}$,

$$(\mathsf{Q}_n E_c)(c') = L^{-4n} L^n L^{2n} \delta_{cc'} = L^{-n}\delta_{cc'}.$$

If $n' < n$, choose the endpoint $y \in \Lambda_n$ of c. Every corridor bond has an endpoint in $B^n(y) \subset \Omega_n \subset \Omega_{n'+1}$. In contrast, every bond in a path contributing to a level-n' row average has both endpoints outside $\Omega_{n'+1}$, by lemma 16.3(iii) when $n' \geq 1$. For $n' = 0$, the same assertion is the defining condition on Λ_0^{b}. Consequently $(\mathsf{Q}_{n'} E_c)(c') = 0$ for $c' \in \Lambda_{n'}^{\mathrm{b}}$.

Given a coarse field $B \in \mathcal{K}^1$, seek $A = \sum_{n=0}^{t} \sum_{c \in \Lambda_n^{\mathrm{b}}} \beta_c E_c$. The equation $\mathsf{Q}A = B$ reads

$$L^{-n'}\beta_{c'} + \sum_{n<n'} \sum_{c \in \Lambda_n^{\mathrm{b}}} \beta_c (\mathsf{Q}_{n'} E_c)(c') = B(c'), \qquad c' \in \Lambda_{n'}^{\mathrm{b}}.$$

Solve first at $n' = 0$ and then at each increasing level. At level n', only coefficients from lower levels appear beside the nonzero diagonal coefficient $L^{-n'}$. This determines a regional field A with $\mathsf{Q}A = B$, proving surjectivity.

Positivity of the vector operator. For $A \in \mathcal{H}^1(\Omega_0)$, the defining quadratic form is

$$\langle A, \Delta_a A\rangle = \|\partial A\|^2 + \|\mathsf{R}\mathrm{d}^* A\|^2 + \langle \mathsf{Q}A, \mathsf{a}\mathsf{Q}A\rangle \geq 0.$$

If it vanishes, then $\partial A = 0$ and $\mathsf{Q}A = 0$, since the bond weight a is positive. The kernel formula gives $A = \mathrm{d}\lambda$ with $\lambda \in N$. By (153), $\mathrm{d}^*\mathrm{d}\lambda = \Delta_\Omega \lambda$, which belongs to the range $\Delta_\Omega N$ of R. The vanishing gauge term therefore gives $\Delta_\Omega \lambda = 0$. The scalar bound (150) on N implies $\lambda = 0$; when $t = 0$, $N = \{0\}$ already. Thus Δ_a and its inverse G are positive definite. Finally, surjectivity of Q makes Q^* injective, and for $B \neq 0$,

$$\langle B, \mathsf{Q}\mathsf{G}\mathsf{Q}^* B\rangle = \langle \mathsf{Q}^* B, \mathsf{G}\mathsf{Q}^* B\rangle > 0.$$

This proves the remaining inverse assertion in (e). □

16.4. **Gauge correction and constrained minimization.**

Proof of proposition 16.2(iii)–(v). For $\lambda \in N$, (153) gives $\mathrm{d}^*\mathrm{d}\lambda = \Delta_\Omega \lambda \in \operatorname{ran} \mathsf{R}$. Consequently

$$\mathsf{R}\mathrm{d}^*(A - \mathrm{d}\lambda) = \mathsf{R}\mathrm{d}^* A - \Delta_\Omega \lambda.$$

Since $\Delta_\Omega : N \to \operatorname{ran} \mathsf{R}$ is bijective and $\mathsf{G}'\mathsf{R}$ maps into N, this expression vanishes exactly for $\lambda_0 = \mathsf{G}'\mathsf{R}\mathrm{d}^* A$:

$$\Delta_\Omega \lambda_0 = \Delta_a' \mathsf{G}'\mathsf{R}\mathrm{d}^* A = \mathsf{R}\mathrm{d}^* A.$$

By proposition 16.1(c), $\partial \mathrm{d}\lambda = 0$ and $\mathsf{Q}\mathrm{d}\lambda = 0$ for every $\lambda \in N$. Thus subtracting $\mathrm{d}\lambda_0$ preserves both the curl and the coarse averages.

To compare full divergences, use $\partial\lambda = \mathrm{d}\lambda$ and the support of $\Delta\lambda$ inside S_0. The parts on S_0 and its complement are orthogonal, so

$$\|\partial^*(A - \partial\lambda)\|^2 = \|(1 - \chi_0)\partial^* A\|^2 + \|\mathrm{d}^* A - \Delta_\Omega \lambda\|^2.$$

Put $f = \mathrm{d}^* A$. Since $\mathsf{R}f = \Delta_\Omega \mathsf{G}'\mathsf{R}f$, the orthogonal decomposition $f = \mathsf{P}f + \mathsf{R}f$ gives

$$\|f - \Delta_\Omega \lambda\|^2 = \langle f, \mathsf{P}f \rangle + \|\Delta_\Omega(\lambda - \mathsf{G}'\mathsf{R}f)\|^2.$$

The first term and the contribution outside S_0 are independent of λ. Injectivity of Δ_Ω on N proves the asserted uniqueness of the minimizer of the ambient divergence norm.

For the inverse identities, take $\lambda \in N$. The curl and average terms in $\Delta_a \mathrm{d}\lambda$ vanish, while $\mathrm{d}^*\mathrm{d}\lambda = \Delta_\Omega \lambda$ lies in $\operatorname{ran} \mathsf{R}$. It follows that

$$\Delta_a \mathrm{d}\lambda = \mathrm{d}\mathsf{R}\Delta_\Omega \lambda = \mathrm{d}\Delta_\Omega \lambda = \mathrm{d}\Delta_a' \lambda.$$

Apply this to $\lambda = \mathsf{G}'\mathsf{R}f$ and then apply G. This gives the first identity in (151); the second is its adjoint. The same identities imply

$$\mathsf{Q}\mathsf{G}\mathrm{d}\mathsf{R} = \mathsf{Q}\mathrm{d}\mathsf{G}'\mathsf{R} = 0, \qquad \mathrm{d}^*\mathsf{G}\mathrm{d}\mathsf{R} = \mathrm{d}^*\mathrm{d}\mathsf{G}'\mathsf{R} = \Delta_a' \mathsf{G}'\mathsf{R} = \mathsf{R}.$$

Here $\mathsf{G}'\mathsf{R}f \in N$ allows the use of both the zero-average gradient identity and (153). Taking adjoints gives $\mathsf{R}\mathrm{d}^*\mathsf{G}\mathsf{Q}^* = 0$ and $\mathsf{R}\mathrm{d}^*\mathsf{G}\mathrm{d} = \mathsf{R}$.

It remains to identify the minimizing lift. Set

$$X = (\partial^*\partial)(\Omega_0) + \mathrm{d}\mathsf{R}\mathrm{d}^*.$$

This operator is symmetric and nonnegative, with $\langle A, XA \rangle = \|\partial A\|^2 + \|\mathsf{R}\mathrm{d}^* A\|^2$. On $\ker \mathsf{Q}$ it is positive definite, since there $\langle Z, XZ \rangle = \langle Z, \Delta_a Z \rangle$. The existence and positivity of $\mathsf{Q}\mathsf{G}\mathsf{Q}^*$ have already been proved in proposition 16.1. The formula for H therefore gives

$$\mathsf{Q}\mathsf{H} = 1, \qquad \Delta_a \mathsf{H} = \mathsf{Q}^*(\mathsf{Q}\mathsf{G}\mathsf{Q}^*)^{-1}.$$

For $Z \in \ker \mathsf{Q}$ and $B \in \mathcal{K}^1$,

$$\begin{aligned} \langle Z, X\mathsf{H}B \rangle &= \langle Z, \Delta_a \mathsf{H}B \rangle - \langle \mathsf{Q}Z, \mathsf{a}B \rangle \\ &= \langle \mathsf{Q}Z, (\mathsf{Q}\mathsf{G}\mathsf{Q}^*)^{-1} B \rangle - \langle \mathsf{Q}Z, \mathsf{a}B \rangle = 0. \end{aligned}$$

Every A with $\mathsf{Q}A = B$ is uniquely of the form $A = \mathsf{H}B + Z$ with $Z \in \ker \mathsf{Q}$, and hence

$$\langle A, XA \rangle = \langle \mathsf{H}B, X\mathsf{H}B \rangle + \langle Z, XZ \rangle.$$

Thus $\mathsf{H}B$ is the unique minimizer of $\langle A, XA \rangle$ on that affine space. This characterization uses only X and Q. Both are independent of a, because R is; consequently H is independent of a.

The inverse identities give

$$\mathsf{R}\mathrm{d}^*\mathsf{H} = \mathsf{R}\mathrm{d}^*\mathsf{G}\mathsf{Q}^*(\mathsf{Q}\mathsf{G}\mathsf{Q}^*)^{-1} = 0.$$

If also $\mathsf{R}\mathrm{d}^* A = 0$, the preceding minimization identity reads $\|\partial A\|^2 \geq \|\partial \mathsf{H}B\|^2$, with equality only at $A = \mathsf{H}B$. For an arbitrary A with $\mathsf{Q}A = B$, use the gauge correction from (iii) and put $A' = A - \mathrm{d}\lambda_0$. Then $\mathsf{Q}A' = B$, $\mathsf{R}\mathrm{d}^* A' = 0$ and $\partial A' = \partial A$. Therefore the minimum curl energy is attained exactly

when $A' = \mathsf{H}B$, or equivalently $A \in \mathsf{H}B + \mathrm{d}N$. Conversely, every field in this affine space has average B and the same curl as $\mathsf{H}B$.

Finally,

$$\mathsf{H}^*\Delta_a\mathsf{H} = (\mathsf{QH})^*(\mathsf{QGQ}^*)^{-1} = (\mathsf{QGQ}^*)^{-1}.$$

Substituting $\Delta_a = X + \mathsf{Q}^*\mathsf{aQ}$, using $\mathsf{QH} = 1$ and $\mathsf{Rd}^*\mathsf{H} = 0$, gives (152). □

17. Coercivity and decay of the inverse

The curl, the projected divergence and the straight block averages control the full first-order norm of a one-form, with constants independent of the fine spacing. The whole-torus form bound at $a = 1$ is Bałaban's [Bał84, Proposition 1.1, (1.90), p. 33]. We prove it in position space by estimating the divergence omitted by the gauge projection. Exponential conjugation, suggested in [Bał84, p. 36], then gives decay for the inverse of each compression to a bond set.

In this section let $L \geq 2$ and $k \geq 0$ be integers, put $N = L^k$ and $\eta = N^{-1}$, and fix $a > 0$. We work in dimension three on the fine torus and its coarse lattice

$$\mathbb{L}_\eta = \eta\mathbb{Z}^3/(\ell_1\mathbb{Z} \times \ell_2\mathbb{Z} \times \ell_3\mathbb{Z}), \qquad \mathbb{L}_1 = \mathbb{Z}^3/(\ell_1\mathbb{Z} \times \ell_2\mathbb{Z} \times \ell_3\mathbb{Z}), \qquad \ell_\mu \in \mathbb{N}, \quad \ell_\mu \geq 1.$$

The distance $|x - x'|_1$ is the least ℓ^1 norm of a lift of $x - x'$, and $\operatorname{dist}(U, U')$ is the induced distance between nonempty sets.

All fields are real. Site functions, tensors $F_{\mu_1\cdots\mu_r}(x)$, one-forms $A_\mu(x)$ and two-forms $F_{\mu\nu}(x)$, $\mu < \nu$, have the inner products $\langle F, F'\rangle = \eta^3 \sum FF'$. A positive bond is indexed by (x, μ), including when $x + \eta e_\mu = x$ on the torus; the direction remains part of its index. For a tensor, multiplication by a site function, and the indicator χ_U of a site set, act at the base point of each component. A bond-set indicator χ_S acts on the indexed one-form components.

Use the forward differences

$$\partial_\mu = \eta^{-1}(\tau_\mu - 1), \qquad (\tau_\mu F)(x) = F(x + \eta e_\mu), \qquad \partial_\mu^* = \eta^{-1}(\tau_\mu^{-1} - 1).$$

The translations are commuting isometries, so these are the adjoint differences for the stated inner products. The exterior derivative is $(\partial f)_\mu = \partial_\mu f$ on scalars and $(\partial A)_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ on one-forms. In particular, $\partial^* A = \sum_\mu \partial_\mu^* A_\mu$. The componentwise gradient and Laplacian are

$$(\nabla F)_{\mu\mu_1\cdots\mu_r} = \partial_\mu F_{\mu_1\cdots\mu_r}, \qquad \Delta = \nabla^*\nabla = \sum_{\mu=1}^{3} \partial_\mu^*\partial_\mu.$$

On scalars, $\nabla = \partial$. Coarse fields have the same conventions with spacing and volume weight one; their differences are denoted by ∂_1.

Identify torus fields with periodic fields on the covering lattice. For $y \in \mathbb{Z}^3$ put

$$B(y) = y + \eta\{0, \ldots, N-1\}^3.$$

Its image is the corner block at the class of y in $\mathbb{L}_1$. Every sum over $B(y)$ or over neighboring blocks below is a sum on the covering lattice. When summing over y, take one coarse period. Thus repeated images of points in two neighboring blocks retain their multiplicities. Define

$$\begin{aligned} (\mathsf{Q}'f)(y) &= \sum_{x \in B(y)} \eta^3 f(x), \\ (\mathsf{Q}A)_\mu(y) &= \sum_{x \in B(y)} \eta^3 \sum_{j=0}^{N-1} \eta\, A_\mu(x + j\eta e_\mu). \end{aligned} \tag{158}$$

Their adjoints use the fine weight η^3 and coarse weight one. Let R be the orthogonal projection of the full scalar space onto $\Delta \ker \mathsf{Q}'$, and put $\mathsf{P} = 1 - \mathsf{R}$. On one-forms define

$$\Delta_a = \partial^*\partial + \partial\mathsf{R}\partial^* + a\mathsf{Q}^*\mathsf{Q}. \tag{159}$$

For a set S of positive bonds, let $\mathcal{H}_S$ be the space of one-forms vanishing off S and set $\Delta_a(S) = \chi_S \Delta_a \chi_S|_{\mathcal{H}_S}$. The projection R in this operator is formed on the whole torus before compression. The inverse, once its existence is established, is denoted by $G(S)$ and extended by zero on $\mathcal{H}_S^\perp$. Every derivative of $G(S)J$ is the ambient derivative of this zero extension.

The constants used in the estimates are

$$C_3 = \frac{30375}{16}, \qquad \gamma' = (C_3 + 1)^{-2}, \tag{160}$$

$$\begin{aligned} \mathsf{b}_{\mathrm{B}} = C_3 + 2, \qquad C_h = \sqrt{\mathsf{b}_{\mathrm{B}}} + 2\sqrt{3}\,\mathsf{b}_{\mathrm{B}}, \\ C'_{\mathrm{B}} = \max\{1 + 24C_h^2,\ 2C_h^2 a^{-1}\}, \qquad \gamma_{0,\mathrm{B}} = (25C'_{\mathrm{B}} + 2a^{-1})^{-1}, \end{aligned} \tag{161}$$

$$c_P = 548208\,\gamma'^{-2}, \qquad c_V = c_P + 6 + 3a, \tag{162}$$

and

$$\delta_0 = \frac{\gamma_{0,\mathrm{B}}}{16 c_V}. \tag{163}$$

Proposition 17.1. Every one-form A on $\mathbb{L}_\eta$ satisfies

$$\begin{aligned} \|\mathsf{P}\partial^* A\| &\le C_h(\|\mathsf{Q}A\| + 2\sqrt{3}\,\|\partial A\|), \\ \langle A, \Delta_a A\rangle &\ge \gamma_{0,\mathrm{B}}(\|\nabla A\|^2 + \|A\|^2). \end{aligned} \tag{164}$$

The constants are positive and depend only on a; in particular, $\gamma_{0,\mathrm{B}} \le 1/25$. ∎

Proposition 17.2. For every set S of positive bonds the following hold.

(a) The operator $\Delta_a(S)$ is symmetric and

$$\langle A, \Delta_a(S) A\rangle \ge \gamma_{0,\mathrm{B}}(\|\nabla A\|^2 + \|A\|^2) \qquad (A \in \mathcal{H}_S).$$

It is invertible on $\mathcal{H}_S$, and its inverse is symmetric and positive definite on that space.

(b) With the zero extension specified above,

$$\|G(S)\|, \quad \|\nabla G(S)\|, \quad \|G(S)\nabla^*\|, \quad \|\nabla G(S)\nabla^*\| \le \gamma_{0,\mathrm{B}}^{-1}.$$

The first two operators have one-form inputs; the last two have rank-two tensor inputs. Their outputs are one-forms or rank-two tensors according to whether the left factor is 1 or ∇.

(c) For $\mathcal{O}_1 \in \{1, \nabla\}$, $\mathcal{O}_2 \in \{1, \nabla^*\}$ and arbitrary nonempty sets U, U' of fine sites,

$$\|\chi_U \mathcal{O}_1 G(S) \mathcal{O}_2 \chi_{U'}\| \le 4\gamma_{0,\mathrm{B}}^{-1} e^{-\delta_0 \operatorname{dist}(U,U')}. \tag{165}$$

The operator domains and targets are those in (b).

The constants are independent of k, L, S and the periods ℓ_1, ℓ_2, ℓ_3. ∎

The proof of proposition 17.1 is in section 17.3. The averaging estimates in section 17.1 compare a face average with the straight average; the scalar estimate in section 17.2 controls the inverse appearing in the projection P. Together they bound $\mathsf{P}\partial^* A$ by the curl and the block averages. For proposition 17.2, the conjugation estimate in section 17.4 retains half the form lower bound, uniformly in the fine spacing. Section 17.5 then applies it on $\mathcal{H}_S$ and chooses a weight determined by the distance to U'.

17.1. **Straight averages and face averages.** The corner blocks partition the fine torus. To see this, choose a lift $\tilde{x} \in \eta\mathbb{Z}^3$ of a site x and take the unique integer vector $\tilde{y}$ with $\tilde{y}_\mu \le \tilde{x}_\mu < \tilde{y}_\mu + 1$. Changing the lift by a period changes $\tilde{y}$ by the same integer period. Thus the class of $\tilde{y}$ is well defined. The N^3 points $\tilde{y} + \eta n$, $0 \le n_\mu < N$, have distinct images: their coordinate differences have modulus less than one, while a nonzero period has a coordinate of modulus at least one. Each block therefore has volume $N^3\eta^3 = 1$. Taking the adjoint of its mean gives

$$(\mathsf{Q}'^{*}\omega)(x) = \omega(y) \qquad (x \in B(y)).$$

Lemma 17.3. On the covering lattice put

$$D_\mu(y) = B(y) \cup B(y + e_\mu), \qquad \mathrm{w}(s) = \min\{s + \eta,\ 2 - \eta - s\}.$$

Then

$$(\mathsf{Q}A)_\mu(y) = \sum_{x' \in D_\mu(y)} \eta^3 \mathrm{w}(x'_\mu - y_\mu) A_\mu(x'). \tag{166}$$

For $s \in \{0, \eta, \dots, 2 - \eta\}$ the weights lie in $[0, 1]$, and

$$\sum_{x' \in D_\mu(y)} \eta^3 \mathrm{w}(x'_\mu - y_\mu) = 1, \qquad \mathrm{w}(s) + \mathrm{w}(s + 1) = 1 \quad (0 \le s \le 1 - \eta,\ s \in \eta\mathbb{Z}).$$

For a coarse one-form H and $x' \in B(y)$, with $s = x'_\mu - y_\mu$ computed in the chosen block lift,

$$(\mathsf{Q}^* H)_\mu(x') = \mathrm{w}(s) H_\mu(y) + \mathrm{w}(s + 1) H_\mu(y - e_\mu).$$

■

Proof. In (158), collect the pairs (x, j) giving the same point $x' = x + j\eta e_\mu$ on the cover. Write $x_\mu - y_\mu = m_0\eta$ and $x'_\mu - y_\mu = m\eta$. The pairs satisfy $m_0 + j = m$, with $0 \le m_0, j < N$. Their number is $\min\{m + 1, 2N - 1 - m\} = \eta^{-1}\mathrm{w}(m\eta)$, for $0 \le m \le 2N - 1$; at the last value it is zero. Each pair has weight η^4, proving (166). There are N^4 pairs in total, so the weights sum to $\eta^4 N^4 = 1$. Their nonnegativity and upper bound follow from the count. For $0 \le s \le 1 - \eta$, $\mathrm{w}(s) = s + \eta$ and $\mathrm{w}(s + 1) = 1 - \eta - s$, proving the two-weight identity.

To compute the adjoint, sum the product of (166) with $H_\mu(y)$ over one coarse period. The summand indexed by (y, x') is invariant when both points are translated by the same period. Each orbit of pairs has exactly one representative with y in a chosen coarse period, and exactly one with x' in a chosen fine period. We may therefore exchange the two sums, retaining multiplicities. If $x' \in B(y_0)$ on the cover, then $x' \in D_\mu(y)$ precisely for $y = y_0$ and $y = y_0 - e_\mu$. Their weights are $\mathrm{w}(s)$ and $\mathrm{w}(s + 1)$. This proves the adjoint formula even when those two coarse sites have the same torus image. □

For the proof of coercivity we also use the outgoing-face average

$$(\widetilde{\mathsf{Q}}A)_\mu(y) = \sum_{\substack{x \in B(y) \\ x_\mu = y_\mu + 1 - \eta}} \eta^2 A_\mu(x).$$

The equality in the summation condition is imposed on the cover.

Lemma 17.4. For a scalar field f and a one-form A the following hold.

(i) $\mathsf{Q}'\mathsf{Q}'^{*} = 1$, $\|\mathsf{Q}'\| = \|\mathsf{Q}'^{*}\| = 1$ and $\|\mathsf{Q}\| \le 1$. The operator $\Pi' = \mathsf{Q}'^{*}\mathsf{Q}'$ is the orthogonal projection onto the block-constant functions.
(ii) $\mathsf{Q}\partial = \partial_1 \mathsf{Q}'$ and $\mathsf{Q}'\partial^* = \partial_1^* \widetilde{\mathsf{Q}}$.
(iii) $\|f - \Pi' f\|^2 \le \frac{3}{2}\|\partial f\|^2$.
(iv) $\|(\mathsf{Q} - \widetilde{\mathsf{Q}})A\|^2 \le 12\|\nabla A\|^2$.
(v) $\|A\|^2 \le 24\|\nabla A\|^2 + 2\|\mathsf{Q}A\|^2$.

■

Proof. The formula for Q'^* and the unit volume of a block give $\mathsf{Q}'\mathsf{Q}'^* = 1$ and $\|\mathsf{Q}'^*\omega\| = \|\omega\|$. Thus Q'^* is an isometry and its adjoint has norm one. The symmetric operator Π' is idempotent and has range $\operatorname{ran}\mathsf{Q}'^*$, the block-constant functions. For the bond average, Cauchy–Schwarz and lemma 17.3 give

$$|(\mathsf{Q}A)_\mu(y)|^2 \le \sum_{x'\in D_\mu(y)} \eta^3 \mathrm{w}(x'_\mu - y_\mu)|A_\mu(x')|^2.$$

Sum over y and exchange covering pairs as in that lemma. Each point receives the two weights $\mathrm{w}(s)$ and $\mathrm{w}(s+1)$, whose sum is one. Summing also over μ proves $\|\mathsf{Q}A\| \le \|A\|$.

For (ii), the straight path telescopes:

$$\begin{aligned}(\mathsf{Q}\partial f)_\mu(y) &= \sum_{x\in B(y)} \eta^3 (f(x+e_\mu) - f(x)) \\ &= (\mathsf{Q}'f)(y+e_\mu) - (\mathsf{Q}'f)(y).\end{aligned}$$

For the divergence, sum its backward difference over a block:

$$(\mathsf{Q}'\partial^* A)(y) = \sum_\mu \sum_{x\in B(y)} \eta^2 (A_\mu(x-\eta e_\mu) - A_\mu(x)).$$

For fixed transverse coordinates the inner sum in direction μ telescopes to the face at $x_\mu = y_\mu - \eta$ minus the face at $x_\mu = y_\mu + 1 - \eta$. These are the outgoing faces of $B(y-e_\mu)$ and $B(y)$, respectively. The result is $\sum_\mu ((\widetilde{\mathsf{Q}}A)_\mu(y-e_\mu) - (\widetilde{\mathsf{Q}}A)_\mu(y)) = (\partial_1^* \widetilde{\mathsf{Q}}A)(y)$.

We prove the remaining estimates by one path calculation on the covering lattice. Let

$$\mathcal{B} = \prod_{\nu=1}^{3} I_\nu, \qquad I_\nu = \{\alpha_\nu, \alpha_\nu + \eta, \dots, \alpha_\nu + \lambda_\nu - \eta\}, \qquad \lambda_\nu \in \{1,2\},$$

and let I_ν^- omit the last point of I_ν. Let v be a scalar function on $\mathcal{B}$. For $x, x' \in \mathcal{B}$ and a permutation σ of $\{1,2,3\}$, replace the coordinates of x by those of x' in the order $\sigma(1), \sigma(2), \sigma(3)$. On the ith segment, write $z^{(i)}(s)$ for the point whose coordinates $\sigma(j)$ are $x'_{\sigma(j)}$ for $j < i$, s for $j = i$, and $x_{\sigma(j)}$ for $j > i$. The increment along this segment is at most $\sum_{s\in I^-_{\sigma(i)}} \eta |\partial_{\sigma(i)} v(z^{(i)}(s))|$. Since $\sum_{s\in I_\nu^-} \eta \le \lambda_\nu$, two applications of Cauchy–Schwarz give

$$|v(x) - v(x')|^2 \le 3 \sum_{i=1}^{3} \lambda_{\sigma(i)} \sum_{s\in I^-_{\sigma(i)}} \eta |\partial_{\sigma(i)} v(z^{(i)}(s))|^2. \tag{167}$$

All the differences are internal to the covering box. Set

$$\|\nabla v\|_{\mathcal{B}}^2 = \sum_\nu \sum_{\substack{z\in\mathcal{B} \\ z_\nu \in I_\nu^-}} \eta^3 |\partial_\nu v(z)|^2.$$

Let $p(x) = \prod_\nu p_\nu(x_\nu)$ and $p'(x') = \prod_\nu p'_\nu(x'_\nu)$ be products of probability weights on the intervals, and suppose

$$p'_{\sigma(j)} \le \eta \quad (j<3), \qquad p_{\sigma(j)} \le \eta \quad (j>1). \tag{168}$$

Multiply (167) by $p(x)p'(x')$ and sum. In its ith term the sums over $x_{\sigma(j)}$ for $j \le i$ and over $x'_{\sigma(j)}$ for $j \ge i$ give one. The two weights left over are among $p'_{\sigma(j)}$ with $j < i$ and $p_{\sigma(j)}$ with $j > i$, so each is at most η. Together with the segment factor η they contribute at most η^3. The remaining variables enumerate each internal bond of direction $\sigma(i)$ once. Hence

$$\sum_{x,x'\in\mathcal{B}} p(x)p'(x')|v(x) - v(x')|^2 \le 3 \max_\nu \lambda_\nu \, \|\nabla v\|_{\mathcal{B}}^2. \tag{169}$$

For (iii), take $\mathcal{B} = B(y)$, $v = f$ and $p_\nu = p'_\nu = \eta$. All interval lengths are one. The left side of (169) is

$$\sum_{x,x' \in B(y)} \eta^6 |f(x) - f(x')|^2 = 2 \sum_{x \in B(y)} \eta^3 |f(x) - (\mathsf{Q}' f)(y)|^2.$$

Summing the resulting bound over disjoint blocks gives (iii), because their internal bonds form a subset of all fine bonds.

For (iv), fix y, μ, take $\mathcal{B} = D_\mu(y)$, $v = A_\mu$ and choose $\sigma(1) = \mu$. Let p_μ be a point mass at $y_\mu + 1 - \eta$, let $p_\nu = \eta$ for $\nu \neq \mu$, and set

$$p'_\mu(s) = \eta\, \mathrm{w}(s - y_\mu), \qquad p'_\nu = \eta \quad (\nu \neq \mu).$$

The weights p' have total mass one and each $p'_\nu \leq \eta$, by lemma 17.3. The only possibly larger weight is p_μ, which is exempt from (168) because $\sigma(1) = \mu$. The two weighted means of v are $(\widetilde{\mathsf{Q}} A)_\mu(y)$ and $(\mathsf{Q}A)_\mu(y)$. Their squared difference is at most the left side of (169), by Cauchy–Schwarz. Since the box has side two in direction μ and side one in the others,

$$|((\mathsf{Q} - \widetilde{\mathsf{Q}})A)_\mu(y)|^2 \leq 6 \|\nabla A_\mu\|^2_{D_\mu(y)}.$$

Sum over y using covering bonds and their base points. An internal bond of a block $B(y_0)$ belongs to $D_\mu(y_0)$ and $D_\mu(y_0 - e_\mu)$. A bond crossing a block face in direction μ belongs only to $D_\mu(y_0)$, and a bond crossing a face in another direction belongs to none of these boxes. Thus each bond is counted at most twice, including with multiplicity when period images coincide. Summing over μ proves (iv).

For (v), keep v, σ, p' as above and replace p by the uniform weight η^3 on $B(y)$, zero on the rest of $D_\mu(y)$. It again satisfies (168). Cauchy–Schwarz applied to the p' mean, followed by (169), yields

$$\sum_{x \in B(y)} \eta^3 |A_\mu(x) - (\mathsf{Q}A)_\mu(y)|^2 \leq 6 \|\nabla A_\mu\|^2_{D_\mu(y)}.$$

The block has volume one, so

$$\|A_\mu\|^2_{B(y)} \leq 12 \|\nabla A_\mu\|^2_{D_\mu(y)} + 2 |(\mathsf{Q}A)_\mu(y)|^2.$$

The same multiplicity bound gives (v) after summing over y and μ. □

17.2. **The scalar inverse and its coarse averages.** For a scalar function or a one-form F, write

$$\|F\|_1^2 = \|\nabla F\|^2 + \|F\|^2.$$

The scalar operator used to represent P has unit averaging coefficient:

$$\Delta' = \Delta + \mathsf{Q}'^* \mathsf{Q}'.$$

The projection formula requires a uniform lower bound for the coarse Gram operator defined below.

Lemma 17.5. The scalar operator Δ' satisfies

$$\langle f, \Delta' f \rangle \geq \tfrac{1}{3} \|f\|_1^2.$$

Its inverse $\mathsf{G}' = (\Delta')^{-1}$ exists and obeys $\|\mathsf{G}' g\|_1 \leq 3 \|g\|$. On coarse scalar functions, the operator

$$\mathsf{M} = \mathsf{Q}' \mathsf{G}'^2 \mathsf{Q}'^*$$

satisfies $\gamma' \leq \mathsf{M} \leq 9$. Moreover,

$$\mathsf{P} = \mathsf{G}' \mathsf{Q}'^* \mathsf{M}^{-1} \mathsf{Q}' \mathsf{G}'. \tag{170}$$

■

Proof. Let $\Pi' = \mathsf{Q}'^{*}\mathsf{Q}'$ be the orthogonal projection onto the block-constant functions. By lemma 17.4(i),(iii),

$$\begin{aligned}\langle f, \Delta' f\rangle &= \|\partial f\|^2 + \|\Pi' f\|^2 \\ &\geq \tfrac{1}{2}\|\partial f\|^2 + \tfrac{1}{3}\|f - \Pi' f\|^2 + \|\Pi' f\|^2 \geq \tfrac{1}{3}\|f\|_1^2.\end{aligned}$$

Thus Δ' is positive definite. For $u = \mathsf{G}' g$,

$$\tfrac{1}{3}\|u\|_1^2 \leq \langle u, \Delta' u\rangle = \langle u, g\rangle \leq \|u\|_1 \|g\|,$$

which proves the inverse bound. The isometry of Q'^{*} then gives $\langle \omega, \mathsf{M}\omega\rangle = \|\mathsf{G}'\mathsf{Q}'^{*}\omega\|^2 \leq 9\|\omega\|^2$.

A normalized block profile. Define

$$\mathsf{g}(t) = \begin{cases} t^2(1-t)^2, & 0 \leq t \leq 1, \\ 0, & t \notin [0,1]. \end{cases}$$

This function is continuously differentiable, $0 \leq \mathsf{g} \leq 1/16$, and its derivative is Lipschitz with constant 2, since $|\mathsf{g}''(t)| = |2 - 12t + 12t^2| \leq 2$ on $[0,1]$. Consequently

$$\begin{aligned}&\bigl|\mathsf{g}(t+\eta) + \mathsf{g}(t-\eta) - 2\mathsf{g}(t)\bigr| \leq \int_0^{\eta} 2(2s)\, ds = 2\eta^2, \\ &m_N := \frac{1}{N}\sum_{j=0}^{N-1} \mathsf{g}\left(\frac{j+\frac{1}{2}}{N}\right) = \frac{1}{30} + \frac{7}{240N^4} \geq \frac{1}{30}.\end{aligned} \tag{171}$$

The identity for m_N follows by expanding $\mathsf{g}(t) = t^2 - 2t^3 + t^4$ and summing the powers of $j + \frac{1}{2}$; it holds also for $N = 1$.

For a lifted coarse site $y \in \mathbb{Z}^3$ and $x \in \eta\mathbb{Z}^3$, put

$$\psi_y(x) = m_N^{-3} \prod_{\mu=1}^{3} \mathsf{g}(x_\mu - y_\mu + \eta/2).$$

The half-mesh displacement makes ψ_y vanish at every fine site outside the covering block $B(y)$. On that block its average is

$$\sum_{x \in B(y)} \eta^3 \psi_y(x) = m_N^{-3} \prod_{\mu=1}^{3} \left(\eta \sum_{j=0}^{N-1} \mathsf{g}((j+\tfrac{1}{2})\eta)\right) = 1.$$

Regard a coarse function ω as a periodic function on $\mathbb{Z}^3$, and define

$$\psi(x) = \sum_{y \in \mathbb{Z}^3} \omega(y)\psi_y(x).$$

At any covering-lattice site, at most one summand is nonzero. For each period v, $\psi_{y+v}(x+v) = \psi_y(x)$, so ψ is periodic and defines a fine-torus function with $\mathsf{Q}'\psi = \omega$.

Second differences and the coarse lower bound. Fix a direction μ and $x \in B(y_0)$, and set

$$t = x_\mu - y_{0,\mu} + \eta/2, \qquad c(x) = m_N^{-3} \prod_{\nu \neq \mu} \mathsf{g}(x_\nu - y_{0,\nu} + \eta/2) \leq 30^3\, 16^{-2}.$$

Along this coordinate line, $\psi = c(x) \sum_{n \in \mathbb{Z}} \omega(y_0 + n e_\mu)\mathsf{g}(t-n)$. A second difference of $\mathsf{g}(t-n)$ can be nonzero only if $-\eta < t - n < 1 + \eta$. For $t \in \{\eta/2, 3\eta/2, \dots, 1 - \eta/2\}$, this allows $n = 0$, $n = 1$ at $t = 1 - \eta/2$, and $n = -1$ at $t = \eta/2$. Hence (171) gives

$$\begin{aligned}|(\partial_\mu^{*}\partial_\mu \psi)(x)| \leq 2c(x)\bigl(&|\omega(y_0)| + \mathbf{1}_{\{t = 1-\eta/2\}}|\omega(y_0 + e_\mu)| \\ &+ \mathbf{1}_{\{t=\eta/2\}}|\omega(y_0 - e_\mu)|\bigr).\end{aligned}$$

Each of the three block-constant functions on the right, with the indicators omitted, has norm $\|\omega\|$. This follows from the isometry of Q'^{*} and invariance of the coarse sum under translation. It

remains true when neighbouring coarse labels coincide modulo a period; the estimate uses their three contributions. The triangle inequality therefore yields

$$\|\partial_\mu^* \partial_\mu \psi\| \le 6\,30^3\,16^{-2}\|\omega\|, \qquad \|\Delta\psi\| \le 18\,30^3\,16^{-2}\|\omega\| = C_3\|\omega\|.$$

Since $\Delta'\psi = \Delta\psi + \mathsf{Q}'^*\omega$, we obtain $\|\Delta'\psi\| \le (C_3+1)\|\omega\|$. The symmetry of Δ' now gives

$$\begin{aligned}\|\omega\|^2 &= \langle \psi, \mathsf{Q}'^*\omega\rangle = \langle \Delta'\psi, \mathsf{G}'\mathsf{Q}'^*\omega\rangle \\ &\le (C_3+1)\|\omega\|\,\|\mathsf{G}'\mathsf{Q}'^*\omega\|.\end{aligned}$$

Thus $\langle \omega, \mathsf{M}\omega\rangle = \|\mathsf{G}'\mathsf{Q}'^*\omega\|^2 \ge (C_3+1)^{-2}\|\omega\|^2 = \gamma'\|\omega\|^2$.

Finally, set $\mathcal{T} = \mathsf{G}'\mathsf{Q}'^*$. The scalar operator Δ' is positive, Q' is onto, and $\Delta' = \Delta$ on $\ker \mathsf{Q}'$. These verify the hypotheses of the range calculation leading to (155). In the present spaces it reads

$$(\operatorname{ran}\mathcal{T})^\perp = \ker(\mathsf{Q}'\mathsf{G}') = \Delta' \ker \mathsf{Q}' = \Delta \ker \mathsf{Q}' = \operatorname{ran}\mathsf{R}.$$

Since $\mathcal{T}^*\mathcal{T} = \mathsf{M} > 0$, the orthogonal projection onto $\operatorname{ran}\mathcal{T}$ is $\mathcal{T}\mathsf{M}^{-1}\mathcal{T}^*$. It equals $1 - \mathsf{R} = \mathsf{P}$, proving (170). □

17.3. **Recovering the omitted divergence.**

Proof of proposition 17.1. The periodic difference operators commute, so the summation-by-parts identity (156) holds with the present spacing and periods:

$$\|\nabla A\|^2 = \|\partial A\|^2 + \|\partial^* A\|^2, \qquad \partial\partial f = 0.$$

The kernel of the scalar Laplacian consists of constants: if $\Delta f = 0$, then $\|\partial f\|^2 = \langle f, \Delta f\rangle = 0$, and a path of bonds joins any two sites of the torus. Since $\langle \partial^* A, 1\rangle = 0$, there is a unique mean-zero function λ_{c} satisfying $\Delta\lambda_{\mathrm{c}} = \partial^* A$. Indeed, the range of the finite-dimensional symmetric operator Δ is the orthogonal complement of its kernel, and its restriction to that complement is injective. The transverse part $A^{\mathrm{T}} = A - \partial\lambda_{\mathrm{c}}$ satisfies

$$\partial^* A^{\mathrm{T}} = 0, \qquad \partial A^{\mathrm{T}} = \partial A, \qquad \|\nabla A^{\mathrm{T}}\| = \|\partial A\|.$$

Put $\varpi = \mathsf{Q}'\lambda_{\mathrm{c}}$. The two identities in lemma 17.4(ii) yield

$$\partial_1 \varpi = \mathsf{Q}A - \mathsf{Q}A^{\mathrm{T}}, \qquad \partial_1^* \widetilde{\mathsf{Q}} A^{\mathrm{T}} = 0.$$

The face average therefore removes the transverse term from the following pairing:

$$\begin{aligned}\|\partial_1\varpi\|^2 &= \langle \partial_1\varpi, \mathsf{Q}A\rangle - \langle \partial_1\varpi, (\mathsf{Q} - \widetilde{\mathsf{Q}})A^{\mathrm{T}}\rangle - \langle \varpi, \partial_1^*\widetilde{\mathsf{Q}}A^{\mathrm{T}}\rangle \\ &\le \|\partial_1\varpi\|\,(\|\mathsf{Q}A\| + 2\sqrt{3}\,\|\nabla A^{\mathrm{T}}\|).\end{aligned}$$

Here lemma 17.4(iv) bounds the difference of the two averages. It follows that

$$\|\partial_1\varpi\| \le \|\mathsf{Q}A\| + 2\sqrt{3}\,\|\partial A\|.$$

To estimate $h = \mathsf{P}\partial^* A$, use the scalar projection formula (170). It gives

$$h = \mathsf{G}'\mathsf{Q}'^*\omega, \qquad \omega = \mathsf{M}^{-1}\mathsf{Q}'\mathsf{G}'\partial^* A, \qquad \|h\|^2 = \langle \omega, \mathsf{M}\omega\rangle \ge \gamma'\|\omega\|^2.$$

Moreover, $\Delta' h = \mathsf{Q}'^*\omega$, so with $\omega' = \omega - \mathsf{Q}'h$ we have

$$\Delta h = \mathsf{Q}'^*\omega', \qquad \|\Delta h\| = \|\omega'\| \le (\gamma'^{-1/2} + 1)\|h\| = \mathsf{b}_{\mathrm{B}}\|h\|, \qquad \|\partial h\|^2 = \langle h, \Delta h\rangle \le \mathsf{b}_{\mathrm{B}}\|h\|^2.$$

We used the isometry of Q'^* and the contraction property of Q' from lemma 17.4(i).

The same identities identify the pairing that controls h:

$$\omega' = \mathsf{Q}'\mathsf{Q}'^*\omega' = \mathsf{Q}'\partial^*\partial h = \partial_1^*\widetilde{\mathsf{Q}}\partial h.$$

Since P is an orthogonal projection and $\partial^* A^{\mathrm{T}} = 0$, this implies

$$\begin{aligned}\|h\|^2 &= \langle \partial^* A, h\rangle \\ &= \langle A^{\mathrm{T}}, \partial h\rangle + \langle \partial\lambda_{\mathrm{c}}, \partial h\rangle \\ &= \langle \lambda_{\mathrm{c}}, \mathsf{Q}'^{*}\omega'\rangle = \langle \varpi, \partial_1^*\widetilde{\mathsf{Q}}\partial h\rangle = \langle \partial_1\varpi, \widetilde{\mathsf{Q}}\partial h\rangle.\end{aligned}$$

The second differences obey

$$\begin{aligned}\|\nabla\partial h\|^2 &= \sum_{\mu,\nu}\|\partial_\mu\partial_\nu h\|^2 \\ &= \sum_{\mu,\nu}\langle h, \partial_\nu^*\partial_\mu^*\partial_\mu\partial_\nu h\rangle = \langle h, \Delta^2 h\rangle = \|\Delta h\|^2,\end{aligned}$$

again by commutation of the translations. Consequently lemma 17.4(i),(iv) gives

$$\begin{aligned}\|\widetilde{\mathsf{Q}}\partial h\| &\le \|\mathsf{Q}\partial h\| + 2\sqrt{3}\,\|\nabla\partial h\| \\ &\le \|\partial h\| + 2\sqrt{3}\,\|\Delta h\| \le C_h\|h\|.\end{aligned}$$

Inserting this in the pairing yields $\|h\| \le C_h\|\partial_1\varpi\|$. Together with the preceding bound on $\partial_1\varpi$, this proves the first inequality in (164).

It remains to estimate the full norm of A. The quadratic form is

$$\mathcal{E} := \langle A, \Delta_a A\rangle = \|\partial A\|^2 + \|\mathsf{R}\partial^* A\|^2 + a\|\mathsf{Q}A\|^2.$$

Orthogonality of R and $\mathsf{P} = 1 - \mathsf{R}$, the ambient identity above, and the estimate just proved give

$$\begin{aligned}\|\nabla A\|^2 &= \|\partial A\|^2 + \|\mathsf{R}\partial^* A\|^2 + \|\mathsf{P}\partial^* A\|^2 \\ &\le (1 + 24C_h^2)\|\partial A\|^2 + \|\mathsf{R}\partial^* A\|^2 + 2C_h^2\|\mathsf{Q}A\|^2 \\ &\le C'_{\mathrm{B}}\mathcal{E}.\end{aligned}$$

The bond-average estimate lemma 17.4(v) now implies

$$\|A\|^2 \le 24\|\nabla A\|^2 + 2\|\mathsf{Q}A\|^2 \le (24C'_{\mathrm{B}} + 2a^{-1})\mathcal{E}.$$

Adding the last two bounds gives $\|\nabla A\|^2 + \|A\|^2 \le \gamma_{0,\mathrm{B}}^{-1}\mathcal{E}$, as required. The constants in (161) depend only on a, and $C'_{\mathrm{B}} \ge 1$ gives $\gamma_{0,\mathrm{B}}^{-1} \ge 25$. □

17.4. **Exponential conjugation.** Let θ be a real function on $\mathbb{L}_\eta$ such that

$$|\theta(x) - \theta(x')| \le \delta|x - x'|_1.$$

On coarse fields use its restriction to $\mathbb{L}_1$. For an operator $\mathcal{O}$ between any of these spaces, put $\mathcal{O}_\theta = e^{-\theta}\mathcal{O}e^{\theta}$, with multiplication on its respective domain and target. In particular, write $\Delta_{a,\theta} = (\Delta_a)_\theta$. Only the Lipschitz constant of θ enters the following estimates.

Lemma 17.6. Suppose $0 < \delta \le 1/8$ and put $\epsilon_0 = 8\delta$.

(i) For

$$\mathcal{O} \in \{\partial, \partial^*, \nabla, \nabla^*, \mathsf{Q}', \mathsf{Q}'^{*}, \mathsf{Q}, \mathsf{Q}^*\}, \qquad \|\mathcal{O}_\theta - \mathcal{O}\| \le \epsilon_0.$$

Here ∂ acts on scalars, ∂^* on one-forms, and ∇ is the componentwise gradient, with its corresponding adjoint.

(ii) If $5\epsilon_0 \le 1/6$, then

$$\|\mathsf{G}'_\theta g\|_1 \le 6\|g\|, \qquad \|\mathsf{G}'_\theta - \mathsf{G}'\| \le 90\epsilon_0, \qquad \|\mathsf{M}_\theta - \mathsf{M}\| \le 1755\epsilon_0.$$

If, in addition, $1755\epsilon_0 \le \gamma'/2$, then

$$\|\mathsf{M}_\theta^{-1}\| \le 2\gamma'^{-1}, \qquad \|\mathsf{M}_\theta^{-1} - \mathsf{M}^{-1}\| \le 3510\gamma'^{-2}\epsilon_0.$$

(iii) If $c_P\epsilon_0 \le 1$, then

$$\|\mathsf{P}_\theta - \mathsf{P}\| \le c_P\epsilon_0, \qquad \|\mathsf{P}_\theta\| \le 2,$$

and for all one-forms A, A',

$$|\langle A', (\Delta_{a,\theta} - \Delta_a)A\rangle| \le c_V\epsilon_0\|A'\|_1\|A\|_1.$$

■

The scalar hypotheses in (ii) will also be useful when the stronger condition in (iii) is unavailable.

Proof. Conjugation preserves products and inverses, and

$$(\mathcal{O}^*)_\theta = (\mathcal{O}_{-\theta})^*.$$

Every indicator commutes with multiplication by e^θ. These identities hold on the finite tori without any restriction on the magnitude of θ.

Differences and averages. For a forward difference,

$$(\partial_\mu)_\theta = \partial_\mu + \beta_\mu\tau_\mu, \qquad \beta_\mu(x) = \eta^{-1}(e^{\theta(x+\eta e_\mu)-\theta(x)} - 1).$$

The quotient distance between the two sites is at most η, so $|\beta_\mu(x)| \le 2\delta$. Here we used $|e^s - 1| \le e^{|s|} - 1 \le 2|s|$ for $|s| \le 1$, and $\delta\eta \le 1$. Since translations preserve the norm,

$$\|(\nabla_\theta - \nabla)F\|^2 \le 12\delta^2\|F\|^2.$$

In particular, $\|\nabla_\theta - \nabla\| \le 2\sqrt{3}\,\delta \le \epsilon_0$; the same holds for ∂ on scalars.

The kernels of Q' and Q are nonnegative. In a scalar block average every contributing site has distance less than 3 from its coarse site. In the straight one-form average the distance is less than 4. The covering formulas remain valid with all multiplicities when some of their images coincide on the torus. Consequently, component by component,

$$|(\mathsf{Q}_\theta - \mathsf{Q})A| \le (e^{4\delta} - 1)\mathsf{Q}|A|,$$
$$|(\mathsf{Q}'_\theta - \mathsf{Q}')f| \le (e^{3\delta} - 1)\mathsf{Q}'|f|.$$

The contraction bounds in lemma 17.4 give $\|\mathsf{Q}_\theta - \mathsf{Q}\| \le 8\delta$ and $\|\mathsf{Q}'_\theta - \mathsf{Q}'\| \le 6\delta$. Applying the same estimates to $-\theta$ and taking adjoints proves (i).

The scalar inverse. For scalar functions f, g,

$$\langle g, \Delta'_\theta f\rangle = \langle \partial_{-\theta}g, \partial_\theta f\rangle + \langle \mathsf{Q}'_{-\theta}g, \mathsf{Q}'_\theta f\rangle.$$

Subtract the unweighted identity. Part (i) and $\|\mathsf{Q}'\| \le 1$ show that the absolute difference is at most

$$\epsilon_0(\|g\|\,\|\partial f\| + \|\partial g\|\,\|f\|) + \epsilon_0^2\|g\|\,\|f\| + (2\epsilon_0 + \epsilon_0^2)\|g\|\,\|f\| \le 5\epsilon_0\|g\|_1\|f\|_1.$$

Indeed, the sum in parentheses is at most $\|g\|_1\|f\|_1$ by the two-dimensional Cauchy–Schwarz inequality, and $\epsilon_0 \le 1$. If $5\epsilon_0 \le 1/6$, lemma 17.5 therefore gives

$$\langle f, \Delta'_\theta f\rangle \ge \tfrac{1}{6}\|f\|_1^2.$$

The inverse is G'_θ by conjugation. Setting $f = \mathsf{G}'_\theta g$ in this inequality yields $\|\mathsf{G}'_\theta g\|_1 \le 6\|g\|$. The resolvent identity reads

$$\mathsf{G}'_\theta - \mathsf{G}' = \mathsf{G}'(\Delta' - \Delta'_\theta)\mathsf{G}'_\theta.$$

Pairing with an arbitrary scalar h, using the symmetry of G', and applying the preceding form estimate, we obtain

$$|\langle h, (\mathsf{G}'_\theta - \mathsf{G}')g\rangle| \le 5\epsilon_0\|\mathsf{G}'h\|_1\|\mathsf{G}'_\theta g\|_1 \le 90\epsilon_0\|h\|\,\|g\|.$$

The coarse inverse and the projection. We retain the order of the factors when estimating products. If $\|\mathcal{O}_j\| \le m_j$, $\|(\mathcal{O}_j)_\theta\| \le 2m_j$ and $\|(\mathcal{O}_j)_\theta - \mathcal{O}_j\| \le \rho_j$, then

$$\prod_{j=1}^{r}(\mathcal{O}_j)_\theta - \prod_{j=1}^{r}\mathcal{O}_j = \sum_{i=1}^{r}\left(\prod_{j<i}(\mathcal{O}_j)_\theta\right)((\mathcal{O}_i)_\theta - \mathcal{O}_i)\left(\prod_{j>i}\mathcal{O}_j\right).$$

Thus

$$\left\|\prod_{j=1}^{r}(\mathcal{O}_j)_\theta - \prod_{j=1}^{r}\mathcal{O}_j\right\| \le \sum_{i=1}^{r} 2^{i-1}\rho_i \prod_{j\neq i} m_j. \tag{172}$$

Apply (172) to $\mathsf{M} = \mathsf{Q}'\mathsf{G}'^2\mathsf{Q}'^*$, using $(m_i) = (1,3,3,1)$ and $(\rho_i) = \epsilon_0(1,90,90,1)$. The preceding bounds give $\|(\mathcal{O}_i)_\theta\| \le 2m_i$, and hence

$$\|\mathsf{M}_\theta - \mathsf{M}\| \le 1701\epsilon_0 \le 1755\epsilon_0.$$

If $1755\epsilon_0 \le \gamma'/2$, then $\mathsf{M} \ge \gamma'$ and the preceding estimate imply $\|\mathsf{M}_\theta\omega\| \ge (\gamma'/2)\|\omega\|$. Thus $\|\mathsf{M}_\theta^{-1}\| \le 2\gamma'^{-1}$, and

$$\mathsf{M}_\theta^{-1} - \mathsf{M}^{-1} = \mathsf{M}_\theta^{-1}(\mathsf{M} - \mathsf{M}_\theta)\mathsf{M}^{-1}$$

gives the inverse-difference bound in (ii).

Now suppose $c_P\epsilon_0 \le 1$. Since $0 < \gamma' \le 1$ and $c_P = 548208\gamma'^{-2}$, both scalar thresholds in (ii) hold. Apply (172) to $\mathsf{P} = \mathsf{G}'\mathsf{Q}'^*\mathsf{M}^{-1}\mathsf{Q}'\mathsf{G}'$ with

$$(m_i) = (3,1,\gamma'^{-1},1,3), \qquad (\rho_i) = \epsilon_0(90,1,3510\gamma'^{-2},1,90).$$

Bounding 2^{i-1} by 16 gives

$$\|\mathsf{P}_\theta - \mathsf{P}\| \le 144\gamma'^{-1}(62 + 3510\gamma'^{-1})\epsilon_0 \le c_P\epsilon_0.$$

The last inequality uses $\gamma' \le 1$. Since P is an orthogonal projection, $\|\mathsf{P}_\theta\| \le 1 + c_P\epsilon_0 \le 2$.

The vector form. By the Hodge identity used in section 17.3,

$$\Delta_a = \nabla^*\nabla - \partial\mathsf{P}\partial^* + a\mathsf{Q}^*\mathsf{Q}.$$

Conjugation and the adjoint rule therefore give

$$\begin{aligned}\langle A', \Delta_{a,\theta}A\rangle = {}& \langle \nabla_{-\theta}A', \nabla_\theta A\rangle \\ & - \langle (\partial^*)_{-\theta}A', \mathsf{P}_\theta(\partial^*)_\theta A\rangle + a\langle \mathsf{Q}_{-\theta}A', \mathsf{Q}_\theta A\rangle.\end{aligned}$$

After subtracting the unweighted terms, the first term contributes at most $(\epsilon_0 + \epsilon_0^2)\|A'\|_1\|A\|_1 \le 2\epsilon_0\|A'\|_1\|A\|_1$. The last contributes at most $a(2\epsilon_0 + \epsilon_0^2)\|A'\|\,\|A\| \le 3a\epsilon_0\|A'\|_1\|A\|_1$. For the middle term set

$$f = \partial^*A, \quad f' = \partial^*A', \qquad r = ((\partial^*)_\theta - \partial^*)A, \quad r' = ((\partial^*)_{-\theta} - \partial^*)A'.$$

The Hodge identity implies $\|f\| \le \|\nabla A\|$, and (i) gives $\|r\| \le \epsilon_0\|A\|$; the analogous bounds hold for f' and r'. Expanding the difference with these four terms bounds it by

$$\begin{aligned}& c_P\epsilon_0\|f'\|\,\|f\| + 2(\|r'\|\,\|f\| + \|f'\|\,\|r\| + \|r'\|\,\|r\|) \\ & \qquad \le (c_P + 2 + 2\epsilon_0)\epsilon_0\|A'\|_1\|A\|_1 \le (c_P + 4)\epsilon_0\|A'\|_1\|A\|_1.\end{aligned}$$

Adding the three estimates gives $c_V = c_P + 6 + 3a$, as claimed. □

17.5. **The compressed inverse and its decay.**

Proof of proposition 17.2. For $A \in \mathcal{H}_S$, compression does not change its quadratic form:

$$\langle A, \Delta_a(S)A\rangle = \langle A, \Delta_a A\rangle \geq \gamma_{0,\mathrm{B}}\|A\|_1^2$$

by proposition 17.1. The operator is symmetric on the finite-dimensional space $\mathcal{H}_S$ and is positive definite there. It therefore has a symmetric positive inverse on that space. This includes $S = \varnothing$, where $\mathcal{H}_S$ is the zero space. Extending the inverse by zero gives

$$G(S) = G(S)\chi_S = \chi_S G(S)\chi_S.$$

For a one-form J, set $A = G(S)J$. Then $\Delta_a(S)A = \chi_S J$, and hence

$$\gamma_{0,\mathrm{B}}\|A\|_1^2 \leq \langle A, J\rangle \leq \|A\|_1\|J\|.$$

For a rank-two tensor J, instead set $A = G(S)\nabla^* J$. The same argument, followed by summation by parts on the whole torus, gives

$$\gamma_{0,\mathrm{B}}\|A\|_1^2 \leq \langle A, \nabla^* J\rangle = \langle \nabla A, J\rangle \leq \|A\|_1\|J\|.$$

Both cases yield $\|A\|_1 \leq \gamma_{0,\mathrm{B}}^{-1}\|J\|$, which proves the four bounds in (b). The derivative in the second case is that of the ambient zero extension; no derivative intrinsic to S occurs.

For the decay estimate, use the rate δ_0 in (163) and set

$$\epsilon_0 = 8\delta_0 = \frac{\gamma_{0,\mathrm{B}}}{2c_V}.$$

Since $\gamma_{0,\mathrm{B}} \leq 1$ and $1 \leq c_P \leq c_V$, we have $\delta_0 \leq 1/8$ and $c_P\epsilon_0 \leq 1$. Let θ be any real function with Lipschitz constant at most δ_0. Multiplication by $e^{\pm\theta}$ preserves $\mathcal{H}_S$ and its orthogonal complement. Thus

$$\Delta_a(S)_\theta = \chi_S \Delta_{a,\theta}\chi_S|_{\mathcal{H}_S}, \qquad G(S)_\theta = e^{-\theta}G(S)e^{\theta}$$

are inverse on $\mathcal{H}_S$, and $G(S)_\theta$ is zero on $\mathcal{H}_S^\perp$. By lemma 17.6(iii), for $A \in \mathcal{H}_S$,

$$\begin{aligned}\langle A, \Delta_a(S)_\theta A\rangle &\geq \langle A, \Delta_a(S)A\rangle - c_V\epsilon_0\|A\|_1^2 \\ &\geq \tfrac{1}{2}\gamma_{0,\mathrm{B}}\|A\|_1^2.\end{aligned}$$

Take $\mathcal{O}_2 = 1$ or ∇^* and set $A = G(S)_\theta(\mathcal{O}_2)_\theta J$, where J is a one-form or a rank-two tensor, respectively. Then $A \in \mathcal{H}_S$ and

$$\Delta_a(S)_\theta A = \chi_S(\mathcal{O}_2)_\theta J.$$

When $\mathcal{O}_2 = 1$, the pairing of the right-hand side with A is at most $\|A\|\,\|J\|$ in absolute value. When $\mathcal{O}_2 = \nabla^*$, the adjoint conjugation rule and lemma 17.6(i) instead give

$$\begin{aligned}\big|\langle A, (\nabla^*)_\theta J\rangle\big| &= \big|\langle \nabla_{-\theta}A, J\rangle\big| \\ &\leq (\|\nabla A\| + \epsilon_0\|A\|)\|J\| \leq \sqrt{2}\,\|A\|_1\|J\|.\end{aligned}$$

Consequently, in both cases, $\|A\|_1 \leq 2\sqrt{2}\,\gamma_{0,\mathrm{B}}^{-1}\|J\|$. For $\mathcal{O}_1 = 1$ or ∇, part (i) also gives

$$\|(\mathcal{O}_1)_\theta A\| \leq \sqrt{2}\,\|A\|_1.$$

We have therefore proved all four conjugated bounds

$$\big\|(\mathcal{O}_1)_\theta G(S)_\theta(\mathcal{O}_2)_\theta\big\| \leq 4\gamma_{0,\mathrm{B}}^{-1}.$$

Finally, for the given nonempty set U' choose

$$\theta(x) = -\delta_0\,\mathrm{dist}(x, U').$$

This function has Lipschitz constant at most δ_0. It vanishes on U' and satisfies $\theta(x) \leq -\delta_0\,\mathrm{dist}(U, U')$ for $x \in U$. Undoing the conjugation gives

$$\chi_U\mathcal{O}_1 G(S)\mathcal{O}_2\chi_{U'} = \chi_U e^{\theta}(\mathcal{O}_1)_\theta G(S)_\theta(\mathcal{O}_2)_\theta e^{-\theta}\chi_{U'}.$$

The two exterior multiplication operators have norms at most $e^{-\delta_0 \operatorname{dist}(U,U')}$ and 1, respectively. The preceding four bounds now imply (165). □

18. Interior estimates for the scalar equation

We next obtain pointwise and Hölder bounds from a local L^2 bound and the lattice Poisson equation. Fix $\xi > 0$ and work on $\xi\mathbb{Z}^3$ with real functions and real fields $F = (F_\mu)_{\mu=1}^3$. The difference operators and the positive Laplacian are

$$\partial_\mu f(x) = \xi^{-1}(f(x+\xi e_\mu) - f(x)), \qquad \partial_\mu^* f(x) = \xi^{-1}(f(x-\xi e_\mu) - f(x)),$$
$$\nabla^* F = \sum_\mu \partial_\mu^* F_\mu, \quad \Delta = \sum_\mu \partial_\mu^* \partial_\mu.$$

We write $\nabla f = (\partial_\mu f)_\mu$, $|x|_1 = \sum_\mu |x_\mu|$ and $|x|_\infty = \max_\mu |x_\mu|$. For a set X of sites put

$$\|f\|_X^2 = \xi^3 \sum_{x\in X} |f(x)|^2, \qquad |f|_X = \sup_{x\in X} |f(x)|.$$

Write $|f| = |f|_{\xi\mathbb{Z}^3}$ and $|F| = \max_\mu |F_\mu|$ for the global suprema. A function on a subset is extended by zero when evaluated outside that subset.

Proposition 18.1. There are an absolute constant $c_I > 0$ and constants $c_I(\alpha) > 0$, depending only on $0 < \alpha < 1$, with the following property. Let $r \geq 16\xi$, let $x_0 \in \xi\mathbb{Z}^3$, and set

$$\mathsf{B}_s = \{x \in \xi\mathbb{Z}^3 : |x - x_0|_\infty \leq s\}.$$

Suppose that u is a function on B_r, that g and the field F are defined on $\xi\mathbb{Z}^3$, and that

$$\Delta u = g + \nabla^* F \qquad \text{on } \mathsf{B}_{r-\xi}.$$

Put

$$\mathsf{N} = r^{-3/2}\|u\|_{\mathsf{B}_r} + r^2 |g|_{\mathsf{B}_r} + r|F|.$$

For all $x, x' \in \mathsf{B}_{r/4}$ the following estimates hold.

(a) For every $0 < \alpha < 1$,

$$|u(x)| \leq c_I \mathsf{N}, \qquad |u(x) - u(x')| \leq c_I(\alpha) \left(\frac{|x-x'|_1}{r}\right)^\alpha \mathsf{N}.$$

(b) If $F = 0$, then for every direction ν and every $0 < \alpha < 1$,

$$r|\partial_\nu u(x)| \leq c_I \mathsf{N}, \qquad r|\partial_\nu u(x) - \partial_\nu u(x')| \leq c_I(\alpha) \left(\frac{|x-x'|_1}{r}\right)^\alpha \mathsf{N}.$$

The constants are independent of ξ, r, x_0, u, g and F. ■

The proof is given in section 18.2. Multiplication by a cutoff produces a finitely supported function whose Laplacian is the prescribed forcing plus terms supported in an annulus away from $\mathsf{B}_{r/4}$. The free kernel constructed below represents both contributions. The kernel and its first differences have the local summability bounds needed for the forcing term. Separation from the annulus permits further differences of the other term, with bounds in terms of $\|u\|_{\mathsf{B}_r}$ and $|F|$. When $F = 0$, the potential of the local forcing also has the derivative bounds in part (b).

18.1. **The free scalar kernel.** For functions k, g on $\xi\mathbb{Z}^3$ write

$$(k * g)(x) = \xi^3 \sum_{z \in \xi\mathbb{Z}^3} k(x - z) g(z)$$

whenever the sum converges absolutely. A mixed difference ∂^β denotes any product of $|\beta|$ operators chosen from the forward and adjoint differences $\partial_\mu, \partial_\mu^*$; the empty product is the identity.

Lemma 18.2. There are an absolute constant $c_K > 0$ and constants $c_K(\alpha) > 0$, depending only on $0 < \alpha < 1$, such that for every $\xi > 0$ there is a function $C_{0,\mathrm{lap}}$ on $\xi\mathbb{Z}^3$ with the following properties.

(i) For every finitely supported function v,

$$C_{0,\mathrm{lap}} * (\Delta v) = v.$$

(ii) For every mixed difference of order $|\beta| \le 3$,

$$|\partial^\beta C_{0,\mathrm{lap}}(x)| \le c_K (|x|_1 + \xi)^{-1-|\beta|} \qquad (x \in \xi\mathbb{Z}^3).$$

(iii) For $s' \ge s \ge \xi$ and $p \in \{1, 2\}$,

$$\sum_{|z|_1 \le s} \xi^3 (|z|_1 + \xi)^{-p} \le c_K s^{3-p},$$

$$\sum_{s < |z|_1 \le s'} \xi^3 |z|_1^{-3} \le c_K (1 + \log(s'/s)).$$

(iv) Suppose that a kernel k satisfies

$$|k(z)| \le (|z|_1 + \xi)^{-2}, \qquad |\partial_\nu k(z)| \le (|z|_1 + \xi)^{-3} \qquad (z \in \xi\mathbb{Z}^3,\ 1 \le \nu \le 3).$$

Let $r \ge \xi$, let $x, x' \in \xi\mathbb{Z}^3$ satisfy $|x - x'|_1 \le 3r$, and let G vanish whenever $|z - x|_1 > 6r$. Then

$$|(k * G)(x)| \le c_K r |G|,$$

$$|(k * G)(x) - (k * G)(x')| \le c_K(\alpha)\, r \left(\frac{|x - x'|_1}{r} \right)^\alpha |G| \qquad (0 < \alpha < 1).$$

■

Proof. The letter c denotes an absolute constant that may change between estimates. We choose c_K large enough to cover the finitely many bounds below.

The one-dimensional heat kernel. Put $\mathrm{i} = \sqrt{-1}$ and $\varpi(k) = 2\xi^{-2}(1 - \cos(\xi k))$. For $a, a' \ge 0$ integers with $a + a' \le 3$, define

$$q_T^{a,a'}(s) = \frac{1}{2\pi} \int_{-\pi/\xi}^{\pi/\xi} e^{\mathrm{i}ks} \left[\xi^{-1}(e^{\mathrm{i}\xi k} - 1)\right]^a \left[\xi^{-1}(e^{-\mathrm{i}\xi k} - 1)\right]^{a'} e^{-T\varpi(k)}\, dk.$$

Here $s \in \xi\mathbb{Z}$ and $T \ge 0$. Write $q_T = q_T^{0,0}$. The forward and adjoint differences on $\xi\mathbb{Z}$ multiply $e^{\mathrm{i}ks}$ by the two factors in brackets, so $q_T^{a,a'} = \partial^a \partial^{*a'} q_T$. Differentiation under the integral on its compact interval gives

$$\frac{d}{dT} q_T = -\partial^* \partial q_T, \qquad q_0(s) = \xi^{-1} \mathbf{1}_{\{0\}}(s).$$

The contour estimate. For $T > 0$ set $\vartheta = \max\{T, \xi^2\}$ and $\sigma = \mathrm{sign}(s)\vartheta^{-1/2}$. The integrand defining $q_T^{a,a'}$ is entire in k and has period $2\pi/\xi$, since s/ξ is an integer. Cauchy's theorem on a rectangle, whose vertical sides cancel, therefore permits the replacement $k \mapsto k + \mathrm{i}\sigma$. Here $|\xi\sigma| \le 1$, $T\sigma^2 \le 1$, and

$$|e^{\mathrm{i}(k + \mathrm{i}\sigma)s}| = e^{-|s|/\sqrt{\vartheta}}.$$

Since $(\cosh b - 1)/b^2$ increases with $|b|$, we have

$$\begin{aligned}\operatorname{Re}\varpi(k+\mathrm{i}\sigma) &= \varpi(k)\cosh(\xi\sigma) - 2\xi^{-2}(\cosh(\xi\sigma)-1)\\ &\ge \varpi(k) - 2(\cosh 1 - 1)\sigma^2.\end{aligned}$$

Consequently $|e^{-T\varpi(k+\mathrm{i}\sigma)}| \le e^2 e^{-T\varpi(k)}$. For either choice of sign,

$$\begin{aligned}\left|\xi^{-1}(e^{\pm\mathrm{i}\xi(k+\mathrm{i}\sigma)}-1)\right| &\le \xi^{-1}(e|e^{\pm\mathrm{i}\xi k}-1| + |e^{\mp\xi\sigma}-1|)\\ &\le e(\varpi(k)^{1/2} + \vartheta^{-1/2}).\end{aligned}$$

For $|k| \le \pi/\xi$,

$$\frac{2}{\pi}|k| \le \varpi(k)^{1/2} = \frac{2}{\xi}\left|\sin\frac{\xi k}{2}\right| \le |k|.$$

It follows that

$$\begin{aligned}|q_T^{a,a'}(s)| &\le e^5 e^{-|s|/\sqrt{\vartheta}} \int_{-\pi/\xi}^{\pi/\xi} (|k| + \vartheta^{-1/2})^{a+a'} e^{-4Tk^2/\pi^2}\,dk\\ &\le c\,\vartheta^{-(1+a+a')/2} e^{-|s|/\sqrt{\vartheta}}.\end{aligned}$$

For the last inequality, when $T \ge \xi^2$ substitute $k = T^{-1/2}k'$ and bound the resulting Gaussian integral over the real line. When $T < \xi^2$, the integrand is at most $((\pi+1)/\xi)^{a+a'}$ on an interval of length $2\pi/\xi$. These bounds are uniform for $a + a' \le 3$.

The three-dimensional heat kernel. Define $p_T(x) = \prod_{\mu=1}^3 q_T(x_\mu)$. Then

$$\frac{d}{dT}p_T = -\Delta p_T, \qquad p_0(x) = \xi^{-3}\mathbf{1}_{\{0\}}(x).$$

A mixed difference acts on the μth factor by $\partial^{a_\mu}\partial^{*a'_\mu}$, where $\sum_\mu (a_\mu + a'_\mu) = |\beta|$. Multiplying the one-dimensional estimates yields

$$\begin{aligned}|\partial^\beta p_T(x)| &\le c\,\vartheta^{-(3+|\beta|)/2} e^{-|x|_1/\sqrt{\vartheta}},\\ \vartheta &= \max\{T, \xi^2\}, \quad |\beta| \le 3.\end{aligned} \tag{173}$$

Time integration and the inverse identity. Define

$$C_{0,\mathrm{lap}}(x) = \int_0^\infty p_T(x)\,dT$$

and put $\varkappa = 3 + |\beta| \in \{3,4,5,6\}$. By (173),

$$\int_0^{\xi^2} |\partial^\beta p_T(x)|\,dT \le c\,\xi^{2-\varkappa} e^{-|x|_1/\xi}.$$

For $x \ne 0$, the substitution $T = |x|_1^2 u^{-2}$ gives

$$\begin{aligned}\int_{\xi^2}^\infty |\partial^\beta p_T(x)|\,dT &\le c\int_0^\infty T^{-\varkappa/2} e^{-|x|_1/\sqrt{T}}\,dT\\ &= 2c|x|_1^{2-\varkappa}\int_0^\infty u^{\varkappa-3}e^{-u}\,du \le c|x|_1^{2-\varkappa}.\end{aligned}$$

For $x = 0$, the same integral is bounded by

$$c\int_{\xi^2}^\infty T^{-\varkappa/2}\,dT = \frac{2c}{\varkappa-2}\xi^{2-\varkappa}.$$

All these integrals converge since $\varkappa \ge 3$. For a nonzero lattice site, $|x|_1 \ge \xi$; moreover, $(1+b)^{\varkappa-2}e^{-b}$ is bounded for $b \ge 0$, uniformly over the four possible values of $\varkappa$. Combining the three estimates

proves part (ii). They also show absolute convergence of the integral defining $C_{0,\mathrm{lap}}$ and of the integrals of its mixed differences, and hence

$$\partial^\beta C_{0,\mathrm{lap}}(x) = \int_0^\infty \partial^\beta p_T(x)\, dT.$$

If v is finitely supported, shifting the summation index gives $(\Delta p_T) * v = p_T * (\Delta v)$, so

$$\int_0^{T_0} p_T * (\Delta v)\, dT = -\int_0^{T_0} \frac{d}{dT}(p_T * v)\, dT = v - p_{T_0} * v.$$

For $T_0 \geq \xi^2$, (173) gives

$$|p_{T_0} * v| \leq c T_0^{-3/2} \xi^3 \sum_z |v(z)| \longrightarrow 0 \qquad (T_0 \longrightarrow \infty).$$

Letting $T_0 \to \infty$ proves part (i).

Lattice shell sums. For $s \geq \xi$, the number of sites with $|z|_1 \leq s$ is at most $(2s/\xi + 1)^3 \leq (3s/\xi)^3$. Thus a shell $s_1 < |z|_1 \leq 2s_1$, with $2s_1 \geq \xi$, contributes at most $216 s_1^{3-p}$ to $\sum_z \xi^3 |z|_1^{-p}$. Cover $0 < |z|_1 \leq s$ by the shells with $s_1 = 2^{-i-1}s$, $i \geq 0$; those with $2s_1 < \xi$ are empty. For $p = 1, 2$,

$$\sum_{i \geq 0} (2^{-i-1}s)^{3-p} \leq s^{3-p}.$$

For nonzero z, replace $(|z|_1 + \xi)^{-p}$ by the larger quantity $|z|_1^{-p}$, and add the contribution $\xi^{3-p} \leq s^{3-p}$ at $z = 0$. This proves the first estimate in part (iii). The annulus $s < |z|_1 \leq s'$ is covered by at most $1 + \log_2(s'/s)$ shells with $s_1 = 2^i s$. Each contributes at most 216 when $p = 3$, which proves the second estimate in part (iii).

The convolution modulus. The first estimate in part (iv) follows from part (iii) with $p = 2$ and $s = 6r$. For the second, the case $x = x'$ is immediate. Otherwise put $\delta = |x - x'|_1 \geq \xi$. For $|z - x|_1 \leq 2\delta$, we also have $|z - x'|_1 \leq 3\delta$. Applying part (iii) to these two balls bounds their contribution to the convolution difference by $5 c_K \delta |G|$.

For $|z - x|_1 > 2\delta$, choose a shortest lattice path $x = x^0, x^1, \ldots, x^{\delta/\xi} = x'$. Every point x'' on this path satisfies

$$|x'' - z|_1 \geq |x - z|_1 - \delta \geq \tfrac{1}{2}|x - z|_1.$$

Each increment $k(x^{i+1} - z) - k(x^i - z)$ equals $\pm \xi \partial_\nu k$ at one of its endpoints. Summing the increments therefore gives

$$|k(x - z) - k(x' - z)| \leq 8\delta |x - z|_1^{-3}.$$

Since $G(z) = 0$ for $|z - x|_1 > 6r$, the second estimate in part (iii), with $s = 2\delta$ and $s' = 6r$, bounds this contribution by

$$8 c_K \delta \big(1 + \log^+(3r/\delta)\big)|G|, \qquad \log^+ t = \max\{\log t, 0\}.$$

Here $2\delta \leq 6r$ by hypothesis. Writing $b = \delta/r \in (0, 3]$, we conclude that

$$\begin{aligned} |(k * G)(x) - (k * G)(x')| &\leq c_K \delta \big[5 + 8(1 + \log^+(3r/\delta))\big]|G| \\ &= c_K r b^\alpha\, b^{1-\alpha} \big[5 + 8(1 + \log^+(3/b))\big]|G|. \end{aligned}$$

For $0 < \alpha < 1$, the last factor depending on b is bounded on $(0, 3]$ by a constant depending only on α. This proves part (iv). □

18.2. **The cutoff argument.**

Proof of proposition 18.1. Use the fixed cutoff η_0 defined by

$$\phi(s) = \begin{cases} e^{-1/s}, & s > 0, \\ 0, & s \le 0, \end{cases} \qquad \vartheta_0(s) = \frac{\phi(s)}{\phi(s) + \phi(1-s)},$$
$$\eta_0(s) = \vartheta_0(4 - 6|s|).$$

Every derivative of $e^{-1/s}$ is a polynomial in $1/s$ times $e^{-1/s}$ and tends to zero as $s \downarrow 0$, so ϕ is smooth. The denominator in ϑ_0 is positive, and the resulting fixed smooth cutoff satisfies $0 \le \eta_0 \le 1$, equals one on $[-1/2, 1/2]$, and vanishes outside $[-2/3, 2/3]$. It is constant near zero, so composition with $|s|$ preserves smoothness. Its derivative bounds are absolute. In the proof, c denotes absolute constants and $c(\alpha)$ constants depending only on α; their values may increase.

The localized equation. Write $(\tau_{\pm\mu} h)(x) = h(x \pm \xi e_\mu)$. Set

$$\zeta(x) = \prod_{\mu=1}^{3} \eta_0((x_\mu - x_{0,\mu})/r).$$

Extend u by zero outside B_r and put $v = \zeta u$. Then $\zeta = 1$ on $\mathsf{B}_{r/2}$ and vanishes outside $\mathsf{B}_{2r/3}$. The mean value theorem and the second-difference formula give

$$|\partial_\mu \zeta| \le \|\eta_0'\|_\infty r^{-1}, \qquad |\Delta \zeta| \le 3\|\eta_0''\|_\infty r^{-2}.$$

Since $\xi \le r/16$, the functions $\partial_\mu \zeta$, $\tau_{-\mu}\partial_\mu \zeta$ and $\Delta\zeta$ vanish outside the annulus

$$\mathsf{Z} = \{z \in \xi\mathbb{Z}^3 : 7r/16 \le |z - x_0|_\infty \le 3r/4\}.$$

We use the product identities with their shifts written explicitly. If $x' \sim x$ denotes the six sites $x \pm \xi e_\mu$, then $\Delta f(x) = \xi^{-2} \sum_{x' \sim x} (f(x) - f(x'))$. For functions h, ψ and a field H this gives

$$\Delta(h\psi)(x) = h(x)\Delta\psi(x) - \xi^{-2} \sum_{x' \sim x} (h(x') - h(x))\psi(x'), \tag{174}$$

and

$$\begin{aligned} \xi^{-2} \sum_{x' \sim x} (h(x') - h(x))\psi(x') = & -\sum_{\mu=1}^{3} \partial_\mu^* \big[(\partial_\mu h)(\psi + \tau_\mu \psi)\big](x) \\ & + \psi(x)\Delta h(x). \end{aligned} \tag{175}$$

To check the second identity, fix μ and write $e = \xi e_\mu$. The corresponding adjoint difference is

$$\begin{aligned} &\partial_\mu^* \big[(\partial_\mu h)(\psi + \tau_\mu \psi)\big](x) \\ &\quad = \xi^{-2} \big[(h(x) - h(x-e))(\psi(x-e) + \psi(x)) \\ &\qquad\qquad - (h(x+e) - h(x))(\psi(x) + \psi(x+e))\big]. \end{aligned}$$

Subtracting the sum of these expressions from $\psi(x)\Delta h(x)$ cancels all terms containing $\psi(x)$ and leaves the left side of (175). The divergence product identity is

$$h\, \partial_\mu^* H_\mu = \partial_\mu^*(hH_\mu) + \tau_{-\mu}\big((\partial_\mu h) H_\mu\big). \tag{176}$$

Indeed, the difference of its first two terms at x is $\xi^{-1}(h(x) - h(x-e))H_\mu(x-e)$, which is its last term.

Apply (174) and (175) with $h = \zeta$, $\psi = u$. On the support of ζ the equation for u holds, since $2r/3 \le r - \xi$. Using (176) for its divergence term gives the equation on the whole lattice

$$\begin{aligned}\Delta v &= \zeta g + G_0 + \sum_{\mu=1}^{3} \partial_\mu^*(\zeta F_\mu + G_\mu),\\ G_0 &= \sum_{\mu=1}^{3} \tau_{-\mu}((\partial_\mu \zeta) F_\mu) - u\Delta\zeta,\\ G_\mu &= (\partial_\mu \zeta)(u + \tau_\mu u).\end{aligned}$$

All terms on the right have finite support. By lemma 18.2(i), and a shift of the convolution sum,

$$C_{0,\mathrm{lap}} * (\partial_\mu^* G) = (\partial_\mu^* C_{0,\mathrm{lap}}) * G \qquad \text{for finitely supported } G.$$

Consequently

$$\begin{aligned}v &= v_1 + v_2,\\ v_1 &= C_{0,\mathrm{lap}} * (\zeta g) + \sum_{\mu=1}^{3} (\partial_\mu^* C_{0,\mathrm{lap}}) * (\zeta F_\mu),\\ v_2 &= C_{0,\mathrm{lap}} * G_0 + \sum_{\mu=1}^{3} (\partial_\mu^* C_{0,\mathrm{lap}}) * G_\mu.\end{aligned}$$

The terms supported in the annulus. The functions G_0, G_μ vanish outside Z. There are at most $(3r/\xi)^3$ sites in B_r, so Cauchy–Schwarz gives

$$\xi^3 \sum_{\mathsf{B}_r} |u| \le (3r)^{3/2} \|u\|_{\mathsf{B}_r}.$$

The same bound applies to the shifted values of u occurring in G_μ, because the sites in question and their shifts belong to B_r. The derivative bounds for ζ therefore imply

$$\begin{aligned}\xi^3 \sum_{z} |G_0(z)| &\le c(r^2|F| + r^{-1/2}\|u\|_{\mathsf{B}_r}) \le cr\,\mathsf{N},\\ \xi^3 \sum_{z} |G_\mu(z)| &\le cr^{1/2}\|u\|_{\mathsf{B}_r} \le cr^2\,\mathsf{N}.\end{aligned}$$

If $x \in \mathsf{B}_{3r/8}$ and $z \in \mathsf{Z}$, then $|x-z|_1 \ge |x-z|_\infty \ge r/16$. Differences commute with these convolutions. For any product ∂^β of at most two forward or adjoint differences, lemma 18.2(ii) gives on this set

$$\begin{aligned}|(\partial^\beta C_{0,\mathrm{lap}})(x-z)| &\le c_K (r/16)^{-1-|\beta|},\\ |(\partial^\beta \partial_\mu^* C_{0,\mathrm{lap}})(x-z)| &\le c_K (r/16)^{-2-|\beta|}.\end{aligned}$$

The second bound uses kernel differences of order $|\beta| + 1 \le 3$. Combining them with the two sum bounds yields

$$r^{|\beta|}|(\partial^\beta v_2)(x)| \le c\,\mathsf{N} \qquad (x \in \mathsf{B}_{3r/8},\ |\beta| \le 2).$$

Join $x, x' \in \mathsf{B}_{r/4}$ by a shortest lattice path inside that box. Summing the first differences of v_2, and then the second differences for $\partial_\nu v_2$, along the path gives

$$|v_2(x) - v_2(x')| + r|\partial_\nu v_2(x) - \partial_\nu v_2(x')| \le c\,\frac{|x-x'|_1}{r}\,\mathsf{N}.$$

Since $|x - x'|_1/r \le 3$, the last factor is at most $3^{1-\alpha}(|x-x'|_1/r)^\alpha$. Thus v_2 and its first differences have the required interior bounds.

The local right-hand side. For $x \in \mathsf{B}_{r/4}$, both ζg and ζF_μ vanish at sites z with $|z-x|_1 > 3r$. The kernel bound and the shell estimate lemma 18.2(ii),(iii), with $p=1$ and $s=3r$, imply

$$|(C_{0,\mathrm{lap}} * (\zeta g))(x)| \le cr^2 |g|_{\mathsf{B}_r}.$$

The normalized kernels $c_K^{-1}\partial_\nu C_{0,\mathrm{lap}}$ and $c_K^{-1}\partial_\mu^* C_{0,\mathrm{lap}}$ satisfy the hypotheses of lemma 18.2(iv), by its mixed-difference bounds. That clause applies with the same radius r: its support requirement is the weaker condition $|z-x|_1 \le 6r$, and $|x-x'|_1 \le 3r$ for $x, x' \in \mathsf{B}_{r/4}$. It gives

$$|\partial_\nu (C_{0,\mathrm{lap}} * (\zeta g))(x)| \le cr|g|_{\mathsf{B}_r},$$

$$|\partial_\nu (C_{0,\mathrm{lap}} * (\zeta g))(x) - \partial_\nu (C_{0,\mathrm{lap}} * (\zeta g))(x')| \le c(\alpha) r \left(\frac{|x-x'|_1}{r}\right)^\alpha |g|_{\mathsf{B}_r}.$$

For the divergence-source terms it gives

$$|((\partial_\mu^* C_{0,\mathrm{lap}}) * (\zeta F_\mu))(x)| \le cr|F|,$$

$$|((\partial_\mu^* C_{0,\mathrm{lap}}) * (\zeta F_\mu))(x) - ((\partial_\mu^* C_{0,\mathrm{lap}}) * (\zeta F_\mu))(x')| \le c(\alpha) r \left(\frac{|x-x'|_1}{r}\right)^\alpha |F|.$$

Summing the gradient bound for $C_{0,\mathrm{lap}} * (\zeta g)$ along a shortest path in $\mathsf{B}_{r/4}$ also gives its difference bound $cr|x-x'|_1|g|_{\mathsf{B}_r}$. Using $|x-x'|_1/r \le 3^{1-\alpha}(|x-x'|_1/r)^\alpha$ once more, and summing the three divergence-source terms, we obtain

$$|v_1(x)| \le c(r^2|g|_{\mathsf{B}_r} + r|F|),$$

$$|v_1(x) - v_1(x')| \le c(\alpha) \left(\frac{|x-x'|_1}{r}\right)^\alpha (r^2|g|_{\mathsf{B}_r} + r|F|).$$

When $F=0$, only $C_{0,\mathrm{lap}} * (\zeta g)$ remains in v_1. Its gradient and gradient-difference estimates above, multiplied by r, are bounded by $cr^2|g|_{\mathsf{B}_r}$ and $c(\alpha)(|x-x'|_1/r)^\alpha r^2|g|_{\mathsf{B}_r}$, respectively.

For $x \in \mathsf{B}_{r/4}$, both x and $x+\xi e_\nu$ belong to $\mathsf{B}_{r/2}$, where $u=v$. Thus $u = v_1 + v_2$ there and $\partial_\nu u = \partial_\nu v_1 + \partial_\nu v_2$ at every observation site. The bounds for v_1 and v_2 give (a), and their first-difference bounds give (b) when $F=0$. □

19. Regularity of the one-scale propagator

Fix integers $L \ge 2$ and $j \ge 0$, put $\xi = L^{-j}$, and let $N \ge 4$ be an even positive multiple of L. We work on

$$\mathsf{T} = \mathbb{R}^3/N\mathbb{Z}^3, \qquad \mathsf{T}_h = h\mathbb{Z}^3/N\mathbb{Z}^3 \quad (h = \xi, 1).$$

Distances $|x-y|_1$ and $|x-y|_\infty$ are the least lifted ℓ^1 and ℓ^∞ distances on this torus. All fields in this section are real. Write $\mathcal{F}_\xi^m$ for the m-component functions on T_ξ, and $\mathcal{A}_\xi = \mathcal{F}_\xi^3$ for the one-forms. We use the ξ^3-weighted inner products, summing over all components, and the associated norm $\|\cdot\|$. In pointwise estimates $|f(x)|$ denotes the maximum of the component moduli, and $|f| = \sup_x |f(x)|$.

The differences, scalar gradient, divergence and curl have the conventions of the preceding sections, now with spacing ξ. Thus $\partial_\mu f(x) = \xi^{-1}(f(x+\xi e_\mu) - f(x))$, ∇ differentiates every component, and $\Delta = \nabla^*\nabla$ acts componentwise. The adjoints of ∇ and $\nabla\nabla$ on one-forms are ∇^* and $\nabla^*\nabla^*$, with domains $\mathcal{F}_\xi^9$ and $\mathcal{F}_\xi^{27}$, respectively; these domains are tensor spaces.

Use the corner blocks and straight averages of (158), with $k=j$, $\eta=\xi$ and all three torus periods equal to N. We write these averages as Q'_j and Q_j, and the blocks as

$$B^j(y) = \mathsf{T}_\xi \cap (y + [0,1)^3), \qquad y \in \mathsf{T}_1.$$

Let R_j be the orthogonal projection in the scalar space onto $\Delta \ker \mathsf{Q}'_j$, and set $\mathsf{P}_j = 1 - \mathsf{R}_j$. The unit-weight operator and its inverse are

$$\Delta_{1,j} = \partial^*\partial + \partial \mathsf{R}_j \partial^* + \mathsf{Q}_j^* \mathsf{Q}_j, \qquad G_j = \Delta_{1,j}^{-1}.$$

The projection is formed on the whole torus. The inverse exists by proposition 17.1, applied with the parameters just specified and $a = 1$.

For $u \in \mathsf{T}$ and $r > 0$ put $U_r(u) = \{x \in \mathsf{T}_\xi : |x - u|_\infty \le r\}$. For $0 \le \alpha < 1$ define

$$\|f\|_\alpha = \sup_{\substack{x,x' \in \mathsf{T}_\xi \\ 0<|x-x'|_1 \le 1}} |x - x'|_1^{-\alpha} |f(x) - f(x')|,$$

with value zero if there are no such pairs. In particular, $\|f\|_0$ is an oscillation seminorm. Let γ_{f} be $\gamma_{0,\mathrm{B}}$ of (161) at $a = 1$, let δ_{f} be δ_0 of (163) at $a = 1$, and let γ' be the scalar constant in (160). Set

$$\delta_P = \frac{\gamma'}{28080}, \qquad \delta_* = \tfrac{1}{2} \min\{\delta_{\mathrm{f}}, \delta_P\}.$$

Proposition 19.1. There are positive constants c, c_α $(0 \le \alpha < 1)$, c'_ε $(0 < \varepsilon < 1)$, and $c''_{\alpha,\varepsilon}$ $(\alpha \ge 0,\ \varepsilon > 0,\ \alpha + \varepsilon < 1)$, depending only on the displayed exponents, with the following properties. In particular, the constants are independent of L, j, N and of the centers below. Let $u, u' \in \mathsf{T}$, let the real function ζ vanish outside $U_1(u)$, and let $J \in \mathcal{A}_\xi$, $J' \in \mathcal{F}^9_\xi$ and $J'' \in \mathcal{F}^{27}_\xi$ vanish outside $U_1(u')$. Write $E = e^{-\delta_* |u - u'|_1}$. Then, for every $x \in U_1(u)$,

(I1) The value, gradient and Laplacian satisfy

$$|(G_j J)(x)|, \quad |(\nabla G_j J)(x)|, \quad |(\Delta G_j J)(x)| \le cE|J|, \qquad |(G_j \nabla^* J')(x)| \le cE|J'|.$$

(I2) For $0 \le \alpha < 1$,

$$\|\zeta \nabla G_j J\|_\alpha \le c_\alpha E(\|\zeta\|_\alpha + |\zeta|)|J|,$$
$$\|\zeta G_j \nabla^* J'\|_\alpha \le c_\alpha E(\|\zeta\|_\alpha + |\zeta|)|J'|.$$

(I3) For $0 < \varepsilon < 1$,

$$|(\nabla G_j \nabla^* J')(x)| \le c'_\varepsilon E(\|J'\|_\varepsilon + |J'|).$$

(I4) If $\alpha \ge 0$, $\varepsilon > 0$ and $\alpha + \varepsilon < 1$, then

$$\|\zeta \nabla G_j \nabla^* J'\|_\alpha \le c''_{\alpha,\varepsilon} E(\|\zeta\|_\alpha + |\zeta|)(\|J'\|_{\alpha+\varepsilon} + |J'|).$$

(I5) The six L^2 estimates are

$$\|\zeta G_j J\|, \quad \|\zeta \nabla G_j J\|, \quad \|\zeta \nabla \nabla G_j J\| \le cE|\zeta| \|J\|,$$
$$\|\zeta G_j \nabla^* J'\|, \quad \|\zeta \nabla G_j \nabla^* J'\| \le cE|\zeta| \|J'\|,$$
$$\|\zeta G_j \nabla^* \nabla^* J''\| \le cE|\zeta| \|J''\|.$$

■

These localized regularity and decay estimates have the form of Bałaban's estimates in [Bał84, Proposition 1.2]. We obtain them from the Hilbert-space bounds and scalar interior estimates proved above. For $f = G_j J$, the componentwise equation is

$$\Delta f = J + \partial \mathsf{P}_j \partial^* f - \mathsf{Q}_j^* \mathsf{Q}_j f.$$

The first task is to control the projection term by the decay of f and its first differences. This gives a local bound for Δf, and hence the second-difference L^2 estimate. Scalar interior estimates then give the pointwise and Hölder bounds. For a divergence source, one further derivative reaches the critical decay of the scalar kernel; the Hölder difference of the source supplies the needed cancellation. The proof of (I5) is in section 19.2; the proof of (I1)–(I4), and the choice of a common c, are completed in section 19.5.

19.1. **Decay of the scalar projection.** We first record the support of an adjoint-average column. It will also localize the term $\mathsf{Q}_j^*\mathsf{Q}_j f$ in the componentwise equation. For a coarse bond $b = \langle y, y+e_\mu\rangle$, let e_b be its unit coordinate vector. The straight average is

$$(\mathsf{Q}_j A)_\mu(y) = \sum_{x\in B^j(y)} \xi^3 \sum_{t=0}^{L^j-1} \xi A_\mu(x + t\xi e_\mu).$$

Collecting the coefficient of $A_\nu(x)$ gives

$$(\mathsf{Q}_j^* e_b)_\nu(x) = \begin{cases} \xi\,\#\{0 \le t < L^j : x - t\xi e_\mu \in B^j(y)\}, & \nu = \mu, \\ 0, & \nu \ne \mu, \end{cases} \tag{177}$$
$$0 \le (\mathsf{Q}_j^* e_b)_\mu(x) \le 1, \qquad \operatorname{supp}(\mathsf{Q}_j^* e_b) \subset y + [0,2)^3, \qquad \|\mathsf{Q}_j^* e_b\| \le 1.$$

The box in this formula is understood as its image on the torus. The coefficient is at most $\xi L^j = 1$; its integral is

$$\sum_x \xi^3 (\mathsf{Q}_j^* e_b)_\mu(x) = \sum_{x\in B^j(y)} \xi^3 \sum_{t=0}^{L^j-1} \xi = 1.$$

Its squared norm is therefore at most one. The support assertion follows directly from the paths in the average.

We use the following constants:

$$\begin{aligned} C_P &= 414\gamma'^{-1}, \qquad \sigma(\delta) = 3\coth^3(\delta/2) \quad (\delta > 0), \\ c_* &= e^{12} + 4\gamma_{\mathrm{f}}^{-1} e^{18}\bigl(13 + 4\sqrt{3} + \sqrt{3} C_P \sigma(\delta_*)\bigr). \end{aligned} \tag{178}$$

These constants and the decay rates depend on none of L, j, N. The explicit formulas for the preceding scalar gap and vector bounds give $\gamma', \gamma_{\mathrm{f}}, \delta_{\mathrm{f}}, \delta_P \le 1$. Here C_P is distinct from the perturbation constant c_P in (162).

Proposition 19.2. For nonempty sets $U, U' \subset \mathsf{T}_\xi$,

$$\begin{aligned} \|\chi_U \partial \mathsf{P}_j \chi_{U'}\| &\le C_P e^{-\delta_P \operatorname{dist}(U,U')}, \\ \|\chi_U \mathsf{P}_j \chi_{U'}\| &\le C_P e^{-\delta_P \operatorname{dist}(U,U')}, \\ \|\chi_U \Delta \mathsf{P}_j \chi_{U'}\| &\le 3 C_P e^{-\delta_P \operatorname{dist}(U,U')}. \end{aligned} \tag{179}$$

The norms are operator norms between the appropriate scalar and one-form spaces. For every scalar function g, the function $\Delta \mathsf{P}_j g$ is constant on each block, and

$$|\Delta \mathsf{P}_j g|_{B^j(y)} = \|\chi_{B^j(y)} \Delta \mathsf{P}_j g\|.$$

Suppose, in addition, that $0 < \delta \le \delta_P/2$, $A \ge 0$, $u' \in \mathsf{T}$ and

$$\|\chi_{B^j(y)} g\| \le A e^{-2\delta |y-u'|_1} \qquad (y \in \mathsf{T}_1).$$

Then, for every $u \in \mathsf{T}$, $r > 0$ and $y \in \mathsf{T}_1$,

$$\begin{aligned} \|\chi_{U_r(u)} \partial \mathsf{P}_j g\| &\le C_P e^{3(r+1)} \sigma(\delta) A e^{-\delta |u-u'|_1}, \\ \|\chi_{B^j(y)} \mathsf{P}_j g\| &\le C_P e^{6} \sigma(\delta) A e^{-\delta |y-u'|_1}, \\ |\Delta \mathsf{P}_j g|_{B^j(y)} &\le 3 C_P e^{6} \sigma(\delta) A e^{-\delta |y-u'|_1}. \end{aligned} \tag{180}$$

■

Proof. Use lemmas 17.4 and 17.5 and (170) with $\mathsf{Q}' = \mathsf{Q}'_j$, and put

$$\Delta' = \Delta + \mathsf{Q}'^{*}_j \mathsf{Q}'_j, \qquad \mathsf{G}' = (\Delta')^{-1}, \qquad \mathsf{M} = \mathsf{Q}'_j \mathsf{G}'^2 \mathsf{Q}'^{*}_j.$$

Then $\mathsf{M} \ge \gamma'$ and $\mathsf{P}_j = \mathsf{G}'\mathsf{Q}'^{*}_j \mathsf{M}^{-1} \mathsf{Q}'_j \mathsf{G}'$. For a δ_P-Lipschitz function θ, use the conjugation convention of section 17.4. Here

$$\epsilon_0 = 8\delta_P = \gamma'/3510, \qquad \delta_P \le \tfrac{1}{8}, \quad 5\epsilon_0 \le \tfrac{1}{6}, \quad 1755\epsilon_0 = \tfrac{1}{2}\gamma'.$$

Thus lemma 17.6(i),(ii) gives

$$\|\partial_\theta - \partial\| \le \epsilon_0, \qquad \|(\mathsf{Q}'_j)_\theta\|, \|(\mathsf{Q}'^{*}_j)_\theta\| \le 2,$$
$$\|\mathsf{G}'_\theta g\|_1 \le 6\|g\|, \qquad \|\mathsf{M}_\theta^{-1}\| \le 2\gamma'^{-1}.$$

Since $\epsilon_0 \le 1$, $\|\partial_\theta \mathsf{G}'_\theta g\| \le \sqrt{2}\|\mathsf{G}'_\theta g\|_1 \le 6\sqrt{2}\|g\|$. Conjugating the formula for P_j and multiplying these bounds yields

$$\|(\partial \mathsf{P}_j)_\theta\| \le 6\sqrt{2}\cdot 2\cdot 2\gamma'^{-1}\cdot 2\cdot 6 = 288\sqrt{2}\gamma'^{-1} \le C_P, \qquad \|(\mathsf{P}_j)_\theta\| \le 288\gamma'^{-1} \le C_P.$$

The Laplacian can be handled without a second derivative estimate for G'. Indeed, $\Delta = \Delta' - \mathsf{Q}'^{*}_j \mathsf{Q}'_j$ and $\Delta'\mathsf{G}' = 1$ give

$$\Delta \mathsf{P}_j = \mathsf{Q}'^{*}_j (\mathsf{M}^{-1}\mathsf{Q}'_j \mathsf{G}' - \mathsf{Q}'_j \mathsf{P}_j). \tag{181}$$

Consequently,

$$\|(\Delta \mathsf{P}_j)_\theta\| \le 2(2\gamma'^{-1}\cdot 2\cdot 6 + 2\cdot 288\gamma'^{-1}) = 1200\gamma'^{-1} \le 3C_P.$$

Choose $\theta(x) = -\delta_P \operatorname{dist}(x, U')$. Undoing conjugation as in section 17.5 multiplies each of these three operator bounds by at most $e^{-\delta_P \operatorname{dist}(U,U')}$. This proves (179). Formula (181) also places $\Delta \mathsf{P}_j g$ in the range of Q'^{*}_j, the functions constant on each block. Since $\xi^3 \# B^j(y) = 1$, its block supremum and block L^2 norm agree.

For the estimates with block inputs we need two elementary consequences of the torus geometry. First,

$$\operatorname{dist}(U_r(u), U_{r'}(u')) \ge |u - u'|_1 - 3(r + r') \tag{182}$$

whenever both sets are nonempty. This is the triangle inequality, since each point in $U_r(u)$ lies at ℓ^1 distance at most $3r$ from u. Second, for every $\lambda > 0$ and every continuous torus point u,

$$\sum_{y \in \mathsf{T}_1} e^{-\lambda |u-y|_1} \le \coth^3(\lambda/2) \le \sigma(\lambda).$$

To see this, choose least-distance integer lifts of the coarse sites; they are distinct, so the torus sum is bounded by the corresponding sum over $\mathbb{Z}^3$. For the fractional part $t \in [0,1]$ of one coordinate,

$$\sum_{n \in \mathbb{Z}} e^{-\lambda|n-t|} = \frac{e^{-\lambda t} + e^{-\lambda(1-t)}}{1 - e^{-\lambda}} \le \coth(\lambda/2).$$

The numerator is convex in t, and hence is largest at an endpoint. Taking the product over the three coordinates gives the assertion. The triangle inequality then gives the convolution bound

$$\sum_{y \in \mathsf{T}_1} e^{-2\delta|u-y|_1 - 2\delta|y-u'|_1}$$
$$\le e^{-\delta|u-u'|_1} \sum_{y \in \mathsf{T}_1} e^{-\delta|u-y|_1} \le \sigma(\delta) e^{-\delta|u-u'|_1}.$$

If $0 < \delta'' \le \delta' \le 1$, (182) implies

$$e^{-\delta' \operatorname{dist}(U_r(u), U_{r'}(y'))} \le e^{3(r+r')} e^{-\delta''|u-y'|_1}.$$

For $|u-y'|_1 \le 3(r+r')$ the right side is at least one; otherwise apply (182) and then $\delta' \ge \delta''$. Decompose $g = \sum_{y'} \chi_{B^j(y')} g$ and apply this inequality with $\delta' = \delta_P$, $\delta'' = 2\delta$ and $r' = 1$, using $B^j(y') \subset U_1(y')$. The first bound in (179) gives

$$\begin{aligned}\|\chi_{U_r(u)} \partial \mathsf{P}_j g\| &\le C_P e^{3(r+1)} A \sum_{y' \in \mathsf{T}_1} e^{-2\delta|u-y'|_1 - 2\delta|y'-u'|_1} \\ &\le C_P e^{3(r+1)} \sigma(\delta) A e^{-\delta|u-u'|_1}.\end{aligned}$$

If $U_r(u)$ is empty the assertion is immediate. Apply the other two bounds in (179) with $r = r' = 1$, and use $B^j(y) \subset U_1(y)$. This proves the remaining bounds in (180), with the block norm replaced by its equal supremum for $\Delta \mathsf{P}_j g$. □

19.2. **Localized second differences and the six L^2 bounds.** Four of the six estimates in (I5) follow directly from the first-order inverse bounds. The new estimate is the bound for $\nabla\nabla G_j$; its adjoint gives the sixth one. We obtain the second differences from the componentwise Laplacian, using a cutoff whose derivative bounds remain valid on the smallest permitted tori.

The specialization of proposition 17.2(c) to the full bond set and $a = 1$ is

$$\|\chi_U \mathcal{O}_1 G_j \mathcal{O}_2 \chi_{U'}\| \le 4\gamma_{\mathrm{f}}^{-1} e^{-\delta_{\mathrm{f}} \operatorname{dist}(U,U')}, \qquad \mathcal{O}_1 \in \{1, \nabla\}, \quad \mathcal{O}_2 \in \{1, \nabla^*\}, \tag{183}$$

for nonempty site sets U, U'. Its hypotheses hold with $k = j$, $\eta = \xi$ and periods N. This also identifies the inverse in that result with G_j.

We first establish a local second-difference estimate for an arbitrary one-form f. If $j = 0$, then

$$\|\chi_{U_1(u)} \nabla\nabla f\| \le 12 \|\chi_{U_3(u)} f\|.$$

If $j \ge 1$, then

$$\|\chi_{U_1(u)} \nabla\nabla f\| \le \|\chi_{U_3(u)} \Delta f\| + 12\|\chi_{U_4(u)} f\| + 4\sqrt{3}\|\chi_{U_4(u)} \nabla f\|. \tag{184}$$

For $j = 0$, the difference

$$\partial_\kappa \partial_\mu f_i(x) = f_i(x + e_\kappa + e_\mu) - f_i(x + e_\kappa) - f_i(x + e_\mu) + f_i(x)$$

uses four values in $U_3(u)$ when $x \in U_1(u)$. Its square is bounded by four times the sum of their squares. Sum over x, the components i, and the nine pairs (κ, μ). Translation of each summation set into $U_3(u)$ gives $\|\chi_{U_1(u)} \nabla\nabla f\|^2 \le 144 \|\chi_{U_3(u)} f\|^2$.

For $j \ge 1$ we have $2\xi \le 1$. Let ϕ be the even function on $\mathbb{R}$ given, for $s \ge 0$, by

$$\phi(s) = \begin{cases} 1, & 0 \le s \le 2, \\ 1 - 2(s-2)^2, & 2 \le s \le 5/2, \\ 2(3-s)^2, & 5/2 \le s \le 3, \\ 0, & s \ge 3. \end{cases}$$

It is continuously differentiable, $|\phi'| \le 2$, and ϕ' is Lipschitz with constant 4. For $t \in \mathbb{R}/N\mathbb{Z}$ write $|t|_N$ for its distance from zero and put $q(t) = \phi(|t|_N)$. The same derivative bounds hold for this periodic function. Indeed, $q = 1$ if $N = 4$. For $N \ge 6$ there is no difficulty at zero, where ϕ is constant; at the antipodal point q vanishes in a neighborhood if $N \ge 8$, while for $N = 6$ it is locally $2(N/2 - |t|_N)^2$. In the latter case its derivative vanishes at the antipode and has Lipschitz constant 4 across it.

Define

$$\Theta(x) = \prod_{\mu=1}^{3} q(x_\mu - u_\mu).$$

Then $0 \le \Theta \le 1$, $\Theta = 1$ on $U_2(u)$ and $\Theta = 0$ outside $U_3(u)$. The derivative bounds give $|\partial_\mu \Theta| \le 2$. Moreover,

$$|q(t+\xi) + q(t-\xi) - 2q(t)| = \left| \int_0^\xi (q'(t+s) - q'(t-s))\, ds \right| \le \int_0^\xi 8s\, ds = 4\xi^2,$$

so $|\partial_\mu^* \partial_\mu \Theta| \le 4$ and $|\Delta\Theta| \le 12$. Since $\xi \le 1$, both $\partial_\mu\Theta$ and $\Delta\Theta$ vanish outside $U_4(u)$.

The second differences at a point of $U_1(u)$ use only values in $U_2(u)$, where $\Theta = 1$. Applying componentwise the identity proved in section 17.3 gives

$$\|\chi_{U_1(u)} \nabla\nabla f\| \le \|\nabla\nabla(\Theta f)\| = \|\Delta(\Theta f)\|.$$

The product rule needed here is

$$\Delta(\Theta f) = \Theta \Delta f + (\Delta\Theta) f - \sum_{\mu=1}^{3} (w_\mu + w_\mu(\cdot - \xi e_\mu)), \qquad w_\mu = (\partial_\mu \Theta)\partial_\mu f. \tag{185}$$

For completeness, if $e = \xi e_\mu$, expand

$$\Theta(x \pm e) f(x \pm e) = \Theta(x) f(x \pm e) + (\Theta(x \pm e) - \Theta(x)) f(x) + (\Theta(x \pm e) - \Theta(x))(f(x \pm e) - f(x)).$$

The final product is $\xi^2 w_\mu(x)$ for the upper sign and $\xi^2 w_\mu(x - e)$ for the lower sign. Substitution in $\xi^2 \Delta F(x) = \sum_\mu (2F(x) - F(x + \xi e_\mu) - F(x - \xi e_\mu))$ proves (185), including both translated terms. Now

$$\|w_\mu\| \le 2\|\chi_{U_4(u)} \partial_\mu f\|, \qquad \|w_\mu(\cdot - \xi e_\mu)\| = \|w_\mu\|,$$

and Cauchy–Schwarz gives $\sum_\mu \|\chi_{U_4(u)} \partial_\mu f\| \le \sqrt{3} \|\chi_{U_4(u)} \nabla f\|$. Taking norms in (185), and using the support and derivative bounds for Θ, proves (184).

Proof of proposition 19.1(I5). We prove all six estimates with the constant c_* in (178). Each $U_1(u)$ is nonempty: every coordinate of a torus point lies within $\xi/2 \le 1$ of a fine site. Since ζ vanishes outside $U_1(u)$,

$$\|\zeta F\| \le |\zeta| \|\chi_{U_1(u)} F\|.$$

Apply (183) and (182) with $r = r' = 1$. For instance,

$$\|\zeta G_j J\| \le 4\gamma_{\mathrm{f}}^{-1} e^{6\delta_{\mathrm{f}}} e^{-\delta_{\mathrm{f}}|u-u'|_1} |\zeta| \|J\|.$$

The same calculation with the other choices of $\mathcal{O}_1, \mathcal{O}_2$ proves the estimates for $\zeta \nabla G_j J$, $\zeta G_j \nabla^* J'$ and $\zeta \nabla G_j \nabla^* J'$. Their constants are at most c_*, and $\delta_{\mathrm{f}} \ge \delta_*$.

It remains to estimate two output derivatives and then take the adjoint. Put $f = G_j J$. For $w \in \mathsf{T}$ and $r > 0$, the same first-order bounds give

$$\|\chi_{U_r(w)} f\|, \quad \|\chi_{U_r(w)} \nabla f\| \le 4\gamma_{\mathrm{f}}^{-1} e^{3(r+1)} e^{-\delta_{\mathrm{f}}|w-u'|_1} \|J\|. \tag{186}$$

If $U_r(w)$ is empty the left sides vanish; otherwise use (182) and $\delta_{\mathrm{f}} \le 1$. When $j = 0$, the local estimate just proved yields

$$\|\chi_{U_1(u)} \nabla\nabla f\| \le 48\gamma_{\mathrm{f}}^{-1} e^{12} e^{-\delta_{\mathrm{f}}|u-u'|_1} \|J\| \le c_* e^{-\delta_*|u-u'|_1} \|J\|.$$

Suppose now that $j \ge 1$. The componentwise equation is

$$\Delta f = J + \partial \mathsf{P}_j \partial^* f - \mathsf{Q}_j^* \mathsf{Q}_j f.$$

We estimate its three terms on $U_3(u)$. The source term vanishes there unless $U_3(u)$ meets $U_1(u')$, in which case $|u - u'|_1 \le 12$. Thus

$$\|\chi_{U_3(u)} J\| \le e^{12} e^{-\delta_*|u-u'|_1} \|J\|.$$

Next, (177) shows that $(\mathsf{Q}_j^* B)(x)$ uses only coarse bonds b for which $x \in b_- + [0,2)^3$. The straight average $(\mathsf{Q}_j f)(b)$ uses values $f(z)$ in the same box. Hence $|z-x|_\infty < 2$, and

$$\chi_{U_3(u)} \mathsf{Q}_j^* \mathsf{Q}_j f = \chi_{U_3(u)} \mathsf{Q}_j^* \mathsf{Q}_j \chi_{U_5(u)} f.$$

Since $\|\mathsf{Q}_j\| \le 1$ by lemma 17.4(i),

$$\|\chi_{U_3(u)} \mathsf{Q}_j^* \mathsf{Q}_j f\| \le \|\chi_{U_5(u)} f\|.$$

Finally, the divergence is

$$(\partial^* f)(x) = -\sum_{\mu=1}^{3} (\partial_\mu f_\mu)(x - \xi e_\mu).$$

The translated block $B^j(y) - \xi e_\mu$ lies in $U_1(y)$. Cauchy–Schwarz in the three components, followed by translation of the sums and (186), gives

$$\begin{aligned} \|\chi_{B^j(y)} \partial^* f\| &\le \sqrt{3} \|\chi_{U_1(y)} \nabla f\| \\ &\le 4\sqrt{3} \gamma_{\mathrm{f}}^{-1} e^6 e^{-\delta_{\mathrm{f}} |y-u'|_1} \|J\|. \end{aligned}$$

Since $2\delta_* \le \delta_{\mathrm{f}}, \delta_P$, (180) applies to $g = \partial^* f$ with $\delta = \delta_*$ and $A = 4\sqrt{3}\gamma_{\mathrm{f}}^{-1} e^6 \|J\|$. At $r = 3$ it gives

$$\|\chi_{U_3(u)} \partial \mathsf{P}_j \partial^* f\| \le 4\sqrt{3} \gamma_{\mathrm{f}}^{-1} C_P e^{18} \sigma(\delta_*) e^{-\delta_* |u-u'|_1} \|J\|.$$

Insert these three estimates into (184), and use (186) at radii 4 and 5 for the remaining terms. This gives

$$\begin{aligned} &\|\chi_{U_1(u)} \nabla \nabla G_j J\| \\ &\quad \le \Big[e^{12} + 4\gamma_{\mathrm{f}}^{-1} \big(e^{18} + \sqrt{3} C_P e^{18} \sigma(\delta_*) + 12 e^{15} + 4\sqrt{3} e^{15} \big) \Big] e^{-\delta_* |u-u'|_1} \|J\| \\ &\quad \le c_* e^{-\delta_* |u-u'|_1} \|J\|, \end{aligned}$$

where the last inequality uses $e^{15} \le e^{18}$. Multiplication by $|\zeta|$ proves the estimate with two output derivatives.

The positive operator G_j is symmetric. Therefore

$$\chi_{U_1(u)} G_j \nabla^* \nabla^* \chi_{U_1(u')} = \big(\chi_{U_1(u')} \nabla \nabla G_j \chi_{U_1(u)} \big)^*.$$

Exchange u and u' in the bound just proved and take the operator norm. This proves the final estimate

$$\|\zeta G_j \nabla^* \nabla^* J''\| \le c_* e^{-\delta_* |u-u'|_1} |\zeta| \|J''\|,$$

and completes (I5). □

19.3. **Hölder divergence data.** We strengthen the derivative estimate in proposition 18.1 by allowing a Hölder continuous divergence source. We use the lattice, differences and norms of section 18, with arbitrary spacing $\xi > 0$, and write

$$\mathsf{B}_s(x_0) = \{x \in \xi\mathbb{Z}^3 : |x - x_0|_\infty \le s\}.$$

For a function H on a set $X \subset \xi\mathbb{Z}^3$, put

$$[H]_{\beta,r;X} = \sup_{\substack{z,z' \in X \\ z \ne z'}} \left(\frac{r}{|z-z'|_1} \right)^\beta |H(z) - H(z')|, \qquad 0 < \beta < 1, \quad r > 0,$$

with value zero if there are no such pairs. Write $[H]_{\beta,r} = [H]_{\beta,r;\xi\mathbb{Z}^3}$ for a function defined on the whole lattice. For fields these seminorms and the local supremum $|F|_X$ are the maxima over the components. In particular, a seminorm with subscript X makes no assertion about the zero extension across the boundary of X.

Proposition 19.3. For every $0 < \beta < 1$ there is a constant $c_{II}(\beta) > 0$ with the following property. Let $r \geq 32\xi$, $x_0 \in \xi\mathbb{Z}^3$ and $\mathsf{B}_s = \mathsf{B}_s(x_0)$. Let u be a function on B_r, and let g and $F = (F_\mu)_{\mu=1}^3$ be defined on $\xi\mathbb{Z}^3$, with

$$\Delta u = g + \nabla^* F \qquad \text{on } \mathsf{B}_{r-\xi}.$$

Put

$$\mathsf{N}_\beta = r^{-3/2}\|u\|_{\mathsf{B}_r} + r^2|g|_{\mathsf{B}_r} + r\big(|F|_{\mathsf{B}_r} + [F]_{\beta,r;\mathsf{B}_r}\big).$$

For all $x, x' \in \mathsf{B}_{r/8}$ and every direction ν,

$$\begin{aligned} r|\partial_\nu u(x)| &\leq c_{II}(\beta)\mathsf{N}_\beta, \\ r|\partial_\nu u(x) - \partial_\nu u(x')| &\leq c_{II}(\beta)\left(\frac{|x-x'|_1}{r}\right)^\beta \mathsf{N}_\beta. \end{aligned} \tag{187}$$

The constant is independent of ξ, r, x_0, u, g, F. ■

The proof, given after lemma 19.4, subtracts the potential of a cutoff of F. A derivative of this potential contains two differences of the free kernel and therefore has a critical singularity of order -3. The following estimate uses its cancellation: summing a difference over a box leaves two boundary faces. Subtracting the value of the source at the observation point then gives a bound uniform in the mesh.

Lemma 19.4. For every $0 < \beta < 1$ there is a constant $c_S(\beta) > 0$ with the following property. Suppose that k_1 is a function on $\xi\mathbb{Z}^3$ satisfying, for every z and all directions λ, λ',

$$\begin{aligned} |k_1(z)| &\leq (|z|_1 + \xi)^{-2}, \\ |\partial_\lambda k_1(z)| &\leq (|z|_1 + \xi)^{-3}, \\ |\partial_{\lambda'}\partial_\lambda k_1(z)| &\leq (|z|_1 + \xi)^{-4}. \end{aligned}$$

Let $k = \partial_\nu k_1$ for some direction ν. If $r \geq \xi$, $x_0 \in \xi\mathbb{Z}^3$, and G is a function on the whole lattice vanishing outside $\mathsf{B}_r(x_0)$, then for all $x, x' \in \mathsf{B}_r(x_0)$,

$$\begin{aligned} |(k*G)(x)| &\leq c_S(\beta)([G]_{\beta,r} + |G|), \\ |(k*G)(x) - (k*G)(x')| &\leq c_S(\beta)\left(\frac{|x-x'|_1}{r}\right)^\beta ([G]_{\beta,r} + |G|). \end{aligned} \tag{188}$$

The constant depends only on β. ■

Proof. In this proof c denotes constants depending only on β. All sums are over $\xi\mathbb{Z}^3$ unless otherwise specified.

Cancellation on boxes. For $w \in \xi\mathbb{Z}^3$ and a positive integer multiple s of ξ, set

$$\mathsf{S}_{s,w}(x) = \xi^3 \sum_{z \in \mathsf{B}_s(w)} k(x-z).$$

We claim that, if $x, x' \in \mathsf{B}_{s/2}(w)$, then

$$|\mathsf{S}_{s,w}(x)| \leq 72, \qquad |\mathsf{S}_{s,w}(x) - \mathsf{S}_{s,w}(x')| \leq 144\,\frac{|x-x'|_1}{s}. \tag{189}$$

Fix the two coordinates of z other than z_ν, and let $a_x(t)$ be $k_1(x-z)$ when $z_\nu = t$. Since $k(x-z) = \xi^{-1}(a_x(z_\nu - \xi) - a_x(z_\nu))$, the sum in the ν direction telescopes:

$$\sum_{t = w_\nu - s, w_\nu - s + \xi, \ldots, w_\nu + s} \big(a_x(t-\xi) - a_x(t)\big) = a_x(w_\nu - s - \xi) - a_x(w_\nu + s).$$

Consequently

$$\mathsf{S}_{s,w}(x) = \xi^2 \sum_{(z_\mu)_{\mu \neq \nu}} \big(a_x(w_\nu - s - \xi) - a_x(w_\nu + s)\big).$$

There are $(2s/\xi+1)^2 \le (3s/\xi)^2$ terms in this remaining sum. At either face the ν coordinate of $x-z$ has modulus at least $s/2$; the lower face is at $z_\nu = w_\nu - s - \xi$. Thus

$$|\mathsf{S}_{s,w}(x)| \le \xi^2 (3s/\xi)^2 \, 2(s/2)^{-2} = 72.$$

Join x to x' by a shortest lattice path whose coordinates change monotonically. The entire path lies in $\mathsf{B}_{s/2}(w)$. A step changes a boundary value of k_1 by $\pm\xi\partial_\lambda k_1$, evaluated at one of its endpoints. The distance from that endpoint to either face is still at least $s/2$. Summing over the $|x-x'|_1/\xi$ steps therefore gives

$$|\mathsf{S}_{s,w}(x) - \mathsf{S}_{s,w}(x')| \le \xi^2 (3s/\xi)^2 \, 2|x-x'|_1 (s/2)^{-3} = 144 \, \frac{|x-x'|_1}{s}.$$

This proves (189).

The two shell sums. For every $s>0$ we shall use

$$\xi^3 \sum_{0<|z|_1\le s} |z|_1^{\beta-3} \le \frac{216}{2^\beta - 1} s^\beta,$$
$$\xi^3 \sum_{|z|_1>s} |z|_1^{\beta-4} \le \frac{216}{1-2^{\beta-1}} s^{\beta-1}.$$

Indeed, the shell $\rho < |z|_1 \le 2\rho$ has at most $(6\rho/\xi)^3$ points if $2\rho \ge \xi$, and is empty otherwise. Its contributions to the two sums are at most $216\rho^\beta$ and $216\rho^{\beta-1}$, respectively. For the first sum take $\rho = 2^{-i-1}s$, $i \ge 0$; for the second take $\rho = 2^i s$. The resulting geometric series have sums $s^\beta/(2^\beta-1)$ and $s^{\beta-1}/(1-2^{\beta-1})$.

The pointwise bound. Let $\hat r$ be the least multiple of ξ not smaller than r. Then $r \le \hat r < r+\xi \le 2r$. Set $s = 2\hat r$. For $x \in \mathsf{B}_r(x_0)$ the support of G is contained in $\mathsf{B}_s(x)$, so

$$(k*G)(x) = \xi^3 \sum_{\substack{z\in\mathsf{B}_s(x)\\ z\ne x}} k(x-z)\big(G(z)-G(x)\big) + G(x)\mathsf{S}_{s,x}(x).$$

The first term is bounded by

$$[G]_{\beta,r} r^{-\beta} \xi^3 \sum_{0<|z-x|_1\le 12r} |z-x|_1^{\beta-3} \le \frac{216\,12^\beta}{2^\beta-1}[G]_{\beta,r},$$

because $|z-x|_1 \le 3s \le 12r$ on the box. The second is at most $72|G|$ by (189). This proves the first estimate in (188), with

$$c_1 = \max\left\{\frac{216\,12^\beta}{2^\beta-1}, 72\right\}.$$

The Hölder bound. Put $\delta = |x-x'|_1$ and suppose $\delta > 0$. If $\delta > r/4$, the pointwise estimate at x and x' gives

$$|(k*G)(x) - (k*G)(x')| \le 2c_1 \, 4^\beta (\delta/r)^\beta \big([G]_{\beta,r} + |G|\big).$$

We may therefore assume $\delta \le r/4$. Set $s_1 = 2\delta$ and $s_2 = 4\hat r$. Both are positive integer multiples of ξ, and $x' \in \mathsf{B}_{s_1/2}(x) \subset \mathsf{B}_{s_2/2}(x)$. Subtracting $G(x)$ on the common box gives

$$(k*G)(x) - (k*G)(x') = \xi^3 \sum_{z\in\mathsf{B}_{s_2}(x)} \big(k(x-z) - k(x'-z)\big)\big(G(z)-G(x)\big) + G(x)\big(\mathsf{S}_{s_2,x}(x) - \mathsf{S}_{s_2,x}(x')\big).$$

Since $s_2 \ge 4r$, the last term is at most $36|G|\delta/r \le 36|G|(\delta/r)^\beta$.

For $z \notin \mathsf{B}_{s_1}(x)$ we have $|z-x|_1 > 2\delta$. Every point x'' on a shortest path from x to x' satisfies $|x''-z|_1 \geq |x-z|_1 - \delta \geq |x-z|_1/2$. Each step changes $k(\,\cdot - z)$ by a first difference evaluated at $x''-z$ for one of its endpoints. The second-difference bound for k_1 therefore gives

$$|k(x-z) - k(x'-z)| \leq 16\delta\,|x-z|_1^{-4}.$$

The far part of the sum is consequently bounded by

$$16\delta[G]_{\beta,r} r^{-\beta}\xi^3 \sum_{|z-x|_1>2\delta} |z-x|_1^{\beta-4} \leq \frac{16\cdot 216\,2^{\beta-1}}{1-2^{\beta-1}}[G]_{\beta,r}(\delta/r)^\beta.$$

For the near part, recenter the terms with $k(x'-z)$ at $G(x')$. This part is $\mathsf{n} - \mathsf{n}'$, where

$$\mathsf{n} = \xi^3 \sum_{\substack{z\in\mathsf{B}_{s_1}(x)\\ z\neq x}} k(x-z)(G(z)-G(x)),$$

$$\mathsf{n}' = \xi^3 \sum_{\substack{z\in\mathsf{B}_{s_1}(x)\\ z\neq x'}} k(x'-z)(G(z)-G(x')) + (G(x')-G(x))\mathsf{S}_{s_1,x}(x').$$

Here $|z-x|_1 \leq 6\delta$ and $|z-x'|_1 \leq 7\delta \leq 9\delta$. The first shell sum gives

$$|\mathsf{n}| \leq \frac{216\,6^\beta}{2^\beta - 1}[G]_{\beta,r}(\delta/r)^\beta;$$

the sum in n' has the same bound with 9^β in place of 6^β. Finally, $x' \in \mathsf{B}_{s_1/2}(x)$, so (189) bounds the remaining term of n' by $72[G]_{\beta,r}(\delta/r)^\beta$. Combining the near, far and boundary terms proves the second estimate in (188). Enlarging the constant to cover both estimates gives $c_S(\beta)$. □

Proof of proposition 19.3. The constants denoted by c below depend only on β, the constants in lemmas 18.2 and 19.4 and proposition 18.1, and the fixed cutoff η_0 of section 18.2. In particular, they are independent of the mesh and radius.

Cutting off the divergence source. Set

$$\zeta(x) = \prod_{\mu=1}^{3} \eta_0((x_\mu - x_{0,\mu})/r).$$

Then $0 \leq \zeta \leq 1$, $\zeta = 1$ on $\mathsf{B}_{r/2}$, and ζ vanishes outside $\mathsf{B}_{2r/3}$. The product inequality for factors in $[0,1]$ and the mean value theorem give

$$|\zeta(z) - \zeta(z')| \leq \|\eta_0'\|_\infty\,|z-z'|_1/r.$$

Define $G_\mu = \zeta F_\mu$ on B_r and $G_\mu = 0$ elsewhere. This is a finitely supported field, with $|G_\mu| \leq |F|_{\mathsf{B}_r}$ and

$$[G_\mu]_{\beta,r} \leq [F]_{\beta,r;\mathsf{B}_r} + \max\{3, \|\eta_0'\|_\infty\}|F|_{\mathsf{B}_r}. \tag{190}$$

To verify the global seminorm in this estimate, put $t = |z-z'|_1/r$. If both points lie in B_r, then

$$|G_\mu(z) - G_\mu(z')| \leq |F_\mu(z) - F_\mu(z')| + |\zeta(z) - \zeta(z')|\,|F|_{\mathsf{B}_r},$$

and

$$|\zeta(z) - \zeta(z')| \leq \min\{1, \|\eta_0'\|_\infty t\} \leq \max\{1, \|\eta_0'\|_\infty\} t^\beta.$$

If one point lies in $\mathsf{B}_{2r/3}$ and the other outside B_r, then $t > 1/3$, so the difference of the two values of G_μ is at most $|F|_{\mathsf{B}_r} \leq 3^\beta t^\beta |F|_{\mathsf{B}_r}$. In every remaining case not already covered, both values vanish. Dividing by t^β proves (190).

The potential of the cutoff field. Put

$$u_1 = \sum_{\mu=1}^{3} (\partial_\mu^* C_{0,\mathrm{lap}}) * G_\mu.$$

For finitely supported fields, forward and adjoint differences commute with convolution by shifting the summation index, as in section 18.2. Hence lemma 18.2(i) gives $\Delta u_1 = \sum_\mu \partial_\mu^* G_\mu$. For $x \in \mathsf{B}_{r/2-\xi}$, the points x and $x - \xi e_\mu$ lie in $\mathsf{B}_{r/2}$, where $G_\mu = F_\mu$. Consequently $u_2 = u - u_1$ satisfies

$$\Delta u_2 = g \qquad \text{on } \mathsf{B}_{r/2-\xi}.$$

Fix μ and take $k_1 = c_K^{-1}\partial_\mu^* C_{0,\mathrm{lap}}$. The mixed differences of orders one, two and three in lemma 18.2(ii) show that k_1 satisfies all three hypotheses of lemma 19.4, and the kernel hypotheses of lemma 18.2(iv). For $x \in \mathsf{B}_r$, the support of G_μ lies within ℓ^1 distance $6r$ of x. The pointwise estimate in part (iv) therefore gives

$$|u_1(x)| \le 3c_K^2 r\,|F|_{\mathsf{B}_r} \qquad (x \in \mathsf{B}_r).$$

There are at most $(2r/\xi)^3$ points in $\mathsf{B}_{r/2}$, since $r \ge \xi$. Hence

$$\|u_1\|_{\mathsf{B}_{r/2}} \le (2r)^{3/2}\, 3c_K^2 r\,|F|_{\mathsf{B}_r}.$$

Moreover,

$$\partial_\nu u_1 = \sum_\mu (\partial_\nu \partial_\mu^* C_{0,\mathrm{lap}}) * G_\mu, \qquad \partial_\nu \partial_\mu^* C_{0,\mathrm{lap}} = c_K \partial_\nu k_1.$$

Apply lemma 19.4 to this k_1 and each G_μ, and then use (190). For $x, x' \in \mathsf{B}_r$ we obtain

$$\begin{aligned} r|\partial_\nu u_1(x)| &\le cr([F]_{\beta,r;\mathsf{B}_r} + |F|_{\mathsf{B}_r}), \\ r|\partial_\nu u_1(x) - \partial_\nu u_1(x')| &\le c\left(\frac{|x-x'|_1}{r}\right)^\beta r([F]_{\beta,r;\mathsf{B}_r} + |F|_{\mathsf{B}_r}). \end{aligned}$$

The remaining scalar equation. Apply proposition 18.1(b) to u_2 on $\mathsf{B}_{r/2}$, with right-hand side g and zero divergence field. Its radius condition holds because $r/2 \ge 16\xi$, and its equation was verified above. Its controlling quantity satisfies

$$\begin{aligned} \mathsf{N} &= (r/2)^{-3/2}\|u_2\|_{\mathsf{B}_{r/2}} + (r/2)^2|g|_{\mathsf{B}_{r/2}} \\ &\le 2^{3/2} r^{-3/2}(\|u\|_{\mathsf{B}_r} + \|u_1\|_{\mathsf{B}_{r/2}}) + r^2|g|_{\mathsf{B}_r} \\ &\le c\mathsf{N}_\beta. \end{aligned}$$

Thus, for $x, x' \in \mathsf{B}_{r/8}$,

$$\begin{aligned} (r/2)|\partial_\nu u_2(x)| &\le c_I c \mathsf{N}_\beta, \\ (r/2)|\partial_\nu u_2(x) - \partial_\nu u_2(x')| &\le c_I(\beta) c \left(\frac{2|x-x'|_1}{r}\right)^\beta \mathsf{N}_\beta. \end{aligned}$$

At these sites the identity $u = u_1 + u_2$ also holds at the forward-shifted sites used by ∂_ν. Adding the estimates for the two potentials proves (187), after increasing $c_{II}(\beta)$. □

19.4. **Lifts and pointwise projection estimates.** The scalar interior estimates apply on the covering lattice. The balls of radius 16 and 32 used below may cover the torus more than once. We therefore count the lifted blocks, retaining repeated images on the torus. Let $\mathsf{p} : \xi\mathbb{Z}^3 \to \mathsf{T}_\xi$ be the covering map and write $f^\uparrow = f \circ \mathsf{p}$. Differences commute with this lift. Fix $w \in \mathsf{T}_\xi$, a lift w_0, and an integer $r \ge 1$. The set

$$\mathsf{Y} = \{\widetilde{y} \in \mathbb{Z}^3 : w_{0,\mu} - r - 1 < \widetilde{y}_\mu \le w_{0,\mu} + r \text{ for } \mu = 1,2,3\}$$

has $(2r+1)^3$ elements. Its unit boxes cover $\mathsf{B}_r(w_0)$, and $|\mathsf{p}\widetilde{y} - w|_1 \le 3(r+1)$. Each lifted block maps bijectively to $B^j(\mathsf{p}\widetilde{y})$, even when different lifted blocks have the same image. Consequently,

$$\|f^{\uparrow}\|_{\mathsf{B}_r(w_0)} \le \sum_{\widetilde{y}\in\mathsf{Y}} \|\chi_{B^j(\mathsf{p}\widetilde{y})} f\|,$$
$$|f^{\uparrow}|_{\mathsf{B}_r(w_0)} \le \max_{\widetilde{y}\in\mathsf{Y}} |f|_{B^j(\mathsf{p}\widetilde{y})}.$$

The first inequality follows by summing the squared block norms and then bounding the square root by their sum. Thus, if $0 < \delta \le 1$ and $\|\chi_{B^j(y)} f\| \le Ae^{-\delta|y-u'|_1}$ for every coarse site y, then

$$\|f^{\uparrow}\|_{\mathsf{B}_r(w_0)} \le (2r+1)^3 e^{3(r+1)} Ae^{-\delta|w-u'|_1}. \tag{191}$$

If instead $|f(x)| \le Ae^{-\delta|x-u'|_1}$ at every fine site, then

$$|f^{\uparrow}|_{\mathsf{B}_r(w_0)} \le e^{3r} Ae^{-\delta|w-u'|_1}. \tag{192}$$

Both estimates follow from the triangle inequality and $\delta \le 1$.

We will also compare nearby pairs. For $z, z' \in \xi\mathbb{Z}^3$ with $|z - z'|_1 \le 1$, every nonzero period translate of $z - z'$ has norm at least $N - |z - z'|_1 \ge 3$. Hence $|\mathsf{p}z - \mathsf{p}z'|_1 = |z - z'|_1$. Conversely, after fixing a lift x_0 of a torus site x, choose a least-length representative of $x' - x$ to obtain a lift x_0' satisfying $|x_0 - x_0'|_1 = |x - x'|_1$.

The Laplacian of $\mathsf{P}_j g$ is constant on each unit block. The block estimates proved above therefore give the bounded right-hand side needed in proposition 18.1(b).

Lemma 19.5. Let $A \ge 0$, $u' \in \mathsf{T}$, and suppose that the real scalar function g satisfies

$$\|\chi_{B^j(y)} g\| \le Ae^{-2\delta_*|y-u'|_1} \qquad (y \in \mathsf{T}_1).$$

There are an absolute constant c_{P1} and constants $c_{P1}(\alpha)$ depending only on $0 < \alpha < 1$ such that

$$\begin{aligned} |(\partial\mathsf{P}_j g)(x)| &\le c_{P1} Ae^{-\delta_*|x-u'|_1}, \\ |(\partial\mathsf{P}_j g)(x) - (\partial\mathsf{P}_j g)(x')| &\le c_{P1}(\alpha)|x - x'|_1^{\alpha} Ae^{-\delta_*|x-u'|_1} \end{aligned} \tag{193}$$

whenever $x, x' \in \mathsf{T}_\xi$ and $|x - x'|_1 \le 1$. ■

Proof. Put $h = \mathsf{P}_j g$ and $A_1 = C_P e^6 \sigma(\delta_*) A$. The block estimates (180) give

$$\|\chi_{B^j(y)} h\| \le A_1 e^{-\delta_*|y-u'|_1}, \qquad |\Delta h|_{B^j(y)} \le 3A_1 e^{-\delta_*|y-u'|_1}.$$

Here $2\delta_* \le \delta_P$, as required in that estimate. Fix a lift x_0 of x. The lifting bounds with $r = 16$ yield

$$\|h^{\uparrow}\|_{\mathsf{B}_{16}(x_0)} \le 33^3 e^{51} A_1 e^{-\delta_*|x-u'|_1},$$
$$|\Delta h^{\uparrow}|_{\mathsf{B}_{16}(x_0)} \le 3e^{51} A_1 e^{-\delta_*|x-u'|_1}.$$

For the second bound, use the block supremum and $|\mathsf{p}\widetilde{y} - x|_1 \le 51$ in the covering above. Apply proposition 18.1(b) to $h^{\uparrow}$, with right-hand side $\Delta h^{\uparrow}$ and zero divergence field. The radius is admissible since $\xi \le 1$, and its data norm is

$$\mathsf{N} \le (33^3/64 + 768) e^{51} A_1 e^{-\delta_*|x-u'|_1}.$$

Choose a lift x_0' at distance $|x - x'|_1 \le 1$ from x_0. Both points lie in $\mathsf{B}_4(x_0)$. The derivative estimates of that proposition give (193), with c_{P1} absolute and $c_{P1}(\alpha)$ depending only on α. □

19.5. **Pointwise and Hölder bounds for the inverse.**

Proof of the pointwise and Hölder conclusions of proposition 19.1. Throughout the proof, c denotes a positive absolute constant which may increase, and $c(\beta)$ may also depend on the displayed exponent. Fix the source center u' and put

$$E(x) = e^{-\delta_* |x-u'|_1}.$$

Since $\delta_* \le 1$, we have $E(z) \le e^{|z-x|_1} E(x)$. Let J and J' vanish outside $U_1(u')$, with the ranks specified in the proposition. The weighted volume of $U_1(u')$ is at most 27: each coordinate interval contains at most $2/\xi + 1 \le 3/\xi$ sites. Thus

$$\|J\| \le 9|J|, \qquad \|J'\| \le 9\sqrt{3}\,|J'|.$$

Consider either

$$(f,n) = (G_j J, |J|) \quad \text{or} \quad (f,n) = (G_j \nabla^* J', |J'|).$$

The first-order decay estimate proposition 17.2(c) and $\operatorname{dist}(U_1(v), U_1(u')) \ge |v-u'|_1 - 6$ imply

$$\|\chi_{U_1(v)} f\|, \quad \|\chi_{U_1(v)} \nabla f\| \le A_0 e^{-\delta_{\mathrm{f}} |v-u'|_1}, \qquad A_0 = 36\sqrt{3}\,\gamma_{\mathrm{f}}^{-1} e^6 n. \tag{194}$$

This estimate holds for every continuous center v.

The componentwise equation. Write $J^\natural = J$ or $\nabla^* J'$ according to the chosen pair. The definition of the inverse gives

$$\Delta f = J^\natural + h - k, \qquad h = \partial \mathsf{P}_j \partial^* f, \qquad k = \mathsf{Q}_j^* \mathsf{Q}_j f. \tag{195}$$

Both lower-order terms have pointwise bounds with the required decay. Indeed, $\partial^* f(x) = -\sum_\mu \partial_\mu f_\mu(x - \xi e_\mu)$ and $B^j(y) - \xi e_\mu \subset U_1(y)$, so

$$\|\chi_{B^j(y)} \partial^* f\| \le \sqrt{3}\,\|\chi_{U_1(y)} \nabla f\| \le \sqrt{3} A_0 e^{-2\delta_* |y-u'|_1}.$$

We used $2\delta_* \le \delta_{\mathrm{f}}$ in the last inequality. Apply lemma 19.5 to obtain

$$|h(x)| \le cnE(x). \tag{196}$$

For the averaging term, let $x \in B^j(y)$. The adjoint formula (177) shows that only the two coarse bonds of direction μ with initial points y and $y - e_\mu$ contribute to $(\mathsf{Q}_j^* B)_\mu(x)$, each with coefficient in $[0,1]$. In fact, $x - t\xi e_\mu$ lies in one of those two blocks for $0 \le t < L^j$. For either initial point y'', the column $\mathsf{Q}_j^* e_b$ has norm at most one and is supported in $y'' + [0,2)^3 \subset U_1(y'' + \mathbf{e})$, where $\mathbf{e} = (1,1,1)$. Hence

$$|(\mathsf{Q}_j f)(b)| = |\langle \mathsf{Q}_j^* e_b, f\rangle| \le A_0 e^{-\delta_{\mathrm{f}} |y'' + \mathbf{e} - u'|_1}.$$

For these two choices $|x - y'' - \mathbf{e}|_1 \le 3$. Since $\delta_* \le \delta_{\mathrm{f}}$, summing their contributions gives

$$|k(x)| \le 2e^3 A_0 E(x) \le cnE(x). \tag{197}$$

An ordinary source. Suppose $f = G_j J$. Where $J(x) \ne 0$, the distance $|x - u'|_1$ is at most three, and $|J(x)| \le e^3 |J| E(x)$. Together with (195)–(197), this gives

$$|(\Delta G_j J)(x)| \le c|J|E(x). \tag{198}$$

Fix a lift x_0 of x and a component f_μ. The block bound obtained from (194) and (191) gives

$$\|f_\mu^\uparrow\|_{\mathsf{B}_{16}(x_0)} \le 33^3 e^{51} A_0 E(x) \le c|J|E(x).$$

Equation (198) and (192) bound $|\Delta f_\mu^\uparrow|_{\mathsf{B}_{16}(x_0)}$ by the same expression with a larger constant. Apply both parts of proposition 18.1 at radius 16, with this right-hand side and zero divergence field. For $|x - x'|_1 \le 1$, choose the nearest lift x_0'; both lifted points lie in $\mathsf{B}_4(x_0)$. The resulting estimates are

$$\begin{aligned} &|(G_j J)(x)|, \quad |(\nabla G_j J)(x)| \le c|J|E(x), \\ &|(\nabla G_j J)(x) - (\nabla G_j J)(x')| \le c(\alpha) |x - x'|_1^\alpha |J| E(x), \qquad 0 < \alpha < 1. \end{aligned} \tag{199}$$

A divergence source. Now let $f = G_j\nabla^* J'$ and fix a component ν. For a lift x_0 of x and $r \in \{16, 32\}$, define $F_\mu = (J'_{\mu\nu})^\uparrow$ on $\mathsf{B}_r(x_0)$ and $F_\mu = 0$ outside that ball. Put $g = (h-k)^\uparrow_\nu$. For $z \in \mathsf{B}_{r-\xi}(x_0)$, both z and $z - \xi e_\mu$ lie in $\mathsf{B}_r(x_0)$. Therefore

$$\Delta f^\uparrow_\nu = g + \nabla^* F \quad \text{on } \mathsf{B}_{r-\xi}(x_0).$$

If F is nonzero, some $z \in \mathsf{B}_r(x_0)$ has $\mathsf{p}z \in U_1(u')$, which implies $|x - u'|_1 \le 3(r+1)$. It follows, also when $F = 0$, that

$$|F| \le e^{3(r+1)}|J'|E(x).$$

The bounds (194), (196), (197) and the lifting estimates give

$$\|f^\uparrow_\nu\|_{\mathsf{B}_r(x_0)} + |g|_{\mathsf{B}_r(x_0)} \le c|J'|E(x) \qquad (r = 16, 32).$$

At radius 16, proposition 18.1(a) applies with the global bound on this restricted field F. Comparing nearest lifts as before, it gives

$$\begin{aligned} |(G_j\nabla^* J')(x)| &\le c|J'|E(x), \\ |(G_j\nabla^* J')(x) - (G_j\nabla^* J')(x')| &\le c(\alpha)|x - x'|_1^\alpha |J'|E(x) \qquad (0 < \alpha < 1). \end{aligned} \tag{200}$$

To estimate the gradient, take $r = 32$ and $0 < \beta < 1$. For two sites of $\mathsf{B}_{32}(x_0)$ at distance at most one, their torus distance equals their lifted distance. The corresponding difference of F is therefore at most $|z - z'|_1^\beta \|J'\|_\beta$. At distances greater than one it is at most $2|J'| \le 2|z - z'|_1^\beta |J'|$. Consequently,

$$[F]_{\beta,32;\mathsf{B}_{32}(x_0)} \le 32^\beta(\|J'\|_\beta + 2|J'|).$$

This seminorm vanishes when $F = 0$. In the other case the preceding support argument supplies the decay factor, so

$$[F]_{\beta,32;\mathsf{B}_{32}(x_0)} \le 32e^{99}(\|J'\|_\beta + 2|J'|)E(x).$$

We may now apply proposition 19.3, since $32 \ge 32\xi$. Its data norm is at most $c(\beta)(\|J'\|_\beta + |J'|)E(x)$. The nearest lifts of x, x' belong to $\mathsf{B}_4(x_0) = \mathsf{B}_{32/8}(x_0)$, and hence

$$\begin{aligned} |(\nabla G_j\nabla^* J')(x)| &\le c(\beta)(\|J'\|_\beta + |J'|)E(x), \\ |(\nabla G_j\nabla^* J')(x) - (\nabla G_j\nabla^* J')(x')| &\le c(\beta)|x - x'|_1^\beta (\|J'\|_\beta + |J'|)E(x). \end{aligned} \tag{201}$$

Multiplication by the observation cutoff. Let ζ vanish outside $U_1(u)$ and put $E = e^{-\delta_*|u-u'|_1}$. For $x \in U_2(u)$, we have $E(x) \le e^6 E$. The pointwise conclusions follow from (198)–(201), using $\beta = \varepsilon$ for the gradient with divergence input. For the Hölder conclusions, let q denote, in turn, $\nabla G_j J$, $G_j\nabla^* J'$ or $\nabla G_j\nabla^* J'$, and put

$$m = |J|, \quad |J'|, \quad \|J'\|_{\alpha+\varepsilon} + |J'|,$$

respectively. For $0 < \alpha < 1$, the preceding estimates give

$$|q(x)| \le c_0 mE, \qquad |q(x) - q(x')| \le c_0|x - x'|_1^\alpha mE$$

when $x \in U_2(u)$ and $|x - x'|_1 \le 1$. For the third choice use $\beta = \alpha + \varepsilon < 1$ and $|x - x'|_1^{\alpha+\varepsilon} \le |x - x'|_1^\alpha$. The constant c_0 depends only on the indicated exponents.

If neither point lies in $U_1(u)$, the difference of ζq vanishes. Otherwise both lie in $U_2(u)$, and

$$\begin{aligned} |\zeta(x)q(x) - \zeta(x')q(x')| &\le |\zeta(x) - \zeta(x')|\,|q(x)| + |\zeta(x')|\,|q(x) - q(x')| \\ &\le c_0|x - x'|_1^\alpha(\|\zeta\|_\alpha + |\zeta|)mE. \end{aligned}$$

This proves the asserted seminorm bounds for $\alpha > 0$, including pairs crossing the support boundary. For $\alpha = 0$ use $\|\zeta q\|_0 \le 2|\zeta| \sup_{U_1(u)} |q|$; in the third case take $\beta = \varepsilon$ in (201). Together with the six L^2 bounds proved in section 19.2, these estimates prove proposition 19.1, after enlarging its common constant c to include the L^2 constant. □

20. Two-scale propagators

Retain the real, three-dimensional setting of section 19: $L \geq 2$ and $j \geq 0$ are integers, $\xi = L^{-j}$, and the torus side N is a positive even multiple of L. In addition to T_ξ and T_1, use $\mathsf{T}_L = L\mathbb{Z}^3/N\mathbb{Z}^3$. Write $\mathcal{F}_h$ for the real scalar functions on T_h and $\mathcal{A}_h = \mathcal{F}_h^3$ for its one-forms. Their inner products have weight h^3. The fine differences are ∂, ∇, Δ. At spacing 1 or L, the first differences carry a subscript 1 or L. The notation Δ^1 denotes the componentwise Laplacian at spacing 1; Δ_1 below denotes the two-scale operator on the fine lattice. Distances, component maximum norms, $U_r(u)$ and the seminorms $\|\cdot\|_\alpha$ retain their preceding meanings.

For $y' \in \mathsf{T}_L$ let

$$B(y') = \mathsf{T}_1 \cap (y' + [0, L)^3).$$

The averages from spacing 1 to spacing L are

$$(\mathsf{Q}'_{1,L} g)(y') = L^{-3} \sum_{y \in B(y')} g(y),$$

$$(\mathsf{Q}_{1,L} B)_\mu(y') = L^{-4} \sum_{y \in B(y')} \sum_{t=0}^{L-1} B_\mu(y + te_\mu).$$

The two indices distinguish this step from $\mathsf{Q}_j : \mathcal{A}_\xi \to \mathcal{A}_1$. Both block families partition their respective fine lattices. Telescoping the straight paths gives

$$\mathsf{Q}_j \partial = \partial_1 \mathsf{Q}'_j, \qquad \mathsf{Q}_{1,L} \partial_1 = \partial_L \mathsf{Q}'_{1,L}.$$

In each $B(y')$ use the coordinate tree $\mathcal{T}(y')$ whose paths from y' change coordinates in the order $3, 2, 1$. Thus its direction-μ bonds have their coordinates of index less than μ equal to those of y'. Every path lies inside its block.

Definition 20.1. Let $\Lambda' \subset \mathsf{T}_L$ be arbitrary and put $\Lambda = \bigcup_{y' \in \Lambda'} B(y')$. Let Λ^* consist of the unit bonds with at least one endpoint in Λ, and let Λ^{c*} consist of those with neither endpoint there. Let Λ'^* consist of the spacing-L bonds with at least one endpoint in Λ'; it includes bonds leaving Λ'. Define

$$N(\mathsf{Q}') = \{\lambda \in \mathcal{F}_\xi : \mathsf{Q}'_j \lambda = 0 \text{ on } \mathsf{T}_1 \setminus \Lambda, \\ \mathsf{Q}'_{1,L} \mathsf{Q}'_j \lambda = 0 \text{ on } \Lambda'\}.$$

Let R be the orthogonal projection onto $\Delta N(\mathsf{Q}')$, and put $\mathsf{P} = 1 - \mathsf{R}$. For $B \in \mathcal{A}_1$ set

$$\mathsf{q}(B) = \sum_{b \in \Lambda^{c*}} |B(b)|^2 + L \sum_{c \in \Lambda'^*} |(\mathsf{Q}_{1,L} B)(c)|^2.$$

The symmetric operator Δ_1 on $\mathcal{A}_\xi$ is defined by

$$\langle A, \Delta_1 A \rangle = \|\partial A\|^2 + \|\mathsf{R} \partial^* A\|^2 + \mathsf{q}(\mathsf{Q}_j A). \tag{202}$$

■

Proposition 20.2. The operator Δ_1 is positive definite. Its inverse $G = \Delta_1^{-1}$ satisfies every estimate (I1)–(I5) of proposition 19.1, with G in place of G_j and $E = e^{-\delta_{\mathrm{per}}|u-u'|_1}$. Here $\delta_{\mathrm{per}} > 0$ depends only on L; the constants c, c_α, c'_ε and $c''_{\alpha,\varepsilon}$ depend only on L and their displayed exponents. They are independent of j, N, Λ', the continuous centers and the fields. The support conditions, tensor source spaces and exponent ranges are exactly those in proposition 19.1, including the oscillation seminorm at $\alpha = 0$ and all six localized L^2 estimates. ■

The quadratic form separates into a fluctuation with zero unit-scale average, an axial field on T_1, and a scalar gauge correction. The fluctuation is controlled by G_j. The other two terms require lower bounds on their constrained spaces; the axial lower bound follows by combining estimates inside the blocks with estimates across their faces. Completing the scalar square then expresses

$G - G_j$ in terms of maps from unit bonds to fine bonds and inverses on the unit lattice. Exponential decay of these inverses and the one-scale regularity estimates give the conclusion. The decomposition and representation are proved in section 20.3; the decay estimates and the proof of proposition 20.2 are completed in section 20.6.

20.1. **Averages and constrained extensions.** The averaging maps control the energy of a prescribed coarse field in both directions. Averaging does not increase the curl energy, while every unit-scale field has a fine-scale extension with uniformly bounded energy. The scalar estimate below gives the gap on fields with zero mean in each L-block.

Lemma 20.3. The averages satisfy the following estimates.

(a) For $A \in \mathcal{A}_\xi$,
$$\|\partial_1 \mathsf{Q}_j A\| \le \|\partial A\|.$$

(b) For $\lambda \in \mathcal{F}_\xi$,
$$\|\Delta^1 \mathsf{Q}'_j \lambda\|^2 \le 3\|\Delta\lambda\|^2.$$

(c) For every $B \in \mathcal{A}_1$ there is $A \in \mathcal{A}_\xi$ such that
$$\mathsf{Q}_j A = B, \qquad \langle A, \Delta A\rangle \le \gamma_1 \|B\|^2, \qquad \gamma_1 = 3\pi^2(\pi/2)^8.$$
In particular, Q_j is onto.

(d) If $\omega \in \mathcal{F}_1$ and $\mathsf{Q}'_{1,L}\omega = 0$, then
$$\|\Delta^1 \omega\| \ge 4L^{-2}\|\omega\|.$$

Moreover,
$$\|\mathsf{Q}_{1,L} B\| \le \|B\|, \qquad B \in \mathcal{A}_1, \tag{203}$$
where the norm on the left has the volume weight L^3. ∎

Proof. Put $n = L^j$, so that $n\xi = 1$. We use the telescoping identity
$$f(x + e_\mu) - f(x) = \sum_{s=0}^{n-1} \xi\, \partial_\mu f(x + s\xi e_\mu).$$
For $\mu < \nu$, insert the path average defining Q_j into the coarse curl and telescope in both directions. Since $B^j(y + e_\mu) = B^j(y) + e_\mu$, this gives
$$(\partial_1 \mathsf{Q}_j A)_{\mu\nu}(y) = \sum_{x \in B^j(y)} \xi^3 \sum_{s,t=0}^{n-1} \xi^2 (\partial A)_{\mu\nu}(x + s\xi e_\mu + t\xi e_\nu).$$
The positive weights sum to one. The Cauchy–Schwarz inequality therefore bounds the square by the same average of $|(\partial A)_{\mu\nu}|^2$. Sum over y. The blocks partition T_ξ, and translation preserves its weighted sum, so
$$\begin{aligned}\sum_y |(\partial_1 \mathsf{Q}_j A)_{\mu\nu}(y)|^2 &\le \sum_{s,t=0}^{n-1} \xi^2 \sum_{x \in \mathsf{T}_\xi} \xi^3 |(\partial A)_{\mu\nu}(x + s\xi e_\mu + t\xi e_\nu)|^2 \\ &= \|(\partial A)_{\mu\nu}\|^2.\end{aligned}$$
Summing over the three pairs proves (a).

For (b), commutation with the scalar average gives
$$\Delta^1 \mathsf{Q}'_j \lambda = \partial_1^* \mathsf{Q}_j \partial\lambda = \sum_{\mu=1}^{3} \partial_{1,\mu}^* (\mathsf{Q}_j \partial\lambda)_\mu.$$

The same double telescoping, now in one direction, gives

$$(\partial^{*}_{1,\mu}(\mathsf{Q}_j\partial\lambda)_\mu)(y)$$
$$= -\sum_{x\in B^j(y-e_\mu)} \xi^3 \sum_{s,t=0}^{n-1} \xi^2(\partial_\mu\partial_\mu\lambda)(x+(s+t)\xi e_\mu).$$

Thus its norm is at most $\|\partial_\mu\partial_\mu\lambda\| = \|\partial^*_\mu\partial_\mu\lambda\|$; the equality uses the translation relating forward and adjoint differences. Consequently,

$$\|\Delta^1\mathsf{Q}'_j\lambda\|^2 \le 3\sum_\mu \|\partial^*_\mu\partial_\mu\lambda\|^2 \le 3\|\Delta\lambda\|^2.$$

For the last inequality, commutation of the differences gives

$$\|\Delta\lambda\|^2 = \sum_{\mu,\nu}\langle\partial^*_\mu\partial_\mu\lambda, \partial^*_\nu\partial_\nu\lambda\rangle = \sum_{\mu,\nu}\|\partial_\mu\partial_\nu\lambda\|^2.$$

To construct the extension in (c), temporarily use complex functions and the Hermitian inner products. Set

$$\mathcal{P}_N = \frac{2\pi}{N}\mathbb{Z}^3 \cap [-\pi,\pi)^3, \qquad \epsilon_p(x) = e^{\mathrm{i}p\cdot x}.$$

The restrictions to T_1 are its N^3 characters. They remain distinct characters on T_ξ and are orthogonal there, with squared norm N^3 on both lattices. Moreover,

$$\Delta\epsilon_p = \Delta(p)\epsilon_p, \qquad \Delta(p) = \sum_{\nu=1}^{3} 4\xi^{-2}\sin^2(\xi p_\nu/2) \le |p|_2^2 \le 3\pi^2.$$

For the one-dimensional average, write

$$g(\theta) = \sum_{k=0}^{n-1} \xi e^{\mathrm{i}\theta k\xi}.$$

We have $g(0) = 1$, and, when $0 < |\theta| \le \pi$,

$$|g(\theta)| = \frac{\xi|\sin(\theta/2)|}{|\sin(\xi\theta/2)|} \ge \frac{2}{\pi}.$$

Here $|\sin(\theta/2)| \ge |\theta|/\pi$ and $|\sin(\xi\theta/2)| \le \xi|\theta|/2$. The block average contributes three factors and the path average one more. If $A_\mu = \epsilon_p$ and the other components vanish, then

$$(\mathsf{Q}_jA)_\mu = w_\mu(p)\epsilon_p|_{\mathsf{T}_1}, \qquad w_\mu(p) = g(p_\mu)\prod_{\nu=1}^{3} g(p_\nu), \qquad |w_\mu(p)| \ge (2/\pi)^4.$$

Expand $B_\mu = \sum_{p\in\mathcal{P}_N}\beta_\mu(p)\epsilon_p|_{\mathsf{T}_1}$ and set

$$A^{\mathbb{C}}_\mu = \sum_{p\in\mathcal{P}_N}\beta_\mu(p)w_\mu(p)^{-1}\epsilon_p.$$

Then $\mathsf{Q}_jA^{\mathbb{C}} = B$, and orthogonality gives

$$\langle A^{\mathbb{C}}, \Delta A^{\mathbb{C}}\rangle = \sum_{\mu,p} N^3\Delta(p)|w_\mu(p)|^{-2}|\beta_\mu(p)|^2$$
$$\le 3\pi^2(\pi/2)^8\sum_{\mu,p} N^3|\beta_\mu(p)|^2 = \gamma_1\|B\|^2.$$

Since Q_j and B are real, $A = \operatorname{Re}A^{\mathbb{C}}$ still has $\mathsf{Q}_jA = B$. Finally, $\langle A, \Delta A\rangle = \|\nabla A\|^2 \le \|\nabla A^{\mathbb{C}}\|^2$, which proves (c).

If $\mathsf{Q}'_{1,L}\omega = 0$, the mean of ω on every $B(y')$ vanishes. The path spectrum computed in the proof of proposition 16.1(a),(b), with three tensor factors, gives the gap $4\sin^2(\pi/(2L)) \ge 4L^{-2}$ for this

box. The blocks have L sites along each edge and the torus period satisfies $N \geq 2L$, so their internal bonds are distinct. Summing the box inequalities and discarding the bonds joining different blocks gives

$$4L^{-2}\|\omega\|^2 \leq \|\partial_1\omega\|^2 = \langle\omega, \Delta^1\omega\rangle \leq \|\omega\|\,\|\Delta^1\omega\|,$$

which proves (d).

For (203), apply Cauchy–Schwarz to the probability weights in $\mathsf{Q}_{1,L}$:

$$|(\mathsf{Q}_{1,L}B)_\mu(y')|^2 \leq \sum_{y\in B(y')} L^{-3}\sum_{t=0}^{L-1} L^{-1}|B_\mu(y+te_\mu)|^2.$$

Multiply by the coarse volume L^3 and sum over y', μ. For each t, the blocks followed by translation by te_μ cover T_1 once. Each squared component therefore occurs L times with weight L^{-1}, giving the claim. □

20.2. **Constrained extensions and the gauge correction.** To separate the two scales, we first extend prescribed unit-scale averages to the fine lattice. The scalar extension minimizes the squared Laplacian; the vector extension minimizes the one-scale gauge-fixed energy. Their Euler equations identify the additional part of the scalar projection and remove the mixed term in the vector energy.

Write

$$N_j = \ker \mathsf{Q}'_j, \qquad D_j = \partial^*\partial + \partial\mathsf{R}_j\partial^*, \qquad \Delta_{1,j} = D_j + \mathsf{Q}_j^*\mathsf{Q}_j.$$

Thus $G_j = \Delta_{1,j}^{-1}$ is the one-scale inverse already constructed. Inverses of operators compressed to a subspace will always be extended by zero on its orthogonal complement; this convention includes the zero subspace.

Lemma 20.4. The following constructions are defined in the setting of section 20.

(a) For every $\varphi \in \mathcal{F}_1$ there is a unique minimizer $H'_j\varphi$ of $\|\Delta\lambda\|^2$ subject to $\mathsf{Q}'_j\lambda = \varphi$. The map H'_j is linear and

$$\mathsf{Q}'_jH'_j = I, \qquad \mathsf{R}_j\Delta H'_j = 0.$$

Set $\Delta'_j = (\Delta H'_j)^*(\Delta H'_j)$. Then

$$\langle\varphi, \Delta'_j\varphi\rangle \geq \tfrac{1}{3}\|\Delta^1\varphi\|^2.$$

(b) The scalar extension satisfies

$$\begin{aligned} \partial H'_j &= G_j\mathsf{Q}_j^*\partial_1, \\ \partial\Delta H'_j &= \Delta G_j\mathsf{Q}_j^*\partial_1, \\ \Delta H'_j &= \partial^* G_j\mathsf{Q}_j^*\partial_1. \end{aligned} \tag{204}$$

(c) Let

$$W = \{\omega \in \mathcal{F}_1 : \omega = 0 \text{ on } \mathsf{T}_1 \setminus \Lambda,\ \mathsf{Q}'_{1,L}\omega = 0\},$$

and let $\iota_W : W \hookrightarrow \mathcal{F}_1$ be the inclusion. On W,

$$\langle\omega, \Delta'_j\omega\rangle \geq \frac{16}{3L^4}\|\omega\|^2.$$

Consequently

$$C' = \iota_W(\iota_W^*\Delta'_j\iota_W)^{-1}\iota_W^*$$

is defined, and

$$\begin{aligned} N(\mathsf{Q}') &= N_j \oplus H'_jW, \\ \mathsf{R} &= \mathsf{R}_j + \Delta H'_jC'H_j'^*\Delta. \end{aligned} \tag{205}$$

The two terms in the second line are orthogonal projections onto the mutually orthogonal spaces ΔN_j and $\Delta H'_jW$, respectively.

(d) Define

$$\mathcal{A}^{\mathrm{ax}} = \{B \in \mathcal{A}_1 : B = 0 \text{ on } \mathcal{T}(y') \text{ for every } y' \in \Lambda'\}.$$

For every $B \in \mathcal{A}_1$ there is a unique $\omega \in W$ such that $B + \partial_1\omega \in \mathcal{A}^{\mathrm{ax}}$.

(e) For $\omega \in W$ and $A \in \mathcal{A}_\xi$,

$$\begin{aligned}\partial(A - \partial H_j'\omega) &= \partial A,\\ \mathsf{R}_j\partial^*(A - \partial H_j'\omega) &= \mathsf{R}_j\partial^* A,\\ \mathsf{q}\big(\mathsf{Q}_j(A - \partial H_j'\omega)\big) &= \mathsf{q}(\mathsf{Q}_j A).\end{aligned}$$

(f) The operator D_j is positive definite on $\ker \mathsf{Q}_j$. Let $\iota : \ker \mathsf{Q}_j \hookrightarrow \mathcal{A}_\xi$ be the inclusion and set

$$\widetilde{G}_j = \iota(\iota^* D_j \iota)^{-1}\iota^*, \qquad S = (\mathsf{Q}_j G_j \mathsf{Q}_j^*)^{-1}.$$

For every $B \in \mathcal{A}_1$ there is a unique minimizer $H_j B$ of $\langle A, D_j A\rangle$ subject to $\mathsf{Q}_j A = B$. Define the symmetric operator Δ_j by $\langle B, \Delta_j B\rangle = \langle H_j B, D_j H_j B\rangle$. Then

$$\begin{aligned}H_j &= G_j \mathsf{Q}_j^* S,\\ \widetilde{G}_j &= G_j - G_j \mathsf{Q}_j^* S \mathsf{Q}_j G_j,\\ \Delta_j &= S - I.\end{aligned} \tag{206}$$

Moreover, with γ_1 as in lemma 20.3,

$$\|\partial_1 B\|^2 \le \langle B, \Delta_j B\rangle \le \gamma_1 \|B\|^2, \qquad I \le S \le (1+\gamma_1) I,$$

and $\langle A', D_j H_j B\rangle = 0$ for $A' \in \ker \mathsf{Q}_j$.

(g) Put

$$F(B) = \mathsf{q}(B) + \langle B, \Delta_j B\rangle, \qquad \gamma_0'' = (1 + 2916 L^4)^{-1}.$$

For $B \in \mathcal{A}^{\mathrm{ax}}$,

$$\mathsf{q}(B) + \|\partial_1 B\|^2 \ge \gamma_0'' \|B\|^2, \tag{207}$$

and hence

$$F(B) \ge \gamma_0'' \|B\|^2. \tag{208}$$

Let T_F be the symmetric operator of the form F, and let $\iota_{\mathrm{ax}} : \mathcal{A}^{\mathrm{ax}} \hookrightarrow \mathcal{A}_1$ be the inclusion. The operator

$$\widetilde{C} = \iota_{\mathrm{ax}}(\iota_{\mathrm{ax}}^* T_F \iota_{\mathrm{ax}})^{-1}\iota_{\mathrm{ax}}^*$$

is defined.

■

Proof. The scalar extension. Since $\mathsf{Q}_j' \mathsf{Q}_j'^* = I$, the affine constraint in (a) is nonempty and has direction space N_j. If $\nu \in N_j$ and $\Delta\nu = 0$, then $\|\partial\nu\|^2 = \langle \nu, \Delta\nu\rangle = 0$, so ν is constant. Its block average is zero, and hence $\nu = 0$. Thus the squared Laplacian is strictly positive on the direction space. Its unique minimizer is characterized by

$$\mathsf{Q}_j' \lambda = \varphi, \qquad \langle \Delta\lambda, \Delta\nu\rangle = 0 \quad (\nu \in N_j).$$

The second condition is $\mathsf{R}_j \Delta\lambda = 0$. These equations also show that the minimizer depends linearly on φ. By lemma 20.3(b),

$$\|\Delta^1 \varphi\|^2 = \|\Delta^1 \mathsf{Q}_j' H_j' \varphi\|^2 \le 3 \|\Delta H_j' \varphi\|^2,$$

which proves the last assertion of (a).

On scalar functions, $\partial\partial = 0$ and $\partial^*\partial = \Delta$. Therefore

$$D_j \partial H_j' \varphi = \partial \mathsf{R}_j \Delta H_j' \varphi = 0, \qquad \mathsf{Q}_j \partial H_j' \varphi = \partial_1 \mathsf{Q}_j' H_j' \varphi = \partial_1 \varphi.$$

It follows that $\Delta_{1,j} \partial H_j' \varphi = \mathsf{Q}_j^* \partial_1 \varphi$. Applying G_j gives the first identity in (204). Commutation of ∂ with Δ gives the second, and application of ∂^* gives the third.

The additional scalar projection. For $\omega \in W$, part (a) and lemma 20.3(d) give

$$\langle \omega, \Delta'_j \omega \rangle \geq \tfrac{1}{3}\|\Delta^1 \omega\|^2 \geq \frac{16}{3L^4}\|\omega\|^2.$$

Thus the compressed inverse defining C' exists. The mixed average constraint is equivalent to $\mathsf{Q}'_j \lambda \in W$: on every block outside Λ, the unit-site averages already vanish, so their coarse average vanishes as well. For $\lambda \in N(\mathsf{Q}')$,

$$\lambda - H'_j \mathsf{Q}'_j \lambda \in N_j.$$

Both N_j and $H'_j W$ lie in $N(\mathsf{Q}')$, and their intersection is zero because $\mathsf{Q}'_j H'_j = I$. This proves the direct sum in (205).

The equation $\mathsf{R}_j \Delta H'_j = 0$ makes its two images under Δ orthogonal. The map $Z = \Delta H'_j \iota_W$ is injective by the lower bound just proved, and $Z^*Z = \iota_W^* \Delta'_j \iota_W$. Consequently the orthogonal projection onto its range is

$$Z(Z^*Z)^{-1}Z^* = \Delta H'_j C' H'^*_j \Delta.$$

Adding the projection onto ΔN_j gives the second identity in (205).

Choosing the axial representative. Write $\Gamma_{y',y}$ for the coordinate-tree path from y' to y, and $B(\Gamma)$ for the signed sum of B along a path. On a block $B(y')$, vanishing on the tree is equivalent to vanishing of these sums along all $\Gamma_{y',y}$: each tree bond extends the root path to its initial endpoint, so its value is the difference of the two root-path sums. Along such a path,

$$(\partial_1 \omega)(\Gamma_{y',y}) = \omega(y) - \omega(y').$$

The required tree condition and the zero block mean therefore force

$$\omega(y') = L^{-3} \sum_{y \in B(y')} B(\Gamma_{y',y}), \qquad \omega(y) = \omega(y') - B(\Gamma_{y',y}).$$

Define ω by these formulas on the blocks with $y' \in \Lambda'$ and by zero outside Λ. Its mean on every block is zero, so $\omega \in W$, and the tree condition holds. The same formulas establish uniqueness, proving (d).

The first two identities in (e) follow from $\partial\partial = 0$ and $\mathsf{R}_j \partial^* \partial H'_j = \mathsf{R}_j \Delta H'_j = 0$. For the last one, the change in the unit-scale average is $\mathsf{Q}_j \partial H'_j \omega = \partial_1 \omega$. It vanishes on Λ^{c*} because both endpoints of each such bond lie outside Λ. Its coarse average also vanishes, since

$$\mathsf{Q}_{1,L} \partial_1 \omega = \partial_L \mathsf{Q}'_{1,L} \omega = 0.$$

These are exactly the two sets of values entering $\mathfrak{q}$, which proves its invariance.

The vector extension. The form of D_j is nonnegative. On $\ker \mathsf{Q}_j$ it agrees with the positive form of $\Delta_{1,j}$, so the compression of D_j is positive definite. By lemma 20.3(c), Q_j is onto. Thus Q_j^* is injective and $\mathsf{Q}_j G_j \mathsf{Q}_j^*$ is positive definite; in particular, S exists.

The constrained-square calculation in section 16.4, with $X = D_j$, $\mathsf{Q} = \mathsf{Q}_j$ and $a = 1$, now applies. Indeed, $A_B = G_j \mathsf{Q}_j^* S B$ satisfies $\mathsf{Q}_j A_B = B$ and $\langle A', D_j A_B \rangle = \langle \mathsf{Q}_j A', S B \rangle = 0$ for $A' \in \ker \mathsf{Q}_j$. Positivity on that kernel makes $A_B = H_j B$ the unique minimizer, and

$$\langle H_j B, D_j H_j B \rangle = \langle H_j B, \Delta_{1,j} H_j B \rangle - \|B\|^2 = \langle B, (S - I) B \rangle.$$

This proves the first and third identities in (206) and the mixed-term identity in (f).

For the second identity, set $C_1 = G_j - G_j \mathsf{Q}_j^* S \mathsf{Q}_j G_j$. Then $\mathsf{Q}_j C_1 = 0$ and, for every $A' \in \ker \mathsf{Q}_j$,

$$\langle A', D_j C_1 J \rangle = \langle A', \Delta_{1,j} C_1 J \rangle = \langle A', J \rangle.$$

These equations uniquely characterize the inverse of the compressed D_j, extended by zero: the difference of two solutions lies in $\ker \mathsf{Q}_j$ and has zero D_j energy. Hence $C_1 = \widetilde{G}_j$.

Curl contraction in lemma 20.3(a) gives

$$\langle B, \Delta_j B \rangle \geq \|\partial H_j B\|^2 \geq \|\partial_1 \mathsf{Q}_j H_j B\|^2 = \|\partial_1 B\|^2.$$

For the upper bound, choose the extension A from lemma 20.3(c). Since $\Delta - D_j = \partial \mathsf{P}_j \partial^* \geq 0$, minimality gives

$$\langle B, \Delta_j B\rangle \leq \langle A, D_j A\rangle \leq \langle A, \Delta A\rangle \leq \gamma_1 \|B\|^2.$$

The bounds on S now follow from $S = I + \Delta_j$.

The axial lower bound. Let $B \in \mathcal{A}^{\mathrm{ax}}$, and write

$$B = B_1 + B_2, \qquad B_1 = \chi_{\Lambda^*} B, \qquad B_2 = \chi_{\Lambda^{c*}} B.$$

Every tree bond in a block belonging to Λ has both endpoints in Λ. Thus B_1 vanishes on the trees and outside Λ^*, and the supported axial estimate lemma 20.6, proved in section 20.4, applies to B_1. Write $\varkappa = (108L^4)^{-1}$ and $F_0(B) = \mathfrak{q}(B) + \|\partial_1 B\|^2$. The weighted coarse contraction gives

$$L \sum_{c \in \Lambda'^*} |(\mathsf{Q}_{1,L} B_2)(c)|^2 \leq L^{-2} \|\mathsf{Q}_{1,L} B_2\|^2 \leq L^{-2} \|B_2\|^2.$$

Here the coarse scalar product has weight L^3. Also

$$\|\partial_1 B_2\|^2 \leq \langle B_2, \Delta^1 B_2\rangle \leq 12 \|B_2\|^2.$$

Apply the supported estimate to $B_1 = B - B_2$, and bound each square of a difference by twice the sum of the two squares. It follows that

$$\begin{aligned} \varkappa \|B_1\|^2 &\leq 2 \left(L \sum_{c \in \Lambda'^*} |(\mathsf{Q}_{1,L} B)(c)|^2 + \|\partial_1 B\|^2 \right) + (2L^{-2} + 24) \|B_2\|^2 \\ &\leq 27 F_0(B). \end{aligned}$$

Indeed, $\|B_2\|^2$ is the first term in $\mathfrak{q}(B)$, and $2 + 2L^{-2} \leq 3$ since $L \geq 2$. The supports of B_1 and B_2 are disjoint, so

$$\|B\|^2 = \|B_1\|^2 + \|B_2\|^2 \leq (1 + 27\varkappa^{-1}) F_0(B) = (1 + 2916 L^4) F_0(B).$$

This proves (207). Part (f) gives $F(B) \geq F_0(B)$, hence (208) and the existence of $\widetilde{C}$. □

20.3. **The two-scale representation.** The constrained extensions give coordinates in which the two-scale energy is a sum of three positive forms. The additional divergence term is a scalar square; shifting that square produces the correction to the one-scale inverse.

Proposition 20.5. Let the operators be those of lemma 20.4.

(a) The map

$$\ker \mathsf{Q}_j \times \mathcal{A}^{\mathrm{ax}} \times W \longrightarrow \mathcal{A}_\xi, \qquad (A', B, \omega) \longmapsto A' + H_j B - \partial H'_j \omega$$

is a linear bijection. In these coordinates,

$$\begin{aligned} \langle A, \Delta_1 A\rangle &= \langle A', D_j A'\rangle + F(B) + \langle \omega - \omega_0, \Delta'_j (\omega - \omega_0)\rangle, \\ \omega_0 &= C' H'^*_j \Delta \partial^* (A' + H_j B). \end{aligned} \tag{209}$$

In particular, Δ_1 is positive definite.

(b) Put $\mathsf{K} = \partial \Delta H'_j C' H'^*_j \partial^*$. The inverse $G = \Delta_1^{-1}$ satisfies

$$\begin{aligned} \langle J, GJ\rangle = \langle J, \partial H'_j C' H'^*_j \partial^* J\rangle \\ + \langle (I - \mathsf{K})J, (\widetilde{G}_j + H_j \widetilde{C} H_j^*)(I - \mathsf{K})J\rangle. \end{aligned} \tag{210}$$

Equivalently,

$$G = \partial H'_j C' H'^*_j \partial^* + (I - \mathsf{K})^* (\widetilde{G}_j + H_j \widetilde{C} H_j^*)(I - \mathsf{K}).$$

(c) Define maps from $\mathcal{A}_1$ to $\mathcal{A}_\xi$ and operators on $\mathcal{A}_1$ by

$$\begin{aligned} X &= G_j \mathsf{Q}_j^*, & Y &= G_j \Delta X, \\ \Theta &= \partial_1 C' \partial_1^*, & V &= -S + S\widetilde{C}S, \\ \mathsf{N}_1 &= X^* \Delta X, & \mathsf{N}_2 &= (\Delta X)^* G_j \Delta X. \end{aligned}$$

Then

$$\begin{aligned} G &= G_j + X U_0 X^* - X\Theta Y^* - Y\Theta X^*, \\ U_0 &= \Theta + V - \Theta \mathsf{N}_1 V - V \mathsf{N}_1 \Theta + \Theta \mathsf{N}_2 \Theta + \Theta \mathsf{N}_1 V \mathsf{N}_1 \Theta. \end{aligned} \tag{211}$$

If $\Lambda' = \varnothing$, then $W = \{0\}$, $C' = 0$, $\widetilde{C} = S^{-1}$, $V = 0$ and $G = G_j$.

■

Proof. The coordinates and the energy. Given $A \in \mathcal{A}_\xi$, choose the unique $\omega \in W$ for which

$$B = \mathsf{Q}_j A + \partial_1 \omega \in \mathcal{A}^{\mathrm{ax}};$$

this is possible by lemma 20.4(d). Set $A' = A + \partial H_j' \omega - H_j B$. Since $\mathsf{Q}_j \partial H_j' = \partial_1$ and $\mathsf{Q}_j H_j = I$, we have $\mathsf{Q}_j A' = \mathsf{Q}_j A + \partial_1 \omega - B = 0$. This proves surjectivity. Conversely, any such representation of A satisfies $\mathsf{Q}_j A + \partial_1 \omega = B \in \mathcal{A}^{\mathrm{ax}}$. The uniqueness of its axial correction determines ω, then B, and finally A'. The map is linear, so it is the stated bijection.

Put $A_0 = A' + H_j B$ and $\mathsf{R}_W = \Delta H_j' C' H_j'^* \Delta$. The projection identity (205) is an orthogonal sum. Hence

$$\langle A, \Delta_1 A \rangle = \|\partial A\|^2 + \|\mathsf{R}_j \partial^* A\|^2 + \mathfrak{q}(\mathsf{Q}_j A) + \|\mathsf{R}_W \partial^* A\|^2.$$

The first three terms are unchanged when $A = A_0 - \partial H_j' \omega$ is replaced by A_0, by lemma 20.4(e). Since $\mathsf{Q}_j A_0 = B$, their sum is

$$\begin{aligned} \langle A_0, D_j A_0 \rangle + \mathfrak{q}(B) &= \langle A', D_j A' \rangle + 2\langle A', D_j H_j B \rangle \\ &\quad + \langle H_j B, D_j H_j B \rangle + \mathfrak{q}(B) = \langle A', D_j A' \rangle + F(B). \end{aligned}$$

The mixed term is zero by the minimizing equation in lemma 20.4(f).

It remains to compute the last divergence term. On scalars $\partial^* \partial = \Delta$, so

$$\partial^* A = \partial^* A_0 - \Delta H_j' \omega, \qquad \mathsf{R}_W \partial^* A_0 = \Delta H_j' \omega_0.$$

The vector ω_0 lies in W. For $\omega \in W$, the compressed inverse satisfies $C' \Delta_j' \omega = \omega$; therefore

$$\mathsf{R}_W \Delta H_j' \omega = \Delta H_j' C' (\Delta H_j')^* (\Delta H_j') \omega = \Delta H_j' \omega.$$

It follows that

$$\|\mathsf{R}_W \partial^* A\|^2 = \|\Delta H_j' (\omega_0 - \omega)\|^2 = \langle \omega - \omega_0, \Delta_j' (\omega - \omega_0) \rangle,$$

which proves (209).

All three terms in that identity are nonnegative. Their lower bounds on the three constrained spaces are given in lemma 20.4(c,f,g). If the sum is zero, then $A' = 0$ and $B = 0$, so $\omega_0 = 0$; the scalar term then forces $\omega = 0$. Thus $A = 0$, proving positivity and the existence of G.

The inverse. We use the following finite-dimensional square completion. If a symmetric operator T is positive definite on a subspace $\mathcal{V}_0$, and C_0 is its compressed inverse extended by zero, then, for $v \in \mathcal{V}_0$,

$$2\langle v, J \rangle - \langle v, Tv \rangle = \langle J, C_0 J \rangle - \langle v - C_0 J, T(v - C_0 J) \rangle.$$

Indeed, $C_0 J \in \mathcal{V}_0$ and $\langle v, T C_0 J \rangle = \langle v, J \rangle$ on that subspace. The supremum over v is therefore $\langle J, C_0 J \rangle$, attained at $v = C_0 J$. Applying this first to $T = \Delta_1$ on $\mathcal{A}_\xi$ gives

$$\langle J, G J \rangle = \sup_{A \in \mathcal{A}_\xi} \big(2\langle A, J \rangle - \langle A, \Delta_1 A \rangle \big).$$

Use the coordinates in (a) and write $\omega = \omega_0 + \omega'$. Symmetry of C' and Δ gives

$$\begin{aligned}\langle \partial H'_j \omega_0, J\rangle &= \langle \partial H'_j C' H'^*_j \Delta \partial^* A_0, J\rangle \\ &= \langle A_0, \partial \Delta H'_j C' H'^*_j \partial^* J\rangle = \langle A_0, \mathsf{K} J\rangle.\end{aligned}$$

The expression to be maximized thus becomes

$$\begin{aligned}&- 2\langle \omega', H'^*_j \partial^* J\rangle - \langle \omega', \Delta'_j \omega'\rangle \\ &+ 2\langle A', (I-\mathsf{K})J\rangle - \langle A', D_j A'\rangle \\ &+ 2\langle B, H^*_j (I-\mathsf{K})J\rangle - F(B).\end{aligned}$$

For fixed A' and B, translation by $\omega_0 \in W$ is a bijection of W. Consequently the three variables range independently over W, $\ker \mathsf{Q}_j$ and $\mathcal{A}^{\mathrm{ax}}$. The three suprema are

$$\begin{gathered}\langle H'^*_j \partial^* J, C' H'^*_j \partial^* J\rangle, \\ \langle (I-\mathsf{K})J, \widetilde{G}_j (I-\mathsf{K})J\rangle, \\ \langle H^*_j (I-\mathsf{K})J, \widetilde{C} H^*_j (I-\mathsf{K})J\rangle.\end{gathered}$$

Their sum is (210). Both sides of its operator version are symmetric and have this quadratic form, so they agree.

Expansion about the one-scale inverse. By (204), $\partial H'_j = X \partial_1$ and $\partial \Delta H'_j = \Delta X \partial_1$. It follows that

$$\partial H'_j C' H'^*_j \partial^* = X \Theta X^*, \qquad \mathsf{K} = \Delta X \Theta X^*.$$

By (206), $H_j = XS$ and $\widetilde{G}_j = G_j - XSX^*$. Thus

$$\widetilde{G}_j + H_j \widetilde{C} H^*_j = G_j + XVX^*.$$

Substitution into (b), using symmetry of Θ, gives

$$G = X\Theta X^* + (I - X\Theta(\Delta X)^*)(G_j + XVX^*)(I - \Delta X \Theta X^*).$$

We keep the order of the factors when multiplying. Since G_j and Δ are symmetric,

$$(\Delta X)^* G_j = Y^*, \qquad (\Delta X)^* X = X^* \Delta X = \mathsf{N}_1, \qquad (\Delta X)^* G_j \Delta X = \mathsf{N}_2.$$

The two cross products and the remaining double product are

$$\begin{aligned}X\Theta(\Delta X)^*(G_j + XVX^*) &= X\Theta Y^* + X\Theta \mathsf{N}_1 V X^*, \\ (G_j + XVX^*)\Delta X \Theta X^* &= Y\Theta X^* + XV\mathsf{N}_1\Theta X^*, \\ X\Theta(\Delta X)^*(G_j + XVX^*)\Delta X\Theta X^* &= X\Theta\mathsf{N}_2\Theta X^* + X\Theta\mathsf{N}_1 V \mathsf{N}_1 \Theta X^*.\end{aligned}$$

The first two are subtracted and the last is added. Collecting the terms with outer factors X and X^* gives exactly (211).

If $\Lambda' = \varnothing$, then $\mathcal{A}^{\mathrm{ax}} = \mathcal{A}_1$ and $\mathsf{q}(B) = \|B\|^2$. Since $\Delta_j = S - I$, we have $F(B) = \langle B, SB\rangle$ and $\widetilde{C} = S^{-1}$, so $V = 0$. Also $W = \{0\}$, hence $C' = 0$, $\Theta = 0$ and $U_0 = 0$. Formula (211) then gives $G = G_j$. □

In (211), the maps X and Y have unit-bond inputs and fine-bond outputs, while the coefficients U_0 and Θ act on unit bonds. The estimates in section 20.6 use this distinction to transfer the one-scale regularity bounds to the three correction terms.

20.4. **The axial-gauge gap.** The tree condition controls bonds inside a block by its curl. For bonds crossing a block face, the curl controls the variation along the face and the coarse average controls the mean. We keep a separate estimate on the tangential face bonds, since these are the error terms in the crossing-face calculation.

Lemma 20.6. Suppose that $B \in \mathcal{A}_1$ vanishes outside Λ^* and on $\mathcal{T}(y)$ for every $y \in \Lambda'$. Then

$$L \sum_{c \in \Lambda'^*} |(\mathsf{Q}_{1,L} B)(c)|^2 + \|\partial_1 B\|^2 \geq \frac{1}{108} L^{-4} \|B\|^2. \tag{212}$$

■

Proof. Put $F = \partial_1 B$. Coordinates in a block or in two adjacent blocks are taken on a lift of their coarse sites. Since $N \geq 2L$, the sites in these coordinate boxes are distinct. For $y \in \mathsf{T}_L$, define

$$\Phi_\mu^-(y) = \{x \in B(y) : x_\mu = y_\mu\},$$
$$\Phi_\mu^+(y) = \{x \in B(y) : x_\mu = y_\mu + L - 1\},$$
$$X_y^\mu = \sum_{(x,\mu) \subset B(y)} |B_\mu(x)|^2, \qquad X_y = \sum_\mu X_y^\mu, \qquad S_y = \sum_{p \subset B(y)} |F(p)|^2.$$

Here a bond or plaquette is inside a block when all its vertices are there. For a face Φ, let Φ° be its bonds, with both endpoints in Φ, and put $E_\Phi = \sum_{b \in \Phi^\circ} |B(b)|^2$.

For a coarse bond $c = (y, \mu)$, write

$$\Phi'(c) = \Phi_\mu^+(c_-), \qquad \Phi''(c) = \Phi_\mu^-(c_+), \qquad B(c) = \{(x, \mu) : x \in \Phi'(c)\}.$$

Thus $B(c)$ consists of the unit bonds crossing from $B(c_-)$ to $B(c_+)$. Translation by e_μ maps $\Phi'(c)$ and its tangential bonds bijectively onto $\Phi''(c)$ and its tangential bonds. Let $\mathcal{P}(c)$ consist of the plaquettes with corners $x, x + e_\mu, x + e_\mu + e_\nu, x + e_\nu$, where $(x, \nu) \in \Phi'(c)^\circ$. There is exactly one such plaquette for each bond of $\Phi'(c)^\circ$. Set

$$f_c(x) = B_\mu(x) \quad (x \in \Phi'(c)), \qquad \bar{f}_c = L^{-2} \sum_{x \in \Phi'(c)} f_c(x),$$
$$Y_c = \sum_{x \in \Phi'(c)} |f_c(x)|^2, \qquad S_c = \sum_{p \in \mathcal{P}(c)} |F(p)|^2,$$
$$D_c = \sum_{(x,\nu) \in \Phi'(c)^\circ} |f_c(x + e_\nu) - f_c(x)|^2.$$

Bonds inside a block. Fix a block $B(y)$ on whose tree B vanishes. For a bond $(x, \iota) \subset B(y)$, put

$$x^{(i)} = (y_1, \dots, y_i, x_{i+1}, \dots, x_3), \qquad 0 \leq i < \iota.$$

All these sites and their translates by e_ι remain in the block. The bond $(x^{(\iota-1)}, \iota)$ lies in the tree, so its value is zero. For $i < \iota$, join $x^{(i)}$ to $x^{(i-1)}$ in direction i. At consecutive sites $w, w + e_i$ on this segment, both (w, i) and $(w + e_\iota, i)$ lie in the tree: their coordinates of index less than i equal those of y. Hence

$$F(w, i, \iota) = B_i(w) + B_\iota(w + e_i) - B_i(w + e_\iota) - B_\iota(w) = B_\iota(w + e_i) - B_\iota(w).$$

Telescope along these segments. If $\varpi(x, \iota)$ denotes the resulting set of plaquettes, then $B_\iota(x) = \sum_{p \in \varpi(x,\iota)} \pm F(p)$ and $\#\varpi(x, \iota) \leq (\iota - 1)(L - 1)$. Therefore

$$|B_\iota(x)|^2 \leq (\iota - 1)(L - 1) \sum_{p \in \varpi(x,\iota)} |F(p)|^2. \tag{213}$$

For $\iota = 1$, the bond already lies in the tree.

For a fixed plaquette (w, i, ι) with $i < \iota$, its occurrence requires $w_m = y_m$ for $m < i$. The coordinates of x with index greater than i must equal those of w; there are at most $L - 1$ choices of

x_i, and the first $i-1$ coordinates are arbitrary. Its multiplicity is thus at most $(L-1)L^{i-1}$, and hence at most $(L-1)L^{\iota-2}$. Summing (213) for $\iota = 2, 3$ gives

$$X_y^\iota \le (\iota-1)(L-1)^2 L^{\iota-2} \sum_{i<\iota} \sum_{(w,i,\iota)\subset B(y)} |F(w,i,\iota)|^2.$$

The plaquette sets for different ι are disjoint, since ι is their larger direction. Consequently,

$$X_y \le 2(L-1)^2 L\, S_y \le 3L^3 S_y. \tag{214}$$

Tangential face bonds. Let $\Phi = \{x \in B(y) : x_\mu = s\}$ be either face normal to direction μ. Its bonds have direction $\iota \neq \mu$. In the preceding count for (w,i,ι), impose $x_\mu = s$. If $\mu < i$, one of the first $i-1$ coordinates is fixed, giving at most $(L-1)L^{i-2} \le L-1$ choices. If $\mu = i$, there are at most $L^{i-1} \le L$ choices. If $\mu > i$, then $i < \iota$, $i < \mu$ and $\iota \neq \mu$ force $i = 1$, so there are at most $L-1$ choices. Thus the multiplicity is at most L in every case. Summing (213) over the face, and again using the disjoint plaquette sets, gives the sharper estimate

$$E_\Phi \le 2(L-1)L\, S_y. \tag{215}$$

Bonds crossing a face. This calculation does not require the tree condition. For $(x,\nu) \in \Phi'(c)^\circ$, the plaquette relation is

$$f_c(x+e_\nu) - f_c(x) = B_\nu(x+e_\mu) - B_\nu(x) \mp F(p), \qquad p \in \mathcal{P}(c).$$

Apply the three-term Cauchy–Schwarz inequality, then sum over the tangential bonds. The translation bijection between the faces gives

$$S_c \ge \tfrac{1}{3} D_c - E_{\Phi'(c)} - E_{\Phi''(c)}. \tag{216}$$

The coarse average also sees the mean of the crossing bonds. Put $\beta = y_\mu$, on the chosen lift. In

$$(\mathsf{Q}_{1,L} B)(c) = L^{-4} \sum_{x \in B(c_-)} \sum_{k=0}^{L-1} B_\mu(x + k e_\mu),$$

a component at coordinate $z_\mu = \beta + t$ occurs $t+1$ times for $0 \le t \le L-1$, and $2L-1-t$ times for $L \le t \le 2L-2$. The terms at $t = L-1$ therefore contribute $L^{-1}\bar f_c$. The other terms are

$$\mathsf{a}_c = L^{-4} \sum_{x \in B(c_-)\setminus\Phi'(c)} (x_\mu - \beta + 1) B_\mu(x),$$
$$\mathsf{b}_c = L^{-4} \sum_{x \in B(c_+)\setminus\Phi^+_\mu(c_+)} (2L - 1 + \beta - x_\mu) B_\mu(x).$$

They involve only direction-μ bonds inside the indicated blocks. Each sum has at most L^3 coefficients, all of modulus at most L^{-3}, so

$$|\mathsf{a}_c|^2 \le L^{-3} X^\mu_{c_-}, \qquad |\mathsf{b}_c|^2 \le L^{-3} X^\mu_{c_+}.$$

Since $L^{-1}\bar f_c = (\mathsf{Q}_{1,L}B)(c) - \mathsf{a}_c - \mathsf{b}_c$, we obtain

$$|(\mathsf{Q}_{1,L}B)(c)|^2 \ge \tfrac{1}{3} L^{-2} |\bar f_c|^2 - L^{-3}(X^\mu_{c_-} + X^\mu_{c_+}). \tag{217}$$

The face graph is the product of two paths with L vertices. Taking two tensor factors in the path spectrum from the proof of proposition 16.1(a),(b) shows that its first positive eigenvalue is

$$\kappa_L = 4 \sin^2 \frac{\pi}{2L} \ge 4L^{-2}.$$

Its constant eigenspace is removed by subtracting the face mean. Hence

$$D_c \geq \kappa_L \sum_{x\in\Phi'(c)} |f_c(x) - \bar{f}_c|^2,$$
$$Y_c = L^2|\bar{f}_c|^2 + \sum_{x\in\Phi'(c)} |f_c(x) - \bar{f}_c|^2.$$

Multiply (216) by $\frac{1}{4}L^{-2}$ and add (217). The inequality $\frac{1}{4}L^{-2}\kappa_L \geq L^{-4}$ makes the coefficients of the mean and the zero-mean part agree. We conclude that

$$\begin{aligned}\tfrac{1}{4}L^{-2}S_c + |(\mathsf{Q}_{1,L}B)(c)|^2 \geq \tfrac{1}{3}L^{-4}Y_c \\ - \tfrac{1}{4}L^{-2}(E_{\Phi'(c)} + E_{\Phi''(c)}) \\ - L^{-3}(X^\mu_{c_-} + X^\mu_{c_+}).\end{aligned} \tag{218}$$

Summation over the region. Every unit bond is either inside one block or crosses from that block to its unique neighbour in the bond direction. Thus it belongs to exactly one of the sets of internal bonds or $B(c)$. Since B vanishes outside Λ^*,

$$X = \sum_{y\in\Lambda'} X_y, \qquad Y = \sum_{c\in\Lambda'^*} Y_c, \qquad \|B\|^2 = X + Y.$$

If $y \notin \Lambda'$, every internal bond of $B(y)$ has both endpoints outside Λ, so X_y and all its E_Φ vanish. Write

$$S_{\mathrm{in}} = \sum_{y\in\Lambda'} S_y, \qquad S_{\mathrm{co}} = \sum_{c\in\Lambda'^*} S_c, \qquad T = L\sum_{c\in\Lambda'^*} |(\mathsf{Q}_{1,L}B)(c)|^2.$$

Each face occurs as $\Phi'(c)$ or $\Phi''(c)$ for exactly one coarse bond. Each block is the initial block of one bond in each direction and the final block of one bond in each direction. Therefore (214) and (215) give

$$\sum_{c\in\Lambda'^*} (E_{\Phi'(c)} + E_{\Phi''(c)}) \leq 12(L-1)L\,S_{\mathrm{in}},$$
$$\sum_{\substack{c\in\Lambda'^* \\ c=(y,\mu)}} (X^\mu_{c_-} + X^\mu_{c_+}) \leq 2X \leq 4(L-1)^2 L\,S_{\mathrm{in}}.$$

The six faces account for the first factor 12; the two occurrences of each directional block sum account for the second estimate. Summing (218) now yields

$$\begin{aligned}\tfrac{1}{4}L^{-2}S_{\mathrm{co}} + L^{-1}T &\geq \tfrac{1}{3}L^{-4}Y - \left(3\frac{L-1}{L} + 4\frac{(L-1)^2}{L^2}\right) S_{\mathrm{in}} \\ &\geq \tfrac{1}{3}L^{-4}Y - 7S_{\mathrm{in}}.\end{aligned}$$

Multiply by $1/36$. Since $L^{-2}/144 \leq 1$, $L^{-1}/36 \leq 1$ and $7/36 \leq 1/4$, it follows that

$$S_{\mathrm{co}} + T \geq \frac{1}{108}L^{-4}Y - \tfrac{1}{4}S_{\mathrm{in}}.$$

On the other hand, the block bound gives

$$\tfrac{3}{4}S_{\mathrm{in}} \geq \tfrac{1}{4}L^{-3}X \geq \frac{1}{108}L^{-4}X.$$

Adding proves

$$T + S_{\mathrm{in}} + S_{\mathrm{co}} \geq \frac{1}{108}L^{-4}(X+Y) = \frac{1}{108}L^{-4}\|B\|^2.$$

Internal plaquettes in different blocks are distinct. A plaquette in $\mathcal{P}(c)$ has two bonds in $B(c)$ and its other two bonds inside the adjacent blocks; it therefore belongs to no other $\mathcal{P}(c')$ and is inside no block. Consequently $S_{\mathrm{in}} + S_{\mathrm{co}} \leq \|F\|^2$, which proves (212). □

20.5. **Exponential kernels and regular columns.** The scalar correction is constrained by a zero mean on each selected block. Its inverse can therefore be estimated with exponential weights that are constant on blocks. We first prove the corresponding matrix estimate, then explain how regularity passes through the maps from unit bonds to fine bonds.

Let $\mathbb{Y}$ and $\mathbb{Y}'$ each be the sites or the positive bonds of T_1. A bond has the position of its initial point. These spaces have counting-measure inner products, so their indicator vectors e_s are orthonormal. For $\delta, c > 0$, write

$$U(s,s') = \langle e_s, U e_{s'} \rangle, \qquad \mathfrak{U}(\delta, c) = \{U : |U(s,s')| \le c e^{-\delta |s-s'|_1} \text{ for all } s, s'\}.$$

All maps in this subsection are real and linear. Recall $\sigma(\delta) = 3\coth^3(\delta/2)$.

Lemma 20.7. (i) For any continuous centers u, u',

$$\sum_{s \in \mathbb{Y}} e^{-\delta|u-s|_1 - \delta|s-u'|_1} \le \sigma(\delta/2) e^{-\delta|u-u'|_1/2}.$$

Consequently, for composable maps,

$$\mathfrak{U}(\delta, c_1)\mathfrak{U}(\delta, c_2) \subset \mathfrak{U}(\delta/2, \sigma(\delta/2) c_1 c_2).$$

(ii) Let T be symmetric and belong to $\mathfrak{U}(\delta_1, c_1)$ on $\ell^2(\mathbb{Y})$. Partition $\mathbb{Y}$ into cells of diameter at most D. Suppose that a subspace $\mathcal{H}$ is invariant under multiplication by every real function constant on the cells, and that

$$\langle v, Tv \rangle \ge \gamma \|v\|^2 \quad (v \in \mathcal{H}), \qquad \gamma > 0.$$

For its inclusion ι, the compressed inverse

$$C = \iota (\iota^* T \iota)^{-1} \iota^*$$

exists, has norm at most γ^{-1}, and satisfies

$$|C(s,s')| \le \frac{2}{\gamma} e^{\theta D} e^{-\theta |s-s'|_1}$$

for a $\theta > 0$ depending only on c_1, δ_1, γ, D. The inverse on a zero-dimensional space is extended by zero.

■

Proof. Each unit site is the position of at most three indices of $\mathbb{Y}$. The scalar torus sum in section 19.1 and the triangle inequality give (i): three times the scalar bound $\coth^3(\delta/4)$ is $\sigma(\delta/2)$. Expanding the matrix product gives the kernel conclusion.

For (ii), let Π project onto $\mathcal{H}$. Fix $s_0 \in \mathbb{Y}$ and put

$$\rho(s) = \min_{z \text{ in the cell of } s} |z - s_0|_1.$$

The cell diameter and triangle inequality imply

$$\rho(s_0) = 0, \qquad \rho(s) \ge |s - s_0|_1 - D, \qquad |\rho(s) - \rho(s')| \le |s - s'|_1 + D.$$

For the last inequality, choose a minimizing z' in the cell of s' and use $\rho(s) \le |s - s_0|_1 \le |s - s'|_1 + D + \rho(s')$, then exchange s, s'. Multiplication $E = e^{\theta\rho}$ and its inverse preserve $\mathcal{H}$. They also preserve its orthogonal complement, since they are symmetric, and hence commute with Π.

The lower bound for T makes $\iota^* T \iota$ injective and thus invertible in finite dimension. If $v = (\iota^* T \iota)^{-1} g$, then $\gamma \|v\|^2 \le \langle v, g \rangle$; this proves $\|C\| \le \gamma^{-1}$. The entries of $ETE^{-1} - T$ are bounded in modulus by

$$c_1 e^{-\delta_1 |s-s'|_1} \big(e^{\theta(|s-s'|_1 + D)} - 1 \big).$$

Both their row sums and their column sums are at most

$$m(\theta) = 3c_1 \sum_{z\in\mathbb{Z}^3} e^{-\delta_1|z|_1}\big(e^{\theta(|z|_1+D)} - 1\big).$$

Indeed, least-distance lifts of distinct positions are distinct, and there are at most three indices at each position. If a matrix B has row sums of $|B|$ at most m_1 and column sums at most m_2, Cauchy–Schwarz in the double sum gives

$$\begin{aligned}|\langle g, Bf\rangle| &\le \left(\sum_{s,s'} |B(s,s')||g(s)|^2\right)^{1/2}\left(\sum_{s,s'} |B(s,s')||f(s')|^2\right)^{1/2}\\ &\le (m_1m_2)^{1/2}\|g\|\|f\|.\end{aligned}$$

Thus $\|ETE^{-1} - T\| \le m(\theta)$. For $0 \le \theta \le \delta_1/2$ the series is dominated by a summable exponential; it is continuous and tends to zero as $\theta \downarrow 0$. Choose $0 < \theta \le \delta_1/2$ with $m(\theta) \le \gamma/2$.

On $\mathcal{H}$ the conjugated form has lower bound $\gamma/2$, so $T_\theta = \iota^* ETE^{-1}\iota$ is invertible with $\|T_\theta^{-1}\| \le 2/\gamma$. Commutation with Π shows that $\iota T_\theta^{-1}\iota^* = ECE^{-1}$. Its (s, s_0) entry is $e^{\theta\rho(s)}C(s, s_0)$, whose modulus is at most $2/\gamma$. The lower bound for $\rho(s)$ proves the stated kernel estimate. □

To retain the fine-lattice regularity, fix $\delta > 0$ and consider a map $Y : \mathcal{A}_1 \to \mathcal{A}_\xi$ and a family $\mathsf{k} = (k, (k_\alpha)_{0\le\alpha<1})$ of positive constants. We write $Y \in \mathfrak{X}(\delta, \mathsf{k})$ if each column Ye_b satisfies, for every $u \in \mathsf{T}$, $x \in U_1(u)$ and real ζ supported in $U_1(u)$,

$$|(Ye_b)(x)|, \quad |(\nabla Ye_b)(x)|, \quad |(\Delta Ye_b)(x)| \le kE_b,$$
$$\|\zeta\nabla Ye_b\|_\alpha \le k_\alpha E_b(\|\zeta\|_\alpha + |\zeta|), \qquad \|\zeta\nabla\nabla Ye_b\| \le kE_b|\zeta|, \qquad E_b = e^{-\delta|u-b_-|_1}.$$

The last norm uses the fine weight ξ^3. Multiplication of a family by a constant below is componentwise. Put

$$\begin{aligned}C_1^{\mathfrak{X}} &= \max\{9\sigma(\delta/2), e^{3\delta}\},\\ C_2^{\mathfrak{X}} &= \sqrt{3}\,\sigma(\delta/2),\\ C_3^{\mathfrak{X}} &= 243e^{3\delta}\sigma(\delta/2).\end{aligned} \tag{219}$$

Lemma 20.8. (i) If $Y \in \mathfrak{X}(\delta, \mathsf{k})$ and $U \in \mathfrak{U}(\delta, c)$ acts on $\mathcal{A}_1$, then

$$YU \in \mathfrak{X}(\delta/2, \sigma(\delta/2)c\,\mathsf{k}).$$

(ii) If $Y, Z \in \mathfrak{X}(\delta, \mathsf{k})$, then Z^*Y, $Z^*\Delta Y$ and $(\partial^* Z)^*\partial^* Y$ belong to $\mathfrak{U}(\delta/2, C_1^{\mathfrak{X}}k^2)$, and $\mathsf{Q}_j Y \in \mathfrak{U}(\delta, C_1^{\mathfrak{X}}k)$.

(iii) Suppose that O satisfies (I1)–(I5) of proposition 19.1 with rate δ and constants $c, c_\alpha, c'_\varepsilon, c''_{\alpha,\varepsilon}$. If $f : \mathcal{A}_1 \to \mathcal{A}_\xi$ is linear and

$$|(fe_b)(x)| \le ke^{-\delta|u-b_-|_1} \quad (x \in U_1(u))$$

for every u and b, then

$$Of \in \mathfrak{X}(\delta/2, C_2^{\mathfrak{X}}k(c, (c_\alpha))).$$

In particular, this applies to $f = \mathsf{Q}_j^*$ with $k = e^{9\delta}$.

(iv) If $Y, Z \in \mathfrak{X}(\delta, \mathsf{k})$, then YZ^* satisfies all of (I1)–(I5) with rate $\delta/2$ and constants

$$c = c'_\varepsilon = C_3^{\mathfrak{X}}k^2, \qquad c_\alpha = c''_{\alpha,\varepsilon} = C_3^{\mathfrak{X}}k(k + k_\alpha).$$

■

Proof. The blocks $B^j(y) \subset U_1(y)$ partition the fine lattice and have weighted volume one. Also

$$\sum_{x \in U_1(u)} \xi^3 \le (2+\xi)^3 \le 27.$$

These volume bounds account for the component factors in the proof.

For (i), expand $YUe_{b'} = \sum_b U(b,b')Ye_b$. Each defining norm or seminorm is subadditive and homogeneous. The convolution bound of lemma 20.7(i) therefore gives all five column estimates, with the same factors of ζ.

For (ii), split the inner product into blocks:

$$\begin{aligned} |(Z^*Y)(b,b')| &\le \sum_{y \in \mathsf{T}_1} \sum_{x \in B^j(y)} \xi^3 3 |(Ze_b)(x)||(Ye_{b'})(x)| \\ &\le 3k^2 \sum_y e^{-\delta|y-b_-|_1 - \delta|y-b'_-|_1}. \end{aligned}$$

The same bound holds with ΔY in place of Y. For the divergence product, the adjoint difference gives

$$|(\partial^* Ye_b)(x)| \le \sum_{\mu=1}^{3} |(\partial_\mu (Ye_b)_\mu)(x - \xi e_\mu)| \le 3k e^{-\delta|y-b_-|_1}, \quad x \in B^j(y),$$

since $x - \xi e_\mu \in U_1(y)$. Its inner product therefore has coefficient $9k^2$. Convolution proves the three kernel bounds. Finally, $(\mathsf{Q}_j Ye_{b'})(b)$ is a probability average of values in $b_- + [0,2)^3$, which is contained in $U_1(b_- + (1,1,1))$. The change of center gives

$$|(\mathsf{Q}_j Ye_{b'})(b)| \le e^{3\delta} k e^{-\delta|b_- - b'_-|_1}.$$

For (iii), decompose $fe_b = \sum_y \chi_{B^j(y)} fe_b$. The term indexed by y has supremum at most $ke^{-\delta|y-b_-|_1}$ and L^2 norm at most $\sqrt{3}\, ke^{-\delta|y-b_-|_1}$. Apply (I1) to its value, gradient and Laplacian, the first estimate of (I2) to its localized gradient seminorm, and the third estimate of (I5) to its localized second differences. Summing by lemma 20.7(i) gives, respectively, the coefficients $\sigma(\delta/2)ck$, $\sigma(\delta/2)c_\alpha k$ and $\sqrt{3}\,\sigma(\delta/2)ck$. They are bounded by the asserted family. For $f = \mathsf{Q}_j^*$, (177) gives a column bounded by one and supported in $b_- + [0,2)^3$. If it is nonzero at an $x \in U_1(u)$, then $|u - b_-|_1 \le 9$, which proves the required hypothesis with $k = e^{9\delta}$.

For (iv), write

$$YZ^*J = \sum_b Ye_b \langle Ze_b, J\rangle.$$

Let the three source fields have the support prescribed in proposition 19.1, and put $E'_b = e^{-\delta|u'-b_-|_1}$. The volume bound and Cauchy–Schwarz give

$$\begin{aligned} |\langle Ze_b, J\rangle| &\le 81 k E'_b |J|, \qquad |\langle Ze_b, J\rangle| \le 9k E'_b \|J\|, \\ |\langle \nabla Ze_b, J'\rangle| &\le 243 k E'_b |J'|, \quad |\langle \nabla Ze_b, J'\rangle| \le 9\sqrt{3}\, k E'_b \|J'\|. \end{aligned}$$

Adjointness identifies the second row with the coefficients for $\nabla^* J'$. For the last source, use the second-difference column bound with the indicator χ of $U_1(u')$:

$$\begin{aligned} |\langle Ze_b, \nabla^*\nabla^* J''\rangle| &= |\langle \chi \nabla\nabla Ze_b, J''\rangle| \\ &\le k E'_b \|J''\|. \end{aligned}$$

No seminorm of this indicator is involved.

On the output side, the defining column bounds give the three pointwise quantities and the localized gradient seminorm, together with

$$\|\zeta Ye_b\|, \ \|\zeta \nabla Ye_b\| \le 9\sqrt{3}\, k E_b |\zeta|, \qquad \|\zeta\nabla\nabla Ye_b\| \le k E_b |\zeta|.$$

We also need the localized value seminorm:

$$\|\zeta Y e_b\|_\alpha \le e^{3\delta} k E_b(\|\zeta\|_\alpha + |\zeta|). \tag{220}$$

To prove it, put $g = Ye_b$ and take $0 < |x - x'|_1 \le 1$. If either cutoff value is nonzero, exchange the points so that $x' \in U_1(u)$. A shortest coordinate path from x' to x lies in the union of the boxes $U_1(u+v)$ with $v \in \{-1,0,1\}^3$. On this path the value and gradient of g are bounded by $e^{3\delta}kE_b$. Telescoping along the path yields

$$|g(x) - g(x')| \le |x - x'|_1 e^{3\delta} k E_b \le |x - x'|_1^\alpha e^{3\delta} k E_b.$$

Apply this and the value bound to

$$\zeta(x)g(x) - \zeta(x')g(x') = (\zeta(x) - \zeta(x'))g(x) + \zeta(x')(g(x) - g(x')).$$

If both cutoff values vanish the difference is zero. This proves (220), including $\alpha = 0$ and pairs crossing the support of ζ.

Every estimate for YZ^* now pairs one of the output bounds with one of the source coefficient bounds and sums over b. For example,

$$(\nabla Y Z^* \nabla^* J')(x) = \sum_b (\nabla Y e_b)(x) \langle \nabla Z e_b, J' \rangle,$$

$$\|\zeta Y Z^* \nabla^* J'\|_\alpha \le \sum_b \|\zeta Y e_b\|_\alpha |\langle \nabla Z e_b, J' \rangle|.$$

Convolution contributes $\sigma(\delta/2)$ and the rate $\delta/2$. For (I1), (I3) and all six members of (I5), the coefficient products are at most $243k^2$. In the two seminorm estimates they are at most $243k \max\{k_\alpha, e^{3\delta}k\}$. For (I3) and (I4) we have used only $|J'|$, which is bounded by the stated seminorm-plus-supremum expression. The constants in the lemma dominate these products. □

20.6. **The two-scale estimates.** The operator of the coarse penalty has finite range. We record its normalization, since the scalar product on $\mathcal{A}_L$ carries the weight L^3.

Lemma 20.9. On $\mathcal{A}_1$ put

$$T_{\mathsf{q}} = \chi_{\Lambda^{c*}} + L^{-2} \mathsf{Q}_{1,L}^* \chi_{\Lambda'^*} \mathsf{Q}_{1,L}. \tag{221}$$

Then $\langle B, T_{\mathsf{q}} B \rangle = \mathsf{q}(B)$, $0 \le T_{\mathsf{q}} \le 1 + L^{-2}$, and $T_{\mathsf{q}} \partial_1 \omega = 0$ for $\omega \in W$. Moreover,

$$\begin{aligned} \Delta_1 &= \partial^* \partial + \partial \mathsf{R} \partial^* + \mathsf{Q}_j^* T_{\mathsf{q}} \mathsf{Q}_j \\ &= \Delta - \partial \mathsf{P} \partial^* + \mathsf{Q}_j^* T_{\mathsf{q}} \mathsf{Q}_j, \qquad T_F = T_{\mathsf{q}} + \Delta_j. \end{aligned}$$

The unit-bond entries of T_{q} lie in $[0, 1 + 2L^{-5}]$ and vanish when $|b_- - b'_-|_1 \ge 4L$. In particular, $T_{\mathsf{q}} \in \mathfrak{U}(\delta, 2e^{4\delta L})$ for every $\delta > 0$. ■

Proof. The second term of the quadratic form of (221) is

$$L^{-2} \|\chi_{\Lambda'^*} \mathsf{Q}_{1,L} B\|^2 = L \sum_{c \in \Lambda'^*} |(\mathsf{Q}_{1,L} B)(c)|^2.$$

The contraction of $\mathsf{Q}_{1,L}$ gives the upper bound. For $\omega \in W$, its gradient vanishes on Λ^{c*} and $\mathsf{Q}_{1,L} \partial_1 \omega = \partial_L \mathsf{Q}'_{1,L} \omega = 0$. This proves the annihilation assertion. The operator identities follow from the quadratic forms and the previously proved lattice identity $\Delta = \partial^* \partial + \partial \partial^*$.

Write $(\mathsf{Q}_{1,L} B)(c) = \sum_b q(c,b) B(b)$. For a coarse bond of direction μ,

$$q(c,b) = L^{-4} \#\{(y,t) : y \in B(c_-),\ 0 \le t < L,\ y + te_\mu = b_-\}$$

when b has direction μ, and the coefficient is zero otherwise. Thus $0 \le q(c,b) \le L^{-3}$. A nonzero coefficient places b_- in $B(c_-)$ or $B(c_+)$, so for fixed b at most two coarse bonds contribute. The entries of the second term of (221) are consequently

$$L \sum_{c \in \Lambda'^*} q(c,b) q(c,b') \in [0, 2L^{-5}].$$

If such an entry is nonzero, the two positions lie in one path-averaging domain: their coordinate differences are at most $2L-2$ in direction μ and $L-1$ in the other directions. Their torus distance is therefore at most $4L-4<4L$. The indicator term is diagonal. Since $1+2L^{-5}\le 2$, the exponential bound follows. □

Proof of proposition 20.2. The decomposition in proposition 20.5 proves positivity and gives (210) and (211). The scalar and axial lower bounds in lemma 20.4 give

$$\|C'\|\le \frac{3L^4}{16}, \qquad \|\widetilde{C}\|\le 1+2916L^4.$$

By proposition 19.1 and lemma 20.8(iii), $X=G_j\mathsf{Q}_j^*$ belongs to a class $\mathfrak{X}$ with an absolute rate and with constants depending only on the indicated exponent. In particular its columns have a pointwise Laplacian bound. Apply the same result to $f=\Delta X$, decreasing the input rate if necessary: it gives $Y=G_j\Delta X$ with the same kind of uniformity. All subsequent rates may be decreased and constants increased when finitely many estimates are combined.

Next, $S^{-1}=\mathsf{Q}_jG_j\mathsf{Q}_j^*=\mathsf{Q}_jX$ has an exponential kernel by lemma 20.8(ii). The bound $S^{-1}\ge(1+\gamma_1)^{-1}$ from lemma 20.4 is absolute. Apply lemma 20.7(ii) to $T=S^{-1}$ on all unit bonds, with singleton cells and $D=0$. It gives an exponential kernel for S with absolute constants. The same follows for $\Delta_j=S-1$.

The entries of ∂_1 and its adjoint are 0 or ±1 and vanish at distance greater than one. By (204),

$$\Delta_j'=\partial_1^*(\partial^*X)^*(\partial^*X)\partial_1.$$

Thus lemma 20.8(ii) and kernel multiplication give exponential decay for Δ_j' with absolute constants. For its compression to W, use the cells $B(y')$, which have diameter $3(L-1)$. Both the support condition and the zero block means defining W are preserved by block-constant multiplication. The lower bound is $16/(3L^4)$. Hence lemma 20.7(ii) gives decay for C' with constants depending only on L. Kernel multiplication gives the same conclusion for $\Theta=\partial_1C'\partial_1^*$.

By lemma 20.9, $T_F=T_{\mathsf{q}}+\Delta_j$ has an exponential kernel with constants depending only on L. The axial space is a coordinate subspace of the unit-bond space, so it is invariant under every multiplication. Apply lemma 20.7(ii) with singleton cells and lower bound $(1+2916L^4)^{-1}$. This gives the required decay of $\widetilde{C}$, and hence of $V=-S+S\widetilde{C}S$.

The two maps $\mathsf{N}_1=X^*\Delta X$ and $Y^*\Delta X$ have exponential kernels with absolute constants by lemma 20.8(ii). Taking the adjoint preserves these bounds, so they also hold for $\mathsf{N}_2=(\Delta X)^*Y=(Y^*\Delta X)^*$. The six terms defining U_0 in (211) are finite products of these kernels and those of Θ, V. Repeated use of lemma 20.7(i) gives U_0 an exponentially decaying kernel with constants depending only on L.

Finally, lemma 20.8(i) puts XU_0, $X\Theta$ and $Y\Theta$ in classes $\mathfrak{X}$ with constants depending only on L and the exponents. After decreasing their rates to a common positive rate, part (iv) gives every estimate (I1)–(I5) for

$$XU_0X^*, \qquad X\Theta Y^*, \qquad Y\Theta X^*.$$

Equation (211) and the estimates for G_j give the same family for G. Signs in that identity do not affect the norm bounds.

Choose $\delta_{\mathrm{per}}>0$ as the minimum of the rates used above and enlarge the estimate constants for G. The kernel estimates and the three inverse estimates use only the unindexed constants, so this rate is independent of the Hölder exponents. All bounds are uniform in j, N, Λ'; their constants depend only on L and the displayed exponents, as stated in proposition 20.2. □

21. The global Dirichlet inverse

We construct the inverse on a separated multiscale region by joining the interior two-scale inverses of section 20 and massive boundary inverses. The gauge projection is nonlocal, so its replacement

by the local projections contributes a separate error. We estimate this error before summing the Neumann series. The resulting bounds are uniform in the lattice spacing, the volume and the number of layers.

Fix an odd integer $L \geq 13$, positive integers K, m, and $0 \leq t \leq K$. In level-t coordinates put

$$\xi = L^{-t}, \qquad \xi_n = L^n \xi, \qquad T = \mathbb{R}^3/(\mathsf{S}\mathbb{Z})^3, \qquad \mathsf{S} = 2L^{m+K-t}.$$

Thus the bare lattice has spacing ξ and $\xi_t = 1$. Let $\mathfrak{M}, R_1$ be positive powers of L and let $\Omega_0 \supset \cdots \supset \Omega_t$, with $\Omega_0 \neq \emptyset$ and $\Omega_{t+1} = \emptyset$, satisfy

$$\Omega_n \text{ is a union of cubes of } \pi[\mathfrak{M}\xi_n] \quad (0 \leq n \leq t),$$
$$\operatorname{dist}_\infty(\Omega_n^c, \Omega_{n+1}) > R_1 \mathfrak{M} \xi_n \quad (0 \leq n < t).$$

Here $\pi[r]$ is the standard partition into cubes of side r centered at the sites of $r\mathbb{Z}^3$ modulo the periods. A complementary region has the convention of section 7. Require also that S be an even multiple of $\mathfrak{M}\xi_{t_*}$, where t_* is the last index with $\Omega_{t_*} \neq \emptyset$. Empty final layers and regions equal to T are allowed.

Write

$$\mathcal{L}_n = \overline{\Omega_n \setminus \Omega_{n+1}}, \qquad \Lambda_n = \xi_n \mathbb{Z}^3 \cap \mathcal{L}_n, \qquad \Lambda_* = \bigcup_{n=0}^{t} \Lambda_n,$$

with the lattice understood modulo the periods of T. For $y \in \Lambda_n$ put $\xi_y = \xi_n$ and let $\Delta(y)$ be the centered cube of side ξ_y; put $\widetilde{\Delta}(y) = \{|x - y|_\infty \leq \xi_y\}$. The bare sites of $\Delta(y)$ form the block $B^n(y)$. These blocks partition the bare sites of Ω_0. Let $y(x)$ be the center of the block containing x. Use the weighted distance

$$d_{\mathbf{B}}(x, x') = \inf_{\Gamma \ni x, x'} \left(\sum_{n=1}^{t} \xi_n^{-1} \operatorname{length}(\Gamma \cap \mathcal{L}_n) + \xi^{-1} \operatorname{length}(\Gamma \setminus \Omega_1) \right),$$

where Γ ranges over connected finite unions of segments, as in section 12.

21.1. **The operator and the estimates.** We index bonds only in the positive orientations. Let $\mathfrak{b}_0$ contain every bare bond with both endpoints in Ω_0, and contain only bonds with at least one endpoint there. The spaces $\mathcal{S}$, $\mathcal{A}$ and $\mathcal{Y}$ consist, respectively, of real functions on the bare sites of Ω_0, on $\mathfrak{b}_0$, and on Λ_*. Extend the first two by zero. Their inner products have weights ξ^3, ξ^3 and ξ_y^3. The spaces $\mathcal{A}^{(r)}$, $r = 1, 2$, consist of functions on all bare bonds with r additional direction indices; their inner products have weight ξ^3 and sum over all indices.

For a positive bond b, its localization point is b_-, except that it is b_+ if only b_+ lies in Ω_0. Support of a bond field and the indicators χ_y of $\Delta(y)$ refer to this point. Multiplication by a site function ζ, however, uses $\zeta(b_-)$. Differences act on the zero extensions: $\nabla_\nu = \xi^{-1}(T_\nu - I)$ and $\Delta = \sum_\nu \nabla_\nu^* \nabla_\nu$. The adjoints ∇^* and $\nabla^*\nabla^*$ take their values in $\mathcal{A}$ by restriction to $\mathfrak{b}_0$. Let $\chi_{\mathcal{S}}$ and $\chi_{\mathcal{A}}$ be the orthogonal projections given by restriction and zero extension onto these spaces. Use the scalar gradient and exterior derivative ∂ from section 17, and set

$$\Delta_{\mathcal{S}} = \chi_{\mathcal{S}} \Delta \chi_{\mathcal{S}}, \qquad \partial_0 = \chi_{\mathcal{A}} \partial \chi_{\mathcal{S}}.$$

The scalar averaging map and the scalar Dirichlet operator are

$$(Q'\lambda)(y) = L^{-3n} \sum_{x \in B^n(y)} \lambda(x) \quad (y \in \Lambda_n), \qquad \Delta_a' = \Delta_{\mathcal{S}} + Q'^* \mathsf{a}' Q', \qquad \mathsf{G}' = (\Delta_a')^{-1},$$

where a' multiplies by $a_n \xi_n^{-2}$ on Λ_n, $a_0 = a_1 = 1$ and $a_{n+1} = a_n/(L^{-2} + a_n)$ for $n \geq 1$. In particular, $1 - L^{-2} \leq a_n \leq 1$. The scalar inverse exists by the block energy estimate proved below. Let R be the orthogonal projection onto $\Delta_{\mathcal{S}} \ker Q'$ and put $\mathsf{P} = I - \mathsf{R}$.

For $n \geq 1$, let Γ_n be the positive level-n bonds with an endpoint in Ω_n and none in Ω_{n+1}; let Γ_0 be the bonds of $\mathfrak{b}_0$ with no endpoint in Ω_1. For $c = (c_-, \mu) \in \Gamma_n$, put

$$(QA)(c) = L^{-4n} \sum_{x \in B^n(c_-)} \sum_{j=0}^{L^n-1} A_\mu(x + j\xi e_\mu). \tag{222}$$

Define the symmetric nonnegative operator Q^*aQ by its form

$$(A, Q^*aQA) = \sum_{n=0}^{t} \xi_n \sum_{c \in \Gamma_n} |(QA)(c)|^2.$$

Here the unit coefficient $a = 1$ is fixed; Q^*aQ denotes the operator defined by this form. The vector operator is

$$\Delta_a(\Omega_0) = \chi_{\mathcal{A}} \partial^* \partial \chi_{\mathcal{A}} + \partial_0 \mathsf{R} \partial_0^* + Q^*aQ. \tag{223}$$

For a site or bond field, use the componentwise supremum norm $|f|$ and the seminorm

$$\|f\|_{\alpha,r} = \sup_{0<|x-x'|_1 \leq r} \left(\frac{r}{|x-x'|_1} \right)^\alpha |f(x) - f(x')|, \qquad \|f\|_\alpha = \|f\|_{\alpha,1}.$$

The comparison is between the same components at translated sites or bonds, using initial points for the bond distances. For $\alpha = 0$ this is the corresponding oscillation seminorm. Write $\|f\|$ for the weighted L^2 norm.

Proposition 21.1. There are positive constants M_0, γ, δ, C, depending only on L, with the following property. Suppose the separated sequence above satisfies $\mathfrak{M} \geq M_0$. Every positive-power choice $R_1 \geq L$ is allowed. Then $\Delta_a(\Omega_0) \geq \gamma I$, and its inverse G is symmetric and positive definite.

Let $y, y' \in \Lambda_*$, put $E = e^{-\delta d_{\mathbf{B}}(y,y')}$, and let x be a bare bond localized in $\Delta(y)$. For J supported in $\Delta(y')$,

$$\begin{aligned} |\mathsf{G}J(x)| &\leq C\xi_y^2 E|J|, \\ |\nabla \mathsf{G}J(x)|, |\mathsf{G}\nabla^* J(x)| &\leq C\xi_y E|J|, \\ |\Delta \mathsf{G}J(x)| &\leq CE|J|. \end{aligned} \tag{224}$$

Here $J \in \mathcal{A}$ except in the divergence-input bound, where $J \in \mathcal{A}^{(1)}$. For $0 \leq \alpha < 1$, and a real site function ζ supported in $\widetilde{\Delta}(y)$,

$$\|\zeta \nabla \mathsf{G}J\|_\alpha, \|\zeta \mathsf{G}\nabla^* J\|_\alpha \leq C_\alpha \xi_y^{1-\alpha} (\|\zeta\|_{\alpha,\xi_y} + |\zeta|) E|J|, \tag{225}$$

with the same respective input spaces and supports.

For $J \in \mathcal{A}^{(1)}$ supported in $\widetilde{\Delta}(y')$, one has

$$|\nabla \mathsf{G}\nabla^* J(x)| \leq C_\varepsilon E(\|J\|_{\varepsilon,\xi_{y'}} + |J|), \qquad 0 < \varepsilon < 1, \tag{226}$$

$$\|\zeta \nabla \mathsf{G}\nabla^* J\|_\alpha \leq C_{\alpha,\varepsilon} \xi_y^{-\alpha} (\|\zeta\|_{\alpha,\xi_y} + |\zeta|) E(\|J\|_{\alpha+\varepsilon,\xi_{y'}} + |J|), \tag{227}$$

where the second bound assumes $0 \leq \alpha < 1$, $\varepsilon > 0$, $\alpha + \varepsilon < 1$, and the same condition on ζ as in (225). The constants with subscripts depend only on L and those exponents.

For J supported in $\Delta(y')$ and a real site function ζ supported in $\Delta(y)$, the six bounds are

$$\begin{aligned} \|\zeta \mathsf{G}J\| &\leq C\xi_y^2 |\zeta| E\|J\|, \\ \|\zeta \nabla \mathsf{G}J\|, \|\zeta \mathsf{G}\nabla^* J\| &\leq C\xi_y |\zeta| E\|J\|, \\ \|\zeta \nabla \mathsf{G}\nabla^* J\|, \|\zeta \nabla \nabla \mathsf{G}J\|, \|\zeta \mathsf{G}\nabla^* \nabla^* J\| &\leq C|\zeta| E\|J\|. \end{aligned} \tag{228}$$

An input with r adjoint differences belongs to $\mathcal{A}^{(r)}$, with $\mathcal{A}^{(0)} = \mathcal{A}$. The same inequalities hold with χ_y in place of ζ and 1 in place of $|\zeta|$.

Finally, there are a finite family of cubes $\mathfrak{D}$, local inverses $\mathsf{G}_\square$, a quadratic partition $(h_\square)$, and remainder blocks $K_{\square',\square}$, constructed below, such that

$$\mathsf{G} = \sum_{i=0}^{\infty} \mathsf{G}_0 \mathcal{R}^i = \sum_{i=0}^{\infty} (\mathcal{R}^*)^i \mathsf{G}_0 = \sum_{i=0}^{\infty} \sum_{\omega \in \mathfrak{D}^{2i+1}} \mathsf{G}_\omega. \tag{229}$$

The series converge in operator norm. Here $\mathsf{G}_0 = \sum_\square h_\square \mathsf{G}_\square h_\square$, $\Delta_a \mathsf{G}_0 = I - \mathcal{R}$, and

$$\mathcal{R} = \sum_{\square',\square} K_{\square',\square} \mathsf{G}_\square h_\square.$$

For $\omega = (\square_0, \dots, \square_{2i})$, set $|\omega| = i$ and define

$$\mathsf{G}_\omega = h_{\square_0} \mathsf{G}_{\square_0} h_{\square_0} \prod_{k=1}^{i} (K_{\square_{2k-1},\square_{2k}} \mathsf{G}_{\square_{2k}} h_{\square_{2k}}).$$

The factors are ordered with increasing k from left to right. For every i, each bound in (224)–(228) also bounds the sum of the corresponding moduli, norms or seminorms over $\omega \in \mathfrak{D}^{2i+1}$, with its right side multiplied by 2^{-i}. Use G_ω for undifferentiated input and for $\nabla \mathsf{G} \nabla^*$; use G^*_ω for $\mathsf{G}\nabla^*$ and $\mathsf{G}\nabla^*\nabla^*$. The same bounds therefore hold for $\mathsf{G}_0 \mathcal{R}^i$ and $(\mathcal{R}^*)^i \mathsf{G}_0$, respectively.

All constants are independent of $K, m, t, \mathfrak{M}, R_1$, the regions, the bond frame, and the number of layer changes. ∎

We first obtain the scalar inverse and the differentiated gauge projection directly from energy estimates. These estimates control the discrepancy between global and local gauge conditions. The local two-scale bounds then give a parametrix whose remainder has norm $O(\mathfrak{M}^{-1})$ in exponentially weighted block norms. The final argument sums individual products of remainder blocks, which gives the absolute walk bounds in (229) as well as the operator estimates.

21.2. **Geometry and summation over cells.** The scalar estimates below will also be used on auxiliary tori. For these estimates allow any bare spacing $\xi > 0$ and torus side S that is an even multiple of $\mathfrak{M}\xi_s$, where s is the last nonempty level. Keep the centered partitions and separation condition above. Discarding empty final regions changes neither the blocks, the operators nor the weighted distance. No normalization $\xi_s = 1$ is imposed in these supporting estimates.

This extension preserves the elementary grid facts used earlier. Indeed, S/ξ_n is an even integer at every nonempty level, the ratios of successive spacings are the odd integer L, and $\mathsf{S} \geq 2\mathfrak{M}\xi_n$. Thus the coordinate and nesting argument in section 4 and the block-partition proof of lemma 16.3 apply: fine sites lie on no cell face, and the bare block at $y \in \Lambda_n$ consists of

$$y + \xi k, \qquad |k_\mu| \leq \tfrac{1}{2}(L^n - 1), \quad k \in \mathbb{Z}^3.$$

Its inner-bond graph is the product of three paths. The proof of the metric properties in lemma 12.3 uses only finite segment covers and the torus triangle inequality, so applies in this setting as well.

Put $\rho(x) = \xi_{y(x)}$ on the bare sites of Ω_0 and $\rho(x) = \xi$ elsewhere. In this subsection write $d = d_{\mathbf{B}}$, and extend the empty regions beyond s.

Lemma 21.2. The following estimates hold uniformly in the number of layers.

(a) For neighboring bare sites x, x',

$$\frac{\rho(x')}{\rho(x)} \in \{L^{-1}, 1, L\}, \qquad d(x,x') \leq \frac{\xi}{2}(\rho(x)^{-1} + \rho(x')^{-1}).$$

In particular, the latter bound is at most $\min\{1, L\xi/\rho(x), L\xi/\rho(x')\}$. For $x \in B^n(y)$, one has $d(x,y) \leq 3/2$.

(b) Suppose $\mathfrak{M} \geq L + 10$ and $R_1\mathfrak{M} \geq L(L + 9)$. The cube

$$C(y) = \{x : |x - y|_\infty \leq (L + 9)\xi_y\}$$

has a lift to a cube in $\mathbb{R}^3$ on which the covering map is injective; its lifted bare sites preserve nearest-neighbor relations. If $y \in \Lambda_n$, then $C(y) \cap \Omega_{n+2} = \emptyset$; if $n \geq 1$, also $C(y) \subset \Omega_{n-1}$. Every bare site $x \in C(y)$ satisfies

$$L^{-1}\xi_y \leq \rho(x) \leq L\xi_y, \qquad d(x, y) \leq L\xi_y^{-1}|x - y|_1.$$

(c) For $y \in \Lambda_j$ and $y' \in \Lambda_{j'}$,

$$d(y, y') \geq R_1\mathfrak{M}(|j - j'| - 1)_+. \tag{230}$$

Consequently, for real q and $\varrho = \log L/(R_1\mathfrak{M})$,

$$(\xi_y/\xi_{y'})^q \leq L^{|q|}e^{|q|\varrho d(y,y')}. \tag{231}$$

(d) Put

$$c_S(\kappa) = 8L^3(2 + 4/\kappa)^3, \qquad \mathsf{r}_S(\kappa) = \frac{2(3\log L + 1)}{\kappa}. \tag{232}$$

For $\kappa > 0$ and $R_1\mathfrak{M} \geq \mathsf{r}_S(\kappa)$, and for $y, y' \in \Lambda_*$ in the second estimate,

$$\sup_{y\in\Lambda_*} \sum_{y'\in\Lambda_*} e^{-\kappa d(y,y')} \leq c_S(\kappa), \tag{233}$$

$$\sum_{z\in\Lambda_*} e^{-2\kappa d(y,z)-2\kappa d(z,y')} \leq c_S(\kappa)e^{-\kappa d(y,y')}. \tag{234}$$

■

Proof. Neighbors and local charts. Give a site outside Ω_0 level zero. If two neighboring sites had levels n and $n' \geq n + 2$, they would belong to Ω_{n+1}^c and Ω_{n+2}, respectively. Separation would give a distance greater than $R_1\mathfrak{M}\xi_{n+1}$, contrary to their distance ξ. This proves the scale comparison. Each half of their connecting bond, apart from its midpoint, lies inside the bare cube of its endpoint, where the weighted length density is $\rho(x)^{-1}$ or $\rho(x')^{-1}$. The midpoint has zero length, so adding the two halves proves the distance bound. A coordinate path from a bare site to its cell center stays in that cell and has length at most $3\xi_y/2$, proving the last assertion of (a).

For (b), $\mathsf{S} \geq 2\mathfrak{M}\xi_y \geq 2(L + 10)\xi_y$. The lift of $C(y)$ lies in an open cube of side $2(L + 10)\xi_y$, on which the covering map is injective. Two lifted sites with neighboring images differ by a bare coordinate vector: any nonzero period would have a coordinate of size at least $\mathsf{S} - \xi > 2(L + 9)\xi_y$. Separation from Ω_{n+2}, and from Ω_{n-1}^c when $n \geq 1$, gives the stated containments. The levels of bare sites in $C(y)$ therefore lie between $n - 1$ and $n + 1$; at level zero, sites outside Ω_0 also have $\rho = \xi_y$. This proves the bounds for ρ. A monotone coordinate path in the lifted cube has $|x - y|_1/\xi$ steps. Apply (a) to these steps to obtain the asserted local metric bound.

Crossing layers. The connecting-set argument in section 12.2 applies with these same partitions and separation. Specifically, if a connected finite union of segments Γ joins $y \in \Lambda_j$ to $y' \in \Lambda_{j'}$, put $n = \max\{i : \Gamma \cap \Omega_i \neq \emptyset\}$. Then $n \geq \max\{j, j'\}$ and

$$|\Gamma|_{\mathbf{B}} \geq \xi_n^{-1}|y - y'|_1, \qquad |\Gamma|_{\mathbf{B}} \geq R_1\mathfrak{M}(n - \min\{j, j'\} - 1)_+. \tag{235}$$

For clarity, the first inequality uses the 1-Lipschitz function $z \mapsto |z - y|_1$ in the interval-image argument there, before the weaker sup-distance bound is taken. For the second, each intermediate separating layer contributes at least $R_1\mathfrak{M}$ to the weighted length: the distance to the next region runs through an interval of length $R_1\mathfrak{M}\xi_i$ within layer i. Taking the infimum over Γ gives (230). Equation (231) follows from $\xi_y/\xi_{y'} = L^{j-j'}$ and $|j - j'| \leq 1 + (|j - j'| - 1)_+$.

The site sum. Take half of each lower bound in (235), exponentiate, and sum over the possible largest visited levels. This gives

$$e^{-\kappa d(y,y')} \le \sum_{n\ge \max\{j,j'\}} e^{-\kappa\xi_n^{-1}|y-y'|_1/2} e^{-\kappa R_1\mathfrak{M}(n-\min\{j,j'\}-1)_+/2}.$$

Choose distinct least-distance lifts of the sites in $\Lambda_{j'}$ to $\xi_{j'}\mathbb{Z}^3$. The elementary one-dimensional bound

$$\sum_{k\in\mathbb{Z}} e^{-\beta|u+hk|} \le \frac{2}{1-e^{-\beta h}} \le 2+\frac{2}{\beta h}$$

implies, by multiplication over the coordinates,

$$\sum_{y'\in\Lambda_{j'}} e^{-\kappa\xi_n^{-1}|y-y'|_1/2} \le (2+4/\kappa)^3 L^{3(n-j')}.$$

Put $p=n-\min\{j,j'\}$. The hypothesis gives $L^3 e^{-\kappa R_1\mathfrak{M}/2}\le e^{-1}$, so the last lattice factor times the layer factor is at most

$$L^3(2+4/\kappa)^3 e^{-(p-1)_+}.$$

For each p, the pair (j',n) is determined by $j'-j$ and p, with $|j'-j|\le p$; hence there are at most $2p+1$ choices. Finally,

$$\sum_{p\ge 0}(2p+1)e^{-(p-1)_+} < 8.$$

This proves (233). For the convolution bound, the triangle inequality gives

$$e^{-2\kappa(d(y,z)+d(z,y'))} \le e^{-\kappa d(y,y')} e^{-\kappa d(y,z)};$$

sum in z and apply the site estimate. □

21.3. **The scalar input.** Use the auxiliary-torus setting of section 21.2. For $y\in\Lambda_n$ write $n(y)=n$ and $B_y=B^n(y)$, and let S_0 denote the bare sites of Ω_0. The scale function ρ is as above. Write $\langle\cdot,\cdot\rangle_{\mathrm{c}}$ for the coarse inner product. The averaging adjoint is $(Q'^*h)(x)=h(y(x))$, since the fine and coarse weights are ξ^3 and ξ_y^3. Thus $Q'Q'^*=I$.

Put $\mathsf{W}_{\mathrm{av}}=Q'^*\mathsf{a}'Q'$. The operator $\Delta'_a=\Delta_{\mathcal{S}}+\mathsf{W}_{\mathrm{av}}$ uses the full scalar Dirichlet form, independently of the bond frame $\mathfrak{b}_0$. In this subsection ∇ and ∇^* denote the ambient scalar gradient and its adjoint; scalar data are restricted by $\chi_{\mathcal{S}}$ after a divergence is taken. All differences use zero extensions.

Proposition 21.3. There are positive constants γ_{lap} and $\delta_{0,\mathrm{lap}}$, depending only on L, with

$$\gamma_{\mathrm{lap}}\le 1-L^{-2}, \qquad 6\delta_{0,\mathrm{lap}}(L^2+1)\le \tfrac14\gamma_{\mathrm{lap}},$$

for which the following assertions hold for every nonempty separated region Ω_0 in the setting above.

(a) The operator Δ'_a is symmetric and positive definite. For every scalar v supported in S_0,

$$\langle v,\Delta'_a v\rangle \ge \|\nabla v\|^2, \qquad \langle v,\Delta'_a v\rangle \ge \gamma_{\mathrm{lap}}\|\rho^{-1}v\|^2. \tag{236}$$

Consequently G' exists and is symmetric and positive definite.

(b) Suppose

$$\mathfrak{M}\ge L+10, \qquad R_1\mathfrak{M}\ge \max\{L(L+9), 5\delta_{0,\mathrm{lap}}^{-1}\log L\}. \tag{237}$$

There is a constant C_{lap}, depending only on L, with the following property. Let $y,y'\in\Lambda_*$, let λ be supported in $B_{y'}$, and let $F_\nu(x)=0$ for $x\notin B_{y'}$. Regard $F_\nu(x)$ as the value of F on $[x,x+\xi e_\nu]$, and put $\nabla^*F=\sum_\nu \partial_\nu^* F_\nu$. For every bare site x with $|x-y|_\infty\le 2\xi_y$ and every ν,

$$\begin{aligned} |(\mathsf{G}'\lambda)(x)| &\le C_{\mathrm{lap}}\xi_y^2 e^{-\delta_{0,\mathrm{lap}}d_{\mathbf{B}}(y,y')/2}|\lambda|,\\ |\partial_\nu(\mathsf{G}'\lambda)(x)| &\le C_{\mathrm{lap}}\xi_y e^{-\delta_{0,\mathrm{lap}}d_{\mathbf{B}}(y,y')/2}|\lambda|,\\ |(\mathsf{G}'\chi_{\mathcal{S}}\nabla^*F)(x)| &\le C_{\mathrm{lap}}\xi_y e^{-\delta_{0,\mathrm{lap}}d_{\mathbf{B}}(y,y')/2}|F|. \end{aligned} \tag{238}$$

The suprema on the right are over sites and, for F, sites and directions. For $\beta \in \{1/2, 3/4\}$ and a real function ζ on the bare torus which vanishes when $|x-y|_\infty \geq \xi_y$, one also has

$$\|\zeta \mathsf{G}' \chi_{\mathcal{S}} \nabla^* F\|_\beta \leq C_{\mathrm{lap}} \xi_y^{1-\beta} (\|\zeta\|_{\beta,\xi_y} + |\zeta|) e^{-\delta_{0,\mathrm{lap}} d_{\mathbf{B}}(y,y')/2} |F|. \tag{239}$$

The constants are independent of ξ, the torus side, the number of layers, $\mathfrak{M}$, R_1, and the regions satisfying these hypotheses. All differences and seminorms use zero extensions, including at the boundary of S_0.

■

An exponential weight first gives an L^2 estimate. The scalar interior estimate then gives the value and gradient near an output cell. The separation between layers absorbs the ratio of the input and output cell sizes into half of the exponential decay.

Proof. Coercivity and the weighted estimate. The path-product calculation in section 16.2 gives, for the mean $\bar{v} = (Q'v)(y)$ on a cell,

$$\xi \sum_{b \subset B_y} |v(b_+) - v(b_-)|^2 \geq 4\xi_y^{-2} \|v - \bar{v}\|_{B_y}^2.$$

For a one-site cell the right side is zero. The same calculation applies here because the inner-bond graph of B_y is the product of three paths of $L^{n(y)}$ vertices. Since

$$\langle v, \mathsf{W}_{\mathrm{av}} v \rangle = \sum_y a_{n(y)} \xi_y |\bar{v}|^2, \qquad \|v\|_{B_y}^2 = \|v - \bar{v}\|_{B_y}^2 + \xi_y^3 |\bar{v}|^2,$$

and $\gamma_{\mathrm{lap}} \leq a_n \leq 1$, summing the cell inequalities proves the second bound in (236). The omitted bond terms are nonnegative. The first bound follows from $\mathsf{W}_{\mathrm{av}} \geq 0$. Positivity and invertibility follow on the finite-dimensional space of functions on S_0.

We next prove a weighted estimate for $u = \mathsf{G}'(f + \chi_{\mathcal{S}} \nabla^* F)$, where f is supported in S_0 and F vanishes on bonds with neither endpoint in S_0. Let φ be a function on the torus satisfying $|\varphi(x) - \varphi(x')| \leq d_{\mathbf{B}}(x, x')$ and let $0 \leq \theta \leq \delta_{0,\mathrm{lap}}$. On a bond meeting S_0, define $\varphi(b)$ at its initial endpoint if that endpoint belongs to S_0, and at its final endpoint otherwise. We claim that

$$\|\rho^{-1} e^{\theta\varphi} u\| \leq \frac{6}{\gamma_{\mathrm{lap}}} (\|\rho e^{\theta\varphi} f\| + \|e^{\theta\varphi} F\|). \tag{240}$$

The last norm uses the bond values $\varphi(b)$ just defined.

Put $v = e^{\theta\varphi} u$, extended by zero. Pairing the equation with $e^{\theta\varphi} v$ and using the ambient adjoint gives

$$\langle e^{\theta\varphi} v, \Delta_a' e^{-\theta\varphi} v \rangle = \langle v, e^{\theta\varphi} f \rangle + \langle \nabla(e^{\theta\varphi} v), F \rangle. \tag{241}$$

We first estimate the conjugated form for any real v supported in S_0 and any $0 \leq \sigma \leq 1$, without the narrower restriction on θ. For an inner bond, write $v_\pm = v(b_\pm)$ and $h = \sigma(\varphi(b_+) - \varphi(b_-))$. Its contribution before the factor ξ is

$$(v_+ - v_-)^2 - 2v_+ v_- (\cosh h - 1).$$

At an exit bond it is the square of the value inside S_0. The neighboring-site bounds in lemma 21.2(a) give

$$|\varphi(b_+) - \varphi(b_-)| \leq \min\{1, L\xi/\rho(b_+), L\xi/\rho(b_-)\}.$$

Thus $\cosh h - 1 \leq h^2$ and the six bonds incident to each site give a total loss at most $6L^2\sigma^2 \|\rho^{-1} v\|^2$ in the Laplacian form.

On B_y, put $\psi = \varphi - \varphi(y)$ and $\epsilon = e^{3\sigma/2} - 1$. Since $|\psi| \leq 3/2$ there,

$$\left| (Q' e^{\sigma\varphi} v)(y) (Q' e^{-\sigma\varphi} v)(y) - (Q'v)(y)^2 \right| \leq \epsilon(2+\epsilon)(Q'|v|)(y)^2.$$

Here $\epsilon(2+\epsilon) = e^{3\sigma} - 1$, and $\xi_y^3 (Q'|v|)(y)^2 \le \|v\|_{B_y}^2$. Adding the two losses gives the estimate

$$\begin{aligned} \langle e^{\sigma\varphi} v, \Delta_a' e^{-\sigma\varphi} v\rangle \ge \langle v, \Delta_a' v\rangle \\ - (6L^2\sigma^2 + e^{3\sigma} - 1)\|\rho^{-1} v\|^2. \end{aligned} \tag{242}$$

Now take $\sigma = \theta \le \delta_{0,\mathrm{lap}}$. The smallness condition gives $e^{3\theta} - 1 \le 6\theta$. Using the mean of the two bounds in (236), we obtain

$$\langle e^{\theta\varphi} v, \Delta_a' e^{-\theta\varphi} v\rangle \ge \tfrac{1}{2}\|\nabla v\|^2 + \tfrac{1}{4}\gamma_{\mathrm{lap}}\|\rho^{-1} v\|^2. \tag{243}$$

For the last term of (241), set $h = \theta(\varphi(b_+) - \varphi(b_-))$ on an inner bond. Then

$$\nabla(e^{\theta\varphi} v)(b) = e^{\theta\varphi(b)} \left(\nabla v(b) + \xi^{-1}(e^h - 1) v_+\right), \qquad \xi^{-1}|e^h - 1| \le e\theta L \rho(b_+)^{-1}.$$

At an exit bond there is only the weighted boundary value, which is already included in $\|\nabla v\|$. A site is the final endpoint of three positive bonds. Since $e\theta L\sqrt{3} \le 1$, Cauchy–Schwarz gives

$$|\langle \nabla(e^{\theta\varphi} v), F\rangle| \le \|e^{\theta\varphi} F\|(\|\nabla v\| + \|\rho^{-1} v\|).$$

Writing $X = \|\rho^{-1} v\|$, $Y = \|\nabla v\|$, $a = \|\rho e^{\theta\varphi} f\|$ and $b = \|e^{\theta\varphi} F\|$, we obtain

$$\tfrac{1}{2}Y^2 + \tfrac{1}{4}\gamma_{\mathrm{lap}} X^2 \le (a+b)X + bY \le \tfrac{1}{8}\gamma_{\mathrm{lap}} X^2 + \frac{2}{\gamma_{\mathrm{lap}}}(a+b)^2 + \tfrac{1}{2}Y^2 + \tfrac{1}{2}b^2.$$

Hence $X^2 \le 32\gamma_{\mathrm{lap}}^{-2}(a+b)^2$, which proves (240).

From local energy to pointwise and Hölder bounds. Assume now (237). Use the cube $C(y)$ of lemma 21.2(b), and put

$$B^*(y) = \{x \in S_0 : |x - y|_\infty \le (L+8)\xi_y\},$$

with norms over bare sites or bonds. Set

$$N_y = \xi_y^{-3/2}\|u\|_{B^*(y)} + \xi_y^2 |f|_{C(y)} + \xi_y |F|_{C(y)}.$$

Here $|F|_{C(y)}$ takes the supremum over bonds with an endpoint in $C(y)$. For x, x' within $2\xi_y$ of y in the supremum distance and $\beta \in \{1/2, 3/4\}$, the local estimates we need are

$$\begin{aligned} |u(x)| &\le C_E N_y, \\ |u(x) - u(x')| &\le C_E(|x - x'|_1/\xi_y)^\beta N_y, \\ \xi_y|\partial_\nu u(x)| &\le C_E N_y \qquad\qquad \text{if } |F|_{C(y)} = 0, \end{aligned} \tag{244}$$

with C_E depending only on L, after taking the maximum of the constants for the two exponents.

If $n(y) = 0$, every site within 3ξ of y either lies outside S_0, where $u = 0$, or belongs to $B^*(y)$. Its value is at most $\xi^{-3/2}\|u\|_{B^*(y)}$. This bounds the value and, by the two endpoint values, ξ times its first difference. Distinct bare sites have distance at least $\xi = \xi_y$, which gives the Hölder bound as well.

If $n(y) \ge 1$, lemma 21.2(b) identifies the bare sites of $C(y)$ with those of a Euclidean cube and places the cube inside Ω_0. Take $r = 8\xi_y$. Then $r \ge 16\xi$, and all neighbors of the cube B_r centered at y lie in $C(y)$. There the equation reads

$$\Delta u = g + \nabla^* F, \qquad g = f - \mathsf{W}_{\mathrm{av}} u.$$

Every cell meeting B_r lies in $B^*(y)$: its side is at most $L\xi_y$. Its side is also at least ξ_y/L, so the averaging inequality gives

$$|\mathsf{W}_{\mathrm{av}} u|_{\mathsf{B}_r} \le L^{7/2}\xi_y^{-7/2}\|u\|_{B^*(y)}.$$

Apply proposition 18.1 after restricting F to the bonds with an endpoint in B_r. This restriction leaves its divergence on B_r unchanged. The quantity N in that proposition is bounded by $C(L)N_y$. Its value and β-Hölder estimates on $\mathsf{B}_{r/4}$ give the first two inequalities of (244); if $|F|_{C(y)} = 0$, its

gradient estimate gives the third. This proves the local estimates, including the boundary case at level zero.

Decay between cells. For this step allow f to be supported in the sites of $B_{y'}$ and their neighbors, and F on bonds with an endpoint in $B_{y'}$; retain the support conditions at S_0 used in (240). Set $\varphi(x) = d_{\mathbf{B}}(x, y')$ and $\theta = \delta_{0,\mathrm{lap}}$. The metric triangle inequality permits this weight. On the support of f, $\varphi \le 5/2$ and $\rho \le L\xi_{y'}$; the same bound for $\varphi(b)$ holds on the support of F. There are at most $7(\xi_{y'}/\xi)^3$ such sites and $6(\xi_{y'}/\xi)^3$ such bonds. Therefore (240) implies

$$\|\rho^{-1} e^{\delta_{0,\mathrm{lap}}\varphi} u\| \le C(\xi_{y'}^{5/2}|f| + \xi_{y'}^{3/2}|F|). \tag{245}$$

Throughout this step C depends only on L.

For $x \in B^*(y)$, the local geometric bounds give $\rho(x) \le L\xi_y$ and $d_{\mathbf{B}}(x, y) \le 3L(L+8)$. Hence

$$\xi_y^{-3/2}\|u\|_{B^*(y)} \le C\left[\xi_y^2\left(\frac{\xi_{y'}}{\xi_y}\right)^{5/2}|f| + \xi_y\left(\frac{\xi_{y'}}{\xi_y}\right)^{3/2}|F|\right] e^{-\delta_{0,\mathrm{lap}} d_{\mathbf{B}}(y,y')}.$$

The layer-crossing bound (230) and (237) imply, for $0 \le p \le 5/2$,

$$\left(\frac{\xi_{y'}}{\xi_y}\right)^p \le L^p L^{p(|n(y)-n(y')|-1)_+} \le L^p e^{\delta_{0,\mathrm{lap}} d_{\mathbf{B}}(y,y')/2}.$$

It follows that the first term in N_y is at most

$$C(\xi_y^2|f| + \xi_y|F|)e^{-\delta_{0,\mathrm{lap}} d_{\mathbf{B}}(y,y')/2}.$$

The remaining terms in N_y vanish unless the support of f, or an endpoint of a bond on which F is nonzero, meets $C(y)$. In that event choose such an endpoint or site x_1. It belongs to $B_{y'}$ or is a neighbor of a site in $B_{y'}$. Thus

$$d_{\mathbf{B}}(y, y') \le d_{\mathbf{B}}(y, x_1) + d_{\mathbf{B}}(x_1, y') \le 3L(L+9) + \tfrac{5}{2}.$$

The smallness condition on $\delta_{0,\mathrm{lap}}$ gives $1 \le e^3 e^{-\delta_{0,\mathrm{lap}} d_{\mathbf{B}}(y,y')/2}$. Consequently all of N_y is bounded by the preceding display, with a larger constant C. Applying (244) proves the value and β-Hölder estimates for u, and its gradient estimate when $F = 0$.

Finally take $(f, F) = (\lambda, 0)$ and then $(f, F) = (0, F)$. In the second application, the support at initial sites in $B_{y'}$ is contained in the incident-bond support used in the proof, and every such bond has an endpoint in S_0. These two applications give all three inequalities in (238). Their range includes both endpoints of every bond meeting B_y, since each such endpoint lies within $\xi_y/2 + \xi \le 2\xi_y$ of y.

For the cutoff estimate put $u = \mathsf{G}'\chi_{\mathcal{S}}\nabla^* F$ and $E = e^{-\delta_{0,\mathrm{lap}} d_{\mathbf{B}}(y,y')/2}$. The local bounds just proved have constants $D_0, D_1 \le C\xi_y E|F|$ for the value and β-Hölder difference. If $\zeta(x) \ne 0$ and $h = |x - x'|_1 \le \xi_y$, both points lie within $2\xi_y$ of y, and the product identity gives

$$|\zeta(x)u(x) - \zeta(x')u(x')| \le (|\zeta|D_1 + \|\zeta\|_{\beta,\xi_y} D_0)(h/\xi_y)^\beta.$$

If both cutoff values vanish there is nothing to prove; if $h > \xi_y$, use $2|\zeta|D_0 \le 2|\zeta|D_0(h/\xi_y)^\beta$. Dividing by h^β and taking the supremum over $0 < h \le 1$ proves (239). □

21.4. **The scalar Gram inverse and differentiated projection.** Here ∂ maps scalar functions to all positive star bonds of Ω_0, with adjoint ∂^* for the fine inner products of weight ξ^3. Let Ξ multiply a fine function by $\xi_x = \rho(x)$ and a coarse function by ξ_y. Put

$$\mathsf{T} = \mathsf{G}'Q'^*, \qquad \mathsf{V} = \mathsf{T}^*\mathsf{T} = Q'\mathsf{G}'^2Q'^*, \qquad \mathsf{C} = \mathsf{V}^{-1}, \qquad S = \Xi^{-2}\mathsf{V}\Xi^{-2}.$$

The block coordinates and weighted pairings are exactly those used in the proofs of proposition 16.1(b) and (155). Since $Q'Q'^* = I$ and G' is positive definite, the same positivity calculation gives $\mathsf{V} > 0$

on the auxiliary tori. The projection calculation also applies: $\Delta'_a = \Delta_{\mathcal{S}}$ on $\ker Q'$. For a coarse operator K, distinguish its counting kernel $K(y,y')$ from its matrix

$$K[y,y'] = \langle e_y, Ke_{y'}\rangle_{\mathrm{c}} = \xi_y^{3/2} K(y,y') \xi_{y'}^{-3/2}, \qquad e_y = \xi_y^{-3/2} \mathbf{1}_{\{y\}}.$$

The normalization by Ξ^{-2} removes the fourth-order scale of V. We prove a uniform gap for S using functions supported on individual blocks with prescribed averages. Exponential conjugation then gives decay of its inverse.

Use the localization point fixed above also for all star bonds. Call a bond attached to y when this point lies in B_y.

Proposition 21.4. There are positive constants $\gamma_{0,\mathrm{gp}}, \delta_1, \delta_2, \mathfrak{r}_4, \mathfrak{m}_5, \mathfrak{r}_5, B_{5,\mathrm{gp}}$, depending only on L, with the following properties for every separated sequence in the stated setting.

(i) The Gram operator satisfies $\mathsf{V} \ge \gamma_{0,\mathrm{gp}} \Xi^4$. If $R_1 \mathfrak{M} \ge \mathfrak{r}_4$, then

$$\begin{aligned} |S^{-1}[y,y']| &\le \frac{2}{\gamma_{0,\mathrm{gp}}} e^{-3\delta_1 d_{\mathbf{B}}(y,y')}, \\ |\mathsf{C}(y,y')| &\le \frac{2}{\gamma_{0,\mathrm{gp}}} \xi_y^{-7/2} \xi_{y'}^{-1/2} e^{-3\delta_1 d_{\mathbf{B}}(y,y')}. \end{aligned} \tag{246}$$

In particular,

$$|\mathsf{C}(y,y')| \le \frac{2L^{1/2}}{\gamma_{0,\mathrm{gp}}} \xi_y^{-4} e^{-2\delta_1 d_{\mathbf{B}}(y,y')}. \tag{247}$$

The weighted kernel $\xi_y^3 \mathsf{C}(y,y')$ is symmetric in y, y'.

(ii) The orthogonal projections $\mathsf{P}, \mathsf{R} = 1 - \mathsf{P}$ satisfy

$$\mathsf{P} = \mathsf{G}' Q'^{*} \mathsf{C} Q' \mathsf{G}', \qquad \operatorname{ran} \mathsf{R} = \Delta_{\mathcal{S}} \ker Q'.$$

The range of R, regarded as functions on the whole bare lattice by zero extension, is supported in Ω_1 and its bare nearest neighbours. If two separated sequences have the same $\Omega_1, \dots, \Omega_t$, their kernels R agree at all bare sites of the intersection of their outer regions.

(iii) Suppose that $\mathfrak{M} \ge \mathfrak{m}_5$ and $R_1 \mathfrak{M} \ge \mathfrak{r}_5$. For $x \in B_y$, $x' \in B_{y'}$, and star bonds b, b' attached to y, y', respectively,

$$\begin{aligned} &|\mathsf{P}(x,x')|, \quad |(\partial \mathsf{P})(b,x')|, \quad |(\mathsf{P}\partial^{*})(x,b')|, \quad |(\partial \mathsf{P} \partial^{*})(b,b')| \\ &\qquad \le B_{5,\mathrm{gp}} \left(\frac{\xi}{\xi_{y'}} \right)^3 [1, \xi_y^{-1}, \xi_y^{-1}, \xi_y^{-2}] e^{-\delta_2 d_{\mathbf{B}}(y,y')}. \end{aligned} \tag{248}$$

The four factors in brackets correspond to the four kernels. All kernels are taken against counting measure, and $\mathsf{R}(x,x') = \mathbf{1}[x = x'] - \mathsf{P}(x,x')$. ∎

The mixed derivative in (248) has output scale ξ_y^{-2} even when the input block has a different size; the scale-ratio estimate absorbs that difference into exponential decay.

Proof. Weighted scalar energy. Put $c_{0,\mathrm{gp}} = 1 - L^{-2}$ and choose $0 < \varepsilon_0 \le 1$, depending only on L, such that

$$6L^2 \varepsilon_0^2 + e^{3\varepsilon_0} - 1 \le \tfrac{1}{4} c_{0,\mathrm{gp}}. \tag{249}$$

Write $A = \Delta'_a$ and $G = \mathsf{G}'$ in this proof. The block argument proving (150) uses only the coordinates and pairings just checked, so $\langle u, Au \rangle \ge c_{0,\mathrm{gp}} \|\Xi^{-1} u\|^2$. Let θ be real and satisfy $|\theta(x) - \theta(x')| \le \varepsilon_0 d_{\mathbf{B}}(x,x')$. Apply (242) with $\sigma = \varepsilon_0$ and $\varphi = \theta/\varepsilon_0$. The loss is at most $c_{0,\mathrm{gp}} \|\Xi^{-1} u\|^2/4$. Use half of the original form to bound $\|\partial u\|^2$ and half to bound $c_{0,\mathrm{gp}} \|\Xi^{-1} u\|^2$. This gives

$$\langle u, e^{\theta} A e^{-\theta} u \rangle \ge \tfrac{1}{2} \|\partial u\|^2 + \tfrac{1}{4} c_{0,\mathrm{gp}} \|\Xi^{-1} u\|^2. \tag{250}$$

If $v = e^{\theta} G e^{-\theta} f$, its pairing with f yields

$$\|\Xi^{-1} v\| \le \frac{4}{c_{0,\mathrm{gp}}} \|\Xi f\|, \qquad \|\partial v\| \le \sqrt{\frac{8}{c_{0,\mathrm{gp}}}} \|\Xi f\|. \tag{251}$$

For f supported in $B_{y'}$, take $\theta = \varepsilon_0 d_{\mathbf{B}}(\,\cdot\,, y')$ and $v = e^{\theta} G f$. Put $r = d_{\mathbf{B}}(y'', y')$. The weight is at most $3\varepsilon_0/2$ on $B_{y'}$, at least $\varepsilon_0(r - 3/2)$ on $B_{y''}$, and at least $\varepsilon_0(r - 5/2)$ at the endpoints in Ω_0 of bonds touching $B_{y''}$. For an internal bond, writing $\theta_{\pm} = \theta(b_{\pm})$,

$$\partial(e^{-\theta} v)(b) = e^{-\theta_-} \left(\partial v(b) - \xi^{-1} (1 - e^{\theta_- - \theta_+}) v(b_+) \right), \qquad \xi^{-1} |1 - e^{\theta_- - \theta_+}| \le 2\varepsilon_0 L \xi_{b_+}^{-1}.$$

For an exit bond there is only the weighted difference term. Each site is the terminal endpoint of three positive bonds. The triangle inequality therefore gives

$$\|\partial G f\|_{\mathrm{st}(B_{y''})} \le e^{-\varepsilon_0 (r - 5/2)} (\|\partial v\| + 2\varepsilon_0 L \sqrt{3}\, \|\Xi^{-1} v\|).$$

Together with (251), applied to $e^{\theta} f$, and $\|\Xi e^{\theta} f\| \le e^{3\varepsilon_0/2} \xi_{y'} \|f\|$, this proves

$$\left(\|\Xi^{-1} G f\|_{B_{y''}}^2 + \|\partial G f\|_{\mathrm{st}(B_{y''})}^2 \right)^{1/2} \le C_{3,\mathrm{gp}} \xi_{y'} e^{-\varepsilon_0 r} \|f\|. \tag{252}$$

Here $C_{3,\mathrm{gp}}$ depends only on L. In particular, $\|G f\|_{B_{y''}} \le C_{3,\mathrm{gp}} \xi_{y''} \xi_{y'} e^{-\varepsilon_0 r} \|f\|$.

Upper bound for the normalized Gram matrix. Let χ_y be the indicator of B_y. Since $Q'^{*} e_y = \xi_y^{-3/2} \chi_y$,

$$S[y, y'] = \xi_y^{-7/2} \xi_{y'}^{-7/2} \langle G\chi_y, G\chi_{y'} \rangle. \tag{253}$$

The blocks partition Ω_0, and $\|\chi_y\| = \xi_y^{3/2}$. Applying the last bound on each block gives, with $r = d_{\mathbf{B}}(y, y'')$ and $r' = d_{\mathbf{B}}(y'', y')$,

$$|S[y, y']| \le C_{3,\mathrm{gp}}^2 \sum_{y''} \frac{\xi_{y''}}{\xi_y} \frac{\xi_{y''}}{\xi_{y'}} e^{-\varepsilon_0 (r + r')}.$$

If $R_1 \mathfrak{M} \ge 8(3 \log L + 1)/\varepsilon_0$, the scale-ratio estimate (231) bounds the quotients by $L^2 e^{\varepsilon_0 (r + r')/4}$. The triangle inequality and (233) at rate $\varepsilon_0/4$ then give

$$|S[y, y']| \le C_S e^{-\varepsilon_0 d_{\mathbf{B}}(y, y')/2}, \qquad C_S = C_{3,\mathrm{gp}}^2 L^2 c_S(\varepsilon_0/4). \tag{254}$$

The uniform Gram gap. Define $g(s) = 1 + \cos(2\pi s)$ for $|s| \le 1/2$ and $g(s) = 0$ otherwise. Then $0 \le g \le 2$, g is continuously differentiable, and g' is Lipschitz with constant $4\pi^2$. For $h > 0$,

$$|g(s + h) + g(s - h) - 2g(s)| \le \int_0^h 4\pi^2 (2r)\, dr = 4\pi^2 h^2.$$

For $n(y) \ge 1$ put $\varphi_y(x) = \prod_{\mu=1}^{3} g((x_\mu - y_\mu)/\xi_y)$, using coordinate representatives of least absolute value on the torus; for $n(y) = 0$ put $\varphi_y = \chi_y$. The torus side is at least $2L\xi_y$, so the choice of representative is immaterial wherever a factor is nonzero. The odd integer $N = \xi_y/\xi$ gives block coordinates $-(N-1)/2, \dots, (N-1)/2$. Points outside the block have some coordinate of modulus at least $(N+1)\xi/2$; thus φ_y is supported on B_y. For $N > 1$ these coordinates form a complete set of residues modulo N, and the sum of $\cos(2\pi m/N)$ over them is zero. Taking the product of the three one-dimensional sums gives

$$(Q' \varphi_y)(y') = \mathbf{1}[y = y']. \tag{255}$$

The same identity holds for the singleton blocks.

Let B_y^{+} consist of B_y and its bare nearest neighbours. Each coordinate second difference of φ_y consists of two factors bounded by 2 and a second difference of g with step ξ/ξ_y. A difference

crossing the chosen torus representative has both relevant factors zero, so the same estimate applies there. For a singleton block use $|\Delta\chi_y| \le 6\xi^{-2}$. Hence

$$|\Delta\varphi_y| \le C_\varphi \xi_y^{-2}, \qquad \operatorname{supp}\Delta\varphi_y \subset B_y^+, \qquad \|\Delta\varphi_y\|^2 \le 7C_\varphi^2 \xi_y^{-1}. \tag{256}$$

Here C_φ is an absolute constant, and the last inequality uses the bound $N^3 + 6N^2 \le 7N^3$ on the number of sites of B_y^+.

Given $\omega' \in \mathcal{Y}$, let $\psi = \sum_y \omega'(y)\varphi_y$. Then $Q'\psi = \omega'$ and

$$A\psi = \chi_{\mathcal{S}}\Delta\psi + \sum_y a_{n(y)}\xi_y^{-2}\omega'(y)\chi_y.$$

This identity uses the full star form: every bond on which ψ has a nonzero difference is included. Each bare site belongs to at most seven sets B_y^+, since its block and the blocks of its six neighbours are the only possibilities. Thus (256) and the disjointness of the blocks imply

$$\|\chi_{\mathcal{S}}\Delta\psi\| \le 7C_\varphi \|\Xi^{-2}\omega'\|_{\mathrm{c}}, \qquad \|A\psi\| \le C_{2,\mathrm{gp}}\|\Xi^{-2}\omega'\|_{\mathrm{c}}.$$

Here $C_{2,\mathrm{gp}} \ge 1$ is an absolute constant. Take $\omega' = \Xi^4\omega$. Symmetry of A and the averaging adjoint identity give

$$\|\Xi^2\omega\|_{\mathrm{c}}^2 = \langle Q'\psi, \omega\rangle_{\mathrm{c}} = \langle A\psi, GQ'^*\omega\rangle \le C_{2,\mathrm{gp}}\|\Xi^2\omega\|_{\mathrm{c}}\|GQ'^*\omega\|.$$

Dividing and squaring, with the zero vector treated separately, proves, for $\gamma_{0,\mathrm{gp}} = C_{2,\mathrm{gp}}^{-2}$,

$$\langle \omega, \mathsf{V}\omega\rangle_{\mathrm{c}} \ge \gamma_{0,\mathrm{gp}}\|\Xi^2\omega\|_{\mathrm{c}}^2, \qquad S \ge \gamma_{0,\mathrm{gp}}. \tag{257}$$

Decay of the inverse. Put $\kappa = \varepsilon_0/2$ and choose $\kappa' > 0$ so that

$$\kappa' \le \kappa/4, \qquad \frac{4\kappa' C_S c_S(\kappa/2)}{e\kappa} \le \tfrac{1}{2}\gamma_{0,\mathrm{gp}}. \tag{258}$$

Choose $0 < 3\delta_1 \le \min\{\kappa', \varepsilon_0/16\}$ and $\mathfrak{r}_4$ large enough that

$$\mathfrak{r}_4 \ge \frac{8(3\log L + 1)}{\varepsilon_0}, \qquad \frac{\log L}{\mathfrak{r}_4} \le \frac{\delta_1}{20}.$$

These constants depend only on L. Under $R_1\mathfrak{M} \ge \mathfrak{r}_4$, the upper Gram bound and the site sum at rate $\kappa/2$ apply. Fix y' and conjugate the matrix of S on counting $\ell^2(\Lambda_*)$ by $\theta(y) = \kappa' d_{\mathbf{B}}(y, y')$; thus $S_\theta = e^\theta S e^{-\theta}$. For $r = d_{\mathbf{B}}(y_1, y_2)$,

$$|(S_\theta - S)[y_1, y_2]| \le \kappa' r e^{\kappa' r} C_S e^{-\kappa r} \le \frac{4\kappa' C_S}{e\kappa} e^{-\kappa r/2}.$$

Here $\kappa' \le \kappa/4$ and $re^{-\kappa r/4} \le 4/(e\kappa)$. The row and column sums are at most $4\kappa' C_S c_S(\kappa/2)/(e\kappa) \le \gamma_{0,\mathrm{gp}}/2$. The Cauchy–Schwarz row-and-column estimate bounds the operator norm by the same number. Thus $\langle v, S_\theta v\rangle \ge (\gamma_{0,\mathrm{gp}}/2)\|v\|_{\ell^2}^2$, and $\|S_\theta^{-1}\| \le 2/\gamma_{0,\mathrm{gp}}$. Undoing the conjugation gives the first bound in (246), since $\kappa' \ge 3\delta_1$. Finally,

$$\mathsf{C}(y, y') = \xi_y^{-7/2}\xi_{y'}^{-1/2} S^{-1}[y, y'] = \xi_y^{-4}(\xi_y/\xi_{y'})^{1/2} S^{-1}[y, y'].$$

The first identity gives the second bound in (246); the scale-ratio estimate and $\varrho/2 \le \delta_1$ give (247). Symmetry of V^{-1} in the weighted coarse inner product gives the asserted weighted symmetry.

Projection and outer-region independence. The projection formula and range are (155), with $D = A$. By (153), every member of $N = \ker Q'$ vanishes outside Ω_1, and its ambient Laplacian is supported in Ω_1 and its bare nearest neighbours. These neighbours lie in Ω_0. Hence $\Delta_{\mathcal{S}} N = \Delta N$ as spaces of functions on the full bare lattice. Moreover N consists exactly of the functions supported in Ω_1 whose averages on the positive-level blocks vanish; this description involves only $\Omega_1, \dots, \Omega_t$. The ambient orthogonal projection onto ΔN is therefore independent of the outer region. Its restriction to $\mathcal{S}$ is R, which proves the assertion about its kernels.

The differentiated kernels. Use the constants $\delta_{0,\mathrm{lap}}, C_{\mathrm{lap}}$ of proposition 21.3. Set

$$\delta_2 = \tfrac{1}{8}\min\{\delta_{0,\mathrm{lap}}, \delta_1\}.$$

Choose $\mathfrak{m}_5, \mathfrak{r}_5$ so their hypotheses imply (237) and

$$\mathfrak{r}_5 \geq \max\{\mathfrak{r}_4, 10(\log L + 1)/\delta_2\}.$$

For a block centre y_1 put $F_0(x; y_1) = (G\chi_{y_1})(x)$ and $F_1(b; y_1) = (\partial G\chi_{y_1})(b)$. The pointwise scalar estimate gives

$$|F_0(x; y_1)| \leq C_{\mathrm{lap}}\xi_y^2 e^{-\delta_{0,\mathrm{lap}} d_{\mathbf{B}}(y,y_1)/2}, \qquad |F_1(b; y_1)| \leq C_{\mathrm{lap}}\xi_y e^{-\delta_{0,\mathrm{lap}} d_{\mathbf{B}}(y,y_1)/2}. \tag{259}$$

For an exit bond the endpoint x in Ω_0 lies outside Ω_1, by separation, and therefore $\xi_y = \xi$. The other endpoint value of $G\chi_{y_1}$ is zero, so $|F_1(b; y_1)| = \xi^{-1}|F_0(x; y_1)|$; this proves the second bound for exit bonds as well.

The right factors require the volume weights. Symmetry of G and the adjoint relation give

$$(Q'G\mathbf{1}_{\{x'\}})(y_2) = (\xi/\xi_{y_2})^3 F_0(x'; y_2), \qquad (Q'G\partial^*\mathbf{1}_{\{b'\}})(y_2) = (\xi/\xi_{y_2})^3 F_1(b'; y_2).$$

For the second identity, pair $G\chi_{y_2}$ with $\partial^*\mathbf{1}_{\{b'\}}$, move ∂ to the first factor, and use the fine bond weight ξ^3. Since $Q'^*\omega = \sum_y \omega(y)\chi_y$, each of the four projection kernels is

$$\sum_{y_1,y_2} F_\alpha(\,\cdot\,; y_1)\,\mathsf{C}(y_1, y_2)(\xi/\xi_{y_2})^3 F_\beta(\,\cdot'\,; y_2), \qquad \alpha, \beta \in \{0, 1\}. \tag{260}$$

Here $\alpha = 1$ denotes the left derivative, $\beta = 1$ the right adjoint derivative, and the corresponding arguments are bonds.

Put $\delta = 2\delta_2$ and $r_1 = d_{\mathbf{B}}(y, y_1)$, $r_2 = d_{\mathbf{B}}(y_1, y_2)$, $r_3 = d_{\mathbf{B}}(y_2, y')$. By (246) and (259), the summand is bounded by $2C_{\mathrm{lap}}^2/\gamma_{0,\mathrm{gp}}$ times

$$\xi^3 \xi_y^{2-\alpha} \xi_{y_1}^{-7/2} \xi_{y_2}^{-7/2} \xi_{y'}^{2-\beta} e^{-2\delta(r_1+r_2+r_3)}.$$

The scale product has the exact factorization

$$\xi_y^{2-\alpha} \xi_{y_1}^{-7/2} \xi_{y_2}^{-7/2} \xi_{y'}^{2-\beta} = \xi_{y'}^{-3} \xi_y^{-\alpha-\beta} \left(\frac{\xi_y}{\xi_{y_1}}\right)^{2+\beta} \left(\frac{\xi_{y'}}{\xi_{y_1}}\right)^{3/2-\beta} \left(\frac{\xi_{y'}}{\xi_{y_2}}\right)^{7/2}. \tag{261}$$

Equation (231) bounds the last three factors by $L^7 e^{5\varrho(r_1+r_2+r_3)}$. Our threshold gives $5\varrho \leq \delta$. By the triangle inequality,

$$e^{-2\delta(r_1+r_2+r_3)} e^{5\varrho(r_1+r_2+r_3)} \leq e^{-\delta d_{\mathbf{B}}(y,y')/2} e^{-\delta r_1/2} e^{-\delta r_2/2}.$$

Sum first over y_2 and then over y_1. Equation (233) at rate $\delta/2 = \delta_2$ applies because $\mathsf{r}_S(\delta_2) \leq 10(\log L + 1)/\delta_2 \leq R_1\mathfrak{M}$. It contributes $c_S(\delta_2)^2$. The remaining factors are $(\xi/\xi_{y'})^3 \xi_y^{-\alpha-\beta}$, proving (248) with $B_{5,\mathrm{gp}}$ depending only on L. □

For the local comparisons below, put $\mathcal{P} = \partial_0 \mathsf{P} \partial_0^*$ and choose common constants

$$\begin{aligned} \delta_{\mathrm{u}} = \delta_2, \qquad B_{\mathrm{u}} &= \max\{1, C_{\mathrm{lap}}, 2L^{1/2}/\gamma_{0,\mathrm{gp}}, B_{5,\mathrm{gp}}\}, \\ \mathfrak{m}_{\mathrm{u}} = \mathfrak{m}_5, \qquad \mathfrak{r}_{\mathrm{u}} &= \mathfrak{r}_5. \end{aligned} \tag{262}$$

The choices of $\mathfrak{m}_5, \mathfrak{r}_5$ imply the scalar and Gram hypotheses, and δ_2 is no larger than their rates. Thus (238), (247) and the fourth kernel bound in (248) hold with the common rate δ_{u} and constant B_{u}. Since $\partial_0 = \chi_{\mathcal{A}}\partial\chi_{\mathcal{S}}$, the kernel of $\mathcal{P}$ is the restriction of $\partial\mathsf{P}\partial^*$ to $\mathfrak{b}_0$ in both variables. The localization points agree, so this restriction preserves the fourth bound.

21.5. **The local models.** We first choose a quadratic partition whose members meet at most two consecutive layers. On its interior supports the local model is a two-scale operator on a larger torus. At the outer boundary we use a compressed curl operator with a mass. Throughout this subsection assume $\mathfrak{M} \geq L^5$; all further lower bounds on $\mathfrak{M}$ will depend only on L.

Use ϑ_0, η_0 from section 18.2, and fix

$$h(s) = \cos\Big(\tfrac{\pi}{2}\vartheta_0(4|s| - \tfrac{3}{2})\Big), \qquad \varpi(v) = \prod_{\mu=1}^{3} \frac{\eta_0(v_\mu)}{\int \eta_0}, \qquad \varpi_r(v) = r^{-3}\varpi(v/r).$$

Thus h is smooth, equals one for $|s| \leq 3/8$, and vanishes for $|s| \geq 5/8$. The identity $\vartheta_0(s)+\vartheta_0(1-s) = 1$ gives $h(s)^2 + h(s-1)^2 = 1$ for $0 \leq s \leq 1$. All other integer translates vanish there, so $\sum_{i\in\mathbb{Z}} h(s-i)^2 = 1$ for every s. The smooth nonnegative function ϖ has integral one and support in $\{|v|_\infty \leq 2/3\}$. Convolutions below use the periodic lift of an indicator function.

For $0 \leq n \leq t_*$ put $W_n = \mathfrak{M}\xi_n$ and $r_n = W_n/(6L)$. Let $\mathcal{Z}_n$ be the corners of the cubes of $\pi[W_n]$. For $z \in \mathcal{Z}_n$ set

$$\square = \{x : |x - z|_\infty \leq W_n\}, \qquad \widetilde{\square}^i = \{x : |x - z|_\infty \leq (1+i)W_n\}, \qquad g_\square(x) = \prod_{\mu=1}^{3} h((x_\mu - z_\mu)/W_n).$$

Coordinate differences in this formula are their periodic representatives. A cube is indexed by (n, z), even if two such cubes have the same point set. This convention matters when the torus side is $2W_n$.

Put $d_n(x) = \operatorname{dist}_\infty(x, \Omega_n^c)$, with values $+\infty$ and 0 for a full and an empty region, respectively, and define

$$\Psi_n = \mathbf{1}_{\{d_n \geq 3r_n/2\}} * \varpi_{r_n/2}, \qquad \sigma_n = \sin(\pi\Psi_n/2), \qquad \tau_n = \cos(\pi\Psi_n/2), \qquad \sigma_0 = 1.$$

The definitions for $n = t_* + 1$ use the empty region. For a cube of level n set

$$h_\square = g_\square \sigma_n \tau_{n+1}, \qquad S_\square = \operatorname{supp} h_\square, \qquad S_\square^{(a)} = \{x : \operatorname{dist}_\infty(x, S_\square) \leq a r_n\}. \tag{263}$$

Let $\mathfrak{D}_n$ contain the cubes for which $h_\square(b_-) \neq 0$ for some $b \in \mathfrak{b}_0$, and put $\mathfrak{D} = \bigcup_n \mathfrak{D}_n$. The two cutoffs are

$$\zeta_\square = \mathbf{1}_{S_\square^{(1/4)}} * \varpi_{r_n/12}, \qquad \theta_\square = \mathbf{1}_{S_\square^{(7/12)}} * \varpi_{r_n/12}.$$

A cube in $\mathfrak{D}_0$ is a boundary cube if $\widetilde{\square}^2 \cap \Omega_1 = \emptyset$. All other cubes in $\mathfrak{D}$ are called interior cubes.

Lemma 21.5. The following properties hold uniformly in the number of layers.

(a) The functions take values in $[0, 1]$, and

$$\sum_{n,z} h_\square(x)^2 = 1 \quad (x \in T), \qquad \sum_{\square\in\mathfrak{D}} h_\square(b_-)^2 = 1 \quad (b \in \mathfrak{b}_0). \tag{264}$$

For $\phi = h_\square, \zeta_\square, \theta_\square$, its first and second derivatives are bounded by $c_\flat r_n^{-1}$ and $c_\flat r_n^{-2}$. The same bounds hold for first and second lattice differences; in particular $|\Delta\phi| \leq 3c_\flat r_n^{-2}$ and $|\phi(x) - \phi(x')| \leq c_\flat r_n^{-1}|x - x'|_1$. The constant $c_\flat$ is absolute. Moreover,

$$\zeta_\square = 1 \text{ on } S_\square^{(1/6)}, \quad \operatorname{supp}\zeta_\square \subset S_\square^{(1/3)}, \qquad \theta_\square = 1 \text{ on } S_\square^{(1/2)}, \quad \operatorname{supp}\theta_\square \subset S_\square^{(2/3)}.$$

(b) For a cube of level n and center z,

$$S_\square^{(5/6)} \subset \{|x - z|_\infty < 5W_n/6\}, \qquad d_{n+1} \leq W_n/2 \text{ on } S_\square^{(5/6)}.$$

For $n \geq 1$, also $d_n \geq r_n/6$ there. If $y \in \Lambda_m$ and $S_\square \cap \{x : |x - y|_\infty \leq 6\xi_y\} \neq \emptyset$, then $n \in \{m-1, m\}$ and $y \in S_\square^{(1/6)}$. For a fixed y there are at most 16 such cubes.

(c) An interior cube of level $n \geq 1$ satisfies $\square \subset \Omega_{n-1}$. For every interior cube,

$$\widetilde{\square}^3 \cap \Omega_{n+2} = \emptyset, \qquad S_{\square}^{(5/6)} \subset \mathcal{L}_n \cup \mathcal{L}_{n+1}, \qquad \operatorname{dist}_\infty(S_{\square}^{(5/6)}, \Omega_0^c) > \mathfrak{M}\xi.$$

If $\widetilde{\square}^2$ meets Ω_{n+1}, then $\widetilde{\square}^3 \subset \Omega_n$ and the torus side is at least $26W_n$. For a boundary cube, $\operatorname{dist}_\infty(\square, \Omega_1) \geq 2\mathfrak{M}\xi$.

(d) Let $r > 0$ and $j \geq 0$. Suppose the open sup ball of radius r about x is disjoint from Ω_{j+1}. Then

$$d_{\mathbf{B}}(x, x') \geq r/\xi_j \quad \text{if } |x - x'|_\infty \geq r. \tag{265}$$

Consequently, if $\square$ is interior, $\Delta(y'') \cap S_\square \neq \emptyset$, and $\Delta(y) \not\subset S_{\square}^{(1/6)}$, then

$$d_{\mathbf{B}}(y, y'') \geq \frac{\mathfrak{M}}{72L^2}. \tag{266}$$

■

Proof. At each level the period ratio S/W_n is an even integer at least two. Periodizing the identity for h and taking its product in the three coordinates gives $\sum_{z \in \mathcal{Z}_n} g_\square^2 = 1$, including the period ratio two. The distance d_n is 1-Lipschitz when finite. Since the convolution radius is at most $r_n/3$, $\sigma_n = 0$ on $\{d_n \leq r_n\}$ and $\sigma_n = 1$ on $\{d_n \geq 2r_n\}$, with the complementary assertions for τ_n. Separation implies $\sigma_n = 1$ on the support of σ_{n+1}. Therefore

$$\sum_z h_\square^2 = \sigma_n^2(1 - \sigma_{n+1}^2) = \sigma_n^2 - \sigma_{n+1}^2.$$

Summing over n telescopes to one. Discarding cubes outside $\mathfrak{D}$ does not change this sum at b_-.

Differentiation of a convolution of an indicator with ϖ_r gives bounds $C_k r^{-k}$ for $k = 1, 2$, by the L^1 norms of the corresponding derivatives of ϖ. The chain and product rules give the asserted bounds for $\sigma_n, \tau_n, g_\square, h_\square$. Only fixed cutoff norms occur; $W_n \geq r_n$ and $r_{n+1} \geq r_n$. Each lattice difference is an average of the corresponding continuous derivative along one or two coordinate intervals. This also proves the bound for $\Delta\phi$. Integrating along a shortest coordinate path proves the Lipschitz bound. The support of $\varpi_{r_n/12}$ is contained in the sup ball of radius $r_n/12$; adding and subtracting this radius from $r_n/4$ and $7r_n/12$ proves the cutoff assertions.

For the support bounds, use

$$S_\square \subset \{|x - z|_\infty \leq 5W_n/8\} \cap \{d_{n+1} \leq 2r_{n+1}\}, \qquad S_\square \subset \{d_n \geq r_n\} \quad (n \geq 1).$$

Enlargement by $5r_n/6$ gives (b), because $2r_{n+1} = W_n/3$ and $r_n = W_n/(6L)$. In particular it does not reach Ω_{n+2}.

To prove the overlap bound, choose $x \in S_\square$ with $|x - y|_\infty \leq 6\xi_m$. Since $y \notin \Omega_{m+1}$, $d_{m+1}(x) \leq 6\xi_m < r_{m+1}$, excluding $n \geq m + 1$; for $n > m + 1$ separation gives the same contradiction. If $n \leq m - 2$, separation at $y \in \Omega_m$ gives $d_{m-1}(x) > (\mathfrak{M} - 6)\xi_m > 2r_{m-1}$. This contradicts the upper bound for d_{n+1} when $n = m - 2$; when $n < m - 2$, it places x in Ω_{n+2}, giving the same contradiction by separation. Thus $n \in \{m - 1, m\}$. Now

$$|z - y|_\infty \leq 5W_n/8 + 6\xi_m < W_n, \qquad 6\xi_m \leq r_n/6.$$

In each coordinate an open interval of length $2W_n$ contains at most two possible corner coordinates. There are at most eight centers at each of the two levels, and the second inequality gives $y \in S_{\square}^{(1/6)}$.

Here is the geometric point needed at an interior cube. If $n \geq 1$, or if $\widetilde{\square}^2$ meets Ω_{n+1}, some W_n-cube with corner z belongs to $\mathcal{L}_n$. Indeed, take $x \in S_\square$ and a W_n-cube through x. It lies in Ω_n: for $n \geq 1$ this follows from $d_n(x) > 0$; in the other case every point of $\widetilde{\square}^3$ is within $7W_n < R_1 W_n$ of a point of Ω_{n+1}, so belongs to Ω_n. If the chosen cube lies in Ω_{n+1}, use $d_{n+1}(x) \leq W_n/3$ to choose a point of Ω_{n+1}^c within $W_n/3$ of x. It is within $23W_n/24$ of z. A W_n-cube through it has corner z, lies in Ω_n by separation, and does not lie in Ω_{n+1}. It is the required cube. The assertion about

its corner follows directly from the half-integer grid: a center within $3W_n/2$ of z differs from z by $\pm W_n/2$ in each coordinate.

Thus $z \in \Omega_n$. For $n \geq 1$, separation from Ω_{n-1}^c gives $\square \subset \Omega_{n-1}$. If $\widetilde{\square}^2$ meets Ω_{n+1}, the argument just given proves $\widetilde{\square}^3 \subset \Omega_n$; also $\Omega_{n+1} \neq \emptyset$ forces $\mathsf{S} \geq 2\mathfrak{M}\xi_{n+1} \geq 26W_n$. Since $d_{n+1}(z) < W_n$, throughout $\widetilde{\square}^3$ one has $d_{n+1} < 5W_n < R_1\mathfrak{M}\xi_{n+1}$; this excludes Ω_{n+2}. The support therefore meets only layers $n, n+1$. For $n \geq 1$ its distance from Ω_0^c is larger than $R_1\mathfrak{M}\xi$. At $n = 0$ an interior cube has $\widetilde{\square}^3 \subset \Omega_0$. Every point of Ω_0^c is at distance at least $4W_0$ from z: otherwise a W_0-cube of the complement through that point has center within $9W_0/2$ of z; the half-integer grid places its center within $7W_0/2$, so the cube lies in $\widetilde{\square}^3$, a contradiction. Subtracting $5W_0/6$ proves the boundary margin in (c). For a boundary cube, the defining exclusion of Ω_1 from $\widetilde{\square}^2$ immediately gives its distance at least $2\mathfrak{M}\xi$ from $\square$.

For (d), fix $0 < r' < r$ and use the clipped distance $f(z) = \min\{|z - x|_\infty, r'\}$. On the portion of a connecting set where $0 < f < r'$, all points lie outside Ω_{j+1} and the weighted length density is at least ξ_j^{-1}. The interval-image calibration in section 12.2, applied to this 1-Lipschitz function, gives weighted length at least r'/ξ_j. Taking the infimum and then $r' \uparrow r$ proves (265). For (266), choose $x'' \in \Delta(y'') \cap S_\square$ and $x \in \Delta(y) \setminus S_\square^{(1/6)}$. The ball of radius $r_n/6$ about x'' is contained in $S_\square^{(1/6)}$ and misses Ω_{n+2}. Thus

$$d_{\mathbf{B}}(x, x'') \geq \frac{r_n}{6\xi_{n+1}} = \frac{\mathfrak{M}}{36L^2}.$$

For any point $z \in \Delta(y)$, the segment from y to z stays in the cell interior except possibly at its endpoint. Its weighted length is $\xi_y^{-1}|z - y|_1 \leq 3/2$; the endpoint has zero length. These cell-to-center bounds give $d_{\mathbf{B}}(y, y'') \geq \mathfrak{M}/(36L^2) - 3 \geq \mathfrak{M}/(72L^2)$, since $\mathfrak{M} \geq L^5$. □

We also need to compare the cell of a nearby bare site with the original cell. If $y \in \Lambda_*$, $x \in \Omega_0$ is a bare site, $|x - y|_\infty \leq 2\xi_y$, and $y_* = y(x)$, then

$$L^{-1}\xi_y \leq \xi_{y_*} \leq L\xi_y, \qquad d_{\mathbf{B}}(y, y_*) \leq 9L. \tag{267}$$

Indeed x lies in the chart $C(y)$ of lemma 21.2(b). Its scale comparison gives the first assertion, and its coordinate-path bound gives $d_{\mathbf{B}}(x, y) \leq L\xi_y^{-1}|x - y|_1 \leq 6L$. The cell-to-center bound is $d_{\mathbf{B}}(x, y_*) \leq 3/2$; the triangle inequality proves the second assertion.

For an interior cube of level n, fix a lift $\widetilde{z}$ of its center, put $W = W_n$, and define

$$T_\square = \mathbb{R}^3/(8W\mathbb{Z})^3, \qquad O = \widetilde{z} + (-W, W)^3, \qquad \iota_\square = \pi_\square \circ (\pi|_O)^{-1}.$$

Let $\Upsilon_\square$ be the union in $T_\square$ of the images of the W-cubes contained in $\widetilde{z} + [-3W, 3W]^3$ whose images in T belong to Ω_{n+1}. Thus this intersection is taken cube-wise. Both covering maps are injective on O; the second is injective on the larger closed cube of side $6W$. The maps identify the lattices and centered partitions of spacings ξ_j, $j \leq n + 1$, within O, since both periods are multiples of these spacings. On this set, $x \in \Omega_{n+1}$ if and only if $\iota_\square x \in \Upsilon_\square$.

Use on $T_\square$ the sequence with members $0, \dots, n$ equal to $T_\square$, member $n + 1$ equal to $\Upsilon_\square$, and all later members empty. Its bare bond frame is the whole torus. Attach the corresponding spaces $\mathcal{S}_\square, \mathcal{Y}_\square, \mathcal{A}_\square$ and operators $Q'_\square, Q_\square, \mathsf{R}_\square, \mathsf{P}_\square, \mathsf{G}'_\square$ to this sequence and put

$$\mathcal{P}_\square = \partial \mathsf{P}_\square \partial^*, \qquad \Delta_{a,\square} = \partial^*\partial + \partial \mathsf{R}_\square \partial^* + Q_\square^* a Q_\square = \Delta - \mathcal{P}_\square + Q_\square^* a Q_\square.$$

Here and below $a = 1$. For a boundary cube, put $T_\square = T$ and $\iota_\square = I$. Let $\mathcal{A}_\square$ consist of the fields on those bonds of $\mathfrak{b}_0$ with both endpoints in $\widetilde{\square}^3$, and let $\chi_\square$ be its orthogonal projection. Set

$$\Delta_{a,\square} = \chi_\square \partial^* \partial \chi_\square + \xi^{-2} I \quad \text{on } \mathcal{A}_\square.$$

Let $\mathcal{S}_\square$ consist of the site functions in $\mathcal{S}$ supported in $\widetilde{\square}^3$. All boundary fields are extended by zero, and their higher tensor source spaces contain all bare bonds.

For either kind write $|o-o'|_\square = \xi_n^{-1}|o-o'|_{1,T_\square}$. A cutoff ϕ supported in $S_\square^{(5/6)}$ defines two mutually adjoint transfers, by

$$(\phi v)(b) = \phi(b_-)v(\iota_\square b), \qquad (\phi A)(\iota_\square b) = \phi(b_-)A(b),$$

with value zero elsewhere. The same convention applies to scalar fields and tensors. The common weight ξ^3 and the one-to-one correspondence of sites and bonds prove adjointness. A transferred smooth cutoff is extended by zero off $\pi_\square(O)$; it remains smooth with the same derivative bounds, since its support stays within the smaller cube of radius $19W/24$. In particular transfers introduce no boundary distribution or change in the cutoff estimates.

Under these identifications, for fields supported in $S_\square^{(2/3)}$ at an interior cube,

$$\begin{gathered}\Delta_{\mathcal{S}} = \Delta, \quad \partial_0 = \partial, \quad \partial_0^* = \partial^*, \quad \chi_{\mathcal{A}}\partial^*\partial\chi_{\mathcal{A}} = \partial^*\partial, \\ Q'^*\mathsf{a}'Q' = Q_\square'^*\mathsf{a}'Q_\square', \qquad Q^*aQ = Q_\square^*aQ_\square.\end{gathered} \tag{268}$$

Their outputs are supported in $S_\square^{(5/6)}$. To check the averages, every cell meeting this larger set has level n or $n+1$, lies within O, and corresponds to the cell of the same level in the local sequence. Indeed the set is inside Ω_n and outside Ω_{n+2}, with the margins in lemma 21.5; its distance to the boundary of $\square$ exceeds the side ξ_{n+1} of such a cell. The converse follows from the same margins and the equality of the membership tests for Ω_{n+1} and $\Upsilon_\square$.

A determining bond whose endpoint blocks meet $S_\square^{(2/3)}$ also has level n or $n+1$. For a smaller level its blocks lie in Ω_n^c. For level $n+2$ their distance to Ω_{n+1}^c is greater than $(R_1\mathfrak{M}-2L)\xi_{n+1} > W/2$; for a larger level their blocks lie in Ω_{n+2}. Both alternatives contradict the support margins. The remaining blocks lie in $S_\square^{(5/6)}$, because $2\xi_{n+1} \le r_n/6$, and their determining bonds satisfy the same endpoint tests in the two models. The straight-path formula (222) is unchanged by $\iota_\square$, so its matrix weights, and hence the quadratic averaging forms, agree. The scalar average agrees cell by cell. Finally the difference operators involve only neighboring sites. Their support enlargement is at most ξ, that of the scalar averaging term at most ξ_{n+1}, and that of the bond averaging term at most $2\xi_{n+1}$. This proves (268) and its support assertion.

We shall also use, for bare sites in the interior of a cube,

$$d_{\mathbf{B}}(x,x') \le L|x-x'|_\square, \qquad |x-x'|_\square \le 6\mathfrak{M}. \tag{269}$$

For an interior cube the coordinate path in O stays in Ω_{n-1} when $n \ge 1$, so every step has weighted length at most $L\xi/\xi_n$ by lemma 21.2(a). At level zero use the bound one per step. On $T_\square$ this lifted path has length $|x-x'|_1$, since its coordinate displacements are less than $2W$, half the local period. For a boundary cube use a shortest coordinate path in T and again the bound one per step. This proves both estimates.

The scalar and Gram estimates apply on each interior local torus with constants independent of the cube. Set $\delta_{\mathrm{pr}} = \delta_{\mathrm{u}}/L$ and $\mathfrak{m}_{\mathrm{pr}} = \mathfrak{m}_{\mathrm{u}}$, using (262), and assume $\mathfrak{M} \ge \max\{L^5, L\mathfrak{m}_{\mathrm{pr}}\}$. Write $\mathsf{q}(o,r) = \{x : |x-o|_\infty \le r\}$ on the torus in question. For local cells $\Delta(w), \Delta(w')$, put $E_{w,w'} = e^{-\delta_{\mathrm{pr}}|w-w'|_\square}$. There is B_{pr}, depending only on L, such that

$$\begin{aligned} |(\mathsf{G}_\square'\lambda)(x)| &\le B_{\mathrm{pr}}\xi_n^2 E_{w,w'}|\lambda|, & |(\nabla\mathsf{G}_\square'\lambda)(x)| &\le B_{\mathrm{pr}}\xi_n E_{w,w'}|\lambda|, \\ |(\mathsf{G}_\square'\nabla^*F)(x)| &\le B_{\mathrm{pr}}\xi_n E_{w,w'}|F|, & |\mathsf{C}_\square(w,w')| &\le B_{\mathrm{pr}}\xi_n^{-4}E_{w,w'}, \\ |\mathcal{P}_\square(b,b')| &\le B_{\mathrm{pr}}(\xi/\xi_{w'})^3\xi_n^{-2}E_{w,w'}. \end{aligned} \tag{270}$$

Here $x \in \Delta(w)$, the site function λ and site–direction function F are supported in $\Delta(w')$, b, b' have localization points in the two respective cells, and

$$\mathsf{C}_\square = (Q_\square'\mathsf{G}_\square'^2Q_\square'^*)^{-1}, \qquad \mathcal{P}_\square = \partial\mathsf{P}_\square\partial^*.$$

The last two kernels use counting measure.

Indeed, the local sequence is separated with block number $\mathfrak{M}' = \mathfrak{M}/L$ and last index $n+1$. Its torus side is $8\mathfrak{M}'\xi_{n+1}$, and $\Upsilon_\square$ is a union of $\mathfrak{M}'\xi_{n+1}$-cubes. All preceding regions are the whole torus. If $\Upsilon_\square$ is empty, discard that final member; the period condition still holds. Thus its separation conditions hold for every gap number; choose an auxiliary R_1' with $R_1'\mathfrak{M}' \geq \mathfrak{r}_{\mathrm{u}}$. Since $\mathfrak{M}' \geq \mathfrak{m}_{\mathrm{u}}$, propositions 21.3 and 21.4 and (262) apply. This auxiliary choice imposes no restriction on the original gap number. The local layers have indices $n, n+1$, whence

$$\xi_n \leq \xi_w \leq L\xi_n, \qquad d_\square(w,w') \geq \xi_{n+1}^{-1}|w-w'|_1 = L^{-1}|w-w'|_\square.$$

The distance bound follows by calibrating a connecting set with the distance from one endpoint, as in lemma 21.2. Substitution gives (270), for example with $B_{\mathrm{pr}} \geq L^2 B_{\mathrm{u}}$.

The vector estimates below allow both adjacent cell sizes. This will let us localize at a cell without changing the local model or distinguishing which of its two layers contains it.

Proposition 21.6. There are $\delta_{\mathrm{loc}} \in (0,1]$ and $B_{\mathrm{l}} \geq 1$, depending only on L, and $B_{\mathrm{l}}(\alpha)$, $B_{\mathrm{l}}'(\varepsilon)$, and $B_{\mathrm{l}}'(\alpha,\varepsilon)$, depending only on L and the displayed exponents, with the following properties. Assume $\mathfrak{M} \geq \max\{L^5, L\mathfrak{m}_{\mathrm{pr}}\}$, and let $\square \in \mathfrak{D}_n$ be of either kind. Then $\Delta_{a,\square}$ is positive definite and its inverse $\mathsf{G}_\square$ is symmetric. Take $\lambda_1, \lambda_2 \in \{\xi_n, \xi_{n+1}\}$, $o, o' \in T_\square$, and put $E = e^{-\delta_{\mathrm{loc}}|o-o'|_\square}$. Let ζ be a site function supported in $\mathsf{q}(o,\lambda_1)$, multiplying through the initial point. Let J, J', J'' belong respectively to $\mathcal{A}_\square, \mathcal{A}_\square^{(1)}, \mathcal{A}_\square^{(2)}$ and be supported in $\mathsf{q}(o',\lambda_2)$. Let $0 \leq \alpha < 1$, $\varepsilon > 0$, and $\alpha + \varepsilon < 1$.

For boundary cubes all one-forms are extended by zero, $\mathcal{A}_\square^{(r)} = \mathcal{A}^{(r)}$, and $\nabla^*, \nabla^*\nabla^*$ are the adjoints of $\nabla, \nabla\nabla$ on $\mathcal{A}_\square$. Localization may use the base point relative either to Ω_0 or to $\Omega_0 \cap \widetilde{\square}^3$. For a bond x in $\mathsf{q}(o, 2\lambda_1)$ we have

$$\begin{aligned}
|(\mathsf{G}_\square J)(x)| &\leq B_{\mathrm{l}}\xi_n^2 E|J|, & |(\nabla\mathsf{G}_\square J)(x)| &\leq B_{\mathrm{l}}\xi_n E|J|,\\
|(\mathsf{G}_\square\nabla^* J')(x)| &\leq B_{\mathrm{l}}\xi_n E|J'|, & |(\Delta\mathsf{G}_\square J)(x)| &\leq B_{\mathrm{l}} E|J|, \qquad \text{(R1)}\\
|(\nabla\mathsf{G}_\square\nabla^* J')(x)| &\leq B_{\mathrm{l}}'(\varepsilon)E(\|J'\|_{\varepsilon,\lambda_2} + |J'|). & & \qquad \text{(R3)}
\end{aligned}$$

The localized Hölder bounds are

$$\begin{aligned}
\xi_n^{-1}\|\zeta\mathsf{G}_\square J\|_{\alpha,\lambda_1}, \quad \|\zeta\nabla\mathsf{G}_\square J\|_{\alpha,\lambda_1} &\leq B_{\mathrm{l}}(\alpha)\xi_n(\|\zeta\|_{\alpha,\lambda_1} + |\zeta|)E|J|,\\
\|\zeta\mathsf{G}_\square\nabla^* J'\|_{\alpha,\lambda_1} &\leq B_{\mathrm{l}}(\alpha)\xi_n(\|\zeta\|_{\alpha,\lambda_1} + |\zeta|)E|J'|, \qquad \text{(R2)}\\
\|\zeta\nabla\mathsf{G}_\square\nabla^* J'\|_{\alpha,\lambda_1} &\leq B_{\mathrm{l}}'(\alpha,\varepsilon)(\|\zeta\|_{\alpha,\lambda_1} + |\zeta|)E(\|J'\|_{\alpha+\varepsilon,\lambda_2} + |J'|). \qquad \text{(R4)}
\end{aligned}$$

Finally,

$$\begin{aligned}
\xi_n^{-2}\|\zeta\mathsf{G}_\square J\|, \quad \xi_n^{-1}\|\zeta\nabla\mathsf{G}_\square J\|, \quad \|\zeta\nabla\nabla\mathsf{G}_\square J\| &\leq B_{\mathrm{l}}|\zeta|E\|J\|,\\
\xi_n^{-1}\|\zeta\mathsf{G}_\square\nabla^* J'\|, \quad \|\zeta\nabla\mathsf{G}_\square\nabla^* J'\| &\leq B_{\mathrm{l}}|\zeta|E\|J'\|,\\
\|\zeta\mathsf{G}_\square\nabla^*\nabla^* J''\| &\leq B_{\mathrm{l}}|\zeta|E\|J''\|. \qquad \text{(R5)}
\end{aligned}$$

All constants are uniform in the spacing, torus, cube, gap number and number of layers. ∎

Proof. For an interior cube we first identify its operator with the two-scale operator of proposition 20.2. First convert that corner-block operator to centered blocks in its normalized coordinates, with spacings $\eta = L^{-n}, 1, L$. There are two translations: one changes the origin of each fine block, and the other changes the origin of the coarse blocks. Put

$$\eta = L^{-n}, \quad s = (1-\eta)/2, \quad s' = (L-1)/2, \quad e = e_1+e_2+e_3, \quad (\tau_v f)(x) = f(x-v), \quad \tau = \tau_{s'e}\tau_{se}.$$

Since L is odd, both shifts belong to their respective lattices. Superscript c denotes centered blocks. Since the centered fine and coarse blocks are the corresponding corner blocks shifted by $-se$ and $-s'e$,

$$\mathsf{Q}_n^{\mathrm{c}} = \mathsf{Q}_n\tau_{se}, \qquad \mathsf{Q}_{1,L}^{\mathrm{c}} = \mathsf{Q}_{1,L}\tau_{s'e}, \qquad \tau_{s'e}\mathsf{Q}_n^{\mathrm{c}} = \mathsf{Q}_n\tau, \qquad \mathsf{Q}_{1,L}^{\mathrm{c}}\mathsf{Q}_n^{\mathrm{c}} = \mathsf{Q}_{1,L}\mathsf{Q}_n\tau.$$

The same identities hold for the scalar averages. The last two use the commutation of the fine average with unit-lattice translations. The coarse set Λ' remains arbitrary; its union of unit blocks becomes $\Lambda_c = \Lambda - s'e$. Thus both its membership tests and the two zero-average conditions are translated by τ. Equivalently, $N^c = \tau^{-1}N$. The Laplacian commutes with τ, so the same relation holds for their Laplacian images; orthogonality gives $\mathsf{R}^c = \tau^{-1}\mathsf{R}\tau$. The coarse penalty satisfies $\mathsf{q}^c(\mathsf{Q}_n^c A) = \mathsf{q}(\mathsf{Q}_n \tau A)$. Translations commute with the differences and preserve the inner products. Hence

$$\Delta_1^c = \tau^{-1}\Delta_1\tau, \qquad G^c = \tau^{-1}G\tau.$$

The estimates of proposition 20.2 pass to G^c: the centers translate along with the supports, while their distance, the norms and the Hölder seminorms are unchanged.

Next rescale by $(\mathscr{S}_n f)(X) = f(\xi_n X)$. The resulting torus has side $8\mathfrak{M}$, an even multiple of L, and bare spacing η. Its coarse subset Λ' consists of the level-$n+1$ sites in $\Upsilon_\square$. The decomposition into centered blocks is exact: in each coordinate every integer $|k| \le (L^{n+1}-1)/2$ has a unique expression

$$k = L^n i + i', \qquad |i| \le (L-1)/2, \qquad |i'| \le (L^n-1)/2.$$

It gives $Q'_\square = \mathsf{Q}_n'^c$ outside the coarse region and $Q'_\square = \mathsf{Q}_{1,L}'^c \mathsf{Q}_n'^c$ on its coarse sites, after rescaling. The analogous bond identity follows by writing each straight-path index uniquely as $L^n i + i'$ with $0 \le i < L$ and $0 \le i' < L^n$. The determining bonds of level n have neither endpoint in the coarse region; those of level $n+1$ have an endpoint in its coarse-site subset. Thus their tests are precisely the tests defining q.

Every scalar product acquires the same factor ξ_n^3, and a difference acquires ξ_n^{-1}. The quadratic averaging penalty becomes

$$\xi_n \sum_{b:\ b_-,b_+\notin\Lambda_c} |(\mathsf{Q}_n^c \mathscr{S}_n A)(b)|^2 + \xi_{n+1} \sum_{c\in\Lambda'^*} |(\mathsf{Q}_{1,L}^c \mathsf{Q}_n^c \mathscr{S}_n A)(c)|^2,$$

where the first sum is over unit bonds. The zero-average spaces and their Laplacian images correspond, so the common scalar-product factor also preserves orthogonal projection. Consequently

$$\mathsf{R}_\square = \mathscr{S}_n^{-1}\mathsf{R}^c\mathscr{S}_n, \qquad \Delta_{a,\square} = \xi_n^{-2}\mathscr{S}_n^{-1}\Delta_1^c\mathscr{S}_n, \qquad \mathsf{G}_\square = \xi_n^2\mathscr{S}_n^{-1}G^c\mathscr{S}_n. \tag{271}$$

In particular the local operator is positive definite. An expression with r differences, including adjoint differences, acquires the factor ξ_n^{2-r}. Moreover

$$|\mathscr{S}_n f| = |f|, \qquad \|\mathscr{S}_n f\|_\alpha = \|f\|_{\alpha,\xi_n}, \qquad \|f\| = \xi_n^{3/2}\|\mathscr{S}_n f\|.$$

The common factor in the last identity cancels in operator bounds. Thus proposition 20.2 gives (R1)–(R5) at radius ξ_n, except for the first entry of (R2), which we derive below. This application uses no restriction on Λ' beyond the one already admitted in that theorem.

To allow the larger radius $L\xi_n$, use a partition at scale ξ_n. For the corners $w \in \xi_n(\mathbb{Z}+\frac12)^3$ on $T_\square$, set

$$\varphi_w(x) = \prod_{\mu=1}^3 h((x_\mu - w_\mu)/\xi_n)^2.$$

The identity for h gives $\sum_w \varphi_w = 1$. Each member lies in $[0,1]$, is supported in $\mathsf{q}(w, 5\xi_n/8)$, and has $|\nabla\varphi_w| \le C_\varphi/\xi_n$. By (277), $\|\varphi_w\|_{\alpha,\xi_n} \le C_\varphi$. Let $W(o,\lambda)$ consist of those members meeting $\mathsf{q}(o, 2\lambda)$. Since the grid spacing is ξ_n and $\lambda \le L\xi_n$, both the number of these members and their centers' normalized distances from o are bounded by constants depending only on L. Decompose $J = \sum_{w'} \varphi_{w'} J$ and $\zeta = \sum_w \varphi_w \zeta$, and likewise for the tensors. The product and change-of-scale inequalities (277)–(278) give

$$\|\varphi_w\zeta\|_{\alpha,\xi_n} \le \|\zeta\|_{\alpha,\lambda_1} + C_\varphi|\zeta|, \qquad \|\varphi_{w'}J'\|_{\varepsilon,\xi_n} \le \|J'\|_{\varepsilon,\lambda_2} + C_\varphi|J'|,$$
$$\|F\|_{\alpha,\lambda_1} \le L\max\{\|F\|_{\alpha,\xi_n}, 2|F|\}.$$

For any rate $\delta \le 1$, the triangle inequality bounds the exponential between w, w' by $C(L)$ times the exponential between o, o'. Applying the radius-ξ_n estimates to all pairs and summing proves (R1), (R3), (R5), (R4), and the last two entries of (R2). The change of seminorm scale also uses their sup-norm bounds; for (R4) apply (R3) with $\alpha + \varepsilon$ in place of ε. Every bond in $\mathsf{q}(o, 2\lambda_1)$ belongs to a radius-ξ_n cube of this partition, which proves the larger output domain.

For the remaining entry of (R2), apply (277) to $X = \mathsf{q}(o, \lambda_1)$. Its enlarged cube is $X^{+\lambda_1} = \mathsf{q}(o, 2\lambda_1)$, and the required coordinate-path estimate applies since $5\lambda_1 \le 5L\xi_n < 8\mathfrak{M}\xi_n$. Thus

$$\|\zeta \mathsf{G}_\square J\|_{\alpha,\lambda_1} \le \|\zeta\|_{\alpha,\lambda_1} \sup_{X^{+\lambda_1}} |\mathsf{G}_\square J| + |\zeta|_{\lambda_1} \sup_{X^{+\lambda_1}} |\nabla \mathsf{G}_\square J|.$$

Now (R1) and $\lambda_1 \le L\xi_n$ give the assertion.

For a boundary cube the mass gives a direct proof. Here $n = 0$. On $\mathcal{A}_\square$ put $\mathsf{t} = \xi^2 \chi_\square \partial^* \partial \chi_\square$. The ambient difference identity (156) and $\|\tau_\mu - I\| \le 2$ imply $0 \le \partial^*\partial \le \Delta \le 12\xi^{-2}$. Consequently

$$\Delta_{a,\square} = \xi^{-2}(13I - (12I - \mathsf{t})) \ge \xi^{-2} I, \qquad \mathsf{G}_\square = \xi^2 \sum_{i \ge 0} 13^{-i-1} (12I - \mathsf{t})^i.$$

For counting kernels $|O(b, b')| \le \|O\|$, since both coordinate vectors have squared norm ξ^3. The entry $\mathsf{t}(b, b')$ vanishes unless the two bonds belong to a common plaquette, and hence unless $|b_- - b'_-|_1 \le 2\xi$. The i-th power therefore vanishes at distances greater than $2i\xi$, while its norm is at most 12^i. Summing the remaining geometric series gives

$$|\mathsf{G}_\square(b, b')| \le \xi^2 (12/13)^{|b_- - b'_-|_1/(2\xi)} = \xi^2 e^{-2\vartheta |b_- - b'_-|_1/\xi}, \qquad \vartheta = \tfrac{1}{4} \log(13/12) > 0. \tag{272}$$

The kernels are extended by zero outside $\mathcal{A}_\square$. Every operator in (R1)–(R5) is obtained by taking at most two differences of this kernel. These differences are sums of at most 36 shifts by at most 2ξ, with the appropriate factor ξ^{-1} or ξ^{-2}. Their kernels are bounded by $C\xi^{2-r} e^{-2\vartheta |b_- - b'_-|_1/\xi}$, where r is the total number of differences.

Either allowed base point differs from the initial point by at most ξ. There are at most $6(4L+1)^3$ bonds whose base points lie in $\mathsf{q}(o, 2L\xi)$. For output and input bonds in the required cubes,

$$|b_- - b'_-|_1/\xi \ge |o - o'|_\square - 3(3L + 2).$$

Summing the kernel over the input bonds gives (R1) and (R3). The row and column bounds and lemma 21.7(e) give all six estimates in (R5). The powers of ξ are unchanged by these finite sums. Finally distinct sites are at least ξ apart and $\lambda_1 \le L\xi$, so $\|F\|_{\alpha,\lambda_1} \le 2L|F|$. If $(\zeta F)(b) \ne 0$, its initial point lies in $\mathsf{q}(o, \lambda_1)$ and its base point in $\mathsf{q}(o, 2\lambda_1)$. Thus the sup-norm estimates just proved give (R2) and (R4) as well.

Choose $\delta_{\mathrm{loc}} \le \min\{1, \delta_{\mathrm{per}}, \vartheta\}$, and enlarge the finitely many constants to cover both kinds of cubes. The interior rate is that of proposition 20.2; the boundary rate is absolute. This gives precisely the stated uniformity. □

21.6. **Combining cell estimates.** For $p \in \{2, \infty\}$ define

$$N_p(O; y, y') = \|\chi_y O \chi_{y'}\|_{p \to p},$$
$$\|O\|_{p,\kappa} = \sup_y \sum_{y' \in \Lambda_*} e^{\kappa d_{\mathbf{B}}(y,y')} N_p(O; y, y'). \tag{273}$$

The domain and range may be $\mathcal{S}$, $\mathcal{A}$, $\mathcal{Y}$, site functions with a direction index, or the subspace of $\mathcal{A}^{(r)}$ supported in Ω_0. An operator with values in an ambient tensor space is restricted to that subspace when its cell norm is used. On each of these spaces the χ_y are mutually orthogonal projections with sum I. Write $O \in \mathfrak{K}_q(\kappa, A)$ when $N_\infty(O; y, y') \le A \xi_y^q e^{-\kappa d_{\mathbf{B}}(y,y')}$. Products below are asserted only when their domains match.

Lemma 21.7. Write $d = d_{\mathbf{B}}$. The following rules hold.

(a) For $\kappa \geq 0$,

$$\|\chi_y O f\|_p \leq \sum_{y'} N_p(O; y, y') \|\chi_{y'} f\|_p,$$
$$N_p(O_1 O_2; y, y') \leq \sum_z N_p(O_1; y, z) N_p(O_2; z, y'),$$
$$\|O_1 O_2\|_{p,\kappa} \leq \|O_1\|_{p,\kappa} \|O_2\|_{p,\kappa}, \qquad N_p(O; y, y') \leq \|O\|_{p,\kappa} e^{-\kappa d(y,y')}.$$

Also $\|I\|_{p,\kappa} = 1$. If O is an endomorphism and $\|O\|_{p,\kappa} \leq 1/2$, then $(I - O)^{-1} = \sum_{i \geq 0} O^i$, with $\|O^i\|_{p,\kappa} \leq 2^{-i}$ and $\|(I - O)^{-1}\|_{p,\kappa} \leq 2$.

(b) If $0 \leq \kappa < \kappa'$, $R_1 \mathfrak{M} \geq \mathsf{r}_S(\kappa' - \kappa)$ and $N_p(O; y, y') \leq A e^{-\kappa' d(y,y')}$, then $\|O\|_{p,\kappa} \leq A c_S(\kappa' - \kappa)$. For $\kappa > 0$ and $R_1 \mathfrak{M} \geq \mathsf{r}_S(\kappa/2)$,

$$\sum_z e^{-\kappa d(y,z) - \kappa d(z,y')} \leq c_S(\kappa/2) e^{-\kappa d(y,y')/2}. \tag{274}$$

(c) If $q \in \mathbb{R}$, $\epsilon > 0$ and $\epsilon R_1 \mathfrak{M} \geq |q| \log L$, then $\xi_{y'}^q \leq L^{|q|} \xi_y^q e^{\epsilon d(y,y')}$.

(d) Suppose $\kappa > 0$, $R_1 \mathfrak{M} \geq \mathsf{r}_S(\kappa/4)$ and $\kappa R_1 \mathfrak{M}/4 \geq |q_2| \log L$. If $O_j \in \mathfrak{K}_{q_j}(\kappa, A_j)$ for $j = 1, 2$, then

$$O_1 O_2 \in \mathfrak{K}_{q_1 + q_2}(\kappa/2, L^{|q_2|} c_S(\kappa/4) A_1 A_2).$$

(e) For maps between fine site or bond spaces, including direction indices,

$$N_2(O; y, y') \leq \left(N_\infty(O; y, y') N_\infty(O^*; y', y)\right)^{1/2},$$
$$N_2(O^*; y', y) = N_2(O; y, y').$$

The same assertions hold for restrictions by arbitrary set indicators, including maps between $\mathcal{A}_\square$ and $\mathcal{A}$.

(f) For multipliers ϕ, ϕ' on the range and domain, and set indicators χ, χ' on fine site or bond spaces,

$$\|\chi(O\phi' - \phi O)\chi'\|_{\infty \to \infty} \leq \|\chi O \chi'\|_{\infty \to \infty} \sup_{\chi(x)\chi'(x') \neq 0} |\phi'(x') - \phi(x)|.$$

(g) If $N_2(O; y, y') \leq k(y, y')$ with $k \geq 0$ symmetric, then $\|O\|_{2 \to 2} \leq \sup_y \sum_{y'} k(y, y')$.

■

Proof. Insert $I = \sum_z \chi_z$ and use the triangle inequality to obtain the first two inequalities in (a). The metric triangle inequality then gives submultiplicativity of the weighted row sums. Each block is bounded by its row sum. Since the cell projections partition each nonzero space, the identity has norm one; the usual geometric series in this finite-dimensional normed algebra proves the inverse assertion.

The first assertion of (b) is (233), and its second is (234) at rate $\kappa/2$. Part (c) is (231). To prove (d), apply (a) and transfer $\xi_z^{q_2}$ to the output scale by (c). The remaining sum is bounded by

$$\sum_z e^{-3\kappa d(y,z)/4 - \kappa d(z,y')} \leq e^{-\kappa d(y,y')/2} \sum_z e^{-\kappa d(y,z)/4} \leq c_S(\kappa/4) e^{-\kappa d(y,y')/2}.$$

For (e), the fine inner products all have weight ξ^3. Thus the adjoint matrix is the transpose, and Cauchy–Schwarz in the double sum bounds its ℓ^2 norm by the square root of the product of its absolute row and column sums. These sums are the two displayed ℓ^∞ norms. This argument also applies after arbitrary restrictions; adjoint equality gives the second assertion. In (f), each matrix entry $O(x, x')$ is multiplied by $\phi'(x') - \phi(x)$, so the absolute row sums give the bound.

Finally, put $\bar{k} = \sup_y \sum_{y'} k(y,y')$. Orthogonality, (a), Cauchy–Schwarz and symmetry give

$$\|Of\|^2 \le \sum_y \Big(\sum_{y'} k(y,y') \|\chi_{y'} f\| \Big)^2 \le \bar{k} \sum_{y,y'} k(y,y') \|\chi_{y'} f\|^2 \le \bar{k}^2 \|f\|^2.$$

□

We also need the product rules with their lattice shifts. Write $f^{\pm\nu} = T_{\pm\nu} f$. For a set X, let $X^{+\lambda} = \{x : \mathrm{dist}_\infty(x,X) \le \lambda\}$, and let $\|\cdot\|_{\alpha,\lambda;X}$ restrict both arguments of the seminorm to X; for bond fields these are the initial points.

Lemma 21.8. (a) On a bare torus, for site or bond fields,

$$\begin{aligned} \nabla_\nu(\phi f) &= \phi^{+\nu} \nabla_\nu f + (\nabla_\nu \phi) f, \\ \phi \nabla_\nu^* f &= \nabla_\nu^*(\phi f) + (\nabla_\nu \phi)^{-\nu} f^{-\nu}. \end{aligned} \tag{275}$$

The commutator $\mathsf{L}(\phi) = \phi\Delta - \Delta\phi$ is antisymmetric and satisfies

$$\begin{aligned} \mathsf{L}(\phi) f &= \sum_\nu \big[(\nabla_\nu \phi) \nabla_\nu f + (\nabla_\nu^* \phi) \nabla_\nu^* f \big] - (\Delta\phi) f \\ &= -\sum_\nu \nabla_\nu^* \big[(\nabla_\nu \phi)(f + f^{+\nu}) \big] + (\Delta\phi) f. \end{aligned} \tag{276}$$

(b) For $0 \le \alpha < 1$ and $0 < \lambda' \le \lambda$,

$$\begin{aligned} \|\phi f\|_{\alpha,\lambda} &\le |f|_{X^{+\lambda}} \|\phi\|_{\alpha,\lambda} + |\phi| \|f\|_{\alpha,\lambda;X^{+\lambda}} \quad (\mathrm{supp}\,\phi \subset X), \\ \|f\|_{\alpha,\lambda;X} &\le \lambda \sup_X |\nabla f|. \end{aligned} \tag{277}$$

The second inequality holds when X is the torus, or a cube of radius λ'' with $2\lambda'' + \lambda$ smaller than the torus side. Also

$$\begin{aligned} \|f\|_{\alpha,\lambda'} &\le \|f\|_{\alpha,\lambda}, \\ \|f\|_{\alpha,\lambda} &\le (\lambda/\lambda')^\alpha \max\{\|f\|_{\alpha,\lambda'}, 2|f|\}. \end{aligned} \tag{278}$$

(c) For $\mathfrak{M} \ge L^5$, $h = h_\square$ and $J \in \mathcal{A}^{(1)}$, transfer to the local model gives

$$h\nabla^* J = \nabla^*(hJ) + \mathsf{m}_h(J), \qquad \mathsf{m}_h(J) = \sum_\nu (\nabla_\nu h)^{-\nu} J_\nu^{-\nu}. \tag{279}$$

On the right, hJ and $\mathsf{m}_h(J)$ are moved by $\iota_\square$ without an additional multiplier. The divergence is the adjoint in $\mathcal{A}_\square$. For a boundary cube the term $\mathsf{m}_h(J)$ is also restricted by $\chi_{\mathcal{A}}$.

■

Proof. Expanding $T_\nu - I$ gives (275). For the commutator, at a site or initial bond point x,

$$(\phi\Delta f - \Delta(\phi f))(x) = \xi^{-2} \sum_{x' \sim x} (\phi(x') - \phi(x)) f(x').$$

Substituting $f(x') = f(x') - f(x) + f(x)$ gives the first form in (276). Applying the second rule of (275) to the divergence in its second form gives the same neighbor sum. Antisymmetry follows because both ϕ and Δ are symmetric.

For the product seminorm, expand $\phi(x)f(x) - \phi(x')f(x')$ into the difference of ϕ times $f(x')$ and $\phi(x)$ times the difference of f. If either point belongs to X, both belong to $X^{+\lambda}$; otherwise the product vanishes. For the gradient bound, a coordinate path of length $|x - x'|_1$ stays in the cube or torus and bounds the difference by $|x - x'|_1 \sup_X |\nabla f|$. The stated period condition ensures that least-distance lifts in the cube have this coordinate displacement. Multiplying by $(\lambda/|x - x'|_1)^\alpha$ proves the bound. Splitting pairs at distance λ' proves (278).

For (c), apply the second identity in (275) in each direction. All terms have initial points within ξ of $S_\square$, and their bonds have endpoints within 2ξ. The support margin $2\xi \le 5r_n/6$ places these

bonds in the faithful local chart. For an interior cube they lie inside Ω_0, so $\iota_\square$ commutes with the required shifts and the divergence is unrestricted. For a boundary cube the bonds lie in $\widetilde{\square}^3$; the global and local restrictions agree there. This gives the compression specified in (c). □

21.7. **Comparison of the global and local gauge terms.** Write $\mathcal{P} = \partial_0 \mathsf{P} \partial_0^*$ and, on an interior local torus, $\mathcal{P}_\square = \partial \mathsf{P}_\square \partial^*$. These operators need not agree on forms supported in the common region. Their difference can nevertheless be expressed through commutators supported where an intermediate cutoff changes. The distance from that set to the input and output gives the following estimate.

Lemma 21.9. There are $c_{\mathrm{F}} > 0$, C_{F} and thresholds depending only on L such that, if $\mathfrak{M}$ and $R_1\mathfrak{M}$ exceed these thresholds, then every interior cube $\square \in \mathfrak{D}_n$ and every $J \in \mathcal{A}$ supported in $S_\square^{(1/3)}$ satisfy

$$\sup_{b \text{ localized in } S_\square^{(1/3)}} |((\mathcal{P} - \mathcal{P}_\square)J)(b)| \le C_{\mathrm{F}} \xi_n^{-2} e^{-c_{\mathrm{F}}\mathfrak{M}} |J|. \tag{280}$$

Here the local expression is transferred by $\iota_\square$. One may take $c_{\mathrm{F}} = \delta_{\mathrm{pr}}/(288L)$. ■

Proof. We use $\mathsf{T} = \mathsf{G}' Q'^*$ and $\mathsf{C} = (\mathsf{T}^*\mathsf{T})^{-1}$, and attach a subscript $\square$ to the corresponding local operators. For a local cell centre w, write B_w for the bare sites of its cell $\Delta(w)$. Let $\theta = \theta_\square$, and let $\Theta : \mathcal{S}_\square \to \mathcal{S}$ be transfer by θ. Its adjoint is the opposite transfer. The cells meeting $\operatorname{supp}\theta \subset S_\square^{(2/3)}$ are common to the two settings. Thus multiplication by $\theta(y)$ at their centres defines $\bar\Theta : \mathcal{Y}_\square \to \mathcal{Y}$; it is zero on all other cells. The opposite transfer is $\bar\Theta^*$, because corresponding cells have the same weight ξ_y^3.

Set

$$K_\theta = \Theta \Delta'_{a,\square} - \Delta'_a \Theta, \qquad Z = \Theta Q'^*_\square - Q'^* \bar\Theta, \qquad Y_\square = Z - K_\theta \mathsf{T}_\square,$$
$$K_\theta^\sharp = \Theta^* \Delta'_a - \Delta'_{a,\square} \Theta^*, \quad Z^\sharp = \Theta^* Q'^* - Q'^*_\square \bar\Theta^*, \quad Y = Z^\sharp - K_\theta^\sharp \mathsf{T}.$$

The maps $Y_\square : \mathcal{Y}_\square \to \mathcal{S}$ and $Y : \mathcal{Y} \to \mathcal{S}_\square$ enter the two resolvent identities

$$\Theta \mathsf{T}_\square = \mathsf{T} \bar\Theta + \mathsf{G}' Y_\square, \qquad \Theta^* \mathsf{T} = \mathsf{T}_\square \bar\Theta^* + \mathsf{G}'_\square Y. \tag{281}$$

For example, multiply $\Delta'_a \Theta = \Theta \Delta'_{a,\square} - K_\theta$ on the left by G' and on the right by $\mathsf{T}_\square$, and use $\Delta'_{a,\square} \mathsf{T}_\square = Q'^*_\square$. Writing $\mathsf{V} = \mathsf{T}^*\mathsf{T} = \mathsf{C}^{-1}$ and $\mathsf{V}_\square = \mathsf{C}_\square^{-1}$, we obtain

$$\mathsf{V}\bar\Theta - \bar\Theta \mathsf{V}_\square = Y^* \mathsf{G}'_\square \mathsf{T}_\square - \mathsf{T}^* \mathsf{G}' Y_\square,$$
$$\mathsf{C}\bar\Theta - \bar\Theta \mathsf{C}_\square = \mathsf{C}(\mathsf{T}^* \mathsf{G}' Y_\square - Y^* \mathsf{G}'_\square \mathsf{T}_\square) \mathsf{C}_\square.$$

Substitute these formulas and $\mathsf{P} = \mathsf{T}\mathsf{C}\mathsf{T}^*$ into the difference of projections. The ordered products give

$$\begin{aligned} \mathsf{P}\Theta - \Theta \mathsf{P}_\square &= \mathsf{T}(\mathsf{C}\bar\Theta - \bar\Theta \mathsf{C}_\square)\mathsf{T}_\square^* + \mathsf{T}\mathsf{C} Y^* \mathsf{G}'_\square - \mathsf{G}' Y_\square \mathsf{C}_\square \mathsf{T}_\square^* \\ &= \mathsf{T}\mathsf{C} Y^* \mathsf{G}'_\square \mathsf{R}_\square - \mathsf{R}\mathsf{G}' Y_\square \mathsf{C}_\square \mathsf{T}_\square^*. \end{aligned} \tag{282}$$

We next locate the commutators. At a bare site $u \in \Omega_0$, put $j = n(y(u))$ and identify the common cells by $\iota_\square$. The local identities (268) give

$$\begin{aligned} (K_\theta f)(u) &= (\mathsf{L}(\theta) f)(u) + a_j \xi_j^{-2} L^{-3j} \sum_{u' \in B_{y(u)}} (\theta(u) - \theta(u')) f(u'), \\ (Z\omega)(u) &= (\theta(u) - \theta(y(u)))\omega(y(u)). \end{aligned} \tag{283}$$

The same formulas hold for the sharp operators, with the domains reversed. Let $\mathcal{T}$ be the set of centres $y(u)$ for which θ is not constant on $B_{y(u)} \cup \{u \pm \xi e_\nu\}_\nu$, and put $T_\theta = \bigcup_{y \in \mathcal{T}} B_y$. All four operators vanish outside T_θ. Since $\theta = 1$ on $S_\square^{(1/2)}$ and vanishes outside $S_\square^{(2/3)}$, the cutoff margins

and $\mathfrak{M} \geq L^5$ imply that these cells are contained in $S_\square^{(5/6)}$ and disjoint from $S_\square^{(5/12)}$. They have levels $n, n+1$. Their number, and the total number of local cells, are at most $m = (8\mathfrak{M})^3$.

The first identity in (276), the cutoff derivative bounds, and $a_j \leq 1$ imply

$$|K_\theta f(u)| \leq C\mathfrak{M}^{-1}(\xi_n^{-1} \sup|\nabla f| + \xi_n^{-2} \sup|f|), \qquad |Z\omega(u)| \leq C\mathfrak{M}^{-1}|\omega(y(u))|. \tag{284}$$

The suprema are over the cell of u and its bare neighbours. Indeed the averaging term in (283) is bounded by $3c_\flat r_n^{-1}\xi_j^{-1} \sup|f|$, and its analogue for Z by $3c_\flat r_n^{-1}\xi_j|\omega(y(u))|$. These estimates apply to the sharp operators as well.

Assume $|J| \leq 1$. For a local cell centre w, let $D(w)$ be its least normalized local distance from a cell meeting the ξ-neighbourhood of $S_\square^{(1/3)}$. Decompose the input into these cells and apply the scalar and Gram estimates of (270). Using $\mathsf{T}_\square^* = Q'_\square \mathsf{G}'_\square$, the sup-norm contraction of $Q'_\square$, and $Q'^*_\square \omega = \sum_w \omega(w)\chi_w$, gives

$$\begin{aligned} |\mathsf{T}_\square^* \partial^* J(w)| &\leq Cm\xi_n e^{-\delta_{\mathrm{pr}} D(w)}, \\ \omega = \mathsf{C}_\square \mathsf{T}_\square^* \partial^* J: \quad |\omega(w)| &\leq Cm^2 \xi_n^{-3} e^{-\delta_{\mathrm{pr}} D(w)}, \\ f = \mathsf{T}_\square \omega: \quad \sup_{B_w}(|f| + \xi_n|\nabla f|) &\leq Cm^3 \xi_n^{-1} e^{-\delta_{\mathrm{pr}} D(w)}, \\ g = \mathsf{G}'_\square \mathsf{R}_\square \partial^* J: \quad \sup_{B_w}|g| &\leq Cm^4 \xi_n e^{-\delta_{\mathrm{pr}} D(w)}. \end{aligned}$$

The last line follows from $g = \mathsf{G}'_\square \partial^* J - \mathsf{G}'_\square f$. Each convolution retains the displayed exponential because $D(w) \leq |w - w'|_\square + D(w')$; bounding the number of cells by m accounts for its additional factor.

A cell with centre in $\mathcal{T}$, or a neighbouring cell needed in (284), is separated from the input cells by at least $r_n/12 - 4\xi_{n+1} \geq r_n/24$ in sup distance. The local torus distances equal those of the common lifts there. Hence $D(w) \geq \mathfrak{M}/(144L)$ on these cells. With $E = e^{-\delta_{\mathrm{pr}}\mathfrak{M}/(144L)}$,

$$|Y_\square \omega| \leq Cm^3 \xi_n^{-3} E, \qquad \sup_{T_\theta}|g| \leq Cm^4 \xi_n E. \tag{285}$$

The global scalar and Gram estimates, with the common constants of (262), give

$$\mathsf{G}', \mathsf{T}, \mathsf{T}^* \in \mathfrak{K}_2(\delta_{\mathrm{u}}, B_{\mathrm{u}}), \qquad \nabla\mathsf{G}', \nabla\mathsf{T} \in \mathfrak{K}_1(\delta_{\mathrm{u}}, B_{\mathrm{u}}), \qquad \mathsf{C} \in \mathfrak{K}_{-4}(\delta_{\mathrm{u}}, B_{\mathrm{u}}).$$

Choose the additional threshold $R_1\mathfrak{M} \geq \mathsf{r}_S(\delta_{\mathrm{u}}/16)$. Then $(\delta_{\mathrm{u}}/16)R_1\mathfrak{M} \geq 2(3\log L + 1)$, so lemma 21.7 applies successively to $\nabla\mathsf{T} \cdot \mathsf{C}$, $\mathsf{T}^* \cdot \mathsf{G}'$, $\mathsf{C} \cdot (\mathsf{T}^*\mathsf{G}')$, and $\nabla\mathsf{T} \cdot (\mathsf{C}(\mathsf{T}^*\mathsf{G}'))$. The resulting orders are $-3, 4, 0, 1$, and their rates are respectively $\delta_{\mathrm{u}}/2, \delta_{\mathrm{u}}/2, \delta_{\mathrm{u}}/4, \delta_{\mathrm{u}}/8$. Thus

$$\nabla\mathsf{T}\mathsf{C} \in \mathfrak{K}_{-3}(\delta_{\mathrm{u}}/2, C), \qquad \nabla\mathsf{R}\mathsf{G}' = \nabla\mathsf{G}' - \nabla\mathsf{T}(\mathsf{C}(\mathsf{T}^*\mathsf{G}')) \in \mathfrak{K}_1(\delta_{\mathrm{u}}/8, C). \tag{286}$$

For $Y[u; y_1] = (Y\mathbf{1}_{\{y_1\}})(u)$, the sharp versions of (283)–(284) now show that

$$|Y[u; y_1]| \leq C\mathfrak{M}^{-1} e^{-\delta_{\mathrm{u}} d_{\mathbf{B}}(y(u), y_1)} \quad (u \in T_\theta), \qquad Y[u; y_1] = 0 \quad (u \notin T_\theta).$$

Here the neighbouring cells remain in $S_\square^{(5/6)}$, have sides at most $L\xi_n$, and have distance at most $9L$ from $y(u)$ by (267). The adjoint uses the two different volume weights:

$$(Y^* g)(y_1) = \xi_{y_1}^{-3} \sum_u \xi^3 Y[u; y_1] g(u).$$

Counting the $(\xi_{y'}/\xi)^3$ sites in a cell and applying (231) with exponent 3 gives

$$|Y^* g(y_1)| \leq C\mathfrak{M}^{-1} \sum_{y' \in \mathcal{T}} e^{-3\delta_{\mathrm{u}} d_{\mathbf{B}}(y', y_1)/4} \sup_{T_\theta}|g|.$$

At the output bonds in (280), and at the endpoints of the input bonds, $\theta = 1$. Consequently $\Theta\partial^* J = \partial_0^* J$ and $\partial\Theta f = \partial f$ at those output bonds. Equation (282) therefore yields

$$(\mathcal{P} - \mathcal{P}_\square)J = \partial\mathsf{T}\mathsf{C}Y^* g - \partial\mathsf{R}\mathsf{G}' Y_\square \omega$$

there. Apply (286), (285), and the site sum (233). The second term has size at most $Cm^4\xi_n^{-2}E$; for the first, summing first over y_1 and then over the at most m centres in $\mathcal{T}$ gives $Cm^5\xi_n^{-2}E$. With $c_{\mathrm{F}} = \delta_{\mathrm{pr}}/(288L)$, $m^5E = (8\mathfrak{M})^{15}e^{-2c_{\mathrm{F}}\mathfrak{M}} \le Ce^{-c_{\mathrm{F}}\mathfrak{M}}$, proving (280). □

21.8. **The cutoff remainder.** For $\square \in \mathfrak{D}_n$, write $h = h_\square$ and $\zeta = \zeta_\square$, with transfers understood as in the local construction. Define maps $\mathcal{A}_\square \to \mathcal{A}$ by

$$\mathsf{K}_\square = h\Delta_{a,\square} - \Delta_a(\Omega_0)h, \qquad \mathsf{K}_\square^{\mathrm{loc}} = \begin{cases} \mathsf{L}(h) + \mathsf{L}_Q(h), & \square \text{ interior}, \\ \chi_{\mathcal{A}}\mathsf{L}'(h), & \square \text{ boundary}, \end{cases}$$

where $\mathsf{L}_Q(h) = hQ_\square^* a Q_\square - Q_\square^* a Q_\square h$ and $\mathsf{L}'(h) = h\partial^*\partial - \partial^*\partial h$. For an interior cube put

$$\begin{aligned} K_{\square,\square} &= \mathsf{K}_\square^{\mathrm{loc}} + \zeta(\mathcal{P} - \mathcal{P}_\square)h + \zeta(\mathcal{P}_\square h - h\mathcal{P}_\square), \\ K_{\square',\square} &= h_{\square'}^2(1-\zeta)\mathcal{P}h \quad (\square' \neq \square). \end{aligned} \tag{287}$$

For a boundary cube set $K_{\square,\square} = \mathsf{K}_\square^{\mathrm{loc}}$ and $K_{\square',\square} = 0$ for $\square' \neq \square$. Thus the error in patching the local inverses consists of a local commutator, a change of gauge projection, a commutator with the local gauge projection, and the part of the global projection outside ζ.

Proposition 21.10. There are $\mathfrak{m}_{\mathrm{H}}, \mathfrak{r}_{\mathrm{H}}$, $\kappa_{\mathrm{H}} > 0$ and $C_{\mathrm{H}} \ge 1$, depending only on L, such that the following holds when $\mathfrak{M} \ge \mathfrak{m}_{\mathrm{H}}$ and $R_1\mathfrak{M} \ge \mathfrak{r}_{\mathrm{H}}$. The operators

$$\mathsf{G}_0 = \sum_\square h_\square \mathsf{G}_\square h_\square, \qquad \mathcal{R} = \sum_{\square',\square} K_{\square',\square}\mathsf{G}_\square h_\square$$

satisfy $\mathsf{G}_0^* = \mathsf{G}_0$ and $\Delta_a(\Omega_0)\mathsf{G}_0 = I - \mathcal{R}$. For $y, y' \in \Lambda_*$ put $e_{\mathrm{H}} = e^{-\kappa_{\mathrm{H}} d_{\mathbf{B}}(y,y')}$. Then:

(H1) For $p = 2, \infty$,

$$\sum_{\square',\square} N_p(K_{\square',\square}\mathsf{G}_\square h_\square; y, y') \le C_{\mathrm{H}}\mathfrak{M}^{-1}e_{\mathrm{H}}.$$

(H2) The adjoint blocks satisfy

$$\sum_{\square',\square} N_\infty(h_\square \mathsf{G}_\square K_{\square',\square}^*; y, y') \le C_{\mathrm{H}}\mathfrak{M}^{-1}e_{\mathrm{H}}.$$

The same bound holds for $\sum_\square N_\infty(h_\square\mathsf{G}_\square\mathsf{K}_\square^*; y, y')$.

(H3) If $0 \le \alpha < 1$, ψ is a function of bare sites supported in $\widetilde{\Delta}(y)$, and $A \in \mathcal{A}$, then

$$\begin{aligned} &\sum_{\square',\square} \|\psi h_\square \mathsf{G}_\square K_{\square',\square}^* A\|_{\alpha,\xi_y} \\ &\qquad \le C_{\mathrm{H}}(\alpha)\mathfrak{M}^{-1}(\|\psi\|_{\alpha,\xi_y} + |\psi|)\sum_{y''} e^{-\kappa_{\mathrm{H}} d_{\mathbf{B}}(y,y'')}|\chi_{y''}A|. \end{aligned}$$

In particular this bounds $\|\psi\mathcal{R}^*A\|_{\alpha,\xi_y}$.

(H4) If $0 < \varepsilon < 1$ and $J \in \mathcal{A}^{(1)}$ is supported in $\widetilde{\Delta}(y')$, then

$$\sum_{\square',\square} |\chi_y K_{\square',\square}\mathsf{G}_\square h_\square \nabla^* J| \le C'_{\mathrm{H}}(\varepsilon)\mathfrak{M}^{-1}\xi_{y'}^{-1}e_{\mathrm{H}}(\|J\|_{\varepsilon,\xi_{y'}} + |J|),$$

$$\sum_{\square',\square} \|\chi_y K_{\square',\square}\mathsf{G}_\square h_\square \nabla^* J\| \le C_{\mathrm{H}}\mathfrak{M}^{-1}\xi_{y'}^{-1}e_{\mathrm{H}}\|J\|.$$

The constants with an exponent as argument depend only on L and that exponent. One may choose

$$\kappa_{\mathrm{H}} = \min\left\{\frac{\delta_{\mathrm{loc}}}{2L}, \frac{\delta_{\mathrm{pr}}}{2L}, \frac{\delta_{\mathrm{u}}}{4}, \frac{c_{\mathrm{F}}}{12L}\right\}. \tag{288}$$

■

Proof. Local cells and transfers. For an interior cube, let w range over its local coarse centres. For a boundary cube let w range over all bare sites of the ambient torus, including those outside Ω_0, with $\xi_w = \xi_n = \xi$. Write B_w for the corresponding bare cell and χ_w for localization to it. In either case $\xi_w \in \{\xi_n, \xi_{n+1}\}$ and $f = \sum_w \chi_w f$ on the local spaces; localization preserves those spaces. Cells of the global sequence meeting $S_\square^{(5/6)}$ lie inside $\square$ and are local cells with the same multiplications. For an interior cube this is the common-cell assertion proved with (268); for a boundary cube the set is disjoint from Ω_1, so only bare cells occur.

For $s \in S_\square^{(5/6)}$, the sup ball of radius $4\xi_{n+1}$ about s lies in the common lifted interior: its distance from the boundary is positive because $4\xi_{n+1} < \mathfrak{M}\xi_n/6$. Two points in this ball differ by at most $8\xi_{n+1}$ in each coordinate, no more than half the side of either torus. The map $\iota_\square$ therefore preserves their ℓ^1 and sup distances. In particular it preserves the scaled seminorms at scales at most ξ_{n+1} of functions supported in $S_\square^{(5/6)}$, supports in $\mathsf{q}(o,\lambda)$ for o in that set and $\lambda \le 4\xi_{n+1}$, and the $2\xi_{n+1}$-neighbourhoods of cells meeting that set.

We also use the following uniform local lattice sum. The centres w are distinct points of the lattice of spacing ξ_n. Choose a lift $\widetilde{o}$ of o and a point $o_0 \in \xi_n\mathbb{Z}^3$ within sup distance $\xi_n/2$ of it. For each w choose a closest lift $\widetilde{w}$. The vectors $v = (\widetilde{w} - o_0)/\xi_n$ are distinct and $|o - w|_\square \ge |v|_1 - 3$. Consequently, for decreasing $F \ge 0$,

$$\sum_w F(|o-w|_\square) \le \sum_{v\in\mathbb{Z}^3} F(\max\{|v|_1 - 3, 0\}), \qquad \sum_w e^{-\mu|o-w|_\square} \le e^{3\mu}\coth^3(\mu/2). \tag{289}$$

The second bound follows by multiplying the three one-dimensional geometric sums.

The identity and the local commutator. For an interior cube, the support of hv is separated from Ω_0^{c}. The local identities (268) and the Hodge identity give

$$\Delta_a(\Omega_0)(hv) = \Delta(hv) - \mathcal{P}(hv) + Q_\square^* a Q_\square(hv), \qquad h\Delta_{a,\square}v = h\Delta v - h\mathcal{P}_\square v + hQ_\square^* a Q_\square v.$$

Indeed $\partial\partial^*(hv) = \partial_0(\mathsf{P}+\mathsf{R})\partial_0^*(hv)$; the term with R cancels the gauge term of Δ_a. Since $\zeta h = h$, subtraction yields

$$\mathsf{K}_\square = \mathsf{K}_\square^{\mathrm{loc}} + (1-\zeta)\mathcal{P}h + \zeta(\mathcal{P} - \mathcal{P}_\square)h + \zeta(\mathcal{P}_\square h - h\mathcal{P}_\square). \tag{290}$$

The bonds occurring in $\Delta(hv)$ have both endpoints in $S_\square^{(5/6)} \subset \Omega_0$, so the displayed local commutators take values in $\mathcal{A}$.

For a boundary cube $n = 0$. The support of hv and the plaquettes meeting it lie within 2ξ of $S_\square$, at distance at least $2\mathfrak{M}\xi - 2\xi$ from Ω_1. They consequently meet no averaging bond of positive level: the level-one blocks lie within $2\xi_1$ of Ω_1, and the blocks at level $j \ge 2$ lie in $\Omega_{j-1} \subset \Omega_1$ by separation. On the remaining bonds Q is the identity, so $Q^* a Q(hv) = \xi^{-2}hv$. The support assertion of proposition 21.4(ii) also gives $\partial_0\mathsf{R}\partial_0^*(hv) = 0$: the divergence of hv vanishes on Ω_1 and its bare nearest neighbours, which contain the range of R. The mass terms therefore cancel between the global and local operators, leaving $\mathsf{K}_\square = \chi_{\mathcal{A}}\mathsf{L}'(h)$. Here the restrictions agree on every plaquette meeting $\operatorname{supp}(hv)$, by the boundary margin in lemma 21.5.

The quadratic partition and $\zeta = 1$ on $S_\square$ now give $\mathsf{K}_\square = \sum_{\square'} K_{\square',\square}$. Thus

$$\begin{aligned}\Delta_a(\Omega_0)\mathsf{G}_0 &= \sum_\square (h_\square \Delta_{a,\square} - \mathsf{K}_\square)\mathsf{G}_\square h_\square \\ &= \sum_\square h_\square^2 - \mathcal{R} = I - \mathcal{R}.\end{aligned}$$

The local inverses are symmetric and opposite transfers by $h_\square$ are adjoint, which proves $\mathsf{G}_0^* = \mathsf{G}_0$.

We will need the local commutator in two forms. Let χ_X^+ indicate the $2\xi_{n+1}$-neighbourhood of X. For $p = 2, \infty$,

$$\begin{aligned} \|\chi_X \mathsf{K}_\square^{\mathrm{loc}} v\|_p &\le C\mathfrak{M}^{-1}(\xi_n^{-1}\|\chi_X^+ \nabla v\|_p + \xi_n^{-2}\|\chi_X^+ v\|_p), \\ (\mathsf{K}_\square^{\mathrm{loc}})^* A &= \nabla^* \mathsf{w}_\square(A) + \mathsf{v}_\square(A), \\ \xi_n\|\chi_X \mathsf{w}_\square(A)\|_p + \xi_n^2\|\chi_X \mathsf{v}_\square(A)\|_p &\le C\mathfrak{M}^{-1}\|\chi_X^+ A\|_p. \end{aligned} \tag{291}$$

All four outputs in this display are supported in the $2\xi_{n+1}$-neighbourhood of $S_\square$, contained in $S_\square^{(1/6)}$. To verify the estimates, write $(Q_\square v)(c) = \sum_b w_c(b)v(b)$. Expanding the averaging and curl commutators gives

$$(\mathsf{L}_Q(h)v)(b) = \xi^{-3} \sum_{j, c\in\Gamma_j} \xi_j w_c(b) \sum_{b'} w_c(b')(h(b_-) - h(b'_-))v(b'), \tag{292}$$

$$(\mathsf{L}'(h)v)(b) = \xi^{-2} \sum_{\mathsf{f}\ni b} \sum_{b'\in\partial\mathsf{f}} \pm(h(b_-) - h(b'_-))v(b'). \tag{293}$$

Only $j = n, n+1$ occur in the first line. The initial point of b belongs to a unique block of that level; a relevant coarse bond has this block centre as one endpoint and the direction of b, so there are at most two choices. The straight-path average (222) has nonnegative weights, total weight one, and $w_c(b) \le L^{-3j}$. Consequently $\xi^{-3}\xi_j w_c(b) \le \xi_j^{-2}$ and $\sum_{b'} w_c(b') = 1$. The two fine bonds in a nonzero summand are at sup distance at most $2\xi_{n+1}$. Their ℓ^1 distance is at most $4\xi_{n+1}$; the Lipschitz bound on h consequently makes the absolute row sum at most $C\mathfrak{M}^{-1}\xi_n^{-2}$. In the second line four plaquettes meet b, with three nontrivial terms each; their initial points are at ℓ^1 distance at most 2ξ. Since $n = 0$ there, its row sum has the same bound. Both matrices are antisymmetric in the equal fine weights, so their column sums obey the same estimate. The Schur bound in lemma 21.7 proves their localized $p = 2, \infty$ estimates.

For $\mathsf{L}(h)$, the first form of (276) gives the first line of (291), since $r_n^{-1} = 6L\mathfrak{M}^{-1}\xi_n^{-1}$. Its second form supplies the required divergence: for interior cubes,

$$\mathsf{w}_\square(A)_\nu = (\nabla_\nu h)(A + A^{+\nu}), \qquad \mathsf{v}_\square(A) = -(\Delta h)A - \mathsf{L}_Q(h)A.$$

For boundary cubes take $\mathsf{w}_\square(A) = 0$ and $\mathsf{v}_\square(A) = -\chi_{\mathcal{A}}\mathsf{L}'(h)A$. The same derivative and row-sum estimates prove the last line of (291). The shifts have length ξ; the averaging matrices have range $2\xi_{n+1}$, and $2\xi_{n+1} \le r_n/6$. This proves the support assertion and also $(\mathsf{K}_\square^{\mathrm{loc}})^* = -\mathsf{K}_\square^{\mathrm{loc}}$ under the local identification.

The three gauge terms. For interior local cells w, w' and $p = 2, \infty$, we claim

$$\|\chi_w(\mathcal{P}_\square h - h\mathcal{P}_\square)\chi_{w'}\|_{p\to p} \le C\mathfrak{M}^{-1}\xi_n^{-2}(1 + |w - w'|_\square)e^{-\delta_{\mathrm{pr}}|w-w'|_\square}. \tag{294}$$

The counting kernel in (270) has the factor $(\xi/\xi_{w'})^3\xi_n^{-2}$. There are $3(\xi_{w'}/\xi)^3$ fine bonds in the input cell, so its row sum is at most $3B_{\mathrm{pr}}\xi_n^{-2}e^{-\delta_{\mathrm{pr}}|w-w'|_\square}$. For points in the two cells, $|x - x'|_1 \le \xi_n(|w - w'|_\square + 3L)$. Multiplication by $h(x') - h(x)$ and $\mathrm{Lip}(h) \le c_{\mathrm{b}}/r_n$ proves (294) for $p = \infty$. The commutator is antisymmetric; its opposite block has the same bound. Schur's estimate gives $p = 2$.

By lemma 21.9, the two mutually adjoint transfers

$$\zeta(\mathcal{P} - \mathcal{P}_\square)h, \qquad h(\mathcal{P} - \mathcal{P}_\square)\zeta$$

have norm at most $C_{\mathrm{F}}\xi_n^{-2}e^{-c_{\mathrm{F}}\mathfrak{M}}$ for $p = \infty$. Their input and output are supported in $S_\square^{(1/3)}$, where initial points are localization points. Applying Schur's estimate to this pair proves the same bound for $p = 2$.

For the global gauge operator, the fourth kernel in (248), the at most $6(\xi_{y''}/\xi)^3$ input bonds in a global cell, and symmetry give

$$\begin{aligned} N_\infty(\mathcal{P}; y, y'') &\le 6B_{\mathrm{u}}\xi_y^{-2}e^{-\delta_{\mathrm{u}}d_{\mathbf{B}}(y,y'')}, \\ N_2(\mathcal{P}; y, y'') &\le 6B_{\mathrm{u}}\xi_y^{-1}\xi_{y''}^{-1}e^{-\delta_{\mathrm{u}}d_{\mathbf{B}}(y,y'')}. \end{aligned}$$

If $(\delta_{\mathrm{u}}/4)R_1\mathfrak{M} \geq 2\log L$, the scale-ratio estimate gives, in either order,

$$N_p(\mathcal{P}; y, y''),\ N_p(\mathcal{P}; y'', y) \leq 6B_{\mathrm{u}}L^2\xi_{y''}^{-2}e^{-3\delta_{\mathrm{u}}d_{\mathbf{B}}(y,y'')/4}. \tag{295}$$

We combine these estimates in a form which can be applied to every local inverse. Suppose $0 < \delta_\natural \leq 1$, $y' \in \Lambda_* \cap S_\square^{(1/6)}$, and $u \in \mathcal{A}_\square$ satisfies

$$\xi_n^{-2}\|\chi_{B_w}^+ u\|_p + \xi_n^{-1}\|\chi_{B_w}^+ \nabla u\|_p \leq Ue^{-\delta_\natural|w-y'|_\square} \tag{296}$$

for every local cell w. Set

$$\kappa_\natural = \min\left\{\frac{\delta_\natural}{2L}, \frac{\delta_{\mathrm{pr}}}{2L}, \frac{\delta_{\mathrm{u}}}{4}, \frac{c_{\mathrm{F}}}{12L}\right\}. \tag{297}$$

Under the preceding thresholds and $R_1\mathfrak{M} \geq \mathsf{r}_S(\kappa_\natural/2)$, we have

$$\sum_{\square'} \|\chi_y K_{\square',\square} u\|_p \leq C(\delta_\natural)\mathfrak{M}^{-1}Ue^{-\kappa_\natural d_{\mathbf{B}}(y,y')}. \tag{298}$$

Here and below the constants depend only on L and the indicated exponent or rate.

The local term vanishes unless B_y meets $S_\square^{(1/6)}$. It is then a common local cell, so (291) and (296) give $C\mathfrak{M}^{-1}Ue^{-\delta_\natural|y-y'|_\square}$. By (269), $d_{\mathbf{B}}(y,y') \leq L|y-y'|_\square$. For the comparison term, decompose u into local cells. The local lattice sum and the preceding operator bound give

$$\|\chi_y\zeta(\mathcal{P} - \mathcal{P}_\square)hu\|_p \leq Ce^{-c_{\mathrm{F}}\mathfrak{M}}U.$$

If this is nonzero, y, y' lie in the interior of $\square$, where $d_{\mathbf{B}}(y,y') \leq 6L\mathfrak{M}$. Thus

$$e^{-c_{\mathrm{F}}\mathfrak{M}} \leq e^{-c_{\mathrm{F}}\mathfrak{M}/2}e^{-c_{\mathrm{F}}d_{\mathbf{B}}(y,y')/(12L)}.$$

For the local gauge commutator, put $\mu = \frac{1}{2}\min\{\delta_{\mathrm{pr}}, \delta_\natural\}$. Equations (294) and (296) give

$$\begin{aligned}
&\|\chi_y\zeta(\mathcal{P}_\square h - h\mathcal{P}_\square)u\|_p \\
&\quad\leq C\mathfrak{M}^{-1}U\sum_w (1+|y-w|_\square)e^{-\delta_{\mathrm{pr}}|y-w|_\square - \delta_\natural|w-y'|_\square} \\
&\quad\leq C\mathfrak{M}^{-1}Ue^{-(\mu/L)d_{\mathbf{B}}(y,y')}.
\end{aligned}$$

Indeed the exponent is at least $\mu|y-y'|_\square + \mu|y-w|_\square$, and $(1+s)e^{-\mu s} \leq (2/\mu)e^{-\mu s/2}$, whose local lattice sum is finite uniformly in the torus.

There are at most 16 cubes $\square'$ whose cutoffs meet a bond localized in B_y, so the remaining sum is at most $16\|\chi_y(1-\zeta)\mathcal{P}hu\|_p$. It vanishes unless $B_y \not\subset S_\square^{(1/6)}$. For an exiting bond this follows as well: its localization point differs from its initial point by ξ, whereas $S_\square^{(5/6)}$ is at distance greater than $\mathfrak{M}\xi$ from Ω_0^{c}. Decompose hu over the cells y'' meeting $S_\square$. They are common cells with $\xi_{y''} \geq \xi_n$, and the buffer estimate (266) gives $d_{\mathbf{B}}(y,y'') \geq \mathfrak{M}/(72L^2)$. Using (295) and (296), their sum is at most

$$CU\sum_{y''} e^{-3\delta_{\mathrm{u}}d_{\mathbf{B}}(y,y'')/4 - (\delta_\natural/L)d_{\mathbf{B}}(y'',y')}.$$

Reserve $\delta_{\mathrm{u}}/4$ of the first rate for the buffer. The convolution estimate (274), at rate $2\kappa_\natural \leq \min\{\delta_{\mathrm{u}}/2, \delta_\natural/L\}$, then gives

$$CUe^{-\delta_{\mathrm{u}}\mathfrak{M}/(288L^2)}c_S(\kappa_\natural)e^{-\kappa_\natural d_{\mathbf{B}}(y,y')}.$$

The exponentially small factors in the comparison and off-support terms are bounded by $C\mathfrak{M}^{-1}$. Every retained spatial rate is at least $\kappa_\natural$; this proves (298).

Application to local inverses. Take $\delta_\natural = \delta_{\mathrm{loc}}$, so that $\kappa_\natural = \kappa_{\mathrm{H}}$. Let $f = \chi_{y'}f$. There are at most 16 cubes with $h_\square f \neq 0$: their supports meet $\widetilde{\Delta}(y')$. For each, $y' \in S_\square^{(1/6)}$ and $\xi_{y'} \in \{\xi_n, \xi_{n+1}\}$. The $2\xi_{n+1}$-neighbourhood of a local cell B_w, including the initial points of its bonds, lies in $\mathfrak{q}(w, 3\xi_{n+1})$. Cover this by 27 cubes $\mathfrak{q}(o_i, \xi_{n+1})$, with $o_i = w + 2\xi_{n+1}k$, $k \in \{-1,0,1\}^3$. Their centres satisfy $|o_i - w|_\square \leq 6L$. Apply the value and gradient estimates of proposition 21.6(R1), or (R5) for $p = 2$,

to $u = \mathsf{G}_\square h_\square f$ on these cubes. Since the input is supported in $\mathsf{q}(y', \xi_{y'})$, we obtain (296) with $U = C\|f\|_p$. Equation (298) bounds the sum over $\square'$ by $C\mathfrak{M}^{-1}e_{\mathrm{H}}\|f\|_p$. To pass to the sum of block norms in (H1), observe that for each of the at most 16 input cubes there are at most 17 nonzero output blocks: the diagonal block, and at most 16 cubes whose supports meet $\widetilde{\Delta}(y)$. Each block norm is bounded by the preceding estimate; summing this fixed number proves (H1).

For (H4), the transferred product rule (279) gives

$$h\nabla^* J = \nabla^*(hJ) + \mathsf{m}_h(J), \qquad \mathsf{m}_h(J) = \sum_\nu (\nabla_\nu h)^{-\nu} J_\nu^{-\nu},$$

with the boundary compression prescribed there. The left side vanishes unless $S_\square$ meets $\mathsf{q}(y', 3\xi_{y'})$; again there are at most 16 such cubes, with $y' \in S_\square^{(1/6)}$ and $\xi_{y'} \in \{\xi_n, \xi_{n+1}\}$. Require $\mathfrak{M} \geq 6L^2 c_\flat$. By (277),

$$\|hJ\|_{\varepsilon,\xi_{y'}} \leq \|J\|_{\varepsilon,\xi_{y'}} + |J|, \qquad \|\mathsf{m}_h(J)\|_p \leq 3c_\flat r_n^{-1}\|J\|_p \leq \xi_n^{-1}\|J\|_p.$$

The form hJ is supported in $\mathsf{q}(y', \xi_{y'})$. The shifted correction is supported in $\mathsf{q}(y', 3\xi_{y'})$; partition its bonds among the 27 cubes of radius $\xi_{y'}$ covering that cube. Use the same 27-cube cover on the output. The divergence value and gradient estimates (R1), (R3) of proposition 21.6, applied to hJ, and its value and gradient estimates applied to the correction pieces, give (296) with

$$U = C(\varepsilon)\xi_n^{-1}(\|J\|_{\varepsilon,\xi_{y'}} + |J|) \quad (p = \infty).$$

For $p = 2$, the four corresponding entries of (R5) give $U = C\xi_n^{-1}\|J\|$. The displaced input and output centres cost at most e^{12L} in these estimates. Now (298), the finite cube count, and $\xi_n^{-1} \leq L\xi_{y'}^{-1}$ prove both assertions of (H4).

The adjoint remainder. For (H2) and (H3), fix an output cube $\square$ for which $h_\square$ meets $\widetilde{\Delta}(y)$. There are at most 16 choices, and for each $y \in S_\square^{(1/6)}$ and $\xi_y \in \{\xi_n, \xi_{n+1}\}$. For local forms $\mathsf{w} \in \mathcal{A}_\square^{(1)}$ and $\mathsf{s} \in \mathcal{A}_\square$, decompose into local cells and apply the value and divergence entries of (R1) and (R2) in proposition 21.6. The product rule gives $\|\psi h\|_{\alpha,\xi_y} \leq \|\psi\|_{\alpha,\xi_y} + |\psi|$ and $|\psi h| \leq |\psi|$. We obtain

$$\begin{aligned} |\chi_y h \mathsf{G}_\square(\nabla^*\mathsf{w} + \mathsf{s})| &\leq C\Sigma(\mathsf{w}, \mathsf{s}), \\ \|\psi h \mathsf{G}_\square(\nabla^*\mathsf{w} + \mathsf{s})\|_{\alpha,\xi_y} &\leq C(\alpha)(\|\psi\|_{\alpha,\xi_y} + |\psi|)\Sigma(\mathsf{w}, \mathsf{s}), \\ \Sigma(\mathsf{w}, \mathsf{s}) &= \sum_w e^{-\delta_{\mathrm{loc}}|y-w|_\square}\big(\xi_n|\chi_w \mathsf{w}| + \xi_n^2|\chi_w \mathsf{s}|\big). \end{aligned} \tag{299}$$

The support and seminorm transfers used here are those verified at the start of the proof.

Let $A = \chi_{y'} A$. Taking adjoints in (287), and using (291), gives

$$\begin{aligned} K_{\square,\square}^* A &= \nabla^* \mathsf{w}_\square(A) + \mathsf{v}_\square(A) + h(\mathcal{P} - \mathcal{P}_\square)\zeta A - (\mathcal{P}_\square h - h\mathcal{P}_\square)\zeta A, \\ K_{\square',\square}^* A &= h\mathcal{P}(1-\zeta)h_{\square'}^2 A \quad (\square' \neq \square). \end{aligned}$$

Only the first two local terms occur for a boundary cube. We estimate the contribution of each term to Σ. For the local terms, (291) gives $\xi_n|\mathsf{w}_\square(A)| + \xi_n^2|\mathsf{v}_\square(A)| \leq C\mathfrak{M}^{-1}|A|$. They are supported within $2\xi_{n+1}$ of both $B_{y'}$ and $S_\square$. Unless $B_{y'} \cap S_\square^{(1/3)} \neq \emptyset$ they vanish. Otherwise $B_{y'}$ is a common cell and the support meets at most $(7L)^3$ local cells, whose centres obey $|w - y'|_\square \leq 12L$. Their contribution is at most

$$C\mathfrak{M}^{-1}e^{-\delta_{\mathrm{loc}}|y-y'|_\square}|A| \leq C\mathfrak{M}^{-1}e^{-(\delta_{\mathrm{loc}}/L)d_{\mathbf{B}}(y,y')}|A|.$$

The comparison term also vanishes unless $B_{y'}$ meets $S_\square^{(1/3)}$. Its norm bound, multiplied by ξ_n^2 in Σ and summed with (289), contributes at most

$$Ce^{-c_{\mathrm{F}}\mathfrak{M}}|A| \leq Ce^{-c_{\mathrm{F}}\mathfrak{M}/2}e^{-c_{\mathrm{F}}d_{\mathbf{B}}(y,y')/(12L)}|A|.$$

For the local gauge commutator, put $\mu = \frac{1}{2}\min\{\delta_{\mathrm{pr}}, \delta_{\mathrm{loc}}\}$. The factor ξ_n^{-2} in (294) cancels the ξ_n^2 in Σ. Its contribution is bounded by

$$C\mathfrak{M}^{-1}|A|\sum_w e^{-\delta_{\mathrm{loc}}|y-w|_\square}(1+|w-y'|_\square)e^{-\delta_{\mathrm{pr}}|w-y'|_\square} \le C\mathfrak{M}^{-1}e^{-(\mu/L)d_{\mathbf{B}}(y,y')}|A|.$$

This is the same local convolution used in (298), with the polynomial factor on the input distance.

Finally put $A' = h_{\square'}^2 A$. Then $|A'| \le |A|$, $A' = \chi_{y'}A'$, and at most 16 cubes $\square'$ give a nonzero A'. The form $h\mathcal{P}(1-\zeta)A'$ is supported in common cells y'' meeting $S_\square$ and vanishes unless $B_{y'} \not\subset S_\square^{(1/6)}$. For these cells (266) gives $d_{\mathbf{B}}(y'',y') \ge \mathfrak{M}/(72L^2)$. Use the reverse-order bound in (295), with $\xi_{y''}^{-2} \le \xi_n^{-2}$. After cancellation against ξ_n^2 in Σ, the contribution is at most

$$\begin{aligned} &C|A|\sum_{y''} e^{-(\delta_{\mathrm{loc}}/L)d_{\mathbf{B}}(y,y'')-3\delta_{\mathrm{u}}d_{\mathbf{B}}(y'',y')/4} \\ &\qquad \le Ce^{-\delta_{\mathrm{u}}\mathfrak{M}/(288L^2)}c_S(\kappa_{\mathrm{H}})e^{-\kappa_{\mathrm{H}}d_{\mathbf{B}}(y,y')}|A|. \end{aligned}$$

Here we again reserve $\delta_{\mathrm{u}}/4$ for the buffer and apply (274) to the remaining two rates.

All four contributions are bounded by $C\mathfrak{M}^{-1}e_{\mathrm{H}}|A|$. Insert this bound into the first line of (299), take the supremum over $A = \chi_{y'}A$ with $|A| \le 1$, and sum the at most $16 \cdot 17$ pairs of cubes. This proves (H2). For (H3), write $A = \sum_{y''}\chi_{y''}A$ and apply the second line of (299) to each component. Sum first over those same finitely many cube pairs and then over y''. The additional assertions for $\mathsf{K}_\square^*$ and $\mathcal{R}^*$ follow by their defining sums and the triangle inequality.

All requirements on $\mathfrak{M}$ used above are the local model and gauge-comparison thresholds together with $\mathfrak{M} \ge 6L^2c_\flat$. Choose their maximum as $\mathfrak{m}_{\mathrm{H}}$. Choose $\mathfrak{r}_{\mathrm{H}}$ to dominate the scalar, Gram and gauge-comparison thresholds and $\mathsf{r}_S(\kappa_{\mathrm{H}}/2)$. The condition needed in (295) is already implied by $\mathsf{r}_S(\delta_{\mathrm{u}}/16)$. These are finitely many constants depending only on L. □

21.9. **Estimates for the approximate inverse.** The local bounds pass through the partition with constants independent of the number of cubes. The point is that only boundedly many cutoffs meet an output cell; derivatives of these cutoffs have the scale needed by the lower-order local estimates.

Lemma 21.11. Suppose $\mathfrak{M} \ge \max\{L^5, L\mathfrak{m}_{\mathrm{pr}}, 6L^2c_\flat\}$. Put $\kappa_{\mathrm{a}} = \delta_{\mathrm{loc}}/L$, let $\widetilde{\chi}_y$ indicate the bonds with localization point in $\mathsf{q}(y, 2\xi_y)$, and write

$$\Sigma_p(f) = \sum_z e^{-\kappa_{\mathrm{a}}d_{\mathbf{B}}(y,z)}\|\chi_z f\|_p, \qquad e_{\mathrm{a}} = e^{-\kappa_{\mathrm{a}}d_{\mathbf{B}}(y,y')}.$$

Let ζ be supported in $\widetilde{\Delta}(y)$, and let $0 \le \alpha < 1$, $\varepsilon > 0$, $\alpha + \varepsilon < 1$. With constants depending only on L and the displayed exponents, the following estimates hold for $f \in \mathcal{A}$:

(G1) $\xi_y^{-2}|\widetilde{\chi}_y\mathsf{G}_0 f|$, $\xi_y^{-1}|\widetilde{\chi}_y\nabla\mathsf{G}_0 f|$ and $|\widetilde{\chi}_y\Delta\mathsf{G}_0 f|$ are at most $C\Sigma_\infty(f)$.
(G2) $\xi_y^{-1}\|\zeta\nabla\mathsf{G}_0 f\|_{\alpha,\xi_y} \le C(\alpha)(\|\zeta\|_{\alpha,\xi_y} + |\zeta|)\Sigma_\infty(f)$.
(G3) $\xi_y^{-2}\|\zeta\mathsf{G}_0 f\|$, $\xi_y^{-1}\|\zeta\nabla\mathsf{G}_0 f\|$ and $\|\zeta\nabla\nabla\mathsf{G}_0 f\|$ are at most $C|\zeta|\Sigma_2(f)$.

For $J \in \mathcal{A}^{(1)}$ supported in $\widetilde{\Delta}(y')$, the differentiated-data estimates are

(G4)

$$\begin{aligned} \xi_y^{-1}|\widetilde{\chi}_y\mathsf{G}_0\nabla^* J| &\le Ce_{\mathrm{a}}|J|, \\ \xi_y^{-1}\|\zeta\mathsf{G}_0\nabla^* J\|_{\alpha,\xi_y} &\le C(\alpha)(\|\zeta\|_{\alpha,\xi_y} + |\zeta|)e_{\mathrm{a}}|J|. \end{aligned}$$

(G5)

$$\begin{aligned} |\widetilde{\chi}_y\nabla\mathsf{G}_0\nabla^* J| &\le C(\varepsilon)e_{\mathrm{a}}(\|J\|_{\varepsilon,\xi_{y'}} + |J|), \\ \|\zeta\nabla\mathsf{G}_0\nabla^* J\|_{\alpha,\xi_y} &\le C(\alpha,\varepsilon)(\|\zeta\|_{\alpha,\xi_y} + |\zeta|)e_{\mathrm{a}}(\|J\|_{\alpha+\varepsilon,\xi_{y'}} + |J|). \end{aligned}$$

(G6) $\xi_y^{-1}\|\zeta \mathsf{G}_0\nabla^* J\|$, $\|\zeta\nabla \mathsf{G}_0\nabla^* J\|$ and $\|\zeta \mathsf{G}_0\nabla^*\nabla^* J\|$ are at most $C|\zeta|e_{\mathrm{a}}\|J\|$. In the third estimate $J \in \mathcal{A}^{(2)}$ with the same support.

Each estimate remains valid if its left side is replaced by the sum over $\square$ of the corresponding expression with $h_\square \mathsf{G}_\square h_\square$ in place of G_0. ■

Proof. Support and scale. For $\square \in \mathfrak{D}_n$, put $h = h_\square$. A difference of order at most two moves an initial point by at most 2ξ; the localization point differs from the initial point by at most ξ. Thus an output in $\mathsf{q}(y, 2\xi_y)$ can occur only if $S_\square$ meets $\mathsf{q}(y, 6\xi_y)$. The local support geometry gives at most 16 such cubes, with

$$\xi_n \le \xi_y \le L\xi_n, \qquad y \in S_\square^{(1/6)}.$$

Likewise, $h\chi_z f$ requires an intersection with $\mathsf{q}(z, 2\xi_z)$, and a term containing $h\nabla^* J$ or $h\nabla^*\nabla^* J$ requires one with $\mathsf{q}(y', 4\xi_{y'})$. The corresponding input scales belong to $\{\xi_n, \xi_{n+1}\}$ and the centers to $S_\square^{(1/6)}$. The local metric comparison then gives

$$e^{-\delta_{\mathrm{loc}}|y-z|_\square} \le e^{-\kappa_{\mathrm{a}} d_{\mathbf{B}}(y,z)},$$

and the same inequality with y' in place of z. The faithful transfer of these supports also preserves the suprema and seminorms used below.

One product calculation. The cutoff bounds show that the functions

$$h, \qquad \xi_n \nabla_\nu h, \qquad \xi_n^2 \nabla_\nu \nabla_\rho h \tag{300}$$

and their shifts have suprema at most 1, under the stated threshold. For the first two, their seminorms at radii at most ξ_{n+1} are also at most 1: use (277) and $r_n^{-1} = 6L\mathfrak{M}^{-1}\xi_n^{-1}$.

Use (275) once or twice on the output and (279) on the input. For the second input divergence, the exact identity is

$$\begin{aligned} h\nabla_\nu^*\nabla_\rho^* J_{\nu\rho} = {} & \nabla_\nu^*\nabla_\rho^*(hJ_{\nu\rho}) \\ & + \nabla_\nu^*((\nabla_\rho h)^{-\rho} J_{\nu\rho}^{-\rho}) + \nabla_\rho^*((\nabla_\nu h)^{-\nu} J_{\nu\rho}^{-\nu}) \\ & + (\nabla_\rho\nabla_\nu h)^{-\nu-\rho} J_{\nu\rho}^{-\nu-\rho}. \end{aligned} \tag{301}$$

This follows by three applications of the second rule in (275); the boundary compression is that of (279).

Consequently every required expression $Oh\mathsf{G}_\square hO'g$ is a sum of a bounded number of terms

$$\xi_n^{-i}\phi\, O_1 \mathsf{G}_\square O_1'(\xi_n^{-i'}\phi' g^{\mathrm{sh}}).$$

Here $O \in \{I, \nabla, \Delta, \nabla\nabla\}$, $O' \in \{I, \nabla^*, \nabla^*\nabla^*\}$, $g = \chi_z f$ or J, and ϕ, ϕ' are shifts of the functions in (300). The differential orders of O_1, O_1' have decreased by i, i', respectively. A backward output difference can occur for $O = \Delta$; use $\nabla_\nu^* u = -(\nabla_\nu u)^{-\nu}$. The input shift is at most 2ξ.

If $O' = I$, the input remains $h\chi_z f$. Otherwise $\phi' J^{\mathrm{sh}}$ is supported in $\mathsf{q}(y', 4\xi_{y'})$. Split this support, bond by bond, among the 64 cubes of radius $\xi_{y'}$ centered at $y' + \xi_{y'}k$, $k \in \{\pm 1, \pm 3\}^3$. The one term for which an input Hölder seminorm is needed has $i' = 0$: its input is hJ, left unsplit, and

$$\|hJ\|_{\beta,\xi_{y'}} \le \|J\|_{\beta,\xi_{y'}} + |J| \qquad (0 < \beta < 1).$$

The shifted inputs of lower order need only their suprema or L^2 norms.

Application of the local estimates. Apply proposition 21.6 with output radius ξ_y and input radius ξ_z or $\xi_{y'}$; these are the two admissible scales just identified. Where a cutoff is required, replace ζ by $\zeta\phi$. For the first two types of coefficients in (300), the product rule gives

$$\|\zeta\phi\|_{\alpha,\xi_y} + |\zeta\phi| \le 2(\|\zeta\|_{\alpha,\xi_y} + |\zeta|).$$

Second derivatives of h occur only in (G1) and (G3), where no seminorm of that coefficient is needed.

The value and derivative bounds (R1), (R3) give (G1) and the supremum parts of (G4), (G5); (R2) gives (G2) and the seminorm in (G4). For (G5), apply (R3), (R4) to the term $i = i' = 0$, and (R1), (R2) to the lower-order terms. The L^2 estimates (R5) give (G3), (G6). For suprema involving backward differences, cover $\mathsf{q}(y, 4\xi_y)$ by the analogous 64 output cubes. Each displaced center has local distance at most $9L$ from the original one, so these covers cost only a constant depending on L.

The powers of ξ_n from the coefficients and local estimates combine to the displayed output power of ξ_y, up to a factor at most L^2. Convert the exponentials by the metric comparison above, sum over the at most 16 contributing cubes, and for (G1)–(G3) also over the input cells z. This proves the stronger assertions with the sum of local expressions on the left. The estimates for G_0 follow by the triangle inequality. □

21.10. **Proof of the global inverse estimates.**

Proof of proposition 21.1. Return to the physical setting of the proposition, in which $\xi_y \le 1$. Fix the constants in the scalar, local-model, gauge-comparison and remainder estimates before choosing $\mathfrak{M}$. Decrease the remainder rate, if necessary, so that $0 < \kappa_{\mathrm{H}} \le \min\{1, \kappa_{\mathrm{a}}/2\}$, and put

$$\kappa = \kappa_{\mathrm{H}}/2, \qquad \delta = \kappa/8 = \kappa_{\mathrm{H}}/16. \tag{302}$$

Choose $M_0 = M_0(L)$ large enough to meet every preceding lower bound on $\mathfrak{M}$, and so that LM_0 meets every preceding lower bound on $R_1\mathfrak{M}$. Require also

$$M_0 \ge \max\{2, 2C_{\mathrm{H}} c_S(\kappa_{\mathrm{H}}/2)\}, \qquad LM_0 \ge \mathsf{r}_S(\kappa_{\mathrm{H}}/8). \tag{303}$$

There are finitely many such conditions and all their constants depend only on L; in particular C_{H} is independent of $\mathfrak{M}$. Thus the choice is not circular. For every $R_1 \ge L$ and $\mathfrak{M} \ge M_0$ all conditions hold. The last one also gives $\delta R_1 \mathfrak{M} \ge 3 \log L + 1 \ge \log L$. Below write $d = d_{\mathbf{B}}$.

Convergence and absolute walk bounds. Set

$$E_{\square',\square} = K_{\square',\square}\mathsf{G}_{\square}h_{\square}, \qquad n_p(y,y') = \sum_{\square',\square} N_p(E_{\square',\square}; y, y'),$$

and define $n_p^*(y,y')$ by replacing each E by E^*. The transfers are adjoint and the local inverses symmetric. Hence $n_2^*(y,y') = n_2(y',y)$, and (H1), (H2) of proposition 21.10 give

$$n_p(y,y'),\ n_p^*(y,y') \le C_{\mathrm{H}}\mathfrak{M}^{-1} e^{-\kappa_{\mathrm{H}} d(y,y')} \qquad (p = 2, \infty).$$

By (233) and (303), their weighted row sums at rate κ are at most $1/2$. Write $n_p^{(i)}$ and $n_p^{*(i)}$ for their matrix powers, with identity matrix at $i = 0$. Submultiplicativity gives

$$n_p^{(i)}(y,y'),\ n_p^{*(i)}(y,y') \le 2^{-i} e^{-\kappa d(y,y')}. \tag{304}$$

Since $\mathcal{R} = \sum E_{\square',\square}$, $\|\mathcal{R}\|_{p,\kappa}$ and $\|\mathcal{R}^*\|_{p,\kappa}$ are at most $1/2$. The block calculus proves convergence of both Neumann series. The identity $\Delta_a(\Omega_0)\mathsf{G}_0 = I - \mathcal{R}$ then gives a right inverse on the finite-dimensional space $\mathcal{A}$, and therefore

$$\mathsf{G} = \Delta_a(\Omega_0)^{-1} = \mathsf{G}_0(I - \mathcal{R})^{-1}.$$

The operator $\Delta_a(\Omega_0)$ is symmetric and nonnegative by its defining form. Invertibility makes it positive definite, and G is symmetric. Taking adjoints gives $\mathsf{G} = (I - \mathcal{R}^*)^{-1}\mathsf{G}_0$. Expanding each finite product gives exactly the walks in (229), and reversing the factors gives G_ω^*. Convergence in the finite-dimensional block norm implies operator-norm convergence.

To carry estimates through this expansion, fix an output cell y and a seminorm p such that

$$\sum_{\square} \mathsf{p}(h_{\square}\mathsf{G}_{\square}h_{\square} f) \le c \sum_{z} e^{-\kappa_{\mathrm{a}} d(y,z)} \|\chi_z f\|_p.$$

Applying the block multiplication rule to the i remainder factors, then (304) and (274), yields

$$\begin{aligned}\sum_{|\omega|=i} \mathsf{p}(\mathsf{G}_\omega f) &\le 2^{-i} c \sum_{z,y'} e^{-\kappa d(y,z)-\kappa d(z,y')} \|\chi_{y'} f\|_p \\ &\le 2^{-i} c\, c_S(\kappa/2) \sum_{y'} e^{-\kappa d(y,y')/2} \|\chi_{y'} f\|_p.\end{aligned} \tag{305}$$

All sums over walks at fixed i are finite. Applying this inequality to the normalized seminorms in (G1)–(G3) of lemma 21.11 proves the value, gradient and Laplacian bounds for undifferentiated input, the three corresponding L^2 bounds, and

$$\|\zeta \nabla \mathsf{G} J\|_{\alpha,\xi_y} \le C(\alpha)\xi_y (\|\zeta\|_{\alpha,\xi_y} + |\zeta|) e^{-\kappa d(y,y')/2} |J|. \tag{306}$$

The same estimates hold for sums of walk seminorms with a factor 2^{-i}, before summing over i. For the versions with χ_y, use the indicator ζ_y of $\widetilde{\Delta}(y)$, acting at initial points: $\|\chi_y F\| \le \|\zeta_y F\|$.

Differentiated data without an output derivative. Let J be supported in $\widetilde{\Delta}(y')$, and take

$$(\mathsf{J}, q, p) = (\nabla^* J, 1, \infty), \quad (\nabla^* J, 1, 2), \quad (\nabla^* \nabla^* J, 0, 2).$$

The last case uses $J \in \mathcal{A}^{(2)}$; the others use $\mathcal{A}^{(1)}$. By (G4), (G6), the sum of local p-norms in cell z is at most $C\xi_z^q e^{-\kappa_{\mathrm{a}} d(z,y')} \|J\|_p$. Use the adjoint expansion and (304), then transfer ξ_z^q to ξ_y^q with loss $e^{\kappa d(y,z)/4}$. The site sum gives

$$\begin{aligned}\sum_{|\omega|=i} \|\chi_y \mathsf{G}_\omega^* \mathsf{J}\|_p &\le 2^{-i} C L \xi_y^q \sum_z e^{-3\kappa d(y,z)/4 - \kappa d(z,y')} \|J\|_p \\ &\le 2^{-i} C L c_S(\kappa/4) \xi_y^q e^{-\kappa d(y,y')/2} \|J\|_p.\end{aligned} \tag{307}$$

This proves the remaining undifferentiated-output supremum and L^2 estimates. For ζ supported in $\Delta(y)$, $\|\zeta F\| \le |\zeta| \|\chi_y F\|$: every retained bond whose initial point is in that cell is localized there.

For the Hölder seminorm use (G4) at $i = 0$. For $i \ge 1$, separate the leftmost adjoint remainder block from G_ω^*, apply (H3), and then apply (307) to the remaining walk of length $i - 1$. Since $\mathfrak{M}^{-1} \le 1/2$, this bounds the sum by

$$\begin{aligned}&2^{-i} C(\alpha)(\|\zeta\|_{\alpha,\xi_y} + |\zeta|) \sum_z e^{-\kappa_{\mathrm{H}} d(y,z)} \xi_z e^{-\kappa d(z,y')/2} |J| \\ &\le 2^{-i} C(\alpha) \xi_y (\|\zeta\|_{\alpha,\xi_y} + |\zeta|) e^{-\kappa d(y,y')/4} |J|.\end{aligned}$$

Here the scale transfer costs $e^{\kappa d(y,z)/4}$, and (274) at rate $\kappa/2$ sums the remainder. Summing in i gives

$$\|\zeta \mathsf{G} \nabla^* J\|_{\alpha,\xi_y} \le C(\alpha)\xi_y (\|\zeta\|_{\alpha,\xi_y} + |\zeta|) e^{-\kappa d(y,y')/4} |J|. \tag{308}$$

The mixed derivative. For $0 < \beta < 1$ put $[J]_\infty = \|J\|_{\beta,\xi_{y'}} + |J|$ and $[J]_2 = \|J\|$. Use in (305) one of the three normalized gradient seminorms

$$\xi_y^{-1} |\widetilde{\chi}_y \nabla F|, \qquad \frac{\|\zeta \nabla F\|_{\alpha,\xi_y}}{\xi_y (\|\zeta\|_{\alpha,\xi_y} + |\zeta|)}, \qquad \frac{\|\zeta \nabla F\|}{\xi_y |\zeta|},$$

with $(p, c) = (\infty, C)$, $(\infty, C(\alpha))$, $(2, C)$, respectively; the zero multiplier is immediate. For $i \ge 1$, separate the rightmost forward remainder block acting on $\nabla^* J$. Apply (305) to the remaining $i - 1$ factors and (H4) to that last block. Convolution at rate $\kappa/2$ gives

$$\sum_{|\omega|=i} \mathsf{p}(\mathsf{G}_\omega \nabla^* J) \le 2^{-i} c C(\beta) \xi_{y'}^{-1} e^{-\kappa d(y,y')/4} [J]_p \le 2^{-i} c C(\beta) \xi_y^{-1} e^{-\kappa d(y,y')/8} [J]_p.$$

For $p = 2$ the constant is independent of β. The last scale transfer uses $\delta R_1 \mathfrak{M} \ge \log L$. For $i = 0$ use (G5), (G6). Take $\beta = \varepsilon$ for the supremum bound, $\beta = \alpha + \varepsilon$ for the Hölder bound, and sum in

i. After restoring the normalizing factors this gives the mixed supremum and L^2 bounds, together with

$$\|\zeta\nabla\mathsf{G}\nabla^*J\|_{\alpha,\xi_y} \le C(\alpha,\varepsilon)(\|\zeta\|_{\alpha,\xi_y}+|\zeta|)e^{-\delta d(y,y')}(\|J\|_{\alpha+\varepsilon,\xi_{y'}}+|J|). \tag{309}$$

The unscaled seminorms and coercivity. Since $\xi_y \le 1$, (278) gives

$$\|F\|_\alpha \le \xi_y^{-\alpha}(\|F\|_{\alpha,\xi_y}+2|F|).$$

For $F=\zeta\nabla\mathsf{G}J$ or $\zeta\nabla\mathsf{G}\nabla^*J$, the supremum is controlled by the corresponding estimate on $\mathsf{q}(y,2\xi_y)$ already proved. For $F=\zeta\mathsf{G}\nabla^*J$, a bond contributing to this product is localized in a cell centered at some $y^\star$ satisfying (267):

$$d(y,y^\star)\le 9L, \qquad L^{-1}\xi_y \le \xi_{y^\star} \le L\xi_y.$$

There are at most $(2L(L+2))^3$ such cells: their disjoint interiors have side at least ξ_y/L and lie in $\mathsf{q}(y,(L+2)\xi_y)$. Applying (307) to them bounds the supremum by $C|\zeta|\xi_y e^{-\kappa d(y,y')/2}|J|$. The same finite sum bounds the sums of walk suprema. Combining these bounds with (306), (308) and (309) proves the asserted ordinary Hölder estimates, with their factors $\xi_y^{1-\alpha}$ and $\xi_y^{-\alpha}$.

The L^2 estimate for G and $\xi_y\le 1$ give $N_2(\mathsf{G};y,y')\le Ce^{-\kappa d(y,y')/2}$. This symmetric bound has uniformly bounded row sums by (233). Thus lemma 21.7(g) gives a uniform bound on $\|\mathsf{G}\|_{2\to2}$; its reciprocal is a positive lower bound γ for $\Delta_a(\Omega_0)$.

Every term estimate above has the factor 2^{-i}, and every decay rate is at least δ. Choose the constants in the proposition to be twice the largest relevant term constants. This choice bounds both the sum over i and each sum over walks of length i. The forward and adjoint orientations are exactly those used in the three preceding parts of the proof. This proves all assertions of the proposition. □

22. Coarse covariance and the minimizing lift

The inverse in section 21 acts on the fine field. Its coarse covariance determines the minimizing lift of prescribed coarse averages. We estimate this covariance and its inverse. The additional ingredient is a right inverse of the averaging map with a uniform energy bound. It gives a spectral gap after normalizing by the local scale; exponential conjugation then gives decay of the inverse covariance.

Keep the real, three-dimensional, level-t setting of section 21, with $L\ge 13$ odd and $a=1$. Here $\mathfrak{b}_0$ is either the inner frame or the star frame. Write $\Gamma=\bigsqcup_{n=0}^t\Gamma_n$ for the rows of Q, put $\xi_c=\xi_n$ for $c\in\Gamma_n$, and give row fields the inner product

$$\langle B,B'\rangle=\sum_{c\in\Gamma}\xi_c^3B(c)B'(c).$$

The adjoint Q^* uses this product and the fine product of weight ξ^3. Let m multiply a row field by ξ_c. Thus the averaging term of (223) is $Q^*\mathsf{m}^{-2}Q$. Set

$$\mathsf{Z}=Q\mathsf{G}Q^*, \qquad \mathsf{H}=\mathsf{G}Q^*\mathsf{Z}^{-1}. \tag{310}$$

These are the operators of proposition 16.2(v) at $a=1$: the row sets and centered averages coincide with (149), and their inner products and Dirichlet projections agree. In particular, $\mathsf{H}B$ is the unique curl-energy minimizer subject to $QA=B$ and $\mathsf{R}\partial_0^*A=0$. With only $QA=B$, the minimizers are $\mathsf{H}B+\partial\ker Q'$.

For $c\in\Gamma_n$, let $y(c)=c_-$ if $c_-\in\Omega_n$, and $y(c)=c_+$ otherwise. Then $y(c)\in\Lambda_n$. Use $d_{\mathbf{B}}(y,c)=d_{\mathbf{B}}(y,y(c))$ and $d_{\mathbf{B}}(c,c')=d_{\mathbf{B}}(y(c),y(c'))$. For an operator K on row fields, distinguish its matrix in the orthonormal basis $e_c=\xi_c^{-3/2}\delta_c$ from its counting kernel:

$$[K]_{cc'}=\langle e_c,Ke_{c'}\rangle, \qquad K(c,c')=(K\delta_{c'})(c)=\left(\frac{\xi_{c'}}{\xi_c}\right)^{3/2}[K]_{cc'}. \tag{311}$$

Proposition 22.1. There are positive constants M_1, γ, θ, C, depending only on L, such that the following holds for every separated sequence in this setting with $\mathfrak{M} \ge M_1$ and every positive power $R_1 \ge L$ of L. The coarse covariance Z is symmetric and positive definite, and

$$\gamma \|\mathsf{m} B\|^2 \le \langle B, \mathsf{Z} B\rangle \le \|\mathsf{m} B\|^2 \qquad (B \in L^2(\Gamma)). \tag{312}$$

Its inverse satisfies

$$\begin{aligned} |[\mathsf{Z}^{-1}]_{cc'}| &\le 2\gamma^{-1} \xi_c^{-1} \xi_{c'}^{-1} e^{-\theta d_{\mathbf{B}}(c,c')}, \\ |\mathsf{Z}^{-1}(c,c')| &\le C \xi_c^{-2} e^{-\theta d_{\mathbf{B}}(c,c')/2}. \end{aligned} \tag{313}$$

The constants are independent of the torus, K, m, t, the regions, their number of layer changes, the frame, and $\mathfrak{M}, R_1$. ∎

22.1. **A quantitative lift of the coarse data.** We first construct a form for each row. Its averages at that level are exact; its only possible error is on the next level, at a row crossing the boundary of that level. This error can be removed in one step.

Proposition 22.2. For the separated sequence and bond frame fixed above, with $\mathfrak{M}, R_1 \ge L$, there are a numerical constant C_E and a linear map E from the coarse row space to $\mathcal{A}$ such that

$$Q\mathsf{E} = I, \qquad \|\nabla \mathsf{E} B\|^2 \le C_E \|\mathsf{m}^{-1} B\|^2 = C_E \sum_{n=0}^{t} \sum_{c \in \Gamma_n} \xi_n |B(c)|^2. \tag{314}$$

Consequently the covariance bound (312) holds with $\gamma = (C_E + 1)^{-1}$. ∎

Proof. The averaging weights. All one-dimensional weights and profiles below have domain $\mathbb{Z}$. Use Q_n for the average (222) on all positive level-n bonds, applied to the zero extension of A. For $c = (y, \mu)$ put $N_n = L^n$ and $h_n = (N_n - 1)/2$. For any odd integer N, write $h = (N-1)/2$ and define

$$\tau_N(s) = \#\{i \in \mathbb{Z} : 0 \le i < N,\ |s - i| \le h\}.$$

Counting the occurrences of each bare bond in (222) gives

$$\begin{aligned} (Q_n A)(c) &= \sum_x w_c(x) A_\mu(x), \\ w_c(y + \xi k) &= N_n^{-4} \tau_{N_n}(k_\mu), \quad -h_n \le k_\mu \le 3h_n + 1, \quad |k_\nu| \le h_n\ (\nu \ne \mu), \end{aligned} \tag{315}$$

with $w_c = 0$ elsewhere. These sites are the two endpoint blocks $B^n(c_-) \cup B^n(c_+)$ and are distinct on the torus: each coordinate range has length less than $2\xi_n$, whereas the period is at least $2\xi_n$. There are N_n^4 terms before collecting equal sites, and $0 \le \tau_{N_n} \le N_n$. Hence

$$0 \le w_c \le N_n^{-3}, \qquad \sum_x w_c(x) = 1.$$

Moreover, τ_N vanishes outside $[-h, 3h]$, so every contributing bare bond has both endpoints in those two blocks.

Let $B_{n+1}(z)$ denote the level-n sites in the centered block of side ξ_{n+1} about z. Splitting each such block into level-n blocks and writing a path index as $i = i_1 N_n + i_0$, with $0 \le i_1 < L$ and $0 \le i_0 < N_n$, gives

$$(Q_{n+1} A)(c') = \sum_{c''} w_{c'}^{(1)}(c'') (Q_n A)(c''), \qquad 0 \le w_{c'}^{(1)} \le L^{-3}, \qquad \sum_{c''} w_{c'}^{(1)}(c'') = 1. \tag{316}$$

Indeed, after this splitting the average is $L^{-4} \sum_{z \in B_{n+1}(c'_-)} \sum_{i_1=0}^{L-1} (Q_n A)_\mu(z + i_1 \xi_n e_\mu)$; the same count with $N = L$ proves the assertions about the weights. A nonzero weight requires that c'' be parallel to c' and that both endpoints of c'' lie in $B_{n+1}(c'_-) \cup B_{n+1}(c'_+)$.

Forms with one prescribed average. For odd $N \geq 3$, define

$$u(s) = \frac{6}{(N+1)(N+2)}\left(\frac{(N+1)^2}{4} - s^2\right) \qquad (|s| \leq h+1),$$
$$u(s) = 0 \qquad (|s| > h+1),$$
$$v^{\pm}(s) = u(s)(1+\beta_{\pm}s), \qquad \beta_+ = \frac{10}{N+3}, \qquad \beta_- = -\frac{10(N+1)}{(N-1)(N+3)}.$$

The profiles vanish at $s = \pm(h+1)$, so their supports are contained in $[-h, h]$. The sums of the second and fourth powers of the integers from $-h$ to h give

$$\sum_s u(s) = N, \qquad \sum_s su(s) = 0, \qquad \sum_s s^2 u(s) = \frac{N(N-1)(N+3)}{20}.$$

Consequently

$$\sum_s v^{\pm}(s) = N, \qquad \sum_s sv^+(s) = \frac{N(N-1)}{2}, \qquad \sum_s sv^-(s) = -\frac{N(N+1)}{2}.$$

For $|s| \leq h$ the triangular weights are $\tau_N(s) = s + (N+1)/2$ and $\tau_N(s+N) = (N-1)/2 - s$. Thus

$$\begin{pmatrix} \sum_s \tau_N(s)v^+(s) & \sum_s \tau_N(s)v^-(s) \\ \sum_s \tau_N(s+N)v^+(s) & \sum_s \tau_N(s+N)v^-(s) \end{pmatrix} = N^2 I_2. \tag{317}$$

The formulas also give, with a numerical constant C,

$$|u| + |v^{\pm}| \leq C, \qquad |u(s+1) - u(s)| + |v^{\pm}(s+1) - v^{\pm}(s)| \leq \frac{C}{N}.$$

Indeed, $|\beta_{\pm}| \leq C/N$ and $|\beta_{\pm}|(h+1) \leq C$; the difference of u is bounded by C/N, and $v(s+1) - v(s) = (u(s+1) - u(s))(1 + \beta(s+1)) + \beta u(s)$. The zero values at $\pm(h+1)$ make these bounds valid also across the support boundary.

At level $n \geq 1$, take $N = N_n$ and let $\varphi^{\pm}_{y,\mu}$ have only its μ-component nonzero, given by

$$(\varphi^{\pm}_{y,\mu})_\mu(y + \xi k) = v^{\pm}(k_\mu) \prod_{\nu \neq \mu} u(k_\nu) \quad (|k|_\infty \leq h_n),$$

and zero off $B^n(y)$. Only the two level-n bonds $c^+ = (y, \mu)$ and $c^- = (y - \xi_n e_\mu, \mu)$ of direction μ have averaging weights meeting this block. They are distinct even when the period is $2\xi_n$. By (315), (317) and $\sum u = N_n$,

$$Q_n \varphi^+_{y,\mu} = \delta_{c^+}, \qquad Q_n \varphi^-_{y,\mu} = \delta_{c^-} \tag{318}$$

on all positive level-n bonds.

For $c \in \Gamma_n$, $n \geq 1$, use the chosen endpoint $y(c)$: take $\varphi_c = \varphi^+_{c_-,\mu}$ if $y(c) = c_-$ and $\varphi_c = \varphi^-_{c_+,\mu}$ otherwise. For $c \in \Gamma_0$ take $\varphi_c = \delta_c$. These forms belong to $\mathcal{A}$: for $n \geq 1$ their nonzero bonds start in $B^n(y(c)) \subset \Omega_n \subset \Omega_1$, and separation places their other endpoints in Ω_0; for $n = 0$ this follows from $c \in \mathfrak{b}_0$.

There is a numerical constant C_φ such that

$$\|\nabla \varphi_c\|^2 \leq C_\varphi \xi_c. \tag{319}$$

To check this including the zero extension, let $n \geq 1$. For each direction ν, the derivative can be nonzero at only $(N_n + 1)N_n^2 \leq \frac{4}{3}N_n^3$ sites. The extra layer of sites is outside $B^n(y(c))$, since $N_n + 1 < 2N_n$ does not wrap around the torus. The product formula and the profile bounds give

$$|\nabla_\nu (\varphi_c)_\mu| \leq C\xi_n^{-1} \qquad (1 \leq \nu \leq 3).$$

Multiplying their squares by the site weight ξ^3 and the number of sites proves (319), uniformly in n. At level zero the energy of δ_c is 6ξ, which is covered by increasing the same constant.

Removing the error between adjacent levels. Call a row $c \in \Gamma_n$ crossing if exactly one endpoint lies in Ω_n. The block geometry of lemma 16.3 shows that the averages of φ_c, for $c \in \Gamma_n$, vanish at every other level except possibly $n+1$. For completeness, if $1 \le n' < n$, the averaging paths of a row in $\Gamma_{n'}$ have their initial points outside $\Omega_{n'+1}$, whereas the support of φ_c starts inside Ω_n. For $n' = 0 < n$, a row has no endpoint in Ω_1. If $n' \ge n+2$, its averaging paths lie inside $\Omega_{n'-1} \subset \Omega_{n+1}$, whereas φ_c has no initial point there. These assertions also cover $n = 0$ because c has no endpoint in Ω_1. At its own level the average is δ_c by (318), or by $Q_0 = I$.

For $c' \in \Gamma_{n+1}$, (316) gives

$$(Q\varphi_c)(c') = w^{(1)}_{c'}(c) =: \mathsf{N}(c', c). \tag{320}$$

If this entry is nonzero, both endpoints of c lie in the two endpoint blocks of c', and neither lies in Ω_{n+1}. Hence they lie in the block of the outer endpoint of c': c' is crossing. By lemma 16.3(ii), that block consists of level-n sites in $\Omega_n \setminus \Omega_{n+1}$, so c is not crossing.

Set $\mathsf{N}(c', c) = 0$ for all other pairs and put

$$\mathsf{E}_0 B = \sum_c B(c)\varphi_c.$$

Then $Q\mathsf{E}_0 = I + \mathsf{N}$. The matrix N uses only noncrossing input rows and has only crossing output rows, so $\mathsf{N}^2 = 0$. Therefore

$$\mathsf{E} = \mathsf{E}_0(I - \mathsf{N}), \qquad Q\mathsf{E} = I.$$

The row sums of N are at most one and its entries at most L^{-3}. Thus, for $c' \in \Gamma_{n+1}$,

$$\xi_{n+1}|(\mathsf{N}B)(c')|^2 \le \xi_{n+1} \sum_c \mathsf{N}(c', c)|B(c)|^2 \le L^{-2} \sum_{c:\, \mathsf{N}(c',c)\neq 0} \xi_n |B(c)|^2.$$

Each c occurs for at most two rows c': its two endpoints lie in one outer block, whose center can be an endpoint of only two bonds parallel to c. Hence

$$\|\mathsf{m}^{-1}\mathsf{N}B\|^2 \le 2L^{-2}\|\mathsf{m}^{-1}B\|^2.$$

Thus $I - \mathsf{N}$ is uniformly bounded in this weighted norm.

Put $\beta = (I - \mathsf{N})B$. At any bare site and in any one component, at most two of the forms φ_c are nonzero, since the blocks partition the bare sites of Ω_0; outside that region only a level-zero form can contribute. A first difference therefore has at most four nonzero summands. By Cauchy–Schwarz and (319),

$$\|\nabla \mathsf{E}B\|^2 \le 4 \sum_c |\beta(c)|^2 \|\nabla\varphi_c\|^2 \le 4C_\varphi \|\mathsf{m}^{-1}\beta\|^2 \le C_E \|\mathsf{m}^{-1}B\|^2$$

for a numerical constant C_E, proving (314).

The covariance gap. The quadratic form of Δ_a and the ambient identity (156) give

$$\|\mathsf{m}^{-1}QA\|^2 \le \langle A, \Delta_a A\rangle \le \|\nabla A\|^2 + \|\mathsf{m}^{-1}QA\|^2, \qquad A \in \mathcal{A}.$$

Indeed, the gauge term is $\|\mathsf{R}\partial_0^* A\|^2 \le \|\partial^* A\|^2$, and $\|\partial A\|^2 + \|\partial^* A\|^2 = \|\nabla A\|^2$. Completing the square about $\mathsf{G}Q^*B$ yields

$$\langle B, \mathsf{Z}B\rangle = \max_{A\in\mathcal{A}} \{2\langle QA, B\rangle - \langle A, \Delta_a A\rangle\}.$$

For $A = s\mathsf{E}\mathsf{m}^2 B$, the expression in braces is at least $[2s - (C_E+1)s^2]\|\mathsf{m}B\|^2$. Taking $s = \gamma = (C_E+1)^{-1}$ proves the lower bound in (312). For every A the same expression is at most

$$2\langle QA, B\rangle - \|\mathsf{m}^{-1}QA\|^2 = \|\mathsf{m}B\|^2 - \|\mathsf{m}^{-1}QA - \mathsf{m}B\|^2 \le \|\mathsf{m}B\|^2,$$

which proves the upper bound. □

22.2. **Decay of the coarse inverse.**

Proof of the decay bounds in proposition 22.1. Write $d = d_{\mathbf{B}}$ and let $\delta_3 > 0$ be the rate in proposition 21.1. The averaging weights in (315) localize each column of Q^* in a bounded number of cells. We first record how their positions and scales compare.

For $c \in \Gamma_n$, $n \geq 1$, the bare sites of $\Delta^n(c_-) \cup \Delta^n(c_+)$ are, by lemma 16.3(ii),(iii), partitioned into two level-n blocks if both endpoints belong to Ω_n, and otherwise into one level-n block and the L^3 level-$(n-1)$ blocks of the outer endpoint. Let $\mathcal{C}(c)$ be their centers. For $n = 0$ put $\mathcal{C}(c) = \{y(c)\}$. Thus

$$|\mathcal{C}(c)| \leq 1 + L^3, \qquad \xi_z \leq \xi_c, \qquad d(z, y(c)) \leq 4L \quad (z \in \mathcal{C}(c)).$$

For the distance bound, join $y(c)$ to the midpoint of c inside its level-n cell. This segment has weighted length $1/2$. If the other endpoint also lies in Ω_n, the second half has the same length. Otherwise, a center $z \in B(c_+)$ or $B(c_-)$ in the outer block can be joined to that midpoint by a segment of ℓ^1 length at most $2\xi_n$. Apart from the midpoint, this segment lies in the interior of the outer level-n cell, which belongs only to layer $n-1$. Its weighted length is therefore at most $2L$. The endpoints have zero length, so the same estimates apply with the closed-layer convention. At level zero the distance vanishes.

Every row with position $y(c) = z$ is one of the six positive bonds incident to z at its level. Hence lemma 21.2(d) gives

$$\sup_{y \in \Lambda_*} \sum_{c \in \Gamma} e^{-\kappa d(y,c)} \leq 6 c_S(\kappa) \quad \text{if } R_1 \mathfrak{M} \geq \mathsf{r}_S(\kappa). \tag{321}$$

In particular this holds with $y = y(c')$ for any row c'.

For $c \in \Gamma_n$ of direction μ, put $J_c = Q^* \delta_c$ and $A^c = \mathsf{G} J_c$. The adjoint is taken for the bare weight ξ^3 and the coarse weight ξ_c^3. Thus (315) gives

$$\langle A, J_c \rangle = \xi_c^3 \sum_x w_c(x) A_\mu(x), \qquad (J_c)_\mu(x) = L^{3n} w_c(x)$$

on the admitted bare bonds; all other components and all bonds outside the frame have value zero. Since $0 \leq w_c \leq L^{-3n}$, we have $0 \leq J_c \leq 1$. For $n \geq 1$, every contributing bond is inner to Ω_0 by lemma 16.3(iii), so its localization point is its initial point. Restricting J_c to the cells just described writes it as a sum of at most $1 + L^3$ fields, each of supremum at most one and supported in a cell $\Delta(z)$ with $z \in \mathcal{C}(c)$. For $n = 0$, $J_c = \delta_c$ and the same assertion holds with one cell.

Apply (224) and (225) to these fields. Since $d(y,z) \geq d(y,c) - 4L$, there are constants $C_1 = C_1(L)$ and $C_{1,\alpha} = C_{1,\alpha}(L,\alpha)$ such that

$$\begin{aligned} |A^c(b)| &\leq C_1 \xi_y^2 e^{-\delta_3 d(y,c)}, \\ |(\partial_\nu A^c_\mu)(x)| &\leq C_1 \xi_y e^{-\delta_3 d(y,c)}, \\ \|\zeta \nabla A^c\|_\alpha &\leq C_{1,\alpha} \xi_y^{1-\alpha} (\|\zeta\|_{\alpha,\xi_y} + |\zeta|) e^{-\delta_3 d(y,c)}. \end{aligned} \tag{322}$$

Here b is localized in $\Delta(y)$, x is a bare site of $\Delta(y)$, and ζ vanishes outside $\widetilde{\Delta}(y)$, as in the statement.

The weights w_c are nonnegative and sum to one. Consequently $(QA)(c)$ is bounded by the largest value of $|A|$ on the contributing bonds. Use (322), $\xi_z \leq \xi_c$ and $d(z,c') \geq d(c,c') - 4L$ in the identity $\mathsf{Z}(c,c') = (QA^{c'})(c)$. This gives a constant $C_{\mathsf{Z}} = C_{\mathsf{Z}}(L)$ with

$$|\mathsf{Z}(c,c')| \leq C_{\mathsf{Z}} \xi_c^2 e^{-\delta_3 d(c,c')}.$$

Normalize the covariance in the orthonormal basis of (311):

$$S = [\mathsf{m}^{-1} \mathsf{Z} \mathsf{m}^{-1}], \qquad S_{cc'} = \xi_c^{-1} \xi_{c'}^{-1} [\mathsf{Z}]_{cc'}.$$

The gap (312) says $\gamma I \leq S \leq I$. Moreover, (311) and the last bound give

$$|S_{cc'}| \leq C_{\mathsf{Z}} (\xi_c / \xi_{c'})^{5/2} e^{-\delta_3 d(c,c')}.$$

The matrix S is symmetric, so the same estimate holds with the reciprocal scale ratio. Taking the smaller of the two bounds yields

$$\gamma I \le S \le I, \qquad |S_{cc'}| \le C_{\mathsf{Z}} e^{-\delta_3 d(c,c')}. \tag{323}$$

We apply the conjugation calculation following (258) to the row distance d. The summation constant here is $B_{\mathrm{row}} = 6c_S(\delta_3/2)$. Choose

$$\theta = \min\left\{\frac{\delta_3}{4}, \frac{\gamma\delta_3}{4C_{\mathsf{Z}}B_{\mathrm{row}}}\right\} > 0. \tag{324}$$

Choose the threshold M_1 in the statement at least as large as M_0 from proposition 21.1, and so that, for every $\mathfrak{M} \ge M_1$ and $R_1 \ge L$,

$$R_1\mathfrak{M} \ge \mathsf{r}_S(\delta_3/4), \qquad \theta R_1 \mathfrak{M} \ge \log L. \tag{325}$$

This requires only a larger number depending on L: it suffices to impose these inequalities with $R_1\mathfrak{M}$ replaced by LM_1.

Fix $c_0 \in \Gamma$, set $\psi(c) = d(c, c_0)$, and put $S_\theta = e^{\theta\psi} S e^{-\theta\psi}$. The triangle inequality gives $|\psi(c) - \psi(c')| \le d(c, c')$. By the cited calculation, (323) and the actual row sum (321),

$$\|S_\theta - S\| \le \frac{4C_{\mathsf{Z}}\theta B_{\mathrm{row}}}{e\delta_3} \le \frac{\gamma}{e} < \frac{\gamma}{2}.$$

Thus $\|S_\theta^{-1}\| \le 2/\gamma$, and undoing the conjugation gives $|(S^{-1})_{cc'}| \le 2\gamma^{-1} e^{-\theta d(c,c')}$.

Undoing the normalization gives $[\mathsf{Z}^{-1}]_{cc'} = \xi_c^{-1}\xi_{c'}^{-1}(S^{-1})_{cc'}$, which is the first bound in (313). By (311), the counting kernel consequently satisfies

$$|\mathsf{Z}^{-1}(c, c')| \le 2\gamma^{-1}\xi_c^{-2}(\xi_{c'}/\xi_c)^{1/2} e^{-\theta d(c,c')}.$$

The scale-ratio estimate (231), with exponent $1/2$, and the second condition in (325) imply

$$(\xi_{c'}/\xi_c)^{1/2} e^{-(\theta/2)d(c,c')} \le L^{1/2}.$$

This proves the second bound in (313), with constant $2\gamma^{-1}L^{1/2}$. □

23. The constrained system at the trivial configuration

The scalar inverse, the fine-field inverse and the coarse covariance solve the linear constrained system at the trivial configuration. We prove a bound for this system which includes its gauge-fixing variables and multipliers. The forcing may contain a divergence; the estimate therefore uses the regularity of the inverses as well as their pointwise decay.

Keep the three-dimensional physical torus and level-t coordinates of section 21: $L \ge 13$ is odd, $K, m \ge 1$, $0 \le t \le K$, $\xi = L^{-t}$ and $\xi_n = L^n\xi$. We use the inner bond frame and take the whole torus as ambient region. Let $\mathfrak{m}$ be a power of L and $\mathfrak{g} \ge L^2$. Assume that $\Omega_0 \supset \cdots \supset \Omega_t$ are unions of the cubes of $\pi[\mathfrak{m}\xi_n]$ at their respective levels and

$$\mathrm{dist}_\infty(\Omega_n^c, \Omega_{n+1}) > \mathfrak{g}\xi_n \qquad (0 \le n < t).$$

Empty regions are allowed; $\Omega_n = \emptyset$ when $\mathfrak{m} > L^{m+K-n}$, and $\Omega_{t+1} = \emptyset$. The cells $\Delta(y)$, tops Λ_*, row set $\Gamma = \bigsqcup_n \Gamma_n$ and distance $d_{\mathbf{B}}$ are those of sections 21 and 22. In particular, the rows are positive and have no endpoint in the next region. Write $d = d_{\mathbf{B}}$.

Let $\mathcal{S}, \mathcal{A}, \mathcal{Y}, \mathcal{R}$ be the spaces of $\mathfrak{g}^c$-valued functions on the bare sites of Ω_0, its positive inner bare bonds, Λ_* and Γ, respectively. Here $\mathfrak{g}^c$ is the space of traceless complex 2×2 matrices, with matrix operator norm $|\cdot|$. Their pairings are the sums of $\frac{1}{2}\mathrm{tr}(XY)$ with weights $\xi^3, \xi^3, \xi_y^3, \xi_c^3$, respectively. They are complex bilinear. All transposes T^{T} below use these pairings.

Apply the real operators of the preceding sections to each Lie algebra coordinate, retaining their names $Q, Q', \mathsf{G}', \mathsf{C}, \mathsf{P}, \mathsf{R}, \mathsf{G}, \mathsf{Z}, \mathsf{H}$ and their weights $a = 1$. Set

$$D = \partial_0, \qquad \mathcal{L} = \chi_{\mathcal{A}}\partial^*\partial\chi_{\mathcal{A}}, \qquad \mathsf{L}_1 = \mathcal{L} + DD^{\mathsf{T}} + Q^{\mathsf{T}}\mathsf{m}^{-2}Q.$$

Here $\mathcal{L}$ is the ambient curl operator compressed to inner bonds, and m multiplies a row field by ξ_c. Scalar fields and bond fields are extended by zero; ∇_ν is $\xi^{-1}(T_\nu - I)$ on these extensions, with its output restricted to the corresponding inner index set. Thus $(D\lambda)(b) = \xi^{-1}(\lambda(b_+) - \lambda(b_-))$ and $D^{\mathsf{T}}A = \sum_\mu \nabla_\mu^{\mathsf{T}} A_\mu$. The scalar Dirichlet operator in Δ_a' includes all six neighbors, including the zero-extension terms at the boundary.

On $\mathcal{X} = \mathcal{A} \times \mathcal{S}^2 \times \mathcal{Y} \times \mathcal{S}^2 \times \mathcal{Y} \times \mathcal{R}$, write $\Psi = (A; v, w, \sigma; p, q, r; \omega)$ and define

$$\mathbb{K}_0^{[5]}\Psi = \begin{pmatrix} \mathsf{L}_1 A - D(w+p) - Q^{\mathsf{T}}\omega \\ \Delta_a' v + w - D^{\mathsf{T}}A \\ \Delta_a' w - Q'^{\mathsf{T}}\sigma \\ Q'v \\ \Delta_a' p + Q'^{\mathsf{T}} r \\ p + \Delta_a' q \\ -Q'q \\ QA \end{pmatrix}. \tag{326}$$

Let $\mathbb{K}_0^{[4]}$ be the coordinate compression obtained by removing ω and the last equation. The other two compressions are

$$\mathbb{K}_0^{[2]}(v,w,\sigma) = (\Delta_a' v + w,\ \Delta_a' w - Q'^{\mathsf{T}}\sigma,\ Q'v), \qquad \mathbb{K}_0^{[1]} w = \Delta_a' w.$$

Denote their spaces by $\mathcal{X}^{[i]}$, with $\mathcal{X}^{[5]} = \mathcal{X}$.

A bare site's position is the top of its cell; the position of an inner bond, or of a bond with an additional direction index, is that of its initial point. A top is its own position, and a row has the position $y(c)$ of section 22. For a function on any of these index sets put

$$|f|_y = \max_{\text{position}(i)=y} |f(i)|, \qquad |f|_{s;y} = \xi_y^{-s}|f|_y, \qquad \max\emptyset = 0.$$

The solution seminorm is

$$\begin{aligned} |\Psi|_y = \max\{&|A|_{-1;y}, |\nabla A|_{-2;y}, |v|_{0;y}, |Dv|_{-1;y}, |w|_{-2;y}, |Dw|_{-3;y}, \\ &|\sigma|_{-4;y}, |p|_{-2;y}, |q|_{0;y}, |Dq|_{-1;y}, |r|_{-4;y}, |\omega|_{-3;y}\}. \end{aligned} \tag{327}$$

For $i = 4, 5$ the forcing space is $\mathcal{F}^{[i]} = \mathcal{X}^{[i]} \times \mathcal{A}$. Writing its elements as $g = (f_1, \dots, f_8; X_5)$ when $i = 5$, set

$$\iota g = (f_1, f_2, f_3, f_4, f_5 + D^{\mathsf{T}}X_5, f_6, f_7, f_8)$$

and

$$\begin{aligned} |g|_y = \max\{&|f_1|_{-3;y}, |f_2|_{-2;y}, |f_3|_{-4;y}, |f_4|_{0;y}, |f_5|_{-4;y}, \\ &|X_5|_{-3;y}, |f_6|_{-2;y}, |f_7|_{0;y}, |f_8|_{-1;y}\}. \end{aligned} \tag{328}$$

For $i = 4$ omit ω, f_8 from these formulas. For $i = 2$ the unknown and forcing are (v, w, σ) and (f_2, f_3, f_4); for $i = 1$ they are w and f_3. In these two cases $\mathcal{F}^{[i]} = \mathcal{X}^{[i]}$, $\iota = I$, and the seminorms retain precisely the entries involving the displayed components.

Proposition 23.1. There are a positive power $\mathfrak{M}_\dagger$ of L and positive constants $\delta_{\mathrm{cs}}, B_{\mathrm{cs}}$, depending only on L, with the following property. For every sequence above with

$$\mathfrak{m} \geq \mathfrak{M}_\dagger, \qquad \mathfrak{g} \geq L\mathfrak{M}_\dagger,$$

all four operators $\mathbb{K}_0^{[i]}$, $i \in \{1, 2, 4, 5\}$, are invertible. Their inverses $\mathbb{E}_0^{[i]} = (\mathbb{K}_0^{[i]})^{-1}$ satisfy

$$|\mathbb{E}_0^{[i]}\iota g|_y \leq B_{\mathrm{cs}} \sup_{y' \in \Lambda_*} e^{-\delta_{\mathrm{cs}} d(y,y')} |g|_{y'} \qquad (g \in \mathcal{F}^{[i]},\ y \in \Lambda_*). \tag{329}$$

The constants are uniform in K, m, t, the lattice spacing, the torus and the sequence, including its number of layers. For $\Omega_0 = \emptyset$ all spaces are zero and the assertion uses the unique map on the zero

space. In particular, the result applies to the separated sequences of section 21 whenever $\mathfrak{M} \geq \mathfrak{M}_\dagger$, for every power $R_1 \geq L$, by taking $\mathfrak{m} = \mathfrak{M}$ and $\mathfrak{g} = R_1\mathfrak{M}$. ∎

The algebraic reduction in section 23.2 proves invertibility of each compression. The estimate is proved in section 23.4, with the constants chosen in section 23.3. Its additional regularity step concerns $p'' = \mathsf{RG}'D^{\mathsf{T}}X_5$: a local half-Hölder bound permits its gradient to enter the mixed estimate for G.

For the local estimates below use the exponents $\beta \in \{1/2, 3/4\}$. Let $\Delta^{3/2}(y) = \{|x-y|_\infty \leq 3\xi_y/4\}$. The base point of a site is the site itself and that of a bond, also with an extra direction index, is its midpoint. Put

$$\begin{aligned} [f]_{\beta,y} &= \sup_{\substack{i \neq i' \text{ like} \\ \mathrm{base}(i),\mathrm{base}(i') \in \Delta^{3/2}(y)}} \left(\frac{\xi_y}{|i-i'|_1}\right)^\beta |f(i') - f(i)|, \\ |f|^\beta_{s;y} &= \xi_y^{-s} \left(\sup_{\mathrm{base}(i) \in \Delta^{3/2}(y)} |f(i)| + [f]_{\beta,y} \right). \end{aligned} \tag{330}$$

Here like bond indices have the same direction and, when present, the same extra direction; $|i-i'|_1$ is the distance between their initial points, equivalently between their midpoints. Empty suprema are zero. Positions in (327)–(328) continue to use initial points.

23.1. **Identification at the trivial configuration.** The averages and the action Hessian at the identity give the operators in (326). We check their normalization and boundary conventions.

Choose a Hermitian basis t_1, t_2, t_3 of $\mathfrak{g}$ with $\frac{1}{2}\operatorname{tr}(t_j t_k) = \delta_{jk}$. The coordinate expression $f = \sum_j f^j t_j$ makes the pairing nondegenerate, by testing against $\sum_j \overline{f^j} t_j$, and gives the transpose identity

$$(K \otimes 1)^{\mathsf{T}} = K^* \otimes 1$$

for every real operator K used here. In particular, its real inverse extends coordinatewise to the inverse on these spaces. The finite-dimensional norm comparison needed below is

$$\left|\tfrac{1}{2}\operatorname{tr}(t_j X)\right| \leq |X|, \qquad |X| \leq \sum_{j=1}^{3} (|\operatorname{Re} X^j| + |\operatorname{Im} X^j|).$$

Indeed $|t_j| = 1$, since its eigenvalues are ± 1. Thus the real pointwise, difference and Hölder estimates apply to these fields, with at most a fixed factor from six real coordinates.

All contours and averages of the identity configuration are the identity. Hence the scalar transported block average is the arithmetic mean, and its iteration is the existing Q'. For the bond average, the endpoint factors in the straightening must also be differentiated. In the fine-lattice coordinates of lemma 14.7, put

$$\sigma_a(y) = L^{-3} \sum_{x \in B(y)} \bar{a}_{\mathsf{m}}(y,x), \qquad v_s(y) = \exp\left(L^{-3} \sum_{x \in B(y)} \log(e^{isa})_{\mathsf{m}}(y,x) \right).$$

For s near zero these logarithms are defined, and the contour derivative in lemma 14.7 gives $v_s(y) = 1 + is\sigma_a(y) + O(s^2)$. The one-step straightened average at the identity is

$$i^{-1} \log \left[v_s(c_-)^{-1} \mathcal{M}(e^{isa})(c) v_s(c_+) \right].$$

By (124), its derivative at $s = 0$ is

$$\mathbf{Q}^{\mathrm{lin}}_{\mathsf{m}} a(c) - \sigma_a(c_-) + \sigma_a(c_+) = L^{-3} \sum_{x \in B(c_-)} a([x,x']). \tag{331}$$

Here $x \mapsto x'$ bijects the two endpoint blocks, so both endpoint path means cancel. Iteration gives

$$L^{-3n} \sum_{x \in B^n(c_-)} a([x, x + \xi_n e_\mu]).$$

To see the induction, decompose $B^n(c_-)$ into the $B^{n-1}(z)$, and each length-ξ_n path into L successive length-ξ_{n-1} paths. Translation carries $B^{n-1}(z)$ onto $B^{n-1}(z + i\xi_{n-1}e_\mu)$, so the inner path sums concatenate before the block sums are taken. Applying this formula to $a = \xi A$ and dividing the output by ξ_n gives the factor L^{-4n} in (222). At level zero the map is the identity. All these paths are inner by lemma 16.3. Thus the straightened linearisation is precisely Q. Its positive rows are the existing Γ_n: removing inward bonds leaves exactly the bonds with an endpoint in Ω_n and none in Ω_{n+1}. The scalar weights start at $a_0 = a_1 = 1$ and obey the existing recursion, while the row weight is ξ_c^{-2}.

The scalar difference at the identity is

$$\Delta^\circ f(x) = \xi^{-2} \sum_{\nu=1}^{3} (2f(x) - f(x + \xi e_\nu) - f(x - \xi e_\nu)),$$

with zero extension. It is therefore $\Delta_{\mathcal{S}}$, and the inner gradient is $D = \partial_0$. At the identity, the exterior derivative (67) is the ambient curl ∂ applied coordinatewise. By proposition 9.2, with $c = 1$ on the whole torus,

$$\left. \frac{d^2}{ds^2} \right|_{s=0} A^\xi(1, e^{is\xi A}) = (\partial A, \partial A)_\xi.$$

Polarization identifies the ambient Hessian with $\partial^*\partial \otimes 1$. Restrict its bilinear form to inner bond fields extended by zero to obtain $\mathcal{L}$. The curl energy still sums over plaquettes crossing the boundary of Ω_0.

These identities identify $\Delta'_a, \mathsf{C}, \mathsf{P}, \mathsf{R}$ with their coordinatewise extensions from section 21.4, and give

$$\mathsf{L}_1 - D\mathsf{P}D^{\mathsf{T}} = \mathcal{L} + D\mathsf{R}D^{\mathsf{T}} + Q^{\mathsf{T}}\mathsf{m}^{-2}Q = \mathsf{G}^{-1}, \qquad Q\mathsf{G}Q^{\mathsf{T}} = \mathsf{Z}. \tag{332}$$

Under the bounds of proposition 23.1, the threshold choice in section 23.3 makes propositions 16.1 and 22.1 applicable. Their inverses exist, and the same formulas give $\mathsf{H} = \mathsf{G}Q^{\mathsf{T}}\mathsf{Z}^{-1}$. No positivity of the complex bilinear form is used.

23.2. **Elimination of the auxiliary variables.** We solve the two scalar constraint blocks before solving for A. Put

$$\mathsf{N} = \mathsf{G}'Q'^{\mathsf{T}}\mathsf{C}, \qquad \mathsf{P} = \mathsf{N}Q'\mathsf{G}', \qquad \mathsf{R} = I - \mathsf{P}.$$

For the three equations

$$\Delta'_a v + w = k_2, \qquad \Delta'_a w - Q'^{\mathsf{T}}\sigma = f_3, \qquad Q'v = f_4,$$

substitution of $w = \mathsf{G}'(f_3 + Q'^{\mathsf{T}}\sigma)$ and $v = \mathsf{G}'(k_2 - w)$ into the last equation gives

$$\mathsf{C}^{-1}\sigma = Q'\mathsf{G}'k_2 - Q'\mathsf{G}'^2 f_3 - f_4.$$

Consequently the unique solution is

$$\begin{aligned} \sigma &= \mathsf{C}(Q'\mathsf{G}'k_2 - Q'\mathsf{G}'^2 f_3 - f_4), \\ w &= \mathsf{P}k_2 + \mathsf{R}\mathsf{G}'f_3 - \mathsf{N}f_4, \qquad v = \mathsf{G}'(k_2 - w). \end{aligned} \tag{333}$$

This proves invertibility of $\mathbb{K}_0^{[2]}$ by taking $k_2 = f_2$; that of $\mathbb{K}_0^{[1]}$ follows from $\mathsf{G}' = (\Delta'_a)^{-1}$.

For the other scalar block, put $h_5 = f_5 + D^{\mathsf{T}}X_5$. Substitute $q = \mathsf{G}'(f_6 - p)$ and $p = \mathsf{G}'(h_5 - Q'^{\mathsf{T}}r)$ into $-Q'q = f_7$. This gives

$$\begin{aligned} r &= \mathsf{C}(Q'\mathsf{G}'^2 h_5 - Q'\mathsf{G}'f_6 - f_7), \\ p &= \mathsf{R}\mathsf{G}'h_5 + \mathsf{P}f_6 + \mathsf{N}f_7, \qquad q = \mathsf{G}'(f_6 - p). \end{aligned} \tag{334}$$

In particular this block is independent of A.

For the full system, apply (333) with $k_2 = f_2 + D^{\mathsf{T}}A$ and put

$$\varphi = \mathsf{P}f_2 + \mathsf{R}\mathsf{G}'f_3 - \mathsf{N}f_4, \qquad F_{\mathrm{eff}} = f_1 + D\varphi + Dp. \tag{335}$$

Then

$$w = \mathsf{P}D^{\mathsf{T}}A + \varphi, \qquad v = \mathsf{G}'(f_2 + D^{\mathsf{T}}A - w), \tag{336}$$

and the first equation, by (332), becomes $\mathsf{G}^{-1}A = F_{\mathrm{eff}} + Q^{\mathsf{T}}\omega$. For $i = 4$ the last term is absent, so $A = \mathsf{G}F_{\mathrm{eff}}$. For $i = 5$, imposing $QA = f_8$ gives

$$\omega = \mathsf{Z}^{-1}(f_8 - Q\mathsf{G}F_{\mathrm{eff}}), \qquad A = \mathsf{G}(F_{\mathrm{eff}} + Q^{\mathsf{T}}\omega). \tag{337}$$

Together with (333)–(336), these formulas determine every component uniquely and solve its equation. They prove invertibility of both remaining compressions.

23.3. **Local estimates for the constrained system.** We use the scalar, vector and coarse inverse estimates already proved, with a common decay rate. Write δ_G for the rate in proposition 21.1, and use $\delta_{0,\mathrm{lap}}$, δ_1 and θ from proposition 21.3, (247) and (313). Set

$$\delta_* = \min\{1, \delta_{0,\mathrm{lap}}/2, 2\delta_1, \delta_G, \theta/2\}, \qquad \delta_{\mathrm{cs}} = \delta_*/2. \tag{338}$$

Choose a power $\mathfrak{M}_\dagger$ of L so large that

$$\begin{aligned} \mathfrak{M}_\dagger &\geq \max\{L^2, L+10, M_0, M_1\}, \\ L\mathfrak{M}_\dagger &\geq \max\{L(L+9), 5\delta_{0,\mathrm{lap}}^{-1}\log L, \mathfrak{r}_4, 16\delta_*^{-1}\log L, \mathsf{r}_S(\delta_*/4)\}. \end{aligned} \tag{339}$$

Here M_0, M_1 are the thresholds in propositions 21.1 and 22.1. If $\mathfrak{m} \geq \mathfrak{M}_\dagger$ and $\mathfrak{g} \geq L\mathfrak{M}_\dagger$, refining the region cubes to side $\mathfrak{M}_\dagger\xi_n$ puts the same sequence in the preceding setting with $(\mathfrak{M}, R_1) = (\mathfrak{M}_\dagger, L)$. The layers, operators and metric are unchanged. All the cited estimates apply, and lemma 21.2 applies with this choice. The empty-region case has only zero spaces; below assume $\Omega_0 \neq \emptyset$. Constants denoted by C depend only on L and may increase. All top indices below belong to Λ_*.

For the finite component spaces and local seminorms above, write $T \in \mathfrak{O}(\delta_{\mathrm{cs}}, C)$ when

$$|Tf|_y \leq C \sup_z e^{-\delta_{\mathrm{cs}} d(y,z)} |f|_z. \tag{340}$$

Composition multiplies these constants without changing the rate: insert the two bounds and use $d(y,z) + d(z,z') \geq d(y,z')$. Finite sums add their constants. A bound by the maximum over $d(y,z) \leq R$ gives (340) with the additional factor $e^{\delta_{\mathrm{cs}}R}$.

Only one summation is needed to pass from the preceding localized estimates to these bounds. Suppose $f = \sum_z h_z$ and $N_y(Th_z) \leq C\xi_y^a e^{-\delta_* d(y,z)} k_z$, where N_y is a target seminorm. For $|s| \leq 4$, the scale comparison and (339) give $(\xi_z/\xi_y)^s \leq L^4 e^{\delta_* d(y,z)/4}$. Consequently (233) gives

$$\begin{aligned} \xi_y^{-s-a} N_y(Tf) &\leq CL^4 \sum_z e^{-3\delta_* d(y,z)/4} \xi_z^{-s} k_z \\ &\leq CL^4 c_S(\delta_*/4) \sup_z e^{-\delta_{\mathrm{cs}} d(y,z)} \xi_z^{-s} k_z. \end{aligned} \tag{341}$$

The witnesses c_S, r_S are those of (232). All losses below are absorbed in C; subsequent compositions do not repeat this summation or reduce the decay rate.

We abbreviate a bound of this kind between weights s, s' by $T : s \to s'$. From propositions 21.1 and 21.3, (247) and (313), (341) gives the bounds used below:

$$\begin{gathered} \mathsf{G}' : -4 \to -2, \quad -2 \to 0, \qquad \nabla\mathsf{G}' : -4 \to -3, \quad -2 \to -1, \\ \mathsf{G}'\nabla^{\mathsf{T}} : -3 \to -2, \quad -1 \to 0, \qquad \mathsf{C} : 0 \to -4, \qquad \mathsf{Z}^{-1} : -1 \to -3, \\ \mathsf{G} : -3 \to -1, \qquad \nabla\mathsf{G} : -3 \to -2, \qquad \mathsf{G}\nabla^{\mathsf{T}} : -2 \to -1. \end{gathered} \tag{342}$$

For C a cell contains just its one top index; for Z^{-1} at most six rows have a given position, by (321). These are counting-kernel bounds, with output factors ξ_y^{-4} and ξ_y^{-2}, respectively.

The scalar average and its weighted transpose act within one cell: Q' takes its arithmetic mean, and $(Q'^{\mathsf{T}}\nu)(x) = \nu(y(x))$. Thus both preserve every weight. The bounds for Q, Q^{T} follow from (315) and the double-block geometry in section 22.2. The averaging weights are nonnegative with sum one, and their weighted-adjoint coefficients are at most one. A fine bond belongs to double blocks of at most two consecutive levels and, at either level, to those of at most six incident rows. Their positions are within distance $C(L)$, and their scales differ by at most L. Hence Q, Q^{T} preserve each weight $|s| \le 4$ in the sense of (340). This also explains why no additional volume factor enters the coarse constraint.

Two local comparisons convert the Hölder estimates to the norms of this section. If a cell meets $\widetilde{\Delta}(y)$, take a bare site of that cell within $\xi/2$ of a point of intersection. It lies within $2\xi_y$ of y, so (267) gives

$$L^{-1}\xi_y \le \xi_z \le L\xi_y, \qquad d(y,z) \le 9L \quad \text{if } \Delta(z) \cap \widetilde{\Delta}(y) \ne \emptyset. \tag{343}$$

For like indices with base points in $\Delta^{3/2}(y)$, the coordinate path between their sites or initial points keeps the corresponding base points in that cube. Its successive differences telescope, using the departure or arrival index according to the sign of the step. If their total length is h, then $(\xi_y/h)^{\beta} h \le (9/2)\xi_y$ for both selected exponents. At nonzero levels these indices are inner by the local chart; at level zero there is no distinct site pair, and the only possible bond pair in one direction is $\langle y - \xi e_\mu, y\rangle$, $\langle y, y + \xi e_\mu\rangle$. Thus

$$[f]_{\beta,y} \le \tfrac{9}{2}\xi_y \max_{\Delta(z)\cap\Delta^{3/2}(y)\ne\emptyset} |\nabla f|_z. \tag{344}$$

At levels zero and one the simpler bound $[f]_{\beta,y} \le 2L \sup_{\Delta^{3/2}(y)} |f|$ follows from the minimum nonzero distance $\xi \ge \xi_y/L$.

Choose a fixed real smooth cutoff ζ_y^* supported in $|x-y|_\infty < 9\xi_y/10$, equal to one for $|x-y|_\infty \le 4\xi_y/5$, with $\|\zeta_y^*\|_{\beta,\xi_y} + |\zeta_y^*| \le C$ for both exponents. Such torus cutoffs are products of even one-dimensional cutoffs in the least coordinate distance; their support stays away from the half-period, so they are smooth. At a positive level, $3\xi_y/4 + \xi/2 \le 4\xi_y/5$, so the cutoff is one at the initial points of all indices with base points in $\Delta^{3/2}(y)$. Consequently

$$\sup_{\Delta^{3/2}(y)} |u| + [u]_{\beta,y} \le \xi_y^{\beta} \|\zeta_y^* u\|_\beta + 3 \sup_{\Delta^{3/2}(y)} |u|. \tag{345}$$

Pairs at distance at most one enter the cutoff seminorm; the remaining pairs are bounded by twice the supremum, since $\xi_y \le 1$. At level zero the left side is at most three times the supremum, by the minimum nonzero distance ξ_y.

For scalar divergence smoothing, let h_z be the input restricted to indices of position z, with zero extension to the bare lattice. The cutoff estimate (239) and its pointwise companion, applied at the nearby cells in (343), bound the enlarged local norm of $\mathsf{G}'\nabla^{\mathsf{T}} h_z$ by $C\xi_y e^{-\delta_* d(y,z)} |h_z|_z$. Use (345) at positive levels and the supremum bound at level zero. Summing by (341) gives

$$\mathsf{G}'\nabla^{\mathsf{T}} : (-3) \longrightarrow (-2, \beta), \tag{346}$$

where the pair denotes the enlarged Hölder norm. The same assertion holds for $\mathsf{G}'D^{\mathsf{T}}$: set $X_A(x,\mu) = A(\langle x, x + \xi e_\mu\rangle)$ on inner bonds and zero otherwise. Then $D^{\mathsf{T}}A = \nabla^{\mathsf{T}} X_A$ and $|X_A|_{s;y} \le |A|_{s;y}$. Similarly every gradient bound for G' restricts to $D\mathsf{G}'$.

For the mixed vector estimate a partition at midpoint base points is needed. Choose smooth $0 \le \phi_y \le 1$, equal to one on $\Delta(y)$ and zero outside the interior of $\Delta^{3/2}(y)$, and put

$$\zeta_y(p) = \frac{\phi_y(p)}{\sum_z \phi_z(p)} \qquad (p \in \Omega_0).$$

The denominator is at least one. If $p \in \Delta^{3/2}(y)$, $p' \in \Omega_0$, $|p - p'|_1 \le \xi_y$, and ϕ_z is nonzero at either point, then $|z - y|_\infty \le 2\xi_y + \xi_z$. Separation excludes levels differing by two or more: the gap to the larger level exceeds L times its spacing, whereas this distance is less than three times that spacing. Counting lattice points at the three remaining levels gives at most $C(L)$ such centers, all with $\xi_z \ge \xi_y/L$. The quotient rule, or $|a/h - a'/h'| \le |a - a'| + |h - h'|$ for $0 \le a, a' \le 1 \le h, h'$, yields

$$|\zeta_y(p) - \zeta_y(p')| \le C(L)|p - p'|_1/\xi_y.$$

Evaluate this partition at bond midpoints. For a field X with one extra direction index, the two-point product identity gives

$$\|\zeta_z X\|_{\beta,\xi_z} + |\zeta_z X| \le C\xi_z^s |X|^\beta_{s;z} \tag{347}$$

on inner indices. Indeed, when both base points are in the support cube use the Hölder difference of X and the displayed Lipschitz bound. If only one is in that cube, the product at the other is zero and the same Lipschitz bound controls the difference; if neither is there, both products vanish.

The initial point of a bond with midpoint in this support lies in the closed $\widetilde{\Delta}(z)$. At positive levels use $3\xi_z/4 + \xi/2 < \xi_z$; at level zero the initial point may be z or $z - \xi e_\mu$. In particular the incoming bond must not be discarded. Extend $\zeta_z X$ by zero to all bare bonds. For a pair at distance at most ξ_z meeting the support, all endpoints lie within $3\xi_z$ of z. At positive levels the gap places them in Ω_0, so the inner-index Hölder bound applies. At level zero a distinct pair has distance exactly ξ_z, and a pair crossing the index boundary contributes at most the supremum. Thus zero extension costs at most a factor two in (347). Since $X = \sum_z \zeta_z X$, apply (226) at exponent $1/2$ to these zero extensions and then (341). Restriction of the output to inner indices gives

$$\nabla \mathsf{G} \nabla^\mathsf{T} : (-2, 1/2) \longrightarrow (-2). \tag{348}$$

The same output cutoff gives the two further bounds needed below:

$$\nabla \mathsf{G} : (-3) \longrightarrow (-2, 1/2), \qquad \nabla \mathsf{G} \nabla^\mathsf{T} : (-2, 3/4) \longrightarrow (-2, 1/2). \tag{349}$$

For the first estimate apply (345), with its level-zero alternative, to $\nabla \mathsf{G} h_z$ and use (225) with exponent $1/2$. Its pointwise companion bounds the enlarged supremum at the nearby cells in (343). Summation by (341) proves the first bound.

For the second, put $J_z = \zeta_z X$, with the zero extension just checked, and $V_z = \nabla \mathsf{G} \nabla^\mathsf{T} J_z$. The partition estimate at $\beta = 3/4$ and (227), with $(\alpha, \varepsilon) = (1/2, 1/4)$, give

$$\xi_y^{1/2} \|\zeta_y^* V_z\|_{1/2} \le C e^{-\delta_* d(y,z)} \xi_z^{-2} |X|^{3/4}_{-2;z}.$$

The enlarged supremum obeys the same bound by (226) at exponent $3/4$, applied at nearby cells. That supremum also handles level zero. Now (341), with $s = -2$ and $a = 0$, proves the second bound. Both estimates retain δ_{cs} and the thresholds (339).

23.4. **Bounds for the solution.**

Quantitative bound in proposition 23.1. Invertibility and the solution formulas were proved in section 23.2. Fix a source g. We say that a component has weight s if its norm $|\cdot|_{s;y}$ is bounded by $C \sup_z e^{-\delta_{\mathrm{cs}} d(y,z)} |g|_z$; Hölder weight $(s, 1/2)$ means the same bound in the enlarged norm. The latter dominates the ordinary norm, since the base point of an index lies in the cell of its position.

Composing (342) and the averaging bounds gives

T	bound for T	bound for ∇T
N	$0 \to -2$	$0 \to -3$
P	$-2 \to -2$	$-2 \to -3$
$\mathsf{P} D^\mathsf{T}$	$-1 \to -2$	$-1 \to -3$
RG'	$-4 \to -2$	$-4 \to -3$.

The last line uses $\mathsf{RG}' = \mathsf{G}' - \mathsf{PG}'$. The same gradient bounds hold after restriction to D.

In addition,

$$\mathsf{R}\mathsf{G}'D^{\mathsf{T}} : (-3) \longrightarrow (-2, \beta). \tag{350}$$

For $\mathsf{G}'D^{\mathsf{T}}$ this is (346). For the subtracted term $f = \mathsf{P}\mathsf{G}'D^{\mathsf{T}}X$, the displayed compositions bound f in weight -2 and ∇f in weight -3. The comb estimate and scale comparison give

$$|f|^{\beta}_{-2;y} \le C \max_{d(y,z)\le 9L} \{|f|_{-2;z}, |\nabla f|_{-3;z}\}.$$

Indeed the supremum term costs at most L^2, and (344) costs at most $(9/2)L^3$ after multiplying by ξ_y^2. The local comparison rule following (340) proves (350) without a scalar mixed-derivative estimate.

For $i = 4, 5$, split the scalar solution as

$$p = p' + p'', \qquad p' = \mathsf{R}\mathsf{G}'f_5 + \mathsf{P}f_6 + \mathsf{N}f_7, \qquad p'' = \mathsf{R}\mathsf{G}'D^{\mathsf{T}}X_5. \tag{351}$$

The source weights and these bounds show that p has weight -2, Dp' weight -3, and p'' Hölder weight $(-2, \beta)$. The formulas for q and r then give weights $0, -1, -4$ for q, Dq, r. Equation (335) likewise gives weights $-2, -3$ for $\varphi, D\varphi$.

The effective vector source is $F_{\mathrm{eff}} = f_1 + D\varphi + Dp' + Dp''$. The derivative of p'' is handled as a divergence, rather than estimated in the supremum norm. For an inner bond $b = \langle x, x + \xi e_\mu \rangle$, define

$$X^{(p)}_\nu(b) = -\delta_{\mu\nu} p''(b_+). \tag{352}$$

If $b - \xi e_\mu$ is inner, the sole nonzero component in the divergence gives

$$(\nabla^{\mathsf{T}} X^{(p)})(b) = \xi^{-1}[p''(b_+) - p''(b_-)] = (Dp'')(b).$$

Otherwise its predecessor value is zero and

$$Z^{(p)}(b) := (Dp'' - \nabla^{\mathsf{T}} X^{(p)})(b) = -\xi^{-1} p''(x).$$

In this case $x \in \Lambda_0$: a site in Ω_1 has all its neighbors in Ω_0 by separation. Thus the position of b is x, $\xi_x = \xi$, and $|Z^{(p)}|_{-3;x} \le |p''|_{-2;x}$. At the other positions this correction vanishes. Hence

$$F_{\mathrm{eff}} = f_1' + \nabla^{\mathsf{T}} X^{(p)}, \qquad f_1' = f_1 + D\varphi + Dp' + Z^{(p)}, \tag{353}$$

where f_1' has weight -3.

We check that the endpoint shift in (352) preserves the required Hölder weight. For a scalar function u, let $X^{(u)}_\nu(b) = -\delta_{\mu\nu} u(b_+)$. If the midpoint of b lies in $\Delta^{3/2}(y)$, both endpoints lie in the closed $\widetilde{\Delta}(y)$. This follows from $3\xi_y/4 + \xi/2 \le \xi_y$ at positive levels; at level zero the possible bonds are the incoming and outgoing bonds at y. Consequently the supremum of $X^{(u)}$ is controlled by the suprema of u in cells meeting $\widetilde{\Delta}(y)$. Their scales and distances obey (343). At levels zero and one the lattice-spacing bound following (344) controls the Hölder seminorm as well.

At a level $j \ge 2$, choose $b^\sharp = b_+$ if that endpoint is in $\Delta^{3/2}(y)$, and $b^\sharp = b_-$ otherwise; the latter endpoint then lies in this cube. For $z = y(b_+)$ both endpoints lie in $\Delta^{3/2}(z)$: the cell of b_+ contains its closed bare cube, and $|b_\pm - z|_\infty \le \xi_z/2 + \xi/2 \le 3\xi_z/4$, since $\xi_z \ge L\xi$. For like bonds b, b' put $r = |b_- - b'_-|_1 \ge \xi$ and split

$$\begin{aligned} u(b'_+) - u(b_+) = {} & [u(b'_+) - u(b'^\sharp)] + [u(b'^\sharp) - u(b^\sharp)] \\ & + [u(b^\sharp) - u(b_+)]. \end{aligned}$$

The middle pair lies in $\Delta^{3/2}(y)$ at distance at most $r + 2\xi \le 3r$. Each outer pair lies in the enlarged cube of the endpoint's position and has distance ξ. Multiplication by $(\xi_y/r)^\beta$ costs at most 3^β for the middle Hölder quotient and at most L^β for each outer quotient. Finally $\xi_y^2 \le L^2 \xi_z^2$. Together with the low-level case this proves

$$|X^{(u)}|^{\beta}_{-2;y} \le C \max_{\Delta(z) \cap \widetilde{\Delta}(y) \neq \emptyset} |u|^{\beta}_{-2;z}. \tag{354}$$

The local comparison rule now transfers the Hölder weight of p'' to $X^{(p)}$.

Apply (342) and (348) to (353). They give weights $-1,-2$ for $\mathsf{G}F_{\mathrm{eff}}$ and $\nabla\mathsf{G}F_{\mathrm{eff}}$. For $i=4$ this is the estimate for A. For $i=5$ use the coarse-constraint formulas (337). Since f_8 and $Q\mathsf{G}F_{\mathrm{eff}}$ have weight -1, ω has weight -3. The bounds for Q^{T}, G and $\nabla\mathsf{G}$ preserve the weights $-1,-2$ of $A,\nabla A$.

It remains to recover the first scalar subsystem. Its formulas give $w=\mathsf{P}D^{\mathsf{T}}A+\varphi$, so w, Dw have weights $-2,-3$. To bound v we must separately check $D^{\mathsf{T}}A$. At a site x, in each direction whose predecessor bond is inner,

$$(\nabla_\mu^{\mathsf{T}}A_\mu)(x)=-(\nabla_\mu A_\mu)(x-\xi e_\mu).$$

If that bond is missing, the term instead equals $-\xi^{-1}A_\mu(x)$, and vanishes if the outgoing bond is also missing. In the nonzero exceptional case $x\in\Lambda_0$, so $\xi_x^2\xi^{-1}|A_\mu(x)|\le|A|_{-1;x}$. In the first case the predecessor's position has scale within a factor L of the position of x, and distance at most $9L$, by (267). Summing the three directions therefore gives

$$|D^{\mathsf{T}}A|_{-2;y}\le C\max_{d(y,z)\le 9L}\{|\nabla A|_{-2;z},|A|_{-1;z}\}.$$

Thus $D^{\mathsf{T}}A$ has weight -2. The elimination formulas and the scalar bounds now give weights $0,-1,-4$ for v, Dv, σ. For $i=2$ the same scalar calculation uses $A=0$ and $w=\varphi$; for $i=1$, $w=\mathsf{G}'f_3$. These are the solutions of the compressed systems, not restrictions of the full inverse.

Every entry in the solution norm has now been bounded at its prescribed weight. Taking the largest of the finitely many constants gives $B_{\mathrm{cs}}=B_{\mathrm{cs}}(L)$ and the estimate in proposition 23.1. The rates and $\mathfrak{M}_\dagger$ were fixed before this constant was chosen. □

24. Complex averages near the identity

We estimate centered straightened averages and their first derivatives. The derivative kernels retain a factor η^3, which compensates the volume factor in the weighted transpose. The endpoint correction and its iteration follow Bałaban's averaging construction [Bał85a, pp. 34–39, (110), (121)–(127)]; we use centered contours. The differentiated iteration follows the method of [Bał85a, pp. 39–43, Propositions 5–7].

For one averaging step, use level-n units, $0\le n<K$, so that fine bonds have length one. We use the centered scheme and write $\mathcal{M}=\mathcal{M}_{\mathsf{m}}$ and $W(y,x)=W_{\mathsf{m}}(y,x)$. For a positive bond c of $\mathbb{T}^{(n+1)}$, let $\mathsf{D}_c=B(c_-)\cup B(c_+)$ and let P_c^+ be its positive inner fine bonds. All field coordinates below are on positive bonds; configurations on reverse bonds are given by inversion. Suprema are over P_c^+ when the output is at c.

Let W be a complex configuration and put $\widetilde{W}=e^{ia}W$. Whenever the contour variables, averages and series logarithms exist, define

$$\begin{aligned}\zeta(W,a)(y,x)&=i^{-1}\log(\widetilde{W}(y,x)W(y,x)^{-1}),\\ \mathsf{v}(W,a)(y)&=\exp\left(iL^{-3}\sum_{x\in B(y)}\zeta(W,a)(y,x)\right),\\ \mathcal{Q}^1(W,a)(c)&=i^{-1}\log[\mathsf{v}(W,a)(c_-)^{-1}\mathcal{M}(\widetilde{W})(c)\mathsf{v}(W,a)(c_+)\mathcal{M}(W)(c)^{-1}],\end{aligned}\tag{355}$$

where $y=c_-$ or c_+ and $x\in B(y)$. Thus

$$\mathcal{M}(e^{ia}W)(c)=\mathsf{v}(W,a)(c_-)\,e^{i\mathcal{Q}^1(W,a)(c)}\mathcal{M}(W)(c)\mathsf{v}(W,a)(c_+)^{-1}.\tag{356}$$

Write $\mathbf{Q}(W)=d_a\mathcal{Q}^1(W,0)$ and $C^1(W,a)=\mathcal{Q}^1(W,a)-\mathbf{Q}(W)a$. For these kernels, $\|T\|_{\mathrm{row}}=\sup_c\sum_e|T(c,e)|$ and $\|T\|_{\mathrm{col}}=\sup_e\sum_c|T(c,e)|$, where the entry norm is the operator norm on $\mathfrak{g}^c$. On a box Y of level-n sites, restrict the coarse indices to $\mathsf{D}_c\subset\mathsf{Y}$ and the fine indices to bonds with both endpoints in Y.

24.1. **Iterated averages.** Fix $1 \le k \le K$ and use level-k units, so that the bare spacing is $\eta = L^{-k}$. For a positive level-k bond c, set

$$Y^{(n)} = B^{k-n}(c_-) \cup B^{k-n}(c_+), \qquad 0 \le n \le k.$$

For $n < k$, the bonds in $Y^{(n)}$ have both endpoints there; at $n = k$ the bond set consists of c alone. Suprema at level n are over these positive bonds. Bare fields are extended by zero when a whole-lattice expression is used.

For a configuration $\mathbf{U}$ on the bare bonds in $Y^{(0)}$, write $W_n = \mathcal{M}^n\mathbf{U}$ whenever these complex averages exist. Define the straightened iterates by

$$a_0 = \eta A, \qquad a_{n+1} = \mathcal{Q}^1(W_n, a_n), \qquad \mathcal{Q}_n(\mathbf{U}; A) = a_n. \tag{357}$$

Thus a_n is a field on the bonds in $Y^{(n)}$; definedness is part of the proposition below. Put

$$Q_k(\mathbf{U}) = \eta\mathbf{Q}(W_{k-1}) \cdots \mathbf{Q}(W_0), \qquad Q'_k(\mathbf{U}) = \mathfrak{q}_{W_{k-1}} \cdots \mathfrak{q}_{W_0}, \tag{358}$$

where

$$(\mathfrak{q}_W\lambda)(y) = L^{-3} \sum_{x \in B(y)} \mathrm{Ad}(W(y,x))\lambda(x).$$

The first product takes its value at c and the second at $y = c_-$ or c_+. Kernels are taken against counting measure. At the identity, these are the straight bond average and the arithmetic site average identified in section 23.1, in the present units.

Proposition 24.1. There are constants $c_{\mathrm{av}} > 0$ and $C_{\mathrm{av}}, C_2, D \ge 1$, depending only on L, with the following properties. Let A', A be $\mathfrak{g}^c$-valued bare fields on $Y^{(0)}$, with $\|A'\|_\infty, \|A\|_\infty < c_{\mathrm{av}}$, and put $\mathbf{U} = e^{i\eta A'}$.

(i) The complex averages W_n are defined for $0 \le n \le k$. They are holomorphic in A' and have the form $W_n = e^{iB_n}$, where B_n is holomorphic, $\mathfrak{g}^c$-valued, and

$$\|B_n\|_\infty \le DL^n\eta\|A'\|_\infty.$$

(ii) The iterates in (357) are defined and jointly holomorphic, and

$$\|a_n\|_\infty \le 2L^n\eta\|A\|_\infty.$$

The derivative of $A \mapsto \mathcal{Q}_k(\mathbf{U}; A)$ at zero is $Q_k(\mathbf{U})$.

(iii) The kernel of $Q_k(\mathbf{U})$ is supported on the bare bonds in $Y^{(0)}$. Writing $a' = \|A'\|_\infty$, one has

$$\begin{aligned} \sum_b |Q_k(\mathbf{U})(c,b)| &\le \min\{2, 1 + C_{\mathrm{av}}a'\}, \\ |Q_k(\mathbf{U})(c,b)| &\le \eta^3 \min\{2, 1 + C_{\mathrm{av}}a'\}, \\ \sum_b |(Q_k(\mathbf{U}) - Q_k(\mathbf{1}))(c,b)| &\le C_{\mathrm{av}}a', \\ |(Q_k(\mathbf{U}) - Q_k(\mathbf{1}))(c,b)| &\le C_{\mathrm{av}}\eta^3 a'. \end{aligned} \tag{359}$$

If x_j is the level-j site with $x \in B^j(x_j)$, then

$$\begin{aligned} T_{\mathbf{U}}(y,x) &= W_{k-1}(y, x_{k-1}) \cdots W_0(x_1, x), \\ (Q'_k(\mathbf{U})\lambda)(y) &= L^{-3k} \sum_{x \in B^k(y)} \mathrm{Ad}(T_{\mathbf{U}}(y,x))\lambda(x), \\ |\,\mathrm{Ad}(T_{\mathbf{U}}(y,x)) - 1| &\le C_{\mathrm{av}}a'. \end{aligned} \tag{360}$$

(iv) For $K_A(c,b) = \partial\mathcal{Q}_k(\mathbf{U}; A)(c)/\partial A(b)$,

$$\begin{aligned} |K_A(c,b)| &\le \eta^3(1 + C_{\mathrm{av}}a' + C_2\|A\|_\infty), \\ |K_A(c,b) - Q_k(\mathbf{U})(c,b)| &\le C_2\eta^3\|A\|_\infty. \end{aligned} \tag{361}$$

The constants are independent of the torus, the bond and k. All maps are determined by the indicated local data. ■

24.2. **One-step estimates.**

Lemma 24.2. There is $r_{\mathrm{st}} > 0$, depending only on L, such that

$$r_{\mathrm{st}} \le \min\left\{r_*, \frac{\omega_M}{16}, \frac{1}{16\kappa_2^{\mathrm{SO}}}\right\}.$$

Put

$$\kappa_{\mathrm{st}} = r_{\mathrm{st}}^{-1}, \qquad K_{\mathrm{st}} = 24L^3 r_{\mathrm{st}}^{-2}, \qquad H_{\mathrm{st}} = 11 r_{\mathrm{st}}^{-2}.$$

For $W = e^{ib}$, $\|b\|_\infty, \|a\|_\infty < r_{\mathrm{st}}$, the maps in (355) and $i^{-1}\log \mathsf{v}(W,a)$ are defined and jointly holomorphic. The three logarithmic maps take values in $\mathfrak{g}^c$, vanish when $a = 0$, and have norm at most $\min\{1, \kappa_{\mathrm{st}}\|a\|_\infty\}$. For real b, a they take values in $\mathfrak{g}$ and $\mathsf{v}(W,a)$ takes values in G; the averages agree with the real small-curvature branch. In particular,

$$W(y,x) = e^{i\zeta_b(y,x)}, \qquad \zeta_b(y,x) = \zeta(\mathbf{1}, b)(y,x), \qquad |\zeta_b(y,x)| \le \kappa_{\mathrm{st}}\|b\|_\infty. \tag{362}$$

For $h : \mathsf{D}_c \to G$, the transformed expressions are defined and satisfy

$$\begin{aligned} \zeta(W^h, \mathrm{Ad}(h)a)(y,x) &= \mathrm{Ad}(h(y))\zeta(W,a)(y,x), \\ \mathsf{v}(W^h, \mathrm{Ad}(h)a)(y) &= h(y)\mathsf{v}(W,a)(y)h(y)^{-1}, \\ \mathcal{Q}^1(W^h, \mathrm{Ad}(h)a)(c) &= \mathrm{Ad}(h(c_-))\mathcal{Q}^1(W,a)(c). \end{aligned} \tag{363}$$

Here $(\mathrm{Ad}(h)a)(e) = \mathrm{Ad}(h(e_-))a(e)$.

Let $\mathbf{S} = \mathbf{Q}(\mathbf{1})$. Then

$$(\mathbf{S}a)(c) = L^{-3} \sum_{x \in B(c_-)} a([x, x']), \qquad \|\mathbf{S}\|_{\mathrm{row}} \le L, \quad \|\mathbf{S}\|_{\mathrm{col}} \le L^{-2},$$

where $x' = x + Le_\mu$ for a bond c in direction μ. For $\|b\|_\infty < r_{\mathrm{st}}/16$ and $\|a\|_\infty \le r_{\mathrm{st}}/2$,

$$\begin{aligned} \|\mathbf{Q}(e^{ib}) - \mathbf{S}\|_{\mathrm{row}}, \quad \|\mathbf{Q}(e^{ib}) - \mathbf{S}\|_{\mathrm{col}} &\le K_{\mathrm{st}}\|b\|_\infty, \\ |C^1(e^{ib}, a)(c)| &\le H_{\mathrm{st}}\|a\|_\infty^2, \\ |\partial_{a(e)} C^1(e^{ib}, a)(c)| &\le H_{\mathrm{st}}\|a\|_\infty. \end{aligned} \tag{364}$$

The two remainder bounds at c use only the supremum on P_c^+. All these kernels vanish unless $e \in \mathsf{P}_c^+$. Their bounds hold on the whole lattice and with the box restrictions above. In particular,

$$\|d_a \mathcal{Q}^1(e^{ib}, a)\|_{\mathrm{col}} \le L^{-2} + K_{\mathrm{st}}\|b\|_\infty + 6H_{\mathrm{st}}\|a\|_\infty. \tag{365}$$

■

Proof. Use the finite-chart argument of lemma 14.7, now for the three centered directions and the two variables (b, a). By lemma 14.2, the defining contour and averaging maps are holomorphic near the identity configuration. At $(b, a) = (0, 0)$, all arguments of logarithms in (355) are the identity. Products, inverses and series logarithms therefore give a common neighborhood on which the three logarithmic maps are holomorphic and bounded by one. Choose r_{st} within this neighborhood and the displayed upper bounds. For fixed b, the three maps vanish at $a = 0$; Cauchy's formula on complex lines gives the bound $r_{\mathrm{st}}^{-1}\|a\|_\infty$.

If b, a are real, telescoping the four unitary factors around a plaquette gives

$$|\partial(e^{ia}e^{ib})(p) - \mathbf{1}| \le 4(\|a\|_\infty + \|b\|_\infty) < 8r_{\mathrm{st}}.$$

The chosen radius and lemmas 14.2 and 14.6 give the real values and branch agreement. The i^{-1} multiples of these logarithms near the identity belong to $\mathfrak{g}$. Their traces are holomorphic functions of (b, a) vanishing on the real slice, so they vanish throughout the joint domain, by the real-slice

uniqueness argument in the proof of proposition 11.2. This proves the $\mathfrak{g}^c$ assertion. Equation (362) follows by exponentiating the first definition at $(\mathbf{1}, b)$.

The identity $(e^{ia}W)^h = e^{i\,\mathrm{Ad}(h)a}W^h$ and (117) conjugate the first logarithm by $h(y)$ and the last by $h(c_-)$. Unitary conjugation preserves the domain and commutes with the series logarithm and exponential; this proves (363). Differentiation gives, in particular,

$$\mathbf{Q}(W^h)(c,e) = \mathrm{Ad}(h(c_-))\mathbf{Q}(W)(c,e)\,\mathrm{Ad}(h(e_-))^{-1},$$

and the corresponding identities for the Taylor coefficients. The conjugating maps are isometries.

The identity derivative is (331). Each pair consisting of $x \in B(c_-)$ and a bond of $[x, x']$ contributes L^{-3} to its kernel. There are $L^3 L$ such pairs for fixed c. For a fixed fine bond e in direction μ, its possible starting points are $x = e_- - je_\mu$, $0 \le j < L$; each belongs to exactly one block $B(c_-)$. Thus there are L pairs for fixed e, giving the row and column bounds for $\mathbf{S}$. Also, D_c has $2L^3$ sites, so $\#\mathsf{P}_c^+ \le 6L^3$. A fine bond lies either inside one block, in which case it belongs to the six incident double blocks, or between two adjacent blocks, in which case it belongs only to their double block. Consequently it belongs to at most six sets P_c^+.

Put $r = r_{\mathrm{st}}$ temporarily. For fixed c, e, the map $b \mapsto \mathbf{Q}(e^{ib})(c,e)$ is holomorphic for $\|b\|_\infty < r$ and bounded by r^{-1}: apply Cauchy's formula in the $a(e)$ direction at zero to the joint map. A second application bounds its derivative in the b direction by $4r^{-2}$ on $\|b\|_\infty \le 3r/4$. Integration from zero to b gives

$$|\mathbf{Q}(e^{ib})(c,e) - \mathbf{S}(c,e)| \le 4r^{-2}\|b\|_\infty.$$

The row and column counts just proved give the first line of (364) with the stated K_{st}.

For fixed b, Cauchy's formula also gives

$$|C^1(e^{ib}, a)(c)| \le \frac{\|a\|_\infty^2}{r(r - \|a\|_\infty)}.$$

On $\|a\|_\infty < 3r/4$, the derivative $\partial_{a(e)}\mathcal{Q}^1$ has norm at most $4/r$, and its value at zero has norm at most $1/r$. Thus $\partial_{a(e)}C^1$ is bounded by $8/r$ there and vanishes at zero. Cauchy's formula on this ball bounds it by $32\|a\|_\infty/(3r^2)$. For $\|a\|_\infty \le r/2$ these bounds are at most the asserted ones with $H_{\mathrm{st}} = 11r^{-2}$. The definitions give the stated support. Summing the last bound over the at most six coarse bonds incident to a fixed fine bond proves (365). Restricting either index set to the stated box can only decrease these sums. □

Proof of proposition 24.1. Write $r = r_{\mathrm{st}}$, $\kappa = \kappa_{\mathrm{st}}$ and $H = H_{\mathrm{st}}$ for the constants of lemma 24.2. Put $\theta_n = L^n\eta$. The sums used below are

$$\sum_{j<n} \theta_j \le 2L^{-1}\theta_n, \qquad \sum_{n<k} \theta_n \le 2L^{-1}. \tag{366}$$

Choose

$$c_0 \le \min\{r/4, L^2 \log 2/(8H)\}, \qquad D = 4(8\kappa + 2), \qquad c_\sharp = \min\{c_0, r/(16D)\}.$$

We first work on $\|A'\|_\infty < c_\sharp$. The smaller radius c_{av} will be chosen below.

The background averages. At the reference configuration $\mathbf{1}$, define $a'_0 = \eta A'$ and $a'_{n+1} = \mathcal{Q}^1(\mathbf{1}, a'_n)$. The identity row bound and the quadratic remainder give, for $t_n = \|a'_n\|_\infty/\theta_n$,

$$t_{n+1} \le t_n(1 + HL^{-1}\theta_n t_n).$$

As long as $t_j \le 2t_0$ for $j \le n$, multiplication and (366) give

$$t_{n+1} \le t_0 \exp(4HL^{-2}t_0) \le 2t_0.$$

Also $2t_0 \le r/2$, so the same one-step domain is available at the next stage. Induction proves the recursion and $\|a'_n\|_\infty \le 2\theta_n\|A'\|_\infty$.

Let $s_n = \mathsf{v}(\mathbf{1}, a'_n) = e^{i\sigma_n}$ and set $v_0 = 1$, $v_{n+1} = v_n s_n$ on the next coarser sites. The one-step estimate gives $\|\sigma_n\|_\infty \le \kappa \|a'_n\|_\infty$. At a positive level-$n$ bond b, the candidate

$$X_n(b) = v_n(b_-) e^{i a'_n(b)} v_n(b_+)^{-1}$$

is a product of $2n+1$ exponentials. The sum of their exponent norms is at most

$$E_n = 2 \sum_{j<n} \|\sigma_j\|_\infty + \|a'_n\|_\infty \le (8\kappa L^{-1} + 2)\theta_n \|A'\|_\infty.$$

Thus $|X_n - 1| \le e^{E_n} - 1 \le 2E_n < 1/2$, and its series logarithm defines $B_n = i^{-1} \log X_n$ with

$$\|B_n\|_\infty \le 4E_n \le D\theta_n \|A'\|_\infty < r/16.$$

These fields are holomorphic. For real A', all factors belong to G, and the logarithms are in $i\mathfrak{g}$; the real-slice uniqueness argument in the proof of proposition 11.2 then gives tracelessness for complex A'.

We now identify the candidate with the actual average. At $n = 0$, $X_0 = e^{i\eta A'}$. If $W_n = X_n = e^{iB_n}$, its restriction to every next-step double block lies in the domain of lemma 24.2; hence W_{n+1} is defined and holomorphic. For real A', the straightening identity and unitary gauge covariance give

$$\mathcal{M}\big((e^{i a'_n})^{v_n}\big) = \big(e^{i a'_{n+1}}\big)^{v_n s_n} = X_{n+1}.$$

Both sides are holomorphic on the same complex ball, so this identity holds there. This proves the induction and (i).

The construction can first be made on the whole lattice with the identity reference and the zero extension of A'. The block identity $B^{n+1}(y) = \bigcup_{z \in B(y)} B^n(z)$ implies that the one-step double block of every bond in the prescribed bond set of $Y^{(n+1)}$ lies in $Y^{(n)}$. Descending through these blocks shows that the restriction uses only A' in $Y^{(0)}$. The same argument applies to all subsequent recursions and their derivatives.

Linear kernels and transports. The row and column norms are submultiplicative, by summing $|(ST)(i,j)| \le \sum_l |S(i,l)||T(l,j)|$. Set $\beta_n = \|B_n\|_\infty$ and $\beta = \sum_{n<k} \beta_n \le 2DL^{-1}a'$. The one-step estimates give

$$\|\mathbf{Q}(W_n)\|_{\mathrm{row}} \le L + K_{\mathrm{st}}\beta_n, \qquad \|\mathbf{Q}(W_n)\|_{\mathrm{col}} \le L^{-2} + K_{\mathrm{st}}\beta_n.$$

Since $\eta L^k = 1$ and $\eta L^{-2k} = \eta^3$, multiplication yields

$$\|Q_k(\mathbf{U})\|_{\mathrm{row}} \le e^{K_{\mathrm{st}} L^{-1}\beta}, \qquad |Q_k(\mathbf{U})(c,b)| \le \eta^3 e^{K_{\mathrm{st}} L^2 \beta}. \tag{367}$$

Choose the positive radius so that

$$c_{\mathrm{av}} \le \min\left\{ \frac{c_\sharp}{4}, \frac{\log 2}{8K_{\mathrm{st}}DL}, \frac{\log 2}{96HL} \right\}. \tag{368}$$

Then the exponents in (367) are at most $\frac{1}{4}\log 2$. Using $e^s \le 1 + 2s$ for $0 \le s \le \log 2$ gives the first two bounds of (359), with $C_{\mathrm{av}} \ge 4K_{\mathrm{st}}DL$. Their holomorphic differences from the value at $A' = 0$ vanish there and are bounded by 4 in the row norm and $4\eta^3$ entrywise. The Schwarz estimate on each complex line gives the other two bounds when $C_{\mathrm{av}} \ge 4/c_{\mathrm{av}}$.

Expanding the product of the scalar means gives the formula in (360): the chains of nested blocks correspond bijectively to $x \in B^k(y)$. The one-step contour estimate writes each factor as $W_n(z, x') = e^{i\zeta_n(z,x')}$, with $|\zeta_n| \le \kappa\beta_n$. Consequently, if $z = \kappa\beta$, then

$$|\operatorname{Ad}(T_{\mathbf{U}}(y,x)) - 1| \le e^{2z} - 1 \le 4z \le 8\kappa DL^{-1}a'.$$

Here $z \le 1/8$ follows from $c_{\mathrm{av}} \le c_\sharp/4 \le r/(64D)$ and $\kappa r = 1$. This proves (iii), after increasing C_{av} if necessary.

The second perturbation. Now use the actual complex bases W_n in (357). For $t_n = \|a_n\|_\infty/\theta_n$, the one-step estimates give

$$t_{n+1} \le t_n(1 + K_{\mathrm{st}}L^{-1}\beta_n + HL^{-1}\theta_n t_n).$$

The previous multiplication argument, with the additional background term, gives

$$t_{n+1} \le t_0 \exp(K_{\mathrm{st}} L^{-1}\beta + 4HL^{-2}t_0) \le 2t_0$$

whenever the preceding bounds hold. The radius choice makes the exponent smaller than $\log 2$ and keeps $\|a_n\|_\infty \le 2c_{\mathrm{av}} < r/2$. This proves the recursion and bound in (ii). Joint holomorphy and tracelessness follow at each step from the one-step result. Since $a_n = 0$ at $A = 0$, the chain rule gives (358).

For the derivative at general A, the same rule gives

$$K_A = \eta(\mathcal{Q}^1)'(W_{k-1}, a_{k-1}) \cdots (\mathcal{Q}^1)'(W_0, a_0).$$

By (365) and $\sum_{n<k} \|a_n\|_\infty \le 4L^{-1}\|A\|_\infty$,

$$|K_A(c,b)| \le \eta^3 \exp(K_{\mathrm{st}} L^2\beta + 24HL\|A\|_\infty).$$

The exponent is at most $\frac{1}{2}\log 2$. The exponential estimate proves the first bound of (361) with $C_2 \ge 48HL$; it also gives $|K_A| \le 2\eta^3$. For fixed A', the holomorphic map $A \mapsto K_A - K_0$ vanishes at zero and is bounded by $4\eta^3$ on the ball of radius c_{av}. The same Schwarz estimate gives the second bound with $C_2 \ge 4/c_{\mathrm{av}}$. All the choices above depend only on L. □

25. The constrained system near a flat configuration

We extend proposition 23.1 to small complex perturbations, allowing independent current and multiplier fields. The perturbation of the scalar equation for p contains a divergence. Keeping that divergence in the forcing space permits the estimate without a derivative of p. The other new term is the curvature remainder in the action Hessian.

Keep the torus, separated sequence, inner frame, spaces, pairings, positions and weights of section 23, and assume throughout that $\mathfrak{m} \ge \mathfrak{M}_\dagger$ and $\mathfrak{g} \ge L\mathfrak{M}_\dagger$. Put $Y = \Omega_0$. Let the ambient region A contain all bare sites within sup distance ξ of Y, and let $\mathbf{U}$ and $\mathbf{J}$ be a complex configuration and a $\mathfrak{g}^c$-valued one-form on its inner bare bonds. Let $\mathsf{j} \in \mathcal{R}$.

25.1. **The operators and the inverse estimate.** For $\mathsf{U}_\nu(x) = \operatorname{Ad}\mathbf{U}(\langle x, x+\xi e_\nu\rangle)$, covariant differences are

$$\nabla_\nu f(x) = \xi^{-1}(\mathsf{U}_\nu(x) f(x+\xi e_\nu) - f(x)), \qquad \nabla_\nu^{\mathsf{T}} f(x) = \xi^{-1}(\mathsf{U}_\nu(x-\xi e_\nu)^{-1} f(x-\xi e_\nu) - f(x)).$$

Here f is a site field or a component of a bond field, indexed by its initial point. A term evaluated outside its index set is zero and uses no transport. In particular, $(Dv)_\mu = \nabla_\mu v$ on inner bonds and $D^{\mathsf{T}}A = \sum_\mu \nabla_\mu^{\mathsf{T}} A_\mu$. The Dirichlet Laplacian has the same omission convention and retains its full diagonal:

$$\Delta^\circ f(x) = \xi^{-2}\left(6f(x) - \sum_{\nu=1}^{3}[\mathsf{U}_\nu(x) f(x+\xi e_\nu) + \mathsf{U}_\nu(x-\xi e_\nu)^{-1} f(x-\xi e_\nu)]\right). \tag{369}$$

At a positive-level row $c \in \Gamma_n$, rescale its double block to level-n units and use the straightened map of section 24. The physical constraint is $\xi_c^{-1}\mathcal{Q}_n(\mathbf{U}; \xi_c A)(c)$. Its linear and quadratic Taylor terms define $QA(c)$ and $C^{(2)}(A)(c)$, respectively. Thus, with $W_j = \mathcal{M}^j\mathbf{U}$,

$$(QA)(c) = \frac{\xi}{\xi_c}(\mathbf{Q}(W_{n-1}) \cdots \mathbf{Q}(W_0)A)(c). \tag{370}$$

At a top $y \in \Lambda_n$, $Q'\lambda(y)$ is the ordered scalar mean of (360), on the single block $B^n(y)$ in these units. At level zero both averages are the identity and $C^{(2)} = 0$. These definitions are made wherever the local maps exist; the proposition below establishes their existence under its smallness hypothesis.

Define symmetric operators by their quadratic forms:

$$\begin{aligned}(A,\Delta_{\mathsf{A}}(\mathbf{U})A)_\xi &= \frac{d^2}{ds^2}\Big|_{s=0}\ \xi^{-1}\sum_{p\subset\mathsf{A}} a_p(e^{is\xi A}\mathbf{U}),\\ (A,\Delta^{(2)}[\mathsf{j}]A)_\xi &= 2(C^{(2)}(A),\mathsf{j})_{\mathbf{B}}.\end{aligned}\tag{371}$$

The first form is on the inner bonds of A; the second is on $\mathcal{A}$. Both are homogeneous quadratic polynomials. Polarization and the nondegenerate bilinear pairings give unique symmetric operators. Compress the first to $\mathcal{A}$ by extending its arguments by zero, and denote the result by $\Delta_Y(\mathbf{U})$. Set

$$\mathsf{L}=\Delta_Y(\mathbf{U})+DD^{\mathsf{T}},\qquad \mathsf{L}_1=\mathsf{L}+Q^{\mathsf{T}}\mathsf{m}^{-2}Q,\qquad \Delta'_a=\Delta^\circ+Q'^{\mathsf{T}}\mathsf{a}'Q'.$$

For a bond field k, let

$$(\mathsf{E}_-^{\mathsf{T}}k)(x)=\sum_{b_-=x}k(b),\qquad (\mathsf{E}^{\mathsf{T}}k)(x)=\sum_{b_-=x}k(b)+\sum_{b_+=x}\mathrm{Ad}(\mathbf{U}(b))^{-1}k(b),$$

where the bonds are positive and inner in Y. Write $v_-(b)=v(b_-)$ and $v_+(b)=\mathrm{Ad}(\mathbf{U}(b))v(b_+)$, and put

$$\begin{aligned}\mathsf{m}_J(v)&=\tfrac{i}{2}[\mathbf{J},v_-+v_+],\\ \mathsf{g}_0(A,v)&=\tfrac12\mathsf{E}^{\mathsf{T}}(i[A,\mathbf{J}])-\tfrac12\mathsf{E}_-^{\mathsf{T}}(i[Dv,\mathbf{J}]),\\ \mathsf{x}(A,v)&=-\tfrac{i}{2}[\mathbf{J},v_-]-\Delta^{(2)}[\mathsf{j}](A-Dv).\end{aligned}\tag{372}$$

Define $\mathbb{K}^{[5]}$ from (326) by replacing its operators with those just defined and subtracting $\mathsf{m}_J(v)+\Delta^{(2)}[\mathsf{j}](A-Dv)$ in the first slot and $\mathsf{g}_0(A,v)+D^{\mathsf{T}}\mathsf{x}(A,v)$ in the fifth. The same coordinate compressions define $\mathbb{K}^{[i]}$ for $i=1,2,4$. In (327) and the forcing injection use these covariant differences; write $|\Psi|_{\mathbf{U},y}$ and $\iota_{\mathbf{U}}$ for the resulting seminorm and injection. The forcing seminorm (328) is unchanged.

Put $\rho=3L+6$ and $\mathsf{N}(y)=\bigcup_{d(y,z)\le\rho}\Delta(z)$. We call $(\mathbf{U},\mathbf{J},\mathsf{j})$ *ϵ-small in the trivial gauge* if $\mathbf{U}(b)=e^{i\xi\mathbf{A}(b)}$ on the positive bare bonds with both endpoints within sup distance ξ of Y, and, for every $y\in\Lambda_*$,

$$\xi_y|\mathbf{A}(b)|\le\epsilon,\qquad \xi_y^2|\mathbf{A}(b+\xi e_\nu)-\mathbf{A}(b)|\le\epsilon\xi,\qquad |\mathbf{J}|_{-3;y},\ |\mathsf{j}|_{-3;y}\le\epsilon.\tag{373}$$

The first two inequalities apply when $b_-\in\mathsf{N}(y)$, or when $b_-\notin Y$ and $\mathrm{dist}_\infty(b_-,\mathsf{N}(y))\le\xi$. Both endpoints of b, and of the shifted bond in the second inequality, must be in the specified collar. We call the family *ϵ-small* if a G-valued gauge on that collar makes it ϵ-small in the trivial gauge. The current transforms by conjugation at b_-, and the row field by conjugation at c_-, also when its position is c_+.

Proposition 25.1. There are constants $\epsilon_*>0$ and $B_*>0$, depending only on L, such that every ϵ_*-small family above defines all the local averages and operators, and each $\mathbb{K}^{[i]}$, $i\in\{1,2,4,5\}$, is invertible. Its inverse $\mathbb{E}^{[i]}$ satisfies

$$|\mathbb{E}^{[i]}\iota_{\mathbf{U}}g|_{\mathbf{U},y}\le B_*\sup_z e^{-\delta_{\mathrm{cs}}d(y,z)}|g|_z\qquad (g\in\mathcal{F}^{[i]},\ y\in\Lambda_*).\tag{374}$$

In the trivial gauge, with $\mathbb{V}^{[i]}$ defined in (384),

$$\begin{aligned}\mathbb{E}^{[i]}\iota_0&=\mathbb{E}_0^{[i]}\iota_0(I-\mathbb{V}^{[i]}\mathbb{E}_0^{[i]}\iota_0)^{-1},\\ \mathbb{V}^{[i]}\mathbb{E}_0^{[i]}\iota_0&\in\mathfrak{O}(\delta_{\mathrm{cs}},1/4).\end{aligned}\tag{375}$$

The constants are uniform in the torus, lattice spacing and separated sequence, including its number of layers. The operators and their inverses depend on $\mathbf{U},\mathbf{J}$ only at bonds with both endpoints in the collar, and on j. For $Y=\emptyset$ the assertions use the unique maps on the zero spaces. ∎

We first prove the local averaging and Hessian estimates. Their combination gives $\mathbb{K}_0 - \mathbb{K} = \iota_0 \mathbb{V}$, with $\mathbb{V}$ small in the forcing norm. The proof of proposition 25.1 is then a geometric series on the forcing space, followed by the change from ι_0 to $\iota_{\mathbf{U}}$.

In the local estimates below the family is ϵ-small in the trivial gauge, with

$$0 < \epsilon \le \epsilon_0 := \min\{1/8, c_{\mathrm{av}}/(2L)\}.$$

Constants denoted by C depend only on L and may increase. Subscript 0 denotes the trivial configuration; in particular, $|\Psi|_{0,y}$ is (327) with ordinary differences.

25.2. **Local averaging estimates.** We first record the local geometry used below and in the coefficient estimates. Write $y(x)$ for the position of a bare site $x \in Y$. The closed bare cube centered at x is contained in $\Delta(y(x))$, by the nested partitions. Thus, if $x, x' \in Y$ and $|x - x'|_\infty \le \xi$, their cells meet. Joining their centers to a point of intersection costs at most $3/2$ in each cell for the weighted length. The layer gap in lemma 21.2 then gives

$$d(y(x), y(x')) \le 3, \qquad L^{-1}\xi_{y(x)} \le \xi_{y(x')} \le L\xi_{y(x)}. \tag{376}$$

In particular, $\Delta(y(x')) \subset \mathsf{N}(y(x))$. For bare sites, the separation of Ω_1 from Y^c, with $\mathsf{g} > 3$, also gives

$$x \in Y, \quad x' \notin Y, \quad |x - x'|_\infty \le 3\xi \quad \Longrightarrow \quad x \in \Lambda_0, \quad \xi_{y(x)} = \xi. \tag{377}$$

For $t = 0$ the latter conclusion holds throughout Y.

The complex bilinear pairing requires a small distinction from a Hilbert space adjoint. For $X, Y \in \mathfrak{g}^c$ and a linear map T on $\mathfrak{g}^c$,

$$|\langle X, Y\rangle| \le |X|\,|Y|, \qquad \tfrac{1}{2}|X| \le \sup_{|Y|\le 1} |\langle X, Y\rangle|, \qquad |T^{\mathsf{T}}| \le 2|T|. \tag{378}$$

Indeed, in Pauli coordinates $X = \sum_a x_a \sigma_a$ one has $|x|_2 \le |X| \le \sqrt{2}|x|_2$, since the eigenvalues of X^*X are $|x|_2^2 \pm |i\bar{x} \times x|$. For $X \ne 0$, take $Y = (\sqrt{2}|x|_2)^{-1} \sum_a \bar{x}_a \sigma_a$ for the lower bound; the upper bound follows from the trace pairing. Applying these two bounds to $\langle T^{\mathsf{T}} X, Y\rangle = \langle X, TY\rangle$ proves the last assertion.

Lemma 25.2. All the local maps defining Q, Q' and $C^{(2)}$ exist. There is a constant $C = C(L)$ such that, for $|s| \le 4$,

$$\begin{aligned} |Tf|_{s-\ell;y} &\le C \max_{d(y,z)\le 8L} |f|_{s;z}, \\ |\delta T f|_{s-\ell;y} &\le C\epsilon \max_{d(y,z)\le 8L} |f|_{s;z}, \end{aligned} \tag{379}$$

where $\delta T = T(\mathbf{U}) - T(\mathbf{1})$, $\ell = 0$ for $T = Q, Q^{\mathsf{T}}, Q', Q'^{\mathsf{T}}$, and $\ell = 2$ for $T = Q^{\mathsf{T}} \mathsf{m}^{-2} Q, Q'^{\mathsf{T}} \mathsf{a}' Q'$. The input f belongs to the corresponding site, top, bond or row space, and all maxima are over $z \in \Lambda_*$. Moreover,

$$\begin{aligned} |\Delta^{(2)}[\mathsf{j}]u|_{-3;y} &\le C\epsilon \max_{d(y,z)\le 8L} |u|_{-1;z}, \\ |\mathsf{m}_J(v)|_{-3;y} &\le C\epsilon \max_{d(y,z)\le 8L} |v|_{0;z}, \\ |\mathsf{g}_0(A, v)|_{-4;y} &\le C\epsilon \max_{d(y,z)\le 8L} \big(|A|_{-1;z} + |Dv|_{-1;z}\big), \\ |\mathsf{x}(A, v)|_{-3;y} &\le C\epsilon \max_{d(y,z)\le 8L} \big(|v|_{0;z} + |A|_{-1;z} + |Dv|_{-1;z}\big). \end{aligned} \tag{380}$$

■

Proof. Let c be a positive level-n bond with an endpoint $y_0 \in \Lambda_n$, $n \ge 1$, and put $\mathsf{Y}_c = B^n(c_-) \cup B^n(c_+)$. Here c may be inward; this will be needed for the scalar mean. Every site of Y_c is within sup distance $3\xi_n/2$ of y_0. Since $\mathsf{g} \ge L^2$, separation puts these sites in Ω_{n-1}. Since $y_0 \notin \Omega_{n+1}$, the

next separation puts them outside Ω_{n+2}, when that set is nonempty. Consequently their levels are among $n-1, n, n+1$, and all bonds used in this double block are inner in Y. The own-cell part of (373) gives

$$\xi_n \sup_{\mathsf{Y}_c} |\mathbf{A}| \le L\epsilon \le c_{\mathrm{av}}/2.$$

After rescaling to level-n units, the bare spacing is $\eta = \xi/\xi_n$ and $\mathbf{U} = e^{i\eta(\xi_n \mathbf{A})}$. Thus proposition 24.1 applies with the flat reference $\mathbf{1}$. It constructs the actual iterated averages and the straightened constraint on this block. In (370) the physical derivative is precisely its Q_n: the input factor ξ_n cancels the output factor ξ_n^{-1}.

For a top $y \in \Lambda_n$, use the full-frame bond $c_y = \langle y, y+\xi_n e_1\rangle$ in this argument. It need not be a row. Every contour used in the scalar mean at y lies in the nested single block $B^n(y)$; the restrictions of the actual averages are determined there, successively from finer levels. The resulting scalar mean is therefore independent of this auxiliary double block. This proves its definedness as well as that of the row maps. At level zero the assertions are immediate.

Write $Q(c,b)$ for the counting kernel. The estimates (359) and (360) give

$$\sum_b |Q(c,b)| \le C, \quad |Q(c,b)| \le C(\xi/\xi_c)^3, \quad \sum_b |\delta Q(c,b)| \le C\epsilon, \quad |\delta Q(c,b)| \le C\epsilon(\xi/\xi_c)^3.$$

The scalar transport has norm at most C and differs from the identity by at most $C\epsilon$. The weighted pairings give

$$(Q^{\mathsf{T}}\omega)(b) = \sum_c (\xi_c/\xi)^3 Q(c,b)^{\mathsf{T}}\omega(c),$$
$$(Q'^{\mathsf{T}}\sigma)(x) = \mathrm{Ad}(T_{\mathbf{U}}(y,x))^{\mathsf{T}}\sigma(y), \qquad x \in B^n(y).$$

In the second line the averaging factor cancels because $\xi_y^3 L^{-3n} = \xi^3$. Transposes of kernel entries are taken for $\langle\cdot,\cdot\rangle$ and obey (378).

Only twelve positive-level rows can contribute at a fixed bare bond. Indeed, lemma 16.3 partitions a reduced row's double block into cells of levels $n, n-1$. If $b_- \in \mathsf{Y}_c$, then $n \in \{n(b_-), n(b_-)+1\}$. At either level the unique level-n block containing b_- forces c to be one of its six incident positive bonds. There is at most one additional identity row at level zero.

For these reduced rows, the cells $\mathcal{C}(c)$ described in section 22.2 have sides ξ_c or ξ_c/L, and their centers are within $4L$ of the row position. Thus any two of their centers are within $8L$. The weighted transpose cancels the kernel factor $(\xi/\xi_c)^3$, and a change of position costs at most L^4 for $|s| \le 4$. This proves the first four bounds in (379). The two scalar maps in fact preserve the position. For the bond potential, expand once as

$$(Q^{\mathsf{T}}\mathsf{m}^{-2}Qf)(b) = \sum_c (\xi_c/\xi)^3 \xi_c^{-2} Q(c,b)^{\mathsf{T}} \sum_{b'} Q(c,b')f(b').$$

Both b and b' belong to the same Y_c, and $\xi_c^{-2} \le \xi_{y(b)}^{-2}$. The same count and scale comparison prove the weight loss of two with radius $8L$. For the scalar potential, all indices have the same position and $0 < \mathsf{a}'(y) \le \xi_y^{-2}$. Finally, expand each potential difference in its two factors, for example

$$\delta(Q^{\mathsf{T}}\mathsf{m}^{-2}Q) = \delta Q^{\mathsf{T}}\mathsf{m}^{-2}Q(\mathbf{U}) + Q(\mathbf{1})^{\mathsf{T}}\mathsf{m}^{-2}\delta Q.$$

Each summand keeps its two fine indices in the same double block and gains the factor ϵ. This proves all of (379) without enlarging the support radius by composition.

It remains to track the physical factor in the quadratic term. For $n \ge 1$ set $f_c(\widetilde{A}) = \mathcal{Q}_n(\mathbf{U};\widetilde{A})(c)$, and let $K_{\widetilde{A}}(c,b)$ be its derivative in $\widetilde{A}(b)$. Define $K'_c(u)(c,b) = \frac{d}{ds}\big|_{s=0} K_{su}(c,b)$. By (361),

$$|K'_c(u)(c,b)| \le C_2(\xi/\xi_c)^3 \sup_{\mathsf{Y}_c} |u|.$$

The second derivative of f_c is symmetric, so its quadratic part is $\frac{1}{2}\sum_b K_c'(u)(c,b)u(b)$. Rescaling this part and then polarizing (371) gives

$$\begin{aligned} C^{(2)}(u)(c) &= \tfrac{1}{2}\xi_c \sum_b K_c'(u)(c,b)u(b), \\ (\Delta^{(2)}[\mathsf{j}]u)(b) &= \xi^{-3} \sum_{\substack{c\in\Gamma_n,\ n\ge 1 \\ b_-,b_+\in \mathsf{Y}_c}} \xi_c^4 K_c'(u)(c,b)^{\mathsf{T}}\mathsf{j}(c). \end{aligned} \tag{381}$$

The factor ξ_c^4 consists of the physical factor ξ_c and the row-pairing weight ξ_c^3. The kernel vanishes when either endpoint of b is outside the double block; it depends on u and $\mathbf{U}$ only on that block. Using (378), each summand in the second line has norm at most $2C_2\xi_c|\mathsf{j}(c)|\sup_{\mathsf{Y}_c}|u|$. For $y = y(b)$,

$$\xi_c|\mathsf{j}(c)| \le \epsilon\xi_y^{-2}, \qquad \sup_{\mathsf{Y}_c}|u| \le L\xi_y^{-1}\max_{d(y,z)\le 8L}|u|_{-1;z}.$$

The twelve-row count proves the first bound in (380). There is no level-zero contribution because $C^{(2)}$ vanishes there.

Finally, $\epsilon \le 1/8$ and the own-cell amplitude bound give $|\operatorname{Ad}(\mathbf{U}(b))^{\pm 1}| \le e^{2\epsilon} < 2$ on inner bonds. Apply $|[X,Y]| \le 2|X|\,|Y|$ to (372). Endpoint positions satisfy (376). For example, a current term at an incoming bond to a site of position y is bounded by $2\epsilon L^4\xi_y^{-4}|A|_{-1;y(b)}$ after transport. There are at most three incoming and three outgoing bonds. The outgoing Dv terms have the same weight; the two terms of m_J have weight -3. These observations prove the bounds for m_J and g_0. The bound for x follows by applying the multiplier estimate to $u = A - Dv$ and estimating its direct current term. This proves (380). □

25.3. **The complex Hessian and its curvature remainder.** The Hessian differs from the covariant curl squared by terms containing the plaquette holonomy minus the identity. Adding the gradient-divergence term gives the componentwise Dirichlet Laplacian with a local curvature remainder. The calculation must be made in the ambient region before compression to the inner bonds of Y.

Lemma 25.3. Let $\mathbf{U}$ be a complex configuration on the inner bonds of A. For an inner plaquette $p = p_{\mu\nu}(x)$, $\mu < \nu$, put $Y_p = \partial\mathbf{U}(p)$, $y_0(p) = \frac{1}{2}\operatorname{tr} Y_p$, and $\mathsf{U}_\mu(x) = \operatorname{Ad}(\mathbf{U}(\langle x, x+\xi e_\mu\rangle))$. For a one-form A on the inner bonds of A, set

$$\begin{aligned} a_1 &= A_\mu(x), & a_2 &= \mathsf{U}_\mu(x)A_\nu(x+\xi e_\mu), \\ a_3 &= \operatorname{Ad}(Y_p)\mathsf{U}_\nu(x)A_\mu(x+\xi e_\nu), & a_4 &= \operatorname{Ad}(Y_p)A_\nu(x). \end{aligned}$$

With $\varsigma = (1,1,-1,-1)$ and $S_p(A) = \sum_i \varsigma_i a_i$, the ambient Hessian satisfies

$$(A, \Delta_{\mathsf{A}}(\mathbf{U})A)_\xi = \xi^3 \sum_p \Big\{ y_0(p)\langle \xi^{-1}S_p(A), \xi^{-1}S_p(A)\rangle + i\sum_{i<j}\varsigma_i\varsigma_j\langle [a_i,a_j], F_{\mathbf{U}}(p)\rangle \Big\}, \tag{382}$$

where the sum is over the inner plaquettes of A.

Suppose A contains every bare site within sup distance ξ of Y. On every plaquette p containing an inner bond of Y, assume $|Y_p - \mathbf{1}| \le 1$ and $|\operatorname{Ad}(\mathbf{U}(b'))^{\pm 1}| \le 2$ on its four bonds. For an inner bond b of Y, put

$$\mathsf{f}^\circ(b) = \xi^{-2}\max_{p\ni b}|Y_p - \mathbf{1}|, \qquad |A|_{(b)} = \max_{\substack{p\ni b,\ b'\in p \\ b' \text{ inner in } Y}} |A(b')|.$$

Then, with an absolute constant C,

$$\mathsf{L} = \Delta_Y(\mathbf{U}) + DD^{\mathsf{T}} = \Delta^\circ + \mathsf{Z}_{\mathrm{curv}}, \qquad |(\mathsf{Z}_{\mathrm{curv}}A)(b)| \le C\mathsf{f}^\circ(b)|A|_{(b)}. \tag{383}$$

Both $\Delta_Y(\mathbf{U})$ and $\mathsf{Z}_{\mathrm{curv}}$ at b depend only on the four plaquettes containing b and their bonds. ■

Proof. Moving the transports to the right in the plaquette product, as in (100), gives the algebraic identity

$$\partial(e^{it\xi A}\mathbf{U})(p) = e^{it\xi a_1}e^{it\xi a_2}e^{-it\xi a_3}e^{-it\xi a_4}Y_p.$$

The quadratic term in an ordered product of exponentials is

$$\frac{1}{2}\sum_i X_i^2 + \sum_{i<j} X_iX_j = \frac{1}{2}\Big(\sum_i X_i\Big)^2 + \frac{1}{2}\sum_{i<j}[X_i, X_j].$$

Taking $X_i = it\xi\varsigma_i a_i$, the coefficient of t^2 in $a_p(e^{it\xi A}\mathbf{U})$ is therefore

$$\frac{\xi^2}{4}\operatorname{tr}\left(S_p(A)^2Y_p + \sum_{i<j}\varsigma_i\varsigma_j[a_i, a_j]Y_p\right).$$

Since S_p is traceless, $S_p^2 = \langle S_p, S_p\rangle\mathbf{1}$. Also $Y_p = y_0\mathbf{1} + i\xi^2F_{\mathbf{U}}(p)$ by (70), and commutators are traceless. Thus this coefficient is $\xi^4/2$ times the brace in (382). The action contributes a factor ξ^{-1}, and its second derivative contributes a factor two, proving that identity for every complex $\mathbf{U}$.

Write $D^{(1)}$ for the ambient curl, $(D^{(1)}A)_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu$. The signed sum has the exact decomposition

$$\begin{aligned}\xi^{-1}S_p(A) &= (D^{(1)}A)_{\mu\nu}(x) + \varrho_p(A),\\ \varrho_p(A) &= \xi^{-1}(I - \operatorname{Ad}(Y_p))\big[\mathsf{U}_\nu(x)A_\mu(x+\xi e_\nu) + A_\nu(x)\big].\end{aligned}$$

Consequently $\Delta_{\mathsf{A}}(\mathbf{U}) = D^{(1)\mathsf{T}}D^{(1)} + Z_1$, where

$$\begin{aligned}(A, Z_1A)_\xi = \xi^3\sum_p\Big\{&(y_0-1)\langle D^{(1)}A, D^{(1)}A\rangle + 2y_0\langle D^{(1)}A, \varrho_p(A)\rangle\\ &+ y_0\langle\varrho_p(A), \varrho_p(A)\rangle + i\sum_{i<j}\varsigma_i\varsigma_j\langle[a_i, a_j], F_{\mathbf{U}}(p)\rangle\Big\}.\end{aligned}$$

Here the curl in a summand is evaluated at p.

Put $\kappa_p = |Y_p - \mathbf{1}|$. For a determinant-one two-by-two matrix, $Y_p^{-1} - \mathbf{1}$ is the adjugate of $Y_p - \mathbf{1}$ and has the same operator norm. Hence, when $\kappa_p \le 1$,

$$|y_0 - 1| \le \kappa_p, \qquad \xi^2|F_{\mathbf{U}}(p)| \le \kappa_p, \qquad |I - \operatorname{Ad}(Y_p^{\pm1})| \le 3\kappa_p.$$

The last inequality follows by writing $Y_pXY_p^{-1} - X = (Y_p - \mathbf{1})XY_p^{-1} + X(Y_p^{-1} - \mathbf{1})$ and using $|Y_p^{\pm1}| \le 2$. If $|A|_{(p)}$ denotes the maximum on the four bonds of p, the transport bounds give

$$|a_i| \le C|A|_{(p)}, \qquad |D^{(1)}A(p)| \le C\xi^{-1}|A|_{(p)}, \qquad |\varrho_p(A)| \le C\xi^{-1}\kappa_p|A|_{(p)}.$$

Polarize the form of Z_1 against a one-form supported at b, of value e with $|e| \le 1$. Each contributing plaquette contains b; there are at most four. Every term is bounded by $C\xi^{-2}\kappa_p|A|_{(p)}$, using $\kappa_p^2 \le \kappa_p$ and $|[X, X']| \le 2|X||X'|$. The factors ξ^3 in the form and the one-bond pairing cancel. The complex bilinear dual bound in (378) then gives

$$|(Z_1A)(b)| \le C\xi^{-2}\max_{p\ni b}\kappa_p\max_{p\ni b}|A|_{(p)}.$$

Only the touching plaquettes require the bounds used here.

It remains to combine curl and divergence. At a bond whose neighboring bonds and plaquettes are available in the ambient region, direct summation gives

$$\big((D^{(1)\mathsf{T}}D^{(1)} + DD^{\mathsf{T}})A\big)_\rho = \sum_\nu\nabla_\nu^{\mathsf{T}}\nabla_\nu A_\rho + \sum_{\nu\ne\rho}[\nabla_\rho, \nabla_\nu^{\mathsf{T}}]A_\nu.$$

Indeed, the off-diagonal terms are $-\nabla_\nu^{\mathsf T}\nabla_\rho A_\nu+\nabla_\rho\nabla_\nu^{\mathsf T}A_\nu$; in the remaining direction, $\nabla_\rho\nabla_\rho^{\mathsf T}=\nabla_\rho^{\mathsf T}\nabla_\rho$. For $\rho\neq\nu$, the mixed commutator is

$$\begin{aligned}([\nabla_\rho,\nabla_\nu^{\mathsf T}]f)(x)=\xi^{-2}\big[&\mathsf U_\rho(x)\mathsf U_\nu(x+\xi e_\rho-\xi e_\nu)^{-1}\\&-\mathsf U_\nu(x-\xi e_\nu)^{-1}\mathsf U_\rho(x-\xi e_\nu)\big]f(x+\xi e_\rho-\xi e_\nu).\end{aligned}$$

The bracket is $(\mathrm{Ad}(Y')-I)\mathsf U_\nu(x-\xi e_\nu)^{-1}\mathsf U_\rho(x-\xi e_\nu)$, where Y' is the holonomy based at x around this plaquette. For its standard holonomy Y_p, $Y'=VY_p^{\pm1}V^{-1}$ with V one of its bond variables. Thus the bracket has norm at most $C\kappa_p$. This plaquette contains both the output bond and the bond where A_ν is evaluated.

Finally extend A by zero from the inner bonds of Y. Every plaquette touching an inner bond b of Y, and every bond in the preceding calculation, lies in $\mathsf A$ by the collar hypothesis. The ambient divergence of A is supported on the sites of Y, so compressing the ambient $DD^{\mathsf T}$ gives the regional operator. The ambient sum $\sum_\nu\nabla_\nu^{\mathsf T}\nabla_\nu A_\rho$ compresses to $\Delta^\circ A_\rho$: its diagonal remains $6\xi^{-2}$, with missing off-diagonal values zero. Adding the mixed-commutator bound to the bound for Z_1 proves (383). The plaquettes crossing the boundary of Y remain in this calculation. Plaquettes containing no inner bond of Y contribute zero after polarization and zero extension. Hence the compressed operators use only the stated touching plaquettes and are unchanged if the ambient region is replaced by another region containing the same collar. □

25.4. **The perturbation in the forcing space.** Write $\delta T=T_{\mathbf U}-T_0$ for each operator depending on the configuration, and put

$$\mathsf e_\nu(x)=\xi^{-1}(\mathsf U_\nu(x)-I),\qquad \mathsf e'_\nu(x)=\xi^{-1}(\mathsf U_\nu(x)^{-1}-I).$$

For a site field f, or a component of a bond field, define

$$X_\nu^{[f]}(x)=\begin{cases}\mathsf e_\nu(x)f(x+\xi e_\nu)-\mathsf e'_\nu(x)f(x), & x+\xi e_\nu \text{ is an index},\\ 0, & \text{otherwise},\end{cases}$$

$$(\mathsf c f)(x)=-\xi^{-1}\left(\sum_\nu{}'\,\mathsf e_\nu(x-\xi e_\nu)+\sum_\nu{}''\,\mathsf e'_\nu(x)\right)f(x).$$

The primed sum retains the directions with $x-\xi e_\nu$ in the index set; the double-primed sum retains those with $x+\xi e_\nu$ in it. For $p\in\mathcal S$, $X^{[p]}$ is a one-form on the inner bonds of Y.

For $\Psi=(A;v,w,\sigma;p,q,r;\omega)$, define $\mathbb V^{[5]}\Psi=(f_1,\dots,f_8;X_5)$ by

$$\begin{aligned}f_1=&-\delta\Delta^\circ A-\mathsf Z_{\mathrm{curv}}A-\delta(Q^{\mathsf T}\mathsf m^{-2}Q)A+\delta D(w+p)\\&+\mathsf m_J(v)+\Delta^{(2)}[\mathsf j](A-Dv)+\delta Q^{\mathsf T}\omega,\\ f_2=&-\delta\Delta'_a v+\delta D^{\mathsf T}A,\qquad f_3=-\delta\Delta'_a w+\delta Q'^{\mathsf T}\sigma,\\ f_4=&-\delta Q'v,\\ f_5=&-\mathsf c p-\delta(Q'^{\mathsf T}\mathsf a'Q')p-\delta Q'^{\mathsf T}r+\mathsf g_0(A,v)+\delta D^{\mathsf T}\mathsf x(A,v),\\ X_5=&-X^{[p]}+\mathsf x(A,v),\\ f_6=&-\delta\Delta'_a q,\qquad f_7=\delta Q'q,\qquad f_8=-\delta QA.\end{aligned}\tag{384}$$

Here $D=D_{\mathbf U}$ and $\delta\Delta'_a=\delta\Delta^\circ+\delta(Q'^{\mathsf T}\mathsf a'Q')$. Delete absent components and terms for the other compressions. Let $\mathsf n g$ have fifth component $\delta D^{\mathsf T}X_5$ and all other components zero; put $\mathsf n=0$ for $i=1,2$.

Lemma 25.4. There is $C_V=C_V(L)\geq1$ such that, for each of the four systems,

$$\mathbb K_0^{[i]}-\mathbb K^{[i]}=\iota_0\mathbb V^{[i]},\qquad |\mathbb V^{[i]}\Psi|_y\leq C_V\epsilon\max_{d(y,z)\leq9L}|\Psi|_{0,z}.\tag{385}$$

The two solution seminorms obey

$$(1+8\epsilon)^{-1}|\Psi|_{0,y} \le |\Psi|_{\mathbf{U},y} \le (1+8\epsilon)|\Psi|_{0,y}, \tag{386}$$

and the injection satisfies

$$\iota_{\mathbf{U}} = \iota_0(I+\mathsf{n}), \qquad |\mathsf{n}g|_y \le 12L^3\epsilon \max_{d(y,z)\le 3} |g|_z. \tag{387}$$

■

Proof. If x lies in the cell at y, each connection bond based within sup distance ξ of x is tested by (373): an initial point in Y lies in a touching cell, while an initial point outside Y lies in the specified exterior collar. The same observation applies to the shifted pairs below. Since $\mathsf{U}_\nu = e^{i\xi \operatorname{ad}\mathbf{A}_\nu}$ and $|\operatorname{ad} X| \le 2|X|$, the exponential series gives

$$\begin{aligned} |\mathsf{e}_\nu|, |\mathsf{e}'_\nu| &\le 4|\mathbf{A}_\nu|, \qquad |\mathsf{e}_\nu + \mathsf{e}'_\nu| \le 8\xi|\mathbf{A}_\nu|^2, \\ |\mathsf{U}_\nu^{\pm 1}| &\le 1 + 4\xi|\mathbf{A}_\nu|, \\ |\mathsf{e}_\nu(x') - \mathsf{e}_\nu(x)| &\le 4|\mathbf{A}_\nu(x') - \mathbf{A}_\nu(x)|. \end{aligned} \tag{388}$$

The last bound holds for e'_ν as well. For example, these inequalities follow from $|e^T - I| \le 2|T|$, $|e^T + e^{-T} - 2I| \le 2|T|^2$ and $|e^T - e^{T'}| \le 2|T - T'|$ when $|T|, |T'| \le 1/4$; here $2\xi|\mathbf{A}_\nu| \le 2\epsilon \le 1/4$.

When $x + \xi e_\nu$ is an index, the same-point identities

$$\nabla_\nu f = \mathsf{U}_\nu \nabla^0_\nu f + \mathsf{e}_\nu f, \qquad \nabla^0_\nu f = \mathsf{U}_\nu^{-1}\nabla_\nu f + \mathsf{e}'_\nu f$$

bound a difference of weight $s-1$ by $1+4\epsilon$ times the other difference, plus 4ϵ times the value of weight s. If that neighbor is absent, the differences agree. Each differentiated field in (327) occurs with its value, proving (386). Also

$$(\delta D f)(b) = \mathsf{e}_\mu(b_-) f(b_+), \qquad (\delta D^{\mathsf{T}} A)(x) = \sum_\mu{}' \mathsf{e}'_\mu(x - \xi e_\mu) A_\mu(x - \xi e_\mu),$$

where the sum retains the inner incoming bonds. These operators lower the weight by one with bound $C\epsilon$ on neighboring cells. At weight -3, the three terms and adjacent-scale ratio give $12L^3\epsilon$, proving (387).

For the Laplacian, subtraction in (369) gives

$$\delta\Delta^\circ f(x) = -\xi^{-1} \sum_\nu \big[\mathsf{e}_\nu(x) f(x + \xi e_\nu) + \mathsf{e}'_\nu(x - \xi e_\nu) f(x - \xi e_\nu)\big],$$

with missing terms omitted. At an index with all six neighbors, insert $f(x+\xi e_\nu) = f(x) + \xi\nabla^0_\nu f(x)$ and its backward counterpart to obtain

$$\begin{aligned} \delta\Delta^\circ f(x) = &- \sum_\nu \big[\mathsf{e}_\nu(x)\nabla^0_\nu f(x) - \mathsf{e}'_\nu(x - \xi e_\nu)\nabla^0_\nu f(x - \xi e_\nu)\big] \\ &- \xi^{-1} \sum_\nu [\mathsf{e}_\nu(x) + \mathsf{e}'_\nu(x - \xi e_\nu)] f(x). \end{aligned} \tag{389}$$

At every index, expansion of the two terms of $\nabla^{0\mathsf{T}}_\nu X^{[f]}_\nu$ instead gives

$$\delta\Delta^\circ f = \sum_\nu \nabla^{0\mathsf{T}}_\nu X^{[f]}_\nu + \mathsf{c} f. \tag{390}$$

The primed sums in c cancel exactly the diagonal terms introduced by this expansion, including at missing neighbors.

The zeroth-order coefficient in (389) is small at the physical scale: split it into $\mathsf{e}_\nu(x) + \mathsf{e}'_\nu(x)$ and the difference of e'_ν at the two neighboring bonds. By (388) and the first-difference hypothesis, its norm after multiplication by ξ^{-1} is at most $C\epsilon\xi_y^{-2}$. The coefficient of $\mathsf{c}f$ has the same bound. If a neighbor is missing, there is a site outside Y within 2ξ of x; hence $\xi_y = \xi$ by (377). Each remaining

coefficient is then bounded directly by $4\epsilon\xi_y^{-2}$. Thus, for $s \in \{-2,-1,0\}$ and with maxima over $d(y,z) \le 3$,

$$\begin{aligned} |\delta\Delta^\circ f|_{s-2;y} &\le C\epsilon \max_z (|f|_{s;z} + |\nabla^0 f|_{s-1;z}), \\ |X^{[f]}|_{s-1;y} + |\mathsf{c}f|_{s-2;y} &\le C\epsilon \max_z |f|_{s;z}. \end{aligned} \tag{391}$$

In the first line, derivatives are needed only where both indices belong to the field's index set. The second line applies to p without a derivative estimate.

For a plaquette touching a bond based at $x \in \Delta(y)$, write its holonomy as $e^{X_1}e^{X_2}e^{X_3}e^{X_4}$, with the four signed exponents $\pm i\xi\mathbf{A}$ in boundary order. The parallel pairs and (373) give

$$\sum_j |X_j| \le 4\epsilon\xi/\xi_y, \qquad \left|\sum_j X_j\right| \le 2\epsilon\xi^2/\xi_y^2.$$

The exponential remainder bound therefore yields

$$|\partial\mathbf{U}(p) - I| \le (2\epsilon + 16\epsilon^2)\xi^2/\xi_y^2 \le 4\epsilon\xi^2/\xi_y^2.$$

The transported coefficients have norm at most two by (388). Applying lemma 25.3, and comparing the touching cells, gives

$$|\mathsf{Z}_{\mathrm{curv}}A|_{-3;y} \le C\epsilon \max_{d(y,z)\le 3} |A|_{-1;z}.$$

In particular $\delta\mathsf{L} = \delta\Delta^\circ + \mathsf{Z}_{\mathrm{curv}}$, since the curvature term vanishes at the identity.

Subtracting the system at $\mathbf{U}$ from the trivial system now gives (384). In its fifth slot, the relevant identity is

$$\begin{aligned} -\delta\Delta'_a p - \delta Q'^{\mathsf{T}} r + \mathsf{g}_0 + D^{\mathsf{T}}\mathsf{x} = {} & -\mathsf{c}p - \delta(Q'^{\mathsf{T}}\mathsf{a}'Q')p - \delta Q'^{\mathsf{T}} r + \mathsf{g}_0 + \delta D^{\mathsf{T}}\mathsf{x} \\ & + D_0^{\mathsf{T}}(-X^{[p]} + \mathsf{x}). \end{aligned}$$

This proves the operator identity in (385) for all four compressions.

The differentiated fields A, v, w, q have weights $-1, 0, -2, 0$; the first line of (391) controls their Laplacian differences in weights $-3, -2, -4, -2$. For p, the second line controls $\mathsf{c}p$ in weight -4 and $X^{[p]}$ in weight -3. The averaging, potential and current estimates of lemma 25.2, together with the first-order bounds above, give precisely the other weights in (328).

For the composed term $\delta D^{\mathsf{T}}\mathsf{x}$ at a site x, every contributing multiplier block contains both endpoints of an incoming bond, hence x itself. Its input positions and the position of x are in the same row block; their distances are at most $8L$. Its weight--3 bound is multiplied by a coefficient of size at most $4\epsilon/\xi_y$, giving weight -4. The direct current term uses neighboring cells. Thus all inputs in the defect lie within $9L$ of the output, with bound $C\epsilon$; here $\epsilon^2 \le \epsilon$. There are only finitely many component terms and the row incidence is bounded independently of the layer count. Their maximum fixes C_V and proves the remaining estimate. □

25.5. **Proof of the inverse estimate.**

Proof of proposition 25.1. First work in the trivial gauge. Enlarge B_{cs} to $\max\{1, B_{\mathrm{cs}}\}$ and choose $0 < \epsilon_* \le \epsilon_0$ so that

$$C_V\epsilon_* B_{\mathrm{cs}} e^{9L\delta_{\mathrm{cs}}} \le \tfrac{1}{4}, \qquad 12L^3\epsilon_* e^{3\delta_{\mathrm{cs}}} \le 1.$$

All local operators are defined by lemma 25.2. Suppress i in the proof. The finite-range and composition bounds following (340), together with proposition 23.1 and lemma 25.4, give

$$T := \mathbb{V}\mathbb{E}_0\iota_0 \in \mathfrak{O}(\delta_{\mathrm{cs}}, 1/4).$$

Consequently $\sum_{k\ge 0} T^k$ converges in the supremum norm. Applying the same decay bound to each term and summing gives $(I-T)^{-1} \in \mathfrak{O}(\delta_{\mathrm{cs}}, 4/3)$.

If $\mathbb{K}\Psi = 0$, then $\Psi = \mathbb{E}_0\iota_0\mathbb{V}\Psi$. Hence $g = \mathbb{V}\Psi$ satisfies $g = Tg$, so $g = 0$ and $\Psi = 0$. Finite dimensionality makes $\mathbb{K}$ invertible. Moreover,

$$\mathbb{K}\mathbb{E}_0\iota_0 g' = \iota_0(I - T)g',$$

which proves (375) by setting $g' = (I - T)^{-1}g$. The injection correction in (387) has decay constant at most one. Combining it with the norm comparison gives (374), since $2 \cdot (4/3)B_{\mathrm{cs}} \cdot 2 \le 16B_{\mathrm{cs}}$. Take $B_* = 16B_{\mathrm{cs}}$.

For a general small family, let u be its trivializing gauge on the bare collar and continue it arbitrarily to A. Conjugation acts at sites, at the initial points of bonds, and at the initial endpoints of rows. The unitary covariance of the straightening maps gives covariance of their derivatives and quadratic Taylor coefficients. In particular, the defined local averages in the trivial gauge, transported back by u, are the actual averages of $\mathbf{U}$: the contour products, their logarithms and the successive averages obey the same conjugation identities. This establishes their domains as well as their values.

Gauge invariance of the ambient action, differentiated twice, gives covariance of its Hessian. The invariant trace pairing preserves transposes and polarization; the differences and the commutators in (372) obey the same covariance. Thus, writing $\mathcal{U} = \mathrm{Ad}(u)$ on the respective component spaces,

$$\mathbb{K}[\mathbf{U}^u, \mathrm{Ad}(u)\mathbf{J}, \mathrm{Ad}(u)\mathrm{j}] = \mathcal{U}\mathbb{K}[\mathbf{U}, \mathbf{J}, \mathrm{j}]\mathcal{U}^{-1}, \qquad \iota_{\mathbf{U}^u}\mathcal{U} = \mathcal{U}\iota_{\mathbf{U}}.$$

Unitarity preserves the local solution and forcing seminorms, so the bound in the trivial gauge proves the stated bound.

The compressed Hessian uses only plaquettes touching inner bonds of Y, all of whose bonds have both endpoints in the collar. The averages and every other coefficient use inner bonds of Y and the row field. Hence the continuation of u has no effect, and $\mathbb{K}$, and therefore its inverse, have the asserted dependence. □

26. Local regular gauges

Small curvature gives a gauge in which the bond variables are close to the identity. To control the first differences of the resulting potential uniformly in the spacing, we also use a bound on the divergence of the plaquette variables. The construction smooths a comb section, proves that the smoothed section has no zero, and normalizes it to obtain the gauge.

Theorem F. Fix odd $L \ge 13$. There are absolute constants $c_* > 0$ and $B_* \ge 1$ with the following property. Use level-k units, where $1 \le k \le K$ and $\eta = L^{-k}$. Let Q be a closed cube of positive integer side l, formed from cubes of the level-k partition, and let $\widetilde{Q}$ be the concentric cube of side $l + 4$, less than half the torus period. Suppose $\alpha > 0$, $l\alpha \le c_*$ and that U is a real configuration on the bare lattice, with

$$|\partial U(p) - \mathbf{1}| < \alpha\eta^2, \qquad |(D_U^{\mathsf{T}}\partial U)(b)| < \alpha\eta^2$$

for every plaquette with a corner in $\widetilde{Q}$ and every bond with an endpoint there, respectively. There are a real gauge u on the sites of $\widetilde{Q}$ and a real potential A_0 on the bonds with both endpoints in $\widetilde{Q}$ such that $U^{u^{-1}} = e^{i\eta A_0}$ there and

$$|A_0|, \quad |\nabla^\eta A_0|, \quad |D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} A_0| < B_* l\alpha \quad \text{at every positive bond with an endpoint in } Q.$$

Here ∇^η is the ordinary first difference, and $D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}}$ is ordinary curl followed by its transpose. ■□

Proof. Reduction to a box. Lift $\widetilde{Q}$ to the covering lattice. Its side is less than half the period, so the lift identifies its sites, bonds and plaquettes with

$$\Omega = o + \eta\{0, \dots, N - 1\}^3, \qquad s = N\eta = l + 4.$$

The faces lie halfway between bare sites because L is odd. We have $0 < \eta \le 1/13$ and $5 \le s \le 5l$; put $\hat{\mu} = \eta e_\mu$. Write $\mathfrak{B}$ for the positive inner bonds and $d(x)$ for the minimum coordinate distance

to an extreme site face. Set $\Omega^{(j)} = \{x : d(x) \geq j\eta\}$. A bond is full if all four plaquettes containing it lie in the box. At such a bond the divergence $D_U^{\mathsf{T}}\partial U$ computed in the box is the given divergence. This is the matrix-valued expression of (68), with the antisymmetric extension of the plaquette variables, rather than the field-strength current $J(U)$.

We prove that $s\alpha \leq c_\square$ gives a gauge with $|A_0| < B_\square s\alpha$ on $\mathfrak{B}$ and $|\nabla^\eta A_0|, |D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} A_0| < B_\square s\alpha$ at base sites of depth at least $3/2$. This suffices with $c_* = c_\square/5$ and $B_* = 5B_\square$: sites of Q have depth at least 2, and a positive bond with an endpoint in Q has base depth at least $2 - \eta \geq 3/2$. The translated bonds in its first differences also lie in the box.

Smoothing the comb section. Use the real algebra $\mathbb{H} = \{a\mathbf{1} + iX : a \in \mathbb{R}, X \in \mathfrak{g}\}$, with scalar part $\mathrm{Re}(a\mathbf{1} + iX) = a$. For $q \in \mathbb{H}$, $q^*q = |q|^2\mathbf{1}$, its unit sphere is G, and $\langle p, q\rangle_{\mathbb{H}} = \mathrm{Re}(p^*q)$ is a real inner product with norm $|q|$. These identities follow from $X^2 = |X|^2\mathbf{1}$ for $X \in \mathfrak{g}$. In particular, multiplication by an element of G is an isometry.

Let $\phi_0(x)$ be transport from x to o, lowering the coordinates in the order $1, 2, 3$. Moving a bond through these paths inserts at most $2(N-1)$ plaquettes, as in the comb calculation of lemma 14.4. Hence

$$|\phi_0(x)^{-1} U(x, x+\hat{\mu})\phi_0(x+\hat{\mu}) - \mathbf{1}| \leq 2s\alpha\eta.$$

On quaternion-valued sections use the left action

$$(\mathcal{D}\phi)_\mu(x) = \eta^{-1}\big(U(x, x+\hat{\mu})\phi(x+\hat{\mu}) - \phi(x)\big).$$

This is a different representation from the adjoint action in D_U. Its adjoint is taken with the real inner product above and sums over the sites and positive inner bonds of the box. In particular,

$$(\mathcal{D}^*\mathcal{D}\phi)(x) = \eta^{-2} \sum_{x' \sim x, x' \in \Omega} \big(\phi(x) - U(x, x')\phi(x')\big).$$

The positive operator $\mathcal{D}^*\mathcal{D} + 1$ is invertible. Let

$$(\mathcal{D}^*\mathcal{D} + 1)\phi = \phi_0, \qquad \psi = \mathcal{D}\phi, \qquad \varepsilon_0 = \|\mathcal{D}\phi_0\|_\infty \leq 2s\alpha. \tag{392}$$

At a maximum of $|\phi|$, the inner product of $\mathcal{D}^*\mathcal{D}\phi$ with ϕ is nonnegative. Equation (392) therefore gives $\|\phi\|_\infty \leq 1$.

The equation for ψ and its boundary terms. Let $n(b)$ count the missing transverse neighbours of a bond $b = (x, \mu)$; it is zero exactly when the bond is full. For its μ component define

$$\begin{aligned}(\mathcal{L}\psi)_\mu(x) = \eta^{-2}\Big[\sum_{\nu \neq \mu} \sum_{\pm : x \pm \hat{\nu} \in \Omega} &\big(\psi_\mu(x) - U(x, x \pm \hat{\nu})\psi_\mu(x \pm \hat{\nu})\big) \\ &+ 2\psi_\mu(x) - U(x, x+\hat{\mu})\psi_\mu(x+\hat{\mu}) - U(x, x-\hat{\mu})\psi_\mu(x-\hat{\mu})\Big].\end{aligned}$$

Bond values outside $\mathfrak{B}$, and every product containing such a value, are zero. Thus the transverse directions have a Neumann boundary condition and the longitudinal direction has a Dirichlet boundary condition.

Write $Y_{\mu\nu}(x)$ for the plaquette holonomy in the order $\mu, \nu, -\mu, -\nu$, minus $\mathbf{1}$, and put $Y^-_{\mu\nu}(x) = \mathrm{Ad}(U(x, x-\hat{\nu}))Y_{\mu\nu}(x-\hat{\nu})$ when that plaquette is present. Commuting the two differences in (392) gives

$$(\mathcal{L} + 1)\psi = \mathcal{D}\phi_0 + M - W. \tag{393}$$

The terms on the right are

$$M_\mu(x) = \eta^{-3}\Big[\sum_{\nu\neq\mu:\,x+\hat\nu\in\Omega} Y_{\mu\nu}(x)\big(\phi(x)+\eta\psi_\nu(x)+\eta U(x,x+\hat\nu)\psi_\mu(x+\hat\nu)\big) - \sum_{\nu\neq\mu:\,x-\hat\nu\in\Omega} Y^-_{\mu\nu}(x)\big(\phi(x)+\eta\psi_\mu(x)\big)\Big],$$

$$W_\mu(x) = \eta^{-2}\sum_{\nu\neq\mu:\,x-\hat\nu\in\Omega} \big((\mathbf{1}+Y^-_{\mu\nu}(x))^{-1}-\mathbf{1}\big) \cdot U(x,x-\hat\nu)U(x-\hat\nu,x-\hat\nu+\hat\mu)\psi_\nu(x-\hat\nu+\hat\mu).$$

For completeness, the forward commutation uses

$$U(x,x+\hat\mu)\psi_\nu(x+\hat\mu)-\psi_\nu(x) = U(x,x+\hat\nu)\psi_\mu(x+\hat\nu)-\psi_\mu(x) + \eta^{-1}Y_{\mu\nu}(x)U(x,x+\hat\nu)U(x+\hat\nu,x+\hat\nu+\hat\mu)\phi(x+\hat\nu+\hat\mu).$$

The backward commutation is this identity at $x-\hat\nu$, transported to x; its reversed plaquette contributes $(\mathbf{1}+Y^-_{\mu\nu})^{-1}-\mathbf{1}$. Together with the longitudinal second difference these are (393), including absent neighbours.

Put $\Sigma = \|\psi\|_\infty$. The curvature bound gives $|Y|, |Y^-|, |(\mathbf{1}+Y^-)^{-1}-\mathbf{1}| \le \alpha\eta^2$. The terms containing ψ in M and W are consequently bounded by $6\alpha\Sigma$ and $2\alpha\Sigma$. For a full bond, the remaining coefficient satisfies

$$\sum_{\nu\neq\mu}(Y_{\mu\nu}-Y^-_{\mu\nu}) = \eta(D_U^{\mathsf T}\partial U)_\mu(x) + 2\sum_{\nu<\mu}\big(c_{\nu\mu}(x)-c_{\nu\mu}(x-\hat\nu)\big)\mathbf{1},$$

where $c_{\nu\mu} = \frac12 \operatorname{tr}\partial U(p_{\nu\mu})$. This follows by replacing an oppositely oriented plaquette g^{-1} by $2c\mathbf{1}-g$. Since $1-c = \frac12|g-\mathbf{1}|^2$, the norm of this coefficient is at most $\alpha\eta^3+2\alpha^2\eta^4$. At a nonfull bond there are at most three remaining plaquettes, so the bound $3\alpha\eta^2 n(b)$ suffices. We have proved

$$|M_\mu(x)| \le \alpha(1+2\alpha\eta)+3\alpha\eta^{-1}n(b)+6\alpha\Sigma, \qquad |W_\mu(x)| \le 2\alpha\Sigma. \tag{394}$$

A uniform bound at the boundary. Let $\mathcal{L}_0$ be $\mathcal{L}$ with all transports replaced by 1, acting on the sites carrying the μ bonds. Its maximum principle and the inequality

$$\langle\psi_\mu, \mathcal{L}\psi_\mu\rangle_{\mathbb H} \ge |\psi_\mu|\mathcal{L}_0|\psi_\mu|$$

give, for $\alpha \le 1/16$,

$$(\mathcal{L}_0+1)|\psi_\mu| \le \varepsilon_0 + 2\alpha + 8\alpha\Sigma + 3\alpha\eta^{-1}n(b).$$

At a zero of ψ_μ the same inequality holds directly because the off-diagonal coefficients of $\mathcal{L}_0$ are nonpositive.

The boundary forcing is controlled by a one-dimensional barrier. Choose $\kappa > 0$ so that $2\sinh(\kappa\eta/2) = \eta$, and set

$$\beta(j) = \frac{\cosh(\kappa\eta(j-(N-1)/2))}{\sinh(\kappa\eta N/2)}.$$

For $0 \le j < N$, $0 < \beta(j) \le 2$: indeed $\kappa \ge 4/5$ and $N\eta \ge 5$. Its second difference plus β vanishes at interior sites, whereas at either endpoint the Neumann second difference plus β equals η^{-1}. For $\nu \neq \mu$ put $\beta_\nu(x) = \beta((x_\nu - o_\nu)/\eta)$. The longitudinal Dirichlet terms are nonnegative, so

$$(\mathcal{L}_0+1)\sum_{\nu\neq\mu}\beta_\nu \ge \eta^{-1}n(b).$$

Comparison with $\varepsilon_0 + 2\alpha + 8\alpha\Sigma + 3\alpha\sum_{\nu\neq\mu}\beta_\nu$ gives $\Sigma \le \varepsilon_0 + 14\alpha + 8\alpha\Sigma$. Consequently,

$$\Sigma \le 2\varepsilon_0 + 28\alpha \le 10\epsilon, \qquad \epsilon = s\alpha. \tag{395}$$

The section does not vanish. In the gauge ϕ_0^{-1} write $\chi = \phi_0^{-1}\phi - \mathbf{1}$, $\widetilde{\psi} = \phi_0^{-1}\psi$ and $\widetilde{U} = U^{\phi_0^{-1}}$. Extend $\widetilde{\psi}$ by zero outside $\mathfrak{B}$, with the same zero-product convention. Equation (392) becomes

$$\chi(x) = \eta^{-1}\sum_{\nu}(\widetilde{\psi}_\nu(x) - \widetilde{\psi}_\nu(x-\hat{\nu})) - \eta^{-1}\sum_{\nu}(\widetilde{U}(x,x-\hat{\nu}) - \mathbf{1})\widetilde{\psi}_\nu(x-\hat{\nu}).$$

The last term is bounded by $3\varepsilon_0\Sigma$, and $|\chi(x+\hat{\nu}) - \chi(x)| \le \eta(\Sigma + \varepsilon_0)$; the latter follows from the definition of ψ and $|\phi| \le 1$. For any site x_0 choose a product K of k consecutive sites in each direction, containing x_0, where $k = \lceil \eta^{-1} \rceil$. It fits in Ω because $k\eta < 1 + \eta \le 2 \le s$. Summing the divergence over K leaves two faces in each direction, hence

$$\left| k^{-3}\sum_{K}\chi \right| \le \frac{6\Sigma}{k\eta} + 3\varepsilon_0\Sigma \le 6\Sigma + 3\varepsilon_0\Sigma.$$

Every site of K is at path length at most $3(k-1)\eta < 3$ from x_0. Comparing its value with the mean yields

$$\|\phi - \phi_0\|_\infty \le 9\Sigma + 3\varepsilon_0 + 3\varepsilon_0\Sigma \le \tfrac{1}{2} \tag{396}$$

when $\epsilon \le 1/1000$. Thus $\rho = |\phi|$ lies in $[1/2, 1]$ and $u = \phi/\rho$ is G-valued.

Ordinary first differences. Put

$$V = U^{u^{-1}}, \qquad \omega_\mu = \phi^{-1}\psi_\mu, \qquad \psi'(x,x') = \eta^{-1}u(x)^{-1}(U(x,x')\phi(x') - \phi(x)).$$

Then ω is quaternion-valued, $|\omega| \le 2\Sigma \le 20\epsilon$, and $|\psi'| \le \Sigma$. The definition gives

$$\rho(x')V(x,x') = \rho(x)\mathbf{1} + \eta\psi'(x,x'), \qquad V(x,x+\hat{\mu}) = \frac{\mathbf{1} + \eta\omega_\mu(x)}{|\mathbf{1} + \eta\omega_\mu(x)|}.$$

At $x \in \Omega^{(2)}$ all six neighbours occur. Substitute $\psi_\mu = \phi\omega_\mu$ into $\mathcal{L}\psi$ and use (392) to obtain

$$\begin{aligned}\rho\Delta^\eta\omega_\mu = {} & u^{-1}(\mathcal{L}\psi)_\mu + (\rho\mathbf{1} - u^{-1}\phi_0)\omega_\mu \\ & + \eta^{-1}\sum_{x'\sim x}\psi'(x,x')(\omega_\mu(x') - \omega_\mu(x)).\end{aligned} \tag{397}$$

Here Δ^η is the positive ordinary Laplacian. The coefficient of $\omega_\mu(x)$ arises from $u^{-1}\mathcal{D}^*\mathcal{D}\phi = u^{-1}\phi_0 - \rho\mathbf{1}$; the remaining differences give the displayed first-order term. Using (393), (394) with $n(b) = 0$, and (395) and (396), we obtain

$$|\Delta^\eta\omega_\mu(x)| \le 55\epsilon + 120\epsilon\max_{x'\sim x}\eta^{-1}|\omega_\mu(x') - \omega_\mu(x)|. \tag{398}$$

We give the absorption argument, so that the estimate is uniform as η tends to zero. If a quaternion-valued function f on $\Omega^{(1)}$ satisfies $|f| \le w$ and $|\Delta^\eta f| \le a + \sigma\max_{x'\sim x}\eta^{-1}|f(x') - f(x)|$ on $\Omega^{(2)}$, let

$$M_f = \max_{x\in\Omega^{(2)},\nu}\min\{1, d(x)\}|\nabla^\eta_\nu f(x)|.$$

Choose a maximizing site x_1 and put $d_1 = \min\{1, d(x_1)\}$. If $d_1 < 64\eta$, then $M_f \le 128w$. Otherwise the cube of radius $r = d_1/4 \ge 16\eta$ about x_1 and its neighbours have depth at least $d_1/2$. All differences in its equation are bounded by $2M_f/d_1$. Apply proposition 18.1(b), with $F = 0$, to the real scalar function $\langle e, f\rangle_{\mathbb{H}}$ for each unit quaternion e. The normalized L^2 norm is at most $6w$. Taking the supremum over e gives, with $c_I \ge 1$,

$$M_f \le 4c_I(6w + a/16) + \tfrac{1}{2}c_I\sigma M_f.$$

Thus, if $c_I\sigma \le 1$, $M_f \le \max\{128w, 8c_I(6w + a/16)\}$. Apply this to (398) with $w = 20\epsilon$, $a = 55\epsilon$, and $\sigma = 120\epsilon$. For $\epsilon \le (120c_I)^{-1}$ it follows that

$$|\nabla^\eta_\nu\omega_\mu(x)| \le 2560c_I\epsilon \qquad (d(x) \ge 1).$$

Finally, for $q = a\mathbf{1} + iX \in \mathbb{H}$ with $a > 0$, set

$$\Theta(q) = a^{-1} X \int_0^1 \frac{dt}{1 + t^2 |X|^2 / a^2}.$$

The elementary formula for e^{iX} shows that $e^{i\Theta(q)} = q/|q|$ and $\Theta(\mathbf{1}) = 0$. The map is smooth on $a > 0$ and has an absolute Lipschitz constant $c_\Theta \geq 1$ on $|q - \mathbf{1}| \leq 1/2$. Set $A_0 = \eta^{-1}\Theta(\mathbf{1} + \eta\omega)$ and $C_M = 2560 c_I$. Then $e^{i\eta A_0} = V$, $|A_0| \leq 20 c_\Theta \epsilon$, and $|\nabla^\eta A_0| \leq c_\Theta C_M \epsilon$ at depth at least 1. The arguments of Θ lie in its stated ball because $20\eta\epsilon \leq 1/2$.

We also need the ordinary curl-divergence of this potential. For $a = (a_1, a_2, a_3, a_4) \in \mathfrak{g}^4$, put

$$P(a) = e^{ia_1} e^{ia_2} e^{-ia_3} e^{-ia_4} - \mathbf{1} - i(a_1 + a_2 - a_3 - a_4).$$

Since $P(0) = P'(0) = 0$, the maximum of $\|P''\|$ on $\max_i |a_i| \leq 1$ gives an absolute $c_P \geq 1$ such that

$$|P(a)| \leq c_P \tau^2, \qquad |P(a) - P(a')| \leq c_P \tau \max_i |a_i - a_i'| \quad (\max_i |a_i|, \max_i |a_i'| \leq \tau \leq 1).$$

Let $b = (x, \mu)$ be full, and let a_0 and a_1 be the maximum of $|A|$ and of the ordinary first differences between bonds in its four plaquettes. For $V = e^{i\eta A}$, expand each plaquette as $\partial V = \mathbf{1} + i\eta^2 D_{\mathbf{1}} A + P$, with the orientation signs in the linear term. In the divergence, the identity terms cancel. At each of the two directions $\nu \neq \mu$, the three errors are bounded by

$$4\eta^3 a_0 a_1, \qquad c_P \eta^3 a_0 a_1, \qquad 2 c_P \eta^3 a_0^3.$$

Indeed, $|D_{\mathbf{1}} A| \leq 2a_1$, $|\operatorname{Ad}(V(x, x - \hat{\nu})) - 1| \leq 2\eta a_0$, and the arguments of P at the two translated plaquettes differ by at most $\eta^2 a_1$. Dividing by η^3 yields

$$|(D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} A)(b)| \leq \eta^{-2} |(D_V^{\mathsf{T}} \partial V)(b)| + 2(4 + c_P) a_0 a_1 + 4 c_P a_0^3, \qquad \eta a_0 \leq 1. \tag{399}$$

The full plaquette divergence transforms by conjugation under a real gauge, so its norm is unchanged.

If $d(x) \geq 3/2$, every initial point of the four-plaquette star has depth at least $3/2 - \eta \geq 1$. Thus for the constructed potential we may use $a_0 \leq 20 c_\Theta \epsilon \leq 1$ and $a_1 \leq c_\Theta C_M \epsilon$ in (399). Since $V = U^{u^{-1}}$, its first term is at most $\alpha \leq \epsilon$. We obtain

$$|D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} A_0| \leq \big[1 + 2(4 + c_P) c_\Theta C_M + 80 c_P c_\Theta\big] \epsilon.$$

Choose

$$c_\square \leq \min\{1/1000, (120 c_I)^{-1}, (20 c_\Theta)^{-1}\},$$

and choose $B_\square$ larger than 1, $20 c_\Theta$, $c_\Theta C_M$ and $1 + 2(4 + c_P) c_\Theta C_M + 80 c_P c_\Theta$. The box estimates follow. With $c_* = c_\square / 5$ and $B_* = 5 B_\square$, they give all three asserted bounds on Q. □

27. The inverse for regular configurations

Bałaban studied regularity and decay of lattice propagators in a background gauge field [Bał85b]. Local gauges reduce the constrained system to the small-field inverse of proposition 25.1. Near the boundary of the ambient region, the local system instead has only the bare scale; its two massive operators can be inverted directly. A partition of unity combines these local inverses. The error is small because its output lies away from the support of the input cutoff, so local exponential decay supplies a factor exponentially small in the block size.

Fix odd $L \geq 13$. Keep the physical scales $\xi = L^{-t}$, $\xi_n = L^n \xi$ and the spaces, pairings, positions and norms of section 25. Here $(\Omega_n)_{n \leq t}$ is separated with block parameter $\mathfrak{M}$ and relative gap parameter $R_1 \geq L$, both powers of L, and the ambient region is $\mathsf{A} = \Omega_0$. We assume

$$\ell \geq 2L\mathfrak{M}\xi_t.$$

The current $\mathbf{J}$ and the row multiplier j remain independent variables. Empty index sets retain the conventions already used. Throughout this section C denotes a constant depending only on L, whose value may increase between estimates.

Definition 27.1. Let $\mathsf{Q}_n^\natural$ consist of the embedded compact cubes which are unions of $\mathfrak{M}$-cubes at level n, have side at most $12\mathfrak{M}\xi_n$, and lie in $\mathcal{L}_n \cup \mathcal{L}_{n+1}$. There is no requirement that such a cube meet $\mathcal{L}_n$. For $\alpha_0, \alpha_1, \gamma_0 > 0$, the class $\mathfrak{P}_t^\natural((\Omega_n); \alpha_0, \alpha_1, \gamma_0)$ consists of pairs $(\mathbf{U}, \mathbf{J})$ on the inner bare bonds of Ω_0 for which $\mathbf{U} = e^{i\xi A'} U$, with U G-valued and A' a $\mathfrak{g}^c$-valued one-form, satisfying the following conditions for every $0 \le n \le t$.

(i) If the corners of a bare plaquette p lie in Ω_0 and its base point lies in $\mathcal{L}_n$, then

$$|\partial U(p) - I| < \alpha_0 \xi^2 \xi_n^{-2}, \qquad |\partial \mathbf{U}(p) - I| < \alpha_0 \xi^2 \xi_n^{-2}.$$

(ii) For each $\mathsf{q} \in \mathsf{Q}_n^\natural$, there are a cubic box $\mathsf{q} \subset \square \subset \Omega_0$, a G-valued gauge u on its bare sites, and a real one-form A on its inner bonds such that

$$U^u = e^{i\xi A}, \qquad \xi_n |A| \le \alpha_0, \qquad \xi_n^2 |\nabla^\xi A| \le \alpha_0.$$

The difference in this display is the ordinary first difference.

(iii) For each positive inner bond b with $b_- \in \mathcal{L}_n$ and each defined covariant difference based at $x \in \mathcal{L}_n$,

$$\xi_n |A'(b)| < \alpha_1, \qquad \xi_n^2 |(\nabla_\mu^U A'_\nu)(x)| < \alpha_1.$$

The second condition is imposed exactly when the three bonds used by that difference belong to the inner bonds of Ω_0.

(iv) At every positive inner bond with $b_- \in \mathcal{L}_n$, $|\mathbf{J}(b)| < \gamma_0 \xi_n^{-3}$.

■

We use the operators of section 25 with this ambient region. Their averaging maps will be shown to exist in lemma 27.6. We suppress $[i]$ for a fixed $i \in \{1, 2, 4, 5\}$, and write $|\Psi|_y$ and ι for the covariant solution seminorm and forcing injection. The forcing space is $\mathcal{F} = \mathcal{F}^{[i]}$.

Theorem G. There are a power $\mathfrak{M}_*$ of L and positive constants $a_{\mathrm{reg}}, \delta_{\mathrm{reg}}, B_{\mathrm{reg}}$, depending only on L, with the following property. Suppose $\mathfrak{M} \ge \mathfrak{M}_*$, $(\mathbf{U}, \mathbf{J}) \in \mathfrak{P}_t^\natural((\Omega_n); \alpha_0, \alpha_1, \gamma_0)$, and

$$0 < \alpha_0, \alpha_1, \gamma_0 \le a_{\mathrm{reg}}, \qquad |\mathsf{j}|_{-3;y} \le \gamma' \le a_{\mathrm{reg}} \quad (y \in \Lambda_*), \qquad \gamma' \ge 0.$$

Then each of the four systems is defined and invertible. For every $g \in \mathcal{F}$ and $y \in \Lambda_*$,

$$\mathbb{E} = \mathbb{K}^{-1}, \qquad |\mathbb{E}\iota g|_y \le B_{\mathrm{reg}} \sup_z e^{-\delta_{\mathrm{reg}} d(y,z)} |g|_z. \tag{400}$$

The constants are uniform in the volume and number of layers.

■□

The proof occupies the rest of the section. The local inverses decay at twice the rate used in (400). One half of this decay is retained in the global estimate; the other half is spent on the distance between the input cutoff and the support of its commutator.

27.1. **Pieces and cutoffs.** Put $s_n = \mathfrak{M}\xi_n$, and let $\mathcal{C}_n$ be the partition into cubes of side s_n. For a cube $\kappa \in \mathcal{C}_n$, write $\kappa^{[r]}$ for its concentric enlargement of side $(1 + 2r)s_n$. The enlargements used below are embedded cubes, since $\ell \ge 2L\mathfrak{M}\xi_t$ and $L \ge 13$. Fix the local parameters

$$\mathfrak{m}_\flat = \mathfrak{M}_\dagger, \qquad \mathfrak{g}_\flat = L\mathfrak{M}_\dagger.$$

In choosing $\mathfrak{M}_*$ we require

$$\mathfrak{M} \ge L\mathfrak{m}_\flat, \qquad 3\mathfrak{g}_\flat + 1 \le \tfrac{1}{2}\mathfrak{M}, \qquad \mathfrak{M} > 30L, \qquad \mathfrak{M} \ge 16(\rho + 1)L^2, \qquad \rho = 3L + 6. \tag{401}$$

These conditions involve only L. Further increases of $\mathfrak{M}_*$ will control the cutoff derivatives and the remainder.

Definition 27.2. For $1 \le n \le t$, divide the cubes of $\mathcal{C}_n$ contained in $\mathcal{L}_n$ according as their distance from Ω_n^c is less than $2s_n$ or at least $2s_n$; denote their unions by Buf_n and Deep_n, respectively. Put $\mathrm{Deep}_0 = \mathcal{L}_0$ and $\mathrm{Buf}_{t+1} = \emptyset$. The pieces of size n are

$$\mathcal{P}_n = \{\kappa \in \mathcal{C}_n : \kappa \subset \mathrm{Deep}_n \cup \mathrm{Buf}_{n+1}\}, \qquad \mathcal{P} = \bigcup_{n=0}^{t} \mathcal{P}_n.$$

A piece has a unique size $n(\kappa)$, since the cube sides are distinct. A size-zero piece with $\kappa^{[2]} \cap \Omega_1 = \emptyset$ is a boundary piece; all other pieces are regular.

Fix a nonincreasing $\chi \in C^\infty(\mathbb{R})$ equal to 1 on $(-\infty, 0]$ and to 0 on $[1, \infty)$. If z and s are the center and side of κ, set

$$h_{\kappa;a,b}(x) = \prod_{\mu=1}^{3} \chi\Big(\frac{|x_\mu - z_\mu| - (1/2 + a)s}{(b-a)s}\Big), \qquad 0 \le a < b,$$

using least coordinate distances on the torus, and put

$$\widetilde{h}_\kappa = h_{\kappa;1/3,2/3}, \qquad \phi_\kappa = h_{\kappa;0,1/12}, \qquad \theta_\kappa = \frac{\phi_\kappa}{\sum_{\kappa'} \phi_{\kappa'}}. \tag{402}$$

The last expression is taken on the open set where its denominator is positive; it contains Ω_0 by lemma 27.5.

For a regular piece $\kappa \in \mathcal{P}_n$, let $e_m = 2\mathsf{g}_\flat \xi_m$ and define

$$\begin{aligned} \Omega_{n+1}^\kappa &= \Omega_{n+1} \cap \kappa^{[1]}, \\ \Omega_n^\kappa &= \kappa^{[1]} + [-e_n, e_n]^3, \\ \Omega_m^\kappa &= \Omega_{m+1}^\kappa + [-e_m, e_m]^3 \quad (0 \le m < n). \end{aligned} \tag{403}$$

Here $\Omega_{t+1} = \emptyset$; if $n = t$ the first line is only this empty auxiliary member. For a boundary piece put $\Omega_0^\kappa = \kappa^{[1]} \cap \Omega_0$ and $\Omega_m^\kappa = \emptyset$ for $m \ge 1$. Intersections in these definitions are cube-wise intersections in $\mathcal{C}_n$ (in $\mathcal{C}_0$ for a boundary piece). Complete each local sequence by empty members to the same top index t, retaining the spacing $\xi = L^{-t}$. Local objects carry a superscript or subscript κ. ∎

For the local system define the row multiplier purely by restriction:

$$\mathsf{j}^\kappa(c) = \begin{cases} \mathsf{j}(c), & \text{the midpoint of } c \text{ lies in } \kappa^{[5/6]}, \\ 0, & \text{otherwise.} \end{cases} \tag{404}$$

The first case uses the common rows in lemma 27.5. Let $\mathbb{K}_\kappa$ be the system for this local sequence, with row multiplier j^κ and the ambient Hessian on Ω_0. Its inverse $\mathbb{E}_\kappa$ is constructed in lemma 27.6.

27.2. **Metric and geometry of the pieces.** The local regions are smaller than the original regions. To compare their distances, we must account for the extra length charged to segments lying on a common face of two closed layers.

Lemma 27.3. For a separated sequence with powers $\mathfrak{m}, R \ge L$ of L as block and relative gap parameters, put

$$i(x) = \max\{m \ge 1 : x \in \Omega_m\}, \qquad g(x) = \xi_{i(x)}^{-1},$$

where $i(x) = 0$ when $x \notin \Omega_1$. Then

$$d_{\mathbf{B}}(x, x') = \inf_{\Gamma \ni x, x'} \int_\Gamma g \, d\ell^1, \tag{405}$$

where the infimum is over connected finite unions of closed segments anywhere in the torus. Consequently, for two such sequences with the same spacings and $\widehat{\Omega}_m \subset \Omega_m$ for every m, their distances satisfy $\widehat{d} \ge d$. ∎

Proof. We first relate the integral to the outer length in (92). Cut any finite segment family into pieces of sup length less than $\ell/4$. Lifts identify the intersection of two pieces with a point or a Euclidean segment. Subdividing at intersection endpoints and removing repeated overlaps gives injectively parametrized segments $p_i + sv_i$, $0 \le s \le 1$, which meet only in finitely many points. The trace of a cube on each segment is a finite union of intervals: only finitely many translates of a lifted cube meet it, and each translate imposes linear inequalities on s. The same holds for regions, layers and their complements.

For finite unions of parameter intervals $J_i \subset [0,1]$, with endpoints included or omitted, set $Z = \bigcup_i \{p_i + sv_i : s \in J_i\}$. We claim that

$$\operatorname{length}(Z) = \sum_i |v_i|_1 |J_i|. \tag{406}$$

The segments over the closures of the intervals give the upper bound. For the reverse bound, fix a coordinate μ. For a segment with $v_\mu \neq 0$, let $n(p, v, J; s)$ count the parameters in J whose μ-coordinate equals s modulo ℓ. By lifting that coordinate and dividing its image into periods,

$$\int_0^\ell n(p, v, J; s)\, ds = |v_\mu|\, |J|.$$

Except at finitely many coordinate values, the number of points of Z in the slice $x_\mu = s$ is the sum of these counts for its nonoverlapping segments. Every finite segment cover (p'_k, v'_k) of Z supplies at least as many parameter hits in that slice. Exclude also the finitely many slices containing a covering segment constant in this coordinate. Integration gives $\sum_i |v_{i,\mu}||J_i| \le \sum_k |v'_{k,\mu}|$. Sum over μ and take the infimum over covers to prove (406).

For the finite-valued densities used here, define the integral on Γ by summing $|v_i|_1 \int_0^1 g(p_i + sv_i)\, ds$ over such a nonoverlapping representation. A common subdivision and an affine change of parameter show that this value is independent of the representation. Formula (406) and (93) give

$$|\Gamma|_{\mathbf{B}} = \int_\Gamma w\, d\ell^1, \qquad w = \sum_{m=1}^{t} \xi_m^{-1} \mathbf{1}_{\mathcal{L}_m} + \xi_0^{-1} \mathbf{1}_{T_\ell \setminus \Omega_1}.$$

For a representation with overlaps the corresponding sum is an upper bound. Integrals of nonnegative densities are monotone under inclusion, by retaining the appropriate subdivided edges.

At every point, g is one of the nonnegative terms defining w. Moreover $w = g$ except on the boundaries of Ω_m, $m \ge 2$: an additional layer contribution at a point of maximum depth lies in $\mathcal{L}_{m-1} \cap \Omega_m \subset \partial\Omega_m$. Thus $w \ge g$ gives the inequality $\ge$ in (405).

For the other inequality, a connected finite segment union containing distinct x, x' contains an injective polygonal path between them. Indeed, subdivide at endpoints and intersections; the resulting finite edge graph is connected, and a chain with the fewest edges has no repeated vertex. Its integral of g is no larger. Cut this path at the endpoints of all its parameter intervals in $q \cap \Omega_m^c$, where $m \ge 2$ and q is a defining $\mathfrak{m}$-cube of Ω_m. Each resulting segment either meets all these faces only at its endpoints or lies in one such set $q \cap \Omega_m^c$.

For a segment $[u, u+v]$ of the latter kind, separation gives $q \cap \Omega_{m+1} = \emptyset$: every point of q is within $\mathfrak{m}\xi_m < R\mathfrak{m}\xi_m$ of Ω_m^c. Hence $g = \xi_m^{-1}$ throughout q, and $w = \xi_m^{-1}$ in its interior. Lift the segment to the Euclidean cube of side $\mathfrak{m}\xi_m$ centered at z. This is possible in one lift because different translates of that cube are positively separated. Put $a = z - u$ and $a' = z - u - v$ in this lift. For $0 < \sigma < 1$ replace the segment by the three successive segments with vectors σa, $(1-\sigma)v$, and $-\sigma a'$. Except at the two original endpoints they lie in the interior of q, and their weighted cost is at most

$$\xi_m^{-1}(\sigma|a|_1 + (1-\sigma)|v|_1 + \sigma|a'|_1) \le \xi_m^{-1}|v|_1 + 3\sigma\mathfrak{m}.$$

The other path segments have equal w- and g-costs. If there are r face segments, the modified connected union therefore has weighted length at most $\int_\Gamma g\, d\ell^1 + 3\sigma\mathfrak{m}r$. Letting $\sigma \downarrow 0$ proves the reverse inequality. Points have zero length, so $x = x'$ causes no exception; if Ω_0 is empty, both

densities are the constant ξ_0^{-1}. Finally, region inclusion decreases the maximum-depth index and increases g, proving the distance comparison. □

Lemma 27.4. For an inner-frame row $c \in \Gamma_n$ let

$$\Delta_c = \Delta^n(c_-) \cup \Delta^n(c_+), \qquad \mathsf{Y}_c = B^n(c_-) \cup B^n(c_+).$$

The row box Δ_c is a union of top cells of side ξ_n or ξ_n/L. The positions and base points of c, of the bare sites in Y_c, and of bonds with both endpoints there belong to this union. Any two points of Δ_c can be joined inside it at weighted cost at most $3L+3$. In particular, $\Delta_c \subset \mathsf{N}(y)$ for every top cell of the union.

For a sequence with gap parameter $\mathsf{g} \geq L^2$, if $d(y,y') \leq \rho = 3L+6$, then

$$L^{-1}\xi_y \leq \xi_{y'} \leq L\xi_y, \qquad |y-y'|_\infty \leq \rho L \xi_y, \qquad \mathsf{N}(y) \subset \{x : |x-y|_\infty \leq (\rho+1)L\xi_y\}. \tag{407}$$

If a real C^1 function h satisfies $\xi_y |\partial_\mu h| \leq \vartheta^{-1}$ on every cell $\Delta(y)$, then, for every connected finite segment union $\Gamma \subset \Omega_0$ and $p, p' \in \Gamma$,

$$|h(p) - h(p')| \leq \vartheta^{-1} |\Gamma|_{\mathbf{B}}. \tag{408}$$

■

Proof. By lemma 16.3(ii), at least one endpoint of c belongs to Λ_n. Its level-n cube is its top cell; the other is either another such cell or a union of level-$(n-1)$ cells. There is at most one outer endpoint, and at level zero both endpoints of an inner row belong to Λ_0. Bare sites have their positions in these cells, and the row box is convex, so it contains all the relevant midpoints. For $p \in \Delta^n(e)$, $p' \in \Delta^n(e')$, with $e, e' \in \{c_-, c_+\}$, use $[p,e] \cup [e,e'] \cup [e',p']$. Each endpoint cube receives at most $3\xi_n$ of ℓ^1 length. A segment issuing from a lattice site meets cell faces only finitely often, since the faces have half-integral bare coordinates and L is odd. The weights are therefore ξ_n^{-1}, or $L\xi_n^{-1}$ in the one possible outer cube. This proves the cost bound and the assertion about $\mathsf{N}(y)$.

Since $\rho < L^2 \leq \mathsf{g}$, the layer-gap estimate (230), with g in place of $R_1\mathfrak{M}$, gives the scale comparison in (407). For $y \in \Lambda_j$, a connecting set of weighted length less than g cannot meet Ω_{j+2}. Otherwise the distance from Ω_{j+2} takes every value between 0 and $\mathsf{g}\xi_{j+1}$ on that set. The corresponding points lie in $\mathcal{L}_{j+1}$, and the interval-image argument in section 12.2 charges length at least $\mathsf{g}\xi_{j+1}$ there, a contradiction. All its points consequently count with weight at least $(L\xi_y)^{-1}$. Taking the infimum gives $|y-y'|_\infty \leq L\xi_y d(y,y')$. Add the radius $\xi_{y'}/2 \leq L\xi_y/2$ of the endpoint cell to obtain the last inclusion.

Finally, extract a polygonal path in Γ as in the preceding proof. At a point of maximum depth m, the derivative bound is $\vartheta^{-1}\xi_m^{-1}$, while the closed-layer density is at least ξ_m^{-1}. Integrating the derivative of h on each segment and summing proves (408). □

Lemma 27.5. There are constants $N_{\mathrm{G}}, C_{\mathrm{G}} \geq 1$, depending only on L, with the following properties.

(a) The pieces cover Ω_0 and have disjoint interiors. For a regular $\kappa \in \mathcal{P}_n$,

$$\kappa^{[2]} \subset \mathcal{L}_n \cup \mathcal{L}_{n+1}, \qquad \operatorname{dist}_\infty(\kappa^{[1]}, \Omega_{n+2}) > 9 s_{n+1}. \tag{409}$$

For a boundary piece, $\kappa^{[2]} \cap \Omega_0 \subset \mathcal{L}_0$, and the same distance inequality holds. Empty final regions are understood throughout. If κ' meets $\kappa^{[1]}$, their sizes differ by at most one. Each of the following numbers is at most N_{G}: pieces meeting $\kappa^{[1]}$, enlargements $\kappa'^{[1]}$ containing a fixed point, and pieces with $\kappa'^{[1]} \cap \kappa^{[1]} \cap \Omega_0 \neq \emptyset$.

(b) A regular local sequence is $(\mathfrak{m}_\flat, \mathsf{g}_\flat)$-separated; its region Ω_0^κ and its bare collar lie in $\kappa^{[3/2]} \subset \Omega_0$. A boundary local sequence is $(\mathfrak{M}, R_1\mathfrak{M})$-separated. For every piece,

$$\Omega_m^\kappa \subset \Omega_m, \qquad d^\kappa \geq d. \tag{410}$$

The local and global indices with basepoint in $\kappa^{[5/6]}$ coincide, with their positions, cells, kinds and weights.

(c) The functions in (402) are smooth on an open neighborhood of Ω_0, and

$$\sum_{\kappa} \theta_\kappa = 1, \qquad 0 \le \theta_\kappa \le 1, \qquad \operatorname{supp}_{\Omega_0} \theta_\kappa \subset \kappa^{[1/12]}.$$

For either the global or the local sequence, for every top y and $h \in \{\theta_\kappa, \widetilde{h}_\kappa\}$,

$$|h| \le 1, \qquad \xi_y |\partial h| \le C_{\mathrm{G}}/\mathfrak{M}, \qquad \xi_y^2 |\partial^2 h| \le C_{\mathrm{G}}/\mathfrak{M} \quad \text{on } \mathsf{N}(y). \tag{411}$$

Here the maxima range over coordinate derivatives, and the neighborhood is taken in the sequence concerned.

(d) Define global sets of tops

$$\begin{aligned} S'_\kappa &= \{y : \mathsf{N}(y) \cap \kappa^{[1/12]} \neq \emptyset\}, \\ S_\kappa &= \{y : \widetilde{h}_\kappa \text{ is not constant on } \mathsf{N}(y)\}. \end{aligned} \tag{412}$$

Both lie in $\kappa^{[5/6]}$, and

$$d(y, y') \ge D_0 := \frac{\mathfrak{M}}{8L} \qquad (y \in S_\kappa,\ y' \in S'_\kappa). \tag{413}$$

If $\mathsf{N}^\kappa(y)$ meets $\kappa^{[2/3]}$, then its tops and cells are global ones and

$$\mathsf{N}^\kappa(y) \subset \kappa^{[19/24]}, \qquad \mathsf{N}^\kappa(y) \subset \mathsf{N}(y). \tag{414}$$

All local indices at these positions have basepoints in $\kappa^{[5/6]}$. In particular a local neighborhood meeting $\kappa^{[1/12]}$, or on which $\widetilde{h}_\kappa$ is not constant, has its top in S'_κ, respectively S_κ.

■

Proof. Cover and margins. By the nested partitions and lemma 12.2, each layer is the union of the cubes of $\mathcal{C}_n$ contained in it. Splitting these cubes into Deep_n and Buf_n, and then subdividing the latter at level $n-1$, proves the covering assertion. Each piece lies either in Deep_n or in Buf_{n+1}, exclusively. In the second case its unique parent in $\mathcal{C}_{n+1}$ lies in Buf_{n+1}. Pieces at different sizes have disjoint interiors: either they lie in different layers, or the smaller is in a buffer parent and the larger in a different deep cube at that parent's level.

For a regular piece there are three cases. If $n \ge 1$ and $\kappa \subset \mathrm{Deep}_n$, its $2s_n$ margin gives $\kappa^{[2]} \subset \Omega_n$, and $\kappa \subset \Omega_{n+1}^c$ separates it from Ω_{n+2} by more than $13s_{n+1}$. If $\kappa \subset \mathrm{Buf}_{n+1}$, its distance from Ω_n^c exceeds $13s_n$. Every point of its parent is within $3s_{n+1}$ of Ω_{n+1}^c, so its distance from Ω_{n+2} exceeds $10s_{n+1}$. Finally, if $n = 0$ and $\kappa \subset \mathcal{L}_0$ is regular, then $\kappa^{[2]}$ meets Ω_1. Every point of κ is within $3s_0$ of Ω_1, and hence more than $10s_0$ from Ω_0^c; its distance from Ω_2 exceeds $13s_1$. These facts prove $\kappa^{[2]} \subset \Omega_n$, exclude Ω_{n+2} there, and give (409). Dividing its cubes according to membership in Ω_{n+1} proves the asserted two-layer containment. For a boundary piece the definition excludes Ω_1 from $\kappa^{[2]}$ and separation gives the same margin from Ω_2.

If a piece κ' meets $\kappa^{[1]}$, a point of intersection is interior to $\kappa^{[2]}$, so its layer has index n or $n+1$. A piece in layer m has size m or $m-1$; thus the sizes differ by at most one. Counting cubes of side s_n/L that meet $\kappa^{[1]}$ bounds the first number in (a). At a point of Ω_0, a piece containing that point shows that all enlargements through it have at most three sizes; at each size their number is bounded by a fixed coordinate count. Outside Ω_0 only size-zero boundary pieces can occur. If two enlargements meet in Ω_0, their sizes differ by at most two, and all the points of κ' lie in the image of a cube concentric with κ of side $(4L^2+3)s_n$. Their disjoint interiors have volumes at least $(s_n/L^2)^3$. The volume ratio bounds the last number by a constant depending only on L, and one such constant N_{G} covers all three counts.

Local sequences and common indices. Since $\mathfrak{M} \ge L\mathfrak{m}_\flat$ and both are powers of L, $\kappa^{[1]}$ is a union of $\mathfrak{m}_\flat$-cubes at every level at most $n+1$. Each e_m is an integer multiple of $\mathfrak{m}_\flat \xi_m$. Thus (403) gives

nested regions of the required grids, with successive gaps at least $e_m = 2\mathsf{g}_\flat\xi_m > \mathsf{g}_\flat\xi_m$. Completing by empty members imposes no further gap condition. Moreover

$$\sum_{m=0}^{n} e_m \le 3\mathsf{g}_\flat\xi_n \le \tfrac{1}{2}s_n - \xi.$$

The region Ω_0^κ and its bare collar therefore lie in $\kappa^{[3/2]} \subset \kappa^{[2]} \subset \Omega_n$. This proves local separation, the collar assertion and $\Omega_m^\kappa \subset \Omega_m$. A boundary local sequence has only its level-zero region, a union of $\mathfrak{M}$-cubes, and is separated with the stated parameters. The distance comparison follows from lemma 27.3.

At every interior point of $\kappa^{[1]}$ the lists of memberships in Ω_m and Ω_m^κ agree. For $m \le n$ this follows from the containments just proved, for $m = n+1$ from the cube-wise intersection, and for larger m from the exclusion of Ω_{n+2}. The boundary case uses the same argument with only $m = 0$. Cube-wise intersection agrees with point-set membership in this interior, since the grids are nested.

Any global row box meeting $\kappa^{[1]}$ has level at most $n+1$. Indeed, a row of level $m \ge n+2$ has an endpoint in Ω_m, and its box lies within $2\xi_m$ of that endpoint, whereas (409) and separation put $\kappa^{[1]}$ more than $9s_{m-1}$ from Ω_m. This is impossible since $2\xi_m = 2L\xi_{m-1} < s_{m-1}$. The same level bound holds for a global cell meeting $\kappa^{[1]}$, and for every local row or top by construction. For an index with basepoint in $\kappa^{[5/6]}$, its sites, endpoints, position and cell thus lie within $\xi/2 + 3\xi_{n+1}/2 < s_n/6$ of that basepoint, inside $\kappa^{[1]}$. Their region memberships agree. This determines the same bare-site, inner-bond, top and row indices, with the same positions, cells, kinds and weights. In particular no point on a partition face is being treated as a bare site.

The partition and its derivatives. Each ϕ_κ is 1 on its piece and vanishes outside $\kappa^{[1/12]}$, so the cover gives $\Phi := \sum \phi_\kappa \ge 1$ on Ω_0. The cutoff factors are constant near the coordinate cusp and the period cut; elsewhere they are smooth. Hence Φ is smooth and positive on an open neighborhood of Ω_0, and the partition and support assertions follow.

For either sequence, let $x \in \mathsf{N}(y)$ lie in the cell of a nearby top of level j. Then $\xi_j \ge \xi_y/L$ by lemma 27.4, and the maximum global depth at x is at least j, also for a local cell because $\Omega_j^\kappa \subset \Omega_j$. If a cutoff belonging to $\kappa' \in \mathcal{P}_{n'}$ is not constant near x, then $x \in \kappa'^{[2]}$ and $x \notin \Omega_{n'+2}$. Consequently

$$\frac{\xi_y}{s_{n'}} \le \frac{L^2}{\mathfrak{M}}.$$

At most N_{G} nonconstant cutoff factors occur at a point. Their first and second derivatives have bounds $C/s_{n'}$ and $C/s_{n'}^2$. Differentiating $\theta_\kappa = \phi_\kappa/\Phi$, with $\Phi \ge 1$, therefore gives bounds $C(L)/\mathfrak{M}$ and $C(L)/\mathfrak{M}^2$ after multiplication by ξ_y and ξ_y^2. This proves (411).

Separated supports. Put $\varrho = (\rho+1)L^2\xi_n \le s_n/16$. For $y \in S_\kappa \cup S'_\kappa$, the set $\mathsf{N}(y)$ meets $\kappa^{[2/3]}$. Such a top has level at most $n+1$. If its level were at least $n+3$, all its nearby cells would lie in Ω_{n+2}; if it were $n+2$, its distance from $\kappa^{[1]}$ would exceed $9s_{n+1}$, whereas $(\rho+1)L\xi_{n+2} < 9s_{n+1}$. Both contradict the intersection. Thus (407) places $\mathsf{N}(y)$ within ϱ of y. For $y' \in S'_\kappa$ some point of this neighborhood lies in $\kappa^{[1/12]}$. For $y \in S_\kappa$ some point lies in $\kappa^{[2/3]} \setminus \operatorname{int}\kappa^{[1/3]}$: otherwise, since the neighborhood has diameter at most $2\varrho < s_n/3$, the cutoff would be identically 0 or 1 there. These observations put both tops in $\kappa^{[5/6]}$.

Let z be the center of κ and $f(x) = \min\{|x - z|_\infty, 3s_n/2\}$. The cube $\kappa^{[1]}$ contains no point of Ω_{n+2}, so every point there counts with weight at least ξ_{n+1}^{-1}. Apply the interval-image argument of section 12.2 to f on a connecting set: between its endpoint values below $3s_n/2$, its points lie in that cube. It gives $|f(y) - f(y')| \le \xi_{n+1}d(y,y')$. For the two tops just considered,

$$f(y) - f(y') \ge \tfrac{1}{4}s_n - 2\varrho \ge \tfrac{1}{8}s_n,$$

which is (413).

Every local top has scale at most ξ_{n+1}. If $\mathsf{N}^\kappa(y)$ meets $\kappa^{[2/3]}$, its points are within $2\varrho \le s_n/8$ of that cube, hence in $\kappa^{[19/24]}$. Its tops and cells are consequently among the common ones already

identified. The inequality $d \le d^\kappa$ gives $\mathsf{N}^\kappa(y) \subset \mathsf{N}(y)$. Indices at these positions have basepoints at most $\xi_{n+1}/2 \le s_n/24$ beyond these cells or tops, so they lie in $\kappa^{[5/6]}$. The last assertions in (d) follow from this inclusion and the cutoff supports. □

27.3. **The actual averages and the local inverses.** The local gauge is used to verify the domains of the averaging maps and to apply proposition 25.1. The maps themselves remain the actual nested averages of $\mathbf{U}$. Near the physical boundary there is only one layer; the mass terms then give a separate inverse without a small gauge potential.

Lemma 27.6. The constants can be chosen so that all the averages defining the global and local systems exist. On the rows and tops identified in $\kappa^{[5/6]}$, their relative constraint germs, linear averages and quadratic coefficients agree. In particular, the entries of the averaging kernels Q, Q', their weighted transposes, and the quadratic coefficient kernels agree at the corresponding indices. For every piece κ and $i \in \{1,2,4,5\}$, the local operator $\mathbb{K}_\kappa^{[i]}$, with ambient region $\mathsf{A} = \Omega_0$, is invertible. Its inverse satisfies

$$|\mathbb{E}_\kappa^{[i]} \iota_\kappa g|_{\mathbf{U},y} \le B_{\mathrm{loc}} \sup_{z \in \Lambda_*^\kappa} e^{-2\delta_{\mathrm{reg}} d^\kappa(y,z)} |g|_z, \qquad y \in \Lambda_*^\kappa. \tag{415}$$

Here the constants depend only on L, and the empty-region convention is unchanged.

For use in the cutoff estimates, set

$$\mathsf{f}(b) = \xi^{-2} \max_p |\partial \mathbf{U}(p) - \mathbf{1}|,$$

where p is an inner plaquette of Ω_0 with a corner within sup distance ξ of b_-. On the global and local systems,

$$|\operatorname{Ad}(\mathbf{U}(b))^{\pm 1}| \le 2, \qquad \xi_y^2 \mathsf{f}(b) \le L^2 \alpha_0 \le 1, \qquad |\mathbf{J}|_{-3;y} \le \gamma_0, \qquad |\mathsf{j}^\kappa|_{-3;y} \le \gamma'. \tag{416}$$

In the curvature inequality b is an inner bond at position y; the adjoint bound holds on all inner ambient bonds. For the global system replace j^κ by j. ■

Proof. A small potential on a regular piece. Let $n = n(\kappa)$. By lemma 27.5, $\kappa^{[2]}$ belongs to the prescribed natural cube family, and the bare ξ-collar of $Y_\kappa = \Omega_0^\kappa \subset \kappa^{[3/2]}$ lies in $\kappa^{[2]} \subset \Omega_0$. The real gauge condition gives a unitary u and a real field A_0 on these bonds such that

$$U^u = e^{i\xi A_0}, \qquad \xi_n |A_0| \le \alpha_0, \qquad \xi_n^2 |\nabla^\xi A_0| \le \alpha_0.$$

Write $A'' = \operatorname{Ad}(u) A'$. The class bounds and unitary covariance give

$$\mathbf{U}^u = e^{i\xi A''} e^{i\xi A_0}, \qquad \xi_n |A''| \le \alpha_1, \qquad \xi_n^2 |\nabla_{U^u} A''| \le \alpha_1.$$

Indeed, $\kappa^{[2]}$ meets only layers $n, n+1$. The difference between the covariant and ordinary derivatives is $\xi^{-1}(e^{i\xi \operatorname{ad} A_{0,\nu}} - I) A''(\cdot + \xi e_\nu)$. Thus, for $\alpha_0 \le 1/8$,

$$\xi_n^2 |\nabla^\xi A''| \le \alpha_1 + 4\alpha_0 \alpha_1 \le 2\alpha_1.$$

Put $Z = e^{i\xi A''} e^{i\xi A_0}$. If $\alpha_0 + \alpha_1 \le 1/8$, then $|Z - I| \le e^{\xi(|A''| + |A_0|)} - 1 \le 1/7$. The series logarithm defines the traceless field $\mathbf{A}_\kappa = (i\xi)^{-1} \log Z$, and

$$\xi_n |\mathbf{A}_\kappa| \le 2(\alpha_0 + \alpha_1).$$

The power-series telescoping identity gives $|\log Z - \log Z'| \le 2|Z - Z'|$ when both matrices lie within $1/2$ of I, and

$$|e^X e^Y - e^{X'} e^{Y'}| \le e^{1/4}(|X - X'| + |Y - Y'|) \quad \text{if } |X|, |Y|, |X'|, |Y'| \le 1/8.$$

Apply these inequalities at two translated bonds. They give $\xi_n^2 |\nabla^\xi \mathbf{A}_\kappa| \le 4(2\alpha_1 + \alpha_0)$, and hence

$$\mathbf{U}^u = e^{i\xi \mathbf{A}_\kappa}, \qquad \xi_{n+1} |\mathbf{A}_\kappa|, \ \xi_{n+1}^2 |\nabla^\xi \mathbf{A}_\kappa| \le 8L^2(\alpha_0 + \alpha_1). \tag{417}$$

The difference bound is asserted only when both translated bonds lie in $\kappa^{[2]}$. Fix this gauge for the rest of the argument.

Domains and agreement of the actual maps. Every global row midpoint and scalar top belongs to a piece. By lemmas 27.4 and 27.5, a row so covered has level $j \le n+1$, and its bare double block lies inside $\kappa^{[1]}$. For a scalar top $y \in \Lambda_j$, the auxiliary full-frame bond $c_y = \langle y, y + \xi_j e_1 \rangle$ may be inward and need not be a row. Its double block lies within $3\xi_j/2$ of y, as in the proof of lemma 25.2; the same piece margins put it inside $\kappa^{[1]}$.

For a regular piece choose the radius so that

$$\alpha_0 + \alpha_1 \le 1/8, \qquad 8L^2(\alpha_0 + \alpha_1) < c_{\mathrm{av}}/2. \tag{418}$$

Rescale a level-j double block by ξ_j. Its fine spacing is $\eta = L^{-j}$, and its potential is $\xi_j \mathbf{A}_\kappa$, of norm less than $c_{\mathrm{av}}/2$. Thus proposition 24.1 defines the actual nested averages and an open relative-constraint germ at zero on this block. The input and output scale factors in (370) cancel in its first derivative. The same argument covers every local row, since its level is at most $n+1$ and its bare block is inner in $Y_\kappa \subset \kappa^{[3/2]}$. On a boundary piece all rows and tops have level zero, so $Q = Q' = I$ and $C^{(2)} = 0$.

Unitary covariance (117) and the straightening covariance (363) return these definitions to the original gauge. They preserve the defining domains and the series logarithms, not only the values. Reversed bond variables and averages are obtained by inversion. For the relative logarithm, $\|a_j\|_\infty \le 2c_{\mathrm{av}} < \log 2$ from the proof of proposition 24.1 keeps both exponentials in the logarithm neighborhood, with logarithms ia_j and $-ia_j$.

By lemma 14.2, the value and definedness of each iterate depend only on its bare double block. Block nesting gives the same successive inputs and domain tests for the global and local data. Thus different covering pieces or gauges evaluate the same recursion. The equality holds on an open relative-constraint neighborhood, so the linear and quadratic coefficients agree too. For Q', all scalar contours remain in $B^j(y)$, and the ordered transport formula (360) shows independence of the auxiliary double block. Finally, common indices have the same levels and pairing weights by lemma 27.5; their weighted transposes therefore agree as well.

Coefficient bounds and the regular inverse. For an inner bond at a local top y of level j, its initial point belongs to $\Omega_j^\kappa \subset \Omega_j$; its global level is consequently at least j. This proves the current bound in (416). The truncated row field has the global values at common rows, with the same weights, and vanishes elsewhere, proving its bound. The base point of a plaquette entering $\mathsf{f}(b)$ is within 2ξ of b_-. For $j \ge 1$, separation places it in Ω_{j-1}, so the complex curvature condition gives

$$|\partial \mathbf{U}(p) - I| \le L^2 \alpha_0 \xi^2 \xi_y^{-2}.$$

For $j = 0$ use α_0 instead. Also $\xi|A'| \le \alpha_1$ on every ambient bond, so $|\operatorname{Ad}(\mathbf{U})^{\pm 1}| \le e^{2\alpha_1} \le 2$. These arguments apply to the global indices as well.

Every local top has $\xi_y \le \xi_{n+1}$. Thus (417) verifies the literal neighborhood conditions (373), including its collar, with

$$\epsilon = \max\{8L^2(\alpha_0 + \alpha_1), \gamma_0, \gamma'\}.$$

Choose a_{reg} so that $\epsilon \le \epsilon_*$. The local separation parameters are $\mathfrak{m}_\flat = \mathfrak{M}_\dagger$ and $\mathfrak{g}_\flat = L\mathfrak{M}_\dagger$, and the collar is contained in Ω_0. Hence proposition 25.1 applies to every regular piece, with rate δ_{cs} and constant B_*, uniformly in the number of empty higher layers. The absolute-kernel estimates proving the first bound of (379) and (381) are preserved by this unitary gauge. They therefore hold for the local operators in the original gauge, with uniform constants. Since those proofs sum absolute kernel norms, the bounds also allow an inserted scalar factor bounded in modulus, as will be used in the cutoff calculation.

The massive boundary inverse. Let now κ be a boundary piece and put $Y = Y_\kappa$. The tops are the bare sites of Y, the rows are its positive inner bare bonds, and all pairing weights are ξ^3. Since there are no positive layers, (93) gives $d^\kappa(y,z) = \xi^{-1}|y - z|_1$, with periodic distance on the ambient

torus. In particular,

$$Q = Q' = I, \qquad \Delta^{(2)}[\mathsf{j}^\kappa] = 0, \qquad \Delta'_a = \Delta^\circ + \xi^{-2}, \qquad \mathsf{L}_1 = \mathsf{L} + \xi^{-2}.$$

The multiplier therefore drops out. The natural-class hypotheses on the whole ambient region imply

$$|\partial U(p) - I| \le \alpha_0, \qquad \xi|A'(b)| \le \alpha_1, \qquad \xi^3|\mathbf{J}(b)| \le \gamma_0 \le 1.$$

We first bound the two massive operators

$$T_1 = I + \xi^2 \Delta_Y(\mathbf{U}), \qquad T_2 = I + \xi^2 \Delta^\circ_{\mathbf{U}}.$$

For real U, the transports in (382) are isometries. Moreover, $0 \le y_0(p) \le 1$ when $\alpha_0 \le 1/2$, $|S_p(A)|^2 \le 4\sum_i |a_i|^2$, and $|F_U(p)| \le \xi^{-2}\alpha_0$. The commutator part obeys

$$\sum_{i<j} |\langle [a_i, a_j], F_U(p)\rangle| \le 3\xi^{-2}\alpha_0 \sum_i |a_i|^2.$$

Every bond lies in at most four plaquettes. Thus, for a real field A on the inner bonds of Y, extended by zero,

$$-12\alpha_0 \|A\|^2_\xi \le (A, \xi^2 \Delta_Y(U) A)_\xi \le (16 + 12\alpha_0)\|A\|^2_\xi. \tag{419}$$

This is the ambient form compressed to Y; plaquettes crossing the boundary of Y remain in the sum. The real Hessian is symmetric, and (369) writes $\xi^2 \Delta^\circ_U = 6I - \mathsf{h}$, where h is symmetric with at most six isometric entries per row and column. Consequently the two real massive operators have spectra in $[1 - 12\alpha_0, 17 + 12\alpha_0]$ and $[1, 13]$, respectively.

For the following norm calculation use the Hermitian pairing obtained by complexifying the real pairing. Its norm and the matrix norm on $\mathfrak{g}^c$ are equivalent with absolute constants, as in (378). Write $T_k = T_{k,0} + E_k$, with $T_{k,0}$ evaluated at U. Moving the unitary transports to the right in a plaquette gives

$$|\partial \mathbf{U}(p) - \partial U(p)| \le e^{4\alpha_1} - 1, \qquad |\operatorname{Ad}(\mathbf{U}(b))^{\pm 1} - \operatorname{Ad}(U(b))^{\pm 1}| \le e^{2\alpha_1} - 1.$$

The entries of $\xi^2\Delta_Y$ and $\xi^2\Delta^\circ$ are fixed polynomials in these transports and holonomies, by (382) and (369). All arguments stay in a fixed bounded set, so $|E_k(i,i')|_2 \le C_E \alpha_1$, with an absolute $C_E \ge 1$. The subscript denotes the induced Hermitian operator norm.

For bonds, let $\varrho(i,i')$ be graph distance through bare bonds of the ambient torus, one step joining bonds in a common plaquette. For sites use the bare nearest-neighbor graph. The matrices $T_{k,0}$ and E_k have range one for these distances, with at most thirteen nonzero entries in each row and column. This follows for the Hessian by polarization of (382), and for the scalar operator from (369). Also $\varrho(i,i') \ge d^\kappa(y(i), y(i'))/2$.

Choose $\max\{\alpha_0, \alpha_1\} \le 1/24$ and $C_E\alpha_1 \le 1/52$. Both $T_{k,0}$ then have spectra in $[1/2, 18]$. The Schur bound gives $\|E_k\|_2 \le 1/4$; with $\tau = 4/37$ we obtain

$$\|I - \tau T_k\|_2 \le \frac{35}{37} + \frac{1}{37} = q_\partial < 1, \qquad q_\partial = \frac{36}{37}.$$

Hence $T_k^{-1} = \tau \sum_{m \ge 0} (I - \tau T_k)^m$. The (i,i') entry vanishes in every term with $m < \varrho(i,i')$, and the remaining terms form a geometric tail. With $\delta_\partial = -\frac{1}{4}\log q_\partial > 0$, this proves

$$|T_k^{-1}(i,i')| \le C q_\partial^{\varrho(i,i')} \le C e^{-2\delta_\partial d^\kappa(y(i), y(i'))}. \tag{420}$$

At each position there are at most three bond indices and one site index. Summing one half of the exponential in (420) over periodic lattice representatives gives

$$|T_k^{-1} f|_y \le C \sup_z e^{-\delta_\partial d^\kappa(y,z)} |f|_z;$$

the constant is bounded by a multiple of $\sum_{z \in \mathbb{Z}^3} e^{-\delta_\partial |z|_1}$, independently of the torus and of Y.

Elimination of the boundary slots. In the following formulas all operators are those of the local system, and set $\mathsf{g}(A,v)=\mathsf{g}_0(A,v)+D^{\mathsf{T}}\mathsf{x}(A,v)$. For $i=4,5$, slots two through seven of $\mathbb{K}_\kappa\Psi=\iota_\kappa g$ give

$$\begin{aligned} &v=f_4, \qquad w=f_2+D^{\mathsf{T}}A-\Delta_a' f_4, \quad \sigma=\Delta_a' w-f_3,\\ &q=-f_7, \quad p=f_6+\Delta_a' f_7, \qquad\qquad r=f_5+D^{\mathsf{T}}X_5+\mathsf{g}(A,v)-\Delta_a' p. \end{aligned} \tag{421}$$

For $i=5$, the last slot gives $A=f_8$, and the first gives

$$\omega=\mathsf{L}_1 A-D(w+p)-\mathsf{m}_J(v)-f_1.$$

For $i=4$, substitution of w into the first slot gives

$$(\Delta_Y(\mathbf{U})+\xi^{-2})A=f_1+D(f_2-\Delta_a' f_4)+Dp+\mathsf{m}_J(f_4),$$

whose inverse is $\xi^2 T_1^{-1}$. The $i=2$ system uses the first line of (421) with $A=0$. For $i=1$, $w=\xi^2 T_2^{-1} f_3$. These formulas solve all four square systems uniquely.

The remaining factors in these formulas have uniformly bounded range in units of ξ. Their kernels have exactly the powers of ξ prescribed by the input and output weights in (327)–(328): a difference contributes ξ^{-1}, a Laplacian ξ^{-2}, and the current ξ^{-3}. The coefficient bound $\xi^3|\mathbf{J}|\le 1$ therefore suffices for $\mathsf{m}_J,\mathsf{g}_0,\mathsf{x}$. Each component, including the differences in the solution norm, is a finite sum of such products with at most one factor $\xi^2 T_k^{-1}$. The weighted-supremum composition estimate of section 23.3 and (420) give the local inverse bound with rate δ_∂ and a constant B_∂.

Choose

$$2\delta_{\mathrm{reg}}=\min\{\delta_{\mathrm{cs}},\delta_\partial,1\}, \qquad B_{\mathrm{loc}}=\max\{B_*,B_\partial\}.$$

The regular and boundary estimates now imply (415). All radius restrictions above are finitely many inequalities on a_{reg} depending only on L. □

Lemma 27.7. Let the global and local systems be those constructed in lemma 27.6, with the common ambient region Ω_0 and the multiplier (404). A field supported at indices with basepoints in $\kappa^{[2/3]}$ is identified uniquely with a field of the other system by zero extension. Under this identification,

$$\mathbb{K}\Psi=\mathbb{K}_\kappa\Psi, \qquad \operatorname{supp}(\mathbb{K}\Psi)\subset\{\text{indices with basepoint in }\kappa^{[5/6]}\}. \tag{422}$$

The two solution seminorms of Ψ agree at common tops; at every other top the respective seminorm is zero. At an output index with basepoint $z\in\kappa^{[5/6]}$, the system reads Ψ only at basepoints within sup distance $s_{n(\kappa)}/6$ of z. ■

Proof. The domain assertion of lemma 27.6 identifies the actual recursive averages and their relative constraint germs, not only their estimates. Thus Q,Q' and $C^{(2)}$ have identical entries at common rows and tops; the weighted transposes agree when the weights do. We now check the remaining support and boundary issues.

The differences, their transposes, the Dirichlet Laplacian and the current terms in (372) use only endpoints and one bare step. Polarizing the ambient plaquette formula shows that the Hessian couples only bonds of a common plaquette, using the configuration on that plaquette. Both compressed Hessians are entries of this one ambient Hessian, including plaquettes crossing the local boundary. The averages connect a row to its double block Y_c, or a top to its bare block. The quadratic multiplier connects two bonds in one Y_c. All their basepoints lie in the row box or cell of lemma 27.4.

Inspection of (326) and (372) therefore shows that an entry of the system is a sum of kernel products along chains passing through at most one such box or cell and at most two additional bare steps. The longest chains occur in $D^{\mathsf{T}}\mathsf{x}$ and in $\Delta^{(2)}[\mathsf{j}](A-Dv)$; taking Dv in the latter stays inside Y_c. If either end of a chain lies in $\kappa^{[5/6]}$, its box meets $\kappa^{[1]}$, hence has level at most $n+1$ by the geometric proof. Its sup diameter is at most $2\xi_{n+1}$. All the objects along the chain are thus within

$2\xi_{n+1}+2\xi \le 4\xi_{n+1} < s_n/6$ of either endpoint. This proves the stated input range and shows that a field supported in $\kappa^{[2/3]}$ has output supported in $\kappa^{[5/6]}$.

For such an output, add the possible $\xi/2$ from a bond midpoint to its endpoint and $\xi_{n+1}/2$ to a position. The total is at most $5\xi_{n+1} < s_n/6$, so all these sites and positions lie inside $\kappa^{[1]}$. Their region memberships, indices, weights and averaging blocks agree. The local current and configuration are restrictions of the same ambient data, and the Hessian entries agree by the common ambient construction.

It remains to check the discarded rows of the multiplier. If a global row midpoint lies outside $\kappa^{[5/6]}$ and its box meets $\kappa^{[1]}$, the row has level at most $n+1$ and the box is within ξ_{n+1} of its midpoint. Its distance from $\kappa^{[2/3]}$ is therefore greater than $s_n/6-\xi_{n+1} > \xi$. If the box does not meet $\kappa^{[1]}$, it does not meet the support either. In both cases A and v, and hence $A-Dv$, vanish on the bare double block of the row. Its contribution to $\Delta^{(2)}[\mathsf{j}](A-Dv)$ is zero. All remaining rows are common rows with the same multiplier value and quadratic coefficient. This proves (422).

For the seminorm assertion, a nonzero value of a component or of one of $\nabla A, Dv, Dw, Dq$ has basepoint within ξ of $\kappa^{[2/3]}$. Every index used by that difference is within 2ξ of this cube and lies in $\kappa^{[5/6]}$. The position of such an index is within $3\xi/2+\xi_{n+1}/2 \le 2\xi_{n+1} < s_n/6$ of the support. It is therefore a common top with the same weight. Shifted bonds are indices in both sequences or in neither, and the transports are those of the same configuration. Each entry of (327) agrees, while entries outside these positions vanish. Finally, the common-index identification in lemma 27.5 gives existence and uniqueness of the zero extensions used throughout. □

27.4. **The cutoff error.** Multiplication by a real function h is always evaluated at index base points, in particular at bond and row midpoints. We write $[T,h]=Th-hT$. The scalar variable p has no derivative in (327); accordingly its Laplacian commutator must be kept in divergence form.

For a scalar field $f(x)$, or a bond component indexed by its initial point x, write $\hat{x}$ for its base point. Put

$$k_\nu(x)=\xi^{-1}\big(h(\hat{x}+\xi e_\nu)-h(\hat{x})\big), \qquad X_\nu^{h[f]}(x)=k_\nu(x)\big(\mathsf{U}_\nu(x)f(x+\xi e_\nu)+f(x)\big)$$

when the shifted index exists, and put both expressions equal to zero otherwise. With the same convention for a missing previous index, define

$$(\mathsf{c}^h f)(x)=\xi^{-1}\sum_\nu \big(k_\nu(x)-k_\nu(x-\xi e_\nu)\big)f(x).$$

Direct expansion of (369) gives, including at the boundary,

$$[\Delta^\circ,h]f=\sum_\nu \nabla_\nu^{\mathsf{T}} X_\nu^{h[f]}+\mathsf{c}^h f. \tag{423}$$

For $f=p$, the first term is $D^{\mathsf{T}}X^{h[p]}$.

For any of the linear maps of (A,v) in (372), use $\mathsf{t}^h(A,v)=\mathsf{t}(hA,hv)-h\mathsf{t}(A,v)$. Define $\mathbb{C}_h\Psi\in\mathcal{F}$ by taking the corresponding components of $[\mathbb{K},h]\Psi$ in all f_j slots except f_5, and replacing that slot by

$$\begin{aligned} f_5 &= \mathsf{c}^h p+[Q'^{\mathsf{T}}\mathsf{a}'Q',h]p+[Q'^{\mathsf{T}},h]r-\mathsf{g}_0^h(A,v)-[D^{\mathsf{T}},h]\mathsf{x}(A,v),\\ X_5 &= X^{h[p]}-\mathsf{x}^h(A,v). \end{aligned} \tag{424}$$

The smaller systems retain just their existing slots.

Lemma 27.8. On a local system from lemma 27.6, let $\vartheta\ge 1$ and let h be real and C^2 on an open neighborhood of its region, with

$$|h|\le 1, \qquad \xi_y|\partial h|\le \vartheta^{-1}, \qquad \xi_y^2|\partial^2 h|\le \vartheta^{-1} \quad \text{on } \mathsf{N}^\kappa(y).$$

There is a constant $C_{\mathbb{C}}$ depending only on L such that

$$[\mathbb{K}_\kappa, h] = \iota_\kappa \mathbb{C}_h, \qquad |\mathbb{C}_h \Psi|_y \le C_{\mathbb{C}} \vartheta^{-1} \max_{d^\kappa(y,z)\le\rho} |\Psi|_z. \tag{425}$$

The left seminorm is zero if h is constant on $\mathsf{N}^\kappa(y)$. Also $|h\Psi|_y \le 2|\Psi|_y$ and $|hg|_y \le |g|_y$. ∎

Proof. We omit the local superscript. Neighboring cells have scale ratios between L^{-1} and L. The line segments joining a site to an incident bond midpoint, or joining neighboring indices of one component, lie in $\mathsf{N}(y)$ at the output position. The derivative assumptions therefore give

$$|k_\nu(x)| \le \vartheta^{-1}\xi_y^{-1}, \qquad \xi^{-1}|k_\nu(x) - k_\nu(x-\xi e_\nu)| \le \vartheta^{-1}\xi_y^{-2}$$

when the two shifted indices exist. If one is absent, its position is at level zero, so $\xi = \xi_y$ gives the same bound with an absolute constant. In the interior, expanding the Laplacian commutator in first differences instead gives

$$[\Delta^\circ, h]f(x) = -\sum_\nu \big(k_\nu(x)\nabla_\nu f(x) - k_\nu(x-\xi e_\nu)\nabla_\nu^{\mathsf{T}} f(x)\big) - \mathsf{c}^h f(x).$$

Consequently, for the weights used in the system,

$$\begin{aligned} |[\Delta^\circ, h]f|_{s-2;y} &\le C\vartheta^{-1} \max_{d(y,z)\le\rho} (|f|_{s;z} + |\nabla f|_{s-1;z}), \\ |X^{h[f]}|_{s-1;y} + |\mathsf{c}^h f|_{s-2;y} &\le C\vartheta^{-1} \max_{d(y,z)\le\rho} |f|_{s;z}. \end{aligned} \tag{426}$$

At missing neighbors the first bound follows directly from $\xi = \xi_y$.

The first-order calculation is equally local. For example,

$$\begin{aligned} [D^{\mathsf{T}}, h]X(x) = \xi^{-1} &\sum_{b_+ = x} (h(\hat b) - h(x))\,\mathrm{Ad}(\mathbf{U}(b))^{-1} X(b) \\ &- \xi^{-1} \sum_{b_- = x} (h(\hat b) - h(x)) X(b). \end{aligned} \tag{427}$$

The analogous formula for $[D, h]$ has the two endpoint values and the midpoint value of h. Since the adjoint transports have norm at most two, these commutators lower the weight by one with a factor $C\vartheta^{-1}$.

All averaging terms can be treated by the same inserted difference. If $T(i, i')$ is one of their kernels, then

$$([T, h]f)(i) = \sum_{i'} T(i, i')(h(\hat{i}') - h(\hat{i}))f(i').$$

The row terms have both indices in a single row box; scalar terms have them in a single cell. By lemma 27.4, these sets lie in the output neighborhood and the difference of h is at most $\rho\vartheta^{-1}$. The kernel bounds in (379), now available by lemma 27.6, show that the commutators with $Q, Q^{\mathsf{T}}, Q', Q'^{\mathsf{T}}$ preserve weights, while those with $Q^{\mathsf{T}}\mathsf{m}^{-2}Q$ and $Q'^{\mathsf{T}}\mathsf{a}'Q'$ lower them by two. Expanding each potential within its common row or cell keeps the same support neighborhood; one does not compose two enlarged neighborhoods.

The formula (381) treats $\Delta^{(2)}$ in the same way. Together with (372) and the current bounds, it gives weights $-3, -4, -3$ for $\mathsf{m}_J^h(v), \mathsf{g}_0^h(A, v), \mathsf{x}^h(A, v)$, respectively, with bounds $C\vartheta^{-1}$ times the neighboring solution seminorms. In the first slot the additional terms are

$$-[\Delta^{(2)}, h](A - Dv) + \Delta^{(2)}[D, h]v.$$

They have weight -3 with the same bound. In the composed terms here and in $[D^{\mathsf{T}}, h]\mathsf{x}$, the common row box contains the output bond or its endpoint, and contains all input indices. Thus the sharp row neighborhood still applies.

For a regular piece its bare collar lies in the ambient region, so (383) gives $[\mathsf{L}, h] = [\Delta^\circ, h] + [\mathsf{Z}_{\mathrm{curv}}, h]$. The curvature kernel joins bonds of a common plaquette and has size at most $C\mathsf{f}(b)$. Its commutator

therefore lowers weight -1 to -3 with the required factor, since $\xi_y^2 \mathsf{f}(b) \le 1$. For a boundary piece every scale is ξ. The Hessian formula (382) bounds its kernel by $C\xi^{-2}$ on pairs of bonds of a common plaquette; the same inserted difference proves the assertion. The kernel of DD^{T} is treated by the two first-order formulas.

Finally, (423) and

$$(D^{\mathsf{T}}\mathsf{x})^h = [D^{\mathsf{T}}, h]\mathsf{x} + D^{\mathsf{T}}\mathsf{x}^h$$

give the fifth component of $[\mathbb{K}, h]\Psi = \iota \mathbb{C}_h \Psi$. The other components are their defining commutators. Apply the first bound of (426) to A, v, w, q and the second to p; the bounds just proved give each forcing weight in (328). Every term contains a difference of values of h within $\mathsf{N}(y)$, proving the vanishing assertion. The product rule for a first difference, and $\vartheta^{-1} \le 1$, give $|h\Psi|_y \le 2|\Psi|_y$; the forcing norm contains no derivatives, so $|hg|_y \le |g|_y$. □

The input cutoff also requires a divergence correction. For $i = 4, 5$, let $h \star g$ agree with hg except in its fifth scalar component, where

$$(h \star g)_5 = hf_5 - [D^{\mathsf{T}}, h]X_5. \qquad \iota(h \star g) = h\iota g. \tag{428}$$

For $i = 1, 2$, set $h \star g = hg$. The identity follows by adding $D^{\mathsf{T}}(hX_5)$ to the displayed component. Formula (427) also proves

$$|\theta_\kappa \star g|_y \le (1 + C\vartheta^{-1}) \max_{z \in S'_\kappa,\ d(y,z) \le 3} |g|_z, \qquad \vartheta = \mathfrak{M}/C_{\mathrm{G}}. \tag{429}$$

Indeed, the midpoint-to-endpoint difference contributes $\xi\vartheta^{-1}\xi_y^{-1}/2$, the factor ξ^{-1} cancels it, and the weight of X_5 is -3. Adjacent positions have scale ratio at most L and distance at most three. A nonzero cutoff value or difference has a site or midpoint in $\kappa^{[1/12]}$; both relevant neighborhoods meet this set. Thus both the output position and the input positions belong to S'_κ. The same statements hold in the local sequence, using (414). Taking $\mathfrak{M}_*$ larger makes the factor in (429) at most two.

27.5. **The parametrix and convergence.** Fix a_{reg} and $\delta = \delta_{\mathrm{reg}}$ as in lemma 27.6. Enlarge $\mathfrak{M}_*$ to satisfy the geometric margins, the cutoff requirements, and

$$\begin{aligned} &C_q = 2C_{\mathbb{C}}C_{\mathrm{G}}B_{\mathrm{loc}}e^{2\rho+8}, \qquad B_T = 4B_{\mathrm{loc}}e^{3\delta}, \\ &q_{\mathfrak{M}} = C_q\mathfrak{M}^{-1}e^{-\delta D_0}, \qquad N_{\mathrm{G}}q_{\mathfrak{M}} \le \tfrac{1}{2}, \qquad B_{\mathrm{reg}} = 2N_{\mathrm{G}}B_T. \end{aligned} \tag{430}$$

The contraction condition follows, for example, from $\mathfrak{M} \ge 2N_{\mathrm{G}}C_q$. All these choices depend only on L.

Define the local remainder on the original system space by

$$\mathsf{R}_\kappa = -[\mathbb{K}_\kappa, \widetilde{h}_\kappa]\mathbb{E}_\kappa\theta_\kappa.$$

Restrictions and zero extensions use the common indices in the enlarged pieces.

For $g \in \mathcal{F}$, restrict its entries to the common indices in $\kappa^{[5/6]}$ and extend by zero to the other local indices; denote this restriction by $g|_\kappa$. Define, with all operators on the right taken in the local system,

$$\begin{aligned} \mathsf{T}_\kappa g &= \widetilde{h}_\kappa \mathbb{E}_\kappa \iota_\kappa(\theta_\kappa \star (g|_\kappa)), \\ \mathsf{R}^\sharp_\kappa g &= -\mathbb{C}_{\widetilde{h}_\kappa} \mathbb{E}_\kappa \iota_\kappa(\theta_\kappa \star (g|_\kappa)). \end{aligned} \tag{431}$$

Their supported outputs are identified with global fields and extended by zero. The input correction uses only indices in $\kappa^{[5/6]}$, by (429). For the output correction, the local neighborhoods meeting the support of $\widetilde{h}_\kappa$ lie in $\kappa^{[19/24]}$. Moving from a site to an incident bond changes the base point by at most ξ, and $\xi \le s_{n(\kappa)}/24$. Thus the divergence injections and their cutoffs still use common indices in $\kappa^{[5/6]}$. In particular,

$$\mathsf{R}_\kappa\iota = \iota\mathsf{R}^\sharp_\kappa, \qquad \widetilde{h}_\kappa\mathbb{E}_\kappa\theta_\kappa\iota = \mathsf{T}_\kappa. \tag{432}$$

These are identities of the actual forcing injections, with the midpoint convention of (427).

Lemma 27.9. For every $g \in \mathcal{F}$,

$$\begin{aligned} |\mathsf{R}^\sharp_\kappa g|_y &\le q_{\mathfrak{M}} \sup_{z \in S'_\kappa} e^{-\delta d(y,z)} |g|_z, \\ |\mathsf{T}_\kappa g|_y &\le B_T \sup_{z \in S'_\kappa} e^{-\delta d(y,z)} |g|_z. \end{aligned} \tag{433}$$

The first seminorm vanishes outside S_κ; the second vanishes at tops outside $\kappa^{[5/6]}$. ■

Proof. Let $u = \mathbb{E}_\kappa \iota_\kappa(\theta_\kappa \star (g|_\kappa))$. The local inverse bound, the cutoff bound and $d^\kappa \ge d$ give, at every common top v,

$$|u|_v \le 2B_{\mathrm{loc}} e^{6\delta} \sup_{z \in S'_\kappa} e^{-2\delta d(v,z)} |g|_z.$$

Indeed the input cutoff moves a position by distance at most three. Applying (425), and then $d(y,v) \le d^\kappa(y,v) \le \rho$ for its output neighborhood, yields

$$|\mathsf{R}^\sharp_\kappa g|_y \le \frac{2C_{\mathbb{C}} C_{\mathrm{G}} B_{\mathrm{loc}}}{\mathfrak{M}} e^{2\delta\rho + 6\delta} \sup_{z \in S'_\kappa} e^{-2\delta d(y,z)} |g|_z.$$

This seminorm vanishes outside S_κ. On that set the support separation gives $d(y,z) \ge D_0$ for every $z \in S'_\kappa$. Hence

$$e^{-2\delta d(y,z)} \le e^{-\delta D_0} e^{-\delta d(y,z)}.$$

Since $\delta \le 1/2$, the first estimate follows from (430).

For T_κ, use the local inverse at rate δ and $|\widetilde{h}_\kappa u|_y \le 2|u|_y$. The input cutoff contributes at most $2e^{3\delta}$, giving B_T. Its values are supported in $\kappa^{[2/3]}$. The derivatives in its solution seminorm can reach adjacent indices, whose positions remain in $\kappa^{[5/6]}$ by (414). This proves the second support assertion and also justifies the comparison of local and global solution seminorms used above. □

Proof of theorem G. Set

$$\mathsf{Par} = \sum_\kappa \widetilde{h}_\kappa \mathbb{E}_\kappa \theta_\kappa, \qquad \mathsf{R} = \sum_\kappa \mathsf{R}_\kappa, \qquad \mathsf{R}^\sharp = \sum_\kappa \mathsf{R}^\sharp_\kappa.$$

At a given top at most N_{G} remainder terms have nonzero seminorm. Thus $\mathsf{R}^\sharp \in \mathfrak{O}(\delta, N_{\mathrm{G}} q_{\mathfrak{M}})$, and (430) makes it a contraction. The local/global operator agreement and the partition of unity give

$$\begin{aligned} \mathbb{K}(\widetilde{h}_\kappa \mathbb{E}_\kappa \theta_\kappa F) &= \theta_\kappa F - \mathsf{R}_\kappa F, \\ \mathbb{K}\mathsf{Par} = I - \mathsf{R}, \qquad \mathsf{R}\iota &= \iota \mathsf{R}^\sharp. \end{aligned} \tag{434}$$

Here $\widetilde{h}_\kappa = 1$ on the support of θ_κ. Given $F \in \mathcal{X}$, take $g = (F;0)$ so that $\iota g = F$; for $i = 1,2$ take $g = F$. The convergent series $g_\infty = \sum_{m \ge 0} (\mathsf{R}^\sharp)^m g$ satisfies $(I - \mathsf{R}^\sharp) g_\infty = g$. Therefore

$$\mathbb{K}\mathsf{Par}\iota g_\infty = \iota(I - \mathsf{R}^\sharp) g_\infty = F.$$

Surjectivity on the finite-dimensional space $\mathcal{X}$ proves invertibility. By (432),

$$\mathbb{E}\iota = \left(\sum_\kappa \mathsf{T}_\kappa \right) \sum_{m \ge 0} (\mathsf{R}^\sharp)^m.$$

At each top, at most N_{G} of the solution seminorms $|\mathsf{T}_\kappa g|_y$ are nonzero, so $\sum_\kappa \mathsf{T}_\kappa \in \mathfrak{O}(\delta, N_{\mathrm{G}} B_T)$. The composition rule following (340) therefore bounds the mth summand by

$$N_{\mathrm{G}} B_T (N_{\mathrm{G}} q_{\mathfrak{M}})^m \sup_z e^{-\delta d(y,z)} |g|_z.$$

Since there are finitely many tops and $\max_y |\cdot|_y$ is a norm, this bound also proves absolute operator convergence. Summing the geometric series gives $2N_{\mathrm{G}} B_T = B_{\mathrm{reg}}$ and proves (400). □

28. The inverse with corner averages

The real background construction uses the corner averages of section 14.6. At the identity, the resulting linear system is a restriction of the flat system already inverted in proposition 23.1. Local gauges and a partition of unity give an inverse at a regular configuration. Positivity requires a further comparison with the positive local inverses.

Work in level-n units, with $1 \le n \le K$, $\eta = L^{-n}$ and $\xi_j = L^j\eta$. The torus has side $2L^{m+K-n}$ in these units. Let $\Omega_0 \supseteq \cdots \supseteq \Omega_n$ be a sequence of regions, with $\Omega_{n+1} = \Omega_{n+2} = \emptyset$. Use its layers, tops, cells, local block side and weighted distance from definition 9.3 and section 12. Put $d = d_{\mathbf{B}}$. Write E_0 for its bare sites, S_α for its positive inner bare bonds, and S_β for the positive bare bonds with at least one endpoint in E_0.

The base site of a bond is its initial endpoint when that endpoint belongs to E_0, and its final endpoint otherwise. A bond difference has the base site of its first bond. A fine index is assigned to the top whose cell contains its base site; a top index is assigned to itself. Thus $|f|_y$ is the supremum of the values assigned to y, with value zero on an empty set. This convention allows six bonds at one base site near the outer boundary. All fine fields are extended by zero.

For a real configuration U on all bare bonds, let D_S be its covariant gradient into either $S = S_\alpha$ or $S = S_\beta$, and let D_S^* be its adjoint for the fine pairing $\eta^3 \sum\langle\cdot,\cdot\rangle$. The scalar Dirichlet Laplacian is $\Delta_U = D_{S_\beta}^* D_{S_\beta}$; it retains all six diagonal terms, including those for exterior neighbors. Covariant differences of a bond field are taken when both translated bonds belong to S.

The scalar and reduced bond averages are

$$(\mathsf{Q}'_{\mathsf{c}}\lambda)(y) = L^{-3j} \sum_{x \in B^j(y)} \mathrm{Ad}(T_U^{\mathsf{c}}(y,x))\lambda(x), \qquad y \in \Lambda_j,$$

$$(\mathsf{Q}_{\mathsf{c}}A)(c) = \sum_{b \in S_\alpha} Q_j^{\mathsf{c}}[U](c,b)A(b), \qquad c \in \mathbf{B}^\flat \cap \Gamma_j^+.$$

Here $\mathbf{B}^\flat$ consists of the positive interior and outward determining bonds, and the transports and kernels are those of section 14.6. The scalar top pairing has weight ξ_y^3 and the bond-row pairing has weight ξ_c^{-1}. Every star below uses these pairings. In particular, $\mathsf{Q}'_{\mathsf{c}}(\mathsf{Q}'_{\mathsf{c}})^* = I$: on each cell the transport is orthogonal and there are L^{3j} sites. At level zero the two averages are I and ηI.

Let $\varkappa_0 = \varkappa_1 = 1$ and $\varkappa_{j+1} = \varkappa_j/(L^{-2} + \varkappa_j)$ for $j \ge 1$, and let a multiply a level-j scalar top by $\varkappa_j\xi_j^{-2}$. Set

$$\mathsf{A}'_{\mathsf{c}} = \Delta_U + (\mathsf{Q}'_{\mathsf{c}})^*\mathsf{a}\mathsf{Q}'_{\mathsf{c}}, \qquad \mathsf{R}_{\mathsf{c}} = \text{orthogonal projection onto } \Delta_U \ker \mathsf{Q}'_{\mathsf{c}}.$$

Let H_S be the Hessian of the ambient Wilson action $\eta^{-1}\sum_p a_p(e^{i\eta A}U)$ at $A = 0$, compressed to fields supported in S. Plaquettes crossing the boundary remain in this definition. Define

$$\mathsf{M}_S = (\mathsf{Q}_{\mathsf{c}})^*\mathsf{Q}_{\mathsf{c}} + \eta^{-2}\chi_{S\setminus S_\alpha}, \qquad \mathsf{A}_{\mathsf{c}}[S] = \mathsf{H}_S + D_S\mathsf{R}_{\mathsf{c}}D_S^* + \mathsf{M}_S.$$

The last mass term acts only on the touching boundary bonds.

For a power M of L, use the pieces of definition 27.2 with $s_j = M\xi_j$. This definition uses only the regions and cubes. A size-zero piece whose double enlargement misses Ω_1 is a boundary piece; all other pieces are regular. The needed cover and cutoff properties are proved below under the hypotheses of the theorem.

Theorem H. Let $L \ge 13$ be odd. There are a power $M_\diamond$ of L and constants $\varepsilon_\diamond, B_\diamond, \delta_\diamond > 0$, depending only on L, with the following property. With the objects just defined, suppose that

(i) $M \ge M_\diamond$ is a power of L, $\rho \ge LM$, each Ω_j is a union of M-cubes at level j, and

$$\mathrm{dist}_\infty(\Omega_j^c, \Omega_{j+1}) > \rho\xi_j \quad (0 \le j < n).$$

Moreover $\Omega_1 \ne \emptyset$, and the level-n torus has at least 6ρ sites in each coordinate direction.

(ii) U is G-valued on all bare bonds, and, for some $0 \le \alpha \le L^{-2}\min\{1, c_{\mathsf c}\}$,

$$|\partial U(p) - \mathbf 1| \le \alpha\eta^2 \mathsf L_{\mathbf B}(p)^{-2} \quad \text{on every inner plaquette of } \Omega_0.$$

(iii) For every regular piece κ of size j, there are a real gauge v_κ on $\kappa^{[1]}$ and a real field A_κ on its inner bonds such that

$$U^{v_\kappa} = e^{i\eta A_\kappa}, \qquad \xi_j |A_\kappa| \le \varepsilon, \qquad \xi_j^2 |\nabla^\eta A_\kappa| \le \varepsilon, \qquad 0 < \varepsilon \le \varepsilon_\diamond.$$

The difference bound is required when both translated bonds belong to $\kappa^{[1]}$.

Then $\mathsf A'_{\mathsf c}$ is positive definite. Its inverse $\mathsf G'_{\mathsf c}$ and its covariant gradient satisfy

$$\begin{aligned} |\mathsf G'_{\mathsf c} f|_y &\le B_\diamond \xi_y^2 \sup_z e^{-\delta_\diamond d(y,z)} |f|_z, \\ |D\mathsf G'_{\mathsf c} f|_y &\le B_\diamond \xi_y \sup_z e^{-\delta_\diamond d(y,z)} |f|_z. \end{aligned} \tag{435}$$

Here D takes values on S_β.

If the plaquette bound in (ii) holds at every ambient bare plaquette having a corner in Ω_0, then, for $S = S_\alpha$ and $S = S_\beta$, the operator

$$\mathsf A_{\mathsf c}[S] = \mathsf H_S + D\mathsf R_{\mathsf c} D^* + \mathsf M_S$$

is positive definite. Writing $\mathsf G_{\mathsf c,S} = \mathsf A_{\mathsf c}[S]^{-1}$, we have

$$\begin{aligned} |\mathsf G_{\mathsf c,S} J|_y &\le B_\diamond \xi_y^2 \sup_z e^{-\delta_\diamond d(y,z)} |J|_z, \\ |\nabla^U \mathsf G_{\mathsf c,S} J|_y &\le B_\diamond \xi_y \sup_z e^{-\delta_\diamond d(y,z)} |J|_z. \end{aligned} \tag{436}$$

The second line uses only pairs (b,μ) for which both b and $b+\eta e_\mu$ belong to S. All suprema range over Λ_*. The constants are independent of the volume, the scale, the regions and the number of layers. ■□

The proof occupies the rest of this section. All constants denoted by C depend only on L and may increase between estimates.

28.1. **The local system and its flat inverse.** The averages used in the theorem are defined under its curvature hypothesis. A level-q double block lies in Ω_0 and meets no layer below $q-1$. Its inner curvature is therefore at most $L^2\alpha\eta^2\xi_q^{-2}$. Since $L^2\alpha \le c_{\mathsf c}$, the folding and double-block construction in the proof of proposition 14.10 applies. Construct the rows in its axial gauge and use covariance to return to U. At level zero they are already the stated identity maps.

Suppress the subscript S on D_S. For $\Psi = (A; v, w, \sigma)$ define

$$\mathbb K_{\mathsf c}^{[3]}\Psi = ((\mathsf H_S + DD^* + \mathsf M_S)A - Dw, \mathsf A'_{\mathsf c} v + w - D^*A, \mathsf A'_{\mathsf c} w - (\mathsf Q'_{\mathsf c})^*\sigma, \mathsf Q'_{\mathsf c} v).$$

The scalar system $\mathbb K_{\mathsf c}^{[2]}$ consists of the last three equations with $A = 0$, and $\mathbb K_{\mathsf c}^{[1]} w = \mathsf A'_{\mathsf c} w$. Use the solution and forcing seminorms

$$\begin{aligned} |\Psi|_y &= \max\{\xi_y|A|_y, \xi_y^2|\nabla^U A|_y, |v|_y, \xi_y|Dv|_y, \\ &\qquad \xi_y^2|w|_y, \xi_y^3|Dw|_y, \xi_y^4|\sigma(y)|\}, \\ |F|'_y &= \max\{\xi_y^3|f_1|_y, \xi_y^2|f_2|_y, \xi_y^4|f_3|_y, |f_4(y)|\}. \end{aligned}$$

Delete the absent entries for the scalar systems, whose gradient norms always use S_β. For the four-component system they use the chosen frame S. Near a regular piece both frames coincide.

We record the elimination needed to recover the vector inverse. If $\mathsf G'_{\mathsf c} = (\mathsf A'_{\mathsf c})^{-1}$ exists, then

$$\mathsf N_{\mathsf c} = \mathsf Q'_{\mathsf c}(\mathsf G'_{\mathsf c})^2(\mathsf Q'_{\mathsf c})^*$$

is positive definite, because $(\mathsf{Q}'_{\mathsf{c}})^*$ is injective. The orthogonal complement of $\Delta_U \ker \mathsf{Q}'_{\mathsf{c}} = \mathsf{A}'_{\mathsf{c}} \ker \mathsf{Q}'_{\mathsf{c}}$ is $\operatorname{ran}(\mathsf{G}'_{\mathsf{c}}(\mathsf{Q}'_{\mathsf{c}})^*)$. Hence

$$I - \mathsf{R}_{\mathsf{c}} = \mathsf{G}'_{\mathsf{c}}(\mathsf{Q}'_{\mathsf{c}})^* \mathsf{N}_{\mathsf{c}}^{-1} \mathsf{Q}'_{\mathsf{c}} \mathsf{G}'_{\mathsf{c}}.$$

Write $e_1 J = (J, 0, 0, 0)$. With this forcing, the last three equations give $w = (I - \mathsf{R}_{\mathsf{c}}) D^* A$, and the first becomes $\mathsf{A}_{\mathsf{c}}[S] A = J$. This identifies the desired vector block once the system is inverted.

Replace the first region by the whole torus and set $U = I$; write $\mathbb{K}_0$ for this reference system. Its distance is unchanged, since weighted length outside Ω_1 has weight η^{-1} for either first region. At the identity, the corner transports are trivial and the dimensionless bond average is $\mathsf{m} Q_0$, where Q_0 is the physical flat average of section 23 and $(\mathsf{m} r)(c) = \xi_c r(c)$. In fact, differentiation of the endpoint factors cancels the transverse comb terms, and concatenation of the remaining straight segments gives the physical average multiplied by ξ_c. The two row pairings then give

$$Q^*_{\mathsf{c},0} = Q_0^{\mathsf{T}} \mathsf{m}^{-3}, \qquad Q^*_{\mathsf{c},0} Q_{\mathsf{c},0} = Q_0^{\mathsf{T}} \mathsf{m}^{-2} Q_0.$$

The scalar averages, scalar weights and projections agree as well. Thus the flat four-component system is the restriction of $\mathbb{K}_0^{[4]}$ in proposition 23.1 to $p = q = r = 0$. For forcing $(f_1, f_2, f_3, f_4, 0, 0, 0)$ and $X_5 = 0$, (334) gives precisely this restriction. The retained seminorms are the ones displayed above. The scalar subsystems are the corresponding scalar systems there. Consequently there are $B_1, \delta_1 > 0$, depending only on L, such that

$$|\mathbb{E}_0 F|^0_y \le B_1 \sup_z e^{-\delta_1 d(y,z)} |F|'_z, \qquad \mathbb{E}_0 = \mathbb{K}_0^{-1},$$

provided M and ρ meet the thresholds of that theorem. The first-source-to-vector block, denoted by G_0, is the positive flat inverse identified in eq. (332) and proposition 21.1.

Call tops z, y near, and write $z \sim y$, if their cells contain bare sites at sup distance at most η, or sites in one reduced-row double block. The cell geometry gives $d(y, z) \le 6L$ and $L^{-1} \le \xi_z / \xi_y \le L$. These are the only comparisons needed in the local estimates.

28.2. **Cutoffs and locality.** For the pieces of definition 27.2, with $s_j = M \xi_j$, use the quadratic partition

$$h_\kappa = \frac{\phi_\kappa}{(\sum_{\kappa'} \phi_{\kappa'}^2)^{1/2}}, \qquad \phi_\kappa = h_{\kappa;0,1/12}, \qquad \widetilde{h}_\kappa = h_{\kappa;1/3,2/3}. \tag{437}$$

Here multiplications act at the base site of an index. In particular, a touching bond whose initial endpoint is outside Ω_0 is multiplied at its final endpoint. The cover and two-layer argument in lemma 27.5 apply to these same pieces. They give

$$\sum_\kappa h_\kappa^2 = 1, \qquad \operatorname{supp} h_\kappa \subset \kappa^{[1/12]}, \qquad \widetilde{h}_\kappa = 1 \text{ on } \kappa^{[1/3]}, \qquad \operatorname{supp} \widetilde{h}_\kappa \subset \kappa^{[2/3]}.$$

At each point at most N_G enlarged pieces $\kappa^{[1]}$ occur. One may take $N_G = 256$: at a point of layer q, their sizes are among $q-2, q-1, q, q+1$, and there are at most 4^3 centers at each size. Differentiating $\phi_\kappa (\sum \phi_{\kappa'}^2)^{-1/2}$, whose denominator is at least one, gives constants $C_G = C_G(L)$ such that, for $\theta = M / C_G \ge L^2$,

$$\begin{aligned} |h(x') - h(x)| &\le \frac{\eta}{\theta \xi(x)} \qquad (|x - x'|_\infty \le \eta), \\ |h(x + \eta e_\nu) + h(x - \eta e_\nu) - 2h(x)| &\le \frac{\eta^2}{\theta \xi(x)^2}. \end{aligned} \tag{438}$$

The sites in either formula belong to Ω_0, and h is either cutoff. For the first estimate, integrate its first derivative along $[x, x'] \subset \Delta_0(x) \cup \Delta_0(x')$; the two scales differ by at most L. Integrating the second derivative against $(\eta - |t|)_+$ gives the second estimate.

For a regular piece of size j, its double enlargement lies in $\mathcal{L}_j \cup \mathcal{L}_{j+1}$ and is more than $8M\eta$ from Ω_0^c. For a boundary piece, $\operatorname{dist}_\infty(\kappa^{[1]}, \Omega_1) \ge M\eta$. The first assertion follows from the three cases in the cover argument: a deep cube of positive size, a subdivided buffer cube, or a size-zero cube

whose double enlargement meets Ω_1. For the second, $\kappa^{[2]}$ and Ω_1 are disjoint unions of M-cubes, and their grid forces the displayed margin.

On a regular $\kappa^{[1]}$, the original and reference indices have the same cells, positions and base sites. A corner double block meeting $\kappa^{[3/4]}$ has level j or $j+1$ and lies in $\kappa^{[1]}$. Indeed, lower-level endpoints lie outside Ω_j, while the gap from Ω^c_{j+1} excludes levels at least $j+2$. Its diameter is less than $2\xi_{j+1} \le s_j/12$ when $M \ge 24L$. Thus either system, acting on a field supported in $\kappa^{[r]}$, $0 \le r \le 2/3$, produces a field supported in $\kappa^{[r+1/12]}$ and uses U only inside $\kappa^{[1]}$. Conversely, its values in $\kappa^{[r]}$ depend only on the field in $\kappa^{[r+1/12]}$. The scalar averages couple one cell, the row energy couples bonds in one double block, and the differential and Hessian terms have smaller range.

28.3. **Comparison on a regular piece.** Extend a local gauge v_κ by the identity outside $\kappa^{[1]}$, and let $\mathcal{U}_\kappa$ act on a fine field by $\mathrm{Ad}(v_\kappa)$ at its initial site. On a level-q scalar top it acts at $o^{(q)}_{\mathsf{c}}(y)$, and on a bond row at $o^{(q)}_{\mathsf{c}}(c_-)$. The position used by a cutoff remains the top or the assigned base site. Covariance of the corner transports follows by cancellation of the intermediate gauge factors. Differentiating the straightened average gives the same covariance for its bond kernel. The folded extensions commute with these gauge changes. Together with gauge invariance of the Wilson action, this proves covariance of $\mathbb{K}_{\mathsf{c}}$ and preservation of supports and pairings. The covariant seminorms are preserved as well.

Put $\widehat{\varepsilon} = L^2\varepsilon$. On a regular piece of size j, all relevant cells have level j or $j+1$, so

$$\xi_q|A_\kappa| \le \widehat{\varepsilon}, \qquad \xi_q^2|\nabla^\eta A_\kappa| \le \widehat{\varepsilon}.$$

In the plaquette product the terms of first order are the two ordinary differences; the remaining terms are bounded by $16\eta^2 \sup|A_\kappa|^2$. Thus, for $\varepsilon \le 1/16$,

$$|\partial U^{v_\kappa}(p) - I| \le (2\varepsilon + 16\varepsilon^2)\eta^2\xi_j^{-2} \le 3\widehat{\varepsilon}\eta^2\xi_q^{-2}.$$

We take

$$\varepsilon_0 \le L^{-2}\min\{(4L^2)^{-1}, \min(1, c_{\mathsf{c}})/3\}.$$

Thus proposition 14.10 applies on every required double block.

For a field supported in $\kappa^{[2/3]}$, identify the original and reference indices there and set

$$V_\kappa = \mathbb{K}_0 - \mathcal{U}_\kappa \mathbb{K}_{\mathsf{c}} \mathcal{U}_\kappa^{-1}.$$

There is $C_V = C_V(L)$ such that

$$|V_\kappa \Psi|'_y \le C_V \varepsilon \max_{z \sim y} |\Psi|^0_z. \tag{439}$$

Its output is supported in $\kappa^{[3/4]}$. Here and below the corresponding scalar estimates are obtained by deleting the absent entries; all differential stencils are inside $\kappa^{[1]}$, where the two frames agree.

We justify the estimate by grouping the differential and averaging terms. The identities of lemma 25.4, with $\xi = \eta$ and $A' = A_\kappa$, apply to the first differences and their adjoints. In the Laplacian, use the cancellation (389). Its zeroth-order coefficient is the sum of $e(x) + e'(x)$ and $e'(x - \eta e_\mu) - e'(x)$; the first is bounded by $4\eta|A_\kappa(x)|^2$ and the second by twice the ordinary increment of A_κ. Consequently the Laplacian defect sends value/gradient weight $(s, s-1)$ to weight $s-2$ with bound $C\widehat{\varepsilon}$, for $s = -1, 0, -2$. This retains the first-difference hypothesis and incurs no negative power of η in the constant. The ambient plaquettes in lemma 25.3 are the same on this interior stencil. That formula and the displayed curvature bound give the same estimate for $\mathsf{H}_S + DD^*$ relative to the componentwise flat Laplacian.

For the scalar average, proposition 14.10 bounds the difference of each transport from the identity by $t \le C\widehat{\varepsilon}$. Since its coefficients have total mass one, both the scalar average and its weighted adjoint differ from their flat counterparts by at most t. Thus the defect of $(\mathsf{Q}'_{\mathsf{c}})^* \mathsf{a} \mathsf{Q}'_{\mathsf{c}}$ is at most $2t\xi_q^{-2}$ on one cell. The linear scalar average and adjoint in the other two equations satisfy the corresponding bounds without this factor.

Write $Q = \mathsf{Q}_{\mathsf{c}}$ and $Q_0 = Q_{\mathsf{c},0}$ in the row calculation. The weighted adjoint gives

$$((Q^*Q - Q_0^*Q_0)A)(b) = \eta^{-3}\sum_c \xi_c^{-1}\big[(Q(c,b) - Q_0(c,b))^{\mathsf{T}}(QA)(c) + Q_0(c,b)^{\mathsf{T}}((Q-Q_0)A)(c)\big].$$

The absolute row and entry bounds from proposition 14.10 are $2\xi_c$ and $2\eta^3\xi_c^{-2}$; their defects have the same scale factors times $C\widehat{\varepsilon}$. A bond in this piece belongs to at most twelve relevant rows, at levels $j, j+1$. Their cells have scales within a factor L of the output scale and tops near its top. The displayed formula is therefore bounded by $C\widehat{\varepsilon}\xi_y^{-3}\max_{z\sim y}|\Psi|_z^0$. The added boundary mass is absent on a regular piece. These estimates prove (439) in every forcing slot.

The difference between covariant and ordinary first differences is e times a neighboring value. Its contribution to a solution seminorm is at most $2L^2\widehat{\varepsilon}$ times a neighboring seminorm. Our cap on ε_0 consequently gives

$$|\mathcal{U}_\kappa^{-1}\Psi|_y \le \tfrac{3}{2}\max_{z\sim y}|\Psi|_z^0, \qquad |\mathcal{U}_\kappa\Psi|_y^0 \le \tfrac{3}{2}\max_{z\sim y}|\Psi|_z.$$

The flat commutator with either cutoff satisfies

$$|[\mathbb{K}_0, h]\Psi|_y' \le C_C\theta^{-1}\max_{z\sim y}|\Psi|_z^0, \qquad \theta = M/C_G, \tag{440}$$

for $C_C = C_C(L)$. To see the second-order cancellation, put $k_\nu(x) = \eta^{-1}(h(x+\eta e_\nu) - h(x))$. Then

$$[\Delta, h]f(x) = -\sum_\nu (k_\nu(x)\nabla_\nu^\eta f(x) + k_\nu(x-\eta e_\nu)\nabla_\nu^\eta f(x-\eta e_\nu)) - \eta^{-1}\sum_\nu (k_\nu(x) - k_\nu(x-\eta e_\nu))f(x).$$

The first and second cutoff differences bound these terms with the required weights. The first-order commutators contain only one cutoff difference. For the averages, the oscillation of h is at most $3/\theta$ on a cell and $4L/\theta$ on a double block, by summing its increments along coordinate paths. Insert these oscillations in the scalar and bond kernels above. At most eighteen flat reduced rows lie over a bond, so their sum has a bound depending only on L. This proves (440); we may enlarge C_C to be at least $300L^4$. The product rule also gives $|h\Psi|_y^0 \le 2\max_{z\sim y}|\Psi|_z^0$.

28.4. **Boundary inverses and positivity.**

Proof of theorem H. Boundary pieces. For a boundary piece κ, let

$$E^\kappa = E_0 \cap \kappa^{[1]}, \qquad S^\kappa = \{b \in S : \{b_-, b_+\} \cap E_0 \subset E^\kappa\}.$$

Let χ_κ restrict the vector component to S^κ and the three scalar components to E^κ, and set $\mathbb{K}_\kappa = \chi_\kappa \mathbb{K}_{\mathsf{c}} \chi_\kappa$ on its range. For the scalar systems delete the absent components. Use the reduced seminorm

$$|\Psi|_y^{\mathrm{r}} = \max\{\eta|A|_y, |v|_y, \eta^2|w|_y, \eta^4|\sigma(y)|\}.$$

There are $B_b, \delta_b > 0$, depending only on L, such that, for M sufficiently large, these compressions are invertible and

$$|\mathbb{E}_\kappa F|_y^{\mathrm{r}} \le B_b \sup_z e^{-\delta_b d(y,z)}|F|_z', \qquad \mathbb{E}_\kappa = \mathbb{K}_\kappa^{-1}\chi_\kappa. \tag{441}$$

For the vector system we use the additional ambient plaquette bound. Its first-source-to-vector block is

$$\mathsf{G}_\kappa \chi_{S^\kappa}, \qquad \mathsf{G}_\kappa = (\chi_{S^\kappa}\mathsf{H}_S\chi_{S^\kappa} + \eta^{-2})^{-1},$$

where G_κ is positive definite. If $|h| \le 1$ and $h = 0$ outside $\kappa^{[2/3]}$, then

$$|h\Psi|_y \le 2\max_{z\sim y}|\Psi|_z^{\mathrm{r}}.$$

If h also satisfies (438), then

$$|[\mathbb{K}_\kappa, h]\Psi|'_y \le 25\theta^{-1} \max_{z\sim y} |\Psi|^{\mathrm{r}}_z.$$

For a field supported in $\kappa^{[r]}$, $0 \le r \le 2/3$, the compression agrees with the full system and its output is supported in $\kappa^{[r+1/12]}$. The reverse dependence statement holds on the same two sets. We prove these local assertions directly. The margin from Ω_1 makes every site of E^κ a level-zero top. There $\mathsf{Q}'_{\mathsf{c}} = I$, $\mathsf{Q}_{\mathsf{c}} = \eta I$, and the row energy, including the touching-bond term, is $\eta^{-2} I$. No positive-level row meets S^κ: its double block is either contained in Ω_1 or lies within $3L\eta/2$ of an endpoint in Ω_1. The definition of S^κ also gives $D^*\chi_{S^\kappa} = \chi_{E^\kappa} D^* \chi_{S^\kappa}$. Put

$$D_\kappa = \chi_{S^\kappa} D \chi_{E^\kappa}, \qquad A'_\kappa = \chi_{E^\kappa} \Delta_U \chi_{E^\kappa} + \eta^{-2}, \qquad W_\kappa = \chi_{S^\kappa} \mathsf{H}_S \chi_{S^\kappa} + \eta^{-2}.$$

Scalar zero extension retains all six diagonal contributions of Δ_U. The compressed vector system is

$$((W_\kappa + D_\kappa D_\kappa^*)A - D_\kappa w, A'_\kappa v + w - D_\kappa^* A, A'_\kappa w - \sigma, v).$$

Consequently its solution is

$$\begin{aligned} v &= f_4, \qquad A = W_\kappa^{-1}[f_1 + D_\kappa(f_2 - A'_\kappa f_4)], \\ w &= f_2 + D_\kappa^* A - A'_\kappa f_4, \qquad \sigma = A'_\kappa w - f_3. \end{aligned} \tag{442}$$

For the three-slot scalar system set $A = 0$; for the one-slot system use $w = (A'_\kappa)^{-1} f_3$.

The plaquette Hessian formula in lemma 25.3 gives $\mathsf{H}_S \ge -12\alpha\eta^{-2}$. Indeed, each of the four bond variables occurs in three commutator pairs in one plaquette, and each bond belongs to four plaquettes. All contributing plaquettes have a corner in Ω_0, which is precisely where the additional hypothesis is used. Since $12\alpha \le 1/2$, we have

$$W_\kappa \ge \tfrac{1}{2}\eta^{-2}, \qquad A'_\kappa \ge \eta^{-2}.$$

The off-diagonal row sum of $\eta^2 W_\kappa$ is at most 13, and a nonzero entry joins base sites at distance $d \le 2$. For $\eta^2 A'_\kappa$ the corresponding bounds are 6 and 1. Exponential conjugation therefore gives, with $t = 1/300$,

$$|W_\kappa^{-1}(b, b')| \le 4\eta^2 e^{-td(\mathsf{x}(b), \mathsf{x}(b'))}, \qquad |(A'_\kappa)^{-1}(x, x')| \le 2\eta^2 e^{-td(x,x')}.$$

Indeed, the conjugation argument proving (246) applies: the error row and column sums are bounded by $13(e^{2t} - 1) \le 1/4$ and $6(e^t - 1) \le 1/2$, respectively, half the corresponding spectral lower bounds.

At most six bonds have a given base site, including the incoming touching bonds. Hence summing the first kernel bound over the bonds at one site costs at most 24, rather than 12. The elementary bounds

$$\eta^2 |A'_\kappa f|_y \le 13 \max_{z\sim y} |f|_z, \quad \eta |D_\kappa f|_y \le 2 \max_{z\sim y} |f|_z, \quad \eta |D_\kappa^* A|_y \le 6 \max_{z\sim y} |A|_z$$

and (442), followed by the weighted summation estimate used in (341), prove (441) with $\delta_b = t/2$. The first inverse block and its positivity also follow from (442).

All the compressed entries couple base sites within η. For $[W_\kappa, h]$, the off-diagonal sum is at most $12\eta^{-2}$ even without a curvature bound: the common plaquette entry is a product of isometries with $Y \mapsto c_p Y \pm \frac{1}{2}\eta^2 i[Y, F_p]$, whose norm is at most one since $c_p^2 + \eta^4 |F_p|^2 = 1$. The corresponding sums for $D_\kappa D_\kappa^*$, A'_κ and D_κ^* are bounded by $12\eta^{-2}, 6\eta^{-2}, 6\eta^{-1}$. In $[D_\kappa, h]$ one endpoint is the base site, so only one difference remains. Multiplication by the source weights bounds the four slots by $25, 12, 6, 0$ times $\theta^{-1} \max_{z\sim y} |\Psi|^{\mathrm{r}}_z$.

The range bound proves the two locality assertions. Differences of $h\Psi$ use only two neighboring values and their unitary transports, giving the factor two in the full seminorm. For touching bonds the base sites of translated bonds are still at sup distance at most η: the only possible longer pair would have an intervening exterior site with both axial neighbors interior. This is excluded because the complementary M-cubes contain at least two bare sites in that direction. Thus the argument covers both frames.

The parametrix. Use the flat inverse $\mathbb{E}_0^{[i]}$, $i = 1, 2, 3$, identified above, and the local defect estimate. For a fixed system suppress $[i]$; for the vector system fix either frame and assume the ambient plaquette bound. Let $B_1, \delta_1 > 0$ be the common flat bounds, let B_b, δ_b be the boundary bounds just proved, and put

$$\delta_2 = \min\{\delta_1, \delta_b\}, \quad B_2 = \max\{1, B_1, B_b\}, \quad k_0 = e^{6L\delta_2}, \quad B_3 = 6N_G k_0^2 B_2.$$

We choose $M_\diamond$ large enough for all the preceding geometric, flat-inverse and summation thresholds, and choose $\varepsilon_\diamond$ small enough for the local comparison. Increase $M_\diamond$ and decrease $\varepsilon_\diamond$ so that

$$C_*(C_V\varepsilon + C_C C_G/M) \le \tfrac{1}{2}, \qquad C_* = 27L^2 N_G k_0^2 B_2 B_3. \tag{443}$$

All these choices depend only on L; in particular we require $\delta_2\rho \ge 8\log L$ whenever $\rho \ge LM$.

For a regular piece let $\mathcal{U}_\kappa$ be its gauge map, $V_\kappa = \mathbb{K}_0 - \mathcal{U}_\kappa \mathbb{K}_{\mathsf{c}} \mathcal{U}_\kappa^{-1}$, and $C_h = [\mathbb{K}_0, h]$. Define

$$P_\kappa = \mathcal{U}_\kappa^{-1} h_\kappa \mathbb{E}_0 h_\kappa \mathcal{U}_\kappa, \quad R_\kappa = \mathcal{U}_\kappa^{-1}(V_\kappa h_\kappa - C_{h_\kappa})\mathbb{E}_0 h_\kappa \mathcal{U}_\kappa, \tag{444}$$

and on a boundary piece set $P_\kappa = h_\kappa \mathbb{E}_\kappa h_\kappa$ and $R_\kappa = -[\mathbb{K}_\kappa, h_\kappa]\mathbb{E}_\kappa h_\kappa$. The local identifications and support margins give

$$\mathbb{K}_{\mathsf{c}} P_\kappa = h_\kappa^2 - R_\kappa, \qquad \mathbb{K}_{\mathsf{c}} P = I - R, \qquad P = \sum_\kappa P_\kappa, \quad R = \sum_\kappa R_\kappa.$$

For example, on a regular piece expand $(\mathbb{K}_0 - V_\kappa)(h_\kappa \Phi)$, where $\Phi = \mathbb{E}_0 h_\kappa \mathcal{U}_\kappa F$. Both remainder terms are supported in $\kappa^{[1/6]}$; the boundary identity follows from the compression and locality proved above.

In the exponential supremum classes of (340), the local estimates give

$$P_\kappa \in \mathfrak{O}(\delta_2, 3k_0^2 B_2), \qquad R_\kappa \in \mathfrak{O}(\delta_2, 2k_0^2 B_2(C_V\varepsilon + C_C/\theta)).$$

The factor three combines the cutoff bound two with the gauge norm comparison $3/2$. Two neighboring-top shifts cost k_0^2, since each has distance at most $6L$. On boundary pieces the corresponding constants are $2k_0 B_b$ and $25k_0 B_b/\theta$, bounded by the displayed choices. A nonzero output seminorm has its top in $\kappa^{[1]}$, so at most N_G pieces contribute at any top. Thus

$$P \in \mathfrak{O}(\delta_2, B_3/2), \qquad R \in \mathfrak{O}(\delta_2, 1/2).$$

The composition rule following (340) now sums the geometric series. Since $\mathbb{K}_{\mathsf{c}} P(I-R)^{-1} = I$ on a finite-dimensional space, the full system is invertible and

$$\mathbb{E}_{\mathsf{c}} = \mathbb{K}_{\mathsf{c}}^{-1} = P(I-R)^{-1}, \qquad |\mathbb{E}_{\mathsf{c}} F|_y \le B_3 \sup_z e^{-\delta_2 d(y,z)} |F|'_z. \tag{445}$$

For $i = 1$, the nonnegative symmetric operator A'_{c} is therefore positive definite. For $i = 2$, this also makes $\mathsf{N}_{\mathsf{c}} = \mathsf{Q}'_{\mathsf{c}}(\mathsf{G}'_{\mathsf{c}})^2(\mathsf{Q}'_{\mathsf{c}})^*$ positive definite, and (445) gives the scalar three-component estimate used below. The elimination identity gives, for $i = 3$,

$$[\mathbb{E}_{\mathsf{c}} e_1 J]_A = \mathsf{G}_{\mathsf{c},S} J.$$

For the scalar inverse, the solution value has factor ξ_y^{-2} and the forcing has factor ξ_z^4 in (445); its gradient has factor ξ_y^{-3}. For the vector inverse these three factors are ξ_y^{-1}, ξ_z^3 and ξ_y^{-2}, respectively. Absorb the input scale pointwise under the supremum using

$$(\xi_z/\xi_y)^s e^{-\delta_2 d(y,z)} \le L^4 e^{-\delta_2 d(y,z)/2}, \qquad |s| \le 4,$$

by the scale-distance bound and $\delta_2\rho \ge 8\log L$. Substitution into (445) proves (435) and (436) with $B_\diamond = L^4 B_3$ and $\delta_\diamond = \delta_2/2$.

It remains to prove positivity of $\mathsf{A}_{\mathsf{c}}[S]$. Let P_A be the first-source-to-vector block of P. Then

$$P_A = \sum_{\kappa \text{ regular}} \mathcal{U}_\kappa^{-1} h_\kappa G_0 h_\kappa \mathcal{U}_\kappa + \sum_{\kappa \text{ boundary}} h_\kappa G_\kappa h_\kappa.$$

The flat G_0 and each G_κ are positive definite; unitary gauge covariance and $\sum h_\kappa^2 = 1$ show that P_A is symmetric positive definite.

Put

$$\|a\|_1 = \max_y\{\xi_y|a|_y, \xi_y^2|\nabla^U a|_y\}, \qquad \|\Psi\| = \max_y |\Psi|_y.$$

If $\Psi = \mathbb{E}_{\mathsf{c}} e_1 J = (a; v, w, \sigma)$, elimination and the scalar three-slot bound imply

$$\|\Psi\| \le 3L^2 B_3 \|a\|_1. \tag{446}$$

At an interior site, D^*a is a sum of three transported backward first differences, so that $\xi_y^2|D^*a|_y \le 3L^2\|a\|_1$. A site with an exterior neighbor, and all its interior neighbors, lie in the zero layer by the gap from Ω_1. There D^*a has at most six incident-bond terms, and $\eta^2|D^*a| \le 6\|a\|_1 \le 3L^2\|a\|_1$. Applying the three-slot inverse to $(D^*a, 0, 0)$ proves (446) for both frames.

The operator $\mathsf{A}_{\mathsf{c}}[S]P_A$ is similar to the symmetric matrix $P_A^{1/2}\mathsf{A}_{\mathsf{c}}[S]P_A^{1/2}$. Let $\mathsf{A}_{\mathsf{c}}[S]P_A J = zJ$ with $J \ne 0$, and set $a = \mathsf{G}_{\mathsf{c},S} J$. Thus $P_A J = za$. On a regular piece apply $\mathbb{E}_0$ to $\mathbb{K}_0(h_\kappa \mathcal{U}_\kappa \widetilde{h}_\kappa \Psi)$ and then multiply by $\mathcal{U}_\kappa^{-1} h_\kappa$. Locality and $\widetilde{h}_\kappa = 1$ near the support of h_κ give

$$h_\kappa^2 \Psi = P_\kappa e_1 J + \Theta_\kappa, \quad \Theta_\kappa = \mathcal{U}_\kappa^{-1} h_\kappa \mathbb{E}_0(h_\kappa V_\kappa + C_{h_\kappa})\mathcal{U}_\kappa \widetilde{h}_\kappa \Psi.$$

For a boundary piece the same identity holds with $\Theta_\kappa = h_\kappa \mathbb{E}_\kappa[\mathbb{K}_\kappa, h_\kappa]\widetilde{h}_\kappa \Psi$. The two cutoff bounds and the local comparison give

$$|\Theta_\kappa|_y \le 9B_2(C_V\varepsilon + C_C/\theta)\|\Psi\|.$$

For boundary pieces use the reduced norm before applying $\mathbb{E}_\kappa$; this gives $50B_b/\theta$, which is bounded by the displayed constant. At most N_G terms have nonzero seminorm at a given top. Summing and taking the vector component yields

$$(1-z)a = \Big[\sum_\kappa \Theta_\kappa\Big]_A, \qquad |1-z|\|a\|_1 \le C_*(C_V\varepsilon + C_C/\theta)\|a\|_1 \le \tfrac{1}{2}\|a\|_1. \tag{447}$$

Since $a \ne 0$, every eigenvalue z is at least $1/2$. The symmetric matrix $P_A^{1/2}\mathsf{A}_{\mathsf{c}}[S]P_A^{1/2}$ and hence $\mathsf{A}_{\mathsf{c}}[S]$ are positive definite. □

29. The inverse on the original region

The action on a region contains only its inner plaquettes. To pass from the ambient inverse to this action, we first apply theorem H on a collar of the positive-level regions, where the two Hessians agree. A second inverse near the outer boundary then gives the inverse on the whole region. Exponential conjugation supplies its decay estimates.

Use the preceding level-n units and a separated sequence with parameters $\mathfrak{M}, R_1$, where $R_1 \ge L$ and both are powers of L. Put $X = \Omega_0 \cap \mathbb{T}^{(0)}$. Let $\mathfrak{b}$ and $\mathfrak{p}$ be the inner positive bonds and inner plaquettes, and let D be the covariant gradient from X to $\mathfrak{b}$. Thus

$$\Delta_U = D^*D + \eta^{-2} n_\partial,$$

where $n_\partial(x)$ counts the missing bare neighbours. Let H_U be the Hessian of $\eta^{-1}A_{\Omega_0}$; only plaquettes in $\mathfrak{p}$ enter this Hessian. Write Q', Q for the corner scalar and reduced bond averages of the preceding section. Their fine, scalar-top and bond-row pairings have weights $\eta^3, \xi_y^3, \xi_c^{-1}$. In particular, the level-zero bond average is ηI.

Let Ξ multiply by the local scale and retain the scalar multiplier $a|_{\Lambda_j} = \varkappa_j \xi_j^{-2}$ from the preceding section. Define $R_\perp$ to be the orthogonal projection onto $\Delta_U \ker Q'$, and set

$$A' = \Delta_U + Q'^* a Q', \qquad A^\perp = H_U + D R_\perp D^* + Q^* Q.$$

Once the indicated inverses exist, put

$$G' = (A')^{-1}, \qquad N = Q'G'^2Q'^*, \quad C = N^{-1},$$
$$P = G'Q'^*CQ'G', \quad R = I - P, \qquad A_0 = H_U + DRD^* + Q^*Q, \qquad G_0 = A_0^{-1}.$$

We will prove that $R = R_\perp$ and $A_0 = A^\perp$.

All fine fields are extended by zero. In $|v|_y$, bonds and differences are assigned by their initial or base site, and $|v|_{[s]} = \sup_y \xi_y^{-s}|v|_y$. For a bond field,

$$(\nabla_\nu^U A_\mu)(x) = \begin{cases} \eta^{-1}(\mathrm{Ad}(U_\nu(x))A_\mu(x+\eta e_\nu) - A_\mu(x)), & \langle x, x+\eta e_\nu\rangle \in \mathfrak{b}, \\ -\eta^{-1}A_\mu(x), & \text{otherwise.} \end{cases}$$

The zero extension also applies when the translated bond is missing. Scalar gradients are evaluated on $\mathfrak{b}$.

Theorem I. Fix odd $L \geq 13$. There are a power M_{ps} of L and constants $g_{\mathrm{p}}, B_{\mathrm{ps}}, a_{\mathrm{ps}} > 0$ and $0 < \delta_{\mathrm{ps}} \leq 1$, depending only on L, with the following property. Suppose

$$\mathfrak{M} \geq M_{\mathrm{ps}}, \qquad R_1\mathfrak{M} \geq g_{\mathrm{p}},$$

and that the level-n torus has at least $6LM_{\mathrm{ps}}$ sites in each direction. Let U be real on the inner bonds. For some $0 < \alpha \leq a_{\mathrm{ps}}$, assume

$$|\partial U(p) - I| < \alpha\eta^2 \mathsf{L}_{\mathbf{B}}(p)^{-2}, \qquad |(D_U^{\mathsf{T}}\partial U)(b)| < \alpha\eta^2 \mathsf{L}_{\mathbf{B}}(b)^{-3}$$

at every inner plaquette p and every full positive bond b. The second expression is the matrix-valued plaquette divergence of (68), including its scalar part. Then A', N, A_0 are positive definite for their stated pairings. For f supported over one top y',

$$|G'f|_y \leq B_{\mathrm{ps}}\xi_y^2 e^{-\delta_{\mathrm{ps}}d(y,y')} \sup|f|, \qquad |DG'f|_y \leq B_{\mathrm{ps}}\xi_y e^{-\delta_{\mathrm{ps}}d(y,y')} \sup|f|,$$
$$|G_0f|_y \leq B_{\mathrm{ps}}\xi_y^2 e^{-\delta_{\mathrm{ps}}d(y,y')} \sup|f|, \qquad |\nabla^U G_0f|_y \leq B_{\mathrm{ps}}\xi_y e^{-\delta_{\mathrm{ps}}d(y,y')} \sup|f|.$$

The scalar and vector lines apply on their respective spaces. The counting kernel of C satisfies

$$|C(y,y')| \leq B_{\mathrm{ps}} e^{-\delta_{\mathrm{ps}}d(y,y')} \min\{\xi_y^{-4}, \xi_y^{-7/2}\xi_{y'}^{-1/2}\}.$$

All constants are uniform in the depth, lattice spacing and regions. ■□

Proof. We use the summation estimates (231) and (233) in the following form. For $a > 0$, put

$$c_{\mathrm{sum}}(a) = L^4 c_S(a/2), \qquad w_{\mathrm{sum}}(a) = \max\{\mathsf{r}_S(a/2), 8a^{-1}\log L\}.$$

If $R_1\mathfrak{M} \geq w_{\mathrm{sum}}(a)$, then, for $|s| \leq 4$,

$$\sum_{y'} e^{-ad(y,y')}(\xi_y/\xi_{y'})^s \leq c_{\mathrm{sum}}(a).$$

Indeed the scale ratio costs at most $L^4 e^{ad/2}$. All occurrences below of a summation threshold refer to w_{sum}; the final choice of the gap includes them.

The collar. Fix a power $M = M_{\mathrm{ps}} \geq \max\{M_\diamond, L^2\}$ of L, and put $\rho = LM$. We will choose a power $g_0 \geq L^2M$ and require $R_1\mathfrak{M} \geq Lg_0$. Suppose first that $\Omega_1 \neq \emptyset$. Let Ω_0^{a} be the union of the level-zero M-cubes at sup distance at most $g_0\eta$ from Ω_1. It contains this neighbourhood and lies within $(g_0 + M)\eta \leq 2g_0\eta$ of Ω_1. Hence

$$\mathrm{dist}_\infty((\Omega_0^{\mathrm{a}})^c, \Omega_1) > g_0\eta, \qquad \mathrm{dist}_\infty(\Omega_0^{\mathrm{a}}, \Omega_0^c) > (L-2)g_0\eta.$$

The auxiliary sequence $(\Omega_0^{\mathrm{a}}, \Omega_1, \dots, \Omega_n)$ satisfies the hypotheses on regions in theorem H with gap $\rho = LM$: the higher regions are unions of M-cubes by refinement, and their original gaps exceed $\rho\xi_j$. The torus floor is 6ρ. Write X_{a} for its bare sites and S_α, S_β for its inner and touching positive bonds.

Every ambient plaquette touching the collar is an original inner plaquette, and every bond in S_β is full and belongs to $\mathfrak{b}$. Also Ω_1 and its first two neighbour layers lie in the collar. Extend U by the identity outside the original inner bonds. On the collar and its neighbouring plaquettes this extension changes neither tested quantity nor the local block side: the positive members of the sequence are unchanged. In particular, the extended field satisfies the ambient curvature hypothesis of theorem H on the collar.

Here is the required local gauge check. A cube $Q \subset \Omega_q$ of side $l\xi_q$, where $q \geq 1$ and l is a multiple of L, has a concentric enlargement of side $(l+4L)\xi_q$. Provided this is smaller than half the torus period, its entire 2η neighbourhood lies in Ω_{q-1}: its points are at distance at most $(2L^2+2)\xi_{q-1}$ from Q, less than the separation. The local block sides of the tested plaquettes and bonds are therefore at least ξ_q/L. In level-q units the hypotheses of theorem F hold with parameter $L^3\alpha$; the divergence acquires a factor ξ_q under this change of units. Apply that theorem on the smaller enlargement of side $(l+4)\xi_q$ and restrict to Q. With its constants B_*, c_*, put

$$B_{\mathrm{p}} = \max\{1, LB_*/21\}, \qquad c_{\mathrm{p}} = \min\{1, 21c_*/L\}.$$

The resulting potential satisfies

$$\xi_q|A|,\ \xi_q^2|\nabla^\eta A|,\ \xi_q^3|D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} A| < 21L^2 B_{\mathrm{p}} l\alpha, \qquad 21L^2 l\alpha \leq c_{\mathrm{p}}. \tag{448}$$

For $q = 0$ use the comb gauge of lemma 14.4 in the enlarged box. Since it has at most $5l$ sites per side, $|U^h - I| \leq 10l\alpha$. Real logarithms $U^h = e^{i\eta A}$ give $\eta|A| < 16l\alpha$ and $\eta^2|\nabla^\eta A| < 32l\alpha$. The ordinary curl–divergence is η^{-2} times four signed sums of four potential values. Its stencil lies in the 2η neighbourhood of Q, hence in the enlarged box. Thus

$$\eta^3|D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} A| < 4 \cdot 4 \cdot 16 l\alpha = 256 l\alpha < 21L^2 B_{\mathrm{p}} l\alpha.$$

This proves (448) also for $q = 0$.

Now let Q have side $sM\xi_j$, $1 \leq s \leq 12$, and meet the auxiliary layer j. For $j = 0$ use $q = 0, l = sM$ above; the enlargement lies within $14M\eta \leq g_0\eta$ of the collar. For $j \geq 1$ use $q = j-1, l = sLM$; the enlargement lies within $14LM\xi_{j-1}$ of Ω_j and hence in Ω_{j-1}. Its side is at most $16M$ in the present units, less than half the period. Changing the two estimates from scale ξ_{j-1} to ξ_j costs at most L^2. Thus these gauges satisfy

$$U^v = e^{i\eta A}, \qquad \xi_j|A|,\ \xi_j^2|\nabla^\eta A| < \kappa_\square \alpha, \qquad \kappa_\square = 252L^5 B_{\mathrm{p}} M,$$

if $252L^3 M\alpha \leq c_{\mathrm{p}}$. A regular piece's $3M\xi_j$ enlargement either meets layer j, or lies wholly in layer $j+1$. In the second case it is contained in a cube of side $2M\xi_{j+1}$ meeting that layer; restrict its gauge. Both bounds at scale ξ_j follow. Choose the radius so that

$$\alpha \leq L^{-2}\min(1, c_{\mathsf{c}}), \qquad 252L^3 M\alpha \leq c_{\mathrm{p}}, \qquad \kappa_\square \alpha \leq \varepsilon_\diamond. \tag{449}$$

These are exactly the local gauge hypotheses of theorem H.

We next identify its operators. Positive-level cells and tops are unchanged. Nested locality from proposition 14.10 identifies their scalar transports and the restrictions of Q'; both scalar Laplacians retain the full diagonal $6\eta^{-2}$. The two kernels of Q' coincide under zero extension: their elements vanish in layer zero. Their Laplacian images are the same subspace, supported in Ω_1 and its bare neighbours. Thus their orthogonal projections agree there.

The positive-level interior and outward rows also coincide. Their double blocks lie in the collar, within $2L\eta$ of Ω_1 at level one, and inside Ω_{j-1} at higher levels. Nested locality identifies their kernels, with the same normalization $\eta\prod_{i<j}\mathbf{Q}_{\mathsf{c}}$ and row energy $\sum_c \xi_c^{-1}|QA(c)|^2$. Every ambient plaquette contributing on S_α or S_β is original inner, so the Hessian forms agree. On the support of the projection all six incident bonds lie in S_α; the projected divergence forms agree as well. The level-zero rows in $S_\beta \setminus S_\alpha$ contribute $\xi_0^{-1}|\eta A(b)|^2 = \eta|A(b)|^2$, precisely the additional mass term in theorem H.

Consequently its scalar operator is the compression A'_{a} of A' to X_{a}, and its two vector operators are the compressions $A_{\mathrm{a}}[S]$ of $A^{\perp}$ to S. They are positive definite. With $B_0 = \max(1, B_\diamond)$, their inverses obey

$$\begin{aligned} |G'_{\mathrm{a}} g|_{[2]} &\le B_0 \|g\|_\infty, \qquad & |DG'_{\mathrm{a}} g|_{[1]} &\le L B_0 \|g\|_\infty, \\ |G_{\mathrm{a},S} g|_{[2]} &\le L^2 B_0 \|g\|_\infty, \qquad & |\nabla^U G_{\mathrm{a},S} g|_{[1]} &\le L B_0 \|g\|_\infty. \end{aligned} \tag{450}$$

Here scalar gradients are taken on S_α, and vector differences where both translated bonds lie in S. These follow directly from the supremum bounds of theorem H. On a touching bond whose initial endpoint is outside the collar both endpoints have scale η, so changing its assigned endpoint does not change the scale. In what follows write G_{a} for G_{a,S_α}.

Two cutoffs and the scalar inverse. Put $d_1(x) = \eta^{-1} \operatorname{dist}_\infty(x, \Omega_1)$ and $t = (g_0 - 5)/2$. Choose ζ_1, ζ_2 with $\zeta_1^2 + \zeta_2^2 = 1$, equal to $(1, 0)$ for $d_1 \le t$ and to $(0, 1)$ for $d_1 \ge 2t$, by using the cosine and sine of $\pi(d_1 - t)/(2t)$ in between. Their difference on a bare bond is at most $2/t$. If Ω_1 is empty, use $(\zeta_1, \zeta_2) = (0, 1)$ and omit the collar terms.

Let A'_1 be the compression to $\{d_1 > t - 2\}$. There $A' = \Delta_U + \eta^{-2}$, so its diagonal is $7\eta^{-2}$ and its off-diagonal row sum is at most $6\eta^{-2}$. The maximum principle gives $\|G'_1 g\|_\infty \le \eta^2 \|g\|_\infty$. Both cutoffs commute with $Q'^* a Q'$. Their commutators with Δ_U are supported in $t - 1 < d_1 < 2t + 1$, where all neighbours are in layer zero and in the collar, and satisfy

$$|[\Delta_U, \zeta_i] v(x)| \le 12 t^{-1} \eta^{-2} \max_{x' \sim x} |v(x')|.$$

Thus

$$P' = \zeta_1 G'_{\mathrm{a}} \zeta_1 + \zeta_2 G'_1 \zeta_2, \qquad A' P' = I + E', \qquad \|E'\|_{\infty \to \infty} \le 12 t^{-1} (B_0 + 1).$$

For $t \ge 24(B_0 + 1)$ this proves invertibility. Since $A' = D^* D + \eta^{-2} n + Q'^* a Q'$ is symmetric and nonnegative, it is positive definite. The product rule for D gives

$$\max\{|G' g|_{[2]}, |DG' g|_{[1]}\} \le b' \|g\|_\infty, \qquad b' = 2(L + 4) B_0. \tag{451}$$

Also $Q' Q'^* = I$. The projection calculation in section 21.4 therefore applies with these weighted adjoints: $N > 0$, P is orthogonal, and $\operatorname{ran} R = \Delta_U \ker Q'$. Every member of $\ker Q'$ vanishes in layer zero. Consequently R is supported in Ω_1 and its bare neighbours, and

$$D R D^* + \sum_{j \ge 1} Q_j^* Q_j$$

is supported, in both variables, where $d_1 \le 2L$.

A smooth distance. Choose a smooth even function ψ, with $0 \le \psi \le 1$, equal to one on $[-1/2, 1/2]$ and zero outside $(-3/4, 3/4)$, and put $c_\psi = \max(1, \|\psi'\|_\infty, \|\psi''\|_\infty)$. For $z \in \Lambda_*$ set

$$\Psi_z(x) = \prod_{\mu=1}^{3} \psi((x - z)_\mu / \xi_z), \qquad \phi_z = \Psi_z / \sum_w \Psi_w, \qquad f_{y'}(x) = \sum_z \phi_z(x) d(z, y').$$

Use least coordinate lifts; the supports make this choice immaterial. The denominator is at least one on Ω_0. At any point within distance η of a layer-j site, only levels $j - 1, j, j + 1$ occur, and at most eight centres occur at each level. Each contributing centre has weighted distance at most $9L/2$ from that point: its supporting cube meets only those three layers, and a coordinate path has length at most $9\xi_z/4$ and weight at most $2L/\xi_z$. The quotient rule bounds their first and second derivatives by $C(L)\xi_j^{-1}$ and $C(L)\xi_j^{-2}$, respectively. Subtract one fixed $d(z_0, y')$ when differentiating $\sum_z \phi_z = 1$. The resulting differences are at most $9L$. Integrating the first and second derivative bounds along bare segments proves

$$\begin{aligned} |f_{y'}(x) - d(x, y')| &\le s_0, \\ |f_{y'}(x') - f_{y'}(x)| &\le s_1 \eta / \xi(x), \\ |f_{y'}(x + \eta e_\mu) + f_{y'}(x - \eta e_\mu) - 2 f_{y'}(x)| &\le s_2 \eta^2 / \xi(x)^2, \end{aligned} \tag{452}$$

where $s_0 = 9L/2$ and s_1, s_2 depend only on L and our fixed bump. The last estimate is used only when all three sites are in X. Two sites in a cell, adjacent bare sites, and two sites in a row double block have weighted distance at most $4L$. Thus the oscillation of $f_{y'}$ on any such set is at most $s_3 = 4L + 2s_0 = 13L$; in general it is at most $d(y(x), y(x')) + s_3$.

For $\varphi = \theta f_{y'}$ with $0 < \theta \le (s_1 + s_3)^{-1}$, Taylor's formula gives the following conjugation bound:

$$\|(e^{\varphi} A' e^{-\varphi} - A')u\|_\infty \le \epsilon' \theta \max\{|u|_{[2]}, |Du|_{[1]}\}, \tag{453}$$

where $\epsilon' = (12L^2 + 2)s_3 + 12Ls_1 + 3(s_2 + 2s_1^2)$. Indeed, outside Ω_1 we have $\xi(x) = \eta$, and every existing neighbour has scale at most $L\eta$; the direct row bound therefore suffices. Inside Ω_1, write each transported neighbour as $u(x)$ plus its covariant difference. The latter terms contribute $12Ls_1$; pairing the two opposite neighbours in each direction bounds the coefficient of $u(x)$ by $3(s_2 + 2s_1^2)$. The average term contributes $2s_3$, using its cell support. The remaining direct neighbour bound is $12L^2 s_3$.

Take $\theta_1 = \min\{(s_1 + s_3)^{-1}, (2\epsilon' b')^{-1}\}$. By (451) the conjugation error composed with G' has supremum norm at most $1/2$. Its Neumann inverse costs a factor two. On the source cell $e^{\varphi} \le 3$, and on the output cell $e^{-\varphi} \le 3e^{-\theta_1 d(y,y')}$. The product rule for $D(e^{-\varphi} v)$ adds at most $2L^2 |v|_{[2]}$ to the normalized derivative. Hence the scalar assertions hold with

$$\delta_1 = \theta_1, \qquad B_1 = 18(1 + 2L^2)b'.$$

The Gram gap. Put $K = Q'G'Q'^*$. Its inverse has the variational formula

$$(\nu, K^{-1}\nu)_{\mathbf{B}} = \min_{Q'\lambda = \nu} (\lambda, A'\lambda)_\eta.$$

On a positive-level cell use $\psi_y(x) = \prod_\mu (1 - 3|(x - y)_\mu|/\xi_y)_+$. It vanishes on the outermost sheet, its mean is at least 8^{-3}, and its first differences are at most $3/\xi_y$. Let $h_y(x)$ be the bare comb transport from y to x. The curvature-only transport estimate in the proof of proposition 14.10, applied to the path from the lowest corner o through y to x, gives

$$\left|\mathrm{Ad}(T_U^{\mathsf{c}}(y, x) h_y(x)^{-1}) - \mathrm{Ad}(U(\Gamma_{o,y}))\right| \le 2\kappa_7^{\mathsf{c}} \alpha \le \tfrac{1}{2}.$$

Consequently

$$S_y = |B^j(y)|^{-1} \sum_{x \in B^j(y)} \psi_y(x) \,\mathrm{Ad}(T_U^{\mathsf{c}}(y, x) h_y(x)^{-1}) \quad \text{satisfies} \quad \|S_y^{-1}\| \le 2 \cdot 8^3.$$

Set $\lambda = \nu$ on layer zero and $\lambda(x) = \psi_y(x) \,\mathrm{Ad}(h_y(x))^{-1} S_y^{-1} \nu(y)$ on the other cells. Then $Q'\lambda = \nu$. The comb gauge bound lemma 14.4 gives $|U^{h_y} - I| \le 2\alpha\eta/\xi_y$, hence $|D\lambda_y| \le 4\xi_y^{-1} |S_y^{-1}\nu(y)|$ if $4\alpha \le 1$. Distinct positive-level test functions have no neighbouring nonzero values, so their energy cross terms vanish. There are at most $3(\xi_y/\eta)^3$ positive bonds per cell. Layer zero contributes at most $12\eta^{-2}$, and the average term at most Ξ^{-2}. Therefore

$$K^{-1} \le C_K \Xi^{-2}, \qquad C_K = 3 \cdot 2^{24} + 13.$$

We use twice the matrix inversion calculation in section 21.4: a positive symmetric matrix with a uniform gap and an exponentially decaying kernel has an exponentially decaying inverse, by conjugating with $e^{\theta d(\cdot, y')}$ and bounding the error by its row and column sums. Remove the coarse weights by $\Phi\mu(y) = \xi_y^{3/2} \mu(y)$. For $\widetilde{K} = \Xi^{-1} K \Xi^{-1}$ we have $\widetilde{K} \ge C_K^{-1}$ and

$$|\widetilde{K}^{\Phi}(y, y')| \le B_1 e^{-\delta_1 d(y,y')}.$$

Indeed, the scalar estimate first gives the extra factor $(\xi_y/\xi_{y'})^{5/2}$; symmetry gives its reciprocal, and the smaller is at most one. The inversion calculation gives constants $B_K, \delta_K > 0$ for its inverse kernel. Absorbing the scale ratio and summing rows and columns gives $\|\Xi^2 K^{-1}\| \le n_1$ for a fixed n_1,

in the weighted coarse Hilbert norm. Since $Q'^{*}Q'$ is an orthogonal projection, $N \geq K^2$ as quadratic forms. Applying the preceding norm bound to $K\Xi^{-2}\mu$ yields

$$\widetilde{N} := \Xi^{-2} N \Xi^{-2} \geq n_1^{-2}.$$

Two applications of the scalar estimate give

$$|\widetilde{N}^{\Phi}(y,y')| \leq B_1^2 L^5 c_{\mathrm{sum}}(\delta_1/4) e^{-\delta_1 d(y,y')/2}.$$

Here the two scale ratios have powers $3/2$ and $7/2$; the same scale-ratio and convolution estimates used in section 21.4 absorb them. Apply the matrix inversion calculation once more. Undoing Φ and Ξ^{-2} gives the factor $\xi_y^{-7/2}\xi_{y'}^{-1/2}$; absorbing the remaining half-power ratio gives the factor ξ_y^{-4} at half the rate. Denote common constants for these two bounds by $B_2, \delta_2 > 0$.

Put $V = DG'Q'^{*}$ and $Y = DPD^{*} = VCV^{*}$. The differentiated scalar bound gives $|Ve_z|_y \leq B_1 \xi_y e^{-\delta_1 d(y,z)}|e|$. Taking its weighted transpose, and counting at most $3\xi_{y'}^3\eta^{-3}$ positive bonds per cell, gives

$$|(V^{*}A)(z)| \leq 3B_1 \xi_z^{-3} \sum_{y'} \xi_{y'}^4 e^{-\delta_1 d(z,y')} |A|_{y'}.$$

In the product VCV^{*} the scale factor is $\xi_y \xi_z^{-4} \xi_{z'}^{-3} \xi_{y'}^4$. It equals $\xi_y^{-2}(\xi_y/\xi_z)^3(\xi_{z'}/\xi_z)(\xi_{y'}/\xi_{z'})^4$. Absorbing these ratios and taking the two exponential sums proves, for fixed $B_Y, \delta_Y > 0$,

$$|YA|_y \leq B_Y \xi_y^{-2} \sum_{y'} e^{-\delta_Y d(y,y')} |A|_{y'}. \tag{454}$$

The regional vector inverse. The one-plaquette Hessian calculation gives $H_U = D^{\mathsf{T}}D + H'_U$, where

$$\sum_{b'} |(D^{\mathsf{T}}D)(b,b')| \leq 16\eta^{-2}, \qquad \sum_{b'} |H'_U(b,b')| \leq 20\alpha \mathsf{L}_{\mathbf{B}}(b)^{-2}.$$

Only bonds in a common inner plaquette contribute. The counts are four plaquettes at a bond, times four entries for the first bound and at most five curvature factors for the second. Also $Q_0^{*}Q_0 = \eta^{-2}$ when $d_1 \geq 2$. Compress A_0 to the bonds whose endpoints satisfy $d_1 > t - 3$. The nonlocal part vanishes there, so $A_{\mathrm{l}} = H_U + \eta^{-2}$ on zero extensions. For $40\alpha \leq 1$, its spectrum lies in $[\frac{1}{2}, 34]\eta^{-2}$. The operator $J = I - (4/69)\eta^2 A_{\mathrm{l}}$ has norm at most $67/69$ and range one in bare sup distance. Its geometric series therefore gives

$$\|G_{\mathrm{l}} g\|_\infty \leq c_{\mathrm{l}} \eta^2 \|g\|_\infty, \qquad c_{\mathrm{l}} = 6 \sum_{z \in \mathbb{Z}^3} (67/69)^{|z|_\infty}.$$

First establish positivity separately. The nonlocal part commutes with each ζ_i, since it is supported where $d_1 \leq 2L < t$. For $C_i = \zeta_i^2 A_0 - 2\zeta_i A_0 \zeta_i + A_0 \zeta_i^2$,

$$A_0 = \sum_i \zeta_i A_0 \zeta_i + \tfrac{1}{2} \sum_i C_i.$$

The kernel of C_i is $(\zeta_i(b_-) - \zeta_i(b'_-))^2 H_U(b,b')$. It is supported in the layer-zero transition strip and has row sum at most $132 t^{-2}\eta^{-2}$. The collar bound in (450), symmetry and positivity imply $A_{\mathrm{a}} \geq (L^2 B_0)^{-1}\Xi^{-2}$: the symmetric positive matrix $\Xi^{-1} G_{\mathrm{a}} \Xi^{-1}$ is similar to $\Xi^{-2} G_{\mathrm{a}}$, whose supremum norm is at most $L^2 B_0$. Combining this with the layer-zero lower bound and $\zeta_1^2 + \zeta_2^2 = 1$, and taking $t \geq 17 L B_0$, gives

$$(A, A_0 A)_\eta \geq (2L^2 B_0)^{-1} (\Xi^{-1}A, \Xi^{-1}A)_\eta.$$

Now form $P_0 = \zeta_1 G_{\mathrm{a}} \zeta_1 + \zeta_2 G_{\mathrm{l}} \zeta_2$. Its error is supported in the same transition strip. The single commutators have row bound $66 t^{-1}\eta^{-2}$ there, so

$$A_0 P_0 = I + E_0, \qquad \|E_0\|_{\infty \to \infty} \leq 66 t^{-1}(L^2 B_0 + c_{\mathrm{l}}).$$

For $t \geq 132(L^2 B_0 + c_{\mathrm{l}})$, the same Neumann and product-rule argument as for G' yields

$$\max\{|G_0 g|_{[2]}, |\nabla^U G_0 g|_{[1]}\} \leq b'' \|g\|_\infty, \qquad b'' = 4(L^2 B_0 + c_{\mathrm{l}}).$$

Use the same smooth weight $\varphi = \theta f_{y'}$. Since $R = I - P$,

$$A_0 = (D^{\mathsf{T}}D + DD^*) + H'_U + Q^*Q - Y.$$

For $x \in \Omega_1$ all neighbours needed in the curl–divergence identity from the proof of lemma 25.3 are inner. Its diagonal part is the scalar covariant Laplacian on each component; its two mixed transports differ by at most $2\alpha\eta^2\xi(x)^{-2}$. Thus the same opposite-neighbour cancellation used in (453) applies to the vector components. In layer zero direct row bounds suffice. For Q^*Q, retain the adjoint factor:

$$(Q^*Qv)(b) = \eta^{-3}\sum_j \xi_j^{-1} \sum_{c\in\Gamma_j^\flat} Q(c,b)^{\mathsf{T}} \sum_{b'} Q(c,b')v(b').$$

The corner product bounds give row sum $2\xi_j$ and level-j column sum $2\eta^3\xi_j^{-2}$. On its double block the weight oscillation is at most $2s_3\theta$ and $|u| \le L^2\xi_j^2|u|_{[2]}$. Hence each of the at most three levels contributes at most $8L^2s_3\theta|u|_{[2]}$. Finally (454), $|e^r - 1| \le |r|e^{|r|}$, and one exponential sum give

$$\|(e^{\varphi}A_0e^{-\varphi} - A_0)u\|_\infty \le \epsilon''\theta \max\{|u|_{[2]}, |\nabla^U u|_{[1]}\},$$

provided $\theta \le \delta_Y/4$, where one may take

$$\epsilon'' = 12Ls_1 + 3(s_2 + 2s_1^2) + 115L^2s_3 + 3B_YL^2(s_3 + 2/\delta_Y)c_{\mathrm{sum}}(\delta_Y/4).$$

Choose additionally $\theta \le (2\epsilon''b'')^{-1}$. The same conjugated Neumann inverse and the same source and output weight bounds as in the scalar case give the vector estimates, with $B_3 = 18(1 + 2L^2)b''$ and rate $\delta_3 = \theta$.

All constants used so far depend only on L and the fixed collar constants. Choose the power g_0 so that $g_0 \ge L^2M$, $t = (g_0 - 5)/2$ exceeds the four bounds used above, and Lg_0 exceeds the finitely many summation and scale-ratio thresholds. Put $g_{\mathrm{p}} = Lg_0$. Choose a_{ps} below the restrictions in (449), $(4\kappa_7^{\mathfrak{c}})^{-1}$, and $1/40$. Finally enlarge B_{ps} and decrease δ_{ps} to cover the three pairs of bounds. The empty-region case is vacuous; the case $\Omega_1 = \emptyset$ was included by omitting the collar terms. □

29.1. **The constrained inverse.** The remaining constraint is imposed on the reduced corner rows. We construct a right inverse with fields supported in the cores of the averaging cells. Its energy bound gives a uniform gap for the row Gram matrix, and hence the constrained inverse and the weighted transpose bounds used in the nonlinear equation.

Keep the regional operators of theorem I. Put

$$M_Q = QG_0Q^*, \qquad H_0 = G_0Q^*M_Q^{-1}, \qquad \widetilde{G}_0 = G_0 - H_0QG_0.$$

The existence of M_Q^{-1} is part of the following proposition. For a bond or plaquette field, let $|f|_{(-s)} = \sup_{\mathfrak{e}} \mathsf{L}_{\mathbf{B}}(\mathfrak{e})^s|f(\mathfrak{e})|$; for sites use $\xi(x)$. A difference uses the largest local scale at the endpoints of its two bonds. Set

$$\|A\|_X = \max\{|A|_{(-1)}, |\nabla^U A|_{(-2)}\}, \qquad \|\omega\|_1^2 = \sum_c \xi_c^{-1}|\omega(c)|^2.$$

Below D on bond fields is the covariant curl into inner plaquettes, whereas D^* on bond fields is the divergence into sites. The region and zero-extension conventions remain those fixed above.

Proposition 29.1. There are constants $0 < a_Q \le a_{\mathrm{ps}}$, $\gamma_Q > 0$, $w_S, c_\delta \ge 1$, and $K_1 \ge 4$, depending only on L, with the following property. Assume the hypotheses of theorem I with $\alpha \le a_Q$ and $R_1\mathfrak{M} \ge w_S$. Then $M_Q \ge \gamma_Q I$ in the row pairing. Put $K_7 = K_1^2c_\delta$. We have $QH_0 = I$, $Q\widetilde{G}_0 = 0$, and

$$\|H_0\beta\|_X \le K_1 \sup|\beta|, \qquad \|\widetilde{G}_0 f\|_X \le K_7|f|_{(-3)}.$$

For real bond fields f, inner-plaquette fields h, site fields g, and row fields ω,

$$\sup|H_0^*f| \le K_1|f|_{(-3)}, \qquad \sup|(DH_0)^*h| \le 3K_1|h|_{(-2)}, \qquad \sup|(D^*H_0)^*g| \le 3K_1|g|_{(-2)},$$
$$|Rg|_{(-2)} \le (1+K_1^3)|g|_{(-2)}, \qquad |Q^*\omega|_{(-3)} \le 2(L^3+2)\sup|\omega|.$$

Adjoints use the fine weight η^3 and row weight ξ_c^{-1}. All bounds are uniform in the volume, lattice spacing, regions and number of layers. ■

Proof. Temporarily retain all positive determining rows, including the inward ones, and write Q^+ for this corner average. The weight of each row is still ξ_c^{-1}. For a level-j row c, choose $y_c = c_-$ when $c_- \in \Lambda_j$ and $y_c = c_+$ otherwise. There are at most six rows with a given top. Put $\vartheta_c = L$ for an outward row and $\vartheta_c = 1$ otherwise. Every plaquette in its double block D_c has curvature at most $\vartheta_c^2\alpha L^{-2j}$, and $\mathsf{D}_c \subset \Omega_0$. Thus the folding, locality, product and path calculations in the proof of proposition 14.10 apply whenever $L^2\alpha \le c_{\mathsf{c}}$.

A lift supported in the cell cores. For $j \ge 1$ put $N = L^j$, $n = (N-1)/2$ and $w = \lfloor N/8 \rfloor$. Use the tents

$$h(P) = (w - |P-n|)_+, \qquad \varphi_\pm(P) = (w - |P-(n\pm w)|)_+,$$

and set

$$g^+ = (n+w)\varphi_+ - (n-w)\varphi_-, \qquad g^- = (n+w+1)\varphi_- - (n-w+1)\varphi_+.$$

Their moments are

$$\sum h = w^2, \qquad \sum g^+(P)(P+1) = \sum g^-(P)(N-1-P) = 2Nw^3,$$
$$\sum g^+(P)(N-1-P) = \sum g^-(P)(P+1) = 0.$$

Indeed, a tent centred at p has mass w^2 and first moment pw^2. Also $N/18 \le w \le N/8$. All three profiles vanish outside $|P-n| \le 2w-1$, and their support, together with one neighbouring sheet, lies in $1 \le P \le N-2$ and $N/L \le P \le N-1-N/L$.

Let z_y be the lowest corner of the cell of $y = y_c$, and write $P_\nu(x) = \eta^{-1}(x-z_y)_\nu$. Let $p_y(x) = U(\Gamma_{z_y,x})^{-1}$ for the bare comb in the cell. If c has direction μ, put $\epsilon_c = +$ and $g_c = I$ when $y = c_-$; when $y = c_+$ put $\epsilon_c = -$ and $g_c = U([z_y - \xi_j e_\mu, z_y])^{-1}$. The latter path lies in D_c. On positive bonds of direction μ set

$$\psi_c(b) = \frac{N^2}{2\eta w^7} g^{\epsilon_c}(P_\mu(b_-)) \prod_{\nu\ne\mu} h(P_\nu(b_-)), \qquad (Z_c e)(b) = \psi_c(b)\,\mathrm{Ad}(p_y(b_-)g_c)e,$$

and set these fields to zero elsewhere. At level zero use $Z_c e = \eta^{-1}\mathbf{1}_{\{c\}}e$. Put $Z\omega = \sum_c Z_c\omega(c)$ and $\Theta = Q^+Z$.

The profile bounds give, with $a_1 = 18^4/2$,

$$\xi_j|\psi_c| \le a_1, \qquad \xi_j^2\eta^{-1}|\psi_c(b+\eta e_\nu) - \psi_c(b)| \le 18^5/2, \qquad \eta N^{-2}\sum_b |\psi_c(b)| \le 18.$$

For $f(x) = \mathrm{Ad}(p_y(x)g_c)e$, the comb comparison in lemma 14.4 gives

$$|U_\nu(x)p_y(x+\eta e_\nu)p_y(x)^{-1} - I| \le 2\alpha/N, \qquad |\nabla_\nu^U f(x)| \le 4\alpha\xi_j^{-1}|e|.$$

The loop uses at most two strips, each of width at most $N-1$, in a cell whose plaquettes have curvature at most αN^{-2}. The product rule therefore yields

$$|Z_c e| \le a_1\xi_j^{-1}|e|, \qquad |\nabla^U Z_c e| \le 18^5\xi_j^{-2}|e|, \tag{455}$$

for $\alpha \le 1$. Every nonzero term, and every site or plaquette entering its first differences, lies in the core $\mathsf{K}(y) = \{n-2w \le P_\nu \le n+2w\}$.

Correction of the row values. Write $\Theta^=$ for the same-level part and $\Theta^\uparrow$ for the part from level j to level $j+1$. Locality gives

$$\Theta = \Theta^= + \Theta^\uparrow, \qquad (\Theta^\uparrow)^2 = 0.$$

Here is the support argument. A row can meet a positive-level core at its own level only through one of its endpoints. A row at a higher level must be outward: its outer block lies in the immediately lower layer. A row at a lower level can enter the cell only in its outermost order-$(j-1)$ sheet, where the core-supported profile vanishes. The level-zero case uses $Z_c e = \eta^{-1}\mathbf{1}_{\{c\}}e$ and gives the same alternatives. Two consecutive upward couplings would place a point of Ω^c_{j+1} and a point of Ω_{j+2} at distance at most $(1+(L-1)/2+L)\xi_{j+1} < L^2\xi_{j+1}$, contradicting the separation of the regions.

The absolute entry bound from proposition 14.10 gives $|\Theta^{\uparrow}(c,c')| \le 36L^{-2}$. There are at most $6L^3$ entries in each row and six in each column. The column sum in the weighted Schur test carries the additional factor $\xi_{c'}/\xi_c = L^{-1}$. Hence

$$\|\Theta^{\uparrow}\| \le 216/L < 17.$$

For the same-level part, a nested straight chain ending at a bond of direction μ in the cell of y occurs $P_\mu + 1$ times from $\langle y, y+\xi_j e_\mu\rangle$ and $N-1-P_\mu$ times from $\langle y-\xi_j e_\mu, y\rangle$. These counts follow by choosing its longitudinal starting site; the transverse coordinates are fixed. The displayed moments consequently give

$$\eta N^{-3}\sum_b \psi_{c'}(b)n_c(b) = \delta_{cc'}, \qquad \eta N^{-3}\sum_b |\psi_{c'}(b)|n_c(b) \le 18.$$

The curvature-only product estimate in the proof of proposition 14.10 writes the row as its leading chain sum plus a remainder of row norm at most $\kappa_S \vartheta_c^2 \alpha \xi_j$, where one may take $\kappa_S = 8\kappa_9^{\mathsf{c}} L^{-3}$. Compare each chain transport with the path from the lowest corner of c_- to $z_{y_{c'}}$, followed by the comb to b_-. It stays in D_c and has at most $4N$ bonds. The same proof bounds its error by $\kappa_7^{\mathsf{c}}\vartheta_c^2\alpha$. After multiplication by $p_{y_{c'}}(b_-)g_{c'}$, the comparison transport is constant in b. On the diagonal it is the identity; off the diagonal its coefficient is the vanishing moment. Therefore

$$|(\Theta^{=} - I)(c,c')| \le \kappa_\Theta L^2 \alpha, \qquad \kappa_\Theta = 36\kappa_7^{\mathsf{c}} + \kappa_S a_1.$$

There are at most twelve same-level entries in a row and six in a column. Thus $\|\Theta^{=} - I\| \le 1/36$ if $324\kappa_\Theta L^2\alpha \le 1$. At level zero the same-level part is exactly the identity. It follows that

$$\Theta = (I+\Theta^{\uparrow})[I + (I-\Theta^{\uparrow})(\Theta^{=}-I)], \qquad \|\Theta^{-1}\| \le 36, \quad \|\Theta\| \le 19.$$

Indeed, the second factor differs from the identity by at most $18/36$, while $(I+\Theta^{\uparrow})^{-1} = I - \Theta^{\uparrow}$.

Energy of the lift. Use the nonnegative form

$$\mathcal{N}(A) = \eta^3 \sum_{p \text{ inner}} |DA(p)|^2 + \eta \sum_{p \text{ crossing}} \Big(\sum_{b\subset\partial p} |A(b)|\Big)^2 + \|D^*A\|_\eta^2 + \|\mathsf{L}_{\mathbf{B}}^{-1}A\|_\eta^2 + \|Q^+A\|_1^2.$$

The crossing sum is over non-inner plaquettes containing an inner bond; it uses the zero extension of A, and no value of U outside the region. A plaquette or site meeting a positive-level profile meets profiles from just one core. All its bonds lie in that cell. Otherwise only level-zero profiles contribute. This follows from the one-sheet core margin and disjointness of the cells.

For a core at y, put $s_\mu = \sum_{c: y_c = y,\, \mathrm{dir}(c)=\mu} |\omega(c)|$. There are at most two summands, so $\sum_\mu s_\mu^2 \le 2\sum_{c:y_c=y}|\omega(c)|^2$. The value and difference bounds (455), with at most N^3 base sites, bound the curl, divergence and weighted-value energies by

$$(8\cdot 18^{10} + 6\cdot 18^{10} + 2a_1^2)\sum_{c:y_c=y}\xi_c^{-1}|\omega(c)|^2.$$

For level zero the corresponding constants are $16, 12, 1$: each bond belongs to four plaquettes and touches two sites, with at most six bonds at a site. The row energy is at most $19^2\|\omega\|_1^2$. Consequently, with $c_Z = 15\cdot 18^{10}$,

$$\mathcal{N}(Z\omega) \le c_Z\|\omega\|_1^2, \qquad Z^\sharp = Z\Theta^{-1}, \qquad Q^+Z^\sharp = I, \qquad \mathcal{N}(Z^\sharp\omega) \le 36^2 c_Z\|\omega\|_1^2.$$

The regional Hessian calculation gives $H_U = D^{\mathsf{T}}D + \Delta'$ and, with $C_H = 20$, the bound

$$|\Delta' A(b)| \le C_H \alpha \mathsf{L}_{\mathbf{B}}(b)^{-2} \max_{b' \in \mathrm{st}(b)} |A(b')|.$$

Only inner plaquettes occur. The star consists of b and the other bonds of these plaquettes, so it is symmetric and has at most thirteen members; their local scales differ by at most L. At an incomplete star the local scale is η, so the same bound follows directly from the one-plaquette formula. Applying $2ab \le a^2 + b^2$ to $a(b) = \mathsf{L}_{\mathbf{B}}(b)^{-1}|A(b)|$ gives, for $\alpha L \le 1$,

$$|(A, \Delta' A)_\eta| \le 13 C_H \|\mathsf{L}_{\mathbf{B}}^{-1} A\|_\eta^2, \qquad (A, G_0^{-1} A)_\eta \le (1 + 13 C_H)\mathcal{N}(A).$$

The second inequality also uses that R is orthogonal and the reduced row energy is a partial sum of the full row energy.

Take $A = Z^\sharp \omega$. Cauchy–Schwarz for the positive operator G_0 yields

$$\|\omega\|_1^2 = (A, Q^{+*}\omega)_\eta \le \gamma_Q^{-1/2} \|\omega\|_1 (\omega, Q^+ G_0 Q^{+*}\omega)_1^{1/2}, \qquad \gamma_Q = [36^2 (1 + 13 C_H) c_Z]^{-1}.$$

This proves the full-row lower bound. If J restricts to the interior and outward rows, J^* is zero extension and $\|J^*\omega\|_1 = \|\omega\|_1$. Hence $M_Q = J Q^+ G_0 Q^{+*} J^* \ge \gamma_Q I$. All smallness requirements hold after choosing

$$a_Q \le \min\{a_{\mathrm{ps}}, L^{-2} \min(1, c_{\mathsf{c}}), (324 \kappa_\Theta L^2)^{-1}\}.$$

Write $d = d_{\mathbf{B}}$, $B = B_{\mathrm{ps}}$ and $\delta = \delta_{\mathrm{ps}}$ for the regional constants. The cells meeting the double block of a reduced row c form a set $\mathsf{N}(c)$ of at most $L^3 + 1$ tops, and

$$L^{-1} \xi_c \le \xi_y \le L \xi_c, \qquad d(y_c, y) \le L + 1 \quad (y \in \mathsf{N}(c)). \tag{456}$$

For an interior row there are just its two endpoint cells. For an outward row there is the inner endpoint cell and the L^3 cells partitioning its outer block. A coordinate path first moving transversely in the inner cell and then along the row has weighted length at most $L + 1$.

The row Schur inverse. For data e_c at one row, the adjoint formula is

$$(Q^* e_c)(b) = \eta^{-3} \xi_c^{-1} Q(c, b)^{\mathsf{T}} e.$$

The corner entry bound in proposition 14.10 gives $|Q^* e_c| \le 2\xi_c^{-3}|e|$, supported over $\mathsf{N}(c)$. Splitting it into its cell parts and applying theorem I therefore gives

$$\begin{aligned} |G_0 Q^* e_c|_y &\le C_1 \xi_y^2 \xi_c^{-3} e^{-\delta d(y, y_c)} |e|, \\ |\nabla^U G_0 Q^* e_c|_y &\le C_1 \xi_y \xi_c^{-3} e^{-\delta d(y, y_c)} |e|, \end{aligned} \tag{457}$$

where $C_1 = 4(L^3 + 1) B e^{(L+1)\delta}$ is sufficient. Using $|QA(c)| \le 2\xi_c \max_{y \in \mathsf{N}(c)} |A|_y$ once more yields, with $C_M = 4L^2 e^{(L+1)\delta} C_1$,

$$|M_Q(c, c')| \le C_M (\xi_c / \xi_{c'})^3 e^{-\delta d(y_c, y_{c'})}.$$

Weighted symmetry says $|M_Q(c, c')| = (\xi_c / \xi_{c'}) |M_Q(c', c)|$. Thus the scale factor can be replaced by $\min\{(\xi_c/\xi_{c'})^3, (\xi_{c'}/\xi_c)^2\}$, which is at most $(\xi_c/\xi_{c'})^{1/2}$.

We use the exponential conjugation argument already proved for positive matrices in section 21.4. For the row pairing, its symmetric normalization is $\omega(c) \mapsto \xi_c^{-1/2} \omega(c)$. Let $c_* = c_{\mathrm{sum}}(\delta/2)$ and choose

$$0 < \delta_* \le \min\{\delta/4, \gamma_Q \delta / (48 C_M c_*)\}.$$

Provided the gap exceeds the summation threshold at $\delta/2$, the perturbation under conjugation by $e^{\delta_* d(y_c, y_{c'})}$ has norm at most $\gamma_Q/2$: there are at most six rows per top and

$$(e^{\delta_* s} - 1) e^{-\delta s} \le (4\delta_*/\delta) e^{-\delta s/2}.$$

The lower bound just proved and the Neumann inverse give

$$|M_Q^{-1}(c, c')| \le 2 \gamma_Q^{-1} (\xi_c / \xi_{c'})^{1/2} e^{-\delta_* d(y_c, y_{c'})}.$$

Insert this in $H_0 = G_0 Q^* M_Q^{-1}$ and use (457). The intermediate scale factor is

$$\xi_y^2 \xi_c^{-3} (\xi_c/\xi_{y'})^{1/2} = \xi_y^{-1} (\xi_y/\xi_c)^{5/2} (\xi_y/\xi_{y'})^{1/2}.$$

The triangle inequality leaves at least $\delta/2$ on $d(y, y_c)$. The scale-ratio sum at exponent $5/2$ is therefore bounded by c_*. The derivative estimate has ξ_y^{-2} in place of ξ_y^{-1}. Taking

$$B_H^\circ = \max\{B, 72\gamma_Q^{-1} C_1 c_*\}, \qquad B_H = L^{1/2} B_H^\circ,$$

we obtain the two bounds with B_H°, rate δ_*, and the remaining factor $(\xi_y/\xi_{y'})^{1/2}$. Choose

$$\delta_S = \min\{\delta_*/2, (L+2)^{-1}\}.$$

If $(\delta_* - \delta_S) R_1 \mathfrak{M} \geq \frac{1}{2} \log L$, the scale-ratio estimate absorbs that factor, giving

$$\begin{aligned} |H_0 \beta|_y &\leq B_H \xi_y^{-1} \sum_{y'} e^{-\delta_S d(y,y')} |\beta|_{y'}, \\ |\nabla^U H_0 \beta|_y &\leq B_H \xi_y^{-2} \sum_{y'} e^{-\delta_S d(y,y')} |\beta|_{y'}. \end{aligned} \tag{458}$$

Here $|\beta|_y$ is the maximum over rows with top y. The identity $QH_0 = I$ follows from the definition of M_Q.

Weighted norms and transposes. Increase w_S to include the preceding gap conditions, the weighted threshold $w_{\mathrm{sum}}(\delta_S/4)$, and $4\delta_S^{-1} \log L$. Put

$$c_\delta = \max\{1, c_{\mathrm{sum}}(\delta_S/4)\}, \qquad \overline{B} = B_H c_\delta, \qquad K_1 = 4L^7 \overline{B}.$$

For $|s| \leq 4$ the sum $\sum_{y'} e^{-\delta_S d(y,y')/4} (\xi_y/\xi_{y'})^s$ is at most c_δ. For an object over y, its local scale lies between ξ_y and $L\xi_y$. For $s \geq 0$, therefore,

$$|f|_y \leq \xi_y^{-s} |f|_{(-s)}, \qquad |f|_{(-s)} \leq L^s \sup_y \xi_y^s |f|_y.$$

The second bound is an equality without L^s for site fields. These comparisons also apply to differences. If one of the translated bonds is not inner, its nonzero neighbour has scale η, and the normalized difference is bounded by $|A|_{(-1)}$. Hence the zero-extension differences give the same X norm as its inner pairs. The regional bounds and (458) now give

$$\|G_0 f\|_X \leq L^2 \overline{B} |f|_{(-3)} \leq \tfrac{1}{4} K_1 |f|_{(-3)}, \qquad \|H_0 \beta\|_X \leq L^2 \overline{B} \sup |\beta| \leq K_1 \sup |\beta|.$$

For the transposes, fix c and put $A = H_0 e_c$. The row pairing gives the factor ξ_c:

$$\langle e, H_0^* f(c) \rangle = \xi_c (A, f)_\eta.$$

There are at most $3(\xi_y/\eta)^3$ positive bonds and the same number of inner plaquettes over y. Using (458) and $|DA|_y \leq 2|\nabla^U A|_y$ gives

$$\begin{aligned} |H_0^* f(c)| &\leq 3 B_H \sum_y \frac{\xi_c}{\xi_y} e^{-\delta_S d(y,y_c)} \xi_y^3 |f|_y, \\ |(DH_0)^* h(c)| &\leq 6 B_H \sum_y \frac{\xi_c}{\xi_y} e^{-\delta_S d(y,y_c)} \xi_y^2 |h|_y. \end{aligned} \tag{459}$$

To estimate $D^* A(x)$, consider the two bonds in each coordinate direction at x. If both are inner, their contribution is a transported first difference at $x - \eta e_\mu$. Its cell scale differs by at most L and its top distance by less than three; since $e^{4\delta_S} < 3/2$, its modulus is at most $\frac{3}{2} L^2 B_H \xi_y^{-2} e^{-\delta_S d(y,y_c)} |e|$. If exactly one is inner, both endpoints of that bond lie in layer zero. The contribution is $\eta^{-1} |A(b)|$

and obeys the same bound by the value estimate. If neither is inner it vanishes. Summing the three directions and then the site pairing gives

$$|(D^*H_0)^*g(c)| \le \tfrac{9}{2}L^2B_H\sum_y \frac{\xi_c}{\xi_y}e^{-\delta_S d(y,y_c)}\xi_y^2|g|_y.$$

The weighted sum at exponent one bounds the right sides by $3\overline{B}|f|_{(-3)}$, $6\overline{B}|h|_{(-2)}$, and $\frac{9}{2}L^2\overline{B}|g|_{(-2)}$, respectively. These are the asserted transpose bounds.

For the scalar projection, write $Pg = G'Q'^*CQ'G'g$. The scalar regional estimates and the same weighted sums give successively

$$\sup|Q'G'g| \le \overline{B}|g|_{(-2)}, \qquad |Q'^*C\mu|_{(-4)} \le \overline{B}\sup|\mu|, \qquad |G'f|_{(-2)} \le \overline{B}|f|_{(-4)}.$$

Here $|Q'\lambda(y)| \le |\lambda|_y$ and $|Q'^*\mu|_y = |\mu(y)|$. Thus $|Pg|_{(-2)} \le \overline{B}^3|g|_{(-2)}$ and the bound for $R = I - P$ follows.

At each level the corner product column bound is $\sum_c |Q(c,b)| \le 2\eta^3\xi_j^{-2}$. If $\mathsf{L}_{\mathbf{B}}(b) = \xi_n$, only levels $n-1, n, n+1$ can contribute. Multiplying by the adjoint weight and summing therefore gives

$$\eta^{-3}\mathsf{L}_{\mathbf{B}}(b)^3\sum_c \xi_c^{-1}|Q(c,b)| \le 2(L^3+2),$$

which proves the bound on Q^*. The row support similarly gives $\sup|QA| \le 2L|A|_{(-1)}$.

Finally $Q\widetilde{G}_0 = 0$ follows from $QH_0 = I$. We also need a local form of its decay. Put $\beta = QG_0f$. A row with top z has support at distance at most $L+1$ from z, cell scales at most $L\xi_z$, and row sum at most $2\xi_z$. The regional inverse bound therefore gives

$$|\beta|_z \le 12L^2B_H\xi_z^3\sum_{y'} e^{-\delta_S d(z,y')}|f|_{y'}.$$

Insert this in (458). The triangle inequality gives

$$d(y,z) + d(z,y') \ge \tfrac{1}{2}d(y,y') + \tfrac{1}{2}d(y,z), \qquad \xi_y^{-1}\xi_z^3 = \xi_y^2(\xi_z/\xi_y)^3.$$

The remaining sum over z is at most c_δ. For the derivative the output factor is ξ_y instead of ξ_y^2. Since $\widetilde{G}_0 = G_0 - H_0QG_0$ and $B_H + 12L^2B_H\overline{B} \le K_1^2$, this proves

$$\begin{aligned} |\widetilde{G}_0f|_y &\le K_1^2\xi_y^2\sum_{y'} e^{-\delta_S d(y,y')/2}|f|_{y'}, \\ |\nabla^U\widetilde{G}_0f|_y &\le K_1^2\xi_y\sum_{y'} e^{-\delta_S d(y,y')/2}|f|_{y'}. \end{aligned} \tag{460}$$

For the global norm, the value part of the regional bound gives

$$\sup|\beta| \le 2L^2\overline{B}|f|_{(-3)} \le \tfrac{1}{2}K_1|f|_{(-3)}.$$

Hence

$$\|\widetilde{G}_0f\|_X \le (\tfrac{1}{4}K_1 + \tfrac{1}{2}K_1^2)|f|_{(-3)} \le K_1^2|f|_{(-3)} \le K_7|f|_{(-3)}.$$

□

30. The nonlinear regional equation

We now construct a critical configuration from nearby reduced averaging data. The regional inverse supplies the linear part. The nonlinear constraint uses centered averages, which must first be compared with the corner averages in that inverse. A gauge correction will then impose the Landau condition while prescribing the induced gauge at every block top.

Keep the separated regional setting of proposition 29.1, write U_0 for the real reference configuration, and abbreviate $\mathsf{L} = \mathsf{L}_{\mathbf{B}}$. The curvature parameter is denoted by α_0. Constants and radii below

depend only on L; the radii may be decreased to satisfy the preceding estimates. For a general sequence, define $\mathfrak{U}'_k(\mathbf{B}, r)$ by the same inner-plaquette bound as (75), together with

$$|(D_U^{\mathsf{T}}\partial U)(b)| < r\eta^2 \mathsf{L}(b)^{-3} \quad \text{at every full bond } b.$$

This agrees with the earlier definition for the constant sequence.

30.1. **The action on the original region.** Let $\mathcal{P}(b)$ be the inner plaquettes containing b, and let $\mathrm{st}(b)$ consist of their positive bonds. At a free scale $\ell \ge \eta$, put

$$s_b^\ell(A) = \ell \max_{\mathrm{st}(b)} |A|, \qquad x_b^\ell(A) = \max\{s_b^\ell(A), \ell^2 \max_{\mathrm{st}(b)} |\nabla^{U_0} A|\}.$$

The second maximum uses only differences determined by that star. Define s_p^ℓ, g_p^ℓ in the same way on a plaquette, with g_p^ℓ the scaled first-difference maximum. Empty maxima are zero. A missing superscript means $\ell = \mathsf{L}(b)$ or $\mathsf{L}(p)$, respectively. The adjacent-layer geometry gives $s_b \le L|A|_{(-1)}$, $x_b \le L^2\|A\|_X$ and $\max\{s_p, g_p\} \le L\|A\|_X$.

Proposition 30.1. Suppose U_0 is real and $|\partial U_0(p) - I| \le \alpha_0 \eta^2 \mathsf{L}(p)^{-2}$ at every inner plaquette, where $0 \le \alpha_0 \le 1/2$. In the fine pairing of weight η^3, set

$$F(A) = \eta^{-1} \sum_{p \in \mathfrak{p}} a_p(e^{i\eta A} U_0), \quad J_0 = F'(0), \quad \Delta' = H_{U_0} - D^*D, \quad \mathcal{V}(A) = F'(A) - J_0 - H_{U_0} A.$$

Here D in D^*D is the curl into the original inner plaquettes. There is an absolute $C_{\mathrm{ec}} \ge 1$ such that, with $\rho_{\mathrm{ec}} = 1/256$,

$$F'(A) = J_0 + D^*DA + \Delta' A + \mathcal{V}(A). \tag{461}$$

The remainder is entire, depends at b only on its star, and satisfies $\mathcal{V}(0) = \mathcal{V}'(0) = 0$. The operators preserve real fields, and

$$\begin{aligned} |\Delta' A|_{(-3)} &\le C_{\mathrm{ec}} L \alpha_0 |A|_{(-1)}, \\ |\mathcal{V}(A)|_{(-3)} &\le C_{\mathrm{ec}} L^3 |A|_{(-1)} \|A\|_X && (L|A|_{(-1)} \le \rho_{\mathrm{ec}}), \\ |\mathcal{V}(A_1) - \mathcal{V}(A_2)|_{(-3)} &\le 4 C_{\mathrm{ec}} L^4 r \|A_1 - A_2\|_X && (\|A_i\|_X \le r \le \rho_{\mathrm{ec}}/(4L^2)). \end{aligned}$$

Locally, if the incident curvature is at most $\alpha_0 \eta^2 \ell^{-2}$, and $\ell = \eta$ for a nonfull bond, these bounds follow from

$$\begin{aligned} \ell^3 |\Delta' A(b)| &\le C_{\mathrm{ec}} \alpha_0 s_b^\ell(A), \\ \ell^3 |\mathcal{V}(A)(b)| &\le C_{\mathrm{ec}} s_b^\ell(A) x_b^\ell(A) \quad (s_b^\ell(A) \le \rho_{\mathrm{ec}}), \\ \ell^3 |\mathcal{V}(A_1)(b) - \mathcal{V}(A_2)(b)| &\le 4 C_{\mathrm{ec}} r x_b^\ell(A_1 - A_2) \quad (x_b^\ell(A_i) \le r \le \rho_{\mathrm{ec}}/4). \end{aligned} \tag{462}$$

For $E = e^{i\eta A} U_0$,

$$F'(A)(b) = \mathsf{f}(-i\eta \operatorname{ad}_{A(b)}) J_{\Omega_0}(E)(b). \tag{463}$$

The current uses only the inner plaquettes. If $\eta|A(b)| \le 1/8$, its norm and the gradient norm differ by factors at most $1 + 4\eta|A(b)|$ and $1 + 2\eta|A(b)|$, respectively. At a full bond $J_{\Omega_0}(U_0) = J(U_0)$; at a nonfull bond $\eta^3|J_0(b)| \le 3\alpha_0$. For real A with $s_p^\ell(A) \le \rho_{\mathrm{ec}}$,

$$|\partial E(p) - I| \le |\partial U_0(p) - I| + [2g_p^\ell(A) + C_{\mathrm{ec}} s_p^\ell(A)^2] \eta^2 \ell^{-2}. \tag{464}$$

■

Proof. The entire local expansion in proposition 13.1 applies to each inner plaquette. Its linear coefficient is the Hessian H_{U_0}, by the one-plaquette calculation in lemma 25.3. That calculation bounds the local row of Δ' by $C_{\mathrm{ec}} \alpha_0 \ell^{-2}$, with an absolute constant also valid for complex variations. No exterior plaquette is added.

For the nonlinear term, apply the local estimate (514) at spacing $\widehat{\eta} = \eta/\ell$ and perturbation $\widehat{A} = \ell A$. The reference transports are isometries, its curvature bound is $\alpha_0 \widehat{\eta}^2$, and an incomplete

star has $\widehat{\eta} = 1$. Thus all hypotheses of that estimate hold. The scaled action is ℓF, while a variation of $\widehat{A}$ contributes ℓ^{-1}; the ratio of pairing weights therefore multiplies its gradient by ℓ^3. This gives the second line of (462). The local Cauchy-disc argument following (514) gives its third line. The layer comparisons preceding the proposition give the three global estimates.

The exponential differential in proposition 13.1 is computed bond by bond, so restricting the plaquette sum gives (463) with its stated factors. The real plaquette calculation there, at the same scaled spacing, gives (464) after enlarging C_{ec}. A nonfull bond has at most three inner plaquettes and has local scale η; each contributes at most $\alpha_0\eta^{-3}$ to J_0. This proves the remaining current bound. Reality follows by differentiation of the real action in real coordinates. □

We record the regularity conversion used after solving the equation. The proof of lemma 9.4 is local to the four plaquettes at a full bond. At scale $\ell = \mathsf{L}(b)$, its scalar term obeys

$$|a(b)| \le r^2\eta^3\ell^{-4} \le r^2\eta^2\ell^{-3},$$

since every incident plaquette has local scale at least ℓ. Thus the same proof gives, for a general separated sequence,

$$\mathfrak{U}'_k(\mathbf{B}, r) \subset \mathfrak{U}_k(\mathbf{B}, r) \subset \mathfrak{U}'_k(\mathbf{B}, r + r^2). \tag{465}$$

Suppose $U_0 \in \mathfrak{U}'_k(\mathbf{B}, \alpha_0)$, A is real, and

$$\max\{\|A\|_X, |D^*DA|_{(-3)}\} \le \alpha'_2, \qquad \alpha_2 = L^2\alpha'_2.$$

If

$$\begin{gathered}\alpha_2 \le \rho_{\mathrm{ec}}, \quad C_{\mathrm{ec}}\alpha_2 \le 1, \quad 4\alpha_2 + C_{\mathrm{ec}}(\alpha_0 + \alpha_2) \le \tfrac{1}{2}, \\ 0 < 2\alpha_0 + 3\alpha_2 \le \tfrac{1}{3},\end{gathered} \tag{466}$$

then

$$e^{i\eta A}U_0 \in \mathfrak{U}'_k(\mathbf{B}, 2(2\alpha_0 + 3\alpha_2)). \tag{467}$$

Indeed the plaquette bound has radius at most $\alpha_0 + 3\alpha_2$. At a full bond the action expansion and current conversion bound the current by $(1 + 4\alpha_2)(1 + C_{\mathrm{ec}}\alpha_2)(\alpha_0 + \alpha_2)\mathsf{L}(b)^{-3} \le (2\alpha_0 + 3\alpha_2)\mathsf{L}(b)^{-3}$. For $r = 2\alpha_0 + 3\alpha_2$, the scalar estimate just proved gives divergence radius at most $r + r^2 < 2r$, as required.

30.2. **The centered average on a fixed ball.** The straightened mean average must be defined on a ball whose radius is independent of the depth. For a set X of level-j sites, put $X_i = B^{j-i}(X)$, $0 \le i \le j$, and let $\mathsf{P}(X_i)$ be its inner positive bonds. Block nesting gives $X_i = B(X_{i+1})$, so every double block needed at one stage lies in the preceding member. These restrictions will be called the tower over X.

Put $\Omega_{-1} = \Omega_0$ and

$$Y_j = \{y \in \mathbb{T}^{(j)} : B^j(y) \subset \Omega_{j-1}\}, \qquad E_j = \mathsf{P}(Y_j).$$

For $y \in Y_j$, set $\vartheta_j(y) = 1$ if $y \in \Omega_j$ and $\vartheta_j(y) = L$ otherwise; at a bond take the maximum of the endpoint values. In particular $\vartheta_c = 1$ on interior reduced rows and L on outward rows. The aligned regions imply that every bond or plaquette in $B^j(X)$, for $X \subset Y_j$, is inner and has scale at least $\xi_j/\max_X \vartheta_j$. Its finer towers lie in the corresponding regions.

Proposition 30.2. There are positive $a_{\mathsf{m}}, c_{\mathsf{m}}$ and $C_{\mathsf{m}}, \kappa_{6,\mathsf{m}} \ge 1$, depending only on L, with the following property. Suppose U_0 is real and

$$|\partial U_0(p) - I| \le \alpha_0\eta^2\mathsf{L}(p)^{-2}, \qquad 0 < \alpha_0 \le L^{-2}a_{\mathsf{m}},$$

at every inner bare plaquette. The actual centered straightened averages $a_j(A)$ and accumulated factors $v_j[A]$ are holomorphic on

$$\mathcal{D}_{\mathsf{m}} = \{A : |A|_{(-1)} < c_{\mathsf{m}}/L\},$$

and are real for real A. For $c \in E_j$, let P_c^j be the inner bonds of its endpoint double block and put $\tau_c(A) = \xi_j \sup_{\mathsf{P}_c^j} |A|$. Then

$$|a_j(A)(c)| \le 2\tau_c(A) \le 2\vartheta_c |A|_{(-1)},$$
$$a_j(A) = Q_j^{\mathsf{m}} A + C_j^{\mathsf{m}}(A), \qquad |C_j^{\mathsf{m}}(A)(c)| \le C_{\mathsf{m}} \tau_c(A)^2,$$
$$\sum_b |Q_j^{\mathsf{m}}(c,b)| \le \xi_j (1 + C_{\mathsf{m}} \vartheta_c^2 \alpha_0) \le 2\xi_j, \qquad \sum_b |Q_j^{\mathsf{m}}(c,b)| \mathsf{L}(b)^{-1} \le 2\vartheta_c.$$

Here $Q_0^{\mathsf{m}} = \eta I$. Write $K_A(c,b) = \partial_{A(b)} a_j(A)(c)$. On $|A|_{(-1)} \le c_{\mathsf{m}}/(4L)$,

$$|(K_A - Q^{\mathsf{m}})\delta A(c)| \le 4 C_{\mathsf{m}} \tau_c(A) \tau_c(\delta A),$$
$$|K_A \delta A(c)| \le \tfrac{8}{3} \vartheta_c |\delta A|_{(-1)}.$$

On the whole complex ball $\mathcal{D}_{\mathsf{m}}$ the kernels are supported in P_c^j and satisfy

$$\sum_{c \in E_j} |K_A(c,b)| \le 2\eta^3 \xi_j^{-2}, \qquad \mathsf{L}(b)^3 \sum_j \xi_j^{-1} \sum_{c \in \Gamma_j^{\flat,+}} |K_A(c,b)| \le 2(L^3+2)\eta^3. \tag{468}$$

The factor $v_j[A](y)$ depends only on A in $B^j(y)$, belongs to G for real A, and satisfies

$$|v_j[A](y)^{\pm 1} - I| \le 4\kappa_{6,\mathsf{m}} \xi_j \sup_{\mathsf{P}(B^j(y))} |A|.$$

On a tower over $X \subset Y_j$, set $\vartheta = \max_X \vartheta_j$, $\alpha = \vartheta^2 \alpha_0$, and $a = \xi_j \sup_{\mathsf{P}(B^j(X))} |A|$. The intermediate reference curvatures and perturbations obey

$$\varpi_i \le 2\alpha L^{-2(j-i)}, \qquad \sup |a_i| \le 2aL^{i-j}. \tag{469}$$

All constants are independent of the number of layers. ∎

Proof. The one-step chart. By lemma 24.2, the centered chart has a fixed complex radius $r_{\mathsf{m}} > 0$ near the identity. For a real W of sufficiently small double-block curvature, the comb gauge of lemma 14.4 gives $W^h = e^{ib}$ with $\sup |b| \le 2\pi L \varpi_c(W) < r_{\mathsf{m}}/2$. The covariance formula (363) therefore gives the same perturbation ball at W. Decrease the radius so that the logarithms have norm at most one. Their vanishing at zero and Cauchy's estimate give constants $\kappa_{5,\mathsf{m}}, \kappa_{6,\mathsf{m}}$ with

$$|\log \mathsf{v}(W,a)|, \ |\mathcal{Q}_{\mathsf{m}}^1(W,a)| \le \kappa_{6,\mathsf{m}} \sup |a|, \quad |C_{\mathsf{m}}^1(W,a)| \le \kappa_{5,\mathsf{m}} (\sup |a|)^2,$$

and $|\partial_{a(b)} C_{\mathsf{m}}^1(W,a)| \le \kappa_{5,\mathsf{m}} \sup |a|$ on the half ball. These maps are real on real inputs.

The leading transport kernel is

$$\mathbf{S}_{\mathsf{m}}(W)(c,b) = L^{-3} \sum_{\substack{x \in B(c_-) \\ b \in [x,x']}} \mathrm{Ad}(W_{\mathsf{m}}(c_-,x) W([x,b_-])), \qquad x' = x + Le_\mu.$$

Its row and column bounds are L, L^{-2}. At the identity the endpoint terms cancel, so $\mathbf{Q}_{\mathsf{m}}(I) = \mathbf{S}_{\mathsf{m}}(I)$. The Schwarz estimate in the proof of proposition 14.10, now using the centered chart, bounds each difference entry by $(r_{\mathsf{m}}^{-1} + 2) r_{\mathsf{m}}^{-1} \sup |b|$. There are at most $6L^3$ entries in one row and six double blocks containing one fine bond. Gauge covariance thus gives a common $\kappa_9 = \kappa_{9,\mathsf{m}}$ such that

$$\begin{aligned} \|\mathbf{Q}_{\mathsf{m}}(W) - \mathbf{S}_{\mathsf{m}}(W)\|_{\mathrm{row,col}} &\le \kappa_9 \varpi, \\ \|d_a \mathcal{Q}_{\mathsf{m}}^1(W,a)\|_{\mathrm{col}} &\le L^{-2} + \kappa_9 \varpi + 6\kappa_5 \sup |a|. \end{aligned} \tag{470}$$

We abbreviate $\kappa_5 = \kappa_{5,\mathsf{m}}$ below. Single-block factors have the same domain: fold into an adjacent block as in proposition 14.10, extend a by zero, and use their dependence only on the original block.

Iteration. Consider a tower with initial curvature at most αL^{-2j} and initial perturbation a_0 with $\tau = L^j \sup |a_0| < c_{\mathsf{m}}$. The local proof of proposition 14.1 gives

$$\varpi_{i+1} \le L^2 \varpi_i + C_0^{\mathrm{SO}} \varpi_i^2, \qquad \varpi_i \le 2\alpha L^{-2(j-i)}, \qquad \sum_{i<j} \varpi_i \le 4\alpha L^{-2},$$

after decreasing a_{m} to stay in the one-step domain. The averages agree with the real branch by the same lemma. Choose the radii also so that

$$2c_{\mathsf{m}} \le r_{\mathsf{m}}/2, \qquad 4\kappa_9 a_{\mathsf{m}} \le \tfrac{1}{2}\log 2, \qquad 24\kappa_5 L c_{\mathsf{m}} \le \tfrac{1}{2}\log 2.$$

For $t_i = L^{-i}\sup|a_i|$, the expansion gives $t_{i+1} \le t_i(1+\kappa_9 L^{-1}\varpi_i + \kappa_5 L^{i-1}t_i)$. If $t_i \le 2t_0$ through level $n<j$, multiplication yields

$$t_{n+1} \le t_0 \exp(4\kappa_9 L^{-3}\alpha + 4\kappa_5 L^{-2}\tau) \le 2t_0.$$

This closes the induction and all its domain conditions. Every composition is holomorphic on the entire initial ball. Restricting to an endpoint tower proves locality.

For the derivative product $T_{j-1}\cdots T_0$, where $T_i = d_a\mathcal{Q}^1_{\mathsf{m}}(W_i,a_i)$, we have

$$\|T_i\|_{\mathrm{col}} \le L^{-2}(1+\varepsilon_i), \qquad \varepsilon_i = L^2(2\kappa_9\alpha L^{-2(j-i)} + 12\kappa_5\tau L^{i-j}).$$

Since $\sum_{i<j}\varepsilon_i \le 4\kappa_9\alpha + 24\kappa_5 L\tau \le \log 2$, submultiplicativity gives a column bound $2L^{-2j}$. The single-block factors are holomorphic compositions on this same domain.

The regional estimates. The geometry above gives the tower hypotheses with $\alpha = \vartheta^2\alpha_0 \le a_{\mathsf{m}}$ and $a_0 = \eta A$, since $\tau \le \vartheta|A|_{(-1)} < c_{\mathsf{m}}$. This proves the domain, value and intermediate bounds. At zero the chain rule gives $Q^{\mathsf{m}}_j = \eta\mathbf{Q}_{\mathsf{m}}(W_{j-1})\cdots\mathbf{Q}_{\mathsf{m}}(W_0)$. Its row sum is at most $\xi_j\exp(4\kappa_9 L^{-3}\vartheta_c^2\alpha_0)$. Choose $C_{\mathsf{m}} \ge 8\kappa_9 L^{-3}$ and shrink a_{m} so that $C_{\mathsf{m}}a_{\mathsf{m}} \le 1$. The row bounds follow, including the weighted bound since $\mathsf{L}(b) \ge \xi_j/\vartheta_c$ on its support.

For the remainder use the local ball $\tau_c(A) < c_{\mathsf{m}}$, where $|a_j(A)(c)| \le 2\tau_c(A)$. Cauchy's estimate gives $|C^{\mathsf{m}}_j(A)(c)| \le 4\tau_c(A)^2/c_{\mathsf{m}}$ on the half ball; outside it the value and linear bounds give $|C^{\mathsf{m}}_j(A)(c)| \le 4\tau_c(A) \le 8\tau_c(A)^2/c_{\mathsf{m}}$. Enlarge C_{m} to at least $8/c_{\mathsf{m}}$, and decrease a_{m} again so that $C_{\mathsf{m}}a_{\mathsf{m}} \le 1$. On the quarter ball apply Cauchy's formula to $C^{\mathsf{m}}_j(A+z\delta A)(c)$ with $|z| < \tau_c(A)/\tau_c(\delta A)$; the quadratic bound on this disc proves the stated derivative estimate. Zero local seminorms are handled by locality. Direct Cauchy estimation of a_j on the global ball of radius c_{m}/L gives the factor $8/3$.

The derivative product on the tower over all Y_j gives $\sum_{c\in E_j}|K_A(c,b)| \le 2\eta L^{-2j}$. If $\mathsf{L}(b) = \xi_n$, only levels $n-1, n, n+1$ contribute to reduced rows; multiplication by $\xi_n^3\xi_j^{-1}$ and summation proves (468). Finally write the single-step factors as e^{iX_i}. Their logarithm estimates give

$$\sum_{i<j}|X_i(y)| \le 4\kappa_{6,\mathsf{m}}L^{j-1}\sup_{\mathsf{P}(B^j(y))}|\eta A|.$$

Choose $4\kappa_{6,\mathsf{m}}c_{\mathsf{m}} \le 1$. The product estimate $|\prod e^{\pm iX_i} - I| \le e^{\sum|X_i|} - 1$, with reversed order for the inverse, then gives the asserted bound for $v_j^{\pm 1}$. □

30.3. **Comparison of the two averages.** The inverse just constructed uses corner averages. The nonlinear constraint uses the centered averages, so its linearization must be compared with the corner operator. There is a curvature-sized error even after their output frames have been aligned.

Keep the regional setting of theorem I. For $y \in Y_j$, let $o_j(y)$ be the lowest corner of its order-j block and put

$$t_j(y) = U_0(\Gamma_{y,o_j(y)}), \qquad (T\mu)(y) = \mathrm{Ad}(t_j(y))\mu(y), \qquad (T\omega)(c) = \mathrm{Ad}(t_j(c_-))\omega(c).$$

The path is the bare comb within the block. Thus $t_0 = I$, and T preserves the sup norms and the scalar-top and row pairings. Write Q'_{m} for the scalar average formed from the accumulated centered transports, and Q_{m} for the derivative Q^{m} in proposition 30.2. The corner averages retain the notation Q', Q. For a reduced row c set $\vartheta_c = 1$ if it is interior and $\vartheta_c = L$ if it is outward.

Lemma 30.3. There are $a_F > 0$ and $\kappa_F \ge 1$, depending only on L, such that, under the regional hypotheses with $\alpha_0 \le a_F$,

$$Q'_{\mathrm{m}} = T(Q' + E'), \qquad Q_{\mathrm{m}} = T(Q+E). \tag{471}$$

Both errors are real, vanish at level zero and have the same cell or double-block support as the corresponding averages. For $y \in \Lambda_j$ and $x \in B^j(y)$,

$$|E'(y,x)| \le \kappa_F \alpha_0 L^{-3j}.$$

For a positive reduced row c of level j,

$$\sum_b |E(c,b)| \le \kappa_F \vartheta_c^2 \alpha_0 \xi_j, \qquad \sum_{c \in \Gamma_j^{\flat,+}} |E(c,b)| \le \kappa_F L^2 \alpha_0 \eta^3 \xi_j^{-2}. \tag{472}$$

Consequently

$$\mathsf{L}_{\mathbf{B}}(b)^3 \sum_j \xi_j^{-1} \sum_{c \in \Gamma_j^{\flat,+}} |E(c,b)| \le \kappa_F L^2 (L^3+2) \alpha_0 \eta^3. \tag{473}$$

The constants are independent of the number of layers. ∎

Proof. We give the comparison for both choices of root in each block. The corner estimate in proposition 14.10 and the centered estimate (470) give a common $\kappa_9 = \kappa_9(L)$ with

$$\|\mathbf{Q}_\star(W) - \mathbf{S}_\star(W)\|_{\mathrm{row,col}} \le \kappa_9 \varpi(W), \qquad \star \in \{\mathrm{m}, \mathrm{c}\}.$$

Choose $a_F \le L^{-2} \min\{a_{\mathsf{m}}, c_{\mathsf{c}}\}$, and decrease it so that both one-step charts are available throughout every contributing tower.

For a level-j row let $\mathsf{D}_c = B^j(c_-) \cup B^j(c_+)$ and $a_c = \vartheta_c^2 \alpha_0$. Every plaquette there has curvature at most $a_c L^{-2j}$. The nested restrictions and folding argument of proposition 14.10, followed by proposition 14.1, give

$$\varpi_i \le 2 a_c L^{-2(j-i)}, \qquad \sum_{i<j} \varpi_i \le 4 a_c L^{-2}.$$

All factors contributing to row c use only D_c. For a fixed input bond, take the union of the towers of rows whose double blocks contain it; the same bounds hold there with $\bar a_j(b) \le L^2 \alpha_0$ in place of a_c.

Write $Q_j^{\star,\mathrm{r}}$ for the difference between $Q_j^\star$ and the product of its leading kernels, multiplied by η. The ordered-product calculation already used in proposition 14.10 gives

$$\sum_b |Q_j^{\star,\mathrm{r}}(c,b)| \le \eta L^j [\exp(4\kappa_9 L^{-3} a_c) - 1],$$
$$\sum_c |Q_j^{\star,\mathrm{r}}(c,b)| \le \eta L^{-2j} [\exp(4\kappa_9 \bar a_j(b)) - 1].$$

Require $4\kappa_9 L^2 a_F \le \log 2$. Since $e^u - 1 \le 2u$ on this interval, these are bounded by $\kappa_S a_c \xi_j$ and $\kappa_S \bar a_j(b) \eta^3 \xi_j^{-2}$, respectively, where $\kappa_S = 8\kappa_9$. This retains the column estimate before any sum over levels is taken.

The leading products in the two schemes have the same nested chains. Compare their transports to a single bare path in D_c. In the centered scheme first replace the contour mean by one of its defining contour transports. The mean estimate in lemma 14.6 bounds this error by $\kappa_1^{\mathrm{SO}} \varpi_i \le 2\kappa_1^{\mathrm{SO}} a_c L^{-2(j-i)}$. Each one-step contour and initial straight segment has length at most $4L$. The path-replacement argument in the proof of proposition 14.10, using (116), therefore costs at most $(4L\kappa_4 + 2\kappa_1^{\mathrm{SO}}) a_c L^{-2(j-i)}$ at level i. Summing these errors and using the comb gauge to compare the resulting bare paths gives a common $\kappa_7(L)$ such that each accumulated transport differs from a same-endpoint bare path of length at most $16L^j$ by at most $\kappa_7 a_c$. For scalar transports use α_0 in place of a_c.

For a scalar output y, take the comb from $o_j(y)$ to x. Prepending the path from y to $o_j(y)$ gives the comparison path for the centered transport. Both lie in $B^j(y)$ and have length at most $5L^j$. After multiplying by $t_j(y)^{-1}$, the two adjoint transports differ by at most $4\kappa_7 \alpha_0$. The scalar averaging factor L^{-3j} gives the asserted kernel estimate for E'.

For the bond kernels take the comb from $o_j(c_-)$ to b_-; its length is at most $4L^j$. The centered comparison path, with the frame path prepended, has length at most $6L^j$. The difference of the aligned adjoint transports is again at most $4\kappa_7 a_c$. There are L^{4j} chains from one row and at most L^j chains ending at one input bond, as counted in the proof of proposition 14.10. Multiplication by ηL^{-3j} therefore gives the respective bounds

$$4\kappa_7 a_c \xi_j, \qquad 4\kappa_7 \bar{a}_j(b)\eta^3 \xi_j^{-2}.$$

Adding the two derivative remainders proves (472) with $\kappa_F = 2\kappa_S + 4\kappa_7$.

If $\mathsf{L}_{\mathbf{B}}(b) = \xi_i$, only levels $i-1, i, i+1$ have a row double block containing b. The last sum is thus bounded by $\kappa_F L^2 \alpha_0 \eta^3 (L^3 + 1 + L^{-3})$, which proves (473). At level zero both schemes are ηI on bonds and I on sites, so their errors vanish. Reality and the support assertions follow from the corresponding real kernels and their nested locality. □

30.4. **Prescribing the induced gauge.** Put $a_{\mathrm{in}} = \min\{a_Q, a_F, L^{-2}a_{\mathsf{m}}, 1/2\}$. Assume the preceding regional geometry and $U_0 \in \mathfrak{U}'_k(\mathbf{B}, \alpha_0)$, where $\alpha_0 \le a_{\mathrm{in}}$. Put

$$\|\lambda\|_1 = \max\{\sup|\lambda|, |D\lambda|_{(-1)}\}, \qquad \|\lambda\|_Y = \max\{\|\lambda\|_1, |\Delta\lambda|_{(-2)}\}.$$

For $w = e^{i\lambda}$, write $e^{i\eta A^w} U_0 = (e^{i\eta A} U_0)^w$, using the small logarithm at each inner bond. Its induced factor at a top $y \in \Lambda_j$ is

$$\mathcal{R}_{\mathbf{B}}(U_0, A; w)(y) = v_j[A^w](y)^{-1} w(y) v_j[A](y).$$

Here v_j is the accumulated factor of the mean chart; at level zero it equals 1.

Lemma 30.4. There are $c_6 > 0$ and K_3, depending only on L, such that, if $\alpha_0 \le c_6$, A_0 is a real bond field with $\|A_0\|_X \le c_6$, and $r : \Lambda_* \to SU(2)$ satisfies $\sup|r-1| \le c_6$, there is a real site function λ for which $w = e^{i\lambda}$ satisfies

$$RD^* A_0^w = 0, \qquad \mathcal{R}_{\mathbf{B}}(U_0, A_0; w) = r \quad \text{on } \Lambda_*,$$

and

$$\|\lambda\|_Y \le K_3\big(|D^* A_0|_{(-2)} + \sup|r-1|\big), \qquad |A_0^w - A_0|_{(-1)} \le 2C_{\mathrm{gc}} \|\lambda\|_Y.$$

In particular, $w = r$ on Λ_0. ■

Proof. The scalar system. Use the regional scalar operators

$$G' = (\Delta + Q'^* a Q')^{-1}, \qquad N = Q' G'^2 Q'^*, \qquad C = N^{-1}, \qquad P = G' Q'^* C Q' G', \quad R = 1 - P.$$

The bare and top pairings have weights η^3 and ξ_y^3, respectively. The real comb transports are isometries, and their pillars partition the bare sites; hence $Q'Q'^* = 1$. In particular, $\operatorname{ran} R = \ker(Q'G')$. Define

$$H' = G' Q'^* a + G'^2 Q'^* C (1 - Q' G' Q'^* a).$$

Then

$$Q'H' = 1, \qquad R\Delta H' = 0, \qquad R\Delta G' f = f \quad (f \in \operatorname{ran} R).$$

Indeed, if $\tau = (1 - Q'G'Q'^* a)\nu$ and $\sigma = C\tau$, then $Q'H'\nu = \nu$ and $\Delta H'\nu = G'Q'^*\sigma$, which R annihilates. For $f \in \ker(Q'G')$, the identity $\Delta G' = 1 - Q'^* a Q' G'$ gives $\Delta G' f = f$. It follows that

$$(f, \nu) \longmapsto G' f + H' \nu$$

is the inverse of $\lambda \mapsto (R\Delta\lambda, Q'\lambda)$: the displayed identities give a right inverse, and the dimensions agree because $Q'G'$ is onto.

Increase the common scalar estimate constant $K_1 \ge 1$ if necessary. The kernel bounds and weighted sums give

$$\|G' f\|_Y \le (1 + K_1)|f|_{(-2)}, \qquad |Rf|_{(-2)} \le (1 + K_1^3)|f|_{(-2)},$$
$$\|H'\nu\|_Y \le K_2 \sup|\nu|, \qquad K_2 = (1 + K_1)^4.$$

The two weight conversions used here are $|G'g|_{(-2)} \le K_1|g|_{(-4)}$ and $|(C\nu)(y)| \le K_1\xi_y^{-4}\sup|\nu|$. They bound P by three factors K_1. For H', its first term is bounded by $1+K_1$; the second is bounded by $(1+K_1)K_1^2(1+K_1)$. Their sum is at most $(1+K_1)^4$. The Δ component of the first estimate follows from $\Delta G'f = f - Q'^* a Q' G' f$.

We need these identities for the inner-bond divergence, also when the level-zero values are prescribed. If $n(x)$ is the number of missing neighbours, then $\Delta = D^*D + \eta^{-2}n$. Let Λ_0° consist of the level-zero sites whose neighbours in Ω_0 are all in Λ_0. For f supported there, $(G')^{-1}f = \Delta f + Q'^* a Q' f$ is supported on Λ_0 and therefore equals $Q'^*\mu$ for a top function μ. Thus $f = G'Q'^*\mu$ and $Rf = 0$. Separation puts every site with $n(x) > 0$ in Λ_0°: it and its neighbours are within 2η of Ω_0^c, while Ω_1 is at distance greater than $R_1\mathfrak{M}\eta \ge L^2\eta$. Consequently

$$RD^*D = R\Delta, \qquad \|\lambda\|_{Y^\circ} \le 7\|\lambda\|_Y, \qquad \|\lambda\|_{Y^\circ} := \max\{\|\lambda\|_1, |D^*D\lambda|_{(-2)}\}.$$

The factor 7 follows from $n \le 6$ and $\xi(x) = \eta$ where $n(x) > 0$.

The divergence remainder. For $|A|_{(-1)}, \|\lambda\|_1 \le \rho_2 = 1/64$, the small bond logarithm is real analytic and satisfies

$$\begin{aligned} D^*A^{e^{i\lambda}} &= D^*A - D^*D\lambda + \mathcal{N}(A,\lambda), \\ |\mathcal{N}(A,\lambda)|_{(-2)} &\le C_{\mathrm{gc}}(\|A\|_X + \|\lambda\|_{Y^\circ})\|\lambda\|_{Y^\circ}, \\ |A^{e^{i\lambda}} - A|_{(-1)} &\le 2C_{\mathrm{gc}}\|\lambda\|_1, \end{aligned} \tag{474}$$

with $C_{\mathrm{gc}} = 93$. Here is the cancellation giving the middle estimate. At a site put $l = \lambda(x)$, and at an adjacent bond put $u = \eta A$ and $d = \eta D\lambda$, transporting incoming values to x. The outgoing and incoming logarithms are

$$\Psi_{\mathrm{out}} = \frac{1}{i}\log(e^{il}e^{iu}e^{-i(l+d)}), \qquad \Psi_{\mathrm{in}} = \frac{1}{i}\log(e^{i(l-d)}e^{iu}e^{-il}).$$

Both have the expansion

$$\Psi = \mathrm{Ad}(e^{il})u - k(l)d + \Psi^{(2)}, \qquad |\Psi^{(2)}| \le 14|d|\max\{|u|,|d|\},$$

where $k(l) = (e^{i\,\mathrm{ad}\,l} - 1)/(i\,\mathrm{ad}\,l)$, defined by its power series at zero. The derivative in (u,d) gives the first two terms; the remainder vanishes when $d = 0$, and its derivative at $(u,d) = 0$ vanishes. The small logarithm estimate and the mixed-vanishing Cauchy bound give the displayed estimate with its factor $|d|$. Also $|\mathrm{Ad}(e^{il}) - 1| \le 3|l|$ and $|k(l) - 1| \le 3|l|$. In the signed sum at x, the linear terms combine into $(\mathrm{Ad}(e^{il}) - 1)D^*A - (k(l) - 1)D^*D\lambda$. The at most six quadratic terms contribute $84\max|D\lambda|\max\{|A|,|D\lambda|\}$. Multiplying by $\xi(x)^2$, using $\xi(x) \le \mathsf{L}(b)$ at every adjacent inner bond and $|D^*A|_{(-2)} \le 3\|A\|_X$, proves the estimate. The same expansion proves the last bound. Only existing inner bonds enter this calculation.

The induced factors. There are $c_{\mathrm{top}} > 0$ and C_{top} depending only on L such that, for $|A|_{(-1)}, \|\lambda\|_1 \le L^{-1}c_{\mathrm{top}}$, all induced top factors are defined and

$$\begin{aligned} T^{-1}\frac{1}{i}\log\mathcal{R}_{\mathbf{B}}(U_0, A; e^{i\lambda}) &= Q'\lambda + q(A,\lambda), \\ \sup|q(A,\lambda)| &\le [\kappa_F\alpha_0 + C_{\mathrm{top}}(|A|_{(-1)} + \|\lambda\|_1)]\|\lambda\|_1. \end{aligned} \tag{475}$$

The maps in this display are real analytic on a neighbourhood of the closed admitted set. We choose $c_{\mathrm{top}} \le c_{\mathsf{m}}/4$. For $w = e^{i\lambda}$ the real angle estimate at a bond gives

$$|A^w|_{(-1)} \le |A|_{(-1)} + \tfrac{\pi}{2}|D\lambda|_{(-1)} < c_{\mathsf{m}}/L.$$

Indeed the angle of $w(b_-)e^{i\eta A(b)}U_0(b)w(b_+)^{-1}U_0(b)^{-1}$ is at most $\eta|A(b)| + (\pi/2)\eta|D\lambda(b)|$; it is in the small logarithm branch. Thus both A and A^w lie in the mean domain of proposition 30.2. We prove the top-factor estimate uniformly in the depth.

On one mean block let r be the incoming gauge factor and put

$$S = \sup|r-1|, \qquad g = \max_b \theta(r(b_-)^{-1}\,\mathrm{Ad}(W(b))r(b_+)).$$

Let ϖ bound the reference plaquettes. The one-step factor satisfies

$$\begin{aligned} S' &\le (1+\kappa\varpi)S + \kappa g, \\ g' &\le Lg + \kappa(\varpi S + g^2 + |a|g), \\ \rho' &= q_W^{\mathsf{m}}\rho + u, \qquad |u| \le \kappa(S+|a|)(g+\varpi S), \end{aligned} \tag{476}$$

where $\rho = (1/i)\log r$ and $\kappa = \kappa(L)$. To see the last estimate, write $\widehat{r}(x) = \mathrm{Ad}(W(y,x))r(x)$ and $\delta_x = (1/i)\log(r(y)^{-1}\widehat{r}(x))$. Comparison with an ordered path in the block gives $|\delta_x| \le C(L)(g+\varpi S)$: the path contributes its length times g, while the change of transport contributes ϖS by the conjugation estimate. The mean-chart formula gives $r'(y) = r(y)e^{i\tau(y)}$ with $\tau(y) = L^{-3}\sum_x \delta_x + O_L(|a|(g+\varpi S))$. Expanding the two logarithms then gives the first and third estimates in (476). For the second, compare the two endpoint formulas along each straight path of length L joining the blocks. Before averaging their logarithms, its transported gauge difference has angle at most $Lg + C(L)\varpi S$. The endpoint corrections are the averages of δ_x, up to the displayed error. The quadratic logarithm errors are bounded by $C(L)(g+\varpi S)^2$; with the fixed small radii this is at most $C(L)(g^2+\varpi S)$. This proves the second estimate with its leading coefficient exactly L. The estimates at a single block follow by reflection into an adjacent double block: reflected bonds retain their values up to reversal, and the joining corridor has identity reference and zero potential. Curvature, gauge-difference and size bounds are preserved, and locality leaves the factor at the original block unchanged.

Apply these estimates in the tower ending at a top $y \in \Lambda_j$. The preceding mean-chart estimates give

$$\varpi_i \le 2\alpha L^{-2(j-i)}, \qquad |a_i| \le 2aL^{i-j}, \qquad a := \xi_j \sup_{B^j(y)} |A| \le |A|_{(-1)},$$

$$\epsilon := \max\{\sup_{B^j(y)} |\lambda|, \xi_j \sup_{B^j(y)} |D\lambda|\} \le \|\lambda\|_1.$$

where initially $S_0 \le \epsilon$ and $g_0 \le (\pi/2)L^{-j}\epsilon$. For the latter inequality use $\theta(e^{-i\lambda(x)}e^{i(\lambda(x)+\eta D\lambda(b))}) \le (\pi/2)\eta|D\lambda(b)|$. Here the bond suprema range over bonds with both ends in $B^j(y)$. At tops the tower lies entirely in the corresponding pillar, so its scale-conversion factor is 1. The reference curvature hypothesis supplies $\alpha \le L^2\alpha_0$. We reduce the reference radius so that $\alpha \le \alpha_* \le a_{\mathsf{m}}$, with α_* chosen below.

Put $G_i = L^{j-i}g_i$ and $S_* = 2 + 32\kappa$. The recurrences become

$$\begin{aligned} G_{i+1} &\le [1 + \kappa L^{-1}L^{i-j}(G_i + 2a)]G_i + 2\kappa\alpha L^{i-j-1}S_i, \\ S_{i+1} &\le (1+\kappa\varpi_i)S_i + \kappa L^{i-j}G_i. \end{aligned}$$

Choose the tower radii so that $20\kappa L^{-2}c_{\mathrm{top}} \le 1/4$, $4\kappa S_* L^{-2}\alpha_* \le 1$, and $4\kappa L^{-2}\alpha_* \le 1/4$, also keeping $2c_{\mathrm{top}}$, $8c_{\mathrm{top}}$ and $S_* c_{\mathrm{top}}$ within the one-step chart radii. Using $\sum_{i<j} L^{i-j} \le 2L^{-1}$, discrete Gronwall closes the simultaneous induction

$$G_i \le 8\epsilon, \qquad S_i \le S_*\epsilon.$$

Indeed the multiplicative errors in each recurrence sum to at most $1/4$, and their inhomogeneous sums are at most ϵ and $16\kappa L^{-1}\epsilon$, respectively; $e^{1/4} < 4/3$ gives the stated bounds. All intermediate logarithms remain in their charts. Iterating the last relation in (476) gives

$$\rho_j = q_{j-1}^{\mathsf{m}} \cdots q_0^{\mathsf{m}}\lambda + C_j', \qquad |C_j'| \le C_{\mathrm{top}}(a+\epsilon)\epsilon.$$

Each real scalar mean average has norm at most one. The sum of its remainders is bounded by

$$\kappa(S_*\epsilon + 2a)\sum_{i<j}(8L^{i-j}\epsilon + 2\alpha L^{-2(j-i)}S_*\epsilon).$$

This proves the asserted bound uniformly in j. To identify this factor, let a_i', v_i' be the averages and accumulated factors of A^w. The factorization (356) and real gauge covariance give inductively

$$e^{ia_i'}W_i = (e^{ia_i}W_i)^{r_i}, \qquad v_i' r_i = w v_i[A].$$

At the next level use $r_{i+1} = \mathsf{v}(W_i, a_i')^{-1} r_i \mathsf{v}(W_i, a_i)$; multiplication proves the second identity at level $i+1$. The small logarithm identifies the first identity with the recursion for a_i'. At $i = j$ the second identity gives $r_j = v_j[A^w]^{-1} w v_j[A]$, precisely the induced factor. Nested locality identifies the tower values with those of the sequence. All means and logarithms satisfy strict domain inequalities at the admitted real pairs; their finite compositions are therefore real analytic on a neighbourhood of those pairs.

The iterated scalar mean average is $Q'_{\mathrm{mean}} = T(Q' + E')$, where the earlier mean/corner comparison gives $|(E'\lambda)(y)| \le \kappa_F \alpha_0 |\lambda|_y$. Thus $q = E'\lambda + T^{-1} C_j'$ on Λ_j, which proves (475). On Λ_0 the induced factor is $e^{i\lambda}$, $Q'\lambda = \lambda$ and $T = 1$, so $q = 0$.

Solving the two equations. Set

$$K' = 1 + K_1 + K_2, \qquad K_3 = 4(1 + K_1^3)K', \qquad K'' = 4K_3.$$

Choose $c_6 \le \min\{a_{\mathrm{in}}, L^{-2}\alpha_*\}$ small enough that

$$K'' c_6 \le \min\{\rho_2, L^{-1} c_{\mathrm{top}}\},$$
$$7 C_{\mathrm{gc}} (1 + K_1^3) K' (1 + 7K'') c_6 \le \tfrac{1}{2},$$
$$[\kappa_F + C_{\mathrm{top}}(1 + K'')] K' c_6 \le \tfrac{1}{2}.$$

All these choices precede the sequence and its depth. Put $m_0 = |D^* A_0|_{(-2)} + \sup|r - 1|$ and $s = 4(1 + K_1^3) m_0$. Since $|D^* A_0|_{(-2)} \le 3 \|A_0\|_X$, we have $m_0 \le 4c_6$ and $K's \le K'' c_6$. On the real max-norm ball

$$f \in \operatorname{ran} R, \quad |f|_{(-2)} \le s, \qquad \sup|\nu| \le s,$$

put $\lambda = G' f + H' \nu$ and define

$$\Psi(f, \nu) = \left(R[D^* A_0 + \mathcal{N}(A_0, \lambda)], T^{-1} \frac{1}{i} \log r - q(A_0, \lambda) \right).$$

The logarithm is defined and its norm is at most $(\pi/2)\sup|r - 1|$. The scalar estimates give $\|\lambda\|_Y \le K's$ and $\|\lambda\|_{Y^\circ} \le 7K's$. The chosen caps and the two remainder estimates imply

$$|\Psi_1|_{(-2)} \le \tfrac{1}{4} s + \tfrac{1}{2} s, \qquad \sup|\Psi_2| \le \tfrac{\pi}{8} s + \tfrac{1}{2} s.$$

This continuous map sends the ball into its strict interior and is C^2 on the open ball.

For completeness, the finite-dimensional fixed-point fact used here follows by a degree integral. If such a map had no fixed point, put $F_t(p) = p - t\Psi(p)$ and choose a C^1 bump ω of integral one supported sufficiently near zero. The functions $\omega(F_t) \det DF_t$ are supported in a common compact subset of the open ball. If C_{ji} are the cofactors of DF_t, then $\sum_i \partial_i C_{ji} = 0$, by equality of mixed derivatives and alternation of the determinant, and

$$\partial_t [\omega(F_t) \det DF_t] = -\sum_i \partial_i \left[\omega(F_t) \sum_j C_{ji} \Psi_j \right].$$

Its integral is constant in t, equal to 1 at $t = 0$ and to 0 at $t = 1$, a contradiction. The zero-radius case is immediate.

Take a fixed point and its $\lambda = G' f + H' \nu$. The scalar identities give $R\Delta\lambda = f$ and $Q'\lambda = \nu$. The first fixed-point equation, (474), and $RD^*D = R\Delta$ give $RD^* A_0^w = 0$. The second equation and (475) identify the two small logarithms and hence the induced top factors. Finally $\|\lambda\|_Y \le K's = K_3 m_0$ and (474) give the bounds. □

30.5. **The nonlinear rows and gauge directions.** Keep the real background U_0, the reduced corner average Q, and the weighted spaces of proposition 29.1. Write $a_j(A) = \mathcal{Q}_j^{\mathsf{m}}(U_0; A)$ for the straightened averages. On the positive interior and outward bonds of $\Gamma_j^{\flat,+}$ put $\mathcal{Q}(A)(c) = a_j(A)(c)$ and $K_A = \mathcal{Q}'(A)$. The frame rotation $(T\omega)(c) = \mathrm{Ad}(t_j(c_-))\omega(c)$ preserves both the row inner product and the supremum norm. Define

$$\mathcal{C}(A) = T^{-1}\mathcal{Q}(A) - QA, \qquad \widehat{K}_A = T^{-1}K_A = Q + \mathcal{C}'(A).$$

The mean/corner comparison gives

$$\mathcal{C}(A) = EA + T^{-1}(C_j(A))_{j\le k}, \qquad \mathcal{C}'(0) = E.$$

In particular the first term is linear and is retained throughout.

Lemma 30.5. Put $C_3 = 68/c_{\mathsf{m}}$ and

$$\epsilon(\alpha, r) = \kappa_F L^2\alpha + 4(C_{\mathsf{m}} + 1)Lr, \qquad \epsilon^*(\alpha, r) = \kappa_F L^2\alpha + C_3 Lr.$$

The map $\mathcal{Q}$ is real-analytic on $|A|_{(-1)} < c_{\mathsf{m}}/L$. If $|A|_{(-1)}, |A_i|_{(-1)} \le r \le c_{\mathsf{m}}/(4L)$, then, with $\epsilon = \epsilon(\alpha_0, r)$ and $\epsilon^* = \epsilon^*(\alpha_0, r)$,

$$\sup|\mathcal{C}(A)| \le L\epsilon|A|_{(-1)}, \qquad \sup|\mathcal{C}'(A)\delta A| \le L\epsilon|\delta A|_{(-1)},$$
$$\sup|\mathcal{C}(A_1) - \mathcal{C}(A_2)| \le L\epsilon|A_1 - A_2|_{(-1)},$$
$$|\mathcal{C}'(A)^*\omega|_{(-3)} \le 3L^3\epsilon^* \sup|\omega|,$$
$$|(\mathcal{C}'(A_1)^* - \mathcal{C}'(A_2)^*)\omega|_{(-3)} \le 2L^4 C_3|A_1 - A_2|_{(-1)} \sup|\omega|.$$

■

Proof. Proposition 30.2 and lemma 30.3 give, for a level-j row c,

$$|\mathcal{C}'(A)\delta A(c)| \le (\kappa_F L^2\alpha_0\vartheta_c + 4C_{\mathsf{m}}\vartheta_c^2 r)|\delta A|_{(-1)}, \qquad \vartheta_c \in \{1, L\}.$$

Indeed $E(c, b)$ is supported in the double block D_c, its row sum is at most $\kappa_F L^2\alpha_0\xi_j$, and every bond there has local scale at least ξ_j/ϑ_c. The remaining term is the derivative of the quadratic remainder C_j. Since $\mathcal{C}(0) = 0$, integration along line segments proves the first three inequalities.

For the transpose estimates use the weighted column norm

$$\|B\|_{\mathrm{col}} = \sup_b \mathsf{L}(b)^3 \sum_c \xi_c^{-1}|B(c, b)|.$$

At a bond of local scale ξ_n, only rows of levels $n-1, n, n+1$ occur. The straightened kernel satisfies $\sum_{c\in\Gamma_j^{\flat,+}} |K_A(c, b)| \le 2\eta^3\xi_j^{-2}$, also on the complex ball $|A|_{(-1)} < c_{\mathsf{m}}/L$. Hence

$$\|T^{-1}K_A\|_{\mathrm{col}} \le 2(L^3 + 2)\eta^3.$$

Apply the Cauchy derivative estimate on the ball of radius $c_{\mathsf{m}}/(2L)$ and integrate on the ball of radius $c_{\mathsf{m}}/(4L)$. With the stated, larger constant C_3 this gives

$$\|\mathcal{C}'(A_1) - \mathcal{C}'(A_2)\|_{\mathrm{col}} \le C_3 L(L^3 + 2)\eta^3|A_1 - A_2|_{(-1)}.$$

The column estimate in the mean/corner comparison gives $\|E\|_{\mathrm{col}} \le \kappa_F L^2\alpha_0(L^3 + 2)\eta^3$. Thus $\|\mathcal{C}'(A)\|_{\mathrm{col}} \le 3L^3\epsilon^*\eta^3$. Finally $|B^*\omega|_{(-3)} \le \eta^{-3}\|B\|_{\mathrm{col}} \sup|\omega|$, and $L(L^3 + 2) \le 2L^4$. □

The straightening leaves an endpoint factor at an outward bond. Put $W_{0,j} = \mathcal{M}^j U_0$ and $V_0 = M_{\mathbf{B}^\flat} U_0$. For an outward level-$j$ bond c with outer endpoint z, let $v_z(q) = \mathsf{v}(W_{0,j-1}, q|_{B(z)})(z)$. The restriction uses only interior rows of level $j-1$. The one-step straightening estimate gives $\theta(v_z(q)) \le \kappa_{6,\mathsf{m}} \sup_{B(z)}|q|$. Define $\Xi(q) = q$ on interior rows and

$$\Xi(q)(c) = \begin{cases} i^{-1}\log[v_z(q)e^{iq(c)}], & z = c_-, \\ i^{-1}\log[e^{iq(c)}W_{0,j}(c)v_z(q)^{-1}W_{0,j}(c)^{-1}], & z = c_+. \end{cases}$$

This is real-analytic when the interior values are less than r_{m} and $|q(c)| + \kappa_{6,\mathsf{m}} \sup_{\text{interior}} |q| < \pi$. Define the gauge-removed row logarithm by $\mathcal{Q}^\flat = \Xi \circ \mathcal{Q}$ and put $K_A^\flat = (\mathcal{Q}^\flat)'(A)$. The straightening recursion gives

$$M_{\mathbf{B}^\flat}(e^{i\eta A}U_0) = (e^{i\mathcal{Q}^\flat(A)}V_0)^{v[A]}, \qquad v[A]|_{\Lambda_j} = v_j[A].$$

The bounds $|a_j(c)| \le 2\vartheta_c |A|_{(-1)}$ place $\mathcal{Q}(A)$ in the indicated open set under the following hypotheses.

Lemma 30.6. If $|A|_{(-1)} \le r \le c_{\mathsf{m}}/(4L)$ and $2(L + \kappa_{6,\mathsf{m}})r \le 1$, then $K_A^\flat$ and K_A have the same kernel, and their adjoints have the same range. Also $\mathcal{Q}^\flat(A) = 0$ if and only if $\mathcal{Q}(A) = 0$. ■

Proof. If $d\Xi(q)\delta q = 0$, its interior components give $\delta q = 0$ there, so all v_z have zero variation. At each outward row the remaining derivative is that of $X \mapsto i^{-1}\log(ve^{iX})$ or $X \mapsto i^{-1}\log(e^{iX}v')$, with fixed $v, v' \in G$. These maps are local diffeomorphisms on the specified logarithm branch. Thus $\delta q = 0$ also on outward rows. Finite dimension makes $d\Xi(q)$ bijective. The chain rule now gives the kernel and transpose-range assertions. Finally $v_z(0) = I$. If $\Xi(q) = 0$, its interior components vanish, hence $v_z(q) = I$, and its outward components then give $q = 0$. This proves the last assertion. □

Let $\delta_\lambda A$ be the derivative at $t = 0$ of the bare gauge change $A^{e^{it\lambda}}$, and set

$$\varrho_A(\lambda) = \frac{1}{i}\,\frac{d}{dt}\mathcal{R}_{\mathbf{B}}(U_0, A; e^{it\lambda})\Big|_{t=0}$$

at the block tops. The scalar operators G', H' satisfy

$$(R\Delta, Q')^{-1}(f, \nu) = G'f + H'\nu \quad (f \in \operatorname{ran} R),$$

with $\|G'f\|_Y \le (1 + K_1)|f|_{(-2)}$ and $\|H'\nu\|_Y \le K_2 \sup|\nu|$. Recall that $R\Delta = RD^*D$ and $\|\lambda\|_{Y^\circ} \le 7\|\lambda\|_Y$.

Lemma 30.7. After decreasing c_6 in lemma 30.4, when A is real and $\alpha_0, \|A\|_X \le c_6$, the map

$$\lambda \longmapsto (RD^*\delta_\lambda A, \varrho_A(\lambda)|_{\Lambda_*})$$

is a linear bijection from the real site functions onto $\operatorname{ran} R$ times the real functions on the tops. In particular,

$$\{RD^*\delta_\lambda A : \varrho_A(\lambda)|_{\Lambda_*} = 0\} = \operatorname{ran} R.$$

■

Proof. The divergence and induced-gauge estimates, differentiated at the identity gauge, give

$$D^*\delta_\lambda A = -D^*D\lambda + \dot N_A\lambda, \quad |\dot N_A\lambda|_{(-2)} \le C_{\mathrm{gc}}\|A\|_X\|\lambda\|_{Y^\circ},$$
$$T^{-1}\varrho_A(\lambda) = Q'\lambda + \dot q_A\lambda, \qquad \sup|\dot q_A\lambda| \le (\kappa_F\alpha_0 + C_{\mathrm{top}}|A|_{(-1)})\|\lambda\|_1.$$

Parametrize the bare functions by $\lambda = -G'f + H'\nu$. Put $K' = 1 + K_1 + K_2$ and give (f, ν) the norm $\max\{|f|_{(-2)}, \sup|\nu|\}$. Then $\|\lambda\|_Y \le K'\|(f, \nu)\|$, and the displayed map, after rotation of its second component, is

$$(f, \nu) \longmapsto (f + R\dot N_A\lambda, \nu + \dot q_A\lambda).$$

Choose $c_6 < \min\{L^{-1}c_{\mathrm{top}}, \rho_2\}$ and

$$[7(1 + K_1^3)C_{\mathrm{gc}} + \kappa_F + C_{\mathrm{top}}]K'c_6 \le \tfrac{1}{2}.$$

The perturbation has norm at most $1/2$, so its Neumann series converges. This proves bijectivity and both stated consequences. □

Lemma 30.8. Suppose the hypotheses of lemmas 30.6 and 30.7 hold, $3K_1L^3\epsilon^*(\alpha_0, r) < 1$, and $E_A = e^{i\eta A}U_0$ satisfies

$$|\partial E_A(p) - I| < L^{-2}a_{\mathsf{m}}\eta^2\mathsf{L}(p)^{-2}$$

at every inner plaquette. Then E_A is critical for the action on its actual reduced averaging fibre if and only if

$$F'(A) \in \operatorname{ran} K_A^* = \operatorname{ran} K_A^{\flat *}.$$

Moreover $(F'(A), \delta_\lambda A)_\eta = 0$ for every real λ, and $K_A \delta_\lambda A = 0$ whenever $\varrho_A(\lambda)$ vanishes on the tops. Criticality is preserved by a fixed real gauge transformation. ■

Proof. The operator $H_0^* \widehat{K}_A^* = I + H_0^* \mathcal{C}'(A)^*$ is invertible by proposition 29.1 and lemma 30.5. Hence K_A^* is injective and K_A is onto. By lemma 30.6, the physical row map $x(A) = e^{i\mathcal{Q}^\flat(A)} V_0$ also has surjective derivative and $\ker dx(A) = \ker K_A$. By proposition 30.2, the curvature condition places the actual averages in their analytic domain. Their factorization is

$$n(A) := M_{\mathbf{B}^\flat}(E_A) = x(A)^{v[A]}, \qquad v[A]|_{\Lambda_j} = v_j[U_0; A].$$

For a variation δA write $\mu = i^{-1}(dv[A]\delta A)v[A]^{-1}$. Differentiation gives

$$dn(A)\delta A = (dx(A)\delta A)^{v[A]} + \operatorname{rot}_{n(A)} \mu.$$

For z a coarse configuration, write $\operatorname{rot}_z \mu = (d/dt) z^{e^{it\mu}}|_{t=0}$. Gauge covariance gives

$$dn(A)\delta_\lambda A = \operatorname{rot}_{n(A)}(\lambda|_{\Lambda_*}), \qquad dx(A)\delta_\lambda A = \operatorname{rot}_{x(A)} \varrho_A(\lambda).$$

The first relation shows that $dn(A)$ is onto: after using surjectivity of $dx(A)$, remove the unwanted rotation by a bare gauge direction equal to μ on the tops.

Let $\mathfrak{G}$ be the space of bare gauge directions. If $K_A \delta A = 0$, the preceding formula for dn and a bare extension of μ give $\delta A \in \ker dn(A) + \mathfrak{G}$. Conversely, if $dn(A)\delta A = 0$, then

$$dx(A)\delta A = -\operatorname{rot}_{x(A)}(\operatorname{Ad}(v[A]^{-1})\mu).$$

By lemma 30.7, choose λ whose induced top values are $\operatorname{Ad}(v[A]^{-1})\mu$. Then $\delta A + \delta_\lambda A \in \ker dx(A) = \ker K_A$. Consequently $\ker dn(A) + \mathfrak{G} = \ker K_A + \mathfrak{G}$.

Gauge invariance of the action implies $F'(A) \perp \mathfrak{G}$. The finite-dimensional submersion theorem realizes every vector of $\ker dn(A)$ as the velocity of a curve in the actual fibre. Thus criticality is equivalent to $F'(A) \perp \ker dn(A)$, and hence to $F'(A) \perp \ker K_A$. This is the stated multiplier condition. The second gauge-covariance identity also proves the claim when the induced top values vanish. Finally a fixed gauge maps each actual averaging fibre diffeomorphically onto the corresponding transformed fibre and preserves the action; it therefore preserves criticality. □

30.6. **Solving the inhomogeneous equation.** The reference need not be critical. Keep the regional action F of proposition 30.1 and write $J_0 = F'(0)$, $j = |J_0|_{(-3)}$. The action remainder in this proof is $V(A) = \mathcal{V}(A)$. Its local and weighted estimates are those of proposition 30.1; in particular the real plaquette bound has radius $\alpha_0 + 3L^2 r$ if $\|A\|_X \le r$, $Lr \le \rho_{\mathrm{ec}}$ and $C_{\mathrm{ec}} L^2 r \le 1$.

Proposition 30.9. There are $c_8, r_0 > 0$, depending only on L, with the following property. Assume the preceding separated regional setting with real U_0 and $\alpha_0 \le c_8$. Put

$$K_8 = K_1(16 + 9K_1^3), \qquad \Theta = 4K_8, \qquad K_5 = 2(K_7 + K_1).$$

For a real row field $B^\natural$, set $\beta = \sup |B^\natural|$ and $r_1 = K_5(j + \beta)$. If $r_1 \le r_0$, the system

$$F'(A) + DRD^*A = K_A^* \omega, \qquad \mathcal{Q}(A) = B^\natural$$

has exactly one real solution with $\|A\|_X \le r_0$. Its multiplier is unique without a separate smallness assumption, and

$$\|A\|_X \le r_1, \qquad \sup|\omega| \le \Theta r_1, \qquad RD^*A = 0, \qquad F'(A) = K_A^* \omega,$$
$$|D^*DA|_{(-3)} \le 3(1 + L^3)\Theta r_1.$$

The configuration $E = e^{i\eta A} U_0$ is critical on its actual reduced averaging fibre.

In particular, suppose B is prescribed on the reduced rows and vanishes below the top level. Take $B^\natural = B$. For any real gauge h satisfying $h = v_j[U_0; A]^{-1}$ on every Λ_j, the configuration E^h is critical on

$$C(\mathbf{B}^\flat, e^{iB} M_{\mathbf{B}^\flat} U_0).$$

The values of h away from these tops may be chosen arbitrarily. ∎

Proof. Write $\|\cdot\| = \|\cdot\|_X$ and suppress the subscript on H_0. We choose $c_8 \le c_6$ and $r_0 \le c_6$ small enough that the action estimates, the row estimates and lemma 30.6 apply on $\|A\| \le r_0$; also choose them so that $C_{\mathrm{ec}} L^3 c_8 \le 1$ and

$$c_8 + 3L^2 r_0 < L^{-2} a_{\mathsf{m}}.$$

All further smallness requirements below concern constants depending only on L and quantities that tend to zero with (c_8, r_0).

Put $\Delta_R = F''(0) + DRD^*$. The exact regional Hessian identity gives $G_0^{-1} = \Delta_R + Q^*Q$. The definitions of H and $\widetilde{G}_0$, and symmetry, imply

$$QH = I, \quad Q\widetilde{G}_0 = 0, \quad \widetilde{G}_0 Q^* = 0, \quad \widetilde{G}_0 \Delta_R = I - HQ, \quad \ker \widetilde{G}_0 = \operatorname{ran} Q^*.$$

For the last identity, $\widetilde{G}_0 f = 0$ implies $G_0 f = G_0 Q^* (Q G_0 Q^*)^{-1} Q G_0 f$, and G_0 is invertible.

Eliminating the multiplier. Let $\widehat{B} = T^{-1} B^\natural$ and put

$$\mathcal{G}(A) = F'(A) + DRD^* A, \qquad S_A = I + H^* \mathcal{C}'(A)^*, \qquad w(A) = S_A^{-1} H^* \mathcal{G}(A).$$

Decrease c_8, r_0 so that, with $\epsilon^* = \epsilon^*(\alpha_0, r_0)$, $3K_1 L^3 \epsilon^* \le 1/8$. Then $\|S_A^{-1}\|_{\infty\to\infty} \le 8/7$ throughout the ball. The field equation becomes

$$\Psi(A) = \widetilde{G}_0[-J_0 - V(A) + \mathcal{C}'(A)^* w(A)] + H[\widehat{B} - \mathcal{C}(A)].$$

More precisely, the original system is equivalent on this ball to $A = \Psi(A)$ and $\omega = T w(A)$. To see both directions, first apply $\widetilde{G}_0$ to the field equation and use the constraint; applying H^* gives $S_A \widehat{\omega} = H^* \mathcal{G}(A)$. Conversely, a fixed point gives $QA = \widehat{B} - \mathcal{C}(A)$. The displayed identities then show that $\mathcal{G}(A) - (Q^* + \mathcal{C}'(A)^*) w(A)$ belongs to $\operatorname{ran} Q^*$. Its image under H^* is zero by the definition of $w(A)$; since $H^* Q^* = I$, it is zero.

The transpose estimates are needed here in addition to the field estimates. Since $|DA|_{(-2)} \le 2\|A\|$ and $|D^*A|_{(-2)} \le 3\|A\|$,

$$H^*\mathcal{G}(A) = H^* J_0 + (DH)^* DA + H^*(\Delta' A + V(A)) + (D^*H)^* R D^* A$$

gives

$$\sup |H^*\mathcal{G}(A)| \le K_8(j + \|A\| + |V(A)|_{(-3)}),$$
$$\sup |H^*(\mathcal{G}(A_1) - \mathcal{G}(A_2))| \le K_8(\|A_1 - A_2\| + |V(A_1) - V(A_2)|_{(-3)}).$$

The contraction. Let $j \le r \le r_0$ and $\|A\|, \|A_i\| \le r$. By further reducing r_0, the action estimates give

$$|V(A)|_{(-3)} \le r/8, \qquad |V(A_1) - V(A_2)|_{(-3)} \le \tfrac{1}{2}\|A_1 - A_2\|.$$

It follows that $\sup |w(A)| \le (17/7) K_8 r \le 3K_8 r$. Writing $w_i = w(A_i)$, subtraction gives

$$S_{A_1}(w_1 - w_2) = H^*(\mathcal{G}(A_1) - \mathcal{G}(A_2)) - H^*(\mathcal{C}'(A_1)^* - \mathcal{C}'(A_2)^*) w_2.$$

Choose $6K_1 L^4 C_3 r_0 \le 1$. The preceding estimates then give

$$\sup |w_1 - w_2| \le \tfrac{8}{7}[\tfrac{3}{2} K_8 + 6K_1 L^4 C_3 K_8 r] \|A_1 - A_2\| \le 3K_8 \|A_1 - A_2\|.$$

With $\epsilon = \epsilon(\alpha_0, r_0)$ and $\epsilon^* = \epsilon^*(\alpha_0, r_0)$, it follows that

$$\|\Psi(A)\| \le K_7 j + K_1 \beta + [K_7(C_{\mathrm{ec}} L^3 r + 9K_8 L^3 \epsilon^*) + K_1 L \epsilon] r,$$
$$\|\Psi(A_1) - \Psi(A_2)\| \le \{K_7[4C_{\mathrm{ec}} L^4 r + 6K_8 L^4 C_3 r + 9K_8 L^3 \epsilon^*] + K_1 L \epsilon\} \|A_1 - A_2\|.$$

Decrease c_8, r_0 once more so that the braces in the second line, at $r = r_0$ and $\alpha_0 = c_8$, are at most $1/2$. The bracket in the first line is then also at most $1/2$. Consequently

$$\|\Psi(A)\| \le \tfrac{1}{2}K_5(j+\beta) + \tfrac{1}{2}r, \qquad \|\Psi(A_1) - \Psi(A_2)\| \le \tfrac{1}{2}\|A_1 - A_2\|.$$

Apply the contraction principle on the balls of radii r_0 and $r_1 = K_5(j+\beta)$. Their fixed points agree, and the multiplier is determined by $w(A)$. If $r_1 = 0$, then $J_0 = B^\natural = 0$ and $A = 0$ is the fixed point directly. In all cases

$$\|A\| \le r_1, \qquad \sup|\omega| \le \tfrac{17}{28}\Theta r_1.$$

The gauge condition and criticality. The curvature estimate and the choice of c_8, r_0 verify the hypotheses of lemma 30.8. For every λ with $\varrho_A(\lambda) = 0$ at the tops, pair the field equation with $\delta_\lambda A$. Both its action term and its multiplier term vanish. Since R is an orthogonal projection,

$$0 = (RD^*A, RD^*\delta_\lambda A)_\eta.$$

By lemma 30.7 the second arguments fill $\operatorname{ran} R$, which contains RD^*A. Thus $RD^*A = 0$, so $F'(A) = K_A^*\omega$ and lemma 30.8 gives criticality on the actual averaging fibre.

For the second derivative estimate, use

$$D^*DA = (Q^* + \mathcal{C}'(A)^*)T^{-1}\omega - J_0 - \Delta' A - V(A).$$

Our choices give $3L^3\epsilon^* \le 1/(8K_1) \le 1/32$, $|\Delta' A|_{(-3)} \le r_1$ and $|V(A)|_{(-3)} \le r_1/8$. The reduced-row bound for Q^* therefore yields

$$|D^*DA|_{(-3)} \le [2(L^3+2) + 1/32]\tfrac{17}{28}\Theta r_1 + \tfrac{17}{8}r_1 \le 3(1+L^3)\Theta r_1.$$

Finally suppose B vanishes below the top level. On the interior rows $a_j = B_j$. Each endpoint factor v_z at an outward level-j row depends only on the interior rows of level $j-1$, where $a_{j-1} = 0$; hence $v_z = I$. The factorization of the actual averages and $h = v_j^{-1}$ at the level-j tops then give $M_{\mathbf{B}^\flat}(E^h) = e^{iB}M_{\mathbf{B}^\flat}U_0$. The top sets are disjoint, so these prescriptions admit a real gauge extension. Criticality follows from gauge invariance, as proved in lemma 30.8. □

30.7. **Regularity on the gauge slice.** Put

$$C_{\mathrm{abs}} = C_{\mathrm{ec}}(L^3 + L) + 3L(\kappa_F L^2 + 4C_{\mathsf{m}}L),$$

and choose

$$0 < c_7 \le \min\left\{a_{\mathrm{in}}, \frac{c_{\mathsf{m}}}{4L}, \frac{1}{(2+C_{\mathrm{ec}})L}, \frac{1}{C_{\mathrm{ec}}L^3}, \frac{2}{K_1 C_{\mathrm{abs}}}, \frac{\rho_{\mathrm{ec}}}{L}\right\}.$$

The constants are fixed before the sequence and the fields.

Lemma 30.10. In the preceding regional setting, let $U_0 \in \mathfrak{U}'_k(\mathbf{B}, \alpha_0)$, $0 < \alpha_0 \le c_7$, and let A be real. Suppose

$$|A|_{(-1)} \le c_7, \qquad RD^*A = 0, \qquad E = e^{i\eta A}U_0 \in \mathfrak{U}'_k(\mathbf{B}, \alpha_0).$$

If $|a_j(A)(c)| \le \beta$ at every positively oriented interior or outward row $c \in \Gamma_j^{\flat,+}$, including level zero, then

$$\|A\|_X \le 2K_1(\alpha_0 + 4\beta), \qquad |D^*DA|_{(-3)} \le 8K_1(\alpha_0 + 4\beta).$$

■

Proof. Put $r = |A|_{(-1)}$. The choice $r \le c_{\mathsf{m}}/(4L)$ lies strictly inside the mean chart. We use the actual corner rows Q restricted to the interior and outward bonds. Their adjoint satisfies

$$|Q^*\omega|_{(-3)} \le 2(1 + L^{-3})\sup|\omega|. \tag{477}$$

To see this, extend ω by zero to inward rows. If $\mathsf{L}(b) = \xi_i$, locality permits only levels $i-1, i, i+1$ in the column at b. A double block of an interior or outward row of level $i-1$ misses Ω_i, whereas an endpoint of b lies in Ω_i; hence that level contributes zero. At each remaining level j the column

bound and the pairings give the factor $\eta^{-3}\xi_j^{-1}(2\eta^3\xi_j^{-2}) = 2\xi_j^{-3}$. Multiplication by ξ_i^3 proves (477), including the endpoint levels by omitting absent indices.

Write $J' = D^*DA$. At a full bond the local action decomposition, the bounds on the currents of U_0 and E, and the chart-gradient estimate give

$$\mathsf{L}(b)^3|J'(b)| \le [2 + (2 + C_{\mathrm{ec}})Lr]\alpha_0 + C_{\mathrm{ec}}L^3 r\|A\|_X \le 3\alpha_0 + C_{\mathrm{ec}}L^3 r\|A\|_X.$$

Here $s_b(A) \le Lr \le \rho_{\mathrm{ec}}$ and $x_b(A) \le L^2\|A\|_X$. At a nonfull inner bond, the bond and every bond of its inner plaquettes are level-zero interior rows. On these bonds $\eta|A| = |a_0(A)| \le \beta$. There are at most three inner plaquettes in the divergence, with four values in each curl. Since the real transports preserve norm,

$$\eta^3|J'(b)| \le 3 \cdot 4\beta = 12\beta.$$

The expression is zero if there is no inner plaquette. Consequently

$$|J'|_{(-3)} \le \max\{3\alpha_0 + C_{\mathrm{ec}}L^3 r\|A\|_X, 12\beta\}. \tag{478}$$

For a row $c \in \Gamma_j^{\flat,+}$ put $\widehat{a}(c) = \mathrm{Ad}(t_j(c_-))^{-1}a_j(A)(c)$. The mean/corner comparison and the mean Taylor remainder give

$$QA = \widehat{a} - \mathcal{C}(A), \qquad \sup|\widehat{a}| \le \beta, \qquad \sup|\mathcal{C}(A)| \le L(\kappa_F L^2\alpha_0 + 4C_{\mathsf{m}}Lr)r.$$

Indeed the row of the linear difference applied to A is at most $\kappa_F L^2\alpha_0\vartheta_c r$, and the mean remainder is at most $C_{\mathsf{m}}\vartheta_c^2 r^2$; $\vartheta_c \le L$ gives the displayed common bound. The rotations preserve norms, and both sides are real.

The slice condition and $H_{U_0} = D^*D + \Delta'$ now give

$$(H_{U_0} + DRD^* + Q^*Q)A = f_A, \qquad f_A = J' + \Delta' A + Q^*(\widehat{a} - \mathcal{C}(A)).$$

Use (478), $|\Delta' A|_{(-3)} \le C_{\mathrm{ec}}L\alpha_0 r$ and (477). Since $\alpha_0, r \le c_7$ and $r \le \|A\|_X$,

$$|f_A|_{(-3)} \le 3\alpha_0 + [12 + 2(1 + L^{-3})]\beta + C_{\mathrm{abs}}c_7\|A\|_X.$$

Apply G_0 and its weighted estimate. The last term contributes at most $\|A\|_X/2$, so it can be absorbed:

$$\|A\|_X \le \frac{K_1}{2}(3\alpha_0 + 16\beta) \le 2K_1(\alpha_0 + 4\beta).$$

Finally $C_{\mathrm{ec}}L^3 r \le 1$ in (478) gives the asserted bound for D^*DA, using $K_1 \ge 1$. □

31. Background inverses and current multipliers

The multiplier in the background operator is obtained from the first constrained inverse. We first estimate this multiplier, then apply theorem G again with that value. The weighted transpose introduces a quotient of local scales; exponential decay absorbs it uniformly in the number of layers.

Keep the geometry, natural regular-pair class, spaces and bilinear pairings of section 27, with all the straightened rows of the determining set. In particular, $\mathsf{A} = Y = \Omega_0$, and write $\Delta(\mathbf{U}) = \Delta_{\Omega_0}(\mathbf{U})$ for the Hessian in (371). It sums over the inner plaquettes of Ω_0. All operators below are evaluated at $\mathbf{U}$. Whenever the indicated inverses exist, set

$$\begin{gathered}\mathsf{G}' = (\Delta_a')^{-1}, \qquad \mathsf{C}' = (Q'\mathsf{G}'^2Q'^{\mathsf{T}})^{-1}, \qquad \mathsf{P} = \mathsf{G}'Q'^{\mathsf{T}}\mathsf{C}'Q'\mathsf{G}', \qquad \mathsf{R} = I - \mathsf{P}, \\ \mathsf{V} = \mathsf{G}'\mathsf{R}D^{\mathsf{T}}, \qquad \mathsf{S} = I - D\mathsf{V}.\end{gathered} \tag{479}$$

The current form and its symmetric representing operator are

$$\begin{aligned}\mathfrak{q}_{\mathbf{J}}(A, v) &= (i[v_-, A] - i[A, v_+] - i[v_-, Dv], \mathbf{J})_\xi, \\ (A, \Delta_\pi'[\mathbf{J}]A)_\xi &= \mathfrak{q}_{\mathbf{J}}(A, \mathsf{V}A).\end{aligned} \tag{480}$$

Here $v_\pm$ are as in (372). The second line is a homogeneous quadratic polynomial in A, so polarization with the nondegenerate pairing defines $\Delta'_\pi[\mathbf{J}]$. For an independent row multiplier j put

$$\mathcal{L}_\mathsf{j} = \Delta(\mathbf{U}) - \Delta'_\pi[\mathbf{J}] - \mathsf{S}^\mathsf{T}\Delta^{(2)}[\mathsf{j}]\mathsf{S} + D\mathsf{R}D^\mathsf{T} + Q^\mathsf{T}\mathsf{m}^{-2}Q. \tag{481}$$

The row weight $\mathsf{m}^{-2}(c) = \xi_c^{-2}$ is unchanged. Define successively

$$\begin{aligned} \mathsf{G} &= \mathcal{L}_0^{-1}, \qquad \mathsf{H} = \mathsf{G}Q^\mathsf{T}(Q\mathsf{G}Q^\mathsf{T})^{-1}, \qquad \mathsf{j}_* = \mathsf{H}^\mathsf{T}\mathbf{J}, \\ \mathsf{G}_1 &= \mathcal{L}_{\mathsf{j}_*}^{-1}, \qquad \mathsf{H}_1 = \mathsf{G}_1Q^\mathsf{T}(Q\mathsf{G}_1Q^\mathsf{T})^{-1}, \qquad \mathsf{j}_{1*} = \mathsf{H}_1^\mathsf{T}\mathbf{J}. \end{aligned} \tag{482}$$

The proposition below establishes existence in this order. In particular, the second inverse is taken only after j_* has been bounded.

Proposition 31.1. For every odd $L \geq 13$ there are positive constants $a_{\mathrm{bg}}, C_{\mathrm{bg}}$ and a power $\mathfrak{M}_{\mathrm{bg}}$ of L with the following property. Let $0 \leq t \leq K$, let $\mathfrak{M} \geq \mathfrak{M}_{\mathrm{bg}}$ and $R_1 \geq L$ be powers of L, and let $(\Omega_n)_{n\leq t}$ be a separated sequence with these parameters on a torus of side $\ell \geq 2L\mathfrak{M}\xi_t$. Suppose

$$(\mathbf{U}, \mathbf{J}) \in \mathfrak{P}_t^\natural((\Omega_n); \alpha_0, \alpha_1, \gamma_0), \qquad 0 < \alpha_0, \alpha_1, \gamma_0 \leq a_{\mathrm{bg}}.$$

Then all inverses in (479)–(482) exist. With $\mathbb{E}^{[i]}(\mathsf{j}) = \mathbb{K}^{[i]}[\mathbf{U}, \mathbf{J}, \mathsf{j}]^{-1}$, one has

$$\mathsf{H} = [\mathbb{E}^{[5]}(0)]_{A,8}, \qquad \mathsf{H}_1 = [\mathbb{E}^{[5]}(\mathsf{j}_*)]_{A,8}, \tag{483}$$

and, for every top y,

$$|\mathsf{j}_*|_{-3;y}, \ |\mathsf{j}_{1*}|_{-3;y} \leq C_{\mathrm{bg}}\gamma_0 \leq a_{\mathrm{reg}}. \tag{484}$$

All four systems at j_* are invertible and satisfy (400) with the same $\delta_{\mathrm{reg}}, B_{\mathrm{reg}}$. The quadratic correction in G_1 obeys

$$(A, \Delta^{(2)}[\mathsf{j}_*]A)_\xi = 2(\mathsf{H}C^{(2)}(A), \mathbf{J})_\xi, \qquad \mathsf{G}_1^{-1} = \mathsf{G}^{-1} - \mathsf{S}^\mathsf{T}\Delta^{(2)}[\mathsf{j}_*]\mathsf{S}.$$

If the class decomposition has $A' = 0$ and $\mathbf{J}$ is real, then H, H_1 and both multipliers are real. All constants are independent of the lattice spacing, volume and number of layers. ∎

31.1. **Elimination of the auxiliary variables.** Write $\mathsf{g}(A, v) = \mathsf{g}_0(A, v) + D^\mathsf{T}\mathsf{x}(A, v)$, with the terms in (372). The scalar substitutions in section 23.2 also apply to the present operators. The current terms enter when the resulting expressions are substituted into the first equation.

Lemma 31.2. Let $(\mathbf{U}, \mathbf{J}, \mathsf{j})$ satisfy the hypotheses of theorem G, with ambient region $\mathsf{A} = Y = \Omega_0$. The scalar inverses G' and C' in (479) exist. The operators $\mathcal{L}_\mathsf{j}$ and $Q\mathcal{L}_\mathsf{j}^{-1}Q^\mathsf{T}$ are invertible, and

$$[\mathbb{E}^{[4]}]_{A,1} = \mathcal{L}_\mathsf{j}^{-1}, \qquad [\mathbb{E}^{[5]}]_{A,8} = \mathcal{L}_\mathsf{j}^{-1}Q^\mathsf{T}(Q\mathcal{L}_\mathsf{j}^{-1}Q^\mathsf{T})^{-1}.$$

Here the inverses $\mathbb{E}^{[i]}$ are taken at this triple. ∎

Proof. The operator $\Delta'_a = \mathbb{K}^{[1]}$ is invertible by theorem G. With $\mathsf{G}' = (\Delta'_a)^{-1}$, the scalar elimination preceding (333), with $k_2 = f_2$, gives

$$Q'\mathsf{G}'^2Q'^\mathsf{T}\sigma = Q'\mathsf{G}'f_2 - Q'\mathsf{G}'^2f_3 - f_4.$$

The same substitutions recover v, w uniquely from σ, so solutions correspond bijectively to solutions of this equation. Its right-hand side ranges over the whole scalar coarse space as f_4 varies. Invertibility of $\mathbb{K}^{[2]}$ therefore proves invertibility of $Q'\mathsf{G}'^2Q'^\mathsf{T}$ and defines C'. The substitutions (333)–(334) now hold with these operators and C' in place of C.

The identity $\langle \mathrm{Ad}(V)X, Y\rangle = \langle X, \mathrm{Ad}(V)^{-1}Y\rangle$ for $V \in G^c$ and reversal of each neighboring pair in (369) show that Δ° is symmetric. The diagonal terms, including those at the boundary, are symmetric as well. Thus Δ'_a, G', C', P and R are symmetric, and

$$\mathsf{V}^\mathsf{T} = D\mathsf{R}\mathsf{G}'.$$

Put $\Delta^{(2)} = \Delta^{(2)}[\mathsf{j}]$ for this proof. Cyclicity of the trace rewrites (480) as

$$\mathfrak{q}_{\mathbf{J}}(A, v) = (v_- + v_+, i[A, \mathbf{J}])_\xi - (v_-, i[Dv, \mathbf{J}])_\xi.$$

Its half derivative in the A direction $\dot{A}$ is $(\dot{A}, \mathsf{m}_J(v))_\xi$. For its half derivative in v, move the endpoint factors to the site pairing. The derivatives of the endpoint factors give $(\dot{v}, \mathsf{g}_0(A, v))_\xi$, while differentiating Dv gives $(D\dot{v}, -\frac{i}{2}[\mathbf{J}, v_-])_\xi$. Adding the quadratic form of $\Delta^{(2)}$ consequently gives

$$\begin{aligned} \frac{1}{2}\, d_{(A,v)}\big[\mathfrak{q}_{\mathbf{J}}(A, v) + (A - Dv, \Delta^{(2)}(A - Dv))_\xi\big](\dot{A}, \dot{v}) \\ = (\dot{A}, \mathsf{m}_J(v) + \Delta^{(2)}(A - Dv))_\xi + (\dot{v}, \mathsf{g}(A, v))_\xi. \end{aligned} \tag{485}$$

In the last term we used $(D\dot{v}, \mathsf{x}(A, v))_\xi = (\dot{v}, D^{\mathsf{T}}\mathsf{x}(A, v))_\xi$.

Let $\mathsf{W}_\mathsf{j} = \Delta'_\pi[\mathbf{J}] + \mathsf{S}^{\mathsf{T}}\Delta^{(2)}\mathsf{S}$. Its quadratic form is the form in (485) evaluated at $(A, \mathsf{V}A)$. Since $\mathsf{S}A = A - D\mathsf{V}A$, the chain rule and $\mathsf{V}^{\mathsf{T}} = D\mathsf{R}\mathsf{G}'$ give

$$\mathsf{W}_\mathsf{j} A = \mathsf{m}_J(\mathsf{V}A) + \Delta^{(2)}\mathsf{S}A + D\mathsf{R}\mathsf{G}'\mathsf{g}(A, \mathsf{V}A). \tag{486}$$

Indeed, pairing the right-hand side with $\dot{A}$ gives half the derivative of this quadratic form. Its symmetric representing operator is unique.

Take a right-hand side with only its first and, for $i = 5$, eighth components possibly nonzero. The scalar substitutions give

$$v = \mathsf{V}A, \qquad w = \mathsf{P}D^{\mathsf{T}}A, \qquad p = \mathsf{R}\mathsf{G}'\mathsf{g}(A, \mathsf{V}A).$$

They also determine $\sigma = \mathsf{C}'Q'\mathsf{G}'D^{\mathsf{T}}A$, $r = \mathsf{C}'Q'\mathsf{G}'^2\mathsf{g}(A, \mathsf{V}A)$ and $q = -\mathsf{G}'p$ uniquely. Substitute these expressions in the first equation and use (486). Since $\mathsf{R} = I - \mathsf{P}$, that equation becomes

$$\mathcal{L}_\mathsf{j} A = f_1 + Q^{\mathsf{T}}\omega,$$

with the last term omitted for $i = 4$. Here the Hessian is $\Delta(\mathbf{U})$: both the ambient region and Y are Ω_0, so no compression changes its plaquette sum.

For $i = 4$, the equivalence just proved recovers every auxiliary component from A. Invertibility of $\mathbb{K}^{[4]}$ therefore gives a unique solution of $\mathcal{L}_\mathsf{j} A = f_1$ for each f_1, proving the first block identity. For $i = 5$, the remaining equation $QA = f_8$ becomes

$$Q\mathcal{L}_\mathsf{j}^{-1}Q^{\mathsf{T}}\omega = f_8 - Q\mathcal{L}_\mathsf{j}^{-1}f_1.$$

Again all components are recovered uniquely from a solution. Invertibility of $\mathbb{K}^{[5]}$, with f_8 arbitrary, proves that the operator on the left is invertible. Taking $f_1 = 0$ gives the second block identity. □

31.2. **The multiplier bound.** Write $\delta = \delta_{\mathrm{reg}}$ and $S_\delta = c_S(\delta/4)$, with c_S from (232). Choose a power $\mathfrak{M}_{\mathrm{bg}} \geq \mathfrak{M}_*$ of L so large that, for every $\mathfrak{M} \geq \mathfrak{M}_{\mathrm{bg}}$ and $R_1 \geq L$,

$$R_1\mathfrak{M} \geq \mathsf{r}_S(\delta/4), \qquad \frac{\log L}{R_1\mathfrak{M}} \leq \frac{\delta}{4}. \tag{487}$$

It suffices to impose these inequalities at $R_1 = L$. Set

$$C_{\mathrm{bg}} = \max\{1, 6LB_{\mathrm{reg}}S_\delta\}, \qquad a_{\mathrm{bg}} = a_{\mathrm{reg}}/C_{\mathrm{bg}}.$$

These choices depend only on L.

Multiplier bound in proposition 31.1. Let j be any independent multiplier with $|\mathsf{j}|_{-3;y} \leq a_{\mathrm{reg}}$ for every top, and put $T = [\mathbb{E}^{[5]}(\mathsf{j})]_{A,8}$. The inverse exists by theorem G, since $a_{\mathrm{bg}} \leq a_{\mathrm{reg}}$. We estimate $T^{\mathsf{T}}\mathbf{J}$ once, then use the estimate at $\mathsf{j} = 0$ and at the induced multiplier.

Fix a positive row c and $e \in \mathfrak{g}^c$. Let e_c equal e at c and zero at the other positive rows. The weighted transpose gives

$$\begin{aligned}\xi_c^3\langle e,(T^{\mathsf{T}}\mathbf{J})(c)\rangle &= (e_c,T^{\mathsf{T}}\mathbf{J})_{\mathbf{B}} = (Te_c,\mathbf{J})_\xi\\ &= \xi^3\sum_b\langle (Te_c)(b),\mathbf{J}(b)\rangle,\end{aligned} \tag{488}$$

where the sum is over positive inner bare bonds. The source $g=(e_8e_c;0)$ satisfies $\iota g = e_8e_c$ and, by (328),

$$|g|_{y(c)} = \xi_c|e|, \qquad |g|_y = 0 \quad (y \neq y(c)).$$

Here $y(c)$ is the position of c and $\xi_{y(c)}=\xi_c$. The solution seminorm dominates $|A|_{-1;y}=\xi_y|A|_y$. Thus (400) gives

$$\xi_y|(Te_c)(b)| \le B_{\mathrm{reg}}e^{-\delta d(y,y(c))}\xi_c|e| \qquad (\mathrm{position}(b)=y).$$

At most $3(\xi_y/\xi)^3$ positive bare bonds have position y, and the regular-pair condition bounds the current there by $\gamma_0\xi_y^{-3}$. Their cell volume cancels the current weight in (488); the input and output weights leave one quotient ξ_c/ξ_y. The upper pairing bound in (378) therefore yields

$$\xi_c^3|\langle e,(T^{\mathsf{T}}\mathbf{J})(c)\rangle| \le 3B_{\mathrm{reg}}\gamma_0|e|\sum_{y\in\Lambda_*}\frac{\xi_c}{\xi_y}e^{-\delta d(y,y(c))}.$$

The scale comparison (231) and (487) imply

$$\frac{\xi_c}{\xi_y} \le L\exp\!\left(\frac{\log L}{R_1\mathfrak{M}}d(y,y(c))\right) \le Le^{\delta d(y,y(c))/4}.$$

Consequently the sum is at most $L\sum_y e^{-\delta d(y,y(c))/4} \le LS_\delta$ by (233). The lower pairing bound in (378) contributes a factor of two when recovering the matrix norm. Taking the maximum over rows of position y, for which $\xi_c=\xi_y$, gives

$$|T^{\mathsf{T}}\mathbf{J}|_{-3;y} \le 6LB_{\mathrm{reg}}S_\delta\gamma_0 \le C_{\mathrm{bg}}\gamma_0. \tag{489}$$

At $\mathsf{j}=0$, lemma 31.2 gives G,H and identifies $T=\mathsf{H}$. Thus the first block identity holds and $|\mathsf{j}_*|_{-3;y} \le C_{\mathrm{bg}}\gamma_0 \le a_{\mathrm{reg}}$. □

31.3. **The second inverse.**

Completion of the proof of proposition 31.1. We have $|\mathsf{j}_*|_{-3;y} \le a_{\mathrm{reg}}$ at every top. Thus theorem G applies to the triple $(\mathbf{U},\mathbf{J},\mathsf{j}_*)$. It gives all four inverses and their claimed decay. Apply lemma 31.2 at this triple to obtain G_1, $(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}$ and the second block identity in (483). The defining weighted transpose gives

$$2(C^{(2)}(A),\mathsf{H}^{\mathsf{T}}\mathbf{J})_{\mathbf{B}} = 2(\mathsf{H}C^{(2)}(A),\mathbf{J})_\xi.$$

Together with (371), this identifies the quadratic correction. Subtracting (481) at $\mathsf{j}=0$ from its value at j_* gives the stated formula for G_1^{-1}.

At this admissible multiplier, take $T=\mathsf{H}_1$ in (489). It gives $|\mathsf{j}_{1*}|_{-3;y} \le C_{\mathrm{bg}}\gamma_0 \le a_{\mathrm{reg}}$, proving the second bound in (484).

Real data. Suppose the class decomposition has $A'=0$ and $\mathbf{J}$ is real. On a regular piece, the fixed unitary gauge in the proof of lemma 27.6 writes $U^u = e^{i\xi A_0}$ with A_0 real. The covering argument there places every global row double block and scalar block in a regular or boundary piece, and identifies the actual relative-constraint germ before returning to the original gauge. In the rescaled block, the proof of proposition 24.1 gives actual bases $W_n = e^{iB_n}$ with B_n real and $\|B_n\|_\infty < r_{\mathrm{st}}/16$. For a sufficiently small real relative perturbation, its iterates start real and satisfy $\|a_n\|_\infty \le 2c_{\mathrm{av}} < r_{\mathrm{st}}/2$. The real assertion of lemma 24.2 therefore applies at each step and keeps a_n real. The first and second Taylor coefficients of the actual constraint are consequently real. The fixed unitary covariance (363) preserves the real form and returns these coefficients to the original

gauge. The agreement of germs in lemma 27.6 identifies them with the global Q and $C^{(2)}$. The scalar transport formula (360) makes Q' real as well. On boundary pieces $Q = Q' = I$ and $C^{(2)} = 0$.

For unitary U, its adjoint transports preserve $\mathfrak{g}$, and $i[X, Y] \in \mathfrak{g}$ for $X, Y \in \mathfrak{g}$. Thus the covariant differences and the current commutators are real. The positive real pairing weights make the transpose of a real map real. The action is real on real perturbations of U, so its Hessian is real; likewise $\Delta^{(2)}[\mathsf{j}]$ is real for real j, by its quadratic form and polarization. It follows from (372) and the system equations that $\mathbb{K}^{[i]}$ is real at every such real triple. Its inverse is real by uniqueness of solutions. At $\mathsf{j} = 0$, the block identity (483) makes H real, and therefore $\mathsf{j}_* = \mathsf{H}^{\mathsf{T}}\mathbf{J}$ is real. Apply the same argument at this real multiplier to obtain real H_1 and $\mathsf{j}_{1*} = \mathsf{H}_1^{\mathsf{T}}\mathbf{J}$.

□

32. The nonlinear response

The nonlinear correction is obtained in two steps. First we solve $X = \mathsf{C}(A' - \mathsf{H}X)$ and denote its solution by $\mathsf{D}(A')$. Its derivative enters the reduced field W defined below. The quadratic bound for this field makes $\mathbf{A} = -\mathsf{GW}(\mathsf{K}B' + \mathbf{A})$ a contraction; the response is then $\mathcal{L}(\mathsf{K}B' + \mathbf{A})$, where $\mathcal{L}(A') = A' - \mathsf{HD}(A')$. We prove this implication under explicit bounds on the nonlinear maps and the four linear operators.

Use the separated layers, cells $\Delta(y)$, reduced positive rows c, and distance $d = d_{\mathbf{B}^\flat}$ of section 12. A bare bond lies over the unique $y \in \Lambda_*$ whose cell contains its initial site; write $y_b = y$. Rows are placed at y_c as in that section, and ξ_c is the spacing of the row. Let $\mathcal{A}$ and $\mathcal{X}$ be the complex Lie-algebra-valued fields on these bare bonds and rows, respectively. In this section the row variables are dimensionless, and the bilinear pairings are

$$(A, f)_\xi = \xi^3 \sum_b \langle A(b), f(b)\rangle, \qquad (X, Y)_1 = \sum_c \xi_c^{-1} \langle X(c), Y(c)\rangle. \tag{490}$$

Every transpose below uses these pairings. In particular, the row weight here is ξ_c^{-1}, whereas the physical row pairing in section 31 has weight ξ_c^3. The letters $Q, \mathsf{H}, \mathsf{S}, \mathsf{K}, \mathsf{G}$ below denote maps satisfying the stated assumptions.

For $A, f \in \mathcal{A}$ and $X \in \mathcal{X}$ put

$$|A|_y = \xi_y \max_{y_b = y} |A(b)|, \qquad |f|_y^\sharp = \xi_y^3 \max_{y_b = y} |f(b)|, \qquad |X|_y = \max_{y_c = y} |X(c)|.$$

An empty maximum is zero. Let $\mathfrak{g}_y$ be finite seminorms on $\mathcal{A}$, and set $\|A\|_y = \max\{|A|_y, \mathfrak{g}_y(A)\}$. Fix $\theta_0 > 0$, put $\rho_0 = 3L + 2$, and define

$$N(y) = \{z : d(y, z) \le 2\rho_0\}, \qquad \mathsf{e} = \exp(2\rho_0\theta_0).$$

Admissible weights are positive functions with $w(y) \le e^{\theta_0 d(y,z)} w(z)$ for all y, z. Write $|A|_w = \sup_y w(y)|A|_y$, and likewise $|f|_w^\sharp$, $|X|_w$, and $\|A\|_w$. At $w = 1$ write $\sup|A|$, $\sup|f|^\sharp$, $\sup|X|$, and $\|A\|$. These are finite-dimensional norms; the zero-dimensional cases are included. The weight comparison we shall use is

$$w(y) \max_{z \in N(y)} h(z) \le \mathsf{e} \sup_z w(z)h(z) \qquad (h \ge 0). \tag{491}$$

32.1. **Hypotheses and the response theorem.** Fix $c_\flat, C_2, C_3, C_{\mathcal{V}} \ge 1$ and $c_4, \rho_1 \in (0, 1]$. Let $Q : \mathcal{A} \to \mathcal{X}$ be linear, and let

$$\mathsf{C} : \{\sup|A| < c_4\} \longrightarrow \mathcal{X}, \qquad \mathcal{V} : \{\sup|A| < 2\rho_1\} \longrightarrow \mathcal{A}$$

be holomorphic. Assume the following local estimates for every y and every argument in the specified domains. All maxima in these hypotheses are over $z \in N(y)$.

(i) The linear map and its transpose satisfy

$$|QA|_y \le c_\flat \max|A|_z, \qquad |Q^{\mathsf{T}}X|_y^\sharp \le c_\flat \max|X|_z.$$

(ii) $\mathsf{C}(0) = \mathsf{C}_0' = 0$, and

$$|\mathsf{C}(A)|_y \le C_2 \max |A|_z^2.$$

If $\sup|A_i| \le r \le c_4/4$, then

$$|\mathsf{C}(A_1) - \mathsf{C}(A_2)|_y \le 4C_2 r \max |A_1 - A_2|_z.$$

For $\sup|A|, \sup|A_i| \le c_4/2$,

$$\begin{aligned} |\mathsf{C}_A'^{\mathsf{T}} X|_y^\sharp &\le C_3 \max|A|_z \max|X|_z, \\ |(\mathsf{C}_{A_1}'^{\mathsf{T}} - \mathsf{C}_{A_2}'^{\mathsf{T}})X|_y^\sharp &\le C_3 \max|A_1 - A_2|_z \max|X|_z. \end{aligned} \tag{492}$$

(iii) $\mathcal{V}(0) = \mathcal{V}'(0) = 0$. For $\sup|A| \le \rho_1$,

$$|\mathcal{V}(A)|_y^\sharp \le C_{\mathcal{V}} \max|A|_z \max\|A\|_z.$$

If $\|A_i\| \le r \le \rho_1/4$, then

$$|\mathcal{V}(A_1) - \mathcal{V}(A_2)|_y^\sharp \le 4C_{\mathcal{V}} r \max\|A_1 - A_2\|_z.$$

Let $\mathsf{a} \ge 0$ act on rows by multiplication, and fix $\mathbf{J} \in \mathcal{A}$ with $\gamma_0 := \sup|\mathbf{J}|^\sharp \le 1$. Assume that the linear maps

$$\mathsf{H}, \mathsf{K} : \mathcal{X} \to \mathcal{A}, \quad \mathsf{S} : \mathcal{X} \to \mathcal{X}, \quad \mathsf{G} : \mathcal{A} \to \mathcal{A}$$

obey, for one finite $\hat{\mathsf{b}} \ge \max\{1, \sup \mathsf{a}\}$, every admissible w, every $X \in \mathcal{X}$, and every $f \in \mathcal{A}$,

$$\begin{aligned} \|\mathsf{H}X\|_w,\ \|\mathsf{K}X\|_w,\ |\mathsf{S}X|_w \le \hat{\mathsf{b}}|X|_w, \qquad |(\mathsf{S} - \mathsf{a})X|_w \le 2\hat{\mathsf{b}}|X|_w, \\ \|\mathsf{G}f\|_w \le \hat{\mathsf{b}}|f|_w^\sharp, \qquad |\mathsf{H}^{\mathsf{T}} f|_w \le 8\hat{\mathsf{b}}|f|_w^\sharp. \end{aligned} \tag{493}$$

Choose the radii by

$$\begin{aligned} c_W &= \min\left\{\frac{c_4}{8}, \frac{1}{32C_2\mathsf{e}}, \frac{1}{32C_3\mathsf{e}}, \frac{\rho_1}{8}\right\}, \\ C_W &= \frac{72C_3\mathsf{e}}{c_W} + 32c_\flat C_2\mathsf{e}^2 + 30c_\flat C_3\mathsf{e}^2 + 40C_{\mathcal{V}}\mathsf{e}, \\ \varepsilon_3 &= \frac{c_W}{\hat{\mathsf{b}}}, \qquad \varepsilon_B = \min\left\{\frac{c_W}{4\hat{\mathsf{b}}^2}, \frac{1}{8C_W\hat{\mathsf{b}}^4}\right\}. \end{aligned} \tag{494}$$

Proposition 32.1. Under the hypotheses above, for every $\sup|A'| \le \varepsilon_3$ there is a unique solution $X = \mathsf{D}(A')$ of

$$X = \mathsf{C}(A' - \mathsf{H}X), \qquad \sup|X| \le \varepsilon_3/\hat{\mathsf{b}}. \tag{495}$$

It satisfies $\sup|\mathsf{D}(A')| \le 4C_2 \sup|A'|^2$ and is holomorphic for $\sup|A'| < \varepsilon_3$. Put $\mathcal{L}(A') = A' - \mathsf{H}\mathsf{D}(A')$. For $\sup|A_i'| \le \varepsilon \le \varepsilon_3$,

$$\begin{aligned} |\mathsf{D}(A_1') - \mathsf{D}(A_2')|_w &\le 16C_2\mathsf{e}\varepsilon|A_1' - A_2'|_w, \\ \tfrac{1}{2}|A_1' - A_2'|_w \le |\mathcal{L}(A_1') - \mathcal{L}(A_2')|_w &\le 2|A_1' - A_2'|_w, \\ \|\mathcal{L}(A_1') - \mathcal{L}(A_2')\|_w &\le 2\|A_1' - A_2'\|_w. \end{aligned} \tag{496}$$

On the open ball, write $\mathfrak{D}(A') = \mathsf{D}'(A')$. With $T = \mathcal{L}(A')$,

$$\mathfrak{D}(A')^{\mathsf{T}} = \mathsf{C}_T'^{\mathsf{T}}(I + \mathsf{H}^{\mathsf{T}}\mathsf{C}_T'^{\mathsf{T}})^{-1}. \tag{497}$$

■

For $\sup|A'| < \varepsilon_3$, abbreviate $\mathsf{D} = \mathsf{D}(A')$, $T = \mathcal{L}(A')$, and $\mathfrak{D} = \mathfrak{D}(A')$, and define

$$\begin{aligned} \mathsf{W}(A') = &-(\mathfrak{D}^{\mathsf{T}} - \mathsf{C}''[A']^{\mathsf{T}})\mathsf{H}^{\mathsf{T}}\mathbf{J} - Q^{\mathsf{T}}(\mathsf{S} - \mathsf{a})\mathsf{D} \\ &- \mathfrak{D}^{\mathsf{T}}(\mathsf{S} - \mathsf{a})QA' + \mathfrak{D}^{\mathsf{T}}(\mathsf{S} - \mathsf{a})\mathsf{D} + \mathcal{V}(T) - \mathfrak{D}^{\mathsf{T}}\mathsf{H}^{\mathsf{T}}\mathcal{V}(T). \end{aligned} \tag{498}$$

Here $\mathsf{C}''[A'] = \frac{d}{dt}\mathsf{C}'_{tA'}|_{t=0}$. The subtraction in the first term removes the linear part of $\mathfrak{D}^{\mathsf{T}}$; its quantitative effect is proved below.

Theorem J. Under the hypotheses of this subsection, for each $B' \in \mathcal{X}$ with $\sup|B'| \le \varepsilon_B$ the system

$$\begin{gathered} X = \mathsf{C}(\mathsf{K}B' + \mathbf{A} - \mathsf{H}X), \qquad \mathbf{A} + \mathsf{GW}(\mathsf{K}B' + \mathbf{A}) = 0, \\ \mathbf{H} = \mathsf{K}B' + \mathbf{A} - \mathsf{H}X \end{gathered} \tag{499}$$

has exactly one solution with $\sup|X| \le \varepsilon_3/\hat{\mathsf{b}}$ and $\|\mathbf{A}\| \le \hat{\mathsf{b}}\varepsilon_B$. Write $\mathbf{A}_0(B')$ and $\mathbf{H}(B')$ for its second and third components. They are holomorphic for $\sup|B'| < \varepsilon_B$, vanish at zero, and satisfy

$$\|\mathbf{A}_0(B')\| \le 8C_W\hat{\mathsf{b}}^5 \sup|B'|^2 \le \hat{\mathsf{b}}\sup|B'|, \qquad \|\mathbf{H}(B')\| \le 4\hat{\mathsf{b}}\sup|B'|. \tag{500}$$

For every admissible w and $\sup|B'_i| \le \varepsilon_B$,

$$\begin{aligned} \|\mathbf{A}_0(B'_1) - \mathbf{A}_0(B'_2)\|_w &\le \hat{\mathsf{b}}|B'_1 - B'_2|_w, \\ \|\mathbf{H}(B'_1) - \mathbf{H}(B'_2)\|_w &\le 4\hat{\mathsf{b}}|B'_1 - B'_2|_w, \\ \|\mathbf{H}(B') - \mathsf{K}B'\|_w &\le 10C_W\hat{\mathsf{b}}^5 \sup|B'|\,|B'|_w. \end{aligned} \tag{501}$$

If a nonempty $\Sigma \subset \Lambda_*$ contains the support of B' in the row positions, then

$$\begin{aligned} \|\mathbf{H}(B')\|_y &\le 4\hat{\mathsf{b}}e^{-\theta_0 d(y,\Sigma)}\sup|B'|, \\ \|\mathbf{H}(B') - \mathsf{K}B'\|_y &\le 10C_W\hat{\mathsf{b}}^5 e^{-\theta_0 d(y,\Sigma)}\sup|B'|^2. \end{aligned} \tag{502}$$

■□

32.2. **The change of variables and its derivative.**

Proof of proposition 32.1. Fix $\sup|A'| \le \varepsilon \le \varepsilon_3$. On the ball $\sup|X| \le \varepsilon/\hat{\mathsf{b}}$,

$$\sup|A' - \mathsf{H}X| \le 2\varepsilon \le c_4/4, \qquad \sup|\mathsf{C}(A' - \mathsf{H}X)| \le 4C_2\varepsilon^2 \le \varepsilon/\hat{\mathsf{b}}.$$

The Lipschitz constant of this map of X is at most $8C_2\hat{\mathsf{b}}\varepsilon \le 1/4$. The contraction theorem gives a unique fixed point in each of these nested balls. Uniqueness in the largest ball identifies them, and the image estimate at $\varepsilon = \sup|A'|$ gives the quadratic bound. Iteration from zero has increments bounded by $4^{-n}\varepsilon_3/\hat{\mathsf{b}}$, uniformly in A' on the open ball. The iterates are holomorphic and remain in the domain of C, so their limit is holomorphic.

Put $X_i = \mathsf{D}(A'_i)$ and $T_i = A'_i - \mathsf{H}X_i$. The local difference estimate and (491) give

$$|X_1 - X_2|_w \le 8C_2\mathsf{e}\varepsilon(|A'_1 - A'_2|_w + \hat{\mathsf{b}}|X_1 - X_2|_w).$$

Absorb the last term, whose coefficient is at most $1/4$. This proves the first bound in (496). Moreover,

$$\|\mathsf{H}(X_1 - X_2)\|_w \le 16C_2\mathsf{e}\hat{\mathsf{b}}\varepsilon|A'_1 - A'_2|_w \le \tfrac{1}{2}|A'_1 - A'_2|_w.$$

The remaining bounds follow by subtracting this correction from $A'_1 - A'_2$.

For $\sup|T| \le 2\varepsilon_3$ set $M_T = \mathsf{C}'_T{}^{\mathsf{T}}$. By (492), for $\sup|T| \le 2\varepsilon \le 2\varepsilon_3$,

$$\begin{aligned} |M_T\varphi|^\sharp_w &\le 2C_3\mathsf{e}\varepsilon|\varphi|_w, \\ |(M_{T_1} - M_{T_2})\varphi|^\sharp_w &\le C_3\mathsf{e}|T_1 - T_2|_w \sup|\varphi|. \end{aligned} \tag{503}$$

The second estimate holds for $\sup|T_i| \le 2\varepsilon_3$. Consequently $\mathsf{H}^{\mathsf{T}}M_T$ has weighted operator norm at most $16C_3\mathsf{e}\hat{\mathsf{b}}\varepsilon_3 \le 1/2$. Its Neumann series gives

$$\Psi_T = (I + \mathsf{H}^{\mathsf{T}}M_T)^{-1}, \qquad |\Psi_T\varphi|_w \le 2|\varphi|_w.$$

Differentiate $\mathsf{D}(A') = \mathsf{C}(A' - \mathsf{H}\mathsf{D}(A'))$ and transpose. This gives $\mathfrak{D}^{\mathsf{T}}(I + \mathsf{H}^{\mathsf{T}}M_T) = M_T$, which proves (497). □

We record within the argument the derivative estimates needed for (498). If $\sup|A_i'| \le \varepsilon < \varepsilon_3$ and $\mathfrak{D}_i = \mathfrak{D}(A_i')$, then

$$\begin{aligned} |\mathfrak{D}_i^{\mathsf{T}}\varphi|_w^\sharp &\le 4C_3\mathsf{e}\varepsilon|\varphi|_w, \\ |(\mathfrak{D}_1^{\mathsf{T}} - \mathfrak{D}_2^{\mathsf{T}})\varphi|_w^\sharp &\le 8C_3\mathsf{e}|A_1' - A_2'|_w \sup|\varphi|. \end{aligned} \tag{504}$$

Indeed, the first follows from $\mathfrak{D}_i^{\mathsf{T}} = M_i\Psi_i$. For the second use

$$\Psi_1 - \Psi_2 = -\Psi_2\mathsf{H}^{\mathsf{T}}(M_1 - M_2)\Psi_1, \quad \mathfrak{D}_1^{\mathsf{T}} - \mathfrak{D}_2^{\mathsf{T}} = (I - M_2\Psi_2\mathsf{H}^{\mathsf{T}})(M_1 - M_2)\Psi_1.$$

Here $M_2\Psi_2\mathsf{H}^{\mathsf{T}}$ has weighted source norm at most $32C_3\mathsf{e}c_W \le 1$, while (503) and $|T_1 - T_2|_w \le 2|A_1' - A_2'|_w$ bound the last two factors by $4C_3\mathsf{e}|A_1' - A_2'|_w \sup|\varphi|$. At zero, $M_0 = 0$, $\Psi_0 = I$, and $\mathcal{L}'(0) = I$; thus $\mathfrak{D}(0) = 0$ and

$$\frac{d}{dt}\mathfrak{D}(tA')^{\mathsf{T}}\Big|_{t=0} = \mathsf{C}''[A']^{\mathsf{T}}. \tag{505}$$

32.3. **The reduced field and the second contraction.**

Proof of theorem J. First we establish the estimate that makes the second contraction possible. The field W is holomorphic on $\sup|A'| < \varepsilon_3$ and vanishes at zero. Indeed, all its factors are holomorphic there, and $\sup|\mathcal{L}(A')| < 2\varepsilon_3 \le \rho_1/4$ keeps the argument in the domain of $\mathcal{V}$. We shall show, for $\|A_i'\| \le \varepsilon < \varepsilon_3$,

$$|\mathsf{W}(A_1') - \mathsf{W}(A_2')|_w^\sharp \le C_W\hat{\mathsf{b}}(1 + \gamma_0\hat{\mathsf{b}})\varepsilon\|A_1' - A_2'\|_w. \tag{506}$$

The case $\varepsilon = 0$ is immediate. Otherwise write $\delta = \|A_1' - A_2'\|_w$, $\delta_0 = |A_1' - A_2'|_w \le \delta$, $X_i = \mathsf{D}(A_i')$, and $T_i = \mathcal{L}(A_i')$. The bounds already proved give

$$\sup|X_i| \le 4C_2\varepsilon^2, \quad |X_1 - X_2|_w \le 16C_2\mathsf{e}\varepsilon\delta_0, \quad \|T_i\| \le 2\varepsilon, \quad \|T_1 - T_2\|_w \le 2\delta.$$

The cancellation in the current term. Put $\mathsf{N}(A') = \mathfrak{D}(A')^{\mathsf{T}} - \mathsf{C}''[A']^{\mathsf{T}}$ and $\varphi = \mathsf{H}^{\mathsf{T}}\mathbf{J}$. Differentiating (492) at zero gives

$$|\mathsf{C}''[A']^{\mathsf{T}}\varphi|_w^\sharp \le C_3\mathsf{e}|A'|_w \sup|\varphi|.$$

Together with (504), this bounds the difference of N at any two points of the open amplitude ball by $9C_3\mathsf{e}|A_1' - A_2'|_w \sup|\varphi|$. By (505), $\mathsf{N}(0) = \mathsf{N}'(0) = 0$. Let $R = \varepsilon_3/\varepsilon > 1$ and set

$$g(z) = (\mathsf{N}(zA_1') - \mathsf{N}(zA_2'))\varphi \qquad (|z| < R).$$

Then $g(z)/z$ extends holomorphically across zero, vanishes there, and has weighted source norm at most $9C_3\mathsf{e}\delta_0 \sup|\varphi|$. Apply scalar Schwarz after each norm-one linear functional and take the supremum. At $z = 1$ this gains the factor $R^{-1} = \hat{\mathsf{b}}\varepsilon/c_W$. Since $\sup|\varphi| \le 8\hat{\mathsf{b}}\gamma_0$, the first term of (498) has difference bounded by

$$\frac{72C_3\mathsf{e}}{c_W}\gamma_0\hat{\mathsf{b}}^2\varepsilon\delta_0. \tag{507}$$

The remaining terms. For terms containing $\mathfrak{D}^{\mathsf{T}}$ use the single product estimate

$$|\mathfrak{D}_1^{\mathsf{T}}\varphi_1 - \mathfrak{D}_2^{\mathsf{T}}\varphi_2|_w^\sharp \le 8C_3\mathsf{e}\delta_0 \sup|\varphi_1| + 4C_3\mathsf{e}\varepsilon|\varphi_1 - \varphi_2|_w. \tag{508}$$

The local bounds for Q, Q^{T} cost at most $c_\flat\mathsf{e}$ in weighted norms and $c_\flat$ at weight one. Applying them, (493), and the displayed bounds for X_i, T_i gives the following estimates for the differences of terms two through six in (498):

term	bound in $\|\cdot\|_w^\sharp$
$-Q^{\mathsf{T}}(\mathsf{S} - \mathsf{a})X$	$32c_\flat C_2\mathsf{e}^2\hat{\mathsf{b}}\varepsilon\delta_0$
$-\mathfrak{D}^{\mathsf{T}}(\mathsf{S} - \mathsf{a})QA'$	$24c_\flat C_3\mathsf{e}^2\hat{\mathsf{b}}\varepsilon\delta_0$
$\mathfrak{D}^{\mathsf{T}}(\mathsf{S} - \mathsf{a})X$	$6C_3\mathsf{e}\varepsilon\delta_0$
$\mathcal{V}(T)$	$16C_{\mathcal{V}}\mathsf{e}\varepsilon\delta$
$-\mathfrak{D}^{\mathsf{T}}\mathsf{H}^{\mathsf{T}}\mathcal{V}(T)$	$24C_{\mathcal{V}}\mathsf{e}\varepsilon\delta$

For the fourth term, (508) first gives $(64+128\mathsf{e})C_2C_3\mathsf{e}\hat{\mathsf{b}}\varepsilon^2\delta_0$; use $32C_2\mathsf{e}\hat{\mathsf{b}}\varepsilon\le 1$. For the last two terms, $\|T_i\|\le 2\varepsilon\le\rho_1/4$ permits the local Lipschitz bound for $\mathcal{V}$ and gives $\sup|\mathcal{V}(T_i)|^\sharp\le 4C_{\mathcal{V}}\varepsilon^2$. The last application of (508) gives $(256+512\mathsf{e})C_3C_{\mathcal{V}}\mathsf{e}\hat{\mathsf{b}}\varepsilon^2\delta$; use $32C_3\mathsf{e}\hat{\mathsf{b}}\varepsilon\le 1$. Adding the table and (507), with $c_\flat,\mathsf{e},\hat{\mathsf{b}}\ge 1$, proves (506). Taking $A_2'=0$ also yields

$$|\mathsf{W}(A')|_w^\sharp\le C_W\hat{\mathsf{b}}(1+\gamma_0\hat{\mathsf{b}})\|A'\|\,\|A'\|_w\qquad(\|A'\|<\varepsilon_3).\tag{509}$$

The fixed point. Let $\sup|B'|\le r\le\varepsilon_B$, and consider $\Phi_{B'}(\mathbf{A})=-\mathsf{GW}(\mathsf{K}B'+\mathbf{A})$ on $\|\mathbf{A}\|\le\hat{\mathsf{b}}r$. Its argument has strong norm at most $2\hat{\mathsf{b}}r\le\varepsilon_3/2$. Since $\gamma_0\le 1\le\hat{\mathsf{b}}$, (506) and (509) imply

$$\begin{aligned}\|\Phi_{B'}(\mathbf{A})\|&\le 2C_W\hat{\mathsf{b}}^3(\hat{\mathsf{b}}\sup|B'|+\|\mathbf{A}\|)^2\le 8C_W\hat{\mathsf{b}}^5r^2\le\hat{\mathsf{b}}r,\\ \|\Phi_{B'}(\mathbf{A}_1)-\Phi_{B'}(\mathbf{A}_2)\|_w&\le 4C_W\hat{\mathsf{b}}^4r\|\mathbf{A}_1-\mathbf{A}_2\|_w\le\tfrac12\|\mathbf{A}_1-\mathbf{A}_2\|_w.\end{aligned}\tag{510}$$

The contraction theorem in the finite-dimensional strong norm gives a unique fixed point. As before, the nested balls have the same fixed point, denoted $\mathbf{A}_0(B')$. Taking $r=\sup|B'|$ proves its bounds in (500). The first equation in (499) then uniquely determines $X=\mathsf{D}(\mathsf{K}B'+\mathbf{A}_0(B'))$, and the last gives $\mathbf{H}(B')=\mathcal{L}(\mathsf{K}B'+\mathbf{A}_0(B'))$. Conversely, any solution in the stated balls must arise this way. The factor-two estimate for $\mathcal{L}$ proves the remaining bound in (500). Holomorphic iteration from zero has successive differences at most $2^{-n}\hat{\mathsf{b}}\varepsilon_B$ in the strong norm. Thus $\mathbf{A}_0$, and consequently $\mathbf{H}$, are holomorphic on the open datum ball.

For two data put $A_i'=\mathsf{K}B_i'+\mathbf{A}_0(B_i')$ and $r=\max_i\sup|B_i'|$. Their fixed-point equations give

$$\|\mathbf{A}_0(B_1')-\mathbf{A}_0(B_2')\|_w\le\tfrac12(\hat{\mathsf{b}}|B_1'-B_2'|_w+\|\mathbf{A}_0(B_1')-\mathbf{A}_0(B_2')\|_w).$$

Absorption proves the first bound in (501), and then $\|A_1'-A_2'\|_w\le 2\hat{\mathsf{b}}|B_1'-B_2'|_w$. Applying (496) proves the bound for $\mathbf{H}$.

Finally set $A'=\mathsf{K}B'+\mathbf{A}_0(B')$. The preceding bounds at the zero datum imply $\|A'\|\le 2\hat{\mathsf{b}}\sup|B'|$ and $\|A'\|_w\le 2\hat{\mathsf{b}}|B'|_w$. Hence (509) and the first-correction bound give

$$\begin{aligned}\|\mathbf{A}_0(B')\|_w&\le 8C_W\hat{\mathsf{b}}^5\sup|B'|\,|B'|_w,\\ \|\mathsf{HD}(A')\|_w&\le 64C_2\mathsf{e}\hat{\mathsf{b}}^3\sup|B'|\,|B'|_w.\end{aligned}$$

Since $64C_2\mathsf{e}\le 2C_W$ and $\mathbf{H}(B')-\mathsf{K}B'=\mathbf{A}_0(B')-\mathsf{HD}(A')$, these prove the last bound in (501). The weight $w(y)=e^{\theta_0 d(y,\Sigma)}$ is admissible by the triangle inequality and equals one on Σ; thus $|B'|_w=\sup|B'|$. This proves (502). □

33. The local nonlinear maps

We verify the local hypotheses of section 32 for the relative averages and the action of a regular pair. The averaging estimates follow from the fixed local gauges used in lemma 27.6. For the action, the plaquette calculation of section 13 must allow a complex reference and the incomplete plaquette stars at the boundary.

Keep the geometry and natural regular-pair class of section 27, with $\mathfrak{M}\ge\mathfrak{M}_*$. Use the dimensionless row pairing, the norms, and $N(y)$ of section 32. For a pair $(\mathbf{U},\mathbf{J})$ define

$$\mathsf{g}_y(A)=\sup_{b,\nu}\xi_y^2|(\nabla_\nu^{\mathbf{U}}A)(b)|,\qquad\|A\|_y=\max\{|A|_y,\mathsf{g}_y(A)\},$$

where b lies over y and the translated bond and connecting bond used by the difference are both inner bare bonds of Ω_0. Let $\mathcal{Q}(A)(c)$ be the relative iterate (357) on the double block of c, starting with $a_0=\xi A$ and evaluated at level $n(c)$. Its definedness is part of the proposition below. Set

$$Q=\mathcal{Q}'(0),\qquad\mathsf{C}(A)=\mathcal{Q}(A)-QA,\qquad F(A)=\xi^{-1}\sum_{p\subset\Omega_0}a_p(e^{i\xi A}\mathbf{U}),$$

where the sum is over the inner bare plaquettes. With the gradient taken in the bare pairing of (490), put

$$\mathcal{V}(A) = F'(A) - F'(0) - F''(0)A.$$

The action and its gradient are entire: each plaquette term is a finite product of matrix exponentials and fixed reference matrices, and the pairing is nondegenerate.

Proposition 33.1. There are constants $a_{\mathrm{nl}} > 0$, $c_4, \rho_1 \in (0,1]$ and $c_\flat, C_2, C_3, C_{\mathcal{V}} \geq 1$, depending only on L, with the following property. If

$$(\mathbf{U}, \mathbf{J}) \in \mathfrak{P}_t^\natural((\Omega_n); \alpha_0, \alpha_1, \gamma_0), \qquad 0 < \alpha_0, \alpha_1, \gamma_0 \leq a_{\mathrm{nl}},$$

then $\mathcal{Q}$ is defined and holomorphic on $\{\sup|A| < c_4\}$. The maps $Q, \mathsf{C}, \mathcal{V}$ satisfy hypotheses (i)–(iii) of section 32 with these constants and the seminorms above. Also $\sup|\mathbf{J}|^\sharp \leq \gamma_0 \leq 1$. The constants are uniform in the volume and number of layers. ∎

Proof. The averaging maps. For a row c let P_c be the positive bare bonds whose endpoints lie in $B^{n(c)}(c_-) \cup B^{n(c)}(c_+)$. The double-block geometry in lemma 27.4 gives

$$\#\mathsf{P}_c \leq 6(\xi_c/\xi)^3, \qquad L^{-1}\xi_c \leq \xi_{y_b} \leq L\xi_c \quad (b \in \mathsf{P}_c). \tag{511}$$

We shall use the slightly sharper distance bound $d(y_c, y_b) \leq (5L+3)/2 < \rho_0$. For $n(c) \geq 1$, join y_c to b_- by coordinate segments inside the double block, of total length at most $5\xi_c/2$. The block lies in $\Omega_{n(c)-1}$, so this path has weighted length at most $5L/2$. The path from b_- to its cell center has weighted length at most $3/2$. The coordinate paths cross cell walls only finitely often, as in lemma 27.4. For $n(c) = 0$ one has $b = c$ and $y_b = y_c$. At each fixed level a bare bond belongs to at most six double blocks: if its endpoints lie in one block, the row must be incident to that block; otherwise its two blocks determine the row. The scale comparison in (511) allows at most three levels. Thus each bare bond lies in at most 18 sets P_c.

Denote by $c_{\mathrm{av}}, C_{\mathrm{av}}, C_{\mathrm{var}}$ the constants of proposition 24.1, writing C_{var} for its constant C_2. Choose

$$a_{\mathrm{nl}} = \min\left\{a_{\mathrm{reg}}, \frac{1}{16}, \frac{c_{\mathrm{av}}}{64L^2}, \frac{1}{2L^2}\right\}, \qquad c_4 = \min\left\{1, \frac{c_{\mathrm{av}}}{2L}\right\}.$$

For a row of positive level choose a covering regular piece and fix its unitary gauge u from lemma 27.6. By (417),

$$\mathbf{U}^u = e^{i\xi\mathbf{A}_\kappa}, \qquad \xi_c \sup_{\mathsf{P}_c}|\mathbf{A}_\kappa| \leq 8L^2(\alpha_0+\alpha_1) \leq c_{\mathrm{av}}/4.$$

On the whole local ball $\mathcal{U}_c = \{\xi_c \sup_{\mathsf{P}_c}|A| < c_{\mathrm{av}}\}$, rescale by ξ_c, with $\eta = \xi/\xi_c$ and $\hat{A} = \xi_c \operatorname{Ad}(u)A$. Proposition 24.1 defines every relative iterate of $e^{i\eta\hat{A}}e^{i\eta\xi_c\mathbf{A}_\kappa}$. Pulling back by u gives the actual recursion at every point of $\mathcal{U}_c$: its first input is ξA, and (117) and (363) identify each successive contour, endpoint correction and series logarithm. Unitary conjugation preserves their domains. Block nesting keeps all inputs within P_c. Different covering pieces consequently evaluate the same recursion, since their ungauged initial data and successive operations coincide. This also proves locality on P_c. For a global field,

$$\xi_c \sup_{\mathsf{P}_c}|A| \leq L \max_{b \in \mathsf{P}_c}|A|_{y_b} \leq L\sup|A| < Lc_4 \leq c_{\mathrm{av}}/2.$$

All rows are therefore defined on the asserted ball. At level zero, including every row of a boundary piece, the recursion is simply $\mathcal{Q}(A)(c) = \xi A(c)$.

The counting-measure derivative kernel $K_A(c,b) = \partial\mathcal{Q}(A)(c)/\partial A(b)$ vanishes off P_c. The input rescaling contributes ξ_c to the kernel in (361); hence its factor is $\xi_c\eta^3 = \xi^3\xi_c^{-2}$. The fixed unitary gauges preserve its norm. Thus, throughout $\mathcal{U}_c$,

$$\begin{aligned} |K_A(c,b)| &\leq C_Q\xi^3\xi_c^{-2}, \\ |K_A(c,b) - K_0(c,b)| &\leq C_Q\xi^3\xi_c^{-2}\,\xi_c \sup_{\mathsf{P}_c}|A|, \end{aligned} \tag{512}$$

where one may take $C_Q = \max\{1, C_{\mathrm{var}}, 1 + C_{\mathrm{av}}c_{\mathrm{av}}/2 + C_{\mathrm{var}}c_{\mathrm{av}}\}$. At level zero the kernel is ξI on its single bond, and its difference is zero. The recursion vanishes at $A = 0$, so $\mathsf{C}(0) = \mathsf{C}_0' = 0$. The physical average in (370) divides this recursion by ξ_c; thus $(QA)(c) = \xi_c(Q_{\mathrm{phys}}A)(c)$.

The row count and (512) give $|QA|_y \le 6C_Q L \max_{N(y)} |A|$. For transposes the pairing is essential:

$$(Q^{\mathsf{T}}X)(b) = \xi^{-3} \sum_{c : b \in \mathsf{P}_c} \xi_c^{-1} K_0(c,b)^{\mathsf{T}} X(c). \tag{513}$$

There are at most 18 terms, and the transpose of a matrix kernel has norm at most twice its norm, by the pairing estimate used in (378). Multiplying by $\xi_{y_b}^3$ gives $|Q^{\mathsf{T}}X|_y^\sharp \le 36C_Q L^3 \max_{N(y)} |X|$. We take $c_\flat = 36C_Q L^3$.

Put $A_s = A_2 + s(A_1 - A_2)$ and integrate the derivative of C along this segment. The second bound in (512) yields

$$|\mathsf{C}(A_1)(c) - \mathsf{C}(A_2)(c)| \le 6C_Q \int_0^1 \xi_c \sup_{\mathsf{P}_c} |A_s|\, ds\, \xi_c \sup_{\mathsf{P}_c} |A_1 - A_2|.$$

For $A_2 = 0$ the integral supplies a factor $1/2$; for $\sup |A_i| \le r$ it is at most Lr. These are the quadratic and difference estimates with $C_2 = 3C_Q L^2$. Applying (513) to $K_A - K_0$ gives

$$|\mathsf{C}_A'^{\mathsf{T}} X|_y^\sharp \le 36C_Q L^4 \max_{N(y)} |A| \max_{N(y)} |X|.$$

Indeed, if $b \in \mathsf{P}_c$, the triangle inequality puts every input bond of P_c over $N(y_b)$. For the difference of two such transposes, use the whole ball $\mathcal{U}_c$. The holomorphic kernel $K_A - K_0$ is bounded there by $C_Q \xi^3 \xi_c^{-2} c_{\mathrm{av}}$. If $\sup |A_i| \le c_4/2$ and $\delta = A_1 - A_2$ does not vanish on P_c, the disc

$$A_s + z\delta, \qquad |z| < \frac{c_{\mathrm{av}}}{2\xi_c \sup_{\mathsf{P}_c} |\delta|}$$

lies in $\mathcal{U}_c$. Cauchy's estimate followed by integration in s gives $|K_{A_1} - K_{A_2}| \le 2C_Q \xi^3 \xi_c^{-2} \xi_c \sup_{\mathsf{P}_c} |\delta|$. The same transpose sum proves the required difference bound with $C_3 = 72C_Q L^4$. If δ vanishes locally, the kernel difference is zero.

The action remainder. We first give the local complex-reference estimate needed for rescaling. At spacing $0 < \eta \le 1$, let S_b be a subset of the plaquettes incident to a positive bond b, and let N_b be their positive bonds. Suppose either that all four incident plaquettes are present or that $\eta = 1$. Assume on these bonds and plaquettes that

$$|\operatorname{Ad}(\mathbf{U}(b'))^{\pm 1}| \le 2, \qquad |\partial\mathbf{U}(p) - I| \le \alpha\eta^2, \qquad 0 \le \alpha \le \tfrac{1}{2}.$$

Let $s = \sup_{\mathsf{N}_b} |A|$, let g be the largest norm of a covariant difference involving only bonds of N_b, and put $\mathsf{n} = \max(s, g)$. Write $r(A)$ for the gradient at b of $\eta^{-1} \sum_{p \in \mathsf{S}_b} a_p(e^{i\eta A}\mathbf{U})$, using the η^3 pairing, and expand $r(tA) = \sum_{q \ge 0} t^q r_q(A)$. We claim that, for an absolute $C_{\mathrm{a}} \ge 1$,

$$\left| \sum_{q \ge 2} r_q(A) \right| \le C_{\mathrm{a}} s\mathsf{n}, \qquad s \le \rho_{\mathrm{a}} := 1/256. \tag{514}$$

Use the transported variables α_j, the logarithmic remainder N_p, and $\Theta_p = DA(p) + N_p$ from section 13.2, now with $\mathbf{U}$ in place of U_0. The transport bounds give $|\alpha_1|, |\alpha_4| \le s$ and $|\alpha_2|, |\alpha_3| \le 2s$. The general complex-matrix calculation leading to (106) gives, for $s \le 1/128$,

$$|N_p| \le 144s^2, \qquad \eta|\Theta_p| \le 9s, \qquad |\Theta_p| \le 2g + 144s^2, \qquad |\mathsf{L}_{p,b}| \le 192s.$$

Here $\mathsf{L}_{p,b}$ is the derivative correction in (109); the last bound includes the possible factor two from transport of the varied bond. Put $c_p = \frac{1}{2} \operatorname{tr} \partial\mathbf{U}(p)$ and $F_p = (i\eta^2)^{-1}(\partial\mathbf{U}(p) - c_p I)$. Cayley–Hamilton

gives $c_p^2 + \eta^4\langle F_p, F_p\rangle = 1$. Also $2i\eta^2 F_p = \partial\mathbf{U}(p) - \partial\mathbf{U}(p)^{-1}$. The Neumann estimate bounds $|\partial\mathbf{U}(p)^{-1} - I|$ by $2\alpha\eta^2$. Together with $|c_p - 1| \le \alpha\eta^2$, these identities give

$$|F_p| \le \tfrac{3}{2}\alpha, \qquad |c_p| \le \tfrac{3}{2}, \qquad |1 + c_p| \ge \tfrac{3}{2}, \qquad |1 - c_p| \le \tfrac{3}{2}\alpha^2\eta^4.$$

Thus the algebraic trace and differential identities (108) and (112) still hold, with

$$|M_p^{\mathrm{r}}| \le \eta^4(2|\Theta_p|^3 + 3|F_p||\Theta_p|^2).$$

Subtracting $r(0) + D^{\mathsf{T}}DA(b)$ gives the same four terms as (113). Here D maps the bonds of N_b to S_b, and its transpose obeys $|D^{\mathsf{T}}X(b)| \le 6\eta^{-1}\sup|X|$. The first, third and fourth terms are bounded, respectively, by $54\alpha s$, $Cs^2(s+\alpha)$ and $Cs(\alpha + \mathsf{n})$, using the displayed bounds and $|\mathsf{L}_{p,b}^{\mathsf{T}}| \le 2|\mathsf{L}_{p,b}|$.

For $D^{\mathsf{T}}(cN)$, suppose first that all four plaquettes are present. Pair opposite incident plaquettes p, p' in the transpose and let v be the connecting inverse bond transport. The identities (101) give

$$|\operatorname{Ad}(v)\alpha_j(p') - \alpha_j(p)| \le 5\eta g \qquad (1 \le j \le 4).$$

For $j = 1, 4$ the bound is $2\eta g$; for $j = 2, 3$ insert those identities, giving the three contributions $2\eta g, 2\eta g, \eta g$. Both the original and transported quadruples have sum of norms at most $12\eta s < 1/8$. The series logarithm commutes with this conjugation on their common domain. Integrating (106) along the segment between the quadruples therefore gives

$$|\operatorname{Ad}(v)N_{p'} - N_p| \le 3840\eta s g.$$

Together with $|c_p - c_{p'}| \le 3\alpha^2\eta^4$, this bounds $D^{\mathsf{T}}(cN)$ by $Cs\mathsf{n}$. If the star is incomplete, $\eta = 1$, and the direct bound $6\sup|cN| \le Cs^2$ suffices. Consequently

$$|r(A) - r(0) - D^{\mathsf{T}}DA(b)| \le Cs(\alpha + \mathsf{n}), \qquad s \le 1/128.$$

For fixed A with $0 < s \le 1/256$, apply the coefficient estimate used after (114) on the circle $|t| = T = (128s)^{-1} \ge 2$. The sum of the coefficients of order at least two is at most $2Cs(\alpha/T + \mathsf{n}) \le 130Cs\mathsf{n}$. These are exactly the coefficients r_q, $q \ge 2$. For $s = 0$ all coefficients vanish by locality. This proves (514).

Return to a bare bond b over y, and set $n = n(y)$. The inner incident plaquettes use only bonds over sites z with $d(y, z) \le 5$ and $|n(z) - n| \le 1$. For the distance, join the two initial sites along at most two bare edges, at cost at most two, and join each to its cell center, at cost at most $3/2$. The adjacent-scale assertion follows from the layer separation, as in lemma 27.4: the corners lie within ξ of b_- and cannot cross two layers. If $n \ge 1$, all four plaquettes are inner, since $b_- \in \Omega_1$ and $\operatorname{dist}_\infty(\Omega_0^c, \Omega_1) > R_1\mathfrak{M}\xi > \xi$. At $n = 0$ the rescaled spacing below is one.

Rescale by ξ_n, putting $\eta = \xi/\xi_n$ and $\hat{A} = \xi_n A$. The regular-pair conditions give

$$|\operatorname{Ad}(\mathbf{U}(b'))^{\pm 1}| \le e^{2\alpha_1} \le 2, \qquad |\partial\mathbf{U}(p) - I| \le L^2\alpha_0\eta^2, \qquad L^2\alpha_0 \le \tfrac{1}{2}.$$

The local estimate thus applies with $\alpha = L^2\alpha_0$. The rescaled local star action is ξ_n times the original one, whereas a variation of $\hat{A}$ varies A by ξ_n^{-1} times as much. These factors cancel. Its derivative at b equals that of the full action, and the ratio of the pairing weights is $(\xi/\eta)^3 = \xi_n^3$. Consequently

$$\xi_n^3 F'(A)(b) = r(\hat{A}), \qquad \xi_n^3\mathcal{V}(A)(b) = \sum_{q \ge 2} r_q(\hat{A}).$$

On this star the local seminorms satisfy

$$s(\hat{A}) \le L\max_{N(y)}|A|, \qquad \mathsf{n}(\hat{A}) \le L^2\max_{N(y)}\|A\|.$$

Indeed, the rescaled differences are ξ_n^2 times the original differences, and every difference on the star is one admitted in the corresponding g_z. It follows from (514) that

$$|\mathcal{V}(A)|_y^\sharp \le C_{\mathrm{a}}L^3\max_{N(y)}|A|\max_{N(y)}\|A\| \quad \text{if } L\sup|A| \le \rho_{\mathrm{a}}.$$

Take $\rho_1 = \rho_{\mathrm{a}} L^{-2}$ and $C_{\mathcal{V}} = C_{\mathrm{a}} L^4$.

Finally, suppose $\|A_i\| \le r \le \rho_1/4$ and put $\delta = A_1 - A_2$. If $\mathsf{n}(\hat\delta) > 0$, on the disc $A_s + z\delta$ of radius $R = L^2 r/\mathsf{n}(\hat\delta)$ one has $\mathsf{n}(\hat A_s + z\hat\delta) \le 2L^2 r \le \rho_{\mathrm{a}}/2$. The local holomorphic remainder is bounded there by $C_{\mathrm{a}}(2L^2 r)^2$. Cauchy's estimate and integration in s give

$$\xi_y^3 |\mathcal{V}(A_1)(b) - \mathcal{V}(A_2)(b)| \le 4 C_{\mathrm{a}} L^2 r\, \mathsf{n}(\hat\delta) \le 4 C_{\mathcal{V}} r \max_{N(y)} \|\delta\|.$$

If $\mathsf{n}(\hat\delta) = 0$, locality makes the difference zero. The identities $\mathcal{V}(0) = \mathcal{V}'(0) = 0$ follow from its definition, and the bound on $\mathbf{J}$ is the regular-pair condition at the initial site of each bond. □

34. The projected propagator

The nonlinear response uses the part of the background inverse left after subtracting its scalar and coarse constraints. After the coarse subtraction the operator is already a block of the constrained inverse. To estimate the scalar subtraction, we place two copies of that system in a triangular system. Its coupling contains a derivative in the first equation. A local Hölder estimate permits this derivative to be treated as a divergence before the local inverses are combined.

Keep the setting and operators of section 31. In this section write Q_{phys} for its physical linear average; the dimensionless average in section 33 is $Q = \mathsf{m} Q_{\mathrm{phys}}$. Define

$$\mathsf{E}_1 = \mathsf{G}_1 D \mathsf{R} D^{\mathsf{T}} \mathsf{G}_1, \qquad \mathfrak{G} = \mathsf{G}_1 - \mathsf{E}_1 - \mathsf{H}_1 Q_{\mathrm{phys}} \mathsf{G}_1. \tag{515}$$

The projection R is the scalar operator in (479).

Proposition 34.1. There are a positive constant a_{pr}, a power $\mathfrak{M}_{\mathrm{pr}}$ of L, and a constant B_{pr}, depending only on L, with the following property. In the geometry of proposition 31.1, suppose $\mathfrak{M} \ge \mathfrak{M}_{\mathrm{pr}}$ and

$$(\mathbf{U}, \mathbf{J}) \in \mathfrak{P}_t^{\natural}((\Omega_n); \alpha_0, \alpha_1, \gamma_0), \qquad 0 < \alpha_0, \alpha_1, \gamma_0 \le a_{\mathrm{pr}}.$$

Then the operators in (515) exist and, for $T = \mathsf{E}_1$ or $\mathfrak{G}$,

$$\max\{|Tf|_{-1;y}, |\nabla T f|_{-2;y}\} \le B_{\mathrm{pr}} \sup_z e^{-\delta_{\mathrm{reg}} d(y,z)} |f|_{-3;z}. \tag{516}$$

Here f is any bare bond field and $y \in \Lambda_*$. The derivative is taken at $\mathbf{U}$. The constants are uniform in the spacing, volume and number of layers. ■

Proof. The two local estimates. We first allow an independent multiplier j and use only the system [4], suppressing this superscript. Set $\rho = 3L + 6$. For a site or bond field, including a bond field with an additional direction, define the transported half-Hölder norm by

$$\begin{aligned} [f]_{\mathbf{U},1/2;y} &= \sup_{i \ne i'} \left(\frac{\xi_y}{|i - i'|_1} \right)^{1/2} |\operatorname{Ad}(\mathbf{U}(\Gamma_{x,x'})) f(i') - f(i)|, \\ |f|^{1/2}_{\mathbf{U},s;y} &= \xi_y^{-s} \left(\sup_{\Delta^{3/2}(y)} |f| + [f]_{\mathbf{U},1/2;y} \right). \end{aligned} \tag{517}$$

The pairs have the same kind and both base points in $\Delta^{3/2}(y)$, as in (330). The points x, x' are their sites or initial endpoints, and $\Gamma_{x,x'}$ is the coordinate comb (62). Above level zero its containing box lies in the preceding region, so all its bonds are inner. At level zero a distinct bond pair consists of the incoming and outgoing bonds in one direction, and the connecting comb is one inner bond. Empty suprema are zero. The subscript 0 denotes the identity configuration.

Let $X_1(b, \nu)$ be a $\mathfrak{g}^c$-valued bond tensor and put

$$\begin{aligned} g^+ &= (g, X_1), \qquad \iota^+ g^+ = \iota g + e_1 \nabla^{\mathsf{T}} X_1, \\ |g^+|^{1/2}_y &= \max\{|g|_y, |X_1|^{1/2}_{\mathbf{U},-2;y}\}, \\ |\Psi|^{+1/2}_y &= \max\{|\Psi|_y, |\nabla A|^{1/2}_{\mathbf{U},-2;y}\}. \end{aligned} \tag{518}$$

The transpose uses the bare bilinear pairing. The additional divergence enters the first equation; ι retains its fifth-slot divergence. We will prove, on every local piece,

$$\begin{aligned} |\mathbb{E}_\kappa \iota_\kappa^+ g^+|_y &\le B_{\mathrm{e}} \sup_z e^{-2\delta_{\mathrm{reg}} d^\kappa(y,z)} |g^+|_z^{1/2}, \\ |\mathbb{E}_\kappa \iota_\kappa g|_y^{+1/2} &\le B_{\mathrm{e}} \sup_z e^{-2\delta_{\mathrm{reg}} d^\kappa(y,z)} |g|_z, \end{aligned} \tag{519}$$

with $B_{\mathrm{e}} \ge 1$ depending only on L, after decreasing the admissible smallness radius.

At the identity these estimates follow from the elimination in sections 23.2 and 23.4. With $p = p' + p''$ as in (351), the effective vector source for g^+ is

$$f_1' + \nabla^{\mathsf{T}}(X^{(p)} + X_1), \qquad f_1' = f_1 + D\varphi + Dp' + Z^{(p)}.$$

The boundary correction $Z^{(p)}$ is exactly the one in (353); it has weight -3. For the first estimate use the half-Hölder bounds for p'' and $X^{(p)}$, and then (342) and (348). These bound A and ∇A in weights $-1, -2$. The remaining components follow from the same scalar substitutions and bounds used in section 23.4. For the second estimate $X_1 = 0$. Now use exponent $3/4$ in (350) and (354), and apply (349) to f_1' and $X^{(p)}$. This bounds the additional half-Hölder norm of ∇A. Both estimates have rate δ_{cs}.

To transfer them to a regular piece, work in its small gauge and suppress κ. Require $\epsilon \le \epsilon_{\mathrm{e}} := \min\{\epsilon_*, 2^{-11}\}$. We record the local geometry needed for the transported norms. Let $\mathsf{M}(y)$ consist of the tops of cells meeting $\widetilde{\Delta}(y)$, together, at level zero, with the positions of $y - 2\xi e_\mu$ when both preceding sites belong to the region. Then

$$z \in \mathsf{M}(y) \quad \Longrightarrow \quad d(y,z) \le \tfrac{3}{2}(L+2) = \rho/2, \qquad L^{-1}\xi_y \le \xi_z \le L\xi_y. \tag{520}$$

For a meeting cell, join its intersection point to y and to its own top. The first segment costs at most $3/2$ inside $\Delta(y)$ and $3L/2$ outside it: the latter part has ℓ^1-length at most $3\xi_y/2$ and lies in the preceding region. The segment in the meeting cell costs at most $3/2$. This uses (405); at level zero the outer cost only improves. The exceptional point is reached by two neighboring cells, at cost at most six. The scale bound is (407). Adding neighbors to $\mathsf{M}(y)$ gives a set $\mathsf{M}^+(y)$ of radius at most $3L/2 + 6$ with the same scale comparison.

The comb in (517) lies in $\widetilde{\Delta}(y) \cap Y \subset \mathsf{N}(y)$. The sum of the norms of its adjoint exponents is at most $2\epsilon|x - x'|_1/\xi_y \le 9\epsilon$. Thus its transport and inverse differ from I by at most $4\epsilon|x - x'|_1/\xi_y$. Comparing the two differences gives

$$(1 + 18\epsilon)^{-1}|f|_{0,s;y}^{1/2} \le |f|_{\mathbf{U},s;y}^{1/2} \le (1 + 18\epsilon)|f|_{0,s;y}^{1/2}. \tag{521}$$

The required shift and product bounds follow directly from this norm. For an ordinary index shift by at most ξ in each coordinate, split pairs at distance $\xi_y/(8L)$. The more distant pairs, and all pairs at levels zero and one, are bounded by $C(L)$ times the supremum. For a closer pair at a higher level, both shifted indices exist and their base points lie in the enlarged cube of the first shifted index's position: their separation is at most $\xi_z/8$ and the first base point lies in $\Delta(z)$. Its Hölder norm applies. The positions lie in $\mathsf{M}^+(y)$, and the scale factors cost only powers of L. Expanding the difference of a product also shows that a coefficient bounded by a_0 with differences bounded by $a_1|x - x'|_1/\xi_y$ has multiplier norm at most $a_0 + 9a_1$. Missing shifted indices occur only in the low-level case already bounded by the supremum.

Put $\mathsf{U}_\nu = \mathrm{Ad}(\mathbf{U}_\nu)$ and $\mathsf{e}_\nu = (\mathsf{U}_\nu - I)/\xi$. The bounds (388), summed along the comb, give

$$|\mathsf{e}_\nu| \le 4\epsilon\xi_y^{-1}, \qquad |\mathsf{e}_\nu(x) - \mathsf{e}_\nu(x')| \le 4\epsilon\xi_y^{-2}|x - x'|_1.$$

When the shifted bond exists, $(\nabla_{\mathbf{U},\nu} - \nabla_{0,\nu})A(b) = \mathsf{e}_\nu(b_-)A(b + \xi e_\nu)$; otherwise the difference is zero. The preceding product and shift bounds, followed by the ordinary comb estimate, bound its half-Hölder norm by $C(L)\epsilon \max_{d(y,z)\le 2\rho} |\Psi|_{0,z}$. Together with (386) and (521), this compares the two enhanced solution norms by a nearby maximum with constant $C(L)$, in either direction.

The injection difference contains no derivative of its input. With $\mathsf{e}'_\nu = (\mathsf{U}_\nu^{-1} - I)/\xi$,

$$((\nabla^{\mathsf{T}}_{\mathbf{U}} - \nabla^{\mathsf{T}}_0)X_1)(b) = \sum_\nu \mathsf{e}'_\nu(b_- - \xi e_\nu)X_1(b - \xi e_\nu, \nu),$$

where missing predecessors contribute zero. Combining this with (387) gives

$$\iota^+_{\mathbf{U}} g^+ = \iota^+_0(g^+ + (\mathsf{n}^+ g^+, 0)), \qquad |\mathsf{n}^+ g^+|_y \le C(L)\epsilon \max_{d(y,z)\le 3} |g^+|^{1/2}_{0,z}.$$

The two nonzero entries have the required weights $-3, -4$, since the coefficients have weight -1 and the tensors X_1, X_5 have weights $-2, -3$.

Use the existing factorization $\mathbb{K}_0 - \mathbb{K} = \iota_0 \mathbb{V}$ and put $\Psi_0 = \mathbb{E}_0 \iota_0^+ g^+$. The identities

$$\mathbb{E}\iota_0^+ g^+ = \Psi_0 + \mathbb{E}\iota_0 \mathbb{V}\Psi_0, \qquad \Psi = \mathbb{E}\iota_0 g = \mathbb{E}_0 \iota_0(g + \mathbb{V}\Psi) \tag{522}$$

transfer the two estimates. In the first identity bound Ψ_0 strongly by the identity-configuration estimate, then apply (385) and the already proved strong bound for $\mathbb{E}\iota_0$. In the second, bound Ψ strongly first and apply the identity-configuration enhanced-output estimate to $g + \mathbb{V}\Psi$. Composition and nearby maxima preserve δ_{cs} by (340). The injection and norm comparisons just proved return both estimates to $\mathbf{U}$. Unitary gauge covariance preserves all transported norms and intertwines the injections.

On a boundary piece $\xi_y = \xi$. The minimum nonzero distance and the bound $|\mathsf{U}^{\pm 1}| \le 2$ imply

$$|f|^{1/2}_{\mathbf{U},s;y} \le C(L) \max_{d(y,z)\le 3} |f|_{s;z}, \qquad |\nabla^{\mathsf{T}} X_1|_{-3;y} \le 9 \max_{d(y,z)\le 3} |X_1|_{-2;z}.$$

Hence $\iota^+ g^+ = \iota \widetilde{g}$ with $|\widetilde{g}|_y \le 10 \max_{d(y,z)\le 3} |g^+|^{1/2}_z$, and $|\Psi|^{+1/2}_y \le C(L) \max_{d(y,z)\le 6} |\Psi|_z$. The massive boundary inverse of lemma 27.6 proves both estimates. Taking the larger constant and the common rate $2\delta_{\mathrm{reg}}$ proves (519).

The coupling of the two systems. For $\Psi' = (A'; v', w', \sigma'; p', q', r')$ set

$$f = D^{\mathsf{T}} A' - w', \qquad \mathbb{J}\Psi' = e_1 Df, \qquad \mathbb{K}^{\mathrm{d}}(\Psi, \Psi') = (\mathbb{K}\Psi - \mathbb{J}\Psi', \mathbb{K}\Psi').$$

The prime here distinguishes the second copy of the system. For $b = \langle x, x + \xi e_\mu \rangle$ put

$$X^{(f)}_\nu(b) = -\delta_{\mu\nu} \mathsf{U}_\mu(x) f(b_+), \qquad c^{(f)}(b) = \begin{cases} -\xi^{-1} f(x), & b - \xi e_\mu \text{ is not inner,} \\ 0, & b - \xi e_\mu \text{ is inner.} \end{cases} \tag{523}$$

Expansion of the transpose gives $Df = \nabla^{\mathsf{T}} X^{(f)} + c^{(f)}$. The correction is supported at level zero, where $\xi_y = \xi$, and therefore $|c^{(f)}|_{-3;y} \le |f|_{-2;y}$. Write $\mathsf{j}^+\Psi'$ for the enhanced forcing with just these two entries. Then

$$\mathbb{J} = \iota^+ \mathsf{j}^+, \qquad |\mathsf{j}^+\Psi'|^{1/2}_y \le C_J \max_{d(y,z)\le \rho} |\Psi'|^{+1/2}_z. \tag{524}$$

When both adjacent bonds exist, a summand of $D^{\mathsf{T}} A'$ is $-\mathsf{U}_\nu(x - \xi e_\nu)^{-1} \nabla_\nu A'_\nu(x - \xi e_\nu)$. Otherwise its site is at level zero and direct endpoint values give the same weight. Thus

$$|f|_{-2;y} \le (12L^2 + 1) \max_{d(y,z)\le 3} |\Psi'|_z, \qquad |X^{(f)}|_{-2;y} \le 2L^2 \max_{d(y,z)\le 3} |f|_{-2;z}.$$

At levels zero and one, the Hölder quotient costs at most $3L$ times the enlarged supremum, so these bounds suffice. At higher levels all the shifted indices are inner, and

$$X^{(f)}_\mu(b) = \sum_\nu \mathsf{U}_\mu(x) \mathsf{U}_\nu(x + \xi e_\mu - \xi e_\nu)^{-1} (\nabla_\nu A'_\nu)(x + \xi e_\mu - \xi e_\nu) + \mathsf{U}_\mu(x) w'(b_+).$$

The coefficients in the sum are bounded by four, with ordinary differences bounded by $16\epsilon_{\mathrm{e}} |x - x'|_1 / \xi_y$. The shift and product bounds above, followed by (521), therefore use exactly the half-Hölder norm of $\nabla A'$. For $\bar{w}(b) = w'(b_+)$, direct subtraction gives

$$(\nabla_\nu \bar{w})(b) = (Dw')(\langle b_+, b_+ + \xi e_\nu \rangle) + \xi^{-1}(\mathsf{U}_\nu(x) - \mathsf{U}_\nu(b_+)) w'(b_+ + \xi e_\nu).$$

The coefficient difference is at most $4\epsilon_{\mathrm{e}}\xi/\xi_z$ at a position z. Consequently the comb bound controls $\bar{w}$ by the existing entries w', Dw'. Let $T = \mathrm{Ad}(\mathbf{U}(\Gamma_{x,x'}))$. For the transported product $\mathsf{U}_\mu\bar{w}$, conjugating the coefficient by this transport changes it by at most $28\epsilon_{\mathrm{e}}|x-x'|_1/\xi_y$; this follows by inserting the two factors $T-I, T^{-1}-I$ and then the ordinary coefficient difference. Since $28\epsilon_{\mathrm{e}} \le 1$, the product estimate applies. The positions in the shifted sum lie in $\mathsf{M}^+(y)$, and those in the w' estimate lie within six of $\mathsf{M}(y)$. Both radii are at most ρ. This proves (524). Boundary pieces use only its low-level case.

The triangular inverse is

$$(\mathbb{K}^{\mathrm{d}})^{-1}(F,F') = (\mathbb{E}F + \mathbb{E}\mathbb{J}\mathbb{E}F', \mathbb{E}F').$$

Use the maximum of the two ordinary forcing norms on $\mathcal{F}^{[4]} \times \mathcal{F}^{[4]}$, and the maximum of the two strong solution norms. Apply the second estimate of (519) to the second copy, then (524), and then the first estimate to the first copy. The local doubled inverse has rate $2\delta_{\mathrm{reg}}$ and constant

$$B_{\mathrm{d,loc}} = B_{\mathrm{e}}(1 + C_J e^{\rho} B_{\mathrm{e}}). \tag{525}$$

The additional cutoff commutator. Take h, ϑ as in lemma 27.8 and put

$$k_b^- = (h(\hat{b}) - h(b_-))/\xi, \qquad k_b^+ = (h(b_+) - h(\hat{b}))/\xi, \qquad \phi_h = [D^{\mathsf{T}}, h]A'.$$

The midpoint convention gives

$$\begin{aligned} [\mathbb{J}, h]\Psi' &= e_1([D,h]f + D\phi_h), \\ ([D,h]f)(b) &= k_b^+ \mathsf{U}(b) f(b_+) + k_b^- f(b_-), \\ \phi_h(x) &= -\sum_{b_+ = x} k_b^+ \mathsf{U}(b)^{-1} A'(b) - \sum_{b_- = x} k_b^- A'(b). \end{aligned} \tag{526}$$

The first term has weight -3 bounded by $C(L)\vartheta^{-1}\max_{d(y,z)\le 6}|\Psi'|_z$. At level zero the direct estimate for the last term is

$$|D\phi_h|_{-3;y} \le \tfrac{27}{2}\vartheta^{-1} \max_{d(y,z)\le 3} |A'|_{-1;z}.$$

At a positive level the cancellation between incoming transports is needed. Write $x^- = x - \xi e_\nu$ for a fixed direction ν. For $b = \langle x, x+\xi e_\mu\rangle$, expansion gives $\xi D\phi_h(b) = -\sum_\nu (T_\nu^- + T_\nu^+)$, where

$$T_\nu^- = \xi k_\nu^-(x+\xi e_\mu)\nabla_\mu A'_\nu(x) + (k_\nu^-(x+\xi e_\mu) - k_\nu^-(x))A'_\nu(x)$$

and

$$\begin{aligned} T_\nu^+ &= \mathsf{U}_\nu(x^-)^{-1}\{\xi k_\nu^+(x^- + \xi e_\mu)\nabla_\mu A'_\nu(x^-) + (k_\nu^+(x^-+\xi e_\mu) - k_\nu^+(x^-))A'_\nu(x^-)\} \\ &\quad + k_\nu^+(x^- + \xi e_\mu)\mathsf{W}_\nu A'_\nu(x^- + \xi e_\mu), \\ \mathsf{W}_\nu &= \mathsf{U}_\mu(x)\mathsf{U}_\nu(x^- + \xi e_\mu)^{-1} - \mathsf{U}_\nu(x^-)^{-1}\mathsf{U}_\mu(x^-). \end{aligned}$$

For $\nu = \mu$ the last coefficient vanishes. Otherwise it is the difference of the transports along the two sides of a plaquette. If its variable is P, the adjoint bounds give $|\mathsf{W}_\nu| \le 48|P-I|$. The small-potential bounds imply

$$|P - I| \le 8\epsilon_{\mathrm{e}}\xi^2\xi_y^{-2}, \qquad |\mathsf{W}_\nu| \le \xi^2\xi_y^{-2}.$$

Indeed the sum of the four exponents is bounded by $2\epsilon_{\mathrm{e}}\xi^2\xi_y^{-2}$ and their quadratic remainder by $16\epsilon_{\mathrm{e}}^2\xi^2\xi_y^{-2}$; $384\epsilon_{\mathrm{e}} \le 1$ gives the second inequality. All these plaquettes lie in the collar where the small-potential bounds hold. The cutoff assumptions give $|k^\pm| \le \vartheta^{-1}\xi_y^{-1}/2$ and adjacent differences at most $\vartheta^{-1}\xi\xi_y^{-2}/2$. Substitution in the expansion therefore yields

$$|D\phi_h(b)| \le \vartheta^{-1}\left(\tfrac{9}{2}\xi_y^{-1}\max|\nabla A'| + 6\xi_y^{-2}\max|A'|\right).$$

The nearby scale comparison proves the required weight -3 bound without a negative power of the bare spacing in its constant.

Let $c_h\Psi'$ be the ordinary forcing whose only entry is $[D,h]f + D\phi_h$. Increase C_J to include this estimate. With the original commutator map $\mathbb{C}_h$ set

$$\mathbb{C}_h^{\mathrm{d}}(\Psi,\Psi') = (\mathbb{C}_h\Psi - c_h\Psi', \mathbb{C}_h\Psi').$$

Writing $\iota^{\mathrm{d}} = \iota \oplus \iota$, we have

$$[\mathbb{K}^{\mathrm{d}}, h] = \iota^{\mathrm{d}}\mathbb{C}_h^{\mathrm{d}}, \qquad |\mathbb{C}_h^{\mathrm{d}}\vec{\Psi}|_y \le (C_{\mathbb{C}} + C_J)\vartheta^{-1} \max_{d(y,z)\le\rho} |\vec{\Psi}|_z. \tag{527}$$

This seminorm vanishes when h is constant on $\mathsf{N}(y)$. The construction is gauge covariant, so it applies in the original local gauges. Its output is ordinary doubled forcing.

The total inverse and the physical multiplier. Use the same pieces and cutoffs as in section 27.5, and write $\delta = \delta_{\mathrm{reg}}$. Choose

$$\begin{aligned} C_q^{\mathrm{d}} &= 2(C_{\mathbb{C}} + C_J)C_{\mathrm{G}}B_{\mathrm{d,loc}}e^{2\rho+8}, \\ B_T^{\mathrm{d}} &= 4B_{\mathrm{d,loc}}e^{3\delta}, \qquad q_{\mathfrak{M}}^{\mathrm{d}} = C_q^{\mathrm{d}}\mathfrak{M}^{-1}e^{-\delta D_0}, \qquad B_{\mathrm{d}} = 2N_{\mathrm{G}}B_T^{\mathrm{d}}. \end{aligned} \tag{528}$$

Enlarge the power threshold to include all earlier thresholds and $2N_{\mathrm{G}}C_q^{\mathrm{d}}$. The ordinary cutoff correction $\theta_\kappa \star g$ acts on each forcing copy. The new coupling uses only w' at the two endpoints and A' on incident bonds. These indices lie within two bare spacings of the support, hence in the existing common-index margins between $\kappa^{[2/3]}$ and $\kappa^{[5/6]}$. Thus the local/global operator agreement and the lifted identities (432) hold on the product space.

The proof of lemma 27.9 now applies with B_{loc} replaced by $B_{\mathrm{d,loc}}$ and $C_{\mathbb{C}}$ by $C_{\mathbb{C}} + C_J$. The commutator support and its separation from the input are unchanged, so its remainder constant is $q_{\mathfrak{M}}^{\mathrm{d}}$. The overlap bound gives $N_{\mathrm{G}}q_{\mathfrak{M}}^{\mathrm{d}} \le 1/2$. The same Neumann argument therefore proves

$$|(\mathbb{K}^{\mathrm{d}})^{-1}\iota^{\mathrm{d}}(g,g')|_y \le B_{\mathrm{d}} \sup_z e^{-\delta d(y,z)} \max\{|g|_z, |g'|_z\}. \tag{529}$$

To apply this at the actual multiplier, take

$$a_{\mathrm{d}} = \min\{a_{\mathrm{reg}}, \epsilon_{\mathrm{e}}/(16L^2)\}, \qquad a_{\mathrm{pr}} = \min\{a_{\mathrm{bg}}, a_{\mathrm{d}}/\max(1, C_{\mathrm{bg}})\}.$$

Include $\mathfrak{M}_{\mathrm{bg}}$ in $\mathfrak{M}_{\mathrm{pr}}$. By (484), $|\mathsf{j}_*|_{-3;y} \le C_{\mathrm{bg}}\gamma_0 \le a_{\mathrm{d}}$. The existing local gauge estimate consequently gives

$$\max\{8L^2(\alpha_0+\alpha_1), \gamma_0, \sup_y |\mathsf{j}_*|_{-3;y}\} \le \epsilon_{\mathrm{e}}.$$

The truncated multiplier has the same bound on every regular piece. On boundary pieces its quadratic coefficient vanishes. Thus all local estimates used in (529) apply at this physical triple.

Finally apply the doubled inverse to $(0, e_1 f)$. The scalar elimination in lemma 31.2 gives $A' = \mathsf{G}_1 f$ and $w' = \mathsf{P}D^{\mathsf{T}}A'$. The first-copy A component is therefore $\mathsf{G}_1 D\mathsf{R}D^{\mathsf{T}}\mathsf{G}_1 f = \mathsf{E}_1 f$. This proves its bound with constant B_{d}. The same elimination for system [5], with eighth input zero, gives

$$[\mathbb{E}^{[5]}(\mathsf{j}_*)]_{A,1} = \mathsf{G}_1 - \mathsf{H}_1 Q_{\mathrm{phys}}\mathsf{G}_1.$$

Its strong bound is (400). Subtracting E_1 proves (516) with $B_{\mathrm{pr}} = B_{\mathrm{reg}} + B_{\mathrm{d}}$. □

34.1. **Application to the nonlinear response.** We can now apply theorem J to a regular pair. The row variables in that theorem are dimensionless, so the physical inverse blocks must first be rescaled. Keep $\mathsf{G}, \mathsf{H}, \mathsf{H}_1$ from section 31, and write * for the transpose with physical row weight ξ_c^3. With $\mathsf{m}X(c) = \xi_c X(c)$, put

$$\begin{aligned} H &= \mathsf{H}\mathsf{m}^{-1}, \qquad K = \mathsf{H}_1\mathsf{m}^{-1}, \\ S &= \mathsf{m}^3(Q_{\mathrm{phys}}\mathsf{G}Q_{\mathrm{phys}}^*)^{-1}\mathsf{m}^{-1}. \end{aligned} \tag{530}$$

The fourth operator is $\mathfrak{G}$ of (515). The superscript ^{T} denotes transpose for the dimensionless pairing (490).

Proposition 34.2. In the geometry of proposition 31.1, suppose $\mathfrak{M} \geq \mathfrak{M}_{\mathrm{pr}}$ and

$$(\mathbf{U}, \mathbf{J}) \in \mathfrak{P}_t^{\natural}((\Omega_n); \alpha_0, \alpha_1, \gamma_0), \qquad 0 < \alpha_0, \alpha_1, \gamma_0 \leq \min\{a_{\mathrm{nl}}, a_{\mathrm{pr}}\}.$$

Use the nonlinear maps and seminorms of proposition 33.1 and set

$$\theta_0 = \delta_{\mathrm{reg}}/2, \qquad \hat{\mathsf{b}} = \max\{1, B_{\mathrm{pr}}, C_{\mathrm{bg}}\}.$$

Define $c_W, C_W, \varepsilon_3, \varepsilon_B$ by (494) with these constants. For $\sup|B'| \leq \varepsilon_B$, the system (499), with

$$(\mathsf{H}, \mathsf{S}, \mathsf{K}, \mathsf{G}, \mathsf{a}) = (H, S, K, \mathfrak{G}, I),$$

has a unique solution $(X, \mathbf{A}_0, \mathbf{H})$ in the balls

$$\sup|X| \leq c_W \hat{\mathsf{b}}^{-2}, \qquad \|\mathbf{A}_0\| \leq \hat{\mathsf{b}}\varepsilon_B.$$

Its components are holomorphic for $\sup|B'| < \varepsilon_B$. The bounds (500), (501) and (502) hold with these constants. Moreover,

$$\mathcal{Q}(\mathbf{H}(B')) = B' + Q\mathbf{A}_0(B'). \tag{531}$$

All constants depend only on L. ∎

Proof. We verify the linear bounds (493); the local nonlinear hypotheses are proposition 33.1. The two row pairings satisfy $(X, Y)_{\mathbf{B}} = (X, \mathsf{m}^4 Y)_1$. Since $Q_{\mathrm{phys}} = \mathsf{m}^{-1}Q$, this gives $Q^*_{\mathrm{phys}} = Q^{\mathsf{T}}\mathsf{m}^3$. Hence

$$S = (Q\mathsf{G}Q^{\mathsf{T}})^{-1}, \qquad H = \mathsf{G}Q^{\mathsf{T}}S, \qquad K = \mathsf{G}_1 Q^{\mathsf{T}}(Q\mathsf{G}_1 Q^{\mathsf{T}})^{-1}.$$

In particular, $QH = QK = I$, and the physical quadratic term $Q^*_{\mathrm{phys}}\mathsf{m}^{-2}Q_{\mathrm{phys}}$ is $Q^{\mathsf{T}}Q$, so the dimensionless multiplier is $\mathsf{a} = I$.

Apply the inverse $\mathbb{E}^{[5]}(0)$ to a forcing with $f_8 = \mathsf{m}^{-1}X$ and all other components zero. Its norm in (328) is exactly $|X|_y$. The elimination in lemma 31.2 gives $A = HX$ and $\mathsf{m}^3\omega = SX$. The corresponding solution norms in (327) dominate $\|HX\|_y$ and $|SX|_y$: the actual seminorm uses only the covariant differences already present there. At the multiplier j_*, the same forcing gives $A = KX$. Thus (400) proves

$$\|HX\|_y,\ \|KX\|_y,\ |SX|_y \leq B_{\mathrm{reg}} \sup_z e^{-\delta_{\mathrm{reg}} d(y,z)} |X|_z.$$

Proposition 34.1 gives the corresponding bound for $\mathfrak{G}$ with constant B_{pr}. Multiplication by an admissible weight proves their weighted bounds, since $\theta_0 \leq \delta_{\mathrm{reg}}$ and $B_{\mathrm{reg}} \leq B_{\mathrm{pr}} \leq \hat{\mathsf{b}}$. Also $|(S - I)X|_w \leq 2\hat{\mathsf{b}}|X|_w$.

For the transpose, repeat the weighted-pairing calculation (488)–(489) with $H = \mathsf{H}\mathsf{m}^{-1}$, row weight ξ_c^{-1} and the current bound γ_0 replaced at each top by $|f|^{\sharp}_y$. The preceding column estimate then gives

$$|H^{\mathsf{T}} f|_{y'} \leq 6B_{\mathrm{reg}} \sum_y \frac{\xi_{y'}}{\xi_y} e^{-\delta_{\mathrm{reg}} d(y,y')} |f|^{\sharp}_y. \tag{532}$$

By (231) and (487), the scale ratio costs at most $Le^{\delta_{\mathrm{reg}} d(y,y')/4}$. The weight costs at most $e^{\theta_0 d(y,y')}$, with $\theta_0 = \delta_{\mathrm{reg}}/2$, leaving decay at rate $\delta_{\mathrm{reg}}/4$. The sum bound (233) therefore yields

$$|H^{\mathsf{T}} f|_w \leq 6LB_{\mathrm{reg}} c_S(\delta_{\mathrm{reg}}/4)|f|^{\sharp}_w \leq C_{\mathrm{bg}}|f|^{\sharp}_w \leq 8\hat{\mathsf{b}}|f|^{\sharp}_w.$$

Theorem J now gives the asserted solution and bounds; proposition 32.1 also gives holomorphy of its first component $X = \mathsf{D}(KB' + \mathbf{A}_0)$. The argument of D has supremum at most $2\hat{\mathsf{b}}\varepsilon_B \leq \varepsilon_3/2$, so it lies in that map's holomorphic domain. Finally $X = \mathsf{C}(\mathbf{H})$ and $\mathbf{H} = KB' + \mathbf{A}_0 - HX$ imply $Q\mathbf{H} = B' + Q\mathbf{A}_0 - X$. Adding $\mathsf{C}(\mathbf{H}) = X$ proves (531). □

35. Constrained critical response

We now identify the response as a constrained critical configuration. The fixed-point equation first gives stationarity on the gauge slice at fixed straightened average. Gauge invariance removes the remaining multiplier in the gauge direction. A second use of gauge invariance converts this equation for the straightened average into criticality for the original block average.

Use the geometry, regular-pair hypotheses and constants of proposition 34.2. The constraint in this section uses the reduced rows $\mathbf{B}^\flat$. Let $M_{\mathbf{B}^\flat}$ be the family of actual iterated averages on these rows, and let $C(\mathbf{B}^\flat, V)$ be its real fiber. The pair is (U_0, J), where U_0 is G-valued and

$$J = F'(0), \qquad J = Q^{\mathsf{T}}\lambda_0 \quad \text{for some } \lambda_0 \in \mathcal{X}. \tag{533}$$

Here F is the inner-plaquette action of section 33; thus its gradient includes the incomplete plaquette stars at the boundary. Membership of (U_0, J) in the stated regular-pair class, including its current bound, remains an assumption. Keep $H, K, S, \mathfrak{G}$ from section 34.1, and write $\Pi = I - D\mathsf{V}$ for the field projection S in (479).

With the constants of lemma 24.2, set

$$r_{\mathrm{b}} = \min\{r_{\mathrm{st}}/2, [16(1+\kappa_{\mathrm{st}})]^{-1}\}, \qquad \varepsilon_{\mathrm{crit}} = \min\{\varepsilon_B, r_{\mathrm{b}}/2\}. \tag{534}$$

These radii depend only on L.

Theorem K. Under these hypotheses the response $T(B') = \mathbf{H}(B')$ satisfies, for $\sup|B'| < \varepsilon_{\mathrm{crit}}$,

$$\mathcal{Q}(T) = B', \qquad \mathsf{R}D^{\mathsf{T}}T = 0, \qquad F'(T) = (Q + \mathsf{C}'_T)^{\mathsf{T}}\lambda(B'). \tag{535}$$

The multiplier is uniquely determined and holomorphic, with

$$\lambda(B') = (I + H^{\mathsf{T}}\mathsf{C}'^{\mathsf{T}}_T)^{-1}H^{\mathsf{T}}F'(T). \tag{536}$$

For real B', both T and λ are real, and $E = e^{i\xi T}U_0$ is a critical configuration of $C(\mathbf{B}^\flat, M_{\mathbf{B}^\flat}(E))$. ■□

The last assertion concerns the actual block averages. Their relation to B' involves the endpoint factors in (548) below; these factors are needed at outward rows.

35.1. **Gauge identities for the linear operators.** We begin with the scalar identities underlying the gauge slice. Let $\mathcal{N} = \ker Q'$ in the space of bare site fields. The formula for P in (479) gives $\mathsf{P}^2 = \mathsf{P}$, $\mathsf{R}^2 = \mathsf{R}$ and $Q'\mathsf{G}'\mathsf{R} = 0$. These operators are symmetric. For $\chi \in \mathcal{N}$ one has $\Delta'_a\chi = \Delta^\circ\chi$, and consequently

$$\mathsf{G}'\Delta^\circ\chi = \chi, \qquad \mathsf{R}\Delta^\circ\chi = \Delta^\circ\chi.$$

Conversely, $\chi_f = \mathsf{G}'\mathsf{R}f$ lies in $\mathcal{N}$ and $\Delta^\circ\chi_f = \mathsf{R}f$. Thus

$$\operatorname{ran}\mathsf{R} = \Delta^\circ\mathcal{N}, \qquad \Delta^\circ\mathsf{G}'\mathsf{R} = \mathsf{R}. \tag{537}$$

The boundary term in the scalar Laplacian is

$$(\Delta^\circ - D^{\mathsf{T}}D)\chi(x) = \xi^{-2}(6 - n(x))\chi(x),$$

where $n(x)$ counts the inner neighbors of x. A site with a missing neighbor lies in Λ_0, where Q' is the identity. Hence this term vanishes for $\chi \in \mathcal{N}$. It follows that $\mathsf{V}D\chi = \chi$ and $\Pi D\chi = 0$ on $\mathcal{N}$. Since V takes values in $\mathcal{N}$, $\mathsf{V}D\mathsf{V} = \mathsf{V}$, so $\Pi^2 = \Pi$. Also $\Pi A = A$ whenever $\mathsf{R}D^{\mathsf{T}}A = 0$.

The other identity we need is

$$QD\chi = 0 \qquad (\chi \in \mathcal{N}). \tag{538}$$

To prove it, consider one averaging step with reference W, in units in which the fine spacing is one. For $h_s = e^{is\rho}$ write $W^{h_s} = e^{ia_s}W$. The logarithm gives $a_s = -s\mathsf{d}_W\rho + O(s^2)$. The first formula in (355) and contour covariance give

$$\zeta(W, a_s)(y, x) = s\{\rho(y) - \mathrm{Ad}(W(y,x))\rho(x)\} + O(s^2).$$

Therefore the induced gauge after one step satisfies

$$\mathsf{v}(W,a_s)(y)^{-1}h_s(y)=\exp(is\mathsf{q}_W\rho(y)+O(s^2)). \tag{539}$$

Substitution into (356), followed by differentiation at $s=0$, proves

$$\mathbf{Q}(W)\mathsf{d}_W\rho=\mathsf{d}_{\mathcal{M}(W)}\mathsf{q}_W\rho.$$

This holds first for real fields and then by complex linearity for every ρ. All maps are defined on the local domains used in proposition 33.1. Iterating, with $W_j=\mathcal{M}^jU_0$ and $\mu_j=\mathsf{q}_{W_{j-1}}\cdots\mathsf{q}_{W_0}\chi$, gives

$$(Q_{\mathrm{phys}}D\chi)(c)=\xi_c^{-1}\{\mathrm{Ad}(W_{n(c)}(c))\mu_{n(c)}(c_+)-\mu_{n(c)}(c_-)\}.$$

At an interior row both scalar means vanish when $Q'\chi=0$. At an outward level-j row, the outer endpoint z has $B(z)\subset\Lambda_{j-1}$. Its level-j mean is an average of vanishing level-$(j-1)$ means, so it also vanishes. The local means agree on overlapping towers by the nested transport formula (360). At level zero the assertion follows from $\chi=0$ on Λ_0. Multiplying by m proves (538).

Next put $\Delta=F''(0)$ and $\Delta_\pi=\Delta-\Delta'_\pi[J]$. For a site field v the gauge coordinate near $(A,v)=(0,0)$ is

$$A^v(b)=(i\xi)^{-1}\log(e^{iv_-}e^{i\xi A(b)}e^{-iv_+}),\qquad v_+=\mathrm{Ad}(U_0(b))v(b_+).$$

Expanding the three exponentials and then the logarithm to second order gives

$$\begin{aligned}A^v&=A-Dv+\tfrac12N(A,v)+O((|A|+|v|)^3),\\ N(A,v)&=i[v_-,A]-i[A,v_+]-i[v_-,Dv].\end{aligned} \tag{540}$$

Indeed the quadratic part of $\log(e^Xe^Ye^Z)$ is $\frac12([X,Y]+[X,Z]+[Y,Z])$; here $v_+-v_-=\xi Dv$. Every inner plaquette transforms by conjugation, so $F(A^v)=F(A)$. Comparing the first two homogeneous terms gives

$$(Dv,J)_\xi=0,\qquad (A-Dv,\Delta(A-Dv))_\xi=(A,\Delta A)_\xi-(N(A,v),J)_\xi.$$

With $v=\mathsf{V}A$, the definition (480) and polarization therefore give

$$\Delta_\pi=\Pi^{\mathsf{T}}\Delta\Pi. \tag{541}$$

For $O=\mathsf{G}$ or G_1, its inverse in (481) is

$$O^{-1}=\Delta_\pi-\Pi^{\mathsf{T}}X_0\Pi+D\mathsf{R}D^{\mathsf{T}}+Q^{\mathsf{T}}Q,$$

where $X_0=0$ or $\Delta^{(2)}[\mathsf{j}_*]$. Equations (538) and (541) imply $O^{-1}D\chi=D\Delta^\circ\chi$ for $\chi\in\mathcal{N}$. Take $\chi=\mathsf{G}'\mathsf{R}f$ to obtain $OD\mathsf{R}=D\mathsf{G}'\mathsf{R}$. Applying Q and $\mathsf{R}D^{\mathsf{T}}$, and transposing the first identity, now yields

$$\begin{aligned}\mathsf{R}D^{\mathsf{T}}H=\mathsf{R}D^{\mathsf{T}}K=0,\qquad Q\mathsf{G}_1D\mathsf{R}=0,\qquad \mathsf{R}D^{\mathsf{T}}\mathsf{G}_1D\mathsf{R}=\mathsf{R},\\ Q\mathfrak{G}=\mathsf{R}D^{\mathsf{T}}\mathfrak{G}=0,\qquad \mathfrak{G}Q^{\mathsf{T}}=\mathfrak{G}D\mathsf{R}=0.\end{aligned} \tag{542}$$

For the last line, substitute the first line into $\mathfrak{G}=\mathsf{G}_1-\mathsf{G}_1D\mathsf{R}D^{\mathsf{T}}\mathsf{G}_1-KQ\mathsf{G}_1$. This operator is symmetric: the middle term is symmetric, and $KQ\mathsf{G}_1=\mathsf{G}_1Q^{\mathsf{T}}(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}Q\mathsf{G}_1$ is symmetric. In particular, with $\mathcal{K}=\ker Q\cap\ker(\mathsf{R}D^{\mathsf{T}})$, one has

$$\mathfrak{G}\mathsf{G}_1^{-1}=I-\mathsf{G}_1D\mathsf{R}D^{\mathsf{T}}-KQ,\qquad \mathfrak{G}\mathsf{G}_1^{-1}\delta=\delta\quad(\delta\in\mathcal{K}). \tag{543}$$

35.2. **The action on the slice.** Call $\ker(\mathsf{R}D^{\mathsf{T}})$ the slice, and put

$$\Delta_1 = \mathsf{G}_1^{-1} - D\mathsf{R}D^{\mathsf{T}} - Q^{\mathsf{T}}Q = \Delta_\pi - \Pi^{\mathsf{T}}\Delta^{(2)}[\mathsf{j}_*]\Pi.$$

For $\sup|A'| < \varepsilon_3$, write $D_0 = \mathsf{D}(A')$, $\mathcal{L}(A') = A' - HD_0$ and $\mathcal{L}' = I - H\mathfrak{D}(A')$. The first contraction and $QH = I$ give

$$\mathcal{Q}(\mathcal{L}(A')) = QA', \qquad (Q + \mathsf{C}'_{\mathcal{L}(A')})\mathcal{L}' = Q, \qquad \mathsf{R}D^{\mathsf{T}}\mathcal{L}' = \mathsf{R}D^{\mathsf{T}}.$$

Differentiating the lower Lipschitz bound in (496) shows that $\sup|\mathcal{L}'\delta| \geq \frac{1}{2}\sup|\delta|$. Thus $\mathcal{L}'$ is invertible and maps $\mathcal{K}$ onto

$$\mathcal{K}_T = \ker(Q + \mathsf{C}'_T) \cap \ker(\mathsf{R}D^{\mathsf{T}}), \qquad T = \mathcal{L}(A').$$

It also maps the slice onto itself, and $\mathcal{L}$ preserves the slice.

Let $\mathsf{C}^{(2)}(A) = \frac{1}{2}\mathsf{C}''[A]A$ be the homogeneous quadratic term of C. In terms of the physical coefficient $C^{(2)}$ in proposition 31.1, $\mathsf{C}^{(2)} = \mathsf{m}C^{(2)}$ and $H\mathsf{C}^{(2)} = \mathsf{H}C^{(2)}$. Let $R_F(A) = F(A) - F(0) - (J, A)_\xi - \frac{1}{2}(A, \Delta A)_\xi$, so that $R'_F = \mathcal{V}$. From $\mathsf{G}^{-1}H = Q^{\mathsf{T}}S$ and (542) we get $\Delta_\pi H = Q^{\mathsf{T}}(S - I)$ and $H^{\mathsf{T}}\Delta_\pi H = S - I$. Define

$$\begin{aligned} V(A') = \; & -(D_0 - \mathsf{C}^{(2)}(A'), H^{\mathsf{T}}J)_1 - (QA', (S-I)D_0)_1 \\ & + \tfrac{1}{2}(D_0, (S-I)D_0)_1 + R_F(\mathcal{L}(A')). \end{aligned} \tag{544}$$

Differentiating its four terms gives exactly the six terms of W in (498): the derivatives of D_0, $\mathsf{C}^{(2)}(A')$ and $\mathcal{L}(A')$ are $\mathfrak{D}$, $\mathsf{C}''[A']$ and $I - H\mathfrak{D}$, respectively, and S is symmetric. Hence $V' = \mathsf{W}$.

For A' in the slice, both A' and $T = \mathcal{L}(A')$ are fixed by Π. Insert $T = A' - HD_0$ into the Taylor expansion defining R_F and use (541). The quadratic term in $(J, HD_0)_\xi$ is

$$(J, H\mathsf{C}^{(2)}(A'))_\xi = \tfrac{1}{2}(A', \Delta^{(2)}[\mathsf{j}_*]A')_\xi$$

by proposition 31.1; it changes Δ_π into Δ_1. The remaining terms are (544), so

$$F(\mathcal{L}(A')) = F(0) + (J, A')_\xi + \tfrac{1}{2}(A', \Delta_1 A')_\xi + V(A'). \tag{545}$$

Differentiation along the slice gives the identity needed below:

$$(F'(T), \mathcal{L}'\delta)_\xi = (J + \Delta_1 A' + \mathsf{W}(A'), \delta)_\xi \qquad (\mathsf{R}D^{\mathsf{T}}\delta = 0). \tag{546}$$

35.3. **Endpoint factors and gauge directions.** The successive endpoint corrections in (356) must be compared at the two different levels of an outward row. On each local tower put

$$a_0 = \xi A, \quad a_{j+1} = \mathcal{Q}^1(W_j, a_j), \quad v_0 = 1, \quad v_{j+1}(x) = v_j(x)\mathsf{v}(W_j, a_j)(x).$$

The local definedness and restriction agreement proved in proposition 33.1 apply for $\sup|A| < c_4/4$. For $y \in \Lambda_n$ set $v[A](y) = v_n(y)$. Each factor uses only the nested block below y, so this definition is independent of the incident row. These finite products are holomorphic and are G-valued at real A. For real A, iteration of (356) and real gauge covariance gives

$$\mathcal{M}^n(e^{i\xi A}U_0)(c) = v_n(c_-)e^{i\mathcal{Q}(A)(c)}V_0(c)v_n(c_+)^{-1}, \qquad V_0 = M_{\mathbf{B}^\flat}(U_0).$$

At an interior row both endpoint factors are $v[A]$. At an outward level-n row with outer endpoint z, $z \in \Lambda_{n-1}$ and

$$v_n(z) = v[A](z)\theta_z(\mathcal{Q}(A)), \qquad \theta_z(\beta) = \mathsf{v}(W_{n-1}, \beta)(z).$$

The last factor sees only the interior level-$(n-1)$ rows in $B(z)$, on which β is its bond input.

The seam factors and the following twist are defined for $\sup|\beta| < r_{\mathrm{b}}$, with r_{b} as in (534). The twist is unchanged at interior rows; at an outward row it is given by the series logarithm of

$$(e^{i\beta^\natural}V_0)(c) = \begin{cases} e^{i\beta(c)}V_0(c)\theta_z(\beta)^{-1}, & z = c_+, \\ \theta_z(\beta)e^{i\beta(c)}V_0(c), & z = c_-. \end{cases} \tag{547}$$

To verify the domain, use the local unitary gauges of proposition 33.1. There $U_0^u = e^{i\xi A_\kappa}$, with $\xi A_\kappa = i^{-1}\log(U_0^u)$ bondwise. The principal logarithm of a unitary matrix is skew-Hermitian, so A_κ is real. The background recursion in proposition 24.1 has real fields and unitary endpoint factors. Its identification with the actual averages puts W_{n-1}, in the same local gauge, in the form $e^{iB_{n-1}}$ with B_{n-1} real and $\|B_{n-1}\|_\infty < r_{\mathrm{st}}/16$. Thus lemma 24.2 gives $\theta_z = e^{i\zeta_z}$ and $|\zeta_z(\beta)| \le \kappa_{\mathrm{st}} \sup|\beta|$ for $\sup|\beta| < r_{\mathrm{st}}$; unitary conjugation preserves this bound. Each matrix whose logarithm occurs in (547), after multiplication by $V_0(c)^{-1}$, is a product of two exponentials whose exponent norms sum to less than $1/16$. Its logarithm is therefore defined, and $|\beta^\natural(c)| < 1/8$. The twist is holomorphic and real at real β.

Write $\mathsf{T}^\natural_\beta = d_\beta(\beta^\natural)$. This derivative is invertible throughout $\sup|\beta| < r_{\mathrm{b}}$. Indeed, freeze ζ_z in each outward-row logarithm. Right or left multiplication by the opposite exponential gives a local inverse in that row variable; all the intervening logarithms remain near the identity. Thus the row-diagonal part T_{d} of the derivative is invertible. The remaining part T_{u} depends only on interior rows and takes values only at outward rows. Hence $(T_{\mathrm{d}}^{-1}T_{\mathrm{u}})^2 = 0$, and

$$(\mathsf{T}^\natural_\beta)^{-1} = (I - T_{\mathrm{d}}^{-1}T_{\mathrm{u}})T_{\mathrm{d}}^{-1}$$

is holomorphic. For real A with $\sup|A| < c_4/4$ and $\sup|\mathcal{Q}(A)| < r_{\mathrm{b}}$, substitution of the seam factors into the iterated average proves

$$M_{\mathbf{B}^\flat}(e^{i\xi A}U_0) = (e^{i\mathcal{Q}(A)^\natural}V_0)^{v[A]}. \tag{548}$$

The left-hand side is the original real block average.

For real A near zero and $w_\tau = e^{i\tau\chi}$, let A^{w_τ} be the local logarithmic coordinate determined by $e^{i\xi A^{w_\tau}}U_0 = (e^{i\xi A}U_0)^{w_\tau}$, and set

$$X_\chi(A) = \partial_\tau A^{w_\tau}|_0, \qquad \varrho(A;\chi)(y) = i^{-1}\partial_\tau[v[A^{w_\tau}](y)^{-1}w_\tau(y)v[A](y)]\big|_0.$$

These maps are real-linear in χ and real-analytic in A. They satisfy

$$X_\chi(0) = -D\chi, \qquad \varrho(0;\chi) = Q'\chi. \tag{549}$$

The first identity follows from the logarithm at the identity. For the second, the induced gauge at level j is $v_j[A^{w_\tau}]^{-1}w_\tau v_j[A]$, by the recursion for the endpoint factors. At $A = 0$, apply (539) at each successive level: its first-order coefficient is carried from ρ to $\mathsf{q}_{W_j}\rho$. Iteration gives $Q'\chi$ at each layer node by (358); at level zero the node factor is 1 and the derivative is χ.

For a row c and $|q| < 1/4$, define the linear map

$$\ell_{c,q}[\sigma_-,\sigma_+] = \frac{d}{d\tau}\Big|_0 i^{-1}\log[e^{i\tau\sigma_-}e^{iq}V_0(c)e^{-i\tau\sigma_+}V_0(c)^{-1}].$$

It is holomorphic in q. Put $K_A = Q + \mathsf{C}'_A$. Gauge covariance of the actual average and (548) show that the twisted datum transforms by $\mathcal{R} = v[A^w]^{-1}wv[A]$. Differentiating gives

$$(\mathsf{T}^\natural_{\mathcal{Q}(A)}K_A X_\chi(A))(c) = \ell_{c,\mathcal{Q}(A)^\natural(c)}[\varrho(A;\chi)(c_-),\varrho(A;\chi)(c_+)]. \tag{550}$$

If $\varrho(A;\chi) = 0$, invertibility of the twist derivative gives $K_A X_\chi(A) = 0$. Plaquette gauge invariance gives $(F'(A), X_\chi(A))_\xi = 0$.

The map

$$\chi \longmapsto (\varrho(A;\chi), \mathsf{R}D^{\mathsf{T}}X_\chi(A)) \tag{551}$$

is a bijection from the real bare-site-field space $\mathcal{S}_{\mathbb{R}}$ onto the product of the real fields on Λ_* and $\mathsf{R}\mathcal{S}_{\mathbb{R}}$, for A in a neighborhood of zero that may depend on the finite data. At zero it is $(Q'\chi, -\mathsf{R}D^{\mathsf{T}}D\chi)$. Its kernel is zero by (537) and $\mathsf{G}'\Delta^\circ\chi = \chi$ on $\mathcal{N}$. The map Q' is onto: its transported means are over disjoint blocks, and one can prescribe each mean by setting a single site value in that block. Moreover $\operatorname{ran}\mathsf{R} = \Delta^\circ\mathcal{N}$, with Δ° injective on $\mathcal{N}$. Thus the domain and target have the same real dimension. The determinant at zero is nonzero, and remains nonzero in a neighborhood by continuity. No uniform size of that neighborhood is needed.

35.4. **Stationarity and the original constraint.**

Proof of theorem K. The response bounds give $\|T\| \le 4\hat{\mathsf{b}}\varepsilon_B \le \varepsilon_3 \le c_4/8$. All coefficients in the construction are real at (U_0, J). Indeed the contour maps and the action are real on real fields, so their Taylor coefficients are real. The scalar operators, their inverses and R are then real; the current form, the two successive inverse operators and their row Gram inverses are real as well. Transposes preserve reality for the real pairings. Thus $H, K, S, \mathfrak{G}$ are real. Both contractions, started at zero, have real iterates for real data, so their unique limits T and $\mathbf{A}_0$ are real.

Put $A' = KB' + \mathbf{A}_0(B')$. Since $\mathbf{A}_0 = -\mathfrak{G}\mathsf{W}(A')$, (542) gives $\mathbf{A}_0 \in \mathcal{K}$, $QA' = B'$ and $\mathsf{R}D^{\mathsf{T}}A' = 0$. Also $T = \mathcal{L}(A')$ and $\sup|A'| \le 2\hat{\mathsf{b}}\varepsilon_B \le \varepsilon_3/2$. The first two equations of (535) follow from the identities for $\mathcal{L}$.

For $\delta \in \mathcal{K}$, the two constraint terms in Δ_1 vanish when paired with δ, and $\mathsf{G}_1^{-1}KB'$ lies in $\operatorname{ran} Q^{\mathsf{T}}$. By symmetry and (543),

$$(\delta, \Delta_1 A')_\xi = (\delta, \mathsf{G}_1^{-1}\mathbf{A}_0)_\xi = -(\mathfrak{G}\mathsf{G}_1^{-1}\delta, \mathsf{W}(A'))_\xi = -(\delta, \mathsf{W}(A'))_\xi.$$

Also $(J, \delta)_\xi = 0$ by (533). Equation (546) and $\mathcal{L}'\mathcal{K} = \mathcal{K}_T$ show that $F'(T)$ annihilates $\mathcal{K}_T$. The annihilator of this kernel is the range of the transposed constraint map, so

$$F'(T) = K_T^{\mathsf{T}}\widetilde{\lambda} + D\mathsf{R}\widetilde{\phi}. \tag{552}$$

Multiplication by H^{T} identifies $\widetilde{\lambda}$ with (536). The inverse there exists by the same Neumann estimate as (497): its perturbation has norm at most $8\hat{\mathsf{b}}C_3 \sup|T| \le 8C_3 c_W \le 1/4$. It is holomorphic and real at real data. In particular K_T^{T} is injective and the asserted multiplier, once stationarity is proved, is unique.

Define the holomorphic residual $z(B') = F'(T) - K_T^{\mathsf{T}}\lambda(B')$. By (552) it lies in $\operatorname{ran} D\mathsf{R}$. For real B' it is real, hence $z = D\mathsf{R}\phi$ for a real ϕ. For sufficiently small real B', $T(B')$ lies in the neighborhood of (551). Choose χ there with

$$\varrho(T;\chi) = 0, \qquad \mathsf{R}D^{\mathsf{T}}X_\chi(T) = \mathsf{R}\phi.$$

The gauge law and action invariance give $(z, X_\chi(T))_\xi = 0$. On the other hand,

$$(z, X_\chi(T))_\xi = (\mathsf{R}\phi, \mathsf{R}D^{\mathsf{T}}X_\chi(T))_\xi = (\mathsf{R}\phi, \mathsf{R}\phi)_\xi.$$

The real site pairing is positive definite, so $z = 0$. A holomorphic function vanishing on the real points of a ball vanishes on its complexification: its Taylor coefficients are obtained by real difference quotients. Applying the one-variable identity theorem to $\zeta \mapsto z(\zeta B')$ then gives $z(B') = 0$ throughout the chosen ball. This proves the third equation of (535).

We next prove criticality with the original block average fixed. For a complex site field σ on Λ_* put

$$Z(B';\sigma) = \Big(\lambda(B'), (\mathsf{T}^\natural_{B'})^{-1}\big(\ell_{c,B'^\natural(c)}[\sigma(c_-), \sigma(c_+)]\big)_c\Big)_1. \tag{553}$$

This is holomorphic in B' and linear in σ. For small real B', the first component of the bijection (551) runs over every real σ. Equations (550) and (535), with action invariance, imply

$$Z(B'; \varrho(T;\chi)) = (\lambda, K_T X_\chi(T))_1 = (F'(T), X_\chi(T))_\xi = 0.$$

Complex linearity in σ and the preceding continuation argument show that $Z(B';\sigma) = 0$ for every σ on the entire chosen ball.

Now fix real B' and a differentiable curve E_τ in $C(\mathbf{B}^\flat, M_{\mathbf{B}^\flat}(E))$ with $E_0 = E = e^{i\xi T}U_0$. For small τ, its local logarithm defines a differentiable real curve A_τ with $E_\tau = e^{i\xi A_\tau}U_0$ and $A_0 = T$. The strict domain bounds above let us apply (548) along the curve. Constancy of the actual averages gives

$$e^{i\mathcal{Q}(A_\tau)^\natural}V_0 = (e^{iB'^\natural}V_0)^{h_\tau}, \qquad h_\tau = v[A_\tau]^{-1}v[T].$$

Set $\sigma = i^{-1}\dot{h}_0$. Differentiating its local logarithm, just as in (550), yields

$$K_T \dot{A}_0 = (\mathsf{T}^{\natural}_{B'})^{-1}\big(\ell_{c,B'^{\natural}(c)}[\sigma(c_-),\sigma(c_+)]\big)_c.$$

Since the physical action is ξF, its derivative along this arbitrary constraint curve is

$$\frac{d}{d\tau}\Big|_0 A_{\Omega_0}(E_\tau) = \xi(F'(T),\dot{A}_0)_\xi = \xi Z(B';\sigma) = 0.$$

This proves the required constrained criticality. □

36. Compatible data at successive depths

We extend a reduced datum through one interface and then repeat the construction down to the bare lattice. Near an interface the extension increases the curvature bound by a fixed factor. The separation of the regions ensures that the next interface still sees the original datum, so this factor is independent of the number of levels.

Use bare lengths, so the spacing at level j is L^j. Let $1 \le k \le K$, let $L \ge 13$ be odd, and suppose that $\mathfrak{M} \ge 1, R_1 \ge L$ are powers of L, with $R_1\mathfrak{M} \ge L^2$. Let $\Omega_0 \supset \cdots \supset \Omega_k$ be a decreasing sequence of regions. Each Ω_j is a union of $\mathfrak{M}$-cubes at level j, with $\Omega_j = \emptyset$ when no such cubes exist, and

$$\operatorname{dist}_\infty(\Omega_j^c, \Omega_{j+1}) > R_1\mathfrak{M}L^j, \qquad 0 \le j < k.$$

Assume also that $\mathbb{T}^{(k)}$ has at least $6L$ sites in each coordinate direction. We allow empty regions and put $\Omega_{-1} = \Omega_0$, $\Omega_{k+1} = \emptyset$. The complementary region Ω_j^c is the closure of the point-set complement; in particular, it contains $\partial\Omega_j$. Distance to the empty set is infinite. Differences below are point-set differences.

36.1. **The descent statement.** Write $\Gamma_j^\flat$ for the reduced members of the original sequence. For $0 \le n \le k$, the reduced determining set of $(\Omega_0,\dots,\Omega_n)$ is

$$\mathbf{B}^{(n),\flat} = \bigcup_{j<n} \Gamma_j^\flat \;\cup\; \Gamma_n^{(n)}, \qquad \Gamma_n^{(n)} = \{c \text{ of level } n :\; c_\pm \in \Omega_0,\; \{c_-,c_+\} \cap \Omega_n \ne \emptyset\}.$$

Indeed, the lower members depend only on $\Omega_0, \Omega_j, \Omega_{j+1}$, while the last region of the truncated sequence has no successor. Thus $\mathbf{B}^{(k),\flat} = \mathbf{B}^\flat$, and $\mathbf{B}^{(0),\flat}$ consists of all inner bare bonds of Ω_0. At a descent from n to $n-1$, the newly assigned bonds are exactly the level-$(n-1)$ bonds with an endpoint in Ω_n. Separation puts both endpoints of these bonds in Ω_{n-1}.

For a reduced determining set at depth n, denote its member of level j by E_j. A plaquette p of level j is *tested* if at least one of its bonds belongs to E_j, and every bond either belongs to E_j or has both endpoints in $\Omega_{j-1} \setminus \Omega_j$. Its holonomy $\partial V(p)$ is the ordered plaquette product using $V_j(c)$ in the first case and $\mathcal{M}(V_{j-1})(c)$ in the second. The two cases are disjoint, since a bond of E_j meets Ω_j.

Definition 36.1. For $\varepsilon > 0$ and $C \ge 1$, let $\mathcal{D}(\mathbf{B}^\flat;\varepsilon,C)$ be the real data whose tested holonomies satisfy $|\partial V(p) - \mathbf{1}| < C\varepsilon$, and satisfy the smaller bound $|\partial V(p) - \mathbf{1}| < \varepsilon$ whenever, for some $1 \le j \le k$, the tested plaquette has level $j-1$ or j and $\operatorname{dist}_\infty(p,\partial\Omega_j) \le 6L^j$. ■

The same definition applies to every truncated sequence. In particular, a real datum whose tested holonomies have distance less than ε from $\mathbf{1}$ belongs to this class.

Proposition 36.2 (Compatible lower data)**.** There are $C_\downarrow \ge 1$ and $\varepsilon_\downarrow > 0$, depending only on L, with the following property. Let $\varepsilon > 0$, $V \in \mathcal{D}(\mathbf{B}^\flat;\varepsilon,C_\downarrow)$ and $C_\downarrow\varepsilon \le \varepsilon_\downarrow$. There are real data $V^{(n)}$ on $\mathbf{B}^{(n),\flat}$, $0 \le n \le k$, with $V^{(k)} = V$, such that:

(i) $V_j^{(n)} = V_j$ on $\Gamma_j^\flat$ for $j \le n$.

(ii) For $1 \le n \le k$ and $c \in \Gamma_n^{(n)}$,

$$|\mathcal{M}(V_{n-1}^{(n-1)})(c)V_n^{(n)}(c)^{-1} - \mathbf{1}| \le C_\downarrow \varepsilon. \tag{554}$$

(iii) Every tested holonomy of $V^{(n)}$ is defined and satisfies $|\partial V^{(n)}(p) - \mathbf{1}| < C_\downarrow\varepsilon$. Except for level-$n$ plaquettes with a corner in Ω_{n+1}, its tested plaquettes are tested for the original datum and $\partial V^{(n)}(p) = \partial V(p)$.

Moreover $V^{(n)} \in \mathcal{D}(\mathbf{B}^{(n),\flat}; \varepsilon, C_\downarrow)$ for $1 \le n \le k$. The constants are independent of k and of the number of changes in the sequence of regions. ■

36.2. **Extension through one interface.** The averages used to define tested holonomies exist for every real datum. A second-case bond has $j \ge 1$. Its endpoints are corners of a plaquette meeting Ω_j, hence lie within L^j of Ω_j. Their blocks lie within $3L^j/2 < L^{j+1}$ of Ω_j, hence in Ω_{j-1} by separation, and are disjoint from Ω_j by block membership in a region. Every fine bond of the double block is therefore an interior bond of E_{j-1}, where V_{j-1} is prescribed. The definition of the real averaging map applies on this double block.

For a site z of level $k-1$, let $[z]$ be its block center at level k. A coarse site $y \in \Omega_k$ is called inner if every coarse site y' with $|y' - y|_\infty = L^k$ also belongs to Ω_k. For a bond or plaquette $\mathfrak{e}$ at level $k-1$ or k, let

$$\varepsilon_V^\sharp(\mathfrak{e}; s) = \max\{|\partial V(p) - \mathbf{1}| : \ p \text{ tested at level } k-1 \text{ or } k,\ \operatorname{dist}_\infty(p, \mathfrak{e}) \le sL^k\}, \qquad \varepsilon_V^\sharp(\mathfrak{e}) = \varepsilon_V^\sharp(\mathfrak{e}; 3).$$

An empty maximum is zero.

Lemma 36.3 (Extension through the last interface)**.** The constants in proposition 36.2 can be chosen so that the following one-step construction is possible. If every tested holonomy of a real datum V on $\mathbf{B}^\flat$ satisfies $|\partial V(p) - \mathbf{1}| \le \varepsilon_\downarrow$, then there is a real datum $V^\downarrow$ on $\mathbf{B}^{(k-1),\flat}$ such that:

(i) $V_j^\downarrow = V_j$ on $\Gamma_j^\flat$ for $j < k$.

(ii) For every $c \in \Gamma_k$,

$$|\mathcal{M}(V_{k-1}^\downarrow)(c)V_k(c)^{-1} - \mathbf{1}| \le C_\downarrow \varepsilon_V^\sharp(c).$$

The discrepancy is zero if $c_-, c_+ \in \Omega_k$.

(iii) All tested holonomies of $V^\downarrow$ are defined. Except for level-$(k-1)$ plaquettes having a corner in Ω_k, their plaquettes are also tested for V, and their holonomies equal those of V. For a level-$(k-1)$ plaquette p having a corner in Ω_k,

$$|\partial V^\downarrow(p) - \mathbf{1}| \le C_\downarrow \varepsilon_V^\sharp(p).$$

If every block center of a corner of p is inner, its holonomy is either $\mathbf{1}$ or $\partial V_k(p')$ for a coarse plaquette p' whose four bonds belong to Γ_k.

■

Proof. Use level-$(k-1)$ lengths for this proof and put $\mathsf{h} = (L-1)/2$. Thus fine bonds have length one and coarse bonds length L.

Construction. Let K consist of the coarse sites outside Ω_k which share a coarse plaquette with a site of $\Lambda_k = \Omega_k \cap \mathbb{T}^{(k)}$, and put $\mathsf{Z} = \Lambda_k \cup \mathsf{K}$. Each site of K lies within L of Ω_k. Its partition cube, of side L, consequently lies within $3L/2 < L^2 \le R_1\mathfrak{M}$ of Ω_k and hence in Ω_{k-1}. Block membership then gives

$$B(\mathsf{K}) \subset \Omega_{k-1} \setminus \Omega_k, \qquad B(\Lambda_k) = \Omega_k \cap \mathbb{T}^{(k-1)}.$$

All fine bonds with endpoints in $B(\mathsf{Z})$ belong to $\Gamma_{k-1}^{(k-1)}$. Those with both endpoints in $B(\mathsf{K})$ are retained interior bonds; the others have an endpoint in $B(\Lambda_k)$ and are new. Conversely every new bond has this form: its endpoint in Ω_k has center in Λ_k, and the other center is the same site or a nearest neighbor and thus belongs to Z.

Define a coarse field on the bonds with endpoints in Z by

$$\mathsf{V}(c) = \begin{cases} V_k(c), & \{c_-, c_+\} \cap \Lambda_k \neq \emptyset, \\ \mathcal{M}(V_{k-1})(c), & c_-, c_+ \in \mathsf{K}. \end{cases}$$

Every coarse plaquette meeting Λ_k has its corners in Z and is tested, with holonomy $\partial\mathsf{V}(p') = \partial V(p')$. On the fine bonds with endpoints in $B(\mathsf{Z})$ put

$$N(b) = \begin{cases} \mathbf{1}, & [b_-] = [b_+], \\ \mathsf{V}(\langle [b_-], [b_+] \rangle), & [b_-] \neq [b_+]. \end{cases}$$

On the corresponding sites define

$$\sigma(z) = \begin{cases} V_{k-1}([z], z), & [z] \in \mathsf{K}, \\ \mathbf{1}, & z \in B(\Lambda_k). \end{cases} \tag{555}$$

The contour variables in this formula are defined when $\varepsilon_\downarrow$ is sufficiently small, by lemma 14.5: all plaquettes inside an outer block are tested fine plaquettes. Likewise the small-curvature formula for $\mathcal{M}(V_{k-1})$ applies on double blocks joining two sites of K, by lemma 14.6. Preserve V_j on every $\Gamma_j^\flat$, $j < k$, and set

$$V_{k-1}^\downarrow(b) = \sigma(b_-)^{-1} N(b) \sigma(b_+) \tag{556}$$

on the new bonds. These prescriptions cover the lower determining set and respect bond reversal.

For any coarse bond with endpoints in Z, the field N is trivial inside its two blocks and constant, equal to $\mathsf{V}(c)$, across its corridor. Its double-block plaquettes have holonomy $\mathbf{1}$. Every contour transport inside either block is $\mathbf{1}$, and every parallel crossing path has transport $\mathsf{V}(c)$. Therefore

$$\varpi_c(N) = 0, \qquad \mathcal{M}(N)(c) = \mathsf{V}(c).$$

For a fine plaquette $p = p_{\mu\nu}(z)$ meeting Ω_k, block centers have the form

$$[z + ae_\mu + a'e_\nu] = [z] + a\alpha L e_\mu + a'\alpha' L e_\nu, \qquad a, a' \in \{0,1\}, \quad \alpha, \alpha' \in \{0,1\}.$$

Here $\alpha = 1$ precisely when $z_\mu - [z]_\mu = \mathsf{h}$, and similarly for α'. Thus $\partial N(p) = \mathbf{1}$ if $\alpha\alpha' = 0$. Otherwise it equals $\partial\mathsf{V}([p]) = \partial V([p])$, where $[p] = p_{\mu\nu}([z])$ is a tested coarse plaquette meeting Λ_k. This also shows that every such fine plaquette has its corners in $B(\mathsf{Z})$.

Comparison with the prescribed outer field. Put $\widetilde{N} = (V_{k-1}^\downarrow)^\sigma$ on these fine bonds. It equals N on the new bonds. We use two consequences of lemmas 14.5 and 14.6. For a real field W on a block with $\varpi_y(W) < \omega_M$, the contour family has diameter at most $\frac{3}{4}(L-1)^2\varpi_y(W) < 1/40$, and proposition 4.3 defines its mean. For a bond inside the block,

$$|W(y, b_-) W(b) W(y, b_+)^{-1} - \mathbf{1}| \leq 2\kappa_1^{\mathrm{SO}} \varpi_y(W).$$

To see this, reverse b if necessary so that a coordinate-ordered contour to b_+ ends with b. Its prefix is a contour to b_-. Each contour differs from its mean by at most $\kappa_1^{\mathrm{SO}}\varpi_y(W)$, and unitary multiplication preserves this distance.

For a corridor bond b of a coarse bond c, assume $\varpi_c(W) < \min\{\omega_M, (\kappa_2^{\mathrm{SO}})^{-1}\}$. Then

$$|W(c_-, b_-) W(b) W(c_+, b_+)^{-1} - \mathcal{M}(W)(c)| \leq \kappa_\downarrow \varpi_c(W), \qquad \kappa_\downarrow = 2\kappa_1^{\mathrm{SO}} + \kappa_2^{\mathrm{SO}} + \kappa_3.$$

Indeed, take c positive, put $x = b_-$, $x^* = b_+$ and $x^+ = x + Le_\mu$. Contours from c_+ to x^* and x^+ chosen with direction μ last have transports satisfying $W(\Gamma^*)W([x^*, x^+]) = W(\Gamma^+)$. Replacing them by their means costs at most $2\kappa_1^{\mathrm{SO}}\varpi_c(W)$. The resulting transport is $W_x^{\mathsf{m}} W_{\mathsf{m}}[c]$, which differs from $W_{\mathsf{m}}[c]$ by at most $\kappa_2^{\mathrm{SO}}\varpi_c(W)$; the latter differs from $\mathcal{M}(W)(c)$ by at most $\kappa_3\varpi_c(W)$.

Apply the first estimate to retained bonds within an outer block and the second to retained bonds between two outer blocks. Their plaquettes are tested and lie within $2L$ of the bond. Consequently, with $K_\downarrow = 2 + 2\kappa_\downarrow$,

$$|\widetilde{N}(b) N(b)^{-1} - \mathbf{1}| \leq K_\downarrow \varepsilon_V^\sharp(b; 2). \tag{557}$$

On bonds with both endpoints in Ω_k the two fields are equal.

Averages and plaquettes. Fix $c \in \Gamma_k$ and let $d_c = \sup_{b\in \mathsf{P}_c} |\widetilde{N}(b)N(b)^{-1} - \mathbf{1}|$. Choosing $\varepsilon_\downarrow$ sufficiently small allows us to write $\widetilde{N} = e^{ia}N$ on P_c, with a real and

$$\|a\|_\infty \le (\pi/2)d_c \le r_{\mathrm{st}}/2.$$

To estimate its average, apply the box gauge of lemma 14.4 to the flat field N on this double block. Its sides are less than half the fine torus period, so there is a real gauge h with $N^h = \mathbf{1}$. On positive bonds $\widetilde{N}^h = e^{i\,\mathrm{Ad}(h)a}$, and conjugation preserves $\|a\|_\infty$. Apply lemma 24.2 at the identity and return by (363). Thus $v_\pm = \mathsf{v}(N,a)(c_\pm)$ lie in G, $Q = \mathcal{Q}^1(N,a)(c)$ lies in $\mathfrak{g}$, and $|i^{-1}\log v_\pm|, |Q| \le \kappa_{\mathrm{st}}\|a\|_\infty$. Unitary conjugation preserves the logarithm domains. Also $\varpi_c(\widetilde{N}) \le 4\|a\|_\infty < \min\{\omega_M, (\kappa_2^{\mathrm{SO}})^{-1}\}$, so the averages agree with the real small-curvature branch. The factorization (356) now gives

$$\mathcal{M}(\widetilde{N})(c) = v_- e^{iQ}\mathcal{M}(N)(c)v_+^{-1}, \qquad |\mathcal{M}(\widetilde{N})(c) - \mathcal{M}(N)(c)| \le 3\kappa_{\mathrm{st}}\|a\|_\infty.$$

The last estimate follows by telescoping the three unitary perturbations. Reversing c inverts both averages and leaves their distance unchanged. Since $\sigma = 1$ at coarse sites, using $V_{k-1}(y,y) = \mathbf{1}$ on K, gauge covariance identifies this discrepancy with that for $V^\downarrow$. Every fine bond of P_c lies within h of c; hence (557) gives $d_c \le K_\downarrow \varepsilon_V^\sharp(c)$. This proves (ii) with $C_\downarrow \ge (3\pi/2)\kappa_{\mathrm{st}}K_\downarrow$. If $c_\pm \in \Omega_k$, both endpoint blocks lie in Ω_k and $\widetilde{N} = N$ on P_c, giving the asserted equality.

If p is a fine plaquette with a corner in Ω_k, its four bonds belong to the new highest member and $\partial V^\downarrow(p)$ is conjugate to $\partial\widetilde{N}(p)$. Telescoping the four unitary factors gives

$$|\partial V^\downarrow(p) - \mathbf{1}| \le 4K_\downarrow\varepsilon_V^\sharp(p;2) + |\partial N(p) - \mathbf{1}| \le (4K_\downarrow + 1)\varepsilon_V^\sharp(p).$$

For the last inequality, a nontrivial $\partial N(p)$ is the tested coarse holonomy of $[p]$, which lies within h of p. If every corner of p has inner block center, all its corners lie in $B(\Lambda_k)$; there $\sigma = 1$ and its bonds are new, so its holonomy equals $\partial N(p)$. In this case all four bonds of $[p]$, when it occurs, belong to Γ_k.

At levels below $k-1$, the members and the regions entering the definition of a tested plaquette are unchanged. At level $k-1$, a plaquette without a corner in Ω_k contains no new bond; its first-case bonds and second-case bonds are also unchanged. On a second-case bond its holonomy uses an average of interior bonds of the preceding member, all of whose values were preserved. Thus these holonomies equal those of V. Every remaining tested plaquette is a fine plaquette meeting Ω_k, already treated above. This proves (iii). All the smallness requirements are satisfied by choosing

$$C_\downarrow = \max\{1 + 4K_\downarrow, (3\pi/2)\kappa_{\mathrm{st}}K_\downarrow\},$$
$$0 < \varepsilon_\downarrow < \min\{\omega_M, (\kappa_2^{\mathrm{SO}})^{-1}, r_{\mathrm{st}}/(\pi K_\downarrow)\}.$$

These choices depend only on L. □

36.3. **Iteration without loss of radius.**

Proof of proposition 36.2. We retain the equality assertion in (iii) while applying lemma 36.3 successively. Separation gives

$$\mathrm{dist}_\infty(\partial\Omega_j, \Omega_{j+1}) > L^{j+2}.$$

All segments used below may be taken in local Euclidean coordinates: the torus side is at least $6L^{k+1} \ge 78L^n$, so sup balls of radius $7L^n$ are isometric to their Euclidean lifts.

Suppose that $V^{(n)}$ has been constructed with (i) and (iii), starting with $V^{(k)} = V$. Its tested holonomies have distance from $\mathbf{1}$ less than $C_\downarrow\varepsilon \le \varepsilon_\downarrow$, so set $V^{(n-1)} = (V^{(n)})^\downarrow$. The truncated sequence satisfies the same geometric hypotheses.

We first bound the local sizes in the nonexact cases of the one-step lemma. Let $\mathfrak{e}$ be either a bond of $\Gamma_n^{(n)}$ with an endpoint that is not inner, or a level-$(n-1)$ plaquette meeting Ω_n with a corner whose block center is not inner. There are $z_0 \in \mathfrak{e} \cap \Omega_n$ and a coarse site $y' \notin \Omega_n$ with

$|z_0 - y'|_\infty \le 2L^n$. For a bond this follows by taking an outside site at distance at most L^n from its noninner endpoint. For a plaquette, take such a site within L^n of the noninner block center; the additional distance to z_0 is at most $\frac{L+1}{2}L^{n-1} < L^n$. The segment from z_0 to y' meets $\partial\Omega_n$ at a point q with $|q - z_0|_\infty \le 2L^n$.

A plaquette p entering $\varepsilon^\sharp_{V^{(n)}}(\mathfrak{e})$ lies within $3L^n$ of $\mathfrak{e}$, and every point of $\mathfrak{e}$ lies within L^n of z_0. Hence

$$\operatorname{dist}_\infty(p, \partial\Omega_n) \le 3L^n + L^n + 2L^n = 6L^n.$$

If p has level n, all its corners are within $7L^n < L^{n+2}$ of this boundary and thus outside Ω_{n+1}. If p has level $n-1$, it is already outside the exceptional family in (iii). In both cases (iii) identifies its holonomy with that of the original datum, and the smaller bound in definition 36.1, with $j = n$, applies. Therefore

$$\varepsilon^\sharp_{V^{(n)}}(\mathfrak{e}) < \varepsilon. \tag{558}$$

The one-step preservation property proves (i) at depth $n-1$. For a coarse bond with both endpoints inner, the discrepancy in (ii) vanishes. Every other coarse bond is covered by (558); the one-step estimate therefore proves (554).

Consider a level-$(n-1)$ plaquette p meeting Ω_n. Separation puts all its bonds in $\Gamma^{(n-1)}_{n-1}$, so it is tested. If all its block centers are inner, its new holonomy is identity or a tested holonomy at depth n, whose distance from $\mathbf{1}$ is less than $C_\downarrow\varepsilon$ by induction. Otherwise the one-step bound and (558) give the same bound. All other tested holonomies are unchanged by the one-step lemma. Their levels are at most $n-1$, so they were not exceptional at depth n and already equal the original holonomies. This proves (iii) and completes the induction.

It remains to check the smaller bound for a truncated datum. A level-n plaquette meeting Ω_{n+1} satisfies

$$\operatorname{dist}_\infty(p, \partial\Omega_n) > L^{n+2} - L^n > 6L^n.$$

It therefore cannot be one of the plaquettes tested for the smaller radius at depth n: its level forces $j = n$ in that condition. Thus every plaquette tested for the smaller radius has unchanged original holonomy, by (iii), and inherits its distance bound $|\partial V^{(n)}(p) - \mathbf{1}| < \varepsilon$. □

The compatibility estimate applies directly to a configuration realizing the lower datum. If $M_{\mathbf{B}^{(n-1),\flat}}(U) = V^{(n-1)}$, then its retained rows of every level $j < n$ equal those of $V^{(n)}$. For $c \in \Gamma^{(n)}_n$, separation puts the fine double block P_c in $\Gamma^{(n-1)}_{n-1}$, so locality gives

$$\mathcal{M}^n U(c) = \mathcal{M}(V^{(n-1)}_{n-1})(c).$$

Consequently its remaining datum discrepancy is bounded by (554).

37. Existence and uniqueness of the real background

37.1. **Improving the regularity radius.** The following improvement keeps the regularity radius uniform when a new level is added to the region sequence. We continue to use the reduced determining set $\mathbf{B}^\flat$.

Proposition 37.1 (Improvement of the regularity radius)**.** There are $B_{\mathrm{rad}} \ge \max\{4, L^2\}$ and a power $\rho_{\mathrm{rad}} \ge L^2$ of L, depending only on L, with the following property. Let $\rho \ge \rho_{\mathrm{rad}}$ be a power of L, and assume the separated-region setting, with

$$\mathfrak{M} \ge M_{\mathrm{ps}}, \qquad R_1 \ge L, \qquad R_1\mathfrak{M} \ge L^2\rho, \qquad \#\mathbb{T}^{(k)} \text{ in each direction} \ge 6\rho.$$

There is $a_{\mathrm{rad}}(\rho) > 0$, depending only on L, ρ, such that the following holds. Let V be real reduced data with all tested holonomies defined and satisfying $|\partial V(p) - \mathbf{1}| < \varepsilon_1$, where $\varepsilon_1 > 0$. If U is a critical configuration in $C(\mathbf{B}^\flat, V) \cap \mathfrak{U}'_k(\varepsilon_0)$, where $0 < \varepsilon_0 \le a_{\mathrm{rad}}(\rho)$, then

$$U \in \mathfrak{U}'_k(B_{\mathrm{rad}}\varepsilon_1).$$

The constants are independent of the number of levels and of the number of strict inclusions between successive regions. ∎

Proof. We prove the improvement

$$U \in \mathfrak{U}'_k\Big(\max\{4\varepsilon_1,\, 3K^\sharp\varepsilon_1 + 3\theta\varepsilon_0\}\Big), \qquad \theta = C_b\rho^2 e^{-\delta\rho/24} + C_c\rho^4\varepsilon_0, \tag{559}$$

when $0 < \varepsilon_1 \le \varepsilon_0$ and ε_0 is sufficiently small as specified below. Here $\delta = \delta_S$ is the exponent in (458), and $K^\sharp, C_b, C_c$ depend only on L.

Fix an inner plaquette or a full bond $\mathfrak{e}$ of positive local index j, and choose a corner $x_0 \in \Omega_j \setminus \Omega_{j+1}$. Put $n = j$ if $\operatorname{dist}_\infty(x_0, \Omega_{j+1}) > 5\rho\xi_j$, and $n = j+1$ otherwise; set

$$\xi = \xi_n, \quad \widehat{y} = [x_0]_n, \quad N = \{|x - \widehat{y}|_\infty \le 4\rho\xi\}, \quad d_\infty(x) = \xi^{-1}|x - \widehat{y}|_\infty, \quad d_1(x) = \xi^{-1}|x - \widehat{y}|_1.$$

Then

$$N \subset \Omega_{n-1}, \qquad N \cap \Omega_{n+1} = \varnothing, \qquad \operatorname{dist}_\infty(N, \Omega_0^c) > 8\rho\xi. \tag{560}$$

Indeed, if $n = j$, separation leaves $(L - \frac{9}{2})\rho\xi > 8\rho\xi$ between N and Ω_{j-1}^c; the defining distance excludes Ω_{j+1}. If $n = j+1$, choose a point of Ω_{j+1} within $5\rho\xi_j$ of x_0. The remaining separation is $(L^2 - 5 - \frac{9}{2}L)\rho\xi_j \ge 8L\rho\xi_j$. Since $x_0 \notin \Omega_{j+1}$, the next separation excludes Ω_{j+2}. This also proves $n \le k$. The torus side, being twice a power of L and at least 6ρ, is at least $2L\rho$ in level-k units. Thus N has a single coordinate chart and side less than one third of the torus side.

Every site in N belongs to layer $n-1$ or n. The weighted distance therefore satisfies

$$\begin{aligned} &d_{\mathbf{B}}(y, y') \ge s' - s && \text{if } d_\infty(y) \le s,\ d_\infty(y') \ge s', \quad 0 \le s \le s' \le 4\rho, \\ &d_{\mathbf{B}}(y, y') \ge \xi^{-1}|y - y'|_1 && \text{if } d_\infty(y), d_\infty(y') \le \rho/3. \end{aligned} \tag{561}$$

For the first inequality, any connecting set crosses the annulus, where both layer weights are at least ξ^{-1}. For the second, a connecting set contained in N has weighted length at least its coordinate ℓ^1 distance. If it leaves N, the first inequality gives length at least $11\rho/3$, greater than the possible ℓ^1 distance 2ρ between the two endpoints. This includes $n = 1$, where the finer part is the zero layer.

A reference configuration trivial near $\widehat{y}$. Choose a cube $\square$ of the $\rho\xi$ partition containing $\widehat{y}$, and let $\square_s$ be its concentric enlargement of side $(\rho + 2s)\xi$. The regional estimate (448), including its ordinary curl-divergence conclusion, applies on $\square_\rho$. Use level n if this cube lies in Ω_n, and level $n-1$ otherwise. The dimensionless side is 3ρ or $3L\rho$; the required enlarged cube lies in N. After extension of the local gauge by the identity, write $U^\sharp = U^{u^{-1}}$. Gauge covariance preserves criticality and the tested-holonomy bounds. On $\square_\rho$,

$$U^\sharp = e^{i\eta A_0}, \qquad \xi|A_0|,\ \xi^2|\nabla^\eta A_0|,\ \xi^3|D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} A_0| < a, \qquad a = 63L^6 B_{\mathrm{p}}\rho\varepsilon_0.$$

The L^3 conversion in the third bound is included in a. The local-gauge hypotheses follow from $63L^3\rho\varepsilon_0 \le c_{\mathrm{p}}$ and from the chosen lower bound on ρ.

Take a product cutoff χ, equal to one on $\square_{\rho/3}$ and zero off $\square_{2\rho/3}$, with first and second differences bounded by $c_\varphi/(\rho\xi)$ and $c_\varphi/(\rho\xi)^2$. Set $A^\natural = \chi A_0$, $A' = (1-\chi)A_0$ on $\square_\rho$, extend $A^\natural$ by zero, and define $U_0 = e^{i\eta A'}$ there, $U_0 = U^\sharp$ elsewhere. The cutoff is zero well before the boundary, and

$$U^\sharp = e^{i\eta A^\natural} U_0.$$

The product rule for $D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}}(fA_0)$ has, for each transverse direction, one term $(\nabla f) D A_0$, two terms $(\nabla f)\nabla A_0$, and two terms $(\nabla^2 f)A_0$, besides $f D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}} A_0$. Their bounds are $2, 1, 1, 1, 1$ times $c_\varphi a \xi^{-3}$. Hence, with $\alpha_2 = (1 + 12c_\varphi)a$,

$$\xi|fA_0| \le a, \qquad \xi^2|\nabla^\eta(fA_0)| \le (1 + c_\varphi)a, \qquad \xi^3|D_{\mathbf{1}}^{\mathsf{T}} D_{\mathbf{1}}(fA_0)| \le \alpha_2, \qquad f = \chi, 1 - \chi.$$

The local real action estimates give curvature and current at most $3\alpha_2$ in their respective scale norms. The full matrix divergence is at most $(1 + 3\alpha_2)3\alpha_2 \le 4\alpha_2$. Outside this calculation's domain

a complete plaquette star agrees with that of $U^\sharp$. Consequently

$$U_0 \in \mathfrak{U}'_k(\alpha'_0), \qquad \alpha'_0 = 6\alpha_2, \qquad |A^\natural|_{(-1)} \le \alpha_2, \quad \|A^\natural\|_X \le 2\alpha_2. \tag{562}$$

For the last bound the difference between the covariant and ordinary derivatives is at most $2a^2\xi^{-2}$. In particular $U_0 = \mathbf{1}$ on $d_\infty \le \rho/3$. By locality its averages through level n are the identity on bonds whose endpoints satisfy $d_\infty \le \rho/3 - 1$. On bare stars in this smaller region,

$$J_0 := F'(0) = 0, \qquad \Delta' = 0. \tag{563}$$

Prescribing the local datum gauge. Let $\rho' = \lfloor \rho/3 \rfloor$, $Y = \{y \in \mathbb{T}^{(n)} : d_\infty(y) \le \rho'\}$, and $Y^\circ = Y \setminus \Omega_n$. For $W_i = \mathcal{M}^i U^\sharp$, define

$$z(y) = W_n(\Gamma_{\widehat{y},y}) \quad (y \in Y), \qquad z(x) = z(y)W_{n-1}(y,x) \quad (x \in B(y),\ y \in Y^\circ).$$

The prescriptions agree at the centers, and prescribe z at tops of levels n and $n-1$: $Y \cap \Omega_n \subset \Lambda_n$ and $B(Y^\circ) \subset \Lambda_{n-1}$. Extend z by the identity at the other tops.

Here are the estimates needed from this construction. A coarse plaquette meeting Ω_n has the holonomy of the datum. For a coarse plaquette outside Ω_n, the fine plaquettes in its four blocks are tested, so one averaging step bounds its holonomy by $L^2\varepsilon_1 + C\varepsilon_1^2 \le 2L^2\varepsilon_1$. The comb estimate therefore gives

$$|W_n^z(c) - \mathbf{1}| \le 2L^2 \max_{\partial c} d_1\, \varepsilon_1.$$

In each fine block $B(y)$, $y \in Y^\circ$, covariance gives $W_{n-1}^z(y,x) = \mathbf{1}$ exactly. Within that block the contour comparison bounds a bond by $C\varepsilon_1$. Across two blocks the same comparison gives

$$|W_{n-1}^z(b) - \mathbf{1}| \le |W_n^z(c) - \mathbf{1}| + C\varepsilon_1,$$

where c joins their centers. To see the latter estimate, compare the two averaged contour transports with paths passing through b. Moving the crossing step through the transverse path uses at most $L-1$ plaquettes. Each contributes at most ε_1; the two contour errors and the coarse-average error are also $C(L)\varepsilon_1$. Thus all the fine and coarse bonds just described have size at most $C(L)(1+d_1)\varepsilon_1$ in this gauge.

The chart bounds before gauging imply that the coarse bonds of W_n here have size at most $\kappa_4 L^2\varepsilon_0 + a$, and the fine bonds inside one block have size at most $\kappa_4\varepsilon_0 + a$. The comb has at most $3\rho'$ bonds and a fine contour at most $3(L-1)/2$ bonds. The contour comparison therefore gives

$$\sup|z - \mathbf{1}| \le \zeta := \rho(\kappa_4 L^2\varepsilon_0 + a) + \tfrac{3}{2}L(\kappa_4\varepsilon_0 + a) + 2\kappa_1^{\mathrm{SO}}\varepsilon_0 \le C(L)\rho a.$$

Apply lemma 30.4 to $A^\natural$, with the values $r(y) = z(y)v_j[A^\natural](y)$ for $y \in \Lambda_j$. Its smallness hypotheses follow from $\sup|r - \mathbf{1}| \le \zeta + C_v\alpha_2$, where $C_v = 4\kappa_{6,\mathsf{m}}$. It gives a real chart A and a gauge w such that

$$E = e^{i\eta A}U_0 = (U^\sharp)^w, \qquad RD^*A = 0, \qquad |A|_{(-1)} \le a_f := \alpha_2 + 2C_{\mathrm{gc}}K_3(6\alpha_2 + \zeta + C_v\alpha_2) \le C(L)\rho a.$$

The induced frame is z at every top. At an outward row its outer-end correction is also trivial in the region under consideration: the finer reference contours are the identity and the corresponding contours of W_{n-1}^z are exactly the identity. Thus the one-step factor is $\mathbf{1}$, not merely small. The reduced chart row $\mathcal{Q}(A)$ consequently satisfies

$$|\mathcal{Q}(A)(c)| \le C(L)(1 + d_1(y_c))\varepsilon_1 \quad \text{if } d_\infty(y_c) \le \rho/3 - 3, \qquad \sup|\mathcal{Q}(A)| \le b_f := 2La_f. \tag{564}$$

The first bound follows by the real logarithm estimate $|\log W| \le (\pi/2)|W - \mathbf{1}|$. Only interior rows at level $n-1$ and interior or outward rows at level n occur here.

Since E and U_0 lie in $\mathfrak{U}'_k(\alpha'_0)$ and $RD^*A = 0$, lemma 30.10 gives, with $B_1 = 8K_1$,

$$\|A\|_X,\ |D^*DA|_{(-3)} \le r_f := B_1(\alpha'_0 + 4b_f) \le C(L)\rho^2\varepsilon_0. \tag{565}$$

The reference current has $|J_0|_{(-3)} \le 3\alpha'_0$: at nonfull bonds the inner star has at most three plaquettes and local scale η. Criticality and the small-chart transversality give $F'(A) = K_A^*\omega$.

Localizing the Euler equation. Use the corner frame of the reduced Schur operators, writing $\widehat{q} = T^{-1}\mathcal{Q}(A)$ and $\widehat{\omega} = T^{-1}\omega$. The map T is an isometry row by row. With $\mathcal{C}$ denoting the nonlinear mean/corner remainder, the equations become

$$J_0 + D^*DA + \Delta' A + \mathcal{V}(A) = (Q^* + \mathcal{C}'(A)^*)\widehat{\omega},$$
$$QA = \widehat{q} - \mathcal{C}(A).$$

Apply H_0^*, using $QH_0 = 1$:

$$\widehat{\omega} = H_0^* J_0 + (DH_0)^* DA + H_0^*(\Delta' A + \mathcal{V}(A) - \mathcal{C}'(A)^*\widehat{\omega}). \tag{566}$$

The action and chart estimates give

$$\sup|\mathcal{C}(A)| \le Cr_f^2, \quad |\mathcal{C}'(A)^*\omega|_{(-3)} \le Cr_f \sup|\omega|, \quad |\Delta' A|_{(-3)} + |\mathcal{V}(A)|_{(-3)} \le Cr_f^2.$$

Here $\alpha_0' \le r_f$ has been used. For r_f sufficiently small, the last multiplier term in (566) is absorbed, and $\sup|\widehat{\omega}| \le Cr_f$. The Schur identities now give

$$A = H_0\widehat{q} - \widetilde{G}_0 J_0 + A_2, \qquad A_2 = \widetilde{G}_0(-\mathcal{V}(A) + \mathcal{C}'(A)^*\widehat{\omega}) - H_0\mathcal{C}(A), \qquad \|A_2\|_X \le Cr_f^2. \tag{567}$$

We spell out how decay improves this global bound. Write $d = d_{\mathbf{B}}$ and put

$$\mathrm{rt}(y,y') = \max\{\xi_y/\xi_{y'}, \xi_{y'}/\xi_y\}, \qquad D_\rho = \rho/6 - 3, \qquad e_\rho = e^{-\delta D_\rho/4} \le 2e^{-\delta\rho/24}.$$

The already proved summation estimate implies

$$\sum_{y'} e^{-\delta d(y,y')}(1 + d(y,y'))\,\mathrm{rt}(y,y')^4 \le C,$$
$$\sum_{d(y,y')\ge D_\rho} e^{-\delta d(y,y')/2}\,\mathrm{rt}(y,y')^4 \le Ce_\rho. \tag{568}$$

Indeed, extract $e^{-\delta D_\rho/4}$ in the second sum; in the first absorb $1 + d$ into three quarters of the exponential. The remaining sums are the same uniform top sums as before.

For $d_\infty(y) \le \rho/6$, split the terms in (567) at $d_\infty(y') = \rho/3 - 3$. In the near part, (561) gives $1 + d_1(y') \le (1 + d_1(y))(1 + d(y,y'))$. In the far part the weighted distance is at least D_ρ. The current J_0 vanishes in the near part by (563). The bounds (458) and (460), followed by (568), yield

$$\varsigma_y := \max\{\xi_y|A|_y, \xi_y^2|\nabla A|_y\} \le C(1 + d_1(y))\varepsilon_1 + Ce_\rho r_f + Cr_f^2. \tag{569}$$

The scale ratio in the $\widetilde{G}_0$ term has power three, which is covered by (568).

Next evaluate (566) at a row with $d_\infty(y_c) \le 3$. Its $H_0^* J_0$ term is a far contribution. For $(DH_0)^*DA$, split at $d_\infty(y) = \rho/6$, use $|DA|_y \le 2|\nabla A|_y$ and (569) in the near part, and the global bound in the far part. The transposed kernels in (459) have the additional factor ξ_c/ξ_y, bounded by $\mathrm{rt}(y_c, y)$; the same sums still apply. The remaining terms are globally $O(r_f^2)$. It follows that

$$|\widehat{\omega}(c)| \le C\varepsilon_1 + Ce_\rho r_f + Cr_f^2 \qquad (d_\infty(y_c) \le 3). \tag{570}$$

The corners of $\mathfrak{e}$ and of its star lie over tops with $d_\infty \le 3$, and their local scales differ from $\mathsf{L}(\mathfrak{e})$ by at most L. If $\mathfrak{e} = b$ is a bond, every row containing b has level $n - 1$ or n and its top satisfies $d_\infty(y_c) < 3$. To check the first assertion, a row of level $i \le n - 2$ would have a top inside $N \subset \Omega_{n-1} \subset \Omega_{i+1}$. For $i \ge n + 1$, support membership first forces $i = n + 1$, and then places that top in $N \cap \Omega_{n+1}$. Both are impossible. By (563), the Euler equation at b therefore gives

$$D^*DA(b) = (Q^* + \mathcal{C}'(A)^*)\widehat{\omega}(b) - \mathcal{V}(A)(b),$$

where the first term uses only the rows just located. The transpose row bound and (570) control this term. Combining with (569) gives

$$\varepsilon_{\mathrm{loc}} \le K^\sharp\varepsilon_1 + C_b\rho^2 e^{-\delta\rho/24}\varepsilon_0 + C_c\rho^4\varepsilon_0^2, \qquad \varepsilon_{\mathrm{loc}} \le L^2 r_f. \tag{571}$$

Here $\varepsilon_{\mathrm{loc}}$ is the maximum of the potential and first-difference sizes on the plaquette or full star, and, in the bond case, of $\mathsf{L}(b)^3|D^*DA(b)|$. The coefficients in (571) depend only on L.

Returning to gauge-invariant quantities. The reference is the identity on the relevant full star. Choose the small radius so that $\varepsilon_{\mathrm{loc}} \le \min\{(2C_{\mathrm{ec}})^{-1}, 2/75\}$. The real action estimates then give

$$|\partial E(p) - \mathbf{1}| \le \tfrac{5}{2}\varepsilon_{\mathrm{loc}}\eta^2\mathsf{L}(\mathfrak{e})^{-2}, \qquad |J(E)(b)| \le \tfrac{5}{2}\varepsilon_{\mathrm{loc}}\mathsf{L}(b)^{-3}.$$

For the second bound, use $F'(A) = D^*DA + \mathcal{V}(A)$ on this star and $|J(E)(b)| \le (1+4\eta|A(b)|)|F'(A)(b)|$. The full-divergence conversion gives

$$|D_E^{\mathsf{T}}\partial E(b)| \le (1 + \tfrac{5}{2}\varepsilon_{\mathrm{loc}})\tfrac{5}{2}\varepsilon_{\mathrm{loc}}\,\eta^2\mathsf{L}(b)^{-3} \le \tfrac{11}{4}\varepsilon_{\mathrm{loc}}\eta^2\mathsf{L}(b)^{-3}.$$

These bounds also hold when $\varepsilon_{\mathrm{loc}} = 0$, by passage to arbitrarily small positive radii. Gauge covariance returns them to U. Since $\varepsilon_1 > 0$, their strict enlargement by $3(K^\sharp\varepsilon_1 + \theta\varepsilon_0)$ proves the required bounds at every positive-index element.

At index zero, a plaquette is a tested plaquette of V_0, so its error is less than ε_1. A full bond of index zero has four plaquettes, of indices zero or one. In $S_U = \eta D_U^{\mathsf{T}}\partial U$ the identity matrices cancel between each forward and backward plaquette. Thus

$$|S_U(b)| < 4\max\{\varepsilon_1,\, 3(K^\sharp\varepsilon_1 + \theta\varepsilon_0)L^{-2}\} \le \max\{4\varepsilon_1,\, 3K^\sharp\varepsilon_1 + 3\theta\varepsilon_0\}.$$

This proves (559).

All smallness conditions used above follow from

$$\varepsilon_0 \le \min\left\{L^{-2}c_{\mathrm{curv}},\ \frac{c_{\mathrm{p}}}{63L^3\rho},\ \frac{m_*}{63L^6B_{\mathrm{p}}\rho^2}\right\}$$

with a sufficiently small $m_*(L) > 0$. Here $c_{\mathrm{curv}} > 0$ is no larger than a_{m}, c_{c} and the real averaging curvature threshold in proposition 14.1. More explicitly, choose m_* so that α_2 satisfies the real-action caps; α_0', $2\alpha_2$ and $\zeta + C_v\alpha_2$ satisfy the chart, gauge and slice caps; a_f, r_f lie in those same fixed chart radii; the multiplier coefficient is at most $1/2$; and $L^2 r_f \le \min\{(2C_{\mathrm{ec}})^{-1}, 2/75\}$. Every quantity in this list is bounded by a fixed multiple of ρa, so this is one positive choice depending only on L.

Choose ρ_{rad} large enough for the preceding geometry, the local gauge and Schur estimates, and for $C_b\rho^2 e^{-\delta\rho/24} \le 1/24$ at every power $\rho \ge \rho_{\mathrm{rad}}$ of L. Define

$$a_{\mathrm{rad}}(\rho) = \min\left\{L^{-2}c_{\mathrm{curv}},\ \frac{c_{\mathrm{p}}}{63L^3\rho},\ \frac{m_*}{63L^6B_{\mathrm{p}}\rho^2},\ \frac{1}{24C_c\rho^4}\right\},$$
$$B_{\mathrm{rad}} = \max\{4, L^2, 12K^\sharp\}.$$

Then $\theta \le 1/12$. If $B_{\mathrm{rad}}\varepsilon_1 \ge \varepsilon_0$ there is nothing to prove. Otherwise (559) improves the radius to $\max\{B_{\mathrm{rad}}\varepsilon_1, \varepsilon_0/2\}$. Repeat while the second entry is larger. After finitely many repetitions the radius is $B_{\mathrm{rad}}\varepsilon_1$, proving the proposition. □

37.2. **Ascent to a critical configuration.** Fix a power ρ of L large enough that

$$\rho \ge \max\{\rho_{\mathrm{rad}}, LM_{\mathrm{ps}}\}, \qquad L^2\rho \ge w_S.$$

Assume $\mathfrak{M} \ge M_{\mathrm{ps}}$, $R_1 \ge L$, $R_1\mathfrak{M} \ge L^2\rho$, and at least 6ρ sites in each direction of $\mathbb{T}^{(k)}$. These conditions hold with the same constants for every truncated sequence $(\Omega_0, \dots, \Omega_n)$: the separation is inherited, and its terminal torus has at least as many sites.

Let K_5, Θ, r_0, c_8 be the constants of the actual critical-point solver, proposition 30.9. Take $c_{\mathrm{ref}} = a_{\mathrm{in}}$ and $r_L = 1/2$; the latter is a fixed real logarithmic datum radius. Set

$$r_s = \min\{r_0, \tfrac{1}{2}K_5 r_L\},$$
$$k_a = 5L^2\max\{1, 4(1+L^3)\Theta\}K_5, \qquad K_9 = 4 + 6k_a,$$
$$c_{\mathrm{asc}} = \min\left\{\tfrac{1}{2}, c_8, c_{\mathrm{ref}}, \frac{r_s}{5K_5}, \frac{\rho_{\mathrm{ec}}}{k_a}, \frac{1}{C_{\mathrm{ec}}k_a}, \frac{1}{2(4k_a + C_{\mathrm{ec}}(1+k_a))}, \frac{1}{3(2+3k_a)}\right\}.$$

All these constants depend only on L.

Proposition 37.2 (Existence of a regular critical configuration)**.** In the preceding geometry, put

$$a_{\mathrm{ex}}(\rho) = \frac{1}{C_\downarrow} \min\left\{\varepsilon_\downarrow,\ \frac{c_{\mathrm{asc}}}{L^3 B_{\mathrm{rad}}},\ \frac{a_{\mathrm{rad}}(\rho)}{K_9 L^3 B_{\mathrm{rad}}}\right\}.$$

If $0 < \varepsilon \le a_{\mathrm{ex}}(\rho)$ and

$$V \in \mathcal{D}(\mathbf{B}^\flat; \varepsilon, C_\downarrow),$$

then $C(\mathbf{B}^\flat, V)$ contains a real critical configuration in $\mathfrak{U}'_k(B_{\mathrm{rad}} C_\downarrow \varepsilon)$. The constants are independent of the number of levels and of changes in the region sequence. In particular, the conclusion holds if every tested holonomy of V has distance less than ε from the identity. ■

Proof. Write $\varepsilon' = C_\downarrow \varepsilon$. Proposition 36.2 supplies data $V^{(n)}$ on $\mathbf{B}^{(n),\flat}$ with $V^{(k)} = V$, agreeing with V on the retained lower rows, satisfying the terminal mismatch estimate, and having every tested holonomy strictly within ε' of the identity.

We first prove the ascent step. Suppose $1 \le n \le k$, and a real configuration U_0 satisfies

$$U_0 \in \mathfrak{U}'_{n-1}(B_{\mathrm{rad}} \varepsilon'),$$
$$\mathcal{M}^j U_0 = V_j^{(n)} \quad (j < n,\ c \in \Gamma_j^\flat),$$
$$|\mathcal{M}^n U_0(c) V_n^{(n)}(c)^{-1} - \mathbf{1}| \le \varepsilon' \quad (c \in \Gamma_n^{(n)}).$$

Let $j_n(\mathfrak{e})$ be the local index for the truncated sequence. Since $j_{n-1}(\mathfrak{e}) = \min\{j_n(\mathfrak{e}), n-1\}$, the index changes by at most one. In scaling-independent notation the regularity bounds are

$$|\partial U(p) - \mathbf{1}| < r L^{-2 j_n(p)}, \qquad |S_U(b)| < r L^{-3 j_n(b)}, \qquad S_U = \eta D_U^{\mathsf{T}} \partial U.$$

Consequently

$$U_0 \in \mathfrak{U}'_n(\alpha_0), \qquad \alpha_0 = L^3 B_{\mathrm{rad}} \varepsilon' \le c_{\mathrm{asc}}.$$

This includes the passage from depth zero to depth one. Use level-n lengths in what follows.

The solver requires a bound for the entire inner-star current $F'(0) = J_{\Omega_0}(U_0)$. At full bonds it is bounded by $\alpha_0 \mathsf{L}^{-3}$ because $\mathfrak{U}'_n(\alpha_0) \subset \mathfrak{U}_n(\alpha_0)$. If an inner bond is not full, it and its incident inner plaquettes have index zero and scale η. Each plaquette contribution is bounded by $\eta^{-3}|\partial U_0(p) - \mathbf{1}|$, and there are at most four contributions. Thus

$$\mathsf{j} := |F'(0)|_{(-3)} \le 4\alpha_0.$$

No plaquette outside the original domain is used.

Put $B_j = 0$ for $j < n$, and at a positively oriented terminal bond define

$$B_n(c) = \frac{1}{i} \log(V_n^{(n)}(c) \mathcal{M}^n U_0(c)^{-1}).$$

The real logarithm exists and

$$\beta := \sup |B_n| \le \tfrac{\pi}{2} \varepsilon' \le \alpha_0, \qquad e^{iB} M_{\mathbf{B}^{(n),\flat}} U_0 = V^{(n)}.$$

The nonlinear datum correction is exactly B. Indeed, at an outward row its additional factor is constructed from $B_{j-1} = 0$, so that factor is the identity. There are no edge rows in the reduced set. Moreover $K_5(\mathsf{j} + \beta) \le 5 K_5 \alpha_0 \le r_s \le r_0$ and $\beta < r_L$. The actual solver therefore provides a real solution with

$$\|A\|_X \le 5 K_5 \alpha_0, \qquad |D^* D A|_{(-3)} \le 15(1 + L^3) \Theta K_5 \alpha_0, \qquad R D^* A = 0.$$

Choose any real gauge h with $h(y) = v_j[A](y)^{-1}$ on every Λ_j. The top sets are disjoint, so these values extend to all bare sites. The solver's constraint and criticality statements give

$$U = (e^{i\eta A} U_0)^h \text{ critical in } C(\mathbf{B}^{(n),\flat}, V^{(n)}).$$

It remains to verify the regularity needed for the reset. The two solver bounds have the common upper bound

$$\alpha_2' = 5\max\{1, 4(1+L^3)\Theta\}K_5\alpha_0.$$

Conversion to the local star sizes costs L^2: $\alpha_2 = L^2\alpha_2' = k_a\alpha_0$. The definition of c_{asc} ensures

$$\alpha_2 \le \rho_{\mathrm{ec}}, \qquad C_{\mathrm{ec}}\alpha_2 \le 1, \qquad 4\alpha_2 + C_{\mathrm{ec}}(\alpha_0+\alpha_2) \le \tfrac{1}{2}, \qquad 2\alpha_0 + 3\alpha_2 \le \tfrac{1}{3}.$$

The regularity transfer (467), at $E = e^{i\eta A}U_0$, consequently gives

$$|\partial E(p) - \mathbf{1}| \le (\alpha_0+3\alpha_2)\eta^2 \mathsf{L}(p)^{-2}, \qquad |J(E)(b)| \le (2\alpha_0+3\alpha_2)\mathsf{L}(b)^{-3}.$$

For $r = 2\alpha_0 + 3\alpha_2 > 0$, the full-divergence conversion gives radius $r + r^2 < 2r$. Thus

$$E \in \mathfrak{U}_n'(K_9\alpha_0), \qquad U \in \mathfrak{U}_n'(K_9\alpha_0).$$

The second inclusion uses gauge invariance, and does not require a small extension of h. This is a preliminary bound; it has not yet restored the induction radius. Since

$$K_9\alpha_0 = K_9L^3B_{\mathrm{rad}}\varepsilon' \le a_{\mathrm{rad}}(\rho),$$

proposition 37.1 applies to the critical configuration U and the tested holonomies of $V^{(n)}$. It gives

$$U \in \mathfrak{U}_n'(B_{\mathrm{rad}}\varepsilon'),$$

which completes the ascent step.

We now iterate. At depth zero the reduced determining set consists of all inner bare bonds, so put $U^{(0)} = V_0^{(0)}$. Its plaquette errors are strictly less than ε'. At a full bond, subtract the identity from each of its four plaquettes in the two transported differences defining $S_{U^{(0)}}$. The identities cancel, and $|S_{U^{(0)}}| < 4\varepsilon'$. Hence $U^{(0)} \in \mathfrak{U}_0'(B_{\mathrm{rad}}\varepsilon')$.

Suppose $U^{(n-1)}$ has been constructed in the required fiber and with this radius. Agreement of the descended data gives $\mathcal{M}^j U^{(n-1)} = V_j^{(n)}$ on the retained rows for $j < n$. For a terminal bond c of level n, separation puts its fine double block inside Ω_{n-1}. Every bond needed for its average therefore belongs to the previous terminal frame, on which $\mathcal{M}^{n-1}U^{(n-1)} = V_{n-1}^{(n-1)}$. Locality and the descent mismatch give

$$\mathcal{M}^n U^{(n-1)}(c) = \mathcal{M}(V_{n-1}^{(n-1)})(c), \qquad |\mathcal{M}^n U^{(n-1)}(c) V_n^{(n)}(c)^{-1} - \mathbf{1}| \le \varepsilon'.$$

The ascent step supplies $U^{(n)}$. At $n = k$ this is the asserted configuration. Every step begins and ends with the same radius $B_{\mathrm{rad}}\varepsilon'$, so no loss accumulates with the depth of the sequence. □

37.3. **Uniqueness of the regular critical orbit.**

Proposition 37.3. Assume the separated-region setting with

$$\mathfrak{M} \ge M_{\mathrm{ps}}, \qquad R_1 \ge L, \qquad R_1\mathfrak{M} \ge w_S, \qquad \#\mathbb{T}^{(k)} \text{ in each direction} \ge 6LM_{\mathrm{ps}}.$$

There is $a_{\mathrm{orb}} > 0$, depending only on L, such that any two real critical configurations in

$$C(\mathbf{B}^\flat, V) \cap \mathfrak{U}_k'(\alpha), \qquad 0 < \alpha \le a_{\mathrm{orb}},$$

belong to the same residual gauge orbit. Here V is any real reduced datum. The constant is independent of the number of levels and of changes in the region sequence. ■

Proof. We first put two regular configurations with the same datum in a common chart. This part does not use criticality. We will prove that, for α sufficiently small, $U, U_0 \in C(\mathbf{B}^\flat, V) \cap \mathfrak{U}_k'(\alpha)$ admit a real A and a gauge g such that

$$U^g = e^{i\eta A}U_0, \qquad g = v_j[A] \text{ on } \Lambda_j, \qquad RD^*A = 0, \qquad \|A\|_X \le C_{\mathrm{sl}}\alpha, \qquad a_j(A) = 0 \text{ on } \Gamma_j^\flat. \tag{572}$$

We take $C_{\mathrm{sl}} \ge 1$ large enough that the previously proved slice estimate reads

$$\|A\|_X \le C_{\mathrm{sl}}(\alpha + 4\beta^\sharp)$$

when both configurations have regularity radius α, the chart lies in its stated small radius, $RD^*A = 0$, and the raw reduced rows satisfy $\sup|a_j(A)| \le \beta^\sharp$.

A preliminary gauge. Write $W_n = \mathcal{M}^n U$, $W_{0,n} = \mathcal{M}^n U_0$. The curvature propagation estimate makes their contour means well defined on every block used below. Define $g_k = \mathbf{1}$ on $\Omega_k^{(k)} = \Lambda_k$, and recursively, for $n = k-1, \dots, 0$, set

$$g_n = \mathbf{1} \quad \text{on } \Lambda_n,$$
$$g_n(x) = W_{0,n}(y,x)^{-1} g_{n+1}(y) W_n(y,x), \quad x \in B(y), \quad y \in \Omega_{n+1}^{(n+1)}.$$

The decomposition into these blocks and Λ_n is disjoint. At a block center both contour transports are the identity, so g_n extends g_{n+1}. Thus $\widehat{g} = g_0$ is defined on all bare sites, equals the identity at all original tops, and satisfies

$$(\mathcal{M}^n U^{\widehat{g}})(y,x) = W_{0,n}(y,x) \quad (y \in \Omega_{n+1}^{(n+1)},\ x \in B(y)). \tag{573}$$

This follows directly from covariance and the recursion.

We need a bound on bonds of every truncated frame, not only on the original reduced rows. Put

$$\delta_n(b) = |\mathcal{M}^n U^{\widehat{g}}(b) W_{0,n}(b)^{-1} - \mathbf{1}|.$$

On original interior and outward bonds $\delta_n = 0$: their endpoints are tops, and the original data agree. At an inward bond of level n, the bond lies in the corridor of an outward bond of level $n+1$. The latter mismatch is zero. The contour defect at the inner endpoint is zero by (573). At the outer endpoint, comparison with a contour of at most $3(L-1)/2$ fine bonds bounds the defect by $4\kappa_1^{\mathrm{SO}}\alpha$, since these fine bonds are interior datum bonds and both fine plaquette sizes are at most 2α. The corridor comparison in the proof of lemma 36.3 therefore gives

$$\delta_n(b) \le C_0 \alpha$$

with $C_0 = 4\kappa_\downarrow + 4\kappa_1^{\mathrm{SO}}$.

For a bond b with both endpoints in Ω_n, let $j_b \ge n$ be the largest level containing both endpoints. Applying the same comparison downwards gives

$$\delta_n(b) \le C_0\alpha + 4\kappa_\downarrow \alpha \sum_{i=n}^{j_b - 1} L^{2(i-j_b)}.$$

Indeed, when the endpoints lie in one block their contour defects vanish and the one-block estimate is at most $8\kappa_1^{\mathrm{SO}}\alpha$. When they lie in different blocks, their centers form a bond with the same value of j_b. The contour defects again vanish, and the extra corridor error is at most $4\kappa_\downarrow \alpha L^{2(n-j_b)}$ by curvature propagation. At level j_b one has an original interior or inward bond, which starts the induction. Since $\sum_{s\ge 1} L^{-2s} \le 2L^{-2}$, this proves

$$\delta_n(b) \le C_{\mathrm{pg}}\alpha \quad \text{for every inner level-}n\text{ bond meeting } \Omega_n, \qquad C_{\mathrm{pg}} = 4(1+2L^{-2})\kappa_\downarrow + 4\kappa_1^{\mathrm{SO}}. \tag{574}$$

Bonds with just one endpoint in Ω_n are original outward bonds and have zero mismatch. All the comparisons use carried double blocks; separation places their fine bonds in Ω_n. The constants thus do not depend on the depth. Set $U' = U^{\widehat{g}}$. It has the same original reduced datum as U and the same regularity radius.

Normalizing one level. We describe the operation used in the upward induction. At depth j, suppose

$$U'^{g_0} = e^{i\eta_j A_0} U_0, \qquad g_0 = v_i[A_0] \text{ on the lower tops } (i < j), \qquad g_0 = v_{j-1}[A_0] \text{ on the terminal tops.}$$

The residual factor $v_i[A_0]^{-1} g_0$ is therefore the identity on lower tops and $\mathsf{v}_{j-1}[A_0]^{-1}$ on terminal tops. Prescribe the induced gauge correction to be

$$r = \mathbf{1} \quad \text{on lower tops}, \qquad r = \mathsf{v}_{j-1}[A_0] \quad \text{on terminal tops.}$$

Every fine block of a terminal top lies in Ω_j. The mean-tower estimate at level $j-1$, followed by the one-step top-factor bound, gives

$$\sup|r-\mathbf{1}| \le (2\kappa/L)\|A_0\|_{X,\mathbf{B}^{(j)}}.$$

Here $\kappa = \kappa_{6,\mathsf{m}}$ is the one-step factor constant from proposition 30.2, enlarged if necessary. The prescribed-top correction supplies $w = e^{i\lambda}$, $A = A_0^w$ and $g = wg_0$ with

$$RD^*A = 0, \qquad g = v_i[A] \text{ on every depth-}j \text{ top},$$
$$|A|_{(-1)} \le k_1\|A_0\|_X, \qquad k_1 = 1 + 2C_{\mathrm{gc}}K_3(3+2\kappa/L).$$

We enlarge k_1 if necessary so that $k_1 \ge \max\{1, 2\kappa/L\}$. The estimate uses $|D^*A_0|_{(-2)} \le 3\|A_0\|_X$. The normalization is exact: the new residual factor is the prescribed r times the old one.

Suppose the reduced data of U' and U_0 at depth j differ by at most γ. Let $B_i(c)$ be the small logarithm of that discrepancy. The normalized chart identities give $a_i(c) = B_i(c)$ at interior bonds. If c is outward with outer end z, they give

$$e^{ia_i(c)} = \begin{cases} e^{iB_i(c)}W_{0,i}(c)\mathsf{v}(W_{0,i-1}, B_{i-1})(z)W_{0,i}(c)^{-1}, & z = c_+, \\ \mathsf{v}(W_{0,i-1}, B_{i-1})(z)^{-1}e^{iB_i(c)}, & z = c_-. \end{cases} \tag{575}$$

To verify this identity, use $\mathcal{M}^iU' = s_i(c_-)^{-1}e^{ia_i(c)}W_{0,i}(c)s_i(c_+)$ with $s_i = v_i[A]^{-1}g$. At the outer endpoint $s_i(z) = \mathsf{v}_{i-1}[A](z)^{-1}s_{i-1}(z) = \mathsf{v}_{i-1}[A](z)^{-1}$. The block $B(z)$ consists of interior rows at level $i-1$, on which $a_{i-1} = B_{i-1}$, so locality replaces this one-step factor by the one displayed in (575).

The logarithm estimate gives $|B_i| \le \pi\gamma/2$. The one-step factor has a logarithm of norm at most $\kappa\sup|B_{i-1}|$ on its block. Subadditivity of the real group angle in (575), and the small branch $|a_i| < \pi$, imply

$$\sup_{\Gamma_i^{(j),\flat}} |a_i| \le C_q\gamma, \qquad C_q = \tfrac{\pi}{2}(1+\kappa).$$

Both $e^{i\eta_j A}U_0 = U'^g$ and U_0 have regularity radius α at depth j. Lemma 30.10 therefore improves the chart to

$$\|A\|_{X,\mathbf{B}^{(j)}} \le C_{\mathrm{sl}}(\alpha + 4C_q\gamma). \tag{576}$$

This bound is independent of the size of the incoming chart, once that chart satisfies the correction caps.

Induction over the levels. Choose

$$k_2 = \max\{\pi C_{\mathrm{pg}},\ C_{\mathrm{sl}}(1 + 4C_qC_{\mathrm{pg}})\}.$$

Choose $\sigma > 0$ smaller than the common weighted chart radius, the gauge-correction radius and the slice radius, and so small that $2(L+\kappa)\sigma < 1/4$. Finally take $c_{\mathrm{cmp}} > 0$ within all reference and curvature thresholds already used, including that of proposition 14.1, and such that

$$k_1k_2L^2c_{\mathrm{cmp}} \le \sigma. \tag{577}$$

These are choices of constants depending only on L.

At depth zero, $\eta_0 = 1$ and every scale equals one. Define

$$A^{(0)}(b) = \frac{1}{i}\log(U'(b)U_0(b)^{-1}), \qquad g^{(0)} = \mathbf{1}.$$

For $\alpha \le c_{\mathrm{cmp}}$, (574) and the real logarithm estimate give $\|A^{(0)}\|_X \le \pi C_{\mathrm{pg}}\alpha \le k_2\alpha$: a first difference contains at most two values of $A^{(0)}$, transported by isometries. The chart is normalized at every depth-zero site.

Suppose a normalized pair $(A^{(j-1)}, g^{(j-1)})$ has been constructed with $\|A^{(j-1)}\|_X \le k_2\alpha$. In level-$j$ lengths put $A_0 = LA^{(j-1)}$. The dimensionless field is unchanged: $\eta_j A_0 = \eta_{j-1}A^{(j-1)}$. Only the new terminal local index changes, by one; thus

$$\|A_0\|_{X,\mathbf{B}^{(j)}} \le L^2 k_2 \alpha.$$

The prior terminal site set contains both the new level-$(j-1)$ tops and the level-j terminal tops. The mean objects depend on ηA, so the incoming pair has exactly the normalization required for the one-level operation.

The reference configurations remain in $\mathfrak{U}'_j(\alpha)$: truncating the local indices weakens the defining scale bounds. The geometry of each truncation satisfies the same operator thresholds, and the terminal torus has at least as many sites. For $j < k$, (574) gives $\gamma = C_{\mathrm{pg}}\alpha$ on its reduced frame. For $j = k$, the original reduced data agree, so $\gamma = 0$. Condition (577) permits the correction, and (576) gives $\|A^{(j)}\|_X \le k_2\alpha$ at intermediate levels and $\|A^{(k)}\|_X \le C_{\mathrm{sl}}\alpha$ at the last level. Thus the bound is restored after every rescaling.

Set $A = A^{(k)}$ and $g = g^{(k)}\widehat{g}$. The preliminary gauge is the identity on the original tops, so the final pair remains normalized there. At the final level all B_i vanish. On every outward block the factor in (575) is consequently the identity, and the small logarithmic branch gives $a_i(c) = 0$. Interior rows vanish as well. This proves (572).

Criticality and uniqueness. Now take $U_0 = U_a$, $U = U_b$, where both are critical in the same reduced fiber. Use (572), and choose a_{orb} so small that

$$a_{\mathrm{orb}} \le \min\{1, c_{\mathrm{cmp}}, c_8, c_6, L^{-2}a_{\mathsf{m}}\},$$
$$C_{\mathrm{sl}}a_{\mathrm{orb}} \le \min\{r_0, c_6\}, \qquad 4K_5 a_{\mathrm{orb}} \le r_0.$$

The choices of c_8, r_0 in proposition 30.9 ensure $3K_1L^3\epsilon^* \le 1/8$ and the applicability of lemma 30.6 throughout this ball. Thus the multiplier equivalence of lemma 30.8 applies at both A and zero.

The configuration $E = U_b^g = e^{i\eta A}U_a$ is critical for its induced datum by gauge covariance. The converse part of the criticality statement gives $F'(A) = K_A^*\omega_b$. Together with $RD^*A = 0$ and $a_j(A) = 0$, this makes (A, ω_b) a solution of the actual solver system with datum $B^\natural = 0$. The same argument at $A = 0$, using the criticality of U_a, gives a solution $(0, \omega_a)$ of that same system. Its forcing satisfies $\mathsf{j} = |F'(0)|_{(-3)} \le 4\alpha$ by the earlier inner-star calculation. The displayed caps ensure both admissibility of the solver and $\|A\|_X \le r_0$. Its uniqueness in that ball therefore gives $A = 0$.

It follows that $U_b^g = U_a$. The normalization in (572) gives $g = v_j[0] = \mathbf{1}$ at all tops. Every node of the reduced determining set is a top, so g is a residual gauge transformation. This proves the proposition. □

Corollary 37.4. In the geometry of proposition 37.2, put

$$a_{\mathrm{can}}(\rho) = \min\left\{a_{\mathrm{ex}}(\rho), \frac{a_{\mathrm{orb}}}{B_{\mathrm{rad}}C_\downarrow}, \frac{a_{\mathrm{rad}}(\rho)}{B_{\mathrm{rad}}C_\downarrow}\right\}.$$

Let V be real reduced data whose tested holonomies are defined and have distance less than ε from the identity, where $0 < \varepsilon \le a_{\mathrm{can}}(\rho)$. The critical configurations in $C(\mathbf{B}^\flat, V) \cap \mathfrak{U}'_k(a_{\mathrm{orb}})$ form one nonempty residual gauge orbit. This orbit lies in $\mathfrak{U}'_k(B_{\mathrm{rad}}\varepsilon)$. ■□

Proof. Proposition 37.2 supplies a critical configuration with radius $B_{\mathrm{rad}}C_\downarrow\varepsilon$. This is within both a_{orb} and $a_{\mathrm{rad}}(\rho)$. Apply proposition 37.1 once more, now using the original tested holonomy bound ε, to obtain radius $B_{\mathrm{rad}}\varepsilon$. Proposition 37.3 puts every critical configuration of radius a_{orb} in this orbit. Gauge invariance preserves the improved radius. □

37.4. **The regular pair supplied by the real construction.** Fix a power $m = m(L)$ of L with

$$m \ge \max\{\mathfrak{M}_{\mathrm{pr}}, L^2\}.$$

We increase the geometric thresholds in the real existence theorem so that $\mathfrak{M} \ge m$ and $\rho \ge Lm$. The sequence can then be regarded as separated with block parameter m and relative gap L. Indeed, every original block is a union of m-blocks, its complement is the same point set, and the original separation exceeds $Lm\xi_q$. The layers, determining rows, local scales and weighted distance are unchanged. In the regular-pair notation we take $t = k$ and $\xi = \eta$. The final lattice has at least $6Lm$ sites in each direction, which also implies the volume hypothesis of proposition 34.2.

With $B_{\mathrm{p}}, c_{\mathrm{p}}$ from (448), put $C_{\mathrm{P}} = 504L^2B_{\mathrm{p}}$. Choose $\varepsilon_{\mathrm{P}} > 0$, depending only on L, so that

$$\varepsilon_{\mathrm{P}} \le \tfrac{1}{3}, \qquad 504L^2m\varepsilon_{\mathrm{P}} \le c_{\mathrm{p}}, \qquad C_{\mathrm{P}}m\varepsilon_{\mathrm{P}} \le \min\{1, a_{\mathrm{nl}}, a_{\mathrm{pr}}\}. \tag{578}$$

Lemma 37.5. Suppose U_0 is real and regular at radius $0 < \varepsilon \le \varepsilon_{\mathrm{P}}$:

$$|\partial U_0(p) - I| < \varepsilon\eta^2\mathsf{L}(p)^{-2}, \qquad |J(U_0)(b)| < \varepsilon\mathsf{L}(b)^{-3},$$

at inner plaquettes and full bonds, respectively. With $\alpha = C_{\mathrm{P}}m\varepsilon$, the fixed witness (m, L) gives

$$(U_0, F'(0)) \in \mathfrak{P}_k^{\natural}((\Omega_q); \alpha, \alpha_1, \alpha) \qquad (\alpha_1 > 0).$$

Here F is the action summed over inner plaquettes. The assertion includes every cube in the natural class, whether or not it meets the lower of its two layers. ∎

Proof. Take $A' = 0$ in the definition of the class. At a plaquette based in layer q, one has $\mathsf{L}(p) \ge \xi_q$, giving the required curvature bound. At a full bond the action gradient is $F'(0)(b) = J(U_0)(b)$. An inner bond which is not full lies at the bare scale, as do all the corners of its inner plaquettes. Each such plaquette contributes at most $\varepsilon\eta^{-3}$ to $F'(0)(b)$: its field strength is bounded by $\varepsilon\eta^{-2}$, and the transpose difference contributes η^{-1}. There are at most four contributions. Thus, for the top y_b of the initial site,

$$\sup_b \xi_{y_b}^3 |F'(0)(b)| \le 4\varepsilon < \alpha.$$

This proves the current condition, including boundary bonds.

It remains to construct the local gauges. For a full bond b the quaternion decomposition of each plaquette gives

$$D_{U_0}^{\mathsf{T}}\partial U_0(b) = a(b)I + i\eta^2 J(U_0)(b), \qquad |a(b)| \le \eta^{-1} \max_{p \ni b} |\partial U_0(p) - I|^2.$$

Since $\mathsf{L}(p) \ge \mathsf{L}(b) \ge \eta$,

$$|D_{U_0}^{\mathsf{T}}\partial U_0(b)| < 2\varepsilon\eta^2\mathsf{L}(b)^{-3}. \tag{579}$$

The scalar part in this calculation is the sum of the two differences of $\frac{1}{2}\operatorname{tr}\partial U_0$; $1 - \frac{1}{2}\operatorname{tr} g = \frac{1}{2}|g - I|^2$ for $g \in SU(2)$ gives the stated bound.

Let $q \ge 1$ and let $\mathfrak{q} \subset \Omega_q$ be a cube of side $l\xi_q \le 12m\xi_q$, aligned with the m-partition. Enlarge it concentrically to side $(l + 4L)\xi_q$. The enlarged cube and its 2η-neighborhood lie in Ω_{q-1}, since

$$2L\xi_q + 2\eta \le (2L^2 + 2)\xi_{q-1} < Lm\xi_{q-1}.$$

Its side is at most $16m$, whereas half the torus side is at least $3Lm > 16m$. Thus the local gauge theorem applies after rescaling by ξ_q and extending U_0 by I outside Ω_0. That extension does not change any plaquette or difference used near the enlarged cube. The curvature and matrix-divergence parameters in these units are bounded by $2L^3\varepsilon$, and

$$2L^3 l\varepsilon \le 24L^3m\varepsilon \le (L/21)c_{\mathrm{p}} \le c_*.$$

Restricting the resulting gauge to $\mathfrak{q}$ gives $U_0^v = e^{i\eta A}$ and

$$\xi_q|A|, \ \xi_q^2|\nabla^\eta A| < 2B_*L^3l\varepsilon \le C_{\mathrm{P}}m\varepsilon = \alpha.$$

Only ordinary first differences occur here. A natural cube is contained in $\mathcal{L}_q \cup \mathcal{L}_{q+1} \subset \Omega_q$, so this applies to every required cube.

At level zero no enlargement is needed. The cube has at most $12m$ bare spacings in each direction. The comb gauge of lemma 14.4, with $s = 12m$, gives $U_0^v = e^{ia^0}$ and $|a^0| \le 12\pi m\varepsilon$; its logarithm is defined since $24m\varepsilon \le c_{\mathrm{p}} < 2$. For $A = \eta^{-1}a^0$,

$$\eta|A| \le 12\pi m\varepsilon < \alpha, \qquad \eta^2|\nabla^\eta A| \le 24\pi m\varepsilon < \alpha.$$

These estimates hold wherever the difference is defined from bonds in the cube. This completes the natural-class conditions. □

38. The canonical holomorphic background

For a real reference U_0, write $W_{0,j} = \mathcal{M}^j U_0$ and $V_0 = M_{\mathbf{B}^\flat} U_0$. The response of theorem K prescribes the straightened average. We first convert the actual datum into that coordinate. Choose $\kappa_{\mathrm{m}} \ge \max\{\kappa_{6,\mathsf{m}}, \kappa_{\mathrm{st}}, 2L\}$. Write R_{m} for the mean-Landau projection R of theorem K; this notation distinguishes it from the corner projection used in the real inhomogeneous solver. For an outward row $c \in \Gamma_j$ with outer endpoint z, put $\theta_z(B) = \mathsf{v}(W_{0,j-1}, B)(z)$, where its argument is restricted to the interior level-$(j-1)$ bonds in $B(z)$. Define $B_\natural = B$ at interior rows and, at outward rows, by

$$B_\natural(c) = \begin{cases} i^{-1}\log[e^{iB(c)}V_0(c)\theta_z(B)V_0(c)^{-1}], & z = c_+, \\ i^{-1}\log[\theta_z(B)^{-1}e^{iB(c)}], & z = c_-. \end{cases} \tag{580}$$

On a ball $\sup|B| < r_{\mathrm{tw}}$, with $r_{\mathrm{tw}} > 0$ depending only on L, this is holomorphic, real at real B, and

$$\sup|B_\natural| \le C_{\mathrm{tw}} \sup|B|, \qquad C_{\mathrm{tw}} \ge 1.$$

Indeed, $\theta_z(B) = e^{i\zeta_z(B)}$ with $|\zeta_z(B)| \le \kappa_{\mathrm{m}} \sup|B|$. Each logarithm is a product of two exponentials whose total exponent norm is at most $(1 + \kappa_{\mathrm{m}})\sup|B|$. Choose this less than $1/16$ and use $|\log(e^X e^Y)| \le s + 4s^2 \le 2s$, $s = |X| + |Y|$. This also proves reality and the bound. Since the twist fixes interior rows, $\theta_z(B_\natural) = \theta_z(B)$; substituting in (547) shows $(B_\natural)^\natural = B$ whenever both arguments belong to the indicated balls.

Proof of theorem E. The real reference. For real B, the value of the datum at a retained bond changes by at most $\beta = \sup|B|$. The averaged value on the other side of a tested plaquette is a product of three straightening factors and the old averaged bond. Each factor changes by at most $\kappa_{\mathrm{m}}\beta$; hence this averaged value changes by at most $3\kappa_{\mathrm{m}}\beta$. The four-factor product estimate for unitary matrices gives

$$|\partial(e^{iB}V_0) - \partial V_0| \le 4\max\{1, 3\kappa_{\mathrm{m}}\}\beta.$$

The local small-field hypothesis for each such average follows because its supporting plaquettes are tested interior plaquettes. Choose C_∂ at least the displayed coefficient and make the radii small enough that these averages remain defined and

$$\varepsilon_{\mathrm{G}}(\rho) + C_\partial r_{\mathrm{G}}(\rho) \le a_{\mathrm{can}}(\rho).$$

The real orbit assertion follows from corollary 37.4; proposition 8.3 supplies its unique axial representative. At $B = 0$ we obtain $U_0 \in \mathfrak{U}'_k(B_{\mathrm{rad}}\varepsilon_1)$. Increase the geometric floors to those of the fixed-witness construction and choose $B_{\mathrm{rad}}\varepsilon_{\mathrm{G}} \le \varepsilon_{\mathrm{P}}$. Then lemma 37.5 puts $(U_0, F'(0))$ in the class required by theorem K. Criticality and the reduced criticality–multiplier equivalence at $A = 0$ give

$$J_0 := F'(0) = Q^{\mathsf{T}}\lambda_0.$$

Thus all hypotheses of that response theorem have now been verified for the reference constructed here.

The response and its multiplier. Set

$$A(B) = \mathbf{H}(B_\natural), \qquad \lambda(B) = (I + H^{\mathsf{T}}\mathsf{C}'_{A(B)}{}^{\mathsf{T}})^{-1}H^{\mathsf{T}}F'(A(B)).$$

Choose $C_{\mathrm{tw}}r_{\mathrm{G}} < \varepsilon_{\mathrm{crit}}$. Then theorem K gives the field, row and gauge equations, holomorphy and reality. The response bound gives $\|A(B)\| \le C\beta$. The field norm here controls $\|A\|_{X,\mathbf{B}}$ with a factor

depending only on L: the scale of either endpoint of a bare bond or of a bond in one of its plaquettes differs from its local scale by at most one power of L, and a curl is a sum of first differences.

We record the multiplier estimate, since its initial value need not vanish. Group the inner plaquettes by their base cells and set

$$E_y(A,\delta) = \eta^3 \sum_{p \text{ over } y} \langle DA(p), D\delta(p)\rangle.$$

There are at most $3(\xi_y/\eta)^3$ plaquettes in this sum, and $DA = \nabla_\mu A_\nu - \nabla_\nu A_\mu$ there. Thus

$$|E_y(A,\delta)| \le 12\xi_y^{-1}\mathsf{g}_y(A)\mathsf{g}_y(\delta). \tag{581}$$

The calculation is bilinear, including for complex fields. The linear coefficient in the local star calculation of proposition 33.1, rescaled by ξ_y, gives $|\Delta' A|_y^\sharp \le C(L)\max_{N(y)}|A|$, where $\Delta' = F''(0) - D^{\mathsf{T}}D$. Indeed, its unscaled bound is $C_{\mathrm{a}}\alpha \sup|\widehat{A}|$, with $\alpha = L^2\alpha_0 \le 1/2$ and $\sup|\widehat{A}| \le L\max_{N(y)}|A|$. Taylor's formula and the exact transpose identity give

$$(F'(A) - J_0, \delta)_\eta = \sum_y E_y(A,\delta) + (\Delta' A + \mathcal{V}(A), \delta)_\eta.$$

Test a row c against a unit vector e and take $\delta = He_c$. The row pairing contributes ξ_c, while the energy bound contributes ξ_y^{-1}. The column estimate for H from proposition 34.2, the local action bounds and the last identity therefore bound the dual pairing by

$$C\sum_y \frac{\xi_c}{\xi_y} e^{-\delta_{\mathrm{reg}} d(y,y_c)} \max_{z\in N(y)} \|A\|_z \le C\|A\|.$$

The last step is precisely the scale-ratio summation in (532). Hence $|H^{\mathsf{T}}(F'(A) - J_0)|_\infty \le C\|A\|$. Also $|H^{\mathsf{T}}J_0|_\infty \le 8\hat{\mathsf{b}}\sup|J_0|^\sharp \le C\varepsilon_1$. The inverse in the formula for λ has norm at most two on this ball. Consequently

$$\sup|\lambda(B)| \le C(\varepsilon_1 + \beta), \qquad \lambda(0) = H^{\mathsf{T}}Q^{\mathsf{T}}\lambda_0 = \lambda_0.$$

The equation for the action gradient now gives

$$D^{\mathsf{T}}DA = (Q + \mathsf{C}'_A)^{\mathsf{T}}\lambda - J_0 - \Delta' A - \mathcal{V}(A). \tag{582}$$

The weighted column and local action bounds control its (-3) norm by $C(\varepsilon_1+\beta)$ on the whole ball.

Complex averages and the axial gauge. We verify definedness of the averages before applying gauge covariance. On a one-step double block, let $\mathcal{O}_c(\theta)$ mean that each contour family has relative diameter less than $\theta\omega_{\mathrm{c}}$ and each loop entering the outer logarithm lies within θ of I. These are open conditions. In the three coordinate templates of a double block, continuity at $(a^0,a) = (0,0)$ gives a radius $r_{\mathrm{c}}(L) > 0$ such that $e^{ia}e^{ia^0} \in \mathcal{O}_c(1/2)$ when $|a^0|, |a| < r_{\mathrm{c}}$. A comb gauge puts any real reference with plaquette deviations at most $r_{\mathrm{c}}/(4\pi L)$ into this form with $|a^0| \le r_{\mathrm{c}}/2$.

For a complex gauge g, relative contour products are conjugated at their initial site. Thus a condition number at most $\varkappa$ multiplies their diameter and the loop distance by at most $\varkappa$. The mean transforms in the same way: its logarithmic mean equation is conjugated, and uniqueness in its defining neighborhood identifies the result. It follows that the average of W^g is defined and equals $\mathcal{M}(W)^g$ whenever $W \in \mathcal{O}_c(1/2)$ and the two endpoint condition numbers are at most two.

The mean tower bounds give, inside a pillar of level j,

$$|a_n| \le 2L^{n-j}|A|_{(-1)}, \qquad \mathsf{v}_i = e^{iY_i}, \quad |Y_i| \le 2\kappa_{\mathrm{m}}L^{i-j}|A|_{(-1)}.$$

On a double block the same bounds hold with $j-1$ in place of j. Hence the accumulated factor $v_n = \mathsf{v}_0\cdots\mathsf{v}_{n-1}$ has condition number at most $4/3$ if $|A|_{(-1)} < \rho_{\mathrm{f}}$, for a sufficiently small $\rho_{\mathrm{f}}(L) > 0$. Shrinking the curvature and field radii also puts every straightened one-step configuration in $\mathcal{O}_c(1/2)$. Induction over the tower then proves, for every gauge h with condition number at most $3/2$,

$$\mathcal{M}^n((e^{i\eta A}U_0)^h) = \mathsf{x}_n^{hv_n}, \qquad \mathsf{x}_n = e^{ia_n}W_{0,n}. \tag{583}$$

At the induction step the endpoint factor is hv_n, whose condition number is at most two; the preceding one-step argument therefore establishes both definedness and the identity.

Use the parent $z(x)$ and level $n(x)$ in each pillar. Starting at its top, define

$$\begin{aligned} k(y) &= I && (y \in \Lambda_*),\\ k(x) &= k(z(x))[\mathsf{v}_{n(x)}(z(x))\cdots\mathsf{v}_{n(z(x))-1}(z(x))]^{-1}\mathsf{x}_{n(x)}(\Gamma_{z(x),x}), \\ h(x) &= k(x)v_{n(x)}(x)^{-1}. \end{aligned} \tag{584}$$

The second line applies only below the top. The parent has higher level, so the recursion is well defined. All factors are holomorphic and unitary at real A. Since U_0 is axial, the reference transport along each comb is I; moving its unitary factors to the right expresses the straightened comb as a product of exponentials. A comb has at most $3(L-1)/2$ bonds. Along an ascent its levels are distinct, and each index in an accumulated factor occurs at most once. The total norm of the exponents in $h(x)$ is therefore at most

$$[3(L-1)+2\kappa_{\mathrm{m}}]\sum_{i<j}L^{i-j}|A|_{(-1)} \le (\kappa_{\mathrm{m}}+5)|A|_{(-1)}.$$

Choose this less than $1/5$. Then $|h-I|<1/4$, $\mathrm{cn}(h)<3/2$, and $h(0)=I$. The recursion and (583) give

$$h = v_j^{-1} \text{ on } \Lambda_j, \qquad h(x) = h(z(x))(\mathcal{M}^{n(x)}(e^{i\eta A}U_0))(\Gamma_{z(x),x}).$$

Set $\Phi_{V_0}(B) = (e^{i\eta A(B)}U_0)^{h(B)}$. For an interior row both endpoint factors hv_j are I, and $a_j = B_\natural = B$ there. At an outward level-j row, the outer endpoint z has level $j-1$, so $h(z)v_j(z) = \mathsf{v}_{j-1}(z)$. The latter factor depends only on the interior rows in $B(z)$, where $a_{j-1} = B_{j-1}$; it is therefore $\theta_z(B)$. Substituting (580) cancels this factor on the appropriate side of the row. This proves $M_{\mathbf{B}^\flat}(\Phi_{V_0}(B)) = e^{iB}V_0$ for complex B, with all averages defined.

Identification at real points. For real B, the response and the gauge are real and unitary. The criticality conclusion of theorem K, followed by gauge invariance, makes $\Phi_{V_0}(B)$ critical in its actual fiber. The bounds on A and (582), together with the regularity conversion for the local action, put $e^{i\eta A}U_0$ in $\mathfrak{U}'_k(a_{\mathrm{orb}})$ after decreasing ε_{G} and r_{G} once more. The class is invariant under the real gauge h. The last identity for h is exactly the outside-node axial condition of proposition 8.3. Thus the unique regular orbit and its unique axial representative give $\Phi_{V_0}(B) = U^\circ(B)$. The identity theorem, applied to matrix entries on the complexification of the real row space, proves uniqueness of the holomorphic continuation.

Bounds. All preceding smallness conditions are finite inequalities between positive constants depending only on L and ρ; they can be met simultaneously. On the resulting whole ball, A, $D^{\mathsf{T}}DA$ and $h-I$ are uniformly bounded and vanish at zero. The Schwarz lemma on each complex line through zero therefore bounds them by $C\beta/r_{\mathrm{G}}$ in their stated norms.

For curvature, the local action estimate gives $|\partial(e^{i\eta A}U_0) - I| \le C\eta^2\mathsf{L}^{-2}$ throughout this ball. Conjugation by h costs at most $3/2$. The holomorphic map

$$B \longmapsto \eta^{-2}\mathsf{L}(p)^2[\partial\Phi_{V_0}(B)(p) - \partial U_0(p)]$$

is thus uniformly bounded and vanishes at zero. For a full bond the action-current comparison gives

$$|J(e^{i\eta A}U_0)(b)| \le \tfrac{4}{3}|F'(A)(b)|, \qquad F'(A) = \mathcal{Q}'(A)^{\mathsf{T}}\lambda.$$

The weighted column bound controls the latter by $C\mathsf{L}(b)^{-3}$. The current transforms by $J(E^h)(b) = \mathrm{Ad}(h(b_-))J(E)(b)$, as follows by cancellation of the endpoint factors in its plaquette formula. Hence

$$B \longmapsto \mathsf{L}(b)^3[J(\Phi_{V_0}(B))(b) - J(U_0)(b)]$$

is also uniformly bounded and vanishes at zero. Applying Schwarz to these two maps and adding the original regularity bound $B_{\mathrm{rad}}\varepsilon_1$ proves the last two estimates. Increase $K_{\mathrm{G}}(\rho)$ to cover all the finite constants above. □

Choose a regularity radius $\varepsilon_{\mathrm{v}} > 0$ and smaller datum radii $\bar{\varepsilon}, r_B > 0$ such that

$$\varepsilon_{\mathrm{v}} + \varepsilon_{\mathrm{v}}^2 \le a_{\mathrm{orb}}, \qquad \bar{\varepsilon} \le \varepsilon_{\mathrm{G}}(\rho), \qquad r_B \le r_{\mathrm{G}}(\rho), \qquad B_{\mathrm{rad}}(\bar{\varepsilon} + C_\partial r_B) \le \varepsilon_{\mathrm{v}}.$$

For V_0 with tested holonomies less than $\bar{\varepsilon}$ and real $\sup|B| < r_B$, the orbit in theorem E lies in $\mathfrak{U}'_k(\varepsilon_{\mathrm{v}}) \subset \mathfrak{U}_k(\varepsilon_{\mathrm{v}})$. Conversely, any critical configuration regular at radius ε_{v} lies in

$$\mathfrak{U}_k(\varepsilon_{\mathrm{v}}) \subset \mathfrak{U}'_k(\varepsilon_{\mathrm{v}} + \varepsilon_{\mathrm{v}}^2) \subset \mathfrak{U}'_k(a_{\mathrm{orb}}),$$

and hence in that orbit. Thus, at regularity radius ε_{v}, the conditional regular-critical set in definition 11.1 is exactly the orbit constructed here. Its axial representative is $U^\circ(B)$, and its conditional holomorphic continuation is the restriction of Φ_{V_0}, by the uniqueness in proposition 11.2. The whole ball $\sup|B| < r_B$ is its domain of defined complex averages, since the determining equation holds there. At real points the real and complex averages agree.

39. Local expansions of the background inverse

Keep the separated-sequence setting, scales and torus bound of section 27. We use the physical fifth system and the natural pairs of definition 27.1.

We use the pieces and local systems from the proof of theorem G. The following carriers record the data needed by each factor. For a cube $\mathsf{q} \in \mathsf{Q}_n^\natural$ and an integer $r \ge 0$, let q^{+r} be the union of the bare $\mathfrak{M}$-cubes contained both in Ω_0 and in the concentric cube of side $s(\mathsf{q}) + 2r\mathfrak{M}\xi_n$. A cell is a nonempty family of at most r_{w} cubes of $\mathsf{Q}^\natural$, connected by common points; its union is also denoted by X, and

$$X^\natural = \bigcup_{\mathsf{q}\in X} \mathsf{q}^{+5}.$$

The family is part of the cell. A tree walk $\omega = (X_0, \dots, X_m; p)$ has

$$p(i) < i, \qquad X_i^\circ \cap X_{p(i)}^\circ \ne \varnothing, \qquad p(i+1) \in \{i, p(i), p^2(i), \dots, 0\}.$$

Its length is m, its carrier is $[\omega] = \bigcup_i X_i^\natural$, and $d^{\mathrm{tr}}(\omega; y, z)$ is the infimum of the weighted lengths of connected finite unions of segments containing y, z and meeting every X_i.

For the piece κ of size n, use the following family for $X(\kappa)$: $\{\kappa^{[1]}\}$ for a regular deep piece; $\{\kappa^{[1]}, \widehat{\kappa}\}$ for a buffer piece, where $\widehat{\kappa}$ is its containing cube at level $n+1$; and the bare $\mathfrak{M}$-cubes contained in $\kappa^{[1]} \cap \Omega_0$ for a layer-zero piece. Set

$$\mathcal{N}(\kappa) = \{\kappa' : \kappa'^{[1]} \cap \kappa^{[1]} \cap \Omega_0 \text{ has nonempty interior}\}, \qquad X^{(2)}(\kappa) = \bigcup_{\kappa'\in\mathcal{N}(\kappa)} X(\kappa'),$$

with the union of the families. These are cells with at most 27 and $27N$ cubes, respectively, where N is the piece-overlap bound. Indeed, all the layer-zero cubes share a point with the central cube, and a buffer cube meets its containing cube. For a neighbour κ', the families of $X(\kappa')$ and $X(\kappa)$ meet at a common interior point of their defining regions. This proves connectedness of the union. The piece geometry also gives

$$\kappa^{[2]} \cap \Omega_0 \subset X(\kappa)^\natural, \qquad X(\kappa')^\natural \subset X^{(2)}(\kappa)^\natural \quad (\kappa' \in \mathcal{N}(\kappa)).$$

Here and below intersections of regions are taken cube by cube; all lattice sites in the corresponding point intersection belong to this region.

For fixed L, r_{w}, a cell overlaps at most $n_1 = n_1(L, r_{\mathrm{w}})$ cells. Indeed, cubes sharing a point have indices differing by at most two, and at each index their side lengths and corner positions have a bounded number of choices. A connected family of at most r_{w} cubes containing a fixed cube is therefore chosen from a bounded set of chains of length at most $r_{\mathrm{w}} - 1$. Put $n_{\mathrm{w}} = 4n_1$. There are at most n_{w}^m tree walks of length m with a prescribed first cell. The depth-first parent maps number at most 4^m: their successive depth drops are nonnegative integers with sum at most $m-1$, encoded by fewer than $2m$ binary symbols. Each subsequent cell has at most n_1 choices.

For a region $W \subset \Omega_0$, define $\mathfrak{P}_t^\natural((\Omega_n);\alpha_0,\alpha_1,\gamma_0) \restriction W$ by definition 27.1, with W in place of Ω_0, $\mathcal{L}_n \cap W$ in place of $\mathcal{L}_n$, and $\{\mathsf{q} \in \mathsf{Q}_n^\natural : \mathsf{q} \subset W\}$ in place of $\mathsf{Q}_n^\natural$. Thus the pair is given on the inner bare bonds of W and has one decomposition $\mathbf{U} = e^{i\xi A'}U$ there, with U real. The curvature bounds are required on inner plaquettes of W; the bounds on A', its covariant differences and $\mathbf{J}$ are required only on the bonds and triples lying in W. For every indicated cube q, the real gauge witness in that definition must have $\mathsf{q} \subset \square \subset W$. All bounds retain their original scale ξ_n and their original strict or non-strict inequalities. The real factor and its gauge witnesses need not extend beyond W. We call this the *local natural class*.

On a region $W \subset \Omega_0$, call a pair *piecewise local* if its restriction to $X(\kappa)^\natural$ lies in the local natural class for every piece whose carrier lies in W. This is an open set of the bond coordinates $(\mathbf{U},\mathbf{J})$. In fact, at a point of a local natural class keep its real factor U fixed and use a local logarithm of $\mathbf{U}U^{-1}$. The finitely many strict conditions on A' and $\mathbf{J}$ persist; the gauge conditions on U remain unchanged. Thus the local class is open, and the piecewise class is the finite intersection of its inverse images under restriction. Restrictions to smaller regions preserve the piecewise condition. A pair in the local natural class is piecewise local.

Use the constants of the regular parametrix as follows:

$$\delta = \delta_{\mathrm{reg}}, \qquad N = N_{\mathrm{G}}, \qquad B_4 = \max\{1, B_T/4\},$$
$$C_q = 2C_{\mathbb{C}}C_{\mathrm{G}}B_{\mathrm{loc}}e^{2\rho+8}, \qquad \rho = 3L+6,$$
$$q = q_{\mathfrak{M}} = C_q\mathfrak{M}^{-1}e^{-\delta\mathfrak{M}/(8L)}.$$

Here $B_T = 4B_{\mathrm{loc}}e^{3\delta}$, as in (430), and $0 < \delta \le 1/2$. Put

$$S = c_S(\delta/4) = 8L^3(2+16/\delta)^3, \qquad q_1 = Sq, \qquad C_j = 96LB_4NS.$$

We increase the lower bound on $\mathfrak{M}$ so that

$$\mathfrak{M} \ge 32L^4, \qquad R_1\mathfrak{M} \ge \mathsf{r}_S(\delta/4), \qquad \frac{\log L}{R_1\mathfrak{M}} \le \frac{\delta}{4}, \qquad Nq_1 \le \tfrac{1}{2}, \qquad q, q_1 \le \mathfrak{M}^{-9}.$$

The summation and scale-ratio conditions are (487); in particular $\sum_z e^{-\delta d(y,z)/4} \le S$. Further lower bounds will be imposed in the proof. All are bounds by constants depending only on L.

Proposition 39.1. For sufficiently small $a_{\mathrm{tr}} > 0$ and sufficiently large $\mathfrak{M}$, the following holds for natural pairs with radii $\alpha_0,\alpha_1,\gamma_0 \le a_{\mathrm{tr}}$. There are local holomorphic functions j_ϖ and local holomorphic operators $\mathbb{E}_\Xi$, defined below, such that

$$\mathsf{j}_* = \mathsf{H}^{\mathsf{T}}\mathbf{J} = \sum_\varpi \mathsf{j}_\varpi, \qquad \mathsf{j}^0 = \sum_{|\varpi|=0} \mathsf{j}_\varpi, \qquad |\mathsf{j}_* - \mathsf{j}^0|_{-3;y} \le 2C_j\gamma_0 Nq_1.$$

For the physical fifth system at the multiplier j_*,

$$\mathbb{E}_* = \mathbb{K}^{[5]}[\mathbf{U},\mathbf{J},\mathsf{j}_*]^{-1} = \sum_\Xi \mathbb{E}_\Xi.$$

Both sums converge absolutely. Each nonzero $\mathbb{E}_\Xi$ has a tree walk $\omega(\Xi)$ of cells with $r_{\mathrm{w}} = 27N$. For every such tree ω,

$$\sum_{\Xi:\,\omega(\Xi)=\omega} |\mathbb{E}_\Xi \iota g|_y \le B_{\mathrm{tr}}(C_{\mathrm{tr}}\mathfrak{M}^{-3})^{|\omega|} \sup_z e^{-\delta d^{\mathrm{tr}}(\omega;y,z)/2}|g|_z, \qquad B_{\mathrm{tr}} = 27N \cdot 8B_4. \tag{585}$$

The terms and this estimate are defined on the piecewise local class on $[\omega]$. The map ι and the seminorms may be taken at any continuation of the local configuration to Ω_0. The operators and their values on their stated input supports do not depend on that continuation. Restriction from a larger local domain gives the same terms, and restriction of a global pair gives the terms in the displayed inverse. ∎

Proof. The local inverse factors. For a piece κ, take the system on $\mathfrak{S}^\kappa$ with ambient region $W_\kappa = X(\kappa)^\natural$. The proof of lemma 27.6 applies to these local data. For a regular piece, $\kappa^{[2]} \in \mathsf{Q}_n^\natural$ and $\kappa^{[2]} \subset W_\kappa$. The local natural-class condition supplies the real gauge on a containing cube inside W_κ. Restrict it to $\kappa^{[2]}$. The logarithm of $e^{i\xi\,\mathrm{Ad}(u)A'}e^{i\xi A}$, as in that proof, has potential and first-difference bounds at scale $n+1$ at most $8L^2(\alpha_0+\alpha_1)$. Also $\Omega_0^\kappa \subset \kappa^{[3/2]}$, so the local region is interior to W_κ. For a layer-zero piece the one-layer inverse applies and the multiplier does not enter. The touching plaquettes in the Hessian have all corners in $\kappa^{[2]} \cap \Omega_0$. Thus enlarging the ambient region from W_κ does not change this local operator. All the other coefficients use bonds in its local sequence and row values with basepoints in $\kappa^{[5/6]}$. The same local decay and commutator estimates therefore hold.

More explicitly, for the lifted source cutoff $\theta_\kappa\star$, the maps

$$T_\kappa = \widetilde{h}_\kappa \mathbb{E}_\kappa \iota(\theta_\kappa \star \cdot), \qquad R_\kappa^\sharp = -\mathbb{C}_{\widetilde{h}_\kappa}\, \mathbb{E}_\kappa \iota(\theta_\kappa \star \cdot)$$

satisfy the estimates used in the parametrix proof: T_κ has constant $4B_4$, and $R_\kappa^\sharp$ has constant q, both at rate δ. The output support of the latter and the input support of both lie in $\kappa^{[5/6]}$; their separation is $\mathfrak{M}/(8L)$. This follows from

$$[\mathbb{K}_\kappa, \widetilde{h}_\kappa] = \iota \mathbb{C}_{\widetilde{h}_\kappa}, \qquad |\mathbb{C}_{\widetilde{h}_\kappa}\Psi|_y \le C\mathfrak{M}^{-1} \max_{d^\kappa(y,z)\le\rho_0} |\Psi|_z$$

and the local inverse bound at rate 2δ, by the same separated-support composition argument. The identities $\theta_\kappa\iota = \iota(\theta_\kappa\star)$ and $R_\kappa\iota = \iota R_\kappa^\sharp$ are unchanged.

These assertions hold simultaneously at any continuation of the local configuration. Indeed, every transport in a source-cutoff commutator, in $\iota R_\kappa^\sharp$, or in a difference of $T_\kappa g$ has both endpoints within $\xi_{n+1} + 2\xi \le s_n/6$ of $\kappa^{[5/6]}$; hence it lies in $\kappa^{[1]} \cap \Omega_0 \subset W_\kappa$. The source seminorm contains no transport. Consequently no comparison of unrelated configurations is used in composing the factors.

For $\varpi = (\kappa_0, \dots, \kappa_m)$, set

$$E_\varpi = \widetilde{h}_{\kappa_0} E_{\kappa_0} \theta_{\kappa_0} R_{\kappa_1} \cdots R_{\kappa_m}.$$

The preceding identities give, with $y_0 = y$,

$$|E_\varpi \iota g|_y \le 8B_4 q^m \sup e^{-\delta \sum_{i=0}^m d(y_i, y_{i+1})} |g|_{y_{m+1}}, \tag{586}$$

where $y_i \in \kappa_{i-1}^{[5/6]} \cap \kappa_i^{[5/6]}$ for $1 \le i \le m$, and $y_{m+1} \in \kappa_m^{[5/6]}$, all points being tops. The operator vanishes unless these intersections contain tops. Its output has basepoints in $\kappa_0^{[2/3]}$, and it uses the input only at basepoints in $\kappa_m^{[1/12]}$. These support assertions follow directly from the first and last cutoffs.

The entries of each local system are holomorphic on the open set of triples: the derivatives and Hessian are polynomial in bond transports, while the averaging coefficients are Taylor coefficients of the holomorphic averaging chart. The inverse is holomorphic where its determinant is nonzero. The finite intersection of these open sets for the pieces of ϖ contains all the prescribed piecewise local data. Thus E_ϖ is holomorphic there. This argument uses the local operators themselves, so restriction to a smaller carrier or to a global pair gives the same factors.

The multiplier terms. In this paragraph take all factors at multiplier zero and set $H_\varpi = [E_\varpi^{[5]}]_{A,8}$. Define $\mathsf{j}_\varpi(c)$ by

$$\xi_c^3 \langle e, \mathsf{j}_\varpi(c)\rangle = \xi^3 \sum_b \langle H_\varpi e_c(b), \mathbf{J}(b)\rangle,$$

where the sum is over bonds with basepoint in $\kappa_0^{[2/3]}$. Their endpoints lie in $X(\kappa_0)^\natural$, so every value of $\mathbf{J}$ is given. The nondegenerate bilinear form determines a unique element. It is holomorphic, and

vanishes unless the basepoint of c lies in $\kappa_m^{[1/12]}$. Put

$$d(\varpi;c)=\inf_{y_i\in\kappa_{i-1}^{[5/6]}\cap\kappa_i^{[5/6]}}\sum_{i=1}^{m}d(y_i,y_{i+1}),\qquad y_{m+1}=\mathsf{y}(c),$$

with value zero for $m=0$. Then

$$\xi_c^3|\mathsf{j}_\varpi(c)|\le N^{-1}C_j\gamma_0q_1^m e^{-\delta d(\varpi;c)/2}. \tag{587}$$

Here is the weighted summation. The eighth-slot source e_c has size $\xi_c|e|$, while the field component has weight ξ_y. There are at most $3(\xi_y/\xi)^3$ bare bonds at y, and $|\mathbf{J}(b)|\le\gamma_0\xi_y^{-3}$. Thus (586) gives

$$|\xi_c^3\langle e,\mathsf{j}_\varpi(c)\rangle|\le 24B_4\gamma_0q^m|e|\sum_{y_0,\dots,y_m}\frac{\xi_c}{\xi_{y_0}}e^{-\delta\sum_{i=0}^m d(y_i,y_{i+1})}.$$

The scale-ratio estimate bounds the quotient by $L\exp(\delta\sum_i d(y_i,y_{i+1})/4)$. Retain rate $\delta/2$ for $d(\varpi;c)$ and sum the remaining rate $\delta/4$, successively, over the $m+1$ tops. Each sum is at most S. Finally $|Z|\le 2\sup_{|e|\le1}|\langle e,Z\rangle|$ gives (587) with the displayed C_j.

For fixed c, at most N^{m+1} nonzero sequences of length m occur, by the terminal support and the piece-neighbour bound. This proves absolute convergence, $|\mathsf{j}_*|_{-3;y}\le 2C_j\gamma_0$, and the asserted tail bound. Summing the already proved inverse series and taking its weighted transpose identifies the sum with $\mathsf{H}^{\mathsf{T}}\mathbf{J}$. Locally, at a row c in $\kappa^{[5/6]}$, define

$$\mathsf{j}^0(c)=\sum_{\kappa'\colon \mathrm{base}(c)\in\kappa'^{[1/12]}}\mathsf{j}_{(\kappa')}(c).$$

All these κ' are in $\mathcal{N}(\kappa)$; their carriers are in $X^{(2)}(\kappa)^\natural$, and at most N occur. Therefore this is a holomorphic local function satisfying the sharper bound $\xi_c^3|\mathsf{j}^0(c)|\le C_j\gamma_0$.

The vertex and the resolvent. For c above level zero let B_c be the symmetric bilinear coefficient defined by $C^{(2)}_{\mathrm{phys}}(A)(c)=\xi_cB_c(A,A)$. Here the physical remainder is $\xi_c^{-1}\mathsf{C}(A)(c)$, where C is the dimensionless remainder of eq. (512). Consequently

$$\mathsf{C}^{(2)}(A)(c)=\xi_c^2B_c(A,A).$$

Polarizing the quadratic action gives

$$\xi^3\langle X,\Delta^{(2)}[\mathsf{j}']A(b)\rangle=2\sum_c\xi_c^4\langle B_c(X_b,A),\mathsf{j}'(c)\rangle.$$

Only bonds in the double block Y_c enter the c-summand. With $F=\Delta^{(2)}[\mathsf{j}'](A-Dv)$, define $\mathbb{W}[\mathsf{j}']\Psi$ to have components $f_1=F, X_5=-F$ and all other components zero. Thus

$$\iota\mathbb{W}[\mathsf{j}']\Psi=(F,0,0,0,-D^{\mathsf{T}}F,0,0,0),\qquad \mathbb{K}[\mathsf{j}+\mathsf{j}']=\mathbb{K}[\mathsf{j}]-\iota\mathbb{W}[\mathsf{j}'].$$

Let $\mathrm{Pos}(c)$ be the positions of the bare sites in its double block Y_c. For an interior row these are its two endpoints; for an outward row they are its inner endpoint and the L^3 tops in its outer block. Their scales lie between ξ_c/L and ξ_c, and their mutual distances are at most $\rho_0=3L+6$. A top belongs to at most twelve such sets: six incident rows at its own level, and six at the next level incident to its unique containing block. The vertex obeys

$$|\mathbb{W}[\mathsf{j}']\Psi|_y\le C_{\mathrm{vert}}\max_{c\colon y\in\mathrm{Pos}(c)}\xi_c^3|\mathsf{j}'(c)|\max_{\substack{c\colon y\in\mathrm{Pos}(c)\\ z\in\mathrm{Pos}(c)}}|\Psi|_z,\qquad d(y,z)\le\rho_0. \tag{588}$$

To verify the estimate, write $\beta_{bb'}(X,X')=B_c(X_b,X'_{b'})$. Use the constant C_Q of eq. (512). Divide its difference estimate for $K_{sA}-K_0$ by $|s|$ and let $s\to0$. Applied to an input X_b, it gives

$$\left|\frac{d}{ds}\right|_{s=0}K_{sA}(c,b)X\Big|\le C_Q\xi^3\xi_c^{-1}|X|\sup_{\mathsf{P}_c}|A|.$$

The derivative on the left is $2\xi_c^2 B_c(X_b, A)$. Division by ξ_c^2, rather than by ξ_c, gives

$$|2B_c(X_b, A)| \le C_Q(\xi/\xi_c)^3 |X| \sup |A|.$$

Choose the phases of the components of $A(b')$ separately in a fixed Pauli basis. The resulting sum of the absolute coordinate coefficients gives $\sum_{b'} |\beta_{bb'}| \le (27/2) n_{\mathfrak{g}}^2 C_Q (\xi/\xi_c)^3$, where $n_{\mathfrak{g}}$ bounds the fixed finite-dimensional coordinate norms. In the polarized identity the factor $\xi^{-3}\xi_c^4(\xi/\xi_c)^3 = \xi_c$. At most twelve rows contribute at a given top, and within each $\mathrm{Pos}(c)$, $\xi_c/L \le \xi_z \le \xi_c$. Applying this to $A - Dv$ proves (588), for example with $C_{\mathrm{vert}} = 1296 n_{\mathfrak{g}}^2 L^3 C_Q$. The same formula proves that the vertex is local to its active double blocks and holomorphic. In particular an active row based in $\kappa'^{[1/12]}$ has $\mathsf{Y}_c \subset \kappa'^{[5/12]} \cap \Omega_0 \subset X(\kappa')^\natural$. No extension enters its coefficients or its nonzero transports.

Choose a_{tr} so that $2C_j a_{\mathrm{tr}}$ is within the local-inverse radius. Write $\delta\mathsf{j} = \mathsf{j}_* - \mathsf{j}^0$ and $\mathbb{E}_0 = \mathbb{K}[\mathsf{j}^0]^{-1}$. By the tail estimate and (588), $\mathbb{W}[\delta\mathsf{j}]\mathbb{E}_0\iota$ has decay rate δ and norm at most

$$2C_{\mathrm{vert}} C_j \gamma_0 N q_1 e^{\rho_0} B_5, \qquad B_5 = 16 N B_4.$$

Increase $\mathfrak{M}$ to make this at most $1/2$. The resolvent identity then gives the convergent series

$$\mathbb{E}_*\iota = \mathbb{E}_0\iota \sum_{n\ge 0} \big(\mathbb{W}[\delta\mathsf{j}]\mathbb{E}_0\iota\big)^n.$$

Decorations and tree decay. For $\Xi = (\varpi^0, \varpi'_1, \varpi^1, \dots, \varpi'_n, \varpi^n)$, with $|\varpi'_l| \ge 1$, define

$$\mathbb{E}_\Xi = E_{\varpi^0}\iota \mathbb{W}[\mathsf{j}_{\varpi'_1}] E_{\varpi^1} \cdots \iota \mathbb{W}[\mathsf{j}_{\varpi'_n}] E_{\varpi^n}.$$

The trunk factors are evaluated at the local j^0 on all their row supports, and zero elsewhere. List first all their $X^{(2)}$-cells in order. Then list the branches $\varpi'_n, \dots, \varpi'_1$, each in reverse order, using their X-cells. Consecutive trunk cells have the preceding trunk cell as parent; the first cell of branch l has the last trunk cell of ϖ^{l-1} as parent, and the other branch cells have the preceding branch cell as parent. This defines $\omega(\Xi)$.

Every factor is defined and holomorphic on the piecewise local class on its carrier. The $X^{(2)}$-cells contain all the extra one-piece carriers needed for j^0. The vertex uses only the last branch piece and its double blocks. This proves the stated local realization of $\mathbb{E}_\Xi$. If the product is nonzero, consecutive pieces within every branch or trunk meet at a top. At a vertex, a contributing active row has its position in the interiors of the enlarged last preceding trunk piece, first succeeding trunk piece and last branch piece. Indeed, its double block has diameter $2\xi_c < s_n/3$ at each of the relevant piece scales; the cutoff supports are $[1/12]$ and $[2/3]$. Thus the two trunk $X^{(2)}$-cells overlap, and the first branch cell is contained in and overlaps its prescribed parent. The parent order is depth first by construction, so this is a tree walk.

Write $m_l = |\varpi^l|$, $m'_l = |\varpi'_l|$. Iterating (586), (587) and (588) gives

$$|\mathbb{E}_\Xi \iota g|_y \le 8B_4 \prod_{l=0}^{n} q^{m_l} \prod_{l=1}^{n} (C_{\mathrm{br}}\gamma_0 q_1^{m'_l}) \sup e^{-\delta\Sigma_{\mathrm{tr}} - \frac{\delta}{2}\sum_l d(\varpi'_l; c_l)} |g|_z, \qquad C_{\mathrm{br}} = 8B_4 C_{\mathrm{vert}} e^{\rho_0} C_j.$$

Here Σ_{tr} is the sum of the successive trunk distances, including the jumps across vertices. Each such jump is at most ρ_0; inserting its exponential cost explains the factor e^{ρ_0} in C_{br}. Join successive tops by segment unions of weighted length arbitrarily close to their distance, and join each vertex top to the position of its active row. Attach the branch chains at that position. Their connected union meets every cell of $\omega(\Xi)$; the condition $m'_l \ge 1$ ensures that the first cell of each branch is met. Subadditivity of weighted length gives

$$d^{\mathrm{tr}}(\omega(\Xi); y, z) \le \Sigma_{\mathrm{tr}} + \sum_l d(\varpi'_l; c_l) + n\rho_0.$$

The exponential above is consequently bounded by $e^{n\rho_0/2}e^{-\delta d^{\mathrm{tr}}/2}$. Moreover

$$|\omega(\Xi)| = \sum_{l=0}^{n} m_l + \sum_{l=1}^{n} m'_l + 2n, \qquad q_1^{m'} \le \mathfrak{M}^{-3(m'+2)} \quad (m' \ge 1).$$

Together with $q^{m_l} \le \mathfrak{M}^{-3m_l}$ and $\gamma_0 \le 1$, this proves

$$|\mathbb{E}_\Xi \iota g|_y \le 8B_4\big(e^{\rho_0}\max\{1, C_{\mathrm{br}}\}\mathfrak{M}^{-3}\big)^{|\omega|} \sup_z e^{-\delta d^{\mathrm{tr}}(\omega;y,z)/2}|g|_z.$$

For a fixed tree with m edges, at most $2^m(27N)^{m+1}$ decorations give that tree. There are 2^m ways to cut its ordered list into the trunk and reversed-branch blocks. A prescribed $X(\kappa)$ has at most 27 preimages; a prescribed $X^{(2)}(\kappa)$ has at most $27N$. For the latter claim the family contains the distinct cube $\kappa^{[1]}$ for a regular piece and κ for a layer-zero piece, which determines the piece. Thus one may take

$$C_{\mathrm{tr}} = 54Ne^{\rho_0}\max\{1, C_{\mathrm{br}}\}$$

in (585). Choose the last lower bound on $\mathfrak{M}$ so that $n_{\mathrm{w}}C_{\mathrm{tr}}\mathfrak{M}^{-3} \le 1/2$, with $r_{\mathrm{w}} = 27N$. There are finitely many first cells, so the tree bound gives absolute convergence in each output seminorm. These seminorms separate the points of the finite-dimensional system, and ι is onto, so the operator series converges absolutely. Expanding $\mathbb{E}_0$ and $\delta\mathsf{j}$ in the resolvent series now gives exactly these decorated products. Absolute convergence justifies the rearrangement and identifies their sum with $\mathbb{E}_*$. □

40. Fluctuation covariances and their local expansion

We measure lengths at level k, where $0 \le k < K$, so that $\xi = L^{-k}$ and $\xi_n = L^{n-k}$. Let $(\Omega_n)_{n\le k}$ be a separated sequence with parameters $\mathfrak{M}, R_1$, and let

$$\Omega_{k+1} = \bigcup_{y\in\Theta'} \Delta(y) \subset \Omega_k, \qquad \mathrm{dist}_\infty(\Omega_{k+1}, \Omega_k^c) > R_1\mathfrak{M},$$

where Ω_{k+1} is a union of $\mathfrak{M}$-cubes at level $k+1$. Write $\mathbf{B}$ and $\mathbf{B}^+$ for the determining sets of the shorter and longer sequences, with all their bonds. The pair $(\mathbf{U}, \mathbf{J})$ belongs to the natural regular class of the longer sequence, with radii $\alpha_0, \alpha_1, \gamma_0$.

At level k the spacing is 1. For a coarse bond $c = \langle c_-, c_- + \epsilon L e_\mu\rangle$, put $h = (L-1)/2$. Its *pivot bond* is

$$b_0(c) = \langle c_- + \epsilon h e_\mu, c_- + \epsilon(h+1)e_\mu\rangle.$$

It has the midpoint of c, and reversal reverses the pivot. Write P_c for the inner bonds of the double block $B(c_-) \cup B(c_+)$. Write $\mathrm{Ax}(B(y))$ for the positive bonds of the coordinate tree $\mathcal{T}(y)$ defined before definition 20.1.

Lemma 40.1. For coarse bonds c, c', one has

$$b_0(c') \in \mathsf{P}_c \quad\Longleftrightarrow\quad c' \in \{c, \bar{c}\}.$$

The pivot of a positive coarse bond is positive. Consequently, for a set Γ of positive coarse bonds, the pivots $b_0(c)$, $c \in \Gamma$, are pairwise distinct, and $b_0(c)$ is the only one of them in P_c. ■

Proof. The endpoints of $b_0(c')$ lie in the distinct blocks $B(c'_-)$ and $B(c'_+)$. If this pivot belongs to P_c, both endpoints lie in $B(c_-) \cup B(c_+)$. The blocks partition the fine lattice, so $\{c'_-, c'_+\} = \{c_-, c_+\}$. Thus $c' = c$ or $c' = \bar{c}$. Conversely, the pivot of either orientation has both endpoints in the double block of c.

The explicit pivot formula gives it the direction and orientation of its coarse bond. If $c, c' \in \Gamma$ and $b_0(c') \in \mathsf{P}_c$, the alternative $c' = \bar{c}$ is excluded by positivity. Hence $c' = c$. In particular equality of two selected pivots implies equality of their coarse bonds. □

Put $\Theta = B(\Theta')$, and let Γ' consist of the positive level-$(k+1)$ bonds with an end point in Θ'. Write $\mathfrak{X}(\Theta)$ for the level-k bonds with an end point in Θ, and $\check{\mathfrak{X}}(\Theta)$ for this set with the pivot bonds $b_0(c)$, $c \in \Gamma'$, and their reverses removed. For any subset $\Theta'' \subset \Theta'$, set

$$Y = \check{\mathfrak{X}}(B(\Theta'')), \qquad Y^{\mathrm{ax}} = Y \setminus \bigcup_{y \in \Theta'} (\mathrm{Ax}(B(y)) \cup \overline{\mathrm{Ax}(B(y))}).$$

The pivots removed in the definition of Y are those associated with the coarse bonds having an end point in Θ''. Nevertheless $Y \subset \check{\mathfrak{X}}(\Theta)$: a pivot associated with any other coarse bond has neither end point in $B(\Theta'')$. No cube condition is imposed on Θ''. All spaces in this section use positive bonds as coordinates; their bilinear pairings and kernels use counting measure.

Let $V = \mathcal{M}^k \mathbf{U}$. The following one-step maps use the plain average at V:

$$\begin{aligned} Q_V B(c) &= \frac{1}{iL} \frac{d}{dt}\Big|_0 \log(\mathcal{M}(e^{itB}V)(c)\mathcal{M}(V)(c)^{-1}), \\ \widetilde{Q}_V(B)(c) &= \frac{1}{i} \log(\mathcal{M}(e^{iB}V)(c)\mathcal{M}(V)(c)^{-1}), \\ C_V^{(2)}(B_1, B_2) &= \tfrac{1}{2}\partial_s \partial_t \Big|_{s=t=0} \widetilde{Q}_V(sB_1 + tB_2). \end{aligned}$$

If $\kappa_V(c)$ is the coefficient of $B(b_0(c))$ in $Q_V B(c)$, the elimination map C_V keeps the nonpivot coordinates and sets

$$(\mathsf{C}_V \check{B})(b_0(c)) = -\kappa_V(c)^{-1} (Q_V \check{B}^0)(c).$$

Thus $Q_V \mathsf{C}_V = 0$. Here and below a superscript 0 on a field means extension by zero. Let

$$\mathsf{b}_V(B)(y,x) = \frac{1}{i} \frac{d}{ds}\Big|_0 \log((e^{isB}V)(y,x)V(y,x)^{-1}), \qquad x \in B(y),$$

where the contour variable is the one used in the mean chart, and define

$$(B_1, \mathsf{G}_V^{(2)} B_2) = \sum_{y \in \Theta'} \sum_{x \in B(y) \setminus \{y\}} \langle \mathsf{b}_V(B_1)(y,x), \mathsf{b}_V(B_2)(y,x) \rangle.$$

The right inverse h_V of $LQ_V = \widetilde{Q}_V'(0)$ is supported on the pivots and has value $L^{-1}\kappa_V(c)^{-1}\nu(c)$ at $b_0(c)$. Put $\mathsf{D}_V^{(2)} = h_V C_V^{(2)}$.

The extension J maps a field on $\mathfrak{X}(\Theta)$ to the row space of $\mathbf{B}$. These are interior top rows, with $\xi_c = \mathsf{a}(c) = 1$, so its transpose is restriction. For a row field j_1, define the symmetric operator $\mathsf{E}^{(2)}[j_1]$ by

$$(B_1, \mathsf{E}^{(2)}[j_1] B_2) = 2(\mathsf{J}\mathsf{D}_V^{(2)}(B_1, B_2), j_1)_{\mathbf{B}}.$$

The operators $Q, \mathsf{G}_1, \mathsf{H}_1$ in the next display are the straightened operators of the shorter sequence. With $j_* = \mathsf{H}^{\mathsf{T}}\mathbf{J}$ and $j_{1*} = \mathsf{H}_1^{\mathsf{T}}\mathbf{J}$, set

$$\Delta_k = \mathsf{J}^{\mathsf{T}} (Q \mathsf{G}_1 Q^{\mathsf{T}})^{-1} \mathsf{J} - 1 - \mathsf{E}^{(2)}[j_{1*}]. \tag{589}$$

For $\mathsf{Y} = Y$ or Y^{ax} let $\mathsf{C} = \mathsf{C}_V \chi_{\mathsf{Y}}^{\mathsf{T}}$, where $\chi_{\mathsf{Y}}^{\mathsf{T}}$ extends by zero, and put

$$T_Y = \mathsf{C}^{\mathsf{T}} (\mathsf{G}_V^{(2)} + \Delta_k) \mathsf{C}, \qquad T_Y^{\mathrm{ax}} = \mathsf{C}^{\mathsf{T}} \Delta_k \mathsf{C}.$$

The map C in the two formulas has domain Y and Y^{ax}, respectively. We denote their inverses by $C^{(k)}(Y)$ and $\widetilde{C}^{(k)}(Y)$. Formula (589) defines the form used here, including away from a critical pair. In particular its first term uses the inverse of $Q\mathsf{G}_1 Q^{\mathsf{T}}$.

Theorem L. There are constants $a_{\mathrm{cv}}, \delta_{\mathrm{cv}}, C_{\mathrm{cv}}, B_0, r_{\mathrm{cv}}^{\mathrm{sep}} > 0$ and a power M_{cv} of L, depending only on L, with the following property. Suppose $\mathfrak{M}, R_1$ are powers of L,

$$\mathfrak{M} \geq M_{\mathrm{cv}}, \quad R_1 \geq L, \quad R_1 \mathfrak{M} \geq r_{\mathrm{cv}}^{\mathrm{sep}}, \quad \ell_k \geq 2L\mathfrak{M}, \quad 0 < \alpha_0, \alpha_1, \gamma_0 \leq a_{\mathrm{cv}}.$$

All the maps above are defined, every $\kappa_V(c)$ is invertible, and T_Y, T_Y^{ax} are invertible. Both covariance operators are symmetric for their bilinear pairings. Either covariance, denoted by $\mathcal{C}$, has a kernel satisfying

$$|\mathcal{C}(b,b')| \le B_0 e^{-\delta_{\mathrm{cv}}|b_- - b'_-|_1}. \tag{590}$$

Its operator norms for the supremum norm, the ℓ^2 norm of matrix norms, and the Hermitian norm of bond fields are at most $B = 6B_0\sigma(\delta_{\mathrm{cv}})^3$.

There is an absolutely convergent expansion

$$\mathcal{C} = \sum_{\omega} \mathcal{C}_\omega$$

over tree walks of cells of the longer sequence. Their families have at most $r_{\mathrm{cv}} = 27N^2$ cubes, where N is the overlap constant of the regular pieces. If $\omega = (X_0, \dots, X_m)$, and X_t is the last cell of its designated trunk, then $\mathcal{C}_\omega(b,b') = 0$ unless $b_- \in X_0$ and $b'_- \in X_t$. Each term is a holomorphic function of the data on its carrier $[\omega]$, on the open piecewise local class defined in the proof. This class contains the local natural class of the longer sequence; restriction of a global pair gives the corresponding global term. Uniformly on that class,

$$|\mathcal{C}_\omega(b,b')| \le B_0 (C_{\mathrm{cv}}\mathfrak{M}^{-1/2})^m \mathfrak{M}^{-m/2} e^{-\delta_{\mathrm{cv}} d^{\mathrm{tr}}_{\mathbf{B}^+}(\omega; y(b), y(b'))/2}, \tag{591}$$

where $y(b) = b_-$ when $b_- \in \Theta$, and $y(b) = b_+$ otherwise. There are at most n_{cv}^m such tree walks with a prescribed first cell, for a constant $n_{\mathrm{cv}} = n_{\mathrm{cv}}(L)$, and the constants can be chosen so that

$$M_{\mathrm{cv}} \ge (2C_{\mathrm{cv}} n_{\mathrm{cv}})^2, \qquad \sum_{|\omega|=m} |\mathcal{C}_\omega(b,b')| \le 2^{-m-1} B_0 e^{-\delta_{\mathrm{cv}}|b_- - b'_-|_1}.$$

Every partial sum over a set of tree walks satisfies (590) and the same three norm bounds. The constants are independent of the number of layers, their depths, Y, the volume, and the pair. ■□

Proof. We first establish a gap at the flat pair. We then append the fluctuation coordinate to the system already inverted in theorem G. Its local inverse has the same cutoff expansion. The final expansion inserts the two current multipliers into that system. Constants denoted by $C(L)$ below may be enlarged; all choices precede the choice of the pair or the regions. Take $r_{\mathrm{cv}}^{\mathrm{sep}}$ at least the previously required background gap thresholds and $\max\{\mathsf{r}_S(\delta_{\mathrm{reg}}/8), 8\log L/\delta_{\mathrm{reg}}\}$.

The one-step coefficients. On a double block $\mathsf{D}_c = B(c_-) \cup B(c_+)$ write P_c for its bonds. The one-step chart for the mean gives, on a fixed ball $|\widetilde{a}| < 3r_*/4$ at $V = e^{i\widetilde{a}}$, the factorization

$$\mathcal{M}(e^{iB}V)(c)\mathcal{M}(V)(c)^{-1} = e^{i\sigma(c_-)} e^{iQ_{\mathrm{str}}(V,B)(c)} e^{-i\,\mathrm{Ad}(\mathcal{M}V(c))\sigma(c_+)}.$$

Each of the first two chart functions is bounded by $|B|/r_*$ on a smaller ball, and the last adjoint has norm at most e^2. For $|B| < r_1 = r_*/40$ the sum of the three exponent norms is at most $10|B|/r_* < 1/4$. The logarithm estimate and Cauchy's inequality therefore give

$$|LQ_V(c,b)|,\ |\mathsf{b}_V(y,x;b)| \le r_1^{-1}, \qquad |C_V^{(2)}(B_1,B_2)(c)| \le 4r_1^{-2} \sup_{\mathsf{P}_c}|B_1| \sup_{\mathsf{P}_c}|B_2|.$$

The coefficients are holomorphic in $\widetilde{a}$. Their differences from the coefficients at 1 are at most $C(L)|\widetilde{a}|$. At 1, $\kappa_1(c) = L^{-3}$ and $\mathsf{b}_1(B)(y,x) = \bar{B}(y,x)$, the arithmetic contour mean. After reducing a fixed radius $s_0(L) > 0$,

$$|\kappa_V(c)^{-1}| \le 2L^3, \qquad |\kappa_V(c)^{-1} - L^3| \le C(L)|\widetilde{a}|.$$

These facts also hold after a real gauge transformation, since its adjoints preserve the norms.

A bond belongs to at most six double blocks, each containing at most $6L^3$ positive bonds. Consequently the kernels of C_V, $\mathsf{C}_V^{\mathsf{T}}$ and $\mathsf{G}_V^{(2)}$ are bounded by a constant $C_K(L)$; the kernel of $\mathsf{E}^{(2)}[j_1]$ is at most $C_K \max|j_1(b_0(c))|$. Every such kernel vanishes unless its two bonds belong to a common

P_c. A column of C_V has at most seven entries and a row at most $6L^3$ entries. These counts follow directly from pivot elimination. For the gauge kernel use

$$\mathsf{G}_V^{(2)}(b,b') = \sum_{y,x} \mathsf{b}_V(y,x;b)^{\mathsf{T}} \mathsf{b}_V(y,x;b');$$

only one block and fewer than L^3 sites contribute. For $\mathsf{E}^{(2)}$ there are at most six contributing coarse bonds. The same formulas give $C(L)|\tilde{a}|$ bounds on the differences of C_V, $\mathsf{C}_V^{\mathsf{T}}$ and $\mathsf{G}_V^{(2)}$ from their flat values.

The flat gap and the additional coordinate. At the flat pair put $M = Q\mathsf{G}_0 Q^{\mathsf{T}}$ and, for $B = \mathsf{C}_1 \check{B}$ extended by zero, put $A = \mathsf{G}_0 Q^{\mathsf{T}} M^{-1} \mathsf{J} B$. Then $QA = \mathsf{J}B$, and the existing elimination identity gives

$$(B, (\mathsf{J}^{\mathsf{T}} M^{-1} \mathsf{J} - 1)B) = (A, \Delta_{\Omega_0}(1)A)_\xi + \|\mathsf{R} D^{\mathsf{T}} A\|_\xi^2. \tag{592}$$

The Hessian here is the inner-plaquette Hessian used to define the constrained inverse. At the flat field its quadratic form equals the sum of squared curls over the inner bare plaquettes of Ω_0.

Let $\mathcal{P}'$ be the level-k plaquettes with a corner at distance at most L from Θ. Their corners lie in Ω_k by the $2L$ margin of the fluctuation structure. Iterating the flat plaquette-average identity gives

$$\sum_{p' \in \mathcal{P}'} |\bar{\partial} Q A(p')|^2 \le \xi^3 \sum_{p \subset \Omega_0} |\partial A(p)|^2 \le (A, \Delta_{\Omega_0}(1)A)_\xi.$$

Indeed, set $a_0 = \xi A$, $a_{n+1} = \mathbf{Q}^s a_n$. Each one-step plaquette estimate contributes a factor L; after k steps its factor is $L^k \xi^4 = \xi^3$. Composed blocks keep all contributing bare plaquettes inside Ω_0. At $k = 0$ this is the identity $QA = A$. On the bonds of $\mathcal{P}'$, $QA = \mathsf{J}B = B$.

On one double block consider the quadratic form

$$n_c(B) = \sum_{p \subset \mathsf{D}_c} |\bar{\partial} B(p)|^2 + \sum_{y = c_\pm} \sum_{x \in B(y) \setminus \{y\}} |\bar{B}(y,x)|^2 + |Q_1 B(c)|^2.$$

For its axial version omit the middle sum and restrict B to vanish on the two box trees. Both forms are positive definite. If their value is zero, plaquette telescoping on the box gives $B = d\nu$. In the first case the contour means, and in the second case the tree paths, make ν constant on each block. The remaining constraint is $Q_1 B(c) = L^{-1}(\nu_+ - \nu_-)$, so $B = 0$. Translation reduces the norm comparison to six finite-dimensional forms: three directions and the ordinary and axial cases. Thus, for some $0 < C_L < \infty$,

$$\sum_{b \in \mathsf{P}_c} |B(b)|^2 \le C_L n_c(B),$$

and the same inequality holds for the axial form on its domain.

For $B = \mathsf{C}_1 \check{B}$ the constraint term vanishes. On a block whose center is outside Θ', B is zero; in the axial case it vanishes on both relevant trees. Every bond is covered by a double block, while each plaquette and each block is counted at most six times. Summing the last inequality and using (592) yields

$$T_{Y,0} \ge \gamma I, \qquad T_{Y,0}^{\mathrm{ax}} \ge \gamma I, \qquad \gamma = (6C_L)^{-1}. \tag{593}$$

We used $\|\mathsf{C}_1 \check{B}\|_2 \ge \|\check{B}\|_2$. The axial inequality has its own null-space argument above; no gauge term is discarded from the ordinary inequality.

Append $\check{B}$ to $\Psi = (A; v, w, \sigma; p, q, r; \omega)$. The first seven equations are those of $\mathbb{K}^{[5]}(j)$, and the last two are

$$QA - \mathsf{J}\mathsf{C}\check{B} = f_8, \qquad \mathsf{C}^{\mathsf{T}}[(\mathsf{G}_V^{(2)} - 1 - \mathsf{E}^{(2)}[j_1])\mathsf{C}\check{B} + \mathsf{J}^{\mathsf{T}}\omega] = f_9. \tag{594}$$

The axial system uses Y^{ax} and omits $\mathsf{G}_V^{(2)}$. We call the systems $\mathbb{K}^{[9]}$ and $\mathbb{K}^{[9a]}$. For either one use

$$|(\Psi, \check{B})|_y = \max(|\Psi|_y, |\check{B}|_{-1;y}), \quad |(g, f_9)|_y = \max(|g|_y, |f_9|_{-3;y}), \quad \iota(g, f_9) = (\iota g, f_9).$$

A new bond has position b_-, scale 1, and midpoint as base point for cutoffs.

The previously proved elimination for $\mathbb{K}^{[5]}$ now leaves only one equation. With $M_\natural = Q\mathsf{G}_\natural Q^{\mathsf{T}}$, it is

$$T(j,j_1)\check{B} = f_9 - \mathsf{C}^{\mathsf{T}}\mathsf{J}^{\mathsf{T}}M_\natural^{-1}(f_8 - Q\mathsf{G}_\natural \mathsf{F}F), \tag{595}$$
$$\Psi = \mathbb{E}^{[5]}(j)(F, f_8 + \mathsf{J}\mathsf{C}\check{B}),$$

where $T(j,j_1) = \mathsf{C}^{\mathsf{T}}(\mathsf{G}_V^{(2)} + \mathsf{J}^{\mathsf{T}}M_\natural^{-1}\mathsf{J} - 1 - \mathsf{E}^{(2)}[j_1])\mathsf{C}$. The axial formula omits the gauge term. Thus the enlarged system is invertible exactly when this Schur operator is, and its $(\check{B},9)$ block is $T(j,j_1)^{-1}$. At (j_*, j_{1*}) the background identities give $\mathsf{G}_\natural = \mathsf{G}_1$, so the Schur operators are precisely T_Y, T_Y^{ax}.

At the flat pair the existing inverse estimate and the finite range of C_1 give $|T_0(b,b')| \le B_T e^{-\delta_T d(b_-,b'_-)}$, uniformly in the structure. This follows by expanding $\mathsf{C}_1^{\mathsf{T}}(G_0^{(2)} - 1 + \mathsf{J}^{\mathsf{T}}M^{-1}\mathsf{J})\mathsf{C}_1$: there are at most 7^2 entries from the two elimination columns, and their positions move a bounded distance $C(L)$. The axial calculation deletes $G_0^{(2)}$.

Apply the conjugation argument proving (246) to $T \ge \gamma I$. Here the error row and column sums are at most $2aB_TS/\delta_T$ for $a \le \delta_T/2$, where $S = \sup_b \sum_{b'} e^{-\delta_T d(b_-,b'_-)/2} < \infty$. Taking also $a \le \gamma\delta_T/(4B_TS)$ gives $|T^{-1}(b,b')| \le (2/\gamma)e^{-ad(b_-,b'_-)}$. The real symmetric gap extends to the Hermitian form on complex fields. The top-lattice summation bound makes S uniform. Insert this estimate in (595) and use the existing $\mathbb{E}_0^{[5]}$ estimate. The two finite-range maps $\mathsf{J}\mathsf{C}_1$ and $\mathsf{C}_1^{\mathsf{T}}\mathsf{J}^{\mathsf{T}}$ have bounded row sums; their rows all have scale 1. We obtain

$$\mathbb{E}_0^{[9]}\iota,\ \mathbb{E}_0^{[9a]}\iota \in \mathfrak{O}(\delta_c, B_c)$$

with δ_c, B_c depending only on L.

Local inversion and gluing. In a small trivial gauge the old eight equations satisfy the perturbation estimate used in proposition 25.1. The additional terms are $\mathsf{J}(\mathsf{C}_V - \mathsf{C}_1)\check{B}$ in slot 8, and the differences of

$$\mathsf{C}_V^{\mathsf{T}}(\mathsf{G}_V^{(2)} - 1)\mathsf{C}_V, \qquad \mathsf{C}_V^{\mathsf{T}}\mathsf{E}^{(2)}[j_1]\mathsf{C}_V, \qquad \mathsf{C}_V^{\mathsf{T}}\mathsf{J}^{\mathsf{T}}$$

in slot 9. The coefficient bounds above give

$$\mathbb{K}_0 - \mathbb{K} = \iota_0 V_c, \qquad |V_c\Phi|_y \le C_V\varepsilon \max_{d(y,z)\le\rho_c} |\Phi|_z. \tag{596}$$

Here the triple is ε-small in the earlier sense, $|j_1|_{-3;y} \le \varepsilon$, and one may take $\rho_c = 19L + 6$. In a product of three new kernels the intermediate indices admit at most $1512L^6$ choices, and the total displacement of their positions is at most $12L$. Thus no count depends on the depth or the number of layers. The already proved averaging estimate sends this small gauge to $V = e^{i\widetilde{a}}$ with $|\widetilde{a}| \le C_{\mathrm{ia}}L\varepsilon$ on each double block; choose ε so that this is at most s_0. Reducing $\varepsilon_c(L) > 0$ further makes $C_V\varepsilon_c B_c e^{\rho_c} \le 1/4$. The same Neumann argument as before, with the unchanged forcing injection in the ninth coordinate, gives a uniform local inverse for both systems.

The cutoff calculation also extends without a new differential estimate. Multiplication by h uses the midpoint of the new bond. In each new kernel the commutator inserts $h(\hat{b}') - h(\hat{b})$; in a product of three kernels there are three such links. Hence, for a cutoff varying on scale ϑ,

$$[\mathbb{K}, h] = \iota\mathbb{C}_h, \qquad |\mathbb{C}_h\Phi|_y \le C_C\vartheta^{-1} \max_{d(y,z)\le\rho_c} |\Phi|_z. \tag{597}$$

It vanishes where h is constant on the old dependency neighborhood together with these three-link neighborhoods. The extension J commutes with h, since its two coordinates have the same midpoint.

We check that the regular-piece construction applies to the present geometry. Truncating the longer sequence at k only merges its last two layers. Its curvature, complex perturbation and current bounds consequently imply those of the shorter sequence. For the real gauge condition the only possible new cube has index $k-1$. Its side is at most $12\mathfrak{M}/L < \mathfrak{M}$. If it is within $12\mathfrak{M}$ of Ω_k^c, the separation excludes Ω_{k+1} and the cube already belongs to the longer class. Otherwise it is

contained in the concentric one-cube enlargement of a level-k $\mathfrak{M}$-cube through one of its points. That enlargement lies in Ω_k and belongs to the longer class. Restrict its gauge; the weights decrease by L^{-1} and L^{-2}. Thus the pair belongs to the shorter natural class with the same radii.

Use the pieces κ and cutoffs of theorem G, with

$$\Theta'^{\kappa} = \{y \in \Theta' : \Delta(y) \subset \kappa^{[1]}\}, \qquad \Theta''^{\kappa} = \Theta'' \cap \Theta'^{\kappa}.$$

If Θ'^{κ} is empty the local system is the previous eight-coordinate system. Otherwise κ has size k and lies in the deep part of Ω_k. Indeed a smaller piece lies either outside Ω_k or within $3\mathfrak{M}$ of its boundary; a point of Ω_{k+1} in its one-cube enlargement would then be within $5\mathfrak{M}$, contrary to the $R_1\mathfrak{M} \geq 13\mathfrak{M}$ margin. For these pieces use the regular local sequence

$$\Omega_k^{\kappa} = \kappa^{[1]} + [-e_k, e_k]^3, \qquad \Omega_m^{\kappa} = \Omega_{m+1}^{\kappa} + [-e_m, e_m]^3, \qquad e_m = 2R_\flat \mathfrak{M}_\flat \xi_m.$$

This prescription also applies when $k = 0$: a fluctuation piece at level zero uses this enlarged regular domain, rather than the compressed bare boundary domain. The torus and size conditions ensure that all these cubes are embedded.

For $\mathfrak{M} \geq 32L^2 R_\flat \mathfrak{M}_\flat$, the following comparisons fit inside $\kappa^{[2]}$. A block $\Delta(y)$ within $10L$ of $\kappa^{[5/6]}$ belongs to Θ'^{κ}; a double block within $8L$ has its coarse bond in Γ'^{κ}. To see this, the preceding boundary argument first forces the piece to have size k, and then $12L < \mathfrak{M}/6$ puts the block in $\kappa^{[1]}$. Membership in $\mathfrak{X}(\Theta), Y, Y^{\mathrm{ax}}$ therefore agrees with local membership in this neighborhood. The pivot condition is the same on both sides, and each tree bond is decided by its own block. The coefficient formulas show that the local and global new kernels agree wherever the cutoffs use them. For $\mathsf{E}^{(2)}[j_1]$ each contributing pivot also lies in $\kappa^{[5/6]}$, where the truncated multiplier equals j_1.

Every $c \in \Gamma'$ occurs in one of these local structures: choose an end point $y \in \Theta'$ and a piece containing it. The preceding comparison makes this a top fluctuation piece, with $y \in \Theta'^{\kappa}$; moreover the double cube of c lies within $3L/2$ of y and hence in $\kappa^{[1]}$. Thus the local mean and pivot charts define all the global one-step coefficients used above.

The natural-class gauge on $\kappa^{[2]}$, together with the same conversion of ordinary to covariant first differences as in the regular inverse proof, makes the local quadruple ε-small with

$$\varepsilon = 8L^2(\alpha_0 + \alpha_1 + \gamma_0 + \gamma').$$

Here both truncated multipliers are bounded by γ'; the global scale is no smaller than the local scale, so their weighted bounds persist. Choose $a_3(L) > 0$ so that radii and multiplier bounds at most a_3 give $\varepsilon \leq \varepsilon_c$; it suffices to require $a_3 \leq \varepsilon_c/(32L^2)$ in addition to the earlier regular-inverse radius restrictions. The local Hessian remains the ambient Hessian used in the existing proof. The bare pieces without fluctuations retain their existing compressed inverse and every touching-frame plaquette term.

Let $\mathbb{E}_\kappa$ be these local inverses. The old partition satisfies $\operatorname{supp}\theta_\kappa \subset \kappa^{[1/12]}$, while $\widetilde{h}_\kappa = 1$ on $\kappa^{[1/3]}$ and is supported in $\kappa^{[2/3]}$. For the forcing product $\star$ already used in the regular inverse proof, set $\theta \star (g, f_9) = (\theta \star g, \theta f_9)$. Then

$$\iota(\theta \star (g, f_9)) = \theta \iota(g, f_9),$$

since the correction $-[D^{\mathsf{T}}, \theta]X_5$ in slot 5 cancels the extra divergence, and the ninth slot is unchanged. The local and global indices, weights, and transports agree on $\kappa^{[5/6]}$. The new commutator is supported there. Thus the same parametrix gives

$$\mathrm{Par} = \sum_{\kappa} \widetilde{h}_\kappa \mathbb{E}_\kappa \theta_\kappa, \qquad \mathbb{K}\,\mathrm{Par} = 1 - \mathcal{R}, \qquad \mathcal{R}\iota = \iota \mathcal{R}^\sharp.$$

The input and commutator supports are separated by $D_0 = \mathfrak{M}/(8L)$ in the multiscale distance. Apply (597) with $\vartheta = \mathfrak{M}/C_G$ and spend half the local inverse decay across this separation. For some $\delta_4, B_8, C_q > 0$,

$$|\mathcal{R}_\kappa^\sharp g|_y \leq q' \sup_{z \in S'_\kappa} e^{-\delta_4 d(y,z)} |g|_z, \qquad q' = C_q \mathfrak{M}^{-1} e^{-\delta_4 D_0/2}.$$

The first factor $\widetilde{h}_\kappa \mathbb{E}_\kappa \iota(\theta_\kappa \star \cdot)$ obeys the same estimate with $4B_8$ instead of q'. The old proof applies verbatim to the remaining coordinates. For the new coordinate it uses only the range $12L$, midpoint cutoffs and scale 1, as checked above.

For $Nq' \le 1/2$, expand $(1-\mathcal{R})^{-1}$. If $\varpi = (\kappa_0, \dots, \kappa_m)$, denote the resulting product by $\mathbb{E}_\varpi$. It is supported in $\kappa_0^{[2/3]}$ at its output and uses input only in $\kappa_m^{[1/12]}$. A nonzero product has consecutive $\kappa_{j-1}^{[5/6]}$ and $\kappa_j^{[5/6]}$ meeting at a top, and

$$|\mathbb{E}_\varpi \iota g|_y \le 8B_8 (q')^m \sup e^{-\delta_4 \sum_{j=0}^m d(y_j, y_{j+1})} |g|_{y_{m+1}}. \tag{598}$$

Here $y_0 = y$, the intermediate y_j lie in these overlaps, and $y_{m+1} \in \kappa_m^{[5/6]}$. Summation over pieces gives $\mathbb{E}\iota \in \mathfrak{O}(\delta_4, 16NB_8)$. This proves invertibility of both enlarged systems for all quadruples with these bounds. By (484), reducing a_{cv} makes the physical quadruple $(\mathbf{U}, \mathbf{J}, j_*, j_{1*})$ one of them. Equations (595) prove existence of the two covariances. Their symmetry follows from symmetry of the gauge form, $\mathsf{E}^{(2)}[j_{1*}]$, and $(Q\mathsf{G}_1 Q^{\mathsf{T}})^{-1}$.

The second multiplier. We use the local expansion from proposition 39.1 for the shorter sequence. In its notation

$$j_* = \sum_\varpi j_\varpi = j^0 + \delta j, \qquad \mathbb{E}^{[5]}(j_*) = \sum_\Xi \mathbb{E}_\Xi.$$

The first sum has its zero-length part j^0; its positive-length terms have the factor $q_1^{|\varpi|}$, where $q_1 \le \mathfrak{M}^{-9}$. In particular $|\delta j|_{-3} \le C(L)\gamma_0 \mathfrak{M}^{-9}$. The second sum is over the first decorated sequences, with trunk cells $X^{(2)}(\kappa)$ and branch cells $X(\kappa)$. With $\delta_b = \delta_{\mathrm{reg}}$, $C_b = B_{\mathrm{tr}}$ and $C_{\mathrm{w},b} = C_{\mathrm{tr}}$, the estimate is

$$\sum_{\omega(\Xi)=\omega} |\mathbb{E}_\Xi \iota g|_y \le C_b (C_{\mathrm{w},b} \mathfrak{M}^{-3})^{|\omega|} \sup_z e^{-\delta_b d^{\mathrm{tr}}_{\mathbf{B}}(\omega; y, z)/2} |g|_z. \tag{599}$$

The factors and this estimate are defined on the open piecewise local class on the carrier. Each j_ϖ is supported on rows in its last piece's $[1/12]$ cube. The operator $\mathbb{E}_\Xi$ uses input only in the last piece's $[1/12]$ cube and has output in the first piece's $[2/3]$ cube.

Define $j_{1,\Xi}$ by its weighted transpose, not an unweighted transpose of the kernel:

$$\xi_c^3 \langle e, j_{1,\Xi}(c) \rangle = ([\mathbb{E}_\Xi]_{A,8} e_c, \mathbf{J})_\xi. \tag{600}$$

The right side is summed over the bare output bonds of the first piece. These bonds have both end points in its carrier, so this defines the same function on local data. It is holomorphic there and vanishes unless the midpoint of c lies in the last piece's $[1/12]$ cube.

The current bound and the bare-to-top count used for the total multiplier give, for g supported in slot 8 at the row c,

$$\xi_c^3 |\langle e, j_{1,\Xi}(c) \rangle| \le 3\gamma_0 \sum_y \xi_y^{-1} |\mathbb{E}_\Xi \iota g|_y, \qquad |g|_{y(c)} = \xi_c |e|.$$

If $\underline{d}(\omega; z)$ denotes the least length of a tree meeting all the cells and containing z, then $d^{\mathrm{tr}}(\omega; y, z)$ dominates both $d(y, z)$ and $\underline{d}(\omega; z)$. Split the exponential in (599) accordingly, and use the existing scale-ratio summation, increasing the fixed floor so that $R_1 \mathfrak{M} \ge \mathsf{r}_S(\delta_b/8)$ and $\log L/(R_1\mathfrak{M}) \le \delta_b/8$:

$$\sum_y \frac{\xi_z}{\xi_y} e^{-\delta_b d(y,z)/4} \le L c_S(\delta_b/8).$$

Taking the three Pauli coordinates in (600) yields

$$\sum_{\omega(\Xi)=\omega} \xi_c^3 |j_{1,\Xi}(c)| \le C_j' \gamma_0 (C_{\mathrm{w},b} \mathfrak{M}^{-3})^{|\omega|} e^{-\delta_b \underline{d}(\omega; y(c))/4}. \tag{601}$$

This proves $j_{1*} = \sum_\Xi j_{1,\Xi}$ by absolute convergence and the already identified block $\mathsf{H}_1 = [\mathbb{E}^{[5]}(j_*)]_{A,8}$. Let j_1^0 be the sum of the zero-length terms and $\delta j_1 = j_{1*} - j_1^0$.

We also need the stronger small tail bound

$$|j^0|_{-3},\ |j_1^0|_{-3} \le C_j\gamma_0, \qquad |\delta j|_{-3} + |\delta j_1|_{-3} \le C_j''\gamma_0\mathfrak{M}^{-9}. \tag{602}$$

For the second tail, write $E_* = \mathbb{E}^{[5]}(j_*)$, $E_0 = \mathbb{E}^{[5]}(j^0)$, and let E_{00} be its sum of one-piece factors. The weighted transpose of

$$E_* - E_{00} = (E_* - E_0) + (E_0 - E_{00})$$

gives δj_1. The resolvent identity bounds the first difference by two total inverses and the old vertex applied to δj; the second is the sum of positive-length cutoff products. For a product of length m, replace the overlap supremum by a sum. The scale ratio is bounded along the entire chain by $L\exp(\delta_b\sum_i d(y_i,y_{i+1})/4)$, with one factor L, independent of m. Summing successively over its $m+1$ intermediate positions gives a bound

$$\xi_c^3|j_{1,(\varpi)}(c)| \le C(L)\gamma_0 q_{\mathfrak{M}}^m c_S(\delta_b/4)^{m+1} \le C(L)\gamma_0 q_1^m, \qquad q_1 = c_S(\delta_b/4)q_{\mathfrak{M}}.$$

There are at most N^{m+1} contributing sequences, counted backwards from the last piece containing the midpoint of c. With $Nq_1 \le 1/2$ their positive-length sum is $O(\gamma_0 q_1)$. The first resolvent difference has the same bound after the weighted-transpose estimate. This proves the second inequality in (602). For $j_1^0(c)$ at a local pair there are at most N contributing one-piece factors, all with midpoint $c \in \kappa^{[1/12]}$. The one-piece inverse estimate and the same summation give the first inequality on the local class as well. Choose a_{cv} so that $C_j a_{\mathrm{cv}} \le a_3/2$.

Two vertices and the tree expansion. Use the local inverse at $q^0 = (\mathbf{U},\mathbf{J},j^0,j_1^0)$ as reference. The old vertex $\mathbb{W}[j']$ is extended by a zero ninth component; the second vertex has only a ninth component,

$$\mathbb{W}_1[j'](\Psi,\check{B}) = (0, \mathsf{C}^{\mathsf{T}}\mathsf{E}^{(2)}[j']\mathsf{C}\check{B}).$$

The equations give the exact identity

$$\mathbb{K}(q_*) = \mathbb{K}(q^0) - \iota\mathbb{W}[\delta j] - \iota\mathbb{W}_1[\delta j_1]. \tag{603}$$

The old vertex has range at most $3L+6$. For the new one the three-kernel product above gives

$$|\mathbb{W}_1[j']\Phi|_y \le C_{W,1} \max_{d(y,y(c))\le 8L} \xi_c^3|j'(c)| \max_{d(y,z)\le 12L} |\Phi|_z. \tag{604}$$

Only pivot rows occur in the first maximum, and they have $\xi_c = 1$. There are at most $(36L^3)^3$ index choices in this product, so $C_{W,1}$ depends only on L. Both vertices therefore have this form of bound, after enlarging the constant for the first one.

Let $\mathcal{N}(\kappa)$ be the pieces whose [1] cubes have a common interior point with $\kappa^{[1]}$ in Ω_0. The local expansion already uses $X^{(2)}(\kappa) = \bigcup_{\kappa'\in\mathcal{N}(\kappa)} X(\kappa')$, with the union of their cube families. For the present expansion use also

$$X^{(3)}(\kappa) = \bigcup_{\kappa'\in\mathcal{N}(\kappa)} X^{(2)}(\kappa').$$

There are at most N neighbors and at most $27N$ cubes in each family of $X^{(2)}$, so $X^{(3)}$ has at most $27N^2$ cubes. The families are connected: each neighbor's $X^{(2)}$ contains the family of $X(\kappa)$. They are therefore cells for $r_{\mathrm{cv}} = 27N^2$. At most $27N^2$ pieces have a prescribed $X^{(3)}$ and family. Indeed that family contains the cube $\kappa^{[1]}$ for a regular piece, or κ for a bare piece; these cubes determine the piece and are distinct across the two kinds.

Every base cell and hence every one of these larger cells is also a cell of the longer sequence, with the same carrier. Only a cube of index $k-1$ requires checking. In a piece's family it meets either $\mathcal{L}_{k-1}$ or the buffer of Ω_k, so its distance to Ω_k^c is less than $3\mathfrak{M}$. The earlier cube comparison puts it in the longer class. Its side determines its index uniquely through $\mathfrak{M}\xi_n \le s < L\mathfrak{M}\xi_n$, so both the index and its five-cube enlargement are unchanged. The usual count of connected cell families and depth-first parent maps gives n_{cv}^m tree walks of length m with a fixed first cell.

Consider products

$$\mathbb{E}_{\Upsilon} = \mathbb{E}_{\varpi^{(0)}} \iota \mathbb{W}_{\beta_1} \mathbb{E}_{\varpi^{(1)}} \cdots \iota \mathbb{W}_{\beta_n} \mathbb{E}_{\varpi^{(n)}}. \tag{605}$$

Here every $\varpi^{(l)}$ is a cutoff sequence for the system at q^0. A vertex label β_l is either a positive-length sequence ϖ' with $\mathbb{W}_{\beta_l} = \mathbb{W}[j_{\varpi'}]$, or a first decoration Ξ' of positive tree length with $\mathbb{W}_{\beta_l} = \mathbb{W}_1[j_{1,\Xi'}]$. Make a trunk from the $X^{(3)}$ cells of the sequences $\varpi^{(0)}, \dots, \varpi^{(n)}$ in their order. Attach a branch for each vertex to the last cell preceding that vertex. A branch of the first kind is the reversed sequence of its X cells. A branch of the second kind has the reversed $X^{(2)}$ trunk of $\omega(\Xi')$, keeping its own X branches attached to the same cells. List the outer branches in reverse vertex order. After each reversed inner trunk, list its branches in increasing label order $k' = 1, \dots, n'$, retaining their internal order and attachment cells. This gives the depth-first parent map p, with $p(i) < i$ and each next parent an ancestor of the preceding index. Its initial uninterrupted branch is exactly the outer trunk.

A nonzero product gives a tree walk. Consecutive cells within a cutoff sequence overlap by (598). At an old vertex this is the overlap proved for the first multiplier expansion. At a new vertex, nonzero entries give bonds b, b_1, b_2, b_3 and a pivot $b_0(c)$, with b linked to b_1, $b_1, b_2 \in \mathsf{P}_c$, and b_2 linked to b_3. Their midpoints obey

$$|\widehat{b_0(c)} - \hat{b}|_\infty \le 4L, \qquad |\widehat{b_0(c)} - \hat{b}_3|_\infty \le 4L.$$

They are all within L of Ω_{k+1}. A piece whose [2] cube contains such a point has size at least $k-1$: otherwise that point would lie in a layer outside the interior of Ω_k. Thus its side is at least $\mathfrak{M}/L \ge 32L^3$. The three cutoff supports and $4L \le (32L^3)/8$ put the pivot midpoint in the interior of all three [1] cubes. The branch's last piece is therefore a neighbor of the preceding trunk piece, its $X^{(2)}$ cell lies in that trunk's $X^{(3)}$ cell, and the two adjacent trunk cells overlap. This proves all required overlaps.

If $m_l = |\varpi^{(l)}|$ and m'_l is the length of the branch label, then $m'_l \ge 1$ and counting nodes gives

$$|\omega(\Upsilon)| = \sum_{l=0}^{n} m_l + \sum_{l=1}^{n} (m'_l + 2). \tag{606}$$

For a fixed tree walk of length m there are at most $10^m (27N^2)^{m+1}$ such products. To check the count, mark each node as an outer-trunk start, a first-kind branch start, a second-kind branch start, an inner branch start, or none. There are at most 5^m choices. The already proved first decoration count contributes at most $2^{m'}$ to each second-kind branch, hence at most 2^m in total. Each prescribed cell has at most $27N^2$ piece preimages. These marks, the parent map and the selected pieces recover the cutoff sequences and vertex labels.

We estimate the product before summing it. Write z_l for the last intermediate point of leg $l-1$ and y_l for the first point of leg l. The vertex bounds use a row c_l with

$$d(z_l, y(c_l)) \le 8L, \qquad d(z_l, y_l) \le 12L.$$

Apply (598) on each leg. A first-kind branch contributes $C_j \gamma_0 q_1^{m'_l} e^{-\delta_b \mathsf{d}(\varpi'; c_l)/2}$, where $\mathsf{d}(\varpi'; c_l)$ is the least length of a chain through its successive piece overlaps to $y(c_l)$. A second-kind branch contributes, by (601),

$$C'_j \gamma_0 (C_{\mathrm{w},b} \mathfrak{M}^{-3})^{m'_l} e^{-\delta_b \underline{d}(\omega(\Xi'); y(c_l))/4}.$$

Join the paths used in the legs, the branch chains or trees, and the two vertex links. Their union is connected, contains the two endpoint positions, and meets every cell. The two links cost at most $20L$ per vertex. Therefore, with $\delta_5 = \min(\delta_4, \delta_b/4, 1)$, the product of all exponentials is bounded by

$$e^{20Ln} e^{-\delta_5 d^{\mathrm{tr}}_{\mathbf{B}}(\omega; y, y')}.$$

This follows by choosing paths within any prescribed error of the defining infima and letting that error tend to zero. Every cell of a positive-length first-kind branch is met by an overlap point; the last point is connected to $y(c_l)$. This is the reason these branches exclude length zero.

Now $q' \le C_q \mathfrak{M}^{-1}$, $q_1 \le \mathfrak{M}^{-9}$, and $3m' \ge m' + 2$ for $m' \ge 1$. Absorb the fixed vertex factors and e^{20L} into $C_\sharp(L)$. Equation (606) gives

$$|\mathbb{E}_\Upsilon \iota g|_y \le 8 B_8 (C_\sharp \mathfrak{M}^{-1})^{|\omega|} \sup_{y'} e^{-\delta_5 d^{\mathrm{tr}}_{\mathbf{B}}(\omega; y, y')} |g|_{y'}.$$

The count of products then gives

$$\sum_{\omega(\Upsilon)=\omega} |\mathbb{E}_\Upsilon \iota g|_y \le 216 N^2 B_8 (C_{\mathrm{cv}} \mathfrak{M}^{-1})^{|\omega|} \sup_{y'} e^{-\delta_5 d^{\mathrm{tr}}_{\mathbf{B}}(\omega; y, y')} |g|_{y'}, \tag{607}$$

where one may take $C_{\mathrm{cv}} = 270 N^2 C_\sharp$.

The sum of these products is the physical inverse. Indeed (602), the two vertex estimates and the total reference inverse imply

$$(\mathbb{W}[\delta j] + \mathbb{W}_1[\delta j_1]) \mathbb{E}(q^0) \iota \in \mathfrak{O}(\delta_4, C(L) \gamma_0 \mathfrak{M}^{-9}).$$

Increase M_{cv} until this bound is at most $1/2$. The resolvent expansion of (603) is then

$$\mathbb{E}(q_*) = \mathbb{E}(q^0) \sum_{n \ge 0} \Big[\iota (\mathbb{W}[\delta j] + \mathbb{W}_1[\delta j_1]) \mathbb{E}(q^0) \Big]^n.$$

Insert the cutoff expansion and the two multiplier expansions. Estimate (607) and the tree count show absolute convergence, so regrouping gives $\mathbb{E}(q_*) = \sum_\Upsilon \mathbb{E}_\Upsilon$. This equality is an identity of matrices: the forcing injection is onto, and the spaces are finite dimensional.

Local terms and the covariance bounds. We give the local meaning of the products, since choosing a global continuation of local data would not define their coefficients intrinsically. Use the piecewise local class of section 39 for the shorter sequence. Its openness was proved there.

The longer local natural class lies in this piecewise class. The pointwise inequalities only weaken on truncation. For the cube condition, a shorter cube not in the longer class must have index $k-1$ and distance greater than $12\mathfrak{M}$ from Ω_k^c. If it lies in $X(\kappa)^\natural$, the carrier geometry forces κ to have size k: for a smaller piece every point of its carrier is within $6\mathfrak{M}$ of the piece, which is outside Ω_k or within $3\mathfrak{M}$ of its boundary. For a top piece the carrier is $\kappa^{[6]} \cap \Omega_0$. In its embedded chart, choose a cube of side $2\mathfrak{M}$, a union of eight top cubes, containing the shorter cube and contained in $\kappa^{[6]}$. This is possible because the shorter side is less than $\mathfrak{M}$. The enlarged cube stays more than $10\mathfrak{M}$ from Ω_k^c, so lies in the longer natural class and in the same carrier. Its real gauge restricts to the original cube with smaller weights. This proves the required local inclusion without assuming a global extension of the pair.

For a given outer decoration take $W = [\omega(\Upsilon)]$. Define j^0, j_1^0 by their local formulas on the rows in the $[5/6]$ cubes of its trunk pieces, and set them to zero on the other rows. Their bounds in (602) make all the local inverse factors well defined. They are holomorphic: the coefficient charts are holomorphic, pivot inverses have nonzero determinant, and each finite local system remains invertible on this open class. Only the second vertex needs an additional restriction. If κ^- is the last piece of the preceding leg, use its local C_κ and $\mathsf{E}^{(2)}_\kappa$, and truncate j' to rows in $\kappa^{-[5/6]}$. Restrict the input to the common fluctuation bonds with midpoint in $\kappa^{-[2/3]}$, and retain output only at common bonds in $\kappa^{-[1/12]}$. Denote the resulting vertex by $\mathbb{W}_{1,\kappa^-}[j']$; it is zero for a piece with no fluctuations.

The three links of its kernel move a midpoint by at most $6L$. Thus, behind θ_{κ^-}, all input, output and coefficient indices are in precisely the common sets checked above, and

$$\theta_{\kappa^-} \iota \mathbb{W}_{1,\kappa^-}[j'] = \theta_{\kappa^-} \iota \mathbb{W}_1[j']$$

at the restriction of a global pair. The local vertex obeys (604). Its coefficients depend only on the given data in $\kappa^{-[2]}$. All factors of (605), with this replacement, are therefore holomorphic on the piecewise local class on W, independent of a continuation, and agree with the global factors on restriction. The overlap and chain arguments just proved use only these local formulas, so (607) holds on this entire class.

Define

$$\mathcal{C}_\omega = \sum_{\omega(\Upsilon)=\omega} [\mathbb{E}_\Upsilon]_{\check{B},9}.$$

The sum is finite for fixed ω. It gives the asserted local holomorphic term, and the global sum is the covariance by (595). For a ninth-coordinate input equal to e at b' and zero elsewhere, $\iota g = g$, $|g|_{b'_-} = |e|$, and all other forcing seminorms vanish. The output satisfies $|\check{B}(b)| \le |(\Psi, \check{B})|_{b_-}$. Thus (607) becomes

$$|\mathcal{C}_\omega(b, b')| \le 216 N^2 B_8 (C_{\mathrm{cv}} \mathfrak{M}^{-1})^m e^{-\delta_5 d^{\mathrm{tr}}_{\mathbf{B}}(\omega; b_-, b'_-)}. \tag{608}$$

The first and last cutoffs put $\hat{b}$ in the first piece's [2/3] cube and $\hat{b}'$ in the last piece's [1/12] cube. These pieces have side at least $32L^3$, as shown above. Moving a midpoint by $1/2$ to its initial endpoint therefore keeps that endpoint in the piece's [1] cube and hence in its X cell. This proves the two endpoint support statements.

The weighted distance for the longer sequence is no larger than that for the shorter sequence, including tree distances. For the latter comparison cut a tree into segments at all intersections and marked points. On each segment the inverse-scale integrand decreases when the last layer is added. Replace that segment by a path within any error of the longer weighted distance between its endpoints; their union remains connected and meets all marked cells. Subadditivity and then passage to the infimum prove the comparison. Each bond here lies in the interior of Ω_k, so $d(b_-, y(b)) \le 1$. Adding the two bond segments to a tree costs at most 2. Taking $\delta_{\mathrm{cv}} = \delta_5/4$ and enlarging B_0 by e^2 therefore turns (608) into (591).

For fixed b at most N first pieces have $\hat{b} \in \kappa^{[2/3]}$. Moreover $d^{\mathrm{tr}}_{\mathbf{B}}(\omega; b_-, b'_-) \ge d(b_-, b'_-) \ge |b_- - b'_-|_1$. Choose $M_{\mathrm{cv}} \ge (2C_{\mathrm{cv}} n_{\mathrm{cv}})^2$ and $B_0 \ge \max(432 N^3 B_8, 216 e^2 N^2 B_8)$. The count of tree walks and (608) give the length-m estimate in the statement. Summing it proves the kernel estimate for the full covariance and for every partial sum.

Finally a site is the initial point of at most three positive bonds, and the unit-spacing torus sum is bounded by the $\mathbb{Z}^3$ sum. Thus every row sum and column sum of the absolute kernel are at most $3B_0\sigma(\delta_{\mathrm{cv}})^3$. This proves the supremum bound and, by the Schur test, the ℓ^2 bound. For a 2×2 matrix $X \in \mathfrak{g}^c$,

$$\tfrac{1}{2}|X|^2 \le \langle X^*, X\rangle \le |X|^2;$$

the Hermitian operator norm is consequently at most $\sqrt{2}$ times that ℓ^2 bound, and hence at most B. All choices used only L and the fixed constants of the earlier inverse and multiplier estimates. □

41. Local nonlinear response with decoupling parameters

41.1. **Local propagators with decoupling parameters.** Use the cells, their carriers, and the piecewise local class of proposition 39.1. Put

$$\delta = \delta_{\mathrm{reg}}, \qquad \mathring{\delta}_0 = \delta/4, \qquad r_{\mathrm{w}} = 27 N_{\mathrm{G}}.$$

The scale is the level-t scale, so that $\xi_t = 1$ and $\xi = \xi_0$. All transposes below use the existing physical pairings: weight ξ^3 on bare sites and bonds, and weight ξ_z^3 on scalar tops and physical rows. Write

$$H_{\mathrm{pr}} = G Q_{\mathrm{pr}}^{\mathsf{T}} (Q_{\mathrm{pr}} G Q_{\mathrm{pr}}^{\mathsf{T}})^{-1}, \quad S_{\mathrm{pr}} = (Q_{\mathrm{pr}} G Q_{\mathrm{pr}}^{\mathsf{T}})^{-1}, \quad P_{\mathrm{f}} = D(1 - \mathsf{R}_{\mathsf{m}}) D^{\mathsf{T}}.$$

Here R_{m} is the projection defined by the scalar mean average. The operators $H_{1,\mathrm{pr}}$ and $\mathfrak{G}$ are those at the physical multiplier $\mathsf{j}_* = H_{\mathrm{pr}}^{\mathsf{T}} \mathbf{J}$.

For a cell X, set

$$X^{\boxplus} = \bigcup_{q\in X}\{x : \operatorname{dist}_\infty(x,q) \le 5\mathfrak{M}\xi_{n(q)}\}.$$

The hull of a walk or tree ω is the union of these sets over its cells; its carrier is $[\omega] = \Omega_0$ intersected with that hull. Let $D_\omega(y,z)$ be the infimum of the weighted lengths of connected graphs containing y, z and meeting every cell of ω. Two cubes of a standard partition are wall adjacent when they share a face; wall components use this adjacency. Write $r_0 = 2(3L+2)$ for the response radius, using the value $\rho_0 = 3L+2$ fixed in the local response estimates.

Proposition 41.1. There are $a_{\mathrm{D}} > 0$, M^{pr}, r^{pr}, C_{w} and B, depending only on L, for which the following holds. Assume the regular-pair frame hypotheses, with

$$\alpha_0, \alpha_1, \gamma_0 \le a_{\mathrm{D}}, \quad \mathfrak{M} \ge M^{\mathrm{pr}}, \quad R_1\mathfrak{M} \ge r^{\mathrm{pr}}, \quad \ell \ge 2L\mathfrak{M},$$

and the additional summation and scale-ratio requirements

$$\frac{\log L}{R_1\mathfrak{M}} \le \frac{\mathring{\delta}_0}{48}, \quad R_1\mathfrak{M} \ge \mathsf{r}_S(\mathring{\delta}_0/16), \quad \mathfrak{M} \ge (8C_{\mathrm{w}}n_{1,\natural})^2.$$

Here $n_{1,\natural} = n_1(L, r_{\mathrm{w}})$ is the cell-neighbour bound from the preceding construction; M^{pr} also exceeds all floors in that construction and in the doubled inverse.

Each of $H_{\mathrm{pr}}, S_{\mathrm{pr}}, H_{1,\mathrm{pr}}, \mathfrak{G}, P_{\mathrm{f}}$ has an absolutely convergent expansion $O = \sum_\omega O_\omega$. Its terms are holomorphic on the piecewise local class on $[\omega]$, depend only on the pair and the sequence there, and agree under restriction of larger local domains. They are gauge covariant and commute with the real structure. A nonzero kernel has both arguments in cells of ω. The local construction includes layer zero and all touching ambient plaquettes.

The terms obey the following bounds with $v_m = (C_{\mathrm{w}}/\mathfrak{M})^m$, $m = |\omega|$:

$$|(K_\omega e_c)(b)| \le Bv_m e^{-\mathring{\delta}_0 D_\omega(y,y_c)}|e|,$$

$$|(\nabla_{\mathbf{U}} K_\omega e_c)(b,\nu)| \le Bv_m \xi_y^{-1} e^{-\mathring{\delta}_0 D_\omega(y,y_c)}|e|, \tag{609}$$

$$|S_{\mathrm{pr},\omega}(c,c')| \le Bv_m \xi_y^{-2} e^{-\mathring{\delta}_0 D_\omega(y,y')}, \tag{610}$$

$$\max\{\xi_y^{-2}|(\mathfrak{G}_\omega f)(b)|, \xi_y^{-1}|(\nabla_{\mathbf{U}}\mathfrak{G}_\omega f)(b,\nu)|\} \le Bv_m e^{-\mathring{\delta}_0 D_\omega(y,y')} \sup|f|, \tag{611}$$

$$|P_{\mathrm{f},\omega}(b,b')| \le Bv_m(\xi/\xi_{y'})^3 \xi_y^{-2} e^{-2\mathring{\delta}_0 D_\omega(y,y')}. \tag{612}$$

In the first line $K = H_{\mathrm{pr}}$ or $H_{1,\mathrm{pr}}$; b lies over y, c' is at y', and f vanishes outside the bare bonds over y'. Kernels use counting measure. For every fixed initial cell,

$$\sum_\omega v_{|\omega|} \le 2, \qquad \sum_\omega (|\omega|+1)v_{|\omega|} \le 4.$$

Fix a real number $\varkappa > 0$ and a standard partition $\mathsf{P} = \pi[\mathsf{w}]$ with at least three cubes in each direction and

$$\mathsf{w} \ge 88 r_{\mathrm{w}}\mathfrak{M}, \qquad \mathring{\delta}_0\mathsf{w} \ge 2176\varkappa.$$

Let $\mathsf{P}(\omega)$ be the cubes meeting the hull, let $\mathsf{Y} \subset \mathsf{P}$, and put $W = \Omega_0 \cap \bigcup\mathsf{Y}$, with the cube-wise intersection. For a local natural pair on W and parameters $s = 0$ off Y, $|s_\Delta| \le e^{\varkappa}$, set

$$O(s) = \sum_{\mathsf{P}(\omega)\subset\mathsf{Y}} \Big(\prod_{\Delta\in\mathsf{P}(\omega)} s_\Delta\Big) O_\omega[p|_{[\omega]}].$$

These series converge absolutely, uniformly on the local class and on bounded parameter sets. They are holomorphic in the pair and in s, and have degree at most one in each s_Δ. The assignment of terms is fixed independently of s and of the local domain. At $s \equiv 1$ on the whole partition they equal the five global operators above.

For $\mathsf{m}(c) = \xi_c$, define

$$(H_s, S_s, K_s, \mathfrak{G}_s) = (H_{\mathrm{pr}}(s)\mathsf{m}^{-1}, \mathsf{m}^3 S_{\mathrm{pr}}(s)\mathsf{m}^{-1}, H_{1,\mathrm{pr}}(s)\mathsf{m}^{-1}, \mathfrak{G}(s)), \qquad P_s = P_{\mathrm{f}}(s).$$

These are the operators of the local response structure on $Y^1(\mathsf{Y}) \subset Y^2(\mathsf{Y}) \subset Y^3(\mathsf{Y})$. They satisfy its strong and column estimates with

$$b = b_0 e^{8\varkappa}, \quad \widehat{b} = b_{\mathrm{D}} e^{8\varkappa}, \quad \mu = 3\mathring{\delta}_0/16, \quad \theta = \mathring{\delta}_0/8, \qquad b_{\mathrm{D}} = b_0 c_S(\mathring{\delta}_0/16),$$

including $|P_s A|_w^\sharp \le \widehat{b}|A|_w$. The same estimates hold for every subsum. The constants b_0, b_{D} depend only on L. Here $Y^j(\mathsf{c})$ consists of the tops within sup distance $j\mathfrak{M}\xi_{n(q)}$ from a cube q of a cell X with $\mathsf{P}(X) \subset \mathsf{c}$. Each operator is carried in both variables by $Y^1(\mathsf{Y})$. It is the sum of the operators belonging to the wall components of $\{s \neq 0\}$, each depending only on the pair and parameters in that component. The sets Y^j belonging to distinct components are separated by more than the response radius r_0, for $j \le 4$. ∎

Proof. The common local trees. The proof of proposition 39.1 already constructs the local fifth-system factors. The same factors at $\mathsf{j} = 0$ give the ordinary walks. Sum their $A, 8$ and $\omega, 8$ blocks to define $H_{\mathrm{pr},\omega}$ and $S_{\mathrm{pr},\omega}$; sum the $A, 8$ blocks of the decorated factors to define $H_{1,\mathrm{pr},\omega}$. The inverse block identities identify their global sums.

For the projected operator use the doubled system. On each piece its coupling is

$$\mathbb{J}\Psi' = e_1 D(D^{\mathsf{T}} A' - w').$$

It is independent of the multiplier and uses only bonds of $\Omega_0^\kappa \subset \kappa^{[2]} \cap \Omega_0$. Thus it is the same operator for every local ambient region containing the carrier. The local regular gauge, the touching-plaquette hypothesis and the layer-zero inverse apply exactly as in the preceding local-tree proof. The half-Hölder estimates used in the proof of proposition 34.1 give the local doubled strong inverse; the commutator (526) gives ordinary forcing and vanishes where the cutoff is constant. Consequently the lifted product proof of (529) applies on these local classes, with constants B_4^{d} and

$$q_{\mathrm{d}} = C_q^{\mathrm{d}} \mathfrak{M}^{-1} e^{-\delta\mathfrak{M}/(8L)}.$$

Take C_q^{d} at least the larger of the global and local commutator constants, and increase the floor until $q_{\mathrm{d}} \le \mathfrak{M}^{-9}$.

In each decorated label Ξ, use the same local zero-length multiplier j^0 and the same branch multipliers $\mathsf{j}_{\varpi'_k}$ as for the fifth system. The doubled vertex is diagonal:

$$\mathbb{W}^{\mathrm{d}}[j](\Psi, \Psi') = (\mathbb{W}[j]\Psi, \mathbb{W}[j]\Psi').$$

Indeed the coupling $\mathbb{J}$ cancels in the difference of the systems at two multipliers. The support argument is unchanged: a row used at a junction has its full averaging set in $\kappa'^{[5/12]}$, and all three pieces at that junction lie in the assigned carrier. The same j^0 and vertex therefore have the same values under restriction or under a continuation outside the carrier.

To check summability, write m_k for the trunk lengths and $m'_k \ge 1$ for the branch lengths. The strong product estimate is bounded by

$$8B_4^{\mathrm{d}} \prod_k q_{\mathrm{d}}^{m_k} \prod_{k\ge 1} (C_b^{\mathrm{d}} \gamma_0 q_1^{m'_k}) \sup_z \exp\{-\delta \textstyle\sum d_{\mathrm{trunk}} - \frac{\delta}{2} \sum d_{\mathrm{branch}}\} \, |g|_z, \quad C_b^{\mathrm{d}} = 8B_4^{\mathrm{d}} C_{\mathrm{vert}} e^{\rho_0} C_j.$$

Here q_1, C_j are those of the first multiplier expansion. The vertex sees only its weak field seminorm, which is bounded by the strong solution seminorm. The junctions have distance at most ρ_0; joining them changes the exponent by at most $\delta\rho_0$ per branch. If $m = \sum m_k + \sum m'_k + 2n$, then $\mathfrak{M}^{-9m'_k} \le \mathfrak{M}^{-3(m'_k+2)}$. There are at most $2^m (27 N_{\mathrm{G}})^{m+1}$ decorated labels for a fixed tree. Thus, after increasing a constant $C_{\mathrm{tr}}^{\mathrm{d}}$,

$$\sum_{\Xi:\omega(\Xi)=\omega} |\mathbb{E}_\Xi^{\mathrm{d}} \iota g|_y \le B_{\mathrm{tr}}^{\mathrm{d}} (C_{\mathrm{tr}}^{\mathrm{d}} \mathfrak{M}^{-3})^m \sup_z e^{-\delta D_\omega(y,z)/2} |g|_z.$$

Decrease a_{D} so that $|j^0|_{-3} \le a_{\mathrm{d}}/2$ and increase the floor so that $4C_{\mathrm{vert}}B_5^{\mathrm{d}}C_jN_{\mathrm{G}}e^{\rho_0}q_1e \le 1$. Here $B_5^{\mathrm{d}} = 16N_{\mathrm{G}}B_4^{\mathrm{d}}$, and a_{D} is at most both the first-tree and doubled-inverse radii. These choices also include the local small-triple restriction of those inverse estimates. The resolvent series in $\mathsf{j}_* - \mathsf{j}^0$ then converges. Its absolute convergence permits the same rearrangement as in the fifth-system proof, identifying the sum with the doubled inverse at j_*.

On this common set of decorated labels put

$$\mathfrak{G}_\omega = \sum_{\Xi:\omega(\Xi)=\omega} \big([\mathbb{E}_\Xi^{[5]}]_{A,1} - [\mathbb{E}_\Xi^{\mathrm{d}}]_{A,1'}\big). \tag{613}$$

The first slot $1'$ is in the second copy and the output A is in the first. The elimination in proposition 34.1 identifies the two sums as $G_1 - H_{1,\mathrm{pr}}Q_{\mathrm{pr}}G_1$ and $G_1D\mathsf{R}_{\mathsf{m}}D^{\mathsf{T}}G_1$. It follows that their difference is $\mathfrak{G}$. Both terms in (613) are defined on the same carrier before summation; the equality therefore also specifies a single local parameter assignment.

Here is the scale conversion in these block estimates. A force supported over y' in slot 1 or $1'$ has forcing seminorm $\xi_{y'}^3 \sup|f|$. The output seminorm contains $\xi_y|A|$ and $\xi_y^2|\nabla A|$. An eighth-slot column has forcing seminorm $\xi_{c'}|e|$, while the last output component has weight ξ_c^3. These give, respectively, the factors $\xi_{y'}^3/\xi_y$, $\xi_{y'}^3/\xi_y^2$, $\xi_{c'}/\xi_y$, $\xi_{c'}/\xi_y^2$ and $\xi_{c'}/\xi_c^3$ before scale comparison. Absorb the ratios of scales, of powers at most three, into half of the rate $\delta/2$. Moving the chosen position of a row to either end changes the distance by at most a fixed constant. This proves (609)– (611), enlarging B and C_{w}. The field columns in (613) have no eighth-slot or row-normalization factor.

The scalar mean projection. Let

$$F_\varpi = -[\mathbb{E}_\varpi^{[2]}]_{w,4}, \qquad Z_{\varpi'} = \mathbb{E}_{\varpi'}^{[1]}Q'^{\mathsf{T}}.$$

Only the values of $Z_{\varpi'}e_z$ at the final supported tops of F_ϖ are used. Take $C_T = 1 + C_{\mathrm{av}}c_{\mathrm{av}}/2$. The local radius (418) and (360) bound the transport adjoint by C_T; undoing its real gauge preserves the norm. The extra factor 2 below allows the bilinear transpose. The scalar transpose column is

$$(Q'^{\mathsf{T}}e_z)(x) = \mathrm{Ad}(T_{\mathbf{U}}(z,x))^{\mathsf{T}}e, \qquad x \in \Delta(z),$$

and is zero elsewhere, with norm at most $2C_T|e|$. The factor L^{-3j} in the average cancels the ratio ξ_z^3/ξ^3 in the transpose. This column depends only on the configuration in $\Delta(z)$; at level zero it is the identity column.

For $\varpi = (\kappa_0, \dots, \kappa_m)$ and $\varpi' = (\kappa'_0, \dots, \kappa'_{m'})$, define

$$P_{\varpi,\varpi'}(b,b') = \sum_{z \in \Lambda_* \cap \kappa_m^{[1/12]}} (\xi/\xi_z)^3 (DF_\varpi)(b,z)(DZ_{\varpi'})(b',z)^{\mathsf{T}}. \tag{614}$$

The last factor is transposed as a map of the Lie algebra. The factor $(\xi/\xi_z)^3$ is required by the physical pairings. A nonzero term forces $\kappa'_{m'} \in \mathcal{N}(\kappa_m)$. Attach it to the walk

$$(X^{(2)}(\kappa_0), \dots, X^{(2)}(\kappa_m), X^{(2)}(\kappa'_{m'-1}), \dots, X^{(2)}(\kappa'_0)),$$

of length $m + m'$. The base cell $X(\kappa'_{m'})$ of the omitted junction piece is contained in $X^{(2)}(\kappa_m)$. All scalar columns $\Delta(z)$ and all local factors in (614) lie in this walk's carrier.

The two column estimates before the sum over z are

$$\xi_y^3|(DF_\varpi e_z)(b)| \le 8B_4q^m e^{-\delta d(y;\varpi;z)}|e|, \quad \xi_{y'}^3|(DZ_{\varpi'}e_z)(b')| \le 16C_TB_4q^{m'}\xi_z^4 e^{-\delta d(y';\varpi';z)}|e|.$$

Write $E(z) = d(y;\varpi;z) + d(y';\varpi';z)$ and $D = D_\omega(y,y')$. Then $E(z) \ge D, d(y,z)$. The weighted transpose sum is bounded by

$$\sum_z (\xi/\xi_z)^3\xi_z^4\xi_y^{-3}\xi_{y'}^{-3}e^{-\delta E(z)} \le L^2c_S(\delta/4)(\xi/\xi_{y'})^3\xi_y^{-2}e^{-\delta D/2}.$$

For completeness, the power outside the exponential on the left is $(\xi/\xi_{y'})^3\xi_y^{-2}(\xi_z/\xi_y)$. Scale comparison bounds the last ratio by $Le^{\lambda d(y,z)}$, where $\lambda = \log L/(R_1\mathfrak{M}) \le \delta/8$. Use $E(z) \ge \frac{1}{2}D + \frac{1}{2}d(y,z)$ and sum the remaining exponential at rate at least $\delta/4$. This is the $a = a' = 3$ case of the weighted scalar gluing estimate; the larger displayed constant also covers both choices of bond endpoint.

The transpose on the Lie algebra costs at most 2. Thus a pair contributes at most

$$B_P q^{m+m'}(\xi/\xi_{y'})^3\xi_y^{-2}e^{-\delta D/2}.$$

For a fixed walk of length n, there are at most $(n+1)(27N_{\mathrm{G}})^{n+1}N_{\mathrm{G}}$ attached pairs. Since $q \le \mathfrak{M}^{-9}$, $n+1 \le 2^n$ and the floor may ensure $54N_{\mathrm{G}}\mathfrak{M}^{-8} \le 1$, their sum gives (612). The scalar inverse identities give

$$\sum_{\varpi} F_{\varpi} = G'Q'^{\mathsf{T}}C', \qquad \sum_{\varpi'} Z_{\varpi'} = G'Q'^{\mathsf{T}}, \qquad 1 - \mathsf{R}_{\mathsf{m}} = G'Q'^{\mathsf{T}}C'Q'G', \quad C' = (Q'G'^2Q'^{\mathsf{T}})^{-1}.$$

Absolute convergence allows multiplication and the two differences term by term. Their sum is P_{f}.

Locality and summation. Every factor just constructed is an inverse of a finite local matrix, a local mean derivative, a difference, or one of the previously constructed multiplier terms. Its entries are holomorphic on the indicated open local class. Equal traces of two sequences on a carrier give the same pieces, row positions and weights, cutoffs, local sequences and neighbours. They therefore give identical matrices and identical finite products, including the attached scalar pairs. In particular none of these assertions requires a regular continuation to the whole torus. Gauge changes intertwine each matrix and preserve the physical pairings; the real structure commutes with the local means and inverses. These properties pass to every displayed term.

The number of walks of length m with a fixed first cell is at most $n_{1,\natural}^m$, and the number of the trees used here is at most $(4n_{1,\natural})^m$. Choose C_{w} to dominate all constants above. The floor makes $4n_{1,\natural}C_{\mathrm{w}}/\mathfrak{M} \le 1/2$. The geometric series and its first moment prove the two mass bounds. For each fixed output top, the supports of the first cutoffs and their first differences permit only a bounded number of first cells. This is the same local overlap bound used in the regular parametrix. Consequently the series converge uniformly on the local classes, in each finite-dimensional operator norm. All these estimates also hold after discarding any set of terms.

The parameters. For a graph Γ meeting the cells of ω, the hull lies within sup distance

$$\rho = (12r_{\mathrm{w}} + 10)\mathfrak{M} + 4 \le 23r_{\mathrm{w}}\mathfrak{M} \le 23\mathsf{w}/88$$

of Γ. The extra 4 permits the first-difference output window. A tube of radius ρ around a connected graph of ordinary length λ_Γ meets at most

$$8\max\{1, 4\lambda_\Gamma/(\mathsf{w} - 2\rho)\}$$

partition cubes. To see this, take a maximal D-separated net in the graph, with $2D < \mathsf{w} - 2\rho$. Its balls of radius $D + \rho$ each meet at most eight cubes. If the net has at least two points, its disjoint balls of radius $D/2$ each contain graph length at least $D/2$: distance to the centre is 1-Lipschitz and takes every value from 0 to $D/2$ on the connected graph. Thus the net has at most $2\lambda_\Gamma/D$ points. Let $2D$ increase to $\mathsf{w} - 2\rho$. The eight-cube assertion follows by lifting to the grid; the torus side is at least $3\mathsf{w}$, so a set of sup diameter less than w meets at most two intervals in each direction.

In the present scale ordinary length is at most weighted length. Taking the infimum over Γ gives

$$\#\mathsf{P}(\omega) \le 8 + \frac{68}{\mathsf{w}}D_\omega(y,y'), \qquad |s^\omega|e^{-aD_\omega(y,y')} \le e^{8\varkappa}e^{-(a-68\varkappa/\mathsf{w})D_\omega(y,y')}.$$

The parameter loss is at most $\mathring{\delta}_0/32$. Each retained rate is at least $\mathring{\delta}_0/2$; we may therefore use $\mathring{\delta}_0/4$ after inserting the parameters and summing the masses and first-cell overlaps. The proof used absolute values, so the same bound holds for arbitrary subsums. Uniform convergence on bounded parameter sets also follows directly from the finite number of partition cubes. Each term

is holomorphic and contains each parameter at most once; grouping by the finitely many sets $\mathsf{P}(\omega)$ proves the asserted polynomial and holomorphy statements.

Normalization and the response bounds. The identity $Q_{\mathrm{pr}} = \mathsf{m}^{-1}Q$ gives $Q_{\mathrm{pr}}^{\mathsf{T}} = Q^{\mathsf{T}}\mathsf{m}^{3}$, since dimensionless rows have weight ξ_c^{-1}. Let $B_\sharp$ absorb the masses and local overlaps. The parameter estimates for the normalized columns are

$$|(H_s e_{c'})(b)|, |(K_s e_{c'})(b)| \le B_\sharp e^{8\varkappa}\xi_{y'}^{-1} e^{-\mathring{\delta}_0 d(y,y')/4}|e|,$$
$$|(\nabla H_s e_{c'})(b,\nu)|, |(\nabla K_s e_{c'})(b,\nu)| \le B_\sharp e^{8\varkappa}\xi_y^{-1}\xi_{y'}^{-1} e^{-\mathring{\delta}_0 d(y,y')/4}|e|,$$
$$|S_s(c,c')| \le B_\sharp e^{8\varkappa}(\xi_y/\xi_{y'}) e^{-\mathring{\delta}_0 d(y,y')/4}.$$

Multiplying the first two lines by ξ_y and ξ_y^2 gives the column bound with the explicit ratio $\xi_y/\xi_{c'}$. The corresponding ratio for $\mathfrak{G}_s f$, with input seminorm $|f|_{y'}^\sharp = \xi_{y'}^3 \sup|f|$, is $(\xi_y/\xi_{y'})^3$. For P_s there are at most $3(\xi_{y'}/\xi)^3$ bare bonds over y', so its kernel bound gives

$$|P_s A|_y^\sharp \le 3B_\sharp e^{8\varkappa}(\xi_y/\xi_{y'}) e^{-\mathring{\delta}_0 d(y,y')/4}|A|_{y'}$$

when A is supported over y'. The scale hypothesis gives

$$(\xi_y/\xi_{y'})^p \le L^3 e^{\mathring{\delta}_0 d(y,y')/16}, \qquad p = 1, 3.$$

There are at most six rows at a top. Summing the column bounds and the bare blocks therefore gives the strong response estimates with $b_0 = 6L^3 B_\sharp$ and $\mu = 3\mathring{\delta}_0/16$. Multiplication by an admissible weight of rate $\theta = \mathring{\delta}_0/8$ leaves $\mu - \theta = \mathring{\delta}_0/16$ for the top sum, proving the displayed value of b_{D}.

The stronger column bounds before scale absorption also give the transpose needed for the local energy. For the dimensionless row pairing, testing $H_s^{\mathsf{T}} f$ at c contributes ξ_c; summing bare bonds over y contributes at most $3\xi_y^3$. The factor ξ_c^{-1} in the first column bound cancels the row factor. Duality in the Lie algebra costs at most 2, and hence

$$|H_s^{\mathsf{T}} f|_{y_c} \le 6B_\sharp e^{8\varkappa} \sum_y e^{-\mathring{\delta}_0 d(y,y_c)/4}|f|_y^\sharp, \qquad |H_s^{\mathsf{T}} f|_w \le \widehat{b}|f|_w^\sharp. \tag{615}$$

After multiplication by the weight, the first inequality leaves rate $\mathring{\delta}_0/8$, so the already imposed summation floor at $\mathring{\delta}_0/16$ suffices.

Finally, if a term has $\mathsf{P}(\omega) \subset \mathsf{Y}$, its whole carrier lies in W, and its endpoint tops lie in $Y^1(\mathsf{Y})$. A path from $Y^j(\mathsf{c})$ to outside $Y^{j+1}(\mathsf{c})$ crosses an annulus of weighted width at least $\mathfrak{M}/L^2$. Increase the floor so that this exceeds r_0. For $j \le 4$ the r_0-neighbourhood is then inside $\operatorname{int}\bigcup \mathsf{c}$. This proves the local margins; for $j \le 2$ the larger distance to the complement is at least $(5-j)\mathfrak{M}/L^2 > 3r_0$ after the same choice. All bonds in the first covariant difference are consequently in W, so its weighted norm is the response seminorm g_y used above.

The hull is connected. The partition cubes meeting a connected set are wall connected: all cubes at a point are joined by changing one grid coordinate at a time, and a separation into two wall components would disconnect the set. Thus $s^\omega \neq 0$ places its hull in one wall component of $\{s \neq 0\}$. Absolute convergence gives the component splitting. The preceding margin shows that sites in Y^j of different components, $j \le 4$, cannot be within distance r_0. Restriction consistency of each local coefficient gives the claimed component dependence. □

41.2. **The nonlinear response and its current.** Keep the hypotheses, parameters and local regions of proposition 41.1, and put $\eta = \xi$. Put

$$(Y, Y^1, Y^2) = (Y^1(\mathsf{Y}), Y^2(\mathsf{Y}), Y^3(\mathsf{Y})).$$

Decrease a_{D} so that

$$a_{\mathrm{D}} \le \min\left\{a_{\mathrm{nl}}, a_{\mathrm{pr}}, \frac{\min(1, c_{\mathrm{av}})}{8(L + 2\pi L^2)}\right\}.$$

We localize the actual maps of proposition 33.1 as follows. On rows over Y^1, use the straightened average $\mathcal{Q} = Q + \mathsf{C}$ and put $\mathsf{a} = 1$; on all other rows set $Q, \mathsf{C}, \mathsf{a}$ equal to zero. Let $F(A) = \eta^{-1} \sum_{p \subset W} a_p(e^{i\eta A}\mathbf{U})$, where the sum is over plaquettes with all corners in W. On bonds over Y, set

$$\mathcal{V}(A) = F'(A) - F'(0) - F''(0)A, \qquad \Delta' A = F''(0)A - D^{\mathsf{T}} D A,$$

and retain the given current there; set these three maps to zero on the other bonds. We continue to write $\mathbf{J}$ for this restricted current. Use the covariant-difference seminorm g_y on Y^2 and set it to zero elsewhere. All row pairings retain weight ξ_c^{-1} and all bare pairings weight η^3.

The local margins put each retained double block and every plaquette touching a bond over Y^2 inside W. To verify the whole averaging ball, fix a row c of positive level and use row units $\eta_c = \xi/\xi_c$. Its double block is a coordinate box with at most $2\eta_c^{-1}$ sites in each direction. The real curvature there is at most $L^2\alpha_0\eta_c^2$. The comb gauge of lemma 14.4, taken entirely in this box, gives $U^h = e^{i\eta_c B_0}$ with $\sup|B_0| \le 2\pi L^2\alpha_0$. If A' is the perturbation in $\mathbf{U} = e^{i\xi A'}U$, then $\sup|\xi_c \operatorname{Ad}(h)A'| \le L\alpha_1$. The product logarithm used in (417) therefore writes $\mathbf{U}^h = e^{i\eta_c B}$, where

$$\sup|B| \le 2(L\alpha_1 + 2\pi L^2\alpha_0) \le c_{\mathrm{av}}/4.$$

Here the sum in parentheses is at most $1/8$, so the same series logarithm is defined. On $\sup|A| < c_4$, the relative field $\xi_c \operatorname{Ad}(h)A$ has norm less than $Lc_4 \le c_{\mathrm{av}}/2$. Apply proposition 24.1 at this complex base. Unitary covariance of its domains and recursion identifies its relative map with the actual straightened average. The counting-kernel rescaling is $\xi_c\eta_c^3 = \xi^3\xi_c^{-2}$, exactly as in (512). At level zero the recursion is the identity. Thus the averaging, action and transpose estimates of proposition 33.1 apply with their existing constants. All gauges and stencils remain in W. Fixing the comb gauge in a neighborhood gives local holomorphy in the complex bond data; the intrinsic recursion identifies the overlapping charts.

Choose the fixed geometric floor so that the carrier margins exceed $3r_0$. Use c_W, C_W from (494), with the original averaging constants. The five operators satisfy (493) with $\widehat{b} = b_{\mathrm{D}}e^{8\varkappa}$: the new row weight satisfies $0 \le \mathsf{a} \le 1$, hence $|(S_s - \mathsf{a})X|_w \le 2\widehat{b}|X|_w$. The reduced field W_s is (498) with these localized maps and parameter operators.

For the multiplier estimate we use the quadratic energy in its bilinear form. If the local structure is carried by $Y \subset Y^1 \subset Y^2$, put

$$E_y(A,\delta) = \eta^3 \sum_{p \text{ over } y} \langle D_{\mathbf{U}} A(p), D_{\mathbf{U}}\delta(p)\rangle \quad (y \in Y^1), \qquad E_y = 0 \quad (y \notin Y^1), \qquad E = \sum_y E_y.$$

All these plaquettes lie in the local region. The count and curl estimate in (581) apply unchanged:

$$|E_y(A,\delta)| \le 12\xi_y^{-1}\mathsf{g}_y(A)\mathsf{g}_y(\delta).$$

This calculation is bilinear and remains valid at a complex regular pair; it asserts no positivity there. For real A at a real pair, $E(A,A) \ge 0$, and in the global case $E(A,A) = (DA, DA)_\eta$.

The local action estimate, at level $n(y)$, gives

$$|\Delta' A|_y^\sharp \le C_E \max_{N(y)} |A|, \qquad \Delta' = F''(0) - D^{\mathsf{T}} D$$

on the retained bonds. In fact its unscaled error is bounded by $C_{\mathrm{a}}\alpha \sup|\widehat{A}|$, where $\widehat{A} = \xi_y A$ and $\alpha = L^2\alpha_0 \le 1/2$; the adjacent-level ratio $\xi_y/\xi_{y'} \le L$ gives the displayed bound with $C_E \ge C_{\mathrm{a}}L$. By the definition of the Taylor remainder,

$$(F'(A) - F'(0), \delta)_\eta = E(A,\delta) + (\Delta' A + \mathcal{V}(A), \delta)_\eta \tag{616}$$

whenever δ vanishes off the bonds over Y. Indeed, first add $\Delta' A = F''(0)A - D^{\mathsf{T}}DA$ and $\mathcal{V}(A) = F'(A) - F'(0) - F''(0)A$ at those bonds. In the resulting pairing, a plaquette contributes to $(DA, D\delta)_\eta$ only if it has a bond over Y. Its base lies in $N_5(Y) \subset Y^1$, so grouping these plaquettes by their bases gives exactly $E(A,\delta)$.

Let H satisfy the column and whole-field transpose estimates used in proposition 34.2, with weighted constant $\widehat{b}$. Define $H^{\circledast}(A)$ by

$$(H^{\circledast}(A), X)_1 = E(A, HX) + (\Delta' A + \mathcal{V}(A), HX)_\eta.$$

The nondegenerate row pairing defines it uniquely. For $\|A_i\| \le 2c_W/\widehat{b}$ and every admitted weight w,

$$|H^{\circledast}(A_1) - H^{\circledast}(A_2)|_w \le c_{\circledast}\widehat{b}\,\|A_1 - A_2\|_w, \qquad c_{\circledast} = 100(1 + C_E + C_{\mathcal{V}})\mathsf{e}. \tag{617}$$

Use the normalized column estimate proved in proposition 41.1, before its scale ratio is absorbed into the exponential. For a test row c over y' and a bond b over y,

$$|(He_c)(b)| \le B_\sharp e^{8\varkappa}\xi_c^{-1} e^{-\mathring{\delta}_0 d(y,y')/4}, \qquad \mathsf{g}_y(He_c) \le B_\sharp e^{8\varkappa}\frac{\xi_y}{\xi_c} e^{-\mathring{\delta}_0 d(y,y')/4}.$$

The row pairing contributes ξ_c. It cancels the factor $\xi_y^{-1}(\xi_y/\xi_c)$ in the energy term. In the lower-order pairing the bare volume and the three bond directions leave the factor $\xi_c\xi_c^{-1}$ as well. Thus no scale quotient remains in either sum. The local Lipschitz estimate for $\mathcal{V}$, the bound on Δ', and complex bilinear duality bound their difference by

$$(24 + 6C_E + 48C_{\mathcal{V}})B_\sharp e^{8\varkappa}\sum_y e^{-\mathring{\delta}_0 d(y,y')/4}\max_{N(y)}\|A_1 - A_2\|.$$

An admissible weight costs at most $e^{\mathring{\delta}_0 d(y,y')/8}$, and the neighborhood maximum costs e. The remaining sum is at most $c_S(\mathring{\delta}_0/16)$. Since $b_{\mathrm{D}} \ge 6L^3 B_\sharp c_S(\mathring{\delta}_0/16)$, this proves (617) with the displayed, larger constant. This choice does not change the averaging constant $c_\flat$ or the contraction radii.

Write $J = F'(0)$ in the global case. Then (616) says $H^{\mathsf{T}}(F'(A) - J) = H^{\circledast}(A)$. Thus, with

$$\Psi_A = (I + H^{\mathsf{T}}\mathsf{C}'_A{}^{\mathsf{T}})^{-1}, \quad \omega(A) = \Psi_A[H^{\mathsf{T}}J + H^{\circledast}(A)], \quad m_\omega = 8\widehat{b}\,\gamma_0 + 2c_{\circledast}c_W,$$

the previously proved inverse and transpose bounds give

$$\sup|\omega(A)| \le 2m_\omega, \tag{618}$$

$$|\omega(A_1) - \omega(A_2)|_w \le (32C_3\mathsf{e}\,m_\omega + 2c_{\circledast})\widehat{b}\,\|A_1 - A_2\|_w. \tag{619}$$

For the second inequality subtract $\Psi_{A_i}\varphi_i$: the inverse difference is at most $32C_3\mathsf{e}\,\widehat{b}\,m_\omega\|A_1 - A_2\|_w$, and $\|\Psi_{A_2}\|_w \le 2$ handles the remaining term by (617). At a critical reference $J = Q^{\mathsf{T}}\lambda_0$ and $QH = I$, one has $\omega(0) = H^{\mathsf{T}}J = \lambda_0$. These estimates also retain the possibly nonzero initial multiplier.

Define the multiplier and current using the energy already introduced:

$$\omega_s(A) = (I + H_s^{\mathsf{T}}\mathsf{C}'_A{}^{\mathsf{T}})^{-1}[H_s^{\mathsf{T}}\mathbf{J} + H_s^{\circledast}(A)], \qquad j_s(A) = f_A^{-1}(Q + \mathsf{C}'_A)^{\mathsf{T}}\omega_s(A) - Q^{\mathsf{T}}H_s^{\mathsf{T}}\mathbf{J}. \tag{620}$$

Here $f_A = f(-i\eta\,\mathrm{ad}\,A)$ acts bondwise, $f(z) = (e^z - 1)/z$, with its removable value at zero. It satisfies $\|f_A^{-1}\| \le 2$ on the present ball. For $\|A_i\| \le 2c_W/\widehat{b}$ one has

$$|j_s(A_1) - j_s(A_2)|_w^\sharp \le C_j\widehat{b}(1 + \gamma_0\widehat{b})\|A_1 - A_2\|_w, \qquad j_s(0) = 0, \tag{621}$$

where $C_j = C_j(L) > 0$ is fixed below. Here and below the constants are uniform in the number of layers.

We include the estimate because differentiating the current pointwise would lose a derivative. Put $g_i = (Q + \mathsf{C}'_{A_i})^{\mathsf{T}}\omega_s(A_i)$ and $m = 8\widehat{b}\gamma_0 + 2c_{\circledast}c_W$. The transpose estimates for Q, C' and the preceding multiplier estimates give, for $\delta = \|A_1 - A_2\|_w$,

$$\sup|g_i|^\sharp \le 2(c_\flat + 1)m, \qquad |g_1 - g_2|_w^\sharp \le (c_\flat\mathsf{e} + 1)|\omega_s(A_1) - \omega_s(A_2)|_w + 2C_3\mathsf{e}m\delta.$$

The series for f gives $\|f(z)-I\| \le 1/14$ and $\|f(z_1)-f(z_2)\| \le (4/7)\|z_1-z_2\|$. The inverse difference identity then gives $|f(z_1)^{-1}-f(z_2)^{-1}| \le |z_1-z_2|$ for $|z_i| \le 1/8$. Since $|\operatorname{ad} X| \le 2|X|$ and $\eta|A_1(b)-A_2(b)| \le |A_1-A_2|_{y_b}$,

$$|j_s(A_1)-j_s(A_2)|_w^\sharp \le 2|g_1-g_2|_w^\sharp + 2\delta \sup|g_2|^\sharp.$$

Inserting the multiplier difference bound gives

$$[4c_\flat \mathsf{e}(32C_3\mathsf{e}\widehat{b}m + 2c_\circledast \widehat{b}) + 4C_3\mathsf{e}m + 8c_\flat m]\delta.$$

Since $c_W \le 1$ and $\widehat{b} \ge 1$, one has $m \le (8+2c_\circledast)(1+\gamma_0\widehat{b})$. A fixed $C_j(L)$ therefore bounds the displayed expression by the right side of (621). The inverse series and the energy formula prove joint holomorphy. At zero $\Psi_0 = I$, $H_s^\circledast(0) = 0$ and $f_0 = I$, which proves $j_s(0) = 0$ with the subtraction in (620).

Theorem M. In the preceding setting put

$$\varepsilon_{\mathrm{D}} = \min\{c_W/(4b_{\mathrm{D}}^2), 1/(8C_W b_{\mathrm{D}}^4), \varepsilon_{\mathrm{crit}}\}.$$

For $\sup|B'| \le \varepsilon_{\mathrm{D}} e^{-32\varkappa}$ there is a unique triple $(X, \mathbf{A}, \mathbf{H})$ in the balls

$$\sup|X| \le c_W \widehat{b}^{-2}, \qquad \|\mathbf{A}\| \le \varepsilon_{\mathrm{D}} b_{\mathrm{D}} e^{-24\varkappa}$$

solving

$$A' = K_s B' + \mathbf{A}, \quad X = \mathsf{C}(A' - H_s X), \quad \mathbf{A} + \mathfrak{G}_s \mathsf{W}_s(A') = 0, \quad \mathbf{H} = A' - H_s X.$$

The two successive iterations converge to it, and

$$\|\mathbf{H}\| \le 4\widehat{b}\sup|B'|, \qquad \sup|X| \le 16C_2\widehat{b}^2(\sup|B'|)^2, \qquad \|\mathbf{A}\| \le 8C_W\widehat{b}^5(\sup|B'|)^2.$$

Set $j(s;B') = j_s(\mathbf{H}(s;B'))$ and $\mathbf{H}^P(s;B') = P_s\mathbf{H}(s;B')$. For every weight $w \in \mathcal{W}_\theta$ the three maps are Lipschitz in B' with respective constants

$$4\widehat{b}, \qquad 8C_j\widehat{b}^3, \qquad 4\widehat{b}^2$$

in $\|\cdot\|_w$, $|\cdot|_w^\sharp$ and $|\cdot|_w^\sharp$. In particular, changing only datum rows over y' gives decay $e^{-\theta d(y,y')}$ at y. All three maps vanish at $B' = 0$.

They are jointly holomorphic in the interior parameter polydisc, in B', and in $(A_0, \mathbf{J})$ on the open local natural class with a fixed real factor $\mathbf{U} = e^{i\eta A_0}U$. They split as sums over the wall components c_i of $\{s \ne 0\}$. The response and $\mathbf{H}^P$ of a component are supported over $Y^1(\mathsf{c}_i)$; its current is supported over $N_{\rho_0}(Y^1(\mathsf{c}_i)) \subset Y^2(\mathsf{c}_i)$. Each depends only on the pair in W_{c_i}, the parameters on c_i, and the datum rows over $Y^1(\mathsf{c}_i)$. It vanishes when those datum rows vanish. The maps are unchanged by extending the pair to a larger local natural class while keeping $s = 0$ outside the old set. They commute with unitary gauge transformations and with complex conjugation; in particular real inputs give real outputs. ■□

Proof. The parameter estimates proved above give

$$\varepsilon_{\mathrm{D}} e^{-32\varkappa} \le \min\{c_W/(4\widehat{b}^2), 1/(8C_W\widehat{b}^4)\}.$$

Thus theorem J applies with the displayed balls. Its weighted estimate gives the response bound; the first fixed-point equation gives $|X| \le 4C_2|A'|^2 \le 16C_2\widehat{b}^2|B'|^2$. Composition with (621), using $\gamma_0 \le 1 \le \widehat{b}$, gives the current constant. Applying P_s gives the third constant. Exponential weights centred at y' give the stated decay. Uniform convergence of both iterations gives joint holomorphy. The domain in $(A_0, \mathbf{J})$ is open: the real-factor condition is fixed and the other conditions are finitely many strict inequalities between continuous functions.

We give the locality argument, including the inverse in W_s. Transposes preserve both-variable carriers because the pairings have diagonal, nonzero weights. The kernels of $Q, \mathsf{C}_T', \mathsf{C}''[T]$ have range ρ_0; at a row or bond near a component their transpose sums therefore use only data within

its $2\rho_0$-neighbourhood. If two fields agree on $N_{r_0}(Y_i)$, their first fixed-point solutions agree on $N_{\rho_0}(Y_i)$. To see this, keep the full solution X on that set and define it elsewhere by $\mathsf{C}(A'' - H_i X)$ for the component system. Since H_i uses only rows in Y_i, the resulting vector solves the component fixed-point equation and remains in the same contraction ball. Uniqueness gives the claimed agreement and agreement of $A' - H_s X$ on $N_{r_0}(Y_i)$. Further,

$$(I + H_s^{\mathsf{T}} \mathsf{C}_T'^{\,\mathsf{T}})^{-1} = \sum_{k\ge 0} (-H_s^{\mathsf{T}} \mathsf{C}_T'^{\,\mathsf{T}})^k.$$

Every term agrees near Y_i with the corresponding term for the component system: $\mathsf{C}_T'^{\,\mathsf{T}}$ uses the ρ_0-neighbourhood and H_s^{T} then restricts to Y_i. Induction proves the assertion for all powers. The defining six terms of W_s therefore agree over Y_i with the component reduced field.

Let $\mathbf{A}_i$ be the component solutions and put $\mathbf{A} = \sum_i \mathbf{A}_i$. A ρ_0-neighbourhood meets at most one Y_i, because different carriers are separated by more than $r_0 = 2\rho_0$. Hence $\|\mathbf{A}\| = \max_i \|\mathbf{A}_i\|$; the same observation controls the first fixed-point arguments. The local agreement just proved gives

$$-\mathfrak{G}_s \mathsf{W}_s(K_s B' + \mathbf{A}) = -\sum_i \mathfrak{G}_i \mathsf{W}_i(K_i B' + \mathbf{A}_i) = \sum_i \mathbf{A}_i.$$

Uniqueness in the second contraction now gives the decomposition of $\mathbf{H}$.

For the current, $H_i^{\circledast}$ uses only E_y with $y \in N_{\rho_0}(Y_i)$ and $\Delta', \mathcal{V}$ over Y_i. The preceding inverse-series comparison therefore identifies the multipliers on each component. The transpose of $Q + \mathsf{C}'$ enlarges its support only to $N_{\rho_0}(Y_i)$; these enlarged supports are disjoint. This proves the current decomposition. The identity $P_i \mathbf{H}_j = 0$ for $i \neq j$ proves the decomposition of $\mathbf{H}^P$.

Finally, restricting the pair to W_{c_i} changes none of the formulas used in these comparisons. In particular all plaquettes touching a bond over Y_i and all relevant averaging blocks lie in that region. Thus Q, C agree near Y_i, $\mathcal{V}, \Delta', \mathbf{J}$ agree over Y_i, and E_y, g_y agree on the enlarged carrier. The same argument proves invariance under extensions of the local pair. Dependence on the datum only through $K_i B'$ proves the remaining assertions, including vanishing at zero. The walk terms, local averages, action derivatives and energy pairings intertwine unitary gauge transformations and conjugation. Both convergent iterations and the inverse series therefore intertwine them as well. □

41.3. **Identification with the critical family.** In this subsection the determining set is the reduced set $\mathbf{B}^{\flat}$, as in theorem K. All actual fibers and gauge factors refer to these rows and their top nodes. At $\mathsf{Y} = \mathsf{P}$ and $s = 1$, all monomials s^{ω} equal one. The complete expansions therefore sum to the operators of proposition 34.2:

$$S = (QGQ^{\mathsf{T}})^{-1}, \qquad H = GQ^{\mathsf{T}}S, \qquad K = G_1 Q^{\mathsf{T}}(QG_1Q^{\mathsf{T}})^{-1},$$
$$\mathfrak{G} = G_1 - G_1 D \mathsf{R}_{\mathsf{m}} D^{\mathsf{T}} G_1 - KQG_1, \qquad P_1 = D(1 - \mathsf{R}_{\mathsf{m}})D^{\mathsf{T}}.$$

Thus the response at one is independent of the partition and its parameter radius. It is defined on the full ball $\sup|B'| < \varepsilon_{\mathrm{D}}$, using the global estimates with $\widehat{b} = b_{\mathrm{D}}$. In this subsection $\mathbf{H}(B')$ denotes that global response, whether or not decoupling parameters are present.

Proposition 41.2. Let U_0 be real, let $(U_0, J^{\circ}(U_0))$ belong to the global natural class with radii at most a_{D}, and assume $J^{\circ}(U_0) = Q^{\mathsf{T}}\lambda_0$. Use the mean-Landau projection R_{m} and the row pairing with weights ξ_c^{-1}. For $\sup|B'| < \varepsilon_{\mathrm{D}}$, $A = \mathbf{H}(B')$ satisfies

$$\mathcal{Q}(A) = B', \qquad \mathsf{R}_{\mathsf{m}} D^{\mathsf{T}} A = 0, \qquad F'(A) = (Q + \mathsf{C}_A')^{\mathsf{T}} \omega(A).$$

Moreover

$$\mathbf{J} + j_1(A) = J^{\circ}(e^{i\eta A} U_0), \qquad P_1 A = DD^{\mathsf{T}} A, \qquad \omega(0) = \lambda_0.$$

The multiplier obeys

$$\sup|\lambda_0| \le 8 b_{\mathrm{D}} \gamma_0, \qquad \sup|\omega(A)| \le 2 m_{\omega},$$
$$|\omega(\mathbf{H}(B_1')) - \omega(\mathbf{H}(B_2'))|_w \le 4(32 C_3 \mathfrak{e} m_{\omega} + 2c_{\circledast}) b_{\mathrm{D}}^2 |B_1' - B_2'|_w.$$

where $m_\omega = 8b_{\mathrm{D}}\gamma_0 + 2c_{\circledast}c_W$.

Conversely, if

$$\|A\| \le r_1 := \frac{\varepsilon_{\mathrm{D}}}{4c_\flat}, \quad \mathsf{R}_{\mathsf{m}} D^{\mathsf{T}} A = 0, \quad F'(A) = (Q + \mathsf{C}'_A)^{\mathsf{T}}\lambda,$$

then $\sup|\mathcal{Q}(A)| < \varepsilon_{\mathrm{D}}$ and

$$A = \mathbf{H}(\mathcal{Q}(A)), \qquad \lambda = \omega(A).$$

In particular this applies to a real A in that ball and slice whenever $e^{i\eta A}U_0$ is a critical configuration for its actual determining datum. ■

Proof. The equations in the first display are theorem K. The energy identity gives $\omega(A) = (I + H^{\mathsf{T}}\mathsf{C}'_A{}^{\mathsf{T}})^{-1}H^{\mathsf{T}}F'(A)$. Since $QH = I$, the multiplier equation implies $\omega(A) = \lambda$, and $Q^{\mathsf{T}}H^{\mathsf{T}}J = J$ at the reference. Consequently the current definition gives $J + j_1(A) = f_A^{-1}F'(A) = J^{\circ}(e^{i\eta A}U_0)$. The last equality is the local action derivative identity, at every bare bond, including incomplete boundary stars. The slice equation gives $P_1 A = DD^{\mathsf{T}}A$. The preceding energy bounds and the response Lipschitz estimate give the three multiplier estimates.

For the converse set

$$X = \mathsf{C}(A), \quad A' = A + HX, \quad B' = \mathcal{Q}(A), \quad \mathbf{A} = A' - KB'.$$

The local estimates and the definition of r_1 give

$$|B'| \le 2c_\flat r_1 = \varepsilon_{\mathrm{D}}/2, \quad |X| \le C_2 r_1^2 \le c_W b_{\mathrm{D}}^{-2}, \quad \|A'\| \le 2r_1 < c_W/b_{\mathrm{D}}, \quad \|\mathbf{A}\| \le b_{\mathrm{D}}\varepsilon_{\mathrm{D}}.$$

The first fixed-point uniqueness gives $X = \mathsf{D}(A')$. The identities $QH = QK = I$ show $QA' = B'$ and $Q\mathbf{A} = 0$. The on-shell identities $\mathsf{R}_{\mathsf{m}}D^{\mathsf{T}}H = \mathsf{R}_{\mathsf{m}}D^{\mathsf{T}}K = 0$ put both A' and $\mathbf{A}$ in the mean-Landau slice. Write $\mathcal{L}(A') = A' - H\mathsf{D}(A')$ and $K_A = Q + \mathsf{C}'_A$. The differentiated first fixed-point equation gives $K_A\mathcal{L}' = Q$. The reduced-action identity used in theorem K therefore yields, for every δ in that slice,

$$(J + \Delta_1 A' + \mathsf{W}_1(A'), \delta)_\eta = (F'(A), \mathcal{L}'\delta)_\eta = (Q^{\mathsf{T}}\lambda, \delta)_\eta.$$

The difference lies in $\mathrm{ran}(D\mathsf{R}_{\mathsf{m}})$. Apply $\mathfrak{G}$, using the already proved on-shell relations

$$\mathfrak{G}D\mathsf{R}_{\mathsf{m}} = 0, \quad \mathfrak{G}Q^{\mathsf{T}} = 0, \quad \mathfrak{G}\Delta_1 A' = A' - KQA'.$$

Since $J = Q^{\mathsf{T}}\lambda_0$, this gives $\mathbf{A} + \mathfrak{G}\mathsf{W}_1(A') = 0$. The triple is in the two contraction balls, so uniqueness identifies $A = \mathbf{H}(B')$ and then $\lambda = \omega(A)$.

It remains to justify the multiplier for a real critical configuration. The estimate $\|\mathsf{C}'_A H\|_\infty \le 4C_2 b_{\mathrm{D}} r_1 \le 1/2$ shows that $K_A H = I + \mathsf{C}'_A H$ is invertible. Thus K_A is onto and K_A^{T} is injective. For any real δ in $\ker K_A$, the finite-dimensional implicit function theorem gives

$$A_\tau = A + \tau\delta + H\beta(\tau), \qquad \mathcal{Q}(A_\tau) = \mathcal{Q}(A), \qquad \beta(0) = \beta'(0) = 0.$$

Let $v[A]$ be the node factors in the actual-average identity. Set $g_\tau = v[A]v[A_\tau]^{-1}$ on the reduced top set Λ_* and $g_\tau = I$ at other bare sites. Each top has one prescribed value, so this defines a real gauge transformation. The formula for the actual averages, including its outward twist, gives

$$M_{\mathbf{B}^\flat}((e^{i\eta A_\tau}U_0)^{g_\tau}) = M_{\mathbf{B}^\flat}(e^{i\eta A}U_0).$$

Criticality and gauge invariance therefore imply $(F'(A), \delta)_\eta = 0$. The real pairings are positive definite; hence the annihilator of $\ker K_A$ is the range of its real adjoint. There is a real λ with $F'(A) = K_A^{\mathsf{T}}\lambda$, unique by injectivity. □

Corollary 41.3. Let $U_0 = \Phi_{V_0}(0)$ be the canonical real background, and write the mean-Landau pair of its holomorphic continuation as

$$e^{i\eta\mathcal{H}(B)}U_0 = \Phi_{V_0}(B)^{u_B^{-1}}, \qquad \mathsf{R}_{\mathsf{m}}D^{\mathsf{T}}\mathcal{H}(B) = 0, \qquad u_B v[\mathcal{H}(B)] = I \quad \text{on } \Lambda_*.$$

Assume its reference pair lies in the natural class with radii at most a_{D}. On the common ball

$$\sup|B| < \min\{r_B, \varepsilon_{\mathrm{D}}/C_{\mathrm{tw}}\},$$

where r_B is the radius of this pair and the inverse datum twist, one has

$$\mathcal{H}(B) = \mathbf{H}(1; B_\natural), \qquad \mathcal{Q}(\mathcal{H}(B)) = B_\natural.$$

For real B on this ball,

$$u_B^{-1} = v[\mathcal{H}(B)] \quad \text{on } \Lambda_*, \qquad M_{\mathbf{B}^\flat}(e^{i\eta\mathcal{H}(B)}U_0) = (e^{iB}V_0)^{v[\mathcal{H}(B)]}.$$

■□

Proof. Criticality of U_0 gives the real initial multiplier by the preceding argument at zero. For real B sufficiently near zero, $\mathcal{H}(0) = 0$ puts $\mathcal{H}(B)$ in the radius-r_1 ball. Gauge covariance preserves criticality, so the proposition identifies it with $\mathbf{H}(\mathcal{Q}(\mathcal{H}(B)))$. Compare its actual-average identity with $M_{\mathbf{B}^\flat}(e^{i\eta\mathcal{H}(B)}U_0) = (e^{iB}V_0)^{v[\mathcal{H}(B)]}$. Cancellation of the node factors and V_0 gives $e^{i\mathcal{Q}(\mathcal{H}(B))^\natural} = e^{iB}$. Both logarithms are in the fixed logarithm ball; hence $\mathcal{Q}(\mathcal{H}(B))^\natural = B$ and $\mathcal{Q}(\mathcal{H}(B)) = B_\natural$. This proves the response identity near real zero. Both sides are holomorphic on the stated common complex ball. Their difference vanishes on a real neighbourhood of zero, hence on its complexification, and then along each complex radial line on the whole ball. The remaining equations follow from the on-shell response identity and the normalization of u_B. □

42. Gaussian normalizers near the trivial configuration

42.1. **Critical germs and constrained Gaussian integrals.** Fix a height $k \geq 0$ and a finite side $\mathfrak{s} \in \{2L^n : n \in \mathbb{N}\}$. In the level-$k$ scale put

$$\Lambda_j = L^{j-k}\mathbb{Z}^3/\mathfrak{s}\mathbb{Z}^3, \qquad \ell_j = L^{j-k}, \qquad \eta = L^{-k}.$$

In this subsection a form B is the dimensionless coordinate in $e^{iB}U$ on the positively oriented bonds. The difference d has no spacing factor, and r denotes restriction of site functions to the target level. The inner products are unweighted sums of $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$ on bonds or sites. Thus $B = \ell_j A$ for the physical field A at level j. The covariant gauge differential is $\mathsf{D}_U\lambda(b) = \operatorname{Ad}(U(b))\lambda(b_+) - \lambda(b_-)$, without a factor ℓ_j^{-1}. Reversed tangent coordinates are continued by $B(\bar{b}) = -\operatorname{Ad}(U(b))^{-1}B(b)$. All assertions near the trivial configuration in this subsection concern germs; their neighbourhoods may depend on k and $\mathfrak{s}$.

Let $\mathcal{V}_j, \mathcal{A}_j, \mathfrak{G}_j$ denote configurations, forms and site functions on Λ_j. Use the existing averaging map $\mathcal{M}$ between successive levels, and set

$$A(U) = \sum_p [1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr}\partial U(p)], \qquad \mathrm{GF}_j(V) = \sum_{y\in\Lambda_j}\sum_{x\in B(y)\setminus\{y\}} [1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} V(y,x)].$$

No spacing factor enters these functions. Write $\mathsf{Q} = \mathcal{M}'(\mathbf{1})$, $\mathsf{Q}^{[j]} = (\mathcal{M}^j)'(\mathbf{1})$, and $\mathsf{L}_V B(x) = \frac{1}{i}\frac{d}{dt}|_0 \log(e^{itB}V)(y,x)$ for $x \in B(y)\setminus\{y\}$. At $V = \mathbf{1}$ this is the mean contour sum $\mathsf{L}B(x) = \bar{B}(y,x)$.

For a surjective linear map $M : E \to F$ of real Euclidean spaces and a positive definite form T on $\ker M$, use the constrained Gaussian integral

$$I(T, M) = \det(MM^\mathsf{T})^{-1/2}\int_{\ker M} e^{-T(B)/2}\, dB, \qquad \mathbb{S}(X, M) = \begin{pmatrix} X & M^\mathsf{T} \\ M & 0 \end{pmatrix}. \tag{622}$$

The Lebesgue measure is that of the induced inner product.

Theorem N. There are unique real-analytic germs $\mathcal{S}_j$ on $\mathcal{V}_j$, $0 \leq j \leq k$, and $p_j : \mathcal{V}_j \to \mathcal{V}_{j-1}$, $1 \leq j \leq k$, with $\mathcal{S}_0 = A$, $\mathcal{S}_j(\mathbf{1}) = 0$, $\nabla\mathcal{S}_j(\mathbf{1}) = 0$, $p_j(\mathbf{1}) = \mathbf{1}$ and the following properties. Put $\mathcal{F}_j = \mathcal{S}_{j-1} + \mathrm{GF}_j$. For W near $\mathbf{1}$, $p_j(W)$ is the unique critical point near $\mathbf{1}$ of $\mathcal{F}_j$ on $\{V : \mathcal{M}(V) = W\}$. Its restricted Hessian $T_{j,W}$ is positive definite; it is a strict local minimum, is axial on Λ_j, and

$$\mathcal{M}(p_j(W)) = W, \qquad \mathcal{S}_j = \mathcal{S}_{j-1}\circ p_j = \mathcal{F}_j \circ p_j \geq 0.$$

These germs are covariant under lattice isometries mapping Λ_j onto itself and, for a coarse gauge transformation v near $\mathbf{1}$ or constant, under its block-constant extension $\tilde{v}$:

$$p_j(W^v) = p_j(W)^{\tilde{v}}, \quad p_j(\mathsf{s}W) = \mathsf{s}p_j(W), \quad \mathcal{S}_j(W^v) = \mathcal{S}_j(\mathsf{s}W) = \mathcal{S}_j(W).$$

Here each equality is read on a common neighbourhood where its two members are defined. Their Hessians h_j at $\mathbf{1}$ have null space $\{B : dB = 0\}$, and

$$h_0(B) = |dB|^2, \qquad h_j(B) = \min_{\mathsf{Q}A=B} \{h_{j-1}(A) + |\mathsf{L}A|^2\}.$$

The minimizer is $p_j'(\mathbf{1})B$.

Set $c_j = p_1 \circ \cdots \circ p_j$, with c_0 the identity. Then

$$\mathcal{M}^j c_j = I, \quad \mathcal{M}^{j-1} c_j = p_j, \quad \mathcal{S}_j = A \circ c_j, \qquad \nabla A(c_j(W)) = (\mathcal{M}^j)'(c_j(W))^{\mathsf{T}} \nabla \mathcal{S}_j(W). \tag{623}$$

In particular $c_j(W)$ is critical on its iterated-average fibre. Let $W_k = W$, $W_{j-1} = p_j(W_j)$, $U = W_0 = c_k(W)$, and $\mathcal{Q}_U^{[j]} = (\mathcal{M}^j)'(U)$. If H_W is the Hessian of A on the fibre $\mathcal{M}^k = W$ at U, put

$$\mathbf{L}_W = (\mathsf{L}_{W_{j-1}} \mathcal{Q}_U^{[j-1]})_{j=1}^k, \quad Z_j(W) = I(T_{j,W}, \mathcal{M}'(p_j(W))), \quad \Phi_j^\natural(W) = \log Z_j(W) - \log Z_j(\mathbf{1}).$$

The form $H_W + |\mathbf{L}_W \cdot|^2$ is positive definite on $\ker \mathcal{Q}_U^{[k]}$, and

$$\mathcal{Z}_k(W) := I(H_W + |\mathbf{L}_W \cdot|^2, \mathcal{Q}_U^{[k]}) = \prod_{j=1}^k Z_j(W_j). \tag{624}$$

Consequently the normalized logarithm $\Psi_k = \log \mathcal{Z}_k - \log \mathcal{Z}_k(\mathbf{1})$, with $\Psi_0 = 0$, satisfies

$$\Psi_k = \Phi_k^\natural + \Psi_{k-1} \circ p_k. \tag{625}$$

All these logarithms are real analytic and invariant under the stated gauge transformations and isometries. In (625), the function Ψ_{k-1} is transported from the tower of height $k-1$ and side $L\mathfrak{s}$ by the dilation $x \mapsto Lx$; forms are relabelled without an amplitude factor. ■□

Proof. The mean map is analytic near $\mathbf{1}$. Its derivative is onto: a form supported on the pivot bonds has $\mathsf{Q}A(c) = L^{-2} A(b_0(c))$, so the values on the distinct pivots prescribe every output independently. Their contours stay inside blocks, hence $\mathsf{L}A = 0$ for such a form. Full rank persists near $\mathbf{1}$, making all the fibres below analytic submanifolds. The gauge-fixing function is nonnegative and vanishes exactly on the axial slice. At an axial point, writing $Y_x(V) = i^{-1} \log V(y, x)$ and using $1 - \cos|Y| = |Y|^2/2 + O(|Y|^4)$ gives $\nabla \mathrm{GF}_j = 0$ and $\mathrm{GF}_j''(V)(B) = |\mathsf{L}_V B|^2$.

Suppose $\mathcal{S}_{j-1}$ has been constructed and its Hessian has the asserted null space. The form $g(A) = h_{j-1}(A) + |\mathsf{L}A|^2$ is positive definite on $\ker \mathsf{Q}$. Indeed, $g(A) = 0$ implies $dA = 0$. A curl-free form on a torus is $d\lambda + a$ with a constant: subtract its three period averages, and integrate the remaining form along paths. Interchanging steps changes the integral by a plaquette curl, and its periods now vanish, so the potential is well defined. If also $\mathsf{L}A = 0$, then, for $x = y + \ell_{j-1} \sum n_\mu e_\mu \in B(y)$,

$$\lambda(x) = \lambda(y) - \sum_\mu n_\mu a_\mu.$$

The identities $\mathsf{Q}d = d\mathsf{r}$ and $\mathsf{Q}a = La$ follow by telescoping the mean paths. Thus $\mathsf{Q}A = 0$ implies $d\mathsf{r}\lambda + La = 0$. Summing around a coarse torus loop gives $a = 0$, after which $\mathsf{r}\lambda$, and then λ, is constant.

Use the decomposition $\mathcal{A}_{j-1} = \ker \mathsf{Q} \oplus (\ker \mathsf{Q})^\perp$. The implicit function theorem parametrizes the fibre by $V = \exp i(B_0 + \beta(B_0, W))$, where $B_0 \in \ker \mathsf{Q}$ and $\beta(0, \mathbf{1}) = 0$, $\beta(B_0, \mathbf{1}) = O(|B_0|^2)$. The Hessian in B_0 of $\mathcal{F}_j(V)$ at $(0, \mathbf{1})$ is $g|_{\ker \mathsf{Q}}$. A second application of that theorem gives its unique analytic critical point $B_0(W)$; positivity on a common convex coordinate ball gives the strict minimum. Define p_j by this point. The gauge transformation equal to $p_j(W)(y, x)$ at $x \in B(y)$ and

to 1 on Λ_j makes it axial, preserves the constraint and $\mathcal{S}_{j-1}$, and can only decrease GF_j. Both configurations lie in the uniqueness neighbourhood for W sufficiently close to $\mathbf{1}$; hence the minimizer was already axial.

Coarse gauge transformations extended blockwise preserve the fibre problem and its gauge-fixing function. The same holds for the lattice isometries in the statement. Indeed, set $(\mathsf{s}V)(\mathsf{s}b) = V(b)$. Such an isometry maps blocks, straight paths and contour families bijectively to their images. Path holonomies and contour means are therefore unchanged after relabelling. For the matrices W_x in the averaging formula, with $x' = x + c_+ - c_-$, reversal gives

$$W^{\bar{c}}_{x'} = V[c]^{-1} W_x^{-1} V[c], \qquad \log W^{\bar{c}}_{x'} = -V[c]^{-1}(\log W_x)V[c].$$

Here all fields are in a sufficiently small neighbourhood of $\mathbf{1}$, where both logarithms are given by their series. Summing and exponentiating gives $\mathcal{M}(V)(\bar{c}) = \mathcal{M}(V)(c)^{-1}$. Thus $\mathcal{M}(\mathsf{s}V) = \mathsf{s}\mathcal{M}(V)$ even when the isometry reverses a positive bond. The contour means also show that GF_j is preserved, and the plaquette trace shows that A is preserved. Induction and local uniqueness give the stated covariance and invariance. Differentiating the constraint and criticality at $\mathbf{1}$ shows that $p_j'(\mathbf{1})B$ is the g-orthogonal lift of B, giving the displayed minimum formula for h_j. A null vector is therefore curl-free. Conversely write a coarse curl-free form as $B = d\gamma + a'$. Put $a = L^{-1}a'$ and $\lambda(x) = \gamma(y) - \sum n_\mu a_\mu$ in each block. Then $A = d\lambda + a$ satisfies $\mathsf{Q}A = B$, $\mathsf{L}A = 0$ and $dA = 0$, proving $h_j(B) = 0$. This closes the induction. The same uniqueness induction also shows that the germs agree under truncation of the tower and the stated dilation.

At $V = p_j(W)$, axiality removes the gradient of GF_j, so constrained criticality and surjectivity give $\nabla\mathcal{S}_{j-1}(V) = \mathcal{M}'(V)^{\mathsf{T}}\mu$. Differentiate $\mathcal{M}p_j = I$ and $\mathcal{S}_j = \mathcal{S}_{j-1}p_j$ to obtain $\mu = \nabla\mathcal{S}_j(W)$. Iterating the chain rule proves (623).

We record the two Gaussian calculations used in the product. For $X|_{\ker M} = T$, an orthonormal decomposition $E = \ker M \oplus (\ker M)^\perp$ gives

$$|\det \mathbb{S}(X, M)| = \det(T)\det(MM^{\mathsf{T}}), \qquad I(T, M) = (2\pi)^{(\dim E - \dim F)/2}|\det \mathbb{S}(X, M)|^{-1/2}.$$

These formulas show analyticity of the normalizers: the restricted Hessian has an analytic extension to the ambient space. Explicitly, if q_V^1 is the second derivative of $i^{-1}\log(\mathcal{M}(e^{iB}V)W^{-1})$, that extension is $\mathcal{F}_j''(V) - (\nabla\mathcal{S}_j(W), q_V^1)$. To verify it, choose a second-order correction C with $\mathcal{M}'(V)C = -q_V^1(B, B)$ for a tangent B; the curve $e^{i(tB + t^2C/2 + O(t^3))}V$ lies in the fibre. Its second derivative of $\mathcal{F}_j$ gives the asserted formula. The bordered determinant is nonzero by positivity.

Next let $M : E \to F$ and $N : F \to F'$ be onto, and let T be positive on $\ker NM$. For $\zeta \in \ker N$ let $\sigma\zeta$ minimize T over $M\xi = \zeta$. Then $T(b + \sigma\zeta) = T_1(b) + T_2(\zeta)$, where $T_1 = T|_{\ker M}$. The Jacobian J of this splitting satisfies

$$J^2 = \det((S^{-1})_{11}), \quad S = MM^{\mathsf{T}}, \qquad \det(NSN^{\mathsf{T}}) = \det(NN^{\mathsf{T}})\det S\, J^2.$$

Here the block 11 is on $\ker N$. The first identity follows by replacing $\sigma\zeta$ by the orthogonal lift $M^{\mathsf{T}}S^{-1}\zeta$, which changes the splitting by a shear of determinant one; the second is the block inverse formula on $F = \ker N \oplus (\ker N)^\perp$. The Gaussian and coarea factors therefore give

$$I(T, NM) = I(T_1, M)I(T_2, N). \tag{626}$$

The Hessians of successive fibre minima have precisely this Schur complement. In submersion coordinates (u, v, w), where $m(u, v, w) = (v, w)$ and $n(v, w) = w$, a critical section satisfies $F_u = 0$, so $f_{vv} = F_{vv} - F_{vu}F_{uu}^{-1}F_{uv}$ for $f(v, w) = F(u(v, w), v, w)$. Adding a function of (v, w) adds its vv Hessian to this Schur complement. Positive fibre Hessians at the two stages therefore give a positive total Hessian and the minimum formula used in (626).

Apply this calculation inductively to $F_i = A + \sum_{j\le i}\mathrm{GF}_j \circ \mathcal{M}^{j-1}$, with $m = \mathcal{M}^{i-1}$, $n = \mathcal{M}$ and the critical section c_{i-1}. It gives a positive Hessian of F_i on the fibre $\mathcal{M}^i = W_i$ at $c_i(W_i)$, and its normalizer is $\prod_{j\le i} Z_j(W_j)$. Since W_{j-1} is axial, each gauge-fixing summand has zero gradient

there and contributes the quadratic form $|\mathsf{L}_{W_{j-1}}\mathcal{Q}_U^{[j-1]}B|^2$. At $i=k$ the total Hessian is therefore $H_W+|\mathbf{L}_W B|^2$, proving (624). Taking normalized logarithms gives (625). Gauge transformations and isometries act orthogonally on these tangent spaces, including the adjoint correction on reversed bonds; they preserve both Gaussian and coarea factors. This proves the remaining invariance assertions. □

42.2. **The determinant in a background gauge.** We specify the correction from the derivative of the mean to the covariant straight average. At one level put

$$(\mathsf{n}_V B)(y)=L^{-3}\sum_{x\in B(y)}\frac{1}{i}\frac{d}{dt}\Big|_0\log[(e^{itB}V)(y,x)V(y,x)^{-1}],$$
$$(\mathfrak{a}_V\lambda)(y)=L^{-3}\sum_{x\in B(y)}\mathrm{Ad}(V(y,x))\lambda(x),\qquad\qquad \mathring{\mathcal{Q}}_V=\mathcal{M}'(V)+\mathsf{D}_{\mathcal{M}(V)}\mathsf{n}_V.$$

For $V_j=\mathcal{M}^j(U)$, use the products $\mathfrak{a}_U^{[j]}=\mathfrak{a}_{V_{j-1}}\cdots\mathfrak{a}_{V_0}$ and $\mathring{\mathcal{Q}}_U^{[j]}=\mathring{\mathcal{Q}}_{V_{j-1}}\cdots\mathring{\mathcal{Q}}_{V_0}$, and the recursion

$$\mathsf{n}_U^{[0]}=0,\qquad \mathsf{n}_U^{[j]}=\mathfrak{a}_{V_{j-1}}\mathsf{n}_U^{[j-1]}+\mathsf{n}_{V_{j-1}}\mathcal{Q}_U^{[j-1]}.$$

The restriction of site functions to Λ_j is $r=\mathsf{r}$; its transpose extends by zero. Define q_U by

$$\frac{1}{i}\log[\mathcal{M}^k(e^{iB}U)\mathcal{M}^k(U)^{-1}]=\mathcal{Q}_U^{[k]}B+\tfrac12 q_U(B,B)+O(|B|^3),$$

and set

$$c_U(\lambda,B)(b)=\tfrac{i}{2}[\lambda(b_-)+\mathrm{Ad}(U(b))\lambda(b_+),B(b)],\qquad \sigma_B=r^{\mathsf{T}}\mathsf{n}_U^{[k]}B,\qquad \tau_U B=B+\mathsf{D}_U\sigma_B.$$

At $U=\mathbf{1}$, abbreviate the scalar mean and correction by $\mathfrak{a}^{[j]}$ and $\mathsf{n}^{[j]}$, and the straight average by $\mathring{\mathsf{Q}}^{[j]}$. Omit [1] for one step.

For independent variables (U,J,μ) near $(\mathbf{1},0,0)$, define the symmetric operator $X_{U,J,\mu}$ by

$$(B,X_{U,J,\mu}B)=A''(U)(B,B)+2(J,c_U(\sigma_B,B))+(J,c_U(\sigma_B,\mathsf{D}_U\sigma_B))-(\mu,q_U(\tau_U B,\tau_U B))$$
$$+|(1-L^{3k}\mathfrak{a}_U^{[k]\mathsf{T}}\mathfrak{a}_U^{[k]})\mathsf{D}_U^{\mathsf{T}}B|^2. \tag{627}$$

All pairings and transposes here are the unweighted ones specified above. Put

$$\mathfrak{F}_{\mathfrak{s}}(U,J,\mu)=-\tfrac12\log|\det\mathbb{S}(X_{U,J,\mu},\mathring{\mathcal{Q}}_U^{[k]})|+\log|\det\mathbb{S}(\mathsf{D}_U^{\mathsf{T}}\mathsf{D}_U,\mathfrak{a}_U^{[k]})|. \tag{628}$$

Proposition 42.1. For fixed $k\ge1$ and finite side $\mathfrak{s}$, both matrices in (628) are invertible at $(\mathbf{1},0,0)$. The function $\mathfrak{F}_{\mathfrak{s}}$ is a real-analytic germ there and is invariant under simultaneous gauge transformation of (U,J,μ), for gauges near $\mathbf{1}$ or constant. Its matrix entries are built by finite compositions and derivatives of local analytic means and the plaquette action, with the same local formulas on finite towers and on the infinite tower. These formulas are covariant under lattice isometries preserving Λ_k, with the tangent action on reversed bonds. They admit complex analytic extensions near the trivial triple.

For the one-step maps there is a common radius $\rho_L\in(0,1]$, depending only on L: on the polydisc $|\mathbf{V}(b)-1|<\rho_L$ for all bonds of a double block, of both orientations, the complex means and contour variables are holomorphic and within $1/2$ of 1. Their left derivatives, including the straightened derivative kernels, are holomorphic on this same open polydisc. In particular $N(B)=i^{-1}\log\mathcal{M}(e^{iB})-\mathsf{Q}B$ has $N(0)=N'(0)=0$.

For W near $\mathbf{1}$,

$$\Psi_k(W)=\mathfrak{F}_{\mathfrak{s}}(c_k(W),\nabla A(c_k(W)),\nabla\mathcal{S}_k(W))-\mathfrak{F}_{\mathfrak{s}}(\mathbf{1},0,0). \tag{629}$$

■

Proof. Differentiating gauge covariance of the mean and of the contour variables gives the one-step identities

$$\mathcal{M}'(V)\mathsf{D}_V = \mathsf{D}_{\mathcal{M}(V)}r, \quad \mathsf{n}_V\mathsf{D}_V = \mathfrak{a}_V - r, \quad \mathring{\mathcal{Q}}_V\mathsf{D}_V = \mathsf{D}_{\mathcal{M}(V)}\mathfrak{a}_V, \quad \mathfrak{a}_V\mathfrak{a}_V^{\mathsf{T}} = L^{-3}I.$$

The last identity follows because blocks are disjoint and adjoint transport preserves the bilinear Lie-algebra pairing. Multiplication of these identities and the recursion for $\mathsf{n}_U^{[j]}$ yield

$$\mathring{\mathcal{Q}}_U^{[j]} = \mathcal{Q}_U^{[j]} + \mathsf{D}_{V_j}\mathsf{n}_U^{[j]}, \quad \mathfrak{a}_U^{[j]} = r + \mathsf{n}_U^{[j]}\mathsf{D}_U, \quad \mathfrak{a}_U^{[j]}\mathfrak{a}_U^{[j]\mathsf{T}} = L^{-3j}I. \tag{630}$$

In particular $L^{3k}\mathfrak{a}_U^{[k]\mathsf{T}}\mathfrak{a}_U^{[k]}$ is the orthogonal projection onto $(\ker\mathfrak{a}_U^{[k]})^\perp$ for real U. At $U = \mathbf{1}$, $\mathfrak{a}^{[k]}$ is the arithmetic mean on a block of L^{3k} sites and $\mathring{\mathsf{Q}}^{[k]}$ is the corresponding mean of straight path sums. Indeed the centred derivative is $\mathsf{Q} = \mathring{\mathsf{Q}} - d\mathsf{n}$, and telescoping gives $\mathring{\mathsf{Q}}d = d\mathfrak{a}$. These identities also show that $\mathring{\mathsf{Q}}^{[k]}$ is onto, by the pivot lift at each level.

The operators at the trivial triple are positive on their constraint kernels. For the scalar operator, $|d\lambda|^2 = 0$ makes λ constant, and $\mathfrak{a}^{[k]}\lambda = 0$ then makes it zero. For the vector operator write $P = L^{3k}\mathfrak{a}^{[k]\mathsf{T}}\mathfrak{a}^{[k]}$; its quadratic form is $|dB|^2 + |(1-P)d^{\mathsf{T}}B|^2$. If it vanishes and $\mathring{\mathsf{Q}}^{[k]}B = 0$, write $B = d\lambda + a_0$ with a_0 constant. The latter constraint is $d\mathfrak{a}^{[k]}\lambda + L^k a_0 = 0$. Torus periods give $a_0 = 0$, and $\mathfrak{a}^{[k]}\lambda$ is constant. Therefore

$$|d\lambda|^2 = (\lambda, Pd^{\mathsf{T}}d\lambda) = L^{3k}(\mathfrak{a}^{[k]}\lambda, \mathfrak{a}^{[k]}d^{\mathsf{T}}d\lambda) = 0,$$

since the last scalar mean has zero total sum. Thus $B = 0$. The bordered determinant formula in the preceding proof gives invertibility of both matrices.

Their entries are analytic. To check the local assertion uniformly over translates, choose a polydisc about the trivial configuration on one double block, small enough that its complex means and contour variables are defined and within $1/2$ of 1. Translation, dilation and coordinate permutation identify every double block with this one; including both orientations makes the same statement valid under reflections. The double blocks lift injectively from the finite towers. Thus one radius depending only on L serves every one-step map. Finite iteration gives the locally defined entries above for fixed k, and the same formulas on the infinite lattice. Derivatives of order at most two still depend on only finitely many bond variables; the resulting matrices have finite range. The determinant and its chosen logarithm are analytic near the trivial triple because their values there are nonzero.

Gauge transformations intertwine the means, their derivatives, D_U, $\mathsf{n}_U^{[k]}$, $\mathfrak{a}_U^{[k]}$ and $\mathring{\mathcal{Q}}_U^{[k]}$. They are orthogonal for the pairings, commute with r^{T} and intertwine q_U, c_U as well. They consequently conjugate both bordered matrices. The same argument applies to lattice isometries, using the tangent adjoint on any reversed bond. It proves the asserted covariance and invariance.

We now relate the two determinants to the fibre integral. The mixed derivative of gauge invariance, computed from

$$(e^{isB}U)^{e^{it\lambda}} = \exp i(sB - t\mathsf{D}_U\lambda + st\,c_U(\lambda, B) + O(t^2) + O((|s|+|t|)^3))U,$$

gives

$$A''(U)(B, \mathsf{D}_U\lambda) = (\nabla A(U), c_U(\lambda, B)). \tag{631}$$

At $U = c_k(W)$, the second-order fibre calculation from the preceding proof gives

$$H_W(B) = A''(U)(B,B) - (\nabla\mathcal{S}_k(W), q_U(B,B)), \qquad B \in \ker\mathcal{Q}_U^{[k]}. \tag{632}$$

Let $\mathfrak{k} = \{\lambda \in \mathfrak{G}_0 : \lambda|_{\Lambda_k} = 0\}$. Gauging a curve in this fibre by $e^{-it\kappa}$ preserves its action and changes its tangent from B to $B + \mathsf{D}_U\kappa$. Hence $H_W(B + \mathsf{D}_U\kappa) = H_W(B)$. Since every intermediate configuration is axial,

$$(\mathbf{L}_W\mathsf{D}_U\kappa)(x) = \kappa(x) - \kappa(y), \qquad x \in \Lambda_{j-1}\setminus\Lambda_j, \quad x \in B(y).$$

Ordering the levels in decreasing order makes this map triangular with identity diagonal. In particular its determinant on $\mathfrak{k}$ is one.

We use the following change-of-constraint calculation. Suppose $D:\mathfrak{G}\to E$ and $D_c:\mathfrak{G}_c\to F$ are injective, $Q:E\to F$ and $r:\mathfrak{G}\to\mathfrak{G}_c$ are onto, and $QD=D_cr$. Let $\mathcal{E}=Q^{-1}(D_c\mathfrak{G}_c)$, and let $H\ge 0$ on $\mathcal{E}$ have null space $D\mathfrak{G}$. Given $n:E\to\mathfrak{G}_c$, set $Q'=Q+D_cn$ and $r'=r+nD$. Suppose r' is onto, and put $\mathfrak{k}'=\ker r'$. Let $L':\ker Q'\to K_g$, where K_g is a Euclidean space of dimension $\dim\mathfrak{k}'$. If $L'D|_{\mathfrak{k}'}$ is invertible, then

$$I(H+|L'\cdot|^2,Q')|\det(L'D|_{\mathfrak{k}'})|\det(r'r'^{\mathsf T})^{1/2} \tag{633}$$

is independent of n,L'. Here is the determinant calculation. The projection $P_\perp:F\to(D_c\mathfrak{G}_c)^\perp$ satisfies $P_\perp Q'=P_\perp Q$; since the range of Q' contains $D_c\mathfrak{G}_c$, Q' is onto and $\mathcal{E}=Q'^{-1}(D_c\mathfrak{G}_c)$. Put $v'=D|_{\mathfrak{k}'}$, $K'=\ker Q'\ominus D\mathfrak{k}'$ and $\mathcal{E}_\perp=\mathcal{E}\ominus D\mathfrak{G}$. Projection $P':K'\to\mathcal{E}_\perp$ is an isomorphism preserving H. Integrating first in the variable κ in $B=a+v'\kappa$ removes L' and contributes

$$(2\pi)^{\dim\mathfrak{k}'/2}|\det L'v'|^{-1}\det(v'^{\mathsf T}v')^{1/2}|\det P'|^{-1}\int_{\mathcal{E}_\perp}e^{-H/2}.$$

To compute the remaining Jacobian put $s'=r'^{\mathsf T}(r'r'^{\mathsf T})^{-1}$ and $N'=Q'Q'^{\mathsf T}$. The parametrization $\Theta(a,\kappa,\gamma)=a+D\kappa+Ds'\gamma$ of $\mathcal{E}$ has, in its two orthogonal decompositions, the determinants

$$\begin{aligned}|\det\Theta|&=|\det P'|\det(D^{\mathsf T}D)^{1/2}\det(r'r'^{\mathsf T})^{-1/2}\\&=\det(v'^{\mathsf T}v')^{1/2}\det(D_c^{\mathsf T}N'^{-1}D_c)^{1/2}.\end{aligned}$$

The first follows by splitting off $D\mathfrak{G}$; for the second the final orthogonal component is $Q'^{\mathsf T}N'^{-1}D_c\gamma$. The block inverse formula gives

$$\det N'\det(D_c^{\mathsf T}N'^{-1}D_c)=\det(D_c^{\mathsf T}D_c)\det(P_\perp QQ^{\mathsf T}P_\perp),$$

where the last determinant is on the range of $P_\perp$. Substitution in (633) leaves only these fixed determinants, $\det(D^{\mathsf T}D)$ and the integral on $\mathcal{E}_\perp$; the dimension $\dim\mathfrak{k}'=\dim\mathfrak{G}-\dim\mathfrak{G}_c$ is fixed as well. This proves the claimed independence.

First assume D_W injective. Then D_U is injective: $\mathcal{Q}_U^{[k]}\mathsf{D}_U=\mathsf{D}_Wr$ implies $r\lambda=0$ for a null vector, and the triangular map $\mathbf{L}_W\mathsf{D}_U|_{\mathfrak{k}}$ then gives $\lambda=0$. On $\mathcal{E}=\ker\mathcal{Q}_U^{[k]}+\mathsf{D}_U\mathfrak{G}_0$ extend H_W by $H(B+\mathsf{D}_U\lambda)=H_W(B)$. Its gauge invariance makes this well defined. Removing $\mathbf{L}_WB$ by the invertible triangular gauge map and using the positive total Hessian shows that $H\ge0$ with null space exactly $\mathsf{D}_U\mathfrak{G}_0$. The change-of-constraint calculation applies.

For its first choice take $n=0,L'=\mathbf{L}_W$. Since $rr^{\mathsf T}=I$ and the gauge determinant is one, (633) equals $\mathcal{Z}_k(W)$. For the second take $n=\mathsf{n}_U^{[k]}$, $r'=\mathfrak{a}_U^{[k]}$, $Q'=\mathring{\mathcal{Q}}_U^{[k]}$ and $L'=(1-L^{3k}r'^{\mathsf T}r')\mathsf{D}_U^{\mathsf T}$. Then $r'r'^{\mathsf T}=L^{-3k}I$ and $L'\mathsf{D}_U|_{\ker r'}=v'^{\mathsf T}v'$ is positive. For $B\in\ker Q'$, $\tau_UB\in\ker\mathcal{Q}_U^{[k]}$, so $H(B)=H_W(\tau_UB)$. Expanding this expression with (631) and (632) identifies $H(B)+|L'B|^2$ with (627) at $J=\nabla A(U)$, $\mu=\nabla\mathcal{S}_k(W)$.

Apply the bordered determinant formula to this vector Gaussian and to the scalar pair $(\mathsf{D}_U^{\mathsf T}\mathsf{D}_U,r')$. The second application gives

$$\det(v'^{\mathsf T}v')=|\det\mathbb{S}(\mathsf{D}_U^{\mathsf T}\mathsf{D}_U,r')|\det(r'r'^{\mathsf T})^{-1}.$$

Comparison of the two choices therefore yields

$$\log\mathcal{Z}_k(W)-\mathfrak{F}_{\mathfrak{s}}(c_k(W),\nabla A(c_k(W)),\nabla\mathcal{S}_k(W))=\tfrac12(\dim\mathcal{A}_0-\dim\mathcal{A}_k)\log(2\pi)+\tfrac32k\dim\mathfrak{G}_k\log L.$$

Both sides extend continuously near $\mathbf{1}$. Configurations with D_W injective are dense there. Indeed the analytic determinant $\det(\mathsf{D}_W^{\mathsf T}\mathsf{D}_W)$ is not identically zero: put $W=e^{i\epsilon t_1}$ in direction 1, $W=e^{i\epsilon t_2}$ in direction 2 and $W=1$ in direction 3, where $0<\mathfrak{s}\epsilon<\pi$. A covariantly constant site function is fixed by the two loop holonomies $e^{i\mathfrak{s}\epsilon t_1}$ and $e^{i\mathfrak{s}\epsilon t_2}$. Their adjoints are rotations through angle $2\mathfrak{s}\epsilon$ about the two distinct Pauli axes, whose fixed subspaces intersect only at zero. This proves

injectivity for arbitrarily small ϵ. The identity consequently holds also at $W = \mathbf{1}$. Subtracting that value removes its constant right-hand side and proves (629). □

43. The accumulated Gaussian coefficient

43.1. **Second derivatives of the Gaussian determinant.** Keep the tower, means and Gaussian normalizers of section 42. The infinite tower uses the same lattices without the periodic quotient. All complex means are their holomorphic branches near the trivial configuration. The neighbourhoods may depend on height and volume.

All Brillouin-zone integrals use normalized measure:

$$\int f(\kappa)\, d\kappa = (2\pi)^{-3} \int_{[-\pi,\pi]^3} f(\kappa)\, d^3\kappa,$$

with the same convention for p and q. We use Bloch symbols with respect to translations by $\mathbb{Z}^3$. Each fine lattice has finitely many indices per unit cell, carrying the pairing weights of the operator's domain and target. The symbol is characterized by

$$T(e^{i\kappa\cdot x} v) = e^{i\kappa\cdot x} \widehat{T}(\kappa) v, \qquad v(x+n) = v(x).$$

For finite-range kernels this is a finite sum of exponentials. A holomorphic symbol on a strip has an exponentially decreasing kernel, obtained by Fourier inversion and a shift of the integration contour. Products, weighted transposes and periodisation commute with this correspondence. Indeed, the product rule is the absolutely convergent convolution of the kernels; periodising this convolution proves the torus rule.

We need only second derivatives of a local operator function $T(\zeta)$. Along $\zeta_\pm = e^{\pm ip\cdot x}\varepsilon_\pm$, record them as the quadruple $\mathbf{T} = (T_{00}, T_{10}, T_{01}, T_{11})$: the first component is $\widehat{T(0)}$, the next two carry momentum κ to $\kappa \pm p$, and the last is the mixed derivative, which preserves momentum. If $S_r = \{\kappa : |\operatorname{Im}\kappa_\mu| \le r\}$, give these four components the sum of their supremum operator norms on, respectively,

$$S_{2\delta}, \quad \{\kappa : \kappa, \kappa + p \in S_{2\delta}\}, \quad \{\kappa : \kappa, \kappa - p \in S_{2\delta}\}, \quad \mathbb{R}^3, \qquad p \in S_\delta.$$

We call this quadruple a jet. Products are compositions with the indicated momentum changes. Equivalently they are products of

$$T_{00} + \epsilon_+ T_{10} + \epsilon_- T_{01} + \epsilon_+\epsilon_- T_{11}, \qquad \epsilon_+^2 = \epsilon_-^2 = 0.$$

The resulting norm is submultiplicative, and transpose preserves it. Three jets with zero 00-component have zero product. Put $\mathbf{T}_0 = (T_{00}, 0, 0, 0)$. Thus a jet is invertible whenever T_{00}^{-1} is bounded on $S_{2\delta}$, and

$$\mathbf{T}^{-1} = (1 - \mathbf{N} + \mathbf{N}^2)\mathbf{T}_0^{-1}, \quad \mathbf{N} = \mathbf{T}_0^{-1}\mathbf{T} - 1, \quad \|\mathbf{T}^{-1}\| \le \iota(1 + \iota\|\mathbf{T}\|)^2, \quad \iota = \sup \|T_{00}^{-1}\|.$$

Set

$$\mathsf{t}(\mathbf{T}) = \int \operatorname{tr} T_{11}(\kappa)\, d\kappa, \qquad \ell(\mathbf{T}) = \mathsf{t}(\mathbf{N} - \tfrac{1}{2}\mathbf{N}^2).$$

Contour shifts in the periodic integrals give trace cyclicity. Expanding the two nilpotent variables then yields

$$\ell(\mathbf{T}) = \int \operatorname{tr}\left[T_{00}(\kappa)^{-1} T_{11}(\kappa) - T_{00}(\kappa)^{-1} T_{10}(\kappa - p) T_{00}(\kappa - p)^{-1} T_{01}(\kappa)\right] d\kappa,$$

$$\ell(\mathbf{TS}) = \ell(\mathbf{T}) + \ell(\mathbf{S}), \qquad \frac{d}{dt}\ell(\mathbf{T}_t) = \mathsf{t}(\mathbf{T}_t^{-1}\dot{\mathbf{T}}_t).$$

The last formula holds for differentiable families with uniformly bounded zero-component inverses. To see additivity directly, conjugate $\mathbf{N_T}$ by $\mathbf{S}_0$, write $\mathbf{N_{TS}} = a + b + ab$, and use the vanishing of cubic products and $\mathsf{t}(ab) = \mathsf{t}(ba)$. The derivative follows by applying additivity to $\mathbf{T}_t^{-1}\mathbf{T}_{t+h} =$

$1 + h\mathbf{T}_t^{-1}\dot{\mathbf{T}}_t + o(h)$. Block triangular jets contribute the sum of their diagonal blocks. Field-independent invertible diagonal factors, including changes of the pairing weights, contribute zero.

Suppose T is a local analytic operator function and $\widehat{T(0)}$ is invertible for real κ. Compactness of one period cube gives an invertible strip. On a sufficiently large torus,

$$D^2 \log\det T(0)[\zeta_+, \zeta_-] = \operatorname{Tr}(T_0^{-1}T^{(2)}) - \operatorname{Tr}(T_0^{-1}T_+^{(1)}T_0^{-1}T_-^{(1)}).$$

Fourier decomposition makes the right side a sum of the integrand defining $\ell(\mathbf{T})$. Its normalisation by the torus volume converges to $\ell(\mathbf{T})$ with error $Ce^{-c\mathfrak{s}}|\varepsilon_+||\varepsilon_-|$, uniformly for real torus momenta p. Indeed, the integrand is periodic and bounded on a strip; its Fourier coefficients decrease exponentially, and the torus Riemann sum retains precisely the coefficients indexed by $\mathfrak{s}\mathbb{Z}^3$. Different external momenta give zero trace unless their sum is in $2\pi\mathbb{Z}^3$. For real kernels the same formula applies to $\log|\det T|$.

Write $\Pi_{\mathfrak{s}}^F = F_{\mathfrak{s}}''(\mathbf{1})$, extended bilinearly to complex forms. For a positive bond b, let δ_b^a equal t_a on b and zero elsewhere. For $p \in (2\pi/\mathfrak{s})\mathbb{Z}^3$ modulo $2\pi\mathbb{Z}^3$, let $e_{p,\mu}^a(b)$ equal $e^{ip\cdot b-}t_a$ on bonds of direction μ and zero on the others. An infinite-lattice bond in such a formula is represented by its image on the torus.

We call a family $F_{\mathfrak{s}}$ tempered when it is real analytic near $\mathbf{1}$, vanishes there, is invariant under gauge transformations near $\mathbf{1}$, constant gauges, and all lattice isometries, and its volume-normalised plane-wave polarisation converges uniformly to $\delta_{aa'}\widehat{\pi}_{\mu\nu}(p)$, where $\widehat{\pi}$ is bounded and holomorphic on a strip. The neighbourhood and strip may depend on the family. Its coefficient is

$$\mathsf{b}[F] = -\partial_\mu\partial_\nu\widehat{\pi}_{\mu\nu}(0), \qquad \mu \neq \nu.$$

We verify that this definition agrees with the kernel coefficient and is independent of the chosen pair.

Constant gauge invariance first gives $\nabla F_{\mathfrak{s}}(\mathbf{1}) = 0$, since the adjoint representation has no nonzero fixed vector. Differentiating gauge invariance gives $\Pi^F[d\lambda, B] = 0$. Translation invariance removes unequal momenta, and Fourier inversion on the tori gives

$$\Pi_{\mathfrak{s}}^F[\delta_x^a, \delta_y^{a'}] \longrightarrow \delta_{aa'}\pi_{\mu\nu}(y_- - x_-), \qquad \pi_{\mu\nu}(u) = \int \widehat{\pi}_{\mu\nu}(p)e^{ip\cdot u}\,dp.$$

The error in this inversion is bounded by the uniform plane-wave error. The strip gives exponential decay of π, and its reality follows from the finite-torus polarisations. Gauge invariance passes to the limit as

$$\sum_\mu (e^{ip_\mu} - 1)\widehat{\pi}_{\mu\nu}(p) = 0.$$

In the bond-midpoint convention $\widetilde{\pi}_{\mu\nu} = e^{i(p_\mu - p_\nu)/2}\widehat{\pi}_{\mu\nu}$, the lattice isometries give $\widetilde{\pi}(Rp) = R\widetilde{\pi}(p)R^{\mathsf{T}}$ for all signed permutation matrices R. In particular $\widetilde{\pi}$ is even. Its constant term is cI; the gauge relation $\sum_\mu \sin(p_\mu/2)\widetilde{\pi}_{\mu\nu} = 0$ forces $c = 0$. Reflection and permutation symmetry leave for its quadratic term only

$$\alpha\delta_{\mu\nu}|p|^2 + \beta p_\mu p_\nu + \gamma\delta_{\mu\nu}p_\mu^2.$$

The cubic term of the gauge relation forces $\beta = -\alpha$ and $\gamma = 0$. Hence

$$\widetilde{\pi}_{\mu\nu}(p) = \mathsf{b}[F](|p|^2\delta_{\mu\nu} - p_\mu p_\nu) + O(|p|^4).$$

Termwise differentiation of its exponentially convergent Fourier series now gives

$$\mathsf{b}[F] = \sum_u \pi_{\mu\nu}(u)u_\mu u_\nu = -\tfrac{1}{2}\sum_u \pi_{\nu\nu}(u)u_\mu^2 \quad (\mu \neq \nu).$$

The coefficient is real, additive, and independent of the pair of distinct directions.

Write $\Phi_{l,\mathfrak{s}}^{\natural}$ for the normalizer above with $j = k = l$ and side $\mathfrak{s}$.

Theorem O. Fix $G = \mathrm{SU}(2)$, dimension three, and odd $L \geq 13$, with the tower means and Gaussian normalizers specified above. For each $l \geq 1$, the one-step Gaussian normalizer $\Phi_l^\natural = (\Phi_{l,\mathfrak{s}}^\natural)_\mathfrak{s}$ is tempered. Its members are real analytic near $\mathbf{1}$, and their second differentials on individual bonds have an infinite-volume limit

$$\lim_{\mathfrak{s}\to\infty} \Pi_\mathfrak{s}^{\Phi_l^\natural}[\delta_{\hat{x}}^a, \delta_{\hat{y}}^{a'}] = \delta_{aa'}\pi_{\mu\nu}^{(l)}(\hat{y}_- - \hat{x}_-),$$

where μ, ν are the bond directions and $\pi^{(l)}$ is real and exponentially decreasing. The coefficient

$$b_l^0 = -\tfrac{1}{2}\sum_{u\in\mathbb{Z}^3} u_\mu^2 \pi_{\nu\nu}^{(l)}(u), \qquad \mu \neq \nu,$$

is absolutely convergent, independent of the two distinct directions, and depends only on l, L. There is a finite constant $c_2(L)$ such that

$$\mathsf{c}_k := \sum_{l=1}^{k} L^{-(k-l)} b_l^0 = \mathsf{b}[\Psi_k], \qquad |\mathsf{c}_k| \leq c_2(L) \quad (k \geq 0).$$

One may take

$$c_2(L) = \sup_{k\geq 0} |\mathsf{b}[\Psi_k]| \leq C_{\mathrm{rem}}(L)\delta_{\mathrm{loop}}^{-2} + C_{\mathrm{uv}}.$$

The constants on the right are those in propositions 43.1 to 43.3. ■□

For complex configurations the plaquette action means its holomorphic extension $A(U) = \sum_p [1 - \frac{1}{2}\operatorname{tr} U_p]$. It agrees with the real action because an SU(2) matrix has real trace. On the infinite lattice its derivatives on finitely supported forms use only the finitely many plaquettes meeting their supports.

For U whose intermediate means lie in the one-step polydisc, set

$$D_U = \eta^{-1}\mathsf{D}_U, \qquad Q_U = \eta\mathring{\mathcal{Q}}_U^{[k]}, \qquad R_U = \mathfrak{a}_U^{[k]}, \qquad H_U = \eta^{-2}A''(U),$$
$$K_U = H_U + D_U D_U^\mathsf{T}, \qquad \Delta_U = D_U^\mathsf{T} D_U,$$
$$X_U = H_U + D_U(1 - R_U^\mathsf{T} R_U)D_U^\mathsf{T}, \quad \Xi_U = X_U + Q_U^\mathsf{T} Q_U, \quad Y_U = \Delta_U + R_U^\mathsf{T} R_U.$$

The transposes T in this display use the physical weights η^3 at the fine level and 1 at the coarse level. Thus $(b, H_U b)_\eta = \eta^{-1}A''(U)(\eta b, \eta b)$. If t denotes an unweighted transpose, then

$$D_U^\mathsf{T} = \eta^{-1}\mathsf{D}_U^t, \qquad R_U^\mathsf{T} = \eta^{-3}\mathfrak{a}_U^{[k]t}, \qquad Q_U^\mathsf{T} = \eta^{-2}\mathring{\mathcal{Q}}_U^{[k]t}.$$

In particular $R_U R_U^\mathsf{T} = I$. The symmetric idempotent $P = R_U^\mathsf{T} R_U$ gives $(1-P)^t(1-P) = 1 - P$. Substitution in the corrected Hessian at $J = \mu = 0$ therefore gives $X_{U,0,0} = \eta^2 X_U$. Consequently, with unweighted transposes on the left and physical transposes on the right,

$$\mathbb{S}(X_{U,0,0}, \mathring{\mathcal{Q}}_U^{[k]}) = \begin{pmatrix} \eta^2 & 0 \\ 0 & \eta^{-1} \end{pmatrix} \mathbb{S}(X_U, Q_U),$$
$$\mathbb{S}(\mathsf{D}_U^t \mathsf{D}_U, \mathfrak{a}_U^{[k]}) = \begin{pmatrix} \eta^2 & 0 \\ 0 & 1 \end{pmatrix} \mathbb{S}(\Delta_U, R_U) \begin{pmatrix} 1 & 0 \\ 0 & \eta \end{pmatrix}.$$

These diagonal factors are independent of the field. They change logarithmic determinants only by constants and hence leave their polarisation jets unchanged.

43.2. **A uniform strip for the free response.** Let $\eta = L^{-k}$, $\ell_j = L^{j-k}$, and use pairing weight ℓ_j^3 at level j. In this subsection d is the exterior derivative divided by the lattice spacing. The straight averages on potentials and sites are denoted by Q and R:

$$Q = \eta\, \mathring{\mathsf{Q}}^{[k]}, \qquad R = \mathfrak{a}^{[k]}, \qquad Qd = dR, \qquad RR^{\mathsf{T}} = 1.$$

Thus R is the mean over a block of side one, and Q is the mean over the pairs consisting of a site in the initial block and a bond on its straight path to the translated site in the final block. Put

$$K = d^{\mathsf{T}}d + dd^{\mathsf{T}}, \qquad \Delta = d^{\mathsf{T}}d,$$
$$\Xi = K - dR^{\mathsf{T}}Rd^{\mathsf{T}} + Q^{\mathsf{T}}Q, \qquad Y = \Delta + R^{\mathsf{T}}R,$$
$$G = (K+1)^{-1}, \qquad g = (\Delta+1)^{-1}.$$

Here K, G act on forms and Δ, Y, g on sites.

For $m \in \mathsf{m} = \{m : |m_\mu| \le (\eta^{-1}-1)/2\}$ and $q_m = \kappa + 2\pi m$, set

$$d_\mu(q) = \eta^{-1}(e^{i\eta q_\mu} - 1), \qquad D(q) = \sum_\mu 2\eta^{-2}(1 - \cos \eta q_\mu),$$
$$s(\tau) = \eta \sum_{|j| \le (\eta^{-1}-1)/2} e^{i\eta j \tau}, \qquad v(q) = \prod_\nu s(q_\nu), \qquad u_\mu(q) = e^{i(1-\eta)q_\mu/2} s(q_\mu) v(q).$$

The functions $e^{2\pi i m\cdot x}$ form an orthonormal basis for a unit cell of the fine lattice. In this basis

$$\widehat{R}\lambda = \sum_m v(q_m)\widehat{\lambda}_m, \qquad (\widehat{Q}b)_\mu = \sum_m u_\mu(q_m)\widehat{b}_{\mu,m},$$

and the transpose coefficients are $v(-q_m)$ and $u_\mu(-q_m)$. The symbols of K, Δ multiply each coefficient by $D(q_m)$. These formulas follow by summing the block and straight-path geometric series; the same sums give

$$u_\mu(q) d_\mu(q) = (e^{iq_\mu} - 1) v(q).$$

The iterated block formulas follow by the unique decomposition of a centered L^n-block into centered L^{n-1}-blocks. They also give, in absorbed lengths,

$$\mathsf{Q}^{[n]} = \mathring{\mathsf{Q}}^{[n]} - d\mathsf{n}^{[n]}, \quad \|\mathring{\mathsf{Q}}^{[n]}\|_{\infty\to\infty} \le L^n, \quad \|\mathsf{n}^{[n]}\|_{\infty\to\infty} \le \tfrac{3}{2}L^n, \quad \|\mathsf{Q}^{[n]}\|_{\infty\to\infty} \le 4L^n.$$

The correction $\mathsf{n}^{[n]}$ is the sum of the successive contour averages followed by restriction. The identity follows by induction from $\mathsf{Q} = \mathring{\mathsf{Q}} - d\mathsf{n}$, $\mathsf{n}d = \mathfrak{a} - \mathsf{r}$. The pivot lift in the proof of theorem N is a right inverse for both one-step means, since its contour correction vanishes. Composing these lifts proves surjectivity of their iterates. The unweighted transpose of restriction is extension by zero, so $\mathsf{r}\mathsf{r}^{\mathsf{T}} = 1$.

Proposition 43.1. There are absolute constants $\delta_{\mathrm{loop}} > 0$ and C_{loop}, independent of k and L, such that the following symbols are bounded by C_{loop} on $S_{2\delta_{\mathrm{loop}}}$: the square-root propagators, their compositions with one d or d^{T}, and Q, R and their transposes. The four symbols

$$G^{1/2}(t\Xi + (1-t)K + 1)G^{1/2}, \quad G^{1/2}(\Xi + \tau)G^{1/2},$$
$$g^{1/2}(tY + (1-t)\Delta + 1)g^{1/2}, \quad g^{1/2}(Y+\tau)g^{1/2}, \qquad 0 \le t, \tau \le 1,$$

and the coarse symbols

$$Q(\Xi+\tau)^{-1}Q^{\mathsf{T}}, \qquad R(Y+\tau)^{-1}R^{\mathsf{T}}$$

are invertible there, with inverse norms at most C_{loop}. ∎

Proof. The essential point is coercivity at real momentum. We give that argument before passing to a strip. Cauchy–Schwarz in the averaging formulas gives $\|Q\|, \|R\| \le 1$. A fine bond occurs in

exactly η^{-1} of the η^{-4} pairs for the straight average, which cancels the physical pairing weights. Site blocks are disjoint. With exponential weights the same count gives

$$\|e^{\theta\cdot x}Qe^{-\theta\cdot x}\| \le e^{2|\theta|_1}, \qquad \|e^{\theta\cdot x}Re^{-\theta\cdot x}\| \le e^{|\theta|_1}.$$

In particular, for real κ, $\sum_m |u_\mu(q_m)|^2 \le 1$ and $\sum_m v(q_m)^2 = 1$.

Take $\kappa \in [-\pi,\pi]^3 \setminus \{0\}$, and write $D_m = D(q_m)$, $v_m = v(q_m)$, $u_{\mu,m} = u_\mu(q_m)$. The sine formula for s gives

$$D_m \ge 4 \quad (m \ne 0), \qquad 4|\kappa|^2/\pi^2 \le D_0 \le |\kappa|^2 \le 3\pi^2,$$
$$v_0 \ge (2/\pi)^3, \qquad |u_{\mu,0}| \ge (2/\pi)^4, \qquad 1 - v_0^2 \le \pi^2 D_0/48.$$

Consequently $QGQ^* \ge c_C I$ and $RgR^* \ge c_C I$, with $c_C = (2/\pi)^8/(3\pi^2+1)$, by retaining the $m=0$ term. For site functions,

$$|\sum_m v_m\lambda_m|^2 \ge \tfrac{1}{2}v_0^2|\lambda_0|^2 - \tfrac{1}{4}\sum_{m\ne 0} D_m|\lambda_m|^2,$$

so $Y \ge c_Y(\Delta+1)$, for an absolute $c_Y > 0$.

For forms in $\ker Q$, decompose

$$b_m = b_m^\perp + d(q_m)\lambda_m, \qquad b_m^\perp \perp d(q_m)\mathfrak{g}^c.$$

Set

$$\chi_m = D_m\lambda_m, \quad \gamma = \sum_m v_m\chi_m, \quad \rho_m = \chi_m - v_m\gamma, \quad \beta = \sum_m v_m\lambda_m.$$

Then $\sum_m v_m\rho_m = 0$, and

$$q(b) := \|db\|^2 + \|(1-R^{\mathsf{T}}R)d^{\mathsf{T}}b\|^2 = \sum_m D_m|b_m^\perp|^2 + \|\rho\|^2,$$
$$\|b\|_1^2 := (b,(K+1)b) = \sum_m (D_m+1)|b_m^\perp|^2 + |\gamma|^2 + \|\rho\|^2 + \sum_m D_m^{-1}|\chi_m|^2.$$

The constraint reads

$$\sum_m u_{\mu,m}b_{\mu,m}^\perp + d_\mu^{\mathrm{c}}(\kappa)\beta = 0.$$

Since $d_\mu^{\mathrm{c}} = u_{\mu,0}d_\mu(\kappa)/v_0$, the distance of $(u_{\mu,0}b_{\mu,0}^\perp)_\mu$ from the coarse gradient space is at least $(2/\pi)^4|b_0^\perp|$. The constraint bounds it by $(\sum_{m\ne0}|b_m^\perp|^2)^{1/2}$. Thus

$$\|b^\perp\|^2 \le C\sum_m D_m|b_m^\perp|^2, \qquad |\beta| \le CD_0^{-1/2}\|b^\perp\|.$$

Moreover, with $S = \sum_m v_m^2 D_m^{-1} \ge v_0^2 D_0^{-1}$,

$$\beta = S\gamma + \sum_{m\ne0} v_m\rho_m(D_m^{-1} - D_0^{-1}).$$

The last sum is bounded by $CD_0^{-1/2}\|\rho\|$, using $1 - v_0^2 \le CD_0$. Hence

$$|\gamma| \le CD_0^{1/2}(\|b^\perp\| + \|\rho\|), \qquad |\rho_0| \le CD_0^{1/2}\|\rho\|.$$

These bounds control $D_0^{-1}|\chi_0|^2$; the remaining inverse weights are at most $1/4$. Substitution in the formula for $\|b\|_1^2$ proves $\|b\|_1^2 \le Cq(b)$ on $\ker Q$.

For general b, split off

$$b_0 = GQ^*(QGQ^*)^{-1}Qb.$$

Its energy is at most $c_C^{-1}|Qb|^2$, and $b - b_0 \in \ker Q$. The triangle inequality for $q^{1/2}$, together with $q \le \|\cdot\|_1^2$, therefore gives

$$c_\Xi(K+1) \le \Xi \le K+1, \qquad c_Y(\Delta+1) \le Y \le \Delta+1$$

with absolute positive constants. Continuity includes $\kappa = 0$.

For $q = q' + i\theta$, $|q'_\mu| \le \pi/\eta$, $|\theta_\mu| \le 1/8$, the cosine formula gives

$$\mathrm{Re}(D(q)+1) \ge \tfrac{9}{10}(D(q')+1), \qquad |d_\mu(\pm q)| \le \tfrac{6}{5}(|d_\mu(q')| + 1/8).$$

Thus the propagator square roots and their first derivatives are uniformly bounded on $S_{1/8}$. The weighted averaging estimates give the same conclusion for Q, R. Each sandwiched operator in the statement is a fixed polynomial in these bounded symbols, so it has an absolute bound on $S_{1/8}$. Cauchy's estimate bounds its first momentum derivatives on $S_{1/16}$, uniformly in k, t, τ. The real coercivity just proved and the Neumann series therefore give its inverse on a smaller absolute strip. The coarse Schur symbols are bounded there by their factorisation through $G^{1/2}$ or $g^{1/2}$. On the real axis they are bounded below by $c_C/2$, since $\Xi + \tau \le 2(K+1)$ and $Y + \tau \le 2(\Delta+1)$. A second Cauchy estimate and Neumann-series argument give their inverses on a smaller strip. Fix such a $\delta_{\mathrm{loop}} \le 1/64$ and a common C_{loop}. All choices were absolute. □

Define

$$C = Q\Xi^{-1}Q^{\mathsf{T}}, \qquad h = \Xi^{-1}Q^{\mathsf{T}}C^{-1}, \qquad \nu = C^{-1} - 1.$$

These are periodic operators with real kernels and symbols holomorphic on a neighbourhood of the closed strip, by compactness at each fixed k. They obey

$$Qh = 1, \quad d^{\mathsf{T}}dh = Q^{\mathsf{T}}\nu, \quad (1 - R^{\mathsf{T}}R)d^{\mathsf{T}}h = 0, \quad d^{\mathsf{T}}\nu = 0.$$

Indeed, $Xh = Q^{\mathsf{T}}\nu$, where $X = K - dR^{\mathsf{T}}Rd^{\mathsf{T}}$. Applying d^{T}, and taking the real-momentum Hermitian pairing with $\rho = (1 - R^{\mathsf{T}}R)d^{\mathsf{T}}hB$, gives $(\rho, \Delta\rho) = 0$. Thus ρ is constant at a zero momentum, and $R\rho = 0$ removes that constant as well. Applying R then gives $d^{\mathsf{T}}\nu = 0$.

For $b = \widehat{h}(p)\varepsilon$, $n = \widehat{\nu}(p)\varepsilon$, and $\gamma = \widehat{R}\,\widehat{d^{\mathsf{T}}}b$, these relations imply

$$D(q_m)b_{\mu,m} = u_\mu(-q_m)n_\mu + d_\mu(q_m)v(-q_m)\gamma.$$

The sandwiched factorisation of h bounds $\|b\|, |n|, |\gamma|$ uniformly by $C|\varepsilon|$. For $m \ne 0$, the sine formula for s and the preceding momentum bounds give

$$|b_m| + |f_m| \le C|m|_\infty^{-1} \prod_\mu \max\{1, 2|m_\mu| - 1\}^{-1}|\varepsilon|,$$

where $d(e^{ipx}b) = e^{ipx}f$. In the formula for $f_{\mu\nu,m}$, the terms containing γ cancel. The displayed bound is summable independently of k, since $|m|_\infty \ge \prod_\mu \max(1, |m_\mu|)^{1/3}$. It follows that the periodic potential and its curl are uniformly bounded in supremum norm on the strip.

Finally, for fixed k and $p \to 0$,

$$b_m = O(|p|^2|\varepsilon|)\ (m \ne 0), \qquad b_0 = \varepsilon + O(|p||\varepsilon|), \qquad n = O(|p|^2|\varepsilon|).$$

To prove this, use the trial field $b^{(0)}_{\mu,0} = \varepsilon_\mu/u_\mu(p)$ and $b^{(0)}_m = 0$ otherwise. It satisfies $Qb^{(0)} = \varepsilon$. Its residual under X is $O(|p|^2|\varepsilon|)$: the zero mode has two derivatives, and every nonzero mode contains $v(-q_m) = O(|p|)$ times a divergence of order $|p|$. The saddle matrix $\mathbb{S}(X, Q)$ is invertible at zero momentum, by coercivity on $\ker Q$ and surjectivity of Q, and hence has bounded inverse near zero. This proves the asserted expansions.

43.3. **The Gaussian coefficient along the critical branch.** We use absorbed bond lengths and unweighted transposes when differentiating the critical branch. Let c_k be its fine representative, $p_k = \mathcal{M}^{k-1}c_k$ its preceding-level representative, and $\mathcal{S}_k$ its critical action. The scaled free response h and multiplier response ν satisfy

$$\begin{aligned} c_k'(\mathbf{1})B - \eta hB &\in d\mathfrak{G}_0, \\ (\nabla\mathcal{S}_k)'(\mathbf{1})B &= \eta\nu B, \\ \partial_t \nabla A(c_k(e^{itB}))\Big|_0 &= \eta\mathring{\mathsf{Q}}^{[k]\mathsf{T}}\nu B, \\ p_k'(\mathbf{1})B - \mathring{\mathsf{Q}}^{[k-1]}(\eta hB) &\in d\mathfrak{G}_{k-1}. \end{aligned}$$

Here d in the displayed gradient spaces is unscaled. We verify these identities, including the distinction between the original and straightened means.

Put $A_1 = c_k'(\mathbf{1})B$, $\mu_1 = (\nabla \mathcal{S}_k)'(\mathbf{1})B$, $A_2 = \eta h B$, and $\mu_2 = \eta \nu B$. Differentiate the constraint and the multiplier equation. The first derivative of the plaquette action at $\mathbf{1}$ is zero, and its Hessian is $d^{\mathsf{T}} d$. Thus the mixed Campbell–Hausdorff term makes no contribution, and

$$\mathsf{Q}^{[k]} A_1 = B, \qquad d^{\mathsf{T}} d A_1 = \mathsf{Q}^{[k]\mathsf{T}} \mu_1.$$

The scaled response equations, after conversion to unweighted pairings, give

$$\mathring{\mathsf{Q}}^{[k]} A_2 = B, \qquad d^{\mathsf{T}} d A_2 = \mathring{\mathsf{Q}}^{[k]\mathsf{T}} \mu_2, \qquad d^{\mathsf{T}} \mu_2 = 0.$$

Use $\mathsf{Q}^{[k]} = \mathring{\mathsf{Q}}^{[k]} - d\mathsf{n}^{[k]}$ and $\mathsf{Q}^{[k]} d = d\mathsf{r}$. The correction $A_2' = A_2 + d\mathsf{r}^{\mathsf{T}} \mathsf{n}^{[k]} A_2$ satisfies the two equations for A_1, with μ_2 in place of μ_1. Consequently $E = A_1 - A_2'$ satisfies

$$\mathsf{Q}^{[k]} E = 0, \qquad |dE|^2 = (\mathsf{Q}^{[k]} E, \mu_1 - \mu_2) = 0.$$

For real B, E is real, so $dE = 0$.

By the curl-free decomposition used in theorem N, write $E = d\lambda + a$ with a constant. Its zero mean constraint becomes $d\mathsf{r}\lambda + L^k a = 0$. Summing around each coarse period gives $a = 0$. Thus $A_1 - A_2$ is a gradient. The mean $\mathsf{Q}^{[k]}$ is onto, so its transpose is injective and $\mu_1 = \mu_2$. Finally differentiate $p_k = \mathcal{M}^{k-1} c_k$; both $\mathsf{Q}^{[k-1]} d$ and $\mathsf{Q}^{[k-1]} - \mathring{\mathsf{Q}}^{[k-1]}$ take values in gradients. This proves all four identities, first for real B, then complex linearly.

Write $T_{\mathfrak{s}}^v, T_{\mathfrak{s}}^g$ for the two saddle matrices in (628). We apply (629) to the critical branch. These are the vector and scalar saddle matrices of proposition 42.1. Their local analytic formulas include the terms linear in J, μ in (627). At fixed height a common neighbourhood makes every intermediate mean and contour well defined. On a torus larger than four times their range, each contour lifts uniquely to the covering lattice. Differentiating the lifted identities, and then taking the finite products and transposes, shows that both full saddle matrices are the periodizations of their infinite-lattice kernels. At real fields they have real matrices in the Pauli basis. This proves the local and periodization hypotheses for the logarithmic determinant calculation, including its multiplier variables.

At zero field their Bloch matrices are invertible for every real momentum: the vector saddle uses the free coercivity on $\ker Q$ and the surjectivity of Q; the scalar saddle uses the corresponding facts for $d^{\mathsf{T}} d$ and R. Compactness gives an invertible strip at each fixed height. Hence the logarithmic determinant polarisation formula applies to the full triple.

Denote its limiting bilinear symbol by

$$\mathsf{p}_{\mathfrak{F}}(p)[\varepsilon_+, \varepsilon_-] = -\tfrac{1}{2}\ell(\mathbf{T}^v) + \ell(\mathbf{T}^g),$$

where the jets are taken along $e^{\pm ipx}\varepsilon_\pm$ in the triple field. Constant gauge invariance implies $D\mathfrak{F}_{\mathfrak{s}}(0) = 0$: the coefficient of each field variable would otherwise be a nonzero fixed vector of the adjoint representation. Differentiating the full gauge covariance, and using this zero first derivative, gives the Ward identity

$$D^2 \mathfrak{F}_{\mathfrak{s}}(0)[(d\lambda, 0, 0), \zeta] = 0.$$

The first derivative of the triple in the Gaussian formula is therefore equivalent, under its second differential, to

$$\mathsf{h} B = (\eta h B,\ \eta \mathring{\mathsf{Q}}^{[k]\mathsf{T}} \nu B,\ \eta \nu B).$$

The periodic operators defining this expression commute with periodization. Write $\mathsf{h}(e^{ipx}\varepsilon) = e^{ipx}\widehat{\mathsf{h}}(p)\varepsilon$. The determinant trace formula then gives

$$\mathfrak{s}^{-3} \Pi_{\mathfrak{s}}^{\Psi_k}[e_{p,\mu}^a, e_{-p,\nu}^{a'}] = \mathsf{p}_{\mathfrak{F}}(p)[\widehat{\mathsf{h}}(p) e_\mu t_a, \widehat{\mathsf{h}}(-p) e_\nu t_{a'}] + O(e^{-c\mathfrak{s}}),$$

uniformly in the real torus momenta, at fixed k. The right side is bounded and holomorphic on a strip. Colour invariance makes it a multiple of $\delta_{aa'}$ on torus momenta; density and holomorphy give

the same conclusion throughout the strip. Together with the gauge and lattice invariances of the Gaussian construction, this proves that Ψ_k is tempered.

Write $\widehat{\mathsf{h}} = (\widehat{\mathsf{h}}^A, \widehat{\mathsf{h}}^\vartheta)$, separating the potential from the two current components. The small momentum estimate gives

$$\widehat{\mathsf{h}}^\vartheta(p) = O(|p|^2), \qquad \widehat{\mathsf{h}}^A(p)\varepsilon = \eta\varepsilon + O(|p|).$$

The Ward identity supplies one more power of momentum in a mixed term. Indeed, apply it to $\lambda = e^{ipx}\varphi$, pass to the limiting symbol, then set $p = \tau w$ and $\varphi = t_a$. Since $\widehat{d}(\tau w)t_a = i\tau\eta(w_\mu t_a)_\mu + O(\tau^2)$, division by τ and passage to zero show

$$\mathsf{p}_{\mathfrak{F}}(0)[(a,0),\varepsilon] = 0$$

for every constant potential a; the same holds in the second argument. Holomorphy and the expansion of $\widehat{\mathsf{h}}^A$ therefore make each mixed potential-current term $O(|p|^3)$, while the current-current term is $O(|p|^4)$. Thus

$$\widehat{\pi}^{\Psi_k}(p) = \widehat{\pi}^0(p) + O(|p|^3), \qquad \mathsf{b}[\Psi_k] = -\partial_1\partial_2\widehat{\pi}^0_{12}(0),$$

where

$$\widehat{\pi}^0_{\mu\nu}(p) = \mathsf{p}_{\mathfrak{F}}(p)[(\widehat{\mathsf{h}}^A(p)e_\mu t_3, 0), (\widehat{\mathsf{h}}^A(-p)e_\nu t_3, 0)].$$

We next record the scale change for a tempered functional F. On a tower of side $\mathfrak{s}$, identify the preceding lattice with the unit lattice of side $L\mathfrak{s}$. Then $F \circ p_k$, meaning $(F_{L\mathfrak{s}} \circ p_k)_{\mathfrak{s}}$, is tempered and

$$\mathsf{b}[F \circ p_k] = L^{-1}\mathsf{b}[F].$$

The vanishing first derivative of F and its Ward identity reduce its composite polarisation to

$$\Pi^F_{L\mathfrak{s}}[\mathring{\mathsf{Q}}^{[k-1]}(\eta h B), \mathring{\mathsf{Q}}^{[k-1]}(\eta h B')].$$

For a plane wave, the straight path formula for the average gives

$$\mathring{\mathsf{Q}}^{[k-1]}(\eta h e^a_{p,\mu}) = L^{-1}\sum_{m'\in\mathcal{M}_L}\sum_\rho H_{m'}(p)_{\rho\mu}e^a_{(p+2\pi m')/L,\rho},$$

where $\mathcal{M}_L$ is the centred set of representatives modulo L, and

$$H_{m'}(p)_{\rho\mu} = \sum_{m\equiv m' \,(\mathrm{mod}\, L)} u'_\rho(p+2\pi m)(\widehat{h}(p)e_\mu)_{\rho,m}.$$

Here $u'_\rho(q)$ is the average of the exponential along the preceding-level straight paths; it is entire, with $u'_\rho(q) = 1 + O(|q|)$ near zero. Only opposite m' pair under the translation-invariant polarisation. The two factors L^{-1} and the volume factor L^3 give

$$\widehat{\pi}^{F\circ p_k}_{\mu\nu}(p) = L\sum_{m'\in\mathcal{M}_L}\sum_{\rho,\sigma} H_{m'}(p)_{\rho\mu}\widehat{\pi}^F_{\rho\sigma}((p+2\pi m')/L)H_{-m'}(-p)_{\sigma\nu}.$$

The finite sum is holomorphic on a strip and inherits the convergence of the finite-volume polarisations. The branch covariance gives the remaining invariances, so the composite is tempered. Moreover

$$H_{m'}(p) = O(|p|^2)\ (m' \neq 0), \qquad H_0(p) = I + O(|p|), \qquad \widehat{\pi}^F(p) = O(|p|^2).$$

It follows that $\widehat{\pi}^{F\circ p_k}(p) = L\widehat{\pi}^F(p/L) + O(|p|^3)$, proving the asserted coefficient identity.

The Gaussian gluing identity now yields, as families,

$$\Phi^\natural_l = \Psi_l - \Psi_{l-1}\circ p_l, \qquad \Psi_0 = 0.$$

Therefore each $\Phi^\natural_l$ is tempered, and

$$\mathsf{b}[\Phi^\natural_l] = \mathsf{b}[\Psi_l] - L^{-1}\mathsf{b}[\Psi_{l-1}], \qquad \sum_{l=1}^{k} L^{-(k-l)}\mathsf{b}[\Phi^\natural_l] = \mathsf{b}[\Psi_k].$$

The latter sum telescopes. The unique near-trivial germs in the Gaussian construction depend only on the tower, so $\mathsf{b}[\Phi_l^\natural]$ depends only on l and L.

43.4. **A uniform determinant remainder.** We return to the physical pairings of weight ℓ_j^3 and to differences d divided by the lattice spacing.

We first bound the field derivatives which enter the determinants. There are numbers $\rho_* > 0$ and $C_* > 0$, depending only on L, such that the scaled straight average Q_U and scalar average R_U, for $U = e^{i\eta a}$, are holomorphic on $\|a\|_\infty < \rho_*$ and have operator norms at most C_*, independently of the height.

Here is the uniform-radius argument. Write the one-step mean in logarithmic coordinates as

$$A_j = \mathsf{Q}A_{j-1} + N(A_{j-1}), \qquad |N(A)|_\infty \le C_N |A|_\infty^2, \qquad A_0 = \eta a.$$

Choose $r_{\log} > 0$ with $e^{r_{\log}} - 1 < \rho_L$. The one-step remainder is holomorphic on that ball and satisfies the displayed quadratic bound on its half-ball, by its double zero and Schwarz's lemma. The folding identity gives $\|\mathsf{Q}^{[n]}\|_{\infty\to\infty} \le 4L^n$. With $a_j = \ell_j^{-1}|A_j|_\infty$, iteration gives

$$a_j \le 4a_0 + 4C_N L^{-2} \sum_{i=1}^{j} \ell_i a_{i-1}^2, \qquad \sum_{i=1}^{j} \ell_i \le \tfrac{3}{2}.$$

Choose ρ_* so that

$$384 C_N L^{-2} \rho_* \le 4, \quad 16\rho_* \le r_{\log}, \quad 8\rho_* \le 1, \quad 32\rho_* < \rho_L,$$

where $r_{\log}, \rho_L$ are the one-step logarithmic and holomorphic radii. Induction gives $a_j \le 8a_0$ throughout the complex ball. Thus every intermediate configuration $V_j = e^{iA_j}$ satisfies

$$|V_j - \mathbf{1}|_\infty \le 16 \ell_j |a|_\infty.$$

These estimates also ensure that all intermediate logarithms are the specified analytic branches.

Define the one-step straightened kernel by

$$(\mathring{\mathcal{Q}}_V B)(c) = \sum_{b \in \mathsf{P}_c^+} k_V(c,b) B(b).$$

The one-step derivative kernels obey

$$|k_V - k_{\mathbf{1}}| + |\operatorname{Ad} V(y,x) - 1| \le C_L |V - \mathbf{1}|_\infty.$$

To see this uniformly on the smaller ball, apply Cauchy's formula first to the left variation $e^{itB}V'$, with $r_L' = \rho_L/16$. The mean derivative, its gauge correction, and their straightened sum are bounded by $1/r_L', 1/r_L', 5/r_L'$, respectively. Apply it next to e^{isA}, with $r_L'' = \log(1 + 3\rho_L/4)$. Since $|\log V| \le 2|V - \mathbf{1}|$, subtraction at $s = 0$ gives the kernel bound $20|A|/(r_L' r_L'')$ and the transport bound $4|A|/r_L''$; when $2|A| > r_L''$, the same bounds follow from the preceding absolute bounds after increasing the constant. Conversion to the fixed Hilbert norm costs only $2\sqrt{2}$.

A double block contains at most $6L^3$ fine bonds, and a fine bond belongs to at most six double blocks. Schur's estimate therefore bounds the difference between a scaled one-step straight average and its free value by $C_L' |V - \mathbf{1}|_\infty$. The scalar average has disjoint blocks and obeys the same estimate. The free factors have norm at most one. Since $\sum_{j=1}^{k} \ell_{j-1} \le 3/2$, the products defining Q_U, R_U have norms at most $\exp(24C_L')$, after decreasing $\rho_* \le 1$. This proves the assertion. Cauchy's formula on the whole ball also gives bounds C_*/ρ_* and $4C_*/\rho_*^2$ for the first and mixed second derivatives. The products retain their block locality. Each intermediate contour lies within its block, and the union of the preceding-level blocks making up a later block is precisely that block. Induction therefore confines the row of Q_U at c and its field dependence to bonds with both endpoints in $B^k(c_-) \cup B^k(c_+)$. The row of R_U at y and its field dependence lie in $B^k(y)$.

We apply these bounds along the plane-wave responses

$$\zeta_\pm = \eta e^{\pm ip\cdot x} \alpha_\pm, \qquad \alpha_\pm = \widehat{h}(\pm p) e_{\mu_\pm} t_3.$$

The preceding free estimate bounds both $\alpha_\pm$ and their covariant curls in supremum norm. We use bold letters for jets along these two fields. All the jets below have width δ_{loop}, with $|p_\mu| \le \delta_{\mathrm{loop}}$.

The Hessian has a useful local form. For an oriented plaquette with bonds $b_1, \dots, b_4$, put

$$T_1 = 1, \quad T_2 = U(b_1), \quad T_3 = U(b_1)U(b_2)U(b_3)^{-1}, \quad T_4 = U_p, \qquad (\varsigma_i) = (+,+,-,-).$$

Moving all four exponentials to the left in $(e^{itB}U)_p$ gives $\prod_i e^{itY_i}U_p$, where $Y_i = \varsigma_i \operatorname{Ad}(T_i)B(b_i)$. Its second derivative uses

$$\partial_t^2 \prod_i e^{itY_i}\Big|_0 = -\Big(\sum_i Y_i\Big)^2 - \sum_{i<j}[Y_i, Y_j].$$

The Pauli identities $Y^2 = \langle Y, Y\rangle I$ and $U_p^{-1} = (\operatorname{tr} U_p)I - U_p$ now give

$$H_U = (d + E_U)^{\mathsf{T}} z_U (d + E_U) + f_U,$$

where

$$E_U B(p) = \eta^{-1} \sum_i \varsigma_i (\operatorname{Ad}(T_i) - 1) B(b_i),$$

$$z_U(p) = \tfrac{1}{2} \operatorname{tr} U_p, \qquad F_U(p) = \frac{U_p - U_p^{-1}}{2i\eta^2},$$

$$(B, f_U B) = \eta^3 \sum_p \tfrac{i}{2} \operatorname{tr} \sum_{i<j} [Y_i, Y_j] F_U(p).$$

Write $e_U = D_U - d$. At $U = e^{i\eta a}$, its first derivative is $i \operatorname{ad} a$, and its second coefficient is $-\eta(\operatorname{ad} a)^2/2$. The derivatives of E_U are sums of at most four such terms. The first derivative of $z_U - 1$ vanishes; its quadratic coefficient is $-\eta^4\langle da, da\rangle/2$. The first derivative of f_U contains one da; its second derivative is bounded by $C(|a|^2 + \eta|a||da|)|B|^2$. Consequently the jets of

$$e_U,\ E_U,\ z_U,\ f_U,\ Q_U,\ R_U$$

are bounded by a constant depending only on L. Indeed, the first four kernels couple indices in one plaquette, with at most thirteen neighbours; their Bloch factors on this strip are uniformly bounded. For the last two kernels use the preceding Cauchy bounds, the double-block overlap of six (one for scalar blocks), and the bounded exponential factor across a block. Their entries, and hence their jets, are holomorphic in p: each entry is a finite sum of local kernel derivatives multiplied by exponentials in the momentum. The 00 components of $e_U, E_U, z_U - 1, f_U$ vanish.

We shall also use the following trace bound:

$$|\mathsf{t}(G\mathbf{B}_1 G\mathbf{B}_2 G\mathbf{B}_3)| \le C_{\mathrm{tr}} \prod_{j=1}^{3} \|\mathbf{B}_j\|.$$

The same bound holds with g in place of G. Expand the 11 component. Every term has three propagators evaluated at κ or $\kappa \pm p$. Schatten Hölder followed by integral Hölder reduces the estimate to the integral of the cube of one free propagator. The Bloch sectors unfold to the fine Brillouin cube, and

$$\int \|\widehat{G}(\kappa + \vartheta)\|_3^3 \, d\kappa \le 9(10/9)^3 (2\pi)^{-3} \int_{\mathbb{R}^3} (1 + 4|q|^2/\pi^2)^{-3} \, dq.$$

Here $|\operatorname{Im} \vartheta_\mu| \le 2\delta_{\mathrm{loop}}$ and the previously proved lower bound for $1 + D(q + \vartheta)$ was used. The integral is finite and independent of η. The finite number of jet components is absorbed in C_{tr}.

Put

$$\mathcal{L}(z) = \ell(\boldsymbol{\Delta} + z) - \tfrac{1}{2}\ell(\mathbf{K} + z), \qquad \mathcal{L}^\sharp = (1 - \partial_z + \tfrac{1}{2}\partial_z^2)\mathcal{L}(z)\big|_{z=1}.$$

For $z > 0$ these jets are invertible at width $\delta_z = \min(\delta_{\mathrm{loop}}, \sqrt{z}/5)$. In fact, on $S_{2\delta_z}$, $\operatorname{Re}(D(q) + z) \ge z/3$. Near $z = 1$ the width is fixed, and differentiation in the jet algebra gives

$$\partial_z \ell(\mathbf{K} + z) = \mathsf{t}((\mathbf{K} + z)^{-1}), \qquad \partial_z^2 \ell(\mathbf{K} + z) = -\mathsf{t}((\mathbf{K} + z)^{-2}),$$

and the same identities for $\boldsymbol{\Delta}$.

Proposition 43.2. Let $\widehat{\pi}^0$ be the polarisation of the Gaussian determinant evaluated on the potential responses $\zeta_\pm$, with the two multiplier components set to zero. Then $\widehat{\pi}^0_{\mu_+\mu_-} - \mathcal{L}^\sharp$ is holomorphic for $|p_\mu| < \delta_{\mathrm{loop}}$ and has absolute value at most $C_{\mathrm{rem}}(L)$, uniformly in k. ∎

Proof. The vector and scalar saddle matrices in the Gaussian formula, up to field-independent changes of pairing weights, are

$$\mathbb{S}(X_U, Q_U), \qquad \mathbb{S}(\Delta_U, R_U), \qquad X_U = \Xi_U - Q_U^{\mathsf{T}} Q_U.$$

Set

$$\mathbf{V} = \mathbf{Q}^{\mathsf{T}}\mathbf{Q} - \mathbf{D}\mathbf{R}^{\mathsf{T}}\mathbf{R}\mathbf{D}^{\mathsf{T}}, \qquad \boldsymbol{\Xi} = \mathbf{K} + \mathbf{V}, \qquad \mathbf{Y} = \boldsymbol{\Delta} + \mathbf{R}^{\mathsf{T}}\mathbf{R}.$$

Multiplication by a triangular jet of determinant one changes $\mathbb{S}(\boldsymbol{\Xi} - \mathbf{Q}^{\mathsf{T}}\mathbf{Q}, \mathbf{Q})$ into $\mathbb{S}(\boldsymbol{\Xi}, \mathbf{Q})$. Its Schur factorisation has diagonal blocks $\boldsymbol{\Xi}$ and $-\mathbf{Q}\boldsymbol{\Xi}^{-1}\mathbf{Q}^{\mathsf{T}}$. The free strip estimates make both 00 blocks invertible; hence the jet factorisation is legitimate. The scalar factorisation is identical. Additivity of ℓ therefore gives

$$\widehat{\pi}^0 = -\tfrac{1}{2}\ell(\boldsymbol{\Xi}) - \tfrac{1}{2}\ell(\mathbf{Q}\boldsymbol{\Xi}^{-1}\mathbf{Q}^{\mathsf{T}}) + \ell(\mathbf{Y}) + \ell(\mathbf{R}\mathbf{Y}^{-1}\mathbf{R}^{\mathsf{T}}).$$

Write $\mathbf{G}_U = (\mathbf{K}+1)^{-1}$ and $\mathbf{P}_\tau = (\boldsymbol{\Xi} + \tau)^{-1}$. Integrate first in the added constraint $\mathbf{V}$ and then in the mass:

$$\ell(\boldsymbol{\Xi}) = \ell(\mathbf{K}+1) + \int_0^1 \mathsf{t}((\mathbf{K}+1+t\mathbf{V})^{-1}\mathbf{V})\,dt - \int_0^1 \mathsf{t}(\mathbf{P}_\tau)\,d\tau.$$

Two uses of the resolvent identity give

$$\begin{aligned}\mathbf{P}_\tau = \mathbf{G}_U + (1-\tau)\mathbf{G}_U^2 + (1-\tau)^2\mathbf{G}_U^2\mathbf{P}_\tau \\ - \mathbf{G}_U\mathbf{V}\mathbf{P}_\tau - (1-\tau)\mathbf{G}_U^2\mathbf{V}\mathbf{P}_\tau.\end{aligned}$$

Thus

$$\ell(\boldsymbol{\Xi}) = (1 - \partial_z + \tfrac{1}{2}\partial_z^2)\ell(\mathbf{K}+z)\big|_{z=1} + \rho_v,$$

where

$$\begin{aligned}\rho_v = &\int_0^1 \mathsf{t}((\mathbf{K}+1+t\mathbf{V})^{-1}\mathbf{V})\,dt \\ &- \int_0^1 \left[(1-\tau)^2\mathsf{t}(\mathbf{G}_U^2\mathbf{P}_\tau) - \mathsf{t}(\mathbf{G}_U\mathbf{V}\mathbf{P}_\tau) - (1-\tau)\mathsf{t}(\mathbf{G}_U^2\mathbf{V}\mathbf{P}_\tau)\right] d\tau.\end{aligned}$$

The same computation for $(\boldsymbol{\Delta}, \mathbf{R}^{\mathsf{T}}\mathbf{R}, \mathbf{Y})$ defines ρ_g. It remains to bound these two remainders and the two coarse Schur determinants.

Introduce bounded jets

$$\boldsymbol{\kappa} = G^{1/2}(\mathbf{K} - K)G^{1/2}, \quad x = \mathbf{Q}G^{1/2}, \quad y = G^{1/2}\mathbf{D}\mathbf{R}^{\mathsf{T}}, \quad \omega = x^{\mathsf{T}}x - yy^{\mathsf{T}}.$$

The local Hessian identity above proves that $\boldsymbol{\kappa}$ is bounded. Each term contains only a bounded jet among e, E, z, f, a square-root propagator, or a propagator with one derivative; for example its Hessian part is

$$\begin{aligned}&(dG^{1/2})^{\mathsf{T}}(z-1)dG^{1/2} + (dG^{1/2})^{\mathsf{T}}zEG^{1/2} + (EG^{1/2})^{\mathsf{T}}zdG^{1/2} \\ &\quad + (EG^{1/2})^{\mathsf{T}}zEG^{1/2} + G^{1/2}fG^{1/2}.\end{aligned}$$

Its remaining part is the expansion of $G^{1/2}[(d+e)(d+e)^{\mathsf{T}} - dd^{\mathsf{T}}]G^{1/2}$. In particular $\kappa_{00} = 0$. Set

$$B_t = (1 + \boldsymbol{\kappa} + t\omega)^{-1}, \qquad B'_\tau = (1 + \boldsymbol{\kappa} + \omega - (1-\tau)G)^{-1}.$$

The free strip estimates bound the inverses of the 00 components uniformly in t, τ; the finite nilpotent inverse formula therefore bounds the full jets uniformly. We have $\mathbf{G}_U = G^{1/2}B_0G^{1/2}$ and $\mathbf{P}_\tau = G^{1/2}B'_\tau G^{1/2}$.

Cyclicity converts the four traces in ρ_v to

$$\begin{aligned}
&\mathsf{t}(xB_t x^{\mathsf{T}}) - \mathsf{t}(y^{\mathsf{T}} B_t y),\\
&\mathsf{t}(xB'_\tau G B_0 x^{\mathsf{T}}) - \mathsf{t}(y^{\mathsf{T}} B'_\tau G B_0 y),\\
&\mathsf{t}(xB'_\tau G B_0 G B_0 x^{\mathsf{T}}) - \mathsf{t}(y^{\mathsf{T}} B'_\tau G B_0 G B_0 y),\\
&\mathsf{t}(G B_0 G B_0 G B'_\tau).
\end{aligned}$$

The first three traces act on a coarse bond or site fibre, of dimension at most nine, and are bounded by the jet norm. The fourth is bounded by the three-propagator estimate. For the scalar remainder use

$$\kappa_g = g^{1/2}(\boldsymbol{\Delta} - \Delta)g^{1/2} = (dg^{1/2})^{\mathsf{T}} e g^{1/2} + (eg^{1/2})^{\mathsf{T}} d g^{1/2} + (eg^{1/2})^{\mathsf{T}} e g^{1/2}, \qquad x_g = \mathbf{R} g^{1/2}.$$

There is no y term. The same proof bounds ρ_g. Finally $\mathbf{Q}\boldsymbol{\Xi}^{-1}\mathbf{Q}^{\mathsf{T}} = xB'_0 x^{\mathsf{T}}$, and the scalar counterpart has the same form. Their jet norms and their 00-inverse norms are bounded; their fibres have dimension at most nine. The finite formula for ℓ therefore bounds their logarithmic polarisations.

It follows that

$$\widehat{\pi}^0 - \mathcal{L}^\sharp = -\tfrac{1}{2}\rho_v + \rho_g - \tfrac{1}{2}\ell(\mathbf{Q}\boldsymbol{\Xi}^{-1}\mathbf{Q}^{\mathsf{T}}) + \ell(\mathbf{R}\mathbf{Y}^{-1}\mathbf{R}^{\mathsf{T}})$$

is uniformly bounded. All jets are holomorphic in p, as are their inverses; the bounds justify the trace and parameter integrals. This also proves the asserted holomorphy. □

43.5. **The massive coefficient.** In this subsection the lattice spacing is one, forms have absorbed lengths, and all pairings are unweighted. Put

$$\psi(q) = \sum_{\mu=1}^{3} \psi^{(\mu)}(q), \quad \psi^{(\mu)}(q) = 2 - 2\cos q_\mu, \qquad \Upsilon_z(q) = (\psi(q) + z)^{-1}.$$

For $z > 0$, let

$$\Delta_U = \mathsf{D}_U^{\mathsf{T}} \mathsf{D}_U, \qquad \mathsf{K}_U = A''(U) + \mathsf{D}_U \mathsf{D}_U^{\mathsf{T}},$$

and define on a finite torus, near the trivial configuration,

$$\mathfrak{L}_{z,\mathfrak{s}}(U) = \Big[\log\det(\Delta_{U'} + z) - \tfrac{1}{2}\log\det(\mathsf{K}_{U'} + z)\Big]_{U'=\mathbf{1}}^{U'=U}.$$

At $U = \mathbf{1}$ the symbols of both operators are $\psi(q) I$. Indeed, the scalar operator is $d^{\mathsf{T}} d$, and the form operator is $d^{\mathsf{T}} d + d d^{\mathsf{T}}$; the mixed terms cancel in their Fourier symbols. Both massive matrices are positive at $\mathbf{1}$, so their real logarithms are real analytic on a neighbourhood of $\mathbf{1}$.

Their kernels are entire local functions of the absorbed potential and commute with translations, gauge transformations and lattice isometries. The logarithmic determinant polarisation formula therefore makes $\mathfrak{L}_z = (\mathfrak{L}_{z,\mathfrak{s}})_{\mathfrak{s}}$ a tempered functional: its zero-field inverse symbols $(\psi + z)^{-1} I$ are holomorphic on a strip, the finite-torus Fourier sums converge exponentially to the corresponding integrals, and colour invariance makes the polarisation diagonal in the Pauli basis. We write $\widetilde{\beta}(z) = \mathsf{b}[\mathfrak{L}_z]$. This is real analytic for $z > 0$, since its polarisation is an integral of products of $(\psi + z)^{-1}$ and the first two derivatives of the local kernels.

Proposition 43.3. There is an absolute constant C_{uv} such that

$$z^{1/2}|\widetilde{\beta}(z)| + z^{3/2}|\widetilde{\beta}'(z)| + z^{5/2}|\widetilde{\beta}''(z)| \le C_{\mathrm{uv}}, \qquad 0 < z \le 1.$$

■

Proof. We first identify the coefficient by a local trace, then compute that trace. For a real bond function θ, put $U_\theta = e^{i\theta t_3}$, and set $\vartheta_2^\circ(x) = x_1$, $\vartheta_1^\circ = \vartheta_3^\circ = 0$. The plaquette angles of $f\vartheta^\circ$ are f in the 12-plane and zero in the other two planes. Let τ_x be the trace on indices based at x. For sufficiently small $|f|$, define

$$\varrho_x(\theta) = \tau_x \log(\Delta_\theta + z) - \tfrac{1}{2}\tau_x \log(\mathsf{K}_\theta + z).$$

These logarithms are well defined even though ϑ° is unbounded.

To verify this and the estimates used below, write $t_\pm = (t_1 \pm it_2)/2$ and $B = B^3 t_3 + w t_+ + \overline{w} t_-$. The shifts $u_\mu w(x) = e^{2i\theta_\mu(x)} w(x + e_\mu)$ are unitary. At a plaquette $p = p_{\mu\nu}(x)$, put

$$(y_1, y_2, y_3, y_4) = (w_\mu, u_\mu w_\nu, -e^{2if_p} u_\nu w_\mu, -e^{2if_p} w_\nu)(x), \qquad f_p = (d\theta)(p).$$

Differentiating its four Wilson factors gives

$$(B, \mathsf{K}_\theta B) = \sum_p \left[\cos f_p \left((dB^3)(p)^2 + \left| \sum_i y_i \right|^2 \right) - 2 \sin f_p \sum_{i<j} \operatorname{Im}(y_i \overline{y}_j) \right] + \sum_x \left[(d^{\mathsf{T}} B^3)(x)^2 + \left| \sum_\mu (u_\mu^* - 1) w_\mu(x) \right|^2 \right].$$

Here $\langle t_+, t_- \rangle = 1/2$ and $[t_+, t_-] = t_3$. Since $\sum_{i<j} |y_i||y_j| \le (3/2) \sum_i |y_i|^2$ and a bond belongs to four plaquettes,

$$\Delta_\theta \ge 0, \qquad \mathsf{K}_\theta \ge -12 \sup_p |\sin f_p| \quad \text{if } \sup_p |f_p| \le \pi/2.$$

In particular $\mathsf{K}_{f\vartheta^\circ} + z \ge z/2$ for small $|f|$. Both kernels have range at most two and uniformly bounded trigonometric-polynomial entries.

Identification by a local density. For either operator T_θ, the resolvent obeys

$$|(T_\theta + z + w)^{-1}(i,j)| \le (z/4 + w)^{-1} e^{-c|x(i) - x(j)|}, \qquad w \ge 0,$$

uniformly under the preceding lower bound. Indeed, conjugation by $e^{\max(-N, \min(N, \sigma \cdot x))}$ changes its finite-range kernel by an operator of norm at most $C(e^{2|\sigma|} - 1)$. Choose $|\sigma| = c$ so this is at most $z/4$, apply the Neumann series, and let $N \to \infty$, choosing the direction of σ along $x(i) - x(j)$.

The identity

$$\log(T + z) = \int_0^\infty \left[(1 + w)^{-1} - (T + z + w)^{-1} \right] dw$$

then gives a local second-derivative kernel $\mathcal{P}(x; b, b') = \partial_{\theta(b)} \partial_{\theta(b')} \varrho_x(0)$, with

$$|\mathcal{P}(x; b, b')| \le C e^{-c(|x - b_-| + |x - b'_-|)}.$$

For completeness, if $R_w = (T_0 + z + w)^{-1}$, the integrand of this derivative is the local trace of

$$R_w T^{(2)}[\delta_b, \delta_{b'}] R_w - R_w T^{(1)}[\delta_b] R_w T^{(1)}[\delta_{b'}] R_w - R_w T^{(1)}[\delta_{b'}] R_w T^{(1)}[\delta_b] R_w.$$

The differentiated kernels have finite support near the indicated bonds. The bound $C(z/4 + w)^{-2}$ is integrable and justifies differentiation under the integral. Summing in x, using trace cyclicity and $\partial_w R_w = -R_w^2$, yields $\operatorname{Tr}(R_0 T^{(2)} - R_0 T^{(1)} R_0 T^{(1)})$. Periodising R_0 gives the finite-torus inverse; its exponential decay shows that these finite-support traces converge to their infinite-lattice values. Consequently

$$\sum_x \mathcal{P}(x; b, b') = \pi_{\mu\nu}(b'_- - b_-).$$

The local trace is invariant under $\theta \mapsto \theta + d\lambda$, by conjugation with $e^{-i\lambda t_3}$, and is even in θ, by conjugation with it_1. Expand the resolvent at $f\vartheta^\circ$ to second order. The Taylor coefficients of the finite-range kernel grow at most quadratically in position, and the remainder is $O(|f|^3(1 + |x|)^3)$.

The resolvent identity with three insertions and the preceding exponential bound make all local sums absolutely convergent, with an integrable $O(|f|^3(z/4+w)^{-2})$ remainder. Thus

$$\varrho_0(f\vartheta^\circ) - \varrho_0(0) = \tfrac{1}{2} f^2 \sum_{b,b'} \mathcal{P}(0;b,b') x_1 x_1' + O(|f|^3),$$

where both bonds have direction two. Gauge invariance with $\lambda(x) = x_2$ gives $\sum_{b'} \mathcal{P}(0;b,b') = 0$. Using symmetry, translation covariance, and $2x_1x_1' = x_1^2 + (x_1')^2 - (x_1 - x_1')^2$, the sum equals

$$-\tfrac{1}{2} \sum_u u_1^2 \pi_{22}(u) = \widetilde{\beta}(z).$$

We have proved

$$\varrho_0(f\vartheta^\circ) - \varrho_0(0) = \tfrac{1}{2}\widetilde{\beta}(z) f^2 + O(|f|^3).$$

The magnetic resolvent calculation. Put $\varphi = 2f$. Then $u_1 u_2 = e^{i\varphi} u_2 u_1$ and u_3 commutes with both. For

$$W_\varphi(n) = e^{-i\varphi n_1 n_2/2} u_1^{n_1} u_2^{n_2} u_3^{n_3},$$

we have

$$W_\varphi(n) W_\varphi(n') = e^{i\varphi(n_1 n_2' - n_2 n_1')/2} W_\varphi(n+n'), \qquad \tau_0 W_\varphi(n) = \delta_{n0}.$$

For a matrix symbol $a(q) = \sum_n a_n e^{in\cdot q}$, write $\operatorname{Op}_\varphi(a) = \sum_n a_n W_\varphi(n)$ and $\|a\|_{(j)} = \sum_n |a_n|(1+|n|)^j$. The product symbol is

$$a \star b = ab - \tfrac{i\varphi}{2}\{a,b\} - \tfrac{\varphi^2}{8}\{\{a,b\}\} + O_{(0)}(|\varphi|^3 \|a\|_{(3)} \|b\|_{(3)}),$$

with remainder constant $1/48$, where $\{a,b\} = a_1 b_2 - a_2 b_1$ and $\{\{a,b\}\} = a_{11}b_{22} - 2a_{12}b_{12} + a_{22}b_{11}$. This follows term by term from the exponential remainder $|e^{is} - 1 - is + s^2/2| \le |s|^3/6$. Also $\|\operatorname{Op}_\varphi(a)\| \le \|a\|_{(0)}$ and $\tau_0 \operatorname{Op}_\varphi(a) = \int \operatorname{tr} a$.

Suppose a Hermitian $N' \times N'$ trigonometric-polynomial matrix has the expansion $\sigma_\varphi = \psi I + \varphi\sigma_1 + \varphi^2\sigma_2 + O_{(3)}(|\varphi|^3)$, and $\operatorname{Op}_\varphi(\sigma_\varphi) + z \ge z/2$. With $U_w = (\psi + z + w)^{-1}$, a right resolvent parametrix is

$$r_0 = U_w I, \qquad r_1 = -U_w^2 \sigma_1, \qquad r_2 = U_w^3 \sigma_1^2 - U_w^2 \sigma_2 + \tfrac{1}{8} U_w \{\{U_w, \psi\}\} I.$$

The coefficients of orders one and two in $(r_0 + \varphi r_1 + \varphi^2 r_2) \star (\sigma_\varphi + z + w) - I$ vanish: $\{U_w, \psi\} = 0$, and $\{-U_w^2\sigma_1, \psi\} + \{U_w, \sigma_1\} = 0$. On a strip depending on z, the symbol norms of these terms are bounded by $C_z(z+w)^{-1}$. The product error is therefore $O_{(0)}(|\varphi|^3(z+w)^{-1})$. Multiplying by the exact inverse bounds the local-trace error by $C_z|\varphi|^3(z+w)^{-1}(z/2+w)^{-1}$, which is integrable in w. Integrating the resolvent gives

$$\begin{aligned}\tau_0 \log(\operatorname{Op}_\varphi(\sigma_\varphi) + z) = {} & \int \operatorname{tr}\log(\psi + z) + \varphi \int \Upsilon_z \operatorname{tr}\sigma_1 \\ & + \varphi^2 \int \Big[-\tfrac{1}{2}\Upsilon_z^2 \operatorname{tr}\sigma_1^2 + \Upsilon_z \operatorname{tr}\sigma_2 + \tfrac{N'}{24}\Upsilon_z^2 \psi_{11}\psi_{22} \Big] + O(|\varphi|^3).\end{aligned}$$

To check the last term, use $\psi_{12} = 0$,

$$\{\{U_w, \psi\}\} = -2U_w^2 \psi_{11}\psi_{22} + 2U_w^3(\psi_1^2\psi_{22} + \psi_2^2\psi_{11}),$$

then integrate in w. Periodic integration by parts gives $\int \Upsilon_z^3 \psi_1^2 \psi_{22} = \frac{1}{2}\int \Upsilon_z^2 \psi_{11}\psi_{22}$, and the same formula with the indices exchanged.

The three colour sectors. The neutral and two charged sectors are invariant. Complex conjugation exchanges the charged sectors, so their local traces agree. The neutral scalar operator is independent of f; either charged scalar operator is $\operatorname{Op}_\varphi(\psi)$. The neutral form operator is

$$d^{\mathsf{T}} d + d d^{\mathsf{T}} + (\cos f - 1) d_{12}^{\mathsf{T}} d_{12}.$$

If σ^{n} is the symbol of $d_{12}^{\mathsf{T}} d_{12}$, then $0 \le \sigma^{\mathrm{n}} \le \psi I$ and $\operatorname{tr}\sigma^{\mathrm{n}} = \psi^{(1)} + \psi^{(2)}$. Its logarithmic trace changes by $-\frac{1}{2} f^2 \int \Upsilon_z(\psi^{(1)} + \psi^{(2)}) + O(f^4)$.

The displayed plaquette quadratic form shows that the charged form operator is a polynomial of degree at most two in the $u_\mu^{\pm 1}$, with coefficients analytic in φ. Reordering these shifts therefore writes it as $\mathrm{Op}_\varphi(\sigma_\varphi)$, where

$$\sigma_\varphi = \psi I + \varphi\sigma_1 + \varphi^2\sigma_2 + O_{(3)}(|\varphi|^3).$$

The matrices σ_1, σ_2 are fixed Hermitian trigonometric polynomials, independent of z. The preceding lower bound verifies the positivity hypothesis of the parametrix calculation. Adding the two charged scalar traces, subtracting one half of the two charged form traces and the neutral form trace, and using $\varphi^2 = 4f^2$, gives

$$\widetilde{\beta}(z) = 8\int \left[\tfrac{1}{2}\Upsilon_z^2 \operatorname{tr}\sigma_1^2 - \tfrac{1}{24}\Upsilon_z^2\psi_{11}\psi_{22} - \Upsilon_z \operatorname{tr}\sigma_2\right] + \tfrac{1}{2}\int \Upsilon_z(\psi^{(1)} + \psi^{(2)}).$$

The linear term vanishes by evenness of the local density.

Thus $\widetilde{\beta}(z) = \int(\Upsilon_z^2 g_2 + \Upsilon_z g_1)$, where g_1, g_2 are fixed bounded trigonometric polynomials. Differentiating under the integral, for $n = 0, 1, 2$,

$$|\widetilde{\beta}^{(n)}(z)| \le C\left[(n+1)!\int \Upsilon_z^{n+2} + n!\int \Upsilon_z^{n+1}\right].$$

Finally $\psi(q) \ge 4|q|^2/\pi^2$ on the period cube. For $j \ge 2$, the substitution $q = \sqrt{z}\,q'$ gives $\int \Upsilon_z^j \le C_j z^{3/2-j}$; $\int \Upsilon_z \le C_1$, because $|q|^{-2}$ is locally integrable in three dimensions. These estimates prove the asserted bound. □

43.6. **The accumulated coefficient.** The scaled massive polarisation is related to the unit-spacing massive coefficient by

$$-\partial_{p_1}\partial_{p_2}\mathcal{L}(z)(0) = \eta\widetilde{\beta}(\eta^2 z).$$

To verify the normalization, use

$$\Delta_U = \eta^{-2}\mathsf{D}_U^{\mathsf{T}}\mathsf{D}_U, \qquad K_U = \eta^{-2}\mathsf{K}_U.$$

The logarithmic determinant combination at mass z therefore differs from the unscaled combination at mass $\eta^2 z$ by a field-independent constant. Under $x \mapsto x/\eta$, the fine torus has side $\mathfrak{s}/\eta$. Expand the two response fields:

$$\zeta_\pm = \eta\sum_{m,\rho} b_{\rho,m}^{\pm} e^{i(\pm p + 2\pi m)x} e_\rho t_3.$$

Momentum conservation pairs only m with $-m$. The two field factors η, and the torus volume ratio η^{-3}, give

$$\mathcal{L}(z)(p) = \eta^{-1}\sum_{m,\rho,\sigma} \widehat{\pi}_{\rho\sigma}^{(\eta^2 z)}(\eta(p + 2\pi m)) b_{\rho,m}^{+} b_{\sigma,-m}^{-}.$$

This identity follows first at the common torus momenta from the finite-volume determinant formula and then near zero by holomorphy. At fixed k,

$$b_{\rho,m}^{\pm} = O(|p|^2) \quad (m \ne 0), \qquad b_{\rho,0}^{\pm} = \delta_{\rho\mu_\pm} + O(|p|).$$

Since the massive polarisation is $O(|p|^2)$ at zero, all nonzero modes contribute $O(|p|^4)$, and

$$\mathcal{L}(z)(p) = \eta^{-1}\widehat{\pi}_{\mu_+\mu_-}^{(\eta^2 z)}(\eta p) + O(|p|^3).$$

Two derivatives in p prove the asserted identity. The massive inverse symbols and the local kernel derivatives are jointly analytic in $z > 0$ and in p near zero. Hence mass and momentum derivatives commute, giving

$$-\partial_1\partial_2\mathcal{L}^\sharp(0) = \eta\left[\widetilde{\beta}(\eta^2) - \eta^2\widetilde{\beta}'(\eta^2) + \tfrac{1}{2}\eta^4\widetilde{\beta}''(\eta^2)\right].$$

Proof of theorem O. Temperedness and the identity for c_k were proved from Gaussian gluing. The kernel limit, reality, exponential decay and the coefficient formula follow from the tempered polarisation calculation. In particular all the kernel sums above converge absolutely.

For $k \geq 1$, Cauchy's estimate applied to the uniform determinant remainder, together with the Ward reduction of the actual Gaussian coefficient, gives

$$\left|\mathfrak{b}[\Psi_k] + \partial_1\partial_2\mathcal{L}^\sharp(0)\right| \leq C_{\mathrm{rem}}(L)\delta_{\mathrm{loop}}^{-2}.$$

Use the preceding scaling identity and set $z = \eta^2$. The massive estimate gives

$$\begin{aligned}\eta\left|\widetilde{\beta}(\eta^2) - \eta^2\widetilde{\beta}'(\eta^2) + \tfrac{1}{2}\eta^4\widetilde{\beta}''(\eta^2)\right| &\leq z^{1/2}|\widetilde{\beta}(z)| + z^{3/2}|\widetilde{\beta}'(z)| + z^{5/2}|\widetilde{\beta}''(z)| \\ &\leq C_{\mathrm{uv}}.\end{aligned}$$

This proves the bound uniformly in k; at $k = 0$ the coefficient is zero. The supremum is therefore finite and is the least possible bound for the displayed accumulated sums.

□

44. Histories and the exact measure decomposition

We decompose the effective measure into finitely many terms,

$$\rho_k\, dV_k = \sum_{h\in\mathcal{H}_k} \tau_h.$$

A history records the failed local tests at successive scales. The tests first select a domain on which to compare the field with a background, then enlarge its complement until the remaining collar passes the required bounds. Gauge factors are inserted on the selected blocks. Their normalization, rather than a pointwise identity for the weighted histories, gives the displayed measure identity.

Fix the odd integer $L \geq 13$. Let $r, \mathfrak{p}_0 \in \mathbb{N}$, $r \geq 2$, $A_0, A_1 > 0$, $\Theta_1 \geq 1$, $\varpi_T \in \mathbb{N}$ and $0 < \theta_c < 1$. The subscript of θ_c is a mark. Take powers of L satisfying

$$R_1 \geq L, \qquad R_1\mathfrak{M} \geq L^2, \qquad M_1 \geq R_1\mathfrak{M}, \qquad M_2 = L^2M_1, \qquad M \geq L^4M_1, \qquad M \geq \varpi_T.$$

For $x_0, \dots, x_K > 1$, set $\beta = x_0$ and

$$\begin{gathered} g_j = x_j^{-1/2}, \qquad y_j = \log x_j, \qquad b_j = L^{-1}x_{j-1} - x_j \quad (1 \leq j \leq K), \\ \varepsilon_j = A_0 g_j y_j^{\mathfrak{p}_0}, \qquad \delta_j = A_1 g_j y_j^{\mathfrak{p}_0}, \qquad \delta_j^\flat = \frac{\delta_j}{4\Theta_1}, \qquad \mathfrak{b}_j = \delta_j/g_j, \\ R(x) = \min\{L^a : a \in \mathbb{Z}_{\geq 0},\ L^a \geq (\log x)^r\}, \qquad \check{x}_j = \min_{i\leq j} x_i, \qquad \check{R}_j = R(\check{x}_j). \end{gathered} \tag{634}$$

Throughout this section assume

$$\varepsilon_j \leq 1, \qquad \delta_j \leq 1, \qquad M\check{R}_j \leq L^{m+K-j} \qquad (0 \leq j \leq K). \tag{635}$$

For the small-coupling regime used later, on $x \geq \gamma^{-2}$ the function $x^{-1/2}(\log x)^{\mathfrak{p}_0}$ decreases if $\log\gamma^{-2} > 2\mathfrak{p}_0$; hence the first two bounds follow from $\max(A_0, A_1)\gamma(\log\gamma^{-2})^{\mathfrak{p}_0} \leq 1$. The later running-coupling estimates impose a bound on $|b_j|$; the partition below uses only (635).

At step $t = k + 1$, lengths are measured at level k. The coarse partition $\mathcal{P}_t$ has side $W_t = LM\check{R}_t$. A test cube $\square'$ has side $L^2M_2\check{R}_t$, and its enlargement $\square'^{\sim a}$ adds a layers of cubes of side $LM_2\check{R}_t$. At level j, a density cube has side $LM_2\check{R}_j$ in level-j units, and its enlargement uses layers of side $M_2\check{R}_j$. Thus a test cube of step t is a density cube of level t. The bounds above ensure that all these are standard torus partitions and that the test partition refines $\mathcal{P}_t$. For coarse regions, complements and differences are taken cube by cube; $X^{\sim a}$ and $X^{\sim -a}$ add or remove a coarse layers. A prime or a hat denotes the $\mathcal{P}_t$-hull. These conventions include the empty region and the whole torus.

We first specify the tests, including their domains. A partial map is *stably defined* at a point if it is defined and continuous on an open neighborhood of that point. Its stable domain is open, and its

restriction there is continuous. For a density cube $\square$ at level $j \geq 1$, use the reduced determining set of the tower of $D = \square^{\sim 4}$ and the datum induced by V_j as in (60). Write $U_{j,\square}(V_j) = U^{\flat}_{j,D}(V_j)$ for its reduced minimal orbit when that orbit exists. Define

$$\mathsf{m}_{j,\square}(V_j) = \max_{p \text{ inner bare plaquette of } \square^{\sim}} |\partial U_{j,\square}(V_j)(p) - \mathbf{1}|.$$

The value is independent of the orbit representative, since plaquette variables change by unitary conjugation. With $\eta_j = L^{-j}$, put

$$\chi_j(\square) = \mathbf{1}_{\{\mathsf{m}_{j,\square} \text{ stably defined, } \mathsf{m}_{j,\square} < \varepsilon_j \eta_j^2\}}. \tag{636}$$

For $j = 0$ use the bare plaquettes of V_0 in $\square^{\sim}$ and the threshold ε_0 directly.

For a test cube at step t, normalize an element of $U_{t,\square'}(V_t)$ so that its level-k contour means equal $\mathbf{1}$ on $Y_{\square'} = (\square'^{\sim 3})^{(t)}$. When the orbit exists and these means exist for one representative, this gives a uniquely determined restriction

$$V^{\mathrm{bg}}_{k,\square'} = (\mathcal{M}^k U_{t,\square'}(V_t))\big|_{\text{level-}k \text{ bonds inner to } \square'^{\sim 3}}.$$

Its conditional well-definedness is checked below. The two cube tests are

$$\begin{aligned} \chi^{\mathrm{P}}_t(\square') &= \chi_t(\square'), \\ \chi^{\mathrm{Q}}_t(\square') &= \mathbf{1}_{\{V^{\mathrm{bg}}_{k,\square'} \text{ stably defined, } \max_{b \in (\square'^{\sim 2})^{(k)*}} |V_k(b) V^{\mathrm{bg}}_{k,\square'}(b)^{-1} - \mathbf{1}| < 2\delta_k\}}. \end{aligned} \tag{637}$$

Here $X^{(k)*}$ consists of the level-k bonds with an endpoint in X, with the pivot bonds of incident level-t bonds removed. Each bond tested in the second line has both endpoints in $\square'^{\sim 3}$.

An input of step t is a nested family $T_\ell \supset \Omega_1 \supset \Lambda_1 \supset \cdots \supset \Omega_k \supset \Lambda_k$, where the members of index j are regions for $\mathcal{P}_j$. The first input is empty. For any finite collection $\mathcal{Q}$ of cubes and binary tests χ, write

$$[P; \chi]_{\mathcal{Q}} = \prod_{q \in P} (1 - \chi(q)) \prod_{q \in \mathcal{Q} \setminus P} \chi(q).$$

Exactly one of these products is one: that of the failed cubes.

For $t \geq 2$, set $Y = \Lambda_k^c$ and select the failed P-tests among test cubes in $(\widehat{Y}^{\sim 4})^c$. Their union is P_t, with indicator $\mathbf{1}^{\mathrm{P}}_t$. Set

$$B(P_t'') = (P_t'^{\sim 1} \cup \widehat{Y}^{\sim 5})^c, \qquad \Omega_t = (Q_t'^{\sim 2} \cup P_t'^{\sim 4} \cup \widehat{Y}^{\sim 8})^c, \tag{638}$$

where P_t'' is the set of level-t sites of the first region, and Q_t is selected by the Q-tests among test cubes in $B(P_t'')^{\sim -1}$. Denote this second product by $\mathbf{1}^{\mathrm{Q}}_t$. At $t = 1$, first select the failed bare density cubes P_0, then the failed P-tests P_1 in $(P_0'^{\sim})^c$; $\mathbf{1}^{\mathrm{P}}_1$ is the product of these two indicators. Use instead

$$B(P_1'') = (P_1'^{\sim 1} \cup P_0'^{\sim 2})^c, \qquad \Omega_1 = (Q_1'^{\sim 2} \cup P_1'^{\sim 4} \cup P_0'^{\sim 5})^c, \tag{639}$$

with Q_1 selected in $B(P_1'')^{\sim -1}$ as before. The domains are coarse regions, and $\Omega_t \subset \Lambda_k$ for $t \geq 2$.

For the domains through step t, form the completed collared sequence

$$\Omega_0^{\diamond} = \Omega_1^{\sim} \supset \Omega_1 \supset \cdots \supset \Omega_t, \tag{640}$$

where the first enlargement uses $\mathcal{P}_1$. Use its reduced determining set $\mathbf{B}_t^{\diamond}$: its level-j bonds have both endpoints in $\Omega_0^{\diamond}$, an endpoint in Ω_j, and no endpoint in Ω_{j+1}, with $\Omega_{t+1} = \emptyset$. The datum is the corresponding restriction of $(V_j)_{j \leq t}$. Let $U_t^{\diamond}$ be its minimal orbit when it exists. Let $\mathfrak{X}_0$ be the level-k bonds with at least one endpoint in Ω_t, and put $\mathfrak{X} = \Omega_t^{(k)*}$. If the orbit exists and its level-k contour means exist on the blocks centered in $\Omega_t^{(t)}$, let

$$V_k^{\mathrm{bg}\diamond} = (\mathcal{M}^k U)\big|_{\mathfrak{X}_0}, \qquad U \in U_t^{\diamond}, \quad \mathcal{M}^k U \text{ in block axial gauge on } \Omega_t^{(t)}. \tag{641}$$

This includes the incident bonds crossing the boundary of Ω_t. The restriction is independent of the choice of U, as proved below. For a positive bond $b \in \mathfrak{X}_0$, define

$$\mathcal{A}_k(b) = \frac{1}{i}\log[V_k(b)V_k^{\mathrm{bg}\diamond}(b)^{-1}] \tag{642}$$

on the stable domain of the background, excluding the value $-\mathbf{1}$ inside the logarithm. The logarithm is intrinsic; extend the field by $\mathcal{A}_k(\bar{b}) = -\mathcal{A}_k(b)$. Elsewhere mark it undefined. This convention defines a partial odd field even when the background does not exist. It is not a logarithm formula imposed again on negative bonds.

Write $\mathfrak{S} = \Omega_t \setminus \Omega_t^{\sim -1}$ and let K be the coarse cubes in $\Omega_t^{\sim -1}$. For $c \in \mathsf{K}$, set $c^\star = (c^{(k)*} \setminus \mathfrak{S}^{(k)*}) \cap \mathfrak{X}$. Define binary tests, with value zero at undefined fields, by

$$\chi_b = \mathbf{1}_{\{|\mathcal{A}_k(b)|<\delta_k\}}, \qquad \chi_b^0 = \mathbf{1}_{\{|\mathcal{A}_k(b)|<\delta_k^\flat\}}, \qquad \chi_b^1 = \chi_b - \chi_b^0, \qquad \mathrm{lo}(c) = \prod_{b\in c^\star} \mathbf{1}_{\{|\mathcal{A}_k(b)|<\theta_c\delta_k\}}.$$

For test cubes $\square' \subset \Omega_t^{\sim -1}$, put $\chi^{(k)}(\square') = \prod_{b\in(\square'^{\sim 2})^{(k)*}} \chi_b$. Select their failed union R. Its complement below is relative to $\Omega_t^{\sim -1}$, and

$$\mathfrak{U}(R) = \bigcup_{\square' \subset R^c} (\square'^{\sim 2})^{(k)*}.$$

The collar consists only of exterior bonds. For a coarse region $\mathcal{Y}$, let $\mathbb{B}(\mathcal{Y})$ be the bonds of $\mathfrak{X}$ with both endpoints in $\mathcal{Y}$, and put

$$\mathrm{Col}(\mathcal{Y}) = \{b \in \mathfrak{X} : \mathrm{e}(b) \cap \mathcal{Y} = \emptyset,\ \min_{b'\in\mathbb{B}(\mathcal{Y})} \mathrm{dist}_\infty(b,b') \le \varpi_T\}.$$

Distances between bonds are distances between their segments, and an empty minimum is $+\infty$. Thus crossing bonds are excluded. For a coarse set R, set $Z[\mathsf{R}] = (\Omega_t^c)^{\sim 2} \cup \mathsf{R}^\sim$. Let $\mathrm{rim}(Z)$ be the cubes of K in Z which touch Z^c, and let $\mathsf{K}(I)$ consist of the cubes of K containing an endpoint of a bond in I.

An answer is a pair $(I^\natural, \mathsf{R}^\natural)$ of subsets of $\mathfrak{X}$ and K. Starting with $\mathsf{R}_0 = R'$, explore the following sets, removing everything already explored:

$$\begin{aligned} Z^{[n]} &= Z[\mathsf{R}_n], & \Lambda^{[n]} &= (Z^{[n]})^c, \\ \mathfrak{B}_n &= \mathrm{Col}(\Lambda^{[n]}) \setminus \bigcup_{i<n} \mathfrak{B}_i, & I_n &= \mathfrak{B}_n \cap I^\natural, \\ \mathfrak{C}_n &= \mathrm{rim}(Z^{[n]}) \setminus \bigcup_{i<n} \mathfrak{C}_i, & \mathsf{R}_n^{\mathrm{lo}} &= \mathfrak{C}_n \cap \mathsf{R}^\natural, \\ \mathsf{R}_{n+1} &= \mathsf{R}_n \cup \mathsf{K}(I_n) \cup \mathsf{R}_n^{\mathrm{lo}}. \end{aligned} \tag{643}$$

Stop at the first n_* with $I_{n_*} = \mathsf{R}_{n_*}^{\mathrm{lo}} = \emptyset$. It exists, since successive nonempty answers consume new members of finite sets. The outcome is $\Xi = (R, I, \mathsf{R}^{\mathrm{lo}})$, with $I = \bigcup_{n<n_*} I_n$ and $\mathsf{R}^{\mathrm{lo}} = \bigcup_{n<n_*} \mathsf{R}_n^{\mathrm{lo}}$. Its explored sets are $E^T = \bigcup_{n\le n_*} \mathfrak{B}_n$ and $E^H = \bigcup_{n\le n_*} \mathfrak{C}_n$. A triple is admissible if its recursion with its own answer $(I, \mathsf{R}^{\mathrm{lo}})$ explores every member of I and R^{lo}. The output and strip are

$$Z_t = (\Omega_t^c)^{\sim 2} \cup (R' \cup \mathsf{K}(I) \cup \mathsf{R}^{\mathrm{lo}})^\sim, \qquad \Lambda_t = Z_t^c, \qquad S_t^{\mathrm{pr}} = \mathfrak{S} \cup R. \tag{644}$$

They satisfy $\Lambda_t \subset \Omega_t^{\sim -2}$ and $S_t^{\mathrm{pr}} \subset \Omega_t \setminus \Lambda_t$. Thus adjoining $\Omega_t \supset \Lambda_t$ gives the next input.

The fluctuation factor attached to an admissible triple is

$$\begin{aligned} \chi^\wedge(\Xi) = \prod_{\square' \subset R} (1 - \chi^{(k)}(\square')) \prod_{b\in\mathfrak{U}(R)\setminus E^T} \chi_b \prod_{b\in I} \chi_b^1 \prod_{b\in E^T\setminus I} \chi_b^0 \\ \times \prod_{c\in\mathsf{R}^{\mathrm{lo}}} (1 - \mathrm{lo}(c)) \prod_{c\in E^H\setminus\mathsf{R}^{\mathrm{lo}}} \mathrm{lo}(c). \end{aligned} \tag{645}$$

The full outcome o_t consists of P_t, Q_t, Ξ_t (and P_0 at the first step), with indicator $\mathbf{1}_t(\mathsf{o}_t) = \mathbf{1}_t^{\mathrm{P}}\mathbf{1}_t^{\mathrm{Q}}\chi^{\wedge}(\Xi_t)$. A history $h = (\mathsf{o}_1, \dots, \mathsf{o}_k) \in \mathcal{H}_k$ successively uses the output of each outcome as its next input. There are finitely many histories, including the empty history at $k = 0$. Write $\mathcal{S}(h) = (\Omega_j, \Lambda_j, S_j^{\mathrm{pr}})_{j \le k}$ for its label.

Evaluate the tests at $V_j = \mathcal{M}^j U$, and put

$$\mathbf{1}_h = \prod_{t \le k} \mathbf{1}_t(\mathsf{o}_t), \qquad \mathfrak{G}_h = \prod_{t \le k} \mathfrak{G}_{t-1}(V_{t-1}; P_t''), \qquad \mathfrak{w}_h = \mathbf{1}_h \mathfrak{G}_h. \tag{646}$$

For a bounded complex Borel function ς on $\mathcal{V}_0$, define

$$\tau_h[\varsigma] = (\mathcal{M}^k)_*(\mathfrak{w}_h \varsigma \, dU), \qquad \tau_h = \tau_h[e^{-\beta A_K}]. \tag{647}$$

Proposition 44.1. The tests and weights above are Borel. At each bare field exactly one history has $\mathbf{1}_h = 1$, and

$$0 \le \mathfrak{w}_h \le \mathfrak{G}_h \le \prod_{t \le k} \mathsf{z}(x_{t-1}, \varepsilon_{t-1})^{-(L^3-1)\#P_t''}.$$

Consequently every $\tau_h[\varsigma]$ is a finite complex measure, positive when $\varsigma \ge 0$. If $\varepsilon_j \le c_2^{\mathrm{SO}}$ for $0 \le j < k$ and ς is gauge invariant, then

$$\sum_{h \in \mathcal{H}_k} \tau_h[\varsigma] = (\mathcal{M}^k)_*(\varsigma \, dU). \tag{648}$$

In particular,

$$\rho_k \, dV_k = \sum_{h \in \mathcal{H}_k} \tau_h, \qquad \sum_{h \in \mathcal{H}_k} \tau_h(\mathcal{V}_k) = Z_{K,\ell,\beta}.$$

Here $\rho_k \, dV_k$ denotes the effective measure of lemma 4.7; no absolute-continuity assertion is implicit in this notation. ∎

44.1. **Domains of the tests and the collar exploration.** We first justify the conditional axial normalizations used above. Suppose an orbit for a reduced determining set is given, and let Y be a set of top sites. No site of $B(Y) \setminus Y$ is a node. If the contour means of $\mathcal{M}^k U$ exist, set

$$u(x) = (\mathcal{M}^k U)(y, x) \quad (x \in B(y) \setminus \{y\},\ y \in Y), \qquad u = \mathbf{1} \text{ at every other bare site.}$$

Disjointness of the blocks makes this consistent, and u fixes every node. Covariance of the contour mean gives $(\mathcal{M}^k U^u)(y, x) = \mathbf{1}$. Conversely, if $U' = U^v$ and both representatives are axial on Y, then $v(y) = \mathbf{1}$ at the top nodes and $\mathbf{1} = v(y)\mathbf{1}v(x)^{-1}$ on each block. Thus $v = \mathbf{1}$ on $B(Y)$ and at all other nodes. The averaged representatives agree on every defined bond whose endpoints lie in this union.

For the tower of $\square'^{\sim 4}$, the margin $M_2 \check{R}_t \ge L^2 M_1$ puts $\square'^{\sim 3}$ in the top member: each of its M_1-cubes is at distance greater than M_1 from the complement, and hence lies in the two-layer interior used to form the tower. Its top sites are nodes by lemma 8.4. Block membership then puts all required bare blocks inside $\square'^{\sim 4}$. The preceding argument defines $V_{k,\square'}^{\mathrm{bg}}$ uniquely. A bond with an endpoint in $\square'^{\sim 2}$ has its other endpoint in $\square'^{\sim 3}$, since its length is $1 < L M_2 \check{R}_t$. This also verifies the bond domain in (637).

For (641), a site within $2L$ of Ω_t lies in Λ_k when $t \ge 2$, and in $\Omega_0^{\diamond}$ when $t = 1$. Indeed, (638) separates Ω_t from Λ_k^c by at least $8W_t > 2L$, while the first collar has width $W_1 > 2L$. Consequently the bare blocks needed for the level-k averages on $\mathfrak{X}_0$ lie in $\Omega_0^{\diamond}$. Each site of $Y = \Omega_t^{(t)}$ has a neighbor inside the same coarse cube, so it is the endpoint of a top datum bond and is a node. The noncentral sites of $B(Y)$ are not nodes of the reduced set. Finally, an endpoint $x \notin \Omega_t$ of a bond of $\mathfrak{X}_0$ has a level-k neighbor x' in its own coarse cube: in the centered integer coordinates of that cube, change the first coordinate by one toward its interior. Both x, x' lie within 2 of Ω_t and outside it. The bond $\langle x, x' \rangle$ therefore belongs to the level-k member of the reduced datum, so x is a node. Thus every endpoint of $\mathfrak{X}_0$ lies in $B(Y)$ or is a node, and the same normalization argument proves uniqueness on all of $\mathfrak{X}_0$, including the crossing bonds. If Ω_t is empty, the assertion concerns the empty restriction.

The stable domain convention now gives the Borel properties without requiring backgrounds at arbitrary fields. The stable restriction of each background or of $\mathsf{m}_{j,\square}$ is continuous. For (642), multiplication and inversion are continuous, $G \setminus \{-\mathbf{1}\}$ is open, and the intrinsic logarithm is continuous there. The passing sets of the density and background tests, of χ_b, χ_b^0, $\mathrm{lo}(c)$ and of the cube test $\chi^{(k)}(\square')$ are therefore open in the finite product of field spaces. The indicator $\chi_b^1 = \chi_b - \chi_b^0$ is Borel as a difference of Borel functions. Complements and finite products of these indicators are Borel as well.

We will also need their covariance. Let v be a gauge transformation at level n and extend it to the bare lattice by making it constant on each order-n block; denote its restrictions by s_j. The covariance of the actual averages, including their domains, gives $\mathcal{M}^j(U^{\tilde{v}}) = (\mathcal{M}^j U)^{s_j}$. For a test of step $t \le n$, s_k is constant on every block centered at level t. Gauge transformation maps the defining critical orbit to that of the transformed datum: it preserves the constraints, criticality and the curvature and current bounds, and conjugates the residual groups. It preserves the axial condition on these blocks. Uniqueness of the restrictions just proved therefore makes the background transform as a configuration. The fluctuation logarithm transforms by $\mathrm{Ad}\, s_k(b_-)$, with its domain unchanged. The same applies to the tower backgrounds, and plaquette norms are unchanged. The gauge transformations are homeomorphisms, so they also preserve the stable domains. All outcome tests of step $t \le n$ are therefore invariant. The earlier gauge factors are invariant as well, since their contour means are conjugated by a common block value. The plaquette part of step $n+1$ is invariant: it depends only on V_{n+1} (and on V_0 through conjugation when $n = 0$).

We next check the only geometric point needed in the fluctuation split. For any stage of the exploration, $R' \subset \mathsf{R}_n$. If a site z is within distance strictly less than W_t of $\Lambda^{[n]}$, its coarse cube touches that region. This cube belongs to $\Omega_t^{\sim -1}$ and not to R_n: otherwise its touching cube would be in $(\Omega_t^c)^{\sim 2} \cup \mathsf{R}_n^{\sim}$. The test cube containing z is consequently outside R, and all incident bonds belong to its two-layer enlargement. In particular, they belong to $\mathfrak{U}(R)$. If b belongs to the collar, a nearest inner bond of $\Lambda^{[n]}$ has its whole segment in that region. Each endpoint of b is therefore within $\varpi_T + 1 < W_t$ of $\Lambda^{[n]}$. This proves

$$E^T \subset \mathfrak{U}(R). \tag{649}$$

The elementary grid facts used here remain valid on the torus: in centered cube coordinates, touching means that each index differs by at most one, and an inner bond stays in its containing coarse region. The bounds on M, M_2 ensure that a one-bond extension lies inside the indicated two test layers.

Proof of the partition and Borel assertions in proposition 44.1. Fix the fields after the choices of P_t, Q_t. Run (643) with R the failed cube set and answer

$$I^\natural = \{b : \chi_b^1 = 1\}, \qquad \mathsf{R}^\natural = \{c : \mathrm{lo}(c) = 0\}.$$

At each stage the recursion asks only for the answer on the newly explored sets. Replacing the answer by the union of its discovered parts therefore leaves every stage unchanged, by induction up to the stopping time. Its outcome is admissible. Every factor in (645) for this outcome is one. For the factors χ_b^0 this uses (649): the collar bonds lie in passed cube tests, hence $\chi_b = 1$, and absence from I gives $\chi_b^1 = 0$, so $\chi_b^0 = 1$.

Conversely, if an admissible triple has $\chi^\wedge = 1$, the factors on R force its cubes to fail. Every bond tested by a cube outside R has $\chi_b = 1$, either from its unexamined factor or from $\chi_b^0 = 1$ or $\chi_b^1 = 1$ on E^T. Thus R is precisely the failed cube set. The remaining factors give the actual answer on E^T and E^H. Induction through the recursion identifies this triple with the outcome just constructed. Hence $\sum_\Xi \chi^\wedge(\Xi) = 1$, with exactly one nonzero term. The elementary failed-set identities for P_0, P_t, Q_t then give one outcome, and induction over the steps gives one history.

All index sets are finite and fixed by the chosen outcome. The Borel assertions proved above therefore apply to each indicator and weight. Finally, each factor of (52) is between zero and z^{-1}

and each block has L^3-1 noncentral sites. This proves the stated bound. Finite pushforwards of the resulting bounded densities give the claimed measures. □

44.2. **Gauge normalization on the selected histories.** A passed density test controls the plaquettes of its block field. Precisely, if $\chi_j(\square)=1$ and $\varepsilon_j \le c_2^{\mathrm{SO}}$, then

$$|\partial V_j(p)-\mathbf{1}|<2\varepsilon_j \quad \text{when } p \text{ has a corner in } \square. \tag{650}$$

For $j=0$, the stronger bound ε_0 follows directly from the test: a one-bond extension of $\square$ lies in $\square^{\sim}$. For $j\ge 1$, all corners of such a plaquette lie in $\square^{\sim}$, and its bonds are top datum bonds of the tower of $\square^{\sim 4}$. Thus its block holonomy is that of $\mathcal{M}^j U$ for an element U of the tested orbit. All bare blocks over its four corners lie in $\square^{\sim}$.

Here the curvature assumption is local, so we spell out its use in the averaging argument. If Z is this union of four bare blocks, call a level-n bond or plaquette covered when the bare blocks over its vertices lie in Z. Extend U by $\mathbf{1}$ outside its original bond domain. The actual averages on covered bonds are unchanged, by locality, and are restrictions of a real configuration at every level. For a covered level-$(n+1)$ plaquette, every level-n plaquette in its four blocks is covered. Thus lemma 14.9 applies at each stage and gives, for the maximal covered curvature,

$$\gamma_{n+1} \le L^2\gamma_n + C_0^{\mathrm{SO}}\gamma_n^2.$$

The small-curvature branch agrees with the actual average there; reversed bonds carry the inverse value. If $\gamma_0 \le \alpha L^{-2j}$ with $0<\alpha\le c_2^{\mathrm{SO}}$, the normalized induction of section 14.5 gives

$$\gamma_n \le \alpha L^{2(n-j)} \prod_{l<n}(1+\tfrac{1}{2}L^{2(l-j)}) < 2\alpha L^{2(n-j)} \qquad (0\le n\le j).$$

Its hypothesis is preserved because $2c_2^{\mathrm{SO}} \le \varpi'_L < \omega_M$ and $C_0^{\mathrm{SO}} c_2^{\mathrm{SO}} \le L^2/4$. Choose $\alpha<\varepsilon_j$ above the finite maximum $L^{2j}\mathsf{m}_{j,\square}(V_j)$. This proves (650) without a global curvature assumption on U.

On the support of a plaquette part of step j, every test cube inside $\Omega_j^{\sim 4}$ passes. Indeed, (638) and (639) put this enlargement in the region where the P-tests were selected, and outside their failed hull. Hence (650) holds at every plaquette with a corner in $\Omega_j^{\sim 4}$. At the first level the bare tests likewise give the bound ε_0 at every plaquette with a corner in P_0^c.

Fix a history h' through step $t-1$ and a plaquette part of step t. Let $y\in P_t''$ and consider the double block of the level-t bond $\langle y, y+Le_1\rangle$ in level-$(t-1)$ units. Its sites lie within $L+(L-1)/2<2L<W_t$ of y. For $t\ge 2$, (638) puts y at distance at least $5W_t$ from the hull of Λ_{t-1}^c, so the double block lies in $\Lambda_{t-1}\subset\Omega_{t-1}^{\sim 4}$. For $t=1$, (639) instead puts it in P_0^c. On the support of $\mathbf{1}_{h'}\mathbf{1}_t^{\mathrm{P}}$, its plaquette curvatures are therefore less than $2\varepsilon_{t-1}\le\omega_M$. The small-curvature contour estimate lemma 4.4 shows that every $V_{t-1}(y,x)$ used by $\mathfrak{G}_{t-1}(\,\cdot\,;P_t'')$ exists. This includes $P_t''=\emptyset$, when the gauge factor is one.

Proof of (648). For fixed h' and plaquette part p, set $P''=P_t''(\mathsf{p})$ and $e_{\mathsf{p}}=\mathfrak{w}_{h'}\mathbf{1}_t^{\mathrm{P}}(\mathsf{p})$, with all fields evaluated at their actual bare iterates. Summing the Q and fluctuation outcomes gives

$$\sum_{\mathsf{o}_t:\,\mathsf{p}(\mathsf{o}_t)=\mathsf{p}} \mathfrak{w}_{(h',\mathsf{o}_t)} = e_{\mathsf{p}}\mathfrak{G}_{t-1}(V_{t-1};P'').$$

Let v vary independently on $B(P'')\setminus P''$ at level $t-1$ and be the identity elsewhere. Extend it constantly on the order-$(t-1)$ bare blocks. Its restriction to level t is the identity. Consequently V_t is unchanged, V_{t-1} transforms by v, and the covariance proved above leaves e_{p} unchanged. The function ς is unchanged by its assumed gauge invariance.

For every bounded Borel function F on $\mathcal{V}_t$, Haar invariance on the bare bonds and Fubini's theorem now give

$$\begin{aligned}&\int F(V_t)e_{\mathsf{p}}\,\mathfrak{G}_{t-1}(V_{t-1};P'')_{\varsigma}\,dU\\&\quad=\int F(V_t)e_{\mathsf{p}\varsigma}\left[\int \mathfrak{G}_{t-1}(V_{t-1}^{v};P'')\,dv\right]dU=\int F(V_t)e_{\mathsf{p}\varsigma}\,dU.\end{aligned}\tag{651}$$

The integrands are bounded Borel functions on finite products. Where $e_{\mathsf{p}}\neq 0$, the domain check just proved permits (55), which makes the bracket one; elsewhere the outside factor vanishes. Summing (651) over the plaquette parts and using their partition of one yields

$$\sum_{h:\,h|_{t-1}=h'}\tau_h[\varsigma]=\mathbf{T}\tau_{h'}[\varsigma].$$

Starting with $\tau_{\emptyset}[\varsigma]=\varsigma\,dU$ proves (648) by induction. Apply it to $\varsigma=e^{-\beta A_K}$ and then take total masses, using lemma 4.7, to obtain the remaining assertions. □

44.3. **Regular backgrounds and the cost of a failed test.** We use the cubes and tests of (636) and (637). The point of the comparison below is that failure of a test costs either plaquette action or gauge-fixing energy, even when the background in the definition of the test is undefined. All estimates are local to the four-layer enlargement of the test cube.

Choose a power $\rho_{\mathrm{s}}\geq\rho_{\mathrm{G}}$ of L and impose the following additional restrictions on the constants of the history construction:

$$\begin{aligned}\mathfrak{M}\geq\mathfrak{M}_{\mathrm{G}},\qquad R_1\mathfrak{M}\geq L^2\rho_{\mathrm{s}},\qquad B_3\geq B_{\mathrm{rad}},\\ \varepsilon_{\mathrm{v}}+\varepsilon_{\mathrm{v}}^2\leq a_{\mathrm{orb}},\qquad 0<\bar\varepsilon\leq\varepsilon_{\mathrm{G}}(\rho_{\mathrm{s}}),\qquad 0<r_B\leq r_{\mathrm{G}}(\rho_{\mathrm{s}}),\\ B_3(\bar\varepsilon+C_{\partial}r_B)\leq\varepsilon_{\mathrm{v}}.\end{aligned}\tag{652}$$

Here ε_{v} is the radius in the definition of the regular critical orbit. These choices are possible: choose the geometric parameters and B_3 first, then ε_{v}, and finally the two datum radii. They are independent of the lattice and the number of steps.

Proposition 44.2 (Regularity of localized backgrounds)**.** Let $1\leq t\leq K$, $k=t-1$, and let $\square'$ be a test cube. Put

$$X=\square'^{\sim 4},\qquad X_3=\square'^{\sim 3},\qquad Y=X_3^{(t)}.$$

Suppose (652) and the standing history conditions hold. If $0<\varepsilon_{\mathrm{r}}\leq\bar\varepsilon$ and

$$|\partial V_t(p')-\mathbf{1}|<\varepsilon_{\mathrm{r}}\quad\text{at every inner level-}t\text{ plaquette of }X,\tag{653}$$

then the orbit $U_{t,\square'}(V_t)$ exists. Every member is regular at radius $B_3\varepsilon_{\mathrm{r}}$, and

$$\mathsf{m}_{t,\square'}(V_t)<B_3\varepsilon_{\mathrm{r}}\eta_t^2.\tag{654}$$

The function on the left is stably defined at every such V_t.

If also $B_3\varepsilon_{\mathrm{r}}\leq c_2^{\mathrm{SO}}$, then $V_{k,\square'}^{\mathrm{bg}}$ is stably defined there, is in block axial gauge on Y, and satisfies

$$|\partial V_{k,\square'}^{\mathrm{bg}}(p)-\mathbf{1}|\leq 2B_3\varepsilon_{\mathrm{r}}L^{-2}\quad\text{for all level-}k\text{ plaquettes with corners in }X_3.\tag{655}$$

For every inner level-t bond c of X_3, its average uses the first branch and

$$\mathcal{M}(V_{k,\square'}^{\mathrm{bg}})(c)=V_t(c).$$

In particular, the P-test passes if $B_3\varepsilon_{\mathrm{r}}\leq\varepsilon_t$; under the additional smallness condition, the Q-test is the strict bond inequality in (637). ■

Proof. First consider induced data on any separated tower of depth t whose initial region is contained in X. The lift in proposition 6.3 has value $\mathbf{1}$ on bonds within an order-$(t-j)$ block and value $V_t(c)$ on bonds crossing over c at level j. On a level-j plaquette, crossing a block boundary in one coordinate is unaffected by a step in the other coordinate. Thus either the four block centers are distinct and the product is a level-t plaquette variable with the same cyclic order and base point, or the factors cancel and the product is $\mathbf{1}$. All these coarse plaquettes lie in X. At a tested plaquette, an edge outside the determining member uses the average of the preceding member; by proposition 6.3 this is the same lifted value. Hence every tested holonomy of the induced datum is $\mathbf{1}$ or an inner plaquette variable of V_t in X. The datum is regular at radius ε_{r}. Its coordinates are copies of $\mathbf{1}$ or of coordinates of V_t, so the map to induced data is continuous.

The tower of X is separated. To check the remaining geometric conditions, measure lengths at level t and put $w = M_2\check{R}_t$. Then $w \geq L^2 M_1$ and $M\check{R}_t \leq L^{m+K-t}$, while $M \geq L^4 M_1$. In particular the torus has at least $6R_1\mathfrak{M}$ sites in each direction. The gap between X_3 and X^c is at least w; proposition 7.6 therefore gives $X_3 \subset \Omega_t$. This remains true if an enlargement is the whole torus. Block nesting gives

$$X_3^{(j)} = B^{t-j}(Y), \qquad 0 \leq j \leq t.$$

Every $y \in Y$ has a nearest neighbor in the same w-cube, since $w \geq L$. Thus it is an endpoint of an inner level-t bond of X_3, which belongs to the top determining member. These sites are nodes, and the sites of $B(Y) \setminus Y$ are not nodes by lemma 7.5.

Apply theorem E to the induced reduced datum, at $\rho = \rho_{\mathrm{s}}$. There are no pinch rows in the reduced setting. The radius comparison following its proof identifies its orbit with the full regular critical set at radius ε_{v}: the constructed orbit lies in $\mathfrak{U}_t'(B_{\mathrm{rad}}\varepsilon_{\mathrm{r}}) \subset \mathfrak{U}_t(\varepsilon_{\mathrm{v}})$, while

$$\mathfrak{U}_t(\varepsilon_{\mathrm{v}}) \subset \mathfrak{U}_t'(\varepsilon_{\mathrm{v}} + \varepsilon_{\mathrm{v}}^2) \subset \mathfrak{U}_t'(a_{\mathrm{orb}}).$$

All members of this orbit have the stated linear regularity bound. For bare plaquettes with corners in X_3, the local block side in level-t units is one. In particular their curvature is less than $B_3\varepsilon_{\mathrm{r}}L^{-2t}$, proving (654).

The existence assertion also supplies local continuous representatives. Indeed, near any induced datum V_0, write $V' = e^{iB(V')}V_0$ on positive rows, where $B(V') = i^{-1}\log(V'V_0^{-1})$ is the logarithm near $\mathbf{1}$. After restricting the neighborhood, $\sup|B(V')| < r_B$. The map $V' \mapsto U^\circ(B(V'))$ is continuous by theorem E, and represents the regular orbit by (652). Composing with the continuous map to induced data gives local selections over V_t. The coarse condition (653) is open. A finite maximum of gauge-invariant plaquette norms evaluated on these selections is continuous and independent of the selection. This proves stable definedness of $\mathsf{m}_{t,\square'}$.

For the asserted level-k bound, put $\alpha = B_3\varepsilon_{\mathrm{r}}L^{-2} \leq c_2^{\mathrm{SO}}$. On $B^t(Y)$ the bare curvature is less than αL^{-2k}. The iteration proof of proposition 14.1 applies on these nested block unions: every double block and every four-block plaquette cone at the next level is contained in the preceding union. More explicitly, if ϖ_j is the maximum curvature there, lemma 14.9 gives

$$\varpi_{j+1} \leq L^2\varpi_j + C_0^{\mathrm{SO}}\varpi_j^2, \qquad \varpi_j \leq \alpha L^{2(j-k)} \prod_{a<j}(1 + \tfrac{1}{2}L^{2(a-k)}) < 2\alpha L^{2(j-k)}.$$

The product is at most $\exp(\frac{1}{2}\sum_{a=0}^{k-1} L^{2(a-k)}) < 2$. The definition of c_2^{SO} closes the induction and keeps each participating average in the first branch. Consequently $W = \mathcal{M}^k U$ has curvature at most 2α on $B(Y)$. For $k = 0$, this statement follows directly from the bare bound. Each $y \in Y$ is an endpoint of a double block in $B(Y)$, so lemma 14.6 defines every contour mean $W(y,x)$ used below.

Normalize by the residual gauge transformation equal to $W(y,x)$ at $x \in B(y) \setminus \{y\}$, $y \in Y$, and equal to $\mathbf{1}$ elsewhere. On inner bonds of X_3 this gives

$$V_{k,\square'}^{\mathrm{bg}}(b) = W([b_-]_t, b_-)\, W(b)\, W([b_+]_t, b_+)^{-1}. \tag{656}$$

The conditional uniqueness of the normalized background identifies this with the field in (637). Gauge covariance preserves its curvature and its coarse average; the top constraint sets the latter equal to V_t. On each local continuous selection, all iterated averages and contour means used in (656) belong to their analytic small-curvature domains. They are therefore continuous. Formula (656) and independence of the selection prove stable definedness. The remaining assertions follow from the definitions of the tests. □

The comparison with a field satisfying the same coarse constraint uses the following reconstruction estimate. It is useful to retain the contour error explicitly.

Lemma 44.3 (Reconstruction on a double block)**.** Set $\kappa_{\mathsf{c}} = 4\kappa_1^{\mathrm{SO}} + \kappa_2^{\mathrm{SO}} + \kappa_3$. Let c be a positive level-$(i+1)$ bond, $0 \le i < K$, and let W, W_0 be real configurations on P_c with $\max\{\varpi_c(W), \varpi_c(W_0)\} \le \varpi \le \varpi_L$. If

$$|W(y,x)W_0(y,x)^{-1} - \mathbf{1}| \le \zeta \quad (y = c_-, c_+,\ x \in B(y)),$$
$$|\mathcal{M}(W)(c)\mathcal{M}(W_0)(c)^{-1} - \mathbf{1}| \le a,$$

then

$$|W(b)W_0(b)^{-1} - \mathbf{1}| \le a + 2\kappa_{\mathsf{c}}\varpi + 2\zeta, \qquad b \in \mathsf{P}_c. \tag{657}$$

■

Proof. Use level-i lengths. The smallness of ϖ_L permits lemma 14.6. For an inner bond b of $B(y)$, orient b away from y in its coordinate direction. Choose a rectilinear path to b_+ which traverses that direction last; its last step is b. Apply the contour estimate to that path and to the path with its last step removed. Unitary multiplication preserves the norm, so

$$|W(b) - W(y,b_-)^{-1}W(y,b_+)| \le 2\kappa_1^{\mathrm{SO}}\varpi.$$

For a corridor bond $b = \langle x, x+e_\mu\rangle$, write $x = c_- + u + \frac{1}{2}(L-1)e_\mu$, with $u \perp e_\mu$, and put $x_\perp = c_- + u$. The path from c_- to x that makes the μ steps last, followed by b and the reversed corresponding path from c_+, is the transverse path to $x_\perp$, the straight length-L segment, and the reversed translated transverse path. Replacing the endpoint paths by their contour means costs $4\kappa_1^{\mathrm{SO}}\varpi$. The relative-loop estimate and the average-versus-straight-transport estimate in lemma 14.6 then give

$$|W(c_-,b_-)W(b)W(c_+,b_+)^{-1} - \mathcal{M}(W)(c)| \le \kappa_{\mathsf{c}}\varpi.$$

Apply these two estimates also to W_0. Subtracting the reconstructed products costs at most 2ζ inside a block and at most $a + 2\zeta$ in the corridor. This proves respectively $4\kappa_1^{\mathrm{SO}}\varpi + 2\zeta$ and the bound (657). Every bond or its reverse is of one of these types, and the latter bound dominates the former. □

Proposition 44.4 (Cost of a failed plaquette test)**.** Assume the setting of proposition 44.2, and suppose $\varepsilon_t \le B_3\bar{\varepsilon}$. If $\chi_t^{\mathrm{P}}(\square') = 0$, then some inner level-$t$ plaquette of $X = \square'^{\sim 4}$ satisfies $|\partial V_t - \mathbf{1}| \ge \varepsilon_t/B_3$. For every bare configuration U' on the inner bonds of X with $\mathcal{M}^t U' = V_t$ on its inner level-t bonds,

$$L^t A_X(U') \ge \min\{c_1^{\mathrm{P}}\varepsilon_t^2, e_*\}, \qquad c_1^{\mathrm{P}} = \tfrac{1}{4}B_3^{-2}. \tag{658}$$

At level zero a failed density test gives $A_{\square^\sim}(V_0) \ge \frac{1}{2}\varepsilon_0^2$. ■

Proof. If all coarse curvatures were less than ε_t/B_3, proposition 44.2 would make the test pass. Thus the stated plaquette exists. Extend U' by $\mathbf{1}$ on other bare bonds. The full bare cone under this plaquette lies in X, since X is a union of level-t cubes. Locality identifies its iterated average with that of U'. Apply proposition 15.1 to this one plaquette; its cone action is at most $A_X(U')$. At level zero use $a_p(V_0) = \frac{1}{2}|\partial V_0(p) - \mathbf{1}|^2$. □

For the second test define

$$\varepsilon_{\mathrm{bg}} = \min\{\bar{\varepsilon}, c_2^{\mathrm{SO}}/B_3\}, \qquad \widetilde{\sigma}_{\mathrm{Q}} = \frac{4}{5L^2}\varepsilon_{\mathrm{bg}}, \qquad \widetilde{c}_2^{\mathrm{Q}} = \frac{2}{5\kappa_{\mathrm{c}}B_3 + 2},$$
$$c_2^{\mathrm{Q}} = \tfrac{1}{4}(\widetilde{c}_2^{\mathrm{Q}})^2, \qquad \sigma_{\mathrm{Q}} = \min\{\tfrac{1}{4}\widetilde{\sigma}_{\mathrm{Q}}^2, e_*\}. \tag{659}$$

These constants depend only on $L, B_3, \bar{\varepsilon}$. Extend the gauge-fixing function by

$$G_k^{\infty}(Y; V_k) = \begin{cases} \dfrac{1}{2}\displaystyle\sum_{y\in Y}\sum_{x\in B(y)\setminus\{y\}} |V_k(y,x) - \mathbf{1}|^2, & \text{if all these contour means are defined,} \\ +\infty, & \text{otherwise.} \end{cases} \tag{660}$$

Proposition 44.5 (Cost of a failed fluctuation test)**.** Assume the setting of proposition 44.2, and impose the local tie

$$V_t(c) = \mathcal{M}(V_k)(c) \quad \text{on every inner level-}t\text{ bond of } X. \tag{661}$$

If $\chi_t^{\mathrm{Q}}(\square') = 0$, set $a_{\mathrm{Q}} = \min\{\widetilde{c}_2^{\mathrm{Q}}\delta_k, \widetilde{\sigma}_{\mathrm{Q}}\}$. Then either a contour mean $V_k(y,x)$, $y \in Y$, $x \in B(y)$, is undefined or has distance at least a_{Q} from $\mathbf{1}$, or an inner level-k plaquette of X has curvature at least a_{Q}. For every bare configuration U' on the inner bonds of X such that $\mathcal{M}^k U' = V_k$ on its inner level-k bonds,

$$L^k A_X(U') + G_k^{\infty}(Y; V_k) \geq \min\{c_2^{\mathrm{Q}}\delta_k^2, \sigma_{\mathrm{Q}}\}. \tag{662}$$

This includes $k = 0$, with $\mathcal{M}^0$ the identity. ∎

Proof. We first establish the comparison that makes the contrapositive quantitative. Suppose all inner level-k plaquettes of X have curvature at most ϖ_0, where $0 < \varpi_0 \leq \widetilde{\sigma}_{\mathrm{Q}}$, and suppose $|V_k(y,x) - \mathbf{1}| \leq \zeta$ on the indicated blocks. The four-block cone under every inner coarse plaquette lies in X. Since $C_0^{\mathrm{SO}}\widetilde{\sigma}_{\mathrm{Q}} \leq 1/5$, lemma 14.9 and (661) give

$$|\partial V_t(p') - \mathbf{1}| \leq (L^2 + \tfrac{1}{5})\varpi_0 < \tfrac{5}{4}L^2\varpi_0 =: \varepsilon_{\mathrm{r}}.$$

Here $\varepsilon_{\mathrm{r}} \leq \varepsilon_{\mathrm{bg}}$. By proposition 44.2 the background is stably defined, has the same average as V_k on inner coarse bonds of X_3, is in block axial gauge on Y, and has curvature at most $\frac{5}{2}B_3\varpi_0$ there. Moreover

$$\tfrac{5}{2}B_3\varpi_0 \leq 2c_2^{\mathrm{SO}}L^{-2} < \varpi_L.$$

Thus lemma 44.3 applies to V_k and this background on each double block of an inner coarse bond of X_3, with $a = 0$. Their contour defect is at most ζ, so

$$|V_k(b)V_{k,\square'}^{\mathrm{bg}}(b)^{-1} - \mathbf{1}| \leq 5\kappa_{\mathrm{c}}B_3\varpi_0 + 2\zeta \tag{663}$$

for every inner level-k bond of X_3. Indeed, a bond whose endpoints are in one block is covered by any inner coarse bond incident to its center; such a bond exists within the same w-cube. Otherwise its two block centers are nearest neighbors in Y, and their double block covers it. In particular (663) covers all bonds tested in $\chi_t^{\mathrm{Q}}(\square')$.

If neither alternative in the statement holds, all the finite maxima of curvature and contour error are strictly below a_{Q}. Choose $0 < \varpi_0 < a_{\mathrm{Q}}$ above the curvature maximum and let $\zeta < a_{\mathrm{Q}}$ be the contour maximum. Then

$$5\kappa_{\mathrm{c}}B_3\varpi_0 + 2\zeta < (5\kappa_{\mathrm{c}}B_3 + 2)a_{\mathrm{Q}} \leq 2\delta_k.$$

Stable definedness and (663) make the test pass, a contradiction. This proves the alternative.

An undefined contour gives infinite gauge-fixing energy; a contour of size at least a_{Q} gives energy at least $\frac{1}{2}a_{\mathrm{Q}}^2$. In the plaquette alternative, extend U' as in the preceding proof and use proposition 15.1 at level $k \geq 1$, obtaining action at least $\min\{\frac{1}{4}a_{\mathrm{Q}}^2, e_*\}$ after multiplication by L^k. For $k = 0$ the individual plaquette action is at least $\frac{1}{2}a_{\mathrm{Q}}^2$, which implies the same lower bound. Both energies are nonnegative. Finally,

$$\min\{\tfrac{1}{4}a_{\mathrm{Q}}^2, e_*\} = \min\{c_2^{\mathrm{Q}}\delta_k^2, \sigma_{\mathrm{Q}}\},$$

proving (662). □

Under the ceilings $c_1^{\mathrm{P}}\varepsilon_t^2 \le e_*$ and $c_2^{\mathrm{Q}}\delta_k^2 \le \sigma_{\mathrm{Q}}$, the minima in (658) and (662) take their first values. If $|b_t| \le B_\sharp$, the identities $L^{-1}x_{t-1} = x_t + b_t$ and $x_j\varepsilon_j^2 = A_0^2 y_j^{2\mathfrak{p}_0}$ give

$$L^{t-1}x_{t-1}A_X(U') \ge c_1^{\mathrm{P}}(1 - B_\sharp g_t^2)A_0^2 y_t^{2\mathfrak{p}_0}, \tag{664}$$

$$x_k(L^k A_X(U') + G_k^{\infty}(Y; V_k)) \ge c_2^{\mathrm{Q}} A_1^2 y_k^{2\mathfrak{p}_0}. \tag{665}$$

Each inequality retains the corresponding failure and averaging hypotheses of the preceding propositions. These costs are supported in the one-layer enlargement of the coarse cell $\mathfrak{c}''$ containing $\square'$. Indeed, in level-t units, every point of X is within $4M_2\check{R}_t$ of $\square' \subset \mathfrak{c}''$, and $4M_2\check{R}_t \le M\check{R}_t$.

44.4. **Profiles and the action reserved for large fields.** The local changes of coupling are supported inside the regular regions. Their smooth profiles have disjoint transition regions; this preserves positivity of the resulting position-dependent coupling. We also identify disjoint sets of plaquettes whose action will pay for the failed tests.

Fix a nonnegative smooth function ψ on $\mathbb{R}^3$, invariant under coordinate permutations and sign changes, with integral one and support in $[-1/4, 1/4]^3$. Put

$$C_{\mathrm{mo}} = \sum_{\mu=1}^{3} \int_{\mathbb{R}^3} |\partial_\mu \psi(u)|\, du.$$

For a set $S \subset T_\ell$ and a physical length $s > 0$, define

$$\varphi[S; s](y) = \int_{\mathbb{R}^3} \psi(u)\mathbf{1}\{\mathrm{dist}_\infty(y - su, S) \ge s/2\}\, du, \tag{666}$$

where the distance from the empty set is $+\infty$.

Lemma 44.6. The profile is smooth, takes values in $[0, 1]$, vanishes where $\mathrm{dist}_\infty(y, S) \le s/4$, and equals one where $\mathrm{dist}_\infty(y, S) \ge 3s/4$. Moreover,

$$\partial^\alpha \varphi[S; s](y) = s^{-|\alpha|} \int (\partial^\alpha \psi)(u)\mathbf{1}\{\mathrm{dist}_\infty(y - su, S) \ge s/2\}\, du. \tag{667}$$

Thus the profile has Lipschitz constant C_{mo}/s for the sup metric. If the sets S_i have mutual distance at least s, then

$$1 - \varphi[\bigcup_i S_i; s] = \sum_i (1 - \varphi[S_i; s]).$$

■

Proof. Lift the distance function to the periodic covering space. The set where it is at least $s/2$ is closed. After the change of variables $w = \widetilde{y} - su$, the integral is convolution with $s^{-3}\psi((\widetilde{y} - w)/s)$. Its derivatives are integrable and have compact support, so differentiation under the integral gives (667). The derivative bounds and the Lipschitz estimate follow by integrating along a shortest lifted segment. On the set where $\psi \ne 0$, one has $|u|_\infty < 1/4$. The Lipschitz property of distance therefore gives both asserted plateaux, including their endpoints. Finally, the open $s/2$-neighbourhoods of the S_i are disjoint. Their indicators add, which proves the last identity. □

For a history h of level k, write

$$s_j = M\check{R}_j a_{K-j}, \qquad \varphi_j = \varphi[Z_j; s_j], \qquad \varphi_{j,D} = \varphi[D; s_j].$$

The last notation is used for a component D of Z_j. The cells have side s_j, and $a_{K-j} \le s_j/4$. Hence $\varphi_j = 1$ on $\Lambda_j^{\sim -1}$, $\mathrm{supp}\,\varphi_j \subset \Lambda_j$, and

$$1 - \varphi_j = \sum_{D \subset Z_j} (1 - \varphi_{j,D}). \tag{668}$$

Indeed distinct components have distance at least s_j. The tents of (760) also satisfy

$$\varphi_j h_z^{(j)} = h_z^{(j)} \qquad (z \in \Lambda_j^{\sim -1} \cap \mathbb{T}^{(j)}). \tag{669}$$

To see this, the cell of z has distance at least s_j from Z_j, whereas a nonzero tent is supported within $a_{K-j} \le s_j/4$ of z.

The position-dependent bare coupling and its level-n version are

$$\beta_n(y) = \beta - \sum_{j=1}^{n} L^j b_j \varphi_j(y), \qquad x_n(y) = L^{-n}\beta_n(y). \tag{670}$$

Here $x_n(\cdot)$ is a function, whereas x_n without an argument is the running coupling already fixed. In this subsection assume comparable cell sizes,

$$\check{R}_j \le L\check{R}_{j+1} \quad (1 \le j < k), \qquad \text{equivalently } s_1 \le \cdots \le s_k. \tag{671}$$

Use the endpoint conventions $\Omega_0 = \Lambda_0 = T_\ell$, $Z_0 = \emptyset$, $\Omega_{k+1} = \Lambda_{k+1} = \emptyset$ and $Z_{k+1} = T_\ell$.

Lemma 44.7. At a point where the profiles have the pattern $1, \dots, 1, \varphi_n, 0, \dots, 0$,

$$\beta_k = (1 - \varphi_n)L^{n-1}x_{n-1} + \varphi_n L^n x_n. \tag{672}$$

Every point has this pattern, or all profiles vanish. Consequently

$$0 < \min_{0 \le n \le k} L^n x_n \le \beta_k(y) \le \max_{0 \le n \le k} L^n x_n.$$

Formula (672) holds throughout the layer $\mathcal{L}_n = \Omega_n \setminus \Omega_{n+1}$ for $1 \le n \le k$; on $\mathcal{L}_0$, $\beta_k = \beta$. For $y \in Z_n$, $\beta_k(y) = \beta_{n-1}(y)$. On $\Lambda_n^{\sim -1} \cap Z_{n+1}$, $\beta_k = L^n x_n$. For every complex bare field $\mathbf{U}$,

$$A(\beta_k(\cdot), \mathbf{U}) + \sum_{j=1}^{k} L^j b_j A(\varphi_j, \mathbf{U}) = \beta A(\mathbf{U}). \tag{673}$$

■

Proof. History separation gives $\operatorname{dist}_\infty(Z_i, \Omega_{i+1}) \ge 8 s_{i+1}$. By (671), $\varphi_i = 1$ on Ω_{i+1}, hence on every later Λ_j and every later profile's support. Thus at most one profile can lie strictly between zero and one. Since $Z_i \subset Z_j$ for $i < j$, all profiles of index at least n vanish on Z_n. Now

$$L^j b_j = L^{j-1}x_{j-1} - L^j x_j, \qquad \beta - \sum_{j=1}^{n} L^j b_j = L^n x_n.$$

Substitution gives the convex combination and positivity. On $\mathcal{L}_n$, earlier profiles equal one and later ones vanish because $\Omega_{n+1}^c \subset Z_{n+1}$. The remaining constant values follow from the plateaux. Finally the weighted action is linear in its weight; applying (670) at each plaquette centre proves (673). □

We next reserve action for each component created by the tests. A *piece* is a pair (t, D), where $1 \le t \le k$ and D is a component of Z_t. Write $\mathrm{Pc}(h)$ for the set of pieces. The inner hull $T^\natural$ at level t is the union of the level-t cells containing a bare cube of T. Hulls and enlargements in the following definition are taken at level t.

Definition 44.8. Set

$$F_t = \begin{cases} P_0^\natural \cup P_1^\natural \cup Q_1^\natural, & t = 1, \\ P_t^\natural \cup Q_t^\natural, & t \ge 2, \end{cases} \qquad N_{t,D} = (F_t \cap D)^\sim.$$

The share of (t, D) is the set $\mathcal{P}(N_{t,D})$ of bare plaquettes whose four corners lie in $N_{t,D}$. For a set of pieces $\mathfrak{u}$, define

$$A_{\mathrm{cr}}(\mathfrak{u}; U) = \sum_{(t,D) \in \mathfrak{u}} L^{t-1} x_{t-1} A_{N_{t,D}}(U), \qquad A_{\mathrm{cr}}(U) = A_{\mathrm{cr}}(\mathrm{Pc}(h); U). \tag{674}$$

■

Proposition 44.9. For each piece, $N_{t,D} \subset D \cap \Omega_t^c$. It contains the closed s_t-neighbourhood of every failed test cube contained in D, including a density cube of P_0 when $t = 1$. If $t \geq 2$, then

$$\operatorname{dist}_\infty(N_{t,D}, \widehat{Z_{t-1}}) \geq 3s_t, \qquad N_{t,D} \subset \Lambda_{t-1}.$$

The regions $N_{t,D}$ are pairwise disjoint. At every point of $N_{t,D}$, the profiles of index less than t equal one and those of index at least t vanish. In particular each plaquette in its share has weight $L^{t-1}x_{t-1}$. Thus for every real bare field U,

$$A(\beta_k(\cdot), U) \geq A_{\mathrm{cr}}(U) \geq A_{\mathrm{cr}}(\mathfrak{u}; U) \geq 0. \tag{675}$$

■

Proof. For any region T,

$$T \subset T^\natural \subset \widehat{T} \subset (T^\natural)^\sim.$$

The first inclusion follows by taking the cell containing a bare cube of T; for the last, a cell meeting T touches one of these cells. For a union of test cubes, the cells of the inner hull are precisely those containing a test cube, since that partition refines the cell partition. The formulas (638) and (639) therefore give $F_t^\sim \subset \Omega_t^c \subset Z_t$. Every cell of $N_{t,D}$ touches a cell of D, so belongs to the same component of Z_t. This proves $N_{t,D} \subset D \cap \Omega_t^c$. For a failed cube $\square \subset D$, its inner hull is contained in $F_t \cap D$. Its one-cell enlargement contains the closed s_t-neighbourhood of $\square$, proving the neighbourhood assertion also for P_0.

For $t \geq 2$, the tests select P_t outside $\widehat{Z_{t-1}}^{\sim 4}$. They select Q_t in $B(P_t'')^{\sim -1}$, where

$$B(P_t'')^c = \widehat{P}_t^{\sim 1} \cup \widehat{Z_{t-1}}^{\sim 5}.$$

Consequently F_t has no cell in $\widehat{Z_{t-1}}^{\sim 4}$. One enlargement leaves a distance of at least $3s_t$ from $\widehat{Z_{t-1}}$, and hence from Z_{t-1}. This proves the asserted distance and inclusion in Λ_{t-1}.

At a fixed level disjointness follows from the distinct components. If $t < t'$, then

$$N_{t,D} \subset Z_t \subset Z_{t'-1} \subset \widehat{Z_{t'-1}},$$

whereas $N_{t',D'}$ has positive distance from that hull. This proves disjointness across levels. At a point of $N_{t,D} \subset Z_t$, all later profiles vanish. For $j < t$, the distance from Z_j is at least $3s_t \geq 3s_j$, so $\varphi_j = 1$. The telescoping identity in lemma 44.7 gives the claimed weight.

The centre of a bare plaquette with a corner in a region of cells lies in that region: it belongs to the bare cube centred at that corner, which is contained in its cell. Thus the weight statement applies at every centre in a share. The shares are disjoint sets of plaquettes and $a_p(U) \geq 0$ for a real field. Split the weighted action into its sums over shares and over the remaining plaquettes. The former is $A_{\mathrm{cr}}(U)$; the latter is nonnegative by lemma 44.7. Removing any shares decreases that sum, which proves (675). □

45. Convergent local cluster sums

A small interaction can be exponentiated after its connected contributions have been summed absolutely. We first prove the finite polymer criterion and then check it for domains of cubes. The criterion is the Kotecký–Preiss condition, in the form recorded by Fernández and Procacci [FP07, §2, (2.15)]. The proof below retains a finite-order bound throughout; in particular, the identity for the logarithm follows only after convergence has been established.

45.1. **The pinned cluster estimate.** Let $\mathcal{P}$ be a finite set with a symmetric, reflexive incompatibility relation $\nsim$. An activity is a function $h : \mathcal{P} \to \mathbb{C}$. Its partition function is

$$\Xi(h) = \sum_{\substack{\mathcal{Q} \subset \mathcal{P} \\ \mathcal{Q} \text{ pairwise compatible}}} \prod_{Z \in \mathcal{Q}} h(Z).$$

The empty subset contributes 1. For $n \geq 1$, let $\mathcal{C}_n$ be the connected graphs on $\{1, \dots, n\}$, and set

$$\rho^T(Z_1, \dots, Z_n) = \sum_{G \in \mathcal{C}_n} \prod_{\{i,j\} \in G} (-\mathbf{1}_{Z_i \nsim Z_j}), \qquad \rho^T(Z) = 1. \tag{676}$$

Repetitions among the Z_i are allowed. The coefficient vanishes unless the incompatibility graph is connected; such a family is called a cluster. Define

$$S = \sum_{n \geq 1} \frac{1}{n!} \sum_{(Z_1, \dots, Z_n) \in \mathcal{P}^n} \rho^T(Z_1, \dots, Z_n) \prod_{i=1}^{n} h(Z_i).$$

Theorem P (The finite polymer criterion)**.** Suppose $a : \mathcal{P} \to [0, \infty)$ satisfies

$$\sum_{Y \nsim Z} |h(Y)| e^{a(Y)} \leq a(Z) \qquad (Z \in \mathcal{P}). \tag{677}$$

Then S converges absolutely, $\Xi(h) = e^S \neq 0$, and

$$\Theta(Z) := \sum_{n \geq 1} \frac{1}{(n-1)!} \sum_{Z_2, \dots, Z_n \in \mathcal{P}} |\rho^T(Z, Z_2, \dots, Z_n)| \prod_{i=2}^{n} |h(Z_i)| \leq e^{a(Z)}. \tag{678}$$

For every nonnegative function f on $\mathcal{P}$,

$$\sum_{n \geq 1} \frac{1}{n!} \sum_{Z_1, \dots, Z_n} |\rho^T(Z_1, \dots, Z_n)| \sum_{l=1}^{n} f(Z_l) \prod_{i \neq l} |h(Z_i)| \leq \sum_{Z} f(Z) e^{a(Z)}. \tag{679}$$

All these assertions hold for a subsystem and for activities whose moduli are smaller. If holomorphic activities on an open set in $\mathbb{C}^N$ have a common majorant satisfying (677), their cluster sum, and every sum over a fixed subfamily of clusters, is holomorphic there. ■□

Proof. Write $\theta_n(Z)$ for the term of order n in (678), and $\Theta_R = \sum_{n=1}^{R} \theta_n$. We prove the recursion

$$\Theta_{R+1}(Z) \leq \exp\Big(\sum_{Y \nsim Z} |h(Y)| \Theta_R(Y) \Big). \tag{680}$$

Remove vertex 1 from a connected graph on $n \geq 2$ vertices. Its remaining components form a partition $\mathcal{B}$ of $\{2, \dots, n\}$, and each block must have a nonempty set of edges to vertex 1. For a fixed block B, the sum over these edge sets is

$$\prod_{i \in B} (1 - \mathbf{1}_{Z_i \nsim Z_1}) - 1 = -\tau_B, \qquad \tau_B = \mathbf{1}_{\{Z_i \nsim Z_1 \text{ for some } i \in B\}}.$$

Summing first within each component therefore gives the exact identity

$$\rho^T(Z_1, \dots, Z_n) = \sum_{\mathcal{B}} \prod_{B \in \mathcal{B}} (-\tau_B \rho^T(Z_B)). \tag{681}$$

After taking absolute values, bound $\tau_B \leq \sum_{i \in B} \mathbf{1}_{Z_i \nsim Z_1}$. Put

$$\beta_k(Z) = \frac{1}{k!} \sum_{Y_1, \dots, Y_k} |\rho^T(Y_1, \dots, Y_k)| \Big(\sum_{i=1}^{k} \mathbf{1}_{Y_i \nsim Z} \Big) \prod_{i=1}^{k} |h(Y_i)|.$$

Symmetry in the Y_i gives $\beta_k(Z) = \sum_{Y \nsim Z} |h(Y)| \theta_k(Y)$. For blocks of ordered sizes $k_1, \dots, k_s$, summation over partitions contributes $(n-1)!/(s! \prod_l k_l!)$. Consequently

$$\theta_n(Z) \le \sum_{s=1}^{n-1} \frac{1}{s!} \sum_{\substack{k_1+\cdots+k_s=n-1 \\ k_l \ge 1}} \prod_{l=1}^{s} \beta_{k_l}(Z).$$

Sum this inequality for $2 \le n \le R+1$, add $\theta_1 = 1$, and enlarge the sum by allowing every $k_l \le R$ independently. The resulting exponential series proves (680). Since $\Theta_1 = 1$, (677) and induction imply $\Theta_R \le e^a$. Monotone convergence proves (678).

Let S_n denote the term of order n in S, and let $\overline{S}_n$ be the same sum with every term replaced by its modulus. Rooting at the first polymer yields

$$\sum_{n \ge 1} \overline{S}_n = \sum_{n \ge 1} \frac{1}{n} \sum_{Z} |h(Z)| \theta_n(Z) \le \sum_{Z} |h(Z)| e^{a(Z)} < \infty.$$

To identify the exponential, expand $\prod_{i<j} (1 - \mathbf{1}_{Z_i \nsim Z_j})$ as a sum over graphs and partition each graph into its connected components. If $\Xi^{(n)}$ is the sum over compatible families of size n, this gives

$$\Xi^{(n)} = \sum_{s=1}^{n} \frac{1}{s!} \sum_{\substack{k_1+\cdots+k_s=n \\ k_l \ge 1}} \prod_{l=1}^{s} S_{k_l}.$$

The sum of the moduli of all products on the right, over all orders, is at most $\exp(\sum_n \overline{S}_n)$. We may thus regroup it by s or by $\sum_l k_l$. The first grouping gives e^S; the second gives Ξ. Reflexivity excludes repeated polymers from a compatible family, so $\Xi^{(n)} = 0$ for $n > \#\mathcal{P}$.

For (679), move the marked position l to the first position. Symmetry and $n/n! = 1/(n-1)!$ identify its left side with $\sum_Z f(Z)\Theta(Z)$. The same estimate bounds the absolute cluster sum over families meeting $\mathcal{Q} \subset \mathcal{P}$ by

$$\sum_{Z \in \mathcal{Q}} |h(Z)| e^{a(Z)}. \tag{682}$$

Indeed, mark each occurrence of a member of $\mathcal{Q}$; every such family is counted at least once. Restriction and domination preserve (677). A common majorant gives a summable majorant for every cluster series, uniformly in the holomorphic parameters. The Weierstrass theorem then proves the last assertion. The empty polymer system has $\Xi = 1$ and $S = 0$. □

45.2. **Domains, entropy, and decay.** Let $L \ge 13$ be odd, $m' \ge 0$ an integer, and j an integer with $0 \le j \le K$. Fix the partition π_j of the torus into cubes of side ML^{j-K}, where $M = L^{m'}$ and $m' < m + K - j$. Two cubes are adjacent when they share a wall. A localization domain is a nonempty union of cubes connected by such adjacencies; their finite set is denoted by $\mathcal{D}_j$. For $X \in \mathcal{D}_j$, write $\#X$ for its number of cubes. Define $d_j(X)$ as the least number of cube edges in a connected union of edges contained in X which meets every cube of X. A single corner is allowed and has size zero. Cycles are allowed in this definition.

Proposition 45.1 (Entropy and joining)**.** For every domain X,

$$\#X \le 8(d_j(X) + 1).$$

For every cube $\square$ and $n \ge 0$,

$$\#\{X \in \mathcal{D}_j : \square \subset X,\ d_j(X) = n\} \le 2048 \cdot 9216^n. \tag{683}$$

In particular, with $c_{\mathrm{E}} = \log 18432$ and $C_{\mathrm{E}} = 4096$,

$$\sum_{X \in \mathcal{D}_j : \square \subset X} e^{-c d_j(X)} \le C_{\mathrm{E}} \qquad (c \ge c_{\mathrm{E}}). \tag{684}$$

If the graph joining indices whose domains have equal or adjacent cubes is connected, their union is a domain and

$$d_j\Big(\bigcup_{i=1}^{n} X_i\Big) + 4 \le \sum_{i=1}^{n} (d_j(X_i) + 4). \tag{685}$$

■

Proof. A domain has a spanning connected edge set: join the edge skeletons of its cubes along a spanning tree of the cube adjacency graph. A connected set with n edges has at most $n+1$ vertices. A cube met by this set contains one of its vertices, and a vertex belongs to at most eight cubes. This proves the first inequality.

Choose a minimizing connected edge set for a domain containing $\square$, and start at a vertex in $\square$. Traversing every edge twice gives a walk of $2n$ steps visiting all its vertices. There are at most eight initial vertices and six choices for each step. At most $8(n+1)$ cubes meet the vertices of this walk. The walk together with the subset of these cubes constituting X determines X, so the number of possibilities is bounded by

$$8 \cdot 6^{2n} \cdot 2^{8(n+1)} = 2048 \cdot 9216^n.$$

Summing the geometric series proves (684).

If two domains have equal or adjacent cubes, choose vertices of their spanning edge sets in those cubes. The two vertices can be joined inside the cubes by at most four edges. The union of the two spanning sets and this path spans both domains. Thus $d_j(X_1 \cup X_2) \le d_j(X_1) + d_j(X_2) + 4$. Adding the domains in an order supplied by a spanning tree of their incompatibility graph proves (685) and connectivity of the union. □

Domains γ, γ' are incompatible if a cube of one is equal or adjacent to a cube of the other. This is a symmetric reflexive relation. Given $\mathcal{P} \subset \mathcal{D}_j$ and activities $z(\gamma)$, regroup the cluster series by its union:

$$\mathsf{L}(X) = \sum_{n \ge 1} \frac{1}{n!} \sum_{\substack{\gamma_1,\dots,\gamma_n \in \mathcal{P} \\ \gamma_1 \cup \cdots \cup \gamma_n = X}} \rho^T(\gamma_1, \dots, \gamma_n) \prod_{i=1}^{n} z(\gamma_i). \tag{686}$$

Let $\overline{\mathsf{L}}(X)$ denote the corresponding sum of moduli.

Theorem Q (Local cluster sums)**.** Suppose $\varepsilon > 0$, $r'' \ge 0$, $r \ge r'' + c_{\mathrm{E}} + 8$, and

$$C_{\mathrm{gas}} = C_{\mathrm{E}} e^{8+4r''}, \qquad 7C_{\mathrm{gas}}\varepsilon \le 1, \qquad |z(\gamma)| \le \varepsilon e^{-r d_j(\gamma)}.$$

Then (686) converges absolutely and vanishes unless $X \in \mathcal{D}_j$. It depends only on the activities of domains contained in X, and

$$\Xi(z) = \exp \sum_{X \in \mathcal{D}_j} \mathsf{L}(X) \ne 0. \tag{687}$$

Moreover, for $X \in \mathcal{D}_j$,

$$|\mathsf{L}(X)| \le \overline{\mathsf{L}}(X) \le C_{\mathrm{gas}} \varepsilon_X e^{-r'' d_j(X)}, \qquad \varepsilon_X = \max_{\substack{\gamma \in \mathcal{P} \\ \gamma \subset X}} |z(\gamma)| e^{r d_j(\gamma)}. \tag{688}$$

The maximum is zero when its index set is empty. For holomorphic activities with the displayed common majorant, every $\mathsf{L}(X)$ is holomorphic. ■□

Proof. Multiply each activity by its required decay weight:

$$\widehat{z}(\gamma) = z(\gamma) e^{r''(d_j(\gamma)+4)}, \qquad a(\gamma) = \#\gamma.$$

The first bound in proposition 45.1 gives

$$|\widehat{z}(\gamma)| e^{\#\gamma} \le \varepsilon e^{8+4r''} e^{-(r-r''-8)d_j(\gamma)}.$$

By (684), its sum over domains containing one cube is at most $C_{\mathrm{gas}}\varepsilon$. Every domain incompatible with γ contains one of the at most $7\#\gamma$ cubes equal or adjacent to a cube of γ. Hence

$$\sum_{\gamma' \nsim \gamma} |\widehat{z}(\gamma')| e^{\#\gamma'} \le 7\#\gamma\, C_{\mathrm{gas}}\varepsilon \le \#\gamma.$$

Thus theorem P applies to both $\widehat{z}$ and z. Absolute convergence permits regrouping by the union. A nonzero connected coefficient has a connected incompatibility graph, so its union is a domain by proposition 45.1. This proves (687) and the support assertion.

Fix a cube $\square \subset X$. Every cluster with union X has a member containing $\square$, and all its members are contained in X. By (685), multiplication of its absolute weight by $e^{r''(d_j(X)+4)}$ is bounded by the same cluster weight formed with $\widehat{z}$. Apply (682) to the members satisfying $\square \subset \gamma \subset X$. It follows that

$$e^{r''(d_j(X)+4)}\overline{\mathsf{L}}(X) \le \sum_{\square \subset \gamma \subset X} |\widehat{z}(\gamma)| e^{\#\gamma} \le C_{\mathrm{gas}}\varepsilon_X.$$

Dropping $e^{-4r''} \le 1$ proves (688). The series indexed by clusters with union X uses only activities inside X, which proves locality. Holomorphy follows from the common-majorant assertion of theorem P. □

46. Gaussian conditioning and local interactions

The local expansion uses a Gaussian field coupled to auxiliary real variables. Its covariance and coupling may be complex. We control the integral by completing the square with a real reference pair. The resulting bound charges only the variables on which the coupling is supported. This will preserve locality when the auxiliary field is defined on the whole torus.

46.1. **Normalization and an integrable majorant.** We use orthonormal coordinates on the real spaces of bond fields, with three coordinates for each positive bond. Pairings are extended bilinearly to the complexification, transposes are bilinear transposes, and operator norms are induced by the Hermitian Euclidean norm. Thus the formulas below apply directly to the field spaces and pairings used earlier.

For a complex symmetric matrix O with $\operatorname{Re} O > 0$, the integral

$$I(O,J) = \int_{\mathbb{R}^d} e^{-B^{\mathsf{T}}OB/2 + J^{\mathsf{T}}B}\, dB$$

converges absolutely and is holomorphic in (O,J). Indeed, locally $\operatorname{Re} O \ge cI$ and $|J| \le C$, so the integrand has the integrable majorant $e^{-c|B|^2/2 + C|B|}$. The set of such symmetric matrices is open and convex, and its members are invertible: if $Ov = 0$, positivity of $\operatorname{Re}(v^* O v)$ implies $v = 0$. At real positive O and real J, completion of the square gives the Gaussian formula. Holomorphic continuation on this connected domain gives

$$I(O,0)^2 \det(O/2\pi) = 1, \qquad I(O,J) = I(O,0) e^{J^{\mathsf{T}}O^{-1}J/2}. \tag{689}$$

In particular, $I(O,0)^{-1}$ is the unique holomorphic square root of $\det(O/2\pi)$ which is positive for real positive O. Uniqueness follows because the quotient of two such roots is a continuous function with values in $\{1,-1\}$. We use this root in

$$d\mu_C(B) = \det{}^{1/2}(C^{-1}/2\pi) e^{-B^{\mathsf{T}}C^{-1}B/2}\, dB, \qquad \int d\mu_C = 1.$$

The zero-dimensional integral and determinant both equal one.

Let C be a symmetric invertible matrix on $\mathbb{C}^d$ with $\operatorname{Re} C^{-1} > 0$, and $\Gamma : \mathbb{C}^D \to \mathbb{C}^d$. For real X, B, write

$$\Theta_{C,\Gamma}(X,B) = \frac{\det{}^{1/2}(C^{-1})}{(2\pi)^{(d+D)/2}} \exp\left[-\frac{|X|^2}{2} - \frac{B^{\mathsf{T}}C^{-1}B}{2} - B^{\mathsf{T}}\Gamma X - \frac{X^{\mathsf{T}}\Gamma^{\mathsf{T}}C\Gamma X}{2} \right]. \tag{690}$$

The root is the preceding one, with the scalar factor $(2\pi)^{-d/2}$ placed outside it. Define

$$\mathfrak{J}(C,\Gamma)[F] = \int_{\mathbb{R}^D}\int_{\mathbb{R}^d} \Theta_{C,\Gamma}(X,B)F(B)\,dB\,dX$$

when the integral of the modulus is finite.

Proposition 46.1 (Gaussian domination)**.** Suppose $0<\gamma\le 1$, $c\ge 1$, and real matrices $C_{\mathrm{r}}=C_{\mathrm{r}}^{\mathsf T}$, Γ_{r} satisfy

$$C_{\mathrm{r}}^{-1}\ge \gamma I, \qquad \|C_{\mathrm{r}}^{-1}\|,\ \|\Gamma_{\mathrm{r}}\|, \|C_{\mathrm{r}}\Gamma_{\mathrm{r}}\|\le c.$$

Assume that each of

$$C^{-1}-C_{\mathrm{r}}^{-1}, \qquad \Gamma-\Gamma_{\mathrm{r}}, \qquad \Gamma^{\mathsf T}C\Gamma - \Gamma_{\mathrm{r}}^{\mathsf T}C_{\mathrm{r}}\Gamma_{\mathrm{r}} \tag{691}$$

has norm at most $\widehat{\varepsilon}\ge 0$. Let P be a real orthogonal coordinate projection on $\mathbb{R}^D$ of rank q, with $\Gamma=\Gamma P$ and $\Gamma_{\mathrm{r}}=\Gamma_{\mathrm{r}}P$. For $\rho\ge 0$, suppose

$$u=\widehat{\varepsilon}+\rho\le \frac{1}{12}\min\{\gamma, c^{-2}\}.$$

Then $\operatorname{Re}C^{-1}\ge (11/12)\gamma I$, and

$$\int\!\!\int |\Theta_{C,\Gamma}(X,B)|e^{\rho|B|^2}\,dB\,dX \le \exp\left(\frac{9}{2}\gamma^{-1}ud+6c^2uq\right). \tag{692}$$

There is also the pointwise bound

$$|\Theta_{C,\Gamma}(X,B)|e^{\rho|B|^2}\le \frac{(2c)^{d/2}}{(2\pi)^{(d+D)/2}}\exp\left[-\frac{|X|^2}{8}-\frac{\gamma}{16c^2}|B|^2\right]. \tag{693}$$

Consequently $\mathfrak{J}(C,\Gamma)[F]$ is defined whenever F is Borel and $|F(B)|\le F_0e^{\rho|B|^2}$. ∎

Proof. The first defect in (691) gives the claimed lower bound on $\operatorname{Re}C^{-1}$. Set

$$\widetilde{B}=B+C_{\mathrm{r}}\Gamma_{\mathrm{r}}X, \qquad a=2\widehat{\varepsilon}+4uc^2\le 6c^2u\le \tfrac12.$$

Completion of the square with the reference pair leaves three error terms. Their absolute values sum to at most

$$\tfrac12\widehat{\varepsilon}|PX|^2+\tfrac12\widehat{\varepsilon}|B|^2+\widehat{\varepsilon}|B|\,|PX|\le \widehat{\varepsilon}(|B|^2+|PX|^2).$$

Use $|B|^2\le 2|\widetilde{B}|^2+2c^2|PX|^2$. The real part of the exponent in $\Theta_{C,\Gamma}e^{\rho|B|^2}$ is therefore at most

$$-\frac{|(I-P)X|^2}{2}-\frac{1-a}{2}|PX|^2-\frac12\widetilde{B}^{\mathsf T}(C_{\mathrm{r}}^{-1}-4uI)\widetilde{B}. \tag{694}$$

This is a real change of variables in the integral. Integrating first in $\widetilde{B}$, and then in the two orthogonal parts of X, gives the bound

$$\left|\frac{\det C^{-1}}{\det C_{\mathrm{r}}^{-1}}\right|^{1/2}\det(I-4uC_{\mathrm{r}})^{-1/2}(1-a)^{-q/2}.$$

The three factors are bounded respectively by

$$e^{\widehat{\varepsilon}d/(2\gamma)}, \qquad e^{4ud/\gamma}, \qquad e^{aq}\le e^{6c^2uq}.$$

For the first, factor $C^{-1}=C_{\mathrm{r}}^{-1}[I+C_{\mathrm{r}}(C^{-1}-C_{\mathrm{r}}^{-1})]$ and use $\|C_{\mathrm{r}}\|\le\gamma^{-1}$. For the second, every eigenvalue of $4uC_{\mathrm{r}}$ lies in $[0,1/3]$. For the third, $0\le a\le 1/2$. Both use $-\log(1-s)\le 2s$ on $[0,1/2]$. This proves (692).

For a majorant independent of the reference, use $P=I$ in (694). Its negative quadratic terms are bounded above by $-|X|^2/4-\gamma|\widetilde{B}|^2/4$. Since $|\widetilde{B}|^2\ge |B|^2/2-c^2|X|^2$, retain the fraction $1/(2c^2)$ of the second term. The result is at most $-|X|^2/8-\gamma|B|^2/(16c^2)$. Finally $\|C^{-1}\|\le 2c$, whence $|\det C^{-1}|^{1/2}\le (2c)^{d/2}$. This proves (693), whose right side is integrable. □

The bound also proves holomorphy under the integral. More precisely, suppose C, C^{-1}, Γ depend holomorphically on a parameter in an open subset of $\mathbb{C}^N$, and have references satisfying proposition 46.1 with the same $\gamma, c, \widehat{\varepsilon}$. The references need not depend holomorphically on the parameter. If F is jointly Borel, holomorphic in that parameter for every real B, and has the common bound $F_0 e^{\rho|B|^2}$, then $\mathfrak{J}(C,\Gamma)[F]$ is holomorphic. The determinant root and the integrand are holomorphic; (693) permits Cauchy's formula, or Morera's theorem, to pass through the integral.

46.2. **The conditioning identity.** The covariance recovered by the conditioning integral is $C + C\Gamma\Gamma^{\mathsf{T}}C$. This identity will identify the local expansion with the whole fluctuation integral. It requires an equality of covariances, in addition to local estimates on the kernels.

Proposition 46.2. Under the hypotheses of proposition 46.1,

$$\mathfrak{J}(C,\Gamma)[e^{if^{\mathsf{T}}B}] = e^{-f^{\mathsf{T}}(C+C\Gamma\Gamma^{\mathsf{T}}C)f/2} \qquad (f \in \mathbb{R}^d). \tag{695}$$

In particular, $\mathfrak{J}(C,\Gamma)[1] = 1$. Suppose also that C_* is symmetric and invertible on $\mathbb{C}^{D_*}$, and $\chi : \mathbb{R}^{D_*} \to \mathbb{R}^d$ is a coordinate restriction, with

$$C + C\Gamma\Gamma^{\mathsf{T}}C = \chi C_* \chi^{\mathsf{T}}. \tag{696}$$

Assume a real positive reference satisfies

$$\gamma I \le C_{*\mathrm{r}}^{-1} \le cI, \qquad \|C_*^{-1} - C_{*\mathrm{r}}^{-1}\| \le \widehat{\varepsilon}.$$

Then, for every Borel F with $|F(B)| \le F_0 e^{\rho|B|^2}$,

$$\mathfrak{J}(C,\Gamma)[F] = \int d\mu_{C_*}(B_*)\, F(\chi B_*). \tag{697}$$

Both integrals converge absolutely. ■

Proof. The majorant justifies integrating first in B. Apply (689) with $O = C^{-1}$ and $J = if - \Gamma X$. Its quadratic ΓX term cancels the corresponding term of (690). The remaining integral is

$$e^{-f^{\mathsf{T}}Cf/2} \int d\mu_I(X) e^{-if^{\mathsf{T}}C\Gamma X}.$$

One more application of (689) proves (695).

For the whole integral,

$$\operatorname{Re} C_*^{-1} - 2\rho I \ge (\gamma - \widehat{\varepsilon} - 2\rho) I \ge \tfrac{5}{6}\gamma I.$$

Its integrand is therefore absolutely integrable. Integrating $\Theta_{C,\Gamma}$ in X gives an integrable complex density on $\mathbb{R}^d$; integrating μ_{C_*} in the complementary coordinates gives another. By (695), (689) and (696), these finite complex measures have identical Fourier transforms. Uniqueness of the Fourier transform of finite measures makes their densities equal almost everywhere. Fubini's theorem, with the two absolute bounds above, now gives (697) for the stated F. □

47. Local Gaussian integration

We now apply the polymer criterion to a Gaussian integral with local interactions and small-field tests. The Gaussian variables are decoupled by parameters attached to the larger cubes. Finite differences in these parameters produce the decay needed for the cluster sum. We first record how the size of a connected region changes under this passage to larger cubes.

47.1. **Enlargement and Gaussian data.** Fix $0 \le k < K$, put $t = k+1$, and assume $m' < m+K-t$. Lengths in this section are measured in the scale of level k. Fine cubes have side $M = L^{m'}$, and big cubes have side LM. For a fine region Y, let $Y^{\uparrow}$ be the union of the big cubes containing a fine cube of Y, and put

$$Y^{\sharp} = (Y^{\sim})^{\uparrow}.$$

For a lattice bond b, let $\square(b)$ be the union of the fine cubes containing its endpoints, and put $b^{\sharp} = \square(b)^{\sharp}$.

Lemma 47.1 (Size after enlargement)**.** For $Y \in \mathcal{D}_k$, $Y^{\sharp} \in \mathcal{D}_t$ and

$$d_t(Y^{\sharp}) \le \frac{d_k(Y)}{L-2} \le \frac{13}{9L} d_k(Y). \tag{698}$$

For every big cube Δ and $\lambda \ge \log 18432$,

$$\sum_{\substack{Y \in \mathcal{D}_k \\ \Delta \subset Y^{\sharp}}} e^{-\lambda d_k(Y)} \le (L+2)^3\, 4096. \tag{699}$$

The regions $\square(b)$ and $b^{\sharp}$ are connected, and both have linear size zero. At most $6((L+2)M)^3$ positive bonds satisfy $\Delta \subset b^{\sharp}$. ■

Proof. Use integer coordinates in units of the fine cube size, with a common origin for the two partitions. A big cube has the form $Q[U] = U + [0,L]^3$, $U \in L\mathbb{Z}^3$, modulo the periods of the torus. It belongs to $Y^{\sharp}$ precisely when Y has a cube of index $u \in U + \{-1,\dots,L\}^3$, modulo these periods. For a fixed fine cube, these big cubes form a rectangular array containing its unique parent. Parent cubes of adjacent fine cubes are equal or adjacent. Thus $Y^{\sharp}$ is wall-connected.

Let Γ be a connected set of fine edges and corners in Y which meets every cube of Y, with $|\Gamma| = d_k(Y)$. A one-corner set is allowed. For each coordinate direction μ, choose $s_\mu \in \{1,\dots,L-2\}$ so that the number $n_\mu(s_\mu)$ of edges of Γ whose initial coordinate has residue s_μ modulo L is minimal. Then

$$\sum_{\mu=1}^{3} n_\mu(s_\mu) \le \frac{|\Gamma|}{L-2}.$$

Map each fine corner by

$$R(v)_\mu = L \left\lceil \frac{v_\mu - s_\mu}{L} \right\rceil.$$

This map respects the periods. A fine edge is contracted to a point unless its initial coordinate has the chosen residue; otherwise it maps to one coarse edge. Corners of a fine cube map to corners of its parent, so the image is a connected coarse edge set in $Y^{\uparrow}$. It has at most $|\Gamma|/(L-2)$ edges.

If $Q[U] \subset Y^{\sharp}$, choose a cube of Y with index $u \in U + \{-1,\dots,L\}^3$. Since Γ meets this cube, it contains one of its corners $v \in u + \{0,1\}^3$. For $U_\mu - 1 \le v_\mu \le U_\mu + L + 1$ and $1 \le s_\mu \le L-2$, the formula for R gives $R(v)_\mu \in \{U_\mu, U_\mu + L\}$. Thus the image meets every cube of $Y^{\sharp}$, proving the first bound in (698). The second follows from $L \ge 13$.

For fixed $\Delta = Q[U]$, every Y in (699) contains one of the at most $(L+2)^3$ fine cubes with indices in $U + \{-1,\dots,L\}^3$. Apply the entropy estimate of Proposition 45.1 to each of these cubes. The endpoints of a nearest-neighbor bond lie in one fine cube or in two adjacent fine cubes. These cubes have a common corner, so $d_k(\square(b)) = 0$; (698) gives $d_t(b^{\sharp}) = 0$. If $\Delta \subset b^{\sharp}$, one endpoint belongs to one of the same $(L+2)^3$ cubes. They contain at most $((L+2)M)^3$ lattice sites, and each site is an endpoint of six positive bonds. This proves the last assertion. □

Let $\mathfrak{X}$ be the nonpivot bonds of a region Ω_t for π_t. Write $\mathfrak{X}\langle Z\rangle$ for the bonds with at least one endpoint in Z, and $\mathfrak{X}^{\circ}\langle Z\rangle$ for those with both endpoints there. Fields are odd under bond reversal.

On a reversal-invariant set $\mathfrak{e}$, use the coordinates on $\mathfrak{e}^+$, with Euclidean norm and measure

$$\|B\|^2 = \sum_{b\in\mathfrak{e}^+} |B(b)|^2, \qquad dB = \prod_{b\in\mathfrak{e}^+} dB(b).$$

There are $3\#\mathfrak{e}^+$ real coordinates. In particular, for a big region Z,

$$N(Z) := 3\#\mathfrak{X}^+\langle Z\rangle \le 18(LM)^3 \#_{\pi_t} Z. \tag{700}$$

Indeed, each bond counted has an endpoint among the $(LM)^3\#_{\pi_t}Z$ sites of Z.

Definition 47.2. Fix a nonempty big region R, and constants $\bar\kappa \ge 1$, $0 < \gamma_{\mathrm r} \le 1$, $c_{\mathrm r} \ge 1$, and $\hat\varepsilon \ge 0$. A decoupled Gaussian structure assigns to each fine region $Z_\circ$ with $Z_\circ^\sharp \subset R$, including the empty region, operators

$$C(Z_\circ, s) : \mathcal{H}(Z_\circ) \longrightarrow \mathcal{H}(Z_\circ), \qquad \Gamma(Z_\circ, s) : \mathcal{H}(R) \longrightarrow \mathcal{H}(Z_\circ).$$

Here $\mathcal{H}(Z) = \mathbb{C}^{3\#\mathfrak{X}^+\langle Z\rangle}$, C is symmetric and invertible, and s ranges over the closed polydisc $|s(\Delta)| \le e^{\bar\kappa}$ on

$$\mathsf{P}(R, Z_\circ) = \{\Delta \subset R : \Delta \not\subset Z_\circ^\sharp\}.$$

The following properties are required.

(St1) The kernels of C, C^{-1}, Γ extend holomorphically to a neighborhood of this closed polydisc.

(St2) Let $Z_\circ^\sharp \subset Z \subset R$, let γ_i be the wall-components of Z, and suppose $s = 0$ outside Z. Put $Z_i = Z_\circ \cap \gamma_i$ and let s_i agree with s in γ_i and vanish elsewhere. Then

$$C(Z_\circ, s) = \bigoplus_i C(Z_i, s_i), \qquad \chi_{\mathcal{H}(Z_i)}\Gamma(Z_\circ, s) = \Gamma(Z_i, s_i), \tag{701}$$

and $\Gamma(Z_i, s_i)$ uses only auxiliary coordinates in $\mathfrak{X}\langle\gamma_i\rangle$.

(St3) For each $Z_\circ$ there are real $C_{\mathrm r} = C_{\mathrm r}^{\mathsf T}$, $\Gamma_{\mathrm r}$, independent of s, such that $\Gamma_{\mathrm r}$ uses only coordinates in $\mathfrak{X}\langle Z_\circ^\sharp\rangle$,

$$C_{\mathrm r}^{-1} \ge \gamma_{\mathrm r} I, \qquad \|C_{\mathrm r}^{-1}\|,\ \|\Gamma_{\mathrm r}\|,\ \|C_{\mathrm r}\Gamma_{\mathrm r}\| \le c_{\mathrm r}, \tag{702}$$

and, throughout the polydisc,

$$\max\{\|C^{-1} - C_{\mathrm r}^{-1}\|, \|\Gamma - \Gamma_{\mathrm r}\|, \|\Gamma^{\mathsf T} C\Gamma - \Gamma_{\mathrm r}^{\mathsf T} C_{\mathrm r}\Gamma_{\mathrm r}\|\} \le \hat\varepsilon. \tag{703}$$

When $R = T_\ell$, the structure is complete if it also has a symmetric invertible whole covariance C_*, with a real reference satisfying

$$\gamma_{\mathrm r} I \le C_{*\mathrm r}^{-1} \le c_{\mathrm r} I, \qquad \|C_*^{-1} - C_{*\mathrm r}^{-1}\| \le \hat\varepsilon,$$

and the covariance identity (696) holds for every $Z_\circ$ at $s = 1$. This is property (St4). ∎

All intersections here are intersections of cube sets. The decompositions in (St2) are disjoint even for bonds crossing the boundary of $Z_\circ$. To see this, a fine bond has endpoints in one big cube or two adjacent big cubes. It cannot have endpoints in different wall-components of Z. Moreover, the touching enlargement of a fine cube is a rectangular array of fine cubes, and its big-cube parents form a wall-connected array. If that enlargement lies in Z, it lies in the component containing the original fine cube. Taking unions gives

$$(Z_\circ \cap \gamma_i)^\sharp = Z_\circ^\sharp \cap \gamma_i, \qquad \mathfrak{X}\langle Z_\circ\rangle = \bigsqcup_i \mathfrak{X}\langle Z_\circ \cap \gamma_i\rangle.$$

The sets $\mathfrak{X}\langle\gamma_i\rangle$ are pairwise disjoint for the same endpoint reason.

Write

$$\mathbb{J}(Z_\circ; s)[F] = \mathfrak{J}(C(Z_\circ, s), \Gamma(Z_\circ, s))[F],$$

with the conditioning integral of Section 46. Under (St2), this integral factorizes over the components whenever $F = \prod_i F_i$, with F_i depending on $\mathcal{H}(Z_i)$, and the corresponding integrals are absolutely convergent. Indeed, both quadratic forms and the mixed term split into the indicated blocks. The

determinant root is multiplicative on these blocks: it is so for positive real matrices, and then throughout the domain by holomorphy. The unused auxiliary coordinates integrate to one. Fubini's theorem gives the asserted product.

47.2. **The local activities and their bounds.** For the rest of this section, $\mathfrak{X}$ is the set of nonpivot bonds of the whole lattice. Let $\mathfrak{b} \in (0,\infty]$ and $t_b \in [0,1]$. A potential family $\mathbb{V}^o = (\mathbb{V}^o(Y;B))$ is indexed by $Y \in \mathcal{D}_k$ with $Y^\sharp \subset R$ and depends only on the field on $\mathfrak{X}^\circ\langle Y\rangle$. It is of sort Σ_{v} if it is a Borel function on $|B(b)| < \mathfrak{b}$, and of sort Σ_{q} if it is Borel on all real fields. The labels o are called origins. We allow at most four origins, and require $t_b = 1$ for all b if any origin has sort Σ_{v}. The abstract sort Σ_{q} does not require a quadratic dependence on B. Set

$$\|\mathbb{V}\|_{\star,r} = \sup_{Y,B} \frac{e^{r d_k(Y)} |\mathbb{V}(Y;B)|}{1 + M^{-3} \sum_{b \in \mathfrak{X}^{\circ+}\langle Y\rangle} |B(b)|^2}, \tag{704}$$

where B ranges over the domain of the family.

We give the constants explicitly because their smallness conditions will be checked for the running interactions. Let $\gamma_2 \ge 0$, $\lambda_{\mathrm{out}} > 0$, and put

$$\lambda_{\mathrm{s}} = \log 18432, \qquad S_{\mathrm{s}} = 4096, \qquad w_0 = (4 S_{\mathrm{s}} L^3)^{-1},$$
$$\mathsf{C}_{\mathrm{ex}} \ge \max\{8e^8, 13/9, 3 + \tfrac{1}{4}\log(7 S_{\mathrm{s}})\}.$$

Define

$$\begin{aligned} u &= \hat{\varepsilon} + 2(LM)^{-3} + \tfrac{1}{2}\gamma_2, \qquad c_G = \mathsf{C}_{\mathrm{ex}}(\gamma_{\mathrm{r}}^{-1} + c_{\mathrm{r}}^2), \\ c_{\mathrm{v}} &= 2 + \mathsf{C}_{\mathrm{ex}} c_G (LM)^3 u, \qquad c_{\mathrm{w}} = c_{\mathrm{v}} + 3, \\ \lambda_1 &= \lambda_{\mathrm{out}} + 8c_{\mathrm{w}} + \mathsf{C}_{\mathrm{ex}} + \lambda_{\mathrm{s}}, \qquad \lambda_2 = \lambda_1 - 8c_{\mathrm{w}}, \\ \lambda_V &= \frac{\mathsf{C}_{\mathrm{ex}}}{L}\lambda_1 + 2\lambda_{\mathrm{s}}, \qquad \lambda_0 = \lambda_V - \lambda_{\mathrm{s}}, \\ w_\star &= \frac{w_0 e^{-4\lambda_1}}{\mathsf{C}_{\mathrm{ex}}(L+2)^3 S_{\mathrm{s}}}, \qquad \mathsf{p} = e^{-\gamma_2 \mathfrak{b}^2/2}, \qquad \mathsf{p}_\star = \mathsf{C}_{\mathrm{ex}}((L+2)M)^3 e^{4\lambda_1}\mathsf{p}, \\ K_E &= \mathsf{C}_{\mathrm{ex}} S_{\mathrm{s}} e^{8c_{\mathrm{w}} - 4\lambda_1}, \qquad K_P^\Pi = K_E, \qquad K_\Pi = K_E (L+2)^3 S_{\mathrm{s}} e^{4\lambda_1}/w_0. \end{aligned} \tag{705}$$

When $\mathfrak{b} = \infty$, set $\mathsf{p} = 0$.

For a reversal-invariant bond set $\mathfrak{e}$, put

$$\mathsf{in}_{\mathfrak{e}}(B) = \prod_{b \in \mathfrak{e}^+} \mathbf{1}_{\{t_b |B(b)| < \mathfrak{b}\}}, \qquad \mathsf{out}_{\mathfrak{e}}(B) = \prod_{b \in \mathfrak{e}^+} \mathbf{1}_{\{t_b |B(b)| \ge \mathfrak{b}\}}.$$

A Mayer index $\iota = (\mathbf{D}, \mathfrak{e}) \ne (\emptyset,\emptyset)$ consists of a set of vertices (Y,o) and a reversal-invariant set of large-field bonds. If Y^{v} is the union of the vertex domains of sort Σ_{v}, we require $\mathfrak{e} \cap \mathfrak{X}\langle Y^{\mathrm{v}}\rangle = \emptyset$ and $b^\sharp \subset R$ for $b \in \mathfrak{e}$. Define

$$\begin{aligned} Z_\circ(\iota) &= \bigcup_{(Y,o)\in\mathbf{D}} Y \ \cup \bigcup_{b\in\mathfrak{e}} \square(b), \\ F_\iota(B) &= (-1)^{\#\mathfrak{e}^+} \mathsf{in}_{\mathfrak{X}\langle Y^{\mathrm{v}}\rangle}(B) \mathsf{out}_{\mathfrak{e}}(B) \prod_{(Y,o)\in\mathbf{D}} (e^{\mathbb{V}^o(Y;B)} - 1). \end{aligned} \tag{706}$$

The function is defined to be zero when its small-field test vanishes, so no potential is evaluated outside its domain. It is Borel and depends only on $\mathfrak{X}\langle Z_\circ(\iota)\rangle$.

For $\gamma \in \mathcal{D}_t$, $\gamma \subset R$, let $\mathcal{I}(\gamma)$ consist of the indices with $Z_\circ(\iota)^\sharp \subset \gamma$, and put $\mathcal{K}(\gamma,\iota) = \gamma \setminus Z_\circ(\iota)^\sharp$, as a set of big cubes. If $f_\iota(s) = \mathbb{J}(Z_\circ(\iota); s)[F_\iota]$, define

$$T(\gamma,\iota) = \sum_{\mathcal{K}' \subset \mathcal{K}(\gamma,\iota)} (-1)^{\#\mathcal{K}(\gamma,\iota) - \#\mathcal{K}'} f_\iota(\mathbf{1}_{\mathcal{K}'}), \qquad \mathsf{z}(\gamma) = \sum_{\iota \in \mathcal{I}(\gamma)} T(\gamma,\iota). \tag{707}$$

The index sets in these sums are finite. With the incompatibility relation of Section 45, set

$$E(X)=\sum_{n\ge 1}\frac{1}{n!}\sum_{\substack{\gamma_1,\dots,\gamma_n\subset R\\ \gamma_i\in\mathcal{D}_t,\ \cup_i\gamma_i=X}}\rho^{T}(\gamma_1,\dots,\gamma_n)\prod_i \mathsf{z}(\gamma_i). \tag{708}$$

Expanding the activities in this expression gives its refined series, indexed also by $\iota_i\in\mathcal{I}(\gamma_i)$.

The estimates keep track of specified vertices or large-field bonds. For bounds $A^o\ge\|\mathbb{V}^o\|_{\star,\lambda_V}$, assign regions and weights to vertices, positive bonds, and big cubes as follows:

e	(Y,o)	b	Δ
$\mathrm{rg}(e)$	$Y^\sharp$	$b^\sharp$	Δ
$w(e)$	$(A^o/w_0)e^{-\lambda_0 d_k(Y)}$	p	$e^{-(\bar\kappa-1)}$

$$\widehat{w}(e)=w(e)e^{\lambda_1(d_t(\mathrm{rg}(e))+4)}. \tag{709}$$

The elements associated with (γ,ι) are $\mathcal{E}(\gamma,\iota)=\mathbf{D}\cup\mathfrak{e}^+\cup\mathcal{K}(\gamma,\iota)$. A set of marks $\mathfrak{m}$ is any set of vertices and positive bonds; it contains no big cubes. Put

$$\overline{T}(\gamma,\iota)=e^{c_{\mathrm{v}}\#\gamma}\prod_{e\in\mathcal{E}(\gamma,\iota)}w(e),\qquad \eta^{\mathfrak{m}}=\sup_{\Delta\subset R}\sum_{\substack{e\in\mathfrak{m}\\ \Delta\subset\mathrm{rg}(e)}}\widehat{w}(e). \tag{710}$$

Here and below $\#\gamma=\#_{\pi_t}\gamma$. Let $\overline{z}^{\mathfrak{m}}(\gamma)$ be the sum of $\overline{T}(\gamma,\iota)$ over indices whose element set meets $\mathfrak{m}$. Let $\overline{E}^{\mathfrak{m}}(X)$ be the refined absolute cluster sum with T replaced by $\overline{T}$, each term multiplied by the number of its polymers whose element set meets $\mathfrak{m}$. Finally, $E^{\mathfrak{m}}(X)$ is the subseries of $E(X)$ consisting of refined terms that meet $\mathfrak{m}$.

Theorem R (Gaussian cluster expansion)**.** Let a Gaussian structure as in Definition 47.2 and at most four potential families be given. Suppose

$$u\le \mathsf{C}_{\mathrm{ex}}^{-1}\min\{\gamma_{\mathrm{r}},c_{\mathrm{r}}^{-2}\},\qquad \bar\kappa\ge 4\lambda_1+1,\qquad \mathsf{p}_\star\le\tfrac12,\qquad \|\mathbb{V}^o\|_{\star,\lambda_V}\le w_\star. \tag{711}$$

All conditioning integrals in (707) converge absolutely, and the refined series (708) does so for every $X\in\mathcal{D}_t$, $X\subset R$. If $\|\mathbb{V}^o\|_{\star,\lambda_V}\le A^o\le w_\star$, then

$$|T(\gamma,\iota)|\le\overline{T}(\gamma,\iota), \tag{712}$$

$$\overline{z}^{\mathfrak{m}}(\gamma)\le\eta^{\mathfrak{m}}e^{8c_{\mathrm{w}}-4\lambda_1}e^{-\lambda_2 d_t(\gamma)}, \tag{713}$$

$$|E^{\mathfrak{m}}(X)|\le\overline{E}^{\mathfrak{m}}(X)\le K_E\eta^{\mathfrak{m}}e^{-\lambda_{\mathrm{out}}d_t(X)}. \tag{714}$$

In particular,

$$|E(X)|\le\left(K_\Pi\sum_o\|\mathbb{V}^o\|_{\star,\lambda_V}+K_P^\Pi\mathsf{p}_\star\right)e^{-\lambda_{\mathrm{out}}d_t(X)}. \tag{715}$$

When $R=T_\ell$ and the structure is complete, the whole integral converges absolutely and satisfies

$$\Xi:=\int d\mu_{C_*}(B)\,\mathsf{in}_{\mathfrak{X}}(B)\exp\left(\sum_o\sum_{Y\in\mathcal{D}_k}\mathbb{V}^o(Y;B)\right)=\exp\left(\sum_{X\in\mathcal{D}_t}E(X)\right)\neq 0. \tag{716}$$

The integrand on the left is zero where its test fails. ■□

Proof. We first bound one Mayer integrand. Put

$$\alpha_Y=w_0e^{-\lambda_{\mathrm{s}}d_k(Y)},\qquad x_Y=1+M^{-3}\sum_{b\in\mathfrak{X}^{\circ+}\langle Y\rangle}|B(b)|^2.$$

The potential bound gives $|\mathbb{V}^o(Y;B)|\le w(Y,o)\alpha_Y x_Y$. Since $w(Y,o)\le 1$, $\alpha_Y x_Y\ge 0$, and $|e^z-1|\le|z|e^{|z|}$,

$$|e^{\mathbb{V}^o(Y;B)}-1|\le w(Y,o)e^{2\alpha_Y x_Y}.$$

Also, for $0\le t_b\le 1$,

$$\mathbf{1}_{\{t_b|B(b)|\ge\mathfrak{b}\}}\le\mathsf{p}\,e^{\gamma_2|B(b)|^2/2}.$$

For infinite threshold the left side is zero, as required.

Suppose $Z_\circ(\iota)^\sharp \subset Z$, with Z a big region. The entropy bound, summed over the fine cubes of Z and at most four origins, gives

$$\sum_{(Y,o)\in\mathbf{D}} \alpha_Y \le \#Z.$$

For a fixed bond in the interior of a vertex domain, select the fine cube containing one of its endpoints. The same entropy bound and $4w_0S_{\mathrm{s}} = L^{-3}$ show $\sum_{(Y,o):\, b\in\mathfrak{X}^\circ\langle Y\rangle} \alpha_Y \le L^{-3}$. Therefore, with $\rho_* = 2(LM)^{-3} + \gamma_2/2$,

$$|F_\iota(B)| \le e^{2\#Z} \prod_{e\in\mathbf{D}\cup\mathfrak{e}^+} w(e) e^{\rho_*\|B\|^2}. \tag{717}$$

Take parameters supported in Z. By (St2), Γ uses only auxiliary coordinates in $\mathfrak{X}\langle Z\rangle$; the same holds for Γ_{r}. Apply Proposition 46.1 with $d \le N(Z)$, $q \le N(Z)$, and $\hat\varepsilon + \rho_* = u$. Its exponential cost is at most $c_G u N(Z)$, since $\mathsf{C}_{\mathrm{ex}} \ge 8e^8$. Using (700) and $\mathsf{C}_{\mathrm{ex}} \ge 18$, we obtain

$$\mathbb{J}^{\mathrm{abs}}(Z_\circ; s)[F_\iota] \le e^{c_{\mathrm{v}}\#Z} \prod_{e\in\mathbf{D}\cup\mathfrak{e}^+} w(e). \tag{718}$$

The same domination proves holomorphy in the decoupling parameters. In applying the finite-dimensional bound, first integrate the unused standard Gaussian coordinates; this keeps its cost proportional to $N(Z)$.

For completeness, if a holomorphic function g on the polydisc of radius $r > 1$ has modulus at most M_0, its finite difference in a coordinate set $\mathcal{K}$ obeys

$$\left|\sum_{A\subset\mathcal{K}} (-1)^{\#\mathcal{K}-\#A} g(\mathbf{1}_A)\right| \le M_0 (r-1)^{-\#\mathcal{K}}. \tag{719}$$

Indeed, one coordinate difference is represented by the Cauchy kernel $1/(z(z-1))$. Integrating these kernels over circles of radius $r' < r$ gives $M_0(r'-1)^{-\#\mathcal{K}}$; let $r' \uparrow r$. Apply this to f_ι, with all coordinates outside $\mathcal{K}(\gamma,\iota)$ set to zero, using (718) with $Z = \gamma$. Since $(e^{\bar\kappa}-1)^{-1} \le e^{-(\bar\kappa-1)}$, this proves (712).

We next sum the element weights through a fixed big cube. Lemma 47.1 and the definition of λ_0 give

$$\sum_{Y:\Delta\subset Y^\sharp} \widehat{w}(Y,o) \le \frac{(L+2)^3 S_{\mathrm{s}} e^{4\lambda_1}}{w_0} A^o \le \mathsf{C}_{\mathrm{ex}}^{-1},$$
$$\sum_{b:\Delta\subset b^\sharp} \widehat{w}(b) \le \mathsf{p}_\star, \qquad \widehat{w}(\Delta) = e^{-(\bar\kappa-1)+4\lambda_1} \le 1.$$

Let $\mathfrak{m}_{\mathrm{all}}$ be all vertices and positive bonds. With at most four origins,

$$\eta^{\mathfrak{m}_{\mathrm{all}}} \le H \sum_o A^o + \mathsf{p}_\star \le 1, \qquad H = \frac{(L+2)^3 S_{\mathrm{s}} e^{4\lambda_1}}{w_0}. \tag{720}$$

The sum through Δ over all three kinds of elements is at most 2. For marks whose vertices have origins in S, the corresponding bound is

$$\eta^{\mathfrak{m}} \le H \sum_{o\in S} A^o + \mathsf{p}_\star \mathbf{1}_{\{\mathfrak{m} \text{ contains a bond}\}}. \tag{721}$$

The regions of the elements of $\mathcal{E}(\gamma,\iota)$ are connected and have union γ. The joining bound of Proposition 45.1 yields

$$d_t(\gamma) + 4 \le \sum_{e\in\mathcal{E}(\gamma,\iota)} (d_t(\mathrm{rg}(e)) + 4).$$

Consequently

$$\overline{T}(\gamma,\iota) \le e^{c_{\mathrm{v}}\#\gamma - 4\lambda_1 - \lambda_1 d_t(\gamma)} \prod_{e\in\mathcal{E}(\gamma,\iota)} \widehat{w}(e).$$

For fixed γ, the map $\iota \mapsto \mathcal{E}(\gamma,\iota)$ is injective: its vertices and positive bonds recover ι. Enlarge the sum to all sets of elements whose regions lie in γ, choose one marked element, and sum the others by a product of $1+\widehat{w}(e)$. Every element has a cube in its region, so

$$\sum_{\substack{\iota\in\mathcal{I}(\gamma)\\ \mathcal{E}(\gamma,\iota)\cap\mathfrak{m}\neq\emptyset}} \prod_{e\in\mathcal{E}(\gamma,\iota)} \widehat{w}(e) \le \eta^{\mathfrak{m}}\#\gamma\, e^{2\#\gamma}.$$

Since $\#\gamma \le e^{\#\gamma}$ and $\#\gamma \le 8(d_t(\gamma)+1)$, this proves (713).

It remains to apply the polymer criterion with enough decay left for the union of a cluster. Set

$$\mu = \lambda_{\mathrm{out}} + 1, \qquad a(\gamma) = \#\gamma, \qquad \overline{h}(\gamma) = \overline{z}^{\mathfrak{m}_{\mathrm{all}}}(\gamma), \qquad \widetilde{h}(\gamma) = \overline{h}(\gamma) e^{\mu(d_t(\gamma)+4)}.$$

The activity bound and entropy estimate give, for every Δ,

$$\sum_{\gamma:\Delta\subset\gamma} \overline{z}^{\mathfrak{m}}(\gamma) e^{\mu(d_t(\gamma)+4)+a(\gamma)} \le \eta^{\mathfrak{m}} S_{\mathrm{s}} e^{8c_{\mathrm{w}} - 4\lambda_1 + 4\mu + 8}. \tag{722}$$

Here $\lambda_2 - \mu - 8 = \mathsf{C}_{\mathrm{ex}} + \lambda_{\mathrm{s}} - 9 \ge \lambda_{\mathrm{s}}$. The exponent on the right is $-24c_{\mathrm{w}} - 4\lambda_{\mathrm{s}} + 12 - 4\mathsf{C}_{\mathrm{ex}}$. There are at most $7\#\gamma$ cubes equal or adjacent to a cube of γ. By (720) and $7S_{\mathrm{s}}e^{12-4\mathsf{C}_{\mathrm{ex}}} \le 1$,

$$\sum_{\gamma'\asymp\gamma} \widetilde{h}(\gamma') e^{a(\gamma')} \le \#\gamma = a(\gamma).$$

Thus Theorem P applies to $\widetilde{h}$, and hence also to $\overline{h}$ and z.

In a refined cluster, summing the number of marked polymers amounts to replacing, in turn, one factor $\overline{h}$ by $\overline{z}^{\mathfrak{m}}$. For a cluster of union X, the joining estimate permits the extraction of $e^{-\mu(d_t(X)+4)}$. Apply the one-mark estimate (679) to

$$f(\gamma) = \mathbf{1}_{\{\gamma\subset X\}} \overline{z}^{\mathfrak{m}}(\gamma) e^{\mu(d_t(\gamma)+4)}.$$

Dropping the condition that the union equal X adds only nonnegative terms. Then (722) gives

$$\overline{E}^{\mathfrak{m}}(X) \le e^{-\mu(d_t(X)+4)} \#X\, \eta^{\mathfrak{m}} S_{\mathrm{s}} e^{8c_{\mathrm{w}} - 4\lambda_1 + 4\mu + 8}.$$

Use $\#X \le 8e^{d_t(X)}$ and $\mathsf{C}_{\mathrm{ex}} \ge 8e^8$ to obtain (714). Every Mayer index contains a vertex or a positive large-field bond. Thus every refined term is marked by $\mathfrak{m}_{\mathrm{all}}$, which proves its absolute convergence. Taking $A^o = \|\mathbb{V}^o\|_{\star,\lambda_V}$ in (720) proves (715).

Finally let $R = T_\ell$ and assume (St4). Expand each $e^{\mathbb{V}}$ as $1+(e^{\mathbb{V}}-1)$ and each small-field indicator outside the selected Σ_{v} domains as $1 - \mathsf{out}_{\{b,\bar{b}\}}$. This finite expansion is exactly

$$\mathsf{in}_{\mathfrak{X}}(B) e^{\sum_{o,Y} \mathbb{V}^o(Y;B)} = 1 + \sum_{\iota} F_\iota(B).$$

The identity also holds where a test fails: for each fixed set of vertices either its retained test is zero, or the sum over its large-field subsets produces the missing zero test. No extension of a small-field potential is used. By (717) and Proposition 46.2, each term is absolutely integrable and its integral is $f_\iota(\mathbf{1}_{\mathsf{P}})$. Finite-difference inversion gives

$$\Xi = 1 + \sum_{\iota} \sum_{\mathcal{K}\subset\mathsf{P}(R,Z_\circ(\iota))} f_\iota[\mathcal{K} \mid \emptyset].$$

In this display the bracket denotes the alternating sum in (707). Let γ_i be the wall-components of $Z_\circ(\iota)^\sharp \cup \mathcal{K}$. Each vertex and each large-field bond belongs to a unique component, since its enlarged

region is connected. Split ι accordingly. Gaussian factorization and the product of the alternating sums show

$$f_\iota[\mathcal{K} \mid \emptyset] = \prod_i T(\gamma_i, \iota_i).$$

If some ι_i is empty, its component consists only of cubes of $\mathcal{K}$. Its factor is the nonempty finite difference of the constant function one, hence zero.

The nonzero terms are in bijection with finite compatible families $\{\gamma_i\}$, each supplied with an index $\iota_i \in \mathcal{I}(\gamma_i)$. The inverse takes the unions of the vertex sets and large-field bonds, and sets $\mathcal{K} = \bigcup_i \gamma_i \setminus Z_\circ(\iota)^\sharp$. Compatibility makes the bond fields disjoint and preserves the exclusion of large-field bonds from the tested vertex domains. Thus

$$\Xi = \sum_{\{\gamma_i\} \text{ compatible}} \prod_i \mathsf{z}(\gamma_i).$$

Theorem P and the absolute convergence already proved identify its logarithm by the cluster series. Grouping by the union gives (716). □

47.3. **Comparison and parameter dependence.** We write $E[\underline{\mathbb{V}}; \mathbf{t}]$ when the potential system and the bond parameters need to be displayed. Let $E^\circ[\underline{\mathbb{V}}]$ be the cluster sum of the Σ_{q} origins alone, with threshold ∞. We call this the Gaussian part; in its application below, these origins will be quadratic forms.

Corollary 47.3. Assume the hypotheses of Theorem R. If $\underline{\mathbb{V}}$ has at most three origins and $\|\mathbb{W}\|_{\star,\lambda_V} \le w_\star$, then

$$|E[\underline{\mathbb{V}} \cup \{\mathbb{W}\}; \mathbf{t}](X) - E[\underline{\mathbb{V}}; \mathbf{t}](X)| \le K_\Pi \|\mathbb{W}\|_{\star,\lambda_V} e^{-\lambda_{\mathrm{out}} d_t(X)}. \tag{723}$$

Here $\mathbb{W}$ has either sort, with sort Σ_{q} required if $\mathbf{t} \neq 1$. For any system of at most four origins,

$$|E(X) - E^\circ(X)| \le \left(K_\Pi \sum_{o \text{ of sort } \Sigma_{\mathrm{v}}} \|\mathbb{V}^o\|_{\star,\lambda_V} + K_P^\Pi \mathsf{p}_\star \right) e^{-\lambda_{\mathrm{out}} d_t(X)}. \tag{724}$$

For a complete structure on the torus,

$$\int d\mu_{C_*}(B) \exp\left(\sum_{o \text{ of sort } \Sigma_{\mathrm{q}}} \sum_Y \mathbb{V}^o(Y; B) \right) = \exp\left(\sum_X E^\circ(X) \right),$$

with an absolutely convergent integral.

Suppose in addition that $\mathbb{W}$ has the sort of an existing origin o_1, and that both $\underline{\mathbb{V}} \cup \{\mathbb{W}\}$ and the system obtained by replacing $\mathbb{V}^{o_1}$ with $\mathbb{V}^{o_1} + \mathbb{W}$ satisfy the theorem's bounds. Their activities and their cluster sums agree. ■□

Proof. Mark the new origin. The refined terms containing no vertex of that origin are exactly those of the old system, with unchanged tests and Gaussian integrals. Absolute convergence therefore identifies the difference in (723) with the marked subseries. Use (721) and (714).

For (724), mark every Σ_{v} vertex and every large-field bond. The unmarked terms are precisely the terms of the Σ_{q} subsystem with no large-field bonds. For that subsystem at infinite threshold, all other terms vanish. The same two estimates prove the bound. The infinite threshold changes only p and $\mathsf{p}_\star$ to zero, so Theorem R applies and also proves the displayed integral identity.

For the last assertion, merge the two origin labels. For each domain, the three nonempty choices of vertices before merging contribute

$$(e^V - 1) + (e^W - 1) + (e^V - 1)(e^W - 1) = e^{V+W} - 1.$$

Their unions, sorts, and tests agree with those of the merged vertex. Thus the Mayer functions sum along the fibers of the merging map to the new Mayer function. The conditioning integral and

finite differences are linear, so the activities agree. Their absolutely convergent cluster sums agree as well. □

Proposition 47.4. Fix the geometry, threshold, bond parameters, origins, and sorts in Theorem R. Let the Gaussian structure and potentials depend on z in an open set $\mathcal{O} \subset \mathbb{C}^N$. Suppose the structure has the same constants for every z, the kernels C, C^{-1}, Γ are jointly holomorphic on an open neighborhood of the closed decoupling polydisc times $\mathcal{O}$, and each potential is jointly Borel in (B, z) and holomorphic in z at fixed B in its domain. If

$$\sup_{z\in\mathcal{O}} \|\mathbb{V}^o_z\|_{\star,\lambda_V} \le A^o \le w_\star,$$

then $T_z(\gamma,\iota)$, $\mathsf{z}_z(\gamma)$, $E_z(X)$, and every fixed marked part $E^{\mathfrak{m}}_z(X)$ are holomorphic in z. Their refined series have the common summable majorants of Theorem R.

For two systems satisfying that theorem on regions R, R', with the same constants and the same remaining discrete data, let $X \subset R \cap R'$. Suppose their potentials agree for $Y^\sharp \subset X$, and, at every $(Z_\circ, \mathbf{1}_A)$ with $Z^\sharp_\circ \subset X$ and $A \subset X \setminus Z^\sharp_\circ$, their operators C agree and their kernels Γ agree on auxiliary coordinates in $\mathfrak{X}\langle X\rangle$. Then all terms supported in X, and hence $E(X)$, agree. ■

Proof. The index sets are independent of z. The tests are fixed, and the zero extension in (706) is across a fixed Borel set. Thus $F_{\iota,z}$ is jointly Borel and is holomorphic in z for every real B. The bound (717) is uniform in z. The common Gaussian majorant in Proposition 46.1 justifies holomorphy under each conditioning integral. The real references need not depend holomorphically on z. Finite differences and the finite activity sums preserve holomorphy. Finally, $\overline{E}^{\mathfrak{m}_{\mathrm{all}}}(X)$ is a common summable majorant of the refined series. Uniform convergence proves the assertion for E_z and each of its fixed marked subseries.

For locality, the Mayer indices with enlarged support in X and their functions coincide. At parameters supported in X, (St2) makes Γ vanish on the auxiliary coordinates outside $\mathfrak{X}\langle X\rangle$. Those coordinates integrate to one, leaving identical conditioning integrals in the two systems. Their finite differences, activities, and cluster sums therefore agree. □

For a system parametrized by backgrounds P that can be restricted to big regions, the locality hypothesis takes the following useful form. Whenever $Z^\sharp_\circ \subset Z \subset R$ and $s = 0$ outside Z, require

$$\begin{aligned} C_R(Z_\circ, s; P) &= C_Z(Z_\circ, s|_{\mathsf{P}(Z,Z_\circ)}; P|_Z), \\ \Gamma_R(Z_\circ, s; P) &= \Gamma_Z(Z_\circ, s|_{\mathsf{P}(Z,Z_\circ)}; P|_Z)\chi_{\mathfrak{X}\langle Z\rangle}. \end{aligned} \tag{725}$$

Together with restriction-compatible potentials, this implies that $E(X;P)$ depends only on $P|_X$.

The expansion also preserves the symmetries of its data. Let r be a lattice isometry preserving the two cube partitions and the nonpivot bond set, and let O_b be real orthogonal maps on $\mathfrak{g}$, for $b \in \mathfrak{X}^+$. They define an orthogonal field map O by

$$(\mathsf{O}B)((\mathsf{r}b)^+) = O_b B(b),$$

with odd extension to reversed bonds. Transform the tests by $\widetilde{t}_{(\mathsf{r}b)^+} = t_b$. If the transformed data satisfy

$$\widetilde{C} = \mathsf{O}C\mathsf{O}^{\mathsf{T}}, \qquad \widetilde{\Gamma} = \mathsf{O}\Gamma\mathsf{O}^{\mathsf{T}}, \qquad \widetilde{\mathbb{V}}^o(\mathsf{r}Y; \mathsf{O}B) = \mathbb{V}^o(Y; B),$$

at the correspondingly transformed decoupling vertices, then $\widetilde{E}(\mathsf{r}X) = E(X)$. In fact r gives bijections of the Mayer indices and compatible polymers; the field maps preserve the tests, Lebesgue measure, and all quadratic forms in the conditioning density. The determinant root is preserved by real orthogonal conjugation, first for positive real matrices and then by holomorphy. Hence the terms agree, and absolute convergence allows their reindexing. This argument includes reflections and orthogonal maps of determinant -1.

Likewise, complex conjugation of the operators at every decoupling vertex and of the potentials on real fields conjugates T, z, E. The tests and finite-difference coefficients are real, the chosen

determinant root commutes with conjugation, and the coefficients ρ^T are integers. Thus real data give real cluster sums and real marked parts.

48. Local variations of the fluctuation integral

The cluster expansion assigns a local term to each connected region. To separate the changes in the interaction from the changes in the cutoff, we shall also assign the first variation to individual bonds. This requires a bound on the integral of the absolute value of a finite-difference kernel.

48.1. **Insertions and changes in the interaction.** Use the setting and constants of Theorem R. Unless stated otherwise, all bond parameters in this subsection equal one. Write $\|\mathbb{W}\| = \|\mathbb{W}\|_{\star,\lambda_V}$ for the potential norm. Let $g(Z_\circ, s; \mathbb{X}, B)$ denote the density $\Theta_{C(Z_\circ,s),\Gamma(Z_\circ,s)}(\mathbb{X}, B)$ of (690). If $Z_\circ^\sharp \subset \gamma$, put

$$g^\gamma(Z_\circ; \mathbb{X}, B) = \sum_{A \subset \mathcal{K}} (-1)^{\#\mathcal{K}-\#A} g(Z_\circ, \mathbf{1}_A; \mathbb{X}, B), \qquad \mathcal{K} = \gamma \setminus Z_\circ^\sharp. \tag{726}$$

Lemma 48.1. Suppose F is Borel and $\mathbb{J}^{\mathrm{abs}}(Z_\circ; s)[F] \le J_0 < \infty$ for all decoupling parameters in the closed polydisc that vanish outside γ. Then

$$\iint |g^\gamma(Z_\circ; \mathbb{X}, B)|\, |F(B)|\, d\mathbb{X}\, dB \le e^{-(\bar\kappa-1)\#\mathcal{K}} J_0. \tag{727}$$

In particular the finite difference of $\mathbb{J}(Z_\circ; s)[F]$ equals $\iint g^\gamma F$. ∎

Proof. The last assertion follows by linearity of the finitely many absolutely convergent integrals. For the bound, apply the Cauchy representation used in (719) to the density before integration. At every fixed $(\mathbb{X}, B)$, it is holomorphic in the decoupling parameters. On circles of radius $1 < r < e^{\bar\kappa}$, each coordinate difference contributes the kernel $(re^{i\theta} - 1)^{-1}$ in angular coordinates. Taking absolute values, multiplying by $|F(B)|$, and applying Tonelli gives

$$\iint |g^\gamma|\, |F| \le (r-1)^{-\#\mathcal{K}} \sup_\theta \mathbb{J}^{\mathrm{abs}}(Z_\circ; s_\theta)[F] \le (r-1)^{-\#\mathcal{K}} J_0.$$

The angular and field integrand is measurable, since the density is continuous in both sets of variables and F is Borel. Let $r \uparrow e^{\bar\kappa}$ and use $(e^{\bar\kappa} - 1)^{-1} \le e^{-(\bar\kappa-1)}$. When $\mathcal{K}$ is empty the assertion is the original bound at $s = 0$. □

Let $\underline{\mathbb{V}}$ have at most three origins and satisfy $\|\mathbb{V}^o\| \le w_\star$. An insertion $\mathbb{W}$ is a potential family of either sort with finite norm, without a smallness requirement. A Mayer index with an insertion is $\iota' = (\mathbf{D}, \mathfrak{e}, Y')$. Its enlarged support includes $Y'^\sharp$, and its tested region includes Y' if $\mathbb{W}$ has sort Σ_{v}. The set $\mathfrak{e}$ avoids the bonds incident to the whole tested region. Unlike an ordinary Mayer index, $(\mathbf{D}, \mathfrak{e})$ may be empty. Put

$$\begin{aligned} Z_\circ(\iota') &= \bigcup_{(Y,o)\in\mathbf{D}} Y \;\cup \bigcup_{b\in\mathfrak{e}} \square(b) \;\cup Y', \\ F'_{\iota'}(B) &= (-1)^{\#\mathfrak{e}^+} \mathsf{in}_{\mathfrak{X}\langle Y^{\mathrm{v}}\rangle}(B) \mathsf{out}_{\mathfrak{e}}(B) \prod_{(Y,o)\in\mathbf{D}} (e^{\mathbb{V}^o(Y;B)} - 1) \mathbb{W}(Y'; B). \end{aligned} \tag{728}$$

Again the function is zero where the test fails. Define $T'(\gamma, \iota')$ by the finite difference of the conditioning integral of $F'_{\iota'}$, and let $\mathsf{z}'[\underline{\mathbb{V}}; \mathbb{W}](\gamma)$ be its finite sum over $Z_\circ(\iota')^\sharp \subset \gamma$. The cluster sum with one insertion is

$$E'[\underline{\mathbb{V}}; \mathbb{W}](X) = \sum_{n\ge1} \frac{1}{(n-1)!} \sum_{\substack{\gamma_1,\dots,\gamma_n \subset R \\ \gamma_i \in \mathcal{D}_t,\ \cup_i \gamma_i = X}} \rho^T(\gamma_1, \dots, \gamma_n) \mathsf{z}'(\gamma_1) \prod_{i=2}^n \mathsf{z}(\gamma_i). \tag{729}$$

Theorem S (First variation)**.** With these hypotheses, all terms in (729) are defined, and its refined series converges absolutely. It is linear in insertions of a fixed sort, and

$$|E'[\underline{\mathbb{V}};\mathbb{W}](X)| \le K_\Pi \|\mathbb{W}\| e^{-\lambda_{\mathrm{out}} d_t(X)}. \tag{730}$$

It vanishes if $\mathbb{W}(Y;\cdot) = 0$ whenever $Y^\sharp \subset X$. For $\epsilon_\star = w_\star/\|\mathbb{W}\|$, with $\epsilon_\star = \infty$ when $\mathbb{W} = 0$, the cluster sum with the additional origin $\epsilon\mathbb{W}$ is holomorphic for $|\epsilon| < \epsilon_\star$, and

$$\frac{d}{d\epsilon} E[\underline{\mathbb{V}} \cup \{\epsilon\mathbb{W}\}](X)\Big|_{\epsilon=0} = E'[\underline{\mathbb{V}};\mathbb{W}](X). \tag{731}$$

The insertion sum has the same locality, orthogonal covariance, and conjugation properties as its Gaussian data and potentials. Under the parameter hypotheses of Proposition 47.4, with a jointly Borel, holomorphic insertion of uniformly bounded norm, it is holomorphic in those parameters.

If an origin o_1 and a family $\mathbb{W}_1$ of its sort satisfy $\|\mathbb{V}^{o_1}\|, \|\mathbb{W}_1\| \le w_\star/2$, then

$$|E'[\underline{\mathbb{V}} +_{o_1} \mathbb{W}_1;\mathbb{W}](X) - E'[\underline{\mathbb{V}};\mathbb{W}](X)| \le \frac{8K_\Pi}{w_\star} \|\mathbb{W}\| \, \|\mathbb{W}_1\| e^{-\lambda_{\mathrm{out}} d_t(X)}. \tag{732}$$

Here $+_{o_1}$ means addition to the named existing origin. ■□

Proof. Use the majorants in (710) with every origin bound equal to $w_\star$. Adjoin a new origin o_+ of the insertion's sort. The map

$$\jmath(\mathbf{D}, \mathfrak{e}, Y') = (\mathbf{D} \cup \{(Y', o_+)\}, \mathfrak{e})$$

is a bijection from insertion indices to ordinary indices having exactly one vertex of the new origin. It preserves the support, tested region, and decoupling cubes. If $\overline{T}^+$ is the four-origin majorant, set

$$\overline{T}'(\gamma, \iota') = w_\star^{-1} \overline{T}^+(\gamma, \jmath\iota').$$

We first prove the stronger integral bound

$$T'(\gamma, \iota') = \iint g^\gamma F'_{\iota'}, \qquad \iint |g^\gamma| \, |F'_{\iota'}| \le \|\mathbb{W}\| \overline{T}'(\gamma, \iota'). \tag{733}$$

There is nothing to prove when $\mathbb{W} = 0$. Otherwise, put $c = w_\star/\|\mathbb{W}\|$ and introduce the Borel potential

$$\mathbb{U}(Y; B) = \log(1 + c|\mathbb{W}(Y; B)|).$$

It has the insertion's domain and sort, and $\|\mathbb{U}\| \le w_\star$. For the system with this fourth origin,

$$|F_{\jmath\iota'}(B)| = c|F'_{\iota'}(B)|.$$

Apply (718) to this admitted system and then Lemma 48.1. Dividing by c proves (733). The same lemma gives $\iint |g^\gamma| \, |F_\iota| \le \overline{T}(\gamma, \iota)$ for ordinary terms.

Let $\mathfrak{m}_+$ mark the new origin. In the absolute marked cluster series for the augmented system, retain only terms having exactly one new-origin vertex. The vertex can occur at any of the n polymers. Permuting that polymer to the first position preserves ρ^T and replaces $1/n!$ by $1/(n-1)!$. The map $\jmath$ and the definition of $\overline{T}'$ therefore give

$$\sum_{\text{refined insertion indices}} \frac{|\rho^T(\gamma_1, \dots, \gamma_n)|}{(n-1)!} \overline{T}'(\gamma_1, \iota'_1) \prod_{i=2}^{n} \overline{T}(\gamma_i, \iota_i) \le \frac{\overline{E}^{+,\mathfrak{m}_+}(X)}{w_\star} \le K_\Pi e^{-\lambda_{\mathrm{out}} d_t(X)}. \tag{734}$$

The last inequality uses $\eta^{\mathfrak{m}_+} \le H w_\star$ from (721). This is a summable majorant independent of the insertion and of the admitted potentials. Together with (733), it proves absolute convergence and (730). Linearity follows first under each absolutely convergent integral and then in the series. The support assertion holds term by term, since $Y'^\sharp \subset \gamma_1 \subset X$.

To identify the derivative, apply Proposition 47.4 to the fourth origin $\epsilon\mathbb{W}$. Its refined series is uniformly dominated on $|\epsilon| < \epsilon_\star$. A term containing p new-origin vertices has the factors

$\prod_{j=1}^{p}(e^{\epsilon\mathbb{W}(Y_j;B)}-1)$. At $\epsilon=0$ its derivative is zero unless $p=1$, in which case the single factor is replaced by $\mathbb{W}(Y';B)$. This differentiation may be performed under the field integral: use Cauchy's formula on a fixed circle $|\epsilon|<\epsilon_\star$, followed by (733)'s ordinary-term counterpart and Fubini. The common summable majorant then permits differentiation of the refined series by the same Cauchy formula. Exactly one of its polymers carries the derivative; the same permutation and factorial count as above give (731). If $\mathbb{W}=0$, the added-origin terms vanish and the identity holds directly.

Locality and orthogonal covariance follow by differentiating the corresponding equalities for the augmented system. For conjugation, differentiate along real ϵ. For holomorphic parameters, the uniform insertion bound provides one positive radius in ϵ; the augmented cluster sum is jointly holomorphic, so its partial derivative at zero is holomorphic as well.

Finally set $a=K_\Pi\|\mathbb{W}\|e^{-\lambda_{\mathrm{out}}d_t(X)}$. When $\mathbb{W}_1\neq 0$, let $r=w_\star/(2\|\mathbb{W}_1\|)\geq 1$ and

$$\phi(v)=E'[\underline{\mathbb{V}}+_{o_1}v\mathbb{W}_1;\mathbb{W}](X).$$

The system is admitted on $|v|\leq r$, so $|\phi(v)|\leq a$; it is holomorphic on $|v|<r$. If $r>1$, Schwarz's lemma applied to $(\phi(rz)-\phi(0))/(2a+\delta)$, followed by $\delta\downarrow 0$, gives $|\phi(1)-\phi(0)|\leq 2a/r$. For $r=1$ this is the triangle inequality. This yields the bound with $4K_\Pi/w_\star$, hence (732). The zero cases follow directly from linearity. □

Consider now a path $(\underline{\mathbb{V}}_v)_{0\leq v\leq 1}$ with fixed origins and sorts, at most three origins, and $\|\mathbb{V}_v^o\|\leq w_\star$. Suppose its potentials are jointly Borel in (v,B), are C^1 in v at every fixed B, and have velocities satisfying

$$V^o:=\sup_{0\leq v\leq 1}\|\dot{\mathbb{V}}_v^o\|<\infty.$$

The derivatives at the endpoints are one-sided.

Proposition 48.2. For this path,

$$E[\underline{\mathbb{V}}_1](X)-E[\underline{\mathbb{V}}_0](X)=\int_0^1\sum_o E'[\underline{\mathbb{V}}_v;\dot{\mathbb{V}}_v^o](X)\,dv. \tag{735}$$

The integrand is Borel and bounded in modulus by $K_\Pi\sum_o V^o\,e^{-\lambda_{\mathrm{out}}d_t(X)}$.

Suppose that the nonzero velocities have one sort and admit a decomposition into insertions $\dot{\mathbb{V}}_{v,b}$ of that sort, jointly Borel in (v,B), such that

$$\sum_o\dot{\mathbb{V}}_v^o(Y;B)=\sum_{b\in\mathfrak{X}^+}\dot{\mathbb{V}}_{v,b}(Y;B),\qquad \dot{\mathbb{V}}_{v,b}(Y;\cdot)=0\quad\text{if } b\notin\mathfrak{X}^\circ\langle Y\rangle.$$

If $\overline{v}=\sup_{v,b}\|\dot{\mathbb{V}}_{v,b}\|<\infty$, then the terms

$$E^\bullet(X,b)=\int_0^1 E'[\underline{\mathbb{V}}_v;\dot{\mathbb{V}}_{v,b}](X)\,dv \tag{736}$$

are defined and satisfy

$$\sum_{b\in\mathfrak{X}^+}E^\bullet(X,b)=E[\underline{\mathbb{V}}_1](X)-E[\underline{\mathbb{V}}_0](X), \tag{737}$$

$$|E^\bullet(X,b)|\leq K_\Pi\overline{v}\,e^{-\lambda_{\mathrm{out}}d_t(X)},\qquad E^\bullet(X,b)=0\quad\text{unless } b^\sharp\subset X. \tag{738}$$

They inherit locality, covariance, and conjugation from the path and its bond insertions. They are holomorphic in additional parameters when the Gaussian structure, path, velocities, and bond insertions satisfy the joint Borel and holomorphic hypotheses above with uniform norm bounds. ■

Proof. The velocities are jointly Borel as limits of difference quotients, using a forward increment except near the right endpoint and a backward increment there. For an ordinary Mayer index the tests are independent of v, and

$$\partial_v(e^{V_v}-1)=\dot{V}_v+\dot{V}_v(e^{V_v}-1).$$

Differentiate its finite product. Each summand is an insertion integrand in one of two ways: remove the differentiated vertex and insert its velocity, or retain that vertex and insert its velocity. The two constructions have the same tested region and support as the original index. Conversely, an insertion index of a fixed origin arises uniquely by the first construction if its vertex is absent, and by the second if its vertex is present.

Use the pointwise fundamental theorem of calculus for these finite products and integrate against g^{γ}. The bound (733), with $\|\mathbb{W}\| \le V^{o}$, justifies Fubini over the path and field variables. Summing the finite index sets gives

$$\mathsf{z}_{v_2}(\gamma) - \mathsf{z}_{v_1}(\gamma) = \int_{v_1}^{v_2} \sum_{o} \mathsf{z}'[\underline{\mathbb{V}}_v; \dot{\mathbb{V}}_v^{o}](\gamma)\, dv.$$

Thus each activity is absolutely continuous with this derivative almost everywhere. Products of the activities are absolutely continuous. In the derivative of a cluster product, the distinguished factor can be put first; symmetry again converts $n/n!$ to $1/(n-1)!$. Equation (734) bounds the resulting absolute series, uniformly in v, by $K_{\Pi} \sum_{o} V^{o} e^{-\lambda_{\mathrm{out}} d_t(X)}$. Fubini now permits summing the integrated derivatives, which proves (735). The integrands are Borel: their field integrands are jointly Borel, their integrals converge absolutely, and their cluster series are pointwise limits of Borel partial sums.

Apply Theorem S to each bond insertion and integrate its bound. Linearity in insertions of one sort and the finite bond sum prove (737). If $b^{\sharp} \not\subset X$, then b cannot lie in $\mathfrak{X}^{\circ}\langle Y \rangle$ for a domain $Y^{\sharp} \subset X$. Every insertion integrand therefore vanishes, proving the support assertion. The identities expressing locality, covariance, and conjugation hold before the path integral. Under the stated uniform parameter bounds the integrand is holomorphic at each v and bounded by an integrable constant, so integration also preserves holomorphy. □

48.2. **Changes in the sharp cutoff.** We now assume $0 < \mathfrak{b} < \infty$. All potentials in this subsection have sort Σ_{q}; their fields are therefore defined without a small-field test. There may be up to four origins, each with norm at most $w_{\star}$. For a bond b, let $\mathbf{t}^{b}(t', t)$ take the value t' at b and t at every other positive bond. The term assigned to this bond will be

$$E^{\chi}(X, b) = \int_0^1 \partial_{t'} E[\underline{\mathbb{V}}; \mathbf{t}^{b}(t', t)](X)\Big|_{t'=t} dt. \tag{739}$$

The derivative in this formula is an almost-everywhere derivative. Its existence requires an argument because the tests are characteristic functions.

Lemma 48.3. Let Ω be a countable disjoint union of finite-dimensional Lebesgue spaces, let $d \in L^{1}(\Omega; \mathbb{C})$, and let $\theta : \Omega \to (0, 1]$, $\theta' : \Omega \to [0, 1]$ be measurable. Suppose $\bar{k} \ge 0$ is integrable on $(0, 1)$,

$$\int_{\Omega} |d| \mathbf{1}_{\{t_1 < \theta \le t_2\}} \le \int_{t_1}^{t_2} \bar{k}(v)\, dv \quad (0 \le t_1 < t_2 \le 1), \qquad \int_{\Omega} |d| \mathbf{1}_{\{\theta = \theta'\}} = 0.$$

Set

$$H(t', t) = \int_{\Omega} d\mathbf{1}_{\{\theta \le t'\}} \mathbf{1}_{\{\theta' \le t\}}, \qquad S(t) = \int_{\Omega} d\mathbf{1}_{\{\theta' \le \theta \le t\}}.$$

Then S is absolutely continuous, $S(0) = 0$, and $\partial_{t'} H(t, t) = S'(t)$ for almost every $t \in (0, 1)$. Consequently,

$$\int_0^1 \partial_{t'} H(t, t)\, dt = \int_{\Omega} d\mathbf{1}_{\{\theta' \le \theta\}}. \tag{740}$$

■

Proof. Put

$$P_n(t) = \int_{\Omega} |d| \mathbf{1}_{\{\theta \le t\}} \mathbf{1}_{\{|\theta - \theta'| \le 1/n\}}.$$

The marginal bound shows that S and every P_n are absolutely continuous and vanish at zero. Moreover, P_n and $P_n - P_{n+1}$ are nondecreasing, so $P_n' \ge P_{n+1}' \ge 0$ almost everywhere. Dominated convergence and the tie hypothesis give

$$\int_0^1 P_n'(t)\,dt = P_n(1) \longrightarrow \int_\Omega |d|\mathbf{1}_{\{\theta=\theta'\}} = 0.$$

Outside one null set, S and all P_n are differentiable and $P_n'(t) \downarrow 0$.

Fix such a t. For $0 < h \le 1/n$, the difference between the increments $H(t+h,t) - H(t,t)$ and $S(t+h) - S(t)$ is supported where $t < \theta \le t+h$ and $t < \theta' \le \theta$. Its modulus is at most $P_n(t+h) - P_n(t)$. For $-1/n \le h < 0$, the corresponding difference is supported where $t+h < \theta \le t$ and $\theta < \theta' \le t$, and its modulus is at most $P_n(t) - P_n(t+h)$. Thus

$$\limsup_{h\to 0} \left| \frac{H(t+h,t) - H(t,t)}{h} - S'(t) \right| \le P_n'(t) \quad \text{for every } n.$$

Letting $n \to \infty$ proves the derivative assertion. The fundamental theorem for S gives (740). □

Theorem T (Cutoff variation)**.** In the preceding setting, (739) is defined for every $X \in \mathcal{D}_t$, $X \subset R$, and every positive bond b. The diagonal function $v \mapsto E[\underline{\mathbb{V}}; (v,\dots,v)](X)$ is absolutely continuous, and

$$E(X) - E^{\circ}(X) = \sum_{b \in \mathfrak{X}^+} E^{\chi}(X,b), \tag{741}$$

$$|E^{\chi}(X,b)| \le K_P^{\Pi} e^{4\lambda_1} \mathfrak{p} e^{-\lambda_{\mathrm{out}} d_t(X)}. \tag{742}$$

The term is zero unless $b^\sharp \subset X$, and it inherits locality, orthogonal covariance, and conjugation from the Gaussian data and the potentials. Under the joint measurability and holomorphy hypotheses of Proposition 47.4, with uniform constants and a fixed threshold, it is holomorphic in the parameters. ■□

Proof. Fix X. A refined cluster index is $\alpha = (n; (\gamma_i); (\iota_i))$, where $\iota_i = (\mathbf{D}_i, \mathfrak{e}_i)$, and its coefficient is $c_\alpha = \rho^T(\gamma_1,\dots,\gamma_n)/n!$. On the product of the real integration spaces, write $\omega = (\mathbb{X}_i, B_i)_{i=1}^n$ and define the density

$$d_\alpha(\omega) = c_\alpha \prod_{i=1}^n g^{\gamma_i}(Z_\circ(\iota_i); \mathbb{X}_i, B_i) F_{\iota_i}^{\mathbf{1}}(B_i).$$

Here $\mathbf{1}$ denotes the family of bond parameters equal to one. Lemma 48.1 and the Mayer majorants give

$$\int |d_\alpha| \le \bar{D}_\alpha := |c_\alpha| \prod_i \overline{T}(\gamma_i, \iota_i), \qquad \sum_\alpha \bar{D}_\alpha < \infty. \tag{743}$$

Let $Q_\alpha = \{(i,b) : b \in \mathfrak{e}_i^+\}$. Define $\theta(y) = \min\{1, \mathfrak{b}/|y|\}$ for $y \neq 0$ and $\theta(0) = 1$, and put $\theta_{i,b} = \theta(B_i(b))$. Since all origins have sort Σ_{q}, the only tests in a Mayer integrand are the out tests. Factoring them from their values at $\mathbf{1}$ yields

$$E[\underline{\mathbb{V}}; \mathfrak{t}](X) = \sum_\alpha \int d_\alpha(\omega) \prod_{(i,b)\in Q_\alpha} \mathbf{1}_{\{\theta_{i,b} \le t_b\}}\, d\omega. \tag{744}$$

In particular the density in this formula is independent of the bond parameters.

We verify the marginal and tie hypotheses of Lemma 48.3. Fix a test pair (i,b) and integrate $|d_\alpha|$ over all variables except $y = B_i(b) \in \mathbb{R}^3$. Denote the resulting nonnegative integrable density by $d_{\alpha,i,b}^{\perp}(y)$. It is supported in $|y| \ge \mathfrak{b}$. Polar coordinates followed by $r = \mathfrak{b}/v$ show that the marginal of $\theta_{i,b}$ has density

$$\bar{k}_{\alpha,i,b}(v) = \mathfrak{b}^3 v^{-4} \int_{S^2} d_{\alpha,i,b}^{\perp}((\mathfrak{b}/v)u)\, d\sigma(u), \qquad 0 < v < 1.$$

Its integral is $\int |d_\alpha|$; the sphere $|y| = \mathfrak{b}$ is null. For distinct test pairs the variables are distinct coordinate factors, even when the physical bonds agree and the cluster indices differ. The set on which their norms agree is null by Fubini, since a sphere in $\mathbb{R}^3$ is null. Thus ties between distinct ratios have zero $|d_\alpha|$-mass.

Let $\mathcal{A}^b(X)$ consist of the indices with at least one test at b. On each such space put

$$\theta^b = \max_{i:(i,b)\in Q_\alpha} \theta_{i,b}, \qquad \theta'^b = \max_{(i,c)\in Q_\alpha,\ c\neq b} \theta_{i,c},$$

where the second maximum is zero if its set is empty. The θ^b-marginal is dominated by $\bar{k}^b = \sum_{\alpha\in\mathcal{A}^b(X)} \sum_{i:(i,b)\in Q_\alpha} \bar{k}_{\alpha,i,b}$. It is integrable: its integral is bounded by the refined absolute cluster sum marked at the single bond b, namely $\overline{E}^{\{b\}}(X)$. The tie $\theta^b = \theta'^b$ is contained in a finite union of the pairwise ties in each integration space, and hence has zero mass.

Apply Lemma 48.3 on the disjoint union of the spaces indexed by $\mathcal{A}^b(X)$, with density d_α, and set

$$S_b(v) = \sum_{\alpha\in\mathcal{A}^b(X)} \int d_\alpha \mathbf{1}_{\{\theta'^b \le \theta^b \le v\}}.$$

The terms of (744) outside $\mathcal{A}^b(X)$ are independent of t'. The lemma therefore proves (739) and the useful formula

$$E^\chi(X,b) = S_b(1) = \sum_{\alpha\in\mathcal{A}^b(X)} \int d_\alpha \mathbf{1}_{\{\theta'^b \le \theta^b\}}. \tag{745}$$

Each S_b is absolutely continuous and vanishes at zero. Outside the null tie sets there is a unique largest ratio among all test pairs. Assigning it to its physical bond in (744) gives

$$E[\underline{\mathbb{V}}; (v,\dots,v)](X) = E^\circ(X) + \sum_b S_b(v).$$

The no-test terms are exactly E°; at zero every term with a test vanishes. This proves absolute continuity and (741).

Every term containing a test at b has $b^\sharp \subset X$, which proves the support assertion. Moreover, (745) and the marked bound give

$$|E^\chi(X,b)| \le \overline{E}^{\{b\}}(X) \le K_E \eta^{\{b\}} e^{-\lambda_{\mathrm{out}} d_t(X)}.$$

The bond weight is $\eta^{\{b\}} \le \mathsf{p} e^{\lambda_1 (d_t(b^\sharp)+4)} = \mathsf{p} e^{4\lambda_1}$, since $d_t(b^\sharp) = 0$. As $K_E = K_P^\Pi$, this is (742). The locality, covariance, and conjugation identities for $E[\mathbf{t}]$ hold for every bond parameter family. They therefore persist under the real difference quotients and the integral in (739).

For holomorphy use (745). The set selected by its indicator is independent of the complex parameters. For each fixed α, the integrand is jointly Borel, pointwise holomorphic in the parameters, and its L^1 norm is bounded by $\bar{D}_\alpha$ uniformly in them. These facts imply holomorphy of its integral. Indeed, on a polydisc of radius $2r$ about a point a, the iterated Cauchy formula expresses its integral at z, $|z_j - a_j| < r$, as an integral over the boundary torus with kernel

$$\prod_{j=1}^N \frac{2re^{i\vartheta_j}}{a_j + 2re^{i\vartheta_j} - z_j}.$$

This kernel has modulus at most 2^N. The uniform L^1 bound justifies Fubini before the field integration. The resulting boundary integral is holomorphic in z, by the same bound on each smaller polydisc. Finally (743) gives uniform convergence of the series of holomorphic integrals. The exceptional set of cutoff parameters where the original derivative fails to exist may vary with the complex parameter; the fixed-set formula (745) removes it from this argument. □

48.3. **Interpolation in the field variables.** Fix a finite threshold $\mathfrak{b}$ and $\mathsf{C}_{\mathcal{A}} \geq 3$. For a fine domain Y, put $J_Y = \mathfrak{X}^{\circ+}\langle Y\rangle$, and let

$$\mathbb{D}^{\mathrm{f}}(Y) = \{B : |B(b)| < \mathsf{C}_{\mathcal{A}}\mathfrak{b} \text{ for every } b \in J_Y\}. \tag{746}$$

The fields are complex and the norm at each bond is the matrix operator norm. For a family holomorphic on these polydiscs define

$$\|\mathbb{V}\|^{\mathrm{f}}_{\star,r} = \sup_{Y} \sup_{B\in\mathbb{D}^{\mathrm{f}}(Y)} \frac{e^{rd_k(Y)}|\mathbb{V}(Y;B)|}{Q_Y(B)}, \qquad Q_Y(B) = 1 + M^{-3}\sum_{b\in J_Y}|B(b)|^2. \tag{747}$$

Its restriction to real fields with $|B(b)| < \mathfrak{b}$ has sort Σ_{v}, with no larger norm.

Suppose $\mathbb{V}^{\mathrm{J}}, \mathbb{V}^{\mathrm{G}}, \mathbb{V}^{\mathrm{A}}, \mathbb{V}'$ are such families. For bond parameters $\mathbf{v} = (v_b) \in [0,1]^{\mathfrak{X}^+}$, define $(\mathbf{v}\cdot B)(b) = v_b B(b)$ at positive bonds, and set

$$\begin{aligned}\mathbb{V}^{\mathrm{int}}_{\mathbf{v}}(Y;B) &= \mathbb{V}^{\mathrm{J}}(Y;\mathbf{v}\cdot B)\\ &\quad + \int_0^1 (1-\sigma)\,\partial_\epsilon^2[\mathbb{V}^{\mathrm{G}} + \tau\mathbb{V}^{\mathrm{A}}](Y;\sigma\mathbf{v}\cdot B + \epsilon B)\Big|_{\epsilon=0}\,d\sigma,\\ \mathbb{V}'_{\mathbf{v}}(Y;B) &= \mathbb{V}'(Y;\mathbf{v}\cdot B).\end{aligned} \tag{748}$$

Write $\mathbb{V}_v$ for the restriction to the diagonal $\mathbf{v} = (v,\dots,v)$, and put

$$\dot{\mathbb{V}}^{\mathrm{int}}_{v,b} = \partial_{v_b}\mathbb{V}^{\mathrm{int}}_{\mathbf{v}}\Big|_{\mathbf{v}=(v,\dots,v)}, \qquad \dot{\mathbb{V}}'_{v,b} = \partial_{v_b}\mathbb{V}'_{\mathbf{v}}\big|_{\mathbf{v}=(v,\dots,v)}.$$

These parameters occur in the potentials; all cutoff parameters in the cluster sums remain equal to one.

Proposition 48.4. Assume the four norms in (747) are finite, $\mathbb{V}^{\mathrm{J}}(Y;0) = \mathbb{V}'(Y;0) = 0$, and $\mathbb{V}^{\mathrm{G}}, \mathbb{V}^{\mathrm{A}}$ vanish through degree two at zero. Put

$$n_{\mathrm{int}} = \|\mathbb{V}^{\mathrm{J}}\|^{\mathrm{f}}_{\star,r} + \|\mathbb{V}^{\mathrm{G}}\|^{\mathrm{f}}_{\star,r} + |\tau|\|\mathbb{V}^{\mathrm{A}}\|^{\mathrm{f}}_{\star,r}, \qquad n' = \|\mathbb{V}'\|^{\mathrm{f}}_{\star,r}. \tag{749}$$

The interpolated potentials are defined and jointly continuous in (v,B), and are C^1 in $v\in[0,1]$ at every admitted real field. They vanish at $v=0$; for $v>0$,

$$\mathbb{V}^{\mathrm{int}}_v(Y;B) = \mathbb{V}^{\mathrm{J}}(Y;vB) + v^{-2}[\mathbb{V}^{\mathrm{G}} + \tau\mathbb{V}^{\mathrm{A}}](Y;vB). \tag{750}$$

For every $v\in[0,1]$, these diagonal expressions are holomorphic on $\mathbb{D}^{\mathrm{f}}(Y)$ and satisfy

$$\|\mathbb{V}^{\mathrm{int}}_v\|^{\mathrm{f}}_{\star,r} \leq n_{\mathrm{int}}, \qquad \|\mathbb{V}'_v\|^{\mathrm{f}}_{\star,r} \leq n'.$$

The insertions are Borel in (v,B), vanish unless $b\in J_Y$, and obey

$$\partial_v\mathbb{V}^{\mathrm{int}}_v = \sum_b \dot{\mathbb{V}}^{\mathrm{int}}_{v,b}, \qquad \partial_v\mathbb{V}'_v = \sum_b \dot{\mathbb{V}}'_{v,b}, \tag{751}$$

$$\|\dot{\mathbb{V}}^{\mathrm{int}}_{v,b}\|_{\star,r} \leq 9n_{\mathrm{int}}, \qquad \|\dot{\mathbb{V}}'_{v,b}\|_{\star,r} \leq 9n', \tag{752}$$

$$\|\partial_v\mathbb{V}^{\mathrm{int}}_v\|_{\star,r} \leq 4\|\mathbb{V}^{\mathrm{J}}\|^{\mathrm{f}}_{\star,r} + \tfrac{9}{8}(\|\mathbb{V}^{\mathrm{G}}\|^{\mathrm{f}}_{\star,r} + |\tau|\|\mathbb{V}^{\mathrm{A}}\|^{\mathrm{f}}_{\star,r}), \tag{753}$$

$$\|\partial_v\mathbb{V}'_v\|_{\star,r} \leq 4n'.$$

The construction is linear in the four input families, with $\tau\mathbb{V}^{\mathrm{A}}$ regarded as one input. It preserves locality, orthogonal covariance, conjugation, and holomorphic dependence on additional parameters under uniform norm bounds. ∎

Proof. Fix Y, a real field below the threshold, and one of the four potentials $\mathbb{V}$. Write

$$f(\mathbf{x},B) = \mathbb{V}(Y;\mathbf{x}\cdot B), \qquad s(\mathbf{x},B) = \partial_\epsilon^2 f(\mathbf{x}+\epsilon\mathbf{1},B)\Big|_{\epsilon=0}, \qquad g_B(z) = \mathbb{V}(Y;zB).$$

These are holomorphic wherever the fields in their definitions belong to the polydisc. In particular f is holomorphic on a neighborhood of $|x_b| \le 3$, since the real field satisfies the strict bound $|B(b)| < \mathfrak{b}$. Put $A = \|\mathbb{V}\|^{\mathrm{f}}_{\star,r} e^{-r d_k(Y)} Q_Y(B)$. For $|x_b| \le c$, $c \ge 1$, the norm bound gives $|f(\mathbf{x}, B)| \le c^2 A$.

For $\mathbf{x} \in [0,1]^{J_Y}$, Cauchy's estimate on $|\zeta| = 1$ in $f(\mathbf{x} + \zeta e_b, B)$ gives $|\partial_b f| \le 4A$. Applied twice in the common direction $\epsilon\mathbf{1}$, it gives $|s| \le 8A$. On $|\epsilon|, |\zeta| \le 1$, all coordinates of $\mathbf{x} + \zeta e_b + \epsilon\mathbf{1}$ have modulus at most three. Cauchy's estimates in these two variables therefore give $|\partial_b s| \le 18A$. For $x \in [0,1]$, the circles of radii one and two about x give

$$|g'_B(x)| \le 4A, \qquad |g'''_B(x)| \le \frac{3!}{2^3} 9A = \frac{27}{4} A. \tag{754}$$

The integral in (748) is thus well defined. On the diagonal, integration by parts gives

$$\int_0^1 (1-\sigma) g''_B(\sigma v)\, d\sigma = v^{-2}\{g_B(v) - g_B(0) - v g'_B(0)\}, \qquad v > 0.$$

This proves (750). At zero the integral is $g''_B(0)/2 = 0$. For a complex field in the polydisc, g_B is holomorphic on a disc of radius greater than one and has modulus at most A on $|z| \le 1$. If it vanishes to order three, successive applications of Schwarz's lemma to $g_B(z), g_B(z)/z, g_B(z)/z^2$ give $|g_B(v)| \le A v^3$. Hence $|v^{-2} g_B(v)| \le Av$, including its zero value at $v = 0$. This proves the two polydisc norm bounds. Holomorphy for $v > 0$ follows by composition, and at $v = 0$ both functions are zero.

The integral expression is continuously differentiable in v, including at zero. Its derivative for either of the cubic inputs is $\int_0^1 (1-\sigma)\sigma g'''_B(\sigma v)\, d\sigma$. Equation (754) and $\int_0^1 (1-\sigma)\sigma\, d\sigma = 1/6$ give the factor $9/8$ in (753). Differentiating in one bond parameter instead replaces g'''_B by $\partial_b s$ in this formula. Its bound gives

$$|\dot{\mathbb{V}}^{\mathrm{int}}_{v,b}(Y; B)| \le 4A_{\mathrm{J}} + 3A_{\mathrm{G}} + 3|\tau| A_{\mathrm{A}}, \qquad |\dot{\mathbb{V}}'_{v,b}(Y; B)| \le 4A'.$$

These bounds are stronger than (752). They are locally bounded in B, so differentiation and continuity under the σ-integral follow by dominated convergence. Thus all insertions are Borel. Parameters outside J_Y do not occur, and the chain rule on the diagonal gives (751).

Linearity and locality hold in each defining operation. For covariance, the complex extension of a real orthogonal map of $\mathfrak{g}$ preserves the matrix norm on $\mathfrak{g}^c$. To check this, write $Z = A + iA'$ with A, A' Hermitian and traceless. Cayley–Hamilton and polarization give

$$|Z|^2 = \langle A, A\rangle + \langle A', A'\rangle + 2\sqrt{\langle A, A\rangle\langle A', A'\rangle - \langle A, A'\rangle^2}.$$

Indeed $Z^* Z = (\langle A, A\rangle + \langle A', A'\rangle) I + i[A, A']$, and the traceless Hermitian last term has eigenvalues equal to the positive and negative of the displayed square-root term. This also shows $|Z^*| = |Z|$. Thus the transformed fields stay in their polydiscs; bond multiplication commutes with the orthogonal maps, after the bonds have been relabeled. Covariance and conjugation now pass through the derivatives and real integrals. Joint holomorphy in further parameters follows from the same Cauchy formulas and the locally uniform bounds. If J_Y is empty, all four inputs vanish at its sole field, and every assertion holds with value zero. □

49. Domains and local terms of the effective action

The fluctuation expansion will be iterated on domains which control both the original field and its backgrounds on smaller towers. We distinguish these analytic cores from the pointwise domains used for the Gaussian construction.

49.1. **Analytic cores and restriction.** Assume $R_1 \ge L$, $M_1 \ge R_1\mathfrak{M}$, and $M \ge M_1$, with these scales powers of L. Let $1 \le j \le K$, $m' < m + K - j + 1$, and $X \in \mathcal{D}_{j-1}$. Use level-j units, so the bare spacing is $\xi = \eta_j = L^{-j}$. A triple is $P = (U, A', \mathbf{J})$ on $\mathrm{Bd}(X)$, with U real and the two forms complex. Its pair is

$$\mathbf{p}(P) = (\mathbf{U}, \mathbf{J}), \qquad \mathbf{U} = e^{i\eta_j A'} U. \tag{755}$$

The forms are specified at positive bonds; the expression for $\mathbf{U}$ is extended by reversal as a configuration. For a real gauge transformation on all endpoints of these bonds put

$$P^u = (U^u, \mathrm{Ad}(u)A', \mathrm{Ad}(u)\mathbf{J}), \qquad P^{\mathfrak{c}} = (U, A'^*, \mathbf{J}^*).$$

Adjoints are taken bondwise at positive bonds. A triple is real when both forms are Hermitian. Its conjugate pair has configuration $(\mathbf{U}^*)^{-1}$.

Definition 49.1. The simple domain $\mathcal{Q}_j(X; b_0, b_1, c_0)$ consists of triples satisfying

(1) $|\partial U - 1| < b_0\eta_j^2$, and every cubic box in X of side at most $\mathrm{o}LM$ admits a real gauge $U^w = e^{i\eta_j A}$ on its inner bonds with $|A|, |\nabla^{\eta_j} A| < \mathrm{o}LMB^{(\mathrm{cg})}b_0$;
(2) $|A'|, |\nabla_U^{\eta_j} A'| < b_1$;
(3) $|\partial\mathbf{U} - 1| < b_0\eta_j^2$ and $|\mathbf{J}| < c_0$.

Except for the explicitly inner-bond gauge bounds, each inequality is required wherever its stencil lies in $\mathrm{Bd}(X)$. Boxes are unions of bare cubes, with an injective rectangular lift of equal side lengths.

The core $\mathcal{U}_j^\circ(X; b_0, b_1, c_0)$ is the subset satisfying the following additional conditions. For every $1 \le p \le j$, every $Z \in \mathcal{D}_{p-1}$ with $Z \subset X$, both tower rules $\mathfrak{r}_1, \mathfrak{r}_2$, and every truncation $1 \le n \le p$, take the reduced determining set of the tower of Z, with all inward bonds omitted. The averaged complex datum must exist. Relative to the averaged real datum, its positive-bond quotients must lie in the logarithm ball $|V'V^{-1} - 1| < 1$, and their logarithms must lie strictly inside a polydisc on which the continuation of the real background exists. Write $U_n^{\mathfrak{r}}(Z; M'(\mathbf{U}))$ for this continued background and $J_n^{\mathfrak{r}}$ for its current at spacing η_n. With

$$c = b_0 L^{-(j-p)}, \qquad Z^{\wr p} = \bigcup\{\square \in \pi_{p-1} : \square^{\sim 2} \subset Z\},$$

require

$$|\partial U_n^{\mathfrak{r}} - 1| < cL^{-2p}, \qquad |J_n^{\mathfrak{r}}| < cL^{2(n-p)} \quad \text{on } Z^{\wr p}. \tag{756}$$

Require the same existence and bounds for the datum from U. The real and complex configurations in these conditions are the ones in the specified decomposition. When $c_0 = b_0$, omit the final radius. A pair lies in the core if it is the pair of a triple in the core. ∎

In these tower conditions the logarithmic perturbation is $B_l(b) = i^{-1}\log[M'(\mathbf{U})_l(b)M'(U)_l(b)^{-1}]$. The strict continuation requirement means that some radius $r > \sup_{l,b}|B_l(b)|$ is admitted. The resulting background is independent of that radius by continuation uniqueness. The conditions retain the existence of the backgrounds; pointwise curvature estimates alone do not assert it.

For $j \le j'$, $X \subset Y$, and $\lambda = L^{-(j'-j)}$, define

$$\rho_X^{j' \to j}(U, A', \mathbf{J}) = (U|_X, \lambda A'|_X, \lambda^3 \mathbf{J}|_X). \tag{757}$$

The pair has unchanged configuration and current scaled by λ^3. At the same level write $P|_X$.

Lemma 49.2. The simple domains and cores increase with each radius and pass to smaller domains at the same level. More generally, if $1 \le j \le j' \le K$, $m' < m + K - j' + 1$, $X \in \mathcal{D}_{j-1}$, $Y \in \mathcal{D}_{j'-1}$, and $X \subset Y$, then

$$P \in \mathcal{U}_{j'}^\circ(Y; b_0, b_1, c_0) \quad \Longrightarrow \quad \rho_X^{j' \to j} P \in \mathcal{U}_j^\circ(X; \lambda b_0, \lambda b_1, \lambda^3 c_0). \tag{758}$$

The same implication holds with $\mathcal{Q}$ in place of $\mathcal{U}^\circ$. ∎

Proof. Increasing a radius weakens each defining inequality; the indices of the tower conditions do not change. For restriction, all stencils and boxes of the smaller domain are stencils and boxes of the larger one, and the smaller tower index set is a subset of the larger.

For scaling put $\xi' = \eta_{j'} = \lambda\eta_j$. The complex configuration is unchanged because $\eta_j \lambda A' = \xi' A'$. Curvature bounds acquire the factor $\lambda^2 \le \lambda$. A box of side at most $\mathfrak{o}LM$ at level j has side at most that at level j'; keep its gauge and replace its potential A by λA. Potentials scale by λ and their derivatives by λ^2. The same calculation applies to A', and the current coordinate has the required factor λ^3. This proves the simple-domain assertion.

For (p, Z) in a lower-level tower condition, the real and complex configurations on $\mathrm{Bd}^\circ(Z)$ are unchanged. Thus the averaged data, logarithms, and continued backgrounds are the same. Their required constant is also unchanged: $(\lambda b_0)L^{-(j-p)} = b_0 L^{-(j'-p)}$. Every required tower condition therefore follows from the corresponding condition at level j'. □

49.2. **Local terms and their norms.**

Definition 49.3. For fixed real U, the slice of a core is $\mathcal{O}_U = \{(A', \mathbf{J}) : (U, A', \mathbf{J}) \in \mathcal{U}_j^\circ(X; b_0, b_1, c_0)\}$. A function on the core is analytic if near each point of every slice it is the restriction of a holomorphic function on an open neighborhood in the vector space of the two forms. It is a function of the pair if its value is independent of the admitted decomposition. It is gauge invariant if $E(P^u) = E(P)$ whenever both triples lie in the core, and conjugation real if $E(P^{\mathfrak{c}}) = \overline{E(P)}$ whenever both triples lie there. ■

Definition 49.4. For $m' < m + K - j$ and $X \in \mathcal{D}_j$, a local term of level j at radii (b_0, b_1) is a bounded, analytic, gauge-invariant function of the pair on $\mathcal{U}_j^\circ(X; b_0, b_1)$. Its norm is $\|E\|_X = \sup_{P \in \mathcal{U}_j^\circ(X; b_0, b_1)} |E(P)|$, with the supremum of an empty set equal to zero. ■

Definition 49.5. A pointed family consists of local terms $E(X, \cdot, z)$ for $X \in \mathcal{D}_j$ and $z \in X \cap \mathbb{T}^{(j)}$. A family without base point consists of local terms $H(X, \cdot)$ for $X \in \mathcal{D}_j$. Their norms are

$$\|E\|_{j,\kappa} = \sup_{X,z} e^{\kappa d_j(X)} \|E(X, \cdot, z)\|_X, \qquad \|H\|_{j,\kappa} = \sup_X e^{\kappa d_j(X)} \|H(X, \cdot)\|_X. \tag{759}$$

Unspecified members are zero. Evaluation at a real configuration means evaluation at its pair $(U, J^{(j)}(U))$, provided this pair lies in the core. Evaluation at $\mathbf{1}$ means evaluation at $(\mathbf{1}, 0)$, with the same requirement. ■

49.3. **Assigning local terms to sites.** For $0 \le j \le K$, use level-j units and put $q_j = 2L^{m+K-j}$. Periodize the product hat function:

$$\hat{h}(u) = \prod_{\nu=1}^{3} (1 - |u_\nu|)_+, \qquad h_z^{(j)}(y) = \sum_{n \in \mathbb{Z}^3} \hat{h}(\hat{y} - \hat{z} - q_j n). \tag{760}$$

Here $\hat{y}, \hat{z}$ are lifts, with $\hat{z} \in \mathbb{Z}^3$. Evaluate a tent at the midpoint of a bond and at the barycentre of a plaquette. At step $k+1$ write $h_z = h_z^{(k+1)}$.

Proposition 49.6. The tents are well defined, take values in $[0, 1]$, and sum to one. On every bond of a level $i \le j$,

$$h_z^{(j)}(b) = h_z^{(j)}(\bar{b}) = \tfrac{1}{2}\{h_z^{(j)}(b_-) + h_z^{(j)}(b_+)\}.$$

They commute with translations by level-j lattice vectors and with signed coordinate permutations. At step $k+1$, in level-k units,

$$h_z(b) \ne 0 \implies |x_b - z|_\infty < L, \quad |b_\pm - z|_\infty \le L, \qquad \sum_{b \in \mathfrak{x}_0^+} h_z(b) = 3L^3.$$

If $M \ge L$, then $h_z(b) \neq 0$ also implies $z \in b^\sharp$. Consequently, for numbers $E(X,b)$ vanishing unless $b^\sharp \subset X$,

$$\sum_{z \in X \cap \mathbb{T}^{(k+1)}} \sum_{b \in \mathfrak{X}^+} h_z(b) E(X,b) = \sum_{b \in \mathfrak{X}^+} E(X,b). \tag{761}$$

■

Proof. The period is at least $2L$, so at most one term in (760) is nonzero. Changing a lift reindexes the sum. In one dimension the translates of $\hat{h}_1(u) = (1 - |u|)_+$ sum to one; taking products proves the partition identity. The hat is affine on each unit interval. A level-i bond, expressed in level-j units, lies in such an interval in its varying coordinate, because L^{j-i} is an integer. This proves the midpoint identity. Evenness and the product formula prove the isometry assertions.

The support gives the strict midpoint distance. Its varying coordinate is a half-integer in level-k units, and the other coordinates are integers. Moving by one half to either endpoint therefore gives the stated endpoint bound. If $M \ge L$, the centers of the fine cubes containing z and b_- are at distance less than $M + L \le 2M$. Their coordinate differences are multiples of M; hence these cubes touch, and $z \in \square(b)^\sim \subset b^\sharp$.

For any positive integer N, summing the two linear pieces of the hat gives

$$\sum_{n \in \mathbb{Z}} \hat{h}_1(n/N) = \sum_{n \in \mathbb{Z}} \hat{h}_1((n + 1/2)/N) = N.$$

At the midpoints of the bonds of a fixed direction, one coordinate uses the shifted sum and two use the unshifted sum, with $N = L$. Their product is L^3. Summing the three directions proves the bond sum. Finally every contributing tent site belongs to X, and its sum over those sites is one. Interchanging the two finite sums proves (761). □

For $z \in \mathbb{T}^{(j)}$, let $\square_z$ be its unique cube in π_j and put $X_z = \square_z^\sim$. For $X \in \mathcal{D}_j$, define

$$\mathsf{A}^{\mathsf{t}}(X,P,z) = \begin{cases} \eta_j^{-1} \displaystyle\sum_p h_z^{(j)}(p) a_p(\mathbf{U}), & X = X_z, \\ 0, & X \neq X_z. \end{cases} \tag{762}$$

Only plaquettes with nonzero tent weight enter the sum.

Lemma 49.7. For $1 \le j \le K$, $m' < m + K - j$, and $M \ge L$, (762) is a pointed family at radii (b_0, b_1), vanishes at the trivial triple, and satisfies

$$\|\mathsf{A}^{\mathsf{t}}\|_{j,\kappa} \le \tfrac{3}{2} b_0^2 e^{7\kappa}.$$

On a whole-lattice configuration at which the terms are defined, their sum over (X,z) is $A^{\eta_j}(\mathbf{U})$. ■

Proof. The cubes touching $\square_z$ are joined to it by changing their coordinate offsets one at a time, so $X_z \in \mathcal{D}_j$. A tree of seven edges joining the eight corners of $\square_z$ meets all these cubes. Hence $d_j(X_z) \le 7$. A plaquette with nonzero tent weight has barycentre within one of z; its corners are within $1 + \eta_j/2 < L \le M$. They therefore lie in X_z, and all four bonds are available there.

For a fixed plaquette direction, the same product calculation as above, now with two shifted coordinates and $N = L^j$, gives total weight L^{3j}. Thus $\sum_p h_z^{(j)}(p) = 3\eta_j^{-3}$. For a complex matrix V of determinant one, $2 - \operatorname{tr} V = \det(V - 1)$ and $|\det(V-1)| \le |V-1|^2$. The core curvature bound therefore gives

$$|a_p(\mathbf{U})| \le \tfrac{1}{2} b_0^2 \eta_j^4, \qquad |\mathsf{A}^{\mathsf{t}}(X_z,P,z)| \le \tfrac{3}{2} b_0^2.$$

This and $d_j(X_z) \le 7$ prove the norm bound. The finite sum is holomorphic in A' on every slice, depends only on $\mathbf{U}$, and is gauge invariant by invariance of the plaquette trace. It is therefore a local term. At the trivial triple every plaquette action vanishes. Finally summing the partition of unity at each plaquette proves the action identity. □

50. Local Gaussian fluctuation fields

The fluctuation form has long range, even when the background is known only locally. We first separate it into connected local contributions. A second expansion, obtained from local resolvents, does the same for its inverse square root. Inserting one parameter for each coarse cube then gives both the covariance and the coupling to an independent Gaussian field. The positivity estimate below is uniform in the region; the final block identity identifies the resulting integral with the original fluctuation integral.

Fix a step $t = k+1$, $0 \le k < K$, and use level-k units: the spacings are $\xi_n = L^{n-k}$, the bare spacing is $\xi = L^{-k}$, and the torus has side $\ell_k = 2L^{m+K-k}$. The sequence is the constant sequence through level t. Write $\mathfrak{X}$ for its nonpivot level-k bonds, including both orientations. All kernels below use counting measure on positive bonds and the pairing $\frac{1}{2}\operatorname{tr}(XY)$. The transpose is bilinear; $|\cdot|_2$ denotes the operator norm for the Hermitian norm on $\mathfrak{g}^c$. Put $|b-b'| = |b_- - b'_-|_\infty$ and

$$\|T\|_\theta^{\mathrm{w}} = \max\left\{\sup_b \sum_{b'} |T(b,b')|_2 e^{\theta|b-b'|}, \sup_{b'} \sum_b |T(b,b')|_2 e^{\theta|b-b'|}\right\}.$$

These norms dominate the Hilbert-space norm and are submultiplicative. Restrictions and extensions by zero are denoted by χ and χ^{T}.

The coarse cubes have side LM, where $M = L^{m'}$ and $m' < m+K-k-1$. Choose a grain w and a mesh h, both powers of L, and a clearance r such that

$$\begin{gathered} R_1 \ge L, \quad M_1 \ge R_1\mathfrak{M}, \quad M_1 \le w, \quad Lw \le h, \quad Lh \le M, \\ \rho_{\mathrm{t}} + \rho_{\mathrm{c}} + L < r, \qquad 4(r+1) \le w, \qquad \delta_{\mathrm{cv}} w \ge 128 L \log(4n_{\mathrm{w}}), \\ \rho_{\mathrm{t}} = \frac{L(L+R_1+1)}{L-1}\mathfrak{M}, \qquad \rho_{\mathrm{c}} = 22 r_{\mathrm{cv}} L\mathfrak{M}. \end{gathered} \tag{763}$$

The covariance floors of theorem L are imposed on $\mathfrak{M}$ and $R_1\mathfrak{M}$. The constant n_{w} bounds the number, to the power $N-1$, of connected sets of N grain cubes containing a specified cube. The mesh has a further lower bound $h \ge h_\sharp(L)$, fixed in the proof. All these choices precede the lattice and the step.

Here and below, enlargement of a cube means enlargement by the indicated number of layers of cubes of its own partition. A region is a union of closed cubes; connectedness of a grain region allows face, edge and corner contacts. Write $\mathfrak{X}\langle Z\rangle$ for the nonpivot bonds with at least one endpoint in Z, and $\mathfrak{X}^\circ\langle Z\rangle$ for those with both endpoints in Z.

Background domain. For a region W, a pair $p = (\mathbf{U}, \mathbf{J})$ is written $\mathbf{U} = e^{i\xi A'}U$, with U real. We use pairs satisfying the following conditions at radii $(\alpha_0, \alpha_1, \alpha_2; a)$:

(1) Both plaquette curvatures are less than $\alpha_0 \xi^2 L^{-2}$.
(2) Every cubic box $q \subset W$ made of level-t $\mathfrak{M}$-cubes, of side at most $12\mathfrak{M}L$, lies in a box $q' \subset W$ on which U has a real gauge $U^v = e^{i\xi A}$ with $L|A|, L^2|\nabla^\xi A| \le \alpha_0$.
(3) $L|A'|, L^2|\nabla^U A'| < \alpha_1$ and $L^3|\mathbf{J}| < \alpha_2$.
(4) Every mesh cube $\square$ with $\square^{\sim 2} \subset W$ has a real gauge on $\square^{\sim 2}$ with $U^v = e^{i\xi A}$ and $L|A|, L^2|\nabla^\xi A| \le a$.

All bonds and stencils in these conditions are contained in W. Let $\mathcal{P}_h(W; \alpha_0, \alpha_1, \alpha_2; a)$ be this domain, and put $\mathcal{P}_{h,*}(W) = \mathcal{P}_h(W; a_{\mathrm{w}}, a_{\mathrm{w}}, a_{\mathrm{w}}; a_{\mathrm{w}})$, where the positive L-dependent radius a_{w} is chosen below. The conditions pass to restrictions. Their existential real-gauge condition is retained under small complex perturbations by keeping the real factor fixed; hence these are open domains when $\alpha_1 \le \pi$.

A level-t triple $P = (U, A'_P, J_P)$ is given on the bare bonds incident to its region. Its level-k reading is

$$p_{k,W}(P) = (e^{i\eta_t A'_P}U, L^{-3}J_P)|_W, \qquad \eta_t = L^{-t}, \qquad (U, A') = (U, L^{-1}A'_P). \tag{764}$$

The simple domain $\mathcal{Q}_t(X;b_0,b_1,c_0)$ of Definition 49.1 is expressed in level-t units. Its bounds imply the preceding pair conditions if

$$12\mathfrak{M} \le \mathsf{o}LM, \qquad 5h \le \mathsf{o}L^2M, \qquad b_0 \le a_{\mathrm{G}}M^{-1}, \quad b_1 \le \alpha_1, \quad c_0 \le \alpha_2,$$

where

$$a_{\mathrm{G}} = \frac{a_{\mathrm{w}}}{2\max\{1, \mathsf{o}LB^{(\mathrm{cg})}\}},$$

with $\alpha_0 = a_{\mathrm{w}}/2$. Indeed the two kinds of boxes in the pair conditions have side at most $\mathsf{o}LM$ after rescaling, while the potential and gradient rescale by L^{-1} and L^{-2}.

The operators. Let A denote the fluctuation form of (589), including the gauge term. The change of units in (764) gives the same form on the same nonpivot coordinates: the defining quadratic form is $(\mathsf{C}_V B, (\mathsf{G}_V^{(2)} + \Delta_k)\mathsf{C}_V B)_1$. Thus the restriction of the form and its inverse agree with those already constructed, including their zero-extension convention. We construct local coefficients a_S and r_ω below, with

$$\mathsf{A} = \sum_{S \text{ connected}} a_S, \qquad \mathsf{A}^{-1/2} = \sum_{\omega} r_\omega$$

on the whole torus. Their domains $X(S)$ and $X(\omega)$ are the coarse cube hulls of their figures. A coefficient is multiplied once by each parameter in its domain:

$$\mathsf{A}(s;p) = \sum_S \prod_{\Delta \subset X(S)} s(\Delta)\, a_S[p], \qquad r(s;p) = \sum_\omega \prod_{\Delta \subset X(\omega)} s(\Delta)\, r_\omega[p].$$

Terms whose parameter monomial vanishes are omitted. These expressions require the background only on the union of the surviving domains.

Let R be a coarse region and let $Z_\circ$ be a level-k region with $Z_\circ^\sharp = (Z_\circ^\sim)^\uparrow \subset R$. Complete parameters on the coarse cubes of $R \setminus Z_\circ^\sharp$ by setting $\bar{s} = 1$ on $Z_\circ^\sharp$ and $\bar{s} = 0$ off R. Put $\chi_\circ = \chi_{\mathfrak{X}\langle Z_\circ\rangle}$ and $E_\circ = \chi_\circ^{\mathsf{T}}\chi_\circ$. Define

$$\mathbf{C} = (\chi_\circ \mathsf{A}(\bar{s};p)\chi_\circ^{\mathsf{T}})^{-1}, \qquad \mathbf{\Gamma} = \chi_\circ \mathsf{A}(\bar{s};p)(1 - E_\circ) r(\bar{s};p)\chi_{\mathfrak{X}\langle R\rangle}^{\mathsf{T}}. \tag{765}$$

The reference operators $\mathbf{C}_{\mathrm{r}}, \mathbf{\Gamma}_{\mathrm{r}}$ are these operators at $s = 0$ and $p_{\mathrm{r}} = (U, 0)$.

Theorem U. There are positive constants $a_{\mathrm{w}}, a_{\mathrm{Str}}, \theta_{\mathrm{Str}}, C_{\mathrm{Str}}$, $0 < \gamma_{\mathrm{r}} \le 1$, and $c_{\mathrm{r}} \ge 1$, depending only on L, with the following property. Assume the geometric conditions above, $\bar{\kappa} \ge 1$, and

$$\theta_{\mathrm{Str}} M \ge 16\bar{\kappa}, \qquad 0 < \alpha_0 \le a_{\mathrm{w}}/2, \qquad \varepsilon_\circ := e^{-\theta_{\mathrm{Str}}M} + \alpha_1 + \alpha_2 \le a_{\mathrm{Str}}. \tag{766}$$

For every $p \in \mathcal{P}_h(R;\alpha_0,\alpha_1,\alpha_2;a_{\mathrm{w}})$, (765) defines a local Gaussian structure with $\widehat{\varepsilon} = C_{\mathrm{Str}}\varepsilon_\circ$:

(1) The kernels of $\mathbf{C}, \mathbf{C}^{-1}, \mathbf{\Gamma}$ are jointly holomorphic in the pair and the parameters on a neighborhood of the closed polydisc $|s(\Delta)| \le e^{\bar{\kappa}}$.

(2) If the parameters vanish off a coarse region Z, the structure is the direct sum of the structures on the wall-connected components of $Z_\circ^\sharp \cup Z$. Each coupling has input supported on the inner bonds of its component. Restricting the background to any region containing all surviving domains leaves these operators unchanged.

(3) The reference covariance is real and symmetric, with $\mathbf{C}_{\mathrm{r}}^{-1} \ge \gamma_{\mathrm{r}} I$, and

$$\|\mathbf{C}_{\mathrm{r}}^{-1}\|, \quad \|\mathbf{\Gamma}_{\mathrm{r}}\|, \quad \|\mathbf{C}_{\mathrm{r}}\mathbf{\Gamma}_{\mathrm{r}}\| \le c_{\mathrm{r}}.$$

Its coupling has input supported in $\mathfrak{X}^\circ\langle Z_\circ^\sharp\rangle$. Each of

$$\mathbf{C}^{-1} - \mathbf{C}_{\mathrm{r}}^{-1}, \qquad \mathbf{\Gamma} - \mathbf{\Gamma}_{\mathrm{r}}, \qquad \mathbf{\Gamma}^{\mathsf{T}}\mathbf{C}\mathbf{\Gamma} - \mathbf{\Gamma}_{\mathrm{r}}^{\mathsf{T}}\mathbf{C}_{\mathrm{r}}\mathbf{\Gamma}_{\mathrm{r}}$$

has norm $\|\cdot\|^{\mathrm{w}}_{4\theta_{\mathrm{Str}}}$ at most $\widehat{\varepsilon}$.

(4) On the whole torus the covariance is $\mathbf{C}_* = \mathsf{A}(p)^{-1}$, and at $s = 1$

$$\mathbf{C} + \mathbf{C}\boldsymbol{\Gamma}\boldsymbol{\Gamma}^{\mathsf{T}}\mathbf{C} = \chi_\circ \mathbf{C}_* \chi_\circ^{\mathsf{T}}. \tag{767}$$

The real whole covariance $\mathbf{C}_{*\mathrm{r}} = \mathsf{A}(p_{\mathrm{r}})^{-1}$ satisfies $\gamma_{\mathrm{r}} I \le \mathbf{C}_{*\mathrm{r}}^{-1} \le c_{\mathrm{r}} I$ and $\|\mathbf{C}_*^{-1} - \mathbf{C}_{*\mathrm{r}}^{-1}\| \le \widehat{\varepsilon}$. Empty cores have the zero-dimensional convention. If, in addition, $e^{-\theta_{\mathrm{Str}} M}, \alpha_1 + \alpha_2 \le (LM)^{-3}$, then $4C_{\mathrm{Str}} + c_{\mathrm{r}} + 3$ is a defect number for the unscaled structure. For $|\tau - 1| \le \frac{1}{2}(LM)^{-3}$, the scaled structure $(\tau^{-1}\mathbf{C}, \tau^{1/2}\boldsymbol{\Gamma})$, with the principal root, has the same references and defect at most

$$\widehat{\varepsilon}_* - m_0, \qquad \widehat{\varepsilon}_* = 2\widehat{\varepsilon} + c_{\mathrm{r}}(LM)^{-3}, \qquad m_0 = \tfrac{1}{2} c_{\mathrm{r}} (LM)^{-3} > 0. \tag{768}$$

Its whole covariance is $\tau^{-1}\mathbf{C}_*$. ■□

The proof occupies the rest of this section. The cancellation which creates the a_S is what makes their parameter dependence local. The mesh resolvents preserve that locality for the inverse square root. Their weighted bounds then separate the exponentially small contribution from outside the core collar from the small complex change of background.

50.1. **Connected coefficients of the form.** For a grain region Y, let $\Omega_0(Y) = Y$ and form its maximal separated tower by retaining at stage $q+1$ the level-$(q+1)$ $\mathfrak{M}$-cubes contained in $\Omega_q(Y)$ and at distance strictly greater than $R_1 \mathfrak{M} \xi_q$ from its complement. Put $S_0(b) = \{\sigma : \mathrm{dist}_\infty(b, \sigma) < r\}$ and $Y^- = \{b \in \mathfrak{X} : S_0(b) \subset Y\}$. The clearance ensures that Y^- consists of fluctuation bonds of this tower. The background is admissible on every such tower, with the same radii. Indeed a site in layer n of the meet lies in an original layer $n' \ge n$, so its curvature, potential, derivative and current bounds only weaken. Each natural gauge cube of the meet is contained in a plain gauge cube of the original sequence lying in Y. Restrict that gauge to the smaller cube; its containing box is then inside Y. For the constant sequence the enclosing box can be seen directly. If the natural cube is already at level t, use that cube. Otherwise its side is at most $12\mathfrak{M}\xi_n < \mathfrak{M}L$. In each coordinate enclose its interval in two adjacent intervals of length $\mathfrak{M}L$: use the two intervals straddling a crossed grid point, or, if there is no crossing, choose the neighboring interval inside the same interval of length $\mathfrak{M}L^2$. Every such coarse interval meets the original interval in a bare coordinate. Taking products, each $\mathfrak{M}L^2$-cube used contains a bare site of the original cube. Since $w > \mathfrak{M}L$ and these sides are powers of L, the grain partition is coarser than this partition; those cubes therefore lie in Y. The resulting box has side $2\mathfrak{M}L$ and lies in Y. Thus the local regularity conditions, including the containing-box condition, hold before any operator is evaluated. Let g_Y be its fluctuation form compressed to Y^- and extended by zero, and put $g_\emptyset = 0$. Define

$$a_S = \sum_{Y \subset S} (-1)^{\#S - \#Y} g_Y. \tag{769}$$

Then $\sum_{S \subset Y} a_S = g_Y$: the coefficient of $g_{Y'}$ is $\sum_{Y' \subset S \subset Y} (-1)^{\#S - \#Y'}$, which is zero unless $Y' = Y$. In particular the whole-torus sum is A.

The tower rule has range $e_n = (L + R_1 + 1)\mathfrak{M} \sum_{q<n} \xi_q$, so $e_{k+1} \le \rho_{\mathrm{t}}$. Indeed membership at stage $q+1$ checks precisely the level-q cubes within $R_1 \mathfrak{M} \xi_q$ of the containing level-$(q+1)$ cube. Their sites are within $(L + R_1 + 1)\mathfrak{M}\xi_q$ of the original site. Induction proves the claim, separation and maximality. Thus changing a grain cube farther than e_{k+1} from a cell hull changes neither the tower nor its fluctuation coordinates there.

We use the form expansion obtained from the same inverse and multiplier terms as theorem L. Its three kinds of terms are

$$\begin{aligned} A_\kappa^{(1)} &= \theta_\kappa \mathsf{C}^{\mathsf{T}} (\mathsf{G}_V^{(2)} - 1 - \mathsf{E}^{(2)}[j_1^0]) \mathsf{C}, \\ A_\Xi^{(2)} &= \mathsf{C}^{\mathsf{T}} \mathsf{J}^{\mathsf{T}} [\mathbb{E}_\Xi]_{\omega,8} \mathsf{J} \mathsf{C}, \\ A_{\kappa,\Xi'}^{(3)} &= -\theta_\kappa \mathsf{C}^{\mathsf{T}} \mathsf{E}^{(2)}[j_{1,\Xi'}] \mathsf{C}. \end{aligned} \tag{770}$$

In the third kind $|\omega(\Xi')| \geq 1$: the zero-length multiplier terms are already included in j_1^0. Here C includes the extension of the coordinates under consideration. The first term is attached to $X^{(3)}(\kappa)$, the second to the tree of Ξ, and the third to $X^{(3)}(\kappa)$ followed by the reversed trunk of Ξ' and its branches. They sum to the form by inserting the existing expansions of $(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}$ and j_{1*} into (589). Absolute convergence permits this regrouping.

The outer kernels link only bonds in a common double block, at most $36L^3$ neighbors per bond. The first and third kinds therefore have both endpoints in the first cell; the second has its input in the last trunk cell. A nonzero third-kind term forces the final multiplier cell to overlap its new first cell. Marking branch starts and reversing the trunk reduces its decoration count to the count already used for the multiplier expansion. Moving endpoints through the outer kernels adds at most $8L$ to the second-kind tree length and $16L$ to the third; the first kind has length at most $12L$. The inverse and multiplier majorants, these finite neighbor sums and the decoration count give

$$|A_{Y,\omega}(b,b')| \leq B_0(C_{\mathrm{cv}}\mathfrak{M}^{-1/2})^{|\omega|}\mathfrak{M}^{-|\omega|/2}e^{-\delta_{\mathrm{cv}}d^{\mathrm{tr}}(\omega;y(b),y(b'))/2}. \tag{771}$$

For the third kind, if the original trunk length is $m' \geq 1$, the new length is $m'+1$; the factor $\mathfrak{M}^{-3m'}$ absorbs $\mathfrak{M}^{-(m'+1)}$, and the at most $2^{m'}$ branch markings are absorbed in C_{cv}. The second kind has the stronger factor $\mathfrak{M}^{-3|\omega|}$. Enlarge the covariance constants and their fixed floor on $\mathfrak{M}$ together, so that the tree count times $C_{\mathrm{cv}}\mathfrak{M}^{-1/2}$ remains at most $1/2$; decrease the rate once if necessary.

The local products are defined as follows. In the first kind use the local $\mathsf{C}_\kappa, \mathsf{E}_\kappa, \mathsf{G}_\kappa$, restrict j_1^0 to $\kappa^{[5/6]}$, cut the input to $\kappa^{[2/3]}$ and multiply the output by θ_κ. In the second kind use the local JC at the last trunk cell, with input cut $\kappa^{[2/3]}$ and row cut $\kappa^{[1/12]}$, followed by $\mathbb{E}_\Xi$ and the local $\mathsf{C}^{\mathsf{T}}\mathsf{J}^{\mathsf{T}}$ at the first cell, with output cut $\kappa_0^{[5/6]}$. In the third kind use the already constructed local solution $W_{1,\kappa}[j_{1,\Xi'}]$ and multiply its ninth component by $-\theta_\kappa$. These are exactly the local restrictions of (770). Their factors are holomorphic on their carriers. Equal tower traces on the cell hulls and equal fluctuation coordinates give equal decorations and equal local matrices, hence equal terms.

Proposition 50.1. The coefficients a_S are symmetric and holomorphic in the background on $S^\sim$. They vanish unless S is connected, and their kernels vanish unless $S_0(b) \cup S_0(b') \subset S$. There are L-dependent constants $C_a < \infty$ and $\delta_a = \delta_{\mathrm{cv}}/(4L) > 0$ such that

$$|a_S(b,b')| \leq C_a\epsilon^{\#S-1}e^{-\delta_a\Lambda_S(b,b')}, \qquad \epsilon = (2n_{\mathrm{w}})^{-1}. \tag{772}$$

Here Λ_S is the least length of a connected finite union of segments through b_-, b'_- meeting $\sigma^\sim$ for every cube $\sigma \subset S$. ∎

Proof. The support and symmetry follow from (769). Expand each $g_Y(b,b')$ in trees. Their weighted length is at least $L^{-1}(\lambda_\omega - 2)$, where λ_ω is the plain tree length through b_-, b'_- and the cells: all weights are at least L^{-1}, and the two unit bonds connect the preferred endpoints. Only a bounded number of first cells contain b_-, and the tree count times $C_{\mathrm{cv}}\mathfrak{M}^{-1/2}$ is at most $1/2$. The remaining majorants thus have a uniformly bounded sum.

Let $N(\omega)$ consist of the grain cubes within e_{k+1} of the walk hull. If S contains a cube outside $S_0(b) \cup S_0(b') \cup N(\omega)$, toggle the first such cube in a fixed ordering. The finite-range observation and local product identity give the same tree and kernel on the toggled tower, with opposite sign in (769). This is an involution; absolute convergence justifies the cancellation.

For a survivor every cube of S lies within $r+1$ of a tree through the endpoints and the cells. For cubes in S_0 this is the definition; for the others use the hull diameter ρ_{c} and $e_{k+1} + \rho_{\mathrm{c}} + L < r$. Hence

$$\lambda_\omega \geq \max\{\Lambda_S(b,b'), (\#S-8)w/32\}.$$

For the second bound, traverse a connected segment cover twice and divide the tour into pieces of length less than $w/2$. Every cube within $r+1 \leq w/4$ of the cover is within $w/2$ of one of at most $4\lambda_\omega/w + 1$ selected points. At most eight grain cubes lie this close to a point, by counting intervals in each coordinate. Approximation of the infimum gives the bound. The cells of a nonzero tree

form a connected set in Y; its intersecting grain cubes form a connected subset of Y, and every surviving cube touches this subset. Thus S is connected.

There are at most $2^{\#S}$ subregions. Spend half the exponential decay on each of the two lower bounds for λ_ω. The grain inequality in (763) absorbs $2^{\#S}$ and gives (772); the cases $\#S \le 8$ are absorbed in C_a. Holomorphy follows from locally uniform convergence of the tree sums and the finite alternating sum. The rooted exploration-tree count of connected grain regions gives

$$\sum_{S \supset S_0(b),\ S \text{ connected}} \epsilon^{\#S-1} \le 2. \tag{773}$$

Indeed a depth-first traversal records one of at most n_{w}^{N-1} connected sets of N cubes through a specified cube, and $n_{\mathrm{w}}\epsilon = 1/2$. □

50.2. **Local resolvents and their strings.** Write $\gamma > 0$ for the flat gap in (593). It applies to every maximal tower above. We first transfer it to local backgrounds, and then construct the inverse square root without extending those backgrounds beyond their domains.

Put $c_{\mathrm{lat}}(v) = 3\sum_{z\in\mathbb{Z}^3} e^{-v|z|_\infty}$. Least lifts of torus sites give the same bound for sums over positive bonds. By (773), with $\theta_a = \delta_a/2$, every sum of coefficients with multipliers of modulus at most κ has weighted norm at most κB_a, where

$$B_a = \max\{\gamma,\, 2\sqrt{2}C_a c_{\mathrm{lat}}(\theta_a)\}.$$

If all its terms have $\Lambda_S \ge D$, its unweighted norm is at most $\kappa B_a e^{-\theta_a D}$. Here $\sqrt{2}$ converts the matrix norm on $\mathfrak{g}^c$ to its Hermitian norm.

The complex comparison uses the disc

$$p_z = (e^{iz\xi A'}U, z\mathbf{J}), \qquad |z| < R = \frac{c_{\mathrm{d}} a}{\alpha_1 + \alpha_2}, \qquad c_{\mathrm{d}} = 1/16. \tag{774}$$

If $\alpha_0 \le a/2$ and $a \le 1$, this disc satisfies the first three pair conditions at radii (a, a, a); its real mesh gauges are unchanged. Only curvature needs verification. At a plaquette based in layer n, put $\vartheta = \xi/\xi_n$ and $\tau = |z|\alpha_1$. Transport its four perturbations to the base point using U. The product is EP, where $P = \partial U$ and $E = e^{iX_1}e^{iX_2}e^{-iX_3}e^{-iX_4}$. The two transverse values are expressed through the covariant differences at the base point. Consequently

$$\sum_j |X_j| \le 6\tau\vartheta, \qquad |X_1 + X_2 - X_3 - X_4| \le \tau\vartheta^2(2 + 4\alpha_0).$$

The latter bound uses $|(1 - \mathrm{Ad}\, P)X| \le 2|P - 1|\,|X|$. The exponential series, with its terms of degrees zero and one removed, has norm at most $\frac{1}{2}s^2e^s$, $s = \sum|X_j|$. Thus

$$|E - 1| \le \vartheta^2\tau(2 + 4\alpha_0 + 18\tau e^{6\tau}) < \tfrac{1}{2}a\vartheta^2. \tag{775}$$

Together with $|P - 1| < \alpha_0\vartheta^2$, this proves the claim. It avoids changing scales at the opposite corners of the plaquette. Schwarz's lemma, applied to scalar matrix elements and then taking the operator norm, now bounds the difference of any holomorphic coefficient from its value at p_0 by $2/R$ times its uniform majorant.

Choose $a_{\mathrm{w}} > 0$ so that $a_{\mathrm{w}} \le c_{\mathrm{d}} a_{\mathrm{cv}}\gamma/(64B_a)$ and the inherited covariance radii hold. A real mesh gauge with bounds a_{w} can be viewed relative to the flat reference, at radii $(a_{\mathrm{w}}, 3a_{\mathrm{w}}/2, a_{\mathrm{w}}/2)$. Apply (774) with $a = a_{\mathrm{cv}}$. The resulting form differs from its flat value by at most $\gamma/16$. The same bound compares a pair in $\mathcal{P}_{h,*}$ with its real reference. Unitary gauge covariance preserves both the Hermitian norm and the form inequality.

For a mesh cube with center $y_\square$, use the periodic product cutoff

$$h_\square(b) = \prod_{\nu=1}^{3} \cos\!\left(\frac{\pi(x_b - y_\square)_\nu}{2h}\right) \quad \text{if } |x_b - y_\square|_\infty < h, \qquad h_\square(b) = 0 \text{ otherwise},$$

where x_b is the midpoint and least lifts are used on the support. The one-dimensional identity $\cos^2 u + \sin^2 u = 1$ gives $\sum_{\square} h_{\square}^2 = 1$. At most eight cutoffs are nonzero at a bond, and $|h_{\square}(b) - h_{\square}(b')| \le (3\pi/2h)(|b - b'| + 1/2)$. Consequently the kernel identity

$$O - \sum_{\square} h_{\square} O h_{\square} = \tfrac{1}{2} \sum_{\square} (h_{\square}(b) - h_{\square}(b'))^2 O(b, b')$$

has norm at most $18\pi^2 h^{-2}(4/\theta^2 + 1/2)\|O\|_{\theta}^{\mathrm{w}}$. This is the only localization loss in the gap estimate.

Let $\mathfrak{X}^{\triangleright}\langle W\rangle$ contain both orientations of those positive bonds whose initial point lies in W. Compress $\sum_{S^{\sim} \subset \square^{\sim 2}} a_S$ to $\mathfrak{X}^{\triangleright}\langle \square^{\sim}\rangle$, obtaining $a^{\square}$. At the flat pair the coefficient sum is the tower form for the union of all grains with $\sigma^{\sim} \subset \square^{\sim 2}$. The compressed bonds are deep in that union, so its gap is γ. The two disc comparisons give a real gap at least $7\gamma/8$ and a complex real-part gap at least $3\gamma/4$ for $a^{\square}$.

More generally, take any reversal-invariant bond set Y and any partial sum $A_{\mathcal{W}} = \sum_{S \in \mathcal{W}} a_S$ that is full near Y: whenever $h_{\square}$ meets Y, its enlarged cube lies in the background domain and $\mathcal{W}$ includes all S with $S^{\sim} \subset \square^{\sim 2}$. For a term excluded from this local sum and a bond starting in $\square^{\sim}$, the length is at least $h - 3w \ge 10h/13$. Indeed choose a point of $S^{\sim}$ outside $\square^{\sim 2}$ and a point of its grain enlargement on a connecting tree; their distance is at most $3w$, while the first point is more than h from $\square^{\sim}$. If $h_{\square}(b') > 0$, the same argument from the midpoint instead gives $\Lambda_S(b, b') \ge 3h/2 - 3w - 1/2 \ge h$. Choose $h_{\sharp}$ so that

$$B_a e^{-10\theta_a h/13} \le \gamma/32, \qquad 18\pi^2 h^{-2}(4/\theta_a^2 + 1/2) B_a \le \gamma/16.$$

Localization by $\sum h_{\square}^2 = 1$ then gives real gap $3\gamma/4$ for the compression to Y; the complex comparison leaves real-part gap $\gamma/2$.

For completeness, the resolvent estimate used next follows by the same exponential conjugation for every compression. If $\|K\|_{\theta_0}^{\mathrm{w}} \le B$ and $\operatorname{Re} K \ge gI$, then

$$|(\mu + K)^{-1}(b, b')|_2 \le (\mu + g/2)^{-1} \exp\!\left(-\frac{\theta_0 g}{2B}|b - b'|\right), \qquad \mu \ge 0. \tag{776}$$

Conjugate by $e^{\theta|b-b'|}$, with $\theta = \theta_0 g/(2B)$. Convexity gives $|e^s - 1| \le (\theta/\theta_0) e^{\theta_0 d}$ when $|s| \le \theta d$, so the conjugated operator differs from K by at most $g/2$ in norm. Its inverse has norm at most $(\mu + g/2)^{-1}$, proving (776).

We now keep the above Y and full partial sum, writing A_Y for its compression. Put $Y^{\square} = Y \cap \mathfrak{X}^{\triangleright}\langle \square^{\sim}\rangle$ and let $E_{\square}$ be its projection. All operators in the following formulas are extended by zero to Y:

$$\begin{aligned} g_{\square}^{\mu} &= (\mu + E_{\square} a^{\square} E_{\square})^{-1} \quad \text{on } Y^{\square}, \\ k_{\square,S} &= 1_Y \big(a_S h_{\square} - 1_{\{S^{\sim} \subset \square^{\sim 2}\}} h_{\square} a_S\big) E_{\square}, \\ \Pi^{\mu} &= \sum_{\square} h_{\square} g_{\square}^{\mu} h_{\square}, \qquad e^{\mu} = -\sum_{\square, S} k_{\square,S} g_{\square}^{\mu} h_{\square}. \end{aligned}$$

Multiplying out, using fullness and $\sum h_{\square}^2 = 1$, gives $(\mu + A_Y)\Pi^{\mu} = 1_Y - e^{\mu}$. The products in $\Pi^{\mu}(e^{\mu})^n$ are indexed by strings $\omega = (\square_0; (S_1, \square_1), \dots, (S_n, \square_n))$: $S_i^{\sim}$ must meet both neighboring mesh enlargements, or the product vanishes. Their figures are $F_{\omega} = \bigcup_i \square_i^{\sim 2} \cup \bigcup_i S_i^{\sim}$. Define

$$g_{\omega}^{\mu} = h_{\square_0} g_{\square_0}^{\mu} h_{\square_0} \prod_{i=1}^{n} (-k_{\square_i, S_i} g_{\square_i}^{\mu} h_{\square_i}), \qquad r_{\omega} = \frac{1}{\pi} \int_0^{\infty} \mu^{-1/2} g_{\omega}^{\mu}\, d\mu. \tag{777}$$

Each product uses the background only on F_{ω}. Its input and output bonds have both endpoints in that figure.

Here is the convergence estimate, including the loss from the commutators. Put $m_S = \sqrt{2}C_a\epsilon^{\#S-1}e^{-\delta_a\Lambda_S}$ on its prescribed support, and zero elsewhere. A majorant for $k_{\square,S}$ is

$$\mathfrak{k}_{\square,S}(c,c') = \begin{cases} \dfrac{3\pi}{2h}(|c-c'|+1/2)1_{\{h_\square(c)+h_\square(c')>0\}}m_S(c,c'), & S^\sim \subset \square^{\sim 2}, \\ 1_{\{h_\square(c')>0\}}m_S(c,c'), & \text{otherwise.} \end{cases}$$

Set $\theta_\square = 3\theta_a\gamma/(8B_a)$ and $\widehat{g}(c,c') = (4/\gamma)e^{-\theta_\square|c-c'|}$. By (776), $|g^\mu_\square| \le \gamma(4\mu+\gamma)^{-1}\widehat{g}$. Choose $0 < \theta_g \le \min(\delta_a/4, \theta_\square/6)$. The preceding support and length bounds, the sixteen possible cutoffs at a pair of bonds, and (773) give

$$\widehat{K}(c,c') := \sum_{\square,S}\mathfrak{k}_{\square,S}(c,c')e^{\theta_g\Lambda_S(c,c')} \le 2\sqrt{2}C_a e^{-\delta_a|c-c'|/2}\left[\frac{24\pi}{h}(|c-c'|+1/2)+8e^{-\delta_a h/4}\right].$$

Its row and column sums are bounded by a number $\epsilon_k(h) \to 0$. The kernel $\widehat{g}(c,c')e^{3\theta_g|c-c'|}$ has both sums at most C_g/γ, where $C_g = 4c_{\mathrm{lat}}(\theta_\square/2)$. Increase $h_\sharp$ so that $\epsilon_k(h)C_g/\gamma \le 1/2$.

Let $\Lambda_\omega(b,b')$ be the least connecting length through the endpoints and every piece $\square_i^{\sim 2}, \sigma^\sim$ of F_ω. Join the intermediate bonds in the ordered product by coordinate paths, and join the pieces by the trees for the S_i. This gives $\Lambda_\omega \le 3|b-c_0| + \sum_i(\Lambda_{S_i}(c_{i-1},c_i') + 3|c_i'-c_i|)$. Replacing the factors of g^μ_ω by $h_{\square_0}\widehat{g}h_{\square_0}$ and $\mathfrak{k}_{\square_i,S_i}\widehat{g}h_{\square_i}$ defines a nonnegative majorant m_ω, for which

$$\sup_b \sum_{b'}\sum_{|\omega|=n} m_\omega(b,b')e^{\theta_g\Lambda_\omega(b,b')} \le \frac{C_g}{\gamma}2^{-n}. \tag{778}$$

The identical column estimate follows by summing products in the opposite order. Moreover $|g^\mu_\omega| \le (\gamma/(4\mu+\gamma))^{n+1}m_\omega$ and $|r_\omega| \le \sqrt{\gamma}\, m_\omega/2$. These bounds prove convergence and holomorphy of the integrals and all string sums, locally uniformly on the background domain.

The error e^μ has row and column sums at most $\frac{1}{2}\gamma/(4\mu+\gamma)$, by the same $\widehat{K}\widehat{g}$ bound. Hence $(\mu+A_Y)\sum_{n=0}^N \Pi^\mu(e^\mu)^n = 1_Y - (e^\mu)^{N+1}$ and $\sum_\omega g^\mu_\omega = (\mu+A_Y)^{-1}$. Integration gives $\sum_\omega r_\omega = A_Y^{-1/2}$. Indeed accretivity implies $\|e^{-tA_Y}\| \le e^{-\gamma t/2}$; the Laplace resolvent formula identifies the integral with $\pi^{-1/2}\int_0^\infty t^{-1/2}e^{-tA_Y}\,dt$. Squaring and using $t+s=u$ gives the beta integral $\int_0^1[v(1-v)]^{-1/2}dv = \pi$, hence its square is $\int_0^\infty e^{-uA_Y}\,du = A_Y^{-1}$. It is symmetric because every resolvent is. This also proves the claimed identity on the whole torus.

50.3. **Parameters, positivity and conditioning.** We finish the proof of theorem U. The same majorants control the form coefficients and the strings. Set $n_S = m_S$, $n_\omega = \sqrt{\gamma}\, m_\omega/2$, and $\theta_\flat = \min(\theta_g, \delta_a/2)$. Then (773) and (778) give

$$\begin{aligned} \sup_b \sum_{b',S} n_S(b,b')e^{\theta_\flat\Lambda_S(b,b')} &\le b_a := 2\sqrt{2}C_a c_{\mathrm{lat}}(\delta_a/2), \\ \sup_b \sum_{b',\omega} n_\omega(b,b')e^{\theta_\flat\Lambda_\omega(b,b')} &\le b_r := C_g/\sqrt{\gamma}. \end{aligned} \tag{779}$$

Both estimates hold with rows and columns exchanged. On the disc (774), the coefficient differences therefore obey

$$|a_S[p]-a_S[p_{\mathrm{r}}]|_2 \le \tfrac{1}{3}\varrho(\alpha_1+\alpha_2)n_S, \qquad |r_\omega[p]-r_\omega[p_{\mathrm{r}}]|_2 \le \tfrac{2}{3}\varrho(\alpha_1+\alpha_2)n_\omega, \quad \varrho = \frac{3}{c_{\mathrm{d}}a_{\mathrm{w}}}. \tag{780}$$

For the first estimate use radius $c_{\mathrm{d}}a_{\mathrm{cv}}/(\alpha_1+\alpha_2)$, with $a_{\mathrm{cv}} \ge 2a_{\mathrm{w}}$; for the second use $c_{\mathrm{d}}a_{\mathrm{w}}/(\alpha_1+\alpha_2)$ and integrate the resolvent difference. Taking $\alpha_1+\alpha_2 \le c_{\mathrm{d}}a_{\mathrm{w}}/3$ keeps these discs larger than one.

We need two geometric facts about the coarse domains $X(\iota)$. Each piece of a figure has diameter at most $5h$; a tree admissible for Λ_ι meets every piece. Thus the figure is contained in the

$5h$-neighborhood of that tree. The segment-cover count used above, now at side LM, gives

$$\#X(\iota) \le 8 + \frac{32}{(L-1)M}\Lambda_\iota. \tag{781}$$

Indeed a doubled tour divided at spacing $(LM-10h)/2$ gives at most $8\max(1, 4\,\mathrm{length}/(LM-10h))$ intersected cubes; $10h \le M$ and passage to the infimum give the display. If a figure leaves $Z_\circ^\sharp$ and the first bond meets $Z_\circ$, its length is at least $M/2$. To see this, a point in a coarse cube outside $Z_\circ^\sharp$ is at distance at least M from $Z_\circ$: otherwise the fine cube containing it would belong to $Z_\circ^\sim$. The initial point of the first bond is within one of $Z_\circ$, and a tree point lies within $5h$ of that figure point. Hence its length is at least $M - 1 - 5h \ge M/2$. For a product $a_S(b,c)r_\omega(c,b')$, the same conclusion holds for $\Lambda_S(b,c) + \Lambda_\omega(c,b')$, since their trees meet at c_-.

Choose

$$\theta_C = \frac{\theta_\flat \gamma}{16 b_a}, \qquad b_C = \frac{4}{\gamma} c_{\mathrm{lat}}(\theta_C/2), \qquad 0 < \theta_{\mathrm{Str}} \le \min(\theta_\flat/18, \theta_C/8).$$

For $|s| \le e^{2\bar\kappa}$, (781) and $\theta_{\mathrm{Str}} M \ge 16\bar\kappa$ imply

$$\left| \prod_{\Delta \subset X(\iota)} \bar s(\Delta) \right| \le \exp\{\theta_{\mathrm{Str}} M + \theta_{\mathrm{Str}} \Lambda_\iota/3\}. \tag{782}$$

Here $L \ge 13$ absorbs $64\bar\kappa/((L-1)M)$. An individual form term outside the core hull has length at least $M/2$, so the factor $e^{\theta_{\mathrm{Str}} M}$ costs at most $e^{2\theta_{\mathrm{Str}}\Lambda_S}$. Keeping another such length factor gives $e^{-\theta_{\mathrm{Str}} M}$; the remaining exponent is within $\theta_\flat$ even after the weight $e^{\theta_\flat |b-b'|/2}$ is inserted. Terms inside the hull have parameter one and are compared by (780). Consequently

$$\|\chi_\circ(\mathsf{A}(\bar s; p) - \mathsf{A}_{\mathrm{r}})\|^{\mathrm{w}}_{\theta_\flat/2} \le \varrho b_a \varepsilon_\circ. \tag{783}$$

The reference sum is full near $\mathfrak{X}\langle Z_\circ\rangle$. In fact $h_\square(b) \ne 0$ and an endpoint $x \in Z_\circ$ imply $|y-x|_\infty < 7h/2 + 1/2 < M$ for every $y \in \square^{\sim 2}$, so every coarse cube meeting that enlarged mesh cube lies in $Z_\circ^\sharp$. The real partial-sum gap proved above therefore gives $\mathbf{C}_{\mathrm{r}}^{-1} \ge 3\gamma I/4$. Choose $a_{\mathrm{Str}} \le \min(c_{\mathrm{d}} a_{\mathrm{w}}/3, \gamma/(4\varrho b_a))$. Then (783) gives $\operatorname{Re}\mathbf{C}^{-1} \ge \gamma I/2$ and $\|\mathbf{C}^{-1}\|^{\mathrm{w}}_{\theta_\flat/2} \le 2b_a$. The exponential resolvent bound yields $|\mathbf{C}(b,b')|_2 \le (4/\gamma)e^{-\theta_C|b-b'|}$. The same estimate holds for $\mathbf{C}_{\mathrm{r}}$, and hence their weighted norms at $4\theta_{\mathrm{Str}}$ are at most b_C. The resolvent identity now gives

$$\|\mathbf{C} - \mathbf{C}_{\mathrm{r}}\|^{\mathrm{w}}_{4\theta_{\mathrm{Str}}} \le \varrho b_a b_C^2 \varepsilon_\circ.$$

To estimate the coupling, expand its two factors as $a_S r_\omega$, with the projection $1 - E_\circ$ between them. If both domains lie in $Z_\circ^\sharp$, subtract the reference product in two terms and use (780). Otherwise their combined length is at least $M/2$. The two parameter factors cost $e^{2\theta_{\mathrm{Str}}M + \theta_{\mathrm{Str}}(\Lambda_S+\Lambda_\omega)/3}$; after retaining $e^{-\theta_{\mathrm{Str}}M}$ and the desired bond weight, the total exponent charged to length is at most $31\theta_{\mathrm{Str}}/3 \le \theta_\flat$. Using (779) in either order proves

$$\|\mathbf{\Gamma} - \mathbf{\Gamma}_{\mathrm{r}}\|^{\mathrm{w}}_{4\theta_{\mathrm{Str}}} \le \varrho b_a b_r \varepsilon_\circ, \qquad \|\mathbf{\Gamma}_{\mathrm{r}}\|^{\mathrm{w}}_{\theta_\flat} \le b_a b_r, \qquad \|\mathbf{\Gamma}\|^{\mathrm{w}}_{4\theta_{\mathrm{Str}}} \le 2 b_a b_r.$$

The last inequality follows also by reducing a_{Str} so that $\varrho\varepsilon_\circ \le 1$. Split $\mathbf{\Gamma}^{\mathsf{T}} C \mathbf{\Gamma} - \mathbf{\Gamma}_{\mathrm{r}}^{\mathsf{T}} C_{\mathrm{r}} \mathbf{\Gamma}_{\mathrm{r}}$ into the three differences, one factor at a time. The preceding bounds give $\varrho b_a^2 b_r^2 b_C(3 + 2b_a b_C)\varepsilon_\circ$. Taking C_{Str} to dominate these finitely many coefficients proves all the stated differences. Take $\gamma_{\mathrm{r}} = \min(1, 3\gamma/4)$ and $c_{\mathrm{r}} \ge \max(1, b_a, b_a b_r, b_a b_r b_C)$. These choices depend only on L.

It remains to check the structural assertions. For fixed background, group all terms according to the subset of coarse parameters in their domains. There are only finitely many such subsets, and their coefficient sums converge absolutely by (779). Thus $\mathsf{A}(s)$ and $r(s)$ are polynomials, of degree at most one in each parameter. Their coefficients are holomorphic by local uniform convergence; inversion is holomorphic where the determinant is nonzero. The estimates were proved on the larger polydisc $|s| \le e^{2\bar\kappa}$, which supplies the required neighborhood.

Each figure is connected. Its coarse hull is wall-connected: for grain enlargements this follows by refining the touching chain, and successive string pieces overlap the neighboring mesh enlargements. If a parameter vanishes off Z, each surviving domain therefore lies in one wall-component of $Z^\sharp_\circ \cup Z$. The inverse of the resulting block-diagonal form has the same blocks. Every string input bond is inner to its figure, so the coupling has precisely the asserted inner-bond support. The definitions of the coefficients and strings also show restriction consistency. Real backgrounds make all local matrices, integrals and coefficients real. Symmetry of $\mathbf{C}$ follows from symmetry of each a_S. These verify the locality, factorization and reference requirements of eqs. (701) to (703).

On the whole torus put $G = \mathsf{A}^{-1}$ and split the coordinates into the core and its complement. Then $\mathbf{C} = A_{00}^{-1}$ and $\mathbf{\Gamma} = A_{01}\chi_1\mathsf{A}^{-1/2}$. Symmetry and $(\mathsf{A}^{-1/2})^2 = G$ give $\mathbf{\Gamma}\mathbf{\Gamma}^{\mathsf{T}} = A_{01}G_{11}A_{10}$. From $GA = AG = 1$, $G_{10} = -G_{11}A_{10}\mathbf{C}$ and $G_{00} = \mathbf{C} + \mathbf{C}A_{01}G_{11}A_{10}\mathbf{C}$, which is (767). The whole-torus gap and difference estimates are the preceding ones with the entire torus as the core. This proves the remaining Gaussian identity in eq. (696).

Finally replace $(\mathbf{C}, \mathbf{\Gamma})$ by $(\tau^{-1}\mathbf{C}, \tau^{1/2}\mathbf{\Gamma})$. The quadratic product $\Gamma^{\mathsf{T}}C\Gamma$ is unchanged; the inverse covariance and the coupling differences are at most $2\widehat{\varepsilon} + c_{\mathrm{r}}|\tau - 1|$, using $|\tau^{1/2} - 1| \le |\tau - 1|$ on the prescribed disc. The whole identity scales by τ^{-1}. Since $|\tau - 1| \le \frac{1}{2}(LM)^{-3}$, this gives (768) with a strictly positive reserve. When the two smallness terms are at most $(LM)^{-3}$, the unscaled defect number is at most $4C_{\mathrm{Str}} + c_{\mathrm{r}} + 3$. For the scaled structure the bound instead reads $(LM)^3\widehat{\varepsilon}_* \le 4C_{\mathrm{Str}} + c_{\mathrm{r}}$, with the strict reserve in (768). This completes the proof.

50.4. **Local charts between lattices.** At height k, write the infinite point lattices as $L^{j-k}\mathbb{Z}^3$, $0 \le j \le k+1$. A finite lattice $\mathfrak{w} = (\ell, K)$ has side $s_{\mathfrak{w}} = 2L^{m+K-k}$ in these units; its covering map is $\mathrm{pr}^{\mathfrak{w}}$. Fix an integer $n_{\mathrm{e}} \ge 1$ and require $s_{\mathfrak{w}} \ge C_L M$, where $C_L \ge 8n_{\mathrm{e}}L$. The normalizer uses $n_{\mathrm{e}} = 1$; a later construction may increase this fixed halo width before choosing the lattices.

A set $\widehat{X} \subset \mathbb{R}^3$ is *near* for $\mathfrak{w}$ if it lies in a cuboid with all side lengths less than $s_{\mathfrak{w}}/2$. Its image is called *non-wrapping*. In the infinite lattice, near means bounded. Set

$$\widehat{X}^+ = \{y : \mathrm{dist}_\infty(y, \widehat{X}) \le n_{\mathrm{e}}LM\}. \tag{784}$$

The halo $X^+ = \mathrm{pr}^{\mathfrak{w}}(\widehat{X}^+)$ does not depend on the near lift. Indeed, two near lifts of a nonempty set differ by one period vector: after matching one point, the coordinate differences between any other matched pair have magnitude less than one period and are periods, hence vanish. The halo lies in a cuboid with side lengths less than $3s_{\mathfrak{w}}/4$, so its covering map is injective. The empty set has empty halo.

If $\widehat{X}$ is near for two lattices and $v \in LM\mathbb{Z}^3$, the corresponding chart is

$$\mathsf{i}(\mathrm{pr}^{\mathfrak{w}'}\hat{y}) = \mathrm{pr}^{\mathfrak{w}''}(\hat{y} + v), \qquad \hat{y} \in \widehat{X}^+. \tag{785}$$

It acts on fields by carrying the sites and oriented bonds, and on operators by relabelling their rows and columns. In a halo, any displacement of norm at most LM is its actual lifted displacement: an alternative differs by a period and would have a coordinate of magnitude at least $s_{\mathfrak{w}} - LM \ge 7s_{\mathfrak{w}}/8$. Thus charts preserve all contained lattice stencils, blocks, contours and pivots. Cubes contained in a halo also lift to cubes: the difference between the inverse chart and a lift on the cube is continuous and period-valued, hence constant.

We compare a local construction only when its entire input domain and the stencils it uses are in these halos. Chart consistency means equality after (785) whenever both inputs are admitted. It gives a unique infinite-lattice value on every finite domain that embeds in some member of the chosen family of tori: choose an embedding, evaluate there, and use the zero-translation chart to compare two choices. This defines values on every finite domain when the family of sides is unbounded. Bounds and slice holomorphy pass to this value, since its evaluation on any fixed finite domain uses one such torus.

50.5. **The Gaussian normalizer.** We localize the normalization before expanding the interaction. The logarithm of the determinant has a convergent series on every deep subregion. Taking its finite difference over coarse cubes leaves only closed chains whose support fills the prescribed region. Short chains then have a large spatial length; long chains are controlled by the spectral gap.

Use the constant sequence, units and localization parameters of section 50, with $t = k + 1$. The coarse cubes have side LM. Choose a further clearance r_N satisfying

$$\rho_{\mathrm{t}} < r_N, \qquad 4h \le r_N, \qquad 4r_N \le LM, \qquad 2r_N + 1 < LM.$$

Keep $12\mathfrak{M} \le \mathsf{o}LM$ and $5h \le \mathsf{o}L^2M$. There will be an additional lower bound $M \ge M_N(L)$. Let Γ be all positive level-t bonds, $\mathfrak{X}_0$ all level-k bonds, and $\mathfrak{X}$ their nonpivot subset. Thus positive bonds remain the independent coordinates. Write $P_0 = (\mathbf{1}, 0, 0)$ and set

$$\alpha_N = \frac{a_N}{\max\{1, \mathsf{o}LMB^{(\mathrm{cg})}\}}, \qquad \mathcal{Q}^N(X) = \mathcal{Q}_t(X; \alpha_N, a_N, a_N), \tag{786}$$

where $a_N > 0$ is chosen below. The level-k reading of P is always (764); in particular the current is $L^{-3}J_P$.

For a level-k bond b, let $S_N(b)$ be the union of the coarse cubes at sup distance less than r_N from its segment. For a coarse region Y, put

$$Y_N^- = \{b \in \mathfrak{X} : S_N(b) \subset Y\}, \qquad A_{Y,N} = \chi_{Y_N^-} \mathsf{A}_Y \chi_{Y_N^-}^{\mathsf{T}}.$$

Here A_Y is the actual form for the maximal tower over Y, not the restriction of the whole-torus form. If $c \in \Gamma$, let $X(c)$ be the union of the coarse cubes containing c_- and c_+. For brevity write $m_c(V) = \kappa_V(c)$ for the already defined pivot coefficient.

Proposition 50.2. There are L-dependent constants $a_N, \delta_N > 0$, $M_N, E_b \ge 1$, and $E_N = 3L^3E_b$, such that the following holds when $M \ge M_N$. For every wall-connected coarse region X there are local terms $N(X, b; P)$, $b \in \mathfrak{X}_0^+$, on $\mathcal{Q}^N(X)$. They are holomorphic on every fixed-real-background slice, depend only on the pair represented by P, are gauge invariant and conjugation real, and vanish at P_0. Their pointed versions are

$$N^{(t)}(X, P, z) = \sum_{b \in \mathfrak{X}_0^+} h_z(b) N(X, b; P). \tag{787}$$

A nonzero summand has $z \in X$. Uniformly in the lattice, k, X, P, b, z,

$$|N(X, b; P)| \le E_b e^{-\delta_N M d_t(X)}, \qquad |N^{(t)}(X, P, z)| \le E_N e^{-\delta_N M d_t(X)}. \tag{788}$$

On the whole torus put $V = \mathcal{M}^k\mathbf{U}$ and $\Phi_t(P) = \sum_X \sum_{z \in X} N^{(t)}(X, P|_X, z)$. Then

$$e^{2\Phi_t(P)} = \frac{\det \mathsf{A}(P_0)}{\det \mathsf{A}(P)} \prod_{c \in \Gamma} \left(\frac{\det m_c(\mathbf{1})}{\det m_c(V)} \right)^2. \tag{789}$$

For real P, the form is positive and all pivot determinants are positive. With Lebesgue measure for the real pairing on nonpivot coordinates, define

$$Z_k[P] = \prod_{c \in \Gamma} |\det m_c(V)|^{-1} \int \exp(-\tfrac{1}{2}(B, \mathsf{A}(P)B)_1)\, dB.$$

Then

$$\Phi_t(P) = \log Z_k[P] - \log Z_k[P_0]. \tag{790}$$

The terms are unchanged by a local lattice chart carrying the background and all the cubes of X. Consequently they define the corresponding family on the infinite lattice, with the same domains and bounds. ∎

Proof. The local series. The simple-domain bounds and the choice of α_N put the level-k pair in $\mathcal{P}_{h,*}(Y)$ for every nonempty coarse $Y \subset X$, if $a_N \le a_{\mathrm{w}}$. Indeed the two real-gauge boxes in that definition fit in the boxes of the simple domain: their sides in level-t units are at most $12\mathfrak{M}$ and $5h/L$. Restriction preserves the curvature, perturbation and current bounds. The all-meet construction in section 50.1 therefore defines A_Y from the data on Y itself.

Since $r_N > r$ and $4h \le r_N$, the bonds of Y_N^- are deep enough for the localization proof in section 50.2. In particular, after decreasing a_N, its real-reference compression has spectrum in $[\gamma/2, B_a]$, and its complex perturbation has norm at most $\gamma/8$. Take

$$c_N = \tfrac{1}{2}(B_a + \gamma/2), \qquad q_N = \frac{B_a - \gamma/4}{B_a + \gamma/2} < 1, \qquad T_Y = 1 - c_N^{-1} A_{Y,N}.$$

The midpoint estimate gives $\|T_Y\| \le q_N$. Thus, for $b \in (Y_N^-)^+$, the series

$$f_b(Y;P) = 3 \log c_N - \sum_{n \ge 1} \frac{1}{n} \operatorname{tr} T_Y^n(b,b) \tag{791}$$

converges normally and has modulus at most $F_N = 3|\log c_N| + 3\log(1/(1-q_N))$. The trace is the trace on the three-dimensional complex Lie algebra. Define

$$E^N(X;P,b) = \sum_{S_N(b) \subset Y \subset X} (-1)^{\#X - \#Y} f_b(Y; P|_Y). \tag{792}$$

All sums over regions here mean sums over their coarse cubes.

The pivot contribution is

$$z_c(P) = -\operatorname{Log}\det(L m_c(\mathcal{M}^k \mathbf{U})).$$

Its domain and branch require a check. The cubes $X(c)$ contain a cube of side $2L$ containing the endpoint double block and all its bare descendants. To see this, in each transverse coordinate choose one adjacent interval of length L inside the same coarse interval; in the direction of c use the two consecutive intervals centered at its endpoints. The coarse intervals contain $M \ge 3$ such intervals. The simple-domain gauge on this cube gives $U^u = e^{i\eta_t A}$ with $|A| < a_N$. The logarithm of $e^{i\eta_t \operatorname{Ad}(u) A'_P} e^{i\eta_t A}$ is therefore $i\xi\phi$ with $\|\phi\|_\infty < 4a_N/L$. By proposition 51.1, its level-k average is $e^{i\widetilde{a}}$, holomorphic in ϕ, with $\|\widetilde{a}\|_\infty \le C_Q \|\phi\|_\infty$.

The pivot coefficient of the one-step derivative is holomorphic in $\widetilde{a}$ and equals L^{-3} at zero. The one-step derivative bound therefore gives $|1 - L^3 m_c(e^{i\widetilde{a}})| \le C_{\mathrm{piv}} \|\widetilde{a}\|_\infty$ on a fixed smaller chart, with C_{piv} depending only on L. Here one may use (364): a variation supported on the pivot changes neither endpoint contour variable, so the straightened derivative at that coordinate is $L m_c$. Choose a_N so that $4C_Q a_N/L$ lies in this chart and $4C_{\mathrm{piv}} C_Q a_N/L \le 1/2$, as well as $4a_N/L < c_{\mathrm{plain}}$. Put $V = \mathcal{M}^k \mathbf{U}$ and $T = L^3 m_c(e^{i\widetilde{a}})$. Gauge covariance gives $T = L^3 \operatorname{Ad}(g) m_c(V) \operatorname{Ad}(g')^{-1}$ for two real group elements g, g'. Their adjoint matrices have determinant one, so $\det T = \det(L^3 m_c(V))$. The bound $|1 - T| \le 1/2$ places the eigenvalues of T in the disk $|z-1| \le 1/2$. Each has argument at most $\pi/6$, so their product has argument at most $\pi/2$ and avoids the negative real axis. Consequently

$$z_c(P) = 6 \log L - \operatorname{Log}\det(L^3 m_c(V)), \qquad |z_c(P) - 6 \log L| \le 3 \log 2. \tag{793}$$

This also proves holomorphy on each slice. All averages used here are the complex averages; at a reversed bond the value is the inverse, and the domain comparison follows from the same fixed real gauge.

Define $N^{\mathrm{un}}(X,b;P)$ to be $-\frac{1}{2}E^N(X;P,b)$ when b is nonpivot and $S_N(b) \subset X$, to be $z_c(P)$ when $b = b_0(c)$ and $X = X(c)$, and to be zero otherwise. Put

$$N(X,b;P) = N^{\mathrm{un}}(X,b;P) - N^{\mathrm{un}}(X,b;P_0). \tag{794}$$

These definitions give pair dependence, holomorphy and trivial vanishing. Gauge covariance conjugates every diagonal matrix in (791); traces and determinants are unchanged. Complex

conjugation preserves the convergent series and the specified logarithm branch, giving conjugation reality.

Decay of the connected terms. The neighborhood of a bond has side lengths at most $2r_N+1 < LM$. It meets at most two coarse intervals in each coordinate; their cubes have a common corner. Thus $S_N(b)$ is wall connected, $\#S_N(b) \le 8$, and $d_t(S_N(b)) = 0$. Likewise $X(c)$ consists of one cube or two adjacent cubes and has size zero.

Expand each factor of T_Y into the identity or $-c_N^{-1}a_S$. A term of $T_Y^n(b,b)$ is a closed chain $b_0 = b, b_1, \dots, b_n = b$, together with the connected grain regions attached to its nonidentity steps. Its coarse support is the union of the $S_N(b_i)$ and of the coarse cubes containing those grain regions. In (792), the sum over all Y containing this support is zero unless the support is exactly X. If $e_n(X,b)$ is the sum of these exact-support chains, then

$$E^N(X;P,b) = 3\log c_N\, \mathbf{1}_{\{X=S_N(b)\}} - \sum_{n\ge1} \frac{e_n(X,b)}{n}. \tag{795}$$

Put $\theta_N = \delta_a/2 = \delta_{\mathrm{cv}}/(8L)$. The connected-form estimate and the ordered row sums used in eq. (779) give a constant $Q_N > 1$, depending only on L, such that the absolute weights of all length-n chains beginning at b, multiplied by $e^{\theta_N\lambda}$, sum to at most $3Q_N^n$. Here λ is the sum of the lengths of their connecting trees. In particular,

$$|e_n(X,b)| \le 3Q_N^n e^{-\theta_N LMd_t(X)/128}, \qquad |e_n(X,b)| \le 3\,2^{\#X} q_N^n. \tag{796}$$

The second estimate follows directly from the finite difference of $\operatorname{tr} T_Y^n(b,b)$.

For the length bound in the first estimate, join the step trees; their union is connected and has length at most λ. Every cube of X lies within r_N+1 of this union. Choose a maximal $LM/8$-separated set on it. If it has $N \ge 2$ points, its length is at least $NLM/16$: on each covering segment, the inverse images of the distance ranges $[0, LM/16)$ about the chosen points are disjoint, and connectedness makes each range have that full length. Every cube of X meets a ball of radius $r_N + 1 + LM/8 < LM/2$ about one chosen point. Such a ball meets at most eight coarse cubes, with a common corner. If $N = 1$ then $d_t(X) = 0$; otherwise $\#X \le 8N \le 128\lambda/(LM)$, hence $LMd_t(X) \le 128\lambda$.

For $d = d_t(X) \ge 1$, use $\#X \le 16d$ and split the sum at $n_* = \lfloor \vartheta_N LMd \rfloor$, where $\vartheta_N = \theta_N/(256\log Q_N)$. The first majorant in (796) gives

$$\sum_{1\le n\le n_*} \frac{|e_n|}{n} \le \frac{3Q_N}{Q_N-1} e^{-\theta_N LMd/256}.$$

The second gives

$$\sum_{n>n_*} \frac{|e_n|}{n} \le \frac{3}{1-q_N} e^{16d\log 2 - \vartheta_N LMd\log(1/q_N)}.$$

Increase M_N so that $\vartheta_N LM\log(1/q_N) \ge 32\log 2$, and take

$$0 < \delta_N \le L\min\{\theta_N/256, \tfrac12\vartheta_N\log(1/q_N)\}.$$

For $d = 0$, there are at most eight cubes and $|E^N| \le 2^8 F_N$. These estimates, together with (793), prove the bond bound after fixing E_b. Subtracting the trivial value costs no extra factor: the nonpivot coefficient has the factor $1/2$ in its definition. The tent support lies at distance less than $L < r_N$ from b. Thus a nonzero term in (787) has $z \in X$, including the pivot case, and $\sum_b h_z(b) = 3L^3$ gives the pointed bound.

Connectedness and the normalization identity. We must also show that finite differences on regions that are not wall connected vanish. Let $Y = Y_1 \cup Y_2$, with no wall adjacency between the two nonempty regions. A bare site lies in one coarse cube, and a bond or plaquette cannot join the two components. Every smaller partition cube whose center lies in Y_i has its interior there. For $x \in Y_1$, a coordinate-monotone coarse-cube path to a cube of Y_2 first leaves Y_1 through a cube of

Y^c; the distance from x does not increase before that exit. Hence $\operatorname{dist}_\infty(x, Y^c) \le \operatorname{dist}_\infty(x, Y_2)$. The defining distance-to-complement test for a tower cube is therefore the same in its component as in Y. Induction through the levels gives $\Omega_q(Y) = \Omega_q(Y_1) \cup \Omega_q(Y_2)$ with disjoint site sets and equal level assignments.

The actual form now splits. Its derivative and Hessian stencils, averaging blocks and contour paths remain in their component. Consequently the scalar and field averages, their weighted transposes, the scalar inverse and projection split. Sums, products, inverses and bilinear polarization preserve this decomposition. Applying these operations to the displayed construction (589), including both induced multipliers and the pivot and gauge forms, gives

$$\mathsf{A}_Y = \mathsf{A}_{Y_1} \oplus \mathsf{A}_{Y_2}.$$

The same holds after deep compression. In (792), $f_b(Y)$ depends only on the component containing $S_N(b)$; summing with alternating signs over any other component gives zero. Thus the finite difference is supported on wall-connected X.

Möbius inversion on the finite set of coarse cubes now gives

$$\sum_{X \supset S_N(b)} E^N(X; P|_X, b) = f_b(T_\ell; P).$$

The tents sum to one, so the sum of all pointed terms is

$$\Phi_t(P) = -\frac{1}{2} \sum_{b \in \mathfrak{X}^+} [f_b(T_\ell; P) - f_b(T_\ell; P_0)] + \sum_{c \in \Gamma} [z_c(P) - 6 \log L].$$

In finite dimension, $\exp(\sum_b f_b(T_\ell; P)) = \det \mathsf{A}(P)$: the logarithm series applies to T_{T_ℓ}, whose norm is less than one, and its trace is the sum of the diagonal traces. Also $\exp z_c = \det(L m_c(V))^{-1}$ and $m_c(\mathbf{1}) = L^{-3}$. This proves (789). For real P, the gap is at least $3\gamma/8$ and the pivot determinants are real and avoid the nonpositive axis. The real Gaussian integral is positive, and its determinant formula, together with $\Phi_t(P) \in \mathbb{R}$, proves (790) with the real logarithms.

Local charts. Take a finite coarse region in the infinite lattice and embed its halo in two tori of side at least $8LM$, without wrapping. The chart between the images preserves levels, directions, bond midpoints and all distances at most LM. The simple sets correspond: plaquette and derivative stencils are carried bijectively, and any compact cube contained in the region lifts to a translate of the same Euclidean cube. The latter assertion follows because the difference between the inverse chart and a lift is a continuous, lattice-valued function on that cube, hence is constant.

The chart therefore carries all tower tests, paths, blocks and pivot bonds. The actual operator construction is a renaming of the same finite coordinates, including every inverse, so $A_{\mathsf{i}Y,N}(\mathsf{i}_* P)(\mathsf{i}b, \mathsf{i}b') = A_{Y,N}(P)(b, b')$. It also carries $S_N(b)$ and $X(c)$. The series (791), the finite differences and the pivot determinants consequently agree term by term. The tents agree because their values are functions of $(x_b - z)/L$ on their support. This proves chart independence. It defines the infinite family on each finite region, preserves slice holomorphy, reality, gauge invariance and the bounds, and is consistent when the region is placed in any larger torus. □

51. The local interaction

The fluctuation integral has three nonlinear terms: the Jacobian of the constraint chart, the remainder of the gauge fixing, and the remainder of the action at its critical configuration. The first two are supported on one or two fine cubes. We localize the third by taking a finite difference in each cube outside the plaquette being observed. Holomorphy in these parameters gives exponential decay in the size of the resulting region.

Use the level-k units and the triple domain of section 50, with $0 \le k < K$. Fix $\bar{\kappa} \ge 1$. The fine cubes have side $M = L^{m'} \ge L^2$, the coarse cubes have side LM, and the torus has side at least $C_L M$, where $C_L \ge 8L$. We write $\mathcal{D}_k$ for the wall-connected fine-cube regions and $d_k(Y)$ for

their linear size. All forms below are specified on positive bonds; negative bonds carry their odd extensions. Kernels use counting measure, and

$$(B_1, B_2)_1 = \sum_{b \in \mathfrak{X}^+} \langle B_1(b), B_2(b) \rangle, \qquad \langle X, Y \rangle = \tfrac{1}{2} \operatorname{tr}(XY).$$

For $P = (U, A', \mathbf{J}) \in \mathcal{Q}_{k+1}(Y; a, a, a)$, put $\mathbf{U} = e^{i\eta_{k+1}A'}U$ and $V = \mathcal{M}^k\mathbf{U}$ on the level-k bonds inside Y. Its existence for small a is proved below. The same bare pair, when used in the level-k response equations, is $(\mathbf{U}, L^{-3}\mathbf{J})$, with decomposition $(U, L^{-1}A')$.

51.1. **A uniform chart for the pivot variables.** We first record the plain-average consequence of the centered iteration.

Proposition 51.1. There are positive $a_{\mathrm{plain}}, c_{\mathrm{plain}}$ and $C_Q \geq 8$, depending only on L, such that, on a tower of height j over a box,

$$\varpi(U_0) \leq a_{\mathrm{plain}} L^{-2j}, \quad \tau = L^j \sup|a_0| < c_{\mathrm{plain}} \quad \Longrightarrow \quad \left| \frac{1}{i} \log\bigl[\mathcal{M}^j(e^{ia_0}U_0)(\mathcal{M}^j U_0)^{-1}\bigr] \right| \leq C_Q \tau. \tag{797}$$

The averages and logarithm are holomorphic in a_0, local to the endpoint double-block tower, and real for real a_0. Here and below a local supremum in this assertion is taken on that tower. ■

Proof. To see the assertion, use the centered iteration of proposition 30.2 and its one-step factorization. Decreasing $a_{\mathrm{plain}}, c_{\mathrm{plain}}$, the reference curvatures and centered perturbations at every intermediate stage remain in the one-step mean chart. The accumulated factors v_i are sufficiently close to the identity that their conjugations preserve the strict logarithm domains. On these domains, the series logarithm commutes with conjugation, so the one-step factorization remains valid for the accumulated complex gauges. Induction in the factorization therefore gives the complex plain average

$$\mathcal{M}^j(e^{ia_0}U_0)(c) = v_j(c_-)e^{ia_j(c)}W_j(c)v_j(c_+)^{-1}, \qquad W_j = \mathcal{M}^j U_0.$$

The centered estimates give $|a_j(c)| \leq 2\tau$ and $|v_j(c_\pm)^{\pm 1} - 1| \leq C_v(L)\tau$. Choose $C_Q \geq 8(1 + C_v)$ and $c_{\mathrm{plain}} \leq C_Q^{-1}$. Since W_j is unitary, the relative product differs from 1 by at most $e^{(2+2C_v)\tau} - 1 \leq C_Q\tau/2 < 1/2$. The logarithm series gives (797). The factorization also proves holomorphy and locality; for real arguments the complex and real means coincide at each stage by uniqueness of the local mean. □

Let $r_1 = r_*/40$ and choose $\varsigma_{0,\mathrm{c}} > 0$ within the one-step coefficient radius established in section 40. Thus, if V on a double block D_c has a unitary gauge $V^h = e^{i\widetilde{a}}$ with $\sup|\widetilde{a}| \leq \varsigma_{0,\mathrm{c}}$, the functions

$$f_V(B) = \frac{1}{i} \log\bigl[\mathcal{M}(e^{iB}V)(c)\mathcal{M}(V)(c)^{-1}\bigr], \qquad x_V(B)(y,x) = \frac{1}{i} \log\bigl[(e^{iB}V)(y,x)V(y,x)^{-1}\bigr]$$

are holomorphic and bounded by 1 for $\sup|B| < r_1$, and vanish at $B = 0$. Write $\ell_V = f_V'(0) = LQ_V$ and $R_V = f_V - \ell_V$. The pivot coefficient is $\mathsf{m}_c(V) = Q_V(\delta_{b_0(c)}\cdot)$. Set $H_V = (L\mathsf{m}_c(V))^{-1}$ and $\mathsf{h}_V D = (H_V D)\delta_{b_0(c)}$. Then

$$|H_V| \leq 2L^2, \qquad \ell_V \mathsf{h}_V = 1, \qquad \sup|\mathsf{C}_V \mathcal{A}| \leq 2L^2 r_1^{-1} \sup|\mathcal{A}|,$$
$$|R_V(B)| \leq 2r_1^{-2}(\sup|B|)^2, \qquad |R_V(B_1) - R_V(B_2)| \leq 8sr_1^{-2} \sup|B_1 - B_2| \quad (\sup|B_i| \leq s \leq r_1/4). \tag{798}$$

The last two bounds are the Cauchy remainder and derivative bounds for f_V; the coefficient bounds and the right inverse were proved with the fluctuation form.

Definition 51.2. Put

$$s_3 = \frac{r_1^2}{32L^2}, \qquad r_3 = \frac{s_3}{2}, \qquad \rho_3 = \frac{s_3}{4L^2}, \qquad C_P = \max\{C_Q, 2L^2/r_1\}, \qquad r_P^\circ = \frac{r_3}{2C_P}. \tag{799}$$

■

For $\sup|B_0| < r_3$, the map $D \mapsto R_V(B_0 - \mathsf{h}_V D)$ maps the closed ball $|D| \le \rho_3$ into its quarter and has Lipschitz constant at most $1/2$: indeed $\sup|B_0 - \mathsf{h}_V D| < s_3$ and $2(s_3/r_1)^2 = \rho_3/4$, while $8 s_3 r_1^{-2} \cdot 2L^2 = 1/2$. Its iterates from zero converge uniformly to a holomorphic map $\mathsf{D}_V(B_0)$. Consequently

$$\begin{aligned} \psi_V^{\mathrm{lin}}(B_0) &= B_0 - \mathsf{h}_V \mathsf{D}_V(B_0), \qquad f_V(\psi_V^{\mathrm{lin}}(B_0)) = \ell_V B_0, \\ |\mathsf{D}_V(B_0)| &\le 4 r_1^{-2} (\sup|B_0|)^2, \qquad \sup|\mathsf{h}_V \mathsf{D}_V(B_0)| \le \tfrac{1}{8} \sup|B_0|. \end{aligned} \tag{800}$$

The first bound follows from the contraction estimate and the value of the map at zero; the second follows from $\sup|B_0| < r_3$. Cauchy's inequality, applied in the pivot direction, gives

$$\mathsf{N}_V(B_0) = H_V \partial_{B_0(b_0(c))} \mathsf{D}_V(B_0), \qquad \mathsf{N}_V(0) = 0, \qquad |\mathsf{N}_V(B_0)| \le \tfrac{1}{4} \quad (\sup|B_0| \le r_3/2). \tag{801}$$

For example, on the ball of radius r_3 the first bound in (800) gives the derivative bound $8 r_3 / r_1^2$ on its half-ball, which after multiplication by $2L^2$ is $1/4$.

All these maps commute with unitary gauges and are real on real data. For a holomorphic family of V presented in a fixed unitary gauge, the coefficients and H_V are holomorphic, and the uniform fixed-point iteration proves joint holomorphy. The same uniqueness proves covariance under translations and under a coordinate chart preserving the double block, its pivot and its ordered paths. These charts preserve the actual holonomies defining the means, hence preserve f_V before the iteration is applied. For a set of coarse bonds, the corrections just constructed assemble componentwise: the only pivot in P_c is $b_0(c)$. Thus h_V and C_V restrict to the one-double-block maps and (800) holds in every row. Any holomorphic constraint chart with the same germ at zero agrees with this one on their common ball, by the identity theorem.

Choose $a_P^\circ > 0$, depending only on L, so small that

$$a_P^\circ \le \min\{a_{\mathrm{plain}}, c_{\mathrm{plain}}, c_2^{\mathrm{SO}}, \varsigma_{0,\mathrm{c}}/(2C_Q + 8\pi)\}. \tag{802}$$

Then every $P \in \mathcal{Q}_{k+1}(Y; a_P^\circ, a_P^\circ, a_P^\circ)$ has the block field V used above. In fact, with $V_0 = \mathcal{M}^k U$,

$$V = e^{iq} V_0, \qquad |q| \le C_Q L^{-1} \sup|A'|, \qquad |\partial V_0 - 1| \le 2 L^{-2} a_P^\circ. \tag{803}$$

Apply (797) to the tower over each endpoint double block with $a_0 = \eta_{k+1} A'$, for which $L^k \sup|a_0| = L^{-1} \sup|A'|$; the real curvature estimate is the tower estimate for U. For $k = 0$ the same conclusions are immediate from $V = \mathbf{U}$. On a double block, the comb gauge of V_0 has a real logarithm of norm at most $4\pi L^{-1} a_P^\circ$. In this fixed gauge, $V = e^{iq^h} e^{ia^0}$, and the logarithm of this product has norm at most $2(C_Q + 4\pi) L^{-1} a_P^\circ \le \varsigma_{0,\mathrm{c}}$. This verifies the chart hypothesis uniformly. It also proves fixed-U holomorphy, since this gauge does not depend on A'.

Definition 51.3. For the restriction $\mathcal{A}_c$ to the nonpivot bonds of P_c, define

$$\beta_c(P, \mathcal{A}) = \psi_V^{\mathrm{lin}}(\mathsf{C}_V \mathcal{A}_c)(b_0(c)), \qquad \mathsf{Jb}_c(P, \mathcal{A}) = \mathsf{N}_V(\mathsf{C}_V \mathcal{A}_c). \tag{804}$$

■

If $\alpha = \sup|\mathcal{A}| < r_P^\circ$, then

$$|\beta_c| \le 2 C_P \alpha \le 1, \qquad |\mathsf{Jb}_c| \le \frac{\alpha}{4 r_P^\circ} < \frac{1}{4}. \tag{805}$$

The second estimate uses (801), its vanishing at zero, and the Schwarz estimate on each complex radial line.

51.2. **Finite differences and the three potentials.** For disjoint subsets $\mathcal{K}, \mathsf{N}$ of a finite parameter set, use the brackets

$$F[\mathcal{K} \mid \mathsf{N}] = \sum_{S \subset \mathcal{K}} (-1)^{\#\mathcal{K} - \#S} F(\mathbf{1}_{S \cup \mathsf{N}}), \qquad F^*[\mathcal{K} \mid \mathsf{N}] = (F - F(0))[\mathcal{K} \mid \mathsf{N}]. \tag{806}$$

If F is holomorphic on $|s(\Delta)| < R$, $R > 1$, and $|F| \le b$, then

$$|F[\mathcal{K} \mid \mathsf{N}]| \le b(R-1)^{-\#\mathcal{K}}, \qquad |F^*[\mathcal{K} \mid \mathsf{N}]| \le 2b(R-1)^{-\#\mathcal{K}}. \tag{807}$$

Indeed the power series in one coordinate gives $|F(s^{\Delta \to 1}) - F(s^{\Delta \to 0})| \le b \sum_{n \ge 1} R^{-n} = b/(R-1)$, uniformly in the remaining coordinates. Apply this successively to the coordinates of $\mathcal{K}$. The starred assertion applies the same estimate to $F - F(0)$. For $R = e^{\bar\kappa}$ with $\bar\kappa \ge 1$, $(R-1)^{-1} \le e^{-(\bar\kappa - 1)}$, and therefore

$$|F[\mathcal{K} \mid \mathsf{N}]| \le b e^{-(\bar\kappa - 1)\#\mathcal{K}}. \tag{808}$$

Expansion of the finite sum also gives

$$F(\mathbf{1}_W) - F(0) = \sum_{\mathcal{K} \subset W \setminus \mathsf{N}} F^*[\mathcal{K} \mid \mathsf{N}] \qquad (\mathsf{N} \subset W). \tag{809}$$

For $k \ge 1$, suppose a shift $\mathfrak{h}_Y(s; P, \mathcal{A}) = (\mathbf{A}_Y, \mathbf{K}_Y)$ is given on

$$P \in \mathcal{Q}_{k+1}(Y; a_h, a_h, a_h), \qquad \sup|\mathcal{A}| < \varepsilon_h, \qquad |s(\Delta)| < e^{\bar\kappa}. \tag{810}$$

We require

$$0 < a_h, \varepsilon_h \le 1, \qquad c_h \ge 1, \tag{811}$$

and the following properties. For fixed U the shift is jointly holomorphic, depends on P only through $(\mathbf{U}, \mathbf{J})$, and satisfies

$$|\mathbf{A}_Y|, |\nabla_{\mathbf{U}}^{\eta_k} \mathbf{A}_Y|, |\mathbf{K}_Y| \le c_h \sup|\mathcal{A}| \tag{812}$$

on every available bond or stencil. It vanishes at $s = 0$. If the support of s has wall-components γ, the shift is the sum of its values at $s\mathbf{1}_\gamma$, and that summand vanishes on bonds not inside γ. Extension of s by zero to a larger region restricts to the same shift. The shift is gauge covariant, real for real inputs, and commutes with translations by $M\mathbb{Z}^3$ and with nonwrapping coordinate charts whenever the transported triple belongs to its domain. These are hypotheses at this stage; the response construction below provides the shift with these properties.

For a level-k bond b, let $\square(b)$ be the union of the fine cubes containing its endpoints. For a coarse bond c, define $\square(c)$ in the same way; it equals $\square(b_0(c))$ and contains D_c. For a bare plaquette p, let $X(p)$ be the union of the fine cubes containing its corners. These regions belong to $\mathcal{D}_k$ and have linear size zero. Indeed $\square(b)$ and $\square(c)$ consist of one or two adjacent cubes, and $X(p)$ consists of one, two or four cubes with a common corner. There are at most $7\#Y$ different $X(p) \subset Y$, and at most $12(M/\eta_k)^3$ plaquettes with any one value of $X(p)$. The first count chooses a base cube and a subset of at most two of the three positive directions; the second chooses one of at most four cubes, a bare site in it, and one of three plaquette directions.

For $\mathsf{t} > 1$, put $g = \mathsf{t}^{-1/2}$ and $\mathcal{A} = gB$. Write

$$\mathcal{R}_3 f(B) = f(B) - \sum_{n=0}^{2} \frac{1}{n!}\, \partial_z^n f(zB)\big|_{z=0}, \qquad \sigma_1(w) = \left(\sum_{n \ge 0} \frac{(-w)^n}{(2n+1)!} \right)^2.$$

All logarithms below are their series at 1. For a single fine cube define

$$\Psi^{\mathrm{G}}(\square; P, B) = \sum_{y \in \square^{(k+1)}} \sum_{x \in B(y) \setminus \{y\}} \left(1 - \tfrac{1}{2} \operatorname{tr}[(e^{igB} V)(y,x) V(y,x)^{-1}]\right),$$

and set this expression to zero on other domains. The Jacobian and gauge potentials are

$$\mathbb{V}^{\mathrm{J}}(Y;P,B)=\sum_{c:\,\square(c)=Y}\{\operatorname{tr}\log(1-\mathsf{Jb}_c)+\log\sigma_1(\langle\beta_c,\beta_c\rangle)\}+\sum_{b\in\mathfrak{X}^+:\,\square(b)=Y}\log\sigma_1(\langle gB(b),gB(b)\rangle),\tag{813}$$

$$\mathbb{V}^{\mathrm{G}}(Y;P,B)=-\mathsf{t}\mathcal{R}_3\Psi^{\mathrm{G}}(Y;P,\cdot)(B).\tag{814}$$

For the action, put $a_p(W)=1-\frac{1}{2}\operatorname{tr}\partial W(p)$ and $W_1(X_1,W)=\eta_k^{-1}\sum_{p:X(p)=X_1}a_p(W)$. If $k\ge 1$, define

$$\Psi^{\mathrm{A}}(Y;P,B)=\mathbb{V}^{\mathrm{act}}(Y;P,gB)=\sum_{X_1\subset Y}\big[s\mapsto W_1(X_1,e^{i\eta_k\mathbf{A}_Y(s;P,gB)}\mathbf{U})\big]^*[Y\setminus X_1\mid X_1],\tag{815}$$

where only regions $X_1=X(p)$ contribute. If $k=0$, put

$$\Psi^{\mathrm{A}}(Y;P,B)=\sum_{p:X(p)=Y}[a_p(e^{iB'}\mathbf{U})-a_p(\mathbf{U})],\quad B'|_{\mathfrak{X}}=gB,\quad B'(b_0(c))=\beta_c(P,gB).\tag{816}$$

Finally set

$$\mathbb{V}^{\mathrm{A}}=-\mathsf{t}\mathcal{R}_3\Psi^{\mathrm{A}},\tag{817}$$

$$\mathbb{V}^{\mathrm{int}}=\mathbb{V}^{\mathrm{J}}+\mathbb{V}^{\mathrm{G}}+\tau\mathbb{V}^{\mathrm{A}}.\tag{818}$$

Proposition 51.4. For $k\ge 1$ set $a_P=\min\{a_P^\circ,a_h\}$ and $\varepsilon_{\mathrm{A}}=\min\{r_P^\circ,\varepsilon_h,c_h^{-1}\}$; for $k=0$ set $a_P=a_P^\circ$ and $\varepsilon_{\mathrm{A}}=r_P^\circ$. Put

$$\varepsilon_P=\varepsilon_{\mathrm{A}}/2,\qquad b_{\mathrm{A}}=6(1+1200)^2M^3,\qquad K_P=224b_{\mathrm{A}}M^3\mathsf{C}_{\mathcal{A}}\varepsilon_{\mathrm{A}}^{-3}.\tag{819}$$

Suppose $\mathsf{C}_{\mathcal{A}}\ge 3$ and $\mathsf{C}_{\mathcal{A}}\delta(\mathsf{t})\le\varepsilon_P$, where $\delta(\mathsf{t})=g\mathfrak{b}(\mathsf{t})$. The three potentials are defined for

$$P\in\mathcal{Q}_{k+1}(Y;a_P,a_P,a_P),\qquad B\in\mathbb{D}^{\mathrm{f}}_{\mathsf{t}}(Y):=\{\sup_{\mathfrak{X}^\circ\langle Y\rangle}|B|<\mathsf{C}_{\mathcal{A}}\mathfrak{b}(\mathsf{t})\}.$$

They are jointly holomorphic on each fixed-U slice, depend only on the pair, are gauge invariant and real on real data, and have the translation and coordinate-chart covariance specified above. The Jacobian potential vanishes at zero; the other two vanish to third order. For $o\in\{\mathrm{J},\mathrm{G},\mathrm{A}\}$,

$$|\mathbb{V}^o(Y;P,B)|\le K_P\delta(\mathsf{t})e^{-(\bar\kappa-2)d_k(Y)}\left(1+M^{-3}\sum_{b\in\mathfrak{X}^{\circ+}\langle Y\rangle}|B(b)|^2\right).\tag{820}$$

All constants are independent of the volume, k and t. The interaction terms are asserted only on this strict cutoff domain; their later use outside it is through the cutoff in the integral. ■

Proof. We estimate the functions before taking their Taylor remainders on the larger ball $g\sup|B|<\varepsilon_{\mathrm{A}}$. Let $\alpha=g\sup|B|$. For $|w|\le 1$, the sine series gives $|\sigma_1(w)-1|\le(2/5)|w|$, hence $|\log\sigma_1(w)|\le|w|$. Also $|\operatorname{tr}\log(1-N)|\le 4|N|$ when $|N|\le 1/4$. Together with (805), these bound each pivot summand in (813) by $2\alpha/r_P^\circ$ and each nonpivot summand by α/r_P°. At most $6(M/L)^3$ coarse bonds and $6M^3$ nonpivot bonds have a prescribed region Y. Thus

$$|\mathbb{V}^{\mathrm{J}}|\le 7M^3(r_P^\circ)^{-1}g\sup|B|,\qquad|\Psi^{\mathrm{G}}|\le M^3/2,\qquad|\Psi^{\mathrm{A}}|\le 112b_{\mathrm{A}}e^{-(\bar\kappa-2)d_k(Y)}.\tag{821}$$

The first function is zero unless $d_k(Y)=0$. For the second inequality, the relative contour matrix X satisfies $|X-1|\le C_P\alpha\le 1$. Since $\det X=1$, $|1-\operatorname{tr}X/2|=|\det(X-1)|/2\le 1/2$. There are $(M/L)^3(L^3-1)<M^3$ contour pairs in a fine cube.

For the action bound we use the following plaquette estimate, valid for complex configurations. If $0<\eta,\rho\le 1$ and the two field components and their two cross derivatives at a plaquette are bounded by ρ, then

$$|\partial(e^{i\eta A}U)-1|\le(1+600\eta\rho)|\partial U-1|+600\rho\eta^2.\tag{822}$$

To verify it, transport the four fields to the base point and write the new plaquette as $e^{a}e^{b'}Ze^{-a'}e^{-b}$, where $Z=\partial U$, $|a|,|b|\le\eta\rho$, $|a'|,|b'|\le 2\eta\rho$, and $|a'-a|,|b'-b|\le\eta^2\rho$. The part containing $Z-1$ is bounded by $e^{6\eta\rho}|Z-1|$. In the remaining part replace a',b' by a,b and then use $|e^{a}e^{b}-e^{b}e^{a}|\le 2(e^{|a|}-1)(e^{|b|}-1)$. The two replacements and the commutator cost at most $[e^6+e^5+2e^2(e-1)^2]\rho\eta^2<600\rho\eta^2$, while $e^{6\eta\rho}\le 1+(e^6-1)\eta\rho$.

For $k\ge 1$, apply (822) with $\rho=c_h\alpha<1$ and $\eta=\eta_k$. The triple bounds imply $|\partial\mathbf{U}-1|\le\eta_k^2$, so the resulting plaquette is within $(1+1200)\eta_k^2$ of 1. The determinant identity above and the plaquette count give

$$|W_1(X_1,e^{i\eta_k\mathbf{A}_Y}\mathbf{U})|\le\eta_k^{-1}\,12(M/\eta_k)^3\frac{(1+1200)^2\eta_k^4}{2}=b_{\mathrm{A}}.$$

The starred-bracket bound is therefore $2b_{\mathrm{A}}e^{-(\bar\kappa-1)n}$, where $n=\#(Y\setminus X_1)$. Since $d_k(X_1)=0$, we have $d_k(Y)\le n$. Indeed enumerate the remaining cubes so that each is adjacent to a preceding cube. An edge tree meeting the preceding region can be extended to the new cube with at most one edge: from a corner of the adjacent old cube, move to its common face if necessary. Starting at a common corner of X_1 proves the bound. Also $\#Y\le 8(d_k(Y)+1)$, because an edge tree of $d_k(Y)$ edges has at most $d_k(Y)+1$ vertices, each in at most eight cubes. There are at most $7\#Y\le 56e^{d_k(Y)}$ roots X_1 to sum, which proves the third bound in (821). For $k=0$, $|B'|\le 2C_P\alpha$ and $|\nabla_{\mathbf{U}}B'|\le 3\sup|B'|$; the small radius gives $6C_P\alpha\le 1$. The same plaquette estimate bounds the difference in (816) by $2b_{\mathrm{A}}$. It is zero unless $d_k(Y)=0$.

All displayed functions are jointly holomorphic on the larger ball. For a bounded holomorphic function f on a ball of radius R, Cauchy's coefficient estimates along the line through B give

$$|\mathcal{R}_3 f(B)|\le 2\sup|f|\,(\sup|B|/R)^3\qquad(\sup|B|\le R/2).$$

Take $R=\varepsilon_{\mathrm{A}}/g$. The cutoff hypothesis puts $\mathbb{D}^{\mathrm{f}}_{\mathsf{t}}(Y)$ in this half-ball. Using $\mathsf{t}g^2=1$ and $g\sup|B|\le\mathsf{C}_{\mathcal{A}}\delta(\mathsf{t})$ gives

$$\mathsf{t}|\mathcal{R}_3 f(B)|\le 2\sup|f|\,M^3\mathsf{C}_{\mathcal{A}}\varepsilon_{\mathrm{A}}^{-3}\delta(\mathsf{t})\,M^{-3}\sum_b|B(b)|^2.$$

This proves (820) for the gauge and action terms using (821); the Jacobian bound follows directly from its first inequality and (819). The vanishing orders follow from the definitions. Pair dependence, reality and gauge invariance follow before taking the brackets or remainders, since the local chart is covariant and traces are invariant. Translations and coordinate charts carry every defining path, pivot, plaquette and cube to its counterpart; restrictions and the finite brackets commute with these bijections. This proves the remaining claims. □

51.3. **The quadratic gauge term and the global sums.** Write the contour derivative as $\mathsf{G}^{(1)}[V](B)(y,x)=x'_V(0)B(y,x)$. It defines a symmetric operator on the nonpivot bonds inside a fine cube by

$$(B_1,\widehat{\mathsf{G}}_{\square}(P)B_2)_1=\sum_{y\in\square^{(k+1)}}\sum_{x\in B(y)\setminus\{y\}}\langle\mathsf{G}^{(1)}[V](B_1)(y,x),\mathsf{G}^{(1)}[V](B_2)(y,x)\rangle.\tag{823}$$

Its kernel is holomorphic on fixed-U slices and has row sums at most

$$c_{\mathsf{G}}=6L^6r_1^{-2}.\tag{824}$$

In fact its kernel is $\sum_{y,x}\mathsf{b}_V(y,x;b)^{\mathsf{T}}\mathsf{b}_V(y,x;b')$, with $|\mathsf{b}_V|\le r_1^{-1}$ and support inside $B(y)$. For fixed b, at most one y, fewer than L^3 values of x, and at most $3L^3$ bonds b' contribute. The transpose costs a factor of at most 2 in the matrix norm, giving (824). Define the free quadratic potential, for arbitrary B, by

$$\mathbb{V}^{\mathrm{q}}(\square;P,B)=-\frac{1-\tau}{2}(B,\widehat{\mathsf{G}}_{\square}(P)B)_1,\tag{825}$$

and set it to zero on other regions. Expansion of $\frac{1}{2}\operatorname{tr} e^{ix} = 1 - \frac{1}{2}\langle x, x\rangle + O(|x|^4)$ at $x = x_V(gB)$ gives

$$\mathbb{V}^{\mathrm{G}}(\square; P, B) = -\mathsf{t}\Psi^{\mathrm{G}}(\square; P, B) + \tfrac{1}{2}(B, \widehat{\mathsf{G}}_{\square}(P)B)_1. \tag{826}$$

We describe exactly when these local expressions recover the global ones. Let U_+ be a real critical background over a coarse field W, choose its averaged field $V = \mathcal{M}^k U_+$ in axial gauge, and assume $P_W = (U_+, 0, J^{(k+1)}(U_+)) \in \mathcal{Q}_{k+1}(T_\ell; a_P, a_P, a_P)$. Suppose that the real fibre chart and the critical background over $V_W(B) = e^{i\psi_V^{\mathrm{lin}}(\mathsf{C}_V gB)}V$ exist for $\sup|B| < \mathfrak{b}(\mathsf{t})$. For $k \geq 1$ require, in addition, that the full-support shift $\mathbf{A} = \mathbf{A}_{T_\ell}(1; P_W, gB)$ satisfy the background identification

$$\begin{aligned} e^{i\eta_k \mathbf{A}} U_+ &\sim_{\mathrm{gauge}} U_k(V_W(B)), \\ J^{(k)}(e^{i\eta_k \mathbf{A}} U_+) &= L^{-3} J^{(k+1)}(U_+) + \mathbf{K}_{T_\ell}(1; P_W, gB). \end{aligned} \tag{827}$$

Here $U_k(V_W(B))$ denotes a representative of the critical background over $V_W(B)$, and $\sim_{\mathrm{gauge}}$ denotes gauge equivalence. The current equality states the normalization needed later; the action identity below uses only the first assertion.

Under these hypotheses, the local Jacobian terms sum to the nonlinear correction to the chart Jacobian, including the exponential-coordinate density. The constant linear-coordinate determinant is separate. To verify the nonlinear correction, let Π retain the pivot coordinates and $N = \mathsf{h}_V \partial \mathsf{D}_V(\mathsf{C}_V gB)$. Since $\Pi N = N$ and each double block has exactly one pivot, $\Pi N \Pi$ is block diagonal with blocks Jb_c. Thus $\operatorname{Tr} N^n = \sum_c \operatorname{tr} \mathsf{Jb}_c^n$, and the absolutely convergent logarithm series splits as in (813). The remaining density factors are precisely $\sigma_1(\langle \beta_c, \beta_c\rangle)$ at pivots and $\sigma_1(\langle gB(b), gB(b)\rangle)$ elsewhere. The one-double-block chart agrees with the global chart by the germ uniqueness proved above, so no second choice of pivot variable occurs.

The gauge terms sum by sorting the pairs (y, x) according to their fine cube. On inner block bonds the pivot elimination and nonlinear correction leave gB unchanged. Since $V(y, x) = 1$ in axial gauge, the sum of Ψ^{G} is the gauge-fixing function $G_W(B)$, and

$$\widehat{\mathsf{G}}_W = \sum_{\square \in \pi_k} \chi_{\square}^{\mathsf{T}} \widehat{\mathsf{G}}_{\square}(P_W|_{\square})\chi_{\square}. \tag{828}$$

Consequently the sum of (826) is $-\mathsf{t}G_W(B) + (B, \widehat{\mathsf{G}}_W B)_1/2$.

Finally let $\mathbb{F}_W(B)$ be the Wilson action of the critical background over $V_W(B)$, in level-k units. Then

$$\sum_{Y \in \mathcal{D}_k} \mathbb{V}^{\mathrm{act}}(Y; P_W|_Y, gB|_Y) = \mathbb{F}_W(B) - \mathbb{F}_W(0). \tag{829}$$

For $k \geq 1$, fix $X_1 = X(p)$ and apply (809) to $F_{X_1}(s) = W_1(X_1, e^{i\eta_k \mathbf{A}_{T_\ell}(s)} U_+)$. Consistency identifies the brackets with those in (815). A support component disjoint from X_1 does not change the fields on its plaquettes; if $\mathcal{K} \cup X_1$ is disconnected, pairing subsets in the alternating sum over any such component makes its bracket zero. The remaining brackets are indexed exactly by connected $Y \supset X_1$. Summing first over Y and then over X_1 therefore gives $A^{\eta_k}(e^{i\eta_k \mathbf{A}_{T_\ell}(1)} U_+) - A^{\eta_k}(U_+)$, since the shift at zero vanishes. Gauge invariance and (827) identify this with the right side of (829); the shift at $B = 0$ vanishes by its bound. For $k = 0$, the constrained background is the field itself, and every plaquette appears exactly once in (816).

The left side of (829) is holomorphic for $g \sup|B| < \varepsilon_{\mathrm{A}}$. If the critical action has the quadratic expansion $\mathbb{F}_W(zB) - \mathbb{F}_W(0) = z^2 g^2 (B, \mathsf{B}B)_1/2 + O(z^3)$ for real B, z, comparison of its first three Taylor coefficients gives

$$\sum_{Y} \mathbb{V}^{\mathrm{A}}(Y; P_W|_Y, B|_Y) = -\mathsf{t}[\mathbb{F}_W(B) - \mathbb{F}_W(0)] + \tfrac{1}{2}(B, \mathsf{B}B)_1. \tag{830}$$

This is the exact action remainder used with the Gaussian fluctuation form; the background and quadratic-expansion hypotheses must be checked at that application.

51.4. **Construction of the shift.** We now supply the shift used above. Assume $k \ge 1$ and the geometric hypotheses of proposition 41.1, with partition width $w = M$ and parameter radius $e^{\bar\kappa}$. In particular,

$$M \ge 88 r_w \mathfrak{M}, \qquad \mathring{\delta}_0 M \ge 2176\bar\kappa.$$

Assume also $12\mathfrak{M} \le \mathsf{o} L^2 M$ and the room condition $C_L \ge 8\mathsf{n}_{\mathrm{e}} L$, where $\mathsf{n}_{\mathrm{e}} \ge 1$ is the number of coarse layers used in coordinate-chart halos. Use the constant frame of height k: all its regions through level k are the torus, and its terminal region is empty. This terminal region belongs to the response frame; the localization step still has its entire torus available. Thus all constraint rows have level k, their weights are 1, and the bare spacing is η_k.

There are $0 < c_{\mathfrak{s}} \le 1 \le C_{\mathfrak{s}}$, depending only on L, for which one may take

$$a_{\mathfrak{h}} = \frac{c_{\mathfrak{s}}}{\max\{1, \mathsf{o} L M B^{(\mathrm{cg})}\}}, \qquad \varepsilon_{\mathfrak{h}} = c_{\mathfrak{s}} \min\{1, \varepsilon_{\mathrm{v}}\} e^{-C_{\mathfrak{s}}\bar\kappa}, \qquad \mathsf{c}_{\mathfrak{h}} = C_{\mathfrak{s}} e^{C_{\mathfrak{s}}\bar\kappa}. \tag{831}$$

Use $a_h = a_{\mathfrak{h}}$, $\varepsilon_h = \varepsilon_{\mathfrak{h}}$ and $c_h = \mathsf{c}_{\mathfrak{h}}$ in the preceding subsection. These choices are uniform in k, the cutoff and the torus. To specify the required smallness, set $b_* = \max\{4b_{\mathrm{D}}, 8C_j b_{\mathrm{D}}^3\}$ and $C_\times = 2C_P b_*$. Choose $C_{\mathfrak{s}} \ge \max\{32, 2C_P b_*\}$ and choose $c_{\mathfrak{s}}$ no larger than

$$\min\{a_P^\circ, r_P^\circ, L a_{\mathrm{D}}, \varepsilon_{\mathrm{D}}/(2C_P), (1200 C_\times)^{-1}, 1\}.$$

Decreasing it further preserves every assertion below.

For $P \in \mathcal{Q}_{k+1}(Y; a_{\mathfrak{h}}, a_{\mathfrak{h}}, a_{\mathfrak{h}})$ and $\sup|\mathcal{A}| < \varepsilon_{\mathfrak{h}}$, define a datum on all positive level-k bonds by

$$B'_Y(P,\mathcal{A})(b) = \begin{cases} \mathcal{A}(b), & b \text{ is a nonpivot bond inside } Y, \\ \beta_c(P,\mathcal{A}), & b = b_0(c),\ \square(c) \subset Y, \\ 0, & \text{otherwise.} \end{cases} \tag{832}$$

Here β_c is the pivot value of $\psi_V^{\mathrm{lin}}(\mathsf{C}_V \mathcal{A})$ from the preceding subsection. The first two cases exhaust the level-k bonds inside Y. In particular,

$$\sup|B'_Y| \le 2C_P \sup|\mathcal{A}| < \varepsilon_{\mathrm{D}} e^{-32\bar\kappa}. \tag{833}$$

Let s vanish off Y. Form the decoupled response on this constant frame with pair $(\mathbf{U}, L^{-3}\mathbf{J})$ restricted to the inner bare bonds of Y, and put

$$\mathfrak{h}_Y(s; P, \mathcal{A}) = (\mathbf{H}(s; B'_Y), j(s; B'_Y))\big|_{\mathrm{Bd}(Y)}. \tag{834}$$

The restriction includes the incident crossing bonds.

Proposition 51.5. The maps (834) satisfy the shift hypotheses (812), including their support, restriction, gauge, reality and coordinate-chart properties, with the constants (831). They commute with complex conjugation. In the real global setting described before (827), they satisfy that identification, provided the backgrounds there exist as single regular critical gauge orbits. ∎

Proof. We first verify that the response is defined. Read the triple in level-k units with decomposition $(U, L^{-1}A')$. Its natural-class radii can be taken to be

$$(\alpha_0, \alpha_1, \gamma_0) = (L^{-1}c_{\mathfrak{s}}, L^{-1}a_{\mathfrak{h}}, L^{-3}a_{\mathfrak{h}}).$$

They are at most a_{D}. The curvature bound of the simple domain is $a_{\mathfrak{h}}\eta_{k+1}^2 = L^{-2}a_{\mathfrak{h}}\eta_k^2$. Every cube needed by the local natural class has side at most $12\mathfrak{M}$ in level-k units, hence at most $12\mathfrak{M}/L \le \mathsf{o} L M$ in level-$(k+1)$ units. The simple domain supplies its own gauge on that cube, with logarithm and derivative bounded by $\mathsf{o} L M B^{(\mathrm{cg})} a_{\mathfrak{h}} \le c_{\mathfrak{s}}$. Changing units multiplies this logarithm by L^{-1} and its derivative by L^{-2}. The intermediate scale factors in the natural-class inequalities are at most 1. The same changes of units apply to A', while the current is multiplied by L^{-3}. These observations verify all the local natural-class inequalities on the inner bonds used by the response. Equation (833) now permits theorem M.

The source is holomorphic at fixed real U, depends only on the pair, and restricts consistently: each pivot value uses just its endpoint double-block tower. The response has these properties by theorem M. Its estimates, since every active scale factor is 1, give

$$\sup|\mathbf{H}|, \quad \sup|\nabla_{\mathbf{U}}^{\eta_k}\mathbf{H}|, \quad \sup|j| \le b_* e^{24\bar{\kappa}} \sup|B'_Y| \le \mathsf{c}_{\mathfrak{h}} \sup|\mathcal{A}|.$$

The derivative estimate also applies at the incident boundary. Indeed, the component response is supported over $Y^1(\gamma)$ and its current over $Y^2(\gamma)$. The margin construction ensures that a bare bond over $Y^j(\gamma)$, $j \le 4$, has both endpoints in $\operatorname{int}\gamma$. It also places the whole differentiation stencil based over $Y^3(\gamma)$ inside that region. Thus a nonzero stencil uses the same pair as the response estimate; outside that margin both terms of the derivative vanish. No choice of a pair outside Y is used in this argument.

For completeness, each endpoint or plaquette corner in this assertion is at intrinsic distance at most $\rho_0 + 5/2 \le r_0$ from its base site. The margin estimate in the local propagator construction puts it in the hull of a cell X with $\mathsf{P}(X) \subset \gamma$. Every partition cube containing this point then belongs to $\mathsf{P}(X)$; their union is a neighborhood of the point. Thus the point lies in $\operatorname{int}\gamma$. The local $8L$ cube used to compare the nonlinear formulas also lies in γ: if the base site belongs to q^{+3}, then $4L\xi_y \le 4L^3\xi_m \le 2\mathfrak{M}\xi_m$, so this cube is contained in $\mathsf{q}^{\boxplus} \subset \bigcup \mathsf{P}(X)$. Consequently the component sum in theorem M gives exactly the support and splitting assertions required here. At $s = 0$ every walk monomial vanishes, since every trail meets a partition cube; the linear operators and both fixed-point iterates are zero, hence so are the shift and current. Extension by zero preserves the source rows used by each component and therefore preserves the response. Unitary gauges preserve all the constructions. Complex conjugation commutes with the local mean series, with the pivot iteration and with the response iteration; it conjugates s as well as the fields. This proves the reality assertions, including conjugation covariance.

We give the coordinate-chart argument because the support at crossing bonds matters. A translation by $M\mathbb{Z}^3$, or a chart of a nonwrapping domain, is on each local lift a translation by a vector in $\mathfrak{M}\mathbb{Z}^3$. It preserves the fine cubes, the underlying lattices, the ordered averaging paths and the endpoint blocks. Restrict first to one component γ of $\{s \ne 0\}$. The map takes each cell trail inside γ to its translated trail with the same label and weight. Its inverse gives the inverse correspondence of trails. Each local inverse and projection in a walk is the inverse or projection of the translated finite matrix, so its kernel is carried to that of the translated walk. Products, sums, weighted transposes and the convergent inverse series preserve this equality. The parameter monomial is unchanged when s is transported with its cubes. Thus the four operators in the response equations are intertwined by the chart.

On the $8L$ cubes around the sites of $Y^3(\gamma)$, the local holonomies, currents and logarithmic averages agree after transport. Their Taylor series and derivatives therefore agree as well. Starting the two fixed-point iterations at zero now proves equality of every iterate, and uniform convergence proves equality of the responses and their currents. The pivot source is carried to the corresponding source by its already proved locality and uniqueness. This proves transport on the bonds over $Y^2(\gamma)$. Both fields vanish on all other incident bonds. There is no extra nonzero value on the other side: preservation of endpoint blocks and injectivity of the chart imply that a transported bond is over the transported margin exactly when the original bond was over the original margin. Summing the components proves the asserted chart covariance. As before, the transported triple must belong to the stated domain.

It remains to identify the real global response. Put $V = \mathcal{M}^k U_+$ and $B' = B'_{T_\ell}(P_W, \mathcal{A})$. The finer constraint set over V is contained in the coarser one over W, so U_+ is critical for the finer constraint. The multiplier argument of proposition 41.2, at $A = 0$, gives $J^{(k)}(U_+) = Q^{\mathsf{T}}\lambda_0$. At $s = 1$ the response is independent of the auxiliary partition and its parameter radius. By proposition 41.2, with $A = \mathbf{H}(1; B')$ and $K_A = \mathcal{Q}'(A)$,

$$\mathcal{Q}(A) = B', \qquad F'(A) = K_A^{\mathsf{T}}\omega(A), \qquad J^{(k)}(e^{i\eta_k A}U_+) = L^{-3}J^{(k+1)}(U_+) + j(1; B'). \tag{835}$$

All rows of this frame are interior. Its actual-average factorization is

$$\mathcal{M}^k(e^{i\eta_k A}U_+) = (e^{iB'}V)^{v[A]}. \tag{836}$$

We check that stationarity in the reduced coordinates implies criticality for this actual average. For a real node function σ let

$$\ell_{B'}\sigma(c) = \frac{d}{dt}\Big|_{t=0} \frac{1}{i}\log\Big[e^{it\sigma(c_-)}e^{iB'(c)}V(c)e^{-it\sigma(c_+)}V(c)^{-1}\Big].$$

The function $z(B';\sigma) = (\omega(A), \ell_{B'}\sigma)_1$ is holomorphic in B' and linear in σ. For a small real gauge variation $w_t = e^{it\chi}$, write $e^{i\eta_k A^{w_t}}U_+ = (e^{i\eta_k A}U_+)^{w_t}$ and set

$$X_\chi(A) = \partial_t A^{w_t}|_0\,, \qquad \rho(A;\chi) = \frac{1}{i}\partial_t\big(v[A^{w_t}]^{-1}w_t v[A]\big)\Big|_0.$$

Differentiating (836) gives $K_A X_\chi(A) = \ell_{B'}\rho(A;\chi)$. At $A = 0$, the derivative of the accumulated mean factors gives $\rho(0;\chi) = Q'\chi$, the scalar block average. It is onto: the blocks $B^k(y)$ are disjoint, and one may choose one site in each block and assign its value so that its transported average is any prescribed node value. In finite dimensions surjectivity persists for A sufficiently close to zero. Gauge invariance of F and (835) therefore give

$$z(B';\rho(A;\chi)) = (F'(A), X_\chi(A))_{\eta_k} = 0$$

for all node values and all real B' near zero. Holomorphy first extends this equality to a complex neighborhood and then, along complex lines, to the whole response ball $\sup|B'| < \varepsilon_{\mathrm{D}}$.

Now let E_t be any differentiable real curve through $E = e^{i\eta_k A}U_+$ with the actual average in (836) fixed. Its local logarithm gives $E_t = e^{i\eta_k A_t}U_+$. Put $h_t = v[A_t]^{-1}v[A]$ and $\sigma = (1/i)\partial_t h_t|_0$. The fixed-average identity yields $K_A\dot{A} = \ell_{B'}\sigma$. Consequently the derivative of its action is $\eta_k(F'(A), \dot{A})_{\eta_k} = \eta_k z(B';\sigma) = 0$. Thus E is critical for its actual average.

Choose a bare gauge whose values on the level-k sites are $v[A]^{-1}$. It makes this average $e^{iB'}V = V_W(\mathcal{A})$. Finally the estimates at $s = 1$ give

$$\sup|A|, \quad \sup|\nabla^{\eta_k}_{U_+}A|, \quad \sup|j(1;B')| \le C_\times \sup|\mathcal{A}| =: \rho, \qquad 600\rho < \tfrac{1}{2}\min\{1, \varepsilon_{\mathrm{v}}\}.$$

The plaquette estimate (822) bounds the curvature of E by $[(3/2)L^{-2} + 1/2]\varepsilon_{\mathrm{v}}\eta_k^2 < \varepsilon_{\mathrm{v}}\eta_k^2$; its current is bounded by $(L^{-3} + 1/2)\varepsilon_{\mathrm{v}} < \varepsilon_{\mathrm{v}}$. These are the regularity inequalities at level k. They are gauge invariant. The assumed uniqueness of the regular critical orbit therefore identifies E with the background over $V_W(\mathcal{A})$. The current identity in (835) proves the other assertion of (827). □

51.5. **Tower estimates and substitution in the analytic domain.** The pointwise bounds defining the simple domain do not by themselves give the same bounds for its constrained backgrounds. We first estimate those backgrounds, including their dependence on complex data. This will give a quantitative margin for substituting the fluctuation into an interaction from an earlier scale.

All determining sets in this subsection are reduced: every inward bond is removed. We write them as $\mathbf{B}^\flat$, including those of truncated towers. Fix the geometric thresholds of theorem E and proposition 34.2, and decrease the regularity threshold and the complex datum radius so that the canonical background belongs to the natural regular-pair class required by proposition 34.2. Write $\bar{\varepsilon}$ and r_{B} for these thresholds. They are uniform in the depth and the volume. We use $B_3 \ge 6$ for the real-background regularity constant, increased if necessary.

Lemma 51.6. For every regular real datum V_0 on $\mathbf{B}^\flat$, put $U_0 = U^\circ_{\mathbf{B}^\flat}(V_0)$. On $\sup|B| < r_{\mathrm{B}}$ there is a holomorphic potential $\mathcal{H}_{\mathbf{B}^\flat,V_0}(B)$, vanishing at zero, such that

$$\Phi_{\mathbf{B}^\flat,V_0}(B) = (e^{i\eta\mathcal{H}(B)}U_0)^{u_B}, \qquad \mathsf{R}_{\mathsf{m}}D^{\mathsf{T}}\mathcal{H}(B) = 0, \qquad u_B(y)v_q[U_0;\eta\mathcal{H}(B)](y) = I \quad (y \in \Lambda_q). \tag{837}$$

The potential and u_B are real and unitary, respectively, at real data. For suitable $B_5 \geq 1$ and $0 < \delta_{\mathrm{tw}} \leq 1$, the derivative $h = D\mathcal{H}(B)e_c$ satisfies

$$\max\{\xi_y|h(b)|,\ \xi_y^2|\nabla_{U_0}h(b)|,\ \xi_y^3|D_{U_0}^{\mathsf{T}}D_{U_0}h(b)|\} \leq B_5 e^{-\delta_{\mathrm{tw}}d(y,y_c)}|e|. \tag{838}$$

Here an endpoint of b lies in $\Delta(y)$, y_c is an endpoint of c in its layer, the first differences use only inner bonds, and the last member is asserted where the four incident plaquettes are inner. The constants are independent of the depth, the sequence of regions and the volume. ■

Proof. Use the potential in the proof of theorem E: $\mathcal{H}(B) = \mathbf{H}(B_\natural)$, with the actual datum converted to the straightened datum by the endpoint factors. The construction of the axial representative gives (837). At a top, the ascent is empty, so the axial gauge factor there is exactly the inverse of the accumulated factor v_q. The identities first hold at real data and extend to the complex row ball by holomorphy. The fixed-point uniqueness and the gauge normalization give uniqueness of the pair near zero, and hence throughout its connected domain.

We give the derivative argument, since the third bound in (838) is needed below. The derivative of $B \mapsto B_\natural$ in a direction e_c is bounded by $C(L)|e|$ and is supported in a bounded number of rows within distance $4L$ of y_c. Indeed, interior rows read $B(c)$ directly; an outward row uses the one-step endpoint factor, which depends only on the interior rows of the adjacent block. Differentiating the local logarithm on half its domain gives the stated bound. There are no edge rows for the reduced determining set.

Take $w(y) = e^{\theta d(y,y_c)}$ with $\theta > 0$ smaller than the rate in (501). Differentiating that estimate bounds the weighted potential and first-difference norm of h by $C(L)|e|$. The multiplier is

$$\lambda(A) = (I + H^{\mathsf{T}}\mathsf{C}_A'^{\mathsf{T}})^{-1}H^{\mathsf{T}}F'(A).$$

Its derivative has the same weighted bound. To see this without estimating a second difference of h in advance, pair $F''(A)h$ against a column of H and use the differentiated energy identity (581). The curl term is paired with the first differences of that column, and the remaining terms have the local bounds of proposition 33.1. The column estimate and (532) then bound $H^{\mathsf{T}}F''(A)h$ in the weighted row norm by $C\|h\|_w$. The derivative of $H^{\mathsf{T}}\mathsf{C}_A'^{\mathsf{T}}$ has the same bound by the local averaging estimate. Its undifferentiated operator has weighted norm less than $1/2$ after reducing the datum radius, so its inverse is bounded by two. Thus $|\dot\lambda|_w \leq C\|h\|_w$.

Differentiate the stationarity equation. Since the gauge equation also differentiates to $\mathsf{R}_{\mathsf{m}}D^{\mathsf{T}}h = 0$,

$$D^{\mathsf{T}}Dh = (Q + \mathsf{C}_A')^{\mathsf{T}}\dot\lambda + [D_A\mathsf{C}_A'^{\mathsf{T}}h]\lambda - \Delta'h - \mathcal{V}'(A)h.$$

Every term on the right has weighted source norm at most $C(|\dot\lambda|_w + \|h\|_w)$: this is the transpose averaging bound and the local action-remainder bound, with their fixed neighborhood enlargement. This proves the last estimate. Changing from the initial endpoint to either endpoint costs at most a factor L^3 in the scale and a fixed exponential factor in the distance. Increase B_5 and decrease δ_{tw} to include these factors. □

For an inner bond b over y, let $N_b(f)$ denote the maximum on the left of (838), omitting the last member when its plaquette star is not inner. The exponential summation bound gives finite constants $B^\sharp \leq B^\flat$ such that

$$N_b(\mathcal{H}(B_1) - \mathcal{H}(B_2)) \leq B^\sharp \sup|B_1 - B_2|, \tag{839}$$

$$N_b(\mathcal{H}(B_1) - \mathcal{H}(B_2)) \leq B^\flat e^{-\delta_{\mathrm{tw}}D/2} \sup|B_1 - B_2| \tag{840}$$

in the second line when the data agree at all rows within distance D of y. The same second estimate holds for the linearization $\mathsf{H} = D\mathcal{H}(0)$. These inequalities follow by integrating the derivative along the segment between the two data and summing (838). Cauchy's estimate on the line zB also gives

$$N_b(\mathcal{H}(B) - \mathsf{H}B) \leq 2B^\sharp r_{\mathrm{B}}^{-1}(\sup|B|)^2, \qquad \sup|B| \leq r_{\mathrm{B}}/2. \tag{841}$$

A pillar $B^q(y)$ of $\mathbf{B}^\flat$ is *free* if $q \geq 1$ and it contains no node of $\mathbf{B}^\flat$ other than its top y.

We need one quantitative comparison between the axial and Landau representatives. On a free pillar of level q, put $h_y = \sup_{B^q(y)} \xi_y |\mathcal{H}(B)|$. For a sufficiently small $r_{\mathrm{G}} \leq r_{\mathrm{B}}$,

$$\Phi(B) = W(B)^{u(B)}, \qquad W(B) = e^{i\eta\mathcal{H}(B)} U_0, \qquad |u(B)(x)^{\pm 1}| \leq e^{c_u h_y} \leq e^{2/25} \tag{842}$$

on the free pillars, provided the base curvature there is in the averaging domain. Here c_u depends only on L. For completeness, the gauge is $u(B)(x) = v_q[U_0; \eta\mathcal{H}(B)](y)^{-1} \mathcal{T}_{W(B)}(x)$, where $\mathcal{T}$ is the product of contour variables along the ascent from x to y. At level $i < q$ the logarithm of its contour factor is bounded by $C(L) L^{i+1-q} h_y$, by proposition 51.1; the accumulated top factor satisfies the same geometric sum bound. Summing $\sum_{i<q} L^{i-q} \leq (L-1)^{-1}$ gives the first bound in (842). Choose r_{G} so that $c_u B^\sharp r_{\mathrm{G}} \leq 2/25$ and every contour logarithm stays in its domain. At real data this is the axial gauge normalization; both sides are holomorphic, so the identity extends to complex data. In particular the condition number of $u(B)(x)$ is less than $6/5$.

We record the local expansions used to pass between the potential, curvature and current. Their constants do not deteriorate under a change of lattice scale.

Lemma 51.7. Let $\eta \leq 1/13$. On a plaquette based at x, number the edges in directions μ, ν, μ, ν, with initial points $x, x + \eta e_\mu, x + \eta e_\nu, x$, respectively. Let $\phi = \max |f|$, let $\mathbf{W}$ be a complex configuration, and suppose $|T||T^{-1}| \leq \varkappa$ for $T = \mathbf{W}_1, \mathbf{W}_4, \partial\mathbf{W}, \partial\mathbf{W}\mathbf{W}_4$. If $|\partial\mathbf{W} - I| \leq 1/4$ and $4\varkappa\eta\phi \leq 1/4$, then $\partial(e^{i\eta f}\mathbf{W}) = R\partial\mathbf{W}$, where

$$|R - I - i\eta^2 D_{\mathbf{W}} f| \leq 11\varkappa^2\eta^2\phi^2 + 6\varkappa\eta\phi|\partial\mathbf{W} - I|. \tag{843}$$

On the four plaquettes incident to a bond, let U_0 be real, $h = \max\{|f|, |\nabla_{U_0} f|\} \leq 1/2$, and $f_0 = \max \eta^{-2}|\partial U_0 - I| \leq 1$. Then

$$|J(e^{i\eta f}U_0) - J(U_0) - D_{U_0}^{\mathsf{T}} D_{U_0} f| \leq 134(h^2 + f_0 h). \tag{844}$$

Only differences between bonds of these plaquettes enter h. ∎

Proof. Number the edges of the positively oriented plaquette so that its product is $W_1 W_2 W_3^{-1} W_4^{-1}$. Moving these transports to the right expresses R as four exponentials with generators

$$f_1, \quad \mathrm{Ad}(W_1) f_2, \quad -\mathrm{Ad}(\partial W\, W_4) f_3, \quad -\mathrm{Ad}(\partial W) f_4.$$

Their sum is

$$\eta D_W f + (1 - \mathrm{Ad}(\partial W))(\mathrm{Ad}(W_4) f_3 + f_4).$$

The second term is at most $6\varkappa\phi|\partial W - I|$. For exponentials whose total exponent norm is $s \leq 1/4$, the terms of degree at least two sum to at most $s^2 e^s/2$. With $s \leq 4\varkappa\eta\phi$ this gives (843).

For the current, retain one more Taylor order at the real base. Writing the four exponents as $Y_1, \ldots, Y_4$, one has $Y_3 = -Y_1 - \delta_1$, $Y_4 = -Y_2 - \delta_2$, with $|\delta_i| \leq (15/13)\eta^2 h$. The quadratic commutator is

$$\frac{1}{2}\sum_{i<i'} [Y_i, Y_{i'}] = [Y_1, Y_2] + \frac{1}{2}(-[Y_1, \delta_1] - 2[Y_2, \delta_1] - [Y_2, \delta_2] + [\delta_1, \delta_2]).$$

The second term is bounded by $5\eta^3 h^2$; replacing $\mathrm{Ad}(U_{0,1}) f_2$ in the first term by f_ν costs $2\eta^3 h^2$. The third-order remainder in $(R - R^{-1})/2$ is at most $15\eta^3 h^3$. Including the base plaquette and its inverse therefore gives

$$F_{e^{i\eta f}U_0} = F_{U_0} + D_{U_0} f + i[f_\mu, f_\nu] + \rho_F, \qquad |\rho_F| \leq 24\eta(h^2 + f_0 h), \qquad |F_{e^{i\eta f}U_0}| \leq 3f_0 + 6h.$$

Insert this into the covariant backward difference defining J. For each of the two transverse directions, replacing its perturbed transport by the real transport costs at most $15(h^2 + f_0 h)$. The difference of the commutator costs at most $4h^2$, since each transported difference of f is bounded by ηh. The two remainders cost at most $48(h^2 + f_0 h)$. The linear terms are exactly $J(U_0) + D_{U_0}^{\mathsf{T}} D_{U_0} f$. Thus the total error is $2(15 + 4 + 48)(h^2 + f_0 h)$, as asserted. □

Proposition 51.8. There are $a^\sharp \in (0, 1/60]$ and $K^\sharp_{\mathrm{tw}} \geq 1$, uniform in the depth and volume, with the following property. In level-n units, let $\eta = L^{-n}$ and let $\mathsf{L}_-(e)$ be the smallest pillar scale among the endpoints or corners of a bond or plaquette. Suppose $\epsilon_0 + \epsilon_1 \leq a^\sharp$ and

$$|\partial U - I|,\ |\partial(e^{i\eta f}U) - I| < \epsilon_0 \eta^2 \mathsf{L}_-^{-2}, \qquad |f| < \epsilon_1 \mathsf{L}_-^{-1},$$

where U is real. The tower configuration at the averaged datum of $e^{i\eta f}U$ exists and is holomorphic in f on an open neighborhood of these data. If the corners of a plaquette, or all corners of the four plaquettes incident to a bond, lie in free pillars of one level with scale ξ, then

$$|\partial U_{\mathbf{B}^\flat} - I| < K^\sharp_{\mathrm{tw}}(\epsilon_0 + \epsilon_1^2)\eta^2\xi^{-2}, \qquad |J(U_{\mathbf{B}^\flat})| < K^\sharp_{\mathrm{tw}}(\epsilon_0 + \epsilon_1^2)\xi^{-3}.$$

The same assertions hold with U in place of $e^{i\eta f}U$. ∎

Proof. Let C_Q be the constant in proposition 51.1, and choose

$$0 < a_{\mathrm{av}} \leq \min\{a_{\mathrm{plain}}, c_{\mathrm{plain}}, c_2^{\mathrm{SO}}, 1\}, \qquad (4C_Q + 2)a_{\mathrm{av}} < \omega_M.$$

This common radius is admissible for the plain average and the real curvature estimate in proposition 14.1. A real configuration with curvature $\alpha\eta^2\mathsf{L}_-^{-2}$ and $L^2\alpha \leq a_{\mathrm{av}}$ induces a $c^\natural\alpha$-regular datum, with $c^\natural = 2L^2$. At a tested level-q plaquette its four factors are the values of $\mathcal{M}^q U$; the averaging corridor lies in Ω_{q-1}, where the bare curvature is bounded by $L^2\alpha L^{-2q}$. The real curvature averaging estimate therefore gives this assertion, including outward tests. At level zero it is the original curvature bound.

Choose

$$r_0 = \frac{1}{2L}\min\{a_{\mathrm{av}}, r_{\mathrm{G}}/C_Q\}, \qquad a^\sharp \leq \min\{1/60, r_0, \bar{\varepsilon}/(22c^\natural), a_{\mathrm{av}}/(22B_3c^\natural)\}.$$

Equip forms with $\|w\| = \sup_b \mathsf{L}_-(b)|w(b)|$. For $\|w\| < 2r_0$, the plain-average logarithm $B(w) = Q(U; \eta w)$ satisfies

$$M_{\mathbf{B}^\flat}(e^{i\eta w}U) = e^{iB(w)}V_0, \qquad V_0 = M_{\mathbf{B}^\flat}U, \qquad \sup|B(w)| \leq C_Q L\|w\| < r_{\mathrm{G}}.$$

Thus the tower is $\Phi_{V_0}(B(w))$ and exists on this entire complex ball. Put $W(w) = e^{i\eta\mathcal{H}(B(w))}U_0$. For the plaquette or bond in the statement, let $q \geq 1$ be the common pillar level and put $\xi = L^{q-n}$. Thus the bare spacing in level-q units is $\eta/\xi = L^{-q} \leq 1/13$. The two maps

$$F^\partial(w) = \eta^{-2}\xi^2(\partial W(w) - I), \qquad F^J(w) = \xi^3 J(W(w))$$

are holomorphic there and bounded by one, after decreasing r_{G} if necessary. Indeed, (839) controls the potential, its first difference and $D^\mathsf{T}D$ in the local scale. The plaquette expansion and the current expansion then bound these maps by

$$B_3c^\natural a^\sharp + 3h + 11h^2, \qquad B_3c^\natural a^\sharp + h + 134(h^2 + B_3c^\natural a^\sharp h),$$

where $h = B^\sharp r_{\mathrm{G}}$ can be chosen at most $1/300$.

Write $f = f_{\mathrm{R}} + if_{\mathrm{I}}$. The projections onto the two real parts are contractions, also after applying D_U. The plaquette expansion gives

$$|D_U f| \leq \mathsf{L}_-^{-2}(2.1\epsilon_0 + 11\epsilon_1^2).$$

Applying the same expansion to f_{R} and f_{I} shows that each real configuration $e^{i\eta f_{\mathrm{R}}}U$ and $e^{i\eta f_{\mathrm{I}}}U$ has curvature at most $22(\epsilon_0 + \epsilon_1^2)\eta^2\mathsf{L}_-^{-2}$. Its induced datum is therefore regular. The real-background bound and (842) imply, for either $F = F^\partial, F^J$,

$$|F(0)| \leq B_3c^\natural\epsilon_0, \qquad |F(f_{\mathrm{R}})|, |F(f_{\mathrm{I}})| \leq \frac{6}{5}\,22B_3c^\natural(\epsilon_0 + \epsilon_1^2).$$

The Taylor terms of order at least two of a holomorphic map bounded by one on $\|w\| < 2r_0$ have sum at most $2\|w\|^2/(2r_0)^2$ on the half ball. Its linear term is complex linear. Subtract it at f, f_{R} and f_{I} to obtain

$$|F(f)| \le 3|F(0)| + |F(f_{\mathrm{R}})| + |F(f_{\mathrm{I}})| + \frac{6\|f\|^2}{4r_0^2} \le (56B_3c^\natural + 2r_0^{-2})(\epsilon_0 + \epsilon_1^2).$$

Finally (842) conjugates these quantities to those of the tower, with loss less than $6/5$. Take $K_{\mathrm{tw}}^\sharp \ge 2(56B_3c^\natural + 2r_0^{-2})$. The argument at $f = 0$ proves the last assertion. All choices precede the depth, the sequence and the volume. □

Proposition 51.9. The following consequences hold with the same constants.

(f) Let $(U, A', \mathbf{J}) \in \mathcal{Q}_p(Z; \alpha_0, \alpha_1, c_0)$ and $\alpha_0 + \alpha_1 \le a^\sharp$. For both tower rules and every truncation $1 \le n \le p$, the tower quantities exist and, on $Z^{\natural p}$, satisfy

$$|\partial U_n^{\mathfrak{r}} - I| < K_{\mathrm{tw}}^\sharp(\alpha_0 + \alpha_1^2)L^{-2p}, \qquad |J_n^{\mathfrak{r}}| < K_{\mathrm{tw}}^\sharp(\alpha_0 + \alpha_1^2)L^{2(n-p)}. \tag{845}$$

They are holomorphic for fixed U on the open simple domain. The estimates also hold for the tower of U. Consequently, if $\alpha_i \le b_i$ and $K_{\mathrm{tw}}^\sharp(\alpha_0 + \alpha_1^2) \le b_0$, then

$$\mathcal{Q}_j(X; \alpha_0, \alpha_1, c_0) \subset \mathcal{U}_j^\circ(X; b_0, b_1, c_0). \tag{846}$$

The trivial triple belongs to every core with positive radii.

(g2) Let $1/40 \le d < 1$, $\vartheta = 1 - d$, and

$$0 < b_0 \le b_1, \qquad b_1^2 \le c_\natural b_0, \qquad b_0 + b_1 \le a^\sharp, \qquad c_\natural \ge 1, \qquad b_2 = \frac{d^2 b_0}{32K_{\mathrm{tw}}^\sharp(1 + c_\natural)}.$$

For $P = (U, A', \mathbf{J}) \in \mathcal{U}_j^\circ(X; \vartheta b_0, \vartheta b_1, c_0)$, put $\mathbf{U} = e^{i\eta_j A'}U$. If $|\mathbf{A}|, |\nabla_{\mathbf{U}}\mathbf{A}| < b_2$ and $|\mathbf{J} + \mathbf{K}| < b_0$, then

$$P \dotplus (\mathbf{A}, \mathbf{K}) := \left(U, \frac{1}{i\eta_j}\log(e^{i\eta_j \mathbf{A}}e^{i\eta_j A'}), \mathbf{J} + \mathbf{K}\right) \in \mathcal{U}_j^\circ(X; b_0, b_1).$$

The logarithm and the substitution are locally holomorphic.

■

Proof. For (f), pass to level-n units and set $\lambda = L^{-(p-n)}$, $f = \lambda A'$. The hypotheses of proposition 51.8 hold with $\epsilon_0 = \lambda^2\alpha_0$ and $\epsilon_1 = \lambda\alpha_1$. On the two-layer interior the relevant plaquettes and their stars lie in free pillars of level n, so their local scale is one. Since $\lambda^2\eta_n^2 = L^{-2p}$, this is (845). For a subtower (p, Z) of a level-j domain, rescale the triple by $(U, A', \mathbf{J}) \mapsto (U, \lambda_p A', \lambda_p^3\mathbf{J})$, $\lambda_p = L^{-(j-p)}$. The bound in (845) is then at most $\lambda_p K_{\mathrm{tw}}^\sharp(\alpha_0 + \alpha_1^2)$, which proves every clause in (846). At the trivial triple the datum, canonical background and Landau potential are trivial, proving the final assertion.

For (g2), put $R = 2K_{\mathrm{tw}}^\sharp(1 + c_\natural)/d$ and $\beta = Rb_2 = db_0/16$. On $|z| \le R$ write

$$A'_z = (i\eta_j)^{-1}\log(e^{iz\eta_j \mathbf{A}}e^{i\eta_j A'}).$$

The logarithm is defined since $\eta_j(\beta + \vartheta b_1) \le 1/780$. The logarithmic remainder has Lipschitz constant $\epsilon = 8\eta_j(\beta + \vartheta b_1) < 0.0103 \le d/2$. Thus

$$|A'_z| < \vartheta b_1 + (1 + \epsilon)\beta < b_1.$$

The change from $\nabla_{\mathbf{U}}$ to ∇_U costs at most $\frac{1}{4}\sup|\mathbf{A}|$. Equivariance and the same Lipschitz bound for the logarithmic remainder give

$$|\nabla_U A'_z| < (1 + \epsilon)(\vartheta b_1 + 5\beta/4) < b_1.$$

For a plaquette, apply the local product expansion with condition number at most 11/10. Its relative increment has norm at most $(5/2)\beta\eta_j^2$, and hence

$$|\partial(e^{iz\eta_j \mathbf{A}}\mathbf{U}) - I| < (1 - d + 5.01d/32)b_0\eta_j^2 < b_0\eta_j^2.$$

The real factor is unchanged and the new current is bounded by hypothesis. The entire disc therefore lies in the simple domain with radii (b_0, b_1).

Fix a subtower (p, Z) and put $\lambda_p = L^{-(j-p)}$. Normalize either its curvature or current by the corresponding power in (845). By (f) the resulting holomorphic function $g(z)$ is bounded on $|z| < R$ by

$$K_{\mathrm{tw}}^\sharp(\lambda_p b_0 + \lambda_p^2 b_1^2) \le K_{\mathrm{tw}}^\sharp(1 + c_\natural)\lambda_p b_0.$$

At zero the core hypothesis gives $|g(0)| < (1 - d)\lambda_p b_0$. Schwarz's lemma applied to $g - g(0)$ yields $|g(1) - g(0)| \le d\lambda_p b_0$. This is the required tower bound. The tower of the unchanged real factor already satisfies the stronger bound at zero. Local holomorphy follows from the convergent matrix-logarithm series. Every argument is local and also applies to finite domains in the infinite lattice. □

Proposition 51.10. Use the core and tower hypotheses of proposition 51.9, at level $j \ge 1$, and put $C_{\mathrm{os}} = 18$. The half-radius form of outer substitution has

$$b_2 = \frac{b_0}{128K_{\mathrm{tw}}^\sharp(1 + c_\natural)}. \tag{847}$$

For $P \in \mathcal{U}_j^\circ(X; b_0/2, b_1/2, c_0)$, with the radii of proposition 51.9(g2),

$$|\mathbf{A}|, |\nabla_{\mathbf{U}}\mathbf{A}| < b_2, \quad |\mathbf{J} + \mathbf{K}| < b_0 \quad \Longrightarrow \quad P \dotplus (\mathbf{A}, \mathbf{K}) \in \mathcal{U}_j^\circ(X; b_0, b_1). \tag{848}$$

More generally, let $0 < \tau < 1$, $c_\natural \ge 1$, and

$$\begin{gathered} 0 < b_0 \le b_1, \qquad b_1^2 \le c_\natural b_0, \qquad 2(b_0 + b_1) \le a^\sharp, \\ b_2 = \frac{\tau b_0}{C_{\mathrm{os}}K_{\mathrm{tw}}^\sharp(1 + c_\natural)}. \end{gathered} \tag{849}$$

If $P \in \mathcal{U}_j^\circ(X; (1 - \tau)b_0, (1 - \tau)b_1, c_0)$ and the three bounds on the left of (848) hold, then the same conclusion holds. In both cases the logarithmic substitution is locally holomorphic, and its pair is $(e^{i\eta_j \mathbf{A}}\mathbf{U}, \mathbf{J} + \mathbf{K})$. ■

Proof. The first assertion is proposition 51.9(g2) with $d = 1/2$. For the second, a complex disc with twice the target radii will give an increment proportional to τ. We first record the pointwise estimate used on that disc. Write $\xi = \eta_j \le 1/13$, $\mathbf{U} = e^{i\xi A'}U$, and suppose the simple input bounds are (β_0, β_1). For $\beta_0, \beta_1, s \le 1/60$ and $|f|, |\nabla_{\mathbf{U}} f| < s$, put

$$A^f = (i\xi)^{-1} \log(e^{i\xi f} e^{i\xi A'}).$$

Then

$$\begin{gathered} |A^f - A'| < \tfrac{11}{10}s, \qquad |\nabla_U(A^f - A')| < \tfrac{11}{10}s, \\ |\partial(e^{i\xi f}\mathbf{U}) - \partial\mathbf{U}| < \tfrac{9}{4}s\xi^2. \end{gathered} \tag{850}$$

All logarithms are defined since $\xi(s + \beta_1) \le 1/390$.

Here are the increment estimates, including the cancellation in the derivative bound. Set $\epsilon_* = 1/380$ and $q_* = (1 + \epsilon_*)/(1 - \epsilon_*) < 51/50$. At a bond, the two matrices $e^{i\xi f}e^{i\xi A'}$ and $e^{i\xi A'}$ lie in the ϵ_*-ball of I. The logarithm series gives $|A^f - A'| \le q_*|f|$. At a derivative stencil, transport the neighboring values with its real bond $u = U_\mu(x)$, and put

$$a_1 = i\xi f_\nu(x), \quad a_2 = i\xi \operatorname{Ad}(u) f_\nu(x + \xi e_\mu), \quad c_1 = i\xi A'_\nu(x), \quad c_2 = i\xi \operatorname{Ad}(u) A'_\nu(x + \xi e_\mu).$$

Changing from $\mathbf{U}$ to U gives

$$|a_2 - a_1| < \xi^2 s(1+\omega_*), \qquad |c_2 - c_1| < \xi^2 \beta_1, \qquad \omega_* = 2\beta_1(1+\epsilon_*) < 1/29.$$

Indeed the difference of the two transported f's is a conjugation by $e^{i\xi A'_\mu(x)}$; its norm is at most $(e^{2\xi\beta_1} - 1)|f|$. Let $E_i = e^{a_i}e^{c_i}$, $H_i = e^{c_i}$, and $D_i = E_i - H_i$. The telescoping identity for powers in the logarithm series gives

$$\begin{aligned} &|(\log E_2 - \log H_2) - (\log E_1 - \log H_1)| \\ &\quad \le \frac{|D_2 - D_1|}{1-\epsilon_*} + \frac{(|E_2 - E_1| + |H_2 - H_1|)|D_1|}{2(1-\epsilon_*)^2}. \end{aligned}$$

For example, write each difference of n-th powers as a sum of n products with one increment. Differencing these products leaves either the increment $D_2 - D_1$ or one increment of E or H; their multiplicities are n and $n(n-1)/2$, respectively. The exponential difference bounds give

$$\begin{aligned} |\Delta| &\le q_*(|a_2 - a_1| + \xi s|c_2 - c_1|) \\ &\quad + \tfrac{1}{2}q_*^2\xi s(|a_2 - a_1| + 2|c_2 - c_1|) \\ &< \xi^2 s\left\{q_*(1+\omega_*) + q_*\xi\beta_1 + \tfrac{1}{2}q_*^2\xi[s(1+\omega_*) + 2\beta_1]\right\} < \tfrac{11}{10}\xi^2 s. \end{aligned}$$

Here Δ is the logarithmic difference on the left; it equals $i\xi^2\nabla_U(A^f - A')$.

For the plaquette bound use eq. (843) with $\varkappa = 101/100$. The required transport condition numbers are at most

$$e^{2\xi\beta_1}\frac{1+\beta_0\xi^2}{1-\beta_0\xi^2} < 1.003 < \varkappa.$$

The two contained derivatives give $|D_{\mathbf{U}}f| < 2s$. The product remainder therefore bounds the plaquette increment by

$$\xi^2 s[2 + 11\varkappa^2 s + 6\varkappa\xi\beta_0](1+\beta_0\xi^2) < \tfrac{9}{4}\xi^2 s.$$

This proves (850). The real-factor gauge conditions are unchanged, so the substituted triple has simple radii $(\beta_0 + \frac{9}{4}s, \beta_1 + \frac{11}{10}s)$, with its current bounded separately.

Now set

$$\beta_\iota = (1-\tau)b_\iota, \qquad \gamma_\iota = 2b_\iota, \qquad s = \tfrac{4}{9}(1+\tau)b_0, \qquad R = s/b_2 > 2.$$

For $|z| < R$, apply (850) with $f = z\mathbf{A}$. The resulting triples, with current set to zero for this tower estimate, have simple radii (γ_0, γ_1):

$$\beta_0 + \tfrac{9}{4}s = 2b_0, \qquad \beta_1 + \tfrac{11}{10}s \le 2b_1, \qquad \gamma_0 + \gamma_1 \le a^\sharp.$$

The logarithms and the tower quantities are holomorphic on this disc by proposition 51.9(f). For a subtower (p, Z), let $\lambda = L^{-(j-p)}$. After division by the curvature or current scale in (845), each tower quantity $F(z)$ is bounded by $\lambda K^\sharp_{\mathrm{tw}}(\gamma_0 + \gamma_1^2)$. At zero the core hypothesis gives $|F(0)| < (1-\tau)\lambda b_0$. Schwarz's lemma applied to $F - F(0)$ gives

$$|F(1) - F(0)| \le 2\lambda K^\sharp_{\mathrm{tw}}(\gamma_0 + \gamma_1^2)\frac{b_2}{s} \le \tau\lambda b_0,$$

because $\gamma_0 + \gamma_1^2 \le 4(1+c_\natural)b_0$ and $C_{\mathrm{os}} = 18$. Thus every tower has the required radius b_0. The real-factor towers are unchanged. At $z = 1$, (850) with $s = b_2$ gives the simple target radii, since $b_2 \le \tau b_0/36$; the current has the required bound by hypothesis. This proves core membership. The logarithm series also proves local holomorphy and the displayed identity of the pair. □

51.6. **Inner substitution on an octet.** An octet $\square$ of level n is the union of the eight cubes of π_{n-1} meeting a vertex. Put $Z = \square^{\sim 5}$. Thus Z has side $12M/L$ in level-n units. We use its $\mathfrak{r}_1$ tower and its truncations, with reduced determining sets $\mathbf{B}_q^{\square}$, $q \le n$. The fields below depend only on the configuration on the inner bonds of Z.

For positive radii α_0, α_1, the window domain on Z consists of pairs (U, A') on $\mathrm{Bd}(Z)$, with U real, whose decomposition $\mathbf{U} = e^{i\eta_n A'} U$ satisfies conditions (i), (ii), and the curvature inequality in (iii) of definition 49.1, with radii α_0, α_1. We also require the tower conditions in that definition for $p = n$, the region Z, and the rule $\mathfrak{r}_1$, at every truncation $1 \le q \le n$, for both the real and complex data, with $c = \alpha_0$ in (756).

There is a convenient way to put the coarse datum near the identity. Let $x_{\square}$ be the centre of the octet and, in level-n units, put $y_0 = x_{\square} + (e_1 + e_2 + e_3)/2$, a level-$n$ site in the octet; let $\Gamma_{y_0,x}$ be its coordinate comb, and set

$$g(x) = \mathbf{U}(\Gamma_{y_0,x}), \qquad W = \mathbf{U}^g, \qquad V = M_{\mathbf{B}_n^{\square}}(W), \qquad v = g^{-1}|_{\mathbf{B}_n^{\square}}.$$

For sufficiently small $a_{\square} = a_{\square}(L) > 0$, if $\mathbf{U} = e^{i\eta_n A'} U$ satisfies the simple bounds (α_0, α_1) on Z and $(LM)^3(\alpha_0 + \alpha_1) \le a_{\square}$, then

$$M_{\mathbf{B}_n^{\square}}(\mathbf{U}) = V^v, \qquad |V - I| < K_{\square} M \alpha_0, \qquad |v(x)|\,|v(x)^{-1}| < e^{K_{\square} M \alpha_1}, \qquad v(y_0) = I, \tag{851}$$

where $K_{\square} = 5C_Q$. These maps are holomorphic for fixed U. They commute with conjugation, constant rotations, translations preserving π_{n-1}, and the corresponding translations in local charts.

Here are the estimates that verify the assertion, including the use of a complex gauge. In bare units write the side of Z as N, so $N\eta_n = 12M/L$. A comb has at most $3N$ edges. Comparing its product with the real comb gives $\mathsf{k}(g) \le e^{72(M/L)\alpha_1} \le 11/10$. In comb gauge the first-direction bonds are trivial; those of the second direction on the initial coordinate plane are trivial; and those of the third direction on the initial line are trivial. Each remaining bond is consequently a product of at most $2N$ plaquette variables or their inverses. If their distance from I is $\epsilon \le 1/100$, the product is at distance at most $e^{(200/99)N\epsilon} - 1$ from I. Hence

$$|W - I| \le 30(M/L)\alpha_0\eta_n, \qquad |a_2| \le 33(M/L)\alpha_0\eta_n, \quad a_2 = i^{-1}\log W.$$

To justify averaging covariance, let u be the real comb and put $g' = gu^{-1} = e^{i\lambda}$. Then

$$|\lambda| \le 54(M/L)\alpha_1 < 1/16, \quad \eta_n^{-1}|\lambda(x + \eta_n e_\mu) - \lambda(x)| \le 235((M/L)\alpha_0 + \alpha_1).$$

The latter estimate follows by writing $W = (\mathbf{U}^u)^{g'}$, taking the local logarithm, and using $|i^{-1}\log(\mathbf{U}^u)| \le \frac{3}{2}(\alpha_1 + 27(M/L)\alpha_0)\eta_n$. Decrease $a_{\square}$ until all these logarithms and their averaging corridors lie in the domain of proposition 51.1. Averaging covariance holds there at real (a_2, λ) and therefore, by the identity theorem on the convex logarithmic domain, at complex arguments. Thus $M^q(W) = g_- M^q(\mathbf{U}) g_+^{-1}$. The bound $|Q_q(a_2)| \le 33C_Q(M/L)\alpha_0 \le 1/10$ gives $|V - I| \le 37C_Q(M/L)\alpha_0 < K_{\square} M\alpha_0$. This proves (851). The construction uses finite products, logarithms and the same local averaging branch; it gives the asserted holomorphy and covariance. For example a real gauge h sends V to $\mathrm{Ad}(h(y_0))V$ and $v(x)$ to $h(x)v(x)h(y_0)^{-1}$.

Fix $0 < \beta_{\mathrm{s}} \le 1/4$. Increase $K_{\mathrm{tw}}^{\sharp}$ so that, with

$$K = K_{\mathrm{tw}}^{\sharp}, \qquad c_h = 2C_Q(B^{\sharp})^2 K_{\square},$$

one has $K \ge \max\{4K_{\square}, c_h, 1\}$. There is a floor $M_\flat = M_\flat(L, B_5, \delta_{\mathrm{tw}}, \beta_{\mathrm{s}})$ and, after $M \ge M_\flat$ is fixed, a positive radius $a^\flat$ independent of the depth and volume with the properties used below. More explicitly, choose the floor so that

$$\delta_{\mathrm{tw}} M \ge 8, \qquad 7C_Q L^3 B^\flat B^{\sharp} K_{\square} M e^{-\delta_{\mathrm{tw}} M/4} \le \beta_{\mathrm{s}}.$$

Let r_{I} be a common smaller radius for the free-pillar gauge comparison, averaging, changes of real base datum and restriction to a truncated tower. Choose $a^\flat \le a^\sharp$ so that

$$\begin{gathered}(LM)^3(1+2\beta_{\mathrm{s}})a^\flat \le a_\square, \qquad 8(C_Q + C_Q L B^\sharp K M)a^\flat \le r_{\mathrm{I}}, \qquad B^\sharp K M a^\flat \le 10^{-4},\\ 11 C_Q^2 (B^\sharp)^3 K_\square^2 M^2 a^\flat \le \beta_{\mathrm{s}} r_{\mathrm{B}}, \qquad 134 (c_h M)^2 a^\flat \le \beta_{\mathrm{s}},\\ [2c_u c_h + 4(c_u + c_w)B^\sharp K_\square + 2K_\square] M a^\flat \le \beta_{\mathrm{s}}/2, \qquad c_w a^\flat \le 1/6.\end{gathered} \tag{852}$$

The constant $c_w = C(L)$ bounds the sum of contour logarithms in a truncated tower, exactly as in (842). These are finitely many upper bounds on a radius chosen after M.

Proposition 51.11. Let $1 \le j < n$, $\nu = n - j$, and let $t_\nu > 0$ satisfy, with $\Theta_s = \sup_{\nu \ge 1} t_\nu L^{-s\nu}$,

$$\Theta_1 \le 1, \quad (1+10\beta_{\mathrm{s}})\Theta_2 + \beta_{\mathrm{s}}/4 + 1/1000 \le 1/8, \quad (1+3\beta_{\mathrm{s}})\Theta_3 + 5\beta_{\mathrm{s}}\Theta_2 + \beta_{\mathrm{s}}/8 + 1/500 \le 1/16. \tag{853}$$

Suppose

$$A_0 + A_1 \le a^\flat, \quad a_0 + a_1 \le a^\flat, \quad 2B_3^2 K M a_0 < a_1, \qquad A_0 \le t_\nu a_0. \tag{854}$$

Let (U, A') belong to the window domain on Z with radii $((1+2\beta_{\mathrm{s}})A_0, (1+2\beta_{\mathrm{s}})A_1)$, and put $\mathbf{U} = e^{i\eta_n A'} U$. In particular its $\mathfrak{r}_1$ tower exists and has the corresponding curvature and current bounds on $\square^{\sim 3}$. Use the datum V above and put

$$B_V = i^{-1} \log V, \qquad H_n = \mathcal{H}_{\mathbf{B}^\square_{n},1}(B_V), \qquad Q_0(c) = Q_q(\eta_n H_n)(c) \quad (c \in \Gamma_q^j).$$

For $\tau \in [0,1]$ and $\sup|B'| < c_3 a_0$, where $c_3 = \beta_{\mathrm{s}}/(16 B^\sharp)$, define $H = \mathcal{H}_{\mathbf{B}^\square_{j},1}(\tau Q_0 + B')$. All these objects exist; for fixed U they are holomorphic in (A', B') on an open neighborhood of the stated domain. Moreover $\sup|B_V| < 6KMA_0/5$ and, on $\square^{\sim 3}$,

$$|H|, |\nabla H| < a_1/2, \qquad |\partial e^{i\eta_j H} - I| < a_0 \eta_j^2/8, \qquad |J^{(j)}(e^{i\eta_j H})| < a_0/16. \tag{855}$$

For every $X \subset \square^{\sim 2}$ in $\mathcal{D}_{j-1}$,

$$(1, H, J^{(j)}(e^{i\eta_j H}))|_X \in \mathcal{U}_j^\circ(X; a_0/4, a_1/2, a_0/4). \tag{856}$$

■

Proof. We give the restriction argument first. Restrict a real regular critical configuration to a subtower whose initial region lies inside a later member of the original tower. A variation preserving the restricted datum extends by the original field outside that region. The crossing plaquettes are fixed because the boundary bonds are level-zero determining bonds. Induction in the averaging order shows that the extended variation preserves every original datum. It is therefore critical. Near the trivial datum uniqueness in theorem E identifies the two backgrounds. The same identity extends to the common complex logarithmic ball. Axial normalization gives its gauge as the product of contour variables, with

$$|w^{\pm 1}| \le e^{c_w \sup|H|}. \tag{857}$$

Each contour logarithm at level $q < j$ is bounded by $C(L) L^{q-j} \sup|H|$; their sum proves this bound. This proves the same statement for truncating a tower without restricting its initial region. Changes of real base datum and small complex node gauges follow in the same way: the identities hold at real arguments by uniqueness and extend on their convex logarithmic domains. In particular the tower of $\mathbf{U}$ is $\Phi_{\mathbf{B}^\square_{n},1}(B_V)^{\tilde{v}}$.

The logarithm estimate in (851) gives $\sup|B_V| \le b_V = (33/20) K_\square M A_0 < 6KMA_0/5$. The tower and its truncation have free pillars over $\square^{\sim 4}$. Hence $h_+ := \sup_{\square^{\sim 4}} |H_n| \le B^\sharp b_V$. Averaging gives

$$q_{\mathrm{out}} := C_Q L B^\sharp b_V \ge \sup|Q_0|, \qquad q_{\mathrm{in}} := C_Q B^\sharp b_V L^{-\nu} \ge \sup_{\mathsf{P}^q(c) \subset \square^{\sim 4}} |Q_0(c)|.$$

Every row in the second supremum has level j. From a bond over $\square^{\sim 3}$ the other rows have weighted distance at least $D = (M/2)L^{\nu-1}$. Apply (840) to these rows and (839) to the near rows. The floor on M gives

$$B^\flat e^{-\delta_{\mathrm{tw}} D/2} q_{\mathrm{out}} \le (\beta_{\mathrm{s}}/4) A_0 L^{-2\nu}, \qquad N_b(\mathcal{H}_j(\tau Q_0)) \le h_1 := c_h M A_0 L^{-\nu}.$$

Splitting Q_0 into the near and far parts also gives, by (841),

$$N_b(\mathcal{H}_j(\tau Q_0) - \tau \mathsf{H} Q_0) \le 2B^\flat e^{-\delta_{\mathrm{tw}} D/2} q_{\mathrm{out}} + 2B^\sharp q_{\mathrm{in}}^2 / r_{\mathrm{B}} \le \beta_{\mathrm{s}} A_0 L^{-2\nu}. \tag{858}$$

At $\tau = 1$, the restriction identity and the free-pillar gauge comparison give

$$\begin{aligned} e^{i\eta_j \mathcal{H}_j(Q_0)} &= (U_Z)^G, \qquad G = u_j^{-1} g_j u_n^{-1} \tilde{v}^{-1}, \\ \mathsf{k}(G) &\le \exp(2c_u h_1 + 2(c_u + c_w) h_+ + K_\square M (1 + 2\beta_{\mathrm{s}}) A_1) \\ &\le e^{\beta_{\mathrm{s}}/2} \le \frac{1 + 3\beta_{\mathrm{s}}}{1 + 2\beta_{\mathrm{s}}}. \end{aligned}$$

Here U_Z is the tower of $\mathbf{U}$; g_j is the contour gauge for the truncation. The window hypothesis therefore bounds the curvature by $(1+3\beta_{\mathrm{s}})A_0 L^{-2\nu}\eta_j^2$ and the current by $(1+3\beta_{\mathrm{s}})A_0 L^{-3\nu}$. Applying (843) and (844) at the trivial base, and using $134 h_1^2 \le \beta_{\mathrm{s}} A_0 L^{-2\nu}$, yields

$$|\partial \mathcal{H}_j(Q_0)| < (1 + 4\beta_{\mathrm{s}}) A_0 L^{-2\nu}, \quad |\partial^{\mathsf{T}} \partial \mathcal{H}_j(Q_0)| < (1 + 3\beta_{\mathrm{s}}) A_0 L^{-3\nu} + \beta_{\mathrm{s}} A_0 L^{-2\nu}.$$

Apply (858) at 1, use linearity of H, and apply it once more at τ. The two bounds become, respectively, $(1 + 8\beta_{\mathrm{s}})A_0 L^{-2\nu}$ and $(1 + 3\beta_{\mathrm{s}})A_0 L^{-3\nu} + 3\beta_{\mathrm{s}} A_0 L^{-2\nu}$. Adding B' changes N_b by less than $\beta_{\mathrm{s}} a_0/16$, and hence by less than the $\beta_{\mathrm{s}} a_0/8$ allowed below. Thus

$$h := \max\{|H|, |\nabla H|\} < h_1 + \beta_{\mathrm{s}} a_0/8, \quad |\partial H| < (1 + 8\beta_{\mathrm{s}}) A_0 L^{-2\nu} + \beta_{\mathrm{s}} a_0/4,$$

and $|\partial^{\mathsf{T}} \partial H|$ is bounded by $(1+3\beta_{\mathrm{s}})A_0 L^{-3\nu} + 3\beta_{\mathrm{s}} A_0 L^{-2\nu} + \beta_{\mathrm{s}} a_0/8$. The first and third assumptions in (854) give $h < a_1/2$. Also $134h^2 \le 2\beta_{\mathrm{s}} A_0 L^{-2\nu} + a_0/500$, with $11h^2 \le 2\beta_{\mathrm{s}} A_0 L^{-2\nu} + a_0/1000$ after the same radius reduction. The local expansions and (853) now give (855).

Finally apply the restriction identity to every subtower (p, Z') of X and both tower rules. Its gauge has condition number at most $e^{c_w a_1} < 6/5$. The curvature bound is therefore at most $(6/5)(a_0/8)L^{-2j}$, smaller than $(a_0/4)L^{-(j-p)}L^{-2p}$. The current bound at truncation $n' \le p$ is at most $(6/5)(a_0/16)L^{-3(j-n')}$, smaller than $(a_0/4)L^{-(j-p)}L^{2(n'-p)}$, since $2j - p - n' \ge 0$. The real factor is trivial, and the potential and current bounds are those already proved. These are all the core conditions, so (856) follows. The strict finite bounds and the common logarithmic balls also prove the stated holomorphy. □

51.7. **Replacing an old interaction by an octet interaction.**

Definition 51.12. A system of radii of height k is $\mathfrak{a} = ((a_{0,j}, a_{1,j}))_{1 \le j \le k+1}$. Put

$$\bar{a}_i = (1 + \beta_{\mathrm{s}}) a_{i,k+1}, \qquad \varrho_j = \max_i \bar{a}_i / a_{i,j}, \qquad s_{\mathfrak{a}} = \delta(\mathfrak{t}) / a_{0,k+1}. \tag{859}$$

For $a^\star, s^\star > 0$ and $c_\natural, c_{\mathrm{sh}} \ge 1$, the system is admissible if

$$\begin{aligned} &\bar{a}_0 + \bar{a}_1 \le a^\star, \qquad c_{\mathrm{sh}} M a_{0,j} < a_{1,j}, \quad a_{1,j}^2 \le c_\natural a_{0,j} \quad (1 \le j \le k+1), \\ &a_{i,k+1} \le (1 + \beta_{\mathrm{s}}) \sqrt{k + 2 - j}\, a_{i,j}, \qquad a_{i,j} \le \bar{a}_i \quad (j \le k), \qquad s_{\mathfrak{a}} \le s^\star. \end{aligned} \tag{860}$$

The two comparisons indexed by $j \le k$ are void at height zero. Write $\mathcal{U}_j(X) = \mathcal{U}_j^\circ(X; a_{0,j}, a_{1,j})$. ■

Fix $0 < \beta_{\mathrm{s}} \le 1/5$ and $0 < \theta_{\mathrm{oc}} < 1$ with $\frac{3}{2}\theta_{\mathrm{oc}}^2 < 1$. The octet radii and the system of height $k-1$ are

$$b_i^{\mathrm{oc}} = \theta_{\mathrm{oc}} a_{i,k+1}, \quad a_{i,k}^{\mathrm{oc}} = b_i^{\mathrm{oc}}/(1 + \beta_{\mathrm{s}}), \quad a_{i,j}^{\mathrm{oc}} = \min\{a_{i,j}, b_i^{\mathrm{oc}}\} \quad (j < k). \tag{861}$$

Its output radii are b_i^{oc}, and it is admissible with $s^\star$ replaced by $(1 + \beta_{\mathrm{s}}) s^\star / \theta_{\mathrm{oc}}$. Indeed shape is preserved by a common contraction and by these minima. For slow growth, if the minimum is $a_{i,j}$

use $\theta_{\mathrm{oc}}^2(k+2-j)/(k+1-j)<1$; otherwise the output bound suffices. Thus $1\le \varrho_j^{\mathrm{oc}}\le \varrho_j$ and $\varrho_j^2 L^{j-k}<1$ for $j<k$, since $\varrho_j^2\le (6/5)^4(k+2-j)$ and $L\ge 13$.

We use the canonical maps $\mathbb{H}_q=\mathcal{H}_{\mathbf{B}_q^{\square},1}$. They are real at real data, commute with adjunction and constant rotations, and commute with the translations of lattice charts used above. Indeed transport the constraint, Landau condition and node normalization in (837). The averaging paths, covariant differences and action are carried to their counterparts; the geometric distances and norms are unchanged. The transported pair solves the same normalized equations. Uniqueness near zero and holomorphy identify the maps on the row ball. At real data their exponentials are gauge transforms of the unique regular critical orbit, by theorem E. Their derivative bound is (838). These are properties of the constructed maps.

Lemma 51.13. Let $(\Omega_q)_{0\le q\le n}$ and $(\Omega'_q)_{0\le q\le n'}$ be separated sequences with $n'\le n$ and $\Omega'_q\subset\Omega_q$ for $0\le q\le n'$. A real regular critical configuration for $(\Omega_q)_{0\le q\le n}$ restricts to a real regular critical configuration for $(\Omega'_q)_{0\le q\le n'}$ at the same regularity radius. This remains true after a real gauge transformation of the restricted configuration. ■

Proof. Glue a configuration with the same restricted datum to the original configuration outside Ω'_0. The gluing preserves the original datum, by induction on averaging order. At level zero the fields agree outside Ω'_0 and on its determining rows. At level q, a row with an endpoint in Ω'_q and none in Ω'_{q+1} is a determining row. If neither endpoint lies in Ω'_q, its entire averaging corridor is outside that region, where the induction hypothesis applies. Original determining rows have no endpoint in Ω_{q+1} and hence none in Ω'_{q+1}, so their averages are unchanged. A plaquette crossing Ω'_0 has no corner in Ω'_1, by separation. Its inner bonds are fixed level-zero determining rows. Thus the action of the glued field minus the action inside Ω'_0 is independent of the variation. Differentiation proves criticality. For a plaquette or complete plaquette star, its highest layer indices satisfy $\lambda'\le\lambda$. Restriction multiplies its bare-scale regularity quantity by $L^{-2(\lambda-\lambda')}$ for curvature and $L^{-3(\lambda-\lambda')}$ for current. These factors are at most one. Real gauges preserve these norms, the fibres and the action. □

For an octet of level k, put $X'=(\square^{\sim 5})^{\uparrow}$ and form, from $\mathbf{U}$ on X', the fields

$$B_V=i^{-1}\log V,\quad A_{\square,k}=\mathbb{H}_k(B_V),\quad Q_{\square,j}=i^{-1}\log M_{\mathbf{B}_j^{\square}}(e^{i\eta_k A_{\square,k}}),\quad A_{\square,j}=\mathbb{H}_j(Q_{\square,j}).$$

For a local term E on $X\subset\square^{\sim 2}$ define

$$E^{\square}(P')=E((1,A_{\square,j},J^{(j)}(e^{i\eta_j A_{\square,j}}))|_X),\qquad \mathbf{U}=\mathbf{U}(P'). \tag{862}$$

Proposition 51.14. There are $a^\star_{\mathrm{oc}}>0$, $c^{\mathrm{oc}}_{\mathrm{sh}}$ and C_{oc}, independent of the height and volume, such that the following holds for an admissible system with $a^\star\le a^\star_{\mathrm{oc}}$ and $c_{\mathrm{sh}}\ge c^{\mathrm{oc}}_{\mathrm{sh}}$. Let $k\ge 2$, $1\le j<k$, and $X\in\mathcal{D}_j$ satisfy $X\subset\square^{\sim 2}$. Then (862) is defined on $\mathcal{U}_k^\circ(X';b_0^{\mathrm{oc}},b_1^{\mathrm{oc}})$, and its inner triple belongs to

$$\mathcal{U}_j^\circ(X;a^{\mathrm{oc}}_{0,j}/4,a^{\mathrm{oc}}_{1,j}/2,a^{\mathrm{oc}}_{0,j}/4). \tag{863}$$

It is a local term there, depends only on $\mathbf{U}$ on the inner bonds of $\square^{\sim 5}$, and satisfies

$$|E^{\square}(P')-E(1,0)|\le C_{\mathrm{oc}}\|E\|_X(d_j(X)+2)^2\varrho_j^2(L\xi_j)^4,\qquad \xi_j=L^{j-k}. \tag{864}$$

Also $\|E^{\square}\|_{X'}\le\|E\|_X$. The map $E\mapsto E^{\square}$ is linear and preserves conjugation reality and chart and translation covariance. If U^* is in an existing regular critical orbit at level k or $k+1$ and its level-k pair on X' belongs to the octet core, then its level-j pair on X belongs to $\mathcal{U}_j(X)$ and

$$E^{\square}(\mathbf{p}^{(k)}_{X'}(U^*))=E(\mathbf{p}^{(j)}_X(U^*)). \tag{865}$$

The same identity holds for the plain triples whenever they belong to the indicated cores. ■

Proof. Take $c_{\mathrm{sh}}^{\mathrm{oc}} = 2B_3^2 K$ and decrease $a_{\mathrm{oc}}^{\star} \le a^{\flat}$ until

$$\varpi := \tfrac{6}{5} K M b_0^{\mathrm{oc}} < \min\{c_{\mathrm{plain}}/(2LB^{\sharp}), r_{\mathrm{B}}/(2C_Q LB^{\sharp}), 1/(120MB^{\sharp})\}.$$

Also require $a_{\mathrm{oc}}^{\star} \le \min\{a_{\mathrm{av}}, \omega_M/2\}$. The real averaging bound then makes its real and complex branches agree at the curvature $b_0^{\mathrm{oc}}\eta_k^2$: at each intermediate order q the curvature is at most $2b_0^{\mathrm{oc}} L^{2(q-k)} \le \omega_M$, so the next complex averaging step agrees with the real step.

Apply proposition 51.11 with $A_i = b_i^{\mathrm{oc}}/(1+2\beta_{\mathrm{s}})$, $a_i = a_{i,j}^{\mathrm{oc}}$, $n = k$, $\tau = 1$, $B' = 0$. The core includes the required window. The sequence $t_\nu = (6/5)\sqrt{1+\nu}$ is admitted: its three suprema are attained at $\nu = 1$, and at $L = 13$, $\beta_{\mathrm{s}} = 1/5$ the last two left sides of (853) are less than 0.082 and 0.039. The octet radius system verifies (854), giving (863). Holomorphy holds near every fixed-real-factor slice. The maps depend only on the pair; a gauge on X' induces the constant rotation by its value at y_0 on the inner triple. This proves locality and invariance. Evaluation and covariance give the norm bound, linearity, reality and transport.

For the gain put $\nu = k - j$, $d = d_j(X)$ and $a^{\psi} = a_{0,j}/(8K)$. The simple-to-core inclusion gives

$$\|A\|_{X,\nabla} < a^{\psi}, \quad |J| < a_{0,j} \quad \Longrightarrow \quad (1, A, J) \in \mathcal{U}_j(X) :$$

the plaquette bound is less than $4a^{\psi}\eta_j^2$, and $K(4a^{\psi} + (a^{\psi})^2) < a_{0,j}$ after the radius reduction. Thus $\Psi(A,J) = E(1,A,J)$ is holomorphic on this product ball and bounded by $\|E\|_X$. Its differential at zero vanishes: constant $SU(2)$ invariance makes each of its linear functionals on the traceless matrices invariant under conjugation, hence zero. Cauchy's estimate on radial lines gives

$$|\Psi(A,J) - \Psi(0,0)| \le 8\|E\|_X (N/a^{\psi})^2, \quad \max\{\|A\|_{X,\nabla}, \sup|J|\} \le N. \tag{866}$$

For $N \ge a^{\psi}/2$ use the bound $2\|E\|_X$ instead.

Put $\sigma = B^{\sharp}\varpi$, $\phi_1 = L^{-\nu}\sigma$, $\phi_2 = L^{-2\nu}\sigma$, and

$$\epsilon_\nu = C_Q B^{\sharp}(B^{\sharp} L^{-\nu} + B^{\flat} L e^{-\delta_{\mathrm{tw}} M L^{\nu-1}/2}).$$

For $\sup|B| < \varpi$, let $f_B = L^{-\nu}\mathbb{H}_k(B)$. The response estimate, scaling and (844) give $|f_B| \le \phi_1$, $|\nabla f_B| \le \phi_2$ on the box of radius $3ML^{\nu-1} + 3\eta_j$ around $x_\square$, and $|J^{(j)}(e^{i\eta_j f_B})| \le 2L^{-3\nu}\sigma$ on X. Its averaged datum on the truncated tower is bounded by $C_Q L B^{\sharp} \sup|B|$ globally and by $C_Q L^{-\nu} B^{\sharp} \sup|B|$ on rows within weighted distance $ML^{\nu-1}$ of that smaller box. Splitting the response into these near and far rows bounds its level-j potential, gradient and curl-transpose-curl by $\epsilon_\nu \sup|B|$. Since $e^{-t} \le 1/t$,

$$\frac{\epsilon_\nu \varpi}{a^{\psi}} \le C_2 \varrho_j \xi_j, \qquad C_2 = \tfrac{48}{5} K^2 M C_Q B^{\sharp} B^{\flat} (1 + 2L^2/(\delta_{\mathrm{tw}} M)).$$

If $C_2 \varrho_j \xi_j < 1$, both potentials lie in the ball of Ψ, and $8\phi_1 \le \epsilon_\nu \varpi < a^{\psi}$. For real B their exponentials are gauge equivalent, by lemma 51.13 and uniqueness at depth j. Gauge invariance identifies their values under E. The identity theorem extends this equality to $\sup|B| < \varpi$. At $B = B_V$ it gives $E^{\square}(P') = \Psi(f, J_f)$, where $f = f_{B_V}$ and $J_f = J^{(j)}(e^{i\eta_j f})$.

Choose a cube centre $z \in X$ and coordinates in the box. For $|\lambda_\mu| < 2\phi_1$ put

$$g_\lambda(x) = \prod_{\mu=1}^{3} e^{i(x_\mu - z_\mu)\lambda_\mu}, \quad T_\mu = \prod_{\kappa > \mu} e^{i(x_\kappa - z_\kappa)\lambda_\kappa}, \quad \tilde{\lambda}_\mu = T_\mu^{-1} \lambda_\mu T_\mu,$$

$$A_\lambda(b) = \mathrm{Ad}(g_\lambda(b_-))\mathcal{L}_b, \quad \mathcal{L}_b = (i\eta_j)^{-1} \log(e^{i\eta_j f(b)} e^{-i\eta_j \tilde{\lambda}_\mu(b_-)}), \quad J_\lambda = \mathrm{Ad}(g_\lambda) J_f.$$

The total exponent norm in g_λ is at most $3M\sigma < 1/40$, so its condition number is less than 1.06. The logarithm has Lipschitz constant 1.3 in these variables. Directly, $g_\lambda(x + \eta_j e_\kappa) = g_\lambda(x) e^{i\eta_j \tilde{\lambda}_\kappa(x)}$ and $|\tilde{\lambda}_\mu(x + \eta_j e_\kappa) - \tilde{\lambda}_\mu(x)| \le 2\eta_j \tilde{l}^2 e^{2\eta_j \tilde{l}}$, where $\tilde{l} \le 2.12\phi_1$. Therefore

$$|\nabla_\kappa A_{\lambda,\mu}| \le 1.06(2.02\tilde{l}|\mathcal{L}_{b'}| + 1.3\phi_2 + 2.63\tilde{l}^2).$$

It follows that $|A_\lambda| \le 5\phi_1$, $|\nabla A_\lambda| \le 2\phi_2$, $|J_\lambda| \le 3\phi_1$. Thus all these pairs remain in the ball of Ψ. At real λ the gauge is unitary and $\Psi(A_\lambda, J_\lambda) = \Psi(f, J_f)$; holomorphy extends this identity to the complex parameter ball.

Set $\lambda_\mu = f_\mu(z)$. A coordinate path gives $|f_\mu(x) - \lambda_\mu| \le r(x)\phi_2$, where $r(x) = \sum_\mu |x_\mu - z_\mu|$. Also $|\tilde\lambda_\mu(x) - \lambda_\mu| \le 2.12\sigma r(x)\phi_2 \le 0.02 r(x)\phi_2$. The preceding difference estimate then gives $|A_\lambda(x)| \le 1.41 r(x)\phi_2$ and $|\nabla A_\lambda| \le 1.5\phi_2$. On bonds incident to X, $r(x) \le 3.1M(d+2)$, so

$$\max\{\|A_\lambda\|_{X,\nabla}, \sup|J_\lambda|\} \le N_X = 5M(d+2)\phi_2, \qquad N_X/a^\psi \le C_3(d+2)\varrho_j\xi_j^2, \quad C_3 = 48K^2M^2B^\sharp.$$

Equation (866) proves the gain with $8C_3^2$. If $C_2\varrho_j\xi_j \ge 1$, use $\varrho_j^2\xi_j < 1$ to obtain $1 < C_2^6\varrho_j^2\xi_j^4$; the trivial bound suffices. Taking $C_{\mathrm{oc}} = \max\{C_2^6, 8C_3^2\}$ and enlarging ξ_j^4 to $(L\xi_j)^4$ proves (864).

For the physical identity, lemma 51.13 applies to every truncation of the octet tower inside the global critical field. The real and complex averages agree. Uniqueness of the regular critical orbits identifies successively $e^{i\eta_k A_{\square,k}}$ and $e^{i\eta_j A_{\square,j}}$ with gauge transforms of $U^*|_Z$. The currents transform by conjugation, so the inverse gauge on X carries the inner pair to $\mathbf{p}_X^{(j)}(U^*)$. This proves (865). For plain triples restriction scales the radii by (ξ_j, ξ_j, ξ_j^3); since $\xi_j b_i^{\mathrm{oc}} < a_{i,j}$, they belong to the required cores and pair dependence gives the same equality. □

51.8. **Substitution of a local term.** Use the shift $(\mathbf{A}_Y, \mathbf{K}_Y)$ constructed in proposition 51.5, with constants $(a_\natural, \varepsilon_\natural, c_\natural)$. Its input at step $k+1$ is read as $(\mathbf{U}, L^{-3}\mathbf{J})$ in level-k units. This factor will also be needed when we identify its values on critical backgrounds. Choose $0 < c_{\mathrm{os}} \le 1$ so that

$$128K_{\mathrm{tw}}^\sharp(1 + c_\natural)c_{\mathrm{os}} \le 1.$$

Let $0 < a_i^+ \le a_\natural$ and $0 < \varepsilon \le \varepsilon_\natural$. For $Y \in \mathcal{D}_k$ write

$$\mathfrak{D}(Y) = \{(P, \mathcal{A}) : P \in \mathcal{U}_{k+1}^\circ(Y; a_0^+, a_1^+), \quad \sup|\mathcal{A}| < \varepsilon\}.$$

The form $\mathcal{A}$ is specified on the nonpivot level-k bonds inside Y. A term with radii is a local term f on $X \in \mathcal{D}_j$, $1 \le j \le k$, satisfying

$$0 < b_0 \le b_1, \quad b_1^2 \le c_\natural b_0, \quad b_0 + b_1 \le a^\sharp, \qquad a_i^+ \le 2\sqrt{1+\nu}\, b_i, \quad \nu = k+1-j. \tag{867}$$

Its amplitude radius is

$$\varrho_{\mathsf{tm}} = \frac{c_{\mathrm{os}} b_0 L^{k-j}}{c_\natural \varepsilon}. \tag{868}$$

For $X^\uparrow \subset Y$, set $\theta = L^{-\nu}$, $\vartheta = L^{-(k-j)}$, and

$$\Phi_Y[f](s, \lambda; P, \mathcal{A}) = f(\rho_X^{k+1\to j} P \dotplus (\lambda\vartheta \mathbf{A}_Y(s), \lambda\vartheta^3 \mathbf{K}_Y(s))|_X). \tag{869}$$

Thus the extra potential scales by ϑ, its first difference by ϑ^2, and its current by ϑ^3. Here s ranges over the decoupling polydisc $|s_\square| < e^{\bar\kappa}$.

Proposition 51.15. For $(P, \mathcal{A}) \in \mathfrak{D}(Y)$ and $|\lambda| < \varrho_{\mathsf{tm}}$, the substituted triple in (869) lies in $\mathcal{U}_j^\circ(X; b_0, b_1)$. The function $\Phi_Y[f]$ is bounded by $\|f\|_X$ and is holomorphic in (s, λ) and locally holomorphic on each fixed-real-U slice in the remaining variables. At $\lambda = 0$, $s = 0$, or $\mathcal{A} = 0$, its value is $f(\rho_X P)$. It is local for the roots given by the cubes of $X^\uparrow$ and consistent under zero extension to larger domains.

Suppose $\varrho_{\mathsf{tm}} > 1$. With the starred bracket of eq. (806), put

$$\mathbb{V}[f](Y; \mathfrak{u}) = [\Phi_Y[f](\cdot, 1; \mathfrak{u})]^*[\mathsf{P}_Y \setminus \mathsf{P}_{X^\uparrow} \mid \mathsf{P}_{X^\uparrow}]. \tag{870}$$

It vanishes at $\mathcal{A} = 0$, is linear in f, kills constant terms, and satisfies, with $m = \#_{\pi_k}(Y \setminus X^\uparrow)$,

$$|\mathbb{V}[f](Y; \mathfrak{u})| \le 2\varrho_{\mathsf{tm}}^{-1}\|f\|_X(e^{\bar\kappa} - 1)^{-m} \le 2\varrho_{\mathsf{tm}}^{-1}\|f\|_X e^{-(\bar\kappa - 1)m}. \tag{871}$$

Both functions inherit pair dependence, gauge invariance, reality, conjugation, translations and local charts from f and the shift. For chart transport both the original and transported points must belong to their specified domains.

For a real block field W, assume its averaged background is defined, real and admissible, with invertible pivots; assume the fibre chart is defined at $\mathcal{A}$, its level-k critical background exists, and $\mathfrak{u}_W = (P_W, \mathcal{A}) \in \mathfrak{D}(T_\ell)$. Then f is defined on both critical orbits, is constant on each, and

$$f(U_k(V_W[\mathcal{A}])) - f(U_{k+1}(W)) = \sum_{Y \supset X^\uparrow} \mathbb{V}[f](Y; \mathfrak{u}_W|_Y). \tag{872}$$

■

Proof. Since $4\sqrt{1+\nu} \le L^\nu$, restriction puts $\rho_X P$ in the half-radius core, with current radius $c_0 = \theta^3 a_0^+ \le b_0/338$. The shift bounds and (868) give

$$|\lambda\vartheta\mathbf{A}|, |\lambda\vartheta^2\nabla_{\mathbf{U}}\mathbf{A}| < c_{\mathrm{os}} b_0 \le b_2, \qquad |\lambda\vartheta^3\mathbf{K}| < c_{\mathrm{os}} b_0 \le b_0/256.$$

Thus the total current has norm less than b_0. Proposition 51.10 proves core membership.

For completeness, local holomorphy does not require the core itself to be open in all its variables. Fix a real U and a point in its slice. The shift has a holomorphic extension to a neighborhood of that point. Its substituted configuration is a product of matrix exponentials and fixed matrices. The logarithm relative to its fixed real factor exists near the point, and provides a holomorphic pair of local coordinates for the slice on which f is holomorphic. Their composition proves the claim. The zero values follow from the zero values of the shift. Every bond incident to X has an endpoint in a root cube. A component of the decoupling support disjoint from all roots therefore cannot affect the substituted triple on X. This proves root locality. Restriction and transport commute with the bond formulas, including reversal, so consistency and the asserted symmetries follow from those of the shift and f.

Subtract $\Phi_Y[f](s,0)$ before taking the starred bracket. Amplitude Cauchy gives the bound $2\varrho_{\mathsf{tm}}^{-1}\|f\|_X$ at $\lambda = 1$; the bracket bound then gives (871). In particular the finite vertex formula is

$$\mathbb{V}[f](Y) = \sum_{X^\uparrow \subset W \subset Y} (-1)^{\#Y - \#W} f(T_W) - \mathbf{1}_{Y = X^\uparrow} f(T_\emptyset), \quad T_W = \rho_X P \dotplus \mathfrak{h}_Y(\mathbf{1}_W)|_X,$$

where the factors ϑ, ϑ^3 in T_W are those of (869). This also proves local holomorphy of the potential and its inheritance assertions.

At full support, the identification of the actual shift gives the level-k background on the fibre. Its current at level j is

$$\vartheta^3(L^{-3} J^{(k+1)}(U_+) + \mathbf{K}) = J^{(j)}(U_k(V_W[\mathcal{A}])).$$

At zero support the pair is the restriction of U_+. Gauge invariance gives the asserted orbit values. Finally sum the finite differences over all subsets of cubes. Components disjoint from the roots cancel in pairs, by root locality; the remaining support together with the connected root set is connected. Zero-extension consistency identifies its coefficient with the corresponding $\mathbb{V}[f](Y)$. The total is $\Phi(\mathbf{1}, 1) - \Phi(0, 1)$, proving (872). □

51.9. **The fourth potential.** Let $\mathfrak{f}_\circ$ be an array of pointed interactions $G^{(j)}(X, P, z)$, counterterms e_j, and unpointed interactions $H^{(j)}(X, P)$, $1 \le j \le k$. Set $H^{(j)} = 0$ off its slot. Write

$$g_j = \sup_{X,P,z} e^{\kappa d_j(X)} |G^{(j)}(X, P, z)|, \quad r_j = \sup_{X,P} e^{\kappa d_j(X)} |H^{(j)}(X, P)|, \quad g_j^{\mathrm{ct}} = \tfrac{3}{2} a_{0,j}^2 |e_j|,$$

and

$$N_{\mathfrak{a},\omega}(\mathfrak{f}_\circ) = \sum_{j=1}^{k} \omega_{k+1-j}(g_j + g_j^{\mathrm{ct}} + r_j). \tag{873}$$

All suprema are over the corresponding cores. The action is

$$S[\mathfrak{f}_\circ](U) = \sum_{j=1}^{k} \Big\{ \sum_{z} \sum_{X \ni z} (G^{(j)}(X,U,z) - G^{(j)}(X,1,z)) - e_j A^{\eta_j}(U) \Big\} + \sum_{j=1}^{k} \sum_{X} (H^{(j)}(X,U) - H^{(j)}(X,1)). \tag{874}$$

Use the tent action to put $E_{X,z} = G^{(j)}(X,\cdot,z) - e_j \mathsf{A}^{\mathsf{t}}(X,\cdot,z)$. It obeys

$$\|E_{X,z}\|_X \le g_j e^{-\kappa d_j(X)} + g_j^{\mathrm{ct}} \mathbf{1}_{X=X_z}, \qquad \#_{\pi_j} X_z = 27, \quad d_j(X_z) \le 7.$$

For $j < k$ and a level-j site z, let $\square(z)$ be the octet centered at the upper corner of the unique level-$(k-1)$ partition cube containing z; put $X'(z) = (\square(z)^{\sim 5})^{\uparrow}$. Thus $z \in X'(z)$, $\#_{\pi_k} X'(z) \le 8$ and $d_k(X'(z)) = 0$. Call X narrow at z if $j < k$ and $X \subset \square(z)^{\sim 2}$, and wide at z otherwise. A domain is narrow if it is nonwrapping and narrow at the lexicographically first site z_X in a near lift. This site is independent of the near lift after projection, since two such lifts differ by a period. All other domains are wide. At level k every domain is wide.

Replace the narrow terms by their octet evaluations. More precisely, form the following four families of local terms:

index and condition	level, domain	$f_{\mathfrak{i}}$
(j,z,X), X wide at z	j, X	$E_{X,z}$
(j,z), $j<k$	$k, X'(z)$	$\displaystyle\sum_{z \in X \subset \square(z)^{\sim 2}} (E_{X,z}^{\square(z)} - E_{X,z}(1))$
(j,X), X wide	j, X	$H^{(j)}(X,\cdot)$
(j,X), X narrow	$k, X'(z_X)$	$H^{(j)}(X,\cdot)^{\square(z_X)} - H^{(j)}(X,1)$.

(875)

The first and third families use $(a_{0,j}, a_{1,j})$; the other two use $(b_0^{\mathrm{oc}}, b_1^{\mathrm{oc}})$. For $k = 1$ the second and fourth families are empty. Define

$$\mathbb{V}'[\mathfrak{f}_\circ](Y;\mathfrak{u}) = \sum_{\mathfrak{i}: X_{\mathfrak{i}}^{\uparrow} \subset Y} \mathbb{V}[f_{\mathfrak{i}}](Y;\mathfrak{u}). \tag{876}$$

At height zero this rule is zero.

Here are the quantitative choices used in the estimate. Let $\kappa_{\mathcal{D}} = \log 9216$ and $C_{\mathrm{E},2} = 2048 \sum_{n \ge 0} (n+2)^2 e^{-n}$. Fix $0 < \delta_{\mathrm{w}} < 1/2$ and require

$$\delta_{\mathrm{w}} \kappa > \kappa_{\mathcal{D}} + 2, \qquad \bar{\kappa} \ge \kappa + 1, \qquad 1 + \beta_{\mathrm{s}} \le 2\sqrt{2}\theta_{\mathrm{oc}}. \tag{877}$$

Put

$$C_\Phi = 3 C_{\mathcal{A}} c_{\mathfrak{h}} / c_{\mathrm{os}}, \qquad C_\Sigma = 9 C_{\mathrm{E},2}, \qquad C_\Lambda = 2 C_\Sigma C', \qquad K_\Lambda = C_\Lambda C_\Phi,$$
$$C' = \max\{27 M^3 e^{7\kappa}, 8 M^3 L^4 (1 + C_{\mathrm{oc}} \max\{C_{\mathrm{E},2}, 81\})\}. \tag{878}$$

For $k = 1$ use $C' = 27 M^3 e^{7\kappa}$. Select ceilings satisfying

$$a^\star \le \min\{a_{\mathfrak{h}}, a^\sharp, a_{\mathrm{oc}}^\star\}, \qquad c_{\mathrm{sh}} \ge c_{\mathrm{sh}}^{\mathrm{oc}}, \qquad 2 C_\Phi s^\star \le 1. \tag{879}$$

At the parameter $\mathtt{t}$, assume $\delta(\mathtt{t}) \le 1$ and $C_{\mathcal{A}} \delta(\mathtt{t}) \le \varepsilon_{\mathfrak{h}}$. The actual localization domain is

$$\mathfrak{D}_k(Y) = \{(P, \mathcal{A}) : P \in \mathcal{U}_{k+1}^{\circ}(Y; \bar{a}_0, \bar{a}_1), \quad \sup|\mathcal{A}| < C_{\mathcal{A}} \delta(\mathtt{t})\}.$$

In particular $\mathcal{A} = gB$ lies here on the open cutoff polydisc $|B| < C_{\mathcal{A}} \mathfrak{b}(\mathtt{t})$.

Proposition 51.16. Assume the tower, shift and octet hypotheses above and let $\mathfrak{a}$ satisfy (860) with the ceilings (879). The function (876) is defined on $\mathfrak{D}_k(Y)$ and is locally holomorphic on every fixed-real-U slice. In particular it is holomorphic in B throughout the cutoff polydisc. It is linear and local in the array, vanishes at $\mathcal{A} = 0$, depends on the input pair, and is gauge invariant. It preserves holomorphic parameters, reality, conjugation, translations and local charts. Constant

changes of an interaction do not change it, nor does adding a zero slot. Restriction from a larger admissible radii system commutes with the construction.

If positive lag weights satisfy

$$\varrho_j^2 L^{-(k-j)} \le c_\omega \omega_{k+1-j} \qquad (1 \le j \le k), \tag{880}$$

then

$$|\mathbb{V}'[\mathfrak{f}_\circ](Y;\mathfrak{u})| \le K_\Lambda c_\omega s_{\mathfrak{a}} N_{\mathfrak{a},\omega}(\mathfrak{f}_\circ) e^{-(1-2\delta_{\mathrm{w}})\kappa d_k(Y)}. \tag{881}$$

The constants depend only on the fixed geometric, decay and shift constants displayed in (878), and on C_{oc} when $k \ge 2$; thus they are uniform in the lattice, the radii, the array and $\mathtt{t}$.

For a real W, assume its averaged background is defined, real and admissible with invertible pivots, its fibre chart is defined throughout the real cutoff, all level-k backgrounds on the real cutoff exist, and $P_W \in \mathcal{U}^\circ_{k+1}(T_\ell; \bar a_0, \bar a_1)$. Then the action is defined and constant on both critical orbits, and for every real field below the cutoff,

$$\sum_Y \mathbb{V}'[\mathfrak{f}_\circ](Y;(P_W, gB)|_Y) = S[\mathfrak{f}_\circ](U_k(V_W(B))) - S[\mathfrak{f}_\circ](U_{k+1}(W)). \tag{882}$$

■

Proof. First check the amplitude radius, which is the source of the small factor $s_{\mathfrak{a}}$. By admissibility, $\bar a_i \le (1+\beta_{\mathrm{s}})^2\sqrt{k+2-j}\, a_{i,j} \le 2\sqrt{k+2-j}\, a_{i,j}$. For the octet terms, whose level is k, the corresponding inequality is $\bar a_i / b_i^{\mathrm{oc}} = (1+\beta_{\mathrm{s}})/\theta_{\mathrm{oc}} \le 2\sqrt{2}$. The radius and shape bounds follow from (879) and (861). Thus every term in (875) satisfies (867). Moreover

$$\varrho_{\mathsf{tm}} C_\Phi s_{\mathfrak{a}} = 3 b_0 L^{k-j} / a_{0,k+1} > 1.$$

For old radii the right side is at least $5/(2\sqrt{2})$, by slow growth and $\beta_{\mathrm{s}} \le 1/5$; for octet radii it equals $3\theta_{\mathrm{oc}} > 3/(2\sqrt{2})$. Hence every radius is at least $(C_\Phi s_{\mathfrak{a}})^{-1} \ge 2$. The one-term proposition now proves definition, holomorphy and the inheritance properties. All sums are finite. In particular, parameter holomorphy is preserved by evaluation at the fixed substituted points. The index sets, substituted points and finite vertex differences are independent of the chosen radii, which proves restriction compatibility.

We give the summation estimate. Write $D_m = \max\{2L^{m-1} - 2, 0\}$. A domain wide at a site z contains a point outside $\square(z)^{\sim 2}$, whose distance from z is greater than $2ML^{k-1-j}$ in level-j units. The diameter bound $|x-z|_\infty \le M(d_j(X)+2)$ gives $d_j(X) > 2L^{k-j-1} - 2$. If X wraps, its diameter bound and the period at least $8LM$ in level-k units give $d_j(X) \ge 2L^{k-j+1} - 2$. Consequently each term at level j has $d_j(X) \ge D_{k-j}$; octet terms are at level k.

Set $b_{j,X} = \sum_{\mathfrak{i}: j_{\mathfrak{i}} = j, X_{\mathfrak{i}} = X} \|f_{\mathfrak{i}}\|_X$. We claim

$$b_{j,X} \le B_j (d_j(X)+1) e^{-\kappa d_j(X)}, \qquad B_j = 8M^3 g_j + r_j \quad (j < k),$$

and

$$\begin{aligned} B_k = {} & 8M^3 g_k + 27 M^3 e^{7\kappa} g_k^{\mathrm{ct}} + r_k \\ & + \sum_{j<k} 8M^3 L^4 C_{\mathrm{oc}} \varrho_j^2 L^{-(k-j)} (C_{\mathrm{E},2} g_j + 81 g_j^{\mathrm{ct}} + C_{\mathrm{E},2} r_j). \end{aligned}$$

For wide pointed terms count at most $8M^3(d_j(X)+1)$ sites; at $j<k$ the tent vanishes because X_z is narrow at z. At $j=k$ its at most $27M^3$ sites give the displayed $e^{7\kappa}$ term. The unpointed term occurs once.

For the narrow terms use (864). The rooted-domain count gives, for $q > \kappa_{\mathcal{D}} + 2$,

$$\sum_{X \ni z,\ d_j(X) \ge D} (d_j(X)+2)^2 e^{-q d_j(X)} \le C_{\mathrm{E},2} e^{-D}. \tag{883}$$

Indeed at most $2048\,9216^n$ rooted domains have $d_j(X) = n$; multiply by $(n+2)^2 e^{-qn}$ and use $e^{-2n} \le e^{-D} e^{-n}$ for $n \ge D$. For each pointed octet term this bounds the sum of norms by

$$C_{\mathrm{oc}} \varrho_j^2 L^{4-4(k-j)} (C_{\mathrm{E},2} g_j + 81 g_j^{\mathrm{ct}}).$$

The unpointed terms obey the same bound with $C_{\mathrm{E},2} r_j$. A fixed target $X'(z)$ contains at most $8M^3 L^{3(k-j)}$ possible level-j sites. Multiplying these two factors gives $L^{4-(k-j)}$, as in B_k. These terms have $d_k(X'(z)) = 0$, completing the claim.

Fix j and put $d = d_j(X)$, $m = \#_{\pi_k}(Y \setminus X^{\uparrow})$ and $N_Y = \#_{\pi_k} Y$. Parent trees and attachment of cubes give $d_k(Y) \le d + m$ and $N_Y \le 8(d+1) + m$. Therefore

$$\begin{aligned} b_{j,X} e^{-(\bar\kappa - 1)m} &\le B_j e^{-(1-\delta_{\mathrm{w}})\kappa d_k(Y)} (d+1) e^{-\delta_{\mathrm{w}}\kappa d} e^{-\delta_{\mathrm{w}}\kappa m} \\ &\le \frac{9 B_j}{N_Y} e^{-(1-\delta_{\mathrm{w}})\kappa d_k(Y)} (d+2)^2 e^{-\delta_{\mathrm{w}}\kappa d}. \end{aligned}$$

For the second inequality insert $1 \le (8(d+1)+m)/N_Y$ and use $m e^{-m} \le 1$. There are $L^{3(k-j)} N_Y$ possible level-j cube centres in Y. Sum first over domains containing such a centre, use (883) with $D = D_{k-j}$, and cancel N_Y. It follows that

$$\sum_{j, X : X^{\uparrow} \subset Y} b_{j,X} e^{-(\bar\kappa - 1)m} \le C_\Sigma \sum_j \varpi_{k-j} B_j e^{-(1-2\delta_{\mathrm{w}})\kappa d_k(Y)}, \qquad \varpi_m = L^{3m} e^{-D_m}. \tag{884}$$

Here $\varpi_0 = 1$ and $\varpi_m \le L^{4-m}$, since $D_m \ge 2(m-1)(L-1) \ge 4(m-1)\log L$ for $m \ge 2$; the case $m = 1$ is equality. Also $\varrho_j \ge 1$. Using (880) in B_j and B_k now gives

$$\sum_j \varpi_{k-j} B_j \le c_\omega C' N_{\mathfrak{a},\omega}(\mathfrak{f}_\circ).$$

Multiply (884) by $2 C_\Phi s_{\mathfrak{a}}$, using the one-term bound and the amplitude-radius estimate. This is (881).

Finally apply (872) to every term of (875). For a narrow term, (865) identifies its octet evaluation with its value on the actual critical background, on both sides of the step. Each old interaction occurs exactly once, as a wide term or in an octet sum. The subtracted constants cancel. The tents sum to A^{η_j}, so the remaining counterterms are exactly those in (874). This proves (882) and the well-definedness of its orbit values. □

The two choices used in the iteration can now be stated without absorbing a change of weights into the smallness hypothesis:

radii	ω_n	K
admissible	$(n+1)L^{-(n-1)}$	$(1+\beta_{\mathrm{s}})^4 K_\Lambda$
$\varrho_j = 1$ $(1 \le j \le k)$	$L^{-(n-1)}$	K_Λ.

(885)

Indeed slow growth gives $\varrho_j^2 \le (1+\beta_{\mathrm{s}})^4 (k+2-j)$, and $\sum_{n \ge 1} (n+1) L^{-(n-1)} = (2 - L^{-1})/(1 - L^{-1})^2 \le 3$. For any one-step rate $\lambda_V \le (1 - 2\delta_{\mathrm{w}})\kappa$, use its normalization

$$\nu(\mathbb{V}') = (K s_{\mathfrak{a}})^{-1} \sup_{Y, \mathfrak{u}} e^{\lambda_V d_k(Y)} |\mathbb{V}'(Y; \mathfrak{u})|, \qquad \check{s}(K) = \min\{s^\star, w_*/(4K N_*)\}. \tag{886}$$

Then $\nu(\mathbb{V}'[\mathfrak{f}_\circ]) \le N_{\mathfrak{a},\omega}(\mathfrak{f}_\circ)$, and $s_{\mathfrak{a}} \le \check{s}(K)$, $N_{\mathfrak{a},\omega} \le N_*$ give the required fourth-potential reserve $4K N_* s_{\mathfrak{a}} \le w_*$.

52. One step of the local fluctuation expansion

We combine the Gaussian construction with the interaction potentials and the change in the earlier local terms. The argument in this section uses their quantitative properties as hypotheses. The subsequent application verifies these properties for the actual fields and identifies the resulting integral with the renormalization step.

Fix $0 \le k < K$, $m' < m + K - k - 1$, and $M \ge M_1 \ge R_1\mathfrak{M} \ge L^2$. Let $\mathtt{t} > 1$, $g = \mathtt{t}^{-1/2}$, and $\tau = x/\mathtt{t}$. The functions $\varepsilon(\mathtt{t}), \delta(\mathtt{t})$ are the running thresholds; write $\mathfrak{b} = \mathtt{t}^{1/2}\delta(\mathtt{t})$. A system of positive radii is $\mathfrak{a} = ((a_{0,j}, a_{1,j}))_{1 \le j \le k+1}$. For $0 < \beta_{\mathrm{s}} \le 1/5$, put

$$\bar{a}_i = (1 + \beta_{\mathrm{s}})a_{i,k+1}, \qquad s_{\mathfrak{a}} = \frac{\delta(\mathtt{t})}{a_{0,k+1}}, \qquad \mathcal{U}^{\circ}(Y) = \mathcal{U}^{\circ}_{k+1}(Y; \bar{a}_0, \bar{a}_1).$$

This core is defined for $Y \in \mathcal{D}_k$, including every $X \in \mathcal{D}_{k+1}$. Require its slices $\mathcal{O}_U(Y)$ to be open. On a slice, analyticity then means holomorphy in $(A', \mathbf{J})$.

52.1. Input bounds and constants.

Definition 52.1. The local input consists of the following data, with constants independent of the domain and the triple.

(1) For every $X \in \mathcal{D}_{k+1}$ and $P \in \mathcal{U}^{\circ}(X)$, a Gaussian structure (C, Γ) on X with constants

$$\bar{\kappa} \ge 1, \quad 0 < \gamma_{\mathrm{r}} \le 1, \quad c_{\mathrm{r}} \ge 1, \quad \hat{\varepsilon}_0 = \tfrac{1}{2}u_{\mathrm{G}}(LM)^{-3}, \quad u_{\mathrm{G}} > 0.$$

It is a function of the pair, jointly holomorphic in the slice and decoupling parameters on a neighborhood of the closed decoupling polydisc, and gauge covariant on nonpivot coordinates. It is consistent with restriction of a whole-torus triple to X. On the whole torus it is complete, with covariance $C_*(P)$.

(2) For each fine cube $\square$, a symmetric gauge-fixing operator $\widehat{\mathsf{G}}_{\square}(P)$ on its inner nonpivot bonds, with row and column sums at most $c_{\mathsf{G}} \ge 1$. It depends only on the pair, is holomorphic on each slice, and transforms by the bondwise adjoint maps under a gauge transformation. Its restrictions agree when the ambient domain changes.

(3) Three potentials $\mathbb{V}^{\mathrm{J}}, \mathbb{V}^{\mathrm{G}}, \mathbb{V}^{\mathrm{A}}$ on $\mathcal{U}^{\circ}(Y) \times \mathbb{D}^{\mathrm{f}}(Y)$. They are jointly holomorphic on every slice and field polydisc, are functions of the pair, and satisfy $\mathbb{V}^{o}(Y; P^u, \mathrm{Ad}(u)B) = \mathbb{V}^{o}(Y; P, B)$. The J potential vanishes at zero and the G and A potentials vanish through degree two. Uniformly in P,

$$\|\mathbb{V}^{o}\|^{\mathrm{f}}_{\star,\lambda_V} \le K_P\delta(\mathtt{t}), \qquad K_P \ge 1. \tag{887}$$

(4) A fourth potential $\mathbb{V}'$, with the same analyticity, pair dependence and gauge invariance, vanishing at zero. For $K_\Lambda \ge 1$, define

$$\nu(\mathbb{V}') = (K_\Lambda s_{\mathfrak{a}})^{-1} \sup_{Y,P,B} e^{\lambda_V d_k(Y)}|\mathbb{V}'(Y; P, B)|, \tag{888}$$

where $P \in \mathcal{U}^{\circ}(Y)$ and $B \in \mathbb{D}^{\mathrm{f}}(Y)$. Require $\nu(\mathbb{V}') \le N_*$.

Families with additional complex parameters are jointly holomorphic on these domains and satisfy the same uniform bounds. Translation covariance and conjugation identities are imposed when their corresponding conclusions are used. ■

All restrictions of triples in this section are at level $k+1$. The actual Gaussian construction reads such a triple at level k by (764); in particular its current coordinate is $L^{-3}\mathbf{J}$.

Use the expansion constants (705) with

$$\hat{\varepsilon}_{\mathrm{S}} = (u_{\mathrm{G}} + c_{\mathrm{r}})(LM)^{-3}, \qquad \gamma_2 = 2(LM)^{-3}, \qquad \lambda_{\mathrm{out}} = (1 + 4\beta_{\mathrm{s}})\kappa. \tag{889}$$

Then $(LM)^3 u = 3 + c_{\mathrm{r}} + u_{\mathrm{G}} =: u^*_{\mathrm{S}}$. In particular $c_{\mathrm{v}}, c_{\mathrm{w}}$ are independent of M. Define

$$K_{\mathsf{G}} = \tfrac{1}{2}c_{\mathsf{G}}M^3, \qquad \sigma_* = \min\{\tfrac{1}{2}(LM)^{-3}, w_*/(2K_{\mathsf{G}})\}, \tag{890}$$

and

$$\mathfrak{b}_*^2 = \frac{2}{\gamma_2}\log\Big(2\mathsf{C}_{\mathrm{ex}}((L+2)M)^3 e^{4\lambda_1}\Big), \tag{891}$$

$$K_{\mathbf{M}} = 3L^3\max\{K_\Pi K_{\mathsf{G}},\ K_P^\Pi e^{4\lambda_1},\ 32K_\Pi K_P,\ 81K_\Pi K_\Lambda\}. \tag{892}$$

Take $\mathsf{C}_{\mathcal{A}} \geq 3$, $N_* \geq 1$, $0 < a^* \leq 1$, and $s^* > 0$, and require

$$\begin{gathered}(LM)^3 \geq \mathsf{C}_{\mathrm{ex}} u_{\mathrm{S}}^* \max\{\gamma_{\mathrm{r}}^{-1}, c_{\mathrm{r}}^2\}, \qquad \bar\kappa \geq 4\lambda_1 + 1,\\ 8K_P s^* \leq w_*, \qquad 4K_\Lambda N_* s^* \leq w_*, \qquad \bar a_0 + \bar a_1 \leq a^*, \qquad s_{\mathfrak{a}} \leq s^*,\\ \varepsilon(\mathsf{t}), \delta(\mathsf{t}) \leq 1, \qquad \mathfrak{b}(\mathsf{t}) \geq \mathfrak{b}_*.\end{gathered} \tag{893}$$

The lower bound on C_{ex} remains the one in the Gaussian cluster expansion. These conditions imply all its smallness assumptions at the present threshold and at threshold infinity.

52.2. **The two local contributions.** For a fine cube define

$$\hat{\mathsf{g}}_b(\square; P, B) = \tfrac{1}{2}\langle B(b), (\widehat{\mathsf{G}}_\square(P)B)(b)\rangle, \qquad \mathbb{V}^{\mathrm{q}}(\square; P, B) = -(1-\tau)\sum_b \hat{\mathsf{g}}_b(\square; P, B).$$

Set these families equal to zero on other fine domains, and set the bond insertion to zero when the bond is not inner to the cube. Use the scaled structure

$$(C_\tau, \Gamma_\tau) = (\tau^{-1}C, \tau^{1/2}\Gamma), \tag{894}$$

with the principal root. Construct the interpolation of $\mathbb{V}^{\mathrm{J}}, \mathbb{V}^{\mathrm{G}}, \mathbb{V}^{\mathrm{A}}, \mathbb{V}'$ by (748). For each X, P, b, with the Gaussian data on $R = X$ and the potentials evaluated at restrictions of P, put

$$\begin{aligned}\mathbf{G}_\tau(X, b; P) &= -(1-\tau)\int_0^1 E'[\{\sigma\mathbb{V}^{\mathrm{q}}\}; \hat{\mathsf{g}}_b](X)\, d\sigma,\\ E^\chi(X, b; P) &= E^\chi[\{\mathbb{V}^{\mathrm{q}}\}](X, b),\\ E^\bullet[\mathbb{V}'](X, b; P) &= \int_0^1 E'[\{\mathbb{V}^{\mathrm{q}}, \mathbb{V}_v^{\mathrm{int}}, \mathbb{V}_v'\}; \dot{\mathbb{V}}_{v,b}^{\mathrm{int}} + \dot{\mathbb{V}}_{v,b}'](X)\, dv.\end{aligned} \tag{895}$$

The first integral has threshold infinity; the other two terms have threshold $\mathfrak{b}(\mathsf{t})$. The G term and the cutoff term do not depend on $\mathbb{V}'$. Distribute these terms to sites by

$$\mathbf{G}_\tau(X, P, z) = \sum_b h_z(b)\mathbf{G}_\tau(X, b; P), \qquad \mathbf{F}[\mathbb{V}'](X, P, z) = \sum_b h_z(b)(E^\chi + E^\bullet[\mathbb{V}'])(X, b; P). \tag{896}$$

Theorem V (Local one-step expansion)**.** Assume the local input of Definition 52.1 and (893). For $x \in D(\mathsf{t}, \sigma_*\mathsf{t})$, all terms in (895) are defined. Their bond terms vanish unless $b^\sharp \subset X$, and their site terms are pointed families at output radii $(\bar a_0, \bar a_1)$. Writing $d = d_{k+1}(X)$, one has

$$|\mathbf{G}_\tau(X, P, z)| \leq K_{\mathbf{M}}|1-\tau|e^{-\lambda_{\mathrm{out}}d}, \tag{897}$$

$$|\mathbf{F}[\mathbb{V}'](X, P, z)| \leq K_{\mathbf{M}}\big(\delta(\mathsf{t}) + s_{\mathfrak{a}}\nu(\mathbb{V}') + e^{-\gamma_2\mathfrak{b}^2/2}\big)e^{-\lambda_{\mathrm{out}}d}. \tag{898}$$

The terms are jointly holomorphic in x, the slice variables, and additional holomorphic input parameters under uniform bounds. They inherit translation covariance and conjugation. Real input at real x gives real output.

For two admitted fourth potentials,

$$|\mathbf{F}[\mathbb{V}'] - \mathbf{F}[\mathbb{V}'']|(X, P, z) \leq K_{\mathbf{M}} s_{\mathfrak{a}}\nu(\mathbb{V}' - \mathbb{V}'')e^{-\lambda_{\mathrm{out}}d}. \tag{899}$$

On the half-radius x-disc, the difference at x, x' is bounded by the right side of (898) times $2|x - x'|/(\sigma_*\mathsf{t})$.

With $\mathbb{V}^{\mathrm{int}} = \mathbb{V}^{\mathrm{J}} + \mathbb{V}^{\mathrm{G}} + \tau\mathbb{V}^{\mathrm{A}}$, the exact local sum rules are

$$\sum_{z} \mathbf{G}_{\tau}(X, P, z) = E^{\circ}[\{\mathbb{V}^{\mathrm{q}}\}](X), \tag{900}$$

$$\sum_{z} \mathbf{F}[\mathbb{V}'](X, P, z) = E[\{\mathbb{V}^{\mathrm{q}}, \mathbb{V}^{\mathrm{int}}, \mathbb{V}'\}](X) - E^{\circ}[\{\mathbb{V}^{\mathrm{q}}\}](X). \tag{901}$$

The site sums run over $X \cap \mathbb{T}^{(k+1)}$. For a whole-torus triple P_*, these formulas give an absolutely convergent integral and the identity

$$\begin{aligned} &\int d\mu_{\tau^{-1}C_*(P_*)}(B) \prod_{b\in\mathfrak{X}^+} \mathbf{1}_{\{|B(b)|<\mathfrak{b}\}} \exp\left[\sum_{Y\in\mathcal{D}_k} (\mathbb{V}^{\mathrm{q}} + \mathbb{V}^{\mathrm{int}} + \mathbb{V}')(Y; P_*|_Y, B|_Y)\right] \\ &\qquad = \exp\left[\sum_{X,z} (\mathbf{G}_{\tau} + \mathbf{F}[\mathbb{V}'])(X, P_*|_X, z)\right] \neq 0. \end{aligned} \tag{902}$$

The integrand is zero where its test fails. Each refined series in (895) has a summable majorant for the integrals of the absolute values of its terms, uniform in x, P and in bounded input families. ■□

Proof. The disc gives $\operatorname{Re}\tau > 1/2$, $|\tau| < 3/2$, and $|1-\tau|K_{\mathsf{G}} \le w_*/2$. For the scaled Gaussian structure, the inverse covariance and coupling defects are at most $2\hat{\varepsilon}_0 + c_{\mathrm{r}}|1-\tau|$; the product $\Gamma^{\mathsf{T}}C\Gamma$ is unchanged. Thus all three defects are at most $\hat{\varepsilon}_{\mathrm{S}} - \frac{1}{2}c_{\mathrm{r}}(LM)^{-3}$. The structure remains local, and its whole covariance is $\tau^{-1}C_*$. This is the scaling calculation at the end of Theorem U. The first numerical condition gives $u \le \mathsf{C}_{\mathrm{ex}}^{-1}\min(\gamma_{\mathrm{r}}, c_{\mathrm{r}}^{-2})$, and the threshold floor gives $\mathsf{p}_* \le 1/2$.

At real fields, $|\langle A, B\rangle| \le |A||B|$. Using $ab \le (a^2+b^2)/2$ and the row and column bounds of $\widehat{\mathsf{G}}_{\square}$ gives

$$\left|\tfrac{1}{2}(B, \widehat{\mathsf{G}}_{\square}B)_1\right| \le \tfrac{1}{2}c_{\mathsf{G}} \sum_{b\in\square} |B(b)|^2.$$

The same upper bound holds for any single bond contribution. Since a fine cube has $d_k = 0$, this proves $\|\hat{\mathsf{g}}_b\|_{\star,\lambda_V} \le K_{\mathsf{G}}$ and $\|\mathbb{V}^{\mathrm{q}}\| \le w_*/2$. Also $a_{0,k+1} \le 1$, so $\delta(\mathsf{t}) = s_{\mathfrak{a}}a_{0,k+1} \le s^*$. Proposition 48.4 therefore yields

$$\begin{aligned} \|\mathbb{V}^{\mathrm{int}}_v\| \le \tfrac{7}{2}K_P\delta \le \tfrac{7}{16}w_*, \qquad \|\mathbb{V}'_v\| \le K_{\Lambda}s_{\mathfrak{a}}\nu \le \tfrac{1}{4}w_*, \\ \|\dot{\mathbb{V}}^{\mathrm{int}}_{v,b}\| \le 32K_P\delta, \qquad \|\dot{\mathbb{V}}'_{v,b}\| \le 9K_{\Lambda}s_{\mathfrak{a}}\nu, \\ \|\dot{\mathbb{V}}^{\mathrm{int}}_{v,b} + \dot{\mathbb{V}}'_{v,b}\| \le 9(7/16 + 1/4)w_* = 99w_*/16. \end{aligned} \tag{903}$$

Here all norms have rate λ_V.

Apply the insertion theorem to the first and third lines of (895), and the cutoff theorem to its second line. They prove existence, support, and the bond bounds

$$|\mathbf{G}_{\tau}(X, b; P)| \le |1-\tau|K_{\Pi}K_{\mathsf{G}}e^{-\lambda_{\mathrm{out}}d},$$

$$|E^{\chi} + E^{\bullet}|(X, b; P) \le (K_P^{\Pi}e^{4\lambda_1}\mathsf{p} + 32K_{\Pi}K_P\delta + 9K_{\Pi}K_{\Lambda}s_{\mathfrak{a}}\nu)e^{-\lambda_{\mathrm{out}}d}.$$

The tent sum over nonpivot bonds is at most $3L^3$. Equation (892) now gives (897) and (898).

Along the first path the sum of its bond velocities is $\mathbb{V}^{\mathrm{q}}$. The path identity gives its cluster difference from the empty system, whose no-test cluster sum is zero. Along the third path it gives the difference between the full system and $\{\mathbb{V}^{\mathrm{q}}\}$ at finite threshold. The cutoff term supplies the difference between the finite-threshold and no-test sums for this last system. The tent identity (761) proves (900) and (901). Restriction consistency identifies the local terms with those computed from the whole-torus data. The complete Gaussian cluster expansion then proves (902).

The uniform insertion majorant (734), together with (903), justifies all exchanges of the path integral and refined sums. For a cutoff term, apply Lemma 48.3 separately to each refined density in (745). Its derivative has integral of absolute value bounded by that density's total variation. These

masses are bounded by the single-bond marked series, independently of the parameters. This gives the asserted common majorant.

Holomorphy follows from the parameter version of the insertion theorem, the fixed-density cutoff formula, and finite tent sums. Pair dependence and gauge invariance hold because the corresponding entries are identical, or related by orthogonal adjoint maps. They pass through every integral. The same argument proves translation covariance and conjugation, using the real covariant tents. For real inputs, holomorphic potentials real on real fields satisfy $V(B^*) = \overline{V(B)}$ throughout the connected field polydisc by the identity theorem. Thus the output is real. The bounds already proved show that these functions are local terms on the output core.

It remains to prove dependence on the fourth input. Put $\Delta V = \mathbb{V}' - \mathbb{V}''$ and $\nu_\Delta = \nu(\Delta V) \le 2N_*$. Linearity of interpolation gives $\|\Delta V_v\| \le K_\Lambda s_{\mathfrak{a}} \nu_\Delta$ and $\|\Delta \dot V_{v,b}\| \le 9K_\Lambda s_{\mathfrak{a}} \nu_\Delta$. Split the difference of insertion sums into the change of insertion and the change of its background system. The first costs at most $9K_\Pi K_\Lambda s_{\mathfrak{a}} \nu_\Delta e^{-\lambda_{\mathrm{out}} d}$. For the second, the fourth-origin base norm is at most $w_*/4$, its change is at most $w_*/2$, and its insertion norm is at most $99w_*/16$. Equation (732) bounds this cost by $(99/2)K_\Pi K_\Lambda s_{\mathfrak{a}} \nu_\Delta e^{-\lambda_{\mathrm{out}} d}$. Integrating and using $9 + 99/2 = 117/2 < 81$, then summing the tents, proves (899). Finally Cauchy's derivative estimate on the half-radius x-disc gives the stated x-dependence. □

For comparison with the decay rate of the input potentials, let $0 < \delta_{\mathrm{w}} < 1/2$, and define

$$L_\Pi = \frac{2\mathsf{C}_{\mathrm{ex}}(1 + 4\beta_{\mathrm{s}})}{1 - 2\delta_{\mathrm{w}}}, \qquad \kappa_\Pi = \frac{2}{1 - 2\delta_{\mathrm{w}}} \left[\frac{\mathsf{C}_{\mathrm{ex}}}{L} (8c_{\mathrm{w}} + \mathsf{C}_{\mathrm{ex}} + \lambda_{\mathrm{s}}) + 2\lambda_{\mathrm{s}} \right]. \tag{904}$$

If $L \ge L_\Pi$ and $\kappa \ge \kappa_\Pi$, the term proportional to κ in λ_V and its remaining terms are each at most $(1 - 2\delta_{\mathrm{w}})\kappa/2$. Hence $\lambda_V \le (1 - 2\delta_{\mathrm{w}})\kappa$. If also $\bar\kappa \ge 4\lambda_1 + 1$, then $\lambda_1 \ge \mathsf{C}_{\mathrm{ex}} + \lambda_{\mathrm{s}} \ge 2$, and hence

$$\bar\kappa - 2 \ge 4\lambda_1 - 1 > \lambda_1 > \lambda_{\mathrm{out}} > \kappa > \lambda_V.$$

These choices are independent of the lattice, step and running coupling.

52.3. **The small-field measure.** We next identify the integral to which the Gaussian expansion applies. All lengths and pairings in this subsection are in level-k units. Put $t = k + 1$, and let Γ, $\mathfrak{X}_0$ and $\mathfrak{X}$ be the whole-lattice sets used above. If N_j is the number of level-j sites, then

$$\#\Gamma = 3N_t, \qquad \#\mathfrak{X}^+ = 3(N_k - N_t), \qquad n_{\mathcal{O}} = 3(N_k - N_t), \qquad n_\perp = 3\#\mathfrak{X}^+ - n_{\mathcal{O}}.$$

Here $n_{\mathcal{O}}$ counts the gauge variables within blocks.

Let W have a real regular critical background $U_t(W)$. Assume that its averaged contours are defined, and choose a representative U_+ for which $p = \mathcal{M}^k U_+$ is axial at the level-t sites. The residual-orbit and axial-representative results make p independent of this choice. We work on an open set where p is defined continuously and $\sup|\partial p - 1| < \varpi_\chi$, with $\varpi_\chi > 0$ sufficiently small depending only on L. Use the actual constraint chart from definition 51.3:

$$V_W[\mathcal{A}] = e^{i\psi_p^{\mathrm{lin}}(\mathsf{C}_p \mathcal{A})} p, \qquad V_W(B) = V_W[gB], \qquad g = \mathsf{t}^{-1/2}.$$

There are $r_\chi > 0$ and $C_\chi \ge 1$, depending only on L, such that this chart is defined when $C_\chi \sup|\mathcal{A}| < r_\chi$, and its full logarithmic field has norm at most $2C_\chi \sup|\mathcal{A}|$. Decrease r_χ, ϖ_χ so that all these real charts lie in the small-curvature branch of the averages and in the radii of proposition 14.11.

Write $b = \mathfrak{b}(\mathsf{t})$, $\delta = gb$, and $\epsilon = \varepsilon(\mathsf{t})$. Assume

$$\epsilon \le 1, \qquad C_\chi \delta \le r_\chi, \qquad C_\chi \delta \le \epsilon. \tag{905}$$

Increasing C_χ if necessary, the last inequality is ensured by $A_0/A_1 \ge C_\chi$. Define $\chi_{\mathsf{t}}^{\mathrm{sf}}(V)$ to be one precisely when $p(\mathcal{M}V)$ is stably defined and

$$\left| \frac{1}{i} \log\bigl(V(b')p(\mathcal{M}V)(b')^{-1}\bigr) \right| < \delta \qquad (b' \in \mathfrak{X}^+);$$

otherwise it is zero. These are intrinsic logarithms, and no pivot variable is tested. Let $\mathfrak{G}_k[\mathsf{t}]$ be the block gauge-fixing factor: for each $y \in \mathbb{T}^{(t)}$ and $x \in B(y) \setminus \{y\}$ its factor is

$$\mathsf{z}(\mathsf{t},\epsilon)^{-1}\mathbf{1}_{\{|V(y,x)-1|<\epsilon\}}\exp\{-\mathsf{t}(1-\tfrac{1}{2}\operatorname{Re}\operatorname{tr}V(y,x))\},$$

where z is the corresponding single-group integral. The product is zero where a contour is undefined. For a bounded Borel function $\chi_{\mathsf{t}}^{\mathrm{sf}}F$, the small-field measure is the pushforward under $\mathcal{M}$ of $\mathfrak{G}_k[\mathsf{t}]\chi_{\mathsf{t}}^{\mathrm{sf}}F\,dV$.

For the chart field $B' = \psi_p^{\mathrm{lin}}(\mathsf{C}_p g B)$ put

$$\mathsf{J}_W(B) = \sum_{c\in\Gamma}\log\det(1-\mathsf{Jb}_c(p,gB)) + \sum_{b'\in\mathfrak{X}_0^+}\log\sigma_1(|B'(b')|^2), \qquad \sigma_1(s) = \left(\frac{\sin\sqrt{s}}{\sqrt{s}}\right)^2. \tag{906}$$

The value at $s=0$ is one; all logarithms here are real. Set

$$\mathsf{pf}_k(\mathsf{t}) = \mathsf{z}(\mathsf{t},\epsilon)^{-(N_k-N_t)}(\sigma_{\mathrm{H}}\mathsf{t}^{-3/2})^{\#\mathfrak{X}^+}L^{-3\#\Gamma}, \qquad \sigma_{\mathrm{H}} = (2\pi^2)^{-1}. \tag{907}$$

At a whole-lattice real background let $\mathsf{B}_W = \mathsf{C}_p^{\mathsf{T}}\Delta_k(U_+,J^\circ)\mathsf{C}_p$ be the action part of (589), and put

$$\widehat{\mathsf{G}}_W = \mathsf{C}_p^{\mathsf{T}}\mathsf{G}^{(2)}[p]\mathsf{C}_p, \qquad \mathsf{A}_W = \mathsf{B}_W + \widehat{\mathsf{G}}_W, \qquad C_W = \mathsf{A}_W^{-1}.$$

The current in this formula is $J^\circ = L^{-3}J^{(t)}(U_+)$, as in (764). Take the background in the radius of theorem U, so that A_W is real positive definite. Let S be gauge invariant on the minimal orbits in question, and assume that $U_k(V_W(B))$ exists for every $\sup|B| < b$. For the Hessian identification, assume in addition that the constant tower at (U_+,J°) satisfies the hypotheses of proposition 34.2 and theorem K, and that there is an open neighbourhood $\mathcal{N}$ of p in the space of all real level-k fields on which $U_k(V)$ exists and its minimal orbit contains the restored critical response at U_+. Thus this hypothesis concerns arbitrary nearby data, not just the constraint fibre. At $k=0$ it means $U_0(V)=V$ on the corresponding open regularity neighbourhood. Write

$$\mathbb{F}_W(B) = A^{\eta_k}(U_k(V_W(B))), \qquad G_W(B) = \sum_y\sum_{x\in B(y)\setminus\{y\}}(1-\tfrac{1}{2}\operatorname{Re}\operatorname{tr}V_W(B)(y,x)).$$

Define the three remainders directly by

$$\begin{aligned}\mathbb{P}^J(B) &= \mathsf{J}_W(B),\\ \mathbb{P}^G(B) &= -\mathsf{t}G_W(B) + \tfrac{1}{2}(B,\widehat{\mathsf{G}}_W B)_1,\\ \mathbb{P}^A(B) &= -\mathsf{t}[\mathbb{F}_W(B)-\mathbb{F}_W(0)] + \tfrac{1}{2}(B,\mathsf{B}_W B)_1,\end{aligned}$$

and $s_W[S](B) = S(U_k(V_W(B))) - S(U_t(W))$. These formulas also specify the normalization of the potentials. Under the global identification hypotheses stated before (827), they equal the whole totals of its three actual local potentials. Put $P_W = (U_+, 0, J^{(t)}(U_+))$ and define

$$Z^{(k)}(W;\mathsf{A}_W) = \prod_{c\in\Gamma}|\det m_c(p)|^{-1}\int_{\mathbb{R}^{3\#\mathfrak{X}^+}} e^{-\frac{1}{2}(B,\mathsf{A}_W B)_1}\,dB = Z_k[P_W].$$

The last equality uses the actual form at P_W in (790).

Proposition 52.2. Let the preceding hypotheses hold, let $\tau = x/\mathsf{t}$ with $\operatorname{Re}\tau > 0$, and suppose $B \mapsto F_{x,S}(V_W(B))$ is bounded Borel on $\sup|B| < b$, where $F_{x,S}(V) = \exp\{-xA^{\eta_k}(U_k(V)) + S(U_k(V))\}$.

The chart density is

$$\mathsf{I}[F_{x,S}](W) = \mathsf{pf}_k(\mathsf{t})\, Z^{(k)}(W;\mathsf{A}_W) e^{-xA^{\eta_k}(U_t(W))+S(U_t(W))}\tau^{-3\#\mathfrak{X}^+/2} \times \int \mu_{\tau^{-1}C_W}(dB)\, \mathbf{1}_{\{\sup|B|<b\}} \exp\left\{ \begin{array}{l} -\dfrac{1-\tau}{2}(B,\widehat{\mathsf{G}}_W B)_1 \\ +\mathbb{P}^J(B)+\mathbb{P}^G(B)+\tau\mathbb{P}^A(B)+s_W[S](B) \end{array} \right\}. \tag{908}$$

The integration is over real fields, and the integral is absolutely convergent. If $\chi_{\mathsf{t}}^{\mathrm{sf}} F_{x,S}$ is bounded Borel on the whole configuration space, this is the Borel density of the small-field pushforward on the open background domain. It is continuous there if the chart integrand is continuous in the background and locally uniformly bounded.

Moreover the uncut quadratic gauge integral satisfies

$$\int \mu_{\tau^{-1}C_W}(dB) e^{-(1-\tau)(B,\widehat{\mathsf{G}}_W B)_1/2} = \tau^{n_{\mathcal{O}}/2}. \tag{909}$$

For the connected quadratic terms of the Gaussian expansion, with the logarithm normalized at zero potential,

$$\sum_{X,z} \mathbf{G}_\tau(X, P_W, z) = \frac{n_{\mathcal{O}}}{2} \operatorname{Log}\tau. \tag{910}$$

The last identity is asserted on the small τ-domain where that expansion is defined. ■

Proof. The fibre and its measure. The chart has average W and unchanged nonpivot logarithms. Conversely, a field of average W passing the nonpivot test has these logarithms as its chart coordinates. On each double block it has the same nonpivot entries as the chart field; proposition 14.11 gives equality of the remaining pivot entry, including across the real averaging branches. Thus the chart parametrizes exactly the tested fibre.

All within-block bonds are nonpivot. A block contour has at most $3(L-1)/2$ bonds, so its transport changes by at most $3L\delta/2$. The small-curvature choice makes both contour families lie within $1/40$ of their means. If their members are V_i, V_i^0, their means are $\bar{V}, 1$, and $e = \max_i |V_i - V_i^0|$, the logarithmic mean equations give

$$|\bar{V}-1|\left(1-\frac{1}{39}-\frac{1}{20}\right) \le \frac{40}{39}e, \qquad |\bar{V}-1| \le \frac{10}{9}e.$$

Indeed, subtract the mean equations and use $|\log X - \log Y - (X-Y)| \le |X-Y|/39$ in the $1/40$ ball; then write $V_i\bar{V}^{-1} - V_i^0 = (\bar{V}^{-1}-1) + (V_i-1)(\bar{V}^{-1}-1) + (V_i - V_i^0)$. Consequently $|V_W(B)(y,x)-1| \le C_\chi\delta$. All gauge indicators are therefore one, and the gauge factor is $\mathsf{z}(\mathsf{t},\epsilon)^{-(N_k-N_t)} e^{-\mathsf{t}G_W(B)}$.

For clarity, the determinant in the change of variables can be calculated in a single double block. Write $q_U(a) = i^{-1}\log(\mathcal{M}(e^{ia}U)\mathcal{M}(U)^{-1})$ and $n_U = \partial_0 q_U(0) = L m_c(U)$, where ∂_0 differentiates the pivot variable. The correction used in the chart obeys $q_U(a - A_c y_U(a)) = \dot{q}_U(a)$. At $\dot{q}_U(a) = 0$, differentiation in that variable gives

$$\partial_0 q_U(a - A_c y_U(a))(1 - \partial_0 y_U(a)) = n_U.$$

For $U' = e^{i(a - A_c y_U(a))}U$ and pivot value β, differentiating the exponential gives

$$n_{U'}\, \mathsf{dE}(\beta)(1 - \partial_0 y_U(a)) = n_U, \qquad \mathsf{dE}(Y)H = \int_0^1 \mathrm{Ad}(e^{isY})H\, ds.$$

On the line spanned by Y, this map is the identity; on its orthogonal plane its determinant is $(\sin|Y|/|Y|)^2$. Thus

$$(\det n_{U'})^{-1} = (\det n_U)^{-1}\sigma_1(|\beta|^2)\det(1-\partial_0 y_U(a)). \tag{911}$$

The factors are positive by the chart bounds.

In exponential coordinates, normalized Haar measure is $\sigma_{\mathrm{H}}\sigma_1(|Y|^2)\,dY$ for $|Y|<\pi$. One can see the normalization and invariance directly: the radial integral is $4\pi\int_0^\pi \sin^2 r\,dr = 2\pi^2$, and differentiating $e^{iT(Y)} = e^{iY}u$ gives $\mathsf{dE}(T(Y))DT(Y) = \mathsf{dE}(Y)$. Hence this density is invariant under right translation away from the two omitted chart points, which have zero measure.

Fix the nonpivot entries. The small-curvature pivot domain is open, and its averaging map is injective by proposition 14.11. A plaquette containing the pivot places this domain in one exponential chart after the radius is decreased. The differential calculation above makes it a local diffeomorphism with Haar Jacobian $\det n_{U'}$. The ordinary change-of-variables formula, followed by (911), therefore integrates out that pivot. Distinct double blocks contain no other pivot, so this calculation factors over Γ. The nonpivot substitution $a = gB$ contributes $(\sigma_{\mathrm{H}}\mathsf{t}^{-3/2})^{\#\mathfrak{x}^+}$. It follows that the tested pushforward has density

$$\mathsf{pf}_k(\mathsf{t})\prod_{c\in\Gamma}|\det m_c(p)|^{-1}\int_{\sup|B|<b} e^{-\mathsf{t}G_W(B)+\mathsf{J}_W(B)}F(V_W(B))\,dB. \tag{912}$$

The factor $L^{-3\#\Gamma}$ in (907) accounts for $\det n_U = L^3\det m_c(U)$.

This argument also gives the Borel assertion without a choice of conditional measures. The tested set is Borel; the intrinsic nonpivot logarithms and the continuous chart make its inverse Borel. For a nonnegative Borel integrand, Tonelli applies to the successive pivot and nonpivot substitutions. Apply the same identity to the absolute value before treating a complex integrand. On every chart fibre the Jacobian is bounded and the field box has finite volume; boundedness of $F(V_W(B))$ therefore gives absolute convergence. The same bounds locally in W, followed by dominated convergence, give the stated continuity.

The action Hessian and gauge directions. The background response identifies $\varphi(\beta) = A^{\eta_k}(U_k(e^{i\beta}p))$ near $\beta = 0$. We record its second differential in the normalization used here. Write the response as $e^{i\xi\widehat{A}(\beta)}U_+$ followed by its restoring gauge, and let $\omega(\beta)$ be its constraint multiplier. On this constant tower all constraint rows have spacing one, and the row injection J has $\mathsf{J}^{\mathsf{T}}\mathsf{J} = I$. The response equations are

$$F'(\widehat{A}) = K_{\widehat{A}}^{\mathsf{T}}\omega, \qquad \mathcal{Q}(\widehat{A}) = \mathsf{J}\beta, \qquad \mathsf{R}D^{\mathsf{T}}\widehat{A} = 0.$$

Differentiating at zero, with $a = \widehat{A}'(0)\nu$ and $\mu = \omega'(0)\nu$, gives the saddle equations

$$(\Delta_1 + D\mathsf{R}D^{\mathsf{T}})a - Q^{\mathsf{T}}\mu = 0, \qquad Qa = \mathsf{J}\nu.$$

Here $\mathsf{G}_1^{-1} = \Delta_1 + D\mathsf{R}D^{\mathsf{T}} + Q^{\mathsf{T}}Q$. Solving these saddle equations gives

$$a = \mathsf{H}_1\mathsf{J}\nu, \qquad \mu = [(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1} - I]\mathsf{J}\nu.$$

The chain rule and differentiated constraint give the envelope identity $\varphi'(\beta)\nu = (\omega(\beta), \mathsf{J}\nu)_{\mathbf{B}}$. Since $\omega(0) = \mathsf{H}_1^{\mathsf{T}}J^\circ$, we obtain

$$D\varphi(0)\nu = (\mathsf{H}_1\mathsf{J}\nu, J^\circ)_\xi, \qquad D^2\varphi(0)(\nu,\nu) = (\nu, \mathsf{Z}_0\nu)_1, \qquad \mathsf{Z}_0 = \mathsf{J}^{\mathsf{T}}(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}\mathsf{J} - I.$$

For $k = 0$, the response is $\widehat{A}(\beta) = \beta$; $Q = \mathsf{H}_1 = I$ and $\mathsf{G}_1^{-1} = \Delta(U_+) + I$ give the same identities.

If $V(s)$ stays in the fibre, its background lift has $\mathcal{M}^{k+1}U(s) = W$ and starts at U_+. Criticality gives $D\varphi(0)B = 0$ for every $B \in \ker Q_p$. Differentiating the constraint twice gives

$$\ddot{\beta}(0) = \kappa - 2\mathsf{D}_p^{(2)}(B,B), \qquad \kappa \in \ker Q_p.$$

The chain rule and the already identified response Hessian therefore give

$$\frac{d^2}{ds^2}\Big|_0\varphi(\beta(s)) = (B, \mathsf{Z}_0B)_1 - (B, \mathsf{E}^{(2)}[\mathsf{H}_1^{\mathsf{T}}J^\circ]B)_1 = (B, \Delta_k(U_+, J^\circ)B)_1.$$

The gauge-fixing term has zero first differential at the axial field and second differential $\mathsf{G}^{(2)}[p]$. Thus the free-coordinate Hessian is precisely A_W, not a separately chosen Gaussian form.

Let $\mathcal{E}^0$ be the real site functions vanishing at the level-t sites. Define $j\rho = \chi_{\mathfrak{X}}\mathfrak{d}_p\rho$ and let $(rB)(x)$ be the first variation of the block contour at (y,x) in the direction C_pB, with $(rB)(y)=0$. Covariance of the actual average gives $Q_p\mathfrak{d}_p\rho = 0$ and $rj\rho = \rho$: the first contour variation is $\rho(x)-\rho(y)$ because $p(y,x)=1$. Moreover

$$(B_1, \widehat{\mathsf{G}}_W B_2)_1 = (rB_1, rB_2)_0.$$

Every real B consequently has the unique decomposition $B = B'' + j\rho$ with $\rho = rB$ and $rB'' = 0$. Gauge invariance along the curves $V(s)^{e^{-isa\rho}}$, for every real a, shows that the action Hessian annihilates $j\rho$. Indeed, comparing the second derivatives gives $2a\,q(B,j\rho) + a^2 q(j\rho, j\rho) = 0$ for every a. The two quadratic forms therefore occupy complementary blocks of dimensions $n_\perp, n_{\mathcal{O}}$. Taking their determinant in an adapted basis, whose change-of-basis determinant cancels in the quotient, gives

$$\det(\tau \mathsf{B}_W + \sigma \widehat{\mathsf{G}}_W) = \tau^{n_\perp}\sigma^{n_{\mathcal{O}}} \det \mathsf{A}_W.$$

Positivity of A_W makes B_W nonnegative, and hence the form on the left has positive real part for $\operatorname{Re}\tau, \operatorname{Re}\sigma > 0$. Continuity from $\tau=\sigma=1$ fixes the square-root branch. The Gaussian determinant formula now proves (909).

The density and its logarithm. Compatibility of the two critical backgrounds gives $\mathbb{F}_W(0) = A^{\eta_k}(U_t(W))$. Insert the definitions of the potentials: their quadratic terms cancel the exponent $-\tau(B,\mathsf{A}_W B)_1/2$, leaving exactly the integrand of (912), times $e^{xA^{\eta_k}(U_t(W))-S(U_t(W))}$. Since

$$\mu_{\tau^{-1}C_W}(dB) = \tau^{3\#\mathfrak{X}^+/2}\left(\int e^{-(B,\mathsf{A}_W B)_1/2}\,dB\right)^{-1} e^{-\tau(B,\mathsf{A}_W B)_1/2}\,dB,$$

this proves (908), with the same absolute convergence already obtained in (912).

Finally multiply the quadratic potential by $\theta \in [0,1]$. The whole Gaussian integral is

$$\exp\left\{\frac{n_{\mathcal{O}}}{2}[\operatorname{Log}\tau - \operatorname{Log}(\tau+\theta(1-\tau))]\right\}.$$

The connected expansion has the same exponential by its whole-integral identity, is continuous in θ, and has zero total at $\theta=0$. Their logarithms thus differ by a continuous $2\pi i\mathbb{Z}$-valued function vanishing at zero. At $\theta=1$ this proves (910). □

52.4. **The actual small-field step.** We now use the actual covariance, potentials and normalizer in the one-step expansion. The only loss in the fourth-potential estimate depends on the chosen weights for the earlier terms; its smallness ceiling must therefore be chosen with those same weights.

Keep the geometric hypotheses of theorem U and proposition 50.2 and the notation of section 52. The shifts and substitutions are those of the fourth-potential construction, with an admissible system of radii $\mathfrak{a}$; retain its octet condition when $k \geq 2$. Choose the common radii so that

$$\begin{gathered}\bar a_0 + \bar a_1 \leq a^\sharp, \qquad \bar a_0 \leq \min\{a_{\mathrm{G}}/M, \alpha_N, a_P, a_N\}, \qquad \bar a_1 \leq \min\{a_P, a_N\},\\ \theta_{\mathrm{Str}} M \geq 16\bar\kappa, \qquad e^{-\theta_{\mathrm{Str}}M} + \bar a_1 + \bar a_0 \leq \min\left\{a_{\mathrm{Str}}, \frac{u_{\mathrm{G}}}{2C_{\mathrm{Str}}}(LM)^{-3}\right\},\\ \mathsf{C}_{\mathcal{A}}\delta(\mathsf{t}) \leq \varepsilon_P, \qquad \lambda_V \leq \bar\kappa - 2.\end{gathered} \tag{913}$$

Here $a^\sharp$ is the tower-continuation radius, and the fourth-potential construction retains its own admissibility, shift and rate inequalities. No existence condition on a tower is replaced by a curvature bound.

When $k \geq 1$, the local action member is

$$\mathsf{W}_\zeta(X,P) = \zeta\eta_k^{-1} \sum_{p:X(p)=X} \left(1 - \tfrac{1}{2}\operatorname{tr}\partial\mathbf{U}(p)\right), \qquad (\mathbf{U},\mathbf{J}) = \mathbf{p}(P). \tag{914}$$

Here $X(p)$ is the union of the cubes containing the corners of p. It has at most four cubes meeting at a corner, so $d_k(X(p)) = 0$. The member is gauge invariant, holomorphic, and zero at the trivial

pair. Indeed $\left|1-\frac{1}{2}\operatorname{tr} V\right| = \frac{1}{2}|\det(V-I)| \le \frac{1}{2}|V-I|^2$, and there are at most $12(M/\eta_k)^3$ plaquettes with a given $X(p)$. Thus $\|\mathsf{W}_\zeta\|_{k,\kappa} \le 6M^3 a_{0,k}^2|\zeta|$ on the input core. Its total is $\zeta A^{\eta_k}(\mathbf{U})$. If the last slot is vacant, let $\mathfrak{g} = \mathfrak{f}_\circ \sqcup \mathsf{W}_\zeta$ denote the array with this member in slot k; otherwise take $\mathfrak{g} = \mathfrak{f}_\circ$ and $\zeta = 0$. At $k = 0$ the input array is empty. Select one of the weight cases (ω, K) of proposition 51.16, so that

$$|\mathbb{V}'[\mathfrak{g}](Y; P, B)| \le \mathsf{K} s_{\mathfrak{a}} N_{\mathfrak{a},\omega}(\mathfrak{g}) e^{-\lambda_V d_k(Y)}. \tag{915}$$

At $k = 0$ the array is empty and this potential is zero. In (892) replace K_Λ by K, and denote the resulting constant by $K_{\mathbf{M}}(\mathsf{K})$. Use

$$\check{s}(\mathsf{K}) = \min\{s^*,\, w_*/(4\mathsf{K}N_*)\}, \qquad s_{\mathfrak{a}} \le \check{s}(\mathsf{K}), \qquad N_{\mathfrak{a},\omega}(\mathfrak{g}) \le N_*. \tag{916}$$

Assume the other conditions of (893), with the same replacement. The radius s^* here is small enough for the shift construction and $8K_P s^* \le w_*$; it need not satisfy the fourth-potential inequality for every case.

The actual output is

$$E^{(k+1)}(X, P, z) = \mathbf{N}^{(k+1)}(X, P, z) + \mathbf{G}_\tau(X, P, z) + \mathbf{F}[\mathbb{V}'[\mathfrak{g}]](X, P, z). \tag{917}$$

The normalizer in this formula is independent of the input array and of t, x.

Theorem W. Under the preceding hypotheses, the actual fields satisfy the local input conditions of definition 52.1. For $x \in D(\mathsf{t}, \sigma_*\mathsf{t})$, the family (917) is defined on every output core $\mathcal{U}^\circ(X)$, $X \in \mathcal{D}_{k+1}$. It is a pointed family, holomorphic in x, the slice variables, and holomorphic input parameters under uniform size bounds. It has the translation and conjugation properties when the input arrays and the selected shifts have them.

Suppose also that

$$4\beta_{\mathsf{S}}\kappa \ge 1, \tag{918}$$

$$\delta_N M \ge \kappa + 1, \tag{919}$$

$$E_0 \ge 2E_N. \tag{920}$$

Then, with $d = d_{k+1}(X)$,

$$\begin{aligned} |E^{(k+1)}(X, P, z)| \le \Big[\frac{E_0}{2} + K_{\mathbf{M}}(\mathsf{K})\big(|1-\tau| + \delta(\mathsf{t}) + s_{\mathfrak{a}} N_{\mathfrak{a},\omega}(\mathfrak{g}) \\ + e^{-\gamma_2 \mathfrak{b}(\mathsf{t})^2/2}\big)\Big] e^{-(\kappa+1)d}. \end{aligned} \tag{921}$$

If, further,

$$K_{\mathbf{M}}(\mathsf{K})\sigma_* \le E_0/4, \tag{922}$$

$$K_{\mathbf{M}}(\mathsf{K})\big(\delta(\mathsf{t}) + s_{\mathfrak{a}} N_* + e^{-\gamma_2 \mathfrak{b}(\mathsf{t})^2/2}\big) \le E_0/4. \tag{923}$$

the bracket in (921) is at most E_0; at $x = \mathsf{t}$ it is at most $3E_0/4$.

For a whole-torus block field W, assume the background and chart hypotheses of proposition 52.2, that its triple P_W lies in $\mathcal{U}^\circ(T_\ell)$, and that the actual potentials have the global identification stated before (827) and the fourth potential sums to $s_W[S]$. In particular, retain the open whole-data response neighbourhood required in proposition 52.2. For the density conclusion also assume that $\chi_{\mathsf{t}}^{\mathrm{sf}} F_{x,S}$ is bounded and Borel on the whole configuration space $\mathcal{V}_k$. Then the small-field pushforward density is

$$\begin{aligned} \rho_{x,S}^{\mathrm{sf}}(W) = \mathsf{pf}_k(\mathsf{t})\, Z_{\mathrm{triv}}^{(k)} \tau^{-3\#\mathfrak{X}^+/2} \\ \times \exp\left\{-xA^{\eta_k}(U_t(W)) + S(U_t(W)) + \sum_{X,z} E^{(k+1)}(X, P_W|_X, z)\right\}. \end{aligned} \tag{924}$$

All integrals in this identity are absolutely convergent. Equivalently, remove $\mathbf{G}_\tau$ from the sum and replace the power of τ by $\tau^{-n_\perp/2}$. The powers use the principal logarithm. ■□

Proof. We first check the common domain. By definition 49.1 and lemma 49.2,

$$\mathcal{U}^\circ(Y) \subset \mathcal{Q}_t(Y;\bar a_0,\bar a_1,\bar a_0) \subset \mathcal{Q}_t(Y;a_P,a_P,a_P)\cap\mathcal{Q}^N(Y). \tag{925}$$

The first inclusion retains all the tower conditions of the core. The second follows coordinate by coordinate from (913). Reading the triple at level k by (764), its potential and derivative acquire the factors L^{-1},L^{-2}, and its current the factor L^{-3}. The gauge boxes required for the actual Gaussian structure are contained in the boxes in the simple-domain definition, by $12\mathfrak{M}\le \mathsf{o}LM$ and $5h\le \mathsf{o}L^2M$. Thus theorem U applies with $(\alpha_0,\alpha_1,\alpha_2)=(a_{\mathrm w}/2,\bar a_1,\bar a_0)$; its defect is at most $u_{\mathrm G}(LM)^{-3}/2$.

The slices used here are open. To see the point that is not contained in the simple inequalities, fix the real factor U. Its own tower conditions are either satisfied or the slice is empty. Under $\bar a_0+\bar a_1\le a^\sharp$, the continued tower configurations and currents are holomorphic in A' on the simple-domain slice, by the tower continuation theorem. There are only finitely many subdomains, depths and stencils. The extra core conditions are strict inequalities on these continuous functions; their intersection with $|\mathbf{J}|<\bar a_0$ is open.

The Gaussian structure's holomorphy, restrictions and whole covariance are theorem U; pair dependence is built into its coefficients. For completeness, its covariance follows through the construction. At each local tower, a unitary gauge conjugates the finite operators at the initial points of their bond coordinates. Products, bilinear transposes, compressions and inverses preserve this rule. The connected coefficient sum uses the same subsets and signs; the mesh resolvent products use scalar cutoffs and the same integration variable. Hence both coefficient families, their absolutely convergent parameter series, and finally $\mathbf{C}$ and $\mathbf{\Gamma}$ have the required conjugation rule. Complex conjugation is the identical calculation with the antilinear map $A\mapsto A^*$ and conjugated parameters; translations relabel the coordinates and cube indices. These maps preserve the real pairings, so they give precisely the covariance relations used in the Gaussian expansion. The gauge-fixing operator and the three potentials are those of eq. (824) and proposition 51.4. Their domain contains the core by (925), and $\mathsf{C}_{\mathcal{A}}\delta(\mathsf{t})\le\varepsilon_P$ puts the entire field polydisc in their domain. Since $\lambda_V\le\bar\kappa-2$, their estimate gives (887). The same positive-height constant works at $k=0$: replacing $\varepsilon_{\mathrm A}$ by the larger radius r_P° only decreases that bound. The fourth-potential theorem supplies its holomorphy, dependence and vanishing, and (915) gives $\nu(\mathbb{V}'[\mathfrak{g}])\le N_{\mathfrak{a},\omega}(\mathfrak{g})$ when ν is defined with K. The ceiling (916) is precisely what checks $4\mathsf{K}N_*s_{\mathfrak{a}}\le w_*$. We can therefore apply theorem V with the actual data.

The normalizer bound is

$$|\mathbf{N}^{(k+1)}(X,P,z)|\le E_N e^{-\delta_N Md}.$$

Add this to (897) and (898). Both decay rates are at least $\kappa+1$, and $E_N\le E_0/2$, proving (921). The two final numerical conclusions follow by inserting (922); at $\tau=1$ the G term is zero. No additional estimate is needed for this rate change.

For the whole-torus assertion, the actual potential sums are the Jacobian, gauge and action remainders in proposition 52.2; the action identity uses the Hessian identified in section 52.3. The fourth sum is $s_W[S]$ by hypothesis. Hence (902) is exactly the integral in (908). By (790),

$$Z^{(k)}(W)=Z^{(k)}_{\mathrm{triv}}\exp\sum_{X,z}\mathbf{N}^{(k+1)}(X,P_W|_X,z).$$

This proves (924). Finally (910) gives $\sum\mathbf{G}_\tau=n_{\mathcal{O}}\operatorname{Log}\tau/2$; since $3\#\mathfrak{X}^+=n_\perp+n_{\mathcal{O}}$, it gives the equivalent expression. Absolute convergence is the one already proved for the actual pushforward and in theorem V. □

52.5. **Consistency on non-wrapping domains.** We require a common input array on different tori to have the same slot and coefficient numbers, and equal local terms after every zero-translation chart on a common bounded domain. These conditions concern the actual input functions, not only their norms. They imply that the fourth potentials agree under these charts, by their local substitution construction.

The core conditions also agree under a chart. Here is the geometric and analytic check. Cubes and their wall adjacency are preserved, so subdomains $Z \subset X$ correspond. In constructing a tower, test the distance from a cube to the complement by containment of its open R-neighborhood. All radii used are at most the halo width: at level $p \le t$ they are $M_1 L^{q+1}$ or $R_1 \mathfrak{M} L^q$, and in level-t units they are at most M. Thus the chart carries each stage of either tower to the corresponding stage. The two-layer interiors are carried along as well, since their radius is $5ML^{p-1}/2 \le ML^t$ in bare units. The reduced determining sets, their nodes, blocks and ordered comb contours therefore correspond.

The one-step averaging formulas then have identical matrix arguments, including their domain tests and their logarithm branches. Induction gives equality of every actual iterated average. Relabelling also preserves the action, the regularity inequalities and the residual gauge group. It consequently identifies the real critical orbits and their tower-axial representatives. A continuation on a polydisc is carried to a holomorphic continuation on the identical coordinate polydisc; the two agree at real points and hence everywhere by uniqueness. The radius is unchanged. This proves

$$\mathsf{i}_* \mathcal{U}_j^{\circ,\mathfrak{w}'}(X; b_0, b_1, c_0) = \mathcal{U}_j^{\circ,\mathfrak{w}''}(\mathsf{i}X; b_0, b_1, c_0), \tag{926}$$

including all the strict tower-current conditions. The same argument applies to the simple domains and gives a complex linear bijection between corresponding slices.

For the Gaussian data it is enough to compare parameters equal to zero or one. On each grain subregion Y, the preceding tower argument identifies the reduced rows, pivot bonds, ordered contours and inner bond spaces. Their finite differential and averaging matrices are relabelled by the chart. Bilinear transposes and inverses are therefore relabelled as well, so the local fluctuation forms agree. The chart bijects the subsets of a grain region and preserves their cardinalities. It consequently preserves every term and sign of the finite connected-coefficient sum. At a pair of bonds outside its indicated support both coefficients vanish; applying the inverse chart checks the support in both directions.

The same argument carries mesh strings and their figures. Their cutoff factors, local resolvents and commutators correspond, so the products correspond for every positive resolvent parameter. Absolute convergence permits integration with the inverse-square-root weight and summation over strings. The parameter monomial is unchanged. Compression, inversion and the product in (765) now give, for every parameter subset $\mathcal{K}$,

$$\begin{aligned} \mathbf{C}''(\mathsf{i}Z_\circ, \mathbf{1}_{\mathsf{i}\mathcal{K}}) &= \mathsf{i}_* \mathbf{C}'(Z_\circ, \mathbf{1}_{\mathcal{K}}), \\ \mathbf{\Gamma}''(\mathsf{i}Z_\circ, \mathbf{1}_{\mathsf{i}\mathcal{K}}) &= \mathsf{i}_* \mathbf{\Gamma}'(Z_\circ, \mathbf{1}_{\mathcal{K}}). \end{aligned} \tag{927}$$

The square-root branch is the same positive-real continuation. The actual block fields, gauge operators and potentials likewise agree, as do the normalizer terms by proposition 50.2.

We spell out why this suffices for the connected terms. The coordinate map on the positive bond fields is a real isometry and preserves Lebesgue measure. Under it the conditioning integrands have the same Gaussian determinant, quadratic pairings and cutoff tests. Their absolute integrals are therefore equal, and so are the integrals themselves. This includes the zero-dimensional core. The chart bijects the Mayer indices and the subsets used in the finite differences. It preserves incompatibility and hence the connected graph coefficients. Corresponding terms of the refined connected series agree, and the series are absolutely summable.

For the interaction insertion the same assertion holds at every interpolation parameter $v \in (0,1)$: scaling the fields and differentiating the indicated bond coordinate commute with relabelling. For

the cutoff insertion compare the functions of its varied bond threshold before differentiating. The deciding cutoff is the same on both tori; where it fails both integrands are zero, so a potential outside its domain is never evaluated. The common integrable majorants in theorem V permit the interpolation integrals. Finally $h_{\mathsf{i}z}(\mathrm{i}b) = h_z(b)$, and the nonzero tent summands lie in the halo. Thus

$$E^{(k+1),\mathfrak{w}''}(\mathrm{i}X, \mathsf{i}_* P, \mathsf{i}z) = E^{(k+1),\mathfrak{w}'}(X, P, z). \tag{928}$$

The same identity holds for each of its three summands.

Let now the sides of the chosen family $\mathfrak{W}$ be unbounded, with one common input array in the preceding sense. Every bounded domain $\widehat{X}$ embeds in some member. By (926) its core is the pullback of the core of any such member, independent of the embedding. Define

$$\widehat{E}^{(k+1)}(\widehat{X}, \hat{P}, \hat{z}) = E^{(k+1),\mathfrak{w}}(\mathrm{pr}^{\mathfrak{w}}\, \widehat{X}, \mathrm{pr}^{\mathfrak{w}}_*\, \hat{P}, \mathrm{pr}^{\mathfrak{w}}\, \hat{z}). \tag{929}$$

Equation (928) makes this well defined and unique. It is a pointed local family: on each finite domain its holomorphy, pair dependence and gauge invariance are those of one finite-dimensional evaluation. It has the same conjugation and translation properties when the input arrays do. These statements remain valid for holomorphic input families under uniform size bounds.

The size of a bounded domain is preserved by the chart. Indeed, the chart and its inverse map cubes and faces of each dimension bijectively. They therefore map the connected edge sets meeting every cube onto each other, preserve the number of edges, and also preserve the zero-edge case of a single common corner. Taking the least edge count gives equality of the two sizes. Therefore (921) passes to the infinite model. More precisely, put

$$\widehat{N}_{\mathfrak{a}}(\widehat{X}) = \inf_{\mathfrak{w}:\, \widehat{X} \text{ near for } \mathfrak{w}} N_{\mathfrak{a},\omega}(\mathfrak{g}^{\mathfrak{w}}).$$

This is the infimum of a nonempty set in $[0, N_*]$. The left side of the bound is independent of the selected embedding and the right side is affine increasing in this size. Taking the infimum proves

$$\begin{aligned} |\widehat{E}^{(k+1)}(\widehat{X}, \hat{P}, \hat{z})| \le \Big[\frac{E_0}{2} + K_{\mathbf{M}}(\mathsf{K})\big(|1-\tau| + \delta(\mathfrak{t}) + s_{\mathfrak{a}}\widehat{N}_{\mathfrak{a}}(\widehat{X}) \\ + e^{-\gamma_2 \mathfrak{b}(\mathfrak{t})^2/2}\big)\Big] e^{-(\kappa+1)d_{k+1}(\widehat{X})}. \end{aligned} \tag{930}$$

The required unbounded family exists at every height: take $K = k+1$ and let the integer m increase until $2L^{m+1} \ge C_L M$, then continue increasing it.

52.6. **The quadratic coefficient of the local output.** We extract the quadratic field-strength term from the output of a step. The estimate must be uniform in the level, whereas the neighbourhood on which a finite-volume critical representative is defined may depend on the volume. We first distinguish these two requirements.

Measure lengths at level $j \ge 1$, so that the coarse lattice has spacing one and the fine lattice has spacing $\xi = L^{-j}$. Write D for the fine curl, $d = \nabla^{\xi}$ for the site gradient, and $\Delta = d^* d$. The straight averages are

$$(Qh)(c) = \xi L^{-3j} \sum_{x \in B^j(c_-)} \sum_{b \in [x, c_+ - c_- + x]} h(b), \qquad (Q'\lambda)(y) = L^{-3j} \sum_{x \in B^j(y)} \lambda(x).$$

Fine pairings have weight ξ^3; coarse pairings have weight one. Thus $Qd = \partial Q'$, with ∂ the coarse site gradient. Let H_N be the free Landau response on the torus of side $N = 2L^a$. Its column $h^N_{c,Y}$, together with its multiplier $\omega^N_{c,Y}$, solves

$$D^{\mathsf{T}} D h = Q^{\mathsf{T}} \omega, \qquad Qh = Y\delta_c, \qquad (d^* h, \Delta\lambda)_{\xi} = 0 \tag{931}$$

for every λ supported in one fine block and having zero sum there. Choose a fixed lower bound N_{H} for the side so that the constant-sequence geometry of lemma 51.6 is admitted. Increase it to meet the torus condition in theorem E and $N_{\mathsf{H}}\delta_{\mathsf{H}} \ge 1$. It depends only on the fixed geometric

constants, not on j. The corresponding infinite-lattice response is denoted by $\mathsf{H}^{[j]}$. Below we prove its existence, periodization, and the uniform bound

$$|\mathfrak{k}^c_\mu(x)|,\quad |\nabla^\xi_\nu \mathfrak{k}^c_\mu(x)|,\quad |(D^{\mathsf{T}} D\mathfrak{k}^c)_\mu(x)| \le C_{\mathsf{H}} e^{-\delta_{\mathsf{H}}|x-c_-|_\infty},\qquad (\mathsf{H}^{[j]}B)(b) = \sum_{c>0} \mathfrak{k}^c(b)B(c). \tag{932}$$

The kernel is real, and $C_{\mathsf{H}} \ge 1$ and $0 < \delta_{\mathsf{H}} \le 1$ are common to every level.

For a localization domain X, let $\mathfrak{V}(X)$ be the space of independent pairs $v = (A, \mathbf{J})$ on its incident fine bonds, with norm

$$\|v\|_X = \max\{\sup|A|, \sup|\nabla^\xi A|, \sup|\mathbf{J}|\}.$$

Only differences defined on those bonds enter the maximum. The trivial-pair chart is $P[v] = (I, A, \mathbf{J})$. The core estimate proposition 51.9 gives a uniform usable radius

$$r_0 = a^{\mathrm{us}}(b_0, b_1) := \min\{b_1, a^\sharp/4, b_0/(4K^\sharp_{\mathrm{tw}})\},\qquad P[\mathfrak{V}_{r_0}(X)] \subset \mathcal{U}^\circ_j(X; b_0, b_1). \tag{933}$$

We verify this inclusion in the proof. In particular the reading $\mathsf{e}_{X,z}(v) = E(X, P[v], z)$ of a local term is holomorphic on this ball and is invariant whenever two points of the ball represent gauge-equivalent pairs. Primes on a reading denote derivatives at $v = 0$.

Fix $M = L^{m'}$, $m' \ge 6$, and an unbounded family $\mathfrak{W}$ of tori carrying level j, all with level-j side $N \ge N_{\mathsf{H}}$. At a common height $h \ge j-1$, require $K \ge j$ and the room and near-domain chart hypotheses of eq. (926); the constant sequences must also satisfy the hypotheses of lemma 51.6 and theorem E, as fixed in the choice of N_{H}. Use the background constants fixed for the cores, in particular $0 < \varepsilon_{\mathrm{v}} \le a_{\mathrm{orb}}$. Let $E^{\mathfrak{w}}$ be pointed local families on these tori and $\widehat{E}$ their companion on the infinite model, at the same positive core radii b_0, b_1. Suppose that

$$|E^{\mathfrak{w}}(X, P, z)| \le \mathcal{A} e^{-\kappa d_j(X)},\qquad |\widehat{E}(\widehat{X}, P, \hat{z})| \le \mathcal{A} e^{-\kappa d_j(\widehat{X})} \tag{934}$$

on their full cores. The hypotheses for the core and chart comparison in proposition 51.9 and eq. (926) are understood, with the same constants at every height. Take $\kappa \ge \kappa_{\mathrm{am}} := 4\log 9216 + 9$. For a real block field W, define the total, whenever its minimal background orbit exists and all displayed pairs lie in the term domains, by

$$\mathcal{P}[E^{\mathfrak{w}}](W) = \sum_z \sum_{X \ni z} E^{\mathfrak{w}}(X, (U_j(W), J^{(j)}(U_j(W)))|_{\mathrm{Bd}(X)}, z). \tag{935}$$

Here $U_j(W)$ denotes the canonical axial representative, and a local term is read as a function of its pair. For the near-trivial totals used below, the usable-ball argument also proves that all representatives are in the term domains, and hence proves representative-independent definedness. For B of polynomial growth put

$$\mathfrak{h}_X B = ((\mathsf{H}^{[j]}B)|_{\mathrm{Bd}(X)}, (D^{\mathsf{T}} D\mathsf{H}^{[j]}B)|_{\mathrm{Bd}(X)}).$$

With t_1, t_2, t_3 an orthonormal Pauli basis, define

$$\Pi(c, c') = \frac{1}{3}\sum_{a=1}^{3} \sum_{\hat{z} \in \mathbb{Z}^3} \sum_{\widehat{X} \ni \hat{z}} \hat{\mathsf{e}}''_{\widehat{X},\hat{z}}[\mathfrak{h}_{\widehat{X}}(t_a\delta_c), \mathfrak{h}_{\widehat{X}}(t_a\delta_{c'})]. \tag{936}$$

All bond sums in this subsection use positive orientations; the kernel extends oddly in either bond.

Theorem X. Under these hypotheses the series (936) is absolutely convergent. Put $\delta_{\mathrm{am}} = \frac{1}{2}\min(\delta_{\mathsf{H}}, \kappa/M)$. There is C_{am}, depending only on $C_{\mathsf{H}}, \delta_{\mathsf{H}}, \kappa, M$, such that

$$|\Pi(c, c')| \le C_{\mathrm{am}} r_0^{-2} \mathcal{A} e^{-\delta_{\mathrm{am}}|c'_- - c_-|_\infty},\qquad \sum_{c'>0} |\Pi(c, c')|(1 + |c'_- - c_-|_\infty)^2 \le C_{\mathrm{am}} r_0^{-2}\mathcal{A}. \tag{937}$$

On every finite torus $\mathfrak{w} \in \mathfrak{W}$, the total $\mathcal{P}[E^{\mathfrak{w}}]$ is defined and analytic near the trivial coarse field. For fixed infinite-lattice bonds c, c',

$$\lim_{\mathfrak{w}\in\mathfrak{W},\ N\to\infty} D^2[\mathcal{P}[E^{\mathfrak{w}}](e^{iB})]_{B=0}[t_a\delta_{\mathrm{pr}_j c}, t_{a'}\delta_{\mathrm{pr}_j c'}] = \delta_{aa'}\Pi(c, c'). \tag{938}$$

The neighbourhood in this assertion need not be uniform in $\mathfrak{w}$.

If the finite-volume totals are invariant under lattice isometries near the trivial field, then $\Pi(c,c') = \Pi_{\mu\nu}(u)$, where $u = c'_- - c_-$, and its moments of orders zero and one vanish. Its second moments have the form

$$\sum_{u\in\mathbb{Z}^3} \Pi_{\mu\nu}(u) u_\rho u_\sigma = \mathsf{cf}(\widehat{E})(\delta_{\mu\rho}\delta_{\nu\sigma} + \delta_{\mu\sigma}\delta_{\nu\rho} - 2\delta_{\mu\nu}\delta_{\rho\sigma}), \tag{939}$$

where

$$\mathsf{cf}(\widehat{E}) = \sum_u \Pi_{\mu\nu}(u) u_\mu u_\nu \quad (\mu \neq \nu), \qquad |\mathsf{cf}(\widehat{E})| \leq C_{\mathrm{am}} r_0^{-2} \mathcal{A}. \tag{940}$$

All these moment series converge absolutely. The same coefficient is obtained from any finite-volume functionals equal to the totals up to constants near the trivial field. If those functionals are real up to constants, the kernel and the coefficient are real. ■□

The proof separates the uniform linear response from the two summations in (936). Gauge invariance removes the second derivative of the background and leaves only its linear response. Exponential decay then controls both the large-domain tail and the periodic images.

The free response. We give the infinite-volume argument, retaining only the three bounds in (932). The finite response is the derivative at zero in lemma 51.6. At the trivial field the nonlinear constraint remainder and action remainder have zero derivative. Differentiating their equations gives (931); the Landau normalization gives its last equation. The estimate (838), and the multiplier estimate in its proof, give uniform exponential bounds for $h, \nabla^\xi h, D^{\mathsf{T}} D h, \omega$. Here every layer site is at the single coarse scale, and the distance is the coarse ℓ^1 distance. The base point of a fine bond is within $1/2$ in sup distance of its block centre. Decreasing the decay rate and increasing the bound therefore gives (932), with constants independent of j.

For clarity, uniqueness uses the local gauge condition, not a choice of a limiting sequence. For a real homogeneous solution, pairing the first equation with h gives $\|Dh\|_\xi^2 = 0$. On a torus the last equation is equivalent to $Rd^*h = 0$, since the blockwise mean-zero functions span $\ker Q'$. Thus h lies in the kernel of the positive free operator $D^{\mathsf{T}}D + dRd^* + Q^{\mathsf{T}}\mathsf{a}Q$, with the positive row weight of proposition 21.1, and is zero. Also $Q^{\mathsf{T}}\omega = 0$ implies $\omega = 0$: at the fine bond crossing the central outgoing face of $B^j(c_-)$ in the direction of c, only the row c contributes, and its weighted transpose coefficient is one. Complex uniqueness follows by separating real and imaginary parts.

On the infinite lattice consider an exponentially decaying homogeneous solution. The same pairing gives $Dh = 0$. Set

$$\phi(x) = -\xi \sum_{n\geq 0} h_1(x + n\xi e_1).$$

Telescoping and $Dh = 0$ give $d\phi = h$. The full line sum $\xi \sum_{n\in\mathbb{Z}} h_1(x + n\xi e_1)$ is constant in every coordinate, again by telescoping, and is zero by transverse decay. Thus $\phi(x) = \xi \sum_{n<0} h_1(x + n\xi e_1)$ as well. Using the first expression for $x_1 \geq 0$ and the second for $x_1 < 0$ shows that ϕ decays exponentially. Since $\partial Q'\phi = Qd\phi = 0$, decay gives $Q'\phi = 0$. Decompose ϕ into its restrictions to the fine blocks. Each restriction is an allowed test function in (931); absolute summability permits their sum. Consequently

$$\|\Delta\phi\|_\xi^2 = \sum_y (d^*h, \Delta(\phi \mathbf{1}_{B^j(y)}))_\xi = 0, \qquad \|d\phi\|_\xi^2 = (\phi, \Delta\phi)_\xi = 0.$$

Therefore $h = 0$, and the same crossing-face argument gives $\omega = 0$.

Lift finite-torus columns to the infinite lattice and let $N \to \infty$. Their values at the countably many bonds and rows are uniformly bounded, so a diagonal subsequence converges pointwise. Every equation and every blockwise gauge test involves finitely many values and passes to the limit. At a fixed row the periodic source is eventually $Y\delta_c$. The uniform exponential bounds pass to the

limit too. The uniqueness just proved identifies all subsequential limits. Real linearity, translation covariance and the fact that the coefficients are scalar in colour give a real kernel $\mathfrak{k}^c$.

Absolute convergence allows periodization through the local operators and through a blockwise gauge test. The periodized column solves the finite-torus system, so uniqueness gives

$$h^N_{\mathrm{pr}_j c,Y}(\mathrm{pr}_j b) = \sum_{n\in N\mathbb{Z}^3} \mathfrak{k}^c(b+n)Y. \tag{941}$$

This holds after applying ∇^ξ or $D^\mathsf{T}D$. The sum over nonzero images is bounded by a convergent geometric shell sum. In particular finite columns satisfy the counterpart of (932) with periodic distance and a fixed larger constant, and converge locally to the infinite columns.

We shall also use the gradient property of this response. If a coarse site function λ has finite support, let $\lambda^\flat$ be constant with value $\lambda(y)$ on $B^j(y)$. Then $Qd\lambda^\flat = \partial\lambda$. For $E = \mathsf{H}^{[j]}\partial\lambda - d\lambda^\flat$ one has $QE = 0$ and $D^\mathsf{T}DE = Q^\mathsf{T}\omega$. Pairing gives $DE = 0$. For polynomially growing λ, regroup the absolutely convergent kernel sum by coarse sites: the coefficient at y is the response to $\partial\delta_y$ and is curl-free by the finite-support case. Applying the four-term curl to the sum proves

$$D\mathsf{H}^{[j]}\partial\lambda = 0, \qquad Q\mathsf{H}^{[j]}B = B. \tag{942}$$

The second identity follows first for a single column and then by absolute summation. A curl-free form on the infinite cubic lattice has a primitive: sum along the three coordinate segments, and use each plaquette relation to interchange adjacent coordinate steps. Thus $\mathsf{H}^{[j]}\partial\lambda = d\theta$, and its $D^\mathsf{T}D$ image vanishes. No decay of this primitive is needed for a polynomial λ.

Local derivatives and summation.

Proof of theorem X. We first verify (933). Write $r = r_0$. The fixed constants have $K^\sharp_{\mathrm{tw}} \geq 1$ and $a^\sharp \leq 1/60$, so $r \leq 1/240$. If $\|v\|_X < r$, the flat real reference admits the identity gauge, and the potential, its first differences and the independent current have size less than r. The only additional simple-domain condition is the plaquette bound. With the four positive link potentials of a plaquette written as $Y_i = i\xi A(b_i)$, one has

$$|Y_i| \leq s := \xi r, \qquad |Y_3 - Y_1|,\ |Y_4 - Y_2| \leq \xi^2 r.$$

Subtract $e^{-Y_1}e^{-Y_2}$ from $e^{-Y_3}e^{-Y_4}$, and commute the two remaining factors. The two difference terms are bounded by $e^{2s}\xi^2 r$; the commutator is bounded by $2e^{2s}\xi^2 r^2$. After multiplication by $e^{Y_1}e^{Y_2}$, this gives

$$|\partial(e^{i\xi A})(p) - I| \leq 2e^{4s}(1+r)r\xi^2 < 3r\xi^2.$$

Thus $P[v] \in \mathcal{Q}_j(X; 3r, r, b_0)$. The inequalities

$$3r \leq b_0, \qquad r \leq b_1, \qquad K^\sharp_{\mathrm{tw}}(3r + r^2) \leq 4K^\sharp_{\mathrm{tw}} r \leq b_0, \qquad 4r \leq a^\sharp$$

give (933) by (846). For an infinite domain use any sufficiently large near lift. The chart comparison eq. (926) preserves these conditions and the norm in both directions.

The same core inclusion holds for a unitary gauge transform of such a triple. Indeed, all curvature, current and covariant difference bounds defining $\mathcal{Q}_j(X; 3r, r, b_0)$ are unchanged by unitary conjugation. A cube gauge w for the original real factor is replaced by wu^{-1}, with the same potential and bounds, for its transform by u. Thus the transformed triple still lies in this simple domain, and (846) puts it in the same core. Whenever $P[v]^u$ and $P[w]$ represent the same pair, with v, w in the usable ball, gauge invariance and pair dependence of the local term therefore give $e(w) = e(v)$.

For each reading $e = \mathsf{e}_{X,z}$, Cauchy's inequality on the bidisc $|s_i| < r_0/(2\|v_i\|_X)$ gives

$$|e''[v_1, v_2]| \leq 4r_0^{-2}\mathcal{A}e^{-\kappa d_j(X)}\|v_1\|_X\|v_2\|_X. \tag{943}$$

Constant gauge rotations preserve this ball. Their invariance implies $e' = 0$, and makes e'' scalar in colour: rotations by π about the Pauli axes kill every invariant linear form and all off-diagonal

entries of an invariant bilinear form; rotations permuting the axes make its diagonal entries equal. Consequently

$$e''[Yf, Y'f'] = \langle Y, Y'\rangle e''[t_1 f, t_1 f']$$

for scalar vectors f, f'.

The Hessian also kills local gauge directions. For a real site function θ on the endpoints of the incident bonds, set

$$u_t = e^{it\theta}, \qquad w(t,s) = \left(\frac{1}{i\xi}\log(u_t(b_-)e^{i\xi s A(b)}u_t(b_+)^{-1}),\ s\,\mathrm{Ad}(u_t(b_-))\mathbf{J}(b)\right).$$

There are finitely many bonds, so the series logarithms and both points $w(t,s), sv$ lie in the ball for sufficiently small s, t. Gauge invariance gives $e(w(t,s)) = e(sv)$. At zero, $\partial_s w = v$ and $\partial_t w = -(d\theta, 0)$. The mixed chain-rule term involving e' vanishes. Thus

$$e''[(d\theta, 0), v] = 0. \tag{944}$$

Splitting a complex θ into its real and imaginary parts extends the identity complex linearly.

Let $\rho_w(X) = \inf_{x\in X}|w - x|_\infty$, with periodic distance on a torus. A base point of an incident bond lies within ξ of X. The three response estimates therefore give

$$\|\mathfrak{h}_X(Y\delta_c)\|_X \le e^{1/13}C_{\mathsf{H}}e^{-\delta_{\mathsf{H}}\rho_{c_-}(X)}|Y|.$$

Periodization gives the same estimate on a torus with a fixed larger constant. The shell count $\#\{u \in \mathbb{Z}^3 : |u|_\infty \le n\} = (2n+1)^3$ shows that the response series for a polynomially growing input converges in every local norm. It may be differentiated by the finite-difference operators and inserted in any local bilinear form term by term.

Here are the domain sums needed for both the kernel and its limit. An edge set realizing $d_j(X)$, together with a segment in each endpoint cube, gives $\mathrm{diam}_\infty X \le M(d_j(X)+2)$. The walk count in proposition 45.1 is unchanged on the infinite lattice. A site belongs to at most eight cubes; hence $\kappa/4 \ge \log 9216 + 9/4$ implies

$$\sum_{\substack{X\ni z\\ d_j(X)\ge D}} e^{-\kappa d_j(X)/4} \le C_{\mathrm{E},0}e^{-D}.$$

For real $D \ge 0$, sum the geometric series starting at $\lceil D\rceil$. These bounds also hold for wrapping domains on the torus. Put

$$T_{w,w'}(X) = e^{-\kappa d_j(X)}e^{-\delta_{\mathsf{H}}(\rho_w(X)+\rho_{w'}(X))}, \qquad S_p(a) = \sum_{u\in\mathbb{Z}^3}(1+|u|_\infty)^p e^{-a|u|_\infty}.$$

All $S_p(a)$ used below are finite by the shell count. For $z \in X$, the diameter bound gives

$$|w - z|_\infty \le \rho_w(X) + M(d_j(X)+2), \qquad |w - w'|_\infty \le \rho_w(X) + \rho_{w'}(X) + M(d_j(X)+2).$$

Since $\delta_{\mathrm{am}} \le \delta_{\mathsf{H}}/2$ and $M\delta_{\mathrm{am}} \le \kappa/2$, distributing these two inequalities in the exponential yields

$$T_{w,w'}(X) \le e^{\kappa}e^{-\delta_{\mathrm{am}}(|w-z|_\infty+|w'-z|_\infty)/2}e^{-\kappa d_j(X)/2}, \tag{945}$$

$$T_{w,w'}(X) \le e^{3\kappa/2}e^{-\delta_{\mathrm{am}}|w-w'|_\infty}e^{-\delta_{\mathrm{am}}|z-w|_\infty/2}e^{-\kappa d_j(X)/4}. \tag{946}$$

The first inequality gives a product of two summable endpoint factors after summation over $X \ni z$. The second gives

$$\sum_z \sum_{\substack{X\ni z\\ d_j(X)\ge D}} T_{w,w'}(X) \le Ce^{-\delta_{\mathrm{am}}|w-w'|_\infty}e^{-D}. \tag{947}$$

Here and below C depends only on $C_{\mathsf{H}}, \delta_{\mathsf{H}}, \kappa, M$. For the torus, choose one nearest lift of each site; its exponential sum is bounded by the corresponding S_0.

Combining (943) with these estimates proves absolute convergence of (936) and its exponential bound. Summing the bound over the three directions at $c'_- = c_- + u$, with weight $(1+|u|_\infty)^2$, proves (937) after enlarging C_{am}. The same calculation with any polynomial weight is finite. For

a fixed base point z, the two endpoint factors in (945) also justify all double response sums with polynomially growing inputs.

The finite-volume Hessian and its limit. Apply theorem E to the constant full-torus sequence and trivial datum. There are no lower-layer rows or inward rows. The auxiliary geometric parameters were fixed before N_{H}. Choose the datum bound and then the row ball small enough that the regular orbit lies in the selected ε_{v} class. Its axial representative has a holomorphic continuation $\Phi_I(B)$ near zero, with $\Phi_I(0) = I$. All bonds are inner. On a smaller finite-volume neighbourhood set

$$A(B) = \frac{1}{i\xi} \log \Phi_I(B), \qquad \mathbf{J}(B) = J(\Phi_I(B)).$$

These maps are holomorphic and vanish at zero. There are only finitely many domains and bonds, so their restrictions $w_X(B) = (A(B), \mathbf{J}(B))|_{\mathrm{Bd}(X)}$ lie in $\mathfrak{V}_{r_0}(X)$ on one common neighbourhood. The background belongs to the actual minimal orbit; independence of its representative therefore identifies the total with

$$\Psi_N(B) = \sum_z \sum_{X \ni z} \mathsf{e}^N_{X,z}(w_X(B)).$$

This finite sum proves the asserted analyticity. No uniform lower bound on this neighbourhood is needed.

The critical-branch calculation in section 43, converted from absorbed links to the present potential units, gives

$$A'(0)(Y\delta_c) = h^N_{c,Y} - d\theta_{c,Y}, \qquad \mathbf{J}'(0)(Y\delta_c) = D^{\mathsf{T}} D h^N_{c,Y}.$$

That calculation already includes the endpoint correction between the original and straightened means. The vanishing of e' removes the second derivative of w_X, and (944) removes $d\theta_{c,Y}$. It follows that

$$D^2\Psi_N(0)[Y\delta_c, Y'\delta_{c'}] = \sum_z \sum_{X \ni z} (\mathsf{e}^N_{X,z})''[\mathfrak{v}_X h^N_{c,Y}, \mathfrak{v}_X h^N_{c',Y'}], \tag{948}$$

where $\mathfrak{v}_X h = (h, D^{\mathsf{T}} D h)|_{\mathrm{Bd}(X)}$. This also holds for wrapping X: $\theta_{c,Y}$ is a site function on the entire torus.

For a near domain $\widehat{X}$ projecting to X, the chart identity for the companion holds throughout the usable ball and preserves every norm stencil. Differentiating it twice identifies the two local Hessians. We compare their sums without dropping the wrapping terms.

Fix c, c' on the infinite lattice, write $u = c'_- - c_-$, and take

$$\mathrm{FD}_N = \{z \in \mathbb{Z}^3 : -N/2 \le (z - c_-)_\mu < N/2,\ \mu = 1, 2, 3\}.$$

Projection maps this set bijectively onto the coarse torus. A non-wrapping pointed domain has exactly one near lift with its point in FD_N. A far or wrapping domain has $d_j(X) \ge D_N := \max\{0, N/(2M) - 2\}$, by the diameter estimate. On each near lift, periodization gives the absolutely convergent bilinear expansion

$$(\mathsf{e}^N_{X,z})''[\mathfrak{v}_X h^N_{\mathrm{pr}_j c,Y}, \mathfrak{v}_X h^N_{\mathrm{pr}_j c',Y'}] = \sum_{n,n' \in N\mathbb{Z}^3} \hat{\mathsf{e}}''_{\widehat{X},\hat{z}}[\mathfrak{h}_{\widehat{X}}(Y\delta_{c-n}), \mathfrak{h}_{\widehat{X}}(Y'\delta_{c'-n'})].$$

The terms $n = n' = 0$ give the infinite sum restricted to near domains and base points in FD_N. There are four errors: the wrapping part of (948), the nonzero image pairs, the infinite base points outside FD_N, and the far infinite domains with base point inside it. By (947), the first and fourth errors are at most $C r_0^{-2} \mathcal{A} |Y||Y'| e^{-D_N}$.

For the other two, put

$$t_N(c'') = \sum_{z \notin \mathrm{FD}_N} e^{-\delta_{\mathrm{am}} |c''_- - z|_\infty / 2}.$$

Splitting the exponential into two equal factors gives

$$t_N(c) \le S_0(\delta_{\mathrm{am}}/4) e^{-\delta_{\mathrm{am}} N/8}, \quad t_N(c') \le S_0(\delta_{\mathrm{am}}/4) e^{-\delta_{\mathrm{am}}(N/2 - |u|_\infty)/4}.$$

The outside-base-point error is at most $Cr_0^{-2}\mathcal{A}|Y||Y'|t_N(c)$, by (945). For a nonzero image pair, either $n \neq 0$ or $n' \neq 0$. The map $(z,n) \mapsto z+n$ is a bijection from $\mathrm{FD}_N \times (N\mathbb{Z}^3 \setminus \{0\})$ onto $\mathbb{Z}^3 \setminus \mathrm{FD}_N$. Sum that endpoint factor first; the other is bounded by $S_0(\delta_{\mathrm{am}}/2)$. Thus the image-pair error is at most

$$Cr_0^{-2}\mathcal{A}|Y||Y'|S_0(\delta_{\mathrm{am}}/2)(t_N(c)+t_N(c')).$$

All four errors tend to zero for fixed c, c'. The colour identity proved above identifies the limiting form with $\langle Y, Y'\rangle \Pi(c,c')$, proving (938).

Gauge cancellation and the moment tensor. Suppose the finite totals are invariant under lattice isometries. Their Hessians then satisfy the corresponding signed bond identity. Pass to the limit in (938). Reversing either bond changes its sign, so the identity holds for all oriented bonds. Translations give $\Pi(c,c') = \Pi_{\mu\nu}(c'_- - c_-)$.

For a polynomially growing coarse site function χ, (942) makes $\mathsf{H}^{[j]}\partial\chi$ a gradient, with zero $D^{\mathsf{T}}D$ image. Every local Hessian annihilates this vector. The summability just proved permits the response and domain sums to be interchanged, giving the Ward identity

$$\sum_{\nu,u}[\chi(u+e_\nu)-\chi(u)]\Pi_{\mu\nu}(u) = 0. \tag{949}$$

Taking $\chi(u) = u_\nu$ proves the vanishing of every zeroth moment.

To handle reflections, use the difference of bond midpoints, $\tilde{u} = u + (e_\nu - e_\mu)/2$. If $re_\alpha = \varsigma_\alpha e_{\pi(\alpha)}$, the signed bond law is

$$\Pi_{\mu\nu}(u) = \varsigma_\mu\varsigma_\nu\Pi_{\pi(\mu)\pi(\nu)}(u'), \qquad u' + \tfrac{1}{2}(e_{\pi(\nu)} - e_{\pi(\mu)}) = r\tilde{u}. \tag{950}$$

Indeed the positive representative of $r\langle x, x+e_\mu\rangle$ starts at $r(x+e_\mu/2) - e_{\pi(\mu)}/2$. This also shows that $u \mapsto u'$ is an integer bijection. For $r = -I$ its midpoint argument is $-\tilde{u}$, while the two orientation signs multiply to one. Hence the first midpoint moment vanishes. The zeroth moment already vanishes, so the first moment in u vanishes too. For the same reason the second moments in u and $\tilde{u}$ agree.

Write these second moments as $P_{\mu\nu,\rho\sigma}$. Reflecting any one axis in (950) shows that a component vanishes unless each index value occurs an even number of times. Coordinate permutations leave just four possible values:

$$P_{\mu\mu,\mu\mu} = q_0, \quad P_{\mu\mu,\rho\rho} = q_1, \quad P_{\mu\nu,\mu\nu} = q_2, \quad P_{\mu\nu,\nu\mu} = q_3 \qquad (\mu \neq \rho,\ \mu \neq \nu).$$

Symmetry of the two moment indices gives $q_3 = q_2$. Now take $\chi(u) = u_\alpha u_\beta u_\gamma$ in (949). Its terms of degrees zero and one vanish after summation, leaving

$$P_{\mu\alpha,\beta\gamma} + P_{\mu\beta,\alpha\gamma} + P_{\mu\gamma,\alpha\beta} = 0. \tag{951}$$

With all four indices equal this gives $q_0 = 0$; with $\alpha = \mu$, $\beta = \gamma = \rho \neq \mu$, it gives $q_1 + 2q_2 = 0$. Thus q_2 is the coefficient in (940), independent of the distinct pair μ, ν, and (939) follows. Its bound is a part of (937). Finally, adding constants to the finite totals does not change their Hessians. If they are real up to constants, those Hessians are real; (938) and the absolutely convergent moment sum give reality of the kernel and coefficient. □

52.7. **The global domain of the step.** The small-field expansion is an actual step on a uniformly small class of block fields. We verify this before iterating it. The verification also identifies the physical backgrounds in its local potentials and in the quadratic subtraction.

Retain the common geometric hypotheses of theorems E and U, lemma 51.6, and propositions 50.2 and 51.5, at every height under consideration. In particular, choose $B_3 \geq B_{\mathrm{rad}}$, $\rho_{\mathrm{s}} \geq \rho_{\mathrm{G}}$, $\mathfrak{M} \geq \mathfrak{M}_{\mathrm{G}}$, $R_1\mathfrak{M} \geq L^2\rho_{\mathrm{s}}$, and

$$\varepsilon_{\mathrm{v}} + \varepsilon_{\mathrm{v}}^2 \leq a_{\mathrm{orb}}, \qquad \bar{\varepsilon} \leq \varepsilon_{\mathrm{G}}(\rho_{\mathrm{s}}), \qquad r_{\mathrm{B}} \leq r_{\mathrm{G}}(\rho_{\mathrm{s}}), \qquad B_3(\bar{\varepsilon} + C_\partial r_{\mathrm{B}}) \leq \varepsilon_{\mathrm{v}}.$$

The level lattices have the size required by these results. We use the same unbounded family of admitted tori as in theorem X, including its floor N_{H}. Increase the room constant to $C_L \geq 2\mathsf{o}L^2+13$. Write $c_{\mathrm{rg}}, B_{\mathrm{rg}}$ for the constants of theorem F, put $l_M = \mathsf{o}LM+2$, and choose

$$B^{(\mathrm{cg})} \geq 2B_{\mathrm{rg}}\left(1+\frac{2}{\mathsf{o}L^2}\right). \tag{952}$$

All these choices precede the height and the volume.

For a real critical configuration at level j, set

$$\mathsf{m}_j(U) = L^{2j}\max_p |\partial U(p) - I|.$$

For $1 \leq j \leq K$ and $0 < e \leq 1$, let $\mathcal{C}_j(b_0, b_1; e)$ be the set of block fields whose minimal regular orbit exists, has $\mathsf{m}_j < e$, and whose pairs on every domain lie in the cores with radii (b_0, b_1). The assertion about pairs is required for every representative. We use $\mathcal{W}_{k+1}(\mathsf{t}; b_0, b_1)$ for the global domain of the actual step: its averaged axial background and the whole cutoff chart are defined; the lower backgrounds over that chart exist; the whole background belongs to the shift domain; the octet restrictions needed in proposition 51.14 have their minimal orbits; and every plain triple of the top background on a domain belongs to the core with radii (b_0, b_1). These are the actual domain requirements of the preceding construction, including the off-fibre response neighbourhood in proposition 52.2.

Proposition 52.3. There is $\varepsilon_{\mathcal{W}} > 0$, independent of height and volume, with the following properties. If $\mathsf{t} > 1$ is real, $\varepsilon(\mathsf{t}) \leq 1$, $0 < b_0, b_1 \leq 1$, $\delta(\mathsf{t}) \leq \varepsilon_{\mathcal{W}}$, and $\varpi(W) < \varepsilon_{\mathcal{W}} b_0$, then $W \in \mathcal{W}_{k+1}(\mathsf{t}; b_0, b_1)$. Its minimal orbit is contained in $\mathfrak{U}'_{k+1}(B_3\varepsilon_{\mathcal{W}} b_0)$. The averaged axial field $V = V_k^{\mathrm{bf}}(W)$ is independent of its construction, depends continuously on W, and satisfies

$$\mathcal{M}V = W, \qquad \varpi(V) \leq 2B_3 L^{-2}\varepsilon_{\mathcal{W}} b_0. \tag{953}$$

For a real cutoff field B, its chart is $V_W(B) = e^{iB'}V$, where

$$\sup|B'| < 2C_{\mathrm{db}}\delta(\mathsf{t}), \qquad \mathcal{M}V_W(B) = W, \qquad \varpi(V_W(B)) < K_2\varepsilon_{\mathcal{W}}, \quad K_2 = 2B_3L^{-2} + 8C_{\mathrm{db}}. \tag{954}$$

If $2b_0 \leq \varepsilon_{\mathcal{W}}$, then

$$\begin{aligned}\mathcal{C}_j(b_0, b_1; e) = \{W : U_j(W) \text{ exists, and its orbit has} \\ \mathsf{m}_j < \min(e, b_0), \quad |J^{(j)}| < b_0\}.\end{aligned} \tag{955}$$

On this set every plain triple belongs to its core. The set is invariant under every isometry of the level-j lattice, and

$$\mathcal{C}_{k+1}(b_0, b_1; e) \subset \mathcal{W}_{k+1}(\mathsf{t}; b_0, b_1).$$

It contains the open neighbourhood of the identity

$$\{W : \varpi(W) < \min\{\bar{\varepsilon}, e/B_3, \varepsilon_{\mathcal{W}} b_0\}\}. \tag{956}$$

■

Proof. Choose $\varepsilon_{\mathcal{W}} \leq \varepsilon_{\mathfrak{h}}$ so small that

$$2L^2 B_3 K_2 \varepsilon_{\mathcal{W}} \leq \bar{\varepsilon}, \qquad L^2 B_3 K_2 \varepsilon_{\mathcal{W}} \leq \varepsilon_{\flat}, \qquad 2B_3 L^{-2}\varepsilon_{\mathcal{W}} \leq \varpi_{\mathrm{db}},$$

$$C_{\mathrm{db}}\varepsilon_{\mathcal{W}} \leq r_{\mathrm{db}}, \qquad l_M B_3 \varepsilon_{\mathcal{W}} \leq c_{\mathrm{rg}}, \qquad B_{\mathrm{rg}} l_M B_3 \varepsilon_{\mathcal{W}} \leq \mathsf{o}LMB^{(\mathrm{cg})} a_{\mathfrak{h}}, \qquad B_3\varepsilon_{\mathcal{W}} \leq a_{\mathfrak{h}} \leq 1.$$

Here $a_{\mathfrak{h}}, \varepsilon_{\mathfrak{h}}$ are the fixed radii in the actual shift construction. Reducing the first constant also imposes all preceding curvature ceilings. These are finitely many positive bounds depending only on the fixed geometric constants.

Theorem E constructs the whole minimal orbit and gives the claimed regularity. By proposition 14.1, its level-k average has curvature at most $2B_3L^{-2}\varepsilon_{\mathcal{W}} b_0$. The contour estimates of lemmas 14.4 and 14.5 therefore put every contour family in its unique-mean domain. For $x \in B(y)$, set $u(x) = (\mathcal{M}^k U)(y, x)$, and set $u = I$ at the other bare sites. Since $u(y) = I$, this is a residual

gauge for the top datum. It makes all these contour variables equal to I. Two representatives so normalized have the same averaged field: their residual gauge is determined by these contour variables and its value I at the block centres. This proves (953) and independence. Restriction of criticality in lemma 51.13, followed by the uniqueness in theorem E, identifies the orbit over V with this normalized subset of the top orbit.

The representative supplied by theorem E varies holomorphically in logarithmic row coordinates near every datum under consideration. The real averages and their contour means are continuous there: a convergent sequence of means has, by compactness of the group, a convergent subsequence, and its limit is the unique mean of the limiting family. Applying the preceding axial gauge proves continuity of V_k^{bf} on the indicated open curvature set.

The pivot chart estimates used in proposition 52.2 now apply to V. Since $\sup|gB| < \delta(\mathsf{t})$, they give the first two conclusions of (954). The value of the chart does not depend on which permitted ball is used: two holomorphic representatives of its correction have the same germ at zero and hence agree on the smaller ball. At spacing one, $|\nabla B'| \le 2\sup|B'|$. The real plaquette product estimate gives

$$|\partial(e^{iB'}V) - I| \le |\partial V - I| + 4\sup|B'|,$$

which proves the last bound in (954). Thus theorem E also constructs every lower background along the cutoff chart. At $k = 0$ the lower configuration is the chart field itself; the current formula gives $|J(V_W(B))| \le 4\varpi(V_W(B))$, within the regularity radius by the same choice of $\varepsilon_{\mathcal{W}}$.

We check the local conditions rather than assuming them from whole-lattice existence. A cube of side at most $\mathsf{o}LM$ is contained in a coordinate cube of integral side $l < l_M$; enlarge the latter by two layers at each face. The room inequality puts this enlargement inside a chart with sides less than half the torus period. If the whole curvature and current have radius α, theorem F supplies there a real gauge with

$$|A|, |\nabla A| < B_{\mathrm{rg}} l_M \alpha.$$

This is precisely the local-gauge condition of the simple set when its right side is at most $\mathsf{o}LMB^{(\mathrm{cg})}b_0$. The other simple-set inequalities follow from curvature and current, and the perturbing field of a plain triple is zero.

For every localized tower used by a core or an octet, apply lemma 51.13 to the restricted real critical field. At level q, its defining averages have curvature at most $2L^2\alpha$, by proposition 14.1. The choice above puts them inside the real-background radius, so theorem E identifies their minimal orbit with the restriction. The complex and real means agree on these real fields. At depth p, curvature and current are multiplied respectively by $L^{-2(k+1-p)}$ and $L^{-3(k+1-p)}$; hence all tower inequalities in the core hold. This proves plain-core membership for the top background and existence of the localized minimal orbits needed along the lower cutoff chart. A residual gauge on an octet extends by I outside it: its boundary nodes are fixed and its outer determining rows are unchanged. Thus the same statements hold for every representative, not only for the chosen axial field. The shift hypotheses now follow from $P_W \in \mathcal{Q}_{k+1}(T_\ell; a_{\mathfrak{h}}, a_{\mathfrak{h}}, a_{\mathfrak{h}})$ and $\sup|gB| < \varepsilon_{\mathfrak{h}}$.

This also discharges the response hypothesis in proposition 52.2. The simple-set gauges just constructed verify the natural-class conditions in the proof of proposition 51.5; in level-k units the current is $L^{-3}J^{(k+1)}$. The actual response of proposition 34.2 and theorem K therefore applies near $p = V$, including variations off its one-step fibre. For $k \ge 1$ take the open set $\{V' : \varpi(V') < \bar{\varepsilon}\}$ about p; theorem E constructs its backgrounds. The response is real regular and critical, and uniqueness identifies it with that constructed family. For $k = 0$ take instead the open set of fields with both curvature and current below the regularity ceilings; the background is the field itself. The off-fibre derivatives and quadratic subtraction in proposition 52.2 are thus derivatives of actual backgrounds.

It remains to check the assertions for $\mathcal{C}_j$. Its core condition immediately gives $\mathsf{m}_j < b_0$ and $|J^{(j)}| < b_0$ on the whole torus. Conversely, these two bounds imply the strengthened regularity

bound $b_0 + b_0^2 \le 2b_0$. The preceding local gauge proof applies with $\alpha = 2b_0$. By (952) and $M \ge L^2$,

$$2B_{\mathrm{rg}} l_M b_0 \le \mathsf{o} L M B^{(\mathrm{cg})} b_0.$$

The restricted critical-tower argument is unchanged. It puts every plain triple in the core, for any $b_1 > 0$, and proves (955). Curvature, current norms, averages and critical orbits are carried to their counterparts by an isometry: transport the paths, and differentiate the invariant action. Consequently this description is isometry invariant. Also $\varpi(W) < 2b_0 \le \varepsilon_{\mathcal{W}}$ by proposition 14.1; applying the already proved domain assertion with radii $(1, 1)$ gives all its conditions except the specified plain-core radii, which we have just checked. Finally theorem E and the first assertion applied at b_0, b_1 prove (956). □

The actual shifts of proposition 51.5, the octet identifications of proposition 51.14, and the substitution identity (872) now hold on this global domain. Hence all three interaction totals and the fourth total in theorem W have their physical meanings there. The positive normalizer is $Z^{(k)}(W) = Z_k[P_W]$, and

$$\mathcal{P}[\mathbf{G}_\tau](W) = \tfrac{1}{2} n_{\mathcal{O}} \operatorname{Log} \tau. \tag{957}$$

Indeed the identified Hessian and gauge quadratic form are those of eq. (910); the latter's gauge Gaussian identity gives this formula. With (905) also imposed, applying theorem V and proposition 52.2 gives the nonzero chart integral and the effective-action value

$$\log(\mathsf{pf}_k(\mathsf{t}) Z^{(k)}(W)) + \mathcal{P}[\mathbf{G}_\tau](W) + \mathcal{P}[\mathbf{F}](W) - \tfrac{3}{2} \#\mathfrak{X}^+ \operatorname{Log} \tau. \tag{958}$$

The zero input array proves (957) even when a parameter family is empty. The chart identity itself needs no whole-space measurability assumption. To regard it as a pushforward density, retain the bounded-Borel hypothesis on $\chi_{\mathsf{t}}^{\mathrm{sf}} F_{x,S}$ from theorem W.

52.8. **Covariance of the totals.** The individual domains of the expansion use a fixed partition. An isometry of the coarse lattice need not preserve that partition. To recover its covariance, first sum the terms carrying one marked bond. Their sum is an unlocalized Gaussian expectation, to which the isometry applies directly. The tents then recover the pointed totals.

Proposition 52.4. Suppose the input pointed totals are covariant under all isometries of their level lattices on $\mathcal{C}_j(a_{0,j}, a_{1,j}; e)$, with a common $e \in (0, 1]$. Use the actual shifts and substitutions and the numerical hypotheses of theorem W. Suppose $2\bar{a}_0 \le \varepsilon_{\mathcal{W}}$, and let $W \in \mathcal{C}_{k+1}(\bar{a}_0, \bar{a}_1; e)$. Then the pointed totals of the actual output are defined for every representative of the minimal orbit, and

$$\mathcal{P}_{\mathsf{r}z}[E^{(k+1)}](\mathsf{r}W) = \mathcal{P}_z[E^{(k+1)}](W) \tag{959}$$

for every isometry r of the level-$(k+1)$ lattice. The same assertion holds separately for the normalizer and the Gaussian term. At height zero the input condition is empty. ■

Proof. By proposition 52.3, both W and $\mathsf{r}W$ belong to the actual global domain. Write $V = V_k^{\mathrm{bf}}(W)$, $\beta b = (\mathsf{r}b)^+$, and define, on positive bonds,

$$(\mathsf{O}B)(\beta b) = O_b B(b), \qquad O_b = \begin{cases} 1, & \mathsf{r}b > 0, \\ -\operatorname{Ad}(V(b)^{-1}), & \mathsf{r}b < 0. \end{cases}$$

Use the same formula on all fine bonds, denoting that map by r_V. The positive pivot bonds are carried to the positive pivot bonds, since the central pivot formula is affine covariant. Thus O is the restriction to the nonpivot coordinates. Each O_b is real orthogonal for the Pauli pairing. Consequently $\mathsf{O}^{\mathsf{T}} = \mathsf{O}^{-1}$, it preserves Lebesgue measure and every bond norm, and

$$\mathsf{O}(\mathbf{t} \cdot B) = (\mathsf{r}\mathbf{t}) \cdot \mathsf{O}B, \qquad (\mathsf{r}\mathbf{t})_{\beta b} = t_b.$$

In particular it preserves the cutoff and complex polydisc $\mathbb{D} = \{\sup |B| < \mathsf{C}_{\mathcal{A}} \mathfrak{b}(\mathsf{t})\}$.

We check its action on the actual chart. Relabelling the path families carries means, constraints and minimal orbits to their counterparts. Normalization of the contour means therefore gives

$V_k^{\mathrm{bf}}(\mathsf{r}W) = \mathsf{r}V$. The identity $e^{i\mathsf{r}_V A}\mathsf{r}V = \mathsf{r}(e^{iA}V)$ follows also on a reversed bond from $V(b)^{-1}e^{-iA(b)} = e^{-i\,\mathrm{Ad}(V(b)^{-1})A(b)}V(b)^{-1}$. Differentiating the actual constraint gives its twisted linear identity. Since pivots are permuted, its pivot elimination and right inverse obey

$$\mathsf{C}_{\mathsf{r}V}\mathsf{O} = \mathsf{r}_V\mathsf{C}_V, \qquad h_{\mathsf{r}V}\mathsf{r}_{\mathcal{M}(V)} = \mathsf{r}_V h_V. \tag{960}$$

The defining correction equations and uniqueness of their holomorphic germ give the analogous identity for $\widetilde{\mathsf{D}}_V$ near zero. Both chart corrections are defined on balls containing $g\mathsf{C}_V B$, and the twist preserves these balls. On their common smaller ball the identity theorem extends the germ identity. Hence

$$V_{\mathsf{r}W}(\mathsf{O}B) = \mathsf{r}V_W(B) \tag{961}$$

throughout the real cutoff chart.

The action and the sum of squared contour logarithms are invariant under this substitution. The Jacobian is invariant as well: differentiating the correction identity conjugates $h_V\partial\widetilde{\mathsf{D}}_V$; its trace-log series is unchanged term by term. Each Haar factor depends only on the squared norm of the corrected bond field, also unchanged. The physical action Hessian and the gauge quadratic form therefore satisfy

$$\mathsf{B}_{\mathsf{r}W} = \mathsf{O}\mathsf{B}_W\mathsf{O}^{\mathsf{T}}, \qquad \widehat{\mathsf{G}}_{\mathsf{r}W} = \mathsf{O}\widehat{\mathsf{G}}_W\mathsf{O}^{\mathsf{T}}.$$

Their sum, inverse and Gaussian measure transform accordingly. The sums of the three local potentials are holomorphic on $\mathbb{D}$. They agree near zero with the physical remainders in proposition 52.2, so uniqueness of holomorphic continuation proves their covariance on all of $\mathbb{D}$. This argument identifies the actual quadratic form; no covariance of a chosen localization is being assumed.

Let $\hat{s}_W = \sum_Y \mathbb{V}'[\mathfrak{f}](Y; P_W|_Y, \cdot)$. We next prove

$$\hat{s}_{\mathsf{r}W}(\mathsf{O}B) = \hat{s}_W(B). \tag{962}$$

At $B = 0$, descent of the top critical orbit gives $\mathsf{m}_k = L^{-2}\mathsf{m}_{k+1} < L^{-2}e$. The constructed response is continuous, so this strict bound persists with $L^{-2}e$ on a neighbourhood of zero. Further descent gives $\mathsf{m}_j < L^{-2(k-j)}L^{-2}e \le e$. The actual substitution and octet identities propositions 51.14 and 51.15 put all corresponding pairs in the input cores. They do so at $\mathsf{r}W$ as well. Thus, for the fields in these two neighbourhoods, the earlier action is the finite sum

$$S[\mathfrak{f}](U) = \sum_{j=1}^{k} \left(\mathcal{P}[E^{(j)}](\mathcal{M}^j U) - c_j - e_j A^{\eta_j}(U) \right), \tag{963}$$

where c_j is its value at the trivial pair. Each earlier total is invariant by hypothesis, and the plaquette action is invariant. Equation (961) and the actual fourth sum therefore give (962) near zero. Both sides are holomorphic on $\mathbb{D}$, so they agree there. At $k = 0$ the sum is empty and both sides are zero.

We now express the marked-bond sums without the partition. Let $\mu = \mu_{\tau^{-1}C_W}$, $q(B) = \frac{1}{2}(B, \widehat{\mathsf{G}}_W B)_1$, and $q_b(B) = \frac{1}{2}\langle B(b), (\widehat{\mathsf{G}}_W B)(b)\rangle$. Write $P_W^{\natural}$ for the total continued potential. The interpolation of (748) has total

$$\begin{aligned}\Psi_W(\mathbf{t}; B) = {} & P_W^J(\mathbf{t}B) + \hat{s}_W(\mathbf{t}B) \\ & + \int_0^1 (1-r) D^2(P_W^G + \tau P_W^A)(r\mathbf{t}B)[B, B]\, dr.\end{aligned} \tag{964}$$

Set

$$Z_G(u) = \int e^{-u(1-\tau)q}\, d\mu, \quad Z_\chi(\mathbf{t}) = \int \chi_{\mathbf{t}} e^{-(1-\tau)q}\, d\mu, \quad Z_\bullet(t) = \int \chi e^{-(1-\tau)q + \Psi_W(t\mathbf{1}; B)}\, d\mu.$$

All these integrals are nonzero by the Gaussian cluster theorem: the quadratic family has size at most $w_*/2$, and the two interpolated families have sizes at most $w_*/2, w_*/4$. The insertion and

cutoff identities used in theorem V give, after summing over the finitely many domains,

$$\sum_X \mathbf{G}_\tau(X,b;P_W|_X) = -(1-\tau)\int_0^1 \frac{\int q_b e^{-u(1-\tau)q}\,d\mu}{Z_G(u)}\,du, \tag{965}$$

$$\sum_X E^\chi(X,b;P_W|_X) = \int_0^1 \frac{\partial_{t_b} Z_\chi(t\mathbf{1})}{Z_\chi(t\mathbf{1})}\,dt, \tag{966}$$

$$\sum_X E^\bullet(X,b;P_W|_X) = \int_0^1 \frac{\int \chi e^{-(1-\tau)q+\Psi_W(t\mathbf{1};B)}\partial_{t_b}\Psi_W(t\mathbf{1};B)\,d\mu}{Z_\bullet(t)}\,dt. \tag{967}$$

Derivatives in the cutoff identity exist almost everywhere; the insertion bounds give integrability in the parameter. The finite sum allows one common null set for all domains. Thus these are equalities of the defined integrals, including for complex τ, not formal differentiated expressions.

Under $B \mapsto \mathsf{O}B$, the measure transforms to its counterpart, q_b to $q_{\beta b}$, $\chi_{\mathbf{t}}$ to $\chi_{\mathbf{rt}}$, and, by (964) and (962), Ψ_W to $\Psi_{\mathsf{r}W}$. The parameter coordinate t_b becomes $t_{\beta b}$, including its exceptional null set. All three bond densities in the preceding displays are therefore covariant. The tent identity

$$h_{\mathsf{r}z}(\beta b) = h_z(b) \tag{968}$$

now proves pointed covariance for $\mathbf{G}_\tau, \mathbf{F}$. Indeed a marked term vanishes unless $b^\sharp \subset X$, whereas $h_z(b) \neq 0$ implies $z \in b^\sharp$. One may therefore drop $X \ni z$ before exchanging the domain and bond sums.

For the normalizer, the finite difference in (792) telescopes to the whole-torus diagonal logarithm. Its nonpivot density is $-\frac{1}{2}\operatorname{tr}[(\log \mathsf{A}_W)(b,b)]$, and its pivot density at $b_0(c)$ is $-\log|\det m_c(V)|$, in each case subtracting the value at the trivial background. Conjugation by O leaves the diagonal trace unchanged and permutes the pivot determinants. Equation (968) gives pointed covariance here too. Adding the three contributions proves (959). □

52.9. **The real progression and its coefficients.** We now iterate the actual step with prescribed coupling increments. Its estimates use the earlier levels only through a summable scalar norm. The covariance just proved allows us to extract the quadratic coefficient of each output, which will determine the intrinsic running coupling.

Choose fixed $\alpha_0, \alpha_1 > 0$ satisfying the radius and rate conditions of theorem W, with

$$\alpha_0 + \alpha_1 \le a^*, \qquad \mathsf{c}_{\mathrm{sh}} M \alpha_0 < \alpha_1, \qquad \alpha_1^2 \le \mathsf{c}_\natural \alpha_0, \qquad a^* \le \min\{\alpha_N, \varepsilon_{\mathcal{W}}/2, a^\sharp\}. \tag{969}$$

At height h use the absolute radii $a_{\iota,j} = \alpha_\iota$ for $0 \le j \le h$ and $a_{\iota,h+1} = \alpha_\iota/(1+\beta_{\mathrm{s}})$, for $\iota = 0,1$. The enlarged output radii are again (α_0,α_1). Use the constant-ratio case of proposition 51.16,

$$\omega_n = L^{-(n-1)}, \qquad \mathsf{K} = K_\Lambda, \qquad K_{\mathbf{M}} = K_{\mathbf{M}}(K_\Lambda),$$

with its ceiling $\check{s}(K_\Lambda)$. All geometric floors and the admitted family of tori in section 52.7 are fixed uniformly in h.

Put $r_\alpha = a^{\mathrm{us}}(\alpha_0,\alpha_1)$ as in (933), and choose

$$E_0 \ge 2E_N, \qquad B_\sharp \ge C_{\mathrm{am}} r_\alpha^{-2} E_0, \qquad N_* \ge \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp). \tag{970}$$

The one-step conditions include $\delta_N M \ge \kappa + 1$ and $4\beta_{\mathrm{s}}\kappa \ge 1$. Fix $A_1 > 0$ and $A_0 \ge C_\chi A_1$, with C_χ as in (905). Write

$$\phi(t) = t^{-1/2}(\log t)^{\mathfrak{p}_0}, \quad \varepsilon(t) = A_0\phi(t), \quad \delta(t) = A_1\phi(t), \quad \mathfrak{b}(t) = A_1(\log t)^{\mathfrak{p}_0},$$

and

$$\Theta_c(t,N) = K_{\mathbf{M}}\left(\delta(t) + \frac{(1+\beta_{\mathrm{s}})\delta(t)}{\alpha_0}N + e^{-\gamma_2 \mathfrak{b}(t)^2/2}\right). \tag{971}$$

Define the real-step majorant by $\Xi(t,N) = \frac{1}{2}E_0 + \Theta_c(t,N)$ for $t > 1$ and $N \geq 0$. The fixed comparison gives $C_\chi \delta(t) \leq \varepsilon(t)$ for every $t > 1$. Choose $T \geq 4$ so that, for all $t \geq T$, the remaining absolute smallness hypotheses of theorem W hold and

$$\begin{aligned} &\varepsilon(t) \leq 1, \qquad \delta(t) \leq \min\{1, \varepsilon_{\mathcal{W}}, r_\chi/C_\chi\}, \qquad \mathfrak{b}(t) \geq \mathfrak{b}_*, \\ &\frac{(1+\beta_{\mathrm{s}})\delta(t)}{\alpha_0} \leq \check{s}(K_\Lambda), \qquad \Theta_c(t, N_*) \leq E_0/2. \end{aligned} \tag{972}$$

Such T exists: $\phi(t) \to 0$, while $\mathfrak{b}(t) \to \infty$. It depends only on the fixed constants.

Let $x_0, \dots, x_k \geq T$ be real, with $b_i = L^{-1}x_{i-1} - x_i$ and $|b_i| \leq B_\sharp$ for $1 \leq i \leq k$. Starting with the empty array, at height h apply the actual step at $t = x = x_h$ to $(E^{(i)}, b_i)_{i \leq h}$, with empty last slot:

$$E^{(h+1)} = \mathbf{N}^{(h+1)} + \mathbf{F}[x_h, x_h; (E^{(i)}, b_i)_{i \leq h}]. \tag{973}$$

The gauge contribution is zero because $\tau = 1$.

Theorem Y. Under these conditions, the real progression (973) is defined through level $k+1$. Its terms are gauge invariant, conjugation real and holomorphic on their cores, and are covariant under the cube-grid translations. On every admitted torus, for $P \in \mathcal{U}_j^\circ(X; \alpha_0, \alpha_1)$,

$$|E^{(j)}[x_\bullet](X, P, z)| \leq E_0 e^{-(\kappa+1)d_j(X)}, \qquad \|\widehat{E}^{(j)}[x_\bullet]\|_{j,\kappa+1} \leq E_0, \quad 1 \leq j \leq k+1. \tag{974}$$

The companions agree with the finite-volume terms on every non-wrapping domain. Their values and domains are unchanged upon restriction to any unbounded subfamily of admitted tori. The level-j terms depend only on $x_0, \dots, x_{j-1}$.

For every $e \in (0,1]$, the pointed totals are covariant under all isometries of their level lattices on $\mathcal{C}_j(\alpha_0, \alpha_1; e)$. Their quadratic coefficients are therefore defined, and

$$e_j := \mathsf{cf}(\widehat{E}^{(j)}) \in \mathbb{R}, \qquad |e_j| \leq C_{\mathrm{am}} r_\alpha^{-2} E_0 \leq B_\sharp. \tag{975}$$

These numbers are independent of the lattice.

On the global small-field class the whole sums satisfy the actual chart and effective-action identities of theorem W and (958). The pushforward-density assertion retains the whole-space bounded-Borel hypothesis of proposition 52.2. ■□

Proof. At height zero the earlier array is empty and the fourth potential vanishes. Suppose the assertion holds through height h. The earlier local terms have norm at most E_0 and the prescribed counterterms have modulus at most $B_\sharp$. Hence the input size satisfies

$$N_{\mathfrak{a}_h^{\mathrm{abs}},\omega}((E^{(i)}, b_i)_{i \leq h}) \leq \sum_{n=1}^{h} L^{-(n-1)}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp) \leq N_*. \tag{976}$$

No radius decreases at an earlier level; the enlarged output radii are precisely the next input radii. The actual shifts and octet substitutions thus satisfy the hypotheses of theorem W, with $s_{\mathfrak{a}} = (1+\beta_{\mathrm{s}})\delta(x_h)/\alpha_0$. The Gaussian and normalizer inputs are theorem U and proposition 50.2, the first three potentials come from proposition 51.5, and the fourth is the substitution of propositions 51.14 and 51.16. Their physical meanings on the global class were established above.

The normalizer contributes at most $(E_0/2)e^{-(\kappa+1)d}$. Since $\tau = 1$, the remaining contribution is at most $\Theta_c(x_h, N_*)e^{-(1+4\beta_{\mathrm{s}})\kappa d}$. The rate condition and (972) prove (974) at the next level. Each operation preserves slice holomorphy, conjugation reality, gauge invariance and covariance under cube-grid translations.

Local charts identify the core, shifts and response, including their domains: the chart carries every contained stencil and all the defining tests in both directions. They therefore identify the constructed term. Two sufficiently large tori give the same answer on a fixed finite domain. This defines the companion, and its bound passes through the chart. Every finite domain fits in some

member of any unbounded admitted subfamily, so that the companion does not depend on this subfamily.

To check the stronger covariance of pointed totals, fix $e \in (0,1]$. The preceding totals have covariance on their classes with this same radius. The inherited curvature allowance at the next step is $L^2 e$, which is at least e. Since $2\alpha_0 \le \varepsilon_{\mathcal{W}}$, proposition 52.4 gives covariance on $\mathcal{C}_{h+1}(\alpha_0, \alpha_1; e)$. The background and every core restriction used here exist by proposition 52.3. Theorem X therefore applies on the near-trivial window (956), with usable radius r_α. It gives (975). The coefficient is real by conjugation of its absolutely convergent series. Its construction from the companion makes it independent of the torus.

The global-domain and substitution identities apply to these earlier terms with their prescribed subtractions. They give the stated whole-sum and chart identities. Finally, every operation at height h uses only the earlier terms and x_h. Induction proves dependence on the stated prefix, and completes the construction. □

52.10. **The running coupling determined by the local output.** The preceding construction allowed prescribed increments $b_j = L^{-1}x_{j-1} - x_j$. We now determine these increments from the quadratic coefficients. Choose $0 < \gamma \le 1/2$ so that the hypotheses of theorem Y hold for every finite sequence with $|b_j| \le B_\sharp$ and $x_j \ge \gamma^{-2}$, and require

$$\gamma^{-2} > \bar{B} := \frac{L}{L-1} B_\sharp, \qquad L^{-1}\gamma^{-2} - B_\sharp > 1. \tag{977}$$

In particular the thresholds $\varepsilon(x), \delta(x)$ are at most one for every $x \ge \gamma^{-2}$.

Given $\beta \ge \gamma^{-2}$, put $x_0 = \beta$. As long as $x_0, \dots, x_j$ have been constructed and $x_j \ge \gamma^{-2}$, form the next local output by (973) and set

$$e_{j+1} = \mathsf{cf}\big(\widehat{E}^{(j+1)}[x_0, \dots, x_j]\big), \qquad x_{j+1} = L^{-1}x_j - e_{j+1}. \tag{978}$$

Stop at the first index J with $x_J < \gamma^{-2}$. The families and the coefficient in this definition use only the indicated prefix, including its already determined increments; the new coupling is not an input to them.

Proposition 52.5. This recursion defines a unique finite sequence of real numbers $x_0, \dots, x_J$, with $J \ge 1$. Its increments are real and $|e_j| \le B_\sharp$. The sequence strictly decreases and satisfies

$$\begin{aligned} x_j - L^{-(j-i)}x_i &= -\sum_{l=i+1}^{j} L^{-(j-l)} e_l, \\ |x_j - L^{-(j-i)}x_i| &\le \bar{B}(1 - L^{-(j-i)}) \qquad (0 \le i \le j \le J). \end{aligned} \tag{979}$$

Moreover,

$$\frac{\beta}{\gamma^{-2} + \bar{B}} < L^J \le \frac{L\beta}{\gamma^{-2} - \bar{B}}, \qquad 1 < L^{-1}\gamma^{-2} - B_\sharp \le x_J < \gamma^{-2}. \tag{980}$$

Every prefix ending before J is an admissible sequence for theorem Y. Its families and coefficients agree with those obtained from any longer such prefix. The coupling sequence is independent of the torus.

More generally, let $(u_0, \dots, u_n)$ be an admissible prescribed sequence with $u_0 = \beta$ and $u_i \ge \gamma^{-2}$ for $0 \le i < n$. It is a prefix of this sequence if and only if

$$L^{-1}u_{j-1} - u_j = \mathsf{cf}\big(\widehat{E}^{(j)}[u_0, \dots, u_{j-1}]\big) \quad (1 \le j \le n). \tag{981}$$

■

Proof. At $j = 0$ the earlier interaction array is empty. Suppose a real prefix has been constructed with all its members at least γ^{-2} and all its increments bounded by $B_\sharp$. The hypotheses of theorem Y apply to this prefix. They give a well-defined next family and a real coefficient of modulus at most

$B_\sharp$. Thus (978) defines the next real member. If it is still above the stopping threshold, it meets the induction hypotheses.

Consistency follows at the same time. At height j the construction reads precisely the old families, the old increments and x_j. Appending a coupling does not change these data. Induction therefore identifies each earlier finite-volume family as a function on its entire core. The companions agree through their local charts, so their coefficients agree as well. In particular the next coefficient requires neither a choice of finite torus nor a radius table at the next coupling. This is the prefix dependence proved in theorem Y, applied at successive heights.

Iteration of the affine recursion gives the first line of (979); the geometric sum of $|e_l| \le B_\sharp$ gives the second. As long as $x_j \ge \gamma^{-2}$, it follows that

$$L^j(\gamma^{-2} - \bar{B}) \le \beta - \bar{B}.$$

The positive coefficient on the left and $L > 1$ exclude an infinite sequence above the threshold. Thus the first passage J exists, is at least one, and the recursion supplies x_J. For $1 \le j \le J$,

$$x_{j-1} - x_j \ge (1 - L^{-1})(x_{j-1} - \bar{B}) > 0.$$

The last step also gives $x_J \ge L^{-1}\gamma^{-2} - B_\sharp > 1$.

For the first-passage bounds use

$$L^{-J}(\beta + \bar{B}) \le x_J + \bar{B} < \gamma^{-2} + \bar{B}$$

and the preceding upper bound at $j = J - 1$. These imply (980). Every earlier prefix has thresholds at most one, members at least γ^{-2}, and bounded increments, so it is admissible.

Finally, equality of a prescribed prefix with the constructed one identifies its next local output and coefficient. Equation (981) then identifies the next coupling. Starting at $u_0 = x_0 = \beta$, induction proves sufficiency. Necessity is (978) and prefix consistency. □

53. Coordinates and tests of a history step

53.1. **Free coordinates and the reference measure.** Fix a history h of length k. All bonds below are taken in positive orientation; the values at reversed bonds are prescribed by inversion for configurations and by a change of sign for Lie algebra fields. Put $\Omega_0 = \mathbb{T}$ and $\Omega_{k+1} = \Lambda_{k+1} = \varnothing$. At level l, write Γ_l^{pr} for the bonds with no endpoint in Ω_{l+1}, and Γ_l^{le} for those with an endpoint in Ω_l and none in Ω_{l+1}. Thus $\Gamma_l^{\mathrm{ti}} = \Gamma_l^{\mathrm{pr}} \setminus \Gamma_l^{\mathrm{le}}$ consists of the bonds with no endpoint in Ω_l. The superscripts distinguish the coordinates that are retained freely from those determined by averaging.

Let $\mathfrak{X}_l^{\mathrm{E}}$ be the nonpivot bonds of step $l+1$ with no endpoint in Λ_{l+1}. Once the strip coordinate of that step has been defined, denote its scaled value by $\mathsf{a}'_l = g_l^{-1}\mathcal{A}'^{\diamond}_l$. The free coordinate space is

$$\mathsf{X}_h^{\mathrm{fr}} = \prod_{l=0}^{k} \mathcal{V}(\Gamma_l^{\mathrm{le}}) \times \prod_{l=0}^{k-1} \mathfrak{g}^{\mathfrak{X}_l^{\mathrm{E}+}}. \tag{982}$$

We write a point as $(\mathbf{v}, \mathfrak{q})$. Its completion $\mathrm{cp}_h(\mathbf{v}, \mathfrak{q}) = (\mathbf{w}, \mathfrak{q})$ retains the free coordinates and sets, successively,

$$w_0 = v_0, \qquad w_l(c) = \begin{cases} v_l(c), & c \in \Gamma_l^{\mathrm{le}}, \\ \mathcal{M}(w_{l-1})(c), & c \in \Gamma_l^{\mathrm{ti}}. \end{cases} \tag{983}$$

The average in the second line uses only $\Gamma_{l-1}^{\mathrm{pr}}$: both blocks over a bond with no endpoint in Ω_l lie outside that region. In particular, completion is a Borel map defined on the whole free space. Restriction rs_h to the free coordinates satisfies $\mathrm{rs}_h\mathrm{cp}_h = \mathrm{id}$. The image of completion is precisely the set on which the finitely many tying equations in (983) hold. It is Borel, since each equation equates Borel maps into the compact metric space G.

Give the configuration coordinates Haar probability measure and the $\mathfrak{q}$-coordinates Lebesgue measure for $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$. Define

$$d\mathscr{M}_h^{\mathrm{fr}} = \prod_l dv_l \prod_{l<k} d\mathfrak{q}_l, \qquad \mathscr{M}_h = (\mathrm{cp}_h)_* \mathscr{M}_h^{\mathrm{fr}}. \tag{984}$$

This measure is positive and σ-finite. Indeed, bounding all the finitely many Lie algebra coordinates gives Borel sets of finite measure which exhaust the completed space up to its null complement; completion and restriction preserve these coordinates. This last observation matters: an arbitrary image of a σ-finite measure need not be σ-finite.

For a bare field U for which all the required strip variables exist, its completed coordinate is

$$\xi_h(U) = \big((\mathcal{M}^l U)|_{\Gamma_l^{\mathrm{pr}}}, (\mathsf{a}'_l(U))_{l<k}\big).$$

It is the completion of the corresponding free coordinate. We use partial pushforwards on their Borel domains: for a finite measure ν and a Borel partial map F, $F_*\nu(E) = \nu(\{U \in \operatorname{dom} F : F(U) \in E\})$. In particular the history measure is $(\xi_h)_*(\mathfrak{w}_h \rho_0\, dU)$.

At the step $t = k+1$, let $\mathfrak{X}_0$ consist of all level-k bonds with an endpoint in Ω_t, including pivot bonds. The old completed coordinates split as

$$\phi = (V^{\mathrm{in}}, \phi^+) \in \mathcal{V}(\mathfrak{X}_0) \times \mathsf{X}^+_{h'}.$$

Here ϕ^+ contains all lower proper coordinates, the level-k coordinates with no endpoint in Ω_t, and all old external fields. New free coordinates are $W \in \mathcal{V}(\Gamma_t^{\mathrm{le}}(h'))$ and $\mathsf{a}' \in \mathfrak{g}^{\mathfrak{X}_k^{\mathrm{E}+}(h')}$. If v_k is the level-k component of ϕ^+, complete the new level by

$$v_t(c) = W(c) \quad (c \in \Gamma_t^{\mathrm{le}}), \qquad v_t(c) = \mathcal{M}(v_k)(c) \quad (c \in \Gamma_t^{\mathrm{ti}}).$$

These formulas define a Borel injection $\imath^+_{h'}(W, \mathsf{a}', \phi^+)$ into the new completed space; its inverse on its image is coordinate projection. For $c \in \Gamma_t^{\mathrm{ti}}$ the double block lies outside Ω_t, so this formula never calls for V^{in}.

Proposition 53.1. Let $\mathscr{M}^+_{h'}$ be the measure on the untouched coordinates obtained from $\mathscr{M}_h$. Then

$$d\mathscr{M}_h = dV^{\mathrm{in}}\, d\mathscr{M}^+_{h'}, \qquad \mathscr{M}_{h'} = (\imath^+_{h'})_*(dW\, d\mathsf{a}'\, d\mathscr{M}^+_{h'}).$$

Both measures in these formulas are σ-finite. ■

Proof. The old free space splits as $\mathcal{V}(\mathfrak{X}_0) \times \mathsf{Y}^+$. Completion has the form $\mathrm{id} \times \Pi^+$: none of the lower tying equations involves the current level-k inner bonds. Thus its image measure has the first displayed product decomposition. The map Π^+ preserves every old Lie algebra coordinate, which proves σ-finiteness of $\mathscr{M}^+_{h'}$ by the same bounded-coordinate exhaustion. Completion of the new free space is

$$\imath^+_{h'} \circ (\mathrm{id}_W, \mathrm{id}_{\mathsf{a}'}, \Pi^+).$$

Its image measure is the second displayed measure. These identities hold first on measurable rectangles and then on Borel sets by uniqueness of the product measure. □

53.2. **The regional fibre chart.** We need the constraint chart with a fixed exterior configuration. Let P'' be a set of level-t sites, Γ the coarse bonds with an endpoint in P'', and $\mathfrak{X}_0$ the fine bonds with an endpoint in $B(P'') = \bigcup_{y \in P''} B(y)$. Let $\mathfrak{X}$ be its nonpivot bonds, and let $\mathfrak{X}_{\mathrm{out}}$ be the remaining fine bonds in the double blocks of Γ. Fix $\eta^+ \in \mathcal{V}(\mathfrak{X}_{\mathrm{out}})$, and write

$$q(V^{\mathrm{in}}, \eta^+) = \mathcal{M}(V^{\mathrm{in}} \uplus \eta^+)|_\Gamma.$$

Choose $V_0 \in \mathcal{V}(\mathfrak{X}_0)$ so that $V = V_0 \uplus \eta^+$ has double-block plaquette deviations $\varpi_c(V) < \varpi_\chi$ for every $c \in \Gamma$. The constants $\varpi_\chi, r_\chi, C_\chi$ are the local chart constants used above, with $0 < r_\chi \le 1$ and $C_\chi \ge 1$, decreased or increased, as appropriate, depending only on L.

Proposition 53.2. Extend a real nonpivot field $\mathcal{A}$ by zero on $\mathfrak{X}_{\mathrm{out}}$. If $C_\chi \sup|\mathcal{A}| < r_\chi$, the chart

$$B' = \psi_V^{\mathrm{lin}}(\mathsf{C}_V\mathcal{A}), \qquad V[\mathcal{A}] = (e^{iB'}V)|_{\mathfrak{X}_0}$$

is defined, is continuous in all its data, and satisfies

$$B'|_{\mathfrak{X}} = \mathcal{A}, \quad B'|_{\mathfrak{X}_{\mathrm{out}}} = 0, \quad \sup|B'| \le 2C_\chi \sup|\mathcal{A}|, \quad q(V[\mathcal{A}], \eta^+) = q(V_0, \eta^+).$$

Conversely, any V^{in} in this fibre whose nonpivot logarithms relative to V_0 satisfy the same bound equals $V[\mathcal{A}]$, where $\mathcal{A}(b) = i^{-1}\log(V^{\mathrm{in}}(b)V_0(b)^{-1})$ on the positive nonpivot bonds. No bound on its pivot entries is required in this converse.

Put $n_c(V) = L\mathsf{m}_c(V)$ and $\mathsf{z}_c(V) = -\log\det n_c(V)$. The determinants are positive, so the numbers $\mathsf{z}_c(V)$ are real. At $\mathcal{A} = gB$ define

$$\mathsf{J}_V[gB] = \sum_{c\in\Gamma} \log\det(1 - \mathsf{Jb}_c(V, gB)) + \sum_{b\in\mathfrak{X}_0^+} \log\sigma_1(|B'(b)|^2). \tag{985}$$

Let $g = \mathsf{t}^{-1/2} > 0$, $C_\chi g r \le r_\chi$, and let $W \mapsto V_{0,W}$, $W \in O \subset G^\Gamma$, be a Borel family of such references with $q(V_{0,W}, \eta^+) = W$. Write $V_W = V_{0,W} \uplus \eta^+$, $V_W(B) = V_W[gB]$, and $\mathsf{J}_W(B) = \mathsf{J}_{V_W}[gB]$. If $\Upsilon(V^{\mathrm{in}}, W) \ge 0$ is Borel and vanishes unless $W \in O$ and

$$\left|\frac{1}{i}\log(V^{\mathrm{in}}(b)V_{0,W}(b)^{-1})\right| < gr \quad (b \in \mathfrak{X}^+),$$

then

$$\begin{aligned}&\int \Upsilon(V^{\mathrm{in}}, q(V^{\mathrm{in}}, \eta^+))\, dV^{\mathrm{in}} \\ &\quad = (\sigma_{\mathrm{H}}\mathsf{t}^{-3/2})^{\#\mathfrak{X}^+} \int_O dW\, e^{\sum_{c\in\Gamma}\mathsf{z}_c(V_W)} \int_{\sup|B|<r} e^{\mathsf{J}_W(B)}\Upsilon(V_W(B), W)\, dB.\end{aligned} \tag{986}$$

The same identity holds for complex Υ if either side with $|\Upsilon|$ is finite. ∎

Proof. The constraint calculation is local to a double block. On P_c, put $U_c = V|_{\mathsf{P}_c}$. The contraction used in definition 51.3 gives

$$\widetilde{\mathsf{D}}_V(\mathsf{B})(c) = n_c(V)y_{U_c}(\mathsf{B}|_{\mathsf{P}_c}), \qquad \psi_V^{\mathrm{lin}}(\mathsf{B})(b_0(c)) = \mathsf{B}(b_0(c)) - y_{U_c}(\mathsf{B}|_{\mathsf{P}_c}).$$

The maps and bounds are unchanged when exterior nonpivot entries are fixed at zero: each double block contains just its own pivot. In particular $\|\mathsf{C}_V\|_{\infty\to\infty} \le C_\chi$, the real solution has full norm at most $2C_\chi \sup|\mathcal{A}|$, and it solves the averaging constraint at every c. Its plaquette deviations are bounded by $\varpi_\chi + 8C_\chi \sup|\mathcal{A}|$. Choose the constants so that this is below the small-curvature ceiling. The same local contraction proves continuous dependence on $V_0, \eta^+, \mathcal{A}$.

The converse uses the global uniqueness in proposition 14.11. The proposed chart and V^{in} agree at every nonpivot entry and have the same averages. For a given c, their restrictions to P_c° are within r_χ/C_χ of the reference. Choose $\varpi_\chi + r_\chi/C_\chi \le \omega_{\mathrm{p}}$. The pivot proposition then says that the same coarse value has exactly one pivot preimage, including the other real branches of the global average. Hence the two configurations agree also at every pivot.

The differential and determinant computation (911) applies to this same double block. It gives

$$(\det n_c(V_W(B) \uplus \eta^+))^{-1} = e^{\mathsf{z}_c(V_W)}\sigma_1(|B'(b_0(c))|^2)\det(1 - \mathsf{Jb}_c(V_W, gB)).$$

All factors are positive; the local bounds give $\|\mathsf{Jb}_c\| \le 1/2$. The pivot subspace is a direct sum over c, so its logarithmic determinant is the sum in (985).

To justify the integral for a merely Borel family, first fix the nonpivot entries w. The permitted pairs (w, W) form a Borel set, since the intrinsic logarithm is continuous on $G \setminus \{-1\}$. The chart supplies its Borel pivot solution $\Pi(w, W)$. On the small-curvature pivot domain, the map from pivots to averages is a diffeomorphism onto its image: injectivity is the pivot proposition, and its differential is invertible. The converse just proved shows that every nonzero value of the integrand

occurs in this domain, even though no pivot cutoff was imposed. Changing the pivot variables to W therefore gives the product of the inverse determinants in the preceding display. Tonelli interchanges this integral with the nonpivot integral. For each W, substitute $w(b) = e^{igB(b)}V_{0,W}(b)$. The nonpivot Haar factors are

$$(\sigma_{\mathrm{H}} g^3)^{\#\mathfrak{X}^+} \prod_{b\in\mathfrak{X}^+} \sigma_1(g^2|B(b)|^2)\, dB.$$

Together with the pivot factors they give precisely (986). All maps used in this computation are Borel, and Tonelli requires no finiteness assumption. Applying the positive formula to $|\Upsilon|$, followed by Fubini and linearity, proves the complex version. □

The gauge factor has the same regional interpretation. If V_0 is axial on P'', then the averaged contour bound gives

$$|V_W(B)(y,x) - 1| \le C_\chi g \max_{b\subset B(y)} |B(b)| \quad (y \in P'',\ x \in B(y)).$$

When these bounds are below ϵ, all gauge cutoffs pass, and the product of the $(L^3-1)\#P''$ factors is

$$\mathfrak{G}(V_W(B); P''; \mathsf{t}, \epsilon) = \mathsf{z}(\mathsf{t},\epsilon)^{-(L^3-1)\#P''} e^{-\mathsf{t}G_k(P''; V_W(B))}. \tag{987}$$

Indeed the block contains no pivot bond; its logarithmic perturbation is just gB, so the local averaged-contour estimate applies there directly. Multiplying the defining single-contour factors proves the identity.

53.3. **The strip substitution.** The substitution changes the reference background of an external bond. It must preserve the tests which determined the history. Write $\mathfrak{S} = \Omega_t \setminus \Omega_t^{\sim -1}$ and $S_t^{\mathrm{pr}} = \mathfrak{S} \cup R$, where R is the union of failed fluctuation-test cubes. With the test-cube enlargements already fixed, set

$$\mathcal{N}_{\mathrm{s}} = \bigcup_{\square' \subset \Omega_t^{\sim -1}\setminus R} (\square'^{\sim 2})^{(k)*}, \qquad \mathcal{N}_\ell = \bigcup_{\square' \subset R} (\square'^{\sim 2})^{(k)*}.$$

The two zones are

$$\mathcal{Z}_\theta = (S_t^{\mathrm{pr}(k)*} \cap \mathcal{N}_\ell) \setminus \mathcal{N}_{\mathrm{s}}, \qquad \mathcal{Z}_1 = S_t^{\mathrm{pr}(k)*} \setminus (\mathcal{N}_{\mathrm{s}} \cup \mathcal{N}_\ell). \tag{988}$$

They are disjoint reversal-invariant sets of external nonpivot bonds; put $\mathcal{Z} = \mathcal{Z}_\theta \cup \mathcal{Z}_1$. The plain zone $\mathcal{Z}_1$ meets no bond used in an external fluctuation test. On $\mathcal{Z}_\theta$ we will leave all fluctuations below the test thresholds unchanged.

Here are the geometric facts used in this assertion. Put $X_\flat = \Omega_t^c \cup \mathfrak{S} \cup R'$, where R' is the coarse hull of R. Every region Z encountered in the exploration contains $(\Omega_t^c)^{\sim 2} \cup (R')^{\sim}$, hence contains $X_\flat^{\sim}$. No coarse cube of $X_\flat$ touches a cube of Z^c. A strip bond has an endpoint in $X_\flat$; its length is one, whereas the coarse side is W_t, so it has no endpoint in Z^c. Moreover every bond of $\mathfrak{X} \setminus \mathcal{N}_{\mathrm{s}}$ has an endpoint in $\mathfrak{S} \cup R$, by considering the test cube of that endpoint. Consequently

$$\mathfrak{X} = \mathcal{N}_{\mathrm{s}} \sqcup \mathcal{Z}, \qquad \mathcal{Z} \subset \mathfrak{X}_{\mathrm{E}}, \qquad \mathrm{dist}_\infty(b, Z^c) \ge W_t - 1 \quad (b \in \mathcal{Z}).$$

The last bound also separates a zone bond from the endpoints of integrated bonds by $W_t - 1$. Since $\varpi_T < W_t - 1$, every explored collar lies in $\mathcal{N}_{\mathrm{s}}$. A rim cube touches Z^c and cannot belong to R'; its tested bonds likewise belong to $\mathcal{N}_{\mathrm{s}}$. This holds at each stage, so $E^T \subset \mathcal{N}_{\mathrm{s}}$ and $c^\star \subset \mathcal{N}_{\mathrm{s}}$ for $c \in E^H$. Finally, an interior test cube belongs either to R or to its complement, so its tested bonds lie in $\mathcal{N}_\ell \cup \mathcal{N}_{\mathrm{s}}$, as do all the sets $c^\star$. By its definition, $\mathcal{Z}_1$ is disjoint from this union. This proves the claimed avoidance of all external test bonds.

First consider one bond. Fix $0 < \delta \le 1$, a smooth cutoff $\vartheta_{\mathrm{co}} : [0,\infty) \to [0,1]$ equal to zero on $[0, 4/3]$ and to one on $[5/3, \infty)$, and $H \in \mathfrak{g}$. In the cutoff case assume $8\|\vartheta_{\mathrm{co}}'\|_\infty |H| \le \delta$, and put

$$\theta(e^{iA'}) = \vartheta_{\mathrm{co}}(|A'|/\delta) \quad (|A'| < \pi), \qquad \theta(-1) = 1.$$

In the plain case put $\theta = 1$, with no restriction on H. Define

$$\Psi_H(W') = W' e^{-i\theta(W')H} e^{iH}, \qquad f_H(A') = \frac{1}{i}\log(e^{iA'} e^{-i\theta(e^{iA'})H}). \tag{989}$$

The logarithm is intrinsic and is defined unless its argument is -1.

Lemma 53.3. The map Ψ_H is a diffeomorphism of G. Its Haar Jacobian is 1 in the plain case and, in the cutoff case,

$$J_H(e^{iA'}) = 1 - \frac{\vartheta'_{\mathrm{co}}(|A'|/\delta)}{\delta|A'|}\langle A', H\rangle \in [7/8, 9/8], \tag{990}$$

where the quotient term is zero at $A' = 0$, and $J_H(-1) = 1$. The map f_H is continuous and injective on its domain in $\{|A'| < \pi\}$; its domain and its image omit at most one point of that ball. For $0 < s \le 4\delta/3$, in the cutoff case,

$$|f_H(A')| < s \quad\Longleftrightarrow\quad |A'| < s, \qquad f_H(A') = A' \text{ there.}$$

In both cases, wherever f_H is defined,

$$\big||f_H(A')| - |A'|\big| \le \theta(e^{iA'})|H|.$$

■

Proof. For $\alpha_{W'}(X) = \frac{d}{ds}|_{s=0}\theta(W'e^{isX})$, differentiating $\frac{1}{2}\operatorname{tr}(e^{iA'}e^{isX})$ gives

$$\alpha_{e^{iA'}}(X) = \frac{\vartheta'_{\mathrm{co}}(|A'|/\delta)}{\delta|A'|}\langle A', X\rangle.$$

The cutoff is locally constant near 1 and -1, so this also proves smoothness at those points. If $T = e^{-i\theta(W')H}e^{iH}$, the right-trivialized differential is

$$X \longmapsto T^{-1}(X - \alpha_{W'}(X)H)T.$$

Conjugation preserves volume, and the rank-one factor has determinant $1 - \alpha_{W'}(H)$. The smallness hypothesis gives $|\alpha_{W'}(H)| \le 1/8$, proving the Jacobian formula. For completeness, global invertibility follows from a contraction. Since $\varphi(W') = 2\arcsin(|W' - 1|/2)$, the cutoff has Lipschitz constant $2\|\vartheta'_{\mathrm{co}}\|_\infty/\delta$: on its nonconstant interval the derivative of $2\arcsin(s/2)$ is less than 2, and elsewhere the cutoff is constant. Moreover $\|\vartheta'_{\mathrm{co}}\|_\infty \ge 3$, so $|H| \le 1/24$. For each $W \in G$, the equation $\Psi_H(W') = W$ is equivalent to

$$W' = T_W(W') := We^{-iH}e^{i\theta(W')H}.$$

The estimate $|e^{icH} - 1| \le 2|c||H|$ for $|c| \le 1$ shows that T_W is a contraction with constant at most $1/2$. It has a unique fixed point on the complete metric space G. Thus Ψ_H is bijective. Its differential is invertible by (990), so its inverse is smooth. The plain map is the identity. Right translation by e^{-iH}, followed by the intrinsic logarithmic chart, gives f_H. The excluded point of either chart is unique, proving the assertions about its domain and image. If $|A'| \le 4\delta/3$, then $\theta(e^{iA'}) = 0$ and $f_H(A') = A'$. Conversely, injectivity gives the same equality whenever the image has norm below that threshold. Finally the group-angle triangle inequality applied to $e^{if_H(A')} = e^{iA'}e^{-i\theta H}$ and its inverse relation gives the last estimate. □

For the step, fix a real shift $\mathfrak{H} = (H_b)_{b\in\mathcal{Z}^+}$ with

$$8\|\vartheta'_{\mathrm{co}}\|_\infty|H_b| \le \delta_k \quad (b \in \mathcal{Z}^+_\theta). \tag{991}$$

Take the cutoff case at $b \in \mathcal{Z}_\theta$, the plain case at $b \in \mathcal{Z}_1$, and put

$$\check\Psi(\mathsf{a}', \mathfrak{H})(b) = \begin{cases} g_k^{-1} f_{H_b}(g_k \mathsf{a}'(b)), & b \in \mathcal{Z}^+, \\ \mathsf{a}'(b), & b \in \mathfrak{X}^{\mathrm{E}+} \setminus \mathcal{Z}. \end{cases} \tag{992}$$

Extend by antisymmetry. Its real domain requires $g_k|\mathsf{a}'(b)| < \pi$ on zone bonds and the definedness of the logarithms. Set

$$\mathsf{Zs}_{\mathrm{J}} = \sum_{b \in \mathcal{Z}_\theta^+} \log J_{H_b}(e^{i g_k \mathsf{a}'(b)}), \tag{993}$$

$$\mathsf{Zs}_{\mathrm{L}} = \sum_{b \in \mathcal{Z}^+} \log \frac{\sigma_1(g_k^2 |\mathsf{a}'(b)|^2)}{\sigma_1(g_k^2 |\check{\Psi}(\mathsf{a}', \mathfrak{H})(b)|^2)}. \tag{994}$$

Proposition 53.4. The strip map is injective and Borel on its domain. Its undefined set in the zone logarithmic charts is Lebesgue-null. For every nonnegative Borel F vanishing unless $g_k|\mathsf{a}(b)| < \pi$ at all external bonds,

$$\int F(\mathsf{a})\, d\mathsf{a} = \int_{\operatorname{dom} \check{\Psi}} e^{\mathsf{Zs}_{\mathrm{J}} + \mathsf{Zs}_{\mathrm{L}}} F(\check{\Psi}(\mathsf{a}', \mathfrak{H}))\, d\mathsf{a}', \tag{995}$$

where the domain on the right is restricted by $g_k|\mathsf{a}'(b)| < \pi$ at every external bond. The external fluctuation-test product takes the same value at $g_k \check{\Psi}(\mathsf{a}', \mathfrak{H})$ and $g_k \mathsf{a}'$. ∎

Proof. On one bond the Haar density in logarithmic coordinates is $\sigma_{\mathrm{H}} \sigma_1(|A|^2)\, dA$. Apply Haar substitution by Ψ_H, and then right translation by e^{-iH}. Dividing by the positive density $\sigma_1(|f_H(A')|^2)$, and scaling $A' = g_k \mathsf{a}'$, gives the Lebesgue factor

$$J_H(e^{i g_k \mathsf{a}'}) \frac{\sigma_1(g_k^2 |\mathsf{a}'|^2)}{\sigma_1(g_k^2 |g_k^{-1} f_H(g_k \mathsf{a}')|^2)}.$$

The factors g_k^3 on the two sides cancel. The product of these factors is $e^{\mathsf{Zs}_{\mathrm{J}} + \mathsf{Zs}_{\mathrm{L}}}$. Apply the one-bond identity successively, using Tonelli for nonnegative functions. Each omitted set fixes one of the finitely many three-dimensional coordinates to a single point and is null. Off the zones the map is the identity, so the restriction to the logarithmic ball at every external bond loses no nonzero integrand. This proves (995) without an integrability assumption.

The tested thresholds are $\delta_k, \delta_k^\flat, \theta_c \delta_k$, all below $4\delta_k/3$. On the cutoff zone their indicators are preserved by Lemma 53.3; off both zones the variables are unchanged. No tested bond lies in the plain zone. Each external test is a product or difference of these indicators, which proves the last assertion. □

53.4. **Localization of the actual strip background.** The shift in the preceding substitution is the difference between two critical backgrounds. We now construct the second background and estimate that difference. The comparison first takes place before averaging, up to gauge. The axial normalization then turns the estimate at the nodes into an estimate for the two averaged configurations.

We use reduced determining sets throughout. Fix the joint averaging radius

$$r_{\mathrm{av}} = \min\{r_{\mathrm{st}}, \kappa_{6,\mathsf{m}}^{-1}\}, \qquad \kappa_6 = r_{\mathrm{av}}^{-1}, \qquad \varpi_{\mathrm{av}} = \min\{\varpi_L', r_{\mathrm{av}}/(4\pi L)\}.$$

Here r_{st} is from lemma 24.2, and $\kappa_{6,\mathsf{m}}$ is the centered contour constant in the proof of proposition 30.2. In particular $\kappa_6 \geq 16$. For a real double-block reference of curvature at most ϖ_{av}, the comb gauge of lemma 14.4 has logarithm of norm at most $r_{\mathrm{av}}/2$. The joint holomorphy, unit bound and contour estimate of lemma 24.2 therefore apply on the product of the reference and perturbation balls of radius r_{av}. Denote by $a_{\mathrm{R}} > 0$ and $\varrho_{\mathrm{R}}^0 > 0$ the regular-reference and datum thresholds for the response in lemma 51.6. Decrease the regular-reference threshold so that $a_{\mathrm{R}} \leq \min\{L^{-2} a_{\mathsf{m}}, \varpi_{\mathrm{av}}/(2L^2)\}$, as well as its earlier ceiling. The regional curvature estimate in proposition 30.2 then bounds the intermediate reference curvatures by ϖ_{av}.

Set

$$c_{\mathcal{Q}} = \min\{c_{\mathsf{m}}/L, r_{\mathrm{av}}/(64L), (8\kappa_6)^{-1}\}.$$

The centered iteration is defined on each pillar ball $\xi_y \sup|a| < c_{\mathcal{Q}}$. The datum conversion in the proof of lemma 51.6 has a uniform bound $\sup|B_\natural| \le C_{\mathrm{dat}} \sup|B|$, with C_{dat} depending only on L. Thus (500) gives $\sup_y \xi_y|\mathcal{H}(B)|_y \le C_H \sup|B|$, where $C_H = 4\hat{\mathfrak{b}} C_{\mathrm{dat}}$. Decrease r_{B} so that $C_H r_{\mathrm{B}} \le c_{\mathcal{Q}}/4$. The response then lies in the pillar quarter-balls used below.

The response constants in lemma 51.6 and eq. (838) can be chosen with a rate $0 < \delta_{\mathrm{lc}} \le 1/8$ and a constant B_5 such that, on the real datum ball of radius $\varrho = \min\{r_{\mathrm{B}}, \varrho_{\mathrm{R}}^0\}$,

$$m_y(B) = \sum_{c>0} e^{-\delta_{\mathrm{lc}} d(y,y_c)}|B(c)|, \qquad \xi_y|\mathcal{H}(B)(b)| \le B_5 m_y(B), \qquad |u_B(y)-1| \le 4\kappa_6 B_5 m_y(B). \tag{996}$$

The bond in the middle estimate has an endpoint in the pillar of y; the last estimate is at a top y. These conclusions require $C_{\mathfrak{P}}\mathfrak{M}B_3\varepsilon_{\mathrm{r}} \le a_{\mathrm{R}}$, as well as the regular-data hypotheses of lemma 51.6. The middle estimate follows by integrating (838) along sB. The same argument, applied to the first-difference estimate in (838), gives

$$\xi_y^2|(\nabla^{U^\circ}\mathcal{H}(B))(x)| \le B_5 m_y(B), \qquad y(x) = y. \tag{997}$$

The two bounds also hold for the difference of responses at B_1, B_2, with $m_y(B_1 - B_2)$ on the right: integrate the derivative along the segment joining the two points in the real datum ball. To verify the last estimate, recall that $u_B(y)^{-1} = v_q[\mathcal{H}(B)](y)$. This factor depends only on the pillar of y. On the ball $\xi_y \sup|a| < c_{\mathcal{Q}}$, its factors have logarithms Y_m with

$$|Y_m| \le 2\kappa_6 L^{m-q}\xi_y \sup|a|, \qquad \sum_{m<q}|Y_m| \le \frac{2\kappa_6}{L-1}\xi_y \sup|a|.$$

These are the one-step contour estimate and the centered averaging bounds used in proposition 51.1. Thus both $v_q[a](y)^{\pm 1} - 1$ are bounded by $2\kappa_6 c_{\mathcal{Q}}$ on this ball. Cauchy's inequality bounds their derivatives by $4\kappa_6$ on the half-ball. The response takes values in the quarter-ball. The chain rule, (838), and integration along sB give the last bound in (996).

Here is the localization statement in the form needed twice below. Let $(\Omega_i)_{i\le n}$ be a separated sequence, and let $\Omega \subset \Omega_0$ have complement $Y_1 \cup \cdots \cup Y_r$, $r = 1$ or 2. Associate a level $l_s \le n$ to each part. The Y_s are regions of M_1-cubes at level l_s, and enlargements of Y_s in this paragraph use that partition. Suppose

$$Y_s^{\sim 4} \setminus Y_s \subset \Omega_{l_s} \setminus \Omega_{l_s+1}, \qquad \mathrm{dist}_\infty(Y_s, Y_{s'}) \ge 10 M_1 L^{\max(l_s, l_{s'})} \quad (s \ne s'), \tag{998}$$

where these lengths are in bare-lattice units. Set

$$C_s = Y_s^{\sim 2} \setminus Y_s, \qquad C_s^{\mathrm{w}} = Y_s^{\sim 3} \setminus Y_s, \qquad \Omega^\natural = \Omega \setminus \bigcup_s C_s.$$

Let $\Omega^{[i]}$ be the $\mathfrak{r}_2$-tower of Ω. The joined sequence and datum are

$$\Omega_i' = (\Omega_i \cap \Omega^\natural) \cup \bigcup_{l_s \ge i} (\Omega^{[i]} \cap C_s), \qquad V_i'(c) = \begin{cases} V_i(c), & c \in \Gamma_i' \cap \Gamma_i, \\ (Q^{s*}_{l_s \to i} V_{l_s})(c), & c \in \Gamma_i' \setminus \Gamma_i. \end{cases} \tag{999}$$

In the second line s is the part of the tower bond c. Assume that this sequence is separated, that each tower bond has both endpoints in one wide collar, and that both data are regular at ε_{r}.

Choose the geometric floors for the response and exponential summation bound, in particular $\delta_{\mathrm{lc}} R_1 \mathfrak{M} \ge 8(3\log L + \log 2)$. Let $A_L = 3(L-1)^2/4$ and let

$$C_{\mathrm{AC}} = 8(L-1)A_L + (1+\pi/3)(L(L-1) + 8A_L), \qquad C_{\mathrm{al}} = \frac{\pi C_{\mathrm{AC}} B_3}{L^2 - 1}.$$

Choose the one-step curvature threshold $\omega_\sharp$ below both the mean threshold divided by A_L and the averaging threshold. It suffices to require

$$\varepsilon_{\mathrm{r}} \le \epsilon_{\mathrm{lc}} := \min\left\{\bar{\varepsilon}, \frac{a_{\mathrm{R}}}{C_{\mathfrak{P}}\mathfrak{M}B_3}, \frac{c_2^{\mathrm{SO}}}{B_3}, \frac{L^2\omega_\sharp}{4B_3}, \frac{\varrho}{2C_{\mathrm{al}}}\right\}. \tag{1000}$$

All thresholds here are the real-background, response, mean, and curvature-iteration thresholds already fixed; none depends on the depth or volume.

Proposition 53.5. Under these hypotheses, and the size condition $2L^{m+K-n} \geq 6R_1\mathfrak{M}$, let U, U' be arbitrary elements of the two minimal orbits. There are gauges w, w' on the bare sites of Ω and a real odd field a on its inner bonds such that

$$(U|_\Omega)^w = e^{ia}(U')^{w'}. \tag{1001}$$

Write $D(z)$ for the weighted distance in the joined sequence from z to the endpoints of its tower bonds, and $D(b) = \min\{D(b_-), D(b_+)\}$. An empty set of tower bonds has distance $+\infty$. For $j(b) = \max\{j : b_- \in \Omega'_j \text{ or } b_+ \in \Omega'_j\}$, the same triple satisfies, outside the wide collars,

$$|a(b)| \leq C_{\mathrm{lc}}\varepsilon_{\mathrm{r}} L^{-j(b)} e^{-\delta_{\mathrm{lc}} D(b)/2}, \tag{1002}$$

$$w'(y) = 1, \qquad |w(y) - 1| \leq C_{\mathrm{lc}}\varepsilon_{\mathrm{r}} e^{-\delta_{\mathrm{lc}} D(y)/2} \quad (y \text{ a node}). \tag{1003}$$

One may take $C_{\mathrm{lc}} = 4\kappa_6 B_5 S(\delta_{\mathrm{lc}}/2) C_{\mathrm{al}}$, where S is the exponential row-sum constant. ∎

Proof. The joined sequence is subordinate to the original sequence. The restriction statement lemma 51.13 shows that $U|_\Omega$ is regular and critical for its own datum V^{rs}, and that $V^{\mathrm{rs}} = V'$ on the common rows. We first align the remaining data in each wide collar.

In the collar of level l, regularity gives bare curvature at most $B_3\varepsilon_{\mathrm{r}} L^{-2l}$. The curvature averaging estimate gives $\varpi_q \leq 2B_3\varepsilon_{\mathrm{r}} L^{2(q-l)}$ for $q \leq l$. Recursively set $v_l = 1$ and

$$v_q(x) = v_{q+1}(z)(\mathcal{M}^q U)(z, x), \qquad x \in B(z).$$

Gauge covariance makes all the one-step averaged contour variables of $\widehat{W}_q = \mathcal{M}^q(U^{v_0})$ equal to one. We recall the resulting elementary double-block estimate. If a configuration has curvature at most a, its path transports in a block differ by at most $A_L a$; when their mean is one each transport is within $4A_L a$ of one. Appending one final edge gives the bound $8A_L a$ on an interior bond. For a crossing bond, commute the L parallel steps past the transverse part of a path. There are at most $L(L-1)$ plaquette exchanges. The line transport therefore differs from the central transport by $(L(L-1) + 8A_L)a$. The logarithmic mean contributes at most $\pi/3$ times this bound, and the other $L-1$ line bonds contribute $8(L-1)A_L a$. Hence

$$|\widehat{W}_q(b)\widehat{W}_{q+1}(c)^{-1} - 1| \leq C_{\mathrm{AC}}\varpi_q$$

when b crosses the blocks of c. Inside one block the lift is one. Descending induction consequently gives

$$|\widehat{W}_q(b)(Q^{s*}_{l\to q}V_l)(b)^{-1} - 1| \leq C_{\mathrm{AC}} \sum_{i=q}^{l-1} \varpi_i \leq \frac{2C_{\mathrm{AC}}B_3}{L^2 - 1}\varepsilon_{\mathrm{r}}.$$

All blocks used are contained in the wide collar. Its top bonds are common rows, so their averages are V_l.

The wide collars are disjoint. Glue these gauges to the identity outside them, obtaining g. At the endpoints of a common row in a wide collar, the endpoint is a level-l_s site and the collar gauge is one. Thus the common datum is unchanged, while the datum of $(U|_\Omega)^g$ is $e^{iB_1}V'$, with

$$B_1 = 0 \text{ on common rows}, \qquad \sup|B_1| \leq C_{\mathrm{al}}\varepsilon_{\mathrm{r}} \leq \varrho/2.$$

The intrinsic logarithm exists because the preceding distance is less than two; its norm is at most $\pi/2$ times that distance.

Let U° be the normalized background at V'. Uniqueness of the regular critical orbit and lemma 51.6 give residual gauges w_0, w_1 with

$$(U|_\Omega)^g = \big((e^{i\eta_n \mathcal{H}(B_1)}U^\circ)^{u_{B_1}}\big)^{w_0}, \qquad U' = (U^\circ)^{w_1}.$$

Put $w = (w_0 u_{B_1})^{-1} g$, $w' = w_1^{-1}$, and $a = \eta_n \mathcal{H}(B_1)$. This proves (1001). Since B_1 is supported on tower rows, exponential summation gives

$$m_y(B_1) \le C_{\mathrm{al}} \varepsilon_{\mathrm{r}} S(\delta_{\mathrm{lc}}/2) e^{-\delta_{\mathrm{lc}} D(y)/2}.$$

For an endpoint of b in layer $j(b)$, its top has scale $\xi_y = L^{j(b)-n}$ and distance at most $3/2$ from that endpoint. Apply (996) and absorb $e^{3\delta_{\mathrm{lc}}/4}$ into $4\kappa_6$. This gives (1002), including the factor $\eta_n/\xi_y = L^{-j(b)}$. At a node outside the collars, $w_0 = w_1 = g = 1$, so $w' = 1$ and $w = u_{B_1}^{-1}$. The last estimate of (996) gives (1003). □

We next verify these hypotheses for the strip. Use level-k lengths, $t = k+1$, and the completed collared sequence (640). Define

$$\Omega^{\mathrm{loc}} = Z_t \cap \Lambda_k \cap \Omega_0^{\diamond}, \qquad (Y_1, l_1) = (\Lambda_t, t), \qquad (Y_2, l_2) = \begin{cases} (Z_k, k), & t \ge 2, \\ ((\Omega_0^{\diamond})^c, 0), & t = 1. \end{cases} \tag{1004}$$

The first-step convention is $\Lambda_0 = \mathbb{T}$. Form its collars, core, joined sequence and datum by (999). For a component $\mathfrak{o}$ of Ω^{loc}, restrict each joined region to $\mathfrak{o}$, and denote the resulting reduced determining set and datum by $\mathbf{B}^{\mathfrak{o}}, V^{\mathfrak{o}}$.

Lemma 53.6. The strip joined sequences, on the full localization region and on each component, are separated and subordinate to the completed collared sequence. They satisfy the geometric hypotheses of proposition 53.5. If the completed datum is regular layer by layer at $(2\varepsilon_i)_{i \le t}$, the joined data are regular at $2\varepsilon_t$. The component minimal orbits exist and their product is the minimal orbit of the joined datum, provided $2\varepsilon_t \le \epsilon_{\mathrm{lc}}$. The component datum uses only the original rows of levels k, t whose endpoints lie in that component. ■

Proof. Write $\sigma_i = M_1 L^{i-k}$ and $\rho_s(x) = \mathrm{dist}_\infty(x, Y_s)$. The domain and exploration rules give

$$\mathrm{dist}_\infty(Y_1, \Omega_t^c) \ge 2W_t, \quad \mathrm{dist}_\infty(Y_2, \Omega_t) \ge W_t, \quad \mathrm{dist}_\infty(Y_1, Y_2) \ge 3W_t.$$

For $t \ge 2$, the second distance is at least $8W_t$; for $t = 1$, it is the one-layer enlargement $\Omega_0^{\diamond} = \Omega_1^{\sim}$. Here Λ_t is obtained by removing the two coarse layers about the collar and explored region; the earlier domain excludes the eight-layer hull of Z_k. Thus $(\Omega^{\mathrm{loc}})^c = Y_1 \cup Y_2$, and $\Omega^{\mathrm{loc}} = Z_t \setminus Z_k$ for $t \ge 2$. The regions are unions of M_1-cubes at level t. Since W_t is larger than $4LM_1$, a point in $Y_s^{\sim 4} \setminus Y_s$ belongs to the old layer l_s. The displayed separation makes the two wide collars disjoint and proves (998).

The tower membership condition at a bare site $x \in \Omega^{\mathrm{loc}}$ is

$$x \in \Omega^{[q]} \iff \rho_s(x) > a_q \sigma_q \ (s = 1, 2), \qquad a_0 = 0,\ a_1 = 1,\ a_q = 2 \ (q \ge 2).$$

Indeed the level-q cube at x is retained precisely when it is at distance at least $a_q \sigma_q$ from the complement. On a collar the old membership condition is $q \le l_s$; in the core the displayed inequalities hold at every old level containing x. Therefore

$$\Omega'_q = \Omega_q^{\diamond} \cap \Omega^{[q]}.$$

For $q \le k$ this is $\Omega^{[q]}$. At the last level it is $\Omega_t \cap \Omega^{\natural}$ if $t \ge 2$; if $t = 1$ it is $\Omega_1 \setminus \Lambda_1^{\sim 1}$, with the latter enlargement in level-1 M_1-cubes. The tower rule gives $\mathrm{dist}_\infty((\Omega'_q)^c, \Omega'_{q+1}) \ge \sigma_{q+1}$. Since $M_1 \ge R_1 \mathfrak{M}$, these are the required separation inequalities.

Membership in a reduced determining set is decided by the endpoints. In the core old and joined memberships agree. If a joined row is not an old row, one endpoint is in C_s; the other is within L, hence in C_s^{w}. Its level is strictly below l_s, for at level l_s both endpoints meet exactly the old-row condition. An endpoint in $C_s^{\mathrm{w}} \setminus C_s$ would belong to the next joined region, which a joined row excludes. Thus both endpoints are in C_s. The lift value is one inside a level-l_s block, or the value of an old level-l_s row whose endpoints remain in that collar. Common rows have levels k, t, since $\Omega^{\mathrm{loc}} \subset \Omega_k^{\diamond}$. This also proves the stated datum dependence.

For completeness, consider a tested joined plaquette. If a corner lies in C_s and its level $q < l_s$, all four corners lie in that collar: a corner outside it would be in Ω'_{q+1}. Every edge value, including one obtained by averaging the finer rows, is the corresponding value of $Q^{s*}_{l_s\to q}V_{l_s}$. Its holonomy is one or an old tested holonomy at level l_s, by the lift calculation in proposition 44.2. Otherwise $q = k$ or t, and it is an old tested plaquette. Its direct edge values are common data. For an edge represented by finer joined rows, those rows are either all common, or are the lift in the collar with $l_s = q$; averaging the lift then recovers V_q. Thus all four edge values agree with the old holonomy. This proves regularity at $2\varepsilon_t$.

Distinct components of a region of level-t M_1-cubes are separated by at least LM_1. A bond or plaquette of a level at most t, with all its vertices in the region, therefore lies in one component. The determining rows, their averaging blocks, and the plaquette action split over components. The constraint set and residual gauge group are products, the action is a sum, and criticality and regularity are componentwise conditions. The regular-background theorem applies to every separated component; their product is the joined minimal orbit. □

Let $S^{\mathrm{pr}}_t = \mathfrak{S} \cup R$. Every strip bond has both endpoints in one component $\mathfrak{o}$. Let $Y^{\mathfrak{o}}_S$ consist of the level-t sites in $\Omega_t \cap \mathfrak{o}$ whose blocks contain an endpoint of such a bond. These sites are nodes of $\mathbf{B}^{\mathfrak{o}}$. No other site of their blocks is a node. An endpoint x of a strip bond either belongs to one of these blocks, or is itself a level-k node. Define its anchor $y(x)$ to be the block center in the first case, and x in the second.

Here are the geometric facts which justify this prescription. The strip is at distance at least W_t from Y_1, Y_2. If a coarse bond c has an endpoint in the strip, every bare site z in $B^t(c_-) \cup B^t(c_+)$ is within $3L/2$ of that endpoint. Therefore it is in the core, in Ω'_k, and in no wide collar. A tower endpoint is within $3LM_1$ of Y_1 or Y_2. Since the largest joined scale is L in the present units, weighted distance is at least L^{-1} times sup distance. Consequently

$$D(z) \geq M\check{R}_t - 3M_1 - 2 \geq \tfrac{1}{2}M\check{R}_t \qquad (z \in B^t(c_-) \cup B^t(c_+)). \tag{1005}$$

The inequalities use the running geometric floors $M_2 = L^2M_1$, $M \geq L^2M_2$, and $\check{R}_t \geq 1$. Every strip bond belongs to one such double block: if it crosses blocks take their common coarse bond; if it stays inside one block take a neighboring center within its test cube. The double block belongs to a single component.

If a strip endpoint x is in Ω_t, its center is in Ω'_t; a neighboring coarse bond in the core shows that center is a node. Otherwise choose a nearest neighbor of x inside its level-t block. Both are in $\Omega'_k \setminus \Omega'_t$, so their bond is a determining row and x is a node. This proves the assertions about the anchors. In particular (1005) holds at both anchors of a strip bond.

For a configuration W on the double block of a strip bond b, put

$$\mathrm{ax}_W(x) = \begin{cases} W(y(x), x), & x \in B(Y^{\mathfrak{o}}_S), \\ 1, & \text{otherwise}, \end{cases} \qquad \mathrm{dr}_b(W) = \mathrm{ax}_W(b_-)W(b)\,\mathrm{ax}_W(b_+)^{-1}.$$

Under a gauge v, its transformation law is

$$\mathrm{dr}_b(W^v) = v(y(b_-))\,\mathrm{dr}_b(W)v(y(b_+))^{-1}.$$

The anchors are fixed by residual gauges. Put

$$c_2^{\mathrm{g}} = \min\{a_{\mathsf{m}}, c_2^{\mathrm{SO}}, \varpi_{\mathrm{av}}/2\}, \qquad c_{\mathrm{strip}} = \min\{c_{\mathsf{m}}, c_{\mathrm{plain}}, r_{\mathrm{av}}/4\}, \qquad C_v = 4\kappa_6.$$

These are curvature, perturbation and gauge-factor constants for the centered iteration in propositions 30.2 and 51.1. For $2L^2B_3\varepsilon_t \leq c_2^{\mathrm{g}}$, curvature averaging on these double-block towers puts every intermediate configuration in the open mean domain. Thus the local strip background is

$$V_k^{\mathrm{bg,loc}\diamond}(b) = \mathrm{dr}_b(\mathcal{M}^k U^{\mathfrak{o}}), \qquad U^{\mathfrak{o}} \in \mathcal{C}(\mathbf{B}^{\mathfrak{o}}, V^{\mathfrak{o}}). \tag{1006}$$

It is independent of the representative. Equivalently, choose one for which $\mathcal{M}^k U^{\circ}$ is axial on Y_S°, and restrict its average to the strip. The normalization is permitted because those centers are nodes and their other block sites are not nodes.

The set of data regular layer by layer is open. Indeed the plain tested holonomies are continuous; on their regularity domain, the finer double blocks used by the other tested holonomies lie in the open averaging domain. All tested holonomies are therefore continuous there, and only finitely many strict inequalities remain. The response lemma 51.6 supplies a continuous local choice of each component representative. Successive averaging and the contour means are continuous on the curvature domain just described. Formula (1006) therefore proves stable definedness and continuity of the local strip background on the regular layer-data domain.

Set

$$c_L = 4 + 2C_v + 4\kappa_6, \qquad C_{\mathrm{xp}} = \pi c_L C_{\mathrm{lc}}, \qquad C_{\mathrm{cu}} = 4C_{\mathrm{lc}}\left(1 + \frac{1 + C_v + 2\kappa_6}{L}\right). \tag{1007}$$

We shall use the following explicit smallness bounds:

$$2\varepsilon_t \le \epsilon_{\mathrm{lc}}, \qquad 2L^2 B_3 \varepsilon_t \le c_2^g, \qquad 2LC_{\mathrm{lc}}\varepsilon_t \le c_{\mathrm{strip}}/2, \qquad 2c_L C_{\mathrm{lc}}\varepsilon_t \le 1. \tag{1008}$$

They are uniform upper bounds on ε_t.

Proposition 53.7. Suppose the completed datum is regular layer by layer at $(2\varepsilon_i)_{i\le t}$, the running geometry and the localization floors above hold, and (1008) holds. At every positive strip bond the intrinsic logarithm

$$H_b = \frac{1}{i}\log\left[V_k^{\mathrm{bg}\diamond}(b) V_k^{\mathrm{bg,loc}\diamond}(b)^{-1}\right] \tag{1009}$$

exists and satisfies

$$|H_b| \le C_{\mathrm{xp}}\varepsilon_t e^{-\delta_{\mathrm{lc}} M \check{R}_t/4}. \tag{1010}$$

If the running parameters also satisfy

$$8\|\vartheta'_{\mathrm{co}}\|_\infty C_{\mathrm{xp}}\sqrt{2L}\frac{A_0}{A_1} e^{-\delta_{\mathrm{lc}} M \check{R}_t/4} \le 1, \tag{1011}$$

then (991) holds and $|H_b| \le \delta_k/24$. Both backgrounds and H_b are continuous on this regular domain, depend only on the datum of the step, and are independent of the internal fine variables. ∎

Proof. Use a representative U of the completed orbit whose level-k average is axial on $\Omega_t^{(t)}$. For the component of b, choose U° axial on Y_S°, and complete it by arbitrary component representatives to an element U' of the joined orbit. The regularity and geometry required in proposition 53.5 hold by lemma 53.6, with $\varepsilon_{\mathrm{r}} = 2\varepsilon_t$. Put

$$s = 2C_{\mathrm{lc}}\varepsilon_t e^{-\delta_{\mathrm{lc}} M \check{R}_t/4}.$$

On the bare double block of b, (1005) and (1002) give $|a| \le L^{-k}s$. At the anchors (1003) gives $w' = 1$ and $|w - 1| \le s$. The joined field has bare curvature at most $2B_3\varepsilon_t L^{-2k}$ on that double block.

Apply the centered averaging factorization to $(U|_\Omega)^w = e^{ia}(U')^{w'}$ on this tower of height t, through level k. Its curvature parameter is $2L^2B_3\varepsilon_t$, and $L^t \sup|a| \le Ls < c_{\mathrm{strip}}$, by (1008). Writing $W_0 = \mathcal{M}^k((U')^{w'})$ and $W_1 = \mathcal{M}^k((U|_\Omega)^w)$, it gives

$$W_1 = (e^{ia_k} W_0)^{f_k}, \qquad \sup|a_k| \le 2s, \quad \sup|a_k| \le r_{\mathrm{av}}/2, \quad |f_k - 1| \le C_v s, \quad \varpi_c(W_0) \le \varpi_{\mathrm{av}}.$$

These are precisely the intermediate estimates in the proof of proposition 51.1; the bounds are local to the double-block tower. For $k = 0$ the factorization is the identity. The one-step contour estimate gives

$$|\operatorname{ax}_{e^{ia_k}W_0}(x)\operatorname{ax}_{W_0}(x)^{-1} - 1| \le 2\kappa_6 s.$$

Write $y_\pm = y(b_\pm)$. Covariance of the dressed values, the two axial normalizations, and $w'(y_\pm) = 1$ give

$$V_k^{\mathrm{bg}\diamond}(b) = w(y_-)^{-1} f_k(y_-)\,\mathrm{ax}_{e^{ia_k}W_0}(b_-)(e^{ia_k}W_0)(b)\,\mathrm{ax}_{e^{ia_k}W_0}(b_+)^{-1} f_k(y_+)^{-1} w(y_+),$$
$$V_k^{\mathrm{bg,loc}\diamond}(b) = \mathrm{ax}_{W_0}(b_-)W_0(b)\,\mathrm{ax}_{W_0}(b_+)^{-1}.$$

All factors are unitary. Comparing their seven factors bounds the distance of the two backgrounds by

$$(2 + 2C_v + 4\kappa_6 + 2)s = c_L s \le 1.$$

The relative product is therefore different from -1. Its intrinsic logarithm has norm at most $\pi c_L s/2$, which proves (1010).

The running relation $\varepsilon_t \le \sqrt{2L}\varepsilon_k = \sqrt{2L}(A_0/A_1)\delta_k$, together with (1011), gives (991). Since $\|\vartheta'_{\mathrm{co}}\|_\infty \ge 3$, it also gives $|H_b| \le \delta_k/24$. Continuity follows from the stable backgrounds and the logarithm. Their defining data use only completed determining rows. No bond of $\mathfrak{X}_0$ belongs to those rows: at level k such a row excludes endpoints in Ω_t. This proves independence of the internal variables. □

The cutoff-zone tests also compare the strip background with a test-cube background. The required estimate follows from the same localization argument:

$$|V_k^{\mathrm{bg}\diamond}(b) V_{k,\square}^{\mathrm{bg}}(b)^{-1} - 1| \le C_{\mathrm{cu}}\varepsilon_t e^{-\delta_{\mathrm{lc}} M_2 \check{R}_t/4}, \quad b_\pm \in \square^{\sim 3}, \quad \square^{\sim 4} \subset \Omega_t. \tag{1012}$$

To verify its applicability, take $\Omega = \square^{\sim 4}$, $Y_1 = \Omega^c$, $l_1 = t$ in proposition 53.5. The whole region lies in the last old layer, so the joined sequence is exactly its $\mathfrak{r}_2$-tower. The common rows are its level-t rows; all lower rows are tower rows in the two-layer collar. The joined datum is the induced lift of V_t, and is regular by proposition 44.2. It thus defines exactly the background appearing in (637). Each point of $\square^{\sim 3}$ has weighted distance at least $M_2\check{R}_t - 2M_1 \ge M_2\check{R}_t/2$ from the tower rows. Indeed its sup distance from Ω^c is at least $LM_2\check{R}_t$, the collar has width $2LM_1$, and the largest weighted scale is L.

On a double block inside $\square^{\sim 3}$, set $s = 2C_{\mathrm{lc}}\varepsilon_t e^{-\delta_{\mathrm{lc}} D/2}$ with this lower bound D. Now the bare field in (1001) has size at most $L^{-t}s$, so at level k the centered factorization gives $\sup|a_k| \le 2L^{-1}s$ and $|f_k - 1| \le C_v L^{-1} s$. The two backgrounds are axial on both coarse block centers. For the gauge $g = (w')^{-1} f_k^{-1} w$, their relation is

$$g(b_-)V(b)g(b_+)^{-1} = e^{i\beta(b)}V'(b), \qquad |\beta| \le 2L^{-1}s, \qquad |g(c_\pm) - 1| \le (1 + C_v/L)s.$$

Their contour means are one. The contour estimate therefore bounds $|g(x) - 1|$ by $(1 + C_v/L)s + \kappa_6 \sup|\beta|$ throughout the double block. Substitution in the bond relation bounds $|VV'^{-1} - 1|$ by

$$\left(2 + \frac{2(1 + C_v + 2\kappa_6)}{L}\right) s.$$

Every bond with endpoints in $\square^{\sim 3}$ lies in such a double block. This proves (1012) with (1007).

Suppose, in addition,

$$8C_{\mathrm{cu}}\sqrt{2L}\frac{A_0}{A_1} e^{-\delta_{\mathrm{lc}} M_2 \check{R}_t/4} \le 1. \tag{1013}$$

If a cutoff-zone bond belongs to the test set of a cube $\square$, then $\square^{\sim 4} \subset \Omega_t$. In fact it is also tested by an explored cube $\square'' \subset \Omega_t^{\sim -1}$; their distance is at most $4LM_2\check{R}_t + 1$, whereas this coarse interior is at distance W_t from Ω_t^c. Adding the diameter of $\square$ and its four-layer enlargement still gives a distance less than W_t, by $M \ge L^2 M_2$. Thus, for $X = V_k^{\mathrm{bg,loc}\diamond}(b) V_{k,\square}^{\mathrm{bg}}(b)^{-1}$,

$$|X - 1| \le |H_b| + \delta_k/8 \le \delta_k/6. \tag{1014}$$

For every $W' \in G$ this implies

$$\mathbf{1}_{\{|\Psi_{H_b}(W')X - 1| < 2\delta_k\}} = \mathbf{1}_{\{|W'X - 1| < 2\delta_k\}}. \tag{1015}$$

If $\theta(W')=1$, the two products agree. Otherwise $|W'-1|<5\delta_k/3$; the second product is within $11\delta_k/6$ of one, and the first is within $(5/3+1/24+1/6)\delta_k=15\delta_k/8$. Both indicators are then one.

The running cutoff supplies $\check R_t\ge y_t^r$ and $y_t\ge -2\log\gamma$. Consequently (1011) and (1013) hold uniformly in the step if $\gamma\le e^{-y_{\mathrm{xp}}/2}$, where

$$\begin{aligned}\rho&=\max\{1,A_0/A_1\},\\ \Upsilon_1&=8\|\vartheta'_{\mathrm{co}}\|_\infty C_{\mathrm{xp}}\sqrt{2L}\rho,\qquad \Upsilon_2=8C_{\mathrm{cu}}\sqrt{2L}\rho,\\ y_{\mathrm{xp}}&=\max\left\{\left(\frac{4\log\Upsilon_1}{\delta_{\mathrm{lc}}M}\right)^{1/r},\left(\frac{4\log\Upsilon_2}{\delta_{\mathrm{lc}}M_2}\right)^{1/r}\right\}.\end{aligned}\tag{1016}$$

This choice is made after the geometric constants and the ratio A_0/A_1; it is independent of the history, volume and level.

53.5. **Stored tests and their readings.** We now specify the factors that remain after a step. Products are indexed by occurrences of tests, so that distinct occurrences are retained even when they define the same function. Assume throughout that $\check R_l\le L\check R_{l+1}$. At step $t=l+1$, split each passing Q-test into its one-bond factors, and each passing rim test into its one-bond inequalities. A failed test remains one factor $1-\chi$. Split the gauge factor at (y,x) into its indicator and its normalized exponential.

The stored factors are the following. Outside Ω_t, retain the density and P-tests, the gauge factors based at $y\notin\Omega_t$, and the Q-tests. Also retain the density tests of level l in $\Omega_l\setminus\Omega_t$. Inside Ω_t, retain the one-bond Q-tests whose tested bond does not belong to $\mathcal{N}_{\mathrm{s}}$, all fluctuation inequalities at external bonds, and the failed rim and cube tests. The fluctuation inequalities at bonds meeting Λ_t will instead remain under the integral. Passing density tests in Ω_t, gauge factors based there, and one-bond Q-tests on $\mathcal{N}_{\mathrm{s}}$ will be treated separately in the support identity.

For a configuration argument (j,b), its *reading* is obtained recursively from the free coordinates. A free bond is read as its coordinate; a tied bond is read by averaging its previous-level double block. On a zone bond of step t, use

$$V_l[\mathbf{v},\mathfrak{q}](b)=e^{ig_l\mathfrak{q}_l(b)}V_l^{\mathrm{bg,loc}\diamond}(b).\tag{1017}$$

Here the local background is evaluated on its determining free coordinates in the component of $\Omega^{\mathrm{loc}\diamond}$ containing b. At a pivot passenger $b_0(c)$, use the Borel pivot rule of proposition 14.11, applied to the readings on P_c° and to the level-t coordinate $V_t(c)$. An equation can be used only after all its arguments have been read and only on the domain of its right side. The rank is zero for free coordinates and one plus the maximum argument rank for a newly read value.

For a test χ of configuration variables, set χ^{wr} equal to the test of its readings when they are all defined and equal to zero otherwise. The written complement is $1-\chi^{\mathrm{wr}}$, including where a reading is undefined. A fluctuation test is written by replacing $|\mathcal{A}_l(b)|$ by $g_l|\mathfrak{q}_l(b)|$. The latter rule uses external coordinates only. Equation (1017) is a coordinate reading; in the cutoff zone its value need not be the physical field $\Psi_H(e^{ig_l\mathfrak{q}_l(b)})V_l^{\mathrm{bg,loc}\diamond}(b)$.

Proposition 53.8. Every stored factor f of step $t=l+1$ is determined by free configuration coordinates of levels at most t and external coordinates $\mathfrak{q}_l$, all located in one component D_f of Z_t. Its written value is Borel and bounded. It takes values in $\{0,1\}$, except for a normalized gauge exponential, which takes values in $[0,\mathsf{z}(x_l,\varepsilon_l)^{-1}]$.

Every configuration argument of f is of one of four kinds: a free bond; a tied bond; a zone bond of step t; or a pivot passenger $b_0(c)$ of that step with

$$\mathsf{P}_c^\circ\subset\Gamma_l^{\mathrm{pr}}\cup\mathcal{Z}_1.$$

Its reading uses only tying equations and zone and pivot equations of step t. In particular no global-background reading and no strip equation of another step enters. For a history of length

k, dependence of a stored factor on a current coordinate $V_k(b)$ is possible only for $t = k$, and then both endpoints of b lie in Z_k. ■

Proof. We give the geometric check, since it is what prevents a stored factor from involving the next integration variables. Use level-l lengths and put $w = LM_2\check{R}_t$. The parameter floors imply

$$6w + (L-1) < W_t = LM\check{R}_t.$$

Every point at distance at most W_t from Ω_t^c lies in Z_t. A test cube contained in $\Omega_t^{\sim -1}$ is at distance at least W_t from Ω_t^c. Consequently a bond whose endpoints are at distance at most ρ from Ω_t^c, with $\rho + 2w < W_t$, cannot belong to $\mathcal{N}_{\mathrm{s}} \cup \mathcal{N}_\ell$. If it meets Ω_t and is nonpivot, it therefore belongs to the plain zone $\mathcal{Z}_1$; if it does not meet Ω_t, it is proper. If it is a pivot $b_0(c)$, the entire double block lies within distance $\rho + L - 1$ of Ω_t^c. Provided $\rho + L - 1 + 2w < W_t$, all its nonpivot bonds are therefore proper or in $\mathcal{Z}_1$. The endpoints of c are outside Λ_t, so c is a free bond of level t.

For each stored factor choose a connected set containing its arguments as follows:

factor	set containing its arguments
bare density test on $\square$	$\square^{\sim}$
P- or Q-test on $\square'$	$\square'^{\sim 4}$
gauge factor based at y	$\Delta_{K-t}(y)$
carried density test on $\square$	$\square^{\sim 4}$
external one-bond inequality at b	$[b_-, b_+]$
failed rim test at c	$c^{\sim}$
failed cube test on $\square'$	$\square'^{\sim 3}$.

For factors outside Ω_t these sets lie within distance $4w$ of Ω_t^c. For a carried density test this uses $\check{R}_l \le L\check{R}_t$. Thus the sets lie in Z_t, and the preceding argument with $\rho = 4w$ classifies all their level-l arguments. Level-t arguments are proper, since their endpoints lie in Z_t, outside Ω_{t+1}.

For an inside one-bond Q-factor, let $\square'$ be its test cube. If $\square'$ were in $\Omega_t^{\sim -1} \setminus R$, all its tested bonds would belong to $\mathfrak{U}(R) = \mathcal{N}_{\mathrm{s}}$. Thus $\square' \subset \mathfrak{S} \cup R$. Its four-layer enlargement lies in Z_t, because $4w < W_t$ and the exploration leaves a coarse collar around $\mathfrak{S} \cup R'$. Its sole level-l argument is either proper or a zone bond; it is nonpivot. The sets for failed rim and cube tests lie in Z_t by the same collar, and all their tested bonds are external. An external bond has both endpoints in Z_t. Each set in the table is connected, hence lies in a unique component D_f.

It remains to follow the equations, including their domains. For a proper bond, each tying step lowers the level and uses only its double block. A region made of the relevant coarse cubes contains these double blocks whenever it contains their endpoints. Induction on the level therefore keeps every entering free coordinate in D_f. A zone equation has rank one: the joined background depends only on its determining free bonds in one component $\mathfrak{o} \subset \Omega^{\mathrm{loc}\diamond}$. Both endpoints of the zone bond lie in $\mathfrak{o}$, so $\mathfrak{o} \subset D_f$. For a pivot equation, $V_t(c)$ has rank zero and its other arguments are proper or plain-zone bonds. Its rank is at most $\max\{l, 1\} + 1$. The double block and all components $\mathfrak{o}$ entering it lie in the same D_f. This proves termination, the four alternatives, and the asserted coordinate dependence.

The zone background has an open domain in its determining coordinates and is continuous there. To see that eliminating irrelevant coordinates preserves this assertion, fix their values: the stable domain is a product with those coordinates, and its section is open, with the same continuous value. The pivot rule is Borel by proposition 14.11; tying maps are Borel. Finite composition therefore gives Borel partial readings. The original tests are Borel functions of their arguments: their stable domains are open, their defining backgrounds are continuous there, and their thresholds are strict. Zero extension and taking $1 - \chi^{\mathrm{wr}}$ preserve measurability. The written fluctuation tests are finite combinations of the inequalities $g_l|\mathfrak{q}_l(b)| < s$. Their ranges, and the bound on the gauge exponential, follow from their formulas. Finally the coordinate dependence just proved has level at most t. Since $t \le k$, a current coordinate can enter only when $t = k$, and then it is located in $D_f \subset Z_k$. □

53.6. **Regularity at the layers of a history.** The global regularity radius does not suffice when the thresholds increase with the level. We first localize the radius in terms of the tested holonomies of the datum. Only reduced determining sets are used in this subsection.

For a bare plaquette or a bare bond at which the current is defined, let $\lambda(\mathfrak{e})$ be the largest index of a corner. In bare units define

$$r_U(p) = L^{2\lambda(p)}|\partial U(p) - 1|, \qquad r_U(b) = L^{3\lambda(b)}|J^{(0)}(U)(b)|.$$

These are the scale-independent local regularity radii. At each bare site x, let $r(x)$ be the maximum of the radii of the elements based there, with empty maximum zero.

Lemma 53.9. There are a power σ of L, a constant $K_{\mathrm{loc}} \geq 4$, and $a_{\mathrm{c}} > 0$, depending only on L, with the following property. Assume the separated region setting, $\mathfrak{M} \geq \mathfrak{M}_{\mathrm{c}}$, $R_1\mathfrak{M} \geq L^2\sigma$, and at least $6R_1\mathfrak{M}$ sites in each terminal-lattice direction. Let U be critical over a reduced datum V, and regular at radius a_{c}. Put $R_{\mathrm{loc}} = 15L\sigma$ and

$$r_R(x) = \max_{d_{\mathbf{B}}(x,z)\leq R_{\mathrm{loc}}} r(z), \qquad v_R(x) = \max_{\substack{p' \text{ tested} \\ d_{\mathbf{B}}(x,p')\leq R_{\mathrm{loc}}}} |\partial V(p') - 1|.$$

Then

$$r(x) \leq K_{\mathrm{loc}} v_R(x) + \tfrac{1}{2} r_R(x). \tag{1018}$$

The constants are uniform in the depth and in the number of changes of the region sequence. ■

Proof. We describe the localization of the proof of proposition 37.1. Its reference configuration must now have a global bound controlled by $r_R(x)$, rather than by the largest radius on the original lattice.

Let j be the layer index of x. If $j = 0$ and x is farther than 5σ bare units from Ω_1, the plaquettes at x, and those in the current at x, are tested level-zero plaquettes. Thus $r(x) \leq 4v_R(x)$. Otherwise choose $q = j$ when $j \geq 1$ and $\mathrm{dist}_\infty(x, \Omega_{j+1}) > 5\sigma L^j$, and choose $q = j + 1$ in the remaining case. Separation implies $1 \leq q \leq n$. In level-q units, the cube of radius 5σ about x lies in Ω_{q-1}, misses both its complement and Ω_{q+1}, and has weighted distance at most $15L\sigma$ from x. Thus only the two layers $q - 1, q$ occur there.

Choose fixed powers $\mathfrak{M}_{\mathrm{c}}, R_{\mathrm{c}}$ large enough for the reduced inverse and slice estimates, and put $\rho_{\mathrm{c}} = R_{\mathrm{c}}\mathfrak{M}_{\mathrm{c}}$. Take $\sigma \geq L^2\rho_{\mathrm{c}}$. Let Q_0 be the cube of side σ containing x, and let Q_s be its enlargement by distance s, in level-q units. Set

$$s_q = \sigma - \rho_{\mathrm{c}}, \qquad s_i = s_q + \sum_{i\leq h<q} (\rho_{\mathrm{c}} + \mathfrak{M}_{\mathrm{c}})L^{h-q}, \qquad \Theta_0 = \mathbb{T}, \quad \Theta_i = Q_{s_i}\ (1 \leq i < q), \quad \Theta_q = Q_{s_q} \cap \Omega_q.$$

The intersections are cube-wise. These are admissible $\mathfrak{M}_{\mathrm{c}}$-regions, since their boundary coordinates differ by multiples of $\mathfrak{M}_{\mathrm{c}}L^{i-q}$. Moreover

$$s_1 < \sigma - \tfrac{5}{6}\rho_{\mathrm{c}} < \sigma - 2, \qquad s_i - s_{i+1} = (\rho_{\mathrm{c}} + \mathfrak{M}_{\mathrm{c}})L^{i-q}.$$

Hence this local sequence satisfies the required separation. Its index is at most the original index on Q_σ, and agrees with it on Q_{s_q}. The relevant cubes have a single torus chart: the side of the torus in these units is at least $2L^3\sigma$.

Write $e = r_R(x)$, $v = v_R(x)$. The cases $e = 0$ and v above a fixed positive threshold give (1018) immediately, after increasing K_{loc}. Otherwise the local curvature and current bounds and the regional gauge estimate provide

$$U^{u^{-1}} = e^{i\eta A_0}, \qquad |A_0|,\ |\nabla A_0| \leq C(L)\sigma e \quad \text{on } Q_\sigma, \qquad \eta = L^{-q}.$$

Indeed, on $Q_{3\sigma}$ the two layer indices give curvature at most $L^2e\eta^2$ and current at most L^3e; the relation between current and the full matrix divergence adds only its quadratic curvature term. The local gauge hypothesis is therefore satisfied once $C(L)\sigma e$ is sufficiently small. Extend A_0 by zero

outside the bonds inside Q_σ, and denote this extension by $\bar{A}_0$. For the reduced determining set of (Θ_i),

$$\|\bar{A}_0\|_X \le C(L)\sigma e, \qquad |J(e^{i\eta\bar{A}_0})|_{(-3)} \le C(L)\sigma e.$$

Inside $Q_{\sigma-2}$ the second bound is the original current bound, since the new local index is no larger. Outside that cube the local index is zero. The four factors of each plaquette then give $\eta^3|J| \le 16\eta \sup|\bar{A}_0|$, which proves the bound also at the abrupt extension.

The extension is critical for the local reduced constraints. To check this, a curve preserving them can vary only on bonds meeting $\Theta_1 \subset Q_{s_1}$. Paste that curve to $U^{u^{-1}}$ outside Q_σ. Inductively in the averaging level, a bond whose endpoints have local index at most its level has unchanged average: at an endpoint of exactly that index this is a local constraint, and at lower indices it follows from the previous level on the double block. Every original reduced row has this property, since $\Theta_i \subset \Omega_i$. Thus the pasted curve preserves the original datum. Every plaquette that varies lies inside Q_σ, and has the same action term in both curves. The first variation of the extended configuration therefore vanishes.

Use the coarse comb and the block contour transports, as in the datum gauge in the proof of proposition 37.1, to prescribe a gauge z on the local tops. On the coarse box within distance $\sigma/4$ of Q_0, the coarse plaquettes meeting Ω_q are tested. Outside Ω_q, their fine plaquettes are tested, and one averaging step bounds the coarse holonomy by $2L^2v$. The comb bound and the fine contour comparison consequently give

$$|W^z(c) - 1| \le C(L)(1 + |y_c - x|_\infty)v$$

on the local rows near x. The original small chart bounds the gauge itself by $\sup|z-1| \le C(L)\sigma^2 e$. The prescribed-top slice correction lemma 30.4, at reference field 1, now gives

$$E = e^{i\eta A} = (e^{i\eta\bar{A}_0})^w, \qquad R\partial^* A = 0, \qquad |A|_{(-1)} \le C(L)\sigma^2 e.$$

Its induced top gauge is z. At an outward row the additional factor at the outer endpoint is exactly one, because the fine block contour transports in the prescribed gauge are one. Thus the reduced nonlinear row $B = \mathcal{Q}(A)$ satisfies

$$\sup|B| \le C(L)\sigma^2 e, \qquad |B(c)| \le C(L)(1 + |y_c - x|_\infty)v \quad \text{if } \operatorname{dist}_\infty(y_c, Q_0) \le \sigma/4 - 2. \tag{1019}$$

There are no inward or edge rows in this reduced case. The logarithm used here is within its intrinsic branch, by the bound on $|A|_{(-1)}$.

For completeness, the slice equations used at this point have the bounds

$$\|A\|_X,\ |\partial^*\partial A|_{(-3)},\ \sup|\omega| \le r_f \le C(L)\sigma^2 e, \tag{1020}$$

where $F'(A) = (Q + \mathcal{C}'(A))^*\omega$. Indeed, with $\alpha = |J(E)|_{(-3)}$, $\beta = \sup|B|$, the equations

$$F'(A) = \partial^*\partial A + \mathcal{V}(A), \qquad QA = B - \mathcal{C}(A), \qquad R\partial^* A = 0$$

and the reduced inverse give

$$\|A\|_X \le C(L)(\alpha + \beta) + C(L)|A|_{(-1)}\|A\|_X.$$

Absorb the last term. Applying H_0^* to the multiplier equation and using the derivative bounds on $\mathcal{C}$ then gives $\sup|\omega| \le C(L)\|A\|_X + C(L)|A|_{(-1)}\sup|\omega|$, which is absorbed in the same way. The action remainder controls $|\partial^*\partial A|_{(-3)}$. These operations use the weighted row pairing $\sum_c \xi_c^{-1}\langle B(c), \omega(c)\rangle$. In particular their constants do not introduce a ratio of different layer scales. The Schur identities, with trivial reference current, are

$$\begin{aligned} A &= H_0 B + \widetilde{G}_0[\mathcal{C}'(A)^*\omega - \mathcal{V}(A)] - H_0\mathcal{C}(A), \\ \omega &= (\partial H_0)^*\partial A + H_0^*[\mathcal{V}(A) - \mathcal{C}'(A)^*\omega]. \end{aligned} \tag{1021}$$

The remainders in these two equations are bounded by $C(L)r_f^2$, respectively in X and the row supremum. This is also the specialization of (566) and (567) to the trivial background.

Apply the decay bounds (458) and (459). In the first equation of (1021), split the row sum at distance $\sigma/4-2$ from Q_0. For $|y-x|_\infty \le \sigma/8$, the near part uses (1019); summability of $(1+d)e^{-\delta d}$ bounds it by $C(L)(1+|y-x|_\infty)v$. The far part is bounded by $C(L)e^{-\delta\sigma/16}\sigma^2 e$. Thus

$$[\![A]\!]_y \le C(L)(1+|y-x|_\infty)v + C(L)e^{-\delta\sigma/16}\sigma^2 e + C(L)\sigma^4 e^2.$$

Use this estimate in the second equation of (1021), splitting now at $|y-x|_\infty = \sigma/8$. The same summation gives, for rows within $4L+2$ of x,

$$|\omega(c)| \le C(L)v + C(L)e^{-\delta\sigma/16}\sigma^2 e + C(L)\sigma^4 e^2.$$

The transpose estimate already incorporates the row weights; no additional power of the local spacing enters this bound. The plaquette estimate for $e^{i\eta A}$ and the local adjoint bound for $K_A^*\omega = F'(A)$ now control the curvature and current at x. The original and local indices agree there, and the neighboring scale conversions cost at most L^2 and L^3. We obtain

$$r(x) \le K^\sharp v + (C_b\sigma^2 e^{-\delta\sigma/16} + C_c\sigma^4 e)e,$$

where all three constants depend only on L. Choose the power σ first so that the exponential coefficient is at most $1/4$, and then choose a_{c} so that $C_c\sigma^4 a_{\mathrm{c}} \le 1/4$ and all preceding chart and slice inequalities hold. Taking $K_{\mathrm{loc}} \ge \max\{4, K^\sharp\}$ proves (1018). □

Corollary 53.10. Under the hypotheses of lemma 53.9, suppose the tested holonomies of level q are smaller than $\varepsilon^{(q)} > 0$. Set

$$\delta_0^{\mathrm{c}} = \frac{\log 2}{30L\sigma}, \qquad \mathsf{c}_{\mathrm{loc}}^{\mathrm{c}} = 10K_{\mathrm{loc}}, \qquad \mathsf{c}_{\mathrm{loc}}^{\mathrm{V}} = e\,\mathsf{c}_{\mathrm{loc}}^{\mathrm{c}}.$$

At every corner x' of a regularity element,

$$r_U(\mathfrak{e}) < \mathsf{c}_{\mathrm{loc}}^{\mathrm{V}} \sup_{\mathcal{L}_q \neq \varnothing} \varepsilon^{(q)} e^{-\delta_0^{\mathrm{c}} d_{\mathbf{B}}(x', \mathcal{L}_q)}. \tag{1022}$$

■□

Proof. Put $P(x) = \max_{p'} e^{-\delta_0^{\mathrm{c}} d_{\mathbf{B}}(x,p')}|\partial V(p')-1|$. The triangle inequality gives $P(z) \le e^{\delta_0^{\mathrm{c}} d_{\mathbf{B}}(x,z)}P(x)$. In particular, within radius R_{loc}, both $v_R(x) \le \sqrt{2}P(x)$ and $P(z) \le \sqrt{2}P(x)$. For $\tau > 0$, maximize $r(x)/(P(x)+\tau)$ on the finite bare lattice and call the maximum S. At a maximizing site, (1018) gives

$$S \le \sqrt{2}K_{\mathrm{loc}} + \frac{S}{\sqrt{2}}, \qquad S \le 2(\sqrt{2}+1)K_{\mathrm{loc}} \le 5K_{\mathrm{loc}}.$$

Let τ decrease to zero. Moving from the base of an element to any of its corners costs weighted distance at most two, so

$$r_U(\mathfrak{e}) \le 10K_{\mathrm{loc}}P(x').$$

A tested plaquette of level q has a corner in $\mathcal{L}_q$, and its other corners are within weighted distance $2L$ of that layer. Hence

$$d_{\mathbf{B}}(x', \mathcal{L}_q) \le d_{\mathbf{B}}(x', p') + 2L.$$

Since $2L\delta_0^{\mathrm{c}} \le 1$, each term of the finite profile is strictly below $e\varepsilon^{(q)}e^{-\delta_0^{\mathrm{c}} d_{\mathbf{B}}(x', \mathcal{L}_q)}$. This proves (1022). If there is no test, the maximum argument gives $r_U = 0$, while the right side is positive at every corner, which belongs to a nonempty layer. □

53.7. **Regularity and support of a step.** We apply the profile estimate to the completed collared sequence. Let

$$\kappa_{\mathrm{run}} = (1+\bar{B}\gamma^2)^{1/2} \le \sqrt{2}, \qquad C_{\mathrm{reg}} = 2L^{1/2}\kappa_{\mathrm{run}}\,\mathsf{c}_{\mathrm{loc}}^{\mathrm{V}}, \qquad \bar{C}_{\mathrm{reg}} = 2(2L)^{1/2}\mathsf{c}_{\mathrm{loc}}^{\mathrm{V}}.$$

Increase the geometric floors so that $16\delta_0^{\mathrm{c}} LM \ge \log L$, and take $\varepsilon_{\mathrm{v}} \le a_{\mathrm{c}}$. The running-coupling inequalities used below are

$$\varepsilon_q \le \varepsilon_j \quad (q \le j), \qquad \varepsilon_q \le L^{(q-j)/2}\kappa_{\mathrm{run}}\varepsilon_j \quad (q > j).$$

They hold on every initial segment of the running sequence.

Proposition 53.11. Suppose the datum of a completed collared sequence of depth n has tested holonomies smaller than $2\varepsilon_q$ at level q. Under the regular-data hypotheses of theorem E, its regular critical orbit exists and is unique. If x' is a corner of a regularity element and $x' \in \mathcal{L}_j$, then

$$r_U(\mathfrak{e}) < C_{\mathrm{reg}}\varepsilon_j \le \bar{C}_{\mathrm{reg}}\varepsilon_j. \tag{1023}$$

In particular, for $c \in \Gamma_j^{\diamond}$, every bare plaquette in $B^j(c_-) \cup B^j(c_+)$ satisfies

$$|\partial U(p) - 1| < C_{\mathrm{reg}}\varepsilon_j L^{-2(j-1)}.$$

■

Proof. Existence and uniqueness follow from theorem E, since $2\varepsilon_q \le 2\varepsilon_n \le \bar{\varepsilon}$. Every path between layers $j < q$ crosses the disjoint gaps between Ω_i^{c} and Ω_{i+1}, $j < i < q$. The gap at level i, measured with the weight L^{-i}, has length at least $8LM\check{R}_{i+1}$. Thus

$$d_{\mathbf{B}}(x', \mathcal{L}_q) \ge 8LM(|q-j|-1)_+. \tag{1024}$$

The same statement for $q < j$ follows by reversing the path. For $q > j$ the geometric floor therefore gives

$$2\varepsilon_q e^{-\delta_0^{\mathrm{c}} d_{\mathbf{B}}(x', \mathcal{L}_q)} \le 2L^{1/2}\kappa_{\mathrm{run}}\varepsilon_j;$$

for $q \le j$ the left side is at most $2\varepsilon_j$. Apply (1022). The endpoints of a row in $\Gamma_j^{\diamond}$, and their j-blocks, lie in layers $j-1$ or j. This follows from block membership for the standard partitions and separation of the sequence. The local regularity index of a plaquette in their union is at least $j-1$. The last assertion follows from the definition of r_U. □

Hereafter the running bounds include

$$\bar{C}_{\mathrm{reg}}\varepsilon_t \le \varepsilon_{\mathrm{v}}, \qquad L^2\bar{C}_{\mathrm{reg}}\varepsilon_t \le c_2^{\mathrm{SO}}, \qquad 2L^2\bar{C}_{\mathrm{reg}}\varepsilon_t \le \bar{\varepsilon}. \tag{1025}$$

We use $\mathsf{PQ}_t(Y)$ for exact averaging on Y and passage of the P, Q tests whose cubes are in Y. The fields in this definition need not arise from a bare field. Put

$$c_{\mathrm{Q}} = \min\{(8c_{\mathrm{piv}})^{-1}, (16L)^{-1}\}, \quad c_{\mathrm{s}} = 2 + 4(2L)^{1/2}\kappa_{\mathsf{c}}\bar{C}_{\mathrm{reg}}, \quad c_{\mathrm{fl}} = \pi(1 + c_{\mathrm{s}}A_0/A_1). \tag{1026}$$

Require $A_1/A_0 \le c_{\mathrm{Q}}$. The smallness conditions used in the next proof are

$$4B_3L^{-2}\varepsilon_t + 2\delta_k \le \omega_{\mathrm{p}}, \quad 12\varepsilon_t + 6(L-1)\delta_k \le 1/40, \quad (1+c_{\mathrm{s}})\varepsilon_t \le 1.$$

These hold, for example, under the ceiling

$$\varepsilon_t \le \min\{\varepsilon_\flat, \bar{\varepsilon}/2, c_2^{\mathrm{SO}}/(2B_3), \omega_{\mathrm{p}}/(2+4B_3L^{-2}), 1/520, (1+c_{\mathrm{s}})^{-1}\}.$$

Proposition 53.12. Let the completed collared datum be regular layer by layer at $(2\varepsilon_q)_{q\le t}$, and suppose $(V_k, V_t) \in \mathsf{PQ}_t(\Omega_t^{\sim 2})$. Then the fluctuation field $\mathcal{A}_k$ is defined on every bond incident to Ω_t, including crossing bonds. On its nonpivot bonds,

$$|\mathcal{A}_k(b)| < c_{\mathrm{fl}}\delta_k. \tag{1027}$$

All fluctuation tests are consequently their stated inequalities in this field; none vanishes through undefinedness. In addition, for $y \in \Omega_t^{(t)}$, $x \in B(y)$,

$$|V_k(y,x) - 1| < 15L\delta_k < \varepsilon_k.$$

■

Proof. Take a test cube $\square' \subset \Omega_t^{\sim}$ and write $G_1 = V_{k,\square'}^{\mathrm{bg}}$. The neighboring P tests and proposition 44.2 give this background in axial gauge, with average V_t, curvature at most $4B_3L^{-2}\varepsilon_t$ on $\square'^{\sim 3}$, and at most $2L^{-2}\varepsilon_t$ on $\square'^{\sim}$. The Q test says that $|V_k G_1^{-1} - 1| < 2\delta_k$ on the nonpivot bonds incident to $\square'^{\sim 2}$. If a pivot bond is in $\square'^{\sim}$, its whole double block is in $\square'^{\sim 2}$. The two configurations have the same average there. The curvature and perturbation bounds meet proposition 14.11; its global uniqueness clause, including the other averaging cases, gives

$$|V_k(b)G_1(b)^{-1} - 1| < 2c_{\mathrm{piv}}\delta_k \quad \text{at the pivot.}$$

Telescoping the four factors of a plaquette yields

$$|\partial V_k(p) - 1| < 8c_{\mathrm{piv}}\delta_k + 2L^{-2}\varepsilon_t \quad (p \subset \square'^{\sim}). \tag{1028}$$

On one block all bonds are nonpivot. Each of the paths defining its averaged contour has length at most $3(L-1)/2$, so its values for V_k, G_1 differ by less than $3(L-1)\delta_k$. The contour family for G_1 has diameter at most $8A_L(2L^{-2}\varepsilon_t)$; that for V_k has diameter at most this number plus $6(L-1)\delta_k$. Both are at most $1/40$. For two such families at pointwise distance ζ, the contraction defining the mean gives $|Y_* - Y'_*| \le (2/5)|Y_* - Y'_*| + (12/5)\zeta$. Thus their logarithmic fixed points differ by at most 4ζ, and their means by at most 5ζ. Since the mean for G_1 is 1, this proves $|V_k(y,x) - 1| < 15(L-1)\delta_k$.

Let $G_2 = p_W^{\diamond}$, the averaged completed background on the incident bonds, completed by V_k outside. For a coarse bond c incident to Ω_t, proposition 53.11 and the curvature averaging estimate give $\varpi_c(G_2) \le 2C_{\mathrm{reg}}\varepsilon_t$. This uses the whole double block, including its part outside Ω_t: there $G_2 = \mathcal{M}^k U$ by the completed datum. Choose $\square' \subset \Omega_t$ containing one endpoint of c; its double block lies in $\square'^{\sim}$. The configurations G_1, G_2 have the same coarse average. Their contour variables agree with 1 on the blocks inside Ω_t; on an outside block the contour of G_2 is that of V_k, bounded above. The double-block comparison lemma 44.3 therefore gives

$$|G_2(b)G_1(b)^{-1} - 1| \le 4\kappa_{\mathsf{c}}C_{\mathrm{reg}}\varepsilon_t + 30(L-1)\delta_k \le c_{\mathrm{s}}\varepsilon_k.$$

Every incident fine bond belongs to one of these double blocks. On a nonpivot bond the triangle inequality gives $|V_k G_2^{-1} - 1| < 2\delta_k + c_{\mathrm{s}}\varepsilon_k$; at a pivot replace $2\delta_k$ by $2c_{\mathrm{piv}}\delta_k$. Both bounds are smaller than 1, by the displayed smallness conditions. Hence the principal logarithm is defined, and on a nonpivot bond

$$|\mathcal{A}_k(b)| \le \frac{\pi}{2}|V_k(b)G_2(b)^{-1} - 1| < \frac{\pi}{2}(2 + c_{\mathrm{s}}A_0/A_1)\delta_k \le c_{\mathrm{fl}}\delta_k.$$

The tests use precisely these nonpivot values. This proves their claimed interpretation and the proposition. □

We also need the converse regularity test for the stored coordinates. At a tied point, the holonomy of a tested plaquette of level q is the ordinary plaquette holonomy of the level-q field. For a first-kind bond this is the datum definition; for a second-kind bond it follows from the tie and the fact that its double block lies in the preceding determining set. Choose a density cube containing the specified corner of the tested plaquette. The cube lies in the level-q small-field region, or is a positive density factor stored when it left that region. Its four-layer body is exactly one of the bodies in proposition 53.8. Thus its written reading gives the same tested holonomy. If $q \ge 1$, passage of this density test bounds the curvature of its defining bare background by $\varepsilon_q L^{-2q}$ on the one-layer enlargement. The q-blocks of the four corners lie there, and the curvature averaging estimate bounds the resulting holonomy by $2\varepsilon_q$. At $q = 0$ the test itself gives ε_0. We have proved the implication

$$\chi_t(\Omega_t) = 1, \quad \mathrm{Stay}_{h'} \neq 0, \quad \text{ties satisfied} \quad \Longrightarrow \quad \mathbf{V}^{\diamond} \text{ is regular at } (\varepsilon_0, 2\varepsilon_1, \dots, 2\varepsilon_t). \tag{1029}$$

Before the step, the same proof uses its positive plaquette factors together with $\mathrm{Stay}_h \neq 0$. For a bare field on the selected history, all ties and all these positive factors hold. This proves the regularity assertion needed to apply proposition 53.12 on its support.

53.8. **Removal of the redundant density tests.** The density tests inside the next small-field region are stronger than what one gets merely by comparing individual bonds. The proof compares the two defining backgrounds on a third, smaller tower. Their discrepancy on its lower rows is far from the plaquette being tested.

Retain the response constants $B_5, \delta_{\mathrm{lc}}, \varrho$ from (996), and put

$$\Theta = 100 B_5 S(\delta_{\mathrm{lc}}/2) C_{\mathrm{AC}} B_3, \qquad D = M_2 \check{R}_k - 2M_1, \qquad \lambda = \sqrt{2} L^{-3/2} < 0.0302.$$

Besides the preceding hypotheses require

$$\frac{A_1}{A_0} \leq \min\{c_{\mathrm{Q}}, (36 c_{\mathrm{piv}})^{-1}, (100\pi c_{\mathrm{piv}} B_5 S(\delta_{\mathrm{lc}}))^{-1}\}, \qquad e^{-\delta_{\mathrm{lc}} D/2} \leq \Theta^{-1}, \\ C_{\mathfrak{P}} \mathfrak{M} B_3 \varepsilon_t \leq a_{\mathrm{R}}, \quad 4 B_3 \varepsilon_t \leq \omega_\sharp, \qquad 2(1 + C_{\mathrm{AC}} B_3)\varepsilon_t \leq \varrho. \tag{1030}$$

These bounds are uniform in the step: $D \geq (L^2 - 2) M_1 y_k^r$, so the exponential condition follows by decreasing the fixed ceiling on the coupling. We also use $\check{R}_k \leq L \check{R}_t$.

Proposition 53.13. If Y is a union of coarse cubes and $(V_k, V_t) \in \mathsf{PQ}_t(Y^\sim)$, then every density cube $\square \subset Y$ passes its level-k density test. For $k \geq 1$ its defining curvature satisfies

$$\mathsf{m}_{k,\square}(V_k) < \tfrac{1}{10} \varepsilon_k \eta_k^2.$$

For $k = 0$ the corresponding bare curvature is smaller than $\varepsilon_0/3$. ∎

Proof. Let $\square'$ be the test cube containing $\square$. The ratio of enlargement scales gives $\square^{\sim a} \subset \square'^{\sim a}$ for $a \leq 4$ and $\square'^{\sim 4} \subset Y^\sim$. By (1028),

$$|\partial V_k - 1| < \left(8 c_{\mathrm{piv}} \frac{A_1}{A_0} + 2\lambda\right) \varepsilon_k < \varepsilon_k/3$$

on $\square'^{\sim 4}$. This proves the assertion for $k = 0$. For $k \geq 1$, proposition 44.2 gives the density background U^{a} on the tower of $\square^{\sim 4}$, with regularity radius $B_3 \varepsilon_k$ and stable dependence on V_k. The passed P test gives the other background U^{b} on the tower of $\square'^{\sim 4}$, of depth t, with curvature smaller than $\varepsilon_t \eta_t^2 \leq \lambda \varepsilon_k \eta_k^2$ on $\square^\sim$.

Restrict both backgrounds to $A = \square^{\sim 2}$ and its tower of depth k. This tower is subordinate to both original towers; $\square^\sim$ lies in its top member. By lemma 51.13 restriction preserves criticality and regularity. Apply the axial alignment used in the proof of proposition 53.5. Its gauge is 1 at level k. Every original node in A is of level at least k; extending the gauge by 1 therefore preserves the original constraints. The aligned data $V^{\mathrm{a}}, V^{\mathrm{b}}$ can be compared with the datum V^{m} induced by $V_k|_A$. At the top, $V^{\mathrm{a}} = V^{\mathrm{m}}$ and $|V^{\mathrm{b}}(V^{\mathrm{m}})^{-1} - 1| \leq 2 c_{\mathrm{piv}} \delta_k$. On the lower rows, summing the curvature bounds of the successive averages gives the additional discrepancies

$$\frac{2 C_{\mathrm{AC}} B_3 \varepsilon_k}{L^2 - 1}, \qquad \frac{4 C_{\mathrm{AC}} B_3 L^{-2} \varepsilon_t}{L^2 - 1},$$

respectively. The middle datum is regular at ε_k by proposition 44.2 and the curvature bound for V_k.

Write $V^\natural = e^{iB^\natural} V^{\mathrm{m}}$. The logarithm bound $|\log X| \leq (\pi/2)|X - 1|$ and (1030) put both $B^\natural$ inside the response ball ϱ. The lower rows have weighted distance at least D from the tops above $\square^\sim$. Exponential summation gives

$$m_y(B) \leq b_{\mathrm{n}} S(\delta_{\mathrm{lc}}) + b_{\mathrm{f}} e^{-\delta_{\mathrm{lc}} D/2} S(\delta_{\mathrm{lc}}/2),$$

where

$$(b_{\mathrm{n}}, b_{\mathrm{f}}) = \begin{cases} (0, \pi C_{\mathrm{AC}} B_3 \varepsilon_k/(L^2 - 1)), & \natural = \mathrm{a}, \\ (\pi c_{\mathrm{piv}} \delta_k, 2\pi C_{\mathrm{AC}} B_3 L^{-2} \varepsilon_t/(L^2 - 1)), & \natural = \mathrm{b}. \end{cases}$$

Consequently $B_5 m_y(B^{\mathrm{a}}) \le \varepsilon_k/100$ and $B_5 m_y(B^{\mathrm{b}}) \le \varepsilon_k/50$.

Let U° represent the middle orbit. Equations (996) and (997) bound both the response and its first differences by ϑ over the plaquette, where ϑ is the appropriate one of these two bounds. The response represents the perturbed orbit up to gauge. Expanding the four factors of its plaquette, as in lemma 51.7, gives

$$|\tau_\natural - \tau_{\mathrm{m}}| \le \eta_k^2(2\vartheta + 11\vartheta^2) + 6\eta_k\vartheta\tau_{\mathrm{m}}, \tag{1031}$$

where $\tau_\natural = |\partial U^\natural(p) - 1|$ and $\tau_{\mathrm{m}} = |\partial U^\circ(p) - 1|$. Indeed $|D_{U^\circ}\mathcal{H}| \le 2\vartheta$; the quadratic remainder is at most $11\eta_k^2\vartheta^2$, and the curvature commutator contributes the last term. The expansion is applicable since $\eta_k \le 1/13$, $\vartheta \le 1/50$, and $\tau_{\mathrm{m}} < \eta_k^2$. Since $\tau_{\mathrm{b}} < 0.0302\varepsilon_k\eta_k^2$, (1031) first gives

$$\tau_{\mathrm{m}} \le \frac{0.0302 + 0.0444}{1 - 6/650}\varepsilon_k\eta_k^2 \le 0.0753\varepsilon_k\eta_k^2.$$

Applying it to a then gives

$$\tau_{\mathrm{a}} \le ((1 + 6/1300)0.0753 + 0.0211)\varepsilon_k\eta_k^2 < 0.1\varepsilon_k\eta_k^2.$$

There are finitely many tested plaquettes. Taking their maximum proves the bound and, by stable definedness, passage of the density test. □

53.9. **The support identity in the new coordinates.** Let Stay_h be the product, with multiplicities, of the written factors stored at all preceding steps. Multiply it by the indicators of the external tiers and call the result $\mathrm{Stay}_h^{\mathrm{T}}$. The tiers at step t are $\mathfrak{b}_k/(4\Theta_1)$ on collar bonds, $\theta_c\mathfrak{b}_k$ on rim bonds, and $C_{\mathrm{xt}}\mathfrak{b}_k$ elsewhere. Choose

$$C_{\mathrm{xt}} \ge c_{\mathrm{fl}}^\sharp := c_{\mathrm{fl}} + c_{\mathrm{H}}A_0/A_1, \qquad c_{\mathrm{H}} = \pi\kappa_{\mathsf{c}}(2L)^{1/2}\max\{2\bar{C}_{\mathrm{reg}}, 4B_3\}.$$

In particular $C_{\mathrm{xt}} \ge c_{\mathrm{fl}} + 1/24$. Assume all preceding smallness conditions, the actual shift bound of proposition 53.7, and

$$2\bar{C}_{\mathrm{reg}}\varepsilon_t < \varpi_\chi, \qquad 2\varepsilon_k < \omega_{\mathrm{p}}, \qquad C_\chi(C_{\mathrm{xt}} + 1)\delta_k \le r_\chi, \qquad C_{\mathrm{xt}}\delta_k < \pi. \tag{1032}$$

Their constants are fixed before the coupling ceiling.

The set of regular bases $d = (W, \phi^+)$ is the open regular-data condition on the completed collared datum, intersected with the Borel set of tied untouched coordinates. For such a base let $V_0[d]$ be the averaged critical background on the incident bonds, in axial gauge inside Ω_t, and use the untouched field on the exterior. Its dependence on the base is Borel, indeed continuous on each regular-data domain. The double-block curvature estimate in the proof of proposition 53.12 and (1032) make it a reference configuration for proposition 53.2. Its average on Γ is W. Set $r = c_{\mathrm{fl}}\mathfrak{b}_k$. The regional chart at g_kB, $B = (B_{\mathrm{I}}, a)$, is defined on the fixed product $|B(b)| < r$; it is a Borel injection with positive Jacobian (985). Its inverse reads the nonpivot logarithms and the average. Its Haar substitution formula is (986).

Let $f_{h'}$ be the factor selecting the outcome of the step, including its gauge weight, and put

$$T = f_{h'}\,\mathrm{Stay}_h^{\mathrm{T}}\,\chi_k(\Omega_k), \qquad S = \mathrm{Stay}_{h'}^{\mathrm{T}}\,\chi_t(\Omega_t);$$

at the first step use $T = f_{h'}$. All products are evaluated as zero when an earlier zero test makes a subsequent partial factor unnecessary. Let R be the product of the integration balls prescribed by the fluctuation split, and let $a = \check{\Psi}_d(a')$.

Proposition 53.14. The actual coordinates and tests satisfy (1357), (1361), and the chart and strip substitution hypotheses (1359) and (1360). Every selected bare field with nonzero new weight has a chart point. Outside the null set of untied untouched coordinates,

$$T(\mathsf{pt}(\Theta(p, B_{\mathrm{I}}))) = \mathbf{1}_R(B_{\mathrm{I}})\,S(\imath p)\,z(x_k, \varepsilon_k)^{-(L^3-1)\#\Omega_t^{(t)}}\,e^{-x_kG_d(B)}. \tag{1033}$$

Moreover, $S(\imath p) > 0$ implies that the base is regular, that a' is in the domain of the strip substitution, and that $|\check{\Psi}_d(a')(b)| < r$ at every external bond. Thus the nonzero support in the integration variables is the product required in (1362). ∎

Proof. The forward coordinate map retains ϕ^+, takes the average of V_k on the new free rows, and records the new strip variable. This is a Borel partial map. Recursion of the recorded strip variables gives equality of its domain with the recursive coordinate domain and $\xi_{h'} = \Sigma\xi_h$. The product of the tests and gauge factors of the selected outcome gives $\mathfrak{w}_{h'} = \mathfrak{w}_h(f_{h'} \circ \xi_h)$; the dependencies of these factors are precisely the retained proper values and the current level-k field. The chart fixes its average, and the inverse strip substitution records a'. Hence $\Sigma\mathsf{pt}\Theta = \imath$. The product-measure assertion proposition 53.1, followed by the regional Haar formula and the fixed-shift Lebesgue formula, proves the two substitution identities, with their positive Borel Jacobians. On a selected bare field the regularity argument following proposition 53.12 applies; that proposition puts its nonpivot logarithms in the chart ball. Global pivot uniqueness then identifies the entire field with the chart point, without any initial bound on its pivot entries.

We prove the test identity carefully, since written readings on a transition zone need not equal physical field values. At a chart point all proper readings equal the physical values. A zone reading is $e^{ig_k a'(b)} V_k^{\mathrm{bg,loc}}(b)$, whereas the physical value is $\Psi_b^{\diamond}(e^{ig_k a'(b)}) V_k^{\mathrm{bg,loc}}(b)$. These agree on $\mathcal{Z}_1$. Every pivot reading used by a stored factor has all of its nonpivot arguments in proper bonds or $\mathcal{Z}_1$, by proposition 53.8. Once the physical double-block curvature is below ω_{p}, proposition 14.11 identifies this reading with the physical pivot. The only remaining reading in $\mathcal{Z}_\theta$ occurs in a one-bond Q test. Equality of that test at the two readings is exactly (1015). All stored fluctuation factors agree by the norm-test invariance of the strip map. Thus every stored factor without a pivot argument agrees with its written version before any extra assumption on the pivot curvature.

Suppose the new interior density tests and these stored factors are nonzero and $B_{\mathrm{I}} \in R$. The internal and external fluctuation factors give the full selected fluctuation split. Its small bonds satisfy $|g_k B(b)| < \delta_k$. For such a bond in $\mathcal{N}_{\mathrm{s}}$, every intersecting Q-test cube has its four-layer enlargement inside Ω_t: the bond meets a two-layer enlargement of a cube in $\Omega_t^{\sim -1}$, and the coarse collar is wider than the sum of these test layers. The test-cube comparison proved with proposition 53.7 gives $|V_k^{\mathrm{bg}}(b)(V_{k,\square'}^{\mathrm{bg}}(b))^{-1} - 1| \le \delta_k$. Since $|e^{ig_k B(b)} - 1| < \delta_k$, this Q test passes. All other Q tests in $\Omega_t^{\sim 2}$ are stored factors without pivot arguments. The P tests there are either the new density tests or stored factors. Together with the exact chart average, this proves $(V_k, V_t) \in \mathsf{PQ}_t(\Omega_t^{\sim 2})$.

Apply proposition 53.12. It gives $|B| < c_{\mathrm{fl}} \mathfrak{b}_k$ even if the chart was initially constructed on the larger ball allowed by (1032). Apply proposition 53.13 with $Y = \Omega_t^{\sim}$. Every density cube meeting a double block over Ω_t lies in this enlargement. Passage of its density test, followed by curvature averaging, gives $\varpi_c(V_k) < 2\varepsilon_k < \omega_{\mathrm{p}}$. This identifies all pivot readings and proves equality of all the stored factors. It also removes the interior density tests; the contour bound in proposition 53.12 removes the interior gauge indicators. The small-bond argument above removes the remaining interior Q tests.

The new external tiers hold at these points. Collar and rim bonds are outside the zones, so the substitution fixes them and the selected fluctuation factors impose their tiers. Elsewhere, $|g_k a'| \le |g_k a| + \delta_k/24$, by the fixed-shift estimate and proposition 53.7. Thus the ordinary tier follows from $C_{\mathrm{xt}} \ge c_{\mathrm{fl}} + 1/24$. Earlier stored factors have exactly the same values in h, h': their determining components lie in $Z_s \subset Z_k$ and all their coordinates are untouched, by proposition 53.8. The same is true of the earlier tier indicators.

Partition the factors of the old test times the step into five disjoint lists: the newly stored factors; the internal fluctuation indicators; the redundant interior density, gauge and Q indicators; the interior gauge exponentials; and the new density tests. This partition keeps occurrences, including carried density tests, rather than identifying equal functions. The second product is $\mathbf{1}_R$, the fifth is

$\chi_t(\Omega_t)$, and the fourth is

$$\prod_{y\in\Omega_t^{(t)}}\prod_{x\in B(y)\setminus\{y\}} z(x_k,\varepsilon_k)^{-1}e^{-x_k(1-\frac12\operatorname{Re}\operatorname{tr}V_k(y,x))}=z(x_k,\varepsilon_k)^{-(L^3-1)\#\Omega_t^{(t)}}e^{-x_kG_d(B)}.$$

If the second or fifth product vanishes, both sides of (1033) vanish. If a newly stored factor without a pivot argument vanishes, it vanishes in both readings, so both sides again vanish. In the remaining case the preceding argument identifies every stored factor, makes the third product and the new tier indicator equal to one, and proves (1033). This argument does not presume that the pivot readings are defined in the cases where a zero factor has already settled the equality.

Finally suppose $S(\imath p)>0$. Equation (1029) gives a regular base. Its tiers and the chart ceiling imply

$$|g_ka'|+|H|<(C_{\mathrm{xt}}+1/24)\delta_k<r_\chi/C_\chi\le 1<\pi.$$

The group-angle bound therefore keeps the logarithm in (989) away from its excluded point. The strip map is defined and gives $|g_ka|<(C_{\mathrm{xt}}+1/24)\delta_k$. Set $B_{\mathrm{I}}=0$ and use the larger chart permitted by (1032). Its nonpivot stored factors agree with the written nonzero factors. The preceding argument now improves $|a|$ to $c_{\mathfrak{fl}}\mathfrak{b}_k=r$. The complement of the tied set has $\mathscr{M}^+$-measure zero by completion. Its product with Haar and Lebesgue coordinates is null, by their σ-finiteness. This proves the asserted support statement and its precise exceptional set. □

For a selected bare field the actual strip backgrounds exist at every step by propositions 53.7 and 53.11. Their logarithms and the strip inverses are defined by the same bounds. Induction therefore puts every such field in $\operatorname{dom}\xi_h$. This also gives the initial support and coordinate-definedness assertion required to iterate the measure identity; empty integration or strip sets are one-point products and do not change the argument.

53.10. **Local terms for the strip substitution.** The strip change of variables produces small boundary terms even when applied to a quadratic form which is not small. We retain the decay of the shift before summing its finite differences.

Use the strip zones $\mathcal{Z}_\theta$, $\mathcal{Z}_1$, $\mathcal{Z}=\mathcal{Z}_\theta\cup\mathcal{Z}_1$, and the component $\mathfrak{o}(b)$ containing a zone bond. Write $\mathsf{P}(b)$ for the one or two adjacent coarse cubes containing its reading blocks, $\mathsf{P}^{\mathrm{loc}}_{\mathfrak{o}}$ for the component's localization cubes, and $\mathsf{P}^{\partial}_{\mathfrak{o}}$ for those containing an endpoint of a tower row. Set

$$\mathsf{Pr}(b)=\{S\in\mathcal{D}_t:\mathsf{P}(b)\subset S\subset\mathsf{P}^{\mathrm{loc}}_{\mathfrak{o}(b)},\ S\cap\mathsf{P}^{\partial}_{\mathfrak{o}(b)}\neq\emptyset\}.$$

For a domain X let $\mathsf{Pr}[X]$ consist of the pairs (b,S) with $b\in\mathcal{Z}^+\langle X\rangle$ and $S\in\mathsf{Pr}(b)$. Here the brackets select bonds with an endpoint in X. For a subset $\mathscr{S}$ put $X(\mathscr{S})=X\cup\bigcup_{(b,S)\in\mathscr{S}}S$.

The geometry of the strip gives

$$\begin{aligned}
d_t(S)+1&\ge\check R_t\quad(S\in\mathsf{Pr}(b)),\\
d_t(X(\mathscr{S}))&\le d_t(X)+\sum_{(b,S)\in\mathscr{S}}(d_t(S)+4),\\
\#X(\mathscr{S})&\le\#X\prod_{(b,S)\in\mathscr{S}}(1+\#S).
\end{aligned}\tag{1034}$$

For the first inequality, a zone endpoint lies farther than $W_t=LM\check R_t$ from any endpoint of a tower row. For rows at Λ_t, their endpoints lie within $3LM_1$ of Λ_t. The zone excludes the two-layer enlargements of all passing test cubes in the coarse one-layer neighborhood of Λ_t, so its endpoints have distance greater than $W_t+2LM_2\check R_t$ from Λ_t. Their distance from a row endpoint is therefore greater than W_t, since $M_2=L^2M_1$. For rows at Z_k, use $\operatorname{dist}_\infty(\Omega_t,Z_k)\ge 8W_t$ and the row collar width $3M_1$ to obtain the lower bound $8W_t-1-3M_1>W_t$. A domain meeting the reading blocks and a boundary cube thus has diameter greater than W_t, whereas its diameter is at most $LM(d_t(S)+2)$. The sizes are integers, giving the first inequality. For the others, each S meets a cube of X. Join

their edge trees in those cubes, using at most four extra edges for each join. The counting bound follows from $\#(X \cup \bigcup S) \le \#X + \sum \#S$ and $\#X \ge 1$. The same argument at a fixed bond gives

$$d_t(\textstyle\bigcup \mathscr{S}) + 1 \le \sum_{S\in\mathscr{S}} (d_t(S) + 4) - 3.$$

All these unions are wall connected. A nonempty union carrying a zone bond contains a cube of Z_t and one of Ω_t.

Suppose the localized shift is given on restriction-stable classes of pairs by $\mathbf{H}(b; S; p)$, vanishes outside $\mathsf{Pr}(b)$, has local holomorphic extensions, is gauge covariant and conjugation real, and satisfies

$$|\mathbf{H}(b; S; p)| \le \bar{h}_S := a_H e^{-\mu_H (d_t(S)+1)}, \qquad a_H > 0. \tag{1035}$$

One family is used on every subregion, by restriction. Let $c_e > 0$, $N_e \ge 1$ be an entropy pair: $\sum_{S\ni\Delta} e^{-c_e d_t(S)} \le N_e$. For $N_b \ge 1$, at most N_b positive zone bonds have an endpoint in any coarse cube. We may take $c_e = c_{\mathrm{E}}$, $N_e = C_{\mathrm{E}}$ and $N_b = 6(LM)^3$.

Fix $\widehat{\kappa} > 0$ and $0 < \rho_{\mathrm{bd}} \le 1/8$ with

$$\mu_H \ge 8(\widehat{\kappa} + 1) + 2c_e, \qquad 128 N_b N_e^2 a_H e^{-\check{R}_t} \le \rho_{\mathrm{bd}}. \tag{1036}$$

Put

$$R_S = 1 + \frac{\rho_{\mathrm{bd}} e^{-c_e d_t(S)}}{2N_e \bar{h}_S}, \qquad r_{\mathrm{bd}} = 2\rho_{\mathrm{bd}}/g_k.$$

A local function $F_X(\mathsf{a}, w)$ depends on the external field only on $\mathfrak{X}_{\mathrm{E}}\langle X\rangle$; w denotes its other arguments. Its domain has the same locality. Assume a is real on $\mathcal{Z}_\theta$, the strip chart is defined at zero shift, and $g_k|\mathsf{a}(b)| \le 1/8$ on $\mathcal{Z}\langle X\rangle$. Assume also that the domain contains the complex ball of radius r_{bd} about a in the directions of these zone bonds, with F_X holomorphic and bounded there by its domain norm $\|F_X\|$.

For parameters $|\lambda(b, S)| < R_S$, define

$$H_\lambda(b) = \sum_S \lambda(b, S)\mathbf{H}(b; S; p), \qquad f_X(\lambda) = F_X(\check{\Psi}(\mathsf{a}, H_\lambda), w).$$

Parameters and shifts vanish off the finite admitted index set. Define the transfer term by

$$F_X^{\mathrm{tr}}(X_1) = \sum_{\substack{\emptyset \ne \mathscr{S} \subset \mathsf{Pr}[X] \\ X(\mathscr{S}) = X_1}} f_X[\mathscr{S} \mid \emptyset], \tag{1037}$$

where the bracket is the finite difference at zero. When evaluated over X_1, only $S \subset X_1$ occur, and the pair is restricted to those supports.

Proposition 53.15. Under these hypotheses the terms in (1037) are defined. If $0 \le \kappa_1 \le \widehat{\kappa} + 1$ and $\kappa' \ge \kappa_1 + 1$, then

$$|F_X^{\mathrm{tr}}(X_1)| \le 2e^{-\check{R}_t} \|F_X\| e^{\kappa' d_t(X)} e^{-\kappa_1 d_t(X_1)}. \tag{1038}$$

If $\|F_X\| \le b e^{-\kappa_{\mathrm{in}} d_t(X)}$ for a family with the same room, $0 \le \kappa_{\mathrm{out}} \le \widehat{\kappa}$ and $\kappa_{\mathrm{in}} \ge \kappa_{\mathrm{out}} + 1 + c_e$, then

$$\left| \sum_{X \subset X_1} F_X^{\mathrm{tr}}(X_1) \right| \le 4N_e e^{-\check{R}_t} b e^{-\kappa_{\mathrm{out}} d_t(X_1)}. \tag{1039}$$

With $H_{\mathrm{tot}}(b) = \sum_S \mathbf{H}(b; S; p)$, the whole sum is

$$\sum_X F_X(\check{\Psi}(\mathsf{a}, H_{\mathrm{tot}}), w) = \sum_X F_X(\mathsf{a}, w) + \sum_{X_1} \sum_{X \subset X_1} F_X^{\mathrm{tr}}(X_1), \tag{1040}$$

provided the room hypotheses hold for every X and $g_k|\mathsf{a}| \le 1/8$ on all zone bonds. Every nonzero transfer is supported on a domain reaching a component boundary and containing a cube of Z_t and one of Ω_t. ∎

Proof. The entropy bound and (1034) give

$$\sum_{S\in\mathsf{Pr}(b)} \bar h_S \le N_e a_H e^{-\check R_t} \le \rho_{\mathrm{bd}}/128, \qquad \sum_{S\in\mathsf{Pr}(b)} R_S \bar h_S \le 65\rho_{\mathrm{bd}}/128 < \rho_{\mathrm{bd}}.$$

Thus $|H_\lambda| < \rho_{\mathrm{bd}}$. The complex chart map $\mathsf{cm}(A,H) = i^{-1}\log(e^{iA}e^{-iH})$ is holomorphic for $|A|,|H| \le 1/8$, and

$$|\mathsf{cm}(A,H) - A| \le \frac{e^{1/4}}{2-e^{1/4}}|H| \le 2|H|.$$

Indeed both matrices in the logarithm comparison are within $e^{1/4}-1$ of one, their difference is at most $e^{1/4}|H|$, and the logarithm series has Lipschitz constant $(2-e^{1/4})^{-1}$ there. The cut-off factor in $\check\Psi$ is fixed when a is fixed. The shifted field therefore lies in the specified complex ball, and f_X is holomorphic with $|f_X| \le \|F_X\|$.

Cauchy's finite-difference estimate gives $|f_X[\mathscr{S} \mid \emptyset]| \le \|F_X\| \prod_{(b,S)\in\mathscr{S}} q(S)$, where $q(S) = (R_S - 1)^{-1}$. For $0 \le \kappa_1 \le \widehat\kappa + 1$ put

$$\omega(S) = (1+\#S)q(S)e^{\kappa_1(d_t(S)+4)}.$$

Writing $n = d_t(S)+1$, we have $1+\#S \le 9n \le 9e^{n-1}$ and $d_t(S)+4 \le 4n$. The rate condition gives

$$\omega(S) \le \frac{18N_e a_H}{e\rho_{\mathrm{bd}}}\, e^{-3\check R_t} e^{-c_e d_t(S)}.$$

After summation and the amplitude condition,

$$\sum_{S\in\mathsf{Pr}(b)} \omega(S) \le (16N_b)^{-1} e^{-2\check R_t}.$$

Let $\Theta(X,X_1)$ be the sum of $\prod_{(b,S)\in\mathscr{S}}(1+\#S)q(S)$ over the nonempty sets in (1037). By the length bound and the product formula for subsets,

$$e^{\kappa_1 d_t(X_1)}\Theta(X,X_1) \le e^{\kappa_1 D}(e^\sigma - 1), \qquad \sigma \le \tfrac12(D+1)e^{-2\check R_t}, \quad D = d_t(X).$$

Here $\#X \le 8(D+1)$ and the bond count supplied the estimate on σ. Since $e^\sigma - 1 \le \sigma e^\sigma \le 2e^{-2\check R_t}e^D$,

$$\Theta(X,X_1) \le 2e^{-2\check R_t} e^{(\kappa_1+1)d_t(X)} e^{-\kappa_1 d_t(X_1)}.$$

The cardinality bound in (1034) also gives the stronger intermediate estimate

$$|F_X^{\mathrm{tr}}(X_1)| \le \|F_X\| \frac{\#X}{\#X_1}\Theta(X,X_1)$$

whenever the index set is nonempty. This proves (1038). To sum over X, write $\#X = \sum_{\Delta\subset X} 1$. The sum at a fixed $\Delta \subset X_1$ is bounded by the entropy estimate because $\kappa_{\mathrm{in}} - \kappa_{\mathrm{out}} - 1 \ge c_e$. Summing over Δ cancels $\#X_1$ and gives $2N_e e^{-2\check R_t} b e^{-\kappa_{\mathrm{out}} d_t(X_1)}$, which implies (1039).

Finally finite Möbius summation gives $f_X(\mathbf{1}) = f_X(0) + \sum_{\emptyset\neq\mathscr{S}} f_X[\mathscr{S} \mid \emptyset]$. Grouping by $X(\mathscr{S})$ and summing the finite family proves (1040). The geometry of each nonempty index gives the support assertion. □

53.11. **The localized strip Jacobian.** For a fixed cut-off-zone bond and a real link potential A, write

$$\mathsf{jc}_\delta(A;H) = \frac{\vartheta'_{\mathrm{co}}(|A|/\delta)}{\delta|A|}\langle A,H\rangle, \qquad j_S = \mathsf{jc}_{\delta_k}(A;\mathbf{H}(b;S;p)), \qquad \bar j_S = \|\vartheta'_{\mathrm{co}}\|_\infty \bar h_S/\delta_k.$$

The first expression is zero at $A = 0$. It is smooth in A, complex linear in the shift, and $|j_S| \le \bar{j}_S$. Define $f_b(\lambda) = \log(1 - \sum_S \lambda(S) j_S)$ with its series branch, and

$$J_b(X_1) = \sum_{\substack{\emptyset \neq \mathscr{S} \subset \mathsf{Pr}(b) \\ \bigcup \mathscr{S} = X_1}} f_b[\mathscr{S} \mid \emptyset]. \tag{1041}$$

In addition to (1036), suppose

$$\begin{gathered} 3\mu_H \ge 16(c_e + 1), \qquad \check{R}_t \ge 12, \qquad 0 < \rho_J \le 1, \\ 128 N_e^2 \|\vartheta'_{\mathrm{co}}\|_\infty a_H e^{-\check{R}_t} \le \rho_J \delta_k. \end{gathered} \tag{1042}$$

Then these terms are defined, and

$$|J_b(X_1)| \le \rho_J e^{-\check{R}_t} e^{-\mu_H (d_t(X_1)+1)/2}. \tag{1043}$$

They vanish unless $X_1 \in \mathsf{Pr}(b)$ and unless $4\delta_k/3 < |A| < 5\delta_k/3$. Their sum is the logarithm of the bond Jacobian:

$$\sum_{X_1} J_b(X_1) = \log(1 - \mathsf{jc}_{\delta_k}(A; H_{\mathrm{tot}}(b))) = \log J_{H_{\mathrm{tot}}(b)}(e^{iA}) \tag{1044}$$

when $H_{\mathrm{tot}}(b)$ is real and $|A| < \pi$. The first equality also holds for a complex shift.

Proof. We have $\sum_S \bar{j}_S \le \rho_J/(128 N_e) \le 1/128$. On the polydisc with radii

$$R_{J,S} = 1 + \frac{e^{-c_e d_t(S)}}{4 N_e \bar{j}_S},$$

the argument of the logarithm differs from one by less than $1/128 + 1/4 < 1/2$. Thus f_b is holomorphic and $|f_b| \le \log 2$. Cauchy's finite-difference estimate gives

$$|f_b[\mathscr{S} \mid \emptyset]| \le \log 2 \prod_{S \in \mathscr{S}} 4 N_e \bar{j}_S e^{c_e d_t(S)}.$$

Multiply by $e^{\mu_H (d_t(X_1)+1)/2}$ and use the fixed-bond union bound in (1034). After dropping the union constraint the result is at most $\log 2(e^\tau - 1)$, where

$$\tau = \sum_{S \in \mathsf{Pr}(b)} 4 N_e \bar{j}_S e^{c_e d_t(S) + \mu_H (d_t(S)+4)/2}.$$

For $n = d_t(S)+1 \ge \check{R}_t \ge 12$, the exponent left after inserting $\bar{j}_S$ is $-\mu_H n/2 + 3\mu_H/2 + c_e(n-1)$. It is at most $-3\mu_H n/8 + c_e(n-1)$, hence at most $-2\check{R}_t - c_e d_t(S)$ by (1042). Therefore $\tau \le \rho_J e^{-\check{R}_t}/32 \le 1/32$. The inequality $\log 2(e^\tau - 1) \le 2\tau$ proves (1043), with the stronger factor $1/16$.

Finite Möbius summation and linearity of jc give the first equality in (1044). The same estimates give $128\|\vartheta'_{\mathrm{co}}\|_\infty |H_{\mathrm{tot}}(b)| \le \rho_J \delta_k$. The actual strip Jacobian formula of lemma 53.3 then gives the second equality; its determinant is positive, so the series is the real logarithm. The support in X_1 follows from the union geometry. The shell support and smoothness follow from the corresponding properties of ϑ'_{co}, which vanishes near zero. □

For the following preservation statements, assume the corresponding properties of the original functions and their domains. In a holomorphic family require, in addition, that $(\zeta, \mathsf{e}) \mapsto F_X(\mathsf{a}_\zeta + \mathsf{e}, w_\zeta)$ is jointly holomorphic on the uniform room. Both constructions are finite combinations of restrictions, the local shift, the chart map and the original functions. They preserve gauge invariance, conjugation reality, locality and Borel measurability. They are holomorphic in holomorphic families of background pairs and other arguments for which the room is uniform and the values of the external field on $\mathcal{Z}_\theta$ are fixed. On $\mathcal{Z}_1$ the chart is holomorphic also in the external field. On $\mathcal{Z}_\theta$ it is smooth in that real field. Indeed the cut-off is then a fixed scalar for the first claim, while in the second it is a

smooth radial function constant near zero. No holomorphic dependence on the real cut-off variable is asserted.

54. Localizing the change of background

The strip substitution uses a difference of two actual backgrounds. We now express this difference as a sum of holomorphic local terms. The construction first solves a reduced response equation on a set of big cubes. Its datum measures the discrepancy of the tower rows. Taking the finite differences in the cube parameters then produces the local terms. At parameters equal to one, criticality and uniqueness identify their sum with the background difference.

54.1. **The reduced response.** Use the reduced determining rows $\mathbf{B}^\flat$, the strong field seminorms of proposition 33.1, and the dimensionless row pairing with weight ξ_c^{-1}. Write $\hat{Q} = \mathcal{Q}'(0)$ in this subsection, so that $Q_{\mathrm{phys}} = \mathsf{m}^{-1}\hat{Q}$, where $\mathsf{m}(c) = \xi_c$. The straightened average and the action are the ones already constructed:

$$\mathsf{C}(A) = \mathcal{Q}(A) - \hat{Q}A, \qquad F(A) = \xi^{-1} \sum_{p \subset \Omega_0} a_p(e^{i\xi A}\mathbf{U}), \qquad \mathcal{V}(A) = F'(A) - F'(0) - \Delta A.$$

Here $\Delta = F''(0)$. In particular, the current of the pair need not equal $F'(0)$.

For the local estimates enlarge the neighbourhood of a top y to $N_L(y)$, consisting of the tops y' for which the associated cubes have distance at most $L\min(\xi_y, \xi_{y'})$. It contains the earlier averaging neighbourhood and has weighted radius $r_{\mathrm{N}} = 3 + 6L^2$. Use

$$|A|_y = \xi_y \sup_{b \text{ over } y} |A(b)|, \qquad \|A\|_y = \max\{|A|_y, \xi_y^2 \sup_{x \text{ over } y} |\nabla_{\mathbf{U}} A(x)|\}, \qquad |f|_y^\sharp = \xi_y^3 \sup_{b \text{ over } y} |f(b)|.$$

The supremum of a covariant difference uses only differences defined in Ω_0. A subscript w denotes the supremum after multiplication by $w(y)$.

The proof of proposition 33.1, with these enlarged neighbourhoods, gives positive c_4, ρ_1 and $C_2, C_3, C_{\mathcal{V}}, C_Q \geq 1$, depending only on L, with the following estimates. All maxima on the right are over $N_L(y)$:

$$\begin{aligned} |\hat{Q}A|_y &\leq C_Q \max |A|_{y'}, & |\hat{Q}^{\mathsf{T}}\varphi|_y^\sharp &\leq C_Q \max |\varphi|_{y'}, \\ |\mathsf{C}(A)|_y &\leq C_2 \max |A|_{y'}^2, & |\mathcal{V}(A)|_y^\sharp &\leq C_{\mathcal{V}} \max |A|_{y'} \max \|A\|_{y'}, \\ |\mathsf{C}_A'^{\mathsf{T}}\varphi|_y^\sharp &\leq C_3 \max |A|_{y'} \max |\varphi|_{y'}. \end{aligned} \tag{1045}$$

The maps vanish to the indicated orders at zero. On a ball of strong radius $r \leq \frac{1}{4}\min(c_4, \rho_1)$, their differences satisfy the respective bounds $4C_2 r \max |A_1 - A_2|$, $4C_{\mathcal{V}} r \max \|A_1 - A_2\|$, and

$$|(\mathsf{C}_{A_1}'^{\mathsf{T}} - \mathsf{C}_{A_2}'^{\mathsf{T}})\varphi|_y^\sharp \leq C_3 \max |A_1 - A_2|_{y'} \max |\varphi|_{y'}.$$

They use only the pair and fields in these neighbourhoods. Indeed the averaging kernel is supported on the double block of a row, that block contains at most $6(\xi_c/\xi)^3$ positive bonds, and a bare bond belongs to at most eighteen such blocks. In the transpose, the derivative's factor ξ_c cancels the row weight ξ_c^{-1}. The kernel estimate (512), its Cauchy derivative, and the plaquette remainder estimate in proposition 33.1 therefore give exactly these bounds. The latter proof also treats a bond next to Ω_0^c: such a bond is in layer zero, where the local spacing equals ξ. No ambient plaquette outside Ω_0 is added to F.

We need two operators in addition to the normalized H, S already constructed. Set

$$\begin{aligned} G_0 = G[\mathbf{U}, 0] &= (\Delta + D\mathsf{R}D^{\mathsf{T}} + \hat{Q}^{\mathsf{T}}\hat{Q})^{-1}, \\ H_0 = G_0\hat{Q}^{\mathsf{T}}(\hat{Q}G_0\hat{Q}^{\mathsf{T}})^{-1}, &\qquad \widetilde{G} = G_0 - H_0\hat{Q}G_0. \end{aligned} \tag{1046}$$

The inverses exist by the retained regular inverse theorem at $(\mathbf{U}, 0)$. These are not the two operators $K, \mathfrak{G}$ in the first nonlinear response.

The local inverse expansion used in proposition 41.1 gives G_0 and the physical-row lift $H_{\mathrm{phys}}[\mathbf{U},0]$. Multiplying the latter on the right by m^{-1} gives H_0. For $\widetilde{G}$ use the terms of G_0 and the products

$$-H_{\mathrm{phys},\omega_1}[\mathbf{U},0]\,Q_{\mathrm{phys}}(\mathbf{U})\,G_{\omega_2}[\mathbf{U},0].$$

Only pairs for which the first cell of ω_2 is linked to a cell of ω_1 contribute. The double block of each row in the middle factor is contained in the first carrier. Thus the product is defined on the union of the two carriers, without a regular extension of the pair.

Here is the summation estimate for these extra products. If a row c contributes, choose a bond b of its double block where the middle average is estimated. A path of weighted length at most $4L^2$ joins $y_c, y(b)$ and the first cell of ω_2. If D is the length for the joined trail, then

$$D \le D_{\omega_1}(y,y_c) + 4L^2 + D_{\omega_2}(y(b),y').$$

Splitting the two rates $\delta/2$ into $3\delta/8$ and $\delta/8$, and using $\delta R_1\mathfrak{M} \ge 512\log L$, absorbs the scale ratio and leaves a row sum at rate $\delta/16$. Write $C_{\mathrm{av,row}}$ for the physical averaging row bound in proposition 33.1. There are at most six rows at a top, so the extra factor is bounded by

$$6C_{\mathrm{av,row}}L^4 e^{3\delta L^2/2}\sum_{r\ge1}(3+2r)(2rL+1)^3 e^{\delta/16-\delta r/32}.$$

The first moment of the first walk expansion controls the number of possible joining cells. Its bound 4, the mass bound 2 for the second expansion, and the fixed number of linked cells give a finite constant depending only on L. This proves the strong field and first-difference estimates for the product. Absolute convergence identifies the sum as $G_0 - H_0\hat{Q}G_0$. The same argument proves local dependence, holomorphy, gauge covariance and compatibility with conjugation.

Assign a parameter once to each big cube meeting the hull of each such term, including a product term. Fix a real number $\varkappa > 0$ and use a partition $\mathsf{P} = \pi[\mathsf{w}]$ with at least three cubes in each direction and

$$\mathring{\delta}_0 R_1\mathfrak{M} \ge 512\log L, \qquad \mathsf{w} \ge 88 r_{\mathrm{w}}\mathfrak{M}, \qquad \mathring{\delta}_0\mathsf{w} \ge 2176\varkappa. \tag{1047}$$

The other block and inverse floors are those of proposition 41.1. Its hull-counting argument applies also to the joined trails. For $|s_\Delta| \le e^{\varkappa}$, the normalized operators and arbitrary subsums satisfy

$$\begin{aligned}
\|H(s)e_{c'}\|_y,\ \|H_0(s)e_{c'}\|_y &\le B_\varkappa \frac{\xi_y}{\xi_{y'}} e^{-3\mathring{\delta}_0 d(y,y')/16}|e|,\\
|S(s)(c,c')| &\le B_\varkappa \frac{\xi_y}{\xi_{y'}} e^{-3\mathring{\delta}_0 d(y,y')/16},\\
\|\widetilde{G}(s)f_{y'}\|_y &\le B_\varkappa \left(\frac{\xi_y}{\xi_{y'}}\right)^3 e^{-3\mathring{\delta}_0 d(y,y')/16}|f_{y'}|^\sharp, \qquad B_\varkappa = B_{\flat,\mathrm{s}}e^{8\varkappa}.
\end{aligned} \tag{1048}$$

Here c, c' are at y, y', and $f_{y'}$ vanishes off the bonds over y'. The terms for H, S are normalized as $H_{\mathrm{phys}}\mathsf{m}^{-1}$ and $\mathsf{m}^3 S_{\mathrm{phys}}\mathsf{m}^{-1}$. Testing $H(s)^{\mathsf{T}}f$ at c' contributes $\xi_{y'}$, while summing bare bonds over y contributes at most $3\xi_y^3$. Consequently the scale factors cancel:

$$|H(s)^{\mathsf{T}}f|_{y'} \le 6B_\varkappa \sum_y e^{-3\mathring{\delta}_0 d(y,y')/16}|f|_y^\sharp. \tag{1049}$$

For the other operators absorb $(\xi_y/\xi_{y'})^p \le L^3 e^{3\mathring{\delta}_0 d(y,y')/512}$, $0 \le p \le 3$. Since at most six rows lie at a top, the four maps

$$(H_0(s), \widetilde{G}(s), H(s)^{\mathsf{T}}, S(s) - I)$$

have exponentially summable local norm kernels with constant $b_{0,\mathrm{s}}e^{8\varkappa}$ and rate $\mu = 11\mathring{\delta}_0/64$, where $b_{0,\mathrm{s}} = 6L^3 B_{\flat,\mathrm{s}} + 1$. The identity in $S(s) - I$ carries no decoupling parameters.

Put

$$\theta_0 = \mathring{\delta}_0/8, \quad e_0 = e^{r_{\mathrm{N}}\theta_0}, \quad S_L(a) = 9L^3 \sum_{r\ge 1} (2r+1)^3 e^{-a(r-1)/2}, \quad \widehat{b}_0 = b_{0,\mathrm{s}} S_L(3\mathring{\delta}_0/64).$$

The weighted top-counting estimate gives operator norm at most $\widehat{b} = \widehat{b}_0 e^{8\varkappa}$ for these four maps whenever $w \in \mathcal{W}_\theta$, $0 \le \theta \le \theta_0$. Choose $a_{\mathrm{D,s}}$ within the common local analytic and inverse radii, and small enough that $2\widehat{b}_0 a_{\mathrm{D,s}} < r_b$, with the seam radius r_b of (534). Set

$$\Gamma = C_2 + C_3 + C_{\mathcal{V}}, \qquad \varepsilon_{\mathrm{D,s}} = \min\left\{ a_{\mathrm{D,s}}, \frac{\min(c_4, \rho_1)}{32\widehat{b}_0}, \frac{1}{32\Gamma e_0 \widehat{b}_0^2} \right\}, \qquad C_{\mathrm{D,s}} = 2\widehat{b}_0. \tag{1050}$$

All these constants depend only on L.

Proposition 54.1. Assume the preceding geometry and let the natural pair have curvature and potential radii at most $a_{\mathrm{D,s}}$ and current radius $\gamma_0 \le \varepsilon_{\mathrm{D,s}} e^{-16\varkappa}$. For $|B'|_\infty \le \varepsilon_{\mathrm{D,s}} e^{-16\varkappa}$ the equations

$$\begin{aligned} T &= H_0(s)(B' - \mathsf{C}(T)) - \widetilde{G}(s)[\mathcal{V}(T) - \mathsf{C}_T'^{\mathsf{T}} \lambda], \\ \lambda &= H(s)^{\mathsf{T}} \mathbf{J} + (S(s) - I)(B' - \mathsf{C}(T)) + H(s)^{\mathsf{T}}[\mathcal{V}(T) - \mathsf{C}_T'^{\mathsf{T}} \lambda] \end{aligned} \tag{1051}$$

have exactly one solution in the ball

$$\max\{\|T\|, |\lambda - H(s)^{\mathsf{T}} \mathbf{J}|_\infty\} \le C_{\mathrm{D,s}} \varepsilon_{\mathrm{D,s}} e^{-8\varkappa}.$$

The solution is holomorphic on the open domains, is gauge covariant, commutes with complex conjugation, and satisfies

$$\max\{\|T(B_1') - T(B_2')\|_w, |\lambda(B_1') - \lambda(B_2')|_w\} \le C_{\mathrm{D,s}} e^{8\varkappa} |B_1' - B_2'|_w.$$

In particular $T(0) = 0$. Setting all parameters outside a set of cubes to zero splits T into its wall components. The component response depends only on the pair in that component and on the datum rows in the neighbourhood of its operator carrier; it vanishes if the datum on that carrier vanishes. The same conclusions hold for local pairs, with the response defined intrinsically by these component equations. ∎

Proof. Set $\lambda^0 = H(s)^{\mathsf{T}} \mathbf{J}$, $\nu = \lambda - \lambda^0$ and $g = |\lambda^0|_\infty \le \widehat{b}\gamma_0$. For $r = |B'|_\infty$ or any larger allowed datum radius use the ball $\max(\|T\|, |\nu|) \le R = 2\widehat{b}r$. Let

$$N(T, \nu) = \mathcal{V}(T) - \mathsf{C}_T'^{\mathsf{T}}(\lambda^0 + \nu).$$

The right sides of (1051), with λ^0 subtracted in the second equation, define a map $\Phi_{B'}$ of (T, ν). For two points in the ball, (1045) gives

$$|\mathsf{C}(T_1) - \mathsf{C}(T_2)|_w + |N(T_1, \nu_1) - N(T_2, \nu_2)|_w^\sharp \le e_0(4\Gamma R + C_3 g) \max(\|T_1 - T_2\|_w, |\nu_1 - \nu_2|_w).$$

In the transpose term split the difference into the change of C_T' and the change of ν; their bounds are $C_3 e_0 (g + R)\|T_1 - T_2\|_w$ and $C_3 e_0 R |\nu_1 - \nu_2|_w$. This explains the coefficient in the displayed inequality. The radius choice gives

$$R \le \tfrac{1}{16} \min(c_4, \rho_1), \qquad 4\widehat{b} e_0 \Gamma R \le \tfrac{1}{4}, \qquad \widehat{b} e_0 C_3 g \le \tfrac{1}{4}.$$

Thus $\Phi_{B'}$ contracts by $1/2$ in every stated weighted norm. Its value at zero has norm at most $\widehat{b}|B'|$, so it preserves the radius-R ball. Iteration proves existence and uniqueness. Comparing two fixed points and absorbing the contraction term proves the weighted estimate. The same uniform iteration proves holomorphy; at zero both nonlinear terms vanish.

For completeness, the component argument must retain the unparameterized identity. Each partial walk sum is carried strictly inside its wall component, with four N_L-neighbourhoods still inside that component. To check this margin, let z be an endpoint in a cube of level n occurring in a term. After $q \le 4$ neighbourhood steps, the scale is at most $L^2 \xi_n$ and the whole top cube is within

$$D_q = ((q+2)L^2 + qL^3)\xi_n$$

of z. Inductively, the next cube has a point within $L^3\xi_n$ of the preceding one. Separation from Ω_{n+3}, since $z \in \Omega_{n+2}^c$, prevents its scale from exceeding $L^2\xi_n$. The retained block floor gives $\mathfrak{M} \geq L^3$; therefore $D_4 + \xi \leq 5L^3\xi_n \leq 5\mathfrak{M}\xi_n$. All four neighbourhoods, including the endpoints of their bare bonds and the rows through the first three, lie in the enlarged cube and hence in the hull. Every partition cube containing a point of the hull is included in its parameter set, so the hull lies in the interior of their union. Carriers belonging to different wall components have disjoint such neighbourhoods. The local nonlinear maps consequently agree with the component maps near each carrier. For a component i, the response is supported on its carrier. Its multiplier has the form $\lambda_i = -B' + \lambda_i^{\mathrm{loc}}$, where λ_i^{loc} is supported in the carrier's neighbourhood. In the combined system use $\lambda = -B' + \sum_i \lambda_i^{\mathrm{loc}}$. On the support of T_i this agrees with λ_i; the local equations therefore combine to (1051). Uniqueness gives the splitting. This also shows independence of unused datum rows. All averaging blocks and plaquettes entering these comparisons lie in the corresponding component. Thus their formulas depend only on its pair, and equal restrictions give the same iteration without choosing a global regular extension. Finally the local matrices, nonlinear maps and pairings intertwine unitary gauge transformations and conjugation, so every iterate does as well. □

54.2. **Intrinsic regional response and criticality.** Here a local natural pair means the restrictions of the inequalities in definition 27.1, including its containing-box gauge condition; a regular continuation outside the region is not required. For $\mathfrak{k} \subset \mathsf{P}$, put $W_{\mathfrak{k}} = \Omega_0 \cap \bigcup \mathfrak{k}$, with cube-wise intersection. In each walk series retain exactly the indices whose parameter sets lie in $\mathfrak{k}$. Denote the resulting operators by $O^{\mathfrak{k}}(s)$. Their terms use only the pair on $W_{\mathfrak{k}}$. Let $Y_{\mathfrak{k}}^{\circ}$ be the union of their top carriers and put $Y_{\mathfrak{k}}^{[q]} = N_L^q(Y_{\mathfrak{k}}^{\circ})$. The preceding margin proof gives $Y_{\mathfrak{k}}^{[4]} \subset W_{\mathfrak{k}}$, with the associated bare bonds and averaging blocks understood.

Retain $\hat{Q}, \mathsf{C}$ only at rows over $Y_{\mathfrak{k}}^{[2]}$. On bonds over this set retain the nonlinear remainder of

$$F_{W_{\mathfrak{k}}}(A) = \xi^{-1} \sum_{p \subset W_{\mathfrak{k}}} a_p(e^{i\xi A}\mathbf{U}).$$

Set these three maps to zero elsewhere; call them $(\hat{Q}_{\mathfrak{k}}, \mathsf{C}_{\mathfrak{k}}, \mathcal{V}_{\mathfrak{k}})$. Replace the four operators in (1051) by their $\mathfrak{k}$-sums and the current by its zero extension. This is the intrinsic regional system:

$$(1051) \quad \text{with } (H_0, \widetilde{G}, H, S; \mathsf{C}, \mathcal{V}, \mathbf{J}) \text{ replaced by } (H_0^{\mathfrak{k}}, \widetilde{G}^{\mathfrak{k}}, H^{\mathfrak{k}}, S^{\mathfrak{k}}; \mathsf{C}_{\mathfrak{k}}, \mathcal{V}_{\mathfrak{k}}, \mathbf{J}^0). \tag{1052}$$

As before the identity subtracted from $S^{\mathfrak{k}}$ is not truncated or parameterized.

Proposition 54.2. The regional system has the existence, uniqueness, estimates and holomorphic dependence of proposition 54.1. Its field component, denoted $\mathbf{H}_{\mathfrak{k}}(s; \mathbf{p}; B')$, is zero off $Y_{\mathfrak{k}}^{\circ}$, uses B' only on $Y_{\mathfrak{k}}^{[1]}$, and vanishes if B' is zero on $Y_{\mathfrak{k}}^{\circ}$. For $\mathfrak{k} \subset \mathfrak{k}'$, zero extension of the parameters and restriction of the pair give the same response. It splits over wall components, intertwines unitary gauges and conjugation, and agrees with the global response when a regular global pair is available. ■

Proof. The margin places every double row block over $Y_{\mathfrak{k}}^{[3]}$, and every plaquette of Ω_0 touching the corresponding bonds, inside $W_{\mathfrak{k}}$. Thus the local formulas used in (1045) are unchanged near the carrier. Multiplication of the row maps by an indicator does not enlarge their transpose estimates. The regional action remainder has the same plaquette formula there. Consequently the truncated triple satisfies the same local estimates.

For a field supported on the carrier, compute its regional first difference by extending the field by zero:

$$\nabla_\mu^W A(b) = \xi^{-1}\{\mathrm{Ad}(\mathbf{U}(\langle b_-, b_- + \xi e_\mu\rangle))A(b + \xi e_\mu) - A(b)\},$$

where the first term is zero when the translated bond is not in W. If both bonds belong to W, the joining bond does also. The margin shows that this norm agrees with the norm for any continuation of $\mathbf{U}$. The partial walk bounds therefore act on this intrinsic norm. The contraction

proof of proposition 54.1 now applies without any regularity assumption on a continuation. It proves uniqueness, and hence independence of any continuation used to write the zero extensions.

For holomorphy use the same finite-dimensional local charts as for the walk terms. In a chart $e^a\mathbf{U}$ with decomposition $\mathbf{U} = e^{i\xi A'}U$, the new potential is $(i\xi)^{-1}\log(e^a e^{i\xi A'})$. If $|a| < 1/5$ and $|\xi A'| < 1/6$, its argument stays within $1/2$ of the identity. The strict local class inequalities persist on a smaller chart. Uniform iteration then proves holomorphy jointly in the pair, the datum and the cube parameters.

Support and dependence on datum rows follow directly from the supported operators and their transposes. For restriction, the surviving walk terms are identical; near their carriers the two truncated triples have the same averaging blocks and plaquettes. The component argument at the end of proposition 54.1 applies, including the term $-B'$ in the multiplier. Uniqueness proves restriction, splitting, and agreement with the global system. The same iteration proves both covariance claims. □

The free-boundary current requires a separate convention. A bond is *full* when all four ambient plaquettes through it belong to Ω_0. This designation refers to Ω_0, even for a pair given on $W_\mathfrak{k}$. Write $J_W(\mathbf{U}) = F'_W(0)$ and set

$$\mathbf{p}^{\mathrm{tc}}_\mathfrak{k} = (\mathbf{U}, \mathbf{J}^{\mathrm{tc}}_\mathfrak{k}), \qquad \mathbf{J}^{\mathrm{tc}}_\mathfrak{k}(b) = \begin{cases} \mathbf{J}(b), & b \text{ full}, \\ J_{W_\mathfrak{k}}(\mathbf{U})(b), & b \text{ not full}. \end{cases} \tag{1053}$$

At a nonfull bond, its initial point and the base points of its adjacent plaquettes of Ω_0 lie in layer zero: they are within 2ξ of a point outside Ω_0, whereas deeper layers have the retained separation. If the curvature radius there is α'_0, the current obeys

$$\xi^3 |J_{W_\mathfrak{k}}(\mathbf{U})(b)| < 24\alpha'_0. \tag{1054}$$

Indeed, each plaquette curvature satisfies $\frac{1}{2}\xi^{-2}|g - g^{-1}| \le \xi^{-2}|g - 1|$. The inverse identity for G^c gives $|g^{-1} - 1| = |g - 1|$; the small-potential bound makes the adjacent adjoint transports smaller than 2. The two transverse directions in the plaquette derivative therefore give at most $6\alpha'_0\xi^{-3}$, which is smaller than the stated bound.

If $24\alpha'_0 \le \gamma_0$, truncation preserves the current inequality of the natural class. Moreover it commutes with restriction for the response:

$$\mathbf{H}_{\mathfrak{k}'}(s^0; \mathbf{p}^{\mathrm{tc}}_{\mathfrak{k}'}; B') = \mathbf{H}_\mathfrak{k}(s; (\mathbf{p}|_{W_\mathfrak{k}})^{\mathrm{tc}}_\mathfrak{k}; B'). \tag{1055}$$

To see this, every plaquette of Ω_0 through a bond of a surviving walk carrier already lies in $W_\mathfrak{k}$. Thus the two regional action currents coincide at precisely the bonds used by those walks and by $H^{\mathfrak{k}\mathsf{T}}\mathbf{J}^0$. The two systems and their uniqueness balls are identical.

At parameters one we will use the actual critical current. The unparameterized walk sums satisfy the same bounds with $\varkappa = 0$; hence the datum radius is $\varepsilon_{\mathrm{D,s}}$ and $\|T\| \le C_{\mathrm{D,s}}|B'|_\infty$.

Proposition 54.3. Let U_0 be real, let $(U_0, J_{\Omega_0}(U_0))$ satisfy the natural pair bounds with curvature and potential radii at most $a_{\mathrm{D,s}}$ and current radius at most $\varepsilon_{\mathrm{D,s}}$, and suppose $J_{\Omega_0}(U_0) = \hat{Q}^{\mathsf{T}}\lambda_0$. For $|B'|_\infty < \varepsilon_{\mathrm{D,s}}$, the reduced response at parameters one satisfies

$$\mathcal{Q}(T) = B', \qquad \mathsf{R}D^{\mathsf{T}}T = 0, \qquad F'(T) = (\hat{Q} + \mathsf{C}'_T)^{\mathsf{T}}\lambda. \tag{1056}$$

For real B', the configuration $e^{i\xi T}U_0$ is critical for the actual row constraint with datum $(e^{iB'^\natural}V_0)^{v[T]}$, where $V_0 = M_{\mathbf{B}^+}U_0$, the outward-row twist is (547), and $v[T]$ is the accumulated node factor. ■

Proof. The inverse identities give

$$\hat{Q}H_0 = I, \quad \hat{Q}\widetilde{G} = 0, \quad \widetilde{G}G_0^{-1} = I - H_0\hat{Q}, \quad \ker\widetilde{G} = \operatorname{ran}\hat{Q}^{\mathsf{T}}, \qquad \hat{Q}H = I.$$

Applying $\hat{Q}$ to the first response equation gives $\mathcal{Q}(T) = B'$. Put $K_T = \hat{Q} + \mathsf{C}'_T$, $z = F'(T) - K_T^{\mathsf{T}}\lambda$ and $\phi = \mathsf{R}D^{\mathsf{T}}T$. The two equation defects are

$$0 = \widetilde{G}(z + D\phi), \qquad 0 = \mathsf{L}G'\phi - H^{\mathsf{T}}(z + D\phi) \tag{1057}$$

for a fixed linear map L. Here is the term accounting for this map. The correction Δ'_π in the Hessian is the polarization of the quadratic gauge expression (540). With $\Upsilon = G'\mathsf{R}D^{\mathsf{T}}$, it has the form

$$(A_1, \Delta'_\pi A_2) = b(\Upsilon A_1, A_2) + b(\Upsilon A_2, A_1) + b'(\Upsilon A_1, \Upsilon A_2).$$

The retained linear identity $\Upsilon H = 0$ gives $H^{\mathsf{T}}\Delta'_\pi T = \mathsf{L}G'\phi$. Also

$$H^{\mathsf{T}}(\Delta - \Delta'_\pi + D\mathsf{R}D^{\mathsf{T}}) = (S - I)\hat{Q}, \qquad H^{\mathsf{T}}J = \lambda_0.$$

Substitution of $F'(T) = J + \Delta T + \mathcal{V}(T)$ and $\hat{Q}T = B' - \mathsf{C}(T)$ proves both defects. Their first equation says $z + D\phi = \hat{Q}^{\mathsf{T}}\sigma$; their second says $\sigma = \mathsf{L}G'\phi$. Thus

$$F'(T) + D\phi = K_T^{\mathsf{T}}\lambda + \hat{Q}^{\mathsf{T}}\mathsf{L}G'\phi.$$

For real data sufficiently near zero, use the gauge bijection (551) to select a gauge tangent $X_\chi(T)$ with node variation zero and $\mathsf{R}D^{\mathsf{T}}X_\chi(T) = \phi$. Its norm is at most a finite constant times $\|\phi\|$. The gauge law gives $K_T X_\chi(T) = 0$, and action invariance gives $(F'(T), X_\chi(T)) = 0$. Pairing the last display with $X_\chi(T)$ therefore yields

$$(\phi, \phi) = -(\mathsf{L}G'\phi, \mathsf{C}'_T X_\chi(T)), \qquad |(\mathsf{L}G'\phi, \mathsf{C}'_T X_\chi(T))| \le C|T|_\infty(\phi, \phi).$$

At this step C may depend on the finite geometry. Since $T \to 0$ with B', it follows that $\phi = 0$ near the real origin. Holomorphy on the connected datum ball gives $\phi = 0$ throughout it. The defects then give $z = 0$, proving (1056).

Finally the actual averaging identity eq. (548) gives the stated row datum. The same node Ward argument as in (553) applies: pair $F'(T) = K_T^{\mathsf{T}}\lambda$ with gauge tangents, use the gauge bijection near zero, and continue the resulting identity holomorphically. It annihilates the node term in the derivative of any curve in the actual constraint, so the derivative of the action along that curve is zero. The radius choice above keeps the seam twist and its invertible derivative in their retained domain. This proves criticality, including outward rows. □

54.3. **The response on a joined component.** Fix a step $t = k + 1$ and a component $\mathfrak{o}$ of the localization region in lemma 53.6. Its given sequence is denoted $\Omega^{\diamond}_{\bullet}$ and its joined sequence $\Omega^{\mathfrak{o}}_{\bullet}$. For the remainder of the section use level-k lengths, so $\xi_n = L^{n-k}$. Let $\mathsf{P}^{\mathrm{loc}}_{\mathfrak{o}}$ be the big cubes meeting $\mathfrak{o}$, and let $\mathsf{P}^{\partial}_{\mathfrak{o}}$ be those containing an endpoint of a tower row. Rows always have the inward rows removed.

We spell out the pair classes needed for the local terms. For $W = \mathsf{S} \cap \mathfrak{o}$ a pair $\mathbf{p} = (e^{i\eta_k A'}U, \mathbf{J})$, with U real, belongs to $\mathfrak{P}^{\mathfrak{o}}(\mathsf{S}; \alpha'_0, \alpha'_1)$ if, for the given sequence, it satisfies the following inequalities. The curvature of both configurations is smaller than $\alpha'_0 L^{-2n}$ at a plaquette based in layer n. The potential and its covariant first differences satisfy $\xi_n|A'| < \alpha'_1$ and $\xi_n^2|\nabla^U A'| < \alpha'_1$ at their initial sites in that layer. The current satisfies $\xi_n^3|\mathbf{J}| < \alpha_2$. For every tested cube $q \in \mathsf{Q}_n$ contained in W, U has a real gauge on a containing cube $q \subset q' \subset W$, in which $\xi_n|A| \le \alpha_0$ and $\xi_n^2|\nabla A| \le \alpha_0$. Here $\alpha_0, \alpha_1, \alpha_2$ are fixed positive radii, and $0 < \alpha'_i \le \alpha_i$, $i = 0, 1$. Define

$$\begin{aligned} \mathfrak{P}^{\mathfrak{o}}_{\mathbf{H}}(\mathsf{S}) &= \bigcup_{\alpha'_0 + \alpha'_1 \le \varrho_{\mathbf{H}}} \mathfrak{P}^{\mathfrak{o}}(\mathsf{S}; \alpha'_0, \alpha'_1), \\ \mathsf{bs}_{\mathsf{S}}(\mathbf{p}) &= \inf\{\alpha'_0 + \alpha'_1 : \mathbf{p} \in \mathfrak{P}^{\mathfrak{o}}(\mathsf{S}; \alpha'_0, \alpha'_1)\}. \end{aligned} \tag{1058}$$

These classes are open: keep the real configuration and its gauges fixed, absorb a nearby configuration into A' with the local series logarithm, and use the strict curvature, potential and current inequalities. They are stable under unitary gauges, conjugation and restriction; restriction can only decrease bs.

Choose the response radii in the following order:

$$\begin{aligned} &0 < \beta_0' \le \beta_0 \le a_{\mathrm{D,s}}, \quad 0 < \beta_1 \le a_{\mathrm{D,s}}, \quad 0 < \beta_2 \le \varepsilon_{\mathrm{D,s}} e^{-16\bar\kappa}, \quad 24\beta_0' \le \beta_2, \\ &0 < \varrho_{\mathrm{rw}} \le \varepsilon_{\mathrm{D,s}} e^{-16\bar\kappa}, \quad \alpha_0 \le \beta_0, \quad \alpha_2 \le \beta_2, \quad \varrho_{\mathbf{H}} \le \min(\beta_0', \beta_1, 1/3). \end{aligned} \tag{1059}$$

Assume the inverse floors above and

$$M \ge 88 r_{\mathrm{w}} \mathfrak{M}, \qquad \mathring{\delta}_0 M \ge 2176 \bar\kappa, \qquad m' < m + K - t.$$

The usual strip floors $M_1 \ge R_1 \mathfrak{M}$ and $M \ge L^4 M_1$ remain in force.

Proposition 54.4. For every pair in $\mathfrak{P}^{\mathfrak{o}}_{\mathbf{H}}(\mathsf{S})$, the truncated pair is a natural local pair for the joined sequence, with radii $(\beta_0, \beta_1, \beta_2)$, and with curvature radius β_0'. Its response is

$$\mathbf{H}^{\mathfrak{o}}_{\mathsf{S}}(s; \mathbf{q}; B') = L^{-1} \mathbf{H}_{\mathsf{S}}(s; (\mathbf{U}, L^3 \mathbf{J}); B'), \qquad |B'|_\infty < \varrho_{\mathrm{rw}}. \tag{1060}$$

The right side uses level-t lengths and the joined sequence. All estimates, holomorphy, restriction, component splitting and covariance properties above apply. In the dimensionless strong norm for the joined sequence,

$$\|\mathbf{H}^{\mathfrak{o}}_{\mathsf{S}}(s; \mathbf{q}; B')\| \le \mathsf{A}_{\mathrm{rs}} |B'|_\infty, \qquad \mathsf{A}_{\mathrm{rs}} = C_{\mathrm{D,s}} e^{8\bar\kappa}. \tag{1061}$$

At parameters one, for the whole component and a real critical pair $(U_0, J_{\mathfrak{o}}(U_0))$, this is the critical response of proposition 54.3. It also satisfies

$$\|T\| \le C_{\mathrm{rs}} |B'|_\infty, \qquad |F'_{\mathfrak{o}}(T) - J_{\mathfrak{o}}(U_0)|^\sharp \le C_{\mathrm{rs}} |B'|_\infty, \qquad C_{\mathrm{rs}} = (C_Q + 1) e_0 C_{\mathrm{D,s}}. \tag{1062}$$

■

Proof. The joined layer index is no greater than the given one, which on $\mathfrak{o}$ is either k or t. Hence every curvature, potential, difference and current inequality remains valid when its given local scale is replaced by the joined local scale. For the gauge condition, let a joined test cube of level n lie in W. If it is contained in Ω_t, enlarge it, when $n < t$, to a cube of side $2\mathfrak{M}\xi_t$ inside $W \cap \Omega_t$. Otherwise enlarge it, when $n < k$, to a cube of side $2\mathfrak{M}\xi_k$ inside W. This is possible because W is a union of level-t M_1-cubes, $12\mathfrak{M}\xi_n < \mathfrak{M}\xi_{n+1}$, and the containing grid has at least twice that side. In the second case the enlarged cube meets layer k; in the first it lies in layer t. The given gauge condition therefore supplies a containing gauge cube inside W. Its stronger scale bounds imply those at level n. This proves the natural gauge condition, also for a joined test cube not meeting its own layer.

For the free boundary use (1054). The current radius becomes at most $\max(\alpha_2, 24\alpha_0') \le \beta_2$, while all other conditions are unchanged. The current is a polynomial in the configuration entries and inverses, so truncation is holomorphic and commutes with gauges and conjugation.

On changing from level k to level t, lengths and forms are multiplied by L^{-1} and L, respectively, and currents by L^3. Thus the dimensionless inequalities and strong norms are unchanged, and truncation commutes with this rescaling. The partition side becomes M. The strict period condition gives at least $2L \ge 3$ cubes per direction. Apply proposition 54.2 and scale the response back. All walks for the whole component have parameter sets in $\mathsf{P}^{\mathrm{loc}}_{\mathfrak{o}}$, so putting its parameters equal to one gives the unparameterized inverse.

To apply proposition 54.3, a critical reference has the required multiplier. At zero $\hat Q H = I$. The implicit function theorem therefore makes every vector in $\ker \hat Q$ tangent to a curve with constant straightened datum. The averaging identity eq. (548) corrects this curve by its node gauge to one with constant actual datum. Gauge invariance and criticality make $F'(0)$ orthogonal to $\ker \hat Q$, whence $F'(0) = \hat Q^{\mathsf{T}} \lambda_0$. Small curvature keeps the real and complex means in the same unique mean chart throughout these curves.

The field estimate at parameters one was proved above. For the current difference write

$$F'(T) - J = \hat Q^{\mathsf{T}} (\lambda - H^{\mathsf{T}} J) + \mathsf{C}_T'^{\mathsf{T}} \lambda.$$

The contraction estimate and (1045), with the constant weight, bound the first term by $C_Q e_0 C_{\mathrm{D,s}}|B'|$. For the second, use $|H^{\mathsf{T}}J| \le \widehat{b}_0\beta_2$, $\|T\|, |\lambda - H^{\mathsf{T}}J| \le C_{\mathrm{D,s}}|B'|$, and the radius choice, which makes its coefficient at most $e_0 C_{\mathrm{D,s}}$. The resulting bound is (1062). Scaling the derivative by L^{-3} preserves its sharp norm. □

54.4. **The tower source and its read-out.** We first record the averaging estimates used at the two ends of the construction. On a double-block tower of height j, let U be real with bare curvature at most αL^{-2j}, and put $\mathbf{U} = e^{ia'}U$, $A = L^j \sup|a'|$. With κ_6 as above, put $C''_{\mathrm{ia}} = 4(8\kappa_6 + 2)$. Choose an auxiliary radius

$$0 < c_\sharp \le \min\{c_{\mathsf{m}}, c_{\mathrm{plain}}, 1\}, \qquad C''_{\mathrm{ia}} c_\sharp < r_{\mathrm{av}}/16.$$

There is

$$0 < c_{\mathrm{ia}} \le \min\{c_\sharp/4, a_{\mathsf{m}}, a_{\mathrm{plain}}, \varpi_{\mathrm{av}}/2\},$$

depending only on L, with the following properties. For $\alpha < c_{\mathrm{ia}}$ and $A < c_\sharp$, the averages and contour variables in the first three lines below are defined and satisfy the displayed bounds. If also $A, B = L^j \sup|a_0| < c_{\mathrm{ia}}$, the relative recursion and accumulated factors below are defined:

$$\begin{aligned} W_{0,n} &= \mathcal{M}^n U, \qquad W_n = \mathcal{M}^n \mathbf{U} = e^{ia''_n} W_{0,n}, \\ \varpi(W_{0,n}) &\le 2\alpha L^{2(n-j)}, \qquad |a''_n| \le C''_{\mathrm{ia}} L^{n-j} A, \\ W_n(y,x) &= e^{i\zeta_n(y,x)} W_{0,n}(y,x), \qquad |\zeta_n| \le \kappa_6 C''_{\mathrm{ia}} L^{n-j} A, \\ a_{n+1} &= \mathcal{Q}^1_{\mathsf{m}}(W_n, a_n), \qquad |a_n| \le 2L^{n-j}B, \qquad v_n = \prod_{i<n} e^{i\lambda_i}, \quad \sum_{i<n} |\lambda_i| \le \frac{2\kappa_6}{L-1} B. \end{aligned} \tag{1063}$$

The reference curvature estimate is for $n < j$. All quantities are holomorphic in the perturbations; when the input is real the two definitions of the mean agree.

Here are the extra estimates needed beyond the real reference iteration in propositions 30.2 and 51.1. For W_n, that iteration first gives its centered perturbation a'_n and factors v'_n. The relative product

$$v'_n(c_-) e^{ia'_n(c)} \operatorname{Ad}(W_{0,n}(c))(v'_n(c_+)^{-1})$$

has total exponent norm at most $(8\kappa_6 L^{-1} + 2)L^{n-j}A$. Its series logarithm gives the bound for a''_n. The choices $2c_{\mathrm{ia}} \le \varpi_{\mathrm{av}}$ and $C''_{\mathrm{ia}} c_\sharp < r_{\mathrm{av}}/16$ put these references and perturbations in the complex one-step chart. Its contour Schwarz bound gives the estimate for ζ_n. For clarity, the one-step estimates at this complex reference follow from the small-gauge estimates as follows. On a double block choose a real gauge $W_0^h = e^{ia^0}$, $\sup|a^0| \le r_{\mathrm{av}}/2$. Let $c_\ell \ge 1$ be an absolute constant such that

$$|\log(e^X e^Z) - X - Z| \le c_\ell |X|\,|Z| \qquad (|X|, |Z| \le 1/16).$$

Indeed this holomorphic remainder vanishes on either coordinate axis; the Schwarz estimate in each variable on the 1/12-ball gives the bound. For $\sup|a'| < r_{\mathrm{c}} = r_{\mathrm{av}}/16$, the logarithm $\widetilde{a} = i^{-1}\log(e^{i\operatorname{Ad}(h)a'} e^{ia^0})$ satisfies

$$\sup|\widetilde{a}| < 3r_{\mathrm{av}}/4, \qquad \sup|\widetilde{a} - a^0| \le 2c_\ell \sup|a'|.$$

By lemma 24.2, the one-step map is jointly holomorphic and bounded by one on the product of the reference and perturbation balls of radius r_{av}. Cauchy's inequality bounds its linear kernel by r_{av}^{-1}, and its derivative in the reference by $4r_{\mathrm{av}}^{-2}$ on the $3r_{\mathrm{av}}/4$-ball. Thus the change of a kernel entry is at most $8c_\ell r_{\mathrm{av}}^{-2} \sup|a'|$. There are at most $6L^3$ entries in a row and six rows containing a bond. Together with the real row bound this gives

$$\|\mathbf{Q}(\mathbf{W}) - \mathbf{S}(W_0)\|_{\mathrm{row}} \le \kappa'_9(\varpi(W_0) + \sup|a'|), \qquad \|\mathbf{S}(W_0)\|_{\mathrm{row}} \le L, \quad \kappa'_9 = \kappa_9 + 48L^3 c_\ell r_{\mathrm{av}}^{-2}.$$

The Taylor remainder on the $r_{\mathrm{av}}/2$-ball is at most $2r_{\mathrm{av}}^{-2}|a|^2$. Its derivative is bounded by $(32/3)r_{\mathrm{av}}^{-2}|a|$: apply Cauchy's inequality on the $3r_{\mathrm{av}}/4$-ball and Schwarz's inequality to the derivative, which

vanishes at zero. We may therefore take $\kappa_5' = 11 r_{\mathrm{av}}^{-2}$. Gauge conjugations here are isometries, so these bounds hold before choosing the gauge. They give, with $\epsilon_n = 2\alpha L^{2(n-j)} + C_{\mathrm{ia}}'' L^{n-j} A$,

$$t_{n+1} \le t_n \{1 + \kappa_9' L^{-1} \epsilon_n + \kappa_5' L^{-1} L^{n-j} t_n\}, \qquad t_n = L^{j-n} \sup |a_n|.$$

Since $\sum \epsilon_n \le 4L^{-2}\alpha + 2C_{\mathrm{ia}}'' L^{-1} A$ and $\sum_{n<j} L^{n-j} \le 2L^{-1}$, decrease c_{ia} so the resulting product is at most two. Induction gives $t_n \le 2B$. The one-step factor bound $|\lambda_n| \le \kappa_6 |a_n|$ then gives the last estimate. These arguments use only the double block: each lower double block is nested in it. The products stay in the strict mean and logarithm domains. Real uniqueness and holomorphic continuation give the factorization for complex perturbations.

The remaining real comparison is with a lifted top average. Put

$$A_L = \tfrac{3}{4}(L-1)^2, \qquad C_2^{\mathrm{AC}} = 8(L-1)A_L + (1+\pi/3)\{L(L-1) + 8A_L\}.$$

If the intermediate curvatures are ϖ_n in the mean chart, there is a gauge, equal to one at the top sites, which makes all averaged block contour variables equal to one and satisfies

$$|\widehat{W}_q(c)(Q_{j\to q}^{s*} W_{0,j})(c)^{-1} - 1| \le C_2^{\mathrm{AC}} \sum_{n=q}^{j-1} \varpi_n. \tag{1064}$$

Indeed define it downwards by $v_q(x) = v_{q+1}(y) W_{0,q}(y,x)$ for $x \in B(y)$. The values agree at the centers and gauge covariance makes every contour mean equal to one. The path exchange estimate bounds the diameter of a contour family by $A_L \varpi_q$; its individual transports are then within $4A_L\varpi_q$ of one. Appending a last step shows that each internal bond is within $8A_L\varpi_q$ of one. For a crossing bond, moving the straight L-step path across an axial contour uses at most $L(L-1)$ plaquettes. Comparing its mean with the straight transport costs at most $(1+\pi/3)\{L(L-1)+8A_L\}\varpi_q$; the other $L-1$ internal bonds cost $8(L-1)A_L\varpi_q$. Thus a crossing bond differs from the next average by at most $C_2^{\mathrm{AC}}\varpi_q$. Descending induction, using unitarity and the composition of lifts, proves (1064).

For a tower row c of level q and part s, let $l = l_s > q$. With $x^{(i)}$ the successive ancestors, define the descent and source by

$$\begin{aligned} d_{q,l}(x) &= \mathbf{U}_{(l-1)}(x^{(l)}, x^{(l-1)}) \cdots \mathbf{U}_{(q)}(x^{(q+1)}, x), \\ \mathsf{src}_{\mathsf{S}}[\mathbf{p}](c) &= \frac{1}{i} \log X_c[\mathbf{U}], \qquad X_c = d_{q,l}(c_-)^{-1} (Q_{l\to q}^{s*} \mathbf{U}_{(l)})(c) d_{q,l}(c_+) \mathbf{U}_{(q)}(c)^{-1}. \end{aligned} \tag{1065}$$

Use this value when both endpoint blocks of order l lie in $\bigcup \mathsf{S}$, and put zero at all other rows. A tower endpoint lies in an M_1-cube of the wide collar, entirely in layer l of the given sequence. Its order-l block and its M_1-cube lie in the same big cube. Consequently the condition just imposed is equivalent to both row endpoints belonging to $\bigcup \mathsf{S}$.

Set $\Sigma_{\mathrm{ia}} = (2\kappa_6 + 1) C_{\mathrm{ia}}''$ and reduce c_{ia} so $\Sigma_{\mathrm{ia}} c_{\mathrm{ia}} < 3/64$. For the real field, (1064) and $\sum_{n<l} 2\alpha_0' L^{2(n-l)} \le \alpha_0'$ give $|X_c[U] - 1| \le C_2^{\mathrm{AC}} \alpha_0'$. Inserting the complex perturbation in both descents, the top average and the inverse lower average gives total exponent norm at most $\Sigma_{\mathrm{ia}} L^l \sup |a'| < \Sigma_{\mathrm{ia}} \alpha_1'$. Thus

$$|X_c[\mathbf{U}] - 1| \le C_2^{\mathrm{AC}} \alpha_0' + 2\Sigma_{\mathrm{ia}} \alpha_1' < 1/2.$$

The logarithm yields, after taking the infimum of the admissible radii,

$$|\mathsf{src}_{\mathsf{S}}[\mathbf{p}]|_\infty \le C_{\mathrm{src}} \mathsf{bs}_{\mathsf{S}}(\mathbf{p}), \qquad C_{\mathrm{src}} = 2\max(C_2^{\mathrm{AC}}, 2\Sigma_{\mathrm{ia}}). \tag{1066}$$

This also proves holomorphy and locality on the two endpoint blocks. Gauge factors cancel through the descent, leaving conjugation by $u(c_-)$; conjugation of the complex matrices commutes with every mean and logarithm. These give covariance and reality of the source.

For a level-k bond b, let $y_\pm$ be the centers of its two containing level-t blocks. Its reading block is $B^t(y_-) \cup B^t(y_+)$. Put $\mathsf{lv}(x) = t$, $\mathsf{y}(x) = y(x)$ for $x \in \Omega_t$, and $\mathsf{lv}(x) = k$, $\mathsf{y}(x) = x$ otherwise. Let $\mathrm{ax}_{\mathbf{U}}(x)$ be the level-k averaged contour variable from $y(x)$ to x inside Ω_t, and one outside.

Superscript ax denotes dressing by these factors. For $\mathbf{U}' = e^{i\eta_k \mathbf{A}}\mathbf{U}$, let $v_\pm$ be the accumulated factors of the relative recursion at $(\mathsf{lv}(b_\pm), \mathsf{y}(b_\pm))$. Define

$$\mathsf{rdr}_b(\mathbf{U};\mathbf{A}) = \mathrm{Ad}(\mathrm{ax}_{\mathbf{U}}(b_-))^{-1}\frac{1}{i}\log\left[(\mathcal{M}^k\mathbf{U})^{\mathrm{ax}}(b)\{v_-^{-1}(\mathcal{M}^k\mathbf{U}')^{\mathrm{ax}}(b)v_+\}^{-1}\right]. \tag{1067}$$

All constituents use only the reading block. At a real configuration its real averages coincide with the complex averages on the domains just proved. The accumulated factors always use the complex recursion.

For $j = t$, write $A = L^t \sup|a'|$ and $B = L^t \sup|a_0|$, $a_0 = \eta_k\mathbf{A}$, and suppose $A, B < c_{\mathrm{ia}}$. The product logarithm gives

$$e^{ia_0}e^{ia'}U = e^{i\widetilde{a}}U, \qquad L^t \sup|\widetilde{a}| \le 2(A+B) < 4c_{\mathrm{ia}} \le c_\sharp.$$

Thus the plain-average and contour bounds in (1063) apply also to this combined perturbation; the relative recursion uses the smaller radius c_{ia}. They show that the argument of the last logarithm is a product of unitary factors whose product is one, with inserted exponentials of total norm at most

$$(2\kappa_6+1)C_{\mathrm{ia}}''L^{-1}\{A+2(A+B)\} + \frac{4\kappa_6}{L-1}B \le \Sigma_{\mathrm{ia}}(A+B) < 3/32.$$

It is therefore within $3/16$ of one. The outer axial conjugation has norm at most $e^{1/8}$, so the read-out has norm less than $3e^{1/8}/13 < 1$. It is holomorphic in a_0 on $B < c_{\mathrm{ia}}$ and zero at $a_0 = 0$. Schwarz's inequality gives

$$|\mathsf{rdr}_b(\mathbf{U};\mathbf{A})| \le L^t \sup|\eta_k\mathbf{A}|/c_{\mathrm{ia}} = C_{\mathrm{rdr}} \sup|\mathbf{A}|, \qquad C_{\mathrm{rdr}} = L/c_{\mathrm{ia}}. \tag{1068}$$

The covariance formula for the dressed averages and the accumulated factors gives

$$\mathsf{rdr}_b(\mathbf{U}^u; \mathrm{Ad}(u)\mathbf{A}) = \mathrm{Ad}(u(b_-))\mathsf{rdr}_b(\mathbf{U};\mathbf{A}).$$

The same constituent formulas give compatibility with complex conjugation.

54.5. **Local terms of the shift.** Set

$$C_{\mathrm{stp}} = \max\{C_{\mathrm{rdr}}C_{\mathrm{src}}, 8L^3\}, \qquad c_{\mathrm{stp}} = \min\{L^{-2}c_{\mathrm{ia}}, (4C_2^{\mathrm{AC}})^{-1}\}.$$

In addition to the response conditions require

$$\varrho_{\mathbf{H}} \le c_{\mathrm{stp}}, \qquad C_{\mathrm{stp}}\varrho_{\mathbf{H}} \le \varrho_{\mathrm{rw}}, \qquad C_{\mathrm{stp}}C_{\mathrm{D,s}}e^{8\bar\kappa}\varrho_{\mathrm{rw}} \le 1, \qquad \bar\kappa \ge 4, \quad \check R_t \ge 19. \tag{1069}$$

For a zone bond b, let $\mathsf{P}\langle b\rangle$ be the one or two big cubes containing its endpoints. Their union contains its reading block. For $\mathsf{S} \supset \mathsf{P}\langle b\rangle$, put

$$\Phi_{b,\mathsf{S}}(s;\mathbf{p}) = \mathsf{rdr}_b\Big(\mathbf{U}; \mathbf{H}^{\mathfrak{o}}_{\mathsf{S}}(s; \mathbf{p}^{\mathrm{tc}}_{\mathsf{S}}; \mathsf{src}_{\mathsf{S}}[\mathbf{p}])\Big). \tag{1070}$$

Here $\mathbf{p}^{\mathrm{tc}}_{\mathsf{S}}$ is the free-boundary truncation in (1053). The source estimate and (1069) put its datum in the response ball; the last inequality puts the response in the read-out ball. Consequently Φ is holomorphic for $|s_\Delta| < e^{\bar\kappa}$, on the entire class $\mathfrak{P}^{\mathfrak{o}}_{\mathbf{H}}(\mathsf{S})$, and

$$|\Phi_{b,\mathsf{S}}(s;\mathbf{p})| \le C_{\mathrm{stp}}C_{\mathrm{D,s}}e^{8\bar\kappa}\mathsf{bs}_{\mathsf{S}}(\mathbf{p}). \tag{1071}$$

This is a composition of holomorphic maps in local matrix charts; in particular, no holomorphic choice of the real reference in a decomposition is required.

Use ordinary finite differences, keeping the anchor $\mathsf{P}\langle b\rangle$ at one:

$$\mathbf{H}(b;\mathsf{S};\mathbf{p}) = \sum_{\mathcal{K}\subset\mathsf{S}\setminus\mathsf{P}\langle b\rangle} (-1)^{\#(\mathsf{S}\setminus\mathsf{P}\langle b\rangle)-\#\mathcal{K}}\Phi_{b,\mathsf{S}}(\mathbf{1}_{\mathcal{K}\cup\mathsf{P}\langle b\rangle};\mathbf{p}). \tag{1072}$$

Set this expression to zero if the anchor is missing.

Proposition 54.5. Under the response floors and radii above, the functions $\mathbf{H}(b;\mathsf{S};\cdot)$ are holomorphic on $\mathfrak{P}^{\mathfrak{o}}_{\mathbf{H}}(\mathsf{S})$, commute with conjugation, and are gauge covariant at b_-. They vanish unless S is connected, contains $\mathsf{P}\langle b\rangle$, and meets the boundary cubes of the localization component. For $\mathsf{S}\in\mathcal{D}_t$,

$$|\mathbf{H}(b;\mathsf{S};\mathbf{p})|\le C_{\mathrm{stp}}C_{\mathrm{D,s}}\mathsf{bs}_{\mathsf{S}}(\mathbf{p})e^{-\bar\kappa(d_t(\mathsf{S})+1)/4}. \tag{1073}$$

Thus their amplitude and decay rate may be taken to be

$$\mathsf{am}_{\mathbf{H}}=C_{\mathrm{stp}}C_{\mathrm{D,s}}\varrho_{\mathbf{H}},\qquad \mu_{\mathbf{H}}=\bar\kappa/4. \tag{1074}$$

If $\bar\kappa/4\ge c_{\mathrm{E}}=\log 18432$, then, with $C_{\mathrm{E}}=4096$ and the supremum of an empty set interpreted as zero,

$$\sum_{\mathsf{S}\subset\mathsf{P}^{\mathrm{loc}}_{\mathfrak{o}}}\ \sup_{\substack{\mathbf{p}\in\mathfrak{P}^{\mathfrak{o}}_{\mathbf{H}}(\mathsf{S})\\ \mathsf{bs}_{\mathsf{S}}(\mathbf{p})\le\varrho_{\mathrm{sz}}}}|\mathbf{H}(b;\mathsf{S};\mathbf{p})|\le C_{\mathrm{stp}}C_{\mathrm{D,s}}C_{\mathrm{E}}\varrho_{\mathrm{sz}}e^{-(\bar\kappa/4-c_{\mathrm{E}})\check R_t}. \tag{1075}$$

■

Proof. Restriction of the pair, the truncated current and the tower source commutes with restriction of the cube parameters. The component property of the response then says that the read-out uses only the component containing the reading block. If S has another component, pairing the subsets in (1072) with and without one cube in that component gives zero. If S misses the boundary cubes, every source row is common to the two sequences; its source is zero. The response and the read-out both vanish at zero source. This proves the support assertions. The same constituent properties prove covariance and conjugation.

Every nonzero term joins a reading cube to a boundary cube. The zone collar gives a distance at least $LM(\check R_t-1)$ between them. Counting big-cube steps yields

$$N:=\#\mathsf{S}\ge d_t(\mathsf{S})+1\ge\check R_t.$$

The Cauchy integral for the ordinary finite difference in (1072) gives, since the anchor has at most two cubes,

$$|\mathbf{H}|\le C_{\mathrm{stp}}C_{\mathrm{D,s}}\mathsf{bs}_{\mathsf{S}}(\mathbf{p})\,e^{8\bar\kappa}(e^{\bar\kappa}-1)^{-(N-2)}.$$

Put $\epsilon_0=-\log(1-e^{-4})<1/50$. As $N\ge 19$, $8\le(N-3)/2$, and hence

$$8\bar\kappa-(N-2)(\bar\kappa-\epsilon_0)\le-\tfrac14\bar\kappa N.$$

This proves (1073). The connected-set count with the prescribed anchor, followed by the geometric sum starting at $d_t+1=\check R_t$, gives (1075) with the displayed entropy constants. □

The finite-difference identity also gives the exact total before any estimate. If $\mathbf{p}$ is given on the whole component, locality identifies every value in a smaller cube set with the corresponding value on the whole component. Möbius inversion gives

$$\sum_{\mathsf{S}\subset\mathsf{P}^{\mathrm{loc}}_{\mathfrak{o}}}\mathbf{H}(b;\mathsf{S};\mathbf{p}|_{\mathsf{S}\cap\mathfrak{o}})=\mathsf{rdr}_b\Big(\mathbf{U};\mathbf{H}^{\mathfrak{o}}_{\mathsf{P}^{\mathrm{loc}}_{\mathfrak{o}}}(\mathbf{1};\mathbf{p}^{\mathrm{tc}};\mathsf{src}[\mathbf{p}])\Big). \tag{1076}$$

All sums are finite. Terms with missing anchor are zero, and the anchored subset identity applies to the remaining terms.

54.6. **The actual background in the local class.** We verify the local gauges needed at the physical pair. This uses the layerwise estimate (1023), not a bound by the largest threshold without its scale factors. Let U be an element of a completed collared background of depth n, with datum regular at $(2\varepsilon_i)_{i\le n}$. Assume the running bounds under which proposition 53.11 holds, and fix $1\le t\le n$, $k=t-1$. If $\lambda(\mathfrak{e})$ is the largest layer index among the corners of a regularity element, then

$$|\partial U(p)-1|=L^{-2\lambda(p)}r_U(p),\qquad |J^{(u)}(U)(b)|=L^{3(u-\lambda(b))}r_U(b).$$

The running inequalities give $\varepsilon_{i+1} \le L^2 \varepsilon_i$; thus $\varepsilon_i L^{-2i}$ and $\varepsilon_i L^{-3i}$ decrease. At a corner of index at least i, it follows that

$$|\partial U(p) - 1| < C_{\mathrm{reg}} \varepsilon_i L^{-2i}, \qquad |J^{(u)}(U)(b)| < C_{\mathrm{reg}} \varepsilon_i L^{3(u-i)}. \tag{1077}$$

Put the current equal to zero on bonds that do not have four plaquettes in $\Omega_0^\diamond$; the same bound holds there.

The matrix divergence required for the gauge theorem also has the correct layerwise bound. For a full bond of index λ, each adjacent plaquette has radius less than both $C_{\mathrm{reg}} \varepsilon_\lambda$ and $\varepsilon_{\mathrm{v}} < 1$. Its squared curvature is therefore smaller than $L^{-4\lambda} C_{\mathrm{reg}} \varepsilon_\lambda$. The scalar/vector decomposition in (579) gives, in level-n lengths,

$$|D_U^{\mathsf{T}} \partial U(b)| < 2 C_{\mathrm{reg}} \varepsilon_\lambda \eta_n^2 \mathsf{L}(b)^{-3}. \tag{1078}$$

For a bare gauge $U^v = e^{ia}$, measure its size of index i by

$$\max\{L^i |a|,\ L^{2i} |a_\nu(x^{+\mu}) - a_\nu(x)|\}.$$

Writing $A = \eta_k^{-1} a$ identifies these with $\xi_i |A|$ and $\xi_i^2 |\nabla^\xi A|$ in level-k lengths. Increasing the index by one costs at most L^2.

Lemma 54.6. Put

$$\mathsf{c}_{\mathrm{lgs}} = 8 B_* L^3 c_*^{-1} \bar{C}_{\mathrm{reg}}, \qquad \ell_{\mathrm{c}} = M / L^2.$$

Assume $c_* \le 1$, the geometric floors

$$R_1 \mathfrak{M} \ge L^3, \qquad M \ge L^4 R_1 \mathfrak{M}, \qquad M \ge L^5 \mathfrak{M},$$

the history period conditions, and $\mathsf{c}_{\mathrm{lgs}} M \varepsilon_t \le 1$. For $0 \le i \le t$, every compact $\mathfrak{M}\xi_i$-aligned cube $q \subset \Omega_i^\diamond$ of side at most $12\mathfrak{M}\xi_i$ has a real gauge of index i at most $\mathsf{c}_{\mathrm{lgs}} M \varepsilon_t$. Every enlarged tiny cube $\mathfrak{y}^\sim$ meeting the bonds of the step has such a gauge of index t. These gauges restrict to any region containing the cube. ∎

Proof. First let $i \ge 1$, and write the side of q as $l\xi_i$, where $1 \le l \le 12\mathfrak{M}$. Its concentric enlargement of side $(l+4)\xi_i$, together with its $2\eta_k$-neighborhood, lies in $\Omega_{i-1}^\diamond$: its distance from q is at most $(2L+2)\xi_{i-1}$, whereas the gap is larger than $L^3 \xi_{i-1}$. Its side is less than half the period, since $l + 4 < M \le L^{m+K-i}$. Extend U by the identity outside $\Omega_0^\diamond$. No plaquette or divergence used in this enlargement changes.

In level-i units the curvature and divergence parameters are bounded by

$$\max\left\{ 2L^3 C_{\mathrm{reg}} \varepsilon_{i-1}, \max_{q' \ge i} 2L^{-2(q'-i)} C_{\mathrm{reg}} \varepsilon_{q'} \right\} = 2L^3 C_{\mathrm{reg}} \varepsilon_{i-1}.$$

Indeed all relevant indices are at least $i-1$, and $\varepsilon_{q'+1} \le L^2 \varepsilon_{q'}$. The smallness needed by theorem F follows from

$$2L^3 C_{\mathrm{reg}} l \varepsilon_{i-1} \le c_* \mathsf{c}_{\mathrm{lgs}} M \varepsilon_t \le c_*.$$

That theorem gives a gauge of size $2B_* L^3 C_{\mathrm{reg}} l \varepsilon_{i-1}$, which is at most $\mathsf{c}_{\mathrm{lgs}} M \varepsilon_t$.

For $i = 0$, write N for the side of the cube in bare units. Its aligned faces have half-integral coordinates, so it contains $N \le 12\mathfrak{M}$ bare sites in each direction. The comb gauge of lemma 14.4 gives $|a| \le \pi(N-1) C_{\mathrm{reg}} \varepsilon_0$; its first differences are bounded by twice this number. Since $N \le 12\mathfrak{M}$ and $(N-1) C_{\mathrm{reg}} \varepsilon_0 < 1$, this gives the same required bound.

An enlarged tiny cube has side $3\ell_{\mathrm{c}}$ in level-k lengths. Each of its points lies within $2\ell_{\mathrm{c}} + 1 < LM$ of Ω_t. The history gaps put it in $\Lambda_k \subset \Omega_k$ when $k \ge 1$, and in $\Omega_0^\diamond$ when $k = 0$. It is a compact cube, since $3\ell_{\mathrm{c}} + 4 < M$. For $k \ge 1$, the preceding gauge argument at index k, followed by the factor L^2 for index t, bounds its size by

$$L^2 \, 2B_* L^3 C_{\mathrm{reg}} (3\ell_{\mathrm{c}}) \varepsilon_{k-1} = 6 B_* L^3 C_{\mathrm{reg}} M \varepsilon_{k-1} \le \mathsf{c}_{\mathrm{lgs}} M \varepsilon_t.$$

At $k = 0$ the comb bound at index one is at most $6\pi C_{\mathrm{reg}} M \varepsilon_0$, with the same conclusion. The stated floors and the definition of $\mathsf{c}_{\mathrm{lgs}}$ give all inequalities used above. Restriction does not alter a gauge on its cube. □

Apply this at the last step $t = n$, and assume

$$\mathsf{c}_{\mathrm{lgs}} M \varepsilon_t \le \min\{\alpha_0, \alpha_2, \varrho_{\mathbf{H}}\}. \tag{1079}$$

For any $\mathsf{S} \subset \mathsf{P}_{\mathfrak{o}}^{\mathrm{loc}}$, use the pair restricted to $W = \mathsf{S} \cap \mathfrak{o}$, with decomposition $(U|_W, 0)$. A tested cube in W is contained in $\Omega_i^{\diamond}$ and has side at most $12\mathfrak{M}\xi_i$, so the containing cube in the pair definition may be the cube itself. The gauge bound is at most α_0. Equation (1077) proves the curvature bound with $\alpha_0' = C_{\mathrm{reg}}\varepsilon_t$ and the current bound with α_2. The perturbation bound holds for every $\alpha_1' > 0$, because the perturbation is zero. Since $C_{\mathrm{reg}} < \mathsf{c}_{\mathrm{lgs}} M$, one can take $0 < \alpha_1' \le \min(\alpha_1, \varrho_{\mathbf{H}} - C_{\mathrm{reg}}\varepsilon_t)$. Letting α_1' decrease to zero proves

$$\mathbf{p}_U^{\mathfrak{o}}|_{\mathsf{S}\cap\mathfrak{o}} \in \mathfrak{P}_{\mathbf{H}}^{\mathfrak{o}}(\mathsf{S}), \qquad \mathsf{bs}_{\mathsf{S}}(\mathbf{p}_U^{\mathfrak{o}}|_{\mathsf{S}\cap\mathfrak{o}}) \le C_{\mathrm{reg}}\varepsilon_t. \tag{1080}$$

All decompositions and gauges here are on the actual restricted field.

Corollary 54.7. Let the completed background have depth $n \ge t$, and retain the preceding running and geometric hypotheses. If $m' < m + K - t$ and

$$\mathsf{c}_{\mathrm{lgs}} M \varepsilon_t \le \tfrac{1}{2} \min\{\alpha_0, \alpha_2\}, \tag{1081}$$

then, on every nonempty region X for π_t, the actual pair of step t, with its decomposition $(U|_X, 0)$, belongs to

$$\mathscr{O}^+_{h|_t}(X; \alpha_0/2, \alpha_1, \alpha_2/2). \tag{1082}$$

■□

Proof. The step-t index at a site is the minimum of its depth-n index and t. Equation (1077), together with the monotonicity of the thresholds, therefore gives the curvature and current conditions with $\alpha_0/2, \alpha_2/2$ on $X \cap \Omega_0^{\diamond}$. The complex part is zero. The containing-box gauges of lemma 54.6 give the natural cube condition with radius $\alpha_0/2$; its tiny-cube gauges give the same radius near every step bond. These are precisely the additional gauges in the definition of $\mathscr{O}^+$. One decomposition satisfies all conditions. □

54.7. **Identification with the background difference.** Let $\mathsf{v} \in \mathsf{F}_t^{\mathrm{reg}}$, and choose an element U of its completed collared background $U_t^{\diamond}(\mathbf{V}^{\diamond+})$. On a localization component $\mathfrak{o}$, take the pair $\mathbf{p}_U^{\mathfrak{o}}$ consisting of U and its given-sequence current. Assume (1079); by (1080) this pair is in the required local class. Use the running and smallness hypotheses of the joined background construction, including $\varepsilon_t \le \epsilon_{\mathrm{H}}$, and require also

$$C_{\mathrm{stp}} C_{\mathrm{rs}} \varrho_{\mathrm{rw}} \le \varepsilon_{\mathrm{v}}, \qquad 2C_{\mathrm{reg}}\varepsilon_t \le \varepsilon_{\mathrm{v}}. \tag{1083}$$

We prove

$$\sum_{\mathsf{S}\subset\mathsf{P}_{\mathfrak{o}}^{\mathrm{loc}}} \mathbf{H}(b; \mathsf{S}; \mathbf{p}_U^{\mathfrak{o}}|_{\mathsf{S}\cap\mathfrak{o}}) = \mathrm{Ad}(\mathsf{ax}_U(b_-))^{-1} \mathbf{H}^{(k)\diamond}(b). \tag{1084}$$

Here the right side is the difference of the two actual axial backgrounds used in the strip map.

Write $U_0 = U|_{\mathfrak{o}}$. The restriction of the completed background is a critical configuration for the joined sequence. The free-boundary rule (1055) makes the truncated pair on the whole component $(U_0, J_{\mathfrak{o}}U_0)$. Put

$$B' = \mathsf{src}[\mathbf{p}_U^{\mathfrak{o}}], \qquad T = \mathbf{H}^{\mathfrak{o}}_{\mathsf{P}^{\mathrm{loc}}_{\mathfrak{o}}}(\mathbf{1}; (U_0, J_{\mathfrak{o}}U_0); B'), \qquad U'' = e^{i\eta_k T} U_0.$$

The source is real and smaller than ϱ_{rw}; proposition 54.3 applies. It makes U'' critical for its new actual datum. Moreover,

$$\|T\|, \quad |F'_{\mathfrak{o},U_0}(T) - J_{\mathfrak{o}}U_0|^{\sharp} \le C_{\mathrm{rs}}|B'| \le \varepsilon_{\mathrm{v}}/C_{\mathrm{stp}}. \tag{1085}$$

We identify the datum before invoking uniqueness. At a joined node x of level q, define

$$w(x) = \mathsf{dsc}_{q,n^\diamond(x)} U_0(x),$$

where $n^\diamond(x)$ is the index of the given completed sequence. The tower definition gives $e^{iB'} V_0 = w^{-1} V^{\mathrm{loc}} w$ on the joined rows. For a common row of level n, an inner endpoint has both indices equal to n, so its descent is the identity. An outer endpoint has joined index $n-1$ and given index in $\{n-1, n\}$. In the first case the descent has no step; in the second it starts at the center of its own block and is again the identity.

The outward twists in the straightened constraint are trivial. To see this at an outer endpoint z of level n, its level-n block has either no tower row or all its lower rows in the same tower. In the first case the source there is zero. In the second, let the tower reach level l, and put $g_1 = \mathsf{dsc}_{n-1,l}$. On the lower block the source gives

$$e^{i\beta(c)} W_0(c) = g_1(c_-)^{-1} g_1(c_+).$$

The lower lift inside this block is the identity. Each perturbed contour from z to x is $g_1(z)^{-1} g_1(x)$. But

$$g_1(x) = \mathsf{dsc}_{n,l}(z) W_0(z, x), \qquad g_1(z) = \mathsf{dsc}_{n,l}(z);$$

the contour therefore equals the original one. Its relative mean is zero, and its outward twist is one. This verifies the seam condition for all joined rows.

Let r be the node factor supplied by proposition 54.3. Set $g = wr^{-1}$ at joined nodes and $g = 1$ at other bare sites. Then $U''' = (U'')^g$ has datum V^{loc} and is critical for it. We next check its regularity, which is not implied by criticality alone.

The completed background has normalized curvature and current at most $C_{\mathrm{reg}} \varepsilon_t \le \varepsilon_{\mathrm{v}}/2$. Adjacent joined indices differ by at most one. For a plaquette of joined index n, the expansion of its four perturbed bonds gives

$$\begin{aligned} |\partial U''(p) - 1| &\le |\partial U_0(p) - 1| + \eta_k^2 (2|\nabla_{U_0} T| + 2|T|^2) \\ &\le |\partial U_0(p) - 1| + 4L^{-2n} \|T\|. \end{aligned}$$

Its maximum corner index is at most $n+1$. After normalization the added term is at most $4L^2 \varepsilon_{\mathrm{v}} / C_{\mathrm{stp}}$. Since $C_{\mathrm{stp}} \ge 8L^3$, the plaquette remains within the ε_{v}-bound.

For the current, differentiating the left exponential gives the exact identity

$$F'_{\mathfrak{o},U_0}(T)(b) = f(Z) J_{\mathfrak{o}}(U'')(b), \qquad f(z) = \frac{e^z - 1}{z}, \qquad Z = -i\eta_k \operatorname{ad} T(b).$$

By (1085), $|Z| \le 2/C_{\mathrm{stp}} \le 1/(4L^3)$. Thus $|f(Z) - 1| \le 1/100$ and $|f(Z)^{-1}| \le 100/99$. The conversion from a bond's index to a corner's index costs at most L^3, so its normalized current is bounded by

$$\frac{100}{99} \left(C_{\mathrm{reg}} \varepsilon_t + L^3 C_{\mathrm{rs}} \varrho_{\mathrm{rw}} \right) \le \frac{100}{99} \left(\frac{1}{2} + \frac{1}{8} \right) \varepsilon_{\mathrm{v}} < \varepsilon_{\mathrm{v}}.$$

These bounds are gauge invariant. The uniqueness of the regular critical orbit therefore identifies U''' with the joined background.

Finally, at the two nodes used by the read-out, the given and joined indices agree in the core. Hence $w = 1$, the node factors r are exactly $v_\pm$ in (1067), and $g = v_\pm^{-1}$. Substituting the identified joined background in that equation gives the right side of (1084). The argument of its logarithm lies in the series-logarithm ball proved above, so that logarithm agrees with the intrinsic real logarithm. Equation (1076) proves (1084). In the axial representative $\mathsf{ax}_U = 1$, its sum is precisely $\mathbf{H}^{(k)\diamond}$.

54.8. **The sum of the strip terms.** We apply the preceding identification to a chain point $y = (W, B, \phi^+)$ of a bare field, with regular base d. Use the adapted representative U_{d} of its completed collared background. Let $\mathsf{U} = U_{\mathsf{d}} \vee v_0$ be the glued configuration, let $\mathfrak{q}$ be the old stored fields, and write $B = (B_{\mathrm{I}}, \mathsf{a})$. The new stored external field is a'. The coordinate and support results give

$$|\mathsf{a}(b)|,\ |\mathsf{a}'(b)| < \mathsf{rad}_k(b) \le C_{\mathrm{xt}}\mathfrak{b}_k, \qquad g_k|\mathsf{a}(b)|,\ g_k|\mathsf{a}'(b)| < 1/8 \tag{1086}$$

under their tier ceiling $C_{\mathrm{xt}}\delta_k \le 1/8$. Both $(\mathfrak{q}, \mathsf{a})$ and $(\mathfrak{q}, \mathsf{a}')$ therefore satisfy the stored-data tiers of h'. The old tiers are the ones of h, since prolongation does not change an earlier stored set or its radius.

Let P_X be the plain triple read from U at level t on X, and let $\mathbf{p}_T$ be the pair of U_{d} restricted to the localization region. Restriction of the triple, its pair and the old stored data gives

$$P_{X_1}|_X = P_X, \qquad \mathbf{p}_X^{\mathrm{loc}}(P_X) = \mathbf{p}_T|_X, \qquad (\mathfrak{q}|_{X_1})|_X = \mathfrak{q}|_X \quad (X \subset X_1). \tag{1087}$$

For the pair identity, the glued configuration is U_{d} on all inner bonds of $\Omega_0^\diamond$. A current is computed there only when all four plaquettes are inner, and is zero otherwise. Thus its reading is the same before and after gluing; the level-t current is multiplied by L^{-3} on passage to level k. This proves the stated pair identity with the correct current normalization.

Equation (1080) applies at every component, including components with no zone bond. It follows that $\mathbf{p}_T$, and every restriction in (1087), belongs to the classes of proposition 54.5. Adaptation at the last layer means that $\mathcal{M}^k U_{\mathsf{d}}$ is axial on $\Omega_t^{(t)}$. Hence (1084) has no conjugating factor, and

$$H_{\mathrm{tot}}[\mathbf{p}_T] = \mathfrak{H}_{\mathsf{d}}, \qquad \mathsf{a} = \check{\Psi}(\mathsf{a}', H_{\mathrm{tot}}[\mathbf{p}_T]). \tag{1088}$$

The first identity is the actual background shift of the base. For the second, the definition of the stored coordinate gives $\mathsf{a} = \mathsf{a}'$ off $\mathcal{Z}$, and on $\mathcal{Z}$ it gives

$$g_k\mathsf{a}(b) = \mathsf{cm}(g_k\mathsf{a}'(b), \vartheta_b\mathfrak{H}_{\mathsf{d}}(b)).$$

These are exactly the values of (992). The smallness proved for the actual shift is $8\|\vartheta'_{\mathrm{co}}\|_\infty|\mathfrak{H}_{\mathsf{d}}(b)| \le \delta_k$, so the real strip map and its positive Jacobians are defined.

The following statement keeps visible the analytic domain conditions on the local potentials. They are the same conditions required to form the boundary terms in section 62.

Proposition 54.8. Use a chain point as above and the local terms of proposition 54.5, with the rate and amplitude conditions (1036) and (1042). Assume P_X belongs to the intermediate graded space $\mathscr{U}_X$ for each X, and that the kept, rejected and cluster families satisfy the domain and restriction hypotheses of section 62 there. In particular the cluster family is defined on the stored tiers and has the indicated complex room in the new zone variables; the two quadratic families are defined for $|\mathsf{a}_1| < 2C_{\mathrm{xt}}\mathfrak{b}_k$. Assume (1380) and all the actual background and chart ceilings used above. Then every term in the following equality is defined, and

$$\begin{aligned}\sum_{X\in\mathcal{D}_t} B^{\mathrm{sr}}_{h'}(X;\mathsf{U};\mathfrak{q},\mathsf{a}') &= \mathsf{Zs}_{\mathsf{d}}(\mathsf{a}') \\ &+ \sum_{X\in\mathcal{D}_t} \Big\{[E^{\mathrm{cl}}_{h'} + B^{\times}_{h'} + K^{\mathrm{kp}}_{h'}](X;\mathsf{U};\mathfrak{q},\mathsf{a}) \\ &\qquad -[E^{\mathrm{cl}}_{h'} + B^{\times}_{h'} + K^{\mathrm{kp}}_{h'}](X;\mathsf{U};\mathfrak{q},\mathsf{a}')\Big\}.\end{aligned} \tag{1089}$$

The quadratic forms in this notation do not depend on $\mathfrak{q}$. The identity holds also at $k = 0$, when the old stored family is empty. ∎

Proof. Both external fields are in the tiers by (1086); thus the two cluster values in (1089) are defined. The quadratic domains contain both fields. The background and stored-data restrictions in (1087) put each evaluation at the point $(P_X; (\mathfrak{q}, \mathsf{a}')|_X)$ used to define the strip terms.

Extend $\mathsf{a}'|_{X_1}$ by zero when forming a term on X_1. Replacing this extension by the global field a' changes neither the domain condition nor the value: each shifted bond used by that term lies in X_1, and each original function over $X \subset X_1$ reads only its external bonds in X. The same locality holds for the cluster function's external domain. Restriction of $\mathbf{p}_T$ therefore replaces the local pair by one global pair in all five rows. Restriction of the old data and the triple does the same for their further arguments.

For the four transfer rows use the families $F_X^7, F_X^8, F_X^9, F_X^{\mathrm{E}}$ in (1381). They are functions over X, including their domains, and their values at either a or a' are the actual local Lebesgue, kept, rejected and cluster terms. The room hypotheses, (1380) and proposition 53.15 show that every finite-difference vertex in their transfer terms is defined.

Here is the exact finite summation, including its locality requirement. For fixed X, put

$$f(\mathscr{S}) = F_X\left(\check{\Psi}\left(\mathsf{a}', \sum_{(b,S)\in\mathscr{S}} \mathbf{H}(b;S;\mathbf{p}_T)\right), w\right), \qquad \mathscr{S} \subset \mathsf{Pr}[X].$$

The same subset gives the same value in every target X_1 containing its supports. Each nonempty subset is assigned exactly once, to $X(\mathscr{S})$, in (1037). Möbius summation thus gives

$$\sum_{X_1} F_X^{\mathrm{tr}}(X_1) = f(\mathsf{Pr}[X]) - f(\varnothing) = F_X(\mathsf{a}, w) - F_X(\mathsf{a}', w).$$

The last equality uses (1088) on the bonds read by F_X, and $\check{\Psi}(\mathsf{a}', 0) = \mathsf{a}'$. If $\mathsf{Pr}[X]$ is empty, both differences are zero. Summing over X proves the identities for rows $8, 9, \mathrm{E}$.

For row 7, a bond is assigned to its one anchor $X = \mathsf{P}\langle b\rangle$, so the same calculation gives

$$\sum_X R^7(X) = \sum_{b\in\mathcal{Z}^+} \{\mathsf{ls}_b(\mathsf{a}) - \mathsf{ls}_b(\mathsf{a}')\} = \mathsf{Zs}_{\mathrm{L}}(\mathsf{a}'; \mathfrak{H}_{\mathsf{d}}).$$

The logarithm is the one in (994); the bounds $g_k|\mathsf{a}|, g_k|\mathsf{a}'| < 1/8$ keep both values on its stated branch. For row 6, (1044) and (1088) give

$$\sum_X R^6(X) = \sum_{b\in\mathcal{Z}_\theta^+} \log J_{\mathfrak{H}_{\mathsf{d}}(b)}(e^{ig_k \mathsf{a}'(b)}) = \mathsf{Zs}_{\mathrm{J}}(\mathsf{a}'; \mathfrak{H}_{\mathsf{d}}).$$

The Jacobians are positive, so the local series logarithms agree with these real logarithms. Adding the five finite sums proves (1089), since $\mathsf{Zs}_{\mathsf{d}} = \mathsf{Zs}_{\mathrm{J}} + \mathsf{Zs}_{\mathrm{L}}$. □

If the strip region is empty, then $\mathcal{Z} = \varnothing$: the Jacobian sum and every transfer sum are empty. Thus the strip terms vanish on their domains. If already $\Omega_1 = \varnothing$, the completed and localization regions, and every stored bond set, are empty. Their pair and stored-data spaces are singletons; the graded conditions on the empty background impose no restriction on a triple. The five rows are then defined empty sums, while the kept, rejected and cluster terms are the zero terms of the empty-region convention. Both sides of (1089) are zero.

55. Gaussian fields in a history step

The integration region and its exterior now share a quadratic form. We first extend the connected coefficients to the collared background sequence. Local inverses then separate the integrated field from its conditional mean; the remaining quadratic form is divided into terms kept on strip components and terms crossing between them.

55.1. **The form and its local coefficients.** Fix a history through step $t = k+1$, with $m' < m + K - t$, and use level-k units. Thus $\xi = L^{-k}$, $\xi_n = L^{n-k}$ and $\ell_c = M/L^2$. The collared sequence is

$$\Omega_\bullet^\diamond = (\Omega_1^\sim, \Omega_1, \dots, \Omega_t). \tag{1090}$$

Its fluctuation bonds $\mathfrak{X}$ are the nonpivot level-k bonds whose endpoints lie in $\Omega_0^\diamond$ and at least one of whose endpoints lies in Ω_t. They split into integration bonds $\mathfrak{X}_{\mathrm{I}}$, having an endpoint in Λ_t, and external bonds $\mathfrak{X}_{\mathrm{E}}$. Keep the grain, clearance and covariance floors of (763), and require $w \le \ell_c$. All kernels and transposes use the positive-bond counting pairing of section 50. Here $a^{\mathrm{cv}} = a_{\mathrm{cv}}$ is the previously fixed covariance radius. Every tiny cube meeting $\mathfrak{X}$ has its enlargement inside $\Omega_0^\diamond$: its points are within $2\ell_c + 2$ of Ω_t, whereas the history collar has width at least LM.

For a coarse region R, a pair $p = (\mathbf{U}, \mathbf{J})$ is given on the inner bare bonds of $R \cap \Omega_0^\diamond$, with one decomposition $\mathbf{U} = e^{i\xi A'} U$, U real. Write $\mathscr{O}(R; \alpha_0, \alpha_1, \alpha_2)$ for the following local class. At every contained stencil based in layer n,

$$|\partial U - 1|, |\partial \mathbf{U} - 1| < \alpha_0 \xi^2 \xi_n^{-2}, \qquad \xi_n |A'| < \alpha_1, \quad \xi_n^2 |\nabla^U A'| < \alpha_1, \quad \xi_n^3 |\mathbf{J}| < \alpha_2. \tag{1091}$$

A plain gauge cube of index n is an embedded cube made of level-n $\mathfrak{M}$-cubes, of side at most $12\mathfrak{M}\xi_n$, contained in the union of layers $n, n+1$ and meeting layer n. Each such cube q contained in $R \cap \Omega_0^\diamond$ must lie in an embedded cube $q' \subset R \cap \Omega_0^\diamond$ with a real gauge $U^v = e^{i\xi A}$ satisfying $\xi_n |A| \le \alpha_0$ and $\xi_n^2 |\nabla^\xi A| \le \alpha_0$. Finally, every tiny cube y containing an integration-bond endpoint, with $y^\sim \subset R \cap \Omega_0^\diamond$, must admit such a gauge on $y^\sim$ at index t. The class $\mathscr{O}^+(R; \alpha_0, \alpha_1, \alpha_2)$ imposes this last condition for tiny cubes containing any fluctuation-bond endpoint. These definitions use the same U throughout. Bounds on derivatives are imposed only where their complete stencil lies in the stated domain. Restriction preserves these classes: restrict each supplied cube gauge to the tested cube itself. Real gauge transformations and conjugation also preserve them, as does passage to the real reference $(U, 0)$. For $\alpha_1 \le \pi$ they are open. Indeed one may fix U and all its gauges, use a local inverse of the exponential at each $i\xi A'(b)$, and preserve the finitely many strict field inequalities.

On the shorter sequence through level k, use the straightened scalar and field averages and the operators $\mathsf{G}_1, \mathsf{H}_1$ already constructed from them. Put $j_{**} = \mathsf{H}_1^{\mathsf{T}} \mathbf{J}$ and $V = \mathcal{M}^k \mathbf{U}$. The one-step pivot maps at V use the ordinary one-step average. For row fields supported on $\mathfrak{X}_0[\Omega_\bullet^\diamond]$, define

$$\begin{aligned} \langle B_1, \Delta_k^\sharp(p) B_2 \rangle = (B_1, (Q\mathsf{G}_1 Q^{\mathsf{T}})^{-1} B_2)_{\mathbf{B}} - (B_1, \mathsf{a} B_2)_{\mathbf{B}} \\ - 2(\mathsf{D}_V^{(2)}(B_1, B_2), j_{**})_{\mathbf{B}}, \end{aligned} \tag{1092}$$

where $\mathsf{a} = 1$ on level-k rows. Extend the inputs by zero before applying the one-step maps, and restrict their output back to these rows. The fluctuation form on the nonpivot coordinates is

$$\mathsf{A}^\sharp[\Omega_\bullet^\diamond](p) = \mathsf{C}_V^{\mathsf{T}} (\mathsf{G}_V^{(2)} + \Delta_k^\sharp(p)) \mathsf{C}_V. \tag{1093}$$

This is the same construction as the constant-sequence form, with its actual row spaces and zero extensions. In particular its domain requires the nested averages, pivot inverses and the scalar and field inverses appearing in this formula. We verify their existence before using it.

For a grain region Y, take the maximal tower $\Omega_\bullet(Y)$ of section 50.1 and form the meet $\Omega_n^\diamond \langle Y \rangle = \Omega_n^\diamond \cap \Omega_n(Y)$. Compress its fluctuation form to the bonds with $S_0(b) \subset Y$, and extend by zero to $\mathfrak{X}$. Denote the resulting form by g_Y and put $g_\emptyset = 0$. Define a_S by (769), with these g_Y.

Proposition 55.1. Assume $\alpha_0, \alpha_1, \alpha_2 \le a^{\mathrm{cv}}$ and the grain conditions above. If $p \in \mathscr{O}(R; \alpha_0, \alpha_1, \alpha_2)$, all the meet operators defining a_S exist whenever $S \subset R$. The coefficients have the symmetry, connected support and estimate of proposition 50.1, with the same constants. In particular, writing $2\theta_a = \delta_a$, $\epsilon_\star = (2n_{\mathrm{w}})^{-1}$,

$$|a_S[p](b, b')| \le C_a \epsilon_\star^{\#S - 1} e^{-2\theta_a \Lambda_S(b, b')} =: \beta_S(b, b'). \tag{1094}$$

They are holomorphic and depend only on the pair on $\Omega_0^\diamond \cap S$ and the sequence in $S^\sim$. Their domains have the same locality. Restriction, real gauge transformations and conjugation commute with their construction.

If $\alpha_0 \le a^{\mathrm{cv}}/2$ and $R_d = c_{\mathrm{d}} a^{\mathrm{cv}}/(\alpha_1+\alpha_2) > 1$, the same coefficients are holomorphic on $p_z = (e^{iz\xi A'}U, z\mathbf{J})$ for $|z| < R_d$ and satisfy (1094) there. For $R_d \ge 2$,

$$|a_S[p] - a_S[(U,0)]| \le (2/R_d)\beta_S. \tag{1095}$$

On the whole torus their finite sum is the actual form $\mathsf{A}^\sharp[\Omega_\bullet^\diamond](p)$. ■

Proof. First check admissibility on a meet. A site in its layer n belongs to an original layer $j \ge n$, so every bound in (1091) weakens under restriction. For the gauge condition it suffices to enclose a candidate cube $q \subset \Omega_n^\diamond \cap Y$ in a plain cube of the original sequence lying in Y. Choose the largest $j \ge n$ with $q \subset \Omega_j^\diamond$. If $j = n$, use q itself. Otherwise the side of q is less than $\mathfrak{M}\xi_j$. In each coordinate enclose its interval in two adjacent intervals of length $\mathfrak{M}\xi_j$: straddle a crossed grid point, or choose the neighboring interval inside the same interval of length $L\mathfrak{M}\xi_j$ if no grid point is crossed. Every coarse interval used meets the original interval in a bare coordinate. Thus each level-j $L\mathfrak{M}$-cube used in the product contains a bare site of q and lies in both $\Omega_j^\diamond$ and Y; the grain partition is coarser than this partition. The enclosing cube has side $2\mathfrak{M}\xi_j$. It contains a bare site outside $\Omega_{j+1}^\diamond$ and cannot meet $\Omega_{j+2}^\diamond$, by separation and its diameter. It is therefore a plain cube of index j. Its supplied gauge, restricted to q, obeys the bounds of index n because $\xi_n \le \xi_j$. This proves the natural gauge condition on every meet, including its containing-box requirement.

The regular-pair inverse and pivot results now define (1092)–(1093) on every meet, with uniform radii. The form expansion (770) applies to these actual operators. Its local matrices and their domains depend only on the carrier, and the tower dependence still has range $e_{k+1} \le \rho_t$. Consequently the proof of proposition 50.1 applies without changing a constant: toggle a grain cube outside the endpoint supports and the e_{k+1}-neighborhood of the tree hull; the same matrix occurs with opposite signs. Each surviving cube lies within $r+1$ of the tree, and the two length bounds used there absorb the finite alternating sum and the grain count. This proves connectedness and (1094). No new summation estimate is needed.

For locality, a meet with $Y \subset S$ only uses the original sequence at bare sites of Y, and the pair on $\Omega_0^\diamond \cap Y$. If two sequences agree in $S^\sim$, all their meets with such Y agree, as do the nonpivot bond sets and the endpoint supports. Each operator in the finite alternating sum is then the same partial map, with the same domain. The gauge and conjugation identities follow through the nested averages, the bilinear transposes, inverses and pivot maps. They give real coefficients at real pairs. Finally the disc estimate (775) verifies the regularity bounds on every meet for p_z; the fixed real gauges are unchanged. Cauchy's estimate gives (1095). Finite inclusion–exclusion gives the whole-torus identity. □

The same enclosing-cube argument, with the coarse region R in place of Y, applies to every candidate cube contained in $R \cap \Omega_n^\diamond$, without the layer-intersection condition. It therefore verifies the gauge condition of the normalizer class in section 56. In particular $\mathscr{O}(R;\alpha_0,\alpha_1,\alpha_2) \subset \mathscr{O}^N(R;a_{\mathrm{nf}})$ whenever $a_{\mathrm{nf}} \ge \max\{\alpha_0,\alpha_1,\alpha_2\}$.

Represent a_S in the orthonormal Pauli coordinates of the positive bonds. Its footprint is $S^\sim$, and its coarse hull $X(S)$ is the union of the LM-cubes meeting this footprint. In a region R, use only indices with $X(S) \subset R$ and set

$$\mathsf{A}_R^{+N}(p) = \sum_{\substack{X(S)\subset R \\ S^\sim \subset N}} !a_S[p]. \tag{1096}$$

This is a finite sum. Restricting the pair and the region retains exactly those indices whose hull remains inside the smaller region; their values and domains are unchanged. The zero entries outside

$S_0(b) \cup S_0(b') \subset S$ are retained in these matrix representations. On the whole torus write A^+ for the sum; it represents (1093).

Splitting across components. If two coarse regions Y_1, Y_2 have no equal or face-adjacent cubes, their meet operators, including their domains, form a direct sum:

$$\mathsf{A}^\sharp_{Y_1 \cup Y_2} = \mathsf{A}^\sharp_{Y_1} \oplus \mathsf{A}^\sharp_{Y_2}. \tag{1097}$$

Edge and corner contacts are allowed. To see the tower splitting, join lifts of points in different parts by a coordinate-monotone chain of coarse cubes. Before reaching the other part the chain passes through a cube of $(Y_1 \cup Y_2)^c$; a point of that cube is no farther from the first point than the second point is. Thus the distance to the complement of one's own part equals the distance to the complement of the union at every stage of the tower recursion. Induction gives the disjoint union of the two towers and hence of the two meets.

No inner bond or plaquette joins the parts. Each averaging block is in the part of its center; a top row and its double block stay in that part by the last tower margin $R_1\mathfrak{M} \geq L^2 > 3L/2$. Therefore the actual nested averages, their derivatives, the gradient and its bilinear transpose, the Hessian form and the pivot maps all split, with identical branch domains. Their sums, products and inverses split as well; an inverse exists precisely when both blocks are invertible. This proves (1097) for the actual form, not only for its expansion. Any compression whose retained endpoint support is wall-connected splits in the same way. In particular the deep integration compression and each diagonal entry of every power in its logarithm series are those of its own component. Empty components carry the unique operator on the zero-dimensional space.

55.2. **A local gap and the conditional inverse.** The tiny-cube gauge condition has a specific purpose: it transfers the flat gap to every local compression used to integrate a history field. The comparison must be made before inverting that compression. Write γ_{tr} for the gap in (593) and put $\gamma_{\mathrm{pos}} = \min\{1, \gamma_{\mathrm{tr}}/2\}$. The counting norm and the coefficient bound give constants

$$B_{\mathrm{w}} = 2\sqrt{2}C_a, \qquad \delta = \delta_{\mathrm{cv}}/(8L), \qquad \varepsilon_{\mathrm{w}} = \frac{2(\alpha_1 + \alpha_2)}{c_{\mathrm{d}} a_{\mathrm{cv}}}.$$

For each coefficient there are fixed majorants m_S and lengths $l_S = \Lambda_S$ such that

$$|a_S(p)| \leq m_S, \qquad |a_S(p) - a_S(U,0)| \leq \varepsilon_{\mathrm{w}} m_S, \qquad \sum_S m_S(b,b') e^{\delta l_S(b,b')} \leq B_{\mathrm{w}}. \tag{1098}$$

This is (1094) summed by (773), with the positive-bond coordinate norm. A connecting tree has endpoints b_-, b'_-; every point of $S^\sim$ is within $3w$ of it.

We henceforth take $L^2 w \leq \ell_c$, enlarge a fixed lower bound on ℓ_c, and reduce a fixed upper bound a_{pos} on α_0. All these choices depend only on L. The quantitative requirements on these choices are given in the proof below.

Proposition 55.2. Let $p \in \mathscr{O}(R; \alpha_0, \alpha_1, \alpha_2)$ and choose one of its admitted real references $(U, 0)$. For every nonempty union W of tiny cubes containing integration-bond endpoints, with $W^\sim \subset R$, and every N containing the closed $2\ell_c/3$-neighborhood of W,

$$\left\langle f, \sum_{\substack{X(S) \subset R \\ S^\sim \subset N}} a_S(U,0) f \right\rangle \geq \gamma_{\mathrm{pos}} |f|^2 \tag{1099}$$

for real f supported on integration bonds with an endpoint in W. For $p \in \mathscr{O}^+(R; \alpha_0, \alpha_1, \alpha_2)$ the same assertion holds with all fluctuation bonds in place of integration bonds.

There are constants $B_{\mathrm{f}} \geq 1$, $\delta_{\mathrm{pl}} > 0$ depending only on L such that, when $\alpha_1 + \alpha_2$ is sufficiently small, the local inverses constructed below have a convergent connected expansion t_J. Its terms are

holomorphic on their actual hulls, commute with restriction, gauge transformations and conjugation, and have fixed majorants satisfying

$$|t_J(p)| \le m_J, \qquad |t_J(p) - t_J(U,0)| \le \varepsilon_* m_J, \qquad \sum_J m_J(b,b') e^{\delta_{\mathrm{pl}} l_J(b,b')} \le B_{\mathrm{f}}. \tag{1100}$$

Here $\varepsilon_* = C_{\mathrm{res}} \mathsf{cn}_{\mathrm{w}} \varepsilon_{\mathrm{w}}$, where $\mathsf{cn}_{\mathrm{w}} = B_{\mathrm{w}} c_{\mathrm{lat}}(\delta/8)/\gamma_{\mathrm{pos}}$; $c_{\mathrm{res}} > 0$ and C_{res} are absolute constants. The same bound holds for the coefficients a_S with these enlarged constants. Every term has its input and output bonds in its footprint, which lies within $\rho_* = 4\ell_c$ of a connecting tree of length l_J. On the whole torus,

$$\sum_J t_J = (A_{\mathrm{II}}^+)^{-1}, \qquad A^+ = \sum_S a_S = \operatorname{vec} A^\sharp. \tag{1101}$$

All assertions include the zero-dimensional case. ∎

Proof. Fix a tiny cube y whose enlargement lies in the pair domain and for which the appropriate tiny-cube gauge is available. In that gauge $U = e^{i\xi A}$, with $L|A| \le \alpha_0$ and $L^2|\nabla^\xi A| \le \alpha_0$. Set

$$\varsigma(x) = \min\{1, \max\{0, 11/3 - 4\operatorname{dist}_\infty(x,y)/\ell_c\}\}, \qquad \widetilde{A}(b) = \varsigma(b_-) A(b)$$

on positive inner bonds of $y^\sim$, extend by zero, and impose antisymmetry on reversed bonds. The cutoff is one within $2\ell_c/3$ of y and zero beyond $11\ell_c/12$. Its Lipschitz constant is $4/\ell_c$. The finite-difference product rule gives $L|\widetilde{A}| \le \alpha_0$ and $L^2|\nabla^\xi \widetilde{A}| \le 2\alpha_0$. This remains true at a stencil crossing the boundary: both initial points then have distance greater than $\ell_c - 2 \ge 11\ell_c/12$ from y, so both cutoff values vanish. The real plaquette estimate proved with the action bounds gives

$$|\partial e^{i\xi \widetilde{A}} - 1| \le 6\alpha_0 \xi^2 L^{-2}.$$

Thus $(e^{i\xi\widetilde{A}}, 0)$ has the flat real reference and is admitted globally at radii $(7\alpha_0, 3\alpha_0, \alpha_0)$. Choose $7\alpha_0 \le a_{\mathrm{cv}}/2$ and $8\alpha_0 \le c_{\mathrm{d}} a_{\mathrm{cv}}$. Its complete form differs from the flat form by at most

$$\varepsilon_\beta B_{\mathrm{w}} c_{\mathrm{lat}}(\delta), \qquad \varepsilon_\beta = 8\alpha_0/(c_{\mathrm{d}} a_{\mathrm{cv}}).$$

On coefficients with $S^\sim$ in the closed $2\ell_c/3$-neighborhood of y, this form is the original one conjugated by the real orthogonal matrix $\operatorname{Ad} v(b_-)$. This follows from exact coefficient locality, including its domain, not from a bound on the total form. The flat form is bounded below by γ_{tr}: (593) applies to the collared sequence, whose fluctuation bonds have the required $2L$ margin. Its proof uses the same actual straightened average and pivot coordinates as (1093).

Here are the two losses in passing from this comparison to a partial sum. Put $\widehat{\delta} = \min(1, \delta)$ and $\lambda = \ell_c/3 - 3w - 1/2$. If a footprint is not contained in that neighborhood and the output midpoint is within $\ell_c/3$ of y, its connecting length is at least λ and at least $|b - b'|$. The Schur bound therefore gives a discarded-tail norm at most $E_1 = B_{\mathrm{w}} c_{\mathrm{lat}}(\delta/2) e^{-\delta\lambda/2}$. For localization use $h_y(x) = \max(0, 1 - 3\operatorname{dist}_\infty(x,y)/\ell_c)$ and $\chi_y = h_y/(\sum_z h_z^2)^{1/2}$ on the relevant field bonds. The denominator is at least one. At most sixteen cutoffs occur at a pair of bonds, and their Lipschitz bound gives

$$\sum_y |\chi_y(b) - \chi_y(b')|^2 \le 144 \ell_c^{-2} (|b - b'| + 1/2)^2.$$

Apply the exact localization identity from section 50.2. Summing the remaining exponential bounds its error by

$$E_2 = 156 B_{\mathrm{w}} c_{\mathrm{lat}}(\delta/4) (\widehat{\delta}\ell_c)^{-2}.$$

Localize a field on W with the cutoffs indexed by tiny cubes in W. Every near footprint is among the retained terms; compare its localized form with the global gauged form and discard the two tails. The resulting lower bound is

$$\gamma_{\mathrm{tr}} - \varepsilon_\beta B_{\mathrm{w}} c_{\mathrm{lat}}(\delta) - 2E_1 - E_2.$$

Choose a_{pos} and the tiny-cube floor so that the last three terms sum to at most $\gamma_{\mathrm{tr}}/2$. Since $3w \le \ell_c/3$ and $L^2 w \le \ell_c$, these choices are uniform in the history and volume. This proves (1099), in its two stated domains.

We give the actual inverse expansion, retaining its localization. For each tiny cube y containing an integration-bond endpoint put

$$\zeta_y(x) = \min\{1, \max\{0, 3 - 3\,\mathrm{dist}_\infty(x,y)/\ell_c\}\}, \quad N_y = \{\mathrm{dist}_\infty(\cdot, y) < 2\ell_c\}, \quad \eta_y(b) = \frac{h_y(x_b)^2}{\sum_z h_z(x_b)^2}.$$

Here x_b is the midpoint. Write $X_y = \{b \in \mathfrak{X}_{\mathrm{I}}^+ : \mathrm{dist}_\infty(x_b, y) < \ell_c\}$, and use a cube in a region R only when every big cube meeting N_y belongs to R. All inverses below act on X_y and are then extended by zero. With $a_S^{\mathrm{I}} = p_{\mathrm{I}} a_S p_{\mathrm{I}}$, set

$$A^y = p_{X_y} \sum_{S^\sim \subset N_y} a_S^{\mathrm{I}} p_{X_y}, \qquad G_y^\mu = (\mu + A^y)^{-1},$$

$$e_{y,S} = \begin{cases} [a_S^{\mathrm{I}}, \zeta_y] p_{X_y}, & S^\sim \subset N_y, \\ a_S^{\mathrm{I}} \zeta_y, & S^\sim \not\subset N_y. \end{cases}$$

The local gap applies to $A^y(U,0)$: take the tiny cubes inside $y^\sim$ that meet integration bonds; their $2\ell_c/3$-neighborhood lies in N_y, and they contain the initial points of X_y. The complex difference has norm at most $\varepsilon_{\mathrm{w}} B_{\mathrm{w}} c_{\mathrm{lat}}(\delta)$. The same argument applies to the affine interpolation of all coefficients between p and $(U,0)$ on the disc

$$|z| \le R_{\mathrm{w}} = \frac{\gamma_{\mathrm{pos}}}{2\varepsilon_{\mathrm{w}} B_{\mathrm{w}} c_{\mathrm{lat}}(\delta)}.$$

Take $\varepsilon_{\mathrm{w}} \mathsf{cn}_{\mathrm{w}} \le c_{\mathrm{res}}$, with $c_{\mathrm{res}} = 1/128$. Then $R_{\mathrm{w}} \ge 64$, the interpolated coefficients are bounded by $2m_S$, and every local form has real-part gap $\gamma_{\mathrm{pos}}/2$. If $\varepsilon_{\mathrm{w}} = 0$, this interpolation is constant. Exponential conjugation, exactly as in (776), gives

$$|G_y^\mu(b,b')| \le \frac{2}{\mu + \gamma_{\mathrm{pos}}/2} e^{-\delta_{\mathrm{loc}}|b-b'|}, \qquad \delta_{\mathrm{loc}} = \frac{\delta}{16\mathsf{cn}_{\mathrm{w}}}.$$

A counted string is $\jmath = (y_0; (y_1, S_1), \dots, (y_n, S_n))$, with $N_{y_{i-1}} \cap S_i^\sim \ne \emptyset \ne S_i^\sim \cap N_{y_i}$. Define

$$t_\jmath^\mu = (-1)^n \zeta_{y_0} G_{y_0}^\mu \eta_{y_0} \prod_{i=1}^n (e_{y_i,S_i} G_{y_i}^\mu \eta_{y_i}). \tag{1102}$$

Its footprint is the union of these N_y and $S_i^\sim$; its hull consists of the big cubes meeting that union. An uncounted string has zero matrix, since a nonzero coefficient entry has both of its bonds in $S_i^\sim$.

We record the convergence calculation, since it also controls the cross-component terms below. Let $g_y(b,b') = e^{-\delta_{\mathrm{loc}}|b-b'|}$ on $X_y \times X_y$, and zero elsewhere. Bound $e_{y,S}$ by $\widehat{e}_{y,S} = 2m_S |\zeta_y(b') - \zeta_y(b)| 1_{X_y}(b')$ in the near case, and by $2m_S \zeta_y(b')$ otherwise. Use $\delta_{\mathrm{pl}} = \delta_{\mathrm{loc}}/8$. For a nonzero leak followed by $\eta_y(b'')$, either a cutoff changes outside distance $2\ell_c/3$, while $x_{b''}$ lies within $\ell_c/3$, or a point of $S^\sim$ lies outside N_y. In the first case $l_S(b,b') + |b' - b''| > \ell_c/3 - 1$; in the second $l_S(b,b') > 2\ell_c/3 - 1/2$. The latter follows by joining that point to a coefficient tree within $3w \le \ell_c/3$. In both cases, after retaining the exponential of the string length, the remaining factors are bounded by

$$e^{\delta l_S} \varsigma_0 e^{-\delta|b-b'|/4} e^{-\delta_{\mathrm{loc}}|b'-b''|/4}, \qquad \varsigma_0 = e^{-\delta_{\mathrm{loc}}(\ell_c/3 - 1)/4}.$$

Here a local connecting segment has length $|b_- - b'_-|_1 \le 3|b - b'|$; this accounts for the ratio $1/8$ in δ_{pl}. The first resolvent contributes a nonnegative array P with $P(b,b') \le 1$ and row sum at most $c_{\mathrm{lat}}(\delta_{\mathrm{loc}}/4)$. Each further leak and resolvent contributes an array Q whose row sum is at most

$$q = \frac{8B_{\mathrm{w}}}{\gamma_{\mathrm{pos}}} \varsigma_0 c_{\mathrm{lat}}(\delta/4) c_{\mathrm{lat}}(\delta_{\mathrm{loc}}/4).$$

Choose the tiny-cube floor so that $q \le 1/12$ and $c_{\mathrm{lat}}(\delta_{\mathrm{loc}}/4)q \le 1/4$. Expanding intermediate bonds gives the weighted majorant $2PQ^n$ for strings of length n. Its sum over all n is less than 3. Dividing by $\mu + \gamma_{\mathrm{pos}}/2$ proves absolute convergence, uniformly also on the interpolation disc. The same Cauchy estimate as in (1095) gives the difference bound in (1100). One may take $C_{\mathrm{res}} = 64$ and enlarge B_{f} to $\max(B_{\mathrm{w}}, C_{\mathrm{res}}/(c_{\mathrm{res}}\gamma_{\mathrm{pos}}))$.

Joining the intermediate segments to the coefficient trees gives a connected tree of the string length. Every point of N_{y_i} is within $4\ell_c$ of its live intermediate bond. This proves the stated footprint bound and endpoint support. All factors are defined on the hull, so the product has exactly the claimed locality. Gauge covariance and conjugation hold factor by factor.

Finally put $\Pi^\mu = \sum_y \zeta_y G_y^\mu \eta_y$ and $E^\mu = \sum_{y,S} e_{y,S} G_y^\mu \eta_y$ on the whole torus. Since $\zeta_y \eta_y = \eta_y$ and $\sum_y \eta_y = 1$, multiplication of the definitions gives $(\mu + A_{\mathrm{II}}^+)\Pi^\mu = 1 + E^\mu$. The preceding absolute majorant sums $\Pi^\mu \sum_{n\ge 0}(-E^\mu)^n$ and identifies its terms with (1102). It is a right inverse and hence the inverse in finite dimension. At $\mu = 0$ this is (1101). Empty integration space gives the unique matrix on the zero-dimensional space and an empty string sum. □

55.3. **Kept forms and terms crossing the strips.** Write $p_{\mathrm{I}}, p_{\mathrm{E}}$ for the coordinate projections, and $p_{\mathfrak{k}}$ for the external coordinates of a strip component. There are two kinds of terms in the conditional quadratic form:

$$d_S = p_{\mathrm{E}} a_S p_{\mathrm{E}}, \qquad d_{(S_1,\mathfrak{J},S_2)} = -p_{\mathrm{E}} a_{S_1} p_{\mathrm{I}} t_{\mathfrak{J}} p_{\mathrm{I}} a_{S_2} p_{\mathrm{E}}. \tag{1103}$$

Their footprint is the union of the footprints of their factors; $X(\mathfrak{z})$ is its big-cube hull. A term is kept on $\mathfrak{k}$ when $\mathsf{kp}_t(X(\mathfrak{z})) = 1$ and $\mathfrak{k} = \mathfrak{k}_{X(\mathfrak{z})}$; retain its compression to $p_{\mathfrak{k}}$. Write $d_{\mathfrak{z}}^{\mathrm{k}}$ for this compression, or zero when the term is not kept, and $d_{\mathfrak{z}}^{\times} = d_{\mathfrak{z}} - d_{\mathfrak{z}}^{\mathrm{k}}$. The geometry of the strip gives two facts used below: a kept hull lies in $\mathfrak{k}^{\sim}$, and every footprint of diameter less than $W_t = LM\check{R}_t$ containing an external bond is kept on its unique strip component. Define

$$D_{\mathfrak{k}} = \sum_{\mathfrak{z}\ \text{kept on}\ \mathfrak{k}} p_{\mathfrak{k}} d_{\mathfrak{z}} p_{\mathfrak{k}}, \qquad \Delta_{\mathfrak{k}}^{\mathrm{loc}} = \tfrac{1}{2}(D_{\mathfrak{k}} + D_{\mathfrak{k}}^{\mathsf{T}}), \tag{1104}$$

$$\varphi_{\mathfrak{k},X}(f;p) = -\tfrac{1}{2} \sum_{\substack{\mathfrak{z}\ \text{kept on}\ \mathfrak{k} \\ X(\mathfrak{z})=X}} \langle f, p_{\mathfrak{k}} d_{\mathfrak{z}}(p) p_{\mathfrak{k}} f\rangle, \tag{1105}$$

$$B^{\times}(X;p,f) = -\tfrac{1}{2} \sum_{X(\mathfrak{z})=X} \langle f, d_{\mathfrak{z}}^{\times}(p) f\rangle, \qquad K^{\mathrm{kp}}(X;p,f) = \sum_{\mathfrak{k}} \varphi_{\mathfrak{k},X}(f;p). \tag{1106}$$

The sums below establish their domains and convergence.

Proposition 55.3. Under the preceding local-gap and inverse hypotheses, put $B_{\mathrm{ch}} = 2c_{\mathrm{lat}}(\delta_{\mathrm{pl}}/4)^2 B_{\mathrm{f}}^3$. For any fixed decay exponent $\kappa > 0$, take $\delta_{\mathrm{pl}} LM \ge C_2(\kappa+2)$, $c_2\delta_{\mathrm{pl}} LM \ge 2$ and $(LM)^6 \le e^{c_2\delta_{\mathrm{pl}} LM}$, where $c_2 > 0, C_2, C_3$ are absolute constants. Set $K_{\mathrm{ch}} = C_3(LM)^6 e^{C_3(\kappa+1)} B_{\mathrm{ch}}$. For $p \in \mathscr{O}(X;\alpha)$ and $\sup|f| \le b_e$, the series in (1105)–(1106) converge absolutely and

$$|K^{\mathrm{kp}}(X;p,f)| \le \tfrac{1}{2} K_{\mathrm{ch}} b_e^2 e^{-\kappa(d_t(X)+1)}, \tag{1107}$$

$$|B^{\times}(X;p,f)| \le \tfrac{1}{2} C_3 B_{\mathrm{ch}} b_e^2 e^{-(\check{R}_t - 2)} e^{-\kappa d_t(X)}. \tag{1108}$$

These functions are finite quadratic polynomials in the external coordinates supported in X, with holomorphic coefficient functions of the pair on X. They commute with restriction, real gauge transformations and conjugation. A nonzero kept term requires $\mathsf{kp}_t(X) = 1$ and its unique component; a nonzero rejected term requires $d_t(X) \ge \check{R}_t - 2$. Both require a cube of X contained in $\Omega_t \cap Z_t$.

On the whole torus set

$$C = (A_{\mathrm{II}}^+)^{-1}, \quad M = -CA_{\mathrm{IE}}^+, \quad \Delta^{\mathrm{cond}} = A_{\mathrm{EE}}^+ - A_{\mathrm{EI}}^+ C A_{\mathrm{IE}}^+. \tag{1109}$$

Then $\Delta^{\mathrm{cond}} = \sum_{\mathfrak{z}} d_{\mathfrak{z}}$ and for $F = F_{\mathrm{I}} + f$,

$$-\tfrac{1}{2}\langle F, A^{+}F\rangle = -\tfrac{1}{2}\langle F_{\mathrm{I}} - Mf, A^{+}_{\mathrm{II}}(F_{\mathrm{I}} - Mf)\rangle - \tfrac{1}{2}\sum_{\mathfrak{k}}\langle f, \Delta^{\mathrm{loc}}_{\mathfrak{k}} f\rangle + \sum_{X} B^{\times}(X; p, f). \tag{1110}$$

Moreover

$$|M(b,b')| \le c_M e^{-\delta_M |b-b'|}, \qquad c_M = c_{\mathrm{lat}}(\delta_{\mathrm{pl}}/2)B_{\mathrm{f}}^2, \quad \delta_M = \delta_{\mathrm{pl}}/2. \tag{1111}$$

■

Proof. For a triple in (1103), multiply its three nonnegative majorants and sum the two intermediate bonds. Define its length as the minimum sum of the three lengths over live intermediate bonds. The coefficient trees and the inverse-string tree join at those bonds; their union contains both endpoints and has the asserted length. Its footprint lies within $\rho_* = 4\ell_c$ of the union. Spending one quarter of δ_{pl} on each intermediate bond sum proves

$$\sum_{\mathfrak{z}} m_{\mathfrak{z}}(b,b') e^{3\delta_{\mathrm{pl}} l_{\mathfrak{z}}(b,b')/4} \le B_{\mathrm{ch}}. \tag{1112}$$

The single-coefficient terms obey the same inequality. Replacing the three factors one at a time gives the reference difference $3\varepsilon_* m_{\mathfrak{z}}$; no factor is required to be globally defined outside its own hull. In particular

$$\|\Delta^{\mathrm{loc}}_{\mathfrak{k}}(p) - \Delta^{\mathrm{loc}}_{\mathfrak{k}}(U,0)\| \le 3c_{\mathrm{lat}}(3\delta_{\mathrm{pl}}/4)\varepsilon_* B_{\mathrm{ch}}. \tag{1113}$$

Here is the conversion from connecting length to hull size. Since $4\rho_* \le LM$, the big-cube hull of a live footprint of length l has $n \le 8 + 64l/(LM)$ cubes, and its diameter is at most $l + 2\rho_*$. For a fixed hull with n cubes there are at most $9(LM)^6 n^2$ pairs of positive bonds. Thus its total quadratic coefficient is bounded by

$$9(LM)^6 n^2 B_{\mathrm{ch}} \exp\{-3\delta_{\mathrm{pl}} \max(0, LM(n-8)/64)/4\}.$$

Taking $\delta_{\mathrm{pl}} LM \ge C_2(\kappa+2)$ absorbs n^2 and gives the first bound, using the connected-cube relation $d_t(X) \le n-1$. If a term is rejected, a live footprint cannot have diameter less than W_t by the keeping rule. Hence $l \ge l_\times := W_t - 2 - 2\rho_*$. Use three quarters of this lower bound and one quarter of $LM(n-8)/64$ in the same exponential. Since $l_\times \ge W_t/2 - 2$, the resulting exponent is at least $c_2\delta_{\mathrm{pl}} W_t + (\kappa+1)n$. This gives the intermediate bound

$$|B^{\times}(X; p, f)| \le \tfrac{1}{2} C_3 B_{\mathrm{ch}} b_e^2 (LM)^6 e^{-c_2\delta_{\mathrm{pl}} W_t} e^{-(\kappa+1)n}.$$

The two remaining size assumptions give (1108). They also hold uniformly at $\check{R}_t = 1$.

All live entries have both endpoint bonds inside the footprint. A kept hull meeting an external bond lies in its unique enlarged strip component. A rejected hull must traverse the coarse separation; counting its consecutive big cubes gives $d_t(X) \ge \check{R}_t - 2$. An external bond has an endpoint in the strip, so its containing big cube lies in $\Omega_t \cap Z_t$ and belongs to X. These are the stated support conditions. Absolute uniform convergence of the coefficient series proves holomorphy. Only finitely many external bonds lie in X, so pairing gives a finite quadratic polynomial. Restriction and the two symmetries hold for every factor and hence for the sums.

Insert (1101) in the Schur complement; absolute convergence allows the two coefficient sums to pass through it, giving $\Delta^{\mathrm{cond}} = \sum d_{\mathfrak{z}}$. Symmetrization does not change a quadratic form, so separating kept and rejected terms gives (1110) by completing the finite square. The two-factor sum $M = -\sum_{\jmath,S} t_\jmath p_{\mathrm{I}} a_S p_{\mathrm{E}}$, with half of the common exponential spent on its intermediate bond, gives (1111). □

55.4. **Positivity of the kept part.** The lower bound uses $\mathscr{O}^+$ rather than $\mathscr{O}$: the test field below contains external as well as integration bonds. This is the only additional local-gauge requirement.

Proposition 55.4. For sufficiently large M and sufficiently small $\alpha_1+\alpha_2$, depending only on L, every $p\in\mathscr{O}^+(\mathfrak{k}^\sim;\alpha)$ satisfies, for every complex field $f\in\operatorname{ran} p_\mathfrak{k}$,

$$\operatorname{Re}\langle \bar f,\Delta^{\mathrm{loc}}_\mathfrak{k}(p)f\rangle \ge \tfrac12\gamma_{\mathrm{pos}}|f|^2. \tag{1114}$$

For a real pair the kept matrix is real symmetric. Consequently, if real pairs $p_\mathfrak{k}$ belong to these classes on a union D of strip components, then, with $\gamma'=\gamma_{\mathrm{pos}}/4$, every real field f on their disjoint external bond sets satisfies

$$\sum_{\mathfrak{k}\subset D}\langle f,\Delta^{\mathrm{loc}}_\mathfrak{k}(p_\mathfrak{k})f\rangle \ge 2\gamma'\sum_b |f(b)|^2. \tag{1115}$$

The right side counts their disjoint external bond sets. The statement is unchanged by extending the history beyond this step. ∎

Proof. Take an admitted decomposition of p near all bonds and work first at its real reference. Put $R=\mathfrak{k}^\sim$, $A^R=\sum_{X(S)\subset R}a_S$ and $G^R=\sum_{X(J)\subset R}t_J$. The latter is an approximate inverse with an exact interior identity. Indeed let $\bar\eta_R$ be the sum of η_y over the cubes usable in R. Group the absolutely convergent strings by their first cube, writing $G^R=\sum_y\zeta_yV_y$. Removing that first cube gives

$$V_y=G_y\eta_yY,\qquad Y=1-\sum_y F_yV_y,\qquad F_y=\sum_{X(S)\subset R}e_{y,S}=A^R_{\mathrm{II}}\zeta_y-\zeta_yA^y.$$

These regroupings are legitimate even after removing ζ_y: it is strictly positive on the finite set X_y, and zero elsewhere. Substitution yields

$$1-A^R_{\mathrm{II}}G^R=(1-\bar\eta_R)Y. \tag{1116}$$

Let $s(b)=\operatorname{dist}_\infty(b_-,\mathfrak{k})$, and let J be the fluctuation bonds meeting the union of tiny cubes whose enlargements lie in R. Then the external bonds of $\mathfrak{k}$ belong to J. On its complement $s\ge W_t-\ell_c$. Also $\bar\eta_R(b)\ne1$ implies $s(b)>W_t-4\ell_c$: a missing cutoff cube supplies a point outside R within $\ell_c/3+\ell_c+2\ell_c+1/2$ of b_-, whereas every cube outside the one-coarse-layer enlargement is at distance at least W_t from $\mathfrak{k}$.

For real external f supported on $\mathfrak{k}$, put $g=A^R_{\mathrm{IE}}f$, $u=G^Rg$ and $h=g-A^R_{\mathrm{II}}u$. Let p_1 project onto $J\cap\mathfrak{X}^+_{\mathrm{I}}$ and $p_2=p_{\mathrm{I}}-p_1$. The test field $F=f-p_1u$ is supported on J. Local positivity gives $\langle F,A^RF\rangle\ge\gamma_{\mathrm{pos}}|F|^2\ge\gamma_{\mathrm{pos}}|f|^2$. For the sum Σ_R of all chains with hull in R, direct expansion of this square gives

$$\langle f,\Sigma_Rf\rangle=\langle F,A^RF\rangle+\langle p_1u,h+A^R_{\mathrm{II}}p_2u\rangle-\langle g,p_2u\rangle. \tag{1117}$$

Here $\Sigma_R=A^R_{\mathrm{EE}}-A^R_{\mathrm{EI}}G^RA^R_{\mathrm{IE}}$. By (1116), h is supported where $s>W_t-4\ell_c$; so is p_2u.

Put $B_h=B_{\mathrm{f}}c_{\mathrm{lat}}(\delta_{\mathrm{pl}}/2)$. Both A^R and G^R have weighted Schur norm at most B_h at rate $\delta_{\mathrm{pl}}/2$. Conjugate by $E(b)=e^{\delta_{\mathrm{pl}}s(b)/2}$; s is 1-Lipschitz and $s\le1$ on the support of f. Thus

$$|Eu|\le B_h^2e^{\delta_{\mathrm{pl}}/2}|f|,\qquad |Eh|\le(B_h+B_h^3)e^{\delta_{\mathrm{pl}}/2}|f|.$$

The two error terms in (1117) are therefore bounded by

$$e_P|f|^2,\qquad e_P=(2B_h^3+2B_h^5)e^{-\delta_{\mathrm{pl}}(W_t-4\ell_c-1)/2}.$$

For external bonds b,b' of $\mathfrak{k}$, any chain in R not assigned to $\mathfrak{k}$ with $m_\mathfrak{z}(b,b')\ne0$ has $l_\mathfrak{z}(b,b')\ge l_\times=W_t-2-2\rho_*$. Splitting the exponential in (1112) between this length and the endpoint distance bounds the difference between $p_\mathfrak{k}\Sigma_Rp_\mathfrak{k}$ and $D_\mathfrak{k}$ by

$$e_\times=B_{\mathrm{ch}}c_{\mathrm{lat}}(3\delta_{\mathrm{pl}}/8)e^{-3\delta_{\mathrm{pl}}l_\times/8}.$$

Choose M so that $e_P+e_\times\le\gamma_{\mathrm{pos}}/4$ even with $W_t=LM$, and choose $\alpha_1+\alpha_2$ so that the right side of (1113) is at most $\gamma_{\mathrm{pos}}/4$. This proves (1114), first for real f at the reference, then for complex f

by splitting its real and imaginary parts and applying the norm difference. Conjugation gives reality at real pairs. Finally the external bond sets of different strip components are disjoint; summing their bounds proves (1115), including the empty union. □

56. The normalizer on the integration region

We localize the determinant of the Gaussian block integrated in a history step. The nested background regions affect this block, including its diagonal entries. Positivity follows by comparing short connected terms with those of the constant sequence.

56.1. **The domain of the normalizer.** Work in level-k units and put $t = k+1$, $\xi = L^{-k}$ and $\xi_i = L^{i-k}$. Assume $L \geq 13$ is odd, $M = L^{m'}$, $m' \geq 2$, and $m' < m + K - t$. Let $\Omega_0 \supset \cdots \supset \Omega_t$ be an admitted separated sequence, and let $\Lambda \subset \Omega_t$ be a union of coarse cubes with $\operatorname{dist}_\infty(\Lambda, \Omega_t^c) \geq LM$. Write $\mathfrak{X}_{\mathrm{I}}$ for its integration bonds and Γ^Λ for the positive level-t bonds with an endpoint in Λ. A bond is in $\mathfrak{X}_{\mathrm{I}}$ precisely when it has an endpoint in Λ and is not a pivot bond. Indeed both endpoints then lie in Ω_t, and the remaining condition in the fluctuation coordinates is the exclusion of pivots. In particular these are also nonpivot bonds for the constant sequence.

Keep the grain size w, clearance r_{gr} and mesh size h of the connected-form construction. Impose

$$\mathfrak{M} \geq \max\{L, M^{\mathrm{cv}}\}, \qquad R_1\mathfrak{M} \geq r^{\mathrm{pr}}, \qquad M \geq R_1\mathfrak{M} \geq 13\mathfrak{M},$$
$$Lw \leq h, \qquad Lh \leq \ell_c := M/L^2, \qquad h \geq h_{\mathrm{nf}},$$
$$4h \leq r_N, \qquad 4(r_N+1) \leq LM, \qquad M \geq M_{\mathrm{nf}}.$$

Here w, h are powers of L, $M_1 \leq w \leq M$, $w \leq L^{m+K-k-1}$, and

$$\rho_t + \rho_{\mathrm{c}} + L < r_{\mathrm{gr}}, \qquad 4(r_{\mathrm{gr}}+1) \leq w, \qquad \delta^{\mathrm{cv}} w \geq 2^7 L \log(4n_{\mathrm{w}}).$$

The tower extent ρ_t is less than r_N. The constants $h_{\mathrm{nf}}, M_{\mathrm{nf}}$ depend only on L.

For a nonempty coarse region X, a pair $p = (\mathbf{U}, \mathbf{J})$ on $X \cap \Omega_0$ belongs to $\mathscr{O}^N(X; a)$ if it has one decomposition $\mathbf{U} = e^{i\xi A'} U$, with U real, satisfying:

(i) At sites of layer i, wherever the expressions are defined on $X \cap \Omega_0$,

$$|\partial U - 1|,\ |\partial \mathbf{U} - 1| < a\xi^2\xi_i^{-2}, \qquad \xi_i|A'| < a, \quad \xi_i^2|\nabla^U A'| < a, \quad \xi_i^3|\mathbf{J}| < a.$$

(ii) Every compact cube $q \subset X \cap \Omega_i$ which is a union of level-i $\mathfrak{M}$-cubes and has side at most $12\mathfrak{M}\xi_i$ admits a real gauge $U^v = e^{i\xi A}$ with $\xi_i|A| \leq a$ and $\xi_i^2|\nabla^\xi A| \leq a$.

(iii) If a tiny cube y contains an integration-bond endpoint and $y^\sim \subset X \cap \Omega_0$, the same conclusion holds on $y^\sim$ with ξ_i replaced by $\xi_t = L$.

The same decomposition must satisfy all three conditions.

These conditions are preserved by restriction, real gauge transformations, conjugation and replacement by $p_{\mathrm{r}} = (U, 0)$. They define an open set when $a \leq \pi$: keep U and its gauges fixed, choose the local inverse of the exponential at each $i\xi A'(b)$, and use the finitely many strict inequalities in (i). The inverses exist since $|\xi A'(b)| < a \leq \pi$.

They imply regularity for every meet $\Omega_\bullet\langle Y\rangle$, $Y \subset X$. A site in layer i of the meet had old layer index $\widetilde{i} \geq i$, so $\xi_{\widetilde{i}} \geq \xi_i$ and all scaled inequalities remain valid. Each gauge cube for the meet lies in $X \cap \Omega_i$ and has the size in (ii); take that cube itself as its enclosing cube. This verifies the natural enlarged gauge-cube condition as well.

Choose $a_{\mathrm{nf}} \leq \min\{a_N, a_{\mathrm{w}}, a^{\mathrm{cv}}/2, 1/2\}$ small enough that

$$R := \frac{c_{\mathrm{d}} a^{\mathrm{cv}}}{2a_{\mathrm{nf}}} \geq \max\{16, 16B_{\mathsf{a}}/\gamma^{\mathrm{tr}}\}.$$

Here γ^{tr} is the flat gap used in theorem U, and a_{w} is its local comparison radius. On $p_z = (e^{iz\xi A'}U, z\mathbf{J})$, $|z| < R$, the fixed real gauges work for every meet, and (775) gives the complex curvature bounds.

Thus the connected coefficients of the general-sequence form are defined and satisfy

$$\begin{aligned} |a_S[p_z](b,b')| &\le C_a \epsilon_\star^{\#S-1} e^{-2\theta_a \Lambda_S(b,b')} =: \beta_S(b,b'), \\ |a_S[p](b,b') - a_S[p_{\mathrm{r}}](b,b')| &\le (2/R)\beta_S(b,b'). \end{aligned} \tag{1118}$$

The second bound is Cauchy's estimate. The coefficients vanish unless S is connected and contains the endpoint grain supports. After conversion to the Hermitian norm, the weighted row and column sums are bounded by $B_{\mathrm{nf}} = 2\sqrt{2} C_a c_{\mathrm{lat}}(\theta_a)$, by (773). Restricting to $\Lambda_S \ge u$ gains $e^{-\theta_a u}$. Set

$$\begin{gathered} B_{\mathsf{a}} = \max\{\gamma^{\mathrm{tr}}, B_{\mathrm{nf}}\}, \qquad \gamma_{\mathrm{nf}} = \gamma^{\mathrm{tr}}/2, \qquad c_{\mathrm{nf}} = (B_{\mathsf{a}} + \gamma_{\mathrm{nf}})/2, \\ q_N = \frac{B_{\mathsf{a}} - \gamma_{\mathrm{nf}}/2}{B_{\mathsf{a}} + \gamma_{\mathrm{nf}}} < 1. \end{gathered} \tag{1119}$$

Increase h_{nf} so that the mesh localization in theorem U applies and $B_{\mathrm{nf}} e^{-\theta_a h} \le \gamma^{\mathrm{tr}}/4$.

56.2. **Positivity of the compressed operator.** Let $\mathcal{B}_Y$ be the integration bonds b with $S_N(b) \subset Y$. Let $A_{Y,N}(p)$ be the compression of the fluctuation form for $\Omega_\bullet\langle Y\rangle$ to these bonds. Finite inclusion–exclusion of the defining coefficients gives

$$A_{Y,N}(p_z) = \sum_{\substack{\emptyset \ne S \subset Y \\ S \text{ connected}}} \chi_Y a_S[p_z] \chi_Y^{\mathsf{T}}.$$

Operators are extended by zero when compared on larger bond spaces.

Proposition 56.1. For $p \in \mathscr{O}^N(Y; a_{\mathrm{nf}})$, $A_{Y,N}(p)$ is complex symmetric, its real reference satisfies

$$\gamma_{\mathrm{nf}} I \le A_{Y,N}(p_{\mathrm{r}}) \le B_{\mathsf{a}} I,$$

and

$$\|A_{Y,N}(p) - A_{Y,N}(p_{\mathrm{r}})\| \le \gamma_{\mathrm{nf}}/4, \qquad \|1 - c_{\mathrm{nf}}^{-1} A_{Y,N}(p)\| \le q_N.$$

The assertions include $\mathcal{B}_Y = \emptyset$. ∎

Proof. Symmetry and reality follow from the form and compression. The row and column estimates give the upper bound and $\|A_{Y,N}(p) - A_{Y,N}(p_{\mathrm{r}})\| \le 2B_{\mathrm{nf}}/R \le \gamma_{\mathrm{nf}}/4$. We prove the real lower bound.

Call a mesh cube $\square$ active if its cosine cutoff $h_\square$ meets $\mathcal{B}_Y$, and put $W = \bigcup_{\square \text{ active}} \square^{\sim 2}$. The cutoff is nonzero at b exactly when its midpoint x_b is within h of the center of $\square$. Every point of W is therefore within $7h/2$ of such a midpoint. Since $4h \le r_N$, $W \subset Y$. An integration bond has an endpoint in Λ and $7h/2 + 1/2 < LM$, so $W \subset \Omega_t$. Conversely the closed $2h$ neighborhood of each x_b, $b \in \mathcal{B}_Y$, lies in W.

For every mesh cube $\square'$ with $q := \square'^{\sim 2} \subset W$, there is a tiny cube y with

$$y \subset \Lambda, \qquad q \subset y^{\sim} \subset Y \cap \Omega_0. \tag{1120}$$

To verify this, lift q to a product of intervals of length $5h$. Each factor overlaps one or two coarse intervals. Every product of overlapped coarse intervals belongs to Y, since it contains a mesh cube of q. Choose a point in each factor at the common coarse boundary if there are two intervals. If there is one, choose the endpoint farther from its nearest outer boundary, or the coarse midpoint if it lies in the factor. The product point has distance at least $5h$ from every coarse cube outside those products, and is within $7h/2 + 1/2 < 5h$ of an endpoint in Λ. Hence an overlapped coarse cube belongs to Λ.

Since $5h < \ell_c$, each factor of q overlaps at most two tiny intervals. At a crossed coarse boundary choose the adjacent tiny interval inside the selected Λ cube; its two neighbors cover the factor. In a single coarse interval choose three consecutive tiny intervals containing the factor and take the middle one. Their product is y. Its enlargement lies in the overlapped coarse cubes, hence in Y, and within $\ell_c < LM$ of Λ, hence in Ω_0. This proves (1120). Every tiny cube in Λ contains an

integration-bond endpoint: from its center choose a unit bond remaining in the same averaging block. It is not a pivot, since pivots join different blocks.

Condition (iii) supplies a gauge on $y^{\sim}$ and hence on q. Conditions (i) and (ii) on W are the constant-sequence conditions at layer t, since $W \subset \Omega_t$. Apply the mesh localization proof of theorem U to the family $\mathcal{V}$ of connected S with $S^{\sim} \subset W$. It contains every term supported in an active $\square^{\sim 2}$, so is full near $\mathcal{B}_Y$. Its real constant-sequence sum has gap $3\gamma^{\mathrm{tr}}/4$. Both sequences agree on $S^{\sim}$ for $S \in \mathcal{V}$; locality identifies their coefficients there.

Every remaining $S \subset Y$ reaches outside W. A connecting tree for a nonzero entry comes within $3w$ of a point of $S^{\sim} \setminus W$ and contains b_-. Because the $2h$ neighborhood of x_b lies in W, its length is greater than $2h - 3w - 1/2 \geq h$. The omitted terms have norm at most $B_{\mathrm{nf}} e^{-\theta_a h} \leq \gamma^{\mathrm{tr}}/4$. The difference leaves gap $\gamma_{\mathrm{nf}} = \gamma^{\mathrm{tr}}/2$.

The real reference is self-adjoint, with spectrum between γ_{nf} and B_{a}. Thus

$$\|1 - c_{\mathrm{nf}}^{-1} A_{Y,N}(p_{\mathrm{r}})\| \leq \frac{B_{\mathsf{a}} - \gamma_{\mathrm{nf}}}{B_{\mathsf{a}} + \gamma_{\mathrm{nf}}}.$$

Adding the complex perturbation divided by c_{nf} gives q_N. □

56.3. **Local determinant terms.** For $b \in \mathcal{B}_Y^+$ define

$$f_b(Y;p) = 3 \log c_{\mathrm{nf}} - \sum_{n \geq 1} \frac{1}{n} \operatorname{tr}(1 - c_{\mathrm{nf}}^{-1} A_{Y,N}(p))^n(b,b). \tag{1121}$$

The series converges normally, with $|f_b| \leq F_{\mathrm{nf}} := 3|\log c_{\mathrm{nf}}| - 3\log(1 - q_N)$. For $S_N(b) \subset X$ put

$$E^N(X;p,b) = \sum_{S_N(b) \subset Y \subset X} (-1)^{\#X - \#Y} f_b(Y; p|_Y), \tag{1122}$$

and

$$\mathrm{Ld}(p) = \sum_{b \in \mathfrak{X}_{\mathrm{I}}^+} f_b(T_\ell; p). \tag{1123}$$

These expressions use the present sequence and integration region; the constant-sequence definitions are their special case.

For $c \in \Gamma^\Lambda$ put $z_c(p) = -\operatorname{Log}\det(Lm_c(\mathcal{M}^k \mathbf{U}))$. Its branch and bound are those of (793). Indeed the endpoint blocks fit in a cube of side $2\mathfrak{M}L$ inside $X(c)$. This cube lies in Ω_t, since its points are within LM of the endpoint in Λ. Condition (ii) at t supplies its gauge. Restriction gives precisely the local gauge used in that calculation, with $L|A| \leq a_{\mathrm{nf}}$ and $L|A'| < a_{\mathrm{nf}}$. Since $a_{\mathrm{nf}} \leq a_N$, the same holomorphic average, pivot coefficient and logarithm apply.

Define

$$N^{\mathrm{un}}(X,b;p) = \begin{cases} -\frac{1}{2} E^N(X;p,b), & b \in \mathfrak{X}_{\mathrm{I}}^+,\ S_N(b) \subset X, \\ z_c(p), & b = b_0(c),\ c \in \Gamma^\Lambda,\ X = X(c), \\ 0, & \text{otherwise.} \end{cases} \tag{1124}$$

The cases are disjoint. With the tents of (787), put

$$N^{\mathrm{un}}(X,p,z) = \sum_b h_z(b) N^{\mathrm{un}}(X,b;p), \tag{1125}$$

$$N^{\partial}(X;p) = \sum_z N^{\mathrm{un}}(X,p,z). \tag{1126}$$

Proposition 56.2. After increasing M_{nf}, constants E_{nf} and $\delta_{\mathrm{nf}} > 0$ depending only on L give, for $p \in \mathscr{O}^N(X; a_{\mathrm{nf}})$ and $d = d_t(X)$,

$$\begin{aligned} |E^N(X; p, b)| &\le 2^{-d} E_{\mathrm{nf}} e^{-\delta_{\mathrm{nf}} M d}, \\ |N^{\mathrm{un}}(X, b; p)| &\le E_{\mathrm{nf}} e^{-\delta_{\mathrm{nf}} M d}, \\ |N^{\mathrm{un}}(X, p, z)| &\le 3L^3 E_{\mathrm{nf}} e^{-\delta_{\mathrm{nf}} M d}, \\ |N^{\partial}(X; p)| &\le 24(LM)^3 E_{\mathrm{nf}} e^{-\delta_{\mathrm{nf}} M d}. \end{aligned}$$

The first expression vanishes when X is not wall connected. All terms are holomorphic, gauge invariant and conjugation real. The pointed term vanishes unless $z \in X$. ■

Proof. The connected-chain proof of proposition 50.2 applies to these matrices. We verify its inputs. For $S \subset Y$ the coefficient is the restriction of the same $a_S[p]$, independent of Y. Its support $S^\sim$ lies within $3w$ of a tree realizing Λ_S up to any positive error. Equation (1118) gives the weighted row and column bounds, and proposition 56.1 the uniform contraction q_N. Changing a coarse cube away from $S^\sim$ changes neither the coefficient nor its entries. On wall-disconnected regions the tower meets, local stencils and averaging blocks split into components; their inverses, bilinear forms and deep compressions do likewise.

Inclusion–exclusion leaves only closed chains of coarse support exactly X. Their degree-n trace coefficient obeys the two bounds (796):

$$|e_n| \le 3Q_{\mathrm{nf}}^n e^{-\theta_a LMd/128}, \qquad |e_n| \le 3\, 2^{\#X} q_N^n, \qquad Q_{\mathrm{nf}} = 1 + B_{\mathrm{nf}}/c_{\mathrm{nf}}.$$

The geometry is unchanged: the support cubes lie within r_N+1 of the joined trees, since $3w < r_N$, and the total tree length is at least $LMd/128$. For $d \ge 1$ split $\sum |e_n|/n$ at $\lfloor \vartheta LMd \rfloor$, $\vartheta = \theta_a/(256 \log Q_{\mathrm{nf}})$. The two parts are at most

$$\frac{3Q_{\mathrm{nf}}}{Q_{\mathrm{nf}} - 1} e^{-\theta_a LMd/256}, \qquad \frac{3}{1 - q_N} e^{16d \log 2 - \vartheta LMd \log(1/q_N)}.$$

Choose δ_{nf} below both positive rates and increase M_{nf} so the remaining factor is at most 2^{-d}. For $d = 0$, at most eight cubes occur, and $2^8 F_{\mathrm{nf}}$ suffices. Enlarge E_{nf} to contain this bound, both geometric-series constants and $6 \log L + 3 \log 2$.

Component splitting makes $f_b(Y)$ depend only on the component containing $S_N(b)$; the alternating sum over any other component vanishes. Pivot terms have $d_t(X(c)) = 0$, giving the bond bound. The tents satisfy $\sum_b h_z(b) = 3L^3$ and $\sum_z h_z(b) = 1$. Their support is within L of the bond, hence in X for a nonzero term. This gives the pointed bound and support. At most $24(d+1)(LM)^3$ bonds contribute to N^∂. For $d \ge 1$ none is a pivot, and $\frac{1}{2} 2^{-d}(d+1) \le 1$ gives the last bound. For $d = 0$ use the bond bound directly.

Normal convergence gives holomorphy. Gauge covariance conjugates diagonal blocks and preserves the pivot determinant. Conjugation preserves the series and its specified branch. Finite sums preserve these properties. □

56.4. **The normalization identity.**

Proposition 56.3. For $p \in \mathscr{O}^N(T_\ell; a_{\mathrm{nf}})$, let $A_{\mathrm{II}}(p)$ be the full block on the integration bonds. Then

$$\sum_X N^\partial(X; p) = -\tfrac{1}{2}\mathrm{Ld}(p) + \sum_{c \in \Gamma^\Lambda} z_c(p), \tag{1127}$$

$$\det A_{\mathrm{II}}(p) = e^{\mathrm{Ld}(p)}, \qquad \operatorname{Re} A_{\mathrm{II}}(p) \ge \tfrac{3}{4}\gamma_{\mathrm{nf}} I, \tag{1128}$$

$$\int e^{-(B, A_{\mathrm{II}}(p) B)_1/2}\, dB = \exp(\nu_{\mathfrak{g}} \# \mathfrak{X}_1^+ - \tfrac{1}{2}\mathrm{Ld}(p)), \quad \nu_{\mathfrak{g}} = \tfrac{3}{2}\log(2\pi). \tag{1129}$$

The sum is over wall-connected coarse domains. Empty integration sets have determinant and integral one. ■

Proof. Finite Möbius summation gives $\sum_{X\supset S_N(b)} E^N(X;p,b) = f_b(T_\ell;p)$. Disconnected terms vanish. Summing tents and pivots proves (1127). On the whole torus the meet is the original sequence, so $A_{T_\ell,N} = A_{\mathrm{II}}$. The trace of its convergent matrix logarithm is Ld, proving the determinant identity. The real gap and the perturbation bound give the stated real-part bound. The complex Gaussian formula applies, with square-root branch $e^{-\mathrm{Ld}/2}$: at the real reference both are positive, and on the disc to p the Neumann logarithm and Gaussian integral are continuous and nonzero. This proves the integral with the stated normalization. □

57. Fluctuation integrals with a conditional mean

In a history step only the bonds touching Λ_t are integrated. The remaining field produces a conditional mean. Its real part stays below the cutoff, but its imaginary part must be removed from the Gaussian exponent without moving the boundary of the small-field integration region.

57.1. **The Gaussian data.** Let $\Lambda_t \subset \Omega_t$ be regions for π_t, $t = k+1$, and let $\mathfrak{X}$ be the non-pivot bonds of Ω_t. Set

$$\mathfrak{X}_{\mathrm{I}} = \{b \in \mathfrak{X} : b \text{ touches } \Lambda_t\}, \qquad \mathfrak{X}_{\mathrm{E}} = \mathfrak{X} \setminus \mathfrak{X}_{\mathrm{I}}. \tag{1130}$$

For a bond set $\mathfrak{Y}$, write $\mathfrak{Y}\langle Z\rangle$ for its bonds touching Z, and $\mathfrak{Y}^\circ\langle Z\rangle$ for those with both endpoints in Z. All bond sets are closed under reversal; positive bonds are independent coordinates. Give the integration bonds reversal-invariant thresholds $1 \le \mathfrak{b}_- \le \mathfrak{b}_b \le \mathfrak{b}_+ < \infty$. For an actual step these are

$$\mathfrak{b}_b = \mathfrak{b}_k \quad (b \in \mathbb{B}(\Lambda_t)), \qquad \mathfrak{b}_b = \theta_c \mathfrak{b}_k \quad (b \in \mathfrak{x}(\Lambda_t)). \tag{1131}$$

Definition 57.1. A Gaussian structure with conditional mean on a region $\mathfrak{R}$ consists of families $(\mathbf{C}, \mathbf{\Gamma}, \mathbf{m})(Z_\circ, s)$, indexed by fine regions $Z_\circ^\sharp \subset \mathfrak{R}$ and the same decoupling polydiscs as before. Put $\mathfrak{f} = \mathfrak{X}_{\mathrm{I}}\langle Z_\circ\rangle$ and $\mathfrak{r} = \mathfrak{X}_{\mathrm{I}}\langle \mathfrak{R}\rangle$. The covariance $\mathbf{C}$ is symmetric and invertible on $\mathfrak{f}$, the operator $\mathbf{\Gamma}$ maps fields on $\mathfrak{r}$ to fields on $\mathfrak{f}$, and $\mathbf{m}$ is a field on $\mathfrak{f}$. Their entries, including those of $\mathbf{C}^{-1}$, are holomorphic near the closed polydisc $|s(\Delta)| \le e^{\bar\kappa}$.

When s vanishes outside a coarse region $Z \supset Z_\circ^\sharp$, the covariance is the direct sum over its wall components, and $\mathbf{\Gamma}$ and $\mathbf{m}$ restrict to the corresponding component families. The connecting operator on a component uses auxiliary bonds touching that component only. In particular $\mathbf{\Gamma}$ vanishes on auxiliary bonds outside $\mathfrak{X}_{\mathrm{I}}\langle Z\rangle$.

For each $Z_\circ$ there are real references $C_{\mathrm{r}}, \Gamma_{\mathrm{r}}, m_{\mathrm{r}}$, independent of s, such that, with $S_{\mathrm{r}} = C_{\mathrm{r}}^{-1}$,

$$S_{\mathrm{r}} \ge \gamma_{\mathrm{r}} > 0, \qquad \|S_{\mathrm{r}}\|,\ \|\Gamma_{\mathrm{r}}\|,\ \|C_{\mathrm{r}}\Gamma_{\mathrm{r}}\| \le c_{\mathrm{r}}, \qquad 0 < \gamma_{\mathrm{r}} \le 1 \le c_{\mathrm{r}}. \tag{1132}$$

Here C_{r} is symmetric, $\bar\kappa \ge 1$ and $\hat\varepsilon, \theta_{\mathbf{m}}, \varepsilon_{\mathbf{m}} \ge 0$. The reference Γ_{r} uses only auxiliary bonds touching $Z_\circ^\sharp$. Throughout the closed polydisc,

$$\|\mathbf{C}^{-1} - S_{\mathrm{r}}\|, \quad \|\mathbf{\Gamma} - \Gamma_{\mathrm{r}}\|, \quad \|\mathbf{\Gamma}^{\mathsf{T}}\mathbf{C}\mathbf{\Gamma} - \Gamma_{\mathrm{r}}^{\mathsf{T}} C_{\mathrm{r}}\Gamma_{\mathrm{r}}\| \le \hat\varepsilon, \tag{1133}$$

and, on every positive integration bond,

$$|m_{\mathrm{r}}(b)| \le \theta_{\mathbf{m}}\mathfrak{b}_b, \qquad |\mathbf{m}(b) - m_{\mathrm{r}}(b)| \le \varepsilon_{\mathbf{m}}\mathfrak{b}_b, \qquad \theta_{\mathbf{m}} + \varepsilon_{\mathbf{m}} < 1. \tag{1134}$$

The operators and mean on an empty integration set have their unique zero-dimensional values. ■

Write $S = \mathbf{C}^{-1}$. The conditioning integral is

$$\begin{aligned}\mathbb{J}^{\mathbf{m}}(Z_\circ; s)[F] = \int d\mu_0(\mathbb{X}) \det{}^{1/2}(S/2\pi) \int dB\, F(B) \\ \times \exp\left\{-\tfrac{1}{2}(B - \mathbf{m} + \mathbf{C}\mathbf{\Gamma}\mathbb{X}, S(B - \mathbf{m} + \mathbf{C}\mathbf{\Gamma}\mathbb{X}))_1\right\}.\end{aligned} \tag{1135}$$

The determinant has the branch fixed on matrices with positive real part. The integral is used only after its absolute convergence and $\operatorname{Re} S > 0$ have been proved. If $\mathfrak{f} = \emptyset$, it means $F(0)$.

Fix $\vartheta_* < 1$ and constants satisfying

$$\theta_{\mathbf{m}} + \varepsilon_{\mathbf{m}} \le \vartheta_*, \qquad \mathsf{C}_{\mathrm{mn}} \ge 18(1 - \vartheta_*)^{-2}. \tag{1136}$$

For $\gamma_2 \geq 0$ put

$$\begin{aligned} \lambda_{\mathrm{r}} &= \min\{\gamma_{\mathrm{r}}/2, (1+2c_{\mathrm{r}}^2)^{-1}\}, \qquad c_{\mathrm{mn}} = \mathsf{C}_{\mathrm{mn}}\lambda_{\mathrm{r}}^{-1}, \\ u_0 &= \mathsf{C}_{\mathrm{mn}}[\hat{\varepsilon} + (1+c_{\mathrm{r}})\varepsilon_{\mathbf{m}}^2 + \gamma_2], \qquad c_{\mathrm{P}} = \mathsf{C}_{\mathrm{mn}}(LM)^3 c_{\mathrm{mn}} u_0, \\ \gamma_P &= \mathsf{C}_{\mathrm{mn}}^{-1}\gamma_2 \mathfrak{b}_-^2. \end{aligned} \tag{1137}$$

The smallness condition for this section is

$$\mathsf{C}_{\mathrm{mn}} u_0 \leq \lambda_{\mathrm{r}}. \tag{1138}$$

In particular $\hat{\varepsilon} \leq \lambda_{\mathrm{r}}/16$ and, writing $d_* = 1 - \vartheta_*$, $\varepsilon_{\mathbf{m}} \leq d_*/4$. Indeed $\mathsf{C}_{\mathrm{mn}}^2[\hat{\varepsilon} + (1+c_{\mathrm{r}})\varepsilon_{\mathbf{m}}^2 + \gamma_2] \leq \lambda_{\mathrm{r}} \leq 1/3$ and $\mathsf{C}_{\mathrm{mn}} \geq 18 d_*^{-2}$ give both inequalities.

57.2. **The contour estimate.** Let $\mathfrak{S}, \mathfrak{e} \subset \mathfrak{f}$ be disjoint. The first set carries strict small-field tests and the second carries failed tests. For $\varepsilon' > \varepsilon_{\mathbf{m}}$, consider the tube where $B(b)$ is real and $|B(b)| \geq \mathfrak{b}_b$ on $\mathfrak{e}$, where $|\operatorname{Im} B(b)| < \varepsilon' \mathfrak{b}_b$ elsewhere, and where additionally $|\operatorname{Re} B(b)| + |\operatorname{Im} B(b)| < \mathfrak{b}_b$ on $\mathfrak{S}$. A function on this tube is admitted if it is Borel and, at fixed real values on $\mathfrak{e}$, holomorphic in all remaining variables. Let D be the real part of the tube and extend its tested integrand by zero off D.

Proposition 57.2. Suppose (1136) and (1138). Let G be admitted on the tube and satisfy $|G| \leq F_0$ there. If s vanishes outside a coarse region $Z_\circ^\sharp \subset Z \subset \mathfrak{R}$, then

$$|\mathbb{J}^{\mathbf{m}}(Z_\circ; s)[\mathbf{1}_D G]| \leq F_0 \exp\left\{ -\mathsf{C}_{\mathrm{mn}}^{-1}\gamma_2 \sum_{b \in \mathfrak{e}^+} \mathfrak{b}_b^2 + c_{\mathrm{mn}} u_0 N_{\mathrm{I}}(Z) \right\}, \tag{1139}$$

where $N_{\mathrm{I}}(Z) = 3\#\mathfrak{X}_{\mathrm{I}}^+\langle Z\rangle$. The defining integrals converge absolutely, with an integrable majorant independent of s on its closed polydisc. For a fixed bounded Borel integrand, the integral is holomorphic in s on a neighbourhood of that polydisc. ∎

Proof. We retain the joint Gaussian form, since estimating the conditional density alone would lose the auxiliary-field normalization. Put

$$Q = \begin{pmatrix} 1 + \mathbf{\Gamma}^{\mathsf{T}}\mathbf{C}\mathbf{\Gamma} & \mathbf{\Gamma}^{\mathsf{T}} \\ \mathbf{\Gamma} & S \end{pmatrix}, \qquad Q_{\mathrm{r}} = \begin{pmatrix} 1 + \Gamma_{\mathrm{r}}^{\mathsf{T}} C_{\mathrm{r}} \Gamma_{\mathrm{r}} & \Gamma_{\mathrm{r}}^{\mathsf{T}} \\ \Gamma_{\mathrm{r}} & S_{\mathrm{r}} \end{pmatrix}. \tag{1140}$$

Completion of the square gives $\det Q = \det S$ and

$$(z, Qz) = (\mathbb{X}, \mathbb{X})_1 + (\beta + \mathbf{C}\mathbf{\Gamma}\mathbb{X}, S(\beta + \mathbf{C}\mathbf{\Gamma}\mathbb{X}))_1, \qquad z = (\mathbb{X}, \beta).$$

For real vectors, $\|\beta\|^2 \leq 2\|\beta + C_{\mathrm{r}}\Gamma_{\mathrm{r}}\mathbb{X}\|^2 + 2c_{\mathrm{r}}^2\|\mathbb{X}\|^2$ proves

$$Q_{\mathrm{r}} \geq \lambda_{\mathrm{r}}, \qquad \|Q - Q_{\mathrm{r}}\| \leq 2\hat{\varepsilon}, \qquad \det Q = \det S. \tag{1141}$$

The middle inequality follows by bounding the four blocks of the difference by $\hat{\varepsilon}$.

For $x = (\mathbb{X}, \beta')$ real and $z = (\mathbb{X}, \beta' - i\eta)$, the real reference and the perturbation bound give

$$\operatorname{Re}[-(z, Qz)/2] \leq -(x, Q_{\mathrm{r}}x)/2 + \hat{\varepsilon}\|x\|^2 + (c_{\mathrm{r}}/2 + \hat{\varepsilon})\|\eta\|^2. \tag{1142}$$

If $B = \beta + i\sigma$ and each $\sigma(b)$ lies on the segment from 0 to $\operatorname{Im}\mathbf{m}(b)$, this implies

$$|e^{-(z,Qz)/2}| \leq e^{c_{\mathrm{r}}\|\mathbf{m}\|^2} e^{-3\lambda_{\mathrm{r}}\|\mathbb{X}\|^2/8 - \lambda_{\mathrm{r}}\|\beta\|^2/8}, \qquad z = (\mathbb{X}, B - \mathbf{m}). \tag{1143}$$

To see this, use $\beta' = \beta - \operatorname{Re}\mathbf{m}$, $\|\beta'\|^2 \geq \|\beta\|^2/2 - \|\operatorname{Re}\mathbf{m}\|^2$ and $\|\eta\| \leq \|\operatorname{Im}\mathbf{m}\|$ in (1142). The coefficient of each mean square is at most c_{r}, since $\lambda_{\mathrm{r}} \leq 1/3$ and $\hat{\varepsilon} \leq \lambda_{\mathrm{r}}/8$. Also $\operatorname{Re} S \geq \gamma_{\mathrm{r}} - \hat{\varepsilon} > 0$, $\|S\| \leq c_{\mathrm{r}} + 1$, and $\|\mathbf{m}\|^2 \leq \sum_b \mathfrak{b}_b^2$. Thus (1143), with its mean factor replaced by $e^{c_{\mathrm{r}}\sum_{b\in\mathfrak{f}^+}\mathfrak{b}_b^2}$ and multiplied by $F_0((c_{\mathrm{r}}+1)/(2\pi))^{3\#\mathfrak{f}^+/2}$, is a common integrable majorant. This proves absolute convergence before any contour change.

The auxiliary field outside Z integrates to one: decoupling makes the corresponding columns of $\mathbf{\Gamma}$ zero. Restrict both connecting operators to $\mathfrak{z} = \mathfrak{X}_{\mathrm{I}}\langle Z\rangle$. The compressed perturbation bounds remain valid, so (1141) holds for the resulting Q^Z, Q^Z_{r}. Their dimension n is at most $2N_{\mathrm{I}}(Z)$.

Set $r_1 = 1 - 3d_*/4$, $r_2 = 1 - d_*/4$ and

$$h(v) = \min\{1, \max\{0, 2d_*^{-1}(r_2 - v)\}\}, \qquad \phi^{\mathfrak{b}}(\beta) = h(|\beta|/\mathfrak{b}).$$

For $\nu = \operatorname{Im}\mathbf{m}$, move each bond by

$$\Theta(\beta)(b) = \beta(b) + i\phi_b(\beta(b))\nu(b), \qquad \phi_b = \begin{cases} 0, & b \in \mathfrak{e}, \\ \phi^{\mathfrak{b}_b}, & b \in \mathfrak{S}, \\ 1, & \text{otherwise.} \end{cases} \tag{1144}$$

This stays in the tube. On the support of the cutoff, $|\beta| < r_2\mathfrak{b}_b$ and $|\nu| \le d_*\mathfrak{b}_b/4$; outside it the contour is unchanged. Its Jacobian factor on a small-field bond is

$$J_b = 1 + i\langle \nu(b), \nabla\phi^{\mathfrak{b}_b}(\beta(b))\rangle, \qquad |J_b| \le e^{2d_*^{-2}\varepsilon_{\mathbf{m}}^2}.$$

The latter estimate follows from $|\nabla\phi^{\mathfrak{b}}| \le 2/(d_*\mathfrak{b})$ and $\sqrt{1+a^2} \le e^{a^2/2}$.

Here the contour identity is elementary but the fixed boundary matters. For one bond, choose the first real coordinate parallel to $\nu(b)$. At fixed transverse coordinates the graph $x_1 \mapsto x_1 + i|\nu(b)|\phi^{\mathfrak{b}_b}(\beta)$ is piecewise smooth, joins the same endpoints as the real interval, and lies in the convex complex slice of the tube. Cauchy's theorem equates the two line integrals; the graph derivative is exactly J_b. The two contours coincide where $|\beta| \ge r_2\mathfrak{b}_b$, so no boundary value of G is needed. On an untested bond use a rectangular contour and let its real endpoints tend to infinity; (1143) makes the vertical sides tend to zero. Iterate over the finitely many bonds. Fubini applies to every intermediate contour by the same Gaussian majorant and the bounded Jacobian factors. The failed-test bonds are left fixed, so a bounded Borel factor in those variables is allowed throughout.

Write $\beta' = \beta - \operatorname{Re}\mathbf{m}$ and $\eta = (1-\phi)\nu$. On a failed-test bond, $|\beta'(b)| \ge d_*\mathfrak{b}_b$. Wherever $\eta(b) \ne 0$, either that bond is failed, or $|\beta(b)| > r_1\mathfrak{b}_b$ and $|\beta'(b)| \ge d_*\mathfrak{b}_b/4$. Consequently

$$\|\eta\|^2 \le 16 d_*^{-2}\varepsilon_{\mathbf{m}}^2\|\beta'\|^2.$$

Put $\Sigma_0 = \hat{\varepsilon} + (1 + c_{\mathrm{r}})\varepsilon_{\mathbf{m}}^2 + \gamma_2$ and

$$a = \hat{\varepsilon} + 8d_*^{-2}(1 + c_{\mathrm{r}})\varepsilon_{\mathbf{m}}^2 + \gamma_2 \le 8d_*^{-2}\Sigma_0.$$

Insert (1142), the Jacobian bound and

$$1 \le \exp\left\{\gamma_2\|\beta'\|^2 - \gamma_2 d_*^2 \sum_{b\in\mathfrak{e}^+} \mathfrak{b}_b^2\right\}.$$

The modulus of the shifted integrand is at most

$$F_0 e^{(2/3)d_*^{-2}\varepsilon_{\mathbf{m}}^2 N_{\mathrm{I}}(Z)} e^{-\gamma_2 d_*^2 \sum_{b\in\mathfrak{e}^+}\mathfrak{b}_b^2} e^{-(x,(Q^Z_{\mathrm{r}} - 2a)x)/2}, \qquad x = (\mathbb{X}, \beta'). \tag{1145}$$

Since $\Sigma_0 \le \lambda_{\mathrm{r}}/\mathsf{C}_{\mathrm{mn}}^2$, $2a < \lambda_{\mathrm{r}}/2$. Thus $A = Q^Z_{\mathrm{r}} - 2a \ge \lambda_{\mathrm{r}}/2$. Translate by the real mean and enlarge the real integration domain. Gaussian integration cancels the powers of 2π and leaves $|\det Q^Z|^{1/2}(\det A)^{-1/2}$. Factor out A; the determinant inequality $|\det(1+D)| \le (1+\|D\|)^n \le e^{n\|D\|}$ gives

$$|\det Q^Z|^{1/2}(\det A)^{-1/2} \le e^{2n(a+\hat{\varepsilon})/\lambda_{\mathrm{r}}} \le e^{4(a+\hat{\varepsilon})N_{\mathrm{I}}(Z)/\lambda_{\mathrm{r}}}.$$

The remaining volume rate is bounded by

$$4(a+\hat{\varepsilon})\lambda_{\mathrm{r}}^{-1} + \tfrac{2}{3}d_*^{-2}\varepsilon_{\mathbf{m}}^2 \le 37 d_*^{-2}\lambda_{\mathrm{r}}^{-1}\Sigma_0 \le c_{\mathrm{mn}}u_0.$$

Finally $d_*^2 \ge \mathsf{C}_{\mathrm{mn}}^{-1}$, proving (1139). Empty integration spaces give the same bound directly.

For holomorphy, extend the matrix entries and the mean to a common neighbourhood of the closed polydisc, symmetrizing the continued S block if necessary. On a smaller open neighbourhood their

joint form is within $2\hat\varepsilon+\lambda_{\mathrm r}/8$ of the fixed real reference and the mean norm is below $(\sum_b \mathfrak b_b^2)^{1/2}$. This neighbourhood exists by continuity and the strict bound $\|\mathbf m\| \le \vartheta_*(\sum_b \mathfrak b_b^2)^{1/2}$. The proof of (1143) applies with $\hat\varepsilon + \lambda_{\mathrm r}/16 \le \lambda_{\mathrm r}/8$ in place of $\hat\varepsilon$. It supplies a common integrable majorant there. The fixed-domain integral is therefore holomorphic by dominated integration. In the zero-dimensional case it is constant. □

57.3. **The terms of the expansion.** The localization uses only the part of a fine region which contains integration bonds. For a fine region $Z_\circ$, let $Z_\circ^{\mathrm I}$ be the union of its fine cubes containing an endpoint of an integration bond, and let $\mathrm{Gd}(Z_\circ)$ be the union of the coarse cubes of $Z_\circ^\sharp$ which meet Λ_t. Then

$$\mathfrak X_{\mathrm I}\langle Z_\circ^{\mathrm I}\rangle = \mathfrak X_{\mathrm I}\langle Z_\circ\rangle, \qquad (Z_\circ^{\mathrm I})^\sharp \subset \mathrm{Gd}(Z_\circ) \subset Z_\circ^\sharp. \tag{1146}$$

The first identity follows by taking the fine cube containing the endpoint. To check the second, take such a cube δ, an endpoint $x \in \delta$, and the endpoint $x' \in \Lambda_t$ of the same bond. Every coarse cube Δ of $\delta^\sharp$ meets δ. For $y \in \Delta \cap \delta$, $|y - x'|_\infty \le M + 1 < LM$. The centres of Δ and the coarse cube of Λ_t containing x' are therefore at distance less than $2LM$. Their distance is an integer multiple of LM, so the cubes touch. Thus Δ meets Λ_t, proving the inclusion.

The free parameters are now the coarse cubes outside this guard. Extend s by 1 on $\mathrm{Gd}(Z_\circ)\setminus(Z_\circ^{\mathrm I})^\sharp$, obtaining the family s^{gd} on $\mathsf P(\mathfrak R, Z_\circ^{\mathrm I})$. This is an affine map between the corresponding closed polydiscs.

Let $\underline{\mathbb V} = (\mathbb V^o)_o$ be a finite system of potentials with

$$|\mathbb V^o(Y;B)| \le \mathsf u^o(Y) \le 1, \qquad \mathsf w^o(Y) \ge e\,\mathsf u^o(Y) \tag{1147}$$

on the complex ranges $|\operatorname{Re} B(b)| + |\operatorname{Im} B(b)| < \mathfrak b_b$ for $b \in \mathfrak X_{\mathrm I}^{\circ+}\langle Y\rangle$. They are Borel and holomorphic there. A vertex (Y,o) has $Y^\sharp \subset \mathfrak R$ and contains a source cube. For a finite set $\mathbf D$ of vertices put $Y^{\mathrm v} = \bigcup_{(Y,o)\in\mathbf D} Y$. Choose a reversal-invariant failed set $\mathfrak e \subset \mathfrak X_{\mathrm I} \setminus \mathfrak X_{\mathrm I}\langle Y^{\mathrm v}\rangle$ with $b^\sharp \subset \mathfrak R$. For $\iota = (\mathbf D, \mathfrak e) \ne (\emptyset,\emptyset)$ set

$$\begin{aligned} Z_\circ(\iota) &= \bigcup_{(Y,o)\in\mathbf D} Y \;\cup \bigcup_{b\in\mathfrak e} \square(b), \\ F_\iota(B) &= (-1)^{\#\mathfrak e^+}\,\mathsf{in}_{\mathfrak X_{\mathrm I}\langle Y^{\mathrm v}\rangle}(B)\,\mathsf{out}_{\mathfrak e}(B) \prod_{(Y,o)\in\mathbf D} (e^{\mathbb V^o(Y;B)} - 1). \end{aligned} \tag{1148}$$

Here in uses the strict tests $|B(b)| < \mathfrak b_b$, and out their complements. The product is assigned zero when an inner test fails, before evaluating any potential. Define

$$f_\iota(s) = \mathbb J^{\mathbf m}(Z_\circ(\iota)^{\mathrm I}; s^{\mathrm{gd}})[F_\iota]. \tag{1149}$$

For a set $\mathcal K$ of free coarse cubes, its term and domain are

$$T(\iota,\mathcal K) = f_\iota[\mathcal K \mid \emptyset], \qquad X(\iota,\mathcal K) = Z_\circ(\iota)^\sharp \cup \bigcup \mathcal K. \tag{1150}$$

Thus the bracket is the alternating sum at the indicators of the subsets of $\mathcal K$.

Proposition 57.3. Under the hypotheses of proposition 57.2, every term is defined and satisfies

$$|T(\iota,\mathcal K)| \le e^{c_{\mathrm P} N_\Lambda(X)} \prod_{(Y,o)\in\mathbf D} \mathsf w^o(Y)\, e^{-\gamma_P \#\mathfrak e^+} e^{-(\bar\kappa-1)\#\mathcal K}. \tag{1151}$$

Here $N_\Lambda(X)$ counts the coarse cubes of X meeting Λ_t. An insertion $\mathbb W(Y_1;B)$ bounded by $\mathsf w'(Y_1)$ on its complex range gives the same bound with the additional factor $\mathsf w'(Y_1)$ and with Y_1 included in the domain and the inner tests. The estimate also holds after multiplying a plain term's integrand by any Borel function of the failed variables of modulus at most one. ■

Proof. On the tube with $\mathfrak{S} = \mathfrak{X}_{\mathrm{I}}\langle Y^{\mathrm{v}}\rangle$ and failed set $\mathfrak{e}$, the untested part of F_ι is admitted. Indeed the variables of each potential lie in its complex range, and $|e^z - 1| \le e|z|$ for $|z| \le 1$ bounds it by $F_0 = \prod_{(Y,o)\in\mathbf{D}} \mathsf{w}^o(Y)$. The restrictions use the same integration variables by (1146).

If s vanishes off $\mathcal{K}$, then s^{gd} vanishes outside $Z = \mathrm{Gd}(Z_\circ) \cup \bigcup \mathcal{K}$. Apply proposition 57.2 there. Each failed threshold is at least $\mathfrak{b}_-$, and

$$N_{\mathrm{I}}(Z) \le 18(LM)^3 N_\Lambda(Z). \tag{1152}$$

For this count, every integration bond touching a coarse cube of Z also has an endpoint in a cube of Λ_t. The same centre-distance argument as above shows that the first cube meets Λ_t. Such a cube contains $(LM)^3$ level-k sites, each an endpoint of at most six positive bonds. Multiplication by the three Lie-algebra coordinates proves (1152). Since $Z \subset X$ and $\mathsf{C}_{\mathrm{mn}} \ge 18$, (1139) yields

$$|f_\iota(s)| \le F_0 e^{c_{\mathrm{P}} N_\Lambda(X)} e^{-\gamma_P \#\mathfrak{e}^+}.$$

The function is holomorphic near its closed polydisc by proposition 57.2 and the affine guard extension. Apply the finite-difference Cauchy estimate on the free coordinates in $\mathcal{K}$, setting all other coordinates to zero. Its factor is $(e^{\bar\kappa} - 1)^{-\#\mathcal{K}} \le e^{-(\bar\kappa - 1)\#\mathcal{K}}$ for $\bar\kappa \ge 1$. This proves (1151).

An insertion simply adds its variables to the inner tests and its bound to F_0; the ordinary vertices are unchanged. A bounded Borel function of the failed variables preserves admittedness because those variables are fixed in the holomorphy assertion and in the contour deformation. These observations prove the other two claims. □

57.4. **The local conditions on a graded field.** Only two local conditions on a graded field enter the Gaussian construction. We state them explicitly, so that the additional response conditions on a graded domain play no role in the following argument. Let $P = (U, A', \mathbf{J})$ be a triple at level t on Y, $\mathbf{U} = e^{i\eta_t A'} U$, and let $\sigma = (t, \mathrm{mid})$. For a site x put $n(x) = \max\{0 \le n \le t : x \in \Omega_n^\diamond\}$. The graded radii have the form

$$R_\sigma^{p,i}(x) = c_\sigma(n(x)) r_i(\sigma; n(x)) L^{-pn(x)}, \qquad 0 < c_\sigma(n) \le 1 + \beta_{\mathrm{s}}.$$

Here $0 < \beta_{\mathrm{s}} \le 1/4$; more explicitly, $c_\sigma(t) = 1 + \beta_{\mathrm{s}}$ and, for $n < t$,

$$c_\sigma(n) = 1 - \beta_{\mathrm{s}}(1 - 2^{-(t-n)}) + \theta_{\mathrm{s}} \beta_{\mathrm{s}} 2^{-(t-n)}, \qquad 0 \le \theta_{\mathrm{s}} < 1.$$

The two required conditions are the following.

(L1) On each contained stencil, with its indicated base point,

$$|\partial U - 1|,\ |\partial \mathbf{U} - 1| < R_\sigma^{2,0}, \quad \eta_t^3 |\mathbf{J}| < R_\sigma^{3,0}, \quad \eta_t |A'| < R_\sigma^{1,1}, \quad \eta_t^2 |\nabla^U A'| < R_\sigma^{2,1}.$$

The bond quantities are based at the initial endpoint. Differences are required only when their entire stencil is in Y.

(L2) For each $n \le t$ and each embedded cube $q \subset Y \cap \Omega_n^\diamond$ of side at most $\mathsf{o}LM$ in level-n units, U has a real gauge $U^v = e^{i\eta_n A}$ on q with

$$|A|,\ |\nabla^{\eta_n} A| < \mathsf{o} LM B^{(\mathrm{cg})} r_0(\sigma; n).$$

Also, every cube $q \subset Y$ made of level-n $\mathfrak{M}$-cubes, contained in $\Omega_n^\diamond$ and of side at most $12\mathfrak{M}$ in level-n units, has a containing embedded cube $q' \subset Y$ on which U has a real gauge of index n at most $a_\flat$.

Thus (L2) includes a containing-box condition, not just a gauge on the tested set. In its first clause we have retained only the current-structure instance of the graded condition. These are the local requirements of the graded domains; we do not use their tower-response clauses here.

Proposition 57.4. Assume the two history gaps and $\mathfrak{o}L^3 \ge 3$. Suppose P satisfies (L1)–(L2), and, for $0 \le n \le t$,

$$\begin{gathered}(1+\beta_{\mathrm{s}})r_0(\sigma;n) \le \min\{\alpha_0', \alpha_2'\}, \qquad (1+\beta_{\mathrm{s}})r_1(\sigma;n) \le \alpha_1', \\ a_\flat \le \alpha_0', \qquad \mathfrak{o}L^3 M B^{(\mathrm{cg})} r_0(\sigma;k) \le \alpha_0'.\end{gathered} \tag{1153}$$

For every coarse region $R \subset Y$, the level-k reading

$$p_k(P) = (\mathbf{U}, L^{-3}\mathbf{J}), \qquad \mathbf{U} = e^{i\xi(L^{-1}A')}U$$

on $R \cap \Omega_0^\diamond$ belongs to $\mathscr{O}^+(R;\alpha_0',\alpha_1',\alpha_2')$ and hence to $\mathscr{O}(R;\alpha_0',\alpha_1',\alpha_2')$. It belongs to the normalizer class $\mathscr{O}^N(R;a)$ whenever $a \ge \max_i \alpha_i'$. ■

Proof. A stencil in layer n has base point of depth $l \ge n$. The bound $c_\sigma(l) \le 1+\beta_{\mathrm{s}}$ and $L^{-pl} \le L^{-pn}$ give the four bounds (1091). The change of units is exact: $\xi L^{-1}A' = \eta_t A'$, $\xi^3 L^{-3}\mathbf{J} = \eta_t^3 \mathbf{J}$, and the corresponding first-difference expression is unchanged. The second part of (L2) gives the plain and natural cube gauges, including their containing boxes.

It remains to check the tiny cubes near all fluctuation bonds. Let y contain an endpoint z of such a bond, and let its other endpoint z' lie in Ω_t. Every point of $y^\sim$ is within $2\ell_c + 1 < 8LM\check{R}_t$ of z'. The second history gap therefore puts $y^\sim$ in Ω_k when $k \ge 1$; for $k = 0$ its required containment in $\Omega_0^\diamond$ is already part of the test. The cube is embedded, since its side $3\ell_c$ is less than the torus side. Its side in level-k units is $3M/L^2 \le \mathfrak{o}LM$. Apply the first part of (L2) with $n = k$. Passing from index k to index t costs at most L^2 in the two gauge bounds. The last inequality in (1153) gives the required index-t gauge. The same real factor U is used throughout. The normalizer assertion follows from the containing-cube argument following proposition 55.1; restriction introduces no new stencil. □

57.5. **The square root on the integration bonds.** For the conditional Gaussian we need the square root of the inverse on $\mathfrak{X}_{\mathrm{I}}$, rather than the inverse strings used in the kept form. We use the mesh, cutoffs $h_\square$ and string construction of section 50.2, with the actual integration bonds as its coordinate set. Choose

$$13w \le h, \qquad L^3 \le h, \qquad Lh \le \ell_c,$$

and enlarge the fixed lower bound for h as specified below. Use the mesh-gauge radius $a_{\mathrm{w}} \le a_{\mathrm{cv}}/64$ fixed in section 50.2, and require $\alpha_0 \le \min\{a_{\mathrm{pos}}, a_{\mathrm{w}}/2\}$ and $2(\alpha_1+\alpha_2) \le c_{\mathrm{d}} a_{\mathrm{w}}$. The following domain check is what permits that construction.

If the mesh neighborhood $\widetilde{\square}^2$ is contained in $R \cap \Omega_0^\diamond$ and has a nonempty local coordinate set $Y_\square$, there is a tiny cube y meeting an integration bond such that

$$\widetilde{\square}^2 \subset y^\sim \subset R \cap \Omega_0^\diamond. \tag{1154}$$

To see this, choose an integration endpoint $x \in \Lambda_t$ belonging to a bond of $Y_\square$ and work in the local chart. Each coordinate interval I of $\widetilde{\square}^2$ has length $5h < \ell_c$ and contains the corresponding coordinate of x in its interior. Let J_0 be the big interval containing that coordinate. If I meets only J_0, choose the tiny interval containing the coordinate and move its index, if necessary, to the range $2, \dots, L^3 - 1$. Its three-interval enlargement covers I and stays in J_0. If I crosses a big-grid face, choose the tiny interval in J_0 adjacent to that face. Its enlargement covers I and lies in the two crossed big intervals. Every big cube so used meets the interior of $\widetilde{\square}^2$ and therefore belongs to R. The product tiny cube is in the big cube of x, hence meets the integration set. The history margin puts its enlargement in $\Omega_0^\diamond$.

In fact the same margin gives $\widetilde{\square}^2 \subset \Omega_t$. On this neighborhood the collared sequence is the constant sequence. The pair restrictions, their domains and every coefficient a_S with $S^\sim \subset \widetilde{\square}^2$ coincide with the constant-sequence ones by proposition 55.1. The gauge supplied by (1154) is precisely the mesh gauge used there. The local form is thus a coordinate compression of that

constant-sequence form, and its real part has gap $3\gamma_{\mathrm{tr}}/4$ on the complex disc used in eq. (774). This assertion concerns the inner-plaquette form and its stated coordinate space.

Here are useful common constants for the form and root terms:

$$B_{\mathrm{w}} = 2\sqrt{2}C_a, \qquad \delta = \theta_a, \qquad \gamma_{\mathrm{flat}} = 3\gamma_{\mathrm{tr}}/8, \qquad \theta_{\square} = \frac{3\theta_a\gamma_{\mathrm{tr}}}{8B_a}.$$

Exponential conjugation, as in (776), gives

$$|g^{\mu}_{\square}(b,b')|_2 \le (\mu+\gamma_{\mathrm{flat}})^{-1}e^{-\theta_{\square}|b-b'|}.$$

Put $c_1 = c_{\mathrm{lat}}(\theta_{\square}/2)$, $c_2 = c_{\mathrm{lat}}(\delta/2)$ and

$$K_0 = B_{\mathrm{w}}c_2\left[24\pi\left(\frac{2}{e\delta}+\frac{1}{2}\right)+\frac{16}{\delta}\right].$$

Take $h \ge 2K_0c_1/\gamma_{\mathrm{flat}}$. The commutator and remote-leg bounds of that proof apply on the present coordinate set. For the latter bound, if $h_{\square}(b) \ne 0$ but $S^{\sim} \not\subset \widetilde{\square}^2$, the connecting length is at least $2.5h - 3w - h - \frac{1}{2} \ge h$. Consequently the error of the local resolvent sum has row norm at most $K_0c_1/[h(\mu+\gamma_{\mathrm{flat}})] \le 1/2$. Its geometric series is the inverse of the actual compressed form, not a separately defined continuation.

Use the ordered strings and the $\mu^{-1/2}/\pi$ integral (777) to define r_ω on these local domains. The same summation proof gives constants

$$\begin{aligned} \delta_{\mathrm{G}} &= \min\{\theta_{\square}/6, \delta\}, \qquad B_{\mathrm{G}} = \max\{B_{\mathrm{w}}, 1, 2c_1\gamma_{\mathrm{flat}}^{-1/2}\}, \\ \sum_{\iota} m_\iota(b,b')e^{\delta_{\mathrm{G}}l_\iota(b,b')} &\le B_{\mathrm{G}}, \qquad |T_\iota(p) - T_\iota(U,0)| \le \varepsilon_{\mathrm{G}}m_\iota, \\ \varepsilon_{\mathrm{G}} &= \frac{2(\alpha_1+\alpha_2)}{c_{\mathrm{d}}a_{\mathrm{w}}}. \end{aligned} \tag{1155}$$

Here T_ι is a form term a_S, a root term r_ω, or its transpose. Their footprints lie within $\rho_{\mathrm{G}} = 5h$ of a connecting tree of length l_ι. The difference estimate follows by Cauchy's estimate on the same disc $|z| < c_{\mathrm{d}}a_{\mathrm{w}}/(\alpha_1+\alpha_2)$, assumed at least 2. This proves holomorphy and locality of the strings as well, since the resolvent majorants are integrable in μ. On the whole torus the sum $R_{\mathrm{I}} = \sum_\omega r_\omega$ obeys

$$R_{\mathrm{I}}R_{\mathrm{I}}^{\mathsf{T}} = \chi_{\mathrm{I}}^{\mathsf{T}}\mathsf{A}_{\mathrm{II}}^{-1}\chi_{\mathrm{I}}. \tag{1156}$$

Indeed the resolvent sum is the actual inverse for every μ; the finite-dimensional square-root identity used in eq. (778) then applies unchanged.

57.6. **The actual conditional Gaussian.** Fix $Z_\circ^\sharp \subset R$ and put $e = \mathfrak{X}_{\mathrm{I}}\langle Z_\circ\rangle$ and $r = \mathfrak{X}_{\mathrm{I}}\langle R\rangle$. Complete the parameter family by $\bar{s} = 1$ on $Z_\circ^\sharp$, by s on the other big cubes of R, and by zero outside R. Multiply a coefficient or string by the product of $\bar{s}$ over its big-cube hull, omitting terms with zero product. Write

$$A(\bar{s};p) = \sum_S \bar{s}(S)a_S(p), \qquad R_{\mathrm{I}}(\bar{s};p) = \sum_\omega \bar{s}(\omega)r_\omega(p). \tag{1157}$$

Here and below the matrices use positive-bond coordinates; χ denotes restriction and $p = \chi^{\mathsf{T}}\chi$ denotes the corresponding projection. Define

$$\begin{aligned} C &= (\chi_e A\chi_e^{\mathsf{T}})^{-1}, & \Gamma &= \chi_e A(p_{\mathrm{I}} - p_e)R_{\mathrm{I}}\chi_r^{\mathsf{T}}, \\ M &= -C\,[\chi_e - \Gamma\chi_r R_{\mathrm{I}}^{\mathsf{T}}]A\chi_{\mathrm{E}}^{\mathsf{T}}, & m &= M\mathsf{a}. \end{aligned} \tag{1158}$$

The external field a is given on external bonds whose two endpoints lie in R, and is extended by zero. For a triple P use its reading $p_k(P)$; the reference objects $C_{\mathrm{r}}, \Gamma_{\mathrm{r}}, M_{\mathrm{r}}$ are obtained by setting $s = 0$ and replacing that reading by $(U,0)$. The reference mean is $m_{\mathrm{r}} = M_{\mathrm{r}}\mathsf{a}^{\mathrm{re}}$, where $\mathsf{a}^{\mathrm{re}} = (\mathsf{a}+\mathsf{a}^*)/2$. Empty coordinate sets have their unique zero-dimensional operators.

The small parameter below accounts separately for terms that leave $Z^\sharp_\circ$ and for the complex displacement of the background. Set

$$\begin{aligned} \gamma_0 &= \gamma_{\mathrm{pos}}, \quad B_{\mathrm{nm}} = B_{\mathrm{G}} c_{\mathrm{lat}}(\delta_{\mathrm{G}}/8), \\ \varepsilon_{\mathrm{dc}} &= e^{64\bar\kappa} e^{-\delta_{\mathrm{G}}(M-1)/128} + 64\varepsilon_{\mathrm{G}}, \qquad \bar\kappa \ge 1, \end{aligned} \tag{1159}$$

and require

$$\rho_{\mathrm{G}} \le M/2, \qquad \delta_{\mathrm{G}} M \ge 64\bar\kappa, \qquad B_{\mathrm{nm}}\varepsilon_{\mathrm{dc}} \le \gamma_0/2. \tag{1160}$$

All preceding grain, clearance, mesh and positivity floors remain in force. Put

$$\begin{aligned} \gamma_{\mathrm{r}} &= \gamma_0, \qquad c_{\mathrm{r}} = B_{\mathrm{nm}}^2/\gamma_0, \qquad \hat\varepsilon = 64 B_{\mathrm{nm}}^5 \gamma_0^{-2} \varepsilon_{\mathrm{dc}}, \\ \delta_M &= \frac{\delta_{\mathrm{G}}\gamma_0}{128 B_{\mathrm{nm}}}, \qquad c_M = 72 B_{\mathrm{nm}}^5 \gamma_0^{-2} c_{\mathrm{lat}}(\delta_{\mathrm{G}}/16) c_{\mathrm{lat}}(\delta_M). \end{aligned} \tag{1161}$$

Theorem Z. Let the above hypotheses hold and let $p_k(P) \in \mathscr{O}(R; \alpha_0, \alpha_1, \alpha_2)$. The matrices (1158), including C^{-1}, are holomorphic for $|s(\Delta)| < e^{\bar\kappa+1}$; C is symmetric and invertible. They have the local restriction and component-decoupling properties of definition 57.1, and their real references satisfy (1132)–(1133) with (1161). Moreover,

$$|M_{\mathrm{r}}(b,b')|_2 \le c_M e^{-\delta_M|b-b'|}, \qquad |M(s;p)(b,b') - M_{\mathrm{r}}(b,b')|_2 \le \varepsilon_{\mathrm{dc}} c_M e^{-\delta_M|b-b'|}. \tag{1162}$$

The second bound holds on $|s(\Delta)| \le e^{\bar\kappa}$.

Suppose, for every $b \in \mathfrak{X}_{\mathrm{I}}^+ \langle R \rangle$,

$$\begin{aligned} c_M \sum_{b'} e^{-\delta_M|b-b'|} |\mathsf{a}^{\mathrm{re}}(b')| &\le \theta\, \mathfrak{b}_b, \\ \sqrt{2} c_M \sum_{b'} e^{-\delta_M|b-b'|} |\mathsf{a}(b') - \mathsf{a}^{\mathrm{re}}(b')| &\le \varepsilon'\, \mathfrak{b}_b, \end{aligned} \tag{1163}$$

where both sums are over positive external bonds with both endpoints in R. Then

$$|m_{\mathrm{r}}(b)| \le \theta \mathfrak{b}_b, \qquad |m(b) - m_{\mathrm{r}}(b)| \le (\sqrt{2}\varepsilon_{\mathrm{dc}}\theta + (1+\varepsilon_{\mathrm{dc}})\varepsilon') \mathfrak{b}_b.$$

If $\theta + \sqrt{2}\varepsilon_{\mathrm{dc}}\theta + (1+\varepsilon_{\mathrm{dc}})\varepsilon' < 1$, these are a local system of Gaussian structures with conditional mean in the sense of definition 57.1. On the whole torus the system is complete, with

$$C_* = \mathsf{A}_{\mathrm{II}}^{-1}, \qquad m_* = -\mathsf{A}_{\mathrm{II}}^{-1} \mathsf{A}_{\mathrm{IE}} \mathsf{a}. \tag{1164}$$

If also $(LM)^3 \varepsilon_{\mathrm{dc}} \le 1$, a defect number is

$$u_{\mathrm{gs}} = 128 B_{\mathrm{nm}}^5 \gamma_0^{-2} + c_{\mathrm{r}} + 3.$$

■□

Proof. We give the common product estimate first. It is needed for the mean, as well as for the covariance. A factor may be A, R_{I} or $R_{\mathrm{I}}^{\mathsf{T}}$, with arbitrary coordinate projections between factors. For $1 \le q \le 4$, define the nonnegative kernel

$$K_q(b,b') = B_{\mathrm{G}}^q \sum_{b_1,\dots,b_{q-1}} \prod_{i=1}^{q} e^{-\delta_{\mathrm{G}}|b_{i-1}-b_i|/4}, \qquad b_0 = b, \quad b_q = b'.$$

Its row and column sums are at most B_{nm}^q, and $K_q(b,b') \le B_{\mathrm{nm}}^q e^{-\delta_{\mathrm{G}}|b-b'|/8}$. These follow by retaining half the distance exponent for the endpoints and summing the remaining factors by $c_{\mathrm{lat}}(\delta_{\mathrm{G}}/8)$.

For an individual factor of length l_ι, the hull count in (781) gives, on the open polydisc under consideration, $|\bar s(\iota)| \le e^{16\bar\kappa + \delta_{\mathrm{G}} l_\iota/8}$. The count applies because its footprint is within ρ_{G} of a connecting tree, $\rho_{\mathrm{G}} \le M/2$, and the second inequality of (1160) absorbs the length contribution. If the product starts on e and one of its footprints leaves $Z^\sharp_\circ$, the connected chain must travel at least

$$D_* = (M-1) - \rho_{\mathrm{G}} \ge M/2 - 1.$$

Indeed an endpoint in a fine cube of $Z_\circ$ has distance at least $M-1$ from a big cube outside its sharp enlargement; the footprint can shorten this distance by at most ρ_{G}. The same argument applies with the order reversed.

Split each product into chains entirely inside $Z_\circ^\sharp$ and chains leaving it. On the former, all parameters are 1, and telescoping the q factors bounds the difference from the real reference by $q\varepsilon_{\mathrm{G}}K_q$. On the latter, retain the chain-length gain before summing the intermediate bonds. The bound is $e^{16q\bar{\kappa}}e^{-5\delta_{\mathrm{G}}D_*/8}K_q$. For $q\le 4$ their sum is bounded by $\varepsilon_{\mathrm{dc}}K_q$. Thus, for every such product Π_q with initial projection on e,

$$|\Pi_{q,\mathrm{r}}(b,b')|\le K_q(b,b'),\qquad |\Pi_q(s;p)(b,b')-\Pi_{q,\mathrm{r}}(b,b')|\le \varepsilon_{\mathrm{dc}}K_q(b,b'). \tag{1165}$$

The corresponding row and column estimates justify all series, locally uniformly on the open polydisc.

The reference compression C_{r}^{-1} has gap γ_0. For precision, take the union W of tiny cubes of $Z_\circ$ meeting integration bonds. Then $e\subset \mathfrak{X}_{\mathrm{I}}\langle W\rangle$, $W^\sim\subset R$, and the set-theoretic interior determined by $Z_\circ^\sharp$ contains the $2\ell_c/3$-neighborhood of W. A footprint belongs to that set exactly when its big-cube hull belongs to $Z_\circ^\sharp$. Hence its partial coefficient sum is $A(\bar{0};U,0)$, and (1099) gives the stated gap. This application uses the chosen reference in R; it does not require choosing it anew on $Z_\circ^\sharp$.

Apply (1165) with $q=1,2$ to C^{-1} and Γ. Their differences have norms at most $B_{\mathrm{nm}}\varepsilon_{\mathrm{dc}}$ and $B_{\mathrm{nm}}^2\varepsilon_{\mathrm{dc}}$. Thus C^{-1} has real-part gap $\gamma_0/2$. The resolvent identity and the three-term expansion of $\Gamma^{\mathsf{T}}C\Gamma-\Gamma_{\mathrm{r}}^{\mathsf{T}}C_{\mathrm{r}}\Gamma_{\mathrm{r}}$ give respectively the remaining bounds (1132)–(1133); the latter difference is at most $9B_{\mathrm{nm}}^5\gamma_0^{-2}\varepsilon_{\mathrm{dc}}$. This is bounded by the chosen $\hat{\varepsilon}$. Symmetry follows from that of the coefficients.

For the mean, put $N=[\chi_e-\Gamma\chi_rR_{\mathrm{I}}^{\mathsf{T}}]A\chi_{\mathrm{E}}^{\mathsf{T}}$, so $M=-CN$. This is the difference of a one-factor product and a four-factor product. Therefore

$$|N_{\mathrm{r}}(b,b')|_2\le 2B_{\mathrm{nm}}^4e^{-\delta_{\mathrm{G}}|b-b'|/8},\qquad |N-N_{\mathrm{r}}|_2\le 2B_{\mathrm{nm}}^4\varepsilon_{\mathrm{dc}}e^{-\delta_{\mathrm{G}}|b-b'|/8}.$$

Exponential conjugation of C^{-1} and C_{r}^{-1}, followed by their resolvent identity, gives, with $\delta'=2\delta_M$,

$$\begin{aligned}|C(b,b')|_2,\ |C_{\mathrm{r}}(b,b')|_2&\le 4\gamma_0^{-1}e^{-\delta'|b-b'|},\\ |C-C_{\mathrm{r}}|_2&\le 16B_{\mathrm{nm}}\gamma_0^{-2}c_{\mathrm{lat}}(\delta'/2)\varepsilon_{\mathrm{dc}}e^{-\delta'|b-b'|/2}.\end{aligned}$$

In the product CN, retain the rate $\delta'/2$ at the endpoints and sum the remaining factor with $c_{\mathrm{lat}}(\delta_{\mathrm{G}}/16)$. The identity $CN-C_{\mathrm{r}}N_{\mathrm{r}}=(C-C_{\mathrm{r}})N_{\mathrm{r}}+C(N-N_{\mathrm{r}})$ proves (1162) with (1161). For the fields, use $m-m_{\mathrm{r}}=(M-M_{\mathrm{r}})\mathsf{a}^{\mathrm{re}}+M(\mathsf{a}-\mathsf{a}^{\mathrm{re}})$. The real reference has equal coordinate and operator norms; the complex comparison costs the factor $\sqrt{2}$. This gives precisely the two asserted mean bounds.

Every surviving coefficient or string has its whole connected hull in a single wall component when the parameters vanish outside a region Z. Its input and output bonds have both endpoints in that component. Consequently A and R_{I} split into their coordinate blocks, and multiplication and inversion in (1158) give the required decoupling of C,Γ,M. In particular an external field is restricted by both endpoints, rather than merely by incidence. The identical local matrices occur under restriction, by proposition 55.1 and the locality of the root strings. This proves the local-system assertions. Restriction preserves (1163), since its nonnegative sums only decrease.

On the torus, (1156) and block multiplication give

$$C+C\Gamma\Gamma^{\mathsf{T}}C=\chi_e\mathsf{A}_{\mathrm{II}}^{-1}\chi_e^{\mathsf{T}},\qquad C[\chi_e-\Gamma\chi_{\mathrm{I}}R_{\mathrm{I}}^{\mathsf{T}}]=\chi_e\mathsf{A}_{\mathrm{II}}^{-1}\chi_{\mathrm{I}}.$$

The second identity, multiplied by $A\chi_{\mathrm{E}}^{\mathsf{T}}\mathsf{a}$, yields $m=\chi_em_*$. These are exactly the completeness identities (1327). Finally, $2(LM)^3\hat{\varepsilon}+c_{\mathrm{r}}+3\le u_{\mathrm{gs}}$ under the stated additional hypothesis, proving the defect assertion. □

57.7. **The external tiers and the history step.** We now verify the mean condition for the external fields used in the history step. The collar bonds are $\mathfrak{X}_{\mathrm{col}} = \mathrm{Col}(\Lambda_t)$. The rim bonds are external bonds incident to a coarse cube of $\mathrm{rim}(Z_t)$; equivalently they belong to one of the corresponding sets $c^\star$. Put

$$\mathrm{rad}_k(b) = \mathfrak{b}_k \min\left(\{C_{\mathrm{xt}}\} \cup \left\{\frac{1}{4\Theta_1} : b \in \mathfrak{X}_{\mathrm{col}}\right\} \cup \{\theta_c : b \in \mathfrak{X}_{\mathrm{rim}}\}\right). \tag{1166}$$

Take $\mathsf{a} = \mathsf{a}' + \mathsf{a}''$, where a' is real and $|\mathsf{a}'(b)| < \mathrm{rad}_k(b)$, while $\mathsf{a}'' = 0$ on the rim and $|\mathsf{a}''| \le \varepsilon_a \mathfrak{b}_k$ elsewhere. For a field on R these conditions are imposed after zero extension from the external bonds with both endpoints in R.

Choose $C_e \ge C_{\mathrm{xt}}$ and define

$$\begin{aligned} \vartheta_{\mathrm{col}} &= \frac{c_M c_{\mathrm{lat}}(\delta_M)}{4\Theta_1 \theta_c}, & \vartheta_{\mathrm{rim}} &= c_M c_{\mathrm{lat}}(\delta_M/2) e^{-\delta_M \varpi_T/2}, \\ \vartheta_{\mathrm{far}} &= c_M c_{\mathrm{lat}}(\delta_M/2) \frac{C_e}{\theta_c} e^{-\delta_M (W_t - 2)/2}, & \vartheta_\mu^{\mathrm{re}} &= \vartheta_{\mathrm{col}} + \vartheta_{\mathrm{rim}} + \vartheta_{\mathrm{far}}(1 + \varepsilon_a/C_e). \end{aligned} \tag{1167}$$

The numbers are uniform in the history and volume, apart from the displayed width W_t. Define also

$$\varepsilon_\mu = \sqrt{2}\varepsilon_{\mathrm{dc}} \vartheta_\mu^{\mathrm{re}} + \sqrt{2}(1 + \varepsilon_{\mathrm{dc}}) \vartheta_{\mathrm{far}} \varepsilon_a / C_e. \tag{1168}$$

Proposition 57.5. The above external fields satisfy (1163) with

$$\theta = \vartheta_\mu^{\mathrm{re}}, \qquad \varepsilon' = \sqrt{2} \vartheta_{\mathrm{far}} \varepsilon_a / C_e,$$

provided $\mathfrak{b}_b \ge \theta_c \mathfrak{b}_k$. Consequently, if (1153) holds at the radii of theorem Z and $\vartheta_\mu^{\mathrm{re}} + \varepsilon_\mu < 1$, the actual history step gives a local Gaussian system with conditional mean $(\theta_{\mathbf{m}}, \varepsilon_{\mathbf{m}}) = (\vartheta_\mu^{\mathrm{re}}, \varepsilon_\mu)$. It has the normalizer and all-bond positivity applicability of proposition 57.4. Its local restrictions obey

$$\begin{aligned} &C_R(Z_\circ, s; P) = C_X(Z_\circ, s_X; P|_X), \\ &\Gamma_R(Z_\circ, s; P) = \Gamma_X(Z_\circ, s_X; P|_X) \chi_{\mathfrak{X}_{\mathrm{I}}\langle X\rangle}, \qquad m_R(Z_\circ, s; P, \mathsf{a}) = m_X(Z_\circ, s_X; P|_X, \mathsf{a}|_X) \end{aligned} \tag{1169}$$

when $Z_\circ^\sharp \subset X \subset R$ and s vanishes outside X. On the whole torus it has the covariance and mean (1164). ∎

Proof. If the integration set is empty there is nothing to prove. For an external bond put $d(b') = \min_{b \in \mathfrak{X}_{\mathrm{I}}^+} |b_- - b'^+_-|_\infty$. The two geometric facts needed are

$$d(b') \le \varpi_T - 1 \implies b' \in \mathfrak{X}_{\mathrm{col}}, \qquad d(b') \le W_t - 2 \implies b' \in \mathfrak{X}_{\mathrm{rim}}.$$

For the first, the minimizing integration bond has an endpoint $z \in \Lambda_t$ at distance at most ϖ_T from an endpoint of b'. There is an inner nonpivot bond of Λ_t at z, by the pivot-incidence property. Thus the external bond is in the collar. For the second, each endpoint of b' is at distance at most $1 + (W_t - 2) + 1 = W_t$ from that z. Its coarse cube is outside Λ_t and touches its coarse enlargement; hence it is a rim cube. These arguments also show why the thresholds are $\varpi_T - 1$ and $W_t - 2$.

The integer-valued distance divides the external bonds into three disjoint sets: $d \le \varpi_T - 1$, $\varpi_T \le d \le W_t - 2$, and $d \ge \varpi_T$ with $d > W_t - 2$. The bounds on $|\mathsf{a}'|$ there are respectively $\mathfrak{b}_k/(4\Theta_1)$, $\theta_c \mathfrak{b}_k$, and $C_e \mathfrak{b}_k$. For a fixed integration bond the least-lift lattice sum is at most $c_{\mathrm{lat}}(\delta_M)$. On the second and third sets retain half the exponential for summation and use the other half at the respective distances ϖ_T and $W_t - 2$. This gives

$$c_M \sum_{b'} e^{-\delta_M |b - b'|} |\mathsf{a}'(b')| \le (\vartheta_{\mathrm{col}} + \vartheta_{\mathrm{rim}} + \vartheta_{\mathrm{far}}) \theta_c \mathfrak{b}_k.$$

The field a'' vanishes unless $d > W_t - 2$; its sum is at most $\vartheta_{\mathrm{far}}(\varepsilon_a/C_e)\theta_c \mathfrak{b}_k$. Finally $|\mathsf{a}^{\mathrm{re}}| \le |\mathsf{a}'| + |\mathsf{a}''|$ and $|\mathsf{a} - \mathsf{a}^{\mathrm{re}}| \le |\mathsf{a}''|$. These prove the claimed two weighted sums, including the factor $\sqrt{2}$ in the second one.

The graded-domain assertion follows from proposition 57.4; the mesh condition follows from (1154). Apply theorem Z. Every nonzero term in (1157) has its hull in X when s vanishes outside X, and its local matrix is unchanged on restriction. This gives (1169) directly from (1158). The torus conclusion is the completeness assertion of theorem Z. □

58. The potentials in a history step

The potentials are evaluated at a background that is defined on a collared region and is glued to an arbitrary field outside it. We first show that the local arguments of the potentials lie strictly inside their analyticity domains. The current in these arguments is the current of the background, at the scale of the term being evaluated. This scale dependence must be retained when a term from an earlier step is inserted.

58.1. **The glued background lies in the analytic cores.** Retain the running and separation hypotheses of proposition 53.11, with a collared sequence of depth n. Thus $\bar{B}\gamma^2 \le 1$, $\kappa_{\mathrm{run}} = (1+\bar{B}\gamma^2)^{1/2} \le \sqrt{2}$, and

$$\varepsilon_i \le \varepsilon_j \quad (i \le j), \qquad \varepsilon_i \le \kappa_{\mathrm{run}} L^{(i-j)/2}\varepsilon_j \quad (i \ge j).$$

Use the regular-gauge constants $B_*, c_{*,\mathrm{rg}}$ of theorem F, and the averaging radius c_2^{SO} of proposition 14.1. Define

$$C_{\mathrm{mb}} = 4L^3 B_* \max\left\{12\mathfrak{M}+2,\ \frac{\mathsf{o}LM+2}{\mathsf{o}LMB^{(\mathrm{cg})}}\right\}, \tag{1170}$$

$$\epsilon_{\mathrm{mb}} = \min\left\{\frac{c_{*,\mathrm{rg}}}{4L^3(\mathsf{o}LM+2)},\ \frac{c_{*,\mathrm{rg}}}{4L^3(12\mathfrak{M}+2)},\ \frac{c_2^{\mathrm{SO}}}{\sqrt{2}L^2},\ \frac{\bar{\varepsilon}}{2\sqrt{2}L^2},\ \frac{\varepsilon_{\mathrm{v}}}{\sqrt{2}}\right\}. \tag{1171}$$

These constants are fixed before the couplings and the history. In particular, C_{mb} depends only on $L, \mathsf{o}, \mathfrak{M}, M, B^{(\mathrm{cg})}$; the additional dependence of ϵ_{mb} is on the two regularity radii. We use the three conditions

$$\mathsf{o}LM + 6 < L^{m+K-n}, \tag{1172}$$

$$C\varepsilon_n \le \epsilon_{\mathrm{mb}}, \tag{1173}$$

$$C_{\mathrm{mb}}C\varepsilon_n \le \alpha_0/2. \tag{1174}$$

The first ensures that the cubes used for gauges embed in the torus. The second controls the gauges and the tower backgrounds. The third places the resulting triples inside half the fixed core.

Proposition 58.1. Let U be a real configuration on $\mathrm{Bd}^\circ(\Omega_0^\diamond)$, critical over its own determining averages, with

$$r_U(\mathfrak{e}) < C\varepsilon_{\lambda(\mathfrak{e})}$$

at each regularity element, where $C > 0$. Assume (1172) and (1173). Let U be any extension of U to the bare torus. For $1 \le j \le n$ and $X \in \mathcal{D}_{j-1}$ with $X \subset \Omega_j$, its plain triple is

$$\mathbf{P}_X^{(j)}(\mathsf{U}) = \big(U|_{\mathrm{Bd}(X)},\ 0,\ J^{(j)}(U)|_{\mathrm{Bd}(X)}\big). \tag{1175}$$

It is independent of the extension, and

$$\mathbf{P}_X^{(j)}(\mathsf{U}) \in \mathcal{U}_j^\circ(X; b_0, b_1, c_0) \quad \text{if} \quad b_0, c_0 \ge C_{\mathrm{mb}}C\varepsilon_j, \quad b_1 > 0. \tag{1176}$$

The region X need not be a box or be contractible. If $1 \le i \le j$ and $X' \in \mathcal{D}_{i-1}$ is contained in X, then

$$\rho_{X'}^{j\to i}\mathbf{P}_X^{(j)}(\mathsf{U}) = \mathbf{P}_{X'}^{(i)}(\mathsf{U}). \tag{1177}$$

Under (1174), all these triples belong to

$$\mathcal{U}_j^\circ(X; \alpha_0/2, \alpha_1/2). \tag{1178}$$

In particular the conclusions hold with $C = C_{\mathrm{reg}}$ for every element of the collared background of proposition 53.11, and for its gluing to any outer field. ■

Proof. We give the gauge and tower arguments, since the pointwise curvature bound by itself would not prove core membership. If a regularity element has a corner in $\Omega_i^\diamond$, its highest layer index λ satisfies $\lambda \geq i$. The running inequalities give

$$L^{-2(\lambda-i)}\varepsilon_\lambda \leq \kappa_{\mathrm{run}} L^{-3(\lambda-i)/2}\varepsilon_i. \tag{1179}$$

By the definition of the local radius, therefore,

$$|\partial U(p) - 1| < \kappa_{\mathrm{run}} C\varepsilon_i L^{-2i}, \tag{1180}$$

$$|J^{(u)}(U)(b)| < \kappa_{\mathrm{run}} C\varepsilon_i L^{-3(i-u)}. \tag{1181}$$

The stronger factors $L^{-3(\lambda-i)/2}$ and $L^{-5(\lambda-i)/2}$, respectively, may be retained. Separation puts every bond incident to $X \subset \Omega_1$, with its full plaquette star, inside $\Omega_0^\diamond$. Thus the current of any extension agrees there with the regional current. This proves (1175).

For a cube $q \subset \Omega_i^\diamond$ of side s in level-i units, $i \geq 1$, choose a cube q^+ of integer side $\ell = \lceil s \rceil + 1 < s+2$ containing q. Its concentric enlargement of side $\ell + 4$ lies within distance 4 of q and embeds whenever $s + 6 \leq L^{m+K-i}$. The history gap places the full plaquette stars in this enlargement in layer at least $i - 1$. Put

$$w = \kappa_{\mathrm{run}} C\varepsilon_{i-1}, \qquad \alpha' = 2L^3 w.$$

There the curvature is smaller than $wL^2\eta_i^2$, and

$$|D_U^{\mathsf{T}}\partial U| \leq \eta_i^{-1}(wL^2\eta_i^2)^2 + \eta_i^2|J^{(i)}(U)| < 2L^3 w\eta_i^2.$$

Here $L\eta_i \leq 1$ and $w < 1$. This is the matrix-divergence calculation already used in (579). If $4L^3(s+2)C\varepsilon_{i-1} \leq c_{*,\mathrm{rg}}$, then $\ell\alpha' < c_{*,\mathrm{rg}}$. Applying the regular cube gauge of theorem F at scale i, and restricting its gauge to q, yields

$$U^v = e^{i\eta_i A}, \qquad |A|, |\nabla^{\eta_i} A| < 4L^3 B_*(s+2)C\varepsilon_{i-1}.$$

The finite differences here are required wherever their stencils lie in the inner bonds of q. For $i = 0$ the comb gauge lemma 14.4 instead gives

$$|A|, |\nabla A| < 2\sqrt{2}\pi s C\varepsilon_0 \quad \text{if} \quad \sqrt{2}sC\varepsilon_0 \leq 1.$$

The two gauge conclusions, with (1170) and (1171), imply

$$|A|, |\nabla^{\eta_i} A| < \mathfrak{o}LMB^{(\mathrm{cg})}C_{\mathrm{mb}}C\varepsilon_i \qquad (s \leq \mathfrak{o}LM), \tag{1182}$$

$$|A|, |\nabla^{\eta_i} A| < C_{\mathrm{mb}}C\varepsilon_i \qquad (s \leq 12\mathfrak{M}). \tag{1183}$$

For the second bound the cube is a union of the level-i $\mathfrak{M}$-cubes. Its enlargement embeds since the standing size condition gives $L^{m+K-i} \geq L^6\mathfrak{M} > 12\mathfrak{M} + 6$. For the first use (1172). The two entries of the minimum in (1171) give the respective smallness conditions. At level zero the bounds follow from $2\sqrt{2}\pi < B^{(\mathrm{cg})}C_{\mathrm{mb}}$ and $24\sqrt{2}\pi\mathfrak{M} < C_{\mathrm{mb}}$.

Consider now a separated subordinate sequence of depth $n' \leq n$, with reduced determining set $\mathbf{B}'$. By lemma 51.13, the restriction U' is critical over $V' = M_{\mathbf{B}'}U'$. If λ' is its local layer index, restriction and (1179) give

$$r_{U'}(\mathfrak{e}) < L^{-2(\lambda-\lambda')}C\varepsilon_\lambda \leq \kappa_{\mathrm{run}} C\varepsilon_{\lambda'} \leq e, \qquad e = \kappa_{\mathrm{run}} C\varepsilon_{n'}.$$

The current has the still smaller restriction factor $L^{-3(\lambda-\lambda')}$. By (1171),

$$e \leq \varepsilon_{\mathrm{v}}, \qquad L^2 e \leq c_2^{\mathrm{SO}}, \qquad 2L^2 e \leq \bar{\varepsilon}.$$

For a tested plaquette of order q, its four endpoint blocks lie in Ω'_{q-1}. Their bare plaquettes have curvature smaller than $(L^2 e)L^{-2q}$. The local four-corner induction in the proof of lemma 66.1, with terminal level q and $\alpha = L^2 e$, bounds its averaged holonomy by $2L^2 e$. Thus all tested holonomies

of V' satisfy this bound. Hence theorem E identifies the restricted configuration with the regular critical orbit of V'. The continuation at the real datum exists on its admitted positive radius; at zero it is a real gauge transform of U'.

The analytic averages used in the tower definition are also defined at U' and agree with the real averages. Indeed, for a determining row c of level l, put $D_q = B^{l-q}(c_-) \cup B^{l-q}(c_+)$, $0 \le q \le l$. Separation puts D_0 in the layer-$(l-1)$ region, and $B(D_{q+1}) = D_q$. Every bare plaquette with corners in D_0 has curvature smaller than $(L^2 e)L^{-2l}$. Apply the local induction in the proof of lemma 66.1 with terminal level l and $\alpha = L^2 e$: the one-step estimate uses only the four subordinate corner blocks, all contained in D_q, and its products are bounded by $\exp(1/[2(L^2-1)]) < 2$. Thus, at intermediate level $q < l$, the averaged curvature is bounded by

$$2(L^2 e)L^{2(q-l)} \le 2L^{-2}c_2^{\mathrm{SO}} < \omega_M.$$

The one-step analytic mean is therefore defined and equals the real mean on that double block. Induction on q proves the assertion up to row c, including the reversed row. The logarithmic perturbation of the real datum is zero, strictly inside the continuation radius.

Apply this argument to either tower of $Z \in \mathcal{D}_{p-1}$, $Z \subset X \subset \Omega_j$, truncated at $n' \le p \le j$. The tower is separated and subordinate to the collared sequence. Its continued configuration is a real gauge transform of $U|_Z$. Every plaquette star in $Z^{\natural p}$ lies inside Z, since the two surrounding coarse layers contain all its corners. Real gauge invariance and (1180), (1181) give

$$|\partial U^{\mathfrak{r}}_{n'} - 1| < \kappa_{\mathrm{run}} C\varepsilon_j L^{-2j}, \qquad |J^{\mathfrak{r}}_{n'}| < \kappa_{\mathrm{run}} C\varepsilon_j L^{-3(j-n')}.$$

These are the core bounds (756) with $c = b_0 L^{-(j-p)}$: use $b_0 \ge C_{\mathrm{mb}} C\varepsilon_j > \kappa_{\mathrm{run}} C\varepsilon_j$ and

$$-3(j-n') = -2(j-p) + 2(n'-p) - (j-n').$$

The simple-domain conditions follow from the first gauge bound, the same pointwise estimates, and $A' = 0$. This proves (1176).

The restriction formula (1177) follows from $J^{(i)} = L^{-3(j-i)}J^{(j)}$. The running monotonicity $\varepsilon_j \le \varepsilon_n$ and (1174) give (1178). Finally proposition 53.11 supplies criticality and the layer bounds with $C = C_{\mathrm{reg}}$ for every collared background. Gluing outside $\Omega_0^\diamond$ does not change any of the quantities just used. □

58.2. **The domains before, during and after a step.** The radii at the intermediate stage leave room for both the fluctuation and the terms inherited from earlier levels. We describe the domains, including their tower conditions. The two pointwise and gauge clauses are those used in section 57.4.

A structure is a stage $\sigma = (s, \phi)$, where $\phi =$ mid or out; at level zero only $(0, \mathrm{out})$ occurs. Order the stages by level and put $(s, \mathrm{mid}) \preccurlyeq (s, \mathrm{out})$. For $0 < \beta_{\mathrm{s}} \le 1/4$ and $0 \le \theta_{\mathrm{s}} < 1$, set

$$\begin{aligned} f_{s,l} &= 1 - \beta_{\mathrm{s}}(1 - 2^{-(s-l)}), \qquad c_{(s,\mathrm{out})}(l) = f_{s,l}, \\ c_{(s,\mathrm{mid})}(s) &= 1 + \beta_{\mathrm{s}}, \qquad c_{(s,\mathrm{mid})}(l) = f_{s,l} + \theta_{\mathrm{s}}\beta_{\mathrm{s}} 2^{-(s-l)} \quad (l < s). \end{aligned} \tag{1184}$$

In particular $1 - \beta_{\mathrm{s}} < c_\sigma(l) \le 1 + \beta_{\mathrm{s}}$. A radius table assigns positive numbers $r_\iota(\sigma; l)$, $\iota = 0, 1$, to all these stages and layers. The running table is $r_\iota(\sigma; l) = \alpha_{\iota,l}$. For $x \in \Omega_0^\diamond$ put

$$m_\sigma(x) = \max\{l \le s : x \in \Omega_l^\diamond\}, \qquad R_\sigma^{p,\iota}(x) = c_\sigma(l) r_\iota(\sigma; l) L^{-pl}, \quad l = m_\sigma(x).$$

At a common top layer use the ambient radius:

$$r_\iota(\sigma''|\sigma; l) = \begin{cases} r_\iota(\sigma; s), & l = s'' = s, \\ r_\iota(\sigma''; l), & \text{otherwise.} \end{cases} \tag{1185}$$

For a region Y, write

$$Y_{\mathcal{S}} = Y \cap \Omega_0^\diamond, \qquad \mathrm{Bd}_{\mathcal{S}}(Y) = \mathrm{Bd}(Y) \cap \mathrm{Bd}^\circ(\Omega_0^\diamond). \tag{1186}$$

Outside this bond set the domain imposes no local curvature or potential bound.

For $\sigma' \preccurlyeq \sigma$, a tower address in $Y \in \mathcal{D}_{s'-1}$ consists of $\sigma'' \preccurlyeq \sigma'$, $1 \le i \le s''$, and $Z \in \mathcal{D}_{i-1}$ with $Z \subset Y$ and $Z_{\mathcal{S}} \neq \emptyset$. At depth $q \le i$ use the sequence

$$\big(\Omega_l^{\diamond} \cap \Omega_l^{\mathfrak{r}}(Z)\big)_{0 \le l \le q}, \qquad \mathfrak{r} \in \{\mathfrak{r}_1, \mathfrak{r}_2\}. \tag{1187}$$

This sequence is separated and subordinate: the complement of an intersection is the union of the two complements, so its gap is at least the smaller of the two original gaps. The first member is $Z_{\mathcal{S}}$. Its tower configuration is the analytic continuation from the real averages, with the strict-radius convention in definition 49.1. A core address replaces Z by $X' \in \mathcal{D}_{i-1}$ with $X' \subset Y \cap \Lambda_i$.

Definition 58.2. Let $s' \ge 1$, $Y \in \mathcal{D}_{s'-1}$, and $P = (U, A', \mathbf{J})$ be a triple at level s' on Y, with $\mathbf{U} = e^{i\eta_{s'} A'} U$. The domain $\tilde{U}_{\sigma'|\sigma}(Y)$ is defined by the following four conditions, all with this same decomposition.

(L1) On $\mathrm{Bd}_{\mathcal{S}}(Y)$, where the indicated stencils are contained,

$$|\partial U - 1|,\ |\partial \mathbf{U} - 1| < R_\sigma^{2,0}, \qquad \eta_{s'}^3 |\mathbf{J}| < R_\sigma^{3,0},$$
$$\eta_{s'} |A'| < R_\sigma^{1,1}, \qquad \eta_{s'}^2 |\nabla_U^{\eta_{s'}} A'| < R_\sigma^{2,1}.$$

Plaquettes and differences use their base point; bond bounds use the initial endpoint.

(L2) For every $\sigma'' \preccurlyeq \sigma'$, $0 \le l \le s''$, and cube $q \subset Y \cap \Omega_l^{\diamond}$ of side at most $\mathfrak{o}LM$ in level-l units, U has a real gauge $U^v = e^{i\eta_l A}$ on its inner bonds with

$$|A|, |\nabla^{\eta_l} A| < \mathfrak{o} L M B^{(\mathrm{cg})} r_0(\sigma''; l).$$

For every $l \le s'$ and cube $q \subset Y_{\mathcal{S}} \cap \Omega_l^{\diamond}$ made of level-l $\mathfrak{M}$-cubes and of side at most $12\mathfrak{M}$, there is a containing cube $q \subset q' \subset Y_{\mathcal{S}}$ with a real gauge of index l at most $a_\flat$.

(L3) At every tower address (σ'', i, Z), both rules, and every $1 \le q \le p \le i$, the tower configurations $U_q^{\mathfrak{r}}$ and their currents exist. On $Z^{\langle p}$, write $\ell = \min\{m_{\sigma''}(x), i\}$, where x is the base point or initial endpoint. Require

$$|\partial U_q^{\mathfrak{r}} - 1| < L^{i-p} c_\sigma(\ell) r_0(\sigma''|\sigma; \ell) L^{-2\ell}, \tag{1188}$$
$$\eta_q^3 |J_q^{\mathfrak{r}}| < L^{i-p} c_\sigma(\ell) r_0(\sigma''|\sigma; \ell) L^{-2\ell - \min(q,\ell)}. \tag{1189}$$

The first inequality is imposed only when the four plaquette bonds lie in $\mathrm{Bd}^\circ(Z_{\mathcal{S}})$; the second only when the entire four-plaquette star does. Require the same existence and bounds with the real factor U in place of $\mathbf{U}$.

(L4) At every core address (σ'', i, X'),

$$\rho_{X'}^{s' \to i} P \in \mathcal{U}_i^{\circ}\big(X'; c_\sigma(i) r_0(\sigma''|\sigma; i), c_\sigma(i) r_1(\sigma''|\sigma; i)\big). \tag{1190}$$

Write $\tilde{U}_\sigma = \tilde{U}_{\sigma|\sigma}$. The superscript $1/2$ means that all entries of the table and $a_\flat$ are halved, with the coefficients c_σ unchanged. ∎

Thus the output and intermediate domains of step t are $\tilde{U}_{(t,\mathrm{out})}$ and $\tilde{U}_{(t,\mathrm{mid})}$, respectively. A plain triple on a label uses the regional current only where it is defined:

$$\mathbf{P}_{Y|\mathcal{S}}^{(j)}(\mathsf{U}) = (\mathsf{U}|_{\mathrm{Bd}(Y)}, 0, \mathbf{J}), \qquad \mathbf{J}(b) = \begin{cases} J^{(j)}(\mathsf{U})(b), & \text{the full star of } b \text{ is in } \mathrm{Bd}^\circ(\Omega_0^{\diamond}), \\ 0, & \text{otherwise.} \end{cases} \tag{1191}$$

In particular the current is zero at the incomplete stars along the collar. This convention commutes with restriction and the scaling of currents.

Proposition 58.3. Under the hypotheses of proposition 58.1, suppose the radius table up to depth n satisfies

$$C_{\mathrm{mb}} C \varepsilon_l \le \tfrac{1}{2}(1 - \beta_{\mathrm{s}}) r_0(\tau; l) \quad (0 \le l \le \mathrm{level}(\tau) \le n),$$
$$C_{\mathrm{mb}} C \varepsilon_n \le a_\flat / 2. \tag{1192}$$

Then, for all $\sigma' \preccurlyeq \sigma$ of levels at most n, $s' \geq 1$, and $Y \in \mathcal{D}_{s'-1}$,

$$\mathbf{P}^{(s')}_{Y|\mathcal{S}}(\mathsf{U}) \in \tilde{U}^{1/2}_{\sigma'|\sigma}(Y). \tag{1193}$$

This includes domains meeting the complement of Ω_0°, domains that wrap around the torus, and both output and intermediate stages. ■

Proof. For every applicable layer and every choice of the two structures,

$$\kappa_{\mathrm{run}} C\varepsilon_l < C_{\mathrm{mb}} C\varepsilon_l \leq \tfrac{1}{2}(1-\beta_{\mathrm{s}}) r_0(\tau;l) < \tfrac{1}{2} c_\sigma(l) r_0(\tau;l). \tag{1194}$$

The same inequality holds with the radius selected by (1185), since it is an entry of the same table. At a plaquette base point of depth $l = m_\sigma(x)$, (1180) and (1194) give the half-radius curvature bound. For a full star, (1181) gives

$$\eta^3_{s'} |J^{(s')}(U)(b)| < \kappa_{\mathrm{run}} C\varepsilon_l L^{-3l} < \tfrac{1}{2} R^{3,0}_\sigma(b_-).$$

For an incomplete star the current is zero. The two potential bounds hold since $A' = 0$. This proves (L1).

The gauges (1182) prove (L2) with half the table. For its containing-cube clause use (1183) on q itself: $C_{\mathrm{mb}} C\varepsilon_l \leq a_\flat/2$. At a tower address the meet (1187) is separated and subordinate. The subordinate-sequence argument in the proof of proposition 58.1 therefore shows that its tower exists and is a real gauge transform of $U|_{Z_\mathcal{S}}$. For $\ell = \min\{m_{\sigma''}(x), i\}$, the pointwise estimates give

$$|\partial U^{\mathfrak{r}}_q - 1| < \kappa_{\mathrm{run}} C\varepsilon_\ell L^{-2\ell}, \qquad \eta^3_q |J^{\mathfrak{r}}_q| < \kappa_{\mathrm{run}} C\varepsilon_\ell L^{-3\ell}.$$

Use (1194), $L^{i-p} \geq 1$, and $L^{-3\ell} \leq L^{-2\ell-\min(q,\ell)}$ to obtain (L3) at half the radii.

For a core address $X' \subset \Lambda_i \subset \Omega_i$, all current stars are complete, and

$$\rho^{s'\to i}_{X'} \mathbf{P}^{(s')}_{Y|\mathcal{S}}(\mathsf{U}) = \mathbf{P}^{(i)}_{X'}(\mathsf{U}).$$

Apply proposition 58.1 and (1194) at layer i; the radius of kind 1 is positive and $A' = 0$. This proves (L4). Every condition uses the same real factor. Empty index sets impose no condition, which also covers $Y_\mathcal{S} = \emptyset$. □

At step $t = k+1$ define the fluctuation box $\mathsf{Bx}_t(Y; C_{\mathrm{xt}}; \vartheta)$, for $0 < \vartheta \leq 1$, by

$$\begin{aligned} g_k^{-1}|\mathcal{A}(b)| &< 3\vartheta \mathfrak{b}_k \qquad (b \in \mathfrak{X}_{\mathrm{I}} \cap \mathfrak{X}^\circ\langle Y\rangle), \\ g_k^{-1}|\mathcal{A}(b)| &< 2\vartheta C_{\mathrm{xt}} \mathfrak{b}_k \qquad (b \in \mathfrak{X}_{\mathrm{E}} \cap \mathfrak{X}^\circ\langle Y\rangle). \end{aligned} \tag{1195}$$

Here $\mathfrak{X}^\circ\langle Y\rangle$ consists of the nonpivot fluctuation bonds with both endpoints in Y. The form is specified at positive bonds and extended by antisymmetry.

Corollary 58.4. Let $h' = (h, \mathsf{o}_t)$, $t = k+1$, and impose the support and smallness hypotheses of section 53.9, together with (1172), (1173) and (1192) at depth t, with $C = C_{\mathrm{reg}}$. Let V be a selected bare field in $\mathsf{Sup}(h') \cap \operatorname{dom} \mathsf{v}_h$, and let U_{ad} be any adapted element of its completed collared background. Set $\mathsf{U} = U_{\mathrm{ad}} \vee V$. Every plain triple of U belongs to the half-radius spaces (1193), including the output and intermediate spaces and the spaces of earlier levels. Under (1174), its triples on $X \subset \Omega_j$ also belong to $\mathcal{U}^\circ_j(X; \alpha_0/2, \alpha_1/2)$.

For every $Y \in \mathcal{D}_k$, the real point

$$\mathfrak{u}_Y = \Big(\mathbf{P}^{(t)}_{Y|\mathcal{S}^+}(\mathsf{U}),\ \mathcal{A}_k|_{\mathfrak{X}^\circ\langle Y\rangle},\ \mathfrak{q}[V]|_Y\Big) \in \tilde{U}^{1/2}_{(t,\mathrm{mid})}(Y) \times \mathsf{Bx}_t(Y; C_{\mathrm{xt}}; 1/2) \times \mathfrak{Q}^{\mathrm{rad}}_h(Y) \tag{1196}$$

is defined. If $C_\mathcal{A} \geq 2C_{\mathrm{xt}}$, then $g_k^{-1}\mathcal{A}_k|_Y$ lies in half the field polydisc $\mathbb{D}^{\mathrm{f}}_{x_k}(Y)$. ■□

Proof. On the selected support, the passed density tests give a completed datum with tested holonomies smaller than $2\varepsilon_j$; this is the regularity argument in section 53.7. Thus proposition 53.11 gives the background and its layer bounds, and propositions 58.1 and 58.3 apply to its gluing. Cutting a history after level j does not change any region, address or radius used at stages of levels

at most j, so it does not change these earlier spaces. The same assertion holds with half the table and half $a_\flat$.

The fluctuation field exists by proposition 53.12 and satisfies

$$g_k^{-1}|\mathcal{A}_k| < c_{\mathrm{fl}}\mathfrak{b}_k \le C_{\mathrm{xt}}\mathfrak{b}_k.$$

Passage of the fluctuation tests improves this to $g_k^{-1}|\mathcal{A}_k| < \mathfrak{b}_k$ on the integration bonds. These two estimates imply (1195) with $\vartheta = 1/2$. They also give $g_k^{-1}|\mathcal{A}_k| < \frac{1}{2}C_{\mathcal{A}}\mathfrak{b}_k$ under the stated polydisc inequality. The old stored variables are defined by $V \in \operatorname{dom}\mathsf{v}_h$. The support argument in section 53.9, applied to each earlier step, puts them in their external tiers. Restriction only deletes coordinates and preserves these inequalities. When $k = 0$ this family is empty. This proves (1196). □

58.3. **The regional shift.** The fluctuation changes only the level-k rows touching Ω_{k+1}. We apply the decoupled response to the shorter sequence $\Omega_0, \dots, \Omega_k$, whose determining set is denoted by $\mathbf{B}$. Throughout this subsection $k \ge 1$, lengths are in level-k units, and $\xi_n = L^{n-k}$. Put

$$W_Y = \Omega_0 \cap Y, \qquad \mathfrak{P}^\natural(Y \mid \Omega_\bullet; a) = \mathfrak{P}_k^\natural((\Omega_n)_{n\le k}; a, a, a) \restriction W_Y, \tag{1197}$$

where the intersection is cube-wise. The response uses the same assignment of walk data at every lattice and sequence of a given height.

In addition to the standing size and room conditions, require

$$R_1 \ge L, \quad \mathfrak{M} \ge L^4, \quad R_1\mathfrak{M} \ge 512\log L/\mathring{\delta}_0, \tag{1198}$$

$$\mathfrak{M} \ge M^{\mathrm{pr}}, \quad R_1\mathfrak{M} \ge \mathfrak{r}^{\mathrm{pr}}, \quad \mathfrak{M} \ge (8C_{\mathrm{w}}\mathsf{n}_{1,\natural})^2, \tag{1199}$$

$$M \ge 88\mathsf{r}_{\mathrm{w}}\mathfrak{M}, \qquad \mathring{\delta}_0 M \ge 2176\bar{\kappa}. \tag{1200}$$

These are exactly the frame and walk-expansion requirements of theorem M. Indeed, truncating a separated sequence preserves separation. The partition has side M, and its period is at least $2L^2M$; hence it has at least three cubes in each direction and its period is at least $2L\mathfrak{M}$.

We fix the constants once, before the heights and couplings. With the response constants of theorem M and the pivot constants of (799), set

$$\begin{aligned} C_{\mathrm{b}} &= \max\{24, 5b_{\mathrm{D}}, 10C_j b_{\mathrm{D}}^3\}, \qquad \rho_* = 5b_{\mathrm{D}}\varepsilon_{\mathrm{D}}, \\ C_{\mathrm{e}} &= \max\{24, L^3, 10L^2(1+\rho_*)b_{\mathrm{D}}, 10L^3C_j b_{\mathrm{D}}^3\}, \\ C_{\min} &= \max\{32, 3/a_{\mathrm{D}}, 1/a_P^\circ, 1/r_P^\circ, 2C_P/\varepsilon_{\mathrm{D}}, 2C_P e^{L/4}C_{\mathrm{b}}, (3+2C_P)C_{\mathrm{e}}\}. \end{aligned} \tag{1201}$$

Choose $C \ge C_{\min}$, enlarging it also to preserve the earlier single-lattice restrictions in (831), and put

$$\begin{aligned} \mu_{\mathfrak{h}} &= \mathring{\delta}_0/8, \qquad \mathsf{c}_{\mathfrak{h}} = Ce^{C\bar{\kappa}}, \qquad a_{\mathrm{m}} = C^{-1}\min\{1, \varepsilon_{\mathrm{v}}\}, \\ \varepsilon_{\mathfrak{h}} &= a_{\mathrm{m}}e^{-C\bar{\kappa}}, \qquad a_{\mathfrak{h}} = a_{\mathrm{m}}/\max\{1, \mathsf{o}LMB^{(\mathrm{cg})}\}. \end{aligned} \tag{1202}$$

Thus the previously constructed single-lattice maps are restricted to smaller domains; their defining formulas are unchanged. The inequalities we will use are

$$\begin{aligned} 3a_{\mathrm{m}} \le a_{\mathrm{D}}, \qquad a_{\mathfrak{h}} \le a_P^\circ, \qquad \varepsilon_{\mathfrak{h}} \le r_P^\circ, \qquad 2C_P\varepsilon_{\mathfrak{h}} \le \varepsilon_{\mathrm{D}}e^{-32\bar{\kappa}}, \\ 2C_P e^{L/4}C_{\mathrm{b}}e^{C_{\mathrm{b}}\bar{\kappa}} \le \mathsf{c}_{\mathfrak{h}}, \qquad C_{\mathrm{e}}(3a_{\mathrm{m}} + 2C_P e^{C_{\mathrm{e}}\bar{\kappa}}\varepsilon_{\mathfrak{h}}) \le \varepsilon_{\mathrm{v}}. \end{aligned} \tag{1203}$$

They follow term by term from (1201). In the last inequality use $C \ge (3+2C_P)C_{\mathrm{e}}$ and $e^{(C_{\mathrm{e}}-C)\bar{\kappa}} \le 1$.

Let $\mathfrak{X}_0$ be the level-k bonds inside Ω_0 with an endpoint in $B(P'')$, and let $\mathfrak{X}$ be their nonpivot bonds. Every such bond has both endpoints in Ω_k, by separation. They are interior rows of Γ_k; in particular their row weight is 1. Moreover

$$\mathfrak{X}_0 = \mathfrak{X} \sqcup \{b_0(c), \overline{b_0(c)} : c \in \Gamma\}.$$

The two endpoints of $b_0(c)$ lie in Y precisely when $\square(c) \subset Y$. All bonds needed for its pivot formula then lie in Y. No bond of $\mathfrak{X}_0$ is an outward row of $\mathbf{B}$ or joins two sites in an outer endpoint block.

A source cube meets Ω_{k+1}. Define $\mathrm{dom}_{\mathfrak{h}}(Y \mid \Omega_\bullet)$ to consist of pairs $(P, \mathcal{A})$, with $P = (U, A', \mathbf{J})$ a level-$(k+1)$ triple on Y, which satisfy the following conditions:

(1) $\sup_{\mathfrak{X}^{\circ+}\langle Y\rangle} |\mathcal{A}| < \varepsilon_{\mathfrak{h}}$.

(2) The pair and its specified decomposition

$$\mathbf{p}^{[k]}(P) = (e^{i\eta_{k+1}A'}U, L^{-3}\mathbf{J})\big|_{\mathrm{Bd}^\circ(W_Y)}, \qquad (U, L^{-1}A'), \tag{1204}$$

serve in the natural local class at radius a_{m}.

(3) For every $Y_1 \subset Y$ all of whose cubes are source cubes, $P|_{Y_1} \in \mathcal{Q}_{k+1}(Y_1; a_{\mathfrak{h}}, a_{\mathfrak{h}}, a_{\mathfrak{h}})$.

Complete $\mathcal{A}$ at the pivots by the formula (832), and extend the resulting field by zero to all other rows of $\mathbf{B}$. Denote it by B'_Y. The regional shift is

$$\mathfrak{h}_{Y|\Omega_\bullet}(s; P, \mathcal{A}) = (\mathbf{A}, \mathbf{K}) = (\mathbf{H}, j)(s; \mathbf{p}^{[k]}(P); B'_Y)\big|_{\mathrm{Bd}(Y)}, \tag{1205}$$

where both forms are extended by zero off the inner bonds of Ω_0.

Proposition 58.5. The map (1205) is defined on $\mathbb{D}(Y) \times \mathrm{dom}_{\mathfrak{h}}(Y \mid \Omega_\bullet)$. It depends on P only through its pair, is holomorphic on each open fibre with U fixed, and is continuous on its Borel domain. The domain is hereditary under restriction and invariant under unitary gauges. The shift has the corresponding restriction, gauge and conjugation covariance.

Let x be a bare site of Ω_0, and put $n = \max\{j \le k : x \in \Omega_j\}$. At a positive bond based at x, or a difference stencil based there and contained in $\mathrm{Bd}(Y) \cap \mathrm{Bd}^\circ(\Omega_0)$, one has

$$\xi_n|\mathbf{A}|, \quad \xi_n^2|\nabla_{\mathbf{U}}^{\eta_k}\mathbf{A}|, \quad \xi_n^3|\mathbf{K}| \;\le\; \mathsf{c}_{\mathfrak{h}} \sup_{b \in \mathfrak{X}^{\circ+}\langle Y\rangle} e^{-\mu_{\mathfrak{h}} d_{\mathbf{B}}(x, b_-)}|\mathcal{A}(b)|. \tag{1206}$$

For a union of wall components of the parameter support, the corresponding component response depends only on the pair and source within that union and is supported on its inner bonds. It vanishes if the union contains no source cube. In particular, the shift vanishes at $s = 0$ or $\mathcal{A} = 0$.

If two sequences have the same traces in Y, including the last trace, their shift domains and maps on Y agree. The same statement holds after a fine translation or a chart which identifies these traces. If

$$12\mathfrak{M} \le \mathsf{o} L^2 M, \tag{1207}$$

then on a constant sequence the domain is exactly (810) with the constants (1202), and the map is the retained single-lattice shift. ∎

Proof. For $c \in \Gamma$, both endpoint fine cubes of $\square(c)$ are source cubes. To see this when only one endpoint is in Ω_{k+1}, the midpoint of c lies in the level-$(k+1)$ cube of that endpoint, hence in Ω_{k+1}, and also in the fine cube of the other endpoint. The last domain condition therefore provides the pivot data on $\square(c)$. The pivot estimate gives

$$|\beta_c| \le 2C_P \max_{b \in \mathfrak{X}^+ \cap \mathsf{P}_c} |\mathcal{A}(b)|, \qquad \sup|B'_Y| \le 2C_P \sup|\mathcal{A}|. \tag{1208}$$

The set in the maximum is nonempty: a bond inside the endpoint block belonging to P'' is a nonpivot bond of P_c. The first reserve in (1203) now admits the completed source into theorem M; the second domain condition admits its pair. This proves existence.

The pivot depends only on the pair in its double-block tower and on the nonpivot values in P_c. This tower is inside $W_{\square(c)}$, by the collar separation. Consequently it restricts consistently. The two contractions in theorem M are also independent of the decomposition of the pair. The hereditary assertion for the natural class includes its gauge witnesses: on restriction to W_{Y_1}, replace a containing witness cube for a natural cube $\mathsf{q} \subset W_{Y_1}$ by q itself. The pointwise estimates restrict, and every source subdomain of Y_1 was already tested in Y. This proves heredity.

At a fixed U, the conditions on its gauge witnesses are fixed and the other domain conditions are finitely many strict continuous inequalities. Its fibre is open, and the holomorphic pivot formula followed by theorem M proves holomorphy. To check the Borel assertion as U varies, fix a witness cube and replace a strict logarithm bound r by $r(1-1/q)$, $q\in\mathbb{N}$. The allowable real forms and unitary gauges then form a compact set. Its continuous image under $(a,u)\mapsto (e^{i\eta a})^{u^{-1}}$ is compact. There are finitely many cubes; finite intersections and unions, followed by this countable exhaustion for strict bounds, prove that the witness conditions are Borel.

For continuity at $P_0=(U_0,A_0',\mathbf{J}_0)$, freeze the real factor U_0. On the finitely many bonds used by the response and the pivot towers put, near P_0,

$$\Lambda(P)=(i\eta_{k+1})^{-1}\log(\mathbf{U}U_0^{-1}).$$

Here $\eta_{k+1}|A_0'|<1/32<\log 2$, so the logarithm is defined near the point and equals A_0' there. The fixed-factor fibres are open. Thus the continuous map $P\mapsto(U_0,\Lambda(P),\mathbf{J})$ stays in them near P_0; it preserves the pair and hence the pivot and response. Their fixed-factor holomorphy proves the claimed continuity.

For the quantitative estimate, let $y(x)$ be the top of the pillar containing x. Then $d_{\mathbf{B}}(x,y(x))\le 3/2$. The weighted bounds in theorem M, first at $y(x)$ and then at x, are bounded by $5b_{\mathrm{D}}e^{8\bar\kappa}$ for the field and its first difference, and by $10C_jb_{\mathrm{D}}^3e^{24\bar\kappa}$ for the current, times

$$S_x(B_Y')=\sup_c e^{-\mu_{\mathfrak{h}}d_{\mathbf{B}}(x,c_-)}|B_Y'(c)|.$$

We used $4e^{3/16}<5$ and $8e^{3/16}<10$. For a pivot row $c'=b_0(c)$, select a bond b attaining the maximum in (1208). The two initial points are in Ω_k, at weighted distance at most $2L-1$. Hence

$$S_x(B_Y')\le 2C_Pe^{L/4}\sup_{b\in\mathfrak{X}^{\circ+}\langle Y\rangle}e^{-\mu_{\mathfrak{h}}d_{\mathbf{B}}(x,b_-)}|\mathcal{A}(b)|.$$

The bound for nonpivot rows is immediate. The reserve (1203) proves (1206). At an incident differentiation stencil the same estimate is valid: if its base lies in the third component margin, the entire stencil lies in the pair domain. Otherwise both field values vanish by the response support property. In this second case the connecting transport can be arbitrary.

The component support and restriction proof of proposition 51.5 applies to this frame with its layer weights retained. The local $8L\xi_y$ cube lies in the cell margin because $4L\xi_y\le 4L^3\xi_m\le 2\mathfrak{M}\xi_m$. Thus the component's pivot data and pair lie inside that component. If it has no source cube, it has no row of $\mathfrak{X}_0$, and the component datum is zero. The two response iterations then vanish there. This proves the support and zero assertions. The pivot formula and both response iterations commute with unitary gauges and complex conjugation.

For two sequences with matching traces in Y, the layer regions, natural cubes contained in Y, source cubes, pivot indices and nonpivot bonds inside Y coincide. Thus the three domain conditions coincide. Their local pair and completed data also coincide. The local comparison in theorem M, applied to each wall component, identifies the responses. The same comparison applies under a chart: its local translations carry the fine cubes, levels, walks and double-block formulas to their counterparts. All lengths, inequalities and gauge witnesses are preserved, so this is equality of the transported domains as well as the maps. The halo contains the neighborhoods used in these comparisons: the natural-cube enlargement has width $5\mathfrak{M}\xi_n<LM$, and the layer cube meeting Y has side $\mathfrak{M}\xi_n\le M$.

For a constant sequence all fine cubes are source cubes. The last domain condition at $Y_1=Y$ is therefore necessary. Conversely, the simple domain on Y implies all its restrictions. It also implies the natural-class condition: curvature acquires a factor L^{-2}, the potential and its derivative acquire factors L^{-1} and L^{-2}, and the current acquires L^{-3}. The natural cubes have side at most $12\mathfrak{M}/L\le \mathsf{o}LM$ in level-$(k+1)$ units, so their gauges are supplied by the simple domain with bound

$\mathsf{o}LMB^{(\mathrm{cg})}a_{\mathfrak{h}} \le a_{\mathrm{m}}$. This proves exact domain equality. The completed sources and the response formulas are then those of (834). □

58.4. **Gauges on source-cube domains.** The local gauges in (L2) are given on cubes at their own scale. A source domain at step $t = k+1$ can contain larger cubes which cross the boundary of Ω_t. We first glue the level-k gauges on such a cube. Alignment with one tree gauge makes the transition functions small, and three coordinate interpolations avoid a loss proportional to the number of small boxes. We call the first and second gauge requirements in (L2) of definition 58.2 its parts (a) and (b), respectively.

Put

$$\varsigma_{\mathsf{o}} = \mathsf{o}LM, \qquad \varkappa[a] = \varsigma_{\mathsf{o}} B^{(\mathrm{cg})} a, \qquad q_* = \frac{13L^2}{24B^{(\mathrm{cg})}} + \frac{7}{2}L.$$

Choose the constant in the cube-gauge ceiling to be

$$\mathsf{K}_{\mathrm{cg}} = \mathsf{K}_{\mathrm{I}} = \max\left\{16L^2,\ 180q_* \max\left(1, \frac{13L}{\varsigma_{\mathsf{o}}}\right)\right\}. \tag{1209}$$

It depends only on the already fixed geometric constants, not on the step, volume or couplings. Thus it is chosen after $L, \mathsf{o}, M, B^{(\mathrm{cg})}$, before the radius ceilings and the choice of γ. For every $0 < \varkappa \le 1$, this choice satisfies

$$\mathsf{K}_{\mathrm{I}} \ge 7q_*, \qquad \frac{5.4L^2}{\mathsf{K}_{\mathrm{I}}} + 9679\varkappa\frac{q_*^2}{\mathsf{K}_{\mathrm{I}}^2} + 660.5\frac{Lq_*}{\varsigma_{\mathsf{o}}\mathsf{K}_{\mathrm{I}}} \le 0.99, \tag{1210}$$

and

$$\varsigma_{\mathsf{o}} < 24/L \quad\Longrightarrow\quad \mathsf{K}_{\mathrm{I}} \ge \frac{16\pi L^3}{\varsigma_{\mathsf{o}} B^{(\mathrm{cg})}}. \tag{1211}$$

Proposition 58.6. Let $L \ge 13$, $2 \le t \le \bar{k}$, $k = t-1$, and $\sigma = (t, \mathrm{mid})$ or (t, out). Use the collared sequence and separated label of definition 58.2, with $m' < m + K - \bar{k}$, $(\ell, K) \in \mathfrak{W}_k$, and the two gaps

$$\mathrm{dist}_\infty(\Lambda_j, \Omega_j^c) \ge 2M\check{R}_j, \qquad \mathrm{dist}_\infty(Z_j, \Omega_{j+1}) \ge 8LM\check{R}_{j+1}$$

in level-j units. Let $Y \in \mathcal{D}_k$ and $P = (U, A', \mathbf{J}) \in \tilde{U}_\sigma(Y)$. Suppose

$$\begin{aligned} (1+\beta_{\mathrm{s}})r_\iota(\sigma;n) \le a_{\mathrm{m}} \quad (0 \le n \le t,\ \iota = 0,1), \qquad a_\flat \le a_{\mathrm{m}}, \\ \max\{\mathsf{K}_{\mathrm{cg}}, L^3\}(1+\beta_{\mathrm{s}})r_0(\sigma;n) \le a_{\mathfrak{h}}, \qquad L^3(1+\beta_{\mathrm{s}})r_1(\sigma;n) \le a_{\mathfrak{h}} \quad (n = k, t). \end{aligned} \tag{1212}$$

Here $a_{\mathrm{m}}, a_{\mathfrak{h}}$ are the radii in eq. (1202); in particular $\varkappa[a_{\mathfrak{h}}] \le 1$. Then the pair (1204), with its specified decomposition, belongs to the natural local class of radius a_{m} on $Y_{\mathcal{S}}$. Moreover,

$$P|_{Y_1} \in \mathcal{Q}_t(Y_1; a_{\mathfrak{h}}, a_{\mathfrak{h}}, a_{\mathfrak{h}}) \tag{1213}$$

for every $Y_1 \in \mathcal{D}_k$, $Y_1 \subset Y$, all of whose fine cubes meet Ω_t. Consequently $(P, \mathcal{A})$ lies in $\mathrm{dom}_{\mathfrak{h}}(Y \mid \Omega_\bullet)$ whenever $\sup|\mathcal{A}| < \varepsilon_{\mathfrak{h}}$. At $\sigma = (t, \mathrm{mid})$, the inequalities (1212) are precisely the selected cube and natural-class ceilings of $(\mathrm{C6})_t$. ■

Here is the gluing estimate used for cubes crossing the last interface. All its differences are ordinary differences on integer boxes.

Lemma 58.7. Let U be an SU(2) configuration on the inner bonds of $\square_N = \{0, \dots, N-1\}^3$, and put $\varpi = \max_p |\partial U(p) - 1|$. Let $n_{\mathrm{p}} \ge 24$ be an integer. Suppose every integer box in $\square_N$ with side lengths at most n_{p} has a gauge W such that

$$\max_b |W(b) - 1| \le \gamma_1, \qquad \max_{x,\mu,\nu} |W_\nu(x + e_\mu) - W_\nu(x)| \le \gamma_2.$$

The second maximum is over the stencils with $x, x+e_\mu, x+e_\nu, x+e_\mu+e_\nu$ in the box, including $\mu=\nu$; empty maxima are zero. If

$$z_* = \frac{13}{24}N\varpi + \frac{7}{2}\gamma_1, \qquad n_{\mathrm{p}} z_* \le \frac{1}{14},$$

then U has on $\square_N$ a gauge of sizes

$$\left(6.55 z_*,\ 5.4\gamma_2 + 433 z_*^2 + \frac{634}{n_{\mathrm{p}}} z_*\right). \tag{1214}$$

■

Proof. Write $\gamma_1^\mu(W)$ for the first size restricted to direction μ. We give the interpolation calculation before constructing the gauges.

For hermitian $\theta, \theta^\flat$ of norm at most $\vartheta < 1$, and $0 \le s, s' \le 1$, set

$$D = \theta - \theta^\flat, \quad Z_s = e^{is\theta} e^{-is\theta^\flat}, \quad E_s = Z_s - 1 - s(Z_1 - 1).$$

The identity

$$Z_s - Z_{s'} = i \int_{s'}^{s} e^{i\tau\theta} D e^{-i\tau\theta^\flat}\, d\tau$$

and unitarity give

$$|Z_s - Z_{s'}| \le |s - s'||D|, \quad |D| \le \frac{|Z_1 - 1|}{1-\vartheta}, \quad |E_s| \le \frac{3}{4}\vartheta|D|, \quad |E_s - E_{s'}| \le 3\vartheta|D|\,|s-s'|.$$

For two such pairs, indexed by $1, 2$, put

$$N_{12} = 2\vartheta|D_1 - D_2| + (2|\theta_1 - \theta_2| + |D_1 - D_2|)|D_2|.$$

Subtracting the integrands, and subtracting their values at $\tau = 0$, gives respectively

$$\begin{aligned} |Z_s^{(1)} - Z_s^{(2)}| &\le s|D_1 - D_2| + \frac{s^2}{2}(2|\theta_1 - \theta_2| + |D_1 - D_2|)|D_2|, \\ (1 - \vartheta - \tfrac{1}{2}|D_2|)|D_1 - D_2| &\le |Z_1^{(1)} - Z_1^{(2)}| + |\theta_1 - \theta_2||D_2|, \\ |E_s^{(1)} - E_s^{(2)}| &\le \tfrac{3}{8} N_{12}. \end{aligned}$$

Indeed the remainder in the integrand is bounded by $2\tau\vartheta|D|$, and its difference by τN_{12}. In E_s the constant part cancels; the remaining integrals have total coefficient $\frac{1}{2}s(1-s)(1+2s) \le 3/8$. These observations also give the four preceding bounds.

Now let $W^1 = (W^0)^h$, where $|h - 1| \le \rho \le 1/10$, and put $\theta = \log(h)/i$, $\vartheta = \pi\rho/3$. The logarithm is the real principal branch, with $|\theta| \le \vartheta$. For a cutoff $\chi \in [0,1]$, let $W^\chi = (W^0)^{e^{i\chi\theta}}$. At a bond in direction μ, set

$$\theta^\flat = W_\mu^0(x)\theta(x+e_\mu)W_\mu^0(x)^{-1}, \quad s = \chi(x), \quad \delta = \chi(x+e_\mu) - \chi(x), \quad \Xi = e^{-i\delta\theta^\flat}.$$

Then

$$\begin{aligned} W_\mu^\chi &= (1-s)W_\mu^0 + sW_\mu^1 + E_s W_\mu^0 + Z_s(\Xi - 1)W_\mu^0, \\ Z_1 W_\mu^0 &= W_\mu^1. \end{aligned} \tag{1215}$$

Suppose both W^j have directional sizes g_1^μ and second size g_2, and write $g_1 = \max_\mu g_1^\mu$. If k_1, k_2 bound the first and second differences of χ, put

$$d = \frac{2g_1}{1-\vartheta}, \quad t_1 = 2g_1\vartheta + d, \quad d_2 = \frac{2g_2 + t_1 d}{1 - \vartheta - d/2}, \quad e = 2\vartheta d_2 + (2t_1 + d_2)d.$$

For a contained difference stencil the preceding estimates give

$$|D| \le d, \quad |\Delta_\nu\theta| \le t_1, \quad |\Delta_\nu D| \le d_2, \quad |\Delta_\nu\theta^\flat| \le t_1 + d_2.$$

Here the first bound uses $Z_1 = W_\mu^1 (W_\mu^0)^{-1}$; the second uses unitary conjugation and $|VTV^{-1} - T| \le 2|V-1||T|$; the third uses the two-pair estimate and $|\Delta_\nu Z_1| \le 2g_2$. Thus the three terms in (1215), grouped as its affine part, $E_s W_\mu^0$, and $Z_s(\Xi - 1)W_\mu^0$, have difference bounds

$$\begin{aligned}
&g_2 + 2k_1 g_1,\\
&3k_1\vartheta d + \tfrac{3}{8}e + \tfrac{3}{4}\vartheta d\, g_2,\\
&k_1\vartheta\{k_1 d + d_2 + \tfrac{1}{2}(2t_1 + d_2)d\} + k_2\vartheta + k_1(t_1 + d_2) + k_1\vartheta g_2.
\end{aligned}$$

The last line follows by differencing its three factors; $|\Delta_\nu \Xi| \le k_2\vartheta + k_1(t_1 + d_2)$. Its undifferenced counterpart is

$$|W_\mu^\chi - 1| \le \left(1 + \frac{3\vartheta}{2(1-\vartheta)}\right) g_1^\mu + |\Delta_\mu \chi|\vartheta.$$

To glue a row, put $\omega = \lfloor n_{\mathrm{p}}/2 \rfloor$ and $s_{\mathrm{p}} = n_{\mathrm{p}} - \omega$. Cover $\{0, \dots, N-1\}$ by the intervals

$$I_j = \{j s_{\mathrm{p}}, \dots, \min(j s_{\mathrm{p}} + n_{\mathrm{p}}, N) - 1\},$$

stopping at the first interval reaching $N-1$. Successive overlaps O_j contain ω integers and are disjoint; non-successive intervals are disjoint. If $o_j = (j+1)s_{\mathrm{p}}$, use on O_j

$$\chi_j(c) = \mathrm{ct}\left(\frac{c - o_j - 1}{\omega - 3}\right), \qquad \mathrm{ct}(s) = \begin{cases} 1, & s \le 0,\\ 1 - 3s^2 + 2s^3, & 0 \le s \le 1,\\ 0, & s \ge 1. \end{cases}$$

For gauges v_j on these boxes, interpolate by $e^{i\chi_j \log(v_j v_{j+1}^{-1})/i} v_{j+1}$. The cutoff is constant on the first and last two layers of each overlap. The constant-gauge and interpolation regions therefore overlap in two layers: every bond and every difference stencil, whose coordinate range is at most two, is contained in one such region. No boundary stencil is omitted.

The derivative bounds of this cubic cutoff give

$$k_1 \le \frac{25}{6n_{\mathrm{p}}}, \qquad k_2 \le \frac{46.3}{n_{\mathrm{p}}^2}.$$

For $g_1 \le 0.042$ and $\rho \le 1/10$, we have $\vartheta \le 0.10472$, $d \le 2.234 g_1$, $t_1 \le 2.4435 g_1$, and $d_2 \le 2.3575 g_2 + 6.4348 g_1^2$; the denominator is at least 0.84836. Substitution in the displayed three difference bounds gives, for gluing in direction 1,

$$\begin{aligned}
&\gamma_1^1 \le 1.18 g_1^1 + 4.364\rho/n_{\mathrm{p}}, && \gamma_1^\mu \le 1.18 g_1^\mu \quad (\mu = 2,3),\\
&\gamma_2 \le 1.75 g_2 + 6.2 g_1^2 + 21.7 g_1/n_{\mathrm{p}} + 48.49\rho/n_{\mathrm{p}}^2.
\end{aligned} \tag{1216}$$

The other directions obey the same bounds.

It remains to align the gauges so that this interpolation can be performed three times. The tree paths from $(\lfloor N/2 \rfloor, \lfloor N/2 \rfloor, \lfloor N/2 \rfloor)$, taken in coordinate order, give a gauge W^{c} with $\gamma_1(W^{\mathrm{c}}) \le N\varpi$. Indeed its third component is 1, its second component is 1 on the central third-coordinate plane, and its first component is 1 on the central coordinate line. The 23, 13 and then 12 plaquettes telescope to bounds $|x_3 - c_0|\varpi$ and $(|x_2 - c_0| + |x_3 - c_0|)\varpi$.

Take the product boxes $Q_i = I_{i_1} \times I_{i_2} \times I_{i_3}$. Multiply each given gauge u_i by a constant so that the new gauge v_i agrees with the tree gauge at the central site x_i of Q_i. The relative gauge $m_i = v_i (u^{\mathrm{c}})^{-1}$ has increments bounded by $\gamma_1 + N\varpi$. For neighboring boxes choose z at the first overlap coordinate and the two shared central coordinates. Its distances from the two centers in the varying direction are at most $1, \omega$. The transition $h = v_i v_{i'}^{-1}$ has increments at most $2\gamma_1$. Every overlap point is at distance at most $\omega - 1 + n_{\mathrm{p}}$ from z, whence

$$|h - 1| \le (\omega + 1)N\varpi + (3\omega - 1 + 2n_{\mathrm{p}})\gamma_1 \le n_{\mathrm{p}} z_* =: \rho_1.$$

Glue in directions $1,2,3$, in that order. The transitions between neighboring glued rows remain small. To see this, write $h_1 = e^{i\theta} h_0 e^{-i\theta'}$. With $\theta^\flat = h_0 \theta' h_0^{-1}$, the earlier identity for E_s gives

$$|e^{is\theta} h_0 e^{-is\theta'} - 1| \le \max(|h_0 - 1|, |h_1 - 1|) + \frac{3\vartheta}{4(1-\vartheta)} |h_1 - h_0|.$$

The cutoffs in neighboring rows are identical. Thus their new transition bounds are $\rho_2 = 1.1755\rho_1$ and $\rho_3 = 1.1755\rho_2 < 1/10$, since $\rho_1 \le 1/14$. This proves at each stage that the next logarithms exist.

Since $\gamma_1 \le (2/7) z_*$ and $z_* \le 1/336$, the directional first sizes after the successive stages, in units of z_*, are bounded by

$$(4.702, 0.338, 0.338), \quad (5.549, 5.529, 0.399), \quad (6.548, 6.525, 6.502).$$

In particular each input size is less than 0.042. Using these three input bounds and $\rho_j / n_{\mathrm{p}} \le (1, 1.1755, 1.3819)_j z_*$ in (1216), the successive second sizes are at most

$$1.75\gamma_2 + 0.507 z_*^2 + 54.7 z_*/n_{\mathrm{p}},$$
$$3.0625\gamma_2 + 137.97 z_*^2 + 254.77 z_*/n_{\mathrm{p}},$$
$$5.36\gamma_2 + 432.4 z_*^2 + 633.3 z_*/n_{\mathrm{p}}.$$

These bounds imply (1214). Only three interpolations were used, irrespective of the number of product boxes. □

Proof of proposition 58.6. First consider the pair on $W = Y_{\mathcal{S}}$, in level-k units. Its potential and current are $L^{-1} A'$ and $L^{-3} \mathbf{J}$. At a stencil of the shorter sequence's layer $n \le k$, its base point has $n' = m_\sigma(x) \ge n$. With $\xi = \eta_k$, $\xi_n = L^{n-k}$, condition (L1) gives

$$|\partial U - 1|, \ |\partial \mathbf{U} - 1| < a_{\mathrm{m}} \xi^2 \xi_n^{-2},$$
$$\xi_n |L^{-1} A'| < a_{\mathrm{m}}, \qquad \xi_n^2 |\nabla_U^{\eta_k} (L^{-1} A')| < a_{\mathrm{m}}, \qquad |L^{-3} \mathbf{J}| < a_{\mathrm{m}} \xi_n^{-3}.$$

The extra factors $L^{-(n'-n)}$, with powers $1,2,3$, are at most one. For each natural cube, (L2)(b) supplies its containing gauge cube inside W. Restrict that gauge to the original cube and change the potential from spacing η_n to η_k: the factors are ξ_n^{-1} and ξ_n^{-2} for the potential and its difference. The resulting index-n bounds are at most $a_\flat \le a_{\mathrm{m}}$. These are all the natural local class conditions, with the specified decomposition.

Now let Y_1 be as in the statement. An endpoint x of a bond in $\mathrm{Bd}(Y_1)$ is within $M + \eta_k$ of a point of Ω_t in level-k units. The second gap therefore excludes $x \in Z_k$. It follows that

$$x \in \Lambda_k, \quad m_\sigma(x) \in \{k, t\}, \qquad Y_1 \subset Y \cap \Omega_k^\diamond, \quad \mathrm{Bd}(Y_1) \subset \mathrm{Bd}_{\mathcal{S}}(Y).$$

For $a = a_\natural$, the L^3 ceilings and $c_\sigma(n) \le 1 + \beta_{\mathrm{s}}$ give

$$R_\sigma^{p,\iota}(x) \le a \eta_t^p \quad (p = 1, 2, 3)$$

at the relevant stencils. Hence, on $\mathrm{Bd}(Y_1)$,

$$|\partial U - 1|, \ |\partial \mathbf{U} - 1| < a \eta_t^2, \qquad |A'|, \ |\nabla_U^{\eta_t} A'|, \ |\mathbf{J}| < a.$$

The only remaining simple-domain condition is a real gauge on every cube $\square \subset Y_1$ of side at most ς_{o} in level-t units.

A closed bare cube has N^3 sites, coordinatized by $\square_N$, and side $N\eta_j$ in level-j units. The chart is injective and its side is at most one bare spacing less than the torus side. If two chart sites are neighbors modulo the torus period, their coordinate difference minus the prescribed unit step has modulus strictly less than that period in every coordinate; it must therefore be zero. Thus inner bonds, plaquettes and all difference stencils correspond exactly to those of $\square_N$, including the repeated-direction stencils. Put $N' = \lfloor \varsigma_{\mathrm{o}} L^k \rfloor$. Then

$$N \le \varsigma_{\mathrm{o}} L^t < L(N' + 1).$$

For $f = \eta_t A$, the desired bounds are exactly

$$\sup|f| < \eta_t \varkappa[a], \qquad |\Delta_\mu f_\nu| < \eta_t^2 \varkappa[a]. \tag{1217}$$

This follows from $\Delta_\mu f_\nu = \eta_t^2 \nabla_\mu^{\eta_t} A_\nu$.

Condition (L2)(a) at level k supplies gauges on every integer box of side lengths at most N': enlarge it to a cube of side $\min(N, N')$ inside $\square_N$, apply (L2), and restrict. Their sizes are bounded by

$$\gamma_1 = \eta_k \varkappa[r_0(\sigma; k)], \qquad \gamma_2 = \eta_k^2 \varkappa[r_0(\sigma; k)].$$

The curvature bound in (L1), and the K_{cg} ceiling, give

$$\gamma_1 \le \eta_t \frac{L\varkappa[a]}{\mathsf{K}_{\mathrm{I}}}, \quad \gamma_2 \le \eta_t^2 \frac{L^2 \varkappa[a]}{\mathsf{K}_{\mathrm{I}}}, \quad \varpi < \frac{aL^2}{\mathsf{K}_{\mathrm{I}}} \eta_t^2.$$

If $N \le N'$, the original level-k gauge suffices: rescaling multiplies its potential by L and its first difference by L^2, while $\mathsf{K}_{\mathrm{I}} \ge L^2$. If $N > N'$ and $N' \le 23$, then $N < 24L$. In this case $\varsigma_{\mathsf{o}} < 24/L$ and (1209) implies $\mathsf{K}_{\mathrm{I}} \ge 16\pi L^3/(\varsigma_{\mathsf{o}} B^{(\mathrm{cg})})$. For $N = 1$ there are no bonds. Otherwise the tree gauge used above has

$$\gamma_1(U^u) \le N\varpi < \frac{3}{2\pi} \varkappa[a] \eta_t^2.$$

Its principal logarithm has size $< \frac{1}{2}\varkappa[a]\eta_t^2$; the difference of two values has size $< \varkappa[a]\eta_t^2$. This proves (1217) in the small-box case.

It remains to consider $N > N' \ge 24$. The gluing parameter satisfies

$$0 < z_* \le \eta_t \varkappa[a] \frac{q_*}{\mathsf{K}_{\mathrm{I}}} \le \frac{\eta_t}{7} \le \frac{1}{1183}. \tag{1218}$$

Take

$$n_{\mathrm{p}} = \min\left\{N', \left\lfloor (14 z_*)^{-1} \right\rfloor\right\}.$$

Then $n_{\mathrm{p}} \ge 24$, $n_{\mathrm{p}} z_* \le 1/14$, and

$$n_{\mathrm{p}}^{-1} \le \frac{25}{24}\left(14 z_* + \frac{L\eta_t}{\varsigma_{\mathsf{o}}}\right).$$

Applying lemma 58.7, the final first size is < 0.0056, and the second size is at most

$$\eta_t^2 \varkappa[a] \left[\frac{5.4 L^2}{\mathsf{K}_{\mathrm{I}}} + 9679\, \varkappa[a] \frac{q_*^2}{\mathsf{K}_{\mathrm{I}}^2} + 660.5\, \frac{L q_*}{\varsigma_{\mathsf{o}} \mathsf{K}_{\mathrm{I}}}\right] \le 0.99\, \eta_t^2 \varkappa[a].$$

Indeed $\varkappa[a] \le 1$, and the three terms in brackets are at most $0.3375, 0.2988, 0.2823$. The same choice gives $\mathsf{K}_{\mathrm{I}} \ge 7 q_*$ and the small-box inequality used above. The unitary logarithm bounds

$$|\log V| \le \frac{\pi}{3}|V - 1|, \qquad |\log V - \log V'| \le \frac{|V - V'|}{1 - \max(|V - 1|, |V' - 1|)}$$

now yield (1217): its first constant is at most $6.86/7 < 1$, and its second is at most $0.99/(1 - 0.0056) < 1$. The logarithm is real and its reversal is its negative, so it defines the required real one-form on the inner bonds of $\square$.

All the pointwise and cube-gauge conditions of the simple set have now been proved. Since Y_1 was arbitrary, (1213) follows. Together with the natural-class conclusion this gives the two background conditions of the regional shift domain; its remaining condition is the asserted strict bound on $\mathcal{A}$. □

58.5. **Identification of the regional shift.** At full coupling, the response must use the reduced current of the regional action, whereas the stored triple sets the current to zero at incomplete stencils. The following argument justifies that substitution.

Lemma 58.8. At $Y = T_\ell$ and $s = \mathbf{1}$, let two admissible pairs have the same first component and currents differing by δ, with $\delta = 0$ on every bond touching Ω_1. Their responses have the same first component $\mathbf{H}$, and their reduced-current increments differ by

$$f_{\mathbf{H}}^{-1}\delta - \delta. \tag{1219}$$

If the first pair is $(U_0, J^\circ(U_0))$, with U_0 real and critical, and the datum is supported on $\mathfrak{X}_0$, then both increments are equal. ■

Proof. Let Γ_0^+ be the bare determining rows, namely the inner bonds with no endpoint in Ω_1. On these rows $\mathcal{Q}(A) = \xi A$; hence the nonlinear averaging remainder C and all its derivatives vanish. The correction $\Delta'_\pi[\delta]$ also vanishes. Indeed, the site field $v = \mathsf{V}A$ in (480) has zero scalar averages and therefore vanishes at the bare sites outside Ω_1. On a bond supporting δ, both v_- and v_+ are zero, so every term in that quadratic form is zero. Polarization proves the assertion.

Consequently G, S, H are unchanged when the current changes by δ. Since $QH = 1$, on a bare row $(HX)(c) = \xi^{-1}X(c)$. With $\zeta = H^{\mathsf{T}}\delta$, the weighted row pairing therefore gives

$$\zeta(c) = \begin{cases} \xi^3\delta(c), & c \in \Gamma_0^+, \\ 0, & \text{otherwise}, \end{cases} \qquad Q^{\mathsf{T}}\zeta = \delta.$$

It also gives $\mathsf{C}'_T{}^{\mathsf{T}}\zeta = \mathsf{C}''[A']^{\mathsf{T}}\zeta = 0$. In particular $\Psi_T\zeta = (1 + H^{\mathsf{T}}\mathsf{C}'_T{}^{\mathsf{T}})^{-1}\zeta = \zeta$. The quadratic form defining the additional Hessian correction changes by $2(\mathsf{C}^{(2)}(A), \zeta)_1 = 0$. Thus $G_1, H_1, \mathfrak{G}$ are unchanged as well.

The first contraction, defining D and $\mathcal{L} = A' - H\mathsf{D}(A')$, uses only C, H. In the six terms defining $\mathsf{W}(A')$, the only explicit current term is

$$-(\mathfrak{D}^{\mathsf{T}} - \mathsf{C}''[A']^{\mathsf{T}})H^{\mathsf{T}}\mathbf{J}.$$

Its change is $-\mathsf{C}'_{\mathcal{L}}{}^{\mathsf{T}}\Psi_{\mathcal{L}}\zeta + \mathsf{C}''[A']^{\mathsf{T}}\zeta = 0$. The second contraction consequently has the same iterates and limit. This proves equality of $\mathbf{H}$. The current formula

$$\omega(A) = \Psi_A[H^{\mathsf{T}}\mathbf{J} + H^{\circledast}(A)], \qquad j(A) = f_A^{-1}(Q + \mathsf{C}'_A)^{\mathsf{T}}\omega(A) - Q^{\mathsf{T}}H^{\mathsf{T}}\mathbf{J}$$

then gives (1219).

For the last assertion, criticality supplies the multiplier at $A = 0$, as in the proof of proposition 41.2. That proposition gives $\mathcal{Q}(\mathbf{H}) = B'$. On Γ_0^+ the right side is zero, so $\mathbf{H} = 0$ there. Since $f_0 = 1$ and δ is supported on these rows, $f_{\mathbf{H}}^{-1}\delta = \delta$. □

Proposition 58.9. Let U_+ be in the regular critical orbit of its datum for $\Omega_0, \dots, \Omega_{k+1}$, and suppose $V = \mathcal{M}^k U_+$ is axial on P''. Let P_+ be its plain triple, at level $k+1$, from any gluing outside Ω_0. If $(P_+, \mathcal{A}) \in \mathrm{dom}_{\mathfrak{h}}(T_\ell \mid \Omega_\bullet)$ is real, define

$$(\mathbf{A}, \mathbf{K}) = \mathfrak{h}_{T_\ell|\Omega_\bullet}(\mathbf{1}; P_+, \mathcal{A}), \qquad E = e^{i\eta_k \mathbf{A}}U_+.$$

Let d be the shorter determining datum equal to $e^{iB'}V$ on $\mathfrak{X}_0$, and to $M_{\mathbf{B}}(U_+)$ elsewhere. If the regular critical orbit for $(\mathbf{B}, \mathsf{d})$ exists, E is a gauge transform of an element of it. At every complete current stencil in Ω_0,

$$J^{(k)}(E) = L^{-3}J^{(k+1)}(U_+) + \mathbf{K}. \tag{1220}$$

Moreover $\mathbf{A} = 0$ on bonds not touching Ω_1, and $\mathbf{K} = 0$ at incomplete current stencils. ■

Proof. Fixing the shorter determining datum imposes all the constraints of the longer one. Thus U_+ is critical for the shorter datum. Its reduced current equals its full current at complete stencils. At

an incomplete stencil no corner of its adjacent plaquettes is in Ω_1, by separation. There are at most three remaining plaquettes, each with curvature less than a_{m}, so the action derivative gives

$$|J^{\circ}(U_+)(b)| < 3a_{\mathrm{m}}\xi_0^{-3}.$$

Consequently $(U_+, J^{\circ}(U_+))$ is in the natural class at radius $3a_{\mathrm{m}} \le a_{\mathrm{D}}$. Its current agrees with $L^{-3}\mathbf{J}_+$ on every bond touching Ω_1; this is precisely the scaling $J^{(k+1)} = L^3 J^{(k)}$. The preceding lemma therefore identifies its response with $(\mathbf{A}, \mathbf{K})$.

The row source is zero except on interior level-k rows. The outward-row correction to a datum uses only that outward row and the bonds inside its outer endpoint block; all these source values vanish. Thus the twisted datum in the actual response theorem is B' itself. By proposition 41.2, E is critical with datum $(e^{iB'}M_{\mathbf{B}}(U_+))^v$, where v consists of the site factors. Choose $g = v^{-1}$ at the determining nodes and $g = 1$ at the other bare sites. Gauge covariance then makes E^g critical with datum d. The current identity in that proposition proves (1220). It also shows $\mathbf{A} = 0$ on Γ_0^+. At an incomplete current stencil all adjacent bonds are of this type, so $E = U_+$ on its plaquettes and its current increment is zero.

We check regularity, rather than infer it from criticality. Put $\beta_* = \sup|B'|$ and $\rho = 5b_{\mathrm{D}}e^{8\bar{\kappa}}\beta_* \le \rho_*$. The response estimates at the layer of the base point give

$$\xi_n|\mathbf{A}|,\ \xi_n^2|\nabla^{U_+}\mathbf{A}| \le \rho, \qquad \xi_n^3|\mathbf{K}| \le 10C_j b_{\mathrm{D}}^3 e^{24\bar{\kappa}}\beta_*.$$

If j is the largest level touched by a bond or a plaquette and n is its base-point level, separation gives $n \ge j-1$. Hence $\xi_n^{-1} \le L\mathsf{L}_{\mathbf{B}}^{-1}$. The elementary plaquette variation estimate, applied to $\xi_n\mathbf{A}$, bounds the new curvature by

$$(3a_{\mathrm{m}} + 2(1+\rho_*)\rho)\eta_k^2\xi_n^{-2}.$$

The new current is bounded by $(3a_{\mathrm{m}} + 10C_j b_{\mathrm{D}}^3 e^{24\bar{\kappa}}\beta_*)\xi_n^{-3}$. Multiplication by L^2 and L^3, respectively, is absorbed by C_{e}; the last inequality of (1203), with $\beta_* \le 2C_P\varepsilon_{\mathfrak{h}}$, puts both strictly within the regularity radius ε_{v}. Regularity is gauge invariant. Thus E^g is in the regular critical orbit whenever that orbit exists. □

58.6. **Pinned levelling of the response.** The accumulated endpoint factors of a response alter both its coarse datum and its stored transports. We remove them by a gauge supported inside the pillars. To control first differences, the prescribed values are extended by smooth turns on smaller cells. The only cells of order zero occur at level-one pillars with a two-step neighborhood containing no higher layer.

Work at step $t = k+1$, $1 \le k < K$, with $\xi = \eta_k = L^{-k}$, $\eta_t = \xi/L$, $\xi_q = L^q\xi$, a separated reduced sequence $\Omega_\bullet = (\Omega_0, \dots, \Omega_k)$, and a localization domain $Y \in \mathcal{D}_k$. The fine partition has $M = L^{m'}$ and $m' < m + K - t$. Put

$$\mathfrak{e}_Y = \mathrm{Bd}(Y) \cap \mathrm{Bd}^{\circ}(\Omega_0), \qquad S_Y = \{\text{endpoints of bonds in } \mathrm{Bd}(Y)\}.$$

For $b \in \mathfrak{e}_Y^+$, let $y(b)$ and $n(b)$ be the top and level of the pillar containing b_-. A top y of level $q \ge 1$ is a top in Y if $B^q(y) \subset Y$. Call it *pinnable* if $q \ge 2$, or if $q = 1$ and every bare site within 2ξ of $B^1(y)$ lies in a pillar of level zero or one. Denote the tops in Y by $\mathsf{Tp}(Y)$. For a pinnable top, use the turning cells $B^{q-1}(a)$, $a \in B(y)$; for a nonpinnable top use the single cell $B^1(y)$. Write $\mathsf{Ct}(Y)$ for their centers, $\mathsf{C}(a)$ for the cell, $o(a)$ for its order, and $y(a)$ for its top. These cells are disjoint and partition the pillars of $\mathsf{Tp}(Y)$.

For a triple $P = (U, A', \mathbf{J})$ at level t, set $\mathbf{U} = e^{i\eta_t A'}U$. We shall require the following conditions, denoted $P \in \mathfrak{T}_k(Y; c)$:

$$\begin{gathered}
\xi_{y(b)}L^{-1}|A'(b)| < c, \qquad \xi_{y(b)}^3 L^{-3}|\mathbf{J}(b)| < c, \qquad \xi_{y(b)}^2 L^{-1}|(\nabla_\mu^U A'_\nu)(b_-)| < c, \\
L^{2q}\varpi_{B^q(y)}(U) \le c \quad (y \in \mathsf{Tp}(Y)), \\
U^w = e^{i\xi A^{\mathrm{g}}} \text{ on } \mathsf{P}(\mathsf{C}(a)), \qquad \xi_{o(a)}|A^{\mathrm{g}}| \le c, \quad \xi_{o(a)}^2|\nabla A^{\mathrm{g}}| \le c \quad (o(a) \ge 1).
\end{gathered} \tag{1221}$$

The first line concerns $b \in \mathfrak{e}_Y^+$ and contained difference stencils. In the last line a real gauge w and a real potential A^{g} are required separately on each cell.

An admitted shift $\mathfrak{h} = (\mathbf{A}, \mathbf{K})$ is a pair of complex one-forms on $\mathrm{Bd}(Y)$, zero off $\mathfrak{e}_Y$. Its local size is

$$|\mathfrak{h}|_{\mathbf{U},Y}(b) = \max\left\{\xi_{y(b)}|\mathbf{A}(b)|,\ \xi_{y(b)}^2 \max_{\mu} |(\nabla_\mu^{\mathbf{U}} \mathbf{A}_\nu)(b_-)|,\ \xi_{y(b)}^3 |\mathbf{K}(b)|\right\}. \tag{1222}$$

Here $b = \langle x, x + \xi e_\nu\rangle$, and the inner maximum uses only stencils contained in $\mathfrak{e}_Y$, including $\mu = \nu$; an empty maximum is zero. Let $\|\mathfrak{h}\|_y$ be the maximum over bonds with $y(b) = y$. The neighbors of a top y are y itself and the pinnable level-one tops y' in Y whose pillars are within 2ξ of its pillar. Put $\|\mathfrak{h}\|_y^+ = \max_{y' \in \mathrm{nb}_Y(y)} \|\mathfrak{h}\|_{y'}$.

For $y \in \mathsf{Tp}(Y)$ of level q, let $\mathsf{Rd}_Y(y) = \mathsf{P}(B^q(y))$, except at pinnable level-one tops, where it consists of the bonds of $\mathfrak{e}_Y$ whose endpoints are within 2ξ of $B^1(y)$. Define

$$\varepsilon_y(\mathbf{A}) = \xi_q \sup_{\mathsf{P}(B^q(y))} |\mathbf{A}|, \qquad \widehat{\varepsilon}_y(\mathbf{A}) = \xi_q \sup_{\mathsf{Rd}_Y(y)} |\mathbf{A}|.$$

A top is *levelled* when $\widehat{\varepsilon}_y(\mathbf{A}) < \epsilon$. Notice that $\mathsf{P}(B^q(y)) \subset \mathsf{Rd}_Y(y) \subset \mathfrak{e}_Y$ and $\varepsilon_y(\mathbf{A}) \le \|\mathfrak{h}\|_y$.

Choose $c_{\mathrm{gf}} > 0$ and $C_{\mathrm{gf}} \ge 1$, depending only on L, as the restricted tower constants in the proof below. Fix an even C^2 function ϕ which vanishes for $|s| \ge 1/3$, with $0 \le \phi \le 1$, $\phi(0) = 1$, and $|\phi'|, |\phi''| \le b_\phi$, where $b_\phi \ge 1$. For example, $\phi(s) = (1 - 9s^2)^3$ for $|s| < 1/3$ and zero elsewhere has these properties with $b_\phi = 54$. Set

$$c_{\mathrm{m}} = 10 C_{\mathrm{gf}}^2 e^{4C_{\mathrm{gf}}}, \qquad c_{\mathrm{rt}} = \min(c_{\mathrm{gf}}, 1/20), \qquad \varepsilon_{\mathrm{m}} = \frac{c_{\mathrm{gf}}}{100 c_{\mathrm{m}}}, \qquad C_{\mathrm{m}} = 200 b_\phi^2 L^2 c_{\mathrm{m}}. \tag{1223}$$

Decrease c_{gf}, if necessary, so $c_{\mathrm{gf}} \le 1/1024$, and fix $0 < \epsilon \le \varepsilon_{\mathrm{m}}$. Write $v_q^\sharp[\mathbf{U}; \mathbf{A}](y)$ for the accumulated endpoint factor of the centered relative recursion over $B^q(y)$.

Proposition 58.10. For $P \in \mathfrak{T}_k(Y; c_{\mathrm{rt}})$ and every admitted shift $\mathfrak{h}$, the construction below defines $u = u_Y^\delta[P; \mathbf{A}]$ on all bare sites, equal to one off S_Y, and an admitted shift $\mathfrak{h}^\delta[P] = (\mathbf{A}^\delta, \mathbf{K}^\delta)$. At a levelled top y of level q,

$$u(y) v_q^\sharp[\mathbf{U}; \mathbf{A}](y) = 1, \qquad |u(x)^{\pm 1} - 1| \le C_{\mathrm{m}} \varepsilon_y(\mathbf{A}) \quad (x \in B^q(y)).$$

At a pinnable levelled top it also preserves every stored transport:

$$\mathrm{ax}_{(e^{i\xi \mathbf{A}} \mathbf{U})^u}(a) = \mathrm{ax}_{\mathbf{U}}(a), \qquad a \in B(y).$$

Moreover $\mathrm{cn}(u) \le 3/2$, and on positive bonds

$$\begin{aligned} e^{i\xi \mathbf{A}^\delta(b)} \mathbf{U}(b) &= u(b_-) e^{i\xi \mathbf{A}(b)} \mathbf{U}(b) u(b_+)^{-1}, \\ L^{-3} \mathbf{J}(b) + \mathbf{K}^\delta(b) &= \mathrm{Ad}(u(b_-))(L^{-3} \mathbf{J}(b) + \mathbf{K}(b)). \end{aligned} \tag{1224}$$

The shift is unchanged when $u(b_-) = u(b_+) = 1$, and

$$|\mathfrak{h}^\delta[P]|_{\mathbf{U},Y}(b) \le C_{\mathrm{m}} \|\mathfrak{h}\|_{y(b)}^+. \tag{1225}$$

The construction is local on the indicated pillars and reading sets, commutes with real gauge transformations and complex conjugation, and is Borel in all its data. For fixed U, its values on specified pillars are holomorphic on the open set where their tops are levelled. Its value at a bond is holomorphic when this condition holds for every top in Y whose pillar contains either endpoint.

If $\mathsf{s} \ge 0$ on the bare sites of Ω_0 satisfies

$$\mathsf{s}(x) \le e^{\delta d_{\mathbf{B}}(x,x')} \mathsf{s}(x'), \qquad |\mathfrak{h}|_{\mathbf{U},Y}(b) \le \mathsf{s}(b_-),$$

then

$$\|\mathfrak{h}\|_{y(b)}^+ \le e^{C_{\mathrm{m}}\delta} \mathsf{s}(b_-), \qquad \widehat{\varepsilon}_{y'}(\mathbf{A}) \le L e^{C_{\mathrm{m}}\delta} \mathsf{s}(b_-) \quad (y' \in \mathrm{nb}_Y(y(b)) \cap \mathsf{Tp}(Y)). \tag{1226}$$

All bounds persist after restriction to $\mathrm{Bd}(X)$, $X^\uparrow \subset Y$, using only the difference stencils contained there. ∎

Proof. The prescribed values. On a pillar $B^q(y)$, use the centered relative recursion of (1063) with initial perturbation $\xi\mathbf{A}$. Write v_i for its accumulated factors and $\mathbf{U}_{(i)} = \mathcal{M}^i\mathbf{U}$. For $a \in B(y)$ put

$$\mathrm{ax}_{\mathbf{W}}(a) = (\mathcal{M}^{q-1}\mathbf{W})(y,a), \qquad \mathbf{E} = e^{i\xi\mathbf{A}}\mathbf{U}, \qquad p(a) = \mathrm{ax}_{\mathbf{U}}(a)^{-1} v_q(y)^{-1}\,\mathrm{ax}_{\mathbf{E}}(a).$$

The single-pillar statement is the restriction of the double-block one. Extend the real configuration by the coordinate folding used in proposition 14.10, and extend the two perturbations by zero to an adjacent double block. Folding preserves the maximal plaquette defect. Every contour and lower double block entering the recursion on the original pillar stays in it, so its values and domains are independent of the extension.

Choose c_{gf} below the radius in (1063), with enough reserve that the contour diameters are less than $\omega_{\mathrm{c}}/2$ and all normalized averaging logarithms are within half their defining balls. This follows from the same contour estimates there, applied also to $i^{-1}\log(e^{i\xi\mathbf{A}}e^{ia'})$; its norm is at most $2(\xi|\mathbf{A}| + |a'|)$. The required data are

$$L^{2q}\varpi_{B^q(y)}(U) \le c_{\mathrm{gf}}, \qquad L^q \sup|a'| < c_{\mathrm{gf}}, \qquad \varepsilon_y(\mathbf{A}) < c_{\mathrm{gf}}.$$

They hold at each levelled top with $a' = \eta_t A'$, by (1221), since $L^q|\eta_t A'| = \xi_q L^{-1}|A'| < c_{\mathrm{rt}}$. Thus the actual complex averages and transports are defined. They are holomorphic in $(a', \mathbf{A})$, equal their real versions for real data, and satisfy

$$|v_i^{\pm 1} - 1| \le C_{\mathrm{gf}} L^{i-q}\varepsilon_y, \qquad S := \mathrm{ax}_{\mathbf{U}}(a) = e^{i\zeta'(a)}\,\mathrm{ax}_U(a), \quad |\zeta'(a)| \le C_{\mathrm{gf}} L^q \sup|a'|.$$

At the last step let $\zeta(a) = \xi_{\mathsf{m}}(\mathbf{U}_{(q-1)}, a_{q-1})(y,a)$ and $\bar\zeta = L^{-3}\sum_{a\in B(y)}\zeta(a)$. The one-step factorization gives

$$\mathrm{ax}_{\mathbf{E}}(a) = v_{q-1}(y)e^{i\zeta(a)}Sv_{q-1}(a)^{-1}, \qquad p(a) = S^{-1}e^{-i\bar\zeta}e^{i\zeta(a)}S\,v_{q-1}(a)^{-1}. \tag{1227}$$

Here $|\zeta(a)| \le C_{\mathrm{gf}}L^{-1}\varepsilon_y$. Consequently

$$|p(a)^{\pm 1} - 1| \le \{2C_{\mathrm{gf}}e^{4C_{\mathrm{gf}}}(1 + C_{\mathrm{gf}}) + C_{\mathrm{gf}}\}\varepsilon_y \le \tfrac{1}{2}c_{\mathrm{m}}\varepsilon_y < 1/200.$$

In particular, $p(y) = v_q(y)^{-1}$, and the pin logarithm $\zeta^{\mathrm{pin}}(a) = i^{-1}\log p(a)^{-1}$ has norm at most $c_{\mathrm{m}}\varepsilon_y$. Put this logarithm at every center over a levelled top and zero at all other centers.

The covariance of the transports also holds under a complex gauge h with $\mathrm{cn}(h) \le 3/2$. Indeed, path factors cancel, and left conjugation multiplies each contour diameter and normalized logarithm bound by at most $3/2$. The preceding half-radius reserve keeps the transformed objects in the same branches. Induction over the actual averages therefore gives $\mathrm{ax}_{\mathbf{E}^h}(a) = h(y)\,\mathrm{ax}_{\mathbf{E}}(a)h(a)^{-1}$. The choice $h(y) = v_q(y)^{-1}$, $h(a) = p(a)$ makes this equal to $\mathrm{ax}_{\mathbf{U}}(a)$.

Extending the prescribed values. On a cell $\mathsf{C}(a) = B^o(a)$, $o \ge 1$, consider real endomorphism fields T, normalized by $T(a) = 1$, satisfying

$$\xi_o|\nabla^U_\varsigma T| \le 1, \qquad \xi_o^2|\nabla^U_{\varsigma'}\nabla^U_\varsigma T| \le 1$$

for all signed directions and all contained stencils. The norm on endomorphisms here is the Euclidean operator norm. Among these fields choose the unique minimizer of the sum of their squared Hilbert–Schmidt norms.

This set is nonempty, with a strict feasible point. Indeed, the cell gauge in (1221) gives $T_0(x) = \mathrm{Ad}(w(x)^{-1}w(a))$. Put $D_\varsigma(x) = \mathrm{Ad}(e^{i\xi A^{\mathrm{g}}_\varsigma(x)}) - 1$. After conjugation by the orthogonal maps $\mathrm{Ad}\,w$, its first difference is D_ς/ξ, and its second difference is

$$\xi^{-2}\big[(1 + D_{\varsigma'}(x))D_\varsigma(x + \xi\varsigma') - D_\varsigma(x)\big].$$

The potential and first-difference bounds give, after multiplication by ξ_o and ξ_o^2, respectively, $2e^c c < 1/2$ and $2e^c c + 4e^{2c}c^2 < 1/2$, where $c = c_{\mathrm{rt}}$. Negative directions follow by translating the same identities. The feasible set is finite-dimensional, closed and convex, and the objective is strictly convex. Its minimizer therefore exists and is unique. A coordinate path of length at most $3(L^o - 1)/2$

from a gives $|T(x)| \le 5/2$. The complex-linear extension has norm at most 5 in the matrix norm used for $\mathfrak{g}_{\mathbb{C}}$, and its first and second differences have bounds $2\xi_o^{-1}$ and $2\xi_o^{-2}$.

This choice is continuous in U. To see this, mix a minimizer with a positive amount of the strict feasible field T_0. That mixture remains feasible for every sufficiently nearby configuration. It gives the upper bound for the limiting minimum; boundedness and uniqueness then identify every subsequential limit. Real gauges act on the feasible set by

$$T(x) \longmapsto \mathrm{Ad}(g(x))T(x)\,\mathrm{Ad}(g(a))^{-1}.$$

They preserve the objective, so the minimizer is gauge covariant. Its definition also commutes with translations of the cell.

Set

$$b_a(x) = \prod_{\mu=1}^{3} \phi((x_\mu - a_\mu)/\xi_o), \qquad \mu(x) = b_a(x)T(x)\zeta^{\mathrm{pin}}(a) \quad (x \in \mathsf{C}(a),\ o \ge 1).$$

On a singleton cell set $\mu(a) = \zeta^{\mathrm{pin}}(a)$, and set $\mu = 0$ elsewhere. Finally put $u = e^{-i\mu}$. For $z = |\zeta^{\mathrm{pin}}(a)|$, the product rule gives

$$|\mu| \le 5z, \qquad |\xi \nabla_\varsigma^U \mu| \le 7b_\phi L^{-o} z, \qquad |\xi^2 \nabla_{\varsigma'}^U \nabla_\varsigma^U \mu| \le 11 b_\phi^2 L^{-2o} z. \tag{1228}$$

For completeness, if $\tau = T\zeta^{\mathrm{pin}}(a)$, all values transported to the same base point, the mixed product difference is

$$b_{12}(\tau_{12} - \tau_1 - \tau_2 + \tau_0) + (b_{12} - b_1)(\tau_1 - \tau_0) + (b_{12} - b_2)(\tau_2 - \tau_0) + (b_{12} - b_1 - b_2 + b_0)\tau_0.$$

The four bounds are 2, $2b_\phi$, $2b_\phi$, and $5b_\phi^2$, times $L^{-2o}z$. They also hold after extension by zero: a nonzero bump lies in the inner third, whose two-step neighborhood is inside the cell, since $L^o \ge 13$. At a singleton, pinnability instead ensures that its two-step neighborhood contains only levels zero and one.

The prescribed logarithms lie in the larger polydisc

$$|\zeta(a)| < 2c_{\mathrm{m}}\varepsilon_{y(a)} \quad \text{at levelled tops with } \varepsilon_{y(a)} > 0, \qquad \zeta(a) = 0 \quad \text{at all other centers.}$$

Throughout this polydisc $z < 1/50$, $|\mu| \le 1/10$, $|u^{\pm 1} - 1| \le 6z$, and $\mathrm{cn}(u) \le 5/4$. If u is nontrivial at an endpoint of a bond, that bond belongs to the reading set of a levelled top; hence $\xi|\mathbf{A}(b)| < \epsilon$. Define, on positive bonds,

$$\mathbf{A}^\delta(b) = \frac{1}{i\xi}\log\bigl(u(b_-)e^{i\xi\mathbf{A}(b)}\mathbf{U}(b)u(b_+)^{-1}\mathbf{U}(b)^{-1}\bigr), \qquad \mathbf{K}^\delta(b) = \mathrm{Ad}(u(b_-))(L^{-3}\mathbf{J} + \mathbf{K})(b) - L^{-3}\mathbf{J}(b).$$

When both endpoint values of u equal one, define $\mathbf{A}^\delta(b) = \mathbf{A}(b)$ instead. This proviso is essential: the original shift need not be small on an untouched bond. Extend both forms oddly to reversed bonds. The logarithm on a touched bond is the series logarithm; the estimate below places its argument within $2/5$ of one. The inner-third support and the singleton neighborhood show that bonds outside $\mathfrak{e}_Y$ are untouched, proving admissibility. At the centers $T(a) = 1$ and $b_a(a) = 1$, so $u(a) = p(a)$. A top or a stored site over a pinnable top is either such a center or outside every turning cell. Thus the prescribed values and the stored-transport identity are independent of the frame away from the centers.

The first-difference estimate. We give the mixed estimate that controls the logarithm. For $|\theta_0|, |\theta_1|, |\theta_2|, |\theta_{12}| \le 1/8$ and $|\alpha_0|, |\alpha_1| \le \rho \le 1/12$, put

$$X_0 = e^{-i\theta_0}e^{i\alpha_0}e^{i\theta_2}, \qquad X_1 = e^{-i\theta_1}e^{i\alpha_1}e^{i\theta_{12}}, \quad d_i = |\theta_i - \theta_0|, \quad s = |\theta_{12} - \theta_1 - \theta_2 + \theta_0|.$$

Then $|X_j - 1| < 2/5$, and

$$\begin{aligned} |\log X_0| &\le 3|\alpha_0| + 3d_2, \\ |\log X_1 - \log X_0| &\le 3|\alpha_1 - \alpha_0| + 5s + 7d_1 d_2 + 6\rho d_1. \end{aligned} \tag{1229}$$

Here is the product calculation, including the term which would be lost by estimating four exponentials separately. Write $u_j = e^{-i\theta_j}$ and $a_j = e^{i\alpha_j}$. The difference $X_1 - X_0$ is the sum

$$\begin{aligned}&[(u_1 - u_{12}) - (u_0 - u_2)]u_{12}^{-1} + (u_0 - u_2)(u_{12}^{-1} - u_2^{-1})\\ &\quad + (u_1 - u_0)(a_1 - 1)u_{12}^{-1} + u_0(a_1 - a_0)u_{12}^{-1} + u_0(a_0 - 1)(u_{12}^{-1} - u_2^{-1}).\end{aligned}$$

In the exponential power series, every word in the mixed difference either contains the mixed increment or contains both first increments. If $r = \max|\theta_j|$, this gives

$$|u_{12} - u_1 - u_2 + u_0| \le e^{3r}s + e^{5r}d_1d_2.$$

The ordinary exponential difference estimate bounds the other four terms. The logarithm is Lipschitz with constant 2 in the 2/5-ball. Substitution of $r \le 1/8$, $\rho \le 1/12$ gives (1229); the same estimates give $|\log X_0| < .52 < \log 2$.

The differences in (1228) use U; the logarithm uses $\mathbf{U} = e^{i\eta_t A'}U$. This change preserves their scale. Indeed, put $E_\mu = \mathrm{Ad}(e^{i\eta_t A'_\mu})$ and let $\widehat{E}_\nu$ be its ν-analogue transported with U_μ. On a cell in a level-q pillar, the first line of (1221) gives

$$|E_\mu| \le 1.008, \qquad |E_\mu - 1| \le \tfrac{1}{8}L^{-q}, \qquad |\widehat{E}_\nu - E_\nu| \le \tfrac{1}{8}L^{-2q}.$$

If hats denote the four U-transported values, the mixed $\mathbf{U}$-transported difference is

$$E_\mu\widehat{E}_\nu\widehat{s} + E_\mu(\widehat{E}_\nu - 1)\widehat{d}_1 + (E_\mu\widehat{E}_\nu - E_\nu)\widehat{d}_2 + [(E_\mu - 1)(\widehat{E}_\nu - 1) + (\widehat{E}_\nu - E_\nu)]\theta_0.$$

Since $o \le q$, this yields

$$d_1, d_2 \le 9b_\phi L^{-o}z, \qquad s \le 15b_\phi^2 L^{-2o}z.$$

Fix a contained difference stencil at b, and put $H = \|\mathfrak{h}\|_{y(b)}$, $H_+ = \|\mathfrak{h}\|^+_{y(b)}$. There are three cases. If μ vanishes on its four sites, the shift is unchanged. If a nonzero value lies in a cell of positive order, the entire stencil lies in that cell. Its level q satisfies $o \le q \le o+1$, and

$$z \le \min(1/50, 2c_{\mathrm{m}}H), \qquad \rho \le 1.008L^{-q}H.$$

Apply (1229) with $\alpha_0 = \xi\mathbf{A}_\nu(b_-)$ and $\alpha_1 = \mathrm{Ad}(\mathbf{U}_\mu)\xi\mathbf{A}_\nu(b_- + \xi e_\mu)$. After multiplying by the output scales, it gives

$$\xi_{y(b)}|\mathbf{A}^\delta(b)| \le (3 + 54b_\phi Lc_{\mathrm{m}})H, \qquad \xi^2_{y(b)}|\nabla^{\mathbf{U}}_\mu \mathbf{A}^\delta_\nu(b_-)| \le 175b_\phi^2L^2c_{\mathrm{m}}H.$$

In the remaining case the nonzero turns are singletons. All four sites have levels at most one, and their tops belong to $\mathrm{nb}_Y(y(b))$. With $z_\bullet \le \min(1/50, 2c_{\mathrm{m}}H_+)$, use

$$|\theta_j| \le 5z_\bullet/4, \quad d_1, d_2 \le 5z_\bullet/2, \quad s \le 5z_\bullet$$

in the same calculation. The corresponding bounds are $(3+15Lc_{\mathrm{m}})H_+$ and $59L^2c_{\mathrm{m}}H_+$. The current formula in (1224), the bound on $u-1$, and $\xi^3_{y(b)}L^{-3}|\mathbf{J}(b)| < c_{\mathrm{rt}}$ give $\xi^3_{y(b)}|\mathbf{K}^\delta(b)| \le 3c_{\mathrm{m}}H_+$. All three bounds are absorbed by C_{m}, proving (1225).

Dependence and profile bounds. The pin reads only the actual tower over its pillar, the levelled test reads only $\mathsf{Rd}_Y(y)$, and the frame reads only the real configuration on its cell. These observations give locality and real gauge covariance. For fixed U, on the open set where the relevant tops remain levelled, the pins and the series logarithms are holomorphic. Where both turns become one the logarithmic formula agrees with the inert definition, since the relevant input is small there. Thus there is no additional branch at that locus. Joint continuity follows on each prescribed set of levelled tops from continuity of the frames and of the actual averages. The finitely many levelled sets form Borel pieces, giving the global Borel assertion. For real data all factors are unitary. Holomorphic reflection therefore gives $u(\overline{P}, \overline{\mathfrak{h}}) = u(P, \mathfrak{h})^c$; the real frames commute with the involution, and $|\log X| < \log 2$ identifies the same logarithm after conjugation. This proves the conjugation statement.

A coordinate path within a level-q pillar has weighted length less than 3. Two bare sites at distance at most 2ξ in the supremum norm are joined by at most six bare steps, of weighted length at most 6. Thus a top neighboring $y(b)$, together with a site in its pillar, is at weighted distance at most $3+6+3=12$ from b_-. The exceptional enlarged reading set costs at most six further steps. Its levels are zero or one, so changing $\xi_{y'}$ to the input scale costs at most L. Since $C_{\mathrm{m}} \geq 18$, these facts prove (1226). Restriction to a smaller bond domain retains precisely a subset of these stencils. □

The locality needed for old terms includes two exact support statements. They apply for all complex values of the parameters; holomorphy still requires the forced-levelled condition above.

Proposition 58.11. Let $\gamma_1, \dots, \gamma_N$ be the wall-components of a set of fine cubes in Y. Suppose that the admitted shifts $\mathfrak{h}_i$ vanish unless both endpoints belong to γ_i, and satisfy the margin condition

$$\mathfrak{h}_i(b) = 0 \quad \text{if an endpoint } p \text{ of } b \text{ is within } (L+2)\xi \text{ of a bare site outside } \gamma_i.$$

For $\mathfrak{h} = \sum_i \lambda_i \mathfrak{h}_i$, $\lambda_i \in \mathbb{C}$, one has

$$u_Y^\delta[P; \mathbf{A}] = \prod_i u_Y^\delta[P; \lambda_i \mathbf{A}_i], \qquad \mathfrak{h}^\delta[P] = \sum_i (\lambda_i \mathfrak{h}_i)^\delta[P].$$

At each site at most one factor differs from one. Each pinned summand still vanishes unless both endpoints lie in its component, and a zero input gives a zero summand.

If $Y_1 \subset Y$, $Y_1 \in \mathcal{D}_k$, and an admitted $\mathfrak{h}$ vanishes unless both endpoints belong to Y_1, then $P|_{Y_1} \in \mathfrak{T}_k(Y_1; c_{\mathrm{rt}})$ and

$$u_Y^\delta[P; \mathbf{A}]\big|_{S_{Y_1}} = u_{Y_1}^\delta[P|_{Y_1}; \mathbf{A}|_{\mathrm{Bd}(Y_1)}], \qquad u_Y^\delta = 1 \quad \text{on } S_Y \setminus Y_1.$$

The corresponding shifts agree on $\mathrm{Bd}(Y_1)$ and are zero on its complement. ■

Proof. Every pillar lies in a single fine cube, and hence either in one component or outside their union. Two bare nearest neighbors cannot lie in different wall-components: their fine cubes are equal or adjacent. On a pillar in γ_i, the summed shift equals $\lambda_i \mathfrak{h}_i$; every other shift vanishes. This also identifies its pin and frame. The only enlarged levelled test is at a pinnable level-one top. Every bond of that reading set lies within 2ξ of its pillar; the margin therefore annihilates all other component shifts there. Thus the levelled test agrees as well. On all other pillars the turn is one. These facts prove the gauge product, and the bond formula (1224) proves the sum, including the inert proviso.

To check the asserted support after turning, consider a bond crossing out of γ_i. A nonzero turn at its inside endpoint cannot lie in the inner third of a positive-order cell, since its nearest neighbor would then lie in that cell and in γ_i. If it is a singleton, every point of its level-one pillar is within $L\xi$ of the outside endpoint. The margin forces the original shift to vanish on that whole pillar. Its pin, and hence its turn, is one. The crossing bond is therefore untouched.

For restriction to Y_1, the tops and cells are exactly those of Y lying in Y_1; their frames and pillar data coincide. The extra bonds of $\mathsf{Rd}_Y(y) \setminus \mathsf{Rd}_{Y_1}(y)$ lie outside $\mathrm{Bd}(Y_1)$, and the shift vanishes on them. Thus the levelled tests coincide. On every pillar outside Y_1 the initial perturbation is zero, so its turn is one. The bond formula gives the remaining assertions. The conditions on $P|_{Y_1}$ involve only a subset of the original stencils, pillars and cells. □

We record the endpoint geometry used when these pins are compared between successive histories. A stored site of a level-q top y means a site $a \in B(y)$; its stored transport is $(\mathcal{M}^{q-1}W)(y,a)$.

Lemma 58.12. Let h be a history through level k, $0 \leq l < k$, and let x be an endpoint of an external bond of level l. Then $x \notin \Lambda_{l+1}$. It belongs to Ω_l if $l \geq 1$, and to $\Omega_1^\sim$ if $l = 0$. Write $y = \mathsf{c}_{l+1}(x)$. For the separated truncations of this history and the one-step extensions arising from

an admissible next outcome, with the same historical regions through level $l+1$, the following alternatives hold.

If $x \notin \Omega_{l+1}$, it is a top of level l. For $l \geq 1$ it is its own stored site; for $l = 0$ it has no stored transport. If $x \in \Omega_{l+1}$, it is a stored site of the pinnable top y of level $l+1$. In either nonempty case, the stored transport is the same actual contour average for every such sequence:

$$\mathrm{ax}_W(x) = (\mathcal{M}^{q-1}W)(y_*, x), \qquad (q, y_*) = \begin{cases} (l, x), & x \notin \Omega_{l+1},\ l \geq 1, \\ (l+1, y), & x \in \Omega_{l+1}. \end{cases} \tag{1230}$$

Definedness in either the real or complex sense is the same.

For a subordinate separated sequence containing x, its layer q' is at most $l+1$. If $q' \leq l$, then x is its own top and $B^{q'}(x) \subset B^{l+1}(y)$. If $q' = l+1$, it has the same pinnable top y and the same transport in (1230). ■

Proof. An external bond is incident on Ω_{l+1} and has neither endpoint in Λ_{l+1}. Its endpoints are within one level-l step of Ω_{l+1}. The historical collars have widths at least $8LM\check{R}_{l+1}$ inside Ω_l, or $LM\check{R}_1$ inside $\Omega_1^{\sim}$ when $l = 0$; these widths exceed one. This proves the first assertions.

The level-$(l+1)$ cube containing x is a cube of $Z_{l+1} = \Lambda_{l+1}^c$. Its distance from the next region is at least $8LM\check{R}_{l+2}$, measured at level $l+1$; the same bound holds for a possible next step after the history. Thus no higher region can contain x. The layer is exactly l or $l+1$ according as $x \notin \Omega_{l+1}$ or $x \in \Omega_{l+1}$. The block decomposition now gives the stated top and stored site. For $l \geq 1$ the latter top is pinnable by its level. For $l = 0$, its containing cube is at distance more than two bare steps from every level-two region. The separated level-one collar also contains that two-step neighborhood. Hence every site in the neighborhood is in a layer of level zero or one, which is exactly pinnability.

A subordinate sequence has smaller regions and inherits the same exclusion of level $l+2$. Its layer $q' \leq l$ puts the level-l site x on its own top; nesting of blocks gives $B^{q'}(x) \subset B^{l+1}(y)$. If $q' = l+1$, the same containing block and two-step argument apply. Finally every contour and lower block used in (1230) is inside the indicated pillar. The restriction argument in the proof of proposition 58.10 therefore identifies both its domain and its value. □

We now apply the construction to the real canonical response. Let U_0 be a real configuration on the ambient torus, let U_0 be its restriction to the inner bonds of Ω_0, and put $V_0 = M_{\mathbf{B}}(U_0)$. Its plain triple at level t is

$$P_0 = (\mathsf{U}_0, 0, \mathbf{J}_+), \qquad \mathbf{J}_+(b) = \begin{cases} J^{(t)}(U_0)(b), & \text{the full current stencil is in } \Omega_0, \\ 0, & \text{otherwise.} \end{cases}$$

Use the scale-k pairs $\mathbf{p}_1 = (U_0, L^{-3}\mathbf{J}_+)$ and $\mathbf{p}_0 = (U_0, J^{\circ}(U_0))$. The condition (m2)[$\alpha$] means that $\mathbf{p}_1$, with decomposition $(U_0, 0)$, satisfies the four local natural-class conditions at radius α. All response constants below are those of the regional response, with their common bound denoted by C.

Proposition 58.13. Suppose that U_0 is critical for its reduced determining datum, that P_0 satisfies (m2)[α], and that B' is a real admissible response datum supported on the interior rows of level k, with

$$\mathsf{b} = \sup |B'| \leq \varepsilon_{\mathrm{r}} := \varepsilon_{\mathrm{D}} e^{-32\bar{\kappa}}.$$

Let $(\mathbf{H}, \mathsf{j})$ be the canonical response at parameters one and pair $\mathbf{p}_1$. Assume

$$\alpha \leq \min(a_{\mathrm{D}}/4, L^{-2}c_{\mathrm{gf}}, c_{\mathrm{rt}}), \qquad L\mathsf{C}e^{\mathsf{C}\bar{\kappa}}\mathsf{b} < \epsilon \leq \varepsilon_{\mathrm{m}}. \tag{1231}$$

Then $P_0 \in \mathfrak{T}_k(T_\ell; c_{\mathrm{rt}})$, every top is levelled, and the pinned shift is real. The configuration

$$U^{\delta} = (e^{i\xi\mathbf{H}}U_0)^{u^{\delta}} = e^{i\xi\mathbf{H}^{\delta}}U_0$$

has actual reduced averages $e^{iB'}V_0$ and is critical for that datum. At every complete inner current stencil,

$$J^{(k)}(U^\delta) = L^{-3}J^{(t)}(U_0) + \mathsf{j}^\delta. \tag{1232}$$

At every stored site of a pinnable top, its actual transport is the transport of U_0. If in addition $\mathsf{C}(4\alpha + e^{\mathsf{C}\bar\kappa}\mathsf{b}) \le \varepsilon_{\mathrm{v}}$, then U^δ is regular at radius ε_{v}; it belongs to the minimal orbit for $e^{iB'}V_0$ whenever that orbit exists. ∎

Proof. The two currents agree on every bond touching Ω_1. At an incomplete star the bond is in the bare layer, and there are at most three remaining inner plaquettes. The natural-class curvature bound gives $|J^\circ(U_0)(b)| < 3\alpha\xi^{-3}$ there. Thus $\mathbf{p}_0$ belongs to the natural class at radius 4α, whereas $\mathbf{p}_1$ belongs at radius α. Both responses are defined. Their equality is lemma 58.8: the current difference is supported on bare determining rows, on which the datum and the response vanish. Criticality at $A = 0$ gives $J^\circ(U_0) = Q^{\mathsf{T}}\lambda_0$ by the constraint-multiplier argument in proposition 41.2. That proposition therefore gives

$$\mathcal{Q}(\mathbf{H}) = B', \qquad E := e^{i\xi\mathbf{H}}U_0 \text{ critical for } M_{\mathbf{B}}(E), \qquad J^{(k)}(E) = L^{-3}J^{(t)}(U_0) + \mathsf{j}$$

at complete stencils. These are the same response and current conventions as in proposition 58.9.

The natural gauges also give the regular-triple conditions. The first line of (1221) is immediate for P_0. Every plaquette in $B^q(y)$ has base-point layer q, so its defect is bounded by αL^{-2q}. For a turning cell in that pillar, take the $\mathfrak{M}\xi_q$-cube containing y. It is contained in the same layer: if any of its finer cubes were in Ω_{q+1}, nesting of the partitions would put y there too. It is therefore a natural cube inside Ω_0. Its containing-box gauge restricts to the turning cell. Since $o \le q$, the bounds at scale ξ_q imply the required bounds at ξ_o. Consequently $P_0 \in \mathfrak{T}_k(T_\ell;\alpha)$.

For a row $c \in \Gamma_q$, its double block $\mathsf{Z} = B^q(c_-) \cup B^q(c_+)$ lies in Ω_{q-1}, with $\Omega_{-1} = \Omega_0$. Every base-point layer there satisfies $n \ge \max(q-1, 0)$, and hence $\xi_q \le L\xi_n$. The natural curvature bound and the regional response estimate give

$$L^{2q}\varpi_{\mathsf{Z}}(U_0) \le L^2\alpha, \qquad \xi_q \sup_{\mathsf{P}(\mathsf{Z})} |\mathbf{H}| \le L\mathsf{C}e^{\mathsf{C}\bar\kappa}\mathsf{b}. \tag{1233}$$

The curvature maximum is zero at $q = 0$. These bounds place the actual double-block recursion in the domain used in the proof of proposition 58.10. The nonlinear response average $\mathcal{Q}_q(U_0, \xi\mathbf{H})$ is its straightened terminal value: at order zero the inputs coincide, and at every later order both definitions apply the same one-step factorization to each contained lower double block. Reversal is imposed after the positive-bond step in both definitions. Induction therefore identifies the domains and the values, not only their derivatives. Its terminal value is $B'(c)$.

On an ordinary pillar reading set $n = q$; on the enlarged reading set of a pinnable level-one top $n \ge 0$. The same estimate consequently gives

$$\widehat{\varepsilon}_y(\mathbf{H}) \le L\mathsf{C}e^{\mathsf{C}\bar\kappa}\mathsf{b} < \epsilon.$$

All tops are levelled. By proposition 58.10, u^δ is real and cancels the accumulated factor $v_{\mathbf{B}}(x)$ at every top x; at a bare top both factors are one.

The cancellation at an outward row deserves attention. The terminal double-block factorization is

$$M_{\mathbf{B}}(E)(c) = v_q(c_-)\, e^{iB'(c)}V_0(c)\, v_q(c_+)^{-1}. \tag{1234}$$

At an endpoint of level q, $v_q = v_{\mathbf{B}}$. At the outer endpoint z of an outward row, the endpoint level is $q-1$, and

$$v_q(z) = v_{\mathbf{B}}(z)\, \mathsf{v}_{\mathsf{m}}(\mathcal{M}^{q-1}U_0, a_{q-1})(z).$$

Every bond inside $B(z)$ is an interior row of Γ_{q-1}. Restriction of the same recursion identifies a_{q-1} there with B', which is zero because $q - 1 < k$. The last factor is therefore exactly one. Thus at either endpoint of every row the factors in (1234) are cancelled by u^δ. Real domain covariance

gives $M_{\mathbf{B}}(U^{\delta}) = e^{iB'}V_0$. Gauge invariance preserves criticality. Current covariance and the second identity of (1224) give (1232); the pin identity gives the stored transports.

Finally, the regularity calculation in the proof of proposition 58.9 applies with the initial natural-class radius 4α. The potential and its first difference are bounded at layer n by $\mathsf{C}e^{\mathsf{C}\bar{\kappa}}\mathsf{b}$ after multiplication by ξ_n, ξ_n^2; the current has the corresponding ξ_n^3 bound. A plaquette or bond meeting a higher layer loses at most one power of L in that scale comparison. The plaquette variation estimate and the current identity thus bound the regularity radius by $\mathsf{C}(4\alpha + e^{\mathsf{C}\bar{\kappa}}\mathsf{b})$. The constants of the regional response already absorb these L^2, L^3 losses. Regularity is unchanged by the real gauge u^{δ}, which proves the final assertion. □

58.7. **Tower bounds in the ambient layer scale.** The tower used by a local term can end before the ambient history does. Its local block size is then smaller than the size against which the historical domain measures curvature. We retain the scale factor in the next estimate. This is the input needed to preserve the tower conditions under a shift. All determining sets in this subsection are reduced.

Fix an ambient separated sequence $\Omega_0^{\flat} \supset \cdots \supset \Omega_i^{\flat}$ and a subordinate separated sequence $\Omega_0' \supset \cdots \supset \Omega_p'$, where $1 \le p \le i$ and $\Omega_q' \subset \Omega_q^{\flat}$. Write $Z = \Omega_0'$, $\mathsf{Z} = \mathrm{Bd}^{\circ}(Z)$, $\xi_q = L^q\eta$, and

$$\lambda^{\flat}(x) = \max\{q \le i : x \in \Omega_q^{\flat}\}, \qquad \lambda'(x) = \max\{q \le p : x \in \Omega_q'\},$$
$$l(x) = \xi_{\lambda^{\flat}(x)}, \quad \xi'(x) = \xi_{\lambda'(x)}, \quad \varsigma(x) = l(x)/\xi'(x).$$

The two weighted distances are denoted by $d^{\flat}, d'$. An inner plaquette or full bond e of Z is called settled when $\lambda'(x_e) = \min\{\lambda^{\flat}(x_e), p\}$. The base of a positively oriented bond is its initial point. All differences below are taken only on stencils in Z.

For a real configuration U, a complex form w, and $a > 0$, let $\mathcal{Q}^{\mathrm{T}}(a)$ be the domain

$$\epsilon_U(e) := \eta^{-2} l(x_e)^2 |\partial U(e) - I| < a, \qquad \mathsf{n}_b(w) := \max\{l(x_b)|w(b)|, l(x_b)^2 |\nabla_U w|_{\mathsf{Z}}(b)\} < a. \quad (1235)$$

Put

$$E_U^{\delta}(x) = \max_e e^{-\delta d^{\flat}(x,x_e)}\epsilon_U(e), \qquad E_w^{\delta}(x) = \max_b e^{-\delta d^{\flat}(x,x_b)}\mathsf{n}_b(w).$$

The tower configuration is

$$U^{\mathrm{tw}}(U,w) = \Phi_{V_0}(B(w)), \qquad V_0 = M_{\mathbf{B}'}U, \qquad B(w)(c) = Q_q(U;\eta w)(c), \quad c \in \Gamma_q'.$$

Thus the datum is the actual average of $e^{i\eta w}U$, expressed relative to V_0; there is no replacement of this average by its linearization.

We specify constants used later in the radii selection. Take the real-background and continuation thresholds of theorem E, the local regularity threshold and profile constants from the proof of corollary 53.10, and the response constants of lemma 51.6. Assume their geometric lower bounds on $\mathfrak{M}$ and $\rho = R_1\mathfrak{M}$, and that the top torus has at least 6ρ sites in each direction. Take

$$\delta = \min\{\delta_{\mathrm{c}}, \delta_{\mathrm{tw}}\}, \qquad \delta\rho \ge 64 \log L,$$
$$\mathsf{S}(a) = 54L^3 \sum_{r\ge 1}(2r+1)^3 e^{-a(r-1)/2}, \qquad B_{\mathrm{cx}}^{\sharp} = \max\{1, B_5\mathsf{S}(\delta)\}.$$

Here δ_{c} is the local regularity rate and $B_5, \delta_{\mathrm{tw}}$ are the constants in (838). Let

$$c^{\natural} = 2L^2, \qquad a_{\mathrm{tw}}^{(\mathrm{ia})} = \min\{a_{\mathrm{plain}}, c_{\mathrm{plain}}\}, \qquad \rho_{\mathrm{f}} = \min\{c_{\mathrm{plain}}, [5(\kappa_6+3)]^{-1}\}.$$

Here $a_{\mathrm{plain}}, c_{\mathrm{plain}}$ are the constants of proposition 51.1; decrease a_{plain}, if needed, so that $a_{\mathrm{plain}} \le c_2^{\mathrm{SO}}$. Choose ε_{T} and r_{T} by

$$\varepsilon_{\mathrm{T}} = \min\left\{\frac{\bar\varepsilon}{2}, \varepsilon_{\mathrm{G}}^{\mathrm{s}}(\rho_{\mathrm{s}}), \frac{a_{\mathrm{R}}}{C_{\mathfrak{P}\mathfrak{M}} B_3}, \frac{a_{\mathrm{c,LI}}}{B_3}, \frac{a_{\mathrm{plain}}}{B_3}, \frac{1}{22L^3B_3}\right\}, \tag{1236}$$

$$r_{\mathrm{T}} = \min\left\{\frac{r_{\mathrm{B}}^{\mathrm{s}}(\rho_{\mathrm{s}})}{2}, \frac{\varrho_{\mathrm{R}}^0}{2}, \frac{\bar\varepsilon}{2C_\partial}, \frac{\rho_{\mathrm{f}}}{2LB_{\mathrm{cx}}^\sharp}, \frac{1}{600L^3B_{\mathrm{cx}}^\sharp}, \frac{1}{20}\right\}. \tag{1237}$$

The entries $a_{\mathrm{R}}, \varrho_{\mathrm{R}}^0$ are the regularity and datum thresholds for the response in lemma 51.6; $C_{\mathfrak{P}\mathfrak{M}}$ is its regularity conversion factor. The entries $a_{\mathrm{c,LI}}, \varepsilon_{\mathrm{G}}^{\mathrm{s}}, r_{\mathrm{B}}^{\mathrm{s}}, C_\partial$ are the local regularity, continuation, and tested-holonomy constants just selected. Thus these choices place the same real critical family simultaneously in the local regularity, continuation, and response domains.

Set

$$\begin{aligned} C_{\mathrm{R}} &= \mathsf{c}_{\mathrm{loc}} c^\natural L^4 e^{7L\delta/2}, \qquad C_{\mathrm{X}} = 900L^3B_5 e^{7\delta/2}, \\ C_{\mathrm{h}} &= \frac{15B_5}{4\rho_{\mathrm{f}}} e^{3\delta/2}, \qquad C_\chi = 1 + L^2, \\ r_{\mathrm{w}} &= \min\{r_{\mathrm{T}}/(\mathsf{c}_Q L), a_{\mathrm{tw}}^{(\mathrm{ia})}/(2L)\}, \qquad \delta_{\mathrm{T}} = \delta/4, \end{aligned} \tag{1238}$$

where $\mathsf{c}_Q = C_Q$ is the averaging constant in proposition 51.1. Finally, let

$$\begin{aligned} C_{\mathrm{far}} &= \frac{8\mathsf{c}_Q L^3}{\delta} e^{3(5L+1)\delta/8} \mathsf{S}(\delta/4), \\ K_{\mathrm{a}} &= 7C_{\mathrm{R}} + (6C_{\mathrm{R}} + 5L^2/r_{\mathrm{w}}) C_\chi e^{9L\delta/4} + (C_{\mathrm{X}} + C_{\mathrm{h}}) C_{\mathrm{far}}, \\ K_{\mathrm{T}} &= (C_{\mathrm{X}} + C_{\mathrm{h}}) C_{\mathrm{far}} + 4K_{\mathrm{a}} C_\chi e^{9L\delta/4}, \\ \bar a &= \min\{\varepsilon_{\mathrm{T}}/(5c^\natural), a_{\mathrm{tw}}^{(\mathrm{ia})}/(5L^2), r_{\mathrm{w}}/2, (11L^2)^{-1}, (2K_{\mathrm{a}})^{-1}\}, \qquad a_{\mathrm{T}} = \frac{\bar a}{5C_\chi^2}. \end{aligned} \tag{1239}$$

All constants are independent of the depth, the volume, and the number of layers. The rate δ_{T} depends only on L; $K_{\mathrm{a}}, K_{\mathrm{T}}$ depend only on $L, \rho_{\mathrm{s}}, \bar\varepsilon$; and a_{T} may also depend on $\mathfrak{M}, B_3$. In particular, $0 < \delta_{\mathrm{T}} \le 1/4$, $\delta_{\mathrm{T}}\rho \ge 16 \log L$, $K_{\mathrm{T}} \ge K_{\mathrm{a}} \ge 1$, and $a_{\mathrm{T}} \le (64L^2)^{-1}$.

Proposition 58.14. On $\mathcal{Q}^{\mathrm{T}}(a_{\mathrm{T}})$, the tower and its current are defined and, for fixed U, holomorphic in w. At a settled element, put $n = \lambda^\flat(x_e)$, $n' = \lambda'(x_e) = \min\{n, p\}$, and

$$G(U, w; e) = \begin{cases} L^{2n}(\partial U^{\mathrm{tw}}(e) - I), & e \text{ a plaquette}, \\ L^{2n+n'}\eta^3 J(U^{\mathrm{tw}})(e), & e \text{ a full bond}. \end{cases}$$

Then

$$|G(U, w; e)| \le K_{\mathrm{a}}\{E_U^{\delta_{\mathrm{T}}}(x_e) + E_w^{\delta_{\mathrm{T}}}(x_e)\} \le 1, \tag{1240}$$

$$|G(U, w_1; e) - G(U, w_0; e)| \le K_{\mathrm{T}} E_{w_1 - w_0}^{\delta_{\mathrm{T}}}(x_e). \tag{1241}$$

At real w, the tower is the real axial representative of its actual averaged datum. It respects gauge changes, with the gauge at each pillar extended from its top, and complex conjugation. ∎

Proof. We first record the geometry that permits the comparison of the two scales. The definitions of the weighted lengths and separation give

$$\begin{aligned} d'(x, z) \ge d^\flat(x, z), \qquad d'(x, z) \ge \xi_p^{-1} |x - z|_\infty, \\ |\lambda^\flat(x) - \lambda^\flat(z)| \le 1 + d^\flat(x, z)/\rho, \qquad |\lambda'(x) - \lambda'(z)| \le 1 + d'(x, z)/\rho, \\ \varsigma(x)/\varsigma(z) \le L^{2+2d'(x,z)/\rho}. \end{aligned} \tag{1242}$$

For the first inequality, subdivide a polygonal path at the faces of both layer partitions. Away from shared faces, its tower density is at least the ambient density. On a shared face, move the segment an arbitrarily small distance into a bare cube of the deeper ambient layer, and join its endpoints by two short segments. The added weighted length tends to zero. Taking the path infimum proves the assertion, including paths running along faces. The lower bound by $\xi_p^{-1}|x-z|_\infty$ follows from the smallest tower density. To cross from layer m to layer $m' > m+1$, a connected path crosses, for each $m < q < m'$, a strip of width $\rho\xi_q$ in layer q. Each strip contributes at least ρ to its weighted length. This proves the two index bounds and then the ratio bound.

If $|x-z|_\infty \le 3l(x)$, separation also gives $|\lambda^\flat(x)-\lambda^\flat(z)| \le 1$ and $d^\flat(x,z) \le 9L$. A coordinate path stays in the preceding layer and proves the latter inequality. A site and its tower top are within distance $3/2$. If $b \in \mathsf{P}^q(c)$, the two endpoint blocks of c lie in Ω'_{q-1}, and their coordinate paths give

$$d'(x_b, y_c) \le 5L/2, \qquad \xi_q \le L\xi'(x_b). \tag{1243}$$

Here is the real estimate. If $\epsilon_W < \alpha$, $L^2\alpha \le a_{\mathrm{tw}}^{(\mathrm{ia})}$, and $c^\natural\alpha \le \varepsilon_{\mathrm{T}}$, curvature averaging makes $M_{\mathbf{B}'}W$ $c^\natural\alpha$-regular. More precisely, a tested plaquette of order q averages bare plaquettes in Ω'_{q-1}. On its four-corner block union their curvatures obey $L^{2q}|\partial W - 1| \le L^2 \max \varsigma^{-2}\epsilon_W < a_{\mathrm{tw}}^{(\mathrm{ia})} \le c_2^{\mathrm{SO}}$. The local four-corner induction used in lemma 66.1, at terminal level q, therefore bounds its holonomy by $c^\natural \max \varsigma^{-2}\epsilon_W$ there. These plaquettes are within tower distance $7L/2$ of a corner of the test. The profile estimate in the proof of corollary 53.10, followed by (1242), therefore gives

$$\varsigma(x_e)^2 \mathsf{e}_{U_0}(e) \le C_{\mathrm{R}} \max_f e^{-\delta d'(x_e,x_f)/2}\epsilon_W(f). \tag{1244}$$

Here U_0 is the real critical representative and e_{U_0} measures curvature and current in the tower block scale. Indeed, the square of the scale ratio costs at most $L^4 e^{4(\log L)d'/\rho}$; the condition $\delta\rho \ge 64\log L$ leaves the displayed decay.

Apply this to $W = U$. By (1236), $V_0 = M_{\mathbf{B}'}U$ lies in the continuation and response domains. Integrating (838) gives, with $\Sigma(\beta;x) = \sum_c e^{-\delta d'(x,y_c)}|\beta(c)|$,

$$N_b(\mathcal{H}(B_1) - \mathcal{H}(B_0)) \le \min\{B_{\mathrm{cx}}^\sharp \sup|B_1 - B_0|, B_5 e^{\delta d'(x,y_b)}\Sigma(B_1 - B_0; x)\}. \tag{1245}$$

Here y_b is the pillar top of the initial point of b. The norm N_b includes $\xi'|\mathcal{H}|$, $(\xi')^2|\nabla\mathcal{H}|$, and, at full bonds, $(\xi')^3|D^{\mathsf{T}}D\mathcal{H}|$.

The gauge factor in $\Phi(B) = (e^{i\eta\mathcal{H}(B)}U_0)^{h(B)}$ depends, at a site, only on the potential in its pillar. For a pillar of level q, its product of ascent factors has total exponent norm at most

$$3(L-1)\tau\sum_{m<q} L^{m-q} + 2\kappa_6\tau\sum_{m<q} L^{m-q} \le (\kappa_6+3)\tau, \qquad \tau = \xi_q \sup_{\mathsf{P}(B^q(y))}|\mathcal{H}|.$$

The first sum counts the comb edges; the second counts each averaging correction once. Thus, on $|\mathcal{H}|_{(-1)} < \rho_{\mathrm{f}}$, $|h - I| < 1/4$, $|h^{-1}| < 5/4$, and the condition number of h is less than $3/2$. Applying Cauchy's estimate to this local pillar map, with radius $\rho_{\mathrm{f}}/2$, yields

$$|h(B_1)(x)h(B_0)(x)^{-1} - I| \le \min\{5/8,\ C_{\mathrm{h}}\Sigma(B_1 - B_0; x)/3\}.$$

The site-to-top displacement costs at most $e^{3\delta/2}$, already included in C_{h}.

Let $\mathcal{F}(B)$ be curvature or current in the tower scale at e, so that $G = \varsigma(x_e)^2\mathcal{F}$. The base curvature and current in each local plaquette star are at most $1/22$, by (1236). Express the potential in that local scale and put $N_{\mathrm{loc}}(f) = \max\{|f|, |\nabla f|, |D^{\mathsf{T}}Df|\}$, omitting the last member for curvature. The expansions (843) and (844) show that the unconjugated curvature or current has norm less than one on $N_{\mathrm{loc}}(f) < 1/300$. At level zero the same bound follows directly from the four exponential factors, giving curvature at most $1/22 + 5/300$ and current at most $4e^{2/300}(1/22 + 5/300) < 1$. Cauchy's estimate on the half ball gives Lipschitz constant 600. The passage from N_b to the local star costs

at most L^3, and its diameter costs $e^{7\delta/2}$. Combining this with the preceding gauge estimate gives

$$|\mathcal{F}(B)| < 3/2, \qquad |\mathcal{F}(B_1) - \mathcal{F}(B_0)| \le [C_{\mathrm{X}} + C_{\mathrm{h}}|\mathcal{F}(B_0)|]\Sigma(B_1 - B_0; x_e). \tag{1246}$$

We now pass from the datum to the original potential. Write $\|w\|' = \max_b \mathsf{L}_-(b)|w(b)|$, where L_- is the smaller endpoint block side in the tower. The averaging theorem proposition 51.1 and (1243) show that $B(w)$ is holomorphic on $\|w\|' < r_{\mathrm{w}}$, with $\sup|B(w)| < r_{\mathrm{T}}$. Cauchy's estimate on the half averaging ball gives

$$|B(w_1)(c) - B(w_0)(c)| \le 2\mathsf{c}_Q L \max_{b \in \mathsf{P}^q(c)} \varsigma(x_b)^{-1} l(x_b)|w_1(b) - w_0(b)|. \tag{1247}$$

This also proves existence and holomorphy of the actual tower on the domain in the statement.

Fix a settled e, and write $x = x_e$, $l = l(x)$, $\varsigma = \varsigma(x)$, and $F(w) = G(U, w; e)$. Let $\chi(b)$ be one for $|x_b - x|_\infty \le l$, zero for $|x_b - x|_\infty \ge 2l$, and linear between. Call bonds within distance $3l$ near. The local geometry above and $|\chi(b^+) - \chi(b)| \le \eta/l$ give

$$\mathsf{n}(\chi v) \le C_\chi \mathsf{n}^{\mathrm{nr}}(v), \qquad \|\chi v\|' \le L\varsigma^{-1}\mathsf{n}(\chi v), \qquad \mathsf{n}^{\mathrm{nr}}(v) \le e^{9L\delta/4} E_v^{\delta_{\mathrm{T}}}(x). \tag{1248}$$

For real g with $\mathsf{n}(g) \le \bar{a}$, (843) gives $\epsilon_{e^{ig}U} \le 2\epsilon_U + 3\mathsf{n}_p(g)$. Consequently (1244) gives

$$|F(g)| \le C_{\mathrm{R}}\{2E_U^{\delta_{\mathrm{T}}}(x) + 3\mathsf{n}(g)\}, \qquad |F(0)| \le C_{\mathrm{R}} E_U^{\delta_{\mathrm{T}}}(x).$$

For a holomorphic map bounded by M_0 on a radius-r ball, Taylor's formula and Cauchy's estimate imply

$$|F(z)| \le 3|F(0)| + |F(\Re z)| + |F(\Im z)| + 6M_0\|z\|^2/r^2 \quad (\|z\| \le r/2). \tag{1249}$$

Indeed, each remainder after the linear term is bounded by $2M_0\|z\|^2/r^2$; substitute the real and imaginary parts into the complex-linear first derivative. Apply this with $M_0 = 3\varsigma^2/2$, $r = r_{\mathrm{w}}$, and $z = \chi w$. The factor ς^2 cancels against $\|\chi w\|'^2$. Since $\mathsf{n}(\chi w) \le \bar{a} \le r_{\mathrm{w}}/2$, we obtain

$$|F(\chi w)| \le 7C_{\mathrm{R}} E_U^{\delta_{\mathrm{T}}}(x) + (6C_{\mathrm{R}} + 5L^2/r_{\mathrm{w}}) C_\chi \mathsf{n}^{\mathrm{nr}}(w). \tag{1250}$$

For a change $u = w - \widetilde{w}$ supported outside the distance-l neighborhood, put $D = d'(x, x_b)$. Settlement gives $\varsigma \le 1 + D$: if $\varsigma > 1$, the tower has ended at p, and $D > l/\xi_p = \varsigma$. Moreover,

$$\frac{\varsigma}{\varsigma(x_b)} \le L^2 e^{\delta D/32}, \qquad d'(x, y_c) \ge D - 5L/2.$$

Use these inequalities in (1247) and (1246). Split off one quarter of the exponential for summation and use $(1 + D)e^{-\delta D/4} \le 4/\delta$. The remaining exponential is at least the decay $\delta/4$ required for the ambient profile. The row sum in proposition 12.1, with the site-to-top cost $e^{3\delta/8}$, yields

$$|F(w) - F(\widetilde{w})| \le [C_{\mathrm{X}} + C_{\mathrm{h}}|F(\widetilde{w})|] C_{\mathrm{far}} \max_b e^{-\delta_{\mathrm{T}} d'(x, x_b)} l(x_b)|u(b)|. \tag{1251}$$

For $C_\chi \mathsf{n}(w) \le \bar{a}$, (1250) is at most one. Apply (1251) with $\widetilde{w} = \chi w$, and then (1248). The definition of K_{a} gives (1240).

Finally write $v = w_1 - w_0$ and split the change of F into the change from $w_0 + \chi v$ to w_1, and the change from w_0 to $w_0 + \chi v$. The first is bounded by $(C_{\mathrm{X}} + C_{\mathrm{h}}) C_{\mathrm{far}} E_v^{\delta_{\mathrm{T}}}(x)$. For the second let $n_v = C_\chi \mathsf{n}^{\mathrm{nr}}(v)$ and $E = E_U^{\delta_{\mathrm{T}}}(x) + E_{w_0}^{\delta_{\mathrm{T}}}(x)$. If $n_v = 0$ it vanishes. Otherwise the disc $|z| \le 1 + E/n_v$ satisfies $C_\chi \mathsf{n}(w_0 + z\chi v) \le 5C_\chi^2 a_{\mathrm{T}} = \bar{a}$. The bound already proved on this disc, followed by the one-variable Schwarz estimate for $F(w_0 + z\chi v) - F(w_0)$, gives a bound $4K_{\mathrm{a}} n_v$. If $E = 0$, the same bound follows directly at the endpoints. Equation (1248) and the definition of K_{T} prove (1241). The reality and covariance statements follow first for real data from uniqueness of the axial representative, and then throughout the connected complex domain by holomorphy. □

Corollary 58.15. Let $\Omega_\bullet^\flat$ have depth t, let $1 \le p \le i \le t$, and let X be a region for π_0. Assume $M_1 \ge \rho$, $M = L^{m'} \ge L^3 M_1$, and $m' \le m + K - p + 1$. Use either tower rule and localize by $\Omega_q' = \Omega_q^\flat \sqcap X_{[q]}$, $q \le p$. Put $n_i(x) = \min\{\lambda_{\Omega_\bullet^\flat}(x), i\}$ and $l_i(x) = \eta L^{n_i(x)}$. On the domain (1235) with

$l = l_i$, the localized real and complex towers exist. If an inner plaquette or full bond e has its base within distance $2\xi_p$ of $X^{\wr p}$, define

$$G_{p,X}(U, A'; e) = \begin{cases} L^{2n_i(x_e)}(\partial U_{p,X}(e) - I), & e \text{ a plaquette}, \\ L^{2n_i(x_e)+\min\{p, n_i(x_e)\}-3p} J_{p,X}(e), & e \text{ a full bond}, \end{cases}$$

where $J_{p,X}$ is in level-p units. Then

$$\begin{aligned} &|G_{p,X}(U, A'_1; e) - G_{p,X}(U, A'_0; e)| \\ &\quad \le e^{3\delta_{\mathrm{T}}} K_{\mathrm{T}} \max_b e^{-\delta_{\mathrm{T}} d_{\mathbf{B}^\flat}(y_e, y_b)} \max\{l_i(x_b)|A'_1 - A'_0|(b), l_i(x_b)^2 |\nabla_U (A'_1 - A'_0)|_{\mathsf{Z}}(b)\}. \end{aligned} \tag{1252}$$

The tops y_e, y_b and the distance in this formula belong to the full ambient sequence of depth t, while $\mathsf{Z} = \mathrm{Bd}^\circ(\Omega_0^\flat \sqcap X)$. In particular the estimate applies when a corner of e lies in $X^{\wr p}$. ■□

Proof. The localized sequence is separated because its complement at level q is the union of the two complementary regions. Under the first tower rule the accumulated inward erosion to level q is at most $2\rho \sum_{r<q} \xi_r \le M_1 \xi_q$; under the second it is at most $2M_1 \sum_{1 \le r \le q} \xi_r \le 3M_1 \xi_q$. On the other hand, $\mathrm{dist}_\infty(X^{\wr p}, X^c) \ge 2M\xi_{p-1}$. Thus a site within $2\xi_p$ of this interior remains in every $X_{[q]}$, $q \le p$, since $(2M/L - 2)\xi_p > 3M_1 \xi_p$. At such sites the localized and ambient memberships agree, so the element is settled. The scale-invariant current is $\eta^3 J$, which gives the displayed power of L. Apply (1241). Passing to the deeper ambient sequence can only decrease the weighted distance. Moving each endpoint to its ambient pillar top costs at most $3/2$, and therefore costs $e^{3\delta_{\mathrm{T}}}$. A corner and the base of an element are within distance $\eta \le 2\xi_p$, proving the last assertion. □

58.8. **The margin for substitution into an earlier domain.** Write $K^\sharp = K^\sharp_{\mathrm{tw}}$, and use $C_{\mathrm{os}} = 18$ from proposition 51.10. Let $t = k + 1 \ge 2$. In this subsection the radii table is independent of the structure: its entries are $\mathsf{r}_\iota(n)$, $\iota = 0, 1$. Assume

$$\begin{aligned} \mathsf{r}_0(n) \le \mathsf{r}_1(n), \qquad \mathsf{r}_1(n)^2 \le \mathsf{c}_\natural \mathsf{r}_0(n), \qquad \mathsf{c}_\natural \ge 1, \\ \mathsf{r}_\iota(n) \le \mathsf{r}_\iota(n+1), \qquad L^{-\nu} 2^\nu \mathsf{r}_\iota(n+\nu) \le \mathsf{r}_\iota(n), \qquad (1+\beta_{\mathrm{s}})\mathsf{r}_\iota(n) \le \hat{a}. \end{aligned} \tag{1253}$$

The indices range over the entire table, with $\nu \ge 1$ and $n + \nu$ in its range. These are the conditions (R0), (R1), (Q5.2), and (Q6). Let $n_i(x) = \max\{q \le i : x \in \Omega_q^\diamond\}$, where $\Omega_0^\diamond = \Omega_1^\sim$, and put

$$\mathsf{c}_0 = \tfrac{1}{2}(1 - \theta_{\mathrm{s}})\beta_{\mathrm{s}}, \qquad \mathsf{w}(n) = \mathsf{c}_0\, 2^{-(k+1-n)} \mathsf{r}_0(n), \qquad \mathsf{v}_k(b) = \mathsf{w}(n_k(b_-)). \tag{1254}$$

For a level-j pair $\mathfrak{h} = (\mathbf{A}, \mathbf{K})$, $\mathbf{U} = e^{i\eta_j A'} U$, and a positive profile v, set

$$\|\mathfrak{h}\|^{\mathcal{S}}_{\mathsf{v},\mathbf{U}} = \max_{b \in \mathrm{Bd}_{\mathcal{S}}(X)} \frac{\max\{L^{n_k(b)}\eta_j |\mathbf{A}(b)|, L^{2n_k(b)}\eta_j^2 |\nabla_{\mathbf{U}} \mathbf{A}|(b), L^{3n_k(b)}\eta_j^3 |\mathbf{K}(b)|\}}{\mathsf{v}(b)}. \tag{1255}$$

Only positively oriented bonds enter the maximum, and only differences defined in $\mathrm{Bd}_{\mathcal{S}}(X)$ enter its second member. In particular this norm imposes no condition outside that bond set.

Write $\sigma^{\mathrm{m}} = (t, \mathrm{mid})$. For $\sigma' \preccurlyeq \sigma^* = (t^*, \phi^*) \preccurlyeq (k, \mathrm{out})$, the coefficients in (1184) satisfy

$$[\mathsf{c}_{\sigma^*}(n) - \mathsf{c}_{\sigma^{\mathrm{m}}}(n)]\mathsf{r}_\iota(n) \ge 2\mathsf{w}(n) \qquad (n \le t^*). \tag{1256}$$

Indeed their difference is at least $\beta_{\mathrm{s}} 2^{-(t-n)}(2^{t-t^*} - 1 - \theta_{\mathrm{s}}) \ge (1 - \theta_{\mathrm{s}})\beta_{\mathrm{s}} 2^{-(t-n)}$. The same gap holds for the pointwise right members:

$$\mathsf{R}^{p,\iota}_{\sigma^*}(x) - \mathsf{R}^{p,\iota}_{\sigma^{\mathrm{m}}}(x) \ge 2\mathsf{w}(n_{t^*}(x)) L^{-pn_{t^*}(x)}, \qquad p = 1, 2, 3. \tag{1257}$$

If $n_t(x) \le t^*$, this is (1256). Otherwise put $n_t(x) = t^* + \nu$. The smaller-scale right member is at most $(5/4)2^{-\nu} L^{-(p-1)\nu} \mathsf{r}_\iota(t^*) L^{-pt^*}$, whereas the other is at least $\mathsf{r}_\iota(t^*) L^{-pt^*}$. Their difference is at least $3\mathsf{r}_\iota(t^*) L^{-pt^*}/8$, and $2\mathsf{w}(t^*) \le \mathsf{r}_0(t^*)/8$.

The profile is increasing, satisfies $\mathsf{w}(n-1) \le \mathsf{w}(n)/2$, and obeys

$$\mathsf{w}(n+\nu) L^{-p(n+\nu)} \le \mathsf{w}(n) L^{-pn}, \qquad p = 1, 2, 3.$$

Consequently the norm bound with index n_k also holds with every truncated index n_i, $i \le k$. Furthermore, with $\omega_{\mathrm{T}} = \delta_{\mathrm{T}}\rho$,

$$e^{-\omega_{\mathrm{T}}(\nu-1)}\mathsf{w}(n+\nu) \le C^*\mathsf{w}(n), \qquad C^* = L. \tag{1258}$$

This follows from (Q5.2) and $\omega_{\mathrm{T}} \ge 16\log L$. If, for some $C_\uparrow \ge 1$,

$$2^\nu e^{-\omega_{\mathrm{T}}(\nu-1)}\mathsf{r}_0(n+\nu) \le C_\uparrow \mathsf{r}_0(n) \qquad (n \ge 0,\ \nu \ge 1,\ n+\nu \le k),$$

where n, ν are integers, one may use $C^* = \min\{L, C_\uparrow\}$.

Proposition 58.16. Assume the geometric hypotheses of corollary 58.15 and of the outer-core substitution proposition 51.10, $M \ge L^4 M_1$, and (1253). Let $\sigma' = (j, \phi') \preccurlyeq \sigma^* \preccurlyeq (k, \mathrm{out})$, $j \ge 1$, $X \in \mathcal{D}_{j-1}$, and $P \in \widetilde{U}_{\sigma'|\sigma^{\mathrm{m}}}(X)$. For a shift $\mathfrak{h}$ put $r = \|\mathfrak{h}\|^{\mathcal{S}}_{\mathsf{v}_k,\mathbf{U}}$. Suppose

$$\hat{a} \le a_{\mathrm{T}}, \quad 4\hat{a} \le a^\sharp, \qquad r \le \tfrac{1}{2}, \qquad 4C^*K_{\mathrm{T}}r \le 1, \qquad L^2 C_{\mathrm{os}} K^\sharp (1+\mathsf{c}_\natural) r \le 3, \qquad L^3 r \le 4. \tag{1259}$$

Then substitution is defined on $\mathrm{Bd}_{\mathcal{S}}(X)$. If it is defined on all of $\mathrm{Bd}(X)$, then

$$P \dotplus \mathfrak{h} \in \widetilde{U}_{\sigma'|\sigma^*}(X).$$

■

Proof. Put $\eta = \eta_j$, $\mathfrak{e} = \mathrm{Bd}_{\mathcal{S}}(X)$, and

$$A'' = (i\eta)^{-1}\log(e^{i\eta\mathbf{A}}e^{i\eta A'}).$$

We first control this logarithm and its first differences. For $\ell(X,Z) = \log(e^X e^Z) - X - Z$, the exponential and logarithm series give $|\ell(X,Z)| \le 2(|X|+|Z|)^2$ when $|X|+|Z| \le 1/4$. Cauchy's estimate on the ball of radius $|X|+|Z|$ about (X,Z) gives a derivative bound $8(|X|+|Z|)$ on the half ball. Since $\ell(0,Z) = 0$, the one-variable Schwarz estimate also gives

$$|\ell(X,Z) - \ell(X,Z')| \le 16|X|\,|Z-Z'| \quad (|X|,|Z|,|Z'| \le 1/16).$$

For the latter assertion apply Schwarz on $|z| < 1/16$ to $\ell(zX/|X|,Z) - \ell(zX/|X|,Z')$, bounded there by $|Z-Z'|$.

These estimates imply

$$\eta|A''-A'|(b) \le [1 + 8\eta(|\mathbf{A}(b)| + |A'(b)|)]\eta|\mathbf{A}(b)|. \tag{1260}$$

For a difference based at x, write $b = \langle x, x+\eta e_\nu\rangle$, $e = \langle x, x+\eta e_\mu\rangle$, and $b^+ = b + \eta e_\mu$. With $s = \eta|A'(b^+)| + \max\{\eta|\mathbf{A}(b)|, \eta|\mathbf{A}(b^+)|\}$, transport all terms by $U(e)$. Splitting the change in ℓ first in its first variable and then in its second gives

$$\begin{aligned}\eta^2|\nabla_U(A''-A')|(b) \le (1+8s)\{\eta^2|\nabla_{\mathbf{U}}\mathbf{A}|(b) + (e^{2\eta|A'(e)|}-1)\eta|\mathbf{A}(b^+)|\}\\ + 16\eta|\mathbf{A}(b)|\eta^2|\nabla_U A'|(b).\end{aligned} \tag{1261}$$

The exponential term changes transport from $\mathbf{U}(e)$ to $U(e)$. This derivation includes $\mu = \nu$.

Now $\hat{a} \le (64L^2)^{-1}$, and $\eta|\mathbf{A}| \le r\hat{a}/16$ on $\mathfrak{e}$. Thus the logarithm is defined there. Neighboring sites have layer indices differing by at most one, by separation. Using the profile relations above in (1260) and (1261), for every $i \le k$ and $n = n_i(x)$, gives

$$\begin{aligned}L^n\eta|A''-A'| \le \tfrac{3}{2}r\mathsf{w}(n), \qquad L^{2n}\eta^2|\nabla_U(A''-A')| \le \tfrac{3}{2}r\mathsf{w}(n),\\ |\partial(e^{i\eta\mathbf{A}}\mathbf{U}) - I| \le |\partial\mathbf{U} - I| + 2.03r\mathsf{w}(n)L^{-2n}.\end{aligned} \tag{1262}$$

For clarity, the coefficient for the derivative is bounded by $(1+10^{-3})(1+1.05L\hat{a}) + 16\hat{a} < 3/2$: the neighboring shift is at most $(L/2)r\mathsf{w}(n)L^{-n}$, and $e^{2\eta|A'(e)|} - 1 \le 2.1\hat{a}L^{-n}$. For the last line use (843) with condition number 2. The linear term is at most $2r\mathsf{w}(n)L^{-2n}$; its two remainders are at most $(11/1024 + 6/832)r\mathsf{w}(n)L^{-2n}$. Multiplication by $|\partial\mathbf{U}| \le 1+\hat{a}$ gives 2.03.

Assume now that substitution is defined on all bonds of X. The resulting triple is $P'' = (U, A'', \mathbf{J}+\mathbf{K})$. The first three increments in (L1) are bounded, in their local units, by $r\mathsf{w}$, $3r\mathsf{w}/2$,

and $3r\mathsf{w}/2$; the curvature increment is bounded by $2.03r\mathsf{w}$. Since $r \le 1/2$, (1257) absorbs all of them. Real curvature is unchanged. This proves (L1), including $\eta|A''| < \hat{a}L^{-n_{t^*}}$ and $\eta^2|\nabla_U A''| < \hat{a}L^{-2n_{t^*}}$. The real gauges in (L2) are unchanged.

For (L3), fix an address of level i, its region Z, a rule, and a depth $q \le i$. The meet determining set is exactly the localization in corollary 58.15: its members are $\Omega_r^\diamond \sqcap Z_{[r]}$, $r \le q$. Its inner bonds lie in $\mathfrak{e}$. The three pairs (U, A'), (U, A''), and $(U, 0)$ belong to $\mathcal{Q}^{\mathrm{T}}(a_{\mathrm{T}})$, by (L1), the bounds just proved, and $n_i \le n_t, n_{t^*}$. The tower constructed there is the tower in (L3): its averaged datum is the same in the real and complex senses wherever the field is real, by proposition 51.1; changing the normalization keeps $\eta A'$ and $\eta^3 J$ fixed.

At a place f of (L3), with $\bar{n} = n_i(x_f)$, the inequality for depth q and interior index $q \le p \le i$ is exactly

$$|G_{q,Z}(U, A'; f)| < L^{i-p}\mathsf{c}_{\sigma^{\mathrm{m}}}(\bar{n})\mathsf{r}_0(\bar{n}). \tag{1263}$$

For currents this uses the power $2\bar{n} + \min\{q, \bar{n}\} - 3q$ in corollary 58.15. The place has a corner in $Z^{\wr p}$, which is contained in $Z^{\wr q}$, so that the localized estimate applies even at its incident boundary stencils.

If $n_i(b) > \bar{n}$, crossing the intervening layers costs at least $\rho(n_i(b) - \bar{n} - 1)$ in weighted distance; moving both sites to their tops costs at most three. If $n_i(b) \le \bar{n}$, monotonicity of the profile suffices. Thus (1258) gives

$$e^{-\delta_{\mathrm{T}} d(y_f, y_b)}\mathsf{w}(n_i(b)) \le e^{3\delta_{\mathrm{T}}} C^* \mathsf{w}(\bar{n}).$$

Equations (1252) and (1262) bound the tower increment by

$$\tfrac{3}{2} e^{6\delta_{\mathrm{T}}} K_{\mathrm{T}} C^* r\mathsf{w}(\bar{n}) \le \tfrac{3}{8} e^{3/2}\mathsf{w}(\bar{n}) < 2\mathsf{w}(\bar{n}).$$

The coefficient gap (1256), with $L^{i-p} \ge 1$, absorbs it in (1263). The real tower is unchanged. This proves all of (L3), including existence.

Finally fix a core address (σ'', i, X'). Set

$$b_\iota = \mathsf{c}_{\sigma^*}(i)\mathsf{r}_\iota(i), \qquad \tau = 1 - \frac{\mathsf{c}_{\sigma^{\mathrm{m}}}(i)}{\mathsf{c}_{\sigma^*}(i)}.$$

Then $\tau b_0 \ge 2\mathsf{w}(i)$, $b_1^2 \le (5\mathsf{c}_\natural/4)b_0$, and $2(b_0 + b_1) \le 4\hat{a} \le a^\sharp$. An incident bond of $X' \subset \Lambda_i$ can start in layer $i - 1$. This boundary case gives the bounds

$$|\mathbf{A}_{(i)}| \le \tfrac{L}{2} r\mathsf{w}(i), \quad |\nabla_{\mathbf{U}}\mathbf{A}_{(i)}| \le \tfrac{L^2}{2} r\mathsf{w}(i), \quad |\mathbf{K}_{(i)}| \le \tfrac{L^3}{2} r\mathsf{w}(i)$$

after restriction to level i; here $\mathsf{w}(i-1) \le \mathsf{w}(i)/2$. The allowable potential radius in proposition 51.10 is at least

$$\frac{8\mathsf{w}(i)}{5C_{\mathrm{os}}K^\sharp(1 + \mathsf{c}_\natural)}.$$

The last two inequalities in (1259) put the potential and derivative strictly inside this radius and put the current increment within τb_0. Apply proposition 51.10 to the core of the restricted input. Restriction commutes with substitution because $\eta_i L^{-(j-i)} = \eta_j$, and the current scales by the cube of this factor. The resulting core is precisely the one required in (L4) for σ^*. □

For later use take

$$\mathsf{c}_{\mathrm{rm}} = (C_{\mathrm{os}}L^3)^{-1}, \qquad \mathsf{r}_\natural = \mathsf{c}_{\mathrm{rm}} \min\{1, K_{\mathrm{T}}^{-1}, [K^\sharp(1 + \mathsf{c}_\natural)]^{-1}\}. \tag{1264}$$

Corollary 58.17. If $\hat{a} \le \mathsf{c}_{\mathrm{rm}} \min\{a_{\mathrm{T}}, a^\sharp\}$, then $\|\mathfrak{h}\|^{\mathcal{S}}_{\mathsf{v}_k, \mathbf{U}} < \mathsf{r}_\natural$ implies the conclusions of proposition 58.16 with $\sigma^* = \sigma'$. In particular, if $P^+ \in \widetilde{U}_{(t,\mathrm{mid})}(Y)$, $X \subset Y$, and $\sigma' = (j, \phi') \preccurlyeq (k, \mathrm{out})$, then

$$\rho_X^{t\to j} P^+ \dot{+} \rho_X^{k\to j}\mathfrak{h}^+ \in \widetilde{U}_{\sigma'}(X)$$

whenever the restricted shift has this norm bound and the substitution is defined on $\mathrm{Bd}(X)$.

For a core on X whose incident bonds have both endpoints in Ω_j, put $\mathsf{v}_j^\sharp(b) = b_0 L^{n_k(b)-j}$. If $P \in \mathcal{U}_j^\circ(X;(1-\tau)b_0,(1-\tau)b_1)$ and the radii satisfy the hypotheses of proposition 51.10, then

$$\|\mathfrak{h}\|^{\mathcal{S}}_{\mathsf{v}_j^\sharp,\mathbf{U}} < \frac{\tau}{C_{\mathrm{os}} K^\sharp (1+\mathsf{c}_\natural)} \tag{1265}$$

implies that substitution is defined and belongs to $\mathcal{U}_j^\circ(X;b_0,b_1)$. ■□

Proof. Use $C^* = L$ in proposition 58.16. The definition (1264) gives $r < L^{-3}$, $4LK_{\mathrm{T}} r < 4L^{-2} < 1$, and $L^2 C_{\mathrm{os}} K^\sharp (1+\mathsf{c}_\natural) r < L^{-1}$, which verify (1259). Restriction of a graded triple preserves (L1)–(L4): $\eta A'$, $\eta^2 \nabla A'$, and $\eta^3 \mathbf{J}$ are unchanged; every new tower and core address was already an address of the larger region; and a containing gauge cube in (L2) restricts to the requested smaller cube. Thus the restricted input lies in $\widetilde{U}_{\sigma'|(t,\mathrm{mid})}(X)$, as required.

For the core assertion write $\nu_b = n_k(b) - j \geq 0$. The norm bound gives $|\mathbf{A}| < b_2$, $|\nabla_{\mathbf{U}} \mathbf{A}| < L^{-\nu_b} b_2 \leq b_2$, and $|\mathbf{K}| < L^{-2\nu_b} b_2 \leq b_2$, where $b_2 = \tau b_0 / [C_{\mathrm{os}} K^\sharp (1+\mathsf{c}_\natural)]$. In particular $b_2 \leq \tau b_0/2$, so the current stays below b_0. Apply proposition 51.10. □

58.9. **Re-expanding an earlier term.** We write $C_{\mathcal{A}} = \mathsf{C}_{\mathcal{A}}$ for the fixed field-cutoff constant. We next apply the substitution margin to a term at an earlier level. There are two domains. A regular term on $X \subset \Lambda_j$ is a function of the pair on the fixed core $\mathcal{U}_j^\circ(X;b_0,b_1)$, where $(b_0,b_1) = (\alpha_0,\alpha_1)$ or

$$b_\iota = b_\iota^{\mathrm{oc}} = \frac{\theta_{\mathrm{oc}}}{1+\beta_{\mathrm{s}}} \alpha_\iota .$$

A boundary term on a layered output space at (j,X), or a formed exponent with $j = l+1$, is a function on $\widetilde{U}_{(j,\mathrm{out})}(X)$, together with its stored coordinates of levels $< j$, respectively level l. We use the plain substitution for a regular term and the pinned substitution for a boundary term on a layered space or a formed exponent. All stored coordinates remain fixed.

Write $X^\uparrow$ for the union of level-k fine cubes meeting X. For $Y \supset X^\uparrow$, put

$$\Phi_Y[f](s,\lambda;P,\mathcal{A},\mathfrak{q}) = f\big(\rho_X^{t\to j} P \mathbin{\dot{+}_{\mathcal{S}}} \lambda \rho_X^{k\to j} \mathfrak{h}_Y(s;P,\mathcal{A});\mathfrak{q}_X\big).$$

In the pinned case replace $\lambda\mathfrak{h}_Y$ by $(\lambda\mathfrak{h}_Y)^\delta[P]$ and denote the function by $\Phi_Y^\delta[f]$. Here $\dot{+}_{\mathcal{S}}$ performs the logarithmic substitution on $\mathrm{Bd}_{\mathcal{S}}(X)$ and leaves the other bond components unchanged. This convention is needed because the graded spaces place no bound on those other components. The restrictions of stored coordinates are $\mathfrak{q}_X = (\mathfrak{q}|_X)|^{<j}$, or $(\mathfrak{q}|_X)|^{(l)}$, in the two cases with stored coordinates.

The localized increment is

$$\mathbb{V}^\star[f](Y;\mathfrak{u}) = [\,Y \setminus X^\uparrow \mid X^\uparrow\,]^* \Phi_Y^\star[f](\,\cdot\,,1;\mathfrak{u}), \tag{1266}$$

where $\star$ indicates the plain or pinned choice. It is zero when $X^\uparrow \not\subset Y$, on the domain of the regional shift. The starred bracket is the one of eq. (806); in particular it subtracts the value at zero parameters.

Let $\mu = \mu_{\mathfrak{h}}$ and $\delta_* = \mu M/4$. In this subsection the profile radii $r_\iota(n) = \mathsf{r}_\iota((k,\mathrm{out});n)$ are those of the previous output structure. Bounds on the input triple use the radii of (t,mid). Retain the comparison and margin hypotheses of corollary 58.17 between these two structures and every earlier output structure. In addition to the substitution conditions above, assume the radius comparison

$$2^\nu e^{-\delta_* L(\nu-1)} r_0(n+\nu) \leq C_\uparrow r_0(n) \quad (0 \leq n < n+\nu \leq k). \tag{1267}$$

For regular terms assume $(1+\beta_{\mathrm{s}})\mathsf{r}_\iota((t,\mathrm{mid});n) \leq (1-\tau)b_\iota$, $0 < b_0 \leq b_1$, $b_1^2 \leq \mathsf{c}_\natural b_0$, and $2(b_0+b_1) \leq a^\sharp$. The source-cube ceilings of proposition 58.6 are imposed with $\mathsf{c}_{\mathrm{KL}} = \max\{\mathsf{K}_{\mathrm{I}}, L^3\}^{-1}$:

$$(1+\beta_{\mathrm{s}})\mathsf{r}_\iota((t,\mathrm{mid});n) \leq a_{\mathrm{m}} \quad (0 \leq n \leq t), \qquad (1+\beta_{\mathrm{s}})\mathsf{r}_\iota((t,\mathrm{mid});n) \leq \mathsf{c}_{\mathrm{KL}} a_{\mathfrak{h}} \quad (n = k,t).$$

Also assume

$$(1+\beta_{\mathrm{s}})\mathsf{r}_{\iota}((t,\mathrm{mid});n)\le c_{\mathrm{rt}},\qquad (1+\beta_{\mathrm{s}})\mathsf{r}_1(\sigma;n)\le \log 2,\qquad a_{\flat}\le \min\{a_{\mathrm{m}},c_{\mathrm{rt}}\},\qquad \mathsf{C}_{\mathcal{A}}\delta_k\le \varepsilon_{\natural}.$$

The first inequalities give the regularity required by the levelling construction; the logarithm bound is imposed at both the input and earlier output structures σ. For example, at a top of level q, the scaled bounds are $\xi_q L^{-1}|A'|$, $\xi_q^2 L^{-1}|\nabla^{\xi}A'|$, and $\xi_q^3 L^{-3}|\mathbf{J}|\le c_{\mathrm{rt}}$, by (L1). For each turning cell of order q, the containing cube in (L2) supplies a gauge with the same bounds; rescale its potential by ξ_q^{-1}. Thus proposition 58.10 applies to these graded triples.

The distance from a term to the new active region is

$$\mathsf{d}_k(X)=M^{-1}d_{\mathbf{B}}(X,\Omega_t).$$

Put $\mathsf{a}=\delta_k/\alpha_0$ for a regular term and $\mathsf{a}=\delta_k/r_0(k)$ for a term on a layered output space. With $\mathsf{c}_0=(1-\theta_{\mathrm{s}})\beta_{\mathrm{s}}/2$, define

$$\mathsf{m}_*=\max\left\{\frac{C_{\mathrm{m}}C_{\uparrow}}{\mathsf{r}_{\natural}\mathsf{c}_0},\frac{L}{\epsilon},\frac{C_{\mathrm{m}}(1+\beta_{\mathrm{s}})C_{\mathrm{os}}K^{\sharp}(1+\mathsf{c}_{\natural})}{\tau\theta_{\mathrm{oc}}}\right\},\tag{1268}$$

$$\mathsf{C}_{\mathrm{sb}}=2\mathsf{C}C_{\Sigma}e^{(2C_{\mathrm{m}}+4)\mu}\mathsf{m}_*,\qquad \mathsf{H}_{\mathrm{sb}}=\mathsf{C}_{\mathrm{sb}}\mathsf{C}_{\mathcal{A}}e^{\mathsf{C}_{\mathrm{sb}}\bar{\kappa}}.\tag{1269}$$

Here $C_{\Sigma}=\mathsf{S}(\mu/2)$ is the finite weighted row-sum constant, and $\mathsf{c}_{\natural}=\mathsf{C}e^{\mathsf{C}\bar{\kappa}}$. For $0<\delta_{\mathrm{sb}}\le\delta_*$, set

$$\varrho_X=[\,\mathsf{H}_{\mathrm{sb}}\mathsf{a}e^{-\delta_{\mathrm{sb}}\mathsf{d}_k(X)}\,]^{-1}.\tag{1270}$$

We interpret $e^{-\infty}=0$, so that $\varrho_X=\infty$ if no active source is reachable. The constant C_{sb} is independent of the height, volume, history, couplings, M, and $\bar{\kappa}$. Its dependence is on the displayed fixed analytic and levelling constants.

Proposition 58.18. Under these conditions, the substituted triple belongs to the domain of f, for every $P\in\widetilde{U}_{(t,\mathrm{mid})}(Y)$, $\sup|\mathcal{A}|<\mathsf{C}_{\mathcal{A}}\delta_k$, $s\in\mathbb{D}(Y)$, and $|\lambda|<\varrho_X$. In the pinned case the substitution is locally holomorphic on each fixed-real-factor slice at its restriction to $\mathrm{Bd}(X)$.

Suppose $\varrho_X>1$, and that f is analytic, bounded by b_f, gauge invariant with its stored coordinates, and has the tie for real triples. Then $\mathbb{V}^{\star}[f]$ is defined on every admitted stored-coordinate tier, is analytic there, and

$$|\mathbb{V}^{\star}[f](Y;\mathfrak{u})|\le 2b_f\varrho_X^{-1}e^{-(\bar{\kappa}-1)\#(Y\setminus X^{\uparrow})}.\tag{1271}$$

It vanishes at $\mathcal{A}=0$ and unless Y contains a source cube. It is gauge invariant and has the tie; it is Borel, conjugation real, or a function of the pair if f has the corresponding property. The construction is linear in f. For a formed exponent, the same assertions hold with $b_f=\operatorname{osc}f$, assuming invariance up to its stored coordinates.

If all restrictions of a point $\mathfrak{u}$ on the torus belong to the indicated tiers, then

$$\sum_{Y\supset X^{\uparrow}}\mathbb{V}^{\star}[f](Y;\mathfrak{u}|_Y)=\Phi^{\star}_{T_\ell}[f](\mathbf{1},1;\mathfrak{u})-f(\rho_X^{t\to j}P;\mathfrak{q}_X).\tag{1272}$$

■

Proof. First consider triples with $\eta_t|A'(b)|<\log 2$ at every bond of $\mathrm{Bd}(X)\setminus\mathrm{Bd}_{\mathcal{S}}(X)$. At such a bond the shift is zero and $\log(e^{i\eta_t A'(b)})=i\eta_t A'(b)$. Thus the ordinary substitution is defined there and agrees with $\dot{+}_{\mathcal{S}}$. On the bonds of the label, corollary 58.17 gives substitution room

$$r_{\mathrm{room}}=\mathsf{r}_{\natural}\quad\text{or}\quad r_{\mathrm{room}}=\frac{\tau}{C_{\mathrm{os}}K^{\sharp}(1+\mathsf{c}_{\natural})}$$

for the graded and fixed-core domains, respectively. Restriction from the mid-level space to an earlier output space preserves (L1)–(L4), as in the proof of corollary 58.17.

We verify the decay needed to use this room. Let T_X be the positive-level tops whose pillars contain an endpoint of a bond in $\mathrm{Bd}(X)$. For every top y, put

$$\mathsf{D}_y = \min_{c\in\mathfrak{X}^+} d_{\mathbf{B}}(y, c_-), \qquad \mathsf{D}(b) = \mathsf{D}_{y(b)}.$$

Call a level-k determining bond live if it has an endpoint in Ω_t. For each such bond c, let y_c^{lv} be its endpoint in Ω_t; if both endpoints lie there, take the initial endpoint in the positive orientation. Define

$$\mathsf{D}^{\mathrm{lv}}(b) = \min_{c \text{ live}} d_{\mathbf{B}}(y(b), y_c^{\mathrm{lv}}).$$

An empty minimum is $+\infty$. For every live bond, $d_{\mathbf{B}}(c_-, y_c^{\mathrm{lv}}) \le 1$. For a bond incident to X, with $n = n_k(b_-)$, the triangle inequality and the pillar length bound give

$$M\mathsf{d}_k(X) \le \tfrac{5}{2} + d_{\mathbf{B}}(y(b), y_c^{\mathrm{lv}}).$$

Crossing the $k-n$ separated layers gives the second bound

$$8LM(k-n) \le \tfrac{3}{2} + d_{\mathbf{B}}(y(b), y_c^{\mathrm{lv}}).$$

Indeed every layer crossing traverses a strip of width $8LM$ in its own units; adding these disjoint pieces is exactly the weighted length defining $d_{\mathbf{B}}$. Taking the mean and then the minimum over live bonds yields

$$e^{-\mu\mathsf{D}^{\mathrm{lv}}(b)/2} \le e^{\mu} e^{-\delta_*\mathsf{d}_k(X)} e^{-8\delta_* L(k-n)}. \tag{1273}$$

Here $\mathsf{D}(b) \ge \mathsf{D}^{\mathrm{lv}}(b) - 1$. Divide by $\mathsf{v}_k(b) = \mathsf{c}_0\, 2^{-(k+1-n)} r_0(n)$ and apply (1267). For a core use $\mathsf{v}_j^\sharp(b) = b_0 L^{n-j} \ge b_0$. For a top whose pillar meets an endpoint of a bond incident to X, choose that bond: its top is at weighted distance less than $3/2 + 13/12 + 3/2 < 4.1$. Consequently

$$\max_{b\in\mathrm{Bd}_{\mathcal{S}}(X)^+} \frac{e^{-\mu\mathsf{D}(b)/2}}{\mathsf{v}_k(b)} \le \frac{C_\uparrow e^{4\mu}}{\mathsf{c}_0 r_0(k)} e^{-\delta_*\mathsf{d}_k(X)}, \tag{1274}$$

$$\max_{b\in\mathrm{Bd}_{\mathcal{S}}(X)^+} \frac{e^{-\mu\mathsf{D}(b)/2}}{\mathsf{v}_j^\sharp(b)} \le b_0^{-1} e^{4\mu} e^{-\delta_*\mathsf{d}_k(X)}, \tag{1275}$$

$$\max_{y\in\mathsf{T}_X} e^{-\mu\mathsf{D}_y/2} \le e^{4\mu} e^{-\delta_*\mathsf{d}_k(X)}. \tag{1276}$$

All these maxima are zero if their sets are empty.

The regional response estimate, summed over source bonds, gives

$$|\mathfrak{h}_Y|_{\mathbf{U},Y}(b) \le c_* e^{-\mu\mathsf{D}(b)/2}, \qquad c_* = 2C_\Sigma \mathsf{c}_{\mathfrak{h}} \mathsf{C}_{\mathcal{A}} \delta_k.$$

For pinning, use the slowly varying profile $c_* e^{3\mu/4} e^{-\mu d_{\mathbf{B}}(x,\mathfrak{X})/2}$ in proposition 58.10. Moving from a site to its top costs at most $3/2$ in the weighted metric. It follows that

$$|(\lambda\mathfrak{h}_Y)^\delta|_{\mathbf{U},Y}(b) \le |\lambda| C_{\mathrm{m}} e^{2C_{\mathrm{m}}\mu} c_* e^{-\mu\mathsf{D}(b)/2}, \tag{1277}$$

$$\widehat{\varepsilon}_y(\lambda\mathbf{A}_Y) \le |\lambda| L e^{2C_{\mathrm{m}}\mu} c_* e^{-\mu\mathsf{D}_y/2}. \tag{1278}$$

The top estimate follows also by applying the profile bound at the positive bare bond beginning at the top inside its pillar.

The three entries of m_* now give, for $|\lambda| < \varrho_X$,

$$\|(\lambda\mathfrak{h}_Y)^\delta|_X\|_{\mathsf{v},\mathbf{U}} < r_{\mathrm{room}}, \qquad \max_{y\in\mathsf{T}_X} \widehat{\varepsilon}_y(\lambda\mathbf{A}_Y) < \epsilon.$$

For graded terms use $\delta_k/\mathsf{a} = r_0(k) \le 1$; for regular terms use $\delta_k/\mathsf{a} = \alpha_0 \le 1$ and $\alpha_0/b_0 \le (1+\beta_{\mathrm{s}})/\theta_{\mathrm{oc}}$. These are exactly the two factors allowed by (1268). The first inequality proves membership, and the second forces every endpoint pillar needed at $\mathrm{Bd}(X)$ to be levelled. The local holomorphy clause of proposition 58.10 therefore applies on those bonds. No holomorphy of the threshold map on the entire region is asserted. One may take local extensions of the response into the linear space of

shifts vanishing off $\mathfrak{e}_Y$; projection onto that space preserves the actual response and keeps these extensions admissible for the pinning map.

For fixed stored coordinates, the resulting function $\Phi^\star_Y[f]$ is bounded by b_f on the λ-disc of radius ϱ_X. Subtract its value at $\lambda = 0$. Schwarz's lemma and the starred Cauchy estimate (807) give

$$|\mathbb{V}^\star[f](Y)| \le \frac{2b_f}{\varrho_X}(e^{\bar\kappa} - 1)^{-\#(Y\setminus X^\uparrow)}.$$

This implies (1271) for $\bar\kappa > \log 2$. If only the oscillation is bounded, subtract $f(P_0; \mathfrak{q}_X)$ for any one admitted P_0. The starred bracket removes this constant. Its choice need not be measurable, since it is used only to prove the bound.

We justify the tie for pinned substitution before passing to the full space. The estimates in (1277) hold throughout the larger center polydisc $|\zeta(a)| < 2c_{\mathrm{m}}\varepsilon_{y(a)}$ of proposition 58.10; thus the same margin admits every substitution with these independent center values. By lemma 58.12, initial sites of stored bonds are tops or stored sites of pinnable tops. Turns agree there for inputs with one pair.

For real input triples with one pair and a real shift, let $u, \widetilde{u}$ be their turns and $Q, \widetilde{Q}$ their substituted triples. Then $w = \widetilde{u}u^{-1}$ is unitary, equals one at every stored initial site, and $\mathbf{p}(\widetilde{Q}) = \mathbf{p}(Q^w)$. Gauge invariance admits Q^w. The tie identifies its value with that at $\widetilde{Q}$; invariance and the identity of w on the stored sites identify it with the value at Q. For fixed real input triples these readings are holomorphic on the connected product of the $s, \lambda, \mathcal{A}$ discs. Equality at real points extends throughout this product.

For a function of the pair, allow complex input triples and use their common independent center variables ζ. The same margin admits both substituted triples for every ζ; their readings are holomorphic in ζ. At real ζ, the unitary w argument uses pair dependence in place of the tie. Analytic continuation gives equality at the common actual pin:

$$\Phi^\delta[f](\widetilde{P}, \mathfrak{h}; \mathfrak{q}) = \Phi^\delta[f](P, \mathfrak{h}; \mathfrak{q}) \quad \text{if } \mathbf{p}(\widetilde{P}) = \mathbf{p}(P). \tag{1279}$$

For a plain local term, with no stored data, the same continuation compares the actual pin with the zero center. Thus plain and pinned readings agree.

For completeness, we remove the restriction on the outer bond components. Let ν_R set A' and $\mathbf{J}$ equal to zero on $\mathsf{Out}(R) = \mathrm{Bd}(R)^+ \setminus \mathrm{Bd}_{\mathcal{S}}(R)$, preserving the real factor. Membership in a graded space is unchanged. The regional response and its domain are unchanged on the label: the natural-class data, all determining averages, source cubes and pivot rows use inner bonds only. The same is true of pinning, since every positive-level pillar and its reading set consist of inner bonds. Thus the substituted triples T_X satisfy

$$T_X(s, \lambda; \nu_Y P, \mathcal{A}) = \nu_X T_{X|h'}(s, \lambda; P, \mathcal{A}), \tag{1280}$$

with simultaneous definedness.

A bounded analytic f on a layered output space does not depend on its outer A' and $\mathbf{J}$ coordinates. Each such coordinate ranges over a whole complex line in that space. The restriction of f, or of $f - f(P_0; \mathfrak{q}_X)$ in the oscillation case, is bounded and entire on that line, and hence constant. Applying this successively to the finitely many outer coordinates proves $f(P; \mathfrak{q}_X) = f(\nu_X P; \mathfrak{q}_X)$. Consequently

$$\mathbb{V}^\star[f](Y; \mathfrak{u}) = \mathbb{V}^\star[f](Y; \nu_Y \mathfrak{u}). \tag{1281}$$

The right side is evaluated at a triple satisfying the earlier $\log 2$ restriction. This proves membership and the bound on the full tiers. Composition with this coordinate projection gives local holomorphic extensions and Borel measurability; the projection commutes with restriction, gauge transformations and conjugation.

There is one additional point for the tie: ν_Y need not preserve equality of pairs. For two real input triples with the same pair, choose at each outer bond $Z_b \in \mathfrak{g}$, $|Z_b| \le \pi$, with $e^{iZ_b} = U_1(b)U_2(b)^{-1}$.

Join their normalized real factors in five steps. The intermediate triples with outer components

$$(e^{inZ_b/5}U_2(b),0,0),\qquad (e^{i(n-1)Z_b/5}U_2(b),\eta_t^{-1}Z_b/5,0)$$

have the same pair for $1\le n\le 5$. They belong to the same graded space, and $\pi/5<\log 2$ keeps the second triples in the already treated set. Equality of the two potential values follows from the tie there; normalization then joins successive steps. This proves the tie on the full space. The same argument with arbitrary input triples proves the pair property when f is a function of the pair.

Finally, decoupling the parameters splits the response into its wall components. Pinning preserves that splitting and the support of each component, by the margin and restriction clauses of proposition 58.11. Components meeting neither the base nor a source can therefore be removed before taking the bracket. This proves the stated vanishing and locality. Summing the finite Möbius differences over connected domains containing the base leaves exactly the full-parameter value minus the zero-parameter value. Domain consistency identifies these values on the torus, and the shift at zero parameters is zero. This proves (1272). Gauge covariance, conjugation, the tie on the restricted set and Borel measurability follow by composing the corresponding response and pinning properties with those of f; finite differences preserve them. Linearity is immediate from the bracket. □

58.10. **Counting earlier domains.** A domain of level $j<k$ is admitted if it meets Ω_{j+1}^{c}; at level k it is admitted if it meets $Z_k^{\sim}$. Let n_X be the largest $n\le k$ for which X and Ω_n share a bare cube, and set $n=\max(j,n_X)$, $p=k-n$, $q=n-j$. A domain is placed with integer $\nu\ge 0$ if $d_j(X)\ge 2L^{n-j-\nu}-2$ whenever $j<n\le k$ and X shares a bare cube with $\mathcal{L}_n$. For $a'\ge 0$, put $D_0=D_1=0$, $D_q=8L^q-2$ for $q\ge 2$, and

$$C_{\square}(L,a')=C_{\mathrm{E}}\left(\sum_{p\ge 0}(1+p)^{a'}L^{-5p}\right)\left(\sum_{q\ge 0}(1+q)^{a'}L^{3q}e^{-D_q}\right). \tag{1282}$$

For $0<\delta<1/2$, put $\kappa'=(1-2\delta)\kappa$, $r_V=(1-2\delta)(\kappa-2)$. Assume

$$a=2\delta\kappa\ge 2(c_{\mathrm{E}}+1),\qquad \bar\kappa-1\ge\kappa,\qquad \delta_{\mathrm{pc}}L\ge 10\log L.$$

Use the smaller distance rate needed in each application for δ_{pc}. Set

$$\begin{aligned}&E_q=\max(2L^{q-\nu}-2,0),\quad u_p=2^p e^{-\delta_{\mathrm{pc}}Lp/2},\quad v_q=e^{-aE_q/2},\\ &\varpi_i=L^{3i}\sum_{p+q=i}u_pv_q,\qquad C_{\mathrm{h}}=8C_{\mathrm{E}}\max\{1,[2(1-2\delta)]^{-1}\},\quad C_{\mathrm{h}}'=3(\nu+1)^2L^{4\nu}.\end{aligned} \tag{1283}$$

Lemma 58.19. Write $m(X,Y)=\#_{\pi_k}(Y\setminus X^{\uparrow})$; all sums below have $X^{\uparrow}\subset Y$. For admitted domains,

$$\sum_{j,X}(1+k-j)^{a'}e^{-\kappa d_j(X)-\delta_{\mathrm{pc}}\mathsf{d}_k(X)-(\bar\kappa-1)m(X,Y)}\le 8C_{\square}(L,a')e^{-(\kappa'-1)d_k(Y)}. \tag{1284}$$

This conclusion needs only $a\ge c_{\mathrm{E}}+1$. For placed domains and nonnegative numbers B_j,

$$\begin{aligned}\sum_{j,X}B_j(d_j(X)+1)e^{-\kappa d_j(X)-\delta_{\mathrm{pc}}\mathsf{d}_k(X)-(\bar\kappa-1)m(X,Y)}&\le C_{\mathrm{h}}\sum_{j=0}^{k}\varpi_{k-j}B_j\,e^{-r_Vd_k(Y)},\\ \sum_{i\ge 0}(i+1)L^i\varpi_i&\le C_{\mathrm{h}}'.\end{aligned} \tag{1285}$$

Under $(\mathrm{CR})_t$, an admitted domain has a point at sup distance at least $7LM\check R_t$ from Ω_t. Thus a domain Y containing its base and a source cube satisfies

$$d_k(Y)\ge 7L\check R_t-2. \tag{1286}$$

■

Proof. Truncate the distance to Ω_r^c at its collar width. Along a connected path from X to Ω_t, this 1-Lipschitz function traverses its whole range; the portion strictly between its endpoint values lies in $\mathcal{L}_r$. Its length is at least the collar width. These layer portions have disjoint interiors, so their weighted lengths add, giving $\mathsf{d}_k(X) \geq 8L(k - n_X) \geq 8Lp$. If X is admitted and $q \geq 2$, its points cross the collar between Ω_{n-1}^c and Ω_n. The diameter bound gives $d_j(X) \geq 8L^q - 2 = D_q$. For a placed domain it gives $d_j(X) \geq E_q$: maximality of n_X ensures that X shares a bare cube with $\mathcal{L}_{n_X}$.

Through a fixed level-j cube, proposition 45.1 bounds the admitted sum by $C_{\mathrm{E}} \sum_{p+q=k-j} L^{-8p} e^{-D_q}$. A fine cube contains $L^{3(k-j)}$ such cubes. Use $(1+p+q)^{a'} \leq (1+p)^{a'}(1+q)^{a'}$ and sum to obtain $C_\square(L, a')$ per fine cube. The attachment inequality $d_k(Y) \leq d_j(X) + m(X, Y)$ gives

$$\kappa d_j(X) + (\bar{\kappa} - 1) m(X, Y) \geq \kappa' d_k(Y) + a d_j(X).$$

Pin X by a small cube in Y, use $\#_{\pi_k} Y \leq 8(d_k(Y) + 1)$, and then $d + 1 \leq e^d$. This proves (1284).

For placed domains absorb $d_j(X) + 1$ into $e^{d_j(X)}$, split $a - 1 = (a/2 - 1) + a/2$, and use $a/2 - 1 \geq c_{\mathrm{E}}$. The rooted sum is bounded by $C_{\mathrm{E}} \sum_{p+q=k-j} u_p v_q$. The same pinning argument with $(d+1)e^{-2(1-2\delta)d} \leq \max\{1, [2(1-2\delta)]^{-1}\}$ proves the first inequality in (1285). For the second, $L^{4p} u_p \leq (2/L)^p$. The terms $q \leq \nu$ in $\sum (q+1) L^{4q} v_q$ total at most $(\nu+1)^2 L^{4\nu}$; the rest is at most $(\nu+1) L^{4\nu} \sum_{s \geq 1} (s+1) e^{-26s}$. Together with $\sum (p+1)(2/13)^p < 1.4$, this gives C'_{h}.

For $j < k$ an admitted domain meets the old excluded region; for $j = k$ it meets $Z_k^\sim$. The next collar, using $(\mathrm{CR})_t$ in the latter case, leaves distance at least $7LM\check{R}_t$. Join that point to a point of a source cube in Y and use $|x - y|_\infty \leq M(d_k(Y) + 2)$. □

58.11. **The four historical potential families.** Use the actual regional response of proposition 58.5, its pinned version of proposition 58.10, and the full graded domains of proposition 58.18. All potentials are restricted to $B \in \mathbb{D}^{\mathrm{f}}_{x_k}(Y)$, $\mathcal{A} = g_k B$, and the prescribed stored tiers, including when a defining sum is empty. Throughout these families the counterterm is the actual running increment $b_j = L^{-1} x_{j-1} - x_j$, with $|b_j| \leq B_\sharp$.

Write

$$\widetilde{E}^{(j)}(X, P, z) = E^{(j)}(X, P, z) - b_j \mathbf{1}_{\{X = X_z\}} A^{\eta_j}(h_z^{(j)}, \mathbf{U}).$$

For $z \in (\Lambda_j^{\sim -1})^{(j)}$, let $n = \mathsf{ol}_j(z)$ be the largest index among j and the $j < n \leq k$ with $X'_n(z) \subset \Omega_{n+1}$. Call $X \ni z$ narrow if $n > j$ and $X \subset \square_n(z)^{\sim 2}$, and wide otherwise. The regular list consists of the wide members $\widetilde{E}^{(j)}(X, \cdot, z)$, the octet members

$$f_{j,z} = \sum_{X:\ z \in X \subset \square_n(z)^{\sim 2}} \left[(\widetilde{E}^{(j)}_{X,z})^{\square_n(z)} - \widetilde{E}^{(j)}_{X,z}(\mathbf{1}, 0) \right], \tag{1287}$$

and the edge tents $-b_j A^{\eta_j}(\varphi_j h_y^{(j)}, \cdot)$, $y \notin \Lambda_j^{\sim -1}$. The octet sum is formed before taking its starred bracket; its domain is the octet core at level n on $X'_n(z)$.

The boundary list has the plain members $E^{(j),\mathrm{ed}}_{\mathcal{S}}(X, \cdot) = \sum_{z \notin \Lambda_j^{\sim -1}} E^{(j)}(X, \cdot, z)$ on fixed cores, and the members $\mathscr{B}_h(j, X, \mathrm{mk})$ on layered spaces with stored levels $< j$. The formed list has the old factor exponents φ_f at $j_f = l_f + 1$ on X_f, with stored level l_f only. Restrict each old partial map to its stated space before forming a member.

For a bare plaquette with base in $\mathcal{L}_m$, let $X(p; m)$ be the union of level-m cubes containing its corners, and put

$$W_w^{(m)}(X, \mathbf{U}) = \eta_k^{-1} \sum_{\substack{x_{\mathrm{b}}(p) \in \mathcal{L}_m \\ X(p;m) = X}} w(p) a_p(\mathbf{U}), \qquad p_w = \sum_{j=1}^{k} L^{j-k} b_j (1 - \varphi_j). \tag{1288}$$

Use plain brackets for Wilson, regular and edge members, and pinned brackets for boundary members on layered spaces and for formed members. With only bases contained in Y, define

$$\begin{aligned} \mathbb{V}^{\mathrm{E}} &= \sum_{\text{regular}} \mathbb{V}[f], & \mathbb{V}^{\mathrm{c}} &= -\sum_{m,X} \mathbb{V}[W_{p_w}^{(m)}(X)], \\ \mathbb{V}^{\mathrm{B}} &= \sum_{\text{ed}} \mathbb{V}[f] + \sum_{\text{mk}} \mathbb{V}^{\delta}[f], & \mathbb{V}^{\Phi} &= \sum_{f} \mathbb{V}^{\delta}[\varphi_f]. \end{aligned} \tag{1289}$$

Let $\mathbb{V}^{\mathrm{act}} = \sum_{m,X} \mathbb{V}[W_1^{(m)}(X)]$ and $\mathbb{V}^{\mathrm{A}} = -x_k \mathcal{R}_3 \mathbb{V}^{\mathrm{act}}$. At $k = 0$ use the direct plaquette difference of the completed pivot field. The Jacobian and gauge potentials are the formulas of eqs. (813) and (814), with the regional nonpivot set and pivot completion, continued by zero on unused bonds. The four origins are

$$\mathbb{V}^{\mathrm{int}} = \mathbb{V}^{\mathrm{J}} + \mathbb{V}^{\mathrm{G}} + \mathbb{V}^{\mathrm{A}}, \qquad \mathbb{V}' = \mathbb{V}^{\mathrm{E}} + \mathbb{V}^{\mathrm{c}}, \qquad \mathbb{V}^{\mathrm{B}}, \qquad \mathbb{V}^{\Phi}. \tag{1290}$$

58.12. **The action and coupling potentials.** Set

$$\epsilon_A = \min\{r_P^{\circ}, \epsilon_{\mathfrak{h}}, (3\mathsf{c}_{\mathfrak{h}})^{-1}\}, \quad \epsilon_P = \epsilon_A/2, \quad \delta_{\mathrm{W}} = M \min\{\mu_{\mathfrak{h}}, 1\},$$

with $\epsilon_A = r_P^{\circ}$ and $\mathsf{c}_{\mathfrak{h}} = 1$ at $k = 0$. Take

$$\begin{aligned} K_{\flat} &= \max\{18/r_P^{\circ} + 12C_P + 6,\ 4C_P/(r_P^{\circ})^2\}, \\ K_A &= \max\{672e(1 + 2\mathsf{c}_{\mathrm{pl}})^2(L^4 + 3), 4512C_P\}, \qquad C_{\mathrm{seg}} = 8C_P. \end{aligned} \tag{1291}$$

Here $\mathsf{c}_{\mathrm{pl}} = 600$ is the plaquette product constant. In addition to the preceding domain and rate conditions, require

$$\begin{aligned} &\mathsf{o}L \geq 1, \qquad L^3(1 + \beta_{\mathrm{s}})\mathsf{r}_{\iota}((t, \mathrm{mid}); n) \leq a_P^{\circ} \quad (n = k, t,\ \iota = 0, 1), \\ &(1 + \beta_{\mathrm{s}})\mathsf{r}_0((t, \mathrm{mid}); n) \leq 1 \quad (0 \leq n \leq k), \qquad C_{\mathcal{A}}\delta_k \leq \epsilon_P. \end{aligned} \tag{1292}$$

Proposition 58.20. On the actual graded domain,

$$|\mathbb{V}^{\mathrm{act}}(Y)| \leq K_A \mathsf{c}_{\mathfrak{h}} M^3 \sup|\mathcal{A}| e^{-r_V d_k(Y)}, \tag{1293}$$

$$|\mathbb{V}^{\mathrm{A}}(Y)| \leq 2K_A \mathsf{c}_{\mathfrak{h}} \epsilon_A^{-2} M^3 x_k \sup|\mathcal{A}|^3 e^{-r_V d_k(Y)}, \tag{1294}$$

$$|\mathbb{V}^{\mathrm{int}}(Y)| \leq K_* \delta_k e^{-r_V d_k(Y)} \left(1 + M^{-3} \sum_{b \in \mathfrak{X}^{\circ +}\langle Y \rangle} |B(b)|^2\right), \tag{1295}$$

where $K_* = (2K_{\flat} + 2K_A \mathsf{c}_{\mathfrak{h}} \epsilon_A^{-2}) M^6 C_{\mathcal{A}}$. The action and gauge remainders vanish to order three, the Jacobian potential to order one. They are analytic, Borel, gauge invariant, conjugation real and functions of the pair, and vanish without a source cube. ∎

Proof. First verify the domain at which the estimates are used. If a plaquette has base in $\mathcal{L}_m$, its graded layer index is m, except that at $m = k$ it may be t. Conditions (L1) and (1292) give $|\partial \mathbf{U}(p) - 1| < L^{-2m}$. On any wall-connected union of at most four fine cubes containing a source cube all bond endpoints have indices k or t. Multiplication of each graded bound of order $q \leq 3$ by L^{qt} costs at most L^3; thus the bounds of $\mathcal{Q}_t(\cdot; a_P^{\circ})$ hold. For its cube gauges, apply (L2)(a) at level k and set $A = LA_k$; its first difference is $L^2 \nabla^{\eta_k} A_k$. The same ceiling supplies the required level-t gauge. Finally the actual regional response is defined for every field of norm less than $\epsilon_{\mathfrak{h}}$, by propositions 58.5 and 58.6. This proves the auxiliary domain assertions for all points of the graded domain, without imposing conditions on their outer coordinates.

An occupied Wilson member contains at most $12M^3 L^{3m}$ plaquettes; its base has at most four fine cubes and linear size zero. For the interpolated action set

$$\rho = e\, \mathsf{c}_{\mathfrak{h}} \sup|\mathcal{A}| e^{-\delta_{\mathrm{W}} \mathsf{d}_k(X)} < 1.$$

The layerwise response bound and the four-factor plaquette estimate give, for $|\lambda|\rho \le 1$,

$$|\partial(e^{i\lambda\eta_k \mathbf{A}}\mathbf{U})(p) - 1| \le (1 + 2\mathsf{c}_{\mathrm{pl}})L^{-2m}.$$

The identity $|1 - \frac{1}{2}\operatorname{tr} Z| \le \frac{1}{2}|Z-1|^2$ for $Z \in \mathrm{SL}(2,\mathbb{C})$ gives the bound $6(1+2\mathsf{c}_{\mathrm{pl}})^2 M^3 L^{k-m}$ on the Wilson member. The starred Cauchy estimate on the disc $|\lambda| < \rho^{-1}$ therefore yields

$$|\mathbb{V}[W_1^{(m)}(X)](Y)| \le 12e(1+2\mathsf{c}_{\mathrm{pl}})^2 \mathsf{c}_{\mathfrak{h}} M^3 \sup|\mathcal{A}| L^{k-m} e^{-\delta_{\mathrm{W}} \mathsf{d}_k(X) - (\bar{\kappa}-1)m(X,Y)}. \tag{1296}$$

When $\rho = 0$, take arbitrarily large discs instead. There are at most $7\#_{\pi_k} Y\, L^{3(k-m)}$ occupied members at level m. Moreover $m(X,Y) \ge d_k(Y)$ and $\mathsf{d}_k(X) \ge 8L\max(k-m-1,0)$. The geometric sum

$$\sum_{i\ge 0} L^{4i - 8\max(i-1,0)} \le L^4 + 3$$

and $\#Y \le 8(d_k(Y)+1)$ prove (1293). At $k=0$, the pivot completion satisfies $\sup|B'| \le 2C_P \sup|\mathcal{A}|$; telescoping the four plaquette factors bounds their action difference by $376 C_P \sup|\mathcal{A}|$. Multiplication by $12M^3$ gives the other term in K_A.

For $\mathcal{A} \ne 0$, the function $\phi(z) = \mathbb{V}^{\mathrm{act}}(Y; P, z\mathcal{A})$ is holomorphic on $|z| < T = \epsilon_A/\sup|\mathcal{A}| \ge 2$ and bounded by $K_A \mathsf{c}_{\mathfrak{h}} M^3 \epsilon_A e^{-r_V d_k(Y)}$. Cauchy's estimate for its tail from degree three gives $2T^{-3}$ times this bound, proving (1294). The regional pivot calculation is the same calculation on each pivot block with the unused nonpivot fields zero; source-cube admission proved above supplies its domain. Hence

$$|\mathbb{V}^{\mathrm{J}}| \le K_{\flat} M^3 \sup|\mathcal{A}|, \qquad |\mathbb{V}^{\mathrm{G}}| \le K_{\flat} M^3 x_k \sup|\mathcal{A}|^3.$$

These terms have $d_k(Y) = 0$ on their support. Since $x_k g_k^2 = 1$, $x_k \sup|\mathcal{A}|^3 \le C_{\mathcal{A}} \delta_k \sum |B|^2$, proving (1295). Finite brackets preserve analyticity and symmetries. Taylor coefficients are limits of finite differences at real scalar multiples of $\mathcal{A}$, so they are Borel as well. The zero-field and no-source claims follow from the response vanishing there. □

For the coupling potential use the enlarged layer radii

$$a_m = (1+\beta_{\mathrm{s}})\mathsf{r}_0((t,\mathrm{mid});m) \quad (m<k), \qquad a_k = (1+\beta_{\mathrm{s}})\max\{\mathsf{r}_0((t,\mathrm{mid});k), L^{-2}\mathsf{r}_0((t,\mathrm{mid});t)\},$$

and put

$$\vartheta^a(\epsilon) = 3\mathsf{c}_{\mathfrak{h}}\epsilon \max_{0\le m\le k} a_m^{-1} e^{-\delta_{\mathrm{W}} L \max(k-m-1,0)/2}. \tag{1297}$$

Assume $a_m \le 1$ and $\vartheta^a(C_{\mathcal{A}}\delta_k) \le 1$. Strict field cutoff then makes the value at $\sup|\mathcal{A}|$ strictly less than one.

Proposition 58.21. Suppose $|b_j| \le B_\sharp$, and write $\bar{B} = B_\sharp L/(L-1)$. With ϖ^c from (1283) at $\nu = 1$, $\delta_{\mathrm{pc}} = \delta_{\mathrm{W}}/2$,

$$|\mathbb{V}^{\mathrm{c}}(Y)| \le K_c \vartheta^a(\sup|\mathcal{A}|) M^3 \bar{B} \sum_{m=0}^{k} \varpi^c_{k-m} L^{k-m} a_m^2 e^{-r_V d_k(Y)}, \quad K_c = 12(1+2\mathsf{c}_{\mathrm{pl}})^2 C_{\mathrm{h}}. \tag{1298}$$

If $c_- \alpha_{0,m} \le a_m \le c_+ \alpha_{0,m}$, $\alpha_{0,m} = C_0 g_m y_m^q$, $x_K \ge 2\bar{B}$, $y_K \ge \log(2L)$, and $4\sqrt{L}\,\mathsf{c}_{\mathfrak{h}} c_-^{-1} C_{\mathcal{A}} \vartheta_{0,k} \le 1$, then

$$|\mathbb{V}^{\mathrm{c}}(Y)| \le K_c' \mathsf{c}_{\mathfrak{h}} \frac{c_+^2}{c_-} C_{\mathcal{A}} \vartheta_{0,k} M^3 \bar{B} C_0^2 g_k^2 y_k^{2q} e^{-r_V d_k(Y)}, \tag{1299}$$

where $K_c' = 8\sqrt{L}\, K_c C_{\mathrm{h}}' \sup_{i\ge 0}(1+i)^{2q-1} L^{-i}$. ■

Proof. The proof of (1296) with curvature radius a_m bounds the interpolation by

$$6(1+2\mathsf{c}_{\mathrm{pl}})^2 M^3 \|p_w\|_\infty L^{k-m} a_m^2.$$

Its scalar radius can be taken as

$$a_m/[e\mathsf{c}_{\mathfrak{h}} \sup|\mathcal{A}| e^{-\delta_{\mathrm{W}} \mathsf{d}_k(X)}].$$

Splitting the distance exponential in half and using (1297) bounds its reciprocal by

$$\vartheta^{a}(\sup|\mathcal{A}|)e^{-\delta_{\mathrm{w}}\mathsf{d}_k(X)/2}.$$

Occupied domains have $d_m(X)=0$, are placed with $\nu=1$, and $\|p_w\|_\infty\le\bar B$. Apply (1285) to obtain (1298).

The running-coupling bounds give $L^{k-m}g_m^2\le 2g_k^2$, $y_m\le(1+k-m)y_k$, and $\alpha_{0,k}/\alpha_{0,m}\le\sqrt{3/2}\,L^{(k-m)/2}$. The distance floor absorbs the latter growth in (1297), leaving $\vartheta^{a}(\sup|\mathcal{A}|)\le 4\sqrt{L}\,\mathsf{c}_{\mathfrak{h}}c_-^{-1}C_{\mathcal{A}}\vartheta_{0,k}$. Use $\sum(i+1)L^i\varpi_i^{c}\le C_{\mathrm{h}}'$ and $(1+i)^{2q}\le[\sup(1+i)^{2q-1}L^{-i}](i+1)L^i$. This proves (1299). □

58.13. **Bounds for the inherited terms.** Put

$$\mathsf{s}_k^{\mathrm{fx}}=\delta_k/\alpha_0,\qquad \vartheta_k^{\mathsf{Tab}}=\delta_k/\mathsf{r}_0((k,\mathrm{out});k),$$

and require $\mathsf{H}_{\mathrm{sb}}\max\{\mathsf{s}_k^{\mathrm{fx}},\vartheta_k^{\mathsf{Tab}}\}\le 1/2$. The radius comparisons in proposition 58.18 are imposed for every earlier output structure used below.

Assume the old pointed terms are analytic local terms on the fixed cores, with $\|E^{(j)}\|_{j,\kappa}\le\mathsf{g}_j$, and $|b_j|\le B_\sharp$. Set

$$\begin{aligned}
\mathsf{g}_j'&=\mathsf{g}_j+\alpha_0^2B_\sharp,\qquad & C_{\mathrm{ert}}^{\flat}&=L^4C_{\mathrm{oc}}\max\{C_{\mathrm{E},2},1176\},\\
C_{\mathrm{ert}}&=8M^3e^{26\kappa}C_{\mathrm{ert}}^{\flat},\qquad & B_m^{\mathrm{E}}&=C_{\mathrm{ert}}\left(\mathsf{g}_m'+\sum_{j<m}L^{-(m-j)}\mathsf{g}_j'\right),\\
\mathsf{size}^{\mathrm{E}}&=\sum_{m=1}^{k}\varpi_{k-m}B_m^{\mathrm{E}}.
\end{aligned}\tag{1300}$$

Here and for the regular list use $\nu=3$, $\delta_{\mathrm{pc}}=\delta_{\mathrm{sb}}$ in the placed count.

Proposition 58.22. Assume $\kappa>\kappa_{\mathcal{D}}+2$ and the preceding rate and domain conditions. The regular potential is defined on the full graded domain. Its wide and octet parts satisfy

$$|\mathbb{V}^{\mathrm{E,wd}}(Y)+\mathbb{V}^{\mathrm{E,oc}}(Y)|\le C_VC_{\mathcal{A}}\mathsf{s}_k^{\mathrm{fx}}\mathsf{size}^{\mathrm{E}}e^{-r_Vd_k(Y)},\qquad C_V=2C_{\mathrm{h}}\mathsf{C}_{\mathrm{sb}}e^{\mathsf{C}_{\mathrm{sb}}\bar\kappa},\tag{1301}$$

where $\mathsf{size}^{\mathrm{E}}\le[L/(L-1)]C_{\mathrm{h}}'C_{\mathrm{ert}}\sup_j\mathsf{g}_j'$. If $(\mathrm{CR})_t$ and $\mathsf{r}_0((j,\mathrm{out});j)^2\le r_\star^2(1+k-j)^{a'}$, its edge-tent part satisfies

$$|\mathbb{V}^{\mathrm{E,et}}(Y)|\le C_V'C_{\mathcal{A}}\vartheta_k^{\mathsf{Tab}}M^3B_\sharp r_\star^2e^{-r_Vd_k(Y)},\tag{1302}$$

with $C_V'=3e^{26\kappa}c(a',L)C_{\mathrm{h}}C_{\mathrm{h}}'\mathsf{C}_{\mathrm{sb}}e^{\mathsf{C}_{\mathrm{sb}}\bar\kappa}$ and $c(a',L)=\sup_{i\ge0}(1+i)^{a'-1}L^{-i}$. The edge-tent part vanishes unless (1286) holds. ■

Proof. The octet provision proposition 51.14 gives, at $n=\mathsf{ol}_j(z)>j$,

$$\|f_{j,z}\|\le C_{\mathrm{ert}}^{\flat}\mathsf{g}_j'L^{-4(n-j)}.$$

Indeed, apply its bound to each term before summing; the pointed entropy sum controls

$$\sum_{X\ni z}e^{-\kappa d_j(X)}(d_j(X)+2)^2,$$

and the subtracted tent has $d_j(X_z)\le 26$, so its contribution is at most

$$(3/2)28^2\alpha_0^2B_\sharp=1176\alpha_0^2B_\sharp.$$

At fixed n,X' there are at most $8M^3L^{3(n-j)}$ choices of z, since $z\in X'$ and X' has at most eight level-n cubes. Together with the wide-term bound this yields

$$\sum_{\substack{\text{wide or octet } f\\ j_f=m,\ X_f=X}}\|f\|\le B_m^{\mathrm{E}}(d_m(X)+1)e^{-\kappa d_m(X)}.\tag{1303}$$

These domains are placed with $\nu = 3$. For clarity, if $n_0 = \mathsf{ol}_j(z) < k$, maximality provides a point of $X'_{n_0+1}(z)$ outside Ω_{n_0+2}, at distance at most $2M$ from z in level-(n_0+1) units. Every point of Ω_n, $n \ge n_0+3$, is therefore at distance at least $7ML^n$ from z in bare units. The diameter bound proves placement for a wide member meeting these layers. If $n \le n_0+2$, either $n_0 = j$, when the required lower bound is nonpositive, or wideness puts a point outside $\square_{n_0}(z)^{\sim 2}$, at distance greater than $2ML^{n_0-1-j}$ from z in level-j units. This proves the remaining bound. An octet member cannot meet Ω_n for $n \ge n_0+4$, since all its points lie within $2ML^{n_0}$ of z; its other placement bounds are nonpositive.

Apply (1271) to each member and then (1285) to (1303). This gives (1301). The size bound follows by summing L^{-i} and using the last inequality of (1285).

An edge tent at $y \notin \Lambda_j^{\sim -1}$ does not meet Ω_{j+1}; it is both admitted and placed. Its base need not lie in Λ_j, so use the layered space, not the fixed-core case. Every plaquette with nonzero weight $\varphi_j h_y^{(j)}$ has its corners in $X_y \cap \Lambda_j$, hence all its bonds belong to $\mathrm{Bd}_{\mathcal{S}}(X_y)$. The function $\widehat{f}(P;\mathfrak{q}) = f(\mathbf{U}_P)$ is therefore an entire function of the complex potential on this layered space, independent of the stored data. The layer-j curvature bound and $\sum_p h_y^{(j)}(p) = 3\eta_j^{-3}$ give $|\widehat{f}| \le (3/2)|b_j|\mathsf{r}_0((j,\mathrm{out});j)^2$. The pinned substituted configuration is a complex gauge transform of the plain shifted configuration on these bonds. The turn factors cancel around each plaquette, leaving conjugation at its base; its trace is unchanged. Thus the pinned reading of $\widehat{f}$ is the plain configuration reading of the edge tent. Apply proposition 58.18 to this layered reading, with $\mathsf{a} = \vartheta_k^{\mathsf{Tab}}$. At a fixed domain X_y there are at most M^3 such sites, and the weighted plaquette estimate bounds the sum of their norms by $(3/2)M^3 B_\sharp \mathsf{r}_0((j,\mathrm{out});j)^2$. Since $d_j(X_y) \le 26$, multiply by $e^{26\kappa}(d_j(X_y)+1)e^{-\kappa d_j(X_y)}$ and apply the same placed count, now with $\mathsf{a} = \vartheta_k^{\mathsf{Tab}}$. The inequality $(1+i)^{a'} \le c(a',L)(i+1)L^i$ proves (1302). The admitted-domain separation gives its support assertion. □

For the boundary terms suppose the old terms on layered spaces are analytic, invariant with their stored coordinates and tied, and the edge sums are analytic local terms on their fixed cores. Suppose each has norm at most $\mathsf{B}' e^{-\kappa d_j(X)}$ on that whole space. For the formed family suppose its members are in their prescribed places and

$$\sum_{f:\, j_f = l+1,\ X_f = X} \operatorname{osc} \varphi_f \le B_{\mathrm{fr}} \mathfrak{b}_l^2 e^{-\kappa d_{l+1}(X)}. \tag{1304}$$

Each exponent is defined and analytic on its entire layered space and stored tier, invariant with those stored coordinates, and tied. Its oscillation is the supremum, at fixed stored datum, of the difference between its values at two triples. Thus additive functions of the stored data are immaterial.

Proposition 58.23. Assume $(\mathrm{CR})_t$ and $x_K \ge 2B_\sharp$. The boundary and formed potentials are defined on all stored tiers. They vanish without a source cube and unless (1286) holds. For $a' \ge 0$,

$$\begin{aligned} y_k^{a'} |\mathbb{V}^{\mathrm{B,mk}}(Y)| &\le C_B(a') \mathsf{B}' C_{\mathcal{A}} \vartheta_k^{\mathsf{Tab}} e^{\mathsf{C}_{\mathrm{sb}} \bar{\kappa}} e^{-r_V d_k(Y)}, \\ y_k^{a'} |\mathbb{V}^{\mathrm{B,ed}}(Y)| &\le C_B(a') \mathsf{B}' C_{\mathcal{A}} \mathsf{s}_k^{\mathrm{fx}} e^{\mathsf{C}_{\mathrm{sb}} \bar{\kappa}} e^{-r_V d_k(Y)}, \end{aligned} \tag{1305}$$

where, with $c = 1 - 2\delta$,

$$C_B(a') = \max\left\{1, \frac{16}{c} C_\square(L,0) \mathsf{C}_{\mathrm{sb}}\, e^{2c} (1+\log(2L))^{a'} \left(\frac{a'}{7ecL}\right)^{a'}\right\}, \qquad 0^0 = 1.$$

The formed potential satisfies

$$|\mathbb{V}^{\Phi}(Y)| \le C_\Phi A_1^2 B_{\mathrm{fr}} C_{\mathcal{A}} \vartheta_k^{\mathsf{Tab}} e^{\mathsf{C}_{\mathrm{sb}} \bar{\kappa}} e^{-L \check{R}_t} e^{-r_V d_k(Y)}, \tag{1306}$$

provided either $\delta \le 1/8$, or $\delta > 1/8$ and $2\delta\kappa \ge c_{\mathrm{E}} + 3$. Here

$$C_\Phi = 16 e^{4/7} (2\mathfrak{p}_0)! (1 + 2\log(2L))^{2\mathfrak{p}_0} C_\square(L, 2\mathfrak{p}_0) \mathsf{C}_{\mathrm{sb}}. \tag{1307}$$

All these potentials inherit analyticity, the tie, gauge invariance with the stored fields, and the zero-field property. They inherit Borel measurability, conjugation reality and dependence on the pair when the input terms have those properties. Their values are linear in the input terms. ■

Proof. The full-domain assertion is proposition 58.18; the old domain and tier are precisely the corresponding layered or fixed-core domain and tier in that proposition. In particular no choice of representative outside those spaces is used. The layered boundary and edge domains are admitted under $(\mathrm{CR})_t$. A formed factor is in its place exactly when its domain is admitted. Thus every nonzero summand has the separation (1286).

For either boundary part, write a for $\vartheta_k^{\mathsf{Tab}}$ or $\mathsf{s}_k^{\mathrm{fx}}$. The single-member bound and the per-fine-cube proof of (1284), before absorbing $d+1$, give

$$\sum |\mathbb{V}^{\star}[f](Y)| \le 16\mathsf{C}_{\mathrm{sb}} C_{\square}(L,0) C_{\mathcal{A}} \mathsf{a} e^{\mathsf{C}_{\mathrm{sb}}\bar{\kappa}} \mathsf{B}'(d+1) e^{-(1-2\delta)\kappa d}, \qquad d = d_k(Y).$$

Split the exponent as $r_V d + cd + cd$. Use $(d+1)e^{-cd} \le 1/c$ and $d \ge 7L\check{R}_t - 2$. The running bounds imply $y_k \le (1+\log(2L))\check{R}_t$; maximizing $R^{a'}e^{-7cLR}$ proves (1305).

For formed terms use the oscillation version of (1271) and group them by $j = l+1, X$. The bound (1304) gives the same sum with coefficient $2\mathsf{H}_{\mathrm{sb}}\vartheta_k^{\mathsf{Tab}} B_{\mathrm{fr}} \mathfrak{b}_{j-1}^2$. Put $\Upsilon_k = y_k + \log(2L)$. Then $\mathfrak{b}_{j-1}^2 \le A_1^2 \Upsilon_k^{2\mathfrak{p}_0}(1+k-j)^{2\mathfrak{p}_0}$. Apply the admitted count with rate split $\widehat{a} = c_{\mathrm{E}} + 1$. Its output rate is $\kappa - c_{\mathrm{E}} - 2$. The difference between that rate and r_V is $2\delta\kappa - c_{\mathrm{E}} - 4\delta$, at least $2/7$ in either stated case. Since $\Upsilon_k \le (1 + 2\log(2L))\check{R}_t$,

$$\Upsilon_k^{2\mathfrak{p}_0} e^{-\frac{2}{7}(7L\check{R}_t - 2)} \le e^{4/7}(2\mathfrak{p}_0)!(1+2\log(2L))^{2\mathfrak{p}_0} e^{-L\check{R}_t}.$$

Here $R^n e^{-R} \le n!$ follows from the exponential series. This proves (1306). Subtracting any one value at fixed stored datum reduces the oscillation estimate to the bounded case; the starred bracket kills the subtracted value. No measurable choice is needed for this estimate. The remaining properties follow termwise from proposition 58.18 and finite summation. □

58.14. **The exact finite sums.** Let U_+ be a real representative of the minimal orbit for the longer collared sequence, let U extend it to all bare bonds, and let $P_+ = (\mathsf{U}, 0, \mathbf{J}_+)$ be its plain triple. Assume its restrictions lie in the graded domains above, that $V = \mathcal{M}^k U_+$ is axial on P'', and that $\mathcal{A}$ is real with $\sup|\mathcal{A}| < \epsilon_P$. At the shorter reduced determining set $\mathbf{B}$, let d_z equal $e^{iB_z'}V$ on the changed top rows and $M_{\mathbf{B}}(U_+)$ elsewhere; B_z' is the actual pivot completion of $z\mathcal{A}$. Averages on these rows are the real averages. Their agreement with the analytic response averages follows from the local double-pillar induction in lemma 66.1, using its curvature hypothesis on each double pillar.

Proposition 58.24. Suppose first that $k \ge 1$, and retain the separated-sequence and regularity ceilings of theorem E, with reduced determining set $\mathbf{B}$ and $\bar{\varepsilon} \le \varepsilon_{\mathrm{G}}(\rho_{\mathrm{s}})$. If the old tested datum is ϵ'-regular and

$$\epsilon' + 8C_P \sup|\mathcal{A}| \le \bar{\varepsilon}, \tag{1308}$$

then the minimal orbit of $(\mathbf{B}, \mathsf{d}_z)$ exists for every real $z \in [-1,1]$. Define $F(z) = A_{\Omega_0^{\diamond}}^{\eta_k}(U_{\mathbf{B}}(\mathsf{d}_z))$ on this interval. It extends holomorphically to a neighborhood of $|z| \le 1$. With the regional pivot Jacobian $\mathsf{J}_V^{\diamond}$, gauge functional $\mathbb{G}$, and gauge Hessian $\widehat{\mathsf{G}}_V$, the exact sums are

$$\sum_Y \mathbb{V}^{\mathrm{J}}(Y; P_+|_Y, B|_Y) = \mathsf{J}_V^{\diamond}[\mathcal{A}], \tag{1309}$$

$$\sum_Y \mathbb{V}^{\mathrm{G}}(Y; P_+|_Y, B|_Y) = -x_k \mathbb{G}[\mathcal{A}] + \tfrac{1}{2}(B, \widehat{\mathsf{G}}_V B)_1, \tag{1310}$$

$$\sum_Y \mathbb{V}^{\mathrm{act}}(Y; P_+|_Y, z\mathcal{A}|_Y) = F(z) - F(0), \tag{1311}$$

$$\sum_Y \mathbb{V}^{\mathrm{A}}(Y; P_+|_Y, B|_Y) = -x_k\{F(1) - F(0) - F'(0) - \tfrac{1}{2}F''(0)\}. \tag{1312}$$

For the actual counterterms $b_j = L^{-1}x_{j-1} - x_j$,

$$\sum_Y \mathbb{V}^{\mathrm{c}}(Y) = -\sum_{p \text{ inner}} [\beta_k(x(p)) - L^k x_k]\,[a_p(U_{\mathbf{B}}(\mathsf{d}_1)) - a_p(U_+)]. \tag{1313}$$

All sums are finite and evaluated at $(P_+|_Y, \mathcal{A}|_Y)$; $B = g_k^{-1}\mathcal{A}$. At $k = 0$, the determining map is the identity and $U_{\mathbf{B}}(\mathsf{d}_z) = \mathsf{d}_z$. With this convention the displayed sum identities hold by the direct first-step calculation. ■

Proof. For $k \geq 1$, the completed field is real and $\sup|B'_z| \leq 2C_P|z|\sup|\mathcal{A}|$. Only first-kind rows at the last level change. A tested plaquette at a lower level is unchanged; at the last level it is a product of four unitary factors, each changing by at most $2C_P \sup|\mathcal{A}|$. Telescoping the product, including reversed bonds, gives a change at most $8C_P \sup|\mathcal{A}|$. The strict original test and (1308) therefore give $\bar{\varepsilon}$-regularity. The reduced determining set, shorter separated sequence and lattice-size hypotheses are those of theorem E. Apply that theorem at datum $V_0 = \mathsf{d}_z$ and perturbation $B = 0$, separately for each real z. It gives the entire real path of regular orbits.

For the Jacobian decompose the bond space into nonpivot and pivot coordinates. Its derivative maps into the pivot coordinates; the pivot-to-pivot block is the direct sum of the individual matrices Jb_c, since another pivot never lies in the block of c. Thus the trace of every power is the sum of the individual traces. Their norms are less than $1/2$, so the logarithmic series converges absolutely. Summing the potentials counts each pivot at $\square(c)$ and each nonpivot Haar factor at $\square(b)$, exactly once. This proves (1309).

Inside $B(y)$, $y \in P''$, there is no pivot and the completion equals $\mathcal{A}$. The contour logarithm $\xi(a) = i^{-1}\log[(e^{ia}V)(y,x)V(y,x)^{-1}]$ has first term $\mathsf{G}^{(1)}[V]a$. Since it is traceless, $1 - \frac{1}{2}\operatorname{tr} e^{i\xi(za)} = \frac{1}{2}z^2\langle \mathsf{G}^{(1)}a, \mathsf{G}^{(1)}a\rangle + O(z^3)$. The linear completion fixes these nonpivot bonds. Summing over the disjoint blocks and using $x_k g_k^2 = 1$ proves (1310).

For a Wilson member the finite starred sum is its full-parameter value minus its zero-parameter value: all disconnected additions vanish by locality, and restriction agrees by zero extension of decoupling parameters. The plaquettes are partitioned by (m, X); hence summing those differences gives the action difference of the actual full response. By proposition 58.9 it is a gauge transform of the minimal orbit of d_z. Each plaquette action is gauge invariant, proving (1311). For $k = 0$, each plaquette is instead counted directly at its $X(p)$, and the same identity holds with the completed field. Each summand is holomorphic for $|z|\sup|\mathcal{A}| < \epsilon_A$, a disc of radius greater than one. Their finite sum extends $F - F(0)$. Taylor subtraction commutes with the finite sum, proving (1312).

The same finite calculation applies with any fixed plaquette weight. For p_w, the weighted action difference enters with a minus sign. The telescoping running-coupling identity gives $L^k p_w(p) = \beta_k(x(p)) - L^k x_k$. This proves (1313); no identification of b_j with an intrinsic local coefficient is used. □

For the remaining sums let $\mathsf{U}^\sharp = e^{i\eta_k \mathbf{A}^\delta}\mathsf{U}$ be the pinned full-response field. Suppose it is real and every old plain triple at $\mathsf{U}^\sharp$ belongs to its layered output space, and to the fixed core when $X \subset \Lambda_j$. The corresponding memberships at U are required as well. For actual glued fields these are the output and old-orbit assertions of corollary 58.4 and proposition 58.27. They are stronger than membership of a substituted triple with real factor U.

Proposition 58.25. Under these physical domain conditions, all old terms at U and $\mathsf{U}^\sharp$ are defined on their prescribed stored tiers, and

$$\mathbf{E}_k(\mathsf{U}^\sharp) - \mathbf{E}_k(\mathsf{U}) = \sum_Y \mathbb{V}^{\mathrm{E}}(Y; \mathfrak{u}_+|_Y), \tag{1314}$$

$$\mathbf{E}_k^{\mathrm{bd}}(\mathsf{U}^\sharp, \mathfrak{q}) - \mathbf{E}_k^{\mathrm{bd}}(\mathsf{U}, \mathfrak{q}) = \sum_Y \mathbb{V}^{\mathrm{B}}(Y; \mathfrak{u}_+|_Y), \tag{1315}$$

$$\mathbf{\Phi}_h(\mathsf{U}^\sharp; \mathfrak{q}) - \mathbf{\Phi}_h(\mathsf{U}; \mathfrak{q}) = \sum_Y \mathbb{V}^{\Phi}(Y; \mathfrak{u}_+|_Y). \tag{1316}$$

The last identity is a separate identity of the old formed exponents; it does not absorb them into the regular or boundary terms. ■

Proof. At full parameters the current identity in proposition 58.13 shows that the pinned substituted triple and the old plain triple of $\mathsf{U}^\sharp$ have the same pair. Both are real and admitted; the tie identifies their values at the same stored datum. At zero parameters, restriction of P_+ is exactly the old plain triple of U. Thus (1272) is the physical difference for every layered boundary or formed member. For a plain local member, the plain and pinned readings agree by the proof of (1279).

For the regular list, every old pointed term is either wide or occurs in exactly one octet sum. The octet identity of proposition 51.14 applies because $X_n'(z) \subset \Omega_{n+1}$ and its five-layer cube lies in the domain of the minimal configuration. It identifies each pulled-back narrow term with its original value. The tents partition the weighted action: deep tents have $\varphi_j h_z = h_z$, and the remaining tents are exactly the edge tents. Consequently the sum of the regular members is $\mathbf{E}_k$ plus a field-independent constant; that constant cancels in the difference. This proves (1314).

For the boundary list apply the single-member difference separately to layered boundary terms, with stored levels $< j$, and edge terms, with no stored argument. Reordering the two finite sums proves (1315). For a formed factor of level $l+1$, restrict the stored argument to $(\mathfrak{q}|_{X_f})|^{(l)}$. The physical domain assumptions put both plain triples in its entire defined space. Thus both old exponents are defined; their difference is the sum of that factor's pinned potentials. Summing first over factors and then over domains gives (1316). Additive functions of the stored datum cancel on both sides. This also explains why the oscillation bound, without an absolute bound on each exponent, suffices. □

58.15. **The actual families in the history expansion.** For a family V, let $\|V\|_{r_V}$ be the supremum of $e^{r_V d_k(Y)}|V(Y)|$ over the full graded field domain. A superscript rad restricts the stored coordinates to their prescribed tiers. Thus the two norms differ only in their stored arguments, not in their field cutoff or graded triple. Put

$$\mathbb{V}^{\mathrm{rg}} = \mathbb{V}^{\mathrm{int}} + \mathbb{V}', \qquad \mathbb{V}^{\mathrm{bd}} = \mathbb{V}^{\mathrm{B}} + \mathbb{V}^{\Phi}, \qquad \vartheta_{0,k} = \frac{\delta_k}{\alpha_{0,k}}, \qquad C_{\mathrm{sz}} = \frac{L}{L-1} C_{\mathrm{h}}'.$$

Suppose the old pointed norm at rate $\kappa+1$ is at most E_0; each term on a layered space and each edge sum has bound $\mathsf{B} e^{-\kappa d_j(X)}$; and the formed family satisfies (1304). The terms are Borel and conjugation real, and have the invariance and tie required above. These are the input bounds on the whole stated spaces, not just at the actual real backgrounds.

For the radius specialization assume

$$\alpha_{\iota,n} = C_\iota g_n y_n^q, \qquad q > \mathfrak{p}_0, \qquad \alpha_{0,k} \le \alpha_0, \qquad c_- \alpha_{0,k} \le \mathsf{r}_0((k, \mathrm{out}); k),$$
$$\mathsf{r}_0((j, \mathrm{out}); j) \le c_+ \alpha_{0,j}, \qquad c_- \alpha_{0,m} \le a_m \le (1+\beta_{\mathrm{s}}) c_+ \alpha_{0,m}.$$

Here the last inequality concerns the enlarged radii a_m used for the coupling potential. Retain all domain, pinning, rate and smallness inequalities of propositions 58.18 and 58.20 to 58.23; for the edge-tent estimate use $a' = 2q$. These are the corresponding radius and cutoff requirements in section 69.1.

Define

$$\begin{aligned} C^{\mathrm{rg}} &= K_\flat M^3 C_{\mathcal{A}} \alpha_0 + K_\flat C_{\mathcal{A}}^3 + 2K_A \mathsf{c}_\mathfrak{h} \epsilon_A^{-2} C_{\mathcal{A}}^3 + C_V C_{\mathrm{sz}} C_{\mathrm{ert}} C_{\mathcal{A}} \\ &\quad + K_c' \mathsf{c}_\mathfrak{h} (1+\beta_{\mathrm{s}})^2 c_+^2 c_-^{-1} C_{\mathcal{A}} M^3 \bar{B} C_0^2 + C_V' C_{\mathcal{A}} c_-^{-1} c_+^2 M^3 B_\sharp C_0^2, \\ C^{\mathrm{bd}} &= C_{\mathcal{A}} e^{\mathsf{C}_{\mathrm{sb}} \bar{\kappa}} [C_B(0)(1+c_-^{-1}) + C_\Phi A_1^2 c_-^{-1}], \\ C_{\mathrm{eff}} &= \max\{1, C^{\mathrm{rg}}, C^{\mathrm{bd}}\}. \end{aligned} \tag{1317}$$

These constants depend on the fixed geometric, analytic, radius and input bounds, but not on the history, height, volume or individual running couplings.

Proposition 58.26. The four actual historical families are defined, analytic and Borel on every prescribed tier, are gauge invariant and conjugation real, and have the tie. The regular families are independent of the stored coordinates and depend only on the pair. All four vanish at zero field and on a domain without a source cube. Moreover

$$\|\mathbb{V}^{\mathrm{int}}\|_{r_V} + \|\mathbb{V}'\|_{r_V} \le C_{\mathrm{eff}} \left[\frac{\delta_k}{\alpha_0} (1 + E_0 + \alpha_0^2 B_\sharp) + M^3 g_k \mathfrak{b}_k^3 + \vartheta_{0,k} g_k^2 y_k^{2q} \right], \tag{1318}$$

$$\|\mathbb{V}^{\mathrm{B}}\|_{r_V}^{\mathrm{rad}} + \|\mathbb{V}^{\Phi}\|_{r_V}^{\mathrm{rad}} \le C_{\mathrm{eff}} (\mathsf{B} + B_{\mathrm{fr}}) \vartheta_{0,k}. \tag{1319}$$

The norms of the combined regular and boundary families are bounded by these respective left sides. The sums of norms also give the per-origin weights required in the history cluster expansion. They hold on its conditional-mean field polydiscs whenever those polydiscs lie in the original field cutoff. For $k = 0$ the inherited families are empty and the regular bound is the direct Jacobian, gauge and action bound of proposition 58.20, with $\epsilon_A = r_P^\circ$ and $\mathsf{c}_\mathfrak{h} = 1$. ∎

Proof. The source-cube argument in proposition 58.20 admits all Jacobian, gauge and action members on the full graded domain. For the earlier terms, proposition 58.18 admits every substituted triple and every stored argument on its correct space. The edge-tent argument in the proof of proposition 58.22 supplies the same conclusion for its plain configuration reading. The bounds on the input terms therefore apply to every summand; taking a finite sum leaves no additional domain condition. This remains true when the sum is empty, since the family was restricted to that domain in its definition.

The invariance, tie and pair dependence follow from the re-expansion proof, including the independent-pin argument (1279), and from the action proof. Borel measurability is preserved by composition and finite summation. At fixed real factor and stored argument each summand has a holomorphic extension to a neighborhood of the point; intersect these finitely many neighborhoods and add the extensions. Restricting the stored variables to their tiers does not change this slice. It also preserves Borel measurability, since the preimage of a Borel set is intersected with the restricted domain. This proves all stated regularity properties on the tiers. The zero-field and no-source assertions were proved for the individual members.

For the norms, the running inequalities give $g_j \le g_k$ and $y_j \le (1 + k - j) y_k$. Consequently the edge-tent hypothesis holds with $r_\star = c_+ C_0 g_k y_k^q$, and

$$\vartheta_k^{\mathsf{Tab}} \le c_-^{-1} \vartheta_{0,k}, \qquad \mathsf{s}_k^{\mathrm{fx}} \le \vartheta_{0,k}.$$

Use $\mathsf{g}_j' \le E_0 + \alpha_0^2 B_\sharp$ in (1300). The wide and octet bound is then $C_V C_{\mathrm{sz}} C_{\mathrm{ert}} C_{\mathcal{A}}$ times $\mathsf{s}_k^{\mathrm{fx}} (E_0 + \alpha_0^2 B_\sharp)$, with the common factor $e^{-r_V d_k(Y)}$. The Jacobian bound is $K_\flat M^3 C_{\mathcal{A}} \alpha_0 \mathsf{s}_k^{\mathrm{fx}}$. The gauge and action bounds are respectively $K_\flat C_{\mathcal{A}}^3$ and $2K_A \mathsf{c}_\mathfrak{h} \epsilon_A^{-2} C_{\mathcal{A}}^3$ times $M^3 g_k \mathfrak{b}_k^3$. The last two coefficients in C^{rg} multiply $\vartheta_{0,k} g_k^2 y_k^{2q}$, by (1299) and (1302). Taking each supremum separately and adding gives (1318); no cancellation between origins is used.

For the boundary sum, take $a' = 0$ in (1305), and use the two ratio inequalities above. For the formed sum use (1306) and $e^{-L \check{R}_t} \le 1$. Both coefficients are bounded by C^{bd}. Taking the two suprema separately gives (1319). The domain restriction of the norm can only decrease these suprema. In the conditional Gaussian application, proposition 57.5 supplies the conditional mean

and the cutoff inclusion; evaluating an already defined family on that smaller field polydisc preserves all the bounds. For the actual families write $\mathsf{B} = B$ and $\mathsf{B}_{\mathrm{fr}}^{\flat} = B_{\mathrm{fr}}^{\flat}$. Then eq. (1398) supplies (1304) with the smaller coefficient $\mathsf{B}_{\mathrm{fr}}^{\flat}$ in place of B_{fr}. Use this coefficient and the actual boundary bound B in (1319). Then the two displayed majorants are precisely $S_{\mathrm{rg}}(x_k)$ and $S_{\mathrm{bd}}(x_k)$ of section 69.1, since $g_k = e^{-y_k/2}$, $\mathfrak{b}_k = A_1 y_k^{\mathfrak{p}_0}$ and $\vartheta_{0,k} = (A_1/C_0) y_k^{-(q-\mathfrak{p}_0)}$. Their smallness is obtained there by the single fixed choice of the coupling ceiling. □

58.16. **The actual shifted field.** We now verify that the local potentials give the change in the old exponent on the support of a history. Let $h' = (h, \mathsf{o}_t)$, $t = k+1$, with $k \geq 1$. Take $U \in \mathsf{Sup}(h')$ where the stored variables are defined, and an adapted representative U_+ of its collared critical orbit at level t. Glue U_+ to U outside $\Omega_0^{\diamond} = \Omega_1^{\sim}$, obtaining U. Let

$$P_+ = (\mathsf{U}, 0, \mathbf{J}_+), \qquad V = \mathcal{M}^k U_+, \qquad p = V_k^{\mathrm{bg}\diamond} \uplus V_k|_{\mathfrak{X}_{\mathrm{out}}}, \qquad B = g_k^{-1}\mathcal{A}.$$

Here $\mathcal{A}$ is the actual free fluctuation and $\mathfrak{q}$ the old stored field. The pair read at level k is $(U_+, L^{-3}\mathbf{J}_+)$ on the inner bare bonds. Write $\mathbf{B}$ for the shorter determining set, through level k.

Use the domain, radius and smallness hypotheses of the actual potential construction and pinned response. The additional chart and segment inequalities are

$$\begin{gathered} 2L^2 C_{\mathrm{reg}} \varepsilon_t \leq \varpi_\chi, \qquad C_\chi \mathsf{C}_{\mathcal{A}} \delta_k < r_\chi, \\ 2L^2 C_{\mathrm{reg}} \varepsilon_t + C_{\mathrm{seg}} \mathsf{C}_{\mathcal{A}} \delta_k \leq \bar{\varepsilon}. \end{gathered} \tag{1320}$$

The pair is required to lie within the radii of the regional response and fluctuation form, and the level-t torus has at least $6R_1\mathfrak{M}$ sites in each direction. These are among the simultaneous requirements in section 69.1.

Let B_z' be the completed pivot field at $(P_+, z\mathcal{A})$. Denote by d_z the shorter datum obtained from $M_{\mathbf{B}}(U_+)$ by replacing its rows in $\mathfrak{X}_0$ by $e^{iB_z'}V$. Set

$$(\mathbf{A}, \mathbf{K}) = \mathfrak{h}_{T_\ell|h'}(\mathbf{1}; P_+, \mathcal{A}), \qquad \mathsf{U}^{\mathrm{pl}} = e^{i\eta_k \mathbf{A}}\mathsf{U}, \qquad \mathsf{U}^{\sharp} = e^{i\eta_k \mathbf{A}^{\delta}}\mathsf{U}.$$

The last field uses the pinned levelling.

Proposition 58.27. For every real $z \in [-1, 1]$, the completed field is defined and agrees with the regional chart:

$$e^{iB_z'}V = V_k^{\mathrm{bg}\diamond}[z\mathcal{A}] \quad \text{on } \mathfrak{X}_0. \tag{1321}$$

In particular,

$$V_k|_{\mathfrak{X}_0} = V_k^{\mathrm{bg}\diamond}[\mathcal{A}], \qquad \mathsf{d}_1 = \xi_h^{\diamond}(U).$$

The regular critical orbit of $(\mathbf{B}, \mathsf{d}_z)$ exists throughout this segment. The pinned field $\mathsf{U}^{\sharp}$ belongs to the old critical orbit on $\Omega_0^{\diamond}$ and agrees with U outside it. At every earlier index (j, X) used by the old sums, its plain triple lies in the old layered output domain and, when $X \subset \Lambda_j$, in the required fixed analytic core.

Let

$$\mathbb{F}(z) = L^k A_{\Omega_0^{\diamond}}(U_{\mathbf{B}}(\mathsf{d}_z)).$$

The action is independent of the representative in this orbit. It has a holomorphic continuation near the segment, and at zero

$$\mathbb{F}'(0) = 0, \qquad x_k \mathbb{F}''(0) + (B, \widehat{\mathsf{G}}_V B)_1 = (B, \mathsf{A}_{h'}^{\sharp}(P_+)B)_1. \tag{1322}$$

The gauge quadratic form is evaluated on the same regional free coordinates as the fluctuation form. ■

Proof. The support and graded-domain results give real $\mathcal{A}$, with $\sup|\mathcal{A}| < \mathsf{C}_{\mathcal{A}}\delta_k$, and put every restricted point $(P_+, \mathcal{A}, \mathfrak{q})|_Y$ in its admitted tier. The shorter datum of U_+ is regular at $2L^2C_{\mathrm{reg}}\varepsilon_t$. The segment estimate for the completed datum in proposition 58.24 bounds its regularity radius by

$$2L^2C_{\mathrm{reg}}\varepsilon_t + C_{\mathrm{seg}}|z|\sup|\mathcal{A}|.$$

The last inequality of (1320) therefore gives the regular orbit at every z. The local response identifies these orbits and gives the asserted holomorphic action near the segment.

The chart. On each double block of a coarse bond in Γ, $|\partial p - 1| < 2L^2C_{\mathrm{reg}}\varepsilon_t$. The first two inequalities of (1320) thus put $z\mathcal{A}$ in the regional fibre chart. Its free coordinates equal $z\mathcal{A}$, and its outer coordinates equal those of p. The block field used by the local potentials is also p on these double blocks. Indeed, $\mathsf{U} = U_+$ on their bare pillars; the regularity estimate there is

$$|\partial U_+ - 1| < C_{\mathrm{reg}}\varepsilon_t L^{-2k}.$$

All lower blocks required by the averaging induction remain in the pillar. The local real and complex averages consequently agree there, by the same small-curvature induction used in proposition 60.2. No bound on the exterior gluing is used.

For each pivot $b_0(c)$, the single-bond completion and the Γ-chart use the same restricted configuration on P_c and the same free field, extended by zero outside $\mathfrak{X}$. The local pivot uniqueness of the fibre chart gives

$$B'_z(b_0(c)) = \psi_p^{\mathrm{lin}}(\mathsf{C}_p z\mathcal{A})(b_0(c)).$$

Equality on the free and pivot bonds proves (1321). The actual field V_k and the reference p have the same next average V_t, and the intrinsic logarithm of their quotient on the free bonds is $\mathcal{A}$. The inverse-chart statement therefore gives $V_k = V_k^{\mathrm{bg}\diamond}[\mathcal{A}]$ on $\mathfrak{X}_0$. All other shorter rows were unchanged. Their values are the restrictions of $\mathcal{M}^j U$, so $\mathsf{d}_1 = \xi_h^\diamond(U)$.

Arrival. The pinned-response result gives a real gauge u, with $\mathsf{U}^\sharp = (\mathsf{U}^{\mathrm{pl}})^u$, restores the shorter datum d_1, and leaves every noninner bond of $\Omega_0^\diamond$ unchanged. The prefix support inclusion places U in $\mathsf{Sup}(h)$, so proposition 53.11 also supplies the old orbit with datum $\xi_h^\diamond(U)$. By proposition 58.9, the plain response is a gauge transform of an element of this orbit. Pinning preserves criticality and regularity and restores the exact constraints. Its restriction is therefore itself an element of the old regular critical orbit. The membership statement corollary 58.4, applied to the shorter history, puts its plain triples in the half-radius output spaces and in the half-radius fixed cores, hence in the required full spaces.

The second variation. We give the regional calculation because its row pairing differs from the one on a constant tower. Let $J = J^\circ(U_+)$ be the actual regional action current, and let J inject the level-k rows into the shorter row space. Write m for the row-spacing multiplication. On the injected rows the spacing and the multiplier a are one:

$$\mathsf{m}^{-1}\mathsf{J} = \mathsf{J}, \qquad \mathsf{J}^{\mathsf{T}}\mathsf{a}\mathsf{J} = I.$$

The regional response at small arbitrary real row data β, followed by its restoring gauge, has action $\varphi(\beta)$. Its equations are the full-parameter response equations eq. (1056), with $\mathcal{Q}(\widehat{A}(\beta)) = \mathsf{J}\beta$. The strict regularity bounds allow a smaller neighbourhood on which this restored response is regular. Whenever the regular orbit is defined it contains this response, by the criticality argument of proposition 58.9.

Differentiate these equations at zero. With $a = \widehat{A}'(0)\nu$ and $\mu = \mathsf{m}^{-3}\widehat{\omega}'(0)\nu$, the saddle equations are

$$(\Delta_1 + D\mathsf{R}D^{\mathsf{T}})a - Q^{\mathsf{T}}\mu = 0, \qquad Qa = \mathsf{J}\nu.$$

The inverse identities already used to construct the response give

$$a = \mathsf{H}_1\mathsf{J}\nu, \qquad \mu = [(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1} - \mathsf{a}]\mathsf{J}\nu.$$

The chain rule and the differentiated constraint give the envelope identity

$$D\varphi(\beta)\nu = (\mathsf{m}^{-3}\widehat{\omega}(\beta), \mathsf{J}\nu)_{\mathbf{B}}.$$

At zero, $\mathsf{m}^{-3}\widehat{\omega}(0) = \mathsf{H}_1^{\mathsf{T}} J$. Consequently

$$D\varphi(0)\nu = (\mathsf{H}_1\mathsf{J}\nu, J)_{\eta_k},$$
$$D^2\varphi(0)(\nu,\nu) = (\nu, [\mathsf{J}^{\mathsf{T}}(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}\mathsf{J} - I]\nu)_1.$$

A curve in the coarse fibre lifts to the fixed longer constraint. Criticality of U_+ thus annihilates $D\varphi(0)$ on $\ker Q_p$. For a fibre curve $e^{i\beta(s)}p$ with velocity v, differentiating the averaging constraint twice gives

$$\ddot{\beta}(0) = \kappa - 2\mathsf{D}_p^{(2)}(v,v), \qquad \kappa \in \ker Q_p.$$

Inserting this into the second derivative of $\varphi(\beta(s))$ removes the κ term. The remaining correction is $-2(\mathsf{J}\mathsf{D}_p^{(2)}(v,v), \mathsf{H}_1^{\mathsf{T}} J)_{\mathbf{B}}$. Together with the preceding quadratic form this is exactly $\langle v, \Delta_k^{\sharp}(U_+, J)v\rangle$ from (1092). The completed chart has velocity $v = g_k\mathsf{C}_pB$. Therefore

$$\mathbb{F}'(0) = 0, \qquad x_k\mathbb{F}''(0) = \langle \mathsf{C}_pB, \Delta_k^{\sharp}(U_+, J)\mathsf{C}_pB\rangle.$$

It remains to check the current in the actual triple. The field $J' = L^{-3}\mathbf{J}_+$ agrees with J at every bond touching Ω_1. Their difference is supported on bare determining rows. The proof of lemma 58.8 shows that $\mathsf{G}_1, \mathsf{H}_1$ are unchanged by this difference, and that its transpose image has only bare-row components. The nonlinear averaging remainder vanishes on those rows. Thus the last term of (1092) is unchanged as well. Both forms have the same defined operators; the graded Gaussian-domain result supplies their admission. Finally the block contours in $\widehat{\mathsf{G}}_V$ use only the double-block field where $V = p$. Equation (1093) therefore gives (1322). □

58.17. **The identity for the old exponent.** Let $A_{k,h}$ and Φ_h be the old exponent and retained sum defined in (1474) and section 62.3. Write $\mathsf{J}_V^{\diamond}[\mathcal{A}]$ for the logarithm of the regional pivot and Haar Jacobian, and $\mathbb{G}[\mathcal{A}]$ for the block gauge-fixing action. These are the local totals of the J and G constructions.

Proposition 58.28. At every support point of proposition 58.27, all terms in the following identity are defined:

$$\begin{aligned} &A_{k,h}(\mathsf{U}^{\sharp}, \mathfrak{q}) + \Phi_h(\mathsf{U}^{\sharp}; \mathfrak{q}) - x_k\mathbb{G}[\mathcal{A}] + \mathsf{J}_V^{\diamond}[\mathcal{A}] \\ &\quad = A_{k,h}(\mathsf{U}, \mathfrak{q}) + \Phi_h(\mathsf{U}; \mathfrak{q}) - \tfrac{1}{2}(B, \mathsf{A}_{h'}^{\sharp}(P_+)B)_1 \\ &\qquad + \sum_{Y \in \mathcal{D}_k} [\mathbb{V}^{\mathrm{int}} + \mathbb{V}' + \mathbb{V}^{\mathrm{B}} + \mathbb{V}^{\Phi}](Y|h'; (P_+, \mathcal{A}, \mathfrak{q})|_Y). \end{aligned} \tag{1323}$$

At the first step the same formula holds with no old regular, boundary or retained terms; the level-zero action and completed fibre field give its direct interpretation. ■

Proof. For an inner plaquette of $\Omega_0^{\diamond}$, the pinned and plain fields have the same trace, and the plain field is gauge equivalent to the critical field with datum d_1. For a plaquette not inner to the collar all its bond values are unchanged. To see the latter assertion also for its inner bonds, take a corner outside $\Omega_1^{\sim}$. Every other corner is within one bare spacing of it. Such a corner cannot lie in Ω_1, since the collar has width at least a cell side. Any inner bond of the plaquette is therefore a level-zero determining row, where d_1 is unchanged. For every plaquette weight w, it follows that

$$A(w, \mathsf{U}^{\sharp}) - A(w, \mathsf{U}) = \sum_{p \text{ inner in } \Omega_0^{\diamond}} w(x(p))[a_p(U_{\mathbf{B}}(\mathsf{d}_1)) - a_p(U_+)]. \tag{1324}$$

In particular, $L^k[A(\mathsf{U}^{\sharp}) - A(\mathsf{U})] = \mathbb{F}(1) - \mathbb{F}(0)$.

The exact totals of J, G and A in proposition 58.24, with their Taylor subtractions, and (1322) yield

$$\sum_{Y} \mathbb{V}^{\mathrm{int}}(Y) = \mathsf{J}^{\diamond}_{V}[\mathcal{A}] - x_k \mathbb{G}[\mathcal{A}] - x_k[\mathbb{F}(1) - \mathbb{F}(0)]$$
$$+ \tfrac{1}{2}(B, \mathsf{A}^{\sharp}_{h'}(P_+)B)_1.$$

The counterterm identity eq. (1313), applied with $w_k = \beta_k(\cdot) - L^k x_k$, gives

$$\sum_{Y} \mathbb{V}^{\mathrm{c}}(Y) = -[A(w_k, \mathsf{U}^{\sharp}) - A(w_k, \mathsf{U})].$$

Here the historical counterterms are b_j, so their dimensionless weight is $p_w = L^{-k} w_k$. The regular-member identity eq. (1314) gives

$$\sum_{Y} \mathbb{V}^{\mathrm{E}}(Y) = \mathbf{E}_k(\mathsf{U}^{\sharp}) - \mathbf{E}_k(\mathsf{U}).$$

Here one may first use the plain response; gauge invariance of the regular members identifies it with the pinned response. Adding the first two action changes and using $\beta_k = w_k + L^k x_k$ supplies the full weighted Wilson-action difference.

For the other origins, the pinned substitution and eq. (1315), (1316) give

$$\sum_{Y} \mathbb{V}^{\mathrm{B}}(Y) = \mathbf{E}^{\mathrm{bd}}_k(\mathsf{U}^{\sharp}, \mathfrak{q}) - \mathbf{E}^{\mathrm{bd}}_k(\mathsf{U}, \mathfrak{q}),$$
$$\sum_{Y} \mathbb{V}^{\Phi}(Y) = \Phi_h(\mathsf{U}^{\sharp}; \mathfrak{q}) - \Phi_h(\mathsf{U}; \mathfrak{q}).$$

The second equality is the old-retained identity (1316); its fixed stored data are the same on both sides. The membership and potential bounds prove definedness before these sums are taken. Each sum is finite. Substitution of (1474) now gives (1323); the vacuum constant cancels.

At $k = 0$ the shifted field is the completed one-step chart field on $\mathfrak{X}_0$ and the original field elsewhere. The direct plaquette Taylor expansion and gauge-fixing quadratic form give the same J, G and A total. There are no E, c, B or Φ members, so the same calculation proves the first-step assertion. □

59. Cluster sums for a history step

We use the Gaussian data and term bounds of Section 57. The integration set may be a proper subset of the non-pivot bonds. Accordingly, the guard in the finite differences remains $\mathrm{Gd}(Z_\circ)$; the domain assigned to a term remains $Z^{\sharp}_{\circ} \cup \bigcup \mathcal{K}$.

59.1. **Factorization and the whole integral.** Call $(\iota, \mathcal{K})$ rooted if every wall component of $\mathrm{Gd}(Z_\circ(\iota)) \cup \mathcal{K}$ contains a cube of the guard. Write $\mathcal{I}^{\mathrm{gd}}(Z)$ for the rooted pairs with domain Z. An empty guard permits only $\mathcal{K} = \emptyset$. For a coarse wall-connected region $\gamma \subset \mathfrak{R}$, set

$$\mathsf{z}(\gamma) = \sum_{(\iota,\mathcal{K}) \in \mathcal{I}^{\mathrm{gd}}(\gamma)} T(\iota, \mathcal{K}).$$

These sums are finite. Define $E(X)$ by the local cluster series (686) with these activities. Its refined series expands every activity into its individual terms. For a set Mk of vertices and integration bonds, $E^{\mathsf{Mk}}(X)$ is the subseries containing at least one vertex or failed bond in Mk.

Proposition 59.1. Terms that are not rooted vanish. If Z has wall components $\gamma_1, \dots, \gamma_n$, restriction to these components is a bijection

$$\mathcal{I}^{\mathrm{gd}}(Z) \longrightarrow \prod_{i=1}^{n} \mathcal{I}^{\mathrm{gd}}(\gamma_i),$$

and

$$T(\iota,\mathcal{K}) = \prod_i T(\iota|_{\gamma_i}, \mathcal{K}\cap\gamma_i). \tag{1325}$$

Consequently the sum of terms with domain Z is $\prod_i \mathsf{z}(\gamma_i)$.

For fixed ι, put $Z_\circ = Z_\circ(\iota)$. If $Z_\circ^\sharp \subset Z \subset \mathfrak{R}$, let $\mathcal{K}_0$ be the cubes of $Z \setminus Z_\circ^\sharp$ and N' the cubes of $Z_\circ^\sharp \setminus \mathrm{Gd}(Z_\circ)$. Then

$$\sum_{\mathcal{K}\colon X(\iota,\mathcal{K})=Z} T(\iota,\mathcal{K}) = f_\iota[\mathcal{K}_0 \mid \mathsf{N}']. \tag{1326}$$

In particular, enlarging the guard within $Z_\circ^\sharp$ without removing $(Z_\circ^{\mathrm{I}})^\sharp$ does not change the activities or the cluster sums, whenever the enlarged guards respect restriction to wall components. ■

Proof. Put $A = Z_\circ^{\mathrm{I}}$ and $G = \mathrm{Gd}(Z_\circ)$. For $W \subset \mathcal{K}$, the Gaussian parameter in $f_\iota(\mathbf{1}_W)$ is the indicator of $(W \cup G) \setminus A^\sharp$. Suppose a wall component c of $\mathcal{K}\cup G$ contains a cube of $\mathcal{K}$ and no cube of $A^\sharp$. By the component property of the Gaussian structure, deleting c leaves the covariance, connecting operator and mean on A unchanged. Thus $f_\iota(\mathbf{1}_W) = f_\iota(\mathbf{1}_{W\setminus\mathsf{c}})$. Pairing the subsets in the alternating difference along any cube of $\mathsf{c}\cap\mathcal{K}$ gives zero. This applies in particular to every pair that is not rooted.

Every enlarged vertex domain $Y^\sharp$ and bond domain $b^\sharp$ is wall-connected. If it lies in Z, it lies in exactly one γ_i. The potential and test factors therefore split between the components. Their integration bonds, and the auxiliary bonds used by their connecting operators, are disjoint. The Gaussian covariance is a direct sum, its determinant branch is the product of the component branches, and its mean splits by restriction. Absolute convergence from the preceding section permits Fubini. The resulting integral is a product of component integrals; taking the finite difference separately in the disjoint sets $\mathcal{K}\cap\gamma_i$ proves (1325). Restriction preserves rootedness. Conversely, gluing rooted component indices preserves the exclusion of failed bonds from tested vertex domains and recovers the original index. Every component is nonvoid: a component with only parameter cubes would contradict rootedness. This proves the bijection.

The sets with $X(\iota,\mathcal{K}) = Z$ are exactly $\mathcal{K} = \mathcal{K}_0 \sqcup \mathcal{K}'$ with $\mathcal{K}' \subset \mathsf{N}'$. The elementary finite difference identity

$$\sum_{\mathcal{K}'\subset\mathsf{N}'} f[\mathcal{K}_0\cup\mathcal{K}' \mid \emptyset] = f[\mathcal{K}_0 \mid \mathsf{N}']$$

proves (1326). Expanding its right side, the parameter at $W \subset \mathcal{K}_0$ is the indicator of $(W \cup Z_\circ^\sharp) \setminus A^\sharp$. This expression does not use the guard. Nonrooted terms may be added since they vanish, so the activity is unchanged. The assertion for cluster sums then follows term by term. □

For $\mathfrak{R} = T_\ell$, call the Gaussian structure complete if there are a symmetric invertible whole covariance $\mathbf{C}_*$ and a whole mean $\mathbf{m}_*$, and a real positive reference $\mathbf{C}_{*\mathrm{r}}$, with

$$\gamma_{\mathrm{r}} \le \mathbf{C}_{*\mathrm{r}}^{-1} \le c_{\mathrm{r}}, \qquad \|\mathbf{C}_*^{-1} - \mathbf{C}_{*\mathrm{r}}^{-1}\| \le \hat{\varepsilon},$$

such that, at $s \equiv 1$ and with χ the restriction to the integration bonds touching $Z_\circ$,

$$\mathbf{C} + \mathbf{C}\boldsymbol{\Gamma}\boldsymbol{\Gamma}^{\mathsf{T}}\mathbf{C} = \chi\mathbf{C}_*\chi^{\mathsf{T}}, \qquad \mathbf{m} = \chi\mathbf{m}_*. \tag{1327}$$

For complete data define

$$d\nu(B) = \det{}^{1/2}(\mathbf{C}_*^{-1}/2\pi) e^{-(B-\mathbf{m}_*,\, \mathbf{C}_*^{-1}(B-\mathbf{m}_*))_1/2}\, dB.$$

This is a finite complex measure of mass one. Indeed $\hat{\varepsilon} \le \gamma_{\mathrm{r}}/2$ implies $\mathrm{Re}\,\mathbf{C}_*^{-1} \ge \gamma_{\mathrm{r}}/2$.

Proposition 59.2. For complete data and bounded Borel F,

$$\mathbb{J}^{\mathbf{m}}(Z_\circ;\mathbf{1})[F] = \int F(\chi B)\, d\nu(B). \tag{1328}$$

Let Vert be the potential indices (Y,o) for which Y contains a fine cube meeting Ω_t. The whole tested integral, with its potential sum over these vertices, satisfies

$$\Xi[\mathbb{V}] := \int d\nu(B)\, \mathsf{in}_{\mathfrak{X}_{\mathrm{I}}}(B) \exp \sum_{(Y,o)\in \mathsf{Vert}} \mathbb{V}^o(Y;B) = \sum_{\mathcal{P}'} \prod_{\gamma\in\mathcal{P}'} \mathsf{z}(\gamma), \tag{1329}$$

where $\mathcal{P}'$ ranges over pairwise compatible coarse domains, including the empty family. ∎

Proof. The convergent Gaussian integral with the linear factor $e^{i(f,B)_1}$ gives

$$\mathbb{J}^{\mathbf{m}}[e^{i(f,B)_1}] = \exp\left\{ i(f,\mathbf{m})_1 - \tfrac{1}{2}(f,(\mathbf{C}+\mathbf{C}\boldsymbol{\Gamma}\boldsymbol{\Gamma}^{\mathsf{T}}\mathbf{C})f)_1 \right\}. \tag{1330}$$

One first integrates B and then the real auxiliary Gaussian; Fubini is justified by the joint Gaussian majorant. The whole measure has Fourier transform $\exp\{i(f,\mathbf{m}_*)_1 - (f,\mathbf{C}_* f)_1/2\}$. By (1327), these transforms agree after restriction. Uniqueness of the Fourier transform for finite complex Borel measures gives (1328), including the case of an empty integration set.

Expand each exponential into $1+(e^{\mathbb{V}^o}-1)$. For every selected vertex retain the small-field tests on its domain; on the remaining bonds expand $\mathsf{in} = 1 - \mathsf{out}$. The resulting nonconstant terms are precisely the F_ι of (1148). They are bounded Borel functions, with zero continuation where a potential is undefined. The expansion is finite. For each index, (1328) and the sum of all finite differences give

$$\int F_\iota(\chi B)\, d\nu(B) = f_\iota(\mathbf{1}) = \sum_{\mathcal{K}} T(\iota,\mathcal{K}).$$

Only rooted pairs contribute. Group them by their domain, use (1325), and identify the wall components of a coarse region with a compatible polymer family. This proves (1329). □

59.2. **The majorant and its decay.** Let $w^o(Y) \ge e\, u^o(Y)$, where $|\mathbb{V}^o(Y)| \le u^o(Y) \le 1$ on its complex range, and assume

$$\sum_o w^o(Y) \le \varepsilon_{\mathsf{w}} e^{-r_{\mathsf{w}} d_k(Y)}. \tag{1331}$$

All parameters $r_H, r_{\mathsf{w}}, \kappa''$ are nonnegative. Choose

$$\mathsf{C}_{\mathrm{cs}} \ge \max\{C'_{\mathrm{col}}, 24 C_{\mathrm{E}}^2 e^{24}\}.$$

Here C'_{col} is the coarsening constant in proposition 45.1, and $c_{\mathrm{E}}, C_{\mathrm{E}}$ are the entropy constants used in theorem Q. Set

$$\begin{aligned} \alpha_{\mathsf{w}} &= L^3 \varepsilon_{\mathsf{w}} + (LM)^3 e^{-\gamma_P}, \\ \varepsilon_H &= \mathsf{C}_{\mathrm{cs}} e^{\mathsf{C}_{\mathrm{cs}}(c_{\mathrm{P}}+r_H)} \alpha_{\mathsf{w}}, \\ r_{\mathrm{pol}} &= r_H - \mathsf{C}_{\mathrm{cs}}(c_{\mathrm{P}}+1), \qquad & C_{\mathrm{pol}} &= \mathsf{C}_{\mathrm{cs}} e^{\mathsf{C}_{\mathrm{cs}}\kappa''}, \\ K_{\mathrm{mk}} &= \mathsf{C}_{\mathrm{cs}} e^{\mathsf{C}_{\mathrm{cs}}(c_{\mathrm{P}}+\kappa'')}, \\ K_{\mathrm{or}} &= \mathsf{C}_{\mathrm{cs}} (LM)^3 e^{\mathsf{C}_{\mathrm{cs}}(c_{\mathrm{P}}+r_H+\kappa'')}. \end{aligned} \tag{1332}$$

The hypotheses for the cluster bounds are

$$\begin{aligned} (\mathsf{C}_{\mathrm{cs}}/L) r_H + c_{\mathrm{E}} \le r_{\mathsf{w}}, \qquad & e^{-(\bar\kappa-1)+\mathsf{C}_{\mathrm{cs}} r_H} \le \mathsf{C}_{\mathrm{cs}}^{-1}, \\ r_{\mathrm{pol}} \ge \kappa'' + c_{\mathrm{E}} + \mathsf{C}_{\mathrm{cs}}, \qquad & C_{\mathrm{pol}} \varepsilon_H \le \tfrac{1}{2}. \end{aligned} \tag{1333}$$

Let $\bar T$ denote the majorant in (1151). Write $\bar{\mathsf{z}}$, $\bar E$ and $\bar E^{\mathsf{Mk}}$ for the corresponding nonnegative activity and refined cluster sums.

Theorem AA. Under (1333), the refined cluster series converge absolutely and

$$|\mathsf{z}(\gamma)| \le \bar{\mathsf{z}}(\gamma) \le \varepsilon_H e^{-r_{\mathrm{pol}} d_t(\gamma)}, \tag{1334}$$

$$|E(X)| \le \bar E(X) \le C_{\mathrm{pol}} \varepsilon_H e^{-\kappa'' d_t(X)}. \tag{1335}$$

Both sums vanish if no coarse cube of their domain meets Ω_t. For complete whole-torus data,

$$\Xi[\underline{\mathbb{V}}] = \exp \sum_{X \in \mathcal{D}_t} E(X) \neq 0. \tag{1336}$$

Suppose Mk is closed under reversal on bonds. Put

$$\eta^{\mathsf{Mk}} = \sup_{\Delta} \Bigg[\sum_{\substack{(Y,o)\in \mathsf{Mk} \\ \Delta \subset Y^\sharp}} w^o(Y) e^{r_H (d_t(Y^\sharp)+4)} + \#\{b \in \mathsf{Mk}^+ : \Delta \subset b^\sharp\} e^{-\gamma_P + 4 r_H} \Bigg]. \tag{1337}$$

Then

$$|E^{\mathsf{Mk}}(X)| \le \bar{E}^{\mathsf{Mk}}(X) \le K_{\mathrm{mk}} \eta^{\mathsf{Mk}} e^{-\kappa'' d_t(X)}. \tag{1338}$$

For an admitted insertion with $|\mathbb{W}(Y)| \le \varepsilon' e^{-r_{\mathsf{w}} d_k(Y)}$, where $\varepsilon' \ge 0$ need not be small,

$$|E'[\underline{\mathbb{V}}; \mathbb{W}](X)| \le \bar{E}'(X) \le K_{\mathrm{or}} \varepsilon' e^{-\kappa'' d_t(X)}. \tag{1339}$$

The insertion sum is linear in $\mathbb{W}$. ■□

Proof. We give the counting estimates, since the guard changes the index sets from those of Section 45. The pieces of a domain are $Y^\sharp$ for its vertices, $b^\sharp$ for its positive failed bonds, and the cubes of $\mathcal{K}$. Coarsening commutes with unions. Every piece is wall-connected, and

$$d_t(Y^\sharp) \le (C'_{\mathrm{col}}/L) d_k(Y), \qquad d_t(b^\sharp) = d_t(\Delta) = 0.$$

The connected-union estimate (685) therefore gives

$$d_t(\gamma) \le \sum_{(Y,o)\in \mathbf{D}} (d_t(Y^\sharp) + 4) + 4 \#\mathfrak{e}^+ + 4 \#\mathcal{K} - 4. \tag{1340}$$

For an insertion, add $d_t(Y_1^\sharp) + 4$ on the right. In particular

$$d_t(\gamma) \le (\mathsf{C}_{\mathrm{cs}}/L) \sum_{(Y,o)\in \mathbf{D}} d_k(Y) + \mathsf{C}_{\mathrm{cs}} (\#\mathbf{D} + \#\mathfrak{e}^+ + \#\mathcal{K}). \tag{1341}$$

Introduce the corrected weights and constants

$$\widehat{w}^o(Y) = w^o(Y) e^{r_H (d_t(Y^\sharp)+4)}, \qquad \widehat{w}_{\mathrm{bd}} = e^{-\gamma_P + 4 r_H}, \qquad q_\flat = e^{-(\bar{\kappa}-1) + 4 r_H},$$
$$\beta_\flat = 12 C_{\mathrm{E}} e^{4 r_H} \alpha_{\mathsf{w}}, \qquad \varepsilon_{\mathrm{ro}} = 12 C_{\mathrm{E}} e^{8 c_{\mathrm{P}} + 16} \alpha_{\mathsf{w}}, \qquad r_\flat = r_H - 8 c_{\mathrm{P}} - 16,$$
$$\mathsf{C}_{\mathrm{gas}} = C_{\mathrm{E}} e^{8 + 4\kappa''}.$$

The four rate conditions imply

$$\begin{aligned} (C'_{\mathrm{col}}/L) r_H + c_{\mathrm{E}} \le r_{\mathsf{w}}, \quad q_\flat \le \tfrac{1}{2}, \quad \beta_\flat \le \tfrac{1}{2}, \\ r_\flat \ge \kappa'' + c_{\mathrm{E}} + 8, \qquad 14 \mathsf{C}_{\mathrm{gas}} \varepsilon_{\mathrm{ro}} \le 1. \end{aligned} \tag{1342}$$

These inequalities remain valid if the amplitude is enlarged so that the new α_{w} is at most twice the original one. To check the numerical reserve, use $\mathsf{C}_{\mathrm{cs}} \ge 24 C_{\mathrm{E}}^2 e^{24}$ and

$$\mathsf{C}_{\mathrm{cs}}^2 e^{\mathsf{C}_{\mathrm{cs}} (\kappa'' + c_{\mathrm{P}} + r_H)} \alpha_{\mathsf{w}} \le \tfrac{1}{2}.$$

This bounds $\beta_\flat$ and $14 \mathsf{C}_{\mathrm{gas}} \varepsilon_{\mathrm{ro}}$, including that factor two. The second rate condition gives $q_\flat \le \mathsf{C}_{\mathrm{cs}}^{-1}$; and $r_\flat - r_{\mathrm{pol}} = (\mathsf{C}_{\mathrm{cs}} - 8) c_{\mathrm{P}} + \mathsf{C}_{\mathrm{cs}} - 16 \ge 0$.

For any $v(Y) \le \varepsilon^\natural e^{-r_{\mathsf{w}} d_k(Y)}$, coarsening and the entropy bound give

$$\sum_{\Delta \subset Y^\sharp} v(Y) e^{r_H d_t(Y^\sharp)} \le 2 C_{\mathrm{E}} L^3 \varepsilon^\natural. \tag{1343}$$

Indeed the exponent is at most $-c_{\mathrm{E}}d_k(Y)$; at most $(L+2)^3$ fine cubes can locate the enlarged domain at Δ, and $(L+2)^3 \le 2L^3$. The number of positive integration bonds with $\Delta \subset b^\sharp$ is at most $6((L+2)M)^3$. Thus the corrected vertex and failed-bond weights pinned at Δ sum to at most $\beta_\flat$. Those contained in γ sum to at most $\beta_\flat \#\gamma$.

Multiply the product in $\bar{T}$ by $e^{r_H d_t(\gamma)}$ and use (1340). The result is at most $e^{c_{\mathrm{P}}\#\gamma - 4r_H}$ times the product of corrected weights. We may now sum over all subsets of the vertices, failed bonds and cubes contained in γ. The first two subset sums are bounded by $e^{\beta_\flat \#\gamma}$ and the last by $e^{q_\flat \#\gamma}$. The vertex/failed-bond pair is nonempty, so the first exponential may be replaced by $e^{\beta_\flat \#\gamma} - 1$. Using $e^u - 1 \le ue^u$, $\#\gamma \le e^{\#\gamma}$ and $\#\gamma \le 8(d_t(\gamma)+1)$ yields

$$\bar{\mathsf{z}}(\gamma) \le \varepsilon_{\mathrm{ro}} e^{-r_\flat d_t(\gamma)}. \tag{1344}$$

The same free subset sums, with one vertex or bond required to be marked, give

$$\bar{\mathsf{z}}^{\mathsf{Mk}}(\gamma) \le e^{8c_{\mathrm{P}}+16} \eta^{\mathsf{Mk}} e^{-r_\flat d_t(\gamma)}.$$

Here the sum of marked weights in γ is at most $\#\gamma\, \eta^{\mathsf{Mk}}$. For an insertion, the vertex/failed-bond pair may be empty. Its extra domain weight is summed by (1343); hence

$$\bar{\mathsf{z}}'(\gamma) \le 2C_{\mathrm{E}} L^3 e^{8c_{\mathrm{P}}+16} \varepsilon' e^{-r_\flat d_t(\gamma)}.$$

Theorem Q, with $r = r_\flat$, $r'' = \kappa''$ and $\varepsilon = \varepsilon_{\mathrm{ro}}$, now bounds the refined cluster sum by $\mathsf{C}_{\mathrm{gas}}\varepsilon_{\mathrm{ro}} e^{-\kappa'' d_t(X)}$. Finite expansion of its activities and Tonelli for the majorants justify the refined form of this statement. For one marked factor use the following consequence of the same theorem. If $h_0 \le \varepsilon_{\mathrm{ro}} e^{-r_\flat d_t}$ and $h_1 \le \varepsilon^\natural e^{-r_\flat d_t}$, the cluster sum with exactly one h_1 and all other factors h_0 is at most $2\mathsf{C}_{\mathrm{gas}}\varepsilon^\natural e^{-\kappa'' d_t(X)}$. For $\varepsilon^\natural > 0$, apply the theorem to $h_0 + \lambda h_1$ with $\lambda = (14\mathsf{C}_{\mathrm{gas}}\varepsilon^\natural)^{-1}$ and retain its nonnegative terms of degree one in λ. Their sum times λ is at most $e^{-\kappa'' d_t(X)}/7$. The zero case is immediate. The indicator that a cluster has a mark is bounded by its number of marked factors. The insertion series has $1/(n-1)!$ in place of $1/n!$; symmetrizing its singled out factor gives the same one-factor estimate.

Finally, the definitions and the choice of C_{cs} give

$$\begin{gathered}
\varepsilon_{\mathrm{ro}} \le \varepsilon_H, \quad r_\flat \ge r_{\mathrm{pol}}, \quad \mathsf{C}_{\mathrm{gas}} \le C_{\mathrm{pol}}, \\
2\mathsf{C}_{\mathrm{gas}} e^{8c_{\mathrm{P}}+16} \le K_{\mathrm{mk}}, \\
12 C_{\mathrm{E}} \mathsf{C}_{\mathrm{gas}} e^{8c_{\mathrm{P}}+16} (LM)^3 \le K_{\mathrm{or}}, \\
4 C_{\mathrm{E}} \mathsf{C}_{\mathrm{gas}} e^{8c_{\mathrm{P}}+16+4r_H} L^3 \le K_{\mathrm{or}}.
\end{gathered}$$

These prove all the stated bounds. Each nonvoid index has a vertex containing a source cube or a failed integration bond touching $\Lambda_t \subset \Omega_t$. Its domain therefore has a coarse cube meeting Ω_t, which proves the vanishing assertions. For complete data, (1329) and theorem Q prove (1336). Linearity of the insertion follows first for each integrable term and then for its absolutely convergent series. □

59.3. **Regular and boundary contributions.** Split the origins into disjoint sets Or_{v} and Or_∂, with respective weight bounds $\varepsilon_{\mathsf{v}} e^{-r_{\mathrm{w}} d_k(Y)}$ and $\varepsilon_\partial e^{-r_{\mathrm{w}} d_k(Y)}$. Let E^{b} be the subseries with a boundary origin and put $E^{\mathrm{r}} = E - E^{\mathrm{b}}$.

Corollary 59.3.

$$\begin{aligned}
E^{\mathrm{r}}(X) &= E[\mathbb{V}|_{\mathsf{Or}_{\mathsf{v}}}](X), \\
|E^{\mathrm{r}}(X)| &\le K_{\mathrm{or}}(\varepsilon_{\mathsf{v}} + e^{-\gamma_P}) e^{-\kappa'' d_t(X)}, \\
|E^{\mathrm{b}}(X)| &\le K_{\mathrm{or}} \varepsilon_\partial e^{-\kappa'' d_t(X)}.
\end{aligned} \tag{1345}$$

■□

Proof. Removing the boundary origins removes exactly the refined indices containing one of them; all remaining terms, guards and domains are unchanged. Absolute convergence proves the identity. Apply (1344) to the restricted system with amplitude $\min(\varepsilon_{\mathsf{v}}, \varepsilon_{\mathsf{w}})$. Its cluster bound is at most

$$12C_{\mathrm{E}}\mathsf{C}_{\mathrm{gas}}e^{8c_{\mathrm{P}}+16}(LM)^3(\varepsilon_{\mathsf{v}} + e^{-\gamma_P})e^{-\kappa'' d_t(X)}.$$

For the boundary part, (1343) gives $\eta^{\partial} \le 2C_{\mathrm{E}}L^3 e^{4r_H}\varepsilon_{\partial}$. Use the sharper marked bound in the preceding proof, $2\mathsf{C}_{\mathrm{gas}}e^{8c_{\mathrm{P}}+16}\eta^{\partial}$, and the last two numerical comparisons there. □

All these series have the same majorant when their Gaussian entries and potentials vary with fixed bounds. For jointly holomorphic data, the fixed-domain Gaussian integral and its common integrable majorant prove holomorphy of each finite-difference term; the convergent majorant series then proves holomorphy of every sum and every marked subseries. For jointly Borel data the parameter integrals are Borel by measurable integration, and the cluster sums are pointwise limits of Borel partial sums. Orthogonal changes of the real integration and auxiliary variables preserve the integrals. Complex conjugation commutes with them and with the fixed determinant branch; real data therefore give real sums.

Locality requires agreement of the thresholds on the integration bonds internal to X, of every potential whose enlarged domain lies in X, and of the Gaussian entries at the 0–1 decoupling parameters supported in X, including the connecting-operator columns touching X. Under these hypotheses, all refined indices with union X, their guards and their terms agree: auxiliary columns outside X vanish and their Gaussian integral is one. Thus the sums and the marked subseries agree. The same statement holds for the majorants when the local weights agree.

59.4. **Insertions and the cutoff terms.** An insertion supported on domains containing a source cube can be recovered by adjoining it as an extra origin. Choose $\epsilon_0 > 0$ such that

$$\epsilon_0 \max_Y \sup |\mathbb{W}(Y)| \le 1, \qquad e\epsilon_0\varepsilon' \le M^3 e^{-\gamma_P}.$$

The weights of $\underline{\mathbb{V}} \cup \{\epsilon\mathbb{W}\}$ then have $\alpha_{\mathfrak{b}} \le 2\alpha_{\mathsf{w}}$ for $|\epsilon| < \epsilon_0$. The reserve (1342) gives one convergent majorant on this disc. In a term with p vertices of the new origin, the product of the new factors is $O(\epsilon^p)$. For $p = 1$ its derivative is the corresponding insertion term, with the same domain, guard and tests. This follows also in the integral norm, since

$$\left|\epsilon^{-1}(e^{\epsilon w} - 1) - w\right| \le e|\epsilon|\,|w|^2 \quad (|\epsilon w| \le 1).$$

Cauchy's formula on a smaller circle permits termwise differentiation of the uniformly majorized series. Exactly one new vertex survives, so symmetrizing its polymer position proves

$$\frac{d}{d\epsilon}E[\underline{\mathbb{V}} \cup \{\epsilon\mathbb{W}\}](X)\Big|_{\epsilon=0} = E'[\underline{\mathbb{V}}; \mathbb{W}](X). \tag{1346}$$

The support condition ensures that every nonzero insertion index corresponds to a permissible new vertex.

For complete data, an arbitrary admitted insertion satisfies

$$\sum_X E'[\underline{\mathbb{V}}; \mathbb{W}](X) = \Xi[\underline{\mathbb{V}}]^{-1} \int d\nu(B)\, \mathsf{in}_{\mathfrak{X}_1}(B) \times e^{\sum_{(Y,o)\in \mathsf{Vert}} \mathbb{V}^o(Y;B)} \sum_{Y_1} \mathbb{W}(Y_1; B). \tag{1347}$$

To see this without a support assumption, perform the finite expansion of proposition 59.2 with a distinguished domain Y_1. Exactly one wall component carries the insertion. Thus its integral is $\sum_\gamma \mathsf{z}'(\gamma)\Xi(\mathcal{P}_0(\gamma))$, where $\mathcal{P}_0(\gamma)$ consists of polymers compatible with γ. To identify this ratio, put $h = \mathsf{z}$ and $\bar{h}(\gamma) = (7C_{\mathrm{E}}e^8)^{-1}e^{-r_{\mathfrak{b}}d_t(\gamma)}$. The preceding estimates give $r_{\mathfrak{b}} \ge c_{\mathrm{E}} + 8$ and $\varepsilon_{\mathrm{ro}} \le (14C_{\mathrm{gas}})^{-1} \le (14C_{\mathrm{E}}e^8)^{-1}$. Thus $\bar{h}$ satisfies the polymer convergence criterion, and h lies strictly

inside its open polydisc. On this polydisc the cluster sum S is holomorphic. Differentiating its uniformly convergent series and using symmetry in the polymer positions gives the series below for $\partial S/\partial h(\gamma)$. On the other hand, splitting compatible families according to whether they contain γ gives $\Xi = \Xi(\mathcal{P} \setminus \{\gamma\}) + h(\gamma)\Xi(\mathcal{P}_0(\gamma))$. Since γ is incompatible with itself, both partition functions on the right are independent of $h(\gamma)$. Differentiating this equality and $\Xi = e^S \neq 0$ proves

$$\frac{\Xi(\mathcal{P}_0(\gamma))}{\Xi} = \sum_{n\geq 1} \frac{1}{(n-1)!} \sum_{\gamma_2,\dots,\gamma_n} \rho^T(\gamma,\gamma_2,\dots,\gamma_n) \prod_{i\geq 2} \mathsf{z}(\gamma_i).$$

Its absolute sum is at most $e^{\#\gamma}$. There are finitely many polymers, so multiplication by $\mathsf{z}'(\gamma)$, summation over γ, and grouping by the union are justified. This proves (1347).

For the empty potential system, allow bond parameters $0 \leq t_b \leq 1$ in the tests: $\mathsf{out}_b^{\mathbf{t}}(B) = \mathbf{1}\{t_b|B(b)| \geq \mathfrak{b}_b\}$. For $t_b = 0$ this is zero. These tests are the original failed tests multiplied by a bounded Borel function of the failed variables. Proposition 57.3 therefore gives the same term majorants for every $\mathbf{t}$.

Proposition 59.4. The empty-system cluster sum admits the bond decomposition

$$E[\emptyset](X) = \sum_{b\in\mathfrak{X}_{\mathrm{I}}^+} E^\chi[\emptyset](X,b), \qquad E^\chi[\emptyset](X,b) = \int_0^1 \partial_{t_b} E[\emptyset;\mathbf{t}](X)|_{\mathbf{t}=(u,\dots,u)}\, du. \tag{1348}$$

The derivative exists for every $0 < u < 1$, is integrable, and

$$|E^\chi[\emptyset](X,b)| \leq \bar{E}^{\{b,\bar{b}\}}(X). \tag{1349}$$

This member vanishes unless $b^\sharp \subset X$. ∎

Proof. For $\iota = (\emptyset, \mathfrak{e})$, integrate the Gaussian density first in all variables except those on $\mathfrak{e}$. The alternating sum over parameter cubes, including the sign $(-1)^{\#\mathfrak{e}^+}$, gives a continuous density $\sigma(x)$ such that

$$T(\iota,\mathcal{K};\mathbf{t}) = \int \sigma(x) \prod_{b\in\mathfrak{e}^+} \mathbf{1}\{t_b|x(b)| \geq \mathfrak{b}_b\}\, dx.$$

It satisfies $|\sigma(x)| \leq Ce^{-c\|x\|^2}$ for some $c > 0$. Indeed the real part of each joint Gaussian exponent is a positive quadratic form plus a linear term; absorbing that term leaves a Gaussian bound. Its marginal has the same property, and the alternating sum is finite. Moreover

$$\int |\sigma(x)| \prod_{b\in\mathfrak{e}^+} \mathbf{1}\{|x(b)| \geq \mathfrak{b}_b\}\, dx \leq \bar{T}(\iota,\mathcal{K}).$$

Apply the bounded-failed-factor conclusion of proposition 57.3 with $\psi = \overline{\sigma}/|\sigma|$ where $\sigma \neq 0$, and zero otherwise.

Use spherical coordinates separately on each three-dimensional bond space, and set $\zeta_b = \mathfrak{b}_b/|x(b)|$. The radial Jacobian is $\mathfrak{b}_b^3\zeta_b^{-4}$. Consequently

$$T(\iota,\mathcal{K};\mathbf{t}) = \int_{\prod_b[0,t_b]} \upsilon(\zeta)\, d\zeta, \qquad \int_{[0,1]^{\mathfrak{e}^+}} |\upsilon|\, d\zeta \leq \bar{T}(\iota,\mathcal{K}).$$

The Gaussian decay makes υ continuous on the closed cube after setting it to zero on its zero faces: each factor $\zeta_b^{-4}e^{-c\mathfrak{b}_b^2/\zeta_b^2}$ tends to zero there.

In a product of n terms take a separate copy $\zeta_{i,b}$ for each factor. Partition this cube, up to null sets, according to the coordinate that is largest. On the part where coordinate (i,b) is largest, integration in that coordinate is precisely the integral from zero to one of the corresponding diagonal t_b derivative. It follows that the sum over bonds of the integrals of the absolute derivatives is at most $\prod_i \bar{T}_i$, and the sum without absolute values equals $\prod_i T_i(\mathbf{1})$. Only bonds occurring in some $\mathfrak{e}_i$ contribute.

For completeness, differentiation of the infinite cluster series is justified before this integration. There are finitely many Mayer indices with cubes. For each one υ is bounded and $\bar{T} > 0$, so $\mathsf{M} = \max \|\upsilon\|_\infty / \bar{T} < \infty$. If that index set is empty, set $\mathsf{M} = 0$; all the cluster sums in question are then empty. The derivative of a product is bounded by $\mathsf{M} n \prod_i \bar{T}_i$. The one-mark argument in theorem AA with $h_1 = h_0 = \bar{\mathsf{z}}$ proves summability of the refined majorant multiplied by n. Thus the derivative series converges uniformly for $0 < t_b < 1$ with other parameters fixed, and on the diagonal. Termwise integration is also justified by the sharper largest-coordinate estimate. Summing the identities over refined indices proves (1348); restricting to indices containing b proves (1349) and the support assertion. □

59.5. **The cluster terms in the interior.** The expansion in a domain contained in Λ_t must reproduce the regular expansion. This requires agreement of the local Gaussian operators and potentials, including their domains, before the cluster series is summed.

Fix a step $t = k + 1$, a history h' of depth t, and $X \in \mathcal{D}_t$ with $X \subset \Lambda_t(h')$. Compare it with the empty history $h_\emptyset$ at the same level, with the same coupling prefix, regular families, constants, and choices in the local constructions. Use the geometric and analytic data of the actual history step. In particular, $m' < m + K - t$. When $k \geq 1$, assume

$$\check{R}_j \leq L \check{R}_{j+1} \qquad (1 \leq j < t).$$

This is (671) through level t.

Write $\mathfrak{X}^{1\mathrm{l}}$ for all nonpivot bonds at level k, and set

$$\mathfrak{c} = \mathfrak{X}^{1\mathrm{l}\circ}\langle X\rangle, \qquad \mathfrak{d} = \mathfrak{X}^{1\mathrm{l}}\langle X\rangle.$$

Thus $\mathfrak{c}$ has both endpoints in X, whereas $\mathfrak{d}$ has at least one. For either history let $F_h(X, P, z)$ denote the pointed term formed from $E_h^{\chi} + E_h^{\bullet}$, with the tents in (896), at $\tau = 1$. The regular potential system is $\underline{\mathbb{V}}^{\mathrm{rg}} = \{\mathbb{V}^{\mathrm{int}}, \mathbb{V}'\}$.

For the sum identity below we require the following analytic bounds at (X, P). The empty-history operators form the Gaussian structure of definition 47.2 on X, and its constants satisfy the first three inequalities of (711). For every $Y^\sharp \subset X$, the four potentials J, G, A, and $\prime$, evaluated at $(P|_Y, g_k B)$, are holomorphic on $\mathbb{D}^{\mathrm{f}}(Y)$, with $\mathsf{C}_{\mathcal{A}} \geq 3$. The J and $\prime$ potentials vanish at zero, the G and A potentials vanish through degree two, and

$$\|\mathbb{V}^{\mathrm{J}}\|^{\mathrm{f}}_{\star,\lambda_V} + \|\mathbb{V}^{\mathrm{G}}\|^{\mathrm{f}}_{\star,\lambda_V} + \|\mathbb{V}^{\mathrm{A}}\|^{\mathrm{f}}_{\star,\lambda_V} \leq w_\star, \qquad \|\mathbb{V}'\|^{\mathrm{f}}_{\star,\lambda_V} \leq w_\star. \tag{1350}$$

These are the hypotheses of the interpolation and cluster estimates already proved. They hold on the regular analytic cores used by the actual step, by section 52.4 and theorem V and the common constants of section 69.1.

Proposition 59.5. On X, the historical covariance and connecting operator are the empty-history operators, as partial maps of P, and the conditional mean is zero wherever those operators are defined. The pointed terms satisfy

$$F_{h'}(X, P, z) = F_\emptyset(X, P, z) \tag{1351}$$

as partial maps for $z \in X^{(t)}$.

If the analytic bounds above hold, all terms in

$$\begin{aligned} E[\underline{\mathbb{V}}^{\mathrm{rg}}]_{h'}(X; P) &= \sum_{b \in \mathfrak{X}_{\mathrm{I}}^+} (E_{h'}^{\chi} + E_{h'}^{\bullet})(X, b; P) \\ &= \sum_{z \in X^{(t)}} F_\emptyset(X, P, z) \end{aligned} \tag{1352}$$

are defined and the equalities hold. If, in addition, the old retained factors have their prescribed indices, then the full history cluster sum has this same value. At $k = 0$ there are no old retained factors and the last conclusion holds without that condition. ■

Proof. We first compare the geometry. A nonpivot bond with an endpoint in X is an integration bond, and no external bond meets X. For every region $Y \subset X$,

$$\mathfrak{X}_{\mathrm{I}}\langle Y\rangle = \mathfrak{X}\langle Y\rangle = \mathfrak{X}^{\mathrm{11}}\langle Y\rangle, \qquad \mathfrak{X}^{\circ}_{\mathrm{E}}\langle X\rangle = \emptyset.$$

The same equalities hold for bonds with both endpoints in Y. Every fine cube of X is a source cube. Each contains a whole block $B(y)$, including the nonpivot bond $\langle y, y+e_1\rangle$. All bonds internal to X have threshold $\mathfrak{b}_k$.

If $Z^{\sharp}_{\circ} \subset X$, every bond touching $Z_{\circ}$ lies inside $Z^{\sharp}_{\circ}$. Consequently

$$Z^{\mathrm{I}}_{\circ} = Z_{\circ}, \qquad \mathrm{Gd}(Z_{\circ}) = Z^{\sharp}_{\circ}, \qquad \mathsf{P}^{\mathrm{gd}}(X, Z_{\circ}) = \mathsf{P}(X, Z_{\circ}).$$

The guarded completion of the decoupling parameters therefore makes no further change. For a Mayer index on a polymer $\gamma \subset X$, the condition $Z^{\sharp}_{\circ} \cup \bigcup \mathcal{K} = \gamma$ forces $\mathcal{K}$ to be exactly the cubes of γ outside $Z^{\sharp}_{\circ}$. The resulting index is rooted: γ is connected and the nonempty guard belongs to it. This applies also to indices with an insertion, since their region contains the insertion domain.

The operators. For a grain region Y with $X(Y) \subset X$, every member of its tower lies in $Y \subset \Lambda_t$. The meet with either collared sequence is therefore the tower itself. The two deep-bond sets Y^{-} also coincide and lie in $\mathfrak{c}$. In the alternating formula for a walk term, the compressed operators are consequently the same:

$$\chi_{Y^-}\mathsf{A}^{\sharp}[\Omega_{\bullet}(Y)]\chi^{\mathsf{T}}_{Y^-}.$$

Their domains agree, since they use the same tower and restricted pair. Summing the finite alternating formula gives equal kernels on $\mathfrak{c}^{+} \times \mathfrak{c}^{+}$, zero off this set, with simultaneous definedness.

This comparison extends to the local square root. For a string whose completed parameter weight is nonzero, its entire figure lies in X. The sets of bonds starting in its enlarged mesh cubes coincide for the two histories and are subsets of $\mathfrak{c}$. The local compressed operators, their resolvents $(\mu + \mathsf{a}^{\square})^{-1}$, the cutoff multiplications, and the commutator factors thus coincide. Invertibility is one condition on one compressed matrix. The string products have identical kernels on $\mathfrak{c}$ and vanish off it. The integral

$$\frac{1}{\pi}\int_0^{\infty} \mu^{-1/2}\mathbf{g}^{\mu}_{\boldsymbol{\omega}}\, d\mu$$

and the subsequent string series have the same conditions of entrywise absolute convergence. It follows that both $\mathsf{A}^{\sharp}$ and $\mathbf{r}_{\mathrm{I}}$, with parameters supported in X, agree as partial maps after restriction to $\mathfrak{c}$, and vanish off that set.

For the Gaussian entries let $\mathfrak{e} = \mathfrak{X}^{\mathrm{11}}\langle Z_{\circ}\rangle \subset \mathfrak{c}$. The completed parameter is supported in X. Both constructions read P at level k, with current $L^{-3}\mathbf{J}$, and use the same matrices in

$$\mathbf{C} = (\chi_{\mathfrak{e}}\mathsf{A}^{\sharp}\chi^{\mathsf{T}}_{\mathfrak{e}})^{-1}, \qquad \mathbf{\Gamma} = \chi_{\mathfrak{e}}\mathsf{A}^{\sharp}(\mathbf{1}_{\mathfrak{X}_{\mathrm{I}}} - \mathbf{1}_{\mathfrak{e}})\mathbf{r}_{\mathrm{I}}\chi^{\mathsf{T}}_{\mathfrak{d}}.$$

On $\mathfrak{c}$, both integration indicators are one. This proves simultaneous definedness and equality of $\mathbf{C}, \mathbf{\Gamma}$. Moreover $\mathsf{A}^{\sharp}\chi^{\mathsf{T}}_{\mathrm{E}} = 0$. All remaining factors of the conditional-mean operator are defined when $\mathbf{C}, \mathbf{\Gamma}$ are, so $\mathbf{M} = \mathbf{m} = 0$.

The conditioning integrals now coincide as well. Indeed, with $v = \mathbf{\Gamma}\mathbb{X}$, symmetry of $\mathbf{C}$ gives

$$(B + \mathbf{C}v, \mathbf{C}^{-1}(B + \mathbf{C}v))_1 = (B, \mathbf{C}^{-1}B)_1 + 2(B, v)_1 + (v, \mathbf{C}v)_1.$$

Thus the historical and single-lattice integrands are identical, with the same determinant branch and the same real integration variables. Their Borel, positivity, and absolute-integrability conditions coincide, by Tonelli's theorem. Fubini's theorem then identifies their values. The preceding bijection of guarded indices identifies each finite-difference term, activity, and cluster series, also with an insertion. Absolute convergence of the corresponding series is the same condition.

The potentials. No stored datum at an earlier level has an endpoint in Λ_t, because it is external to $\Lambda_{j+1} \supset \Lambda_t$. The spaces of stored data on $Y \subset X$ are therefore singletons. For $k \geq 1$, the shift domains and maps on Y are those of the constant sequence. To check the trace condition in section 58.3, use $\mathrm{dist}_{\infty}(\Lambda_t, \Omega_t^{c}) \geq 2LM$ in level-k scale. The blocks meeting Y and the fivefold

enlargements required there lie within distance $5\mathfrak{M} < 2LM$ of Λ_t; all belong to every member of the collared sequence. The last trace also agrees, since $Y \subset \Omega_t$.

The margin from every earlier large-field region is

$$\operatorname{dist}_\infty(\Lambda_t, Z_j) \geq 8LM\check{R}_{j+1} \geq 8M\check{R}_j \qquad (1 \leq j < t),$$

in level-j scale. Write s_j for the cell side in a common length scale. Every level-j cell meeting Y is inside $\Lambda_j^{\sim -1}$. Also $\varphi_j = 1$ at distance at most s_j from Y: translating such a point by a mollifier argument of norm at most $s_j/4$ leaves distance at least $(8 - 1 - \frac{1}{4})s_j$ from Z_j. In particular the excess Wilson weight vanishes on plaquettes meeting Y.

All sites of Y are in the top layer k. The regular indices with base in Y are precisely the common wide indices and the octets whose level is k. Their functions, domains, and subtraction at the trivial pair agree. No edge-tent or previously created boundary index has base in Y, since such an index must meet the complement of an earlier $\Lambda_j^{\sim -1}$. Nor can a retained-factor base lie in Y when the retained factors have their prescribed indices: an older factor must meet some Ω_{l+2}^c, and a factor of the last old step must meet $Z_k^{\sim}$. The first possibility contradicts the nesting, and the second the distance $8s_k > s_k$ from Z_k.

These facts identify the four regular potentials as partial maps. For J, the pivot paths lie within the double blocks of the coarse bonds, hence within their local domains; the free coordinates and pivot solutions agree. For G, the cube, block field, and gauge-fixing coordinates agree. For the action potential, the common shift is substituted into the same top-layer Wilson members; all lower-layer members vanish. The Taylor subtraction giving A therefore agrees. Finally the regular and counterterm members have the same plain substituted triples, indices, and starred brackets. This identifies $\mathbb{V}'$, with the same restriction to the shift domain and field polydisc. At $k = 0$, $\mathbb{V}' = 0$ and the completed pivot field gives the same action comparison directly. None of these four potentials uses the pinned substitution for boundary members.

The B and Φ potentials are zero because their index sets are empty. For $k \geq 1$, their domain restrictions remain: they are defined on the common shift domain with $g_k^{-1}\mathcal{A} \in \mathbb{D}^{\mathrm{f}}(Y)$. At $k = 0$ they are zero functions restricted only by $g_0^{-1}\mathcal{A} \in \mathbb{D}^{\mathrm{f}}(Y)$. Under the stated retained-index condition this holds for both potentials. A Mayer index carrying one of these origins consequently has a factor $e^0 - 1$. Wherever the full cluster sum is defined, deleting such zero terms gives the regular cluster sum.

Pointing. The four identical potential families give the same interpolation and bond insertions. Together with the Gaussian comparison, this identifies $E^\chi(X, b; P)$ and $E^\bullet(X, b; P)$ for every common bond, as partial maps. There is a small domain issue for the other bonds. If $b^\sharp \not\subset X$, the empty-system cluster sum is independent of its cutoff parameter t_b. Its derivative is zero wherever defined; if a cutoff term for any other bond is defined, the diagonal cluster sum is defined for almost every parameter, which also defines this zero term. For the potential interpolation the insertion at such a bond is identically zero on every local domain. Its domain contains the domain of any other bond insertion. A defined insertion term for another bond supplies all the Gaussian operators and unmarked terms; replacing its insertion by zero gives absolutely convergent zero integrals and a zero series. Thus here too definedness at one bond supplies definedness and vanishing at every bond with $b^\sharp \not\subset X$.

For $z \in X^{(t)}$, the bond $b_z = \langle z, z + e_1 \rangle$ is in $\mathfrak{c}^+$ and $h_z(b_z) \geq 1 - (2L)^{-1} > 0$. A pointed term therefore includes at least one common bond. The preceding observation removes all additional bonds without changing its domain. On the remaining bonds the weights, values, and domains agree. This proves (1351).

Summation. Apply proposition 48.4 with $\tau = 1$ to the four families in (1350). Its path has two origins, each of norm at most $w_\star$, and endpoints

$$\underline{\mathbb{V}}_1 = \{\mathbb{V}^{\mathrm{J}} + \mathbb{V}^{\mathrm{G}} + \mathbb{V}^{\mathrm{A}}, \mathbb{V}'\}, \qquad \underline{\mathbb{V}}_0 = \{0, 0\}.$$

The velocities have norms at most $4w_\star$; the sum of the two bond insertions has norm at most $18w_\star$ and is Borel. Thus proposition 48.2 applies and gives

$$\sum_b E^\bullet(X,b) = E[\underline{\mathbb{V}}_1](X) - E[\underline{\mathbb{V}}_0](X).$$

Origins with zero potential make no contribution, so $E[\underline{\mathbb{V}}_0] = E[\emptyset]$. The cutoff identity of theorem T gives

$$\sum_b E^\chi[\emptyset](X,b) = E[\emptyset](X) - E^\circ[\emptyset](X).$$

The last term is zero: at infinite threshold and with no potential vertices every Mayer function contains a vanishing outside test. The Gaussian cluster estimates remain valid at infinite threshold since the cutoff error is then zero. Combining these two identities proves the first equality in (1352). All bond terms vanish unless $b^\sharp \subset X$. The tent partition of unity therefore gives the second equality, with no sites outside X.

Finally, the four analytic families are defined on the whole field polydisc. For $k \geq 1$, definedness of the $\prime$ family includes membership of the common shift domain. The B and Φ families are consequently defined and zero at every real field below the threshold. For $k = 0$, these two families are zero on the field polydisc by definition, so the same conclusion holds without a shift-domain condition. The full system has four origins with norms at most $w_\star$, so theorem R proves its absolute convergence. Deleting the zero boundary vertices identifies it with the regular system. This proves the last assertion on the stated analytic domain. □

60. Evaluation on critical orbits

The density is a function of the determining variables and the stored fields. Its local terms are written using a critical configuration. We show that their values do not depend on this choice, and that the configuration produced by the shift gives the same old density.

60.1. **Adapted representatives.**

Definition 60.1. Let $1 \leq n \leq K$, and let $\Omega_0 \supset \cdots \supset \Omega_n$ be a separated collared sequence, with reduced determining set $\mathbf{B}$. A top $y \in \Lambda_q(\mathbf{B})$, $q \geq 1$, has stored sites $a \in B(y)$. The transport of a configuration U' at such a site is

$$\mathsf{ax}_{U'}(a) = (\mathcal{M}^{q-1}U')(y,a).$$

We call U' adapted if all these averaged contour variables are defined and equal to 1. Adaptedness at a set of sites requires this only at its stored sites. In particular, adaptedness at layer q means that $\mathcal{M}^{q-1}U'$ is in block axial gauge on $\Lambda_q(\mathbf{B})$. ■

Proposition 60.2. Under the hypotheses of proposition 53.11 and (1025), every regular critical orbit of a completed collared sequence has an adapted representative. Such a representative can be chosen continuously, and hence Borel measurably, on the set of data that pass the strict regularity tests. Every adapted representative gives the same averaged background on the bonds of the last step. ■

Proof. The pillars, parent $z(x)$, and level $n(x)$ of a non-top site are as in definition 8.2. Write $q(x)$ for its pillar's level. The parent belongs to the same pillar and has a strictly higher level. The nodes of the reduced determining set are its tops: every incident row endpoint is a top, and every top has an incident row in its layer. For the latter assertion, the layer is a union of the separated partition cubes; a neighboring site within the same cube gives the row. The argument for the bare layer also covers every endpoint of an inner bare bond.

Put $e = \bar{C}_{\mathrm{reg}}\varepsilon_n$. The layer estimate gives

$$|\partial U'(p) - 1| < eL^{-2q}$$

on bare plaquettes in a pillar of level q. Fix such a pillar $B^q(y)$, and set $Y_{q-1} = B(y)$ and $Y_l = B^{q-1-l}(Y_{q-1})$. Thus $Y_0 = B^q(y)$ and $B(Y_{l+1}) = Y_l$. Finiteness and the strict curvature inequalities permit $0 < e' < e$ such that all bare plaquette bounds in this pillar hold with e' in place of e. Apply the local induction in the proof of lemma 66.1, with terminal index $q-1$ and $\alpha = e'L^{-2}$. Every double block needed for a coarse plaquette in Y_{l+1} lies in Y_l, so no curvature bound outside the pillar is used. The induction gives

$$|\partial(\mathcal{M}^l U')(p') - 1| \le 2e'L^{2(l-q)} < 2eL^{2(l-q)}, \qquad 0 \le l < q.$$

All block contours used below are therefore in the unique analytic mean chart. Here the bound $e \le c_2^{\mathrm{SO}}$ follows from (1025).

For every non-top bare site define

$$\tau_{U'}(x) = (\mathcal{M}^{n(x)}U')(z(x), x).$$

Unlike the comb transports of proposition 8.3, these are averaged contour variables. They are real analytic in a neighborhood of U', and satisfy

$$\tau_{U'^w}(x) = w(z(x))\tau_{U'}(x)w(x)^{-1}.$$

Indeed, every contour acquires these same endpoint factors, and uniqueness of the group mean preserves them. Define $u_{U'} = 1$ at the tops and recursively set

$$u_{U'}(x) = u_{U'}(z(x))\tau_{U'}(x).$$

The recursion terminates at a top. It fixes all nodes, so $U'^{u_{U'}}$ remains in the same critical orbit, and all its τ's equal 1. The stored sites at layer q, other than the top itself, are precisely the sites with $q(x) = q$ and $n(x) = q-1$. Thus this representative is adapted.

If two representatives in the same residual orbit have all their τ's equal to 1, the covariance formula gives $w(x) = w(z(x))$. Since $w = 1$ at the tops, induction gives $w = 1$ on every pillar. The construction consequently depends only on the datum. Locally, choose the real analytic critical configuration supplied by theorem E, and apply the finite recursion. This proves continuity of the chosen representative. The set of strictly regular data is open: the pure tested plaquettes are products of datum coordinates; on the set where these tests pass, the other tested bonds are analytic one-step averages of those coordinates. All remaining tests are finitely many strict inequalities of continuous functions. Hence the selection is Borel.

Finally let U' and $U'' = U'^w$ be adapted representatives. Every endpoint of a bond in the last step is either a node or a stored site of a last-layer top. The transport covariance gives $w = 1$ at these endpoints. Covariance of $\mathcal{M}^{n-1}$ then gives equality of the averaged backgrounds, including at bonds crossing the last region's boundary. □

60.2. **Rigidity at the stored sites.**

Lemma 60.3. Let $U'' = U'^w$ lie in the same residual orbit for a separated sequence, so that $w = 1$ at its nodes. If the transports of U' and U'' are defined and equal at a stored site a, then $w(a) = 1$.

Consequently, let $f(P; \mathfrak{q})$ be a jointly gauge-invariant local term whose stored fields transform only at a set N_f of nodes and stored sites. If the two configurations have equal transports at the stored sites in N_f, and their plain triples belong to the domain of f, then, at fixed stored data,

$$f(P(U''); \mathfrak{q}) = f(P(U'); \mathfrak{q}). \tag{1353}$$

■

Proof. If y is the top of a, covariance gives

$$\mathsf{ax}_{U'^w}(a) = w(y)\mathsf{ax}_{U'}(a)w(a)^{-1}. \tag{1354}$$

The top is a node, so $w(y) = 1$. Cancellation of the common transport gives $w(a) = 1$. At $a = y$ the transport itself is always 1.

The first assertion gives $w = 1$ on N_f, so $w \cdot \mathfrak{q} = \mathfrak{q}$. Plain triples are covariant: their configuration is restricted, their second component is zero, and their current is covariant at complete stencils and zero at the other stencils. Joint invariance therefore proves (1353). □

For a history h of depth n, let

$$\mathsf{Sd}_h = \{b_- : b \in \mathfrak{X}_l^{\mathrm{E}+}(h),\ 0 \le l < n\}.$$

These are the only sites where a gauge transformation acts on its stored data: $(w \cdot \mathfrak{q})_l(b) = \mathrm{Ad}(w(b_-))\mathfrak{q}_l(b)$.

Lemma 60.4. Every site of Sd_h is a node or a stored site of a top that can be pinned. If $h' = (h, \mathsf{o}_t)$, $t = k + 1$, then $\mathsf{Sd}_h \cap \Omega_t = \emptyset$, and the old stored sites have the same tops, levels and transports for the shorter and longer collared sequences. ■

Proof. Use the external-endpoint geometry of lemma 58.12. For an external bond of level l, its initial point x lies outside Λ_{l+1}. If $x \in \Omega_{l+1}$, it is stored at the pinnable top $\mathsf{c}_{l+1}(x)$ of level $l + 1$. Otherwise it is a top of level l, and a node; at $l = 0$ it is the endpoint of an inner bare bond and hence a bare node.

For $l < k$, the inclusion $\Omega_t \subset \Lambda_k \subset \Lambda_{l+1}$ puts x outside Ω_t. All earlier layers of the two collared sequences coincide. At level k, their top sets differ only inside Ω_t. Membership in this region is constant on each block of order at most t, so the top of an old stored site cannot enter the changed part. The transport in either sequence is the same variable $(\mathcal{M}^{q-1}U')(y, x)$. □

60.3. **The evaluations before and after a step.** Fix a bare field U in the support of a history h. The evaluation configurations are

$$\mathsf{Ev}_h[U] = \{U' \vee U : U' \in U_n^{\diamond}(\xi_h^{\diamond}(U)) \text{ is adapted}\}.$$

At depth zero this set is $\{U\}$. The gluing uses U' on inner bonds of the first collared region and U on the outer bonds.

Proposition 60.5. Assume the actual domain, response and levelling hypotheses of section 58. Let U be in the support of $h' = (h, \mathsf{o}_t)$, and use its actual stored fields $\mathfrak{q}^+$. Then the following assertions hold.

Every local term in the exponent and retained-factor sum of h' is defined at every $\mathsf{U}_+ \in \mathsf{Ev}_{h'}[U]$. Each of these two sums has one value on that set.

For $k \ge 1$, the shifted configuration $\mathsf{U}^\sharp$ associated with any such U_+ belongs to the old critical orbit on the inner bonds and equals U on the outer bonds. It is adapted at Sd_h for the shorter sequence. Both old sums are defined there, and for every $\mathsf{U}^\flat \in \mathsf{Ev}_h[U]$,

$$\begin{aligned} A_{k,h}(\mathsf{U}^\sharp, \mathfrak{q}) &= A_{k,h}(\mathsf{U}^\flat, \mathfrak{q}), \\ \Phi_h(\mathsf{U}^\sharp; \mathfrak{q}) &= \Phi_h(\mathsf{U}^\flat; \mathfrak{q}). \end{aligned} \tag{1355}$$

Here $\mathfrak{q}$ is the restriction of $\mathfrak{q}^+$ to levels less than k. The statement requires no adaptedness of $\mathsf{U}^\sharp$ at the other old sites. ■

Proof. The actual membership result corollary 58.4 puts the plain triples of every glued critical representative in the required core and layered output spaces, and its stored fields in their tiers. The same result at each earlier prefix gives the domains of the old entries. Thus every summand is defined, by the domain statements for the regular, boundary and retained families.

Two critical representatives differ by a residual gauge. Extend it by 1 outside the first collared region. An outer bond has no endpoint in Ω_1; an endpoint inside the collar is therefore a bare top and a node. The extension fixes both endpoints of every outer bond, so the glued configurations differ by this same gauge. By lemma 60.4, all sites acting on the stored data are nodes or stored sites. Both representatives are adapted. Apply lemma 60.3 to each boundary entry and each canonical

retained factor, which are jointly invariant. The regular entries and the weighted Wilson action are invariant without stored data. The finite sums consequently have one value.

For the old evaluation, use the actual arrival verification in proposition 58.27. Its datum is exactly $\xi_h^{\diamond}(U)$: the completed chart recovers the actual level-k field on the changed rows and leaves all other rows unchanged. Since U also belongs to the support of the prefix h, proposition 53.11 supplies this old orbit. The response identification proposition 58.9 gives a regular critical configuration up to gauge, and the pinned levelling restores the exact constraints. Thus $\mathsf{U}^{\sharp}$ is a gluing from that orbit. Its plain triples satisfy the old domain bounds by corollary 58.4.

The pinned shift preserves the transports at the pinnable old tops. By lemma 60.4, the old stored sites have the same transports before the shift as in the longer sequence. Those transports are 1, and hence remain 1 after the shift. At old top sites the transport is 1 identically. This proves adaptedness at Sd_h. An adapted representative of the old orbit exists by proposition 60.2. Compare it with the shifted one by lemma 60.3, term by term as above. This proves both equalities in (1355) separately.

For the first step there is no old critical choice: the determining map at level zero is the identity, the old field is U, and there are no old retained factors. For an empty first region the same statements reduce to this unchanged-field evaluation. □

60.4. **The canonical retained factors.**

Proposition 60.6. At every evaluation configuration of h', the retained-factor sum obtained from the one-step integration is the canonical sum of proposition 62.3. This includes the first step. The equality is an equality of the exponents themselves. ■

Proof. An old factor has the same domain, tier and formula after the history is extended. The first collared region is unchanged, so its plain triple is unchanged as well; and the old components of $\mathfrak{q}^+$ are $\mathfrak{q}$. This identifies every old term separately.

For the new terms, let $\mathsf{a}' = \mathfrak{q}_k^+$. Actual output membership and the tier bounds give all the domains in (1390). The kept-form expansion and the definitions yield

$$\sum_{X\in\mathcal{D}_t} \mathsf{K}_{h'}^{\mathrm{kp}}(X;\mathsf{U}_+;\mathsf{a}') + \sum_{c\in\Gamma_t^{\partial}} z_c(\mathsf{U}_+) = \sum_{f \text{ formed at } t} \varphi_f(\mathsf{U}_+;\mathfrak{q}^+). \tag{1356}$$

Indeed, the level-k pair in the kept form is exactly the pair read from the level-t plain triple. The zero-extended external field is the same on its domain. For an elimination term, both expressions use the same configuration on P_c^t; the pivot logarithm depends only on these bonds. Grouping these finitely many terms by $X(c)$ gives the second family in (1390).

The canonical family is the disjoint union of the old family and the factors just formed. Adding the preceding identities proves the claim. At the first step the old family is empty, so the same identity proves it directly. If the first region is empty, all these sums are empty. □

61. The measure identity for one history step

The local expansion must represent the finite history measure. We first give the measure argument, separating the two changes of variables from the identities for their integrands. This also fixes the order in which integrability is proved.

All coordinate spaces below are finite products of G and $\mathfrak{g}$, with positive bonds as independent coordinates. They are Polish spaces. The reference measures in free coordinates are products of normalized Haar measures and Lebesgue measures. The completed-coordinate measures are their images under the specified Borel injections. All are σ-finite. A partial map is Borel when its domain and its inverse images of Borel sets are Borel. A density given by a partial map means a Borel

function off a Borel null set with integrable absolute value; its values on that null set are immaterial. A product with a zero test is defined to be zero, without evaluating its remaining factors.

We use two facts about Borel images. A Borel image of a Borel subset of a Polish space is analytic and is measurable for the completion of every finite Borel measure. It therefore lies between two Borel sets whose difference has measure zero. If the map is injective, its image is Borel and its inverse on that image is Borel. These are the analytic-image and Luzin–Souslin theorems; see [Tse26, Proposition 13.2, Theorem 14.3 and Corollary 15.10].

61.1. **Coordinates and the hypotheses of a step.** Fix a history h at level k and its extension h' at $t = k + 1$. Write $\nu_h, \nu_{h'}$ for their measures with stored coordinates. At this stage the coordinate maps and the analytic data are inputs. Their required properties are as follows.

The bare space carries the finite positive measure $\rho_0\, dU$. Borel partial coordinate maps $\xi_h, \xi_{h'}$ and bounded nonnegative history weights give

$$\nu_h = (\xi_h)_*(\mathfrak{w}_h \rho_0\, dU), \qquad \nu_{h'} = (\xi_{h'})_*(\mathfrak{w}_{h'} \rho_0\, dU).$$

There are a bounded nonnegative Borel function f on the old coordinate space X_h and a Borel partial map $\Sigma : \mathsf{X}_h \rightharpoonup \mathsf{X}_{h'}$ such that

$$\mathfrak{w}_{h'} = \mathfrak{w}_h (f \circ \xi_h), \qquad \xi_{h'} = \Sigma \circ \xi_h, \tag{1357}$$

including equality of the domains in the second identity. The first identity is required on $\operatorname{dom} \xi_h$. Suppose that the preceding density is already known:

$$\nu_h = \varrho_h \mathscr{M}_h, \qquad \varrho_h = s_h \exp(A_h + \Phi_h), \qquad \int |\varrho_h|\, d\mathscr{M}_h < \infty. \tag{1358}$$

Here $s_h \geq 0$ is Borel, and A_h, Φ_h are defined and Borel off a null set wherever $s_h \neq 0$. Put $T = f s_h$.

The old coordinates split as (V, ϕ^+), with reference measure $dV\, d\mathscr{M}^+(\phi^+)$. The new preliminary coordinates are

$$p = (W, a', \phi^+), \qquad d\mathscr{M}^{\mathrm{pre}}(p) = dW\, da'\, d\mathscr{M}^+(\phi^+).$$

There is an injective Borel map $\imath : \mathsf{X}^{\mathrm{pre}} \to \mathsf{X}_{h'}$ with Borel inverse on its image, and $\mathscr{M}_{h'} = \imath_* \mathscr{M}^{\mathrm{pre}}$.

For a Borel set of bases $d = (W, \phi^+) \in \mathsf{O}$, a chart $V_d(B)$ is defined on a fixed bounded open product of balls R_r in the real field $B = (B_{\mathrm{I}}, a)$. Let

$$\mathsf{Y} = \{(W, B, \phi^+) : d \in \mathsf{O},\ B \in \mathsf{R}_r\}, \qquad \mathsf{pt}(W, B, \phi^+) = (V_d(B), \phi^+).$$

Require pt to be Borel and injective. Its Jacobian j_c is a finite positive Borel function, and for every nonnegative Borel g one has

$$\int_{\mathsf{pt}(\mathsf{Y})} g\, d\mathscr{M}_h = \int_{\mathsf{Y}} j_c(y) g(\mathsf{pt}(y))\, d\lambda(y), \qquad d\lambda = dW\, dB\, d\mathscr{M}^+. \tag{1359}$$

Every bare field in $\operatorname{dom} \xi_h$ with $\mathfrak{w}_{h'} \neq 0$ must have a chain point: $\xi_h(U) = \mathsf{pt}(y)$ for some $y \in \mathsf{Y}$.

The strip substitution is an injective Borel partial map $a = \check{\Psi}_d(a')$, with finite positive Jacobian $j_s(d, a')$. Set

$$\mathsf{Q} = \{(p, B_{\mathrm{I}}) : d \in \mathsf{O},\ a' \in \operatorname{dom} \check{\Psi}_d,\ (B_{\mathrm{I}}, \check{\Psi}_d(a')) \in \mathsf{R}_r\},$$

and

$$\Theta(p, B_{\mathrm{I}}) = (W, (B_{\mathrm{I}}, \check{\Psi}_d(a')), \phi^+), \qquad d\kappa = d\mathscr{M}^{\mathrm{pre}}\, dB_{\mathrm{I}}.$$

The domains and maps are Borel. Require the exact substitution formula and coordinate identity

$$\int_{\mathsf{Y}} g\, d\lambda = \int_{\mathsf{Q}} j_s (g \circ \Theta)\, d\kappa, \tag{1360}$$

$$\Sigma(\mathsf{pt}(\Theta(p, B_{\mathrm{I}}))) = \imath(p). \tag{1361}$$

The first holds for every nonnegative Borel g; the second includes definedness at every point of Q.

Let $S \ge 0$ be the new Borel test on $\mathsf{X}_{h'}$, and let R be the product of the strict integration balls $|B_{\mathrm{I}}(b)| < \mathfrak{b}_b$, with $0 < \mathfrak{b}_b \le r$. Define

$$\mathsf{Pe} = \{p : S(\imath p) > 0,\ d \in \mathsf{O},\ a' \in \operatorname{dom} \check{\Psi}_d,\ |\check{\Psi}_d(a')| < r\}.$$

An empty set of integration bonds gives a one-point space of volume one. Require $\mathscr{M}^{\mathrm{pre}}(\{S \circ \imath > 0\} \setminus \mathsf{Pe}) = 0$ and, outside a Borel κ-null set in Q,

$$T(\mathsf{pt}(\Theta(p, B_{\mathrm{I}}))) = \mathbf{1}_{\mathsf{R}}(B_{\mathrm{I}}) S(\imath p) c_z e^{-x_k G_d(B)}, \qquad a = \check{\Psi}_d(a'). \tag{1362}$$

Here $c_z > 0$ and $G_d(B)$ is real Borel and finite. In particular the set where the first two factors on the right are nonzero is exactly $\mathsf{Pe} \times \mathsf{R}$.

The remaining hypotheses concern the analytic identities. They are required whenever $y = \Theta(p, B_{\mathrm{I}})$ is the chain point of a bare field with nonzero $\mathfrak{w}_{h'}$ and defined new coordinates. The attached representative $\mathsf{U} = \mathsf{U}[d]$ and stored fields $\mathfrak{q}$ are Borel functions of the base. Write $\phi = \mathsf{pt}(y)$. Evaluation of the old exponents is independent of the representative, and

$$\begin{aligned} A_h(\phi) - x_k G_d(B) + J_d(B) &= A_h(\mathsf{U}, \mathfrak{q}) - \tfrac{1}{2}(B, \mathsf{A}_d B)_1 + V_d^{\mathrm{ex}}(B), \\ \Phi_h(\phi) &= \Phi_h(\mathsf{U}, \mathfrak{q}) + V_d^{\Phi}(B). \end{aligned} \tag{1363}$$

The old exponents at ϕ are used here wherever the old density defines them. Equivalently the identities can first be stated at an admitted representative of ϕ. The chart Jacobian has the form

$$j_c(W, B, \phi^+) = c_H\, e^{S_\Gamma(d)} e^{J_d(B)}, \qquad c_H > 0,$$

where S_Γ, J_d are real Borel functions. The symmetric real matrix A_d has $\mathsf{A}_{\mathrm{II}} > 0$, and $V_d = V_d^{\mathrm{ex}} + V_d^{\Phi}$ is Borel and bounded on R for fixed p.

We state the finite sum identities needed for collecting the new action. They are the identities of the local normalizer, conditional form, cluster expansion and strip substitution, respectively. Put

$$\Delta^{\mathrm{cond}} = \mathsf{A}_{\mathrm{EE}} - \mathsf{A}_{\mathrm{EI}} \mathsf{A}_{\mathrm{II}}^{-1} \mathsf{A}_{\mathrm{IE}}, \qquad \nu_{\mathfrak{g}} = \tfrac{3}{2} \log(2\pi), \qquad n_{\mathrm{I}} = \#\mathfrak{X}_{\mathrm{I}}^{+}.$$

Let $\mu_{d,a}$ be the real Gaussian probability measure with covariance $\mathsf{A}_{\mathrm{II}}^{-1}$ and mean $-\mathsf{A}_{\mathrm{II}}^{-1}\mathsf{A}_{\mathrm{IE}} a$. The local totals $N, K, B^{\times}, E, B^{\mathrm{sr}}$ are finite Borel values at their indicated arguments and satisfy

$$S_\Gamma = S_\Lambda + S_\partial, \qquad N = -\tfrac{1}{2} \log\det \mathsf{A}_{\mathrm{II}} + S_\Lambda, \tag{1364}$$

$$K(a) + B^{\times}(a) = -\tfrac{1}{2}(a, \Delta^{\mathrm{cond}} a)_1, \tag{1365}$$

$$\int \mathbf{1}_{\mathsf{R}} e^{V_d(B_{\mathrm{I}}, a)}\, d\mu_{d,a} = e^{E(a)}, \tag{1366}$$

$$B^{\mathrm{sr}}(a') = \log j_s + [E + B^{\times} + K](a) - [E + B^{\times} + K](a'). \tag{1367}$$

Require these identities and the following definitions to have all their entries defined at such a realized point:

$$\begin{aligned} n_t &= \log c_H + \log c_z + \nu_{\mathfrak{g}} n_{\mathrm{I}}, \\ A^+(p) &= A_h(\mathsf{U}, \mathfrak{q}) + n_t + N + E(a') + B^{\times}(a') + B^{\mathrm{sr}}(a'), \\ \Phi^+(p) &= \Phi_h(\mathsf{U}, \mathfrak{q}) + K(a') + S_\partial. \end{aligned} \tag{1368}$$

The right sides depend only on p, are Borel partial maps, and give the same values for every admitted representative of $\imath p$. In particular one admitted representative exists whenever these conditions apply. Collecting the finite local sums into the next regular and boundary families must preserve these identities as complex numbers.

61.2. **Positive fibre densities and integrability.**

Theorem AB (The measure identity)**.** Under the preceding hypotheses, the new exponents are defined and single-valued at $\mathscr{M}_{h'}$-almost every point where S is nonzero. The density

$$\varrho_{h'} = S \exp(A^+ + \Phi^+) \tag{1369}$$

is Borel, nonnegative and integrable after assigning zero on a Borel null set, and $\nu_{h'} = \varrho_{h'} \mathscr{M}_{h'}$. It is strictly positive almost everywhere on $\{S > 0\}$. For every bounded complex Borel F,

$$\int F \, d\nu_{h'} = \int F(\imath p) S(\imath p) e^{A_h(\mathsf{U},\mathfrak{q}) + \Phi_h(\mathsf{U},\mathfrak{q})} \mathsf{I}(d, a') \, d\mathscr{M}^{\mathrm{pre}}(p), \tag{1370}$$

where

$$\mathsf{I}(d, a') = c_H c_z e^{S_\Gamma(d)} j_s(d, a') \int_{\mathsf{R}} e^{-(B, \mathsf{A}_d B)_1/2 + V_d(B)} \, dB_{\mathrm{I}}. \tag{1371}$$

The integrals are absolute. Undefined values on the exceptional null set are assigned zero. ■□

Proof. First make the old density nonnegative everywhere. Its integral over every Borel set is real and nonnegative, because ν_h is positive. Applying this to the sets where its imaginary part has either sign and where its real part is negative shows that it is nonnegative almost everywhere. Enlarge the exceptional Borel null set to N_h and set the density to zero there. Denote the result by $\bar{\varrho}_h$. It vanishes wherever $s_h = 0$ and is strictly positive outside N_h wherever $s_h > 0$. Consequently $f\bar{\varrho}_h$ vanishes where $T = 0$, and equals $T e^{A_h + \Phi_h} > 0$ outside N_h where $T > 0$.

The bare identities (1357) give $\nu_{h'} = \Sigma_*(f\nu_h)$. Every selected bare field has a chain point, so $f\nu_h$ is concentrated on the Borel set $\mathsf{pt}(\mathsf{Y})$. Both changes of variables transport null sets backwards: substitute the indicator of a null set in (1359) or (1360); the integral of a strictly positive Jacobian over its inverse image is zero, so that inverse image has measure zero. Equation (1361) and (1360) also show that $\mathsf{pt}(y)$ lies in $\operatorname{dom}\Sigma$ for almost every $y \in \mathsf{Y}$.

We next justify applying the analytic identities, which were stated at realized bare fields. Let B be the Borel set of bare fields with $\mathfrak{w}_{h'} > 0$ and defined $\xi_{h'}$. The finite measure $\nu = (f\nu_h)|_{\operatorname{dom}\Sigma}$ is the ξ_h-image of the corresponding bare measure. The analytic-image theorem supplies Borel sets $D' \subset \xi_h(\mathsf{B}) \subset D''$ with $\nu(D'' \setminus D') = 0$. The bare formula gives $\nu(\mathsf{X}_h \setminus D'') = 0$, so ν is concentrated on D'. Since its density is strictly positive on $\{T > 0\} \cap \operatorname{dom}\Sigma$ outside N_h, that set outside $D' \cup N_h$ is $\mathscr{M}_h$-null. Pulling the null sets back through the chart gives a Borel set $\mathsf{Y}^* \subset \{T \circ \mathsf{pt} > 0\}$ of full measure in that set, each of whose points is the chain point of a field in B and avoids N_h.

Remove also the exceptional set of (1362) and pull Y^* back through Θ. Its intersection Q^* with $\mathsf{Pe} \times \mathsf{R}$ has full product measure there. Indeed the test identity puts every other point of that product in $\Theta^{-1}\{T \circ \mathsf{pt} > 0\}$, and the inverse image of its omitted null set is null. By Tonelli, the set

$$\mathsf{Pe}^* = \{p \in \mathsf{Pe} : \mathrm{vol}\{B_{\mathrm{I}} \in \mathsf{R} : (p, B_{\mathrm{I}}) \notin \mathsf{Q}^*\} = 0\}$$

is Borel and has full measure in Pe. For each such p the good fibre has the positive volume of R. Choose one point in that fibre. Every hypothesis that is independent of B_{I}, including definedness of (1368), therefore holds at p. The identities involving the integration field hold almost everywhere in its fibre. No measurable choice of this one point is needed.

Define the nonnegative Borel function on Q

$$D(p, B_{\mathrm{I}}) = j_s(d, a') j_c(\Theta(p, B_{\mathrm{I}}))(f\bar{\varrho}_h)(\mathsf{pt}(\Theta(p, B_{\mathrm{I}}))). \tag{1372}$$

Using the two substitutions for nonnegative functions and then the commuting identity gives

$$\int F \, d\nu_{h'} = \int_{\mathsf{Q}} F(\imath p) D(p, B_{\mathrm{I}}) \, d\kappa \qquad (F \geq 0 \text{ Borel}). \tag{1373}$$

All these equalities hold with values in $[0, \infty]$. By the test identity, $D = 0$ almost everywhere off $\mathsf{Pe} \times \mathsf{R}$.

At a good fibre point, insert (1362) and (1363) into (1372). The result is

$$D(p,B_{\mathrm{I}}) = S(\imath p)e^{A_h(\mathsf{U},\mathfrak{q})+\Phi_h(\mathsf{U},\mathfrak{q})} \times c_H c_z e^{S_\Gamma(d)} j_s(d,a') e^{-(B,\mathsf{A}_d B)_1/2+V_d(B)}. \tag{1374}$$

For fixed $p \in \mathsf{Pe}^*$ the last expression is bounded on the bounded field box: V_d is bounded and the quadratic form is continuous. It is therefore integrable. Its integral is the integrand of (1370), without F, and is strictly positive, since $D > 0$ on the full-measure good fibre.

Complete the square in A_d:

$$(B,\mathsf{A}_d B)_1 = (B_{\mathrm{I}} + \mathsf{A}_{\mathrm{II}}^{-1}\mathsf{A}_{\mathrm{IE}}a, \mathsf{A}_{\mathrm{II}}(B_{\mathrm{I}} + \mathsf{A}_{\mathrm{II}}^{-1}\mathsf{A}_{\mathrm{IE}}a))_1 + (a,\Delta^{\mathrm{cond}}a)_1.$$

Gaussian normalization and (1366) give

$$\int_{\mathsf{R}} e^{-(B,\mathsf{A}_d B)_1/2+V_d(B)}\, dB_{\mathrm{I}} = \exp\{\nu_{\mathfrak{g}} n_{\mathrm{I}} - \tfrac{1}{2}\log\det \mathsf{A}_{\mathrm{II}} + K(a) + B^{\times}(a) + E(a)\}.$$

Use (1367) to replace a by a' and (1364) to combine the determinant and pivot factors. The exponent in the fibre integral of (1374) becomes exactly $A^+(p) + \Phi^+(p)$ from (1368). These are identities between complex numbers, before taking exponentials.

Set $\Phi_{\mathrm{step}}(p) = S(\imath p)e^{A^+(p)+\Phi^+(p)}$ on Pe^* and zero elsewhere. It is Borel, nonnegative, and equals the fibre integral of D almost everywhere. Tonelli in (1373) yields

$$\int F\, d\nu_{h'} = \int F(\imath p)\Phi_{\mathrm{step}}(p)\, d\mathscr{M}^{\mathrm{pre}} \qquad (F \geq 0 \text{ Borel}).$$

Taking $F = 1$ proves $\int \Phi_{\mathrm{step}}\, d\mathscr{M}^{\mathrm{pre}} = \nu_{h'}(\mathsf{X}_{h'}) < \infty$. Decomposing a bounded complex F into four nonnegative parts now proves (1370) with absolute convergence.

Finally $\imath(\mathsf{Pe}^*)$ is Borel. The set $\{S > 0\} \setminus \imath(\mathsf{Pe}^*)$ is $\mathscr{M}_{h'}$-null, by $\mathscr{M}_{h'} = \imath_*\mathscr{M}^{\mathrm{pre}}$ and the assumed full measure of Pe. On its complement, the representative-independent exponents descend through the Borel inverse of $\imath$. Their density pulls back to Φ_{step}. The pushforward identity above therefore gives (1369), its positivity and integrability. If collecting the finite sums identifies A^+ with the exponent $A_{t,h'}$ of the next format, the same equality proves that format without a further integrability assumption. □

62. Boundary terms and their propagation

At each step we retain the regular expansion in the interior. The remaining terms are attached to their level and domain. We first estimate the terms created by one integration and then show that the same bound suffices at every level.

62.1. **The terms created at the boundary.** Fix a history step $h' = (h, \mathsf{o}_t)$, $t = k+1$. Write $\mathscr{U}_X$ for its graded space at the intermediate stage, and $\mathfrak{Q}^{\mathrm{rad}}_{h'}(X)$ for the stored data with the prescribed tier bounds. A point of this product is written $(P;\mathfrak{q},\mathfrak{a})$: $\mathfrak{q}$ contains the old data and $\mathfrak{a}$ the new external field, extended by zero off its bonds in X. In particular $|\mathfrak{a}| < C_{\mathrm{xt}}\mathfrak{b}_k$. The background pair read from P in level-k units is $p_X(P)$, and its further restriction to the joined strip region is $p_X^{\mathrm{loc}}(P)$.

For this subsection assume that these readings belong to the normalizer class, the local form class and the classes of the localized strip shift, respectively. Assume also that restriction to $X_0 \subset X$ preserves these memberships and the stored-data tiers. On each fixed real-U slice, $\mathscr{U}_X$ is open, its readings are holomorphic, and they are continuous on the full space of triples. The necessary

estimates for the three analytic ingredients are

$$|N_h^{\partial}(X;p)| \le B_N e^{-\delta_N M d_t(X)}, \tag{1375}$$

$$|K_h^{\mathrm{kp}}(X;p,\mathsf{a}_1)| \le c_{\mathrm{kp}} b_{\mathrm{e}}^2 e^{-\kappa'' d_t(X)},$$

$$|B_h^{\times}(X;p,\mathsf{a}_1)| \le c_{\mathrm{rj}} b_{\mathrm{e}}^2 e^{-(\check{R}_t-2)} e^{-\kappa'' d_t(X)} \quad (|\mathsf{a}_1| \le b_{\mathrm{e}}), \tag{1376}$$

$$|E_h^{\mathrm{cl}}(X;\mathsf{a}_1;P;\mathfrak{q})| \le b_{\mathrm{cs}} e^{-\kappa'' d_t(X)}. \tag{1377}$$

The two forms are quadratic polynomials in the external field, with coefficients analytic in p, and involve only external bonds inside X. The cluster function is local in the same bonds and is holomorphic jointly in the complex coordinates of P and in the perturbations e of a tier field, supported on zone bonds of X, with

$$|\mathsf{e}| < r_{\mathrm{bd}} := 2\rho_{\mathrm{bd}}/g_k.$$

All three ingredients are Borel measurable, gauge invariant, conjugation real and functions of the background pair. These are assumptions on their full indicated domains, including every such external perturbation.

We use the following support and interior identities. The normalizer vanishes unless X contains a cube of Λ_t. The cluster function vanishes unless X and Ω_t have a bare cube in common. The rejected form vanishes unless X contains a cube of $\Omega_t \cap Z_t$ and $d_t(X) \ge \check{R}_t - 2$. For $X \subset \Lambda_t$ the local normalizer and pointed cluster terms are those of the empty history:

$$N_h^{\mathrm{un}}(X,p,z) = N_{\emptyset}^{\mathrm{un}}(X,p,z), \qquad E_h^{\mathrm{cl}}(X;\mathsf{a}_1;P;\mathfrak{q}) = \sum_z F_{\emptyset}(X,P,z). \tag{1378}$$

The normalizer equality includes equality of its natural domains. The cluster equality holds throughout the admitted analytic domain specified above; its right side is independent of the stored data and external field.

Set

$$\kappa_T = (1+4\beta_{\mathrm{s}})\kappa, \qquad \kappa'' = \kappa_T + 1 + c_e. \tag{1379}$$

Use the localized shift of section 53.10 with $\widehat{\kappa} = \kappa_T$ and impose

$$C_{\mathrm{xt}}\delta_k \le \tfrac{1}{8}, \qquad 0 < \rho_{\mathrm{bd}} \le \tfrac{1}{2} C_{\mathrm{xt}}\delta_k, \tag{1380}$$

together with (1036) and (1042). The normalizer bound is supplied by proposition 56.2 with $B_N = 24(LM)^3 E_{\mathrm{nf}}$ and $\delta_N = \delta_{\mathrm{nf}}$. The other inputs will be verified for the actual history construction before this estimate is applied.

For a target domain X_1, restrict the background and old data to each $X_0 \subset X_1$. Define four families over X_0:

$$\begin{aligned} F_{X_0}^{\mathrm{E}} &= E_h^{\mathrm{cl}}(X_0;\,\cdot\,;P|_{X_0};\mathfrak{q}|_{X_0}),\\ F_{X_0}^{8} &= K_h^{\mathrm{kp}}(X_0;p_{X_0}(P),\,\cdot\,),\\ F_{X_0}^{9} &= B_h^{\times}(X_0;p_{X_0}(P),\,\cdot\,),\\ F_{X_0}^{7}(\mathsf{a}_1) &= \sum_{\substack{b\in\mathcal{Z}^+\\ \mathsf{P}(b)=X_0}} \mathsf{ls}_b(\mathsf{a}_1). \end{aligned}$$

The first family is taken on its external perturbation domain. For the second and third, restrict to $|\mathsf{a}_1(b)| < 2C_{\mathrm{xt}}\mathfrak{b}_k$ on the external bonds meeting X_0. Families with $X_0 \not\subset X_1$ are zero, except for F^7, which is independent of the target domain. Here

$$\mathsf{ls}_b(\mathsf{a}_1) = -\log\sigma_1\Big(g_k^2\langle \mathsf{a}_1(b),\mathsf{a}_1(b)\rangle\Big), \qquad \sigma_1(w) = \left(\sum_{n\ge 0}\frac{(-w)^n}{(2n+1)!}\right)^2,$$

with its series logarithm and domain $g_k|\mathsf{a}_1(b)| < 1$. The reading block $\mathsf{P}(b)$ consists of the one or two coarse cubes at the endpoints of b, so $d_t(\mathsf{P}(b)) = 0$.

The five strip terms are

$$R^6(X_1) = \sum_{b \in \mathcal{Z}_\theta^+} J_b(X_1), \qquad R^\nu(X_1) = \sum_{X_0 \subset X_1} (F^\nu_{X_0})^{\mathrm{tr}}(X_1), \quad \nu = 7, 8, 9, \mathrm{E}. \tag{1381}$$

In R^6, J_b is evaluated at the link argument $A = g_k \mathsf{a}(b)$. The local pair used here is $p^{\mathrm{loc}}_{X_1}(P)$. We suppress it, the tier field and the old data from the notation. Define

$$\begin{aligned} B^{\mathrm{nm}}(X) &= N_h^\partial(X; p_X(P)) - \mathbf{1}_{\{X \subset \Lambda_t\}} \sum_z N_\emptyset^{\mathrm{un}}(X, p_X(P), z), \\ B^{\mathrm{cs}}(X) &= E_h^{\mathrm{cl}}(X; \mathsf{a}; P; \mathsf{q}) - \mathbf{1}_{\{X \subset \Lambda_t\}} \sum_z F_\emptyset(X, P, z), \\ B^{\mathrm{rj}}(X) &= B_h^\times(X; p_X(P), \mathsf{a}), \\ B^{\mathrm{sr}}(X) &= \sum_{\nu = 6,7,8,9,\mathrm{E}} R^\nu(X). \end{aligned} \tag{1382}$$

A term multiplied by the indicator is omitted when the indicator is zero. This convention imposes no domain condition on an unused expression. The boundary term created at step t is

$$B_t^{\mathrm{mk}}(X) = B^{\mathrm{nm}}(X) + B^{\mathrm{cs}}(X) + B^{\mathrm{rj}}(X) + B^{\mathrm{sr}}(X). \tag{1383}$$

Proposition 62.1. Under the assumptions above, all the terms in (1383) are defined on $\mathscr{U}_X \times \mathfrak{Q}^{\mathrm{rad}}_{h'}(X)$. They are holomorphic in the complex coordinates of P on each fixed real-U slice, Borel measurable, gauge invariant, conjugation real and functions of the pair. Put

$$\begin{aligned} b_{\mathrm{rj}} &= c_{\mathrm{rj}} C_{\mathrm{xt}}^2 \mathfrak{b}_k^2 e^{-(\check{R}_t - 2)}, \\ b_{\mathrm{sr}} &= 4 e^{-\check{R}_t} \left[N_b \rho_J + N_e N_b + N_e b_{\mathrm{cs}} + 4 N_e C_{\mathrm{xt}}^2 \mathfrak{b}_k^2 (c_{\mathrm{kp}} + c_{\mathrm{rj}} e^{-(\check{R}_t - 2)}) \right]. \end{aligned} \tag{1384}$$

Then, with $d = d_t(X)$,

$$|B_t^{\mathrm{mk}}(X)| \le B_N e^{-\delta_N M d} + (b_{\mathrm{cs}} + b_{\mathrm{rj}}) e^{-\kappa'' d} + b_{\mathrm{sr}} e^{-\kappa_T d}.$$

If

$$\delta_N M \ge \kappa_T, \qquad b_{\mathrm{cs}} + b_{\mathrm{rj}} + b_{\mathrm{sr}} \le 1, \tag{1385}$$

this gives

$$|B_t^{\mathrm{mk}}(X)| \le B_T e^{-\kappa_T d_t(X)}, \qquad B_T = 24 (LM)^3 E_{\mathrm{nf}} + 1. \tag{1386}$$

The term vanishes when $X \subset \Lambda_t$, and unless X and Ω_t share a bare cube. Every strip term vanishes when the strip is empty. ∎

Proof. We check the room needed by each transfer. For F^{E} it is part of the cluster hypothesis. For the quadratic terms,

$$r_{\mathrm{bd}} \le C_{\mathrm{xt}} \delta_k / g_k = C_{\mathrm{xt}} \mathfrak{b}_k.$$

Thus every permitted perturbation of a tier field remains in the radius $2 C_{\mathrm{xt}} \mathfrak{b}_k$ ball. Polynomial dependence on that field gives holomorphy. Locality permits extension by zero outside the bonds meeting X_0 when using (1376). Consequently their norms are bounded by

$$4 c_{\mathrm{kp}} C_{\mathrm{xt}}^2 \mathfrak{b}_k^2 e^{-\kappa'' d_t(X_0)}, \qquad 4 c_{\mathrm{rj}} C_{\mathrm{xt}}^2 \mathfrak{b}_k^2 e^{-(\check{R}_t - 2)} e^{-\kappa'' d_t(X_0)}.$$

For the Haar term, let $s(w) = \sum_{n \ge 0} (-w)^n / (2n+1)!$. For $|w| < 1$, the successive absolute terms after the first have ratio at most $1/20$, so $|s(w) - 1| \le 10/57 < 0.18$. Hence $|\sigma_1(w) - 1| < 2/5$ and $|\log \sigma_1(w)| \le (2/5)/(1 - 2/5) = 2/3$. The series is normally convergent there. On the perturbation ball, $g_k|\mathsf{a} + \mathsf{e}| < 1/8 + 2\rho_{\mathrm{bd}} \le 3/8$, so every one-bond function has the required room. For a fixed

nonempty reading block $\mathsf{P}(b)$ there are at most N_b such bonds, and its size is zero. It follows that $\|F^7_{X_0}\| \le N_b e^{-\kappa'' d_t(X_0)}$.

Equation (1039) now gives

$$\begin{aligned} |R^7(X)| &\le 4N_e N_b e^{-\check{R}_t} e^{-\kappa_T d}, \\ |R^{\mathrm{E}}(X)| &\le 4N_e b_{\mathrm{cs}} e^{-\check{R}_t} e^{-\kappa_T d}, \\ |R^8(X)| &\le 16 N_e c_{\mathrm{kp}} C_{\mathrm{xt}}^2 \mathfrak{b}_k^2 e^{-\check{R}_t} e^{-\kappa_T d}, \\ |R^9(X)| &\le 16 N_e c_{\mathrm{rj}} C_{\mathrm{xt}}^2 \mathfrak{b}_k^2 e^{-(\check{R}_t - 2)} e^{-\check{R}_t} e^{-\kappa_T d}. \end{aligned} \tag{1387}$$

For R^6, at most $N_b \# X \le 8N_b(d+1)$ bonds contribute. Since $\mu_H/2 \ge \kappa_T + 1$, (1043) and $(d+1)e^{-(d+1)} \le e^{-1}$ give

$$|R^6(X)| \le 4N_b \rho_J e^{-\check{R}_t} e^{-\kappa_T d}.$$

Their sum is bounded by $b_{\mathrm{sr}} e^{-\kappa_T d}$.

The first two differences in (1382) vanish inside Λ_t by (1378). Outside it, the normalizer and cluster estimates apply directly. The tier bound in (1376) gives $|B^{\mathrm{rj}}| \le b_{\mathrm{rj}} e^{-\kappa'' d}$. This proves the asserted sum bound, and (1385) gives (1386).

The support statements for the first three kinds follow from the assumed support and interior identities. For a nonzero strip term there is a zone bond with $\mathsf{P}(b) \subset X$ and a localization set reaching the component boundary. The strip geometry puts a cube of $\Omega_t \cap Z_t$ in X. Thus a strip term cannot occur inside Λ_t or away from Ω_t. If the strip is empty there are no Jacobian terms and no nonempty difference sets in the transfer sums.

Restriction of pairs, triples and stored fields is continuous and linear in their complex coordinates. The normalizer and form coefficients are analytic functions of those pairs. For the cluster term use its joint holomorphy with the external perturbation set to zero. The transfer construction preserves these properties: the values of the real tier field on the cut-off zone are fixed on each holomorphic slice. The finite sums also preserve Borel measurability and the three symmetry and pair-dependence identities. This proves the stated regularity. □

The total local exponent before the interior subtraction is

$$N_h^{\partial}(X; p_X(P)) + E_h^{\mathrm{cl}}(X; \mathsf{a}; P; \mathfrak{q}) + B^{\mathrm{rj}}(X) + B^{\mathrm{sr}}(X).$$

If the regular family is $E^{(t)}(X,P,z) = N_\emptyset^{\mathrm{un}}(X, p_X(P), z) + F_\emptyset(X,P,z) - n_{\mathrm{tr}}(X,z)$, subtracting $\mathbf{1}_{\{X \subset \Lambda_t\}} \sum_z (E^{(t)} + n_{\mathrm{tr}})$ gives exactly (1383). All terms in this identity are defined by the interior identities above; the constants n_{tr} cancel. Moreover the finite sum identities (1040) and (1044) show that the five strip terms sum to the actual changes of the Haar density, the two quadratic forms and the cluster exponent, together with the logarithm of the strip Jacobian.

62.2. **Persistence of the boundary bound.** A boundary entry has an index (j, X, o), where $o = \mathrm{mk}$ denotes a term created by an integration and $o = \mathrm{ed}$ an edge part of a regular family. At (j, X, mk) its domain is the layered output space at level j, with the stored data up to j. At (j, X, ed) its domain is the level-j analyticity core, with the same stored-data convention. These domains depend only on the history truncated at j and the radii at levels at most j.

The permitted indices are

$$\begin{aligned} \mathsf{Ix}_j^{\mathrm{mk}} &= \{X \in \mathcal{D}_j : \ X \text{ and } \Omega_j \text{ share a bare cube}, \ X \not\subset \Lambda_j^{\sim -1}\}, \\ \mathsf{Ix}_j^{\mathrm{ed}} &= \{X \in \mathcal{D}_j : \ X \subset \Lambda_j, \ X^{(j)} \setminus \Lambda_j^{\sim -1} \neq \emptyset\}. \end{aligned}$$

For each permitted index we keep at most one entry. An entry is holomorphic on each fixed-real slice, Borel measurable, gauge invariant, conjugation real and dependent only on the pair and its stated stored data. Its norm is the supremum on this domain.

For a pointed regular family with $\|E^{(j)}\|_{j,\kappa+1} \le E_0$, define its edge part by

$$E^{(j),\mathrm{ed}}(X,P) = \sum_{z \in X^{(j)} \setminus \Lambda_j^{\sim -1}} E^{(j)}(X,P,z). \tag{1388}$$

It is independent of the stored data and has the same regularity and symmetries as its summands. The number of base points is at most $M^3 \# X \le 8M^3(d_j(X)+1) \le 8M^3 e^{d_j(X)}$. Therefore

$$\|E^{(j),\mathrm{ed}}(X)\| \le 8M^3 E_0 e^{-\kappa d_j(X)}. \tag{1389}$$

For $X \subset \Lambda_j$ outside $\mathsf{lx}_j^{\mathrm{ed}}$ the sum is empty.

Proposition 62.2. Suppose the one-step hypotheses above hold at every step up to n, with the same constants, and the regular families satisfy $\|E^{(j)}\|_{j,\kappa+1} \le E_0$. Suppose also that each layered output space is contained in the intermediate graded space used for proposition 62.1. Start with no boundary entries. At step t retain the earlier entries, add $B_t^{\mathrm{mk}}(X)$ restricted to the output space for $X \in \mathsf{lx}_t^{\mathrm{mk}}$, and add (1388) for $X \in \mathsf{lx}_t^{\mathrm{ed}}$. Then every entry at level $j \le n$ satisfies

$$\|B(j,X,o)\| \le B e^{-\kappa d_j(X)}, \qquad B = \max\{1, B_T, 8M^3E_0\}.$$

Its domain and values are unchanged at subsequent levels. ∎

Proof. At step t, the new entries have level t and distinct origins. Their indices are disjoint from those of all earlier entries, so each index still has at most one term. The old domains, tiers and index sets are determined by the history truncated at their level; they do not change. An old entry therefore retains its norm and its properties. Restricting a newly created term to the layered output space preserves its properties and can only decrease its supremum. Its bound is $B_T e^{-\kappa_T d_t(X)} \le B e^{-\kappa d_t(X)}$. For an edge entry use (1389). These three cases prove the assertion by induction. The constant is not accumulated over the levels, because distinct entries remain separately indexed. □

62.3. **The retained factors.** The kept quadratic forms need not be small on the stored field. We retain their exponentials, together with the elimination factors at external pivot bonds. At step $t = l+1$, the factors with domain X have exponents

$$\varphi_{t,X}^{\mathrm{kp}}(P;\mathfrak{q}) = \varphi_{\mathfrak{k}_X,X}(\mathsf{a}[\mathfrak{q}]; p_X(P)), \qquad \varphi_{t,X}^{\mathrm{el}}(P;\mathfrak{q}) = \sum_{\substack{c \in \Gamma_t^{\partial} \\ X(c)=X}} z_c(p_X(P)). \tag{1390}$$

The first is included only when the keeping predicate $\mathsf{kp}_t(X)$ is one, and the second only when its index set is nonempty. The field $\mathsf{a}[\mathfrak{q}]$ equals the stored field of level l on external bonds inside X and is zero elsewhere. The pair is read in level-l units:

$$p_X(P) = (e^{i\eta_t A'} U, L^{-3}\mathbf{J})\big|_{\mathrm{Bd}^\circ(X \cap \Omega_0^\diamond)}.$$

We restrict each exponent to the layered output space at level t and its level-l stored-data tiers. Its name is (ν,t,X), with $\nu = \mathrm{kp}$ or $\nu = \mathrm{el}$. There is at most one factor of each kind for a given (t,X).

Assume the hypotheses of the kept-form expansion on $\mathscr{O}(X;\alpha_0,\alpha_1,\alpha_2)$, including its rate condition $\delta_\star^{\mathrm{pl}} LM \ge C_2(\kappa+1)$, and suppose that the layered-space reading belongs to this class. In particular the kept exponent is a finite quadratic polynomial in the external field, with holomorphic coefficients, and its estimate is

$$|\varphi_{\mathfrak{k}_X,X}(\mathsf{a};p)| \le \tfrac{1}{2} K_{\mathrm{ch}} C_{\mathrm{xt}}^2 \mathfrak{b}_l^2 e^{-\kappa(d_t(X)+1)} \quad \text{if } |\mathsf{a}| \le C_{\mathrm{xt}} \mathfrak{b}_l. \tag{1391}$$

Here $K_{\mathrm{ch}} = C_3 (LM)^6 e^{C_3\kappa} B_{\mathrm{ch}}$. We require also

$$L \max\{\alpha_0, \alpha_1\} \le a_{\mathrm{nf}}. \tag{1392}$$

The pivot estimate applies also to coarse bonds crossing the boundary of Ω_t. We verify this point. Let $c \in \Gamma_t$ and choose an endpoint $y_1 \in \Omega_t$. Separation gives $\mathrm{dist}_\infty(\Omega_t, (\Omega_{t-1}^\diamond)^c) > R_1 \mathfrak{M}$. If both

endpoints lie in Ω_t, the two averaging blocks fit in a cube of side $2L\mathfrak{M}$ inside $X(c)\cap\Omega_t$. The layer-t local gauge gives $L|A|\le\alpha_0$ and $L|A'|<\alpha_1$ there. If the second endpoint is outside Ω_t, the blocks fit in a cube of side $2\mathfrak{M}$ inside $X(c)$. Every point of this cube lies within $2\mathfrak{M}<R_1\mathfrak{M}$ of y_1, so the cube lies in the last two layers of $\Omega_\bullet^\diamond$. The layer-$(t-1)$ gauge and the bounds in either layer give $L|A|\le L\alpha_0$ and $L|A'|<L\alpha_1$. In both cases the enclosing cube and its gauge are contained in $X(c)\cap\Omega_0^\diamond$.

Put $a=L\max\{\alpha_0,\alpha_1\}$. In this gauge the logarithmic potential of $e^{i\xi A'}e^{i\xi A}$ has norm less than $4a/L$. The averaging and pivot estimates used in (793) give a matrix $E=L^3m_c-1$ with $|E|\le 4C_{\mathrm{piv}}C_Qa/L\le 1/2$. The trace logarithm satisfies $|\operatorname{tr}\log(1+E)|\le 6|E|$. Its exponential is the determinant and its imaginary part has modulus at most $\pi/2$, so it is the specified principal determinant logarithm. Thus, with a constant C_z depending only on L,

$$|z_c(p)-6\log L|\le C_za. \tag{1393}$$

For example $C_z=24C_{\mathrm{piv}}C_Q/L$ suffices. The local fixed-gauge formula also proves holomorphy; real gauge transformations preserve the determinant and conjugation preserves its branch.

If an elimination factor is present, X is one coarse cube or two adjacent cubes, so $d_t(X)=0$. Writing $N_X=\#\{c\in\Gamma_t^\partial : X(c)=X\}$, we have $1\le N_X\le 6M^3$: there are at most $2M^3$ coarse sites in X, with three positive bond directions. Consequently

$$|\varphi^{\mathrm{el}}_{t,X}(P;\mathfrak{q})-6N_X\log L|\le 6M^3C_za_{\mathrm{nf}}. \tag{1394}$$

Proposition 62.3. Under the class membership and form hypotheses above, the factors (1390), over all steps of a history, form a finite family with distinct names. Each exponent is defined throughout its stated layered space and stored-data tiers, is Borel measurable, holomorphic in the complex coordinates of a fixed-real slice, gauge invariant, conjugation real and dependent only on the pair. The family at level $k+1$ retains every factor of the history truncated at k, with the same domain and exponent.

Define the oscillation at fixed stored data by

$$\operatorname{osc}\varphi=\sup_{\mathfrak{q}}\sup_{P,P'}|\varphi(P;\mathfrak{q})-\varphi(P';\mathfrak{q})|.$$

Then

$$\sum_{f\ \text{at}\ (t,X)}\operatorname{osc}\varphi_f\le B^\flat_{\mathrm{fr}}\max\{1,\mathfrak{b}_l^2\}e^{-\kappa d_t(X)},\qquad B^\flat_{\mathrm{fr}}=K_{\mathrm{ch}}C_{\mathrm{xt}}^2+12M^3C_za_{\mathrm{nf}}. \tag{1395}$$

If $\mathfrak{b}_l\ge 1$, the maximum can be replaced by $\mathfrak{b}_l^2$. Each factor contains a cube of Z_t; at level $k>t$ its domain therefore meets Ω_{t+1}^c. ∎

Proof. The class membership gives definedness of the kept quadratic coefficients and the pivot logarithms at every pair under consideration. Their compositions with the pair and stored-field readings give the stated regularity. Those readings commute with gauge transformations and conjugation, so the same is true of the exponents. Equality of pairs gives equality of the values, including equality of their domains.

A nonzero kept coefficient is assigned to a component $\mathfrak{k}_X$ of $\Omega_t\cap Z_t$ and its domain contains a cube of this component. For an elimination factor the endpoints of its external coarse bond lie outside Λ_t, so $X(c)\subset Z_t$. If $t=k$ this places the factor at the boundary of the current region. If $t<k$, separation of Ω_{t+1} from Z_t by at least $8LM$ puts the same cube in Ω_{t+1}^c.

The construction at step t uses only the history truncated at t. Its layered space uses the radii at levels at most t, and its stored-data tiers use only the external bonds of step t. None changes on extending the history or extending the table of radii. Thus old names, domains and exponents agree exactly. At every step each domain has at most the two distinct names in (1390); finiteness follows from finiteness of the lattice and the number of steps.

For a fixed stored datum, its zero extension obeys $|\mathsf{a}[\mathfrak{q}]| \le C_{\mathrm{xt}}\mathfrak{b}_l$. Twice (1391) bounds the oscillation of a kept exponent by $K_{\mathrm{ch}}C_{\mathrm{xt}}^2\mathfrak{b}_l^2 e^{-\kappa d_t(X)}$. For an elimination exponent, subtract the same constant $6N_X \log L$ at both backgrounds and use (1394). Its oscillation is at most $12M^3C_z a_{\mathrm{nf}}$ and its domain has size zero. Adding the at most two bounds proves (1395). □

62.4. **Radii independent of the number of steps.** The radii table associated with a coupling sequence is $r_\iota(\sigma;n) = \alpha_{\iota,n} = C_\iota x_n^{-1/2}(\log x_n)^q$ for either stage σ containing level n. Suppose $x_n \ge \gamma^{-2}$ and $\log\gamma^{-2} > 2q$. The function $F_q(x) = x^{-1/2}(\log x)^q$ is decreasing on this interval, so $r_\iota(\sigma;n) \le C_\iota F_q(\gamma^{-2})$. The following conditions therefore ensure the required ceiling bounds at every level:

$$\begin{gathered} \mathfrak{o}L^3 \ge 3, \qquad a_\flat \le \alpha_0, \\ (1+\beta_{\mathrm{s}})C_0F_q(\gamma^{-2}) \le \min\{\alpha_0,\alpha_2\}, \\ (1+\beta_{\mathrm{s}})C_1F_q(\gamma^{-2}) \le \alpha_1, \qquad \mathfrak{o}L^3MB^{(\mathrm{cg})}C_0F_q(\gamma^{-2}) \le \alpha_0. \end{gathered} \tag{1396}$$

Explicitly, these bound $(1+\beta_{\mathrm{s}})r_0((j,\mathrm{out});n)$ by $\min\{\alpha_0,\alpha_2\}$, $(1+\beta_{\mathrm{s}})r_1((j,\mathrm{out});n)$ by α_1, and $\mathfrak{o}L^3MB^{(\mathrm{cg})}r_0((j,\mathrm{mid});j-1)$ by α_0. Restriction of the sequence restricts this same table; hence the factor families in proposition 62.3 agree when consecutive levels are described with their own truncated tables.

Likewise the condition

$$A_1(\log\gamma^{-2})^{\mathfrak{p}_0} \ge 1 \tag{1397}$$

gives $\mathfrak{b}_l \ge 1$ at every level and turns (1395) into

$$\sum_{f \text{ at } (l+1,X)} \operatorname{osc}\varphi_f \le B_{\mathrm{fr}}^\flat \mathfrak{b}_l^2 e^{-\kappa d_{l+1}(X)}. \tag{1398}$$

The constant depends on $L, M, \kappa, C_{\mathrm{xt}}$ alone. It may be enlarged uniformly to

$$B_{\mathrm{fr}} = C_3(LM)^6 e^{C_3(\kappa+1)}B_{\mathrm{ch}}C_{\mathrm{xt}}^2 + 12M^3C_z a_{\mathrm{nf}}.$$

This choice also accommodates the one-unit rate reserve in the later locality estimates.

63. Vacuum constants and their large-field deficits

The field-independent part of each integration step has a volume contribution and a correction supported near the large-field region. We compute both from the same local normalizer. In this section $N_j = \#\mathbb{T}^{(j)}$, $y_j = \log x_j$, and $\#_j(Y) = \#(Y \cap \mathbb{T}^{(j)})$. Complements and the operations $Y^\sim, Y^{\sim-1}$ are taken in the cell partition at the indicated level.

63.1. **The constants of one step.** We first record the unnormalized value of the Gaussian terms in section 50.5. At step $j = k+1$, with its notation, put

$$\mathsf{n}^{(j)}(X,b) = \begin{cases} -\frac{1}{2}E^N(X;P_0,b), & b \in \mathfrak{X}^+,\ S_N(b) \subset X, \\ 6\log L, & b = b_0(c),\ X = X(c), \\ 0, & \text{otherwise}, \end{cases} \qquad \mathsf{n}^{(j)}(X,z) = \sum_b h_z(b)\mathsf{n}^{(j)}(X,b). \tag{1399}$$

All the parameters and clearances of section 50.5 are retained. We may enlarge its L-dependent constant E_b so that $E_b \ge 6\log L$, and keep $E_N = 3L^3E_b$. The proof of the normalizer gives

$$\mathsf{n}^{(j)}(X,z) \in \mathbb{R}, \qquad |\mathsf{n}^{(j)}(X,z)| \le E_N e^{-\delta_N M d_j(X)}. \tag{1400}$$

Indeed, P_0 belongs to every normalizer domain. The nonpivot term is bounded by $E_b e^{-\delta_N M d_j(X)}/2$ by the chain estimates in that proof. In the pivot case the one or two cubes of $X(c)$ have a common corner, so $d_j(X(c)) = 0$. The tents are nonnegative and sum over bonds to $3L^3$. These facts give the bound in (1400). The real logarithm series and the real pivot value give reality. The same change

of coordinates used for the normalizer transports these constants between local lattice charts. In particular, their definition does not introduce a choice of a larger torus.

Let the real local families $E^{(j)}$ of the progression be defined on their analytic cores, including the trivial triple, and suppose

$$|E^{(j)}(X,P,z)| \le E_0 e^{-(\kappa+1)d_j(X)}, \qquad \delta_N M \ge \kappa, \quad E_0 \ge 2E_N, \quad \kappa > \log 9216 + 2. \tag{1401}$$

Here and below reality means conjugation reality, so $E^{(j)}(X,\mathbf{1},z)$ is real. Define the two local vacuum families

$$\mathsf{vl}^{(j)}(X,z) = E^{(j)}(X,\mathbf{1},z), \qquad \mathsf{vn}^{(j)}(X,z) = \mathsf{n}^{(j)}(X,z).$$

Both are bounded by $E_0 e^{-\kappa d_j(X)}$.

The three extensive coefficients are

$$\mathsf{co}_j^{\mathrm{b}} = \log \sigma_{\mathrm{H}} - \tfrac{3}{2} y_j, \quad \mathsf{co}_j^{\mathrm{s}} = -(L^3-1)\log \mathsf{z}(x_j,\varepsilon_j), \quad \mathsf{co}_j^{\mathrm{i}} = \tfrac{3}{2}\log(2\pi), \qquad \sigma_{\mathrm{H}} = (2\pi^2)^{-1}. \tag{1402}$$

Assume $x_j > 1$, $0 < \varepsilon_j \le 1$ and $x_j \varepsilon_j^2 \ge 1$ at all levels used here. The last condition holds for $\varepsilon_j = A_0 x_j^{-1/2} y_j^{\mathfrak{p}_0}$ as soon as $A_0(\log \gamma^{-2})^{\mathfrak{p}_0} \ge 1$ and $x_j \ge \gamma^{-2}$. The radial Haar formula, with $w = |u - \mathbf{1}|$, is

$$\mathsf{z}(x,\varepsilon) = \frac{2}{\pi}\int_0^{\varepsilon} e^{-xw^2/2} w^2 \sqrt{1 - w^2/4}\, dw.$$

It follows by writing a unit quaternion as $(\cos\theta, \sin\theta\,\omega)$ and substituting $w = 2\sin(\theta/2)$. Extending the upper integral to $[0,\infty)$, and restricting the lower integral to $[0, x^{-1/2}]$, gives

$$\frac{1}{20} x^{-3/2} \le \mathsf{z}(x,\varepsilon) \le 8 x^{-3/2}. \tag{1403}$$

For the lower bound use $e^{-xw^2/2} \ge e^{-1/2}$ and $\sqrt{1-w^2/4} \ge \sqrt{3}/2$; the resulting coefficient is $1/(\pi\sqrt{3e}) > 1/20$. The upper Gaussian integral has coefficient $\sqrt{2/\pi} < 8$. Consequently, with

$$C_{\mathrm{co}} = \max\{\log(2\pi^2), (L^3-1)\log 20, \tfrac{3}{2}\log(2\pi)\}, \tag{1404}$$

the coefficients in (1402) satisfy

$$|\mathsf{co}_j^{\mathrm{b}} + \tfrac{3}{2} y_j| \le C_{\mathrm{co}}, \quad |\mathsf{co}_j^{\mathrm{s}} - \tfrac{3}{2}(L^3-1) y_j| \le C_{\mathrm{co}}, \quad |\mathsf{co}_j^{\mathrm{i}}| \le C_{\mathrm{co}}.$$

For $\Omega \supset \Lambda$, unions of cells of level $j+1$, let $B_j(Y)$ be the number of positive nonpivot level-j bonds of Y. Write

$$\mathsf{N}^{(j)}(\Omega,\Lambda) = \mathsf{co}_j^{\mathrm{b}} B_j(\Omega) + \mathsf{co}_j^{\mathrm{s}} \#_{j+1}(\Omega) + \mathsf{co}_j^{\mathrm{i}} B_j(\Lambda), \tag{1405}$$

$$\mathsf{E}^{(j)}(\Omega,\Lambda) = \mathsf{N}^{(j)}(\Omega,\Lambda) + \sum_{\substack{X \in \mathcal{D}_{j+1} \\ X \subset \Lambda}} \left\{ \sum_{z \in X \cap \Lambda^{\sim -1}} \mathsf{vl}^{(j+1)}(X,z) + \sum_{z \in X} \mathsf{vn}^{(j+1)}(X,z) \right\}. \tag{1406}$$

The sites in these sums are level-$j+1$ sites. The distinction between the two ranges of z is essential: the regular terms use the interior base points, whereas the Gaussian constants use every site of X. For a label $\mathcal{S} = (\Omega_n, \Lambda_n, S_n)_{n \le k}$ put

$$\begin{aligned} \mathcal{E}_{k,\mathcal{S}} &= -\sum_{j<k} \mathsf{E}^{(j)}(\Omega_{j+1}, \Lambda_{j+1}), \\ E_{(k)} &= \sum_{j<k} \mathsf{E}^{(j)}(T_\ell, T_\ell), \qquad \mathfrak{E}_k^{\geq} = \sum_{j=k}^{K-1} \mathsf{E}^{(j)}(T_\ell, T_\ell), \\ \mathfrak{E}'_n(\mathcal{S}) &= \mathsf{E}^{(n-1)}(T_\ell, T_\ell) - \mathsf{E}^{(n-1)}(\Omega_n, \Lambda_n). \end{aligned} \tag{1407}$$

The last-level sum $\mathfrak{E}_k^{\geq}$ presupposes that the local families have been constructed through level K.

63.2. **Volume and deficit bounds.** Set

$$C_{\mathrm{E},p} = 2048 \sum_{q \ge 0} (q+2)^p e^{-q}, \qquad E_0' = C_{\mathrm{E},0} E_0, \qquad E_0'' = 8 C_{\mathrm{E},1} E_0,$$

and define

$$\begin{aligned} C_{\mathfrak{E}} &= \frac{(6L^3-5)C_{\mathrm{co}} + 2E_0'}{L^3-1}, \\ C_{\log} &= 3(L^3-1), \qquad C_{\mathrm{loc}} = (6L^3-5)C_{\mathrm{co}} + E_0' + 2E_0'', \qquad C_\flat = \max\{C_{\log}, C_{\mathrm{loc}}\}. \end{aligned} \tag{1408}$$

Proposition 63.1. Assume the preceding normalizer and local-family hypotheses at every level in question, and $m' < m + K - j$ whenever $\mathcal{D}_j$ is used. Suppose the couplings satisfy $x_j = L^{-1} x_{j-1} - b_j$, $|b_j| \le B_\sharp$, and $x_K \ge \max\{4, 2B_\sharp\}$. All constants in (1407) are real, and

$$E_{(k)} + \mathcal{E}_{k,\mathcal{S}} = \sum_{n=1}^{k} \mathfrak{E}_n'(\mathcal{S}). \tag{1409}$$

Writing $Z_n = \Lambda_n^c$, one has

$$|\mathfrak{E}_n'(\mathcal{S})| \le C_\flat (1 + y_{n-1})\, \#_n(Z_n^\sim). \tag{1410}$$

For the label with $\Omega_n = \Lambda_n = T_\ell$ every deficit is zero. If the families are given through K, then

$$E_{(K)} = E_{(k)} + \mathfrak{E}_k^{\ge}, \qquad -(3y_k + C_{\mathfrak{E}}) N_k \le \mathfrak{E}_k^{\ge} \le C_{\mathfrak{E}} N_k. \tag{1411}$$

If in addition $\check{R}_n \ge y_n^r$, $r \ge 2$, and $\mathsf{r}_n = M^3 \check{R}_n^4$, the cell count $|Z|_n$ satisfies

$$(1 + y_{n-1}) \#_n(Z^\sim) \le 27(2 + \log(2L))\, \mathsf{r}_n |Z|_n. \tag{1412}$$

All these constants are independent of the number of steps, the torus and the couplings. ∎

Proof. The entropy bound (683) gives, for every site z,

$$\sum_{\substack{X \ni z \\ d_j(X) \ge D}} e^{-\kappa d_j(X)} (d_j(X) + 2)^p \le C_{\mathrm{E},p} e^{-D}.$$

Indeed, at size q the product of entropy and decay is at most $2048 e^{-2q}$, and for $q \ge D$ this is at most $2048 e^{-D} e^{-q}$. Thus each of the local vacuum families $v = \mathsf{vl}^{(j)}, \mathsf{vn}^{(j)}$ obeys

$$\sum_{X \ni z} |v(X,z)| \le E_0', \qquad \sum_{X \supset \square} \sum_{z \in X} |v(X,z)| \le M^3 E_0''. \tag{1413}$$

For the second inequality use $\#_j(X) \le 8M^3 (d_j(X) + 1)$ and choose a site of the fixed cube $\square$.

There are $3(L^3-1)$ positive nonpivot bonds per coarse block: of its $3L^3$ outgoing bonds, precisely three are pivots. Hence $B_j(T_\ell) = 3(L^3-1) N_{j+1}$. A missing bond of a region is an inner bond of its complement, by the bond convention. Its initial point lies in a coarse block of that complement. Therefore

$$0 \le B_j(T_\ell) - B_j(Y) \le 3(L^3-1) \#_{j+1}(Y^c). \tag{1414}$$

The pivot assertion used in this count follows from the corridor incidence: a pivot is a bond of Y exactly when its associated coarse bond is a bond of Y.

On the whole torus the logarithmic coefficients combine to $-3(L^3-1) y_j N_{j+1}$. The remaining coefficients and the two sums in (1413) give

$$|\mathsf{E}^{(j)}(T_\ell, T_\ell) + 3(L^3-1) y_j N_{j+1}| \le [(6L^3-5)C_{\mathrm{co}} + 2E_0'] N_{j+1}.$$

Now $\sum_{j=k}^{K-1} N_{j+1} \le N_k/(L^3-1)$. Also $x_{j-1} = L(x_j + b_j) \ge L x_j/2 > x_j$: backward induction starts from $x_K \ge 2B_\sharp$. Thus $0 < y_j \le y_k$ for $j \ge k$. Summing the last display proves (1411).

To estimate a deficit, put $\Omega = \Omega_n$, $\Lambda = \Lambda_n$, and $Z = \Lambda^c$. Its logarithmic part is

$$-\tfrac{3}{2} y_{n-1} [B_{n-1}(T_\ell) - B_{n-1}(\Omega) - (L^3-1) \#_n(\Omega^c)].$$

By (1414), its modulus is at most $3(L^3-1)y_{n-1}\#_n(\Omega^c)$. The three remaining coefficient contributions total at most $(6L^3-5)C_{\mathrm{co}}\#_n(Z)$.

There are two reasons for a local term to be missing from the restricted sum. Either $X \not\subset \Lambda$, or $X \subset \Lambda$ and its base point lies outside $\Lambda^{\sim -1}$. In the first case X contains a cube of Z. Summing the cube estimate in (1413) over all such cubes bounds each vacuum family by $E_0''\#_n(Z)$. The second case occurs only for $\mathsf{v}|$ and has $z \in Z^\sim \setminus Z$; the site estimate bounds it by $E_0'[\#_n(Z^\sim) - \#_n(Z)]$. These two cases are disjoint. Since $\Omega^c \subset Z \subset Z^\sim$, their sum proves (1410) with (1408). Identity (1409) is the telescoping sum of the defining deficits; reality follows from that of each coefficient and local value.

Finally $x_{n-1} \le (3L/2)x_n \le 2Lx_n$, so $1+y_{n-1} \le (2+\log(2L))\check{R}_n$ when $y_n > 1$ and $\check{R}_n \ge y_n^r$. A cell has at most 27 touching cells, including itself. Hence $\#_n(Z^\sim) \le 27\#_n(Z)$, while $\#_n(Z) = M^3\check{R}_n^3|Z|_n$. This proves (1412). □

64. The sum of the regular and boundary interactions

A uniform bound on each local interaction would give a contribution at every finer lattice site. The regularity of a field in a later layer is stronger. Gauge invariance makes its local interaction quadratic at the trivial field; the resulting fourth power of the scale ratio compensates for the number of finer sites. We establish this estimate before summing over the layers.

Use the constants of proposition 51.9, and assume

$$M\alpha_0 \le \alpha_1, \qquad \alpha_0 + \alpha_1 \le a^\sharp, \qquad \kappa \ge \log 9216 + 2.$$

In this section $1 \le j \le k \le K$, $m' < m + K - k$, and all local families have the fixed radii (α_0, α_1). Write

$$\rho_{\mathrm{sc}} = \frac{\alpha_0}{16K_{\mathrm{tw}}^\sharp}, \qquad C_\natural^{\mathrm{sc}} = 8\max\{16K_{\mathrm{tw}}^\sharp B_*, c_*^{-1}\}^2,$$

where B_*, c_* are the regional-gauge constants in theorem F, with $c_* \le 1$. The constants below are independent of the depth and volume and of the fixed radii.

64.1. The quadratic gain near a regular field.

Lemma 64.1. Use level-j lengths. For $z \in X^{(j)}$, put

$$d = d_j(X), \qquad l = 2M(d+2)+1, \qquad Q = N_j(\{z\}; l/2), \quad \widetilde{Q} = N_j(\{z\}; l/2+2).$$

Suppose $l + 4 < L^{m+K-j}$, and let a real field U have curvature less than $\theta\eta_j^2$ and current less than θ on $\widetilde{Q}$, including every plaquette with a corner there and every bond with an endpoint there. If its pair on X belongs to the analyticity core, then each local term E satisfies

$$|E(U) - E(\mathbf{1})| \le C_\natural^{\mathrm{sc}}\|E\|_X(\theta/\alpha_0)^2 l^2. \tag{1415}$$

■

Proof. The spanning-edge geometry gives $|x - z|_\infty \le M(d+2)$ for $x \in X$. Thus Q contains both endpoints of every bond of $\mathrm{Bd}(X)$. It is a cube of odd integer side formed from level-j unit cubes, and $\widetilde{Q}$ lies in one torus chart.

First suppose $2l\theta \le c_*$ and $a = 2B_*l\theta \le \rho_{\mathrm{sc}}$. The current-to-divergence identity used in theorem F gives, with $\eta = \eta_j \le 1/13$,

$$|D_U^{\mathsf{T}}\partial U| \le \eta^{-1}(\theta\eta^2)^2 + \eta^2|J^{(j)}(U)| < 2\theta\eta^2.$$

Here $\theta \le 1/2$. Apply the regional gauge theorem with radius 2θ. On the bonds meeting Q it gives $U^u = e^{i\eta A}$, with $|A|, |\nabla^\eta A| < a$. Restrict to $\mathrm{Bd}(X)$, put $J_1 = \mathrm{Ad}(u)J^{(j)}(U)$, and consider

$$\gamma(\zeta) = (e^{i\zeta\eta A}, \zeta J_1), \qquad |\zeta| \le R := \rho_{\mathrm{sc}}/a.$$

We have $|J_1| < \theta \le a$ and $R \ge 1$.

This entire disc belongs to the fixed core. To see the margin explicitly, take the real reference field to be one. For $\phi = |\zeta| a \le \rho_{\mathrm{sc}} \le 1/16$, the four exponential plaquette factors give

$$|\partial(e^{i\zeta\eta A}) - 1| \le (2\phi + 11\phi^2)\eta^2 < 4\rho_{\mathrm{sc}}\eta^2.$$

The linear part is the curl of A; the remaining terms are bounded by the exponential-product remainder. The perturbation and its first differences are at most $\rho_{\mathrm{sc}} < 2\rho_{\mathrm{sc}}$, and the current is smaller than $4\rho_{\mathrm{sc}}$. The real reference has trivial cube gauges. These are the simple-domain conditions at radii $(4\rho_{\mathrm{sc}}, 2\rho_{\mathrm{sc}}, 4\rho_{\mathrm{sc}})$. Moreover

$$6\rho_{\mathrm{sc}} \le a^{\sharp}, \quad 4\rho_{\mathrm{sc}} \le \alpha_0, \quad 2\rho_{\mathrm{sc}} \le \alpha_1, \quad K^{\sharp}_{\mathrm{tw}}(4\rho_{\mathrm{sc}} + 4\rho_{\mathrm{sc}}^2) \le \alpha_0/2.$$

Thus proposition 51.9 and radius monotonicity put the disc in the core at (α_0, α_1).

The function $f(\zeta) = E(\gamma(\zeta))$ is holomorphic near the closed disc and bounded by $\|E\|_X$ there. Its derivative at zero vanishes. Indeed conjugation by the four constant quaternions $1, i\sigma_1, i\sigma_2, i\sigma_3$ preserves E. Their adjoint actions sum to zero on $\mathfrak{su}(2)^{\mathbb{C}}$. Apply the differential of $E(e^{i\eta A}, J)$ at $(0,0)$ to the sum of these four conjugates of (A, J_1); it equals both zero and $4f'(0)$. The maximum principle for $(f(\zeta) - f(0))/\zeta^2$ gives

$$|E(U) - E(\mathbf{1})| \le 2\|E\|_X (a/\rho_{\mathrm{sc}})^2.$$

To apply gauge invariance here, both representatives must belong to the core. The triple representing $\gamma(1)$ lies in the simple domain at radii $(4\rho_{\mathrm{sc}}, 2\rho_{\mathrm{sc}}, 4\rho_{\mathrm{sc}})$. Its transform by u^{-1} lies in the same domain: unitary conjugation preserves the curvature, perturbation, and current bounds, and a cube gauge w is replaced by wu, with the same gauge-fixed field. The preceding application of proposition 51.9 therefore puts both triples in the fixed core. Gauge invariance and pair dependence now identify $E(U)$ with $f(1)$.

If either smallness inequality fails, use instead $|E(U) - E(\mathbf{1})| \le 2\|E\|_X$. In the case $2l\theta > c_*$, multiply this bound by $(2l\theta/c_*)^2 > 1$; in the case $a > \rho_{\mathrm{sc}}$, multiply it by $(a/\rho_{\mathrm{sc}})^2 > 1$. All three cases are bounded by

$$2\|E\|_X (2l\theta)^2 \max\{B_*/\rho_{\mathrm{sc}}, c_*^{-1}\}^2.$$

Since $\alpha_0 \le 1$, this proves (1415). □

64.2. **Summation at a site and over the layers.** Fix a history of length k, with $\Omega_0 = \mathbb{T}$, $\Omega_{k+1} = \varnothing$, and layers $\mathcal{L}_n = \Omega_n \setminus \Omega_{n+1}$ in the level-n unit partition. Let U be a real bare field with

$$\mathrm{rad}_{\mathsf{U}}(\mathfrak{e}) < C_{\mathrm{ly}}\varepsilon_{\lambda(\mathfrak{e})} \quad \text{on the collared region } \Omega_0^{\diamond}, \qquad C_{\mathrm{ly}} > 0. \tag{1416}$$

Use the curvature and current radii of section 53.6. Assume its pair belongs to the fixed core for every $X \subset \Lambda_j$, including the domains used by the edge terms. The coupling and geometric conditions used here are

$$\begin{gathered} |b_j| \le B_\sharp, \qquad \varepsilon_q \le \varepsilon_n \quad (q \le n), \qquad \varepsilon_q \le \sqrt{2} L^{(q-n)/2}\varepsilon_n \quad (q \ge n), \\ \check{R}_n \ge L, \qquad M\check{R}_n \le L^{m+K-n}, \qquad C_{\mathrm{ly}}\varepsilon_k \le \alpha_0. \end{gathered} \tag{1417}$$

These are the running bounds already used for the history backgrounds, together with the displayed fixed-radius ceiling.

Let $\|E^{(j)}\|_{j,\kappa+1} \le E_0$. For each permitted created boundary index, take a number $\mathsf{b}_j(X)$ satisfying $|\mathsf{b}_j(X)| \le \mathsf{B}e^{-\kappa d_j(X)}$. Such numbers are the evaluations of the boundary entries when their fields lie

in the stated domains. Use the edge sums (1388), and define

$$\mathbf{E}_k(\mathsf{U}) = \sum_{j=1}^{k} \left\{ \sum_{\substack{X\subset\Lambda_j \\ z\in X^{(j)}\cap\Lambda_j^{\sim -1}}} [E^{(j)}(X,\mathsf{U},z) - E^{(j)}(X,\mathbf{1},z)] - b_j L^j A(\varphi_j,\mathsf{U}) \right\},$$
$$\mathbf{E}_k^{\mathrm{bd}}(\mathsf{U}) = \sum_{j=1}^{k} \left\{ \sum_{X\in \mathsf{lx}_j^{\mathrm{mk}}} \mathsf{b}_j(X) + \sum_{X\in \mathsf{lx}_j^{\mathrm{ed}}} E^{(j),\mathrm{ed}}(X,\mathsf{U}) \right\}. \tag{1418}$$

Here φ_j is the smoothed interior profile: $0 \le \varphi_j \le 1$, and it vanishes at distance at most one quarter of a level-j cell side from Z_j. Only these two properties of the profile enter this estimate.

For the entropy constants $C_{\mathrm{E},p}$ from section 63, set

$$\begin{aligned} C_{\mathrm{f}}' &= 110592, \qquad C_{\mathrm{sh}}^{\mathrm{sc}} = 18 C_{\natural}^{\mathrm{sc}} C_{\mathrm{E},2} M^2 L^4, \qquad C_{\mathrm{a}}^{\mathrm{sc}} = C_{\mathrm{sh}}^{\mathrm{sc}} + 2C_{\mathrm{E},0}, \\ C_{\mathrm{m}}^{\mathrm{sc}} &= \tfrac{13}{12} C_{\mathrm{a}}^{\mathrm{sc}} + 432 C_{\mathrm{f}}' + \tfrac{13}{4}, \qquad C_5 = C_{\mathrm{m}}^{\mathrm{sc}}(E_0 + B_\sharp + M^{-3}\mathsf{B}). \end{aligned} \tag{1419}$$

Proposition 64.2. Under these assumptions, both sums in (1418) are defined, and the sum of the moduli of all their individual summands is bounded by

$$C_5 \sum_{n=0}^{k} \#_n(\mathcal{L}_n) \le C_5 \left(N_k + L^3 \sum_{j=1}^{k} \#_j(Z_j) \right). \tag{1420}$$

In particular this bounds $|\mathbf{E}_k(\mathsf{U})| + |\mathbf{E}_k^{\mathrm{bd}}(\mathsf{U})|$. ∎

Proof. The trivial pair is in every core by proposition 51.9. The membership assumption therefore makes all local and edge terms defined.

Fix $z \in \Lambda_j \cap \mathcal{L}_n$, $j \le n$, and put

$$n' = \max\{n-1, j\}, \quad w = M\check{R}_n L^{n-j}, \quad \theta = \sqrt{2} C_{\mathrm{ly}} \varepsilon_{n'} L^{-2(n'-j)}.$$

We use level-j lengths. The cell-size condition gives $w \le L^{m+K-j}$, half the torus side. The ball of radius $2w$ about z lies in $\Omega_{n'}$: for $n = j$ use the $2M\check{R}_j$ gap from Λ_j to Ω_j^c; for $n > j$ use the $8M\check{R}_n L^{n-j}$ gap from Ω_n to Ω_{n-1}^c. On a neighborhood of radius at most $2w-1$, all plaquette and current stencils thus lie in $\Omega_0^\diamond$ and have layer index $\lambda \ge n'$. Equation (1416) and the coupling ratio imply

$$|\partial\mathsf{U} - 1| < \sqrt{2} C_{\mathrm{ly}} \varepsilon_{n'} L^{-2n'} = \theta \eta_j^2, \qquad |J^{(j)}(\mathsf{U})| < \sqrt{2} C_{\mathrm{ly}} \varepsilon_{n'} L^{-3(n'-j)} \le \theta.$$

The extra factors $L^{-3(\lambda-n')/2}$ and $L^{-5(\lambda-n')/2}$ have been bounded by one.

Call $X \ni z$ short if $2M(d_j(X)+2)+5 < w$. The cubes in lemma 64.1 then fit into the preceding regularity neighborhood and into a torus chart. Since $l \le 3M(d+2)$, that lemma bounds a short term by

$$18 C_{\natural}^{\mathrm{sc}} M^2 L^4 E_0 (C_{\mathrm{ly}} \varepsilon_{n'}/\alpha_0)^2 L^{-4(n-j)} (d+2)^2 e^{-(\kappa+1)d}.$$

The entropy estimate in (1413) and its preceding weighted bound sum the last two factors to at most $C_{\mathrm{E},2}$. For a long domain, $d \ge \frac{1}{2}\check{R}_n L^{n-j} - 3 > 0$. Use the bound $2E_0 e^{-(\kappa+1)d}$ and the same entropy tail to obtain at most $2e^3 C_{\mathrm{E},0} E_0 e^{-\check{R}_n L^{n-j}/2}$. Writing $s = L^{n-j} \ge 1$, the inequalities $\check{R}_n \ge 13$ and $e^{-13s/4} \le 24(4/13)^4 s^{-4} < s^{-4}$ give

$$e^{-\check{R}_n s/2} \le e^{-\check{R}_n/4} s^{-4}.$$

The fixed-radius ceiling and $e^{3-\check{R}_n/4} \le 1$ therefore yield the site estimate

$$\sum_{X:\ z\in X\subset\Lambda_j} |E^{(j)}(X,\mathsf{U},z) - E^{(j)}(X,\mathbf{1},z)| \le C_{\mathrm{a}}^{\mathrm{sc}} E_0 L^{-4(n-j)}. \tag{1421}$$

There are $L^{3(n-j)}\#_n(\mathcal{L}_n)$ level-j sites in $\mathcal{L}_n$. Summing gives $C_{\mathrm{a}}^{\mathrm{sc}}E_0\sum_{n=j}^{k}L^{-(n-j)}\#_n(\mathcal{L}_n)$.

For the action term, a contributing plaquette has all corners in Λ_j. Indeed its center is farther than a quarter-cell side from Z_j, whereas its corners are only a half bare spacing from its center. If its base lies in $\mathcal{L}_n$, then $\lambda(p)\ge n$, and the layer bound gives

$$a_p(\mathsf{U})=\tfrac12|\partial\mathsf{U}(p)-1|^2<(C_{\mathrm{ly}}\varepsilon_n)^2L^{-4n}.$$

There are three orientations and $L^{3n}\#_n(\mathcal{L}_n)$ bare base points. Consequently

$$|b_j|L^jA(\varphi_j,\mathsf{U})\le 3B_\sharp\sum_{n=j}^{k}L^{-(n-j)}(C_{\mathrm{ly}}\varepsilon_n)^2\#_n(\mathcal{L}_n). \tag{1422}$$

Every created-boundary domain has a cube in $Z_j^{\sim}$, since it is not contained in $\Lambda_j^{\sim-1}=(Z_j^{\sim})^c$. The edge indices are among these indices. For a fixed cube, (684) bounds the sum of $e^{-\kappa d}$ by 4096. Even allowing the 27 cubes touching each cube of $Z_j^{\sim}$, summation therefore gives

$$\sum_{X\in\mathsf{lx}_j^{\mathrm{mk}}}e^{-\kappa d_j(X)}\le C_{\mathrm{f}}'M^{-3}\#_j(Z_j^{\sim}).$$

The edge bound (1389) and $\#_j(Z_j^{\sim})\le 27\#_j(Z_j)$ bound the boundary contribution by

$$27C_{\mathrm{f}}'(M^{-3}\mathsf{B}+8E_0)\#_j(Z_j).$$

To sum these bounds, put $\mathsf{N}=\sum_{n=0}^{k}\#_n(\mathcal{L}_n)$. The regular and action sums use $\sum_{a\ge0}L^{-a}\le 13/12$ and $C_{\mathrm{ly}}\varepsilon_n\le\alpha_0\le1$. For the boundary sum note that $x\in Z_j$ implies that its layer index is at most j, since $\Omega_{j+1}\subset\Lambda_j$. Counting bare sites and rescaling gives

$$\#_j(Z_j)\le\sum_{n=0}^{j}L^{-3(j-n)}\#_n(\mathcal{L}_n),\qquad\sum_{j=1}^{k}\#_j(Z_j)\le 2\mathsf{N}.$$

Thus the sum of moduli is at most

$$\Big[\Big(\tfrac{13}{12}C_{\mathrm{a}}^{\mathrm{sc}}+432C_{\mathrm{f}}'\Big)E_0+\tfrac{13}{4}B_\sharp+54C_{\mathrm{f}}'M^{-3}\mathsf{B}\Big]\mathsf{N}\le C_5\mathsf{N}.$$

Finally $\mathcal{L}_n\subset\Omega_{n+1}^c\subset Z_{n+1}$ at bare sites for $n<k$, so $\#_n(\mathcal{L}_n)\le L^3\#_{n+1}(Z_{n+1})$; the top count is at most N_k. These inequalities prove the second part of (1420). □

65. Integrable bounds for the large-field components

A failed test must force a definite contribution either to the Wilson action, to the gauge-fixing exponent, or to a retained quadratic form. The shares of proposition 44.9 prevent the same action from being used at different steps. Within one step, a finite covering multiplicity allows us to sum these lower bounds over the newly large cells.

Use the failed-test estimates of propositions 44.4 and 44.5, with their local averaging ties and curvature ceilings. Write $c_{\mathrm{P}}=c_1^{\mathrm{P}}=\frac14B_3^{-2}$ and $c_{\mathrm{Q}}=c_2^{\mathrm{Q}}$, where the latter and σ_{Q} are defined in (659). The extended gauge energy is (660). The weighted costs are (664) and (665); the failed density test at level zero costs $\frac12\varepsilon_0^2$ in the Wilson action.

65.1. **Averages on the reserved regions.**

Proposition 65.1. Let the collared background of a history h exist at the data of a bare field U, and let $\mathsf{U}=U_{\mathrm{el}}\vee U$ be any glued representative. For every piece (t,D), the averages of orders $t-1$ and t of U equal those of U on the inner bonds of $N_{t,D}$ at their respective levels. On any region N of level-t cubes contained in $N_{t,D}$, these averages on the inner bonds of N can be computed from the restriction of U to the inner bare bonds of N. In particular these assertions hold on $\square'^{\sim4}$ for every failed P_t- or Q_t-cube $\square'\subset D$. If $\check R_0\le L\check R_1$, the glued field also equals U on $\square^{\sim}$ for each density cube $\square\subset P_0\cap D$. ■

Proof. For $t \geq 2$, proposition 44.9 places $N_{t,D}$ inside Ω_{t-1} and outside Ω_t. Its level-$(t-1)$ bonds are therefore among the collared constraints $\Gamma^{\diamond}_{t-1}$. The constrained average of U_{el} equals that of U. All bare bonds needed for this average lie inside the collar; gluing changes none of them. For $t = 1$, an inner bare bond of $N_{1,D}$ is either outside the collar, where gluing leaves U unchanged, or belongs to $\Gamma^{\diamond}_0$, where the constraint gives the same conclusion.

The averaging blocks centered at sites of a region of level-t cubes remain in that region. The double block of every inner level-t bond therefore lies in $N_{t,D}$, so one more averaging step proves the order-t assertion. The same block property proves the assertion for a restriction. The four-layer enlargement of a test cube has width $4M_2\check{R}_t a_{K-t} \leq s_t$, so lies in its share region by proposition 44.9. For a density cube the width is $M_2\check{R}_0 a_K \leq M\check{R}_1 a_{K-1} = s_1$. The order-zero assertion then identifies the bare fields and their plaquette actions there. □

65.2. **The failed tests near a created cell.** The created cells at step t are the cells of

$$\mathcal{W}_t^{\wedge} = P'_t \cup Q'_t \cup R'_t \cup \mathsf{K}(I_t) \cup \mathsf{R}_t^{\mathrm{lo}},$$

with P'_0 added when $t = 1$. Write $\mathrm{Cr}(t, D)$ for the pairs $(t, \mathfrak{c})$ with $\mathfrak{c} \subset D$ a created cell, and $\mathrm{Cr}(\mathfrak{u})$ for their union over a set of pieces $\mathfrak{u}$. By (644), $\mathfrak{c}^{\sim}$ lies in Z_t for every created cell, hence in its unique component D.

For a piece (t, D), with $j = t - 1$, set

$$\mathsf{Y}(t, D) = P''_t \cap D \cap \Omega_t^c, \qquad n^{\mathrm{g}}_{t,D} = (L^3 - 1)\#\mathsf{Y}(t, D).$$

Let $\mathfrak{B}(t, D)$ be the level-j bonds whose endpoints lie in one block $B(y)$, $y \in \mathsf{Y}(t, D)$, and let $\mathfrak{X}^{\mathrm{st}}(t, D)$ be the external nonpivot bonds of step t with both endpoints in D. The variables of this piece range over

$$\mathsf{X}^{\mathrm{fr}}_{t,D} = \mathcal{V}(\mathfrak{B}(t, D)) \times \mathfrak{g}^{\mathfrak{X}^{\mathrm{st}+}(t,D)}, \tag{1423}$$

with product Haar and Lebesgue measure $m_{t,D}$. At a bare field U, the entries are the restriction of $V_j = \mathcal{M}^j U$ and the stored field a'_j.

Lemma 65.2. Suppose the actual history indicator is one, the actual backgrounds and strip substitutions exist, and the hypotheses of lemma 53.3 hold for these substitutions. If $t = 1$, assume also $\check{R}_0 \leq L\check{R}_1$. For a created cell $\mathfrak{c} \subset D$, one of the following tests fails in $\mathfrak{c}^{\sim 2}$.

(1) If $\mathfrak{c} \subset P'_t$ or Q'_t, there is a failed test cube $\square'$ in the corresponding set, contained in a cell $\mathfrak{c}''$ equal to or touching $\mathfrak{c}$, with

$$\square'^{\sim 4} \subset (\mathfrak{c}'')^{\sim} \subset N_{t,D} \cap \mathfrak{c}^{\sim 2}.$$

In the Q'_t case, $Y_{\square'} \subset \mathsf{Y}(t, D)$.

(2) If $t = 1$ and $\mathfrak{c} \subset P'_0$, a failed density cube has $\square^{\sim} \subset N_{1,D} \cap \mathfrak{c}^{\sim 2}$.

(3) Otherwise there is a positive stored bond b with endpoints in $\mathfrak{c}^{\sim 2}$ such that its new variable is undefined or

$$|\mathsf{a}'_j(b)| \geq \varkappa_{\mathrm{w}}(A_1/A_0)p_0(g_j), \qquad \varkappa_{\mathrm{w}} = \min\{(4\Theta_1)^{-1}, \theta_c\}. \tag{1424}$$

■

Proof. A cell of the hull of P_t or Q_t meets one of its test cubes. The cell $\mathfrak{c}''$ containing that cube touches $\mathfrak{c}$; its one-cell enlargement is in the same component and in $N_{t,D}$. The test-cube enlargement has width at most s_t, so lies there and in $\mathfrak{c}^{\sim 2}$. For Q_t, the cube is selected in $B(P''_t)^{\sim -1}$; hence $(\mathfrak{c}'')^{\sim} \subset B(P''_t) \cap \Omega_t^c \cap D$. This contains $Y_{\square'}$. The argument for P_0 is the same, using $M_2\check{R}_0 a_K \leq s_1$. The factors of the history indicator say precisely that these tests fail.

For a cell in R'_t, choose a failed cube $\square' \subset R_t$ meeting it and test the bonds of $(\square'^{\sim 2})^{(j)*}$. A cell of $\mathsf{K}(I_t)$ contains an endpoint of a bond in I_t; use such a bond. For $\mathsf{R}_t^{\mathrm{lo}}$, use the bonds of its failed

rim test. Each tested bond has both endpoints in $\mathfrak{c}^{\sim} \subset D$, and is a nonpivot bond incident to Ω_t. For the cube test this follows from

$$(L+2)M_2\check{R}_t a_{K-t} + a_{K-j} \le s_t;$$

for the other two tests it follows from $a_{K-j} \le s_t$. Thus all these bonds are stored. They belong to $\mathcal{N}_{\mathrm{s}} \cup \mathcal{N}_{\ell}$, so none is in the plain zone $\mathcal{Z}_1$; a bond in I_t belongs to $\mathcal{N}_{\mathrm{s}}$ and is not in $\mathcal{Z}_\theta$.

Failure gives an undefined old logarithm or one of norm at least δ_j, $\delta_j/(4\Theta_1)$, or $\theta_c\delta_j$, respectively. Outside the strip zone the variable is unchanged. In the cutoff zone the inverse change of variables preserves each threshold at most $4\delta_j/3$, by lemma 53.3. If the new logarithm is defined, the old logarithm cannot be undefined or above such a threshold while the new one is below it: below the threshold the change is the identity. The only excluded case is the point -1 of the new logarithmic chart. Divide by g_j, choose the positive orientation, and use $\delta_j/g_j = (A_1/A_0)p_0(g_j)$. □

65.3. **The elementary integrals.** For $x\varepsilon^2 \ge 1$, the normalizer estimates in section 63 give

$$\mathsf{z}_\infty(x/2) \le 8\,2^{3/2}x^{-3/2} \le 500\,\mathsf{z}(x,\varepsilon).$$

Here z_∞ is the same single-group integral with its indicator removed. For any set Y of level-$(j+1)$ sites, define

$$\mathsf{g}_Y(V) = \mathsf{z}(x,\varepsilon)^{-(L^3-1)\#Y} e^{-xG_j^\infty(Y;V)}.$$

Let $\mathsf{g}_Y^{1/2}$ denote the same expression with half the exponent and the full normalization.

Lemma 65.3. The functions just defined are Borel and

$$\int \mathsf{g}_Y^{1/2}(V)\,dV \le 500^{(L^3-1)\#Y}. \tag{1425}$$

There is also a number $\varepsilon_{\mathsf{z}} = r_*^2/(20\pi L^3) > 0$, depending only on L, such that curvature at most ε_{z} on a double block implies

$$\mathsf{m}_c(V) \text{ is invertible}, \qquad |\det \mathsf{m}_c(V)|^{-1} \le 2L^9. \tag{1426}$$

Here r_* is the local analytic radius in lemma 14.7. ■

Proof. The contour means and their domains are Borel. Average over gauge transformations equal to one at the centers $y \in Y$, with independent Haar values at every other site in the blocks. The domain where all contour means exist is gauge invariant, and each contour variable becomes $V(y,x)u(x)^{-1}$. Haar invariance and Tonelli therefore make its integral the measure of that domain, at most one, times

$$\left[\frac{\mathsf{z}_\infty(x/2)}{\mathsf{z}(x,\varepsilon)}\right]^{(L^3-1)\#Y}.$$

This proves (1425), without assuming existence of the means on a set of full measure.

For the pivot bound, a perturbation supported on the pivot bond changes no contour variable inside either block. Thus the local analytic map of lemma 14.7 gives

$$F_c(a^0,X) = \frac{1}{i}\log\left[\mathcal{M}(e^{iA_{c,X}}e^{ia^0})(c)\mathcal{M}(e^{ia^0})(c)^{-1}\right], \qquad |F_c| \le 1$$

for $\sup|a^0| < r_*$, $|X| < r_*$, and $D_X F_c(a^0,0) = L\mathsf{m}_c(e^{ia^0})$. Apply Schwarz in a^0 and Cauchy in X to $F_c(a^0,X) - F_c(0,X)$. Since $\mathsf{m}_c(1) = L^{-3}$, this gives

$$|\mathsf{m}_c(e^{ia^0}) - L^{-3}| \le \frac{2\sup|a^0|}{Lr_*^2}.$$

The comb gauge on a double block of side at most $2L$ gives $V^u = e^{ia^0}$ with

$$\sup|a^0| < 2\pi L\varepsilon_{\mathsf{z}} = r_*^2/(10L^2) < r_*.$$

It follows that $|L^3\mathsf{m}_c(V^u)-1|<1/5$, so $|\det \mathsf{m}_c(V^u)|>(4/5)^3L^{-9}$. Differentiating gauge covariance of the average shows that $\mathsf{m}_c(V^u)$ is obtained from $\mathsf{m}_c(V)$ by orthogonal adjoint maps on its two ends. Their determinant moduli are one. Thus the same lower bound holds for $|\det \mathsf{m}_c(V)|$, proving (1426). □

65.4. **The product bound.** Fix $0<\gamma_{\mathrm{kp}}\le 1$. At each piece let $\mathscr{Q}_{t,D}$ be a real quadratic form satisfying

$$\mathscr{Q}_{t,D}(a')\ge 2\gamma_{\mathrm{kp}}\sum_{b\in\mathfrak{X}^{\mathrm{st}+}(t,D)}|a'(b)|^2. \tag{1427}$$

Put $N_D=\#_t(D)$, and use $x=x_j$, $\varepsilon=\varepsilon_j$ and $Y=\mathsf{Y}(t,D)$ in the preceding gauge weights. Define

$$\bar{\mathsf{u}}_{t,D}(V,a')=(2L^9)^{6N_D}\mathsf{g}_Y^{1/2}(V)\exp\left(-\frac{\gamma_{\mathrm{kp}}}{4}\sum_b|a'(b)|^2\right). \tag{1428}$$

The constants used below are

$$\begin{aligned} c_\star&=5^{-3}\min\left\{\tfrac12,\tfrac12 c_{\mathrm{P}},\tfrac12 c_{\mathrm{Q}}(A_1/A_0)^2,\tfrac14\gamma_{\mathrm{kp}}\varkappa_{\mathrm{w}}^2(A_1/A_0)^2\right\},\\ C'_\star&=6\log(2L^9)+L^3\left(\log 500+\tfrac92\log(4\pi/\gamma_{\mathrm{kp}})\right). \end{aligned} \tag{1429}$$

Proposition 65.4. Assume the hypotheses of lemma 65.2, existence of the collared background, and defined stored variables. Suppose

$$\begin{gathered} x_k\ge 2B_\sharp,\quad p_0(g_k)\ge 1,\quad \check R_l\le L\check R_{l+1}\ (0\le l<k),\\ \varepsilon_t\le B_3\bar\varepsilon,\quad c_{\mathrm{P}}\varepsilon_t^2\le e_*\ (1\le t\le k),\quad c_{\mathrm{Q}}\delta_j^2\le\sigma_{\mathrm{Q}}\ (0\le j<k). \end{gathered}$$

Let U be any glued representative and suppose its level-j curvature is at most ε_{z} on the double blocks of a set $\Gamma'(t,D)$ of positive coarse bonds incident to D. For any set of pieces $\mathfrak{u}$, write

$$z_{t,D}=\prod_{c\in\Gamma'(t,D)}|\det \mathsf{m}_c(\mathcal{M}^j\mathsf{U})|^{-1}.$$

Then

$$\begin{aligned} &e^{-A_{\mathrm{cr}}(\mathfrak{u};\mathsf{U})}\prod_{(t,D)\in\mathfrak{u}}\left[\mathsf{g}_Y(V_j)z_{t,D}e^{-\mathscr{Q}_{t,D}(\mathsf{a}'_j)/2}\right]\\ &\qquad\le\exp\left[-c_\star\sum_{(t,\mathfrak{c})\in\mathrm{Cr}(\mathfrak{u})}p_0(g_t)^2\right]\prod_{(t,D)\in\mathfrak{u}}\bar{\mathsf{u}}_{t,D}(V_j,\mathsf{a}'_j). \end{aligned} \tag{1430}$$

The same bound holds with the displayed pivot determinants $\prod_c\det(L\mathsf{m}_c)^{-1}$ in modulus. For each piece the majorant is Borel and

$$\int\bar{\mathsf{u}}_{t,D}\,dm_{t,D}\le e^{C'_\star\#_t(D)}. \tag{1431}$$

■

Proof. On a piece abbreviate the three exponents by

$$P_{\mathrm{W}}=L^jx_jA_{N_{t,D}}(\mathsf{U}),\qquad P_{\mathrm{G}}=x_jG_j^\infty(\mathsf{Y}(t,D);V_j),\qquad P_{\mathrm{F}}=\mathscr{Q}_{t,D}(\mathsf{a}'_j)/2.$$

Assign the amount $L^jx_ja_p(\mathsf{U})$ to each plaquette in its share, half the gauge exponent at each site of $\mathsf{Y}(t,D)$, and $\gamma_{\mathrm{kp}}|\mathsf{a}'_j(b)|^2/4$ to each stored bond. The sum of these amounts is

$$E_{t,D}=P_{\mathrm{W}}+\tfrac12P_{\mathrm{G}}+\tfrac14\gamma_{\mathrm{kp}}\sum_b|\mathsf{a}'_j(b)|^2.$$

For each created cell sum only the items whose supports lie in $\mathfrak{c}^{\sim2}$. The hypotheses imply $p_0(g_j)\ge p_0(g_t)$ and $1-B_\sharp g_t^2\ge1/2$. The failed-test energy estimates of propositions 44.4 and 44.5, applied using proposition 65.1 and lemma 65.2, give a lower bound $5^3c_\star p_0(g_t)^2$ for this local sum, in each of

the density, P, Q, and fluctuation cases. Every item contains a lattice site, which lies in one level-t cell. At most 5^3 two-cell enlargements contain that site. Thus each item occurs at most 5^3 times, and

$$E_{t,D} \geq c_\star \# \operatorname{Cr}(t,D)\, p_0(g_t)^2. \tag{1432}$$

If a contour mean is undefined, its gauge weight is zero and the product bound is immediate. Otherwise all these exponents are finite.

There are at most $6N_D$ positive coarse bonds incident to D. By (1426), $z_{t,D} \leq (2L^9)^{6N_D}$. After paying $E_{t,D}$, the remaining fluctuation exponent is

$$P_{\mathrm{F}} - \tfrac{1}{4}\gamma_{\mathrm{kp}} \sum_b |\mathsf{a}_j'(b)|^2 \geq \tfrac{3}{4}\gamma_{\mathrm{kp}} \sum_b |\mathsf{a}_j'(b)|^2 \geq \tfrac{1}{4}\gamma_{\mathrm{kp}} \sum_b |\mathsf{a}_j'(b)|^2.$$

The remaining gauge exponent is $P_{\mathrm{G}}/2$. This proves the bound for one piece using (1432). The shares are disjoint by proposition 44.9, so their product proves (1430). The displayed determinant has modulus $L^{-3\#\Gamma'} z_{t,D} \leq z_{t,D}$, proving its version.

For the integral, $\mathsf{Y}(t,D) \subset D$ gives $n^{\mathrm{g}}_{t,D} \leq L^3 N_D$. Every positive stored bond is determined by its initial level-j site in D and one of three directions; hence there are at most $3L^3 N_D$ such bonds. Use (1425), $x_j \varepsilon_j^2 = p_0(g_j)^2 \geq 1$, and the three-dimensional Gaussian integral:

$$\int \bar{\mathsf{u}}_{t,D}\, dm_{t,D} \leq (2L^9)^{6N_D} 500^{L^3 N_D} (4\pi/\gamma_{\mathrm{kp}})^{(9/2)L^3 N_D}.$$

This is (1431). Borel measurability follows from that of the gauge factor and continuity of the Gaussian. □

66. The mass of a history

We now apply the component majorant to the history measure. There are two identifications to make. The retained elimination factors must be the pivot determinants of a glued configuration, and the retained quadratic forms must act on exactly the stored coordinates of the components. Once these are established, the positive density can be bounded on a Borel set of full measure.

Fix a history h of depth $1 \leq k \leq K$. Throughout this section assume the running bounds, support and strip hypotheses of section 53.9, lemma 65.2, and proposition 65.4. The collared background is the critical orbit of proposition 53.11. For $U \in \mathsf{Sup}(h)$, write $\mathsf{U} = U' \vee U$ for a glued representative, with $U' \in U_k^{\diamond}(\xi_h^{\diamond}(U))$. The free coordinate map is denoted by v_h; thus $\xi_h = \mathrm{cp}_h \mathsf{v}_h$. The preceding support results give its Borel definedness on $\mathsf{Sup}(h)$.

66.1. **The elimination factors at the background.** Let Γ_t^{∂} consist of the positive level-t bonds with an endpoint in Ω_t and no endpoint in Λ_t. Both endpoints belong to one component D of Z_t; denote the corresponding subset by $\Gamma^{\partial}(t,D)$. Assume $\alpha_0, \alpha_1 \leq 1$ and $m' < m + K - k$, so that $m' < m + K - t$ for $1 \leq t \leq k$. Also assume, for these t,

$$L^2 C_{\mathrm{reg}} \varepsilon_t \leq c_2^{\mathrm{SO}}, \qquad 2C_{\mathrm{reg}} \varepsilon_t < \min\{\varepsilon_{\mathsf{z}}, \varpi_\chi\}. \tag{1433}$$

Here ϖ_χ is the reference radius of proposition 53.2; the strict ceiling keeps its logarithm on the specified branch.

Lemma 66.1. For $c \in \Gamma_t^{\partial}$, the averages needed to evaluate z_c on the restriction to $X(c)$ are defined on the analytic branch, agree with the full block field, and

$$\varpi_c(\mathcal{M}^{t-1}\mathsf{U}) \leq 2C_{\mathrm{reg}} \varepsilon_t. \tag{1434}$$

Moreover

$$e^{z_c(\mathsf{U})} = \det\bigl(L\mathsf{m}_c(\mathcal{M}^{t-1}\mathsf{U})\bigr)^{-1} > 0. \tag{1435}$$

The determinant is independent of the glued representative. ■

Proof. Put $Y_n = B^{t-n}(c_-) \cup B^{t-n}(c_+)$, $0 \le n \le t$. Then $B(Y_{n+1}) = Y_n$. In level-$(t-1)$ units each point of Y_{t-1} is within $2L$ of Ω_t. The separation of the regions puts these points in Λ_{t-1} when $t \ge 2$, and in $\Omega_0^\diamond$ when $t = 1$. The blocks below them remain in that region. Thus $Y_0 \subset \Omega_0^\diamond$ and all its inner bare bonds belong to $X(c)$. The endpoint conditions also give $c \in \Gamma_t^\diamond$. By proposition 53.11, every bare plaquette with corners in Y_0 has curvature at most αL^{-2t}, where $\alpha = L^2 C_{\mathrm{reg}} \varepsilon_t \le c_2^{\mathrm{SO}}$.

For completeness, the averaging estimate can be iterated on these blocks alone. Let ω_n be the maximum plaquette deviation with corners in Y_n, and set

$$P_n = \prod_{l=0}^{n-1} (1 + \tfrac{1}{2} L^{2(l-t)}).$$

The one-step proof of lemma 14.9 uses only the blocks below the four corners of the coarse plaquette. Their union is contained in Y_n. Hence induction gives

$$\omega_n \le \alpha L^{2(n-t)} P_n, \qquad P_n \le \exp\!\left(\frac{1}{2(L^2-1)}\right) < 2.$$

Indeed, for $n < t$, the right side is at most $2\alpha L^{-2}$, below the one-step curvature radius, and

$$\omega_{n+1} \le L^2 \omega_n + C_0^{\mathrm{SO}} \omega_n^2 \le L^2 \omega_n (1 + \tfrac{1}{2} L^{2(n-t)}).$$

Here $C_0^{\mathrm{SO}} \alpha \le L^2/4$. All double blocks required at the next step lie in Y_n; their curvature is below ω_M, so the real and analytic definitions of the average agree. This also proves by induction that restriction to the inner bonds of $X(c) \cap \Omega_0^\diamond$ changes no required value. At $n = t-1$ the bound is (1434). The real logarithm and positivity in proposition 53.2 now give (1435).

If $U_2' = (U_1')^u$ are two elements of the critical orbit, extend u by one outside the collar. The glued fields are gauge related on its inner bonds, which contain every bond needed by the preceding averages. Locality and covariance therefore relate their block fields on the double block of c. Differentiating covariance gives

$$\mathsf{m}_c(V^u) = \mathrm{Ad}(u(c_-)) \mathsf{m}_c(V) \,\mathrm{Ad}(u(b_0(c)_-))^{-1}.$$

The two adjoint maps have determinant one on $\mathfrak{g}$. This proves independence of the determinant; invertibility also follows from (1426). □

66.2. **The stored forms.** Assume the hypotheses and the fixed-radius ceiling of corollary 54.7 for depth k and each step $t \le k$, and those of proposition 55.4. The radius for the local gauge neighborhoods can be taken to be $r_{\mathsf{N}} = 4R_1\mathfrak{M}$. It satisfies

$$\frac{L(L+R_1+1)}{L-1}\mathfrak{M} < r_{\mathsf{N}}, \qquad 2r_{\mathsf{N}} + 1 < LM,$$

since $L \ge 13$, $R_1 \ge L$, and $M \ge L^4 M_1 \ge L^4 R_1 \mathfrak{M}$. The actual pair read from U' at step $t-1$ therefore lies in $\mathscr{O}^+(\mathfrak{k}^\sim)$ at radii $(\alpha_0/2, \alpha_1, \alpha_2/2)$, for every strip component $\mathfrak{k}$ of that step. Set $\gamma_{\mathrm{kp}} = \gamma_{\mathrm{pos}}/4$.

Each strip component belongs to a unique component D of Z_t, and its positive external bond set $E_{\mathfrak{k}}^+$ is contained in $\mathfrak{X}^{\mathrm{st}+}(t, D)$. These sets partition $\mathfrak{X}^{\mathrm{st}+}(t, D)$. In fact an external bond has no endpoint in Λ_t, so both endpoints belong to a single component of Z_t. It also has an endpoint in its strip component. The component of Z_t containing this bond must therefore be the one containing that strip. The converse follows by the definition of the external bonds.

Consequently the canonical form on a piece is

$$Q_{t,D}(a; U') = \sum_{\mathfrak{k} \subset D} \langle a, \Delta_{\mathfrak{k}}^{\mathrm{loc}}(p_{\mathfrak{k}}(U')) a \rangle. \tag{1436}$$

Extend a by zero off the stored bonds of the piece. The matrices are real symmetric, and proposition 55.4 gives

$$Q_{t,D}(a;U') \geq 2\gamma_{\mathrm{kp}} \sum_{b\in\mathfrak{X}^{\mathrm{st}+}(t,D)} |a(b)|^2. \tag{1437}$$

The depth-k pair is used here at every earlier step; the coefficients and keeping predicates at step t are those of the truncated history $h|_t$.

Suppose now that the formed factors are precisely those of section 62.3, retained from steps $1,\dots,k$. Their total exponent $\Phi_h(\phi)$ is defined when its evaluation on every admitted glued representative is defined and has the same value. The grouping by (t,X) in (1390) then gives, at $\phi=\xi_h(U)$,

$$\Phi_h(\phi) = \sum_{t=1}^{k} \sum_{c\in\Gamma_t^{\partial}} z_c(U') - \frac{1}{2}\sum_{(t,D)} Q_{t,D}(\mathsf{a}'_{t,D}(U);U').$$

All sums here are finite except for the normally convergent coefficient series already used to define the kept forms. Thus their regrouping preserves their values. By lemma 66.1,

$$e^{\Phi_h(\xi_h(U))} = \prod_{t=1}^{k}\prod_{c\in\Gamma_t^{\partial}} \det\bigl(L\mathsf{m}_c(\mathcal{M}^{t-1}\mathsf{U})\bigr)^{-1} e^{-s(U)/2}, \tag{1438}$$

where

$$s(U) = \sum_{(t,D)} Q_{t,D}(\mathsf{a}'_{t,D}(U);U'), \qquad n(U) = \sum_{(t,D)}\sum_{b\in\mathfrak{X}^{\mathrm{st}+}(t,D)} |\mathsf{a}'_{t-1}(b)(U)|^2.$$

The determinant product is positive and independent of U. Since $s(U)$ is real, injectivity of the real exponential in (1438) shows that it too is independent of the chosen admitted representative. It satisfies $s(U)\geq 2\gamma_{\mathrm{kp}} n(U)$, and $n(U)=0$ implies $s(U)=0$.

The following form of this observation avoids choosing a background as a function of U. Where Φ_h is defined and $n(U)>0$, set $\lambda(U)=s(U)/n(U)$; otherwise set $\lambda(U)=2\gamma_{\mathrm{kp}}$. The real quadratic forms

$$Q^{\mathrm{sc}}_{t,D}[U](a) = \lambda(U) \sum_{b\in\mathfrak{X}^{\mathrm{st}+}(t,D)} |a(b)|^2 \tag{1439}$$

have the lower bound (1437). At the actual stored fields their sum equals $s(U)$, including the case $n(U)=0$. Hence (1438) holds with these forms and with every glued representative in its determinant product.

66.3. **The density on the free coordinates.** Let

$$\mathcal{V}_0[h] = \{U : \mathfrak{w}_h(U) > 0\}, \qquad \check{\tau}_h = (\mathsf{v}_h)_*(\mathfrak{w}_h\rho_0\, dU).$$

These are a Borel set and a finite positive Borel measure, respectively. The history mass is $\mathsf{t}(h)=\int \mathfrak{w}_h\rho_0\, dU$. All of $\mathcal{V}_0[h]$ lies in the domain of v_h, so this is also the total mass of $\check{\tau}_h$.

Assume the history has an integrable density in the completed coordinates of the form

$$\mathsf{f}_h(\phi) = \mathrm{Stay}_h^{\mathrm{T}}(\phi)\chi_k(\Omega_k) e^{A_h(\phi)+\Phi_h(\phi)} \tag{1440}$$

outside a Borel $\mathscr{M}_h$-null set. The exponents are partial Borel functions, and where this expression is nonzero their admitted evaluation set is nonempty. Assume that for every admitted glued evaluation,

$$\mathrm{Re}\, A_h(\phi) = -A(\beta_k(\cdot),\mathsf{U}) + \mathrm{Re}\{\mathbf{E}_k(\mathsf{U}) + \mathbf{E}_k^{\mathrm{bd}}(\mathsf{U},\mathfrak{q})\} - \mathcal{E}_{k,\mathcal{S}(h)}, \tag{1441}$$

with real $\mathcal{E}_{k,\mathcal{S}(h)}$. These are the exponents obtained by collecting the terms in (1368). Finally assume the core membership, layer bounds and family estimates of proposition 64.2.

Proposition 66.2. Under these hypotheses, $\check{\tau}_h$ has a nonnegative Borel density f_h with respect to $\mathscr{M}_h^{\mathrm{fr}}$. There is a Borel set $S' \subset \mathsf{v}_h(\mathcal{V}_0[h])$ of full $\check{\tau}_h$-measure such that, for every $v \in S'$, one can choose $U \in \mathcal{V}_0[h]$, $\mathsf{v}_h(U) = v$, and a glued representative U for which

$$f_h(v) \le e^{-\mathcal{E}_{k,\mathcal{S}(h)} + C_5 \sum_{n=0}^{k} \#_n(\mathcal{L}_n) - A(\beta_k(\cdot),\mathsf{U})} \times \prod_{(t,D)} \left[\mathsf{g}_{\Upsilon(t,D)}(V_{t-1}) z_{t,D}^{\mathrm{dsp}}(\mathcal{M}^{t-1}\mathsf{U}) e^{-Q_{t,D}^{\mathrm{sc}}[U](\mathsf{a}'_{t,D}(U))/2} \right], \tag{1442}$$

where $z_{t,D}^{\mathrm{dsp}}$ is the positive determinant product over $\Gamma^{\partial}(t,D)$ in (1438). ∎

Proof. If N is the exceptional set in the completed coordinates, $N' = \mathrm{cp}_h^{-1}(N)$ is $\mathscr{M}_h^{\mathrm{fr}}$-null by (984). Pullback gives the integrable density $d = \mathsf{f}_h \circ \mathrm{cp}_h$ off N'. Indeed completion sends $\check{\tau}_h$ to the completed history measure, and $\mathrm{rs}_h \mathrm{cp}_h = \mathrm{id}$; apply these identities to the indicator of a Borel set in the free coordinates. Since $\check{\tau}_h$ is positive, integrating over the sets where $\operatorname{Im} d$ has either sign, and then where $\operatorname{Re} d < 0$, shows that d is nonnegative almost everywhere. Its modulus, extended by zero on N', is therefore a nonnegative Borel density f_h.

The image of the Borel set $\mathcal{V}_0[h]$ under the Borel map v_h is analytic. The analytic-image argument used in theorem AB supplies a Borel subset S_0 of this image of full $\check{\tau}_h$-measure. The density vanishes outside the external tiers, because $\mathrm{Stay}_h^{\mathrm{T}}$ does. Intersect S_0 with the tiers and remove N' to obtain S'; points with zero density may also be removed. At every remaining v choose a bare preimage U. Put $\phi = \mathrm{cp}_h(v) = \xi_h(U)$.

At this point $\chi_k(\Omega_k) = 1$. Each factor in that test is a passed P_k-test of the selected history. Furthermore,

$$\mathrm{Stay}_h(v) = \prod_{(t,D)} \mathsf{g}_{\Upsilon(t,D)}(V_{t-1}) > 0. \tag{1443}$$

To check the identity, regard stored factors as entries in their defining lists, so that multiplicities are preserved. Every stored indicator, including a carried density test, equals one at U. The written tests have the same values: the test transport preserves their required averages and logarithms, while the contour tests use only internal bonds of $B(y)$, which remain free. The only remaining factors are $\mathfrak{G}_{t-1}^{\mathrm{e}}(V_{t-1}; y, x)$, with $y \in \Upsilon(t,D)$ and $x \in B(y) \setminus \{y\}$. Each such pair occurs once. Their product is precisely the right side of (1443); their contour variables are defined since $\mathfrak{w}_h(U) > 0$.

The tiers now give $\mathrm{Stay}_h^{\mathrm{T}}(\phi) = \mathrm{Stay}_h(v)$. Thus the nonzero density formula has defined exponents and an admitted glued evaluation. Choose one. Equations (1438) and (1439) identify its formed factors. Take the modulus in (1440), use (1441), and apply (1420) to the regular and boundary sums. This proves (1442). All choices were made at a single point; the final bound below will be a function of v alone. □

66.4. **Integration over the components.** For a piece (t,D), let $\mathrm{pr}_{t,D}$ select $V_{t-1}|_{\mathfrak{B}(t,D)}$ and the stored variables on $\mathfrak{X}^{\mathrm{st}+}(t,D)$ from the free coordinates. It is a continuous coordinate projection. The internal bonds of a block $B(y)$, $y \in \Upsilon(t,D)$, are free: the cell containing y lies outside Ω_t; for $t \ge 2$ it avoids the enlarged old large-field region, so $B(y) \subset \Lambda_{t-1}$. For $t = 1$ use $\Omega_0 = T_\ell$. Thus both endpoints belong to Ω_{t-1} and neither to Ω_t, which is the required free bond condition.

Different pieces use disjoint coordinate factors. Different steps carry different level indices. At one step the blocks $B(y)$ assigned to distinct components cannot share a fine lattice site, because its containing coarse block is unique. Every stored bond belongs to a unique component of Z_t, as shown above. In particular every Lebesgue coordinate occurs in exactly one piece; all remaining coordinates carry Haar probability measure. Regrouping the finite product (984) and using Tonelli therefore gives

$$\int \prod_{(t,D)} \bar{\mathsf{u}}_{t,D}(\mathrm{pr}_{t,D}\, v)\, d\mathscr{M}_h^{\mathrm{fr}}(v) = \prod_{(t,D)} \int \bar{\mathsf{u}}_{t,D}\, dm_{t,D} \le e^{C'_\star \sum_{t=1}^{k} \#_t(Z_t)}. \tag{1444}$$

The product measures are σ-finite, and the integrands are nonnegative Borel functions. The equality also holds for no pieces, when only a Haar probability space remains.

Proposition 66.3. Under the hypotheses of this section, set

$$C_{\mathrm{maj}} = \max\{C_5, L^3 C_5 + C'_\star\}.$$

Then

$$e^{\mathcal{E}_{k,\mathcal{S}(h)}} \mathfrak{t}(h) \le \exp\left[C_{\mathrm{maj}} \left(N_k + \sum_{t=1}^{k} \#_t(Z_t(h))\right)\right] \exp\left[-c_\star \sum_{t=1}^{k} p_0(g_t)^2 n_t^{\mathrm{cr}}(h)\right]. \tag{1445}$$

The constants are independent of the volume, the depth, the running couplings and the history. ■

Proof. Apply proposition 65.4 at a point $v \in S'$ with $\Gamma'(t,D) = \Gamma^\partial(t,D)$ and the forms (1439). The background and strip hypotheses hold on the support. Every elimination bond has its endpoints in D, and (1434) verifies its curvature bound. The stored fields are defined at the chosen bare preimage, and (1437) is the required lower bound for the forms. The action reserved for all created cells is at most $A(\beta_k(\cdot), \mathsf{U})$ by proposition 44.9. The pieces partition the created cells. Hence (1442) becomes

$$f_h(v) \le e^{-\mathcal{E}_{k,\mathcal{S}(h)} + C_5 \sum_n \#_n(\mathcal{L}_n) - c_\star \sum_t p_0(g_t)^2 n_t^{\mathrm{cr}}(h)} \prod_{(t,D)} \overline{\mathsf{u}}_{t,D}(\mathrm{pr}_{t,D}\, v).$$

The right side depends only on v and the fixed history. Thus no measurable selection of its auxiliary backgrounds or forms is needed. Integrate on S', which has full history measure, and enlarge the nonnegative integral to the whole free space. Equation (1444) and the count $\sum_n \#_n(\mathcal{L}_n) \le N_k + L^3 \sum_t \#_t(Z_t)$ from (1420) prove (1445). □

67. Summing the large-field histories

A failed test creates a cell whose suppression must cover the volume it produces at every later scale. We first locate these later cells near their creations. We then count the choices of tests inside the created cells and sum over arbitrary subsets of the cell partitions.

Throughout this section $L \ge 13$, $r \ge 2$, and $\mathfrak{p}_0$ is a positive integer. The geometric constants and partitions are those of (635); in particular $M_2 = L^2 M_1 \ge L^4$, $M \ge L^6$, and $M/(LM_2)$ is a power of L at least L. Suppose

$$x_{j-1} = L(x_j + b_j), \qquad |b_j| \le B_\sharp, \qquad y_j = \log x_j, \qquad p_0(g_j) = A_0 y_j^{\mathfrak{p}_0}.$$

Write $N_j = \#\mathbb{T}^{(j)}$ and $s_j = M \check{R}_j a_{K-j}$ for the physical side of a cell of $\mathcal{P}_j$. Define

$$Y_1 = \log \max\{e, 2B_\sharp\}, \qquad Y_0 = \max\left\{Y_1, \frac{2\log(2L)}{L^{1/r} - 1}\right\}.$$

Lemma 67.1. If $y_K \ge Y_1$, then x_j and y_j decrease strictly with j, $\check{R}_j = R(x_j)$, and

$$1 \le y_j, \qquad y_j^r \le \check{R}_j < L y_j^r, \qquad \log(L/2) \le y_{j-1} - y_j \le \log(2L), \qquad K - j + 1 \le y_j.$$

If $y_K \ge Y_0$, then

$$\check{R}_j \le \check{R}_{j'} \le L \check{R}_j \qquad (1 \le j' \le j,\ j - j' \le 2),$$
$$s_{j'} \le L^{-\lfloor (j-j')/2 \rfloor} s_j \quad (1 \le j' \le j).$$

■

Proof. Put $\bar{B} = LB_\sharp/(L-1)$. Since $x_K \ge 2B_\sharp > \bar{B}$ when $B_\sharp > 0$, and $x_K > 0 = \bar{B}$ otherwise, the inequality

$$x_{j-1} - x_j \ge (L-1)(x_j - \bar{B})$$

gives strict decrease by descending induction. The minimum defining $\check{x}_j$ is therefore x_j. Minimality of the power of L defining $R(x_j)$ gives $y_j^r \le R(x_j) < L y_j^r$, since $y_j \ge 1$. Also $|b_j|/x_j \le 1/2$, so taking

logarithms in $x_{j-1}/x_j = L(1 + b_j/x_j)$ gives the two increment bounds. Summing the lower bound, and using $\log(L/2) > 1$, gives $K - j + 1 \le y_j$.

For $j - j' \le 2$, the upper increment bound gives

$$y_{j'} \le y_j + 2\log(2L) \le L^{1/r} y_j.$$

Thus $y_{j'}^r \le L\check{R}_j$, and the minimality of $R(x_{j'})$ gives the claimed comparison. Since $s_{j'}/s_j = L^{j'-j}\check{R}_{j'}/\check{R}_j$, one backward step costs at most one and two backward steps cost at most L^{-1}. Iteration proves the last inequality. □

67.1. **Later volume assigned to created cells.** For a history h, put

$$V_t = R'_t \cup \mathsf{K}(I_t) \cup \mathsf{R}^{\mathrm{lo}}_t, \qquad W_t = [\mathcal{W}^{\wedge}_t(h)]_t.$$

Here $[A]_t$ denotes the cells of a level-t region A. Thus W_t is the union of the cell sets of P'_t, Q'_t, V_t, and also of P'_0 when $t = 1$. Write $n^{\mathrm{cr}}_t(h) = \#W_t$. For a final component X of $Z_k(h)$, let $n^{\mathrm{cr}}_t(h; X)$ count those cells of W_t contained in X.

The definitions of Ω_t and (644) give

$$Z_1 = P_0'^{\sim 7} \cup P_1'^{\sim 6} \cup Q_1'^{\sim 4} \cup V_1^{\sim},$$
$$Z_t = \widehat{Z}_{t-1}^{\sim 10} \cup P_t'^{\sim 6} \cup Q_t'^{\sim 4} \cup V_t^{\sim} \quad (t \ge 2).$$

In particular $Z_t \subset Z_{t+1}$, and a component at one level lies in a unique component at the next.

Proposition 67.2. Suppose $y_K \ge Y_0$. For every component X of $Z_k(h)$ and every choice $r_n \ge 0$,

$$\sum_{n=1}^{k} r_n |Z_n(h) \cap X|_n \le 48^3 \sum_{t=1}^{k} n^{\mathrm{cr}}_t(h; X) \sum_{n=t}^{k} r_n. \tag{1446}$$

Every such component contains a created cell. The same inequality holds for the whole regions $Z_n(h)$, with $n^{\mathrm{cr}}_t(h)$ on the right. ■

Proof. A chain through a layers of cells of side s joins any point of $A^{\sim a}$ to a point of A at distance at most as; the chain stays in $A^{\sim a}$. Apply this observation to the displayed formulas for Z_n. A point newly covered at level n lies within $7s_n$ of a created cell in its component. A point covered from $\widehat{Z}_{n-1}$ lies within $10s_n$ of that hull, and within a further s_n of Z_{n-1}. Induction therefore places every $x \in Z_n$ within

$$\rho_{t,n} = 7s_t + 11 \sum_{j=t+1}^{n} s_j$$

of a created cell of some level $t \le n$ in the same component. The chains and the nesting of components preserve this last assertion at each step. It also proves that every nonempty final component contains a creation.

Assign each cell of $Z_n \cap X$, by its centre, to one such creation. The centre of a cell assigned to a level-t cell lies within $\rho_{t,n} + s_t/2$ of that cell's centre. An interval of length $2a$ contains at most $\lfloor 2a/s_n \rfloor + 1$ points of a lattice of spacing s_n. Applying this in the three coordinates, with lifts on the torus, bounds the number assigned to a fixed creation by

$$\left(\left\lfloor \frac{2\rho_{t,n} + s_t}{s_n} \right\rfloor + 1 \right)^3.$$

Now $2\rho_{t,n} + s_t \le 22 \sum_{j=t}^{n} s_j$, and lemma 67.1 gives

$$\sum_{j=t}^{n} s_j \le s_n \sum_{u \ge 0} L^{-\lfloor u/2 \rfloor} = \frac{2L}{L-1} s_n.$$

Since $44L/(L-1) < 48$, the displayed count is at most 48^3. Consequently

$$|Z_n \cap X|_n \le 48^3 \sum_{t \le n} n_t^{\mathrm{cr}}(h; X).$$

Multiply by r_n and interchange the finite sums to obtain (1446). Distinct final components partition both the earlier regions and the creations, so summing over them gives the whole-region inequality. □

67.2. **Counting the outcomes in fixed created cells.** Set

$$N_\tau = \left(\frac{M}{LM_2}\right)^3, \qquad N_\delta = \left(\frac{M}{M_2}\right)^3, \qquad n_t^{\mathrm{bit}} = 3N_\tau + N_\delta + 1 + 6L^3(M\check{R}_t)^3, \qquad \lambda_L = (6L^3+1)\log 2.$$

Lemma 67.3. For arbitrary sets of cells $W_t \subset \mathcal{P}_t$,

$$\#\{h : [\mathcal{W}_t^{\wedge}(h)]_t \subset W_t \text{ for all } t\} \le \prod_{t=1}^{k} 2^{n_t^{\mathrm{bit}} \# W_t}, \qquad n_t^{\mathrm{bit}} \le (6L^3+1)(M\check{R}_t)^3. \tag{1447}$$

■

Proof. Each of P_t, Q_t, R_t is a union of test cubes. Its hull lies in W_t, and each cell contains exactly N_τ test cubes. These three choices therefore use at most $3N_\tau \# W_t$ binary choices. The set $\mathsf{R}_t^{\mathrm{lo}}$ is a subset of W_t, using $\# W_t$ more.

For the selected collar bonds, we must locate the initial site in a cell of W_t. At the stage at which a bond $b \in I_t$ is selected, it belongs to the collar of a region $\mathcal{Y}$ containing no cell of $(\Omega_t^c)^{\sim 2}$. In level-$(t-1)$ units every endpoint of b lies within $\varpi_T + 3 \le M + 3$ of a site in $\mathcal{Y}$. The cells have side $LM\check{R}_t \ge 13M$. Their centres lie on the cell lattice, so the cell containing that endpoint touches a cell of $\mathcal{Y}$. It belongs to $\Omega_t^{\sim -1}$: otherwise the touched cell would lie in $(\Omega_t^c)^{\sim 2}$. Thus the cell of b_- belongs to $\mathsf{K}(I_t) \subset W_t$. There are $(LM\check{R}_t)^3$ sites per cell and at most six directed bonds per initial site. Hence I_t uses at most $6L^3(M\check{R}_t)^3 \# W_t$ binary choices.

At the first step, the density cubes defining P_0 need not refine $\mathcal{P}_1$. If they do, each cell contains at most

$$\left(\frac{M\check{R}_1}{M_2\check{R}_0}\right)^3 \le N_\delta$$

of them. If the cell partition is finer, every chosen density cube contains a cell of W_1, and distinct density cubes contain disjoint sets of cells. There are then at most $\# W_1 \le N_\delta \# W_1$ eligible density cubes. Both cases cost at most $N_\delta \# W_1$ choices.

These sets determine the history, including its recursively explored sets. Allowing all the subsets just counted can only increase their number, which proves the first bound. Finally

$$3N_\tau + N_\delta + 1 \le M^3(3L^{-15} + L^{-12} + L^{-18}) < M^3 \le (M\check{R}_t)^3$$

proves the second. □

67.3. **Absorbing volume and entropy.** We use two weights:

$$\omega_j = y_j \quad \text{in case II}, \qquad \omega_j = \check{R}_j \quad \text{in case I}.$$

Case II requires $2\mathfrak{p}_0 > 3r + 2$; case I requires $2\mathfrak{p}_0 > 4r + 1$. For $a, C > 0$, put

$$\begin{aligned} Y^{\mathrm{II}}(a,C) &= \max\left\{Y_0, \left(\frac{4}{aA_0}\right)^{1/\mathfrak{p}_0}, \left(\frac{2CL^3M^3}{aA_0^2}\right)^{1/(2\mathfrak{p}_0 - 3r - 2)}\right\}, \\ Y^{\mathrm{I}}(a,C) &= \max\left\{Y_0, \left(\frac{4}{aA_0}\right)^{1/\mathfrak{p}_0}, \left(\frac{2CL^4M^3}{aA_0^2}\right)^{1/(2\mathfrak{p}_0 - 4r - 1)}\right\}. \end{aligned} \tag{1448}$$

Lemma 67.4. In either case $\varrho \in \{\mathrm{I}, \mathrm{II}\}$, if $y_K \geq Y^{\varrho}(a, C)$, then

$$ap_0(g_t)^2 \geq 2p_0(g_t) + C\sum_{n=t}^{K} \omega_n (M\check{R}_n)^3 \quad (0 \leq t \leq K).$$

■

Proof. Write $y = y_t$. The second entry of the maximum ensures $ap_0(g_t)^2/2 \geq 2p_0(g_t)$. By lemma 67.1, the sum has at most y terms, $y_n \leq y$, and $\check{R}_n < Ly^r$. It is at most $L^3M^3y^{3r+2}$ in case II and $L^4M^3y^{4r+1}$ in case I. The last entry of the corresponding maximum bounds C times this expression by the other half of $ap_0(g_t)^2$. □

For $C \geq 0$, $a > 0$, define

$$\Pi_k(C, a) = e^{CN_k} \prod_{t=1}^{k} (1 + e^{-ap_0(g_t)^2})^{\#\mathcal{P}_t}. \tag{1449}$$

Proposition 67.5. Let $c > 0$, $D \geq 0$, $0 < \eth < 1$, and suppose

$$y_K \geq Y^{\varrho}(\eth c, 48^3(D + \lambda_L)).$$

If a nonnegative function F on $\mathcal{H}_k$ satisfies

$$F(h) \leq \exp\left[D\sum_{j=1}^{k} \omega_j \#_j(Z_j(h)) - c\sum_{t=1}^{k} p_0(g_t)^2 n_t^{\mathrm{cr}}(h)\right],$$

then

$$\sum_{h \in \mathcal{H}_k} F(h) \leq \Pi_k(0, (1 - \eth)c). \tag{1450}$$

■

Proof. Apply (1446) with $r_j = (D + \lambda_L)\omega_j (M\check{R}_j)^3$. Since $\#_j(Z_j) = (M\check{R}_j)^3 |Z_j|_j$, lemma 67.4 gives

$$(D + \lambda_L)\sum_j \omega_j \#_j(Z_j) \leq \sum_t [\eth c\, p_0(g_t)^2 - 2p_0(g_t)] n_t^{\mathrm{cr}}(h).$$

Also $\omega_j \geq 1$, $n_j^{\mathrm{cr}} \leq |Z_j|_j$, and lemma 67.3 imply

$$\lambda_L \sum_j \omega_j \#_j(Z_j) \geq \sum_t n_t^{\mathrm{bit}} \log 2\, n_t^{\mathrm{cr}}(h).$$

Subtract this last inequality. With $\zeta_t = e^{-(1-\eth)cp_0(g_t)^2 - 2p_0(g_t)}$, the assumed bound becomes

$$F(h) \leq \prod_t (2^{-n_t^{\mathrm{bit}}} \zeta_t)^{n_t^{\mathrm{cr}}(h)}.$$

Sort histories according to their exact sets W_t of created cells. Equation (1447) bounds the multiplicity by $\prod_t 2^{n_t^{\mathrm{bit}} \#W_t}$, cancelling the binary factors. Summing over all $W_t \subset \mathcal{P}_t$ therefore gives

$$\sum_h F(h) \leq \prod_t \sum_{W_t \subset \mathcal{P}_t} \zeta_t^{\#W_t} = \prod_t (1 + \zeta_t)^{\#\mathcal{P}_t} \leq \Pi_k(0, (1 - \eth)c).$$

□

For later use, set

$$Y^{\mathrm{S}}(a) = \left(\frac{3\log L + \log 2}{2a\mathfrak{p}_0 A_0^2 \log(L/2)}\right)^{1/(2\mathfrak{p}_0 - 1)}.$$

If $y_K \geq \max\{Y_1, Y^{\mathrm{S}}(a)\}$, convexity of $y \mapsto y^{2\mathfrak{p}_0}$ and lemma 67.1 give

$$a[p_0(g_t)^2 - p_0(g_k)^2] \geq 2a\mathfrak{p}_0 A_0^2 y_K^{2\mathfrak{p}_0 - 1}(k - t)\log(L/2) \geq (3\log L + \log 2)(k - t).$$

Consequently

$$\sum_{t=1}^{k} L^{3(k-t)} e^{-a p_0(g_t)^2} \le 2 e^{-a p_0(g_k)^2}.$$

Since $\#\mathcal{P}_t = N_t/(M\check{R}_t)^3 \le M^{-3} L^{3(k-t)} N_k$, the inequality $\log(1+u) \le u$ now yields

$$\Pi_k(C,a) \le \exp[(C + 2M^{-3} e^{-a p_0(g_k)^2}) N_k] \le e^{(C+2)N_k}. \tag{1451}$$

67.4. **Vacuum normalisation and weighted sums.** Let $f : \mathcal{H}_k \to [0,\infty)$ and let e be real on admissible labels. Assume

$$\begin{aligned} f(h) &\le e^{C_{\mathrm{maj}}(N_k + \sum_j \#_j(Z_j(h)))} \prod_t e^{-c p_0(g_t)^2 n_t^{\mathrm{cr}}(h)}, \\ |e(\mathcal{S})| &\le C_\flat \sum_j (1 + y_{j-1}) \#_j(Z_j(\mathcal{S})^\sim). \end{aligned} \tag{1452}$$

Here $c > 0$ and $C_{\mathrm{maj}}, C_\flat \ge 0$. Define

$$D_0 = C_{\mathrm{maj}} + 27(2 + \log(2L)) C_\flat.$$

The constants and free parameters below are fixed independently of the lattice and the coupling sequence.

Proposition 67.6. Assume $2\mathfrak{p}_0 > 3r + 2$ and $0 < \eth < 1$. If

$$y_K \ge \max\{Y^{\mathrm{II}}(\eth c, 48^3(D_0 + \lambda_L)), Y^{\mathrm{S}}((1-\eth)c)\},$$

then

$$\sum_h f(h) e^{e(\mathcal{S}(h))} \le \Pi_k(C_{\mathrm{maj}}, (1-\eth)c) \le e^{(C_{\mathrm{maj}}+2)N_k}. \tag{1453}$$

More generally, for $0 \le \theta < c$, $\varkappa \ge 0$, if

$$y_K \ge Y^{\mathrm{II}}(\eth(c-\theta), 48^3(D_0 + \varkappa + \lambda_L)),$$

then, with $\mathsf{N}(\mathcal{S})$ the number of components of $Z_k(\mathcal{S})$,

$$\sum_h f(h) e^{e(\mathcal{S}(h))} e^{\theta p_0(g_k)^2 \mathsf{N}(\mathcal{S}(h)) + \varkappa \#_k(Z_k(h))} \le \Pi_k(C_{\mathrm{maj}}, (1-\eth)(c-\theta)). \tag{1454}$$

■

Proof. Enlarging a cell region by one layer multiplies its cell count by at most 27. Also

$$1 + y_{j-1} \le (2 + \log(2L)) y_j$$

by lemma 67.1. Thus $F_0(h) = e^{-C_{\mathrm{maj}} N_k} f(h) e^{e(\mathcal{S}(h))}$ satisfies

$$F_0(h) \le \exp\left[D_0 \sum_j y_j \#_j(Z_j) - c \sum_t p_0(g_t)^2 n_t^{\mathrm{cr}} \right].$$

Apply proposition 67.5 in case II, then (1451), to obtain (1453).

Each final component contains a creation, and the creations of distinct components are disjoint. Since $p_0(g_t) \ge p_0(g_k)$ for $t \le k$,

$$p_0(g_k)^2 \mathsf{N}(\mathcal{S}(h)) \le \sum_t p_0(g_t)^2 n_t^{\mathrm{cr}}(h).$$

Also $\#_k(Z_k) \le \sum_j y_j \#_j(Z_j)$. The extra exponent therefore replaces c by $c - \theta$ and D_0 by $D_0 + \varkappa$ in the bound for F_0. A second application of proposition 67.5 gives (1454). □

The same argument controls the additional weights attached to the history of a component. For a final component X, define

$$\mathrm{Dep}_k(X;h) = c_{\mathrm{Dep}} \sum_t p_0(g_t)^2 n_t^{\mathrm{cr}}(h;X),$$
$$\mathrm{Vol}_k(X;h) = C_{\mathrm{vol}} \sum_j \check{R}_j \#_j(Z_j(h) \cap X),$$

and let

$$\Theta^{(k)}(h) = \sum_X [w\,\mathrm{Dep}_k(X;h) + c_a\,\mathrm{Vol}_k(X;h)], \quad 0 \le w < 1, \quad c_a, C_{\mathrm{vol}} \ge 0, \quad 0 < c_{\mathrm{Dep}} \le c.$$

The earlier components partition $Z_j \cap X$; thus this volume is also the sum of the weights $C_{\mathrm{vol}} M^3 \check{R}_j^4 |D|_j$ over the pieces (j, D) contained in X.

Put $c'' = (1-\eth)(1-w)c$. If $c_a = 0$, then

$$\sum_h f(h) e^{e(\mathcal{S}(h)) + \Theta^{(k)}(h)} \le \Pi_k(C_{\mathrm{maj}}, c'') \le e^{(C_{\mathrm{maj}}+2)N_k} \tag{1455}$$

provided

$$y_K \ge \max\{Y^{\mathrm{II}}(\eth(1-w)c, 48^3(D_0 + \lambda_L)), Y^{\mathrm{S}}(c'')\}.$$

If $c_a > 0$, assume $2\mathfrak{p}_0 > 4r+1$. Then

$$\sum_h f(h) e^{e(\mathcal{S}(h)) + \Theta^{(k)}(h)} \le \Pi_k(C_{\mathrm{maj}} + \lambda_L, c'') \le e^{(C_{\mathrm{maj}}+\lambda_L+2)N_k} \tag{1456}$$

provided

$$y_K \ge \max\{Y^{\mathrm{I}}(\eth(1-w)c, 48^3(D_0 + c_a C_{\mathrm{vol}} + \lambda_L)), Y^{\mathrm{S}}(c'')\}.$$

Indeed the component partitions give

$$\Theta^{(k)}(h) \le wc \sum_t p_0(g_t)^2 n_t^{\mathrm{cr}}(h) + c_a C_{\mathrm{vol}} \sum_j \check{R}_j \#_j(Z_j).$$

For $c_a = 0$, apply proposition 67.5 in case II with $(1-w)c, D_0$. For $c_a > 0$, use $y_j \le \check{R}_j$ and apply it in case I with $(1-w)c, D_0 + c_a C_{\mathrm{vol}}$. This gives the bound with $\Pi_k(C_{\mathrm{maj}}, c'')$ also in the second case; increasing its first argument and applying (1451) gives (1456).

68. A lower bound from the empty history

The history without large-field regions contributes a positive measure. Its density gives the lower bound once the regular interaction has been bounded. We first estimate the small-field integral; the passage from a bound on bare history fields to an almost-everywhere bound on the coarse density is treated separately.

Let $1 \le k \le K$, $N_k = \#\mathbb{T}^{(k)}$, and $g_k = x_k^{-1/2}$. Retain the geometric and regularity assumptions of proposition 44.2, and suppose

$$\varepsilon_k \le \varepsilon_\flat, \qquad 2\varepsilon_k \le \bar{\varepsilon}, \qquad p_0(g_k) \ge 4B_3, \qquad g_k \le \bar{\varepsilon}/4. \tag{1457}$$

Write $\mathcal{V}_k^\chi = \{V : \chi_k(\mathbb{T})(V) = 1\}$. On this open set let $U_k(V)$ denote the whole-torus regular critical orbit over V, and put

$$\mathsf{sf}_k(V) = \begin{cases} \exp[-L^k x_k A_K(U_k(V))], & V \in \mathcal{V}_k^\chi, \\ 0, & V \notin \mathcal{V}_k^\chi. \end{cases} \tag{1458}$$

The action is constant on the orbit, so this definition does not require a choice of representative.

Proposition 68.1. The orbit in (1458) exists on $\mathcal{V}_k^\chi$. The function sf_k is Borel, strictly positive there, and bounded by one. For

$$0 < s \le \tfrac{1}{4}\min\{\bar\varepsilon, B_3^{-1}\varepsilon_k\}, \qquad \mathsf{B}_k(s) = \{V : |V(b) - 1| < s \text{ for every positive bond}\},$$

all density tests pass on $\mathsf{B}_k(s)$, and

$$L^k A_K(U_k(V)) < 24 B_3^2 s^2 N_k \quad (V \in \mathsf{B}_k(s)). \tag{1459}$$

Consequently

$$\int \mathsf{sf}_k \, dV \ge \exp\left[-\left(3\log\frac{7}{s^3} + 24B_3^2 s^2 x_k\right) N_k\right]. \tag{1460}$$

In particular $s = g_k$ is allowed under (1457), and the coefficient in parentheses then equals $\frac{9}{2}\log x_k + 3\log 7 + 24B_3^2$. ∎

Proof. Passage of the density test on the cube containing a corner of a coarse plaquette gives $|\partial V - 1| < 2\varepsilon_k$: average the defining background through k levels, as in the support proof for eq. (648). The ceiling $\varepsilon_\flat$ is the curvature-iteration ceiling used there. For a whole-torus datum with coarse curvature less than $\varepsilon_{\mathrm{r}} \le \bar\varepsilon$, use the constant tower $\Omega_j = \mathbb{T}$. Its induced lift has tested holonomies equal to one or to coarse plaquette holonomies, by the lift calculation in the proof of proposition 44.2. The regular-background theorem theorem E therefore gives the orbit, with bare curvature less than $B_3\varepsilon_{\mathrm{r}}L^{-2k}$. Its local continuous representatives also show that its action is continuous. Apply this with $\varepsilon_{\mathrm{r}} = 2\varepsilon_k$. The set $\mathcal{V}_k^\chi$ is open because there are finitely many stably defined strict density tests. Extending the positive continuous function on that set by zero proves the Borel assertion.

If $V \in \mathsf{B}_k(s)$, telescoping the four unitary plaquette factors gives $|\partial V - 1| < 4s$. The same whole-torus argument now gives bare curvature less than $4B_3 s L^{-2k}$. There are $3N_0 = 3L^{3k}N_k$ positive bare plaquettes, and their action contributions are $\frac{1}{2}|\partial U - 1|^2$. Multiplication by L^k gives (1459). For each density cube, apply proposition 44.2 with $\varepsilon_{\mathrm{r}} = 4s$. Its defining curvature is less than $4B_3 s L^{-2k} \le \varepsilon_k L^{-2k}$, so its test passes.

The one-bond Haar ball has measure

$$m(s) = \frac{2}{\pi}\int_0^s w^2\sqrt{1 - w^2/4}\, dw \ge \frac{s^3}{7} \qquad (0 < s \le 1).$$

Indeed the integrand's square-root factor is at least $\sqrt{3}/2$, and $\sqrt{3}/(3\pi) > 1/7$. There are $3N_k$ positive coarse bonds. Integrating over their product ball and using (1459) proves (1460). Finally $g_k p_0(g_k) = \varepsilon_k$ and $p_0(g_k) \ge 4B_3$ put g_k in the stated range. □

68.1. **The density and the bound on its exponent.** Let $h_\emptyset$ be the empty history at level k. Its regions Ω_j, Λ_j, $j \le k$, are the whole torus. Its layers are empty below k, and its last layer is $\mathbb{T}$; its stored external fields and its boundary factors are empty. Thus its reference measure is dV, and its regular exponent is the total of the regular local families, with their vacuum values subtracted and the running-coupling action terms added. Denote this exponent, evaluated on a bare background U', by $\mathbf{E}_k(U')$.

The following statement isolates the exact inputs needed from the history format and the estimate on the regular sum. The vacuum constant $E_{(k)}$ is that of section 63.

Proposition 68.2. Suppose the empty history has a finite positive density $f_k\, dV = \tau_{h_\emptyset}$. Suppose there is a Borel Haar-null set $\mathcal{N}$ such that $f_k = 0$ off $\mathcal{V}_k^\chi$, and, for $V \in \mathcal{V}_k^\chi \setminus \mathcal{N}$, its admissible background set is nonempty and contained in $U_k(V)$. Assume the value $\mathbf{e}_k(V) = \mathbf{E}_k(U')$ is independent of the admissible background and

$$f_k(V) = \mathsf{sf}_k(V)\exp[\mathbf{e}_k(V) + E_{(k)}] \qquad (V \in \mathcal{V}_k^\chi \setminus \mathcal{N}). \tag{1461}$$

Finally suppose that, whenever a bare field has nonzero empty-history weight and U' belongs to its coarse regular critical orbit,

$$|\mathbf{E}_k(U')| \le C_5 N_k. \tag{1462}$$

Then there is a real Borel function rg_k on $\mathcal{V}_k^{\chi}$, bounded by $C_5 N_k$ almost everywhere, such that

$$d\tau_{h_\emptyset} = \mathsf{sf}_k \exp[\mathsf{rg}_k + E_{(k)}]\, dV. \tag{1463}$$

It follows that

$$e^{-E_{(k)}} \rho_k \, dV \;\ge\; e^{-E_{(k)}} d\tau_{h_\emptyset} \;\ge\; e^{-C_5 N_k} \mathsf{sf}_k \, dV. \tag{1464}$$

■

Proof. Take a nonnegative Borel representative of the density. On $\mathcal{V}_k^{\chi} \setminus \mathcal{N}$, (1461) makes it strictly positive. Since both it and sf_k are positive real numbers and $E_{(k)}$ is real, taking absolute values gives

$$\operatorname{Re} \mathbf{e}_k = \log f_k - \log \mathsf{sf}_k - E_{(k)}.$$

Use this Borel expression to define rg_k off $\mathcal{N}$, and set it equal to zero on $\mathcal{N} \cap \mathcal{V}_k^{\chi}$. Null sets do not affect the density, so (1463) holds.

It remains to transfer the support bound to Haar measure. Set

$$B = \{V \in \mathcal{V}_k^{\chi} \setminus \mathcal{N} : |\mathsf{rg}_k(V)| > C_5 N_k\}.$$

For a bare field U with nonzero empty-history weight, if $\mathcal{M}^k U$ lies outside $\mathcal{N}$ then an admissible $U' \in U_k(\mathcal{M}^k U)$ exists. Equation (1462) shows that $\mathcal{M}^k U \notin B$. The positive pushforward formula therefore gives $\tau_{h_\emptyset}(B) = 0$. The density in (1463) is strictly positive on B; hence $dV(B) = 0$. For example, exhaust B by the sets on which this density is at least $1/n$. On $\mathcal{N}$ the exponent was set to zero, proving the almost-everywhere bound on all of $\mathcal{V}_k^{\chi}$. The first inequality in (1464) follows from the positive history decomposition eq. (648); the second follows by integrating $\mathsf{rg}_k \ge -C_5 N_k$. □

Write $\mathsf{t}(h) = \tau_h(\mathcal{V}_k)$. Under the hypotheses just proved,

$$e^{-E_{(k)}} Z_{K,\ell,x_0} = \sum_{h \in \mathcal{H}_k} e^{-E_{(k)}} \mathsf{t}(h) \ge e^{-C_{\mathrm{low}}(x_k) N_k}, \qquad C_{\mathrm{low}}(x) = C_5 + \tfrac{9}{2} \log x + 3 \log 7 + 24 B_3^2. \tag{1465}$$

Indeed retain the single nonnegative term $h_\emptyset$ and use propositions 68.1 and 68.2. If the vacuum constants are supplied through level K, then $E_{(K)} = E_{(k)} + \mathfrak{E}_k^{\ge}$. The upper vacuum estimate in (1411) consequently gives the same lower bound after normalization by $e^{-E_{(K)}}$, with coefficient $C_{\mathrm{low}}(x_k) + C_{\mathfrak{E}}$.

69. The density at every level

We now collect the local terms and apply the measure identity to the objects constructed above. The induction starts with the finite Wilson measure. At each later level, integrability follows from the positive change of variables in theorem AB; it is not an additional condition on the new local expansion.

69.1. **A common choice of constants.** The parameters can be fixed before the number of steps, the couplings and the torus. The order matters: the geometry and the fixed analytic radii are chosen first, the bounds for the regular terms and their quadratic coefficients next, and the upper bound on the coupling last. The constants furnished by an earlier estimate always retain the values given by that estimate.

Proposition 69.1. There are choices of the blocking factor, the geometric parameters, the analytic radii, the decay rates, and numbers $E_0, B_\sharp > 0$ and $0 < \gamma \le 1/2$ for which the numerical hypotheses of the actual small-field step, the strip and boundary constructions, the historical Gaussian and normalizer estimates, and the history-mass estimates hold simultaneously. These choices are independent of the height, the length of the coupling sequence and the torus. All the numerical

hypotheses involving a coupling hold uniformly for $x \ge \gamma^{-2}$; after the other constants are fixed, γ can be decreased.

In particular, the regular progression has the bound E_0, its intrinsic increments have absolute value at most $B_\sharp$, and the three contributions to the new boundary term satisfy, at every step,

$$\mathsf{b}_{\mathrm{cs}}^{\mathrm{th}} \le \tfrac{1}{3}, \qquad \mathsf{b}_{\mathrm{rj}}^{\mathrm{th}} + \mathsf{b}_{\mathrm{sr}}^{\mathrm{th}} \le \tfrac{1}{3}. \tag{1466}$$

The assertion concerns numerical hypotheses. The room required of a finite torus is specified separately below. ∎

Proof. We give the order of the choices and the inequalities which couple different parts of the construction.

First choose integer exponents with

$$r \ge 2, \qquad \mathfrak{p}_0 \ge 6r + 1, \qquad q \ge 41\mathfrak{p}_0 + 1,$$

and the fixed reserve fractions. We may take $0 < \beta_{\mathrm{s}} \le 1/5$ and then choose

$$\frac{1 + \beta_{\mathrm{s}}}{2\sqrt{2}} \le \theta_{\mathrm{oc}} < \sqrt{2/3}.$$

This interval is nonempty. Choose the Gaussian reserve $1/16 \le \vartheta_* < 1$, and fix $\mathsf{C}_{\mathrm{mn}} = 18(1 - \vartheta_*)^{-2}$ and the absolute cluster constant C_{cs} used in (1332). The other reserve fractions are chosen strictly inside their prescribed intervals. Choose the halo width and then an odd $L \ge 13$ above the finitely many combinatorial and analytic floors, including

$$L \ge \frac{2\mathsf{C}_{\mathrm{cs}}(1 + 4\beta_{\mathrm{s}})}{1 - 2\delta_{\mathrm{sp}}}.$$

All constants depending only on L, including the covariance and normalizer decay rates and gaps, are now fixed. With the constants of (1161) and (1137), put

$$N_{\mathrm{gs}} = 64 B_{\mathrm{nm}}^5 \gamma_0^{-2} + c_{\mathrm{r}} + 2, \qquad \bar{c}_P = \mathsf{C}_{\mathrm{mn}}^3 \lambda_{\mathrm{r}}^{-1} N_{\mathrm{gs}}.$$

These numbers depend only on L and the Gaussian reserve.

For the rim estimates, let c_M, δ_M have the values in (1161), and let $B_0, \delta_{\mathrm{cv}}$ be the constants of theorem L. Fix

$$\begin{aligned} C_{\mathsf{T}} &= \max\left\{ c_M c_{\mathrm{lat}}(\delta_M/2),\ B_0^2 c_{\mathrm{lat}}(\delta_{\mathrm{cv}}/2) c_{\mathrm{lat}}(\delta_{\mathrm{cv}}/4) \right\}, \\ \mu_{\mathrm{ch}} &= \tfrac{1}{2}\min\{\delta_M, \delta_{\mathrm{cv}}/2\}, \qquad \epsilon_{\mathrm{ch}} = 96^{-1/2}, \\ C_{\mathrm{pre}} &= c_{\mathrm{coe}} = \lambda_- = C_\flat^{\mathrm{ch}} = 1. \end{aligned}$$

These are positive finite constants fixed before Θ_1, ϖ_T and M. In particular, $C_{\mathsf{T}} \ge c_M c_{\mathrm{lat}}(\delta_M/2)$ and $\mu_{\mathrm{ch}} \le \delta_M/2$.

Next choose the response parameters and the powers $\mathfrak{M}, R_1$ so that the background and local-response constructions apply, with $\rho_{\mathrm{gap}} = L^{-2} R_1 \mathfrak{M}$. Choose their small radii in the order used in those constructions. In particular, choose ε_{v} in the nonempty background window and then $\bar{\varepsilon}, r_{\mathrm{B}}$ small enough that

$$\varepsilon_{\mathrm{v}} + \varepsilon_{\mathrm{v}}^2 \le a_{\mathrm{orb}}, \qquad B_3(\bar{\varepsilon} + C_\partial r_{\mathrm{B}}) \le \varepsilon_{\mathrm{v}}.$$

Here the response radii may impose further positive upper bounds on $\bar{\varepsilon}, r_{\mathrm{B}}$. They involve only parameters already fixed. For bounds depending on the regularity constant, use its γ-independent upper bound $\bar{C}_{\mathrm{reg}}$. The eventual restriction on $\bar{B}\gamma^2$ gives $C_{\mathrm{reg}} \le \bar{C}_{\mathrm{reg}}$, so this choice does not depend on the coupling chosen at the end.

Choose κ above the entropy and cluster floors, including $4\beta_{\mathrm{s}}\kappa \ge 1$ and

$$\tfrac{1}{2}(1 - 2\delta_{\mathrm{sp}})\kappa \ge 2(1 - 2\delta_{\mathrm{sp}}) + c_{\mathrm{E}} + \frac{\mathsf{C}_{\mathrm{cs}}}{L}\left[1 + 2c_{\mathrm{E}} + \mathsf{C}_{\mathrm{cs}}(\bar{c}_P + 2)\right].$$

Set

$$r_{\mathsf{w}} = (1-2\delta_{\mathrm{sp}})(\kappa-2), \qquad \kappa'' = (1+4\beta_{\mathrm{s}})\kappa + 1 + c_{\mathrm{E}}, \qquad \bar{r}_H = \kappa'' + c_{\mathrm{E}} + \mathsf{C}_{\mathrm{cs}}(\bar{c}_P + 2).$$

Choose $\bar{\kappa}$ above the required coarsened rates and require $\bar{\kappa} \geq 1 + \mathsf{C}_{\mathrm{cs}}\bar{r}_H + \log \mathsf{C}_{\mathrm{cs}}$. Then choose M_1, the clearance, the grain and the mesh, followed by the local response radii, the collar factor Θ_1 and the rim width ϖ_T. Include the requirements

$$\Theta_1 > 3C_{\mathsf{T}}/\epsilon_{\mathrm{ch}}, \qquad C_{\mathsf{T}} e^{-\mu_{\mathrm{ch}}\varpi_T} \leq \epsilon_{\mathrm{ch}}/12.$$

Use this same clearance, grain and mesh in the regular and historical normalizers, and use the same r_{N} below. Their coefficient bound and real gap are also identified as in the interior normalizer identity; thus $c_{\mathrm{nf}} = c_{\mathsf{N}}$. The local inverse estimates are applied before these lengths are enlarged; their positive rates are thus fixed. Choose the power $M = L^{m'}$ last among these lengths. All its requirements have the form

$$M \geq C, \qquad cM \geq C + p\log M, \qquad CM^p e^{-cM} \leq \epsilon, \qquad c, \epsilon > 0,$$

with fixed coefficients. They hold for all sufficiently large powers of L. This includes the local-to-global normalizer error, the resolvent error, and the polynomial losses in the historical Gaussian construction. Also require $(LM)^3 \geq \mathsf{C}_{\mathrm{mn}}^{-1}\bar{c}_P$. The normalizer radius and the boundary rate can then be chosen in

$$4\,\mathsf{ms} \leq r_{\mathsf{N}} \leq LM/4 - 1, \qquad L^{-1}C_{\mathrm{lev}}\log L \leq \delta_\star^{\mathrm{sb}} \leq \mu_{\mathfrak{h}} M/4.$$

The lower bounds on M make both intervals nonempty.

For clarity, two of the requirements included here are $\mathsf{ms} \geq L^3$, $L\mathsf{ms} \leq M/L^2$. The small radii are subsequently reduced so that the complex perturbation of the historical Gaussian form is bounded by its available gap. In the notation of its construction this means

$$B_{\mathrm{nm}}\varepsilon_{\mathrm{dc}} \leq \gamma_0/2, \qquad (LM)^3\varepsilon_{\mathrm{dc}} \leq 1.$$

The exponentially small part of $\varepsilon_{\mathrm{dc}}$ is controlled by the choice of M; its remaining part is controlled by the later choices of α_1, α_2. Use the actual mean data $(\theta_{\mathbf{m}}, \varepsilon_{\mathbf{m}}) = (\vartheta_\mu^{\mathrm{re}}, \varepsilon_\mu)$ of proposition 57.5, with $C_e = \varepsilon_a = C_{\mathrm{xt}}$ in (1167). The choice of θ_c below and the rim-width condition give $\vartheta_{\mathrm{col}}, \vartheta_{\mathrm{rim}} \leq \epsilon_{\mathrm{ch}}/12$. The far term is made at most $\epsilon_{\mathrm{ch}}/12$ by the final coupling choice. Thus $\theta_{\mathbf{m}} \leq \theta_{\mathrm{xt}} := \epsilon_{\mathrm{ch}}/3 < 1/16$. Impose also

$$(1+c_{\mathrm{r}})\varepsilon_{\mathbf{m}}^2 \leq (LM)^{-3}, \qquad \varepsilon_{\mathbf{m}} \leq \vartheta_* - \theta_{\mathrm{xt}}.$$

In (1168) the collar and rim terms are multiplied by $\varepsilon_{\mathrm{dc}}$. The remaining far term contains the factor

$$\exp[-\delta_M(LMR(x) - 2)/2].$$

The preceding choices of M and the small radii control the former terms, and the final choice of γ controls the latter. These requirements give $\theta_{\mathbf{m}} + \varepsilon_{\mathbf{m}} \leq \vartheta_*$. Increasing the torus or the number of steps makes no new choice necessary.

Now choose A_1, A_0, C_0, C_1 in this order, retaining $A_0 \geq C_\chi A_1$ with the chart constant in (905), the other threshold comparisons, and $C_1 \geq C_0$ required by the tests and the radii. Choose α_1 sufficiently small and then α_0 sufficiently small, retaining

$$\alpha_0 + \alpha_1 \leq a^*, \qquad \mathsf{c}_{\mathrm{sh}} M\alpha_0 < \alpha_1.$$

The shape constant is determined afterwards:

$$\mathsf{c}_\natural = \max\{1, \alpha_1^2/\alpha_0\}.$$

Consequently $\alpha_1^2 \leq \mathsf{c}_\natural\alpha_0$ imposes no lower bound on α_0. Reduce the remaining radii $\alpha_2, \alpha_3, \alpha_4, a_\flat$ in their order. Their upper bounds are positive. Where such an upper bound contains a difference, the preceding radius was chosen with a strict reserve; for example $2\alpha_1 < a^\flat$ leaves room for $2(\alpha_0 + \alpha_1) \leq a^\flat$.

The rim fraction θ_c has two lower bounds,

$$u_a = \frac{3C_{\mathsf{T}}}{\epsilon_{\mathrm{ch}}\Theta_1}, \qquad u_b = \frac{C_{\mathrm{pre}}(LM)^{3/2}}{\min\{\lambda_-, c_{\mathrm{coe}}\}\epsilon_{\mathrm{ch}}}\, e^{C_\flat^{\mathrm{ch}}\bar{\kappa}-\mu_{\mathrm{ch}}LM/4}.$$

The earlier choices of Θ_1 and M make $u_a, u_b < 1$. Choose $\max\{u_a, u_b\} \le \theta_c < 1$. These are the only lower bounds to be reconciled with its upper bound. The normalizer constants are now evaluated by their defining formulas.

Choose E_0 large enough for the normalizer and regular-step bounds, and then choose

$$E_0 \ge 2E_N, \qquad B_\sharp \ge \mathsf{G}_\sharp := C_{\mathrm{am}} r_\alpha^{-2} E_0, \qquad \bar{B} = \frac{L}{L-1} B_\sharp. \tag{1467}$$

Here r_α is the usable radius of the coefficient theorem. The norm of the earlier array is bounded by $L(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp)/(L-1)$. Thus its bound is fixed before the ceiling on the coupling. The actual-potential constants, including C_{eff} in (1317), are now evaluated with these choices of $E_0, B_\sharp, \bar{B}$. They too are fixed before γ. Choose the history-weight parameters in their permitted ranges; in particular $0 < c_{\mathrm{Dep}} \le c_\star$ and $0 \le \theta < c_\star$. Their mass and entropy thresholds are finite numbers at this stage.

It remains to choose γ. Put $y = \log x$ and $F_p(x) = x^{-1/2}(\log x)^p$. The coupling-dependent upper bounds in the construction are finite sums and products of

$$F_p(x), \qquad (\log x)^{-(q-\mathfrak{p}_0)}, \qquad x^a(\log x)^p e^{-c(\log x)^r}, \qquad e^{-c(\log x)^{2\mathfrak{p}_0}}, \quad c > 0, \tag{1468}$$

with fixed exponents. They tend to zero as $x \to \infty$; the third assertion uses $r \ge 2$. Lower bounds are finite thresholds on y, on positive powers of y, or on $R(x)$. Take γ sufficiently small for these inequalities to hold for every $x \ge \gamma^{-2}$, including

$$\begin{gathered}
\gamma^{-2} \ge 2B_\sharp, \qquad \gamma^{-2} > \bar{B}, \qquad \gamma^{-2} > L(1+B_\sharp), \\
\log\gamma^{-2} > 2\mathfrak{p}_0, \qquad (L^{1/r}-1)\log\gamma^{-2} \ge \log(2L), \\
C_0(\log\gamma^{-2})^{q-\mathfrak{p}_0} \ge 2\bar{C}_{\mathrm{reg}} A_0.
\end{gathered}$$

This also imposes the small chart and logarithm radii, the graded-field and strip inequalities, and the lower bounds on $\mathfrak{b}(x) = A_1(\log x)^{\mathfrak{p}_0}$, including $\theta_c\mathfrak{b}(x) \ge 1$. For example the common local-gauge and chart requirements are

$$\begin{gathered}
\mathsf{c}_{\mathrm{lgs}} M\varepsilon(x) \le \min\{\alpha_0/2, \alpha_1, \alpha_2/2, a_{\mathrm{nf}}/2, a^{\mathrm{pc}}, a^{\mathrm{cv}}, \varrho_{\mathbf{H}}, 1\}, \\
2L^2\bar{C}_{\mathrm{reg}}\varepsilon(x) + C_{\mathrm{seg}}\mathsf{C}_{\mathcal{A}}\delta(x) \le \bar{\varepsilon}, \\
\mathsf{C}_{\mathcal{A}}\delta(x) \le \min\{\varepsilon_P^\diamond, \varepsilon_{\mathfrak{h}}, r_P^\circ/2\}, \\
C_\chi \max\{\mathsf{c}_{\mathfrak{fl}}, \mathsf{C}_{\mathrm{xt}}+1, \mathsf{C}_{\mathcal{A}}\}\delta(x) < r_\chi.
\end{gathered}$$

The graded radii additionally require lower bounds on $(\log x)^{q-\mathfrak{p}_0}$ and upper bounds on $F_q(x)$, including $C_1^2 C_0^{-1} F_q(x) \le \mathsf{c}_\natural$. The strip errors have the form $C(\log x)^{q-\mathfrak{p}_0}e^{-(\log x)^r}$. These are among (1468) and the corresponding positive-power lower bounds. Both fourth-potential ceilings retain their actual values:

$$\frac{(1+\beta_{\mathrm{s}})\delta(x)}{\alpha_0} \le \min\{\check{s}(K_\Lambda), \check{s}((1+\beta_{\mathrm{s}})^4 K_\Lambda)\}.$$

The regular progression uses the first of these; the nonconstant radius profiles use the second. The bound $\Theta_c(x, N_*) \le E_0/2$ in (972) is included.

We spell out the bounds for the history step, since they close the iteration. Its regular and boundary size majorants have the forms

$$S_{\mathrm{rg}}(x) = C_{\mathrm{eff}}\left[\frac{A_1}{\alpha_0}(1+E_0+\alpha_0^2 B_\sharp)F_{\mathfrak{p}_0}(x) + M^3A_1^3F_{3\mathfrak{p}_0}(x) + \frac{A_1}{C_0}F_q(x)F_{\mathfrak{p}_0}(x)\right],$$
$$S_{\mathrm{bd}}(x) = C_{\mathrm{eff}}(\mathsf{B}+\mathsf{B}_{\mathrm{fr}}^\flat)\frac{A_1}{C_0}(\log x)^{-(q-\mathfrak{p}_0)},$$
$$S_0(x) = M^3K_\flat \mathsf{C}_{\mathcal{A}}A_1F_{\mathfrak{p}_0}(x) + M^3(K_\flat + 2K_{\mathrm{A}}(r_P^\circ)^{-2})\mathsf{C}_{\mathcal{A}}^3A_1^3F_{3\mathfrak{p}_0}(x).$$

Here $\mathsf{B}_{\mathrm{fr}}^\flat \le \mathsf{B}_{\mathrm{fr}}$: their formulas differ by the factors $e^{C_3\kappa}$ and $e^{C_3(\kappa+1)}$. The zero-height majorant is S_0. All three tend to zero. For the historical Gaussian use (1137) with

$$\gamma_2 = (LM)^{-3}, \qquad (\mathfrak{b}_-,\mathfrak{b}_+) = (\theta_c\mathfrak{b}_k,\mathfrak{b}_k).$$

Equations (1161) and (1137), together with the defect and mean bounds above, give

$$u_0 \le \mathsf{C}_{\mathrm{mn}}(LM)^{-3}N_{\mathrm{gs}}, \qquad c_P = \mathsf{C}_{\mathrm{mn}}^2\lambda_{\mathrm{r}}^{-1}(LM)^3u_0 \le \bar{c}_P.$$

The lower bound on $(LM)^3$ therefore implies $\mathsf{C}_{\mathrm{mn}}u_0 \le \lambda_{\mathrm{r}}$, as required in (1138). Choose $r_H = \kappa'' + c_{\mathrm{E}} + \mathsf{C}_{\mathrm{cs}}(c_P+2)$. Then $r_H \le \bar{r}_H$ and $r_{\mathrm{pol}} = \kappa'' + c_{\mathrm{E}} + \mathsf{C}_{\mathrm{cs}}$. The preceding floors on $L,\kappa,\bar{\kappa}$ give

$$(\mathsf{C}_{\mathrm{cs}}/L)\bar{r}_H + c_{\mathrm{E}} \le r_{\mathsf{w}}, \qquad e^{-(\bar{\kappa}-1)+\mathsf{C}_{\mathrm{cs}}\bar{r}_H} \le \mathsf{C}_{\mathrm{cs}}^{-1}.$$

Thus the three rate conditions in (1333) hold. For its remaining size condition, fix

$$K_{\mathrm{ap}} = \mathsf{C}_{\mathrm{cs}}^2 e^{\mathsf{C}_{\mathrm{cs}}(\kappa''+\bar{c}_P+\bar{r}_H)}, \qquad \Gamma_P(x) = \mathsf{C}_{\mathrm{mn}}^{-1}(LM)^{-3}\theta_c^2A_1^2(\log x)^{2\mathfrak{p}_0}.$$

Decreasing γ imposes, uniformly for $x \ge \gamma^{-2}$,

$$\begin{aligned} S_{\mathrm{rg}}(x), S_{\mathrm{bd}}(x) &\le (12eL^3K_{\mathrm{ap}})^{-1}, \\ S_0(x) \le (6eL^3K_{\mathrm{ap}})^{-1}, \qquad (LM)^3e^{-\Gamma_P(x)} &\le (6K_{\mathrm{ap}})^{-1}. \end{aligned} \tag{1469}$$

At a step $t = k+1$, the actual cutoff penalty is $\gamma_P = \Gamma_P(x_k)$. Therefore

$$\mathfrak{b}_{\mathrm{cs}}^{\mathrm{th}} \le K_{\mathrm{ap}}\left(e(S_{\mathrm{rg}}(x_k)+S_{\mathrm{bd}}(x_k))L^3 + (LM)^3e^{-\Gamma_P(x_k)}\right) \le \tfrac{1}{3}.$$

For $k=0$ replace the sum of the two sizes by S_0; the same conclusion holds. Apply (1107) and (1108) at the target rate κ'', and put

$$c_{\mathrm{kp}} = c_{\mathrm{kp}}^{\mathrm{th}} = \tfrac{1}{2}C_3(LM)^6e^{C_3(\kappa''+1)}B_{\mathrm{ch}}, \qquad c_{\mathrm{rj}} = c_{\mathrm{rj}}^{\mathrm{th}} = \tfrac{1}{2}C_3B_{\mathrm{ch}}.$$

In (1384) take $N_e = C_{\mathrm{E}}$, $N_b = 6(LM)^3$ and $\rho_J \le 1$. Define

$$c_3^\flat = 4[6(LM)^3(1+C_{\mathrm{E}}) + C_{\mathrm{E}}/2],$$
$$c_4^\flat = C_{\mathrm{xt}}^2[e^2c_{\mathrm{rj}} + 16C_{\mathrm{E}}(c_{\mathrm{kp}} + e^2c_{\mathrm{rj}})].$$

These constants are fixed before γ. Using $\mathfrak{b}_{\mathrm{cs}}^{\mathrm{th}} \le 1/3 \le 1/2$ and $e^{-(\check{R}_t-2)} \le e^2$ in (1384) gives

$$\mathfrak{b}_{\mathrm{rj}}^{\mathrm{th}} + \mathfrak{b}_{\mathrm{sr}}^{\mathrm{th}} \le e^{-\check{R}_t}(c_3^\flat + c_4^\flat\mathfrak{b}_k^2).$$

Writing $y = \log x_t$, the flow bound gives $x_k \le (3L/2)x_t$; after $y \ge \log(2L)$, $\mathfrak{b}_k^2 \le A_1^2(2y)^{2\mathfrak{p}_0}$. Since $\check{R}_t \ge y^r \ge y$ and $(2y)^{2\mathfrak{p}_0}e^{-y/2} \le (8\mathfrak{p}_0/e)^{2\mathfrak{p}_0}$, their sum is at most

$$\gamma\left[c_3^\flat + c_4^\flat A_1^2(8\mathfrak{p}_0/e)^{2\mathfrak{p}_0}\right].$$

Make this at most $1/3$. This proves (1466); the total is at most $2/3$, leaving the required boundary reserve.

Finally impose the finitely many thresholds from the history-mass and history-counting estimates. They are numbers already determined by the preceding constants. In particular the threshold for the kept form is absorbed in $\log\gamma^{-2}$. All remaining strict inequalities are imposed with a positive

reserve. No choice has used the height, a coupling sequence, or a torus. Since every condition on γ was imposed on the entire interval $x \ge \gamma^{-2}$, decreasing γ preserves them. □

Finite tori and the intrinsic coupling. Let $x_0, \dots, x_k$ be an admissible sequence, prolonged to $u_0, \dots, u_K$, where $K \ge \max\{k, 1\}$. The room required for the history step is precisely

$$MR(u_K) \le L^m. \tag{1470}$$

The adjacent-scale inequality $\check{R}_j \le L^{K-j}\check{R}_K$ then gives $M\check{R}_j \le L^{m+K-j}$ at every level. Our thresholds also give $\check{R}_j \ge \max\{L^2, \mathfrak{o}L + 1\}$ and $M > 6$, hence

$$L^2 M \le L^{m+K-j}, \qquad \mathfrak{o}LM + 6 < L^{m+K-j}, \qquad m' < m + K - j.$$

Thus the mesh, grain, containing boxes and response charts fit on every admissible torus. No further room condition enters the numerical iteration.

The larger local-chart assertion requires

$$LMR(u_K) \le L^m. \tag{1471}$$

It then holds at every earlier level as well. It is not a consequence of (1470): equality in that condition is possible and violates (1471). Let $C_{\mathrm{s}} \ge 2$ be a fixed geometric constant with $2LC_{\mathrm{s}} \ge C_L$, where C_L is the room constant of section 52.7. It is chosen before the couplings and the torus. A sufficient choice is

$$m \ge 1 + m' + \max\{\log_L R(u_K + \bar{B}), \lceil \log_L C_{\mathrm{s}} \rceil\}.$$

This floor depends on the terminal coupling, but not on K.

With the step constructed, (1466) and the uniform numerical bounds give the same induction hypotheses at each new level. This is the constants-only implication in corollary 69.4. For the intrinsic flow, (1467) and theorem Y give, in order, a regular next family, its lattice-independent real coefficient b_{k+1}, and

$$x_{k+1} = L^{-1}x_k - b_{k+1}, \qquad |b_{k+1}| \le \mathsf{G}_\sharp \le B_\sharp.$$

The flow is used while $x_k \ge \gamma^{-2}$. The bound $\gamma^{-2} > L(1 + B_\sharp)$ ensures that the first term below this threshold is still greater than 1. Moreover $|x_j - L^{-j}x_0| \le \bar{B}(1 - L^{-j})$ and $\gamma^{-2} > \bar{B}$ show that this first passage occurs in finitely many steps. Thus the uniform construction applies to every preceding truncation.

69.2. **The exponent and its two expressions.** Use the regular progression of theorem Y, the boundary terms of section 62, and the retained factors of section 62.3. Their constants and the geometric data are required to satisfy the quantitative hypotheses of those constructions and of the actual history potentials and Gaussian systems. The simultaneous choice is made in section 69.1. In this section it suffices to keep those inequalities as hypotheses. In particular, $m' < m + K - j$ at every level used.

Write $\mathcal{S} = \mathcal{S}(h)$ for the label of a history of depth k. A regular index at level j is a pair (z, X) with $X \in \mathcal{D}_j$, $X \subset \Lambda_j$, and $z \in X^{(j)} \cap \Lambda_j^{\sim -1}$. Denote this set by $\mathsf{Ix}_j^{\mathrm{rs}}(\mathcal{S})$. The edge and newly created boundary indices are those of proposition 62.2. At a glued field U, with stored fields $\mathfrak{q}$, define

$$\mathbf{E}_k(\mathsf{U}) = \sum_{j=1}^{k} \Big\{ \sum_{(z,X) \in \mathsf{Ix}_j^{\mathrm{rs}}} [E^{(j)}(X, \mathsf{U}, z) - E^{(j)}(X, \mathbf{1}, z)] - b_j L^j A(\varphi_j, \mathsf{U}) \Big\}. \tag{1472}$$

Here the regular term is evaluated at the level-j pair of U. Put

$$\mathbf{E}_k^{\mathrm{bd}}(\mathsf{U}, \mathfrak{q}) = \sum_{j=1}^{k} \Big\{ \sum_{X \in \mathsf{Ix}_j^{\mathrm{mk}}} B_j^{\mathrm{mk}}(X; \mathbf{P}_{X|\mathcal{S}}^{(j)}(\mathsf{U}), (\mathfrak{q}|_X)^{<j}) + \sum_{X \in \mathsf{Ix}_j^{\mathrm{ed}}} E^{(j),\mathrm{ed}}(X, \mathsf{U}) \Big\}. \tag{1473}$$

The first terms are restricted to their layered output spaces. The boundary sum has no subtraction at the trivial field and no profile term. The exponent is

$$A_{k,h}(\mathsf{U},\mathfrak{q}) = -A(\beta_k(\cdot),\mathsf{U}) + \mathbf{E}_k(\mathsf{U}) + \mathbf{E}_k^{\mathrm{bd}}(\mathsf{U},\mathfrak{q}) - \mathcal{E}_{k,\mathcal{S}}. \tag{1474}$$

Each of these finite sums is defined precisely when every term used in it is defined. Terms with empty index sets are omitted, without requiring an evaluation on an unused domain. At level zero, $A_{0,\emptyset} = -\beta A$.

For later use there is a second expression, before the profile terms and the trivial values are separated:

$$\begin{aligned} A_{k,h}(\mathsf{U},\mathfrak{q}) = & -\beta A(\mathsf{U}) + \sum_{j=1}^{k} \sum_{X\subset\Lambda_j} \sum_{z\in X^{(j)}} E^{(j)}(X,\mathsf{U},z) \\ & + \sum_{j=1}^{k} \sum_{X\in \mathsf{Ix}_j^{\mathrm{mk}}} B_j^{\mathrm{mk}}(X; \mathbf{P}_{X|\mathcal{S}}^{(j)}(\mathsf{U}), (\mathfrak{q}|_X)^{<j}) \\ & + \sum_{j=1}^{k} [\mathsf{N}^{(j-1)}(\Omega_j,\Lambda_j) + \mathsf{vn}^{(j)}\langle\Lambda_j\rangle]. \end{aligned} \tag{1475}$$

The sum $X \subset \Lambda_j$ is over $\mathcal{D}_j$. Indeed, the regular and edge indices partition all the pointed terms on these domains. The profile identity gives

$$-A(\beta_k(\cdot),\mathsf{U}) - \sum_{j=1}^{k} b_j L^j A(\varphi_j,\mathsf{U}) = -\beta A(\mathsf{U}).$$

In (1406), the vl sum is exactly the sum of the trivial values subtracted in (1472). These two cancellations prove (1475) as an identity of partial functions, with the same domain. The trivial values are defined: the zero potential and current give the trivial triple in every analytic core, and the regular progression is defined on that core.

Let $\Phi_{k,h}(\mathsf{U};\mathfrak{q}) = \Phi_h(\mathsf{U};\mathfrak{q})$ be the sum of all the retained exponents (1390) from levels $1,\dots,k$. At a completed coordinate ϕ, evaluate $A_{k,h}$ and $\Phi_{k,h}$ on the glued adapted representatives. A value at ϕ is defined when the evaluation set is nonempty, the stored fields lie in their tiers, and the corresponding finite sum is defined and has the same value at every such representative. The two sums are evaluated separately. Gauge invariance of their constituents therefore identifies the exponents themselves, rather than just their exponentials. Set the density equal to zero wherever $\mathsf{Stay}_h^{\mathrm{T}}\chi_k(\Omega_k) = 0$.

69.3. **Collecting one step.** Let $h' = (h,\mathsf{o}_t)$, $t = k+1$. The normalizer has the following exact interior comparison. It concerns its local functions and their domains, not only the sum of its values on a real background.

Lemma 69.2. Use the same grain, clearance, mesh and normalizer clearance in the historical and empty-history constructions. If $X \subset \Lambda_t$, their normalizer classes on X coincide, and

$$N_{h'}^{\mathrm{un}}(X,p,z) = N_{\emptyset}^{\mathrm{un}}(X,p,z), \qquad N_{h'}^{\partial}(X;p) = N_{\emptyset}^{\partial}(X;p) \tag{1476}$$

as partial functions of the pair. On the analytic core, the latter empty-history functions are the constant-sequence functions of proposition 50.2, before the subtraction at the trivial pair. ∎

Proof. Every layer region of the collared sequence contains Λ_t. Thus its cube-wise trace on X is X, as is the trace of the constant sequence. The real factor, its perturbation and current are read on the same bond set. The layer tests and containing-box gauges defining $\mathscr{O}^N(X;a)$ are identical. The tiny-cube tests are identical too: if $y^{\sim} \subset X$ and y meets an integration bond, its other endpoint lies in $y^{\sim}$, so both endpoints lie in X. Whether such a nonpivot bond is an integration bond

depends only on whether an endpoint belongs to Λ_t, which is the same test in both settings. For $S_N(b) \subset X$ both endpoints belong to X; the same argument identifies the compression coordinates. For $X(c) \subset X$ it identifies the contributing pivot bonds.

For every $Y \subset X$, its determining tower $\Omega_\bullet(Y)$ lies in Y. Hence

$$\Omega_n^\diamond \cap \Omega_n(Y) = \Omega_n(Y) = T_\ell \cap \Omega_n(Y).$$

The two meets, their natural operator domains, and their compressed matrices $A_{Y,N}$ are therefore identical. This uses the actual operator locality of proposition 55.1, including its domain assertion. The logarithm series (1121), their finite Möbius differences, and the three cases in (1124) consequently agree. A pivot logarithm uses only the descendants of its endpoint double block, all contained in $X(c)$; it is the same function on the same branch. The tents also agree. This proves (1476).

For the comparison with proposition 50.2, the reading of the triple is $(\mathbf{U}, L^{-3}\mathbf{J})$, and its restriction to Y is the reading used to construct the constant-sequence operator there. The clearances and coordinate sets are the same. The flat gap in section 50.2 is γ^{tr}. Both constructions therefore use

$$B_{\mathsf{a}} = B_a = \max\{\gamma^{\mathrm{tr}}, 2\sqrt{2}C_a c_{\mathrm{lat}}(\theta_a)\},$$

and their logarithm expansion points satisfy $c_{\mathrm{nf}} = c_N$. Thus the two logarithm series and finite sums are literally the same expressions. At the trivial triple their common pointed value is $\mathsf{n}^{(t)}(X, z)$. The analytic-core bounds put the reading in the stated normalizer class, as checked in the local normalizer construction. □

The regular family at level t is the one already constructed from the empty-history expansion:

$$E^{(t)}(X, P, z) = N_\emptyset^{\mathrm{un}}(X, P, z) + F_\emptyset(X, P, z) - \mathsf{n}^{(t)}(X, z).$$

It depends on the coupling prefix, t, X, z, and the pair, and is independent of h'. For $X \subset \Lambda_t$, the same equality holds with the local normalizer and cluster terms of h'. For the normalizer this is lemma 69.2; for the pointed cluster terms it is proposition 59.5. These pointed identities include equality of their natural domains. The same proposition identifies the full cluster sum with the sum of its pointed terms on the admitted analytic domain.

The normalizer and cluster sums in the new exponent therefore split as

$$\begin{aligned} N + E(a') = {} & \sum_{X \subset \Lambda_t} \sum_{z \in X^{(t)}} E^{(t)}(X, \mathsf{U}, z) + \mathsf{vn}^{(t)}\langle \Lambda_t \rangle \\ & + \sum_X [B^{\mathrm{nm}}(X) + B^{\mathrm{cs}}(X)]. \end{aligned} \tag{1477}$$

This is a finite-sum identity using (1382). The bounds and support there restrict the last sum to the admitted newly created boundary indices. Adding $B^\times(a')$ and $B^{\mathrm{sr}}(a')$ supplies the other two terms of B_t^{mk}.

The constant n_t in (1368) is $\mathsf{N}^{(k)}(\Omega_t, \Lambda_t)$. To check the normalization, the regional fibre chart contributes

$$\log c_H = (\log \sigma_{\mathrm{H}} + 3 \log g_k) B_k(\Omega_t) = (\log \sigma_{\mathrm{H}} - \tfrac{3}{2} \log x_k) B_k(\Omega_t).$$

The gauge factor (987) contributes

$$\log c_z = -(L^3 - 1) \log \mathsf{z}(x_k, \varepsilon_k) \#_{k+1}(\Omega_t),$$

and Gaussian integration contributes $\nu_{\mathfrak{g}} B_k(\Lambda_t)$. These are the three terms of (1405). Thus the first sum in (1368) is precisely

$$\begin{aligned} A_{h'}^+(\mathsf{U}, \mathfrak{q}^+) = {} & A_{k,h}(\mathsf{U}, \mathfrak{q}) + \mathsf{N}^{(k)}(\Omega_t, \Lambda_t) + \mathsf{vn}^{(t)}\langle \Lambda_t \rangle \\ & + \sum_{X \subset \Lambda_t} \sum_{z \in X^{(t)}} E^{(t)}(X, \mathsf{U}, z) + \sum_{X \in \mathsf{Ix}_t^{\mathrm{mk}}} B_t^{\mathrm{mk}}(X; \mathbf{P}^{(t)}_{X|\mathcal{S}(h')}(\mathsf{U}), (\mathfrak{q}^+|_X)^{<t}). \end{aligned} \tag{1478}$$

Comparison with (1475) gives

$$A^{+}_{h'} = A_{t,h'}. \tag{1479}$$

This comparison includes equality of domains. Old terms are the same functions on the same domains; only their evaluation at the new glued field has changed.

For the other exponent, the definitions of the kept factors and external pivot factors give

$$\Phi_{t,h'}(\mathsf{U};\mathfrak{q}^{+}) = \Phi_{k,h}(\mathsf{U};\mathfrak{q}) + \sum_{X} K^{\mathrm{kp}}_{h'}(X;\mathsf{U},\mathfrak{a}') + \sum_{c\in\Gamma^{\partial}_{t}} z_c(\mathsf{U}). \tag{1480}$$

Each retained coefficient is assigned once to its strip component. Each external pivot is assigned once to $X(c)$. Thus this is also an equality of finite sums, and it is the second identity in (1368). The actual historical output membership gives every factor its required domain at every admitted glued representative.

Proposition 69.3. Suppose the quantitative hypotheses of this section hold through level t, and suppose the history h has the integrable density

$$\mathsf{Stay}^{\mathrm{T}}_{h}\chi_k(\Omega_k)e^{A_{k,h}+\Phi_{k,h}}$$

with respect to $\mathscr{M}_h$. Then the exponents of h' are defined at every point of $\{\mathsf{Stay}^{\mathrm{T}}_{h'}\chi_t(\Omega_t) > 0\}$ outside a Borel $\mathscr{M}_{h'}$-null set, and

$$\hat{\tau}^{\mathrm{co}\prime}_{h'} = (\mathsf{Stay}^{\mathrm{T}}_{h'}\chi_t(\Omega_t)e^{A_{t,h'}+\Phi_{t,h'}})\mathscr{M}_{h'}. \tag{1481}$$

The density is Borel, nonnegative and integrable, and is strictly positive almost everywhere on that set. Its local families obey the same bounds E_0, B, B_{fr} as at the preceding levels. ∎

Proof. First suppose $\Omega_1(h') \neq \emptyset$. We verify the identities used by theorem AB. The free and completed coordinates of proposition 53.1, the regional fibre chart, and the strip substitution give the two changes of variables and their coordinate commutation. The support identity of section 53.9 gives the factorization of the tests, including the strict integration balls and the new tiers. The fibre chart gives $j_c = c_H e^{S_\Gamma} e^{J}$, while (987) gives the gauge term. All these identities hold on the actual Borel domains.

The on-shell potential identity proposition 58.28 gives the two equalities (1363). In particular its separate identity for the old retained factors is needed here; an identity only for the total exponent would not identify the two new sums. The shifted field lies in the old critical orbit and has the required transports at the old stored sites. By proposition 60.5, each of its two old sums equals the sum evaluated at any adapted representative. This gives the old coordinate evaluation required in (1363).

The actual Gaussian system is supplied by proposition 57.5. On the whole torus its covariance and mean are (1164). The potential bounds give the hypotheses of theorem AA, so (1366) is the convergent cluster identity. The normalizer identity is proposition 56.3. At a real background its logarithm is the real determinant logarithm, hence it gives exactly (1364). The kept and rejected forms sum to the conditional Schur form, as in section 55.1. Finally proposition 54.8 gives the full strip sum (1367), with the actual strip substitution. Each statement includes definedness on the historical domains just verified.

Equations (1477)–(1480) identify the collected expressions with the canonical exponents. The actual core and output membership give their constituent domains for every glued representative. The comparison in proposition 60.5 makes their two sums representative independent, and proposition 60.6 identifies the retained family at every such representative, including at the first step. All partial functions are Borel: their restrictions are continuous, local series converge with common majorants, and the adapted representatives can be chosen Borel. These are precisely the remaining evaluation hypotheses of theorem AB. It proves (1481), its null-set assertion, and integrability.

There is no extra assumption that almost every coordinate point is realized by a bare field. The proof of theorem AB supplies this where needed. More explicitly, the image of the Borel domain of

the bare coordinate map contains a Borel set of full history measure. Off that set the integrable density has integral zero on every Borel subset. Its nonvanishing on the new support therefore makes the complement of the realized image a $\mathscr{M}_{h'}$-null set there. Every representative at a remaining point has the same exponents by the preceding orbit comparison.

If $\Omega_1(h') = \emptyset$, all Ω_j, Λ_j are empty. There are no integration, external or pivot coordinates, no retained factors, and no local or boundary entries used in the exponent. The profiles and extensive constants vanish. The evaluation set is $\{v_0\}$, and $A_{t,h'} = -\beta A(v_0)$, $\Phi_{t,h'} = 0$. The two changes of variables reduce to their zero-dimensional versions. Thus the same theorem gives the density $\mathsf{Stay}_{h'}^{\mathrm{T}} \chi_t(\Omega_t) e^{-\beta A(v_0)}$.

The regular norm bound is (974). The boundary bound follows from proposition 62.2; the newly created entries have the extra rate reserve, and the old entries keep their values and domains. For retained factors use (1398). None of these constants is summed over the levels. □

69.4. **Induction and compatibility.**

Theorem AC. Under the same quantitative hypotheses through level K, the canonical construction gives, for every $0 \le k \le K$ and every $h \in \mathcal{H}_k$,

$$\hat{\tau}_h^{\mathrm{co}\prime} = \big(\mathsf{Stay}_h^{\mathrm{T}} \chi_k(\Omega_k) e^{A_{k,h} + \Phi_{k,h}}\big) \mathscr{M}_h. \tag{1482}$$

The right side is a nonnegative integrable Borel density after assigning zero on a Borel null set. The regular, boundary and retained-factor bounds are uniform in k, h and the volume. Truncation of a history or of the coupling sequence preserves every earlier entry, its domain and its value. ■□

Proof. At level zero the reference measure is product Haar measure, the tests and the retained-factor exponential are one, and $A_{0,\emptyset} = -\beta A$. This is the finite Wilson measure. Suppose the assertion is known through level k. For each $h' \in \mathcal{H}_{k+1}$, the initial history $h = h'|_k$ has the asserted density. Its functions are exactly the functions used as old data in the next step: the regular progression uses only the coupling prefix; a boundary entry of level j uses only the history through j; and the retained factor of level j has its fixed layered domain and its stored fields from that step. Restricting the table of radii gives this same domain, by proposition 62.3. The completed coordinate space, reference measure, tests and evaluation set likewise depend only on h, not on its later extension. Thus the induction hypothesis is exactly the old density hypothesis of proposition 69.3.

Apply that proposition to every extension, including the empty-region case. This proves the next level, with the same bounds. Finiteness of the level range completes the induction. Each local construction is specified by its canonical series, fixed chart and fixed branch. Equality on a common prefix follows inductively from these same formulas; no coupling or family at a later level is used to construct an earlier one. □

Corollary 69.4. Fix the constants of proposition 69.1. Let $x_0, \dots, x_K$ be admissible and suppose $MR(x_K) \le L^m$. Then all conclusions of theorem AC hold at every level $0 \le k \le K$, with these same constants. ■□

Proof. The constant choice imposes each coupling inequality on the entire interval $x \ge \gamma^{-2}$. The adjacent-scale comparisons therefore apply at every step. The room argument following proposition 69.1 gives the required containing boxes, meshes and strict inequality $m' < m + K - k$. The actual potential, Gaussian and strip estimates then have the numerical hypotheses used in proposition 69.3. In particular, (1466) bounds the sum of the cluster, rejected and strip boundary contributions by $2/3$. Together with $\delta_N M \ge \kappa_T$, this gives $B_T = 24(LM)^3 E_{\mathrm{nf}} + 1$ in (1386). Thus the same regular, boundary and retained-factor bounds apply at each step, and theorem AC applies. □

70. Bounds for the constructed history measures

We apply the history estimates to the density constructed in theorem AC. Throughout this section the constants satisfy section 69.1, the couplings are admissible through K, and the torus satisfies

$$MR(x_K) \le L^m.$$

Write $y_j = \log x_j$, $g_j = x_j^{-1/2}$, $N_j = \#\mathbb{T}^{(j)}$, and $\mathtt{t}(h) = \tau_h(\mathcal{V}_k)$. All vacuum constants below are those of the same canonical regular progression.

70.1. The mass and the sum over histories.

Proposition 70.1. For every $1 \le k \le K$ and every history $h \in \mathcal{H}_k$,

$$e^{\mathcal{E}_{k,\mathcal{S}(h)}}\mathtt{t}(h) \le \exp\left[C_{\mathrm{maj}}\left(N_k + \sum_{j=1}^{k} \#_j(Z_j(h))\right) - c_\star \sum_{t=1}^{k} p_0(g_t)^2 n_t^{\mathrm{cr}}(h)\right].$$

The constants $C_{\mathrm{maj}}, c_\star > 0$ are independent of the history, the lattice, the coupling sequence and k. ■

Proof. The density hypothesis of proposition 66.2 is (1482). Equation (1474) gives (1441) at every admitted glued evaluation. For such a field the actual core membership and layer estimates of section 58 give the field hypotheses of proposition 64.2. The regular families satisfy (974), and the boundary families satisfy proposition 62.2, on precisely those domains. Thus the regular and boundary sums have the bound used in proposition 66.2, with the same C_5.

The retained factors are exactly those of proposition 62.3; they are not replaced by absolute-value estimates for arbitrary quadratic polynomials. The actual all-bond gauge class is supplied by corollary 54.7, with its fixed radius margins. The pivot ceilings and the kept-form positivity hypotheses of section 66 follow from the chosen constants. Therefore lemma 66.1 identifies the pivot determinants, and the stored-form argument identifies the coercive component forms. The history support and failed-test hypotheses are those of section 53.9. All the hypotheses of the integrated mass estimate proposition 66.3 are now satisfied, proving the displayed bound.

There is no change of vacuum data on truncating at k. The regular families through k, their trivial values, and the numbers $\mathsf{n}^{(j)}$ are prefix consistent. The three coefficients of (1402) likewise use only their own level. Consequently every term in (1406) with index $j < k$ is the same whether the construction ends at k or at K. This proves the claimed identification of $\mathcal{E}_{k,\mathcal{S}(h)}$. □

The vacuum deficit estimate now converts the preceding bound to the fixed normalization $e^{-E_{(k)}}$. Set

$$f(h) = e^{\mathcal{E}_{k,\mathcal{S}(h)}}\mathtt{t}(h), \qquad e(\mathcal{S}) = -E_{(k)} - \mathcal{E}_{k,\mathcal{S}}.$$

Then

$$f(h)e^{e(\mathcal{S}(h))} = e^{-E_{(k)}}\mathtt{t}(h), \qquad e(\mathcal{S}) = -\sum_{j=1}^{k} \mathfrak{E}'_j(\mathcal{S}). \tag{1483}$$

By proposition 63.1, this is a pair satisfying (1452) with $c = c_\star$. Thus proposition 67.6 gives

$$\sum_{h\in\mathcal{H}_k} e^{-E_{(k)}}\mathtt{t}(h)e^{\theta p_0(g_k)^2 \mathsf{N}(\mathcal{S}(h)) + \varkappa \#_k(Z_k(h))} \le \Pi_k(C_{\mathrm{maj}}, (1-\eth)(c_\star - \theta)) \tag{1484}$$

for $0 \le \theta < c_\star$, $\varkappa \ge 0$, $0 < \eth < 1$, under the threshold

$$y_K \ge Y^{\mathrm{II}}\Big(\eth(c_\star - \theta), 48^3(D_0 + \varkappa + \lambda_L)\Big), \qquad D_0 = C_{\mathrm{maj}} + 27(2 + \log(2L))C_\flat.$$

The stronger finite-product bound (1451) holds when its additional Y^{S} threshold is imposed. These free parameters are fixed before decreasing γ; the inequality is not asserted uniformly over all their values at a single threshold.

For the mark $\Theta^{(k)}$ defined before (1455), use the same substitution (1483). It gives, with $c'' = (1-\eth)(1-w)c_\star$,

$$\begin{aligned} \sum_h e^{-E_{(k)}}\mathsf{t}(h)e^{\Theta^{(k)}(h)} &\le \Pi_k(C_{\mathrm{maj}}, c'') \le e^{(C_{\mathrm{maj}}+2)N_k}, && c_a = 0,\\ \sum_h e^{-E_{(k)}}\mathsf{t}(h)e^{\Theta^{(k)}(h)} &\le \Pi_k(C_{\mathrm{maj}}+\lambda_L, c'') \le e^{(C_{\mathrm{maj}}+\lambda_L+2)N_k}, && c_a > 0. \end{aligned} \tag{1485}$$

The parameter ranges and the respective thresholds are exactly those of (1455) and (1456), with $c = c_\star$. In particular the second line uses $2\mathfrak{p}_0 > 4r+1$.

70.2. **The partition function.**

Proposition 70.2. Put

$$C_{\mathrm{low}}(x) = C_5 + \tfrac{9}{2}\log x + 3\log 7 + 24B_3^2.$$

For the constructed measures,

$$e^{-C_{\mathrm{low}}(x_k)N_k} \le e^{-E_{(k)}}\mathsf{t}(h_\emptyset) \le e^{-E_{(k)}}Z_{K,\ell,\beta} \le e^{(C_{\mathrm{maj}}+2)N_k}. \tag{1486}$$

As measures on the coarse fields,

$$e^{-E_{(k)}}\rho_k\, dV \ge e^{-E_{(k)}}\tau_{h_\emptyset} \ge e^{-C_5N_k}\mathsf{sf}_k\, dV. \tag{1487}$$

With the full vacuum constant one has

$$e^{-[C_{\mathrm{low}}(x_k)+C_{\mathfrak{E}}]N_k} \le e^{-E_{(K)}}Z_{K,\ell,\beta} \le e^{[3y_k+C_{\mathfrak{E}}+C_{\mathrm{maj}}+2]N_k}. \tag{1488}$$

■

Proof. For the empty history all regular regions are the torus. Its external fields, boundary terms and retained factors are empty, and its completed reference measure is coarse product Haar measure. The evaluation set is the regular critical orbit of the coarse field, with its adapted representatives. It is nonempty off the exceptional set furnished by theorem AC. Formula (1474) reduces to (1461), since $\beta_k = L^k x_k$ and $\mathcal{E}_{k,\mathcal{S}(h_\emptyset)} = -E_{(k)}$. The sum $\mathbf{E}_k$ is representative independent by the same evaluation rule.

On the support of an empty history, proposition 64.2 has only its final layer: the earlier layers and all boundary contributions are empty. It gives $|\mathbf{E}_k| \le C_5N_k$. The actual core membership verifies its hypotheses for every admitted critical background. Hence proposition 68.2 applies and gives (1487); integrating and using proposition 68.1 gives the first inequality of (1486). The positive history decomposition eq. (648) gives the middle inequality and

$$Z_{K,\ell,\beta} = \sum_{h\in\mathcal{H}_k}\mathsf{t}(h).$$

Apply (1453) to (1483). Its threshold is part of the chosen small-coupling ceiling. This gives the upper bound.

Finally $E_{(K)} = E_{(k)} + \mathfrak{E}_k^{\ge}$. Multiplication by $e^{-\mathfrak{E}_k^{\ge}}$ and the two bounds in (1411) prove (1488). □

71. Proof of the finite renormalisation theorem

Proof of theorem D. Choose the constants as in section 69.1. They are fixed before K, m and the coupling sequence. The running-coupling comparison gives $\check{R}_j \le L^{K-j}R(x_K)$. Consequently the single torus hypothesis implies

$$M\check{R}_j \le L^{m+K-j} \qquad (0 \le j \le K).$$

The lower radius bounds in the constant choice also give $m' < m+K-j$. Thus every finite-volume construction and domain used above is available at these levels. This argument uses no stronger chart-size hypothesis.

The bare Wilson density is continuous, strictly positive and at most one on a compact product of groups, so its mass belongs to $(0,1]$. Its gauge invariance and the chosen curvature ceilings allow eq. (648); this proves the positive history decomposition. The exact density and its evaluation assertions are theorem AC. Its marginal on v_k is τ_h, by the definition of the completed coordinates. The regular, boundary and retained-factor bounds are respectively theorem Y and propositions 62.2 and 62.3, with the actual domain verifications used in that induction.

Proposition 70.1 proves the individual history bound. Proposition 70.2 proves the partition and measure bounds, with $c_{\mathrm{low}} = C_5 + 3\log 7 + 24B_3^2$. All constants are independent of the number of steps, the history, the volume and the coupling prefix. The weighted estimates follow separately from the same mass bound under their stated thresholds.

Finally the coefficient-determined recursion and its finite first passage are proved in section 52.10. Every prefix before first passage has $x_j \geq \gamma^{-2}$ and $|b_j| \leq B_\sharp$. It therefore satisfies the hypotheses already used, and each coefficient depends only on the preceding prefix. □

72. Local source energies and stability

At level j, let $\mathcal{D}_j$ be the nonempty wall-connected unions of cubes of side ML^{j-K}. For such a union X, $d_j(X)$ is the least number of cube edges in a connected union of edges contained in X and meeting every cube of X; a single corner is allowed and has size zero. Write $X^{\sim 1}$ for its enlargement by one layer of cubes. These are the localization domains and their size in Proposition 45.1.

The following statement fixes the source ball before the cutoff. The effective-density construction that proves it also applies to arbitrary admissible finite real coupling sequences.

Theorem AD (Local source energies and stability)**.** There are choices of the averaging map and the constants preceding the coupling threshold, including an odd integer $L \geq 13$, integers $r \geq 2$ and $m' \geq 0$, and constants

$$M = L^{m'}, \quad B_\sharp \geq 0, \quad C_{\mathrm{s}} \geq 2, \quad E_0 > 0, \quad \kappa_e > \log 9216 + 2, \quad C_{\mathrm{S}} > 0$$

with the following property. For every $\varrho > 0$, there is $\gamma_\varrho \in (0, \frac{1}{2}]$ such that, after fixing the canonical auxiliary data at this threshold, the assertions below hold for $0 < g \leq \gamma_\varrho$. The recursion and all constructed objects use this same threshold assignment.

For every $\beta > X_g$, the coefficient-determined recursion has a finite first-passage index $K \geq 1$, and $g^{-2} < x_K(\beta) \leq X_g$. Define, for $x > 1$,

$$R(x) = \min\{L^u : u \in \mathbb{Z}_{\geq 0},\ L^u \geq (\log x)^r\}.$$

Choose any integer m satisfying

$$m \geq 1 + m' + \max\left\{\log_L R(L(g^{-2} + \bar{B})), \lceil \log_L C_{\mathrm{s}} \rceil\right\}, \qquad \ell = 2L^m. \tag{1489}$$

There are a real vacuum constant $E_{K,\ell,\beta}$, local source energies $e_j(X;w)$, $1 \leq j \leq K$, $X \in \mathcal{D}_j$, and a function $\widetilde{Z}(w)$, defined for $\|w\|_{\flat;T_\ell} < 2\varrho$, such that

$$Z_{K,\ell,\beta}[w] = \exp\left(E_{K,\ell,\beta} + \sum_{j=1}^{K} \sum_{X \in \mathcal{D}_j} e_j(X;w)\right) \widetilde{Z}(w). \tag{1490}$$

They have the following properties.

(i) The functions $e_j(X;\cdot)$ and $\widetilde{Z}$ are holomorphic on the source ball. Each $e_j(X;w)$ depends on w only on $X^{\sim 1}$, and $e_j(X;0) = 0$. They satisfy

$$e_j(X;\bar{w}) = \overline{e_j(X;w)}, \qquad \widetilde{Z}(\bar{w}) = \overline{\widetilde{Z}(w)}.$$

(ii) The source energies and the remaining partition function obey

$$|e_j(X;w)| \le 2M^3E_0\frac{\|w\|_{\flat;T_\ell}}{2\varrho}e^{-\kappa_e d_j(X)} \le C_{\mathrm{S}}e^{-\kappa_e d_j(X)}, \tag{1491}$$

$$|\widetilde{Z}(w)| \le e^{C_{\mathrm{S}}\ell^3}, \tag{1492}$$

$$\widetilde{Z}(0) \ge e^{-C_{\mathrm{S}}(1+\log g^{-2})\ell^3}. \tag{1493}$$

(iii) For two side lengths ℓ, ℓ' satisfying (1489), with the same K, β and coupling sequence,

$$\left|\frac{E_{K,\ell,\beta}}{\ell^3} - \frac{E_{K,\ell',\beta}}{(\ell')^3}\right| \le C_{\mathrm{S}}.$$

All constants displayed before ϱ can be fixed independently of $\varrho, g, K, \beta, \ell, w$, of the level and of the localization domain. The ceiling γ_ϱ may decrease when the source radius increases. ■□

The theorem bounds each local source energy separately. It does not bound their sum uniformly in K. The volume bound applies to $\widetilde{Z}$, after the vacuum energies have been extracted. The positive lower bound is at zero source. No limit as $K \to \infty$ or zero-free region for complex sources is asserted.

The preceding construction supplies the critical backgrounds, the unsourced Gaussian expansion and the positive history measures of theorem D. A varying complex source changes the quadratic form as well as the interaction. Its quadratic Taylor term must therefore be retained in the local factors, with a relative form bound that preserves damping. The linear and higher-order terms enter a cluster expansion about the complex conditional mean.

The proof first controls this source decomposition for one step, including the separate first step. It then constructs the source-dependent regular terms, boundary terms and retained factors through all levels. An identity of complex measures identifies the resulting density with the sourced bare measure after block averaging. The large-field estimates give the bound on $\widetilde{Z}$; the lower bound at zero source uses the positive construction of theorem D. Finally, the first-passage estimate gives the stated range of bare couplings and torus sizes.

73. Sources and finite history measures

The source changes the plaquette coefficient in the bare density. The coupling sequence, tests, history labels, coordinate maps and reference measures remain those of the real construction. We first compare the source with the coupling at every scale. We then record the finite measure identities which the sourced expansion must represent.

73.1. **The source class.** All distances in a source norm are physical distances on $T_\ell = (\mathbb{R}/\ell\mathbb{Z})^3$, where $\ell = 2L^m$. Put $a_i = L^{-i}$. A unit of length at level k has physical length a_{K-k}. The number $M = L^{m'}$ is fixed independently of K and ℓ. For a nonempty subset $S \subset T_\ell$, let

$$|w|_S = \sup_{x\in S}|w(x)|, \qquad \mathrm{Lip}_\infty(w;S) = \sup_{x\ne y\in S}\frac{|w(x)-w(y)|}{d_\infty(x,y)}.$$

The second supremum is zero when S is a singleton. Write $\mathsf{Lp}(S)$ for the complex functions with finite norm

$$\|w\|_{\flat;S} = \max\left\{|w|_S, \frac{40M}{\sqrt{3}}\,\mathrm{Lip}_\infty(w;S)\right\}. \tag{1494}$$

For $\varrho_0, \varrho_1 > 0$, set

$$\mathsf{Src}_S(\varrho_0,\varrho_1) = \left\{w \in \mathsf{Lp}(S) : |w|_S < \varrho_0, \quad \frac{40M}{\sqrt{3}}\,\mathrm{Lip}_\infty(w;S) < \varrho_1\right\}. \tag{1495}$$

It is the open unit ball of the norm

$$\|w\|_{\mathsf{Src}} = \max\left\{\frac{|w|_S}{\varrho_0}, \frac{40M}{\sqrt{3}\varrho_1}\,\mathrm{Lip}_\infty(w;S)\right\}. \tag{1496}$$

In particular, equal radii give $\|w\|_{\flat;S} < \varrho$.

Definition 73.1. A function Φ on $\mathsf{Src}_S(\varrho_0, \varrho_1)$ is *holomorphic* if, for every $w_0, f_1, \dots, f_n \in \mathsf{Lp}(S)$,

$$z \longmapsto \Phi\left(w_0 + \sum_{i=1}^{n} z_i f_i\right)$$

is holomorphic on the open set where its argument belongs to the class. Let Bc be the set of bare plaquette barycentres. If $\mathsf{Bc} \subset S$, a function depends on the source *only on* Y when its value is unchanged by changing w away from the barycentres in Y. ■

Restriction to a subset is contractive. The source class is convex, balanced and invariant under conjugation; taking a real or imaginary part does not increase its norm. These facts follow by applying the triangle inequality to values and differences. They also show that every finite dimensional affine slice in the definition is open. For finite S, the definition is ordinary holomorphy on an open subset of $\mathbb{C}^S$. Thus a holomorphic function of the finitely many plaquette values pulls back to a holomorphic function of sources on the torus.

The Euclidean norm used for presented sources is compatible with (2). Namely, with

$$\|f\|_{\sharp;S} = \max\{|f|_S, 40M \operatorname{Lip}_2(f; S)\},$$

the inequalities $d_\infty \le d_2 \le \sqrt{3} d_\infty$ give

$$\|f\|_{\flat;S} \le \|f\|_{\sharp;S} \le \sqrt{3}\|f\|_{\flat;S}.$$

Consequently a presented source $w = c + \sum_i z_i f_i$ satisfies

$$\|w\|_{\flat;S} \le |c| + \sum_i |z_i| \|f_i\|_{\sharp;S}.$$

Lemma 73.2. Suppose that Φ is holomorphic on a source class and $|\Phi| \le C$. If $\Phi(0) = 0$, then

$$|\Phi(w)| \le C\|w\|_{\mathsf{Src}}.$$

Without the condition at zero, $|\Phi(w) - \Phi(0)| \le 2C\|w\|_{\mathsf{Src}}$. ■

Proof. For $w \neq 0$, restrict to $\lambda \mapsto \Phi(\lambda w)$ on $|\lambda| < \|w\|_{\mathsf{Src}}^{-1}$ and apply the scalar Schwarz lemma. The second assertion applies the first to $\Phi - \Phi(0)$. □

73.2. **Size of the source at each scale.** Fix real couplings $x_0, \dots, x_K$, with $x_0 = \beta$ and

$$x_j = L^{-1} x_{j-1} - b_j, \qquad |b_j| \le B_\sharp, \qquad \bar{B} = \frac{L B_\sharp}{L - 1}, \qquad x_K > \bar{B}.$$

These are the couplings of the real construction. For a source on the torus or on Bc, define

$$x_j^w = x_j - L^{K-j} w, \qquad v_j^w = -\frac{L^{K-j} w}{x_j}. \tag{1497}$$

For a plaquette weight v, write

$$\operatorname{sw}_j(v) = \max_{d_\infty(x(p), x(p')) \le M a_{K-j}} |v(p) - v(p')|.$$

This comparison uses the physical size of a level-j block.

For a history h of level k, its real profiles φ_j satisfy $0 \le \varphi_j \le 1$. The associated bare plaquette coefficients are

$$\beta_k = \beta - \sum_{j=1}^{k} L^j b_j \varphi_j, \qquad \beta_k^w = \beta_k - L^K w. \tag{1498}$$

The profiles and increments are independent of the source.

Proposition 73.3. For $0 \le j \le K$ and $0 \le k \le K$, the preceding definitions give

$$x_j \ge L^{K-j}(x_K - \bar{B}), \qquad |v_j^w| \le \frac{|w|_S}{x_K - \bar{B}}, \qquad \mathrm{sw}_j(v_j^w) \le \frac{M \operatorname{Lip}_\infty(w; S)}{x_j}. \tag{1499}$$

The last two inequalities concern the restriction to Bc. In particular, for $w \in \mathsf{Src}_S(\varrho_0, \varrho_1)$ their right sides are strictly smaller than $\varrho_0/(x_K - \bar{B})$ and $\sqrt{3}\varrho_1/(40x_j)$. Moreover,

$$\beta_k(y) \ge L^k(x_k - \bar{B}) \ge L^K(x_K - \bar{B}) > 0. \tag{1500}$$

Writing $\mathsf{a}(w) = |w|_S/(x_K - \bar{B})$, we have

$$\begin{aligned} |\beta_k^w - \beta_k| &\le \mathsf{a}(w)\beta_k, \\ \operatorname{Re}\beta_k^w \ge (1 - \mathsf{a}(w))\beta_k, &\qquad |\operatorname{Im}\beta_k^w| \le \mathsf{a}(w)\beta_k. \end{aligned} \tag{1501}$$

For real gauge fields U,

$$\left| e^{-A(\beta_k^w, U)} \right| \le e^{-(1-\mathsf{a}(w))A(\beta_k, U)}. \tag{1502}$$

The profile cancellation is the exact identity

$$-A(\beta_k^w, \mathbf{U}) - \sum_{j=1}^{k} b_j L^j A(\varphi_j, \mathbf{U}) = -A(\beta - L^K w, \mathbf{U}), \tag{1503}$$

also for complex gauge fields wherever the plaquette actions are read. ∎

Proof. Iterating the recursion from j to K gives

$$|x_K - L^{-(K-j)}x_j| \le \bar{B}(1 - L^{-(K-j)}).$$

Hence $x_j - \bar{B} \ge L^{K-j}(x_K - \bar{B})$, which proves the coupling bound. If two barycentres have physical distance at most Ma_{K-j}, then $L^{K-j}|w(p) - w(p')| \le M \operatorname{Lip}_\infty(w; S)$. This proves the two source estimates.

The same recursion gives $\beta = L^k x_k + \sum_{j=1}^k L^j b_j$. Thus

$$|\beta_k - L^k x_k| \le B_\sharp \sum_{j=1}^{k} L^j = \bar{B}(L^k - 1),$$

and (1500) follows. Since $|\beta_k^w - \beta_k| = L^K|w|$, the floor proves (1501). Each real plaquette action lies in $[0, 2]$; multiply its value by the real-part bound and sum to obtain (1502). Finally, substituting (1498) and using linearity of the weighted action proves (1503). □

On a region where all the profiles $\varphi_1, \dots, \varphi_k$ equal one, the coefficient in (1498) is simply $\beta_k^w = L^k x_k^w$. In the real history geometry this includes the prescribed one-layer interior of Λ_k under its collar condition. No change in that condition is made by adding w.

73.3. **Mirror extensions.** The main theorem also applies to sources specified on one cube. Let $Q = \Delta_i(y_*)$ have side a_i, with $i \ge -m$, in the periodic cube grid. There are $\ell/a_i = 2L^{m+i}$ cubes in each coordinate direction. For $s \in [n - \frac{1}{2}, n + \frac{1}{2}]$, put $\mathrm{tri}(s) = (-1)^n(s - n)$. This continuous, two-periodic function defines the fold

$$\mathrm{fo}(x) = y_* + a_i \left(\mathrm{tri}\left(\frac{\widetilde{x}_\mu - \widetilde{y}_{*,\mu}}{a_i} \right) \right)_{\mu=1}^{3} \in Q.$$

The even number of cubes makes this independent of the lift. On each grid cube it is a reflection followed by a translation onto Q. A cube with centre $y_* + a_i n$, $n \in \mathbb{Z}^3$, has parity $(-1)^{n_1+n_2+n_3}$.

If $v, v' \in \mathsf{Lp}(Q)$ agree on ∂Q, define $[v \mid v']$ to be $v \circ \mathrm{fo}$ on even cubes and $v' \circ \mathrm{fo}$ on odd cubes. These prescriptions agree on cube faces. In particular,

$$v^{\mathrm{mp}} = [v \mid v], \qquad v^{\mathrm{cmp}} = [v \mid \bar{v}] \quad \text{when } \operatorname{Im} v|_{\partial Q} = 0.$$

Proposition 73.4. For the preceding extensions,

$$\|[v \mid v']\|_{\flat;T_\ell} = \max\{\|v\|_{\flat;Q}, \|v'\|_{\flat;Q}\}. \tag{1504}$$

Thus mirror extension and, when defined, conjugate mirror extension preserve the source norm. The map $(v, v') \mapsto [v \mid v']$ is complex linear on pairs with matching boundary values. If real Lipschitz functions $f_1, \dots, f_n$ on Q vanish on ∂Q, then

$$[c + \textstyle\sum_q z_q f_q \mid c + \sum_q z'_q f_q] = c + \sum_q z_q [f_q \mid 0] + \sum_q z'_q [0 \mid f_q]. \tag{1505}$$

For a holomorphic function Φ on a source class on the torus, its value at (1505) is holomorphic in (c, z, z') on the open set where both cube sources belong to the corresponding cube class. ∎

Proof. On each cube the fold is an isometry. Consequently the supremum norm and each local Lipschitz constant are at most the corresponding maximum for v, v'. Join two torus points by a straight segment between least lifts and cut it at the finitely many grid walls. The values agree at the cut points, and the lengths of the pieces sum to the original length. The triangle inequality therefore gives the same Lipschitz bound globally. Both an even and an odd cube map onto Q, so restriction to those cubes gives the reverse inequalities for the supremum and Lipschitz seminorms. This proves (1504). Linearity gives (1505); the asserted holomorphy is then the definition on a finite dimensional slice. □

For real c, the specialization $z' = \bar{z}$ gives the conjugate mirror extension. Holomorphy in the preceding proposition concerns the independent variables z, z'.

73.4. **Bare density and history measures.** Let $\mathcal{V}_0$ be the finite product of SU(2) over positive bare bonds, with product Haar probability measure dU. For a positive bare plaquette p, write

$$a_p(U) = 1 - \tfrac{1}{2} \operatorname{tr} \partial U(p), \qquad A(w, U) = \sum_p w(x(p)) a_p(U).$$

Thus $0 \le a_p(U) \le 2$. For $\mathsf{b} = L^{-K}\beta$, define

$$\rho_0^w(U) = e^{-A(\beta - L^K w, U)} = \rho_0(U) e^{L^K A(w,U)}, \qquad Z_{K,\ell,\beta}[w] = \int \rho_0^w \, dU, \qquad \rho_0 = e^{-\beta A(1,U)}. \tag{1506}$$

These definitions require no smallness assumption on the finitely many source values.

Fix an admissible real history construction as in Theorem D. Its finite set of histories at level k is $\mathcal{H}_k$, its bounded nonnegative Borel weights are $\mathfrak{w}_h$, and its coarse map is $\mathcal{M}^k$. In completed coordinates write $\mathsf{X}_h^{\mathrm{co}\prime}$ for the history space and $\xi_h : \mathcal{V}_0 \rightharpoonup \mathsf{X}_h^{\mathrm{co}\prime}$ for its Borel partial coordinate map. It is independent of w, as are $\mathfrak{w}_h$ and all tests and stored variables. For a bounded complex Borel density ζ, put

$$\tau_h[\zeta] = (\mathcal{M}^k)_*(\mathfrak{w}_h \zeta \, dU), \qquad \widehat{\tau}_h[\zeta] = (\xi_h)_*(\mathfrak{w}_h \zeta \, dU). \tag{1507}$$

The second image is taken on $\operatorname{dom} \xi_h$; this is the completed history measure denoted $\hat{\tau}_h^{\mathrm{co}\prime}$ in theorem AC. Set $\tau_h^w = \tau_h[\rho_0^w]$ and $\widehat{\tau}_h^w = \widehat{\tau}_h[\rho_0^w]$.

The coordinate construction has

$$\{U : \mathfrak{w}_h(U) \neq 0\} \subset \operatorname{dom} \xi_h, \qquad p_k \xi_h(U) = \mathcal{M}^k U, \tag{1508}$$

where p_k projects onto the last coarse field. The domain inclusion follows from the actual chart construction in Proposition 53.14 and the paragraph following its proof. In statements about other coordinate data below it is an explicit hypothesis.

Proposition 73.5. The measures in (1507) are finite complex measures. For a bounded Borel function F on the completed space,

$$\int F \, d\widehat{\tau}_h^w = \int_{\operatorname{dom} \xi_h} F(\xi_h(U)) \mathfrak{w}_h(U) \rho_0^w(U) \, dU. \tag{1509}$$

If (1508) holds, then

$$\tau_h^w = (p_k)_* \widehat{\tau}_h^w, \qquad \widehat{\tau}_h^w(\mathsf{X}_h^{\mathrm{co}\prime}) = \tau_h^w(\mathcal{V}_k) = \int \mathfrak{w}_h \rho_0^w \, dU. \tag{1510}$$

For the real construction, with its thresholds satisfying $\varepsilon_j \le c_2^{\mathrm{SO}}$ for $j < k$,

$$\sum_{h \in \mathcal{H}_k} \tau_h^w = (\mathcal{M}^k)_*(\rho_0^w \, dU), \qquad \sum_{h \in \mathcal{H}_k} \tau_h^w(\mathcal{V}_k) = Z_{K,\ell,\beta}[w]. \tag{1511}$$

The expectation in (1509) is entire in the plaquette values $w(p)$. For every multi-index α,

$$\partial_w^\alpha \int F \, d\widehat{\tau}_h^w = \int_{\operatorname{dom} \xi_h} F(\xi_h(U)) \mathfrak{w}_h(U) \prod_p (L^K a_p(U))^{\alpha_p} \rho_0^w(U) \, dU. \tag{1512}$$

It is therefore holomorphic on every source class, including classes of sources on the torus. The same assertions hold for coarse history expectations and for $Z[w]$.

If $s \ge |\operatorname{Re} w|_{\mathsf{Bc}}$, then

$$\begin{aligned} |\widehat{\tau}_h^w| &\le \widehat{\tau}_h[|\rho_0^w|] \le \widehat{\tau}_h[\rho_0^{1-s/\mathsf{b}}], \\ |\tau_h^w| &\le \tau_h[|\rho_0^w|] \le \tau_h[\rho_0^{1-s/\mathsf{b}}]. \end{aligned} \tag{1513}$$

Here $\rho_0^{1-s/\mathsf{b}}$ is a bounded positive density even if $1 - s/\mathsf{b}$ is negative. Finally,

$$\rho_0^{\bar{w}} = \overline{\rho_0^w}, \qquad Z[\bar{w}] = \overline{Z[w]}, \qquad \widehat{\tau}_h^{\bar{w}} = \overline{\widehat{\tau}_h^w}, \qquad \tau_h^{\bar{w}} = \overline{\tau_h^w}. \tag{1514}$$

■

Proof. The density ρ_0^w is continuous on a compact finite product. The weights are bounded and the coordinate maps are Borel. Consequently the pushforwards are finite complex measures. Extend $F \circ \xi_h$ by zero outside $\operatorname{dom} \xi_h$; the extension is bounded and Borel. The pushforward formula gives (1509). The two properties in (1508) then give (1510). The sourced density is gauge invariant, so the finite history partition identity (648) of proposition 44.1 applies to ρ_0^w and proves (1511).

For analyticity, put $X_p = L^K a_p$ and $\Psi = (F \circ \xi_h)^\natural \mathfrak{w}_h \rho_0$, where the superscript denotes the zero extension. Then

$$\int F \, d\widehat{\tau}_h^w = \int \Psi \exp\left(\sum_p w(p) X_p \right) dU, \qquad |X_p| \le 2L^K.$$

On every compact set of source values the exponential power series and each of its differentiated series are uniformly absolutely bounded by a constant times $\exp(2L^K \sum_p |w(p)|)$. Integrating these series proves entire dependence and (1512). Restriction to barycentres is complex linear, proving the assertion for torus sources.

For any Borel partition (E_i) of a Borel set E, nonnegativity of $\mathfrak{w}_h$ gives

$$\sum_i \left| \int_{\xi_h^{-1}(E_i)} \mathfrak{w}_h \zeta \, dU \right| \le \int_{\xi_h^{-1}(E)} \mathfrak{w}_h |\zeta| \, dU.$$

Taking the supremum over partitions bounds total variation by the image of $|\zeta|$. The same proof applies to the coarse map. Also

$$|\rho_0^w| = \rho_0 e^{L^K A(\operatorname{Re} w, U)} \le e^{-(\beta - L^K s) A(1,U)} = \rho_0^{1-s/\mathsf{b}}.$$

This proves (1513). Conjugating the defining integrals, whose weights and reference measure are real, proves (1514). □

73.5. **Evaluation of the sourced density.** We use the completed-coordinate convention of Section 61: a partial density defines a measure when it is defined and Borel off a Borel reference-measure null set, and its absolute value is integrable there. Values on the exceptional set are immaterial. A zero test makes the density zero without requiring an evaluation of the other factors.

Fix data of the history format at level k. They consist of the same kinds of regular terms, boundary terms and retained factors as in the real construction, but are not assumed to have been constructed by a sourced induction. Write $A_{k,h}$ for their exponent and Φ_h for the sum of their retained exponents. The bare-field exponent has the form

$$A_{k,h}(\mathsf{U},\mathfrak{q}) = -A(\beta_k,\mathsf{U}) + \mathbf{E}_k(\mathsf{U}) + \mathbf{E}_k^{\mathrm{bd}}(\mathsf{U},\mathfrak{q}) - \mathcal{E}_{k,\mathcal{S}};$$

see equation (1474). Define

$$A_{k,h}^{w}(\mathsf{U},\mathfrak{q}) = A_{k,h}(\mathsf{U},\mathfrak{q}) + L^K A(w,\mathsf{U}). \tag{1515}$$

Thus its Wilson term is $-A(\beta_k^w,\mathsf{U})$.

At $\phi = ((v_n)_{n\le k},\mathfrak{q})$, the evaluation set $\mathsf{Ev}_h(\phi)$ consists of the adapted representatives of its collared critical orbit, glued to the stored exterior field. The value $A_{k,h}^w(\phi)$ is defined only if this set is nonempty, ϕ lies in its tiers, and the finite sum in (1515) is defined and has the same value at every representative. The retained sum $\Phi_h(\phi)$ is evaluated separately by the same convention.

Let Stay_h be the product of the stored factors, $\mathbf{1}_h^{\mathrm{T}}$ the tier indicator, and $\chi_k(\Omega_k)$ the density test. Set

$$g_{k,h}^{w}(\phi) = \begin{cases} 0, & \mathsf{Stay}_h \mathbf{1}_h^{\mathrm{T}} \chi_k(\Omega_k) = 0, \\ \mathsf{Stay}_h(\phi) e^{\Phi_h(\phi)+A_{k,h}^w(\phi)}, & \text{otherwise, when both sums are defined.} \end{cases} \tag{1516}$$

On the remaining points this is a partial map. The sourced format measure is $\mathsf{Fm}_k^w(h) = g_{k,h}^w \mathscr{M}_h$ when it satisfies the preceding definition of an integrable partial density. The assertion that a history measure has this format means both that this measure is defined and that

$$\widehat{\tau}_h^w = \mathsf{Fm}_k^w(h). \tag{1517}$$

The subsequent sourced induction proves this assertion for its constructed data.

Lemma 73.6. For fixed format data, $A_{k,h}^w(\phi)$ is defined exactly where $A_{k,h}(\phi)$ is defined. On this domain,

$$A_{k,h}^w(\phi) = A_{k,h}(\phi) + L^K A(w,\mathsf{U}), \qquad \mathsf{U} \in \mathsf{Ev}_h(\phi), \tag{1518}$$

where the second term is independent of the chosen evaluation representative and has absolute value at most $2L^K \sum_p |w(p)|$. The partial densities for (df, w) and $(\mathsf{df}, 0)$ have the same domain. At level zero,

$$\widehat{\tau}_\emptyset^w = \rho_0^w \, dU = \mathsf{Fm}_0^w(\emptyset). \tag{1519}$$

■

Proof. Two adapted representatives of the same collared orbit differ by a residual gauge transformation. Extend that transformation by the identity outside the collar. The fixed endpoints of the exterior bonds are nodes, so this extension takes one glued representative to the other; this is the gauge comparison used in Proposition 60.5. Every plaquette action, and hence every weighted plaquette action, is therefore constant on the evaluation set. Its stated bound follows from $0 \le a_p \le 2$.

Adding this everywhere defined, constant-on-representatives quantity changes neither the domain of a summand nor the condition that the exponent sum have a single value. The nonempty-set and tier conditions are unchanged. This proves (1518) and the domain assertion for the exponent. In (1516), the test and Φ_h are unchanged, so the density domains agree too. At level zero there are no stored factors or tests, $A_{0,\emptyset} = -\beta A(1,U)$, $\xi_\emptyset$ is the identity and $\mathscr{M}_\emptyset = dU$. Equation (1) proves (1519). □

This lemma compares two sources for the same format data. In particular it makes no assertion about pointwise evaluation of the newly constructed retained or quadratic exponent sums. Those require their own domain and invariance arguments.

Proposition 73.7. Let $k \geq 1$, fix $h \in \mathcal{H}_k$, and let $w^{\mathrm{off}}(p)$ equal $w(p)$ when no corner of p lies in $\Omega_1(h)$ and zero otherwise. Put $w^{\mathrm{on}} = w - w^{\mathrm{off}}$. Then

$$\widehat{\tau}_h^{w} = e^{L^K A(w^{\mathrm{off}}, v_0)} \widehat{\tau}_h^{w^{\mathrm{on}}}. \tag{1520}$$

For fixed format data the same multiplier relates $\mathsf{Fm}_k^{w}(h)$ and $\mathsf{Fm}_k^{w^{\mathrm{on}}}(h)$: one is defined exactly when the other is, and their measures then satisfy (1520). Consequently the format identity (1517) holds for w exactly when it holds for w^{on}. If $\Omega_1(h) = \emptyset$ and the zero-source format identity holds, it follows for every source w. ∎

Proof. Every bond of a plaquette without a corner in $\Omega_1(h)$ is a proper level-zero bond. The coordinate v_0 records U on these bonds. Thus

$$A(w^{\mathrm{off}}, U) = A(w^{\mathrm{off}}, v_0) \quad \text{at } \xi_h(U).$$

The right side is a continuous function of a compact group coordinate. Its exponential and inverse are bounded. Inserting $\rho_0^{w} = e^{L^K A(w^{\mathrm{off}}, U)} \rho_0^{w^{\mathrm{on}}}$ into (1509) proves the measure identity.

The same proper bonds are fixed in every evaluation representative. Indeed, an inner such bond is part of the fixed collared datum, and every other such bond is part of the exterior field used in gluing. Hence $A(w^{\mathrm{off}}, \mathsf{U}) = A(w^{\mathrm{off}}, v_0)$ throughout $\mathsf{Ev}_h(\phi)$. The two sourced exponents and therefore their partial densities differ by the same multiplier. The multiplier and its inverse are bounded Borel functions, so almost everywhere definition, Borel measurability and absolute integrability hold for either density precisely when they hold for the other. Multiplication of measures proves the format identity and its converse. If $\Omega_1(h)$ is empty, $w^{\mathrm{on}} = 0$, giving the last assertion. □

The truncated plaquette weights w^{on} and w^{off} are used here only in finite measure identities; no preservation of a uniform Lipschitz source class is needed for this reduction.

74. First passage and the vacuum normalization

The finite renormalisation theorem applies to prescribed sequences of couplings. The coefficient recursion selects one such sequence from a bare inverse coupling. We first stop that sequence at a prescribed reference coupling and choose torus sizes that work for every resulting cutoff.

Fix the constants preceding the threshold in Theorem D and Proposition 69.1. In particular, $L \geq 13$ is odd, $r \geq 2$, $M = L^{m'}$, and $B_\sharp > 0$. Put

$$\bar{B} = \frac{L}{L-1} B_\sharp, \qquad R(x) = \min\{L^a : a \in \mathbb{Z}_{\geq 0},\ L^a \geq (\log x)^r\} \quad (x > 1).$$

We use the fixed geometric constant $C_{\mathrm{s}} \geq 2$ in the torus floor of Section 69.1, after (1471). The choices are made before the couplings and the lattice. Decrease γ, if necessary, so that

$$0 < \gamma \leq \tfrac{1}{2}, \qquad \gamma^{-2} \geq 2B_\sharp, \qquad \gamma^{-2} > \bar{B}, \qquad L^{-1}\gamma^{-2} - B_\sharp > 1, \qquad (L^{1/r} - 1)\log \gamma^{-2} \geq \log(2L). \tag{1521}$$

These requirements are compatible with the preceding choices by Proposition 69.1. At this threshold fix the canonical auxiliary data. Every construction in this section uses that one assignment, including when the reference coupling or stopping index changes.

An admissible finite sequence is a real sequence $u_0, \dots, u_K$ with

$$u_j \geq \gamma^{-2} \quad (0 \leq j \leq K), \qquad u_j = L^{-1} u_{j-1} - b_j, \quad |b_j| \leq B_\sharp \quad (1 \leq j \leq K). \tag{1522}$$

For $K, m \geq 1$ and $\ell = 2L^m$, it gives an admissible lattice when $MR(u_K) \leq L^m$. These are the sequence and torus hypotheses of Theorem D; prescribed increments need not be the coefficients of the constructed regular terms.

For $\beta \ge \gamma^{-2}$, let $x_j = x_j(\beta)$ instead denote the sequence defined by

$$x_0 = \beta, \qquad b_{j+1} = \mathsf{cf}(\widehat{E}^{(j+1)}[x_0, \dots, x_j]), \qquad x_{j+1} = L^{-1}x_j - b_{j+1}. \tag{1523}$$

Here $\widehat{E}^{(j+1)}$ is the regular family of Theorem Y, and cf is its second-moment coefficient. The family and its coefficient are defined from the displayed prefix before x_{j+1} is chosen. By Proposition 52.5, this recursion gives a unique finite, strictly decreasing real sequence $x_0, \dots, x_J$, with $|b_j| \le B_\sharp$,

$$1 < x_J < \gamma^{-2}, \qquad x_j \ge \gamma^{-2} \quad (0 \le j < J).$$

The sequence and its coefficients are independent of the torus. We call its prefixes the canonical coupling sequences. They agree at common indices when the reference coupling or stopping index changes at this fixed threshold assignment.

For $0 < g \le \gamma$, set

$$X_g = L(g^{-2} + B_\sharp), \qquad X_g + \bar{B} = L(g^{-2} + \bar{B}). \tag{1524}$$

A pair (K, β) is a first-passage datum at g if the least index $j \ge 0$ at which $x_j(\beta) \le X_g$ is $K \ge 1$. Its label is $\mathsf{b} = L^{-K}\beta$. Define

$$I(g) = (g^{-2} + \bar{B},\ X_g - \bar{B}], \tag{1525}$$

and choose sides from

$$\begin{aligned} m_{\mathfrak{L}}(g) &= 1 + m' + \max\left\{\log_L R(L(g^{-2} + \bar{B})), \lceil \log_L C_{\mathrm{s}} \rceil\right\}, \\ \mathfrak{L}(g) &= \{2L^m : m \in \mathbb{Z}_{\ge 1},\ m \ge m_{\mathfrak{L}}(g)\}. \end{aligned} \tag{1526}$$

The floor is an integer, so $\mathfrak{L}(g)$ is nonempty. It depends only on g and the fixed construction constants.

Proposition 74.1 (First passage)**.** Let $0 < g \le \gamma$ and $\beta > X_g$. There is a unique first-passage datum (K, β) at g, with $1 \le K < J$. Its canonical sequence is admissible and satisfies

$$g^{-2} < x_K \le X_g < x_{K-1} < \cdots < x_0 = \beta. \tag{1527}$$

For $0 \le i \le j \le K$,

$$|x_j - L^{-(j-i)}x_i| \le \bar{B}(1 - L^{-(j-i)}). \tag{1528}$$

The label and the terminal coupling obey

$$g^{-2} + B_\sharp \frac{L-2}{L-1} < \mathsf{b} \le L(g^{-2} + \bar{B}) - \bar{B}L^{-K}, \tag{1529}$$

$$\log g^{-2} < \log x_K \le \log g^{-2} + \log \frac{3L}{2}. \tag{1530}$$

Every $\ell = 2L^m \in \mathfrak{L}(g)$ gives an admissible lattice for this sequence. More precisely,

$$LMR(x_j) \le L^{m+K-j} \qquad (0 \le j \le K). \tag{1531}$$

Thus the finite renormalisation theorem and its larger local-chart condition apply at every level. One set $\mathfrak{L}(g)$ serves all first-passage data at g. ∎

Proof. The terminal value $x_J < \gamma^{-2} \le g^{-2} < X_g$ ensures that a first crossing exists. Since $x_0 = \beta > X_g$, its index K is positive. At that crossing,

$$x_K = L^{-1}x_{K-1} - b_K > L^{-1}X_g - B_\sharp = g^{-2}.$$

Consequently $K < J$. Strict decrease gives (1527), and the prefix satisfies (1522). The affine estimate (1528) is (979). Its applications at $(0, K)$ and $(0, K-1)$ give

$$\begin{aligned} \mathsf{b} &\le x_K + \bar{B}(1 - L^{-K}) \le X_g + \bar{B} - \bar{B}L^{-K}, \\ L\mathsf{b} &\ge x_{K-1} - \bar{B}(1 - L^{-(K-1)}) > X_g - \bar{B}. \end{aligned}$$

Since $L^{-1}(X_g-\bar B)=g^{-2}+B_\sharp(L-2)/(L-1)$, these are (1529). The inequality $B_\sharp\le g^{-2}/2$ gives $x_K\le X_g\le (3L/2)g^{-2}$ and proves (1530).

For the side estimate we use the following consequence of the last condition in (1521). If $X\ge\gamma^{-2}$, $s\in\mathbb{Z}_{\ge0}$ and $1<X'\le L^sX$, then

$$R(X')\le L^sR(X). \tag{1532}$$

Indeed, with $Y=\log X$ one has $\log L\le (L^{1/r}-1)Y$, whence induction gives $Y+s\log L\le L^{s/r}Y$. Thus $(\log X')^r\le L^s(\log X)^r$, and $L^sR(X)$ is an allowed power of L in the definition of $R(X')$.

Put $X=X_g+\bar B=L(g^{-2}+\bar B)$. Rearranging (1528) at (j,K) yields

$$x_j+\bar B\le L^{K-j}(x_K+\bar B)\le L^{K-j}X.$$

Apply (1532) to obtain $R(x_j)\le L^{K-j}R(X)$. The floor in (1489) gives $LMR(X)\le L^m$, which proves (1531). In particular, $MR(x_K)\le L^m$ is the room condition of Theorem D, and the stronger inequality $LMR(x_K)\le L^m$ is its local-chart condition (1471). □

Corollary 74.2 (A common interval of labels)**.** For fixed $0<g\le\gamma$, the sets

$$\mathcal{W}_K(g)=\{\beta:(K,\beta)\text{ is a first-passage datum at }g\},\qquad K\ge1,$$

are pairwise disjoint and have union (X_g,∞). Moreover,

$$L^KI(g)\subset\mathcal{W}_K(g)\subset L^K\left(g^{-2}+B_\sharp\frac{L-2}{L-1},L(g^{-2}+\bar B)\right]. \tag{1533}$$

The interval $I(g)$ has positive length. Thus every $\mathsf{b}\in I(g)$ gives a first-passage datum $(K,L^K\mathsf{b})$ for every $K\ge1$. ■□

Proof. The first assertion follows from the existence and uniqueness of the first crossing. The outer inclusion is (1529). The length of $I(g)$ is

$$(L-1)g^{-2}+B_\sharp\frac{L(L-3)}{L-1}>0.$$

Take $\mathsf{b}\in I(g)$ and put $\beta=L^K\mathsf{b}$. Then $\beta\ge L\mathsf{b}>L(g^{-2}+\bar B)=X_g+\bar B$, so the first crossing K' exists. Its label $\mathsf{b}'=L^{-K'}\beta=L^{K-K'}\mathsf{b}$ satisfies

$$L\mathsf{b}'>X_g-\bar B,\qquad \mathsf{b}'\le X_g+\bar B$$

by (1529). If $K'\ge K+1$, then $L\mathsf{b}'\le\mathsf{b}\le X_g-\bar B$, a contradiction. If $K'\le K-1$, then $\mathsf{b}'\ge L\mathsf{b}>X_g+\bar B$, also a contradiction. Hence $K'=K$, proving the inner inclusion. □

Corollary 74.3 (Stopped data)**.** Let (K,β) be a first-passage datum at $0<g\le\gamma$ and let $\ell=2L^m\in\mathfrak{L}(g)$. Put $g_{(0)}=g$, and, for $1\le i\le K-1$, define

$$g_{(i)}^{-2}=L^{-1}x_{K-i}(\beta)-B_\sharp. \tag{1534}$$

For $0\le i\le K-1$, set $k=K-i$. Then (k,β) is a first-passage datum at $g_{(i)}$, with canonical sequence $(x_0,\dots,x_k)$ and label $L^i\mathsf{b}$. Moreover,

$$L^i\ell\in\mathfrak{L}(g_{(i)}),$$

and the lattice with cutoff k and side $L^i\ell$ is admissible and satisfies the larger local-chart condition. For $i\ge1$, $g_{(i)}<g$ and $X_{g_{(i)}}=x_k$. ■□

Proof. There is nothing to prove when $i=0$. For $i\ge1$, $x_k>X_g$ gives $g_{(i)}^{-2}>g^{-2}>0$. The definition implies $X_{g_{(i)}}=x_k<x_{k-1}$, so the first crossing at $g_{(i)}$ is exactly k. Prefix consistency of (1523) identifies the canonical sequence, and $L^{-k}\beta=L^i\mathsf{b}$ gives its label.

The affine estimate gives

$$X_{g_{(i)}}+\bar B=x_k+\bar B\le L^i(x_K+\bar B)\le L^i(X_g+\bar B).$$

Consequently (1532) yields $m_{\mathfrak{L}}(g_{(i)}) \le i + m_{\mathfrak{L}}(g)$, including the unchanged term $\lceil \log_L C_{\mathrm{s}} \rceil$ in the maximum. Hence $L^i \ell \in \mathfrak{L}(g_{(i)})$. Apply proposition 74.1 to this datum. Its finest side exponent is $(m+i)+k = m+K$, as required for the stopped lattice. □

74.1. **The zero-source partition function.** For an admissible lattice $\bar{\mathfrak{w}} = ((\ell, K), u_\bullet)$, write $E_{(k)}[\bar{\mathfrak{w}}]$ for the accumulated vacuum constant of its first k steps, as defined in (1407). These are real numbers attached to the constructed regular terms. Let

$$Z_{K,\ell,\beta} = \int e^{-\beta A_K(U)}\, dU, \qquad \beta = u_0,$$

with the Wilson action and normalized product Haar measure used in Theorem D. The effective measures at all levels have this same positive total mass.

For a canonical sequence, define

$$E_{K,\ell,\beta} = E_{(K)}[\bar{\mathfrak{w}}], \qquad \widetilde{Z}(0) = e^{-E_{K,\ell,\beta}} Z_{K,\ell,\beta}. \tag{1535}$$

Once the construction constants and auxiliary choices are fixed, $E_{K,\ell,\beta}$ depends on K, ℓ, β; a reference coupling does not enter its definition.

Proposition 74.4 (Vacuum stability)**.** Let $u_0, \dots, u_K$ satisfy (1522), and assume $MR(u_K) \le L^m$, where $K, m \ge 1$ and $\ell = 2L^m$. For every $1 \le k \le K$,

$$e^{-C_{\mathrm{low}}(u_k)(L^{K-k}\ell)^3} \le e^{-E_{(k)}[\bar{\mathfrak{w}}]} Z_{K,\ell,u_0} \le e^{(C_{\mathrm{maj}}+2)(L^{K-k}\ell)^3}, \qquad C_{\mathrm{low}}(x) = c_{\mathrm{low}} + \tfrac{9}{2}\log x. \tag{1536}$$

The constants $c_{\mathrm{low}}, C_{\mathrm{maj}}$ are fixed before γ, the couplings, the cutoff and the torus.

In particular, there is a constant C_{fp} with the same independence such that every first-passage datum (K, β) at $0 < g \le \gamma$ and every $\ell \in \mathfrak{L}(g)$ satisfy

$$e^{-C_{\mathrm{fp}}(1+\log g^{-2})\ell^3} \le \widetilde{Z}(0) \le e^{(C_{\mathrm{maj}}+2)\ell^3}. \tag{1537}$$

■

Proof. The sequence and room hypotheses are precisely those of Theorem D. Its level-k vacuum constant is $E_{(k)}[\bar{\mathfrak{w}}]$, and the level-$k$ torus has $(L^{K-k}\ell)^3$ sites. Thus (1536) is the partition bound of Proposition 70.2, (1486).

For canonical first-passage data, proposition 74.1 verifies the hypotheses. By (1530),

$$C_{\mathrm{low}}(x_K) \le c_{\mathrm{low}} + \tfrac{9}{2}\log \tfrac{3L}{2} + \tfrac{9}{2}\log g^{-2}.$$

Take

$$C_{\mathrm{fp}} = \max\left\{ \tfrac{9}{2}, c_{\mathrm{low}} + \tfrac{9}{2}\log \tfrac{3L}{2} \right\}. \tag{1538}$$

Since $\log g^{-2} > 0$, the right side is at most $C_{\mathrm{fp}}(1 + \log g^{-2})$. The case $k = K$ of (1536) proves (1537). □

74.2. **Comparison of vacuum constants on two tori.** The vacuum constant depends on the torus through the local terms. Terms whose domains lift into both tori have the same value. The remaining domains have diameter comparable to the smaller torus, so their exponential decay controls the difference per unit volume.

Keep one threshold assignment, one admissible sequence $u_0, \dots, u_K$, and its real regular progression. Let κ be the decay rate used for the vacuum estimates in Section 63, and require

$$\kappa \ge 2\kappa_{\mathcal{D}} + 5, \qquad \kappa_{\mathcal{D}} = \log 9216.$$

The constants can be chosen with this rate. Let $C_L \ge 8L$ be the room constant for the local lattice charts in that construction.

Proposition 74.5. Suppose that $\bar{\mathfrak{w}}_\iota = ((\ell_\iota, K), u_\bullet)$, $\ell_\iota = 2L^{m_\iota}$, $\iota = 1, 2$, are admissible lattices for the same sequence, and that

$$LMR(u_K) \le L^{m_\iota}, \qquad \ell_\iota \ge C_L M.$$

Put $\ell_* = \min(\ell_1, \ell_2)$ and

$$C_{\mathrm{E},0} = 2048 \sum_{n \ge 0} e^{-n}, \qquad C_{\mathrm{tt}} = 2C_{\mathrm{E},0}, \qquad C'_{\mathrm{tt}} = 2.$$

Then

$$\left| \frac{E_{(K)}[\bar{\mathfrak{w}}_1]}{\ell_1^3} - \frac{E_{(K)}[\bar{\mathfrak{w}}_2]}{\ell_2^3} \right| \le 2C_{\mathrm{tt}} C'_{\mathrm{tt}} E_0 e^{-\kappa \ell_* / (5M)}. \tag{1539}$$

In particular the right side is at most $2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0$. The proposition permits $\ell_1 = \ell_2$. ■

Proof. At level j, the cube side in physical coordinates is $W_j = ML^{j-K}$. Call an infinite-grid domain a near lift for a torus of side ℓ if it is contained in a box with all side lengths strictly less than $\ell/2$. A torus domain is common if it is the projection of a lift near for both tori. The other domains will be called noncommon.

We first record the geometry used below. Projection of a near lift identifies its cubes, their incidences and their edges, and preserves d_j. Two such lifts of the same domain differ by a period translation. For a specified lift of a site z in the domain there is exactly one near lift containing that site. These assertions follow by starting with its cube and continuing across shared faces. No two points separated by a nonzero period can occur in the box, even after comparing its two possible lifts, since each of its side lengths is less than half a period.

A connected union of d cube edges lies in a box of width at most dW_j in each direction. Every cube it meets lies in the box enlarged by W_j at either end. Hence a lifted domain has width at most $(d_j + 2)W_j$. The same assertion gives a near lift of a torus domain whenever $(d_j + 2)W_j < \ell/2$: lift its edges in a connected order, choosing each new edge to share the lifted endpoint of an earlier edge, and lift the cubes meeting them. The lifted union is contained in the enlarged box and projects onto the domain. This argument also applies when the edge union is a single corner. Consequently every noncommon domain on either torus satisfies

$$d_j(X) \ge \frac{\ell_*}{2W_j} - 2. \tag{1540}$$

Indeed, a domain with a smaller size has a lift near for its own torus and of width less than $\ell_*/2$, hence near for both.

The two vacuum families at level j are

$$\mathsf{vl}_\iota^{(j)}(X, z) = E^{(j), \mathfrak{w}_\iota}(X, \mathbf{1}, z), \qquad \mathsf{vn}_\iota^{(j)}(X, z) = \mathsf{n}^{(j), \mathfrak{w}_\iota}(X, z).$$

They are real and each is bounded by $E_0 e^{-\kappa d_j(X)}$; see Section 63.1 and Proposition 63.1. They have the same values on a common lift. For the regular family, dilate that lift by $\lambda_j = L^{K-j+1}$ to read it in the level-$(j-1)$ coordinates of the infinite companion in Theorem Y. The trivial triple $(\mathbf{1}, 0, 0)$ belongs to every analytic core and projects to the trivial triple on both tori. The companion identity therefore gives one value for $\mathsf{vl}^{(j)}$ on the common lift. For $\mathsf{vn}^{(j)}$, use its local logarithm series and pivot term, (1399) and the local-chart identity following (1400). Their definitions are transported by the same chart. All cubes, nonpivot bonds and tents in these definitions are carried by the common lift, so the Gaussian vacuum values agree as well.

Write the full-torus contribution of step $j-1$ as $\mathsf{N}_\iota^{(j-1)} + \mathsf{S}_\iota^{(j)}$, where N is its counting term and

$$\mathsf{S}_\iota^{(j)} = \sum_{z \in \mathbb{T}_\iota^{(j)}} \sum_{X \ni z} (\mathsf{vl}_\iota^{(j)}(X, z) + \mathsf{vn}_\iota^{(j)}(X, z)).$$

There are $3(L^3-1)$ positive nonpivot bonds per level-j site. Thus

$$\ell_\iota^{-3}\mathsf{N}_\iota^{(j-1)} = L^{3(K-j)}\bigl[3(L^3-1)(\mathsf{co}_{j-1}^{\mathrm{b}}+\mathsf{co}_{j-1}^{\mathrm{i}})+\mathsf{co}_{j-1}^{\mathrm{s}}\bigr].$$

The three coefficients in (1402) depend on the common coupling sequence and fixed constants alone. Their contributions per volume therefore cancel.

The common-domain contributions to $\mathsf{S}^{(j)}$ also cancel per volume. To see this without assuming an extra translation law, lift a site to the infinite grid and sum over the near lifts containing it. The preceding uniqueness and common-value statements give one function of the lifted site which factors through both torus coverings. Let $\ell^* = \max(\ell_1,\ell_2)$; it is an integer multiple $q_\iota\ell_\iota$ of each side. In a half-open cube of side ℓ^* each site of the ιth torus has exactly q_ι^3 lifts. Summing the common function on that cube therefore proves equality of the two common contributions divided by their volumes. If $\mathsf{S}_\iota^{(j),\mathrm{nc}}$ denotes the remaining sum, we have

$$\frac{E_{(K)}[\bar{\mathfrak{w}}_1]}{\ell_1^3}-\frac{E_{(K)}[\bar{\mathfrak{w}}_2]}{\ell_2^3}=\sum_{j=1}^{K}\left(\frac{\mathsf{S}_1^{(j),\mathrm{nc}}}{\ell_1^3}-\frac{\mathsf{S}_2^{(j),\mathrm{nc}}}{\ell_2^3}\right). \tag{1541}$$

It remains to bound these sums. Every level-j site lies in the interior of one cube: M is odd and the cube walls have half-integer coordinates in level-j units. The entropy bound of Proposition 45.1 therefore gives, for $D\ge 0$,

$$\sum_{X\ni z;\ d_j(X)\ge D} e^{-\kappa d_j(X)} \le 2048\sum_{n\ge D} e^{-(\kappa-\kappa_{\mathcal{D}})n} \le C_{\mathrm{E},0}e^{-\kappa D/2}.$$

The last inequality uses $\kappa/2-\kappa_{\mathcal{D}}>1$. Put $t_j=\ell_* L^{K-j}/M$. The side condition and the integral power form of the sides imply $m_\iota\ge m'+2$, whence $t_j\ge 2L^2\ge 338$. In particular,

$$\frac{1}{2}\left(\frac{t_j}{2}-2\right)\ge\frac{t_j}{5}.$$

Use (1540) in the entropy tail, multiply by the two-family bound and by the number of sites $\ell_\iota^3 L^{3(K-j)}$, and obtain

$$\ell_\iota^{-3}|\mathsf{S}_\iota^{(j),\mathrm{nc}}| \le C_{\mathrm{tt}}E_0 L^{3(K-j)}e^{-\kappa t_j/5}.$$

Finally put $\alpha=\kappa\ell_*/(5M)>4$. The ratio of successive terms in $\sum_{s\ge 0}L^{3s}e^{-\alpha L^s}$ is at most

$$\exp\bigl(3\log L-\alpha L^s(L-1)\bigr)\le\tfrac{1}{2},$$

so the sum is at most $2e^{-\alpha}$. Summing the preceding estimate over j and the two tori in (1541) proves (1539). □

For first-passage data, the stronger room follows from proposition 74.1. Choosing $C_{\mathrm{s}}\ge C_L$ in the torus floor also gives $\ell\ge C_L M$. Hence the proposition applies to every pair of sides in $\mathfrak{L}(g)$ for the same first-passage datum and the same threshold assignment.

75. Expansion about the conditional mean

The source produces a potential with quadratic growth in the fluctuation. Its natural argument is the fluctuation minus the conditional mean. This distinction matters in the Gaussian estimate: bounding that potential in terms of the unshifted field would introduce the cutoffs on all integration bonds into the volume bound. We first prove the estimate with growth about the mean, then use it for the local expansion.

75.1. **The Gaussian estimate with quadratic growth.** Use the local Gaussian structure of Definition 57.1 at a step $t = k+1$, where $0 \le k < K$, $M = L^{m'}$ and $m' < m + K - t$. The ambient region $\mathfrak{R}$ is nonempty and is a union of coarse cubes. The geometric constants satisfy $R_1 \ge L$, $M_1 \ge R_1\mathfrak{M}$ and $M \ge M_1$, and $\bar\kappa \ge 1$. Thus $\Lambda_t \subset \Omega_t$, the integration bonds $\mathfrak{X}_{\mathrm{I}}$ are the nonpivot bonds touching Λ_t, and the other nonpivot bonds are external. For a fine region $Z_\circ$ with $Z_\circ^\sharp \subset \mathfrak{R}$, write $\mathfrak{f} = \mathfrak{X}_{\mathrm{I}}\langle Z_\circ\rangle$ and $\mathfrak{r} = \mathfrak{X}_{\mathrm{I}}\langle\mathfrak{R}\rangle$. Brackets mean that a bond touches the region; a superscript $\circ$ requires both endpoints in it. Positive bonds are the independent coordinates. The norm $|\cdot|$ on $\mathfrak{su}(2)^{\mathbb{C}}$ is the matrix operator norm, whereas

$$\|z\|^2 = \sum_{b\in\mathfrak{f}^+} (|\operatorname{Re} z(b)|^2 + |\operatorname{Im} z(b)|^2)$$

is the Euclidean coordinate norm.

The covariance, connecting operator and mean are denoted $(C,\Gamma,m)(Z_\circ,s)$. They are holomorphic near the closed decoupling polydisc $|s(\Delta)| \le e^{\bar\kappa}$ and have the restriction and component properties in the cited definition. Their real references satisfy

$$S_{\mathrm{r}} = C_{\mathrm{r}}^{-1} \ge \gamma_{\mathrm{r}} > 0, \qquad \|S_{\mathrm{r}}\|, \|\Gamma_{\mathrm{r}}\|, \|C_{\mathrm{r}}\Gamma_{\mathrm{r}}\| \le c_{\mathrm{r}}, \qquad 0 < \gamma_{\mathrm{r}} \le 1 \le c_{\mathrm{r}},$$

and each of $C^{-1} - S_{\mathrm{r}}$, $\Gamma - \Gamma_{\mathrm{r}}$ and $\Gamma^{\mathsf{T}}C\Gamma - \Gamma_{\mathrm{r}}^{\mathsf{T}}C_{\mathrm{r}}\Gamma_{\mathrm{r}}$ has norm at most $\hat\varepsilon$. The reference connecting operator uses auxiliary bonds touching $Z_\circ^\sharp$ only. The reversal-invariant thresholds satisfy $1 \le \mathfrak{b}_- \le \mathfrak{b}_b \le \mathfrak{b}_+ < \infty$, and

$$|m_{\mathrm{r}}(b)| \le \theta_m\mathfrak{b}_b, \qquad |m(b) - m_{\mathrm{r}}(b)| \le \varepsilon_m\mathfrak{b}_b, \qquad \theta_m + \varepsilon_m \le \vartheta_* < 1.$$

In this section the number $\rho \ge 0$ is a quadratic-growth coefficient, distinct from the source radius ϱ. Put

$$\begin{gathered} d_* = 1 - \vartheta_*, \qquad C_{\mathrm{mn}} \ge 18 d_*^{-2}, \\ \lambda_{\mathrm{r}} = \min\{\gamma_{\mathrm{r}}/2, (1+2c_{\mathrm{r}}^2)^{-1}\}, \qquad c_{\mathrm{mn}} = C_{\mathrm{mn}}\lambda_{\mathrm{r}}^{-1}, \\ \Sigma_\rho = \hat\varepsilon + (1+c_{\mathrm{r}})\varepsilon_m^2 + \gamma_2 + \rho, \qquad u_\rho = C_{\mathrm{mn}}\Sigma_\rho, \qquad \gamma_2 \ge 0. \end{gathered} \tag{1542}$$

We require

$$C_{\mathrm{mn}}u_\rho \le \lambda_{\mathrm{r}}. \tag{1543}$$

For disjoint reversal-invariant sets $\mathfrak{S}, \mathfrak{e} \subset \mathfrak{f}$, the first carrying small-field tests and the second failed tests, use the tube of Section 57.2. On $\mathfrak{e}$ the field is real with $|B(b)| \ge \mathfrak{b}_b$; off $\mathfrak{e}$ its imaginary part satisfies $|\operatorname{Im} B(b)| < \varepsilon'\mathfrak{b}_b$, where $\varepsilon' > \varepsilon_m$; on $\mathfrak{S}$ also $|\operatorname{Re} B(b)| + |\operatorname{Im} B(b)| < \mathfrak{b}_b$. A function is admitted on this tube if it is Borel and, for fixed real failed variables, holomorphic in the other variables. Let D be its real part. Tested integrands are extended by zero outside D. The conditioning integral is the normalized joint integral

$$\mathbb{J}^m(Z_\circ;s)[F] = \int d\mu_0(\mathbb{X}) \det{}^{1/2}(C^{-1}/2\pi) \int dB\, F(B) e^{-\frac12 (B-m+C\Gamma\mathbb{X}, C^{-1}(B-m+C\Gamma\mathbb{X}))_1}. \tag{1544}$$

The determinant branch is the one on matrices with positive real part. Empty integration sets have the value $F(0)$.

Proposition 75.1. Assume (1543). Suppose s vanishes outside a coarse region Z with $Z_\circ^\sharp \subset Z \subset \mathfrak{R}$. If G is admitted on the preceding tube and, at this value of s,

$$|G(B)| \le F_0 e^{\rho\|B - m(Z_\circ,s)\|^2}, \qquad F_0 \ge 0, \tag{1545}$$

then the conditioning integral converges absolutely and

$$|\mathbb{J}^m(Z_\circ;s)[\mathbf{1}_D G]| \le F_0 \exp\left[-C_{\mathrm{mn}}^{-1}\gamma_2 \sum_{b\in\mathfrak{e}^+} \mathfrak{b}_b^2 + c_{\mathrm{mn}}u_\rho N_{\mathrm{I}}(Z)\right], \qquad N_{\mathrm{I}}(Z) = 3\#\mathfrak{X}_{\mathrm{I}}^+\langle Z\rangle. \tag{1546}$$

There is also an integrable majorant for the original joint integrands which is uniform in s and in all G satisfying (1545) with the same F_0, ρ. The majorant may depend on the fixed finite thresholds. ∎

Proof. We use the contour of Proposition 57.2, keeping the additional growth factor in its estimate. Write $S = C^{-1}$ and

$$Q = \begin{pmatrix} 1 + \Gamma^{\mathsf{T}} C \Gamma & \Gamma^{\mathsf{T}} \\ \Gamma & S \end{pmatrix}.$$

Completion of the square and the reference bounds give $Q_{\mathrm{r}} \geq \lambda_{\mathrm{r}}$, $\|Q - Q_{\mathrm{r}}\| \leq 2\hat{\varepsilon}$ and $\det Q = \det S$, as in (1141). The smallness assumption implies

$$\Sigma_\rho \leq d_*^4 \lambda_{\mathrm{r}}/324, \qquad \hat{\varepsilon} + 3\rho \leq \lambda_{\mathrm{r}}/108, \qquad 16 d_*^{-2} \varepsilon_m^2 \leq \tfrac{1}{2}.$$

In particular $\operatorname{Re} S > 0$.

Let $\mathfrak{b}_\Sigma^2 = \sum_{b \in \mathfrak{f}^+} \mathfrak{b}_b^2$. The mean bound gives $\|m\| \leq \vartheta_* \mathfrak{b}_\Sigma$. For real B, (1545) is at most $F_0 e^{2\rho \mathfrak{b}_\Sigma^2} e^{2\rho \|B\|^2}$. The real-part calculation leading to (1143), now including $2\rho\|B\|^2$, gives the common majorant

$$F_0 e^{(c_{\mathrm{r}} + 2\rho)\mathfrak{b}_\Sigma^2} \left(\frac{c_{\mathrm{r}} + 1}{2\pi} \right)^{3 \# \mathfrak{f}^+/2} e^{-3\lambda_{\mathrm{r}} \|\mathbb{X}\|^2/8 - \lambda_{\mathrm{r}} \|B\|^2/8}. \tag{1547}$$

The inequality $\hat{\varepsilon} + 3\rho \leq \lambda_{\mathrm{r}}/108$ leaves the displayed Gaussian decay. This proves absolute convergence before any change of contour.

The auxiliary variables outside Z integrate to one, by the component property of Γ. After their removal the joint form, denoted Q^Z, has the same reference and perturbation bounds, and its dimension n is at most $2N_{\mathrm{I}}(Z)$. Put $\nu = \operatorname{Im} m$. The contour is $\Theta(\beta)(b) = \beta(b) + i\phi_b(\beta(b))\nu(b)$. On failed bonds $\phi_b = 0$, on untested bonds $\phi_b = 1$, and on small-field bonds

$$\phi_b(\beta) = \min\{1, \max\{0, 2d_*^{-1}(1 - d_*/4 - |\beta|/\mathfrak{b}_b)\}\}.$$

It agrees with the real contour near the boundary of each small-field ball and stays in the tube. Its Jacobian factors satisfy $|J_b| \leq e^{2d_*^{-2}\varepsilon_m^2}$. The contour identity in the cited proof applies to the present integrand: along every intermediate contour, $\|B - m\|^2 \leq 2\|B\|^2 + 2\|m\|^2$, and the same Gaussian calculation used in (1547) dominates the growth. Thus the vertical sides on untested variables vanish at infinity, and Fubini remains valid. Failed variables stay real.

Set $\beta' = \beta - \operatorname{Re} m$ and $\eta = (1 - \phi)\nu$. The cutoff geometry gives

$$|\beta'(b)| \geq d_* \mathfrak{b}_b \quad (b \in \mathfrak{e}^+), \qquad \|\eta\|^2 \leq 16 d_*^{-2} \varepsilon_m^2 \|\beta'\|^2.$$

Since $\Theta(\beta) - m = \beta' - i\eta$, the growth factor on the moved contour is at most

$$F_0 e^{\rho(\|\beta'\|^2 + \|\eta\|^2)} \leq F_0 e^{3\rho\|\beta'\|^2/2}. \tag{1548}$$

This is where centering removes the dependence on the mean size from the useful estimate. Put

$$a = \hat{\varepsilon} + 8 d_*^{-2}(1 + c_{\mathrm{r}})\varepsilon_m^2 + \gamma_2 + \tfrac{3}{2}\rho \leq 8 d_*^{-2} \Sigma_\rho. \tag{1549}$$

Multiplying the Gaussian bound, (1548), the Jacobian bound and

$$1 \leq \exp\left(\gamma_2 \|\beta'\|^2 - \gamma_2 d_*^2 \sum_{b \in \mathfrak{e}^+} \mathfrak{b}_b^2 \right)$$

bounds the moved integrand by

$$F_0 e^{(2/3) d_*^{-2} \varepsilon_m^2 N_{\mathrm{I}}(Z)} e^{-\gamma_2 d_*^2 \sum_{b \in \mathfrak{e}^+} \mathfrak{b}_b^2} e^{-\frac{1}{2}(x, (Q_{\mathrm{r}}^Z - 2a)x)}, \qquad x = (\mathbb{X}, \beta').$$

Here $2a < \lambda_{\mathrm{r}}/2$, so $A = Q_{\mathrm{r}}^Z - 2a$ is positive. Translate by the real mean and enlarge the real integration domain. Gaussian integration leaves the determinant ratio, for which

$$|\det Q^Z|^{1/2} (\det A)^{-1/2} \leq e^{2n(a + \hat{\varepsilon})/\lambda_{\mathrm{r}}} \leq e^{4(a + \hat{\varepsilon}) N_{\mathrm{I}}(Z)/\lambda_{\mathrm{r}}}.$$

This follows by factoring out A and using $|\det(1 + D)| \leq e^{n\|D\|}$. The remaining volume coefficient is bounded by

$$4(a + \hat{\varepsilon})\lambda_{\mathrm{r}}^{-1} + \tfrac{2}{3} d_*^{-2} \varepsilon_m^2 \leq 37 d_*^{-2} \lambda_{\mathrm{r}}^{-1} \Sigma_\rho \leq c_{\mathrm{mn}} u_\rho.$$

Also $d_*^2 \ge C_{\mathrm{mn}}^{-1}$. These inequalities prove (1546). For an empty integration set the assertion reduces to $|G(0)| \le F_0$. □

Lemma 75.2. Let $\vartheta' > \vartheta_*$ and suppose that $G(B;n)$ is admitted in B on the fixed tube for every $|n(b)| < \vartheta' \mathfrak{b}_b$, is holomorphic in n for every real B in the tube, and satisfies

$$|G(B;n)| \le F_0 e^{\rho \|B-n\|^2}.$$

Under (1543),

$$s \longmapsto \mathbb{J}^m(Z_\circ; s)[\mathbf{1}_D G(\cdot; m(Z_\circ, s))]$$

is holomorphic on a neighbourhood of the closed decoupling polydisc. It satisfies (1546). The same statement holds after the affine extension of the parameters over a guard. ■

Proof. The mean lies strictly inside the indicated set of centers. The integral and its estimate follow from proposition 75.1 at each s. To prove holomorphy, continue the finitely many entries of C, C^{-1}, Γ, m to a common open neighbourhood and symmetrize the continued S block in Q. Shrink the neighbourhood so that $\|Q - Q_{\mathrm{r}}\| < 2\hat{\varepsilon} + \lambda_{\mathrm{r}}/8$, $\|m\| < \mathfrak{b}_\Sigma$ and $|m(b)| < \vartheta' \mathfrak{b}_b$. It contains the closed polydisc by the strict mean bound and continuity. The same joint Gaussian calculation now uses $\hat{\varepsilon} + \lambda_{\mathrm{r}}/16$ in place of $\hat{\varepsilon}$. Its growth condition remains valid since

$$\hat{\varepsilon} + 3\rho + \lambda_{\mathrm{r}}/16 \le (1/108 + 1/16)\lambda_{\mathrm{r}} < \lambda_{\mathrm{r}}/8.$$

Thus there is a common integrable majorant on this neighbourhood. For every real B the integrand is holomorphic in s; outside D it is identically zero. Dominated integration proves the claim. Guard extension is affine and hence preserves holomorphy. The zero-dimensional case is the constant $G(0;0)$. □

75.2. **Local potentials and their Mayer terms.** Keep the fine and coarse cube partitions, enlargement $Y^\sharp$, guard and source-cube convention of Sections 57.3 and 59. Besides ordinary potentials $V^o(Y;B)$, bounded by $0 \le u^o(Y) \le 1$ on their complex small-field range, allow potentials $V^o(Y;z,\mu)$ holomorphic on

$$|z(b)| < \varsigma_{\mathrm{ce}} \mathfrak{b}_b, \qquad |\mu(b)| < \theta_{\mathrm{sl}} \mathfrak{b}_b, \qquad b \in \mathfrak{X}_{\mathrm{I}}^{\circ+}\langle Y \rangle. \tag{1550}$$

There are finitely many origins o, partitioned into ordinary and centered origins. Require

$$\vartheta_{\mathrm{ad}} = \min\{\varsigma_{\mathrm{ce}} - 1, \theta_{\mathrm{sl}}\} > \vartheta_*. \tag{1551}$$

The name centered refers to their later evaluation at $(z,\mu) = (B - m, m)$; it imposes no vanishing assumption. For each centered potential assume

$$q_Y(z) = M^{-3} \sum_{b \in \mathfrak{X}_{\mathrm{I}}^{\circ+}\langle Y \rangle} |z(b)|^2, \qquad |V^o(Y;z,\mu)| \le a^o(Y)(1 + q_Y(z)), \quad a^o(Y) \ge 0. \tag{1552}$$

Choose tilts $0 < \tau^o(Y) \le 1$ with

$$\sum_{o \text{ centered}} \sum_{Y \supset \square} (\tau^o(Y) + a^o(Y)) \le \tfrac{1}{2} L^{-3} \quad \text{for each fine cube } \square. \tag{1553}$$

All domains in these sums satisfy $Y^\sharp \subset \mathfrak{R}$. The weights must satisfy

$$w^o(Y) \ge e\, u^o(Y) \quad (o \text{ ordinary}), \qquad w^o(Y) \ge e\, a^o(Y)/\tau^o(Y) \quad (o \text{ centered}). \tag{1554}$$

Use the Mayer indices of the ordinary system with the same origins and zero potentials. Thus $\iota = (\mathbf{D}, \mathfrak{e})$ is a finite set of distinct potential vertices and failed bonds, $Y^{\mathrm{v}} = \bigcup_{(Y,o)\in\mathbf{D}} Y$, and no failed bond touches Y^{v}. The region $Z_\circ(\iota)$ is the union of the vertex domains and the fine bond cubes. Its integration part $Z_\circ^{\mathrm{I}}$ contains exactly the fine cubes with an endpoint of an integration bond. The guard $G_\circ$ consists of the coarse cubes in $Z_\circ^\sharp$ meeting Λ_t. One has $(Z_\circ^{\mathrm{I}})^\sharp \subset G_\circ \subset Z_\circ^\sharp$ and the same integration bonds for $Z_\circ$ and $Z_\circ^{\mathrm{I}}$, by Section 57.3.

For $|n(b)| < \vartheta_{\mathrm{ad}}\mathfrak{b}_b$, define

$$F^{\mathrm{ce}}_{\iota}(B;n) = (-1)^{\#\mathfrak{e}^+}\mathsf{in}_{\mathfrak{X}_1\langle Y^{\mathrm{v}}\rangle}(B)\mathsf{out}_{\mathfrak{e}}(B) \times \prod_{\substack{(Y,o)\in\mathbf{D}\\ o\ \text{ordinary}}} (e^{V^o(Y;B)}-1) \prod_{\substack{(Y,o)\in\mathbf{D}\\ o\ \text{centered}}} (e^{V^o(Y;B-n,n)}-1). \tag{1555}$$

A failed inner test makes this function zero before any potential is evaluated. Extend the free parameters by one on $G_\circ \setminus (Z^{\mathrm{I}}_\circ)^\sharp$, writing s^{gd}, and put

$$f^{\mathrm{ce}}_{\iota}(s) = \mathbb{J}^m(Z^{\mathrm{I}}_\circ; s^{\mathrm{gd}})[F^{\mathrm{ce}}_{\iota}(\cdot; m(Z^{\mathrm{I}}_\circ, s^{\mathrm{gd}}))]. \tag{1556}$$

For a finite set $\mathcal{K}$ of free cubes, set

$$T^{\mathrm{ce}}(\iota,\mathcal{K}) = \sum_{W\subset\mathcal{K}} (-1)^{\#\mathcal{K}-\#W} f^{\mathrm{ce}}_{\iota}(\mathbf{1}_W), \qquad X(\iota,\mathcal{K}) = Z^\sharp_\circ \cup \bigcup\mathcal{K}. \tag{1557}$$

Each summand uses its own conditional mean.

Put

$$\rho_+ = (LM)^{-3}, \qquad u_0^+ = u_{\rho_+}, \qquad c^+_{\mathrm{P}} = C_{\mathrm{mn}}(LM)^3 c_{\mathrm{mn}} u_0^+, \qquad \gamma_P = C^{-1}_{\mathrm{mn}}\gamma_2\mathfrak{b}_-^2. \tag{1558}$$

In the rest of this section require

$$C_{\mathrm{mn}}u_0^+ \le \lambda_{\mathrm{r}}. \tag{1559}$$

Proposition 75.3. Every term in (1557) is defined, and f^{ce}_{ι} is holomorphic near its closed polydisc. If $N_\Lambda(X)$ counts the coarse cubes of X meeting Λ_t, then

$$|T^{\mathrm{ce}}(\iota,\mathcal{K})| \le \bar{T}^+(\iota,\mathcal{K}) := e^{c^+_{\mathrm{P}}N_\Lambda(X)} \prod_{(Y,o)\in\mathbf{D}} w^o(Y) e^{-\gamma_P \#\mathfrak{e}^+} e^{-(\bar\kappa-1)\#\mathcal{K}}. \tag{1560}$$

The same bound without its last factor holds for $|f^{\mathrm{ce}}_{\iota}(s)|$ when s vanishes off $\mathcal{K}$. ∎

Proof. Remove the tests from (1555), retaining their sign, and call the resulting product $G_\iota(B;n)$. On the tube of lemma 75.2, every vertex variable obeys $|B(b)| \le |\operatorname{Re} B(b)| + |\operatorname{Im} B(b)| < \mathfrak{b}_b$. Thus $B-n$ and n lie in (1550). The product is admitted in B and holomorphic in n.

For an ordinary factor, $|e^V-1| \le e|V| \le w^o(Y)$. For a centered factor put $q = q_Y(B-n)$, $a = a^o(Y)$ and $\tau = \tau^o(Y)$. Its domain contains a source cube, so (1553) gives $a \le 1$. Since $1+q \le \tau^{-1}e^{\tau q}$,

$$|e^V-1| \le a(1+q)e^{a(1+q)} \le (ea/\tau)e^{(a+\tau)q} \le w^o(Y)e^{(a+\tau)q}. \tag{1561}$$

In the sum of the last exponents, a positive bond b can occur only for domains containing the unique fine cube of its initial site. Its coefficient is therefore at most $L^{-3}/2$ by (1553). For a complex Lie algebra element, $|z(b)|^2 \le 2(|\operatorname{Re} z(b)|^2 + |\operatorname{Im} z(b)|^2)$. Consequently

$$|G_\iota(B;n)| \le \prod_{(Y,o)\in\mathbf{D}} w^o(Y)\, e^{\rho_+\|B-n\|^2}. \tag{1562}$$

The factor $1/2$ in (1553) compensates for this comparison of the operator and coordinate norms.

Apply lemma 75.2 with $\rho = \rho_+$, the inner tests of Y^{v}, the failed set $\mathfrak{e}$, and $\vartheta' = \vartheta_{\mathrm{ad}}$. For s supported on $\mathcal{K}$, its guarded extension is supported on $G_\circ \cup \bigcup\mathcal{K} \subset X$. The bond count (1152) is $N_{\mathrm{I}}(Z) \le 18(LM)^3 N_\Lambda(Z)$; since $C_{\mathrm{mn}} \ge 18$, (1546) gives the asserted bound on f^{ce}_{ι}. The finite-difference Cauchy estimate on the free coordinates has factor $(e^{\bar\kappa}-1)^{-\#\mathcal{K}} \le e^{-(\bar\kappa-1)\#\mathcal{K}}$. This proves (1560). □

75.3. **Factorization and convergent cluster sums.** A pair $(\iota,\mathcal{K})$ is rooted if every wall component of $G_\circ \cup \bigcup \mathcal{K}$ contains a cube of $G_\circ$. For a coarse connected domain γ, define

$$\mathsf{z}^{\mathrm{ce}}(\gamma) = \sum_{\substack{(\iota,\mathcal{K}) \text{ rooted} \\ X(\iota,\mathcal{K})=\gamma}} T^{\mathrm{ce}}(\iota,\mathcal{K}). \tag{1563}$$

The sum is finite. Compatibility means that the coarse domains do not touch. With the corresponding truncated function ρ^T, put

$$E^{\mathrm{ce}}(X) = \sum_{n\ge 1} \frac{1}{n!} \sum_{\gamma_1\cup\cdots\cup\gamma_n = X} \rho^T(\gamma_1,\dots,\gamma_n) \prod_{i=1}^{n} \mathsf{z}^{\mathrm{ce}}(\gamma_i). \tag{1564}$$

Expanding each activity gives the refined series. Its part $E^{\mathrm{ce},\mathsf{Mk}}(X)$ through a specified set of vertices and failed bonds consists of the terms with at least one such vertex or bond. A marked bond set is closed under reversal.

Proposition 75.4. Under the preceding Gaussian and potential hypotheses, nonrooted terms vanish, and a term whose domain has wall components γ_i factors into the corresponding terms on those components. For a complete Gaussian structure on the torus, with whole covariance C_* and mean m_* as in (1327), the whole tested integral is

$$\begin{aligned} \Xi^{\mathrm{ce}} &= \int d\nu(B)\, \mathsf{in}_{\mathfrak{X}_\mathrm{I}}(B) \exp\left(\sum_{o \text{ ordinary},Y} V^o(Y;B) + \sum_{o \text{ centered},Y} V^o(Y;B-m_*,m_*) \right) \\ &= \sum_{\mathcal{P}'} \prod_{\gamma\in\mathcal{P}'} \mathsf{z}^{\mathrm{ce}}(\gamma). \end{aligned} \tag{1565}$$

Here ν is the normalized complex Gaussian with covariance C_* and mean m_*, the vertex sums run over domains containing a source cube, and $\mathcal{P}'$ runs over compatible polymer families, including the empty family. A zero test is read before the potential sum. ∎

Proof. The proof of Proposition 59.1 applies once the center is included in the restriction argument. If a component of the parameter cubes is disjoint from the integration guard, deleting it leaves C,Γ,m on the integration bonds unchanged. The potential product uses the same restricted mean. The alternating sum in any cube of that component therefore vanishes. This proves the nonrooted assertion.

When the domain splits into wall components, every enlarged vertex domain and failed-bond domain belongs to a unique component. The Gaussian covariance is a direct sum, the connecting operators use disjoint auxiliary coordinates, and the mean restricts to the component mean. Thus the centered arguments restrict as well. The integrands, determinant branches and finite differences factor. Absolute convergence from proposition 75.1 permits Fubini. Restriction of indices to the components is a bijection: conversely, their union preserves the failed-bond exclusions and rootedness. This proves the factorization of terms and of their sums.

For the whole integral, first freeze the center at m_*. On the real small-field domain each ordinary potential is bounded by its prescribed size. The centered potentials are bounded by their quadratic-growth estimates, since the thresholds are finite and the center is fixed. Their tested exponentials are therefore bounded Borel functions. The identity between the conditioning integral and the marginal of ν in Proposition 59.2 applies to each such function. Completeness gives $m(Z_\circ,\mathbf{1}) = m_*|_{\mathfrak{f}}$. Expand the finitely many factors into $1+(e^V-1)$ and the remaining tests into $1-\mathsf{out}$. The resulting functions are exactly $F^{\mathrm{ce}}_\iota(B;m_*|_{\mathfrak{f}})$. Their conditioning integrals at all parameters equal to one are $f^{\mathrm{ce}}_\iota(\mathbf{1})$. Summing all finite differences recovers this value, even though each difference has its own center. Remove the nonrooted terms and group by wall components. This gives (1565). □

Suppose, for rates $r_{\mathsf{w}}, r_H, \kappa'' \geq 0$,

$$\sum_o w^o(Y) \leq \varepsilon_{\mathsf{w}} e^{-r_{\mathsf{w}} d_k(Y)}. \tag{1566}$$

Let $c_{\mathrm{E}} = \log 18432$, $C_{\mathrm{E}} = 4096$ and $C_{\mathrm{cs}} \geq \max\{C'_{\mathrm{col}}, 24C_{\mathrm{E}}^2 e^{24}\}$, where $C'_{\mathrm{col}} = 13/9$ is the coarsening constant. Define

$$\begin{aligned} r^+_{\mathrm{pol}} &= r_H - C_{\mathrm{cs}}(c^+_{\mathrm{P}} + 1), & C_{\mathrm{pol}} &= C_{\mathrm{cs}} e^{C_{\mathrm{cs}}\kappa''}, \\ \varepsilon^+_H &= C_{\mathrm{cs}} e^{C_{\mathrm{cs}}(c^+_{\mathrm{P}} + r_H)}(L^3\varepsilon_{\mathsf{w}} + (LM)^3 e^{-\gamma_P}), & & \\ K^+_{\mathrm{mk}} &= C_{\mathrm{cs}} e^{C_{\mathrm{cs}}(c^+_{\mathrm{P}} + \kappa'')}, & K^+_{\mathrm{or}} &= C_{\mathrm{cs}}(LM)^3 e^{C_{\mathrm{cs}}(c^+_{\mathrm{P}} + r_H + \kappa'')}. \end{aligned} \tag{1567}$$

Require the four inequalities

$$\begin{aligned} (C_{\mathrm{cs}}/L) r_H + c_{\mathrm{E}} &\leq r_{\mathsf{w}}, & e^{-(\bar\kappa - 1) + C_{\mathrm{cs}} r_H} &\leq C_{\mathrm{cs}}^{-1}, \\ r^+_{\mathrm{pol}} &\geq \kappa'' + c_{\mathrm{E}} + C_{\mathrm{cs}}, & C_{\mathrm{pol}}\varepsilon^+_H &\leq \tfrac{1}{2}. \end{aligned} \tag{1568}$$

Define the nonnegative activity and refined cluster majorants by replacing each term T^{ce} with $\bar T^+$ and ρ^T with $|\rho^T|$. Write them as $\bar{\mathsf{z}}^+$, $\bar E^+$ and $\bar E^{\mathsf{Mk},+}$. Set

$$\begin{aligned} \eta^{\mathsf{Mk},+} = \sup_\Delta \Bigg[&\sum_{\substack{(Y,o)\in \mathsf{Mk} \\ \Delta \subset Y^\sharp}} w^o(Y) e^{r_H(d_t(Y^\sharp)+4)} \\ &+ \#\{b \in \mathsf{Mk}^+ : \Delta \subset b^\sharp\} e^{-\gamma_P + 4r_H} \Bigg]. \end{aligned} \tag{1569}$$

Theorem AE. Under (1559), (1553), (1554), (1566) and (1568), the refined series converges absolutely and

$$|\mathsf{z}^{\mathrm{ce}}(\gamma)| \leq \bar{\mathsf{z}}^+(\gamma) \leq \varepsilon^+_H e^{-r^+_{\mathrm{pol}} d_t(\gamma)}, \tag{1570}$$

$$|E^{\mathrm{ce}}(X)| \leq \bar E^+(X) \leq C_{\mathrm{pol}}\varepsilon^+_H e^{-\kappa'' d_t(X)}, \tag{1571}$$

$$|E^{\mathrm{ce},\mathsf{Mk}}(X)| \leq \bar E^{\mathsf{Mk},+}(X) \leq K^+_{\mathrm{mk}} \eta^{\mathsf{Mk},+} e^{-\kappa'' d_t(X)}. \tag{1572}$$

The sums vanish when no coarse cube of their domain meets Ω_t. For complete whole-torus data,

$$\Xi^{\mathrm{ce}} = \exp \sum_X E^{\mathrm{ce}}(X) \neq 0. \tag{1573}$$

These are assertions about this fluctuation integral. ■□

Proof. Use an auxiliary ordinary system with the same origins and all potentials zero, the weights $w^o(Y)$ of both kinds, and the larger Gaussian error $\hat\varepsilon + \rho_+$. The Gaussian structure still satisfies all its hypotheses: only upper bounds are enlarged. Its value of u_0 is exactly u_0^+, and its Gaussian volume constant is c^+_{P}. Its index sets, domains and majorants are precisely those just defined. The weight decay and the four numerical hypotheses are those of Theorem AA at these constants. The counting proof of that theorem therefore gives the three majorant bounds above. By proposition 75.3 the actual refined terms are dominated term by term by these majorants. Absolute convergence permits all regroupings, proving the activity, cluster and marked bounds. This use of the ordinary theorem concerns its majorant series, not its zero activities. The geometric vanishing statement follows from the same index sets. Finally proposition 75.4 identifies the whole integral with the compatible polymer sum, and the convergent cluster identity gives (1573). □

Proposition 75.5. The activities and cluster sums restrict to a smaller region when the Gaussian structure, thresholds and potential members restrict there. They depend only on the potential members with $Y^\sharp \subset X$ and the Gaussian data used by the corresponding indices. They are holomorphic in finite dimensional holomorphic families of these data satisfying the same bounds,

and Borel in Borel families. They inherit gauge invariance, conjugation, and dependence only on the represented pair whenever these properties hold for the Gaussian structure and the potential members. ■

Proof. Each term is a finite difference of conditional integrals. Restriction identifies their integration bonds, the guard, the local matrices and the restricted means. It consequently identifies the centered arguments, integrals and terms. Holomorphic parameter families give holomorphic terms by lemma 75.2; locally one may choose its open neighbourhood with a common integrable Gaussian majorant. The cluster majorant is independent of the external parameters, so the refined sums converge locally uniformly and remain holomorphic. For Borel families, the joint integrands are Borel and the same majorants give Borel parameter integrals and limits of partial sums. A gauge change acts by real orthogonal adjoint maps on the integration variables and carries the mean by the same maps. Their Jacobians, the tests and the Gaussian branch are unchanged. Complex conjugation conjugates the matrices, mean and potentials; the branch based at real positive matrices conjugates with them. These changes preserve the finite differences and the absolutely convergent sums. Finally, if the Gaussian data and potential members agree for two representatives of one pair, their integrals agree term by term. This last assertion presupposes the stated agreement; it does not supply it for arbitrary retained quadratic factors. □

76. The centered expansion on a history

We apply the preceding expansion at a step $t = k + 1$ of a fixed history h'. The Gaussian system is the canonical one of Theorem Z and Proposition 57.5. Its graded triple domain is $\tilde{U}_{\sigma_t^{\mathrm{mid}}}(Y)$, and $\mathfrak{Q}_h^{\mathrm{rad}}(Y)$ is the space of stored fields in their prescribed tiers. Define

$$\mathfrak{D}_k^{\mathrm{rad}}(Y) = \tilde{U}_{\sigma_t^{\mathrm{mid}}}(Y) \times \{\mathcal{A} : |g_k^{-1}\mathcal{A}(b)| < C_{\mathcal{A}}\mathfrak{b}_k\} \times \mathfrak{Q}_h^{\mathrm{rad}}(Y). \tag{1574}$$

The field inequality is imposed on all positively oriented bonds of $\mathfrak{X}^{\circ}\langle Y\rangle$. A centered potential has, in addition, an argument

$$\mu \in \mathbb{S}_k^{\mathrm{rad}}(Y) := \{\mu : |\mu(b)| < \theta_{\mathrm{sl}}\mathfrak{b}_k,\ b \in \mathfrak{X}_{\mathrm{I}}^{\circ+}\langle Y\rangle\}. \tag{1575}$$

An empty set of bonds contributes a one-point space. These domains use the running cutoff $\mathfrak{b}_k$, rather than the individual integration thresholds.

Analyticity on these domains means local holomorphic extension in $(A', \mathbf{J}, \mathcal{A}, \mu)$ at fixed real U and fixed stored argument. A parameter family is analytic when these extensions are jointly holomorphic in its complex parameters. Borel measurability is with respect to the trace of the Borel sigma algebra of the finite dimensional ambient product. The ordinary norm is the supremum norm with weight $e^{rd_k(Y)}$. The centered norm is

$$\|\mathbb{W}\|_{\mathrm{I}\star,r}^{\mathrm{rad}} = \sup_{Y,\mathfrak{u},\mu} \frac{e^{rd_k(Y)}|\mathbb{W}(Y;\mathfrak{u},\mu)|}{1 + M^{-3}\sum_{b\in\mathfrak{X}_{\mathrm{I}}^{\circ+}\langle Y\rangle} g_k^{-2}|\mathcal{A}(b)|^2}. \tag{1576}$$

Here $\mathfrak{u} = (P, \mathcal{A}, \mathfrak{q})$ ranges over (1574), and μ over (1575). In particular the centered norm is a supremum norm in the center, the external field and the stored argument. Its permitted growth is confined to the integration field.

For $X_1 \in \mathcal{D}_t$, write $\mathfrak{z} = (\mathsf{a}; P; \mathfrak{q})$ for a point with $P \in \tilde{U}_{\sigma_t^{\mathrm{mid}}}(X_1)$, $\mathfrak{q} \in \mathfrak{Q}_h^{\mathrm{rad}}(X_1)$ and $|\mathsf{a}(b)| < C_{\mathcal{A}}\mathfrak{b}_k$ on its external bonds. Require also the external mean bounds of Theorem Z, at parameters $(\theta_{\mathrm{xt}}, \varepsilon_{\mathrm{xt}})$. We denote this point set by $\mathsf{Pt}_{h'}\langle X_1\rangle$. In the canonical application these bounds are supplied by Proposition 57.5. For $Y^{\sharp} \subset X_1$, put

$$\mathfrak{u}_Y(z;\mathfrak{z}) = (P|_Y, g_k(z \oplus \mathsf{a}|_Y), \mathfrak{q}|_Y), \qquad \mathbb{W}\langle\mathfrak{z}\rangle^{\mathrm{ce}}(Y;z,\mu) = \mathbb{W}(Y;\mathfrak{u}_Y(z;\mathfrak{z}),\mu). \tag{1577}$$

The same substitution without μ evaluates an ordinary potential.

76.1. **Constants and admissible families.** Use C_{mn} and C_{cs} from the preceding section. The constants denoted by C_{mn} and C_{cs} in formulas for the history are these same constants. Set

$$\rho_+ = (LM)^{-3}, \qquad u_0^+ = u_0 + C_{\mathrm{mn}}\rho_+, \qquad c_{\mathrm{P}}^+ = c_{\mathrm{P}} + C_{\mathrm{mn}}^3\lambda_{\mathrm{r}}^{-1}. \tag{1578}$$

For the canonical Gaussian constants define

$$N_{\mathrm{gs}} = C_{\mathrm{res}}B_{\mathrm{nm}}^5\gamma_{\mathrm{pos}}^{-2} + c_{\mathrm{r}} + 2, \qquad \bar{c}_{\mathrm{P}}^+ = C_{\mathrm{mn}}^3\lambda_{\mathrm{r}}^{-1}(N_{\mathrm{gs}} + 1). \tag{1579}$$

The bound B_{nm} and the positive gap γ_{pos} are those of the real reference matrices in theorem Z. Here $C_{\mathrm{res}} = 64$, the coefficient in its Gaussian error bound, and we specialize

$$\gamma_2 = (LM)^{-3}, \qquad (\mathfrak{b}_-, \mathfrak{b}_+) = (\theta_c\mathfrak{b}_k, \mathfrak{b}_k), \qquad \theta_m = \theta_{\mathrm{xt}}, \qquad \varepsilon_m = \sqrt{2}\varepsilon_{\mathrm{dc}}\theta_{\mathrm{xt}} + (1 + \varepsilon_{\mathrm{dc}})\varepsilon_{\mathrm{xt}}.$$

For $\kappa > 2$, $0 < \delta_{\mathrm{sp}} < 1/2$ and $\beta_{\mathrm{s}} > 0$, put

$$r_V = (1 - 2\delta_{\mathrm{sp}})(\kappa - 2), \qquad \kappa'' = (1 + 4\beta_{\mathrm{s}})\kappa + 1 + c_{\mathrm{E}}. \tag{1580}$$

Define

$$\begin{aligned} r_{\mathsf{w}}^+ &= r_V - c_{\mathrm{E}}, \qquad r_H^+ = \kappa'' + c_{\mathrm{E}} + C_{\mathrm{cs}}(c_{\mathrm{P}}^+ + 2), \\ \bar{r}_H^+ &= \kappa'' + c_{\mathrm{E}} + C_{\mathrm{cs}}(\bar{c}_{\mathrm{P}}^+ + 2), \qquad K_{\mathrm{ap}}^+ = C_{\mathrm{cs}}^2 e^{C_{\mathrm{cs}}(\kappa'' + \bar{c}_{\mathrm{P}}^+ + \bar{r}_H^+)}. \end{aligned} \tag{1581}$$

The constants r_{pol}^+, K_{mk}^+ and K_{or}^+ are those of (1567) at $r_H = r_H^+$. The useful ceilings are

$$w_0 = (4C_{\mathrm{E}}L^3)^{-1}, \qquad \varepsilon^{\mathrm{O}} = (16eL^3K_{\mathrm{ap}}^+)^{-1}, \qquad c_M^{\mathrm{O}} = w_0\varepsilon^{\mathrm{O}}. \tag{1582}$$

We use the following strengthened history hypotheses. The geometric, graded-radius and covariance hypotheses of Theorem Z hold on the stated point sets; in particular the local Gaussian structure, its restrictions and its parameter dependence are defined there. In addition,

$$\begin{gathered} 1 \le \theta_c\mathfrak{b}_k \le \mathfrak{b}_b \le \mathfrak{b}_k, \qquad \varepsilon_{\mathrm{dc}} \le (LM)^{-3}, \\ \theta_m + \varepsilon_m \le \vartheta_* < 1, \qquad (LM)^3\varepsilon_m^2 \le 1, \qquad 1 + \vartheta_* < C_{\mathcal{A}}, \quad \vartheta_* < \theta_{\mathrm{sl}}, \\ (LM)^3 \ge C_{\mathrm{mn}}^{-1}\bar{c}_{\mathrm{P}}^+, \qquad L \ge \frac{2C_{\mathrm{cs}}(1 + 4\beta_{\mathrm{s}})}{1 - 2\delta_{\mathrm{sp}}}, \\ \tfrac{1}{2}(1 - 2\delta_{\mathrm{sp}})\kappa \ge 2(1 - 2\delta_{\mathrm{sp}}) + 2c_{\mathrm{E}} + \frac{C_{\mathrm{cs}}}{L}[1 + 2c_{\mathrm{E}} + C_{\mathrm{cs}}(\bar{c}_{\mathrm{P}}^+ + 2)], \\ \bar{\kappa} \ge 1 + C_{\mathrm{cs}}\bar{r}_H^+ + \log C_{\mathrm{cs}}, \qquad (LM)^3e^{-\gamma_P} \le (4K_{\mathrm{ap}}^+)^{-1}. \end{gathered} \tag{1583}$$

The first inequality is required even if there are no integration bonds. These are the strengthened Gaussian, range and rate conditions used below.

Definition 76.1. A frame with a center consists of finitely many analytic ordinary families $\mathbb{W}^o$, $o \in \mathsf{Or}^{\mathrm{f}}$, an onto map $p : \mathsf{Or}^{\mathrm{f}} \longrightarrow \mathsf{Or}^{\mathrm{c}}$, and finitely many analytic centered families indexed by a disjoint set $\mathsf{Or}^{\mathrm{ce}}$. Put

$$\mathbb{W}^{\mathrm{c},o'} = \sum_{p(o)=o'} \mathbb{W}^o, \qquad \mathbb{W}^{\mathrm{ce}} = \sum_{o \in \mathsf{Or}^{\mathrm{ce}}} \mathbb{W}^o.$$

It includes sizes $s^{o'} \in [0,1]$, $s^{\mathrm{ce}} \in [0, w_0]$ with

$$\|\mathbb{W}^{\mathrm{c},o'}\|_{r_V}^{\mathrm{rad}} \le s^{o'}, \qquad \|\mathbb{W}^{\mathrm{ce}}\|_{\mathrm{I}\star,r_V}^{\mathrm{rad}} \le s^{\mathrm{ce}}, \qquad \varepsilon_{\mathsf{w}}^{\mathfrak{F}+} := e\sum_{o'} s^{o'} + ew_0^{-1}s^{\mathrm{ce}} \le (4K_{\mathrm{ap}}^+L^3)^{-1}.$$

If there is no centered origin, require $s^{\mathrm{ce}} = 0$. The fine system consists of all the members. The coarse system consists of the ordinary sums $\mathbb{W}^{\mathrm{c},o'}$ and, when present, the single centered sum $\mathbb{W}^{\mathrm{ce}}$. The individual members need not satisfy the bounds on these sums. ∎

Define the coarse-system weights by

$$\begin{aligned} u^{o'}(Y) &= s^{o'} e^{-r_V d_k(Y)}, & w^{o'}(Y) &= e u^{o'}(Y), \\ a^{\mathrm{ce}}(Y) &= s^{\mathrm{ce}} e^{-r_V d_k(Y)}, & \tau^{\mathrm{ce}}(Y) &= w_0 e^{-c_{\mathrm{E}} d_k(Y)}, \\ w^{\mathrm{ce}}(Y) &= e w_0^{-1} s^{\mathrm{ce}} e^{-r_{\mathsf{w}}^+ d_k(Y)}, & b_{\mathrm{cs}}^{\mathfrak{F}+} &= C_{\mathrm{pol}} \varepsilon_H^{\mathfrak{F}+}, \end{aligned} \tag{1584}$$

where $\varepsilon_H^{\mathfrak{F}+}$ is (1567) at $r_H = r_H^+$ and amplitude $\varepsilon_{\mathsf{w}}^{\mathfrak{F}+}$.

Proposition 76.2. Under (1583), the coarse system of every frame with a center, evaluated at any $\mathfrak{z} \in \mathsf{Pt}_{h'}\langle X_1\rangle$, satisfies the hypotheses of theorem AE. The fine system has the same Gaussian data and defined analytic conditional integrals. Moreover

$$b_{\mathrm{cs}}^{\mathfrak{F}+} \le K_{\mathrm{ap}}^+ [L^3 \varepsilon_{\mathsf{w}}^{\mathfrak{F}+} + (LM)^3 e^{-\gamma_P}] \le \tfrac{1}{2}. \tag{1585}$$

The bounds and majorants are independent of the point and of the individual thresholds within their stated ranges. ■

Proof. For an integration argument $|z(b)| < C_{\mathcal{A}} \mathfrak{b}_b$, the inequality $\mathfrak{b}_b \le \mathfrak{b}_k$ and the external field bound place $\mathfrak{u}_Y(z;\mathfrak{z})$ in (1574). Similarly $|\mu(b)| < \theta_{\mathrm{sl}} \mathfrak{b}_b$ puts the center in (1575). The substitution is linear in the fields. To check joint analyticity, extend μ by zero to the full set of integration bonds, treat that extension as an extra parameter, and compose the local holomorphic extension with restriction. This gives a jointly holomorphic function on the required product domain. The norm (1576) now gives

$$|\mathbb{W}^{\mathrm{ce}}\langle\mathfrak{z}\rangle^{\mathrm{ce}}(Y; z, \mu)| \le a^{\mathrm{ce}}(Y)(1 + q_Y(z)),$$

since $\mathcal{A} = g_k z$ at the integration bonds.

The entropy estimate of Proposition 45.1 gives

$$\sum_{Y \supset \square} [\tau^{\mathrm{ce}}(Y) + a^{\mathrm{ce}}(Y)] \le C_{\mathrm{E}}(w_0 + s^{\mathrm{ce}}) \le \tfrac{1}{2} L^{-3}.$$

Here $r_V \ge c_{\mathrm{E}}$ follows from (1583). Thus the tilts fit, and $w^{\mathrm{ce}} = e a^{\mathrm{ce}}/\tau^{\mathrm{ce}}$. The ordinary bounds are immediate from their norms. Both types of weights are bounded in sum by $\varepsilon_{\mathsf{w}}^{\mathfrak{F}+} e^{-r_{\mathsf{w}}^+ d_k(Y)}$.

The Gaussian error formula and the mean bound give

$$u_0^+ \le C_{\mathrm{mn}} (LM)^{-3} (N_{\mathrm{gs}} + 1), \qquad c_{\mathrm{P}}^+ \le \bar{c}_{\mathrm{P}}^+.$$

Consequently $C_{\mathrm{mn}} u_0^+ \le \lambda_{\mathrm{r}}$ by the lower bound on $(LM)^3$. The inequalities for L and κ give

$$\frac{C_{\mathrm{cs}}}{L} r_H^+ + c_{\mathrm{E}} \le \frac{1}{2}(1 - 2\delta_{\mathrm{sp}})\kappa + c_{\mathrm{E}} + \frac{C_{\mathrm{cs}}}{L}[1 + 2c_{\mathrm{E}} + C_{\mathrm{cs}}(\bar{c}_{\mathrm{P}}^+ + 2)] \le r_V - c_{\mathrm{E}}.$$

The bound on $\bar{\kappa}$ gives the second rate inequality in (1568), and $r_{\mathrm{pol}}^+ = \kappa'' + c_{\mathrm{E}} + C_{\mathrm{cs}}$ gives the third. Finally $c_{\mathrm{P}}^+ \le \bar{c}_{\mathrm{P}}^+$ and $r_H^+ \le \bar{r}_H^+$ bound the exponential constant by K_{ap}^+, proving (1585) and the last rate inequality. The center is admitted because $\vartheta_* < \min(C_{\mathcal{A}} - 1, \theta_{\mathrm{sl}})$. The canonical Gaussian hypotheses supply integrability, restriction and completeness. These checks establish every input to the abstract expansion.

For the fine system, each individual centered member is holomorphic on the same domain, and each conditional integral uses finitely many members on a compact subset of that domain. The Gaussian integrability argument therefore applies. No separate uniform size for an individual member is claimed. All majorant formulas use the coarse sizes and fixed Gaussian constants; none uses the point or an individual threshold. □

76.2. **Expansion and perturbation estimates.**

Theorem AF. Under the hypotheses of proposition 76.2, the fine and coarse systems have the same polymer activities and cluster sums. Write the latter as $E[\mathfrak{F}^+]_{h'}(X|X_1;\mathfrak{z})$. They satisfy

$$|E[\mathfrak{F}^+]_{h'}(X|X_1;\mathfrak{z})| \le \bar{E}^+(X) \le b_{\mathrm{cs}}^{\mathfrak{F}+} e^{-\kappa'' d_t(X)}. \tag{1586}$$

They vanish if X contains no coarse cube meeting Ω_t. Restriction to X leaves the cluster sum at X unchanged. Its value depends only on the potential members with $Y^\sharp \subset X$ and the restricted Gaussian data and thresholds. On the whole torus,

$$\begin{aligned}\int d\nu(B)\, \mathsf{in}_{\mathfrak{X}_{\mathrm{I}}}(B) \exp\Big[\sum_{Y,o\in \mathsf{Or}^{\mathrm{f}}} \mathbb{W}^o\langle\mathfrak{z}\rangle(Y;B) + \sum_{Y,o\in \mathsf{Or}^{\mathrm{ce}}} \mathbb{W}^o\langle\mathfrak{z}\rangle^{\mathrm{ce}}(Y;B-m_*,m_*)\Big] \\ = \exp\sum_X E[\mathfrak{F}^+]_{h'}(X;\mathfrak{z}|_X) \ne 0.\end{aligned} \tag{1587}$$

The sums over Y contain the potential vertices of the expansion, and the integrand is zero before evaluating its exponent wherever a small-field test fails.

With fixed sizes, the activities and cluster sums are holomorphic in holomorphic parameter families and Borel in Borel families. Gauge invariance and conjugation are inherited from the members. At fixed external field a and stored argument $\mathfrak{q}$, equal represented pairs give equal values if the members depend only on the pair. It suffices instead that the two triples are real and the members have equal values at twins. The marked estimate (1572) holds for the coarse system. ■□

Proof. Apply theorem AE and proposition 75.5 to the coarse data admitted by proposition 76.2. To identify fine and coarse activities, group the ordinary origins over each fiber of p and all centered origins into one. At any fixed domain and fixed conditional mean, the finite algebraic identity

$$e^{\sum_{j=1}^n V_j} - 1 = \sum_{\emptyset\ne I\subset\{1,\dots,n\}} \prod_{j\in I} (e^{V_j}-1)$$

expands each merged vertex. Replacing it by the chosen fine vertices changes neither the union of their domains, the guard, the failed-bond exclusions nor the conditional mean. Thus the conditional integrals and finite differences agree after this finite regrouping, at every decoupling parameter. Rootedness and connected domains are preserved. Summing gives equal activities, hence equal convergent cluster sums. This argument requires bounds only on the merged system.

The restriction identities of the canonical Gaussian system are given in proposition 57.5, equation (1169). Restriction also identifies $\mathfrak{u}_Y$ and the center arguments. Therefore the abstract restriction result applies term by term, including to the majorants. Completeness identifies the whole mean with each local mean at full coupling, and theorem AE gives (1587). The finite grouping leaves its integrand unchanged.

For parameter dependence the substitutions in (1577) are jointly holomorphic at fixed real U and fixed stored argument. The Gaussian structure has this same property by theorem Z. The common cluster majorant therefore proves holomorphy of the sum. The substitutions are continuous in ambient coordinates, so they also preserve trace Borel measurability. Gauge changes rotate z and μ by the same real orthogonal adjoint map; conjugation acts on both arguments and on the Gaussian mean. The abstract covariance statements apply. Finally, the canonical Gaussian matrices and mean depend only on the pair. For two triples with that pair, the points in (1577) differ only in the triple. The asserted equality of the potential values therefore identifies every integral. In the real case the triples are twins, so the tie suffices. No such conclusion for arbitrary complex twins is used here. □

Proposition 76.3. Split the ordinary coarse origins into old and new ones, and regard every centered origin as new. Let $\mathfrak{F}_0$ be the old ordinary subsystem and set

$$n_+ = e \sum_{o' \text{ new}} s^{o'} + e w_0^{-1} s^{\mathrm{ce}}, \qquad K_{\mathrm{Lip}}^+ = 2C_{\mathrm{E}} L^3 e^{4r_H^+} K_{\mathrm{mk}}^+. \tag{1588}$$

Then $E[\mathfrak{F}^+] - E[\mathfrak{F}_0]$ is exactly the part of the refined cluster series meeting a new coarse vertex, and

$$|E[\mathfrak{F}^+]_{h'}(X) - E[\mathfrak{F}_0]_{h'}(X)| \le \min(K_{\mathrm{or}}^+, K_{\mathrm{Lip}}^+) n_+ e^{-\kappa'' d_t(X)}. \tag{1589}$$

If the new members vanish for every $Y^\sharp \subset X$, the difference at X vanishes.

For two frames with the same old members and the same named sizes, suppose each new coarse sum has at most half its named size. Let $d^{o'}$ and d^{ce} be the corresponding norms of the differences of those sums. Put

$$n_{12} = e \sum_{o' \text{ new}} d^{o'} + e w_0^{-1} d^{\mathrm{ce}}, \qquad n = \max\left\{ \max_{o' \text{ new}} \frac{d^{o'}}{s^{o'}}, \frac{d^{\mathrm{ce}}}{s^{\mathrm{ce}}} \right\}. \tag{1590}$$

Here $0/0 = 0$ and an empty maximum is zero. Then $0 \le n \le 1$, and

$$|E[\mathfrak{F}_1^+]_{h'}(X) - E[\mathfrak{F}_2^+]_{h'}(X)| \le 4 \min(K_{\mathrm{or}}^+, K_{\mathrm{Lip}}^+) n_{12} e^{-\kappa'' d_t(X)}, \tag{1591}$$

$$|E[\mathfrak{F}_1^+]_{h'}(X) - E[\mathfrak{F}_2^+]_{h'}(X)| \le 4 b_{\mathrm{cs}}^{\mathfrak{F}+} n e^{-\kappa'' d_t(X)} \le 2n e^{-\kappa'' d_t(X)}. \tag{1592}$$

■

Proof. In the refined series a term without a new vertex is a term of the old subsystem. Removing all such terms leaves exactly the marked part. The origin-class bound in the counting proof of Theorem AA, with the same majorants as in theorem AE, gives the constant K_{or}^+. For the second bound, coarsening and the first inequality in (1568) give

$$e^{-r_{\mathrm{w}}^+ d_k(Y)} e^{r_H^+ d_t(Y^\sharp)} \le e^{-c_{\mathrm{E}} d_k(Y)}.$$

For a coarse cube Δ, at most $(L+2)^3$ fine cubes can anchor a Y with $\Delta \subset Y^\sharp$. The entropy bound gives

$$\sum_{Y : \Delta \subset Y^\sharp} e^{-c_{\mathrm{E}} d_k(Y)} \le (L+2)^3 C_{\mathrm{E}} \le 2L^3 C_{\mathrm{E}}.$$

There are no marked failed bonds here. Thus $\eta^{\mathrm{new},+} \le 2C_{\mathrm{E}} L^3 e^{4r_H^+} n_+$. The marked estimate proves (1589). If the new members vanish locally, every marked Mayer factor is zero.

Write $D = \min(K_{\mathrm{or}}^+, K_{\mathrm{Lip}}^+) e^{-\kappa'' d_t(X)}$. If $n_{12} = 0$, the coarse systems coincide. Otherwise set $R = n_+/(2n_{12})$. If $R \le 1$, the triangle inequality through the common old subsystem gives $|E_1 - E_2| \le 2Dn_+ \le 4Dn_{12}$. For $R > 1$, interpolate the new members by $\mathbb{W}_1 + \lambda(\mathbb{W}_2 - \mathbb{W}_1)$, $|\lambda| < R$. Admit this family with sizes $s^{o'}/2 + Rd^{o'}$ and $s^{\mathrm{ce}}/2 + Rd^{\mathrm{ce}}$. Their total amplitude is n_+; each ordinary size is at most $n_+/e \le 1$, and the centered size is at most $w_0 n_+/e \le w_0$. The frame bound is therefore unchanged. The cluster sum is holomorphic in λ. Its difference from its value at zero has modulus at most $2Dn_+$ throughout the disk. Schwarz's lemma at $\lambda = 1$ gives $2Dn_+/R = 4Dn_{12}$.

For the relative bound, if $n \ge 1/2$, the triangle inequality and (1586) give

$$|E_1 - E_2| \le 2 b_{\mathrm{cs}}^{\mathfrak{F}+} e^{-\kappa'' d_t(X)} \le 4 b_{\mathrm{cs}}^{\mathfrak{F}+} n e^{-\kappa'' d_t(X)}.$$

If $0 < n < 1/2$, the same interpolation is admitted at the original named sizes for $|\lambda| < (2n)^{-1}$. Apply Schwarz's lemma with the uniform bound $2 b_{\mathrm{cs}}^{\mathfrak{F}+} e^{-\kappa'' d_t(X)}$. This proves (1592); the case $n = 0$ was already covered. □

77. The source perturbation on one lattice

Inside the small-field region the history expansion agrees with the expansion on one lattice. We need this agreement with an extra potential, together with a bound on the change of the pointed interaction. The Gaussian data are unchanged.

Use the one-lattice Gaussian structure of Definition 47.2, with finite threshold $\mathfrak{b}$, at $t = k+1$, $0 \le k < K$. Assume $L \ge 13$ is odd, $M = L^{m'}$, $1 \le m' < m + K - k - 1$, and let R be a nonempty union of coarse cubes. Write $\mathfrak{X}$ for the nonpivot bonds. For $Y^\sharp \subset R$ put

$$\mathbb{D}^{\mathrm{f}}(Y) = \{B : |B(b)| < C_{\mathcal{A}}\mathfrak{b}\}, \qquad Q_Y(B) = 1 + M^{-3} \sum_{b \in \mathfrak{X}^{\circ+}\langle Y\rangle} |B(b)|^2, \qquad C_{\mathcal{A}} \ge 3,$$

and use the norm $\|V\|^{\mathrm{f}}_{\star,r} = \sup_{Y,B} e^{r d_k(Y)} |V(Y;B)|/Q_Y(B)$. The Gaussian constants and the quantities $u, \lambda_1, \lambda_V, w_\star, K_\Pi, K_P^\Pi, \mathsf{p}_\star$ are those of (705). Require

$$u \le C_{\mathrm{ex}}^{-1} \min(\gamma_{\mathrm{r}}, c_{\mathrm{r}}^{-2}), \qquad \bar\kappa \ge 4\lambda_1 + 1, \qquad \mathsf{p}_\star \le \tfrac{1}{2}. \tag{1593}$$

Thus the Gaussian and numerical hypotheses of Theorems R–T hold. All constants in this section refer to this one set of Gaussian data.

Let $V^{\mathrm{J}}, V^{\mathrm{G}}, V^{\mathrm{A}}, V^{\mathrm{iv}}$ be holomorphic on the field polydiscs. Require $V^{\mathrm{J}}(0) = V^{\mathrm{iv}}(0) = 0$, and require $V^{\mathrm{G}}, V^{\mathrm{A}}$ to vanish through degree two. Set

$$n_{\mathrm{int}} = \|V^{\mathrm{J}}\|^{\mathrm{f}}_{\star,\lambda_V} + \|V^{\mathrm{G}}\|^{\mathrm{f}}_{\star,\lambda_V} + \|V^{\mathrm{A}}\|^{\mathrm{f}}_{\star,\lambda_V}, \qquad n_{\mathrm{iv}} = \|V^{\mathrm{iv}}\|^{\mathrm{f}}_{\star,\lambda_V},$$

and suppose both are at most $w_\star$. The interpolation and bond insertions are those of Proposition 48.4 at $\tau = 1$. Denote its path by $\underline{V}_v = \{V_v^{\mathrm{int}}, V_v^{\mathrm{iv}}\}$ and its bond insertion by $\dot V_{v,b}$. Define

$$E^\bullet(X,b) = \int_0^1 E'[\underline{V}_v; \dot V_{v,b}](X)\, dv, \qquad \mathcal{E}(X,b) = E^\chi[\emptyset](X,b) + E^\bullet(X,b). \tag{1594}$$

Here $E^\chi[\emptyset]$ is the cutoff contribution of the empty potential system. It depends only on the Gaussian data.

Lemma 77.1. The integrals in (1594) are defined, and both terms vanish unless $b^\sharp \subset X$. Moreover

$$E[\{V^{\mathrm{J}} + V^{\mathrm{G}} + V^{\mathrm{A}}, V^{\mathrm{iv}}\}](X) = \sum_{b \in \mathfrak{X}^+} \mathcal{E}(X,b), \tag{1595}$$

$$|E^\bullet(X,b)| \le 9 K_\Pi (n_{\mathrm{int}} + n_{\mathrm{iv}}) e^{-\lambda_{\mathrm{out}} d_t(X)}, \tag{1596}$$

$$|E^\chi[\emptyset](X,b)| \le K_P^\Pi e^{4\lambda_1} e^{-\gamma_2 \mathfrak{b}^2/2} e^{-\lambda_{\mathrm{out}} d_t(X)}. \tag{1597}$$

These objects are holomorphic in holomorphic families of the Gaussian data and potentials with the same bounds. They preserve orthogonal covariance and conjugation. ∎

Proof. Proposition 48.4 gives an admitted path with two origins, bounded velocities, and bond insertions of norm at most $9(n_{\mathrm{int}} + n_{\mathrm{iv}})$. The insertion theorem and pointed-path identity in Section 48 therefore give a bounded Borel integrand, the estimate (1596), and

$$\sum_b E^\bullet(X,b) = E[\underline{V}_1](X) - E[\underline{V}_0](X).$$

The two zero potentials in $\underline{V}_0$ can be removed: the added-potential bound in Theorem R has right side zero. The cutoff identity of Theorem T gives $E[\emptyset](X) = \sum_b E^\chi[\emptyset](X,b)$, since at infinite threshold its empty-system cluster sum is bounded by zero. This proves (1595). Theorem T also gives (1597). The pointed insertion and cutoff terms vanish unless the marked bond cube lies in X. The same theorems give holomorphy of the integrands with common majorants, so dominated integration gives holomorphy of $E^\bullet$. The interpolation is linear in the four input families and commutes with rotations and conjugation. The Gaussian change of variables proves the remaining assertions. □

Let h_z be the tents of Proposition 49.6 and set

$$F(X,z)=\sum_{b\in\mathfrak{X}^+}h_z(b)\mathcal{E}(X,b).$$

The tents are nonnegative. Their total over all fine bonds is $3L^3$; each of the six coarse bonds incident to z has a different pivot bond and contributes $1/2$ there. Thus

$$\sum_{b\in\mathfrak{X}^+}h_z(b)=3L^3-3=:N_L. \tag{1598}$$

The distinctness follows because a fine bond belongs to at most one coarse corridor. The support property of the tents and $m'\ge 1$ show that $F(X,z)=0$ for $z\notin X$ and

$$\sum_{z\in X\cap\mathbb{T}^{(t)}}F(X,z)=E[\{V^{\mathrm{J}}+V^{\mathrm{G}}+V^{\mathrm{A}},V^{\mathrm{iv}}\}](X). \tag{1599}$$

Proposition 77.2. Let W be holomorphic on the same field polydiscs, with $W(0)=0$ and $n_W=\|W\|^{\mathrm{f}}_{\star,\lambda_V}$. Write F^+ for the pointed term with fourth family $V^{\mathrm{iv}}+W$. If $n_{\mathrm{iv}}<w_\star$ and $n_{\mathrm{iv}}+n_W\le w_\star$, then

$$|F^+(X,z)-F(X,z)|\le N_L\frac{18K_\Pi(n_{\mathrm{int}}+w_\star)}{w_\star-n_{\mathrm{iv}}}n_We^{-\lambda_{\mathrm{out}}d_t(X)}. \tag{1600}$$

If $n_{\mathrm{iv}},n_W\le w_\star/2$, one also has

$$|F^+(X,z)-F(X,z)|\le 9N_LK_\Pi\left(1+\frac{8(n_{\mathrm{int}}+n_{\mathrm{iv}})}{w_\star}\right)n_We^{-\lambda_{\mathrm{out}}d_t(X)}. \tag{1601}$$

■

Proof. The fourth family $V^{\mathrm{iv}}+\lambda W$ is admitted for $|\lambda|\le R=(w_\star-n_{\mathrm{iv}})/n_W$ when $n_W>0$. For a fixed bond, its $E^\bullet$ term is holomorphic on $|\lambda|<R$ and bounded by $A=9K_\Pi(n_{\mathrm{int}}+w_\star)e^{-\lambda_{\mathrm{out}}d_t(X)}$. If $R=1$, the triangle inequality bounds its difference at 1 and 0 by $2A$. If $R>1$, Schwarz's lemma applied to that difference gives $2A/R$. The cutoff term is unchanged. Multiplication by the tent mass N_L proves (1600). If $n_W=0$, the difference is zero.

For (1601), linearity of the interpolation gives $\underline{V}^+_v=\underline{V}_v+_{\mathrm{iv}}W_v$ and $\dot V^+_{v,b}=\dot V_{v,b}+\dot W_{v,b}$, with $\|W_v\|\le n_W$ and $\|\dot W_{v,b}\|\le 9n_W$. Subtract the two insertion terms. The change in the insertion is bounded by $9K_\Pi n_We^{-\lambda_{\mathrm{out}}d_t(X)}$. The change in the system, with its insertion fixed, is bounded by the insertion Lipschitz estimate of Theorem S by

$$\frac{8K_\Pi}{w_\star}\,9(n_{\mathrm{int}}+n_{\mathrm{iv}})n_We^{-\lambda_{\mathrm{out}}d_t(X)}.$$

The half-size assumptions are exactly those needed for this estimate. Integrate over $v\in(0,1)$ and sum with the tents. □

77.1. **Identification with the history terms.** An escort on one lattice is a family $W^{11}(Y;P,B)$, holomorphic on the full analytic triple core and on $\mathbb{D}^{\mathrm{f}}(Y)$, vanishing at $B=0$. Its norm $\|W^{11}\|^{\mathrm{co}}_{\star,\lambda_V}$ is the supremum of the preceding weighted quadratic norm also over that core. Write this norm as θ.

Proposition 77.3. Suppose $k\ge 1$ and the unsourced one-lattice step is defined on the full analytic core with the Gaussian hypotheses (1593). Suppose its three internal families and its regular fourth family satisfy

$$n_{\mathrm{int}}\le 3K_P\delta_k\le\tfrac{3}{8}w_\star,\qquad n_{\mathrm{iv}}\le K_\Lambda s_{\mathfrak{a}}N[\mathscr{L}]\le\tfrac14 w_\star.$$

Here $s_{\mathfrak{a}}=(1+\beta_{\mathrm{s}})\delta_k/\alpha_0$, and $N[\mathscr{L}]$ is the weighted size of the old regular list. If

$$K_\Lambda s_{\mathfrak{a}}N[\mathscr{L}]+\theta\le w_\star, \tag{1602}$$

then the step with fourth family $V' + W^{1l}$ is defined on the same core and

$$|F^+(X,P,z) - F(X,P,z)| \le 33(3L^3-3)K_\Pi \theta e^{-\lambda_{\mathrm{out}} d_t(X)}. \tag{1603}$$

It obeys the sum rule (1599), is holomorphic in the triple and source parameters, and inherits gauge invariance, pair dependence and conjugation from the escort. For a history region $X \subset \Lambda_t$, the same pointed term is obtained whenever the unperturbed history potentials agree there with their one-lattice counterparts and the history escort restricts to W^{1l}. ■

Proof. At fixed P, restriction of the escort to $(P|_Y, B)$ is holomorphic on the field polydiscs and has norm at most θ. The fourth-family room (1602) therefore admits the data of lemma 77.1. In (1600),

$$18\frac{n_{\mathrm{int}} + w_\star}{w_\star - n_{\mathrm{iv}}} \le 18\frac{1+3/8}{1-1/4} = 33.$$

This proves the bound. The sum rule and the parameter properties follow from lemma 77.1. Pair dependence identifies the Gaussian matrices and all four potential families, hence every term. For conjugation one uses the conjugate escort and conjugate triple together.

On $X \subset \Lambda_t$, the interior comparison in Proposition 59.5 identifies the integration bonds, matrices, thresholds and ordinary potentials with those on one lattice. Under the stated escort agreement the fourth family is also identical. The conditional integrals, bond insertions and tents therefore give the same pointed term. This comparison uses the full equality of the old regular potentials, including their domains; it is not inferred from an equality of their values at one field. □

Proposition 77.4. At the first step $k = 0$, suppose the same Gaussian and analyticity hypotheses hold with $V' = 0$ and $n_{\mathrm{int}} \le 3w_\star/8$. Let W^{1l} be an escort with $\theta \le w_\star/2$. Then

$$|F^+(X,P,z) - F(X,P,z)| \le 36(3L^3-3)K_\Pi \theta e^{-\lambda_{\mathrm{out}} d_1(X)} = 108(L^3-1)K_\Pi \theta e^{-\lambda_{\mathrm{out}} d_1(X)}. \tag{1604}$$

Suppose also that a centered history frame is admitted on $X \subset \Lambda_1$, its local thresholds are $\mathfrak{b}_0$, and its ordinary and centered additional members, evaluated at zero center, sum to this escort on all the field polydiscs. Then

$$E[\mathfrak{F}^+]_{h'}(X;\emptyset;P;\mathfrak{q}) = \sum_{z \in X \cap \mathbb{T}^{(1)}} F^+(X,P,z). \tag{1605}$$

The centered frame is assumed to contain the canonical first-step system, whose inherited regular and boundary families are empty. The same identities hold in holomorphic source families with these bounds. ■

Proof. Use (1601) with $n_{\mathrm{iv}} = 0$: $9(1 + 8n_{\mathrm{int}}/w_\star) \le 36$. Inside Λ_1 there is no external field on X. The conditional mean is therefore zero for every decoupling parameter. Substituting this value in every centered Mayer factor replaces $V(Y; B-m, m)$ by $V(Y; B, 0)$. The resulting ordinary system has exactly the same terms and activities. The interior comparison of Proposition 59.5 identifies its Gaussian system and thresholds with those on one lattice. Merge the additional members into the escort, using the finite exponential identity in the proof of theorem AF. The remaining families are J, G and A, since the inherited ones are empty at $k = 0$. Equation (1599) proves (1605). All identifications hold at every point of the specified domains, so they also hold in the stated parameter families. □

78. Weighted potentials and their centered expansion

A plaquette source contributes a quadratic form at the conditional mean and a potential for the centered integration field. The latter must retain the oscillation of the weight: a bound by its supremum alone would lose the small factor in the linear term. We obtain both parts from the localized Wilson action.

78.1. **The localized weight and its Taylor coefficients.** Fix $1 \le k < K$, $t = k+1$, a history h', and the canonical shift system of the step. All its localization domains, restrictions and radii are those of Section 58. For a weight w on the bare plaquettes, replace the Wilson member of layer m by

$$W_w^{(m)}(X,\mathbf{U}) = L^k \sum_{p \in \mathsf{Pl}(m,X)} w(p) a_p(\mathbf{U}).$$

Here $\mathsf{Pl}(m,X)$ consists of the plaquettes with base in layer m and smallest level-m domain $X(p;m) = X$. Let $\mathsf{Pl}\langle Y\rangle$ be their union over $m \le k$ and $X^\uparrow \subset Y$, and write $|w|_Y = \sup_{\mathsf{Pl}\langle Y\rangle} |w|$.

The localized potential is the finite difference

$$\mathbb{V}^w(Y;P,\mathcal{A}) = \sum_{m=0}^{k} \sum_{X^\uparrow \subset Y} \sum_{D \subset Y \setminus X^\uparrow} (-1)^{m(X,Y)-\#D} \big[W_w^{(m)}(X, \mathbf{U}_{D \cup X^\uparrow}) - W_w^{(m)}(X,\mathbf{U}) \big], \tag{1606}$$

where D is a set of level-k cubes, $m(X,Y) = \#_{\pi_k}(Y \setminus X^\uparrow)$, $\mathbf{U} = e^{i\eta_t A'} U_{\mathrm{r}}$, and $\mathbf{U}_D = e^{i\eta_k \mathbf{A}_Y(1_D;P,\mathcal{A})}\mathbf{U}$. Every term is read only on X. The domain of (1606) is the domain of the localized shift $(P,\mathcal{A})$, including when the weight or the sum is zero. Stored fields do not enter it.

For positive radii $a_m \le 1$, let $\mathbb{U}_k^a(Y|h')$ consist of triples for which the shift is defined for every $\sup|\mathcal{A}| < \varepsilon_{\mathfrak{h}}$ and

$$|\partial \mathbf{U}(p) - 1| < a_m L^{-2m}$$

at every inner bare plaquette of layer m whose bonds lie in Y. This specifies the domains used below. Set $c_E = \log 18432$ and put

$$\delta_* = M \min(\mu_{\mathfrak{h}}, 1), \qquad r = (1-2\delta)(\kappa_* - 2), \qquad r_0 = \min\{\varepsilon_{\mathfrak{h}}, (3c_{\mathfrak{h}})^{-1}\},$$

and assume

$$0 < \delta < \tfrac{1}{2}, \quad \kappa_* > 0, \quad 2\delta\kappa_* \ge 2(c_E+1), \quad \bar\kappa - 1 \ge \kappa_*, \quad \delta_* L \ge 20 \log L. \tag{1607}$$

Let ϖ_i^c and ϖ_i^w be the coefficients of equation (1283), with $\nu = 1$, $a = 2\delta\kappa_*$, and distance rates $\delta_*/2$ and δ_*, respectively. Equivalently, they are the coefficients ϖ_i in the placed-domain estimate of Lemma 58.19 at those rates. Set

$$K_c = 12(1+2c_{\mathrm{pl}})^2 C_h(L,\delta,1), \quad c_{\mathrm{pl}} = 600, \qquad K_0 = K_c C_h'(L,\delta,1) M^3,$$

$$a^\Sigma = \sum_{m=0}^{k} \varpi_{k-m}^w L^{k-m} a_m.$$

The constants C_h, C_h' are those of the same counting lemma; in particular $C_h'(L,\delta,1) = 12L^4$.

Proposition 78.1. For $P \in \mathbb{U}_k^{\mathbf{1}}(Y|h')$, the function $\mathcal{A} \mapsto \mathbb{V}^w(Y;P,\mathcal{A})$ is holomorphic on $\sup|\mathcal{A}| < r_0$, vanishes at zero, and obeys

$$|\mathbb{V}^w(Y;P,\mathcal{A})| \le K_0 |w|_Y e^{-r d_k(Y)}. \tag{1608}$$

Define its homogeneous Taylor parts and symmetric quadratic form by

$$T_n[w](Y;P;\mathcal{A}) = \frac{1}{n!}\, \partial_z^n \mathbb{V}^w(Y;P,z\mathcal{A})\big|_{z=0}, \tag{1609}$$

$$T_2[w](\mathcal{A}_1,\mathcal{A}_2) = \tfrac{1}{2}\{T_2[w](\mathcal{A}_1+\mathcal{A}_2) - T_2[w](\mathcal{A}_1) - T_2[w](\mathcal{A}_2)\}. \tag{1610}$$

Then $T_0 = 0$, each T_n is a homogeneous polynomial defined for all fields, and

$$|T_n[w](\mathcal{A})| \le K_0 |w|_Y r_0^{-n} \sup|\mathcal{A}|^n e^{-r d_k(Y)}, \tag{1611}$$

$$|T_{\ge N}[w](\mathcal{A})| \le 2K_0 |w|_Y r_0^{-N} \sup|\mathcal{A}|^N e^{-r d_k(Y)} \quad (\sup|\mathcal{A}| \le r_0/2), \tag{1612}$$

$$|2T_2[w](\mathcal{A}_1,\mathcal{A}_2)| \le 4K_0 |w|_Y r_0^{-2} \sup|\mathcal{A}_1| \sup|\mathcal{A}_2| e^{-r d_k(Y)}. \tag{1613}$$

Here $T_{\ge N} = \mathbb{V}^w - \sum_{n<N} T_n[w]$ on the radius-r_0 ball. If $P \in \mathbb{U}_k^a(Y|h')$, there is also the sharper linear bound

$$|T_1[w](\mathcal{A})| \le eK_c c_{\mathfrak{h}} M^3 |w|_Y a^{\Sigma} \sup|\mathcal{A}| e^{-r d_k(Y)}. \tag{1614}$$

All these functions are linear and local in the weight. They vanish at domains containing no source cube, where a source cube means a fine cube meeting Ω_t. They depend on the pair of the triple and inherit its gauge, conjugation and Borel properties. At every fixed real factor their coefficients admit local holomorphic extensions in the remaining triple variables. ■

Proof. The finite sum (1606) proves linearity, locality and independence of the stored datum. A plaquette used in a term lies in $X \subset X^{\uparrow} \subset Y$. The shift vanishes at $\mathcal{A} = 0$ and at domains without a source cube, so each difference then vanishes. For a fixed real factor, take local holomorphic extensions of the finitely many shift values at 1_D and intersect their neighborhoods. Substituting them in (1606) gives a holomorphic extension of the potential. Pair dependence, gauge covariance, conjugation and measurability pass through this same finite formula.

For completeness, the weighted estimate uses the following two bounds for a single Wilson member at points where $\vartheta^a(\sup|\mathcal{A}|) < 1$:

$$|\mathbb{V}[W_w^{(m)}(X)](Y)| \le 12(1+2c_{\mathrm{pl}})^2 \vartheta^a(\sup|\mathcal{A}|) M^3 s_m(w;Y) L^{k-m} a_m^2 e^{-\delta_* \mathsf{d}_k(X)/2 - (\bar\kappa - 1) m(X,Y)},$$
$$|\mathbb{V}[W_w^{(m)}(X)](Y)| \le 12e(1+2c_{\mathrm{pl}})^2 c_{\mathfrak{h}} \sup|\mathcal{A}| M^3 s_m(w;Y) L^{k-m} a_m e^{-\delta_* \mathsf{d}_k(X) - (\bar\kappa - 1) m(X,Y)}.$$

Here $s_m(w;Y)$ is the supremum of the weight on the plaquettes of layer m used in Y, and ϑ^a is the radius ratio of equation (1297). To keep this supremum local, replace w in each Wilson member by its restriction to that member's plaquettes; the localized member is unchanged. Here is the weighted calculation. Put $\rho = ec_{\mathfrak{h}} \sup|\mathcal{A}| e^{-\delta_* \mathsf{d}_k(X)}$. The plaquette product estimate of equation (822), applied with the layer radius, bounds the interpolated Wilson member for $|\lambda|\rho \le a_m$ by

$$b = 6(1+2c_{\mathrm{pl}})^2 M^3 s_m(w;Y) L^{k-m} a_m^2.$$

Its constant value at $\lambda = 0$ cancels in the finite difference. The rooted Cauchy estimate therefore bounds that difference by $2bR^{-1} e^{-(\bar\kappa-1)m(X,Y)}$ for every $1 < R \le a_m/\rho$. If $\rho = 0$, let $R \to \infty$. Otherwise take $R = a_m/\rho$. The distance from an occupied layer-m base to the source gives

$$\rho/a_m \le (e/3)\vartheta^a(\sup|\mathcal{A}|) e^{-\delta_* \mathsf{d}_k(X)/2} < 1.$$

Using this estimate yields the first displayed bound; substituting ρ without weakening its exponential yields the second. The argument is the weighted version of the plaquette and rooted Cauchy calculations in the proofs of Propositions 58.20 and 58.21. The placed-domain estimate of Lemma 58.19 applies: occupied Wilson bases have $d_m(X) = 0$, are placed with $\nu = 1$, and the smaller distance rate $\delta_*/2$ satisfies its hypotheses by (1607). Summing gives

$$|\mathbb{V}^w(Y;P,\mathcal{A})| \le K_c \vartheta^a(\sup|\mathcal{A}|) M^3 |w|_Y \sum_{m=0}^{k} \varpi_{k-m}^c L^{k-m} a_m^2 e^{-r d_k(Y)},$$
$$|\mathbb{V}^w(Y;P,\mathcal{A})| \le eK_c c_{\mathfrak{h}} \sup|\mathcal{A}| M^3 |w|_Y a^{\Sigma} e^{-r d_k(Y)}.$$

For $a_m = 1$, the radius ratio is $3c_{\mathfrak{h}} \sup|\mathcal{A}|$. It is less than one on the radius-r_0 ball. The sum in the first bound is at most $C_h'(L,\delta,1)$, proving (1608).

Apply the one-variable Cauchy formula to $z \mapsto \mathbb{V}^w(Y;P,z\mathcal{A})$ on the disc $|z|\sup|\mathcal{A}| < r_0$. This proves (1611); summing the geometric tail proves (1612). Polarization at normalized fields, using $4T_2(B,B') = T_2(B+B') - T_2(B-B')$, proves (1613). For a fixed family a, the condition $\vartheta^a(|z|\sup|\mathcal{A}|) < 1$ holds for sufficiently small $|z|$. Divide the second potential bound at $z\mathcal{A}$ by $|z|$ and let $z \to 0$ to prove (1614). The Cauchy formula on a common small circle gives local holomorphic extensions of the Taylor coefficients. Homogeneity extends those coefficients to arbitrary fields. Taylor limits at real scalar multiples and polarization preserve measurability and the two symmetries. □

78.2. **Three source terms and the oscillation bound.** We now apply the preceding proposition at a stronger rate $r^\sharp = (1-2\delta)(\kappa^\sharp - 2)$. Assume (1607) for these parameters, and that every middle-domain triple lies in $\mathbb{U}_k^a(Y|h')$ with the enlarged layer radii $a_m \le 1$. Let

$$g_k^{-1}\mathcal{A} = z \oplus a, \qquad \phi = \mu \oplus a, \qquad |z|, |a| < C_{\mathcal{A}}\mathfrak{b}_k, \qquad |\mu| < \theta_{\mathrm{sl}}\mathfrak{b}_k,$$

where inequalities are bondwise and the center μ is on the integration bonds. Require

$$(C_{\mathcal{A}} + \theta_{\mathrm{sl}})\delta_k \le r_0/2, \qquad \delta_k = g_k\mathfrak{b}_k. \tag{1615}$$

Then both $g_k(z+\phi)$ and $g_k\phi$ lie in the Taylor ball. For the canonical construction take $p_b = p_{12}(z_\square)$, where $z_\square$ is the centre of the cube of π_k containing b_-. Its four corners lie in that cube: their distance from $z_\square$ is at most the bare spacing $\eta_k \le 1 \le M/2$. Set $v_b = v - v(p_b)\mathbf{1}$. The bounds below also hold for any choice of p_b whose four corners lie in the same cube. The canonical choice obeys

$$p_{b+\mathsf{v}} = p_b + \mathsf{v} \qquad (\mathsf{v} \in M\mathbb{Z}^3), \tag{1616}$$

because translation takes the containing cube with centre $z_\square$ to the containing cube with centre $z_\square + \mathsf{v}$. With all Taylor parts taken at $(Y;P)$, define

$$\mathbb{W}[v] = -\{\mathbb{V}^{x_k v}(g_k(z+\phi)) - \mathbb{V}^{x_k v}(g_k\phi)\} + \sum_{b \in \mathfrak{X}_{\mathrm{I}}^{\circ+}\langle Y\rangle} x_k v(p_b)\{T_1[\mathbf{1}](g_k z_b) + 2T_2[\mathbf{1}](g_k z_b, g_k\phi)\}, \tag{1617}$$

$$\mathbb{W}^1[v] = -\sum_{b \in \mathfrak{X}^{\circ+}\langle Y\rangle} T_1[x_k v_b](g_k\phi_b),$$

$$\mathbb{W}^3[v] = -T_{\ge 3}[x_k v](g_k\phi). \tag{1618}$$

The canonical centered member is their sum $\mathbb{W}^{\mathrm{S}}[v] = \mathbb{W}[v] + \mathbb{W}^1[v] + \mathbb{W}^3[v]$. The three terms are analytic on the stated products of domains, linear in v, local in the plaquettes of Y, and independent of the stored fields. They inherit the pair, gauge, conjugation and Borel properties. All vanish without a source cube. The first vanishes at $z = 0$; the other two are independent of z and vanish at $\phi = 0$.

Put $q_Y(z) = M^{-3}\sum_{b \in \mathfrak{X}_{\mathrm{I}}^{\circ+}\langle Y\rangle} |z(b)|^2$. For such families define

$$\|F\|_{\mathrm{I}\star,r}^{\mathrm{rad}} = \sup_{Y,(P,\mathcal{A},\mathfrak{q}),\mu} \frac{e^{r d_k(Y)}|F(Y;P,\mathcal{A},\mathfrak{q};\mu)|}{1+q_Y(z)},$$

where the points and centers range over the stated tiers. Let D be the least $d_k(Y)$ over $Y \not\subset \Omega_k$ containing an integration bond, and let D' be the corresponding minimum for any bond of $\mathfrak{X}$; empty minima are $+\infty$. Let $\Theta = \max(C_{\mathcal{A}}, \theta_{\mathrm{sl}})$, and set

$$C^{\mathrm{src}} = 64\max\{3eK_c c_{\mathfrak{h}} M^6, 12K_0 r_0^{-2}\Theta M^3, 6K_0 r_0^{-2} M^6, 4K_0 r_0^{-3}(C_{\mathcal{A}} + \theta_{\mathrm{sl}})^3\}.$$

Proposition 78.2. Suppose $|v|_\infty \le \sigma$ and $x_k|v(p) - v(p')| \le \rho'$ for inner plaquettes of Ω_k whose barycentres are at distance at most M, in level-k units. For $0 \le r \le r^\sharp - 2$, the three norms are bounded by

$$N_W(r) = C^{\mathrm{src}}\{\sigma(1 + g_k\mathfrak{b}_k^3) + \rho'(g_k a^\Sigma + g_k\delta_k) + \sigma(g_k^{-1}a^\Sigma + \mathfrak{b}_k)e^{-(r^\sharp - r - 2)D}\}, \tag{1619}$$

$$N_1(r) = 384eK_c c_{\mathfrak{h}} M^6\Theta\{\rho' a^\Sigma\delta_k + \sigma g_k^{-1}a^\Sigma\mathfrak{b}_k e^{-(r^\sharp - r - 2)D'}\}, \tag{1620}$$

$$N_3(r) = 2K_0 r_0^{-3}\Theta^3\sigma g_k\mathfrak{b}_k^3. \tag{1621}$$

Consequently their sum has norm at most $\mathsf{sz}^{\mathrm{src}} = N_W(r_V) + N_1(r_V) + N_3(r_V)$ at any $0 \le r_V \le r^\sharp - 2$. If an unsourced frame has coarse sizes $s_{o'}$ and

$$\mathsf{sz}^{\mathrm{src}} \le w_0, \qquad eL^3\left(\sum_{o'} s_{o'} + w_0^{-1}\mathsf{sz}^{\mathrm{src}}\right) \le (4K_{\mathrm{ap}}^+)^{-1}, \tag{1622}$$

adding this centered member gives a frame to which theorem AF applies. The statement is uniform over affine holomorphic families of weights obeying the same two bounds. ■

Proof. Taylor expansion and $x_k g_k^2 = 1$ give

$$\begin{aligned}\mathbb{W}[v] = &-\sum_b T_1[x_k v_b](g_k z_b) - 2\sum_b T_2[v_b](z_b,\phi) - T_2[v](z)\\ &- \{T_{\ge 3}[x_k v](g_k(z+\phi)) - T_{\ge 3}[x_k v](g_k\phi)\}.\end{aligned} \tag{1623}$$

The first two sums are over integration bonds. Let $n_Y = \#_{\pi_k} Y$. The entropy geometry and Cauchy–Schwarz give

$$n_Y \le 8e^{d_k(Y)}, \qquad \sum_b |z(b)| \le \tfrac{3}{2} M^3 n_Y (1 + q_Y(z)), \qquad (\sum_b |z(b)|)^2 \le 3M^6 n_Y (1 + q_Y(z)).$$

Let $\omega_Y = \sup_{p,b} |v(p) - v(p_b)|$, with p among the plaquettes used by the potential and b among the bonds in Y. Using (1611)–(1614), the four terms in (1623), after division by $(1 + q_Y(z))e^{-r^\sharp d_k(Y)}$, are bounded respectively by

$$\begin{aligned}&\tfrac{3}{2} e K_c c_{\mathfrak{h}} M^6 n_Y a^\Sigma x_k g_k \omega_Y, \quad 6K_0 r_0^{-2} \Theta M^3 n_Y \mathfrak{b}_k \omega_Y,\\ &6K_0 r_0^{-2} M^6 n_Y |v|_Y, \quad 4K_0 r_0^{-3} (C_{\mathcal{A}} + \theta_{\mathrm{sl}})^3 g_k \mathfrak{b}_k^3 |v|_Y.\end{aligned}$$

The two additional terms satisfy

$$|\mathbb{W}^1[v]| \le 3eK_c c_{\mathfrak{h}} M^6 \Theta n_Y x_k \omega_Y a^\Sigma \delta_k e^{-r^\sharp d_k(Y)}, \quad |\mathbb{W}^3[v]| \le 2K_0 r_0^{-3} \Theta^3 |v|_Y g_k \mathfrak{b}_k^3 e^{-r^\sharp d_k(Y)}.$$

For $Y \subset \Omega_k$, choose p, b in the definition of ω_Y, and let Q, Q' be fine cubes containing a corner of p and the reference plaquette p_b, respectively. Join Q to Q' by a nonrepeating adjacent-cube path in Y, with at most $n_Y - 1$ edges, and use the plaquette at each cube centre. Consecutive barycentres, including the two endpoint links to p and p_b, are at distance at most M. Hence $x_k \omega_Y \le (n_Y + 1)\rho' \le 2n_Y \rho'$; if either index set is empty, $\omega_Y = 0$. For $Y \not\subset \Omega_k$, use $\omega_Y \le 2\sigma$ and the lower bounds $d_k(Y) \ge D$ or D'. Since $n_Y^2 \le 64 e^{2d_k(Y)}$, two units of decay absorb these counts. Substitution gives (1619)–(1621). Subadditivity gives the bound for the sum. The two inequalities in (1622) are precisely the size and amplitude conditions of the centered expansion. Finally each local formula is a finite linear combination of the coordinates of the weight with locally holomorphic coefficients. This proves joint holomorphy and keeps the same norm bounds over the parameter set. □

78.3. **The regional Hessian.** The identification of the quadratic source requires the Hessian of the constrained regional action. We use the reduced determining set of the shorter sequence, including its outward rows. The row pairing is

$$\langle \omega, \nu \rangle_{\mathfrak{R}} = \sum_c \xi_c^{-1} \langle \omega(c), \nu(c) \rangle, \qquad (\omega, \nu)_{\mathbf{B}} = \sum_c \xi_c^3 \langle \omega(c), \nu(c) \rangle.$$

Write $\mathsf{m}\nu(c) = \xi_c \nu(c)$ and let J extend a level-k field on $\mathfrak{X}(\Theta)$ by zero to the shorter rows. Thus $\mathsf{m}\mathsf{J} = \mathsf{J}$. All response operators below use this shorter sequence and these pairings. The following argument is also the regional second-variation calculation in Proposition 58.27.

Proposition 78.3. Consider a separated regional step $k+1$, with $R_1 \ge L$, $R_1 \mathfrak{M} \ge L^2$, and, when $k \ge 1$, at least $6R_1\mathfrak{M}$ sites in each direction of the level-k torus. Let p be a real point of the coarse fibre, admissible on every averaging double block, and axial on Θ'. Suppose the regular critical orbit for its shorter datum $\mathsf{d}(p)$ exists, and take an element U of this orbit. Assume that the pair $(U, J^\circ(U))$ is in the natural class of the regional fluctuation construction, that all its response and fluctuation operators are defined, and that its real row map and node gauge maps are analytic near zero. The last condition holds at the regularity radius used in Propositions 41.2 and 58.27. Assume

also that, along every differentiable curve $V(s)$ in the coarse fibre through p, the shorter regular critical orbit exists for small $|s|$, and

$$\left.\frac{d}{ds}\right|_{s=0} A_{\Omega_0}(U_{\mathbf{B}^\flat}(\mathsf{d}(V(s)))) = 0. \tag{1624}$$

Then every real $B \in \ker Q_p$ is the velocity of a real analytic fibre curve through p. Along every twice differentiable such curve with velocity B, the first variation of the regional action vanishes and

$$\left.\frac{d^2}{ds^2}\right|_{s=0} L^k A_{\Omega_0}(U_{\mathbf{B}^\flat}(\mathsf{d}(V(s)))) = \langle B, \Delta_k(U, J^\circ(U))B\rangle. \tag{1625}$$

The form on the right is real and symmetric and is independent of the choice of U in that orbit among the admitted pairs. In free coordinates the full fluctuation operator is

$$\mathsf{A}(U, J^\circ(U)) = \mathsf{C}_p^{\mathsf{T}} \Delta_k(U, J^\circ(U))\mathsf{C}_p + \widehat{\mathsf{G}}_p. \tag{1626}$$

Here $\widehat{\mathsf{G}}_p$ is the quadratic block-contour form of the same regional step. ∎

Proof. Put $J = J^\circ(U)$ and $F(A) = L^k A_{\Omega_0}(e^{i\xi A}U)$, with $\xi = L^{-k}$. Let $\mathcal{Q}(A)$ be the straightened shorter row map and $K_A = \mathcal{Q}'(A)$. At zero its first two derivatives are

$$K_0 = \mathsf{m}Q, \qquad \mathcal{Q}''(0)[A_1, A_2] = 2\mathsf{m}C^{(2)}(A_1, A_2).$$

The adjoint for the dimensionless row pairing is $K_A^\dagger = K_A^{\mathsf{T}}\mathsf{m}^{-4}$. In particular $K_0^\dagger = Q^{\mathsf{T}}\mathsf{m}^{-3}$.

Criticality of U and surjectivity of Q give a unique real multiplier ν_0 with $J = Q^{\mathsf{T}}\nu_0$. Indeed a tangent vector in the kernel of K_0 can be integrated to a local curve with fixed straightened rows. Restoring the node gauges gives a curve with fixed actual datum. The action is unchanged by this restoration, so its derivative vanishes on $\ker K_0$. The real annihilator of this kernel is the range of $K_0^\dagger$. Since $Q\mathsf{H} = Q\mathsf{H}_1 = 1$,

$$\nu_0 = \mathsf{H}^{\mathsf{T}}J = \mathsf{H}_1^{\mathsf{T}}J, \qquad \omega_0 = \mathsf{m}^3\nu_0.$$

These are the multiplier identities of Proposition 41.2 with the row-spacing factors written out.

We recall why the response linearization uses the constrained Hessian. Set $\Delta_1 = \Delta(U) - \Delta^{(2)}$, where

$$(A_1, \Delta^{(2)}A_2)_\xi = 2(C^{(2)}(A_1, A_2), \nu_0)_{\mathbf{B}}.$$

Differentiating gauge invariance and row covariance at U gives, for $\lambda \in \ker Q'$,

$$(A, \Delta(U)D\lambda)_\xi = \langle \omega_0, \mathcal{Q}''(0)[A, D\lambda]\rangle_{\mathfrak{R}} = (A, \Delta^{(2)}D\lambda)_\xi.$$

Thus Δ_1 annihilates these gauge directions. The projection $\mathsf{S} = 1 - D\mathsf{G}'\mathsf{R}D^{\mathsf{T}}$ differs from the identity by one of these directions, so $\Delta_1\mathsf{S} = \Delta_1 = \mathsf{S}^{\mathsf{T}}\Delta_1$. The definitions of the corrected inverse therefore give

$$\mathsf{G}_1^{-1} = \Delta_1 + D\mathsf{R}D^{\mathsf{T}} + Q^{\mathsf{T}}\mathsf{a}Q.$$

Consequently the saddle system

$$(\Delta_1 + D\mathsf{R}D^{\mathsf{T}})x - Q^{\mathsf{T}}\mu = f, \qquad Qx = g$$

has exactly one solution. Explicitly,

$$\mu' = (Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}(g - Q\mathsf{G}_1 f), \quad x = \mathsf{G}_1 f + \mathsf{G}_1 Q^{\mathsf{T}}\mu', \quad \mu = \mu' - \mathsf{a}g.$$

The inverses here are among the assumed defined operators.

For $k \geq 1$, apply the real analytic implicit function theorem to

$$\Psi(A, \omega; \beta) = (F'(A) + D\mathsf{R}D^{\mathsf{T}}A - K_A^\dagger\omega, \mathcal{Q}(A) - \mathsf{J}\beta).$$

At $(0, \omega_0; 0)$ it vanishes, and its derivative in (A, ω) is

$$(x, \varpi) \longmapsto ((\Delta_1 + D\mathsf{R}D^{\mathsf{T}})x - Q^{\mathsf{T}}\mathsf{m}^{-3}\varpi, \mathsf{m}Qx).$$

The saddle calculation proves invertibility. We obtain real analytic $\widehat{A}(\beta), \widehat{\omega}(\beta)$ near zero.

We next remove the auxiliary gauge term and restore the actual rows; outward rows require this step. Let $\Sigma_A\lambda$ be the derivative of the gauge action $A \mapsto A^{e^{it\lambda}}$, and let $\Pi_A\lambda$ be the corresponding derivatives of the iterated node gauges at the tops. At zero,

$$\Sigma_0\lambda = -D\lambda, \qquad \Pi_0\lambda = Q'\lambda.$$

The map $\lambda \mapsto (\Pi_A\lambda, \mathsf{R}D^{\mathsf{T}}\Sigma_A\lambda)$ is bijective near zero. At zero its kernel is trivial by the scalar inverse on $\ker Q'$. To obtain an arbitrary target (τ, f), first choose $Q'\lambda_1 = \tau$, then put

$$g = -f - \mathsf{R}D^{\mathsf{T}}D\lambda_1, \qquad \lambda_2 = \mathsf{G}'\mathsf{R}g.$$

The scalar inverse identities give $Q'\lambda_2 = 0$ and $\mathsf{R}D^{\mathsf{T}}D\lambda_2 = g$. Thus $\lambda_1 + \lambda_2$ has the required image. Continuity of the determinant proves the nearby assertion.

The derivative of a row is determined by the derivatives of its endpoint rotations. At an outward endpoint z of a level-q row, the extra node factor has the form

$$\mathsf{v}_{q-1}[A](z) = \mathsf{f}_z\big(\mathcal{Q}(A)|_{B(z)}\big), \qquad \mathsf{f}_z(0) = 1,$$

where every bond of $B(z)$ is a level-$(q-1)$ row. The map f_z is holomorphic: it is the exponential of the average of the logarithms of the block contours. For two constrained curves with equal rotation derivatives at the tops, equality of row derivatives at level zero is immediate. Equality at level $q-1$ gives equal derivatives of f_z, hence equal rotations at the outward endpoints of level q. Differentiating the row logarithm then gives equal row derivatives at level q. Induction proves that $\Pi_A\lambda = 0$ implies $K_A\Sigma_A\lambda = 0$.

For $A = \widehat{A}(\beta)$, put $f = \mathsf{R}D^{\mathsf{T}}A$ and choose λ whose image under the preceding bijection is $(0, f)$. Pairing the first equation $\Psi = 0$ with $\Sigma_A\lambda$ gives $(f, f)_\xi = 0$: the action term vanishes by gauge invariance and the multiplier term by $K_A\Sigma_A\lambda = 0$. Since f is real, $f = 0$. Hence

$$\mathcal{Q}(\widehat{A}) = \mathsf{J}\beta, \qquad F'(\widehat{A}) = K_{\widehat{A}}^{\dagger}\widehat{\omega}, \qquad \mathsf{R}D^{\mathsf{T}}\widehat{A} = 0. \tag{1627}$$

Define $\widehat{h}$ to be $v_q[\widehat{A}]^{-1}$ at the tops of level q and the identity at other bare sites. At an outward endpoint the factor f_z equals one, since the lower rows of $\mathsf{J}\beta$ are zero. Thus $\widehat{h}\, v_q = 1$ at both endpoints of every row, and

$$U_\beta = (e^{i\xi\widehat{A}(\beta)}U)^{\widehat{h}(\beta)}$$

has exactly the datum $\mathsf{d}(e^{i\beta}p)$. It is regular after shrinking the neighbourhood, by the strict regularity inequalities at U. For a curve through U_β with this datum fixed, undo $\widehat{h}$ and take its logarithm, giving a curve A_s. Its endpoint rotations have the same top derivatives as some gauge curve, by surjectivity of Π_{A_0}. The row induction above then gives $K_{A_0}\dot{A} = K_{A_0}\Sigma_{A_0}\lambda$. Equation (1627) and gauge invariance show

$$(F'(A_0), \dot{A})_\xi = \langle\widehat{\omega}, K_{A_0}\dot{A}\rangle_{\mathfrak{R}} = (F'(A_0), \Sigma_{A_0}\lambda)_\xi = 0.$$

Therefore U_β is critical, and belongs to the regular critical orbit whenever that orbit exists.

Put $\varphi(\beta) = F(\widehat{A}(\beta))$. Differentiating (1627) at zero and using the saddle inverse gives

$$\widehat{A}'(0) = \mathsf{H}_1\mathsf{J}, \qquad \mathsf{m}^{-3}\widehat{\omega}'(0) = [(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1} - \mathsf{a}]\mathsf{J}.$$

The envelope identity is $\varphi'(\beta)\nu = \langle\widehat{\omega}(\beta), \mathsf{J}\nu\rangle_{\mathfrak{R}}$. Since $\mathsf{m}\mathsf{J} = \mathsf{J}$ and $\mathsf{J}^{\mathsf{T}}\mathsf{a}\mathsf{J} = 1$, it follows that

$$\begin{aligned} \varphi'(0)\nu &= (\mathsf{H}_1\mathsf{J}\nu, J)_\xi, \\ \varphi''(0)[\nu, \nu] &= (\nu, [\mathsf{J}^{\mathsf{T}}(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}\mathsf{J} - 1]\nu). \end{aligned} \tag{1628}$$

For $k = 0$ the same identities follow directly from $\widehat{A}(\beta) = \mathsf{J}\beta$, $\widehat{h} = 1$, $Q = \mathsf{H}_1 = 1$, $\mathsf{R} = 0$, and $\mathsf{G}_1^{-1} = \Delta(U) + 1$.

It remains to restrict this unconstrained Taylor formula to the curved coarse fibre. Its analytic chart is

$$\beta_B(s) = sB - \mathsf{h}_p\widetilde{\mathsf{D}}_p(sB), \qquad B \in \ker Q_p,$$

with $\widetilde{\mathsf{D}}_p(0) = \widetilde{\mathsf{D}}_p'(0) = 0$. Thus every such B is a fibre velocity. By the assumed nearby existence and (1624), $\varphi'(0)$ annihilates $\ker Q_p$. For any twice differentiable fibre curve write $V(s) = e^{i\beta(s)}p$ using the local real logarithm. The constraint has derivatives LQ_p and $2C_p^{(2)}$, so differentiating twice gives

$$\ddot\beta(0) = \kappa - 2\mathsf{D}_p^{(2)}(B,B), \qquad \kappa \in \ker Q_p.$$

The chain rule and (1628) now give

$$\begin{aligned}\frac{d^2}{ds^2}\varphi(\beta(s))\big|_{s=0} &= (B, [\mathsf{J}^{\mathsf{T}}(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}\mathsf{J} - 1]B)\\ &\quad - 2(\mathsf{J}\mathsf{D}_p^{(2)}(B,B), \mathsf{H}_1^{\mathsf{T}}J)_{\mathbf{B}}\\ &= \langle B, \Delta_k(U,J)B\rangle.\end{aligned}$$

This is precisely the regional fluctuation definition, including its curved-constraint correction. The function $\varphi(\beta(s))$ is the orbit action by nearby existence and the preceding criticality argument, which proves (1625).

Finally the axial condition gives

$$G_k(\Theta'; e^{i\beta}p) = \tfrac{1}{2}\mathsf{G}_p^{(2)}(\beta,\beta) + O(|\beta|^3).$$

The first derivative of this function is zero. Adding its Hessian and pulling back by pivot elimination proves (1626). Reality and symmetry follow by polarization of the real second derivative. The action and its constrained Hessian depend on the orbit, so their representing operators do not depend on the chosen admitted representative. □

78.4. **Why the Wilson mixed term vanishes.** The conditional mean is defined by the full fluctuation form, which includes gauge fixing. To use the source split we must separate its two parts. This requires the gauge degeneracy of the Wilson Hessian, as well as the conditional-mean equation.

Let p be a real admissible configuration of the regional fibre, with invertible pivot coefficients. Let Θ' denote the blocks on which the gauge term is imposed and $\Theta'' \subset \Theta'$ the integration blocks. The integration nonpivot bonds are precisely those meeting $B(\Theta'')$. Write C_p for pivot elimination, $D_p\rho$ for the gauge direction, and $\check D_p\rho$ for its nonpivot restriction. Site functions ρ vanish off $B(\Theta'') \setminus \Theta''$. Set

$$j\rho = \chi_{\mathrm{I}}\check D_p\rho, \qquad (bB)(x) = \mathrm{Ad}(p(y,x))^{-1}G_p^{(1)}(B^0)(y,x), \quad x \in B(y)\setminus\{y\}, \quad y \in \Theta''.$$

Here B^0 is the zero extension, and bB is zero at the block centers and outside these blocks. Pairings are complex bilinear; transposes below refer to them.

Lemma 78.4. Let f be the real background action on the regional fibre. At a point satisfying Proposition 78.3, its criticality and Hessian have the form below. More generally, suppose f is invariant under the block gauge transformations, is critical at p, and has Hessian

$$(B_1,B_2) \longmapsto \langle C_pB_1, \Delta_k(U_+, J^\circ(U_+))C_pB_2\rangle = (B_1, HB_2)_1.$$

Here the Hessian assertion means that every real tangent field is the velocity of a twice differentiable fibre curve, and the second derivative along each such curve is the displayed quadratic form. Suppose the actual fluctuation operator is $A = H + \widehat G_p$, with H real symmetric and $\widehat G_p$ the block-contour gauge form. If A_{II} is invertible, put $m = -A_{\mathrm{II}}^{-1}A_{\mathrm{IE}}a$ and $\bar m = m \oplus a$. Then, for every complex external field a,

$$bm = 0, \qquad \chi_{\mathrm{I}}\widehat G_p\bar m = 0, \qquad \chi_{\mathrm{I}}H\bar m = 0. \tag{1629}$$

Consequently, when the localized Wilson first and second variations are respectively zero and H, the two cancellation hypotheses in (1631) hold. At $k=0$ the same conclusion holds for $F^{\mathbf{1}}_{Y,1}$ and its polarized quadratic parts. ■

Proof. We first record the local geometric identities. Differentiating the covariance of the block average and of the contour gives

$$LQ_pD_p\rho(c)=\mathrm{Ad}(\mathcal{M}(p)(c))\rho(c_+)-\rho(c_-)=0,\qquad G_p^{(1)}(D_p\rho)(y,x)=\mathrm{Ad}(p(y,x))\rho(x).$$

The endpoints of the coarse bonds are block centers, where $\rho=0$. Pivot elimination is the unique solution of $Q_pB=0$ with its given nonpivot values, so $C_p\check{D}_p\rho=D_p\rho$ and $bj=1$. The gauge directions supported in $B(\Theta'')$ have no external nonpivot component.

Every contour path in a block uses only the nonpivot bonds of that block; changing pivot values does not change its first variation. Therefore the gauge form has no mixed integration–external block, and

$$\widehat{G}_{\mathrm{IE}}=0,\qquad \widehat{G}_{\mathrm{II}}=b^{\mathsf{T}}b.$$

Indeed its bilinear form is the sum of the pairings of the first contour variations. Applying the inverse adjoint transports in the definition of b leaves these pairings unchanged. This proves the second identity.

We next use gauge invariance of f. For a real tangent vector B, choose a twice differentiable curve $V(s)$ in the fibre with this velocity. For a real site function ρ supported in the block interiors and a real number c, the curve $V(s)^{\exp(-isc\rho)}$ remains in the fibre and has velocity $B+cD_p\rho$. Gauge invariance gives the same second derivative along the two curves. Thus the Hessian q on the constrained bond fields satisfies

$$2c\,q(B,D_p\rho)+c^2q(D_p\rho,D_p\rho)=0\quad(c\in\mathbb{R}).$$

Both coefficients vanish. Surjectivity of pivot elimination onto the tangent space and symmetry imply $H\check{D}_p\rho=0$. Splitting complex arguments into real and imaginary parts gives the same identity over $\mathbb{C}$. In particular,

$$H_{\mathrm{II}}j=H_{\mathrm{EI}}j=0,\qquad j^{\mathsf{T}}H_{\mathrm{II}}=j^{\mathsf{T}}H_{\mathrm{IE}}=0.$$

The conditional-mean equation is $b^{\mathsf{T}}bm+H_{\mathrm{II}}m+H_{\mathrm{IE}}a=0$. Apply j^{T} and use $bj=1$ to obtain $bm=0$. The gauge part of the equation is therefore zero by itself, and so is the Wilson part. This proves (1629). Finally, $2\sum_Y T_2[\mathbf{1}](Z|_Y,\bar{m}|_Y)=(Z,H\bar{m})_1=0$ for integration-supported Z; at the first step the factor two is already included in the polarized form $F_{Y,2}$. □

For the regional action, the invariance used here follows by extending a block gauge to the bare sites, taking it to be the identity at all other sites. Covariance of the averages identifies the two shorter constraint sets, preserves criticality and regularity, and maps their residual gauge orbits onto one another. The action is gauge invariant, so its value on those orbits is unchanged. Proposition 78.3 supplies its Hessian at the same point as the Gaussian. At an actual history the agreement of the regional pair with the pair used to define that Gaussian is still part of the applicability check.

78.5. **The exact source split.** For an arbitrary reference function rf on positive bonds, let $\mathcal{A}_b=\mathcal{A}1_b$. For a field ζ vanishing on the external bonds, define

$$\begin{aligned}\mathcal{W}[w,\mathsf{rf}](Y;P;\mathcal{A},\zeta)=&\sum_{b\in\mathfrak{X}^{\circ+}\langle Y\rangle}T_1[w-\mathsf{rf}(b)](\mathcal{A}_b)+T_{\ge3}[w](\mathcal{A})+T_2[w](\zeta)\\&+2\sum_{b\in\mathfrak{X}_{\mathrm{I}}^{\circ+}\langle Y\rangle}T_2[w-\mathsf{rf}(b)](\zeta_b,\mathcal{A}-\zeta).\end{aligned}\tag{1630}$$

It is defined for $P\in\mathbb{U}^{\mathbf{1}}_k(Y|h')$ and $\sup|\mathcal{A}|<r_0$; no bound on ζ is needed for the polynomial terms. The notation distinguishes this positive-source remainder from the centered member $\mathbb{W}^{\mathrm{S}}[v]$.

Lemma 78.5. Let P be a triple on the torus whose restrictions lie in the Taylor domains. Let a be an external field, $m = \mathsf{M}a$ its unsourced conditional mean, $\bar{m} = m \oplus a$, and suppose

$$\sum_Y T_1[\mathbf{1}](B|_Y) = 0, \qquad \sum_Y T_2[\mathbf{1}](Z|_Y, \bar{m}|_Y) = 0 \tag{1631}$$

for all fields B and all Z vanishing on the external bonds. For an integration field B_{I} put $\mathcal{A}_Y = g_k(B_{\mathrm{I}} \oplus a)|_Y$ and $\zeta_Y = g_k((B_{\mathrm{I}} - m) \oplus 0)|_Y$. Whenever $\sup|\mathcal{A}_Y| < r_0$ for every Y,

$$\sum_Y \mathbb{V}^w(Y; P|_Y, \mathcal{A}_Y) = \sum_Y \mathcal{W}[w, \mathsf{rf}](Y; P|_Y; \mathcal{A}_Y, \zeta_Y) + \sum_Y T_2[w](g_k\bar{m}|_Y). \tag{1632}$$

At $v = -x_k^{-1}w$ and $\mathsf{rf}(b) = w(p_b)$, the canonical centered member evaluated at $z = B_{\mathrm{I}} - m$, $\mu = m$ is exactly the corresponding term $\mathcal{W}[w, \mathsf{rf}]$. Thus the second term in (1632) is the quadratic source at the conditional mean, localized in theorem AG. ■

Proof. Expand the potential as $T_1 + T_2 + T_{\geq 3}$. Decompose its linear term bond by bond using $w = (w - \mathsf{rf}(b)) + \mathsf{rf}(b)\mathbf{1}$. After summing over Y, every reference contribution vanishes by the first identity in (1631), applied to the field supported on that bond. Next write $\mathcal{A}_Y = \zeta_Y + g_k\bar{m}|_Y$ and polarize the quadratic term. Split its mixed term in the weight in the same way. Its reference terms vanish by the second identity, again bond by bond. What remains is (1630) and the quadratic part at $g_k\bar{m}$. All sums are finite. To obtain the last assertion, add (1623) and (1618), substitute $x_k v = -w$, and use linearity in both the weight and the field. The first-order sums join over all bonds, the two cubic remainders at $g_k\phi$ cancel, and the remaining terms are exactly (1630). □

The cancellation hypotheses in this lemma are checked at the actual chain points in the measure argument. In particular the mixed cancellation concerns the Wilson Hessian, not merely the full Hessian with its gauge term.

78.6. **The escort on the one-lattice cores.** The one-lattice family is needed to identify the terms whose supports remain inside the next small-field region. Write $h_\emptyset$ for the history without large-field regions and $\mathcal{Q}_{\mathfrak{h}}(Y) = \mathbb{U}_k^{\mathbf{1}}(Y|h_\emptyset)$. On this history every bond is an integration bond. Define

$$\mathbb{W}^{1\mathrm{l}}[v](Y; P, B) = -\mathbb{V}^{x_k v}(Y|h_\emptyset; P, g_k B) + \sum_b x_k v(p_b) T_1[\mathbf{1}](Y|h_\emptyset; P; g_k B_b). \tag{1633}$$

Its domain is precisely $P \in \mathcal{Q}_{\mathfrak{h}}(Y)$, $g_k \sup|B| < r_0$. The sum is over the positive one-lattice nonpivot bonds in Y.

Proposition 78.6. Assume the strengthened localization rates used above, the core inclusion $\mathcal{U}^\circ(Y) \subset \mathcal{Q}_{\mathfrak{h}}(Y)$, and $C_{\mathcal{A}}\delta_k \leq r_0/2$. On $\mathcal{U}^\circ(Y) \times \mathbb{D}^{\mathrm{f}}(Y)$, (1633) is holomorphic in B, vanishes at $B = 0$, depends only on the pair of P, and is gauge invariant. For fixed real U, it is jointly holomorphic in the weight, $(A', \mathbf{J})$ and B on every open core slice. Its conjugate is $\mathbb{W}^{1\mathrm{l}}[\bar{v}]$. For $0 \leq r \leq r^\sharp - 2$,

$$\|\mathbb{W}^{1\mathrm{l}}[v]\|^{\mathrm{co}}_{\star,r} \leq N^\circ[v], \tag{1634}$$

where

$$\begin{aligned} N^\circ[v] = 64\big[&3eK_c c_{\mathfrak{h}} M^6 L^{-2} \alpha_0 g_k x_k \mathrm{sw}_k(v) \\ &+ (6K_0 r_0^{-2} M^6 + 2K_0 r_0^{-3} C_{\mathcal{A}}^3 g_k \mathfrak{b}_k^3)|v|_\infty\big]. \end{aligned} \tag{1635}$$

The norm is taken after restricting $P \in \mathcal{U}^\circ(X)$ to Y with $Y^\sharp \subset X$, and dividing by $1 + q_Y(B)$. Thus this is one escort on all the one-lattice cores.

Suppose also that the canonical shifts restrict to the one-lattice shift inside $\Lambda_t(h')$. For $Y \subset \Lambda_t(h')$ and every $(P, \mathcal{A})$ in the one-lattice domain,

$$\mathbb{W}[v](Y|h'; P, \mathcal{A}; 0) = \mathbb{W}^{1\mathrm{l}}[v](Y; P, g_k^{-1}\mathcal{A}), \qquad \mathbb{W}^1[v](Y|h'; P, \mathcal{A}; 0) = \mathbb{W}^3[v](Y|h'; P, \mathcal{A}; 0) = 0. \tag{1636}$$

This identity holds on the actual middle tiers whenever their restrictions lie in the one-lattice domain. It gives one escort for all such histories, with the same choice of reference plaquettes. ■

Proof. The core inclusion and the field room put $g_k B$ in the half-radius Taylor ball. On the constant history there are no plaquettes of layers $m < k$. Apply the linear Taylor estimate with $a_k = L^{-2}\alpha_0$ and $a_m = \epsilon$ for $m < k$, then let $\epsilon \downarrow 0$. Since $\varpi_0^w = 1$, this gives

$$|T_1[w](\mathcal{A})| \le eK_c c_{\mathfrak{h}} M^3 L^{-2}\alpha_0 |w|_Y \sup|\mathcal{A}| e^{-r^\sharp d_k(Y)}.$$

The Taylor expansion of minus (1633) is

$$\sum_b T_1[x_k(v - v(p_b))](g_k B_b) + T_2[v](B) + T_{\ge 3}[x_k v](g_k B).$$

Put $d = d_k(Y)$. The cube path defining the local oscillation has at most $d+2$ links, so $|v - v(p_b)|_Y \le (d+2)\mathrm{sw}_k(v)$. Also $\#_{\pi_k} Y \le 8(d+1)$ and $\sum_b |B(b)| \le 12M^3(d+1)(1+q_Y(B))$. Using $d+1 \le e^d$ and $d+2 \le 2e^d$, the first term is bounded by

$$24eK_c c_{\mathfrak{h}} M^6 L^{-2}\alpha_0 g_k x_k \mathrm{sw}_k(v)(1+q_Y(B))e^{-(r^\sharp - 2)d}.$$

The other terms are bounded respectively by $K_0 r_0^{-2} M^3 |v|_\infty q_Y(B) e^{-r^\sharp d}$ and $2K_0 r_0^{-3} C_{\mathcal{A}}^3 g_k \mathfrak{b}_k^3 |v|_\infty e^{-r^\sharp d}$. Their sum is at most the stated bound.

The local formulas and their Taylor coefficients have holomorphic extensions at fixed U. On an open core slice, these local extensions agree with the given function on open intersections, so the function is holomorphic there. Linearity in the finitely many weight coordinates makes holomorphy joint. Pair dependence and both symmetries follow term by term. Restriction to $Y^\sharp \subset X$ preserves the core, by Lemma 49.2; it therefore preserves the bound and all the stated properties.

For $Y \subset \Lambda_t$, there are no external bonds and the stored datum on Y is empty. The canonical shift and each localized Wilson member agree with their one-lattice counterparts. At center zero, $\phi = 0$. Hence the two terms in (1618) vanish, the mixed Taylor term in (1617) vanishes, and the remaining expression is (1633). This proves (1636). □

78.7. **The first step.** At $k = 0$ the shift is the regional pivot completion itself. We give its weighted calculation separately. Let $\mathsf{Pl}_0(Y) = \{p \in \mathcal{P}(\Omega_0^\circledast) : X(p) = Y\}$, and set $|v|_{0,Y} = \sup_{\mathsf{Pl}_0(Y)} |v|$, with value zero for an empty set. Require the following local admission condition:

$$P|_{Y_1} \in \mathcal{Q}_1(Y_1; a_P^\circ) \quad \text{whenever } Y_1 \subset Y, \quad \#_{\pi_0} Y_1 \le 4, \quad Y_1 \text{ contains a source cube.} \tag{1637}$$

The constants $a_P^\circ, r_P^\circ, C_P$ are those of Definition 51.2. On $\sup|\mathcal{A}| < r_P^\circ$, complete the nonpivot field by setting the omitted nonpivot coordinates to zero on each pivot block and applying its analytic pivot chart. Write this completion as $B'(P, \mathcal{A})$, extended by zero outside its regional bonds. Define

$$F_Y^v[P](\mathcal{A}) = \sum_{p \in \mathsf{Pl}_0(Y)} v(p)[a_p(e^{iB'(P,\mathcal{A})}\mathbf{U}) - a_p(\mathbf{U})]. \tag{1638}$$

This remains a partial map with the completion's domain when the sum is empty. Put $K^0 = 4512 C_P M^3 r_P^\circ$. Let $F_{Y,n}^v$ denote its homogeneous Taylor parts, and use the following polarization convention at this step:

$$F_{Y,2}^v(A_1, A_2) = F_{Y,2}^v(A_1 + A_2) - F_{Y,2}^v(A_1) - F_{Y,2}^v(A_2).$$

In particular, $F_{Y,2}^v(A, A) = 2F_{Y,2}^v(A)$.

Proposition 78.7. Under (1637), F_Y^v is holomorphic on the radius-r_P° ball, vanishes at zero, and has modulus at most $K^0|v|_{0,Y}$. It vanishes unless Y contains a source cube and $\mathsf{Pl}_0(Y) \ne \emptyset$; in the

latter case $d_0(Y) = 0$. Its Taylor parts satisfy

$$|F^v_{Y,n}(A)| \le K^0 |v|_{0,Y} (\sup |A|/r_P^\circ)^n, \tag{1639}$$

$$|\mathcal{R}_N F^v_Y(A)| \le 2K^0 |v|_{0,Y} (\sup |A|/r_P^\circ)^N \quad (\sup |A| \le r_P^\circ/2), \tag{1640}$$

$$|F^v_{Y,2}(A_1, A_2)| \le 4K^0 |v|_{0,Y} (r_P^\circ)^{-2} \sup |A_1| \sup |A_2|. \tag{1641}$$

If $|\partial \mathbf{U}(p) - 1| \le \varepsilon_\partial$ on $\mathsf{Pl}_0(Y)$, then

$$|F^v_{Y,1}(A)| \le 192 C_P M^3 |v|_{0,Y} \varepsilon_\partial \sup |A|. \tag{1642}$$

The potential and its parts are linear and local in the weight, depend on P only through $\mathbf{U}$, and have the gauge and conjugation laws of the Wilson action. They are Borel on the admitted domains. At fixed real U, they have holomorphic extensions on the open set where all the required pivot blocks are admitted, jointly with the field and the weight. ■

Proof. Each pivot block contains at most two fine cubes and contains a source cube. Thus (1637) admits its triple to the pivot chart. The construction of Section 51.1, with the unused coordinates set to zero, gives the same local chart and the bound $\sup |B'| \le 2C_P \sup |\mathcal{A}| \le 1$. Every pivot uses only its block; restriction, gauge covariance, conjugation and dependence on the configuration pass to the regional completion. At fixed U, intersection of the finitely many open pivot domains is open. On it the completion is holomorphic jointly with $\mathcal{A}$ and continuous on its admitted domain.

A nonempty $\mathsf{Pl}_0(Y)$ requires $Y = X(p)$ for a plaquette. Such a region consists of one, two or four fine cubes with a common corner; it has $d_0(Y) = 0$ and at most $12M^3$ plaquettes. If it contains a source cube, admission at Y gives $|\mathbf{U}(b)^{\pm 1}| \le e^{1/L}$. Put $\epsilon = 2C_P \sup |\mathcal{A}| \le 1$ and telescope the four plaquette factors. The product difference is at most $4(e^\epsilon - 1)e^{3\epsilon + 4/L} \le 188\epsilon$. Since $|\operatorname{tr} Z| \le 2|Z|$, summing the weighted action differences gives $|F^v_Y| \le 4512 C_P M^3 |v|_{0,Y} \sup |\mathcal{A}|$. This proves the uniform bound and holomorphy. If there is no source cube, the regional field is empty and its extension is zero. Vanishing at zero is immediate from the completion. The one-variable Cauchy formula proves the coefficient and tail bounds. On the bidisc $|s_i| < r_P^\circ/(2 \sup |A_i|)$, the coefficient of $s_1 s_2$ in $F^v_Y(s_1 A_1 + s_2 A_2)$ is the polarized quadratic part; the bidisc Cauchy bound proves (1641).

For the sharper derivative, write $W = \partial \mathbf{U}(p)$. Differentiation in a field C gives

$$D_p a_p(C) = -\tfrac{i}{2} \operatorname{tr}(\Xi W) = -\tfrac{i}{2} \operatorname{tr}(\Xi (W - 1)),$$

where Ξ is the sum of the four transported signed bond fields. Its trace is zero and $|\Xi| \le (1 + e^{2/L} + e^{6/L} + e^{8/L}) \sup |C| \le 8 \sup |C|$ for $L \ge 13$. The derivative of $B'(P, sA)$ at zero has supremum at most $2C_P \sup |A|$. The chain rule and the plaquette count give $12M^3 \cdot 8 \cdot 2C_P |v|_{0,Y} \varepsilon_\partial \sup |A|$, which is (1642). All structural assertions follow from the same finite formula and its Cauchy coefficients. Borel measurability can also be read from the limits of finite differences at real scalars. □

Let $z, a, \mu, \phi = \mu \oplus a$ be as above at $k = 0$, and assume

$$(C_{\mathcal{A}} + \theta_{\mathrm{sl}})\delta_0 \le r_P^\circ/2. \tag{1643}$$

Define the three centered members by

$$\mathbb{W}_0[v] = -[F_Y^{x_0 v}(g_0(z + \phi)) - F_Y^{x_0 v}(g_0 \phi)] + \sum_{b \in \mathfrak{X}_1^{\circ +} \langle Y \rangle} x_0 v(p_b) [F^{\mathbf{1}}_{Y,1}(g_0 z_b) + F^{\mathbf{1}}_{Y,2}(g_0 z_b, g_0 \phi)], \tag{1644}$$

$$\mathbb{W}_0^{\mathrm{src},1}[v] = -\sum_{b \in \mathfrak{X}^{\circ +} \langle Y \rangle} F^{x_0 (v - v(p_b))}_{Y,1}(g_0 \phi_b),$$

$$\mathbb{W}_0^{\mathrm{src},3}[v] = -\mathcal{R}_3 F_Y^{x_0 v}(g_0 \phi). \tag{1645}$$

The mixed term in (1644) has no factor two, by the first-step polarization convention.

Proposition 78.8. Suppose all triples under consideration satisfy (1637), the plaquette bound $|\partial \mathbf{U}-1|\le a_0^{\mathsf{Tab}}$, and (1643). If $|v|_\infty\le\sigma$ and $x_0\mathrm{sw}_0(v)\le\rho'$, the three members are defined on the entire stated tier times the center ball. They are linear and local in the weight, analytic and Borel, depend on the pair, and obey the gauge and conjugation laws. They vanish unless $d_0(Y)=0$. The last two are independent of z and vanish at $\phi=0$. At every rate $r\ge0$ their sum has norm at most

$$\begin{aligned} s_0^{\mathrm{ce}} &= C_0^{\mathrm{src}}[\sigma(1+g_0\mathfrak{b}_0^3)+\rho'(g_0a_0^{\mathsf{Tab}}+g_0\delta_0)] \\ &\quad + 9216C_PM^6\Theta\rho' a_0^{\mathsf{Tab}}\delta_0 + 2K^0(r_P^\circ)^{-3}\Theta^3\sigma g_0\mathfrak{b}_0^3, \end{aligned} \tag{1646}$$

where $\Theta=\max(C_{\mathcal{A}},\theta_{\mathrm{sl}})$ and

$$C_0^{\mathrm{src}}=\max\{4608C_PM^6,96K^0(r_P^\circ)^{-2}\Theta M^3,24K^0(r_P^\circ)^{-2}M^6,4K^0(r_P^\circ)^{-3}(C_{\mathcal{A}}+\theta_{\mathrm{sl}})^3\}. \tag{1647}$$

The norm has denominator $1+M^{-3}\sum_{b\in\mathfrak{X}_{\mathrm{I}}^{\circ+}}|z(b)|^2$. Joint holomorphy holds for affine holomorphic weight families with the same bounds. If the first unsourced frame has size $s_0^{\mathrm{rg}}\le1$ and

$$es_0^{\mathrm{rg}}+ew_0^{-1}s_0^{\mathrm{ce}}\le(4L^3K_{\mathrm{ap}}^+)^{-1},$$

these members satisfy the size conditions for its centered expansion. ∎

Proof. The two fields $g_0(z+\phi)$ and $g_0\phi$ lie in the half-radius ball. On a supporting domain there are at most $12M^3$ positive bonds. If $N=1+q_Y(z)$, then $\sum_b|z(b)|\le6M^3N$ and $(\sum_b|z(b)|)^2\le12M^6N$. A plaquette in $\mathsf{Pl}_0(Y)$ can be joined to any reference plaquette p_b by at most four steps of length M: pass through the centers of the at most four cubes forming $X(p)$. Thus $x_0\sup_{p,b}|v(p)-v(p_b)|\le4\rho'$. Taylor expansion decomposes (1644) into

$$-\sum_b F_{Y,1}^{x_0(v-v(p_b))}(g_0z_b)-\sum_b F_{Y,2}^{v-v(p_b)}(z_b,\phi)-F_{Y,2}^{v}(z)-[\mathcal{R}_3F_Y^{x_0v}(g_0(z+\phi))-\mathcal{R}_3F_Y^{x_0v}(g_0\phi)].$$

Using (1639)–(1642), the four terms, divided by N, are bounded by the four summands

$$4608C_PM^6\rho' g_0a_0^{\mathsf{Tab}},\quad 96K^0(r_P^\circ)^{-2}\Theta M^3\rho' g_0\delta_0,\quad 24K^0(r_P^\circ)^{-2}M^6\sigma,\quad 4K^0(r_P^\circ)^{-3}(C_{\mathcal{A}}+\theta_{\mathrm{sl}})^3\sigma g_0\mathfrak{b}_0^3.$$

The same derivative bound, summed over all bonds, gives the bound $9216C_PM^6\Theta\rho' a_0^{\mathsf{Tab}}\delta_0$ for the first member of (1645). Its last member is bounded by $2K^0(r_P^\circ)^{-3}\Theta^3\sigma g_0\mathfrak{b}_0^3$. Since $d_0(Y)=0$ on every supporting domain, no decay is lost at any rate. These inequalities prove (1646). The structural assertions follow from the preceding proposition; in particular their holomorphic extensions are defined on the open set where the pivot blocks and both shifted fields are admitted. The displayed amplitude condition implies $s_0^{\mathrm{ce}}<w_0$ and is the required centered-frame condition. □

The exact first-step split is as follows. Define

$$\tfrac12(B,\Upsilon_0(P;v)B)_1=\sum_Y F_{Y,2}^{v}[P|_Y](B|_Y). \tag{1648}$$

At a triple on the torus, let m be the unsourced conditional mean and $\bar m=m\oplus a$. Suppose its first and mixed quadratic Wilson derivatives cancel:

$$\sum_Y F_{Y,1}^{\mathbf{1}}(B|_Y)=0,\qquad \sum_Y F_{Y,2}^{\mathbf{1}}(Z|_Y,\bar m|_Y)=0\quad(Z|_{\mathfrak{x}_{\mathrm{E}}}=0).$$

Whenever all the terms are admitted and $z=B_{\mathrm{I}}'-m$ is in the centered field polydisc, finite Taylor expansion gives

$$\begin{aligned} -\sum_Y F_Y^{x_0v}[P|_Y](g_0(B_{\mathrm{I}}'\oplus a)|_Y) &= -\tfrac12(\bar m,\Upsilon_0(P;v)\bar m)_1 \\ &\quad+\sum_Y[\mathbb{W}_0+\mathbb{W}_0^{\mathrm{src},1}+\mathbb{W}_0^{\mathrm{src},3}][v](Y;P|_Y,z|_Y;m|_Y). \end{aligned} \tag{1649}$$

Indeed the reference first derivatives cancel bond by bond; the reference mixed derivatives cancel in the same manner. The remaining quadratic mean term is (1648), since $x_0g_0^2=1$. The two cubic

terms at $g_0\bar{m}$ cancel between the first and third centered members. This is the first-step version of lemma 78.5; verification of its cancellation hypotheses at the actual chain points is part of the measure argument.

Proposition 78.9. Suppose $\alpha_0, \alpha_1 \le a_P^\circ$ and $C_{\mathcal{A}}\delta_0 \le r_P^\circ/2$. On the history without large-field regions define

$$\mathbb{W}_0^{\mathrm{1l}}[v](Y;P,B) = -F_Y^{x_0 v}[P](g_0 B) + \sum_b x_0 v(p_b) F_{Y,1}^{\mathbf{1}}[P](g_0 B_b). \tag{1650}$$

This is an escort on every core $\mathcal{U}_1^\circ(Y;\alpha_0,\alpha_1)\times\mathbb{D}^{\mathrm{f}}(Y)$. It is linear and local in the weight, depends on the pair, is gauge invariant, vanishes at $B=0$, and is jointly holomorphic in the weight, field and open fixed-real core slices. Its conjugate is $\mathbb{W}_0^{\mathrm{1l}}[\bar{v}]$. For every $r \ge 0$,

$$\begin{aligned}\|\mathbb{W}_0^{\mathrm{1l}}[v]\|_{\star,r}^{\mathrm{co}} \le N_0^\circ = {} & 4608 C_P M^6 \rho' g_0 L^{-2}\alpha_0 + 24 K^0 (r_P^\circ)^{-2} M^6 \sigma \\ & + 2K^0 (r_P^\circ)^{-3} C_{\mathcal{A}}^3 \sigma g_0 \mathfrak{b}_0^3\end{aligned} \tag{1651}$$

if $|v|_\infty \le \sigma$, $x_0 \mathrm{sw}_0(v) \le \rho'$. For any history and any $Y \subset \Lambda_1$ on which the triple satisfies (1637), the sum of its three centered source members at center zero equals $\mathbb{W}_0^{\mathrm{1l}}[v]$ on their common field domain. ■

Proof. The core restricts into the simple triple domain at radii $(\alpha_0,\alpha_1,\alpha_0)$ by Lemma 49.2. Radius monotonicity gives (1637), and its plaquette bound is $L^{-2}\alpha_0$. The field room admits $g_0 B$ to the half-radius ball. Expanding (1650) gives the first, unmixed quadratic and cubic terms in the preceding proof, with $\phi = 0$, $\varepsilon_\partial = L^{-2}\alpha_0$ and field bound $C_{\mathcal{A}}\mathfrak{b}_0$. This gives exactly (1651). All supporting domains have $d_0(Y) = 0$. The finite formula and the open pivot extensions prove the analyticity and symmetry claims; restriction to $Y^\sharp \subset X$ preserves the core and the bound. Inside Λ_1 all bonds are integration bonds, the stored datum is empty, and the regional pivot completion is the one-lattice completion. At center zero $\phi = 0$, so the two members of (1645) vanish and (1644) reduces to (1650). □

79. The quadratic part of the source

The quadratic part of the plaquette source is evaluated at the conditional mean of the unsourced Gaussian. It therefore changes the retained quadratic factors while leaving that Gaussian and its conditional mean fixed. The resulting correction is linear in the source. We first construct its local terms, with bounds on their domains of definition; the measure argument will later justify their evaluation on the histories.

79.1. **Local quadratic forms and their matrices.** Fix a step $t = k+1$ and its integration and external bonds $\mathfrak{X} = \mathfrak{X}_{\mathrm{I}} \sqcup \mathfrak{X}_{\mathrm{E}}$. Coordinate fields are indexed by positively oriented bonds and take values in $\mathbb{C}^3$. All quadratic pairings in this section are complex bilinear. We use the canonical coefficient matrices a_S, the local inverse terms t_J, and their footprints from Proposition 55.2. On the whole torus their sums are

$$A = \sum_S a_S, \qquad T = \sum_J t_J = A_{\mathrm{II}}^{-1}, \qquad \mathsf{M} = -T A_{\mathrm{IE}}.$$

The last identity defines the conditional-mean operator. These identities are used only at points in the admitted pair class of proposition 55.2. Every individual coefficient is evaluated on its own hull.

Let $\mathbb{U}(Y)$ be a set of triples $P = (U_{\mathrm{r}}, A', \mathbf{J})$ at level t on $Y \in \mathcal{D}_k$. Assume that symmetric complex bilinear forms $q_Y(P;B,B')$ satisfy

$$|q_Y(P;B,B)| \le k_\upsilon e^{-r_\upsilon d_k(Y)} \sup_{b \in \mathfrak{X}^\circ\langle Y\rangle} |B(b)|^2, \qquad k_\upsilon > 0. \tag{1652}$$

The supremum over an empty set is zero. On each fixed real-factor slice, the coefficients of q_Y have local holomorphic extensions at every point of $\mathbb{U}(Y)$. This is the precise analyticity requirement; the sets $\mathbb{U}(Y)$ themselves need not be open.

To obtain open domains for the matrices, fix a real configuration U on the bare torus. For every region R of the level-t partition, let $\mathscr{P}_U(R)$ be the interior of the set of pairs $(A',\mathbf{J})$ such that

$$\begin{aligned}\Psi_R(A',\mathbf{J}) &:= (e^{i\eta_t A'}U, L^{-3}\mathbf{J})\big|_{\mathrm{Bd}^\circ(R\cap\Omega_0^\diamond)} \in \mathscr{O}(R;\alpha),\\ (U,A',\mathbf{J})|_Y &\in \mathbb{U}(Y) \quad \text{whenever the level-}t\text{ hull of } Y \text{ lies in } R.\end{aligned} \tag{1653}$$

Restriction maps these interiors into one another: the coordinate restriction is a surjective linear map and hence open, and the two conditions in (1653) are preserved by restriction. The map Ψ_R is holomorphic and commutes with restriction.

For $Y\in\mathcal{D}_k$, let $X(Y)$ be its level-t hull. Define a symmetric matrix υ_Y on $\mathscr{P}_U(X(Y))$ by

$$\langle \operatorname{vec} B, \upsilon_Y(P)\operatorname{vec} B'\rangle = -2q_Y(P|_Y;B,B'). \tag{1654}$$

Here $P=(U,A',\mathbf{J})$; the notation leaves the fixed real factor implicit. Let

$$c_E=\log 18432, \qquad C_E=4096, \qquad \beta_\upsilon = 8C_E e^{6\delta_\upsilon M}k_\upsilon, \qquad r_\upsilon-\delta_\upsilon M\ge c_E, \tag{1655}$$

where $0<\delta_\upsilon\le\delta_{\mathrm{pl}}$, the common rate of the canonical coefficients and inverse terms. Lengths are measured at level k.

Proposition 79.1. The matrices υ_Y are well defined and holomorphic. They are supported on $\mathfrak{X}^{\circ+}\langle Y\rangle$ in both indices and admit the majorants and connecting lengths

$$m_Y(b,b')=8k_\upsilon e^{-r_\upsilon d_k(Y)}1_{\{b,b'\in\mathfrak{X}^{\circ+}\langle Y\rangle\}}, \qquad l_Y=M(d_k(Y)+6). \tag{1656}$$

They satisfy

$$\sum_Y m_Y(b,b')e^{\delta_\upsilon l_Y}\le\beta_\upsilon. \tag{1657}$$

For every live pair of bonds there is a connected graph of length at most l_Y, containing their initial points and lying within distance M of every point of the footprint Y. ∎

Proof. Write $c_Y=k_\upsilon e^{-r_\upsilon d_k(Y)}$ and $N_Y(B)=\sup_{\mathfrak{X}^\circ\langle Y\rangle}|B|$. Polarization and rescaling give

$$|q_Y(P;B,B')|\le 2c_Y N_Y(B)N_Y(B').$$

If one seminorm vanishes, apply polarization to (sB,B') and let $s\to\infty$; the corresponding mixed value vanishes as well. The matrix in (1654) is therefore supported on Y. For a coordinate field with value z at one bond, $N_Y(\mathrm{vec}^{-1}z)\le\sqrt{2}|z|$. Applying the last inequality at two single-bond fields gives the block bound $2\cdot 2c_Y\cdot\sqrt{2}\cdot\sqrt{2}=8c_Y$. Its coefficients are the values of q_Y on coordinate basis fields. On the interior in (1653), their local holomorphic extensions agree with those values on an open set; hence they define holomorphic coefficients there.

A minimal edge tree meeting every M-cube of Y has length $Md_k(Y)$. Join the initial points of the two bonds to that tree inside their cubes, at a cost at most $3M$ each. Every point of Y is within M of the original tree. This proves the geometric assertion. For a fixed bond b, every domain contributing to (1657) contains the M-cube of b_-. The connected domain estimate of Proposition 45.1

$$\sum_{Y\supset\square} e^{-cd_k(Y)}\le C_E \qquad (c\ge c_E)$$

then gives

$$\sum_Y m_Y(b,b')e^{\delta_\upsilon l_Y}\le 8k_\upsilon e^{6\delta_\upsilon M}\sum_{Y\supset\square}e^{-(r_\upsilon-\delta_\upsilon M)d_k(Y)}\le\beta_\upsilon.$$

□

Lemma 79.2 (Quadratic parts of the plaquette weight)**.** Let $|v|_\infty \le x_k s_0$. For $k \ge 1$, use the weighted potential and its Taylor parts of proposition 78.1, and set

$$\mathbb{U}(Y) = \mathbb{U}_k^{\mathbf{1}}(Y|h'), \qquad q_Y^v(P;B,B') = T_2[v](Y|h';P;g_kB|_Y, g_kB'|_Y).$$

For $k = 0$, use the first-step potential $F_Y^v[P]$ of proposition 78.7 on $\mathbb{U}_0^\circ(Y)$, and set

$$q_Y^v(P;B,B') = \tfrac{1}{2}F_{Y,2}^v[P](g_0B|_Y, g_0B'|_Y).$$

The polarization convention for $F_{Y,2}$ has no factor $1/2$; its value on (B,B) is twice its homogeneous quadratic part. Then (1652) and its local holomorphic-extension requirement hold with

	k_υ	r_υ
$k \ge 1$	$K_0 r_0^{-2} s_0$	r_V
$k = 0$	$K^0 (r_P^\circ)^{-2} s_0$	any real number.

(1658)

Here r_V is the rate used for the weighted-potential bound, and

$$r_0 = \min\{\varepsilon_\mathfrak{h}, (3c_\mathfrak{h})^{-1}\}, \quad K_0 = K_c C_h'(L, \delta_V, 1) M^3, \quad K^0 = 4512 C_P M^3 r_P^\circ.$$

The forms are linear in v, vanish at $v = 0$, and depend only on plaquettes with all corners in Y. They have the pair and gauge properties of the weighted potential; Borel measurability and conjugation follow under the corresponding hypotheses on the shift system. At the first step those two properties require no shift hypothesis. ■

Proof. For $k \ge 1$, insert $g_kB|_Y$ in (1611) with $n = 2$. The factor $x_k g_k^2 = 1$ cancels the bound $x_k s_0$ for the weight, giving (1652). The coefficient functions have local holomorphic extensions by the Taylor construction; applying polarization to the extensions at $g_k(B + B')|_Y$, $g_kB|_Y$ and $g_kB'|_Y$ gives the required extension for $q_Y^v(P;B,B')$. At the first step, the same calculation uses (1639) and $x_0 g_0^2 = 1$. The potential there vanishes unless $d_0(Y) = 0$, so the bound holds with every r_υ. Linearity, pair dependence and covariance pass from the potential to its Taylor coefficients. Every plaquette used by the localized potential on Y lies in a lower-scale domain whose level-k hull is contained in Y; thus all its corners are in Y. This gives locality in the weight. The first-step sum has the same property directly. The Borel and conjugation assertions pass through the Taylor limits and polarization. □

In particular the matrix bound for these forms is

$$\beta_\upsilon = 8 C_E e^{6\delta_\upsilon M} s_0 \begin{cases} K_0 r_0^{-2}, & k \ge 1, \\ K^0 (r_P^\circ)^{-2}, & k = 0. \end{cases} \tag{1659}$$

At the first step choose $r_\upsilon = c_E + \delta_\upsilon M$. At later steps the inequality $r_V - \delta_\upsilon M \ge c_E$ is a substantive rate requirement.

79.2. **The correction at the conditional mean.** We describe the localization with its numerical losses. A local matrix family with constants (B, δ, ρ) has holomorphic members O_ι on their hulls, nonnegative block majorants m_ι and connecting lengths l_ι such that

$$|O_\iota(p)(b,b')| \le m_\iota(b,b'), \qquad \sum_\iota m_\iota(b,b') e^{\delta l_\iota(b,b')} \le B.$$

Every live pair has a connected graph of length at most $l_\iota(b,b')$ containing both initial points; the footprint is within distance ρ of this graph and contains the two bonds. All restrictions of the family use the same indices, majorants and lengths. These are the properties of the canonical families proved in Proposition 55.2. They also hold for the source vertices by proposition 79.1, with $\rho = M$. The canonical slack is $4M/L^2 \le M$, and $4M \le LM$.

In the following construction the two canonical families have constants $(B_{\mathrm f},\delta_\upsilon,\rho_\upsilon)$, the source family has constants $(\beta_\upsilon,\delta_\upsilon,\rho_\upsilon)$, and $B_{\mathrm f}\ge 1$, $4\rho_\upsilon\le LM$. Write $p_{\mathrm I},p_{\mathrm E}$ for the coordinate projections. There are four kinds of matrix products with exactly one source vertex:

$$\begin{aligned} d_Y &= p_{\mathrm E}\upsilon_Y p_{\mathrm E},\\ d_{Y\jmath S} &= -p_{\mathrm E}\upsilon_Y p_{\mathrm I}t_\jmath p_{\mathrm I}a_S p_{\mathrm E},\\ d_{S\jmath Y} &= -p_{\mathrm E}a_S p_{\mathrm I}t_\jmath p_{\mathrm I}\upsilon_Y p_{\mathrm E},\\ d_{S_1\jmath_1 Y\jmath_2 S_2} &= p_{\mathrm E}a_{S_1}p_{\mathrm I}t_{\jmath_1}p_{\mathrm I}\upsilon_Y p_{\mathrm I}t_{\jmath_2}p_{\mathrm I}a_{S_2}p_{\mathrm E}. \end{aligned} \tag{1660}$$

Each index includes its kind. The footprint of a product is the union of the footprints of its factors, and its hull is denoted by $X(\mathfrak z)$. Its majorant is the corresponding product summed over intermediate bonds; its length is the minimum sum of the constituent lengths over live intermediate bonds. Products whose majorant vanishes are omitted.

Use the keeping rule of Section 55.2: a product is kept on $\mathfrak k$ when $\mathsf{kp}_t(X(\mathfrak z))=1$ and $\mathfrak k=\mathfrak k_{X(\mathfrak z)}$. Let $d_{\mathfrak z}^{\mathrm k}$ be its compression to that component, or zero if it is rejected, and put $d_{\mathfrak z}^{\times}=d_{\mathfrak z}-d_{\mathfrak z}^{\mathrm k}$. Define

$$\widehat D_{\mathfrak k}=\sum_{\mathfrak z\ \text{kept on}\ \mathfrak k} p_{\mathfrak k}d_{\mathfrak z}p_{\mathfrak k},\qquad \widehat\Upsilon_{\mathfrak k}=\tfrac12(\widehat D_{\mathfrak k}+\widehat D_{\mathfrak k}^{\mathsf T}), \tag{1661}$$

$$\widehat\varphi_{\mathfrak k,X}(f;P)=-\tfrac12\sum_{\substack{\mathfrak z\ \text{kept on}\ \mathfrak k\\ X(\mathfrak z)=X}}\langle f,p_{\mathfrak k}d_{\mathfrak z}(P)p_{\mathfrak k}f\rangle, \tag{1662}$$

$$\widehat B^{\times}(X;P,f)=-\tfrac12\sum_{X(\mathfrak z)=X}\langle f,d_{\mathfrak z}^{\times}(P)f\rangle. \tag{1663}$$

Set $c_{\mathrm{lat}}(s)=3\sum_{z\in\mathbb Z^3}e^{-s|z|_\infty}$ and

$$\begin{aligned} c_\upsilon &= c_{\mathrm{lat}}(\delta_\upsilon/4), & B_{\mathrm{ch}}^\sharp &= 16c_{\mathrm{lat}}(\delta_\upsilon/8)^2B_{\mathrm f}^3,\\ C_\Upsilon &= c_\upsilon^2B_{\mathrm f}c_{\mathrm{lat}}(3\delta_\upsilon/8)B_{\mathrm{ch}}^\sharp, & K_{\mathrm{ch}}^\sharp &= C_3(LM)^6e^{C_3\kappa'}B_{\mathrm{ch}}^\sharp, \end{aligned} \tag{1664}$$

where $C_2=342$, $C_3=9e^8$ and $\kappa'>0$.

Theorem AG (Localization of a quadratic source)**.** Assume the standing geometric bounds $W_t\ge LM\ge 13^7$. The preceding series converge absolutely and locally uniformly on their domains. Their coefficients are holomorphic, the matrices $\widehat\Upsilon_{\mathfrak k}$ are symmetric, and

$$\sum_{\mathfrak z}m_{\mathfrak z}(b,b')\le c_\upsilon^2B_{\mathrm f}\beta_\upsilon B_{\mathrm{ch}}^\sharp e^{-3\delta_\upsilon|b-b'|/8},\qquad \|\widehat\Upsilon_{\mathfrak k}\|\le C_\Upsilon\beta_\upsilon. \tag{1665}$$

If

$$\tfrac12\delta_\upsilon LM\ge C_2(\kappa'+1), \tag{1666}$$

then, for $\sup|f|\le b_e$ and $n=\#_{\pi_t}X$,

$$\begin{aligned} |\widehat\varphi_{\mathfrak k,X}(f;P)| &\le \tfrac12c_\upsilon^2B_{\mathrm f}\beta_\upsilon K_{\mathrm{ch}}^\sharp b_e^2e^{-\kappa' n},\\ |\widehat B^{\times}(X;P,f)| &\le \tfrac12c_\upsilon^2B_{\mathrm f}\beta_\upsilon C_3B_{\mathrm{ch}}^\sharp b_e^2(LM)^6e^{-\delta_\upsilon W_t/8}e^{-\kappa' n}. \end{aligned} \tag{1667}$$

These are quadratic polynomials in the external coordinates in X. A kept term is zero unless $\mathsf{kp}_t(X)=1$ and $\mathfrak k=\mathfrak k_X$. A rejected term is zero unless $d_t(X)\ge\check R_t-2$. Either kind requires a big cube of X in $\Omega_t\cap Z_t$.

On the whole torus put $\Upsilon=\sum_Y\upsilon_Y$ and $F=(\mathsf M f,f)$. Then

$$-\tfrac12\langle F,\Upsilon F\rangle=\sum_{\mathfrak k,X}\widehat\varphi_{\mathfrak k,X}(f;P|_X)+\sum_X\widehat B^{\times}(X;P|_X,f), \tag{1668}$$

and

$$\sum_X\widehat\varphi_{\mathfrak k,X}(f;P|_X)=-\tfrac12\langle f,\widehat\Upsilon_{\mathfrak k}(P)f\rangle. \tag{1669}$$

The corrections are linear in the source vertices and depend only on vertices with hull contained in their displayed domain. They commute with compatible gauge transformations and conjugation. ■□

Proof. We give the majorant argument, including the loss in the decay rate. For two local families, multiply their majorants and sum an intermediate bond. Spending $\delta-\delta'$ of the decay on that bond gives a product family with bound $c_{\mathrm{lat}}(\delta-\delta')BB'$ at rate δ'. The live coefficient trees meet at the intermediate bond, so their union is connected, has the sum of their lengths, and retains the larger slack. In particular the composite family

$$-t_{\jmath_1}p_{\mathrm{I}}\upsilon_Y p_{\mathrm{I}}t_{\jmath_2}$$

has bound $c_\upsilon^2 B_{\mathrm{f}}^2\beta_\upsilon$ at rate $\delta_\upsilon/2$.

Set $\epsilon_0=(c_\upsilon^2 B_{\mathrm{f}}\beta_\upsilon)^{-1}$. For $R>1$, adjoin $\epsilon\upsilon_Y$ to the legs λa_S, and adjoin $-\epsilon t_{\jmath_1}p_{\mathrm{I}}\upsilon_Y p_{\mathrm{I}}t_{\jmath_2}$ to the inverse terms $\lambda t_\jmath$, with $|\lambda|<R$, $|\epsilon|<R\epsilon_0$. Each enlarged family has bound $2RB_{\mathrm{f}}$ at rate $\delta_\upsilon/2$. The three-factor estimate for its chains is therefore $R^3B_{\mathrm{ch}}^\sharp$ at rate $3\delta_\upsilon/8$. The chains containing exactly one adjoined member are precisely the four kinds in (1660). Their majorants are $R^{1+r}\epsilon_0 m_\mathfrak{z}$, with $r=0$ or 2. Retaining only those nonnegative terms and letting $R\downarrow 1$ proves the first inequality in (1665).

The conversion of connecting length to hull size uses the geometry in the proof of Proposition 55.3. We give the numerical calculation needed at the present rate. A live footprint of length l and slack ρ_υ, with $4\rho_\upsilon\le LM$, has a hull with $n\le 8+64l/(LM)$ cubes. A fixed hull has at most $9(LM)^6n^2$ ordered pairs of positive bonds. A rejected entry has $l\ge W_t-2-2\rho_\upsilon$. Applying these bounds to the enlarged families, at rate $\delta_\upsilon/2$, gives under (1666)

$$\epsilon_0\sum_{b,b'}\sum_{X(\mathfrak{z})=X} m_\mathfrak{z}(b,b')\le K_{\mathrm{ch}}^\sharp e^{-\kappa' n},$$

$$\epsilon_0\sum_{b,b'}\sum_{\substack{X(\mathfrak{z})=X\\ l_\mathfrak{z}(b,b')\ge W_t-2-2\rho_\upsilon}} m_\mathfrak{z}(b,b')\le C_3B_{\mathrm{ch}}^\sharp(LM)^6e^{-\delta_\upsilon W_t/8}e^{-\kappa' n}.$$

Indeed, put $\delta=\delta_\upsilon/2$, $l_1=\max\{0,LM(n-8)/64\}$ and $l_\times=W_t-2-2\rho_\upsilon$. Before summing the bond pairs, the two bounds are respectively $B_{\mathrm{ch}}^\sharp e^{-3\delta l_1/4}$ and $B_{\mathrm{ch}}^\sharp e^{-3\delta\max(l_1,l_\times)/4}$. The hypothesis $\delta LM\ge 342(\kappa'+1)$ gives

$$\tfrac{3}{4}\delta l_1\ge(\kappa'+1)(n-8).$$

Since $n^2e^{-n}<1$, the first total is at most $9e^8(LM)^6e^{8\kappa'}B_{\mathrm{ch}}^\sharp e^{-\kappa' n}$. For the second, use $\max(l_1,l_\times)\ge l_1/4+3l_\times/4$, $2\rho_\upsilon\le LM/2\le W_t/2$, and $W_t\ge LM\ge 13^7$:

$$\tfrac{3}{4}\delta\max(l_1,l_\times)\ge\delta\left(\tfrac{9}{16}l_\times-\tfrac{3}{128}LM\right)+\tfrac{3}{1024}\delta LMn\ge\tfrac{1}{4}\delta W_t+(\kappa'+1)n.$$

The second total is therefore at most $9(LM)^6B_{\mathrm{ch}}^\sharp e^{-\delta W_t/4}e^{-\kappa' n}$. These are the displayed bounds, since $C_3=9e^8\ge 8$. Pairing with f gives (1667). The first estimate in (1665), summed over one bond index, gives the norm bound by the Schur estimate. Compression and symmetrization do not increase it.

Every coefficient in a product is holomorphic on its own hull. The uniform majorants prove local uniform convergence. Both live endpoint bonds lie in the footprint, so the sums are quadratic polynomials supported in X. The keeping rule gives the stated support properties and the unique component of a kept term.

Absolute convergence permits multiplication and regrouping of the three families. Summing the four kinds gives

$$\Upsilon_{\mathrm{EE}}-\Upsilon_{\mathrm{EI}}TA_{\mathrm{IE}}-A_{\mathrm{EI}}T\Upsilon_{\mathrm{IE}}+A_{\mathrm{EI}}T\Upsilon_{\mathrm{II}}TA_{\mathrm{IE}}.$$

Since $T=A_{\mathrm{II}}^{-1}$ and $\mathsf{M}=-TA_{\mathrm{IE}}$, this is the matrix of $\langle(\mathsf{M}f,f),\Upsilon(\mathsf{M}f,f)\rangle$. Splitting its terms into kept and rejected parts proves (1668); symmetrization does not change a quadratic form and

gives (1669). Each product has one source vertex and uses only restrictions to its hull. This proves linearity and locality. Compatible gauge transformations commute with the coordinate projections, and conjugation commutes with the products and the absolutely convergent sums, proving the two symmetry assertions. □

79.3. **Triples, pairs, and retained factors.** We now use the graded and layered triple domains of a history step. Write $\mathscr{U}_X^{\mathrm{m}}$ for the graded domain and $\mathscr{U}_X^{\mathrm{o}}$ for the layered output domain. The numerical domain hypotheses needed below are the following explicit inclusions and chart properties:

(i) $\mathscr{U}_X^{\mathrm{o}} \subset \mathscr{U}_X^{\mathrm{m}}$; every fixed-real-factor slice of $\mathscr{U}_X^{\mathrm{m}}$ is open, and restriction to a smaller domain preserves it.

(ii) If $P \in \mathscr{U}_X^{\mathrm{m}}$, its inner pair $p_X(P) = (e^{i\eta_t A'} U_{\mathrm{r}}, L^{-3}\mathbf{J})|_{\mathrm{Bd}^\circ(X\cap\Omega_0^\circ)}$ lies in $\mathscr{O}^+(X;\alpha)$, and $P|_Y \in \mathbb{U}(Y)$ for every Y whose level-t hull lies in X.

(iii) In these slices, $\eta_t|A'(b)| < \log 2$ on the inner bonds.

They are imposed at each step, with a single table of radii and its truncations. These hypotheses distinguish membership in the analytic domains from the later question of evaluating a history measure. No independence of an evaluation representative is assumed here.

Proposition 79.3. Under these domain hypotheses, the source corrections are defined on $\mathscr{U}_X^{\mathrm{m}}$ and their restrictions to $\mathscr{U}_X^{\mathrm{o}}$ are analytic functions with stored data. For $\mathsf{kp}_t(X) = 1$, put

$$\widehat{\varphi}_X(P;\mathfrak{q}) = \widehat{\varphi}_{\mathfrak{k}_X,X}(\operatorname{vec} a[\mathfrak{q}]; P), \tag{1670}$$

where $a[\mathfrak{q}]$ is the stored external field, extended by zero away from X. If the stored tiers obey $\sup|\operatorname{vec} a[\mathfrak{q}]| < C_{\mathrm{xt}}\mathfrak{b}_k$, then under (1666)

$$\operatorname{osc}\widehat{\varphi}_X \le 2\sup|\widehat{\varphi}_X| \le b_{\mathrm{fr}}^{\mathrm{S}}\mathfrak{b}_k^2 e^{-\kappa' d_t(X)}, \quad b_{\mathrm{fr}}^{\mathrm{S}} = c_\upsilon^2 B_{\mathrm{f}}\beta_\upsilon K_{\mathrm{ch}}^\sharp C_{\mathrm{xt}}^2 e^{-\kappa'}. \tag{1671}$$

The corrections inherit pair dependence, gauge covariance, Borel measurability and conjugation whenever the local forms have the corresponding properties. They are linear and local in these forms. ■

Proof. Extend the real factor of a triple on X to a real configuration U on the torus. The open slice containing the triple lies in $\mathscr{P}_U(X)$: this follows directly from (i), (ii) and the definition as an interior. Thus proposition 79.1 and theorem AG apply. Every chain uses only the restricted triple on its hull, so its value is independent of the extension U. The uniform majorants prove holomorphy on the open slice. A stored field enters through finitely many coordinates; hence the restriction is analytic with stored data. Substitute $b_e = C_{\mathrm{xt}}\mathfrak{b}_k$ into (1667) and use $\#_{\pi_t}X \ge d_t(X)+1$ to obtain (1671). The same argument works on the factor domain in every later history, since the domains, radii and stored coordinates at this step are unchanged. Each coefficient is a finite product of coefficients with the asserted dependence or symmetry. The convergent sums preserve those properties; for measurability use their Borel partial sums. □

If the forms depend only on the inner pair, so do the corrections. This assertion alone does not establish analyticity as a function of that pair. The following local chart supplies it.

Lemma 79.4. Let $c^\sharp$ be holomorphic on every fixed-real-factor slice of $\mathscr{U}_X^{\mathrm{m}}$ and constant on triples having the same inner pair. It descends to a function on

$$\mathscr{O}^{\mathrm{S}}(X) = \{p_X(P) : P \in \mathscr{U}_X^{\mathrm{m}}\}, \tag{1672}$$

which admits a holomorphic extension near each point of this set. ■

Proof. Choose $P_0 = (U_0, A_0', \mathbf{J}_0)$ over a pair p_0. On the inner bonds define, for pairs $(\mathbf{U},\mathbf{j})$ near p_0,

$$A'(b) = (i\eta_t)^{-1}\log(\mathbf{U}(b)U_0(b)^{-1}), \qquad \mathbf{J}(b) = L^3\mathbf{j}(b),$$

and keep the other coordinates equal to those of P_0. By (iii) the series logarithm is defined near p_0 and recovers A_0' there. This gives a holomorphic local section Θ of p_X. Shrink its neighborhood

so that Θ remains in the open slice of P_0. Then $c^\sharp \circ \Theta$ is the required extension; constancy on the fibers proves agreement with the descended function. The empty inner-bond set gives a single pair and a constant function. □

Corollary 79.5. Suppose the local forms depend only on the inner pair. Then the functions

$$\widehat{K}(X;p,a)=\sum_{\mathfrak{k}}\widehat{\varphi}_{\mathfrak{k},X}(\operatorname{vec}a;P),\qquad \widehat{B}^{\times}(X;p,a)=\widehat{B}^{\times}(X;P,\operatorname{vec}a),\quad p=p_X(P), \tag{1673}$$

are well defined on $\mathscr{O}^{\mathrm{S}}(X)$ and are finite quadratic polynomials in the external coordinates in X, with coefficients analytic on that set. They have the locality, support and symmetry properties of theorem AG. Under (1666), for $\sup|a|\le b_e$,

$$\begin{aligned}|\widehat{K}(X;p,a)|&\le c^{\mathrm{S}}_{\mathrm{kp}}b_e^2e^{-\kappa' d_t(X)},\\ |\widehat{B}^{\times}(X;p,a)|&\le c^{\mathrm{S}}_{\mathrm{rj}}b_e^2e^{-(\check{R}_t-2)}e^{-\kappa' d_t(X)},\\ c^{\mathrm{S}}_{\mathrm{kp}}&=\tfrac12 c_v^2B_{\mathrm{f}}\beta_v K^{\sharp}_{\mathrm{ch}}e^{-\kappa'},\\ c^{\mathrm{S}}_{\mathrm{rj}}&=\tfrac12 c_v^2B_{\mathrm{f}}\beta_v C_3B^{\sharp}_{\mathrm{ch}}(LM)^6e^{-\kappa'}.\end{aligned} \tag{1674}$$

■□

Proof. Apply lemma 79.4 to the finitely many quadratic coefficients given by proposition 79.3. For the bound use $|\operatorname{vec}a|\le|a|$, $\#_{\pi_t}X\ge d_t(X)+1$, and $\delta_vLM/8\ge1$, which follows from (1666). Since $W_t=LM\check{R}_t$, $e^{-\delta_vW_t/8}\le e^{-(\check{R}_t-2)}$. Only the unique component $\mathfrak{k}_X$ contributes to the first sum. □

For a sequence of histories, add (1670) to every retained quadratic exponent and leave the elimination exponents unchanged. Thus

$$\varphi^{v}_f=\begin{cases}\varphi_f+\widehat{\varphi}_{X_f}, & f\text{ is a retained quadratic factor},\\ \varphi_f, & f\text{ is an elimination factor}.\end{cases} \tag{1675}$$

Corollary 79.6. Assume that h has positive level and that the preceding domain hypotheses hold at all its steps, with one table of radii and its truncations. Suppose the unsourced factor family has oscillation bound $B^{\mathrm{z}}\mathfrak{b}_{t-1}^2e^{-\kappa d_t(X)}$ at step t, where $0<\kappa\le\kappa'$. Then the family in (1675) has the same names, domains and factor spaces, and oscillation bound

$$(B^{\mathrm{z}}+\bar{b}^{\mathrm{S}}_{\mathrm{fr}})\mathfrak{b}_{t-1}^2e^{-\kappa d_t(X)},\qquad \bar{b}^{\mathrm{S}}_{\mathrm{fr}}=\max_{t\le\mathrm{level}(h)}b^{\mathrm{S},(t)}_{\mathrm{fr}}.$$

It preserves prefix consistency and the analytic, Borel, pair and gauge properties verified above. For quadratic parts linear in a plaquette weight, the added exponent is linear in that weight and depends only on plaquettes with all corners in X_f. ■□

Proof. At a fixed step the layered space and stored tiers are unchanged under extension of the history. The definitions therefore give the same partial functions before and after that extension. Proposition 79.3 supplies their values and analytic domains. The triangle inequality for oscillation, (1671) and $\kappa\le\kappa'$ give the bound. Every remaining assertion holds factor by factor; the elimination factors were unchanged. □

79.4. **Damping on defined history values.** The last estimate of this section concerns actual evaluations. Let h have level $n\ge1$, and let $U\in\mathcal{V}_0[h]\cap\operatorname{dom}\xi^{\mathrm{co}\prime}_h$. Let $f_t=\operatorname{vec}\mathfrak{q}[U]_{t-1}$ be its stored real field, extended by zero away from the external bonds at step t. Assume the fixed-radius and kept-positivity hypotheses of Section 66.2, and assume that an admitted evaluation representative $\mathsf{U}^\flat$ places every factor triple in its layered domain. The stored-form calculation in equations (1436)–(1438) gives real symmetric kept matrices and

$$\sum_{t=1}^{n}\sum_{\mathfrak{k}\in\mathfrak{K}_t(h)}\langle f_t,\Delta^{\mathrm{loc}}_{\mathfrak{k}}(p_t)f_t\rangle\ge2\gamma_{\mathrm{kp}}\mathfrak{n}(U),\qquad \mathfrak{n}(U)=\sum_{t=1}^{n}\|f_t\|^2. \tag{1676}$$

The external bond sets $E_{\mathfrak{k}}^{+}$ partition the stored bonds piece by piece, so this is also the sum of the squared norms of $f_t|_{E_{\mathfrak{k}}^{+}}$ over t and $\mathfrak{k}$. The determinant factors of the elimination steps have the same value on every glued configuration of the fields of U by Lemma 66.1.

Proposition 79.7. Assume the hypotheses just stated and the quadratic construction at each step, and put

$$\bar{v} = C_{\Upsilon} \max_{1 \le t \le n} \beta_{v}^{(t)}.$$

At every admitted evaluation representative, the product of the modified factors is bounded by

$$\left| e^{\Phi_h^v(\mathsf{U}^\flat;\mathfrak{q}[U])} \right| \le \prod_{t=1}^{n} \prod_{c \in \Gamma_t^{\partial}(h)} |\det(Lm_c(\mathcal{M}^{t-1}\mathsf{U}))|^{-1} \exp\left[-\left(\gamma_{\mathrm{kp}} - \tfrac{1}{2}\bar{v}\right) \mathsf{n}(U) \right], \tag{1677}$$

where U is any glued configuration of the fields of U. The same estimate holds for $\Phi_h^v(\phi)$ whenever that value is defined, meaning that the exponent sum is defined and constant on the evaluation set of ϕ. ∎

Proof. For a component $\mathfrak{k}$ at step t, sum the source chains at the triple supplied by the evaluation representative on each individual hull:

$$\widehat{D}_{\mathfrak{k}}^{\mathrm{ev}}(\mathsf{U}^\flat) = \sum_{\mathfrak{z} \text{ kept on } \mathfrak{k}} p_{\mathfrak{k}} d_{\mathfrak{z}}(P_{X(\mathfrak{z})}(\mathsf{U}^\flat)) p_{\mathfrak{k}}. \tag{1678}$$

These are the compressions in (1662); $p_{\mathfrak{k}}$ projects onto $E_{\mathfrak{k}}^{+}$. Thus the matrix is supported on $E_{\mathfrak{k}}^{+} \times E_{\mathfrak{k}}^{+}$. Every displayed value is defined by the domain hypothesis. The majorants in (1665) are independent of the triple, so the series converges absolutely and its row and column sums give

$$\|\widehat{D}_{\mathfrak{k}}^{\mathrm{ev}}(\mathsf{U}^\flat)\| \le C_{\Upsilon} \beta_{v}^{(t)} \le \bar{v}.$$

This argument does not evaluate all chains at one unverified larger-domain triple. The live endpoint bonds of each chain lie inside its hull, so the stored field restricted to that hull has the same values as f_t in its pairing. Absolute convergence therefore gives the exact product

$$e^{\Phi_h^v} = \prod_{t,c} \det(Lm_c(\mathcal{M}^{t-1}\mathsf{U}))^{-1} \exp\left[-\tfrac{1}{2} \sum_{t,\mathfrak{k}} \langle f_t, (\Delta_{\mathfrak{k}}^{\mathrm{loc}}(p_t) + \widehat{D}_{\mathfrak{k}}^{\mathrm{ev}}) f_t \rangle \right].$$

The component support and the norm bound give

$$\left| \langle f_t, \widehat{D}_{\mathfrak{k}}^{\mathrm{ev}} f_t \rangle \right| \le \bar{v} \|f_t|_{E_{\mathfrak{k}}^{+}}\|^2.$$

Summing over the partition of the stored bonds shows that the real part of the added sum is at least $-\bar{v}\mathsf{n}(U)$. Combining this with (1676) and taking the modulus proves (1677). If the value at ϕ is defined, choose any admitted representative and apply that bound to its common value. No constancy on an evaluation set follows from the norm estimate itself. □

79.5. **The Hessian at an actual history.**

Lemma 79.8. At an admitted real chain point of step $t = k+1$, let U_+ be the collared minimal background in the required axial gauge and let $J_+ = J^{\circ}(U_+)$. Put $p = \mathcal{M}^k U_+$. The quadratic operator in the actual conditional Gaussian is $A = \widehat{G} + \widetilde{A}$, where $\widetilde{A} = C_p^{\mathsf{T}} \Delta_k(U_+, J_+) C_p$ is the Hessian of the regional constrained action. If $\overline{m} = \mathbf{m} \oplus \mathsf{a}$ is its conditional mean, then

$$\chi_{\mathrm{I}} \widehat{G}\, \overline{m} = 0, \qquad \chi_{\mathrm{I}} \widetilde{A}\, \overline{m} = 0.$$

The unit-weight local potentials satisfy

$$\sum_{Y} T_1 V_Y^{\mathrm{act}}(B) = 0, \qquad \sum_{Y} T_{1,1} V_Y^{\mathrm{act}}(B_1, B_2) = (B_1, \widetilde{A} B_2)_1$$

for complex directions as well as real ones. These assertions include the first step. ∎

Proof. The shorter determining set at this point is the reduced determining set of the collared sequence before the step. Adding $\Omega_t^{(t)}$ gives the longer sequence. The actual background is the minimal orbit for this longer sequence; choosing its axial representative does not change the local action values. The real background and shift construction gives its strict $2B_3\varepsilon_t$ regularity and its natural-class representation with zero perturbation. We verify the nearby-orbit and stationarity requirements of proposition 78.3 separately. The shorter datum $d(p)$ is regular at $2L^2C_{\mathrm{reg}}\varepsilon_t \le \bar{\varepsilon}$. The separated shorter sequence satisfies the geometric hypotheses of Theorem E, with the radii fixed by the real choices. For a differentiable fibre curve $V(s)$ through p, set

$$B(s)_k(b) = -i\log(V(s)(b)p(b)^{-1}), \qquad B(s)_j = 0 \quad (j<k).$$

On a sufficiently short interval the logarithm is defined, $B(s)$ lies in the selector's row ball, and $e^{iB(s)}d(p) = d(V(s))$. The real analytic selector of that theorem therefore gives a differentiable representative U_s of the shorter regular orbit. Its constraint set is contained in the fixed longer constraint set. At $s=0$ this orbit is contained in the longer critical orbit, so differentiation of the action along U_s gives zero. The action has the same value on every representative of the shorter orbit. This proves the required stationarity. When $k=0$ the shorter orbit is a singleton; its existence near p follows from the finitely many strict regularity inequalities and continuity. The same longer-constraint argument gives stationarity in that case. The regional criticality and regularity conclusions of Proposition 58.9 now apply. The current is the regional action current; its extension at the boundary is harmless by Lemma 58.8. Thus the regional calculation in Proposition 58.27, restated in proposition 78.3, applies at this point.

The integration bonds are exactly the nonpivot bonds in $B(\Lambda_t^{(t)})$; the complementary nonpivot bonds are external. These are also the two blocks used by the conditional Gaussian. Its internal matrix is invertible by the real conditional Gaussian gap. The gauge-degeneracy identity for the regional Hessian therefore permits lemma 78.4 to be applied to this very block decomposition. It gives the two separate mean identities displayed above.

By Proposition 58.24, the finite sum of local action potentials along a sufficiently short real fibre ray is the change of the regional constrained action. The first derivative vanishes by criticality, and the second derivative is the Hessian just identified. Each local potential is holomorphic near zero and there are finitely many of them. Comparing coefficients along real rays therefore gives the two identities for small real directions. Homogeneity extends them to every real direction, and equality of the resulting complex polynomials extends them to complex directions. Polarization yields the displayed bilinear identity. The same argument at $k=0$ uses the first-step local action potential and its radius r_P°; its unit-weight potential is the actual action potential. Thus the mean cancellation used at the first step has the same Hessian and the same internal block. □

80. The source-dependent interactions

Fix the real construction of Theorem D, including its couplings, histories, collars, tables of radii, test functions and coordinate maps. These data are independent of the source. Throughout this section the source budgets of lemma 82.4 are imposed. We use them through the bounds displayed below; their simultaneous verification is the subject of lemma 82.5. Write $t = k+1$ and $h' = (h, \mathsf{o}_t)$ for a history step. Unless stated otherwise, $\Omega_1(h') \neq \emptyset$.

The regular terms are functions on the fixed cores $\mathcal{U}_t^\circ(X;\alpha_0,\alpha_1)$. The boundary terms are defined on

$$\mathfrak{G}_X(h') = \mathscr{U}_X(h') \times \mathfrak{Q}_{h'}^{\mathrm{rad}}(X),$$

where $\mathscr{U}_X$ is the graded space of triples of the real construction and the second factor imposes the stored-field tiers. A retained exponent formed at level $j = l+1$ has domain

$$\mathscr{P}_{X_f}(h'|_j) = \mathsf{Sp}(\mathcal{S}(h'|_j);(j,X_f,\mathrm{mk})) \times \mathfrak{Q}_{h'|_j}^{(l),\mathrm{rad}}(X_f). \tag{1679}$$

Its name, support and domain are those of the corresponding real factor. Restriction to these domains is part of the construction. We make no assertion about maximal analytic continuation beyond them.

80.1. **One integration and the stored terms.** Let $\mathscr{L}_k^{w\flat}$ contain the regular, boundary and retained entries already constructed through level k. Its counterterms are the real counterterms $b_1,\dots,b_k$. At step t use the centered source family and its one-lattice companion from propositions 78.2 and 78.6; at $t=1$ use their first-step versions proposition 78.9. The quadratic terms of corollary 79.5 are added to the real conditional forms:

$$K^{\mathrm{kp,w}} = K^{\mathrm{kp}} + K^{\mathrm{kp},+}, \qquad B^{\times,w} = B^{\times} + B^{\times,+}. \tag{1680}$$

Here the first correction vanishes unless the domain is retained. The conditional Gaussian, its covariance and its mean are those of the unsourced construction. Their domains and positivity are unchanged.

We denote the centered cluster function by $E_h^{\mathrm{cl},w}(X;\mathsf{a};P;\mathfrak{q})$, the pointed term of the empty history by $F_\emptyset^w(X,P,z)$, and the localized normalizer by $N_\emptyset^{\mathrm{un}}(X,p_X(P),z)$. The new regular term is

$$E^{(t),w}(X,P,z) = N_\emptyset^{\mathrm{un}}(X,p_X(P),z) + F_\emptyset^w(X,P,z) - n_{\mathrm{tr}}^{(t)}(X,z). \tag{1681}$$

Equivalently, $E^{(t),w} = \mathbf{N}^{(t)} + F_\emptyset^w$; $\mathbf{N}^{(t)}$ includes the displayed subtraction and is independent of w. Only the old regular entries, the counterterms and the one-lattice companion enter $F_\emptyset^w$. In particular (1681) does not depend on boundary entries or on a choice of strip representation.

For the boundary contribution use the five localized strip terms $R^{6,w}, R^{7,w}, R^{8,w}, R^{9,w}, R^{\mathrm{E},w}$ of Section 62, inserting $K^{\mathrm{kp,w}}$, $B^{\times,w}$ and $E_h^{\mathrm{cl},w}$ in the last three places. The first two, which come from the strip Jacobian and Haar density, are unchanged. Set

$$\begin{aligned}
B^{\mathrm{nm}}(X) &= N_h^{\partial}(X;p_X(P)) - \mathbf{1}_{\{X\subset\Lambda_t\}}\sum_z N_\emptyset^{\mathrm{un}}(X,p_X(P),z),\\
B^{\mathrm{cs},w}(X) &= E_h^{\mathrm{cl},w}(X;\mathsf{a};P;\mathfrak{q}) - \mathbf{1}_{\{X\subset\Lambda_t\}}\sum_z F_\emptyset^w(X,P,z),\\
B^{\mathrm{rj},w}(X) &= B^{\times,w}(X;p_X(P),\mathsf{a}),\\
B^{\mathrm{sr},w}(X) &= R^{6,w}(X)+R^{7,w}(X)+R^{8,w}(X)+R^{9,w}(X)+R^{\mathrm{E},w}(X),\\
B_t^{\mathrm{mk},w}(X) &= B^{\mathrm{nm}}(X)+B^{\mathrm{cs},w}(X)+B^{\mathrm{rj},w}(X)+B^{\mathrm{sr},w}(X).
\end{aligned} \tag{1682}$$

As usual an expression multiplied by a zero indicator is omitted. Thus its domain need not be imposed when it is unused.

The new retained exponents are

$$\varphi_f^w = \begin{cases} \varphi_f + \widehat{\varphi}_{X_f}^{\widehat{w}_k}, & f \text{ of kind kp},\\ \varphi_f, & f \text{ of kind el}, \end{cases} \qquad \widehat{w}_k(p) = L^{K-k}w(x(p)). \tag{1683}$$

All old entries are prolonged by ignoring the newly stored coordinates. We restrict the three newly formed families to their core, graded-tier and retained domains, respectively. Starting with $\mathscr{L}_0^{w\flat} = (x_0)$, this defines

$$\mathscr{L}_t^{w\flat} = \mathrm{Step}_t^{+}[\mathscr{L}_k^{w\flat};\mathsf{Ss}^w]\Big|_{\text{cores, graded tiers and retained domains}}. \tag{1684}$$

This notation refers to the explicit operations (1680)–(1683) and the centered expansion; it does not presume that the output has the required domains or bounds.

Proposition 80.1. The recursion (1684) is defined at every level. Its regular entries are defined on the whole fixed core, its created boundary entries on every $\mathfrak{G}_X$, and its retained exponents on (1679). On these domains the entries are local, Borel measurable, analytic on each complex slice,

and gauge invariant. Their dependence on the source is holomorphic on finite-dimensional source slices and is local to the indicated domain. Conjugation takes the entry at w to the entry at $\bar{w}$ and conjugates its arguments.

With the notation of the source budgets,

$$\|E^{(j),w}\|_{j,\kappa+1} \le E_0^{\mathrm{S}}, \qquad |B_j^{\mathrm{mk},w}(X)| \le B_T^{\mathrm{th}} e^{-\kappa_T d_j(X)} \le B^{\mathrm{S}} e^{-\kappa d_j(X)}, \tag{1685}$$

where $\kappa_T = (1 + 4\beta_{\mathrm{s}})\kappa$. The boundary term is zero on $X \subset \Lambda_j$ and off the boundary index set. The retained exponents have the same names and domains as at zero source, and their oscillation bound is at most $B_{\mathrm{fr}}^{\flat} + \bar{b}_{\mathrm{fr}}^{\mathrm{S}} \le B_{\mathrm{fr}}^{\mathrm{S}}$. At $w = 0$ all these functions are restrictions of the canonical real functions. The construction commutes with cutting a history and the data at an earlier level. ∎

Proof. We first explain why complex entries are admissible in the potential construction of Sections 58–59. Keep all geometric data and couplings fixed and compare the list with the null list, whose interaction entries vanish. The dependence of the inherited regular potential on the regular entries is complex linear: its operations are restriction, subtraction of the constant and quadratic parts, finite localization, and sums. The inherited boundary and retained potentials have the same property in their respective entries. The action, gauge and Jacobian potentials are independent of these entries. Thus the exact finite action identity for the null list, plus the three linear differences, is the identity for the complex list. The estimates of Proposition 58.22 use absolute values and Cauchy bounds on the same domains. They apply to these differences without a reality assumption. The conjugation identity relates the conjugate lists, rather than declaring their values real.

Suppose the entries through level k have been constructed. Their bounds put the old list in the required box. In particular the edge contribution is bounded by $8M^3E_0^{\mathrm{S}}$, while the boundary and retained boxes are large enough because

$$B^{\mathrm{S}} \ge \max(8M^3E_0^{\mathrm{S}}, B_T^{\mathrm{th}}), \qquad B_{\mathrm{fr}}^{\flat} + \bar{b}_{\mathrm{fr}}^{\mathrm{S}} \le B_{\mathrm{fr}}^{\mathrm{S}}.$$

The ordinary potential bounds therefore hold on the full graded domains and stored-field tiers. The weighted potentials have these domains by proposition 78.2 and lemma 82.5; the budgets in lemma 82.4 put the sum of ordinary and centered potentials within the size allowed by theorem AF. The latter theorem supplies the cluster function on its full external-field perturbation domain, including the tier values. It also supplies its analytic and locality properties and the interior identity with the empty history. Consequently the right side of (1681) is defined on the whole core. The regular bound is verified below independently of the boundary entries.

The quadratic corrections are polynomials in the external field, with analytic coefficients on the pair image $\mathscr{O}^{\mathrm{S}}(X)$, by corollary 79.5. They involve only external bonds in X. We use the following complex version of the boundary argument of Proposition 62.1. Keep the geometric data and domains fixed. For an ingredient F define $F^{\mathrm{cj}}(P;\mathfrak{q}) = \overline{F(P^{\mathfrak{c}};\mathfrak{q})}$, and form $(F + F^{\mathrm{cj}})/2$ and $(F - F^{\mathrm{cj}})/(2i)$. The resulting real and imaginary systems satisfy the real conjugation identities and the same modulus bounds on the same domains. Also take the system with the normalizer, pointed, cluster and quadratic ingredients zero and with the same geometric shift. The proposition applies to these three systems. The rows $8, 9, \mathrm{E}$ are linear in their respective ingredients; rows $6, 7$ are fixed by the geometry. Consequently, for the full boundary term and for the strip term,

$$B = B^{\mathrm{re}} + i(B^{\mathrm{im}} - B^0), \qquad B^0 = R^6 + R^7. \tag{1686}$$

For the other kinds and for rows $8, 9, \mathrm{E}$ the analogous identity has no B^0 term. Thus definedness on the full graded-tier product, holomorphy, measurability, support and the joint gauge and pair identities pass from the three real systems to the complex system. This argument also handles the fixed affine contribution of the strip density.

To obtain the modulus bounds without enlarging their constants, apply the finite transfer estimate to the complex ingredients themselves. Their domains are those of the real and imaginary systems, and their bounds are the original sourced bounds. To check the analytic room explicitly, the allowed

external perturbation has radius $r_{\mathrm{bd}} = 2\rho_{\mathrm{bd}}/g_k \le C_{\mathrm{xt}}\mathfrak{b}_k$. Adding it to a tier field keeps the field in the ball of radius $2C_{\mathrm{xt}}\mathfrak{b}_k$, on which the two quadratic estimates hold. The cluster function is defined on that perturbation domain by the preceding paragraph. The Haar term has $g_k|\mathsf{a}+\mathsf{e}| < 3/8$; hence its fixed series logarithm is analytic there. The localized transfer is a finite sum of values on precisely these domains. Its modulus estimates give

$$|B_t^{\mathrm{mk},w}(X)| \le B_N e^{-\delta_N M d_t(X)} + (b_{\mathrm{cs}} + b_{\mathrm{rj}})e^{-\kappa'' d_t(X)} + b_{\mathrm{sr}} e^{-\kappa_T d_t(X)}.$$

Here b_{rj} and b_{sr} are the expressions in Proposition 62.1, with the enlarged sourced form constants. The source budgets give $\delta_N M \ge \kappa_T$, $\kappa'' \ge \kappa_T$ and $b_{\mathrm{cs}} + b_{\mathrm{rj}} + b_{\mathrm{sr}} \le 1$; hence the boundary bound in (1685) follows. The same proof gives the support assertions. Inside Λ_t the normalizer and cluster subtractions cancel, while no rejected or strip term occurs. Off the boundary index set each of the four terms vanishes separately.

The decomposition (1686) proves the domain and structural assertions, whereas direct complex transfer proves the bounds. Replacing the source and all arguments by their conjugates in each formula proves the required conjugation identity. Restriction and transfer preserve finite-slice holomorphy and Borel measurability. For the cluster terms these properties also follow from the locally uniform majorants in theorem AF. Each source plaquette read in a term over X lies in X, so locality in the source follows through the same operations.

For the new retained factors use corollary 79.6 on the full product (1679). The quadratic correction is defined there and has oscillation at most $\bar{b}^{\mathrm{S}}_{\mathrm{fr}}$ in the normalized factor bound. Adding the real oscillation bound gives the stated box. Old factors are prolonged by the same prefix rule as in the real construction, so none acquires a new condition on later stored fields. This proves all assertions at level t once the regular bound below is inserted.

At level zero the list has no interaction entries. The first source family and escort of proposition 78.9 have the same properties on the first-step domains; the preceding construction therefore starts without an induction assumption. At source zero all added potentials, companions and quadratic forms vanish. Every remaining operation and every restriction is the real one, which proves the zero-source identification. Finally the step at level i uses only data through i. All domains are unchanged when a longer history is cut at i, including the filing spaces and stored tiers. Induction on i gives equality as partial maps under every cut. This also proves persistence of the bounds for older boundary entries. □

80.2. **Control of the regular terms.** The estimate needed for the regular recursion depends only on the difference of two old regular arrays and on the companion. Let $\mathfrak{f}, \widetilde{\mathfrak{f}}$ be these arrays, with the same counterterms, and put

$$\theta = K_\Lambda \mathsf{s}_{\mathfrak{a}} N_{\mathfrak{a},\omega}(\mathfrak{f} - \widetilde{\mathfrak{f}}) + \|\mathbb{W}^{1\mathrm{l}}\|^{\mathrm{co}}_{\star,\lambda_V}, \qquad \mathsf{s}_{\mathfrak{a}} = (1+\beta_{\mathrm{s}})\delta_k/\alpha_0. \tag{1687}$$

The weights of this array norm are $L^{-(k-j)}$ for the old level-j entries.

Lemma 80.2. Let $1 \le k < K$. Suppose the two old lists satisfy the regular box bound, $K_\Lambda \mathsf{s}_{\mathfrak{a}} N_\star \le w_\star/4$, and $\theta \le 3w_\star/4$. Then their pointed terms, with the companion attached to the first list, are defined on the whole next core and satisfy

$$|F_\emptyset[\mathscr{L}; \mathbb{W}^{1\mathrm{l}}](X,P,z) - F_\emptyset[\widetilde{\mathscr{L}}; 0](X,P,z)| \le 33(3L^3-3)K_\Pi \theta e^{-\lambda_{\mathrm{out}} d_t(X)}.$$

■

Proof. The regular localization rule is linear in its array. Hence

$$\widetilde{\mathbb{W}} = \mathbb{V}'[\mathfrak{f} - \widetilde{\mathfrak{f}}] + \mathbb{W}^{1\mathrm{l}}$$

is an escort of size at most θ. The two constructions $F_\emptyset[\mathscr{L}; \mathbb{W}^{1\mathrm{l}}]$ and $F_\emptyset[\widetilde{\mathscr{L}}; \widetilde{\mathbb{W}}]$ have identical local potentials, domains, Gaussian, cutoffs and pointing. They are equal as partial maps. The regular potential uses at most $w_\star/4$ of the allowed size; adding the escort therefore leaves both constructions

within the allowed ball. The one-lattice difference estimate proposition 77.3 gives the asserted bound. This argument also identifies their domains; equality of formal expansion coefficients alone would not suffice. □

Proposition 80.3. Put $D_0^{\mathrm{S}} = E_0^{\mathrm{S}}/4$. Let n_k bound the one-lattice companion at step $k+1$, with zero companion at source zero. Suppose

$$\begin{aligned} 108(L^3-1)K_\Pi n_0 &\le D_0^{\mathrm{S}}, \\ \theta_k^{\mathrm{S}} := K_\Lambda(1+\beta_{\mathrm{s}})\frac{\delta_k}{\alpha_0}\frac{L}{L-1}D_0^{\mathrm{S}} + n_k &\le \min\left(\frac{1}{2}w_\star, \frac{D_0^{\mathrm{S}}}{33(3L^3-3)K_\Pi}\right) \quad (1 \le k < K). \end{aligned} \tag{1688}$$

Then, for every $1 \le j \le K$,

$$\|E^{(j),w} - E^{(j),0}\|_{j,\kappa+1} \le D_0^{\mathrm{S}}\|w\|_{\mathsf{Src}}. \tag{1689}$$

The families have the usual translation covariance, with the source translated in physical coordinates along with the domain. ■

Proof. First omit the factor $\|w\|_{\mathsf{Src}}$ from the claim. The first-step escort estimate proposition 77.4 and the first inequality of (1688) give the base case. Suppose the bound holds through k. The counterterms cancel between the two arrays, and therefore

$$N_{\mathfrak{a},\omega}(\mathfrak{f}^w - \mathfrak{f}^0) = \sum_{j=1}^{k} L^{-(k-j)}\|E^{(j),w} - E^{(j),0}\|_{j,\kappa} \le \frac{L}{L-1}D_0^{\mathrm{S}}.$$

In (1681) the normalizer also cancels. Lemma 80.2 now bounds the new difference by $33(3L^3-3)K_\Pi\theta_k^{\mathrm{S}}$ in rate $\lambda_{\mathrm{out}} \ge \kappa+1$. This is at most D_0^{S}, closing the induction. For the zero-source terms the actual one-step bound is at most $\Xi(x_k, N_\star) \le \Xi(\gamma^{-2}, N_\star) \le 3E_0^{\mathrm{S}}/4$, the stronger numerical budget imposed in lemma 82.4. This is the estimate used in the proof of Theorem Y, with this stronger budget; the statement of that theorem alone only gives E_0. Hence the sourced regular bound in (1685) follows at the same time. This is the regular estimate used in the preceding induction, and involves no boundary estimate.

For fixed j, X, P, z the difference is holomorphic in source slices and vanishes at zero. Applying the source Schwarz estimate lemma 73.2 gives (1689). Taking suprema preserves the estimate.

For translation covariance use the translated old entries and companion in the pointed expansion. The canonical reference plaquette obeys (1616). Thus the weight differences $v - v(p_b)$ translate along with the localized potentials in propositions 78.6 and 78.9; their finite formulas give the translated companion. Restriction, localization, Gaussian integration and pointing commute with the allowed lattice translations. Induction makes the pointed terms equal. The possible difference of the source-independent normalizers is zero at source zero by the real covariance, so (1681) gives covariance for every source. The same physical translation has level-i coordinates $L^{j-i}\mathsf{v}$ when its level-j coordinates are v. These belong to the required $LM\mathbb{Z}^3$ at each earlier step. □

81. The exact sourced history step

The analytic families above must be evaluated at the real history coordinates. This requires both domain information and a comparison of representatives. We first work at a chain point, where the coordinate identities are pointwise, and then pass to the reference measure.

Let $U \in \mathcal{V}_0[h'] \cap \operatorname{dom}\xi_{h'}$ be a bare field with nonzero history weight, and $y = (W, B, \phi^+)$ a chain point of U. Write $B = (B_{\mathrm{I}}, \mathsf{a})$, let d be its base datum, and let U be an adapted representative. Let $\mathsf{a} = \check{\Psi}_{\mathsf{d}}(\mathsf{a}')$ be the real strip change of variables and $\mathsf{q}^+ = (\mathsf{q}, \mathsf{a}')$ the stored data after the step. Put $P = P_+[\mathsf{d}]$ and $P_X = P|_X$.

81.1. **Domains and exact identities at chain points.**

Lemma 81.1. For every such chain point and every source in the class, all the following values are defined. The cluster function and the two conditional forms can be evaluated at both a and a'. For every B_{I} in the integration box Rg the ordinary and weighted potentials can be evaluated at

$$\mathfrak{u}_Y = (P|_Y, g_k(B_{\mathrm{I}} \oplus \mathsf{a})|_Y, \mathfrak{q}|_Y).$$

Their finite sum $\mathsf{V}^w_{\mathsf{d}}(B_{\mathrm{I}}, \mathsf{a})$ is bounded on that box. If m is the real conditional mean, then the centered source splitting gives

$$\sum_Y \mathbb{V}^{\widehat{w}_k}(Y; \mathfrak{u}_Y) = \sum_Y \mathcal{W}[\widehat{w}_k, \mathsf{rf}^w](Y; P|_Y; \mathcal{A}_Y, \zeta_Y) + \mathsf{Kd}^w_{\mathsf{d}}(\mathsf{a}). \tag{1690}$$

Here $\mathsf{rf}^w(b) = \widehat{w}_k(p_b)$, $\mathcal{A}_Y = g_k(B_{\mathrm{I}} \oplus \mathsf{a})|_Y$ and $\zeta_Y = g_k((B_{\mathrm{I}} - \mathsf{m}) \oplus 0)|_Y$. By lemma 78.5, each remainder is the canonical centered member $\mathbb{W}^{\mathrm{S}}[v^w_k]$ evaluated at $z = B_{\mathrm{I}} - \mathsf{m}$, $\mu = \mathsf{m}$, with external field a. The term $\mathsf{Kd}^w_{\mathsf{d}}$ is the sum of the quadratic Taylor terms at $g_k(\mathsf{m} \oplus \mathsf{a})$; in particular it is independent of B_{I}. ∎

Proof. The domain assertions are the actual geometric assertions of Propositions 57.4, 57.5 and 60.6. At the chain point $P_X \in \mathscr{U}_X$ and $(P_X; \mathfrak{q}^+|_X) \in \mathfrak{G}_X$. The strip budget is $C_{\mathrm{xt}}\delta_k/2$; both the old and new external fields lie in the allowed external domain. Restriction to X maps these points into the parameter domain of the centered cluster function. Thus the functions just constructed are defined at both external fields.

For $B_{\mathrm{I}} \in \mathsf{Rg}$, the real box bounds and Proposition 57.5 give $\mathfrak{u}_Y \in \mathfrak{D}^{\mathrm{rad}}_k(Y)$. They also give $P|_Y \in \mathbb{U}^{\mathbf{1}}_k(Y)$ and $\sup|g_k B| \le C_{\mathcal{A}}\delta_k < r_0/2$. The conditional mean belongs to the prescribed mean domain, and $B_{\mathrm{I}} - \mathsf{m}$ belongs to its centered-field domain. The source radius conditions leave the room required by the weighted Taylor expansion. Consequently all ordinary, weighted and centered members are defined. Their estimates, summed over the finite set of Y, give boundedness on Rg.

Apply the weighted Taylor splitting at each Y with the reference coefficient $v^w_k(p_b)$ at bond b. The linear terms at constant weight cancel in the sum by the first variation of the Wilson action at the minimizing field. For the mixed quadratic terms, lemma 78.4 separates the gauge and Wilson parts of the conditional-mean equation: $\chi_{\mathrm{I}}\widehat{G}_{\mathsf{d}}\bar{m} = 0$ and $\chi_{\mathrm{I}}H_{\mathsf{d}}\bar{m} = 0$, where $H_{\mathsf{d}} = \mathsf{A}_{\mathsf{d}} - \widehat{G}_{\mathsf{d}}$. The regional Hessian identity has the normalization $2\sum_Y T_2[\mathbf{1}](Z|_Y, \bar{m}|_Y) = (Z, H_{\mathsf{d}}\bar{m})_1$. Its right side vanishes for every integration-supported Z, including fields supported on one bond. This cancels the reference-weight mixed terms. The remaining pure mean quadratic terms give $\mathsf{Kd}^w_{\mathsf{d}}$; the remaining terms are exactly the three centered source members. This is the sum-rule identity of lemma 78.5, and gives (1690). No integration or choice of logarithm is used in this identity. At $k = 0$ the first-step Taylor sum rules and the source family of proposition 78.9 give the same formula. □

Proposition 81.2. Let $\mu_{\mathsf{d},\mathsf{a}}$ be the conditional Gaussian with the real covariance and mean of the history step. Then

$$\int \chi^{\mathfrak{b}}(B_{\mathrm{I}}) e^{\mathsf{V}^w_{\mathsf{d}}(B_{\mathrm{I}},\mathsf{a}) - \mathsf{Kd}^w_{\mathsf{d}}(\mathsf{a})}\, d\mu_{\mathsf{d},\mathsf{a}} = \exp \sum_{X \in \mathcal{D}_t} E^{\mathrm{cl},w}_{h'}(X; \mathsf{a}; P_X; \mathfrak{q}|_X). \tag{1691}$$

The integrand is zero off Rg. Moreover

$$-\tfrac{1}{2}(\mathsf{a}, \Delta^{\mathrm{cond}}_{\mathsf{d}}\mathsf{a})_1 + \mathsf{Kd}^w_{\mathsf{d}}(\mathsf{a}) = \sum_X [K^{\mathrm{kp,w}} + B^{\times,w}](X; p_X(P_X), \mathsf{a}), \tag{1692}$$

$$\sum_X B^{\mathrm{sr},w}(X; P_X; \mathfrak{q}, \mathsf{a}') = \mathsf{Zs}_{\mathsf{d}}(\mathsf{a}') + \sum_X [E^{\mathrm{cl},w} + B^{\times,w} + K^{\mathrm{kp,w}}](X; \mathsf{a}) - \sum_X [E^{\mathrm{cl},w} + B^{\times,w} + K^{\mathrm{kp,w}}](X; \mathsf{a}'). \tag{1693}$$

All values in these identities are defined at the chain point. ∎

Proof. The complete local Gaussian structure of Theorem Z has the covariance and mean of $\mu_{\mathsf{d},\mathsf{a}}$ and the same integration cutoff. Apply theorem AF at $(\mathsf{a}; P; \mathfrak{q})$, whose membership in its parameter domain was checked above. The ordinary vertices are the four old potentials; the only additional origin is the centered source origin. By lemma 81.1 their sum is the exponent on the left of (1691). A domain containing no integration or external source cube has empty relevant bond set; its ordinary terms and its centered source term at $(0,0)$ vanish. Thus the sum in the centered expansion agrees with the full sum over Y. This proves (1691) with its stated domains and cutoff convention.

The real conditional form satisfies the identity without Kd^w, by Section 61. The exact quadratic localization of corollary 79.5 identifies

$$\mathsf{Kd}^w_{\mathsf{d}}(\mathsf{a}) = \sum_X [K^{\mathrm{kp},+} + B^{\times,+}](X; p_X(P_X), \mathsf{a}).$$

Adding it gives (1692). Each summand is defined on the pair image $\mathscr{O}^{\mathrm{S}}(X)$ and for every external field; no regularity of the datum outside that image is needed.

For (1693) sum the five finite strip transfer identities, equations (1040), (1044), for the sourced forms and cluster function. Their domains were checked in the proof of proposition 80.1. The Jacobian and Haar contributions combine into the logarithm $\mathsf{Zs}_{\mathsf{d}} = \log j_{\mathrm{s}}$ of the Jacobian of the strip change in the external Lie-algebra coordinates. The other three contributions are precisely their differences at a and a'. This proves the identity without exchanging an infinite sum or an integral.

The normalizer and cluster interior identities together with (1682) also show that their total local exponent equals the new vacuum constant plus the regular, edge and created-boundary terms. The constants $n^{(t)}_{\mathrm{tr}}$ cancel by (1681). This is an identity of defined values, including the terms whose boundary indicator is zero. Empty integration regions cause no exception: their Gaussian is the point mass at zero and the same formulas are finite evaluations. □

81.2. **Borel domains and evaluation representatives.**

Proposition 81.3. The evaluations of the regular, boundary and retained entries are Borel partial maps with domains independent of w. More explicitly, regular evaluation at (j, X, z) has domain

$$\{\mathsf{U} : \mathbf{P}^{(j)}_X(\mathsf{U}) \in \mathcal{U}^{\circ}_j(X; \alpha_0, \alpha_1)\};$$

boundary evaluation has the inverse image of

$$\mathsf{Sp}(\mathcal{S}; (j, X, \mathrm{mk})) \times \mathfrak{Q}^{\mathrm{rad}}_{h'|_j}(X)$$

under $(\mathsf{U}, \mathfrak{q}) \mapsto (\mathbf{P}^{(j)}_{X|\mathcal{S}}(\mathsf{U}), \mathfrak{q}|_X)$; and retained evaluation has the analogous inverse image of (1679) with its stored prefix.

Put $D_X = \{\mathsf{U} : P_X(\mathsf{U}) \in \mathscr{U}_X\}$. The conditional forms have domain $D_X \times \mathsf{E}_{h'}$, and the cluster and strip terms have domain

$$\{(\mathsf{U}, \mathfrak{q}, \mathsf{a}_1) : \mathsf{U} \in D_X,\ (\mathfrak{q}, \mathsf{a}_1)|_X \in \mathfrak{Q}^{\mathrm{rad}}_{h'}(X)\}. \tag{1694}$$

The finite fluctuation potential has the Borel domain

$$\{(\mathsf{d}, B) \in \mathsf{Ba}^{\mathrm{reg}}_{h'} \times \mathsf{Rg} : \mathfrak{u}_Y(\mathsf{d}, B) \in \mathfrak{D}^{\mathrm{rad}}_k(Y) \text{ for every } Y\}.$$

■

Proof. The plain-triple maps are continuous and the cores and graded spaces are Borel; the tiers are open, being defined by finitely many strict inequalities. These facts belong to the coordinate construction in Sections 60–61 and do not use an interaction list. By proposition 80.1, each entry is defined on the entire indicated core or product and is trace-measurable there. Thus there is no extra source-dependent domain condition. Composition with the continuous reading maps gives exactly the Borel domains displayed above.

For the cluster and strip terms the continuous reading $(P;\mathfrak{q},\mathsf{a}_1)\mapsto(\mathsf{a}_1^{\uparrow};P;\mathfrak{q})$ maps the graded-tier product into the cluster domain, with strip budget $C_{\mathrm{xt}}\delta_k/2$. The full-domain result of proposition 80.1 therefore gives (1694). A conditional form is a finite polynomial in a_1 with analytic pair coefficients. Pair reading is continuous on $\mathscr{U}_X$, so its evaluation is Borel on $D_X\times\mathsf{E}_{h'}$.

Finally $\mathsf{d}\mapsto P_+[\mathsf{d}]$ is Borel, field restriction is linear, and the old stored coordinates are coordinate projections. Hence all $\mathfrak{u}_Y$ are Borel maps. All five potentials are defined and trace-measurable on $\mathfrak{D}_k^{\mathrm{rad}}(Y)$, so their finite sum is Borel on the displayed intersection of inverse images. The auxiliary quantity Kd^w is used only where its conditional objects and weighted Taylor parts are defined. At chain points this has already been proved; no assertion that its maximal domain is Borel is needed below. □

Lemma 81.4. Let U be in the actual support of h', and let $X\in\mathsf{Ix}_t^{\mathrm{mk}}(\mathcal{S}(h'))$. Let f_X be a complex function on the whole graded-tier product $\mathfrak{G}_X$ satisfying the joint gauge identity

$$f_X(P^u;u\cdot\mathfrak{q}^+)=f_X(P;\mathfrak{q}^+)$$

for every bare gauge transformation. At the actual stored field $\mathfrak{q}^+[U]$, f_X is defined and has one value on all adapted evaluation representatives of U. ■

Proof. Actual support membership puts each representative's plain triple in the graded space and its stored field in the tiers; these memberships are the ones in the proof of Corollary 58.4. Two representatives differ by a residual gauge on the first collared region. Extend it by the identity outside that region. At each outer bond the endpoints in the collar are bare nodes, so this extension relates the two glued configurations as well. It fixes nodes. It also fixes every stored site: the transport of an adapted representative from its stored site a to the corresponding node y obeys

$$\mathsf{ax}_{U'^u}(a)=u(y)\mathsf{ax}_{U'}(a)u(a)^{-1},$$

and both transports and $u(y)$ are the identity. All sites acting on the stored fields are nodes or these stored sites, by Lemmas 60.3 and 60.4. Thus $u\cdot\mathfrak{q}^+[U]=\mathfrak{q}^+[U]$. The plain-triple maps are gauge covariant. Apply the joint gauge identity to the two triples and the fixed stored field. The claimed equality follows. This proof uses no reality condition on f_X. □

Proposition 81.5. At a realized chain point the new action exponent and the new retained exponent sum are separately defined and constant on the evaluation set $\mathsf{Ev}_{h'}(\xi_{h'}U)$. The retained sum made directly by the integration step equals the sum read from $\mathscr{L}_t^{wb}$ on this set. ■

Proof. All representatives in this evaluation set satisfy the graded and tier conditions just used. The action exponent consists of regular and boundary values with the actual gauge invariance and pair dependence, together with the weighted Wilson action. The adapted-gauge comparison of Proposition 60.5 makes these values independent of the representative; the weighted action has the same property by its plaquette gauge invariance.

Consider the additional kept form at X. If $\mathsf{kp}_t(X)=0$, it vanishes. Otherwise X contains a coarse cube of $\Omega_t\cap Z_t$ and belongs to the boundary index set. On the graded-tier product define

$$f_X(P;\mathfrak{q}^+)=K^{\mathrm{kp},+}(X;p_X(P),(\mathfrak{q}_k^+)^{\uparrow}).$$

The quadratic locality theorem shows that this depends only on the external bonds in X; zero extension elsewhere is immaterial. Its pair covariance and the simultaneous action on the stored field give the boundary gauge identity. Apply lemma 81.4, whose hypotheses were just checked. Consequently f_X has the same value at every adapted representative, with the fixed stored field. This proves representative independence of each added kept term at this realized point.

The retained sum produced by the step is the old retained sum, the real newly retained conditional terms, the elementary new factors, and $\sum_X K^{\mathrm{kp},+}(X;p_X,\mathsf{a}')$. The first three parts have the actual representative identity of the real construction, whose proof uses their gauge identities and full

domains, not their real-valuedness. The last part has the identity just proved. Thus the sum is separately well defined.

To compare it with the list, old factors read the identical plain triples and the identical stored prefixes. The new elementary factors are unchanged. For each new kept factor, corollary 79.5 identifies its correction in (1683) with $K^{\mathrm{kp},+}(X_f; p_{X_f}, \mathsf{a}')$; the field may be extended by zero because of external-bond locality. There is one such factor for each domain with $\mathsf{kp}_t(X) = 1$, and the added form vanishes on the other domains. Summing therefore gives exactly the step's retained sum. This proves equality at every representative of the realized point. The argument does not assert representative independence at arbitrary points of the coordinate space. □

81.3. **The measure identity for complex densities.** We give the measure argument because the positivity of the real source-free density cannot be imposed on a complex source. Its geometric ingredients are those of Theorem AB and its proof.

Theorem AH. Suppose the old sourced format has an integrable partial density ρ_h^w and equals $\hat{\tau}_h[\rho_0^w]$. Suppose the sourced analytic families have the domains, chain-point identities and representative identities proved above. Then the density made by the next integration step is an integrable partial density and

$$\hat{\tau}_{h'}[\rho_0^w] = \mathsf{Stay}^{\mathrm{T}}_{h'}\chi_t(\Omega_t)e^{A^w_{t,h'}+\Phi^{\mathrm{made,w}}_{h'}}\mathscr{M}_{h'}. \tag{1695}$$

Each exponent on the right is separately defined almost everywhere where the test is nonzero. There is the variation bound

$$\int |\rho_{h'}^{\mathrm{made,w}}|\, d\mathscr{M}_{h'} \le \|f\|_\infty \int |\rho_h^w|\, d\mathscr{M}_h, \tag{1696}$$

where f is the bounded nonnegative step weight. ■□

Proof. Write $\nu_h = \hat{\tau}_h[\rho_0^w]$ and suppress w from the notation temporarily. The coordinate commutation and weight identity imply, for bounded Borel test functions,

$$\nu_{h'} = \Sigma_*(f\nu_h).$$

This is a finite complex-measure identity, since the bare source density is bounded and the real history weights are nonnegative and bounded. Extend ρ_h by zero on a Borel null set containing its undefined points. Where the old test is nonzero and the exponents are defined, its value is nonzero: it is a positive test times an exponential. This observation does not assert positivity of ρ_h.

We first justify using chain-point identities almost everywhere in the coordinate change. Let ν_+ be the positive measure obtained from the bare real history weight for h' by forgetting the source density and pushing to the old coordinates. The analytic-image argument in the proof of Theorem AB gives Borel sets $D' \subset D''$ such that the actual image lies between them, $\nu_+(D'' \setminus D') = 0$, and ν_+ is concentrated on D'. The measure $f\nu_h$ is dominated in variation by $\|\rho_0^w\|_\infty \nu_+$, so it too vanishes on every Borel subset outside D'. On the nonzero old test outside the old null set its density $f\rho_h$ is nonzero where $f > 0$. If that set outside D' had positive $\mathscr{M}_h$ measure, the complex function $f\rho_h$ could not have integral zero on every Borel subset of it. Thus it is a reference null set. The same argument removes the part outside the domain of Σ. This is the required realization statement; cancellation of the total mass of a set has not been used as a positivity argument.

Use the injective chart and its positive Jacobian j_{c}, then the strip map and its positive Jacobian j_{s}. Their identities are equations (1359), (1360). Write $p = (W, \mathsf{a}', \phi^+)$ for the remaining variables and $\Theta(p, B_{\mathrm{I}})$ for the old chart point. The pullback integrand is the Borel function

$$D(p, B_{\mathrm{I}}) = j_{\mathrm{s}}(p) j_{\mathrm{c}}(\Theta(p, B_{\mathrm{I}}))(f\rho_h)(\mathrm{pt}\,\Theta(p, B_{\mathrm{I}})).$$

It is set equal to zero outside the chart and its nonzero test. Apply the two change-of-variables formulas first to the nonnegative function $|f\rho_h|$. This proves

$$\int |D| \le \|f\|_\infty \int |\rho_h|\, d\mathscr{M}_h < \infty.$$

Complex Fubini is now available. The chart realization proved above and positivity of both Jacobians imply that almost every nonzero-test chart point is a chain point. The inherited test identity puts $\mathsf{Pe} \times \mathsf{Rg}$ in this nonzero-test chart set outside a product null set, as in equation (1362). Pulling back the omitted chart null set through the two changes of variables therefore leaves a set of full product measure in this product. Tonelli gives a Borel set of full measure of $p \in \mathsf{Pe}$ whose good fibre has the full volume of Rg. This box has positive finite volume; in dimension zero its volume is one. Thus each such good fibre is nonempty.

Fix such a p. Choose one admissible B_{I} to verify the domain and representative assertions for the new exponents. They depend on p, not on that choice of B_{I}. Their values obtained through a fixed Borel adapted selection are Borel by proposition 81.3; the representative identity makes them their evaluation values. No measurable choice of this auxiliary B_{I} is required.

The old action identity, the chart Jacobian identity and the test identity give, at almost every point of this fibre, the same finite algebra as equations (1362), (1363): the B-dependent part is

$$\exp\{-\tfrac{1}{2}(B, \mathsf{A}_{\mathsf{d}} B)_1 + \mathsf{V}^w_{\mathsf{d}}(B_{\mathrm{I}}, \mathsf{a})\}.$$

The weighted action enters this identity by its exact weighted finite sum; the old interaction potentials enter by the linear extension established in proposition 80.1. Gaussian conditioning uses the positive real block $(\mathsf{A}_{\mathsf{d}})_{\mathrm{II}}$. Equation (1691) evaluates its fluctuation integral as the exponential of the cluster sum plus $\mathsf{Kd}^w_{\mathsf{d}}(\mathsf{a})$. Equation (1692) combines this last term with the conditional quadratic form. Equation (1693) then replaces a by a' and absorbs j_{s}. The interior subtraction (1682) identifies the remaining local terms with $A^w_{t,h'}$ and the new retained sum. Thus the fibre integral of D is exactly the pullback of the right side of (1695).

The last coordinate embedding is Borel and injective and pushes its reference measure to $\mathscr{M}_{h'}$ on the support of the test. Descending the preceding identity proves (1695), including almost everywhere definedness of both exponents and integrability. The bound on $\int |D|$ and the triangle inequality for its fibre integrals prove (1696). Empty integration boxes have zero-dimensional Gaussian measure and are covered by the same argument. □

81.4. **The stored family and induction through the levels.**

Theorem AI. At every real history step there is a Borel set $N_{h'} \subset \mathsf{X}^{\mathrm{co}\prime}_{h'}$, $\mathscr{M}_{h'}(N_{h'}) = 0$, independent of the source, such that on the nonzero-test set outside $N_{h'}$ the made retained sum and the retained sum read from $\mathscr{L}_t^{w\flat}$ are separately defined and equal for every source in the class. If the old sourced format identity holds, then

$$\hat{\tau}_{h'}[\rho_0^w] = \mathsf{Stay}^{\mathrm{T}}_{h'} \chi_t(\Omega_t) e^{A^w_{t,h'} + \Phi^{\mathscr{L},w}_{h'}} \mathscr{M}_{h'}. \tag{1697}$$

The right side is an integrable partial density and is independent of the Borel choice of adapted representatives. ■□

Proof. Apply the real format theorem and the realization argument in Theorem AB at source zero. Its positive density is nonzero on the test after deletion of a Borel reference null set. Hence outside a Borel set $N_{h'}$ every nonzero-test coordinate point is $\xi_{h'} U$ for a bare field with a chain point. This construction involves only the real histories, coordinates and tests, so the same $N_{h'}$ works for every source.

At each such point proposition 81.5 identifies the two retained sums for every w. The old sourced identity and theorem AH make the made density integrable and exact. The list density has the same other factors and agrees with it off $N_{h'}$ wherever the test is nonzero. On the zero-test set both densities are defined to be zero without evaluating either exponent. Their integrable partial-density classes are therefore equal, which proves (1697). Each exponent is an evaluation value, already independent of the representative; the asserted independence of the measure follows. □

Proposition 81.6. The output density in (1697) is exactly the old density used at the next step, as a partial map, with the same reference measure. At the first step the old identity holds without any earlier construction:

$$\hat{\tau}_{\emptyset}[\rho_0^w] = \rho_0^w \, dU.$$

Consequently the sourced format identity holds at every level and for every prefix of the construction. ■

Proof. We first check the base step. The level-zero test is one, there are no retained or boundary entries, and its action is the bare real action plus $L^K A(w, U)$. Thus its density is exactly ρ_0^w, which is bounded on the compact bare configuration space. The first-step Gaussian and coordinate identities are the real ones. The first-step weighted Taylor splitting, quadratic decomposition and companion are those of proposition 78.9; their actual arguments satisfy the first-step small-field domains. The proof of proposition 81.2 therefore applies with $k = 0$. There are no inherited interaction potentials or old retained exponents to check. The first regular and created boundary families have the domains proved in proposition 80.1, so their trivial values exist and the interior normalizer subtraction is legitimate. The complex measure argument and the source-independent real null set now give (1697) at level one.

For compatibility, cut all data of step $t + 1$ at level t. The recursion (1684) through that level is unchanged: it uses the same couplings, companions, strip rows and domain restrictions. The coordinate space, reference measure, test and evaluation sets are also unchanged. An old retained factor reads its formation-level plain triple and the same stored prefix. A boundary term reads the same filing space and stored tier. The regular values, counterterms, vacuum values and source Wilson term are likewise identical. Thus the two densities have exactly the same factors and the same domains, including the convention on zero tests. This is equality of partial maps, not just equality almost everywhere. The level-one output is therefore the hypothesis needed at level two, and the same argument applies thereafter.

If $\Omega_1 = \emptyset$, the real history format is multiplied by the bounded, nonvanishing source factor read from the stored outer field, as proved in proposition 73.7. There are no new source-dependent families in this case. This settles the remaining histories and completes the induction. □

The retained-factor damping estimate is now applied to defined evaluation values. Specifically proposition 79.7 with $\bar{\upsilon} \le \gamma'$ and $\gamma^\dagger = \gamma'/2$ gives the sourced damping bound off the same real realization null set. It is not being used to establish that the retained sum has an evaluation value. That domain assertion was proved before the estimate.

82. A common choice of constants

The source estimates require extra room in the decay rates before the coupling is made small. We first fix that room, then choose a relative source radius, and finally choose the upper bound on the coupling. The choices are independent of the number of steps and of the torus. Throughout this section the constants of the real construction have their common values in Proposition 69.1. In particular, $\lambda_{\mathrm{s}} = c_{\mathrm{E}} = \log 18432$, $S_{\mathrm{s}} = C_{\mathrm{E}} = 4096$, $C_2 = 342$, and $C_3 = 9e^8$. The quantities furnished by the Gaussian, normalizer and background estimates retain those values when the regular and boundary size bounds are enlarged.

82.1. **Rates and fixed radii.** Set $d = 1 - 2\delta_{\mathrm{sp}} > 0$ and use the shifted pair

$$\kappa^\sharp = \kappa + 3/d, \qquad r^\sharp = d(\kappa^\sharp - 2) = r_V + 3, \qquad r_V = d(\kappa - 2). \tag{1698}$$

The quantities $\kappa'' = (1+4\beta_{\mathrm{s}})\kappa + 1 + c_{\mathrm{E}}$, $\bar{c}_P$, $\bar{r}_H$ and K_{ap} refer to the real Gaussian construction. Define the uniform centered majorants by

$$\bar{c}_P^{+} = C_{\mathrm{mn}}^{3}\lambda_{\mathrm{r,gs}}^{-1}(N_{\mathrm{gs}}+1), \qquad \bar{r}_H^{+} = \kappa'' + c_{\mathrm{E}} + C_{\mathrm{cs}}(\bar{c}_P^{+}+2), \tag{1699}$$
$$K_{\mathrm{ap}}^{+} = C_{\mathrm{cs}}^{2} e^{C_{\mathrm{cs}}(\kappa''+\bar{c}_P^{+}+\bar{r}_H^{+})}, \qquad \theta_{\mathrm{sl}} = (1+\vartheta_*)/2.$$

Here $C_{\mathrm{mn}} = 18(1-\vartheta_*)^{-2}$ and $C_{\mathrm{cs}} = C_{\mathrm{cs},0}$ are the constants of the centered expansion. The number $\bar{c}_P^{+}$ depends on L and the reserve fractions, but not on κ, $\bar{\kappa}$ or M.

In the order of the choices in Proposition 69.1, impose the following additional floors:

$$L \geq \frac{2C_{\mathrm{ex}}(1+4\beta_{\mathrm{s}})}{d}, \tag{1700}$$
$$\tfrac{1}{2}d\kappa \geq 2d + 2c_{\mathrm{E}} + \frac{C_{\mathrm{cs}}}{L}[1 + 2c_{\mathrm{E}} + C_{\mathrm{cs}}(\bar{c}_P^{+}+2)], \tag{1701}$$
$$\tfrac{1}{2}d\kappa \geq 2d + 3c_{\mathrm{E}} + \frac{C_{\mathrm{ex}}}{L}(8c_{\mathrm{w}} + C_{\mathrm{ex}} + c_{\mathrm{E}}), \tag{1702}$$
$$\tfrac{1}{2}d\kappa \geq 2d + 2c_{\mathrm{E}} + 2 + \frac{C_{\mathrm{cs}}}{L}[1 + 2c_{\mathrm{E}} + C_{\mathrm{cs}}(\bar{c}_P+2)], \tag{1703}$$
$$\bar{\kappa} \geq 1 + C_{\mathrm{cs}}\bar{r}_H^{+} + \log C_{\mathrm{cs}}, \tag{1704}$$
$$(LM)^3 \geq C_{\mathrm{mn}}^{-1}\bar{c}_P^{+}, \qquad \delta_{\mathrm{res}} LM \geq 2C_2(\kappa''+1). \tag{1705}$$

The real construction already requires $L \geq 2C_{\mathrm{cs}}(1+4\beta_{\mathrm{s}})/d$ and $4\beta_{\mathrm{s}}\kappa \geq 1$; both requirements are retained. Each new right side involves only constants chosen earlier than the constant on its left. In particular the M floors are imposed after κ'' and δ_{res} have been fixed; M is still allowed to run through arbitrarily large powers of L.

Lemma 82.1. These floors are compatible with the real choices. Put

$$\kappa' = \kappa'', \qquad \delta_v = \delta^{\mathrm{S}} = \frac{2C_2(\kappa''+1)}{LM}. \tag{1706}$$

Then

$$0 < \delta^{\mathrm{S}} < \delta_{\mathbf{M}} \leq \min(\delta_{\mathrm{res}}, \delta_*^{\mathrm{pl}}), \qquad \tfrac{1}{2}\delta^{\mathrm{S}} LM = C_2(\kappa'+1), \tag{1707}$$
$$\lambda_V \leq r_V - c_{\mathrm{E}} \leq r_V - 2, \qquad r^{\sharp} - \delta^{\mathrm{S}} M \geq 2c_{\mathrm{E}} + 5, \qquad \lambda_V \leq r^{\sharp} - 2.$$

The three weighted-potential conditions hold at $(\delta_{\mathrm{sp}}, \kappa^{\sharp})$:

$$2\delta_{\mathrm{sp}}\kappa^{\sharp} \geq a_{\mathrm{pl}}, \quad \bar{\kappa} - 1 \geq C_\kappa \kappa^{\sharp}, \quad M\min(\mu_{\mathfrak{h}}, 1)L \geq 2C_{\mathrm{lev}}\log L. \tag{1708}$$

The centered expansion has its fixed geometric and rate hypotheses, and $\alpha_0, \alpha_1 \leq \frac{1}{2}\min(a_{\mathfrak{h}}, a_P^{\circ})$. Its Gaussian growth condition holds at $\rho_1 = \rho_1^{\mathrm{S}} = (LM)^{-3}$. ∎

Proof. The new conditions are finite lower bounds at the four upward choices $L, \kappa, \bar{\kappa}, M$. Taking the maxima of the old and new floors therefore preserves the construction in the cited proposition. No bound on a constant already chosen is imposed at a later stage.

Write $\bar{r}_H = (1+4\beta_{\mathrm{s}})\kappa + 1 + 2c_{\mathrm{E}} + C_{\mathrm{cs}}(\bar{c}_P+2)$. The real L floor and (1703) give

$$\frac{C_{\mathrm{cs}}}{L}\bar{r}_H \leq r_V - 2c_{\mathrm{E}} - 2.$$

Since $C_{\mathrm{cs}} \geq 2C_2$ and $\bar{r}_H \geq \kappa'' + 1$, $\delta^{\mathrm{S}} M \leq C_{\mathrm{cs}}\bar{r}_H/L$. This proves $r_V - \delta^{\mathrm{S}} M \geq 2c_{\mathrm{E}} + 2$ and hence the asserted inequality for $r^{\sharp}$. For the other rate, insert $\lambda_1 = (1+4\beta_{\mathrm{s}})\kappa + 8c_{\mathrm{w}} + C_{\mathrm{ex}} + c_{\mathrm{E}}$ into $\lambda_V = C_{\mathrm{ex}}\lambda_1/L + 2c_{\mathrm{E}}$. Equations (1700) and (1702) then give $\lambda_V \leq d(\kappa-2) - c_{\mathrm{E}}$.

For the range of δ^{S}, the real resolvent construction gives

$$\delta_{\mathbf{M}} = \frac{c_{\mathrm{res}}\delta_*^{\mathrm{pl}}\gamma_{\mathrm{pos}}}{B_{\mathrm{nm}}} \leq \min(\delta_{\mathrm{res}}, \delta_*^{\mathrm{pl}}), \qquad \delta_{\mathbf{M}} LM > 8\bar{\kappa}.$$

The last inequality follows from its chain condition: $\mu_{\mathrm{ch}}LM/4 > \bar{\kappa}$ and $\mu_{\mathrm{ch}} \le \delta_{\mathbf{M}}/2$. Also $\bar{\kappa} \ge C_{\mathrm{cs}}\bar{r}_H \ge 2C_2(\kappa''+1)$. Thus $\delta^{\mathrm{S}}LM < \delta_{\mathbf{M}}LM$. All divisions are by positive numbers.

The first and third inequalities in (1708) are the corresponding real weighted-potential inequalities, with the rate increased. For the second, the canonical value is $C_\kappa = 1$. The new $\bar{\kappa}$ floor gives $\bar{\kappa} - 1 \ge C_{\mathrm{cs}}\bar{r}_H^{+} \ge 2\kappa'' \ge 2\kappa$, whereas the κ floor gives $\kappa \ge 4/d$. Consequently $\bar{\kappa} - 1 \ge \kappa + 3/d = \kappa^{\sharp}$. The fixed radius bounds are among the real analytic-radius conditions. Finally, $\vartheta_* < \theta_{\mathrm{sl}} < 1$ and $1 + \vartheta_* < 2 < C_{\mathcal{A}}$, so the center ball fits strictly inside the field domain. Equations (1701), (1704) and (1705) are its remaining fixed rate and Gaussian-growth conditions.

The Gaussian condition at ρ_1^{S} uses the defect bound $\varepsilon_{\mathrm{dc}} \le (LM)^{-3}$ and the exact formula

$$(LM)^{3/2}\varepsilon_{\mathbf{m}} = \sqrt{2}(LM)^{3/2}\varepsilon_{\mathrm{dc}}\theta_{\mathrm{xt}} + (1+\varepsilon_{\mathrm{dc}})/4 < 0.25115, \qquad \theta_{\mathrm{xt}} = 96^{-1/2}/3.$$

Thus $(LM)^3\varepsilon_{\mathbf{m}}^2 < 0.0631$. Since $c_{\mathrm{r,gs}} \ge 1$, this implies

$$(LM)^3[\widehat{\varepsilon} + (1+c_{\mathrm{r,gs}})\varepsilon_{\mathbf{m}}^2 + \gamma_{2,\mathrm{gs}} + \rho_1^{\mathrm{S}}] \le C_{\mathrm{res}}B_{\mathrm{nm}}^5\gamma_{\mathrm{pos}}^{-2} + c_{\mathrm{r,gs}} + 2 = N_{\mathrm{gs}}.$$

Here $\gamma_{2,\mathrm{gs}} = (LM)^{-3}$ and $(LM)^3\widehat{\varepsilon} \le C_{\mathrm{res}}B_{\mathrm{nm}}^5\gamma_{\mathrm{pos}}^{-2}$. Consequently the actual centered coefficient satisfies $c_P^{+} \le C_{\mathrm{mn}}^3\lambda_{\mathrm{r,gs}}^{-1}N_{\mathrm{gs}} = \bar{c}_P < \bar{c}_P^{+}$, and the growth inequality follows from $(LM)^3 \ge C_{\mathrm{mn}}^{-1}\bar{c}_P^{+}$. □

82.2. **Source sizes and the final coupling choice.** The constants in the Taylor estimates are denoted by (K_0, r_0) at later steps and by $(\mathsf{K}^0, r_P^{\circ})$ at the first step. Thus

$$r_0 = \min\{\varepsilon_{\mathfrak{h}}, (3c_{\mathfrak{h}})^{-1}\}, \quad \mathsf{K}_0 = 12L^4M^3K_{\mathrm{c}}, \quad \mathsf{K}^0 = 4512C_PM^3r_P^{\circ}.$$

Use the coefficients C^{src} and C_0^{src} in the general and first-step centered bounds, and set

$$\mathsf{k}_{\odot} = 49L^5C_{\mathcal{A}}\max(C^{\mathrm{src}}, C_0^{\mathrm{src}}), \qquad \varepsilon^{\mathrm{O}} = (16eL^3K_{\mathrm{ap}}^{+})^{-1}, \quad c_M^{\mathrm{O}} = w_0\varepsilon^{\mathrm{O}}, \quad \epsilon_{\times}^{+} = \varepsilon^{\mathrm{O}}/3,$$
$$\mathsf{m}_{\odot} = \min\{c_M^{\mathrm{O}}, w_\star/2, E_0/(324L^3K_{\Pi})\}. \tag{1709}$$

For the one-source chains let $c_\upsilon, B_{\mathrm{ch}}^{\sharp}, K_{\mathrm{ch}}^{\sharp}, C_\Upsilon$ be the quantities of eq. (1664) at (1706) and at the real resolvent bound $B_{\mathrm{f}}^{\mathrm{res}}$. Put

$$\mathsf{K}_*^{\mathrm{S}} = \max\{\mathsf{K}_0r_0^{-2}, \mathsf{K}^0(r_P^{\circ})^{-2}\}, \quad b_\upsilon^{\mathrm{S}} = 64S_{\mathrm{s}}e^{6\delta^{\mathrm{S}}M}\mathsf{K}_*^{\mathrm{S}}, \quad m^{\mathrm{S}} = c_\upsilon^2B_{\mathrm{f}}^{\mathrm{res}}b_\upsilon^{\mathrm{S}},$$
$$\rho_1^{\mathrm{S}} = (LM)^{-3}, \qquad \lambda_0^{\mathrm{S}} = \rho_1^{\mathrm{S}}/(16S_{\mathrm{s}}),$$
$$\eta^{\mathrm{S}} = \min\left\{\frac{\lambda_0^{\mathrm{S}}}{12eL^3K_{\mathrm{ap}}}, \frac{w_\star}{4M^3}, \frac{E_0}{2808L^3M^3K_{\Pi}}\right\}, \qquad \bar{n}_{\mathrm{co}}^{\mathrm{S}} = M^3\eta^{\mathrm{S}}, \tag{1710}$$
$$\mathsf{m}_{\odot}^{\mathrm{S}} = \min\{\mathsf{m}_{\odot}, E_0/(81L^3K_{\Pi})\}, \qquad \epsilon_{\mathrm{ap}}^{\mathrm{S}} = \min\{[24e(1+\sqrt{2})L^3K_{\mathrm{ap}}]^{-1}, \epsilon_{\times}^{+}\}.$$

All these numbers are positive and fixed before γ. The fixed size bounds are

$$E_0^{\mathrm{S}} = E_0, \qquad B^{\mathrm{S}} = B, \qquad B_{\mathrm{fr}}^{\mathrm{S}} = B_{\mathrm{fr}}^{\flat} + 1, \qquad \bar{b}_{\mathrm{fr}}^{\mathrm{S}} = 1, \qquad \gamma^{\dagger} = \gamma_{\mathrm{kp}}/2. \tag{1711}$$

In particular $B \ge \max\{1, 8M^3E_0, 24(LM)^3E_{\mathrm{nf}} + 1\}$.

Here is one explicit choice of the two small relative radii. Use the positive stability radius $\bar{\sigma}$ defined below and set

$$\sigma^{\mathrm{S}} = \min\Big\{\frac{1}{2}, \frac{\eta^{\mathrm{S}}}{8\mathsf{K}_*^{\mathrm{S}}}, \frac{\gamma_{\mathrm{kp}}}{C_\Upsilon b_\upsilon^{\mathrm{S}}}, \frac{e^{\kappa''}}{m^{\mathrm{S}}K_{\mathrm{ch}}^{\sharp}C_{\mathrm{xt}}^2}, \frac{c_{\mathrm{kp}}^{\mathrm{th}}e^{\kappa''}}{m^{\mathrm{S}}K_{\mathrm{ch}}^{\sharp}},$$
$$\frac{c_{\mathrm{rj}}^{\mathrm{th}}e^{\kappa''}}{m^{\mathrm{S}}C_3B_{\mathrm{ch}}^{\sharp}(LM)^6}, \frac{\sigma_*}{16}, \bar{\sigma}, \frac{\sqrt{3}}{40}\bar{\sigma}, \frac{\mathsf{m}_{\odot}^{\mathrm{S}}}{4\mathsf{k}_{\odot}}, \frac{M^3\eta^{\mathrm{S}}}{4\mathsf{k}_{\odot}}, \frac{\sigma_{\mathrm{src}}}{2}\Big\}, \tag{1712}$$

and

$$\sigma_{\mathrm{L}}^{\mathrm{S}} = \min\left\{\frac{\eta^{\mathrm{S}}}{2K_{\mathrm{lp}}^{\mathrm{S}}}, \frac{\eta^{\mathrm{S}}}{2K_{\mathrm{lp}}^{\mathrm{S},0}}, \frac{10\mathsf{m}_{\odot}^{\mathrm{S}}}{\sqrt{3}(1+A_1)\mathsf{k}_{\odot}}, \frac{\min(\mathsf{m}_{\odot}, M^3\eta^{\mathrm{S}})}{4(1+A_1)\mathsf{k}_{\odot}}, \frac{20\mu_*}{\sqrt{3}K_{\mathrm{L}}^{\vee}}\right\}. \tag{1713}$$

The constants $K^{\mathrm{S}}_{\mathrm{lp}}, K^{\mathrm{S},0}_{\mathrm{lp}}$ are

$$K^{\mathrm{S}}_{\mathrm{lp}} = \mathsf{K}_0\left[\frac{24\sqrt{3}M^3C_{\mathcal{A}}A_1}{5r_0} + \frac{48M^{3/2}(C_{\mathcal{A}}A_1/4+\sqrt{2}/2)}{5r_0^2}\right],$$
$$K^{\mathrm{S},0}_{\mathrm{lp}} = \mathsf{K}^0\left[\frac{3\sqrt{3}M^3C_{\mathcal{A}}A_1}{5r_P^\circ} + \frac{6\sqrt{2}M^{3/2}(C_{\mathcal{A}}A_1/4+\sqrt{2}/2)}{5(r_P^\circ)^2}\right].$$

The progression constants in these minima are

$$\varepsilon^{\mathrm{P}}_{\mathrm{src}} = \min\{\varepsilon_{\mathfrak{h}}, c_{\mathrm{os}}\alpha_0/(2c_{\mathfrak{h}})\}, \qquad b_{\mathrm{A}} = 6\cdot 1201^2M^3,$$
$$K^{\vee}_{\mathrm{Q}} = \max\{672M^6(\alpha_0/\varepsilon^{\mathrm{P}}_{\mathrm{src}})^2, 2b_{\mathrm{A}}M^3/(r_P^\circ)^2\},$$
$$K^{\vee}_{\mathrm{L}} = \max\{8064M^6\alpha_0/\varepsilon^{\mathrm{P}}_{\mathrm{src}}, 72b_{\mathrm{A}}M^3/r_P^\circ\},$$
$$K^{\mathrm{f}}_{\mathbf{M}} = 3L^3\max\{K_Ee^{4\lambda_1}, 32K_\Pi K_P, (117/2)K_\Pi\},$$
$$\mu_* = \min\{w_\star/32, E_0/(16K^{\mathrm{f}}_{\mathbf{M}})\}, \qquad \sigma_{\mathrm{src}} = \mu_*/K^{\vee}_{\mathrm{Q}}.$$

To specify $\bar\sigma$, put $\bar N_{\mathbf{M}} = 2c_{\mathbf{M}}c_{\mathrm{lat}}(\delta_{\mathbf{M}})$ and

$$s_{\mathrm{st}}(K,r) = \frac{r^2}{4KS_{\mathrm{s}}}\min\left\{\gamma^{\mathrm{pc}}, \frac{\gamma_{\mathrm{pos}}}{4(1+\bar N^2_{\mathbf{M}})}, \frac{\gamma_{\mathrm{pos}}}{32(L+2)^3S_{\mathrm{s}}\max(1,2\bar N^2_{\mathbf{M}})}\right\}.$$

For $(K,r) = (\mathsf{K}_0, r_0), (\mathsf{K}^0, r_P^\circ)$ let

$$C^+_{\mathrm{kp}}[K,r] = 4c_v^2B^{\mathrm{res}}_{\mathrm{f}}C_{\mathrm{E}}e^{6\delta^{\mathrm{S}}M}Kr^{-2}K^\sharp_{\mathrm{ch}}e^{-\kappa''}, \quad C^+_{\mathrm{rj}}[K,r] = C^+_{\mathrm{kp}}[K,r]\frac{C_3B^\sharp_{\mathrm{ch}}(LM)^6}{K^\sharp_{\mathrm{ch}}}.$$

Write $\bar C^+_{\mathrm{kp}}, \bar C^+_{\mathrm{rj}}$ for their maxima over the two choices, and take

$$\bar\sigma = \min\left\{\frac{\mathsf{m}_\odot}{4\mathsf{k}_\odot}, s_{\mathrm{st}}(\mathsf{K}_0,r_0), s_{\mathrm{st}}(\mathsf{K}^0,r_P^\circ), \frac{c^{\mathrm{th}}_{\mathrm{kp}}}{2\bar C^+_{\mathrm{kp}}}, \frac{c^{\mathrm{th}}_{\mathrm{rj}}}{2\bar C^+_{\mathrm{rj}}}, \frac{1}{2C^2_{\mathrm{xt}}\bar C^+_{\mathrm{kp}}}\right\}. \tag{1714}$$

These are minima of finitely many positive numbers. The stability ceilings therefore do not force the source radius to vanish.

Proposition 82.2. Fix the preceding constants, and let $R>0$. There is $\gamma>0$ such that all the real numerical conditions hold and, uniformly for $x\ge\gamma^{-2}$,

$$R \le \sigma^{\mathrm{S}}(x-\bar B), \qquad RF_{\mathfrak{p}_0}(x) \le \sigma^{\mathrm{S}}_{\mathrm{L}}, \qquad F_a(x) = x^{-1/2}(\log x)^a. \tag{1715}$$

The following bounds can be imposed at the same choice:

$$\delta(x),\ x^{-1/2}\mathfrak{b}(x)^3 \le \epsilon^+_\times, \quad (C_{\mathcal{A}}+\theta_{\mathrm{sl}})\delta(x) \le \tfrac12\min(r_0,r_P^\circ),$$
$$S_0(x) \le \epsilon^+_\times, \quad (LM)^3e^{-\Gamma_P(x)} \le (6K^+_{\mathrm{ap}})^{-1}, \tag{1716}$$
$$\mathsf{sz}^{\mathrm{S}}_{\mathrm{rg}}(x), \mathsf{sz}^{\mathrm{S}}_{\mathrm{bd}}(x) \le \epsilon^{\mathrm{S}}_{\mathrm{ap}}.$$

We also impose the stronger regular-output reserve

$$K_{\mathbf{M}}\left(\delta(x) + \frac{(1+\beta_{\mathrm{s}})\delta(x)}{\alpha_0}N_\star + e^{-\gamma_2\mathfrak{b}(x)^2/2}\right) \le E_0/4 \qquad (x\ge\gamma^{-2}). \tag{1717}$$

The same effective-potential bounds hold at the enlarged triple $(\sqrt2E^{\mathrm{S}}_0, \sqrt2B^{\mathrm{S}}, \sqrt2B^{\mathrm{S}}_{\mathrm{fr}})$, with the corresponding majorants used in the complex boundary construction. Any prescribed fixed lower threshold on $\log\gamma^{-2}$ from the history-mass estimates can be included. Decreasing γ preserves all these assertions. ∎

Proof. Only the last, downward choice of the real construction remains. The first condition in (1715) holds when $\gamma \le (\bar{B} + R/\sigma^{\mathrm{S}})^{-1/2}$. The second holds for all sufficiently small γ, since $F_{\mathfrak{p}_0}(x) \to 0$. This is a condition on the entire interval $[\gamma^{-2}, \infty)$, so no monotonicity near its finite endpoint is being presumed.

Every size in (1716) is bounded by a finite sum of the functions F_a, negative powers of $\log x$, and positive multiples of $e^{-c(\log x)^{2\mathfrak{p}_0}}$ with $c > 0$. The negative logarithmic power in the boundary size is $(\log x)^{-(q-\mathfrak{p}_0)}$, where $q > \mathfrak{p}_0$. The left side of (1717) also tends to zero. Thus the stronger ceiling $E_0/4$, used below, can be imposed in this final choice. The source sizes use the same expressions as the real sizes, with E_0, B, B_{fr} replaced by the fixed triple in (1711). For the complex decomposition use the triple multiplied by $\sqrt{2}$. The ten admission inequalities for the effective-potential estimate contain none of these three sizes. They therefore still hold. Its majorants at the enlarged triple tend to zero, and making them at most $\epsilon_{\mathrm{ap}}^{\mathrm{S}}$ gives the required estimates for both the enlarged and original triples. Likewise $\Gamma_P(x) = C_{\mathrm{mn}}^{-1}(LM)^{-3}\theta_c^2 A_1^2 (\log x)^{2\mathfrak{p}_0} \to \infty$. There are finitely many requirements, all valid on a sufficiently late interval. Their intersection with the real conditions and the fixed history thresholds contains such an interval. This proves the claim, including independence of K, m and of any source function. □

82.3. **Uniform bounds at every step.** Let $u_\bullet = (x_0, \dots, x_K)$ be an admissible sequence and set

$$s_0 = R/(x_K - \bar{B}), \qquad R' = \sqrt{3}R/40.$$

Use the source class of radius R in physical lengths. The relative weights obey $|v_k^w|_\infty < s_0$ and $x_k \mathrm{sw}_k(v_k^w) < R'$ by proposition 73.3.

Proposition 82.3. At the preceding choices all numerical hypotheses of the general and first sourced steps hold on their full analytic domains. More precisely, for $t = k+1$, $0 \le k < K$, both the centered member and its one-lattice escort have size at most

$$U_k = \mathsf{k}_\odot [s_0(1 + g_k \mathfrak{b}_k^3 + x_t^{-1/2}) + R'(g_k + \delta_k)] \le \tfrac{3}{4} \min(\mathsf{m}_\odot^{\mathrm{S}}, M^3 \eta^{\mathrm{S}}). \tag{1718}$$

In particular the escort size is at most $\bar{n}_{\mathrm{co}}^{\mathrm{S}}$. Writing n_k for that size, the first-step reserve is explicitly $n_0 \le \bar{n}_{\mathrm{co}}^{\mathrm{S}} \le w_\star/4 \le w_\star/2$. The centered expansion and the boundary construction satisfy

$$\varepsilon_{\mathsf{w}}^{\mathrm{S}} L^3 \le (6K_{\mathrm{ap}}^+)^{-1}, \quad b_{\mathrm{cs}}^{\mathrm{S}} \le \tfrac{1}{3}, \quad b_{\mathrm{rj}}^{\mathrm{S}} + b_{\mathrm{sr}}^{\mathrm{S}} \le \tfrac{1}{2}, \tag{1719}$$

$$C_\Upsilon \beta_\upsilon^{(t)} \le \gamma_{\mathrm{kp}}/8, \quad b_{\mathrm{fr}}^{\mathrm{S},(t)} \le \tfrac{1}{8}, \quad c_{\mathrm{kp}}^{\mathrm{S}} \le c_{\mathrm{kp}}^{\mathrm{th}}/16, \quad c_{\mathrm{rj}}^{\mathrm{S}} \le c_{\mathrm{rj}}^{\mathrm{th}}/16. \tag{1720}$$

The regular one-lattice comparison has all its unperturbed reserves:

$$n_{\mathrm{int}} \le 3K_P \delta_k \le 3w_\star/8, \qquad n_{\mathrm{iv}} \le K_\Lambda s_{\mathfrak{a}}(k) N_\star \le w_\star/4. \tag{1721}$$

The extra escort formed by absorbing the old regular difference has norm at most

$$K_\Lambda s_{\mathfrak{a}}(k) \frac{L}{L-1} \frac{E_0}{4} + \bar{n}_{\mathrm{co}}^{\mathrm{S}} \le \min\{w_\star/2, E_0/(1404 L^3 K_\Pi)\}. \tag{1722}$$

Thus the one-lattice differences with coefficients $33(3L^3 - 3)K_\Pi$ and $108(L^3 - 1)K_\Pi$ apply. ■

Proof. The table cut at t is the same whether computed from $u_\bullet$ or from $u_\bullet|_t$. Its enlarged radii satisfy $a_m^{\mathsf{Tab}} = (1 + \beta_{\mathrm{s}})\alpha_{0,m} \le 1$. The weighted layer bound gives $a^\Sigma \le 12L^4$. Insert this bound in the general centered and escort estimates, and $a_0^{\mathsf{Tab}} \le 1$ in the first-step estimates. The defining maxima of C^{src} and C_0^{src} bound each coefficient. For the far terms use $\mathfrak{b}_k \le g_k^{-1}$ and $x_k \le (3L/2)x_t$; the threshold for $\check{R}_t$ gives $g_k^{-1} e^{-8L\check{R}_t} \le \sqrt{3L/2}\, x_t^{-1/2}$ and $g_k^{-2} e^{-8L\check{R}_t} \le (3L/2) x_t^{-1/2}$. The two far contributions have coefficients at most $13L^5 C^{\mathrm{src}}$ and $36L^5 C_{\mathcal{A}} C^{\mathrm{src}}$. Their sum and the other coefficients are at most $49L^5 C_{\mathcal{A}} \max(C^{\mathrm{src}}, C_0^{\mathrm{src}})$. This proves the bound by U_k for the centered members, their history restrictions, and the two one-lattice escorts.

For either $m = \mathsf{m}_{\odot}^{\mathrm{S}}$ or $m = M^3\eta^{\mathrm{S}}$, the four terms of U_k are bounded respectively by $m/4$, $\epsilon_{\times}^{+}m/4$, $m/8$, and $m/4$. Indeed $s_0 \le \sigma^{\mathrm{S}}$, $g_k\mathfrak{b}_k^3 \le \epsilon_{\times}^{+} < 1/2$ and $x_t^{-1/2} \le 1/2$, while $R'(g_k+\delta_k) \le \sqrt{3}(1+A_1)\sigma_{\mathrm{L}}^{\mathrm{S}}/40$. Equations (1712) and (1713) give the four claims. Their sum is at most $3m/4$.

The ordinary source sizes are at most $\epsilon_{\times}^{+}$. Thus their contribution to the amplitude, multiplied by L^3, is at most $(24K_{\mathrm{ap}}^{+})^{-1}$. The centered contribution is at most $(16K_{\mathrm{ap}}^{+})^{-1}$, since $\mathsf{m}_{\odot} \le w_0\varepsilon^{\mathrm{O}}$. Their sum is at most $(6K_{\mathrm{ap}}^{+})^{-1}$. Also $(LM)^3e^{-\gamma_P} \le (6K_{\mathrm{ap}}^{+})^{-1}$, so the centered cluster estimate yields $b_{\mathrm{cs}}^{\mathrm{S}} \le 1/3$.

For the quadratic correction,

$$\beta_{\upsilon}^{(t)} = 8C_{\mathrm{E}}e^{6\delta^{\mathrm{S}}M}\mathsf{K}^{(t)}(\mathsf{R}^{(t)})^{-2}s_0 \le \tfrac{1}{8}b_{\upsilon}^{\mathrm{S}}\sigma^{\mathrm{S}},$$

where $(\mathsf{K}^{(t)},\mathsf{R}^{(t)})$ is either of the two pairs used above. Substituting the corresponding four members of (1712) into the one-source-chain bounds proves (1720). In particular each old kept or rejected coefficient grows by at most a factor $3/2$. In the strip formula the cluster coefficient can be any number in $[0,1/2]$; since $N_{\mathrm{b}}^{\mathrm{th}} \ge 1$, its source-independent part is also at most $3/2$ times the old strip bound. The real bound (1466), $b_{\mathrm{rj}}^{\mathrm{th}} + b_{\mathrm{sr}}^{\mathrm{th}} \le 1/3$, therefore gives $b_{\mathrm{rj}}^{\mathrm{S}} + b_{\mathrm{sr}}^{\mathrm{S}} \le 1/2$. The total new boundary budget is at most $5/6$.

We verify the one-lattice room separately. The real choices and (1717) give

$$N_\star \ge \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp), \qquad s_{\mathfrak{a}}(k) \le s^\star \le \min\{w_\star/(8K_P), w_\star/(4K_\Lambda N_\star)\},$$
$$K_{\mathbf{M}}(\delta_k + s_{\mathfrak{a}}(k)N_\star + e^{-\gamma_2\mathfrak{b}_k^2/2}) \le E_0/4, \qquad K_{\mathbf{M}} \ge 243L^3K_\Pi K_\Lambda.$$

Here $\delta_k \le s_{\mathfrak{a}}(k)$ since $\alpha_0 \le 1+\beta_{\mathrm{s}}$. These inequalities prove (1721) and

$$K_\Lambda s_{\mathfrak{a}}(k)\frac{L}{L-1}\frac{E_0}{4} \le \min\{w_\star/16, E_0/(3888L^3K_\Pi)\}.$$

Adding (1710) proves (1722); the numerical inequality used is $3888^{-1} + 2808^{-1} < 1404^{-1}$. The unperturbed fourth family and the extra escort therefore have sum of norms at most $3w_\star/4$. At the first step there is no old regular fourth family. Its escort satisfies the still smaller bound $\bar{n}_{\mathrm{co}}^{\mathrm{S}} \le w_\star/4$. The Gaussian hypotheses and the full analytic core are the real ones, so the exact hypotheses of propositions 77.3 and 77.4 now hold. □

82.4. **Application to the actual histories.**

Lemma 82.4. Fix $R = 2\varrho$ in the preceding construction and let the lattice and coupling sequence be admissible for the real finite construction. At every step, for every history, the numerical hypotheses used in the sourced recursion hold with the data

$$(E_0^{\mathrm{S}}, B^{\mathrm{S}}, B_{\mathrm{fr}}^{\mathrm{S}}, \bar{b}_{\mathrm{fr}}^{\mathrm{S}}, \bar{n}_{\mathrm{co}}^{\mathrm{S}}, \delta_\upsilon, \kappa', \theta_{\mathrm{sl}}) = (E_0, B, B_{\mathrm{fr}}^{\flat}+1, 1, M^3\eta^{\mathrm{S}}, \delta^{\mathrm{S}}, \kappa'', (1+\vartheta_*)/2).$$

The conclusion is uniform in K, m and in $w \in \mathsf{Src}_{T_\ell}(2\varrho, 2\varrho)$. ■

Proof. The real fixed-domain and finite-step hypotheses follow from Proposition 69.1 and Theorem D. The sourced sizes and rate conditions were established above, with the same constants at every occurrence. In particular $\kappa' \ge \kappa > 2$, and the box conditions follow from (1711). The real regular majorant satisfies

$$\Xi(\gamma^{-2}, N_\star) \le 3E_0/4 :$$

its normalizer contribution is at most $E_0/2$, and the remaining three terms are bounded by $E_0/4$ in the preceding proof. This is the reserve required by the difference induction; the weaker bound E_0 alone would not suffice.

The escort bound may also be read directly at its final ceiling:

$$U_k \le \left(\frac{5}{8} + \frac{\sqrt{3}}{160}\right) M^3\eta^{\mathrm{S}} < M^3\eta^{\mathrm{S}} \le \min\{w_\star/4, E_0/(2808L^3K_\Pi)\}.$$

It follows that $108(L^3-1)K_\Pi \bar{n}^{\mathrm{S}}_{\mathrm{co}} \le E_0/26 < E_0/4$. For the general step, (1722) gives the corresponding bound with coefficient $33(3L^3-3)K_\Pi$. The regular output room is bounded by

$$\Xi(x_k, N_\star) + 27(L^3-1)K_\Pi \bar{n}^{\mathrm{S}}_{\mathrm{co}} \le E_0.$$

The chain, formed-factor and damping requirements follow from (1707) and (1720), with $\gamma' = \gamma_{\mathrm{kp}}$ and $\gamma^\dagger = \gamma'/2$. All remaining real coercivity, strip and history-counting inequalities retain their original values.

Finally (1712) and (1713) imply

$$2R \le \sigma_{\mathrm{src}}(x_K - \bar{B}), \qquad 2K_{\mathrm{L}}^{\vee} R\gamma \le 40\mu_*/\sqrt{3}.$$

These are stronger than the two progression requirements at radius R. Thus the source progression and the history recursion use one common choice of constants. □

Lemma 82.5. For these constants the first-step and general-step constructions apply on the actual source-independent domains and at every prefix of an admissible history. Their tables, Gaussian and background data, and analytic radii agree on overlapping prefixes. The source class is a complex domain, stable under conjugation and under the admitted translations, and

$$\mathsf{a}(w) = \frac{|w|_{T_\ell}}{x_K - \bar{B}} < \sigma^{\mathrm{S}} \le \frac{1}{2}.$$

The original density is bounded and continuous, and its exact partial history measures and the evaluation maps have the domains used in the sourced recursion. ■

Proof. The real construction supplies the same canonical table on each prefix, the canonical shifts with their analytic and Borel properties, and the Gaussian structures and strip maps on the full domains. The preceding lemma verifies all additional numerical conditions. The weighted Taylor estimates use $(\delta_{\mathrm{sp}}, \kappa^\sharp)$; the quadratic corrections use $(\delta^{\mathrm{S}}, \kappa'')$. We check their domain admission. For a middle-tier triple, the real table's cube and natural-class ceilings are those of Proposition 58.6. That proposition places the pair in the natural class and admits the restriction to every union of source cubes to the pivot domain. The canonical shift is consequently defined on the required middle tier, and is Borel, holomorphic on its open slices, and hereditary under restriction by Proposition 58.5. Use the enlarged radii

$$a_m = (1+\beta_{\mathrm{s}})\mathsf{r}_0((t,\mathrm{mid});m) \quad (m<k), \qquad a_k = (1+\beta_{\mathrm{s}})\max\{\mathsf{r}_0((t,\mathrm{mid});k), L^{-2}\mathsf{r}_0((t,\mathrm{mid});t)\}.$$

The layer bounds of the same table give membership in $\mathbb{U}^a_k(Y|h')$, with $a_m \le 1$ and $a^\Sigma \le 12L^4$. The field room imposed above admits both $g_k(z+\phi)$ and $g_k\phi$ to the half-radius Taylor ball. Thus proposition 78.2 applies on the full tiers; its quadratic coefficient also has the domain stated in lemma 79.2.

For the far-domain estimate, a bond in $\mathfrak{X}^\circ\langle Y\rangle$ has an endpoint in Ω_t. Its containing fine cube lies in both Y and Ω_t. If $Y \not\subset \Omega_k$, another cube of Y lies in the closure of Ω_k^{c}. The history separation puts these cubes at distance at least $8LM\check{R}_t$. An edge tree with $d_k(Y)$ edges meeting the cubes connects two suitable points at distance at most $Md_k(Y)$. Hence $d_k(Y) \ge 8L\check{R}_t$, and therefore $D \ge D' \ge 8L\check{R}_t$. This supplies the far exponential used in (1718). At the first step the table's source-cube ceilings directly give (1637) for the pivot completion. The permitted auxiliary rate is $r_\upsilon = c_{\mathrm{E}} + \delta^{\mathrm{S}}M$, giving the required rate inequality with equality; at later steps it is $r^\sharp$. The centered family has one origin and no additional ordinary origin. Its companion agrees on the small-field core by propositions 78.6 and 78.9. The centered expansion and boundary construction consequently apply with the uniform bounds just proved. The unit-weight subtraction and mean cancellation at an actual chain point are justified by lemma 79.8; in particular they use the regional Hessian, not the Hessian on a constant tower. The complex inherited and boundary families are the explicit conjugate-pair extensions used in the recursion, so no real-valued input theorem is being applied directly to a complex source.

For the relative size at every scale, the real flow equation gives

$$x_{j-1} - \bar{B} \geq L(x_j - \bar{B}), \qquad L^j(x_j - \bar{B}) \geq L^K(x_K - \bar{B}) > 0.$$

Thus $|v_j^w|_\infty \leq \mathsf{a}(w) < \sigma^{\mathrm{S}}$, and the source profile has real part at least half of its positive real profile. The domain and symmetry assertions about sources, and the bounded continuity of the density, follow from the source-class construction. The partial history domains are contained in the domains of the real coarse evaluation maps; they involve only the history tests and charts. The exact complex measure construction proved in the recursion section therefore applies on those domains. Its common null set is supplied by the zero-source realization, and is independent of w.

We record the conditions used by the history estimates. In the last coupling choice put $y = \log \gamma^{-2}$ and $\varepsilon_\gamma = A_0 \gamma y^{\mathfrak{p}_0}$, and impose

$$\begin{gathered} y > 2\mathfrak{p}_0, \quad A_0 y^{\mathfrak{p}_0} \geq 1, \quad (L^{1/r} - 1)y \geq \log(2L), \\ \varepsilon_\gamma \leq \min\{B_3\bar{\varepsilon}, \bar{\varepsilon}/2, \epsilon_{\mathrm{xp}}, \varepsilon_{\mathfrak{b}}\}, \quad c_1^{\mathrm{P}}\varepsilon_\gamma^2 \leq e_*, \quad c_2^{\mathrm{Q}}(A_1/A_0)^2\varepsilon_\gamma^2 \leq \sigma_{\mathrm{Q}}, \\ C_{\mathrm{reg}}\varepsilon_\gamma \leq \min\{\varepsilon_{\mathrm{v}}, L^{-2}c_2^{\mathrm{SO}}, L^{-2}\bar{\varepsilon}/2, \varepsilon_{\mathsf{z}}/2\}, \\ 2C_{\mathrm{reg}}\varepsilon_\gamma \leq \varepsilon_{\mathsf{z}}^+, \quad c_{\mathrm{lgs}} M \varepsilon_\gamma \leq \tfrac{1}{2}\min(\alpha_0, \alpha_2), \quad C_{\mathrm{ly}}\varepsilon_\gamma \leq \alpha_0. \end{gathered}$$

These requirements fit the final choice: $\varepsilon_\gamma \to 0$, and C_{reg} has the fixed upper bound used in the real choices. Differentiation shows that $\varepsilon(x) = A_0 x^{-1/2}(\log x)^{\mathfrak{p}_0}$ is decreasing for $\log x > 2\mathfrak{p}_0$, so $\varepsilon_t \leq \varepsilon_\gamma$. The flow inequality $x_{t-1} < 2Lx_t$ and the third condition give $\check{R}_{t-1} \leq L\check{R}_t$. Thus $p_0(g_t) \geq 1$, $\varepsilon_t \leq B_3\bar{\varepsilon}$, $c_1^{\mathrm{P}}\varepsilon_t^2 \leq e_*$ and $c_2^{\mathrm{Q}}\delta_t^2 \leq \sigma_{\mathrm{Q}}$. These are the remaining majorant conditions ($\mathrm{R}_{\mathrm{maj}}$). Together with the unchanged layer comparison and torus room they give the fixed-radius hypotheses of Proposition 64.2. Its bound applies to the sourced sums by the explicit complex extension in the history section.

The stronger positivity used in that section holds on actual histories. The real local-gauge construction gives the depth-k pair at every earlier strip component at radii $(\alpha_0/2, \alpha_1, \alpha_2/2)$ by Corollary 54.7; the preceding smallness conditions are the ones needed for this use. The local lower bound of Proposition 55.4 therefore applies at each pair. Every stored external bond belongs to exactly one strip component in its large-field piece. Extending a piece field by zero and summing the local bounds proves

$$Q_{t,D}(a; U') \geq 2\gamma_{\mathrm{kp}} \sum_{b \in \mathfrak{X}^{\mathrm{st}+}(t,D)} |a(b)|^2.$$

This is (1437) at the realized pair. The depth-k pair is used at every earlier step, while the keeping predicates are those of $h|_t$. The source-chain estimate is combined with this bound only after the evaluation has been defined.

Use the numbers $C_5^{\mathrm{S}}, c_*^{\mathrm{S}}, C_{\mathrm{maj}}^{\mathrm{S}}, C_{\mathfrak{b}}^{\mathrm{S}}$ of eqs. (1723) and (1724), with the coefficient bound C_{co} fixed by (1404). They satisfy $C_5^{\mathrm{S}} = C_5$ and $c_*^{\mathrm{S}} = c_*/2$. In the entropy threshold put $(c, C_{\mathrm{maj}}, C_{\mathfrak{b}}) = (c_*^{\mathrm{S}}, C_{\mathrm{maj}}^{\mathrm{S}}, C_{\mathfrak{b}}^{\mathrm{S}})$, $D_0 = C_{\mathrm{maj}} + 27(2 + \log(2L))C_{\mathfrak{b}}$, and $\lambda_L = (6L^3 + 1)\log 2$. Then

$$\begin{aligned} Y_0 &= \max\{\log \max(e, 2B_\sharp), 2\log(2L)/(L^{1/r} - 1)\}, \\ Y_{\mathrm{b}}^{\mathrm{S}} &= \max\Bigg\{Y_0, \left(\frac{4}{\eth c A_0}\right)^{1/\mathfrak{p}_0}, \left(\frac{2 \cdot 48^3 (D_0 + \lambda_L) L^3 M^3}{\eth c A_0^2}\right)^{1/(2\mathfrak{p}_0 - 3r - 2)}, \\ &\qquad \left(\frac{3\log L + \log 2}{2(1 - \eth)c\mathfrak{p}_0 A_0^2 \log(L/2)}\right)^{1/(2\mathfrak{p}_0 - 1)}\Bigg\}. \end{aligned}$$

All denominators are positive, since $\eth \in (0,1)$, $c > 0$, $L \geq 13$ and $2\mathfrak{p}_0 - 3r - 2 > 0$. This number is finite and fixed before γ. We include $\log \gamma^{-2} \geq Y_{\mathrm{b}}^{\mathrm{S}}$ in the final choice; then every y_t meets the required entropy threshold. The relative action loss has already been bounded by $1/2$. □

83. History bounds and the source energies

We now use the constructed densities to estimate the partition function. The complex source leaves half of the real action and half of the retained Gaussian decay available. These two reserves pay for the creation of a large-field cell. All estimates in this section concern the fixed assignment and auxiliary data of lemma 82.5.

Write $N_k = (L^{K-k}\ell)^3$ and $y_j = \log x_j$. Set

$$C_5^{\mathrm{S}} = C_{\mathrm{m}}^{\mathrm{sc}}(E_0 + B_\sharp + M^{-3}B^{\mathrm{S}}), \qquad \gamma^\dagger = \tfrac{1}{2}\gamma_{\mathrm{kp}}, \quad c_\star^{\mathrm{S}} = \tfrac{1}{2}c_\star, \tag{1723}$$

where $C_{\mathrm{m}}^{\mathrm{sc}}$ is the constant in equation (1419), and $c_\star, C'_\star$ are the constants in equation (1429). Define

$$\begin{aligned} C_\star'^{\mathrm{S}} &= C'_\star + \tfrac{9}{2}L^3\log 2, \\ \mathsf{C}_{\mathrm{maj}}^{\mathrm{S}} &= \max\{C_5^{\mathrm{S}}, L^3C_5^{\mathrm{S}} + C_\star'^{\mathrm{S}}\}, \\ C_\flat^{\mathrm{S}} &= \max\{3(L^3-1), (6L^3-5)C_{\mathrm{co}} + (C_{\mathrm{E},0} + 16C_{\mathrm{E},1})E_0\}. \end{aligned} \tag{1724}$$

Here C_{co} is the coefficient bound in equation (1404), and $C_{\mathrm{E},p} = 2048\sum_{n\ge 0}(n+2)^p e^{-n}$. These constants are fixed before the source radius and the coupling threshold.

83.1. **The effective density.** For a history $h \in \mathcal{H}_k$, let $\mathcal{S} = \mathcal{S}(h)$ and let $\mathfrak{q}$ denote its stored fields. The regular sum, boundary sum and label vacuum constant are formed by the finite sums of Sections 63–64, with $E^{(j),w}$ and the sourced boundary families in place of the corresponding real families. Put

$$A_{k,h}^w(\mathsf{U},\mathfrak{q}) = -A(\beta_k^w,\mathsf{U}) + \mathbf{E}_k^w(\mathcal{S};\mathsf{U}) + \mathbf{E}_k^{\mathrm{bd},w}(h;\mathsf{U},\mathfrak{q}) - \mathcal{E}_{k,\mathcal{S}}^w, \qquad \beta_k^w = \beta_k - L^K w. \tag{1725}$$

Evaluation at a history point requires that all terms be defined on every adapted representative and have the same value there. The retained sum $\mathbf{\Phi}_h^w$ has its own evaluation requirement. These two requirements are imposed separately.

Theorem AJ (The sourced effective densities)**.** For every admissible finite coupling sequence, every torus with the room required in lemma 82.5, every $0 \le k \le K$ and every $\|w\|_{\flat;T_\ell} < 2\varrho$, the sourced recursion gives finite complex history measures. Their sum is the pushforward of $\rho_0^w\,dU$ by the k-fold block average. For $k \ge 1$ the density on the history variables is

$$\mathsf{Stay}_h^{\mathrm{T}}\chi_k(\Omega_k)\exp(\mathbf{\Phi}_h^w)\exp(A_{k,h}^w) \tag{1726}$$

with the zero-test convention. There is one reference null set outside which, whenever $\mathsf{Stay}_h^{\mathrm{T}}\chi_k(\Omega_k)$ is nonzero, the two exponents are defined for all sources. The local regular terms satisfy

$$\begin{aligned} |E^{(j),w}(X,P,z)| &\le E_0 e^{-(\kappa+1)d_j(X)}, \\ |E^{(j),w}(X,P,z) - E^{(j),0}(X,P,z)| &\le \tfrac{1}{4}E_0\frac{\|w\|_{\flat;T_\ell}}{2\varrho}e^{-(\kappa+1)d_j(X)}. \end{aligned} \tag{1727}$$

The created boundary terms obey $|B_j^{\mathrm{mk},w}(X)| \le B^{\mathrm{S}}e^{-\kappa d_j(X)}$; the sum of the oscillations of retained exponents created at $(i+1,X)$ is at most $B_{\mathrm{fl}}^{\mathrm{S}}\mathfrak{b}_i^2 e^{-\kappa d_{i+1}(X)}$. The functions are Borel measurable, analytic on their stated field domains, and holomorphic in the source. They satisfy their respective gauge identities, with the stored fields transformed jointly with the active fields for retained exponents. Regular and created boundary terms depend on the source only on $X^{\sim 1}$, and a retained exponent only on its domain X_f. They satisfy the conjugation laws of proposition 80.1. The regular terms satisfy the translation law of proposition 80.3. At zero source they agree with the real construction using the same assignment. Restricting the construction to an earlier level gives the construction for that prefix.

On every defined realized evaluation, the retained factors obey

$$|e^{\mathbf{\Phi}_h^w(\xi_h(U))}| \le \prod_{t=1}^{k} \prod_{c\in\Gamma_t^\partial(h)} |\det(L\mathsf{m}_c(\mathcal{M}^{t-1}\mathsf{U}))|^{-1} \exp\left(-\gamma^\dagger \sum_{(t,D)} \sum_{b\in\mathfrak{X}^{\mathrm{st}+}(t,D)} |\mathsf{a}'_{t-1}(b)(U)|^2\right). \quad (1728)$$

Here $U \in \mathcal{V}_0[h]$, and U is any glued representative. For every realized history point the regular and boundary sums are defined and satisfy

$$|\mathbf{E}_k^w| + |\mathbf{E}_k^{\mathrm{bd},w}| \le C_5^{\mathrm{S}} \sum_{n=0}^{k} \#_n(\mathcal{L}_n) \le C_5^{\mathrm{S}} \left(N_k + L^3 \sum_{j=1}^{k} \#_j(Z_j)\right). \quad (1729)$$

■□

Proof. The simultaneous choice in lemma 82.5 gives the domains, rate inequalities and size reserves needed at each step. Apply propositions 80.1 and 80.3 and the exact measure identities theorem AI and proposition 81.6. At level zero the density is ρ_0^w; the output of each step is exactly the partial density required at the next. The inherited real tests partition the finite bare measure, so proposition 73.5 identifies the sum with the block pushforward. The realization argument in theorem AI supplies the single source-independent null set. The formula for the action is (1725); its counterterms and profiles use the real coupling sequence.

The regular size and the uniform difference bound $E_0/4$ are proposition 80.3 at the reserves of lemma 82.5. For fixed j, X, P, z, apply lemma 73.2 to their difference, which is holomorphic and vanishes at zero source. This gives the second line of (1727). The boundary, retained and covariance statements follow from proposition 80.1. Dependence only on the source in X implies the stated dependence on the source in $X^{\sim 1}$. The retained quadratic estimate proposition 79.7, applied after the evaluation values have been established, gives (1728). The determinant product is independent of the glued representative: the pivot bonds partition into the boundaries of the pieces, and their elimination factors are the fixed coordinate Jacobians of the real history.

It remains to check the sum over scales. Use Proposition 64.2 with regular size E_0, boundary size B^{S} and regular rate $\kappa + 1$. Its proof uses moduli of analytic local terms and their quadratic vanishing at the trivial pair, not reality of their values. The sourced families have the same gauge invariance and analytic cores, so the same argument applies. Its domain hypotheses hold on every realized representative: the cores and boundary spaces at half radii are properties of the real tests. For clarity, the summation has three parts. The regular and action terms at a site in layer n have bounds proportional to $L^{-4(n-j)}$; multiplication by the $L^{3(n-j)}$ sites gives $L^{-(n-j)}$. Created boundary domains meet $Z_j^\sim$, and their entropy sum is at most $C'_{\mathrm{f}} M^{-3} \#_j(Z_j^\sim)$. Finally

$$\sum_{j=1}^{k} \#_j(Z_j) \le 2 \sum_{n=0}^{k} \#_n(\mathcal{L}_n), \qquad \#_n(\mathcal{L}_n) \le L^3 \#_{n+1}(Z_{n+1}) \quad (n < k).$$

These are the geometric counts in that proposition and give (1729) with precisely (1723). □

83.2. **The modulus of a history contribution.** We give the loss calculation explicitly, since positivity of the sourced measure is not available. Put $\theta = 1/2$. For a piece (t, D), write $S = \sum_b |\mathsf{a}'(b)|^2$, $\mathsf{g} = \nu e^{-\Pi}$, where $\nu > 0$ and $\Pi \ge 0$, and $\zeta = |z_{t,D}^{\mathrm{dsp}}|$. The determinant estimate of Proposition 65.4 gives

$$0 < \zeta \le A_D := (2L^9)^{6\#_t(D)}.$$

If Q is a real form with $Q \ge 2\theta\gamma_{\mathrm{kp}} S$, then $Q' = \theta^{-1} Q$ satisfies the original lower bound. The product bound of that proposition, applied to Q', is

$$e^{-A_{\mathrm{cr}}} \prod_{(t,D)} \nu e^{-\Pi} \zeta e^{-Q'/2} \le e^{-c_\star \Sigma_{\mathrm{cr}}} \prod_{(t,D)} A_D \nu e^{-\Pi/2} e^{-\gamma_{\mathrm{kp}} S/4}, \quad \Sigma_{\mathrm{cr}} = \sum_t p_0(g_t)^2 \mathsf{n}_t^{\mathrm{cr}}(h).$$

Raise it to the power θ and multiply by $\prod(\nu e^{-\Pi}\zeta)^{1-\theta}$. Since $\zeta \le A_D$ and $\Pi \ge 0$, this gives

$$e^{-\theta A_{\mathrm{cr}}} \prod_{(t,D)} \nu e^{-\Pi}\zeta e^{-Q/2} \le e^{-c_\star^{\mathrm{S}}\Sigma_{\mathrm{cr}}} \prod_{(t,D)} \bar{\mathsf{u}}_{t,D}^{\mathrm{S}}, \quad \bar{\mathsf{u}}_{t,D}^{\mathrm{S}} = A_D \nu e^{-\Pi/2} e^{-\gamma^\dagger S/4}. \tag{1730}$$

If a gauge penalty is infinite, the left side vanishes; the preceding calculation applies to all other cases. The piece integral uses at most $9L^3 \#_t(D)$ real Gaussian coordinates. Replacing γ_{kp} by $\gamma^\dagger = \gamma_{\mathrm{kp}}/2$ multiplies its bound by at most $2^{(9/2)L^3\#_t(D)}$. Thus

$$\int \bar{\mathsf{u}}_{t,D}^{\mathrm{S}}\, dm_{t,D} \le e^{C_\star'^{\mathrm{S}}\#_t(D)}. \tag{1731}$$

All curvature and failed-test hypotheses here are those of the real history. They hold by lemma 82.5; only the quadratic lower bound has been replaced.

Proposition 83.1. For every $h \in \mathcal{H}_k$, $1 \le k \le K$, and every source in the ball,

$$e^{\operatorname{Re}\mathcal{E}^w_{k,\mathcal{S}(h)}} |\hat{\tau}_h^{\mathrm{co}\prime}[\rho_0^w]|(\mathsf{X}_h^{\mathrm{co}\prime}) \le e^{\mathsf{C}_{\mathrm{maj}}^{\mathrm{S}}(N_k + \sum_{j=1}^k \#_j(Z_j))} e^{-c_\star^{\mathrm{S}}\Sigma_{\mathrm{cr}}}. \tag{1732}$$

■

Proof. Pass to the free history coordinates, using the mutually inverse coordinate maps on their supports from Section 66. Pulling back the density (1726) gives an integrable complex Borel density f relative to the free product measure. Extend it by zero on the reference null set. The two coordinate pushforwards are inverse on the supports, so total variation has the same mass in the two coordinate systems.

The real history measure is concentrated on a Borel set of realized free coordinates. The sourced measure is dominated in total variation by $\sup|\rho_0^w|/\min\rho_0$ times that measure, by the bare integral formula of proposition 73.5; the denominator is positive by compactness. It therefore vanishes on every Borel subset of the complement. Since f is its density, $|f|$ has zero integral on that complement. Intersect with the tiers and remove the reference null set. At every remaining point the nonzero test implies that both exponents are defined. Choose a bare preimage and any adapted representative at that point.

By proposition 73.3 and lemma 82.5, $\operatorname{Re}\beta_k^w \ge \theta\beta_k > 0$. The representative is real and $a_p(\mathsf{U}) \ge 0$; hence $\operatorname{Re} A(\beta_k^w, \mathsf{U}) \ge \theta A(\beta_k, \mathsf{U}) \ge \theta A_{\mathrm{cr}}$. Combine (1725), (1729) and (1728), using on each piece the real form $Q = 2\gamma^\dagger S$. The stay factors provide $\nu e^{-\Pi}$, and the determinants provide ζ. Equation (1730) then gives

$$|f(v)| \le e^{-\operatorname{Re}\mathcal{E}^w_{k,\mathcal{S}} + C_5^{\mathrm{S}}\sum_n \#_n(\mathcal{L}_n) - c_\star^{\mathrm{S}}\Sigma_{\mathrm{cr}}} \prod_{(t,D)} \bar{\mathsf{u}}_{t,D}^{\mathrm{S}}(\mathrm{pr}_{t,D}\, v).$$

The right side depends only on the free coordinates; no measurable choice of the bare preimage is required. Different pieces use disjoint coordinate factors, every Lebesgue coordinate belongs to one piece, and the remaining coordinates carry Haar probability measure. Tonelli and (1731), equivalently the product calculation in equation (1444), bound the integral of this product by $e^{C_\star'^{\mathrm{S}}\sum_j \#_j(Z_j)}$. Finally use (1729) and (1724). □

83.3. **Vacuum deficits and the sum over histories.** Define the sourced vacuum data by $\mathsf{vl}^{(j),w}(X,z) = E^{(j),w}(X,\mathbf{1},z)$, keeping the Gaussian constants $\mathsf{vn}^{(j)}$ and the extensive coefficients of Section 63. Use the finite formulas (1405)–(1407) to define $E^w_{(k)}$, $\mathcal{E}^w_{k,\mathcal{S}}$ and $\mathfrak{E}'^w_n(\mathcal{S})$. The two ranges of base points in those formulas are unchanged: regular terms use the interior base points, while Gaussian constants use every base point of the domain.

Proposition 83.2. All these vacuum numbers are defined. For every label $\mathcal{S}$ of level k,

$$E^w_{(k)} + \mathcal{E}^w_{k,\mathcal{S}} = \sum_{n=1}^k \mathfrak{E}'^w_n(\mathcal{S}), \qquad |\mathfrak{E}'^w_n(\mathcal{S})| \le C_\flat^{\mathrm{S}}(1+y_{n-1})\#_n(Z_n^\sim). \tag{1733}$$

Put

$$e_j(X;w)=\sum_{z\in X\cap\mathbb{T}^{(j)}}[E^{(j),w}(X,\mathbf{1},z)-E^{(j),0}(X,\mathbf{1},z)]. \tag{1734}$$

Then

$$E^w_{(k)}=E_{(k)}+\sum_{j=1}^{k}\sum_{X\in\mathcal{D}_j}e_j(X;w),\qquad \widetilde{Z}_k(w):=e^{-E^w_{(k)}}Z_{K,\ell,\beta}[w] \tag{1735}$$

defines a holomorphic function with

$$|\widetilde{Z}_k(w)|\le\sum_{h\in\mathcal{H}_k}e^{-\operatorname{Re}E^w_{(k)}}|\hat{\tau}^{\mathrm{co}\prime}_h[\rho^w_0]|(\mathsf{X}^{\mathrm{co}\prime}_h)\le e^{(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}+2)N_k}. \tag{1736}$$

The functions e_j satisfy (1491) with $\kappa_e=\kappa$, vanish at zero source and depend on the source only on $X^{\sim 1}$. They and $\widetilde{Z}_k$ satisfy conjugation reality. ■

Proof. The trivial pair belongs to every fixed analytic core. The local vacuum values are thus defined, and their moduli have the same bound as in the real construction. We repeat the part of Proposition 63.1 that does not require their reality. The domain entropy gives

$$\sum_{X\ni z}|\mathsf{vl}^{(j),w}(X,z)|\le C_{\mathrm{E},0}E_0,\qquad \sum_{X\supset\square}\sum_{z\in X}|\mathsf{vl}^{(j),w}(X,z)|\le 8M^3C_{\mathrm{E},1}E_0.$$

The same estimates hold for $\mathsf{vn}^{(j)}$. In a restricted vacuum sum, either a missing domain contains a cube of Z_n, or its domain lies in Λ_n and its regular base point belongs to $Z_n^{\sim}\setminus Z_n$. The cube estimate bounds each of the two vacuum families in the first case; the site estimate bounds the second case. The extensive logarithmic part is bounded by $3(L^3-1)y_{n-1}\#_n(\Omega_n^c)$, and its remaining coefficients by $(6L^3-5)C_{\mathrm{co}}\#_n(Z_n)$. Adding these bounds proves the inequality in (1733). Its identity is the telescoping sum of the defining deficits, valid for complex values.

At zero source the data are the canonical real data. The source changes only vl, so subtraction of the finite whole-torus sums gives (1735). By (1727) and the geometric bound $\#_j(X)\le 8M^3(d_j(X)+1)$,

$$|e_j(X;w)|\le 2M^3E_0\frac{\|w\|_{b;T_\ell}}{2\varrho}(d_j(X)+1)e^{-(\kappa+1)d_j(X)}.$$

Use $(d+1)e^{-d}\le 1$ to obtain the asserted decay. Holomorphy, locality and conjugation follow by finite summation. The same properties for $\widetilde{Z}_k$ follow from proposition 73.5 and its defining exponential.

To sum the histories set

$$f(h)=e^{\operatorname{Re}\mathcal{E}^w_{k,\mathcal{S}(h)}}|\hat{\tau}^{\mathrm{co}\prime}_h[\rho^w_0]|(\mathsf{X}^{\mathrm{co}\prime}_h),\qquad e(\mathcal{S})=-\operatorname{Re}\sum_n\mathfrak{E}'^w_n(\mathcal{S}).$$

Proposition 83.1 gives the creation bound for f, and (1733) gives the deficit bound for e. Their product is exactly $e^{-\operatorname{Re}E^w_{(k)}}|\hat{\tau}^{\mathrm{co}\prime}_h[\rho^w_0]|(\mathsf{X}^{\mathrm{co}\prime}_h)$. Apply Proposition 67.6 to this nonnegative function and this real deficit, with constants $(c^{\mathrm{S}}_\star,\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}},C^{\mathrm{S}}_\flat,\eth)$. Its hypotheses $2\mathfrak{p}_0>3r+2$ and the corresponding threshold $y_K\ge\mathsf{Y}_{\mathrm{b}}$ are included in lemma 82.5. It bounds the sum by

$$e^{\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}N_k}\prod_{t=1}^{k}\prod_{\mathfrak{c}\in\mathcal{P}_t}(1+e^{-(1-\eth)c^{\mathrm{S}}_\star p_0(g_t)^2})\le e^{(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}+2)N_k}.$$

The triangle inequality and the finite history decomposition give the first inequality of (1736). □

84. Proof of the source factorisation theorem

Proof of theorem AD. Choose the constants as in lemma 82.5, and put $\kappa_e = \kappa$. They precede the choice of ϱ. After fixing that radius, choose γ_ϱ and the canonical auxiliary data at that threshold. For $0 < g \le \gamma_\varrho$ and $\beta > X_g$, proposition 74.1 gives a unique finite positive first-passage index and $g^{-2} < x_K \le X_g$. The side condition (1489) supplies all the lattice room required in lemma 82.5; arbitrarily large integral m satisfy it.

Apply theorem AJ and proposition 83.2 at $k = K$. The real constant $E_{(K)}$ is $E_{K,\ell,\beta}$ by (1535). The definition $\widetilde{Z} = \widetilde{Z}_K$ and the vacuum splitting give (1490). The local source estimates, holomorphy and conjugation laws are those of proposition 83.2. At $k = K$ we have $N_K = \ell^3$, so its stability estimate bounds $|\widetilde{Z}|$ by $e^{(\mathsf{C}^{\mathrm{S}}_{\mathrm{maj}}+2)\ell^3}$.

At zero source the local source energies vanish, and proposition 74.4 gives

$$\widetilde{Z}(0) = e^{-E_{K,\ell,\beta}} Z_{K,\ell,\beta} \ge e^{-[c_{\mathrm{low}}+(9/2)\log x_K]\ell^3}.$$

The threshold choice includes $B_\sharp \le \frac{1}{2}\gamma_\varrho^{-2}$; therefore $X_g \le (3L/2)g^{-2}$. Choose C_{S} at least

$$\max\{2M^3E_0, \mathsf{C}^{\mathrm{S}}_{\mathrm{maj}} + 2, |c_{\mathrm{low}}| + \tfrac{9}{2}\log(3L/2), \tfrac{9}{2}, 2C_{\mathrm{tt}}C'_{\mathrm{tt}}E_0\}.$$

Since $g \le 1/2$, this yields the lower bound (1493). For two allowed tori, the sequence and assignment are the same; proposition 74.5 gives the asserted bound on their normalized vacuum constants. Every member of the displayed maximum precedes the source radius and threshold. This proves the stated uniformity and completes the proof. □

For a stopped datum from corollary 74.3, the same argument applies to $((L^i\ell, K-i), u_\bullet|_{K-i})$ with reference coupling $g_{(i)} < g$ and the same threshold assignment. The constants and source radius are unchanged. This is an application on the stopped torus; it requires no identification of sources under dilation. The affine and mirror-periodic source families of proposition 73.4 lie in the same ball, so the theorem applies to them as well.

85. Plaquette observables and reflection positivity

The source derivatives give correlations of a scalar observable. To estimate them uniformly in the volume, we first prove the reflection inequality for the Wilson measure. Rectangular tori are needed here: the comparison changes one side at a time.

Let $a > 0$, let $N_\mu \ge 3$ be integers, and put

$$T = \prod_{\mu=1}^{3} \mathbb{R}/(aN_\mu\mathbb{Z}), \qquad \Lambda = \prod_{\mu=1}^{3} a\mathbb{Z}/(aN_\mu\mathbb{Z}).$$

Positive bonds and plaquettes have the usual coordinate orientations. As before, $U(\bar{b}) = U(b)^{-1}$ and $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p)$. For $\beta \ge 0$ define

$$A(U) = \sum_p a_p(U), \qquad \mathcal{O}_a(f) = a^{-1}\sum_p f(x(p))a_p(U), \qquad d\mu_{a,T,\beta} = Z_{a,T,\beta}^{-1} e^{-\beta A(U)}\, dU. \tag{1737}$$

Here dU is product Haar probability measure and $Z_{a,T,\beta}$ is the integral of $e^{-\beta A}$. Since $0 \le a_p \le 2$, this normalization is strictly positive, and $\mathcal{O}_a(f)$ is bounded for every function f on the finite set of plaquette centres. For a bounded Borel function F of the configuration, write

$$\mathcal{Z}_{a,T,\beta}(F) = \int F(U) e^{-\beta A(U)}\, dU, \qquad \langle F\rangle_{a,T,\beta} = Z_{a,T,\beta}^{-1}\mathcal{Z}_{a,T,\beta}(F).$$

Thus, on the cubic lattice with $a = L^{-K}$, $Z_{K,\ell,\beta}[w] = \mathcal{Z}_{a,T_\ell,\beta}(e^{\mathcal{O}_a(w)})$.

For $n \ge 1$ the joint cumulant is

$$S^T_{n;a,T,\beta}(f_1,\dots,f_n) = \partial_{z_1}\cdots\partial_{z_n} \log\left\langle e^{\sum_i z_i\mathcal{O}_a(f_i)}\right\rangle_{a,T,\beta}\Big|_{z=0}. \tag{1738}$$

The expectation is entire and equals one at zero, so the logarithm in this formula exists near zero. If $\mathcal{P}_n$ denotes the set of partitions of $\{1,\dots,n\}$, differentiating the logarithm gives

$$S_n^T(f_1,\dots,f_n)=\sum_{\pi\in\mathcal{P}_n}(-1)^{|\pi|-1}(|\pi|-1)!\prod_{B\in\pi}\left\langle\prod_{i\in B}\mathcal{O}_a(f_i)\right\rangle. \tag{1739}$$

Conversely, the joint moment is the sum over $\pi\in\mathcal{P}_n$ of the products of the cumulants indexed by its blocks. Both identities follow by comparing the coefficient of $z_1\cdots z_n$ in the exponential identity relating the expectation to its logarithm. In particular, cumulants are symmetric and multilinear. For $n\ge 2$ the cumulant of the random variables $\mathcal{O}_a(f_i)$ is unchanged if a scalar constant is added to any of these random variables: such a change adds a linear function of z to the logarithm.

A lattice translation or coordinate reflection acts on configurations by transporting oriented bonds. It permutes the independent Haar variables, possibly replacing some by their inverses. Product Haar measure is unchanged. The plaquette holonomy is carried to a conjugate of the original holonomy or its inverse, both of which have the same trace in SU(2). Hence the Wilson measure is invariant, and $\mathcal{O}_a(f)$ transforms by the corresponding pullback of f.

85.1. **The operator across one layer.** Fix a coordinate direction, write N for the number of sites in that direction, and let Σ be the two-dimensional cross-section. Let $\mathsf{B}(\Sigma)$ be its positive bonds and set $\mathcal{X}=\mathrm{SU}(2)^{\mathsf{B}(\Sigma)}$. A configuration consists of the transverse fields $V^t\in\mathcal{X}$ and the bonds $S^t\in\mathrm{SU}(2)^{\Sigma}$ joining layers t and $t+1$, with t taken modulo N. Write $A_\Sigma(V)$ for the sum of the plaquette actions within a transverse layer and put

$$A_\times(V,S,W)=\sum_{b\in\mathsf{B}(\Sigma)}\left(1-\tfrac12\operatorname{tr}V_bS_{b_+}W_b^{-1}S_{b_-}^{-1}\right). \tag{1740}$$

The two types of plaquettes partition all plaquettes, giving

$$A(U)=\sum_{t\bmod N}\left[A_\Sigma(V^t)+A_\times(V^t,S^t,V^{t+1})\right].$$

On $\mathcal{H}=L^2(\mathcal{X},dV)$ define the integral operator

$$\mathsf{K}(W,V)=\int e^{-\beta A_\times(V,S,W)}\,dS.$$

The kernel is continuous, symmetric and bounded by one. Symmetry follows by replacing S by S^{-1} and using invariance of the trace under cyclic permutations and inversion. Thus K is self-adjoint and Hilbert–Schmidt. Its positivity needs a separate argument.

Proposition 85.1. The crossing operator satisfies $\mathsf{K}\ge 0$. ∎

Proof. Put $\mathsf{k}(u)=e^{-\beta(1-\operatorname{tr}u/2)}$ and $\mathsf{K}_0(V,W)=\prod_b\mathsf{k}(V_bW_b^{-1})$. For $R\in\mathrm{SU}(2)^{\Sigma}$ let $(V^R)_b=R_{b_-}V_bR_{b_+}^{-1}$. The crossing weight is $\mathsf{K}_0(V,W^S)$, and $\mathsf{K}_0(V^R,W^R)=\mathsf{K}_0(V,W)$. Haar invariance therefore gives

$$\mathsf{K}(W,V)=\int dR\,dS\,\mathsf{K}_0(V^R,W^S). \tag{1741}$$

For $v,w\in\mathrm{SU}(2)$, $\operatorname{tr}(vw^{-1})=\sum_{i,j}v_{ij}\overline{w_{ij}}$. Expanding the exponential at every bond consequently yields

$$\mathsf{K}_0(V,W)=\sum_\alpha c_\alpha\phi_\alpha(V)\overline{\phi_\alpha(W)},\qquad c_\alpha\ge 0.$$

Here α specifies a nonnegative integer n_b and n_b pairs of matrix indices at each bond,

$$c_\alpha=e^{-\beta|\mathsf{B}(\Sigma)|}\prod_b\frac{(\beta/2)^{n_b}}{n_b!},\qquad \phi_\alpha(V)=\prod_b\prod_{s=1}^{n_b}(V_b)_{i_{b,s}j_{b,s}}.$$

Every matrix entry has modulus at most one. There are 4^{n_b} choices of indices at bond b, so

$$\sum_{\alpha} c_\alpha = e^{-\beta|\mathsf{B}(\Sigma)|} \prod_b \sum_{n\ge 0} \frac{(2\beta)^n}{n!} = e^{\beta|\mathsf{B}(\Sigma)|} < \infty.$$

This proves absolute uniform convergence. Insert the series into (1741) and set $\widehat{\phi}_\alpha(V) = \int \phi_\alpha(V^R)\, dR$. Then

$$\mathsf{K}(W,V) = \sum_{\alpha} c_\alpha \widehat{\phi}_\alpha(V)\overline{\widehat{\phi}_\alpha(W)}.$$

Since dV is a probability measure, every $\psi \in \mathcal{H}$ is integrable. Absolute convergence permits termwise integration, giving

$$\langle \psi, \mathsf{K}\psi\rangle = \sum_{\alpha} c_\alpha \left| \int \widehat{\phi}_\alpha(V)\psi(V)\, dV \right|^2 \ge 0.$$

□

85.2. **Slabs and the trace formula.** For an integer $q \ge 2$, a slab of q layers has variables $(V^1,\dots,V^q;S^1,\dots,S^{q-1})$ and weight

$$w_q = \exp\left[-\beta \sum_{t=1}^{q} A_\Sigma(V^t) - \beta \sum_{t=1}^{q-1} A_\times(V^t,S^t,V^{t+1})\right].$$

For a bounded Borel function Φ of these variables define

$$\mathfrak{S}[\Phi](W,V) = \int \prod_{1<t<q} dV^t \prod_{1\le t<q} dS^t\, (\Phi w_q)\big|_{V^1=V,V^q=W}, \qquad \mathfrak{Y}[\Phi] = \mathsf{K}^{1/2}\mathfrak{S}[\Phi]\mathsf{K}^{1/2}. \tag{1742}$$

These are Hilbert–Schmidt operators, since the first kernel is bounded by $\|\Phi\|_\infty$ and $\mathsf{K}^{1/2}$ is bounded. Reverse the order of the layers, invert the joining bonds, and conjugate the value of Φ; denote the resulting function by Φ^Θ. This change of variables preserves Haar measure and w_q, and exchanges the two endpoint fields. Consequently

$$\mathfrak{S}[\Phi^\Theta] = \mathfrak{S}[\Phi]^*, \qquad \mathfrak{Y}[\Phi^\Theta] = \mathfrak{Y}[\Phi]^*. \tag{1743}$$

Proposition 85.2 (Trace formula)**.** Partition the N layers cyclically into $r \ge 2$ consecutive slabs of lengths $q_s \ge 2$. Let Φ_s be a bounded Borel function of the inner bonds of slab s. Then

$$\mathcal{Z}_{a,T,\beta}\left(\prod_{s=1}^{r}\Phi_s\right) = \operatorname{Tr}\big(\mathfrak{S}[\Phi_r]\mathsf{K}\cdots\mathfrak{S}[\Phi_1]\mathsf{K}\big) = \operatorname{Tr}\big(\mathfrak{Y}[\Phi_r]\cdots\mathfrak{Y}[\Phi_1]\big). \tag{1744}$$

All the operators attached to a slab depend on its length, the cross-section and β, but not on the total length N. ■

Proof. Keep the first and last transverse fields V_s, W_s of each slab. Every plaquette either lies inside one slab or crosses from its last layer to the first layer of the next slab. Integrating all other variables gives

$$\int \prod_{s=1}^{r} dV_s\, dW_s \prod_{s=1}^{r} \mathfrak{S}[\Phi_s](W_s,V_s)\mathsf{K}(V_{s+1},W_s), \qquad V_{r+1}=V_1.$$

The integrands are bounded and all measures are finite, so Fubini applies. The kernel formula for a product, followed by the Hilbert–Schmidt identity $\operatorname{Tr}(CD) = \iint C(y,x)D(x,y)\,dx\,dy$, gives the first trace. There are at least two Hilbert–Schmidt factors, so the product is trace class. Splitting each K into two square roots and moving the last square root around the trace gives the second identity. The remaining factors form a trace-class operator, which justifies this cyclic permutation. The definitions contain no total layer number N, proving the last assertion. □

Corollary 85.3 (Reflection at bond midpoints)**.** Suppose that N is even. Let θ be reflection in a coordinate plane at a half-integer multiple of a, and let T_+ be either open half-torus bounded by this plane and its opposite parallel plane. For bounded Borel functions F, H depending only on bonds whose two endpoints lie in T_+, the form $\langle \overline{\theta F} H\rangle_{a,T,\beta}$ is positive semidefinite. In particular,

$$|\langle \overline{\theta F} H\rangle|^2 \le \langle \overline{\theta F} F\rangle \langle \overline{\theta H} H\rangle. \tag{1745}$$

■□

Proof. The two halves contain $q = N/2 \ge 2$ layers each. Their inner bond variables identify F, H with slab functions Φ, Ψ. Reflection sends the first slab to the second and reverses its layer order, so $\overline{\theta F}$ is the placed function Φ^{Θ}. By (1743) and proposition 85.2,

$$\mathcal{Z}_{a,T,\beta}(\overline{\theta F} H) = \operatorname{Tr}(\mathfrak{Y}[\Phi]^* \mathfrak{Y}[\Psi]).$$

This is the Hilbert–Schmidt inner product. Its positivity and Cauchy–Schwarz inequality survive division by $Z_{a,T,\beta} > 0$. □

86. The chessboard estimate and comparison of volumes

The reflection estimate bounds a local insertion by its reflected copies. We need the resulting bound on any even number of cells, and we need to compare it on two tori. The trace formula makes both statements consequences of inequalities for powers of one positive operator.

86.1. **Cells and reflected copies.** Fix $a > 0$, an odd integer $\lambda \ge 3$, even integers $\mathsf{n}_\mu \ge 2$, and an offset $o \in a\mathbb{Z}^3$. We call these data a board. Its torus has sides $\ell_\mu = \mathsf{n}_\mu \mathsf{l}$, where $\mathsf{l} = \lambda a$. Write

$$\mathsf{Z} = \prod_{\mu=1}^{3} \mathbb{Z}/\mathsf{n}_\mu\mathbb{Z}, \qquad \mathsf{N} = \prod_\mu \mathsf{n}_\mu, \qquad h_\lambda = (\lambda-1)/2. \tag{1746}$$

The cell $\square_\kappa$ is the image in the torus of $o + \mathsf{l}\tilde\kappa + [-\mathsf{l}/2, \mathsf{l}/2]^3$, with $\tilde\kappa$ any integer representative of $\kappa \in \mathsf{Z}$. Its sites are $o + \mathsf{l}\tilde\kappa + an$, $n \in \mathsf{C}_\lambda := \{-h_\lambda, \dots, h_\lambda\}^3$. The cell faces contain no sites: their varying coordinate is a half-integer multiple of a. Thus the cells partition the sites, their interiors are disjoint, and a plaquette whose centre lies in a cell interior has all its corners in that cell. A plaquette that is not of this kind has its centre on a cell face.

A function of a cell is a bounded complex Borel function of its inner bond variables, where an inner bond has both endpoints in the cell. Transporting the positive bonds of C_λ to $\square_\kappa$ identifies such a function with a unique model function F_0 on the finite product $\mathrm{SU}(2)^{\mathsf{B}^+(\mathsf{C}_\lambda)}$; its placement in $\square_\kappa$ is denoted by $F_0^{(\kappa)}$. Uniqueness follows because arbitrary values of the inner bond variables extend to a configuration on the torus. This also shows that placing a model function preserves its supremum norm.

For $\kappa, \kappa' \in \mathsf{Z}$ set $\epsilon_\mu = (-1)^{\tilde\kappa'_\mu - \tilde\kappa_\mu}$ and define the diagonal isometry

$$\mathsf{r}_{\kappa'\kappa}(x) = o + \mathsf{l}\tilde\kappa' + \epsilon(x - o - \mathsf{l}\tilde\kappa).$$

It carries $\square_\kappa$ to $\square_{\kappa'}$. The evenness of each n_μ makes the signs independent of the representatives; changing a representative changes the formula only by a torus period. Transport configurations and observables by the action used in section 85, including inversion when a bond orientation is reversed. For a function F of $\square_\kappa$, define

$$F^{[\kappa']} = \begin{cases} \mathsf{r}_{\kappa'\kappa} F, & \prod_\mu \epsilon_\mu = 1, \\ \overline{\mathsf{r}_{\kappa'\kappa} F}, & \prod_\mu \epsilon_\mu = -1, \end{cases} \qquad \operatorname{Fold} F = \prod_{\kappa' \in \mathsf{Z}} F^{[\kappa']}. \tag{1747}$$

These are functions of the indicated cells. The composition rule $\mathsf{r}_{\kappa''\kappa'}\mathsf{r}_{\kappa'\kappa} = \mathsf{r}_{\kappa''\kappa}$, together with multiplication of the parity signs, gives

$$(F^{[\kappa']})^{[\kappa'']} = F^{[\kappa'']}, \qquad \operatorname{Fold}(F^{[\kappa']}) = \operatorname{Fold} F. \tag{1748}$$

Copies preserve products and sums and commute with conjugation. Exactly half the cells have each parity, since changing one cell index by one interchanges the parities. Consequently

$$\operatorname{Fold}(cF) = |c|^{\mathsf{N}} \operatorname{Fold} F, \qquad \operatorname{Fold} \bar{F} = \overline{\operatorname{Fold} F}, \qquad |\operatorname{Fold} F| \le \|F\|_\infty^{\mathsf{N}}. \tag{1749}$$

86.2. **Multiple reflections.** We first give the trace inequality needed for an arbitrary even number of slabs. For a Hilbert–Schmidt operator A put $\|A\|_{2k} = (\operatorname{Tr}(A^*A)^k)^{1/(2k)}$. This is finite; cyclicity of the trace gives $\|A^*\|_{2k} = \|A\|_{2k}$, and the spectral theorem for A^*A shows that this number vanishes only when $A = 0$.

Lemma 86.1 (Hölder inequality for traces)**.** For $k \ge 1$ and Hilbert–Schmidt operators $A_1, \dots, A_{2k}$, the product is trace class and

$$|\operatorname{Tr}(A_1 \cdots A_{2k})| \le \prod_{s=1}^{2k} \|A_s\|_{2k}. \tag{1750}$$

■

Proof. The trace-class property follows from the presence of two Hilbert–Schmidt factors. Unless one factor is zero, normalize each A_s to have $\|A_s\|_{2k} = 1$. Let $\mathcal{A}$ be the finite set consisting of these normalized operators and their adjoints, and maximize $|\operatorname{Tr}(B_1 \cdots B_{2k})|$ over $\mathcal{A}^{2k}$; denote the maximum by m. An alternating tuple $(C, C^*, \dots, C, C^*)$ has trace one, so $m \ge 1$.

For $X = B_1 \cdots B_k$ and $Y = B_{k+1} \cdots B_{2k}$, Hilbert–Schmidt Cauchy–Schwarz gives

$$|\operatorname{Tr}(XY)|^2 \le \operatorname{Tr}(XX^*) \operatorname{Tr}(Y^*Y).$$

Both traces on the right are nonnegative and correspond to tuples in $\mathcal{A}^{2k}$. If the original tuple maximizes the modulus, both reflected tuples also maximize it. Among all maximizing tuples choose one with a longest string of consecutive entries alternating between an operator and its adjoint, counting positions cyclically. Let its length be j. If $j \le k$, cyclically shift it to end at position k; if $k < j < 2k$, shift it to start at position one. Reflect the first k entries to obtain $(B_1, \dots, B_k, B_k^*, \dots, B_1^*)$. This is again a maximizer and has an alternating string of length $2\min\{j, k\} > j$, a contradiction. Thus $j = 2k$ and the maximizing trace is one. Rescaling proves the inequality. □

Let $\mathbf{F} = (F_\kappa)_{\kappa \in \mathsf{Z}}$ be a family with F_κ a function of $\square_\kappa$, and abbreviate

$$Z[\mathbf{F}] = \mathcal{Z}_{a,T,\beta}\left(\prod_\kappa F_\kappa\right).$$

For a direction μ and an index $t \bmod \mathsf{n}_\mu$, write $\kappa|t$ for κ with its μth component replaced by t. Define the reflected family by

$$(R_{\mu,t}\mathbf{F})_\kappa = (F_{\kappa|t})^{[\kappa]}. \tag{1751}$$

The cells with $\kappa_\mu = t$ lie in a slab of λ layers, starting just after layer $\tau_0(t) = o_\mu/a + \lambda t - h_\lambda - 1$. Their product is a bounded Borel function $\Phi^{(t)}$ of the inner slab bonds. Here $2 \le \lambda \le \lambda\mathsf{n}_\mu - 1$, so these are slabs to which proposition 85.2 applies.

Proposition 86.2. For every board, every $\beta \ge 0$, and every such family,

$$Z[R_{\mu,t}\mathbf{F}] = \operatorname{Tr}(\mathfrak{Y}[\Phi^{(t)}]^*\mathfrak{Y}[\Phi^{(t)}])^{\mathsf{n}_\mu/2} = \|\mathfrak{Y}[\Phi^{(t)}]\|_{\mathsf{n}_\mu}^{\mathsf{n}_\mu} \ge 0, \tag{1752}$$

$$|Z[\mathbf{F}]| \le \prod_{t \bmod \mathsf{n}_\mu} Z[R_{\mu,t}\mathbf{F}]^{1/\mathsf{n}_\mu}. \tag{1753}$$

■

Proof. The trace formula expresses $Z[\mathbf{F}]$ as the trace of the product of the n_μ slab operators, in reverse cyclic order. In $R_{\mu,t}\mathbf{F}$, an even displacement from slab t is a translation of its slab function; an odd displacement is a translation of the reflected, conjugated function $(\Phi^{(t)})^{\Theta}$. Indeed, in the varying coordinate the map between two slabs is respectively a translation by $(t'-t)\mathsf{l}$ or reflection followed by translation, with reflection formula $x_\mu \mapsto 2o_\mu + (t+t')\mathsf{l} - x_\mu$. The other coordinates are unchanged. Thus the operators in this trace alternate between $Y = \mathfrak{Y}[\Phi^{(t)}]$ and Y^*, by (1743). Since n_μ is even, their trace is $\operatorname{Tr}(Y^*Y)^{\mathsf{n}_\mu/2}$, independently of the initial parity. This proves (1752). Apply lemma 86.1 to the original slab product and use (1752) for each factor to obtain (1753). □

Composition of copies in (1748) implies that $R_{3,t_3}R_{2,t_2}R_{1,t_1}\mathbf{F}$ is the family of all copies of $F_{(t_1,t_2,t_3)}$. In particular its product is $\operatorname{Fold} F_{(t_1,t_2,t_3)}$. For the family of all copies of one F, applying $R_{1,t}$ does not change the family. Equation (1752) therefore shows that $\mathcal{Z}_{a,T,\beta}(\operatorname{Fold} F) \geq 0$. We may consequently define

$$\|F\|_{\mathrm{ch}} = \langle \operatorname{Fold} F \rangle_{a,T,\beta}^{1/\mathsf{N}}. \tag{1754}$$

Proposition 86.3 (Chessboard estimate)**.** For every board, every $\beta \geq 0$, and every family of bounded Borel functions F_κ of their respective cells,

$$\left| \left\langle \prod_{\kappa \in \mathsf{Z}} F_\kappa \right\rangle_{a,T,\beta} \right| \leq \prod_{\kappa \in \mathsf{Z}} \|F_\kappa\|_{\mathrm{ch}}. \tag{1755}$$

■

Proof. Apply (1753) in directions 1, 2, and 3, in that order. The first reflected integrals are nonnegative, so the inequality can be applied successively under their positive roots. The composition of copies just proved gives

$$|Z[\mathbf{F}]| \leq \prod_{t_1,t_2,t_3} \mathcal{Z}_{a,T,\beta}(\operatorname{Fold} F_{(t_1,t_2,t_3)})^{1/\mathsf{N}}.$$

Divide by $Z_{a,T,\beta} > 0$, distributing its Nth root among the N factors. □

Proposition 86.4. On functions of one cell, $\|\cdot\|_{\mathrm{ch}}$ is a seminorm, is unchanged by conjugation or taking a mirror copy, and satisfies

$$\|1\|_{\mathrm{ch}} = 1, \qquad |\langle F \rangle_{a,T,\beta}| \leq \|F\|_{\mathrm{ch}} \leq \|F\|_\infty.$$

For a fixed model function it is independent of the chosen cell and of the offset of the board. ■

Proof. The homogeneity, conjugation, and supremum assertions follow from (1748)–(1749) and the nonnegativity of the folded integral. Set the other cell functions equal to one in proposition 86.3 to bound $|\langle F \rangle|$. For the triangle inequality, expand

$$\operatorname{Fold}(F+H) = \sum_{J \subset \mathsf{Z}} \prod_{\kappa \in J} F^{[\kappa]} \prod_{\kappa \notin J} H^{[\kappa]}.$$

The chessboard estimate and invariance under copies bound the modulus of the mean of the Jth term by $\|F\|_{\mathrm{ch}}^{|J|}\|H\|_{\mathrm{ch}}^{\mathsf{N}-|J|}$. Summing and taking the Nth root gives the triangle inequality. Finally, changing (o,κ) to (o',κ') for the same model function translates its entire fold by $o' - o + \mathsf{l}(\tilde{\kappa}' - \tilde{\kappa}) \in a\mathbb{Z}^3$. The Wilson integral is translation invariant. □

86.3. **Sources and change of volume.** Let w vanish outside the interior of $\square_\kappa$. Define its copies by (1747), acting now on functions of position, and put

$$w^{\mathrm{ch}} = \sum_{\kappa'} w^{[\kappa']} = w^+ + (\bar{w})^-, \tag{1756}$$

where w^+ and w^- are the sums of the un-conjugated transports over respectively even and odd cells. The copies have disjoint interiors and vanish on every cell face. In $\mathcal{O}_a(w)$, only inner plaquettes of

$\square_\kappa$ therefore occur, so $e^{\mathcal{O}_a(w)}$ is a function of that cell. Covariance of the plaquette action and its real values give

$$(e^{\mathcal{O}_a(w)})^{[\kappa']} = e^{\mathcal{O}_a(w^{[\kappa']})}, \qquad \operatorname{Fold} e^{\mathcal{O}_a(w)} = e^{\mathcal{O}_a(w^{\mathrm{ch}})}.$$

In particular,

$$Z_{a,T,\beta}[w^{\mathrm{ch}}] \geq 0, \qquad \|e^{\mathcal{O}_a(w)}\|_{\mathrm{ch}}^{\mathsf{N}} = \frac{Z_{a,T,\beta}[w^{\mathrm{ch}}]}{Z_{a,T,\beta}}. \tag{1757}$$

Here $Z_{a,T,\beta}[w] = \mathcal{Z}_{a,T,\beta}(e^{\mathcal{O}_a(w)})$. If $w = \sum_q z_q v_q$ for real sources v_q of this cell, then $w^{\mathrm{ch}} = \sum_q (z_q v_q^+ + \bar{z}_q v_q^-)$. The dependence on z and $\bar{z}$ is kept explicit.

Proposition 86.5 (Comparison of volumes)**.** Fix $a > 0$, an odd $\lambda \geq 3$, and $\beta \geq 0$. Let $\mathsf{n}_\mu, \mathsf{n}'_\mu$ be even integers with $2 \leq \mathsf{n}'_\mu \leq \mathsf{n}_\mu$ in each direction. For the same bounded Borel model function F_0, placed in any cell and at any offset on the respective boards,

$$0 \leq \mathcal{Z}_{\underline{\mathsf{n}}}(\operatorname{Fold} F_0) \leq \mathcal{Z}_{\underline{\mathsf{n}}'}(\operatorname{Fold} F_0)^{\prod_\mu \mathsf{n}_\mu/\mathsf{n}'_\mu}. \tag{1758}$$

The subscript on $\mathcal{Z}$ specifies the torus with sides $a\lambda\mathsf{n}_\mu$; a, λ, and β are unchanged. ■

Proof. By proposition 86.4, take offset and cell index zero. First change only n_μ. The cross-section, the slab space of λ layers, and the operators K and $\mathfrak{Y}$ are the same on the two tori, by proposition 85.2. The product of the folded cell functions in slab zero is also the same function Φ of those slab variables. To see the last point, index its cells by the two transverse coordinates $\varkappa$. Their reflected model functions have signs $\epsilon_\mu = 1$, $\epsilon_\nu = (-1)^{\varkappa_\nu}$ for $\nu \neq \mu$. A model coordinate n_μ is placed in slab layer $n_\mu + h_\lambda + 1$. These signs and coordinate placements use only the transverse cell counts, λ, and F_0, and hence do not involve the varying length.

Consequently (1752), applied to the fold, gives the two integrals as $\operatorname{Tr} B^k$ and $\operatorname{Tr} B^{k'}$ for the same positive trace-class operator $B = \mathfrak{Y}[\Phi]^*\mathfrak{Y}[\Phi]$, where $k = \mathsf{n}_\mu/2 \geq k' = \mathsf{n}'_\mu/2 \geq 1$. If $b_i \geq 0$ are the eigenvalues of B and $z = \sum_i b_i^{k'} > 0$, then $\rho_i = b_i z^{-1/k'}$ satisfies $\sum_i \rho_i^{k'} = 1$. In particular $0 \leq \rho_i \leq 1$, whence $\sum_i \rho_i^k \leq 1$. Thus

$$\operatorname{Tr} B^k \leq (\operatorname{Tr} B^{k'})^{k/k'}.$$

If $z = 0$, both sides vanish. This proves the comparison in one direction. Change the three directions successively; their exponents multiply to the exponent in (1758). □

For cubic tori with spacing $a_K = L^{-K}$, write

$$p_\ell = \ell^{-3} \log Z_{K,\ell,\beta}.$$

For a board of cell side $\mathsf{l} = a_i$, let $\|F_0\|_{\mathrm{ch};\ell}$ be the seminorm of the model function F_0 placed in any cell at any allowed offset.

Corollary 86.6 (Two tori)**.** Let $\ell = 2L^m$ and $\ell' = 2L^{m'}$, where $m \geq m'$ are integers, let $K + m' \geq 1$, and let $\beta \geq 0$. Then

$$p_\ell \leq p_{\ell'} \leq 0. \tag{1759}$$

If moreover $-m' \leq i \leq K - 1$ and $\mathsf{l} = a_i$, then every bounded Borel model function with $\lambda = L^{K-i}$ satisfies

$$\|F_0\|_{\mathrm{ch};\ell} \leq \|F_0\|_{\mathrm{ch};\ell'} e^{\mathsf{l}^3(p_{\ell'} - p_\ell)}. \tag{1760}$$

Let $\tilde{w}$ be a bounded complex function on $\mathbb{R}^3$ vanishing outside the interior of one cell of side l of the board with offset $o \in a_K\mathbb{Z}^3$. Periodize it on the two tori to obtain $w_\ell, w_{\ell'}$. Their mirror continuations lift to one function on $\mathbb{R}^3$, and

$$0 \leq Z_{K,\ell,\beta}[w_\ell^{\mathrm{ch}}] \leq Z_{K,\ell',\beta}[w_{\ell'}^{\mathrm{ch}}]^{(\ell/\ell')^3}. \tag{1761}$$

■□

Proof. For an allowed i, $\lambda = L^{K-i} \geq L \geq 13$ is odd and the cell counts $\ell/\mathfrak{l}$, $\ell'/\mathfrak{l}$ are even integers at least two. Thus proposition 86.5 applies. For the pressure assertion choose $i = -m'$, which is allowed because $K + m' \geq 1$. Apply the proposition to the constant model function one and use $0 < Z_{K,\ell,\beta} \leq 1$. For general F_0, substitute $\mathcal{Z}(\operatorname{Fold} F_0) = Z_{K,\ell,\beta} \|F_0\|_{\mathrm{ch};\ell}^{(\ell/\mathfrak{l})^3}$ in (1758) and take roots to obtain (1760).

For the source assertion, $2\mathfrak{l} \leq \ell'$, so the cell interior embeds in both tori, and both periodizations give the same model function for $e^{\mathcal{O}_a(w)}$. Reflecting the lifted source through all cells of $\mathbb{R}^3$ gives a single function $\tilde{w}^{\mathrm{ch}}$. It has period $2\mathfrak{l}$ in each coordinate: shifting a cell index by an even integer preserves its reflection sign. Thus it descends to both tori, where it equals w_ℓ^{ch} and $w_{\ell'}^{\mathrm{ch}}$. Equations (1757) and (1758) give (1761). □

87. Analytic bounds for cumulants

The local vacuum energies may diverge as the cutoff is removed. Their mixed derivatives at separated supports will be estimated directly. The remaining part of a cumulant is a derivative of the logarithm of a bounded holomorphic function. The following estimates keep these two contributions separate and require a logarithm only near zero.

For $c_i \geq 0$ set $|z|_c = \sum_i c_i |z_i|$ and $D_c = \{z \in \mathbb{C}^n : |z|_c < 1\}$. We use the polydisc Cauchy bound

$$|\partial_1 \cdots \partial_n F(0)| \leq B \prod_i r_i^{-1} \quad \text{if } |F| \leq B \text{ on } \{|z_i| < r_i\}. \tag{1762}$$

Apply the one-variable Cauchy formula successively on circles of radii smaller than r_i and then increase the radii to r_i. This proves the displayed bound, including for unequal radii.

Lemma 87.1. Let $\Gamma, \mathcal{E}$ be holomorphic on D_c, with $\Gamma(0) = 1$ and $\mathcal{E}(0) = 0$. Suppose that $h = e^{-\mathcal{E}}\Gamma$ satisfies $|h| \leq Q$ there. Then $Q \geq 1$ and

$$|h(z) - 1| \leq (Q+1)|z|_c. \tag{1763}$$

In particular h has a holomorphic logarithm on $|z|_c < (Q+1)^{-1}$, and

$$|\partial_1 \cdots \partial_n (\log \Gamma - \mathcal{E})(0)| \leq (\log 2)[2(Q+1)n]^n \prod_i c_i. \tag{1764}$$

The logarithm of Γ in this formula is its germ with value zero at the origin. If $\Gamma(z) = \mathbb{E} e^{\sum_i z_i X_i}$ for bounded random variables, the mixed derivative of $\log \Gamma$ at zero is their joint cumulant. ■

Proof. First suppose that all c_i are positive. For $z \neq 0$ put $s = |z|_c$. The function $\zeta \mapsto h(\zeta z/s) - 1$ vanishes at zero and is bounded by $Q + 1$ on the unit disc. Schwarz's lemma at $\zeta = s$ proves (1763). On $|z|_c \leq [2(Q+1)]^{-1}$ it gives $|h - 1| \leq 1/2$, whence the power series for $\log(1 + (h-1))$ gives $|\log h| \leq \log 2$. The identity $\Gamma = e^{\mathcal{E}} h$ and the values at zero imply $\log \Gamma = \mathcal{E} + \log h$ in a neighbourhood of zero. The polydisc with radii $r_i = [2(Q+1)nc_i]^{-1}$ lies in this smaller domain. Applying (1762) proves (1764).

For weights that may vanish, replace c_i by $c_i^\epsilon = \max(c_i, \epsilon)$. The corresponding domains are contained in D_c, the bound Q is unchanged, and the derivatives at zero do not depend on ϵ. Letting $\epsilon \downarrow 0$ proves both assertions. In particular the mixed derivative vanishes if any $c_i = 0$. The assertion about cumulants follows from their definition. □

For estimates in the number of insertions, it is useful to control the generating function separately on every coordinate subspace.

Lemma 87.2 (Bounds by the number of active variables)**.** Let h be holomorphic on D_c, with $h(0) = 1$. Suppose that for every nonempty $B \subset \{1, \ldots, n\}$,

$$|h(z)| \leq Q^{|B|} \quad \text{when } z_i = 0 \ (i \notin B), \qquad Q \geq 1.$$

Then

$$|\partial_1 \cdots \partial_n \log h(0)| \le (n-1)!(2^n-1)(eQ)^n \prod_i c_i. \tag{1765}$$

If $h = e^{-\mathcal{E}}\Gamma$ with the normalizations of lemma 87.1, the derivative on the left is that of $\log \Gamma - \mathcal{E}$.

For bounded random variables $|X_i| \le x_i$, one also has

$$|\langle X_1; \dots; X_n \rangle_\nu| \le (n-1)!(2^n-1) \prod_i x_i. \tag{1766}$$

If two probability laws ν, ν' on the same space satisfy

$$|\nu(Y) - \nu'(Y)| \le \epsilon \|Y\|_\infty, \qquad Y = \prod_{i \in J} X_i, \quad \emptyset \ne J \subset \{1, \dots, n\},$$

then

$$|\langle X_1; \dots; X_n \rangle_\nu - \langle X_1; \dots; X_n \rangle_{\nu'}| \le \epsilon \left(\sum_{\pi \in \mathcal{P}_n} |\pi|! \right) \prod_i x_i. \tag{1767}$$

■

Proof. For $b = |B|$ and positive c_i, apply (1762) on the B-coordinate polydisc with radii $(bc_i)^{-1}$. This gives

$$|\partial_B h(0)| \le Q^b b^b \prod_{i \in B} c_i \le (eQ)^b b! \prod_{i \in B} c_i.$$

The last inequality follows from the bth term in the exponential series for e^b. The partition formula for the logarithm, the same coefficient calculation as in (1739), reduces the desired bound to

$$\sum_{\pi \in \mathcal{P}_n} (|\pi| - 1)! \prod_{B \in \pi} |B|! = (n-1)!(2^n - 1). \tag{1768}$$

To verify this identity, first fix $k = |\pi|$. Ordering the k blocks and then their elements gives

$$k! \sum_{|\pi| = k} \prod_{B \in \pi} |B|! = n! \binom{n-1}{k-1}.$$

Multiply this identity by $(k-1)!/k! = 1/k$ and sum over k; the summands on the right become $(n-1)!\binom{n}{k}$. This proves (1768) and hence (1765). Vanishing weights follow by the approximation used in lemma 87.1.

Every moment indexed by B has modulus at most $\prod_{i \in B} x_i$. The partition formula and (1768), using $|B|! \ge 1$, give (1766). To compare two laws, write the difference of the products of moments for a partition with k blocks as a telescoping sum of k terms. Each term is bounded by $\epsilon \prod_i x_i$. Multiplying by the coefficient $(k-1)!$ in the partition formula and summing proves (1767). □

88. Separated derivatives of the source energies

The sum of the source energies need not be bounded uniformly in the cutoff. Its mixed derivatives behave differently when two source supports are separated. A contributing domain must meet both supports, so at fine scales its size is large. The exponential decay in the domain size then makes the sum over scales convergent. Combining this estimate with the normalized partition function gives bounds for the truncated correlations on a fixed physical torus.

Fix the constants and the source radius of theorem AD, one reference coupling $0 < g \le \gamma_\varrho$, and a first-passage datum (K, β). The side ℓ satisfies (1489). We use the assignment in the proof of

lemma 82.5, in particular $\kappa_e = \kappa > \kappa_{\mathcal{D}} + 2$, where $\kappa_{\mathcal{D}} = \log 9216$. Put $e_0(X; w) = 0$. For a real Lipschitz function on T_ℓ set

$$\|f\|_\sharp = \max\{\|f\|_\infty, 40M \operatorname{Lip} f\}, \qquad \mathbb{D}_{\rho_*}(f_1, \dots, f_n) = \left\{ z : \sum_{q=1}^{n} |z_q| \|f_q\|_\sharp < \rho_* \right\}, \qquad \rho_* = 2\varrho. \tag{1769}$$

Here Lip uses Euclidean distance. Since $|x-y| \le \sqrt{3}|x-y|_\infty$, the norm in (2) satisfies $\|f\|_{\flat;T_\ell} \le \|f\|_\sharp$. Thus every source $w(z) = \sum_q z_q f_q$ with $z \in \mathbb{D}_{\rho_*}$ belongs to the source ball. The source energies are holomorphic there, vanish at zero, depend only on the source in $X^{\sim 1}$, and satisfy

$$|e_j(X; w)| \le C_e e^{-\kappa_e d_j(X)}, \qquad C_e = C_{\mathrm{S}}. \tag{1770}$$

These are precisely the properties of theorem AD needed for the derivative estimate.

88.1. **The size and number of contributing domains.** At depth $s = K - j$, the cubes of π_j have physical side $W_s = ML^{-s}$. For $Y \subset T_\ell$ let

$$N_s(Y) = \#\{\Delta \in \pi_{K-s} : \Delta \cap Y \ne \varnothing\}, \qquad S(q) = \frac{2^{12}}{1 - e^{\kappa_{\mathcal{D}} - q}} \quad (q > \kappa_{\mathcal{D}}).$$

Lemma 88.1. If $X \in \mathcal{D}_{K-s}$ and $X^{\sim 1}$ meets both Y and Y', then

$$d_{K-s}(X) \ge \frac{L^s \operatorname{dist}_\infty(Y, Y')}{M} - 4. \tag{1771}$$

For $q > \kappa_{\mathcal{D}}$ and $D \in \mathbb{R}$,

$$\sum_{\substack{X \in \mathcal{D}_{K-s}:\ X^{\sim 1} \cap Y \ne \varnothing, \\ d_{K-s}(X) \ge D}} e^{-q d_{K-s}(X)} \le 27 N_s(Y) S(q) e^{-(q - \kappa_{\mathcal{D}})[D]_+}. \tag{1772}$$

Moreover,

$$N_s(Y) \le (\ell L^s/M)^3, \qquad N_s(Y) \le (2RL^s/M + 2)^3 \quad \text{if } Y \subset \{x : |x - x_0|_\infty \le R\}. \tag{1773}$$

If r is a lattice isometry induced by a signed coordinate permutation and a translation, then $N_s(rY) \le 27 N_s(Y)$; if the second hypothesis of (1773) holds, its second bound also holds for rY. ∎

Proof. A connected set of $d_j(X)$ cube edges meeting all cubes of X has diameter at most $d_j(X)W_s$. Every point of X is within W_s of a vertex of this set. Consequently $\operatorname{diam}_\infty X \le (d_j(X) + 2)W_s$. Every point of $X^{\sim 1}$ is within W_s of X: its cube touches a cube of X, and a point of their intersection has this property. Two points in $X^{\sim 1}$ are therefore at distance at most $(d_j(X) + 4)W_s$, which proves (1771).

Each cube meeting Y touches at most 27 cubes. If $X^{\sim 1}$ meets Y, the domain X contains one of these at most $27 N_s(Y)$ cubes. For a fixed cube $\square$, the entropy bound of Proposition 45.1 gives

$$\#\{X \supset \square : d_j(X) = n\} \le 2048\, 9216^n.$$

Summing its geometric tail, with $n \ge \lceil [D]_+ \rceil$, gives

$$\sum_{\substack{X \supset \square \\ d_j(X) \ge D}} e^{-q d_j(X)} \le S(q) e^{-(q - \kappa_{\mathcal{D}})[D]_+}.$$

Summing over the anchor cubes proves (1772).

There are $(\ell/W_s)^3$ cubes in the torus. To prove the local count, lift the sup-ball into a box with side $2R$. A cube meeting the projection of a box with side lengths r_1, r_2, r_3 has a lift whose lower corner in each coordinate lies in an interval of length $r_\mu + W_s$. This interval contains at most $r_\mu/W_s + 2$ points of the cube grid. Choose one such lift per cube; the chosen corner determines

the cube. Hence the count is at most $\prod_{\mu}(r_{\mu}/W_s+2)$, giving (1773). A signed permutation and translation sends a cube to the projection of a box with all side lengths W_s, so its image meets at most 27 partition cubes. Apply this to the cubes meeting Y. The image of a sup-ball is a sup-ball of the same radius, which proves the last assertion. □

We will use two elementary scale sums. For $\delta>0$ and $c\ge 0$ put

$$C'_{\Sigma}(\delta,c)=\frac{27}{26}(c+1)^3+\left(\frac{6}{e\delta}\right)^3\frac{e^{\delta c/2}}{1-e^{-\delta}},\qquad C''_{\Sigma}(\delta)=1+\frac{1}{1-e^{-2\delta}}. \tag{1774}$$

For $D>0$, $b\ge 0$, and $L\ge 3$,

$$\sum_{s\ge 0}(DL^s)^3e^{-\delta[DL^s-c]_+}\le C'_{\Sigma}(\delta,c), \tag{1775}$$

$$\sum_{s\ge 0}(bL^s+2)^3e^{-\delta[DL^s-c]_+}\le 32\left(C''_{\Sigma}(\delta)+\log_L^+\frac{c+1}{D}\right)+4(b/D)^3C'_{\Sigma}(\delta,c). \tag{1776}$$

To verify these bounds, split at the first s_0 for which $t_s=DL^s>c+1$. The preceding indices number at most $1+\log_L^+((c+1)/D)$, and their cubic sum is at most $\frac{27}{26}(c+1)^3$. For $s=s_0+i$, $t_s>(c+1)L^i$ and $L^i\ge 1+2i$. Thus

$$\sum_{s\ge s_0}e^{-\delta(t_s-c)}\le\frac{1}{1-e^{-2\delta}}.$$

Using $t^3e^{-\delta t/2}\le(6/(e\delta))^3$ gives

$$\sum_{s\ge s_0}t_s^3e^{-\delta(t_s-c)}\le\left(\frac{6}{e\delta}\right)^3\frac{e^{\delta c/2}}{1-e^{-\delta}},$$

which proves (1775). Finally, $(bL^s+2)^3\le 4(b/D)^3t_s^3+32$ gives (1776) from the preceding estimates.

88.2. **Mixed source derivatives.** For each j choose any subset $\mathcal{F}_j\subset\mathcal{D}_j$. Let $\mathcal{E}_{\mathcal{F}}(z)$ be the sum over these subsets of terms of either of the forms

$$e_j(X;\textstyle\sum_q z_qf_q),\qquad \overline{e_j(X;\sum_q\bar z_qf_q)}.$$

The choice can vary with (j,X). Both forms are holomorphic and satisfy (1770); this follows for the second by conjugating the local power series. In the canonical family the two forms agree by the conjugation law, but the estimates below do not need that equality. Every such sum is finite, holomorphic on $\mathbb{D}_{\rho_*}$, and zero at the origin.

Proposition 88.2. Let $n\ge 2$ and suppose the real Lipschitz functions f_q are nonzero, with $d_{\infty}=\operatorname{dist}_{\infty}(\operatorname{supp}f_1,\operatorname{supp}f_2)>0$. Put $\delta_e=\kappa_e-\kappa_{\mathcal{D}}>0$. Then

$$|\partial_1\cdots\partial_n\mathcal{E}_{\mathcal{F}}(0)|\le\left(\frac{n}{\rho_*}\right)^n\prod_q\|f_q\|_{\sharp}\,27C_eS(\kappa_e)\sum_{s=0}^{K}N_s(\operatorname{supp}f_1)e^{-\delta_e[d_{\infty}L^s/M-4]_+}. \tag{1777}$$

The sum can be restricted to depths with $\mathcal{F}_{K-s}\neq\varnothing$. It satisfies

$$\sum_{s=0}^{K}N_s(\operatorname{supp}f_1)e^{-\delta_e[d_{\infty}L^s/M-4]_+}\le C'_{\Sigma}(\delta_e,4)(\ell/d_{\infty})^3. \tag{1778}$$

If $\operatorname{supp}f_1$ lies in a sup-ball of radius R, it also satisfies the volume-independent bound

$$32\left(C''_{\Sigma}(\delta_e)+\log_L^+\frac{5M}{d_{\infty}}\right)+4\left(\frac{2R}{d_{\infty}}\right)^3C'_{\Sigma}(\delta_e,4). \tag{1779}$$

The bounds remain valid for a finite sum $\sum_{\iota}c_{\iota}\mathcal{E}_{\mathcal{F}^{\iota}}$ formed at the tuples $(r_{\iota}f_1,\dots,r_{\iota}f_n)$, where $c_{\iota}\ge 0$, $\sum_{\iota}c_{\iota}\le 1$, and each r_{ι} is a lattice isometry, with N_s in (1777) replaced by $\max_{\iota}N_s(r_{\iota}\operatorname{supp}f_1)$. ■

Proof. If $X^{\sim 1}$ misses $\operatorname{supp} f_q$, locality makes the corresponding energy independent of z_q on $\mathbb{D}_{\rho_*}$. Its full mixed derivative therefore vanishes. In particular only domains seeing both first supports contribute. By lemma 88.1, these satisfy $d_j(X) \geq d_\infty L^{K-j}/M - 4$.

The polydisc with radii $r_q = \rho_*/(n\|f_q\|_\sharp)$ lies in $\mathbb{D}_{\rho_*}$. Iterated one-variable Cauchy inequalities, first at smaller radii and then in the limit, give for either term

$$|\partial_1 \cdots \partial_n \widetilde{e}_j(X;0)| \leq C_e e^{-\kappa_e d_j(X)} \left(\frac{n}{\rho_*}\right)^n \prod_q \|f_q\|_\sharp.$$

Apply (1772) to the contributing domains at each depth, then sum over the depths. This proves (1777), including its restricted-sum version. For (1778), use the first bound of (1773) and (1775) with $D = d_\infty/M$ and $c = 4$. For (1779), use the second bound of (1773) and (1776) with $b = 2R/M$. Lattice isometries preserve both source norms and the distance between the first two supports. Apply the established bound to each transformed tuple and take the weighted sum. The last assertions of lemma 88.1 give the stated bounds for the maximum, so (1778) and (1779) remain unchanged. □

88.3. **Truncated correlations on a fixed torus.** For the Wilson probability measure at (K, ℓ, β) define

$$\mathcal{O}_K(f) = L^K \sum_p f(x(p)) a_p(U), \qquad \Gamma(z) = \left\langle e^{\sum_q z_q \mathcal{O}_K(f_q)} \right\rangle = \frac{Z_{K,\ell,\beta}[\sum_q z_q f_q]}{Z_{K,\ell,\beta}[0]}.$$

The variables are bounded at finite cutoff, so Γ is entire and $\Gamma(0) = 1$. Its logarithm is defined near zero. The joint cumulant, or truncated correlation, is

$$S^T_{n;K,\ell,\beta}(f_1, \dots, f_n) = \partial_1 \cdots \partial_n \log \Gamma(0).$$

Proposition 88.3. Set

$$Q_\ell = \exp(C_{\mathrm{S}}(2 + \log g^{-2})\ell^3).$$

For $n \geq 2$ and real Lipschitz functions whose first two supports have Euclidean distance at least $d > 0$,

$$|S^T_{n;K,\ell,\beta}(f_1, \dots, f_n)| \leq \left(\frac{n}{2\varrho}\right)^n \left[\log 2\,(2Q_\ell + 2)^n + 27 C_{\mathrm{S}} S(\kappa_e) C'_\Sigma(\delta_e, 4) \left(\frac{\sqrt{3}\ell}{d}\right)^3\right] \prod_q \|f_q\|_\sharp. \quad (1780)$$

All constants on the right are independent of K and β. ■

Proof. If one function is zero, the mixed derivative defining the cumulant is zero. Assume all are nonzero, and put $\mathcal{E}(z) = \sum_{j,X} e_j(X; \sum_q z_q f_q)$ and $c_q = \|f_q\|_\sharp/(2\varrho)$. The factorization and stability bounds of theorem AD give

$$h(z) := e^{-\mathcal{E}(z)}\Gamma(z) = \frac{\widetilde{Z}(\sum_q z_q f_q)}{\widetilde{Z}(0)}, \qquad h(0) = 1, \qquad |h(z)| \leq Q_\ell \quad \left(\sum_q c_q |z_q| < 1\right).$$

Apply lemma 87.1 with these weights and the bound Q_ℓ. It gives

$$|\partial_1 \cdots \partial_n (\log \Gamma - \mathcal{E})(0)| \leq \log 2\,(2Q_\ell + 2)^n \left(\frac{n}{2\varrho}\right)^n \prod_q \|f_q\|_\sharp.$$

The source-energy contribution is bounded by (1777)–(1778); the first two supports have sup-distance at least $d/\sqrt{3}$. The triangle inequality now gives (1780). □

The estimate just proved is uniform in the ultraviolet cutoff on a fixed torus. Its dependence on ℓ is still exponential. To pass to growing tori or infinite-volume states, the normalized partition function must instead be estimated on cells. The mixed derivative bound (1779) is already independent of the volume and will also apply to the cell normalisers.

89. Cell normalization and volume-independent bounds

The bound on a fixed torus contains its volume because it uses the stability estimate for the entire normalized partition function. Reflection positivity permits a different normalization. We reflect a source supported in one cell through a board, remove the source-energy terms that see only one reflected copy, and estimate the remaining terms by the distance between copies. The resulting bound depends on the cell size. It can therefore be used on every larger torus.

We retain the constants and first-passage data of section 88. Write

$$E_+ = C_{\mathrm{S}}, \qquad C_{\mathrm{low}} = C_{\mathrm{S}}(1+\log g^{-2}), \qquad C'_\Sigma = C'_\Sigma(\delta_e, 4), \qquad C''_\Sigma = C''_\Sigma(\delta_e).$$

The following argument uses the source-energy properties (1770), together with

$$|\widetilde{Z}_{\ell'}(w)| \le e^{E_+ \ell'^3}, \qquad Z_{K,\ell,\beta} \ge e^{E_\ell - C_{\mathrm{low}}\ell^3}.$$

It thus applies with any constants satisfying these inequalities. In this section $\mathcal{O} = \mathcal{O}_{a_K}$. Here E_ℓ is the vacuum energy in theorem AD on the indicated torus. When comparing $\ell \ge \ell'$, put

$$\Delta_E = \frac{E_{\ell'}}{\ell'^3} - \frac{E_\ell}{\ell^3}, \qquad C_{\mathrm{gap}} \ge \Delta_E.$$

For the systems constructed in theorem AD we may take $C_{\mathrm{gap}} = C_{\mathrm{S}}$; when $\ell = \ell'$, we take $C_{\mathrm{gap}} = 0$. We do not identify the source energies on different tori.

89.1. **The reflected source and its normalizer.** Fix $\ell = 2L^m \ge \ell' = 2L^{m''}$ with $m'' > m'$, a cell side $h = a_i = L^{-i}$, where $-m'' \le i \le K-1$, and an offset $o \in a_K\mathbb{Z}^3$. Both sides satisfy (1489), so the source systems used above exist on them. The cells in $\mathbb{R}^3$ are

$$Q_k = o + hk + [-h/2, h/2]^3, \qquad k \in \mathbb{Z}^3.$$

Thus h/a_K is an odd integer at least three and ℓ'/h is an even integer. A closed carrier A lies in one cell Q_{k_0} at sup-distance at least $D > 0$ from its boundary. Let $\tilde{v}_1, \dots, \tilde{v}_q$ be real Lipschitz functions vanishing outside A. Write v_i and v'_i for their periodizations on T_ℓ and $T_{\ell'}$.

These periodizations preserve the sharp norm in (1769), where the same definition is used on $\mathbb{R}^3$. Indeed, the cell has side at most half either torus side, so distances between points in it agree with their torus distances. For a pair of points not in the same copy of the support, a shortest joining segment meets the boundaries of the copies containing its endpoints, where the function vanishes. The triangle inequality, applied to the two end segments, proves the Lipschitz bound in the other direction. The same argument permits zero extension from a cell whenever the support has positive distance from its boundary.

Reflect A and each $\tilde{v}_i$ into every cell. On the reference torus denote the resulting carriers by A_c and the real functions by $v_i^{[c]}$, where c runs over the $N' = (\ell'/h)^3$ cells. For $w = \sum_i z_i v_i$, its chessboard continuation has value $\sum_i z_i v_i^{[c]}$ in an even cell and $\sum_i \bar{z}_i v_i^{[c]}$ in an odd cell. Parity is the parity of the number of coordinate reflections from k_0 to c. The continuation is Lipschitz with the same supremum bound and Lipschitz bound as $\sum_i z_i \tilde{v}_i$: on each cell the reflection is an isometry, and a segment crossing cell boundaries can be split at those boundaries, where the function is zero. Consequently it belongs to the source ball when $\sum_i |z_i| \|v_i\|_\sharp < \rho_*$. Since the source vanishes on the cell boundary, every contributing plaquette has its center in the cell interior and all its corners in that cell. Thus $e^{\mathcal{O}(w)}$ is a function of the inner cell bonds, as required for the chessboard estimate.

For a domain X on $T_{\ell'}$, set

$$\mathcal{C}(X) = \{c : X^{\sim 1} \cap A_c \ne \varnothing\}.$$

If $\mathcal{C}(X) = \{c\}$, define the holomorphic function

$$\tilde{e}_j(X; z) = \begin{cases} e_j(X; \sum_i z_i v_i^{[c]}), & c \text{ even}, \\ e_j(X; \sum_i \bar{z}_i v_i^{[c]}), & c \text{ odd}. \end{cases}$$

Our normalizer is

$$\mathcal{E}^{\circ}(z) = \frac{1}{N'} \sum_{j=0}^{K} \sum_{X:\,|\mathcal{C}(X)|=1} \tilde{e}_j(X; z). \tag{1781}$$

It is holomorphic on $\mathbb{D}_{\rho_*}(v_1, \dots, v_q)$ and vanishes at zero. It is determined by the reference system, the board, the carrier, and the function $\sum_i z_i \tilde{v}_i$. In particular it is unchanged if that function is represented by a different tuple on the same carrier. Neither the normalizer nor its domain depends on the larger side ℓ.

For a subset $B \subset \{1, \dots, q\}$ put

$$S_c(B) = \bigcup_{i \in B} \operatorname{supp} v_i^{[c]}, \qquad \bar{N}_s(B) = \max_c N_s(S_c(B)).$$

Two distinct carriers $A_c, A_{c'}$ have sup-distance at least $2D$, including across the periodic boundary. To see this, in a coordinate in which their cell indices differ, their lifts lie in intervals with centers differing by a nonzero multiple of h and half-length $h/2 - D$. Every possible period shift retains a nonzero cell-index difference, and the distance is at least $h - 2(h/2 - D) = 2D$. Distances between two reflected supports in a single cell equal the original distances in $\mathbb{R}^3$.

The cube counts can also be read in $\mathbb{R}^3$. A cube of side $W_s = ML^{-s}$ meeting one cell projects to a cube meeting its image, and every such torus cube has a lift meeting the cell. Two different meeting cubes cannot have the same image: their centers differ in each coordinate by at most $h + W_s$, whereas equal images would make these differences multiples of ℓ'. Since $h \le \ell'/2$ and $W_s \le M \le \ell'/(2L)$, they must all vanish. Thus $\bar{N}_s(B)$ is the maximum of the corresponding Euclidean cube counts over all reflected copies: shifting a copy by a period of $T_{\ell'}$ preserves its reflection parity and the cube grid, since ℓ'/h is even and ℓ'/W_s is an integer. This maximum does not depend on ℓ' or ℓ.

Define

$$\Phi_B = C_e S(\kappa_e) \sum_{s=0}^{K} \min\left\{ \left(\frac{hL^s}{M}\right)^3, 27 \bar{N}_s(B) e^{-\delta_e [2DL^s/M - 4]_+} \right\}. \tag{1782}$$

The estimates below will be expressed using

$$\mathsf{U}(R, d) = 27 C_e S(\kappa_e) \left[32 \left(C''_\Sigma + \log_L^+ \frac{5\sqrt{3}M}{d} \right) + 4 C'_\Sigma \left(\frac{2\sqrt{3}R}{d} \right)^3 \right] \tag{1783}$$

and

$$\Phi_1(R, D; h) = C_e S(\kappa_e) \left\{ \frac{27}{26} + 27 \left[32 \left(C''_\Sigma + \log_L^+ \frac{5 \min\{h, M\}}{2D} \right) + 4 C'_\Sigma \left(\frac{R}{D} \right)^3 \right] \right\}. \tag{1784}$$

Both are nonnegative. The first increases with R and decreases with d; the second increases with R, h, decreases with D, and is at most $\Phi_1(R, D; M)$.

Proposition 89.1 (The cell bound)**.** For $z \in \mathbb{D}_{\rho_*}(v_1, \dots, v_q)$ with $z_i = 0$ outside B,

$$\left\| e^{\mathcal{O}(w)} \right\|_{\mathrm{ch};\ell} \le \exp\left(\operatorname{Re} \mathcal{E}^{\circ}(z) + (E_+ + C_{\mathrm{low}} + \Delta_E) h^3 + \Phi_B \right), \qquad E_+ + C_{\mathrm{low}} + \Delta_E \ge 0. \tag{1785}$$

If each $\operatorname{supp} \tilde{v}_i$ is contained in a sup-ball of radius R, then $\Phi_B \le |B| \Phi_1(R, D; h)$. If one such ball contains the whole carrier, the factor $|B|$ may be omitted. Under these radius hypotheses, suppose also that $q \ge 2$ and the first two supports have Euclidean distance at least $d > 0$. Then

$$\begin{aligned} |\partial_1 \cdots \partial_q \mathcal{E}^{\circ}(0)| &\le \left(\frac{q}{\rho_*}\right)^q \prod_i \|v_i\|_\sharp \, 27 C_e S(\kappa_e) \sum_{s=0}^{K} \bar{N}_s(\{1\}) e^{-\delta_e [d_\infty L^s/M - 4]_+} \\ &\le \left(\frac{q}{\rho_*}\right)^q \mathsf{U}(R, d) \prod_i \|v_i\|_\sharp, \end{aligned} \tag{1786}$$

where d_∞ is the sup-distance between the first two supports. ∎

Proof. Let $N = (\ell/h)^3$. The chessboard identity and the comparison of tori in corollary 86.6 give

$$\left\|e^{\mathcal{O}(w)}\right\|_{\mathrm{ch};\ell}^{N} = \frac{Z_{K,\ell,\beta}[w^{\mathrm{ch}}]}{Z_{K,\ell,\beta}} \le \frac{Z_{K,\ell',\beta}[(w')^{\mathrm{ch}}]^{(\ell/\ell')^3}}{Z_{K,\ell,\beta}}.$$

The numerator before taking the power is nonnegative. The reference factorization and upper bound, followed by the floor on T_ℓ, therefore yield

$$\left\|e^{\mathcal{O}(w)}\right\|_{\mathrm{ch};\ell} \le \exp\left\{\frac{1}{N'}\operatorname{Re}\sum_{j,X} e_j(X;(w')^{\mathrm{ch}}) + (E_+ + C_{\mathrm{low}} + \Delta_E)h^3\right\}.$$

This argument does not require the sourced partition function to be strictly positive. At zero source the comparison of pressures gives $p_\ell \le p_{\ell'}$; the lower and upper vacuum bounds imply $0 \le p_{\ell'} - p_\ell \le E_+ + C_{\mathrm{low}} + \Delta_E$.

A domain seeing no carrier contributes zero by locality. A domain seeing only A_c has the same real part as $\tilde{e}_j(X;z)$, also in an odd cell. It remains to bound, at depth $s = K - j$,

$$T_j = \frac{1}{N'}\sum_{X:\,|\mathcal{C}(X)|\ge 2} |e_j(X;(w')^{\mathrm{ch}})|.$$

Every domain contains a cube of π_j. Sum its bound first over domains containing that cube, using the rooted count in lemma 88.1, and then over all cubes. This bounds T_j by $C_e S(\kappa_e)(hL^s/M)^3$. For a second bound, a nonzero summand must meet some $S_c(B)$. It also meets a different carrier, whose distance from A_c is at least $2D$. Hence (1771) gives $d_j(X) \ge 2DL^s/M - 4$. The anchored tail estimate, first for each c and then summed over the N' cells, gives

$$T_j \le 27 C_e S(\kappa_e)\bar{N}_s(B) e^{-\delta_e[2DL^s/M-4]_+}.$$

The factor N' cancels the denominator defining T_j. Taking the smaller bound at each scale proves (1785).

We record explicitly the scale estimate needed here. For $u, D_0 > 0$, $b, c \ge 0$ and $\delta > 0$,

$$\begin{aligned}\sum_{s\ge 0} &\min\{(uL^s)^3, c(bL^s+2)^3 e^{-\delta[D_0L^s-4]_+}\} \\ &\le \frac{27}{26} + c\left[32\left(C''_\Sigma(\delta) + \log_L^+\frac{5\min\{u,1\}}{D_0}\right) + 4\left(\frac{b}{D_0}\right)^3 C'_\Sigma(\delta,4)\right].\end{aligned} \tag{1787}$$

For $uL^s < 1$, sum the first term as a geometric progression; its sum is at most $27/26$. For the other scales use $(bL^s+2)^3 \le 4(b/D_0)^3(D_0L^s)^3 + 32$. The cubic terms are controlled by (1775). Among these scales, those with $D_0L^s \le 5$ lie in the interval

$$\max\{0, -\log_L u\} \le s \le \log_L(5/D_0),$$

so there are at most $1 + \log_L^+(5\min\{u,1\}/D_0)$ of them. The remaining exponentials have sum at most $(1 - e^{-2\delta})^{-1}$, as in the proof of (1776). This proves (1787).

By the ball count in lemma 88.1, $\bar{N}_s(B) \le |B|(2RL^s/M + 2)^3$. For nonempty B, apply (1787) with $u = h/M$, $b = 2R/M$, $D_0 = 2D/M$, and $c = 27|B|$. Its right side is at most $|B|\Phi_1(R,D;h)/(C_eS(\kappa_e))$, with the assertion interpreted directly from (1782) if $C_e = 0$. If B is empty, $\Phi_B = 0$. If the whole carrier is in one ball, the same argument uses $c = 27$.

If $v_i = 0$ for some i, the normalizer is independent of z_i, so both sides of (1786) vanish. Otherwise, (1781) is an average of N' sub-sums of source energies at reflected tuples, with the holomorphic conjugate used in odd cells. Apply proposition 88.2 to these sub-sums. The sharp norms and within-cell distances are preserved, and the first support meets at most $(2RL^s/M + 2)^3$ cubes at depth s. Its sup-distance from the second support is at least $d/\sqrt{3}$. Equation (1776), with $b = 2R/M$ and $D_0 = d/(\sqrt{3}M)$, gives exactly (1786). □

89.2. **Several occupied cells.** Suppose that each nonzero member of a real Lipschitz tuple $\tilde f_1,\dots,\tilde f_n$ on $\mathbb{R}^3$ is supported in a sup-ball of radius R, in one cell of the board, at distance at least $D>0$ from its boundary. Let I_k be the indices in cell Q_k, omitting zero functions. For each occupied cell take the carrier $\bigcup_{i\in I_k}\operatorname{supp}\tilde f_i$ and construct (1781) on the same reference torus. Set

$$\mathcal{E}_{\underline f}(z)=\sum_{k\ \text{occupied}}\mathcal{E}^{\circ}_k((z_i)_{i\in I_k}). \tag{1788}$$

Let f_i denote periodization on T_ℓ and put $w_k=\sum_{i\in I_k}z_i f_i$.

Proposition 89.2. The function (1788) is holomorphic on $\mathbb{D}_{\rho_*}(f_1,\dots,f_n)$ and vanishes at zero. For $z_i=0$ outside $B\subset\{1,\dots,n\}$,

$$|e^{-\mathcal{E}_{\underline f}(z)}|\prod_{k\ \text{occupied}}\left\|e^{\mathcal{O}(w_k)}\right\|_{\mathrm{ch};\ell}\le\exp\bigl((E_++C_{\mathrm{low}}+C_{\mathrm{gap}})h^3+\Phi_1(R,D;h)\bigr)^{|B|}. \tag{1789}$$

If different occupied cells have different images on T_ℓ, the left side bounds

$$\left|e^{-\mathcal{E}_{\underline f}(z)}\left\langle e^{\sum_i z_i\mathcal{O}(f_i)}\right\rangle_{K,\ell,\beta}\right|. \tag{1790}$$

The same normalizer works for all larger tori satisfying the stated vacuum floor and gap bound. For $n\ge 2$, its full mixed derivative vanishes if two cells are occupied or one f_i is zero. If all functions occupy one cell and the first two supports have Euclidean distance at least $d>0$, then

$$|\partial_1\cdots\partial_n\mathcal{E}_{\underline f}(0)|\le(n/\rho_*)^n\mathsf{U}(R,d)\prod_i\|f_i\|_\sharp. \tag{1791}$$

■

Proof. Apply proposition 89.1 in each occupied cell, with the active indices $B\cap I_k$. A cell with no active index contributes one. Every other cell contributes at most

$$\exp\bigl((E_++C_{\mathrm{low}}+C_{\mathrm{gap}})h^3+|B\cap I_k|\Phi_1(R,D;h)\bigr).$$

There are at most $|B|$ such cells, and $E_++C_{\mathrm{low}}+C_{\mathrm{gap}}\ge 0$ by (1785); multiplying proves (1789). For distinct torus cells, the chessboard inequality proposition 86.3 bounds the expectation of the product by the product of its seminorms. Linearity of $\mathcal{O}$ identifies that product with the exponential in (1790). The normalizers use only the fixed reference system and the Euclidean carriers, so the claim about larger sides follows. If two cells are occupied, every summand of (1788) omits a variable. The same is true if some $f_i=0$. Otherwise (1791) is (1786). □

89.3. **Placement and the bound for separated supports.**

Lemma 89.3 (Placement by a lattice translation)**.** Let $h=a_i$ with $-m\le i\le K-1$, and suppose that the support of each $f_q\in\mathrm{Lip}(T_\ell)$ lies in a sup-ball of radius R. If $D>0$ and

$$n(2R+2D+a_K)\le h, \tag{1792}$$

one lattice translation places the whole tuple in cells with margin D. The translated functions have compactly supported lifts $\tilde g_q$ in representative cells with pairwise distinct torus images, one representative for each occupied cell. Periodization preserves their sharp norms. The translation preserves all truncated correlations. When two functions lie in one cell, the distance between the supports of their lifts equals the torus distance. ■

Proof. Put $a=a_K$ and $\lambda=h/a$. Choose lifts of the centers $x_1,\dots,x_n$. For a fixed coordinate consider the λ possible offsets $0,a,\dots,(\lambda-1)a$. An offset is bad for x_q if that coordinate has distance less than $R+D$ from one of the corresponding mirrors. Representatives of the bad offsets inject into points of $a\mathbb{Z}$ in an open interval of length $2(R+D)$. There are fewer than $2(R+D)/a+1$ of them. By (1792), fewer than λ offsets are bad for some q. Choose a good offset in each coordinate.

This gives a board offset $o' \in a\mathbb{Z}^3$ for which all the balls have margin D. Translation by $o - o'$ places them on the prescribed board.

Lift the translated balls to cells with indices in $\{0, \dots, \ell/h - 1\}^3$, and extend each function by zero outside its cell. The positive margin makes these extensions Lipschitz, with the norm preservation proved above. Different representative indices have different torus images. Since $h \le \ell/2$, distances within a representative cell agree with torus distances. Finally translation by a vector of $a_K\mathbb{Z}^3$ permutes bonds and plaquettes and preserves Haar measure and the Wilson action. It therefore preserves the generating functional and all its cumulants. □

For $n \ge 2$ and $R > 0$ set

$$h(n,R) = \min\{L^b : b \in \mathbb{N}, L^b \ge 8n(R+1)\}, \tag{1793}$$

$$Q(n,R) = \exp((E_+ + C_{\mathrm{low}})h(n,R)^3 + \Phi_1(R, R+1; M)), \tag{1794}$$

$$C_n^\infty(d;R) = (n/\rho_*)^n \left[\log 2\,(2Q(n,R)^n + 2)^n + \mathsf{U}(R,d)\right]. \tag{1795}$$

The minimum exists and $8n(R+1) \le h(n,R) < 8Ln(R+1)$.

Proposition 89.4. Let $n \ge 2$, $R, d > 0$, and $\ell \ge 16n(R+1)$. Suppose each real Lipschitz f_q on T_ℓ has support in a sup-ball of radius R, and the first two supports have Euclidean distance at least d. Then

$$|S^T_{n;K,\ell,\beta}(f_1, \dots, f_n)| \le C_n^\infty(d;R) \prod_q \|f_q\|_\sharp. \tag{1796}$$

The constant is independent of K, β, and ℓ. ■

Proof. Zero functions give zero cumulant. Otherwise use the board of side $h = h(n,R)$. Since $\ell/2 = L^m \ge 8n(R+1)$, the minimal power h is at most $\ell/2$, so it is an allowed cell side. Take $D = R + 1$. As $a_K < 1$, $n(2R + 2D + a_K) < 8n(R+1) \le h$. Apply lemma 89.3, and use the torus itself as the reference torus in proposition 89.2, so $C_{\mathrm{gap}} = 0$. For the translated tuple let Γ be its generating function and $\mathcal{E}$ its cell normalizer. Then

$$h_0 = e^{-\mathcal{E}}\Gamma, \qquad h_0(0) = 1, \qquad |h_0| \le Q(n,R)^n \quad \text{on } \mathbb{D}_{\rho_*}.$$

Here (1789) and $\Phi_1(R, D; h) \le \Phi_1(R, D; M)$ give the upper bound. Lemma 87.1 bounds the full derivative of $\log\Gamma - \mathcal{E}$ by the first term of (1795), times the product of sharp norms. If the tuple occupies two cells, the full derivative of $\mathcal{E}$ is zero. Otherwise its bound is (1791); the required separation is preserved by the placement. Adding the bounds and undoing the translation proves (1796). □

89.4. **Factorial growth in the number of insertions.** A finer use of the coordinate-slice bound avoids the rapid growth of $Q(n,R)$ in (1795). Define

$$p = \frac{864 C_e S(\kappa_e)}{\log L}, \tag{1797}$$

$$\varkappa_\star = 5 + p, \tag{1798}$$

$$C_Q^0 = \exp\left[E_+ + C_{\mathrm{low}} + C_e S(\kappa_e)\left\{\frac{27}{26} + 27[32(C''_\Sigma + \log_L 5) + 4C'_\Sigma]\right\}\right], \tag{1799}$$

$$A(d) = 1 + \mathsf{U}(1/10, d), \tag{1800}$$

$$B(R) = \max\left\{\frac{27(4R+1)^3\, 417M\, e^{5+p}}{\rho_*}(2e\, 3^p C_Q^0 + 1),\ 288e^4(48R+1)^3\right\}. \tag{1801}$$

In particular $p \ge 0$, $C_Q^0 \ge 1$, and $B(R) \ge 1$. For later use put

$$A_{\mathrm{PL}} = 1 + 27 C_e S(\kappa_e)\left[32\left(C''_\Sigma + \log_L(5\sqrt{3}M) + \frac{1}{\log L}\right) + 4C'_\Sigma(\sqrt{3}/5)^3\right]. \tag{1802}$$

Then

$$A(d) \le A_{\mathrm{PL}} \max\{1, d^{-4}\}. \tag{1803}$$

Indeed, for $t = \max\{1, d^{-1}\}$, $\log_L^+(5\sqrt{3}M/d) \le \log_L(5\sqrt{3}M) + t/\log L$ and $d^{-3} \le t^3 \le t^4$.

We first specialize the cell bound to a unit board. If a tuple is placed with radius and margin $\eta \le 1/4$, then

$$e^{E_+ + C_{\mathrm{low}} + \Phi_1(\eta,\eta;1)} = C_Q^0 (2\eta)^{-p} =: Q_\eta.$$

This follows from $M \ge 1$ and $\log_L(5/(2\eta)) = \log_L 5 + \log(1/(2\eta))/\log L$. Thus proposition 89.2, with the same torus as reference, gives $|e^{-\mathcal{E}}\Gamma| \le Q_\eta^{|B|}$ on every coordinate slice indexed by B. Applying the graded analytic estimate lemma 87.2 and the bound for $\partial_1 \cdots \partial_n \mathcal{E}(0)$ yields

$$|S_n^T(v_1, \dots, v_n)| \le \left[(n-1)!2^n \left(\frac{eQ_\eta}{\rho_*}\right)^n + \left(\frac{n}{\rho_*}\right)^n \mathsf{U}(\eta, d)\right] \prod_q \|v_q\|_\sharp, \tag{1804}$$

provided different occupied cells have different torus images and one pair of supports has distance at least d.

Proposition 89.5. Let $n \ge 2$ and $R, d > 0$. On any allowed torus, let $f_1, \dots, f_n$ be real Lipschitz functions, each supported in a sup-ball of radius R, with one pair of supports at Euclidean distance at least d. Then

$$|S_{n;K,\ell,\beta}^T(f_1, \dots, f_n)| \le A(d) B(R)^n (n!)^{\varkappa_\star} \prod_q \|f_q\|_\sharp. \tag{1805}$$

All constants are independent of K, β, and ℓ. ∎

Proof. By symmetry of cumulants, label a separated pair by $1, 2$. Assume all functions are nonzero. We treat coarse and fine lattices separately.

If $a_K > 1/(24n)$, the number of positive plaquette barycenters in a sup-ball of radius R is at most $3(2R/a_K + 1)^3$. For each of the three orientations, choose a lift of every barycenter in the coordinate box about a lift of the ball center; these choices inject into the corresponding shifted lattice. The interval count in each coordinate gives the assertion, even if the box is larger than the torus. Since $0 \le a_p \le 2$,

$$|\mathcal{O}(f)| \le 6a_K^{-1}(2R/a_K + 1)^3 \|f\|_\infty \le 144n^4(48R + 1)^3 \|f\|_\sharp.$$

The bounded-variable part of lemma 87.2 now gives

$$|S_n^T(f_1, \dots, f_n)| \le (n-1)!\,[288(48R+1)^3]^n n^{4n} \prod_q \|f_q\|_\sharp.$$

Using $n^{4n} \le e^{4n}(n!)^4$, this is bounded by the second term in (1801) to the power n, times $(n!)^5 \prod_q \|f_q\|_\sharp$.

Suppose now that $a_K \le 1/(24n)$, and put $\varepsilon = 1/(5n)$. The ratio $\ell/\varepsilon = 5n\ell$ is an integer. On the periodic grid $\varepsilon\mathbb{Z}^3/\ell\mathbb{Z}^3$, define

$$\chi_y(x) = \prod_{\mu=1}^3 \max\{0, 1 - |x_\mu - y_\mu|_\ell/\varepsilon\}.$$

In each coordinate the adjacent linear tents sum to one, so $\sum_y \chi_y = 1$. Their products have values in $[0,1]$, support in the sup-ball of radius ε about y, and Lipschitz constant at most $\sqrt{3}/\varepsilon$: telescoping the product bounds its change by $\varepsilon^{-1}\sum_\mu |x_\mu - x'_\mu|_\ell$. Consequently

$$\|\chi_y f\|_\sharp \le (1 + 40\sqrt{3}M/\varepsilon)\|f\|_\sharp. \tag{1806}$$

The support of a piece is contained in the support of f. If f is supported in a sup-ball of radius R, a nonzero piece has its node within $R + \varepsilon$ of the center. Choosing lifts and counting each coordinate shows that there are at most $(2R/\varepsilon + 3)^3$ such pieces.

Expand every $f_q = \sum_y \chi_y f_q$ and use multilinearity of the cumulant. For a nonzero tuple of pieces $v_1, \dots, v_n$, the first two supports remain separated by d. Moreover

$$n(4\varepsilon + a_K) \le \frac{4}{5} + \frac{1}{24} < 1.$$

Thus lemma 89.3 places the tuple on the unit board with radius and margin ε, with distinct representatives for distinct occupied cells. Apply (1804) to that tuple. Since $1/(6n) < \varepsilon \le 1/10$,

$$\begin{aligned} (2R/\varepsilon + 3)^3 &\le [3n(4R+1)]^3, \\ 1 + 40\sqrt{3}M/\varepsilon &\le 417Mn, \\ Q_\varepsilon \le C_Q^0\, 3^p n^p, &\qquad \mathsf{U}(\varepsilon, d) \le \mathsf{U}(1/10, d). \end{aligned}$$

The second inequality follows from $M \ge 1$ and $240\sqrt{3} < 416$; the bounds intentionally use the weaker $\varepsilon > 1/(6n)$. Put $c_R = 27(4R+1)^3\, 417M$. Summing the estimates for all tuples of pieces yields

$$|S_n^T(f_1, \dots, f_n)| \le c_R^n n^{4n} \left[n! \left(\frac{2e\, 3^p C_Q^0}{\rho_*} \right)^n n^{pn} + \left(\frac{n}{\rho_*} \right)^n \mathsf{U}(1/10, d) \right] \prod_q \|f_q\|_\sharp.$$

For $t \ge 0$, the elementary inequality $n! \ge (n/e)^n$ gives $n^{tn} \le e^{tn}(n!)^t$. Use it with $t = 4 + p$ in the first term and $t = 5$ in the second. Both terms are bounded by the first expression in (1801) to the power n, times $(n!)^{5+p} \prod_q \|f_q\|_\sharp$, with the respective remaining factors 1 and $\mathsf{U}(1/10, d)$. Their sum is (1805). □

90. Compactness on separated test functions

Uniform estimates at separated supports give one subsequence for all orders and all test tuples. The support restriction matters in this argument: approximating the entries must preserve their separation. We give the compactness statement on both a fixed torus and $\mathbb{R}^3$, since the latter will also be used after the volume limit.

Let $\mathbb{E} = T_\ell$ or $\mathbb{R}^3$. For an open set $O \subset \mathbb{E}$, let $\mathfrak{S}_n(O)$ be the tuples $(f_1, \dots, f_n)$ of real functions in $C_c^1(O)$ with pairwise disjoint supports. Zero entries are allowed. Put $\mathfrak{S}_0(O) = \{\varnothing\}$ and $\mathfrak{S}(O) = \bigcup_{n \ge 0} \mathfrak{S}_n(O)$. A tuple has type $(d; R)$ if distinct supports are at distance at least $d > 0$ and each support lies in a ball of radius $R > 0$; the centres of these balls may differ. Every separated tuple has such a type, since its supports are compact.

A map on $\mathfrak{S}_n(O)$ is called multilinear if its restriction to $\prod_i C_c^1(O_i)$ is multilinear whenever the open sets $O_i \subset O$ have pairwise disjoint closures. We use $\|f\|_\sharp = \max(\|f\|_\infty, 40M \operatorname{Lip} f)$ also on $\mathbb{R}^3$. With $\|f\|_{C^1} = \max(\|f\|_\infty, \|\nabla f\|_\infty)$, integration along a shortest line segment, lifted on the torus, gives $\|f\|_\sharp \le 40M \|f\|_{C^1}$.

Fix $n_0 \ge 1$ and finite constants $C_n(d; R) \ge 0$ for $n \ge n_0$ and $d, R > 0$. A family $\Phi = (\Phi_n)_{n \ge n_0}$ obeys these bounds if

$$|\Phi_n(f_1, \dots, f_n)| \le C_n(d; R) \prod_i \|f_i\|_\sharp \tag{1807}$$

on every tuple of type $(d; R)$. We impose no bound as d tends to zero.

Proposition 90.1 (Simultaneous subsequence extraction)**.** Let $O_j \subset \mathbb{E}$ be open sets such that every compact subset of $\mathbb{E}$ is contained in O_j for all sufficiently large j. Suppose that the real multilinear families $\Phi^{(j)}$ on $\mathfrak{S}(O_j)$ satisfy (1807) for $n \ge n_0$, with the same constants, once $j \ge j_0(n, R)$. Then there is a subsequence along which $\Phi_n^{(j)}(X)$ converges for every $n \ge n_0$ and every $X \in \mathfrak{S}_n(\mathbb{E})$. The subsequence can also be chosen so that any prescribed countable collection of bounded real sequences converges along it. The limiting family is multilinear and satisfies the same bounds. Symmetry is preserved. ■

Proof. A countable family with controlled supports. Let $\mathfrak{O}$ consist of finite unions of rational open boxes, using their projections on the torus. These sets have compact closures and form a countable basis under finite unions. For each $W \in \mathfrak{O}$ choose a countable C^1-dense subset $\mathfrak{f}(W) \subset C^1_c(W)$.

Here is the separability argument with the support fixed. For $f \in C^1_c(W)$ consider $g_f = (f, \partial_1 f, \partial_2 f, \partial_3 f)$ on $\overline{W}$. The C^1 distance of two functions is the uniform distance of these maps for the norm $|(v_0, v')|_* = \max(|v_0|, |v'|)$. For each integer $p \geq 1$ choose a finite $1/p$-net in $\overline{W}$. Classify the functions by a common modulus bound $|g_f(x) - g_f(y)|_* \leq 1/q$ for $|x - y| \leq 1/p$, and by rational approximations within $1/q$ to their values at the points of the net. There are countably many classes. Each has diameter at most $4/q$, by comparison through a nearest net point. Uniform continuity of g_f on $\overline{W}$ places every f in such a class for each q, with p sufficiently large. Choosing one member of each nonempty class therefore gives the required dense subset. For $W = \varnothing$ take $\mathfrak{f}(W) = \{0\}$.

Take all tuples of all lengths with entries in $\mathfrak{f}(W_i)$, where the W_i have pairwise disjoint closures. This is a countable family; enumerate it by $(X^k)_{k \geq 1}$. Given a tuple $X = (f_1, \dots, f_n)$ of type $(d; R)$, compactness of each support permits a choice of such sets with

$$\operatorname{supp} f_i \subset W_i \subset \{x : \operatorname{dist}(x, \operatorname{supp} f_i) < d/3\}. \tag{1808}$$

Their mutual distances are at least $d/3$, and each lies in a ball of radius $R + d/3$. The union of their closures is compact. For a zero entry choose $W_i = \varnothing$.

Uniform approximation of evaluations. If f_i, h_i have supports in W_i, multilinearity gives

$$\Phi_n(f_1, \dots, f_n) - \Phi_n(h_1, \dots, h_n) = \sum_{i=1}^{n} \Phi_n(h_1, \dots, h_{i-1}, f_i - h_i, f_{i+1}, \dots, f_n).$$

Every tuple on the right retains separation $d/3$. Choose $h_i \in \mathfrak{f}(W_i)$ with $\|f_i - h_i\|_{C^1} \leq \epsilon \leq 1$. For every family obeying (1807) at $(n, R + d/3)$, the preceding identity gives

$$|\Phi_n(X) - \Phi_n(h_1, \dots, h_n)| \leq C_n(d/3; R + d/3)\, n(40M)^n \epsilon (1 + \max_i \|f_i\|_{C^1})^{n-1}. \tag{1809}$$

The constant does not depend on the approximating family or its domain, provided the latter contains all W_i.

Extraction and extension. For each X^k of length at least n_0, its evaluation by $\Phi^{(j)}$ is defined and bounded for all sufficiently large j. Assign arbitrary values to the finitely many undefined entries. Apply the diagonal subsequence argument to these countably many bounded real sequences and to the additional sequences in the statement. Each selected evaluation then converges. For a fixed arbitrary X, use (1809) to approximate it by one X^k, with an error uniform in all sufficiently large j. The triangle inequality shows that the evaluations at X are Cauchy. This proves convergence on every separated tuple along the same subsequence.

Each multilinearity or symmetry identity involves only finitely many evaluations, all eventually defined, so it passes to the limit. The same is true of (1807) for every fixed tuple and type. □

Proposition 90.2. The set of symmetric multilinear families on $\mathfrak{S}(\mathbb{E})$ satisfying (1807) is compact and metrizable for pointwise convergence on separated tuples. ■

Proof. Use the determining family X^k in the preceding proof and let n_k be the length of X^k. A metric is

$$d(\Phi, \Psi) = \sum_{k : n_k \geq n_0} 2^{-k} \min\{1, |\Phi_{n_k}(X^k) - \Psi_{n_k}(X^k)|\}. \tag{1810}$$

The triangle inequality follows term by term. If this expression vanishes, (1809) makes every evaluation of the two families equal, so it is a metric. That same estimate shows that every evaluation is continuous for the metric: approximate its tuple by one X^k, uniformly over all the families. Conversely, the tail of the series is uniformly bounded by $\sum_{k > N} 2^{-k}$, and its first N terms are continuous for pointwise convergence. The two topologies thus agree. By proposition 90.1, every

sequence has a convergent subsequence whose limit still belongs to the set. Sequential compactness in this metric proves compactness. □

90.1. **The fixed-torus application.**

Proposition 90.3. Fix the source assignment, $0 < g \le \gamma_{\varrho}$, and a side ℓ satisfying (1489). Every sequence of first-passage data (K_j, β_j) with $K_j \to \infty$ has a subsequence along which

$$S_n(X) = \lim_j S^T_{n;K_j,\ell,\beta_j}(X)$$

exists simultaneously for all $n \ge 2$ and all $X \in \mathfrak{S}_n(T_\ell)$. The same subsequence may be chosen so that $L^{-K_j}\beta_j$ converges. The limit is real, symmetric and multilinear, and inherits (1780). Such sequences, and hence such limits, exist. ■

Proof. The restrictions of the lattice cumulants to separated C^1 tuples are real, symmetric and multilinear. The coefficient multiplying $\prod_i \|f_i\|_\sharp$ in (1780) is finite and independent of the cutoff and bare coupling. It supplies the constants in (1807), independent of R, on the constant domains $O_j = T_\ell$. Moreover proposition 74.1 bounds the physical labels $L^{-K_j}\beta_j$ in a fixed compact interval. Apply proposition 90.1 with this one additional bounded sequence. Every fixed-tuple inequality passes to the limit, giving the stated bound. Finally corollary 74.2 supplies the nonempty interval $I(g) = (g^{-2} + \bar B, X_g - \bar B]$ and shows that $(K, L^K \mathsf{b})$ is a first-passage datum for every $K \ge 1$ and every $\mathsf{b} \in I(g)$. This proves existence. □

For a family $S = (S_n)_{n\ge 2}$ define its centred moments by

$$\sigma_S(f_1, \dots, f_n) = \sum_{\substack{\pi \in \mathcal{P}_n \\ |B| \ge 2 \ (B\in\pi)}} \prod_{B\in\pi} S_{|B|}((f_i)_{i\in B}), \qquad \sigma_S(\varnothing) = 1. \tag{1811}$$

There is no term for $n = 1$, so the one-point function vanishes. The moment–cumulant identities show that the connected functions of σ_S are precisely S_n, $n \ge 2$. At finite cutoff this is the expectation of the product of $\mathcal{O}_a(f_i) - \langle \mathcal{O}_a(f_i)\rangle$. Since each partition sum is finite, convergence of all the cumulants on separated tuples also gives convergence of these centred moments.

91. Infinite-volume lattice states

We first take the volume limit at a fixed spacing. The configuration space is compact, but the finite periodic measures do not themselves satisfy the infinite-lattice Gibbs equations. We therefore check those equations on each fixed set of bonds before passing to a limit.

Let $a > 0$, let $\mathbb{L}_a = a\mathbb{Z}^3$, and let Bd be its positively oriented nearest-neighbour bonds. A configuration is an element of

$$\mathcal{V}(\mathbb{L}_a) = \mathrm{SU}(2)^{\mathsf{Bd}}, \qquad U(\bar b) = U(b)^{-1}.$$

We use the product topology. This is a compact metrizable space. For $B \subset \mathsf{Bd}$, write $\mathcal{F}(B)$ for the bounded Borel functions depending only on U_B, and $\mathcal{C}_{\mathrm{loc}}$ for the continuous functions depending on finitely many bonds. For a spatial set Q, $\mathsf{Bd}(Q)$ consists of bonds whose closed segments lie in Q. The notation $\mathcal{C}_{\mathrm{loc}}(Q)$ requires a finite bond support in $\mathsf{Bd}(Q)$.

Enumerating the bonds gives a product metric with summable coordinate weights. Replacing all but the first N coordinates by the identity approximates the identity map uniformly in this metric. Uniform continuity therefore shows that $\mathcal{C}_{\mathrm{loc}}$ is uniformly dense in $C(\mathcal{V}(\mathbb{L}_a))$. In particular, its integrals determine a Borel probability measure. We call such a measure a law. The laws form a compact metrizable space in the weak topology.

For a plaquette p, retain the notation $a_p(U) = 1 - \frac{1}{2}\operatorname{tr}\partial U(p) \in [0, 2]$. For bounded compactly supported f on $\mathbb{R}^3$, put

$$\mathcal{O}_a(f) = a^{-1} \sum_{p\in\mathsf{Pl}} f(x(p))a_p. \tag{1812}$$

The sum is finite, so $\mathcal{O}_a(f) \in \mathcal{C}_{\mathrm{loc}}$ even when f is merely bounded. For a law ν define

$$S_n^T[\nu](f_1, \dots, f_n) = \langle \mathcal{O}_a(f_1); \dots; \mathcal{O}_a(f_n) \rangle_\nu. \tag{1813}$$

If $X = (f_1, \dots, f_n)$ and ρ, ν are two laws, write

$$X_a^\nu = \prod_{q=1}^{n} (\mathcal{O}_a(f_q) - \nu(\mathcal{O}_a(f_q))), \qquad \sigma_{\rho,\nu}(X) = \rho(X_a^\nu), \qquad \sigma_\nu = \sigma_{\nu,\nu}. \tag{1814}$$

The empty product is one. The first index specifies the integrating law, the second the centring law. These expressions are symmetric and multilinear in the entries. Lattice isometries $\mathsf{r} \in a\mathbb{Z}^3 \rtimes W_3$ act by pullback on configurations and functions, with inversion when a bond orientation is reversed. Plaquette traces are unchanged by cyclic permutation and inversion, hence

$$\mathsf{r}\mathcal{O}_a(f) = \mathcal{O}_a(\mathsf{r}f), \qquad \sigma_{\mathsf{r}_*\rho,\mathsf{r}_*\nu}(\mathsf{r}X) = \sigma_{\rho,\nu}(X).$$

The same covariance holds for the cumulants.

91.1. **Gibbs equations and periodic limits.** A period lattice is a finite-index subgroup $\Gamma \subset a\mathbb{Z}^3$ with $\mathrm{sys}(\Gamma) := \min_{0 \neq v \in \Gamma} |v|_\infty \geq 3a$. Choose one representative of every positive bond class modulo Γ. Independent Haar variables at these representatives, extended periodically, define dU_Γ on the closed space $\mathcal{V}_\Gamma$ of periodic configurations. Changing representatives only permutes the variables. If D contains one site of each class, then the plaquettes based at sites of D represent the plaquette classes once. Thus

$$S_\Gamma = \sum_{p \in \mathsf{Pl}/\Gamma} a_p, \quad Z_\Gamma[\Phi] = \int e^{-\beta S_\Gamma + \Phi}\, dU_\Gamma, \quad d\mu_\Gamma^\Phi = Z_\Gamma[\Phi]^{-1} e^{-\beta S_\Gamma + \Phi}\, dU_\Gamma \tag{1815}$$

are independent of representatives; here $\beta \geq 0$ and $\Phi \in C(\mathcal{V}_\Gamma)$ is real. Write $\mu_{\Gamma,\beta}$ when $\Phi = 0$. The action and Haar measure show that $\mathsf{r}_*\mu_\Gamma^\Phi = \mu_\Gamma^{\mathsf{r}\Phi}$ whenever $\mathsf{r}^\circ\Gamma = \Gamma$.

For finite $B \subset \mathsf{Bd}$, let $\mathsf{Pl}[B]$ be the plaquettes containing a bond of B, and set

$$H_B = -\beta \sum_{p \in \mathsf{Pl}[B]} a_p, \qquad (\gamma_B F)(U) = \frac{\int F(V \vee U_{B^c}) e^{H_B(V \vee U_{B^c})}\, dV_B}{\int e^{H_B(V \vee U_{B^c})}\, dV_B}. \tag{1816}$$

Every bond is in four plaquettes, so $\#\mathsf{Pl}[B] \leq 4\#B$. The denominator lies in $[e^{-2\beta \#\mathsf{Pl}[B]}, 1]$. The kernel is positive, preserves one and contracts the supremum norm. Integration over the finite set B preserves continuity; if F is local, the remaining support lies in the union of its support and the bonds of $\mathsf{Pl}[B]$, with B removed. Thus $\gamma_B \mathcal{C}_{\mathrm{loc}} \subset \mathcal{C}_{\mathrm{loc}}$.

A Gibbs state is a law satisfying

$$\nu(\gamma_B F) = \nu(F) \quad (B \text{ finite},\ F \in \mathcal{C}_{\mathrm{loc}}). \tag{1817}$$

This equality then holds for every bounded Borel F: the law $E \mapsto \nu(\gamma_B 1_E)$ has the same integrals as ν on $\mathcal{C}_{\mathrm{loc}}$, so the two laws agree. Write $\mathcal{G}_{a,\beta}$ for the Gibbs states and $\mathcal{G}_{a,\beta}^{\mathrm{per}}$ for those invariant under a finite-index subgroup of $a\mathbb{Z}^3$. Equivalently, a periodic state has a period number $P \in \mathbb{N}$, meaning invariance under $Pa\mathbb{Z}^3$; a finite-index subgroup contains such a subgroup, as the orders of the three coordinate classes in the finite quotient have a common multiple.

We also need locally finite perturbations. Let $\mathfrak{A} = (A_i)_{i \in I}$ be real local continuous functions, each with a nonempty finite support B_i, such that each bond lies in only finitely many B_i, counting indices. Replace H_B in (1816) by

$$H_B^{\mathfrak{A}} = H_B + \sum_{i : B_i \cap B \neq \emptyset} A_i$$

to define $\gamma_B^{\mathfrak{A}}$ and the corresponding Gibbs equations.

Proposition 91.1. The Gibbs states form a convex weakly compact set. The subset invariant under any specified subgroup of lattice translations is also convex and weakly compact. Lattice isometries carry Gibbs states to Gibbs states and preserve period numbers. If $t_j \to t$ and $\nu_j \to \nu$ weakly, with ν_j Gibbs for $t_j\mathfrak{A}$, then ν is Gibbs for $t\mathfrak{A}$. ■

Proof. Each Gibbs equation is a closed affine condition on laws, since $\gamma_B F - F$ is continuous. Invariance is likewise expressed by $\nu(F \circ \mathsf{r} - F) = 0$. Intersect these conditions in the compact space of laws. Relabelling bonds, and inverting the reversed ones, preserves product Haar measure and gives $\gamma_{\mathsf{r}B}(\mathsf{r}F) = \mathsf{r}(\gamma_B F)$.

For the last assertion put $A_B = \sum_{i : B_i \cap B \neq \emptyset} A_i$ and $\delta_j = |t_j - t| \|A_B\|_\infty$. The ratio of the two normalized kernel densities lies between $e^{-2\delta_j}$ and $e^{2\delta_j}$. Consequently

$$\|\gamma_B^{t_j\mathfrak{A}} F - \gamma_B^{t\mathfrak{A}} F\|_\infty \le (e^{2\delta_j} - 1)\|F\|_\infty \longrightarrow 0.$$

The fixed kernel applied to $F \in \mathcal{C}_{\mathrm{loc}}$ is continuous. Pass to the limit in the equations for ν_j. □

Proposition 91.2 (Periodic approximations)**.** Let Γ_j be period lattices with $\mathrm{sys}(\Gamma_j) \to \infty$. Suppose that Γ_j acts on the indices of $\mathfrak{A}$, with $A_{v\cdot i} = \tau_v A_i$ and $B_{v\cdot i} = B_i + v$. Put $\Phi_j = \sum_{[i] \in I/\Gamma_j} A_i$ on $\mathcal{V}_{\Gamma_j}$. Every weak limit of $\mu_{\Gamma_j}^{\Phi_j}$ is Gibbs for $\mathfrak{A}$. In particular, limits of $\mu_{\Gamma_j,\beta}$ are translation-invariant Gibbs states.

At $a = a_K$, let $\widetilde{\mu}_{K,\ell,\beta}$ be the periodic lift of the cubic torus measure and define

$$\mathcal{G}_{K,\beta}^{\mathrm{tor}} = \bigcap_{m_0 \ge 1} \overline{\{\widetilde{\mu}_{K,2L^m,\beta} : m \ge m_0\}}.$$

This is a nonempty compact subset of $\mathcal{G}_{K,\beta}^{\mathrm{per}}$. Every member is invariant under $a_K\mathbb{Z}^3 \rtimes W_3$ and has period number one. ■

Proof. The index action is free: $B_i + v = B_i$ with B_i finite nonempty forces $v = 0$. Every orbit has an index whose support meets a fixed bond transversal, so local finiteness makes I/Γ_j finite. Periodicity makes Φ_j independent of the representatives.

Fix a finite B and a local continuous F. Choose a closed cube Q containing its bond support, all plaquettes of $\mathsf{Pl}[B]$, and all B_i meeting B. For large j, its side is smaller than $\mathrm{sys}(\Gamma_j)$. Distinct bonds and plaquettes in Q are then incongruent modulo Γ_j. Choose a bond transversal containing $\mathsf{Bd}(Q)$ and write its variables as (V_B, W). In these variables,

$$-\beta S_{\Gamma_j} + \Phi_j = H_B^{\mathfrak{A}} + H',$$

where H' is independent of V_B. Indeed, a remaining plaquette class, or perturbation orbit, depending on a bond of B would have a representative already included in $H_B^{\mathfrak{A}}$. For functions supported in Q, replacing V_B in a periodic configuration agrees on Q with the nonperiodic replacement in the definition of $\gamma_B^{\mathfrak{A}}$. Integrating first over V_B therefore gives

$$\mu_{\Gamma_j}^{\Phi_j}(\gamma_B^{\mathfrak{A}} F) = \mu_{\Gamma_j}^{\Phi_j}(F).$$

Both functions are continuous and independent of j. The equality passes to every weak limit.

For the empty perturbation family, periodic Haar measure and the action are invariant under every lattice translation. Invariance passes to weak limits. For the cubic sequence, the same is true of every signed coordinate permutation. Its nested tail closures are nonempty compact sets in the space of laws, so their intersection is nonempty and compact. Metrizability identifies this intersection with the subsequential limits. These limits are Gibbs by the first part, proving all assertions. □

Lemma 91.3 (Gauge invariance)**.** Every Gibbs state is invariant under every gauge transformation. Two Gibbs states agreeing on gauge-invariant local continuous functions are equal. ■

Proof. First let a gauge transformation differ from the identity only at one site x. In the kernel for the six incident bonds, it is a left or right translation of each Haar variable and preserves every plaquette trace. Thus $\gamma_{B_x}(F \circ u) = \gamma_{B_x} F$, and the Gibbs equation gives invariance. Compose these changes for finitely many sites. On a local function, an arbitrary gauge transformation uses only its values at the endpoints of a finite bond support; hence invariance holds there and therefore for the law.

Averaging a local continuous F over the gauge variables at those endpoints gives a gauge-invariant local continuous F^{inv} with the same integral in every Gibbs state. Agreement on such averages implies agreement on $\mathcal{C}_{\mathrm{loc}}$, hence equality of the laws. □

91.2. **Reflection positivity for adapted periods.** The diagonal reflections require period lattices that split in the normal direction. Let

$$\mathfrak{N} = \{e_\mu : 1 \le \mu \le 3\} \cup \{e_\mu \pm e_\nu : \mu < \nu\}, \qquad \Sigma(n, c) = \{x : n \cdot x = c\}.$$

For coordinate normals take $c \in \frac{a}{2}\mathbb{Z}$, and for diagonal normals take $c \in a\mathbb{Z}$. These are exactly the positions in these families for which $\theta_{n,c}x = x - 2(n \cdot x - c)n/|n|^2$ preserves the lattice. The coordinate mirrors are site mirrors when $c \in a\mathbb{Z}$ and bond mirrors otherwise.

A period lattice is adapted to n if, for some $h > 0$,

$$\Gamma = \Gamma_\perp \oplus \mathbb{Z} v_\Gamma, \qquad \Gamma_\perp = \Gamma \cap n^\perp, \qquad v_\Gamma = hn/|n|^2. \tag{1818}$$

Then $n \cdot \Gamma = h\mathbb{Z}$, and the linear reflection preserves Γ. If n is coordinate, $h \in a\mathbb{N}$ and $h \ge 3a$; if n is diagonal, $h \in 2a\mathbb{N}$ and $h/2 \ge 3a$. Thus $c + h/2$ is also admissible. The reflections in c and $c + h/2$ differ by translation through v_Γ and act identically on $\mathcal{V}_\Gamma$.

Proposition 91.4 (Two-mirror positivity)**.** Let $a > 0$, $\beta \ge 0$, $n \in \mathfrak{N}$, and let Γ be a period lattice adapted to n, with height period h. For admissible c, put $\theta = \theta_{n,c}$ and $\mathbb{S}^+ = \{c \le n \cdot x \le c + h/2\}$. Then

$$\mu_{\Gamma,\beta}(\overline{\theta F} F) \ge 0 \qquad \big(F \in \mathcal{F}(\mathsf{Bd}(\mathbb{S}^+))\big). \tag{1819}$$

The law is θ-invariant. Both assertions hold also for μ_Γ^Φ if $\Phi = (\Phi_+ + \theta\Phi_+)|_{\mathcal{V}_\Gamma}$ with Φ_+ real continuous and dependent only on the bonds of $\mathbb{S}^+$. No gauge-invariance assumption on F is imposed; bonds lying in either mirror are included. ■

Proof. We retain the boundary variables instead of integrating them into either half. Write $t(x) = n \cdot x - c$. Modulo $\Gamma_\perp$, choose representatives R^0 of the bonds lying in $t = 0$ or $t = h/2$, R^+ of the other bonds contained in $0 \le t \le h/2$, and $R^\times$ of the bonds crossing either boundary. The remaining representatives are the reflected bonds $(\theta b)^+$, $b \in R^+$. For a bond or plaquette, use its midpoint or barycentre to measure height. Translation by jv_Γ changes this height by jh. Every object has a representative in the two half-slabs, their boundaries, or the crossing sets, with height in $(-h/2, h/2]$. Two such representatives congruent modulo Γ differ by an element of $\Gamma_\perp$, since their height difference has modulus less than h. This proves that the chosen representatives are disjoint and exhaustive. There are finitely many of them.

Consequently the coordinates

$$U_0 = U_{R^0}, \quad U_+ = U_{R^+}, \quad U'_+ = (\theta U)_{R^+}, \quad U_\times = U_{R^\times}$$

are independent Haar variables. The necessary inversions in U'_+ preserve Haar measure. On the boundary bonds θ fixes the coordinates modulo Γ, and it interchanges U_+ and U'_+. Thus $F(U) = \widehat{F}(U_0, U_+)$ and $(\theta F)(U) = \widehat{F}(U_0, U'_+)$.

Choose plaquette representatives in the same way. Those lying in the boundaries contribute a weight $W_0(U_0) > 0$. Those in the positive half but not in its boundaries contribute $W_+(U_0, U_+) > 0$, and their reflected copies contribute $W_+(U_0, U'_+)$. The remaining plaquettes meet both sides of a mirror. Their form follows by listing the corner heights

$$t(x), \quad t(x) + an_\mu, \quad t(x) + a(n_\mu + n_\nu), \quad t(x) + an_\nu.$$

A site mirror has none. At a diagonal mirror the only possibility has two opposite corners z, z' in the plane and two reflected corners off it, at heights $\pm a$ relative to the plane. With y the corner in $\mathbb{S}^+$, set $v_p(U) = U(\langle z, y\rangle)U(\langle y, z'\rangle)$. These two bonds belong to the positive half and not to its boundary; the inequality $a < h/2$ ensures that they do not reach the other mirror. Tracing the plaquette boundary gives

$$e^{-\beta a_p(U)} = k_\beta(v_p(U_+)v_p(U'_+)^{-1}), \qquad k_\beta(g) = e^{-\beta}e^{(\beta/2)\,\mathrm{tr}\,g}.$$

At a bond mirror, let x, y be neighbouring sites in the adjacent layer of $\mathbb{S}^+$. Put $u_b = U(\langle x, y\rangle)$, $u'_b = (\theta U)(\langle x, y\rangle)$ and $w_x = U(\langle x, \theta^{(k)}x\rangle)$, where $\theta^{(k)}$ is reflection in the boundary under consideration. The four-bond loop gives

$$e^{-\beta a_p(U)} = k_\beta(u_b w_y (u'_b)^{-1} w_x^{-1}).$$

The crossing variables at the two mirrors are disjoint: their midpoint heights differ by $h/2$, not by a multiple of h. Their Haar integral at each boundary is therefore

$$K(u, u') = \int \prod_{b=(x,y)} k_\beta(u_b (w \cdot u')_b^{-1})\, dw, \qquad (w \cdot u')_b = w_x u'_b w_y^{-1}.$$

Each resulting kernel has a uniformly absolutely convergent expansion $\sum_\alpha c_\alpha \Phi_\alpha(V)\overline{\Phi_\alpha(V')}$ with $c_\alpha \geq 0$ and $\sum_\alpha c_\alpha \|\Phi_\alpha\|_\infty^2 < \infty$. Indeed, the matrix-entry expansion used in proposition 85.1 gives

$$k_\beta(gg'^{-1}) = e^{-\beta} \sum_{N\geq 0} \frac{(\beta/2)^N}{N!} \sum_{I,J\in\{1,2\}^N} g_{IJ}\overline{g'_{IJ}}, \qquad \sum_{N,I,J} e^{-\beta}\frac{(\beta/2)^N}{N!} = e^{\beta}.$$

Products and pullbacks preserve this property. For a bond boundary, simultaneous gauge invariance and Haar invariance replace the single average in K by independent averages on its two entries. The expansion then has features $\Phi_\alpha^{\mathrm{av}}(u) = \int \Phi_\alpha(w \cdot u)\, dw$; their supremum norms do not increase. This proves the assertion also for the product $\mathcal{K}$ of the two boundary kernels.

After integrating $U_\times$, the full Wilson weight is

$$W_0(U_0)W_+(U_0, U_+)W_+(U_0, U'_+)\mathcal{K}(U_+, U'_+).$$

Termwise integration is justified by the summable supremum bounds and gives the unnormalized quadratic form

$$\int dU_0\, W_0(U_0) \sum_\alpha c_\alpha \left| \int \widehat{F}(U_0, U_+)W_+(U_0, U_+)\Phi_\alpha(U_+)\, dU_+ \right|^2 \geq 0.$$

Division by $Z_\Gamma[0] > 0$ proves (1819). For the perturbed measure apply this calculation to Fe^{Φ_+}. Finally θ preserves Haar measure and the action, and the stated perturbation is θ-invariant, proving invariance. □

91.3. **The torus-limit states.** Periodization $f^{\langle\ell\rangle}(x) = \sum_{v\in\ell\mathbb{Z}^3} f(\widetilde{x}+v)$ is well defined for compactly supported f. Grouping the plaquettes by period class gives the exact identity

$$\mathcal{O}_K(f)(\mathrm{per}_\ell\, U) = \mathcal{O}_{a_K}(f^{\langle\ell\rangle})(U). \tag{1820}$$

If the supports lie in $(-\ell/4, \ell/4)^3$, periodization preserves their stated separation and per-entry radius bounds, and their $\|\cdot\|_\sharp$ norms. To check the distance assertion, each coordinate difference of two points in this cube has modulus less than $\ell/2$, so it is already the shortest representative of its class modulo ℓ. The same argument and local lifting give the supremum and Lipschitz norm identities. Periodization commutes with the lattice isometries whenever the source and its image are considered on the same quotient.

Proposition 91.5. Let $K \geq 1$, $\beta \geq 0$ and $\nu \in \mathcal{G}^{\mathrm{tor}}_{K,\beta}$. There are sides $\ell_j \to \infty$ with $\widetilde{\mu}_{K,\ell_j,\beta} \to \nu$ weakly. Along every such sequence, for bounded compactly supported test functions and $w = \sum_q z_q f_q$,

$$\nu(e^{\mathcal{O}_K(w)}) = \lim_j \frac{Z_{K,\ell_j,\beta}[w^{\langle \ell_j \rangle}]}{Z_{K,\ell_j,\beta}[0]}, \tag{1821}$$

$$S^T_n[\nu](X) = \lim_j S^T_{n;K,\ell_j,\beta}(X^{\langle \ell_j \rangle}), \qquad \sigma_\nu(X) = \lim_j \sigma_{K,\ell_j,\beta}(X^{\langle \ell_j \rangle}). \tag{1822}$$

Any bound on the right uniform for tuples in $(-\ell_j/4, \ell_j/4)^3$ with prescribed separation and per-entry radius passes to the left for tuples anywhere in $\mathbb{R}^3$. The state ν is reflection positive at every coordinate site and bond mirror, for bounded local functions of the closed positive half-space. ■

Proof. The sequence exists by the tail-closure description in proposition 91.2. Every product of the observables and their exponentials is local continuous. Weak convergence and (1820) therefore give convergence of all moments and the first formula. Cumulants and centred moments are fixed finite polynomials in these moments, proving (1822). A fixed compact union of supports eventually lies in the central cube, so each uniform bound passes to the limit.

The cubic period lattice is adapted to every coordinate normal, with height period ℓ_j. A fixed finite bond support in $\{x_\mu \geq c\}$ eventually lies in $\{c \leq x_\mu \leq c + \ell_j/2\}$. Apply proposition 91.4 and pass to the weak limit for continuous local F. To extend the resulting inequality to bounded Borel F with the same finite support, approximate its function of the finite bond coordinates in L^2 of the corresponding marginal of ν by continuous functions. Reflection invariance gives the same L^2 convergence for their reflected functions. Cauchy–Schwarz then passes the quadratic form to the limit. Reversing the side follows by reflection invariance. □

The last assertion does not supply diagonal positivity. For $n = e_\mu \pm e_\nu$, the heights of $\ell\mathbb{Z}^3$ are $\ell\mathbb{Z}$, whereas its shortest nonzero pure-normal vector is ℓn, of height 2ℓ. Thus the cubic period lattice does not satisfy (1818) for a diagonal normal.

91.4. **Periodic arrays and their reflection.** The comparison of Gibbs states below uses two finite-volume inequalities. The first compares a periodic array with its expectation in a Gibbs state; the second increases its partition function by reflecting the array across successive slabs.

For a site transversal D of Γ, write Bd_D for the positive bonds starting in D, and let

$$\partial_r D = \{x \in D : x + w \notin D \text{ for some } w \in a\mathbb{Z}^3, |w|_\infty \leq ra\}.$$

If $\Gamma \subset \Gamma' \subset a\mathbb{Z}^3$ have finite index, the array of a local function A is $\mathbb{A} = \sum_{v \in \Gamma'/\Gamma} \tau_v A$ on $\mathcal{V}_\Gamma$, with $|\mathbb{A}| = [\Gamma' : \Gamma]$.

Lemma 91.6 (Lower tangent)**.** Let Γ be a period lattice, $\Gamma' \supset \Gamma$ a finite-index subgroup, D a site transversal of Γ, and $\nu \in \mathcal{G}_{a,\beta}$ invariant under Γ'. Let $A \in \mathcal{C}_{\mathrm{loc}}$ be real and $t \in \mathbb{R}$. If N_{in} is the number of classes in Γ'/Γ having a representative v for which $\tau_v A$ has a support in Bd_D, then

$$\log \frac{Z_\Gamma[t\mathbb{A}]}{Z_\Gamma[0]} \geq t|\mathbb{A}|\nu(A) - 24\beta \#\partial_1 D - 2|t|\|A\|_\infty(|\mathbb{A}| - N_{\mathrm{in}}). \tag{1823}$$

If A has a bond support in $x_0 + [0, ra]^3$, with $x_0 \in a\mathbb{Z}^3$ and integer $r \geq 1$, then

$$|\mathbb{A}| - N_{\mathrm{in}} \leq \#\partial_r D \leq (2r-1)^3 \#\partial_1 D.$$

■

Proof. Put $B = \mathsf{Bd}_D$. Plaquettes whose four bonds lie in B give identical action terms for $V \vee U_{B^c}$ and the periodic extension V^{per}. They represent distinct period classes. Let n_1 count the remaining plaquettes meeting B, and n_2 the period classes not represented by an interior plaquette. Then

$$\sum_{p \in \mathsf{Pl}[B]} a_p(V \vee U_{B^c}) - S_\Gamma(V^{\mathrm{per}}) = \Sigma_1 - \Sigma_2, \qquad 0 \leq \Sigma_i \leq 2n_i.$$

If a positive bond starts at $x \notin \partial_1 D$, all plaquettes containing it have their bonds in B. There are at most nine plaquettes meeting the three positive bonds starting at a specified x, and three plaquettes based at x. Hence $n_1 \le 9\#\partial_1 D$ and $n_2 \le 3\#\partial_1 D$.

Choose one good representative for each class counted by N_{in}, and set $\Phi_B = t\sum_{\mathrm{good}} \tau_v A$ and $e = |t|\|A\|_\infty(|\mathbb{A}| - N_{\mathrm{in}})$. The preceding action comparison, applied to numerator and denominator, gives for every exterior configuration

$$\frac{Z_\Gamma[\Phi_B]}{Z_\Gamma[0]} \ge e^{-2\beta(n_1+n_2)}\gamma_B(e^{\Phi_B}).$$

Jensen's inequality for the normalized kernel, followed by the Gibbs equation and Γ'-invariance, yields

$$\log\frac{Z_\Gamma[\Phi_B]}{Z_\Gamma[0]} \ge tN_{\mathrm{in}}\nu(A) - 2\beta(n_1+n_2).$$

On periodic configurations $|t\mathbb{A} - \Phi_B| \le e$; also $|t(|\mathbb{A}| - N_{\mathrm{in}})\nu(A)| \le e$. The two errors give (1823).

For the last estimate, assign each array class the unique point of $x_0 + v + \Gamma$ in D. If that point is outside $\partial_r D$, the translated cube and all its bonds lie in Bd_D. Different bad classes therefore give different points of $\partial_r D$. Finally, from $x \in \partial_r D$ follow a lattice path to a point outside D, moving at most a in each coordinate at each step and using at most r steps. The point immediately before its first exit from D lies in $\partial_1 D$ and differs from x by a vector of sup norm at most $(r-1)a$. Thus $\partial_r D$ is contained in the union of $(2r-1)^3$ translates of $\partial_1 D$. □

Fix admissible c_0, $h > 0$ with $v_h = hn/|n|^2 \in a\mathbb{Z}^3$, and put

$$\Sigma_j = \Sigma(n, c_0 + jh), \quad \theta_j = \theta_{n,c_0+jh}, \quad \mathbb{S}_j = \{(j-1)h \le n\cdot x - c_0 \le jh\}.$$

The maps $\mathsf{w}_{2k+1} = \tau_{2kv_h}$ and $\mathsf{w}_{2k} = \theta_k$ carry $\mathbb{S}_1$ to $\mathbb{S}_j$. Define $f^{[j]} = \mathsf{w}_j f$ for odd j and $f^{[j]} = \overline{\mathsf{w}_j f}$ for even j. Writing $\vartheta = \theta_{n,c_0+h/2}$, the reflection formulas give

$$\theta_m \mathsf{w}_j = \mathsf{w}_{2m+1-j}, \quad \mathsf{w}_{j+2N} = \tau_{2Nv_h}\mathsf{w}_j, \quad \tau_{v_h}\mathsf{w}_j = \mathsf{w}_{j+1}\vartheta. \tag{1824}$$

All these maps preserve the lattice. The group generated by the θ_j consists of them and the translations τ_{2kv_h}.

Proposition 91.7 (Reflection of arrays)**.** Suppose Γ is adapted to n with height period $2Nh$, $N \ge 1$. If a law μ on $\mathcal{V}_\Gamma$ is reflection positive at every pair Σ_j, Σ_{j+N}, then

$$\left|\mu\left(\prod_{j=1}^{2N} f_j^{[j]}\right)\right| \le \prod_{j=1}^{2N} \mu\left(\prod_{k=1}^{2N} f_j^{[k]}\right)^{1/(2N)} \tag{1825}$$

for bounded Borel functions of the bonds in $\mathbb{S}_1$; each quantity under a root is nonnegative.

Suppose additionally that $\Gamma' \supset \Gamma$ has finite index and the splitting (1818) with height period h. No lower bound on its systole is required. For real $A \in \mathcal{C}_{\mathrm{loc}}(\mathbb{S}_1)$, let T represent $(\Gamma' \cap n^\perp)/(\Gamma \cap n^\perp)$, and set

$$A_T = \sum_{v\in T}\tau_v A, \qquad \mathbb{A} = \sum_{j=1}^{2N}\tau_{(j-1)v_h}A_T, \qquad \mathbb{A}^{\mathrm{f}} = \sum_{j=1}^{2N}\mathsf{w}_j A_T.$$

These define functions on $\mathcal{V}_\Gamma$ independent of representatives, and $|\mathbb{A}| = 2N\#T$. For every real t,

$$Z_\Gamma[t\mathbb{A}] \le Z_\Gamma[t\mathbb{A}^{\mathrm{f}}]. \tag{1826}$$

The law $\mu_\Gamma^{t\mathbb{A}^{\mathrm{f}}}$ is reflection positive at each pair Σ_j, Σ_{j+N}, invariant under the wall group and under $\Gamma' \cap n^\perp$, and satisfies

$$\mu_\Gamma^{t\mathbb{A}^{\mathrm{f}}}(\mathbb{A}^{\mathrm{f}}) = |\mathbb{A}|\mu_\Gamma^{t\mathbb{A}^{\mathrm{f}}}(A). \tag{1827}$$

■

Proof. Let $\mathrm{cy}(f_1,\dots,f_{2N})$ denote the integral on the left of (1825). The copies are cyclic modulo $2N$ on $\mathcal{V}_\Gamma$. Reflection in Σ_m changes index j to $2m+1-j$ and conjugates the copy. Cauchy–Schwarz for the positive form at that pair therefore bounds $|\mathrm{cy}|^2$ by the product of the two integrals obtained by reflecting either half of the cyclic array into the other half. Those two integrals are nonnegative. In particular constant arrays have nonnegative integrals.

Assume first that the constant-array roots $\|f_j\|$ are positive. Among the finitely many arrays with entries chosen from $f_1,\dots,f_{2N}$ maximize $|\mathrm{cy}(\underline{f})|/\prod_j \|f_j\|$. The maximum is at least one and is unchanged by either half-reflection: the product bound and maximality force both reflected arrays to attain it. Choose a maximizing array with a longest constant cyclic string, of length k. Reflect across its terminal boundary. If $k \le N$, this doubles the string; if $N < k < 2N$, it makes the whole array constant. Maximality of k therefore forces $k = 2N$, so the maximum is one. If a root vanishes, replace every f_j by $f_j + \varepsilon$ with real ε. Each constant-array integral is a nonnegative polynomial of degree $2N$ with leading coefficient one. Choose $\varepsilon \to 0$ avoiding their finitely many zeros and pass to the limit. This proves (1825).

For the array comparison take $f = e^{tA_T}$. The splitting of both period groups gives the displayed expression for $\mathbb{A}$. For odd j, $\tau_{(j-1)v_h} = \mathsf{w}_j$; for even j it is $\mathsf{w}_j\vartheta$. Thus

$$Z_\Gamma[t\mathbb{A}]/Z_\Gamma[0] = \mathrm{cy}(f,\vartheta f,\dots,f,\vartheta f), \quad Z_\Gamma[t\mathbb{A}^{\mathrm{f}}]/Z_\Gamma[0] = \mathrm{cy}(f,\dots,f).$$

The unperturbed measure has the required positivity by proposition 91.4. Translation by v_h takes the fold of f to the fold of ϑf, by (1824); translation invariance therefore gives $\|\vartheta f\| = \|f\|$. Inequality (1825) proves (1826).

For any m, the folded perturbation is $\Phi_+ + \theta_m\Phi_+$, where $\Phi_+ = t\sum_{j=m+1}^{m+N} \mathsf{w}_j A_T$. Apply the perturbed assertion of proposition 91.4. This gives positivity and invariance under every wall reflection. Translations in $\Gamma' \cap n^\perp$ permute the transverse classes and commute with the w_j, so they preserve the perturbation and the law. All $2N\#T$ summands in $\mathbb{A}^{\mathrm{f}}$ consequently have the same expectation, which proves (1827). □

The corresponding infinite folded family is $(\mathsf{w}_j\tau_v A)_{j\in\mathbb{Z},\, v\in\Gamma'\cap n^\perp}$. If B_A is a finite nonempty support in the first slab, give its members the supports $\mathsf{w}_j(B_A+v)$. A fixed bond meets only the finitely many slabs containing it; within each such slab only finitely many translates of the finite set B_A contain that bond. Thus this is a locally finite perturbation family, to which the periodic-approximation proposition applies when the index action and period lattice are specified.

91.5. **Domination of periodic Gibbs states.** Fix $K \ge 1$, $a = a_K$ and $\beta > 0$. A board on $a\mathbb{Z}^3$ has cells

$$\square_k = o + lk + [-l/2, l/2]^3, \qquad l = a_i, \quad i \le K-1, \quad o \in a\mathbb{Z}^3.$$

The number $l/a = L^{K-i}$ is odd. Its walls are consequently bond-midpoint mirrors. For $\ell \in \mathfrak{L}$ with $\ell \ge 2l$, the quotient board has an even number ℓ/l of cells in every direction. A function of the inner bonds of $\square_k$ defines a function of the corresponding torus cell; write $\|F\|_{\mathrm{ch},\ell}$ for its chessboard seminorm. This identification commutes with products, mirror copies and complex conjugation. In particular all the conclusions of eq. (1754) and propositions 86.3 and 86.4 apply. Translation by a vector in $2l\mathbb{Z}^3$ is a mirror copy without conjugation.

The following surface estimate removes the boundary condition.

Proposition 91.8. Let $\ell \in \mathfrak{L}$, $y \in a(\mathbb{Z}+1/2)^3$, $D = (y+[0,\ell]^3) \cap a\mathbb{Z}^3$, and let W be the set of positive bonds whose initial point lies in D. For every nonnegative bounded Borel function Ψ of W and every exterior configuration ξ,

$$e^{-36\beta(\ell/a)^2}\widetilde{\mu}_{K,\ell,\beta}(\Psi) \le \gamma_W\Psi(\xi) \le e^{36\beta(\ell/a)^2}\widetilde{\mu}_{K,\ell,\beta}(\Psi). \tag{1828}$$

The same inequalities hold with $\gamma_W\Psi(\xi)$ replaced by $\nu(\Psi)$ for any Gibbs state ν. For every real bounded function H of W,

$$\nu(H) \le \log \widetilde{\mu}_{K,\ell,\beta}(e^H) + 36\beta(\ell/a)^2. \tag{1829}$$

■

Proof. Use the bonds of W as Haar coordinates for both the Gibbs kernel and the torus measure, and put $n = \ell/a$. In each of the three plaquette orientations, $n(n-1)^2$ plaquettes have all their bonds in W and occur with the same action in both measures. There are n^3 torus plaquettes and $n^3 + 2n^2$ plaquettes touching W in that orientation. Thus, after cancellation, the Gibbs action has $N_1 = 12n^2 - 3n$ terms and the torus action has $N_2 = 6n^2 - 3n$ terms. Since $0 \le a_p \le 2$, their difference is between $-2N_2$ and $2N_1$. Comparing both numerators and both normalizing denominators gives a density ratio between $e^{-2\beta(N_1+N_2)}$ and $e^{2\beta(N_1+N_2)}$; $2(N_1+N_2) = 36n^2 - 12n \le 36n^2$ proves (1828). Integrate the kernel inequality against ν and use its Gibbs equation. Finally Jensen's inequality in the kernel gives $\gamma_W H \le \log \gamma_W e^H$; apply (1828) and integrate once more to obtain (1829). □

Proposition 91.9. Let ν be a periodic Gibbs state, let $\mathcal{K}_0$ be a finite nonempty set of cell indices, and let F_k be bounded Borel functions of the inner bonds of $\square_k$. For every sequence $\ell_j \in \mathfrak{L}$ tending to infinity,

$$\left|\nu\left(\prod_{k\in\mathcal{K}_0} F_k\right)\right| \le \liminf_{j\to\infty} \prod_{k\in\mathcal{K}_0} \|F_k\|_{\mathrm{ch},\ell_j}. \tag{1830}$$

More precisely, choose a period number P of ν, an even positive multiple q of P, and k_0 so that $\mathcal{K}_0 \subset k_0 + \{0,\dots,q-1\}^3$. Set $\mathsf{P} = ql$, $m = \prod_k \|F_k\|_\infty$ and

$$N_\ell = \lfloor \ell/\mathsf{P} \rfloor^3, \qquad \epsilon_\ell = 36\beta(\ell/a)^2/N_\ell \le 288\beta\mathsf{P}^3 a^{-2}\ell^{-1} \quad (\ell \ge \mathsf{P}).$$

For $\ell \ge \mathsf{P}$, if $t > 0$ and $2tm \le 1$,

$$\left|\nu\left(\prod_k F_k\right)\right| \le \prod_k \|F_k\|_{\mathrm{ch},\ell} + tm^2 + \epsilon_\ell/t. \tag{1831}$$

For a real bounded function H of one cell, the same packing, with $\ell \ge \mathsf{P}$, gives

$$\nu(H) \le \log \|e^H\|_{\mathrm{ch},\ell} + \epsilon_\ell. \tag{1832}$$

If ν is invariant under every reflection in the walls of the board, then for every $\ell \ge 2l$,

$$\nu(H) \le \log \|e^H\|_{\mathrm{ch},\ell} + 36\beta l^3 a^{-2}\ell^{-1}. \tag{1833}$$

■

Proof. The vectors

$$V_\ell = \{v \in \mathsf{P}\mathbb{Z}^3 : 0 \le v_\mu \le \ell - \mathsf{P}\}$$

are periods of ν, since l/a is an integer and $P \mid q$. They belong to $2l\mathbb{Z}^3$, so $\tau_v F_k$ is the mirror copy in cell $k + v/l$. All these cell indices are distinct and fit in one ℓ-period box: division with remainder by q separates the translated blocks. Their inner bonds lie in the set W of proposition 91.8, with corner $y = o + l(k_0 - (1/2,1/2,1/2))$. Applying (1829) to the sum of the N_ℓ translates of a real local function Φ gives

$$N_\ell \nu(\Phi) \le 36\beta(\ell/a)^2 + \log \widetilde{\mu}_{K,\ell,\beta}\left(\prod_{v\in V_\ell} e^{\tau_v\Phi}\right). \tag{1834}$$

Here $\#V_\ell = N_\ell$, and $\lfloor \ell/\mathsf{P} \rfloor \ge \ell/(2\mathsf{P})$ proves the displayed bound on ϵ_ℓ.

Write $F = \prod_k F_k$, and choose $|\zeta| = 1$ so that $\zeta\nu(F) = |\nu(F)|$. If $m = 0$ there is nothing to prove. Otherwise put $G = \mathrm{Re}(\zeta F)$ and $\Phi = \log(1 + tG)$. Since $|tG| \le 1/2$ and $\log(1+x) \ge x - x^2$ on $[-1/2, 1/2]$,

$$\nu(\Phi) \ge t|\nu(F)| - t^2 m^2.$$

Expand $\prod_{v\in V_\ell}(1+t\tau_v G)$ over subsets $Y\subset V_\ell$. Each product $\prod_{v\in Y}\tau_v G$ is $2^{-|Y|}$ times a sum of $2^{|Y|}$ products of mirror copies of the F_k or $\overline{F_k}$, with coefficients of modulus one. The finite chessboard estimate and invariance of its seminorm under copying and conjugation bound each such mean by $\Pi_\ell^{|Y|}$, where $\Pi_\ell=\prod_k\|F_k\|_{\mathrm{ch},\ell}$. Consequently

$$\widetilde{\mu}_{K,\ell,\beta}\left(\prod_{v\in V_\ell}e^{\tau_v\Phi}\right)\le(1+t\Pi_\ell)^{N_\ell}\le e^{N_\ell t\Pi_\ell}.$$

Insert this in (1834) and divide by $N_\ell t$ to obtain (1831). First let $\ell=\ell_j\to\infty$ at fixed t, and then let $t\downarrow 0$, proving (1830). Applying (1834) directly to H and using the chessboard bound $\|e^H\|_{\mathrm{ch},\ell}^{N_\ell}$ proves (1832). The seminorm is positive because its defining integrand is bounded below by a positive number.

For (1833), sum all the mirror copies of H in an ℓ-period box. The mean of this sum is $(\ell/l)^3\nu(H)$, while the torus exponential mean is exactly $\|e^H\|_{\mathrm{ch},\ell}^{(\ell/l)^3}$. Use (1829) and divide by $(\ell/l)^3$. □

91.6. **A volume threshold for local comparison.** The next threshold depends on the cutoff and coupling. No uniform estimate for its size is asserted. Let W_K be the positive bonds with both endpoints in $[-K,K]^3$, and let ∂W_K be the exterior bonds of plaquettes touching W_K. Denote the continuous density of γ_{W_K} by $k_K(V,\xi_{\partial W_K})$ and put

$$q_\nu(V)=\int k_K(V,\xi_{\partial W_K})\,d\nu(\xi),\qquad \varepsilon_{K,\beta}(\ell)=\inf_{\nu\in\mathcal{G}_{K,\beta}^{\mathrm{tor}}}\left\|\frac{q_\nu}{q_{\widetilde{\mu}_{K,\ell,\beta}}}-1\right\|_\infty.$$

Proposition 91.10. One has $\varepsilon_{K,\beta}(\ell)\to 0$ as $\ell\to\infty$ in $\mathfrak{L}$. Thus there is a least $\ell_\flat(K,\beta)\in\mathfrak{L}$, greater than $4(K+1)$, such that $\varepsilon_{K,\beta}(\ell)<e^{-L^K}$ for all $\ell\ge\ell_\flat(K,\beta)$. For each such ℓ one can choose a single $\nu\in\mathcal{G}_{K,\beta}^{\mathrm{tor}}$ satisfying

$$|\nu(Y)-\widetilde{\mu}_{K,\ell,\beta}(Y)|\le e^{-L^K}\widetilde{\mu}_{K,\ell,\beta}(|Y|)$$

for every bounded Borel function Y of W_K. If $0<r\le K-1$ and $f_1,\dots,f_n$ are bounded functions supported in $[-r,r]^3$, this same state satisfies

$$\left|S^T_{n;K,\ell,\beta}(f_1^{\langle\ell\rangle},\dots,f_n^{\langle\ell\rangle})-S^T_n[\nu](f_1,\dots,f_n)\right|\le c_n^{\mathrm{law}}e^{-L^K}\left(6L^K(2rL^K+1)^3\right)^n\prod_q\|f_q\|_\infty,\qquad(1835)$$

where $c_n^{\mathrm{law}}=\sum_{\pi\in\mathrm{Part}(n)}(\#\pi)!$. ■

Proof. If M is the number of plaquettes touching W_K, then $e^{-2\beta M}\le k_K\le e^{2\beta M}$. The kernel is uniformly continuous on the compact space of its finite interior and exterior coordinates. Thus the functions q_ν have these same positive bounds and are equicontinuous. If $\nu_j\to\nu$ weakly, then $q_{\nu_j}(V)\to q_\nu(V)$ for every V; a finite net in the interior coordinate space and equicontinuity upgrade this to uniform convergence.

The Gibbs equation shows that q_ν is the W_K-marginal density for every torus-limit state. It is also the marginal density for $\widetilde{\mu}_{K,\ell,\beta}$ when $\ell>4(K+1)$: the plaquettes touching W_K fit in $[-K-a,K+a]^3$, whose side is less than the period systole minus a, so the local periodic Gibbs equation applies.

If $\varepsilon_{K,\beta}(\ell)$ did not tend to zero, take a diverging sequence of sides on which it stays bounded below. A subsequence of the corresponding laws converges to a member ν_* of $\mathcal{G}_{K,\beta}^{\mathrm{tor}}$. Uniform convergence of the marginal densities and their common positive lower bound then give $\|q_{\nu_*}/q_{\widetilde{\mu}_{K,\ell,\beta}}-1\|_\infty\to 0$, a contradiction. The asserted state exists by the strict inequality in the definition of the threshold, and the first estimate follows by integrating the density ratio.

For the cumulant estimate, $X_q=\mathcal{O}_K(f_q)$ depends only on W_K, since every plaquette centered in $[-r,r]^3$ lies in $[-K,K]^3$. Counting at most $3(2rL^K+1)^3$ plaquette centers and using $a_p\le 2$

gives $\|X_q\|_\infty \le 6L^K(2rL^K+1)^3\|f_q\|_\infty$. The density-ratio bound applies to every product of the X_q. Expand the two cumulants over partitions. In a partition with k blocks, telescope the product of its k moments; the error is at most $ke^{-L^K}\prod_q \|X_q\|_\infty$ and the cumulant coefficient has modulus $(k-1)!$. Summing over partitions and using (1820) proves (1835). □

91.7. **Transfer of cell normalizers.** Let $f_1,\dots,f_n \in \mathrm{Lip}_c(\mathbb{R}^3;\mathbb{R})$ be nonzero and suppose that each support lies in the interior of one board cell, indexed by $k(q)$. Supports in the same cell may overlap. Put

$$J_k=\{q : k(q)=k\},\quad w_k(z)=\sum_{q\in J_k} z_q f_q,\quad w(z)=\sum_q z_q f_q,\quad B(z)=\{q : z_q\neq 0\},$$

$$\Pi_\ell(z)=\prod_{k:J_k\neq\varnothing}\|e^{\mathcal{O}_K(w_k(z))}\|_{\mathrm{ch},\ell},\qquad \mathbb{D}_\rho(\underline{f})=\left\{z:\sum_q |z_q|\|f_q\|_\sharp<\rho\right\}. \tag{1836}$$

A holomorphic E on $\mathbb{D}_\rho(\underline{f})$ with $E(0)=0$ is a *normalizer with bound* Q on a set of sides $\mathfrak{l}\subset\{\ell\in\mathfrak{L}:\ell\ge 2l\}$ if $\Pi_\ell(z)\le|e^{E(z)}|Q$ there. It is a normalizer *for periodic states* if there is one sequence $\ell_j\to\infty$ such that

$$\liminf_{j\to\infty}\Pi_{\ell_j}(z)\le|e^{E(z)}|Q\quad\text{for every } z\in\mathbb{D}_\rho(\underline{f}). \tag{1837}$$

A *graded* normalizer replaces Q by $Q^{\#B(z)}$. In particular a normalizer on an unbounded set of sides is a normalizer for periodic states. The sequence in (1837) is chosen before z; no uniform convergence of Π_{ℓ_j} is required.

Proposition 91.11. A normalizer for periodic states gives, for every periodic Gibbs state ν,

$$|e^{-E(z)}\nu(e^{\mathcal{O}_K(w(z))})|\le Q. \tag{1838}$$

A normalizer on $\mathfrak{l}$ gives the same bound with ν replaced by $\widetilde{\mu}_{K,\ell,\beta}$ if $\ell\in\mathfrak{l}$ and the occupied cell indices are distinct modulo $(\ell/l)\mathbb{Z}^3$. Both assertions hold with $Q^{\#B(z)}$ for a graded normalizer.

For any law $\mathfrak{s}$ satisfying (1838),

$$\left|S_n^T[\mathfrak{s}](\underline{f})-\partial_1\cdots\partial_n E(0)\right|\le\log 2\,(2Q+2)^n(n/\rho)^n\prod_q\|f_q\|_\sharp. \tag{1839}$$

Under the graded bound this improves to

$$\left|S_n^T[\mathfrak{s}](\underline{f})-\partial_1\cdots\partial_n E(0)\right|\le(n-1)!(2^n-1)(eQ/\rho)^n\prod_q\|f_q\|_\sharp. \tag{1840}$$

If two laws satisfy the same ungraded normalizer bound, their cumulants differ by at most twice the right side of (1839), including for $n=1$ and for overlapping supports. ■

Proof. Every $e^{\mathcal{O}_K(w_k)}$ depends on the inner bonds of its cell, since a plaquette with center inside a cell has all its bonds inside that cell. Apply (1830) to their product, then (1837), obtaining (1838).

For a torus, move each occupied cell by an $\ell\mathbb{Z}^3$ translation into one period box. The images are distinct by hypothesis. Such translations belong to $2l\mathbb{Z}^3$, hence give mirror copies with unchanged chessboard seminorms; they also leave the periodic configurations unchanged. The finite chessboard estimate therefore bounds the exponential mean by $\Pi_\ell(z)$. This proves the torus assertion, with the identical argument for graded bounds. Finally the exponential moment is entire, equals one at zero, and its logarithmic mixed derivative is the cumulant. Apply lemmas 87.1 and 87.2 to $e^{-E(z)}\mathfrak{s}(e^{\mathcal{O}_K(w(z))})$ with weights $\|f_q\|_\sharp$. This gives the two derivative estimates. For two laws the common derivative of E cancels. □

91.8. **Uniform normalizers and cumulant bounds.** Fix the source radius ϱ and $0 < g \le \gamma_\varrho$ as in theorem AD. Write

$$\mathfrak{d}(g) = \{(K, \beta) : K \ge 1,\ \beta \in \mathcal{W}_K(g)\}. \tag{1841}$$

The admissible sides form the unbounded set $\mathfrak{L}(g)$ of (1489). Throughout this subsection take

$$C_e = C_{\mathrm{S}}, \quad \kappa_e = \kappa, \quad E_+ = C_{\mathrm{S}}, \quad C_{\mathrm{low}} = C_{\mathrm{S}}(1 + \log g^{-2}), \quad C_{\mathrm{gap}} = C_{\mathrm{S}}.$$

In particular the constants are independent of (K, β) and of the side. Put $A_0 = E_+ + C_{\mathrm{low}} + C_{\mathrm{gap}}$ and

$$\mathsf{Q}_1 = \exp\{A_0 L^3 + \Phi_1(2, 1; L)\}, \qquad Q^\natural(n, R) = \exp\{A_0 h(n, R)^3 + \Phi_1(R, R+1; M)\}. \tag{1842}$$

Here $h(n, R)$ is (1793). With p and C_Q^0 as in (1797) and (1799), set $C_Q = e^{C_{\mathrm{gap}}} C_Q^0$ and $Q_s(\eta) = C_Q (2\eta)^{-p}$.

Proposition 91.12. For every $(K, \beta) \in \mathfrak{d}(g)$ the following normalizers have radius ϱ and work on all indicated sides, hence also for every periodic Gibbs state.

(1) If all supports lie in one ball $\mathbb{B}(x_0, 2)$, on a board of side L with x_0 at distance at least 3 from its walls, there is a normalizer with bound Q_1 on $\mathfrak{L}(g)$, for any number of entries and any overlap.
(2) If $n \ge 2$, the supports are disjoint, and $\operatorname{supp} f_q \subset \mathbb{B}(x_q, R)$ on a board of side $h(n, R)$ whose walls have distance at least $2R + 1$ from every x_q, there is a graded normalizer with base $Q^\natural(n, R)$ on the sides $\ell \in \mathfrak{L}(g)$ with $\ell \ge 2h(n, R)$.
(3) If $n \ge 2$, $0 < \eta \le 1/4$, the supports are disjoint, and $\operatorname{supp} f_q \subset \mathbb{B}(y_q, \eta)$ on a board of side one whose walls have distance at least 2η from every y_q, there is a graded normalizer with base $Q_s(\eta)$ on $\mathfrak{L}(g)$.

The normalizers in the last two cases satisfy

$$|\partial_1 \cdots \partial_n E(0)| \le (n/\varrho)^n \mathsf{U}(R, d) \prod_q \|f_q\|_\sharp, \tag{1843}$$

with R replaced by η in the third case and $d = \min_{q \ne q'} \operatorname{dist}(\operatorname{supp} f_q, \operatorname{supp} f_{q'}) > 0$. ∎

Proof. For a board of side l, choose the reference side ℓ' to be the least member of $\mathfrak{L}(g)$ with $\ell' \ge 2l$. The source system on $T_{\ell'}$ is supplied by theorem AD at radius 2ϱ. The same theorem gives $Z_{K,\ell,\beta} \ge e^{E_\ell - C_{\mathrm{low}} \ell^3}$ and $E_{\ell'} \ell'^{-3} - E_\ell \ell^{-3} \le C_{\mathrm{gap}}$ for every larger admissible side. These are precisely the floor and gap hypotheses of proposition 89.2. Periodization preserves the sharp norms of functions in a cell interior, so the parameter polydiscs agree on all these tori.

For each occupied cell construct $\mathcal{E}_k^\circ$ on the reference torus, as in (1788), and sum over the cells. This is one holomorphic function, independent of the larger side. If the supports lie in balls of radius R with wall margin D, then (1789) gives

$$|e^{-E(z)} \Pi_\ell(z) \le \exp\{A_0 l^3 + \Phi_1(R, D; l)\}^{\#B(z)}.$$

Each summand of E involves only the variables in its cell. Thus its full mixed derivative vanishes when at least two cells are occupied. When exactly one cell is occupied, (1791) gives (1843), initially with 2ϱ in place of ϱ.

For the second case take $D = R + 1$ and use $\Phi_1(R, R+1; l) \le \Phi_1(R, R+1; M)$. For the third take $l = 1$ and $R = D = \eta$. Directly from the formula for Φ_1,

$$\Phi_1(\eta, \eta; 1) = \Phi_1(1/2, 1/2; 1) + p \log(1/(2\eta)),$$

so the base is $Q_s(\eta)$. In the first case take the single carrier $x_0 + [-2, 2]^3$, with wall margin one, and use the single-carrier estimate of proposition 89.1. Its error is $\Phi_1(2, 1; L)$ once for the carrier; it has no factor equal to the number of entries. This gives Q_1. All admissible sides are at least $2L$, so the first and third cases cover the whole of $\mathfrak{L}(g)$.

Finally restrict these functions from $\mathbb{D}_{2\varrho}(\underline{f})$ to $\mathbb{D}_{\varrho}(\underline{f})$. All inequalities remain valid, and the derivative estimate only becomes weaker. Each indicated set of sides is unbounded, which proves the assertion about periodic states. □

Define

$$C_1^g(R) = 2\log 2\,(2\mathsf{Q}_1+2)\varrho^{-1}(1+40\sqrt{3}M)(2R+3)^3, \tag{1844}$$

$$C_n^{\mathrm{per}}(d;R) = (n/\varrho)^n\left[\log 2\,(2Q^\natural(n,R)^n+2)^n + \mathsf{U}(R,d)\right], \tag{1845}$$

$$C_B = \max\{28^3\cdot 961\cdot 2e^5(8e)^p C_Q M/\varrho,\ 384\cdot 64^3 e^4\}, \tag{1846}$$

$$A^{\mathrm{per}}(d) = 1+\mathsf{U}(1,d), \qquad B^{\mathrm{per}}(R) = C_B(R+1)^3, \tag{1847}$$

$$C_n^g(d;R) = \min\{C_n^{\mathrm{per}}(d;R), A^{\mathrm{per}}(d)B^{\mathrm{per}}(R)^n(n!)^{5+p}\}. \tag{1848}$$

Proposition 91.13. For every $(K,\beta)\in\mathfrak{d}(g)$ and every two periodic Gibbs states ν,ν', a real Lipschitz function supported in a ball of radius R satisfies

$$|\nu(\mathcal{O}_K(f)) - \nu'(\mathcal{O}_K(f))| \le C_1^g(R)\|f\|_\sharp. \tag{1849}$$

For $n\ge 2$ and a tuple of type $(d;R)$,

$$|S_n^T[\nu](\underline{f})| \le C_n^{\mathrm{per}}(d;R)\prod_q \|f_q\|_\sharp, \tag{1850}$$

$$|S_n^T[\nu](\underline{f})| \le A^{\mathrm{per}}(d)B^{\mathrm{per}}(R)^n(n!)^{5+p}\prod_q \|f_q\|_\sharp. \tag{1851}$$

The one-point comparison remains true when either or both laws are periodically extended torus measures with side in $\mathfrak{L}(g)$. The two cumulant bounds hold for the torus measure if all supports lie in $(-\ell/4,\ell/4)^3$; the first also requires $\ell\ge 2h(n,R)$, equivalently $\ell\ge 16n(R+1)$, whereas the second has no additional side restriction. All constants are independent of cutoff, state, side and position of the supports. ■

Proof. We first record the placement step. Points $x_1,\dots,x_n$ can be put at distance at least $r+D$ from the walls of a board of side l by an offset in $a_K\mathbb{Z}^3$ whenever

$$2n(r+D)+na_K\le l.$$

This is the offset lemma used in the proof of proposition 89.4: in each coordinate the forbidden offsets are n intervals of total length at most $2n(r+D)$; the extra na_K allows their endpoints to be rounded to the lattice. For one ball of radius two use $l=L\ge 13$, $r+D=3$. For n balls of radius R use $l=h(n,R)\ge 8n(R+1)$ and $r+D=2R+1$. For n balls of radius η use $l=1$ and $r+D=2\eta$, provided $4n\eta+na_K\le 1$.

For functions in one ball of radius two, the first normalizer in proposition 91.12 and the two-law part of proposition 91.11 give

$$|S_n^T[\nu](\underline{f}) - S_n^T[\nu'](\underline{f})| \le 2\log 2\,(2\mathsf{Q}_1+2)^n(n/\varrho)^n\prod_q\|f_q\|_\sharp.$$

Only one cell is occupied, so this holds as well for any two of the stated torus or periodic Gibbs laws. For a function supported in a ball of radius R, decompose it using the unit-mesh tensor-product tents. There are at most $(2R+3)^3$ nonzero pieces, each supported in a ball of radius $\sqrt{3}<2$ and with sharp norm at most $(1+40\sqrt{3}M)\|f\|_\sharp$. Use the preceding estimate at $n=1$ and sum. This proves (1849).

For a separated tuple of type $(d;R)$, place the balls on the board of side $h(n,R)$. Apply the second normalizer with its ungraded bound $Q^\natural(n,R)^n$. Adding (1843) to (1839) proves (1850).

For the growth estimate first suppose $a_K>1/(32n)$. Counting plaquettes in a radius-R ball gives

$$\|\mathcal{O}_K(f_q)\|_\infty \le 6a_K^{-1}(2R/a_K+1)^3\|f_q\|_\infty \le 192\cdot 64^3 n^4(R+1)^3\|f_q\|_\sharp.$$

The bounded-variable estimate of lemma 87.2 and $n^{4n} \le e^{4n}(n!)^4$ give

$$|S_n^T[\mathfrak{s}](\underline{f})| \le [384 \cdot 64^3 e^4 (R+1)^3]^n (n!)^5 \prod_q \|f_q\|_\sharp$$

for every law $\mathfrak{s}$. This is bounded by (1851).

Now suppose $a_K \le 1/(32n)$. Put $\eta = 1/(8n)$ and $\varepsilon = \eta/\sqrt{3}$. Decompose each f_q into tents of mesh ε. Each piece has support in a ball of radius η, contained in the original support, and norm at most $(1 + 960Mn)\|f_q\|_\sharp$. There are at most $(16\sqrt{3}nR + 3)^{3n}$ tuples of pieces. For each such tuple, $4a_K \le \eta$ and $4n\eta + na_K \le 1$, so the side-one board and the third normalizer are available. Its graded transfer bound gives

$$|S_n^T[\nu](v_1, \dots, v_n)| \le \varrho^{-n} \left[(n-1)!(2^n - 1)(eQ_s(\eta))^n + n^n \mathsf{U}(\eta, d)\right] \prod_q \|v_q\|_\sharp.$$

Use $Q_s(\eta) \le C_Q(8n)^p$, $\mathsf{U}(\eta, d) \le \mathsf{U}(1, d)$ and sum the pieces. To see the resulting exponent explicitly, bound the number of pieces by $[28n(R+1)]^{3n}$ and their norm factor by $(961Mn)^n$. These account for four powers of $n!$, using $n^n \le e^n n!$. The first term in the square brackets is bounded by $[2eC_Q(8e)^p]^n (n!)^{1+p}$; the second is bounded by $e^n (n!)^{1+p} \mathsf{U}(1, d)$. The first term in C_B dominates both resulting constants. This proves (1851).

For the torus assertions use proposition 91.11. Distinct cells meeting $(-\ell/4, \ell/4)^3$ are incongruent modulo the torus period whenever $\ell \ge 2l$: if their indices differed by a nonzero period, two points in these cells would have some coordinate difference at least $\ell - l \ge \ell/2$. Tent pieces retain this central-cube support. Thus the preceding proofs apply to the torus at the asserted side restrictions; the coarse-lattice bound already applies to every law. Finally, for $\ell = 2L^m$, the definition of $h(n, R)$ makes $\ell \ge 2h(n, R)$ equivalent to $\ell \ge 16n(R+1)$. □

92. Continuum limits on separated tuples

The bounds just proved concern compact supports of arbitrary position. They therefore give a common subsequence for all separated test tuples. They do not select a unique continuum limit on $\mathbb{R}^3$.

For $(K, \beta) \in \mathfrak{d}(g)$ the physical label $a_K\beta$ lies in the compact interval

$$I^{\mathrm{lab}}(g) = [g^{-2} + B_\sharp - L^{-1}\bar{B},\ L(g^{-2} + \bar{B})] \subset (0, \infty). \tag{1852}$$

This follows from proposition 74.1. Moreover corollary 74.2 shows that for every $b \in I(g) = (g^{-2} + \bar{B}, X_g - \bar{B}]$, the pair $(K, L^K b)$ lies in $\mathfrak{d}(g)$ for every $K \ge 1$.

A *limit along growing tori* is a family $S = (S_n)_{n \ge 2}$ obtained from $(K_j, \beta_j) \in \mathfrak{d}(g)$, $K_j \to \infty$, and $\ell_j \in \mathfrak{L}(g)$, $\ell_j \to \infty$, by

$$S_n(X) = \lim_j S^T_{n; K_j, \ell_j, \beta_j}(X^{\langle \ell_j \rangle}) \qquad (X \in \mathfrak{S}_n(\mathbb{R}^3)).$$

A *limit through periodic states* replaces these cumulants by $S_n^T[\nu_j](X)$ with $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j, \beta_j}$. If each $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{K_j, \beta_j}$, call it a *volume-first limit*. A defining sequence has label b when $a_{K_j}\beta_j \to b$. Labels describe defining sequences; no uniqueness of a label from the limit family is assumed.

Proposition 92.1. Every growing-torus sequence as above, and every sequence through periodic states, has a subsequence on which all cumulants of order at least two converge simultaneously on $\mathfrak{S}(\mathbb{R}^3)$. One can also require the labels to converge. The limit is real, symmetric and multilinear and satisfies

$$|S_n(X)| \le C_n^g(d; R) \prod_q \|f_q\|_\sharp \quad \text{for every tuple of type } (d; R). \tag{1853}$$

Limits along growing tori and volume-first limits exist with every label $b \in I(g)$. All these limit families lie in the common compact metrizable set of symmetric multilinear families satisfying (1853), with the topology of pointwise convergence on separated tuples.

More generally, for arbitrary periodic states $\rho_j, \nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ one can choose the same subsequence so that

$$\delta_j(f) = \rho_j(\mathcal{O}_{K_j}(f)) - \nu_j(\mathcal{O}_{K_j}(f)) \longrightarrow \delta(f), \qquad \sigma_{\rho_j,\nu_j}(X) \longrightarrow \sigma(X)$$

for all compactly supported C^1 functions f and all separated tuples X. Here δ is real linear, $|\delta(f)| \le C_1^g(R)\|f\|_\sharp$ on radius-R balls, and $S_n^\sigma = S_n$ for $n \ge 2$, while $S_1^\sigma = \delta$. ■

Proof. For growing tori take the domains $O_j = (-\ell_j/4, \ell_j/4)^3$. Every compact set is eventually contained in O_j. For fixed n, R, the side condition $\ell_j \ge 2h(n,R)$ also holds eventually, independently of the position of a tuple in O_j. Thus proposition 91.13 gives the bound table $C_n^g(d;R)$ required by proposition 90.1. For periodic states use the constant domains $\mathbb{R}^3$ and the same table at every index. Include the bounded sequence $a_{K_j}\beta_j$ in the extraction. This proves simultaneous convergence and inheritance of (1853).

For the last assertion include δ_j as the form of order one, with the table $C_1(d;R) = C_1^g(R)$ supplied by (1849). The moment–cumulant identity is

$$\sigma_{\rho_j,\nu_j}(X) = \sum_{\pi\in\mathrm{Part}(n)} \prod_{\{q\}\in\pi} \delta_j(f_q) \prod_{B\in\pi,\,|B|\ge 2} S^T_{|B|}[\rho_j](X_B). \tag{1854}$$

This finite polynomial passes to the limit on each tuple. The inverse partition identity gives its asserted cumulants. If $\rho_j = \nu_j$, this limit is exactly the centered moment functional σ_S of (1811).

For existence at a prescribed $b \in I(g)$, choose $K_j = j$, $\beta_j = L^j b$, and either any admissible sides tending to infinity or any $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{j,L^j b}$. The latter sets are nonempty by proposition 91.2. Extraction preserves the label b.

The compactness assertion is proposition 90.2 applied to the table $C_n^g(d;R)$. □

92.1. **Comparison of the orders of limits.**

Proposition 92.2. Every volume-first limit is a limit along growing tori with the same pairs (K_j,β_j) and the same label. The sides can be chosen above any prescribed sequence λ_j. Conversely, a convergent growing-torus sequence with $\ell_j \ge \ell_\flat(K_j,\beta_j)$ has the same limit and label as a sequence through torus-limit states at the same pairs, without passing to a further subsequence. ■

Proof. Fix K, β and $\nu \in \mathcal{G}^{\mathrm{tor}}_{K,\beta}$, and take a sequence $\ell_i \to \infty$ whose periodic laws converge to ν. For a fixed ball $\mathbb{B}(0,r)$ there are only finitely many plaquette centers in it. Weak convergence gives convergence of every joint cumulant of these finitely many plaquette variables, with repetitions allowed. For $n \le N$, expand

$$S_n^T[\mathfrak{s}](\underline{f}) = a_K^{-n} \sum_{p_1,\dots,p_n} \prod_q f_q(x(p_q))\langle a_{p_1};\dots;a_{p_n}\rangle_{\mathfrak{s}}.$$

The finite sums show that for all bounded functions supported in this ball the torus–state difference is at most $\epsilon\prod_q \|f_q\|_\infty$ once i is large enough, uniformly for $n \le N$.

At the jth pair choose $N = r = j$ and $\epsilon = 1/j$, and choose ℓ_j from a sequence defining ν_j with $\ell_j \ge \max\{\lambda_j, j, 2L^{m_{\mathfrak{L}}(g)}\}$. The resulting torus–state difference tends to zero for every fixed tuple, along the full sequence of indices. This proves the first assertion and retains the label because the pairs are unchanged.

For the converse choose the state ν_j of proposition 91.10 for each j. For a fixed tuple supported in $[-r,r]^3$, eventually $r \le K_j - 1$, so (1835) bounds the difference by a constant times $e^{-L^{K_j}} L^{4nK_j}$. This tends to zero. The one chosen state works for all tuples, and the pairs again remain unchanged. □

92.2. **Translation invariance and the one-point constant.**

Uniform bounds permit the test functions to move with the cutoff. More precisely, suppose all entries $f_q^{(j)}$ and f_q lie in fixed pairwise

disjoint compact sets and $\|f_q^{(j)} - f_q\|_\sharp \to 0$. Telescoping one entry at a time and using the common bound table gives

$$\Phi_n^{(j)}(f_1^{(j)}, \dots, f_n^{(j)}) \longrightarrow \Phi_n(f_1, \dots, f_n). \tag{1855}$$

The norms of the moving entries remain bounded. The same conclusion holds for centered moments, by their finite partition formula, and for the two-state moments in (1854) using the order-one bound. Enlarging the fixed compact supports slightly gives a useful instance: if $v_j \to v$, then $\tau_{v_j} f \to \tau_v f$ in the sharp norm for $f \in C_c^1$, since f and its first derivatives are uniformly continuous.

Proposition 92.3. Every growing-torus limit and every volume-first limit has $\mathbb{R}^3 \rtimes W_3$-invariant cumulants and centered moments. The same holds for a limit through states whenever both the integrating and centering laws are invariant under $a_{K_j}\mathbb{Z}^3 \rtimes W_3$.

In the two-state case the one-point function is $\sigma(f) = c \int f$, for a constant $c \in \mathbb{R}$. Its higher cumulants are S_n and its centered functional is σ_S. If the states obey (1849), then

$$|c| \le \frac{120M}{\pi} C_1^g(1). \tag{1856}$$

■

Proof. For $v \in \mathbb{R}^3$ choose $v_j \in a_{K_j}\mathbb{Z}^3$ with $v_j \to v$. The finite-cutoff cumulants and, when applicable, two-state moments are invariant under τ_{v_j}. Use (1855) to pass to τ_v. In the growing-torus case all supports and their small translates lie in the central cube eventually. The signed coordinate permutations in W_3 preserve every lattice, so their invariance passes directly to the limit. Torus-limit states have these lattice invariances by proposition 91.2.

For the one-point function, lattice isometries act transitively on plaquettes. Thus

$$c_j = 3a_{K_j}^{-4}(\rho_j(a_p) - \nu_j(a_p))$$

does not depend on p, and exactly

$$\delta_j(f) = c_j r_j(f), \qquad r_j(f) = \frac{a_{K_j}^3}{3} \sum_p f(x(p)). \tag{1857}$$

For each orientation its centers form a translate of $a_{K_j}\mathbb{Z}^3$. Partition space into half-open cubes of side a_{K_j} centered there. On each cube the pointwise error from its center is at most $(\sqrt{3}/2)a_{K_j} \operatorname{Lip} f$. Only cubes in the closed $\sqrt{3}a_{K_j}$-neighborhood $N_{a_{K_j}}(f)$ of the support contribute. Averaging over the three orientations gives

$$|r_j(f) - \int f| \le (\sqrt{3}/2)a_{K_j} \operatorname{Lip} f \, |N_{a_{K_j}}(f)| \to 0. \tag{1858}$$

Choose one f_* of integral one. Convergence of $\delta_j(f_*)$ and $r_j(f_*) \to 1$ gives $c_j \to c = \delta(f_*)$; then (1857) proves $\delta(f) = c \int f$ for every f. The moment formula (1854) now identifies σ as the shift of σ_S by this constant.

For completeness, the best sharp norm of a unit-integral C^1 function supported in the open unit ball is $120M/\pi$. Indeed if $\lambda = \operatorname{Lip} f$, then $|f(x)| \le \lambda(1 - |x|)_+$ and $\int (1 - |x|)_+ \, dx = \pi/3$, so $\|f\|_\sharp \ge 40M\lambda \ge 120M/\pi$. Conversely, mollify $(r - |x|)_+$ inside a ball of radius less than one and normalize by its integral $\pi r^4/3$. The resulting functions have sharp norm at most $120M/(\pi r^4)$; let $r \uparrow 1$. Applying (1849) to these functions and passing to the limit proves (1856). □

92.3. **Reflection positivity in the limit.** For a reflection θ in a plane and its open positive half-space, reflection positivity of a functional means

$$\sum_{i,k} \overline{z_i} z_k \, \sigma(\theta X_i \cdot X_k) \ge 0 \tag{1859}$$

for every finite collection of separated tuples X_i supported in that half-space. The concatenated tuples here are separated: the reflected and unreflected supports lie on opposite sides.

Proposition 92.4. The centered functional of every growing-torus limit or volume-first limit is reflection positive at every coordinate plane, from either side. The same holds for a two-state limit as in proposition 92.3 if its integrating laws are eventually reflection positive at one coordinate site mirror, while its centering laws are invariant under the lattice isometries.

If, in addition, the integrating laws are eventually reflection positive at the diagonal lattice mirror $x_1 = x_2$, then the limit functional is reflection positive at every plane with normal $(\pm e_\mu \pm e_\nu)/\sqrt{2}$, $\mu < \nu$, from either side. ■

Proof. At a lattice mirror let X_i be supported strictly on its positive side. A plaquette centered in this side is contained in its closure: for a coordinate mirror this follows from the half-lattice spacing of the centers. For a diagonal mirror, plaquettes spanning both directions in its normal have center height in $a\mathbb{Z}$ and height variation a from center to corner; those spanning only one have center height in $a(\mathbb{Z} + 1/2)$ and height variation $a/2$. In either case a strictly positive center has nonnegative corner heights. Therefore the functions $(X_i)_a^\nu$ depend only on positive-half bonds. Reflection invariance of the centering law gives $\theta(X_i)_a^\nu = (\theta X_i)_a^\nu$. Lattice reflection positivity applied to $F = \sum_i z_i (X_i)_a^\nu$ yields (1859) at finite cutoff.

For a growing torus use the coordinate plane $x_\mu = a_1/2$. This is a bond-midpoint mirror at every cutoff: $a_1/a_K = L^{K-1}$ is odd. Every fixed positive-side support fits in its positive torus half-slab for sufficiently large side, and the exact periodization identity identifies the finite moments with the lattice expressions. Their limits therefore satisfy (1859) at this plane. For a volume-first sequence use the site mirror $x_\mu = 0$, where the states are reflection positive by proposition 91.5. The same argument applies to the assumed coordinate mirror in the general two-state case. Translation invariance and the action of W_3 then give all coordinate planes and both sides.

For the diagonal assertion use $\theta(x_1, x_2, x_3) = (x_2, x_1, x_3)$, which is a lattice isometry at every cutoff. The centering laws are invariant under it. Apply the finite-cutoff argument using the assumed diagonal positivity of the integrating law and pass to the limit in the finite sum. Translations and signed coordinate permutations give all stated diagonal planes. □

The diagonal hypothesis in this proposition is additional. The cubic-period construction of $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ has so far supplied only coordinate reflection positivity. The adapted-period comparison argument below will supply the diagonal assertion for the continuum functionals to which it applies.

The same limit argument admits moving mirrors. If $c_j \in (a_{K_j}/2)\mathbb{Z}$, $c_j \to c$, and the integrating laws are reflection positive at $x_\mu = c_j$, use $\theta_j = \tau_{2(c_j - c)e_\mu}\theta$. Fixed positive-side supports eventually stay positive for θ_j; their reflected supports fit in disjoint compact neighborhoods of their limiting supports. Equation (1855) passes the finite positive matrix to the limit. More generally, any inequality involving finitely many such moment evaluations passes to the limit whenever its admissible values form a closed set.

93. Distributional extensions and growth

The bounds on separated tests define distributions away from coincident points. Their power dependence on the separation also controls Schwartz functions flat on the diagonals. We prove both facts by one local Fourier construction. The dyadic partition used below keeps the supports of the individual Fourier factors separated; flatness then makes its fine-scale sum converge.

Write $E^n_{\neq} = \{(x_1, \dots, x_n) : x_i \neq x_j \ (i \neq j)\}$, where $E = \mathbb{R}^3$ or T_ℓ, and set $m(x) = \min_{i \neq j} |x_i - x_j|_\infty$, with $m = \infty$ for $n = 1$. We complexify multilinear forms by complex linearity in each entry. Thus a bound by $c \prod_i \|f_i\|_\sharp$ on real entries gives the bound $c \prod_i (\|\operatorname{Re} f_i\|_\sharp + \|\operatorname{Im} f_i\|_\sharp)$ on complex entries.

93.1. **A partition adapted to the diagonals.** Put $v(t)=e^{-1/t}$ for $t>0$ and $v(t)=0$ for $t\le 0$, $\vartheta(t)=v(1-16t^2)$, and $c=\int\vartheta$. The two cutoffs

$$\chi(t)=c^{-1}\int_{t-1/2}^{t+1/2}\vartheta(s)\,ds,\qquad \widetilde{\chi}(t)=c^{-1}\int_{t-1}^{t+1}\vartheta(s)\,ds \tag{1860}$$

have values in $[0,1]$. The support of χ is $[-3/4,3/4]$, $\sum_{z\in\mathbb{Z}}\chi(t-z)=1$, and at most two summands are nonzero. The second cutoff equals one on $[-3/4,3/4]$, vanishes outside $[-5/4,5/4]$, and satisfies $|\widetilde{\chi}'|\le 16$. We shall use

$$|v^{(k)}|\le 8^k(k!)^2,\qquad |\vartheta^{(k)}|\le 200^k(k!)^2,\qquad |\chi^{(k)}|\le 200^k(k!)^2. \tag{1861}$$

Here is the derivative estimate. On $t>0$, $v^{(k)}(t)=P_k(1/t)e^{-1/t}$, where $P_{k+1}(s)=s^2(P_k(s)-P_k'(s))$ and $P_0=1$. The weighted coefficient sum $A_k=\sum_j|[s^j]P_k|j!$ satisfies $A_{k+1}\le 2(2k+1)^2A_k\le 8(k+1)^2A_k$; also $s^je^{-s}\le j!$. All derivatives tend to zero at $t=0$, so this proves the first bound globally. For the second, the chain rule gives

$$\vartheta^{(k)}(t)=\sum_{0\le 2j\le k}\frac{k!}{j!(k-2j)!}(-32t)^{k-2j}(-16)^jv^{(k-j)}(1-16t^2).$$

On $|t|<1/4$, use $|32t|\le 8$, $k!/(j!(k-2j)!)\le 3^kj!$, and $j!((k-j)!)^2\le(k!)^2$. The resulting sum is bounded by $(32/31)192^k(k!)^2\le 200^k(k!)^2$ for $k\ge 1$. Outside this interval it vanishes. Finally $c\ge\frac14e^{-4/3}>1/16$, and differentiating (1860) proves the last bound and the stated properties of the cutoffs.

Let $h_q=2^{-q}/16$, $q\ge 0$. On $\mathbb{R}^3$ use the grid $h_q\mathbb{Z}^3$; on T_ℓ use its quotient and periodize the cutoffs. Here $\ell=2L^m$, $m\ge 0$, so ℓ/h_q is an integer and the cutoff supports embed in the torus. Define

$$\chi_{q,z}(y)=\prod_{\mu=1}^{3}\chi(y_\mu/h_q-z_\mu),\qquad \widetilde{\chi}_{q,z}(y)=\prod_{\mu=1}^{3}\widetilde{\chi}(y_\mu/h_q-z_\mu),$$

with the periodized interpretation on the torus. Let $Q_{q,z}$ and $\widetilde{Q}_{q,z}$ be the closed cubes of sup radii $3h_q/4$ and $5h_q/4$ about h_qz. An index tuple $\mathbf{z}=(z_1,\dots,z_n)$ is good when $D(\mathbf{z}):=\min_{i\ne j}|z_i-z_j|_\infty\ge 6$. Write $\chi_{q,\mathbf{z}}=\bigotimes_i\chi_{q,z_i}$ and

$$G_q=\sum_{\mathbf{z}\text{ good}}\chi_{q,\mathbf{z}},\qquad G_{-1}=0,\qquad \Upsilon_q=G_q(1-G_{q-1}). \tag{1862}$$

The enlarged cubes in a good tuple have mutual distance at least $(D(\mathbf{z})-5/2)h_q\ge 7h_q/2$ and each lies in a ball of radius less than one. The grid identities give

$$G_q=0\quad(m\le 9h_q/2),\qquad G_q=1\quad(m\ge 15h_q/2), \tag{1863}$$

$$G_qG_{q-1}=G_{q-1},\qquad \Upsilon_q=G_q-G_{q-1}\ge 0,\qquad \sum_{q\ge 0}\Upsilon_q=1\quad\text{on } E^n_{\ne}. \tag{1864}$$

Indeed, a nonzero grid factor has $|x_i-h_qz_i|_\infty<3h_q/4$, so the distance of two centres differs from that of their points by less than $3h_q/2$. This proves the two plateaux. If $G_{q-1}\ne 0$, then $m>9h_q$, which proves the product identity. In particular, for $q\ge 1$,

$$\operatorname{supp}\Upsilon_q\subset\{9h_q/2\le m\le 15h_q\}.$$

The partition is locally finite on $E^n_{\ne}$.

For a pure derivative in any coordinate, the product rule and (1861) yield

$$|\partial_{i\mu}^k(\chi_{q,\mathbf{z}}(1-G_{q-1})F)|\le 2\cdot 600^k(k!)^2\max_{0\le j\le k}h_q^{j-k}|\partial_{i\mu}^jF|. \tag{1865}$$

For the derivatives of G_q, only two terms in the differentiated coordinate can contribute; the sums in every other coordinate are one. Thus $|\partial_{i\mu}^k G_q| \le 2h_q^{-k}200^k(k!)^2$. For mixed derivatives of a sum of any subfamily of the grid tensors, the corresponding bound is $2^{3n}h_q^{-|\alpha|}200^{|\alpha|}(|\alpha|!)^2$.

One count will supply summability over positions. For $q \ge 1$ and each unit box $\mathbf{w}+[0,1)^{3n}$, there are at most

$$\binom{n}{2}\,33^3 h_q^{-(3n-3)} \tag{1866}$$

good $\mathbf{z}$ with $h_q\mathbf{z}$ in that box and $Q_{q,\mathbf{z}}$ meeting $\operatorname{supp}\Upsilon_q$. At an intersection point some pair has distance at most $15h_q$; the two centres therefore differ by at most $16h_q$ in each coordinate. Choose the pair, the other $n-1$ centres, and one of 33^3 differences. At $q=0$ the unrestricted count is 16^{3n}.

93.2. **The local Fourier construction.**

Proposition 93.1 (Kernel on separated points)**.** Let T be a real multilinear form on separated smooth compactly supported n-tuples over E. Suppose that for every $d>0$ there is $c_T(d)<\infty$ such that

$$|T(f_1,\dots,f_n)| \le c_T(d)\prod_i \|f_i\|_\sharp \tag{1867}$$

whenever the supports have mutual distance at least d and each lies in a ball of radius one. There is a unique distribution $\widetilde{T}$ on $E^n_{\neq}$ such that $\widetilde{T}(\bigotimes_i f_i)=T(f_1,\dots,f_n)$. It has order at most $4n+1$ on compact sets, with a bound depending only on that compact set, M, and the function c_T. Symmetries of T under permutations or isometries are inherited by $\widetilde{T}$. ∎

Proof. Fix a good box at scale q and put $h=h_q$, $\lambda=5h/2$. Lift its enlarged cubes to cubes $C_i\subset\mathbb{R}^3$ of side λ. For $\omega\in\mathbb{Z}^3$ let $e_\omega(t)=e^{2\pi i\omega\cdot t/\lambda}$ and let $E_{i,\omega}$ be $\widetilde{\chi}_{q,z_i}e_\omega$, interpreted through the lift and periodized if necessary. For a smooth F supported in $Q_{q,\mathbf{z}}$, extend its lift by zero and set

$$\widehat{F}(\omega)=\lambda^{-3n}\int_{\prod_i C_i} F(t)\prod_i \overline{e_{\omega_i}(t_i)}\,dt, \qquad [F]_{q,k}=\max\left\{\|F\|_\infty,(\lambda/2\pi)^k\max_{i,\mu}\|\partial_{i\mu}^k F\|_\infty\right\}.$$

Integration by parts in a coordinate attaining $|\omega|_\infty$ gives

$$|\widehat{F}(\omega)|\le 2^k(1+|\omega|_\infty)^{-k}[F]_{q,k}.$$

The real and imaginary parts of $E_{i,\omega}$ have sharp norm at most $1109Mh^{-1}(1+|\omega|_\infty)$: the cutoff gradient is at most $16\sqrt{3}/h$, and differentiating the exponential adds at most $16\sqrt{3}|\omega|_\infty/h$. Also, grouping by $|\omega|_\infty$ gives $\sum_{\omega\in\mathbb{Z}^{3n}}(1+|\omega|_\infty)^{-3n-1}\le 3n8^n$. With $k=4n+1$ these bounds prove absolute convergence of

$$T_{q,\mathbf{z}}(F)=\sum_{\omega\in(\mathbb{Z}^3)^n}\widehat{F}(\omega)T(E_{1,\omega_1},\dots,E_{n,\omega_n}) \tag{1868}$$

and the estimate

$$|T_{q,\mathbf{z}}(F)|\le(2\cdot10^6)^n M^n c_\square h^{-n}[F]_{q,4n+1}. \tag{1869}$$

Here $c_\square$ can be either $c_T(7h/2)$ or $c_T((D(\mathbf{z})-5/2)h)$, whenever the argument is finite; for $n=1$ any valid uniform one-entry bound serves. The numerical factor before $M^n c_\square h^{-n}[F]$ is $6n\,283904^n\le(2\cdot10^6)^n$. Changing a lift multiplies the entries by phases and the Fourier coefficient by their inverse product; hence (1868) is independent of the lifts.

The Fourier partial sums, multiplied by $\bigotimes_i\widetilde{\chi}_{q,z_i}$, converge to F with all derivatives and have support in $\widetilde{Q}_{q,\mathbf{z}}$. For completeness, integration by parts makes the Fourier series of every derivative absolutely summable. Its sum is that derivative: the product Fejér kernels are nonnegative approximate identities, so their convolution converges uniformly to a continuous periodic function and also to the sum of an absolutely convergent Fourier series. The cutoff equals one on the support of F, which proves the claimed convergence after multiplication. If $F=\bigotimes_i f_i$ with $\operatorname{supp} f_i\subset Q_{q,z_i}$,

its coefficients factor. The partial sums therefore give T of the individual Fourier approximations; their convergence in the sharp norm and the telescoping estimate prove $T_{q,\mathbf{z}}(F) = T(f_1, \dots, f_n)$.

Define

$$\widetilde{T}(F) = \sum_{q \ge 0} \sum_{\mathbf{z}\ \text{good}} T_{q,\mathbf{z}}(\chi_{q,\mathbf{z}}(1 - G_{q-1})F). \tag{1870}$$

For F in a fixed compact subset of $E_{\neq}^n$, only finitely many scales and positions occur. Combining (1865) with (1869) gives the asserted distribution bound of order $4n + 1$. For a tensor $F = \bigotimes_i f_i$, each piece in (1870) is the finite sum

$$\bigotimes_i (\chi_{q,z_i} f_i) - \sum_{\mathbf{z}'\ \text{good at } q-1} \bigotimes_i (\chi_{q,z_i}\chi_{q-1,z_i'} f_i),$$

where the second sum is absent at $q = 0$. All entries stay inside the original separated supports. Multilinearity and (1864) therefore give $\widetilde{T}(F) = T(f_1, \dots, f_n)$. In making this computation, the linear form induced on finite sums of tensors is well defined: on the finite-dimensional spans of their factors, point evaluations span the duals, so a tensor representing the zero function has all its coefficients zero.

Finally, the same partition and the local Fourier approximation approximate every test in $C_c^\infty(E_{\neq}^n)$ by finite sums of separated tensors, in all derivatives with supports in one compact subset of $E_{\neq}^n$. This proves uniqueness. Applying uniqueness to the pullback by a symmetry proves the last assertion. □

If the form is originally defined on separated C_c^1 tuples with the same bounds, these values are determined by its kernel. Indeed, cut each entry by the $q = 0$ partition, mollify the finitely many pieces, and let the mollifier radius tend to zero. Their supports retain positive separation, their sharp norms converge, and the telescoping bound recovers the original value.

93.3. **Flat tests and the global bound.** For $\varphi \in \mathcal{S}(\mathbb{R}^{3n})$ put

$$|\varphi|_k = \sup_{x,\, |\alpha| \le k} (1 + |x|^2)^{k/2} |\partial^\alpha \varphi(x)|. \tag{1871}$$

Let $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ consist of Schwartz functions all of whose derivatives vanish whenever two entries coincide. Let $\mathcal{S}_+$ be the subspace additionally flat outside $\{x_{i3} > 0 \text{ for all } i\}$.

Lemma 93.2. Finite sums of tensors of separated smooth compactly supported tests are dense in $\mathcal{S}_{\neq}$ for every norm $|\cdot|_k$. In $\mathcal{S}_+$ the approximating entries may all be supported in the open positive half-space. For $k \ge 4$ and $A \subset \{1, \dots, n\}$, partial integration $\psi_A(y) = \int \varphi(y, x_A)\, dx_A$ maps $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ into $\mathcal{S}_{\neq}(\mathbb{R}^{3(n-|A|)})$ and satisfies

$$|\psi_A|_{k(n-|A|)} \le C_{\int}(k)^{|A|} |\varphi|_{kn}, \qquad C_{\int}(k) = \int_{\mathbb{R}^3} (1 + |x|^2)^{-k/2}\, dx. \tag{1872}$$

In particular $C_{\int}(4) = \pi^2$ and $C_{\int}(10) = 5\pi^2/64$. Also $|\varphi \otimes \psi|_k \le |\varphi|_k |\psi|_k$. If both factors belong to $\mathcal{S}_{\neq}$ and are flat outside the negative and positive half-spaces respectively, their tensor is flat on every diagonal. ■

Proof. Flatness gives the following Taylor estimate. If φ is flat at x^*, $|x - x^*| \le r$ with $r \ge 1$, and $N \ge |\alpha| + s$, then

$$|\partial^\alpha \varphi(x)| \le \frac{|x - x^*|_1^s}{s!} (1 + 2r^2)^{N/2} |\varphi|_N (1 + |x|^2)^{-N/2}. \tag{1873}$$

Apply Taylor's formula with integral remainder to $t \mapsto \partial^\alpha \varphi(x^* + t(x - x^*))$; along this segment $1 + |x|^2 \le (1 + 2r^2)(1 + |x^* + t(x - x^*)|^2)$. When $m(x)$ is small, replace a closest pair by its midpoint. This gives a diagonal point x^* with $|x - x^*|_1 \le 3m(x)$ and $|x - x^*| \le \sqrt{3/2}\, m(x)$. In the half-space case one may instead replace a smallest third coordinate by its minimum with zero.

For the half-space partition restrict the good indices further by $z_{i3} \geq 6$, and denote the resulting sum by G_q^+. The plateau and derivative estimates above hold with $m^+(x) = \min\{m(x), x_{13}, \dots, x_{n3}\}$. Taylor's estimate with $s = k+1$, followed by the mixed derivative bound for the cutoffs, gives

$$|(1 - G_Q)\varphi|_k \leq C(n,k) h_Q |\varphi|_{2k+1},$$

and the same bound with G_Q^+ for $\varphi \in \mathcal{S}_+$. Indeed, the derivatives of the cutoff cost at most h_Q^{-k}, whereas Taylor's remainder contributes h_Q^{k+1}. Fix Q to make this error small. Among the remaining finitely many scales, discard grid centres outside a large box. Their derivatives are bounded with Q fixed, while one extra Schwartz weight makes their $|\cdot|_k$ contribution tend to zero. Only finitely many boxes remain. Approximate each by its local Fourier tensors as in the preceding proof. Their enlarged supports are separated, and in the half-space case have third coordinate at least $19h_q/4$. This proves both density assertions.

For partial integrals, use

$$(1 + |x|^2)^{kn/2} \geq (1 + |y|^2)^{k(n-|A|)/2} \prod_{i \in A} (1 + |x_i|^2)^{k/2}.$$

This permits differentiation under the integral, proves (1872), and preserves flatness on the remaining diagonals. Polar coordinates give the two stated constants. The tensor bound follows from $1 + |x|^2 + |y|^2 \leq (1 + |x|^2)(1 + |y|^2)$. At a diagonal joining the negative and positive groups, one factor and all its derivatives vanish; diagonals inside a group are already covered by its flatness. □

Proposition 93.3. Suppose that T satisfies (1867) on $\mathbb{R}^3$ with $c_T(d) = c_T \max\{1, d^{-a}\}$, where $a \in \mathbb{N}_0$. Its kernel has a unique extension to $\mathcal{S}_{\neq}$ bounded for a Schwartz norm. With $C_{\neq} = 10^{53}$,

$$|\widetilde{T}(\varphi)| \leq C_{\neq}^{n+a} M^n (n!)^8 c_T |\varphi|_{8n+a-1}. \tag{1874}$$

■

Proof. Use the series (1870), now summed over all scales and positions. Set $k = 4n+1$, $s = a+4n-2$, and $N = k + s$. For $F_{q,\mathbf{z}} = \chi_{q,\mathbf{z}}(1 - G_{q-1})\varphi$, (1865) and $\lambda/(2\pi) < 2h_q/5$ give

$$[F_{q,\mathbf{z}}]_{q,k} \leq 2 \cdot 240^k (k!)^2 \sup_{\operatorname{supp} F_{q,\mathbf{z}}} \max_{i,\mu,\, 0 \leq j \leq k} h_q^j |\partial_{i\mu}^j \varphi|. \tag{1875}$$

For $q \geq 1$, this support has $m \leq 15h_q$. The midpoint choice in (1873) has $|x - x^*|_1 \leq 45h_q$ and $|x - x^*| < 1$. Since $N \geq 4n$, it follows that

$$|\partial_{i\mu}^j \varphi(x)| \leq \frac{(45h_q)^s}{s!} 3^{N/2} |\varphi|_N \prod_{l=1}^{n} (1 + |x_l|^2)^{-2}, \qquad j \leq k.$$

To sum the positions, assign $h_q\mathbf{z}$ to its unit box $\mathbf{w} + [0,1)^{3n}$. Its support lies in $\mathbf{w} + [-1/20, 21/20]^{3n}$, and

$$\sum_{\mathbf{w}} \sup_{x \in \mathbf{w} + [-1/20, 21/20]^{3n}} \prod_{l} (1 + |x_l|^2)^{-2} \leq (81\pi^2)^n.$$

For one factor, compare a point in the enlarged unit box with a point in its unit cube through the common centre: the squared weights differ by at most 81. Integrate over that cube and sum; then take the product in the n entries. Combining this estimate, (1866), (1869), and (1875) bounds the sum at scale $q \geq 1$ by

$$\Sigma_q \leq (2 \cdot 10^6)^n M^n c_T (2/7)^a 2 \cdot 240^k (k!)^2 \frac{45^s}{s!} 3^{N/2} \binom{n}{2} 33^3 (81\pi^2)^n h_q^{s-a-n-(3n-3)} |\varphi|_N. \tag{1876}$$

The exponent of h_q is one. Thus the fine-scale sum converges.

At $q = 0$, split the centres into $D(\mathbf{z}) \geq 19$ and $6 \leq D(\mathbf{z}) \leq 18$. In the first group the enlarged supports are separated by more than one, so $c_\square = c_T$; use the Schwartz bound without Taylor's formula and the count 16^{3n}. In the second group a pair in the box has distance at most $39/32$ in

the sup norm. Its midpoint has $|x-x^{*}|_{1}\le 117/32$ and $|x-x^{*}|\le 3/2$; hence (1873) supplies the factor $(117/32)^{s}(11/2)^{N/2}/s!$, and $c_{\square}=c_{T}(32/7)^{a}$. These give, respectively,

$$\Sigma_{0}^{\mathrm{A}}\le (2\cdot 10^{6})^{n}M^{n}c_{T}16^{4n}2\cdot 240^{k}(k!)^{2}(81\pi^{2})^{n}|\varphi|_{N},$$

$$\Sigma_{0}^{\mathrm{B}}\le (2\cdot 10^{6})^{n}M^{n}c_{T}16^{4n}2\cdot 240^{k}(k!)^{2}(81\pi^{2})^{n}(32/7)^{a}\frac{(117/32)^{s}}{s!}(11/2)^{N/2}|\varphi|_{N}.$$

For $n=1$ the second group and every $q\ge 1$ contribution are empty. In general use $(k!)^{2}\le 25^{n}4^{8n}(n!)^{8}$, $(2/7)^{a}3^{N/2}\le 81^{n}$, and $45^{s}/s!\le 3^{45}$ in the fine-scale bound, with $\sum_{q\ge 1}h_{q}=1/16$. For the second coarse group use $(32/7)^{a}(11/2)^{a/2}\le 11^{a}\le 11^{s}$ and $(11\cdot 117/32)^{s}/s!\le 3^{41}$. The three sums, divided by $M^{n}(n!)^{8}c_{T}|\varphi|_{N}$, are at most $10^{28}(10^{28})^{n}$, $10^{3}(10^{30})^{n}$, and $10^{23}(10^{33})^{n}$. For $n\ge 2$ their sum is at most $3(10^{45})^{n}$; for $n=1$ the sole coarse sum is at most 10^{33}. Both are bounded by $C_{\neq}^{n}$.

The series is therefore absolutely convergent, linear, and satisfies the stronger bound with $C_{\neq}^{n}$ in place of $C_{\neq}^{n+a}$. On compactly supported tests away from the diagonals it is the previous finite sum. Uniqueness follows from the density in lemma 93.2. □

Proposition 93.4 (Growth of the Schwinger functions)**.** Suppose a symmetric family $S=(S_{n})_{n\ge 2}$ on separated C_{c}^{1} tuples satisfies

$$|S_{n}(f_{1},\dots,f_{n})|\le A\max\{1,d^{-a}\}B^{n}(n!)^{\varkappa}\prod_{i}\|f_{i}\|_{\sharp}$$

when the separation is at least d and each support lies in a ball of radius one, where $a\in\mathbb{N}$, $A,B\ge 1$, and $\varkappa\ge 0$. Its centred moments σ_{S} extend uniquely to functionals $\mathsf{S}_{n}^{\sigma_{S}}$ on $\mathcal{S}_{\neq}$, with $\mathsf{S}_{0}^{\sigma_{S}}=1$, $\mathsf{S}_{1}^{\sigma_{S}}=0$, and

$$|\mathsf{S}_{n}^{\sigma_{S}}(\varphi)|\le C_{c}^{n}(n!)^{\varkappa+9}|\varphi|_{kn},\qquad k=8+\lceil a/2\rceil,\qquad C_{c}=C_{\neq}^{1+a/2}MB\sqrt{A}.\tag{1877}$$

If a functional σ has these connected functions and one-point function $c\int f$, with $|c|\le\bar{c}$, the same conclusion holds with C_{c} replaced by $C=C_{c}+\bar{c}\,C_{\int}(k)$. ■

Proof. In the moment partition sum, there are at most $\lfloor n/2\rfloor$ blocks and no singleton blocks. Also $\prod_{J\in\pi}|J|!\le n!$, and the number of partitions is at most $n!$: each partition is the cycle partition of a permutation, choosing increasing cycles. Thus

$$|\sigma_{S}(f_{1},\dots,f_{n})|\le A^{\lfloor n/2\rfloor}\max\{1,d^{-a\lfloor n/2\rfloor}\}B^{n}(n!)^{\varkappa+1}\prod_{i}\|f_{i}\|_{\sharp}.$$

Apply proposition 93.3 with singularity order $a\lfloor n/2\rfloor$. The derivative order is $8n+a\lfloor n/2\rfloor-1\le kn$, and its bound is (1877). For $n=1$ the functional is zero. For an uncentred functional use the finite expansion

$$\mathsf{S}_{n}^{\sigma}(\varphi)=\sum_{A_{0}\subset\{1,\dots,n\}}c^{|A_{0}|}\mathsf{S}_{n-|A_{0}|}^{\sigma_{S}}(\psi_{A_{0}}).\tag{1878}$$

It agrees with the original moments on separated tensors. By (1872), each integrated entry costs at most $C_{\int}(k)$, so the binomial formula gives C^{n}. Uniqueness again follows from density. □

For the actual limits of proposition 92.1, put

$$A_{g}=1+\mathsf{U}(1,1)+p,\qquad C_{E,\mathrm{c}}^{g}=C_{\neq}^{3}MB^{\mathrm{per}}(1)\sqrt{A_{g}}.$$

Then (1877) gives the common bound

$$|\mathsf{S}_{n}^{\sigma_{S}}(\varphi)|\le(C_{E,\mathrm{c}}^{g})^{n}(n!)^{14+p}|\varphi|_{10n}.\tag{1879}$$

Indeed, (1783) is nonincreasing in d, so $1+\mathsf{U}(1,d)\le A_g$ when $d\ge 1$. For $d<1$, put $t=d^{-1}$. The logarithmic term in (1783) increases by $p\log t$, and the remaining terms are bounded by $t^3\mathsf{U}(1,1)$. Since $\log t\le t$ and $1,t,t^3\le t^4$,

$$A^{\mathrm{per}}(d)\le A_g\max(1,d^{-4}).$$

Apply proposition 93.4 with $a=4$, $B=B^{\mathrm{per}}(1)$ and $\varkappa=5+p$, using (1851) and its inherited bound (1853). This applies to every growing-torus limit as well as every periodic-state limit; no additional rotation hypothesis is used. A constant one-point function $c\int f$, $|c|\le\bar c$, replaces $C^g_{E,\mathrm{c}}$ by $C^g_{E,\mathrm{c}}+\bar c C_{\int}(10)$.

Proposition 93.5. Suppose the moments of a real symmetric functional σ extend to continuous linear functionals S_n on $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$, agreeing with σ on separated smooth tensors and satisfying

$$|\mathsf{S}_n(\varphi)|\le c_E C_E^n(n!)^{\kappa}|\varphi|_{k_E n}$$

for fixed $k_E\in\mathbb{N}$ and $c_E,C_E,\kappa\ge 0$. These extensions preserve reality and symmetry. Every Euclidean isometry under which σ is invariant also leaves them invariant. If σ is reflection positive, then for $\varphi_n\in\mathcal{S}_+(\mathbb{R}^{3n})$, $0\le n\le n_0$, with $\mathcal{S}_+(\mathbb{R}^0)=\mathbb{C}$,

$$\sum_{n,n'=0}^{n_0}\mathsf{S}_{n+n'}(\Theta\bar\varphi_n\otimes\varphi_{n'})\ge 0. \tag{1880}$$

Here Θ reflects every time coordinate. ■

Proof. Conjugation and permutation preserve the Schwartz seminorms and flatness on diagonals. The respective identities hold on separated smooth tensors and extend by lemma 93.2. For an isometry $rx=\rho x+v$, its pullback preserves the flatness condition and satisfies

$$|\varphi\circ(r^{-1})^{\times n}|_k\le[6(1+n|v|^2)]^{k/2}|\varphi|_k. \tag{1881}$$

Indeed each row of ρ has sum of absolute values at most $\sqrt{3}$, while $1+|x|^2\le 2(1+n|v|^2)(1+|(r^{-1})^{\times n}x|^2)$. The chain rule gives the estimate and shows that the pullback is Schwartz and flat on the diagonals. Density now transfers the isometry identity.

For positivity, approximate each φ_n by finite sums $\varphi_n^{(j)}$ of separated positive-time smooth tensors in the seminorm of order $k=2k_{\mathrm{E}}n_0$; leave φ_0 unchanged. Reflection and conjugation preserve this seminorm. The tensor estimate and the flatness assertion in lemma 93.2 give

$$|\Theta\bar\varphi_n\otimes\varphi_{n'}-\Theta\bar\varphi_n^{(j)}\otimes\varphi_{n'}^{(j)}|_k\le|\varphi_n-\varphi_n^{(j)}|_k|\varphi_{n'}|_k+|\varphi_n^{(j)}|_k|\varphi_{n'}-\varphi_{n'}^{(j)}|_k\longrightarrow 0.$$

Since $k_{\mathrm{E}}(n+n')\le k$, continuity permits this limit in every term of (1880). For each j, expansion of the finite tensor sums expresses the result as one reflection-positive quadratic form for σ, including the empty tuple. Its limit is nonnegative. □

94. Reflection positivity and the translation spectrum

The bounds on separated correlations suffice to construct the Hilbert space of a reflection. The additional positivity at a diagonal plane then restricts the joint energy–momentum spectrum. This construction uses neither coincident-point moments nor a probability measure on continuum fields.

Let σ be a real, symmetric, normalized functional on $\mathfrak{S}(\mathbb{R}^3)$, multilinear on products of test spaces with disjoint closures. Assume that its one-point function is $\sigma(f)=c\int f$ and that its connected functions satisfy a bound table $C_n(d;R)$ as in section 90. Thus, on tuples of separation at least d whose individual supports lie in balls of radius R,

$$\sigma(X)=\sum_{A\subset\{1,\dots,n\}}\prod_{i\in A}\Big(c\int f_i\Big)\sum_{\pi\in\mathrm{Part}_{\ge 2}(A^{\mathrm{c}})}\prod_{J\in\pi}S^{\sigma}_{|J|}(X_J). \tag{1882}$$

In particular the absolute value is bounded by the same finite sum with $|c|\|f_i\|_1$ in place of $c\int f_i$ and $C_{|J|}(d;R)\prod_{i\in J}\|f_i\|_\sharp$ in place of each connected function. Multilinear telescoping shows that σ is continuous when the entries converge in the sharp norm and their supports remain in fixed separated compact sets.

For a unit vector u, put $E_+(u)=\{x : u\cdot x>0\}$, let $\theta_u x = x-2(u\cdot x)u$, and let $\mathcal{D}_u=\mathbb{C}\mathfrak{S}(E_+(u))$ be the free complex vector space on the separated tuples in this half-space. Concatenation is used only when the resulting tuple is separated. The product $\theta_u\bar F\cdot G$ is always defined for $F,G\in\mathcal{D}_u$. Reflection positivity means

$$\sigma(\theta_u\bar F\cdot F)\ge 0 \qquad (F\in\mathcal{D}_u).$$

It makes

$$\langle F,G\rangle_{\sigma,u}=\sigma(\theta_u\bar F\cdot G) \tag{1883}$$

a positive semidefinite Hermitian form, antilinear in its first argument. Quotient by its null space and complete to obtain $\mathcal{H}(\sigma,u)$. Write $[F]$ for a class and $\Omega=[\varnothing]$, so that $\|\Omega\|=1$.

Proposition 94.1 (The translation operators)**.** Suppose that σ is translation invariant and reflection positive at $u\cdot x=0$. Then $\mathcal{H}(\sigma,u)$ is separable, and the maps

$$T_t[F]=[\tau_{tu}F]\quad(t\ge0),\qquad U(v)[F]=[\tau_vF]\quad(v\in u^\perp)$$

extend to a strongly continuous semigroup of positive self-adjoint contractions and a strongly continuous unitary group. They commute and fix Ω. There are strongly commuting self-adjoint operators $H\ge0, P_1, P_2$ such that

$$T_t=e^{-tH},\qquad U(v)=e^{i(v_1P_1+v_2P_2)},\qquad T_tU(v)=\int e^{-tE+iv\cdot q}\,d\mathcal{P}(E,q), \tag{1884}$$

where coordinates are taken in any fixed orthonormal basis of $u^\perp$, and $\mathcal{P}$ is carried by $[0,\infty)\times\mathbb{R}^2$. Moreover $\ker T_t=0$, $H\Omega=P_1\Omega=P_2\Omega=0$, and $T_t\to P_{\ker H}$ strongly as $t\to\infty$.

If an orthogonal transformation r fixes u and leaves σ invariant, $[F]\mapsto[rF]$ extends to a unitary $U(r)$ fixing Ω. It commutes with H and satisfies $U(r)U(v)U(r)^*=U(rv)$. The joint spectrum is invariant under $(E,q)\mapsto(E,rq)$. ∎

Proof. On $\mathcal{D}_u$ write $T_t^0=\tau_{tu}$ and $U^0(v)=\tau_v$. The identities $\theta_u\tau_{tu}=\tau_{-tu}\theta_u$ and $\theta_u\tau_v=\tau_v\theta_u$, together with translation invariance, give

$$\langle F,T_t^0G\rangle=\langle T_t^0F,G\rangle,\qquad \langle U^0(v)F,U^0(v)G\rangle=\langle F,G\rangle.$$

For each $F=\sum_i z_iX_i$ the number

$$c_F=\sup_{t\ge0}|\langle F,T_t^0F\rangle|$$

is finite. Indeed, choose common $d,R>0$ for the finitely many tuples $\theta_uX_i\cdot X_j$. Moving their positive entries by tu does not decrease a distance across the mirror:

$$|y+tu-x|^2=|y-x|^2+2t\,u\cdot(y-x)+t^2\ge|y-x|^2\quad(u\cdot x<0<u\cdot y).$$

The sharp and L^1 norms and the radii of individual supports are unchanged. The bound following (1882), summed with coefficients $|z_iz_j|$, therefore bounds c_F.

Cauchy–Schwarz for the semidefinite form and the semigroup law give the decisive contraction estimate

$$\|T_t^0F\|^2=\langle F,T_{2t}^0F\rangle\le\|F\|\,\|T_{2t}^0F\|. \tag{1885}$$

After j iterations,

$$\|T_t^0F\|\le\|F\|^{1-2^{-j}}\|T_{2^jt}^0F\|^{2^{-j}}\le\|F\|^{1-2^{-j}}c_F^{2^{-j-1}}.$$

If $\|F\|=0$, the first inequality already gives zero. Otherwise letting $j\to\infty$ proves $\|T_t^0F\|\le\|F\|$. Thus these maps preserve the null space and extend as contractions. The maps $U^0(v)$ are invertible isometries and extend as unitaries. The algebraic identities extend by density; symmetry gives self-adjointness, and $T_t=T_{t/2}^2$ gives positivity.

For small translations, enlarge the finitely many supports by less than $d/4$. They remain separated by $d/2$. Translations of a compactly supported C^1 function converge in the sharp norm, by uniform continuity of the function and its first derivatives. Hence the continuity following (1882) gives $\langle F, T_t^0 F\rangle \to \|F\|^2$ and $\langle F, U^0(v)F\rangle \to \|F\|^2$ at the origin. The identities

$$\|(T_t-1)[F]\|^2 = \langle F, T_{2t}^0 F\rangle - 2\,\mathrm{Re}\langle F, T_t^0 F\rangle + \|F\|^2, \qquad \|(U(v)-1)[F]\|^2 = 2\|F\|^2 - 2\,\mathrm{Re}\langle F, U^0(v)F\rangle$$

prove strong continuity on the dense set of classes; the contraction bounds extend it to the completion. If $T_t\psi = 0$, then $\|T_{t/2}\psi\|^2 = \langle \psi, T_t\psi\rangle = 0$. Iteration and continuity at zero give $\psi = 0$.

The spectral measure of T_1 is carried by $(0,1]$. Define $H = -\log T_1$ by spectral calculus. The uniqueness of positive square roots gives $T_{2^{-j}} = T_1^{2^{-j}}$; the semigroup law and strong continuity then give $T_t = e^{-tH}$ for all $t \geq 0$. This also proves uniqueness of H. Every $U(v)$ commutes with T_1 and its spectral projections, so $e^{isH}U(v)$ is a strongly continuous unitary representation of $\mathbb{R} \times u^\perp$. Its joint spectral measure is $\mathcal{P}$ in (1884); its coordinate generators are H, P_1, P_2. The invariance of Ω implies that its spectral measure is concentrated at $(0,0)$. Dominated convergence of e^{-tE} gives the asserted limit of T_t.

For separability use the countable family of separated test tuples constructed in section 90. Given $X \in \mathfrak{S}(E_+(u))$, choose $d > 0$ smaller than all separations in $\theta_u X \cdot X$. Rational neighborhoods of its positive supports of width less than $d/3$ still lie in $E_+(u)$, since $u \cdot x \geq d/2$ on those supports. Approximate each entry in C^1 inside these neighborhoods. The four terms in

$$\|[X]-[Y]\|^2 = \sigma(\theta_u X \cdot X) - \sigma(\theta_u X \cdot Y) - \sigma(\theta_u Y \cdot X) + \sigma(\theta_u Y \cdot Y)$$

then converge to the same value. Rational complex linear combinations of the resulting countable set of classes are dense. Finally r commutes with θ_u and preserves the form. Its unitary extension has the asserted covariance on the dense set of classes, hence everywhere. Spectral uniqueness gives the covariance of $\mathcal{P}$. □

94.1. **Changing the one-point constant.** For real b define, on a separated tuple $X = (f_1, \dots, f_n)$,

$$\mathsf{sh}_b X = \sum_{A \subset \{1,\dots,n\}} \prod_{i \in A} \Big(b \int f_i\Big)\, X_{A^c}. \tag{1886}$$

The map extends linearly, preserves every open-set tuple space, commutes with isometries and conjugation, and respects every defined concatenation. Expanding the subsets gives $\mathsf{sh}_b \mathsf{sh}_{b'} = \mathsf{sh}_{b+b'}$. Thus it is invertible with inverse sh_{-b}. The moment–cumulant partition formula shows that $\sigma' = \sigma \circ \mathsf{sh}_b$ has one-point constant $c + b$ and the same connected functions of order at least two.

Proposition 94.2 (Recentering and reflection)**.** The functionals σ and σ' have the same isometry invariances and the same reflection positivities. They cluster simultaneously. In the setting of proposition 94.1, the map

$$[F]_{\sigma',u} \longmapsto [\mathsf{sh}_b F]_{\sigma,u}$$

extends to a unitary which fixes the vacuum and intertwines the translation operators and their joint spectral measures. ■

Proof. The algebraic identities just established imply

$$\sigma'(\theta_u \bar{F} \cdot G) = \sigma(\theta_u \overline{\mathsf{sh}_b F} \cdot \mathsf{sh}_b G). \tag{1887}$$

They prove reflection positivity in both directions, using sh_{-b} for the converse, and give a surjective isometry on the quotients. The same commutation identities intertwine translations and isometries. Uniqueness of the spectral measure proves the last assertion. Invariance follows also directly from the subset expansion. Each shifted clustering difference is a finite sum of unshifted clustering differences of subtuples, with constant integral factors; the converse uses the inverse shift. □

94.2. **Positivity at a diagonal plane.** The following measure estimate is the analytic step in comparing the Hilbert spaces of two reflections.

Lemma 94.3. Let μ be a finite positive measure on $[0,\infty)\times\mathbb{R}$, and set $w=(E-iq)/\sqrt{2}$. Suppose that, on a Hilbert space $\mathcal{K}$, there are vectors a,b and self-adjoint operators $H_+\ge 0, Q$ with

$$\int e^{-sw-s'\bar w}\,d\mu(E,q)=\langle a,e^{-sH_+}e^{-is'Q}b\rangle \qquad (s,s'\ge 0). \tag{1888}$$

Then $\mu(\{|q|>E\})=0$. No commutation of H_+ and Q is required. ∎

Proof. For $c\in L^1(0,\infty)$ put $p(z)=\int_0^\infty c(s)e^{-sz}\,ds$ on $\operatorname{Re} z\ge 0$. Multiply (1888) by $c(s)\overline{c(s')}$ and integrate. Fubini is justified by the bound $|e^{-sw-s'\bar w}|\le 1$, the finiteness of μ, and the contraction and unitary bounds on the right. Spectral calculus in the indicated order gives

$$\int |p(w)|^2\,d\mu=\langle a,p(H_+)p^\sharp(Q)b\rangle\le \|a\|\|b\|\sup_{x\ge0}|p(x)|\sup_{y\in\mathbb{R}}|p(iy)|,\qquad p^\sharp(y)=\overline{p(-iy)}. \tag{1889}$$

Take $z_0=\varepsilon+i\kappa$, where $\varepsilon>0$ and $\kappa\in\mathbb{R}$, and $c(s)=\Gamma(z_0)^{-1}s^{z_0-1}e^{-\eta s}$ with $\eta>0$. The Gamma integral gives $p(z)=(z+\eta)^{-z_0}$, with the principal logarithm. Consequently

$$\int |w+\eta|^{-2\varepsilon}e^{2\kappa\arg(w+\eta)}\,d\mu\le\|a\|\|b\|\eta^{-2\varepsilon}e^{|\kappa|\pi/2}.$$

Fatou's lemma first as $\varepsilon\downarrow 0$ and then as $\eta\downarrow 0$ yields

$$\int_{w\ne0} e^{2\kappa\arg w}\,d\mu\le\|a\|\|b\|e^{|\kappa|\pi/2}. \tag{1890}$$

For every $\delta>0$, the measure of $\{\arg w\ge\pi/4+\delta\}$ is at most $\|a\|\|b\|e^{-2\kappa\delta}$ for all $\kappa>0$, and hence is zero. Taking $\kappa<0$ proves the corresponding assertion for $\arg w\le-\pi/4-\delta$. Their countable union is exactly $\{|q|>E\}$. □

Proposition 94.4 (The spectral bound from two reflections)**.** Let σ satisfy the bounds and translation invariance of proposition 94.1. Fix $k\in\{1,2\}$ and $\epsilon\in\{1,-1\}$. If σ is reflection positive for the normals e_3 and $e_+=(e_3+\epsilon e_k)/\sqrt{2}$, then its spectral measure in $\mathcal{H}(\sigma,e_3)$ satisfies

$$\mathcal{P}\{(E,q):E<|q_k|\}=0.$$

We denote this spectral-support assertion by $H\ge|P_k|$. ∎

Proof. Put $e_-=(e_3-\epsilon e_k)/\sqrt{2}$ and $W=\{x:x_3>|x_k|\}$. For $F\in\mathbb{C}\mathfrak{S}(W)$, both F and $\theta_{e_+}\theta_{e_3}F$ lie in the positive half-space for e_+. Let μ be the image of the positive spectral measure $\langle[F],\mathcal{P}(\cdot)[F]\rangle$ under $(E,q)\mapsto(E,\epsilon q_k)$. The same correlation has two representations:

$$\begin{aligned}\int e^{-sw-s'\bar w}\,d\mu&=\sigma(\theta_{e_3}\bar F\cdot\tau_{se_++s'e_-}F)\\&=\big\langle[\theta_{e_+}\theta_{e_3}F]_+,T_s^+U^+(s'e_-)[F]_+\big\rangle_+.\end{aligned} \tag{1891}$$

Indeed $se_++s'e_-=(s+s')e_3/\sqrt{2}+\epsilon(s-s')e_k/\sqrt{2}$, which gives the first equality by (1884); the second is the defining reflection form in $\mathcal{H}(\sigma,e_+)$. There $T_s^+=e^{-sH_+}$ and $U^+(s'e_-)=e^{-is'Q}$ for the negative of the momentum generator along e_-. Hence lemma 94.3 shows that the spectral projection of $\{E<|q_k|\}$ annihilates $[F]$.

For arbitrary $F\in\mathcal{D}_{e_3}$, a sufficiently large positive time translation puts all its finitely many supports in W. The same projection therefore annihilates $T_r[F]$. It commutes with T_r, whose kernel is zero by proposition 94.1, and so annihilates $[F]$. Density proves that the projection is zero. □

95. Slab operators and analytic translations

We now isolate the hypotheses used to recover rotations from the cubic symmetries. The insertion estimate constructs bounded operators across a slab. Together with the spectral bound, these operators will give holomorphic correlation kernels. A separate two-point growth estimate will exclude the nonconstant angular harmonics.

Throughout this section, complex test functions are expanded multilinearly into real and imaginary parts. On them we use $\|h\|_\sharp = \max\{\|h\|_\infty, 40M \operatorname{Lip} h\}$; each of the real and imaginary parts has norm at most $\|h\|_\sharp$. Such multilinear identities hold also for Hilbert-space classes: pairing their difference with every $[G]$ reduces the assertion to the multilinearity of σ, and density makes the difference zero.

Write $y = (x_3, x_1)$, leaving x_2 as the third coordinate, and let R_t rotate this plane by $(y^0, y^1) \mapsto (y^0 \cos t - y^1 \sin t, y^0 \sin t + y^1 \cos t)$. Put $n_\upsilon = (\cos\upsilon, \sin\upsilon)$, so $n_0 = (1,0)$ corresponds to e_3. A plate is $f(x) = \alpha(y)\varphi(x_2)$, where $\varphi \in C_c^1(\mathbb{R})$ is real and $\alpha \in C_c^1(\mathbb{R}^2;\mathbb{C})$. Its label is φ; denote the length of the smallest interval containing $\operatorname{supp}\varphi$ by λ_φ, with $\lambda_0 = 0$. Set

$$\operatorname{Prof}_t = \{\alpha \in C_c^1(\mathbb{R}^2) : \operatorname{supp}\alpha \subset [t,2t]\times[-t,t],\ |\alpha| \le 3,\ \operatorname{Lip}\alpha \le 12/t\}. \tag{1892}$$

Definition 95.1 (Hypotheses for the rotation argument)**.** For a real normalized functional σ on separated tuples, the hypotheses are as follows.

(R0) $\sigma(f) = 0$ for every $f \in C_c^1(\mathbb{R}^3)$.

(R1) σ is symmetric and invariant under translations and the cubic group W_3 of signed coordinate permutations.

(R2) For each $n \ge 1$ and $d, R > 0$ there is a finite $C_n^\sigma(d;R)$, non-increasing in d, such that

$$|\sigma(f_1,\dots,f_n)| \le C_n^\sigma(d;R)\prod_i \|f_i\|_\sharp$$

for every tuple with separation at least d and each support in some ball of radius R; the centers are arbitrary.

(R3) σ is reflection positive at $x_3 = 0$.

(R4) In the corresponding Hilbert space, $H \ge |P_1|$ in the sense of spectral support.

(R5) There are $0 < c_{\mathrm{I}} \le 1/16$, $C_{\mathrm{I}} > 0$ and $p_{\mathrm{I}} \ge 0$ such that, for $0 < \eta \le 1$, every real h supported in $B(z, c_{\mathrm{I}}\eta)$ with $|z_3| \le \eta/8$, and $F, G \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta/4\})$,

$$|\sigma(\theta\bar F \cdot (h)\cdot G)| \le C_{\mathrm{I}}\eta^{-p_{\mathrm{I}}}\|h\|_\sharp \|[\tau_{-(\eta/4)e_3}F]\|\,\|[\tau_{-(\eta/4)e_3}G]\|. \tag{1893}$$

(R6) For the diagonal reflection with normal $(e_1+e_3)/\sqrt{2}$, the two-point form is positive:

$$\sum_{a,b}\bar z_a z_b \sigma(\theta_\Sigma f_a, f_b) \ge 0 \quad (f_a \in C_c^1(\{x_1 + x_3 > 0\})).$$

(R7) There is an exponent $a_\Gamma < 4$ such that for every label φ there is $C_\varphi < \infty$ with

$$\sigma(\theta f, f) \le C_\varphi t^{-a_\Gamma} \quad (0 < t \le 1,\ f = \alpha\otimes\varphi,\ \alpha \in \operatorname{Prof}_t). \tag{1894}$$

■

The Hilbert construction of proposition 94.1 applies under (R1)–(R3). Its proof uses only the uniform moment bound in (R2); alternatively the partition formula bounds each connected function by a finite sum of products of those bounds. Cubic symmetry transports the construction to each coordinate normal. The map $[F]^{(\mu)} \mapsto [wF]^{(\nu)}$ is unitary whenever $we_\mu = e_\nu$, since it preserves the reflection form. It intertwines the time and tangential translations. Thus (R4) gives, for every ordered pair $\mu \ne \nu$,

$$H^{(\mu)} \ge |P_\nu^{(\mu)}|. \tag{1895}$$

95.1. **The holomorphic translation semigroup.** For $z \in \mathbb{C}^3$ and $\xi \in \mathbb{C}^2$ put

$$d_3(z) = \operatorname{Re} z_3 - |\operatorname{Im} z_1| - |\operatorname{Im} z_2|, \qquad \mathsf{Tb}_3 = \{d_3 > 0\}, \qquad z^* = (-\bar{z}_1, -\bar{z}_2, \bar{z}_3), \tag{1896}$$

and

$$d_{\mathsf{Tb}}(\xi) = \operatorname{Re} \xi^0 - |\operatorname{Im} \xi^1|, \quad \mathsf{Tb} = \{d_{\mathsf{Tb}} > 0\}, \quad \xi^* = (\bar{\xi}^0, -\bar{\xi}^1), \quad \xi^{\pm} = \xi^0 \pm i\xi^1. \tag{1897}$$

Complex rotations are defined by $(R_\zeta \xi)^{\pm} = e^{\pm i\zeta}\xi^{\pm}$. Both tubes are convex cones, stable under their respective involutions. The map $\xi \mapsto (\xi^1, 0, \xi^0)$ embeds Tb in Tb_3.

Lemma 95.2. Under (R1)–(R4),

$$\mathsf{T}_3(z) = \int e^{-z_3 E + i(z_1 q_1 + z_2 q_2)}\, d\mathcal{P}(E, q) \quad (z \in \overline{\mathsf{Tb}_3}) \tag{1898}$$

is a contraction semigroup, strongly continuous on the closed tube and holomorphic in operator norm on its interior. It satisfies $\mathsf{T}_3(z)^* = \mathsf{T}_3(z^*)$, $\mathsf{T}_3(z)\Omega = \Omega$, and $\mathsf{T}_3(x)[F] = [\tau_x F]$ for real x with $x_3 \geq 0$. Its restriction is denoted $\mathsf{T}(\xi) = \mathsf{T}_3(\xi^1, 0, \xi^0)$. ■

Proof. On the support of $\mathcal{P}$, (1895) gives $|q_1|, |q_2| \leq E$, so the integrand has modulus at most $e^{-d_3(z)E} \leq 1$. Spectral calculus gives the semigroup, adjoint, vacuum and real-translation identities. Dominated convergence in each scalar spectral measure proves strong continuity. On a compact subset of the open tube there is $\delta > 0$ with $d_3 \geq \delta$. Every derivative of degree m in the integrand is bounded by $E^m e^{-\delta E}$. Taylor's formula on a smaller neighborhood, with this uniform bound, permits differentiation in operator norm. This proves holomorphy. □

95.2. **Insertion across a slab.**

Proposition 95.3 (Slab operators)**.** Under (R1)–(R3) and (R5), for every $l > 0$ and $h \in C_c^1(\{0 < x_3 < l\}; \mathbb{C})$ there is a unique bounded operator $B_l(h)$ such that

$$B_l(h)[F] = [(h) \cdot \tau_{l e_3} F]. \tag{1899}$$

It is complex linear in h. For every compact subset K of the slab there is $C(K, l) < \infty$ with $\|B_l(h)\| \leq C(K, l)\|h\|_\sharp$ when $\operatorname{supp} h \subset K$. It has the identities

$$B_l(h) = T_c B_{c'-c}(\tau_{-ce_3} h) T_{l-c'}, \qquad \operatorname{supp} h \subset \{c < x_3 < c'\}, \tag{1900}$$

$$B_l(h)^* = B_l(\overline{\theta_{l/2} h}), \qquad U(v) B_l(h) U(-v) = B_l(\tau_v h), \tag{1901}$$

where $0 \leq c < c' \leq l$ and $v \perp e_3$. For real h supported in $B(z, c_{\mathrm{I}}\eta)$ with $\eta/8 \leq z_3 \leq 3\eta/8$, $0 < \eta \leq 1$,

$$\|B_{\eta/2}(h)\| \leq C_{\mathrm{I}} \eta^{-p_{\mathrm{I}}} \|h\|_\sharp. \tag{1902}$$

If $0 = c_0 < \cdots < c_n$ and $\operatorname{supp} h_j \subset \{c_{j-1} < x_3 < c_j\}$, then

$$[(h_1, \ldots, h_n)] = \prod_{j=1}^{n} B_{c_j - c_{j-1}}(\tau_{-c_{j-1} e_3} h_j)\Omega, \tag{1903}$$

with increasing j from left to right. ■

Proof. For a small piece as in (1902), put $h_0 = \tau_{-(\eta/4)e_3} h$. Apply (R5) to h_0 and $\tau_{(\eta/4)e_3} F, \tau_{(\eta/4)e_3} G$. Translation invariance changes the resulting form into

$$\sigma(\theta \bar{F} \cdot (h) \cdot \tau_{(\eta/2)e_3} G),$$

whose modulus is at most $C_{\mathrm{I}} \eta^{-p_{\mathrm{I}}} \|h\|_\sharp \|[F]\| \|[G]\|$. It descends to the quotient and extends continuously to the Hilbert space. The bounded-form representation theorem gives an operator with this bound. Pairing with every class identifies its value on $[G]$ with $[(h) \cdot \tau_{(\eta/2)e_3} G]$.

For general compact K in the slab, set $\delta = \min_{x\in K} \min(x_3, l - x_3) > 0$, $\eta = \min(1, 4\delta)$ and $b = c_{\mathrm{I}}\eta/2$. Use the C^1 partition

$$\chi_{b,k}(x) = \prod_{\mu=1}^{3} \chi(x_\mu/b - k_\mu), \quad \chi(s) = \begin{cases} \cos^2(\pi s/2), & |s| \le 1, \\ 0, & |s| > 1. \end{cases}$$

The identity $\chi(s) + \chi(s-1) = 1$ for $0 \le s \le 1$ gives $\sum_k \chi_{b,k} = 1$. Each support is a cube of radius b, and $\operatorname{Lip} \chi_{b,k} \le \sqrt{3}\pi/(2b)$. Only finitely many of these cubes meet K. Write $h_k = \chi_{b,k} h$ and let c_k be the projection of $bk_3 - \eta/4$ onto $[0, l - \eta/2]$. Because $b \le \eta/32$ and $\eta/4 \le \delta$, the center of $\tau_{-c_k e_3} h_k$ has third coordinate between $7\eta/32$ and $9\eta/32$. Its support lies in a ball of radius $\sqrt{3}b < c_{\mathrm{I}}\eta$. The small-piece construction therefore applies, and

$$B_l(h) = \sum_k T_{c_k} B_{\eta/2}(\tau_{-c_k e_3} h_k) T_{l - c_k - \eta/2}$$

has the required value on classes. The product estimate

$$\|h_k\|_\sharp \le \left(1 + \frac{40\sqrt{3}\pi M}{c_{\mathrm{I}}\eta}\right) \|h\|_\sharp$$

gives the compact-support bound, with an extra factor two for the real and imaginary parts. The construction is independent of the partition because its values on the dense set of classes are prescribed.

Translation of the tuple in (1899) proves (1900) and tangential covariance. Reflection in $x_3 = l/2$ interchanges its left and right entries; symmetry and conjugation give the adjoint formula. Finally, iterating (1899) through the ordered slabs gives (1903). □

95.3. **Removing coincident planar projections.**

Proposition 95.4. Assume (R1)–(R4). Fix $n \ge 2$ and a real angle t. If $\sigma(R_t X) = \sigma(X)$ for every n-tuple whose supports have pairwise disjoint projections onto the (x_3, x_1) plane, the identity holds for all separated n-tuples. ■

Proof. First suppose that two groups of entries lie respectively in $x_2 < c - \delta$ and $x_2 > c + \delta$, where $\delta > 0$. In the Hilbert space with normal e_2, their correlation after a relative translation su, $u \perp e_2$, $|u| = 1$, is a matrix element of $T_{2\delta}^{(2)} U^{(2)}(su)$. For its finite complex spectral measure m this is

$$\Phi(s) = \int e^{-2\delta E + is(u_1 q_1 + u_3 q_3)}\, dm(E, q).$$

Since $|q_1|, |q_3| \le E$ and $|u_1| + |u_3| \le \sqrt{2}$, this is holomorphic for $|\operatorname{Im} s| < \sqrt{2}\delta$. Thus the relative planar translation is analytic whenever there is such an x_2 gap.

Let A_j be the planar projections of the supports and I_j their x_2 interval hulls. Suppose first that every pair of intersecting A_j has a level separating its two intervals and lying outside all the I_j. Induct on the number of intersecting pairs. Choose a separating level for one pair, and translate all entries above it by se_1. For all sufficiently large s, the planar projections on opposite sides no longer meet, and the number of intersecting pairs strictly decreases. The interval hulls and their separating levels are unchanged. The induction hypothesis gives rotation invariance for these large s. Both correlations are holomorphic on a common strip: after rotation, the relative translation is $sR_t e_1$, still perpendicular to e_2. The identity theorem gives equality also at $s = 0$.

For a general tuple of separation $d > 0$, choose $0 < \varepsilon < d/n$ and partition each entry in the x_2 coordinate with $\chi(2x_2/\varepsilon - k)$. Multilinearity gives a finite sum of tuples whose interval hulls have length at most ε. If two planar projections of a nonzero summand meet, points over their common projection are at x_2 distance at least d. The gap between their two interval hulls has length at least $d - 2\varepsilon > (n-2)\varepsilon$. The other $n - 2$ interval hulls cannot cover this gap, so it contains a level outside

all hulls. The preceding induction applies to each summand. The partition commutes with R_t, and summing proves the assertion. Tuples with a zero entry vanish by multilinearity. □

96. The two-point rotation argument

For a plate label φ, reflection positivity gives a positive spectral measure on the energy–momentum half-plane. Its transform represents the two-point function. The plate growth bound leaves only the angular frequencies $0, 4, -4$. Positivity at the coordinate and diagonal planes gives opposite signs for the coefficient at frequency four.

96.1. **The spectral measure of a label.** Put $Y = [0,\infty)\times\mathbb{R}$, $V = \{(E,q) : E \ge |q|\}$, and, for a profile supported in $y^0 > 0$, set

$$\widetilde{\alpha}(E,q) = \int \alpha(y) e^{-y^0 E + i y^1 q}\, dy. \tag{1904}$$

This transform is continuous and bounded by $\|\alpha\|_1 e^{-\varepsilon E}$ if the support has $y^0 \ge \varepsilon > 0$. Convolution of profiles multiplies their transforms.

We first work with any symmetric bilinear two-point form s on separated supports which is translation invariant, satisfies the order-two bound in (R2), and is reflection positive at $y^0 = 0$. The form $s(\theta\bar f, h)$ on $C_c^1(\{x_3 > 0\};\mathbb{C})$ defines a one-field Hilbert space. The proof of proposition 94.1, with single test functions instead of tuples and with no distinguished vacuum, gives positive time translations, tangential unitaries and a joint spectral measure $\mathcal{P}^s$ on Y.

Proposition 96.1. For every real label φ there is a unique positive Borel measure $\rho[s,\varphi]$ on Y, finite on compact sets, such that all profile transforms are in $L^2(\rho)$ and

$$s\big(\theta(\bar\alpha\otimes\varphi), \alpha'\otimes\varphi\big) = \int \overline{\widetilde{\alpha}}\,\widetilde{\alpha}'\, d\rho. \tag{1905}$$

More precisely, for every Borel set $B \subset Y$,

$$\langle [\alpha\otimes\varphi], \mathcal{P}^s(B)[\alpha'\otimes\varphi]\rangle = \int_B \overline{\widetilde{\alpha}}\,\widetilde{\alpha}'\, d\rho. \tag{1906}$$

■

Proof. Write $\Psi_\alpha = [\alpha\otimes\varphi]$ and $\mu_{\alpha,\alpha'}(B) = \langle \Psi_\alpha, \mathcal{P}^s(B)\Psi_{\alpha'}\rangle$. Riemann sums for convolution converge in C^1, with supports in a fixed compact subset of the positive half-plane. The two-point bound therefore gives

$$\Psi_{\alpha'*\alpha} = \widetilde{\alpha}'(\mathcal{P}^s)\Psi_\alpha. \tag{1907}$$

On the right, the same Riemann sums converge weakly by the spectral representation of translations and dominated convergence; this identifies their norm limit on the left. Commutativity of convolution implies

$$|\widetilde{\alpha}'|^2 \mu_{\alpha,\alpha} = |\widetilde{\alpha}|^2 \mu_{\alpha',\alpha'}. \tag{1908}$$

On $O_\alpha = \{\widetilde{\alpha} \ne 0\}$ define $d\rho_\alpha = |\widetilde{\alpha}|^{-2} d\mu_{\alpha,\alpha}$. Equation (1908) makes these local measures agree on overlaps. The sets O_α cover Y: at (E_0,q_0) take $\alpha(y) = \chi(y^0-2)\chi(y^1)e^{-iy^1 q_0}$, whose transform there is strictly positive. Compact exhaustion of Y gives a countable subcover. Partitioning it into disjoint Borel pieces defines a positive measure ρ with the prescribed restrictions. It is locally finite, since near a point one transform is bounded away from zero.

Equation (1908) also shows that $\mu_{\alpha,\alpha}$ vanishes on the zeros of $\widetilde{\alpha}$: intersect that zero set with each member of the countable cover and use the nonzero transform there. Hence $d\mu_{\alpha,\alpha} = |\widetilde{\alpha}|^2 d\rho$ on all of Y. Polarization proves (1906), and taking $B = Y$ gives (1905).

For uniqueness, a finite complex measure m on Y is determined by its Laplace–Fourier transform. Indeed, vanishing of $\int e^{-tE+ivq} dm$ at all integers $t \ge 0$ and all $v \in \mathbb{R}$ first gives vanishing against $p(e^{-E})e^{ivq}$ for polynomials p. Uniform polynomial approximation on $[0,1]$, followed by Fourier

uniqueness in q, gives vanishing against $g(e^{-E})\mathbf{1}_B(q)$ for continuous g and Borel B. Approximation of indicators of closed subsets of $[0,\infty)$ and the monotone class theorem then give $m=0$. Thus vanishing against every $\widetilde{\alpha}$ also implies $m=0$, by testing the continuous Laplace–Fourier transform against profiles. If two proposed measures satisfy (1905), replace α' by $\alpha''*\alpha$. The preceding uniqueness applies to the finite measure $|\widetilde{\alpha}|^2(\rho'-\rho'')$ and makes it zero. Division on O_α and the countable cover give $\rho'=\rho''$. □

Apply this to $s=\sigma|_{\mathfrak{S}_2}$ under (R1)–(R4) and write $\rho_\varphi=\rho[s,\varphi]$. The one-field space embeds isometrically into $\mathcal{H}(\sigma)$ through $[f]\mapsto[(f)]$. It intertwines translations, so $|\widetilde{\alpha}|^2\rho_\varphi$ is the image of the full spectral measure of $[(\alpha\otimes\varphi)]$ under $(E,q_1,q_2)\mapsto(E,q_1)$. Consequently ρ_φ is carried by V. Reflection of x_1 and uniqueness in proposition 96.1 make it invariant under $q\mapsto -q$.

96.2. **From plate bounds to a kernel bound.** Let $a_0(s)=1+\cos(2\pi(s-3/2))$ on $[1,2]$, zero elsewhere, and let $b_1=(3/2)\mathbf{1}_{[-1/3,1/3]}$, $b=b_1*b_1*b_1$. For $1/2\le\lambda\le 1$ define $\alpha_{t,\lambda}(y)=a_0(y^0/t)b(y^1/(\lambda t))$. These profiles belong to Prof_t: $\int a_0=\int b=1$, $0\le a_0\le 2$, $|a_0'|\le 2\pi$, $0\le b\le 9/8$, and $|b'|\le 9/4$. The last bounds follow by writing $b=(3/2)\int_{u-1/3}^{u+1/3}T(v)dv$ with $T(v)=(9/4)(2/3-|v|)_+$.

There is an absolute $C_b<\infty$ such that every positive locally finite measure ρ carried by V satisfies

$$\Gamma[\rho](5t):=\int e^{-5tE}\,d\rho\le C_b t^{-4}\sup_{1/2\le\lambda\le 1}\int|\widetilde{\alpha}_{t,\lambda}|^2d\rho. \tag{1909}$$

Here the inequality is valid also for infinite integrals. To see it, the transform of b is $\widehat{b}(q)=(\sin(q/3)/(q/3))^3$, and $\widehat{a}_0(s)\ge e^{-2s}$ for $s\ge 0$. Thus

$$|\widetilde{\alpha}_{t,\lambda}(E,q)|^2\ge(t^4/4)e^{-4tE}\widehat{b}(\lambda tq)^2.$$

Let $c_6>0$ be the infimum of the means of $\sin^6$ over intervals of length at least $1/2$. It is positive by periodicity, compactness for lengths between $1/2$ and π, and counting full periods for larger lengths. Set $c_b=\min(\sin^6 1,3^6c_6)$. For $|s|\le 3$, concavity of sine gives $\widehat{b}(\lambda s)^2\ge\sin^6 1$; for $|s|\ge 3$, integration over λ gives the mean of $\sin^6$ on an interval of length $|s|/6$. Therefore

$$\int_{1/2}^1 2\widehat{b}(\lambda s)^2d\lambda\ge c_b(1+|s|)^{-6}.$$

On V this is at least $c_b6^{-6}e^5e^{-tE}$ when $s=tq$, since $(1+x)^{-6}\ge 6^{-6}e^5e^{-x}$. Tonelli proves (1909) with $C_b=4\cdot 6^6e^{-5}/c_b$.

For ρ_φ, the supremum in (1909) is finite at each fixed t by (R2), so $\Gamma_\varphi(t)$ is finite for all $t>0$. Under (R7), it also gives

$$\Gamma_\varphi(t)\le C'_\varphi\max(1,t^{-1})^{4+a_\Gamma},\qquad C'_\varphi=5^{4+a_\Gamma}C_bC_\varphi. \tag{1910}$$

Define

$$k_\varphi(\xi)=\int_V e^{-\xi^0E+i\xi^1q}\,d\rho_\varphi. \tag{1911}$$

It is holomorphic on Tb and bounded by $\Gamma_\varphi(d_{\mathsf{Tb}}(\xi))$. The integrable majorant follows from $|q|\le E$; on smaller tubes it also bounds all derivatives. It is even in ξ^1 and obeys $\overline{k_\varphi(\xi)}=k_\varphi(\xi^*)$.

Proposition 96.2 (The two-point kernels)**.** Under (R1)–(R4), k_φ extends by quarter turns to a real analytic function $\widetilde{k}_\varphi$ on $\mathbb{R}^2\setminus\{0\}$. For disjointly supported profiles,

$$\sigma(\alpha\otimes\varphi,\alpha'\otimes\varphi)=\iint\alpha(y)\alpha'(y')\widetilde{k}_\varphi(y'-y)\,dy\,dy'. \tag{1912}$$

For two labels set $k_{\varphi,\psi} = (k_{\varphi+\psi} - k_{\varphi-\psi})/4$, and likewise for $\widetilde{k}$. Then

$$\sigma(\alpha \otimes \varphi, \alpha' \otimes \psi) = \sigma(\alpha \otimes \psi, \alpha' \otimes \varphi) = \iint \alpha(y)\alpha'(y')\widetilde{k}_{\varphi,\psi}(y' - y)\, dy\, dy'. \tag{1913}$$

■

Proof. If the profiles lie on opposite sides of $y^0 = c$, translate to $c = 0$ and apply (1905). Fubini gives (1912) with k_φ on the positive half-plane. For $v^0, v^1 > 0$, take profiles in small neighborhoods of $0, v$ and rotate both through $-\pi/2$. Cubic invariance equates their correlations, and the integral representation gives $k_\varphi(v) = k_\varphi(R_{-\pi/2}v)$ by shrinking the neighborhoods. This equality makes the quarter-turned definitions agree on overlaps. They cover the punctured plane and give $\widetilde{k}_\varphi$. Reality follows from the evenness in q.

For arbitrary disjoint profile supports of distance $d > 0$, partition each into squares of diameter less than $d/4$ using the functions χ above. Any pair of nonzero pieces is ordered along one of the signed coordinate axes, since one coordinate of a vector between them has modulus at least $d/\sqrt{2}$. The ordered formula applies after a quarter turn, and its sum is (1912).

Polarization in the labels gives the sum of the two mixed correlations in (1913). To see that they are equal, use an even approximate identity j_δ on the plane. Translation invariance and Riemann sums give

$$\sigma((\alpha * j_\delta) \otimes \varphi, (\alpha' * j_\delta) \otimes \psi) = \iint \alpha(u)\alpha'(u')h_\delta(u' - u)\, du\, du',$$

where $h_\delta(v) = \sigma(j_\delta \otimes \varphi, (\tau_v j_\delta) \otimes \psi)$ and δ is smaller than a quarter of the separation. Apply the same identity to $\alpha'(-u), \alpha(-u)$ and change both integration variables to their negatives. Its right side is unchanged. Rotation by π and permutation of the two entries turn its left side into the expression with the labels exchanged. As $\delta \downarrow 0$, convolution converges in C^1 on fixed separated neighborhoods. The order-two bound permits the limit and proves the equality of the mixed correlations. □

96.3. **Eliminating the fourth harmonic.**

Proposition 96.3. Under (R1)–(R4) and (R7), for each $r > 0$ the function $k_{\varphi,r}(\zeta) = k_\varphi(R_\zeta(r, 0))$, initially defined for $|\operatorname{Re}\zeta| < \pi/2$, extends to an entire function with period $\pi/2$. There are real $c_0(r), c_1(r)$ such that

$$k_{\varphi,r}(\zeta) = c_0(r) + 2c_1(r)\cos 4\zeta, \qquad \widetilde{k}_\varphi(r\cos t, r\sin t) = c_0(r) + 2c_1(r)\cos 4t. \tag{1914}$$

■

Proof. For $\zeta = t + is$ in the initial strip, $d_{\mathsf{Tb}}(R_\zeta(r, 0)) = re^{-|s|}\cos t$. The quarter-turn identity of proposition 96.2 agrees on overlapping strips by holomorphic uniqueness, so translates by $j\pi/2$ give one entire periodic function. Move t modulo $\pi/2$ into $[-\pi/4, \pi/4]$. Then

$$|k_{\varphi,r}(t + is)| \le \Gamma_\varphi(re^{-|s|}/\sqrt{2}) \le C_r e^{A|s|}, \qquad A = \max(0, 4 + a_\Gamma) < 8.$$

For an entire $\pi/2$-periodic function with this bound, shifting the contour of its Fourier coefficient from the real axis to height s shows that the coefficient of $e^{4im\zeta}$ vanishes if $4|m| > A$. Only $m = 0, 1, -1$ remain. Evenness in ξ^1 equates the last two coefficients, and reality on real points makes both remaining coefficients real. This proves the formulas. □

Proposition 96.4 (Rotation invariance at two plates)**.** Under (R1)–(R4), (R6) and (R7), $c_1(r) = 0$ for all labels and all $r > 0$. Thus $\widetilde{k}_{\varphi,\psi}(v)$ depends only on $|v|$, and correlations of two plates with disjoint planar supports are invariant under every R_t. Moreover, for $v^0 > 0$ and real s,

$$k_{\varphi,\psi}(R_{is}v) = k_{\varphi,\psi}(v). \tag{1915}$$

■

Proof. For imaginary angles the integral in (1911) is

$$k_{\varphi,r}(is) = \int_V e^{-r(E\cosh s + q\sinh s)}\, d\rho_\varphi \geq 0.$$

Dividing $c_0(r) + 2c_1(r)\cosh 4s$ by $\cosh 4s$ and letting $s \to \infty$ gives $c_1(r) \geq 0$.

For the opposite sign, turn the two-point form through $\pi/4$: $s^{\mathrm{d}}(f,h) = \sigma(R_{\pi/4}f, R_{\pi/4}h)$. It satisfies translation invariance and the same local bound; (R6) gives its coordinate reflection positivity. Hence proposition 96.1 constructs a positive measure ρ_{d} on Y. At this point no cone support for ρ_{d} is assumed.

Let $b_s(E,q) = (E\cosh s + q\sinh s, q\cosh s + E\sinh s)$. It preserves V, has inverse b_{-s}, and its first coordinate is at least $e^{-|s|}E$ on V. Thus $(b_s)_*\rho_\varphi$ has finite Laplace transforms, and its kernel at real $v^0 > 0$ is $k_\varphi(R_{is}v)$. The three harmonics in (1914) give

$$k_\varphi(R_i v) + k_\varphi(R_{-i}v) = (\cosh 4 + 1)k_\varphi(v) - (\cosh 4 - 1)\widetilde{k}_\varphi(R_{\pi/4}v).$$

Multiply by $\overline{\alpha(u)}\alpha'(u')$ at $v = u' - \theta u$ and integrate. The last term represents the turned form by (1912); the other terms are the spectral representations for the boosted measures. Uniqueness in proposition 96.1 therefore gives the identity of positive measures

$$(\cosh 4 - 1)\rho_{\mathrm{d}} + (b_1)_*\rho_\varphi + (b_{-1})_*\rho_\varphi = (\cosh 4 + 1)\rho_\varphi. \tag{1916}$$

In particular $0 \leq \rho_{\mathrm{d}} \leq (\cosh 4 + 1)(\cosh 4 - 1)^{-1}\rho_\varphi$. This proves both its cone support and finiteness of its Laplace transforms. Its kernel equals $\widetilde{k}_\varphi(R_{\pi/4}v)$ on real $v^0 > 0$, by the integral identity against profiles. Analytic continuation in the angle thus gives

$$k[\rho_{\mathrm{d}}](R_{is}(r,0)) = c_0(r) - 2c_1(r)\cosh 4s \geq 0.$$

The same limit proves $c_1(r) \leq 0$. Hence it vanishes. Polarization in $\varphi \pm \psi$ gives the radiality of the mixed kernel. Formula (1913) proves invariance of the two-plate correlations. Holomorphic uniqueness in the initial strip gives (1915). □

97. Point fields and plate kernels

The insertion estimate first gives operators smeared inside a slab. Holomorphy of the translation semigroup improves this to a pointwise operator bound. The essential step is a norm Cauchy estimate for mollified insertions; weak compactness alone would not give the products needed below. Throughout this section we assume (R1)–(R5).

Write $B_d^{\mathrm{c}}(h) = B_{2d}(\tau_{de_3}h)$ and put

$$\wp = p_{\mathrm{I}} + 4, \qquad C_{\mathrm{plt}} = 480 \cdot 8^3 m_2 c_{\mathrm{I}}^{-4} C_{\mathrm{I}} M, \qquad C_{\mathrm{pt}} = 320 \cdot 8^4 m_3 c_{\mathrm{I}}^{-4} C_{\mathrm{I}} M. \tag{1917}$$

Here the absolute constants are obtained from the radial weight

$$m(s) = \begin{cases} c_m, & 0 \leq s \leq 1/3, \\ c_m \cos^2(3\pi(s - 1/3)/4), & 1/3 \leq s \leq 1, \\ 0, & s \geq 1, \end{cases} \qquad c_m = \frac{9\pi}{5\pi^2 - 8},$$

by setting $m_0 = c_m$, $m_1 = 3\pi c_m/4$, $m_2 = \sqrt{2}m_0 m_1 + 2m_0^2$ and $m_3 = m_0^3 + \sqrt{3}m_0^2 m_1$. The function m is C^1, nonnegative and constant near zero; direct integration gives $2\pi\int_0^1 m(s)s\,ds = 1$. Let $j_\delta^{(N)}$, $N = 2, 3$, be $m(2|x|/\delta)$ divided by its integral. These are nonnegative radial C^1 mollifiers supported in $|x| \leq \delta/2$.

For $r > 0$, the product weight

$$\alpha_{r,t}^{(N)}(x) = \prod_{i=1}^{N} r^{-2} m(r^{-1}\sqrt{x_i^2 + t_i^2})$$

has supremum at most $m_0^N r^{-2N}$ and Lipschitz constant at most $\sqrt{N} m_0^{N-1} m_1 r^{-2N-1}$. Iterating the one-variable mean-value formula in polar coordinates gives

$$u(0) = \int_{(-r,r)^N} dt \int dx\, \alpha_{r,t}^{(N)}(x) u(x+it) \tag{1918}$$

whenever u is holomorphic on a neighborhood of the closed polydisc of radius r. All operator integrals below converge in norm; their integrands are continuous with compact support.

97.1. **Removing the smearing.**

Proposition 97.1 (Point plates)**.** For every real label $\varphi \in C_c^1(\mathbb{R})$ there is a norm-holomorphic family $B_\varphi(\xi,\xi')$, linear in φ, on $\mathsf{Tb} \times \mathsf{Tb}$ such that

$$\|B_\varphi(\xi,\xi')\| \le C_{\mathrm{plt}}(\lambda_\varphi + 1)\|\varphi\|_{C^1} \min\{1, d_{\mathsf{Tb}}(\xi), d_{\mathsf{Tb}}(\xi')\}^{-\wp}. \tag{1919}$$

For $\kappa, \kappa' \in \overline{\mathsf{Tb}}$ it satisfies

$$\mathsf{T}(\kappa) B_\varphi(\xi,\xi') \mathsf{T}(\kappa') = B_\varphi(\kappa+\xi, \xi'+\kappa'), \tag{1920}$$

$$B_\varphi(\xi,\xi')^* = B_\varphi(\xi'^*, \xi^*). \tag{1921}$$

If $\operatorname{supp}\alpha \subset \{|y^0| < d\}$, then

$$B_d^{\mathrm{c}}(\alpha \otimes \varphi) = \int \alpha(y) B_\varphi(dn_0 + y, dn_0 - y)\, dy. \tag{1922}$$

These properties determine the family already among weakly continuous families satisfying the shift law and the integral identity for real profiles. ∎

Proof. Put $j_\delta = j_\delta^{(2)}$ and

$$A_\delta = B_\delta^{\mathrm{c}}(j_\delta \otimes \varphi), \qquad B^\delta(\xi,\xi') = \mathsf{T}(\xi - \delta n_0) A_\delta \mathsf{T}(\xi' - \delta n_0).$$

This is defined when both tube distances are at least δ, and holomorphic when they are larger. Radiality of the mollifier and the slab adjoint formula give $A_\delta^* = A_\delta$. The shift and adjoint laws follow at once. Slab localization and tangential covariance give

$$B^\delta(dn_0 + y, dn_0 - y) = B_d^{\mathrm{c}}((\tau_y j_\delta) \otimes \varphi).$$

Consequently, for continuous α supported in $|y^0| < d - \delta$,

$$\int \alpha(y) B^\delta(dn_0 + y, dn_0 - y)\, dy = B_d^{\mathrm{c}}((\alpha * j_\delta) \otimes \varphi). \tag{1923}$$

Indeed, convolution Riemann sums converge in C^1 with one compact support, and the slab bound permits passage to the limit.

We first bound B^δ independently of δ. Set $\eta = \min(1, d_{\mathsf{Tb}}(\xi), d_{\mathsf{Tb}}(\xi'))$, $r = c_{\mathrm{I}}\eta/8$ and $0 < \delta \le r$. For unit vectors v, v', apply (1918) to $u(z) = \langle v, B^\delta(\xi + z, \xi' - z) v' \rangle$. The polydisc of radius $2r$ lies in its domain because $\eta \ge 128 r$. Move the imaginary shift into the vectors by the two contractions with arguments $\xi - \eta n_0/4 + it$ and $\xi' - \eta n_0/4 - it$. The remaining inner integral is a matrix element of

$$B_{\eta/4}^{\mathrm{c}}(\gamma_t \otimes \varphi), \qquad \gamma_t = \alpha_{r,t}^{(2)} * j_\delta.$$

Its profile is supported in $[-2r, 2r]^2$, with supremum $m_0^2 r^{-4}$ and Lipschitz constant $\sqrt{2} m_0 m_1 r^{-5}$.

Partition the label as $\varphi_k(s) = \chi(s/(4r) - k)\varphi(s)$, using the partition in the proof of proposition 95.3. At most $3(\lambda_\varphi + 1)/(c_{\mathrm{I}}\eta)$ terms are nonzero. Each $\gamma_t \otimes \varphi_k$ is supported in $B(4rke_2, c_{\mathrm{I}}\eta)$ and has sharp norm at most $40 M m_2 r^{-5} \|\varphi\|_{C^1}$. The centered version of (1902), followed by integration over $(-r,r)^2$, therefore gives

$$\|B^\delta(\xi,\xi')\| \le (2r)^2 \frac{3(\lambda_\varphi + 1)}{c_{\mathrm{I}}\eta} C_{\mathrm{I}} \eta^{-p_{\mathrm{I}}} 40 M m_2 r^{-5} \|\varphi\|_{C^1}.$$

This is (1919) with B^δ in place of B_φ.

To obtain norm convergence, fix $\epsilon > 0$ and set

$$\delta_\epsilon = \frac{c_{\mathrm{I}}}{16}\min(1,\epsilon), \qquad b_\epsilon = C_{\mathrm{plt}}(\lambda_\varphi + 1)\|\varphi\|_{C^1}(\tfrac{1}{2}\min(1,\epsilon))^{-\wp}.$$

For $\delta \le \delta_\epsilon$, the holomorphic map $z \mapsto B^\delta(\epsilon n_0 + z, \epsilon n_0 - z)$ is bounded by b_ϵ on the polydisc of radius $\epsilon/4$. Cauchy's derivative estimate gives the bound $8b_\epsilon|y|/\epsilon$ for its change from $z = 0$ to a real y with $|y| \le \epsilon/8$. Integrate against $j_{\delta'}$ and use (1923); interchange δ, δ' and use commutativity of convolution. The result is

$$\|B^\delta(\epsilon n_0, \epsilon n_0) - B^{\delta'}(\epsilon n_0, \epsilon n_0)\| \le \frac{4b_\epsilon}{\epsilon}(\delta + \delta').$$

Thus $B_{[\epsilon]} = \lim_{\delta \downarrow 0} B^\delta(\epsilon n_0, \epsilon n_0)$ exists in norm. For $0 < \epsilon \le \min(d_{\mathsf{Tb}}(\xi), d_{\mathsf{Tb}}(\xi'))$, define

$$B_\varphi(\xi, \xi') = \mathsf{T}(\xi - \epsilon n_0) B_{[\epsilon]} \mathsf{T}(\xi' - \epsilon n_0). \tag{1924}$$

This is the limit of $B^\delta(\xi, \xi')$, uniformly when both tube distances exceed ϵ, so it does not depend on the chosen ϵ. Choosing a smaller ϵ locally proves norm holomorphy directly from this formula. The bound, shifts, adjoints and linearity pass to the limit. In (1923), uniform norm convergence on the support and $\alpha * j_\delta \to \alpha$ in C^1 prove (1922).

For uniqueness, the integral identity against all real profiles and weak continuity identify the matrix elements at $(dn_0 + y, dn_0 - y)$, in particular at (dn_0, dn_0). The shift law then identifies every pair of tube points. □

Proposition 97.2 (Point fields)**.** There is a unique weakly continuous family $B(z, z')$, $(z, z') \in \mathsf{Tb}_3^2$, satisfying the three-dimensional shift law and the following slab integral formula. It is norm holomorphic and obeys

$$\|B(z, z')\| \le C_{\mathrm{pt}} \min\{1, d_3(z), d_3(z')\}^{-\wp}, \tag{1925}$$

$$\mathsf{T}_3(\kappa) B(z, z') \mathsf{T}_3(\kappa') = B(\kappa + z, z' + \kappa'), \tag{1926}$$

$$B(z, z')^* = B(z'^*, z^*), \tag{1927}$$

$$B_d^{\mathrm{c}}(h) = \int h(x) B(de_3 + x, de_3 - x)\, dx. \tag{1928}$$

Here $\kappa, \kappa' \in \overline{\mathsf{Tb}_3}$, and in the last identity $h \in C_c^1(\{|x_3| < d\}; \mathbb{C})$. ■

Proof. Use $j_\delta^{(3)}$ in the preceding construction, with $A_\delta = B_\delta^{\mathrm{c}}(j_\delta^{(3)})$ and T_3 in place of T. The convolution identity and the shift and adjoint laws have the same proofs. For completeness, the change in the uniform estimate is as follows. Put $\eta = \min(1, d_3(z), d_3(z'))$ and $r = c_{\mathrm{I}}\eta/8$. The polydisc of radius $2r$ loses at most $6r$ in tube distance. The three-variable mean-value formula, after moving the imaginary shifts into contractions, gives the centered insertion of $\gamma_t = \alpha_{r,t}^{(3)} * j_\delta^{(3)}$. Its support lies in $[-2r, 2r]^3 \subset B(0, c_{\mathrm{I}}\eta)$, and $\|\gamma_t\|_\sharp \le 40 M m_3 r^{-7}$. No label partition is needed. Hence the bound is

$$(2r)^3 C_{\mathrm{I}} \eta^{-p_{\mathrm{I}}} 40 M m_3 r^{-7} = C_{\mathrm{pt}} \eta^{-\wp}.$$

At $(\epsilon e_3, \epsilon e_3)$ use a polydisc of radius $\epsilon/6$. With $b_\epsilon = C_{\mathrm{pt}}(\frac{1}{2}\min(1,\epsilon))^{-\wp}$, Cauchy's estimate and the symmetric convolution argument give

$$\|B^\delta(\epsilon e_3, \epsilon e_3) - B^{\delta'}(\epsilon e_3, \epsilon e_3)\| \le 6b_\epsilon(\delta + \delta')/\epsilon.$$

Norm convergence, extension to arbitrary tube pairs, holomorphy, the integral formula and uniqueness now follow exactly by the limit and shift arguments above. □

Corollary 97.3. For $0 \le d' < d$ and $\operatorname{supp} h \subset \{|x_3| \le d'\}$,

$$\|B_d^{\mathrm{c}}(h)\| \le C_{\mathrm{pt}} \min(1, d - d')^{-\wp} \|h\|_{L^1}. \tag{1929}$$

If $F, F' \in \mathbb{C}\mathfrak{S}(\{x_3 > d\})$, the function

$$v_d(x) = \langle [\tau_{-de_3} F], B(de_3 + x, de_3 - x)[\tau_{-de_3} F'] \rangle \tag{1930}$$

is real analytic for $|x_3| < d$ and satisfies

$$\sigma(\theta \bar{F} \cdot (h) \cdot F') = \int h(x) v_d(x)\, dx, \tag{1931}$$

$$|v_d(x)| \le C_{\mathrm{pt}} \min(1, d - |x_3|)^{-\wp} \|[\tau_{-de_3} F]\| \|[\tau_{-de_3} F']\|. \tag{1932}$$

Moreover, identifying $\xi = (\xi^0, \xi^1)$ with $(\xi^1, 0, \xi^0)$,

$$B_\varphi(\xi, \xi') = \int \varphi(s) B(\xi + s e_2, \xi' - s e_2)\, ds. \tag{1933}$$

■□

Proof. Integrate (1925) to obtain (1929). The slab form gives (1931); its density is the restriction of a holomorphic matrix element on $|\operatorname{Re} z_3| + |\operatorname{Im} z_1| + |\operatorname{Im} z_2| < d$, and the same point bound gives (1932). The right side of (1933) is weakly continuous, satisfies the plate shift law, and, by Fubini and (1928), represents the slab operator of $\alpha \otimes \varphi$. Uniqueness in proposition 97.1 proves the identity. □

97.2. **Products and their analytic continuation.** At an interior junction of two point plates, only the sum of the two adjoining arguments matters. More precisely, for $\kappa_1, \kappa_2 \in \mathsf{Tb}$, $\kappa_0 \in \overline{\mathsf{Tb}}$ and $0 < \epsilon \le \min(d_{\mathsf{Tb}}(\kappa_1), d_{\mathsf{Tb}}(\kappa_2))$,

$$B_\varphi(\xi, \kappa_1) \mathsf{T}(\kappa_0) B_\psi(\kappa_2, \xi') = B_\varphi(\xi, \epsilon n_0) \mathsf{T}(\kappa_1 + \kappa_0 + \kappa_2 - 2\epsilon n_0) B_\psi(\epsilon n_0, \xi'). \tag{1934}$$

This follows by extracting the two semigroup factors with (1920). Thus one may replace the adjoining arguments by half their sum. An outer argument next to Ω may be changed arbitrarily, since $\mathsf{T}(\kappa)\Omega = \Omega$. The same statements hold for point fields and T_3.

For $\xi_1, \dots, \xi_{n-1} \in \mathsf{Tb}$, define

$$K_n(\xi; \underline{\varphi}) = \langle \Omega, B_{\varphi_1}(\kappa, \xi_1/2) B_{\varphi_2}(\xi_1/2, \xi_2/2) \cdots B_{\varphi_n}(\xi_{n-1}/2, \kappa) \Omega \rangle\,. \tag{1935}$$

Here $\kappa \in \mathsf{Tb}$ is arbitrary; the preceding outer argument identity makes the answer independent of it. Set $K_0 = 1$ and $K_1(\varphi) = \langle \Omega, B_\varphi(\kappa, \kappa)\Omega \rangle$. For

$$\mathsf{Tb}^{(n)} = \{x : x_1 \in \mathsf{Tb},\ x_{j+1} - x_j \in \mathsf{Tb}\},$$

let $\Psi(x; \underline{\varphi})$ be the operator product in (1935), applied to Ω, with successive differences $\xi_j = x_{j+1} - x_j$ and first outer argument x_1 in place of κ; set $\Psi(\varnothing) = \Omega$.

Proposition 97.4. The kernels K_n are holomorphic, and Ψ is norm holomorphic on $\mathsf{Tb}^{(n)}$. They are multilinear in the labels, and for $n \ge 2$,

$$|K_n(\xi; \underline{\varphi})| \le \prod_{a=1}^{n} (2^\wp C_{\mathrm{plt}}(\lambda_{\varphi_a} + 1) \|\varphi_a\|_{C^1}) \min\{1, \min_a d_{\mathsf{Tb}}(\xi_a)\}^{-\wp n}. \tag{1936}$$

For profiles whose supports are strictly ordered in time,

$$\sigma(\alpha_1 \otimes \varphi_1, \dots, \alpha_n \otimes \varphi_n) = \int \prod_a \alpha_a(y_a) K_n(y_2 - y_1, \dots, y_n - y_{n-1}; \underline{\varphi})\, dY. \tag{1937}$$

If the first support also lies in $y^0 > 0$, the class of this tuple is

$$[(\alpha_1 \otimes \varphi_1, \dots, \alpha_n \otimes \varphi_n)] = \int \prod_a \alpha_a(y_a) \Psi(Y; \underline{\varphi})\, dY. \tag{1938}$$

The vectors satisfy

$$\begin{aligned}\langle \Psi(x;\underline{\varphi}), \Psi(z;\underline{\psi})\rangle = K_{n+n'}\big((x_n - x_{n-1})^*, \dots, (x_2 - x_1)^*, \\ x_1^* + z_1, z_2 - z_1, \dots, z_{n'} - z_{n'-1}; \varphi_n, \dots, \varphi_1, \psi_1, \dots, \psi_{n'}\big).\end{aligned} \tag{1939}$$

When one side is empty, its formula is obtained by removing the first outer argument of the other vector; when both sides are empty, the value is one. In addition,

$$\overline{K_n(\xi;\underline{\varphi})} = K_n(\bar{\xi};\underline{\varphi}), \qquad K_2(\xi;\varphi,\psi) = k_{\varphi,\psi}(\xi). \tag{1940}$$

Under (R0), $K_1 = 0$. ■

Proof. Holomorphy follows by taking products and matrix elements of the norm-holomorphic point plates. Choose $\kappa = n_0$ and apply (1919) to each factor to get (1936). Taking adjoints reverses the first product; at its junction with the second, (1934) replaces x_1^*, z_1 by half their sum. This gives the Gram formula, including its empty-vector variants.

For the integral formulas choose separating times $0 = c_0 < c_1 < \cdots < c_n$ for positive-time profiles. The slab product (1903) and (1922) express the class as the integral of

$$\prod_a \alpha_a(y_a) \prod_a B_{\varphi_a}(y_a - c_{a-1}n_0, c_a n_0 - y_a)\Omega.$$

At each junction the arguments sum to $y_{a+1} - y_a$. Replacing them by half this sum gives (1938). Taking the inner product with Ω gives (1937). A common translation reduces arbitrary time-ordered supports to this case.

Reality of σ on real tests shows that the imaginary part of the continuous kernel vanishes on real time-ordered configurations: test it against arbitrary products of real profiles in small disjoint neighborhoods. The holomorphic identity theorem gives the first identity in (1940). Applying the same argument to the two integral representations of the two-point function identifies K_2 with $k_{\varphi,\psi}$ first on $\mathbb{R}^2_+$ and then on Tb. Finally a profile of integral one identifies $K_1(\varphi)$ with the one-point value, which vanishes under (R0). □

For a permutation $\mathfrak{s}$ and a frame $\upsilon \in \frac{\pi}{2}\mathbb{Z}$, set

$$\begin{aligned}\mathcal{Z}(\mathfrak{s},\upsilon) &= \{Z : z_{\mathfrak{s}(a+1)} - z_{\mathfrak{s}(a)} \in R_\upsilon \mathsf{Tb},\ a < n\}, \\ K_\upsilon^{\mathfrak{s}}(Z;\underline{\varphi}) &= K_n\big(R_{-\upsilon}\partial(\mathfrak{s}Z);\mathfrak{s}\underline{\varphi}\big).\end{aligned} \tag{1941}$$

Here $\partial Z = (z_{a+1} - z_a)_{a<n}$, and permutations act on points and labels in the same order. Let $\mathcal{Z}_n$ be the union of these tubes and $\mathcal{O}_n = \mathcal{Z}_n \cap (\mathbb{R}^2)^n$.

Lemma 97.5. The functions $K_\upsilon^{\mathfrak{s}}$ agree on overlaps and define one holomorphic function $\mathcal{K}_n(Z;\underline{\varphi})$ on the connected open set $\mathcal{Z}_n$. This set is stable under common complex translations, permutations, quarter turns, conjugation, and R_{is} for real s. The kernel is invariant under the first three operations, with labels permuted together with points, and obeys $\overline{\mathcal{K}_n(Z;\underline{\varphi})} = \mathcal{K}_n(\bar{Z};\underline{\varphi})$. Its real domain is

$$\mathcal{O}_n = \{Y : \text{all } y_a^0 \text{ are distinct, or all } y_a^1 \text{ are distinct}\}.$$

For profiles whose product support lies in one tube, integration of $K_\upsilon^{\mathfrak{s}}$ against their product gives $\sigma(\alpha_1 \otimes \varphi_1, \dots, \alpha_n \otimes \varphi_n)$. ■

Proof. Each tube is a nonempty open convex cone, the preimage of $(R_\upsilon \mathsf{Tb})^{n-1}$ under a surjective linear difference map. A real configuration belongs to it precisely when $n_\upsilon \cdot y_{\mathfrak{s}(a)}$ is strictly increasing. Permuting the entries and making the quarter turn therefore reduces its integral formula to (1937), by cubic invariance of σ.

Every nonempty intersection of two tubes is convex and contains the real part of each of its points. On its real open subset, the two kernels have equal integrals against all products of small supported profiles. Their difference therefore vanishes there. The holomorphic identity theorem on

the connected intersection proves agreement everywhere. This proves that the kernels glue without a choice of continuation path.

The stated stability follows from the difference inequalities defining the tubes; in light-cone coordinates R_{is} multiplies the two coordinates by positive real numbers. The kernel symmetries follow from its definition and (1940). There are at most $4n!$ tubes. The complement of $\mathcal{O}_n$ is the finite union

$$\{y_a^0 = y_b^0,\ y_k^1 = y_l^1\}, \qquad a \neq b,\ k \neq l.$$

Each subspace has codimension two, so its finite complement is connected. Every tube is connected and meets this real set. Their union $\mathcal{Z}_n$ is therefore connected. □

98. Rotation invariance of all correlations

We now assume (R0)–(R5). The polynomial growth of the point plates implies that their kernels have only finitely many angular frequencies. Positivity imposes an arithmetic restriction on the highest frequency; this reduces the rotation problem to the two-point function.

98.1. **Angular frequencies and their phases.**

Lemma 98.1. Put $\mathcal{M}_n = \{m \in \mathbb{Z} : 4|m| \le \wp n\}$. For each tuple of $n \ge 2$ labels there are unique holomorphic functions c_m on $\mathcal{Z}_n$, zero for $m \notin \mathcal{M}_n$, such that

$$\mathcal{K}_n(R_{is}Z; \underline{\varphi}) = \sum_{m \in \mathcal{M}_n} c_m(Z; \underline{\varphi}) e^{-4ms}. \tag{1942}$$

They inherit permutation and translation invariance and satisfy

$$c_m(R_{is}Z) = e^{-4ms} c_m(Z), \qquad \overline{c_m(X)} = c_{-m}(X) \quad (X \in \mathcal{O}_n).$$

They extend uniquely to real-analytic functions $\widetilde{c}_m$ on configurations of distinct planar points, with

$$\widetilde{c}_m(v + R_t(X - v); \underline{\varphi}) = e^{4imt} \widetilde{c}_m(X; \underline{\varphi}). \tag{1943}$$

Here a common vector v is subtracted from every point. ■

Proof. First let the consecutive differences of a real configuration X make angles of modulus less than $\pi/8$ with n_0. For every complex $\zeta = t + is$, choose a quarter-turn frame υ with $|t - \upsilon| \le \pi/4$. All the differences of $R_\zeta X$ lie in $R_\upsilon \mathsf{Tb}$, with tube distances at least

$$\min_a |x_{a+1} - x_a| \cos(3\pi/8) e^{-|s|}.$$

Thus $h_X(\zeta) = \mathcal{K}_n(R_\zeta X)$ is entire, $\pi/2$-periodic, and bounded by $C_X e^{\wp n|s|}$. Shifting the Fourier contour vertically shows that its coefficient at $e^{4im\zeta}$ vanishes if $4|m| > \wp n$. Fourier uniqueness on the real circle then gives a finite expansion, valid throughout the complex plane.

To define the coefficients at every Z, let $\mu = \#\mathcal{M}_n$ and invert the Vandermonde matrix $V_{lm} = e^{-4ml}$, $0 \le l < \mu$. If $W = V^{-1}$, put

$$c_m(Z) = \sum_{l=0}^{\mu-1} W_{ml} \mathcal{K}_n(R_{il}Z).$$

These functions are holomorphic. On each tube, permuted quarter-turned configurations of the preceding narrow angular type form a nonempty real open set. On that set the formula recovers the Fourier coefficients. The holomorphic identity theorem on the tube proves (1942) everywhere. The same Vandermonde comparison proves uniqueness and all the stated symmetry identities.

We give the phase argument explicitly because $\mathcal{O}_n$ need not contain an entire real rotation orbit. If X is a real point of a fixed tube, write its successive differences in that frame as $r_a n_{\vartheta_a}$, $|\vartheta_a| < \pi/2$. For

$$t \in I_X = \bigcap_a (-\pi/2 - \vartheta_a, \pi/2 - \vartheta_a),$$

the full vertical strip $R_{t+is}X$ remains in this tube. Analytically continuing (1942) in the parameter and comparing coefficients gives $c_m(R_tX) = e^{4imt}c_m(X)$ on I_X.

Consider next configurations with no coincident points and no perpendicular pair of distinct difference vectors. They form a rotation-invariant dense open set contained in $\mathcal{O}_n$. For density, the product of all squared differences and all such scalar products is a polynomial nonzero at $((a, a^2))_{a=1}^n$; on the line segment from any configuration to this one it has only finitely many zeros. For these configurations the preceding local phase identity holds along the entire orbit, so $t \mapsto e^{-4imt}c_m(R_tX)$ is locally constant on $\mathbb{R}$, hence constant. Continuity extends the identity to every $X, R_tX \in \mathcal{O}_n$.

For any configuration of distinct points, all but finitely many angles modulo π give distinct time coordinates. Choose such a t and define

$$\widetilde{c}_m(X) = e^{-4imt}c_m(R_tX).$$

The preceding identity makes this independent of t. Near each configuration one may keep t fixed, giving a holomorphic local extension and hence real analyticity. Translation invariance and the same definition give (1943), including rotations about arbitrary centers. □

Set $\bar{m}(0) = \bar{m}(1) = 0$. For $n \geq 2$, let $\bar{m}(n)$ be the largest $|m|$ for which c_m is nonzero for some labels, or zero if all kernels vanish. Then $0 \leq \bar{m}(n) \leq \wp n/4$. A nonzero harmonic is nonzero at some strictly time-ordered real configuration, after permuting labels: otherwise the identity theorem on a tube would make it zero. Conjugation shows that, if $\bar{m}(n) > 0$, the coefficient $c_{-\bar{m}(n)}$ is nonzero at such a configuration. Write $\widetilde{K}_n = \sum_m \widetilde{c}_m$.

98.2. **Positivity of the highest harmonic.**

Lemma 98.2. For all $a, b \geq 0$,

$$2\bar{m}(a+b) \leq \bar{m}(2a) + \bar{m}(2b). \tag{1944}$$

For $a \geq 1$, let $t_a = \widetilde{c}^{(2a)}_{-\bar{m}(2a)}$ on labelled configurations, where a labelled configuration is the unordered set of its distinct points and their labels. On configurations of a points in $\mathbb{R}^2_+$,

$$G_a(S, T) = t_a(\theta S \cup T)$$

is a positive Hermitian kernel. If L is any line, and S, T lie strictly on one side, then

$$|t_a(\theta_{\mathsf{L}} S \cup T)|^2 \leq |t_a(\theta_{\mathsf{L}} S \cup S)||t_a(\theta_{\mathsf{L}} T \cup T)|. \tag{1945}$$

■

Proof. For strictly time-ordered positive configurations x, z, the Gram formula and $(R_{is}x)^* = R_{is}x^*$ give

$$\langle \Psi(R_{is}x), \Psi(R_{is}z)\rangle = \mathcal{K}_{a+b}(R_{is}(\theta x_a, \dots, \theta x_1, z_1, \dots, z_b)).$$

An empty configuration gives Ω. If $\bar{m}(a+b) = 0$, (1944) follows from nonnegativity. Otherwise, by symmetry, take $b \geq 1$. Choose a time-ordered configuration where $c_{-\bar{m}(a+b)} \neq 0$ and translate a separating time between its first a and last b points to zero; when $a = 0$, choose this time below the first point. Cauchy–Schwarz for the two resulting vectors bounds the squared expansion with leading exponent $8\bar{m}(a+b)s$ by a product whose maximal exponent is $4(\bar{m}(2a) + \bar{m}(2b))s$. Letting $s \to +\infty$ proves (1944).

For a finite family of a-point positive configurations, multiply its Gram matrix at R_{is} by $e^{-4\bar{m}(2a)s}$ and let $s \to +\infty$. Its limit is the matrix of G_a, hence positive Hermitian. If some times agree, perturb the kth time by $k\epsilon$, avoiding the finitely many exceptional ϵ, and use continuity of $\widetilde{c}_m$.

If the positive normal to L has angle υ and $v \in \mathsf{L}$, the map $\iota y = R_{-\upsilon}(y - v)$ takes its selected side to $\mathbb{R}^2_+$. The phase law gives

$$t_a(\theta_{\mathsf{L}} S \cup T) = e^{-4i\bar{m}(2a)\upsilon}G_a(\iota S, \iota T).$$

Cauchy–Schwarz for this positive kernel proves (1945); the common phase has modulus one. □

Proposition 98.3. One has

$$\bar{m}(2a) = a\bar{m}(2), \qquad \bar{m}(n) \le \frac{n}{2}\bar{m}(2). \tag{1946}$$

■

Proof. We first prove $a \mid 2\bar{m}(2a)$. Let $\mathcal{N}_a(S)$ mean that t_a vanishes on every $2a$-point labelled configuration containing S.

We need a halving observation. Suppose S' has $j' \le a$ points on one side of a line L, and $S \subset \theta_{\mathsf{L}} S' \cup S'$ satisfies $\mathcal{N}_a(S)$. Fix the points and labels of $\theta_{\mathsf{L}} S'$ and arbitrary labels on another $2a - j'$ points. The resulting value of t_a is real analytic in these free points on the complement of their coincidence sets. This domain is connected, since the excluded affine subspaces have codimension two. On the nonempty open subset where $a - j'$ free points lie on the negative side and a on the positive side, write the configuration as $\theta_{\mathsf{L}} T \cup T'$ with $S' \subset T$. The diagonal value $t_a(\theta_{\mathsf{L}} T \cup T)$ vanishes by $\mathcal{N}_a(S)$; (1945) makes the mixed value zero. The real-analytic identity theorem therefore gives $\mathcal{N}_a(\theta_{\mathsf{L}} S')$ on all configurations. Exchanging the two sides proves $\mathcal{N}_a(S')$ as well.

Suppose now $a \nmid 2\bar{m}(2a)$, and fix any label φ. Give all vertices of a regular $2a$-gon this label. Rotation through π/a preserves the labelled configuration, whereas the phase law multiplies its top harmonic by $e^{-4i\bar{m}(2a)\pi/a} \ne 1$. Thus $\mathcal{N}_a$ holds for every full polygon. We descend from arcs of length $j \ge 2$ to arcs of length $j' = \lceil j/2 \rceil$. Write the vertices as $P_k = p + \rho n_{\omega + k\pi/a}$. Reflection in the line through p of direction $\omega + (k_0 + 1/2)\pi/a$ sends P_{k_0+k} to P_{k_0+1-k}. The arc $P_{k_0+1}, \dots, P_{k_0+j'}$ lies strictly on one side, and its reflected union is an arc of length $2j'$ containing an arc of length j. The halving observation proves $\mathcal{N}_a$ for the shorter arc. Iteration reaches a singleton at any prescribed point. Since the label was arbitrary, t_a then vanishes on every configuration. This contradicts the definition of the nonzero top harmonic. Hence the divisibility holds.

Put $\nu(a) = \bar{m}(2a)$. By (1944), ν is a convex integer sequence with $\nu(0) = 0$ and $\nu(a) \le \wp a/2$. Its increments are increasing bounded integers, hence eventually equal to an integer d. Thus $\nu(a) = da - c$ eventually, where $c = \sum_k (d - (\nu(k) - \nu(k-1))) \ge 0$ is a finite integer. Since $a \mid 2\nu(a)$, every sufficiently large a divides $2c$, forcing $c = 0$. All increments equal $d = \nu(1)$. This proves the even identity. Apply (1944) with $a = \lfloor n/2 \rfloor$ and $b = n - a$ to obtain the second one. □

98.3. **From kernels to separated tests.**

Proposition 98.4. For arbitrary profiles with pairwise disjoint supports,

$$\sigma(\alpha_1 \otimes \varphi_1, \dots, \alpha_n \otimes \varphi_n) = \int \prod_a \alpha_a(y_a) \widetilde{K}_n(Y; \underline{\varphi})\, dY. \tag{1947}$$

If $\bar{m}(n) = 0$, the functional is invariant under R_t on all n-tuples with pairwise disjoint planar projections. ■

Proof. Subtract the right side of (1947) from the left and call the difference $D(\underline{\alpha})$. It is multilinear when each profile is restricted to its own separated compact support. We first use real profiles; complexification then gives the general statement.

Let A_a^0, A_a^1 be the time and space interval hulls of the supports. Suppose each pair whose A_a^1 intersect has a separating time level outside every A_e^0. Induct on the number of such pairs. When there are none, all supports lie in one permuted quarter-turned tube, so lemma 97.5 gives $D = 0$. For the induction step choose a level c and split the profiles below and above it. Translate the upper group by $sn_{\pi/2}$. Both terms of D are real analytic in s. For the correlation this follows from the matrix element of $\mathsf{T}(2\delta, s)$, where $2\delta > 0$ is a time gap about c; it is holomorphic on $|\operatorname{Im} s| < 2\delta$. For the kernel integral, the compact product support remains in the distinct-point domain for every real s. Local holomorphic extensions on a finite cover of this compact set justify integration and give real analyticity. For large s all cross-group spatial intervals are disjoint, so the number of intersecting pairs decreases. The induction hypothesis and the identity theorem imply $D = 0$ also at $s = 0$.

For general profiles with separation δ, partition the plane into the C^1 squares obtained from $\chi(2y^0/\epsilon - j^0)\chi(2y^1/\epsilon - j^1)$, with $0 < \epsilon < \delta/(n+1)$. Each nonzero tuple of pieces has interval lengths at most ϵ. If two spatial intervals overlap, their time intervals have a gap at least $\sqrt{\delta^2 - 4\epsilon^2} > (n-2)\epsilon$. The other $n-2$ time intervals cannot cover this gap, so there is a level outside all intervals. The preceding argument applies to every tuple of pieces; multilinearity proves (1947).

If $\bar{m}(n) = 0$, the phase law makes $\widetilde{K}_n = \widetilde{c}_0$ invariant under all real rotations. Changing variables in (1947) proves invariance for plates. For a general tuple with disjoint planar projections, convolve each entry with the product mollifier

$$J_\epsilon(x) = \prod_{\mu=1}^{3} \frac{2}{\epsilon}\chi(2x_\mu/\epsilon).$$

For fixed sufficiently small ϵ, the convolution Riemann sums are finite sums of translates of J_ϵ, each of which is a plate. Their planar supports stay in fixed mutually disjoint neighborhoods. Multilinearity proves rotation invariance for these sums. They converge in C^1 to the convolutions, which converge in C^1 to the original entries. The telescoping consequence of (R2),

$$|\sigma(g_1, \dots, g_n) - \sigma(g'_1, \dots, g'_n)| \le C_n^\sigma(d; R) \sum_a \|g_a - g'_a\|_\sharp \prod_{b \neq a} \max(\|g_b\|_\sharp, \|g'_b\|_\sharp),$$

on fixed separated compact supports permits both limits. □

Theorem AK (Rotation invariance)**.** Every functional satisfying (R0)–(R7) is invariant under O(3). ■□

Proof. The two-point result proposition 96.4 and the identification $K_2 = k$ show that $K_2(R_{is}\xi)$ is independent of s for real $\xi \in \mathbb{R}^2_+$. Comparing the coefficients in (1942), and then applying the identity theorem, gives $\bar{m}(2) = 0$. By (1946), all $\bar{m}(n)$ vanish. Proposition 98.4 gives invariance under R_t on tuples with disjoint planar projections, and proposition 95.4 gives it on every separated tuple. The empty and one-point values are invariant by normalization and (R0).

Cubic conjugation gives rotations about the other coordinate axes. These generate SO(3): rotate an arbitrary unit vector to e_3 by two coordinate-plane rotations, then the remaining transformation fixes e_3 and is a rotation in its perpendicular plane. A coordinate reflection already belongs to the cubic group, so adjoining it gives all of O(3). □

99. Uniform clustering on a lattice

The estimate needed for decomposition into continuum components must be uniform as the spacing tends to zero. A decay estimate involving the supremum of a lattice polynomial would not suffice. We instead bound correlations by the reflection norms of their local factors. The argument uses coordinate transfer operators, the spectral restriction imposed by a diagonal reflection, and a Fourier cutoff which removes zero energy.

Throughout this section, ν is a translation-invariant probability law on the configurations over $a\mathbb{Z}^3$, $a > 0$. We assume RP at the six coordinate mirrors $\Sigma(e_\mu, 0)$ and $\Sigma(e_\mu, a/2)$, and at one diagonal mirror $\Sigma(e_\mu + \epsilon e_\lambda, c)$ in each coordinate plane, where $\epsilon \in \{1, -1\}$ and $c \in a\mathbb{Z}$. Translations move these diagonal mirrors to height zero and move the coordinate mirrors to every height in $(a/2)\mathbb{Z}$. In particular these hypotheses hold for every periodic Gibbs state by Proposition 103.4.

99.1. **Transfer operators and the lattice spectral restriction.** For a mirror $\Sigma(n, c)$ and a positive normal lattice vector v, suppose RP also holds at $\Sigma(n, c + n \cdot v/2)$. Complete the bounded local functions supported in the upper half-space in the form $\langle [F], [G] \rangle = \nu(\overline{\theta_{n,c}F}G)$, after quotienting its null space. This is the reflection Hilbert space at the mirror. Coordinate pictures use $n = e_\mu$ and $v = ae_\mu$; diagonal pictures use $n = e_\mu + \epsilon e_\lambda$ and $v = an$.

Lemma 99.1. In these pictures the normal translation and the tangential translations induce commuting operators

$$T[F] = [\tau_v F], \qquad V(w)[F] = [\tau_w F] \quad (n \cdot w = 0), \qquad 0 \le T \le 1,$$

where the $V(w)$ are unitary. Moreover

$$\langle [F], T^k V(w)[G] \rangle = \nu(\overline{\theta_{n,c} F}\, \tau_{kv+w} G), \qquad k \ge 0, \tag{1948}$$

and $\|[F]\|^2 \le \nu(|F|^2) \le \|F\|_\infty^2$. In a coordinate picture there is a joint spectral measure $\mathfrak{E}$ on $[0,1] \times (S^1)^2$ such that

$$T^k V(ay) = \int t^k \zeta^y \, d\mathfrak{E}(t, \zeta), \qquad y_\mu = 0. \tag{1949}$$

■

Proof. Polarization of RP, followed by a translation of an arbitrary local function into a half-space, gives invariance under the mirror. Ordinary Cauchy–Schwarz therefore bounds the reflection norm by the $L^2(\nu)$ norm. Normal translation is symmetric in the reflection form. Reflection Cauchy–Schwarz gives

$$\|[\tau_v F]\|^2 \le \|[F]\| \, \|[\tau_{2v} F]\|.$$

Iteration and the uniform bound by $\|F\|_\infty$ show $\|[\tau_v F]\| \le \|[F]\|$. Thus translation descends through the null space to a self-adjoint contraction. The identity $\theta_{n,c} = \theta_{n,c+n\cdot v/2} \tau_v$ and RP at the second mirror show $\langle [F], T[F] \rangle \ge 0$. Tangential translations preserve the form and have inverse translations, hence induce unitaries. Their commutation and (1948) follow on local classes and extend by continuity. The joint spectral theorem for the commuting normal operators $T, V(ae_\lambda), V(ae_{\lambda'})$ gives (1949). □

The following elementary complex-variable estimate isolates the spectral consequence of comparing a coordinate picture with a diagonal picture.

Lemma 99.2. Suppose a finite positive measure η on the closed unit disc satisfies, for every polynomial q,

$$\int |q|^2 \, d\eta \le C \sup_{[0,1]} |q| \sup_{S^1} |q|. \tag{1950}$$

Then $\eta\{|1+z|^2 < 4|z|\} = 0$. ■

Proof. Uniform Taylor approximation extends (1950) to functions holomorphic on a disc of radius greater than one. On the unit disc put

$$\Lambda(\xi) = \log \frac{1+\xi}{1-\xi}, \qquad g_s(z) = \cos\bigl(s\Lambda(\sqrt{z})\bigr), \qquad s > 0.$$

The principal logarithm is analytic because its argument is in the right half-plane. Since Λ is odd and cosine is even, g_s is a single-valued holomorphic function of z. Write $A(z) = |\operatorname{Im} \Lambda(\sqrt{z})|$. Then

$$\sinh^2(sA(z)) \le |g_s(z)|^2 \le \cosh^2(sA(z)), \qquad 0 \le A(z) < \pi/2,$$

and $A = 0$ on $[0,1)$. Applying (1950) to $g_s(rz)$, $0 < r < 1$, gives

$$\int \sinh^2(sA(rz)) \, d\eta(z) \le C \cosh(s\pi/2).$$

Extend A to the unit circle by $A = \pi/2$ away from 1 and $A(1) = 0$; these are its radial limits. Bounded convergence allows $r \uparrow 1$. For every $\delta > 0$ the last inequality implies

$$\eta\{A \ge \pi/4 + \delta\} \le 16 C e^{-2s\delta}$$

for all sufficiently large s, so this set is null. Inside the disc,

$$\tan A(z) = \frac{\sqrt{2(|z| - \operatorname{Re} z)}}{1 - |z|}.$$

Consequently $A(z) > \pi/4$ is equivalent to $|1+z|^2 < 4|z|$, with the same equivalence at the boundary by the radial values. A countable union over $\delta > 0$ proves the claim. □

Proposition 99.3. In every coordinate picture and for each tangential direction $\lambda \neq \mu$,

$$(1-T)^2 \geq T(2 - V(ae_\lambda) - V(ae_\lambda)^*). \tag{1951}$$

Equivalently, its joint spectral measure is supported where $(1-t)^2 \geq t|1-\zeta_\lambda|^2$. In particular, with

$$\mathsf{k} = \frac{2}{\pi}\log(1+\sqrt{2}) > 0, \tag{1952}$$

the spectral measure is supported where

$$t \leq \exp\{-\mathsf{k}|\arg\zeta_\lambda|\}. \tag{1953}$$

■

Proof. Use the available diagonal normal $n = e_\mu + \epsilon e_\lambda$ and write $n^\perp = e_\mu - \epsilon e_\lambda$. For a local upper-half-space function F, let ς be the image of its spectral measure $\langle [F], \mathfrak{E}(\cdot)[F]\rangle$ under $(t,\zeta) \mapsto z = t\zeta_\lambda^\epsilon$. Expansion of a polynomial $q(z) = \sum q_k z^k$ gives

$$\int |q(z)|^2 |z|^{2N}\, d\varsigma = \sum_{k,l} q_k \overline{q_l}\, \nu\big(\overline{\theta F}\, \tau_{kan+lan^\perp} \tau_{2Nae_\mu} F\big). \tag{1954}$$

Choose a diagonal mirror $\Sigma(n,c)$ beyond every endpoint of the bonds supporting θF, then choose N so that every endpoint supporting $\tau_{2Nae_\mu}F$ lies above that mirror. Both choices depend only on F. Set $F_1 = \theta_{n,c}\theta F$ and $F_2 = \tau_{2Nae_\mu}F$. Both functions lie in its upper half-space. In the diagonal picture, the right side of (1954) equals

$$\langle [F_1], q(T_\parallel)\overline{q}(V_\perp)[F_2]\rangle,$$

where $\overline{q}$ denotes coefficient conjugation. Its modulus is at most $\|F\|_\infty^2 \sup_{[0,1]}|q| \sup_{S^1}|q|$. Lemma 99.2 applied to $|z|^{2N}d\varsigma$ shows that ς gives no mass to $|1+z|^2 < 4|z|$: the weight is strictly positive there, since $z = 0$ is outside this set. The identity

$$|1 + t\zeta_\lambda^\epsilon|^2 - 4t = (1-t)^2 - t|1-\zeta_\lambda|^2$$

proves the spectral restriction for every local vector, hence for the whole spectral measure by density. Functional calculus gives (1951).

For $t = e^{-x} > 0$ and $\zeta_\lambda = e^{iy}$, the restriction is $\sinh(x/2) \geq |\sin(y/2)|$. The function $2\operatorname{arcsinh}(\sin(y/2))$ is concave on $[0,\pi]$: its derivative is $\cos(y/2)(1+\sin^2(y/2))^{-1/2}$, a product of nonnegative decreasing functions. It therefore lies above the chord $\mathsf{k}y$. This proves (1953); the case $t = 0$ is immediate. □

99.2. **The invariant part of a correlation.** Let $\mathcal{I}$ be the sigma-algebra of translation-invariant Borel events, completed modulo ν-null sets. In a coordinate picture let Π_0 be the orthogonal projection onto $\ker(1-T)$. It describes the nondecaying part of a correlation.

Proposition 99.4. One has $T^n \to \Pi_0$ strongly and $\Pi_0 V(w) = \Pi_0$ for tangential translations. Moreover

$$\langle [F], \Pi_0[G]\rangle = \nu\big(\overline{E_\nu[F \mid \mathcal{I}]} E_\nu[G \mid \mathcal{I}]\big). \tag{1955}$$

Conditional expectation onto $\mathcal{I}$ is unchanged by translations or coordinate reflections. In particular, ν is ergodic precisely when $\ker(1-T) = \mathbb{C}[1]$. ■

Proof. By the spectral theorem, $\Pi_0 = \mathfrak{E}(\{t=1\})$ and $T^n \to \Pi_0$ strongly. Proposition 99.3 in both tangential directions says that $t = 1$ forces $\zeta = (1,1)$, proving $\Pi_0 V(w) = \Pi_0$. Let P_μ be the $L^2(\nu)$ projection onto the functions fixed by τ_{ae_μ}. The mean ergodic theorem and (1948) give, for upper-half-space functions,

$$\langle [F], \Pi_0[G]\rangle = \langle \theta F, P_\mu G\rangle_{L^2(\nu)}. \tag{1956}$$

Since $\Pi_0 V(w) = \Pi_0$, the right side is unchanged on replacing G by $\tau_w G$. For arbitrary local F_1, F_2, choose $u = kae_\mu$ large enough that $F = \theta\tau_{-u}F_1$ and $G = \tau_u F_2$ are in the upper half-space. Substitution in this identity, using $P_\mu \tau_{\pm u} = P_\mu$, gives

$$\langle F_1, P_\mu \tau_w F_2\rangle_{L^2(\nu)} = \langle F_1, P_\mu F_2\rangle_{L^2(\nu)}.$$

Local functions are dense in $L^2(\nu)$, so $P_\mu\tau_w = P_\mu$ and, by adjoints, $\tau_w P_\mu = P_\mu$. Thus a function fixed by the one normal translation is fixed by every lattice translation. The invariant subspace is $L^2(\mathcal{I}, \nu)$, so $P_\mu = E_\nu[\,\cdot \mid \mathcal{I}]$.

Reflection preserves that subspace and its orthogonal complement, hence commutes with P_μ. Positivity of the left side of (1956) at $G = F$ implies $\langle \theta P_\mu F, P_\mu F\rangle_{L^2(\nu)} \ge 0$. The vectors $P_\mu F$ with F in the upper half-space are dense in the invariant subspace: translate each local function there without changing its projection. Reflection is consequently a positive involution on that subspace and must be the identity; its negative eigenspace is zero. Formula (1956) now becomes (1955). If ν is ergodic, the right side is $\overline{\nu(F)}\nu(G)$, so Π_0 has range $\mathbb{C}[1]$. Conversely that range identity implies $\|E_\nu[F \mid \mathcal{I}]\|_2 = |\nu(F)|$, first for local F and then for all L^2 functions. For $F = 1_E$, $E \in \mathcal{I}$, it says $\nu(E)^{1/2} = \nu(E)$, proving ergodicity. □

99.3. **A bounded density after removing zero energy.** Write $Q_r = [-ra, ra]^3$, where $r \ge 1$ is an integer, and $\theta_\mu = \theta_{e_\mu,0}$. For a bounded Borel function A of the bonds in Q_r, use the site picture to set

$$\psi_A^\mu = [\tau_{rae_\mu} A]_\mu.$$

If S_8 is the group of coordinate sign changes, define

$$\mathcal{N}_r(A) = \max_{\mu,\, s\in\mathsf{S}_8} \|\psi_{sA}^\mu\|^2, \tag{1957}$$

$$\mathcal{N}'_{r,\mu}(A) = \max\left\{\|\psi_A^\mu\|^2, \max_{\lambda\ne\mu} \|\psi_{\theta_\lambda A}^\lambda\|\|\psi_A^\lambda\|\right\}. \tag{1958}$$

Then $\mathcal{N}'_{r,\mu}(A) \le \mathcal{N}_r(A) \le \nu(|A|^2) \le \|A\|_\infty^2$. The square root of $\mathcal{N}_r$ is a seminorm, invariant under S_8. Consequently the parity parts $A_\pm = (A \pm \theta_\mu A)/2$ satisfy $\mathcal{N}_r(A_\pm) \le \mathcal{N}_r(A)$. Translation in the reflection form also gives

$$\nu(\overline{A}\tau_x B) = \langle \psi_{\theta_\mu A}^\mu, T_\mu^{x_\mu/a-2r} V_\mu(x_\perp)\psi_B^\mu\rangle, \qquad x_\mu \ge 2ra. \tag{1959}$$

The correlation at $-x$ is the conjugate of $\nu(\overline{B}\tau_x A)$; coordinate reflection changes x_μ to $-x_\mu$ and reflects both functions. These identities allow the same correlation to be estimated in any coordinate picture.

We require a scalar weight which vanishes at $t = 1$ but controls $1 - t$ from below. Let z_m be the even, period-four sequence $(z_0, z_1, z_2, z_3) = (0, 1, -2, 1)$, and let

$$B(u) = 8(1 - 4|u|/\pi)_+ \quad (-\pi \le u \le \pi)$$

be extended with period 2π. Its Fourier coefficients are

$$b_0 = 1, \qquad b_m = \left(\frac{8\sin(m\pi/8)}{m\pi}\right)^2 \quad (m \ne 0).$$

Integration by parts gives this formula; in particular the Fourier series converges absolutely. Put

$$B_z(u) = \sum_{m\in\mathbb{Z}} z_m b_m e^{imu}, \qquad \Psi(t) = 2\sum_{m\ge1} z_m b_m t^{m-1}, \quad 0 \le t \le 1. \tag{1960}$$

The convention $0^0 = 1$ is used here.

Lemma 99.5. The above series converge uniformly. One has

$$\sum_m |z_m| b_m = 8, \qquad |B_z| \le 8, \qquad B_z(u) = 0 \quad \text{if } \operatorname{dist}(u, 2\pi\mathbb{Z}) \le \pi/4.$$

Moreover $0 \le \Psi \le 8$, $\Psi(1) = 0$, $\Psi(0) = 2b_1 > 0$, and

$$\Psi(e^{-x}) \ge c_\Psi \tanh x \quad (x > 0), \qquad c_\Psi = (1+\sqrt{2})^{-1}. \tag{1961}$$

■

Proof. The identities $|z_m| = 1 - \cos(m\pi/2)$ and $z_m = \cos(m\pi/2) - \cos(m\pi)$ give

$$\sum_m |z_m| b_m = B(0) - B(\pi/2) = 8, \qquad B_z(u) = \tfrac{1}{2}B(u+\pi/2) + \tfrac{1}{2}B(u-\pi/2) - B(u+\pi).$$

The three translated triangles have disjoint interiors of their supports and vanish for $\operatorname{dist}(u, 2\pi\mathbb{Z}) \le \pi/4$. This proves the stated bounds and, at $u = 0$, $\Psi(1) = 0$. Absolute summability gives the uniform convergence. For $0 < t < 1$, the Poisson kernel $P_t(u) = (1-t^2)/(1 - 2t\cos u + t^2)$ yields

$$t\Psi(t) = \frac{1}{2\pi}\int_{-\pi/4}^{\pi/4} B(v)\left[\tfrac{1}{2}P_t(v-\pi/2) + \tfrac{1}{2}P_t(v+\pi/2) - P_t(v-\pi)\right] dv.$$

Writing $s = 1 + t^2$, the bracket is at least

$$(1-t^2)\left[\frac{1}{s} - \frac{1}{s+\sqrt{2}t}\right] = \frac{(1-t^2)\sqrt{2}t}{s(s+\sqrt{2}t)}:$$

the average of $(s \mp 2t\sin v)^{-1}$ is at least $1/s$, whereas $\cos v \ge 1/\sqrt{2}$. Since $(2\pi)^{-1}\int B = 1$, division by t gives

$$\Psi(t) \ge \frac{1-t^2}{1+t^2}\frac{\sqrt{2}}{1+t^2+\sqrt{2}t} \ge c_\Psi \frac{1-t^2}{1+t^2}.$$

This is (1961). Together with $\Psi(0) > 0$ and $\Psi(1) = 0$ it proves positivity on $[0, 1)$. □

Set

$$C_{\mathrm{dn}} = \frac{1}{4\pi^2}\left[128 + 64\left(\frac{16}{\pi\mathsf{k}} + 2 + \frac{64}{(\pi\mathsf{k})^2}\right)\right]. \tag{1962}$$

Lemma 99.6. If Φ is supported in Q_r and $\theta_\mu\Phi = \epsilon\Phi$, $\epsilon \in \{1, -1\}$, let $\varrho_\psi = \langle\psi, \mathfrak{E}_\mu(\cdot)\psi\rangle$, where $\psi = \psi^\mu_\Phi$. The image of $\Psi(t^{2r})d\varrho_\psi(t,\zeta)$ on $(S^1)^2$ has a continuous density h relative to unnormalized angular Lebesgue measure $d\alpha$ on $(-\pi,\pi]^2$, and

$$0 \le h \le C_{\mathrm{dn}} r^2 \mathcal{N}'_{r,\mu}(\Phi). \tag{1963}$$

■

Proof. Write $u(v) = \nu(\overline{\theta_\mu\Phi}\tau_v\Phi) = \epsilon\nu(\overline{\Phi}\tau_v\Phi)$ and

$$K(y) = \sum_{m\in\mathbb{Z}} z_m b_m\, u(a(2rme_\mu + y)), \qquad y_\mu = 0.$$

The series is absolutely convergent. Reflection parity makes $u(a(-ke_\mu + y)) = u(a(ke_\mu + y))$. By (1959), for $k \ge 2r$,

$$u(a(ke_\mu + y)) = \langle\psi, T_\mu^{k-2r} V_\mu(ay)\psi\rangle.$$

Pairing m with $-m$, and observing $z_0 = 0$, therefore gives

$$K(y) = \int \Psi(t^{2r})\zeta^y\, d\varrho_\psi(t,\zeta), \qquad |K(y)| \le 8\|\psi\|^2. \tag{1964}$$

Thus K is the sequence of Fourier coefficients of the weighted momentum measure.

To bound large y, choose a tangential coordinate λ with $|y_\lambda| = |y|_\infty$. Since $K(-y) = \overline{K(y)}$, assume $y_\lambda \geq 2r$. Applying (1959) in the λ picture and summing its absolutely convergent series gives

$$K(y) = \epsilon \langle \psi^{\lambda}_{\theta_\lambda \Phi}, X \psi^{\lambda}_{\Phi} \rangle, \quad X = \int t^{y_\lambda - 2r} B_z(2r \arg \zeta_\mu) \zeta_{\lambda'}^{y_{\lambda'}} \, d\mathfrak{E}_\lambda.$$

Where $B_z(2r \arg \zeta_\mu)$ is nonzero, $|\arg \zeta_\mu| > \pi/(8r)$. The lattice spectral restriction (1953) then gives $t \leq e^{-\pi \mathsf{k}/(8r)}$. Consequently

$$|K(y)| \leq 8 e^{-\pi \mathsf{k}(|y|_\infty - 2r)/(8r)} \mathcal{N}'_{r,\mu}(\Phi), \qquad |y|_\infty \geq 2r. \tag{1965}$$

There are at most $16r^2$ points with $|y|_\infty < 2r$ and $8j$ points with $|y|_\infty = j$. For $c = \pi \mathsf{k}/(8r)$,

$$\sum_{j \geq 2r} 64 j e^{-c(j-2r)} = 64 \left[\frac{2r}{1 - e^{-c}} + \frac{e^{-c}}{(1 - e^{-c})^2} \right] \leq 64 \left[2r + \frac{2r}{c} + \frac{1}{c^2} \right].$$

Here $(1 - e^{-c})^{-1} \leq 1 + c^{-1}$ and $e^{-c}(1 - e^{-c})^{-2} \leq c^{-2}$. Together with (1964), this proves

$$\sum_y |K(y)| \leq 4\pi^2 C_{\mathrm{dn}} r^2 \mathcal{N}'_{r,\mu}(\Phi).$$

The Fourier series $h(\zeta) = (2\pi)^{-2} \sum_y K(y) \zeta^{-y}$ is therefore uniformly convergent and bounded by the right side of (1963). Its angular measure has the same Fourier coefficients as the weighted momentum measure by (1964). Density of trigonometric polynomials gives equality of the measures. Positivity of the weighted measure and continuity of h imply $h \geq 0$ everywhere. □

99.4. **The clustering estimate.**

Proposition 99.7. Under the nine-mirror hypotheses of this section, put

$$C_{\mathrm{dc}} = 16 c_\Psi^{-1} \mathsf{k}^{-2} C_{\mathrm{dn}}. \tag{1966}$$

For every A, B supported in Q_r and every integer $n \geq 1$,

$$0 \leq \langle \psi^\mu_A, T^n_\mu (1 - \Pi_0) \psi^\mu_A \rangle \leq C_{\mathrm{dc}} \mathcal{N}_r(A) \left(\frac{r}{n} + 2 \frac{r^2}{n^2} \right). \tag{1967}$$

If $x \in a\mathbb{Z}^3$ and $n = |x|_\infty / a - 2r \geq 1$, then

$$\left| \nu(\overline{A} \tau_x B) - \nu(\overline{E_\nu[A \mid \mathcal{I}]} E_\nu[B \mid \mathcal{I}]) \right| \leq C_{\mathrm{dc}} \mathcal{N}_r(A)^{1/2} \mathcal{N}_r(B)^{1/2} \left(\frac{r}{n} + 2 \frac{r^2}{n^2} \right). \tag{1968}$$

In particular, for $|x| \geq 4\sqrt{3}\, ra$,

$$\left| \nu(\overline{A} \tau_x B) - \nu(\overline{E_\nu[A \mid \mathcal{I}]} E_\nu[B \mid \mathcal{I}]) \right| \leq 4\sqrt{3} C_{\mathrm{dc}} \, \mathcal{N}_r(A)^{1/2} \mathcal{N}_r(B)^{1/2} \frac{ra}{|x|}. \tag{1969}$$

The constants are independent of the spacing, the law, and the observables. For an ergodic law the subtracted term is $\overline{\nu(A)} \nu(B)$. ■

Proof. Set $D_n = T^n_\mu (1 - \Pi_0) = \int t^n 1_{t<1} \, d\mathfrak{E}_\mu$. It is positive and commutes with tangential translations. First suppose $\theta_\mu \Phi = \pm \Phi$ and let $\psi = \psi^\mu_\Phi$. By Lemma 99.6, the weighted momentum measure has a bounded density. It gives no mass to $\zeta = (1, 1)$; since $\Psi(t^{2r}) > 0$ for $t < 1$, the same is true of ϱ_ψ on

$\{t<1,\zeta=(1,1)\}$. For $0<t<1$ write $s=-\log t$. The two tangential versions of (1953) imply $s\ge \mathsf{k}|\alpha|_\infty$, where $\zeta=e^{i\alpha}$. Since $s\mapsto e^{-ns}/\tanh(2rs)$ is decreasing and $\Psi(t^{2r})\ge c_\Psi\tanh(2rs)$,

$$t^n\le c_\Psi^{-1}\frac{e^{-n\mathsf{k}|\alpha|_\infty}}{\tanh(2r\mathsf{k}|\alpha|_\infty)}\,\Psi(t^{2r})$$

away from the null point $\alpha=0$. At $t=0$, the integrand t^n is zero. Integrating against the bounded density and using $1/\tanh u\le 1+1/u$ gives

$$\begin{aligned}\langle\psi,D_n\psi\rangle&\le c_\Psi^{-1}C_{\mathrm{dn}}r^2\mathcal{N}'_{r,\mu}(\Phi)\int_{(-\pi,\pi]^2}\frac{e^{-n\mathsf{k}|\alpha|_\infty}}{\tanh(2r\mathsf{k}|\alpha|_\infty)}\,d\alpha\\&\le c_\Psi^{-1}C_{\mathrm{dn}}\mathsf{k}^{-2}\mathcal{N}'_{r,\mu}(\Phi)\left(\frac{8r^2}{n^2}+\frac{4r}{n}\right).\end{aligned}$$

Indeed the square integral equals $8\int_0^\pi ue^{-n\mathsf{k}u}/\tanh(2r\mathsf{k}u)\,du$; extend the upper limit to infinity to obtain the displayed bound. For arbitrary $A=A_+ + A_-$, positivity yields $\langle\psi_A,D_n\psi_A\rangle\le 2\langle\psi_{A_+},D_n\psi_{A_+}\rangle+2\langle\psi_{A_-},D_n\psi_{A_-}\rangle$. Use $\mathcal{N}'_{r,\mu}(A_\pm)\le\mathcal{N}_r(A)$ to obtain (1967) with (1966).

For (1968), choose μ with $|x_\mu|=|x|_\infty$ and first suppose $x_\mu>0$. Insert $T_\mu^n=\Pi_0+D_n$ into (1959). Proposition 99.4 identifies the Π_0 term with the subtracted conditional expectation. Cauchy–Schwarz for D_n, its commutation with $V_\mu(x_\perp)$, and (1967) bound the remainder as claimed. For $x_\mu<0$, conjugate and interchange A,B. If $|x|_\infty\ge 4ra$, then $n\ge|x|_\infty/(2a)\ge 2r$, and $r/n+2r^2/n^2\le 4ra/|x|_\infty$. Finally $|x|_\infty\ge|x|/\sqrt{3}$ proves (1969). The ergodic assertion follows from Proposition 99.4. □

100. Decomposition by invariant conditional expectations

A continuum limit of invariant states need not cluster. We construct its components from conditional expectations on the lattice. The essential estimate is uniform in the cutoff: two local products approach the product of their invariant conditional expectations at rate inverse distance. This identifies all moments of the component measure before a component is selected.

At cutoff K, let $\mathcal{I}$ be the Borel sigma-field invariant under all lattice translations. Consider a nonempty class $\mathcal{G}$ of laws which are invariant under lattice isometries and reflection positive at every lattice mirror. Require also that

$$\nu\in\mathcal{G},\quad h\ge 0\text{ bounded and }\mathcal{I}\text{-measurable},\quad \nu(h)>0\quad\Longrightarrow\quad \nu_h:=\frac{h\nu}{\nu(h)}\in\mathcal{G}. \tag{1970}$$

For a sequence $K_j\to\infty$ and such classes $\mathcal{G}_j$, assume common finite positive bounds

$$|S_n^T[\rho](X)|\le C_n^\infty(d;R)\|X\|_\sharp \qquad (n\ge 2,\ X\text{ of type }(d;R)), \tag{1971}$$

$$|\rho(\mathcal{O}_K(f))-\nu(\mathcal{O}_K(f))|\le C_1(R)\|f\|_\sharp, \qquad (\rho,\nu\in\mathcal{G}). \tag{1972}$$

In the second line $\operatorname{supp} f$ lies in a ball of radius R. We take C_1 nondecreasing, and C_n^∞ nonincreasing in $d\in(0,\infty]$ and nondecreasing in R.

A *frame* is a choice $\nu_j\in\mathcal{G}_j$. It converges if

$$\sigma(X)=\lim_j \nu_j(A_j(X)),\qquad A_j(X):=X_{K_j}^{\nu_j}, \tag{1973}$$

exists for every separated tuple. A limit in this frame is, more generally, $\lim_j\rho_j(A_j(X))$ with $\rho_j\in\mathcal{G}_j$. The centring law remains ν_j when the integrating law changes. Every such limit is normalized, real, symmetric and multilinear on separated supports.

Write $d(X)$ for the separation and $R(X)$ for the largest of the minimal radii of balls containing individual supports. Set $d(X)=\infty$ for a tuple of length at most one. Define

$$c_n(d;R)=\sum_{\pi\in\mathrm{Part}(n)}\prod_{J\in\pi}b_{|J|},\qquad b_1=C_1(R),\quad b_k=C_k^\infty(d;R)\ (k\ge 2),\quad c_0=1. \tag{1974}$$

The moment bound and the bound used in reflections are

$$C_{\mathrm{mom}}(X) = c_{|X|}(d(X); R(X))\|X\|_\sharp, \tag{1975}$$

$$\delta(X;\bar R) = \operatorname{dist}_\infty(\operatorname{supp} X, \mathbb{R}^3 \setminus (-\bar R, \bar R)^3), \tag{1976}$$

$$N(X;\bar R) = c_{2|X|}(\min(d(X), 2\delta(X;\bar R)); R(X))\|X\|_\sharp^2. \tag{1977}$$

Here the support is contained in the indicated open cube. Both bounds are zero if a tuple has a zero entry, and both are one on the empty tuple. These conventions avoid evaluating c_n at $R = 0$. Extend C_{mom} to finite formal sums by the triangle inequality. Put

$$k_1 = 4\sqrt{3}, \qquad k_0 = 4\sqrt{3}\, C_{\mathrm{dc}}, \tag{1978}$$

where C_{dc} is the absolute constant in the lattice clustering estimate of proposition 99.7.

Proposition 100.1 (Uniform conditional bounds and clustering)**.** For $\rho, \nu \in \mathcal{G}$ one has $|\rho(X_K^\nu)| \le C_{\mathrm{mom}}(X)$. If $\operatorname{supp} f_i \cup \operatorname{supp} f_i' \subset Q_i$, the sets Q_i lie in balls of radius R and are mutually at distance at least d, then

$$|\rho(X_K^\nu - X_K'^\nu)| \le c_n(d;R) \sum_i \|f_i - f_i'\|_\sharp \prod_{l\ne i} \max(\|f_l\|_\sharp, \|f_l'\|_\sharp). \tag{1979}$$

The same bounds hold almost surely for conditional expectations given $\mathcal{I}$.

Suppose $\bar R \in a_K\mathbb{N}$, both tuples are supported in $(-\bar R, \bar R)^3$, and $a_K \le 2\min(\delta(X;\bar R), \delta(Y;\bar R))$. For $v \in a_K\mathbb{Z}^3$, $|v| \ge k_1\bar R$,

$$|\rho((X\cdot\tau_v Y)_K^\nu) - \rho(E_\rho[X_K^\nu \mid \mathcal{I}]\, E_\rho[Y_K^\nu \mid \mathcal{I}])| \le k_0 \sqrt{N(X;\bar R)N(Y;\bar R)}\, \frac{\bar R}{|v|}. \tag{1980}$$

All constants are independent of the cutoff and of the two laws. ∎

Proof. The moment–cumulant identity for two centring laws is

$$\rho(X_K^\nu) = \sum_{\pi\in\mathrm{Part}(n)} \prod_{J\in\pi} \kappa_J, \qquad \kappa_{\{i\}} = \rho(\mathcal{O}_K(f_i)) - \nu(\mathcal{O}_K(f_i)), \quad \kappa_J = S_{|J|}^T[\rho](X_J)\ (|J|\ge 2). \tag{1981}$$

Applying (1971)–(1972) block by block gives $c_n(d;R)\|X\|_\sharp$ on any tuple of type $(d;R)$. Telescope the product defining $X_K^\nu - X_K'^\nu$ one factor at a time. Each resulting tuple remains supported in the same sets Q_i; the preceding bound gives (1979).

For a bounded Borel function A and an invariant density h, every version $e = E_\rho[A \mid \mathcal{I}]$ remains a version under ρ_h, since

$$\rho_h(kA) = \rho(hkA)/\rho(h) = \rho(hke)/\rho(h)$$

for every bounded invariant k. If $\rho'(A) \le c$ for every $\rho' \in \mathcal{G}$, condition on $\{E_\rho[A \mid \mathcal{I}] > c + \varepsilon\}$. A positive probability for this event would contradict that bound, by (1970). Applying this observation to both signs of X_K^ν and of its difference from $X_K'^\nu$ proves the conditional estimates. A version can be changed to zero on its exceptional null set to make the absolute bound hold everywhere.

Put $r = \bar R/a_K$. A plaquette lies within sup-norm distance $a_K/2$ of its barycentre. The margin assumption therefore makes $A = X_K^\nu$ and $B = Y_K^\nu$ functions of the bonds in $[-ra_K, ra_K]^3$. Lattice covariance of $\mathcal{O}_K$ and invariance of ν give

$$(\mathsf{r}X)_K^\nu = \mathsf{r}(X_K^\nu), \qquad (X\cdot\tau_v Y)_K^\nu = A\,\tau_v B.$$

The concatenation is separated because $|v|_\infty \ge 4\bar R$.

The reflection size of the real function A is the maximum, over coordinate directions μ and coordinate sign changes g, of

$$\rho((\theta_\mu \mathsf{g}A)(\tau_{2\bar R e_\mu}\mathsf{g}A)).$$

Within either reflected tuple the separation is $d(X)$. Between the two tuples it is at least $2\delta(X;\bar R)$, since their μ-coordinates lie respectively below $\bar R - \delta$ and above $\bar R + \delta$. The moment bound thus

makes this reflection size at most $N(X;\bar R)$, and the same argument applies to Y. The coordinate and diagonal mirror hypotheses of proposition 99.7 hold for ρ. Its estimate, with $ra_K=\bar R$, is precisely (1980). □

The estimates also give continuity under small translations of individual entries, uniformly in K,ρ,ν. Indeed, $\|\tau_u f-f\|_\sharp\to0$ as $u\to0$, because ∇f is uniformly continuous. The analogous fact holds for $\|\mathsf R f-f\|_\sharp$ as $\mathsf R\to1$ in $\mathrm O(3)$: on a fixed ball the chain rule gives $\nabla(\mathsf R f)(x)=\mathsf R\nabla f(\mathsf R^{-1}x)$. Enlarge each support by one quarter of $\min(1,d(X))$ and apply (1979); the enlarged supports have at least half the original separation. This supplies a common local modulus of continuity for all the translations used below.

100.1. **Limits at spatial infinity.** For a converging frame put

$$e_j(X)=E_{\nu_j}[A_j(X)\mid\mathcal I],\qquad m_j(X^1,\dots,X^q)=\nu_j\biggl(\prod_{i=1}^{q}e_j(X^i)\biggr).$$

Choose real versions with $|e_j(X)|\le C_{\mathrm{mom}}(X)$. Versions do not affect m_j. Limits below are iterated: the shift v_2 tends to infinity first, with all later shifts fixed.

Proposition 100.2 (Iterated limits)**.** For any separated tuples $X^1,\dots,X^q$, not necessarily separated from one another, $m_j(X^1,\dots,X^q)$ converges and

$$\lim_j m_j(X^1,\dots,X^q)=\lim_{|v_q|\to\infty}\cdots\lim_{|v_2|\to\infty}\sigma(X^1\cdot\tau_{v_2}X^2\cdots\tau_{v_q}X^q).\tag{1982}$$

Every intermediate limit is continuous in the remaining shifts on the open set where the remaining concatenation is separated.

More precisely, put $X'_p=X^1\cdot\tau_{v_{p+1}}X^{p+1}\cdots\tau_{v_q}X^q$ and denote the intermediate limits by $G^{(p)}$, so that $G^{(1)}$ is the unmodified correlation. If $\bar R'>0$ belongs to $\bigcup_K a_K\mathbb Z$ and $\operatorname{supp}X'_p\cup\operatorname{supp}X^p\subset(-\bar R',\bar R')^3$, then

$$|G^{(p-1)}(v_p,\dots,v_q)-G^{(p)}(v_{p+1},\dots,v_q)|\le k_0\prod_{i=2}^{p-1}C_{\mathrm{mom}}(X^i)\sqrt{N(X'_p;\bar R')N(X^p;\bar R')}\,\frac{\bar R'}{|v_p|}\tag{1983}$$

for $|v_p|\ge k_1\bar R'$. ■

Proof. Define

$$G^{(p)}_j=\nu_j\biggl(\prod_{i=2}^{p}e_j(X^i)\,A_j(X'_p)\biggr).$$

For $p=1$ these converge by the definition of the frame. For $p=q$ conditional expectation gives $G^{(q)}_j=m_j(X^1,\dots,X^q)$.

A real invariant weight w with $|w|\le\bar w$ gives

$$\nu_j(wF)=c_+\rho_+(F)-c_-\rho_-(F),\qquad c_\pm=\nu_j(w_\pm),\quad c_++c_-\le\bar w.\tag{1984}$$

Terms with $c_\pm=0$ are omitted; otherwise $\rho_\pm=(\nu_j)_{w_\pm}\in\mathcal G_j$. The conditional expectations $e_j(X)$ remain versions for $\rho_\pm$. Apply (1980) to these two laws, centred at ν_j, with $w=\prod_{i=2}^{p-1}e_j(X^i)$. It bounds $G^{(p-1)}_j-G^{(p)}_j$ by the right side of (1983), whenever the shifts and the cube side belong to the lattice and j is sufficiently large. Using (1979) in the same signed decomposition also gives a local modulus of continuity of $G^{(p)}_j$ in the remaining shifts, independent of j.

Assume $G^{(p-1)}_j$ converges. Fix the later shifts and $\bar R'$. A vector v_p in $(\bigcup_K a_K\mathbb Z)^3$ belongs to every sufficiently fine lattice. Passing to upper and lower limits in the preceding estimate bounds their difference for $G^{(p)}_j$ by twice its right side. Letting $v_p=ze_1$, $z\in\mathbb N$, tend to infinity makes this difference zero. Thus $G^{(p)}_j$ converges. The common modulus proves continuity of its limit and

extends the estimate from the dense set of lattice-compatible shifts to all real shifts. The right side tends to zero as $|v_p| \to \infty$, identifying the resulting limit as $G^{(p)}$. Induction proves every assertion, including (1982) at $p = q$. □

The constants in (1983) depend on the later shifts through X'_p and $\bar{R}'$. No joint limit with all shifts tending to infinity is asserted.

100.2. **Construction of the component measure.** A decomposition of the following kind is determined by the values at spatial infinity. Its space X is compact and metrizable, its probability measure λ has full support, and each component σ_ξ is a normalized, real, symmetric, multilinear functional on separated tuples. Require that each evaluation $\xi \mapsto \sigma_\xi(X)$ be continuous, that distinct points give distinct functionals, and that

$$\int_{\mathsf{X}} \prod_{i=1}^{q} \sigma_\xi(X^i)\, \lambda(d\xi) = \lim_{|v_q|\to\infty} \cdots \lim_{|v_2|\to\infty} \sigma(X^1 \cdot \tau_{v_2} X^2 \cdots \tau_{v_q} X^q). \tag{1985}$$

For $q = 1$ this is the decomposition of σ itself.

Theorem AL (Existence of the component measure)**.** Every converging frame in classes satisfying (1970)–(1972) has such a decomposition. Each component is a limit in the same frame. More precisely, for every $\xi \in \mathsf{X}$ there are invariant events E_j with $\nu_j(E_j) > 0$ such that $\rho_j = (\nu_j)_{1_{E_j}}$ obeys

$$\| E_{\rho_j}[A_j(X) \mid \mathcal{I}] - \sigma_\xi(X) \|_{L^\infty(\rho_j)} \longrightarrow 0 \quad \text{for every separated } X. \tag{1986}$$

Every component satisfies $|\sigma_\xi(X)| \le C_{\mathrm{mom}}(X)$ and the difference estimate (1979). Moreover

$$\#\mathsf{X} \le \liminf_{j} \dim L^2(\mathcal{V}(\mathbb{L}_{K_j}), \mathcal{I}, \nu_j),$$

with infinity allowed on either side. ■□

Proof. Let $\mathfrak{F} = \{X_1, X_2, \ldots\}$ be the countable family with controlled supports constructed in section 90. For every nonempty tuple $X = (f_1, \ldots, f_n)$ and $0 < \varepsilon \le 1$, choose $X^{[\varepsilon]} \in \mathfrak{F}$ whose entries differ from those of X by at most ε in $\|\cdot\|_\sharp$. The construction permits both supports of each entry to lie in a set contained in a ball of radius $R(X) + 1/4$, with different sets at distance at least $\frac{1}{2}\min(1, d(X))$. Indeed, take the open neighbourhoods within distance $\frac{1}{4}\min(1, d(X))$ of the original supports and use their countable dense test sets. The estimate (1979), also for conditional expectations, then gives the error

$$\varepsilon_X(\varepsilon) = c_n\Big(\tfrac{1}{2}\min(1, d(X)); R(X) + \tfrac{1}{4}\Big)\, n\varepsilon \prod_i (\|f_i\|_\sharp + 1) \longrightarrow 0. \tag{1987}$$

Choose the approximation using the fixed enumeration, so that it is independent of the law and of ξ.

Consider the compact metrizable product

$$\Pi = \prod_{X \in \mathfrak{F}} [-C_{\mathrm{mom}}(X), C_{\mathrm{mom}}(X)].$$

The map $y_j = (e_j(X))_{X\in\mathfrak{F}}$ is $\mathcal{I}$-measurable, and $\lambda_j = (y_j)_* \nu_j$ is a probability measure on Π. By proposition 100.2, the integral against λ_j of every monomial in the coordinates converges. Coordinate polynomials contain the constants and separate points; the real Stone–Weierstrass theorem makes them uniformly dense in $C(\Pi)$. Consequently $\lambda_j(\phi)$ converges for every $\phi \in C(\Pi)$: approximate ϕ by a polynomial and use $|\lambda_j(\phi)| \le \|\phi\|_\infty$. The limiting positive functional, equal to one on constants, is represented by a probability measure λ. Let $\mathsf{X} = \operatorname{supp}\lambda$. The countable base of Π shows $\lambda(\mathsf{X}) = 1$, so this is a compact metrizable space with full support.

For every open $V \subset \Pi$, $\liminf_j \lambda_j(V) \ge \lambda(V)$. One may verify this directly by increasing the continuous functions $\min(1, s\,\mathrm{dist}(\,\cdot\,, \Pi \setminus V))$ to 1_V. Fix $\xi \in \mathsf{X}$ and set

$$V_s = \{\zeta : |\zeta_{X_k} - \xi_{X_k}| < 1/s, \quad 1 \le k \le s\}.$$

Then $\lambda(V_s) > 0$, and hence $\lambda_j(V_s) > 0$ for all sufficiently large j. Choose increasing j_s with this property for every $j \ge j_s$. For $j_s \le j < j_{s+1}$ put $E_j = y_j^{-1}(V_s)$; for the finitely many earlier indices use the whole space. These invariant events have positive measure, and $\rho_j = (\nu_j)_{1_{E_j}} \in \mathcal{G}_j$. The unchanged versions of conditional expectation give

$$\|E_{\rho_j}[A_j(X_k) \mid \mathcal{I}] - \xi_{X_k}\|_\infty \le 1/s \qquad (k \le s,\ j_s \le j < j_{s+1}).$$

For an arbitrary nonempty X, comparison with $X^{[\varepsilon]}$ now yields

$$\limsup_j \|E_{\rho_j}[A_j(X) \mid \mathcal{I}] - \xi_{X^{[\varepsilon]}}\|_\infty \le \varepsilon_X(\varepsilon).$$

In particular the numbers $\xi_{X^{[\varepsilon]}}$ are Cauchy as $\varepsilon \downarrow 0$, with the difference between two of them at most the sum of the two errors. Define their limit to be $\sigma_\xi(X)$ and set $\sigma_\xi(\varnothing) = 1$. Then

$$|\sigma_\xi(X) - \xi_{X^{[\varepsilon]}}| \le \varepsilon_X(\varepsilon).$$

This bound and the preceding limsup prove (1986). Integration shows $\rho_j(A_j(X)) \to \sigma_\xi(X)$. It follows that σ_ξ is a functional of the asserted kind and inherits the moment and difference estimates. For $X \in \mathfrak{F}$ it equals the coordinate ξ_X.

The last approximation is uniform in ξ. Thus every evaluation $\sigma_\xi(X)$ is continuous, as a uniform limit of coordinates. Equality of all evaluations implies equality of all coordinates, so the component map is injective.

For tuples in $\mathfrak{F}$, convergence of coordinate monomials proves (1985). For general tuples replace each X^i by $(X^i)^{[\varepsilon]}$. Telescoping the product bounds both the component integral error and the error in m_j by

$$\sum_i \varepsilon_{X^i}(\varepsilon) \prod_{l \ne i} (C_{\mathrm{mom}}(X^l) + \varepsilon_{X^l}(\varepsilon)).$$

This tends to zero independently of j. Empty tuples contribute a factor one. Passing first in j and then in ε proves the required identity.

Finally, if the displayed liminf dimension is a finite integer N, it equals N on infinitely many indices. If X had $N+1$ distinct points, choose disjoint open neighbourhoods of them in Π. Each has positive λ-measure and therefore positive λ_j-measure for all sufficiently large j. Their inverse images under y_j give $N+1$ nonzero orthogonal invariant indicator functions, contradicting dimension N on those indices. □

Every frame has a converging subsequence: the moment bound and proposition 90.1, with $n_0 = 1$, give simultaneous convergence on all nonempty tuples, and the empty tuple has value one. The component construction itself uses the full sequence of a converging frame.

101. Clustering components and the vacuum space

We first show that the preceding decomposition is intrinsic to the limit functional. Reflection positivity will then force the symmetries of each component, and the uniform estimate (1980) will give clustering for every component.

Proposition 101.1 (Uniqueness of the decomposition)**.** Two decompositions satisfying (1985) and the stated compactness, continuity, injectivity and full-support conditions are related by a unique measure-preserving homeomorphism which identifies their component functionals.

If an isometry r preserves σ, it induces a unique measure-preserving homeomorphism h_{r} of X with

$$\sigma_{h_{\mathsf{r}}(\xi)} = \sigma_\xi \circ \mathsf{r}^{-1}, \qquad h_{\mathsf{r}\mathsf{r}'} = h_{\mathsf{r}} h_{\mathsf{r}'}. \tag{1988}$$

For $s > 0$ define

$$(\mathrm{dil}_s \sigma)(f_1, \dots, f_n) = s^{-n} \sigma(f_1(s\,\cdot), \dots, f_n(s\,\cdot)). \tag{1989}$$

The dilated components on the same probability space decompose $\mathrm{dil}_s \sigma$. ■

Proof. For any finite collection of tuples, the two component measures have the same moments in their evaluation coordinates, by (1985), allowing repeated tuples. Both coordinate images lie in a compact box. Polynomial approximation and uniqueness of the representing measure show that their finite-dimensional distributions agree.

Embed the two component spaces in $\mathbb{R}^{\mathfrak{S}(\mathbb{R}^3)}$ by all evaluations. They are compact subsets of this Hausdorff product, and their embedding maps are homeomorphisms onto their images. Every basic neighbourhood of a point in the first image is determined by finitely many evaluations. Its inverse image is a nonempty open set and hence has positive measure. Equality of the finite-dimensional distributions makes it meet the second image. Since that image is closed, it contains the point. Interchanging the two spaces proves equality of the images. Their identification is the required homeomorphism; it is unique by injectivity of the evaluations. Polynomials in these evaluations are uniformly dense on either component space, so equality of their integrals also proves preservation of the measures.

For an isometry, the functions $\sigma_\xi(\mathsf{r}^{-1}X)$ satisfy the decomposition identity: conjugating translations changes v by the orthogonal linear part of r, and therefore preserves the iterated limits $|v| \to \infty$. Apply uniqueness to obtain h_{r}. Composing its defining identities proves the group law and $h_{\mathsf{r}^{-1}} = h_{\mathsf{r}}^{-1}$. For dilation, $f(s\,\cdot)$ is translated by v/s when f is translated by v. Each iterated limit is thus rescaled by $s^{-\sum_i |X^i|}$. This is exactly the product of the factors in (1989). Continuity and injectivity are preserved because dilation of test tuples is invertible. □

Proposition 101.2 (Reflection invariance of the components)**.** Suppose a real symmetric functional σ is translation invariant, is reflection positive at a mirror $\Sigma = (u, c)$, and has a decomposition as above. Then σ and every component are invariant under θ_Σ. Each component is also invariant under $\mathsf{g}\theta_\Sigma\mathsf{g}^{-1}$ whenever the isometry g preserves σ.

Consequently a Euclidean-invariant σ which is reflection positive at one mirror has Euclidean-invariant components. ■

Proof. On the positive half-space, positivity for the two formal vectors $\varnothing, X$ implies $\sigma(\theta_\Sigma X) = \sigma(X)$: use coefficients $1, i$ and equate imaginary parts. Move any tuple into this half-space by a translation parallel to u and use $\theta_\Sigma\tau_{tu} = \tau_{-tu}\theta_\Sigma$. Translation invariance proves the same identity for all tuples.

Let F be a finite formal sum of tuples and put $F_t = \tau_{tu/2}F$. For large t, F_t lies in the positive half-space. Translation invariance and the case $q = 2$ of the decomposition identity give

$$0 \le \lim_{t\to\infty} \sigma(\theta_\Sigma \overline{F_t} \cdot F_t) = \int_{\mathsf{X}} \overline{\sigma_\xi(\theta_\Sigma F)}\sigma_\xi(F)\,\lambda(d\xi). \tag{1990}$$

Indeed, term by term the expression before the limit is $\sigma(\theta_\Sigma \bar{F} \cdot \tau_{tu}F)$, whose large translation limit is prescribed by (1985). Replace F by $F - \theta_\Sigma F$. The last integral is then

$$-\int_{\mathsf{X}} |\sigma_\xi(F - \theta_\Sigma F)|^2\,\lambda(d\xi).$$

It must vanish. The integrand is continuous, and λ has full support; hence every component is fixed by θ_Σ.

Equivalently h_{θ_Σ} in (1988) is the identity. Its conjugate by h_{g} is also the identity, proving the assertion for the conjugated mirror. If σ is Euclidean invariant, every mirror is such a conjugate. Products of two parallel reflections give translations. Every orthogonal map is a product of at most three reflections: reflect its image of a basis vector back to that vector, preserving the previously fixed vectors, and repeat. Thus every component is Euclidean invariant. □

101.1. **Properties inherited from the frame.** The following conclusions apply to every component, including points of measure zero in the component space. Their lattice approximants are the conditioned laws in (1986).

Proposition 101.3. Every limit σ' in a frame satisfying (1970)–(1972) is invariant under translations and cubic isometries, and is reflection positive at every coordinate and diagonal mirror. Its cumulants of order at least two are the limits of the cumulants of its integrating laws and obey (1971). Its one-point function is

$$\sigma'(f) = c_{\sigma'} \int f, \qquad c_{\sigma'} = \lim_j 3a_{K_j}^{-4}(\rho_j(a_p) - \nu_j(a_p)), \qquad |c_{\sigma'}| \le \bar{c}, \tag{1991}$$

where

$$\bar{c} = C_1(1) \inf\{\|f\|_\sharp : f \in C_c^1(B(0,1)),\ \int f = 1\}.$$

Its off-diagonal distributions have order at most $4n+1$ on compact sets. The coordinate reflection Hilbert spaces and their translation semigroups are therefore defined.

In the decomposition of a converging frame, $\xi \mapsto c_\xi$ is continuous, $|c_\xi| \le \bar{c}$, and

$$\sigma(f) = 0, \qquad \int_{\mathsf{X}} c_\xi \, \lambda(d\xi) = 0. \tag{1992}$$

If $\kappa = \int c_\xi^2 \, d\lambda$, then for disjoint f, f',

$$S_2^\sigma(f, f') = \int_{\mathsf{X}} S_2^{\sigma_\xi}(f, f') \, \lambda(d\xi) + \kappa \int f \int f'. \tag{1993}$$

■

Proof. All moment and difference estimates pass to the limit by proposition 100.1. The moment–cumulant polynomial identifies the limiting cumulants with those of σ'. Translations on the lattice-compatible dense subgroup pass directly to the limit; the common continuity estimate extends them to all translations. Cubic symmetries pass directly. The coordinate and diagonal reflection arguments in section 92 apply to the integrating laws ρ_j, with the invariant centring laws ν_j. They give positivity at all the stated mirrors.

The same lattice symmetry makes $\rho_j(a_p) - \nu_j(a_p)$ independent of the plaquette. The barycentre Riemann sum used in section 92 expresses the one-point function as this difference times $3a_{K_j}^{-4} r_j(f)$, where $r_j(f) \to \int f$. Taking a unit-integral test first proves convergence of the coefficient; every other test then gives (1991). Apply (1972) to tests in the unit ball and take the infimum to obtain $\bar{c}$. The distribution construction of section 93 uses precisely these cumulant and one-point bounds, so its order estimate applies. Translation invariance and coordinate positivity supply the hypotheses of section 94.

For a unit-integral f_0, $c_\xi = \sigma_\xi(f_0)$ is a continuous evaluation. Every one-point moment of the frame is zero by its own centring, and the decomposition identity gives (1992). Finally,

$$S_2^{\sigma_\xi}(f, f') = \sigma_\xi(f, f') - c_\xi^2 \int f \int f'.$$

Integrating this equality gives (1993). □

Proposition 101.4 (Uniform clustering of the components)**.** For the decomposition of a converging frame, let $\bar{R} > 0$ belong to $\bigcup_K a_K \mathbb{Z}$ and contain $\operatorname{supp} X \cup \operatorname{supp} Y$ in its open cube. Then, for every $\xi \in \mathsf{X}$ and $|v| \ge k_1 \bar{R}$,

$$|\sigma_\xi(X \cdot \tau_v Y) - \sigma_\xi(X)\sigma_\xi(Y)| \le k_0 \sqrt{N(X;\bar{R}) N(Y;\bar{R})} \, \frac{\bar{R}}{|v|}. \tag{1994}$$

The bound is uniform over all components. ■

Proof. Use the conditioned approximants ρ_j for the given component. For a lattice-compatible v, all the hypotheses of (1980) hold eventually. Its first term converges to $\sigma_\xi(X \cdot \tau_v Y)$. By (1986), the conditional expectations of $A_j(X)$ and $A_j(Y)$ converge in their respective $L^\infty(\rho_j)$ norms to the

constants $\sigma_\xi(X)$ and $\sigma_\xi(Y)$. If their errors are $\varepsilon_j(X)$ and $\varepsilon_j(Y)$, the error in the integral of their product is at most

$$\varepsilon_j(X)(|\sigma_\xi(Y)| + \varepsilon_j(Y)) + |\sigma_\xi(X)|\varepsilon_j(Y) \longrightarrow 0.$$

Thus (1994) follows on the dense subgroup. Approximate any remaining v by such vectors with $|v_j| \geq k_1 \bar{R}$; the continuity under translations already proved gives the assertion. The right side contains no component-dependent quantity. □

101.2. **Zero energy and the component space.** For a translation-invariant, reflection-positive functional σ in the preceding distribution class, let $\mathcal{H}(\sigma, u)$ be its reflection Hilbert space, $\Omega = [\varnothing]$, $T_t = e^{-tH}$, and $P_0 = 1_{\{0\}}(H)$. The spectral theorem and $H \geq 0$ give $T_t \to P_0$ strongly. Consequently

$$\lim_{t\to\infty} \sigma(\theta\bar{F} \cdot \tau_{tu} F') = \langle [F], P_0[F'] \rangle. \tag{1995}$$

Here F, F' are finite sums of tuples in the positive half-space. Also $\langle \Omega, [F] \rangle = \sigma(F)$ and $\sigma(\theta\bar{F}) = \overline{\sigma(F)}$. Since these classes are dense,

$$\ker H = \mathbb{C}\Omega \quad \Longleftrightarrow \quad \lim_{t\to\infty} \sigma(\theta\bar{F} \cdot \tau_{tu} F') = \overline{\sigma(F)}\sigma(F') \quad \text{for all } F, F'. \tag{1996}$$

Indeed, the right side identifies the matrix elements of P_0 with those of the orthogonal projection onto $\mathbb{C}\Omega$.

Applied to each component of a frame, (1994) and its reflection invariance give the right side of (1996). Thus each coordinate Hamiltonian has a one-dimensional kernel.

Theorem AM (The vacuum space of the mixture)**.** Suppose σ has a decomposition satisfying (1985). Assume its components are translation invariant and invariant under the reflection θ defining $\mathcal{H}(\sigma, u)$, and that

$$\sup_\xi |\sigma_\xi(X \cdot \tau_{tu} Y) - \sigma_\xi(X)\sigma_\xi(Y)| \longrightarrow 0 \quad (t \to \infty)$$

for positive-half-space tuples X, Y. Then

$$\langle P_0[F], P_0[F'] \rangle = \int_{\mathsf{X}} \overline{\sigma_\xi(F)}\sigma_\xi(F')\, \lambda(d\xi). \tag{1997}$$

The assignment $P_0[F] \mapsto (\xi \mapsto \sigma_\xi(F))$ extends to a unitary map $\ker H \to L^2(\mathsf{X}, \lambda)$, carrying Ω to 1. The number of components equals $\dim \ker H$ when either is finite; if X is infinite, the kernel is a separable infinite-dimensional Hilbert space.

These conclusions apply to every coordinate reflection Hilbert space of the limit of a converging frame. ■□

Proof. For tuples X, Y in the positive half-space, the two-factor decomposition identity and reflection invariance give

$$\lim_{t\to\infty} \sigma(\theta X \cdot \tau_{tu} Y) = \int_{\mathsf{X}} \sigma_\xi(X)\sigma_\xi(Y)\, d\lambda.$$

Combine this with (1995) and extend sesquilinearly to obtain (1997). It makes the proposed map well defined and isometric. The vectors $P_0[F]$ are dense in $\ker H$, so it extends to an isometry with closed range, and sends Ω to 1.

To prove surjectivity, put $\psi_F(\xi) = \sigma_\xi(F)$ and take the uniform closure of their span in $C(\mathsf{X})$, initially for positive-half-space F. It contains constants and is closed under conjugation. The uniform clustering assumption gives

$$\psi_{X \cdot \tau_{tu} Y} \longrightarrow \psi_X \psi_Y \quad \text{uniformly on } \mathsf{X}.$$

Thus the closure is an algebra. It separates points: any tuple distinguishing two components may be translated into the positive half-space without changing its values. Stone–Weierstrass makes the closure $C(\mathsf{X})$. Since continuous functions are dense in $L^2(\lambda)$, the isometry has dense as well as closed range.

Full support implies that disjoint nonempty open sets give nonzero orthogonal indicator functions in $L^2(\lambda)$. Any n distinct points have n such neighbourhoods. Hence a finite component space has exactly that dimension, and an infinite component space has infinite dimension. Separability follows from that of the reflection Hilbert space. For frame limits, propositions 101.3 and 101.4 supply both invariance and uniform clustering required above. □

102. Two-layer exchange and comparison states

The comparison argument for diagonal reflections requires more than positivity of a finite-volume measure. We first prove an exchange identity for two neighbouring layers of every periodic Gibbs state. The proof inserts a unitary operator in the transfer matrix and compares the resulting partition function with a Gibbs expectation. Strict positivity of the Fourier coefficients of the plaquette weight then turns this operator identity into an identity for arbitrary continuous functions of the two layers.

102.1. **The plaquette kernel and its square root.** Throughout this subsection $G = \mathrm{SU}(2)$ has normalized Haar measure and $\beta > 0$. Set

$$k_\beta(g) = \exp\{-\beta(1 - \tfrac{1}{2}\operatorname{tr} g)\}, \qquad k_S(u,u') = \prod_{b\in S} k_\beta(u(b)u'(b)^{-1})$$

for a finite set S. We write K_S for the integral operator with kernel k_S on $L^2(G^S)$. Inner products are antilinear in the first argument.

For $l \in \frac{1}{2}\mathbb{N}_0$, let D^l be the representation on the homogeneous polynomials of degree $2l$ in two variables, acting by $\phi(z) \mapsto \phi(zg)$, with an invariant inner product obtained by averaging any positive inner product over G. Choose an orthonormal basis and write D^l_{mn} for its matrix coefficients and $\chi_l = \operatorname{tr} D^l$ for its character. Put $\chi_{-1/2} = 0$. The normalized products

$$e_\alpha(u) = \prod_{b\in S} (2l_b+1)^{1/2} D^{l_b}_{m_b n_b}(u(b))$$

are indexed by $\alpha = (\underline{l}, \underline{m}, \underline{n})$. Their finite linear combinations form $\mathcal{T}(G^S)$.

Lemma 102.1. The coefficients satisfy

$$\int_G D^l_{mn}(g)\overline{D^{l'}_{m'n'}(g)}\,dg = \frac{\delta_{ll'}\delta_{mm'}\delta_{nn'}}{2l+1}. \tag{1998}$$

The functions e_α are an orthonormal basis of $L^2(G^S)$. The algebra $\mathcal{T}(G^S)$ is invariant under conjugation and uniformly dense in $C(G^S)$. Each fixed-spin coefficient space is invariant under conjugation. Moreover, χ_l is real, $\chi_l(g^{-1}) = \chi_l(g)$, $|\chi_l| \le 2l+1$, and

$$(\operatorname{tr} g)^N = \sum_l m_{N,l}\chi_l(g), \quad m_{N,l} \ge 0, \quad m_{2l,l} = 1, \quad \sum_l (2l+1)m_{N,l} = 2^N, \tag{1999}$$

where $m_{N,l} = 0$ for $2l > N$. ■

Proof. In the monomial basis $z_1^j z_2^{2l-j}$, a diagonal group element with entries $e^{i\omega}, e^{-i\omega}$ acts with weights $2j - 2l$. Taking $\omega = \pi/(2l+1)$ makes these eigenvalues distinct. An invariant subspace therefore contains every nonzero monomial component of each of its elements, by polynomial spectral projections. The matrix $2^{-1/2}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ sends any monomial to a polynomial with a nonzero z_1^{2l} coefficient, and sends z_1^{2l} to a polynomial with every monomial coefficient nonzero. Thus the representation is irreducible. An intertwiner has invariant kernel and image; it is zero between different dimensions, and is scalar in one irreducible representation by subtracting an eigenvalue. Averaging $D^l(g)BD^{l'}(g)^{-1}$ and taking B to be a matrix unit proves (1998).

Let $\mathcal{P}_N$ be the restrictions to G of polynomials of degree at most N in its four matrix entries. The coefficients with $2l \le N$ span a subspace of dimension $\sum_{j=0}^{N}(j+1)^2 = (N+1)(N+2)(2N+3)/6$. The

identity $g_{11}g_{22}-g_{12}g_{21}=1$ expresses every monomial of degree at most N as a sum of monomials of degrees N and $N-1$. The dimensions of these two homogeneous polynomial spaces add to the same number. Consequently $\mathcal{P}_N$ is exactly the orthogonal sum of these coefficient spaces. Conjugating an entry of an SU(2) matrix gives another entry, up to sign. Hence $\mathcal{P}_N$ is conjugation invariant, as is its orthogonal complement to $\mathcal{P}_{N-1}$, the spin-$N/2$ space. Products over S now give a self-adjoint algebra which contains the constants and separates points. Stone–Weierstrass and (1998) prove the asserted density and completeness.

Every $g\in G$ is conjugate in G to $\operatorname{diag}(e^{i\omega},e^{-i\omega})$, so $\chi_l(g)=\sum_{j=0}^{2l}e^{i(2j-2l)\omega}$. This proves the reality and bounds and gives $\chi_{1/2}\chi_l=\chi_{l+1/2}+\chi_{l-1/2}$. Starting with $m_{0,l}=\delta_{0l}$, the recursion $m_{N+1,l}=m_{N,l-1/2}+m_{N,l+1/2}$ proves (1999); evaluation at the identity gives its last equality. □

Lemma 102.2. Define

$$\lambda_l=\frac{1}{2l+1}\int_G k_\beta(g)\chi_l(g)\,dg,\qquad \lambda_\alpha=\prod_{b\in S}\lambda_{l_b}.$$

Then

$$\frac{e^{-\beta}(\beta/2)^{2l}}{(2l)!(2l+1)}\le\lambda_l\le\frac{\beta^{2l}}{(2l+1)^2(2l)!},\qquad \sum_l(2l+1)^2\lambda_l=1. \tag{2000}$$

The operator K_S is positive, injective and trace class, with $K_Se_\alpha=\lambda_\alpha e_\alpha$ and $\operatorname{Tr}K_S=1$. Its positive square root has the continuous real symmetric kernel

$$q_S(u,u')=\prod_{b\in S}q_\beta(u(b)u'(b)^{-1}),\qquad q_\beta=\sum_l(2l+1)\sqrt{\lambda_l}\chi_l.$$

Writing $q_u=q_S(u,\cdot)$, we have

$$\langle q_u,q_{u'}\rangle=k_S(u,u'),\qquad \|q_u\|=1. \tag{2001}$$

Both K_S and its square root map each fixed-spin space to itself by a strictly positive scalar. In particular, K_S maps $\mathcal{T}(G^S)$ onto itself. If a finite complex measure σ on $G^S\times Y$, with Y compact metrizable, annihilates $(K_S\varphi)(u)\psi(y)$ for every $\varphi\in\mathcal{T}(G^S)$ and a uniformly dense collection of $\psi\in C(Y)$, then $\sigma=0$. ■

Proof. Expanding the exponential and using (1999) gives

$$(2l+1)\lambda_l=e^{-\beta}\sum_{N\ge2l}\frac{(\beta/2)^N}{N!}m_{N,l}.$$

The term $N=2l$ gives the lower bound. For the upper bound use $m_{N,l}\le 2^N/(2l+1)$ and $\sum_{N\ge j}\beta^N/N!\le\beta^je^\beta/j!$. The character expansion $k_\beta=\sum_l(2l+1)\lambda_l\chi_l$ converges absolutely uniformly, since the exponential expansion has a summable majorant. At the identity it gives the sum rule in (2000). Schur orthogonality gives the stated eigenvalues of K_S; their sum with multiplicities is 1. Positivity, injectivity and the trace assertions follow.

The upper bound in (2000) gives $\sum_l(2l+1)^2\sqrt{\lambda_l}<\infty$, by the ratio test. Thus the series defining q_β converges uniformly. Applying orthogonality once more shows that its product kernel has eigenvalues $\sqrt{\lambda_\alpha}$, and convolution of two copies gives k_S. This proves (2001). Finally, division by the nonzero eigenvalues is legitimate on every finite coefficient sum. The assertion about measures follows from this surjectivity, density of products in $C(G^S\times Y)$, and uniqueness of representing measures. □

For a finite bond scheme with vertex set X and bond set S, put $(w\cdot u)(b)=w(b_-)u(b)w(b_+)^{-1}$ and

$$(R_w\varphi)(u)=\varphi(w^{-1}\cdot u),\qquad P\varphi=\int R_w\varphi\,dw.$$

Haar invariance shows that R_w is unitary and P is the orthogonal projection onto gauge invariant functions. Conjugation invariance of q_β gives $q_S(w\cdot u, w\cdot u') = q_S(u,u')$. Consequently K_S and $K_S^{1/2}$ commute with P and every R_w. If C has continuous kernel c, then CP has kernel $\int c(u, w\cdot u')\,dw$.

102.2. **Operator insertions and the finite transfer formula.** Use the third direction as time. A configuration is specified by horizontal layers u_t on the positive bonds of $a\mathbb{Z}^2$ and vertical variables $w_t(y)$ between heights ta and $(t+1)a$. The vertical plaquette over a horizontal bond b has weight

$$k_\beta(u_t(b)[(w_t\cdot u_{t+1})(b)]^{-1}). \tag{2002}$$

Indeed, its holonomy is $u_t(b)w_t(b_+)u_{t+1}(b)^{-1}w_t(b_-)^{-1}$. Horizontal plaquette actions are unchanged by a planar gauge transformation.

For a bounded operator X on $L^2(G^S)$ define

$$\kappa_X(u,u') = \langle q_u, Xq_{u'}\rangle, \qquad A_X(u,u') = \frac{\kappa_X(u,u')}{k_S(u,u')}, \qquad A_X^{(t)}(U) = A_X(u_t|_S, (w_t\cdot u_{t+1})|_S). \tag{2003}$$

The last expression uses the horizontal copies of S at heights t and $t+1$ and the vertical bonds at their endpoints. In particular, it is a continuous local function supported in that closed two-layer slab.

Proposition 102.3. The map $X \mapsto A_X$ is linear, $A_1 = 1$, and

$$\|A_X\|_\infty \le e^{2\beta \# S}\|X\|.$$

It preserves reality, tensor products on disjoint bond sets, and relabeling of bonds. The function κ_X is a continuous kernel of $K_S^{1/2}XK_S^{1/2}$.

Let $n_1, n_2, s \ge 3$ be integers and $\Gamma = a(n_1\mathbb{Z}\times n_2\mathbb{Z}\times s\mathbb{Z})$. Let S_0 and X_0 be the horizontal bonds and vertices modulo the first two periods. On $L^2(G^{S_0})$, let E multiply by $e^{-\beta V}$, where V is the sum of the horizontal plaquette actions in one layer, and set

$$Q = K_{S_0}^{1/2}, \qquad M = QPEQ.$$

Then $M \ge 0$, M is trace class and $Z_\Gamma[0] = \operatorname{Tr} M^s$. At each height t, insert operators $X_{t,j}$ on finitely many horizontal bond sets which are pairwise disjoint modulo the horizontal periods. Relabel them on S_0, and let X_t be their tensor product with the identity on the remaining bonds. Then

$$\mu_{\Gamma,\beta}\left(\prod_{t=1}^{s}\prod_j A_{X_{t,j}}^{(t)}\right) = \frac{\operatorname{Tr}(X_1M\cdots X_sM)}{\operatorname{Tr} M^s}, \qquad \left|\mu_{\Gamma,\beta}\left(\prod_{t,j}A_{X_{t,j}}^{(t)}\right)\right| \le \prod_{t,j}\|X_{t,j}\|. \tag{2004}$$

Heights in this formula are understood modulo s. ∎

Proof. The map $u \mapsto q_u$ is continuous in L^2, by uniform continuity of the kernel. Hence κ_X is continuous and $|\kappa_X| \le \|X\|$. Since $k_S \ge e^{-2\beta\#S}$, the norm bound follows. Applying the root kernel on both sides of X gives the kernel assertion, first against continuous test functions and then in L^2. Formula (2001) gives $A_1 = 1$. Product Haar measure and the product formula for q_S prove the tensor and relabeling assertions. They also prove preservation of reality.

The torus Haar measure is the product of the measures $du_t\,dw_t$, and (2002) gives

$$e^{-\beta S_\Gamma(U)} = \prod_{t=1}^{s} e^{-\beta V(u_t)}k_{S_0}(u_t, w_t\cdot u_{t+1}), \qquad u_{s+1} = u_1.$$

Multiplication by the insertions replaces the tth kernel by κ_{X_t}. Integrating w_t gives the continuous kernel of $F_t = EQX_tQP$. Integrating the remaining cyclic product of kernels therefore gives

$\operatorname{Tr}(F_1\cdots F_s)$. Each factorization needed for cyclicity contains two Hilbert–Schmidt factors Q; moving EQ from the beginning to the end yields

$$Z_\Gamma[0]\mu_{\Gamma,\beta}\left(\prod_{t,j}A^{(t)}_{X_{t,j}}\right)=\operatorname{Tr}(X_1M\cdots X_sM).$$

Since V is gauge invariant, P commutes with $E^{1/2}$ and $M=(E^{1/2}PQ)^*(E^{1/2}PQ)$ is positive trace class. Empty insertions give $Z_\Gamma[0]=\operatorname{Tr}M^s>0$.

For completeness, if $L\ge 0$ is trace class and $B_1,\dots,B_s$ are bounded, the trace Hölder inequality gives

$$|\operatorname{Tr}(B_1L\cdots B_sL)|\le\prod_t\|B_t\|\operatorname{Tr}L^s. \tag{2005}$$

Indeed, apply Lemma 86.1 to the $2s$ factors $B_tL^{1/2},L^{1/2}$. For positive trace class $A\le B$, $\operatorname{Tr}A^s\le\operatorname{Tr}B^s$: in an eigenbasis of A, bound each eigenvalue by $\langle e,Be\rangle$, and then use scalar Jensen in an eigenbasis of B. Applying this to $L^{1/2}B_t^*B_tL^{1/2}\le\|B_t\|^2L$ bounds the Schatten norm of $B_tL^{1/2}$ by $\|B_t\|(\operatorname{Tr}L^s)^{1/(2s)}$. This proves (2005), and hence the second part of (2004). □

102.3. **Exchange of two neighbouring layers.**

Proposition 102.4. Let $a>0$, $\beta>0$, and let ν be a periodic Gibbs state. For a finite horizontal bond set S, an integer t, and every $H\in C(G^S\times G^S)$,

$$\nu H(u_t|_S,(w_t\cdot u_{t+1})|_S)=\nu H((w_t\cdot u_{t+1})|_S,u_t|_S).$$

More precisely, if D is a bounded real operator on $L^2(G^S)$ with transpose $D^{\mathsf T}=-D$, then $\nu(A^{(t)}_D)=0$. ■

Proof. For a real operator, $D^{\mathsf T}=D^*$, so $X_\varepsilon=e^{\varepsilon D}$ is real unitary for real ε. Its remainder after $1+\varepsilon D$ has norm at most $\frac12\varepsilon^2\|D\|^2e^{|\varepsilon|\|D\|}$. By Proposition 102.3, for sufficiently small $|\varepsilon|$ the continuous local insertion $A^{(t)}_{X_\varepsilon}$ lies in $[1/2,3/2]$. Set $A=\log A^{(t)}_{X_\varepsilon}$. Then

$$\|A\|_\infty\le\log 2,\qquad \|A-\varepsilon A^{(t)}_D\|_\infty\le C_D\varepsilon^2. \tag{2006}$$

Choose a period number N_0 of ν and a multiple $N'\ge 3$ of N_0 large enough that distinct translates of S by $N'a\mathbb{Z}^2$ are disjoint. For each $i\ge 1$, take $\Gamma'=N'a\mathbb{Z}^3$, $\Gamma=iN'a\mathbb{Z}^3$ and the array $\mathbb{A}$ of A over Γ'/Γ. On the torus, $e^{\mathbb{A}}$ is a product of unitary insertions: at heights $t+N'k_3$ use the i^2 translates by $N'a(k_1,k_2)$, $0\le k_1,k_2<i$. Their bond sets are pairwise incongruent modulo the horizontal periods: a congruence would give a nonzero $N'a\mathbb{Z}^2$ translation intersecting S, or equal representatives. Thus (2004) gives

$$\log\frac{Z_\Gamma[\mathbb{A}]}{Z_\Gamma[0]}\le 0.$$

Enclose the support of A in a lattice cube of side ra. The lower tangent inequality 91.6, applied on the coordinate domain of side $iN'a$, gives

$$0\ge i^3\nu(A)-\bigl(24\beta+2\log 2(2r-1)^3\bigr)\#\partial_1D_i.$$

Since $\#\partial_1D_i\le 6i^2N'^2$, division by i^3 and $i\to\infty$ yield $\nu(A)\le 0$. Using (2006) for both signs of ε and then letting $\varepsilon\to 0$ proves $\nu(A^{(t)}_D)=0$.

It remains to recover arbitrary functions of the layers. Let Φ be a real antisymmetric trigonometric polynomial on $G^S\times G^S$, and write its finite expansion as

$$\Phi(u,u')=\sum_{\alpha,\alpha'}c_{\alpha\alpha'}e_\alpha(u)\overline{e_{\alpha'}(u')}.$$

Divide its coefficients by $\sqrt{\lambda_\alpha\lambda_{\alpha'}}$ to obtain Φ'. This operation preserves reality and antisymmetry: each fixed pair of spin spaces is conjugation invariant, and swapping the two spaces does not change

the positive divisor. The finite-rank operator D with kernel Φ' is therefore real and skew-adjoint. Applying $K_S^{1/2}$ in both variables gives $\kappa_D = \Phi$, and hence

$$\nu\left[\frac{\Phi}{k_S}(u_t|_S, (w_t \cdot u_{t+1})|_S)\right] = 0.$$

For real continuous H, the function $k_S(u,u')[H(u,u') - H(u',u)]$ is real and antisymmetric. Approximate it uniformly by trigonometric polynomials, then take real parts and antisymmetrize. Division by $k_S \geq e^{-2\beta \# S}$ preserves uniform convergence, proving the desired exchange identity for real H. Complex linearity finishes the proof. □

102.4. **Comparison with a state positive at a prescribed mirror.**

Proposition 102.5. Let $a > 0$, $\beta \geq 0$, $n \in \mathfrak{N}$ and $c \in \mathrm{Adm}(n)$. For every periodic Gibbs state ν and every real $A \in \mathcal{C}_{\mathrm{loc}}(\mathbb{H}^+(n,c))$ there exists a periodic Gibbs state ν_0, reflection positive at $\Sigma(n,c)$, such that $\nu(A) \leq \nu_0(A)$. ■

Proof. Choose a period number N_0 of ν, a nonempty finite bond support $B_A \subset \mathsf{Bd}(\mathbb{H}^+(n,c))$, and a lattice cube $Q = x_0 + [0, ra]^3$ containing its bonds. Choose $h \in 2N_0 a\mathbb{N}$ so that B_A lies in $\mathbb{S}_1 = \{c \leq n \cdot x \leq c + h\}$, and put $v_h = hn/|n|^2$. If $n = e_\mu$, let $\mathfrak{T} = N_0 a(\mathbb{Z}e_{\mu'} \oplus \mathbb{Z}e_{\mu''})$. If $n = e_\mu + s e_\nu$, $s = \pm 1$, let $\mathfrak{T} = N_0 a\mathbb{Z}(e_\mu - s e_\nu) \oplus N_0 a\mathbb{Z}e_\kappa$, where κ is the remaining direction. Set

$$\Gamma' = \mathfrak{T} \oplus \mathbb{Z}v_h, \qquad \iota = [a\mathbb{Z}^3 : \Gamma'], \qquad \Gamma_i = i\mathfrak{T} \oplus \mathbb{Z}(2iv_h), \quad i \geq 3.$$

These groups are adapted to n, with respective height periods h and $2ih$. Moreover $\Gamma' \subset N_0 a\mathbb{Z}^3$, so ν is Γ' invariant, and $\mathrm{sys}(\Gamma_i) \geq ia$.

In the coordinate case choose the domain

$$D_i = \{x \in a\mathbb{Z}^3 : 0 \leq x_\mu < 2ih,\ 0 \leq x_{\mu'}, x_{\mu''} < iN_0 a\}.$$

In the diagonal case choose

$$D_i = \{x \in a\mathbb{Z}^3 : 0 \leq x_\mu - s x_\nu < 2iN_0 a, \quad 0 \leq x_\kappa < iN_0 a, \quad 0 \leq x_\mu + s x_\nu < 2ih\}.$$

The two diagonal coordinates divided by a have the same parity. Reducing them modulo $2iN_0$ and $2ih/a$ preserves this parity and gives one representative per class. Thus these sets are domains of Γ_i. Counting the two end faces in each of the three integer coordinates gives

$$\frac{\#\partial_1 D_i}{\#D_i} \leq \frac{2}{i}\left(\frac{2}{N_0} + \frac{a}{h}\right).$$

For the diagonal coordinates a nearest-neighbour step changes each by at most one, which is the face condition used in this bound.

Let $\mathbb{A}$ be the array of A over Γ'/Γ_i, and $\mathbb{A}^{\mathrm{f}}$ its folded array from Section 91. Its number of terms is $|\mathbb{A}| = \#D_i/\iota$. For $t > 0$, the lower tangent inequality, folding inequality, and Jensen's inequality in the folded measure give, in that order,

$$\begin{aligned} t|\mathbb{A}|\nu(A) - \left(24\beta + 2t\|A\|_\infty (2r-1)^3\right)\#\partial_1 D_i &\leq \log \frac{Z_{\Gamma_i}[t\mathbb{A}]}{Z_{\Gamma_i}[0]} \\ &\leq \log \frac{Z_{\Gamma_i}[t\mathbb{A}^{\mathrm{f}}]}{Z_{\Gamma_i}[0]} \\ &\leq t\mu_{\Gamma_i}^{t\mathbb{A}^{\mathrm{f}}}(\mathbb{A}^{\mathrm{f}}) = t|\mathbb{A}|\mu_{\Gamma_i}^{t\mathbb{A}^{\mathrm{f}}}(A). \end{aligned}$$

The last equality is the wall symmetry of the folded measure. Hence

$$\nu(A) \leq \mu_{\Gamma_i}^{t\mathbb{A}^{\mathrm{f}}}(A) + \iota\left(\frac{24\beta}{t} + 2\|A\|_\infty (2r-1)^3\right)\frac{\#\partial_1 D_i}{\#D_i}. \tag{2007}$$

We keep t fixed while taking the volume limit. To identify its Gibbs specification, write the infinite folded family with its indices:

$$I = \mathbb{Z} \times \mathfrak{T}, \qquad A_{(j,v)} = t w_j \tau_v A, \qquad B_{(j,v)} = w_j(B_A + v). \tag{2008}$$

Here w_j are the alternating translations and reflections which send the first slab to the jth slab. Every bond belongs to at most $2\#B_A$ of these supports. For $g = g_\perp + 2ikv_h \in \Gamma_i$, define $g \cdot (j, v) = (j + 2ik, v + g_\perp)$. The relations $w_{j+2ik} = \tau_{2ikv_h} w_j$ and $\tau_{g_\perp} w_j = w_j \tau_{g_\perp}$ show that this free index action translates both the functions and their supports. Representatives of I/Γ_i are $\{1, \dots, 2i\} \times R_i$, where R_i represents $\mathfrak{T}/i\mathfrak{T}$, and their sum on the torus is $t\mathbb{A}^{\mathrm{f}}$. Thus the finite-volume DLR limit applies to one fixed locally finite perturbation family.

By compactness, a subsequence of the folded torus measures converges to a Gibbs state ν_t for this family. The finite measures, and hence their limit, are invariant under $\mathfrak{T} \oplus \mathbb{Z}(2v_h)$. A fixed continuous local F in $\mathbb{H}^+(n, c)$ lies in $\{c \le n \cdot x \le c + ih\}$ for all sufficiently large i. The two-mirror positivity of the folded torus measure then gives $\mu_{\Gamma_i}^{t\mathbb{A}^{\mathrm{f}}}(\overline{\theta F} F) \ge 0$, and this inequality passes to ν_t. The error in (2007) tends to zero, so $\nu(A) \le \nu_t(A)$.

Finally take a weak limit of a subsequence of $\nu_{1/k}$. Continuity of the local DLR kernels as the strength of the fixed family tends to zero makes its limit ν_0 an unperturbed Gibbs state. The common period group, reflection positivity, and the expectation inequality all pass to this limit. The period group has finite index, so ν_0 is periodic, as required. □

103. Symmetries of periodic Gibbs states

We first use the two-layer identity to obtain a common translation group for all periodic Gibbs states. This makes their class compact and convex. The comparison states can then test membership in its reflection-positive subclass by separation.

103.1. **Temporal gauge and determination by two layers.** Fix an integer t_0. Given the vertical variables w_t, define

$$\hat{w}_{t_0} = 1, \qquad \hat{w}_{t+1} = \hat{w}_t w_t, \qquad \tilde{u}_t = \hat{w}_t \cdot u_t.$$

This defines a continuous map $\mathtt{tg} : \mathcal{V}(\mathbb{L}_a) \to (\mathcal{V}^{(2)})^{\mathbb{Z}}$: each of its coordinates uses finitely many vertical variables and one horizontal bond. The gauge transformation $\hat{w}$ makes all vertical bonds equal to 1. Write $U[\tilde{u}]$ for the resulting configuration, with horizontal layers $\tilde{u}$ and vertical variables 1. For a gauge invariant local continuous function F,

$$F(U) = \tilde{F}(\mathtt{tg}(U)), \qquad \tilde{F}(\tilde{u}) = F(U[\tilde{u}]).$$

For finite horizontal S, let $V_S(u)$ be the sum of the planar plaquette actions which meet S. It contains at most $2\#S$ terms, so $0 \le V_S \le 4\#S$. Define the conditional layer kernel

$$\tilde{k}_{S,t}(v; \tilde{u}) = k_S(\tilde{u}_{t-1}|_S, v) e^{-\beta V_S(v \vee \tilde{u}_t|_{S^c})} k_S(v, \tilde{u}_{t+1}|_S). \tag{2009}$$

The kernel $\tilde{\gamma}_{S,t}$ replaces $\tilde{u}_t|_S$ by v and integrates with the normalized density $\tilde{k}_{S,t}(v; \tilde{u})\, dv$. This density and its normalizer are bounded below by $e^{-8\beta\#S}$ before normalization. They depend continuously on finitely many coordinates. A layer law obeys the chain equations if it is fixed by every $\tilde{\gamma}_{S,t}$ on continuous local functions.

Lemma 103.1. The temporal-gauge image of every Gibbs state obeys the chain equations. The equations are invariant under each index reflection $(\tilde{\theta}_m \tilde{u})_t = \tilde{u}_{m-t}$. If $\mathtt{tg}_* \nu$ is invariant under $\tilde{\theta}_{2t_0+1}$, then ν is invariant under the bond reflection $\theta_{e_3,(t_0+1/2)a}$. ■

Proof. The vertical plaquette formula becomes $1 - \frac{1}{2}\operatorname{tr}(\tilde{u}_t(b)\tilde{u}_{t+1}(b)^{-1})$, and the horizontal plaquette action is unchanged by $\hat{w}_t$. If only the horizontal bonds $B = \{b^{(t)} : b \in S\}$ are integrated in the DLR kernel, the variables $\hat{w}_s$ do not change. The substitution $v = \hat{w}_t \cdot U_B$ preserves product Haar

measure. The plaquettes meeting B consist of the planar plaquettes meeting S, and one vertical plaquette above and one below each bond of S. Their weight is precisely (2009). Thus

$$\gamma_B(\psi \circ \mathsf{tg}) = (\tilde{\gamma}_{S,t}\psi) \circ \mathsf{tg},$$

which proves the chain equations.

Symmetry of k_S exchanges the two neighbouring layers in (2009); hence $(\tilde{\gamma}_{S,m-t}\psi) \circ \tilde{\theta}_m = \tilde{\gamma}_{S,t}(\psi \circ \tilde{\theta}_m)$. This proves covariance of the equations. With $\theta = \theta_{e_3,(t_0+1/2)a}$ one has $\theta U[\tilde{u}] = U[\tilde{\theta}_{2t_0+1}\tilde{u}]$. Consequently invariance of the layer law implies that ν and $\theta_*\nu$ agree on all gauge invariant local continuous functions. Both measures are Gibbs states, so the gauge averaging lemma 91.3 makes them equal. □

Lemma 103.2. For $\beta > 0$, two laws obeying the chain equations and agreeing on two or more consecutive entire layers agree on all layers. ■

Proof. Suppose that the two laws agree on layers $m, \dots, n$, with $m < n$. We extend agreement to $n+1$. Fix a nonempty finite horizontal set S, and use the coordinates

$$u' = \tilde{u}_{n+1}|_S, \qquad v = \tilde{u}_n|_S, \qquad c = (\tilde{u}_n|_{S^c}, \tilde{u}_{n-1}, \dots, \tilde{u}_m) \in Y.$$

Here Y is a compact countable product of copies of G. Put

$$\Phi(v;c) = k_S(\tilde{u}_{n-1}|_S, v)e^{-\beta V_S(v \vee \tilde{u}_n|_{S^c})}, \qquad N(u',c) = \int k_S(v,u')\Phi(v;c)\,dv.$$

The assumption $m < n$ ensures that $\tilde{u}_{n-1}$ is among the coordinates of c. Both functions are continuous and local, with $\Phi \geq e^{-6\beta \# S}$ and $N \geq e^{-8\beta \# S}$. Let ρ'_i be the image of the ith law on (u',c) and set $\sigma_i = N^{-1}\rho'_i$. The chain equation at (S,n) reads

$$\rho_i(\psi) = \int \sigma_i(du', dc) \int k_S(v,u')\Phi(v;c)\psi(u',v,c)\,dv \tag{2010}$$

for every continuous local ψ. Take $\psi = \varphi(v)\chi(c)/\Phi(v;c)$, with $\varphi \in \mathcal{T}(G^S)$ and χ continuous local. This is a function of the already equal layers. Subtracting the two identities gives

$$\int (K_S\varphi)(u')\chi(c)\,d(\sigma_1 - \sigma_2) = 0.$$

The measure injectivity in Lemma 102.2 yields $\sigma_1 = \sigma_2$. Equation (2010) then gives agreement for all continuous local functions of (u', v, c). Enlarging S to contain the finitely many coordinates of any test on layer $n+1$ proves agreement on layers $m, \dots, n+1$.

Reflecting the layer indices about $m+n$ and using the covariance in Lemma 103.1 proves extension to layer $m-1$. Iterating both extensions gives agreement on every finite block of layers. Continuous local functions determine laws on this compact product, so the two laws are equal. □

Proposition 103.3. Every periodic Gibbs state is invariant under every coordinate bond reflection, and consequently under translations by $2a\mathbb{Z}^3$. The periodic Gibbs states are exactly the Gibbs states invariant under $2a\mathbb{Z}^3$; their set is nonempty, compact and convex. ■

Proof. For $\beta > 0$, take the temporal gauge based at t_0. Its layers t_0, t_0+1 are u_{t_0} and $w_{t_0} \cdot u_{t_0+1}$. Proposition 102.4 says that their joint law is invariant under exchange. The layer law and its $\tilde{\theta}_{2t_0+1}$ reflection both obey the chain equations and agree on these two layers. Lemma 103.2 makes them equal, and Lemma 103.1 gives the bond reflection invariance of ν. The same reasoning in each coordinate direction follows by applying it to the coordinate permutation of ν, which is again a periodic Gibbs state.

At $\beta = 0$, the DLR equation on the support of a local function sets its expectation equal to its product Haar integral. This determines a unique Gibbs state, which is invariant under all lattice isometries.

The product of the reflections at $x_\mu = a/2$ and $x_\mu = 3a/2$ is translation by $2ae_\mu$. Thus every periodic state is invariant under $2a\mathbb{Z}^3$. Conversely, this group has finite index, so a Gibbs state invariant under it is periodic. Compactness and convexity follow from the DLR equations and the fixed common period group; nonemptiness follows from the periodic volume construction. □

103.2. **All lattice reflections.**

Proposition 103.4. For every $a > 0$ and $\beta \geq 0$, every periodic Gibbs state is reflection positive at all coordinate site, coordinate bond, and diagonal lattice mirrors, and is invariant under $a\mathbb{Z}^3 \rtimes W_3$. In particular, the periodic Gibbs states are exactly the translation invariant Gibbs states. Their class is nonempty, compact and convex, and contains every cubic torus-limit state. ■

Proof. Fix a mirror $\Sigma(n, c)$, $n \in \mathfrak{N}$, and write $\theta = \theta_{n,c}$. Let $\mathcal{R}$ be the periodic Gibbs states reflection positive at this mirror. It is nonempty by Proposition 102.5 with $A = 0$. It is compact and convex: Proposition 103.3 gives those properties for the ambient class, and each condition $\rho(\overline{\theta F} F) \in [0, \infty)$, with F continuous local in the positive half-space, is closed and affine.

Suppose $\nu \notin \mathcal{R}$. There is a real continuous f and $\delta > 0$ such that $\nu(f) \geq \rho(f) + 3\delta$ for every $\rho \in \mathcal{R}$. Here separation can be obtained in a finite-dimensional space. Indeed, if every finite family of continuous test integrals of ν were attained by an element of $\mathcal{R}$, compactness and the finite intersection property would give an element with all the same integrals, hence $\nu \in \mathcal{R}$. Otherwise the finite vector of integrals of ν lies outside the corresponding compact convex set. Its nearest point separates it by the vector joining that point to the given vector, which gives f and a positive gap.

Choose real local A_0 with $\|f - A_0\|_\infty \leq \delta$. Then $\nu(A_0) \geq \rho(A_0) + \delta$ for every $\rho \in \mathcal{R}$. Translate A_0 by $2kan$, with k so large that its support lies in $\mathbb{H}^+(n, c)$, and call the result A. All the states in question are invariant under this translation by Proposition 103.3, so the same strict inequality holds for A. Proposition 102.5 now supplies $\nu_0 \in \mathcal{R}$ with $\nu(A) \leq \nu_0(A)$, a contradiction. This proves positivity for continuous local functions.

The same positivity holds for bounded Borel local functions. On their finite compact coordinate space, approximate such a function in L^2 for the sum of the marginal laws of ν and $\theta_*\nu$ by continuous functions. Cauchy–Schwarz bounds the change in $\nu(\overline{\theta F} F)$, so positivity passes to the limit. Replacing F by the conjugate of its reflection also gives positivity on the negative half-space and for the opposite normal.

To obtain reflection invariance, positivity makes the form $\nu(\overline{\theta F} F')$ Hermitian. Taking one entry equal to 1 gives $\nu(F) = \nu(\theta F)$ for local F in the positive half-space. Any local F_0 can be translated into that half-space by $v = 2kan$. Since both v and $-v$ are periods of ν and $\theta\tau_v = \tau_{-v}\theta$, the same equality holds for F_0. Thus $\theta_*\nu = \nu$ on all continuous local tests and hence on all Borel sets. The coordinate and diagonal lattice reflections generate $a\mathbb{Z}^3 \rtimes W_3$: adjacent site and bond mirrors give unit translations, coordinate mirrors give sign changes, and $x_\mu = x_\nu$ gives coordinate transpositions. This proves the symmetry assertion. The description and compactness of the state class then follow from the translation-invariant DLR equations, and the torus-limit inclusion is already known. □

Corollary 103.5. Every volume-first continuum limit of the centered moments of cubic torus-limit Gibbs states is reflection positive at diagonal as well as coordinate mirrors. ■□

Proof. Proposition 103.4 supplies diagonal positivity at each lattice spacing. The separated-test limit argument of Section 92, using the moving-entry lemma and the plaquette support geometry, passes these inequalities to the continuum limit. □

104. Insertion estimates for periodic Gibbs states

Fix $K \geq 1$ and $\beta > 0$. For a board (i, o) of side $l = a_i$, a cell function D, and a sequence $\ell_j \to \infty$ of admissible torus sides, write

$$\|D\|_{\mathrm{ch},\ell_\bullet} = \liminf_{j\to\infty} \|D\|_{\mathrm{ch},\ell_j}.$$

The domination inequality (1830) holds for every periodic Gibbs state and every such sequence, on every board. We use it together with Proposition 103.4. All functions in the following argument may be bounded Borel local functions; the application to source observables uses continuous ones.

104.1. **Multiple reflections with an insertion.** We need the following consequence of the finite chessboard estimate. Call (i', o') coarser than (i, o) if $i' \le i$ and $o' - o \in l\mathbb{Z}^3$. Then $d = a_{i'}/a_i = L^{i-i'}$ is odd, and each coarse cell is the union of d^3 fine cells.

Lemma 104.1. For a coarse cell C' and functions D_k of its fine cells,

$$\left\| \prod_{C_k \subset C'} D_k \right\|_{\mathrm{ch},\ell}^{\mathrm{coarse}} \le \prod_{C_k \subset C'} \|D_k\|_{\mathrm{ch},\ell}^{\mathrm{fine}}, \qquad \ell \ge 2a_{i'}. \tag{2011}$$

■

Proof. Writing $z = (o' - o)/l$, the fine cells of coarse cell j have indices

$$z + dj + \{-(d-1)/2, \dots, (d-1)/2\}^3.$$

Every coarse wall is a fine wall. Thus a coarse mirror copy sends each fine cell to a fine cell by the corresponding fine mirror map; its parity is unchanged, since parity is the determinant of that map. The disseminated coarse product is consequently a product over all fine cells in a torus period box. Apply the fine chessboard inequality to this product. Every original factor has one copy for each coarse cell, and its fine seminorm is unchanged by copying. Taking the root whose order is the number of coarse cells proves (2011). □

For a coordinate bond mirror of height c, use the form $\langle F', F\rangle_{\rho,c} = \rho(\overline{\theta_c F'} F)$ on the local positive-half-space algebra, and its seminorm $\|F\|_{\rho,c}$. Reflection positivity gives the Hermitian symmetry and Cauchy–Schwarz inequality for this form.

Proposition 104.2. Let ρ be a periodic Gibbs state. Suppose that a cell C of side l lies between two consecutive board walls $x_3 = c$ and $x_3 = c + l$. For a function D of this cell and local functions F, F' in $\{x_3 \ge c\}$,

$$\left|\rho(\overline{\theta_c F'} D \tau_{le_3} F)\right| \le \|D\|_{\mathrm{ch},\ell_\bullet} \|F'\|_{\rho,c} \|F\|_{\rho,c}. \tag{2012}$$

■

Proof. Put $\widehat{D}F = D\tau_{le_3}F$ and $D^\sharp = \overline{\theta_{c+l/2} D}$. The middle reflection preserves the cell. Translation invariance and $\theta_c \theta_{c+l/2} = \tau_{-le_3}$ give

$$\langle F', \widehat{D} F\rangle_{\rho,c} = \langle \widehat{D^\sharp} F', F\rangle_{\rho,c}.$$

Let $F_n = (\widehat{D^\sharp}\widehat{D})^n F$ and $s_n = \langle F, F_n\rangle_{\rho,c}$. Moving adjoints between the two entries gives $s_{2p} = \|F_p\|^2$, $s_{2p+1} = \|\widehat{D} F_p\|^2$, and $s_n^2 \le s_{n-1} s_{n+1}$. Hence

$$s_1^n \le s_0^{n-1} s_n. \tag{2013}$$

For positive s_1 this follows from the increasing ratios s_{n+1}/s_n; the case $s_1 = 0$ is immediate.

Expanding the iterates yields

$$s_n = \rho\left(\overline{\theta_c F} \prod_{r=0}^{2n-1} D_r \, \tau_{2nle_3} F \right), \qquad D_r = \begin{cases} \tau_{rle_3} D^\sharp, & r \text{ even}, \\ \tau_{rle_3} D, & r \text{ odd}. \end{cases}$$

These are successive mirror copies, so all their seminorms equal $\|D\|_{\mathrm{ch},\ell}$. Choose an odd coarsening factor $d = L^{r_0}$ large enough that F is supported in a coarse cell directly above the wall c. Such a board has side $l' = dl$ and center obtained from that of C by adding $(d-1)le_3/2$; its bottom wall

remains c. For $n = n'd$, group the $2n$ functions D_r into $2n'$ consecutive coarse cells, with F and its reflected copy in the two end cells. Chessboard domination and (2011) give

$$s_n \le \|F\|_\infty^2 \|D\|_{\mathrm{ch},\ell_\bullet}^{2n}.$$

To justify the lower limit here, choose infinitely many sides with $\|D\|_{\mathrm{ch},\ell_j}$ within any given positive error of its lower limit; all other factors are bounded on that same subsequence by the displayed product. Combining this estimate with (2013), taking nth roots, and letting $n' \to \infty$ gives $\|\widehat{D}F\| \le \|D\|_{\mathrm{ch},\ell_\bullet}\|F\|$. If $\|F\| = 0$, use any $n \ge 2$ first to obtain $s_1 = 0$. Cauchy–Schwarz now proves (2012). □

Proposition 104.3. Let ρ, ν be periodic Gibbs states, and let f be a real Lipschitz function compactly supported in the interior of the cell C above. Suppose (f) has a normalizer E of radius ϱ and bound $Q \ge 1$ along $\ell_\bullet$. Then

$$\left|\rho\left(\overline{\theta_c F'}[\mathcal{O}_K(f) - \nu\mathcal{O}_K(f)]\tau_{l e_3} F\right)\right| \le \frac{2Q}{\varrho}\|f\|_\sharp \|F'\|_{\rho,c}\|F\|_{\rho,c}. \tag{2014}$$

■

Proof. There is nothing to prove for $f = 0$. Otherwise put $X = \mathcal{O}_K(f)$ and $D_z = e^{zX - E(z)}$, for $|z| < \varrho/\|f\|_\sharp$. The normalizer gives $\|D_z\|_{\mathrm{ch},\ell_\bullet} \le Q$. The expectation with D_z inserted in (2012) is holomorphic: the exponential series in the bounded local variable X converges uniformly on every bounded z-disc and may be integrated term by term. Cauchy's estimate at zero therefore bounds the insertion of $X - E'(0)$ by $Q\varrho^{-1}\|f\|_\sharp\|F'\|\|F\|$. By (1838), $|\nu(D_z)| \le Q$, so a second Cauchy estimate gives $|\nu X - E'(0)| \le Q\varrho^{-1}\|f\|_\sharp$. Finally (2012) with $D = 1$ bounds the remaining expectation by $\|F'\|\|F\|$. Adding these two contributions proves (2014). □

104.2. **Uniform constants and thin plates.** Suppose the pair (K, β) has the source systems used in the cell bound, on an unbounded set $\mathfrak{l}$ of torus sides, with the common constants

$$(2\varrho; C_e, \kappa_e; E_+, C_{\mathrm{low}}; C_{\mathrm{gap}}), \qquad \varrho > 0, \quad \kappa_e > \kappa_{\mathcal{D}},$$

where the other constants are nonnegative. Here the last constant means that their vacuum terms satisfy $E_{\ell'}\ell'^{-3} - E_\ell \ell^{-3} \le C_{\mathrm{gap}}$ whenever $\ell \ge \ell'$ in $\mathfrak{l}$. Define

$$Q_{\mathrm{mid}}(l) = \exp\left\{(E_+ + C_{\mathrm{low}} + C_{\mathrm{gap}})l^3 + \Phi_1(l/4, l/4; l)\right\}. \tag{2015}$$

The explicit formula for Φ_1 gives

$$\sup_{0<l\le 1} Q_{\mathrm{mid}}(l) \le \bar{Q}_{\mathrm{mid}} := \exp\left\{E_+ + C_{\mathrm{low}} + C_{\mathrm{gap}} + C_e S(\kappa_e)\left[\frac{27}{26} + 27\{32(C_\Sigma'' + \log_L 10) + 4C_\Sigma'\}\right]\right\}. \tag{2016}$$

Lemma 104.4. If all entries of a tuple are supported in the middle cube of one cell, namely the cube of sup-radius $l/4$ about its center, then there is one normalizer E°, holomorphic on the source disc of radius 2ϱ, such that

$$\|e^{\mathcal{O}_K(\sum z_q f_q)}\|_{\mathrm{ch},\ell} \le |e^{E^\circ(z)}| Q_{\mathrm{mid}}(l), \qquad \ell \in \mathfrak{l}, \quad \ell \ge 2l, \quad z \in \mathbb{D}_\varrho(\underline{f}). \tag{2017}$$

For $n \ge 2$, if the first two supports have positive sup-distance d_∞, its mixed derivative obeys

$$|\partial_1 \cdots \partial_n E^\circ(0)| \le \left(\frac{n}{2\varrho}\right)^n \prod_q \|f_q\|_\sharp 27 C_e S(\kappa_e) \sum_{s=0}^{K} \bar{N}_s(\{1\}) e^{-\delta_e[d_\infty L^s/M - 4]_+}, \quad \delta_e = \kappa_e - \kappa_{\mathcal{D}}. \tag{2018}$$

Here $\bar{N}_s(\{1\})$ is the maximum count of cubes of side ML^{-s} meeting a mirror copy of the first support. Consequently (2014) holds with $Q = Q_{\mathrm{mid}}(l)$ for every periodic pair ρ, ν and every f in that middle cube. ■

Proof. Choose the least reference side $\ell' \in \mathfrak{l}$ with $\ell' \ge 2l$, and construct E° there, with carrier the whole middle cube and margin $l/4$. The vacuum floor and comparison of vacuum densities give $\Delta_E \le C_{\mathrm{gap}}$ on every larger side in $\mathfrak{l}$. The cell bound 89.1, using the single carrier rather than counting its entries separately, gives (2017). Its normalizer depends on the reference system, the fixed board and functions, and is independent of the larger side. Periodization preserves the sharp norms and identifies the lattice and torus cell seminorms. Thus this is one normalizer on all the stated sides, not a choice depending on ℓ. The first inequality of (1786), with source radius 2ϱ, gives (2018). An exhausting sequence in $\mathfrak{l}$, after discarding its finitely many sides below $2l$, now satisfies the hypotheses of Proposition 104.3. □

For $0 < t \le 1$ and $\lambda > 0$ set

$$B_\pm(t,\lambda) = \{|x_1| \le t,\ |x_2| \le \lambda/2,\ t \le \pm x_3 \le 2t\}, \qquad l_\lambda = \min\{L^r : r \in \mathbb{N},\ L^r \ge 8 + 2\lambda\}.$$

Both boxes lie in the middle of the central cell of side l_λ, and their sup-distance is at least $2t$. For $\delta > 0$ define the finite number

$$C_{\mathrm{dep}}(\delta, c_0) = e^{\delta c_0} \sup_{x \ge 0} (x+2)(2x+2)e^{-\delta x}(4 + 4/\delta + 4/\delta^2). \tag{2019}$$

Proposition 104.5. Under the source-system assumptions above, for every periodic Gibbs state ρ and real f supported in $B_+(t,\lambda)$,

$$|S_2^T[\rho](\theta f, f)| \le \frac{\|f\|_\sharp^2}{\varrho^2}\left[4\log 2(2Q_{\mathrm{mid}}(l_\lambda)+2)^2 + 27 C_e S(\kappa_e) C_{\mathrm{dep}}(\delta_e, 4)\frac{\lambda + 1 + m'}{t}\right]. \tag{2020}$$

If $f = \alpha \otimes \varphi$ is a plate of Section 95, with $\alpha \in \mathrm{Prof}_t$ and $\operatorname{supp}\varphi \subset [-\lambda/2, \lambda/2]$, then

$$|S_2^T[\rho](\theta f, f)| \le \left(\frac{600 M \|\varphi\|_{C^1}}{\varrho}\right)^2 \left[4\log 2(2Q_{\mathrm{mid}}(l_\lambda)+2)^2 + 27 C_e S(\kappa_e) C_{\mathrm{dep}}(\delta_e, 4)(\lambda + 1 + m')\right] t^{-3}. \tag{2021}$$

■

Proof. Apply Lemma 104.4 to $(\theta f, f)$ in the central cell of side l_λ. The cumulant transfer estimate (1839) bounds its difference from $\partial_1\partial_2 E^\circ(0)$ by $4\log 2(2Q_{\mathrm{mid}}(l_\lambda)+2)^2\|f\|_\sharp^2/\varrho^2$. At depth s, put $x_s = tL^s/M$. A cube grid of side W has at most $\prod_\mu (s_\mu/W + 2)$ cubes meeting any translate of a box of edge lengths s_μ: in each coordinate the admissible integer centers lie in an interval of length $s_\mu/W + 1$. Every mirror copy of the first support lies in a box with edge lengths $(2t, \lambda, t)$. Thus

$$\bar N_s(\{1\}) \le (2x_s + 2)(\lambda x_s/t + 2)(x_s + 2).$$

In (2018), $d_\infty \ge 2t$. Its depth sum is bounded by

$$\sum_{s=0}^{K} (x_s+2)(2x_s+2)(\lambda x_s/t + 2)e^{-\delta_e[2x_s - 4]_+} \le C_{\mathrm{dep}}(\delta_e, 4)\frac{\lambda + 1 + m'}{t}.$$

Here is the scale summation. After extracting $e^{4\delta_e}\sup_{x\ge 0}(x+2)(2x+2)e^{-\delta_e x}$, it remains to bound $(\lambda/t)\sum_s x_s e^{-\delta_e x_s} + 2\sum_s e^{-\delta_e x_s}$. Since successive x_s grow by $L \ge 2$, $\sum_{x_s \le 1} x_s \le 2$ and $\sum_{x_s > 1} x_s^{-1} \le 2$. The inequalities $u^2 e^{-u} \le 2$ and $u e^{-u} \le 1$ give

$$\sum_s x_s e^{-\delta_e x_s} \le 2 + 4/\delta_e^2, \qquad \sum_s e^{-\delta_e x_s} \le 1 + m' + 1/t + 2/\delta_e.$$

The number of scales with $x_s \le 1$ is at most $1 + m' + \log_L(1/t)$, and $\log_L(1/t) \le 1/t$ was used in the second inequality. These bounds give the stated depth estimate and prove (2020).

The profile bounds $\|\alpha\|_\infty \le 3$ and $\operatorname{Lip}\alpha \le 12/t$ yield $\|f\|_\infty \le 3\|\varphi\|_\infty$ and $\operatorname{Lip} f \le 15\|\varphi\|_{C^1}/t$. Consequently $\|f\|_\sharp \le 600 M \|\varphi\|_{C^1}/t$. Substitute this in (2020) and use $t \le 1$ in its constant term to obtain (2021). □

105. The rotation hypotheses for volume-first limits

Let $K_j \to \infty$, $\beta_j > 0$, and let ρ_j, ν_j be periodic Gibbs states at (K_j, β_j). Suppose that these pairs have source systems on unbounded sets of sides with one set of constants $(2\varrho; C_e, \kappa_e; E_+, C_{\mathrm{low}}; C_{\mathrm{gap}})$ as above. Suppose also that their separated moments have uniform bound tables and that

$$\sigma(X) = \lim_{j\to\infty} \rho_j(X_{K_j}^{\nu_j})$$

exists for every separated tuple. The canonical sequences over $\mathfrak{d}(g)$ satisfy the uniform assumptions; their concrete constants are specified below.

Proposition 105.1. The limit σ satisfies the insertion bound (R5) with

$$c_{\mathrm{I}} = \frac{1}{40L}, \qquad p_{\mathrm{I}} = 0, \qquad C_{\mathrm{I}} = \frac{2\bar{Q}_{\mathrm{mid}}}{\varrho}. \tag{2022}$$

It has the inherited properties (R1)–(R3). ∎

Proof. The moving-entry argument of Section 92 gives translation and cubic invariance, the uniform moment bounds, and coordinate reflection positivity. The resulting Hilbert space therefore has the contraction semigroup of Proposition 94.1.

Fix the data in (R5): $0 < \eta \le 1$, a real h supported in $B(y, \eta/(40L))$ with $|y_3| \le \eta/8$, and $F, G \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta/4\})$. Choose a scale and a point, independently of j, with

$$\eta/(5L) < l = a_i \le \eta/5, \qquad y^* \in a_{i+2}\mathbb{Z}^3, \quad |y - y^*|_\infty \le l/(2L^2), \qquad c = y_3^* - l/2. \tag{2023}$$

Since $L \ge 13$, these choices give

$$\operatorname{supp} h \subset \{|x - y^*|_\infty < l/4\}, \qquad -\eta/4 < c < c + l < \eta/4.$$

Indeed, the first radius is less than $l/8 + l/(2L^2) < l/4$, while $|y_3^*| + l/2 \le \eta(1/8 + 1/1690 + 1/10) < \eta/4$. Put $F_1 = \tau_{2ce_3} F$ and $F_2 = \tau_{-le_3} G$. Their finitely many compact supports lie a positive distance above c; furthermore $\theta_c F_1 = \theta F$ and $\tau_{le_3} F_2 = G$.

For all sufficiently large j, (i, y^*) is a lattice board and c is one of its bond walls. In fact $K_j \ge i+2$ ensures $y^* \in a_{K_j}\mathbb{Z}^3$, and the odd integer l/a_{K_j} makes c/a_{K_j} a half-integer. The centered lattice polynomials associated with F_1, F_2 are supported in the closed half-space above c: a plaquette extends by at most $a_{K_j}/2$ in its third coordinate from its center. Apply (2014), with its uniform middle-cell normalizer, to these two polynomials and h. Covariance of the centering law ν_j gives

$$\left|\sigma_{\rho_j,\nu_j}(\theta\bar{F} \cdot (h) \cdot G)\right| \le \frac{2Q_{\mathrm{mid}}(l)}{\varrho} \|h\|_\sharp \sigma_{\rho_j,\nu_j}(\theta_c \bar{F}_1 \cdot F_1)^{1/2} \sigma_{\rho_j,\nu_j}(\theta_c \bar{F}_2 \cdot F_2)^{1/2}.$$

All the tuples on this line are separated and fixed independently of j. Taking the limit gives the same inequality for σ.

Translation invariance identifies the two limiting square roots with $\|[\tau_{-ce_3} F_1]\|$ and $\|[\tau_{-ce_3} F_2]\|$. They are respectively

$$\|e^{-(c+\eta/4)H}[\tau_{-(\eta/4)e_3} F]\|, \qquad \|e^{-(\eta/4-c-l)H}[\tau_{-(\eta/4)e_3} G]\|.$$

Both semigroup times are positive, so contractivity bounds these by the norms in (R5). Finally $l \le 1$ and $Q_{\mathrm{mid}}(l) \le \bar{Q}_{\mathrm{mid}}$. This proves (R5) with (2022). □

Proposition 105.2. The limit σ satisfies (R7) with exponent $a_\Gamma = 3$. If $\sigma(f) = c_\sigma \int f$, one may take $C_0 = 0$ and, for a nonzero plate label φ of support length λ_φ,

$$C_\varphi = \left(\frac{600M\|\varphi\|_{C^1}}{\varrho}\right)^2 \left[4\log 2(2Q_{\mathrm{mid}}(l_{\lambda_\varphi}) + 2)^2 + 27C_e S(\kappa_e) C_{\mathrm{dep}}(\delta_e, 4)(\lambda_\varphi + 1 + m')\right] + (6c_\sigma \lambda_\varphi \|\varphi\|_\infty)^2. \tag{2024}$$

In the self-centered case $\rho_j = \nu_j$, the last term is zero. ∎

Proof. Translate the label in the e_2 direction so that its support interval is centered at zero. This leaves the limiting two-point reflection form unchanged. For $f = \alpha \otimes \varphi$, the moment–cumulant identity is

$$\sigma_{\rho_j,\nu_j}(\theta f, f) = S_2^T[\rho_j](\theta f, f) + \delta_j(\theta f)\delta_j(f), \quad \delta_j(f) = \rho_j \mathcal{O}_{K_j}(f) - \nu_j \mathcal{O}_{K_j}(f).$$

The first term is bounded by (2021), uniformly in j. The one-point limit from Section 92 gives $\delta_j(f) \to c_\sigma \int f$. The box containing f has volume $2\lambda_\varphi t^2$ and $\|f\|_\infty \le 3\|\varphi\|_\infty$, so $|c_\sigma \int f|^2 \le (6c_\sigma \lambda_\varphi \|\varphi\|_\infty)^2 t^4$. For $t \le 1$ this is bounded by the same constant times t^{-3}. Passing to the limit proves the upper bound, and reflection positivity gives its nonnegativity. For equal laws all δ_j vanish, giving the last assertion. □

Theorem AN. Fix $0 < g \le \gamma_\varrho$. Let $(K_j, \beta_j) \in \mathfrak{d}(g)$, $K_j \to \infty$, and let ρ_j, ν_j be periodic Gibbs states. Suppose that their moments centered at ν_j converge on every separated tuple to σ. Then their self-centered moments converge to σ^{c}, and σ^{c} satisfies (R0)–(R7). The uncentered limit σ satisfies (R1)–(R7) and

$$\sigma(f_1, \dots, f_n) = \sum_{A \subset \{1,\dots,n\}} \prod_{i \in A} \left(c_\sigma \int f_i \right) \sigma^{\mathrm{c}}((f_i)_{i \notin A}). \tag{2025}$$

Both functionals are invariant under all Euclidean motions and reflection positive at every Euclidean mirror. Their joint energy–momentum spectrum is contained in the forward cone $\{(E, q) : E \ge |q|\}$.

In particular these conclusions apply to every volume-first continuum subsequence of cubic torus-limit states, with $\rho_j = \nu_j$ and $c_\sigma = 0$. ■□

Proof. The source theorem and the uniform normalizer construction provide systems on $\mathfrak{L}(g)$ at radius 2ϱ, with

$$C_e = C_{\mathrm{S}}, \quad \kappa_e = \kappa, \quad E_+ = C_{\mathrm{S}}, \quad C_{\mathrm{low}} = C_{\mathrm{S}}(1 + \log g^{-2}), \quad C_{\mathrm{gap}} = C_{\mathrm{S}}.$$

These are the constants already verified in Section 91; none depends on j or on the side. Propositions 103.4 and 91.9 supply all lattice symmetries, all mirror positivities, and chessboard domination along any exhausting sequence in $\mathfrak{L}(g)$.

Let $C_n^{\uparrow}(d; R)$ be the non-increasing-in-d envelope of the uniform cumulant table from Proposition 91.12 and the following cumulant bounds. The moment–cumulant formula gives the finite moment tables

$$C_n^{\mathrm{m}}(d; R) = \sum_{\pi \in \mathrm{Part}(n)} \prod_{J \in \pi} C_{|J|}^{\uparrow}(d; R), \qquad C_n^{\mathrm{c}}(d; R) = \sum_{\pi \in \mathrm{Part}_{\ge 2}(n)} \prod_{J \in \pi} C_{|J|}^{\uparrow}(d; R). \tag{2026}$$

For $C_1^{\uparrow}$, use the two-state one-point bound; in the second sum singleton blocks are absent. Each separated subtuple has the same lower separation and radius bounds, so these tables apply uniformly to all the indicated states.

Cumulants are finite polynomials in moments. Thus convergence of σ_{ρ_j,ν_j} gives convergence of every cumulant of order at least two of ρ_j; centering changes none of them. Their partition sums without singleton blocks give convergence of the self-centered moments to σ^{c} and (R0). The one-point limit is $c_\sigma \int f$; the partition expansion gives (2025). The moving-entry and positivity arguments give (R1)–(R3), with the moment tables (2026). Proposition 105.1 gives (R5), and Proposition 105.2 gives (R7), for both sequences.

For (R6), if the entries of F have compact support in $\{x_1 + x_3 > 0\}$, their plaquette polynomials have all bonds in that half-space once a_{K_j} is smaller than the minimum positive height of these supports. The maximal variation of $x_1 + x_3$ over a plaquette from its center is a_{K_j}. Lattice diagonal positivity then gives $\sigma_{\rho_j,\nu_j}(\theta_\Sigma \bar{F} \cdot F) \ge 0$; covariance of ν_j ensures the exact centered identity. Taking the limit gives full diagonal reflection positivity and hence (R6). Together with the moment bounds and coordinate RP, Proposition 94.4 yields $H \ge |P_1|$, which is (R4). There is only one Hilbert space

and one joint spectral measure here: both descriptions complete the same reflection form and have matrix elements $\sigma(\theta\bar{F}\cdot\tau_{te_3+v}G)$. Uniqueness of their Laplace–Fourier representation identifies the spectral measures if the constructions are made separately.

The rotation theorem AK now applies to σ^{c}. Formula (2025) transfers its rotational invariance to σ, because $\int f$ is Euclidean invariant. Transporting coordinate RP by rotations and translations gives every mirror. The same transport turns the component cone bounds into the full forward cone: apply the spectral inequality in a countable dense set of spatial directions and intersect the corresponding closed half-cones. □

106. Conditioning periodic Gibbs states

The decomposition argument changes a lattice law by a bounded translation-invariant density. Such a change preserves the Gibbs equations, including their fixed coupling; this is the closure condition (1970) required of the classes.

Proposition 106.1. Let $\nu\in\mathcal{G}^{\mathrm{per}}_{a,\beta}$ and let $h\ge 0$ be bounded and $\mathcal{I}$-measurable, with $\nu(h)=1$. Then $h\nu\in\mathcal{G}^{\mathrm{per}}_{a,\beta}$. The extreme points of $\mathcal{G}^{\mathrm{per}}_{a,\beta}$ are exactly its ergodic members. ■

Proof. For a finite bond set B and a continuous local function F, choose a continuous local h' with $\nu(|h-h'|)<\varepsilon$. Translate the finite support of h' away from B, obtaining $\tau_v h'$ supported in B^c. Translation invariance of h and ν gives $\nu(|h-\tau_v h'|)<\varepsilon$. The Gibbs equation and the fact that the kernel γ_B leaves outside functions fixed give

$$\nu(\tau_v h'\,\gamma_B F)=\nu\big(\gamma_B(\tau_v h' F)\big)=\nu(\tau_v h' F).$$

Since $\|\gamma_B F\|_\infty\le\|F\|_\infty$, replacing $\tau_v h'$ by h changes each side by at most $\varepsilon\|F\|_\infty$. Letting $\varepsilon\downarrow 0$ proves $(h\nu)(\gamma_B F)=(h\nu)(F)$. The density and the original law are translation invariant, so the new Gibbs law is translation invariant and hence periodic.

If an ergodic ν is a nontrivial convex combination of periodic Gibbs laws, each summand has a bounded Radon–Nikodym density with respect to ν. Invariance of the two measures makes that density translation invariant almost everywhere. It has an invariant version and is constant by ergodicity; its integral is one, so the summand equals ν. Conversely, a nontrivial invariant event E decomposes ν into its normalized restrictions to E and E^c. These are distinct periodic Gibbs states by the first part. □

Proposition 106.2. There is a regular conditional distribution $(\nu^U)_U$ of $\nu\in\mathcal{G}^{\mathrm{per}}_{a,\beta}$ given $\mathcal{I}$. For any such distribution,

$$\nu(F)=\int \nu^U(F)\,\nu(dU) \tag{2027}$$

for bounded Borel F, and ν^U is an ergodic periodic Gibbs state for ν-almost every U. ■

Proof. The configuration space is compact metrizable. Choose a countable, uniformly dense rational vector space of real continuous functions containing 1, and choose conditional expectations for its members. Outside one invariant null set, these conditional expectations satisfy all rational linear relations, preserve 1, and have norm at most the supremum norm. At each remaining U they extend to a norm-one unital real functional on the continuous functions. It is positive: for $0\le F\le 1$, $|1-\Lambda_U(F)|=|\Lambda_U(1-F)|\le 1$. Riesz representation gives a probability measure ν^U. On the exceptional set use ν itself. Uniform approximation and the functional monotone-class theorem show that, for every bounded Borel F, the function $U\mapsto\nu^U(F)$ is a version of $E_\nu[F\mid\mathcal{I}]$. This constructs the distribution and also gives (2027).

Fix a countable real family $\mathcal{C}_0$ of continuous local functions, dense on each finite set of bonds and hence determining finite Borel measures on the configuration space. There are countably many translations and finite bond sets. The identities

$$\nu(1_E\tau_v F)=\nu(1_E F),\qquad \nu(1_E\gamma_B F)=\nu(1_E F),\qquad E\in\mathcal{I},$$

hold for every $F \in \mathcal{C}_0$: the first uses invariance, and the second uses Proposition 106.1 for the normalized restriction to E when $\nu(E) > 0$. Conditional expectation and one countable union of null sets therefore give

$$\nu^U(\tau_v F) = \nu^U(F), \qquad \nu^U(\gamma_B F) = \nu^U(F)$$

simultaneously for all these tests, translations and finite sets. Uniform approximation, and the supremum-norm contraction of γ_B, extend the identities to all continuous local tests. Thus ν^U is a translation-invariant Gibbs state almost surely.

For ergodicity let A_N average translations over $\{0, a, \dots, (N-1)a\}^3$. For any invariant probability law, these averages converge strongly in L^2 to conditional expectation onto $\mathcal{I}$. Indeed A_N is the product of the three commuting one-dimensional Cesaro averages, whose limiting projections commute; their product projects onto the common invariant subspace. Write $g_F(U) = \nu^U(F)$ for $F \in \mathcal{C}_0$. Disintegration and this convergence give

$$\int \nu^U(|A_N F - g_F|^2)\, \nu(dU) = \nu(|A_N F - g_F|^2) \longrightarrow 0.$$

A common subsequence N_i makes the integrand tend to zero for all $F \in \mathcal{C}_0$ outside a single null set. Applying the conditional-distribution identity to the countably many invariant functions g_F and g_F^2 gives $\nu^U(|g_F - g_F(U)|^2) = 0$ almost surely. Thus, in $L^2(\nu^U)$, $A_{N_i} F \to \nu^U(F)$. The mean ergodic theorem for the already invariant law ν^U identifies this limit with $E_{\nu^U}[F \mid \mathcal{I}]$. For any invariant event E, the finite measures $1_E \nu^U$ and $\nu^U(E)\nu^U$ consequently agree on $\mathcal{C}_0$ and are equal. Evaluating at E gives $\nu^U(E) = \nu^U(E)^2$. This holds for every E on the same full-measure set, because $\mathcal{C}_0$ determines measures. Hence ν^U is ergodic. □

106.1. **The classes used in the continuum decomposition.** For the fixed source parameters and $0 < g \le \gamma$, retain the functions C_1^g, C_n^g from (1844) and (1848). For finite $d, R > 0$ write

$$C_n^{g\uparrow}(d; R) = \inf_{0<d'\le d,\ R'\ge R} C_n^g(d'; R'), \qquad C_1^{g\uparrow}(d; R) = C_1^g(R),$$

and define the strictly positive bound system

$$\widehat{C}_1^g(R) = 1 + C_1^g(2R), \qquad \widehat{C}_n^g(d; R) = 1 + C_n^{g\uparrow}(d; R), \quad \widehat{C}_n^g(\infty; R) = 1 + \inf_{d>0} C_n^{g\uparrow}(d; R). \tag{2028}$$

Proposition 106.3. For every $(K, \beta) \in \mathfrak{d}(g)$ the set $\mathcal{G}^{\mathrm{per}}_{K,\beta}$ satisfies the closure condition (1970) and obeys the bound system $\widehat{\mathsf{C}}^g = (\widehat{C}_1^g, (\widehat{C}_n^g)_{n\ge 2})$. Every sequence $K_j \to \infty$, $(K_j, \beta_j) \in \mathfrak{d}(g)$, and $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ therefore defines a frame in a class system with these same bounds. ■

Proof. The class is nonempty by periodic-volume compactness. Its full lattice invariance and RP are Proposition 103.4; its closure under the normalized invariant densities is Proposition 106.1. These are precisely the two class conditions. Proposition 91.13 gives the two-state one-point bound and both cumulant bounds uniformly over the class. A tuple of type $(d; R)$ is also of type $(d'; R')$ whenever $d' \le d$ and $R' \ge R$, so its cumulant bound may be replaced by the infimum defining $C_n^{g\uparrow}$. The infimum is finite and nonnegative; adding one ensures strict positivity and preserves monotonicity. Its extension at $d = \infty$ is nonincreasing in d and nondecreasing in R for the same reason. For the one-point bound, a function supported in a closed radius-R ball is admissible with radius $2R$; (1849) therefore gives the bound $\widehat{C}_1^g(R)\|f\|_\sharp$. All constants were fixed before choosing the cutoff, the states, or the frame. □

107. Point insertions and energy bounds for reconstruction

The reconstruction uses more than reflection positivity and bounds on separated moments. Its additional input is an L^1 estimate for one insertion between two positive-time vectors. We first construct point forms with polynomial energy bounds, then use them to define smeared fields on a common domain.

Let σ be a real, normalized, symmetric multilinear functional on separated tuples. Throughout this section assume:

(Rec1) σ is invariant under all Euclidean isometries.

(Rec2) It is reflection positive at $x_3 = 0$. Its reflection Hilbert space $\mathcal{H}$ is separable, and its translations are represented by e^{-tH} and $\mathcal{U}(v)$, with $H \geq 0$, the stated strong continuity, and joint spectral measure $\mathcal{P}$:

$$e^{-tH}[F] = [\tau_{te_3}F], \qquad \mathcal{U}(v)[F] = [\tau_{(v,0)}F], \qquad e^{i\tau H}\mathcal{U}(v) = \int e^{i(\tau E + v\cdot q)}\, d\mathcal{P}(E,q). \tag{2029}$$

(Rec3) $\mathcal{P}$ is carried by $\{(E,q) : E \geq |q|\}$.

(Rec4) For some $C_\wp, \wp \geq 0$ and $0 < c_0 \leq 1/2$, put $\mathrm{Sl}_\eta = \{|x_3| < c_0\eta\}$. For $0 < \eta \leq 1$, $h \in C_c^1(\mathrm{Sl}_\eta)$ and $F, F' \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta\})$, assume

$$|\sigma(\theta\bar{F}\cdot(h)\cdot F')| \leq C_\wp \eta^{-\wp}\|h\|_1 \|[\tau_{-\eta e_3}F]\|\,\|[\tau_{-\eta e_3}F']\|. \tag{2030}$$

We refer to these four conditions as (Rec). Vacuum uniqueness, $\ker H = \mathbb{C}\Omega$, will be a separate assumption when needed. The numerical triple in (Rec4) is not part of the definition of the resulting field.

Write $\vec{z} = (z_1, z_2)$ and define

$$d_\circ(z) = \operatorname{Re} z_3 - |\operatorname{Im}\vec{z}|, \qquad \mathrm{Tb}_\circ = \{d_\circ > 0\}, \qquad \mathcal{T}(z) = \int e^{-z_3 E + i\vec{z}\cdot q}\, d\mathcal{P}. \tag{2031}$$

The last integral is defined on the closed tube. The cone bound gives $|e^{-z_3E+i\vec{z}\cdot q}| \leq e^{-d_\circ(z)E}$. Consequently $\mathcal{T}$ consists of contractions, is strongly continuous on the closed tube and norm holomorphic in its interior, and satisfies

$$\mathcal{T}(z)\mathcal{T}(w) = \mathcal{T}(z+w), \qquad \mathcal{T}(z)^* = \mathcal{T}(z^*), \qquad z^* = (-\bar{z}_1, -\bar{z}_2, \bar{z}_3).$$

For norm holomorphy, expand the exponential around a point with positive margin; the factor $e^{-d_\circ(z)E}$ bounds every coefficient and the quadratic remainder uniformly on a smaller ball. Dominated convergence against scalar spectral measures proves strong continuity at the boundary. In the interior, $E^k e^{-d_\circ(z)E}$ is bounded for every k, so the range of $\mathcal{T}(z)$ lies in

$$\mathcal{D}_\infty = \bigcap_{k\geq 1} \operatorname{dom} H^k.$$

The vacuum is fixed by $\mathcal{T}(z)$ and its adjoint. For Minkowski coordinates $x = (t, \vec{x})$ put

$$\iota(t,\vec{x}) = (\vec{x}, -it). \tag{2032}$$

This identifies the forward tube $\mathbb{R}^{1+2} + i\{y : y^0 > |\vec{y}|\}$ with $\mathrm{Tb}_\circ$, and $\mathcal{T}(\iota x) = e^{itH}\mathcal{U}(\vec{x})$ is unitary.

107.1. **Densities of bounded insertions.**

Lemma 107.1. For $0 < \eta \leq 1$ there is a unique complex linear map $h \mapsto B_\eta^{\mathrm{in}}(h)$ from $L^1(\mathrm{Sl}_\eta)$ to bounded operators such that

$$\|B_\eta^{\mathrm{in}}(h)\| \leq C_\wp \eta^{-\wp}\|h\|_1, \tag{2033}$$

$$B_\eta^{\mathrm{in}}(h)[F] = [(\tau_{\eta e_3}h)\cdot\tau_{2\eta e_3}F] \quad (h \in C_c^1(\mathrm{Sl}_\eta)). \tag{2034}$$

For real h, its adjoint is $B_\eta^{\mathrm{in}}(\theta h)$. Each scalar matrix element has an L^∞ density of norm at most $C_\wp\eta^{-\wp}\|\psi\|\|\psi'\|$. ∎

Proof. Apply (2030) after translating both formal vectors by ηe_3. The bounded sesquilinear form

$$\langle [F], B_\eta^{\mathrm{in}}(h)[F']\rangle = \sigma(\theta\overline{\tau_{\eta e_3}F}\cdot(h)\cdot\tau_{\eta e_3}F') \tag{2035}$$

vanishes on null classes and therefore defines a bounded operator by Riesz representation. Translate the whole tuple by ηe_3 to obtain (2034). The support inequalities $c_0 \leq 1/2$ make that tuple

separated. Real linearity and the norm bound extend this map from real C_c^1 functions to real L^1 functions.

For fixed vectors the real and imaginary parts of the matrix element are bounded real functionals on L^1, hence have a complex density b. The same constant bounds $|b|$: apply the norm estimate to normalized indicators of small balls and use Lebesgue differentiation. Thus $\|b\|_\infty \le C_\wp \eta^{-\wp} \|\psi\| \|\psi'\|$. Complex linear extension now preserves this constant, since $|\int hb| \le \|h\|_1 \|b\|_\infty$. Reflection invariance, symmetry and reality in (2035) give the adjoint relation on C_c^1, then on L^1 by approximation. The class identity, density of classes, and L^1 continuity prove uniqueness. □

Proposition 107.2 (Point insertions)**.** There is a unique family of bounded operators $B(s, s')$, $s, s' > 0$, with the shift law

$$T_t B(s, s') T_{t'} = B(s + t, s' + t') \qquad (t, t' \ge 0) \tag{2036}$$

and the density identity

$$\langle \psi, B_\eta^{\mathrm{in}}(h) \psi' \rangle = \int h(x) \langle \psi, \mathcal{U}(\vec{x}) B(\eta + x_3, \eta - x_3) \mathcal{U}(-\vec{x}) \psi' \rangle \, dx. \tag{2037}$$

It satisfies

$$\|B(s, s')\| \le C_\wp \min(s, s', 1)^{-\wp}, \qquad B(s, s')^* = B(s', s). \tag{2038}$$

It is norm continuous and commutes with spatial orthogonal transformations acting on $\mathcal{H}$. The operator density

$$b_\eta(x) = \mathcal{U}(\vec{x}) B(\eta + x_3, \eta - x_3) \mathcal{U}(-\vec{x})$$

is norm continuous on Sl_η, has norm at most $C_\wp \eta^{-\wp}$ there, and has real-analytic scalar matrix elements. ■

Proof. For $0 < \eta' < \eta \le 1$, $h \in L^1(\mathrm{Sl}_{\eta'})$ and real w with $|w_3| < c_0(\eta - \eta')$, translation covariance of (2035) gives

$$B_\eta^{\mathrm{in}}(\tau_w h) = \mathcal{T}((\eta - \eta')e_3 + w) \, B_{\eta'}^{\mathrm{in}}(h) \, \mathcal{T}((\eta - \eta')e_3 - w). \tag{2039}$$

The two semigroup arguments have positive third coordinate, and the translated support stays in Sl_η. It suffices to prove the identity on C_c^1 tests and dense classes; L^1 continuity then proves the assertion as written.

Fix ψ, ψ' and let b be their scalar density at margin η. Take nonnegative continuous mollifiers j_ε of integral one, supported in $B(0, \varepsilon)$, with $\varepsilon < c_0 \eta'$. The functions

$$b_\varepsilon(w) = \langle \psi, \mathcal{T}((\eta - \eta')e_3 + w) B_{\eta'}^{\mathrm{in}}(j_\varepsilon) \mathcal{T}((\eta - \eta')e_3 - w) \psi' \rangle$$

are holomorphic and bounded by $C_\wp \eta'^{-\wp} \|\psi\| \|\psi'\|$ on

$$\{w \in \mathbb{C}^3 : |\operatorname{Re} w_3| + |\operatorname{Im} \vec{w}| < \eta - \eta'\}.$$

On the smaller real slab $|w_3| < c_0(\eta - \eta')$, (2039) identifies them with $\int j_\varepsilon(x - w) b(x) \, dx$. Montel's theorem gives a locally uniformly convergent subsequence with holomorphic limit. Mollification also converges to b in L^1 on compact real subsets, so this limit agrees with b almost everywhere. As $\eta' \downarrow 0$ the smaller real slabs exhaust Sl_η. Their continuous representatives agree on overlaps and give one real-analytic representative of b on the whole slab.

Linearity in ψ' and antilinearity in ψ hold everywhere for these representatives: they hold almost everywhere for the original densities, and both sides are continuous. The original L^∞ bound likewise holds at every point. Riesz representation thus defines bounded operators $b_\eta(x)$ with $\|b_\eta(x)\| \le C_\wp \eta^{-\wp}$. This constructs the operator density from scalar densities.

Integrate (2039) against arbitrary tests. Uniqueness of the continuous scalar densities gives

$$b_\eta(x + w) = \mathcal{T}((\eta - \eta')e_3 + w) \, b_{\eta'}(x) \, \mathcal{T}((\eta - \eta')e_3 - w). \tag{2040}$$

For $0 < \eta \le \min(s, s', 1)$ define

$$B(s, s') = T_{s-\eta} \, b_\eta(0) \, T_{s'-\eta}. \tag{2041}$$

The case $x = w = 0$ of (2040) proves independence of η. It also proves the shift law; choosing $\eta = \min(s, s', 1)$ gives the norm bound. For $x \in \mathrm{Sl}_\eta$, choose $\eta' > 0$ with $|x_3| < c_0(\eta - \eta')$ and apply (2040) at $(0, x)$. Factoring spatial translations from the semigroup gives the stated formula for $b_\eta(x)$ and hence (2037).

In (2040) choose a fixed smaller margin locally. Its semigroup factors have positive margins and are norm holomorphic, proving norm continuity of b_η. The same reasoning in (2041) proves norm continuity of B. The insertion adjoint identity makes $b_\eta(x)^* = b_\eta(\theta x)$, so $b_\eta(0)$ is self-adjoint. This proves the adjoint relation for B. Spatial rotations in (2035) give covariance of b_η and invariance of $b_\eta(0)$; they commute with T_t, proving the remaining assertion.

For uniqueness, any bounded-operator family with the shift law has continuous scalar densities in (2037): factor it through $B(\eta/2, \eta/2)$. The density identity then determines its value at $x = 0$, hence $B(\eta, \eta)$. The shift law determines all other values. The same argument shows that changing the admissible triple $C_\wp, \wp, c_0$ does not change the operators, by first restricting insertions to the common smaller slab. □

107.2. **Resolvent regularization of the point form.** Put $R_H = (1 + H)^{-1}$. For real $l > \wp$ define

$$A_l = \Gamma(l)^{-2} \int_0^\infty \int_0^\infty (ss')^{l-1} e^{-s-s'} B(s, s')\, ds\, ds'. \tag{2042}$$

Lemma 107.3. The integral is norm integrable and defines a bounded self-adjoint operator with

$$\|A_l\| \le C_\wp \left(1 + \frac{2\Gamma(l - \wp)}{\Gamma(l)}\right). \tag{2043}$$

For $a > 0$ and $s, s' > 0$,

$$R_H^a A_l R_H^a = A_{l+a}, \qquad T_s A_l T_{s'} = R_H^l B(s, s') R_H^l. \tag{2044}$$

Consequently

$$q_0(\psi, \psi') = \langle (1 + H)^l \psi, A_l (1 + H)^l \psi' \rangle \tag{2045}$$

defines a Hermitian form on $\bigcup_{l > \wp} \operatorname{dom} H^l$, independent of the admissible exponent used for a given pair of vectors. It satisfies the energy bound

$$|q_0(\psi, \psi')| \le \|A_l\|\, \|(1 + H)^l \psi\|\, \|(1 + H)^l \psi'\|$$

on $\operatorname{dom} H^l$, and

$$q_0(T_s \psi, T_{s'} \psi') = \langle \psi, B(s, s') \psi' \rangle. \tag{2046}$$

■

Proof. By (2038), $\|B(s, s')\| \le C_\wp(1 + s^{-\wp} + s'^{-\wp})$. The two Gamma integrals therefore converge for $l > \wp$ and give (2043). Norm continuity makes the operator integrand measurable. Self-adjointness follows by interchanging s, s' and using $B(s, s')^* = B(s', s)$.

Functional calculus and the scalar Gamma integral give

$$R_H^a = \Gamma(a)^{-1} \int_0^\infty t^{a-1} e^{-t} T_t\, dt$$

weakly for $a > 0$. Insert this on both sides of A_l and use the shift law. Fubini is justified by the integrable bound just proved and contractivity. Set $s = r + t$ and $s' = r' + t'$. The two inner integrals are

$$\int_0^s t^{a-1} (s - t)^{l-1}\, dt = s^{a+l-1} \frac{\Gamma(a)\Gamma(l)}{\Gamma(a + l)}.$$

This proves the first identity in (2044). For the second, shift $B(r, r')$ by (s, s'), then write $B(r + s, r' + s') = T_r B(s, s') T_{r'}$ and use the same resolvent integral.

If $l' > l > \wp$ and both vectors lie in $\operatorname{dom} H^{l'}$, the first identity and $R_H^{l'-l}(1 + H)^{l'} = (1 + H)^l$ prove equality of the two definitions of q_0. The domains are nested, so a common admissible exponent

exists for every pair in the stated union. Hermiticity and the bound follow from those of A_l. Finally $(1+H)^l T_s$ is bounded for $s>0$. Apply the second identity at $(s/2, s'/2)$ and cancel the resolvent factors against these bounded regularized powers. The shift law gives (2046). □

107.3. **Recovering an insertion bound after centering.** The hypotheses (Rec) are also available for a functional with nonzero one-point function.

Lemma 107.4. Suppose σ satisfies (Rec1)–(Rec2), its spectrum obeys $E \geq |q_1|$, and $\sigma(f) = c\int f$. If its centered functional satisfies

$$|\sigma^{\mathrm{c}}(\theta\bar{F}\cdot(h)\cdot F')| \leq C_\wp \eta^{-\wp}\|h\|_1 \|[\tau_{-\eta e_3}F]\|_{\sigma^{\mathrm{c}}}\|[\tau_{-\eta e_3}F']\|_{\sigma^{\mathrm{c}}}, \tag{2047}$$

with the supports in (Rec4), then σ satisfies (Rec) with constants $(C_\wp + |c|, \wp, c_0)$. ■

Proof. Spatial rotations conjugate the joint spectral measure, so $E \geq |n\cdot q|$ for every unit spatial vector n. A countable dense set of such vectors gives $E \geq |q|$. The shift identities of Section 94 give $\sigma = \sigma^{\mathrm{c}} \circ \mathsf{sh}_c$ and an isometry $[F]_\sigma \mapsto [\mathsf{sh}_c F]_{\sigma^{\mathrm{c}}}$. The shift commutes with translations, conjugation and reflection, preserves supports, and is multiplicative. Writing $F_c = \mathsf{sh}_c F$, therefore,

$$\sigma(\theta\bar{F}\cdot(h)\cdot F') = \sigma^{\mathrm{c}}(\theta\bar{F}_c\cdot(h)\cdot F'_c) + c\int h\,\sigma^{\mathrm{c}}(\theta\bar{F}_c\cdot F'_c). \tag{2048}$$

The first term has the bound (2047) with the σ norms. Cauchy–Schwarz and contractivity of T_η bound the second by $|c|\|h\|_1\|[\tau_{-\eta e_3}F]\|\|[\tau_{-\eta e_3}F']\|$. Since $\eta \leq 1$ and $\wp \geq 0$, their sum has the required bound. Centering preserves the other stated symmetries and reflection positivity by the same shift identities. □

108. Analytic continuation of the Euclidean kernels

The point-insertion bound gives a product estimate without a factorial. Rotation of the ordering direction then covers all distinct Euclidean points, including configurations with equal third coordinates.

Call $(h_1,\dots,h_n)$ *η-ordered along u* at heights $t_1,\dots,t_n$ if $u \in S^2$, $0<\eta\leq 1$, $t_{k+1}-t_k \geq 2\eta$, and

$$\operatorname{supp} h_k \subset \{|u\cdot x - t_k| < c_0\eta\}.$$

Such tuples are separated by at least η. For $u = e_3$, set $g_k = \tau_{-t_k e_3}h_k$ and $s_k = t_{k+1}-t_k-2\eta \geq 0$.

Proposition 108.1. For an η-ordered tuple along e_3,

$$\sigma(h_1,\dots,h_n) = \langle \Omega, B^{\mathrm{in}}_\eta(g_1)T_{s_1}B^{\mathrm{in}}_\eta(g_2)\cdots T_{s_{n-1}}B^{\mathrm{in}}_\eta(g_n)\Omega\rangle. \tag{2049}$$

If $t_1 \geq \eta$, then

$$[(h_1,\dots,h_n)] = T_{t_1-\eta}B^{\mathrm{in}}_\eta(g_1)T_{s_1}\cdots B^{\mathrm{in}}_\eta(g_n)\Omega, \qquad \|[(h_1,\dots,h_n)]\| \leq \prod_k C_\wp\eta^{-\wp}\|h_k\|_1. \tag{2050}$$

■

Proof. Starting at the right, (2034) proves by downward induction that the product beginning with $B^{\mathrm{in}}_\eta(g_k)$ represents $[\tau_{(\eta-t_k)e_3}(h_k,\dots,h_n)]$. Indeed the intervening translation has height $s_k+\eta-t_{k+1} = -t_k-\eta$, and the insertion adds 2η to the remaining tuple. All these classes have supports in $\{x_3 > (1-c_0)\eta\}$. Taking the vacuum component and using translation invariance proves (2049). If $t_1 \geq \eta$, translate the class forward by $t_1-\eta$ to obtain (2050); contractivity and (2033) give its norm bound. □

For later use, the scalar density formula can be iterated: for $g_k \in L^1(\mathrm{Sl}_\eta)$ and bounded $A_0,\dots,A_n$,

$$\begin{aligned}&\langle\psi, A_0B^{\mathrm{in}}_\eta(g_1)A_1\cdots B^{\mathrm{in}}_\eta(g_n)A_n\psi'\rangle\\ &\quad= \int\prod_k g_k(y_k)\langle\psi, A_0 b_\eta(y_1)A_1\cdots b_\eta(y_n)A_n\psi'\rangle\,dy.\end{aligned} \tag{2051}$$

The second factor is continuous and bounded by $(C_\wp \eta^{-\wp})^n \|\psi\| \|\psi'\| \prod_j \|A_j\|$. Thus the integrand is absolutely integrable. Apply (2037) to the first insertion, then to the remaining ones; Fubini proves the formula.

108.1. **Ordered complex configurations.** For $\zeta_1, \dots, \zeta_{n-1} \in \mathrm{Tb}_\circ$ define

$$\mathbf{W}_n(\zeta_1, \dots, \zeta_{n-1}) = \langle \Omega, B(\eta,\eta)\mathcal{T}(\zeta_1 - 2\eta e_3)B(\eta,\eta)\cdots\mathcal{T}(\zeta_{n-1} - 2\eta e_3)B(\eta,\eta)\Omega\rangle, \tag{2052}$$

where $0 < \eta \le \min(1, \frac{1}{2}\min_k d_\circ(\zeta_k))$. For $n = 1$ this is the real number $\mathbf{W}_1 = \langle \Omega, B(1,1)\Omega\rangle$. On

$$\mathrm{Tb}_\circ^{(n)} = \{z : z_1 \in \mathrm{Tb}_\circ,\ z_{k+1} - z_k \in \mathrm{Tb}_\circ\}$$

put

$$\eta(z) = \min\left(1, d_\circ(z_1), \tfrac{1}{2}\min_{k<n} d_\circ(z_{k+1} - z_k)\right), \tag{2053}$$

$$\Psi_n(z) = \mathcal{T}(z_1 - \eta e_3)B(\eta,\eta)\mathcal{T}(z_2 - z_1 - 2\eta e_3)B(\eta,\eta)\cdots\mathcal{T}(z_n - z_{n-1} - 2\eta e_3)B(\eta,\eta)\Omega, \tag{2054}$$

using any $0 < \eta \le \eta(z)$; the inner minimum is omitted when $n = 1$.

These definitions are independent of η. To compare $\eta' < \eta$, substitute

$$B(\eta,\eta) = T_{\eta-\eta'}B(\eta',\eta')T_{\eta-\eta'}$$

at every insertion. The semigroup law absorbs the adjacent factors into the differences, and $T_t\Omega = \Omega$ removes the final factor. For the scalar product it removes the initial factor as well.

Proposition 108.2. The function $\mathbf{W}_n$ is holomorphic and Ψ_n is holomorphic in norm. They satisfy

$$|\mathbf{W}_n(\zeta)| \le (C_\wp \eta^{-\wp})^n \qquad \text{if } d_\circ(\zeta_k) \ge 2\eta,\ 0 < \eta \le 1, \tag{2055}$$

$$\|\Psi_n(z)\| \le (C_\wp \eta(z)^{-\wp})^n, \tag{2056}$$

$$\mathcal{T}(w)\Psi_n(z) = \Psi_n(z_1 + w, \dots, z_n + w),$$

$$\langle \Omega, \Psi_n(z)\rangle = \mathbf{W}_n(z_2 - z_1, \dots, z_n - z_{n-1}) \tag{2057}$$

for $w \in \overline{\mathrm{Tb}_\circ}$. For ordered tuples,

$$\sigma(h_1, \dots, h_n) = \int \mathbf{W}_n(x_2 - x_1, \dots, x_n - x_{n-1}) \prod_k h_k(x_k)\, dx. \tag{2058}$$

If also $t_1 \ge \eta$, then

$$\langle \psi, [(h_1, \dots, h_n)]\rangle = \int \langle \psi, \Psi_n(x)\rangle \prod_k h_k(x_k)\, dx. \tag{2059}$$

■

Proof. Contractivity and (2038) give the bounds. Locally choose a fixed η strictly smaller than every allowed margin. All semigroup factors are then norm holomorphic; the independence of η proves holomorphy of the defined functions. The semigroup law and the vacuum identities give (2057).

The density can be written with half the margin:

$$b_\eta(y) = \mathcal{T}(y + \tfrac{\eta}{2}e_3)B(\tfrac{\eta}{2},\tfrac{\eta}{2})\mathcal{T}(-y + \tfrac{\eta}{2}e_3). \tag{2060}$$

Both arguments have positive margin, since $|y_3| < c_0\eta \le \eta/2$. This follows directly from the shift law and the formula for b_η. Insert (2060) in (2051), applied to the products (2049)–(2050). With $x_k = y_k + t_k e_3$, consecutive semigroup factors combine into $\mathcal{T}(x_{k+1} - x_k - \eta e_3)$. At the right the factor fixes Ω. At the left it either disappears in the vacuum component or becomes $\mathcal{T}(x_1 - \eta e_3/2)$. These are precisely (2052) and (2054) with margin $\eta/2$, proving the two integral formulas. The supports are compact and stay inside the respective open tubes, so their continuous integrands are integrable. □

108.2. **Changing the ordering direction.** For $u \in S^2$ put

$$d^u_\circ(z) = \operatorname{Re}(u \cdot z) - |\operatorname{Im}(z - (u \cdot z)u)|, \qquad \mathrm{Tb}^{u,\pi}_n = \{z : d^u_\circ(z_{\pi(k+1)} - z_{\pi(k)}) > 0,\ k < n\}.$$

If $R \in \mathrm{SO}(3)$ satisfies $Re_3 = u$, then $d^u_\circ(z) = d_\circ(R^{-1}z)$. Each tube is open and convex; it contains $\operatorname{Re} z$ whenever it contains z. For $n = 1$ it is all of $\mathbb{C}^3$. Define on it

$$\mathbf{S}^{R,\pi}_n(z) = \mathbf{W}_n(R^{-1}(z_{\pi(2)} - z_{\pi(1)}), \dots, R^{-1}(z_{\pi(n)} - z_{\pi(n-1)})). \tag{2061}$$

Any n distinct real points admit a separating direction. More quantitatively, if their mutual distances are at least $\delta > 0$, there is $u \in S^2$ with

$$|u \cdot (x_i - x_j)| \geq \delta/n^2 \quad (i \neq j). \tag{2062}$$

Indeed the spherical zone $|u \cdot (x_i - x_j)/|x_i - x_j|| < n^{-2}$ has area $4\pi/n^2$. The union over the $\binom{n}{2}$ pairs has area less than 2π, so cannot cover S^2. Ordering the projections shows that the real parts of the tubes cover $(\mathbb{R}^3)^n_{\neq}$.

Proposition 108.3. There is a unique continuous kernel s_n on $(\mathbb{R}^3)^n_{\neq}$ such that

$$\sigma(f_1, \dots, f_n) = \int \mathsf{s}_n(x) \prod_i f_i(x_i)\, dx \tag{2063}$$

for all separated tuples. It is real analytic, symmetric and invariant under Euclidean isometries. It extends uniquely to a holomorphic function $\mathbf{S}_n$ on $\mathrm{Tb}^{\mathrm{all}}_n = \bigcup_{u,\pi} \mathrm{Tb}^{u,\pi}_n$, whose restriction to each tube is (2061), independently of the choice of R. This extension is symmetric and obeys

$$|\mathbf{S}_n(z)| \leq (C_\wp \eta^{-\wp})^n \quad \text{if } d^u_\circ(z_{\pi(k+1)} - z_{\pi(k)}) \geq 2\eta, \quad 0 < \eta \leq 1. \tag{2064}$$

In particular,

$$\mathsf{s}_n(x) = \mathbf{W}_n(x_2 - x_1, \dots, x_n - x_{n-1}) \quad (x_{13} < \dots < x_{n3}), \qquad \mathsf{s}_1 = \mathbf{W}_1. \tag{2065}$$

■

Proof. For a tuple ordered along u, rotate it by R^{-1}, permute its entries, and apply (2058). Euclidean invariance and symmetry give the kernel formula with $\mathbf{S}^{R,\pi}_n$. If a real configuration belongs to two ordering regions, choose sufficiently small balls about its entries so that every product of tests in those balls is ordered both ways, with one common positive margin. The two kernel formulas have the same value on every such product. Continuous functions are determined by these product tests, for example by taking products of approximate identities. Thus the kernels agree on each real overlap.

The real ordering regions cover all distinct points by (2062). They therefore define one real-analytic function s_n. To obtain (2063) for arbitrary separated supports, let their least distance be $d > 0$ and choose $\eta = \min(d/(4n^2), 1)$. Partition each test into finitely many pieces supported in balls of radius $c_0\eta/2$. Choose a point of the original support in every nonzero piece. Equation (2062) orders these chosen points with gaps at least 4η; each piece is supported in a $c_0\eta$ slab about its chosen point. Each resulting product tuple is therefore ordered. Apply the formula to each product and sum by multilinearity. For $n = 1$ the same partition uses $\eta = 1$. This proves the formula on every separated tuple. Products of arbitrarily small supported tests also prove uniqueness. Applying uniqueness to permuted or isometrically transported tests gives the claimed real symmetries.

A nonempty complex overlap of two tubes is convex, and contains the real part of each of its points. It consequently has a nonempty open real part. The two holomorphic pieces agree there by the preceding argument. The identity theorem on the connected overlap makes them equal throughout it. They therefore glue to one holomorphic function on the union. The same identity theorem on each tube proves uniqueness and independence of R; it also extends the permutation symmetry. The bound is (2055) after rotation. All constructions depend on σ, not on the chosen admissible numerical triple. □

108.3. **Growth on tests flat at the diagonals.** The extension to the full space of tests used for Schwinger functions requires a bound near every diagonal, not only for products with fixed separated supports.

Proposition 108.4. Put $k_{\mathrm{E}} = \lceil \wp \rceil + 4$ and $C_{\mathrm{E}} = C_\wp 2^\wp e^{2\wp} 3^{k_{\mathrm{E}}/2} \pi^2$. Then $|\mathsf{s}_1| \le C_\wp$, and for $n \ge 2$,

$$|\mathsf{s}_n(x)| \le C_\wp^n (2n^2)^{\wp n} \min(1, \min_{i \ne j} |x_i - x_j|)^{-\wp n}. \tag{2066}$$

For every $\varphi \in \mathcal{S}_{\ne}(\mathbb{R}^{3n})$,

$$\int |\mathsf{s}_n(x)\varphi(x)|\, dx \le 2 C_{\mathrm{E}}^n (n!)^{2\wp} |\varphi|_{k_{\mathrm{E}} n}. \tag{2067}$$

Consequently the Schwinger functions on this space are

$$\mathsf{S}_n^\sigma(\varphi) = \int \mathsf{s}_n(x)\varphi(x)\, dx. \tag{2068}$$

■

Proof. For a configuration with least Euclidean distance ρ, choose the direction in (2062) and use (2064) with $\eta = \min(\rho/(2n^2), 1)$. This proves (2066); the one-point bound follows from $B(1,1)$.

Let $k = k_{\mathrm{E}} n$, $N = \lceil \wp n \rceil$. Then $N \le k$ and $k \ge 4n$. If $\rho \le 1$ and $N \ge 1$, move a closest pair to its midpoint, obtaining a diagonal point x^* with $|x - x^*| \le 1$ and $|x - x^*|_1 \le \sqrt{3}\rho$. Flatness at x^* and Taylor's remainder, with the weighted seminorm of Section 93, give

$$|\varphi(x)| \le \frac{(\sqrt{3}\rho)^N}{N!}\, 3^{k/2} |\varphi|_k (1 + |x|^2)^{-k/2} \le 2\, 3^{k/2} \rho^N |\varphi|_k (1 + |x|^2)^{-k/2}.$$

The factor $3^{k/2}$ accounts for moving at most unit distance along the Taylor segment. Since $N \ge \wp n$, this absorbs the singular factor. When $\rho > 1$ or $\wp = 0$, the unmodified seminorm bound suffices. Finally

$$n^n \le e^n n!, \qquad \int_{\mathbb{R}^{3n}} (1 + |x|^2)^{-k/2}\, dx \le \prod_{i=1}^{n} \int_{\mathbb{R}^3} (1 + |x_i|^2)^{-2}\, dx_i = \pi^{2n}.$$

Combining these estimates proves (2067). The integral functional agrees with σ on separated smooth tensors by (2063). The tensor-density and uniqueness result of Section 93 identifies it with the Schwinger function. □

109. Smeared fields and their boundary distributions

We continue under (Rec). Minkowski space has coordinates $x = (t, \vec{x})$, product $x \cdot y = ts - \vec{x} \cdot \vec{y}$, and forward cone $V_+ = \{t > |\vec{x}|\}$. Write $U(x) = \mathcal{T}(\iota x)$ and fix an integer $l > \wp$. For $f \in \mathcal{S}(\mathbb{R}^{1+2})$ put

$$c_N(f) = \|A_l\| \sum_{k=0}^{N} \binom{N}{k} \|\partial_t^k f\|_1. \tag{2069}$$

The Schwartz seminorms used above give $\|\partial^\alpha F\|_1 \le \pi^{2n} |F|_{|\alpha|+4n}$ on $\mathbb{R}^{3n}$, by comparison with $\prod_i (1 + |x_i|^2)^{-2}$. In particular

$$c_N(f) \le \|A_l\| 2^N \pi^2 |f|_{N+4}. \tag{2070}$$

We use the graph seminorms $\|(1 + H)^j \psi\|$ on $\mathcal{D}_\infty$. The spectral projections $P_N = \mathcal{P}(\{E \le N\})$ have ranges in $\mathcal{D}_\infty$ and converge strongly to the identity. Bounded joint spectral multipliers preserve every $\operatorname{dom} H^j$ and commute there with $(1 + H)^j$. We will also use the following domain criterion:

$$|\langle (1 + H)^j \psi, \chi \rangle| \le C \|\psi\| \quad (\psi \in \mathcal{D}_\infty) \implies \chi \in \operatorname{dom} H^j, \quad \|(1 + H)^j \chi\| \le C. \tag{2071}$$

Indeed, insert $\psi = (1 + H)^j P_N \chi$ to bound $\int_{E \le N} (1 + E)^{2j}\, dm_\chi$ by C^2, then let $N \to \infty$.

109.1. **Construction on a common domain.**

Proposition 109.1. There is a unique complex linear map $f \mapsto \phi(f)$ from $\mathcal{S}(\mathbb{R}^{1+2})$ to linear operators $\mathcal{D}_\infty \to \mathcal{D}_\infty$ with

$$\langle \psi, \phi(f)\psi' \rangle = \int f(x)\, q_0(U(-x)\psi, U(-x)\psi')\, dx. \tag{2072}$$

It is independent of l and of the numerical triple in (Rec). For $j \geq 0$,

$$\|(1+H)^j \phi(f)\psi\| \leq c_{l+j}(f) \|(1+H)^{2l+j}\psi\|. \tag{2073}$$

On $\mathcal{D}_\infty$ it satisfies

$$\begin{aligned} \langle \psi, \phi(f)\psi' \rangle &= \langle \phi(\bar{f})\psi, \psi' \rangle, \\ U(a)\phi(f)U(-a) = \phi(f(\cdot - a)), &\qquad [H, \phi(f)] = i\phi(\partial_t f). \end{aligned} \tag{2074}$$

■

Proof. The integrand in (2072) equals

$$f(x) \langle U(-x)(1+H)^l \psi, A_l U(-x)(1+H)^l \psi' \rangle.$$

Its second factor is bounded and continuous. Denote the resulting sesquilinear form by q_f. It obeys

$$|q_f(\psi, \psi')| \leq \|f\|_1 \|A_l\| \|(1+H)^l \psi\| \|(1+H)^l \psi'\|. \tag{2075}$$

To turn this form into an operator, move energy powers from the first vector to the second. The map $t \mapsto e^{-itH}\xi$ is norm differentiable for $\xi \in \operatorname{dom} H$, with derivative $-ie^{-itH}H\xi$, by dominated convergence in the spectral calculus. Differentiating both vectors in the preceding matrix element, then integrating by parts in t, gives

$$q_f(H\psi, \psi') - q_f(\psi, H\psi') = i q_{\partial_t f}(\psi, \psi').$$

The boundary terms vanish because f is Schwartz and the matrix element and its derivative are bounded. Iteration yields

$$q_f((1+H)^N \chi, \psi') = \sum_{k=0}^{N} \binom{N}{k} i^k q_{\partial_t^k f}(\chi, (1+H)^{N-k}\psi'). \tag{2076}$$

Choose $\chi = R_H^l \psi$ and $N = l + j$. Using (2075) and monotonicity of graph norms,

$$|q_f((1+H)^j \psi, \psi')| \leq c_{l+j}(f) \|\psi\| \|(1+H)^{2l+j}\psi'\|. \tag{2077}$$

At $j = 0$, Riesz representation gives a unique vector $\phi(f)\psi'$. For every j, the domain criterion (2071) gives its membership in $\operatorname{dom} H^j$ and the bound (2073). Thus it belongs to $\mathcal{D}_\infty$.

Linearity, uniqueness and independence of the auxiliary choices follow from the defining form. Hermiticity follows from that of q_0. Changing variables in (2072) proves translation covariance. The identity obtained by integration by parts, tested against dense vectors in $\mathcal{D}_\infty$, proves the commutator formula. □

109.2. **Regularized products and their continuity.**

Lemma 109.2. Let $w, w' \in \mathrm{Tb}_\circ$ and $0 < s \leq \min(1, d_\circ(w), d_\circ(w'))$. Then

$$\mathcal{T}(w)\phi(f)\mathcal{T}(w') = \int f(x)\mathcal{T}(w - se_3 + \imath x) B(s,s) \mathcal{T}(w' - se_3 - \imath x)\, dx \tag{2078}$$

in matrix elements on $\mathcal{H}$. The integrand after removal of f is strongly continuous and bounded by $C_\wp s^{-\wp}$. For $w_1, \dots, w_n \in \mathrm{Tb}_\circ$, set $z_k(x) = \sum_{r \leq k} w_r + \imath x_k$. Then

$$\begin{aligned} &\langle \psi, \mathcal{T}(w_1)\phi(f_1) \cdots \mathcal{T}(w_n)\phi(f_n)\Omega \rangle \\ &\qquad = \int \prod_k f_k(x_k) \langle \psi, \Psi_n(z(x)) \rangle\, dx. \end{aligned} \tag{2079}$$

■

Proof. The positive margins put $\mathcal{T}(w)^*\psi, \mathcal{T}(w')\psi'$ in $\mathcal{D}_\infty$. Apply (2072) and split T_s from each semigroup. Equation (2046) changes the form between these two factors into $B(s,s)$; commuting the remaining spectral multipliers gives (2078). The margin is unchanged by addition of ιx, so all remaining factors are contractions. Strong continuity and the point bound give the stated properties.

For the product choose $s = \min(1, d_\circ(w_1), \frac{1}{2}\min_{k\ge 2} d_\circ(w_k))$. The inner minimum is omitted when $n = 1$. Split each intermediate semigroup into equal halves, and append T_s at the right, where it fixes Ω. This writes the product as a product of bounded regularized fields, each covered by (2078). Successively insert those formulas. The uniform point bounds and $\prod_k |f_k| \in L^1$ justify Fubini. Consecutive semigroup arguments combine to $z_{k+1}(x) - z_k(x) - 2se_3$; the first is $z_1(x) - se_3$, and the final spatial-time translation fixes Ω. The resulting product is $\Psi_n(z(x))$ with margin s, proving (2079). □

Lemma 109.3. Suppose $c_N(f_k^\epsilon - f_k^0) \to 0$ for every N, and C_k^ϵ are contractive joint spectral multipliers converging strongly to C_k^0. Then

$$C_1^\epsilon \phi(f_1^\epsilon) \cdots C_n^\epsilon \phi(f_n^\epsilon)\Omega \longrightarrow C_1^0 \phi(f_1^0) \cdots C_n^0 \phi(f_n^0)\Omega$$

in every graph norm of H. ■

Proof. Work from right to left. If $\chi^\epsilon \to \chi^0$ in all graph norms, then

$$\begin{aligned}&\|(1+H)^j(\phi(f^\epsilon)\chi^\epsilon - \phi(f^0)\chi^0)\| \\ &\quad \le c_{l+j}(f^\epsilon - f^0)\|(1+H)^{2l+j}\chi^\epsilon\| + c_{l+j}(f^0)\|(1+H)^{2l+j}(\chi^\epsilon - \chi^0)\| \longrightarrow 0.\end{aligned}$$

The spectral multipliers commute with the graph powers. Their strong convergence and uniform norm bound preserve the just-proved convergence. Induction starts at $\chi^\epsilon = \Omega$. □

109.3. **Tempered boundary values.** Here we give the boundary argument explicitly; pointwise polynomial growth near the tube boundary alone does not justify passing a limit through a test integral.

Lemma 109.4. Let $\mathbf{F}$ be holomorphic on an open set containing

$$z(x;\eta) = (\iota x_j + \eta h_j e_3)_{j=1}^n, \qquad x \in K, \quad 0 < \eta \le \eta_0 \le 1,$$

where K is compact, and suppose $|\mathbf{F}(z(x;\eta))| \le b\eta^{-p}$. For $g \in C_c^\infty$ supported in K, put $I_g(\eta) = \int g(x)\mathbf{F}(z(x;\eta))\,dx$ and $D_h = \sum_j h_j \partial_{t_j}$. If the integer $N \ge p + 1/2$, then

$$\sup_{0<\eta\le\eta_0} |I_g(\eta)| \le 2^N \eta_0^{-(N-1/2)} b \sum_{r=0}^{N} \|D_h^r g\|_1. \tag{2080}$$

The limit at $\eta = 0$ exists. ■

Proof. On each compact interval of positive heights, holomorphy permits differentiation under the integral. The Cauchy–Riemann equations and integration by parts give $I_g' = I_{-iD_h g}$. Set $J_r = I_{(-iD_h)^r g}$. For $0 \le r \le N$, these functions obey $J_r' = J_{r+1}$ when $r < N$, and $|J_r(\eta)| \le b\|D_h^r g\|_1 \eta^{-(N-1/2)}$. Integrate successively from η_0 toward η. Every integration of η^{-a} with $a \ge 3/2$ lowers the singular exponent by one at a cost at most 2; after $N-1$ integrations the exponent is $1/2$. Its final integral is bounded by 2. The values at η_0 contribute the same bound with the corresponding $\|D_h^r g\|_1$, giving (2080). Apply that estimate once more to $-iD_h g$. Then I_g' is bounded uniformly down to zero, so I_g is Lipschitz there and has a limit. □

Proposition 109.5. There is a unique tempered distribution $\mathcal{W}_n^\sigma$ on $\mathbb{R}^{3n}$ satisfying

$$\mathcal{W}_n^\sigma(f_1 \otimes \cdots \otimes f_n) = \langle \Omega, \phi(f_1) \cdots \phi(f_n)\Omega \rangle. \tag{2081}$$

For arbitrary $\alpha_1, \dots, \alpha_{n-1} \in V_+$,

$$\begin{aligned} &\mathcal{W}_n^\sigma(f_1 \otimes \cdots \otimes f_n) \\ &\quad = \lim_{\epsilon \downarrow 0} \int \prod_k f_k(x_k) \mathbf{W}_n\big(\iota(x_2 - x_1 + i\epsilon\alpha_1), \dots, \iota(x_n - x_{n-1} + i\epsilon\alpha_{n-1})\big)\, dx. \end{aligned} \tag{2082}$$

For $n = 1$, $\mathcal{W}_1^\sigma(f) = \mathbf{W}_1 \int f$. ■

Proof. First take the vacuum component of (2079), with $w_1 = e_3$ and $w_{k+1} = \iota(i\epsilon\alpha_k)$. It identifies the integral on the right of (2082) with

$$\langle \Omega, \phi(f_1)\mathcal{T}(w_2)\phi(f_2)\cdots\mathcal{T}(w_n)\phi(f_n)\Omega\rangle.$$

Its integrand is bounded by $(C_\wp \eta^{-\wp})^n \prod_k |f_k(x_k)|$, with $2\eta = \min(2, \epsilon \min_k(\alpha_k^0 - |\vec{\alpha}_k|))$. The spectral contractions tend strongly to the identity. Lemma 109.3 therefore proves the boundary formula on tensors of Schwartz functions.

To obtain a distribution on general tests, apply Lemma 109.4 to $\mathbf{F}(z) = \mathbf{W}_n(z_2 - z_1, \dots, z_n - z_{n-1})$ with $h_j = 2j$, $\eta_0 = 1$, $b = C_\wp^n$, $p = \wp n$, and $N = \lceil \wp n \rceil + 1$. Every consecutive margin is 2η, so the hypothesis follows from (2055). The resulting limit functional on C_c^∞ has bound

$$|\mathcal{L}(F)| \le 2^N (N+1) C_\wp^n (n(n+1))^N \pi^{2n} |F|_{N+4n}.$$

Here $\sum_j h_j = n(n+1)$ bounds the coefficients in D_h^r, and the L^1 derivative estimate gives the Schwartz norm.

For a smooth cutoff χ equal to one near the origin, $\chi(\cdot/R)F \to F$ in every Schwartz seminorm. Thus the bound extends $\mathcal{L}$ continuously to $\mathcal{S}(\mathbb{R}^{3n})$. It agrees with (2081) first on compactly supported tensors by the boundary identity, then on Schwartz tensors by Lemma 109.3. For uniqueness, finite sums of compactly supported smooth tensors are dense in each fixed Schwartz seminorm: cut off the test, convolve with a product mollifier, and approximate the resulting compact integral by Riemann sums of translates of that product mollifier. All derivatives up to the fixed order converge uniformly on a common compact set. A tempered functional vanishing on those tensors must therefore vanish. □

109.4. **Cyclicity.** Let $\mathcal{D}_0$ be the span of Ω and all $\phi(f_1)\cdots\phi(f_n)\Omega$. It lies in $\mathcal{D}_\infty$, is invariant under the fields and translations, and is dense.

To prove the last assertion, let $\psi \perp \mathcal{D}_0$. Translation covariance first gives

$$\langle \psi, U(a_1)\phi(f_1)\cdots U(a_n)\phi(f_n)\Omega\rangle = 0.$$

Replace the translations successively by $\mathcal{T}(w_j)$, $w_j \in \mathrm{Tb}_\circ$. For one replacement, move all preceding regularized factors to the first vector using Hermiticity. The first positive-margin semigroup puts that vector in $\mathcal{D}_\infty$, so every move is legitimate. The resulting expression is $\langle \chi, U(a)\chi'\rangle$. Its vanishing for all a says that the Fourier transform of the finite joint spectral measure $m_{\chi,\chi'}$ vanishes. Uniqueness of Fourier transforms of finite measures gives $m_{\chi,\chi'} = 0$, and hence $\langle \chi, \mathcal{T}(w)\chi'\rangle = 0$. Induction proves vanishing for every fully regularized product.

By (2079) and product approximate identities, $\langle \psi, \Psi_n(z)\rangle = 0$ throughout $\mathrm{Tb}_\circ^{(n)}$. Equation (2059) then makes ψ orthogonal to classes of time-ordered positive-time tuples. We must still approximate arbitrary positive-time tuples; their supports need not be time ordered.

Fix such a tuple $X = (f_1, \dots, f_n)$. Partition its entries with nonnegative functions $\chi_{h,k}$ supported in cubes of side $2h$ about hk, summing to one. Call a product of pieces good when all their center heights differ by at least $4h/c_0$. For small h, every good product, after permutation, is ordered with $\eta = 2h/c_0$ and first height at least η. Its class is orthogonal to ψ.

Let ξ_h be the sum of the remaining classes. On the compact set consisting of the original positive supports and their reflected negative supports, the continuous kernel s_{2n} is bounded by some M_X.

The kernel formula gives

$$\|\xi_h\|^2 \le M_X \left(\int \prod_i |f_i(x_i)| \sum_{\mathbf{k}\ \mathrm{bad}} \prod_i \chi_{h,k_i}(x_i)\, dx \right)^2 . \tag{2083}$$

The sum is at most one and vanishes unless some two heights differ by less than $(4/c_0 + 2)h$. Its integral is $O_X(h)$: integrate one height first over an interval of that length, bound that test by its supremum norm on its fixed spatial support, then integrate the remaining tests in L^1. Thus $\|\xi_h\| \to 0$ and $\langle \psi, [X] \rangle = \langle \psi, \xi_h \rangle = 0$. All tuple classes span a dense subspace of the reflection Hilbert space, so $\psi = 0$ and $\mathcal{D}_0$ is dense.

The translation representation already has joint spectrum $E \ge |q|$. In the convention $U(x) = \int e^{ip\cdot x}\, d\mathsf{E}(p)$ its momentum measure is the image of $\mathcal{P}$ under $(E, q) \mapsto (E, -q)$, and is carried by $\overline{V_+}$. If $\ker H = \mathbb{C}\Omega$, the only translation-invariant vectors are multiples of Ω: invariance under e^{itH} forces spectral support at $E = 0$. No vacuum-uniqueness assertion is used in the preceding construction.

110. Lorentz covariance and locality

Let $\mathcal{L} = \mathrm{SO}^+(1, 2)$. Every element is a product of spatial rotations and the boosts

$$B(s)(t, x_1, x_2) = (t \cosh s + x_1 \sinh s,\ x_1 \cosh s + t \sinh s,\ x_2).$$

For completeness, Λe_0 is a future unit timelike vector. A spatial rotation and a boost send e_0 to it. Their inverse followed by Λ fixes e_0 and hence acts by an element of $\mathrm{SO}(2)$ on its orthogonal spatial plane. This proves the factorization. These matrices have determinant one and preserve V_+.

Proposition 110.1. There is a unique strongly continuous unitary representation $U(\Lambda)$ of $\mathcal{L}$ fixing Ω and satisfying

$$U(\Lambda)\mathcal{D}_0 = \mathcal{D}_0, \qquad U(\Lambda)\phi(f)U(\Lambda)^{-1}\psi = \phi(f \circ \Lambda^{-1})\psi \quad (\psi \in \mathcal{D}_0). \tag{2084}$$

Together with translations it defines the strongly continuous representation

$$U(a, \Lambda) = U(a)U(\Lambda), \qquad U(a, \Lambda)U(a', \Lambda') = U(a + \Lambda a', \Lambda\Lambda'). \tag{2085}$$

Its spatial-rotation subgroup agrees with the representation on the reflection Hilbert space. ∎

Proof. We first prove invariance of the analytic functions:

$$\mathbf{W}_n(\iota\Lambda\zeta_1, \dots, \iota\Lambda\zeta_{n-1}) = \mathbf{W}_n(\iota\zeta_1, \dots, \iota\zeta_{n-1}) \quad (\zeta_k \in \mathbb{R}^{1+2} + iV_+). \tag{2086}$$

The complex linear map $\widehat{\Lambda} = \iota\Lambda\iota^{-1}$ preserves $\mathrm{Tb}_\circ$. Both sides, viewed as functions of $\iota\zeta_k$, are holomorphic on its connected product. It suffices to compare them at real differences ξ_k of positive third coordinate. Choose $x_1 = 0$, $x_{k+1} = x_k + \xi_k$. Then $\mathbf{W}_n(\xi) = \mathsf{s}_n(x)$. Spatial-rotation invariance follows from that of the Euclidean kernel.

For a boost, use the complex rotation

$$r_z(y_1, y_2, y_3) = (y_1 \cos z + y_3 \sin z,\ y_2,\ y_3 \cos z - y_1 \sin z). \tag{2087}$$

It satisfies $r_{is} = \widehat{B(s)}$. Fix real positive-time differences ξ_k. The set of z for which every $r_z\xi_k$ lies in $\mathrm{Tb}_\circ$ is open and contains the imaginary axis, since real boosts preserve the forward tube. A sufficiently small neighborhood of the compact segment from 0 to is lies in this set and is connected. For small real z, r_z is an ordinary Euclidean rotation preserving the ordering of these points. Thus $\mathbf{W}_n(r_z\xi_1, \dots, r_z\xi_{n-1})$ is constant there by Euclidean invariance. The identity theorem continues this equality to $z = is$. The identity theorem in all difference variables, then the rotation–boost factorization, proves (2086).

In (2082) change variables $x_j = \Lambda y_j$ and use (2086). The approach vectors become $\Lambda^{-1}\alpha_k \in V_+$, which are allowed in that boundary formula. Hence

$$\mathcal{W}_n^\sigma(F \circ (\Lambda^{-1})^{\times n}) = \mathcal{W}_n^\sigma(F). \tag{2088}$$

This first holds on tensors; continuity under invertible linear changes of Schwartz variables and tensor density extend it to all tests.

On a polynomial vector define

$$U(\Lambda)\phi(f_1)\cdots\phi(f_n)\Omega = \phi(f_1\circ\Lambda^{-1})\cdots\phi(f_n\circ\Lambda^{-1})\Omega.$$

Inner products of two such vectors are Wightman values: move the first string across using Hermiticity, reversing its order and conjugating its tests. Equation (2088) preserves every such inner product. The rule therefore respects all linear relations, is isometric, and has inverse given by Λ^{-1}. Density of $\mathcal{D}_0$ gives a unitary extension; the group law and covariance hold on its spanning vectors and hence as stated. This also proves uniqueness.

As $\Lambda\to 1$, the time derivatives of $f\circ\Lambda^{-1}-f$ tend to zero in L^1. The chain rule expresses them as finite sums of derivatives of f after the linear change; dominated convergence, using Schwartz decay uniformly for Λ near 1, proves the assertion. Lemma 109.3 gives strong continuity first on polynomial vectors, then on all of $\mathcal{H}$ by unitarity and density. On polynomial vectors the conjugation $U(\Lambda)U(a)U(\Lambda)^{-1}$ translates every test by Λa. This proves (2085) and continuity of the combined representation.

Finally, the original spatial rotations commute with H and $B(s,s')$, hence with A_l, and preserve q_0. They conjugate $U(x)$ to $U((1\oplus R)x)$. Changing variables in (2072) therefore gives the same field transformation as $U(1\oplus R)$. Both fix the vacuum, so they agree on $\mathcal{D}_0$ and on its closure. □

110.1. **Interchanging spacelike-separated insertions.**

Proposition 110.2. If $f,h\in\mathcal{S}(\mathbb{R}^{1+2})$ have spacelike separated supports, then

$$\phi(f)\phi(h)\psi=\phi(h)\phi(f)\psi\qquad(\psi\in\mathcal{D}_0).\tag{2089}$$

■

Proof. First suppose all tests have compact support. Testing against polynomial vectors and using Hermiticity reduces the assertion to invariance of $\mathcal{W}_n^{\sigma}(f_1\otimes\cdots\otimes f_n)$ under interchange of adjacent entries $k,k+1$ whose supports are spacelike separated. Set $K=\prod_j \operatorname{supp} f_j$; zero entries are trivial. On K, write $\xi=\vec{x}_{k+1}-\vec{x}_k$, $\tau=t_{k+1}-t_k$, and put

$$\delta=\min_K(|\xi|-|\tau|)>0,\quad A=4/\delta,\quad \eta_0=\min((2A)^{-1},1/2),$$
$$h_*=2A\max_{x\in K,\ j<n}(|\vec{x}_{j+1}-\vec{x}_j|+|t_{j+1}-t_j|)+6.$$

We deform the two adjacent heights past each other:

$$h_k^\lambda=-\lambda/2,\quad h_{k+1}^\lambda=\lambda/2,$$
$$h_j^\lambda=\tfrac12+(j-k-1)h_*\quad(j>k+1),\qquad h_j^\lambda=-\tfrac12-(k-j)h_*\quad(j<k),\tag{2090}$$
$$z_j(x;\lambda,\eta)=\iota x_j+\eta h_j^\lambda e_3,\qquad -1\le\lambda\le 1,\quad 0<\eta\le\eta_0.$$

Although the vertical ordering fails during this deformation, the points remain in the union of tilted tubes. Indeed choose $\sin\theta=A\eta$ and

$$u_\theta=\cos\theta\,e_3+\sin\theta\,(\xi/|\xi|,0).$$

Then $\cos\theta\ge\sqrt{3}/2$, and direct calculation gives

$$\begin{aligned}d_\circ^{u_\theta}(z_{j+1}-z_j)&=\eta(h_{j+1}^\lambda-h_j^\lambda)\cos\theta\\&\quad+A\eta\,\frac{\xi}{|\xi|}\cdot(\vec{x}_{j+1}-\vec{x}_j)-A\eta|t_{j+1}-t_j|.\end{aligned}\tag{2091}$$

For $j=k$ this is at least $-\eta+A\eta\delta=3\eta$. For $j\neq k$ the height gap is at least h_*; the definition of h_* makes the same lower bound hold. Consequently

$$z(x;\lambda,\eta)\in\mathrm{Tb}_n^{\mathrm{all}},\qquad|\mathbf{S}_n(z(x;\lambda,\eta))|\le(C_\wp\eta^{-\wp})^n\tag{2092}$$

uniformly in x, λ, η in these ranges. The witnessing direction may depend on x and η; the holomorphic function $\mathbf{S}_n$ is the same on all of these tubes.

For a smooth g supported in K, let $I_g(\lambda,\eta) = \int g(x)\mathbf{S}_n(z(x;\lambda,\eta))\,dx$. The Cauchy–Riemann equations and integration by parts give

$$\partial_\eta I_g = I_{-iD_\lambda g}, \qquad \partial_\lambda I_g = \eta I_{-iD' g}, \qquad D_\lambda = \sum_j h_j^\lambda \partial_{t_j}, \quad D' = \tfrac{1}{2}(\partial_{t_{k+1}} - \partial_{t_k}). \tag{2093}$$

Compactness at every positive height justifies differentiation. Lemma 109.4, with $N = \lceil \wp n \rceil + 1$, gives the uniform bound

$$|I_g(\lambda,\eta)| \le Q(g) := 2^N \eta_0^{-(N-1/2)} C_\wp^n \sum_{r=0}^{N} \sup_{|\lambda|\le 1} \|D_\lambda^r g\|_1. \tag{2094}$$

The coefficients of D_λ are uniformly bounded, so $Q(g) < \infty$.

Take $g = \bigotimes_j f_j$. At $\lambda = 1$ the heights increase in the original order, and (2082) identifies the limit $I_g(1,\eta)$ with the original Wightman value. At $\lambda = -1$ they increase in the order with $k, k+1$ interchanged. The symmetry of the single function $\mathbf{S}_n$ and the same boundary formula identify $I_g(-1,\eta)$ with the interchanged value. But (2093)–(2094) imply

$$|I_g(1,\eta) - I_g(-1,\eta)| \le 2\eta Q(D'g) \longrightarrow 0.$$

This proves the adjacent interchange for compact tests.

For polynomial vectors with other Schwartz entries, cut off those entries and use Lemma 109.3 to pass to the limit in both products. Density of polynomial test vectors gives (2089) for compactly supported f, h. Finally cut off f, h themselves inside their original supports. Their spacelike separation is preserved, and graph-norm continuity passes the equality to the original Schwartz functions. □

Theorem AO (Scalar reconstruction)**.** Under (Rec), the reflection Hilbert space carries a Hermitian scalar field ϕ with the following properties:

(1) The proper orthochronous Poincaré group acts by a strongly continuous unitary representation U fixing the unit vector Ω. Its translation spectrum lies in $\overline{V_+}$.
(2) The dense polynomial domain $\mathcal{D}_0$ contains Ω and is invariant under U and every $\phi(f)$. For $\psi, \psi' \in \mathcal{D}_0$, $f \mapsto \langle \psi, \phi(f)\psi' \rangle$ is tempered, and the Hermiticity relation in (2074) holds.
(3) The field is covariant:

$$U(a,\Lambda)\phi(f)U(a,\Lambda)^{-1} = \phi(f(\Lambda^{-1}(\,\cdot - a))) \quad \text{on } \mathcal{D}_0,$$

and fields with spacelike separated test supports commute there.
(4) The vacuum is cyclic. It is the unique translation-invariant vector up to scalar multiples if $\ker H = \mathbb{C}\Omega$.

The vacuum distributions are $\mathcal{W}_n^\sigma$. Their analytic functions are $\mathbf{W}_n$, with boundary values (2082), and their Euclidean continuations recover σ on every separated C_c^1 tuple by (2063). On $\mathcal{S}_{\neq}(\mathbb{R}^{3n})$ the Schwinger functions are the absolutely convergent integrals (2068). Thus, with the additional vacuum-uniqueness hypothesis, these objects form a scalar Wightman theory. ■□

Proof. The domain, spectrum, cyclicity and Hermiticity were proved in Section 109. Equation (2073), followed by (2070), gives tempered matrix elements. Propositions 110.1 and 110.2 give covariance and locality on $\mathcal{D}_0$. The boundary and Euclidean identities follow from (2082), (2063) and (2065). On differences with third coordinate at least $\epsilon > 0$ and real spatial coordinates, (2055) bounds $\mathbf{W}_n$ by $(C_\wp \min(1, \epsilon/2)^{-\wp})^n$; in particular the Euclidean continuation has the required polynomial growth. Finally (2067) proves the asserted absolute convergence. □

111. Reconstruction of the Gibbs limits

We now verify the reconstruction hypotheses for the lattice limits. The estimates must hold for every limit obtained by conditioning a frame: the canonical components need not be centered, even when the original limit is centered. This is why the two-state estimates and closure of the Gibbs class under invariant conditioning were retained in the preceding arguments.

Fix the sourced construction at radius 2ϱ, and fix $0 < g \le \gamma_\varrho$. In the notation of (1917) and (2022), put

$$C^g_{\mathrm{pt}} = 320 \cdot 8^4 m_3 (40L)^4 \frac{2\bar{Q}_{\mathrm{mid}}}{\varrho} M, \qquad C^g_{\wp} = 16 C^g_{\mathrm{pt}}. \tag{2095}$$

All these constants are fixed before choosing a sequence of states. Let $\bar{c}_g$ denote the number $\bar{c}$ in (1991), evaluated at the bound system $\widehat{\mathsf{C}}^g$ of Proposition 106.3.

Proposition 111.1. Let $K_j \to \infty$, $(K_j, \beta_j) \in \mathfrak{d}(g)$ and $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$. Every limit σ' in the frame $(K_j, \nu_j)_j$ satisfies (Rec) with

$$(C_\wp, \wp, c_0) = (C^g_\wp + |c_{\sigma'}|, 4, 1/2), \qquad |c_{\sigma'}| \le \bar{c}_g.$$

In particular the same choice $(C^g_\wp + \bar{c}_g, 4, 1/2)$ works for every such limit. If the frame itself converges to σ, one can use $(C^g_\wp, 4, 1/2)$ for σ. ∎

Proof. The Gibbs classes form a class system with the stated uniform bounds by Proposition 106.3. A limit in the frame has approximants ρ_j in these same classes; the centering law remains ν_j. Theorem AN therefore applies to σ' and its centered functional. It gives full Euclidean invariance, reflection positivity, the translation semigroup and its joint spectral measure, and the cone $E \ge |q|$. These are (Rec1)–(Rec3). Separability is part of the Hilbert-space construction in Section 94. The one-point function and its bound are (1991).

For (Rec4), take $0 < \eta \le 1$, $F, F' \in \mathbb{C}\mathfrak{S}(\{x_3 > \eta\})$ and $h \in C^1_c(\{|x_3| < \eta/2\})$. The centered functional has (R1)–(R5) with the constants (2022). Corollary 97.3, with $d = \eta$, gives

$$\sigma'^{\mathrm{c}}(\theta\bar{F} \cdot (h) \cdot F') = \int h(x) v_\eta(x)\, dx,$$

where

$$|v_\eta(x)| \le C^g_{\mathrm{pt}} \min(1, \eta - |x_3|)^{-4} \|[\tau_{-\eta e_3} F]\|_{\sigma'^{\mathrm{c}}} \|[\tau_{-\eta e_3} F']\|_{\sigma'^{\mathrm{c}}}.$$

On the support of h the first minimum exceeds $\eta/2$. Integration thus proves

$$|\sigma'^{\mathrm{c}}(\theta\bar{F} \cdot (h) \cdot F')| \le C^g_\wp \eta^{-4} \|h\|_1 \|[\tau_{-\eta e_3} F]\|_{\sigma'^{\mathrm{c}}} \|[\tau_{-\eta e_3} F']\|_{\sigma'^{\mathrm{c}}}. \tag{2096}$$

These are the norms in the coordinate reflection Hilbert space, as required in (2047). Lemma 107.4 transfers the bound to σ' with the additional constant $|c_{\sigma'}|$. For the limit of the frame itself the integrating and centering laws agree, so its one-point function is zero. □

The passage from clustering of separated tuples to clustering of Schwartz tests uses the reflection Hilbert space. Directly translating a Schwartz approximation would give bounds depending on the translation; the Hilbert-space argument keeps them uniform.

Proposition 111.2. Suppose a symmetric, normalized separated-tuple functional σ has the bounds of Section 94, is Euclidean invariant and reflection positive, and has Schwinger functions satisfying

$$|\mathsf{S}_n(\varphi)| \le c_E C_E^n (n!)^\kappa |\varphi|_{kn}, \qquad \varphi \in \mathcal{S}_{\neq}(\mathbb{R}^{3n}),$$

where $k \in \mathbb{N}$ and $c_E, C_E, \kappa \ge 0$. Assume also that

$$\sigma(X \cdot \tau_v Y) \longrightarrow \sigma(X)\sigma(Y) \quad (|v| \to \infty)$$

for all separated tuples X, Y, with the concatenation defined for sufficiently large $|v|$. Then the Schwinger functions satisfy Euclidean invariance, permutation symmetry, reflection positivity (1880), the displayed growth bound, and

$$\lim_{s\to\infty} \mathsf{S}_{n+n'}(\Theta\bar{\varphi}\otimes\tau_{sv}\varphi') = \mathsf{S}_n(\Theta\bar{\varphi})\mathsf{S}_{n'}(\varphi') \tag{2097}$$

for $\varphi \in \mathcal{S}_+(\mathbb{R}^{3n})$, $\varphi' \in \mathcal{S}_+(\mathbb{R}^{3n'})$ and $0 \neq v \in \mathbb{R}^2 \times \{0\}$. We refer to these five properties as (E0′)–(E4), respectively growth, Euclidean invariance, reflection positivity, symmetry and clustering. The test spaces are the flat Schwartz spaces defined in Section 93. ■

Proof. Proposition 93.5 gives the first four properties. For the last, form the algebraic direct sum $\underline{\mathcal{S}}_+ = \bigoplus_{n\geq 0} \mathcal{S}_+(\mathbb{R}^{3n})$, with the positive form

$$\langle \underline{\varphi}, \underline{\psi}\rangle_{\mathcal{S}} = \sum_{n,n'} \mathsf{S}_{n+n'}(\Theta\bar{\varphi}_n \otimes \psi_{n'}).$$

Let $\mathcal{H}^{\mathcal{S}}$ be its completion modulo the null space. If both sequences vanish above degree n_0, their form is bounded in absolute value by

$$\max(1, c_E)\max(1, C_E)^{2n_0}((2n_0)!)^{\kappa}\left(\sum_{n\leq n_0} |\varphi_n|_{2kn_0}\right)\left(\sum_{n\leq n_0} |\psi_n|_{2kn_0}\right). \tag{2098}$$

This follows term by term from the tensor-product seminorm bound; the degree-zero term is ordinary multiplication.

Send a smooth separated positive-time tuple X to the sequence ιX with $\bigotimes X$ in its degree and zero elsewhere. For finite sums of such tuples,

$$\langle \iota F, \iota F'\rangle_{\mathcal{S}} = \sigma(\theta\bar{F}\cdot F'). \tag{2099}$$

Lemma 93.2 and (2098) show that their classes are dense in $\mathcal{H}^{\mathcal{S}}$. They are also dense in $\mathcal{H}(\sigma)$: mollify each entry of a C_c^1 tuple with a common radius tending to zero, smaller than its distance to the reflection plane and half its least support separation. The resulting tuples remain separated in the positive half-space. The moving-entry continuity estimate gives convergence of the three terms in

$$\|[X] - [X_j]\|^2 = \sigma(\theta X\cdot X) - 2\,\mathrm{Re}\,\sigma(\theta X\cdot X_j) + \sigma(\theta X_j\cdot X_j).$$

Consequently (2099) extends to a unitary $V : \mathcal{H}(\sigma) \to \mathcal{H}^{\mathcal{S}}$ carrying Ω to the constant sequence. For a fixed tangential translation, the seminorm estimate (1881) shows that the same approximations converge after translation. Thus

$$V\mathcal{U}(v')V^{-1}[\underline{\varphi}] = [\tau_{(v',0)}\underline{\varphi}].$$

Clustering of tuples gives, first for tuple classes,

$$\langle \chi, \mathcal{U}(sv')\chi'\rangle \longrightarrow \langle\chi,\Omega\rangle\langle\Omega,\chi'\rangle, \qquad v' \neq 0.$$

The difference between the two sides has modulus at most $2\|\chi\|\|\chi'\|$, uniformly in s, since $\mathcal{U}(sv')$ is unitary. Density extends this limit to all vectors. Apply it to the inverse images under V of the classes with φ and φ' in degrees n, n'. The form identity gives exactly (2097). □

Theorem AP (Gibbs limits and scalar fields)**.** Fix the construction and g as above. Every sequence $K_j \to \infty$, $(K_j, \beta_j) \in \mathfrak{d}(g)$ and $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ has a subsequence whose self-centered moments converge on all separated tuples. Let σ be any such limit. It has the canonical decomposition of Theorem AL, unique in the sense of Proposition 101.1. Every component σ_ξ is Euclidean invariant, reflection positive at every mirror, and has the uniform clustering estimate (1994). Its one-point constant may be nonzero, with $|c_\xi| \leq \bar{c}_g$ and $\int c_\xi\, d\lambda = 0$.

Each component satisfies (Rec) with $(C_\wp^g+\bar c_g, 4, 1/2)$ and $\ker H_{\sigma_\xi} = \mathbb{C}\Omega_{\sigma_\xi}$. Its Schwinger functions satisfy (E0′)–(E4) with the common growth estimate

$$|\mathsf{S}_n^{\sigma_\xi}(\varphi)| \le 2(C_E^g)^n (n!)^8 |\varphi|_{8n}, \qquad C_E^g = 16e^8 3^4 \pi^2 (C_\wp^g + \bar c_g). \tag{2100}$$

On its reflection Hilbert space there is a scalar Wightman theory with unique vacuum, common invariant polynomial domain, and the Schwinger functions of σ_ξ, as specified in Theorem AO.

The original limit σ has the same reconstruction and growth conclusions, using $C_\wp^g$ in place of $C_\wp^g + \bar c_g$, except that neither clustering nor vacuum uniqueness is asserted. Its zero-energy space is canonically $L^2(\mathsf{X}, \lambda)$, with Ω corresponding to 1. All stated constants are independent of the sequence, its subsequence and its components. ■□

Proof. The cumulant compactness result, Proposition 92.1, and the moment–cumulant formula give a common convergent subsequence. Proposition 106.3 makes it a converging frame with the fixed bound system $\widehat{\mathsf{C}}^g$. Theorem AL and Proposition 101.1 give its canonical decomposition and uniqueness. By Theorem AL, each component is a limit in this frame, obtained along the full chosen sequence by conditioning ν_j on invariant events of positive probability. Proposition 106.1 keeps the conditioned laws in the periodic Gibbs classes. The centering law is still ν_j, so Proposition 111.1 applies with the possibly nonzero constant c_ξ. Full Euclidean invariance and reflection positivity follow from Theorem AN for these same approximants.

The uniform component clustering estimate is (1994); the supports of X and $\tau_v Y$ are disjoint for the large translations in that estimate. Equation (1996) therefore gives the simple zero-energy space of each component. Substituting $\wp = 4$ in Proposition 108.4 gives (2100); its integral extension is the Schwinger extension by uniqueness on dense separated tensors. Proposition 111.2 now supplies all five Euclidean properties. Theorem AO supplies the Wightman theory.

For σ, the one-point constant is zero and the same arguments apply without the clustering step. The last assertion on its vacuum space is Theorem AM. □

112. A rotation criterion along growing tori

The lattice rotation identity has a bulk term and a term supported where a spatial cutoff changes. Their sum has a continuum limit. Separate control of the two terms gives a sufficient condition for rotation invariance; no separate limit will be assumed in deriving the identity.

112.1. **The finite-lattice identity.** Fix $K, m \ge 1$, $a = a_K$, $\ell = 2L^m$ and $\beta > 0$. We use the central lifts in $[-\ell/2, \ell/2)^3$ when multiplying a position by $\omega \in \mathfrak{so}(3)$. A function has margin r if its support is contained in $(-\ell/2 + r, \ell/2 - r)^3$. Local displacements across a plaquette always use its local lift. Let x_p and x_b be the plaquette barycentre and bond midpoint. In sums $b \subset \partial p$, each of the four underlying bonds has its positive coordinate orientation.

For $b = \langle x, x + ae_\mu\rangle$ and a signed direction $\rho \ne \pm\mu$, let $W_x(\mu, \rho)$ be the holonomy along the four sides of the square starting at x in directions $\mu, \rho, -\mu, -\rho$. Its underlying plaquette is denoted by $p(b, \rho)$. If p contains b, $W_{b,p}$ is the holonomy around p starting with b; the other signed direction is determined uniquely. With $\langle A, B\rangle = \frac12 \operatorname{tr}(AB)$, put

$$\mathbf{w}(V) = \frac{V - V^{-1}}{2i}, \qquad \mathbf{u}_{b,\rho} = -\frac12\{\mathbf{w}(W_x(\mu,\rho)) - \mathbf{w}(W_x(\mu,-\rho))\}, \qquad \mathbf{u}_{b,\mu} = 0. \tag{2101}$$

The indices ρ, λ in sums below range over $1, 2, 3$. Define

$$\begin{aligned} d_{b,p,\rho} &= \langle \mathbf{u}_{b,\rho}, \mathbf{w}(W_{b,p})\rangle, & t_\rho(p) &= \sum_{b\subset\partial p} d_{b,p,\rho}, \\ s_{\lambda\rho}(p) &= \sum_{b\subset\partial p} a^{-1}(x_b - x_p)_\lambda d_{b,p,\rho}, & r_\rho(p) &= t_\rho(p) - \tfrac12(a_{p+ae_\rho} - a_{p-ae_\rho}). \end{aligned} \tag{2102}$$

For functions on the torus, set

$$\nabla^{\mathrm{c}}_{\rho} h(y) = a^{-1}\{h(y+\tfrac{a}{2}e_\rho) - h(y-\tfrac{a}{2}e_\rho)\}, \qquad \nabla^{\mathrm{s}}_{\omega} h(y) = \sum_{\rho} (\omega y)_\rho \frac{h(y+ae_\rho)-h(y-ae_\rho)}{2a}.$$

The latter formula applies to plaquette weights as well, with $p \pm ae_\rho$ retaining the directions of p.

Let χ have margin $2a$. For a plaquette weight v define

$$\mathfrak{S}_{\omega,\chi}[v] = \sum_{p} v(p)\chi(x_p) \sum_{\rho,\lambda} \omega_{\rho\lambda} s_{\lambda\rho}(p), \qquad \mathfrak{K}_{\omega,\chi}[v] = \sum_{p} a^{-1} v(p)\chi(x_p) \sum_{\rho} (\omega x_p)_\rho r_\rho(p). \tag{2103}$$

Write $\mathfrak{S}_\omega[v]$ and $\mathfrak{K}_\omega[v]$ for the same expressions without χ when v has margin $2a$. The shell insertion is

$$\begin{aligned} h_{\omega,\chi}(p) &= a^{-1} \sum_{b \subset \partial p, \rho} \{\chi(x_b) - \chi(x_p)\}(\omega x_b)_\rho d_{b,p,\rho}, \\ \mathfrak{B}_{\omega,\chi} &= \beta \sum_p a_p \nabla^{\mathrm{s}}_{\omega} \chi(x_p) - \beta \sum_p h_{\omega,\chi}(p) \\ &\quad - \tfrac{3}{2} \sum_{p = p_{\mu\nu}(x),\, \mu<\nu} a_p \{(\omega x_p)_\mu \nabla^{\mathrm{c}}_{\mu}\chi(x_p) + (\omega x_p)_\nu \nabla^{\mathrm{c}}_{\nu}\chi(x_p)\}. \end{aligned} \tag{2104}$$

These are bounded polynomial functions of the bond variables. We say that χ covers $f_1, \dots, f_n$ if it equals one at x_p and all four x_b whenever some $f_i(x_p) \neq 0$. All cumulants in the next display are taken in the Wilson measure at (K, ℓ, β), and $\mathcal{O}(f) = a^{-1}\sum_p f(x_p) a_p$. Define the bulk term by

$$\begin{aligned} \mathfrak{U}_{\omega,\chi}(\underline{f}) &= -\beta \langle (\mathfrak{S}_{\omega,\chi} + \mathfrak{K}_{\omega,\chi})[1]; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle \\ &\quad + a^{-1} \sum_{i=1}^{n} \langle (\mathfrak{S}_\omega + \mathfrak{K}_\omega)[f_i]; (\mathcal{O}(f_l))_{l \neq i}\rangle. \end{aligned} \tag{2105}$$

Proposition 112.1 (Finite-lattice rotation identity)**.** If χ covers the real sources $f_1, \dots, f_n$, then

$$\sum_{i=1}^{n} S^{T}_{n;K,\ell,\beta}(f_1, \dots, \nabla^{\mathrm{s}}_{\omega} f_i, \dots, f_n) = \mathfrak{U}_{\omega,\chi}(\underline{f}) + \langle \mathfrak{B}_{\omega,\chi}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle. \tag{2106}$$

The identity also holds for complex sources. ∎

Proof. Let σ_A, $1 \le A \le 3$, be the Pauli basis and let $\partial^A_b F$ denote differentiation after replacing $U(b)$ by $e^{it\sigma_A} U(b)$ at $t = 0$. For a vector field η set

$$X_b = a^{-1} \sum_{\rho} \eta_\rho(x_b) \mathbf{u}_{b,\rho}, \quad \mathfrak{v}_\eta F = \sum_{b,A} \langle X_b, \sigma_A\rangle \partial^A_b F, \quad \operatorname{div} \mathfrak{v}_\eta = \sum_{b,A} \partial^A_b \langle X_b, \sigma_A\rangle.$$

Haar invariance in the b factor gives $\int \partial^A_b F \, dU = 0$. Differentiation under the integral is valid because the configuration space is a finite product of compact groups and the functions used here have continuous derivatives. The product rule therefore gives

$$\int \mathfrak{v}_\eta F \, dU = -\int F \operatorname{div} \mathfrak{v}_\eta \, dU. \tag{2107}$$

For $b \subset \partial p$, its based holonomy changes to $e^{it\sigma_A} W_{b,p}$. Since $\sigma_A^2 = 1$ and $a_p = 1 - \tfrac{1}{2}\operatorname{tr} W_{b,p}$, direct differentiation yields

$$\partial^A_b a_p = \langle \sigma_A, \mathbf{w}(W_{b,p})\rangle, \qquad \partial^A_b \langle \sigma_A, \mathbf{w}(W_{b,p})\rangle = 1 - a_p.$$

There is no sum over A in the second identity. If b does not belong to p, the first derivative is zero. Thus, for $\rho \neq \mu_b$,

$$\sum_{A} \partial^A_b \langle \sigma_A, \mathbf{u}_{b,\rho}\rangle = \tfrac{3}{2}\{a_{p(b,\rho)} - a_{p(b,-\rho)}\}.$$

Summing the first derivative over the Pauli basis, and regrouping the last identity by plaquettes, gives

$$\begin{aligned} \mathfrak{v}_\eta a_p &= a^{-1} \sum_{b \subset \partial p, \rho} \eta_\rho(x_b) d_{b,p,\rho}, \\ \operatorname{div} \mathfrak{v}_\eta &= -\tfrac{3}{2} \sum_{p = p_{\mu\nu}(x),\, \mu<\nu} a_p (\nabla^{\mathrm{c}}_\mu \eta_\mu + \nabla^{\mathrm{c}}_\nu \eta_\nu)(x_p). \end{aligned} \tag{2108}$$

Indeed, the two bonds of p perpendicular to direction ρ have midpoints $x_p \mp \frac{a}{2} e_\rho$, and contribute the difference $\eta_\rho(x_p - \frac{a}{2}e_\rho) - \eta_\rho(x_p + \frac{a}{2}e_\rho)$.

For a plaquette weight v, write $A(v) = \sum_p v(p) a_p$. Take $\eta(y) = \chi(y)\omega y$. Split $\chi(x_b) = \chi(x_p) + [\chi(x_b) - \chi(x_p)]$, and use $\omega x_b = \omega x_p + \omega(x_b - x_p)$ in the first part. The margin of χ makes this a valid equality of lifts wherever that part is nonzero. The resulting three terms are $h_{\omega,\chi}(p)$, $a^{-1}\chi(x_p)\sum_\rho (\omega x_p)_\rho t_\rho(p)$, and $\chi(x_p)\sum_{\rho,\lambda} \omega_{\rho\lambda} s_{\lambda\rho}(p)$. Insert the definition of r_ρ and sum the translated plaquette actions by parts. Since $\omega_{\rho\rho} = 0$, $(\omega(x_p \pm a e_\rho))_\rho = (\omega x_p)_\rho$ wherever the translated cutoff is nonzero. We obtain

$$\mathfrak{v}_{\chi\omega y} A(v) = (\mathfrak{S}_{\omega,\chi} + \mathfrak{K}_{\omega,\chi})[v] + \sum_p v(p) h_{\omega,\chi}(p) - A(\nabla^{\mathrm{s}}_\omega(v\chi)). \tag{2109}$$

The same zero diagonal of ω reduces the divergence in (2108) to the last sum of (2104).

Put $w_z = \sum_i z_i f_i$. The covering condition gives $w_z \chi = w_z$ and $w_z(p) h_{\omega,\chi}(p) = 0$. Consequently

$$\begin{aligned} \mathfrak{v}_{\chi\omega y} A(\beta - a^{-1} w_z) - \operatorname{div} \mathfrak{v}_{\chi\omega y} &= (\mathfrak{S}_{\omega,\chi} + \mathfrak{K}_{\omega,\chi})[\beta - a^{-1} w_z] \\ &\quad + \mathcal{O}(\nabla^{\mathrm{s}}_\omega w_z) - \mathfrak{B}_{\omega,\chi}. \end{aligned} \tag{2110}$$

Its integral against $e^{-A(\beta - a^{-1}w_z)} dU$ is zero by (2107). The partition function is entire in z and positive at zero, so we may normalize in a neighborhood of zero. For any affine insertion $Y_z = Y_0 + \sum_i z_i Y_i$, differentiation of its normalized mean gives

$$\partial_1 \cdots \partial_n \langle Y_z \rangle_{w_z}\big|_0 = \langle Y_0; \mathcal{O}(f_1); \ldots; \mathcal{O}(f_n) \rangle + \sum_i \langle Y_i; (\mathcal{O}(f_l))_{l \neq i} \rangle.$$

To see this, introduce an auxiliary source u for a fixed insertion, differentiate $\partial_u \log \int e^{uY + \sum_i z_i \mathcal{O}(f_i)} d\mu$, and then apply the product rule to $z_i \langle Y_i \rangle_{w_z}$. Apply this identity to the right side of (2110), whose normalized mean vanishes. The constant insertion is $\beta(\mathfrak{S}_{\omega,\chi} + \mathfrak{K}_{\omega,\chi})[1] - \mathfrak{B}_{\omega,\chi}$ and its ith linear coefficient is $\mathcal{O}(\nabla^{\mathrm{s}}_\omega f_i) - a^{-1}(\mathfrak{S}_\omega + \mathfrak{K}_\omega)[f_i]$. Rearrangement gives (2106). □

112.2. **The continuum rotation defect.**

Lemma 112.2 (Discretization error)**.** Let $0 < a = a_K \le 1$, $\omega \in \mathfrak{so}(3)$ and $\operatorname{supp} f \subset \{|y| < \mathcal{R}/2\}$, with $f \in C^3_c(\mathbb{R}^3)$ real. Write $|\omega| = \max_{\rho,\lambda} |\omega_{\rho\lambda}|$ and $\|f\|_{C^3} = \max_{|\alpha| \le 3} \|\partial^\alpha f\|_\infty$. Then

$$D_f = \nabla^{\mathrm{s}}_\omega f - (\omega y) \cdot \nabla f$$

has support in the closed a-neighborhood of $\operatorname{supp} f$, and

$$\begin{aligned} \|D_f\|_\infty &\le |\omega|(1+\mathcal{R}) a^2 \|f\|_{C^3}, \\ \operatorname{Lip} D_f &\le 4\sqrt{3}|\omega|(1+\mathcal{R}) a \|f\|_{C^3}, \\ \|D_f\|_\sharp &\le 160\sqrt{3} M |\omega|(1+\mathcal{R}) a \|f\|_{C^3}. \end{aligned} \tag{2111}$$

If $\mathcal{R}/2 + 2a < \ell/2$, the same statements hold after periodization to T_ℓ, with unchanged norms; the periodized D_f is the difference of the corresponding torus expressions. ■

Proof. Put $r_\rho(y) = (2a)^{-1}[f(y+ae_\rho)-f(y-ae_\rho)]-\partial_\rho f(y)$. Taylor's formula with integral remainder gives

$$r_\rho(y) = \frac{a^2}{4}\int_0^1 (1-s)^2[\partial_\rho^3 f(y+sae_\rho)+\partial_\rho^3 f(y-sae_\rho)]\,ds.$$

Thus $|r_\rho| \le a^2\|f\|_{C^3}/6$. Applying the first-order Taylor formula to $\partial_\sigma f$ gives, without using a fourth derivative,

$$\partial_\sigma r_\rho(y) = \frac{a}{2}\int_0^1 (1-s)[\partial_\rho^2\partial_\sigma f(y+sae_\rho)-\partial_\rho^2\partial_\sigma f(y-sae_\rho)]\,ds, \qquad |\partial_\sigma r_\rho| \le \tfrac{a}{2}\|f\|_{C^3}.$$

Outside the stated neighborhood every term in $D_f = \sum_\rho (\omega y)_\rho r_\rho$ vanishes. Inside it, $|y| \le \mathcal{R}/2 + a$, so $|(\omega y)_\rho| \le 2|\omega|(\mathcal{R}/2+a)$. The product rule therefore gives

$$|D_f| \le |\omega|(\mathcal{R}/2+a)a^2\|f\|_{C^3}, \qquad |\partial_\sigma D_f| \le |\omega|a\|f\|_{C^3}(a/3+3\mathcal{R}/2+3a).$$

Since $a \le 1$, these imply the first two bounds. The third follows from

$$\|h\|_\sharp = \max(\|h\|_\infty, 40M\,\mathrm{Lip}\,h).$$

For the torus statement, the support of D_f has margin a and that of f has margin $2a$. The periodic lift is locally a single translate, or zero, because these translates of the closed supports are disjoint and locally finite. Its supremum and gradient supremum are unchanged. The Lipschitz constant is the gradient supremum both on $\mathbb{R}^3$ and on the flat torus. Finally, shifts by $\pm ae_\rho$ cannot introduce a wraparound term from f, which has margin $2a$. This proves the identity for the periodized difference. □

Consider a sequence $K_j \to \infty$, $\ell_j \to \infty$, $\beta_j > 0$, with $K_j, m_j \ge 1$ and $\ell_j = 2L^{m_j}$. For $n \ge 2$ and separated tuples $X = (f_1, \dots, f_n)$ of real C_c^1 functions in the central cube $(-\ell_j/4, \ell_j/4)^3$, put

$$\Phi_n^{(j)}(X) = S^T_{n;K_j,\ell_j,\beta_j}(X^{\langle \ell_j\rangle}).$$

Assume that these forms converge pointwise to a family $(S_n)_{n\ge 2}$ on every separated tuple and obey a bound table: for every $n \ge 2$ and $R > 0$ there is $j_0(n,R)$ such that, for all $j \ge j_0(n,R)$ and all $d > 0$,

$$|\Phi_n^{(j)}(X)| \le C_n(d;R)\prod_i \|f_i\|_\sharp \tag{2112}$$

whenever the supports have mutual distance at least d and each lies in a ball of radius R. In particular, S obeys the same table. By proposition 92.3, S is translation and W_3 invariant.

Fix one smooth function $\chi_1 : [0,\infty) \to [0,1]$ equal to one on $[0,1]$ and zero on $[2,\infty)$, and put $\chi_{\mathcal{R}}(y) = \chi_1(|y|/\mathcal{R})$. For a separated smooth tuple $\underline{f}$ supported in $\{|y| < \mathcal{R}/2\}$, define

$$\begin{aligned} \mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(\underline{f}) &= \mathfrak{U}_{\omega,\chi_{\mathcal{R}}}(\underline{f}^{\langle \ell_j\rangle}), \\ \mathsf{FI}^{(j)}_{\omega,\mathcal{R}}(\underline{f}) &= \langle \mathfrak{B}_{\omega,\chi_{\mathcal{R}}}; \mathcal{O}(f_1^{\langle \ell_j\rangle}); \dots; \mathcal{O}(f_n^{\langle \ell_j\rangle})\rangle_{K_j,\ell_j,\beta_j}. \end{aligned} \tag{2113}$$

They are defined for all sufficiently large j. Indeed, writing $d_1 = \min(1, d(\underline{f}))$, we may require

$$2\mathcal{R} + 2a_{K_j} < \ell_j/2, \qquad 2\sqrt{3}a_{K_j} \le \mathcal{R}/2, \qquad a_{K_j} \le d_1/4. \tag{2114}$$

The first condition gives the cutoff margin and the second ensures that $\chi_{\mathcal{R}}$ covers the sources. The third keeps the supports of the difference insertions separated.

Proposition 112.3 (Limit of the bulk and shell sum)**.** Under these assumptions, $t \mapsto S_n(e^{t\omega}\underline{f})$ is differentiable, where $(e^{t\omega}f)(y) = f(e^{-t\omega}y)$, and

$$\dot S_n(\omega;\underline{f}) := \frac{d}{dt}S_n(e^{t\omega}\underline{f})\big|_{t=0} = -\sum_i S_n(f_1, \dots, (\omega y)\cdot\nabla f_i, \dots, f_n). \tag{2115}$$

For every such tuple and radius,

$$\lim_{j\to\infty}[\mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(\underline{f})+\mathsf{Fl}^{(j)}_{\omega,\mathcal{R}}(\underline{f})]=-\dot{S}_n(\omega;\underline{f}). \tag{2116}$$

In particular, this limit is independent of $\mathcal{R}$. ■

Proof. For each i, take the closed $d_1/4$-neighborhood A_i of $\operatorname{supp} f_i$. The A_i have mutual distance at least $d_*=d_1/2$ and each lies in a ball of radius $R_*=R(\underline{f})+1/4$. For large j they lie in the central cube. Every f_i, $(\omega y)\cdot\nabla f_i$, D_{f_i}, and $\nabla^{\mathrm{s}}_\omega f_i$ is supported in A_i by lemma 112.2. Applying (2112) gives

$$|\Phi^{(j)}_n(f_1,\dots,D_{f_i},\dots,f_n)|\le C_n(d_*;R_*)160\sqrt{3}M|\omega|(1+\mathcal{R})a_{K_j}\|f_i\|_{C^3}\prod_{l\neq i}\|f_l\|_\sharp.$$

This tends to zero. Pointwise convergence on the fixed tuple with entry $(\omega y)\cdot\nabla f_i$ now proves

$$\sum_i\Phi^{(j)}_n(f_1,\dots,\nabla^{\mathrm{s}}_\omega f_i,\dots,f_n)\longrightarrow\sum_i S_n(f_1,\dots,(\omega y)\cdot\nabla f_i,\dots,f_n).$$

The finite identity (2106) proves the first equality claimed in (2116).

To identify it with an orbit derivative, let $g_i(t,y)=f_i(e^{-t\omega}y)$. On any compact interval of t, the derivatives $\partial_t^k\partial_y^\alpha g_i$ for $k\le 2$, $|\alpha|\le 1$ are bounded and supported in one fixed ball. Taylor's formula thus shows that

$$\frac{g_i(t+h)-g_i(t)}{h}\longrightarrow-(\omega y)\cdot\nabla g_i(t)\quad\text{in }\|\cdot\|_\sharp.$$

For h sufficiently small all these supports lie in fixed, pairwise disjoint neighborhoods of the supports at time t: rotation moves a point of radius r by at most $3|\omega|r|h|$. Expand the difference of S_n by a multilinear telescope and use its bound table on these neighborhoods. Each quotient converges to the corresponding derivative entry, and the remaining entries converge in the sharp norm. This proves the product rule (2115) at every t, with the tuple rotated by $e^{t\omega}$. It differentiates S_n itself; no derivative is interchanged with the cutoff limit. The expression does not involve $\mathcal{R}$. □

Lemma 112.4 (One rotation group and cubic symmetry)**.** A subgroup of $\mathrm{O}(3)$ containing W_3 and $\{e^{t\omega_0}:t\in\mathbb{R}\}$ for some nonzero $\omega_0\in\mathfrak{so}(3)$ is $\mathrm{O}(3)$. ■

Proof. The given one-parameter group consists of all rotations about some unit axis n_0. A signed coordinate permutation takes n_0 to an axis $n_1\neq\pm n_0$: change one nonzero coordinate if there are two, and otherwise permute the nonzero coordinate. Conjugation therefore gives every rotation about n_1 as well. Let H be the subgroup generated by the two axis groups, $O=Hn_0\subset S^2$, and $A=\{\angle(n_0,x):x\in O\}$. Since H contains all rotations about n_0, a point x belongs to O precisely when $\angle(n_0,x)\in A$. By conjugating this assertion, for each $x\in O$ a point y belongs to O precisely when $\angle(x,y)\in A$.

Rotating n_0 about n_1 from angle zero to π traces a continuous path whose final point differs from n_0. Thus $[0,\vartheta_0]\subset A$ for some $\vartheta_0>0$. If $\vartheta,\vartheta'\in A$ and $\vartheta+\vartheta'\le\pi$, choose points on one great circle at angular distances ϑ and $\vartheta+\vartheta'$ from n_0. The preceding characterization first places the former in O and then the latter, so $\vartheta+\vartheta'\in A$. Every angle in $[0,\pi]$ is a finite sum of angles in $[0,\vartheta_0]$; hence $O=S^2$. Given $R\in\mathrm{SO}(3)$, choose $h\in H$ with $hn_0=Rn_0$. Then $h^{-1}R$ fixes n_0 and is a rotation about that axis, so $R\in H$. Finally $-1\in W_3$, giving all of $\mathrm{O}(3)$. □

For a fixed ω, impose the following two conditions on the defining sequence, with the one function χ_1 fixed above:

$$(\mathrm{UV})_\omega:\quad\mathfrak{U}^{(j)}_{\omega,\mathcal{R}}(\underline{f})\longrightarrow 0\quad\text{for every separated smooth tuple and every admitting radius,}\tag{2117}$$

$$(\text{IR-fl})_\omega:\quad\liminf_{\mathcal{R}\to\infty}\limsup_{j\to\infty}|\mathsf{Fl}^{(j)}_{\omega,\mathcal{R}}(\underline{f})|=0\quad\text{for every separated smooth tuple.}\tag{2118}$$

An admitting radius means $\operatorname{supp} f_i\subset\{|y|<\mathcal{R}/2\}$ for every i. These conditions concern the sequence itself; neither the limiting family nor the bound table occurs in their definitions.

Theorem AQ (Rotation criterion)**.** For a convergent sequence obeying (2112), the following assertions hold.

(1) If $(\mathrm{UV})_\omega$ holds, then
$$\lim_j \mathsf{Fl}^{(j)}_{\omega,\mathcal{R}}(\underline{f}) = -\dot{S}_n(\omega;\underline{f}). \tag{2119}$$
In this case $(\mathrm{IR\text{-}fl})_\omega$ holds if and only if all these orbit derivatives vanish.
(2) If both conditions hold for one nonzero ω_0, then S is invariant under $\mathrm{O}(3)$ on separated C^1_c tuples.
(3) Without either condition, S is $\mathrm{O}(3)$ invariant if and only if the combined limit in (2116) vanishes for every tuple and admitting radius for one nonzero ω_0. Equivalently, it vanishes for every ω.
(4) If S is $\mathrm{O}(3)$ invariant and $(\mathrm{UV})_\omega$ holds, then the shell term tends to zero at each fixed admitting radius. In particular $(\mathrm{IR\text{-}fl})_\omega$ holds with ordinary limits in place of both upper and lower limits.

■□

Proof. Under $(\mathrm{UV})_\omega$, subtract the vanishing bulk term from (2116). The limit in (2119) is independent of $\mathcal{R}$, so the double limit condition is equivalent to $|\dot{S}_n(\omega;\underline{f})| = 0$. If all derivatives vanish for a fixed ω_0, apply (2115) at every rotated tuple. The differentiable function $t \mapsto S_n(e^{t\omega_0}\underline{f})$ is constant. This proves invariance on smooth tuples. For a separated C^1_c tuple, convolution with a smooth approximate identity converges in the sharp norm and has supports in fixed, slightly enlarged, pairwise disjoint neighborhoods. The bound table and a multilinear telescope pass the invariance to that tuple. The invariance group of a family is a subgroup; it already contains W_3 by proposition 92.3. Lemma 112.4 gives $\mathrm{O}(3)$. Conversely, invariance makes every orbit derivative zero. Combining these observations with (2116) proves the third assertion, and with (2119) proves the fourth. □

The bound table and convergence used here hold for the constructed growing-torus limits. They establish (2116); they do not establish either (2117) or (2118) separately. For a volume-first limit, rotation invariance proved by the independent volume-first argument makes the combined limit zero along every torus sequence defining that limit.

113. Classical comparison between consecutive cutoffs

The comparison uses two maps between the lattices. Block averaging gives the lower action bound. For the reverse bound we refine a coarse configuration while preserving its straight bond products. The refinement must control the action to first order in the largest plaquette angle; the estimate below provides this control without a gauge choice.

For $W \in G$, put
$$\varphi(W) = \arccos\big(\tfrac{1}{2}\operatorname{tr} W\big), \qquad a(W) = 1 - \tfrac{1}{2}\operatorname{tr} W = \tfrac{1}{2}|W - \mathbf{1}|^2.$$
Thus $|W - \mathbf{1}| = 2\sin(\varphi(W)/2)$, and
$$\tfrac{1}{2}\varphi(W)^2\big(1 - \varphi(W)^2/12\big) \le a(W) \le \tfrac{1}{2}\varphi(W)^2. \tag{2120}$$
The inequalities follow from $u^2/2 - u^4/24 \le 1 - \cos u \le u^2/2$ for $0 \le u \le \pi$. The lower bound follows by integrating $\sin u \ge u - u^3/6$, itself obtained by integrating $\cos u \ge 1 - u^2/2$.

We will also use the following logarithmic estimate. If $X_1, \dots, X_r \in \mathfrak{g}$ and $s = \sum_j |X_j| \le 1/4$, then $W = \prod_j e^{iX_j} \ne -\mathbf{1}$ and
$$Y = \tfrac{1}{i}\log W \in \mathfrak{g}, \qquad |Y| \le s, \qquad \Big|Y - \sum_j X_j\Big| \le s^2 e^s. \tag{2121}$$

Here and below the logarithm is intrinsic. To verify the first bound, note that $\varphi(W_1W_2) \le \varphi(W_1)+\varphi(W_2)$. When the sum on the right is less than π, the identity

$$W_1W_2-\mathbf{1}=\tfrac12(W_1-\mathbf{1})(W_2+\mathbf{1})+\tfrac12(W_1+\mathbf{1})(W_2-\mathbf{1})$$

and $|W+\mathbf{1}|=2\cos(\varphi(W)/2)$ bound the left side by $2\sin((\varphi(W_1)+\varphi(W_2))/2)$. Monotonicity of sine on $[0,\pi/2]$ proves the assertion; when the sum is at least π, the assertion is immediate. Since $\varphi(e^{iX})\le|X|$ and $|Y|=\varphi(W)$, iteration gives $|Y|\le s$. Expanding the product of exponentials in its absolutely convergent series bounds the terms of total degree at least two by $e^s-1-s\le s^2e^s/2$. Apply the same bound to e^{iY} and subtract the two first-order expansions to obtain (2121).

Let $\mathbb{Z}_{\underline{n}}=\prod_{\mu=1}^{3}\mathbb{Z}/n_\mu\mathbb{Z}$, where $n_\mu\ge2$. A configuration is a family $U_\mu(x)\in G$. Write

$$P_{\alpha\beta}(x)=U_\alpha(x)U_\beta(x+e_\alpha)U_\alpha(x+e_\beta)^{-1}U_\beta(x)^{-1},\qquad E_{\alpha\beta}(U)=\sum_x a(P_{\alpha\beta}(x)),$$

and set $E(U)=E_{12}(U)+E_{13}(U)+E_{23}(U)$. The reversal of the two directions inverts $P_{\alpha\beta}$ and does not change its angle or action. Let $\vartheta(U)$ be the largest plaquette angle. For a site x, let $c(x)$ be the largest angle of the six faces of its coordinate cube, namely the plaquettes $P_{\alpha\beta}(x)$ and $P_{\alpha\beta}(x+e_\gamma)$ with $\alpha<\beta$ and $\gamma\notin\{\alpha,\beta\}$.

Fix an integer $L\ge2$ and a direction λ. Enlarge the λ side by L, and let ι_λ multiply that coordinate by L. Every site of the enlarged box has a unique expression $x^{(j)}=\iota_\lambda x+je_\lambda$, $0\le j<L$. If $\vartheta(U)<\pi$, define $\Theta_{\lambda\kappa}(x)=\frac1i\log P_{\lambda\kappa}(x)$ and set

$$\begin{aligned}&U'_\lambda(x^{(j)})=\mathbf{1} && (j<L-1),\quad U'_\lambda(x^{(L-1)})=U_\lambda(x),\\ &U'_\kappa(x^{(j)})=e^{i(j/L)\Theta_{\lambda\kappa}(x)}U_\kappa(x) && (\kappa\ne\lambda).\end{aligned}\tag{2122}$$

We denote this refinement by $U'=\mathsf{Rfn}_\lambda U$.

Lemma 113.1 (Refinement in one direction)**.** Suppose $\vartheta_0=\vartheta(U)\le10^{-4}$, and let μ,ν be the directions other than λ. The refinement preserves the straight coarse bond products and the transverse bonds at embedded sites:

$$\prod_{j=0}^{L-1}U'_\lambda(x^{(j)})=U_\lambda(x),\qquad U'_\kappa(\iota_\lambda x)=U_\kappa(x)\quad(\kappa\ne\lambda).$$

Its plaquettes containing the refined direction satisfy $P'_{\lambda\kappa}(x^{(j)})=e^{i\Theta_{\lambda\kappa}(x)/L}$. For the transverse plaquettes, put $Y_0=\frac1i\log P_{\mu\nu}(x)$ and $Y_1=\mathrm{Ad}(U_\lambda(x))\frac1i\log P_{\mu\nu}(x+e_\lambda)$. Then

$$\varphi(P'_{\mu\nu}(x^{(j)}))\le|(1-j/L)Y_0+(j/L)Y_1|+54c(x)^2.\tag{2123}$$

Consequently,

$$\begin{aligned}E_{\lambda\kappa}(U')&\le L^{-1}(1+\vartheta_0^2)E_{\lambda\kappa}(U),\\ E_{\mu\nu}(U')&\le L(1+\vartheta_0^2)E_{\mu\nu}(U)+221L\vartheta_0E(U),\\ \vartheta(U')&\le\vartheta_0+54\vartheta_0^2\le2\vartheta_0.\end{aligned}\tag{2124}$$

■

Proof. The bond identities follow directly from (2122). For $j<L-1$, the two λ bonds of a plaquette equal the identity, so its two transverse factors give $e^{i\Theta_{\lambda\kappa}(x)/L}$. At $j=L-1$, the same product is $P_{\lambda\kappa}(x)e^{-i(L-1)\Theta_{\lambda\kappa}(x)/L}$, which has the same value.

For a transverse plaquette write

$$\begin{gathered}A_1=\Theta_{\lambda\mu}(x),\quad A_2=\Theta_{\lambda\nu}(x+e_\mu),\quad A_3=\Theta_{\lambda\mu}(x+e_\nu),\quad A_4=\Theta_{\lambda\nu}(x),\\ u_\mu=U_\mu(x),\quad u_\nu=U_\nu(x),\quad u'_\nu=U_\nu(x+e_\mu),\quad u'_\mu=U_\mu(x+e_\nu).\end{gathered}$$

For $0\le t\le1$ consider

$$W(t)=e^{itA_1}u_\mu e^{itA_2}u'_\nu u'^{-1}_\mu e^{-itA_3}u_\nu^{-1}e^{-itA_4}.$$

Then $W(j/L) = P'_{\mu\nu}(x^{(j)})$. Moving the unitary factors past the exponentials gives

$$W(t) = e^{itA_1} e^{it\,\mathrm{Ad}(u_\mu)A_2} e^{iY_0} e^{-it\,\mathrm{Ad}(u_\nu)A_3} e^{-itA_4}.$$

The five exponents and Y_1 have norms at most $c = c(x)$. By (2121),

$$\tfrac{1}{i}\log W(t) = Y_0 + t(A_1 + \mathrm{Ad}(u_\mu)A_2 - \mathrm{Ad}(u_\nu)A_3 - A_4) + e(t), \qquad |e(t)| \le 25e^{5c}c^2 \le 27c^2.$$

At $t = 1$, substitution of the four defining plaquette products cancels the three successive pairs of λ bonds and gives $W(1) = U_\lambda(x)P_{\mu\nu}(x + e_\lambda)U_\lambda(x)^{-1}$. Thus $\frac{1}{i}\log W(1) = Y_1$. Subtracting this identity from the preceding expression gives $\frac{1}{i}\log W(t) = (1-t)Y_0 + tY_1 + e(t) - te(1)$. This proves (2123) and the angle bound in (2124).

For an old plaquette angle $\varphi \le \vartheta_0$, (2120) implies

$$\tfrac{1}{2}\varphi^2 \le (1 + \vartheta_0^2)a. \tag{2125}$$

Indeed, $(1+y)(1-y/12) \ge 1$ for $0 \le y \le 1$. Each plaquette containing λ becomes L plaquettes of angle φ/L, which proves its energy estimate. For a transverse plaquette put $p = |(1-t)Y_0 + tY_1|$. Since $p \le c \le 10^{-4}$,

$$a(W(t)) \le \tfrac{1}{2}(p + 54c^2)^2 \le \tfrac{1}{2}((1-t)|Y_0|^2 + t|Y_1|^2) + 55c^3.$$

The last bound uses $54c^2p + 1458c^4 \le 55c^3$. Summing over $j = 0, \dots, L-1$ and then x, the quadratic term is $\frac{L}{2}\sum_x |Y_0(x)|^2$: the sums of $|Y_0|^2$ and $|Y_1|^2$ coincide by translation and unitary invariance. It is at most $L(1+\vartheta_0^2)E_{\mu\nu}(U)$ by (2125). Each plaquette belongs to two of the coordinate cubes, and hence

$$\sum_x c(x)^3 \le \vartheta_0 \sum_x \sum_{\text{faces of } x} \varphi(P)^2 \le 4(1+\vartheta_0^2)\vartheta_0 E(U).$$

The remaining contribution is at most $220(1+10^{-8})L\vartheta_0 E(U) \le 221 L\vartheta_0 E(U)$. □

Proposition 113.2 (Prolongation)**.** Let $\vartheta_0 = \vartheta(U) \le \frac{1}{4}\,10^{-4}$ and define

$$\widetilde{U} = \mathsf{Prl}_L U = \mathsf{Rfn}_3\mathsf{Rfn}_2\mathsf{Rfn}_1 U$$

on $\mathbb{Z}_{L\underline{n}}$. Then

$$\prod_{j=0}^{L-1} \widetilde{U}_\mu(Lx + je_\mu) = U_\mu(x), \qquad LE(\widetilde{U}) \le (1 + 4000L^4\vartheta_0)E(U), \qquad \vartheta(\widetilde{U}) \le 8\vartheta_0. \tag{2126}$$

The bond product is ordered by increasing j. ■

Proof. After i refinements the largest angle is at most $2^i\vartheta_0$. Thus the input angle of each of the three steps is at most 10^{-4}, and lemma 113.1 applies. At the step refining direction μ, the straight product becomes the old μ bond. At the other two steps, the transverse bonds at embedded sites are preserved. Applying these identities in their order proves the first assertion.

Write $E^{(i)}_{\alpha\beta}$ and $E^{(i)}$ for the energies after i steps, and put $\epsilon = 16\vartheta_0^2 \le 10^{-8}$. The energy inequalities in (2124) give, at step i,

$$E^{(i)}_{i\kappa} \le \frac{1+\epsilon}{L} E^{(i-1)}_{i\kappa}, \qquad E^{(i)}_{\mu\nu} \le L(1+\epsilon)E^{(i-1)}_{\mu\nu} + 221 \cdot 2^{i-1} L\vartheta_0 E^{(i-1)},$$

where $\mu, \nu \ne i$. In particular $E^{(i)} \le 2LE^{(i-1)}$. For the pairs 12, 13, and 23, the step with the error term is, respectively, the third, second, and first. The displayed recurrence therefore yields

$$\begin{aligned}
E^{(3)}_{12} &\le (1+\epsilon)^3 L^{-1} E^{(0)}_{12} + 3536 L^3 \vartheta_0 E^{(0)},\\
E^{(3)}_{13} &\le (1+\epsilon)^3 L^{-1} E^{(0)}_{13} + 884(1+\epsilon) L\vartheta_0 E^{(0)},\\
E^{(3)}_{23} &\le (1+\epsilon)^3 L^{-1} E^{(0)}_{23} + 221(1+\epsilon)^2 L^{-1}\vartheta_0 E^{(0)}.
\end{aligned}$$

After summing and multiplying by L, use $(1+\epsilon)^3-1\le 4\epsilon\le \vartheta_0$, $884(1+\epsilon)\le 885$, $221(1+\epsilon)^2\le 222$, and $885L^2+223\le 464L^4$. These give the coefficient $4000L^4$ in (2126). □

113.1. **Weighted averaging.** For the comparison of smeared actions we need averaging on arbitrary sets of plaquettes. Suppose that a fine lattice of spacing h/L is averaged to a lattice of spacing h. For a coarse plaquette p', let $\mathcal{N}(p')$ be the fine plaquettes all of whose corners lie in the union of the four fine blocks centered at the corners of p'; put $\mathcal{N}(S)=\bigcup_{p'\in S}\mathcal{N}(p')$. The one-step bound (147) is

$$\sum_{p'\in S} a_{p'}(\mathcal{M}W)\le L(1+C_{\mathrm{en}}\omega)^2\sum_{p\in\mathcal{N}(S)} a_p(W),\qquad \omega=\max_{p\in\mathcal{N}(S)}|\partial W(p)-\mathbf{1}|\le \varpi'_L. \tag{2127}$$

This holds for every set S, not only all plaquettes. For clarity, the reason is that the positive linear curvature kernel of Propositions 14.8–14.9 has counting ℓ^2 norm at most $\sqrt{L}$ after restriction to $\mathcal{N}(S)$, while its nonlinear remainder at p' is at most $C_0^{\mathrm{SO}}\omega_{p'}^2$. A fine plaquette belongs to at most twelve coarse cones: its base lies in one coarse block, whose center must be a corner of each of the coarse plaquettes. Consequently

$$\sum_{p'\in S}\omega_{p'}^4\le 12\omega^2\sum_{p\in\mathcal{N}(S)}|\partial W(p)-\mathbf{1}|^2.$$

The triangle inequality gives the curvature bound with factor $\sqrt{L}(1+C_{\mathrm{en}}\omega)$, where $C_{\mathrm{en}}=\sqrt{12}\,C_0^{\mathrm{SO}}L^{-1/2}$; squaring and using $2a_p=|\partial W(p)-\mathbf{1}|^2$ proves (2127).

For a plaquette p, let x_p denote its barycenter. If $p\in\mathcal{N}(p')$, choose a corner z of p in the fine block of a corner y of p'. In the torus distance,

$$d_\infty(x_p,x_{p'})\le d_\infty(x_p,z)+d_\infty(z,y)+d_\infty(y,x_{p'})<\frac{h}{2L}+h\le\frac{3}{2}h. \tag{2128}$$

This estimate uses only the defining union of blocks of the cone.

Lemma 113.3 (Weighted one-step action)**.** Let W have curvature at most $\omega\le\varpi'_L$ on the fine lattice. Write η and η/L for the two dimensionless spacings, and set

$$A^\eta(c,V)=\eta^{-1}\sum_p c(x_p)a_p(V),\qquad q=(1+C_{\mathrm{en}}\omega)^2.$$

For every nonnegative real Lipschitz function c on the torus,

$$A^\eta(c,\mathcal{M}W)\le q\left[A^{\eta/L}(c,W)+\tfrac{3}{2}h\,\mathrm{Lip}_\infty(c)A^{\eta/L}(1,W)\right]. \tag{2129}$$

■

Proof. Put $v=\frac{3}{2}h\,\mathrm{Lip}_\infty(c)$ and $S_\lambda=\{p':c(x_{p'})>\lambda\}$. By (2128), $\mathcal{N}(S_\lambda)\subset\{p:c(x_p)+v>\lambda\}$. Apply (2127) to S_λ and integrate over $\lambda\ge0$. The elementary identity $c(x)=\int_0^\infty\mathbf{1}_{\{c(x)>\lambda\}}\,d\lambda$ gives

$$\sum_{p'}c(x_{p'})a_{p'}(\mathcal{M}W)\le Lq\sum_p(c(x_p)+v)a_p(W).$$

Multiplication by η^{-1} proves the assertion. □

113.2. **The action on a fibre and the excess of a competitor.** We specify the uniform global hypotheses used in this comparison. Fix the geometric constants, the actual shift systems, and the octet data of the sourced construction. We impose the hypotheses on these data in Theorem W and Proposition 52.3, including the global real-background hypotheses and $\alpha_0\le a_{\mathrm{H}}$. They are required at the radii (α_0,α_1), for every calibration $\tau\ge\gamma^{-2}$. The localization floors for the real response equations (996)–(997) are also imposed. In particular the covariance comparison has error at most $\gamma_{\mathrm{r}}/2$, the two interaction potentials have decay $\lambda_V>\log 9216+2$, and $\mathsf{C}_{\mathcal{A}}\mathfrak{b}(\tau)\ge3$, where

$$\varepsilon(\tau)=A_0\tau^{-1/2}(\log\tau)^{\mathfrak{p}_0},\qquad \delta(\tau)=A_1\tau^{-1/2}(\log\tau)^{\mathfrak{p}_0},\qquad \mathfrak{b}(\tau)=\sqrt{\tau}\,\delta(\tau).$$

We use the range on which ε, δ are continuous, decreasing, and tend to zero. Assume

$$K \ge 1, \qquad \ell = 2L^m, \qquad 2L^{m+1} \ge C_L M. \tag{2130}$$

Then each step below is in the global setting of the cited results. Indeed the side in level-n units is $2L^{m+K-n} \ge 8M$; the constant sequence of depth $n \le K$ is separated and its level-n lattice has at least $8R_1\mathfrak{M}$ sites in each direction. These are the geometric hypotheses of the background and response results. The actual whole-data domain in Proposition 52.3 includes the open response neighborhood and identifies the actual shifts with the constructed critical backgrounds. Thus its global sumrules, rather than an abstract provision alone, apply here.

Write $\varpi(V) = \max_p |\partial V(p) - \mathbf{1}|$. For the constant sequence of depth n, denote its real critical background by $U_n(V)$, with its residual gauge freedom understood, and put

$$\mathsf{ab}_n(c, V) = A^{\eta_n}(c, U_n(V)), \qquad \mathsf{ab}_n(V) = \mathsf{ab}_n(1, V), \qquad \eta_n = L^{-n}.$$

These actions are independent of the representative. They are also invariant under a gauge of the level-n datum: extend that gauge arbitrarily to the bare sites, use covariance of the constraint and critical orbit, and then gauge invariance of each plaquette action. At level zero $\mathsf{ab}_0(c, V) = \sum_p c(x_p) a_p(V)$ whenever the identity background is regular. The fields $V_k^{\mathrm{bf}}(W)$ and $V_W[\mathcal{A}]$ are the axial averaged background and its fibre chart from Propositions 52.2–52.3; the variable $\mathcal{A}$ is prescribed on the nonpivot bonds.

Let $C_{\mathrm{db}} = 8L^3$, let $\varepsilon_{\mathcal{W}}$ be the uniform radius of Proposition 52.3, and set

$$\begin{gathered} A_L = \tfrac{3}{4}(L-1)^2, \qquad C_{\mathrm{AC}} = 8(L-1)A_L + (1+\pi/3)\big(L(L-1) + 8A_L\big), \\ C_{\mathrm{ax}} = \max\{8A_L, C_{\mathrm{AC}}\}, \qquad C_{\mathrm{pr}} = 8A_L(L-1) + C_{\mathrm{AC}}, \qquad C'_{\mathrm{pr}} = 1 + 16\pi C_{\mathrm{pr}}, \\ C_{\mathrm{cp}} = B_3 \max\{2C'_{\mathrm{pr}}, 4\pi L^2\}, \qquad C_{\mathrm{cub}} = 50 K_P C_{\mathrm{E},1}/\mathsf{C}_{\mathcal{A}}. \end{gathered}$$

Here K_P is the actual potential bound, and $C_{\mathrm{E},1} = 2048 \sum_{j\ge 0}(j+2)e^{-j}$ is the rooted domain-sum bound of Proposition 45.1. We use the response constants δ_{lc} and $B_5 \ge 8$ of equations (996)–(997). Let S_{e} bound both row and column sums of its kernel $e^{-\delta_{\mathrm{lc}} d(y, c_-)}$, and put

$$\begin{gathered} \rho_{\mathrm{no}} = \min\{r_{\mathrm{B}}, \varrho_{\mathrm{R}}^0\}, \qquad C_{\mathrm{nrg}} = 108 B_5^2 S_{\mathrm{e}}^2 (1 + 24 C_{\mathrm{db}}^2), \\ \varepsilon_{\mathrm{cp}} = \min\left\{ \varepsilon_{\mathcal{W}} \alpha_0, \frac{a_{\mathrm{R}}}{C_{\mathfrak{P}} \mathfrak{M} B_3}, B_3^{-1} \right\}. \end{gathered} \tag{2131}$$

All these constants are independent of the cutoff, level, and volume.

Put $K_{\mathrm{B}}^{\mathrm{s}} = K_{\mathrm{B}}^{\mathrm{s}}(\rho_{\mathrm{s}})$, $C_H = 4C_{\mathrm{db}} \max\{2, K_{\mathrm{B}}^{\mathrm{s}}\}$, and

$$c_\rho = \max\left\{ 2B_3, \frac{C_H A_1}{(1 - L^{-3/2}) A_0} \right\}.$$

Let $\tau_\natural \ge \gamma^{-2}$ be the least calibration at which

$$2c_\rho \varepsilon(\tau_\natural) \le \min\{\varepsilon_{\mathcal{W}} \alpha_0, c_2^{\mathrm{SO}}, \varepsilon_{\mathrm{v}}/4\}, \qquad \delta(\tau_\natural) \le \varepsilon_{\mathcal{W}}, \quad 4C_{\mathrm{db}} \delta(\tau_\natural) \le \min\{1, r_{\mathrm{B}}, r_{\mathrm{B}}^{\mathrm{s}}/2\}. \tag{2132}$$

This minimum exists by continuity, monotonicity, and decay of the thresholds. The same conditions hold at larger calibrations, where also $\varepsilon(\tau) \le 1$ by the global step hypotheses. Here $r_{\mathrm{B}}^{\mathrm{s}} = r_{\mathrm{B}}^{\mathrm{s}}(\rho_{\mathrm{s}})$. Define the competitor radius

$$\begin{aligned} e_{\mathrm{cp}} = \min\Bigg\{ & \frac{\omega_M}{2}, \frac{\varepsilon_{\mathrm{cp}}}{2B_3}, \frac{\gamma_{\mathrm{r}} L^2}{4\pi C_{\mathrm{cub}} C_{\mathrm{ax}}(1 + 2B_3)}, \\ & \frac{L^2 \min\{\rho_{\mathrm{no}}, (66 B_5 S_{\mathrm{e}})^{-1}\}}{2\pi C_{\mathrm{db}} C_{\mathrm{ax}}(1 + 2B_3)}, \frac{L^2 \delta(\tau_\natural)}{\pi C_{\mathrm{ax}}(1 + 2B_3)} \Bigg\}, \qquad \omega_M = (200L^2)^{-1}. \end{aligned} \tag{2133}$$

For $0 < e \le e_{\mathrm{cp}}$, set $h = \frac{\pi}{2} C_{\mathrm{ax}}(1 + 2B_3)L^{-2}e$. The definition gives

$$\begin{gathered} 2B_3 e \le \varepsilon_{\mathrm{cp}}, \qquad C_{\mathrm{cub}} h \le \gamma_{\mathrm{r}}/8, \qquad 2C_{\mathrm{db}} h \le \rho_{\mathrm{no}}/2, \\ B_5 S_{\mathrm{e}}\, 2C_{\mathrm{db}} h \le 1/132, \qquad h \le \delta(\tau_\natural)/2. \end{gathered} \tag{2134}$$

The global radius also satisfies $\varepsilon_{\mathcal{W}} \le \bar{\varepsilon}$ and $4K_2\varepsilon_{\mathcal{W}} \le \varepsilon_{\mathrm{v}}$, where $K_2 = 2B_3L^{-2} + 8C_{\mathrm{db}} \ge 8$. Consequently $4e < \varepsilon_{\mathrm{v}}$.

We first record two consequences of the actual chart. For two axial configurations V^1, V^2 with the same block average and respective curvature bounds $a_1, a_2 < \omega_M$,

$$\varphi(V^1(b)V^2(b)^{-1}) \le \tfrac{\pi}{2} C_{\mathrm{ax}}(a_1 + a_2). \tag{2135}$$

To see this, use the double-block estimates in the proof of Proposition 53.5, and equation (1064). An internal bond differs from the identity by at most $8A_L a_i$. A crossing bond differs from its coarse average by at most $C_{\mathrm{AC}} a_i$. Subtract the two bounds, using equality of the coarse averages in the second case, and use $\varphi(W) \le \frac{\pi}{2}|W - \mathbf{1}|$.

If $\varpi(W) < \varepsilon_{\mathcal{W}}\alpha_0$ and $\sup|\mathcal{A}| < \delta(\tau_\natural)$, write $V_W[\mathcal{A}] = e^{iB'} V_k^{\mathrm{bf}}(W)$. Then

$$B'|_{\mathfrak{X}} = \mathcal{A}, \qquad \sup|B'| \le 2C_{\mathrm{db}} \sup|\mathcal{A}|, \qquad \|B'\|_2^2 \le (1 + 24C_{\mathrm{db}}^2)\|\mathcal{A}\|_2^2. \tag{2136}$$

Here the counting norm on either bond set uses positively oriented bonds. For the last two bounds, fix a coarse bond c. Its pivot value depends only on the nonpivot field in the double block P_c. Replace $\mathcal{A}$ by its restriction to that double block and zero outside. At every calibration $\tau \ge \tau_\natural$ for which the supremum of this restriction is less than $\delta(\tau)$, the chart bound (954) gives pivot size less than $2C_{\mathrm{db}}\delta(\tau)$. The chart is independent of calibration. By continuity of δ and its limit zero, taking the infimum over these calibrations bounds that pivot by $2C_{\mathrm{db}} \max_{b \in \mathfrak{X}^+ \cap \mathsf{P}_c} |\mathcal{A}(b)|$. A nonpivot bond belongs to at most six double blocks, since its initial point belongs to one block and that block is incident to six coarse bonds. Squaring and summing the pivot bound proves (2136). Equality on the nonpivot bonds is the defining property of the fibre chart; the pivot variables solve its constraint.

Proposition 113.4 (Coercivity on the fibre)**.** Let $0 \le k < K$ and $\tau \ge \gamma^{-2}$, and suppose that W belongs to the actual global domain $\mathcal{W}_{k+1}(\tau; \alpha_0, \alpha_1)$ of Proposition 52.3. Put $g = \tau^{-1/2}$ and take a real nonpivot field B with $\sup|B| < \mathfrak{b}(\tau)$. For

$$F_W(B) = \mathsf{ab}_k(V_W[gB]), \qquad G_W(B) = \sum_y \sum_{x \in B(y)\setminus\{y\}} \left[1 - \tfrac{1}{2} \operatorname{Re}\operatorname{tr} V_W[gB](y,x)\right],$$

one has

$$\tau\big(F_W(B) - F_W(0) + G_W(B)\big) \ge (\gamma_{\mathrm{r}}/4 - C_{\mathrm{cub}} g \sup|B|)\|B\|_2^2. \tag{2137}$$

Moreover $F_W(0) = L^{-1}\mathsf{ab}_{k+1}(W)$, $G_W(B) \ge 0$, and $G_W(B) = 0$ when the chart field is axial. ∎

Proof. The global action and gauge sumrules, with their actual background identification, give

$$\tau\big(F_W(B) - F_W(0) + G_W(B)\big) = \tfrac{1}{2}(B, \mathsf{A}_W B)_1 - \sum_{\natural \in \{\mathrm{A},\mathrm{G}\}} \sum_{Y \in \mathcal{D}_k} \mathbb{V}^\natural(Y; P_W|_Y, B|_Y). \tag{2138}$$

Here P_W is the plain triple of the axial top background and $\mathsf{A}_W = \mathbf{C}_*(P_W)^{-1}$; see equations (827), (829)–(830), Theorem W and Proposition 52.3. All the triples on the right belong to their cores by that proposition. The reference covariance satisfies $(B, \mathbf{C}_{*\mathrm{r}}^{-1}B)_1 \ge \gamma_{\mathrm{r}}\|B\|_2^2$, and $\|\mathsf{A}_W - \mathbf{C}_{*\mathrm{r}}^{-1}\| \le \gamma_{\mathrm{r}}/2$. The real part of the quadratic term is therefore at least $\gamma_{\mathrm{r}}\|B\|_2^2/4$.

It remains to use the third-order vanishing of the two potentials, which is stronger than their uniform smallness. For a domain Y, put $d = d_k(Y)$, $b_Y = \max_{b \in \mathfrak{X}^{\circ+}\langle Y\rangle} |B(b)|$, and $r = \mathsf{C}_{\mathcal{A}}\mathfrak{b}(\tau) \ge 3$. There are at most $24M^3(d+1)$ bonds in this domain. If $b_Y = 0$, its potential is zero. Otherwise $z \mapsto \mathbb{V}^\natural(Y; P_W|_Y, zB|_Y)$ is holomorphic for $|z| < r/b_Y$, vanishes to order three, and, by the actual potential bound (887), has modulus at most

$$K_P\delta(\tau)e^{-\lambda_V d}(1 + 24(d+1)r^2) \le 25K_P\delta(\tau)(d+1)r^2 e^{-\lambda_V d}.$$

Apply the maximum principle to the function divided by z^3 on smaller concentric discs and let their radii tend to r/b_Y. This gives

$$|\mathbb{V}^\natural(Y;P_W|_Y,B|_Y)| \le \frac{25K_P}{\mathsf{C}_{\mathcal{A}}} g \sup|B|(d+1)e^{-\lambda_V d} \sum_{b\in\mathfrak{X}^{\circ+}\langle Y\rangle} |B(b)|^2.$$

For a fixed bond, the domains in this sum contain its initial point. Their weighted sum is at most $C_{\mathrm{E},1}$ by Proposition 45.1, since $\lambda_V > \log 9216 + 2$. Summing over both potentials gives $C_{\mathrm{cub}} g \sup|B| \|B\|_2^2$, proving (2137).

The background over $V_k^{\mathrm{bf}}(W)$ is a representative of the background over W, by restriction of criticality and uniqueness as proved in Proposition 52.3. Changing the action spacing from η_k to η_{k+1} contributes the factor L, so $F_W(0) = L^{-1}\mathsf{ab}_{k+1}(W)$. Every term defining G_W is nonnegative and vanishes in axial gauge. □

Call a bare configuration U' a competitor of radius e at level t if $1 \le t \le K$ and

$$\varpi(\mathcal{M}^k U') \le eL^{-2(t-k)} \qquad (0 \le k \le t). \tag{2139}$$

Its excess is $\mathsf{exc}_t(U') = A^{\eta_t}(U') - \mathsf{ab}_t(\mathcal{M}^t U')$.

Proposition 113.5 (Small competitors)**.** If $0 < e \le e_{\mathrm{cp}}$ and U' satisfies (2139), all backgrounds $U_k(V_k')$, $V_k' = \mathcal{M}^k U'$, exist. There are real nonpivot fields $\mathcal{A}_k$, $0 \le k < t$, with

$$\sup|\mathcal{A}_k| \le h_k := \tfrac{\pi}{2} C_{\mathrm{ax}}(1 + 2B_3) eL^{-2(t-k)} \le h, \tag{2140}$$

such that the axial representative of V_k' equals $V_{V_{k+1}'}[\mathcal{A}_k]$. One has

$$\mathsf{ab}_k(V_k') - L^{-1}\mathsf{ab}_{k+1}(V_{k+1}') \ge \tfrac{1}{8}\gamma_{\mathrm{r}} \|\mathcal{A}_k\|_2^2 \tag{2141}$$

and

$$\mathsf{exc}_t(U') \ge \tfrac{1}{8}\gamma_{\mathrm{r}} \sum_{k<t} L^{t-k} \|\mathcal{A}_k\|_2^2 \ge 0. \tag{2142}$$

In particular $A^{\eta_t}(U_k(V_k'))$ is nonincreasing in k. ■

Proof. Since $e \le \varepsilon_{\mathcal{W}}\alpha_0/2 < \bar{\varepsilon}$, the real-background theorem constructs $U_k(V_k')$ for $k \ge 1$. At $k = 0$ the identity background is regular: its curvature is at most e and its current at most $4e < \varepsilon_{\mathrm{v}}$. Fix $k < t$, put $W = V_{k+1}'$, and normalize V_k' to axial gauge by its contour means. This gauge equals the identity on the coarse sites and preserves the average W. Proposition 52.3 puts W in the actual global domain at $\tau_\natural$ and constructs $V_0 = V_k^{\mathrm{bf}}(W)$. Its curvature estimate, applied with any radius strictly above $\varpi(W)$ and then taking the infimum, gives

$$\varpi(V_0) \le 2B_3 L^{-2}\varpi(W) \le 2B_3 eL^{-2(t-k)} \le L^{-2}\varepsilon_{\mathrm{cp}}.$$

The two axial fields have the same average. Apply (2135); their relative intrinsic logarithm is defined and has norm at most $h_k \le \delta(\tau_\natural)/2$. Its restriction to the nonpivot bonds is $\mathcal{A}_k$. The inverse property of the fibre chart identifies the axial configuration with $V_W[\mathcal{A}_k]$.

Apply proposition 113.4 with $B = \sqrt{\tau_\natural}\mathcal{A}_k$. The gauge term vanishes, and $C_{\mathrm{cub}} \sup|\mathcal{A}_k| \le \gamma_{\mathrm{r}}/8$ by (2134). Divide by $\tau_\natural$ and use gauge invariance of the background action to obtain (2141). Finally $A^{\eta_t}(U_k(V_k')) = L^{t-k}\mathsf{ab}_k(V_k')$. Multiply (2141) by L^{t-k} and sum in k; the left sides telescope to the excess, proving (2142). □

Lemma 113.6 (Plaquette differences between successive backgrounds)**.** Under the hypotheses of proposition 113.5, for each $k < t$ one can choose representatives $\widehat{U} \in \mathcal{C}(\mathbf{B}_k, V_k')$ and $\check{U} \in \mathcal{C}(\mathbf{B}_{k+1}, V_{k+1}')$ and elements $u_p, R_p \in G$ such that

$$\partial\widehat{U}(p) = u_p R_p \partial\check{U}(p) u_p^{-1}, \qquad \eta_k^{-1} \sum_p |R_p - \mathbf{1}|^2 \le C_{\mathrm{nrg}} \|\mathcal{A}_k\|_2^2. \tag{2143}$$

■

Proof. Put $V_0 = V_k^{\mathrm{bf}}(V'_{k+1})$ and use (2136) to write the axial version of V'_k as $e^{iB'}V_0$. Then

$$\sup|B'| \le 2C_{\mathrm{db}}h, \qquad \|B'\|_2^2 \le (1+24C_{\mathrm{db}}^2)\|\mathcal{A}_k\|_2^2.$$

For $k \ge 1$, the real response applies at datum radius $\varepsilon_{\mathrm{cp}}$: the base has curvature at most $L^{-2}\varepsilon_{\mathrm{cp}} < \varepsilon_{\mathrm{cp}}$, $C_{\mathfrak{P}}\mathfrak{M}B_3\varepsilon_{\mathrm{cp}} \le a_{\mathrm{R}}$, and $\sup|B'| \le \rho_{\mathrm{no}}/2$. The constant sequence has all pillars of level k and size one in level-k units. Thus equations (996)–(997) give representatives related, up to gauge, by $e^{i\eta_k H}$, where

$$|H(b)| \le B_5 m_{y(b)}, \qquad |\nabla_{\check U} H(x)| \le B_5 m_{y(x)}, \qquad m_y = \sum_{c>0} e^{-\delta_{\mathrm{lc}} d(y,c_-)}|B'(c)|.$$

In particular $\sup|H| \le B_5 S_{\mathrm{e}}\, 2C_{\mathrm{db}}h \le 1/132$. Let ϕ_p be the maximum of $|H|$ on the four bonds of p. The real plaquette formula of Lemma 51.7, with $\eta_k \le 1/13$ and base curvature at most η_k^2, gives

$$|R_p - \mathbf{1}| \le \eta_k^2 |D_{\check U}H(p)| + 11\eta_k^2\phi_p^2 + 6\eta_k\phi_p|\partial\check U(p) - \mathbf{1}| \le 3\eta_k^2 B_5 \max_{x \text{ corner of } p} m_{y(x)}.$$

Here the derivative and curl are covariant with respect to the unperturbed background $\check U$. Indeed $|D_{\check U}H(p)| \le 2B_5 \max m_y$, $\phi_p \le B_5 \max m_y$, $\phi_p \le 1/132$, and $2 + 11/132 + 6/13 < 3$. The product-expansion hypotheses hold since $4\eta_k\phi_p < 1/4$ and $\eta_k^2 < 1/4$. Every bare site is a corner of twelve plaquettes and each pillar contains η_k^{-3} sites. The row and column sum bounds on the response kernel therefore imply

$$\sum_p \max_{x \text{ corner of } p} m_{y(x)}^2 \le 12\eta_k^{-3}\sum_y m_y^2 \le 12\eta_k^{-3}S_{\mathrm{e}}^2\|B'\|_2^2.$$

After multiplying the squared plaquette bound by η_k^{-1}, this is (2143). The unperturbed representative is also in the top background orbit, by the restriction identity used in proposition 113.4. Undoing the level-k axial gauge and the response gauge gives the stated representatives and the conjugations u_p.

At $k = 0$, take $\widehat U = U'$ and $\check U = V_0$. The four-factor telescoping identity gives $|R_p - \mathbf{1}| \le \sum_{b\in\partial p}|B'(b)|$. Each bond occurs in four plaquettes, so $\sum_p |R_p - \mathbf{1}|^2 \le 16\|B'\|_2^2$. Since $C_{\mathrm{nrg}} \ge 16(1+24C_{\mathrm{db}}^2)$, the same estimate holds. □

Proposition 113.7 (Weighted excess)**.** For a competitor as in proposition 113.5 and any bounded complex plaquette weight c,

$$\left|A^{\eta_t}(c,U') - \mathsf{ab}_t(c,\mathcal{M}^tU')\right| \le 4\sup|c|\left[C_{\mathrm{nrg}}\gamma_{\mathrm{r}}^{-1}t\,\mathsf{exc}_t(U')A^{\eta_t}(U')\right]^{1/2}. \tag{2144}$$

■

Proof. For two successive backgrounds use lemma 113.6 and set $s_k = \eta_k^{-1}\sum_p|R_p - \mathbf{1}|^2$. Unitary invariance and the difference of two squares give

$$|a_p(\widehat U) - a_p(\check U)| \le \tfrac12|R_p - \mathbf{1}|(|\partial\widehat U(p) - \mathbf{1}| + |\partial\check U(p) - \mathbf{1}|).$$

By Cauchy–Schwarz and the monotonicity in proposition 113.5, the difference of their weighted actions, both in level-t units, is at most $\sup|c|\,[2L^{t-k}s_kA^{\eta_t}(U')]^{1/2}$. Telescope these weighted actions in k and apply Cauchy–Schwarz once more. Equations (2143) and (2142) give

$$\sum_{k<t} L^{t-k}s_k \le C_{\mathrm{nrg}}\sum_{k<t}L^{t-k}\|\mathcal{A}_k\|_2^2 \le 8C_{\mathrm{nrg}}\gamma_{\mathrm{r}}^{-1}\mathsf{exc}_t(U').$$

The resulting coefficient is $\sqrt{2\cdot 8} = 4$. □

113.3. **Comparison of the two background actions.** We use the holomorphic representative constants $r_{\mathrm{B}}^{\mathrm{s}}$ and $K_{\mathrm{B}}^{\mathrm{s}}$ at the fixed separation ρ_{s}. In the notation of Theorem E and its proof, these are $r_{\mathrm{G}}(\rho_{\mathrm{s}})$ and a corresponding bound $K_{\mathrm{G}}(\rho_{\mathrm{s}})$ for the holomorphic curvature difference. In particular, when $\varpi(V) < \epsilon \le \bar{\varepsilon}$ and B is real with $\sup|B| < \min\{r_{\mathrm{B}}, r_{\mathrm{B}}^{\mathrm{s}}/2\}$, the two backgrounds exist, and their canonical representatives satisfy

$$|\partial U_t(e^{iB}V)(p) - \partial U_t(V)(p)| \le 2K_{\mathrm{B}}^{\mathrm{s}}\eta_t^2 \sup|B|.$$

Together with $|\partial U_t(V)(p) - \mathbf{1}| < B_3\epsilon\eta_t^2$, the difference of squares and the count $3\eta_t^{-3}N_t$ of bare plaquettes give

$$\left|\mathsf{ab}_t(c, e^{iB}V) - \mathsf{ab}_t(c, V)\right| \le 6K_{\mathrm{B}}^{\mathrm{s}}N_t \sup|c| \sup|B|(B_3\epsilon + K_{\mathrm{B}}^{\mathrm{s}} \sup|B|). \tag{2145}$$

Indeed the per-plaquette difference is at most $2K_{\mathrm{B}}^{\mathrm{s}}\eta_t^4 \sup|B|(B_3\epsilon + K_{\mathrm{B}}^{\mathrm{s}} \sup|B|)$, and the action has the prefactor η_t^{-1}.

Consider the two cutoffs $\mathrm{A} = (\ell, K)$ and $\mathrm{B} = (\ell, K+1)$ on the same torus, with $K \ge 2$ and (2130). The constants and auxiliary data are the same in both constructions. Their level-j and level-$(j+1)$ point lattices coincide:

$$\mathbb{T}^{(j),\mathrm{A}} = \mathbb{T}^{(j+1),\mathrm{B}} = \Lambda_{K-j,\ell}.$$

At these levels the blocks, contour means, averaging map, pivot sets, Haar measure, and gauge action agree. Their backgrounds and fibre charts need not agree, since the bare lattices differ; we retain the superscripts on these objects.

Fix $b \ge \gamma^{-2}$ and put $x_j^{\mathrm{A}} = L^{K-j}b$ and $x_j^{\mathrm{B}} = L^{K+1-j}b$. Thus $x_{j+1}^{\mathrm{B}} = x_j^{\mathrm{A}}$. Choose $0 < c_2' \le \min\{a_{\mathrm{plain}}, c_2^{\mathrm{SO}}\}$ and $0 < c_4 \le c_{\mathrm{plain}}$, using the uniform curvature and field radii in Proposition 51.1 and its proof. Its proof supplies the centered estimate with coefficient two, as well as the plain-average estimate stated there. Define $\Theta_\star \ge \tau_\natural$ as the least calibration for which $\epsilon = 2c_\rho\varepsilon(\tau)$ obeys all of

$$\begin{gathered} C_{\mathrm{cp}}\epsilon \le e_{\mathrm{cp}}, \qquad C_{\mathrm{pr}}'B_3\epsilon \le c_2', \qquad B_3\epsilon \le \varpi_L', \\ \tfrac{\pi}{2}B_3\epsilon \le \tfrac{1}{4}\,10^{-4}, \qquad 8\pi B_3\epsilon \le \omega_M, \\ 8\pi^2 C_{\mathrm{pr}}B_3\epsilon \le \min\{c_4, r_{\mathrm{B}}, r_{\mathrm{B}}^{\mathrm{s}}/2\}, \\ C_{\mathrm{en}}B_3\epsilon \le 2/5, \qquad \epsilon \le \min\{1, \bar{\varepsilon}\}. \end{gathered} \tag{2146}$$

Each right side is positive. Since ε is continuous, decreasing, and tends to zero, such calibrations form a closed half-line, proving that $\Theta_\star$ exists. It depends only on the fixed constants.

Choose $1 \le t \le K$ with $x_t^{\mathrm{A}} \ge \Theta_\star$, put $\epsilon = 2c_\rho\varepsilon(x_t^{\mathrm{A}})$, and take $V \in \mathcal{V}_t^{\mathrm{A}} = \mathcal{V}_{t+1}^{\mathrm{B}}$ with $\varpi(V) < \epsilon$. Let $U^{\mathrm{A}} = U_t^{\mathrm{A}}(V)$, $U^{\mathrm{B}} = U_{t+1}^{\mathrm{B}}(V)$, and abbreviate

$$a^{\mathrm{A}} = A^{\eta_t}(U^{\mathrm{A}}), \quad a^{\mathrm{B}} = A^{\eta_{t+1}}(U^{\mathrm{B}}), \quad N_t = (2L^{m+K-t})^3,$$
$$j_t = C_{\mathrm{en}}B_3\epsilon L^{-2(t+1)}, \quad l_t = 4\pi^2 C_{\mathrm{pr}}B_3\epsilon\eta_t.$$

Both backgrounds exist by the real-background theorem. They have curvature less than $B_3\epsilon\eta_t^2$ and $B_3\epsilon\eta_{t+1}^2$, respectively, so

$$0 \le a^{\mathrm{A}}, a^{\mathrm{B}} \le \tfrac{3}{2}B_3^2\epsilon^2 N_t. \tag{2147}$$

Lemma 113.8 (The two one-sided comparisons)**.** With these data, $U' = \mathcal{M}U^{\mathrm{B}}$ is a competitor at level t in A, of admitted radius $2B_3\epsilon$, and $\mathcal{M}^t U' = V$. Moreover

$$a^{\mathrm{A}} \le A^{\eta_t}(U') \le (1 + j_t)^2 a^{\mathrm{B}}. \tag{2148}$$

The reverse comparison is

$$a^{\mathrm{B}} \le (1 + 4000L^4\vartheta_0)a^{\mathrm{A}} + K_{\mathrm{up}}\epsilon^2\eta_t N_t, \qquad \vartheta_0 = \vartheta(U^{\mathrm{A}}), \tag{2149}$$

where

$$K_{\mathrm{up}} = 24\pi^2 K_{\mathrm{B}}^{\mathrm{s}} C_{\mathrm{pr}} B_3^2 (1 + 4\pi^2 K_{\mathrm{B}}^{\mathrm{s}} C_{\mathrm{pr}}).$$

■

Proof. The curvature iteration of Proposition 14.1 applies because $B_3\epsilon \le c_2^{\mathrm{SO}}$ and gives

$$\varpi(\mathcal{M}^{k+1} U^{\mathrm{B}}) \le 2B_3\epsilon L^{-2(t-k)} \qquad (k < t).$$

At $k = t$ use $\mathcal{M}^{t+1} U^{\mathrm{B}} = V$. Since $2B_3\epsilon \le C_{\mathrm{cp}}\epsilon \le e_{\mathrm{cp}}$, proposition 113.5 proves the first inequality in (2148). The arbitrary-set averaging bound with all plaquettes proves the second one, since the input curvature is at most $B_3\epsilon L^{-2(t+1)}$.

For the other direction, the angle of U^{A} is at most $\vartheta_0 \le \frac{\pi}{2} B_3 \epsilon \eta_t^2 \le 10^{-4}/4$. Set $\widetilde{U} = \mathsf{Prl}_L U^{\mathrm{A}}$. By proposition 113.2,

$$A^{\eta_{t+1}}(\widetilde{U}) \le (1 + 4000 L^4 \vartheta_0) a^{\mathrm{A}}, \qquad \varpi(\widetilde{U}) \le 4\pi \varpi(U^{\mathrm{A}}). \tag{2150}$$

Put $U_1 = \mathcal{M}\widetilde{U}$ on the bare lattice of A. Normalize $\widetilde{U}$ to axial gauge by a gauge equal to one at the coarse sites. Its straight products remain $U^{\mathrm{A}}(c)$. Each of the $L-1$ internal bonds in such a product differs from one by at most $8A_L \varpi(\widetilde{U})$, and its crossing bond differs from $U_1(c)$ by at most $C_{\mathrm{AC}} \varpi(\widetilde{U})$. The product telescoping bound therefore gives

$$|U_1(c) - U^{\mathrm{A}}(c)| \le C_{\mathrm{pr}} \varpi(\widetilde{U}), \qquad \varpi(U_1) \le C'_{\mathrm{pr}} \varpi(U^{\mathrm{A}}). \tag{2151}$$

The axial estimates apply because $\varpi(\widetilde{U}) \le 4\pi B_3 \epsilon \eta_t^2 < \omega_M$. The second bound uses four bond differences in a plaquette and (2150).

At level zero in B, (2150) gives the competitor radius $4\pi L^2 B_3 \epsilon$. At levels $k \ge 1$, apply the curvature iteration to U_1 using $C'_{\mathrm{pr}} B_3 \epsilon \le c_2' \le c_2^{\mathrm{SO}}$; this gives radius $2C'_{\mathrm{pr}} B_3 \epsilon$. Thus $\widetilde{U}$ is a competitor of radius $C_{\mathrm{cp}}\epsilon \le e_{\mathrm{cp}}$ at level $t+1$ of B.

The field

$$A(c) = \eta_t^{-1} \tfrac{1}{i} \log(U_1(c) U^{\mathrm{A}}(c)^{-1})$$

is real and has supremum at most $2\pi^2 C_{\mathrm{pr}} B_3 \epsilon \eta_t = l_t/2 < c_4$. The real centered averaging iteration gives

$$\mathcal{M}^t U_1 = (e^{i a_t} V)^u, \qquad \sup |a_t| \le 2 \sup |A| \le l_t,$$

where u is a real gauge on the top lattice. This is the factorization in the proof of Proposition 51.1, also recorded in equation (1063); the reference curvature and field radii are exactly those in (2146). Minimality for the competitor $\widetilde{U}$ and gauge invariance now imply

$$\mathsf{ab}_{t+1}^{\mathrm{B}}(e^{i a_t} V) \le A^{\eta_{t+1}}(\widetilde{U}).$$

By (2146), $l_t < \min\{r_{\mathrm{B}}, r_{\mathrm{B}}^{\mathrm{s}}/2\}$. The datum estimate (2145) in B, with weight one, bounds the difference between this action and a^{B} by

$$6K_{\mathrm{B}}^{\mathrm{s}} N_t l_t (B_3\epsilon + K_{\mathrm{B}}^{\mathrm{s}} l_t) \le K_{\mathrm{up}} \epsilon^2 \eta_t N_t.$$

Combining this with (2150) proves (2149). □

Set

$$\begin{aligned} K_u &= K_{\mathrm{up}} + 10^4 L^4 B_3^3 + 5 C_{\mathrm{en}} B_3^3, \\ K_w &= 6\left[2K_u + 5B_3^2 + 10(C_{\mathrm{nrg}} \gamma_{\mathrm{r}}^{-1} K_u B_3^2)^{1/2}\right], \qquad \|c\|_{(t)} = \sup|c| + a_{K-t} \operatorname{Lip}_\infty(c). \end{aligned} \tag{2152}$$

Proposition 113.9 (Classical action comparison)**.** For the data above, the two background actions satisfy

$$|a^{\mathrm{B}} - a^{\mathrm{A}}| \le K_u \epsilon^2 N_t L^{-t}, \tag{2153}$$

$$|A^{\eta_{t+1}}(c, U^{\mathrm{B}}) - A^{\eta_t}(c, U^{\mathrm{A}})| \le K_w \|c\|_{(t)} \epsilon^2 N_t \sqrt{t L^{-t}} \tag{2154}$$

for every complex Lipschitz function c on T_ℓ. The constants K_u, K_w depend only on the fixed construction constants, not on K, m, b, t, V, or c. ■

Proof. Write $q = (1 + j_t)^2$. Since $j_t \le 2/5$, one has $1 \le q \le 2$, $q - 1 \le 3j_t$, and

$$(q-1)a^{\mathrm{B}} \le 5C_{\mathrm{en}}B_3^3\epsilon^2\eta_t N_t. \tag{2155}$$

Here we used (2147), $\epsilon \le 1$, and $L^{-2(t+1)} \le \eta_t$. The lower comparison (2148) bounds $a^{\mathrm{A}} - a^{\mathrm{B}}$ by this quantity. In the upper comparison (2149), the prolongation error is at most

$$4000L^4 \cdot \tfrac{\pi}{2}B_3\epsilon\eta_t^2 \cdot \tfrac{3}{2}B_3^2\epsilon^2 N_t \le 10^4 L^4 B_3^3\epsilon^2\eta_t N_t.$$

This proves (2153).

For the weighted bound first suppose $0 \le c \le c_{\max}$ and put $L_c = a_{K-t}\,\mathrm{Lip}_\infty(c)$, $v = \frac{3}{2}\eta_t L_c$. Apply lemma 113.3 to U^{B} with physical coarse spacing a_K. Its curvature satisfies the required bound, and its factor is at most q. Thus, for $X = A^{\eta_{t+1}}(c, U^{\mathrm{B}})$ and $Y = A^{\eta_t}(c, U')$,

$$Y \le q(X + va^{\mathrm{B}}).$$

Apply the same bound to $c_{\max} - c$, and use $A^{\eta_t}(U') \ge a^{\mathrm{A}}$. Since $0 \le X \le c_{\max}a^{\mathrm{B}}$, the two inequalities give

$$-(q-1)c_{\max}a^{\mathrm{B}} - 2va^{\mathrm{B}} \le X - Y \le c_{\max}(a^{\mathrm{B}} - a^{\mathrm{A}}) + (q-1)c_{\max}a^{\mathrm{B}} + 2va^{\mathrm{B}}.$$

Equations (2153), (2155), and (2147) yield

$$|X - Y| \le (2K_u c_{\max} + \tfrac{9}{2}B_3^2 L_c)\epsilon^2\eta_t N_t. \tag{2156}$$

For the competitor U', the same inequalities give

$$\mathsf{exc}_t(U') \le 2K_u\epsilon^2\eta_t N_t, \qquad A^{\eta_t}(U') \le 3B_3^2\epsilon^2 N_t.$$

Apply proposition 113.7 to obtain

$$|Y - A^{\eta_t}(c, U^{\mathrm{A}})| \le 10c_{\max}(C_{\mathrm{nrg}}\gamma_{\mathrm{r}}^{-1}K_u B_3^2)^{1/2}\epsilon^2 N_t\sqrt{t\eta_t}.$$

Since $\eta_t \le \sqrt{t\eta_t}$, summing this with (2156) bounds the desired difference by $K_w(c_{\max}+L_c)\epsilon^2 N_t\sqrt{t\eta_t}/6$.

For a real signed weight, apply this result to $c + \sup|c|$ and subtract the constant weight using (2153). Since $K_u \le K_w/6$, the resulting bound is at most $K_w\|c\|_{(t)}\epsilon^2 N_t\sqrt{t\eta_t}/2$. Apply this separately to the real and imaginary parts of a complex weight and add their bounds. This proves (2154). □

113.4. **Field restrictions and comparison of the fibre origins.** Let $1 \le t_* \le K - 1$, with $x_{t_*} \ge \tau_\natural$, and put

$$e_k^{\mathrm{c}} = c_\rho\varepsilon(x_k), \qquad \mathsf{In}_k = \{V : \varpi(V) < 2\varepsilon(x_k)\}.$$

At level zero set $\mathsf{Cut}_0 = \{V : \varpi(V) < e_0^{\mathrm{c}}\}$. For $1 \le k \le t_*$ let Cut_k consist of the fields whose background exists and satisfies $\mathsf{m}_k(U_k(V)) < e_k^{\mathrm{c}}$, where $\mathsf{m}_k(U) = \eta_k^{-2}\max_p|\partial U(p) - \mathbf{1}|$. This number is independent of the representative of the critical orbit.

Lemma 113.10. These sets are open and gauge invariant. For $0 \le k \le t_*$,

$$\mathsf{In}_k \subset \mathsf{Cut}_k \subset \{\varpi < 2e_k^{\mathrm{c}}\} \subset \{\varpi < \varepsilon_{\mathcal{W}}\alpha_0\}. \tag{2157}$$

For $1 \le k \le t_*$, every field in Cut_k is in the actual global step domain at every real parameter $\tau \ge \tau_\natural$. If $k < t_*$, $W \in \mathsf{Cut}_{k+1}$, and $|B(b)| < \mathfrak{b}(x_k)$ on the nonpivot bonds, then

$$V_W(B) \in \mathsf{Cut}_k, \qquad \varpi(V_k^{\mathrm{bf}}(W)) < 2L^{-2}e_{k+1}^{\mathrm{c}}. \tag{2158}$$

For $1 \le k \le t_*$, a background over Cut_k is regular at level k with radius α_0, and

$$|A^{\eta_k}(c, U_k(V))| \le \tfrac{3}{2}(e_k^{\mathrm{c}})^2 N_k \sup|c|. \tag{2159}$$

■

Proof. The choice of $\tau_\natural$ gives $2e_k^{\mathrm{c}} \le \min(\varepsilon_{\mathcal{W}}\alpha_0, c_2^{\mathrm{SO}}, \varepsilon_v/4)$. The real background theorem, Theorem E, gives $\mathsf{m}_k(U_k(V)) \le B_3\varpi(V)$, so $c_\rho \ge 2B_3$ proves the first inclusion. The small-curvature averaging estimate in the proof of Proposition 51.1 gives $\varpi(\mathcal{M}^k U) < 2\mathsf{m}_k(U)$ in this range. This proves the other inclusions, including the level-zero case. Proposition 52.3 now supplies the actual global domain. It also gives continuity of the background in local logarithmic coordinates. Thus the strict curvature condition is open inside $\{\varpi < \bar{\varepsilon}\}$, which contains it by the preceding inclusions. Gauge covariance of the background and unitary invariance of the plaquette norm prove gauge invariance.

The restriction identity in that proposition identifies the orbit over $V_0 = V_k^{\mathrm{bf}}(W)$ with the axial representatives of the orbit over W. Its curvature therefore has $\mathsf{m}_k < L^{-2}e_{k+1}^{\mathrm{c}}$, and averaging proves the second assertion of (2158). On the real cutoff chart $V_W(B) = e^{iB'}V_0$ one has $\sup|B'| < 2C_{\mathrm{db}}\delta(x_k)$. The representative curvature-difference bound preceding (2145), with the last condition in (2132), gives

$$\mathsf{m}_k(U_k(V_W(B))) \le L^{-2}\mathsf{m}_{k+1}(U_{k+1}(W)) + 4C_{\mathrm{db}}K_{\mathrm{B}}^{\mathrm{s}}\delta(x_k) \qquad (k \ge 1).$$

For $k = 0$ the elementary plaquette product bound gives the same inequality with $8C_{\mathrm{db}}$ in place of $4C_{\mathrm{db}}K_{\mathrm{B}}^{\mathrm{s}}$. Both are bounded by C_H. Finally

$$L^{-2}e_{k+1}^{\mathrm{c}} + C_H\delta(x_k) \le \left(c_\rho L^{-3/2} + C_H A_1/A_0\right)\varepsilon(x_k) \le e_k^{\mathrm{c}}.$$

The first term before this estimate is strict, proving downward stability of the cut. The regularity assertion follows from the plain-core membership in Proposition 52.3: both the scaled curvature and current are less than α_0. There are $3\eta_k^{-3}N_k$ bare plaquettes, and each has $a_p(U) < \frac{1}{2}(e_k^{\mathrm{c}})^2\eta_k^4$. Multiplication by $\eta_k^{-1}\sup|c|$ gives (2159). □

Proposition 113.11. Use the two partner lattices above, with top levels t_* and t_*+1, and assume $x_{t_*}^{\mathrm{A}} \ge \Theta_\star$. Let $1 \le k < t_*$ and $W \in \mathsf{Cut}_{k+1}^{\mathrm{A}} \cap \mathsf{Cut}_{k+2}^{\mathrm{B}}$. Then, on every common nonpivot bond,

$$\varphi\left(V_{k+1}^{\mathrm{bf,B}}(W)(b)V_k^{\mathrm{bf,A}}(W)(b)^{-1}\right) \le 8c_\rho\varepsilon(x_k)\sqrt{K_u\gamma_{\mathrm{r}}^{-1}L^{-k}N_k}. \tag{2160}$$

■

Proof. Write $V^{\mathrm{A}}, V^{\mathrm{B}}$ for the two axial backgrounds and $\epsilon_j = 2c_\rho\varepsilon(x_j)$. Both average to W and, by lemma 113.10, have curvature less than $L^{-2}\epsilon_{k+1} \le L^{-3/2}\epsilon_k$. The axial comparison (2135) therefore bounds their relative intrinsic logarithm by $h' = \pi C_{\mathrm{ax}}L^{-3/2}\epsilon_k$. The first inequality in (2146), together with $C_{\mathrm{cp}} \ge 4\pi L^2$, puts h' below the chart and cubic-remainder thresholds in proposition 113.4. Thus V^{B} lies in the chart over W based at V^{A}. If A is its nonpivot logarithm, that proposition yields

$$\tfrac{1}{8}\gamma_{\mathrm{r}}\|A\|_2^2 \le \mathsf{ab}_k^{\mathrm{A}}(V^{\mathrm{B}}) - L^{-1}\mathsf{ab}_{k+1}^{\mathrm{A}}(W).$$

The restriction identity on lattice B gives $\mathsf{ab}_{k+1}^{\mathrm{B}}(V^{\mathrm{B}}) = L^{-1}\mathsf{ab}_{k+2}^{\mathrm{B}}(W)$. Apply (2153) at level k to V^{B} and at level $k+1$ to W. These applications are valid since their curvatures are below $\epsilon_k, \epsilon_{k+1}$. The right side is at most

$$K_u\left(\epsilon_k^2L^{-k}N_k + L^{-1}\epsilon_{k+1}^2L^{-(k+1)}N_{k+1}\right) \le 2K_u\epsilon_k^2L^{-k}N_k,$$

where $N_{k+1} = L^{-3}N_k$ and $\epsilon_{k+1}^2 \le L\epsilon_k^2$. Consequently $\|A\|_2 \le 4\epsilon_k\sqrt{K_u\gamma_{\mathrm{r}}^{-1}L^{-k}N_k}$. Each bond norm is bounded by this square-sum norm, and the intrinsic logarithm has norm equal to the angle. This is (2160); reversing a bond leaves that angle unchanged. □

113.5. **Older interactions at a regular background.** Write $K_{\mathrm{A}} = K_{\mathrm{tw}}^\sharp$ and $a_{\mathrm{A}} = a^\sharp$ for the simple-domain constants of Proposition 51.9, and B_*, c_* for the regional-gauge constants of

Theorem F, with $c_* \le 1$. Assume $M\alpha_0 \le \alpha_1$ and $\alpha_0 + \alpha_1 \le a_{\mathrm{A}}$. Set

$$\begin{aligned} c_{\mathrm{A}} &= \min\{c_*/4, a_{\mathrm{A}}/(64K_{\mathrm{A}}B_*)\}, \qquad C_{\mathrm{A}}^{\mathrm{old}} = 2(64K_{\mathrm{A}}B_*/a_{\mathrm{A}})^2, \\ C'_{\mathrm{A}} &= 6e^{\kappa_{\mathcal{D}}}M^2C_{\mathrm{A}}^{\mathrm{old}}, \qquad \rho_{\mathrm{A}} = a_{\mathrm{A}}\alpha_0/(16K_{\mathrm{A}}), \\ l_0 &= \lceil \log_L(M/c_{\mathrm{A}})\rceil, \qquad C_{\mathrm{lg}} = \sup_{l\in\mathbb{N}_0} L^{4l}e^{-c_{\mathrm{A}}L^l}, \\ C_{\mathrm{old}} &= C'_{\mathrm{A}} + 2C_{\mathrm{E},0}(C_{\mathrm{lg}} + L^{4l_0}). \end{aligned} \tag{2161}$$

Here $C_{\mathrm{E},0} = 2048\sum_{n\ge 0} e^{-n}$ is a permissible rooted-domain entropy constant. In particular $l_0 \ge 1$ and $C_{\mathrm{lg}} \le (4/(ec_{\mathrm{A}}))^4$.

Lemma 113.12. Let $1 \le i \le t \le K$, $16M \le L^{m+K-t}$, and let U be regular at level t with radius α_0 on the torus. Suppose its pairs belong to the fixed cores on all domains used below. If $D^{(i)}$ is a pointed family at radii (α_0, α_1) and $l = t - i$, then

$$\sum_{z\in\mathbb{T}^{(i)}} \sum_{X\in\mathcal{D}_i : X\ni z} |D^{(i)}(X, U, z) - D^{(i)}(X, \mathbf{1}, z)| \le C_{\mathrm{old}}N_tL^{-l}\|D^{(i)}\|_{i,\kappa+1}.$$

■

Proof. We first prove the local quadratic gain that supplies the factor L^{-4l}. Put $\vartheta = L^{-l}$. For a domain of size $d = d_i(X) \le c_{\mathrm{A}}\vartheta^{-1}$, assume $M\alpha_0\vartheta \le c_{\mathrm{A}}\alpha_0$. The same bare configuration is regular at level i with radius $\alpha = \alpha_0\vartheta^2$: curvature rescales by ϑ^2, and current by $\vartheta^3 \le \vartheta^2$. One has $(d+1)\vartheta \le 1$. Indeed this is immediate for $l = 0$, when $d = 0$, while for $l \ge 1$ its left side is at most $c_{\mathrm{A}} + L^{-1} < 1$.

A connected edge set of size d meeting all cubes of X puts X in the cube of radius $(d+1)M$ about one of its corners. Enlarge this cube by two unit layers. Its inner side is $q = 2(d+1)M$, and

$$q + 4 \le 6M\vartheta^{-1} < 16M\vartheta^{-1} \le L^{m+K-i}.$$

Thus it lies in a torus chart. Also $2q\alpha \le 4c_{\mathrm{A}}\alpha_0 \le c_*$. The regional gauge theorem, applied with radius 2α, gives on the inner cube a gauge in which $U = e^{i\eta_i A}$ and $|A|, |\nabla^{\eta_i}A| < \beta = 4B_*(d+1)M\alpha$. Its divergence hypothesis follows directly from

$$|D_U^{\mathsf{T}}\partial U| \le \eta_i^{-1}(\alpha\eta_i^2)^2 + \eta_i^2|J^{(i)}(U)| < 2\alpha\eta_i^2.$$

The gauge-transformed current J_1 also has norm less than β, and $\beta \le \rho_{\mathrm{A}}$.

Consider the complex disc of pairs $\Gamma(\zeta) = (e^{i\zeta\eta_i A}, \zeta J_1)$ for $|\zeta| \le R = \rho_{\mathrm{A}}/\beta$. With $\phi = |\zeta|\beta \le \rho_{\mathrm{A}} \le 1/16$, the exponential-product estimate gives

$$|\partial(e^{i\zeta\eta_i A}) - \mathbf{1}| \le (2\phi + 11\phi^2)\eta_i^2 < 4\rho_{\mathrm{A}}\eta_i^2.$$

The other simple-domain bounds have radii $(4\rho_{\mathrm{A}}, 2\rho_{\mathrm{A}}, 4\rho_{\mathrm{A}})$. Moreover

$$6\rho_{\mathrm{A}} \le a_{\mathrm{A}}, \quad 4\rho_{\mathrm{A}} \le \alpha_0, \quad 2\rho_{\mathrm{A}} \le \alpha_1, \quad K_{\mathrm{A}}(4\rho_{\mathrm{A}} + 4\rho_{\mathrm{A}}^2) \le \alpha_0.$$

The core inclusion (846) therefore puts the whole disc in the fixed core, with strict room in the defining simple-domain bounds. The same holds after undoing the real gauge. Pair dependence and gauge invariance apply there.

For a local term D, the function $f(\zeta) = D(\Gamma(\zeta))$ is holomorphic near the closed disc and bounded by $\|D\|_X$. Its derivative at zero vanishes: the four constant conjugations by $1, i\sigma_1, i\sigma_2, i\sigma_3$ preserve D, and their adjoint actions sum to zero on each copy of the complexified Lie algebra. Applying the linear differential of D at the trivial pair gives $4f'(0) = 0$. The maximum principle applied to $(f(\zeta) - f(0))/\zeta^2$ now gives

$$|D(U) - D(\mathbf{1})| \le 2\|D\|_X R^{-2} = C_{\mathrm{A}}^{\mathrm{old}}\|D\|_X M^2(d+1)^2\vartheta^4.$$

This is the fixed-radius argument also used in the proof of Lemma 64.1; here the displayed radius choices give the constants in (2161).

Let $H_i = \|D^{(i)}\|_{i,\kappa+1}$. At each site the norm bound is $\|D^{(i)}(X,\cdot,z)\|_X \le H_i e^{-(\kappa+1)d}$. There are at most $e^{\kappa_{\mathcal{D}}(n+1)}$ domains of size n containing the site. Since $\kappa + 1 - \kappa_{\mathcal{D}} > 1$,

$$\sum_{X \ni z} (d_i(X)+1)^2 e^{-(\kappa+1)d_i(X)} \le e^{\kappa_{\mathcal{D}}} \sum_{n \ge 0} (n+1)^2 e^{-n} < 6e^{\kappa_{\mathcal{D}}}.$$

If $l \ge l_0$, the preceding quadratic bound applies to every domain with $d_i(X) \le c_{\mathrm{A}} L^l$. Summing it over the L^{3l} sites of each level-t block and then its N_t blocks gives a contribution at most $C'_{\mathrm{A}} N_t L^{-l} H_i$.

For the remaining domains use simply $|D^{(i)}(X,U,z) - D^{(i)}(X,\mathbf{1},z)| \le 2H_i e^{-(\kappa+1)d_i(X)}$. The rooted entropy estimate, with $\kappa + 1 > \kappa_{\mathcal{D}} + 2$, gives, for any $D \ge 0$,

$$\sum_{z} \sum_{X \ni z:\ d_i(X) \ge D} |D^{(i)}(X,U,z) - D^{(i)}(X,\mathbf{1},z)| \le 2C_{\mathrm{E},0} e^{-D} L^{3l} N_t H_i.$$

For $l \ge l_0$ choose $D = c_{\mathrm{A}} L^l$ and use $L^{3l} e^{-c_{\mathrm{A}} L^l} \le C_{\mathrm{lg}} L^{-l}$. For $l < l_0$, choose $D = 0$ and use $L^{3l} \le L^{4l_0} L^{-l}$. Adding the bounds gives the asserted constant C_{old}. □

Proposition 113.13. For $1 \le t \le t_*$ and $V \in \mathsf{Cut}_t$, let $U = U_t(V)$ and let $\mathfrak{f} = ((D^{(i)},0)_{1 \le i \le t}, \emptyset)$ be a bare array at the fixed radii. Then its action is defined on the whole critical orbit, is constant there, and

$$|S[\mathfrak{f}](U)| \le C_{\mathrm{old}} N_t \sum_{i=1}^{t} L^{-(t-i)} \|D^{(i)}\|_{i,\kappa+1}. \tag{2162}$$

■

Proof. Plain-core membership of the level-t background, its tower restrictions, and radius monotonicity put every level-i pair in the fixed core. The defining sum for the action is therefore

$$S[\mathfrak{f}](U) = \sum_{i=1}^{t} \sum_{z} \sum_{X \ni z} [D^{(i)}(X,U,z) - D^{(i)}(X,\mathbf{1},z)].$$

Each summand is gauge invariant. The regularity statement of lemma 113.10 and the room assumption permit lemma 113.12 at every i. Sum its estimates. □

114. Intermediate integration on one lattice

We use the same real backgrounds and Gaussian structures for all sources. Only the fourth potential is changed. This permits a comparison of two choices of the earlier interactions before the last integrations are performed.

At height $k \ge 1$ put

$$r_k^{\mathrm{src}} = \min\{\varepsilon_{\mathfrak{h}}, c_{\mathrm{os}} \alpha_0/(2c_{\mathfrak{h}})\},$$

and set $r_0^{\mathrm{src}} = r_P^{\circ}$ at height zero. Require $C_{\mathcal{A}} \delta(x_k) \le r_k^{\mathrm{src}}/2$. For $k \ge 1$ define the localized weighted action by

$$\mathbb{V}_v^{\mathrm{act}}(Y;P,A) = \sum_{X \subset Y} \mathbb{V}[W_v(X,\cdot)](Y;P,A), \qquad W_v(X,U) = \eta_k^{-1} \sum_{p : X(p) = X} v(p) a_p(U),$$

using the actual shift and vertex brackets of Proposition 51.15. At $k = 0$ use instead

$$\mathbb{V}_v^{\mathrm{act}}(Y;P,A) = \sum_{p : X(p) = Y} v(p) [a_p(e^{iB'(P,A)}\mathbf{U}) - a_p(\mathbf{U})],$$

where B' is the pivot completion. These are the same weighted Wilson members and shifts used in the source calculation above. All sums here are taken on their specified full one-lattice cores.

For $g = x_k^{-1/2}$, subtract the linear reference value at each nonpivot bond and set

$$\mathbb{W}^{\mathrm{src}}[v](Y;P,B) = x_k \mathbb{V}_v^{\mathrm{act}}(Y;P,gB) - x_k \sum_b v(p_b) D_A \mathbb{V}_1^{\mathrm{act}}(Y;P,0)[gB_b]. \tag{2163}$$

Here $B_b = B(b)\delta_b$, and p_b is the fixed reference plaquette at the centre of the fine cube containing b_-. Its corners lie in that cube. The reference subtraction is made on every core, before specializing to a critical field.

Let $K_{\mathrm{Q}}^\vee, K_{\mathrm{L}}^\vee$ be at least the maxima of

	K_{Q}	K_{L}
$k \ge 1$	$672M^6(\alpha_0/r_k^{\mathrm{src}})^2$	$8064M^6\alpha_0/r_k^{\mathrm{src}}$
$k = 0$	$2b_{\mathrm{A}}M^3(r_P^\circ)^{-2}$	$72b_{\mathrm{A}}M^3(r_P^\circ)^{-1}$

(2164)

with $b_{\mathrm{A}} = 6 \cdot 1201^2 M^3$. For a region R put

$$\mathsf{sz}_{k,R}(v) = K_{\mathrm{Q}}^\vee(1 + 2C_{\mathcal{A}}\delta(x_k)/r_k^{\mathrm{src}})|v|_R + K_{\mathrm{L}}^\vee x_k^{1/2}\mathrm{sw}_k(v).$$

The oscillation sw_k is measured between plaquettes whose barycentres are at distance at most M in level-k units.

Lemma 114.1. The source family (2163) is linear in v, holomorphic in the source, the open triple slices, and the field polydisc. It vanishes at $B = 0$, is gauge invariant, depends on the background pair, and is local in the plaquettes of Y. Its quadratic-growth norm satisfies

$$\mathsf{n}_X^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v]) \le \mathsf{sz}_{k,X}(v), \qquad \mathsf{n}^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v]) \le \mathsf{sz}_{k,T_\ell}(v) \le 2K_{\mathrm{Q}}^\vee|v|_\infty + K_{\mathrm{L}}^\vee x_k^{1/2}\mathrm{sw}_k(v). \tag{2165}$$

The norm uses the rate $\lambda_V \le \bar\kappa - 2$ and $1 + q_Y(B)$, with $q_Y(B) = M^{-3}\sum_b |B(b)|^2$. On a real fibre over $W \in \mathsf{Cut}_{k+1}$,

$$\sum_Y \mathbb{W}^{\mathrm{src}}[v](Y; P_W|_Y, B|_Y) = x_k\{A^{\eta_k}(v, U_k(V_W(B))) - A^{\eta_k}(v, U_{k+1}(W))\}. \tag{2166}$$

■

Proof. For $k \ge 1$, each nonzero root $X = X(p)$ has at most four fine cubes and size zero. The determinant identity for a complex unimodular plaquette matrix and the core curvature bound give $\|W_v(X,\cdot)\|_X \le 6M^3\alpha_0^2|v|_Y$. Its amplitude radius in Proposition 51.15 is $c_{\mathrm{os}}\alpha_0/(c_{\mathfrak{h}}r_k^{\mathrm{src}}) \ge 2$. If $N = \#_{\pi_k}Y$, there are at most $7N$ roots, and the number of nonroot cubes is at least $\max(d_k(Y), N-4)$. The sharper first bound in that proposition, together with $e^{\bar\kappa} - 1 \ge e^{\bar\kappa - 1/2}$, therefore gives

$$e^{r' d_k(Y)} N \sup_{|A| < r_k^{\mathrm{src}}} |\mathbb{V}_v^{\mathrm{act}}(Y;P,A)| \le 42N^2 e^{-(N-4)_+/2} M^3\alpha_0^2|v|_Y \le 672M^3\alpha_0^2|v|_Y$$

for $0 \le r' \le \bar\kappa - 1$. The last inequality uses $N^2 e^{-(N-4)_+/2} \le 16$. At $k = 0$, the pivot plaquette calculation in Proposition 51.4 and its proof gives instead $\sup|\mathbb{V}_v^{\mathrm{act}}| \le 2b_{\mathrm{A}}|v|_Y$; the term is zero unless $d_0(Y) = 0$ and $N \le 4$. The same calculation is valid with a weight inside the finite plaquette sum, since it bounds the sum of the plaquette moduli.

Write $F_Y(v)$ for these bounds before multiplying by $e^{r'd}N$. Cauchy's formula gives homogeneous coefficients $|c_j[v](B)| \le F_Y(v)(\sup|B|/r_k^{\mathrm{src}})^j$. Taylor expansion of (2163) is

$$g^{-1}\sum_b c_1[v - v(p_b)](B_b) + c_2[v](B) + x_k \sum_{j \ge 3} c_j[v](gB).$$

For the quadratic term use $\sup|B|^2 \le M^3 q_Y(B)$. For the remainder use the half-radius bound

$$x_k \left| \sum_{j \ge 3} c_j[v](gB) \right| \le 2F_Y(v)(r_k^{\mathrm{src}})^{-2} \sup|B|^2 C_{\mathcal{A}}\delta(x_k)/r_k^{\mathrm{src}}.$$

For the linear term use $\sum_b |B(b)| \le \frac{3}{2}M^3N(1+q_Y(B))$. A cube path from a plaquette to p_b has at most $d+2$ links, so $|v-v(p_b)|_Y \le (d+2)\mathrm{sw}_k(v)$; $e^{-d}(d+2) \le 2$ absorbs one unit of decay. For $k \ge 1$ the resulting coefficients are $672M^6(\alpha_0/r_k^{\mathrm{src}})^2$ for the quadratic term and at most $2016M^6\alpha_0/r_k^{\mathrm{src}}$ for the linear term. For $k=0$ they are respectively $2b_{\mathrm{A}}M^3(r_P^\circ)^{-2}$ and at most $24b_{\mathrm{A}}M^3(r_P^\circ)^{-1}$. Thus (2164) bounds all three terms and proves (2165).

The local shift and pivot formulas, their finite vertex brackets, and Cauchy's formula give the stated parameter properties. The reference subtraction uses only $v(p_b)$ with $p_b \subset Y$, so it preserves locality. The actual-shift identity of Proposition 51.15, summed over the plaquette roots, gives the weighted action difference. The subtracted linear terms have sum zero bond by bond at P_W: they are the derivative of the unweighted action on its averaging fibre, which vanishes at the critical background. This is precisely the criticality calculation of lemma 78.4, including the first-step pivot case. Hence (2166) holds for every weight, not just constant weights. □

114.1. **The recursion and its density.** In addition to the global hypotheses above, assume

$$E_0^{\mathrm{s}} \ge 2E_{\mathsf{N}}, \qquad N_\star^{\mathrm{s}} \ge \frac{L}{L-1}(E_0^{\mathrm{s}} + \tfrac{3}{2}\alpha_0^2 B_\sharp), \tag{2167}$$

$$K_{\mathbf{M}}\left(\delta(\gamma^{-2}) + \mathsf{s}_\gamma N_\star^{\mathrm{s}} + e^{-\gamma_2 \mathfrak{b}(\gamma^{-2})^2/2}\right) \le E_0^{\mathrm{s}}/4, \qquad 8K_\Lambda \mathsf{s}_\gamma N_\star^{\mathrm{s}} \le w_\star, \qquad 8K_P \mathsf{s}_\gamma \le w_\star. \tag{2168}$$

Here $\mathsf{s}_\gamma = (1+\beta_{\mathrm{s}})\delta(\gamma^{-2})/\alpha_0$. Require also $4\beta_{\mathrm{s}}\kappa \ge 1$, and put

$$K^{\mathrm{d}} = \tfrac{351}{2}L^3K_\Pi, \qquad K_{\mathbf{M}}^{\mathrm{f}} = 3L^3\max\{K_E e^{4\lambda_1}, 32K_\Pi K_P, \tfrac{117}{2}K_\Pi\}, \quad \mu_\star = \min\{w_\star/32, E_0^{\mathrm{s}}/(16K_{\mathbf{M}}^{\mathrm{f}})\}.$$

Thus $K^{\mathrm{d}} \le K_{\mathbf{M}}^{\mathrm{f}}$. Set $\sigma_{\mathrm{src}} = \mu_\star/K_{\mathrm{Q}}^\vee$ and choose $\varrho > 0$ with

$$2\varrho \le \sigma_{\mathrm{src}}(\mathfrak{b}' - \bar{B}), \qquad 2K_{\mathrm{L}}^\vee \varrho\gamma \le 40\mu_\star/\sqrt{3}. \tag{2169}$$

The sources belong to $\mathsf{Src} = \mathsf{Src}_{\mathsf{Bc}}(2\varrho, 2\varrho)$. For the line $x_j = L^{K-j}\mathfrak{b}'$ write

$$x_j^w = x_j - L^{K-j}w, \qquad v_j^w = -w/\mathfrak{b}', \qquad \mathsf{s}_j = (1+\beta_{\mathrm{s}})\delta(x_j)/\alpha_0.$$

The source bounds give $|v_j^w|_\infty \le \sigma_{\mathrm{src}}$ and $x_j^{1/2}\mathrm{sw}_j(v_j^w) \le \mu_\star/K_{\mathrm{L}}^\vee$. In particular, lemma 114.1 gives $\mathsf{n}^{\mathrm{f}}(\mathbb{W}^{\mathrm{src}}[v_j^w]) \le 3\mu_\star$.

An early system at t_0 consists of pointed families $E^{(i)}[w]$, $1 \le i \le t_0$, on the fixed pair cores, with $\|E^{(i)}[w]\|_{i,\kappa+1} \le E_0^{\mathrm{s}}$. They are jointly holomorphic in w and every open pair slice, and their member at X depends on w only through the plaquettes in X. For $j \ge t_0$ define

$$\mathfrak{f}_j[w] = ((E^{(i)}[w], 0)_{1\le i\le j}, \emptyset), \qquad E^{(j+1)}[w] = \mathbf{N}^{(j+1)} + \mathbf{F}[\mathbb{V}'[\mathfrak{f}_j[w]] - \mathbb{W}^{\mathrm{src}}[v_j^w]]. \tag{2170}$$

All objects in a step use its real Gaussian structure at $x = \mathsf{t} = x_j$, so that $\tau = 1$. The chain starts at $t_0 = 0$ with the empty array; a classical start at t_0 sets all its early families equal to zero. A hat denotes the latter flow.

Proposition 114.2. Every early system continues through level K by (2170). Its new families have the same norm, holomorphy and source-locality properties as the early families. At each step the fourth family has norm at most $7w_\star/32 < w_\star/4$.

For $t_0 \le k \le t_*$ and $V \in \mathsf{Cut}_k$ put

$$\Phi_k^w(V) = -A^{\eta_k}(x_k^w, U_k(V)) + S[\mathfrak{f}_k[w]](U_k(V)), \qquad F_k^w = \mathbf{1}_{\mathsf{Cut}_k} e^{\Phi_k^w}.$$

These are bounded Borel functions. For $t_0 \le k < t_*$ set

$$c_{k+1}(w) = \log\{\mathsf{pf}_k(x_k)\mathsf{Z}^{(k)}[\mathbf{1}]\} + \sum_{X,z} E^{(k+1)}[w](X, \mathbf{1}, z), \tag{2171}$$

where the logarithm is real. For every $W \in \mathsf{Cut}_{k+1}$,

$$\mathsf{I}[F_k^w](W) = e^{c_{k+1}(w) + \Phi_{k+1}^w(W)} \ne 0. \tag{2172}$$

Consequently, the finite complex measures

$$\nu_{t_0}^{w} = F_{t_0}^{w}\, dV, \qquad \nu_{k+1}^{w} = \mathbf{1}_{\mathsf{Cut}_{k+1}} \mathcal{M}_*[\mathfrak{G}_k[x_k]\chi_k^{\mathrm{sf}} \nu_k^{w}]$$

satisfy

$$\nu_k^{w} = \exp\left(\sum_{t_0 < i \le k} c_i(w) \right) \mathbf{1}_{\mathsf{Cut}_k} e^{\Phi_k^{w}}\, dV. \tag{2173}$$

The chain restarted with its own first t_0 families is the same chain. ■

Proof. Suppose the families through j have been constructed. Their weighted array size is at most $LE_0^{\mathrm{s}}/(L-1) \le N_\star^{\mathrm{s}}$. The actual old-potential rule of Theorem W therefore has norm at most $K_\Lambda \mathsf{s}_j N_\star^{\mathrm{s}} \le w_\star/8$. The source contributes at most $3\mu_\star \le 3w_\star/32$. The three internal potentials have total norm at most $3K_P\delta(x_j) \le 3w_\star/8$. Thus lemma 77.1 applies on the full core. Its unperturbed bound, the tent mass, and the normalization bound of Theorem W give respectively

$$\|\mathbf{F}[\mathbb{V}'[\mathfrak{f}_j]]\|_{j+1,\kappa+1} \le K_{\mathbf{M}} \left(\delta(x_j) + \mathsf{s}_j N_\star^{\mathrm{s}} + e^{-\gamma_2 \mathfrak{b}(x_j)^2/2} \right) \le E_0^{\mathrm{s}}/4, \qquad \|\mathbf{N}^{(j+1)}\|_{j+1,\kappa+1} \le E_0^{\mathrm{s}}/2.$$

Here the output rates exceed $\kappa+1$ by $4\beta_{\mathrm{s}}\kappa \ge 1$ and the normalizer decay assumption. In (1601), the coefficient is at most $162L^3K_\Pi < K^{\mathrm{d}}$ whenever $n_{\mathrm{int}} \le 3w_\star/8$, $n_{\mathrm{iv}} \le w_\star/4$ and $n_W \le w_\star/2$. The source perturbation therefore costs at most $3K^{\mathrm{d}}\mu_\star \le 3E_0^{\mathrm{s}}/16$. The new norm is at most $15E_0^{\mathrm{s}}/16$. The parameter statement of lemma 77.1, the linear old-potential rule and lemma 114.1 give holomorphy. If two sources agree on X, every earlier member used inside X agrees, as does the source family; the same local construction then gives the same new member. This proves the induction, including the first step, whose old array is empty.

For the density identity fix $W \in \mathsf{Cut}_{k+1}$. The cut theorem puts W in the actual global sumrule class. The old-potential sumrule and (2166) identify the fourth-family sum with the change along the fibre of

$$\mathcal{S}_k(U) = S[\mathfrak{f}_k[w]](U) - x_k A^{\eta_k}(v_k^{w}, U).$$

The three internal sumrules and the identity $\mathbf{C}_*(P_W) = \mathsf{A}_W^{-1}$ identify the chart integral with its truncated Gaussian integral. Its cluster sum is the exponential of the sum of the pointed terms by (1599). The Gaussian normalization, localized by $\mathbf{N}^{(k+1)}$, contributes the first logarithm in (2171); this is the actual normalization identity of Theorem W, with the hypotheses verified in Proposition 52.3. Hence the chart value is

$$\exp\left\{ \log(\mathsf{pf}_k \mathsf{Z}^{(k)}[\mathbf{1}]) + \sum_{X,z} E^{(k+1)}[w](X, U_+, z) - A^{\eta_k}(x_k^{w}, U_+) + S[\mathfrak{f}_k[w]](U_+) \right\}.$$

Subtract and add the trivial-pair values in this finite sum. The old members and the new centered member are precisely the action of $\mathfrak{f}_{k+1}[w]$; the weighted action rescales because $x_k^{w}/L = x_{k+1}^{w}$. This proves (2172), including its nonvanishing.

All these actions are defined on the critical orbit by the cut theorem, and are constant there by gauge invariance. Local analytic representatives of that orbit show that Φ_k^{w} is Borel on the cut. Its absolute value is bounded there by the cut action estimate and proposition 113.13. Extending its exponential by zero gives the asserted bounded Borel function. The physical image-density formula applies to it. Every chart over Cut_{k+1} lies in Cut_k, so the inserted cut is identically one in the chart. Induction with (2172) gives (2173). Equality upon restarting follows directly from the deterministic recursion. □

114.2. **Decay of the early interactions.** Put $c_{\log} = \log L/\log \gamma^{-2}$ and

$$\Sigma_\phi = \sum_{a \ge 1} L^{-a/2}(1 + a c_{\log})^{\mathfrak{p}_0}, \qquad C_{\mathrm{fg}} = \exp\{\tfrac{13}{72}(L-1)\Sigma_\phi\}.$$

The series converges. The explicit logarithmic cutoff gives

$$\mathsf{s}_j \le \mathsf{s}_\gamma L^{-(K-j)/2}(1+(K-j)c_{\log})^{\mathfrak{p}_0}. \tag{2174}$$

Indeed $x_j = L^{K-j}\mathsf{b}'$, $\mathsf{b}' \ge \gamma^{-2}$, and $x^{-1/2}(\log x)^{\mathfrak{p}_0}$ decreases on this range. Also $K_{\mathbf{M}} \ge 243 L^3 K_\Pi K_\Lambda$ and (2167)–(2168) imply

$$K^{\mathrm{d}} K_\Lambda \mathsf{s}_\gamma \frac{L}{L-1} \le 13/72. \tag{2175}$$

Proposition 114.3. For two early systems at t_0 and the same source, let

$$\Delta_i = \|E^{(i)}[w] - \widetilde{E}^{(i)}[w]\|_{i,\kappa+1}, \qquad n_j = \sum_{i\le j} L^{-(j-i)}\Delta_i.$$

For $j \ge t_0$,

$$n_j \le C_{\mathrm{fg}} L^{-(j-t_0)} n_{t_0} \le C_{\mathrm{fg}} L^{-(j-t_0)} \frac{L}{L-1} \max_{i\le t_0} \Delta_i. \tag{2176}$$

In particular, the chain and classical start obey this bound with $\max \Delta_i \le E_0^{\mathrm{s}}$. ■

Proof. The normalizers and source families cancel in the difference of the two recursions. Linearity of the old-potential rule and its weighted-lag estimate give $\mathsf{n}^{\mathrm{f}}(\mathbb{V}'[\mathfrak{f}_j - \widetilde{\mathfrak{f}}_j]) \le K_\Lambda \mathsf{s}_j n_j$. Both fourth families have norm below $w_\star/4$, so their difference has norm below $w_\star/2$. Equation (1601) therefore gives $\Delta_{j+1} \le K^{\mathrm{d}} K_\Lambda \mathsf{s}_j n_j$. Consequently

$$n_{j+1} \le (L^{-1} + K^{\mathrm{d}} K_\Lambda \mathsf{s}_j) n_j, \qquad n_j \le L^{-(j-t_0)} n_{t_0} \exp\left(L K^{\mathrm{d}} K_\Lambda \sum_{i=t_0}^{j-1} \mathsf{s}_i \right).$$

Equations (2174) and (2175) bound the last exponential by C_{fg}. The geometric sum bounds n_{t_0}, proving the result. If $t_0 = 0$ the flows coincide. □

Define

$$\epsilon_{\mathrm{fg}}(t,t_0) = C_{\mathrm{old}} C_{\mathrm{fg}} \frac{L}{L-1} E_0^{\mathrm{s}} N_t L^{-(t-t_0)}, \qquad \mathsf{m}_t^S = C_{\mathrm{old}} \frac{L}{L-1} E_0^{\mathrm{s}} N_t. \tag{2177}$$

Combining propositions 113.13 and 114.3, for $t_0 \le t \le t_*$ and $V \in \mathsf{Cut}_t$ we have

$$|\Phi_t^w(V) - \widehat{\Phi}_t^w(V)| \le \epsilon_{\mathrm{fg}}(t,t_0), \qquad |S[\mathfrak{f}_t[w]](U_t(V))| \le \mathsf{m}_t^S. \tag{2178}$$

The same action bound holds for every early system. The weighted Wilson actions cancel in the first difference; this estimate concerns the field-dependent exponents only.

114.3. **A uniform fluctuation tail.** Set

$$\kappa_{\mathrm{b}} = \frac{w_\star}{32 M^3} e^{-\lambda_V}, \qquad C_{\mathrm{sh}} = \tfrac{1}{4} K^{\mathrm{d}} w_\star M^3 C_{\mathrm{E},1}, \quad C_{\mathrm{E},1} = 2048 \sum_{n\ge 0} (n+2) e^{-n}. \tag{2179}$$

These constants are independent of γ and of the cutoff, torus, level and source.

Proposition 114.4. Let $t_0 \le k < t_*$, let b be a nonpivot bond at level k, and let $g = x_k^{-1/2}$. Let $\Sigma \in \mathbb{C}$ be a scalar such that $e^{\Sigma} F_k^w$ is real and nonnegative on Cut_k. Then, for $W \in \mathsf{Cut}_{k+1}$ and $a \ge 0$,

$$0 \le \mathsf{I}[e^{\Sigma} F_k^w \mathbf{1}\{|\mathcal{A}_k^\diamond(b)| \ge g a\}](W) \le e^{-\kappa_{\mathrm{b}} a^2 + C_{\mathrm{sh}}} \mathsf{I}[e^{\Sigma} F_k^w](W), \qquad \mathsf{I}[e^{\Sigma} F_k^w](W) > 0. \tag{2180}$$

The indicator is set to zero where the intrinsic fluctuation is undefined; with this convention it is Borel. ■

Proof. On the stable background domain the map $(V,W)\mapsto V(b)V_k^{\mathrm{bf}}(W)(b)^{-1}$ is continuous. The intrinsic logarithm is continuous away from $-\mathbf{1}$. Taking its norm superlevel set and composing with the Borel map $V\mapsto(V,\mathcal{M}V)$ proves the Borel assertion.

Let $\mathbb{Q}_b(Y;P,B)=\frac12\operatorname{tr}(B(b)^2)$ if $Y=\square(b)$, and zero otherwise. This is holomorphic, gauge invariant and independent of P. Since $d_k(\square(b))=0$ and $|B(b)|^2\le M^3q_{\square(b)}(B)$,

$$\mathsf{n}_X^{\mathrm{f}}(\kappa_{\mathrm{b}}\mathbb{Q}_b)\le w_\star/32,\qquad \mathsf{n}_X^{\mathrm{f}}(\kappa_{\mathrm{b}}\mathbb{Q}_b)=0\quad\text{unless }\square(b)^\sharp\subset X.$$

Adding $\lambda\mathbb{Q}_b$, for $\lambda=0,\kappa_{\mathrm{b}}$, leaves the fourth-family norm at most $w_\star/4$. The Gaussian sumrule used in proposition 114.2 now gives

$$\mathsf{I}[F_k^w e^{\lambda x_k|\mathcal{A}_k^\diamond(b)|^2}](W)=p_W\Xi_\lambda,\qquad \Xi_\lambda=\exp\sum_{X,z}(\mathbf{G}_1+\mathbf{F}_\lambda)(X,P_W|_X,z).\tag{2181}$$

The value assigned to the extra exponential off its domain is immaterial. Indeed every chart lies inside the cut, where $\mathcal{A}_k^\diamond(b)=gB(b)$, so the insertion is $e^{\lambda|B(b)|^2}$. In the Gaussian calculation

$$p_W=\mathsf{pf}_k(x_k)Z^{(k)}(W;\mathsf{A}_W)\exp[-x_k\mathbb{F}_W(0)+\mathcal{S}_k(U_{k+1}(W))]\neq0$$

is independent of λ. The three internal sums are $\mathbb{P}^J,\mathbb{P}^G,\mathbb{P}^A[\mathsf{B}_W]$, the fourth sum is $\mathsf{s}_W[\mathcal{S}_k]+\lambda|B(b)|^2$, and the covariance is A_W^{-1}. At $\tau=1$ the quadratic correction potential vanishes. These identities give (2181) directly from the physical Gaussian change of variables.

The local form of (1601) gives

$$|(\mathbf{F}_{\kappa_{\mathrm{b}}}-\mathbf{F}_0)(X,P,z)|\le K^{\mathrm{d}}w_\star e^{-(\kappa+1)d_{k+1}(X)}/32,$$

and the difference vanishes unless $\square(b)^\sharp\subset X$. All contributing domains contain one fixed level-$(k+1)$ site z_b: choose it in the partition cube containing b_-. Each such domain has at most $8M^3(d_{k+1}(X)+1)$ base points. The rooted entropy bound therefore gives

$$|\log(\Xi_{\kappa_{\mathrm{b}}}/\Xi_0)|\le\frac{K^{\mathrm{d}}w_\star}{32}\sum_{X\ni z_b}8M^3(d_{k+1}(X)+1)e^{-(\kappa+1)d_{k+1}(X)}\le C_{\mathrm{sh}},$$

where the logarithm denotes the displayed cluster-sum difference. Thus $|\Xi_{\kappa_{\mathrm{b}}}/\Xi_0|\le e^{C_{\mathrm{sh}}}$.

The chart measure is positive, and its fields lie in Cut_k. Apply $\mathbf{1}\{|B(b)|\ge a\}\le e^{-\kappa_{\mathrm{b}}a^2}e^{\kappa_{\mathrm{b}}|B(b)|^2}$ under that measure with the nonnegative factor $e^{\Sigma}F_k^w$. The denominator is nonzero by (2181) and nonnegative by the same chart formula, hence positive. The quotient of the tilted and untilted integrals is real and nonnegative, and its modulus is at most $e^{C_{\mathrm{sh}}}$. This proves (2180). □

114.4. **Source radii and field-independent terms.** Let $f_1,\dots,f_n$ be nonzero real Lipschitz functions on T_ℓ and $w_z=\sum_i z_if_i$. Write

$$D_R^\flat=\{z:\sum_i|z_i|\|f_i\|_{\flat;T_\ell}<R\},\qquad r_i(t)=(8nM_t^{\mathrm{rd}}\|f_i\|_{\flat;T_\ell})^{-1},$$
$$M_t^{\mathrm{rd}}=\max\{\varrho^{-1},\tfrac32(e_t^{\mathrm{c}})^2L^{K-t}N_t,\varrho^{-1}\mathsf{m}_t^S\}.$$

The polydisc $D_t^{\mathrm{rd}}=\{|z_i|<r_i(t)\}$ lies in $D^\flat_{\varrho/8}$. Recall that $0<\|f_i\|_\flat\le\|f_i\|_\sharp$ and $\|w_z\|_\flat\le\sum_i|z_i|\|f_i\|_\flat$.

Lemma 114.5. For any flow beginning at $t_0<t$, $1\le t\le t_*$, and $V\in\mathsf{Cut}_t$, the function $z\mapsto\Phi_t^{w_z}(V)$ is holomorphic on $D^\flat_{2\varrho}$, and

$$|\Phi_t^{w_z}(V)-\Phi_t^0(V)|\le1/4\qquad(z\in D_t^{\mathrm{rd}}).\tag{2182}$$

■

Proof. Fix a critical representative $U = U_t(V)$, independent of z. The finite sum $\Psi(w) = S[\mathfrak{f}_t[w]](U)$ is holomorphic and bounded by m_t^S on Src. The exponent difference is $L^{K-t}A^{\eta_t}(w_z, U)+\Psi(w_z)-\Psi(0)$. The cut action bound makes the first term at most $M_t^{\mathrm{rd}}\|w_z\|_\flat < 1/8$. Schwarz's lemma on the complex line through w_z, whose source radius is $2\varrho/\|w_z\|_\flat$, bounds the second term by $\mathsf{m}_t^S\|w_z\|_\flat/\varrho < 1/8$. Their sum proves the assertion. □

For $n \geq 2$ define the physical support spread

$$d_\infty(\underline{f}) = \min_{x_i \in \operatorname{supp} f_i} \max_{i,i'} |x_i - x_{i'}|_\infty.$$

Compactness shows that the minimum is attained and that it is positive exactly when the supports have no common point. For families with the norm and locality properties of the flows, put

$$c_*(z) = \sum_{j=j_0}^{j_1} \sum_{X,z'} E_*^{(j)}[w_z](X, \mathbf{1}, z'), \qquad 1 \leq j_0 \leq j_1 \leq K.$$

Lemma 114.6. The function c_* is holomorphic on $D_{2\varrho}^\flat$ and $|c_*| \leq 2E_0^{\mathrm{s}} C_{\mathrm{E},0} N_{j_0}$ there. If $n \geq 2$ and $d_\infty L^{K-j_1} \geq 6M$, then

$$|\partial_1 \cdots \partial_n c_*(0)| \leq \epsilon_{\mathrm{mx}}(j_1), \qquad \epsilon_{\mathrm{mx}}(j_1) = 2e^3 E_0^{\mathrm{s}} C_{\mathrm{E},0} N_{j_1} e^{-d_\infty L^{K-j_1}/M} \prod_i \frac{n\|f_i\|_\flat}{2\varrho}. \tag{2183}$$

■

Proof. Each member is bounded by $E_0^{\mathrm{s}} e^{-(\kappa+1)d_j(X)}$. Sum first over domains containing a fixed site and then over levels, using $N_j = L^{-3(j-j_0)}N_{j_0}$, to obtain the first bound. All sums are finite and holomorphic.

Apply Cauchy's formula on the polydisc with radii $2\varrho/(n\|f_i\|_\flat)$. A differentiated member vanishes unless, for every i, X contains a plaquette at whose barycentre f_i is nonzero: otherwise locality makes it independent of that coefficient. Choose one such plaquette for each i. Their corners are in X, are within $\eta_j/2$ of their barycentres, and have mutual sup distance at most $M(d_j(X)+2)$ in level-j units. Hence a contributing domain satisfies

$$d_j(X) \geq d_\infty L^{K-j}/M - 3 =: D_j \geq 3.$$

The rooted tail and Cauchy's formula bound its level sum by

$$T_j = E_0^{\mathrm{s}} C_{\mathrm{E},0} N_j e^{3-d_\infty L^{K-j}/M} \prod_i \frac{n\|f_i\|_\flat}{2\varrho}.$$

Since $T_{j-1}/T_j \leq L^3 e^{-6(L-1)} \leq 1/2$, the sum is at most $2T_{j_1}$, proving (2183). □

115. Discarding the large-field histories

We now use the calibrated constants and canonical auxiliary data of Theorem D together with the source conditions above. Fix $0 < \eth < 1$ before decreasing γ, and set

$$c' = (1-\eth)c_\star, \quad C_{\mathrm{maj}}^\Sigma = C_{\mathrm{maj}} + (6L^3+1)\log 2, \quad E_0^{\mathrm{s}} = E_0, \quad N_\star^{\mathrm{s}} = \frac{L}{L-1}(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp),$$

$$\varrho = \min\left\{ \frac{20\mu_\star}{\sqrt{3}K_{\mathrm{L}}^\vee \gamma}, \frac{\sigma_{\mathrm{src}}}{2}(\gamma^{-2} - \bar{B}) \right\} > 0.$$

Require $\log \gamma^{-2}$ to exceed the case-II threshold of Proposition 67.6 at $c = c_\star, \eth$ and the summability threshold $\max\{Y_1, Y^{\mathrm{S}}(c')\}$ preceding (1451). All these choices precede K, m, b'. The data are

$$K \geq 2, \qquad \mathsf{b}' \geq \gamma^{-2}, \qquad MR(\mathsf{b}') \leq L^m, \qquad \ell = 2L^m, \qquad 1 \leq t_* \leq K-1, \quad x_{t_*} \geq \tau_\natural.$$

The calibration also includes the global radius, localization and smallness conditions used in the preceding sections.

Lemma 115.1. For the real constructed history masses and $1 \le t \le K$,

$$\sum_{h\in\mathcal{H}_K:\, Z_t(h)\neq\emptyset} e^{-E_{(K)}}\mathsf{t}(h) \le 2M^{-3}N_t e^{-c'p_0(g_t)^2}\Pi_K(C^{\Sigma}_{\mathrm{maj}}, c'). \tag{2184}$$

■

Proof. The actual masses satisfy the summation input (1452), (1483). In the proof of Proposition 67.5, they are bounded, after multiplication by $e^{-C_{\mathrm{maj}}N_K}$, by

$$\prod_{j=1}^{K}(2^{-n_j^{\mathrm{bit}}}\zeta_j)^{\#W_j}, \qquad \zeta_j = e^{-c'p_0(g_j)^2-2p_0(g_j)},$$

where W_j is the exact set of created cells. The entropy bound (1447) cancels the binary factors even when the sum is restricted to prescribed creation sets. The region Z_t is nonempty exactly when some W_j with $j \le t$ is nonempty: creations persist in later regions, and tracing any nonempty region backwards through the hull recursion reaches a creation. Use the union bound over these possible marked cells in the finite subset product. The restricted sum is at most

$$\Pi_K(C_{\mathrm{maj}}, c')\sum_{j\le t}\#\mathcal{P}_j e^{-c'p_0(g_j)^2}.$$

The proof of (1451), stopped at t, gives

$$\sum_{j\le t}\#\mathcal{P}_j e^{-c'p_0(g_j)^2} \le M^{-3}N_t\sum_{j\le t}L^{3(t-j)}e^{-c'p_0(g_j)^2} \le 2M^{-3}N_t e^{-c'p_0(g_t)^2}.$$

Increase the first argument of Π_K to $C^{\Sigma}_{\mathrm{maj}}$ to obtain (2184). □

Write $V_j = \mathcal{M}^j U$ and define the bare-field weight

$$\mathfrak{w}_t^{\mathrm{sf}}(U) = \mathbf{1}_{\mathsf{In}_t}(V_t)\prod_{j=0}^{t-1}[\mathbf{1}_{\mathsf{Cut}_j}\mathfrak{G}_j[x_j]\chi_j^{\mathrm{sf}}](V_j).$$

It is bounded, nonnegative and Borel. Put

$$R_t = 2M^{-3}N_t\exp\left\{-c'A_0^2(\log x_t)^{2\mathfrak{p}_0} + (C^{\Sigma}_{\mathrm{maj}} + 2 + C_{\mathrm{low}}(\mathsf{b}'))\ell^3\right\}. \tag{2185}$$

Here $C_{\mathrm{low}}(x) = c_{\mathrm{low}} + (9/2)\log x$ is the same lower-bound constant as in proposition 74.4.

Proposition 115.2. For $1 \le t \le t_*$ and every admitted source,

$$(\mathcal{M}^t)_*(\mathfrak{w}_t^{\mathrm{sf}}\rho_0^w\, dU) = \mathbf{1}_{\mathsf{In}_t}\nu_t^w. \tag{2186}$$

For every bounded, nonnegative, gauge-invariant F,

$$\int F\mathfrak{w}_{h_\emptyset^{(t)}}\rho_0\, dU \le \int F\mathfrak{w}_t^{\mathrm{sf}}\rho_0\, dU \le \int F\rho_0\, dU. \tag{2187}$$

Consequently, with $Z = \int\rho_0\, dU$,

$$0 \le Z - \int\mathfrak{w}_t^{\mathrm{sf}}\rho_0\, dU \le R_t Z. \tag{2188}$$

Let $\mathsf{op}_n = \sum_\pi(\#\pi)!$, the sum over partitions of $\{1,\dots,n\}$. If $R_t \le 1/2$ and $T(z) = \int\mathfrak{w}_t^{\mathrm{sf}}\rho_0^{w_z}\, dU$, then T is entire, $T(0) \ge Z/2$, and

$$\left|S^T_{n;K,\ell,L^K\mathsf{b}'}(\underline{f}) - \partial_1\cdots\partial_n\log T(0)\right| \le 4\mathsf{op}_n R_t\prod_i(6\ell^3L^{4K}\|f_i\|_\infty). \tag{2189}$$

■

Proof. The image identity follows by induction from the recursion for ν_j^w and the rule for pushing forward a measure multiplied by a function of its image. At level zero the cut density is $\mathbf{1}_{\mathsf{Cut}_0}\rho_0^w$; at the final level use $\mathsf{In}_t \subset \mathsf{Cut}_t$.

The empty-history weight is the displayed product for $\mathfrak{w}_t^{\mathrm{sf}}$ with three additional zero-one density and compatibility tests at each step. If that weight is nonzero, the density tests imply $\varpi(V_j) < 2\varepsilon(x_j)$ for $j \geq 1$ and $\varpi(V_0) < \varepsilon(x_0)$. Hence all its fields satisfy the cuts and its last field satisfies In_t. The two weights then coincide. This proves the first inequality of (2187) pointwise.

For the second inequality first remove all cuts and fluctuation tests. Let

$$J_j = \int F\rho_0 \prod_{k<j} \mathfrak{G}_k[x_k](\mathcal{M}^k U)\, dU.$$

Gauge transformations on the noncentral sites of the level-$(j-1)$ blocks lift to transformations constant on the corresponding bare blocks. They preserve $F\rho_0$ and every preceding gauge factor. Average the last gauge factor over their product Haar measure. Its defining normalization makes the average one on its averaging domain and zero off that domain. Thus $J_j \leq J_{j-1}$. Fubini applies to bounded nonnegative Borel functions on finite compact products. Iteration gives $J_t \leq J_0$, as asserted; no pointwise bound $\mathfrak{w}_t^{\mathrm{sf}} \leq 1$ is needed.

The history masses add to Z, and the descendants of the empty history through level t add to its mass. Those are exactly the histories with $Z_t = \emptyset$. Apply lemma 115.1, $\Pi_K(C_{\mathrm{maj}}^{\Sigma}, c') \leq e^{(C_{\mathrm{maj}}^{\Sigma}+2)\ell^3}$, and the lower bound $Z \geq e^{E_{(K)} - C_{\mathrm{low}}(\mathsf{b}')\ell^3}$. The vacuum constants cancel, proving (2188).

On the algebra of bounded gauge-invariant Borel functions, the two functionals defined by ρ_0 and $\mathfrak{w}_t^{\mathrm{sf}}\rho_0$ are positive, and their difference is positive by (2187). Therefore their difference on F has modulus at most $R_t Z\|F\|_\infty$. Their normalized moments differ by at most $4R_t\|F\|_\infty$ when $R_t \leq 1/2$. For the bounded observables $Y_i = \mathcal{O}(f_i)$ one has $|Y_i| \leq 6\ell^3 L^{4K}\|f_i\|_\infty$. In each partition term of the cumulant formula, telescope the product of normalized moments; there are $\#\pi$ differences, and every remaining factor is bounded by the product of its observable bounds. Multiplication by $(\#\pi - 1)!$ and summation proves (2189). Finally $\rho_0^{w_z} = \rho_0 e^{\sum_i z_i Y_i}$ with bounded Y_i, so dominated differentiation on compact coefficient sets proves entireness and identifies the logarithmic derivative with this cumulant. Equation (2186) gives $T(z) = \nu_t^{w_z}(\mathsf{In}_t)$ on the source class. □

116. Comparison of adjacent cutoffs

Fix ℓ and a label b in the calibrated range. The lattices $\mathrm{A} = (\ell, K)$ and $\mathrm{B} = (\ell, K+1)$ use the lines $x_j^{\mathrm{A}} = L^{K-j}\mathsf{b}$ and $x_j^{\mathrm{B}} = L^{K+1-j}\mathsf{b}$. At equal depth d their configuration lattices, Haar measures, averaging maps, gauge factors and fluctuation thresholds coincide; their critical backgrounds need not coincide. Write

$$x_{[d]} = L^d\mathsf{b}, \qquad N_{[d]} = \ell^3 L^{3d}, \qquad y_{[d]} = (\log x_{[d]})^{2\mathsf{p}_0}.$$

Choose integers $1 \leq u < s < K$, put $t_0 = K - s$, $t_1 = K - u$, and require $x_{[u]} \geq \Theta_\star$. The partner of a level k of A is $k+1$ of B; a subscript (k) refers to this common depth.

On each lattice run the chain and a classical start at depth s to depth u. On the common final configuration space $\mathcal{X} = \mathcal{V}(\Lambda_{u,\ell})$ set

$$I^\bullet(z) = \int_{\mathsf{In}_{[u]}} e^{\Phi^{\bullet,w_z}}\, dV, \quad \widehat{I}^\bullet(z) = \int_{\mathsf{In}_{[u]}} e^{\widehat{\Phi}^{\bullet,w_z}}\, dV, \quad T^\bullet = e^{c^\bullet} I^\bullet, \quad \widehat{T}^\bullet = e^{\widehat{c}^\bullet}\widehat{I}^\bullet.$$

Here $c^\bullet$ sums all the chain increments through the final level, while $\widehat{c}^\bullet$ sums the increments strictly after the classical start. These identities are (2173). Define the contact function

$$\mathfrak{c}^\bullet(z) = \sum_{j=1}^{t_1^\bullet} \sum_{X,z'} E^{(j),\bullet}[w_z](X, \mathbf{1}, z') - \sum_{j=t_0^\bullet+1}^{t_1^\bullet} \sum_{X,z'} \widehat{E}^{(j),\bullet}[w_z](X, \mathbf{1}, z').$$

Then $c^\bullet - \widehat{c}^\bullet - \mathfrak{c}^\bullet$ is independent of z: it is the sum of the real Gaussian normalization logarithms up to the starting level.

Use the common radii $r_i = r_i(t_1)$ of lemma 114.5, and put

$$\epsilon_{\mathrm{fg}} = C_{\mathrm{old}} C_{\mathrm{fg}} \tfrac{L}{L-1} E_0 N_{[u]} L^{-(s-u)}, \tag{2190}$$

$$\epsilon_{\mathrm{cl}} = 4K_w c_\rho^2 A_0^2 (1 + 2\varrho/\mathsf{b}) y_{[K]} N_{[s]} \sqrt{K} L^{-(K-s)/2}, \tag{2191}$$

$$\epsilon_{\mathrm{sh}} = 3(s-u) N_{[s]} \exp\{C_{\mathrm{sh}} - \kappa_{\mathrm{b}} A_1^2 y_{[u]}/4\}, \tag{2192}$$

$$\epsilon_3 = 4\epsilon_{\mathrm{cl}} + 6\epsilon_{\mathrm{sh}}, \qquad M_c = 4E_0 C_{\mathrm{E},0} N_{[s]}. \tag{2193}$$

In this section impose

$$\epsilon_{\mathrm{fg}} \le 1/8, \quad \epsilon_{\mathrm{cl}}, \epsilon_{\mathrm{sh}} \le 1/20, \quad 256 K_u c_\rho^2 A_0^2 N_{[s]} L^{-(K-s)} \le \gamma_{\mathrm{r}} A_1^2. \tag{2194}$$

Proposition 116.1. At every real point x of the polydisc $|z_i| < r_i$, $\widehat{T}^{\mathrm{A}}(x), \widehat{T}^{\mathrm{B}}(x) > 0$ and

$$\left| \widehat{T}^{\mathrm{B}}(x)/\widehat{T}^{\mathrm{A}}(x) - 1 \right| \le \epsilon_3 \le 1/2. \tag{2195}$$

■

Proof. On $\mathcal{Y} = \mathcal{V}(\Lambda_{s,\ell})$ write $W_k = \mathcal{M}^{k-t_0} W$, and let

$$k^\bullet(W) = \mathbf{1}_{\mathsf{In}_{[u]}}(W_{t_1}) \prod_{k=t_0}^{t_1-1} [\mathbf{1}_{\mathsf{Cut}^\bullet_{(k)}} \mathfrak{G}_{(k)} \chi^\bullet_{(k)}](W_k),$$

$$\widehat{F}^\bullet_z(W) = \mathbf{1}_{\mathsf{Cut}^\bullet_{(t_0)}}(W) \exp[-A^{\eta_{t_0}^\bullet}(L^s(\mathsf{b} - w_z), U^\bullet_{t_0}(W))].$$

Iterating the image formula gives $\widehat{T}^\bullet(z) = \int k^\bullet \widehat{F}^\bullet_z \, dW$. For real x the start is strictly positive on its cut. The positive chart kernel propagates nonnegativity, while the nonzero exponential density formula makes the final density positive on its cut. Since the common inner set is open and contains $\mathbf{1}$, it has positive Haar measure. Thus both final integrals are strictly positive.

We first bound the mass where the two weights differ. Suppose $k^\bullet(W) > 0$ and $k^\circ(W) = 0$. Starting from $W_{t_1} \in \mathsf{In}_{[u]}$, cut stability shows that there is a last index k with $\chi^\circ_{(k)}(W_k) = 0$, and its next field lies in both next-level cuts. Otherwise all the other cuts would follow downwards from their fluctuation tests, and k° would be positive. By proposition 113.11 and (2194), at each nonpivot bond the two background origins differ by an angle at most

$$8 c_\rho \varepsilon(x_{(k)}) (K_u \gamma_{\mathrm{r}}^{-1} L^{-k} N_k)^{1/2} \le \delta(x_{(k)})/2.$$

If all fluctuations of lattice $\bullet$ were smaller than $\delta(x_{(k)})/2$, the angular triangle inequality would make every fluctuation of lattice $\circ$ defined and smaller than $\delta(x_{(k)}) < \pi$. Its test would equal one, a contradiction. Thus some bond has a fluctuation at least half the threshold.

Insert the indicator of this event at its step in the positive classical-start measure. By proposition 114.4 with $a = \mathfrak{b}(x_{(k)})/2$, the next measure is bounded by $\exp(C_{\mathrm{sh}} - \kappa_{\mathrm{b}} \mathfrak{b}(x_{(k)})^2/4)$ times the ordinary next measure. All later positive cut-and-image operations preserve this inequality. There are $s-u$ steps, at most $3N_{[s]}$ nonpivot bonds at each step, and $\mathfrak{b}(x_{(k)})^2 \ge A_1^2 y_{[u]}$. The union bound consequently gives both estimates

$$\int k^\bullet \mathbf{1}_{\{k^\circ = 0\}} \widehat{F}^\bullet_x \, dW \le \epsilon_{\mathrm{sh}} \widehat{T}^\bullet(x). \tag{2196}$$

This argument uses the image identity also for the marked measure, obtained by the same induction as the unmarked one; hence it bounds the final mass, without an assumption that the later gauge factors are at most one.

Where both weights are positive they equal the product of the common gauge factors. The two starts there satisfy $\widehat{F}^{\mathrm{B}}_x = \widehat{F}^{\mathrm{A}}_x e^{q_x}$. Apply proposition 113.9 at $t = t_0$ with $\varepsilon_{\mathrm{r}} = 2c_\rho \varepsilon(x_{[s]})$ and $c_x = L^s(\mathsf{b} - w_x)$. Since $\|w_x\|_\flat < \varrho/8$, $\|c_x\|_{(t_0)} \le L^s(\mathsf{b} + 2\varrho)$. The comparison estimate gives

$$|q_x| \le K_w \|c_x\|_{(t_0)} \varepsilon_{\mathrm{r}}^2 N_{[s]} \sqrt{t_0} L^{-t_0/2} \le \epsilon_{\mathrm{cl}}.$$

Split the two integrals into their common positive set and their two exceptional sets. The latter have masses bounded by (2196); on the common set use $|e^{q_x} - 1| \le 2\epsilon_{\mathrm{cl}}$. First,

$$(1 - \epsilon_{\mathrm{sh}}) \widehat{T}^{\mathrm{B}}(x) \le e^{\epsilon_{\mathrm{cl}}} \widehat{T}^{\mathrm{A}}(x), \qquad \widehat{T}^{\mathrm{B}}(x) \le 2 \widehat{T}^{\mathrm{A}}(x).$$

Then the total variation of the two weighted densities is at most $(2\epsilon_{\mathrm{cl}} + 3\epsilon_{\mathrm{sh}}) \widehat{T}^{\mathrm{A}}(x)$. This implies the weaker bound (2195), with the stated constants. □

The real estimate does not by itself control derivatives in complex sources. The following elementary two-constants argument supplies the required passage.

Lemma 116.2. If f is holomorphic on the polydisc $|z_i| < r_i$, bounded there by B, and bounded at its real points by $0 < \epsilon \le B$, then, for $0 < \sigma \le 1$,

$$|\partial_1 \cdots \partial_n f(0)| \le \epsilon^{1-\sigma} B^\sigma (4n/(\pi\sigma))^n \prod_i r_i^{-1}. \tag{2197}$$

■

Proof. First take a one-variable unit disc. In its upper half put $h(\zeta) = \frac{2}{\pi i} \operatorname{Log} \frac{1+\zeta}{1-\zeta}$. Its real part lies between zero and one, is zero on the diameter and tends to one on the circular arc. For $\nu > 0$ apply the maximum principle to

$$f(\zeta) \epsilon^{-1} e^{-\log(B/\epsilon) h(\zeta)} ((1 - \zeta^2)/2)^\nu.$$

The boundary upper limits of its modulus are at most one; the last factor makes those at ± 1 zero. Let $\nu \downarrow 0$. For $|\zeta| \le \tau < 1$,

$$\Re h(\zeta) = \frac{2}{\pi} \arctan \frac{2\Im\zeta}{1 - |\zeta|^2} \le \delta_\tau := \frac{4}{\pi} \arctan \tau.$$

Reflection gives the same lower-half bound, so $|f| \le \epsilon^{1-\delta_\tau} B^{\delta_\tau}$ on the smaller disc. Iterating in the n variables gives $|f| \le \epsilon^{(1-\delta_\tau)^n} B^{1-(1-\delta_\tau)^n}$ on $|z_i| \le \tau r_i$. Choose $\tau = \pi\sigma/(4n)$. Then $\delta_\tau \le \sigma/n$ and $(1 - \delta_\tau)^n \ge 1 - \sigma$. Cauchy's formula on the smaller polydisc proves the result. The maximum principle with boundary upper limits follows by taking a maximizing sequence in the bounded domain: a limit on the boundary obeys the assumed bound, whereas an interior maximum forces constancy. □

Proposition 116.3. On the common source polydisc the two chain functions are zero-free. Moreover, for $0 < \sigma \le 1$,

$$\begin{aligned} &\left| \partial^n \log T^{\mathrm{B}}(0) - \partial^n \log T^{\mathrm{A}}(0) - \partial^n (\mathfrak{c}^{\mathrm{B}} - \mathfrak{c}^{\mathrm{A}})(0) \right| \\ &\quad \le \left[8\epsilon_{\mathrm{fg}} + 2^{n+1} (4n/(\pi\sigma))^n \epsilon_3^{1-\sigma} (1 + 2M_c)^\sigma \right] \prod_i r_i^{-1}. \end{aligned} \tag{2198}$$

Here $\partial^n = \partial_1 \cdots \partial_n$. ■

Proof. The integrals are holomorphic by dominated differentiation: the quarter-disc bound supplies the integrable majorant $e^{1/4} |e^{\Phi^0}|$ for each of them. The positive classical-start final density at zero shows that $e^{\widehat{\Phi}^0}$ is a fixed nonzero complex scalar times a positive function. Thus $\widehat{I}(z)/\widehat{I}(0)$ is an average of $e^{\widehat{\Phi}^{w_z} - \widehat{\Phi}^0}$ with respect to a probability measure. The quarter-disc bound puts it in the convex region

$$\Re\zeta \ge e^{-1/4} \cos(1/4), \qquad |\zeta| \le e^{1/4}, \qquad |\Im\zeta| \le \tan(1/4) \Re\zeta.$$

In this region $|\operatorname{Log}\zeta| \le 1/2$. Also (2178) implies

$$|I/\widehat{I} - 1| \le \frac{e^{1/2}}{\cos(1/4)}(e^{\epsilon_{\mathrm{fg}}} - 1) \le 2\epsilon_{\mathrm{fg}}.$$

Consequently the holomorphic functions $h^\bullet = \operatorname{Log}(I^\bullet/\widehat{I}^\bullet)$ and $l^\bullet = \operatorname{Log}(\widehat{I}^\bullet/\widehat{I}^\bullet(0))$ satisfy $|h^\bullet| \le 4\epsilon_{\mathrm{fg}}$, $|l^\bullet| \le 1/2$, and $\partial^n \log T^\bullet = \partial^n(c^\bullet + h^\bullet + l^\bullet)$.

The contact bound lemma 114.6 on the classical-start families gives $|\widehat{c}^\bullet(z) - \widehat{c}^\bullet(0)| \le M_c$. Define

$$H = (\widehat{c}^{\mathrm{B}} - \widehat{c}^{\mathrm{B}}(0)) - (\widehat{c}^{\mathrm{A}} - \widehat{c}^{\mathrm{A}}(0)) + l^{\mathrm{B}} - l^{\mathrm{A}}.$$

Then $H(0) = 0$, $|H| \le 1 + 2M_c$, and on the real polydisc

$$H(x) = \log(\widehat{T}^{\mathrm{B}}(x)/\widehat{T}^{\mathrm{A}}(x)) - \log(\widehat{T}^{\mathrm{B}}(0)/\widehat{T}^{\mathrm{A}}(0)).$$

Indeed their exponentials agree, and the continuous difference takes values in $2\pi i\mathbb{Z}$ and is zero at the origin. Equation (2195) bounds this real value by $4\epsilon_3$. Apply lemma 116.2 with $\epsilon = 4\epsilon_3$ and $B = \max(2, 1 + 2M_c)$. The resulting constant satisfies $4^{1-\sigma}2^\sigma \le 2^{n+1}$. Finally Cauchy's formula bounds $\partial^n(h^{\mathrm{B}} - h^{\mathrm{A}})$ by $8\epsilon_{\mathrm{fg}}\prod_i r_i^{-1}$. The source-independent difference between $c^\bullet - \widehat{c}^\bullet$ and $\mathfrak{c}^\bullet$ has zero mixed derivative. Adding the estimates proves (2198). □

117. Summation of the cutoff differences

Write $S^T_{n;K}(\underline{f}) = S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f})$ and retain the calibrated setting of section 115. For the comparison at cutoffs $K, K+1$, choose

$$u = \left\lceil \frac{K}{180(3n+4)} \right\rceil, \qquad s = \lceil K/9 \rceil, \qquad \mathsf{p}_n = 2\mathfrak{p}_0(n+1) + \tfrac{1}{2}.$$

At this point the label is in a compact interval $J = [\mathsf{b}_-, \mathsf{b}_+]$ satisfying

$$\mathsf{b}_- \ge \gamma^{-2}, \qquad MR(\mathsf{b}_+) \le L^m, \qquad \ell = 2L^m, \quad m \ge 1. \tag{2199}$$

All constants chosen before the cutoff, including the source radius ϱ, are fixed throughout J.

Theorem AR. For an arbitrary nonzero Lipschitz tuple of order $n \ge 1$, assume the comparison hypotheses of proposition 116.1 and $R_{t_1} \le 1/2$, where $t_1 = K - u$. Then

$$\begin{aligned} &\left|S^T_{n;K+1}(\underline{f}) - S^T_{n;K}(\underline{f}) - \partial^n(\mathfrak{c}^{\mathrm{B}} - \mathfrak{c}^{\mathrm{A}})(0)\right| \\ &\quad \le \left[8\epsilon_{\mathrm{fg}} + q_n\epsilon_3^{79/80}(1 + 2M_c)^{1/80}\right]\prod_i r_i(t_1)^{-1} \\ &\qquad + 8\mathsf{o}\mathsf{p}_n R_{t_1}\prod_i(6\ell^3 L^{4(K+1)}\|f_i\|_\infty). \end{aligned} \tag{2200}$$

Here $q_n = 2^{n+1}(320n/\pi)^n$. If also $n \ge 2$ and $d_\infty(\underline{f})L^u \ge 6M$, then $|\partial^n\mathfrak{c}^\bullet(0)| \le 2\epsilon_{\mathrm{mx}}(t_1)$ for each $\bullet$.

There are constants $C_n(J) > 0$, $K_0(\ell, n, J)$, and, for every $d > 0$, $K_1(\ell, n, J, d) \ge K_0$, such that all the preceding smallness hypotheses hold for $K \ge K_0$, uniformly in $\mathsf{b} \in J$. For $n \ge 2$, $K \ge K_1$, and all nonzero tuples with $d_\infty(\underline{f}) \ge d$,

$$|S^T_{n;K+1}(\underline{f}) - S^T_{n;K}(\underline{f})| \le C_n(J)\ell^{3(n+1)}K^{\mathsf{p}_n}L^{-K/10}\prod_i \|f_i\|_\sharp. \tag{2201}$$

Here $C_n(J)$ depends only on the fixed constants, n, and the endpoints of J. The thresholds have precisely the additional dependencies displayed, and K_1 may be chosen nonincreasing in d. ■□

Proof. Apply proposition 115.2 at the common final depth u to both lattices. Their discarding factors are the same, since the final coupling, site count, and physical volume are the same. The two errors add to the last term of (2200). Apply the holomorphic derivative estimate (2198) with $\sigma = 1/80$. Its contact term is exactly the one displayed here, since the omitted Gaussian normalization logarithms

are source independent. This proves (2200). Each $\mathfrak{c}^{\bullet}$ is a difference of two contact sums ending at depth u; lemma 114.6 bounds its derivative by $2\epsilon_{\mathrm{mx}}(t_1)$.

We give the uniform estimates, including the choice of thresholds. Put $y_{[j]} = (\log(L^j \mathsf{b}))^{2\mathfrak{p}_0}$, $N_{[j]} = \ell^3 L^{3j}$, and

$$\overline{C}_M = \max\left\{\varrho^{-1}, \frac{3c_\rho^2 A_0^2}{2\mathsf{b}_-}, \varrho^{-1} C_{\mathrm{old}} \frac{L}{L-1} E_0\right\}, \qquad \overline{y}_K = (\log(L^K \mathsf{b}_+))^{2\mathfrak{p}_0}.$$

The source radii of lemma 114.5 satisfy

$$r_i(t_1)^{-1} \le 8n \overline{C}_M \overline{y}_K N_{[u]} \|f_i\|_\sharp. \tag{2202}$$

Indeed the three entries in $M_{t_1}^{\mathrm{rd}}$ are bounded by $\overline{C}_M y_{[K]} N_{[u]}$: for its classical entry use $(e_{t_1}^{\mathrm{c}})^2 L^u = c_\rho^2 A_0^2 y_{[u]}/\mathsf{b}$, and for the old-action entry use $m_{t_1}^S = C_{\mathrm{old}} L(L-1)^{-1} E_0 N_{[u]}$. Also $1 \le y_{[u]} \le y_{[K]} \le \overline{y}_K$ and $\|f_i\|_\flat \le \|f_i\|_\sharp$.

Replace $y_{[K]}$ by $\overline{y}_K$ and $1 + 2\varrho/\mathsf{b}$ by $1 + 2\varrho/\mathsf{b}_-$ in ϵ_{cl} to define $\overline{\epsilon}_{\mathrm{cl}}$. Define the other two majorants by

$$\overline{\epsilon}_{\mathrm{sh}} = 3(s-u) N_{[s]} \exp\left[C_{\mathrm{sh}} - \tfrac{1}{4}\kappa_b A_1^2 (\log(L^u \mathsf{b}_-))^{2\mathfrak{p}_0}\right],$$

$$\overline{R} = 2M^{-3} N_{[u]} \exp\left[-c' A_0^2 (\log(L^u \mathsf{b}_-))^{2\mathfrak{p}_0} + (C_{\mathrm{maj}}^{\Sigma} + 2 + C_{\mathrm{low}}(\mathsf{b}_+))\ell^3\right].$$

They dominate the corresponding errors throughout J; the last assertion uses that $C_{\mathrm{low}}(b)$ is increasing. Since $\log(L^u \mathsf{b}_-) \ge K \log L/[180(3n+4)]$ and $2\mathfrak{p}_0 \ge 2$, both majorants tend to zero after multiplication by any $K^a e^{a'K}$.

The ceiling bounds give

$$\begin{gathered} u(3n+4) - s \le -19K/180 + 3n + 4, \\ 3s - \tfrac{79}{160}(K-s) + 3un \le -K/10 + 7/2 + 3n, \\ 4u - s \le -K/10 + 4, \qquad (7s-K)/2 \le -K/9 + 7/2, \qquad 4s - K \le -5K/9 + 4. \end{gathered} \tag{2203}$$

For example, in the second inequality the coefficient of s is $559/160$; replacing s by $K/9+1$ gives $-19K/180 + 559/160$, while $3un < K/180 + 3n$. The others follow directly from $K/[180(3n+4)] \le u < K/[180(3n+4)] + 1$ and $K/9 \le s < K/9 + 1$. Thus $1 \le u < s < K$, $L^u \mathsf{b}_- \ge \Theta_\star$, $\epsilon_{\mathrm{fg}} \le 1/8$, and the axial comparison condition hold eventually. The latter's left side is a constant times L^{4s-K}. Also $\overline{\epsilon}_{\mathrm{cl}}$ is bounded by a constant times $K^{2\mathfrak{p}_0+1/2} L^{-K/9}$, so that $\overline{\epsilon}_{\mathrm{cl}}, \overline{\epsilon}_{\mathrm{sh}} \le 1/20$ and $\overline{R} \le 1/2$ eventually.

Here is an explicit constant for the asserted rate. With $h = \log(L\mathsf{b}_+) > 1$, take

$$\begin{aligned} C_n(J) = {} & 4(8n\overline{C}_M)^n h^{2\mathfrak{p}_0(n+1)} L^{3n+4} \\ & \times \left[8C_{\mathrm{old}} C_{\mathrm{fg}} \frac{L}{L-1} E_0 + q_n \max\{1, 16 K_w c_\rho^2 A_0^2 (1 + 2\varrho/\mathsf{b}_-)\}(1 + 8E_0 C_{\mathrm{E},0})\right]. \end{aligned} \tag{2204}$$

Let $P = (8n\overline{C}_M \overline{y}_K N_{[u]})^n$. Split the right side of (2200), after removing $\prod_i \|f_i\|_\sharp$, into

$$8\epsilon_{\mathrm{fg}} P, \quad q_n (4\overline{\epsilon}_{\mathrm{cl}})^{79/80} (1 + 2M_c)^{1/80} P, \quad q_n (6\overline{\epsilon}_{\mathrm{sh}})^{79/80} (1 + 2M_c)^{1/80} P, \quad 8\mathsf{o}\mathsf{p}_n \overline{R} (6\ell^3 L^{4(K+1)})^n.$$

This is valid because $(a+b)^{79/80} \le a^{79/80} + b^{79/80}$. The first two terms together are at most $\frac{1}{4} C_n(J) \ell^{3(n+1)} K^{\mathfrak{p}_n} L^{-K/10}$. For the first, use the first inequality of (2203) and $\overline{y}_K \le h^{2\mathfrak{p}_0} K^{2\mathfrak{p}_0}$. For the second, use

$$(1 + 2M_c)^{1/80} \le (1 + 8E_0 C_{\mathrm{E},0}) \ell^{3/80} L^{3s/80}.$$

This combines with the $79/80$ power of $\overline{\epsilon}_{\mathrm{cl}}$ to give exactly $\ell^3 L^{3s - 79(K-s)/160}$; the second inequality of (2203) gives the required exponential, and $\overline{y}_K^{n+1} K^{1/2} \le h^{2\mathfrak{p}_0(n+1)} K^{\mathfrak{p}_n}$ gives its polynomial.

Choose K_0 so that all preceding smallness conditions hold for every $K \ge K_0$, and the last two terms of the four-term display are at most a further quarter of the rate. Such a K_0 exists by the

superexponential decay just proved. This bounds (2200) by half the rate, uniformly on J and in the tuple. For $n \ge 2$ choose $K_1 \ge K_0$ so that, for every $K \ge K_1$,

$$dL^u \ge 6M, \qquad 8e^3 E_0 C_{\mathrm{E},0} \left(\frac{n}{2\varrho}\right)^n N_{[u]} e^{-dL^u/M} \le \tfrac{1}{2} C_n(J) \ell^{3(n+1)} K^{\mathfrak{p}_n} L^{-K/10}.$$

The left side decays doubly exponentially in K apart from $N_{[u]}$, so this choice exists. It bounds $4\epsilon_{\mathrm{mx}}(t_1)$ after the same norm product is removed. Taking the least eventual threshold makes K_1 nonincreasing in d. This proves the theorem. □

Corollary 117.1. Under (2199), fix $n \ge 2$ and $d > 0$. Put

$$C_n'(J) = C_n(J) \sum_{j \ge 0} (1+j)^{\mathfrak{p}_n} L^{-j/10}.$$

For $K' \ge K \ge K_1(\ell, n, J, d)$ and every nonzero tuple with $d_\infty(\underline{f}) \ge d$,

$$|S^T_{n;K'}(\underline{f}) - S^T_{n;K}(\underline{f})| \le C_n'(J) \ell^{3(n+1)} K^{\mathfrak{p}_n} L^{-K/10} \prod_i \|f_i\|_\sharp. \tag{2205}$$

Thus the whole cutoff sequence has a limit, uniformly in the label on J and on every set of tuples satisfying $d_\infty(\underline{f}) \ge d$ and a fixed upper bound on $\prod_i \|f_i\|_\sharp$. The limit is continuous in the label. The same limit is obtained whenever $K_j \to \infty$ and $a_{K_j}\beta_j \in J$ tends to the specified label. ■□

Proof. For $k = K + j$ one has $k \le K(1+j)$, whence

$$\sum_{k \ge K} k^{\mathfrak{p}_n} L^{-k/10} \le K^{\mathfrak{p}_n} L^{-K/10} \sum_{j \ge 0} (1+j)^{\mathfrak{p}_n} L^{-j/10}.$$

The last series converges by geometric domination. Telescoping (2201) proves (2205), and the uniform Cauchy criterion gives the limit. At fixed cutoff, the partition function and its source derivatives are entire functions of the bare coupling: the configuration space is compact and all plaquette actions are bounded. The real partition function is strictly positive. Hence every finite-cutoff cumulant is continuous in the positive real label, and its uniform limit on J is continuous there. Finally write $b_j = a_{K_j}\beta_j$. The difference from the asserted limit is bounded by the uniform cutoff error at b_j, plus the difference of the two limiting values at b_j and its limit. Both tend to zero. □

118. The limit at every positive physical coupling

Lemma 118.1. The constants can be chosen so that the calibrated setting used in theorem AR and corollary 117.1 holds on every label interval satisfying (2199). The choices precede the torus, the cutoff, and the labels. ■

Proof. Use the actual canonical auxiliary data and the ordered choice of Proposition 69.1. Its real background, response, Gaussian, and normalizer hypotheses are the global hypotheses imposed above. The fixed radii are chosen with $\alpha_0 \le a_{\mathrm{H}}$, and the decay and normalizer floors include $4\beta_{\mathrm{s}}\kappa \ge 1$ and the rates used in the rooted sums. These choices precede γ. In particular (1467) gives $E_0 \ge 2E_{\mathsf{N}}$. Take $E_0^{\mathrm{s}} = E_0$ and $N_\star^{\mathrm{s}} = L(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp)/(L-1)$.

Every coefficient in (2168) is now fixed. Its three small quantities are constant multiples of

$$\gamma(\log \gamma^{-2})^{\mathfrak{p}_0}, \qquad \exp[-\gamma_2 A_1^2 (\log \gamma^{-2})^{2\mathfrak{p}_0}/2],$$

and tend to zero as $\gamma \downarrow 0$. Thus the three inequalities hold after decreasing γ. Impose the direct-source half-radius requirement at the same stage; its left side is $C_{\mathcal{A}}\delta(\gamma^{-2})$. The response and classical calibration requirements are positive fixed-radius ceilings on ε and δ, so they hold uniformly above a finite threshold. The derived numbers $\tau_\natural, \Theta_\star$ are then finite; they are thresholds on the final intermediate coupling, which tends to infinity in the cutoff comparison.

Fix $\eth$ and the history constants before this last choice. The case-II and summability thresholds used in lemma 115.1 are finite and independent of γ. They can therefore be imposed on $\log \gamma^{-2}$ simultaneously. Also require $\gamma^{-2} > \bar{B}$ and $R(\gamma^{-2}) \geq C_L/(2L)$. If $MR(\mathsf{b}') \leq L^m$ and $\mathsf{b}' \geq \gamma^{-2}$, monotonicity of R gives $2L^{m+1} \geq C_L M$, as required in (2130). The line $x_j = L^{K-j}\mathsf{b}'$ has zero increments, is admissible, and has precisely the torus room required in Theorem D. The actual mass bounds and vacuum lower bound used in the discarding proof consequently apply to this same line and the same vacuum constant.

Finally $K_{\mathrm{Q}}^{\vee}, K_{\mathrm{L}}^{\vee}, w_\star, E_0$ and $K_{\mathbf{M}}^{\mathrm{f}}$ are positive and finite. Thus $\mu_\star, \sigma_{\mathrm{src}} > 0$, and the stated minimum defining ϱ is positive. Its two defining inequalities give (2169) for every $\mathsf{b}' \geq \gamma^{-2}$. At zero source the added family vanishes identically; the sourced recursion and the canonical real progression therefore coincide step by step. This verifies the interface used in the history comparison. The constants and this source radius involve no finite-lattice datum. □

For $s \in \mathbb{Z}$ and $m + s \geq 1$, let $f^{(s)}(x') = f(L^{-s}x')$ on $T_{L^s\ell}$. Dilation multiplies both the Euclidean and sup distances by L^s. Consequently

$$\|f^{(s)}\|_\infty = \|f\|_\infty, \qquad \operatorname{Lip} f^{(s)} = L^{-s} \operatorname{Lip} f, \qquad d_\infty(\underline{f}^{(s)}) = L^s d_\infty(\underline{f}). \tag{2206}$$

Supports are carried onto their dilates. In particular the absence of a common support point is preserved, and $\|f^{(s)}\|_\sharp \leq \|f\|_\sharp$ when $s \geq 0$.

Lemma 118.2. For $K, m \geq 1$, $-m < s < K$, $\beta > 0$, and every Lipschitz tuple, one has

$$S^T_{n;K,\ell,\beta}(\underline{f}) = L^{sn} S^T_{n;K-s,L^s\ell,\beta}(\underline{f}^{(s)}), \qquad a_{K-s}\beta = L^s a_K \beta. \tag{2207}$$

For every compact interval $J = [b_-, b_+] \subset (0, \infty)$, there is an integer $s \geq 0$ for which

$$L^s b_- \geq \gamma^{-2}, \qquad MR(L^s b_+) \leq L^{m+s}. \tag{2208}$$

One may take any sufficiently large integer satisfying

$$L^s \geq \max\{\gamma^{-2}/b_-, b_+\}, \qquad L^{m+s/2} \geq LM(4r/e)^r. \tag{2209}$$

■

Proof. For $s \geq 0$, (2207) is Lemma 3.3. Explicitly, dilation is a bijection of the bonds and plaquettes and preserves their orientations, plaquette products, and product Haar measure. Corresponding source coefficients agree because $f^{(s)}(L^s x_p) = f(x_p)$ and $a_{K-s}^{-1} L^s = a_K^{-1}$. Thus

$$Z_{K,\ell,\beta}\left[\sum_i z_i f_i\right] = Z_{K-s,L^s\ell,\beta}\left[\sum_i L^s z_i f_i^{(s)}\right].$$

Differentiating local logarithms at zero gives the factor L^{sn}. If $s < 0$, apply the same nonnegative-dilation identity with exponent $-s$ to the data $(K - s, m + s, \beta, \underline{f}^{(s)})$; its dilated functions are the original f_i. This proves the identity in both directions. Equation (2206) follows by taking infima over lifts for the distances and then the minimum over the product of supports.

For (2208), put $x = L^s b_+$ and suppose (2209). Then $x \geq \gamma^{-2} > e$. The least-power definition of R gives $R(x) < L(\log x)^r$. Also $\log x \leq 2s \log L = 4y$, where $y = s \log L/2 > 0$. Since $y^r \leq (r/e)^r e^y$, we obtain

$$MR(x) < LM(4r/e)^r L^{s/2} \leq L^{m+s}.$$

The other inequality in (2208) is already in (2209). Both conditions in (2209) hold for all sufficiently large s, since their right sides are fixed. □

Theorem AS. Fix $\ell = 2L^m$, $m \geq 1$. For every $\mathsf{b} > 0$, $n \geq 2$, and real Lipschitz functions $f_1, \dots, f_n$ with $\bigcap_i \operatorname{supp} f_i = \emptyset$, the limit

$$S_n^{(\ell,\mathsf{b})}(\underline{f}) = \lim_{K\to\infty} S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f}) \tag{2210}$$

exists along all positive integer cutoffs. Convergence is locally uniform in $\mathsf{b} \in (0, \infty)$.

More precisely, fix $J = [b_-, b_+] \subset (0,\infty)$, $d > 0$, and an integer $s \geq 0$ satisfying (2208). Put

$$C'_s = C'_n([L^s b_-, L^s b_+]), \qquad k_s = K_1(L^s\ell, n, [L^s b_-, L^s b_+], L^s d).$$

For every nonzero tuple with $d_\infty(\underline{f}) \geq d$, every $\mathsf{b} \in J$, and $K \geq k_s + s$,

$$\left|S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f}) - S^{(\ell,\mathsf{b})}_n(\underline{f})\right| \leq L^{sn} C'_s (L^s\ell)^{3(n+1)} (K-s)^{\mathsf{p}_n} L^{-(K-s)/10} \prod_i \|f_i\|_\sharp. \tag{2211}$$

For all $K \geq 1$, the finite-cutoff cumulant and its limit are bounded in modulus by

$$\mathsf{op}_n \prod_i (6\ell^3 L^{4(k_s+s)} \|f_i\|_\infty) + L^{sn} C'_s (L^s\ell)^{3(n+1)} k_s^{\mathsf{p}_n} L^{-k_s/10} \prod_i \|f_i\|_\sharp. \tag{2212}$$

The constants are independent of the tuple except for its lower spread bound d.

For every integer $s > -m$, the scaling law is

$$S^{(\ell,\mathsf{b})}_n(\underline{f}) = L^{sn} S^{(L^s\ell, L^s\mathsf{b})}_n(\underline{f}^{(s)}). \tag{2213}$$

If $K_j \to \infty$, $\beta_j > 0$, and $a_{K_j}\beta_j \to \mathsf{b} > 0$, then

$$S^T_{n;K_j,\ell,\beta_j}(\underline{f}) \longrightarrow S^{(\ell,\mathsf{b})}_n(\underline{f}). \tag{2214}$$

No monotonicity of the sequence K_j is required. ■□

Proof. If an entry is zero, multilinearity makes every cumulant zero. Otherwise compactness of the supports gives $d_\infty(\underline{f}) > 0$. For the dilated tuple, (2206) gives spread at least $L^s d$ and norms no larger than the original ones. The interval $[L^s b_-, L^s b_+]$ satisfies (2199) on $T_{L^s\ell}$. Apply corollary 117.1 there with cutoff $k = K - s$ and multiply its estimate by L^{sn} using (2207). This is the Cauchy estimate with the right side of (2211). It tends to zero uniformly in the label, so the whole sequence converges, and passing the second cutoff to infinity proves (2211).

For $K \geq k_s + s$, compare to cutoff $k_s + s$ by the same Cauchy estimate. The finite-cutoff moment bound used in proposition 115.2 bounds that term by the first term of (2212). For $K < k_s + s$ the same finite bound is even smaller. This proves the uniform bound for all cutoffs and, by passage to the limit, for the limit itself. Taking $J = \{\mathsf{b}\}$ proves existence at every positive label. Taking any compact interval proves local uniformity and hence continuity, since each finite-cutoff cumulant is continuous.

For any fixed integer $s > -m$, (2207) holds for all sufficiently large K. The index $K - s$ also runs through all sufficiently large integers, so passage to the limit gives (2213), including negative s. For the last assertion put $b_j = a_{K_j}\beta_j$. Eventually $b_j \in [\mathsf{b}/2, 2\mathsf{b}]$; subtract the limit at b_j and use the uniform cutoff estimate there. Continuity then replaces b_j by b. □

Proposition 118.3. The limits in theorem AS are real, symmetric, and multilinear on separated tuples. They are invariant under every cubic lattice isometry at any fixed scale. They are smooth in the positive physical inverse coupling, with

$$\partial^\nu_{\mathsf{b}} S^{(\ell,\mathsf{b})}_n(\underline{f}) = (-1)^\nu S^{(\ell,\mathsf{b})}_{n+\nu}(\underbrace{1,\dots,1}_{\nu}, \underline{f}). \tag{2215}$$

All derivatives of finite-cutoff cumulants converge to these derivatives locally uniformly in b, with (2211) at order $n + \nu$ and the same spread bound d. ■

Proof. Reality, symmetry, and multilinearity hold at each finite cutoff and pass to the limit. The domain is preserved under the asserted linear combinations: if only entry i changes, the common intersection of the other supports is disjoint from each of its two possible supports and hence from their union. A lattice isometry at a fixed scale permutes bonds and plaquettes at every sufficiently fine cutoff. Haar measure and the Wilson action are invariant; reindexing the source sum gives finite-cutoff invariance, and then invariance of the limit.

Adding constant entries 1 leaves the support intersection empty and does not change the norm product or the spread. For the last fact, the enlarged diameter is at least the old minimum; conversely choose a minimizing old point tuple and put all new points at its first point. At finite cutoff, differentiation in b inserts minus the constant-source observable because

$$\exp[-A^{a_K}(\mathsf{b}-w_z,U)] \quad \text{depends on } \mathsf{b} \text{ and a constant source } t \text{ through } \mathsf{b}-t.$$

The partition function and its derivatives are entire, and its real value at zero source is positive. On a local zero-free neighborhood its logarithm therefore gives

$$\partial_{\mathsf{b}}^{\nu} S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f}) = (-1)^{\nu} S^T_{n+\nu;K,\ell,L^K\mathsf{b}}(1,\dots,1,\underline{f}).$$

Both this sequence and its next derivative converge uniformly on each compact positive interval by theorem AS. Integrating the derivative identity on such an interval and passing to the limit proves that the first limit has the second as its derivative. Iterating proves (2215) and the stated uniform convergence with its rate. □

Corollary 118.4. Let $0 < g \le \gamma$ and let $\ell = 2L^m$, $m \ge 1$. The cutoff limits for separated tuples converge uniformly for $\mathsf{b} \in [g^{-2} - \bar{B}, X_g + \bar{B}]$. In particular this holds on every side in $\mathfrak{L}(g)$. For $\mathsf{b} \in I(g)$ the pairs $(K, L^K\mathsf{b})$ are first-passage data at g for every K. More generally, a sequence of first-passage data whose labels converge has the limit (2214). ■□

Proof. The condition $\gamma^{-2} \ge 2B_\sharp$ gives $\bar{B} \le L[2(L-1)]^{-1} g^{-2} < g^{-2}$. The displayed label interval is consequently a compact subset of $(0,\infty)$, so theorem AS applies. The assertion about $I(g)$ is corollary 74.2. The same result bounds every first-passage label between $g^{-2} + B_\sharp - L^{-1}\bar{B}$ and $X_g + \bar{B}$; the lower endpoint is positive. Thus a convergent sequence of those labels has positive limit, as required in (2214). □

These conclusions concern fixed tori with $m \ge 1$ and tuples with no common support point. The comparison (2200) retains a contact subtraction for general tuples; it does not assert convergence of the unrenormalized one-point function or of coincident-support cumulants.

119. Dilation and the range of torus sizes

For this section a side is $\ell = 2L^m$ with $m \in \mathbb{Z}$, and a cutoff for that side is an integer K with $K + m \ge 1$. The finite Wilson model uses spacing $a_K = L^{-K}$ and $N = 2L^{K+m}$ sites in each coordinate direction. In particular $N \ge 2L \ge 26$, so its bonds and plaquettes have the same incidence properties as before. For a fixed side every sufficiently large integer K is an allowed cutoff. The value $K = -m$, which would give only two sites in each direction, is excluded.

For $z \in \mathbb{Z}$, write

$$\delta_z : T_\ell \longrightarrow T_{L^z\ell}, \qquad \delta_z x = L^z x, \qquad f^{[z]} = f \circ \delta_z^{-1}.$$

On $\mathbb{R}^3$ the same notation means $f^{[z]}(x) = f(L^{-z}x)$. This agrees with the dilation notation used in lemma 118.2 and eq. (2206).

Proposition 119.1 (Finite dilation for all allowed cutoffs)**.** Let $K, m, z \in \mathbb{Z}$, $K + m \ge 1$, $\ell = 2L^m$ and $\beta > 0$. Then $K - z$ is a cutoff for $L^z\ell$, and

$$Z_{K,\ell,\beta}[w] = Z_{K-z,L^z\ell,\beta}[L^z w^{[z]}], \tag{2216}$$

for every complex function w on the torus. For $n \ge 1$,

$$S^T_{n;K,\ell,\beta}(\underline{f}) = L^{zn} S^T_{n;K-z,L^z\ell,\beta}(\underline{f}^{[z]}). \tag{2217}$$

The label changes by $a_{K-z}\beta = L^z a_K \beta$. Distances, support radii and separations are multiplied by L^z, and, with the same M on both tori,

$$\|f^{[z]}\|_\infty = \|f\|_\infty, \qquad \operatorname{Lip} f^{[z]} = L^{-z} \operatorname{Lip} f, \qquad \|f^{[z]}\|_\sharp \le \max(1, L^{-z})\|f\|_\sharp. \tag{2218}$$

■

Proof. Multiplication by L^z takes the coset $x + \ell\mathbb{Z}^3$ onto $L^z x + L^z \ell\mathbb{Z}^3$. It therefore defines a homeomorphism with inverse δ_{-z}. Taking the minimum over lifts proves the distance identity; taking suprema of difference quotients proves (2218). Supports are carried to their images, since this is a homeomorphism. The same facts hold on $\mathbb{R}^3$.

The new spacing is $a_{K-z} = L^z a_K$, and the number of sites is $2L^{(K-z)+(m+z)} = N$. Dilation thus bijects positively oriented bonds and plaquettes, preserves their incidence and cyclic order, and takes x_p to $x_{\delta_z p}$. Set $U^\delta(\delta_z b) = U(b)$. This is a permutation of the factors of the configuration space and preserves product Haar measure. The plaquette holonomy, and hence a_p, is unchanged. Therefore the action is unchanged, while

$$\mathcal{O}_{K-z}(L^z w^{[z]})(U^\delta) = a_{K-z}^{-1} L^z \sum_p w(x_p) a_p(U) = \mathcal{O}_K(w)(U).$$

Changing variables in the integral proves (2216). Taking mixed derivatives of its normalized logarithm at zero proves (2217). Equivalently, each observable has acquired the factor L^z and joint cumulants are multilinear. □

Corollary 119.2 (The full cutoff limit on every torus side)**.** Let $m \in \mathbb{Z}$, $\ell = 2L^m$, $\mathsf{b} > 0$, $n \geq 2$, and let $f_1, \dots, f_n$ be real Lipschitz functions on T_ℓ such that $\bigcap_i \operatorname{supp} f_i = \emptyset$. Then

$$S_n^{(\ell,\mathsf{b})}(\underline{f}) = \lim_{\substack{K\to\infty \\ K+m\geq 1}} S^T_{n;K,\ell,L^K\mathsf{b}}(\underline{f})$$

exists along the whole sequence, locally uniformly for $\mathsf{b} \in (0,\infty)$. For every integer z,

$$S_n^{(\ell,\mathsf{b})}(\underline{f}) = L^{zn} S_n^{(L^z\ell, L^z\mathsf{b})}(\underline{f}^{[z]}). \tag{2219}$$

For any $K_j \to \infty$ and $\beta_j > 0$ with $a_{K_j}\beta_j \to \mathsf{b} > 0$, the same limit is obtained from $S^T_{n;K_j,\ell,\beta_j}(\underline{f})$. ■□

Proof. For $m \geq 1$ this is theorem AS. For $m \leq 0$ set $z = 1 - m$. Then $L^z\ell = 2L$, and the dilated functions remain real and Lipschitz, with empty common support intersection. At every allowed K, (2217) identifies the desired cumulant with L^{zn} times the cumulant on T_{2L} at cutoff $K - z$ and label $L^z\mathsf{b}$. This cutoff runs through all sufficiently large integers. Theorem AS gives convergence and local uniformity on the dilated label interval, and hence on the original interval. Its moving-label conclusion gives the last assertion in the same way. Now both sides of (2217) have limits for every integer z, so that identity yields (2219). □

119.1. **Dilation of limits on $\mathbb{R}^3$.** On the infinite lattice define $U^{(z)}(L^z b) = U(b)$ and let $\nu^{(z)}$ be the pushforward of a law ν under this map. It is a homeomorphism of the product configuration spaces, with inverse given by $-z$.

Proposition 119.3 (Torus states under dilation)**.** For any law ν on the lattice of spacing a_K and compactly supported bounded functions f_i,

$$S_n^T[\nu](\underline{f}) = L^{zn} S_n^T[\nu^{(z)}](\underline{f}^{[z]}). \tag{2220}$$

For every allowed finite torus,

$$(\widetilde{\mu}_{K,\ell,\beta})^{(z)} = \widetilde{\mu}_{K-z,L^z\ell,\beta}. \tag{2221}$$

If $K, K - z \geq 1$, dilation maps $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$ bijectively onto $\mathcal{G}^{\mathrm{tor}}_{K-z,\beta}$. ■

Proof. Only finitely many plaquettes meet the support of each f_i. Relabelling them gives $\mathcal{O}_K(f)(U) = L^z \mathcal{O}_{K-z}(f^{[z]})(U^{(z)})$, exactly as in the finite proof. Pushforward preserves the joint law of these corresponding observables, so multilinearity of cumulants proves (2220).

The finite configuration map in proposition 119.1 commutes with periodic lifting: both ways assign to $L^z b$ the variable at the class of b modulo $\ell\mathbb{Z}^3$. Since that map pushes forward the finite Wilson measure, it gives (2221). If $\widetilde{\mu}_{K,2L^{m_i},\beta} \to \nu$ weakly with $m_i \to \infty$, the configuration homeomorphism implies weak convergence of the pushforwards. By (2221), these are $\widetilde{\mu}_{K-z,2L^{m_i+z},\beta}$; eventually

$m_i + z \geq 1$. Thus $\nu^{(z)} \in \mathcal{G}^{\mathrm{tor}}_{K-z,\beta}$ by proposition 91.2. Applying the same argument with $-z$ proves surjectivity, and the configuration inverse gives injectivity. □

To state the limiting conclusion without imposing a reference coupling, consider the following two classes. A limit along tori is a family $S = (S_n)_{n\geq 2}$ obtained on all separated C^1_c tuples from a sequence (K_j, ℓ_j, β_j) with $K_j \to \infty$, $\ell_j = 2L^{m_j} \to \infty$, $m_j \in \mathbb{Z}$, $K_j + m_j \geq 1$ and $\beta_j > 0$. A limit through torus states is obtained on the same tuples from $S^T_n[\nu_j]$, where $K_j \to \infty$, $K_j \geq 1$, $\beta_j > 0$ and $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$. In either case a label is a limit point of $a_{K_j}\beta_j$ along the defining sequence. For any family set

$$S^{[z]}_n(X) = L^{-zn} S_n(X^{[-z]}). \tag{2222}$$

Proposition 119.4 (Dilation of continuum limits)**.** For either of these two classes and every $z \in \mathbb{Z}$, $S \mapsto S^{[z]}$ is a bijection from limits with label b to limits with label $L^z\mathsf{b}$. For limits along tori the defining sequence becomes $(K_j - z, L^z\ell_j, \beta_j)$. For limits through torus states it becomes $(K_j - z, \beta_j, \nu^{(z)}_j)$, after discarding the finite initial segment with $K_j - z < 1$. The inverse is dilation by $-z$. If S obeys the table $C_n(d; R)$, then $S^{[z]}$ obeys

$$C^{[z]}_n(d; R) = \max(1, L^{-z})^n C_n(L^{-z}d; L^{-z}R). \tag{2223}$$

■

Proof. For a compactly supported function, changing the periodization index from $v \in \ell\mathbb{Z}^3$ to $L^z v$ gives

$$(f^{\langle \ell\rangle})^{[z]} = (f^{[z]})^{\langle L^z\ell\rangle}.$$

Use (2217) with the tuple $X^{[-z]}$ and pass to the limit. The transformed indices satisfy $(K_j - z) + (m_j + z) = K_j + m_j \geq 1$, and both the cutoff and the side still tend to infinity. This proves the assertion along tori. For the other class, proposition 119.3 gives the required membership of $\nu^{(z)}_j$ and its cumulant identity gives the same limit. Multiplication by L^z carries the label sequence to $a_{K_j-z}\beta_j$, preserving subsequential limits in both directions. Finally $(X^{[-z]})^{[z]} = X$ and the two scalar factors in (2222) cancel, proving the asserted inverse.

For the bound, a tuple of type $(d; R)$ is taken by $[-z]$ to one of type $(L^{-z}d; L^{-z}R)$, and $\|f^{[-z]}\|_\sharp \leq \max(1, L^z)\|f\|_\sharp$ by (2218). Thus the scalar multiplying the original table is $L^{-zn}\max(1, L^z)^n = \max(1, L^{-z})^n$. Symmetry and multilinearity are preserved as well. □

The constructed growing-torus and volume-first limits belong to these classes. The proposition describes their dilates and labels. A shifted sequence need not be a first-passage sequence at the original reference coupling. Properties stated with constants at a fixed reference coupling therefore require their own hypotheses at that coupling after dilation.

120. Curvature projections and the Gaussian comparison

The short-distance calculation uses the curvature of the Gaussian fluctuation in an actual renormalization step. Its covariance is an orthogonal projection, and the projections at successive scales are nested. This gives an exact telescoping identity before any estimate is made. We establish that identity first, including its relation to the gauge-fixed covariance, and then estimate its kernels.

120.1. **The nested projections.** Fix a lattice (ℓ, K). In this subsection cochains are real, their pairings are unweighted sums, and d is the coboundary given by signed sums along edges and plaquette boundaries. Write Ch^i_n for the degree-i cochains on the level-n lattice and put $\eta_n = L^{-n}$.

The level-n lattice has spacing L^n in bare units. The straight averages from the bare lattice are

$$(Q_n^{\mathrm{si}}\lambda)(y) = L^{-3n} \sum_{x\in B^n(y)} \lambda(x),$$

$$(Q_n^{s}a)_\mu(y) = L^{-3n} \sum_{x\in B^n(y)} \sum_{i=0}^{L^n-1} a_\mu(x+ie_\mu),$$

$$(Q_n^{\mathrm{pl}}F)_{\mu\nu}(y) = L^{-3n} \sum_{x\in B^n(y)} \sum_{i,i'=0}^{L^n-1} F_{\mu\nu}(x+ie_\mu+i'e_\nu).$$

For an oriented plaquette the last formula has the corresponding orientation sign. At $n=0$ all three averages are the identity. The same formulas define the one-step averages, denoted by boldface $\mathbf{Q}^{\mathrm{si}}$, $\mathbf{Q}^{s}$, and $\mathbf{Q}^{\mathrm{pl}}$.

Let $\mathsf{E} = d\mathsf{Ch}_0^1$ and define

$$\mathsf{Fl}_n = \{F\in\mathsf{E} : Q_n^{\mathrm{pl}}F = 0\}, \qquad \mathsf{Bg}_n = \mathsf{E}\ominus\mathsf{Fl}_n, \qquad \Pi_n = \mathrm{proj}_{\mathsf{Bg}_n}. \tag{2224}$$

In particular, $\mathsf{Bg}_0 = \mathsf{E}$. We use

$$\mathcal{C}_{(j)} = \Pi_{j-1} - \Pi_j, \qquad C_K^{\mathrm{free}} = \Pi_0, \qquad C_{K,s}^{\mathrm{bg}} = \Pi_{K-s}, \qquad C_{K,s}^{\mathrm{fl}} = \Pi_0 - \Pi_{K-s} = \sum_{j=1}^{K-s} \mathcal{C}_{(j)}. \tag{2225}$$

Kernels here and below are matrices in the oriented positive plaquette basis; they include no lattice-volume weights.

Lemma 120.1. The averages satisfy the chain identities

$$Q_n^s d = dQ_n^{\mathrm{si}}, \qquad dQ_n^s = Q_n^{\mathrm{pl}} d, \qquad Q_{n+1}^{\mathrm{pl}} = \mathbf{Q}^{\mathrm{pl}} Q_n^{\mathrm{pl}},$$

and the analogous composition identities for the other two averages. Moreover,

$$\|Q_n^{\mathrm{pl}}F\|^2 \le L^n \|F\|^2, \qquad \mathsf{Fl}_n = d\ker Q_n^s. \tag{2226}$$

The spaces Fl_n increase with n; hence $\Pi_n\Pi_m = \Pi_{\max\{m,n\}}$, and $\mathcal{C}_{(j)}$ is the orthogonal projection onto $\mathsf{Bg}_{j-1}\cap\mathsf{Fl}_j$. ■

Proof. Telescoping the edge and face sums proves the chain identities. The entries of Q_n^{pl} are nonnegative after fixing an orientation, with row sum L^{2n} and column sum L^{-n}. Cauchy–Schwarz in each row and summation give the first inequality in (2226).

If $Q_n^{\mathrm{pl}}da = 0$, the coarse cochain $Q_n^s a$ is a cocycle. On a periodic cubic lattice every real cocycle is $d\lambda + c$, where c is constant: subtract its three periods and integrate the remaining cochain along paths. Set $\theta(x) = \lambda(y)$ for $x\in B^n(y)$ and let $c'' = L^{-n}c$. Then $d(d\theta + c'') = 0$ and $Q_n^s(d\theta + c'') = d\lambda + c$. Subtracting this cocycle from a proves the reverse inclusion in $\mathsf{Fl}_n = d\ker Q_n^s$. The forward inclusion and the remaining assertions follow from the chain identities. □

For $B\in\mathsf{Ch}_n^1$, let $H_n^{\mathrm{cv}}B$ be the unique $F\in\mathsf{Bg}_n$ satisfying $Q_n^{\mathrm{pl}}F = dB$. Existence uses surjectivity of Q_n^s: the pivot lift for the one-step straight average is a right inverse, and composition gives a right inverse for Q_n^s; see the proof of the Gaussian critical-point construction in Theorem N. Equivalently,

$$H_n^{\mathrm{cv}}B = \Pi_n da \quad (Q_n^s a = B), \qquad \|H_n^{\mathrm{cv}}B\|^2 = \min_{Q_n^s a = B} \|da\|^2 \ge \eta_n \|dB\|^2. \tag{2227}$$

Indeed, subtracting two lifts changes their curvatures by Fl_n; lemma 120.1 then gives the formula and its uniqueness. The inequality follows from (2226). This response depends only on dB, has range Bg_n, and $H_0^{\mathrm{cv}} = d$.

The response in (2227) is the curvature of the free Landau response of equations (931)–(932). More precisely, in level-n units its fine mesh is η_n, its fine curl is D, and its straight potential average is Q. If

$$D^*Dh = Q^*\omega, \qquad Qh = B,$$

then

$$\eta_n^2 Dh = H_n^{\mathrm{cv}} B. \tag{2228}$$

To check the factors, $d(\eta_n h) = \eta_n^2 Dh$, $Q_n^s(\eta_n h) = Qh$, and the fine weighted pairing is η_n^3 times the unweighted one. Testing the equation against $a \in \ker Q_n^s$ makes $\eta_n^2 Dh$ orthogonal to $d\ker Q_n^s = \mathsf{Fl}_n$. Formula (2227) now proves (2228).

120.2. **The covariance of one step.** At level $k < K$, let $Q^{\mathrm{lin}} = \mathbf{Q}^s - d\kappa$ be the linearized gauge-covariant average at the trivial field. Here κB is the mean of the block-path integrals used in the one-step gauge; in particular,

$$Q^{\mathrm{lin}} d\theta = d(\theta|_{\mathbb{T}^{(k+1)}}).$$

Let $V = \ker Q^{\mathrm{lin}}$. The pivot parametrization $C_{\mathbf{1},\mathbb{R}}$ identifies the independent bond coordinates with V: it keeps every nonpivot coordinate and sets the pivot at a coarse bond c equal to $-L^2 \sum_{b \neq b_0(c)} w_c(b) B(b)$. The gauge-fixing form is

$$G(B, B') = \sum_y \sum_{x \in B(y) \setminus \{y\}} \overline{B}(y,x)\overline{B'}(y,x),$$

where the bar denotes the average of the coordinate-path integrals, not complex conjugation. Put

$$B_k^{\mathrm{cv}} = \eta_k^{-1}(H_k^{\mathrm{cv}} C_{\mathbf{1},\mathbb{R}})^* H_k^{\mathrm{cv}} C_{\mathbf{1},\mathbb{R}}, \quad \widehat{G} = C_{\mathbf{1},\mathbb{R}}^* G C_{\mathbf{1},\mathbb{R}}, \quad A_k^{\mathrm{cv}} = B_k^{\mathrm{cv}} + \widehat{G}. \tag{2229}$$

Lemma 120.2. One has

$$H_k^{\mathrm{cv}} V = \mathsf{Bg}_k \cap \mathsf{Fl}_{k+1}, \qquad \ker(H_k^{\mathrm{cv}}|_V) = \{d\theta : \theta|_{\mathbb{T}^{(k+1)}} = 0\}.$$

On V, the kernels of G and H_k^{cv} are complementary. ■

Proof. The chain identity gives $Q_{k+1}^{\mathrm{pl}} H_k^{\mathrm{cv}} B = d\mathbf{Q}^s B = dQ^{\mathrm{lin}} B$, so the first range is contained in the second. Conversely, write $F = H_k^{\mathrm{cv}} B$ for $F \in \mathsf{Bg}_k \cap \mathsf{Fl}_{k+1}$. Then $Q^{\mathrm{lin}} B = d\lambda + c'$. Choose the constant $c'' = L^{-1}c'$ and a site function with coarse values $\theta = \lambda + \kappa c''$. The cocycle $c'' + d\theta$ has the same Q^{lin}-average as B, so subtracting it puts B in V without changing $H_k^{\mathrm{cv}} B$.

If $H_k^{\mathrm{cv}} B = 0$, (2227) gives $dB = 0$. Thus $B = d\theta + c$; the periods of $Q^{\mathrm{lin}} B = 0$ force $c = 0$, and its gradient part makes θ constant on the coarse sites. Subtract that constant. Finally, for $B \in V$ put $\theta(x) = \overline{B}(y,x)$ in $B(y)$ and $\theta(y) = 0$. The cochain $B'' = B - d\theta$ has all its averaged block-path integrals zero. Thus $G(B'', B'') = 0$, whereas $d\theta$ lies in $\ker H_k^{\mathrm{cv}}$. If both properties hold, the same path formula forces $\theta = 0$. This proves the direct-sum assertion. □

The following elementary calculation makes the disappearance of the gauge-fixing form from curvature covariances explicit. Suppose $T : V \to E$ is linear, $q \geq 0$, and $V = \ker q \oplus \ker T$. In any coordinates i on V, let M_τ represent $\tau\eta^{-1}\langle Tv, Tw\rangle + q(v,w)$. For $\tau \neq 0$ this form is nondegenerate and

$$\eta^{-1} T i M_\tau^{-1} (Ti)^* = \tau^{-1} \operatorname{proj}_{\operatorname{ran} T}. \tag{2230}$$

Indeed, the two summands of V are orthogonal for this form. Solving the equation with right side $\langle F, Tw\rangle$ first against $w \in \ker T$ removes the $\ker T$ component. The remaining equation says that Tv is $\eta\tau^{-1} \operatorname{proj}_{\operatorname{ran} T} F$.

Proposition 120.3. The flat gap $\gamma_{\mathrm{tr}} > 0$ of the one-step construction satisfies $A_k^{\mathrm{cv}} \geq \gamma_{\mathrm{tr}} I$. For $\tau \neq 0$,

$$\eta_k^{-1} H_k^{\mathrm{cv}} C_{\mathbf{1},\mathbb{R}} (\tau B_k^{\mathrm{cv}} + \widehat{G})^{-1} C_{\mathbf{1},\mathbb{R}}^* (H_k^{\mathrm{cv}})^* = \tau^{-1} \mathcal{C}_{(k+1)}. \tag{2231}$$

At the trivial pair, the Hessian and covariance of a canonical renormalization step are the matrices in (2229), with the identity on the N^2-1 colour components, where $N=2$ denotes the matrix size of the gauge group. In particular, the curvature covariance of that step is $\mathcal{C}_{(k+1)} \otimes I_{N^2-1}$. ■

Proof. The flat coercivity estimate from the one-step construction in equation (593) is $\|dB\|^2 + G(B,B) \geq \gamma_{\mathrm{tr}} \|\check{B}\|^2$ for $B = C_{\mathbf{1},\mathbb{R}} \check{B}$. Combining this with (2227) proves the gap. Apply (2230) and lemma 120.2 to get (2231).

For the operator identification, differentiate the constrained critical-point equations at the trivial pair. The differentiated background is the Landau response in (2228). The action has zero first variation there; its second variation is consequently the squared norm of this linearized curvature, namely $\eta_k^{-1}(H_k^{\mathrm{cv}})^* H_k^{\mathrm{cv}}$ in level-k coordinates. The linearized fluctuation constraint is exactly $Q^{\mathrm{lin}} B = 0$, and the second variation of the gauge term is G. Restriction by the pivot parametrization therefore gives $A_k^{\mathrm{cv}} \otimes I_{N^2-1}$, whose inverse is the whole flat covariance. These are the constrained Hessian and fluctuation-coordinate identities proved in Proposition 52.2 and equation (909); no continuation away from the trivial pair is used here. □

This identification is used only where the canonical step exists with its prescribed constant-sequence background and octet properties. These hypotheses are part of the renormalization construction; the finite-dimensional projection identity itself does not establish them.

120.3. **Uniform kernel bounds.** Fix the constants N_{H}, C_{H}, and δ_{H} in the uniform free-response estimate of equations (931)–(932). Increase the first to $L^{m_{\mathrm{tp}}}$, with $m_{\mathrm{tp}} \geq 2$, and decrease δ_{H} if needed. Their choice is independent of the level and the volume. At an admitted side, (2228) and that estimate give, in level-n units,

$$|H_n^{\mathrm{cv}}(p,c)| \leq C_{\mathrm{tp}}^{\mathrm{k}} \eta_n^2 e^{-\delta_{\mathsf{H}} |z(p) - c_-|_\infty}, \qquad |H_n^{\mathrm{cv}} B(p)| \leq C_{\mathrm{tp}}^{\mathrm{cv}} \eta_n^2 \|B\|_\infty. \tag{2232}$$

The first bound also holds at $n=0$ on every side because $H_0 = d$. The second follows by summing the first: if $s(\delta) = e^{\delta} \sum_{u \in \mathbb{Z}^3} e^{-\delta |u|_\infty}$, then $\sum_y e^{-\delta |x-y|_\infty} \leq s(\delta)$ on every torus, and we can take $C_{\mathrm{tp}}^{\mathrm{cv}} = 3 C_{\mathrm{tp}}^{\mathrm{k}} s(\delta_{\mathsf{H}})$.

Lemma 120.4. If $d\mathbf{Q}^s B = 0$, there is B_2 with

$$dB_2 = dB, \qquad \mathbf{Q}^s B_2 = 0, \qquad \|B_2\|_\infty^2 \leq 384 L^5 \|dB\|^2.$$

■

Proof. First subtract a gradient and a constant as in lemma 120.1 to arrange $\mathbf{Q}^s B = 0$. In each block integrate B from its centre along the ordered coordinate path, subtract the mean of this site function in the block, and subtract its gradient from B. This preserves its straight average. Write $\sigma(c)$ for the sum of $|dB|$ over the plaquettes in the double block belonging to a coarse bond c. Exchanging two successive segments of a coordinate path changes its integral by a sum of curvatures over the strip between them. For an internal bond the resulting strips are disjoint and lie in its block. Its new value is therefore bounded by $\sigma(c)$.

For two neighbouring parallel bonds crossing the corridor of c, the intervening plaquette and the two internal side bonds give a difference at most $3\sigma(c)$. The corridor has diameter $2(L-1)$ in these moves, so any two such values differ by at most $6(L-1)\sigma(c)$. Each of the L^3 straight paths averaged by $\mathbf{Q}^s$ crosses the corridor once and otherwise has $L-1$ internal bonds. Every corridor bond is used L times. Since the average is zero, the corridor mean has modulus at most $(L-1)\sigma(c)$. Every corridor value is thus bounded by $7(L-1)\sigma(c)$. These and the internal bonds exhaust the lattice bonds. There are at most $6L^3$ plaquettes in a double block, so $\sigma(c)^2 \leq 6L^3 \|dB\|^2$; finally $49 \cdot 6 < 384$. □

For an orthogonal projection P, $P(p,p) = \sup_{\|F\|=1, F\in \operatorname{ran} P} |F(p)|^2$ and $|P(p,q)|^2 \le P(p,p)P(q,q)$. Applying lemma 120.4 to $F = H_k^{\mathrm{cv}} B$ in $\mathsf{Bg}_k \cap \mathsf{Fl}_{k+1}$, then using (2227) and (2232), gives

$$\mathcal{C}_{(k+1)}(p,p) \le 384 L^5 (C_{\mathrm{tp}}^{\mathrm{cv}})^2 \eta_k^3 \tag{2233}$$

when $k \ge 1$ and its side is admitted. At $k = 0$ the bound by 1 suffices. On an admitted side N at level n one also has

$$\Pi_n(p,p) \le \tfrac{1}{16} N^2 (C_{\mathrm{tp}}^{\mathrm{cv}})^2 \eta_n^3.$$

To see this, choose the least-norm solution $dB = Q_n^{\mathrm{pl}} F$. It is orthogonal to the cocycles. In Fourier variables its nonzero modes satisfy $\sum_{\mu<\nu} |k_\mu \widehat{B}_\nu - k_\nu \widehat{B}_\mu|^2 = |k|^2 |\widehat{B}|^2$, where $k_\mu = e^{i\theta_\mu} - 1$. The least nonzero $|k|$ is at least $4/N$; hence $\|B\| \le (N/4)\|dB\|$. Now use (2226) and (2232).

Proposition 120.5. There is $C_{\mathrm{tp}}^{\mathrm{bg}}$ depending only on the fixed free-response constants and L such that, on every side,

$$|C_{K,s}^{\mathrm{bg}}(p,q)| \le C_{\mathrm{tp}}^{\mathrm{bg}} (a_K/a_s)^3. \tag{2234}$$

■

Proof. Put $n = K - s$ and write $\ell = 2L^m$. Relabel the same bare periodic graph as $(2L^{m_{\mathrm{tp}}}, \widetilde{K})$, where $\widetilde{K} = m + K - m_{\mathrm{tp}}$. Straight averages and the projections through any common level are unchanged under this relabeling. If $\widetilde{K} \le 0$, then $n \le m_{\mathrm{tp}}$, and $\Pi_n(p,p) \le 1$ proves the result after increasing the constant. Otherwise set $\widetilde{n} = \min(n, \widetilde{K})$. All levels through $\widetilde{K}$ on the relabeled lattice have admitted sides, and

$$\Pi_n(p,p) \le \widetilde{\Pi}_{\widetilde{n}}(p,p) = \widetilde{\Pi}_{\widetilde{K}}(p,p) + \sum_{j=\widetilde{n}+1}^{\widetilde{K}} \widetilde{\mathcal{C}}_{(j)}(p,p).$$

The preceding diagonal bounds and a geometric sum bound this by $(C_{\mathrm{tp}}^{\mathrm{cv}})^2 (768 L^5 + L^{2m_{\mathrm{tp}}}/4) L^{-3\widetilde{n}}$; the case $\widetilde{n} = 0$ is covered by the bound 1. Since $n - \widetilde{n} \le m_{\mathrm{tp}}$, we can take this constant times $L^{3m_{\mathrm{tp}}}$ for $C_{\mathrm{tp}}^{\mathrm{bg}}$. Projection Cauchy–Schwarz gives the off-diagonal bound. □

Proposition 120.6. There are $C_{\mathrm{tp}}, C_{\mathrm{tp}}^{\mathrm{row}} < \infty$ and $\delta_{\mathrm{tp}} > 0$, depending only on the fixed free-response constants and L, such that for the physical sup distance $r = |x(p) - x(q)|_\infty$,

$$|\mathcal{C}_{(j)}(p,q)| \le C_{\mathrm{tp}} (a_K/a_{K-j+1})^3 e^{-\delta_{\mathrm{tp}} r / a_{K-j+1}}. \tag{2235}$$

If $\delta_{\mathrm{tp}} L(L-1) r \ge (3 \log L + \log 2) a_s$, then

$$|C_{K,s}^{\mathrm{fl}}(p,q)| \le 2 L^3 C_{\mathrm{tp}} (a_K/a_s)^3 e^{-\delta_{\mathrm{tp}} L r / a_s}. \tag{2236}$$

For each fixed $t_0 > 0$, the same bound holds when $r \ge t_0 a_s$ with $2L^3 C_{\mathrm{tp}}$ replaced by

$$C_{\mathrm{tp}}^{\mathrm{fl}}(t_0) = C_{\mathrm{tp}} \sum_{i \ge 1} L^{3i} e^{-\delta_{\mathrm{tp}} t_0 (L^i - L)} < \infty.$$

At $k = 0$, or at an admitted side of level k,

$$|(A_k^{\mathrm{cv}})^{-1}(b,b')| \le \frac{2}{\gamma_{\mathrm{tr}}} e^{-\delta_{\mathrm{tp}} |b_- - b'_-|_\infty}, \qquad \sum_b |(H_k^{\mathrm{cv}} C_{\mathbf{1},\mathbb{R}})(p,b)| \le C_{\mathrm{tp}}^{\mathrm{row}} \eta_k^2. \tag{2237}$$

The distances in (2237) are in level-k units. ■

Proof. We give the inverse estimate and keep track of which bounds are uniform in the side. By (2232) and the exponential lattice sum,

$$|\eta_k^{-1} (H_k^{\mathrm{cv}})^* H_k^{\mathrm{cv}}(c,c')| \le 3 (C_{\mathrm{tp}}^{\mathrm{k}})^2 s(\delta_{\mathsf{H}}/2) e^{-\delta_{\mathsf{H}} |c_- - c'_-|_\infty / 2}.$$

The gauge matrix has entries at most L^3 and vanishes unless the two bonds lie in one block. A column of $C_{\mathbf{1},\mathbb{R}}$ has at most seven nonzero entries, of modulus at most $2L^2$, within distance $2L$ of its

bond. Thus $|A_k^{\mathrm{cv}}(b,b')| \le B_{\mathrm{tp}} e^{-\delta\rho(b,b')}$ for fixed B_{tp} and $\delta = \delta_{\mathsf{H}}/2$. Also $\sup_b \sum_{b'} e^{-\delta\rho(b,b')/2} \le S_{\mathrm{tp}} = 3s(\delta_{\mathsf{H}}/4)$.

For completeness, conjugate $A = A_k^{\mathrm{cv}}$ by the diagonal weight $e^{\epsilon\rho(b,b_0)}$. For $0 \le \epsilon \le \delta/2$, convexity gives

$$e^{\epsilon\rho} - 1 \le \frac{2\epsilon}{\delta}(e^{\delta\rho/2} - 1).$$

The triangle inequality therefore bounds the change in each matrix entry by $(2\epsilon/\delta)B_{\mathrm{tp}}e^{-\delta\rho(b,b')/2}$. The row and column sum bound gives a change in operator norm of at most $2\epsilon B_{\mathrm{tp}}S_{\mathrm{tp}}/\delta$. Choose

$$\delta_{\mathrm{tp}} = \min\{\delta/2, \gamma_{\mathrm{tr}}\delta/(4B_{\mathrm{tp}}S_{\mathrm{tp}})\}.$$

The gap in proposition 120.3 and the Neumann series give norm at most $2/\gamma_{\mathrm{tr}}$ for the conjugated inverse. Its (b, b_0) entry proves the first bound in (2237). The pivot rows have sum at most $1 + 12L^5$; summing (2232) gives its second bound.

Insert the inverse estimate in (2231). The pivot rows move bonds by at most $2L$ and have bounded sums, whereas the two response kernels contribute η_k^4. The prefactor η_k^{-1} leaves η_k^3. The triangle inequality extracts $e^{-\delta_{\mathrm{tp}}|z(p)-z(q)|_\infty}$, and the remaining two bond sums converge uniformly by $\delta_{\mathrm{tp}} \le \delta_{\mathsf{H}}/4$. The change from base points to barycentres costs a fixed factor. This proves (2235) at admitted sides and at $k = 0$. At the other sides the level-k distance is at most $N_{\mathsf{H}}/2$; proposition 120.5 and $\mathcal{C}_{(k+1)}(p,p) \le \Pi_k(p,p)$ prove the same bound after a fixed increase of C_{tp}.

Finally write $t = r/a_s$ in the sum in (2225). Its terms are bounded by $C_{\mathrm{tp}}(a_K/a_s)^3 L^{3i} e^{-\delta_{\mathrm{tp}} t L^i}$, $i \ge 1$. Under the displayed condition their successive ratios are at most $1/2$, which proves (2236). For $t \ge t_0$ extract $e^{-\delta_{\mathrm{tp}} L t}$ instead; the remaining convergent series is the stated constant. □

121. The dipole kernel and its lattice approximation

For $u \ne 0$ put

$$\mathbb{K}_3(u) = \frac{3\widehat{u}\widehat{u}^{\mathsf{T}} - I}{4\pi|u|^3} = \nabla^2 \frac{1}{4\pi|u|}, \qquad \mathcal{K}_{\mathrm{free}}(f,h) = \iint f(x)h(y)|x-y|^{-6}\,dx\,dy. \tag{2238}$$

On a torus the distance in the integral is its least-lift distance. We will only use pairs supported together in a ball of radius $\ell/4$, where the integral agrees with the integral of their compactly supported lifts. The eigenvalues of $3\widehat{u}\widehat{u}^{\mathsf{T}} - I$ are $2, -1, -1$, so

$$\sum_{\lambda,\lambda'} \mathbb{K}_3(u)^2_{\lambda\lambda'} = \frac{3}{8\pi^2}|u|^{-6}. \tag{2239}$$

121.1. **The free curvature kernel.** Represent a plaquette by (z, λ), with λ its dual coordinate direction, choosing the cyclic orientation. Its barycentre is $x(z,\lambda) = z + \frac{1}{2}(e_1 + e_2 + e_3 - e_\lambda)$ in bare units. Relative to the original positive plaquette orientation let the sign be $\epsilon_p \in \{\pm 1\}$. For $q_\mu(k) = e^{ik_\mu} - 1$ define

$$F_{\lambda\lambda'}(k) = \frac{\overline{q_\lambda(k)}q_{\lambda'}(k)}{|q(k)|^2}, \qquad P(k) = I - F(k) \quad (k \ne 0), \qquad P(0) = 0.$$

Let Π_∞ and Π_N be the curl projections on $\mathbb{Z}^3$ and $(\mathbb{Z}/N\mathbb{Z})^3$. Their Fourier formulas are

$$\begin{aligned} \Pi_\infty((z,\lambda),(z',\lambda')) &= \delta_{pq} - \int_{[-\pi,\pi]^3} e^{ik\cdot(z-z')} F_{\lambda\lambda'}(k) \frac{dk}{(2\pi)^3}, \\ \Pi_N((z,\lambda),(z',\lambda')) &= \delta_{pq} - N^{-3}\delta_{\lambda\lambda'} - N^{-3} \sum_{k \in \frac{2\pi}{N}\mathbb{Z}^3 \cap (-\pi,\pi]^3 \setminus \{0\}} e^{ik\cdot(z-z')} F_{\lambda\lambda'}(k). \end{aligned} \tag{2240}$$

Indeed, curl has symbol $q\times$, whose range is the Hermitian orthogonal complement of $\bar{q}$. Its orthogonal projection is $P(k)$. On the infinite lattice, cutting out $|k| \le \varepsilon$ in the right inverse $-|q|^{-2}\bar{q}\times P(k)$

shows that the range of this multiplier is the closure of the curl range. The cutoff tends to the identity on that range in L^2. Fourier inversion proves (2240). The finite formula has $P(0) = 0$ because an exact cochain has no constant mode. In particular, $\epsilon_p \epsilon_q C_K^{\mathrm{free}}(p,q) = \Pi_N(p,q)$ after rescaling the bare lattice to spacing one, with $N = \ell/a_K$.

Lemma 121.1. There are absolute constants $C'_{\mathrm{lat}}, C''_{\mathrm{lat}}$ such that, for distinct plaquettes on $\mathbb{Z}^3$,

$$|\Pi_\infty(p,q) - \mathbb{K}_3(x(p)-x(q))_{\lambda(p)\lambda(q)}| \le C'_{\mathrm{lat}} |x(p)-x(q)|^{-5}. \tag{2241}$$

If p, q are distinct plaquettes on the N-torus and $\widetilde{p}, \widetilde{q}$ are lifts whose barycentre difference has sup norm at most $N/2$, then

$$|\Pi_N(p,q) - \Pi_\infty(\widetilde{p},\widetilde{q})| \le C''_{\mathrm{lat}} N^{-3}. \tag{2242}$$

■

Proof. Fix the two dual directions, put $y = z - z'$ and $s = (e_{\lambda'} - e_\lambda)/2$, so that $u = y + s$ is the barycentre difference. Multiplication by the barycentre phase removes the linear lattice error:

$$m(k) = e^{-ik\cdot s} F_{\lambda\lambda'}(k) = \frac{4\sin(k_\lambda/2)\sin(k_{\lambda'}/2)}{4\sum_\mu \sin^2(k_\mu/2)}, \qquad m^0(k) = \frac{k_\lambda k_{\lambda'}}{|k|^2}.$$

Taylor's formula for sine and its derivatives, and $\sum_\mu 4\sin^2(k_\mu/2) \ge 4|k|^2/\pi^2$ on $[-\pi,\pi]^3$, give, for $|\alpha| \le 6$,

$$|\partial^\alpha F(k)| \le C_\alpha |k|^{-|\alpha|}, \qquad |\partial^\alpha (m - m^0)(k)| \le C_\alpha |k|^{2-|\alpha|}.$$

These derivative bounds follow by differentiating the quotient: the numerator of the second difference vanishes to order six before division by $|k|^2 \sum_\mu 4\sin^2(k_\mu/2)$, and differentiating an inverse preserves its corresponding homogeneous bound.

Choose a radial smooth cutoff χ equal to one on $|k| \le 1/2$ and zero on $|k| \ge 1$. Split the Fourier coefficient of F into $(1-\chi)F$, $\chi(m - m^0)$ with phase $e^{ik\cdot u}$, and χm^0 with that phase. The first is a smooth periodic symbol; five integrations by parts give $O(|u|^{-5})$. For the second, use annuli of radius 2^{-j}. Their Fourier integrals are bounded both by $C 2^{-5j}$ and, after six integrations by parts, by $C|u|^{-6} 2^j$. Splitting the sum at $2^j = |u|$ gives $O(|u|^{-5})$.

For the last term put $E(u) = \int e^{ik\cdot u} \chi(k) |k|^{-2}\, dk/(2\pi)^3$. The heat-kernel identity for $|k|^{-2}$, with Fubini justified by the integrability at zero and the rapid decay of $\check{\chi}$, gives $E = (4\pi|\cdot|)^{-1} * \check{\chi}$. Write $\check{\chi}(v) = \varphi(|v|)$. This function is radial, $4\pi \int_0^\infty \varphi(t) t^2 dt = 1$, and $|\varphi(t)| \le C(1+t)^{-7}$. The mean of $|u - v|^{-1}$ over $|v| = t$ is $1/\max(|u|, t)$, hence

$$E(u) = \frac{1}{4\pi|u|} - D(|u|), \qquad D(r) = \int_r^\infty \varphi(t) t^2 (r^{-1} - t^{-1})\, dt.$$

Direct differentiation gives $D'(r) = O(r^{-6})$ and $D''(r) = O(r^{-7})$. The Fourier transform of χm^0 is $-\partial_\lambda \partial_{\lambda'} E$; it therefore equals $-\mathbb{K}_3(u)_{\lambda\lambda'} + O(|u|^{-7})$. The minus sign in (2240) proves (2241) for $|u| > 4$. For smaller nonzero barycentre differences, $|u| \ge 2^{-1/2}$, and the projection bound by one covers the remaining cases.

For the periodic comparison take $N \ge 4$ and remove the ball $|k| \le N^{-1}$ from F, using a smooth cutoff supported in $|k| \le 2N^{-1}$ and its 2π-periodic translates. Call the resulting smooth periodic symbol F_N. It equals F at every nonzero Fourier grid point and is zero at zero. The derivative bounds above give $\int |\partial_\mu^4 F_N| \le CN$; thus $|\check{F}_N(y')| \le CN |y'|_\infty^{-4}$ for $y' \ne 0$. Its Fourier series is absolutely summable, and summing it over the grid gives the exact identity

$$N^{-3} \sum_{k \ne 0} e^{ik\cdot y} F(k) = \sum_{n \in \mathbb{Z}^3} \check{F}_N(y + Nn).$$

The term $n = 0$ differs from $\check{F}(y)$ by at most CN^{-3}, the volume of the removed ball. Since $|y|_\infty \le N/2 + 1/2$, for $n \ne 0$ one has $|y + Nn|_\infty \ge N|n|_\infty/4$. The other terms are bounded in sum

by $CN^{-3}\sum_{n\neq 0}|n|_\infty^{-4}<\infty$. The zero-mode correction in (2240) is also $O(N^{-3})$. This proves (2242); $N\le 3$ follows by increasing the constant and using the projection bound. □

Rescaling the last lemma gives, for a least lift u of the physical barycentre difference of distinct plaquettes,

$$\left|\epsilon_p\epsilon_q C_K^{\mathrm{free}}(p,q)-a_K^3\mathbb{K}_3(u)_{\lambda(p)\lambda(q)}\right|\le C_{\mathrm{lat}}a_K^3(a_K^2|u|^{-5}+\ell^{-3}),\tag{2243}$$

and consequently

$$|C_K^{\mathrm{free}}(p,q)|\le(1+3C_{\mathrm{lat}})a_K^3|u|^{-3}.\tag{2244}$$

Here C_{lat} is absolute and $2^{-1/2}a_K\le|u|\le\sqrt{3}\ell/2$.

121.2. **Separated test pairs and squared kernels.** For the test pairs $\mathfrak{T}_E(d;\vartheta)$ defined in the introduction, the free kernel has the following useful bounds. The two supports lie together in a ball of radius $5d$; all their cross distances belong to $[d,10d]$. Thus

$$10^{-6}\vartheta^2\|f\|_\infty\|h\|_\infty\le\mathcal{K}_{\mathrm{free}}(f,h)\le(4\pi/3)^2\|f\|_\infty\|h\|_\infty.\tag{2245}$$

For a plaquette kernel k define the squared-kernel sum

$$\mathcal{K}[k](f,h)=\sum_{p,q}f(x(p))h(x(q))k(p,q)^2,\qquad \mathcal{K}_{K,s}=\mathcal{K}[C_{K,s}^{\mathrm{bg}}].\tag{2246}$$

There are no additional mesh factors in this sum.

Proposition 121.2. Suppose $8a_K\le d$, $20d\le\ell$, and $(f,h)\in\mathfrak{T}_{T_\ell}(d;\vartheta)$. Then

$$\left|\mathcal{K}[k](f,h)-\frac{3}{8\pi^2}\mathcal{K}_{\mathrm{free}}(f,h)\right|\le C_{\mathrm{tm}}\left[\frac{a_K}{\vartheta d}+(d/\ell)^3+e\right]\|f\|_\infty\|h\|_\infty\tag{2247}$$

for $k=C_K^{\mathrm{free}}$ and $e=0$. The same estimate holds for $k=C_{K,s}^{\mathrm{bg}}$ with

$$t=d/a_s,\qquad e=t^3e^{-\delta_{\mathrm{tp}}Lt/\sqrt{3}},$$

provided $t\ge1$, or provided $\delta_{\mathrm{tp}}L(L-1)t/\sqrt{3}\ge3\log L+\log 2$. The constant is independent of the mesh, side, pair, s, and ϑ; the free case admits an absolute constant, and the background case depends only on the constants of proposition 120.6. ■

Proof. Lift both functions to their common ball of radius $5d$. Their cross distances are unchanged. For a fixed direction the barycentres form a translate of $a_K\mathbb{Z}^3$. Disjoint cubes of side a_K about the contributing points lie in a ball of radius $d+\sqrt{3}a_K/2$, so there are at most $18(d/a_K)^3$ contributing plaquettes for each function. If on their product $|k_1-k_2|\le A_1a_K^3d^{-3}$ and $|k_2|\le A_2a_K^3d^{-3}$, it follows that

$$|\mathcal{K}[k_1]-\mathcal{K}[k_2]|\le 324A_1(2A_2+A_1)\|f\|_\infty\|h\|_\infty.\tag{2248}$$

Use (2243) to replace C_K^{free} by $a_K^3\epsilon_p\epsilon_q\mathbb{K}_3(x(p)-x(q))_{\lambda(p)\lambda(q)}$. Equations (2243)–(2248) bound the error by a constant times $[(a_K/d)^2+(d/\ell)^3]\|f\|_\infty\|h\|_\infty$.

The resulting sum is a Riemann sum over nine products of barycentre grids. On every product of cubes that contributes, all cross distances are at least $d-2\sqrt{3}a_K\ge d/2$. The squared dipole entry is bounded by Cd^{-6} and its gradient by Cd^{-7}. The Lipschitz bounds on f,h thus bound the difference between the integrand and its grid value by $Ca_K\vartheta^{-1}d^{-7}\|f\|_\infty\|h\|_\infty$. The union of contributing cubes for either function has volume at most $8d^3$. Integration and summation over the nine directions give error at most $8\cdot10^4a_K(\vartheta d)^{-1}\|f\|_\infty\|h\|_\infty$. Equation (2239) identifies the integral.

For the background kernel, $C_{K,s}^{\mathrm{bg}}-C_K^{\mathrm{free}}=-C_{K,s}^{\mathrm{fl}}$. All contributing sup distances are at least $d/\sqrt{3}$. The appropriate bound in proposition 120.6 gives $|C_{K,s}^{\mathrm{fl}}(p,q)|\le Ce\,a_K^3d^{-3}$. Since $t^3e^{-\delta_{\mathrm{tp}}Lt/\sqrt{3}}$ is bounded for $t>0$, (2248) and (2244) add an error at most $Ce\|f\|_\infty\|h\|_\infty$. This proves the result. □

122. Differentiating the small-field iteration

We next introduce the two plaquette weights into the actual renormalization maps. The counterterms and the maps themselves are kept fixed. Differentiation then separates two effects: the covariance created in a single step and the transport of derivatives created at earlier steps.

Use the canonical one-lattice datum of theorem D, including its absolute radii, bracket localization, and sum rule. Throughout this section it is assumed to satisfy its full small-coupling hypotheses. Thus every level under consideration is an admitted place; at heights at least two this includes the octet property on all the lattices in its defining class. Write α_0, α_1 for the common output radii, κ for the polymer decay rate, and $\omega_n = (n+1)L^{-(n-1)}$ for the array weights. Use the admissible-radii row of equation (885): our constant K_Λ is $(1+\beta_{\mathrm{s}})^4$ times the constant denoted by K_Λ there. The normalized fourth-potential size is

$$\nu(\mathbb{V}') = (K_\Lambda s_j)^{-1} \sup_{Y,\mathfrak{u}} e^{\lambda_V d_j(Y)} |\mathbb{V}'(Y;\mathfrak{u})|.$$

Choose a uniform ordinary-array ceiling $N_\star \geq 3(E_0 + \frac{3}{2}\alpha_0^2 B_\sharp)$ and impose

$$s_j \leq s_\omega^\star := \min\{s^\star, w_\star/(4K_\Lambda N_\star)\},$$

as in equation (886). Indeed $\sum_{n\geq 1}\omega_n \leq 3$, so the ordinary progression has this weighted array bound even if its original iteration used the smaller weights $L^{-(n-1)}$. The smaller ceiling is imposed in the final coupling choice. The constant $K_{\mathbf{M}}$ is evaluated with this weighted K_Λ.

Fix a lattice and an integer $k < K$ through which these places exist. For now the calibrations $t_j \geq \gamma^{-2}$ may be independent, and the coefficients obey $|x_j/t_j - 1| < \sigma_\star$. The ordinary progression is $(E^{(j)}, e_j)$, with $\|E^{(j)}\|_{j,\kappa} \leq E_0$ and $|e_j| \leq B_\sharp$. Write $\|\cdot\|_{j,+}$ for the same pointed-family norm at rate $\kappa^+ = (1+4\beta_s)\kappa$, and N_j for the bare-array norm at height j. The localized step map is St_j. The fourth-potential bound and (1601) give

$$\|\mathsf{St}_j(f) - \mathsf{St}_j(f')\|_{j+1,+} \leq \theta_j N_j(f-f'), \qquad s_j = \frac{(1+\beta_s)\delta(t_j)}{\alpha_0}, \quad \theta_j = K_{\mathbf{M}} s_j, \quad N_{\star,j} = \frac{w_\star}{4K_\Lambda s_j}, \tag{2249}$$

for arrays of size at most $N_{\star,j}$. The map is jointly holomorphic with the field variables, real on real data, and its member on X depends only on the restriction of the input array to X. Bracket localization is required for this last assertion. These statements apply with the same canonical provision and counterterms at every parameter value. To check the coefficient, each fourth-potential norm is at most $w_\star/4$, their difference is at most $w_\star/2$, and the sum of the intrinsic norms is at most $3w_\star/8$. The cluster difference bound therefore costs at most $162L^3 K_\Pi K_\Lambda s_j N_j(f-f')$ at rate $\lambda_{\mathrm{out}} = \kappa^+$. This is bounded by the right side of (2249), since $K_{\mathbf{M}} \geq 243 L^3 K_\Pi K_\Lambda$. The existence, joint holomorphy and actual-input identification are those of Theorems V–W; locality follows through the bracket localization before applying the cluster map.

At $N_j(f) \leq N_{\star,j}/2$, Cauchy's estimate and (2249) give

$$\|D\mathsf{St}_j(f)g\|_{j+1,+} \leq \theta_j N_j(g), \qquad \|D^2\mathsf{St}_j(f)[g_1,g_2]\|_{j+1,+} \leq \frac{2\theta_j}{N_{\star,j}} N_j(g_1)N_j(g_2). \tag{2250}$$

For the second inequality apply the first to g_1 at $f + zg_2$, and use the circle of radius $N_{\star,j}/(2N_j(g_2))$ in z. A limit from smaller circles covers the boundary of the stated half-ball. For holomorphic arrays the ordinary chain rule is valid in these norms: it follows by the Cauchy formula for each local member, with the uniform exponentially weighted bounds.

122.1. **Existence, locality, and the first derivatives.** Let w_1, w_2 be real nonzero arrays on the bare plaquettes, $w_\lambda = \lambda_1 w_1 + \lambda_2 w_2$, and $w_\lambda^{[j]} = L^{K-j} w_\lambda$. Use the weight member $\mathsf{W}_{w_\lambda^{[j]}}$ in the slot

$\{j\}$ of the bare array. At height zero use the bracketed plaquette action, with the same weight, in the fourth place of the first step. Define

$$f_{\lambda,j} = ((E_\lambda^{(i)}, e_i)_{1\le i\le j}, \emptyset) \sqcup \mathsf{W}_{w_\lambda^{[j]}}, \qquad E_\lambda^{(j+1)} = E^{(j+1)} + \mathsf{St}_j(f_{\lambda,j}) - \mathsf{St}_j(f_{0,j}). \tag{2251}$$

The first output has the identical difference formula with the fourth-place family. No source-dependent counterterm is subtracted in this definition.

Here are explicit parameter domains for the iteration. Put

$$m_{j,a} = 12M^3\alpha_0^2 L^{K-j}|w_a|_\infty, \qquad c^\flat = 81C_P^2C_{\mathcal{A}}^2/K_\Lambda, \qquad d_{1,a} = c^\flat\theta_0 m_{0,a},$$

$$n_{j,a} = m_{j,a} + \sum_{i=1}^{j} \omega_{j+1-i} d_{i,a}, \qquad d_{j+1,a} = \theta_j n_{j,a},$$

$$N_j^{\mathrm{rm}} = N_{\star,j} - \Big(\sum_{n=1}^{j} \omega_n\Big)(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp).$$

The recursion for $n_{j,a}$ and $d_{j+1,a}$, and the definition of N_j^{rm}, use $1 \le j \le k$. The canonical smallness hypotheses make $N_j^{\mathrm{rm}} > 0$. Set

$$\Lambda_0 = \{\lambda : c^\flat \sum_a |\lambda_a| m_{0,a} < N_{\star,0}\}, \qquad \Lambda_j = \{\lambda \in \Lambda_{j-1} : \sum_a |\lambda_a| n_{j,a} < N_j^{\mathrm{rm}}\}. \tag{2252}$$

Proposition 122.1. The weighted progression exists through $j+1$ on Λ_j. Each Λ_j is an open convex neighbourhood of zero, invariant under conjugation and coordinatewise contraction. The local members are jointly holomorphic in the parameters and the field variables, and

$$\|E_\lambda^{(j)} - E_{\lambda'}^{(j)}\|_{j,+} \le \sum_a |\lambda_a - \lambda_a'| d_{j,a}. \tag{2253}$$

They depend on w_λ only through its values on the plaquettes in their domain X. Thus $\dot E_a^{(j)} = \partial_a E_\lambda^{(j)}|_0$ vanishes unless X meets $\operatorname{supp} w_a$, and $\ddot E^{(j)} = \partial_1\partial_2 E_\lambda^{(j)}|_0$ vanishes unless it meets both supports. Their norms and gauge-invariance properties are those of pointed families; in particular $\|\dot E_a^{(j)}\|_{j,+} \le d_{j,a}$. If $x_0, \dots, x_{j-1}$ are real, all source derivatives of $E_\lambda^{(j)}$ at zero are real on real configurations. The added action on a bare configuration is exactly

$$A^{\eta_j}(w_\lambda^{[j]}, U) = a_K^{-1} \sum_p w_\lambda(p) a_p(U), \tag{2254}$$

independent of j. The canonical exact sum rule applies to (2251) at every admitted block field. ∎

Proof. The first-step bracket is linear in the weight and supported on the unique elementary domain assigned to a plaquette. In its chart the plaquette displacement is bounded by $\rho \le 2C_P C_{\mathcal{A}} s_0 \alpha_0$. The bracketed plaquette action is bounded by $19\rho^2 + 6\rho\alpha_0 L^{-2}$, which is at most $81C_P^2C_{\mathcal{A}}^2 s_0\alpha_0^2$ because $s_0 \le 1$ and $L \ge 13$. There are at most $12M^3$ plaquettes assigned to one elementary domain. Divide this raw member bound by $K_\Lambda s_0$, as required by the normalized fourth-potential size. The resulting size is at most $c^\flat \sum_a |\lambda_a| m_{0,a}$. The first-step theorem gives existence on Λ_0, its Lipschitz estimate with $d_{1,a}$, and locality. The chart restrictions used here are $\alpha_\iota \le a_P^\circ$ and $C_{\mathcal{A}}\delta(t_0) \le r_P^\circ/2$; these are supplied by the canonical place conditions.

Assume the assertion through level j. The norm of a weight member is at most $6M^3\alpha_0^2 L^{K-j}|w_a|_\infty$. Since its array weight is $\omega_1 = 2$, the induction gives

$$N_j(f_{\lambda,j} - f_{\lambda',j}) \le \sum_a |\lambda_a - \lambda_a'| n_{j,a}.$$

The ordinary progression has array norm at most the subtracted term in N_j^{rm}. Thus $\lambda \in \Lambda_j$ puts $f_{\lambda,j}$ strictly inside the ball of radius $N_{\star,j}$, and (2249) gives existence and (2253) at the next level.

Holomorphy follows from that of the step. Equality of two weights on X gives equality of every earlier member used inside X; locality of the step completes the locality induction.

The parameter domains are finite intersections of strict inequalities $\sum_a c_a|\lambda_a| < c$, with $c > 0$. Cauchy's formula, uniform in the local core and weighted by $e^{\kappa^+ d_j(X)}$, proves the derivative assertions. Gauge invariance passes through the derivatives. When $x_0, \dots, x_{j-1}$ are real, the conjugation identity for $E_\lambda^{(j)}$ also passes through source derivatives at zero. The identity (2254) follows from $\eta_j^{-1} L^{K-j} = a_K^{-1}$ and the weight-member sum rule. Adding it to the existing action gives the input to the canonical exact step. Its bounded Borel hypothesis holds: at height zero the added factor is the exponential of a finite sum of continuous plaquette functions on a compact product of groups, and at later heights it is the action of an admitted bare array. The canonical rule and interaction sum rules therefore give the exact sourced step. □

For the rest of the section take real running coefficients, $t_j = x_j$, set $y_j = \log x_j$, and assume

$$|x_j - L^{-(j-i)} x_i| \le \bar{B} \quad (i \le j \le k), \qquad x_k \ge 4\bar{B}, \qquad y_k \ge \max(2\mathfrak{p}_0, 1 + \log L). \tag{2255}$$

Put $C_0^\flat = \max(2, c^\flat)$, $C_1^\flat = \max(2, (c^\flat)^2/8)$ and

$$S_A = (4/3)^{1/2} \sum_{n \ge 1} (n+1)^{\mathfrak{p}_0 + 1} L^{1-n/2}, \qquad S_B = (4/3)^{1/2} \sum_{n \ge 0} (n+1)^{\mathfrak{p}_0} L^{-n/2}.$$

If $C_0^\flat S_A \theta_k \le 1$, the recursion above implies

$$\begin{gathered} n_{j,a} \le 2m_{j,a}, \qquad d_{j+1,a} \le 2\theta_j m_{j,a} \quad (1 \le j \le k), \\ d_{j,a} \le C_0^\flat \theta_{j-1} m_{j-1,a} \quad (1 \le j \le k+1). \end{gathered} \tag{2256}$$

To check this induction, (2255) gives $\frac{3}{4} L^{j-i} x_j \le x_i \le \frac{5}{4} L^{j-i} x_j$ and

$$\delta(x_i) \le (4/3)^{1/2} L^{-(j-i)/2} (j - i + 1)^{\mathfrak{p}_0} \delta(x_j).$$

Insert this and $m_{i,a} = L^{j-i} m_{j,a}$ in the convolution with ω_{j+1-i}. Its sum is bounded by $C_0^\flat S_A \theta_j m_{j,a} \le m_{j,a}$. The same comparison gives $\sum_{i \le j} \delta(x_i) \le S_B \delta(x_j)$.

122.2. **Separation of the mixed derivative.** Let D_j be the least sup distance, in level-j units, between the corners of plaquettes in the two supports, and put

$$\nu_j = \frac{32 K_{\mathbf{M}} K_\Lambda}{w_\star} s_j^2 m_{j,1} m_{j,2}, \qquad T(u) = \sum_{n \ge 1} (n+1)^{2\mathfrak{p}_0 + 1} e^{-4\beta_s \kappa [L^{n-1} u - 2]_+} \quad (u > 0).$$

The series converges for $u > 0$; set $T(0) = +\infty$.

Proposition 122.2. Assume (2255), $C_0^\flat S_A \theta_k \le 1$, and $s_j \le s_\omega^\star/2$ for $1 \le j \le k$. With $\ddot{d}_j = \|\ddot{E}^{(j)}\|_{j,+}$, one has

$$\|\ddot{E}^{(j)}\|_{j,\kappa} \le e^{-4\beta_s \kappa [D_j/M - 2]_+} \ddot{d}_j, \tag{2257}$$

$$\ddot{d}_{j+1} \le \theta_j \sum_{i=1}^{j} \omega_{j+1-i} e^{-4\beta_s \kappa [D_i/M - 2]_+} \ddot{d}_i + \nu_j. \tag{2258}$$

If also $\frac{4}{3} C_1^\flat L \theta_j T(D_j/M) \le 1$ for $1 \le j \le k$, then $\ddot{d}_j \le C_1^\flat \nu_{j-1}$, and for $1 \le \bar{k} \le k+1$,

$$\sum_{j \le \bar{k}} \sum_{X, z} \|\ddot{E}^{(j)}(X, \cdot, z)\|_X \le 8 C_1^\flat M^3 C_{E,1} N_{\bar{k}}(w_1) \sum_{j \le \bar{k}} L^{3(\bar{k}-j)} \nu_{j-1} e^{-[L^{\bar{k}-j} D_{\bar{k}}/M - 2]_+}. \tag{2259}$$

Here $N_{\bar{k}}(w_1)$ counts the level-$\bar{k}$ M-cubes containing a corner of its support. ■

Proof. Locality forces a nonzero mixed member on X to meet both supports. The diameter bound for a polymer gives $D_j \le M(d_j(X)+2)$, proving (2257). The second derivative of (2251) is

$$\ddot{E}^{(j+1)} = D^2\mathsf{St}_j(f_{0,j})[\dot f_{1,j},\dot f_{2,j}] + D\mathsf{St}_j(f_{0,j})\ddot f_j.$$

The weighted ceiling gives

$$N_j(f_{0,j}) \le N_\star \le \frac{w_\star}{4K_\Lambda s_\omega^\star} \le \frac{w_\star}{8K_\Lambda s_j} = \tfrac12 N_{\star,j}.$$

Thus the ordinary array lies in the half-ball, and (2256) gives $N_j(\dot f_{a,j}) \le 2m_{j,a}$. Equations (2250) and (2257) prove (2258). The same Cauchy estimate at the first step gives $\ddot d_1 \le (c^\flat)^2\nu_0/8$.

For $n = j+1-i$, the running comparison gives $\nu_{i-1} \le \frac43 L^n(n+1)^{2\mathfrak{p}_0}\nu_j$ and $D_i = L^{n-1}D_j$. If $\ddot d_i \le C_1^\flat\nu_{i-1}$, the convolution in (2258) is therefore at most $\frac43 C_1^\flat L\theta_j T(D_j/M)\nu_j \le \nu_j$. This proves the induction. Finally every contributing domain meets one of at most $L^{3(\bar k-j)}N_{\bar k}(w_1)$ level-j cubes. The anchored polymer sum, including its base points, is bounded by $8M^3C_{E,1}e^{-D}$ when $d_j(X) \ge D$. Use $D = [L^{\bar k-j}D_{\bar k}/M - 2]_+$ and sum over j. □

122.3. **Values at a regular background.** The local-term estimate needed below is quadratic in the curvature of the background. Suppose U is regular of level T at radius ε, all its lower-level pairs belong to their cores, and $M\alpha_0 \le \alpha_1$, $\alpha_0+\alpha_1 \le a^\sharp$. Set

$$d_*(j;T,\varepsilon) = \min\left\{\frac{c_A\alpha_0L^{2(T-j)}}{M\varepsilon}, \frac{L^{m+K-j}-4}{2M}\right\} - 1. \tag{2260}$$

For a local term E of level $j \le T$, the regular-gauge estimate of theorem F and the quadratic Cauchy argument in the proof of lemma 64.1 give

$$|E(U)-E(\mathbf{1})| \le \|E\|_X\left[C_A(\varepsilon/\alpha_0)^2L^{-4(T-j)}M^2(1+d_j(X))^2 + 2\mathbf{1}_{\{d_j(X)\ge d_*(j;T,\varepsilon)\}}\right]. \tag{2261}$$

Indeed, lowering the level changes the regularity radius to $\varepsilon L^{-2(T-j)}$. The regular-gauge estimate applies if $(1+d_j(X))M$ times that radius is at most $c_A\alpha_0$ and $2(1+d_j(X))M+4 < L^{m+K-j}$. Failure of either inequality forces the indicated lower bound on $d_j(X)$; the bounded local-term estimate then gives the second summand.

If $d_* = d_*(T;T,\varepsilon) \ge 3$, summing (2261) for the first derivative families gives

$$\sum_{j\le T}\sum_{X,z}|\dot E_a^{(j)}(X,U,z) - \dot E_a^{(j)}(X,\mathbf{1},z)| \le C_f N_T(w_a)[\varepsilon^2/\alpha_0^2 + e^{-d_*}]\alpha_0\delta(x_{T-1})\frac{|w_a|_\infty}{a_{K-T}}. \tag{2262}$$

Here C_f depends only on the fixed construction constants. For clarity, the level-j anchored polymer sum in the first term costs $L^{3(T-j)}$ and its regularity factor gains $L^{-4(T-j)}$, leaving $L^{-(T-j)}d_{j,a}$. For the second term, $d_*(j;T,\varepsilon) \ge L^{T-j}(d_*+1)-1$ and $d_* \ge 3$ give $L^{3(T-j)}e^{-d_*(j;T,\varepsilon)} \le L^{-(T-j)}e^{-d_*}$. Now use (2256) and $\sum_{j\le T}\delta(x_{j-1}) \le S_B\delta(x_{T-1})$. This proves (2262) with a constant uniform in the level and the volume.

123. Covariance transport along the small-field history

Use the canonical real progression and history decomposition, with $x_j = L^{-1}x_{j-1} - e_j$. A level is used only while the canonical density tests and background sum rule are available. Denote by h_t the history with no large-field region through level t, by $\mathcal{M}$ the one-step block map, and by g_t its nonnegative step factor. For bounded Borel functions F of the original field put

$$m_t[F] = \tau_{h_t}[F\rho_0], \qquad m_t = m_t[1], \qquad \Phi_t[F] = \frac{dm_t[F]}{dm_t}.$$

The positive history measure gives $|m_t[F]| \le \|F\|_\infty m_t$, so the derivatives exist with $|\Phi_t[F]| \le \|F\|_\infty$. All quotient identities below are identities modulo m_t-null sets.

123.1. **The sourced density and the conditional derivatives.** On configurations whose pairs belong to all the cores through level t, set

$$D_t^\lambda(U) = \mathcal{O}(w_\lambda)(U) + \sum_{i=1}^{t} \sum_{X,z} (E_\lambda^{(i)} - E^{(i)})(X, U, z), \qquad \mathcal{O}(w) = a_K^{-1} \sum_p w(p) a_p. \tag{2263}$$

Let $\mathcal{V}_t^\chi$ be the open set on which the full small-field density test at level t passes. The canonical test bounds place it inside the class of the exact sum rule; its minimal background $U_t(W)$ exists, belongs to all the lower cores, and is regular of level t at radius $2B_3\varepsilon_t$. Local terms and the plaquette action are constant on the gauge orbit defining this background.

Proposition 123.1. For $\lambda \in \Lambda_{t-1}$,

$$m_t[e^{\mathcal{O}(w_\lambda)}] = \exp\{D_t^\lambda(U_t(W))\}\, m_t(dW). \tag{2264}$$

Both sides are carried by $\mathcal{V}_t^\chi$; the expression on the right is set to zero off that set. The measure m_t has a strictly positive bounded Borel density there. Consequently, for $F_a = \mathcal{O}(w_a)$,

$$\Phi_t[F_a] = c_{t,a} + X_{t,a} + I_{t,a}, \qquad \Phi_t[F_1 F_2] - \Phi_t[F_1]\Phi_t[F_2] = \sum_{i=1}^{t} \sum_{X,z} \ddot{E}^{(i)}(X, U_t, z), \tag{2265}$$

where

$$c_{t,a} = \sum_{i \le t} \sum_{X,z} \dot{E}_a^{(i)}(X, \mathbf{1}, z), \quad X_{t,a} = \mathcal{O}(w_a)(U_t), \quad I_{t,a} = \sum_{i \le t} \sum_{X,z} [\dot{E}_a^{(i)}(X, U_t, z) - \dot{E}_a^{(i)}(X, \mathbf{1}, z)].$$

■

Proof. Let cs_j be the logarithm of the positive flat normalizing factor of step $j+1$. The exponent of the sourced small-field density at height t is

$$A_t^\lambda(U) = -x_0 A^{\eta_0}(U) + \mathcal{O}(w_\lambda)(U) + \sum_{i \le t} \sum_{X,z} E_\lambda^{(i)}(X, U, z) + \sum_{j<t} \mathsf{cs}_j.$$

Indeed, $A^{\eta_i} = L^i A^{\eta_0}$ and the coupling recursion give $L^t x_t + \sum_{i \le t} L^i e_i = x_0$. In the action of the bare array, the flat values $E_\lambda^{(i)}(X, \mathbf{1}, z)$ cancel those in its vacuum factor. The remaining terms are exactly A_t^λ. The first sourced step and then the exact sum rule in proposition 122.1, applied inductively, show that $m_t[e^{\mathcal{O}(w_\lambda)}]$ has density $e^{A_t^\lambda(U_t(W))}$ on $\mathcal{V}_t^\chi$ and zero elsewhere. The sum rule also proves Borel measurability through its chart integral; it does not require a global continuous choice of U_t. The local cores and the finite number of local terms give boundedness. At $\lambda = 0$ every term of the exponent is real, and the density is strictly positive. Dividing it out gives (2264).

For every fixed finite lattice,

$$|D_t^\lambda(U_t(W))| \le \sum_a |\lambda_a| \left(2a_K^{-1} \sum_p |w_a(p)| + \sum_{i \le t} \#\mathcal{Q}_i\, d_{i,a} \right),$$

uniformly in W. Cauchy's formula or dominated differentiation therefore differentiates the measure identity at zero. The first derivative of the exponential is the first derivative of D_t, and its mixed derivative is $\partial_{12} D_t + \partial_1 D_t \partial_2 D_t$. Subtracting the product gives (2265). Difference quotients of the Borel density show that these versions of the quotients are Borel. The reality of the derivatives follows from real source parameters. □

In particular, on the passing set,

$$|X_{t,a}| \le X_{t,a}^* := 2B_3^2 \varepsilon_t^2 L^{-4t} a_K^{-1} \sum_p |w_a(p)|, \qquad |I_{t,a}| \le I_{t,a}^*, \tag{2266}$$

where $I^*_{t,a}$ is the right side of (2262) with $T = t$ and $\varepsilon = 2B_3\varepsilon_t$. The first bound is the plaquette estimate at a regular background. The second requires the hypotheses of (2262), including $d_*(t; t, 2B_3\varepsilon_t) \geq 3$.

123.2. **Splitting at the first large-field region.** Let g_t^{w} be the sum of the nonnegative step factors over the outcomes different from the all-small-field outcome, with input h_{t-1}, and put

$$w_t = Z^{-1} \int g_t^{\mathrm{w}}\, dm_{t-1}, \qquad g_{j,T} = \prod_{j<i\leq T} g_i \circ \mathcal{M}^{i-1-j}, \qquad m_{j|T} = Z^{-1} g_{j,T} m_j.$$

The factors depend only on the current field because the earlier exposed-bond sets of h_{t-1} are empty. The history product formula therefore gives

$$m_t[F] = \mathcal{M}_*(g_t m_{t-1}[F]). \tag{2267}$$

Provided $\varepsilon_j \leq \varepsilon_\flat$ through the levels used, the test partition and its gauge-invariant extension identity give, for bounded gauge-invariant F,

$$\langle F\rangle = Z^{-1} \int \Phi_T[F]\, dm_T + Z^{-1} \sum_{t=1}^{T} \int \Phi_{t-1}[F] g_t^{\mathrm{w}}\, dm_{t-1}. \tag{2268}$$

To obtain this identity, partition the histories according to their first outcome different from the all-small-field outcome. Sum all later extensions of each such prefix; the normalized test partition leaves precisely its prefix mass. The sole remaining prefix is h_T. Thus $w_t \geq 0$, all the measures $m_{j|T}$ have the same mass $1 - \sum_{t\leq T} w_t$, and

$$\sum_{t\leq T} w_t = 1 - \mathsf{t}(h_T)/Z = \widetilde{\nu}_K\{Z_T \neq \emptyset\}. \tag{2269}$$

The last equality uses consistency of the history measures; the region Z_T is empty exactly for prefixes equal to h_T.

Define a step quotient by

$$\mathsf{Qt}_j[F] = \frac{d\mathcal{M}_*(F g_j m_{j-1})}{dm_j}, \qquad \mathsf{Qt}_j[F; G] = \mathsf{Qt}_j[FG] - \mathsf{Qt}_j[F]\mathsf{Qt}_j[G].$$

Positivity gives $|\mathsf{Qt}_j[F]| \leq \|F\|_\infty$ and $|\mathsf{Qt}_j[F; G]| \leq 2\|F\|_\infty \|G\|_\infty$. The defining pushforward gives

$$\int \mathsf{Qt}_j[F]\, dm_{j|T} = \int F\, dm_{j-1|T}. \tag{2270}$$

Put $\Phi_{j,12} = \Phi_j[F_1F_2] - \Phi_j[F_1]\Phi_j[F_2]$. The same pushforward gives the tower identities

$$\Phi_j[F] = \mathsf{Qt}_j[\Phi_{j-1}[F]], \qquad \Phi_{j,12} = \mathsf{Qt}_j[\Phi_{j-1,12}] + \mathsf{Qt}_j[\Phi_{j-1}[F_1]; \Phi_{j-1}[F_2]]. \tag{2271}$$

These are identities of Radon–Nikodym derivatives, obtained by testing against bounded functions on level j. They require no choice of conditional probability kernels.

Proposition 123.2. Fix $2 \leq \mathsf{k} \leq T$ in the admitted range. Assume the running and regularity hypotheses for (2266) through T, and the mixed derivative hypotheses of proposition 122.2 through k. Then

$$\langle \mathcal{O}(w_1); \mathcal{O}(w_2)\rangle = \sum_{\mathsf{k}<j\leq T} \int \mathsf{Qt}_j[X_{j-1,1}; X_{j-1,2}]\, dm_{j|T} + \mathrm{Rest}_T. \tag{2272}$$

Here the error satisfies

$$\begin{aligned} |\mathrm{Rest}_T| \leq M_{\mathsf{k}} + 2 \sum_{\mathsf{k}\leq j<T} (X^*_{j,1} I^*_{j,2} + I^*_{j,1} X^*_{j,2} + I^*_{j,1} I^*_{j,2}) + 2Y_{T,1}Y_{T,2} \\ + \rho_{12} + \rho_1 Y_{T,2} + \rho_2 Y_{T,1} + \rho_1\rho_2. \end{aligned} \tag{2273}$$

The quantities in this estimate are defined entirely by bounds already obtained:

$$Y_{j,a} = X^*_{j,a} + I^*_{j,a}, \qquad B^T_{j,a} = Y_{j,a} + \sum_{j<i\le T}\sum_{X,z} |\dot E^{(i)}_a(X,\mathbf{1},z)|,$$

$$M'_j = M_j \ (j \le \mathsf{k}), \qquad M'_j = M_{\mathsf{k}} + 2\sum_{\mathsf{k}\le i<j} Y_{i,1}Y_{i,2} \ (j > \mathsf{k}),$$

$$\rho_a = \sum_{t\le T} B^T_{t-1,a} w_t, \qquad \rho_{12} = \sum_{t\le T} (M'_{t-1} + B^T_{t-1,1}B^T_{t-1,2}) w_t.$$

M_j is the right side of (2259) at $\bar k = j$; at level zero set $M_0 = I^*_{0,a} = 0$ and $X^*_{0,a} = 2a_K^{-1}\sum_p |w_a(p)|$. ■

Proof. Equation (2265) gives $|\Phi_j[F_a] - c_{j,a}| \le Y_{j,a}$ and, through k, $|\Phi_{j,12}| \le M_j$. The tower identity and the step covariance bound propagate the latter to $|\Phi_{j,12}| \le M'_j$ at every later level. Center the two observables by $c_{T,a}$ and apply (2268) to each centered observable and to their product. The omitted first-large-field contributions to the means are bounded by ρ_a, and the one to the product by ρ_{12}. At the top the remaining centered means are bounded by $Y_{T,a}$. Its measure has mass at most one, so its product-minus-product-of-means term is at most $2Y_{T,1}Y_{T,2}$.

Integrate the second tower identity against $m_{j|T}$ and use (2270). Summation from $\mathsf{k}+1$ to T leaves the initial mixed derivative bounded by M_{k}. In each later covariance substitute $\Phi_{j-1}[F_a] = c_{j-1,a} + X_{j-1,a} + I_{j-1,a}$. Constants cancel; the three terms involving an I are bounded by twice the corresponding products in (2273). Their measures have mass at most one. Collecting these estimates proves the formula and its bound. □

123.3. **Carrying a single-step covariance to a common level.** For a pointed family f of level $i' \le i$, let $f_i[f]$ be the bare array with that member and with all other members and counterterms zero. Then

$$N_i(f_i[f]) = \omega_{i+1-i'}\|f\|_{i',\kappa}, \qquad S[f_i[f]](U) = \mathsf{Sum}[f](U) - \mathsf{Sum}[f](\mathbf{1}),$$

where Sum denotes the finite sum over domains and base points. Put

$$\dot{\mathsf{St}}_i[f] = D\mathsf{St}_i(f_{0,i})f_i[f]. \tag{2274}$$

This is linear, preserves vanishing on domains disjoint from a fixed set of sites, and

$$\|\dot{\mathsf{St}}_i[f]\|_{i+1,+} \le \theta_i\omega_{i+1-i'}\|f\|_{i',\kappa}. \tag{2275}$$

It follows from (2250) and locality; the insertion parameter is restricted to a disc whose product with $N_i(f_i[f])$ is smaller than the remaining array room.

Let $K^{(j)}$, $j = i+1 \ge 2$, be the second derivative of the step with only the two plaquette weight members inserted, leaving its input families equal to their ordinary values:

$$K^{(i+1)} = D^2\mathsf{St}_i(f_{0,i})[f_i[\mathsf{W}_{L^{K-i}w_1}], f_i[\mathsf{W}_{L^{K-i}w_2}]]. \tag{2276}$$

It vanishes unless its domain meets both supports, and $\|K^{(j)}\|_{j,+} \le \nu_{j-1}/4$ when $s_{j-1} \le s^\star_\omega/2$. Differentiating the exact one-step identity gives

$$\mathsf{Qt}_j[X_{j-1,1}; X_{j-1,2}](W) = \mathsf{Sum}[K^{(j)}](U_j(W)). \tag{2277}$$

Indeed, the sourced quotient is the exponential of $\sum_a \lambda_a X_{j,a}$ plus the change in the sum of the output family. Its logarithm's mixed derivative is exactly the displayed sum.

Starting with $K^{(j)\to j} = K^{(j)}$, define

$$K^{(j)\to(i+1)} = \sum_{i'=j}^{i} \dot{\mathsf{St}}_i[K^{(j)\to i'}], \qquad S^{(j)}_n = \sum_{i'=j}^{n} \mathsf{Sum}[K^{(j)\to i'}]. \tag{2278}$$

Set $k_{i'}^{(j)} = \|K^{(j)\to i'}\|_{i',+}$. Equation (2275) gives

$$k_{i+1}^{(j)} \le \theta_i \sum_{i'=j}^{i} \omega_{i+1-i'} k_{i'}^{(j)}. \tag{2279}$$

If $S_\omega \theta_i \le 1$, where $S_\omega = \sum_n \omega_n$, then $k_{i'}^{(j)} \le k_j^{(j)}$ and $k_{i+1}^{(j)} \le S_\omega \theta_i k_j^{(j)}$. Each family still vanishes on domains disjoint from either set of source corners.

Proposition 123.3. For $j \le T$ in the admitted range,

$$\int \mathsf{Qt}_j[X_{j-1,1}; X_{j-1,2}]\, dm_{j|T} = \int S_T^{(j)}(U_T(W))\, dm_{T|T}(W). \tag{2280}$$

If also $\theta_i \le \theta_{T-1}$ and $S_\omega \theta_i \le 1$ for $j \le i < T$, then

$$\begin{aligned} &|S_T^{(j)}(U_T) - S_T^{(j)}(\mathbf{1})| \\ &\qquad \le C_{\mathrm{ft}} k_j^{(j)} \Bigg\{ N_j(w_1) \left[\frac{4B_3^2 \varepsilon_T^2}{\alpha_0^2} L^{-4(T-j)} + e^{-[d_*(j;T,2B_3\varepsilon_T)]_+} \right] \\ &\qquad + S_\omega \theta_{T-1} N_T(w_1) \sum_{i'=j+1}^{T} L^{3(T-i')} \left[\frac{4B_3^2 \varepsilon_T^2}{\alpha_0^2} L^{-4(T-i')} + e^{-[d_*(i';T,2B_3\varepsilon_T)]_+} \right] \Bigg\}. \end{aligned} \tag{2281}$$

This holds at every passing top-level field. The same right side bounds the error in replacing the right side of (2280) by $S_T^{(j)}(\mathbf{1})\, m_{T|T}(\mathcal{V}_T)$. ■

Proof. Insert μf into the bare array at height i and differentiate its exact sourced step. Its input action is $\mu(\mathsf{Sum}[f](U) - \mathsf{Sum}[f](\mathbf{1}))$. The derivative of the output logarithm is the same subtracted action at U_{i+1} plus $\mathsf{Sum}[\dot{\mathsf{St}}_i[f]](U_{i+1})$. Adding back the constant gives the step quotient identity

$$\mathsf{Qt}_{i+1}[\mathsf{Sum}[f](U_i)] = \mathsf{Sum}[f](U_{i+1}) + \mathsf{Sum}[\dot{\mathsf{St}}_i[f]](U_{i+1}).$$

These identities have bounded Borel versions, by difference quotients of the exact chart density as in proposition 123.1. Summing them over $i' = j, \dots, i$ in (2278) shows that $S_{i+1}^{(j)}(U_{i+1})$ is the step quotient of $S_i^{(j)}(U_i)$. Equations (2270) and (2277) telescope to (2280).

Apply (2261) to each carried family at the regular top background. The anchored polymer sum at level i' costs at most $8M^5 \max(C_A C_{E,3}, 2C_{E,1}) N_{i'}(w_1)$ times its norm and the two brackets displayed above. Keep the bound $k_j^{(j)}$ for the first level. At later levels use $k_{i'}^{(j)} \le S_\omega \theta_{T-1} k_j^{(j)}$ and $N_{i'}(w_1) \le L^{3(T-i')} N_T(w_1)$. This proves (2281) with $C_{\mathrm{ft}} = 8M^5 \max(C_A C_{E,3}, 2C_{E,1})$. The final assertion follows because $m_{T|T}$ has mass at most one. □

124. The fluctuation measure near the trivial field

The evaluation at the trivial top still contains the nonlinear background maps at all intermediate levels. We follow these maps downwards, keeping both the potential and its current in their analyticity cores. Their linearized curvatures give the covariance slices of proposition 120.3.

124.1. **Critical curves and the actual shift.** We first record the finite-dimensional consequence of criticality needed below. All pairings and difference quotients in this paragraph are in level-n units, with fine mesh $\xi = \eta_n$.

Lemma 124.1. Suppose U_t is a curve of configurations on a finite torus, $U_0 = \mathbf{1}$, differentiable at zero bond by bond, and that each U_t, $t \ne 0$, is critical on its fibre $\mathcal{M}^n U = \mathcal{M}^n U_t$. Write $U_t = \mathbf{1} + it\xi \dot{A} + o(t)$ and $\mathcal{M}^n U_t = \mathbf{1} + itB + o(t)$. At an admitted side,

$$\dot{A} = H_N B - \nabla^\xi \theta, \qquad \xi^2 D\dot{A} = H_n^{\mathrm{cv}} B$$

for a real site function θ. ■

Proof. Use logarithmic coordinates $U = e^{ia}$ near the identity, and write $\Upsilon(a) = -i \log \mathcal{M}^n(e^{ia})$. The derivative $X = D\Upsilon(0) = Q^{\mathrm{lin}}_{0\to n}$ is onto. Indeed, the one-step right inverse sets a field to zero on bonds within each block and to the prescribed coarse value on bonds crossing the corresponding face. Each averaging path crosses that face once, and all its internal contour terms vanish. Composing these lifts gives a common real right inverse E of the straight and linearized averages. Thus $Z(a) = E[D\Upsilon(a)E]^{-1}$ is defined near zero.

For $v \in \ker X$, put $v_t = v - Z(a_t)D\Upsilon(a_t)v$, where $a_t = -i \log U_t$. Then v_t is tangent to the fibre and $v_t \to v$. If $S(a)$ is the action, criticality gives $DS(a_t)v_t = 0$. Since $DS(0) = 0$ and $a_t/t \to \xi\dot{A}$, division by t gives $D^2S(0)(\xi\dot{A}, v) = 0$ for every $v \in \ker X$. Factoring this linear functional through X gives a unique multiplier $\dot{\omega}$ such that

$$(D\dot{A}, DA')_\xi = (\dot{\omega}, Q^{\mathrm{lin}}_{0\to n}(\xi A'))_1.$$

Testing gradients shows that $\dot{\omega}$ annihilates coarse gradients. The identity $Q^{\mathrm{lin}}_{0\to n}(\xi A') = QA' - \partial\varkappa_n(\xi A')$ therefore yields $D^*D\dot{A} = Q^*\dot{\omega}$.

For completeness the gauge normalization can be imposed without altering this equation. Let θ_1 be constant on each fine block, with block mean $-\varkappa_n(\xi\dot{A})$. Write $\Theta_0 = \ker Q'$ and decompose $f = d^*(\dot{A} + \nabla^\xi\theta_1)$ orthogonally as $f = f' + f''$, with $f' \in \Delta\Theta_0$ and $f'' \perp \Delta\Theta_0$. Choose $\vartheta \in \Theta_0$ with $\Delta\vartheta = -f'$ and put $\theta = \theta_1 + \vartheta$. Then $Q'\theta = -\varkappa_n(\xi\dot{A})$ and $d^*(\dot{A} + \nabla^\xi\theta) = f''$ annihilates every local Landau test. Consequently $h = \dot{A} + \nabla^\xi\theta$ satisfies $Qh = B$ and the Landau system. Uniqueness in equations (931)–(932) gives $h = H_N B$. Taking the curl and using (2228) proves the assertion. □

At height $i \geq 1$, let $\mathfrak{h}$ be the canonical shift system, with constants $(a_\mathfrak{h}, \varepsilon_\mathfrak{h}, c_\mathfrak{h})$. Its full-parameter value at a block-field background is the actual lower background. More precisely, for W in the class of the whole-torus sum rule, U_+ a representative of its minimal orbit, and $B \in \mathsf{Rg}_{x_i}$, put

$$(\mathbf{A}, \mathbf{K}) = \mathfrak{h}_{T_\ell}(\mathbf{1}; P_W, g_i B).$$

These fields are real and

$$\begin{aligned} |\mathbf{A}|,\ |\nabla^{\eta_i}_{U_+}\mathbf{A}|, |\mathbf{K}| &\leq c_\mathfrak{h} \sup |g_i B|, \\ e^{i\eta_i \mathbf{A}} U_+ = U'^u, &\qquad U' \in \mathcal{C}(\mathbf{B}_i, V_W(B)), \\ J^{(i)}(e^{i\eta_i\mathbf{A}}U_+) &= L^{-3}J^{(i+1)}(U_+) + \mathbf{K}. \end{aligned} \tag{2282}$$

Here the constants defining the class of W are those of this same shift system. To check the statement, the class provides the real background, invertible pivot matrices and the entire fluctuation chart, including the existence of $U_i(V_W(B))$. Its triple and $g_i B$ lie in the shift domain. The parameter $\mathbf{1}$ lies strictly inside the decoupling polydisc, since $e^{\bar{\kappa}} > 1$. The reality and field bounds of the shift therefore apply. At this parameter its background identity applies with precisely $V_W(B)$; its current identity is the last equality in (2282). These are the global shift properties in the construction of Propositions 52.2–52.3.

124.2. **Descent inside the analytic cores.** Put

$$\begin{aligned} \rho_{\mathrm{ch}} &= \min\{a_\mathfrak{h}/3, r_\alpha, 1/60\}, &\qquad \rho_{\mathrm{fl}} &= \min\{\varepsilon_\mathfrak{h}, (60c_\mathfrak{h})^{-1}\}, \\ M_{\mathrm{pl}} &= \tfrac{9}{4}c_\mathfrak{h}\rho_{\mathrm{fl}} + 3L^{-2}\rho_{\mathrm{ch}}, &\qquad M_{\mathrm{gr}} &= L^{-1}\rho_{\mathrm{ch}} + \tfrac{11}{10}c_\mathfrak{h}\rho_{\mathrm{fl}}. \end{aligned} \tag{2283}$$

For a chart point $v = (A, \mathbf{J})$ at level $i+1$ and an unscaled fluctuation $\mathcal{A}$, with $\|v\| < \rho_{\mathrm{ch}}$ and $\sup|\mathcal{A}| < \rho_{\mathrm{fl}}$, let $(\mathbf{A}^\mathfrak{h}, \mathbf{K}^\mathfrak{h}) = \mathfrak{h}_{T_\ell}(\mathbf{1}; P[v], \mathcal{A})$ and define

$$\begin{aligned} A^{\mathrm{lw}} &= (i\eta_i)^{-1}\log(e^{i\eta_i\mathbf{A}^\mathfrak{h}}e^{i\eta_{i+1}A}), \\ v^{\mathrm{lw}} &= (A^{\mathrm{lw}}, L^{-3}\mathbf{J} + \mathbf{K}^\mathfrak{h}). \end{aligned} \tag{2284}$$

The logarithm is taken bond by bond on the identity branch. The shift and substitution estimates give holomorphy, reality at real arguments, and

$$\begin{aligned} |A^{\mathrm{lw}}|,\ |\nabla^{\eta_i} A^{\mathrm{lw}}| &\le L^{-1}\|v\| + \tfrac{11}{10} c_{\mathfrak{h}} \sup|\mathcal{A}|, \\ |\partial(e^{i\eta_i A^{\mathrm{lw}}}) - \mathbf{1}| &\le \eta_i^2 (3L^{-2}\|v\| + \tfrac{9}{4} c_{\mathfrak{h}} \sup|\mathcal{A}|). \end{aligned} \tag{2285}$$

Indeed $P[v]$ has field size at most $3\rho_{\mathrm{ch}}$ and current and potential size at most ρ_{ch}, so it lies both in the shift domain and in the usable core. In the substitution estimate its three parameters are $L^{-1}\|v\|$, $3L^{-2}\|v\|$ and $c_{\mathfrak{h}} \sup|\mathcal{A}|$, each less than $1/60$. This proves the domain assertion as well as the displayed bounds.

Let $\mathsf{Cc}_n(r)$ consist of real potentials A such that

$$\|(A, J^{(n)}(e^{i\eta_n A}))\| \le r, \qquad e^{i\eta_n A} = U^u, \quad U \in \mathcal{C}(\mathbf{B}_n, W), \quad \varpi(W) < \varepsilon_{\mathcal{W}} \alpha_0.$$

The second condition ensures that this small chart point is an actual background. Write $v_A^{[n]} = (A, J^{(n)}(e^{i\eta_n A}))$.

Proposition 124.2. Suppose

$$r < \rho_{\mathrm{ch}}, \qquad \delta_i < \rho_{\mathrm{fl}}, \qquad 6L^{-2} r + 8 C_{\mathrm{db}} \delta_i < \varepsilon_{\mathcal{W}} \alpha_0.$$

For $A \in \mathsf{Cc}_{i+1}(r)$ and $B \in \mathsf{Rg}_{x_i}$, the lowering map at $(v_A^{[i+1]}, g_i B)$ satisfies

$$A^{\mathrm{lw}} \in \mathsf{Cc}_i(L^{-1} r + \tfrac{11}{10} c_{\mathfrak{h}} \delta_i), \qquad v^{\mathrm{lw}} = v_{A^{\mathrm{lw}}}^{[i]}. \tag{2286}$$

With $\widetilde{U} = e^{i\eta_{i+1} A}$ and $\widetilde{U}^{\mathrm{lw}} = e^{i\eta_i A^{\mathrm{lw}}}$, every admitted bare array $\mathfrak{f}$ has the exact identity

$$(\mathbb{V}'[\mathfrak{f}])^{\mathrm{tot}}(P[v_A]; B) = S[\mathfrak{f}](\widetilde{U}^{\mathrm{lw}}) - S[\mathfrak{f}](\widetilde{U}). \tag{2287}$$

In particular this is the difference of the two pointed totals for an inserted family, and for a plaquette weight it is

$$a_K^{-1} \sum_p w(p) [a_p(\widetilde{U}^{\mathrm{lw}}) - a_p(\widetilde{U})]. \tag{2288}$$

Also $0 \in \mathsf{Cc}_n(0)$. ∎

Proof. Choose the axial representative U_+ of the background over W, and write $\widetilde{U} = U_+^u$. Gauge covariance carries B to $\check{B} = \mathrm{Ad}(u)^{-1} B$, preserving its allowed radius. The pair of $P[v_A]$ is the pair of P_W^u; the shift depends only on that pair. Thus (2282) at $(P_W, g_i \check{B})$, conjugated by u, gives both the lower minimal orbit and the current identity in (2286). Equation (2285) bounds the potential and its derivative, and the current is bounded by $L^{-3} r + c_{\mathfrak{h}} \delta_i$.

The upper background has regularity number less than $3r'$ for any $r' > r$ sufficiently close to r. The block-background estimate bounds its averaged plaquette size by $6L^{-2} r'$. The fluctuation chart has the form $V_W(\check{B}) = e^{iB'} V_i^{\mathrm{bf}}(W)$ with $\sup|B'| < 2 C_{\mathrm{db}} \delta_i$. Telescoping the four unitary plaquette factors gives $\varpi(V_W(\check{B})) < 6L^{-2} r' + 8 C_{\mathrm{db}} \delta_i$, which is less than $\varepsilon_{\mathcal{W}} \alpha_0$. This verifies the lower class condition, not only its norm bound.

The whole-torus sum rule for the fourth potential at P_W is the difference of the array actions at the actual two backgrounds. Gauge invariance of the actions and covariance and pair dependence of the local potentials carry it to $P[v_A]$. This proves (2287); the definition of the plaquette array gives (2288). The identity field is flat, critical and has zero current, proving the last assertion. □

If $r_T = 0$ and $r_i = L^{-1} r_{i+1} + \tfrac{11}{10} c_{\mathfrak{h}} \delta_i$, then under the running-coupling bounds of (2255), extended through T,

$$r_i \le \tfrac{8}{5} c_{\mathfrak{h}} \delta_i. \tag{2289}$$

Indeed $\delta_{i+n} \le (5L/4)^{n/2}\delta_i$, and unwinding the recursion gives the geometric series

$$\frac{11}{10}\sum_{n\ge 0}(5/(4L))^{n/2} < 8/5.$$

Thus the preceding proposition can be iterated whenever its three inequalities hold with $r = r_{i+1}$.

124.3. **The quadratic curvature and its covariance.** Define

$$\operatorname{ang}_p^{[n]}(A) = \eta_n(dA)(p), \qquad T_i^{\mathfrak{h}}\mathcal{A} = \eta_i d(D_{\mathcal{A}}\mathbf{A}^{\mathfrak{h}}(0,0)\mathcal{A}), \qquad T_i^{\mathrm{cv}} = H_i^{\mathrm{cv}}C_{\mathbf{1},\mathbb{R}}.$$

Lemma 124.3. Suppose $\ell \ge N_{\mathsf{H}}$. There is a sign $\sigma_i \in \{1,-1\}$ such that $T_i^{\mathfrak{h}} = \sigma_i(T_i^{\mathrm{cv}} \otimes I)$. ■

Proof. Constant gauge covariance and reality imply that each matrix entry of the linearized shift is a real scalar on the colour space. In fact it commutes with the adjoint action of $SU(2)$: the three half-turns about the Pauli axes make it diagonal, and an adjoint rotation cyclically permuting those axes makes the three diagonal entries equal.

For a real fluctuation $\mathcal{A}$, apply (2282) over $W = \mathbf{1}$ to $t\mathcal{A}$. It gives a differentiable curve U_t through $\mathbf{1}$ of critical configurations, with coarse field $V_t^{w_t}$, where $V_t = \exp(itC_{\mathbf{1}}\mathcal{A} + o(t))$. Absorb the part of the gauge transformation equal to the identity on coarse sites into the minimal orbit. The remaining gauge is its extension from w_t on the coarse sites. By lemma 124.1, $T_i^{\mathfrak{h}}\mathcal{A} = (H_i^{\mathrm{cv}} \otimes I)B$, where $V_t^{w_t} = \mathbf{1} + itB + o(t)$.

Fix a coarse site y_0, put $g_t = w_t(y_0)$ and $\check{w}_t = g_t^{-1}w_t$. The bond identity for $V_t^{w_t}$ implies $\check{w}_t(c_-) - \check{w}_t(c_+) = O(t)$. The finite coarse graph is connected, so $\check{w}_t = \mathbf{1} + O(t)$ everywhere. The real site functions $\lambda_t = (it)^{-1}\log \check{w}_t$ are bounded. Along a subsequence $g_t \to g_*$ and $\lambda_t \to \lambda_*$, and expanding the bond identity gives

$$B = \mathrm{Ad}(g_*)(C_{\mathbf{1}}\mathcal{A} - d\lambda_*).$$

The curvature response kills gradients, whence $T_i^{\mathfrak{h}}\mathcal{A} = \mathrm{Ad}(g_*)(T_i^{\mathrm{cv}} \otimes I)\mathcal{A}$.

Apply this to $\mathcal{A} = t_1 a$ for a real scalar field a. The two scalar response matrices, say S, T, satisfy $Sa = \sigma(a)Ta$ with $\sigma(a) \in \{1,-1\}$ when $Ta \ne 0$, and $Sa = 0$ otherwise. If Ta, Ta' are independent, linearity applied to $a + a'$ forces their signs to agree. If they are dependent, subtract a multiple and use $\ker T \subset \ker S$. Thus one sign works for every a and, by complex linearity, for every colour-valued field. Its value is immaterial below. □

Lemma 124.4. Let $v, \mathcal{A}$ be real, $\|v\| \le r$, $\sup|\mathcal{A}| \le a$, and $R = \min(\rho_{\mathrm{ch}}/r, \rho_{\mathrm{fl}}/a) \ge 2$. A quotient with zero denominator is interpreted as ∞, and $\infty^{-n} = 0$ for $n > 0$. With $\Phi_p = \operatorname{ang}_p^{[i+1]}(A) + T_i^{\mathfrak{h}}\mathcal{A}(p)$,

$$\begin{aligned} |a_p(e^{i\eta_i A^{\mathrm{lw}}}) - \tfrac{1}{2}|\Phi_p|^2| &\le M_{\mathrm{pl}}^2\eta_i^4 R^{-3}, \\ |a_p(e^{i\eta_{i+1}A}) - \tfrac{1}{2}|\operatorname{ang}_p^{[i+1]}(A)|^2| &\le M_{\mathrm{pl}}^2\eta_i^4 R^{-3}, \\ |\operatorname{ang}_p^{[i]}(A^{\mathrm{lw}}) - \Phi_p| &\le 4M_{\mathrm{gr}}\eta_i^2 R^{-2}. \end{aligned} \tag{2290}$$

Also $|\operatorname{ang}_p^{[i+1]}(A)| \le 2L^{-2}\eta_i^2 r$ and $|T_i^{\mathfrak{h}}\mathcal{A}(p)| \le 2c_{\mathfrak{h}}\eta_i^2 a$. ■

Proof. Scale both arguments by ζ, $|\zeta| < R$. The lowered potential is holomorphic and has first derivative $L^{-1}A + D_{\mathcal{A}}\mathbf{A}^{\mathfrak{h}}(0,0)\mathcal{A}$; the derivative in v vanishes since the shift is zero when $\mathcal{A} = 0$. If $E(\zeta)$ is its plaquette variable minus I, (2285) gives $|E(\zeta)| \le \eta_i^2 M_{\mathrm{pl}}$. For 2×2 unimodular matrices, $1 - \frac{1}{2}\operatorname{tr}(I+E) = \frac{1}{2}\det E$. Thus this holomorphic scalar function has modulus at most $\frac{1}{2}\eta_i^4 M_{\mathrm{pl}}^2$ and its quadratic coefficient is $\frac{1}{2}\det(i\Phi_p) = \frac{1}{2}|\Phi_p|^2$. Cauchy's estimate and $R \ge 2$ bound the terms of order at least three by $M_{\mathrm{pl}}^2\eta_i^4 R^{-3}$. Set $\mathcal{A} = 0$ for the second assertion. Apply the same argument to the linear curl of the lowered potential, whose modulus is at most $2\eta_i^2 M_{\mathrm{gr}}$ and first coefficient is Φ_p, to get the quadratic remainder. The last two estimates follow by differentiating the shift's gradient estimate at zero and by taking the curl of A. □

124.4. **Four fluctuation cumulants.** At $P[v]$ let C_v be the whole covariance of the canonical Gaussian structure. Write $V_i^{\mathrm{int}} = V_i^J + V_i^G + V_i^A$ and $V_i^0 = V_i'[f_{0,i}]$. The probability measure used in one real step is

$$d\nu_{i,v}(B) = \Xi_i(v)^{-1}\mathbf{1}_{\{|B(b)|<\mathfrak{b}(x_i)\ \forall b\}}\exp\{(V_i^{\mathrm{int}}+V_i^0)^{\mathrm{tot}}(P[v];B)\}\,d\mu_{C_v}(B). \tag{2291}$$

The cluster identity (716) of theorem R makes $\Xi_i(v)$ nonzero throughout the complex core. The intrinsic and ordinary-array potentials are both of the bounded-potential sort and satisfy

$$\|V_i^{\mathrm{int}}\|_{\star,\lambda_V} \le 3K_P\delta_i, \qquad \|V_i^0\|_{\star,\lambda_V} \le K_\Lambda s_i N_\star.$$

Their sum is bounded by

$$n_i^V = \left(\tfrac{7}{2}K_P + K_\Lambda(1+\beta_s)N_\star/\alpha_0\right)\delta_i.$$

The canonical place conditions ensure that these two norms are at most $w_\star/2$ and $w_\star/4$, respectively. Define

$$\begin{aligned} n_f &= \tfrac{1}{2}M^{3/2}, \qquad c_C = 2/\gamma_r,\\ p_{\star,i} &= C_{\mathrm{ex}}((L+2)M)^3 e^{4\lambda_1}e^{-\gamma_2\mathfrak{b}(x_i)^2/2},\\ \epsilon_i^{\mathrm{cu}} &= C_{E,0}(K_\Pi n_i^V + K_P^\Pi p_{\star,i})(4n_f/w_\star)^4. \end{aligned} \tag{2292}$$

Proposition 124.5. For real v with $\|v\| < r_\alpha$, $\nu_{i,v}$ is a probability measure with positive density on the fluctuation box. For components $B_a = B^{\alpha_a}(b_a)$, repetitions allowed, the cumulants of orders at most four obey

$$\begin{aligned} |\langle B_1;\cdots;B_n\rangle_{\nu_{i,v}}| &\le \epsilon_i^{\mathrm{cu}} \quad (n=1,3,4),\\ |\langle B_1;B_2\rangle_{\nu_{i,v}} - C_v(1,2)| &\le \epsilon_i^{\mathrm{cu}}. \end{aligned} \tag{2293}$$

The covariance is holomorphic in the complex chart ball and

$$|C_v(a,b)| \le c_C, \qquad |C_v(a,b) - C_0(a,b)| \le 2c_C\|v\|/r_\alpha. \tag{2294}$$

Its value at zero satisfies

$$C_0 = (A_i^{\mathrm{cv}})^{-1}\otimes I, \qquad T_i^{\mathrm{cv}}(A_i^{\mathrm{cv}})^{-1}(T_i^{\mathrm{cv}})^* = \eta_i\mathcal{C}_{(i+1)}. \tag{2295}$$

These component estimates do not claim spatial decay. ∎

Proof. The real reference inverse in the Gaussian structure has gap γ_r, and its perturbation has norm at most $\widehat{\epsilon} \le \gamma_r/2$. Hence

$$\mathrm{Re}(B^*, C_v^{-1}B) \ge \tfrac{1}{2}\gamma_r\|B\|^2, \qquad \|C_v\| \le 2/\gamma_r.$$

Conjugation covariance makes C_v and its inverse real at real v; they are symmetric, so the displayed inequality makes them positive definite. The potentials are real there. Thus the unnormalized density in (2291) is strictly positive on an open box containing zero. Its integral is positive, proving the probability assertion.

Place the insertion $B^\alpha(b)$ in the elementary domain $\square(b)$ and set all its other members to zero. Its norm is at most n_f, since $t/(1+M^{-3}t^2) \le M^{3/2}/2$. Add $\sum_{a=1}^n z_a B_a$ as a single quadratic-sort origin, on the polydisc $|z_a| < w_\star/(nn_f)$. The resulting system has three origins and is admitted by the Gaussian cluster theorem. Its connected terms and the whole integral are holomorphic in z, and the nonzero-integral identity identifies their sum with the logarithm of that integral. After division by its value at zero, the mixed derivative is exactly the joint cumulant.

Remove the cutoff and the two bounded-potential origins, retaining the insertion origin. The Gaussian-part comparison (724) bounds the difference of connected terms on X by $(K_\Pi n_i^V + K_P^\Pi p_{\star,i})e^{-\lambda_{\mathrm{out}}d_{i+1}(X)}$ throughout the insertion polydisc. Cauchy's formula adds the factor $(nn_f/w_\star)^n$. The mixed derivative vanishes unless X contains the entry block of every insertion; anchoring at one such block bounds the sum over X by $C_{E,0}$. For $1 \le n \le 4$ this proves (2293), because the uncut generating function is $\exp(\frac{1}{2}\sum_{a,b} z_a z_b C_v(a,b))$.

The whole covariance is the finite matrix $C + C\Gamma\Gamma^{*}C$ from the conditioning identity; its entries are holomorphic on the chart ball. For $v \neq 0$ apply Schwarz's lemma to $C_{\zeta v}(a,b) - C_0(a,b)$ on $|\zeta| < r_\alpha/\|v\|$, where its modulus is at most $2c_C$. This gives (2294). Finally, the canonical provision at the trivial pair has the actual constrained Hessian used in proposition 120.3. That proposition and its curvature covariance identity give (2295) with the counting pairings used here. □

Differentiating the normalized one-step integral also supplies the insertions used in section 123:

$$\begin{aligned} \mathsf{Sum}[\dot{\mathsf{St}}_i[f]](P[v]) &= \nu_{i,v}((V_i'[f])^{\mathrm{tot}}), \\ \mathsf{Sum}[K^{(i+1)}](P[v]) &= \langle (V_i'[w_1])^{\mathrm{tot}}; (V_i'[w_2])^{\mathrm{tot}} \rangle_{\nu_{i,v}}. \end{aligned} \tag{2296}$$

The normalization term is independent of the inserted array, so its derivative vanishes. The integral is nonzero on a common parameter neighbourhood and the fluctuation box is bounded; thus differentiation of its logarithm is legitimate. Equations (2287)–(2288) identify these insertions with the actual lower-background differences.

125. Gaussian evaluation of the transported covariance

Take $2 \le \mathsf{k} < T \le K$, with the preceding canonical data and exact covariance reduction through T. Suppose $w_a(p) = f_a(x(p))$, where each f_a is real Lipschitz and supported in a ball of radius $d \ge a_K$, and the two nonempty plaquette supports have corner separation $D_{\mathsf{k}} \ge 3M$ in level-k units. Put

$$\begin{aligned} \widehat{d}_i &= d/a_{K-i}, \quad P_i = 81\widehat{d}_i^{\,3}, \quad \upsilon_i = (a_{K-i+1}x_{i-1})^{-1}, \\ u_a^{[i]} &= \upsilon_i \|w_a\|_\infty, \quad \widehat{m}_a = u_a^{[\mathsf{k}+1]}, \quad Y_i = 1 + A_1 y_i^{\mathfrak{p}_0}. \end{aligned} \tag{2297}$$

Assume the running bounds through T:

$$|x_j - L^{-(j-i)}x_i| \le \bar{B} \quad (i \le j \le T), \qquad x_T \ge 4\bar{B}, \quad y_T \ge \max(2\mathfrak{p}_0, 1 + \log L). \tag{2298}$$

Put $n_a = \#\operatorname{supp} w_a$. Each support contains at most $81(d/a_K)^3$ plaquettes: for each of three directions its barycentres lie in a shifted grid with at most $(2d/a_K + 1)^3 \le 27(d/a_K)^3$ points in the enclosing cube. Consequently $n_a \eta_i^3 \le P_i$. The same coordinate count for the source corners gives

$$N_i(w_a) \le (2\widehat{d}_i/M + 3)^3, \qquad N_i(w_a) \le L^{3(i'-i)} N_{i'}(w_a) \quad (i \le i').$$

The coupling bounds give, for $n = j - i$,

$$\begin{aligned} \tfrac{3}{4}L^n x_j \le x_i \le \tfrac{5}{4}L^n x_j, \quad y_j \le y_i \le (n+1)y_j, \quad \delta_i \le \delta_j, \quad \varepsilon_i \le \varepsilon_j, \\ \upsilon_j \le \tfrac{5}{4}\upsilon_{\mathsf{k}+1}, \qquad |\upsilon_j - \upsilon_{\mathsf{k}+1}| \le \upsilon_{\mathsf{k}+1}\bar{B}/x_{j-1} \quad (\mathsf{k} < j \le T). \end{aligned} \tag{2299}$$

The first line follows directly by comparison with x_j; monotonicity of $x^{-1/2}(\log x)^{\mathfrak{p}_0}$ holds for $\log x \ge 2\mathfrak{p}_0$. For the last line divide $|x_{j-1} - L^{-(j-1-\mathsf{k})}x_{\mathsf{k}}| \le \bar{B}$ by the corresponding physical inverse coupling.

Let r_i be the descending radii in (2289), and set

$$c_R = \max\left\{ \tfrac{8}{5}\sqrt{5L/4}\, c_{\mathfrak{h}}/\rho_{\mathrm{ch}}, \rho_{\mathrm{fl}}^{-1} \right\}, \qquad e_i^{\mathrm{co}} = \epsilon_i^{\mathrm{cu}} + \tfrac{16}{5}\sqrt{5L/4}\, c_C c_{\mathfrak{h}} \delta_i / r_\alpha.$$

In addition to the place conditions assume

$$\begin{aligned} \ell \ge N_{\mathsf{H}}, \qquad c_R\delta_i \le \tfrac{1}{2}, \qquad e_i^{\mathrm{co}} \le 1, \\ [\tfrac{48}{5}\sqrt{5/4}L^{-3/2}c_{\mathfrak{h}} + 8C_{\mathrm{db}}]\delta_i < \varepsilon_{\mathcal{W}}\alpha_0 \quad (\mathsf{k} \le i < T). \end{aligned} \tag{2300}$$

These imply all three hypotheses of the descent proposition with $r = r_{i+1}$, and the Taylor radius there has inverse at most $c_R\delta_i$.

For a colour-valued plaquette field F, define

$$Q^{(j)}(F) = \upsilon_j a_K^{-1} \sum_{p,q} w_1(p) w_2(q) \mathcal{C}_{(j)}(p,q) \langle F_p, F_q \rangle. \tag{2301}$$

The Gaussian contribution after transport from j through i is

$$\begin{aligned} n_{j,i}^{G} = {} & \tfrac{N^2-1}{2} \upsilon_j^2 \sum_{p,q} w_1(p) w_2(q) \mathcal{C}_{(j)}(p,q)^2 \\ & + (N^2-1) \sum_{j<l\le i} \upsilon_j \upsilon_l \sum_{p,q} w_1(p) w_2(q) \mathcal{C}_{(j)}(p,q) \mathcal{C}_{(l)}(p,q). \end{aligned} \tag{2302}$$

Here $N = 2$; retaining $N^2 - 1$ displays the colour multiplicity. There are fixed finite constants C_F, C_F' for which the errors are defined by

$$\begin{aligned} e_{j,j} &= C_F Y_{j-1}^4 P_{j-1}^2 (\delta_{j-1} + e_{j-1}^{\mathrm{co}}) u_1^{[j]} u_2^{[j]}, \\ e_{j,i+1} &= e_{j,i} + C_F' Y_i^2 P_{j-1} P_i (\delta_i + e_i^{\mathrm{co}}) u_1^{[j]} u_2^{[i+1]}. \end{aligned} \tag{2303}$$

Their values are fixed by the estimates in the following proof, independently of i, j, K, ℓ and the sources.

Proposition 125.1. Under these hypotheses, for $\mathsf{k} < j \le i \le T$ and $A \in \mathsf{Cc}_i(r_i)$,

$$\left| S_i^{(j)}(e^{i\eta_i A}) - n_{j,i}^{G} - Q^{(j)}(\mathrm{ang}^{[i]} A) \right| \le e_{j,i}. \tag{2304}$$

In particular $|S_T^{(j)}(\mathbf{1}) - n_{j,T}^{G}| \le e_{j,T}$. ∎

Proof. At height i, the lowered potential lies in $\mathsf{Cc}_i(r_i)$ for every fluctuation in the support of $\nu = \nu_{i,v_A}$. This is a probability measure. Write $F = \mathrm{ang}^{[i+1]} A$ and $T = T_i^{\mathrm{cv}}$. The linearized lower angle is $F + \sigma_i g_i T B$. The bounds already proved give

$$\sum_b |T(p,b)| \le C_{\mathrm{tp}}^{\mathrm{row}} \eta_i^2, \quad |F_p| \le 2 L^{-2} \eta_i^2 r_{i+1}, \quad g_i |TB(p)| \le 2 c_{\mathfrak{h}} \eta_i^2 \delta_i.$$

The means and third and fourth cumulants of the components of B are at most ϵ_i^{cu}; their second cumulants differ from C_0 by at most $e = e_i^{\mathrm{co}} \le 1$. Furthermore $|C_0(a,b)| \le c_C$.

For the initial step take $i = j - 1$. The insertion identities give $S_j^{(j)}(\widetilde{U}) = \langle \Sigma_1; \Sigma_2 \rangle_\nu$, where Σ_a is (2288). Split it as $\Sigma_a = L_a + Q_a + R_a$, with

$$L_a = \sigma_i g_i a_K^{-1} \sum_p w_a(p) \langle F_p, TB(p) \rangle, \qquad Q_a = \tfrac{1}{2} g_i^2 a_K^{-1} \sum_p w_a(p) |TB(p)|^2.$$

If $L_a = \sum c_a(b,\alpha) B^\alpha(b)$ and $Q_a = \frac{1}{2} \sum K_a(b,b') \sum_\alpha B^\alpha(b) B^\alpha(b')$, the row bound, $n_a \eta_i^3 \le P_i$ and $a_K^{-1} \eta_i = x_i \upsilon_{i+1}$ give

$$\begin{aligned} \|R_a\|_\infty &\le 2 M_{\mathrm{pl}}^2 c_R^3 \delta_i^3 x_i P_i u_a^{[j]}, \\ \|L_a\|_\infty &\le c_{\mathfrak{h}}^2 \delta_i^2 x_i P_i u_a^{[j]}, & \|Q_a\|_\infty &\le 2 c_{\mathfrak{h}}^2 \delta_i^2 x_i P_i u_a^{[j]}, \\ \|c_a\|_1 &\le C_{\mathrm{tp}}^{\mathrm{row}} c_{\mathfrak{h}} A_1 y_i^{\mathfrak{p}_0} P_i u_a^{[j]}, & \|K_a\|_1 &\le (C_{\mathrm{tp}}^{\mathrm{row}})^2 P_i u_a^{[j]}. \end{aligned}$$

The covariance bound $2\|G\|_\infty \|H\|_\infty$ handles every term containing R_a. Using $c_R \delta_i \le 1/2$, their sum is at most

$$[24 c_{\mathfrak{h}}^2 M_{\mathrm{pl}}^2 c_R^3 + 4 M_{\mathrm{pl}}^4 c_R^5] \delta_i (\delta_i^2 x_i)^2 P_i^2 u_1^{[j]} u_2^{[j]}.$$

The remaining calculation uses only four cumulants. Differentiating $\exp(\log \nu(e^{\sum z_a B_a}))$ expresses each moment as the sum over partitions of the component indices. For a covariance of two products, exactly the partitions with a block crossing the two products remain. Thus

$$|\langle B_1; B_2 B_3 \rangle_\nu| \le (3 + 2 c_C) \epsilon_i^{\mathrm{cu}},$$

and the covariance of B_1B_2 with B_3B_4 differs from $C_0(1,3)C_0(2,4)+C_0(1,4)C_0(2,3)$ by at most $(11+8c_C)e$. Indeed the eleven crossing partitions give one fourth cumulant, four third-cumulant times mean terms, two covariance products and four covariance times two means; the error is bounded by $\epsilon+4\epsilon^2+4(c_C+1)\epsilon^2+2e(2c_C+e)$, with $\epsilon\le e\le 1$.

Replacing the covariance in $\langle L_1;L_2\rangle$ by C_0 gives $Q^{(j)}(F)$ with error at most $e\|c_1\|_1\|c_2\|_1$. The two mixed linear-quadratic covariances cost at most $\frac{3}{2}(3+2c_C)\epsilon(\|c_1\|_1\|K_2\|_1+\|c_2\|_1\|K_1\|_1)$. The quadratic-quadratic covariance has error at most $\frac{9}{4}(11+8c_C)e\|K_1\|_1\|K_2\|_1$. Its two Gaussian contractions each use $TC_0T^*=\eta_i\mathcal{C}_{(j)}\otimes I$; the colour sum is N^2-1, and their total is $n^G_{j,j}$. Here $g_i^2\eta_i/a_K=\upsilon_j$ and $\sigma_i^2=1$. Since $x_i\delta_i^2\le Y_i^2$, these estimates prove the initial case with

$$C_F=24c_{\mathfrak{h}}^2M_{\mathrm{pl}}^2c_R^3+4M_{\mathrm{pl}}^4c_R^5+(C_{\mathrm{tp}}^{\mathrm{row}})^2c_{\mathfrak{h}}^2 \\ +3(3+2c_C)(C_{\mathrm{tp}}^{\mathrm{row}})^3c_{\mathfrak{h}}+\tfrac{9}{4}(11+8c_C)(C_{\mathrm{tp}}^{\mathrm{row}})^4.$$

For the induction step, the first insertion identity and the exact action difference imply

$$S_{i+1}^{(j)}(\widetilde{U})=\int S_i^{(j)}(\widetilde{U}^{\mathrm{lw}})\,d\nu.$$

The constant old total cancels the subtracted action in the new carried family. Integrating the induction hypothesis therefore costs only $e_{j,i}$. Let $\overline{Q}^{(j)}$ be the absolute coefficient sum in (2301). For any two fields,

$$|Q^{(j)}(F')-Q^{(j)}(F'')|\le\overline{Q}^{(j)}(\|F'\|_\infty+\|F''\|_\infty)\|F'-F''\|_\infty.$$

The angle remainder in (2290) then bounds replacement of the true lower angle by $F+\sigma_ig_iTB$ by $22M_{\mathrm{gr}}c_{\mathfrak{h}}c_R^2\delta_i^3\overline{Q}^{(j)}\eta_i^4$. Its linear expectation costs at most $C_{\mathrm{tp}}^{\mathrm{row}}c_{\mathfrak{h}}g_i\delta_ie\overline{Q}^{(j)}\eta_i^4$. The second moment of two field components differs from C_0 by at most $e+\epsilon^2\le 2e$; hence its quadratic expectation equals $n^G_{j,i+1}-n^G_{j,i}$ with error at most $6(C_{\mathrm{tp}}^{\mathrm{row}})^2g_i^2e\overline{Q}^{(j)}\eta_i^4$. The slice bound gives

$$\overline{Q}^{(j)}\eta_i^4\le C_{\mathrm{tp}}P_{j-1}P_ix_iu_1^{[j]}u_2^{[i+1]}.$$

Since $x_i\delta_i^3\le Y_i^2\delta_i$, $x_ig_i\delta_i\le Y_i$ and $x_ig_i^2=1$, the induction follows with

$$C'_F=C_{\mathrm{tp}}[22M_{\mathrm{gr}}c_{\mathfrak{h}}c_R^2+C_{\mathrm{tp}}^{\mathrm{row}}c_{\mathfrak{h}}+6(C_{\mathrm{tp}}^{\mathrm{row}})^2].$$

At T, the identity potential belongs to $\mathsf{Cc}_T(0)$ and its angle vanishes, proving the final assertion. □

125.1. **The seven errors in the covariance.** Assume in addition $S_\omega\theta_i\le 1$, $s_i\le s_\omega^\star/2$ and $\theta_i\le\theta_{T-1}$ for $\mathsf{k}\le i<T$, and choose $\mathsf{k}<j_1\le T$. Let $\mathrm{Ft}_T^{(j)}$ be the right side of (2281), $W_T=1-m_T(\mathcal{V}_T)/Z$, $C'_{\mathrm{tp}}=C_{\mathrm{tp}}/(1-L^{-3})$, and

$$\overline{n}_j^G=\tfrac{25}{16}(N^2-1)C_{\mathrm{tp}}C'_{\mathrm{tp}}P_{j-1}^2\widehat{m}_1\widehat{m}_2.$$

Define the nonnegative errors

$$E_2=2\sum_{j_1<j\le T}X^*_{j-1,1}X^*_{j-1,2},\qquad E_3=\sum_{\mathsf{k}<j\le j_1}\mathrm{Ft}_T^{(j)},$$
$$E_4=W_T\sum_{\mathsf{k}<j\le j_1}(\overline{n}_j^G+e_{j,T}),\qquad E_5=\sum_{\mathsf{k}<j\le j_1}e_{j,T},\qquad E_6=\sum_{j_1<j\le T}\overline{n}_j^G,$$
$$E_7=\tfrac{N^2-1}{2}\tfrac{9}{4}(C'_{\mathrm{tp}})^2P_{\mathsf{k}}^2\frac{\bar{B}}{x_T}\widehat{m}_1\widehat{m}_2,$$
$$E_8=\tfrac{N^2-1}{2}C_{\mathrm{tp}}^{\mathrm{bg}}P_T(2C'_{\mathrm{tp}}P_{\mathsf{k}}+C_{\mathrm{tp}}^{\mathrm{bg}}P_T)\widehat{m}_1\widehat{m}_2.\tag{2305}$$

Theorem AT. Under these hypotheses,

$$\left|\langle \mathcal{O}(f_1);\mathcal{O}(f_2)\rangle - \tfrac{N^2-1}{2}\upsilon_{\mathsf{k}+1}^2 \mathcal{K}_{K,K-\mathsf{k}}(f_1,f_2)\right| \le |\mathrm{Rest}_T| + \sum_{r=2}^{8} E_r. \tag{2306}$$

■□

Proof. The slice estimate, $u_a^{[j]} \le 5\widehat{m}_a/4$ and $P_{l-1} = L^{-3(l-j)}P_{j-1}$ give $|n_{j,T}^G| \le \overline{n}_j^G$ by a geometric sum. In the exact covariance reduction (2272), the steps $j > j_1$ cost E_2 by positivity and the covariance supremum bound. For $j \le j_1$, transport to level T, replace its background by the identity using (2281), and apply proposition 125.1. Since the top measure has mass $1 - W_T$, these three changes cost E_3, E_4, E_5. Restoring the Gaussian terms for $j > j_1$ costs E_6.

The complete Gaussian sum is exactly

$$\sum_{\mathsf{k}<j\le T} n_{j,T}^G = \tfrac{N^2-1}{2}\sum_{p,q} w_1(p)w_2(q)\left(\sum_{\mathsf{k}<j\le T} \upsilon_j \mathcal{C}_{(j)}(p,q)\right)^2.$$

Put $D = \Pi_{\mathsf{k}} - \Pi_T$ and $\upsilon = \upsilon_{\mathsf{k}+1}$. By (2299) the weighted slice sum differs from υD by at most $\upsilon(\bar{B}/x_T)C'_{\mathrm{tp}}\eta_{\mathsf{k}}^3$; the sum of their absolute values is at most $\frac{9}{4}\upsilon C'_{\mathrm{tp}}\eta_{\mathsf{k}}^3$. The identity $a^2 - b^2 = (a-b)(a+b)$ and the plaquette counts give E_7. Finally $C^{\mathrm{bg}}_{K,K-\mathsf{k}} = D + C^{\mathrm{bg}}_{K,K-T}$. The bound $|C^{\mathrm{bg}}_{K,K-T}| \le C_{\mathrm{tp}}^{\mathrm{bg}}\eta_T^3$ and the same square identity give E_8. The resulting square is precisely the kernel sum in (2306). □

125.2. **Summing the errors.** Write $p = \mathfrak{p}_0$, $y = y_{\mathsf{k}}$, $q = \widehat{d}_{\mathsf{k}}$, $D = D_{\mathsf{k}}$ and $n_T = T - \mathsf{k}$ in this subsection. Constants denoted by c or C depend only on the fixed construction constants. They do not depend on the sources, the levels, the coupling sequence or the torus. We continue to assume $q \ge M$, $D \ge 3M$ and all the hypotheses of theorem AT. Here $d_*(t) = d_*(t;t,2B_3\varepsilon_t)$, and the symbols ρ_a^T, ρ_{12}^T below denote the ρ_a, ρ_{12} of (2273) at the final level T. The regularity assumptions include $\varepsilon_t \le c_A\alpha_0/(8MB_3)$ and $L^{m+K-t} \ge 8M+4$ for $1 \le t \le T$. They make both entries defining $d_*(t)+1$ at least 4, so $d_*(t) \ge 3$.

Lemma 125.2. There are constants $C_{\mathrm{sz}}, C_{\mathsf{M}}, c_{\mathrm{up}}, c_I$ with the following properties. Put $r_{i,a} = \sum_{X,z}|\dot{E}_a^{(i)}(X,\mathbf{1},z)|$. For $j \le \mathsf{k}$, $n = \mathsf{k} - j$,

$$\begin{aligned} X_{j,a}^* + I_{j,a}^* + \sum_{j<i\le\mathsf{k}} r_{i,a} &\le C_{\mathrm{sz}} q^3 L^{4n}(n+1)^{2p} x_{\mathsf{k}} y^{2p}\widehat{m}_a, \\ M_j &\le C_{\mathsf{M}} q^3 L^{4n}(n+1)^{2p} x_{\mathsf{k}} y^{2p} e^{-(D/M-2)}\widehat{m}_1\widehat{m}_2. \end{aligned} \tag{2307}$$

For $\mathsf{k} \le j \le T$,

$$\begin{aligned} d_*(j) \ge d_*(T) \ge 3, \qquad X_{j,a}^* &\le c_X y^{2p}\widehat{d}_j^3\widehat{m}_a, \qquad c_X = \tfrac{405}{2}B_3^2A_0^2, \\ I_{j,a}^* &\le i_T\widehat{m}_a, \qquad i_T = c_I q^3 y^{3p}\max\{x_T^{-1/2}, \sqrt{x_{\mathsf{k}}}e^{-d_*(T)}\}. \end{aligned} \tag{2308}$$

Define

$$b_n = (C_{\mathrm{sz}} + c_{\mathrm{up}}n_T)q^3L^{4n}(n+1)^{2p}x_{\mathsf{k}}y^{2p}, \qquad m_* = C_{\mathsf{M}}q^3x_{\mathsf{k}}y^{2p}.$$

Then, for $0 \le j \le T$ and $n = [\mathsf{k}-j]_+$,

$$Y_{j,a} \le B_{j,a}^T \le b_n\widehat{m}_a, \qquad M_j' \le [m_*L^{4n}(n+1)^{2p} + 2n_Tb_0^2]\widehat{m}_1\widehat{m}_2.$$

■

Proof. Locality of $\dot{E}_a^{(i)}$ requires its polymer to meet a plaquette in the source support. Anchor it in a cell containing a corner of that plaquette. There are $N_i(w_a)$ possible cells, and at most $8M^3(d_i(X)+1)$ choices of its marked site. The entropy sum at the output decay rate therefore gives

$$r_{i,a} \le 8M^3C_{\mathrm{E},1}N_i(w_a)d_{i,a}.$$

The first-derivative recurrence bounds $d_{i,a}$ by a fixed multiple of $\delta_{i-1}|w_a|_\infty/a_{K-i+1}$. For $i=\mathsf{k}-n'$ this is at most

$$CL^{(n'+1)/2}(n'+2)^p\sqrt{x_{\mathsf{k}}}y^p\widehat{m}_a;$$

the number of anchors is at most $125M^{-3}q^3L^{3n'}$. For $i>\mathsf{k}$, the corresponding product is at most $c_{\mathrm{up}}q^3\sqrt{x_{\mathsf{k}}}y^p\widehat{m}_a$. The sum over $0\le n'<n$ is bounded by $Cq^3L^{7n/2}(n+1)^p\sqrt{x_{\mathsf{k}}}y^p\widehat{m}_a$.

For $j=\mathsf{k}-n\ge 1$, the plaquette count and the coupling comparison give $X^*_{j,a}\le 216B_3^2A_0^2q^3L^{3n}(n+1)^{2p}y^{2p}\widehat{m}_a$. At $j=0$ the bound is $162q^3L^{4n}x_{\mathsf{k}}\widehat{m}_a$. In the first-derivative remainder, its bracket is at most 2 because $d_*(j)\ge 3$ and $2B_3\varepsilon_j\le\alpha_0$. It is consequently bounded by $Cq^3L^{7n/2}(n+1)^p\sqrt{x_{\mathsf{k}}}y^p\widehat{m}_a$. These estimates prove the first line of (2307).

For the second line, put $u=D_j/M=L^nD/M\ge 3$ in the mixed-derivative sum. Its term at distance h below j has $\nu_{j-h-1}/\nu_j\le\frac43L^{h+1}(h+2)^{2p}$. Since $L^hu-2\ge(u-2)+3(L^h-1)$, its sum is at most

$$\tfrac43\nu_je^{-(u-2)}\sum_{h\ge0}L^{4h+1}(h+2)^{2p}e^{-3(L^h-1)}.$$

The series converges. Moreover $\nu_j\le CL^n(n+1)^{2p}x_{\mathsf{k}}y^{2p}\widehat{m}_1\widehat{m}_2$ and $N_j(w_1)\le125M^{-3}q^3L^{3n}$. This proves the required estimate, including $M_0=0$.

Above k, both entries defining $d_*(j)$ decrease as j increases. The direct bound for X^* gives the first estimate in (2308). Substitution into the remainder estimate gives

$$I^*_{j,a}\le C(2\widehat{d}_j/M+3)^3[x_j^{-1/2}y_j^{3p}+e^{-d_*(j)}\sqrt{x_j}y_j^p]\widehat{m}_a.$$

Use $(2\widehat{d}_j/M+3)^3\le125M^{-3}q^3$ and $x_T\le x_j\le x_{\mathsf{k}}$. This proves the displayed bound for i_T. The same formula with its original bracket bounded by 2 gives $I^*_{j,a}\le Cq^3\sqrt{x_{\mathsf{k}}}y^p\widehat{m}_a$. Finally $B^T_{j,a}=X^*_{j,a}+I^*_{j,a}+\sum_{j<i\le T}r_{i,a}$, and the definition of M'_j adds at most $2n_Tb_0^2$ above k. Increasing the fixed constants proves the last two assertions. □

Fix the parameter $0<\eth<1$ used in lemma 115.1, and put

$$c'=(1-\eth)c_\star,\qquad B(x)=C^\Sigma_{\mathrm{maj}}+2+c_{\mathrm{low}}+\tfrac92\log x. \tag{2309}$$

We call $\ell=2L^m$ small over the running sequence if

$$MR(x_K)\le L^m,\qquad LM\le L^m,\qquad \ell\ge N_{\mathsf{H}},\qquad B(x_K)\ell^3\le\tfrac12c'A_0^2y_K^{2p}. \tag{2310}$$

Lemma 125.3. Assume the actual history hypotheses of lemma 115.1 and a small side. Then

$$w_t\le W_t=\sum_{i=1}^t w_i\le 2M^{-3}\ell^3L^{3(K-t)}e^{-c'A_0^2y_t^{2p}/2}. \tag{2311}$$

■

Proof. The normalized history measure gives $W_t=\widetilde{\nu}_K\{Z_t\ne\varnothing\}$ by (2269). Divide (2184) by the partition-function lower bound (1486), and use the product bound (1485). This gives

$$W_t\le 2M^{-3}\ell^3L^{3(K-t)}\exp[-c'A_0^2y_t^{2p}+B(x_K)\ell^3].$$

Now $y_t\ge y_K$ and the last condition of (2310) absorb half of the negative exponent. □

Lemma 125.4. Assume also $y_K\ge2\log(1+\bar{B})$ and $c'pA_0^2y_K^{2p-1}\ge13$. Set

$$c_p=\sup_{a\ge0}(a+2)^{4p}L^{-a},\qquad t_T=2M^{-3}\ell^3L^{3(K-\mathsf{k})}e^{-c'A_0^2y_T^{2p}/2},\qquad z_T=c_Xy^{2p}\widehat{d}_T^3+i_T.$$

Then

$$\begin{aligned}&W_T\le t_T,\qquad \rho^T_a\le t_T(2L^4c_p+n_T)b_0\widehat{m}_a,\\ &\rho^T_{12}\le t_T(2L^8c_p+n_T)[m_*+(2n_T+1)b_0^2]\widehat{m}_1\widehat{m}_2.\end{aligned} \tag{2312}$$

In particular $|\mathrm{Rest}_T| \le R_T \widehat{m}_1 \widehat{m}_2$, where

$$\begin{aligned} R_T &= m_* e^{-(D/M-2)} + 2n_T(2c_X y^{2p} q^3 i_T + i_T^2) + 2z_T^2 \\ &\quad + t_T(2L^8 c_p + n_T)[m_* + (2n_T+1)b_0^2 + 2b_0 z_T + t_T(2L^8 c_p + n_T)b_0^2]. \end{aligned} \tag{2313}$$

■

Proof. The running-coupling rate estimate, with exponent 10, is

$$L^{3a} e^{-c' A_0^2 y_{\mathsf{k}-a}^{2p}/2} \le L^{-10a} e^{-c' A_0^2 y_{\mathsf{k}}^{2p}/2}.$$

Here the loss in the logarithmic coupling increment must be controlled before using the power $2p$. For $a \ge 1$, put $\Delta = y_{\mathsf{k}-a} - y_{\mathsf{k}}$ and $v = \bar{B}/x_{\mathsf{k}} \le 1/4$. The running-coupling bound gives

$$\Delta \ge a \log L + \log(1-v) > 2a, \qquad a \log L - \Delta \le \tfrac{4}{3} \bar{B} e^{-y_{\mathsf{k}}}.$$

Since $y_K \ge 2\log(1+\bar{B})$ and $z e^{-z/2} \le 2/e$ for $z > 0$, it follows that

$$y_{\mathsf{k}}(a \log L - \Delta) \le \frac{8}{3e} \frac{\bar{B}}{1+\bar{B}} < 1.$$

Set $N = 2p \ge 2$. The linear and quadratic terms of the power expansion give

$$\begin{aligned} y_{\mathsf{k}-a}^N - y_{\mathsf{k}}^N &\ge N y_{\mathsf{k}}^{N-1} a \log L \\ &\quad + N y_{\mathsf{k}}^{N-2} \left[\frac{N-1}{2} \Delta^2 - y_{\mathsf{k}}(a \log L - \Delta) \right] \ge N y_{\mathsf{k}}^{N-1} a \log L. \end{aligned}$$

The bracket is positive because $\Delta > 2a$ and $a \ge 1$. The case $a = 0$ is immediate. Multiplication by $c' A_0^2/2$ gives a rate at least $c' p A_0^2 y_K^{2p-1} \ge 13$; the volume factor L^{3a} leaves the stated rate 10. For $t \le \mathsf{k}$, put $a = \mathsf{k} - t$. The preceding lemma gives $w_t \le t_T L^{-10a}$, while lemma 125.2 gives

$$B_{t-1,b}^T \le b_0 L^4 c_p L^{5a} \widehat{m}_b, \quad M_{t-1}' + B_{t-1,1}^T B_{t-1,2}^T \le [m_* + (2n_T+1)b_0^2] L^8 c_p L^{9a} \widehat{m}_1 \widehat{m}_2.$$

Sum these products using $\sum_{a \ge 0} L^{-a} \le 2$. For the remaining n_T levels use $w_t \le t_T$ and $n = 0$ in lemma 125.2. This proves (2312). Insert these estimates, (2307) at $j = \mathsf{k}$, and (2308) into the exact remainder (2273). The n_T cross terms give the second term of (2313); the top covariance gives $2z_T^2$; the linear and quadratic tail terms give its last line. □

We now choose the two upper levels:

$$j_1 = \mathsf{k} + \lceil \log_L(q y^p) \rceil, \qquad T = \mathsf{k} + \lceil \log_L(q y^{3p}) \rceil, \qquad n_T = T - \mathsf{k}. \tag{2314}$$

They are used only when $T \le K$. They satisfy $\widehat{d}_{j_1} \le y^{-p}$, $\widehat{d}_T \le y^{-3p}$, $L^{-(T-j_1)} < L y^{-2p}$ and $q y^{3p} \le L^{n_T} < L q y^{3p}$.

Lemma 125.5. There are fixed constants $C_2, C_3, C_3', C_3'', C_5, C_6, C_7, C_8$ such that, after division by $\widehat{m}_1 \widehat{m}_2$, the seven errors in (2305) satisfy

$$\begin{aligned} &E_2 \le C_2 y^{-2p}, && E_6 \le C_6 y^{-6p}, \\ &E_5 \le C_5 n_T^2 (1 + A_1 y^p)^4 q^6 \bar{e}_T, && E_7 \le C_7 q^6 / x_T, \\ &E_8 \le C_8 (q^3 + 1) y^{-9p}, && \\ &E_4 \le W_T [C_6 n_T q^6 + C_5 n_T^2 (1 + A_1 y^p)^4 q^6 \bar{e}_T], && \\ &E_3 \le C_3 (n_T y^{-5p} + y^{-2p}) + C_3' n_T q y^{8p} x_T^{-1/2} + C_3'' n_T q^3 y^{11p} x_{\mathsf{k}} e^{-d_*(T)}, && \end{aligned} \tag{2315}$$

where $\bar{e}_T = \max_{\mathsf{k} \le i < T} (\delta_i + e_i^{\mathrm{co}})$. ■

Proof. For E_2, sum $\widehat{d}_j^{\,6}$ over $j \ge j_1$ in (2308); its first term is at most y^{-6p}, leaving $y^{4p}y^{-6p}$. The identical geometric sum in the Gaussian bound gives E_6. The recurrence for $e_{j,i}$ has at most n_T^2 terms, each bounded by $C(1+A_1y^p)^4q^6\bar{e}_T$; this proves E_5 and, together with the Gaussian size, E_4. Substituting $P_{\mathsf{k}} = 81q^3$, $P_T \le 81y^{-9p}$ into E_7, E_8 gives their displayed bounds.

For E_3, use $k_j \le \nu_{j-1}/4 \le Cx_{j-1}y^{2p}\widehat{m}_1\widehat{m}_2$ in (2281). Its initial quadratic term contains $x_{j-1}\varepsilon_T^2 L^{-4(T-j)}N_j(w_1)$. Use $x_{j-1} \le \frac{5}{4}L^{T-j+1}x_T$, $\varepsilon_T^2 x_T \le A_0^2y^{2p}$, $N_j(w_1) \le (2\widehat{d}_j/M+3)^3$ and $L^{-(T-j)}\widehat{d}_j = \widehat{d}_T$. The cell-dependent term is bounded, after summing over $j \le j_1$, by Cn_Ty^{-5p}. The constant term uses

$$\sum_{j\le j_1} L^{-3(T-j)} \le (1-L^{-3})^{-1}L^3y^{-6p}$$

and is bounded by Cy^{-2p}. The initial exceptional-polymer term is at most $Cq^3x_{\mathsf{k}}y^{2p}e^{-d_*(T)}$ per level. For the carried quadratic term use $N_T(w_1) \le 125$, $\sum_{h\ge 0}L^{3h}L^{-4h} \le (1-L^{-1})^{-1}$, $\theta_{T-1} \le Cx_T^{-1/2}y^p$ and $x_{j-1}\varepsilon_T^2 \le Cqy^{5p}$. It is at most $Cqy^{8p}x_T^{-1/2}$ per level. For the carried exceptional term use $S_\omega\theta_{T-1} \le 1$ and $\sum_{0\le h<n_T-1}L^{3h} < q^3y^{9p}$; this sum is empty when $n_T = 1$. It is bounded by $Cq^3x_{\mathsf{k}}y^{11p}e^{-d_*(T)}$ per level. These four estimates prove the stated bound for E_3. □

Corollary 125.6. Set

$$s_T = \max\{x_T^{-1/2}, x_{\mathsf{k}}e^{-d_*(T)}, x_{\mathsf{k}}e^{-(D/M-2)}, \bar{e}_T, x_{\mathsf{k}}^2t_T\}. \tag{2316}$$

If $s_T \le 1$, there is a fixed C_B such that

$$\begin{aligned}\Big|\langle\mathcal{O}(f_1);\mathcal{O}(f_2)\rangle - \tfrac{N^2-1}{2}\upsilon_{\mathsf{k}+1}^2\mathcal{K}_{K,K-\mathsf{k}}(f_1,f_2)\Big| \\ \le \Big[(C_2+C_3)y^{-2p} + C_3n_Ty^{-5p} + C_6y^{-6p} \\ + C_8(q^3+1)y^{-9p} + 4c_X^2y^{-14p} + C_Bn_T^4q^6y^{15p}s_T\Big]\widehat{m}_1\widehat{m}_2.\end{aligned} \tag{2317}$$

■□

Proof. Insert (2315) and (2313) into (2306). All terms in the seven-error bound other than the first four displayed terms of (2317) are bounded by a fixed multiple of $n_T^4q^6y^{15p}s_T$. Here $x_T^{-1} \le s_T^2 \le s_T$, $W_T \le t_T \le s_T$ and $1+A_1y^p \le (1+A_1)y^p$. For the remainder use $i_T \le c_Iq^3y^{3p}s_T$, $z_T \le c_Xy^{-7p}+i_T$ and $b_0 \le Cn_Tq^3x_{\mathsf{k}}y^{2p}$. The top term satisfies $2z_T^2 \le 4c_X^2y^{-14p}+4i_T^2$. In the last line of (2313), its square bracket is at most $Cn_T^3q^6y^{5p}x_{\mathsf{k}}^2$; the remaining factor is at most Cn_Tt_T. Thus $x_{\mathsf{k}}^2t_T \le s_T$ controls precisely the volume-dependent tail. The other terms are smaller monomials with one of the five factors in (2316). Increasing C_B proves the assertion. □

126. The short-distance law on a small torus

We now verify the preceding hypotheses for the actual first-passage construction. Write $p = \mathfrak{p}_0$ throughout this section. All structural constants and canonical maps are fixed as in proposition 74.1 and Theorem D. In the weighted iteration we use the admissible-radii row and the constants fixed in section 122. In particular K_Λ and $K_{\mathbf{M}}$ have their weighted values. Increasing the lower bound on the inverse reference coupling does not change these maps or their counterterms.

126.1. **The witness depth and the working levels.** Let $0 < g \le \gamma$. The first-passage endpoint is $X_g = L(g^{-2}+B_\sharp)$, and put

$$C_{W,0} = \max\{1, 4L^2M/A_0^2\}, \qquad c_W = C_{W,0}A_0^2. \tag{2318}$$

For integers $s \ge 0$ define

$$\begin{aligned}\bar{x}_s = L^{s+1}(g^{-2}+\bar{B}), \qquad Y_s = \log\bar{x}_s, \qquad C_W(s) = c_WY_s^{2p}, \\ s(d) = \min\{s\in\mathbb{Z}_{\ge 0} : L^sd \ge C_W(s)\}, \qquad \check{d} = L^{s(d)}d.\end{aligned} \tag{2319}$$

The minimum exists because an exponential eventually exceeds a polynomial. For $0 < d < 1$, its value is positive and

$$C_W(s(d)) \le \check{d} < LC_W(s(d)).$$

Indeed minimality gives $L^{s-1}d < C_W(s-1) \le C_W(s)$. The witness depth is non-increasing in d and tends to infinity as $d \downarrow 0$.

For a first-passage datum (K, β) with physical label $\mathsf{b} = a_K\beta$, assume $s = s(d) \le K$ and set

$$\mathsf{k} = \max\{K - s, 2\}, \qquad s_{\mathsf{k}} = K - \mathsf{k} = \min\{s, K-2\}, \qquad \mathsf{L}_s = L^2 C_W(s) Y_s^{3p}. \tag{2320}$$

Thus the construction always uses a height at least two, including when $K = s$ or $K = s+1$.

Lemma 126.1. For $0 < d < 1$ and $s(d) \le K$, one has $s \ge 3$, $2 \le \mathsf{k} \le K-1$ and $s-2 \le s_{\mathsf{k}} \le s$. With $q = d/a_{s_{\mathsf{k}}}$,

$$4MY_s^{2p} \le L^{-2}C_W(s) \le q < LC_W(s), \qquad a_K/d \le C_W(s)^{-1}, \qquad d < LC_W(s)L^{-s}. \tag{2321}$$

On a side with $LM \le L^m$ one has $20d \le \ell$. For a test pair in $\mathfrak{T}_{T_\ell}(d;\vartheta)$ with nonzero sampled weights,

$$D_{\mathsf{k}} \ge q/\sqrt{3} - 1 \ge 3M, \qquad D_{\mathsf{k}}/M - 2 \ge \tfrac{4}{\sqrt{3}} Y_s^{2p} - 3.$$

Furthermore

$$L^{s_{\mathsf{k}}} g^{-2} < x_{\mathsf{k}} \le \bar{x}_s, \quad Y_s - \log(2L^3) < y_{\mathsf{k}} \le Y_s, \quad |(\mathsf{b}\upsilon_{\mathsf{k}+1})^2 - 1| \le 3\bar{B}g^2 L^{2-s}, \quad \upsilon_{\mathsf{k}+1}^2 \le \tfrac{9}{8}\mathsf{b}^{-2}. \tag{2322}$$

If j_1, T are chosen by (2314), then $L^{n_T} < \mathsf{L}_s$. If $\mathsf{L}_s \le L^{s-3}$ and $g^{-2} \ge 4\bar{B}$, then

$$\mathsf{k} < j_1 \le T \le K-2, \qquad n_T \le s-4, \qquad L^{K-T} > L^{s-2}/\mathsf{L}_s, \qquad x_T > \frac{4L^{s-2}g^{-2}}{5\mathsf{L}_s}. \tag{2323}$$

■

Proof. The normalization conditions give $g^{-2} \ge 4$, $\bar{B}g^2 \le 13/24$, $L \ge 13$ and $Y_0 > 2$. Since $L^{-2}C_W(0) > 1 > d$, no index $s \le 2$ can be a witness. The formula for k gives $s-2 \le s_{\mathsf{k}} \le s$; multiplying the bounds on $\check{d}$ by $L^{s_{\mathsf{k}}-s}$ proves (2321). Also $\ell \ge 2LM > 20d$. The two supports have distance at least d. Their plaquette corners have sup distance at least $d/\sqrt{3} - a_K$. Divide by $a_{s_{\mathsf{k}}} \ge a_K$ and use $q \ge 4MY_s^{2p}$ to obtain the two separation bounds.

The first-passage condition gives $x_{K-1} > X_g$. The backwards running estimate gives

$$x_{\mathsf{k}} - \bar{B} \ge L^{s_{\mathsf{k}}-1}(x_{K-1} - \bar{B}) > L^{s_{\mathsf{k}}} g^{-2},$$

because $LB_\sharp - \bar{B} = (L-2)\bar{B} \ge 0$. The upper running estimate gives $x_{\mathsf{k}} \le \bar{x}_{s_{\mathsf{k}}} \le \bar{x}_s$. Since $\bar{B} < g^{-2}$ and $s_{\mathsf{k}} \ge s-2$, $\bar{x}_s < 2L^3 x_{\mathsf{k}}$, proving the logarithmic bound. Finally

$$\upsilon_{\mathsf{k}+1}^{-1} = a_{s_{\mathsf{k}}} x_{\mathsf{k}}, \quad |\upsilon_{\mathsf{k}+1}^{-1} - \mathsf{b}| \le \bar{B} a_{s_{\mathsf{k}}}, \quad \upsilon_{\mathsf{k}+1} < g^2.$$

Thus $|\mathsf{b}\upsilon_{\mathsf{k}+1} - 1| \le \bar{B}g^2 L^{2-s} \le 1/24$; squaring gives (2322). For the last assertion, $L^{n_T} < Lqy_{\mathsf{k}}^{3p} < \mathsf{L}_s$. Under its stated upper bound $n_T < s-3$, whence $T \le K-2$ and the bound on L^{K-T} follows. The running estimate $x_T \ge \frac{4}{5} L^{-n_T} x_{\mathsf{k}}$ proves the last inequality. □

126.2. **A single choice of the small-coupling threshold.** Let $\varepsilon(t) = A_0 t^{-1/2}(\log t)^p$, $\delta(t) = A_1 t^{-1/2}(\log t)^p$ and $\theta(t) = K_{\mathbf{M}}(1+\beta_s)\delta(t)/\alpha_0$. Define $e^{\mathrm{co}}(t)$ by the formulas for e_i^{co} and (2292), replacing x_i by t. In particular its exponential term is a fixed multiple of $\exp[-(LM)^{-3}A_1^2(\log t)^{2p}]$. These functions are continuous, decrease on $[e^{2p}, \infty)$ and tend to zero.

Choose $x_W \ge \max\{\gamma^{-2}, 4\bar{B}, e^{2p}, eL\}$ so that the following hold for every $t \ge x_W$:

$$\begin{gathered} 2\varepsilon(t) \le \varepsilon_{\mathcal{W}}\alpha_0, \quad C_0^\flat S_A\theta(t) \le 1, \quad \tfrac{4}{3}C_1^\flat LT(3)\theta(t) \le 1, \\ (1+\beta_s)\delta(t)/\alpha_0 \le \tfrac{1}{2}s_\omega^\star, \quad \varepsilon(t) \le c_A\alpha_0/(8MB_3), \quad 3K_P\delta(t) \le 3w_\star/8, \\ S_\omega\theta(t) \le 1, \quad c_R\delta(t) \le \tfrac{1}{2}, \quad e^{\mathrm{co}}(t) \le 1, \\ 2[\tfrac{48}{5}\sqrt{5/4}\,L^{-3/2}c_{\mathfrak{h}} + 8C_{\mathrm{db}}]\delta(t) \le \varepsilon_{\mathcal{W}}\alpha_0, \\ c'pA_0^2(\log t)^{2p-1} \ge 13, \quad 2M(\log t)^r \ge N_{\mathsf{H}}, \\ 4\delta_{\mathrm{tp}}L(L-1)M(\log t)^{2p} \ge \sqrt{3}(3\log L + \log 2). \end{gathered} \tag{2324}$$

Require also $\log t \ge \max\{1+\log L, 2\log(1+\bar{B})\}$ and $(\log t)^r \ge C_{\mathrm{s}}$, and the history-summation thresholds preceding lemma 115.1 at the fixed parameter $\eth$. All these requirements can be imposed simultaneously: the quantities to be made small tend to zero, the positive powers of $\log t$ tend to infinity, and $S_A, S_\omega, T(3)$ are finite. Set $\gamma_W^{\mathrm{ref}} = x_W^{-1/2}$.

Lemma 126.2. Let $g \le \gamma_W^{\mathrm{ref}}$, let (K,β) be a first-passage datum with $K \ge 3$, and suppose $MR(x_K) \le L^m$. The canonical construction satisfies all the place, progression, density and regularity hypotheses used above through level K. Consequently, for $2 \le \mathsf{k} < T \le K$, real Lipschitz weights supported in balls of radius $d \ge a_K$, $\widehat{d}_{\mathsf{k}} \ge M$, $D_{\mathsf{k}} \ge 3M$ and $\mathsf{k} < j_1 \le T$, the conclusions of theorem AT and lemma 125.2 apply. The conclusion of lemma 125.5 applies when j_1, T are chosen by (2314). If the side is small, lemma 125.4 also applies, and corollary 125.6 applies at these attached levels when its remaining hypothesis $s_T \le 1$ holds. ∎

Proof. By proposition 74.1, $x_0 > \cdots > x_K > g^{-2} \ge x_W$; the sequence and every initial segment are admissible. The torus condition $MR(x_K) \le L^m$ makes the same finite lattice admissible over every segment. The canonical one-lattice maps, radii and response therefore give the actual places of Theorems V and W at all these heights. Their auxiliary datum is common to the heights, and the octet property holds on every required smaller lattice, as required by the canonical-place hypotheses of those theorems.

The ordinary families satisfy $\|E^{(j)}\| \le E_0$ and $|e_j| \le B_\sharp$. Their weighted bare arrays consequently have norm at most

$$(E_0 + \tfrac{3}{2}\alpha_0^2 B_\sharp)\sum_{n\ge1}(n+1)L^{-(n-1)} < N_\star.$$

The weighted radii and ceiling in section 122, together with (2324), put the fourth potential below $w_\star/4$ and leave the required perturbation room. The intrinsic potential is at most $3w_\star/8$. Thus the Lipschitz and Cauchy derivative bounds proved there apply to these same actual maps; no change of counterterms is involved. Consistency of the ordinary recursion identifies $x_{j+1} = L^{-1}x_j - e_{j+1}$ with the counterterm used in the weighted step. The Gaussian family is the same localized family in the recursion and in the exact density identity, so their flat terms cancel as in the proof of Proposition 123.1.

The first two lines of (2324) give the regular background and first-derivative hypotheses. Since $D_j \ge D_{\mathsf{k}} \ge 3M$ for $j \le \mathsf{k}$, the decreasing function $T(\cdot)$ satisfies $T(D_j/M) \le T(3)$; hence the mixed derivative hypotheses hold. Also $d_*(t) \ge 3$: its analytic-radius entry is at least 4, and $L^{m+K-t} \ge LM \ge 8M+4$ controls its geometric entry. The next two lines give the chain-radius, covariance and strict lowering-domain conditions in (2300); the factor 2 makes that domain inequality strict. The weighted transport hypothesis $S_\omega\theta_i \le 1$ is explicit. The running bounds follow from proposition 74.1. Finally $x_K > e$ implies $R(x_K) \ge L$, so $L^m \ge LM$, and $\ell \ge 2MR(x_K) \ge 2My_K^r \ge N_{\mathsf{H}}$.

The actual history construction has the same running sequence and the finite-family tests required by lemma 115.1. Its small-field cutoffs obey $\varepsilon_j \le \varepsilon_\flat$ by the fixed construction calibration. The

additionally imposed history thresholds and the last two rate conditions verify lemmas 125.3 and 125.4. This proves all the assertions. □

Put $C_{\mathrm{sd}} = 8L^6(C^{\Sigma}_{\mathrm{maj}} + 2 + c_{\mathrm{low}} + 9/2)$. Choose Y_{sd} so that for all $y \ge Y_{\mathrm{sd}}$,

$$C_{\mathrm{sd}} y\,[MR(e^y)]^3 \le \tfrac{1}{2} c' A_0^2 y^{2p}.$$

Such a choice exists since $R(e^y) < Ly^r$ and $2p > 3r + 2$. Define

$$\gamma_W^{\mathrm{sd}} = \min\{\gamma_W^{\mathrm{ref}}, e^{-Y_{\mathrm{sd}}/2}\}. \tag{2325}$$

Lemma 126.3. If $g \le \gamma_W^{\mathrm{sd}}$, each side satisfying $MR(x_K) \le L^m \le L^2 MR(x_K)$ is small over the first-passage sequence. This includes both the least admissible side $2L^{m_{\mathfrak{L}}(g)}$ and the fixed reference side $2MR(X_g)$. ■

Proof. The first three conditions in (2310) follow from lemma 126.2. The upper side bound gives $B(x_K)\ell^3 \le C_{\mathrm{sd}} y_K [MR(x_K)]^3$, and $y_K > \log g^{-2} \ge Y_{\mathrm{sd}}$ gives the fourth.

For the two specific sides recall $g^{-2} < x_K \le X_g$ and the normalization inequalities $X_g + \bar{B} < 2Lx_K$ and $\log(2L) \le (L^{1/r} - 1)\log g^{-2}$. Thus

$$R(x_K) \le R(X_g) \le R(X_g + \bar{B}) \le LR(x_K).$$

The least side is determined by

$$L^{m_{\mathfrak{L}}(g)} = LM \max\{R(X_g + \bar{B}), L^{\lceil \log_L C_{\mathrm{s}} \rceil}\}.$$

The imposed condition $C_{\mathrm{s}} \le (\log x_W)^r$ gives $L^{\lceil \log_L C_{\mathrm{s}} \rceil} < LC_{\mathrm{s}} \le LR(x_K)$. This proves the required upper and lower bounds for both sides. Neither side depends on K or β. □

126.3. **Errors depending only on the physical separation.** For $s \ge 3$ set

$$\begin{aligned} Y_s^- &= Y_s - \log(2L^3), \qquad x_s^- = 4L^{s-2} g^{-2}/(5\mathsf{L}_s), \\ d_s^- &= \min\left\{ \frac{c_A \alpha_0 \sqrt{x_s^-}}{2MB_3 A_0 Y_s^p}, \frac{L^{s-1}}{2\mathsf{L}_s} - \frac{2}{M} \right\} - 1. \end{aligned} \tag{2326}$$

Where $x_s^- \ge e^{2p}$, put

$$\begin{aligned} \sigma_g(s) = \max\{ &(x_s^-)^{-1/2}, \bar{x}_s e^{-d_s^-}, \bar{x}_s e^{3 - 4Y_s^{2p}/\sqrt{3}}, \delta(x_s^-) + e^{\mathrm{co}}(x_s^-), \\ &M^{-3} c' A_0^2 Y_s^{2p} L^{3s} \bar{x}_s^2 e^{-c' A_0^2 (\log x_s^-)^{2p}/2} \}, \end{aligned} \tag{2327}$$

and

$$\begin{aligned} \epsilon_g(s) = {} & (C_2 + C_3)(Y_s^-)^{-2p} + C_3 s (Y_s^-)^{-5p} + C_6 (Y_s^-)^{-6p} \\ & + C_8[(LC_W(s))^3 + 1](Y_s^-)^{-9p} + 4c_X^2 (Y_s^-)^{-14p} \\ & + C_B s^4 (LC_W(s))^6 Y_s^{15p} \sigma_g(s). \end{aligned} \tag{2328}$$

There is $s_1(g) \ge 3$ such that for every $s \ge s_1(g)$,

$$Y_s^- > 0, \qquad \mathsf{L}_s \le L^{s-3}, \qquad x_s^- \ge e^{2p}, \qquad \sigma_g(s) \le 1. \tag{2329}$$

Indeed Y_s grows linearly, L_s is a polynomial of degree $5p$ in Y_s, and x_s^- grows exponentially divided by that polynomial. Both entries defining d_s^- grow faster than every polynomial. The first and fourth terms of σ_g decay exponentially times a polynomial; the remaining terms have a negative exponent whose magnitude grows at least as Y_s^{2p}, with $2p > 1$, overcoming the factors exponential in s. Thus $s^a \sigma_g(s) \to 0$ for every $a \ge 0$, and

$$\sup_{s \ge s_1(g)} Y_s^{2p} \epsilon_g(s) < \infty. \tag{2330}$$

Proposition 126.4. Let $g \le \gamma_W^{\mathrm{ref}}$, let (K,β) be a first-passage datum and let ℓ be small over its canonical sequence. If $s_1(g) \le s(d) \le K$ and $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$, then

$$\left|\langle \mathcal{O}(f);\mathcal{O}(h)\rangle_{K,\ell,\beta} - \tfrac{N^2-1}{2}\upsilon_{\mathsf{k}+1}^2 \mathcal{K}_{K,s_{\mathsf{k}}}(f,h)\right| \le \epsilon_g(s(d))\upsilon_{\mathsf{k}+1}^2 \|f\|_\infty \|h\|_\infty. \tag{2331}$$

■

Proof. If a sampled weight vanishes, the observable and kernel sum involving it both vanish. Otherwise lemmas 126.1 and 126.2 verify all hypotheses of corollary 125.6 at the attached levels, with $T \le K-2$. We check its five small factors rather than infer their smallness from the final cutoff. The first obeys $x_T^{-1/2} < (x_s^-)^{-1/2}$ by (2323). For the second, $\varepsilon_T \le A_0 (x_s^-)^{-1/2} Y_s^p$ and $L^{m+K-T} > ML^{s-1}/\mathsf{L}_s$ give $d_*(T) \ge d_s^-$. The separation estimate in lemma 126.1 controls the third. Monotonicity of δ and e^{co} controls the fourth since $x_i > x_T > x_s^-$ for $i < T$. Finally smallness of the side and $B(x_K) \ge 2$ imply

$$2M^{-3}\ell^3 \le M^{-3} c' A_0^2 y_K^{2p}.$$

Using $K - \mathsf{k} = s_{\mathsf{k}} \le s$, $y_K \le Y_s$, $y_T > \log x_s^-$ and $x_{\mathsf{k}} \le \bar{x}_s$ bounds $x_{\mathsf{k}}^2 t_T$ by the fifth entry of (2327). Thus $s_T \le \sigma_g(s) \le 1$. In (2317) use $Y_s^- < y_{\mathsf{k}} \le Y_s$, $n_T \le s$ and $q < LC_W(s)$. Its bracket is at most $\epsilon_g(s)$, and $\widehat{m}_1 \widehat{m}_2 \le \upsilon_{\mathsf{k}+1}^2 \|f\|_\infty \|h\|_\infty$. □

Put $c_2 = 3(N^2-1)/(16\pi^2)$ and define

$$\begin{aligned}\varrho_{g,\vartheta}(s) = 3\pi^2\, 10^6 \vartheta^{-2} C_{\mathrm{tm}} \left[\frac{1}{\vartheta C_W(s)} + (LC_W(s)L^{-s})^3 + \sup_{t \ge L^{-2}C_W(s)} t^3 e^{-\delta_{\mathrm{tp}} L t/\sqrt{3}}\right] \\ + 3\bar{B} g^2 L^{2-s} + \tfrac{9}{8}\, 10^6 \vartheta^{-2} c_2^{-1} \epsilon_g(s).\end{aligned} \tag{2332}$$

Choose $s_0(g,\vartheta) \ge s_1(g)$ so that $\varrho_{g,\vartheta}(s) \le 1/2$ for every $s \ge s_0$, and put

$$d_0^{\mathrm{ref}}(g,\vartheta) = L^{-s_0(g,\vartheta)-1}, \qquad \bar{\varepsilon}_{\mathrm{ref}}(d;g,\vartheta) = \sup_{s' \ge s(d)} \varrho_{g,\vartheta}(s'). \tag{2333}$$

These quantities exist because (2330) and the explicit formula give $\sup_{s \ge s_0} Y_s^{2p} \varrho_{g,\vartheta}(s) < \infty$. The function $\bar{\varepsilon}_{\mathrm{ref}}$ is non-decreasing in d, takes values in $[0,1/2]$ on $(0, d_0^{\mathrm{ref}}]$, and satisfies

$$\bar{\varepsilon}_{\mathrm{ref}}(d;g,\vartheta) \le C(g,\vartheta)(\log(1/d))^{-2p}. \tag{2334}$$

Indeed $d \le L^{-s_0-1}$ implies $s(d) > s_0$, while $L^{s(d)} d \ge C_W(s(d)) \ge 1$ implies $Y_{s(d)} \ge \log(1/d)$; apply the preceding uniform bound at every $s' \ge s(d)$.

Theorem AU (Short-distance two-point law on a small torus)**.** Let $0 < g \le \gamma_W^{\mathrm{ref}}$ and $0 < \vartheta < 1$. For every first-passage datum (K,β) at g, with $\mathsf{b} = a_K \beta$, every side small over its running sequence, every $0 < d \le d_0^{\mathrm{ref}}(g,\vartheta)$ with $s(d) \le K$, and every $(f,h) \in \mathfrak{T}_{T_\ell}(d;\vartheta)$,

$$\left|\langle \mathcal{O}(f);\mathcal{O}(h)\rangle_{K,\ell,\beta} - c_2 \mathsf{b}^{-2} \mathcal{K}_{\mathrm{free}}(f,h)\right| \le \bar{\varepsilon}_{\mathrm{ref}}(d;g,\vartheta) c_2 \mathsf{b}^{-2} \mathcal{K}_{\mathrm{free}}(f,h). \tag{2335}$$

The constants and error function are independent of K, β, ℓ and the test pair. ■□

Proof. Set $s = s(d)$ and $\upsilon = \upsilon_{\mathsf{k}+1}$. Subtract and add the lattice kernel with coefficient $(N^2-1)\upsilon^2/2$, and then the free kernel with that coefficient. The three errors are

$$\epsilon_g(s)\upsilon^2 \|f\|_\infty \|h\|_\infty, \quad \tfrac{N^2-1}{2}\upsilon^2 |\mathcal{K}_{K,s_{\mathsf{k}}} - \tfrac{3}{8\pi^2}\mathcal{K}_{\mathrm{free}}|, \quad c_2 \mathsf{b}^{-2} |(\mathsf{b}\upsilon)^2 - 1| \mathcal{K}_{\mathrm{free}}.$$

The first is bounded by proposition 126.4. To apply proposition 121.2 to the second, observe that $8a_K \le d$ by (2321) and $20d \le \ell$ by lemma 126.1. Its background-tail condition holds because $q \ge 4MY_s^{2p}$ and the last inequality of (2324) holds at x_W. That theorem therefore bounds the kernel error by

$$C_{\mathrm{tm}} \left[\frac{1}{\vartheta C_W(s)} + (LC_W(s)L^{-s})^3 + \sup_{t \ge L^{-2}C_W(s)} t^3 e^{-\delta_{\mathrm{tp}} L t/\sqrt{3}}\right] \|f\|_\infty \|h\|_\infty.$$

For the third use (2322). Finally $\mathcal{K}_{\mathrm{free}} \geq 10^{-6}\vartheta^2\|f\|_\infty\|h\|_\infty$ and $v^2 \leq 9\mathsf{b}^{-2}/8$. After division by $c_2\mathsf{b}^{-2}\mathcal{K}_{\mathrm{free}}$, the three bounds add to (2332); here $(9/8)(N^2-1)/(2c_2) = 3\pi^2$. Use (2333) to conclude. □

Corollary 126.5. If $0 < g \leq \gamma_W^{\mathrm{sd}}$, the conclusions (2331) and (2335) hold for every first-passage datum and every admitted test pair and depth on the least torus at g, on the reference torus of side $2MR(X_g)$, and on every side satisfying $MR(x_K) \leq L^m \leq L^2 MR(x_K)$. ■□

Proof. These sides are small by lemma 126.3. Apply proposition 126.4 and theorem AU with their respective depth conditions. In particular the reference torus is the same for all first-passage data at g. □

127. Transfer of the two-point law

The short-distance calculation on one torus can be transferred to every admissible torus and every periodic Gibbs state. The required comparison gains a factor equal to the volume of the support. After accounting for the two Lipschitz norms, its error is of order d for test pairs at distance d.

Fix the source radius ϱ and the construction constants of theorem AD, and let $0 < g \leq \gamma_\varrho$. For $(K,\beta) \in \mathfrak{d}(g)$, a *state at* g will mean either $\widetilde{\mu}_{K,\ell,\beta}$ with $\ell \in \mathfrak{L}(g)$, or a periodic Gibbs state at (K,β). Every such law is invariant under $a_K\mathbb{Z}^3$: for the finite torus this follows by relabelling bonds, and for a periodic Gibbs state it is proposition 103.4. The covariance $S_2^T[\mathfrak{s}](f,h)$ is consequently invariant under simultaneous lattice translations. Periodic lifting gives

$$S_2^T[\widetilde{\mu}_{K,\ell,\beta}](f,h) = S_{2;K,\ell,\beta}^T(f^{\langle\ell\rangle}, h^{\langle\ell\rangle}) \tag{2336}$$

by (1820).

Use the number Q_1 of (1842), which is independent of (K,β) and the torus side, and put

$$c_2^{\mathrm{tr}} = 4\log 2\,(2\mathsf{Q}_1+2)^2\varrho^{-2}, \qquad C_2^{\mathrm{tr}}(\vartheta) = 2(50/3)^3 c_2^{\mathrm{tr}}(40M/\vartheta)^2. \tag{2337}$$

Proposition 127.1 (Comparison between states)**.** Let $\mathfrak{s},\mathfrak{s}'$ be two states at g for the same $(K,\beta) \in \mathfrak{d}(g)$. If $a_K \leq r \leq 1/20$ and $f,h \in \mathrm{Lip}_c(\mathbb{R}^3)$ vanish off $\mathbb{B}(x_0,r)$, then

$$|S_2^T[\mathfrak{s}](f,h) - S_2^T[\mathfrak{s}'](f,h)| \leq 2c_2^{\mathrm{tr}}(r/0.3)^3\|f\|_\sharp\|h\|_\sharp. \tag{2338}$$

In particular, if $a_K \leq 5d \leq 1/20$, both functions vanish off one ball of radius $5d$, and $\mathrm{Lip}\, f \leq \|f\|_\infty/(\vartheta d)$, $\mathrm{Lip}\, h \leq \|h\|_\infty/(\vartheta d)$ with $0 < \vartheta \leq 1$, then

$$|S_2^T[\mathfrak{s}](f,h) - S_2^T[\mathfrak{s}'](f,h)| \leq C_2^{\mathrm{tr}}(\vartheta)d\|f\|_\infty\|h\|_\infty. \tag{2339}$$

No separation or sign condition on f,h is required. ■

Proof. First let F,H vanish off $\mathbb{B}(0,1)$. The board of side L centered at zero has walls at distance $L/2 \geq 6.5$ from zero. If neither entry is zero, proposition 91.12 gives a single normalizer for (F,H), of radius ϱ and bound Q_1, on the unbounded set $\mathfrak{L}(g)$. Both entries occupy the same cell, so the distinct-cell condition of proposition 91.11 is vacuous. That proposition applies to all the torus laws and all periodic Gibbs states. In (1839), the normalizer derivative is the same for both laws and cancels on subtraction. Thus the bilinear form $D = S_2^T[\mathfrak{s}] - S_2^T[\mathfrak{s}']$ satisfies

$$|D(F,H)| \leq 2c_2^{\mathrm{tr}}\|F\|_\sharp\|H\|_\sharp. \tag{2340}$$

If an entry is zero the same statement follows by bilinearity.

We pack translates of $\mathbb{B}(x_0,r)$ into this unit ball using lattice translations. Set $\delta = a_K\lceil 2r/a_K\rceil$, so $2r \leq \delta < 3r$ and $\delta\mathbb{Z}^3 \subset a_K\mathbb{Z}^3$. Let P consist of $v \in \delta\mathbb{Z}^3$ such that every coordinate of x_0+v has modulus at most $0.575 - r$. The balls $\mathbb{B}(x_0+v,r)$ have disjoint interiors and lie in the unit ball, since $3(0.575)^2 < 1$. Each interval of length $1.15 - 2r \geq 1.05$ contains at least $\lfloor (1.15-2r)/\delta\rfloor > 0.35/r - 1 \geq 0.3/r$ points of a prescribed translate of $\delta\mathbb{Z}$. Taking the product over the coordinates yields

$$N := \#P \geq (0.3/r)^3.$$

For independent signs $\epsilon_v = \pm 1$ define $F_\epsilon = \sum_{v\in P} \epsilon_v \tau_v f$ and $H_\epsilon = \sum_{v\in P} \epsilon_v \tau_v h$. Their supremum norms are no larger than those of f, h, and their Lipschitz constants are no larger either. To verify the latter assertion, note that each function vanishes on the boundary of its supporting ball. On a segment joining points in different balls, bound the endpoint values by the distances to the respective exit and entry points. Their sum is at most the segment length. The same-ball and outside-ball cases give the same Lipschitz bound. Thus $\|F_\epsilon\|_\sharp \le \|f\|_\sharp$ and $\|H_\epsilon\|_\sharp \le \|h\|_\sharp$. Bilinearity and $\mathbb{E}(\epsilon_v \epsilon_w) = \mathbf{1}_{v=w}$ give

$$\mathbb{E} D(F_\epsilon, H_\epsilon) = \sum_{v\in P} D(\tau_v f, \tau_v h) = N D(f, h),$$

where the last equality uses lattice translation invariance. Apply (2340) to every choice of signs and divide by N to prove (2338). For the last assertion put $r = 5d$. Since $d \le 1/100$, $\vartheta \le 1$ and $M \ge 1$, each sharp norm is bounded by $40M/(\vartheta d)$ times the corresponding supremum norm. Substitution in (2338) gives (2339). □

127.1. **The estimate on every admissible volume.** Write $\mathfrak{T}^1_E(d;\vartheta)$ for the pairs in $\mathfrak{T}_E(d;\vartheta)$ whose entries are C^1. For $b > 0$, say that a covariance Y obeys the two-point estimate at (f, h) with label b and tolerance ϵ if

$$|Y - \mathsf{c}_2 b^{-2} \mathcal{K}_{\mathrm{free}}(f, h)| \le \epsilon \mathsf{c}_2 b^{-2} \mathcal{K}_{\mathrm{free}}(f, h), \qquad \mathsf{c}_2 = \frac{3(N^2-1)}{16\pi^2} = \frac{9}{16\pi^2}. \tag{2341}$$

Here $N = 2$ is fixed by the gauge group. The positive lower bound in (2245) makes this equivalent to the relative-error form of the short-distance calculation.

Lemma 127.2 (Lifting and dilation of test pairs)**.** If $20d < \ell$, periodization takes a pair in $\mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$ to a pair in $\mathfrak{T}_{T_\ell}(d;\vartheta)$ with the same free kernel. Conversely, every pair in the latter set has such a lift. Both assertions preserve C^1 regularity. For $f_t(x) = f(x/t)$, $h_t(x) = h(x/t)$, $t > 0$,

$$(f_t, h_t) \in \mathfrak{T}_{\mathbb{R}^3}(td;\vartheta), \qquad \mathcal{K}_{\mathrm{free}}(f_t, h_t) = \mathcal{K}_{\mathrm{free}}(f, h). \tag{2342}$$

If $n \ge 0$, $\ell = L^n \ell'$ and $t = L^{-n}$, then

$$S_2^T[\widetilde{\mu}_{K+n,\ell',\beta}](f_t, h_t) = L^{2n} S_2^T[\widetilde{\mu}_{K,\ell,\beta}](f, h).$$

Thus the estimate with label $L^{-n}b$ on the left transfers, with the same tolerance, to the estimate with label b on the right. ■

Proof. The two supports lie in one ball of radius $5d$. Since $10d < \ell/2$, that ball lifts isometrically to $\mathbb{R}^3$; all distances between points in the two supports are preserved. The entries vanish before the boundary of a larger lifting chart, so lifting and periodizing preserve supremum norms, Lipschitz constants, integrals and C^1 regularity. This proves the first assertions and the equality of the free kernels. Under dilation, the integrals gain t^3, the Lipschitz constants gain t^{-1}, and distances gain t. In the free kernel the factor t^6 from the two integrations cancels t^{-6} from the kernel. Finally, (2220) and (2221), with $z = -n$, give the displayed covariance identity. Since $(L^{-n}b)^{-2} = L^{2n}b^{-2}$, division by L^{2n} proves the last assertion. □

We use the room constant of (2318) and the witness depth of (2319). To make their dependence on the reference coupling explicit, write

$$\bar{x}_s[g] = L^{s+1}(g^{-2} + \bar{B}), \qquad C_W(s)[g] = C_{W,0} A_0^2 (\log \bar{x}_s[g])^{2\mathfrak{p}_0},$$
$$s(d)[g] = \min\{s \ge 0 : L^s d \ge C_W(s)[g]\},$$

where the minimum is over integers. The exponential L^s eventually exceeds the polynomial in s on the right, so this minimum is finite. We abbreviate $s(d)[g]$ to $s(d)$ when the reference coupling is fixed. Recall that $C_{W,0} A_0^2 \ge 4L^2 M$ and $X_g = L(g^{-2} + B_\sharp)$.

For $0 < g \le \gamma$, $0 < \vartheta < 1$ and $0 < d \le 1/100$, let $\mathsf{e}_{g,\vartheta}(d)$ be the infimum of the tolerances for which (2341) holds on the torus $\ell = 2MR(X_g)$, for every $(K, \beta) \in \mathfrak{d}(g)$ with $s(d) \le K$ and every

pair of type $(d;\vartheta)$, with label $a_K\beta$. The infimum of the empty set is infinity. If the infimum is finite, it itself is an admissible tolerance: for every datum and pair the defining inequality is closed in ϵ and has a positive coefficient on its right-hand side.

By corollary 126.5, the set

$$\mathfrak{G} = \{\gamma' \in (0,\gamma] : \mathsf{e}_{g,\vartheta}(d) \longrightarrow 0 \text{ as } d \to 0 \text{ for every } 0 < g \le \gamma',\ 0 < \vartheta < 1\}$$

is nonempty. It is downward closed. Put

$$\widehat{\gamma}_W = \tfrac{1}{2}\sup \mathfrak{G}. \tag{2343}$$

An element of $\mathfrak{G}$ exceeds $\widehat{\gamma}_W$, so $\widehat{\gamma}_W \in \mathfrak{G}$ and $0 < \widehat{\gamma}_W \le \gamma/2$. Define

$$\gamma_W = \Big[L^2\Big(\bar{B} + \max\{\widehat{\gamma}_W^{-2}, \gamma^{-4}, e^{C_{\mathrm{s}}}\}\Big)\Big]^{-1/2}, \qquad \check{g} = (L^{-2}g^{-2} - \bar{B})^{-1/2} \quad (0 < g \le \gamma_W). \tag{2344}$$

These choices give

$$\begin{gathered} \check{g}^{-2} + \bar{B} = L^{-2}g^{-2}, \qquad X_{\check{g}} = L^{-1}g^{-2} - \bar{B}, \qquad \check{g} \le \widehat{\gamma}_W \le \gamma, \\ \check{g}^{-2} \ge \gamma^{-4}, \qquad \log \check{g}^{-2} \ge C_{\mathrm{s}}, \qquad g^{-2} > L^2\bar{B}, \qquad g < \check{g}, \qquad X_{\check{g}} \le X_g. \end{gathered} \tag{2345}$$

Indeed, $\check{g}^{-2}$ is at least each term in the maximum in (2344); the identity for $X_{\check{g}}$ uses $X_u + \bar{B} = L(u^{-2} + \bar{B})$. In particular $0 < \gamma_W \le \gamma$.

Lemma 127.3 (Changing the reference coupling)**.** Let $0 < g \le \gamma_W$ and $(K,\beta) \in \mathfrak{d}(g)$. Then $K' = K(\beta,\check{g})$ exists and $n = K' - K \in \{2,3\}$. The datum $(K+n,\beta)$ belongs to $\mathfrak{d}(\check{g})$ and its label is $L^{-n}a_K\beta$. Moreover,

$$L^n 2MR(X_{\check{g}}) \in \mathfrak{L}(g) \quad (n \ge 2), \qquad s(L^{-n}d)[\check{g}] \le s(d)[g] + n + 1 \quad (0 < d < 1,\ n = 2,3). \tag{2346}$$

■

Proof. Since $\beta > X_g \ge X_{\check{g}}$, proposition 74.1 gives both first-passage indices, with identical running couplings on their common prefix. By (1527) and (1528),

$$\begin{gathered} g^{-2} < x_K \le X_g, \qquad x_{K'} \le X_{\check{g}} < x_j \ (j < K'), \\ |x_j - L^{-(j-i)}x_i| \le \bar{B}(1 - L^{-(j-i)}) \quad (0 \le i \le j \le K'). \end{gathered}$$

If $K' \le K$, monotonicity on the common prefix gives $x_{K'} \ge x_K > g^{-2} > X_{\check{g}}$, a contradiction. At the next step, $x_{K+1} \ge L^{-1}x_K - \bar{B}(1 - L^{-1}) > X_{\check{g}}$, so $K' \ne K+1$. If $K' \ge K+4$, then

$$X_{\check{g}} < x_{K+3} \le L^{-3}X_g + \bar{B} = L^{-2}(g^{-2} + B_\sharp) + \bar{B}.$$

As $B_\sharp \le \bar{B}$, this implies $(L-1)g^{-2} < (2L^2+1)\bar{B} \le L^2(L-1)\bar{B}$, contrary to (2345). This proves the index assertion.

For the side assertion put $X = X_{\check{g}}$ and $X' = X_g + \bar{B}$. Then $X \ge \gamma^{-4} \ge 16$ and $\log X \ge 2\log\gamma^{-2}$. Using $\bar{B} = B_\sharp L/(L-1)$ and $L \ge 13$ gives

$$X' = L^3\check{g}^{-2} + (L^3+L)\bar{B} \le 4L^3(\check{g}^{-2} + B_\sharp) = 4L^2X.$$

The standing calibration $(L^{1/r} - 1)\log\gamma^{-2} \ge \log(2L)$ therefore yields $\log X' \le L^{1/r}\log X$. Since R is the smallest power of L not less than the rth power of the logarithm,

$$R(X') \le LR(X), \qquad R(X) \ge (\log X)^r \ge \log X \ge \log\check{g}^{-2} \ge C_{\mathrm{s}}.$$

If $R(X) = L^u$, the definition of $m_{\mathfrak{L}}(g)$ gives $m_{\mathfrak{L}}(g) \le m' + u + 2$. Hence $L^n 2MR(X) = 2L^{n+m'+u} \in \mathfrak{L}(g)$ for every $n \ge 2$.

Finally, let $s = s(d)[g]$ and $Y = \log \bar{x}_s[g]$. Then $L^s d \ge C_W(s)[g]$ and $Y \ge 2\log\gamma^{-2} > 4\mathfrak{p}_0$ by the standing calibration $\log\gamma^{-2} > 2\mathfrak{p}_0$. For $n = 2,3$,

$$\bar{x}_{s+n+1}[\check{g}] = L^{s+n}g^{-2} \le L^2\bar{x}_s[g],$$

and consequently

$$(\log \bar{x}_{s+n+1}[\check{g}])^{2\mathfrak{p}_0} \le Y^{2\mathfrak{p}_0} \exp(4\mathfrak{p}_0 \log L/Y) \le L Y^{2\mathfrak{p}_0}.$$

It follows that $C_W(s+n+1)[\check{g}] \le L C_W(s)[g] \le L^{s+n+1} L^{-n} d$. Thus $s+n+1$ is an admissible integer in the definition of $s(L^{-n}d)[\check{g}]$. □

For $0 < g \le \gamma_W$ and $0 < \vartheta < 1$, put

$$\begin{aligned} \mathsf{C}_{\mathrm{law}}(g,\vartheta) &= \frac{10^6 C_2^{\mathrm{tr}}(\vartheta)\bar{x}_0[g]^2}{\mathsf{c}_2 \vartheta^2}, \\ \bar{\varepsilon}_{g,\vartheta}(d) &= \sup_{0<d'\le L^{-2}d} \mathsf{e}_{\check{g},\vartheta}(d') + \mathsf{C}_{\mathrm{law}}(g,\vartheta) d, \qquad 0 < d \le 1/100, \end{aligned} \tag{2347}$$

and

$$d_0(g,\vartheta) = \tfrac{1}{2} \sup\{d \in (0, 1/100] : \bar{\varepsilon}_{g,\vartheta}(d) \le 1/3\}. \tag{2348}$$

Since $\check{g} \le \widehat{\gamma}_W$, the supremum in (2347) tends to zero with d. The tolerance is nondecreasing and tends to zero. The set in (2348) is therefore nonempty, $0 < d_0(g,\vartheta) \le 1/200$, and $\bar{\varepsilon}_{g,\vartheta}(d) \le 1/3$ for $d \le d_0$: there is an element of that set strictly exceeding its half-supremum, and monotonicity gives the claim. All these quantities are independent of cutoff, volume, state, pair and position.

Proposition 127.4 (An admissible reference torus)**.** Let $0 < g \le \gamma_W$, $(K,\beta) \in \mathfrak{d}(g)$, $0 < \vartheta < 1$ and $0 < d \le 1/100$, with $s(d) < K$. Put $n = K(\beta,\check{g}) - K$, $\ell' = 2MR(X_{\check{g}})$ and $\ell = L^n \ell'$. Then $\ell \in \mathfrak{L}(g)$. If $\epsilon_0 = \sup_{0<d'\le L^{-2}d} \mathsf{e}_{\check{g},\vartheta}(d')$ is finite, the lifted torus law at (K,ℓ,β) obeys (2341) for every pair of type $(d;\vartheta)$, with label $a_K\beta$ and tolerance ϵ_0. ■

Proof. The side assertion is lemma 127.3. Put $d' = L^{-n}d$. Since $n \in \{2,3\}$, $d' \le L^{-2}d$ and

$$s(d')[\check{g}] \le s(d)[g] + n + 1 \le K + n.$$

The finite number $\mathsf{e}_{\check{g},\vartheta}(d')$ is an admissible tolerance for $(K+n,\beta) \in \mathfrak{d}(\check{g})$ on ℓ'. The lifting and dilation statements of lemma 127.2 transfer this estimate to (K,ℓ,β) at distance d, with label $a_K\beta$. Its tolerance does not exceed ϵ_0. □

Theorem AV (Uniform two-point estimate)**.** Let $0 < g \le \gamma_W$, $0 < \vartheta < 1$, $(K,\beta) \in \mathfrak{d}(g)$, and $0 < d \le d_0(g,\vartheta)$ with $s(d) < K$. For every state $\mathfrak{s}$ at g and every $(f,h) \in \mathfrak{T}_{\mathbb{R}^3}(d;\vartheta)$,

$$S_2^T[\mathfrak{s}](f,h) = \mathsf{c}_2 b^{-2} \mathcal{K}_{\mathrm{free}}(f,h)(1+\eta), \qquad b = a_K\beta, \qquad |\eta| \le \bar{\varepsilon}_{g,\vartheta}(d) \le 1/3. \tag{2349}$$

The same assertion holds on every torus $\ell \in \mathfrak{L}(g)$ for pairs in $\mathfrak{T}_{T_\ell}(d;\vartheta)$. In particular every such covariance is positive. ■□

Proof. The reference state of proposition 127.4 is available, since the first term of (2347) is finite. The witness inequality and $C_{W,0}A_0^2 \ge 4L^2M$ give $a_K < a_{s(d)} < d$, whereas $5d \le 1/40 < 1/20$. The pair is supported in one ball of radius $5d$, so (2339) bounds the difference from the reference covariance by $C_2^{\mathrm{tr}}(\vartheta) d \|f\|_\infty \|h\|_\infty$. Dividing by the lower bound in (2245) shows that the added relative error is at most

$$\frac{10^6 C_2^{\mathrm{tr}}(\vartheta) b^2 d}{\mathsf{c}_2 \vartheta^2} \le \mathsf{C}_{\mathrm{law}}(g,\vartheta) d,$$

because $b \le \bar{x}_0[g]$ by (1852). This is precisely the remaining term of (2347). Lifting proves the torus assertion. Finally $1 + \eta \ge 2/3$ and the free kernel is positive. □

127.2. **The label in continuum limits.** We first record that smooth test pairs exist for every allowed shape parameter, including pairs separated by the reflection plane.

Lemma 127.5 (Smooth reflection pairs)**.** For every $0 < \vartheta < 1$ and $d > 0$ there is $f \in C_c^\infty(\mathbb{R}^3)$ supported in $\mathbb{B}(2de_3, d)$ such that $(\theta f, f)$ belongs to $\mathfrak{T}^1_{\mathbb{R}^3}(d;\vartheta)$, where $\theta(x_1,x_2,x_3) = (x_1,x_2,-x_3)$. ■

Proof. Put $x = 2de_3$ and

$$r_0 = \max\{\vartheta, (3\vartheta/\pi)^{1/3}\}, \quad r = (1 + r_0)/2, \quad \epsilon = \tfrac{1}{2}\min\{1 - r, r - \vartheta\}.$$

Then $r_0 < r < 1$, $0 < \epsilon < r$, $r + \epsilon < 1$, $r - \epsilon > \vartheta$ and $(\pi/3)r^3 > \vartheta$. Convolve $g_r(y) = \max\{0, 1 - |y - x|/(rd)\}$ with a nonnegative smooth mollifier of integral one supported in $\mathbb{B}(0, \epsilon d)$. The resulting function f satisfies

$$0 \le f \le 1, \quad f(x) \ge 1 - \epsilon/r > 0, \quad \operatorname{Lip} f \le (rd)^{-1}, \quad \int f = (\pi/3)r^3 d^3, \quad \operatorname{supp} f \subset \mathbb{B}(x, (r + \epsilon)d).$$

The integral follows by polar integration of the cone; convolution preserves it. Since $\|f\|_\infty \ge (r - \epsilon)/r > \vartheta/r$, the required Lipschitz inequality holds, as does $\int f \ge \vartheta \|f\|_\infty d^3$. Reflection preserves these bounds. The two supports lie in balls of radius d centered at $\pm x$, so their distance is at least $2d$; it is at most $4d$ because $f(x) > 0$. All conditions defining the test pair are satisfied. □

Lemma 127.6 (Limits and recovery of the label)**.** For fixed (f, h), inequality (2341) passes to the limit if $Y_j \to Y$ and $b_j \to b > 0$, provided it holds along an infinite subsequence with a fixed tolerance.

More generally, fix $(f, h) \in \mathfrak{T}_{\mathbb{R}^3}(d_1; \vartheta)$, $t_i \to 0$ with $t_i > 0$, and $\epsilon_i \to 0$ with $0 \le \epsilon_i < 1$. Suppose for each i that the covariances $Y_{j,i}$ of (f_{t_i}, h_{t_i}) converge to a finite y_i, and for all sufficiently large j obey (2341) with label $b_j > 0$ and tolerance ϵ_i. Then

$$\lim_j b_j^{-2} = \lim_i \frac{y_i}{\mathsf{c}_2 \mathcal{K}_{\mathrm{free}}(f, h)} = u \in [0, \infty). \tag{2350}$$

The limit u is positive if and only if $\sup_j b_j < \infty$; in this case $b_j \to u^{-1/2}$. ■

Proof. The first assertion follows by taking limits on both sides of (2341) along the stated subsequence. For the second put $k = \mathsf{c}_2 \mathcal{K}_{\mathrm{free}}(f, h) > 0$ and $u_j = b_j^{-2}$. Dilation invariance of the free kernel gives, for each fixed i and all sufficiently large j,

$$(1 - \epsilon_i) k u_j \le Y_{j,i} \le (1 + \epsilon_i) k u_j.$$

Taking an upper limit in the left inequality and a lower limit in the right yields

$$(1 - \epsilon_i) k \limsup_j u_j \le y_i \le (1 + \epsilon_i) k \liminf_j u_j.$$

The first inequality makes the upper limit finite. Letting $i \to \infty$ shows that the upper and lower limits coincide, at a number $u \ge 0$. The same inequalities then give $y_i \to ku$. If the b_j are bounded above, u_j have a positive lower bound and $u > 0$. Conversely a positive limit of the positive u_j gives a positive lower bound for the entire sequence, so $\sup_j b_j < \infty$ and $b_j \to u^{-1/2}$. □

Theorem AW (The two-point estimate in the continuum)**.** Let $0 < g \le \gamma_W$. Let S be a limit at g on a fixed torus T_ℓ with $\ell \in \mathfrak{L}(g)$, a limit along growing tori, or a limit through periodic Gibbs states. For the fixed torus, defining sequences at g are required to consist of first-passage data in $\mathfrak{d}(g)$. Write $E = T_\ell$ in the first case and $E = \mathbb{R}^3$ otherwise. Then there is a single label $b(S) \in I^{\mathrm{lab}}(g)$, and $a_{K_j}\beta_j \to b(S)$ along every defining sequence at g. For every $0 < \vartheta < 1$, $0 < d \le d_0(g, \vartheta)$ and $(f, h) \in \mathfrak{T}^1_E(d; \vartheta)$,

$$S_2(f, h) = \mathsf{c}_2 b(S)^{-2} \mathcal{K}_{\mathrm{free}}(f, h)(1 + \eta), \qquad |\eta| \le \bar{\varepsilon}_{g,\vartheta}(d) \le 1/3. \tag{2351}$$

In particular,

$$S_2(f, h) \ge \tfrac{2}{3}\, 10^{-6} \vartheta^2 \mathsf{c}_2 b(S)^{-2} \|f\|_\infty \|h\|_\infty > 0. \tag{2352}$$

For any $(f, h) \in \mathfrak{T}^1_{\mathbb{R}^3}(d_0(g, \vartheta); \vartheta)$,

$$b(S)^{-2} = \lim_{t \to 0} \frac{S_2\langle f_t, h_t\rangle}{\mathsf{c}_2 \mathcal{K}_{\mathrm{free}}(f, h)}. \tag{2353}$$

Here $S_2\langle f,h\rangle = S_2(f,h)$ on $\mathbb{R}^3$, and $S_2\langle f,h\rangle = S_2(f^{\langle\ell\rangle},h^{\langle\ell\rangle})$ on the torus. Thus the two-point function determines the label. If the same family on the same space is also a limit at $0 < g_1 \le \gamma_W$, its label at g_1 equals $b(S)$. ■□

Proof. Given any defining sequence, use its lifted torus laws or its periodic Gibbs laws, denoted by $\mathfrak{s}_j$. For every fixed C^1 test pair, their covariances converge to $S_2\langle f,h\rangle$. On a fixed torus this follows from (2336) and lemma 127.2; in the other two cases it is the definition of convergence on separated tuples. For each fixed $d > 0$, $s(d) < K_j$ eventually. Hence theorem AV applies with tolerance $\bar{\varepsilon}_{g,\vartheta}(d)$, while all labels $b_j = a_{K_j}\beta_j$ lie in the compact positive interval $I^{\mathrm{lab}}(g)$.

Choose one smooth test pair at $d_1 = d_0(g,\vartheta)$ using lemma 127.5, and choose any sequence $t_i \to 0$ in $(0,1]$. The errors $\epsilon_i = \bar{\varepsilon}_{g,\vartheta}(t_i d_1)$ are at most $1/3$ and tend to zero. The second assertion of lemma 127.6 shows that b_j converges to a positive b and recovers b^{-2} from these limiting covariances. That recovery uses only S_2 and the fixed pair, so it gives the same b for every defining sequence. Compactness of $I^{\mathrm{lab}}(g)$ puts this number in that interval.

The first assertion of lemma 127.6 now passes the finite-cutoff estimate to every fixed test pair. On the torus, lift an arbitrary test pair by lemma 127.2. This proves (2351); its lower bound and (2245) give (2352). Apply (2351) to the dilates of any pair at d_1. Since their free kernels agree and their error tends to zero, this gives the full limit (2353). If a second coupling g_1 is available, rescale a single pair to a radius not exceeding either threshold. The two recovery formulas then have identical right-hand sides, so the positive labels coincide. □

The restriction to a small reference coupling does not restrict the separation of positive labels on a fixed torus. Dilation puts any two positive labels into the range just treated.

Corollary 127.7 (Positive labels distinguish torus limits)**.** Let $\ell = 2L^m$, $m \in \mathbb{Z}$. A limit point of the cutoff cumulants on T_ℓ has at most one positive label, allowing all defining sequences with $K_j \to \infty$ and $a_{K_j}\beta_j \to b > 0$. Consequently two such limit points with different positive labels are different. ■□

Proof. We give the first-passage reduction explicitly. For $b > 0$ put $g_z = (L^z b/2)^{-1/2}$. For all sufficiently large integers $z \ge 0$, $g_z \le \gamma_W$. Since $\gamma^{-2} > \bar{B}$, this implies

$$g_z^{-2} + \bar{B} < L^z b < X_{g_z} - \bar{B} = (L/2)L^z b + (L-2)\bar{B}.$$

Thus $L^z b$ lies in the interior of $I(g_z)$. We also have $L^z\ell \in \mathfrak{L}(g_z)$ for all sufficiently large z. To check the volume condition, set $c = (L/2)b + L\bar{B} > 0$ and $u = L(g_z^{-2} + \bar{B}) \le cL^z$. The inequality $\log u \le 2ru^{1/(2r)}$ gives

$$(\log u)^r \le (2r)^r c^{1/2} L^{z/2}.$$

Choose z so large that

$$L^{z/2} \ge L^{2-m} M \max\{(2r)^r c^{1/2}, C_{\mathrm{s}}\}.$$

Then $L^{m+z} \ge L^2 M \max\{(\log u)^r, C_{\mathrm{s}}\}$. From (1489), $R(u) \le L(\log u)^r$ and $L^{\lceil \log_L C_{\mathrm{s}} \rceil} \le LC_{\mathrm{s}}$, the least admissible side is at most $2L^2 M \max\{(\log u)^r, C_{\mathrm{s}}\}$. This proves the asserted membership.

Suppose now that S has label b along (K_j,β_j). For fixed z as above, $L^z a_{K_j}\beta_j \to L^z b$ lies eventually in $I(g_z)$, and $K_j - z \ge 1$ eventually. By (1533), $(K_j - z, \beta_j)$ is then a first-passage datum at g_z. Exact finite dilation (2217) gives a continuum limit

$$S_n^{[z]}(Y) = L^{-zn} S_n(Y^{[-z]}) \quad \text{on } T_{L^z\ell},$$

whose label at g_z is $L^z b$. If S also has positive label b', choose one sufficiently large z that works for both labels. The same family $S^{[z]}$ then has labels $L^z b$ and $L^z b'$ at the two reference couplings g_z and g_z', both at most γ_W. The coupling independence in theorem AW gives $L^z b = L^z b'$, hence $b = b'$. The last assertion follows by applying this conclusion to a putatively equal pair of limits. □

127.3. **Periodic-state limits and their components.** The covariance estimate is unaffected by the centering law. This observation is needed because the component approximation uses one law for integration and another for centering.

Proposition 127.8 (Transfer with an independent centering law)**.** Let $0 < g \le \gamma_W$, $(K_j, \beta_j) \in \mathfrak{d}(g)$, $K_j \to \infty$ and $\rho_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$. If the covariances of ρ_j converge on every separated C^1 pair, their limit has a unique label $b = \lim_j a_{K_j}\beta_j \in I^{\mathrm{lab}}(g)$ and satisfies (2351), (2352) and (2353). In particular, this holds for the second cumulant of any functional

$$\sigma(X) = \lim_j \rho_j(X_{K_j}^{\nu_j}) \quad \text{for every separated tuple } X,$$

where ν_j are arbitrary laws on the corresponding lattices. The label depends only on the sequence (K_j, β_j). ■

Proof. The proof of theorem AW used only convergence of the covariances, the uniform estimate and the compact label interval; it therefore proves the first assertion. For the second, write $A_j(f) = \mathcal{O}_{K_j}(f)$ and $c_j(f) = \nu_j(A_j(f))$. At each cutoff the exact identity

$$\rho_j((A_j(f) - c_j(f))(A_j(h) - c_j(h))) - \rho_j(A_j(f) - c_j(f))\rho_j(A_j(h) - c_j(h)) = S_2^T[\rho_j](f,h)$$

cancels all centering terms. The assumed convergence on the one- and two-entry tuples makes its left side tend to $\sigma(f,h) - \sigma(f)\sigma(h) = S_2^\sigma(f,h)$. Thus the first assertion applies. In particular the recovered label is the limit of the prescribed sequence of physical labels. □

Lemma 127.9 (A vector orthogonal to the vacuum)**.** Let σ be a normalized real functional with reflection positivity at $x_3 = 0$. If $f \in C_c^1(\{x_3 > 0\})$ and $S_2^\sigma(\theta f, f) > 0$, its reflection Hilbert space has dimension at least two. ■

Proof. Write $\Omega = [\varnothing]$, so $\|\Omega\| = 1$. The positive reflection form is Hermitian, and σ is real, so $\sigma(\theta f) = \sigma(f) = c \in \mathbb{R}$. The vector $v = [(f)] - c\Omega$ therefore satisfies

$$\langle \Omega, v\rangle = 0, \qquad \|v\|^2 = \sigma(\theta f, f) - c^2 = S_2^\sigma(\theta f, f) > 0.$$

Thus Ω and v are nonzero orthogonal vectors. □

Corollary 127.10 (Nontriviality)**.** For every limit on $\mathbb{R}^3$ covered by theorem AW, the centered functional σ_S has a reflection Hilbert space of dimension at least two. ■□

Proof. The centered functional is translation invariant and reflection positive at a coordinate mirror by proposition 92.4; for periodic Gibbs limits its hypotheses follow from proposition 103.4. Take the smooth reflection pair of lemma 127.5 with $d = d_0(g, \vartheta)$. Its covariance is positive by (2352). Since $S_2^{\sigma_S} = S_2$, lemma 127.9 applies. □

Corollary 127.11 (The same label in every component)**.** Let $0 < g \le \gamma_W$, $(K_j, \beta_j) \in \mathfrak{d}(g)$, $K_j \to \infty$ and $\nu_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$. Suppose the frame $\mathfrak{j} = (K_j, \nu_j)_j$ converges to σ. Then $a_{K_j}\beta_j \to b \in I^{\mathrm{lab}}(g)$, and every limit σ' in this frame satisfies the two-point estimate with this same label and the same tolerance $\bar\varepsilon_{g,\vartheta}(d)$. For every canonical decomposition $(\mathsf{X}, \lambda, (\sigma_\xi)_{\xi\in\mathsf{X}})$ of σ, every $\xi \in \mathsf{X}$, every $0 < \vartheta < 1$, $0 < d \le d_0(g,\vartheta)$ and $(f,h) \in \mathfrak{T}^1_{\mathbb{R}^3}(d;\vartheta)$,

$$S_2^{\sigma_\xi}(f,h) = \mathsf{c}_2 b^{-2}\mathcal{K}_{\mathrm{free}}(f,h)(1+\eta_\xi), \qquad |\eta_\xi| \le \bar\varepsilon_{g,\vartheta}(d) \le 1/3. \tag{2354}$$

Every component recovers b by (2353), has the positive lower bound (2352), and has a reflection Hilbert space of dimension at least two. For a volume-first limit S, this label is $b(S)$. ■□

Proof. By proposition 106.3, the periodic Gibbs classes form a class system with the required uniform bound system. Every limit in the frame has approximants $\rho_j \in \mathcal{G}^{\mathrm{per}}_{K_j,\beta_j}$ with $\sigma'(X) = \lim_j \rho_j(X_{K_j}^{\nu_j})$ along the full sequence. Apply proposition 127.8; since σ itself is such a limit, the physical labels converge, and their limit is the same for every σ'.

By theorem AL there is a canonical decomposition whose every component is a limit in this frame along its full sequence. Uniqueness, proposition 101.1, identifies the component functionals of any canonical decomposition with those of this one, at every point of the compact component space. Thus the preceding application of proposition 127.8 proves (2354), the recovery formula and positivity for every component. Each is reflection positive at the coordinate mirror by proposition 101.3. The smooth reflection pair and lemma 127.9 give the dimension assertion.

For a volume-first limit, the centered moments of its defining states converge to σ_S: in (1811) each moment is a finite sum of products of convergent cumulants. Its defining states therefore give the stated convergent frame. The label identified above equals $b(S)$ by theorem AW. □

Appendix A. Relation to earlier constructions

The observable in this paper is the centered, rescaled scalar plaquette energy. Its continuum correlations are defined on the diagonal-flat test spaces of theorem 0. To compare this construction with earlier work, one must distinguish the observable, the estimates used to remove the cutoffs, and the symmetry and reconstruction conclusions. These distinctions also separate the fixed-torus convergence of theorem A, the subsequential infinite-volume construction of theorem B, and the two-point asymptotic of theorem C.

A.1. **Renormalization and ultraviolet stability.** Bałaban's work supplies the lattice gauge-theoretic renormalization framework used here: block averaging, regular gauge configurations, background fields and propagators [Bał85a; Bał85c; Bał85b], together with effective-action expansions [Bał87; Bał88]. The construction and use of these objects occupy a substantial part of the present proof. Their role is to retain locality through successive integrations and to control the dependence of the resulting effective interactions on the background field.

The ultraviolet stability theorem for three-dimensional pure gauge theory [Bał85d, Theorem 1 and equations (5)–(6)] gives upper and lower bounds on the renormalized densities and, after stopping at a fixed coarse scale, cutoff-uniform bounds on the normalized partition function. The normalization, including the constant terms in the action, is part of that statement. These bounds control the size of the functional integral; they do not by themselves assert convergence of the correlations of a specified continuum observable.

Here that further task is carried out with local complex sources for the plaquette energy. Differentiation in several separated sources selects terms whose regions of dependence connect their supports. The locality estimates then make the fine-scale contributions summable. Comparison of adjacent cutoffs gives convergence on a fixed torus. Contact subtractions enter this comparison, while the limiting correlation distributions are formulated on diagonal-flat test spaces. Thus the source estimates and cutoff comparison in sections 72, 117 and 118 are essential to theorem A. Ultraviolet stability alone would not replace them.

A.2. **Abelian Higgs models and QED.** King's construction of the Abelian Higgs model in dimensions two and three also distinguishes stability from convergence. The finite-volume continuum limit is obtained by comparing effective actions and making source-dependent generating functionals Cauchy [Kin86a, pp. 649, 655–657]. The infinite-volume paper proves continuum and volume limits with its specified boundary conditions, and establishes translation invariance, reflection positivity and entire dependence on finitely many source parameters [Kin86b, Theorems 2.1–2.3]. The paragraph following its Theorem 2.3 explicitly leaves ergodicity, rotation invariance and regularity as further requirements for the full Osterwalder–Schrader conclusion. Its comparison of rotated lattices is a separate step; the ensuing passage to full rotations uses an expected shape-independent periodic infinite-volume limit [Kin86b, Theorem 2.4 and equation (2.24)]. The matter field, photon mass and boundary conditions in these results are part of their hypotheses; the present nonabelian pure-gauge construction is not an application of those theorems.

The fixed-positive-photon-mass scalar-electrodynamics preprint [SQED, Theorem 1.1(A)] concerns the Abelian curvature as a two-form. Its Gaussian bound permits a distribution-valued law with moments tested on the full Schwartz space. The rotation supplement [SQED-rot, Theorem A] compares two lattice orientations through tori with mass walls. In the present paper, rotations are obtained instead from coordinate and diagonal reflection positivity, a spectral restriction, and analytic continuation; see section 105. The scalar-electrodynamics result keeps the photon mass positive and leaves nontriviality open [SQED, §§ 1.4 and 23.3].

The fermionic QED3 construction in [QEM1, Theorem A] likewise concerns an Abelian field-strength tensor, under its weak-charge and positive-reference-mass assumptions for the stated chamber-fermion lattice model. Its Gaussian estimate controls all polynomial moments on the full Schwartz space; its non-Gaussianity conclusion uses a separate fourth-cumulant estimate. Neither estimate transfers to the scalar Yang–Mills observable by a change of notation. The latter has a quadratic classical counterpart, and the present theorem leaves contact terms on the coincidence diagonals unspecified. The two constructions therefore require different estimates even before their different matter and gauge content is taken into account.

A.3. **Four-dimensional Yang–Mills comparisons.** The conditional four-dimensional $SU(2)$ construction [YM4-cond, Theorems 1.1 and 1.3] also uses scalar plaquette densities and separated correlations. It assumes a coupling law and uniform separated-point bounds. An additional hypothesis comparing partition functions on cubic and twisted tori supplies diagonal reflection positivity for selected volume-first limits. Its all-moment rotation conclusion additionally assumes holomorphic continuation, growth bounds and positivity of the continued Gram kernels. These hypotheses isolate several tasks also present here, but do not establish the corresponding four-dimensional source estimates or analytic continuation. The clustering conclusion is toward the projection onto invariant vectors; the rank of that projection is not proved to be one.

A different four-dimensional construction treats bounded gauge-invariant block observables at a fixed physical block scale [YM4-blocks, Theorem 1.1 and § 1.6]. Boundedness gives joint subsequential laws for arbitrary coupling sequences. The conclusions include hyperoctahedral symmetry, coordinate reflection positivity for observables whose entire regions of dependence lie in the relevant half-space, and a positive Hamiltonian. Removing the physical block scale, obtaining full rotations, proving nontriviality and establishing vacuum uniqueness remain separate problems there.

The fixed block size is consequential: these are bounded extended observables, whereas the normalization of our plaquette-energy field depends on the cutoff. Compactness of the link space does not control that normalization. In the present three-dimensional construction, theorems A to C provide, respectively, the continuum correlations, the component structure and the short-distance information. They make no four-dimensional assertion.

A.4. **Reconstruction and nontriviality.** The passage from suitable Euclidean correlations to a relativistic quantum field is the Osterwalder–Schrader reconstruction theorem [OS73; OS75]. Diagonal-flat test functions and the associated freedom in contact terms already occur in that framework [OS73, §§ 2–3]. Reconstruction requires the appropriate regularity and growth bounds, covariance and positivity; reflection positivity by itself is insufficient. Clustering supplies the additional unique-vacuum conclusion [OS75, § IV.1].

The reconstruction sections of this paper verify these requirements for the scalar correlations constructed here. Their substantive inputs include polynomial energy bounds, tempered boundary values, Euclidean covariance and clustering of the canonical components. The unique-vacuum conclusion applies to these components; it does not assert that an arbitrary invariant mixture has a unique vacuum or that the infinite-volume limit is unique.

Finally, under the additional small-reference-coupling hypothesis of theorem C, the nonzero two-point asymptotic proves nontriviality and recovers the physical inverse coupling. This is stronger

than the existence of a possibly trivial reconstructed space, but it does not prove non-Gaussianity or a mass gap. Nor does the scalar theorem construct a tensor-valued nonabelian field strength. These distinctions specify the scope of the present result independently of the general reconstruction theorem.

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